



Thesis Project by Group CLS

THESIS 1-3

Thesis Project

Group Members:

- Celles, Aaron Kent M.
- Leviste, Lee Ryan A.
- Santeco, Enrique S.

Section 1. Introduction

1.1 Problem Overview

Large-scale global optimization (LSGO) presents a critical challenge in computational intelligence, where the "curse of dimensionality" leads to an explosive increase in local optima and a high risk of premature convergence. While foundational algorithms like Differential Evolution (DE) are robust, their dependence on crossover and random mutation may restrict guided exploration — reducing particle diversity over time and leading to premature convergence.

This notebook implements and evaluates the proposed **Diversity-Preserving Level-Based Learning Differential Evolution (DP-LLDE)** algorithm, tested against the **CEC 2017 Large-Scale Global Optimization Benchmark** across 29 benchmark functions (F1-F30, excluding deprecated F2) at **10 dimensions**.

1.2 Research Objectives

1. Design and implement DP-LLDE, integrating sequential relative particle mutation and dynamic level-based learning into the Differential Evolution framework.
2. Evaluate DP-LLDE on the CEC 2017 benchmark suite (F1-F30) at 10 dimensions.
3. Examine convergence behavior, solution quality, and the effect of diversity-preserving strategies across unimodal, multimodal, hybrid,

and composition functions.

1.3 Algorithm Overview - DP-LLDE

DP-LLDE extends LBLDE (Qiao et al., 2022) with three core contributions:

1. **Sequential Level Assignment** — each individual is assigned a learning source level drawn randomly from $\{2, \dots, NL\}$ on the first generation, decaying toward the elite level (level 1) in subsequent generations.
2. **Diversity-Aware Exemplar Selection** — if two exemplars e_1, e_2 fall below a diversity threshold, one is replaced with the most distant candidate in that level, preventing exemplar collapse.
3. **Normalized Mutation Pressure (Eq. 10):**

$$V_i = e_1 + F_i \times (e_2 - X_i) \times \frac{NL - k}{NL}$$

The factor $(NL-k)/NL$ decays to zero as $k \rightarrow NL$, naturally reducing mutation pressure as learning quality improves.

Parameter	Value
Population Size (NP)	100
Number of Levels (NL)	4
Initial μ_{CR}	0.35
Max FES	$10,000 \times D$
Dimensions (D)	10
Seed	42

Section 2. Benchmark Function Description

2.1 Overview

The **CEC 2017 Large-Scale Global Optimization Benchmark** (Awad et al., 2017) is a standardized suite of 30 test functions designed to evaluate optimization algorithms on complex, high-dimensional search spaces. The benchmark tests an

algorithm's ability to avoid false local optima, convergence speed, and solution accuracy.

All functions are defined over the search space $[-100, 100]^D$ with their respective global optima offset by a function-specific bias value.

Library used: `cec2017` by Duncan Tilley — provides the official shift vectors, rotation matrices, and shuffle permutations from the CEC 2017 specification.

2.2 Unimodal Function Overview

Unimodal functions contain a **single global optimum and no local optima**. They are used to test the convergence speed and accuracy of an algorithm in a straightforward landscape.

No.	Function Name	Global Optimum
F1	Shifted and Rotated Bent Cigar	100
F2	Shifted and Rotated Sum of Different Power (<i>Deprecated</i>)	200
F3	Shifted and Rotated Zakharov	300

Note: F2 is officially deprecated from the CEC 2017 benchmark but is included here for completeness as it remains implemented in the `cec2017` library.

2.3 Multimodal Function Overview

Multimodal functions contain **multiple local optima**, making them significantly harder to solve. They are used to evaluate an algorithm's capacity to avoid premature convergence and escape false local optima.

No.	Function Name	Global Optimum
F4	Shifted and Rotated Rosenbrock's	400
F5	Shifted and Rotated Rastrigin's	500
F6	Shifted and Rotated Expanded Schaffer's F6	600
F7	Shifted and Rotated Lunacek Bi-Rastrigin	700
F8	Shifted and Rotated Non-Continuous Rastrigin's	800

No.	Function Name	Global Optimum
F9	Shifted and Rotated Levy	900
F10	Shifted and Rotated Schwefel's	1000

2.4 Hybrid Function Overview

Hybrid functions divide the search space into several regions, each governed by a **different basic function**. They assess an algorithm's adaptability to heterogeneous landscapes and mixed search behaviors.

No.	Component Functions	Global Optimum
F11	Zakharov; Rosenbrock; Rastrigin	1100
F12	Elliptic; Schwefel; Bent Cigar	1200
F13	Bent Cigar; Rosenbrock; Bi-Rastrigin	1300
F14	Elliptic; Ackley; Schaffer F7; Rastrigin	1400
F15	Bent Cigar; HGBat; Rastrigin; Rosenbrock	1500
F16	Schaffer F6; HGBat; Rosenbrock; Schwefel	1600
F17	Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin	1700
F18	Elliptic; Ackley; Rastrigin; HGBat; Discus	1800
F19	Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer F6	1900
F20	HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer F7	2000

2.5 Composite Function Overview

Composition functions **combine multiple shifted and rotated basic functions** into a single search space, creating highly complex and deceptive optimization landscapes. They are the most challenging category in the CEC 2017 suite and significantly test an algorithm's overall robustness and global search capability.

No.	Component Functions	Global Optimum
F21	Rosenbrock; Elliptic; Rastrigin	2100
F22	Rastrigin; Griewank; Schwefel	2200
F23	Rosenbrock; Ackley; Schwefel; Rastrigin	2300

No.	Component Functions	Global Optimum
F24	Ackley; Elliptic; Griewank; Rastrigin	2400
F25	Rastrigin; HappyCat; Ackley; Discus; Rosenbrock	2500
F26	Schaffer F6; Schwefel; Griewank; Rosenbrock; Rastrigin	2600
F27	HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer F6	2700
F28	Ackley; Griewank; Discus; Rosenbrock; HappyCat; Schaffer F6	2800
F29	F15; F16; F17	2900
F30	F15; F18; F19	3000

Section 3. List of Required Libraries

3.1 List of Libraries and Versions

The following libraries and modules were used for optimization, benchmarking, and visualization in this project:

Library / Module	Purpose	Version
<code>python</code>	Core programming language	3.10+
<code>numpy</code>	Numerical computing and array operations	2.0.2
<code>matplotlib</code>	Plotting and saving convergence graphs	3.10.0
<code>cec2017</code>	CEC 2017 benchmark functions (simple, hybrid, composition)	Latest (pip)
<code>cec2017.simple</code>	Simple benchmark functions F1-F10	—
<code>cec2017.hybrid</code>	Hybrid benchmark functions F11-F20	—
<code>cec2017.composition</code>	Composition benchmark functions F21-F30	—
<code>dplldc</code>	DP-LLDE optimization algorithm (uploaded locally)	Custom
<code>google.colab.files</code>	File upload utility for loading <code>dplldc.py</code> into the Colab environment	Built-in
<code>sys</code>	Clears cached module before re-importing <code>dplldc.py</code>	Built-in

Section 4. Environment Setup and Validation

4.1 Installation

Install `cec2017`

```
In [15]: # !pip install cec2017 --quiet
```

4.2 File Validation

Import `simple.py`, `hybrid.py`, and `composition.py` python files for benchmark functions

```
In [16]: from cec2017 import simple, hybrid, composition
print('cec2017 loaded!')
```

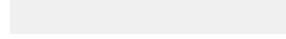
cec2017 loaded!

Import `sys` control over warning messages for verification that `lblde.py` exists

```
In [17]: from google.colab import files
import sys

uploaded = files.upload()

if 'lblde.py' in uploaded:
    if 'lblde' in sys.modules:
        del sys.modules['lblde']
    import lblde
    from lblde import LBLDE
    print('lblde.py uploaded and imported successfully!')
else:
    print('ERROR: Please upload a file named lblde.py')
```

 Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

ERROR: Please upload a file named lblde.py

```
In [18]: import lblde
from lblde import LBLDE
```

Section 5. Configuration and Helper Functions

5.1 Library Imports

Import libraries `numpy`, `matplotlib`, and `cec2017` for numerical computation, visualization, and CEC 2017 benchmark functions.

```
In [19]: import numpy as np
import matplotlib.pyplot as plt
from cec2017 import simple, hybrid, composition
```

5.2 Global Parameters

Define global parameters

```
In [20]: D = 100 # Change this when testing different dimensions
```

```
In [21]: bounds = np.array([[-100.0, 100.0]] * D)
MAX_FES = 10000 * D
NP, NL, MU_CR = 100, 4, 0.35
SEED = 42
```

Store all results for summary table under `all_results` dict

```
In [22]: all_results = {}
```

5.3 Function Definitions

Define `make_wrapper` to adapt CEC 2017 functions to accept a 1D input array from DP-LLDE, returning a large penalty value if the output is invalid (nan/inf).

```
In [23]: def make_wrapper(fn):
    """Wrap a cec2017 function (expects (1,D) input) to accept (D,) input."""
    def wrapper(x):
        val = fn(x.reshape(1, -1))
        if hasattr(val, '__len__'):
            val = float(val[0])
        else:
            val = float(val)
        return val if np.isfinite(val) else 1e18
    return wrapper
```

Define `run_and_plot` to execute a single DP-LLDE optimization run on a given benchmark function, print the results, store them for the summary table, and save the convergence graph.

```
In [24]: def run_and_plot(fn_name, fn_eval, f_global, fn_label):
    """Run LBLDE_s on a function and plot convergence + diversity graphs."""
    print('=' * 80)
    print(f'LBLDE_s on CEC 2017 {fn_name} – Single Run')
    print('=' * 80)
    print(f'Function      : {fn_label} ({D}D)')
    print(f'Bounds        : [-100, 100]')
    print(f'Global optimum : {f_global}')

    opt = LBLDE(
        objective_func=fn_eval,
        bounds=bounds,
        NP=NP, NL=NL, mu_CR_ini=MU_CR,
        max_fes=MAX_FES,
        seed=SEED
    )
    # ← Unpack 4 values now
    best_sol, best_fit, history, diversity_history = opt.optimize(verbose=True)

    error = abs(best_fit - f_global)
    print(f'\n{"="*80}')
    print('Results:')
    print(f'  Best Fitness      : {best_fit:.6e}')
    print(f'  Global Optimum    : {f_global:.6e}')
    print(f'  Error (|best-optimum|) : {error:.6e}')
    print(f'  Best Solution (5 dim) : {best_sol[:5]}')
    print(f'  D^(0) (init. diversity): {opt.D_0:.4e}')
    print(f'{"="*80}')

    # Store for summary
    all_results[fn_name] = {
        'label': fn_label,
        'best_fit': best_fit,
        'f_global': f_global,
        'error': error
    }

    fes_axis = np.linspace(0, MAX_FES, len(history))
    gen_axis = np.arange(len(diversity_history)) # Generations (0, 1, 2, ...)

    # — Two-panel figure —
    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 10))

    # Convergence curve (top) – FEs on x-axis
    ax1.plot(fes_axis, history, color='steelblue', linewidth=1.5)
    ax1.set_ylabel('Best Fitness', fontsize=12)
    ax1.set_xlabel('Function Evaluations (FEs)', fontsize=12)
    ax1.set_title(
        f'LBLDE – CEC 2017 {fn_name}: {fn_label} ({D}D)',
```



```

        fontsize=12
    )
    ax1.grid(True, which='both', linestyle='--', alpha=0.6)

    # Diversity curve (bottom) – Generations on x-axis
    ax2.plot(gen_axis, diversity_history, color='darkorange', linewidth=1.5)
    ax2.set_xlabel('Generation', fontsize=12)
    ax2.set_ylabel('Population Diversity\n(Mean Std Dev)', fontsize=12)
    ax2.set_title(f'Diversity Across Generations – {fn_name}', fontsize=12)
    ax2.grid(True, which='both', linestyle='--', alpha=0.6)

    plt.tight_layout()
    fname = f'convergence_{fn_name}.png'
    plt.savefig(fname, dpi=150)
    plt.show()
    print(f'Graph saved as {fname}\n')

```

Section 6. CEC2017 Benchmark Functions

6.1 Unimodal Functions (F1-F3)

```

In [25]: # =====
# F1 – Shifted and Rotated Bent Cigar
# =====
run_and_plot('F1', make_wrapper(simple.f1), 100.0, 'Shifted and Rotated Bent C

```

```

=====
=
LBLDE_s on CEC 2017 F1 – Single Run
=====
=
Function      : Shifted and Rotated Bent Cigar (100D)
Bounds       : [-100, 100]
Global optimum : 100.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 6.191709e+12 | Diversity = 5.7220e+01
Generation 10: Best = 2.892597e+12 FES = 1100 Diversity = 3.6510e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.692616e+12 FES = 2100 Diversity = 2.4022e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 1.071150e+12 FES = 3100 Diversity = 1.5729e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 8.060984e+11 FES = 4100 Diversity = 1.1213e+01 k_t
= [2, 2, 2, 4]
Generation 50: Best = 5.465411e+11 FES = 5100 Diversity = 8.9014e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 4.483450e+11 FES = 6100 Diversity = 7.1022e+00 k_t
= [2, 4, 3, 4]
Generation 70: Best = 3.379072e+11 FES = 7100 Diversity = 5.9128e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.588013e+11 FES = 8100 Diversity = 5.4426e+00 k_t
= [4, 3, 2, 3]
Generation 90: Best = 1.961992e+11 FES = 9100 Diversity = 4.8479e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.457357e+11 FES = 10100 Diversity = 4.2468e+00 k_t
= [3, 3, 2, 3]
Generation 110: Best = 1.135324e+11 FES = 11100 Diversity = 3.9444e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 8.426841e+10 FES = 12100 Diversity = 3.6921e+00 k_t
= [4, 3, 2, 2]
Generation 130: Best = 6.308995e+10 FES = 13100 Diversity = 3.0868e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 5.149139e+10 FES = 14100 Diversity = 2.7369e+00 k_t
= [2, 2, 3, 3]
Generation 150: Best = 3.464341e+10 FES = 15100 Diversity = 2.4770e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.010283e+10 FES = 16100 Diversity = 2.0969e+00 k_t
= [3, 4, 3, 4]
Generation 170: Best = 2.184821e+10 FES = 17100 Diversity = 1.8590e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.699486e+10 FES = 18100 Diversity = 1.6913e+00 k_t
= [3, 3, 2, 2]
Generation 190: Best = 1.343564e+10 FES = 19100 Diversity = 1.3934e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.029309e+10 FES = 20100 Diversity = 1.1737e+00 k_t
= [3, 4, 3, 4]
Generation 210: Best = 8.355496e+09 FES = 21100 Diversity = 1.0274e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 6.601925e+09 FES = 22100 Diversity = 8.7607e-01 k_t
= [3, 4, 2, 4]

```

Generation 230:	Best = 5.114467e+09	FES = 23100	Diversity = 7.5901e-01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 3.819610e+09	FES = 24100	Diversity = 6.7233e-01	k_t
= [4, 2, 4, 3]				
Generation 250:	Best = 3.085710e+09	FES = 25100	Diversity = 5.5738e-01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 2.653564e+09	FES = 26100	Diversity = 5.0537e-01	k_t
= [2, 2, 3, 4]				
Generation 270:	Best = 2.118113e+09	FES = 27100	Diversity = 4.6859e-01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 1.655357e+09	FES = 28100	Diversity = 4.4568e-01	k_t
= [4, 2, 3, 4]				
Generation 290:	Best = 1.395087e+09	FES = 29100	Diversity = 4.0857e-01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 1.130456e+09	FES = 30100	Diversity = 3.4605e-01	k_t
= [4, 2, 4, 4]				
Generation 310:	Best = 9.138063e+08	FES = 31100	Diversity = 3.0268e-01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 7.372223e+08	FES = 32100	Diversity = 2.7332e-01	k_t
= [3, 3, 2, 4]				
Generation 330:	Best = 5.926447e+08	FES = 33100	Diversity = 2.4703e-01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 4.743355e+08	FES = 34100	Diversity = 2.0666e-01	k_t
= [3, 2, 4, 4]				
Generation 350:	Best = 3.923254e+08	FES = 35100	Diversity = 1.8902e-01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 3.001292e+08	FES = 36100	Diversity = 1.8705e-01	k_t
= [4, 4, 4, 2]				
Generation 370:	Best = 2.449513e+08	FES = 37100	Diversity = 1.6292e-01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 1.914948e+08	FES = 38100	Diversity = 1.4298e-01	k_t
= [2, 2, 2, 2]				
Generation 390:	Best = 1.508393e+08	FES = 39100	Diversity = 1.3081e-01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 1.165381e+08	FES = 40100	Diversity = 1.2275e-01	k_t
= [4, 4, 4, 2]				
Generation 410:	Best = 8.505342e+07	FES = 41100	Diversity = 1.0457e-01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 6.761867e+07	FES = 42100	Diversity = 8.8634e-02	k_t
= [3, 4, 2, 2]				
Generation 430:	Best = 5.722606e+07	FES = 43100	Diversity = 8.2293e-02	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 4.455704e+07	FES = 44100	Diversity = 7.5334e-02	k_t
= [4, 2, 4, 4]				
Generation 450:	Best = 3.466837e+07	FES = 45100	Diversity = 6.2457e-02	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 2.920424e+07	FES = 46100	Diversity = 5.3898e-02	k_t
= [4, 4, 2, 3]				
Generation 470:	Best = 2.381051e+07	FES = 47100	Diversity = 4.4969e-02	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 2.071605e+07	FES = 48100	Diversity = 4.1433e-02	k_t
= [3, 4, 2, 3]				
Generation 490:	Best = 1.747851e+07	FES = 49100	Diversity = 3.9231e-02	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 1.489337e+07	FES = 50100	Diversity = 3.3150e-02	k_t = [4, 4, 4, 2]
Generation 510:	Best = 1.286600e+07	FES = 51100	Diversity = 2.9231e-02	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.091570e+07	FES = 52100	Diversity = 2.6383e-02	k_t = [3, 4, 2, 3]
Generation 530:	Best = 8.843048e+06	FES = 53100	Diversity = 2.5874e-02	k_t = [1, 1, 1, 1]
Generation 540:	Best = 7.322119e+06	FES = 54100	Diversity = 2.3146e-02	k_t = [4, 2, 3, 2]
Generation 550:	Best = 6.355813e+06	FES = 55100	Diversity = 2.1555e-02	k_t = [1, 1, 1, 1]
Generation 560:	Best = 5.153594e+06	FES = 56100	Diversity = 2.1341e-02	k_t = [4, 2, 2, 2]
Generation 570:	Best = 4.361259e+06	FES = 57100	Diversity = 1.8629e-02	k_t = [1, 1, 1, 1]
Generation 580:	Best = 3.669362e+06	FES = 58100	Diversity = 1.6795e-02	k_t = [3, 4, 4, 2]
Generation 590:	Best = 2.939885e+06	FES = 59100	Diversity = 1.5218e-02	k_t = [1, 1, 1, 1]
Generation 600:	Best = 2.380653e+06	FES = 60100	Diversity = 1.4953e-02	k_t = [4, 4, 4, 2]
Generation 610:	Best = 2.000905e+06	FES = 61100	Diversity = 1.3393e-02	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.609453e+06	FES = 62100	Diversity = 1.2475e-02	k_t = [3, 3, 4, 4]
Generation 630:	Best = 1.394787e+06	FES = 63100	Diversity = 1.1662e-02	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.152100e+06	FES = 64100	Diversity = 9.8891e-03	k_t = [3, 4, 3, 3]
Generation 650:	Best = 1.012413e+06	FES = 65100	Diversity = 8.5289e-03	k_t = [1, 1, 1, 1]
Generation 660:	Best = 8.449333e+05	FES = 66100	Diversity = 7.3957e-03	k_t = [2, 4, 4, 4]
Generation 670:	Best = 7.182967e+05	FES = 67100	Diversity = 6.8989e-03	k_t = [1, 1, 1, 1]
Generation 680:	Best = 6.002413e+05	FES = 68100	Diversity = 6.3418e-03	k_t = [3, 3, 2, 3]
Generation 690:	Best = 5.116567e+05	FES = 69100	Diversity = 5.7945e-03	k_t = [1, 1, 1, 1]
Generation 700:	Best = 4.492001e+05	FES = 70100	Diversity = 5.6252e-03	k_t = [2, 3, 2, 4]
Generation 710:	Best = 3.883013e+05	FES = 71100	Diversity = 5.4090e-03	k_t = [1, 1, 1, 1]
Generation 720:	Best = 3.332775e+05	FES = 72100	Diversity = 5.0126e-03	k_t = [4, 4, 2, 3]
Generation 730:	Best = 2.736768e+05	FES = 73100	Diversity = 4.5312e-03	k_t = [1, 1, 1, 1]
Generation 740:	Best = 2.487945e+05	FES = 74100	Diversity = 3.9067e-03	k_t = [3, 4, 3, 3]
Generation 750:	Best = 2.133330e+05	FES = 75100	Diversity = 3.5883e-03	k_t = [1, 1, 1, 1]
Generation 760:	Best = 1.814187e+05	FES = 76100	Diversity = 3.4234e-03	k_t = [2, 3, 4, 4]

Generation 770:	Best = 1.559940e+05	FES = 77100	Diversity = 3.4249e-03	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.257437e+05	FES = 78100	Diversity = 3.3579e-03	k_t
= [3, 3, 4, 3]				
Generation 790:	Best = 9.987217e+04	FES = 79100	Diversity = 3.3676e-03	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 8.372389e+04	FES = 80100	Diversity = 2.7137e-03	k_t
= [4, 2, 3, 4]				
Generation 810:	Best = 6.641346e+04	FES = 81100	Diversity = 2.4270e-03	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 5.180251e+04	FES = 82100	Diversity = 2.2556e-03	k_t
= [4, 4, 4, 2]				
Generation 830:	Best = 4.640915e+04	FES = 83100	Diversity = 1.8278e-03	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 3.965237e+04	FES = 84100	Diversity = 1.5221e-03	k_t
= [4, 3, 4, 3]				
Generation 850:	Best = 3.375635e+04	FES = 85100	Diversity = 1.3884e-03	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.902216e+04	FES = 86100	Diversity = 1.3442e-03	k_t
= [2, 4, 4, 2]				
Generation 870:	Best = 2.450896e+04	FES = 87100	Diversity = 1.2310e-03	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.175647e+04	FES = 88100	Diversity = 1.0344e-03	k_t
= [3, 2, 3, 3]				
Generation 890:	Best = 1.997352e+04	FES = 89100	Diversity = 9.7484e-04	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.729706e+04	FES = 90100	Diversity = 9.3617e-04	k_t
= [2, 4, 4, 2]				
Generation 910:	Best = 1.475889e+04	FES = 91100	Diversity = 9.3013e-04	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.340821e+04	FES = 92100	Diversity = 8.1795e-04	k_t
= [2, 3, 4, 2]				
Generation 930:	Best = 1.161234e+04	FES = 93100	Diversity = 7.0328e-04	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 9.939470e+03	FES = 94100	Diversity = 6.9658e-04	k_t
= [2, 4, 4, 2]				
Generation 950:	Best = 8.620500e+03	FES = 95100	Diversity = 6.7605e-04	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 7.856870e+03	FES = 96100	Diversity = 6.2892e-04	k_t
= [2, 2, 2, 2]				
Generation 970:	Best = 7.051583e+03	FES = 97100	Diversity = 5.4457e-04	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 6.337094e+03	FES = 98100	Diversity = 5.2139e-04	k_t
= [4, 3, 3, 3]				
Generation 990:	Best = 5.846827e+03	FES = 99100	Diversity = 4.8307e-04	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 5.483677e+03	FES = 100100	Diversity = 4.2575e-04	k_t
= [4, 4, 2, 2]				
Generation 1010:	Best = 5.120774e+03	FES = 101100	Diversity = 3.9946e-04	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 4.780975e+03	FES = 102100	Diversity = 3.8099e-04	k_t
= [2, 3, 3, 2]				
Generation 1030:	Best = 4.459113e+03	FES = 103100	Diversity = 3.4830e-04	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 4.220440e+03 = [4, 3, 4, 4]	FES = 104100	Diversity = 3.2658e-04	k_t
Generation 1050: Best = 4.047511e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 2.4840e-04	k_t
Generation 1060: Best = 3.928043e+03 = [2, 3, 4, 2]	FES = 106100	Diversity = 2.2888e-04	k_t
Generation 1070: Best = 3.804415e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 2.1072e-04	k_t
Generation 1080: Best = 3.708628e+03 = [3, 2, 4, 4]	FES = 108100	Diversity = 1.9910e-04	k_t
Generation 1090: Best = 3.642835e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.8024e-04	k_t
Generation 1100: Best = 3.576026e+03 = [3, 2, 2, 4]	FES = 110100	Diversity = 1.5509e-04	k_t
Generation 1110: Best = 3.528034e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.3088e-04	k_t
Generation 1120: Best = 3.486618e+03 = [4, 4, 4, 3]	FES = 112100	Diversity = 1.1296e-04	k_t
Generation 1130: Best = 3.454913e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.0654e-04	k_t
Generation 1140: Best = 3.420148e+03 = [3, 2, 4, 2]	FES = 114100	Diversity = 1.0560e-04	k_t
Generation 1150: Best = 3.385877e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.0355e-04	k_t
Generation 1160: Best = 3.357548e+03 = [4, 3, 3, 4]	FES = 116100	Diversity = 9.4445e-05	k_t
Generation 1170: Best = 3.340100e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 8.5892e-05	k_t
Generation 1180: Best = 3.324525e+03 = [2, 3, 4, 2]	FES = 118100	Diversity = 7.4178e-05	k_t
Generation 1190: Best = 3.309063e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 6.8308e-05	k_t
Generation 1200: Best = 3.300651e+03 = [4, 3, 3, 4]	FES = 120100	Diversity = 6.1118e-05	k_t
Generation 1210: Best = 3.292662e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 5.6429e-05	k_t
Generation 1220: Best = 3.284657e+03 = [4, 3, 2, 2]	FES = 122100	Diversity = 5.0935e-05	k_t
Generation 1230: Best = 3.280899e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 4.4016e-05	k_t
Generation 1240: Best = 3.277407e+03 = [3, 4, 4, 2]	FES = 124100	Diversity = 3.7104e-05	k_t
Generation 1250: Best = 3.274273e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 3.1875e-05	k_t
Generation 1260: Best = 3.272424e+03 = [2, 4, 3, 2]	FES = 126100	Diversity = 2.8247e-05	k_t
Generation 1270: Best = 3.270145e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 2.7181e-05	k_t
Generation 1280: Best = 3.268407e+03 = [3, 4, 2, 2]	FES = 128100	Diversity = 2.6778e-05	k_t
Generation 1290: Best = 3.266828e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 2.4401e-05	k_t
Generation 1300: Best = 3.265246e+03 = [2, 4, 3, 2]	FES = 130100	Diversity = 2.3798e-05	k_t

Generation 1310: Best = 3.263500e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 2.3935e-05	k_t
Generation 1320: Best = 3.261890e+03 = [3, 3, 3, 2]	FES = 132100	Diversity = 2.4117e-05	k_t
Generation 1330: Best = 3.260396e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 2.3029e-05	k_t
Generation 1340: Best = 3.259459e+03 = [2, 3, 2, 4]	FES = 134100	Diversity = 2.0006e-05	k_t
Generation 1350: Best = 3.258431e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.8315e-05	k_t
Generation 1360: Best = 3.257436e+03 = [3, 2, 4, 2]	FES = 136100	Diversity = 1.7909e-05	k_t
Generation 1370: Best = 3.256625e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 1.7582e-05	k_t
Generation 1380: Best = 3.255719e+03 = [3, 4, 2, 3]	FES = 138100	Diversity = 1.7480e-05	k_t
Generation 1390: Best = 3.254913e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 1.6830e-05	k_t
Generation 1400: Best = 3.254414e+03 = [2, 3, 3, 3]	FES = 140100	Diversity = 1.5336e-05	k_t
Generation 1410: Best = 3.254071e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 1.2827e-05	k_t
Generation 1420: Best = 3.253788e+03 = [3, 4, 2, 2]	FES = 142100	Diversity = 1.2060e-05	k_t
Generation 1430: Best = 3.253479e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 1.0971e-05	k_t
Generation 1440: Best = 3.253274e+03 = [2, 4, 4, 4]	FES = 144100	Diversity = 9.9091e-06	k_t
Generation 1450: Best = 3.252990e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 8.8306e-06	k_t
Generation 1460: Best = 3.252863e+03 = [3, 4, 4, 3]	FES = 146100	Diversity = 7.7853e-06	k_t
Generation 1470: Best = 3.252728e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 6.8611e-06	k_t
Generation 1480: Best = 3.252669e+03 = [4, 4, 3, 3]	FES = 148100	Diversity = 5.2569e-06	k_t
Generation 1490: Best = 3.252600e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 4.6407e-06	k_t
Generation 1500: Best = 3.252543e+03 = [3, 2, 4, 2]	FES = 150100	Diversity = 4.5543e-06	k_t
Generation 1510: Best = 3.252503e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 4.0984e-06	k_t
Generation 1520: Best = 3.252460e+03 = [3, 3, 2, 4]	FES = 152100	Diversity = 3.8326e-06	k_t
Generation 1530: Best = 3.252420e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 3.7275e-06	k_t
Generation 1540: Best = 3.252388e+03 = [3, 4, 2, 2]	FES = 154100	Diversity = 3.1133e-06	k_t
Generation 1550: Best = 3.252372e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 2.8210e-06	k_t
Generation 1560: Best = 3.252352e+03 = [2, 4, 2, 2]	FES = 156100	Diversity = 2.6304e-06	k_t
Generation 1570: Best = 3.252339e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 2.2120e-06	k_t

Generation 1580: Best = 3.252330e+03 = [4, 2, 2, 2]	FES = 158100	Diversity = 2.0962e-06	k_t
Generation 1590: Best = 3.252320e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 1.8514e-06	k_t
Generation 1600: Best = 3.252314e+03 = [2, 4, 3, 2]	FES = 160100	Diversity = 1.6994e-06	k_t
Generation 1610: Best = 3.252308e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 1.5268e-06	k_t
Generation 1620: Best = 3.252303e+03 = [4, 4, 4, 2]	FES = 162100	Diversity = 1.3680e-06	k_t
Generation 1630: Best = 3.252300e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.2186e-06	k_t
Generation 1640: Best = 3.252296e+03 = [4, 2, 4, 4]	FES = 164100	Diversity = 1.0978e-06	k_t
Generation 1650: Best = 3.252294e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.0230e-06	k_t
Generation 1660: Best = 3.252291e+03 = [2, 4, 4, 2]	FES = 166100	Diversity = 9.9475e-07	k_t
Generation 1670: Best = 3.252289e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 9.6908e-07	k_t
Generation 1680: Best = 3.252286e+03 = [2, 2, 2, 2]	FES = 168100	Diversity = 9.8690e-07	k_t
Generation 1690: Best = 3.252283e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 9.7757e-07	k_t
Generation 1700: Best = 3.252282e+03 = [2, 3, 4, 3]	FES = 170100	Diversity = 8.7597e-07	k_t
Generation 1710: Best = 3.252280e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 7.5289e-07	k_t
Generation 1720: Best = 3.252279e+03 = [4, 4, 2, 3]	FES = 172100	Diversity = 6.4642e-07	k_t
Generation 1730: Best = 3.252278e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 5.6137e-07	k_t
Generation 1740: Best = 3.252278e+03 = [4, 3, 3, 4]	FES = 174100	Diversity = 5.1481e-07	k_t
Generation 1750: Best = 3.252277e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 4.5769e-07	k_t
Generation 1760: Best = 3.252277e+03 = [2, 4, 4, 3]	FES = 176100	Diversity = 4.0728e-07	k_t
Generation 1770: Best = 3.252276e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 3.7995e-07	k_t
Generation 1780: Best = 3.252276e+03 = [4, 3, 2, 2]	FES = 178100	Diversity = 3.8547e-07	k_t
Generation 1790: Best = 3.252275e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 4.0812e-07	k_t
Generation 1800: Best = 3.252275e+03 = [3, 4, 3, 3]	FES = 180100	Diversity = 4.4307e-07	k_t
Generation 1810: Best = 3.252274e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 4.1917e-07	k_t
Generation 1820: Best = 3.252274e+03 = [4, 4, 2, 4]	FES = 182100	Diversity = 3.6643e-07	k_t
Generation 1830: Best = 3.252274e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 3.1005e-07	k_t
Generation 1840: Best = 3.252274e+03 = [3, 3, 2, 4]	FES = 184100	Diversity = 2.9320e-07	k_t

Generation 1850: Best = 3.252273e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 2.7008e-07	k_t
Generation 1860: Best = 3.252273e+03 = [3, 4, 4, 3]	FES = 186100	Diversity = 2.5700e-07	k_t
Generation 1870: Best = 3.252273e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 2.4900e-07	k_t
Generation 1880: Best = 3.252273e+03 = [3, 3, 3, 4]	FES = 188100	Diversity = 2.2447e-07	k_t
Generation 1890: Best = 3.252273e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 2.0649e-07	k_t
Generation 1900: Best = 3.252273e+03 = [2, 3, 2, 4]	FES = 190100	Diversity = 1.8732e-07	k_t
Generation 1910: Best = 3.252272e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 1.7672e-07	k_t
Generation 1920: Best = 3.252272e+03 = [3, 4, 2, 3]	FES = 192100	Diversity = 1.8738e-07	k_t
Generation 1930: Best = 3.252272e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 2.0571e-07	k_t
Generation 1940: Best = 3.252272e+03 = [3, 3, 3, 2]	FES = 194100	Diversity = 2.3337e-07	k_t
Generation 1950: Best = 3.252272e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 2.4967e-07	k_t
Generation 1960: Best = 3.252272e+03 = [2, 4, 3, 2]	FES = 196100	Diversity = 2.2944e-07	k_t
Generation 1970: Best = 3.252272e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 2.0496e-07	k_t
Generation 1980: Best = 3.252272e+03 = [4, 3, 3, 4]	FES = 198100	Diversity = 1.8574e-07	k_t
Generation 1990: Best = 3.252271e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.5681e-07	k_t
Generation 2000: Best = 3.252271e+03 = [3, 3, 2, 3]	FES = 200100	Diversity = 1.5078e-07	k_t
Generation 2010: Best = 3.252271e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 1.6232e-07	k_t
Generation 2020: Best = 3.252271e+03 = [4, 3, 2, 3]	FES = 202100	Diversity = 1.6561e-07	k_t
Generation 2030: Best = 3.252271e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 1.5571e-07	k_t
Generation 2040: Best = 3.252271e+03 = [4, 4, 3, 2]	FES = 204100	Diversity = 1.3957e-07	k_t
Generation 2050: Best = 3.252271e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.2979e-07	k_t
Generation 2060: Best = 3.252271e+03 = [3, 4, 2, 3]	FES = 206100	Diversity = 1.4745e-07	k_t
Generation 2070: Best = 3.252271e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.4937e-07	k_t
Generation 2080: Best = 3.252271e+03 = [2, 4, 4, 4]	FES = 208100	Diversity = 1.7543e-07	k_t
Generation 2090: Best = 3.252270e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 2.0293e-07	k_t
Generation 2100: Best = 3.252270e+03 = [3, 4, 4, 4]	FES = 210100	Diversity = 2.0834e-07	k_t
Generation 2110: Best = 3.252270e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 2.1162e-07	k_t

Generation 2120: Best = 3.252270e+03 = [2, 2, 4, 3]	FES = 212100	Diversity = 2.4505e-07	k_t
Generation 2130: Best = 3.252270e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 2.6510e-07	k_t
Generation 2140: Best = 3.252269e+03 = [4, 2, 3, 3]	FES = 214100	Diversity = 2.7566e-07	k_t
Generation 2150: Best = 3.252269e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 3.0047e-07	k_t
Generation 2160: Best = 3.252269e+03 = [2, 2, 4, 4]	FES = 216100	Diversity = 3.1333e-07	k_t
Generation 2170: Best = 3.252269e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 3.0454e-07	k_t
Generation 2180: Best = 3.252268e+03 = [3, 3, 4, 4]	FES = 218100	Diversity = 3.3304e-07	k_t
Generation 2190: Best = 3.252268e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 3.0362e-07	k_t
Generation 2200: Best = 3.252268e+03 = [4, 2, 2, 4]	FES = 220100	Diversity = 2.5590e-07	k_t
Generation 2210: Best = 3.252268e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 2.3766e-07	k_t
Generation 2220: Best = 3.252267e+03 = [4, 3, 3, 2]	FES = 222100	Diversity = 2.2234e-07	k_t
Generation 2230: Best = 3.252267e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 2.1034e-07	k_t
Generation 2240: Best = 3.252267e+03 = [4, 4, 2, 3]	FES = 224100	Diversity = 2.2925e-07	k_t
Generation 2250: Best = 3.252267e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 2.2539e-07	k_t
Generation 2260: Best = 3.252267e+03 = [3, 3, 2, 3]	FES = 226100	Diversity = 2.0604e-07	k_t
Generation 2270: Best = 3.252267e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.9927e-07	k_t
Generation 2280: Best = 3.252267e+03 = [4, 3, 3, 3]	FES = 228100	Diversity = 1.7659e-07	k_t
Generation 2290: Best = 3.252267e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.8748e-07	k_t
Generation 2300: Best = 3.252266e+03 = [3, 3, 4, 3]	FES = 230100	Diversity = 1.6427e-07	k_t
Generation 2310: Best = 3.252266e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.5805e-07	k_t
Generation 2320: Best = 3.252266e+03 = [2, 3, 3, 2]	FES = 232100	Diversity = 1.4296e-07	k_t
Generation 2330: Best = 3.252266e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 1.5853e-07	k_t
Generation 2340: Best = 3.252266e+03 = [3, 2, 4, 3]	FES = 234100	Diversity = 1.6677e-07	k_t
Generation 2350: Best = 3.252266e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 1.7680e-07	k_t
Generation 2360: Best = 3.252266e+03 = [2, 4, 2, 2]	FES = 236100	Diversity = 1.7769e-07	k_t
Generation 2370: Best = 3.252266e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 1.4283e-07	k_t
Generation 2380: Best = 3.252266e+03 = [2, 2, 2, 2]	FES = 238100	Diversity = 1.4158e-07	k_t

Generation 2390: Best = 3.252266e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 1.5968e-07	k_t
Generation 2400: Best = 3.252266e+03 = [4, 4, 2, 4]	FES = 240100	Diversity = 1.5860e-07	k_t
Generation 2410: Best = 3.252266e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 1.5279e-07	k_t
Generation 2420: Best = 3.252266e+03 = [4, 4, 4, 4]	FES = 242100	Diversity = 1.2953e-07	k_t
Generation 2430: Best = 3.252265e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 1.4389e-07	k_t
Generation 2440: Best = 3.252265e+03 = [2, 3, 2, 2]	FES = 244100	Diversity = 1.4358e-07	k_t
Generation 2450: Best = 3.252265e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 1.2772e-07	k_t
Generation 2460: Best = 3.252265e+03 = [2, 3, 2, 3]	FES = 246100	Diversity = 1.5242e-07	k_t
Generation 2470: Best = 3.252265e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 1.8056e-07	k_t
Generation 2480: Best = 3.252265e+03 = [3, 4, 2, 2]	FES = 248100	Diversity = 1.9556e-07	k_t
Generation 2490: Best = 3.252265e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 2.0601e-07	k_t
Generation 2500: Best = 3.252264e+03 = [4, 3, 4, 4]	FES = 250100	Diversity = 2.7016e-07	k_t
Generation 2510: Best = 3.252264e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 2.3553e-07	k_t
Generation 2520: Best = 3.252264e+03 = [4, 3, 3, 3]	FES = 252100	Diversity = 1.9692e-07	k_t
Generation 2530: Best = 3.252264e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.8680e-07	k_t
Generation 2540: Best = 3.252264e+03 = [4, 2, 2, 3]	FES = 254100	Diversity = 1.7920e-07	k_t
Generation 2550: Best = 3.252264e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.5944e-07	k_t
Generation 2560: Best = 3.252264e+03 = [2, 2, 2, 4]	FES = 256100	Diversity = 1.4233e-07	k_t
Generation 2570: Best = 3.252264e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.2394e-07	k_t
Generation 2580: Best = 3.252264e+03 = [4, 3, 2, 4]	FES = 258100	Diversity = 1.1179e-07	k_t
Generation 2590: Best = 3.252264e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.2118e-07	k_t
Generation 2600: Best = 3.252264e+03 = [4, 3, 4, 2]	FES = 260100	Diversity = 1.2421e-07	k_t
Generation 2610: Best = 3.252264e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.3115e-07	k_t
Generation 2620: Best = 3.252264e+03 = [2, 4, 2, 2]	FES = 262100	Diversity = 1.2434e-07	k_t
Generation 2630: Best = 3.252263e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.1842e-07	k_t
Generation 2640: Best = 3.252263e+03 = [2, 4, 3, 2]	FES = 264100	Diversity = 1.2739e-07	k_t
Generation 2650: Best = 3.252263e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 1.4794e-07	k_t

Generation 2660: Best = 3.252263e+03 = [3, 3, 4, 2]	FES = 266100	Diversity = 1.6411e-07	k_t
Generation 2670: Best = 3.252263e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 1.7041e-07	k_t
Generation 2680: Best = 3.252263e+03 = [3, 4, 4, 3]	FES = 268100	Diversity = 1.8285e-07	k_t
Generation 2690: Best = 3.252263e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.9453e-07	k_t
Generation 2700: Best = 3.252263e+03 = [2, 2, 3, 3]	FES = 270100	Diversity = 2.4120e-07	k_t
Generation 2710: Best = 3.252262e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 2.7724e-07	k_t
Generation 2720: Best = 3.252262e+03 = [4, 2, 3, 3]	FES = 272100	Diversity = 2.5784e-07	k_t
Generation 2730: Best = 3.252262e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 2.5999e-07	k_t
Generation 2740: Best = 3.252262e+03 = [2, 3, 3, 3]	FES = 274100	Diversity = 2.4002e-07	k_t
Generation 2750: Best = 3.252262e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 2.3262e-07	k_t
Generation 2760: Best = 3.252261e+03 = [4, 3, 2, 2]	FES = 276100	Diversity = 2.0810e-07	k_t
Generation 2770: Best = 3.252261e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.8763e-07	k_t
Generation 2780: Best = 3.252261e+03 = [3, 4, 2, 4]	FES = 278100	Diversity = 1.8590e-07	k_t
Generation 2790: Best = 3.252261e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.9191e-07	k_t
Generation 2800: Best = 3.252261e+03 = [3, 4, 4, 3]	FES = 280100	Diversity = 1.9773e-07	k_t
Generation 2810: Best = 3.252261e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 2.0979e-07	k_t
Generation 2820: Best = 3.252261e+03 = [3, 3, 2, 4]	FES = 282100	Diversity = 2.4245e-07	k_t
Generation 2830: Best = 3.252260e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 2.4235e-07	k_t
Generation 2840: Best = 3.252260e+03 = [4, 2, 4, 3]	FES = 284100	Diversity = 2.2936e-07	k_t
Generation 2850: Best = 3.252260e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 2.6639e-07	k_t
Generation 2860: Best = 3.252260e+03 = [3, 3, 3, 3]	FES = 286100	Diversity = 2.8167e-07	k_t
Generation 2870: Best = 3.252259e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 2.9215e-07	k_t
Generation 2880: Best = 3.252259e+03 = [4, 4, 4, 3]	FES = 288100	Diversity = 2.9669e-07	k_t
Generation 2890: Best = 3.252259e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 3.1860e-07	k_t
Generation 2900: Best = 3.252258e+03 = [2, 4, 3, 4]	FES = 290100	Diversity = 3.0651e-07	k_t
Generation 2910: Best = 3.252258e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 3.0240e-07	k_t
Generation 2920: Best = 3.252258e+03 = [2, 3, 2, 2]	FES = 292100	Diversity = 2.7198e-07	k_t

Generation 2930: Best = 3.252257e+03 = [1, 1, 1, 1]	FES = 293100	Diversity = 3.4039e-07	k_t
Generation 2940: Best = 3.252257e+03 = [2, 3, 2, 2]	FES = 294100	Diversity = 3.4809e-07	k_t
Generation 2950: Best = 3.252257e+03 = [1, 1, 1, 1]	FES = 295100	Diversity = 3.1405e-07	k_t
Generation 2960: Best = 3.252257e+03 = [3, 3, 2, 3]	FES = 296100	Diversity = 2.8149e-07	k_t
Generation 2970: Best = 3.252256e+03 = [1, 1, 1, 1]	FES = 297100	Diversity = 2.8218e-07	k_t
Generation 2980: Best = 3.252256e+03 = [2, 2, 4, 4]	FES = 298100	Diversity = 2.5400e-07	k_t
Generation 2990: Best = 3.252256e+03 = [1, 1, 1, 1]	FES = 299100	Diversity = 2.5217e-07	k_t
Generation 3000: Best = 3.252256e+03 = [2, 3, 2, 2]	FES = 300100	Diversity = 2.3340e-07	k_t
Generation 3010: Best = 3.252256e+03 = [1, 1, 1, 1]	FES = 301100	Diversity = 2.2165e-07	k_t
Generation 3020: Best = 3.252256e+03 = [3, 4, 3, 3]	FES = 302100	Diversity = 2.2555e-07	k_t
Generation 3030: Best = 3.252255e+03 = [1, 1, 1, 1]	FES = 303100	Diversity = 2.3452e-07	k_t
Generation 3040: Best = 3.252255e+03 = [3, 3, 4, 4]	FES = 304100	Diversity = 2.3632e-07	k_t
Generation 3050: Best = 3.252255e+03 = [1, 1, 1, 1]	FES = 305100	Diversity = 2.3248e-07	k_t
Generation 3060: Best = 3.252255e+03 = [3, 4, 2, 4]	FES = 306100	Diversity = 2.3679e-07	k_t
Generation 3070: Best = 3.252255e+03 = [1, 1, 1, 1]	FES = 307100	Diversity = 2.3201e-07	k_t
Generation 3080: Best = 3.252254e+03 = [4, 4, 2, 2]	FES = 308100	Diversity = 2.4347e-07	k_t
Generation 3090: Best = 3.252254e+03 = [1, 1, 1, 1]	FES = 309100	Diversity = 2.3433e-07	k_t
Generation 3100: Best = 3.252254e+03 = [4, 4, 2, 4]	FES = 310100	Diversity = 2.1698e-07	k_t
Generation 3110: Best = 3.252254e+03 = [1, 1, 1, 1]	FES = 311100	Diversity = 2.0012e-07	k_t
Generation 3120: Best = 3.252254e+03 = [3, 2, 2, 2]	FES = 312100	Diversity = 1.9130e-07	k_t
Generation 3130: Best = 3.252254e+03 = [1, 1, 1, 1]	FES = 313100	Diversity = 1.9397e-07	k_t
Generation 3140: Best = 3.252254e+03 = [2, 4, 4, 2]	FES = 314100	Diversity = 1.9173e-07	k_t
Generation 3150: Best = 3.252254e+03 = [1, 1, 1, 1]	FES = 315100	Diversity = 2.0079e-07	k_t
Generation 3160: Best = 3.252253e+03 = [2, 2, 4, 2]	FES = 316100	Diversity = 2.2084e-07	k_t
Generation 3170: Best = 3.252253e+03 = [1, 1, 1, 1]	FES = 317100	Diversity = 2.1250e-07	k_t
Generation 3180: Best = 3.252253e+03 = [4, 3, 2, 3]	FES = 318100	Diversity = 2.3634e-07	k_t
Generation 3190: Best = 3.252253e+03 = [1, 1, 1, 1]	FES = 319100	Diversity = 2.4619e-07	k_t

Generation 3200: Best = 3.252253e+03 = [4, 3, 4, 3]	FES = 320100	Diversity = 2.2454e-07	k_t
Generation 3210: Best = 3.252252e+03 = [1, 1, 1, 1]	FES = 321100	Diversity = 2.3186e-07	k_t
Generation 3220: Best = 3.252252e+03 = [3, 4, 4, 4]	FES = 322100	Diversity = 2.2344e-07	k_t
Generation 3230: Best = 3.252252e+03 = [1, 1, 1, 1]	FES = 323100	Diversity = 2.1239e-07	k_t
Generation 3240: Best = 3.252252e+03 = [2, 2, 4, 2]	FES = 324100	Diversity = 1.9473e-07	k_t
Generation 3250: Best = 3.252252e+03 = [1, 1, 1, 1]	FES = 325100	Diversity = 1.8211e-07	k_t
Generation 3260: Best = 3.252252e+03 = [4, 4, 2, 4]	FES = 326100	Diversity = 1.7267e-07	k_t
Generation 3270: Best = 3.252252e+03 = [1, 1, 1, 1]	FES = 327100	Diversity = 1.8049e-07	k_t
Generation 3280: Best = 3.252252e+03 = [2, 3, 4, 2]	FES = 328100	Diversity = 1.5803e-07	k_t
Generation 3290: Best = 3.252252e+03 = [1, 1, 1, 1]	FES = 329100	Diversity = 1.5369e-07	k_t
Generation 3300: Best = 3.252252e+03 = [3, 4, 2, 2]	FES = 330100	Diversity = 1.5415e-07	k_t
Generation 3310: Best = 3.252251e+03 = [1, 1, 1, 1]	FES = 331100	Diversity = 1.5811e-07	k_t
Generation 3320: Best = 3.252251e+03 = [4, 4, 4, 4]	FES = 332100	Diversity = 1.4633e-07	k_t
Generation 3330: Best = 3.252251e+03 = [1, 1, 1, 1]	FES = 333100	Diversity = 1.5169e-07	k_t
Generation 3340: Best = 3.252251e+03 = [2, 3, 4, 4]	FES = 334100	Diversity = 1.6276e-07	k_t
Generation 3350: Best = 3.252251e+03 = [1, 1, 1, 1]	FES = 335100	Diversity = 1.8864e-07	k_t
Generation 3360: Best = 3.252251e+03 = [2, 2, 3, 3]	FES = 336100	Diversity = 2.2812e-07	k_t
Generation 3370: Best = 3.252251e+03 = [1, 1, 1, 1]	FES = 337100	Diversity = 2.5925e-07	k_t
Generation 3380: Best = 3.252251e+03 = [3, 4, 2, 4]	FES = 338100	Diversity = 2.3748e-07	k_t
Generation 3390: Best = 3.252250e+03 = [1, 1, 1, 1]	FES = 339100	Diversity = 2.3756e-07	k_t
Generation 3400: Best = 3.252250e+03 = [3, 4, 2, 2]	FES = 340100	Diversity = 2.3162e-07	k_t
Generation 3410: Best = 3.252250e+03 = [1, 1, 1, 1]	FES = 341100	Diversity = 2.3079e-07	k_t
Generation 3420: Best = 3.252250e+03 = [3, 4, 2, 4]	FES = 342100	Diversity = 2.1890e-07	k_t
Generation 3430: Best = 3.252250e+03 = [1, 1, 1, 1]	FES = 343100	Diversity = 2.0094e-07	k_t
Generation 3440: Best = 3.252250e+03 = [2, 2, 4, 4]	FES = 344100	Diversity = 2.0896e-07	k_t
Generation 3450: Best = 3.252249e+03 = [1, 1, 1, 1]	FES = 345100	Diversity = 2.0829e-07	k_t
Generation 3460: Best = 3.252249e+03 = [2, 4, 2, 2]	FES = 346100	Diversity = 2.2501e-07	k_t

Generation 3470: Best = 3.252249e+03 = [1, 1, 1, 1]	FES = 347100	Diversity = 2.2328e-07	k_t
Generation 3480: Best = 3.252249e+03 = [2, 2, 3, 4]	FES = 348100	Diversity = 2.1801e-07	k_t
Generation 3490: Best = 3.252249e+03 = [1, 1, 1, 1]	FES = 349100	Diversity = 2.3951e-07	k_t
Generation 3500: Best = 3.252249e+03 = [3, 2, 3, 2]	FES = 350100	Diversity = 2.3811e-07	k_t
Generation 3510: Best = 3.252248e+03 = [1, 1, 1, 1]	FES = 351100	Diversity = 2.4790e-07	k_t
Generation 3520: Best = 3.252248e+03 = [2, 3, 2, 3]	FES = 352100	Diversity = 2.4661e-07	k_t
Generation 3530: Best = 3.252248e+03 = [1, 1, 1, 1]	FES = 353100	Diversity = 2.2681e-07	k_t
Generation 3540: Best = 3.252248e+03 = [4, 4, 2, 4]	FES = 354100	Diversity = 2.2872e-07	k_t
Generation 3550: Best = 3.252248e+03 = [1, 1, 1, 1]	FES = 355100	Diversity = 2.3392e-07	k_t
Generation 3560: Best = 3.252247e+03 = [2, 3, 3, 3]	FES = 356100	Diversity = 2.4658e-07	k_t
Generation 3570: Best = 3.252247e+03 = [1, 1, 1, 1]	FES = 357100	Diversity = 2.4776e-07	k_t
Generation 3580: Best = 3.252247e+03 = [3, 2, 4, 3]	FES = 358100	Diversity = 2.7538e-07	k_t
Generation 3590: Best = 3.252247e+03 = [1, 1, 1, 1]	FES = 359100	Diversity = 2.5908e-07	k_t
Generation 3600: Best = 3.252246e+03 = [3, 2, 2, 2]	FES = 360100	Diversity = 2.3995e-07	k_t
Generation 3610: Best = 3.252246e+03 = [1, 1, 1, 1]	FES = 361100	Diversity = 2.1181e-07	k_t
Generation 3620: Best = 3.252246e+03 = [4, 4, 2, 2]	FES = 362100	Diversity = 2.2141e-07	k_t
Generation 3630: Best = 3.252246e+03 = [1, 1, 1, 1]	FES = 363100	Diversity = 2.2280e-07	k_t
Generation 3640: Best = 3.252246e+03 = [4, 4, 3, 2]	FES = 364100	Diversity = 2.5853e-07	k_t
Generation 3650: Best = 3.252245e+03 = [1, 1, 1, 1]	FES = 365100	Diversity = 2.4776e-07	k_t
Generation 3660: Best = 3.252245e+03 = [3, 4, 3, 3]	FES = 366100	Diversity = 2.3931e-07	k_t
Generation 3670: Best = 3.252245e+03 = [1, 1, 1, 1]	FES = 367100	Diversity = 2.3403e-07	k_t
Generation 3680: Best = 3.252245e+03 = [4, 3, 3, 3]	FES = 368100	Diversity = 2.0075e-07	k_t
Generation 3690: Best = 3.252245e+03 = [1, 1, 1, 1]	FES = 369100	Diversity = 1.8643e-07	k_t
Generation 3700: Best = 3.252245e+03 = [4, 3, 4, 4]	FES = 370100	Diversity = 1.9306e-07	k_t
Generation 3710: Best = 3.252245e+03 = [1, 1, 1, 1]	FES = 371100	Diversity = 2.3339e-07	k_t
Generation 3720: Best = 3.252244e+03 = [3, 4, 4, 4]	FES = 372100	Diversity = 2.4081e-07	k_t
Generation 3730: Best = 3.252244e+03 = [1, 1, 1, 1]	FES = 373100	Diversity = 2.4705e-07	k_t

Generation 3740: Best = 3.252244e+03 = [3, 3, 3, 3]	FES = 374100	Diversity = 2.3708e-07	k_t
Generation 3750: Best = 3.252244e+03 = [1, 1, 1, 1]	FES = 375100	Diversity = 2.4816e-07	k_t
Generation 3760: Best = 3.252243e+03 = [4, 3, 4, 4]	FES = 376100	Diversity = 2.3968e-07	k_t
Generation 3770: Best = 3.252243e+03 = [1, 1, 1, 1]	FES = 377100	Diversity = 2.5208e-07	k_t
Generation 3780: Best = 3.252243e+03 = [4, 3, 4, 2]	FES = 378100	Diversity = 3.1682e-07	k_t
Generation 3790: Best = 3.252242e+03 = [1, 1, 1, 1]	FES = 379100	Diversity = 2.9082e-07	k_t
Generation 3800: Best = 3.252242e+03 = [2, 2, 4, 4]	FES = 380100	Diversity = 2.7039e-07	k_t
Generation 3810: Best = 3.252242e+03 = [1, 1, 1, 1]	FES = 381100	Diversity = 2.5033e-07	k_t
Generation 3820: Best = 3.252242e+03 = [2, 2, 2, 2]	FES = 382100	Diversity = 2.2811e-07	k_t
Generation 3830: Best = 3.252242e+03 = [1, 1, 1, 1]	FES = 383100	Diversity = 2.1292e-07	k_t
Generation 3840: Best = 3.252242e+03 = [3, 4, 3, 2]	FES = 384100	Diversity = 1.7548e-07	k_t
Generation 3850: Best = 3.252242e+03 = [1, 1, 1, 1]	FES = 385100	Diversity = 1.5341e-07	k_t
Generation 3860: Best = 3.252242e+03 = [4, 2, 4, 2]	FES = 386100	Diversity = 1.4914e-07	k_t
Generation 3870: Best = 3.252241e+03 = [1, 1, 1, 1]	FES = 387100	Diversity = 1.5824e-07	k_t
Generation 3880: Best = 3.252241e+03 = [2, 3, 2, 3]	FES = 388100	Diversity = 1.8034e-07	k_t
Generation 3890: Best = 3.252241e+03 = [1, 1, 1, 1]	FES = 389100	Diversity = 2.0336e-07	k_t
Generation 3900: Best = 3.252241e+03 = [2, 3, 3, 3]	FES = 390100	Diversity = 2.0376e-07	k_t
Generation 3910: Best = 3.252241e+03 = [1, 1, 1, 1]	FES = 391100	Diversity = 1.9295e-07	k_t
Generation 3920: Best = 3.252241e+03 = [2, 2, 3, 3]	FES = 392100	Diversity = 2.2258e-07	k_t
Generation 3930: Best = 3.252240e+03 = [1, 1, 1, 1]	FES = 393100	Diversity = 2.4459e-07	k_t
Generation 3940: Best = 3.252240e+03 = [4, 2, 2, 4]	FES = 394100	Diversity = 2.0976e-07	k_t
Generation 3950: Best = 3.252240e+03 = [1, 1, 1, 1]	FES = 395100	Diversity = 1.9403e-07	k_t
Generation 3960: Best = 3.252240e+03 = [4, 2, 4, 3]	FES = 396100	Diversity = 2.1477e-07	k_t
Generation 3970: Best = 3.252240e+03 = [1, 1, 1, 1]	FES = 397100	Diversity = 2.3465e-07	k_t
Generation 3980: Best = 3.252240e+03 = [4, 4, 4, 4]	FES = 398100	Diversity = 2.1843e-07	k_t
Generation 3990: Best = 3.252239e+03 = [1, 1, 1, 1]	FES = 399100	Diversity = 2.5770e-07	k_t
Generation 4000: Best = 3.252239e+03 = [4, 2, 3, 3]	FES = 400100	Diversity = 2.8997e-07	k_t

Generation 4010: Best = 3.252238e+03 = [1, 1, 1, 1]	FES = 401100	Diversity = 3.5851e-07	k_t
Generation 4020: Best = 3.252238e+03 = [4, 4, 4, 3]	FES = 402100	Diversity = 3.4440e-07	k_t
Generation 4030: Best = 3.252238e+03 = [1, 1, 1, 1]	FES = 403100	Diversity = 3.0778e-07	k_t
Generation 4040: Best = 3.252237e+03 = [4, 4, 2, 2]	FES = 404100	Diversity = 2.7303e-07	k_t
Generation 4050: Best = 3.252237e+03 = [1, 1, 1, 1]	FES = 405100	Diversity = 2.4930e-07	k_t
Generation 4060: Best = 3.252237e+03 = [3, 2, 2, 3]	FES = 406100	Diversity = 2.2356e-07	k_t
Generation 4070: Best = 3.252237e+03 = [1, 1, 1, 1]	FES = 407100	Diversity = 1.8346e-07	k_t
Generation 4080: Best = 3.252237e+03 = [3, 3, 3, 2]	FES = 408100	Diversity = 1.4784e-07	k_t
Generation 4090: Best = 3.252237e+03 = [1, 1, 1, 1]	FES = 409100	Diversity = 1.3353e-07	k_t
Generation 4100: Best = 3.252237e+03 = [4, 2, 2, 4]	FES = 410100	Diversity = 1.1553e-07	k_t
Generation 4110: Best = 3.252237e+03 = [1, 1, 1, 1]	FES = 411100	Diversity = 1.0466e-07	k_t
Generation 4120: Best = 3.252237e+03 = [4, 2, 4, 4]	FES = 412100	Diversity = 1.0742e-07	k_t
Generation 4130: Best = 3.252237e+03 = [1, 1, 1, 1]	FES = 413100	Diversity = 1.7177e-07	k_t
Generation 4140: Best = 3.252236e+03 = [4, 4, 2, 4]	FES = 414100	Diversity = 2.0002e-07	k_t
Generation 4150: Best = 3.252236e+03 = [1, 1, 1, 1]	FES = 415100	Diversity = 1.8767e-07	k_t
Generation 4160: Best = 3.252236e+03 = [2, 4, 4, 2]	FES = 416100	Diversity = 1.6383e-07	k_t
Generation 4170: Best = 3.252236e+03 = [1, 1, 1, 1]	FES = 417100	Diversity = 1.5249e-07	k_t
Generation 4180: Best = 3.252236e+03 = [4, 2, 4, 3]	FES = 418100	Diversity = 1.6865e-07	k_t
Generation 4190: Best = 3.252236e+03 = [1, 1, 1, 1]	FES = 419100	Diversity = 1.5863e-07	k_t
Generation 4200: Best = 3.252236e+03 = [2, 2, 2, 2]	FES = 420100	Diversity = 1.4605e-07	k_t
Generation 4210: Best = 3.252236e+03 = [1, 1, 1, 1]	FES = 421100	Diversity = 1.3993e-07	k_t
Generation 4220: Best = 3.252236e+03 = [2, 3, 4, 3]	FES = 422100	Diversity = 1.5718e-07	k_t
Generation 4230: Best = 3.252235e+03 = [1, 1, 1, 1]	FES = 423100	Diversity = 1.6238e-07	k_t
Generation 4240: Best = 3.252235e+03 = [2, 2, 2, 2]	FES = 424100	Diversity = 1.7628e-07	k_t
Generation 4250: Best = 3.252235e+03 = [1, 1, 1, 1]	FES = 425100	Diversity = 1.8512e-07	k_t
Generation 4260: Best = 3.252235e+03 = [4, 4, 4, 4]	FES = 426100	Diversity = 2.1638e-07	k_t
Generation 4270: Best = 3.252235e+03 = [1, 1, 1, 1]	FES = 427100	Diversity = 2.1752e-07	k_t

Generation 4280: Best = 3.252235e+03 = [2, 4, 4, 2]	FES = 428100	Diversity = 2.3145e-07	k_t
Generation 4290: Best = 3.252234e+03 = [1, 1, 1, 1]	FES = 429100	Diversity = 2.1225e-07	k_t
Generation 4300: Best = 3.252234e+03 = [3, 4, 4, 4]	FES = 430100	Diversity = 2.0343e-07	k_t
Generation 4310: Best = 3.252234e+03 = [1, 1, 1, 1]	FES = 431100	Diversity = 1.7760e-07	k_t
Generation 4320: Best = 3.252234e+03 = [4, 3, 2, 4]	FES = 432100	Diversity = 1.7516e-07	k_t
Generation 4330: Best = 3.252234e+03 = [1, 1, 1, 1]	FES = 433100	Diversity = 1.6222e-07	k_t
Generation 4340: Best = 3.252234e+03 = [3, 4, 2, 2]	FES = 434100	Diversity = 1.8380e-07	k_t
Generation 4350: Best = 3.252234e+03 = [1, 1, 1, 1]	FES = 435100	Diversity = 1.8783e-07	k_t
Generation 4360: Best = 3.252233e+03 = [3, 2, 4, 2]	FES = 436100	Diversity = 2.3449e-07	k_t
Generation 4370: Best = 3.252233e+03 = [1, 1, 1, 1]	FES = 437100	Diversity = 2.6891e-07	k_t
Generation 4380: Best = 3.252233e+03 = [3, 3, 4, 3]	FES = 438100	Diversity = 2.7837e-07	k_t
Generation 4390: Best = 3.252233e+03 = [1, 1, 1, 1]	FES = 439100	Diversity = 2.4876e-07	k_t
Generation 4400: Best = 3.252233e+03 = [2, 2, 4, 3]	FES = 440100	Diversity = 2.0677e-07	k_t
Generation 4410: Best = 3.252232e+03 = [1, 1, 1, 1]	FES = 441100	Diversity = 2.0117e-07	k_t
Generation 4420: Best = 3.252232e+03 = [2, 4, 3, 3]	FES = 442100	Diversity = 2.2352e-07	k_t
Generation 4430: Best = 3.252232e+03 = [1, 1, 1, 1]	FES = 443100	Diversity = 2.0178e-07	k_t
Generation 4440: Best = 3.252232e+03 = [4, 3, 2, 3]	FES = 444100	Diversity = 1.8385e-07	k_t
Generation 4450: Best = 3.252232e+03 = [1, 1, 1, 1]	FES = 445100	Diversity = 1.7517e-07	k_t
Generation 4460: Best = 3.252232e+03 = [2, 3, 2, 3]	FES = 446100	Diversity = 1.8186e-07	k_t
Generation 4470: Best = 3.252232e+03 = [1, 1, 1, 1]	FES = 447100	Diversity = 1.6644e-07	k_t
Generation 4480: Best = 3.252231e+03 = [4, 2, 4, 4]	FES = 448100	Diversity = 1.7639e-07	k_t
Generation 4490: Best = 3.252231e+03 = [1, 1, 1, 1]	FES = 449100	Diversity = 1.5859e-07	k_t
Generation 4500: Best = 3.252231e+03 = [3, 4, 3, 3]	FES = 450100	Diversity = 1.4453e-07	k_t
Generation 4510: Best = 3.252231e+03 = [1, 1, 1, 1]	FES = 451100	Diversity = 1.3537e-07	k_t
Generation 4520: Best = 3.252231e+03 = [4, 3, 4, 4]	FES = 452100	Diversity = 1.2346e-07	k_t
Generation 4530: Best = 3.252231e+03 = [1, 1, 1, 1]	FES = 453100	Diversity = 1.1860e-07	k_t
Generation 4540: Best = 3.252231e+03 = [2, 2, 4, 4]	FES = 454100	Diversity = 1.1944e-07	k_t

Generation 4550: Best = 3.252231e+03 = [1, 1, 1, 1]	FES = 455100	Diversity = 1.0897e-07	k_t
Generation 4560: Best = 3.252231e+03 = [3, 4, 2, 2]	FES = 456100	Diversity = 9.9813e-08	k_t
Generation 4570: Best = 3.252231e+03 = [1, 1, 1, 1]	FES = 457100	Diversity = 1.0084e-07	k_t
Generation 4580: Best = 3.252231e+03 = [3, 4, 3, 4]	FES = 458100	Diversity = 1.0303e-07	k_t
Generation 4590: Best = 3.252231e+03 = [1, 1, 1, 1]	FES = 459100	Diversity = 1.0497e-07	k_t
Generation 4600: Best = 3.252231e+03 = [4, 2, 2, 3]	FES = 460100	Diversity = 1.2685e-07	k_t
Generation 4610: Best = 3.252231e+03 = [1, 1, 1, 1]	FES = 461100	Diversity = 1.3256e-07	k_t
Generation 4620: Best = 3.252231e+03 = [4, 4, 3, 4]	FES = 462100	Diversity = 1.2602e-07	k_t
Generation 4630: Best = 3.252230e+03 = [1, 1, 1, 1]	FES = 463100	Diversity = 1.3132e-07	k_t
Generation 4640: Best = 3.252230e+03 = [3, 2, 3, 2]	FES = 464100	Diversity = 1.2680e-07	k_t
Generation 4650: Best = 3.252230e+03 = [1, 1, 1, 1]	FES = 465100	Diversity = 1.3831e-07	k_t
Generation 4660: Best = 3.252230e+03 = [2, 2, 2, 2]	FES = 466100	Diversity = 1.3048e-07	k_t
Generation 4670: Best = 3.252230e+03 = [1, 1, 1, 1]	FES = 467100	Diversity = 1.3591e-07	k_t
Generation 4680: Best = 3.252230e+03 = [2, 3, 4, 2]	FES = 468100	Diversity = 1.4640e-07	k_t
Generation 4690: Best = 3.252230e+03 = [1, 1, 1, 1]	FES = 469100	Diversity = 1.5303e-07	k_t
Generation 4700: Best = 3.252230e+03 = [2, 3, 4, 3]	FES = 470100	Diversity = 1.5310e-07	k_t
Generation 4710: Best = 3.252230e+03 = [1, 1, 1, 1]	FES = 471100	Diversity = 1.1954e-07	k_t
Generation 4720: Best = 3.252230e+03 = [3, 3, 4, 3]	FES = 472100	Diversity = 1.2154e-07	k_t
Generation 4730: Best = 3.252230e+03 = [1, 1, 1, 1]	FES = 473100	Diversity = 1.2046e-07	k_t
Generation 4740: Best = 3.252230e+03 = [2, 4, 4, 3]	FES = 474100	Diversity = 1.2308e-07	k_t
Generation 4750: Best = 3.252230e+03 = [1, 1, 1, 1]	FES = 475100	Diversity = 1.2853e-07	k_t
Generation 4760: Best = 3.252229e+03 = [3, 2, 2, 4]	FES = 476100	Diversity = 1.3878e-07	k_t
Generation 4770: Best = 3.252229e+03 = [1, 1, 1, 1]	FES = 477100	Diversity = 1.2161e-07	k_t
Generation 4780: Best = 3.252229e+03 = [2, 3, 3, 4]	FES = 478100	Diversity = 1.1648e-07	k_t
Generation 4790: Best = 3.252229e+03 = [1, 1, 1, 1]	FES = 479100	Diversity = 1.2172e-07	k_t
Generation 4800: Best = 3.252229e+03 = [3, 2, 4, 2]	FES = 480100	Diversity = 1.1415e-07	k_t
Generation 4810: Best = 3.252229e+03 = [1, 1, 1, 1]	FES = 481100	Diversity = 1.1000e-07	k_t

Generation 4820: Best = 3.252229e+03 = [2, 2, 2, 2]	FES = 482100	Diversity = 1.1332e-07	k_t
Generation 4830: Best = 3.252229e+03 = [1, 1, 1, 1]	FES = 483100	Diversity = 9.8504e-08	k_t
Generation 4840: Best = 3.252229e+03 = [3, 4, 3, 4]	FES = 484100	Diversity = 9.1624e-08	k_t
Generation 4850: Best = 3.252229e+03 = [1, 1, 1, 1]	FES = 485100	Diversity = 8.9586e-08	k_t
Generation 4860: Best = 3.252229e+03 = [3, 2, 3, 4]	FES = 486100	Diversity = 9.3309e-08	k_t
Generation 4870: Best = 3.252229e+03 = [1, 1, 1, 1]	FES = 487100	Diversity = 9.3736e-08	k_t
Generation 4880: Best = 3.252229e+03 = [2, 4, 3, 2]	FES = 488100	Diversity = 1.2368e-07	k_t
Generation 4890: Best = 3.252229e+03 = [1, 1, 1, 1]	FES = 489100	Diversity = 1.6518e-07	k_t
Generation 4900: Best = 3.252229e+03 = [2, 2, 2, 4]	FES = 490100	Diversity = 1.7522e-07	k_t
Generation 4910: Best = 3.252228e+03 = [1, 1, 1, 1]	FES = 491100	Diversity = 2.0382e-07	k_t
Generation 4920: Best = 3.252228e+03 = [2, 3, 4, 3]	FES = 492100	Diversity = 2.3648e-07	k_t
Generation 4930: Best = 3.252228e+03 = [1, 1, 1, 1]	FES = 493100	Diversity = 1.9951e-07	k_t
Generation 4940: Best = 3.252228e+03 = [4, 3, 2, 3]	FES = 494100	Diversity = 1.9712e-07	k_t
Generation 4950: Best = 3.252228e+03 = [1, 1, 1, 1]	FES = 495100	Diversity = 1.6769e-07	k_t
Generation 4960: Best = 3.252228e+03 = [3, 4, 3, 2]	FES = 496100	Diversity = 1.7593e-07	k_t
Generation 4970: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 497100	Diversity = 1.3005e-07	k_t
Generation 4980: Best = 3.252227e+03 = [3, 2, 4, 3]	FES = 498100	Diversity = 1.2361e-07	k_t
Generation 4990: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 499100	Diversity = 9.8875e-08	k_t
Generation 5000: Best = 3.252227e+03 = [2, 3, 2, 3]	FES = 500100	Diversity = 9.1960e-08	k_t
Generation 5010: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 501100	Diversity = 9.7864e-08	k_t
Generation 5020: Best = 3.252227e+03 = [2, 3, 3, 2]	FES = 502100	Diversity = 8.3622e-08	k_t
Generation 5030: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 503100	Diversity = 8.4999e-08	k_t
Generation 5040: Best = 3.252227e+03 = [4, 2, 4, 2]	FES = 504100	Diversity = 8.1695e-08	k_t
Generation 5050: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 505100	Diversity = 6.5645e-08	k_t
Generation 5060: Best = 3.252227e+03 = [3, 4, 4, 4]	FES = 506100	Diversity = 5.5733e-08	k_t
Generation 5070: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 507100	Diversity = 5.0516e-08	k_t
Generation 5080: Best = 3.252227e+03 = [3, 3, 3, 3]	FES = 508100	Diversity = 4.5569e-08	k_t

Generation 5090: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 509100	Diversity = 4.3087e-08	k_t
Generation 5100: Best = 3.252227e+03 = [4, 3, 2, 2]	FES = 510100	Diversity = 5.6624e-08	k_t
Generation 5110: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 511100	Diversity = 6.4260e-08	k_t
Generation 5120: Best = 3.252227e+03 = [3, 4, 2, 4]	FES = 512100	Diversity = 5.4917e-08	k_t
Generation 5130: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 513100	Diversity = 4.0671e-08	k_t
Generation 5140: Best = 3.252227e+03 = [3, 3, 4, 3]	FES = 514100	Diversity = 3.4453e-08	k_t
Generation 5150: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 515100	Diversity = 3.3725e-08	k_t
Generation 5160: Best = 3.252227e+03 = [4, 2, 2, 3]	FES = 516100	Diversity = 3.3652e-08	k_t
Generation 5170: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 517100	Diversity = 3.7282e-08	k_t
Generation 5180: Best = 3.252227e+03 = [2, 2, 4, 4]	FES = 518100	Diversity = 3.7185e-08	k_t
Generation 5190: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 519100	Diversity = 3.8969e-08	k_t
Generation 5200: Best = 3.252227e+03 = [4, 2, 3, 2]	FES = 520100	Diversity = 3.4738e-08	k_t
Generation 5210: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 521100	Diversity = 3.3577e-08	k_t
Generation 5220: Best = 3.252227e+03 = [3, 2, 4, 3]	FES = 522100	Diversity = 2.9361e-08	k_t
Generation 5230: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 523100	Diversity = 3.0273e-08	k_t
Generation 5240: Best = 3.252227e+03 = [4, 2, 2, 4]	FES = 524100	Diversity = 3.0901e-08	k_t
Generation 5250: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 525100	Diversity = 3.1673e-08	k_t
Generation 5260: Best = 3.252227e+03 = [3, 3, 2, 4]	FES = 526100	Diversity = 2.7275e-08	k_t
Generation 5270: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 527100	Diversity = 3.1124e-08	k_t
Generation 5280: Best = 3.252227e+03 = [2, 3, 2, 2]	FES = 528100	Diversity = 3.0569e-08	k_t
Generation 5290: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 529100	Diversity = 3.7204e-08	k_t
Generation 5300: Best = 3.252227e+03 = [2, 4, 2, 4]	FES = 530100	Diversity = 4.4767e-08	k_t
Generation 5310: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 531100	Diversity = 4.6817e-08	k_t
Generation 5320: Best = 3.252227e+03 = [3, 3, 4, 4]	FES = 532100	Diversity = 4.5160e-08	k_t
Generation 5330: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 533100	Diversity = 4.9152e-08	k_t
Generation 5340: Best = 3.252227e+03 = [2, 2, 3, 3]	FES = 534100	Diversity = 5.5092e-08	k_t
Generation 5350: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 535100	Diversity = 6.4471e-08	k_t

Generation 5360: Best = 3.252227e+03 = [4, 4, 3, 2]	FES = 536100	Diversity = 6.7721e-08	k_t
Generation 5370: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 537100	Diversity = 8.9481e-08	k_t
Generation 5380: Best = 3.252227e+03 = [4, 2, 2, 2]	FES = 538100	Diversity = 1.3965e-07	k_t
Generation 5390: Best = 3.252227e+03 = [1, 1, 1, 1]	FES = 539100	Diversity = 2.0170e-07	k_t
Generation 5400: Best = 3.252226e+03 = [3, 4, 3, 3]	FES = 540100	Diversity = 2.1786e-07	k_t
Generation 5410: Best = 3.252226e+03 = [1, 1, 1, 1]	FES = 541100	Diversity = 2.5067e-07	k_t
Generation 5420: Best = 3.252226e+03 = [4, 2, 4, 4]	FES = 542100	Diversity = 2.0644e-07	k_t
Generation 5430: Best = 3.252226e+03 = [1, 1, 1, 1]	FES = 543100	Diversity = 2.0984e-07	k_t
Generation 5440: Best = 3.252226e+03 = [4, 2, 4, 4]	FES = 544100	Diversity = 2.2941e-07	k_t
Generation 5450: Best = 3.252225e+03 = [1, 1, 1, 1]	FES = 545100	Diversity = 2.2721e-07	k_t
Generation 5460: Best = 3.252225e+03 = [4, 2, 2, 3]	FES = 546100	Diversity = 2.3094e-07	k_t
Generation 5470: Best = 3.252225e+03 = [1, 1, 1, 1]	FES = 547100	Diversity = 1.7859e-07	k_t
Generation 5480: Best = 3.252225e+03 = [3, 4, 3, 4]	FES = 548100	Diversity = 1.8999e-07	k_t
Generation 5490: Best = 3.252225e+03 = [1, 1, 1, 1]	FES = 549100	Diversity = 1.7469e-07	k_t
Generation 5500: Best = 3.252225e+03 = [2, 2, 4, 4]	FES = 550100	Diversity = 1.6957e-07	k_t
Generation 5510: Best = 3.252225e+03 = [1, 1, 1, 1]	FES = 551100	Diversity = 1.3914e-07	k_t
Generation 5520: Best = 3.252224e+03 = [4, 4, 3, 4]	FES = 552100	Diversity = 1.2267e-07	k_t
Generation 5530: Best = 3.252224e+03 = [1, 1, 1, 1]	FES = 553100	Diversity = 1.4475e-07	k_t
Generation 5540: Best = 3.252224e+03 = [3, 3, 4, 3]	FES = 554100	Diversity = 1.4715e-07	k_t
Generation 5550: Best = 3.252224e+03 = [1, 1, 1, 1]	FES = 555100	Diversity = 1.3092e-07	k_t
Generation 5560: Best = 3.252224e+03 = [4, 2, 3, 2]	FES = 556100	Diversity = 1.1330e-07	k_t
Generation 5570: Best = 3.252224e+03 = [1, 1, 1, 1]	FES = 557100	Diversity = 1.1458e-07	k_t
Generation 5580: Best = 3.252224e+03 = [2, 3, 3, 2]	FES = 558100	Diversity = 1.1081e-07	k_t
Generation 5590: Best = 3.252224e+03 = [1, 1, 1, 1]	FES = 559100	Diversity = 1.2080e-07	k_t
Generation 5600: Best = 3.252224e+03 = [2, 2, 2, 2]	FES = 560100	Diversity = 1.1415e-07	k_t
Generation 5610: Best = 3.252224e+03 = [1, 1, 1, 1]	FES = 561100	Diversity = 9.9668e-08	k_t
Generation 5620: Best = 3.252224e+03 = [3, 4, 3, 4]	FES = 562100	Diversity = 9.9086e-08	k_t

Generation 5630: Best = 3.252224e+03 = [1, 1, 1, 1]	FES = 563100	Diversity = 9.4134e-08	k_t
Generation 5640: Best = 3.252224e+03 = [2, 2, 4, 2]	FES = 564100	Diversity = 1.0439e-07	k_t
Generation 5650: Best = 3.252224e+03 = [1, 1, 1, 1]	FES = 565100	Diversity = 1.1503e-07	k_t
Generation 5660: Best = 3.252224e+03 = [2, 3, 2, 2]	FES = 566100	Diversity = 1.0082e-07	k_t
Generation 5670: Best = 3.252224e+03 = [1, 1, 1, 1]	FES = 567100	Diversity = 9.7566e-08	k_t
Generation 5680: Best = 3.252223e+03 = [4, 2, 4, 3]	FES = 568100	Diversity = 1.0204e-07	k_t
Generation 5690: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 569100	Diversity = 1.1953e-07	k_t
Generation 5700: Best = 3.252223e+03 = [2, 3, 4, 2]	FES = 570100	Diversity = 1.2915e-07	k_t
Generation 5710: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 571100	Diversity = 1.5106e-07	k_t
Generation 5720: Best = 3.252223e+03 = [4, 3, 3, 3]	FES = 572100	Diversity = 1.2184e-07	k_t
Generation 5730: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 573100	Diversity = 1.1836e-07	k_t
Generation 5740: Best = 3.252223e+03 = [3, 2, 2, 4]	FES = 574100	Diversity = 1.2965e-07	k_t
Generation 5750: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 575100	Diversity = 1.0457e-07	k_t
Generation 5760: Best = 3.252223e+03 = [2, 3, 4, 3]	FES = 576100	Diversity = 7.6048e-08	k_t
Generation 5770: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 577100	Diversity = 6.3517e-08	k_t
Generation 5780: Best = 3.252223e+03 = [4, 3, 3, 4]	FES = 578100	Diversity = 6.4504e-08	k_t
Generation 5790: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 579100	Diversity = 5.9795e-08	k_t
Generation 5800: Best = 3.252223e+03 = [3, 2, 3, 4]	FES = 580100	Diversity = 5.0900e-08	k_t
Generation 5810: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 581100	Diversity = 3.7022e-08	k_t
Generation 5820: Best = 3.252223e+03 = [4, 3, 2, 2]	FES = 582100	Diversity = 3.4901e-08	k_t
Generation 5830: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 583100	Diversity = 4.0908e-08	k_t
Generation 5840: Best = 3.252223e+03 = [2, 3, 3, 4]	FES = 584100	Diversity = 3.6937e-08	k_t
Generation 5850: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 585100	Diversity = 4.0860e-08	k_t
Generation 5860: Best = 3.252223e+03 = [4, 2, 2, 2]	FES = 586100	Diversity = 3.7888e-08	k_t
Generation 5870: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 587100	Diversity = 3.1811e-08	k_t
Generation 5880: Best = 3.252223e+03 = [3, 2, 3, 2]	FES = 588100	Diversity = 2.9552e-08	k_t
Generation 5890: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 589100	Diversity = 2.7197e-08	k_t

Generation 5900: Best = 3.252223e+03 = [3, 3, 4, 3]	FES = 590100	Diversity = 2.3501e-08	k_t
Generation 5910: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 591100	Diversity = 2.3177e-08	k_t
Generation 5920: Best = 3.252223e+03 = [4, 2, 2, 4]	FES = 592100	Diversity = 2.2974e-08	k_t
Generation 5930: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 593100	Diversity = 1.9586e-08	k_t
Generation 5940: Best = 3.252223e+03 = [2, 3, 4, 2]	FES = 594100	Diversity = 1.9139e-08	k_t
Generation 5950: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 595100	Diversity = 2.1168e-08	k_t
Generation 5960: Best = 3.252223e+03 = [4, 4, 3, 4]	FES = 596100	Diversity = 2.2219e-08	k_t
Generation 5970: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 597100	Diversity = 2.6672e-08	k_t
Generation 5980: Best = 3.252223e+03 = [2, 3, 4, 4]	FES = 598100	Diversity = 3.0919e-08	k_t
Generation 5990: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 599100	Diversity = 3.7227e-08	k_t
Generation 6000: Best = 3.252223e+03 = [2, 4, 3, 2]	FES = 600100	Diversity = 4.0508e-08	k_t
Generation 6010: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 601100	Diversity = 4.3585e-08	k_t
Generation 6020: Best = 3.252223e+03 = [3, 4, 3, 3]	FES = 602100	Diversity = 5.4986e-08	k_t
Generation 6030: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 603100	Diversity = 6.1356e-08	k_t
Generation 6040: Best = 3.252223e+03 = [2, 4, 4, 3]	FES = 604100	Diversity = 6.5647e-08	k_t
Generation 6050: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 605100	Diversity = 6.1301e-08	k_t
Generation 6060: Best = 3.252223e+03 = [3, 3, 2, 4]	FES = 606100	Diversity = 5.5220e-08	k_t
Generation 6070: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 607100	Diversity = 5.4219e-08	k_t
Generation 6080: Best = 3.252223e+03 = [3, 2, 2, 3]	FES = 608100	Diversity = 5.1612e-08	k_t
Generation 6090: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 609100	Diversity = 5.1525e-08	k_t
Generation 6100: Best = 3.252223e+03 = [2, 2, 3, 3]	FES = 610100	Diversity = 5.5940e-08	k_t
Generation 6110: Best = 3.252223e+03 = [1, 1, 1, 1]	FES = 611100	Diversity = 6.0240e-08	k_t
Generation 6120: Best = 3.252223e+03 = [4, 4, 2, 4]	FES = 612100	Diversity = 5.4460e-08	k_t
Generation 6130: Best = 3.252222e+03 = [1, 1, 1, 1]	FES = 613100	Diversity = 4.8267e-08	k_t
Generation 6140: Best = 3.252222e+03 = [3, 2, 2, 4]	FES = 614100	Diversity = 5.1767e-08	k_t
Generation 6150: Best = 3.252222e+03 = [1, 1, 1, 1]	FES = 615100	Diversity = 5.3732e-08	k_t
Generation 6160: Best = 3.252222e+03 = [3, 3, 2, 2]	FES = 616100	Diversity = 5.7683e-08	k_t

Generation 6170: Best = 3.252222e+03 = [1, 1, 1, 1]	FES = 617100	Diversity = 6.4943e-08	k_t
Generation 6180: Best = 3.252222e+03 = [3, 4, 3, 4]	FES = 618100	Diversity = 8.2077e-08	k_t
Generation 6190: Best = 3.252222e+03 = [1, 1, 1, 1]	FES = 619100	Diversity = 9.1425e-08	k_t
Generation 6200: Best = 3.252222e+03 = [2, 2, 4, 4]	FES = 620100	Diversity = 1.1421e-07	k_t
Generation 6210: Best = 3.252222e+03 = [1, 1, 1, 1]	FES = 621100	Diversity = 1.1587e-07	k_t
Generation 6220: Best = 3.252222e+03 = [4, 3, 4, 3]	FES = 622100	Diversity = 1.1442e-07	k_t
Generation 6230: Best = 3.252222e+03 = [1, 1, 1, 1]	FES = 623100	Diversity = 1.2886e-07	k_t
Generation 6240: Best = 3.252222e+03 = [4, 2, 3, 4]	FES = 624100	Diversity = 1.6428e-07	k_t
Generation 6250: Best = 3.252222e+03 = [1, 1, 1, 1]	FES = 625100	Diversity = 1.7640e-07	k_t
Generation 6260: Best = 3.252222e+03 = [4, 2, 4, 2]	FES = 626100	Diversity = 1.8726e-07	k_t
Generation 6270: Best = 3.252221e+03 = [1, 1, 1, 1]	FES = 627100	Diversity = 2.0494e-07	k_t
Generation 6280: Best = 3.252221e+03 = [3, 3, 4, 2]	FES = 628100	Diversity = 1.8396e-07	k_t
Generation 6290: Best = 3.252221e+03 = [1, 1, 1, 1]	FES = 629100	Diversity = 1.6548e-07	k_t
Generation 6300: Best = 3.252221e+03 = [2, 4, 2, 3]	FES = 630100	Diversity = 1.7222e-07	k_t
Generation 6310: Best = 3.252221e+03 = [1, 1, 1, 1]	FES = 631100	Diversity = 1.5182e-07	k_t
Generation 6320: Best = 3.252221e+03 = [3, 2, 4, 2]	FES = 632100	Diversity = 1.4591e-07	k_t
Generation 6330: Best = 3.252221e+03 = [1, 1, 1, 1]	FES = 633100	Diversity = 1.3818e-07	k_t
Generation 6340: Best = 3.252221e+03 = [3, 4, 3, 3]	FES = 634100	Diversity = 1.1629e-07	k_t
Generation 6350: Best = 3.252221e+03 = [1, 1, 1, 1]	FES = 635100	Diversity = 9.9329e-08	k_t
Generation 6360: Best = 3.252221e+03 = [2, 3, 3, 3]	FES = 636100	Diversity = 9.5760e-08	k_t
Generation 6370: Best = 3.252221e+03 = [1, 1, 1, 1]	FES = 637100	Diversity = 7.9002e-08	k_t
Generation 6380: Best = 3.252221e+03 = [2, 3, 2, 3]	FES = 638100	Diversity = 9.2376e-08	k_t
Generation 6390: Best = 3.252220e+03 = [1, 1, 1, 1]	FES = 639100	Diversity = 1.0664e-07	k_t
Generation 6400: Best = 3.252220e+03 = [4, 4, 2, 3]	FES = 640100	Diversity = 1.3820e-07	k_t
Generation 6410: Best = 3.252220e+03 = [1, 1, 1, 1]	FES = 641100	Diversity = 1.5318e-07	k_t
Generation 6420: Best = 3.252220e+03 = [4, 2, 2, 3]	FES = 642100	Diversity = 1.4338e-07	k_t
Generation 6430: Best = 3.252220e+03 = [1, 1, 1, 1]	FES = 643100	Diversity = 1.1938e-07	k_t

Generation 6440: Best = 3.252220e+03 = [3, 2, 3, 2]	FES = 644100	Diversity = 1.1007e-07	k_t
Generation 6450: Best = 3.252220e+03 = [1, 1, 1, 1]	FES = 645100	Diversity = 8.2098e-08	k_t
Generation 6460: Best = 3.252220e+03 = [3, 3, 2, 4]	FES = 646100	Diversity = 9.5732e-08	k_t
Generation 6470: Best = 3.252220e+03 = [1, 1, 1, 1]	FES = 647100	Diversity = 1.1428e-07	k_t
Generation 6480: Best = 3.252220e+03 = [3, 4, 3, 2]	FES = 648100	Diversity = 1.1436e-07	k_t
Generation 6490: Best = 3.252220e+03 = [1, 1, 1, 1]	FES = 649100	Diversity = 1.3636e-07	k_t
Generation 6500: Best = 3.252220e+03 = [2, 2, 3, 2]	FES = 650100	Diversity = 1.5420e-07	k_t
Generation 6510: Best = 3.252220e+03 = [1, 1, 1, 1]	FES = 651100	Diversity = 1.5038e-07	k_t
Generation 6520: Best = 3.252219e+03 = [4, 4, 4, 3]	FES = 652100	Diversity = 1.4408e-07	k_t
Generation 6530: Best = 3.252219e+03 = [1, 1, 1, 1]	FES = 653100	Diversity = 1.4156e-07	k_t
Generation 6540: Best = 3.252219e+03 = [2, 2, 3, 2]	FES = 654100	Diversity = 1.3779e-07	k_t
Generation 6550: Best = 3.252219e+03 = [1, 1, 1, 1]	FES = 655100	Diversity = 1.2210e-07	k_t
Generation 6560: Best = 3.252219e+03 = [2, 3, 4, 3]	FES = 656100	Diversity = 1.0334e-07	k_t
Generation 6570: Best = 3.252219e+03 = [1, 1, 1, 1]	FES = 657100	Diversity = 1.1437e-07	k_t
Generation 6580: Best = 3.252219e+03 = [3, 4, 3, 3]	FES = 658100	Diversity = 1.2001e-07	k_t
Generation 6590: Best = 3.252219e+03 = [1, 1, 1, 1]	FES = 659100	Diversity = 1.3266e-07	k_t
Generation 6600: Best = 3.252219e+03 = [4, 3, 2, 4]	FES = 660100	Diversity = 1.4024e-07	k_t
Generation 6610: Best = 3.252219e+03 = [1, 1, 1, 1]	FES = 661100	Diversity = 1.3042e-07	k_t
Generation 6620: Best = 3.252219e+03 = [3, 4, 2, 4]	FES = 662100	Diversity = 1.0769e-07	k_t
Generation 6630: Best = 3.252219e+03 = [1, 1, 1, 1]	FES = 663100	Diversity = 1.1359e-07	k_t
Generation 6640: Best = 3.252218e+03 = [4, 4, 2, 3]	FES = 664100	Diversity = 1.2300e-07	k_t
Generation 6650: Best = 3.252218e+03 = [1, 1, 1, 1]	FES = 665100	Diversity = 1.3048e-07	k_t
Generation 6660: Best = 3.252218e+03 = [2, 2, 3, 4]	FES = 666100	Diversity = 1.1640e-07	k_t
Generation 6670: Best = 3.252218e+03 = [1, 1, 1, 1]	FES = 667100	Diversity = 9.5316e-08	k_t
Generation 6680: Best = 3.252218e+03 = [3, 4, 3, 4]	FES = 668100	Diversity = 1.0788e-07	k_t
Generation 6690: Best = 3.252218e+03 = [1, 1, 1, 1]	FES = 669100	Diversity = 1.2578e-07	k_t
Generation 6700: Best = 3.252218e+03 = [2, 2, 2, 4]	FES = 670100	Diversity = 1.3514e-07	k_t

Generation 6710: Best = 3.252218e+03 = [1, 1, 1, 1]	FES = 671100	Diversity = 1.4463e-07	k_t
Generation 6720: Best = 3.252218e+03 = [2, 2, 4, 4]	FES = 672100	Diversity = 1.3587e-07	k_t
Generation 6730: Best = 3.252218e+03 = [1, 1, 1, 1]	FES = 673100	Diversity = 1.4789e-07	k_t
Generation 6740: Best = 3.252218e+03 = [3, 2, 4, 3]	FES = 674100	Diversity = 1.3998e-07	k_t
Generation 6750: Best = 3.252218e+03 = [1, 1, 1, 1]	FES = 675100	Diversity = 1.2974e-07	k_t
Generation 6760: Best = 3.252218e+03 = [3, 4, 2, 2]	FES = 676100	Diversity = 1.3287e-07	k_t
Generation 6770: Best = 3.252217e+03 = [1, 1, 1, 1]	FES = 677100	Diversity = 1.1114e-07	k_t
Generation 6780: Best = 3.252217e+03 = [2, 4, 4, 4]	FES = 678100	Diversity = 9.9345e-08	k_t
Generation 6790: Best = 3.252217e+03 = [1, 1, 1, 1]	FES = 679100	Diversity = 1.1925e-07	k_t
Generation 6800: Best = 3.252217e+03 = [3, 2, 3, 2]	FES = 680100	Diversity = 1.4067e-07	k_t
Generation 6810: Best = 3.252217e+03 = [1, 1, 1, 1]	FES = 681100	Diversity = 1.2596e-07	k_t
Generation 6820: Best = 3.252217e+03 = [3, 4, 2, 4]	FES = 682100	Diversity = 1.1586e-07	k_t
Generation 6830: Best = 3.252217e+03 = [1, 1, 1, 1]	FES = 683100	Diversity = 1.1198e-07	k_t
Generation 6840: Best = 3.252217e+03 = [2, 2, 2, 4]	FES = 684100	Diversity = 9.6286e-08	k_t
Generation 6850: Best = 3.252217e+03 = [1, 1, 1, 1]	FES = 685100	Diversity = 9.7510e-08	k_t
Generation 6860: Best = 3.252217e+03 = [4, 3, 2, 3]	FES = 686100	Diversity = 1.0545e-07	k_t
Generation 6870: Best = 3.252217e+03 = [1, 1, 1, 1]	FES = 687100	Diversity = 9.5699e-08	k_t
Generation 6880: Best = 3.252217e+03 = [4, 2, 2, 2]	FES = 688100	Diversity = 8.3016e-08	k_t
Generation 6890: Best = 3.252217e+03 = [1, 1, 1, 1]	FES = 689100	Diversity = 8.3304e-08	k_t
Generation 6900: Best = 3.252217e+03 = [2, 2, 3, 4]	FES = 690100	Diversity = 8.3950e-08	k_t
Generation 6910: Best = 3.252217e+03 = [1, 1, 1, 1]	FES = 691100	Diversity = 9.0055e-08	k_t
Generation 6920: Best = 3.252217e+03 = [2, 4, 4, 2]	FES = 692100	Diversity = 1.0207e-07	k_t
Generation 6930: Best = 3.252217e+03 = [1, 1, 1, 1]	FES = 693100	Diversity = 1.1718e-07	k_t
Generation 6940: Best = 3.252216e+03 = [4, 4, 4, 3]	FES = 694100	Diversity = 1.2626e-07	k_t
Generation 6950: Best = 3.252216e+03 = [1, 1, 1, 1]	FES = 695100	Diversity = 1.2590e-07	k_t
Generation 6960: Best = 3.252216e+03 = [3, 4, 3, 3]	FES = 696100	Diversity = 1.2507e-07	k_t
Generation 6970: Best = 3.252216e+03 = [1, 1, 1, 1]	FES = 697100	Diversity = 1.4209e-07	k_t

Generation 6980: Best = 3.252216e+03 = [2, 2, 3, 3]	FES = 698100	Diversity = 1.6121e-07	k_t
Generation 6990: Best = 3.252216e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 1.7033e-07	k_t
Generation 7000: Best = 3.252216e+03 = [4, 2, 2, 4]	FES = 700100	Diversity = 1.6580e-07	k_t
Generation 7010: Best = 3.252216e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 1.6179e-07	k_t
Generation 7020: Best = 3.252216e+03 = [2, 2, 2, 4]	FES = 702100	Diversity = 1.5049e-07	k_t
Generation 7030: Best = 3.252216e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 1.4177e-07	k_t
Generation 7040: Best = 3.252215e+03 = [2, 3, 2, 4]	FES = 704100	Diversity = 1.3789e-07	k_t
Generation 7050: Best = 3.252215e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 1.1676e-07	k_t
Generation 7060: Best = 3.252215e+03 = [2, 3, 4, 2]	FES = 706100	Diversity = 1.4211e-07	k_t
Generation 7070: Best = 3.252215e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 1.3675e-07	k_t
Generation 7080: Best = 3.252215e+03 = [3, 2, 2, 4]	FES = 708100	Diversity = 1.3855e-07	k_t
Generation 7090: Best = 3.252215e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 1.3233e-07	k_t
Generation 7100: Best = 3.252215e+03 = [4, 3, 4, 3]	FES = 710100	Diversity = 1.2766e-07	k_t
Generation 7110: Best = 3.252215e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 1.3677e-07	k_t
Generation 7120: Best = 3.252215e+03 = [4, 3, 2, 2]	FES = 712100	Diversity = 1.0955e-07	k_t
Generation 7130: Best = 3.252215e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 1.3275e-07	k_t
Generation 7140: Best = 3.252215e+03 = [2, 2, 3, 2]	FES = 714100	Diversity = 1.2640e-07	k_t
Generation 7150: Best = 3.252215e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 1.0683e-07	k_t
Generation 7160: Best = 3.252214e+03 = [4, 4, 2, 2]	FES = 716100	Diversity = 1.0704e-07	k_t
Generation 7170: Best = 3.252214e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 1.1843e-07	k_t
Generation 7180: Best = 3.252214e+03 = [2, 2, 4, 3]	FES = 718100	Diversity = 1.3271e-07	k_t
Generation 7190: Best = 3.252214e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 1.2519e-07	k_t
Generation 7200: Best = 3.252214e+03 = [2, 2, 2, 2]	FES = 720100	Diversity = 1.6482e-07	k_t
Generation 7210: Best = 3.252214e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 1.6921e-07	k_t
Generation 7220: Best = 3.252214e+03 = [2, 2, 4, 2]	FES = 722100	Diversity = 1.7553e-07	k_t
Generation 7230: Best = 3.252214e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 1.7212e-07	k_t
Generation 7240: Best = 3.252214e+03 = [2, 2, 2, 2]	FES = 724100	Diversity = 1.4463e-07	k_t

Generation 7250: Best = 3.252214e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 1.4344e-07	k_t
Generation 7260: Best = 3.252213e+03 = [3, 3, 2, 4]	FES = 726100	Diversity = 1.5121e-07	k_t
Generation 7270: Best = 3.252213e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 1.5501e-07	k_t
Generation 7280: Best = 3.252213e+03 = [3, 2, 2, 2]	FES = 728100	Diversity = 1.3105e-07	k_t
Generation 7290: Best = 3.252213e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 1.1781e-07	k_t
Generation 7300: Best = 3.252213e+03 = [4, 4, 3, 2]	FES = 730100	Diversity = 1.0386e-07	k_t
Generation 7310: Best = 3.252213e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 9.6554e-08	k_t
Generation 7320: Best = 3.252213e+03 = [2, 4, 2, 4]	FES = 732100	Diversity = 1.0235e-07	k_t
Generation 7330: Best = 3.252213e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 1.0297e-07	k_t
Generation 7340: Best = 3.252213e+03 = [4, 2, 4, 3]	FES = 734100	Diversity = 1.0205e-07	k_t
Generation 7350: Best = 3.252213e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 1.1659e-07	k_t
Generation 7360: Best = 3.252213e+03 = [3, 4, 2, 4]	FES = 736100	Diversity = 1.0270e-07	k_t
Generation 7370: Best = 3.252213e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 9.3689e-08	k_t
Generation 7380: Best = 3.252213e+03 = [4, 4, 3, 2]	FES = 738100	Diversity = 1.0006e-07	k_t
Generation 7390: Best = 3.252213e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 9.6322e-08	k_t
Generation 7400: Best = 3.252213e+03 = [2, 3, 3, 3]	FES = 740100	Diversity = 8.5806e-08	k_t
Generation 7410: Best = 3.252213e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 9.6785e-08	k_t
Generation 7420: Best = 3.252212e+03 = [4, 3, 2, 4]	FES = 742100	Diversity = 1.1579e-07	k_t
Generation 7430: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 1.6142e-07	k_t
Generation 7440: Best = 3.252212e+03 = [2, 3, 2, 2]	FES = 744100	Diversity = 1.1264e-07	k_t
Generation 7450: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 1.0033e-07	k_t
Generation 7460: Best = 3.252212e+03 = [2, 3, 3, 3]	FES = 746100	Diversity = 9.3228e-08	k_t
Generation 7470: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 8.3578e-08	k_t
Generation 7480: Best = 3.252212e+03 = [4, 4, 2, 4]	FES = 748100	Diversity = 6.9973e-08	k_t
Generation 7490: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 5.6523e-08	k_t
Generation 7500: Best = 3.252212e+03 = [3, 3, 3, 3]	FES = 750100	Diversity = 5.7918e-08	k_t
Generation 7510: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 7.6768e-08	k_t

Generation 7520: Best = 3.252212e+03 = [2, 3, 3, 4]	FES = 752100	Diversity = 8.6639e-08	k_t
Generation 7530: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 753100	Diversity = 1.0650e-07	k_t
Generation 7540: Best = 3.252212e+03 = [3, 2, 3, 4]	FES = 754100	Diversity = 9.1775e-08	k_t
Generation 7550: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 755100	Diversity = 8.9816e-08	k_t
Generation 7560: Best = 3.252212e+03 = [4, 4, 3, 4]	FES = 756100	Diversity = 7.9573e-08	k_t
Generation 7570: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 757100	Diversity = 7.7585e-08	k_t
Generation 7580: Best = 3.252212e+03 = [3, 2, 3, 3]	FES = 758100	Diversity = 6.0151e-08	k_t
Generation 7590: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 759100	Diversity = 5.4991e-08	k_t
Generation 7600: Best = 3.252212e+03 = [4, 3, 4, 2]	FES = 760100	Diversity = 4.4991e-08	k_t
Generation 7610: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 761100	Diversity = 3.7618e-08	k_t
Generation 7620: Best = 3.252212e+03 = [3, 3, 2, 3]	FES = 762100	Diversity = 3.3202e-08	k_t
Generation 7630: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 763100	Diversity = 4.0092e-08	k_t
Generation 7640: Best = 3.252212e+03 = [2, 2, 4, 4]	FES = 764100	Diversity = 4.2933e-08	k_t
Generation 7650: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 765100	Diversity = 4.7229e-08	k_t
Generation 7660: Best = 3.252212e+03 = [3, 3, 3, 3]	FES = 766100	Diversity = 5.2968e-08	k_t
Generation 7670: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 767100	Diversity = 6.7437e-08	k_t
Generation 7680: Best = 3.252212e+03 = [3, 4, 4, 2]	FES = 768100	Diversity = 6.6340e-08	k_t
Generation 7690: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 769100	Diversity = 6.6507e-08	k_t
Generation 7700: Best = 3.252212e+03 = [4, 4, 3, 2]	FES = 770100	Diversity = 5.5544e-08	k_t
Generation 7710: Best = 3.252212e+03 = [1, 1, 1, 1]	FES = 771100	Diversity = 6.6480e-08	k_t
Generation 7720: Best = 3.252211e+03 = [3, 2, 3, 2]	FES = 772100	Diversity = 5.9596e-08	k_t
Generation 7730: Best = 3.252211e+03 = [1, 1, 1, 1]	FES = 773100	Diversity = 5.4504e-08	k_t
Generation 7740: Best = 3.252211e+03 = [4, 2, 3, 4]	FES = 774100	Diversity = 5.0945e-08	k_t
Generation 7750: Best = 3.252211e+03 = [1, 1, 1, 1]	FES = 775100	Diversity = 4.4563e-08	k_t
Generation 7760: Best = 3.252211e+03 = [4, 3, 2, 3]	FES = 776100	Diversity = 4.8342e-08	k_t
Generation 7770: Best = 3.252211e+03 = [1, 1, 1, 1]	FES = 777100	Diversity = 5.8187e-08	k_t
Generation 7780: Best = 3.252211e+03 = [4, 3, 4, 3]	FES = 778100	Diversity = 5.6054e-08	k_t

Generation 7790: Best = 3.252211e+03	FES = 779100	Diversity = 7.6512e-08	k_t
= [1, 1, 1, 1]			
Generation 7800: Best = 3.252211e+03	FES = 780100	Diversity = 9.3848e-08	k_t
= [2, 4, 3, 2]			
Generation 7810: Best = 3.252211e+03	FES = 781100	Diversity = 1.1526e-07	k_t
= [1, 1, 1, 1]			
Generation 7820: Best = 3.252211e+03	FES = 782100	Diversity = 1.1289e-07	k_t
= [3, 3, 3, 2]			
Generation 7830: Best = 3.252211e+03	FES = 783100	Diversity = 1.0187e-07	k_t
= [1, 1, 1, 1]			
Generation 7840: Best = 3.252211e+03	FES = 784100	Diversity = 1.1461e-07	k_t
= [2, 3, 4, 4]			
Generation 7850: Best = 3.252211e+03	FES = 785100	Diversity = 1.1133e-07	k_t
= [1, 1, 1, 1]			
Generation 7860: Best = 3.252211e+03	FES = 786100	Diversity = 1.1975e-07	k_t
= [2, 2, 3, 3]			
Generation 7870: Best = 3.252211e+03	FES = 787100	Diversity = 1.0713e-07	k_t
= [1, 1, 1, 1]			
Generation 7880: Best = 3.252211e+03	FES = 788100	Diversity = 1.1134e-07	k_t
= [4, 2, 4, 2]			
Generation 7890: Best = 3.252211e+03	FES = 789100	Diversity = 1.1399e-07	k_t
= [1, 1, 1, 1]			
Generation 7900: Best = 3.252211e+03	FES = 790100	Diversity = 1.2661e-07	k_t
= [2, 3, 2, 4]			
Generation 7910: Best = 3.252211e+03	FES = 791100	Diversity = 1.2862e-07	k_t
= [1, 1, 1, 1]			
Generation 7920: Best = 3.252210e+03	FES = 792100	Diversity = 1.2925e-07	k_t
= [2, 3, 4, 3]			
Generation 7930: Best = 3.252210e+03	FES = 793100	Diversity = 1.4073e-07	k_t
= [1, 1, 1, 1]			
Generation 7940: Best = 3.252210e+03	FES = 794100	Diversity = 1.3099e-07	k_t
= [2, 4, 3, 4]			
Generation 7950: Best = 3.252210e+03	FES = 795100	Diversity = 1.1484e-07	k_t
= [1, 1, 1, 1]			
Generation 7960: Best = 3.252210e+03	FES = 796100	Diversity = 1.0351e-07	k_t
= [3, 3, 3, 3]			
Generation 7970: Best = 3.252210e+03	FES = 797100	Diversity = 8.9183e-08	k_t
= [1, 1, 1, 1]			
Generation 7980: Best = 3.252210e+03	FES = 798100	Diversity = 1.0264e-07	k_t
= [3, 3, 3, 2]			
Generation 7990: Best = 3.252210e+03	FES = 799100	Diversity = 8.7043e-08	k_t
= [1, 1, 1, 1]			
Generation 8000: Best = 3.252210e+03	FES = 800100	Diversity = 8.4223e-08	k_t
= [3, 3, 2, 3]			
Generation 8010: Best = 3.252210e+03	FES = 801100	Diversity = 1.0630e-07	k_t
= [1, 1, 1, 1]			
Generation 8020: Best = 3.252210e+03	FES = 802100	Diversity = 1.0573e-07	k_t
= [3, 2, 3, 3]			
Generation 8030: Best = 3.252210e+03	FES = 803100	Diversity = 8.0912e-08	k_t
= [1, 1, 1, 1]			
Generation 8040: Best = 3.252210e+03	FES = 804100	Diversity = 6.6485e-08	k_t
= [4, 2, 3, 4]			
Generation 8050: Best = 3.252210e+03	FES = 805100	Diversity = 6.4361e-08	k_t
= [1, 1, 1, 1]			

Generation 8060: Best = 3.252210e+03 = [2, 3, 3, 4]	FES = 806100	Diversity = 5.7891e-08	k_t
Generation 8070: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 807100	Diversity = 5.9371e-08	k_t
Generation 8080: Best = 3.252210e+03 = [2, 4, 2, 3]	FES = 808100	Diversity = 5.2423e-08	k_t
Generation 8090: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 809100	Diversity = 5.3385e-08	k_t
Generation 8100: Best = 3.252210e+03 = [3, 3, 3, 2]	FES = 810100	Diversity = 4.6657e-08	k_t
Generation 8110: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 811100	Diversity = 4.4178e-08	k_t
Generation 8120: Best = 3.252210e+03 = [2, 4, 2, 2]	FES = 812100	Diversity = 5.0451e-08	k_t
Generation 8130: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 813100	Diversity = 4.2191e-08	k_t
Generation 8140: Best = 3.252210e+03 = [3, 3, 3, 3]	FES = 814100	Diversity = 3.7436e-08	k_t
Generation 8150: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 815100	Diversity = 3.8876e-08	k_t
Generation 8160: Best = 3.252210e+03 = [2, 3, 4, 3]	FES = 816100	Diversity = 4.1367e-08	k_t
Generation 8170: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 817100	Diversity = 3.4392e-08	k_t
Generation 8180: Best = 3.252210e+03 = [2, 2, 4, 3]	FES = 818100	Diversity = 3.2152e-08	k_t
Generation 8190: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 819100	Diversity = 3.1344e-08	k_t
Generation 8200: Best = 3.252210e+03 = [2, 2, 4, 3]	FES = 820100	Diversity = 2.8944e-08	k_t
Generation 8210: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 821100	Diversity = 2.8142e-08	k_t
Generation 8220: Best = 3.252210e+03 = [2, 4, 2, 4]	FES = 822100	Diversity = 2.7049e-08	k_t
Generation 8230: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 823100	Diversity = 3.0414e-08	k_t
Generation 8240: Best = 3.252210e+03 = [4, 3, 3, 3]	FES = 824100	Diversity = 3.0081e-08	k_t
Generation 8250: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 825100	Diversity = 2.9249e-08	k_t
Generation 8260: Best = 3.252210e+03 = [4, 2, 4, 2]	FES = 826100	Diversity = 2.7379e-08	k_t
Generation 8270: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 827100	Diversity = 2.8532e-08	k_t
Generation 8280: Best = 3.252210e+03 = [2, 2, 4, 4]	FES = 828100	Diversity = 2.9352e-08	k_t
Generation 8290: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 829100	Diversity = 3.5031e-08	k_t
Generation 8300: Best = 3.252210e+03 = [2, 2, 4, 2]	FES = 830100	Diversity = 2.6439e-08	k_t
Generation 8310: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 831100	Diversity = 2.6142e-08	k_t
Generation 8320: Best = 3.252210e+03 = [2, 3, 4, 2]	FES = 832100	Diversity = 2.5033e-08	k_t

Generation 8330: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 833100	Diversity = 2.9583e-08	k_t
Generation 8340: Best = 3.252210e+03 = [3, 2, 4, 3]	FES = 834100	Diversity = 2.9036e-08	k_t
Generation 8350: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 835100	Diversity = 2.9962e-08	k_t
Generation 8360: Best = 3.252210e+03 = [3, 2, 4, 3]	FES = 836100	Diversity = 2.7169e-08	k_t
Generation 8370: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 837100	Diversity = 2.5404e-08	k_t
Generation 8380: Best = 3.252210e+03 = [3, 3, 2, 4]	FES = 838100	Diversity = 2.2879e-08	k_t
Generation 8390: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 839100	Diversity = 2.8724e-08	k_t
Generation 8400: Best = 3.252210e+03 = [3, 3, 4, 4]	FES = 840100	Diversity = 2.7173e-08	k_t
Generation 8410: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 841100	Diversity = 2.9807e-08	k_t
Generation 8420: Best = 3.252210e+03 = [3, 3, 4, 3]	FES = 842100	Diversity = 2.6138e-08	k_t
Generation 8430: Best = 3.252210e+03 = [1, 1, 1, 1]	FES = 843100	Diversity = 2.5219e-08	k_t
Generation 8440: Best = 3.252210e+03 = [3, 2, 4, 2]	FES = 844100	Diversity = 2.3840e-08	k_t
Generation 8450: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 845100	Diversity = 2.4466e-08	k_t
Generation 8460: Best = 3.252209e+03 = [3, 2, 4, 3]	FES = 846100	Diversity = 1.9881e-08	k_t
Generation 8470: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 847100	Diversity = 1.6585e-08	k_t
Generation 8480: Best = 3.252209e+03 = [4, 2, 2, 2]	FES = 848100	Diversity = 1.3733e-08	k_t
Generation 8490: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 849100	Diversity = 1.1260e-08	k_t
Generation 8500: Best = 3.252209e+03 = [4, 2, 2, 3]	FES = 850100	Diversity = 1.0287e-08	k_t
Generation 8510: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 851100	Diversity = 9.0638e-09	k_t
Generation 8520: Best = 3.252209e+03 = [4, 3, 4, 2]	FES = 852100	Diversity = 9.1773e-09	k_t
Generation 8530: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 853100	Diversity = 9.4438e-09	k_t
Generation 8540: Best = 3.252209e+03 = [2, 2, 2, 2]	FES = 854100	Diversity = 8.7927e-09	k_t
Generation 8550: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 855100	Diversity = 9.3590e-09	k_t
Generation 8560: Best = 3.252209e+03 = [2, 3, 4, 3]	FES = 856100	Diversity = 7.0879e-09	k_t
Generation 8570: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 857100	Diversity = 6.5480e-09	k_t
Generation 8580: Best = 3.252209e+03 = [4, 3, 3, 2]	FES = 858100	Diversity = 5.8248e-09	k_t
Generation 8590: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 859100	Diversity = 5.9869e-09	k_t

Generation 8600: Best = 3.252209e+03 = [4, 2, 2, 3]	FES = 860100	Diversity = 6.0584e-09	k_t
Generation 8610: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 861100	Diversity = 5.6599e-09	k_t
Generation 8620: Best = 3.252209e+03 = [4, 3, 2, 2]	FES = 862100	Diversity = 5.3660e-09	k_t
Generation 8630: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 863100	Diversity = 6.1251e-09	k_t
Generation 8640: Best = 3.252209e+03 = [3, 4, 4, 3]	FES = 864100	Diversity = 5.4013e-09	k_t
Generation 8650: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 865100	Diversity = 5.4956e-09	k_t
Generation 8660: Best = 3.252209e+03 = [4, 4, 4, 4]	FES = 866100	Diversity = 4.5651e-09	k_t
Generation 8670: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 867100	Diversity = 3.9790e-09	k_t
Generation 8680: Best = 3.252209e+03 = [4, 4, 3, 2]	FES = 868100	Diversity = 3.9645e-09	k_t
Generation 8690: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 869100	Diversity = 3.9515e-09	k_t
Generation 8700: Best = 3.252209e+03 = [4, 2, 2, 4]	FES = 870100	Diversity = 3.7415e-09	k_t
Generation 8710: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 871100	Diversity = 3.8962e-09	k_t
Generation 8720: Best = 3.252209e+03 = [3, 4, 3, 4]	FES = 872100	Diversity = 3.1697e-09	k_t
Generation 8730: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 873100	Diversity = 2.4331e-09	k_t
Generation 8740: Best = 3.252209e+03 = [2, 2, 4, 3]	FES = 874100	Diversity = 2.3482e-09	k_t
Generation 8750: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 875100	Diversity = 2.1426e-09	k_t
Generation 8760: Best = 3.252209e+03 = [2, 2, 3, 2]	FES = 876100	Diversity = 1.8557e-09	k_t
Generation 8770: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 877100	Diversity = 1.7542e-09	k_t
Generation 8780: Best = 3.252209e+03 = [4, 4, 2, 4]	FES = 878100	Diversity = 2.0841e-09	k_t
Generation 8790: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 879100	Diversity = 1.7457e-09	k_t
Generation 8800: Best = 3.252209e+03 = [2, 2, 4, 4]	FES = 880100	Diversity = 1.7782e-09	k_t
Generation 8810: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 881100	Diversity = 2.0777e-09	k_t
Generation 8820: Best = 3.252209e+03 = [3, 3, 4, 4]	FES = 882100	Diversity = 2.0419e-09	k_t
Generation 8830: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 883100	Diversity = 2.1244e-09	k_t
Generation 8840: Best = 3.252209e+03 = [4, 2, 2, 2]	FES = 884100	Diversity = 2.5260e-09	k_t
Generation 8850: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 885100	Diversity = 3.2421e-09	k_t
Generation 8860: Best = 3.252209e+03 = [4, 3, 2, 2]	FES = 886100	Diversity = 3.6950e-09	k_t

Generation 8870: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 887100	Diversity = 3.8500e-09	k_t
Generation 8880: Best = 3.252209e+03 = [2, 2, 2, 3]	FES = 888100	Diversity = 3.5269e-09	k_t
Generation 8890: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 889100	Diversity = 3.0595e-09	k_t
Generation 8900: Best = 3.252209e+03 = [2, 2, 2, 3]	FES = 890100	Diversity = 3.6935e-09	k_t
Generation 8910: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 891100	Diversity = 3.5569e-09	k_t
Generation 8920: Best = 3.252209e+03 = [3, 2, 4, 3]	FES = 892100	Diversity = 3.8332e-09	k_t
Generation 8930: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 893100	Diversity = 4.5587e-09	k_t
Generation 8940: Best = 3.252209e+03 = [2, 4, 3, 3]	FES = 894100	Diversity = 5.0635e-09	k_t
Generation 8950: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 895100	Diversity = 4.1341e-09	k_t
Generation 8960: Best = 3.252209e+03 = [3, 3, 2, 2]	FES = 896100	Diversity = 3.0941e-09	k_t
Generation 8970: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 897100	Diversity = 3.1587e-09	k_t
Generation 8980: Best = 3.252209e+03 = [2, 2, 4, 3]	FES = 898100	Diversity = 2.3096e-09	k_t
Generation 8990: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 899100	Diversity = 2.1633e-09	k_t
Generation 9000: Best = 3.252209e+03 = [2, 4, 3, 3]	FES = 900100	Diversity = 1.8429e-09	k_t
Generation 9010: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 901100	Diversity = 1.7165e-09	k_t
Generation 9020: Best = 3.252209e+03 = [4, 4, 3, 3]	FES = 902100	Diversity = 1.7185e-09	k_t
Generation 9030: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 903100	Diversity = 1.5252e-09	k_t
Generation 9040: Best = 3.252209e+03 = [2, 4, 3, 2]	FES = 904100	Diversity = 1.3546e-09	k_t
Generation 9050: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 905100	Diversity = 1.1919e-09	k_t
Generation 9060: Best = 3.252209e+03 = [2, 2, 2, 2]	FES = 906100	Diversity = 1.0817e-09	k_t
Generation 9070: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 907100	Diversity = 1.1044e-09	k_t
Generation 9080: Best = 3.252209e+03 = [2, 3, 3, 2]	FES = 908100	Diversity = 1.3569e-09	k_t
Generation 9090: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 909100	Diversity = 1.4468e-09	k_t
Generation 9100: Best = 3.252209e+03 = [3, 4, 2, 2]	FES = 910100	Diversity = 9.7813e-10	k_t
Generation 9110: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 911100	Diversity = 8.5283e-10	k_t
Generation 9120: Best = 3.252209e+03 = [2, 3, 4, 2]	FES = 912100	Diversity = 7.6263e-10	k_t
Generation 9130: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 913100	Diversity = 1.0394e-09	k_t

Generation 9140: Best = 3.252209e+03 = [3, 4, 4, 3]	FES = 914100	Diversity = 9.1178e-10	k_t
Generation 9150: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 9.2507e-10	k_t
Generation 9160: Best = 3.252209e+03 = [3, 4, 4, 3]	FES = 916100	Diversity = 7.6690e-10	k_t
Generation 9170: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 6.7964e-10	k_t
Generation 9180: Best = 3.252209e+03 = [4, 2, 3, 4]	FES = 918100	Diversity = 5.9946e-10	k_t
Generation 9190: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 5.9863e-10	k_t
Generation 9200: Best = 3.252209e+03 = [3, 2, 2, 2]	FES = 920100	Diversity = 6.3803e-10	k_t
Generation 9210: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 5.9308e-10	k_t
Generation 9220: Best = 3.252209e+03 = [4, 3, 4, 3]	FES = 922100	Diversity = 8.2013e-10	k_t
Generation 9230: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 1.2720e-09	k_t
Generation 9240: Best = 3.252209e+03 = [3, 4, 3, 4]	FES = 924100	Diversity = 1.4268e-09	k_t
Generation 9250: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 1.3522e-09	k_t
Generation 9260: Best = 3.252209e+03 = [4, 2, 4, 4]	FES = 926100	Diversity = 1.5223e-09	k_t
Generation 9270: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 1.4494e-09	k_t
Generation 9280: Best = 3.252209e+03 = [4, 3, 4, 2]	FES = 928100	Diversity = 1.4856e-09	k_t
Generation 9290: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 1.8171e-09	k_t
Generation 9300: Best = 3.252209e+03 = [4, 2, 4, 3]	FES = 930100	Diversity = 1.6526e-09	k_t
Generation 9310: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 1.5730e-09	k_t
Generation 9320: Best = 3.252209e+03 = [3, 3, 3, 4]	FES = 932100	Diversity = 1.3328e-09	k_t
Generation 9330: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 1.3446e-09	k_t
Generation 9340: Best = 3.252209e+03 = [4, 3, 3, 3]	FES = 934100	Diversity = 1.3281e-09	k_t
Generation 9350: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 1.6434e-09	k_t
Generation 9360: Best = 3.252209e+03 = [3, 4, 2, 4]	FES = 936100	Diversity = 1.3758e-09	k_t
Generation 9370: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 1.2109e-09	k_t
Generation 9380: Best = 3.252209e+03 = [3, 3, 4, 3]	FES = 938100	Diversity = 1.2467e-09	k_t
Generation 9390: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 1.2678e-09	k_t
Generation 9400: Best = 3.252209e+03 = [4, 3, 2, 3]	FES = 940100	Diversity = 1.3252e-09	k_t

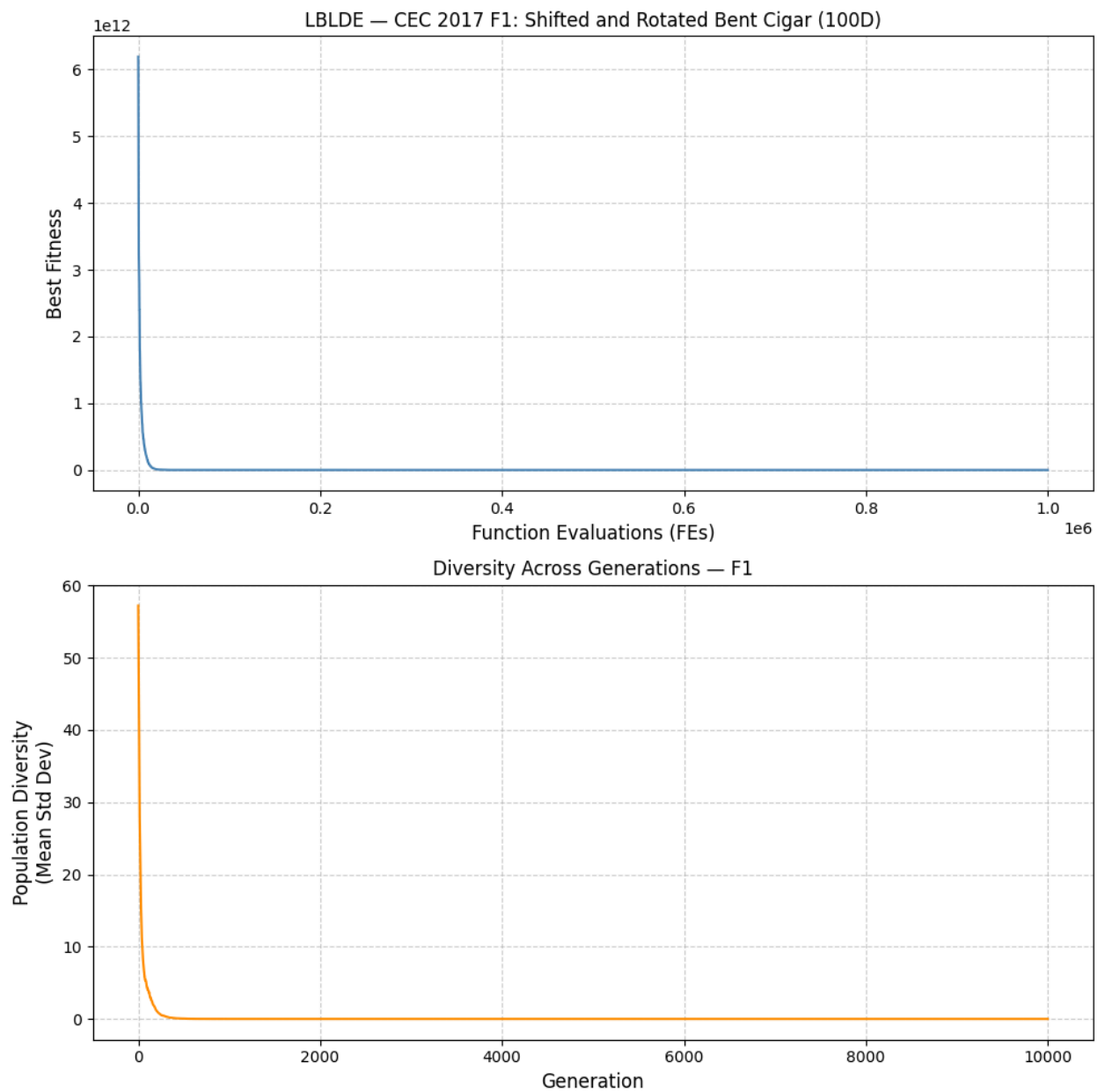
Generation 9410: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 1.3171e-09	k_t
Generation 9420: Best = 3.252209e+03 = [3, 2, 4, 4]	FES = 942100	Diversity = 1.4288e-09	k_t
Generation 9430: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 1.4050e-09	k_t
Generation 9440: Best = 3.252209e+03 = [2, 4, 3, 4]	FES = 944100	Diversity = 1.2308e-09	k_t
Generation 9450: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 1.5414e-09	k_t
Generation 9460: Best = 3.252209e+03 = [3, 2, 3, 4]	FES = 946100	Diversity = 1.3365e-09	k_t
Generation 9470: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 1.3056e-09	k_t
Generation 9480: Best = 3.252209e+03 = [2, 4, 3, 4]	FES = 948100	Diversity = 1.8634e-09	k_t
Generation 9490: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 1.7988e-09	k_t
Generation 9500: Best = 3.252209e+03 = [2, 2, 4, 3]	FES = 950100	Diversity = 1.9120e-09	k_t
Generation 9510: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 1.3516e-09	k_t
Generation 9520: Best = 3.252209e+03 = [4, 4, 3, 4]	FES = 952100	Diversity = 7.9873e-10	k_t
Generation 9530: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 6.1971e-10	k_t
Generation 9540: Best = 3.252209e+03 = [4, 2, 3, 2]	FES = 954100	Diversity = 4.8122e-10	k_t
Generation 9550: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 4.6659e-10	k_t
Generation 9560: Best = 3.252209e+03 = [4, 2, 2, 2]	FES = 956100	Diversity = 4.9817e-10	k_t
Generation 9570: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 4.9202e-10	k_t
Generation 9580: Best = 3.252209e+03 = [4, 4, 3, 2]	FES = 958100	Diversity = 4.9984e-10	k_t
Generation 9590: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 4.9505e-10	k_t
Generation 9600: Best = 3.252209e+03 = [4, 3, 3, 3]	FES = 960100	Diversity = 4.1984e-10	k_t
Generation 9610: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 4.6193e-10	k_t
Generation 9620: Best = 3.252209e+03 = [3, 3, 2, 3]	FES = 962100	Diversity = 3.8728e-10	k_t
Generation 9630: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 3.7241e-10	k_t
Generation 9640: Best = 3.252209e+03 = [2, 3, 4, 2]	FES = 964100	Diversity = 3.0327e-10	k_t
Generation 9650: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 2.9930e-10	k_t
Generation 9660: Best = 3.252209e+03 = [3, 2, 2, 3]	FES = 966100	Diversity = 2.5804e-10	k_t
Generation 9670: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 2.6346e-10	k_t

Generation 9680: Best = 3.252209e+03 = [3, 2, 4, 2]	FES = 968100	Diversity = 3.0024e-10	k_t
Generation 9690: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 969100	Diversity = 3.1714e-10	k_t
Generation 9700: Best = 3.252209e+03 = [2, 3, 3, 3]	FES = 970100	Diversity = 3.8667e-10	k_t
Generation 9710: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 971100	Diversity = 3.7422e-10	k_t
Generation 9720: Best = 3.252209e+03 = [2, 2, 4, 2]	FES = 972100	Diversity = 3.7546e-10	k_t
Generation 9730: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 973100	Diversity = 3.5475e-10	k_t
Generation 9740: Best = 3.252209e+03 = [4, 4, 4, 3]	FES = 974100	Diversity = 3.6130e-10	k_t
Generation 9750: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 975100	Diversity = 3.8022e-10	k_t
Generation 9760: Best = 3.252209e+03 = [2, 2, 4, 3]	FES = 976100	Diversity = 4.9155e-10	k_t
Generation 9770: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 977100	Diversity = 5.0928e-10	k_t
Generation 9780: Best = 3.252209e+03 = [4, 2, 3, 4]	FES = 978100	Diversity = 5.5008e-10	k_t
Generation 9790: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 979100	Diversity = 6.6915e-10	k_t
Generation 9800: Best = 3.252209e+03 = [2, 3, 4, 4]	FES = 980100	Diversity = 9.3166e-10	k_t
Generation 9810: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 981100	Diversity = 9.5295e-10	k_t
Generation 9820: Best = 3.252209e+03 = [3, 3, 3, 2]	FES = 982100	Diversity = 9.2376e-10	k_t
Generation 9830: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 983100	Diversity = 7.1892e-10	k_t
Generation 9840: Best = 3.252209e+03 = [4, 2, 2, 3]	FES = 984100	Diversity = 7.5133e-10	k_t
Generation 9850: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 985100	Diversity = 7.6110e-10	k_t
Generation 9860: Best = 3.252209e+03 = [3, 4, 2, 2]	FES = 986100	Diversity = 6.8296e-10	k_t
Generation 9870: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 987100	Diversity = 6.0483e-10	k_t
Generation 9880: Best = 3.252209e+03 = [4, 4, 3, 2]	FES = 988100	Diversity = 6.2657e-10	k_t
Generation 9890: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 989100	Diversity = 6.5329e-10	k_t
Generation 9900: Best = 3.252209e+03 = [2, 2, 4, 4]	FES = 990100	Diversity = 7.3475e-10	k_t
Generation 9910: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 991100	Diversity = 6.4040e-10	k_t
Generation 9920: Best = 3.252209e+03 = [3, 3, 4, 4]	FES = 992100	Diversity = 6.3407e-10	k_t
Generation 9930: Best = 3.252209e+03 = [1, 1, 1, 1]	FES = 993100	Diversity = 5.6630e-10	k_t
Generation 9940: Best = 3.252209e+03 = [4, 4, 4, 4]	FES = 994100	Diversity = 6.0327e-10	k_t

Generation 9950: Best = 3.252209e+03 FES = 995100 Diversity = 6.9040e-10 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 3.252209e+03 FES = 996100 Diversity = 6.3539e-10 k_t
= [3, 2, 4, 4]
Generation 9970: Best = 3.252209e+03 FES = 997100 Diversity = 5.5713e-10 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 3.252209e+03 FES = 998100 Diversity = 5.1080e-10 k_t
= [2, 4, 3, 4]
Generation 9990: Best = 3.252209e+03 FES = 999100 Diversity = 4.1585e-10 k_t
= [1, 1, 1, 1]

Optimization completed!
Final Best Fitness = 3.252209e+03
Total Function Evaluations = 1000000

=====
=
Results:
Best Fitness : 3.252209e+03
Global Optimum : 1.000000e+02
Error (|best-optimum|) : 3.152209e+03
Best Solution (5 dim) : [-52.20110716 -70.42955972 -2.1834003 -58.32676328
22.08960188]
D^(0) (init. diversity): 5.7220e+01
=====



Graph saved as convergence_F1.png

```
In [26]: # =====
# F2 – Shifted and Rotated Sum of Different Power
# =====
run_and_plot('F2', make_wrapper(simple.f2), 200.0, 'Shifted and Rotated Sum of
```



```

=====
=
LBLDE_s on CEC 2017 F2 – Single Run
=====
=
Function      : Shifted and Rotated Sum of Different Power (Deprecated) (100D)
Bounds       : [-100, 100]
Global optimum : 200.0
WARNING: f2 has been deprecated from the CEC 2017 benchmark suite
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 1.048956e+189 | Diversity = 5.7220e+01
Generation 10: Best = 1.274910e+169 FES = 1100 Diversity = 4.8328e+01
k_t = [1, 1, 1, 1]
Generation 20: Best = 4.679036e+153 FES = 2100 Diversity = 4.1018e+01
k_t = [4, 2, 2, 3]
Generation 30: Best = 4.253245e+151 FES = 3100 Diversity = 3.4733e+01
k_t = [1, 1, 1, 1]
Generation 40: Best = 4.695650e+145 FES = 4100 Diversity = 2.9637e+01
k_t = [2, 3, 4, 3]
Generation 50: Best = 3.567969e+141 FES = 5100 Diversity = 2.5917e+01
k_t = [1, 1, 1, 1]
Generation 60: Best = 1.268862e+138 FES = 6100 Diversity = 2.1757e+01
k_t = [4, 3, 4, 2]
Generation 70: Best = 4.018855e+133 FES = 7100 Diversity = 1.9098e+01
k_t = [1, 1, 1, 1]
Generation 80: Best = 3.383763e+129 FES = 8100 Diversity = 1.6638e+01
k_t = [2, 4, 3, 4]
Generation 90: Best = 3.145800e+125 FES = 9100 Diversity = 1.3794e+01
k_t = [1, 1, 1, 1]
Generation 100: Best = 9.595760e+121 FES = 10100 Diversity = 1.1657e+01
k_t = [3, 4, 2, 2]
Generation 110: Best = 1.097928e+119 FES = 11100 Diversity = 1.0140e+01
k_t = [1, 1, 1, 1]
Generation 120: Best = 2.381444e+118 FES = 12100 Diversity = 9.2365e+00
k_t = [3, 3, 2, 3]
Generation 130: Best = 6.095243e+117 FES = 13100 Diversity = 8.7267e+00
k_t = [1, 1, 1, 1]
Generation 140: Best = 1.080769e+116 FES = 14100 Diversity = 7.7295e+00
k_t = [3, 4, 4, 3]
Generation 150: Best = 8.017693e+114 FES = 15100 Diversity = 7.0605e+00
k_t = [1, 1, 1, 1]
Generation 160: Best = 1.167542e+114 FES = 16100 Diversity = 6.6360e+00
k_t = [2, 2, 4, 4]
Generation 170: Best = 4.901191e+110 FES = 17100 Diversity = 6.1947e+00
k_t = [1, 1, 1, 1]
Generation 180: Best = 2.026556e+110 FES = 18100 Diversity = 5.9518e+00
k_t = [3, 3, 3, 3]
Generation 190: Best = 2.174841e+109 FES = 19100 Diversity = 5.9181e+00
k_t = [1, 1, 1, 1]
Generation 200: Best = 2.432751e+108 FES = 20100 Diversity = 5.7950e+00
k_t = [2, 2, 2, 2]
Generation 210: Best = 7.805621e+105 FES = 21100 Diversity = 5.6144e+00
k_t = [1, 1, 1, 1]
Generation 220: Best = 3.068864e+104 FES = 22100 Diversity = 5.3508e+00

```

```

k_t = [3, 3, 3, 2]
Generation 230: Best = 1.433362e+104 FES = 23100 Diversity = 5.0998e+00
k_t = [1, 1, 1, 1]
Generation 240: Best = 1.025467e+104 FES = 24100 Diversity = 4.9058e+00
k_t = [2, 2, 4, 4]
Generation 250: Best = 2.427629e+103 FES = 25100 Diversity = 4.8463e+00
k_t = [1, 1, 1, 1]
Generation 260: Best = 1.898035e+102 FES = 26100 Diversity = 4.5470e+00
k_t = [3, 2, 3, 3]
Generation 270: Best = 3.871196e+101 FES = 27100 Diversity = 4.4528e+00
k_t = [1, 1, 1, 1]
Generation 280: Best = 8.048434e+100 FES = 28100 Diversity = 4.1457e+00
k_t = [4, 2, 4, 3]
Generation 290: Best = 4.221871e+99 FES = 29100 Diversity = 4.0683e+00 k_t
= [1, 1, 1, 1]
Generation 300: Best = 2.988759e+99 FES = 30100 Diversity = 3.9987e+00 k_t
= [2, 4, 2, 3]
Generation 310: Best = 1.419666e+99 FES = 31100 Diversity = 3.9535e+00 k_t
= [1, 1, 1, 1]
Generation 320: Best = 6.187615e+98 FES = 32100 Diversity = 3.8783e+00 k_t
= [4, 4, 4, 3]
Generation 330: Best = 1.117617e+98 FES = 33100 Diversity = 3.8272e+00 k_t
= [1, 1, 1, 1]
Generation 340: Best = 1.749924e+97 FES = 34100 Diversity = 3.6742e+00 k_t
= [4, 3, 3, 4]
Generation 350: Best = 7.184519e+96 FES = 35100 Diversity = 3.6702e+00 k_t
= [1, 1, 1, 1]
Generation 360: Best = 6.921683e+95 FES = 36100 Diversity = 3.5445e+00 k_t
= [3, 3, 3, 2]
Generation 370: Best = 1.424968e+95 FES = 37100 Diversity = 3.4122e+00 k_t
= [1, 1, 1, 1]
Generation 380: Best = 2.676233e+94 FES = 38100 Diversity = 3.4345e+00 k_t
= [2, 3, 3, 4]
Generation 390: Best = 5.007081e+93 FES = 39100 Diversity = 3.4988e+00 k_t
= [1, 1, 1, 1]
Generation 400: Best = 2.580594e+93 FES = 40100 Diversity = 3.3705e+00 k_t
= [4, 4, 4, 4]
Generation 410: Best = 4.118336e+92 FES = 41100 Diversity = 3.4778e+00 k_t
= [1, 1, 1, 1]
Generation 420: Best = 6.751205e+91 FES = 42100 Diversity = 3.4894e+00 k_t
= [4, 4, 4, 4]
Generation 430: Best = 1.358486e+91 FES = 43100 Diversity = 3.5139e+00 k_t
= [1, 1, 1, 1]
Generation 440: Best = 1.759594e+90 FES = 44100 Diversity = 3.4578e+00 k_t
= [2, 3, 4, 4]
Generation 450: Best = 5.017848e+89 FES = 45100 Diversity = 3.3332e+00 k_t
= [1, 1, 1, 1]
Generation 460: Best = 1.365352e+89 FES = 46100 Diversity = 3.1065e+00 k_t
= [4, 3, 2, 2]
Generation 470: Best = 3.559668e+88 FES = 47100 Diversity = 3.0485e+00 k_t
= [1, 1, 1, 1]
Generation 480: Best = 1.716042e+88 FES = 48100 Diversity = 2.9303e+00 k_t
= [4, 4, 4, 2]
Generation 490: Best = 5.585117e+87 FES = 49100 Diversity = 2.7966e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 1.375744e+87 FES = 50100 Diversity = 2.6804e+00 k_t
= [3, 2, 3, 2]
Generation 510: Best = 2.692025e+86 FES = 51100 Diversity = 2.5020e+00 k_t
= [1, 1, 1, 1]
Generation 520: Best = 1.085530e+86 FES = 52100 Diversity = 2.4359e+00 k_t
= [3, 2, 4, 2]
Generation 530: Best = 3.354171e+85 FES = 53100 Diversity = 2.4656e+00 k_t
= [1, 1, 1, 1]
Generation 540: Best = 9.392352e+84 FES = 54100 Diversity = 2.3895e+00 k_t
= [4, 4, 4, 4]
Generation 550: Best = 4.463017e+84 FES = 55100 Diversity = 2.3630e+00 k_t
= [1, 1, 1, 1]
Generation 560: Best = 1.318057e+84 FES = 56100 Diversity = 2.3595e+00 k_t
= [2, 3, 2, 3]
Generation 570: Best = 4.202428e+83 FES = 57100 Diversity = 2.3264e+00 k_t
= [1, 1, 1, 1]
Generation 580: Best = 1.507279e+83 FES = 58100 Diversity = 2.2005e+00 k_t
= [4, 4, 2, 3]
Generation 590: Best = 7.140539e+82 FES = 59100 Diversity = 1.9261e+00 k_t
= [1, 1, 1, 1]
Generation 600: Best = 4.039408e+82 FES = 60100 Diversity = 1.9507e+00 k_t
= [2, 2, 3, 4]
Generation 610: Best = 1.784845e+82 FES = 61100 Diversity = 1.8930e+00 k_t
= [1, 1, 1, 1]
Generation 620: Best = 5.266007e+81 FES = 62100 Diversity = 1.8417e+00 k_t
= [2, 3, 3, 2]
Generation 630: Best = 1.613349e+81 FES = 63100 Diversity = 1.8393e+00 k_t
= [1, 1, 1, 1]
Generation 640: Best = 1.025896e+81 FES = 64100 Diversity = 1.7544e+00 k_t
= [3, 4, 2, 3]
Generation 650: Best = 5.418850e+80 FES = 65100 Diversity = 1.7428e+00 k_t
= [1, 1, 1, 1]
Generation 660: Best = 2.820962e+80 FES = 66100 Diversity = 1.6393e+00 k_t
= [3, 4, 2, 4]
Generation 670: Best = 1.170730e+80 FES = 67100 Diversity = 1.6565e+00 k_t
= [1, 1, 1, 1]
Generation 680: Best = 9.712147e+79 FES = 68100 Diversity = 1.6344e+00 k_t
= [4, 4, 2, 3]
Generation 690: Best = 4.525270e+79 FES = 69100 Diversity = 1.5912e+00 k_t
= [1, 1, 1, 1]
Generation 700: Best = 2.327366e+79 FES = 70100 Diversity = 1.6090e+00 k_t
= [2, 2, 2, 3]
Generation 710: Best = 1.061103e+79 FES = 71100 Diversity = 1.5569e+00 k_t
= [1, 1, 1, 1]
Generation 720: Best = 6.317023e+78 FES = 72100 Diversity = 1.4397e+00 k_t
= [4, 3, 3, 3]
Generation 730: Best = 4.522103e+78 FES = 73100 Diversity = 1.3376e+00 k_t
= [1, 1, 1, 1]
Generation 740: Best = 2.276120e+78 FES = 74100 Diversity = 1.2014e+00 k_t
= [4, 2, 3, 4]
Generation 750: Best = 1.837385e+78 FES = 75100 Diversity = 1.0690e+00 k_t
= [1, 1, 1, 1]
Generation 760: Best = 1.282241e+78 FES = 76100 Diversity = 1.0828e+00 k_t

```

```

= [3, 4, 3, 4]
Generation 770: Best = 6.902582e+77 FES = 77100 Diversity = 1.0589e+00 k_t
= [1, 1, 1, 1]
Generation 780: Best = 4.734469e+77 FES = 78100 Diversity = 1.0045e+00 k_t
= [2, 4, 4, 3]
Generation 790: Best = 2.516424e+77 FES = 79100 Diversity = 1.0031e+00 k_t
= [1, 1, 1, 1]
Generation 800: Best = 1.643648e+77 FES = 80100 Diversity = 9.8771e-01 k_t
= [3, 2, 2, 3]
Generation 810: Best = 1.115688e+77 FES = 81100 Diversity = 1.0212e+00 k_t
= [1, 1, 1, 1]
Generation 820: Best = 5.795639e+76 FES = 82100 Diversity = 1.0664e+00 k_t
= [2, 4, 3, 2]
Generation 830: Best = 4.332442e+76 FES = 83100 Diversity = 1.0748e+00 k_t
= [1, 1, 1, 1]
Generation 840: Best = 3.035938e+76 FES = 84100 Diversity = 1.0782e+00 k_t
= [3, 2, 3, 4]
Generation 850: Best = 2.032522e+76 FES = 85100 Diversity = 1.0975e+00 k_t
= [1, 1, 1, 1]
Generation 860: Best = 1.204504e+76 FES = 86100 Diversity = 1.1549e+00 k_t
= [2, 3, 3, 4]
Generation 870: Best = 8.521455e+75 FES = 87100 Diversity = 1.1839e+00 k_t
= [1, 1, 1, 1]
Generation 880: Best = 5.395501e+75 FES = 88100 Diversity = 1.2140e+00 k_t
= [2, 2, 3, 3]
Generation 890: Best = 3.703364e+75 FES = 89100 Diversity = 1.1751e+00 k_t
= [1, 1, 1, 1]
Generation 900: Best = 2.096070e+75 FES = 90100 Diversity = 1.1881e+00 k_t
= [3, 4, 4, 2]
Generation 910: Best = 1.295087e+75 FES = 91100 Diversity = 1.1440e+00 k_t
= [1, 1, 1, 1]
Generation 920: Best = 7.530082e+74 FES = 92100 Diversity = 1.1240e+00 k_t
= [3, 4, 4, 4]
Generation 930: Best = 4.423085e+74 FES = 93100 Diversity = 1.0833e+00 k_t
= [1, 1, 1, 1]
Generation 940: Best = 3.057528e+74 FES = 94100 Diversity = 1.0296e+00 k_t
= [4, 4, 2, 3]
Generation 950: Best = 1.410444e+74 FES = 95100 Diversity = 1.0151e+00 k_t
= [1, 1, 1, 1]
Generation 960: Best = 1.021204e+74 FES = 96100 Diversity = 1.0140e+00 k_t
= [2, 3, 3, 2]
Generation 970: Best = 6.138701e+73 FES = 97100 Diversity = 9.8094e-01 k_t
= [1, 1, 1, 1]
Generation 980: Best = 3.940085e+73 FES = 98100 Diversity = 9.4418e-01 k_t
= [4, 3, 3, 3]
Generation 990: Best = 2.600947e+73 FES = 99100 Diversity = 9.0478e-01 k_t
= [1, 1, 1, 1]
Generation 1000: Best = 1.730220e+73 FES = 100100 Diversity = 9.8057e-01 k_t
= [2, 4, 3, 2]
Generation 1010: Best = 9.546453e+72 FES = 101100 Diversity = 9.8067e-01 k_t
= [1, 1, 1, 1]
Generation 1020: Best = 6.278728e+72 FES = 102100 Diversity = 9.7780e-01 k_t
= [3, 3, 3, 2]
Generation 1030: Best = 4.126642e+72 FES = 103100 Diversity = 9.9700e-01 k_t

```

```

= [1, 1, 1, 1]
Generation 1040: Best = 2.723683e+72  FES = 104100  Diversity = 1.0520e+00  k_t
= [4, 4, 4, 3]
Generation 1050: Best = 1.738492e+72  FES = 105100  Diversity = 1.0381e+00  k_t
= [1, 1, 1, 1]
Generation 1060: Best = 1.488612e+72  FES = 106100  Diversity = 1.0507e+00  k_t
= [3, 2, 4, 4]
Generation 1070: Best = 6.812260e+71  FES = 107100  Diversity = 1.1339e+00  k_t
= [1, 1, 1, 1]
Generation 1080: Best = 4.514108e+71  FES = 108100  Diversity = 1.1413e+00  k_t
= [3, 2, 3, 2]
Generation 1090: Best = 2.890299e+71  FES = 109100  Diversity = 1.1566e+00  k_t
= [1, 1, 1, 1]
Generation 1100: Best = 1.843767e+71  FES = 110100  Diversity = 1.2076e+00  k_t
= [3, 2, 3, 4]
Generation 1110: Best = 1.146243e+71  FES = 111100  Diversity = 1.1742e+00  k_t
= [1, 1, 1, 1]
Generation 1120: Best = 8.347871e+70  FES = 112100  Diversity = 1.1174e+00  k_t
= [2, 2, 3, 4]
Generation 1130: Best = 4.534459e+70  FES = 113100  Diversity = 1.1135e+00  k_t
= [1, 1, 1, 1]
Generation 1140: Best = 3.370661e+70  FES = 114100  Diversity = 1.0930e+00  k_t
= [2, 3, 2, 4]
Generation 1150: Best = 1.654070e+70  FES = 115100  Diversity = 1.1038e+00  k_t
= [1, 1, 1, 1]
Generation 1160: Best = 1.096234e+70  FES = 116100  Diversity = 1.1351e+00  k_t
= [4, 3, 4, 4]
Generation 1170: Best = 6.117813e+69  FES = 117100  Diversity = 1.1436e+00  k_t
= [1, 1, 1, 1]
Generation 1180: Best = 3.339448e+69  FES = 118100  Diversity = 1.1624e+00  k_t
= [2, 2, 4, 4]
Generation 1190: Best = 2.199915e+69  FES = 119100  Diversity = 1.1154e+00  k_t
= [1, 1, 1, 1]
Generation 1200: Best = 1.229448e+69  FES = 120100  Diversity = 1.1641e+00  k_t
= [4, 2, 2, 3]
Generation 1210: Best = 9.073805e+68  FES = 121100  Diversity = 1.1527e+00  k_t
= [1, 1, 1, 1]
Generation 1220: Best = 4.741658e+68  FES = 122100  Diversity = 1.1333e+00  k_t
= [2, 2, 3, 4]
Generation 1230: Best = 2.990865e+68  FES = 123100  Diversity = 1.1256e+00  k_t
= [1, 1, 1, 1]
Generation 1240: Best = 1.876245e+68  FES = 124100  Diversity = 1.1076e+00  k_t
= [4, 3, 3, 3]
Generation 1250: Best = 1.373368e+68  FES = 125100  Diversity = 1.1630e+00  k_t
= [1, 1, 1, 1]
Generation 1260: Best = 6.749133e+67  FES = 126100  Diversity = 1.2021e+00  k_t
= [2, 3, 2, 3]
Generation 1270: Best = 3.368760e+67  FES = 127100  Diversity = 1.1593e+00  k_t
= [1, 1, 1, 1]
Generation 1280: Best = 2.820330e+67  FES = 128100  Diversity = 1.1236e+00  k_t
= [3, 3, 3, 2]
Generation 1290: Best = 2.336290e+67  FES = 129100  Diversity = 1.0801e+00  k_t
= [1, 1, 1, 1]
Generation 1300: Best = 1.388075e+67  FES = 130100  Diversity = 1.0471e+00  k_t

```

```

= [4, 2, 4, 2]
Generation 1310: Best = 1.011493e+67 FES = 131100 Diversity = 1.0125e+00 k_t
= [1, 1, 1, 1]
Generation 1320: Best = 6.330627e+66 FES = 132100 Diversity = 1.0308e+00 k_t
= [4, 3, 3, 4]
Generation 1330: Best = 3.756799e+66 FES = 133100 Diversity = 1.0421e+00 k_t
= [1, 1, 1, 1]
Generation 1340: Best = 2.766020e+66 FES = 134100 Diversity = 1.0372e+00 k_t
= [2, 2, 4, 4]
Generation 1350: Best = 1.614952e+66 FES = 135100 Diversity = 1.0087e+00 k_t
= [1, 1, 1, 1]
Generation 1360: Best = 7.377412e+65 FES = 136100 Diversity = 1.0186e+00 k_t
= [4, 3, 4, 4]
Generation 1370: Best = 5.980337e+65 FES = 137100 Diversity = 1.0154e+00 k_t
= [1, 1, 1, 1]
Generation 1380: Best = 3.099849e+65 FES = 138100 Diversity = 9.7462e-01 k_t
= [4, 3, 4, 2]
Generation 1390: Best = 2.033511e+65 FES = 139100 Diversity = 9.8920e-01 k_t
= [1, 1, 1, 1]
Generation 1400: Best = 1.915198e+65 FES = 140100 Diversity = 9.8617e-01 k_t
= [3, 4, 3, 3]
Generation 1410: Best = 1.207611e+65 FES = 141100 Diversity = 9.5015e-01 k_t
= [1, 1, 1, 1]
Generation 1420: Best = 7.775042e+64 FES = 142100 Diversity = 9.5545e-01 k_t
= [3, 3, 4, 4]
Generation 1430: Best = 5.081566e+64 FES = 143100 Diversity = 9.6845e-01 k_t
= [1, 1, 1, 1]
Generation 1440: Best = 3.510688e+64 FES = 144100 Diversity = 9.6164e-01 k_t
= [2, 3, 4, 4]
Generation 1450: Best = 2.209230e+64 FES = 145100 Diversity = 9.9338e-01 k_t
= [1, 1, 1, 1]
Generation 1460: Best = 1.541241e+64 FES = 146100 Diversity = 1.0183e+00 k_t
= [3, 3, 4, 4]
Generation 1470: Best = 8.570145e+63 FES = 147100 Diversity = 1.0385e+00 k_t
= [1, 1, 1, 1]
Generation 1480: Best = 6.973504e+63 FES = 148100 Diversity = 1.0551e+00 k_t
= [4, 2, 2, 3]
Generation 1490: Best = 4.889995e+63 FES = 149100 Diversity = 1.0636e+00 k_t
= [1, 1, 1, 1]
Generation 1500: Best = 2.303403e+63 FES = 150100 Diversity = 1.0406e+00 k_t
= [3, 3, 2, 2]
Generation 1510: Best = 1.758798e+63 FES = 151100 Diversity = 1.0676e+00 k_t
= [1, 1, 1, 1]
Generation 1520: Best = 1.570484e+63 FES = 152100 Diversity = 1.0423e+00 k_t
= [3, 4, 3, 2]
Generation 1530: Best = 1.214274e+63 FES = 153100 Diversity = 1.0413e+00 k_t
= [1, 1, 1, 1]
Generation 1540: Best = 7.586827e+62 FES = 154100 Diversity = 1.0018e+00 k_t
= [4, 3, 3, 3]
Generation 1550: Best = 5.695872e+62 FES = 155100 Diversity = 9.5729e-01 k_t
= [1, 1, 1, 1]
Generation 1560: Best = 3.731430e+62 FES = 156100 Diversity = 9.7349e-01 k_t
= [4, 2, 2, 4]
Generation 1570: Best = 2.723020e+62 FES = 157100 Diversity = 9.7227e-01 k_t

```

```

= [1, 1, 1, 1]
Generation 1580: Best = 1.664376e+62 FES = 158100 Diversity = 1.0035e+00 k_t
= [2, 2, 2, 3]
Generation 1590: Best = 1.092559e+62 FES = 159100 Diversity = 1.0279e+00 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 7.739876e+61 FES = 160100 Diversity = 1.0614e+00 k_t
= [4, 2, 2, 4]
Generation 1610: Best = 4.551738e+61 FES = 161100 Diversity = 1.0541e+00 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 3.477216e+61 FES = 162100 Diversity = 1.0536e+00 k_t
= [2, 4, 3, 3]
Generation 1630: Best = 2.002183e+61 FES = 163100 Diversity = 1.0696e+00 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 1.420724e+61 FES = 164100 Diversity = 1.0734e+00 k_t
= [4, 3, 3, 2]
Generation 1650: Best = 1.002030e+61 FES = 165100 Diversity = 1.0717e+00 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 6.981530e+60 FES = 166100 Diversity = 1.1111e+00 k_t
= [2, 3, 2, 2]
Generation 1670: Best = 2.793402e+60 FES = 167100 Diversity = 1.0977e+00 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 2.565811e+60 FES = 168100 Diversity = 1.0934e+00 k_t
= [2, 3, 4, 3]
Generation 1690: Best = 2.353255e+60 FES = 169100 Diversity = 1.0533e+00 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 1.371025e+60 FES = 170100 Diversity = 1.0681e+00 k_t
= [3, 4, 3, 3]
Generation 1710: Best = 8.414535e+59 FES = 171100 Diversity = 1.0661e+00 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 6.474408e+59 FES = 172100 Diversity = 1.0826e+00 k_t
= [4, 2, 4, 2]
Generation 1730: Best = 3.988794e+59 FES = 173100 Diversity = 1.0814e+00 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 2.545733e+59 FES = 174100 Diversity = 1.0643e+00 k_t
= [4, 2, 3, 2]
Generation 1750: Best = 2.034221e+59 FES = 175100 Diversity = 1.0500e+00 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 1.091751e+59 FES = 176100 Diversity = 1.0962e+00 k_t
= [3, 3, 3, 2]
Generation 1770: Best = 5.737060e+58 FES = 177100 Diversity = 1.0802e+00 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 4.025400e+58 FES = 178100 Diversity = 1.0482e+00 k_t
= [2, 3, 3, 2]
Generation 1790: Best = 2.207746e+58 FES = 179100 Diversity = 1.0502e+00 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 1.171814e+58 FES = 180100 Diversity = 1.0595e+00 k_t
= [2, 3, 4, 3]
Generation 1810: Best = 6.986547e+57 FES = 181100 Diversity = 1.0286e+00 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 5.657256e+57 FES = 182100 Diversity = 1.0282e+00 k_t
= [4, 4, 4, 2]
Generation 1830: Best = 4.155039e+57 FES = 183100 Diversity = 1.0051e+00 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 3.109743e+57 FES = 184100 Diversity = 9.9833e-01 k_t

```

```

= [3, 2, 3, 4]
Generation 1850: Best = 2.473375e+57 FES = 185100 Diversity = 9.4851e-01 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 1.701786e+57 FES = 186100 Diversity = 9.1991e-01 k_t
= [2, 2, 4, 4]
Generation 1870: Best = 1.238771e+57 FES = 187100 Diversity = 9.2952e-01 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 7.769007e+56 FES = 188100 Diversity = 9.7078e-01 k_t
= [4, 2, 2, 3]
Generation 1890: Best = 4.851341e+56 FES = 189100 Diversity = 9.4954e-01 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 3.906798e+56 FES = 190100 Diversity = 9.6325e-01 k_t
= [3, 2, 3, 3]
Generation 1910: Best = 2.724650e+56 FES = 191100 Diversity = 9.7299e-01 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 1.973457e+56 FES = 192100 Diversity = 9.8277e-01 k_t
= [2, 3, 2, 2]
Generation 1930: Best = 1.350535e+56 FES = 193100 Diversity = 9.9631e-01 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 9.957921e+55 FES = 194100 Diversity = 1.0029e+00 k_t
= [3, 2, 2, 3]
Generation 1950: Best = 7.415011e+55 FES = 195100 Diversity = 1.0089e+00 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 5.562975e+55 FES = 196100 Diversity = 9.9367e-01 k_t
= [2, 4, 3, 3]
Generation 1970: Best = 4.556641e+55 FES = 197100 Diversity = 9.5703e-01 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 3.594563e+55 FES = 198100 Diversity = 9.0195e-01 k_t
= [3, 3, 2, 4]
Generation 1990: Best = 2.802435e+55 FES = 199100 Diversity = 8.6863e-01 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 2.192111e+55 FES = 200100 Diversity = 8.5328e-01 k_t
= [3, 4, 4, 4]
Generation 2010: Best = 1.723817e+55 FES = 201100 Diversity = 8.2113e-01 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 1.505166e+55 FES = 202100 Diversity = 7.8802e-01 k_t
= [3, 3, 4, 2]
Generation 2030: Best = 1.080793e+55 FES = 203100 Diversity = 7.7594e-01 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 8.654081e+54 FES = 204100 Diversity = 7.5481e-01 k_t
= [3, 4, 3, 4]
Generation 2050: Best = 6.394174e+54 FES = 205100 Diversity = 7.5459e-01 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 5.054397e+54 FES = 206100 Diversity = 7.6246e-01 k_t
= [3, 4, 3, 2]
Generation 2070: Best = 3.878018e+54 FES = 207100 Diversity = 7.8633e-01 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 2.887726e+54 FES = 208100 Diversity = 7.6442e-01 k_t
= [4, 3, 2, 4]
Generation 2090: Best = 2.312636e+54 FES = 209100 Diversity = 7.6131e-01 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 1.730542e+54 FES = 210100 Diversity = 7.8490e-01 k_t
= [4, 3, 2, 3]
Generation 2110: Best = 1.466640e+54 FES = 211100 Diversity = 8.2384e-01 k_t

```



```

= [1, 1, 1, 1]
Generation 2120: Best = 1.194295e+54 FES = 212100 Diversity = 8.1801e-01 k_t
= [4, 3, 4, 3]
Generation 2130: Best = 9.165141e+53 FES = 213100 Diversity = 8.2861e-01 k_t
= [1, 1, 1, 1]
Generation 2140: Best = 7.773260e+53 FES = 214100 Diversity = 8.4622e-01 k_t
= [4, 2, 3, 3]
Generation 2150: Best = 6.155312e+53 FES = 215100 Diversity = 8.6507e-01 k_t
= [1, 1, 1, 1]
Generation 2160: Best = 4.767188e+53 FES = 216100 Diversity = 8.6803e-01 k_t
= [2, 2, 2, 3]
Generation 2170: Best = 3.926046e+53 FES = 217100 Diversity = 8.5427e-01 k_t
= [1, 1, 1, 1]
Generation 2180: Best = 2.852621e+53 FES = 218100 Diversity = 8.7392e-01 k_t
= [3, 4, 2, 3]
Generation 2190: Best = 2.121427e+53 FES = 219100 Diversity = 8.7669e-01 k_t
= [1, 1, 1, 1]
Generation 2200: Best = 1.639448e+53 FES = 220100 Diversity = 9.2194e-01 k_t
= [3, 4, 4, 2]
Generation 2210: Best = 1.246170e+53 FES = 221100 Diversity = 9.1177e-01 k_t
= [1, 1, 1, 1]
Generation 2220: Best = 8.300831e+52 FES = 222100 Diversity = 9.1277e-01 k_t
= [2, 3, 4, 2]
Generation 2230: Best = 6.973642e+52 FES = 223100 Diversity = 9.0530e-01 k_t
= [1, 1, 1, 1]
Generation 2240: Best = 5.341944e+52 FES = 224100 Diversity = 9.0168e-01 k_t
= [2, 4, 4, 3]
Generation 2250: Best = 3.746874e+52 FES = 225100 Diversity = 8.7128e-01 k_t
= [1, 1, 1, 1]
Generation 2260: Best = 3.272641e+52 FES = 226100 Diversity = 8.5281e-01 k_t
= [3, 4, 3, 3]
Generation 2270: Best = 2.713289e+52 FES = 227100 Diversity = 8.1183e-01 k_t
= [1, 1, 1, 1]
Generation 2280: Best = 1.819374e+52 FES = 228100 Diversity = 8.0327e-01 k_t
= [3, 2, 3, 4]
Generation 2290: Best = 1.560276e+52 FES = 229100 Diversity = 8.1098e-01 k_t
= [1, 1, 1, 1]
Generation 2300: Best = 1.102174e+52 FES = 230100 Diversity = 8.1610e-01 k_t
= [2, 2, 3, 2]
Generation 2310: Best = 8.769834e+51 FES = 231100 Diversity = 8.0350e-01 k_t
= [1, 1, 1, 1]
Generation 2320: Best = 7.765876e+51 FES = 232100 Diversity = 8.2429e-01 k_t
= [2, 3, 3, 4]
Generation 2330: Best = 5.494101e+51 FES = 233100 Diversity = 8.0767e-01 k_t
= [1, 1, 1, 1]
Generation 2340: Best = 4.410115e+51 FES = 234100 Diversity = 7.9044e-01 k_t
= [4, 4, 3, 4]
Generation 2350: Best = 3.770777e+51 FES = 235100 Diversity = 7.6528e-01 k_t
= [1, 1, 1, 1]
Generation 2360: Best = 3.000500e+51 FES = 236100 Diversity = 7.7183e-01 k_t
= [4, 3, 4, 3]
Generation 2370: Best = 2.388319e+51 FES = 237100 Diversity = 7.6756e-01 k_t
= [1, 1, 1, 1]
Generation 2380: Best = 2.096877e+51 FES = 238100 Diversity = 7.8049e-01 k_t

```

```

= [2, 2, 2, 3]
Generation 2390: Best = 1.794086e+51  FES = 239100  Diversity = 8.0003e-01  k_t
= [1, 1, 1, 1]
Generation 2400: Best = 1.579950e+51  FES = 240100  Diversity = 8.2318e-01  k_t
= [3, 3, 3, 3]
Generation 2410: Best = 1.298890e+51  FES = 241100  Diversity = 8.6563e-01  k_t
= [1, 1, 1, 1]
Generation 2420: Best = 1.129311e+51  FES = 242100  Diversity = 8.6359e-01  k_t
= [4, 3, 3, 4]
Generation 2430: Best = 9.705157e+50  FES = 243100  Diversity = 8.3864e-01  k_t
= [1, 1, 1, 1]
Generation 2440: Best = 8.362581e+50  FES = 244100  Diversity = 8.8227e-01  k_t
= [2, 4, 3, 2]
Generation 2450: Best = 7.570176e+50  FES = 245100  Diversity = 9.1767e-01  k_t
= [1, 1, 1, 1]
Generation 2460: Best = 6.343328e+50  FES = 246100  Diversity = 9.5035e-01  k_t
= [4, 4, 4, 2]
Generation 2470: Best = 5.774454e+50  FES = 247100  Diversity = 9.1275e-01  k_t
= [1, 1, 1, 1]
Generation 2480: Best = 5.463143e+50  FES = 248100  Diversity = 9.3165e-01  k_t
= [3, 2, 3, 2]
Generation 2490: Best = 4.606078e+50  FES = 249100  Diversity = 9.4144e-01  k_t
= [1, 1, 1, 1]
Generation 2500: Best = 4.171456e+50  FES = 250100  Diversity = 9.7347e-01  k_t
= [3, 4, 3, 2]
Generation 2510: Best = 3.791920e+50  FES = 251100  Diversity = 1.0320e+00  k_t
= [1, 1, 1, 1]
Generation 2520: Best = 3.274787e+50  FES = 252100  Diversity = 1.0375e+00  k_t
= [4, 3, 2, 4]
Generation 2530: Best = 2.956807e+50  FES = 253100  Diversity = 1.0377e+00  k_t
= [1, 1, 1, 1]
Generation 2540: Best = 2.697759e+50  FES = 254100  Diversity = 1.0369e+00  k_t
= [2, 3, 3, 2]
Generation 2550: Best = 2.401845e+50  FES = 255100  Diversity = 1.0379e+00  k_t
= [1, 1, 1, 1]
Generation 2560: Best = 2.222563e+50  FES = 256100  Diversity = 1.0563e+00  k_t
= [4, 3, 2, 4]
Generation 2570: Best = 1.958940e+50  FES = 257100  Diversity = 1.0946e+00  k_t
= [1, 1, 1, 1]
Generation 2580: Best = 1.670242e+50  FES = 258100  Diversity = 1.0774e+00  k_t
= [4, 4, 3, 2]
Generation 2590: Best = 1.561322e+50  FES = 259100  Diversity = 1.0557e+00  k_t
= [1, 1, 1, 1]
Generation 2600: Best = 1.333204e+50  FES = 260100  Diversity = 1.0238e+00  k_t
= [2, 2, 2, 2]
Generation 2610: Best = 1.236717e+50  FES = 261100  Diversity = 1.0012e+00  k_t
= [1, 1, 1, 1]
Generation 2620: Best = 1.009742e+50  FES = 262100  Diversity = 1.0168e+00  k_t
= [4, 4, 4, 3]
Generation 2630: Best = 9.211353e+49  FES = 263100  Diversity = 1.0426e+00  k_t
= [1, 1, 1, 1]
Generation 2640: Best = 8.558400e+49  FES = 264100  Diversity = 1.0347e+00  k_t
= [4, 2, 3, 3]
Generation 2650: Best = 7.261734e+49  FES = 265100  Diversity = 9.9275e-01  k_t

```

```

= [1, 1, 1, 1]
Generation 2660: Best = 6.995444e+49 FES = 266100 Diversity = 9.6699e-01 k_t
= [4, 4, 4, 4]
Generation 2670: Best = 6.068337e+49 FES = 267100 Diversity = 9.7956e-01 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 5.372312e+49 FES = 268100 Diversity = 9.7452e-01 k_t
= [4, 3, 4, 3]
Generation 2690: Best = 4.760981e+49 FES = 269100 Diversity = 9.3690e-01 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 4.222922e+49 FES = 270100 Diversity = 9.4640e-01 k_t
= [4, 4, 4, 3]
Generation 2710: Best = 3.882000e+49 FES = 271100 Diversity = 9.3360e-01 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 3.655420e+49 FES = 272100 Diversity = 9.2348e-01 k_t
= [2, 4, 4, 2]
Generation 2730: Best = 3.337391e+49 FES = 273100 Diversity = 9.2673e-01 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 2.830583e+49 FES = 274100 Diversity = 9.8986e-01 k_t
= [2, 2, 3, 3]
Generation 2750: Best = 2.774580e+49 FES = 275100 Diversity = 9.6885e-01 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 2.605419e+49 FES = 276100 Diversity = 9.8465e-01 k_t
= [2, 4, 3, 4]
Generation 2770: Best = 2.373391e+49 FES = 277100 Diversity = 9.6113e-01 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 2.191553e+49 FES = 278100 Diversity = 9.7379e-01 k_t
= [3, 4, 3, 4]
Generation 2790: Best = 1.935391e+49 FES = 279100 Diversity = 9.5373e-01 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 1.641205e+49 FES = 280100 Diversity = 9.4715e-01 k_t
= [4, 3, 3, 4]
Generation 2810: Best = 1.409348e+49 FES = 281100 Diversity = 9.8616e-01 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 1.106288e+49 FES = 282100 Diversity = 1.0410e+00 k_t
= [4, 4, 3, 2]
Generation 2830: Best = 9.368462e+48 FES = 283100 Diversity = 1.0442e+00 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 7.757140e+48 FES = 284100 Diversity = 1.0646e+00 k_t
= [3, 2, 3, 4]
Generation 2850: Best = 7.233511e+48 FES = 285100 Diversity = 1.0257e+00 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 6.357066e+48 FES = 286100 Diversity = 1.0043e+00 k_t
= [2, 3, 3, 3]
Generation 2870: Best = 5.661638e+48 FES = 287100 Diversity = 1.0236e+00 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 4.607102e+48 FES = 288100 Diversity = 1.0454e+00 k_t
= [3, 3, 3, 2]
Generation 2890: Best = 4.202485e+48 FES = 289100 Diversity = 1.0388e+00 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 3.827323e+48 FES = 290100 Diversity = 1.0151e+00 k_t
= [4, 4, 4, 4]
Generation 2910: Best = 3.558690e+48 FES = 291100 Diversity = 9.8475e-01 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 2.872789e+48 FES = 292100 Diversity = 9.6051e-01 k_t

```

```

= [2, 3, 2, 2]
Generation 2930: Best = 2.557560e+48 FES = 293100 Diversity = 9.3828e-01 k_t
= [1, 1, 1, 1]
Generation 2940: Best = 2.370847e+48 FES = 294100 Diversity = 9.3105e-01 k_t
= [2, 4, 4, 3]
Generation 2950: Best = 2.055561e+48 FES = 295100 Diversity = 9.0303e-01 k_t
= [1, 1, 1, 1]
Generation 2960: Best = 1.842142e+48 FES = 296100 Diversity = 9.1453e-01 k_t
= [3, 4, 3, 3]
Generation 2970: Best = 1.646776e+48 FES = 297100 Diversity = 9.6373e-01 k_t
= [1, 1, 1, 1]
Generation 2980: Best = 1.388715e+48 FES = 298100 Diversity = 9.9051e-01 k_t
= [4, 3, 3, 3]
Generation 2990: Best = 1.212379e+48 FES = 299100 Diversity = 9.6379e-01 k_t
= [1, 1, 1, 1]
Generation 3000: Best = 1.078764e+48 FES = 300100 Diversity = 9.5573e-01 k_t
= [4, 2, 3, 4]
Generation 3010: Best = 9.498104e+47 FES = 301100 Diversity = 9.2503e-01 k_t
= [1, 1, 1, 1]
Generation 3020: Best = 7.951318e+47 FES = 302100 Diversity = 9.2819e-01 k_t
= [2, 3, 2, 2]
Generation 3030: Best = 7.186939e+47 FES = 303100 Diversity = 9.3072e-01 k_t
= [1, 1, 1, 1]
Generation 3040: Best = 6.647486e+47 FES = 304100 Diversity = 9.5330e-01 k_t
= [3, 2, 2, 3]
Generation 3050: Best = 5.859983e+47 FES = 305100 Diversity = 9.5843e-01 k_t
= [1, 1, 1, 1]
Generation 3060: Best = 4.846808e+47 FES = 306100 Diversity = 1.0031e+00 k_t
= [4, 3, 4, 2]
Generation 3070: Best = 4.248363e+47 FES = 307100 Diversity = 9.6928e-01 k_t
= [1, 1, 1, 1]
Generation 3080: Best = 3.603006e+47 FES = 308100 Diversity = 1.0018e+00 k_t
= [3, 2, 4, 4]
Generation 3090: Best = 3.055658e+47 FES = 309100 Diversity = 1.0456e+00 k_t
= [1, 1, 1, 1]
Generation 3100: Best = 2.727679e+47 FES = 310100 Diversity = 1.0612e+00 k_t
= [3, 2, 3, 3]
Generation 3110: Best = 2.327609e+47 FES = 311100 Diversity = 1.0634e+00 k_t
= [1, 1, 1, 1]
Generation 3120: Best = 1.926269e+47 FES = 312100 Diversity = 1.0469e+00 k_t
= [4, 4, 4, 4]
Generation 3130: Best = 1.780380e+47 FES = 313100 Diversity = 1.0374e+00 k_t
= [1, 1, 1, 1]
Generation 3140: Best = 1.558907e+47 FES = 314100 Diversity = 1.0097e+00 k_t
= [3, 4, 2, 4]
Generation 3150: Best = 1.351131e+47 FES = 315100 Diversity = 1.0304e+00 k_t
= [1, 1, 1, 1]
Generation 3160: Best = 1.170468e+47 FES = 316100 Diversity = 1.0130e+00 k_t
= [2, 3, 4, 4]
Generation 3170: Best = 1.016812e+47 FES = 317100 Diversity = 1.0070e+00 k_t
= [1, 1, 1, 1]
Generation 3180: Best = 8.095509e+46 FES = 318100 Diversity = 1.0078e+00 k_t
= [2, 3, 3, 2]
Generation 3190: Best = 6.771111e+46 FES = 319100 Diversity = 1.0246e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 3200: Best = 6.163037e+46 FES = 320100 Diversity = 9.9316e-01 k_t
= [3, 3, 3, 2]
Generation 3210: Best = 5.300509e+46 FES = 321100 Diversity = 9.7576e-01 k_t
= [1, 1, 1, 1]
Generation 3220: Best = 4.446410e+46 FES = 322100 Diversity = 9.6677e-01 k_t
= [3, 4, 4, 3]
Generation 3230: Best = 3.850271e+46 FES = 323100 Diversity = 9.5992e-01 k_t
= [1, 1, 1, 1]
Generation 3240: Best = 3.310953e+46 FES = 324100 Diversity = 9.7958e-01 k_t
= [4, 4, 2, 3]
Generation 3250: Best = 2.887698e+46 FES = 325100 Diversity = 9.7832e-01 k_t
= [1, 1, 1, 1]
Generation 3260: Best = 2.589168e+46 FES = 326100 Diversity = 9.8241e-01 k_t
= [2, 4, 4, 3]
Generation 3270: Best = 2.246028e+46 FES = 327100 Diversity = 9.4635e-01 k_t
= [1, 1, 1, 1]
Generation 3280: Best = 1.897639e+46 FES = 328100 Diversity = 9.2237e-01 k_t
= [4, 3, 2, 3]
Generation 3290: Best = 1.568513e+46 FES = 329100 Diversity = 9.2961e-01 k_t
= [1, 1, 1, 1]
Generation 3300: Best = 1.469669e+46 FES = 330100 Diversity = 9.3525e-01 k_t
= [4, 3, 2, 4]
Generation 3310: Best = 1.256479e+46 FES = 331100 Diversity = 9.2255e-01 k_t
= [1, 1, 1, 1]
Generation 3320: Best = 1.083249e+46 FES = 332100 Diversity = 9.3778e-01 k_t
= [4, 2, 3, 3]
Generation 3330: Best = 8.466467e+45 FES = 333100 Diversity = 9.4635e-01 k_t
= [1, 1, 1, 1]
Generation 3340: Best = 7.785899e+45 FES = 334100 Diversity = 9.4779e-01 k_t
= [3, 4, 2, 2]
Generation 3350: Best = 7.261150e+45 FES = 335100 Diversity = 9.7059e-01 k_t
= [1, 1, 1, 1]
Generation 3360: Best = 6.594056e+45 FES = 336100 Diversity = 9.6588e-01 k_t
= [4, 2, 2, 3]
Generation 3370: Best = 5.722997e+45 FES = 337100 Diversity = 9.4152e-01 k_t
= [1, 1, 1, 1]
Generation 3380: Best = 5.048015e+45 FES = 338100 Diversity = 9.6070e-01 k_t
= [3, 4, 3, 2]
Generation 3390: Best = 4.609348e+45 FES = 339100 Diversity = 9.8836e-01 k_t
= [1, 1, 1, 1]
Generation 3400: Best = 4.019440e+45 FES = 340100 Diversity = 1.0096e+00 k_t
= [2, 4, 2, 2]
Generation 3410: Best = 3.522609e+45 FES = 341100 Diversity = 9.9626e-01 k_t
= [1, 1, 1, 1]
Generation 3420: Best = 2.989660e+45 FES = 342100 Diversity = 9.6788e-01 k_t
= [3, 4, 4, 4]
Generation 3430: Best = 2.866086e+45 FES = 343100 Diversity = 9.5391e-01 k_t
= [1, 1, 1, 1]
Generation 3440: Best = 2.621019e+45 FES = 344100 Diversity = 9.5598e-01 k_t
= [4, 4, 4, 4]
Generation 3450: Best = 2.321274e+45 FES = 345100 Diversity = 9.7645e-01 k_t
= [1, 1, 1, 1]
Generation 3460: Best = 2.224547e+45 FES = 346100 Diversity = 9.7027e-01 k_t

```

```

= [4, 3, 4, 3]
Generation 3470: Best = 2.015810e+45  FES = 347100  Diversity = 9.5284e-01  k_t
= [1, 1, 1, 1]
Generation 3480: Best = 1.714860e+45  FES = 348100  Diversity = 9.6821e-01  k_t
= [2, 3, 4, 2]
Generation 3490: Best = 1.572202e+45  FES = 349100  Diversity = 9.7342e-01  k_t
= [1, 1, 1, 1]
Generation 3500: Best = 1.511896e+45  FES = 350100  Diversity = 9.5847e-01  k_t
= [3, 4, 2, 2]
Generation 3510: Best = 1.377575e+45  FES = 351100  Diversity = 9.4106e-01  k_t
= [1, 1, 1, 1]
Generation 3520: Best = 1.232173e+45  FES = 352100  Diversity = 9.5403e-01  k_t
= [3, 4, 3, 3]
Generation 3530: Best = 1.107520e+45  FES = 353100  Diversity = 9.6242e-01  k_t
= [1, 1, 1, 1]
Generation 3540: Best = 1.054662e+45  FES = 354100  Diversity = 9.8135e-01  k_t
= [3, 4, 4, 2]
Generation 3550: Best = 9.361748e+44  FES = 355100  Diversity = 9.7996e-01  k_t
= [1, 1, 1, 1]
Generation 3560: Best = 8.890830e+44  FES = 356100  Diversity = 9.8181e-01  k_t
= [2, 2, 4, 4]
Generation 3570: Best = 8.439004e+44  FES = 357100  Diversity = 9.4397e-01  k_t
= [1, 1, 1, 1]
Generation 3580: Best = 7.493435e+44  FES = 358100  Diversity = 9.5903e-01  k_t
= [2, 2, 2, 2]
Generation 3590: Best = 6.986570e+44  FES = 359100  Diversity = 1.0130e+00  k_t
= [1, 1, 1, 1]
Generation 3600: Best = 6.230955e+44  FES = 360100  Diversity = 1.0123e+00  k_t
= [2, 4, 3, 2]
Generation 3610: Best = 6.081547e+44  FES = 361100  Diversity = 1.0178e+00  k_t
= [1, 1, 1, 1]
Generation 3620: Best = 5.567482e+44  FES = 362100  Diversity = 1.0053e+00  k_t
= [3, 3, 4, 3]
Generation 3630: Best = 5.161956e+44  FES = 363100  Diversity = 1.0007e+00  k_t
= [1, 1, 1, 1]
Generation 3640: Best = 4.671225e+44  FES = 364100  Diversity = 9.9751e-01  k_t
= [4, 3, 3, 3]
Generation 3650: Best = 4.527381e+44  FES = 365100  Diversity = 1.0025e+00  k_t
= [1, 1, 1, 1]
Generation 3660: Best = 4.156573e+44  FES = 366100  Diversity = 9.9824e-01  k_t
= [2, 3, 3, 4]
Generation 3670: Best = 3.981612e+44  FES = 367100  Diversity = 9.9650e-01  k_t
= [1, 1, 1, 1]
Generation 3680: Best = 3.665765e+44  FES = 368100  Diversity = 1.0107e+00  k_t
= [2, 4, 3, 2]
Generation 3690: Best = 3.443030e+44  FES = 369100  Diversity = 1.0080e+00  k_t
= [1, 1, 1, 1]
Generation 3700: Best = 3.248440e+44  FES = 370100  Diversity = 9.6907e-01  k_t
= [3, 2, 2, 3]
Generation 3710: Best = 3.029432e+44  FES = 371100  Diversity = 9.5885e-01  k_t
= [1, 1, 1, 1]
Generation 3720: Best = 2.746141e+44  FES = 372100  Diversity = 9.5592e-01  k_t
= [2, 4, 3, 4]
Generation 3730: Best = 2.577113e+44  FES = 373100  Diversity = 9.8077e-01  k_t

```

```

= [1, 1, 1, 1]
Generation 3740: Best = 2.386403e+44 FES = 374100 Diversity = 1.0071e+00 k_t
= [4, 3, 4, 3]
Generation 3750: Best = 2.131163e+44 FES = 375100 Diversity = 1.0321e+00 k_t
= [1, 1, 1, 1]
Generation 3760: Best = 1.999332e+44 FES = 376100 Diversity = 1.0060e+00 k_t
= [4, 4, 3, 3]
Generation 3770: Best = 1.799246e+44 FES = 377100 Diversity = 1.0218e+00 k_t
= [1, 1, 1, 1]
Generation 3780: Best = 1.700839e+44 FES = 378100 Diversity = 1.0187e+00 k_t
= [2, 4, 2, 3]
Generation 3790: Best = 1.506668e+44 FES = 379100 Diversity = 1.0323e+00 k_t
= [1, 1, 1, 1]
Generation 3800: Best = 1.442216e+44 FES = 380100 Diversity = 1.0065e+00 k_t
= [3, 2, 3, 4]
Generation 3810: Best = 1.324057e+44 FES = 381100 Diversity = 9.7020e-01 k_t
= [1, 1, 1, 1]
Generation 3820: Best = 1.266777e+44 FES = 382100 Diversity = 9.8771e-01 k_t
= [4, 3, 2, 2]
Generation 3830: Best = 1.175706e+44 FES = 383100 Diversity = 9.5746e-01 k_t
= [1, 1, 1, 1]
Generation 3840: Best = 1.121855e+44 FES = 384100 Diversity = 9.6480e-01 k_t
= [3, 2, 3, 3]
Generation 3850: Best = 1.020485e+44 FES = 385100 Diversity = 9.7431e-01 k_t
= [1, 1, 1, 1]
Generation 3860: Best = 9.011770e+43 FES = 386100 Diversity = 9.9577e-01 k_t
= [2, 4, 3, 3]
Generation 3870: Best = 8.302509e+43 FES = 387100 Diversity = 1.0119e+00 k_t
= [1, 1, 1, 1]
Generation 3880: Best = 7.801907e+43 FES = 388100 Diversity = 1.0137e+00 k_t
= [3, 4, 4, 2]
Generation 3890: Best = 7.109592e+43 FES = 389100 Diversity = 1.0677e+00 k_t
= [1, 1, 1, 1]
Generation 3900: Best = 6.751381e+43 FES = 390100 Diversity = 1.0539e+00 k_t
= [4, 3, 3, 4]
Generation 3910: Best = 6.085196e+43 FES = 391100 Diversity = 1.0570e+00 k_t
= [1, 1, 1, 1]
Generation 3920: Best = 5.956016e+43 FES = 392100 Diversity = 1.0804e+00 k_t
= [2, 2, 2, 4]
Generation 3930: Best = 5.490190e+43 FES = 393100 Diversity = 1.0726e+00 k_t
= [1, 1, 1, 1]
Generation 3940: Best = 5.158535e+43 FES = 394100 Diversity = 9.9577e-01 k_t
= [2, 4, 3, 3]
Generation 3950: Best = 4.897034e+43 FES = 395100 Diversity = 9.8792e-01 k_t
= [1, 1, 1, 1]
Generation 3960: Best = 4.722131e+43 FES = 396100 Diversity = 9.6622e-01 k_t
= [3, 4, 3, 4]
Generation 3970: Best = 4.219812e+43 FES = 397100 Diversity = 9.6728e-01 k_t
= [1, 1, 1, 1]
Generation 3980: Best = 3.810744e+43 FES = 398100 Diversity = 9.9599e-01 k_t
= [3, 4, 4, 2]
Generation 3990: Best = 3.601472e+43 FES = 399100 Diversity = 1.0149e+00 k_t
= [1, 1, 1, 1]
Generation 4000: Best = 3.277568e+43 FES = 400100 Diversity = 9.7761e-01 k_t

```

```

= [3, 2, 2, 3]
Generation 4010: Best = 2.981362e+43  FES = 401100  Diversity = 1.0073e+00  k_t
= [1, 1, 1, 1]
Generation 4020: Best = 2.819913e+43  FES = 402100  Diversity = 9.9687e-01  k_t
= [3, 3, 3, 3]
Generation 4030: Best = 2.606129e+43  FES = 403100  Diversity = 9.7435e-01  k_t
= [1, 1, 1, 1]
Generation 4040: Best = 2.353201e+43  FES = 404100  Diversity = 9.8992e-01  k_t
= [4, 3, 4, 4]
Generation 4050: Best = 2.196933e+43  FES = 405100  Diversity = 1.0072e+00  k_t
= [1, 1, 1, 1]
Generation 4060: Best = 1.950283e+43  FES = 406100  Diversity = 1.0436e+00  k_t
= [4, 2, 3, 4]
Generation 4070: Best = 1.818297e+43  FES = 407100  Diversity = 1.0716e+00  k_t
= [1, 1, 1, 1]
Generation 4080: Best = 1.690772e+43  FES = 408100  Diversity = 1.0324e+00  k_t
= [4, 3, 2, 3]
Generation 4090: Best = 1.600017e+43  FES = 409100  Diversity = 1.0134e+00  k_t
= [1, 1, 1, 1]
Generation 4100: Best = 1.403174e+43  FES = 410100  Diversity = 9.8747e-01  k_t
= [2, 3, 4, 2]
Generation 4110: Best = 1.325010e+43  FES = 411100  Diversity = 1.0140e+00  k_t
= [1, 1, 1, 1]
Generation 4120: Best = 1.160265e+43  FES = 412100  Diversity = 1.0203e+00  k_t
= [4, 4, 3, 4]
Generation 4130: Best = 1.031268e+43  FES = 413100  Diversity = 1.0742e+00  k_t
= [1, 1, 1, 1]
Generation 4140: Best = 9.747138e+42  FES = 414100  Diversity = 1.0659e+00  k_t
= [2, 4, 3, 3]
Generation 4150: Best = 8.781619e+42  FES = 415100  Diversity = 1.0380e+00  k_t
= [1, 1, 1, 1]
Generation 4160: Best = 8.322343e+42  FES = 416100  Diversity = 1.0195e+00  k_t
= [4, 4, 3, 3]
Generation 4170: Best = 7.761902e+42  FES = 417100  Diversity = 1.0085e+00  k_t
= [1, 1, 1, 1]
Generation 4180: Best = 6.907708e+42  FES = 418100  Diversity = 1.0207e+00  k_t
= [3, 4, 3, 4]
Generation 4190: Best = 6.603981e+42  FES = 419100  Diversity = 1.0119e+00  k_t
= [1, 1, 1, 1]
Generation 4200: Best = 6.059776e+42  FES = 420100  Diversity = 1.0113e+00  k_t
= [3, 4, 2, 2]
Generation 4210: Best = 5.766615e+42  FES = 421100  Diversity = 1.0110e+00  k_t
= [1, 1, 1, 1]
Generation 4220: Best = 5.287528e+42  FES = 422100  Diversity = 9.9024e-01  k_t
= [2, 2, 3, 3]
Generation 4230: Best = 4.843521e+42  FES = 423100  Diversity = 9.7913e-01  k_t
= [1, 1, 1, 1]
Generation 4240: Best = 4.429282e+42  FES = 424100  Diversity = 1.0041e+00  k_t
= [3, 4, 3, 3]
Generation 4250: Best = 4.227024e+42  FES = 425100  Diversity = 9.8943e-01  k_t
= [1, 1, 1, 1]
Generation 4260: Best = 3.798969e+42  FES = 426100  Diversity = 1.0137e+00  k_t
= [3, 3, 4, 3]
Generation 4270: Best = 3.601688e+42  FES = 427100  Diversity = 9.6892e-01  k_t

```



```

= [1, 1, 1, 1]
Generation 4280: Best = 3.360157e+42  FES = 428100  Diversity = 9.9530e-01  k_t
= [2, 2, 2, 4]
Generation 4290: Best = 3.086712e+42  FES = 429100  Diversity = 9.9003e-01  k_t
= [1, 1, 1, 1]
Generation 4300: Best = 2.760185e+42  FES = 430100  Diversity = 9.6885e-01  k_t
= [3, 3, 3, 2]
Generation 4310: Best = 2.326956e+42  FES = 431100  Diversity = 1.0170e+00  k_t
= [1, 1, 1, 1]
Generation 4320: Best = 2.109091e+42  FES = 432100  Diversity = 1.0270e+00  k_t
= [2, 3, 4, 3]
Generation 4330: Best = 1.943711e+42  FES = 433100  Diversity = 1.0665e+00  k_t
= [1, 1, 1, 1]
Generation 4340: Best = 1.773739e+42  FES = 434100  Diversity = 1.0595e+00  k_t
= [3, 2, 3, 2]
Generation 4350: Best = 1.455090e+42  FES = 435100  Diversity = 1.0454e+00  k_t
= [1, 1, 1, 1]
Generation 4360: Best = 1.449307e+42  FES = 436100  Diversity = 1.0338e+00  k_t
= [2, 2, 2, 2]
Generation 4370: Best = 1.239451e+42  FES = 437100  Diversity = 1.0446e+00  k_t
= [1, 1, 1, 1]
Generation 4380: Best = 1.156931e+42  FES = 438100  Diversity = 1.0467e+00  k_t
= [4, 3, 2, 2]
Generation 4390: Best = 9.950521e+41  FES = 439100  Diversity = 1.0561e+00  k_t
= [1, 1, 1, 1]
Generation 4400: Best = 9.719326e+41  FES = 440100  Diversity = 1.0608e+00  k_t
= [4, 3, 3, 2]
Generation 4410: Best = 8.507661e+41  FES = 441100  Diversity = 1.0522e+00  k_t
= [1, 1, 1, 1]
Generation 4420: Best = 8.199087e+41  FES = 442100  Diversity = 1.0439e+00  k_t
= [3, 4, 2, 4]
Generation 4430: Best = 7.012653e+41  FES = 443100  Diversity = 1.0166e+00  k_t
= [1, 1, 1, 1]
Generation 4440: Best = 6.372787e+41  FES = 444100  Diversity = 1.0188e+00  k_t
= [2, 4, 3, 4]
Generation 4450: Best = 5.926309e+41  FES = 445100  Diversity = 9.7368e-01  k_t
= [1, 1, 1, 1]
Generation 4460: Best = 5.662182e+41  FES = 446100  Diversity = 9.9232e-01  k_t
= [3, 4, 4, 3]
Generation 4470: Best = 5.155976e+41  FES = 447100  Diversity = 9.7317e-01  k_t
= [1, 1, 1, 1]
Generation 4480: Best = 4.701736e+41  FES = 448100  Diversity = 1.0244e+00  k_t
= [4, 4, 4, 4]
Generation 4490: Best = 4.372181e+41  FES = 449100  Diversity = 1.0651e+00  k_t
= [1, 1, 1, 1]
Generation 4500: Best = 3.860507e+41  FES = 450100  Diversity = 1.0702e+00  k_t
= [4, 3, 2, 4]
Generation 4510: Best = 3.615290e+41  FES = 451100  Diversity = 1.0476e+00  k_t
= [1, 1, 1, 1]
Generation 4520: Best = 3.342483e+41  FES = 452100  Diversity = 1.0238e+00  k_t
= [4, 4, 3, 2]
Generation 4530: Best = 3.049698e+41  FES = 453100  Diversity = 1.0121e+00  k_t
= [1, 1, 1, 1]
Generation 4540: Best = 2.710718e+41  FES = 454100  Diversity = 1.0177e+00  k_t

```

```

= [3, 2, 4, 4]
Generation 4550: Best = 2.498422e+41 FES = 455100 Diversity = 1.0163e+00 k_t
= [1, 1, 1, 1]
Generation 4560: Best = 2.298939e+41 FES = 456100 Diversity = 1.0360e+00 k_t
= [2, 2, 2, 2]
Generation 4570: Best = 2.155234e+41 FES = 457100 Diversity = 1.0141e+00 k_t
= [1, 1, 1, 1]
Generation 4580: Best = 2.109043e+41 FES = 458100 Diversity = 1.0455e+00 k_t
= [4, 2, 4, 3]
Generation 4590: Best = 1.976927e+41 FES = 459100 Diversity = 1.0382e+00 k_t
= [1, 1, 1, 1]
Generation 4600: Best = 1.873620e+41 FES = 460100 Diversity = 1.0432e+00 k_t
= [4, 3, 3, 3]
Generation 4610: Best = 1.808746e+41 FES = 461100 Diversity = 1.0406e+00 k_t
= [1, 1, 1, 1]
Generation 4620: Best = 1.731244e+41 FES = 462100 Diversity = 1.0350e+00 k_t
= [3, 4, 4, 3]
Generation 4630: Best = 1.621506e+41 FES = 463100 Diversity = 1.0115e+00 k_t
= [1, 1, 1, 1]
Generation 4640: Best = 1.545253e+41 FES = 464100 Diversity = 1.0109e+00 k_t
= [4, 4, 2, 4]
Generation 4650: Best = 1.422932e+41 FES = 465100 Diversity = 1.0419e+00 k_t
= [1, 1, 1, 1]
Generation 4660: Best = 1.359392e+41 FES = 466100 Diversity = 1.0812e+00 k_t
= [2, 2, 4, 3]
Generation 4670: Best = 1.293828e+41 FES = 467100 Diversity = 1.0780e+00 k_t
= [1, 1, 1, 1]
Generation 4680: Best = 1.250114e+41 FES = 468100 Diversity = 1.0871e+00 k_t
= [2, 4, 2, 2]
Generation 4690: Best = 1.144668e+41 FES = 469100 Diversity = 1.0261e+00 k_t
= [1, 1, 1, 1]
Generation 4700: Best = 1.083369e+41 FES = 470100 Diversity = 9.9985e-01 k_t
= [3, 3, 2, 3]
Generation 4710: Best = 1.036503e+41 FES = 471100 Diversity = 1.0034e+00 k_t
= [1, 1, 1, 1]
Generation 4720: Best = 1.018803e+41 FES = 472100 Diversity = 1.0010e+00 k_t
= [4, 2, 3, 2]
Generation 4730: Best = 9.564383e+40 FES = 473100 Diversity = 1.0226e+00 k_t
= [1, 1, 1, 1]
Generation 4740: Best = 9.094557e+40 FES = 474100 Diversity = 1.0201e+00 k_t
= [2, 2, 3, 4]
Generation 4750: Best = 8.891727e+40 FES = 475100 Diversity = 1.0150e+00 k_t
= [1, 1, 1, 1]
Generation 4760: Best = 8.395996e+40 FES = 476100 Diversity = 1.0142e+00 k_t
= [3, 4, 3, 3]
Generation 4770: Best = 8.352830e+40 FES = 477100 Diversity = 9.8574e-01 k_t
= [1, 1, 1, 1]
Generation 4780: Best = 7.898788e+40 FES = 478100 Diversity = 9.6475e-01 k_t
= [2, 3, 4, 2]
Generation 4790: Best = 7.767295e+40 FES = 479100 Diversity = 9.2213e-01 k_t
= [1, 1, 1, 1]
Generation 4800: Best = 7.473749e+40 FES = 480100 Diversity = 9.0016e-01 k_t
= [2, 3, 3, 4]
Generation 4810: Best = 7.253529e+40 FES = 481100 Diversity = 8.9336e-01 k_t

```

```

= [1, 1, 1, 1]
Generation 4820: Best = 6.941054e+40 FES = 482100 Diversity = 9.3725e-01 k_t
= [2, 4, 3, 4]
Generation 4830: Best = 6.446088e+40 FES = 483100 Diversity = 9.5940e-01 k_t
= [1, 1, 1, 1]
Generation 4840: Best = 6.209017e+40 FES = 484100 Diversity = 1.0139e+00 k_t
= [4, 2, 2, 2]
Generation 4850: Best = 6.003303e+40 FES = 485100 Diversity = 1.0299e+00 k_t
= [1, 1, 1, 1]
Generation 4860: Best = 5.788541e+40 FES = 486100 Diversity = 1.0183e+00 k_t
= [4, 2, 4, 4]
Generation 4870: Best = 5.575871e+40 FES = 487100 Diversity = 1.0279e+00 k_t
= [1, 1, 1, 1]
Generation 4880: Best = 5.306280e+40 FES = 488100 Diversity = 1.0083e+00 k_t
= [4, 2, 3, 2]
Generation 4890: Best = 5.009191e+40 FES = 489100 Diversity = 1.0105e+00 k_t
= [1, 1, 1, 1]
Generation 4900: Best = 4.915391e+40 FES = 490100 Diversity = 9.8054e-01 k_t
= [4, 3, 3, 2]
Generation 4910: Best = 4.635490e+40 FES = 491100 Diversity = 9.7865e-01 k_t
= [1, 1, 1, 1]
Generation 4920: Best = 4.387228e+40 FES = 492100 Diversity = 1.0048e+00 k_t
= [2, 4, 2, 3]
Generation 4930: Best = 4.285210e+40 FES = 493100 Diversity = 1.0294e+00 k_t
= [1, 1, 1, 1]
Generation 4940: Best = 4.151026e+40 FES = 494100 Diversity = 1.0504e+00 k_t
= [4, 4, 4, 3]
Generation 4950: Best = 3.967748e+40 FES = 495100 Diversity = 1.0518e+00 k_t
= [1, 1, 1, 1]
Generation 4960: Best = 3.795902e+40 FES = 496100 Diversity = 1.0587e+00 k_t
= [2, 4, 2, 4]
Generation 4970: Best = 3.659158e+40 FES = 497100 Diversity = 1.0627e+00 k_t
= [1, 1, 1, 1]
Generation 4980: Best = 3.589257e+40 FES = 498100 Diversity = 1.0344e+00 k_t
= [2, 3, 4, 4]
Generation 4990: Best = 3.438979e+40 FES = 499100 Diversity = 1.0106e+00 k_t
= [1, 1, 1, 1]
Generation 5000: Best = 3.335009e+40 FES = 500100 Diversity = 1.0099e+00 k_t
= [2, 2, 3, 2]
Generation 5010: Best = 3.281760e+40 FES = 501100 Diversity = 1.0050e+00 k_t
= [1, 1, 1, 1]
Generation 5020: Best = 3.160621e+40 FES = 502100 Diversity = 9.9970e-01 k_t
= [2, 2, 3, 3]
Generation 5030: Best = 3.058843e+40 FES = 503100 Diversity = 1.0065e+00 k_t
= [1, 1, 1, 1]
Generation 5040: Best = 2.874523e+40 FES = 504100 Diversity = 1.0620e+00 k_t
= [3, 3, 3, 4]
Generation 5050: Best = 2.732081e+40 FES = 505100 Diversity = 1.0853e+00 k_t
= [1, 1, 1, 1]
Generation 5060: Best = 2.630371e+40 FES = 506100 Diversity = 1.0885e+00 k_t
= [2, 4, 4, 4]
Generation 5070: Best = 2.510581e+40 FES = 507100 Diversity = 1.0580e+00 k_t
= [1, 1, 1, 1]
Generation 5080: Best = 2.368972e+40 FES = 508100 Diversity = 1.0572e+00 k_t

```

```

= [4, 2, 3, 4]
Generation 5090: Best = 2.302929e+40 FES = 509100 Diversity = 1.0503e+00 k_t
= [1, 1, 1, 1]
Generation 5100: Best = 2.202710e+40 FES = 510100 Diversity = 1.0665e+00 k_t
= [4, 3, 4, 4]
Generation 5110: Best = 2.063324e+40 FES = 511100 Diversity = 1.0723e+00 k_t
= [1, 1, 1, 1]
Generation 5120: Best = 2.024283e+40 FES = 512100 Diversity = 1.0359e+00 k_t
= [4, 3, 4, 2]
Generation 5130: Best = 1.973327e+40 FES = 513100 Diversity = 1.0163e+00 k_t
= [1, 1, 1, 1]
Generation 5140: Best = 1.901124e+40 FES = 514100 Diversity = 1.0029e+00 k_t
= [2, 3, 4, 2]
Generation 5150: Best = 1.807955e+40 FES = 515100 Diversity = 1.0103e+00 k_t
= [1, 1, 1, 1]
Generation 5160: Best = 1.739497e+40 FES = 516100 Diversity = 9.7886e-01 k_t
= [3, 3, 2, 3]
Generation 5170: Best = 1.629801e+40 FES = 517100 Diversity = 9.8489e-01 k_t
= [1, 1, 1, 1]
Generation 5180: Best = 1.557179e+40 FES = 518100 Diversity = 9.9828e-01 k_t
= [2, 3, 3, 2]
Generation 5190: Best = 1.465918e+40 FES = 519100 Diversity = 9.7544e-01 k_t
= [1, 1, 1, 1]
Generation 5200: Best = 1.333963e+40 FES = 520100 Diversity = 1.0008e+00 k_t
= [4, 4, 3, 2]
Generation 5210: Best = 1.283787e+40 FES = 521100 Diversity = 1.0053e+00 k_t
= [1, 1, 1, 1]
Generation 5220: Best = 1.210988e+40 FES = 522100 Diversity = 1.0146e+00 k_t
= [2, 4, 4, 4]
Generation 5230: Best = 1.154424e+40 FES = 523100 Diversity = 1.0261e+00 k_t
= [1, 1, 1, 1]
Generation 5240: Best = 1.043399e+40 FES = 524100 Diversity = 1.0474e+00 k_t
= [4, 2, 3, 3]
Generation 5250: Best = 9.923845e+39 FES = 525100 Diversity = 1.0286e+00 k_t
= [1, 1, 1, 1]
Generation 5260: Best = 9.498388e+39 FES = 526100 Diversity = 1.0227e+00 k_t
= [4, 4, 2, 2]
Generation 5270: Best = 8.737496e+39 FES = 527100 Diversity = 1.0041e+00 k_t
= [1, 1, 1, 1]
Generation 5280: Best = 8.220283e+39 FES = 528100 Diversity = 9.9703e-01 k_t
= [2, 2, 4, 2]
Generation 5290: Best = 7.780118e+39 FES = 529100 Diversity = 1.0159e+00 k_t
= [1, 1, 1, 1]
Generation 5300: Best = 6.922782e+39 FES = 530100 Diversity = 9.9953e-01 k_t
= [3, 4, 2, 4]
Generation 5310: Best = 6.519352e+39 FES = 531100 Diversity = 9.9469e-01 k_t
= [1, 1, 1, 1]
Generation 5320: Best = 6.049264e+39 FES = 532100 Diversity = 1.0195e+00 k_t
= [4, 2, 2, 3]
Generation 5330: Best = 5.330615e+39 FES = 533100 Diversity = 1.0221e+00 k_t
= [1, 1, 1, 1]
Generation 5340: Best = 4.813033e+39 FES = 534100 Diversity = 1.0576e+00 k_t
= [3, 2, 2, 2]
Generation 5350: Best = 4.372233e+39 FES = 535100 Diversity = 1.1175e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 5360: Best = 3.660266e+39 FES = 536100 Diversity = 1.1130e+00 k_t
= [3, 4, 4, 2]
Generation 5370: Best = 3.416455e+39 FES = 537100 Diversity = 1.1158e+00 k_t
= [1, 1, 1, 1]
Generation 5380: Best = 2.972057e+39 FES = 538100 Diversity = 1.1211e+00 k_t
= [4, 3, 2, 3]
Generation 5390: Best = 2.614283e+39 FES = 539100 Diversity = 1.0812e+00 k_t
= [1, 1, 1, 1]
Generation 5400: Best = 2.446124e+39 FES = 540100 Diversity = 1.0817e+00 k_t
= [4, 3, 4, 4]
Generation 5410: Best = 2.126434e+39 FES = 541100 Diversity = 1.0774e+00 k_t
= [1, 1, 1, 1]
Generation 5420: Best = 1.956632e+39 FES = 542100 Diversity = 1.0665e+00 k_t
= [2, 3, 4, 3]
Generation 5430: Best = 1.813642e+39 FES = 543100 Diversity = 1.0214e+00 k_t
= [1, 1, 1, 1]
Generation 5440: Best = 1.726409e+39 FES = 544100 Diversity = 1.0255e+00 k_t
= [3, 2, 3, 2]
Generation 5450: Best = 1.486645e+39 FES = 545100 Diversity = 1.0211e+00 k_t
= [1, 1, 1, 1]
Generation 5460: Best = 1.454245e+39 FES = 546100 Diversity = 9.7673e-01 k_t
= [4, 4, 2, 2]
Generation 5470: Best = 1.397635e+39 FES = 547100 Diversity = 9.8842e-01 k_t
= [1, 1, 1, 1]
Generation 5480: Best = 1.311039e+39 FES = 548100 Diversity = 9.4513e-01 k_t
= [2, 2, 2, 3]
Generation 5490: Best = 1.230808e+39 FES = 549100 Diversity = 9.2000e-01 k_t
= [1, 1, 1, 1]
Generation 5500: Best = 1.143304e+39 FES = 550100 Diversity = 9.3016e-01 k_t
= [4, 2, 3, 4]
Generation 5510: Best = 1.060129e+39 FES = 551100 Diversity = 9.5604e-01 k_t
= [1, 1, 1, 1]
Generation 5520: Best = 1.027462e+39 FES = 552100 Diversity = 9.4912e-01 k_t
= [4, 2, 3, 3]
Generation 5530: Best = 9.906483e+38 FES = 553100 Diversity = 9.6717e-01 k_t
= [1, 1, 1, 1]
Generation 5540: Best = 9.515300e+38 FES = 554100 Diversity = 9.5421e-01 k_t
= [3, 2, 3, 2]
Generation 5550: Best = 9.154063e+38 FES = 555100 Diversity = 9.2210e-01 k_t
= [1, 1, 1, 1]
Generation 5560: Best = 8.543775e+38 FES = 556100 Diversity = 9.2980e-01 k_t
= [4, 3, 3, 4]
Generation 5570: Best = 8.121665e+38 FES = 557100 Diversity = 9.0418e-01 k_t
= [1, 1, 1, 1]
Generation 5580: Best = 7.568096e+38 FES = 558100 Diversity = 8.9153e-01 k_t
= [2, 2, 4, 3]
Generation 5590: Best = 7.441439e+38 FES = 559100 Diversity = 9.0591e-01 k_t
= [1, 1, 1, 1]
Generation 5600: Best = 7.101542e+38 FES = 560100 Diversity = 9.3730e-01 k_t
= [3, 2, 4, 3]
Generation 5610: Best = 6.982021e+38 FES = 561100 Diversity = 9.6378e-01 k_t
= [1, 1, 1, 1]
Generation 5620: Best = 6.412691e+38 FES = 562100 Diversity = 1.0092e+00 k_t

```

```

= [4, 4, 2, 3]
Generation 5630: Best = 6.265673e+38 FES = 563100 Diversity = 1.0166e+00 k_t
= [1, 1, 1, 1]
Generation 5640: Best = 6.086162e+38 FES = 564100 Diversity = 9.9077e-01 k_t
= [4, 4, 2, 4]
Generation 5650: Best = 5.770873e+38 FES = 565100 Diversity = 9.8088e-01 k_t
= [1, 1, 1, 1]
Generation 5660: Best = 5.483674e+38 FES = 566100 Diversity = 9.8743e-01 k_t
= [4, 3, 3, 2]
Generation 5670: Best = 5.309427e+38 FES = 567100 Diversity = 9.6459e-01 k_t
= [1, 1, 1, 1]
Generation 5680: Best = 5.126151e+38 FES = 568100 Diversity = 9.6975e-01 k_t
= [3, 2, 4, 3]
Generation 5690: Best = 4.941713e+38 FES = 569100 Diversity = 9.3485e-01 k_t
= [1, 1, 1, 1]
Generation 5700: Best = 4.902671e+38 FES = 570100 Diversity = 9.1188e-01 k_t
= [4, 2, 4, 2]
Generation 5710: Best = 4.772789e+38 FES = 571100 Diversity = 8.9367e-01 k_t
= [1, 1, 1, 1]
Generation 5720: Best = 4.656494e+38 FES = 572100 Diversity = 8.8187e-01 k_t
= [2, 3, 2, 3]
Generation 5730: Best = 4.535109e+38 FES = 573100 Diversity = 8.8755e-01 k_t
= [1, 1, 1, 1]
Generation 5740: Best = 4.429371e+38 FES = 574100 Diversity = 8.9665e-01 k_t
= [3, 3, 4, 2]
Generation 5750: Best = 4.323174e+38 FES = 575100 Diversity = 8.8699e-01 k_t
= [1, 1, 1, 1]
Generation 5760: Best = 4.258818e+38 FES = 576100 Diversity = 8.8664e-01 k_t
= [2, 3, 4, 4]
Generation 5770: Best = 4.155755e+38 FES = 577100 Diversity = 8.5904e-01 k_t
= [1, 1, 1, 1]
Generation 5780: Best = 4.115633e+38 FES = 578100 Diversity = 8.7394e-01 k_t
= [2, 3, 2, 3]
Generation 5790: Best = 4.044425e+38 FES = 579100 Diversity = 8.8210e-01 k_t
= [1, 1, 1, 1]
Generation 5800: Best = 3.961415e+38 FES = 580100 Diversity = 9.0848e-01 k_t
= [4, 3, 3, 2]
Generation 5810: Best = 3.887851e+38 FES = 581100 Diversity = 9.5997e-01 k_t
= [1, 1, 1, 1]
Generation 5820: Best = 3.782519e+38 FES = 582100 Diversity = 9.7442e-01 k_t
= [3, 4, 4, 2]
Generation 5830: Best = 3.722045e+38 FES = 583100 Diversity = 9.9734e-01 k_t
= [1, 1, 1, 1]
Generation 5840: Best = 3.656023e+38 FES = 584100 Diversity = 9.7630e-01 k_t
= [2, 4, 4, 3]
Generation 5850: Best = 3.582462e+38 FES = 585100 Diversity = 9.7985e-01 k_t
= [1, 1, 1, 1]
Generation 5860: Best = 3.498768e+38 FES = 586100 Diversity = 1.0004e+00 k_t
= [4, 4, 3, 4]
Generation 5870: Best = 3.395854e+38 FES = 587100 Diversity = 1.0097e+00 k_t
= [1, 1, 1, 1]
Generation 5880: Best = 3.321309e+38 FES = 588100 Diversity = 9.8866e-01 k_t
= [2, 2, 3, 2]
Generation 5890: Best = 3.232917e+38 FES = 589100 Diversity = 9.6304e-01 k_t

```

```

= [1, 1, 1, 1]
Generation 5900: Best = 3.216739e+38  FES = 590100  Diversity = 1.0096e+00  k_t
= [3, 3, 2, 3]
Generation 5910: Best = 3.107391e+38  FES = 591100  Diversity = 1.0260e+00  k_t
= [1, 1, 1, 1]
Generation 5920: Best = 3.078634e+38  FES = 592100  Diversity = 1.0279e+00  k_t
= [4, 2, 2, 4]
Generation 5930: Best = 3.029459e+38  FES = 593100  Diversity = 9.6997e-01  k_t
= [1, 1, 1, 1]
Generation 5940: Best = 2.982343e+38  FES = 594100  Diversity = 9.4776e-01  k_t
= [3, 4, 2, 3]
Generation 5950: Best = 2.938958e+38  FES = 595100  Diversity = 9.4270e-01  k_t
= [1, 1, 1, 1]
Generation 5960: Best = 2.806362e+38  FES = 596100  Diversity = 9.4425e-01  k_t
= [2, 3, 2, 3]
Generation 5970: Best = 2.767740e+38  FES = 597100  Diversity = 9.4649e-01  k_t
= [1, 1, 1, 1]
Generation 5980: Best = 2.716958e+38  FES = 598100  Diversity = 9.6991e-01  k_t
= [2, 4, 2, 3]
Generation 5990: Best = 2.613951e+38  FES = 599100  Diversity = 9.7814e-01  k_t
= [1, 1, 1, 1]
Generation 6000: Best = 2.613799e+38  FES = 600100  Diversity = 9.5894e-01  k_t
= [4, 4, 2, 4]
Generation 6010: Best = 2.552103e+38  FES = 601100  Diversity = 9.1436e-01  k_t
= [1, 1, 1, 1]
Generation 6020: Best = 2.514559e+38  FES = 602100  Diversity = 9.2693e-01  k_t
= [4, 4, 4, 4]
Generation 6030: Best = 2.503376e+38  FES = 603100  Diversity = 9.2475e-01  k_t
= [1, 1, 1, 1]
Generation 6040: Best = 2.450348e+38  FES = 604100  Diversity = 9.0072e-01  k_t
= [4, 2, 4, 4]
Generation 6050: Best = 2.413345e+38  FES = 605100  Diversity = 9.0442e-01  k_t
= [1, 1, 1, 1]
Generation 6060: Best = 2.375705e+38  FES = 606100  Diversity = 8.5864e-01  k_t
= [4, 3, 3, 4]
Generation 6070: Best = 2.337545e+38  FES = 607100  Diversity = 8.5078e-01  k_t
= [1, 1, 1, 1]
Generation 6080: Best = 2.315967e+38  FES = 608100  Diversity = 8.7990e-01  k_t
= [4, 2, 2, 3]
Generation 6090: Best = 2.242382e+38  FES = 609100  Diversity = 8.7044e-01  k_t
= [1, 1, 1, 1]
Generation 6100: Best = 2.205082e+38  FES = 610100  Diversity = 9.0153e-01  k_t
= [4, 3, 4, 3]
Generation 6110: Best = 2.179202e+38  FES = 611100  Diversity = 9.1072e-01  k_t
= [1, 1, 1, 1]
Generation 6120: Best = 2.121828e+38  FES = 612100  Diversity = 8.6808e-01  k_t
= [3, 2, 4, 4]
Generation 6130: Best = 2.086009e+38  FES = 613100  Diversity = 9.0584e-01  k_t
= [1, 1, 1, 1]
Generation 6140: Best = 1.983406e+38  FES = 614100  Diversity = 9.3461e-01  k_t
= [4, 2, 3, 3]
Generation 6150: Best = 1.945107e+38  FES = 615100  Diversity = 9.4869e-01  k_t
= [1, 1, 1, 1]
Generation 6160: Best = 1.905859e+38  FES = 616100  Diversity = 9.7043e-01  k_t

```

```

= [4, 3, 4, 3]
Generation 6170: Best = 1.846201e+38 FES = 617100 Diversity = 1.0150e+00 k_t
= [1, 1, 1, 1]
Generation 6180: Best = 1.808312e+38 FES = 618100 Diversity = 1.0370e+00 k_t
= [4, 3, 3, 3]
Generation 6190: Best = 1.752806e+38 FES = 619100 Diversity = 9.5528e-01 k_t
= [1, 1, 1, 1]
Generation 6200: Best = 1.739562e+38 FES = 620100 Diversity = 9.2347e-01 k_t
= [4, 4, 2, 4]
Generation 6210: Best = 1.698973e+38 FES = 621100 Diversity = 9.1634e-01 k_t
= [1, 1, 1, 1]
Generation 6220: Best = 1.664156e+38 FES = 622100 Diversity = 8.9678e-01 k_t
= [2, 3, 2, 3]
Generation 6230: Best = 1.625070e+38 FES = 623100 Diversity = 8.9950e-01 k_t
= [1, 1, 1, 1]
Generation 6240: Best = 1.579148e+38 FES = 624100 Diversity = 8.8363e-01 k_t
= [4, 4, 2, 3]
Generation 6250: Best = 1.552413e+38 FES = 625100 Diversity = 8.5357e-01 k_t
= [1, 1, 1, 1]
Generation 6260: Best = 1.498071e+38 FES = 626100 Diversity = 8.8249e-01 k_t
= [2, 4, 3, 3]
Generation 6270: Best = 1.445772e+38 FES = 627100 Diversity = 9.3699e-01 k_t
= [1, 1, 1, 1]
Generation 6280: Best = 1.429076e+38 FES = 628100 Diversity = 9.6671e-01 k_t
= [4, 3, 3, 4]
Generation 6290: Best = 1.399337e+38 FES = 629100 Diversity = 9.6166e-01 k_t
= [1, 1, 1, 1]
Generation 6300: Best = 1.370996e+38 FES = 630100 Diversity = 9.1028e-01 k_t
= [4, 3, 2, 2]
Generation 6310: Best = 1.346824e+38 FES = 631100 Diversity = 9.3731e-01 k_t
= [1, 1, 1, 1]
Generation 6320: Best = 1.320559e+38 FES = 632100 Diversity = 9.1443e-01 k_t
= [2, 2, 3, 4]
Generation 6330: Best = 1.292109e+38 FES = 633100 Diversity = 9.2511e-01 k_t
= [1, 1, 1, 1]
Generation 6340: Best = 1.275406e+38 FES = 634100 Diversity = 8.9816e-01 k_t
= [3, 2, 3, 2]
Generation 6350: Best = 1.230793e+38 FES = 635100 Diversity = 9.3657e-01 k_t
= [1, 1, 1, 1]
Generation 6360: Best = 1.177999e+38 FES = 636100 Diversity = 9.4392e-01 k_t
= [4, 4, 4, 2]
Generation 6370: Best = 1.148337e+38 FES = 637100 Diversity = 9.4053e-01 k_t
= [1, 1, 1, 1]
Generation 6380: Best = 1.103045e+38 FES = 638100 Diversity = 9.3018e-01 k_t
= [2, 3, 4, 2]
Generation 6390: Best = 1.062497e+38 FES = 639100 Diversity = 9.2784e-01 k_t
= [1, 1, 1, 1]
Generation 6400: Best = 1.029314e+38 FES = 640100 Diversity = 9.3611e-01 k_t
= [4, 4, 2, 4]
Generation 6410: Best = 9.908704e+37 FES = 641100 Diversity = 9.0104e-01 k_t
= [1, 1, 1, 1]
Generation 6420: Best = 9.317898e+37 FES = 642100 Diversity = 8.9381e-01 k_t
= [2, 4, 2, 3]
Generation 6430: Best = 9.064787e+37 FES = 643100 Diversity = 9.0723e-01 k_t

```



```

= [1, 1, 1, 1]
Generation 6440: Best = 8.751891e+37 FES = 644100 Diversity = 9.1809e-01 k_t
= [4, 3, 3, 3]
Generation 6450: Best = 8.492621e+37 FES = 645100 Diversity = 8.9837e-01 k_t
= [1, 1, 1, 1]
Generation 6460: Best = 8.069972e+37 FES = 646100 Diversity = 9.5998e-01 k_t
= [2, 2, 4, 3]
Generation 6470: Best = 7.774355e+37 FES = 647100 Diversity = 9.4621e-01 k_t
= [1, 1, 1, 1]
Generation 6480: Best = 7.521078e+37 FES = 648100 Diversity = 9.5842e-01 k_t
= [3, 2, 4, 2]
Generation 6490: Best = 7.192911e+37 FES = 649100 Diversity = 9.5854e-01 k_t
= [1, 1, 1, 1]
Generation 6500: Best = 6.815682e+37 FES = 650100 Diversity = 9.4003e-01 k_t
= [4, 2, 3, 2]
Generation 6510: Best = 6.691824e+37 FES = 651100 Diversity = 9.3104e-01 k_t
= [1, 1, 1, 1]
Generation 6520: Best = 6.438659e+37 FES = 652100 Diversity = 9.7122e-01 k_t
= [2, 2, 2, 2]
Generation 6530: Best = 6.046763e+37 FES = 653100 Diversity = 9.8968e-01 k_t
= [1, 1, 1, 1]
Generation 6540: Best = 5.796497e+37 FES = 654100 Diversity = 1.0124e+00 k_t
= [2, 3, 2, 4]
Generation 6550: Best = 5.589794e+37 FES = 655100 Diversity = 1.0495e+00 k_t
= [1, 1, 1, 1]
Generation 6560: Best = 5.288984e+37 FES = 656100 Diversity = 1.0359e+00 k_t
= [2, 3, 3, 3]
Generation 6570: Best = 5.238196e+37 FES = 657100 Diversity = 9.8463e-01 k_t
= [1, 1, 1, 1]
Generation 6580: Best = 5.016087e+37 FES = 658100 Diversity = 9.6323e-01 k_t
= [3, 3, 4, 3]
Generation 6590: Best = 4.735701e+37 FES = 659100 Diversity = 9.8748e-01 k_t
= [1, 1, 1, 1]
Generation 6600: Best = 4.579497e+37 FES = 660100 Diversity = 9.9059e-01 k_t
= [2, 3, 4, 4]
Generation 6610: Best = 4.418522e+37 FES = 661100 Diversity = 9.7217e-01 k_t
= [1, 1, 1, 1]
Generation 6620: Best = 4.313175e+37 FES = 662100 Diversity = 9.7823e-01 k_t
= [4, 4, 3, 3]
Generation 6630: Best = 4.168204e+37 FES = 663100 Diversity = 9.6341e-01 k_t
= [1, 1, 1, 1]
Generation 6640: Best = 4.089938e+37 FES = 664100 Diversity = 9.8190e-01 k_t
= [3, 3, 4, 4]
Generation 6650: Best = 3.765990e+37 FES = 665100 Diversity = 9.6333e-01 k_t
= [1, 1, 1, 1]
Generation 6660: Best = 3.522433e+37 FES = 666100 Diversity = 9.5668e-01 k_t
= [4, 2, 4, 4]
Generation 6670: Best = 3.390576e+37 FES = 667100 Diversity = 9.6711e-01 k_t
= [1, 1, 1, 1]
Generation 6680: Best = 3.183292e+37 FES = 668100 Diversity = 9.5806e-01 k_t
= [4, 3, 2, 3]
Generation 6690: Best = 3.089345e+37 FES = 669100 Diversity = 9.7703e-01 k_t
= [1, 1, 1, 1]
Generation 6700: Best = 2.937901e+37 FES = 670100 Diversity = 9.8321e-01 k_t

```

```

= [3, 4, 4, 2]
Generation 6710: Best = 2.815977e+37  FES = 671100  Diversity = 9.7532e-01  k_t
= [1, 1, 1, 1]
Generation 6720: Best = 2.648182e+37  FES = 672100  Diversity = 9.9478e-01  k_t
= [3, 3, 3, 4]
Generation 6730: Best = 2.466936e+37  FES = 673100  Diversity = 1.0231e+00  k_t
= [1, 1, 1, 1]
Generation 6740: Best = 2.437346e+37  FES = 674100  Diversity = 9.9960e-01  k_t
= [3, 3, 4, 2]
Generation 6750: Best = 2.290194e+37  FES = 675100  Diversity = 9.9596e-01  k_t
= [1, 1, 1, 1]
Generation 6760: Best = 2.205061e+37  FES = 676100  Diversity = 1.0182e+00  k_t
= [3, 2, 4, 3]
Generation 6770: Best = 2.107936e+37  FES = 677100  Diversity = 9.9704e-01  k_t
= [1, 1, 1, 1]
Generation 6780: Best = 1.978143e+37  FES = 678100  Diversity = 9.7596e-01  k_t
= [4, 3, 3, 4]
Generation 6790: Best = 1.884313e+37  FES = 679100  Diversity = 9.3628e-01  k_t
= [1, 1, 1, 1]
Generation 6800: Best = 1.817093e+37  FES = 680100  Diversity = 9.5646e-01  k_t
= [4, 2, 3, 3]
Generation 6810: Best = 1.709667e+37  FES = 681100  Diversity = 9.7031e-01  k_t
= [1, 1, 1, 1]
Generation 6820: Best = 1.599429e+37  FES = 682100  Diversity = 9.6160e-01  k_t
= [3, 3, 2, 4]
Generation 6830: Best = 1.460564e+37  FES = 683100  Diversity = 9.8176e-01  k_t
= [1, 1, 1, 1]
Generation 6840: Best = 1.370627e+37  FES = 684100  Diversity = 9.8286e-01  k_t
= [2, 2, 3, 2]
Generation 6850: Best = 1.314507e+37  FES = 685100  Diversity = 9.8764e-01  k_t
= [1, 1, 1, 1]
Generation 6860: Best = 1.245517e+37  FES = 686100  Diversity = 9.6424e-01  k_t
= [3, 2, 2, 3]
Generation 6870: Best = 1.182741e+37  FES = 687100  Diversity = 9.3749e-01  k_t
= [1, 1, 1, 1]
Generation 6880: Best = 1.145489e+37  FES = 688100  Diversity = 9.3364e-01  k_t
= [4, 2, 3, 2]
Generation 6890: Best = 1.089595e+37  FES = 689100  Diversity = 9.5329e-01  k_t
= [1, 1, 1, 1]
Generation 6900: Best = 1.076174e+37  FES = 690100  Diversity = 9.1238e-01  k_t
= [3, 4, 3, 3]
Generation 6910: Best = 1.025464e+37  FES = 691100  Diversity = 8.8316e-01  k_t
= [1, 1, 1, 1]
Generation 6920: Best = 9.902776e+36  FES = 692100  Diversity = 9.2955e-01  k_t
= [2, 2, 4, 4]
Generation 6930: Best = 9.597664e+36  FES = 693100  Diversity = 9.2870e-01  k_t
= [1, 1, 1, 1]
Generation 6940: Best = 9.227135e+36  FES = 694100  Diversity = 9.3076e-01  k_t
= [2, 3, 2, 3]
Generation 6950: Best = 8.871399e+36  FES = 695100  Diversity = 9.7695e-01  k_t
= [1, 1, 1, 1]
Generation 6960: Best = 8.375795e+36  FES = 696100  Diversity = 9.8640e-01  k_t
= [3, 4, 2, 3]
Generation 6970: Best = 8.231437e+36  FES = 697100  Diversity = 9.6999e-01  k_t

```

```

= [1, 1, 1, 1]
Generation 6980: Best = 7.920897e+36 FES = 698100 Diversity = 9.6272e-01 k_t
= [3, 3, 3, 4]
Generation 6990: Best = 7.615246e+36 FES = 699100 Diversity = 9.6317e-01 k_t
= [1, 1, 1, 1]
Generation 7000: Best = 7.351154e+36 FES = 700100 Diversity = 9.4292e-01 k_t
= [2, 2, 4, 4]
Generation 7010: Best = 7.073402e+36 FES = 701100 Diversity = 9.3797e-01 k_t
= [1, 1, 1, 1]
Generation 7020: Best = 6.550553e+36 FES = 702100 Diversity = 9.0057e-01 k_t
= [2, 3, 2, 3]
Generation 7030: Best = 6.262250e+36 FES = 703100 Diversity = 9.0423e-01 k_t
= [1, 1, 1, 1]
Generation 7040: Best = 5.938646e+36 FES = 704100 Diversity = 8.7549e-01 k_t
= [3, 2, 3, 3]
Generation 7050: Best = 5.859786e+36 FES = 705100 Diversity = 8.7714e-01 k_t
= [1, 1, 1, 1]
Generation 7060: Best = 5.538843e+36 FES = 706100 Diversity = 8.5279e-01 k_t
= [3, 2, 3, 3]
Generation 7070: Best = 5.387011e+36 FES = 707100 Diversity = 8.6743e-01 k_t
= [1, 1, 1, 1]
Generation 7080: Best = 5.227670e+36 FES = 708100 Diversity = 8.4119e-01 k_t
= [2, 4, 2, 4]
Generation 7090: Best = 4.989295e+36 FES = 709100 Diversity = 8.5641e-01 k_t
= [1, 1, 1, 1]
Generation 7100: Best = 4.816030e+36 FES = 710100 Diversity = 8.6516e-01 k_t
= [4, 2, 4, 4]
Generation 7110: Best = 4.633294e+36 FES = 711100 Diversity = 8.8983e-01 k_t
= [1, 1, 1, 1]
Generation 7120: Best = 4.412573e+36 FES = 712100 Diversity = 8.9132e-01 k_t
= [2, 2, 2, 4]
Generation 7130: Best = 4.347573e+36 FES = 713100 Diversity = 8.6739e-01 k_t
= [1, 1, 1, 1]
Generation 7140: Best = 4.129716e+36 FES = 714100 Diversity = 8.8111e-01 k_t
= [4, 4, 2, 3]
Generation 7150: Best = 3.999161e+36 FES = 715100 Diversity = 8.6389e-01 k_t
= [1, 1, 1, 1]
Generation 7160: Best = 3.884240e+36 FES = 716100 Diversity = 8.7840e-01 k_t
= [3, 3, 3, 4]
Generation 7170: Best = 3.792154e+36 FES = 717100 Diversity = 8.8432e-01 k_t
= [1, 1, 1, 1]
Generation 7180: Best = 3.625191e+36 FES = 718100 Diversity = 9.1984e-01 k_t
= [3, 4, 3, 3]
Generation 7190: Best = 3.502740e+36 FES = 719100 Diversity = 8.8783e-01 k_t
= [1, 1, 1, 1]
Generation 7200: Best = 3.327234e+36 FES = 720100 Diversity = 8.9523e-01 k_t
= [3, 3, 3, 2]
Generation 7210: Best = 3.259208e+36 FES = 721100 Diversity = 8.9310e-01 k_t
= [1, 1, 1, 1]
Generation 7220: Best = 3.147023e+36 FES = 722100 Diversity = 9.0252e-01 k_t
= [3, 4, 3, 4]
Generation 7230: Best = 3.084280e+36 FES = 723100 Diversity = 9.0758e-01 k_t
= [1, 1, 1, 1]
Generation 7240: Best = 2.967763e+36 FES = 724100 Diversity = 9.2395e-01 k_t

```

```

= [2, 3, 4, 4]
Generation 7250: Best = 2.831025e+36 FES = 725100 Diversity = 8.9544e-01 k_t
= [1, 1, 1, 1]
Generation 7260: Best = 2.654024e+36 FES = 726100 Diversity = 9.3580e-01 k_t
= [4, 3, 2, 3]
Generation 7270: Best = 2.462230e+36 FES = 727100 Diversity = 9.4657e-01 k_t
= [1, 1, 1, 1]
Generation 7280: Best = 2.231720e+36 FES = 728100 Diversity = 9.1699e-01 k_t
= [3, 3, 4, 4]
Generation 7290: Best = 2.020867e+36 FES = 729100 Diversity = 9.5103e-01 k_t
= [1, 1, 1, 1]
Generation 7300: Best = 1.936551e+36 FES = 730100 Diversity = 9.1916e-01 k_t
= [3, 2, 4, 3]
Generation 7310: Best = 1.848716e+36 FES = 731100 Diversity = 9.2409e-01 k_t
= [1, 1, 1, 1]
Generation 7320: Best = 1.764966e+36 FES = 732100 Diversity = 9.4335e-01 k_t
= [4, 3, 4, 2]
Generation 7330: Best = 1.630929e+36 FES = 733100 Diversity = 9.3243e-01 k_t
= [1, 1, 1, 1]
Generation 7340: Best = 1.540123e+36 FES = 734100 Diversity = 9.2926e-01 k_t
= [4, 3, 3, 3]
Generation 7350: Best = 1.460074e+36 FES = 735100 Diversity = 9.3214e-01 k_t
= [1, 1, 1, 1]
Generation 7360: Best = 1.391836e+36 FES = 736100 Diversity = 9.2440e-01 k_t
= [2, 4, 3, 4]
Generation 7370: Best = 1.330473e+36 FES = 737100 Diversity = 9.0045e-01 k_t
= [1, 1, 1, 1]
Generation 7380: Best = 1.235084e+36 FES = 738100 Diversity = 9.0535e-01 k_t
= [4, 4, 4, 3]
Generation 7390: Best = 1.174943e+36 FES = 739100 Diversity = 8.6370e-01 k_t
= [1, 1, 1, 1]
Generation 7400: Best = 1.063466e+36 FES = 740100 Diversity = 8.7598e-01 k_t
= [4, 2, 4, 3]
Generation 7410: Best = 1.030739e+36 FES = 741100 Diversity = 9.1216e-01 k_t
= [1, 1, 1, 1]
Generation 7420: Best = 9.753758e+35 FES = 742100 Diversity = 9.2801e-01 k_t
= [3, 3, 4, 3]
Generation 7430: Best = 9.371486e+35 FES = 743100 Diversity = 8.9989e-01 k_t
= [1, 1, 1, 1]
Generation 7440: Best = 8.598886e+35 FES = 744100 Diversity = 9.3588e-01 k_t
= [4, 3, 4, 4]
Generation 7450: Best = 7.886165e+35 FES = 745100 Diversity = 9.6098e-01 k_t
= [1, 1, 1, 1]
Generation 7460: Best = 7.368003e+35 FES = 746100 Diversity = 9.4687e-01 k_t
= [2, 3, 3, 4]
Generation 7470: Best = 7.054972e+35 FES = 747100 Diversity = 9.3005e-01 k_t
= [1, 1, 1, 1]
Generation 7480: Best = 6.474544e+35 FES = 748100 Diversity = 8.9838e-01 k_t
= [2, 2, 2, 2]
Generation 7490: Best = 5.888525e+35 FES = 749100 Diversity = 8.9232e-01 k_t
= [1, 1, 1, 1]
Generation 7500: Best = 5.419318e+35 FES = 750100 Diversity = 9.1723e-01 k_t
= [3, 3, 2, 4]
Generation 7510: Best = 4.993327e+35 FES = 751100 Diversity = 8.8324e-01 k_t

```

```

= [1, 1, 1, 1]
Generation 7520: Best = 4.716132e+35  FES = 752100  Diversity = 8.5832e-01  k_t
= [2, 2, 4, 2]
Generation 7530: Best = 4.411036e+35  FES = 753100  Diversity = 8.4940e-01  k_t
= [1, 1, 1, 1]
Generation 7540: Best = 4.226287e+35  FES = 754100  Diversity = 8.5071e-01  k_t
= [2, 2, 2, 4]
Generation 7550: Best = 3.863811e+35  FES = 755100  Diversity = 8.6133e-01  k_t
= [1, 1, 1, 1]
Generation 7560: Best = 3.514763e+35  FES = 756100  Diversity = 8.4107e-01  k_t
= [3, 2, 2, 2]
Generation 7570: Best = 3.434786e+35  FES = 757100  Diversity = 8.5624e-01  k_t
= [1, 1, 1, 1]
Generation 7580: Best = 2.967303e+35  FES = 758100  Diversity = 8.5004e-01  k_t
= [4, 3, 3, 3]
Generation 7590: Best = 2.509943e+35  FES = 759100  Diversity = 8.7716e-01  k_t
= [1, 1, 1, 1]
Generation 7600: Best = 2.277231e+35  FES = 760100  Diversity = 9.1260e-01  k_t
= [2, 3, 2, 2]
Generation 7610: Best = 1.980098e+35  FES = 761100  Diversity = 9.1901e-01  k_t
= [1, 1, 1, 1]
Generation 7620: Best = 1.910216e+35  FES = 762100  Diversity = 9.1076e-01  k_t
= [2, 3, 4, 2]
Generation 7630: Best = 1.797291e+35  FES = 763100  Diversity = 9.0644e-01  k_t
= [1, 1, 1, 1]
Generation 7640: Best = 1.747023e+35  FES = 764100  Diversity = 8.7160e-01  k_t
= [4, 4, 4, 2]
Generation 7650: Best = 1.690928e+35  FES = 765100  Diversity = 8.3136e-01  k_t
= [1, 1, 1, 1]
Generation 7660: Best = 1.568164e+35  FES = 766100  Diversity = 8.5918e-01  k_t
= [4, 3, 2, 3]
Generation 7670: Best = 1.480213e+35  FES = 767100  Diversity = 8.5296e-01  k_t
= [1, 1, 1, 1]
Generation 7680: Best = 1.454613e+35  FES = 768100  Diversity = 8.4273e-01  k_t
= [4, 4, 4, 2]
Generation 7690: Best = 1.401408e+35  FES = 769100  Diversity = 8.5527e-01  k_t
= [1, 1, 1, 1]
Generation 7700: Best = 1.327721e+35  FES = 770100  Diversity = 8.6766e-01  k_t
= [3, 2, 3, 3]
Generation 7710: Best = 1.275739e+35  FES = 771100  Diversity = 8.6156e-01  k_t
= [1, 1, 1, 1]
Generation 7720: Best = 1.245922e+35  FES = 772100  Diversity = 8.2558e-01  k_t
= [2, 2, 4, 4]
Generation 7730: Best = 1.175414e+35  FES = 773100  Diversity = 8.1751e-01  k_t
= [1, 1, 1, 1]
Generation 7740: Best = 1.103366e+35  FES = 774100  Diversity = 8.4260e-01  k_t
= [3, 3, 2, 2]
Generation 7750: Best = 1.061244e+35  FES = 775100  Diversity = 8.4750e-01  k_t
= [1, 1, 1, 1]
Generation 7760: Best = 1.013257e+35  FES = 776100  Diversity = 8.4581e-01  k_t
= [4, 4, 4, 2]
Generation 7770: Best = 9.263987e+34  FES = 777100  Diversity = 8.5962e-01  k_t
= [1, 1, 1, 1]
Generation 7780: Best = 8.652116e+34  FES = 778100  Diversity = 8.6183e-01  k_t

```

```

= [2, 3, 3, 2]
Generation 7790: Best = 8.090834e+34 FES = 779100 Diversity = 9.0060e-01 k_t
= [1, 1, 1, 1]
Generation 7800: Best = 7.370990e+34 FES = 780100 Diversity = 8.8446e-01 k_t
= [4, 4, 4, 2]
Generation 7810: Best = 6.857506e+34 FES = 781100 Diversity = 8.7678e-01 k_t
= [1, 1, 1, 1]
Generation 7820: Best = 6.540465e+34 FES = 782100 Diversity = 8.8370e-01 k_t
= [4, 4, 2, 2]
Generation 7830: Best = 5.834226e+34 FES = 783100 Diversity = 8.7973e-01 k_t
= [1, 1, 1, 1]
Generation 7840: Best = 5.641751e+34 FES = 784100 Diversity = 9.0167e-01 k_t
= [2, 2, 3, 3]
Generation 7850: Best = 5.252094e+34 FES = 785100 Diversity = 8.9758e-01 k_t
= [1, 1, 1, 1]
Generation 7860: Best = 4.969984e+34 FES = 786100 Diversity = 9.1970e-01 k_t
= [3, 4, 3, 3]
Generation 7870: Best = 4.489523e+34 FES = 787100 Diversity = 9.2308e-01 k_t
= [1, 1, 1, 1]
Generation 7880: Best = 4.223316e+34 FES = 788100 Diversity = 9.6840e-01 k_t
= [4, 2, 2, 4]
Generation 7890: Best = 3.825607e+34 FES = 789100 Diversity = 9.8313e-01 k_t
= [1, 1, 1, 1]
Generation 7900: Best = 3.747885e+34 FES = 790100 Diversity = 9.4358e-01 k_t
= [2, 4, 4, 3]
Generation 7910: Best = 3.460947e+34 FES = 791100 Diversity = 9.4157e-01 k_t
= [1, 1, 1, 1]
Generation 7920: Best = 2.996917e+34 FES = 792100 Diversity = 9.5997e-01 k_t
= [2, 3, 3, 4]
Generation 7930: Best = 2.808650e+34 FES = 793100 Diversity = 9.3934e-01 k_t
= [1, 1, 1, 1]
Generation 7940: Best = 2.493813e+34 FES = 794100 Diversity = 8.9148e-01 k_t
= [4, 4, 3, 2]
Generation 7950: Best = 2.245253e+34 FES = 795100 Diversity = 8.3828e-01 k_t
= [1, 1, 1, 1]
Generation 7960: Best = 2.003253e+34 FES = 796100 Diversity = 8.7508e-01 k_t
= [3, 3, 4, 4]
Generation 7970: Best = 1.878236e+34 FES = 797100 Diversity = 8.4093e-01 k_t
= [1, 1, 1, 1]
Generation 7980: Best = 1.789328e+34 FES = 798100 Diversity = 8.4213e-01 k_t
= [3, 3, 3, 4]
Generation 7990: Best = 1.686167e+34 FES = 799100 Diversity = 8.4207e-01 k_t
= [1, 1, 1, 1]
Generation 8000: Best = 1.589490e+34 FES = 800100 Diversity = 8.6043e-01 k_t
= [2, 3, 3, 2]
Generation 8010: Best = 1.499016e+34 FES = 801100 Diversity = 8.4568e-01 k_t
= [1, 1, 1, 1]
Generation 8020: Best = 1.418663e+34 FES = 802100 Diversity = 8.7211e-01 k_t
= [3, 2, 4, 3]
Generation 8030: Best = 1.283339e+34 FES = 803100 Diversity = 8.5769e-01 k_t
= [1, 1, 1, 1]
Generation 8040: Best = 1.241177e+34 FES = 804100 Diversity = 8.5090e-01 k_t
= [4, 3, 2, 4]
Generation 8050: Best = 1.137210e+34 FES = 805100 Diversity = 8.2880e-01 k_t

```

```

= [1, 1, 1, 1]
Generation 8060: Best = 1.081839e+34 FES = 806100 Diversity = 8.1852e-01 k_t
= [2, 3, 4, 2]
Generation 8070: Best = 1.023008e+34 FES = 807100 Diversity = 8.0703e-01 k_t
= [1, 1, 1, 1]
Generation 8080: Best = 9.384188e+33 FES = 808100 Diversity = 8.0775e-01 k_t
= [3, 4, 3, 4]
Generation 8090: Best = 8.729458e+33 FES = 809100 Diversity = 8.0383e-01 k_t
= [1, 1, 1, 1]
Generation 8100: Best = 7.941609e+33 FES = 810100 Diversity = 8.0989e-01 k_t
= [3, 2, 3, 2]
Generation 8110: Best = 7.421277e+33 FES = 811100 Diversity = 7.8390e-01 k_t
= [1, 1, 1, 1]
Generation 8120: Best = 6.976720e+33 FES = 812100 Diversity = 7.9355e-01 k_t
= [3, 3, 4, 3]
Generation 8130: Best = 6.506406e+33 FES = 813100 Diversity = 8.0297e-01 k_t
= [1, 1, 1, 1]
Generation 8140: Best = 6.114061e+33 FES = 814100 Diversity = 8.1495e-01 k_t
= [2, 4, 2, 2]
Generation 8150: Best = 5.707996e+33 FES = 815100 Diversity = 8.2238e-01 k_t
= [1, 1, 1, 1]
Generation 8160: Best = 5.215005e+33 FES = 816100 Diversity = 8.4300e-01 k_t
= [4, 2, 2, 3]
Generation 8170: Best = 4.816122e+33 FES = 817100 Diversity = 8.0713e-01 k_t
= [1, 1, 1, 1]
Generation 8180: Best = 4.720124e+33 FES = 818100 Diversity = 7.9577e-01 k_t
= [3, 3, 3, 3]
Generation 8190: Best = 4.335658e+33 FES = 819100 Diversity = 8.1937e-01 k_t
= [1, 1, 1, 1]
Generation 8200: Best = 4.150396e+33 FES = 820100 Diversity = 8.3878e-01 k_t
= [2, 2, 2, 2]
Generation 8210: Best = 3.914053e+33 FES = 821100 Diversity = 8.6023e-01 k_t
= [1, 1, 1, 1]
Generation 8220: Best = 3.811503e+33 FES = 822100 Diversity = 8.5271e-01 k_t
= [2, 4, 4, 3]
Generation 8230: Best = 3.700618e+33 FES = 823100 Diversity = 8.4517e-01 k_t
= [1, 1, 1, 1]
Generation 8240: Best = 3.572076e+33 FES = 824100 Diversity = 8.4837e-01 k_t
= [3, 3, 3, 2]
Generation 8250: Best = 3.447786e+33 FES = 825100 Diversity = 8.5231e-01 k_t
= [1, 1, 1, 1]
Generation 8260: Best = 3.293746e+33 FES = 826100 Diversity = 8.7873e-01 k_t
= [3, 2, 4, 2]
Generation 8270: Best = 3.230430e+33 FES = 827100 Diversity = 8.5891e-01 k_t
= [1, 1, 1, 1]
Generation 8280: Best = 3.138921e+33 FES = 828100 Diversity = 8.4341e-01 k_t
= [4, 2, 4, 2]
Generation 8290: Best = 3.027449e+33 FES = 829100 Diversity = 8.6479e-01 k_t
= [1, 1, 1, 1]
Generation 8300: Best = 2.904276e+33 FES = 830100 Diversity = 8.5760e-01 k_t
= [4, 4, 2, 3]
Generation 8310: Best = 2.707228e+33 FES = 831100 Diversity = 8.6569e-01 k_t
= [1, 1, 1, 1]
Generation 8320: Best = 2.612987e+33 FES = 832100 Diversity = 8.6408e-01 k_t

```

```

= [3, 2, 4, 2]
Generation 8330: Best = 2.552543e+33  FES = 833100  Diversity = 8.6809e-01  k_t
= [1, 1, 1, 1]
Generation 8340: Best = 2.437844e+33  FES = 834100  Diversity = 8.6268e-01  k_t
= [4, 2, 2, 4]
Generation 8350: Best = 2.361906e+33  FES = 835100  Diversity = 8.4555e-01  k_t
= [1, 1, 1, 1]
Generation 8360: Best = 2.287303e+33  FES = 836100  Diversity = 8.7044e-01  k_t
= [4, 2, 2, 2]
Generation 8370: Best = 2.179759e+33  FES = 837100  Diversity = 8.6725e-01  k_t
= [1, 1, 1, 1]
Generation 8380: Best = 2.088505e+33  FES = 838100  Diversity = 8.7880e-01  k_t
= [4, 4, 3, 2]
Generation 8390: Best = 1.993098e+33  FES = 839100  Diversity = 8.4765e-01  k_t
= [1, 1, 1, 1]
Generation 8400: Best = 1.953169e+33  FES = 840100  Diversity = 8.5716e-01  k_t
= [4, 4, 3, 2]
Generation 8410: Best = 1.869585e+33  FES = 841100  Diversity = 8.3367e-01  k_t
= [1, 1, 1, 1]
Generation 8420: Best = 1.847645e+33  FES = 842100  Diversity = 8.1551e-01  k_t
= [3, 4, 2, 4]
Generation 8430: Best = 1.788254e+33  FES = 843100  Diversity = 8.1139e-01  k_t
= [1, 1, 1, 1]
Generation 8440: Best = 1.719705e+33  FES = 844100  Diversity = 8.3048e-01  k_t
= [2, 2, 4, 3]
Generation 8450: Best = 1.677100e+33  FES = 845100  Diversity = 8.3054e-01  k_t
= [1, 1, 1, 1]
Generation 8460: Best = 1.631141e+33  FES = 846100  Diversity = 8.2283e-01  k_t
= [3, 4, 3, 3]
Generation 8470: Best = 1.548684e+33  FES = 847100  Diversity = 8.2258e-01  k_t
= [1, 1, 1, 1]
Generation 8480: Best = 1.443407e+33  FES = 848100  Diversity = 8.0827e-01  k_t
= [4, 4, 3, 4]
Generation 8490: Best = 1.332577e+33  FES = 849100  Diversity = 8.0017e-01  k_t
= [1, 1, 1, 1]
Generation 8500: Best = 1.218367e+33  FES = 850100  Diversity = 7.8453e-01  k_t
= [4, 2, 2, 3]
Generation 8510: Best = 1.103247e+33  FES = 851100  Diversity = 8.0218e-01  k_t
= [1, 1, 1, 1]
Generation 8520: Best = 1.021448e+33  FES = 852100  Diversity = 7.9771e-01  k_t
= [3, 4, 4, 3]
Generation 8530: Best = 9.657671e+32  FES = 853100  Diversity = 8.0160e-01  k_t
= [1, 1, 1, 1]
Generation 8540: Best = 8.766838e+32  FES = 854100  Diversity = 7.9602e-01  k_t
= [3, 2, 3, 3]
Generation 8550: Best = 8.300100e+32  FES = 855100  Diversity = 8.1226e-01  k_t
= [1, 1, 1, 1]
Generation 8560: Best = 8.006189e+32  FES = 856100  Diversity = 8.1376e-01  k_t
= [2, 2, 3, 4]
Generation 8570: Best = 7.271752e+32  FES = 857100  Diversity = 8.0216e-01  k_t
= [1, 1, 1, 1]
Generation 8580: Best = 7.052368e+32  FES = 858100  Diversity = 7.7228e-01  k_t
= [4, 4, 3, 4]
Generation 8590: Best = 6.718914e+32  FES = 859100  Diversity = 7.8936e-01  k_t

```



```

= [1, 1, 1, 1]
Generation 8600: Best = 6.318959e+32  FES = 860100  Diversity = 8.0568e-01  k_t
= [3, 2, 4, 3]
Generation 8610: Best = 5.852718e+32  FES = 861100  Diversity = 8.1016e-01  k_t
= [1, 1, 1, 1]
Generation 8620: Best = 5.554097e+32  FES = 862100  Diversity = 8.1244e-01  k_t
= [4, 2, 2, 4]
Generation 8630: Best = 5.281308e+32  FES = 863100  Diversity = 8.1136e-01  k_t
= [1, 1, 1, 1]
Generation 8640: Best = 5.141638e+32  FES = 864100  Diversity = 8.1496e-01  k_t
= [2, 4, 2, 4]
Generation 8650: Best = 4.819523e+32  FES = 865100  Diversity = 8.2400e-01  k_t
= [1, 1, 1, 1]
Generation 8660: Best = 4.509288e+32  FES = 866100  Diversity = 7.9066e-01  k_t
= [4, 2, 2, 2]
Generation 8670: Best = 4.319808e+32  FES = 867100  Diversity = 7.9337e-01  k_t
= [1, 1, 1, 1]
Generation 8680: Best = 4.080479e+32  FES = 868100  Diversity = 7.7898e-01  k_t
= [3, 2, 2, 3]
Generation 8690: Best = 3.813572e+32  FES = 869100  Diversity = 7.6567e-01  k_t
= [1, 1, 1, 1]
Generation 8700: Best = 3.593008e+32  FES = 870100  Diversity = 8.1243e-01  k_t
= [4, 2, 2, 3]
Generation 8710: Best = 3.399280e+32  FES = 871100  Diversity = 7.9277e-01  k_t
= [1, 1, 1, 1]
Generation 8720: Best = 3.264415e+32  FES = 872100  Diversity = 7.7154e-01  k_t
= [2, 4, 3, 4]
Generation 8730: Best = 3.008810e+32  FES = 873100  Diversity = 7.8862e-01  k_t
= [1, 1, 1, 1]
Generation 8740: Best = 2.806386e+32  FES = 874100  Diversity = 8.1689e-01  k_t
= [3, 3, 2, 2]
Generation 8750: Best = 2.583124e+32  FES = 875100  Diversity = 8.1990e-01  k_t
= [1, 1, 1, 1]
Generation 8760: Best = 2.508818e+32  FES = 876100  Diversity = 8.4166e-01  k_t
= [3, 4, 4, 3]
Generation 8770: Best = 2.310868e+32  FES = 877100  Diversity = 8.5691e-01  k_t
= [1, 1, 1, 1]
Generation 8780: Best = 2.234078e+32  FES = 878100  Diversity = 8.5394e-01  k_t
= [4, 2, 3, 2]
Generation 8790: Best = 2.015520e+32  FES = 879100  Diversity = 8.7356e-01  k_t
= [1, 1, 1, 1]
Generation 8800: Best = 1.939480e+32  FES = 880100  Diversity = 8.6738e-01  k_t
= [2, 4, 2, 3]
Generation 8810: Best = 1.730855e+32  FES = 881100  Diversity = 8.5894e-01  k_t
= [1, 1, 1, 1]
Generation 8820: Best = 1.541073e+32  FES = 882100  Diversity = 8.5679e-01  k_t
= [3, 2, 4, 3]
Generation 8830: Best = 1.526639e+32  FES = 883100  Diversity = 8.6030e-01  k_t
= [1, 1, 1, 1]
Generation 8840: Best = 1.408941e+32  FES = 884100  Diversity = 8.3107e-01  k_t
= [2, 3, 4, 2]
Generation 8850: Best = 1.213742e+32  FES = 885100  Diversity = 8.3725e-01  k_t
= [1, 1, 1, 1]
Generation 8860: Best = 1.174699e+32  FES = 886100  Diversity = 8.4685e-01  k_t

```

```

= [3, 3, 3, 2]
Generation 8870: Best = 1.099521e+32 FES = 887100 Diversity = 8.5815e-01 k_t
= [1, 1, 1, 1]
Generation 8880: Best = 1.029418e+32 FES = 888100 Diversity = 8.4812e-01 k_t
= [2, 2, 3, 3]
Generation 8890: Best = 9.481413e+31 FES = 889100 Diversity = 8.3127e-01 k_t
= [1, 1, 1, 1]
Generation 8900: Best = 8.805415e+31 FES = 890100 Diversity = 8.2276e-01 k_t
= [4, 3, 4, 2]
Generation 8910: Best = 8.320056e+31 FES = 891100 Diversity = 8.1158e-01 k_t
= [1, 1, 1, 1]
Generation 8920: Best = 7.413634e+31 FES = 892100 Diversity = 8.0004e-01 k_t
= [2, 2, 2, 2]
Generation 8930: Best = 6.746128e+31 FES = 893100 Diversity = 8.0321e-01 k_t
= [1, 1, 1, 1]
Generation 8940: Best = 6.153956e+31 FES = 894100 Diversity = 8.1330e-01 k_t
= [2, 4, 3, 3]
Generation 8950: Best = 5.776012e+31 FES = 895100 Diversity = 8.2637e-01 k_t
= [1, 1, 1, 1]
Generation 8960: Best = 5.445984e+31 FES = 896100 Diversity = 8.2954e-01 k_t
= [4, 4, 4, 2]
Generation 8970: Best = 4.778784e+31 FES = 897100 Diversity = 7.9536e-01 k_t
= [1, 1, 1, 1]
Generation 8980: Best = 4.371623e+31 FES = 898100 Diversity = 8.0680e-01 k_t
= [4, 2, 3, 2]
Generation 8990: Best = 4.093597e+31 FES = 899100 Diversity = 8.0520e-01 k_t
= [1, 1, 1, 1]
Generation 9000: Best = 3.828452e+31 FES = 900100 Diversity = 8.1255e-01 k_t
= [2, 3, 4, 2]
Generation 9010: Best = 3.553576e+31 FES = 901100 Diversity = 8.0825e-01 k_t
= [1, 1, 1, 1]
Generation 9020: Best = 3.305907e+31 FES = 902100 Diversity = 8.1025e-01 k_t
= [2, 2, 2, 4]
Generation 9030: Best = 3.068355e+31 FES = 903100 Diversity = 8.0072e-01 k_t
= [1, 1, 1, 1]
Generation 9040: Best = 2.809772e+31 FES = 904100 Diversity = 7.9457e-01 k_t
= [4, 4, 2, 2]
Generation 9050: Best = 2.484244e+31 FES = 905100 Diversity = 7.8555e-01 k_t
= [1, 1, 1, 1]
Generation 9060: Best = 2.276426e+31 FES = 906100 Diversity = 7.9693e-01 k_t
= [4, 3, 2, 3]
Generation 9070: Best = 2.051900e+31 FES = 907100 Diversity = 8.1027e-01 k_t
= [1, 1, 1, 1]
Generation 9080: Best = 1.980855e+31 FES = 908100 Diversity = 8.0043e-01 k_t
= [4, 2, 4, 2]
Generation 9090: Best = 1.843797e+31 FES = 909100 Diversity = 8.0823e-01 k_t
= [1, 1, 1, 1]
Generation 9100: Best = 1.751469e+31 FES = 910100 Diversity = 7.7661e-01 k_t
= [3, 3, 2, 2]
Generation 9110: Best = 1.660658e+31 FES = 911100 Diversity = 7.6117e-01 k_t
= [1, 1, 1, 1]
Generation 9120: Best = 1.595055e+31 FES = 912100 Diversity = 7.9214e-01 k_t
= [2, 2, 2, 2]
Generation 9130: Best = 1.521132e+31 FES = 913100 Diversity = 7.9659e-01 k_t

```

```

= [1, 1, 1, 1]
Generation 9140: Best = 1.448579e+31 FES = 914100 Diversity = 7.7702e-01 k_t
= [2, 3, 2, 4]
Generation 9150: Best = 1.372856e+31 FES = 915100 Diversity = 7.3534e-01 k_t
= [1, 1, 1, 1]
Generation 9160: Best = 1.284745e+31 FES = 916100 Diversity = 7.3569e-01 k_t
= [3, 3, 3, 3]
Generation 9170: Best = 1.245367e+31 FES = 917100 Diversity = 7.2664e-01 k_t
= [1, 1, 1, 1]
Generation 9180: Best = 1.189036e+31 FES = 918100 Diversity = 7.4026e-01 k_t
= [2, 4, 4, 3]
Generation 9190: Best = 1.104828e+31 FES = 919100 Diversity = 7.5583e-01 k_t
= [1, 1, 1, 1]
Generation 9200: Best = 1.025959e+31 FES = 920100 Diversity = 7.7386e-01 k_t
= [2, 4, 2, 2]
Generation 9210: Best = 9.954979e+30 FES = 921100 Diversity = 7.8965e-01 k_t
= [1, 1, 1, 1]
Generation 9220: Best = 9.169686e+30 FES = 922100 Diversity = 7.9760e-01 k_t
= [4, 4, 4, 4]
Generation 9230: Best = 8.590035e+30 FES = 923100 Diversity = 7.9924e-01 k_t
= [1, 1, 1, 1]
Generation 9240: Best = 8.062625e+30 FES = 924100 Diversity = 7.7974e-01 k_t
= [2, 4, 3, 3]
Generation 9250: Best = 7.424311e+30 FES = 925100 Diversity = 7.7245e-01 k_t
= [1, 1, 1, 1]
Generation 9260: Best = 7.265389e+30 FES = 926100 Diversity = 7.6259e-01 k_t
= [2, 3, 4, 3]
Generation 9270: Best = 6.967698e+30 FES = 927100 Diversity = 7.4278e-01 k_t
= [1, 1, 1, 1]
Generation 9280: Best = 6.436111e+30 FES = 928100 Diversity = 7.6017e-01 k_t
= [4, 4, 3, 2]
Generation 9290: Best = 5.746173e+30 FES = 929100 Diversity = 7.5469e-01 k_t
= [1, 1, 1, 1]
Generation 9300: Best = 5.596119e+30 FES = 930100 Diversity = 7.4657e-01 k_t
= [3, 3, 2, 3]
Generation 9310: Best = 5.160773e+30 FES = 931100 Diversity = 7.4996e-01 k_t
= [1, 1, 1, 1]
Generation 9320: Best = 4.892688e+30 FES = 932100 Diversity = 7.3765e-01 k_t
= [3, 4, 4, 2]
Generation 9330: Best = 4.448390e+30 FES = 933100 Diversity = 7.1862e-01 k_t
= [1, 1, 1, 1]
Generation 9340: Best = 4.310523e+30 FES = 934100 Diversity = 7.2463e-01 k_t
= [3, 3, 4, 4]
Generation 9350: Best = 4.050484e+30 FES = 935100 Diversity = 7.4966e-01 k_t
= [1, 1, 1, 1]
Generation 9360: Best = 3.771708e+30 FES = 936100 Diversity = 7.3338e-01 k_t
= [2, 4, 3, 2]
Generation 9370: Best = 3.395699e+30 FES = 937100 Diversity = 7.5729e-01 k_t
= [1, 1, 1, 1]
Generation 9380: Best = 3.242738e+30 FES = 938100 Diversity = 7.7726e-01 k_t
= [4, 2, 4, 4]
Generation 9390: Best = 2.982021e+30 FES = 939100 Diversity = 7.5403e-01 k_t
= [1, 1, 1, 1]
Generation 9400: Best = 2.753224e+30 FES = 940100 Diversity = 7.4820e-01 k_t

```

```

= [3, 2, 4, 2]
Generation 9410: Best = 2.672316e+30  FES = 941100  Diversity = 7.4130e-01  k_t
= [1, 1, 1, 1]
Generation 9420: Best = 2.541934e+30  FES = 942100  Diversity = 7.5074e-01  k_t
= [3, 3, 4, 3]
Generation 9430: Best = 2.486238e+30  FES = 943100  Diversity = 7.3240e-01  k_t
= [1, 1, 1, 1]
Generation 9440: Best = 2.399762e+30  FES = 944100  Diversity = 7.1571e-01  k_t
= [4, 2, 2, 2]
Generation 9450: Best = 2.178337e+30  FES = 945100  Diversity = 7.2954e-01  k_t
= [1, 1, 1, 1]
Generation 9460: Best = 2.066753e+30  FES = 946100  Diversity = 7.2559e-01  k_t
= [2, 2, 4, 2]
Generation 9470: Best = 2.027679e+30  FES = 947100  Diversity = 7.1873e-01  k_t
= [1, 1, 1, 1]
Generation 9480: Best = 1.905778e+30  FES = 948100  Diversity = 7.4191e-01  k_t
= [3, 4, 2, 3]
Generation 9490: Best = 1.811785e+30  FES = 949100  Diversity = 7.3526e-01  k_t
= [1, 1, 1, 1]
Generation 9500: Best = 1.659292e+30  FES = 950100  Diversity = 7.7673e-01  k_t
= [2, 4, 3, 4]
Generation 9510: Best = 1.586923e+30  FES = 951100  Diversity = 7.8089e-01  k_t
= [1, 1, 1, 1]
Generation 9520: Best = 1.490082e+30  FES = 952100  Diversity = 7.8521e-01  k_t
= [2, 3, 2, 3]
Generation 9530: Best = 1.437751e+30  FES = 953100  Diversity = 7.7717e-01  k_t
= [1, 1, 1, 1]
Generation 9540: Best = 1.330581e+30  FES = 954100  Diversity = 7.9549e-01  k_t
= [3, 3, 2, 4]
Generation 9550: Best = 1.256849e+30  FES = 955100  Diversity = 7.7078e-01  k_t
= [1, 1, 1, 1]
Generation 9560: Best = 1.197083e+30  FES = 956100  Diversity = 7.3242e-01  k_t
= [2, 4, 3, 2]
Generation 9570: Best = 1.121348e+30  FES = 957100  Diversity = 7.0821e-01  k_t
= [1, 1, 1, 1]
Generation 9580: Best = 1.054325e+30  FES = 958100  Diversity = 6.9708e-01  k_t
= [2, 3, 4, 4]
Generation 9590: Best = 9.960462e+29  FES = 959100  Diversity = 6.9845e-01  k_t
= [1, 1, 1, 1]
Generation 9600: Best = 9.536969e+29  FES = 960100  Diversity = 6.8331e-01  k_t
= [4, 4, 4, 4]
Generation 9610: Best = 9.365529e+29  FES = 961100  Diversity = 6.8273e-01  k_t
= [1, 1, 1, 1]
Generation 9620: Best = 8.551309e+29  FES = 962100  Diversity = 6.9471e-01  k_t
= [4, 4, 3, 3]
Generation 9630: Best = 8.097438e+29  FES = 963100  Diversity = 7.0254e-01  k_t
= [1, 1, 1, 1]
Generation 9640: Best = 7.508835e+29  FES = 964100  Diversity = 7.3278e-01  k_t
= [4, 3, 3, 2]
Generation 9650: Best = 7.171221e+29  FES = 965100  Diversity = 7.2627e-01  k_t
= [1, 1, 1, 1]
Generation 9660: Best = 6.373570e+29  FES = 966100  Diversity = 7.3713e-01  k_t
= [4, 4, 3, 3]
Generation 9670: Best = 6.192653e+29  FES = 967100  Diversity = 7.3554e-01  k_t

```

```

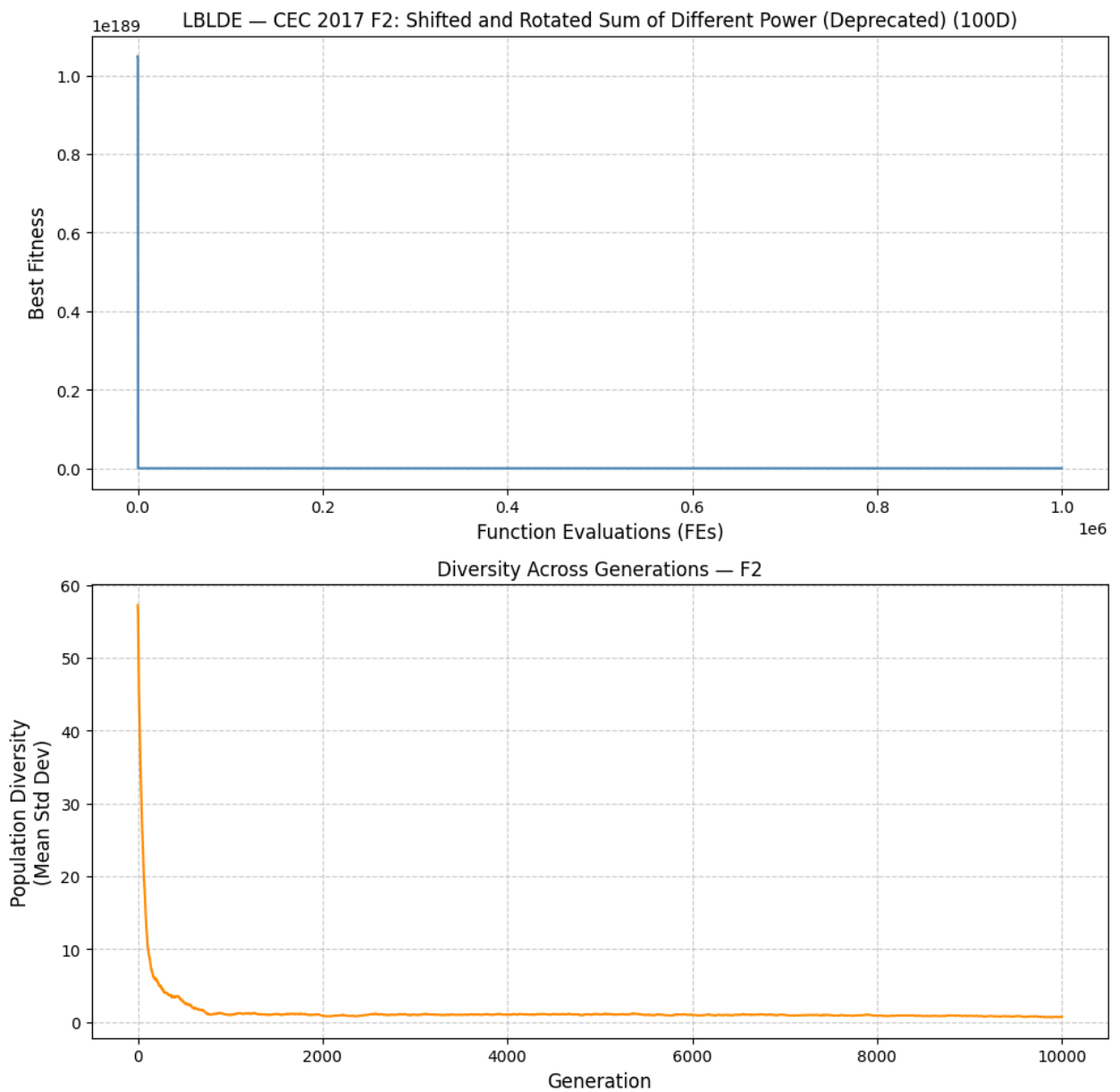
= [1, 1, 1, 1]
Generation 9680: Best = 5.589178e+29  FES = 968100  Diversity = 7.4172e-01  k_t
= [3, 4, 3, 2]
Generation 9690: Best = 5.249218e+29  FES = 969100  Diversity = 7.3194e-01  k_t
= [1, 1, 1, 1]
Generation 9700: Best = 4.723390e+29  FES = 970100  Diversity = 7.2584e-01  k_t
= [3, 4, 3, 4]
Generation 9710: Best = 4.540475e+29  FES = 971100  Diversity = 7.4421e-01  k_t
= [1, 1, 1, 1]
Generation 9720: Best = 4.359595e+29  FES = 972100  Diversity = 7.6022e-01  k_t
= [4, 2, 3, 2]
Generation 9730: Best = 4.186861e+29  FES = 973100  Diversity = 7.4640e-01  k_t
= [1, 1, 1, 1]
Generation 9740: Best = 3.913639e+29  FES = 974100  Diversity = 6.9437e-01  k_t
= [3, 3, 3, 4]
Generation 9750: Best = 3.776988e+29  FES = 975100  Diversity = 7.0207e-01  k_t
= [1, 1, 1, 1]
Generation 9760: Best = 3.629217e+29  FES = 976100  Diversity = 6.8745e-01  k_t
= [3, 3, 3, 2]
Generation 9770: Best = 3.419068e+29  FES = 977100  Diversity = 6.7054e-01  k_t
= [1, 1, 1, 1]
Generation 9780: Best = 3.301646e+29  FES = 978100  Diversity = 6.8200e-01  k_t
= [3, 2, 4, 2]
Generation 9790: Best = 3.103378e+29  FES = 979100  Diversity = 6.8133e-01  k_t
= [1, 1, 1, 1]
Generation 9800: Best = 3.019238e+29  FES = 980100  Diversity = 6.5151e-01  k_t
= [4, 4, 3, 4]
Generation 9810: Best = 2.870673e+29  FES = 981100  Diversity = 6.4467e-01  k_t
= [1, 1, 1, 1]
Generation 9820: Best = 2.814047e+29  FES = 982100  Diversity = 6.4311e-01  k_t
= [4, 4, 3, 4]
Generation 9830: Best = 2.694121e+29  FES = 983100  Diversity = 6.4518e-01  k_t
= [1, 1, 1, 1]
Generation 9840: Best = 2.497867e+29  FES = 984100  Diversity = 6.6006e-01  k_t
= [2, 3, 4, 2]
Generation 9850: Best = 2.376332e+29  FES = 985100  Diversity = 6.5587e-01  k_t
= [1, 1, 1, 1]
Generation 9860: Best = 2.260291e+29  FES = 986100  Diversity = 6.4843e-01  k_t
= [4, 4, 3, 3]
Generation 9870: Best = 2.101613e+29  FES = 987100  Diversity = 6.4177e-01  k_t
= [1, 1, 1, 1]
Generation 9880: Best = 2.016678e+29  FES = 988100  Diversity = 6.3829e-01  k_t
= [3, 2, 4, 3]
Generation 9890: Best = 1.868993e+29  FES = 989100  Diversity = 6.2844e-01  k_t
= [1, 1, 1, 1]
Generation 9900: Best = 1.758030e+29  FES = 990100  Diversity = 6.2504e-01  k_t
= [4, 2, 4, 3]
Generation 9910: Best = 1.630651e+29  FES = 991100  Diversity = 6.6154e-01  k_t
= [1, 1, 1, 1]
Generation 9920: Best = 1.425949e+29  FES = 992100  Diversity = 6.6373e-01  k_t
= [2, 3, 3, 4]
Generation 9930: Best = 1.380804e+29  FES = 993100  Diversity = 6.9044e-01  k_t
= [1, 1, 1, 1]
Generation 9940: Best = 1.244087e+29  FES = 994100  Diversity = 6.7512e-01  k_t

```

```
= [2, 4, 4, 4]
Generation 9950: Best = 1.173742e+29  FES = 995100  Diversity = 6.7207e-01  k_t
= [1, 1, 1, 1]
Generation 9960: Best = 1.070282e+29  FES = 996100  Diversity = 6.6210e-01  k_t
= [2, 4, 2, 3]
Generation 9970: Best = 9.548828e+28  FES = 997100  Diversity = 6.4996e-01  k_t
= [1, 1, 1, 1]
Generation 9980: Best = 8.237659e+28  FES = 998100  Diversity = 6.5522e-01  k_t
= [3, 3, 3, 3]
Generation 9990: Best = 7.536489e+28  FES = 999100  Diversity = 6.7722e-01  k_t
= [1, 1, 1, 1]
```

Optimization completed!
Final Best Fitness = 6.557252e+28
Total Function Evaluations = 1000000

```
=====
=
Results:
  Best Fitness          : 6.557252e+28
  Global Optimum        : 2.000000e+02
  Error (|best-optimum|) : 6.557252e+28
  Best Solution (5 dim) : [-53.34698243 -39.53592583 -40.90475874  66.51904086
13.52221209]
  D^(0) (init. diversity): 5.7220e+01
=====
=
```



Graph saved as convergence_F2.png

```
In [27]: # =====
# F3 – Shifted and Rotated Zakharov
# =====
run_and_plot('F3', make_wrapper(simple.f3), 300.0, 'Shifted and Rotated Zakharov')
```

```

=====
=
LBLDE_s on CEC 2017 F3 – Single Run
=====
=
Function      : Shifted and Rotated Zakharov (100D)
Bounds       : [-100, 100]
Global optimum : 300.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 7.602487e+06 | Diversity = 5.7220e+01
Generation 10: Best = 8.402860e+05 FES = 1100 Diversity = 5.4395e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 6.462855e+05 FES = 2100 Diversity = 5.0579e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 6.462855e+05 FES = 3100 Diversity = 4.7891e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 6.333519e+05 FES = 4100 Diversity = 4.5163e+01 k_t
= [4, 4, 3, 2]
Generation 50: Best = 5.398094e+05 FES = 5100 Diversity = 4.1341e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 4.811412e+05 FES = 6100 Diversity = 3.6621e+01 k_t
= [4, 3, 4, 3]
Generation 70: Best = 4.714241e+05 FES = 7100 Diversity = 2.7074e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 4.065385e+05 FES = 8100 Diversity = 2.0539e+01 k_t
= [3, 4, 4, 4]
Generation 90: Best = 3.938765e+05 FES = 9100 Diversity = 1.5081e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.634403e+05 FES = 10100 Diversity = 1.1489e+01 k_t
= [3, 3, 3, 2]
Generation 110: Best = 3.634403e+05 FES = 11100 Diversity = 9.7667e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.526116e+05 FES = 12100 Diversity = 8.5067e+00 k_t
= [4, 3, 2, 4]
Generation 130: Best = 3.511634e+05 FES = 13100 Diversity = 7.3671e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.461581e+05 FES = 14100 Diversity = 5.9487e+00 k_t
= [2, 2, 4, 2]
Generation 150: Best = 3.438761e+05 FES = 15100 Diversity = 4.9923e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.423604e+05 FES = 16100 Diversity = 4.0568e+00 k_t
= [3, 3, 3, 3]
Generation 170: Best = 3.395982e+05 FES = 17100 Diversity = 3.5433e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.383629e+05 FES = 18100 Diversity = 3.0357e+00 k_t
= [2, 3, 2, 4]
Generation 190: Best = 3.366685e+05 FES = 19100 Diversity = 2.7428e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.353469e+05 FES = 20100 Diversity = 2.5445e+00 k_t
= [3, 2, 4, 3]
Generation 210: Best = 3.337448e+05 FES = 21100 Diversity = 2.4371e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.321685e+05 FES = 22100 Diversity = 2.6839e+00 k_t
= [3, 2, 3, 3]

```


Generation 230:	Best = 3.292912e+05	FES = 23100	Diversity = 3.0578e+00	k_t = [1, 1, 1, 1]
Generation 240:	Best = 3.292912e+05	FES = 24100	Diversity = 2.8557e+00	k_t = [3, 4, 3, 3]
Generation 250:	Best = 3.251149e+05	FES = 25100	Diversity = 2.6469e+00	k_t = [1, 1, 1, 1]
Generation 260:	Best = 3.225657e+05	FES = 26100	Diversity = 2.9867e+00	k_t = [4, 3, 4, 2]
Generation 270:	Best = 3.215072e+05	FES = 27100	Diversity = 2.9524e+00	k_t = [1, 1, 1, 1]
Generation 280:	Best = 3.176222e+05	FES = 28100	Diversity = 3.2018e+00	k_t = [4, 4, 2, 2]
Generation 290:	Best = 3.166375e+05	FES = 29100	Diversity = 3.1414e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 3.137936e+05	FES = 30100	Diversity = 3.3285e+00	k_t = [3, 4, 4, 4]
Generation 310:	Best = 3.128727e+05	FES = 31100	Diversity = 3.1272e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 3.119854e+05	FES = 32100	Diversity = 2.8404e+00	k_t = [4, 2, 2, 3]
Generation 330:	Best = 3.105019e+05	FES = 33100	Diversity = 2.7539e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 3.104253e+05	FES = 34100	Diversity = 2.8282e+00	k_t = [3, 2, 2, 2]
Generation 350:	Best = 3.080485e+05	FES = 35100	Diversity = 2.8054e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 3.026298e+05	FES = 36100	Diversity = 3.6339e+00	k_t = [4, 2, 2, 4]
Generation 370:	Best = 2.992426e+05	FES = 37100	Diversity = 3.6612e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 2.972617e+05	FES = 38100	Diversity = 3.5869e+00	k_t = [3, 4, 4, 4]
Generation 390:	Best = 2.972617e+05	FES = 39100	Diversity = 3.6630e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 2.962715e+05	FES = 40100	Diversity = 3.4643e+00	k_t = [2, 3, 2, 2]
Generation 410:	Best = 2.938636e+05	FES = 41100	Diversity = 3.2995e+00	k_t = [1, 1, 1, 1]
Generation 420:	Best = 2.923148e+05	FES = 42100	Diversity = 3.2823e+00	k_t = [4, 2, 4, 2]
Generation 430:	Best = 2.908175e+05	FES = 43100	Diversity = 3.1277e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 2.895871e+05	FES = 44100	Diversity = 2.9476e+00	k_t = [4, 4, 4, 2]
Generation 450:	Best = 2.875398e+05	FES = 45100	Diversity = 2.7482e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 2.847016e+05	FES = 46100	Diversity = 2.9041e+00	k_t = [3, 2, 2, 3]
Generation 470:	Best = 2.834750e+05	FES = 47100	Diversity = 2.9526e+00	k_t = [1, 1, 1, 1]
Generation 480:	Best = 2.805206e+05	FES = 48100	Diversity = 2.8144e+00	k_t = [2, 2, 2, 4]
Generation 490:	Best = 2.798301e+05	FES = 49100	Diversity = 2.9598e+00	k_t = [1, 1, 1, 1]

Generation 500:	Best = 2.785977e+05	FES = 50100	Diversity = 3.1517e+00	k_t
= [4, 3, 3, 2]				
Generation 510:	Best = 2.774208e+05	FES = 51100	Diversity = 3.1094e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 2.755456e+05	FES = 52100	Diversity = 3.1766e+00	k_t
= [2, 4, 4, 2]				
Generation 530:	Best = 2.732219e+05	FES = 53100	Diversity = 3.1043e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 2.715853e+05	FES = 54100	Diversity = 3.1167e+00	k_t
= [2, 4, 3, 4]				
Generation 550:	Best = 2.709695e+05	FES = 55100	Diversity = 3.1607e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 2.703087e+05	FES = 56100	Diversity = 2.9429e+00	k_t
= [3, 4, 3, 4]				
Generation 570:	Best = 2.694552e+05	FES = 57100	Diversity = 2.6056e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 2.688306e+05	FES = 58100	Diversity = 2.4064e+00	k_t
= [2, 2, 4, 3]				
Generation 590:	Best = 2.670637e+05	FES = 59100	Diversity = 2.2580e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 2.668763e+05	FES = 60100	Diversity = 2.0718e+00	k_t
= [3, 4, 4, 3]				
Generation 610:	Best = 2.659629e+05	FES = 61100	Diversity = 1.9303e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 2.650387e+05	FES = 62100	Diversity = 1.9080e+00	k_t
= [4, 3, 4, 2]				
Generation 630:	Best = 2.631561e+05	FES = 63100	Diversity = 1.8096e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 2.630905e+05	FES = 64100	Diversity = 1.9455e+00	k_t
= [2, 4, 3, 3]				
Generation 650:	Best = 2.616191e+05	FES = 65100	Diversity = 1.8482e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 2.608044e+05	FES = 66100	Diversity = 1.9269e+00	k_t
= [4, 2, 4, 3]				
Generation 670:	Best = 2.594472e+05	FES = 67100	Diversity = 1.9111e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 2.594472e+05	FES = 68100	Diversity = 2.2182e+00	k_t
= [3, 4, 2, 3]				
Generation 690:	Best = 2.585094e+05	FES = 69100	Diversity = 2.2546e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 2.571404e+05	FES = 70100	Diversity = 2.1413e+00	k_t
= [4, 4, 3, 2]				
Generation 710:	Best = 2.552481e+05	FES = 71100	Diversity = 2.2324e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 2.533035e+05	FES = 72100	Diversity = 2.4497e+00	k_t
= [4, 3, 3, 2]				
Generation 730:	Best = 2.521253e+05	FES = 73100	Diversity = 2.4117e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 2.489649e+05	FES = 74100	Diversity = 2.3896e+00	k_t
= [2, 3, 4, 3]				
Generation 750:	Best = 2.489649e+05	FES = 75100	Diversity = 2.6488e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 2.488145e+05	FES = 76100	Diversity = 2.6886e+00	k_t
= [4, 3, 3, 2]				

Generation 770:	Best = 2.475319e+05	FES = 77100	Diversity = 2.4972e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.463345e+05	FES = 78100	Diversity = 2.4621e+00	k_t
= [2, 4, 4, 3]				
Generation 790:	Best = 2.451849e+05	FES = 79100	Diversity = 2.4567e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.446119e+05	FES = 80100	Diversity = 2.3375e+00	k_t
= [2, 3, 2, 4]				
Generation 810:	Best = 2.432094e+05	FES = 81100	Diversity = 2.1992e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.421966e+05	FES = 82100	Diversity = 2.0248e+00	k_t
= [2, 4, 2, 3]				
Generation 830:	Best = 2.408285e+05	FES = 83100	Diversity = 2.0828e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.381059e+05	FES = 84100	Diversity = 2.2898e+00	k_t
= [4, 4, 4, 2]				
Generation 850:	Best = 2.372816e+05	FES = 85100	Diversity = 2.3920e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.365788e+05	FES = 86100	Diversity = 2.4029e+00	k_t
= [2, 3, 3, 3]				
Generation 870:	Best = 2.345728e+05	FES = 87100	Diversity = 2.1683e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.332470e+05	FES = 88100	Diversity = 2.2838e+00	k_t
= [3, 4, 2, 2]				
Generation 890:	Best = 2.327651e+05	FES = 89100	Diversity = 2.3609e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.319831e+05	FES = 90100	Diversity = 2.2930e+00	k_t
= [3, 3, 2, 4]				
Generation 910:	Best = 2.306798e+05	FES = 91100	Diversity = 2.2457e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.295718e+05	FES = 92100	Diversity = 2.4460e+00	k_t
= [2, 2, 4, 3]				
Generation 930:	Best = 2.289034e+05	FES = 93100	Diversity = 2.5517e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.276811e+05	FES = 94100	Diversity = 2.2862e+00	k_t
= [2, 3, 2, 3]				
Generation 950:	Best = 2.256119e+05	FES = 95100	Diversity = 2.2004e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.243448e+05	FES = 96100	Diversity = 2.4154e+00	k_t
= [4, 2, 2, 4]				
Generation 970:	Best = 2.220279e+05	FES = 97100	Diversity = 2.4798e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.200606e+05	FES = 98100	Diversity = 2.6921e+00	k_t
= [3, 4, 3, 2]				
Generation 990:	Best = 2.200606e+05	FES = 99100	Diversity = 2.5506e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 2.200606e+05	FES = 100100	Diversity = 2.6587e+00	k_t
= [2, 4, 2, 2]				
Generation 1010:	Best = 2.175049e+05	FES = 101100	Diversity = 2.4364e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 2.175049e+05	FES = 102100	Diversity = 2.5634e+00	k_t
= [2, 2, 2, 4]				
Generation 1030:	Best = 2.155299e+05	FES = 103100	Diversity = 2.3907e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 2.154289e+05 = [2, 3, 4, 3]	FES = 104100	Diversity = 2.4643e+00	k_t
Generation 1050: Best = 2.146539e+05 = [1, 1, 1, 1]	FES = 105100	Diversity = 2.2357e+00	k_t
Generation 1060: Best = 2.135870e+05 = [4, 2, 3, 4]	FES = 106100	Diversity = 2.0148e+00	k_t
Generation 1070: Best = 2.119841e+05 = [1, 1, 1, 1]	FES = 107100	Diversity = 2.0516e+00	k_t
Generation 1080: Best = 2.119841e+05 = [4, 2, 4, 3]	FES = 108100	Diversity = 2.1647e+00	k_t
Generation 1090: Best = 2.110469e+05 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.8972e+00	k_t
Generation 1100: Best = 2.107020e+05 = [3, 2, 2, 3]	FES = 110100	Diversity = 1.9607e+00	k_t
Generation 1110: Best = 2.096884e+05 = [1, 1, 1, 1]	FES = 111100	Diversity = 2.0545e+00	k_t
Generation 1120: Best = 2.086756e+05 = [2, 2, 2, 2]	FES = 112100	Diversity = 1.9617e+00	k_t
Generation 1130: Best = 2.070051e+05 = [1, 1, 1, 1]	FES = 113100	Diversity = 2.2734e+00	k_t
Generation 1140: Best = 2.051001e+05 = [2, 4, 2, 2]	FES = 114100	Diversity = 2.2389e+00	k_t
Generation 1150: Best = 2.030964e+05 = [1, 1, 1, 1]	FES = 115100	Diversity = 2.6487e+00	k_t
Generation 1160: Best = 2.026138e+05 = [3, 4, 4, 4]	FES = 116100	Diversity = 2.8822e+00	k_t
Generation 1170: Best = 2.010055e+05 = [1, 1, 1, 1]	FES = 117100	Diversity = 2.5782e+00	k_t
Generation 1180: Best = 1.992494e+05 = [4, 4, 2, 4]	FES = 118100	Diversity = 2.5881e+00	k_t
Generation 1190: Best = 1.982986e+05 = [1, 1, 1, 1]	FES = 119100	Diversity = 2.7845e+00	k_t
Generation 1200: Best = 1.973328e+05 = [2, 4, 4, 3]	FES = 120100	Diversity = 2.8962e+00	k_t
Generation 1210: Best = 1.962615e+05 = [1, 1, 1, 1]	FES = 121100	Diversity = 2.7208e+00	k_t
Generation 1220: Best = 1.938340e+05 = [3, 3, 2, 3]	FES = 122100	Diversity = 2.6309e+00	k_t
Generation 1230: Best = 1.930238e+05 = [1, 1, 1, 1]	FES = 123100	Diversity = 2.6458e+00	k_t
Generation 1240: Best = 1.919636e+05 = [2, 4, 2, 2]	FES = 124100	Diversity = 2.5777e+00	k_t
Generation 1250: Best = 1.915760e+05 = [1, 1, 1, 1]	FES = 125100	Diversity = 2.4394e+00	k_t
Generation 1260: Best = 1.894686e+05 = [3, 2, 2, 4]	FES = 126100	Diversity = 2.3727e+00	k_t
Generation 1270: Best = 1.880728e+05 = [1, 1, 1, 1]	FES = 127100	Diversity = 2.3310e+00	k_t
Generation 1280: Best = 1.880728e+05 = [3, 4, 4, 4]	FES = 128100	Diversity = 2.3148e+00	k_t
Generation 1290: Best = 1.873697e+05 = [1, 1, 1, 1]	FES = 129100	Diversity = 2.0237e+00	k_t
Generation 1300: Best = 1.862382e+05 = [4, 3, 3, 2]	FES = 130100	Diversity = 1.7977e+00	k_t

Generation 1310: Best = 1.833425e+05 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.8448e+00	k_t
Generation 1320: Best = 1.833425e+05 = [4, 3, 3, 3]	FES = 132100	Diversity = 2.3110e+00	k_t
Generation 1330: Best = 1.823412e+05 = [1, 1, 1, 1]	FES = 133100	Diversity = 2.2078e+00	k_t
Generation 1340: Best = 1.808967e+05 = [3, 2, 4, 4]	FES = 134100	Diversity = 2.4191e+00	k_t
Generation 1350: Best = 1.801714e+05 = [1, 1, 1, 1]	FES = 135100	Diversity = 2.2051e+00	k_t
Generation 1360: Best = 1.799367e+05 = [3, 3, 2, 4]	FES = 136100	Diversity = 2.0625e+00	k_t
Generation 1370: Best = 1.793891e+05 = [1, 1, 1, 1]	FES = 137100	Diversity = 1.7578e+00	k_t
Generation 1380: Best = 1.783502e+05 = [4, 3, 3, 2]	FES = 138100	Diversity = 1.5918e+00	k_t
Generation 1390: Best = 1.774748e+05 = [1, 1, 1, 1]	FES = 139100	Diversity = 1.4139e+00	k_t
Generation 1400: Best = 1.771788e+05 = [2, 3, 3, 2]	FES = 140100	Diversity = 1.4889e+00	k_t
Generation 1410: Best = 1.770251e+05 = [1, 1, 1, 1]	FES = 141100	Diversity = 1.5034e+00	k_t
Generation 1420: Best = 1.760975e+05 = [4, 3, 2, 3]	FES = 142100	Diversity = 1.3426e+00	k_t
Generation 1430: Best = 1.758181e+05 = [1, 1, 1, 1]	FES = 143100	Diversity = 1.3573e+00	k_t
Generation 1440: Best = 1.751696e+05 = [3, 4, 2, 3]	FES = 144100	Diversity = 1.4589e+00	k_t
Generation 1450: Best = 1.716042e+05 = [1, 1, 1, 1]	FES = 145100	Diversity = 1.7567e+00	k_t
Generation 1460: Best = 1.711042e+05 = [2, 4, 4, 4]	FES = 146100	Diversity = 2.0093e+00	k_t
Generation 1470: Best = 1.708836e+05 = [1, 1, 1, 1]	FES = 147100	Diversity = 2.2151e+00	k_t
Generation 1480: Best = 1.691745e+05 = [2, 3, 4, 2]	FES = 148100	Diversity = 2.4766e+00	k_t
Generation 1490: Best = 1.674477e+05 = [1, 1, 1, 1]	FES = 149100	Diversity = 2.3808e+00	k_t
Generation 1500: Best = 1.664187e+05 = [3, 2, 2, 2]	FES = 150100	Diversity = 2.4050e+00	k_t
Generation 1510: Best = 1.652886e+05 = [1, 1, 1, 1]	FES = 151100	Diversity = 2.4440e+00	k_t
Generation 1520: Best = 1.637301e+05 = [3, 3, 3, 3]	FES = 152100	Diversity = 2.4581e+00	k_t
Generation 1530: Best = 1.637301e+05 = [1, 1, 1, 1]	FES = 153100	Diversity = 2.2432e+00	k_t
Generation 1540: Best = 1.624872e+05 = [3, 3, 4, 4]	FES = 154100	Diversity = 2.2618e+00	k_t
Generation 1550: Best = 1.622350e+05 = [1, 1, 1, 1]	FES = 155100	Diversity = 2.3177e+00	k_t
Generation 1560: Best = 1.606274e+05 = [4, 4, 2, 3]	FES = 156100	Diversity = 2.2081e+00	k_t
Generation 1570: Best = 1.596230e+05 = [1, 1, 1, 1]	FES = 157100	Diversity = 2.0719e+00	k_t

Generation 1580: Best = 1.596230e+05 = [2, 4, 2, 3]	FES = 158100	Diversity = 2.1268e+00	k_t
Generation 1590: Best = 1.584206e+05 = [1, 1, 1, 1]	FES = 159100	Diversity = 2.0815e+00	k_t
Generation 1600: Best = 1.579630e+05 = [3, 4, 2, 4]	FES = 160100	Diversity = 1.9804e+00	k_t
Generation 1610: Best = 1.575636e+05 = [1, 1, 1, 1]	FES = 161100	Diversity = 1.8310e+00	k_t
Generation 1620: Best = 1.569386e+05 = [4, 3, 3, 3]	FES = 162100	Diversity = 1.6656e+00	k_t
Generation 1630: Best = 1.565090e+05 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.5610e+00	k_t
Generation 1640: Best = 1.557012e+05 = [3, 3, 3, 3]	FES = 164100	Diversity = 1.4931e+00	k_t
Generation 1650: Best = 1.556164e+05 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.2850e+00	k_t
Generation 1660: Best = 1.548832e+05 = [3, 3, 4, 3]	FES = 166100	Diversity = 1.2748e+00	k_t
Generation 1670: Best = 1.541253e+05 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.3341e+00	k_t
Generation 1680: Best = 1.539890e+05 = [2, 3, 3, 2]	FES = 168100	Diversity = 1.4413e+00	k_t
Generation 1690: Best = 1.535479e+05 = [1, 1, 1, 1]	FES = 169100	Diversity = 1.4893e+00	k_t
Generation 1700: Best = 1.518439e+05 = [4, 2, 2, 4]	FES = 170100	Diversity = 1.5654e+00	k_t
Generation 1710: Best = 1.518292e+05 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.7588e+00	k_t
Generation 1720: Best = 1.512791e+05 = [4, 2, 2, 2]	FES = 172100	Diversity = 1.7899e+00	k_t
Generation 1730: Best = 1.507977e+05 = [1, 1, 1, 1]	FES = 173100	Diversity = 1.7265e+00	k_t
Generation 1740: Best = 1.502154e+05 = [3, 4, 4, 4]	FES = 174100	Diversity = 1.6846e+00	k_t
Generation 1750: Best = 1.502154e+05 = [1, 1, 1, 1]	FES = 175100	Diversity = 1.5162e+00	k_t
Generation 1760: Best = 1.497491e+05 = [3, 2, 2, 4]	FES = 176100	Diversity = 1.3974e+00	k_t
Generation 1770: Best = 1.488024e+05 = [1, 1, 1, 1]	FES = 177100	Diversity = 1.3355e+00	k_t
Generation 1780: Best = 1.481581e+05 = [4, 3, 3, 4]	FES = 178100	Diversity = 1.4485e+00	k_t
Generation 1790: Best = 1.479825e+05 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.5094e+00	k_t
Generation 1800: Best = 1.476738e+05 = [3, 4, 2, 3]	FES = 180100	Diversity = 1.3774e+00	k_t
Generation 1810: Best = 1.473676e+05 = [1, 1, 1, 1]	FES = 181100	Diversity = 1.4300e+00	k_t
Generation 1820: Best = 1.469512e+05 = [4, 3, 4, 2]	FES = 182100	Diversity = 1.2335e+00	k_t
Generation 1830: Best = 1.464426e+05 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.2617e+00	k_t
Generation 1840: Best = 1.461062e+05 = [4, 4, 4, 2]	FES = 184100	Diversity = 1.2915e+00	k_t

Generation 1850: Best = 1.453752e+05 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.3423e+00	k_t
Generation 1860: Best = 1.450718e+05 = [3, 2, 2, 4]	FES = 186100	Diversity = 1.4517e+00	k_t
Generation 1870: Best = 1.435302e+05 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.8264e+00	k_t
Generation 1880: Best = 1.430785e+05 = [3, 2, 4, 2]	FES = 188100	Diversity = 2.0872e+00	k_t
Generation 1890: Best = 1.424372e+05 = [1, 1, 1, 1]	FES = 189100	Diversity = 2.1078e+00	k_t
Generation 1900: Best = 1.418834e+05 = [2, 2, 4, 2]	FES = 190100	Diversity = 1.9995e+00	k_t
Generation 1910: Best = 1.408333e+05 = [1, 1, 1, 1]	FES = 191100	Diversity = 1.9358e+00	k_t
Generation 1920: Best = 1.399266e+05 = [4, 4, 3, 2]	FES = 192100	Diversity = 1.9856e+00	k_t
Generation 1930: Best = 1.396713e+05 = [1, 1, 1, 1]	FES = 193100	Diversity = 1.9997e+00	k_t
Generation 1940: Best = 1.391067e+05 = [2, 2, 2, 3]	FES = 194100	Diversity = 1.8557e+00	k_t
Generation 1950: Best = 1.388636e+05 = [1, 1, 1, 1]	FES = 195100	Diversity = 1.7650e+00	k_t
Generation 1960: Best = 1.379261e+05 = [4, 3, 4, 4]	FES = 196100	Diversity = 1.7024e+00	k_t
Generation 1970: Best = 1.373968e+05 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.7961e+00	k_t
Generation 1980: Best = 1.370701e+05 = [4, 3, 3, 3]	FES = 198100	Diversity = 1.6447e+00	k_t
Generation 1990: Best = 1.368186e+05 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.4706e+00	k_t
Generation 2000: Best = 1.366005e+05 = [3, 2, 2, 4]	FES = 200100	Diversity = 1.5106e+00	k_t
Generation 2010: Best = 1.356774e+05 = [1, 1, 1, 1]	FES = 201100	Diversity = 1.4383e+00	k_t
Generation 2020: Best = 1.352689e+05 = [2, 2, 2, 3]	FES = 202100	Diversity = 1.4715e+00	k_t
Generation 2030: Best = 1.350772e+05 = [1, 1, 1, 1]	FES = 203100	Diversity = 1.4533e+00	k_t
Generation 2040: Best = 1.347975e+05 = [4, 2, 3, 4]	FES = 204100	Diversity = 1.4218e+00	k_t
Generation 2050: Best = 1.343119e+05 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.4608e+00	k_t
Generation 2060: Best = 1.342214e+05 = [3, 3, 4, 4]	FES = 206100	Diversity = 1.5068e+00	k_t
Generation 2070: Best = 1.336402e+05 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.4309e+00	k_t
Generation 2080: Best = 1.336402e+05 = [2, 3, 2, 3]	FES = 208100	Diversity = 1.2781e+00	k_t
Generation 2090: Best = 1.333623e+05 = [1, 1, 1, 1]	FES = 209100	Diversity = 1.2236e+00	k_t
Generation 2100: Best = 1.327379e+05 = [4, 2, 3, 4]	FES = 210100	Diversity = 1.2174e+00	k_t
Generation 2110: Best = 1.321787e+05 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.2668e+00	k_t

Generation 2120: Best = 1.312116e+05	FES = 212100	Diversity = 1.4857e+00	k_t
= [3, 2, 2, 2]			
Generation 2130: Best = 1.305791e+05	FES = 213100	Diversity = 1.8869e+00	k_t
= [1, 1, 1, 1]			
Generation 2140: Best = 1.296698e+05	FES = 214100	Diversity = 2.0387e+00	k_t
= [4, 2, 2, 3]			
Generation 2150: Best = 1.289074e+05	FES = 215100	Diversity = 2.0952e+00	k_t
= [1, 1, 1, 1]			
Generation 2160: Best = 1.281310e+05	FES = 216100	Diversity = 2.0113e+00	k_t
= [4, 3, 4, 2]			
Generation 2170: Best = 1.248301e+05	FES = 217100	Diversity = 1.9634e+00	k_t
= [1, 1, 1, 1]			
Generation 2180: Best = 1.248301e+05	FES = 218100	Diversity = 2.4042e+00	k_t
= [2, 4, 2, 3]			
Generation 2190: Best = 1.234653e+05	FES = 219100	Diversity = 2.4753e+00	k_t
= [1, 1, 1, 1]			
Generation 2200: Best = 1.228079e+05	FES = 220100	Diversity = 2.3852e+00	k_t
= [4, 4, 4, 4]			
Generation 2210: Best = 1.223787e+05	FES = 221100	Diversity = 2.2812e+00	k_t
= [1, 1, 1, 1]			
Generation 2220: Best = 1.213609e+05	FES = 222100	Diversity = 2.1246e+00	k_t
= [4, 3, 2, 4]			
Generation 2230: Best = 1.203144e+05	FES = 223100	Diversity = 2.1975e+00	k_t
= [1, 1, 1, 1]			
Generation 2240: Best = 1.198530e+05	FES = 224100	Diversity = 2.1275e+00	k_t
= [2, 4, 3, 2]			
Generation 2250: Best = 1.191925e+05	FES = 225100	Diversity = 2.0422e+00	k_t
= [1, 1, 1, 1]			
Generation 2260: Best = 1.186711e+05	FES = 226100	Diversity = 1.8100e+00	k_t
= [2, 4, 4, 2]			
Generation 2270: Best = 1.184252e+05	FES = 227100	Diversity = 1.7381e+00	k_t
= [1, 1, 1, 1]			
Generation 2280: Best = 1.177686e+05	FES = 228100	Diversity = 1.7054e+00	k_t
= [4, 2, 3, 4]			
Generation 2290: Best = 1.172668e+05	FES = 229100	Diversity = 1.7547e+00	k_t
= [1, 1, 1, 1]			
Generation 2300: Best = 1.167261e+05	FES = 230100	Diversity = 1.6881e+00	k_t
= [2, 2, 3, 3]			
Generation 2310: Best = 1.165606e+05	FES = 231100	Diversity = 1.5123e+00	k_t
= [1, 1, 1, 1]			
Generation 2320: Best = 1.163242e+05	FES = 232100	Diversity = 1.3413e+00	k_t
= [3, 4, 3, 2]			
Generation 2330: Best = 1.154654e+05	FES = 233100	Diversity = 1.2106e+00	k_t
= [1, 1, 1, 1]			
Generation 2340: Best = 1.152368e+05	FES = 234100	Diversity = 1.2291e+00	k_t
= [3, 4, 3, 2]			
Generation 2350: Best = 1.150001e+05	FES = 235100	Diversity = 1.3258e+00	k_t
= [1, 1, 1, 1]			
Generation 2360: Best = 1.148225e+05	FES = 236100	Diversity = 1.3421e+00	k_t
= [3, 2, 4, 4]			
Generation 2370: Best = 1.142024e+05	FES = 237100	Diversity = 1.2881e+00	k_t
= [1, 1, 1, 1]			
Generation 2380: Best = 1.130198e+05	FES = 238100	Diversity = 1.4805e+00	k_t
= [4, 2, 4, 4]			

Generation 2390: Best = 1.125084e+05 = [1, 1, 1, 1]	FES = 239100	Diversity = 1.6317e+00	k_t
Generation 2400: Best = 1.123551e+05 = [2, 4, 3, 3]	FES = 240100	Diversity = 1.7180e+00	k_t
Generation 2410: Best = 1.119129e+05 = [1, 1, 1, 1]	FES = 241100	Diversity = 1.6682e+00	k_t
Generation 2420: Best = 1.112712e+05 = [2, 2, 3, 4]	FES = 242100	Diversity = 1.6378e+00	k_t
Generation 2430: Best = 1.110167e+05 = [1, 1, 1, 1]	FES = 243100	Diversity = 1.6410e+00	k_t
Generation 2440: Best = 1.103890e+05 = [4, 3, 2, 2]	FES = 244100	Diversity = 1.6508e+00	k_t
Generation 2450: Best = 1.102819e+05 = [1, 1, 1, 1]	FES = 245100	Diversity = 1.6404e+00	k_t
Generation 2460: Best = 1.098684e+05 = [3, 4, 2, 2]	FES = 246100	Diversity = 1.4956e+00	k_t
Generation 2470: Best = 1.091930e+05 = [1, 1, 1, 1]	FES = 247100	Diversity = 1.3920e+00	k_t
Generation 2480: Best = 1.090987e+05 = [4, 2, 4, 4]	FES = 248100	Diversity = 1.3644e+00	k_t
Generation 2490: Best = 1.086125e+05 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.3706e+00	k_t
Generation 2500: Best = 1.083805e+05 = [3, 3, 4, 4]	FES = 250100	Diversity = 1.4348e+00	k_t
Generation 2510: Best = 1.079608e+05 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.4279e+00	k_t
Generation 2520: Best = 1.078214e+05 = [3, 4, 4, 3]	FES = 252100	Diversity = 1.3800e+00	k_t
Generation 2530: Best = 1.072841e+05 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.1937e+00	k_t
Generation 2540: Best = 1.072283e+05 = [3, 2, 2, 3]	FES = 254100	Diversity = 1.1414e+00	k_t
Generation 2550: Best = 1.069513e+05 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.0635e+00	k_t
Generation 2560: Best = 1.065495e+05 = [3, 3, 4, 3]	FES = 256100	Diversity = 9.8910e-01	k_t
Generation 2570: Best = 1.064450e+05 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.0483e+00	k_t
Generation 2580: Best = 1.057010e+05 = [2, 2, 4, 3]	FES = 258100	Diversity = 1.1913e+00	k_t
Generation 2590: Best = 1.057010e+05 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.1318e+00	k_t
Generation 2600: Best = 1.052795e+05 = [2, 2, 3, 2]	FES = 260100	Diversity = 1.0850e+00	k_t
Generation 2610: Best = 1.052795e+05 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.0681e+00	k_t
Generation 2620: Best = 1.047149e+05 = [4, 2, 3, 4]	FES = 262100	Diversity = 1.0956e+00	k_t
Generation 2630: Best = 1.047149e+05 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.1328e+00	k_t
Generation 2640: Best = 1.044582e+05 = [3, 2, 4, 4]	FES = 264100	Diversity = 1.1372e+00	k_t
Generation 2650: Best = 1.036095e+05 = [1, 1, 1, 1]	FES = 265100	Diversity = 1.0239e+00	k_t

Generation 2660: Best = 1.029747e+05 = [3, 3, 2, 2]	FES = 266100	Diversity = 1.2219e+00	k_t
Generation 2670: Best = 1.022055e+05 = [1, 1, 1, 1]	FES = 267100	Diversity = 1.3756e+00	k_t
Generation 2680: Best = 1.016379e+05 = [3, 3, 2, 4]	FES = 268100	Diversity = 1.6128e+00	k_t
Generation 2690: Best = 1.012735e+05 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.7824e+00	k_t
Generation 2700: Best = 9.994236e+04 = [2, 4, 3, 3]	FES = 270100	Diversity = 1.7927e+00	k_t
Generation 2710: Best = 9.915668e+04 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.8800e+00	k_t
Generation 2720: Best = 9.851392e+04 = [4, 4, 2, 4]	FES = 272100	Diversity = 2.0465e+00	k_t
Generation 2730: Best = 9.851392e+04 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.8322e+00	k_t
Generation 2740: Best = 9.795555e+04 = [2, 3, 4, 2]	FES = 274100	Diversity = 1.5747e+00	k_t
Generation 2750: Best = 9.722307e+04 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.4245e+00	k_t
Generation 2760: Best = 9.699159e+04 = [4, 4, 4, 4]	FES = 276100	Diversity = 1.4354e+00	k_t
Generation 2770: Best = 9.636320e+04 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.3216e+00	k_t
Generation 2780: Best = 9.520495e+04 = [2, 3, 2, 3]	FES = 278100	Diversity = 1.5047e+00	k_t
Generation 2790: Best = 9.441254e+04 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.7500e+00	k_t
Generation 2800: Best = 9.368939e+04 = [2, 2, 2, 3]	FES = 280100	Diversity = 1.6742e+00	k_t
Generation 2810: Best = 9.282572e+04 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.5146e+00	k_t
Generation 2820: Best = 9.214639e+04 = [2, 2, 3, 2]	FES = 282100	Diversity = 1.6626e+00	k_t
Generation 2830: Best = 9.214639e+04 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.5990e+00	k_t
Generation 2840: Best = 9.141467e+04 = [2, 2, 2, 2]	FES = 284100	Diversity = 1.5169e+00	k_t
Generation 2850: Best = 9.100000e+04 = [1, 1, 1, 1]	FES = 285100	Diversity = 1.4804e+00	k_t
Generation 2860: Best = 9.072230e+04 = [2, 3, 4, 2]	FES = 286100	Diversity = 1.3482e+00	k_t
Generation 2870: Best = 9.034223e+04 = [1, 1, 1, 1]	FES = 287100	Diversity = 1.3153e+00	k_t
Generation 2880: Best = 8.980817e+04 = [4, 2, 2, 3]	FES = 288100	Diversity = 1.3189e+00	k_t
Generation 2890: Best = 8.960713e+04 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.2609e+00	k_t
Generation 2900: Best = 8.923336e+04 = [4, 3, 3, 2]	FES = 290100	Diversity = 1.1268e+00	k_t
Generation 2910: Best = 8.857332e+04 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.2513e+00	k_t
Generation 2920: Best = 8.826096e+04 = [3, 4, 4, 3]	FES = 292100	Diversity = 1.2351e+00	k_t

Generation 2930: Best = 8.772800e+04 = [1, 1, 1, 1]	FES = 293100	Diversity = 1.2589e+00	k_t
Generation 2940: Best = 8.720714e+04 = [2, 4, 2, 3]	FES = 294100	Diversity = 1.4299e+00	k_t
Generation 2950: Best = 8.689244e+04 = [1, 1, 1, 1]	FES = 295100	Diversity = 1.4533e+00	k_t
Generation 2960: Best = 8.601237e+04 = [3, 4, 3, 3]	FES = 296100	Diversity = 1.4135e+00	k_t
Generation 2970: Best = 8.575621e+04 = [1, 1, 1, 1]	FES = 297100	Diversity = 1.5116e+00	k_t
Generation 2980: Best = 8.500031e+04 = [3, 2, 3, 4]	FES = 298100	Diversity = 1.4462e+00	k_t
Generation 2990: Best = 8.450562e+04 = [1, 1, 1, 1]	FES = 299100	Diversity = 1.5508e+00	k_t
Generation 3000: Best = 8.356264e+04 = [4, 3, 4, 4]	FES = 300100	Diversity = 1.8256e+00	k_t
Generation 3010: Best = 8.326767e+04 = [1, 1, 1, 1]	FES = 301100	Diversity = 1.7681e+00	k_t
Generation 3020: Best = 8.289692e+04 = [2, 4, 4, 4]	FES = 302100	Diversity = 1.6705e+00	k_t
Generation 3030: Best = 8.249281e+04 = [1, 1, 1, 1]	FES = 303100	Diversity = 1.6189e+00	k_t
Generation 3040: Best = 8.225101e+04 = [3, 4, 4, 3]	FES = 304100	Diversity = 1.4697e+00	k_t
Generation 3050: Best = 8.186702e+04 = [1, 1, 1, 1]	FES = 305100	Diversity = 1.4285e+00	k_t
Generation 3060: Best = 8.120041e+04 = [4, 4, 2, 4]	FES = 306100	Diversity = 1.3737e+00	k_t
Generation 3070: Best = 8.120041e+04 = [1, 1, 1, 1]	FES = 307100	Diversity = 1.2970e+00	k_t
Generation 3080: Best = 8.084670e+04 = [4, 3, 3, 2]	FES = 308100	Diversity = 1.3284e+00	k_t
Generation 3090: Best = 8.075446e+04 = [1, 1, 1, 1]	FES = 309100	Diversity = 1.2844e+00	k_t
Generation 3100: Best = 8.017143e+04 = [3, 2, 3, 4]	FES = 310100	Diversity = 1.2258e+00	k_t
Generation 3110: Best = 7.976219e+04 = [1, 1, 1, 1]	FES = 311100	Diversity = 1.2652e+00	k_t
Generation 3120: Best = 7.953269e+04 = [3, 4, 3, 2]	FES = 312100	Diversity = 1.4010e+00	k_t
Generation 3130: Best = 7.919754e+04 = [1, 1, 1, 1]	FES = 313100	Diversity = 1.2728e+00	k_t
Generation 3140: Best = 7.855156e+04 = [3, 3, 2, 3]	FES = 314100	Diversity = 1.3193e+00	k_t
Generation 3150: Best = 7.727185e+04 = [1, 1, 1, 1]	FES = 315100	Diversity = 1.5294e+00	k_t
Generation 3160: Best = 7.727185e+04 = [2, 4, 2, 3]	FES = 316100	Diversity = 1.5466e+00	k_t
Generation 3170: Best = 7.671612e+04 = [1, 1, 1, 1]	FES = 317100	Diversity = 1.5776e+00	k_t
Generation 3180: Best = 7.637001e+04 = [4, 4, 2, 3]	FES = 318100	Diversity = 1.6373e+00	k_t
Generation 3190: Best = 7.601490e+04 = [1, 1, 1, 1]	FES = 319100	Diversity = 1.5596e+00	k_t

Generation 3200: Best = 7.556670e+04 = [3, 4, 4, 4]	FES = 320100	Diversity = 1.4229e+00	k_t
Generation 3210: Best = 7.485786e+04 = [1, 1, 1, 1]	FES = 321100	Diversity = 1.2551e+00	k_t
Generation 3220: Best = 7.425023e+04 = [3, 4, 4, 2]	FES = 322100	Diversity = 1.3646e+00	k_t
Generation 3230: Best = 7.395055e+04 = [1, 1, 1, 1]	FES = 323100	Diversity = 1.4635e+00	k_t
Generation 3240: Best = 7.281168e+04 = [4, 2, 3, 4]	FES = 324100	Diversity = 1.5445e+00	k_t
Generation 3250: Best = 7.280226e+04 = [1, 1, 1, 1]	FES = 325100	Diversity = 1.4159e+00	k_t
Generation 3260: Best = 7.199134e+04 = [3, 2, 4, 2]	FES = 326100	Diversity = 1.5266e+00	k_t
Generation 3270: Best = 7.154602e+04 = [1, 1, 1, 1]	FES = 327100	Diversity = 1.6212e+00	k_t
Generation 3280: Best = 7.096983e+04 = [4, 3, 4, 4]	FES = 328100	Diversity = 1.5886e+00	k_t
Generation 3290: Best = 7.052587e+04 = [1, 1, 1, 1]	FES = 329100	Diversity = 1.5182e+00	k_t
Generation 3300: Best = 6.966373e+04 = [3, 2, 4, 2]	FES = 330100	Diversity = 1.5216e+00	k_t
Generation 3310: Best = 6.905953e+04 = [1, 1, 1, 1]	FES = 331100	Diversity = 1.6487e+00	k_t
Generation 3320: Best = 6.788810e+04 = [4, 2, 2, 2]	FES = 332100	Diversity = 1.8685e+00	k_t
Generation 3330: Best = 6.634251e+04 = [1, 1, 1, 1]	FES = 333100	Diversity = 2.2552e+00	k_t
Generation 3340: Best = 6.586527e+04 = [3, 3, 2, 3]	FES = 334100	Diversity = 2.1844e+00	k_t
Generation 3350: Best = 6.521968e+04 = [1, 1, 1, 1]	FES = 335100	Diversity = 2.1269e+00	k_t
Generation 3360: Best = 6.488598e+04 = [3, 4, 3, 3]	FES = 336100	Diversity = 2.0775e+00	k_t
Generation 3370: Best = 6.327876e+04 = [1, 1, 1, 1]	FES = 337100	Diversity = 2.0940e+00	k_t
Generation 3380: Best = 6.327876e+04 = [3, 3, 2, 4]	FES = 338100	Diversity = 1.9206e+00	k_t
Generation 3390: Best = 6.288300e+04 = [1, 1, 1, 1]	FES = 339100	Diversity = 1.6328e+00	k_t
Generation 3400: Best = 6.269187e+04 = [3, 3, 4, 4]	FES = 340100	Diversity = 1.3464e+00	k_t
Generation 3410: Best = 6.253792e+04 = [1, 1, 1, 1]	FES = 341100	Diversity = 1.1920e+00	k_t
Generation 3420: Best = 6.181713e+04 = [3, 3, 2, 4]	FES = 342100	Diversity = 1.1561e+00	k_t
Generation 3430: Best = 6.148411e+04 = [1, 1, 1, 1]	FES = 343100	Diversity = 1.2257e+00	k_t
Generation 3440: Best = 6.102161e+04 = [4, 2, 4, 4]	FES = 344100	Diversity = 1.3893e+00	k_t
Generation 3450: Best = 6.035030e+04 = [1, 1, 1, 1]	FES = 345100	Diversity = 1.5634e+00	k_t
Generation 3460: Best = 5.983311e+04 = [4, 4, 3, 3]	FES = 346100	Diversity = 1.5800e+00	k_t

Generation 3470: Best = 5.957562e+04 = [1, 1, 1, 1]	FES = 347100	Diversity = 1.5432e+00	k_t
Generation 3480: Best = 5.869162e+04 = [4, 4, 4, 2]	FES = 348100	Diversity = 1.5529e+00	k_t
Generation 3490: Best = 5.864039e+04 = [1, 1, 1, 1]	FES = 349100	Diversity = 1.6608e+00	k_t
Generation 3500: Best = 5.821612e+04 = [2, 2, 4, 2]	FES = 350100	Diversity = 1.5135e+00	k_t
Generation 3510: Best = 5.740853e+04 = [1, 1, 1, 1]	FES = 351100	Diversity = 1.4900e+00	k_t
Generation 3520: Best = 5.733620e+04 = [2, 3, 3, 3]	FES = 352100	Diversity = 1.3691e+00	k_t
Generation 3530: Best = 5.613249e+04 = [1, 1, 1, 1]	FES = 353100	Diversity = 1.1826e+00	k_t
Generation 3540: Best = 5.548767e+04 = [2, 3, 2, 2]	FES = 354100	Diversity = 1.3527e+00	k_t
Generation 3550: Best = 5.539832e+04 = [1, 1, 1, 1]	FES = 355100	Diversity = 1.3467e+00	k_t
Generation 3560: Best = 5.534047e+04 = [4, 2, 4, 3]	FES = 356100	Diversity = 1.3625e+00	k_t
Generation 3570: Best = 5.498895e+04 = [1, 1, 1, 1]	FES = 357100	Diversity = 1.3340e+00	k_t
Generation 3580: Best = 5.485990e+04 = [4, 2, 2, 4]	FES = 358100	Diversity = 1.2672e+00	k_t
Generation 3590: Best = 5.414882e+04 = [1, 1, 1, 1]	FES = 359100	Diversity = 1.2891e+00	k_t
Generation 3600: Best = 5.386796e+04 = [3, 3, 3, 3]	FES = 360100	Diversity = 1.3470e+00	k_t
Generation 3610: Best = 5.336872e+04 = [1, 1, 1, 1]	FES = 361100	Diversity = 1.4074e+00	k_t
Generation 3620: Best = 5.282421e+04 = [3, 3, 2, 4]	FES = 362100	Diversity = 1.5295e+00	k_t
Generation 3630: Best = 5.182353e+04 = [1, 1, 1, 1]	FES = 363100	Diversity = 1.7073e+00	k_t
Generation 3640: Best = 5.163202e+04 = [2, 3, 2, 4]	FES = 364100	Diversity = 1.6892e+00	k_t
Generation 3650: Best = 5.162441e+04 = [1, 1, 1, 1]	FES = 365100	Diversity = 1.5813e+00	k_t
Generation 3660: Best = 5.124973e+04 = [4, 4, 2, 4]	FES = 366100	Diversity = 1.5193e+00	k_t
Generation 3670: Best = 5.105244e+04 = [1, 1, 1, 1]	FES = 367100	Diversity = 1.4462e+00	k_t
Generation 3680: Best = 5.063282e+04 = [3, 3, 3, 3]	FES = 368100	Diversity = 1.4778e+00	k_t
Generation 3690: Best = 5.051754e+04 = [1, 1, 1, 1]	FES = 369100	Diversity = 1.4080e+00	k_t
Generation 3700: Best = 5.015714e+04 = [2, 2, 2, 2]	FES = 370100	Diversity = 1.2916e+00	k_t
Generation 3710: Best = 4.958938e+04 = [1, 1, 1, 1]	FES = 371100	Diversity = 1.2676e+00	k_t
Generation 3720: Best = 4.944370e+04 = [3, 2, 3, 4]	FES = 372100	Diversity = 1.3998e+00	k_t
Generation 3730: Best = 4.931087e+04 = [1, 1, 1, 1]	FES = 373100	Diversity = 1.2696e+00	k_t

Generation 3740: Best = 4.877564e+04 = [2, 3, 2, 4]	FES = 374100	Diversity = 1.2635e+00	k_t
Generation 3750: Best = 4.869349e+04 = [1, 1, 1, 1]	FES = 375100	Diversity = 1.1694e+00	k_t
Generation 3760: Best = 4.852264e+04 = [3, 4, 3, 3]	FES = 376100	Diversity = 1.1532e+00	k_t
Generation 3770: Best = 4.827744e+04 = [1, 1, 1, 1]	FES = 377100	Diversity = 1.0988e+00	k_t
Generation 3780: Best = 4.789538e+04 = [2, 2, 4, 4]	FES = 378100	Diversity = 1.0677e+00	k_t
Generation 3790: Best = 4.725628e+04 = [1, 1, 1, 1]	FES = 379100	Diversity = 1.2304e+00	k_t
Generation 3800: Best = 4.709776e+04 = [3, 4, 4, 4]	FES = 380100	Diversity = 1.2331e+00	k_t
Generation 3810: Best = 4.683110e+04 = [1, 1, 1, 1]	FES = 381100	Diversity = 1.1464e+00	k_t
Generation 3820: Best = 4.639889e+04 = [4, 3, 3, 4]	FES = 382100	Diversity = 1.1471e+00	k_t
Generation 3830: Best = 4.639889e+04 = [1, 1, 1, 1]	FES = 383100	Diversity = 1.1587e+00	k_t
Generation 3840: Best = 4.612713e+04 = [3, 2, 4, 3]	FES = 384100	Diversity = 1.0712e+00	k_t
Generation 3850: Best = 4.528857e+04 = [1, 1, 1, 1]	FES = 385100	Diversity = 1.1023e+00	k_t
Generation 3860: Best = 4.523933e+04 = [3, 3, 2, 4]	FES = 386100	Diversity = 1.1980e+00	k_t
Generation 3870: Best = 4.493550e+04 = [1, 1, 1, 1]	FES = 387100	Diversity = 1.2941e+00	k_t
Generation 3880: Best = 4.478043e+04 = [4, 3, 4, 3]	FES = 388100	Diversity = 1.3596e+00	k_t
Generation 3890: Best = 4.417998e+04 = [1, 1, 1, 1]	FES = 389100	Diversity = 1.3062e+00	k_t
Generation 3900: Best = 4.417998e+04 = [2, 2, 2, 3]	FES = 390100	Diversity = 1.3240e+00	k_t
Generation 3910: Best = 4.390692e+04 = [1, 1, 1, 1]	FES = 391100	Diversity = 1.1482e+00	k_t
Generation 3920: Best = 4.370522e+04 = [3, 2, 4, 4]	FES = 392100	Diversity = 1.0354e+00	k_t
Generation 3930: Best = 4.370522e+04 = [1, 1, 1, 1]	FES = 393100	Diversity = 9.2144e-01	k_t
Generation 3940: Best = 4.320704e+04 = [2, 3, 4, 2]	FES = 394100	Diversity = 9.5541e-01	k_t
Generation 3950: Best = 4.306861e+04 = [1, 1, 1, 1]	FES = 395100	Diversity = 1.1022e+00	k_t
Generation 3960: Best = 4.280950e+04 = [2, 2, 2, 2]	FES = 396100	Diversity = 1.1432e+00	k_t
Generation 3970: Best = 4.264556e+04 = [1, 1, 1, 1]	FES = 397100	Diversity = 1.0921e+00	k_t
Generation 3980: Best = 4.256287e+04 = [2, 4, 4, 3]	FES = 398100	Diversity = 1.0479e+00	k_t
Generation 3990: Best = 4.248273e+04 = [1, 1, 1, 1]	FES = 399100	Diversity = 1.0088e+00	k_t
Generation 4000: Best = 4.226230e+04 = [3, 4, 2, 2]	FES = 400100	Diversity = 9.4734e-01	k_t

Generation 4010: Best = 4.182298e+04 = [1, 1, 1, 1]	FES = 401100	Diversity = 1.0092e+00	k_t
Generation 4020: Best = 4.173557e+04 = [3, 2, 3, 3]	FES = 402100	Diversity = 1.0987e+00	k_t
Generation 4030: Best = 4.148785e+04 = [1, 1, 1, 1]	FES = 403100	Diversity = 1.0723e+00	k_t
Generation 4040: Best = 4.116385e+04 = [4, 2, 2, 2]	FES = 404100	Diversity = 1.0519e+00	k_t
Generation 4050: Best = 4.116385e+04 = [1, 1, 1, 1]	FES = 405100	Diversity = 1.0048e+00	k_t
Generation 4060: Best = 4.022081e+04 = [2, 3, 3, 4]	FES = 406100	Diversity = 1.1057e+00	k_t
Generation 4070: Best = 3.969846e+04 = [1, 1, 1, 1]	FES = 407100	Diversity = 1.4197e+00	k_t
Generation 4080: Best = 3.933056e+04 = [3, 3, 4, 2]	FES = 408100	Diversity = 1.4524e+00	k_t
Generation 4090: Best = 3.910249e+04 = [1, 1, 1, 1]	FES = 409100	Diversity = 1.4011e+00	k_t
Generation 4100: Best = 3.894000e+04 = [4, 2, 2, 4]	FES = 410100	Diversity = 1.3186e+00	k_t
Generation 4110: Best = 3.860725e+04 = [1, 1, 1, 1]	FES = 411100	Diversity = 1.1332e+00	k_t
Generation 4120: Best = 3.853854e+04 = [2, 3, 2, 3]	FES = 412100	Diversity = 1.1079e+00	k_t
Generation 4130: Best = 3.742183e+04 = [1, 1, 1, 1]	FES = 413100	Diversity = 1.1470e+00	k_t
Generation 4140: Best = 3.742183e+04 = [3, 4, 3, 2]	FES = 414100	Diversity = 1.2575e+00	k_t
Generation 4150: Best = 3.730666e+04 = [1, 1, 1, 1]	FES = 415100	Diversity = 1.2845e+00	k_t
Generation 4160: Best = 3.716983e+04 = [4, 2, 3, 2]	FES = 416100	Diversity = 1.2619e+00	k_t
Generation 4170: Best = 3.680547e+04 = [1, 1, 1, 1]	FES = 417100	Diversity = 1.1959e+00	k_t
Generation 4180: Best = 3.672477e+04 = [4, 3, 2, 3]	FES = 418100	Diversity = 1.0683e+00	k_t
Generation 4190: Best = 3.656988e+04 = [1, 1, 1, 1]	FES = 419100	Diversity = 1.0768e+00	k_t
Generation 4200: Best = 3.633633e+04 = [3, 3, 4, 2]	FES = 420100	Diversity = 9.7037e-01	k_t
Generation 4210: Best = 3.623362e+04 = [1, 1, 1, 1]	FES = 421100	Diversity = 9.2978e-01	k_t
Generation 4220: Best = 3.616479e+04 = [4, 3, 2, 2]	FES = 422100	Diversity = 8.2211e-01	k_t
Generation 4230: Best = 3.609504e+04 = [1, 1, 1, 1]	FES = 423100	Diversity = 7.5751e-01	k_t
Generation 4240: Best = 3.581879e+04 = [3, 4, 2, 3]	FES = 424100	Diversity = 6.8142e-01	k_t
Generation 4250: Best = 3.573868e+04 = [1, 1, 1, 1]	FES = 425100	Diversity = 8.3557e-01	k_t
Generation 4260: Best = 3.549729e+04 = [4, 2, 2, 3]	FES = 426100	Diversity = 7.9651e-01	k_t
Generation 4270: Best = 3.544606e+04 = [1, 1, 1, 1]	FES = 427100	Diversity = 7.6712e-01	k_t

Generation 4280: Best = 3.533409e+04 = [3, 2, 4, 3]	FES = 428100	Diversity = 7.7334e-01	k_t
Generation 4290: Best = 3.522485e+04 = [1, 1, 1, 1]	FES = 429100	Diversity = 7.4081e-01	k_t
Generation 4300: Best = 3.518476e+04 = [2, 2, 4, 2]	FES = 430100	Diversity = 7.6363e-01	k_t
Generation 4310: Best = 3.495559e+04 = [1, 1, 1, 1]	FES = 431100	Diversity = 6.9112e-01	k_t
Generation 4320: Best = 3.486860e+04 = [3, 3, 4, 3]	FES = 432100	Diversity = 7.5662e-01	k_t
Generation 4330: Best = 3.469701e+04 = [1, 1, 1, 1]	FES = 433100	Diversity = 8.3907e-01	k_t
Generation 4340: Best = 3.451800e+04 = [4, 2, 4, 3]	FES = 434100	Diversity = 8.5017e-01	k_t
Generation 4350: Best = 3.434787e+04 = [1, 1, 1, 1]	FES = 435100	Diversity = 8.0275e-01	k_t
Generation 4360: Best = 3.432921e+04 = [4, 3, 3, 2]	FES = 436100	Diversity = 7.7093e-01	k_t
Generation 4370: Best = 3.415799e+04 = [1, 1, 1, 1]	FES = 437100	Diversity = 7.2607e-01	k_t
Generation 4380: Best = 3.405558e+04 = [3, 4, 4, 4]	FES = 438100	Diversity = 7.2366e-01	k_t
Generation 4390: Best = 3.397010e+04 = [1, 1, 1, 1]	FES = 439100	Diversity = 6.9381e-01	k_t
Generation 4400: Best = 3.386123e+04 = [4, 4, 3, 3]	FES = 440100	Diversity = 6.9723e-01	k_t
Generation 4410: Best = 3.360468e+04 = [1, 1, 1, 1]	FES = 441100	Diversity = 6.6559e-01	k_t
Generation 4420: Best = 3.341392e+04 = [4, 3, 2, 4]	FES = 442100	Diversity = 6.6458e-01	k_t
Generation 4430: Best = 3.339974e+04 = [1, 1, 1, 1]	FES = 443100	Diversity = 7.6011e-01	k_t
Generation 4440: Best = 3.324593e+04 = [4, 3, 4, 2]	FES = 444100	Diversity = 6.9035e-01	k_t
Generation 4450: Best = 3.313062e+04 = [1, 1, 1, 1]	FES = 445100	Diversity = 6.3713e-01	k_t
Generation 4460: Best = 3.307880e+04 = [3, 3, 2, 3]	FES = 446100	Diversity = 6.6279e-01	k_t
Generation 4470: Best = 3.299930e+04 = [1, 1, 1, 1]	FES = 447100	Diversity = 6.0585e-01	k_t
Generation 4480: Best = 3.297850e+04 = [3, 4, 4, 3]	FES = 448100	Diversity = 6.0172e-01	k_t
Generation 4490: Best = 3.276405e+04 = [1, 1, 1, 1]	FES = 449100	Diversity = 6.7186e-01	k_t
Generation 4500: Best = 3.251009e+04 = [3, 4, 3, 2]	FES = 450100	Diversity = 7.5780e-01	k_t
Generation 4510: Best = 3.235688e+04 = [1, 1, 1, 1]	FES = 451100	Diversity = 8.1889e-01	k_t
Generation 4520: Best = 3.222698e+04 = [3, 2, 4, 3]	FES = 452100	Diversity = 7.9798e-01	k_t
Generation 4530: Best = 3.202284e+04 = [1, 1, 1, 1]	FES = 453100	Diversity = 7.9403e-01	k_t
Generation 4540: Best = 3.192168e+04 = [3, 4, 4, 3]	FES = 454100	Diversity = 8.1946e-01	k_t

Generation 4550: Best = 3.181756e+04 = [1, 1, 1, 1]	FES = 455100	Diversity = 7.9373e-01	k_t
Generation 4560: Best = 3.150162e+04 = [4, 4, 3, 4]	FES = 456100	Diversity = 7.6141e-01	k_t
Generation 4570: Best = 3.146132e+04 = [1, 1, 1, 1]	FES = 457100	Diversity = 7.3693e-01	k_t
Generation 4580: Best = 3.140741e+04 = [3, 4, 4, 4]	FES = 458100	Diversity = 7.4624e-01	k_t
Generation 4590: Best = 3.125962e+04 = [1, 1, 1, 1]	FES = 459100	Diversity = 7.6353e-01	k_t
Generation 4600: Best = 3.110222e+04 = [2, 4, 3, 2]	FES = 460100	Diversity = 7.6420e-01	k_t
Generation 4610: Best = 3.099426e+04 = [1, 1, 1, 1]	FES = 461100	Diversity = 8.1059e-01	k_t
Generation 4620: Best = 3.082452e+04 = [4, 4, 4, 2]	FES = 462100	Diversity = 8.3006e-01	k_t
Generation 4630: Best = 3.073791e+04 = [1, 1, 1, 1]	FES = 463100	Diversity = 7.5106e-01	k_t
Generation 4640: Best = 3.065376e+04 = [2, 4, 4, 3]	FES = 464100	Diversity = 6.7142e-01	k_t
Generation 4650: Best = 3.036101e+04 = [1, 1, 1, 1]	FES = 465100	Diversity = 6.1685e-01	k_t
Generation 4660: Best = 3.025402e+04 = [3, 4, 2, 3]	FES = 466100	Diversity = 7.0715e-01	k_t
Generation 4670: Best = 3.008461e+04 = [1, 1, 1, 1]	FES = 467100	Diversity = 7.3262e-01	k_t
Generation 4680: Best = 3.004300e+04 = [2, 2, 4, 3]	FES = 468100	Diversity = 7.4898e-01	k_t
Generation 4690: Best = 2.983264e+04 = [1, 1, 1, 1]	FES = 469100	Diversity = 7.1495e-01	k_t
Generation 4700: Best = 2.970424e+04 = [2, 4, 4, 4]	FES = 470100	Diversity = 7.3095e-01	k_t
Generation 4710: Best = 2.962961e+04 = [1, 1, 1, 1]	FES = 471100	Diversity = 7.8704e-01	k_t
Generation 4720: Best = 2.955596e+04 = [3, 2, 4, 2]	FES = 472100	Diversity = 7.8568e-01	k_t
Generation 4730: Best = 2.931555e+04 = [1, 1, 1, 1]	FES = 473100	Diversity = 7.6307e-01	k_t
Generation 4740: Best = 2.899524e+04 = [4, 2, 2, 3]	FES = 474100	Diversity = 7.6096e-01	k_t
Generation 4750: Best = 2.899524e+04 = [1, 1, 1, 1]	FES = 475100	Diversity = 9.4798e-01	k_t
Generation 4760: Best = 2.891998e+04 = [3, 2, 4, 2]	FES = 476100	Diversity = 9.2821e-01	k_t
Generation 4770: Best = 2.882665e+04 = [1, 1, 1, 1]	FES = 477100	Diversity = 8.2610e-01	k_t
Generation 4780: Best = 2.876706e+04 = [3, 2, 3, 3]	FES = 478100	Diversity = 7.3642e-01	k_t
Generation 4790: Best = 2.844384e+04 = [1, 1, 1, 1]	FES = 479100	Diversity = 7.9807e-01	k_t
Generation 4800: Best = 2.824755e+04 = [2, 3, 4, 4]	FES = 480100	Diversity = 8.4477e-01	k_t
Generation 4810: Best = 2.811468e+04 = [1, 1, 1, 1]	FES = 481100	Diversity = 8.8383e-01	k_t

Generation 4820: Best = 2.800698e+04 = [4, 4, 4, 4]	FES = 482100	Diversity = 8.6203e-01	k_t
Generation 4830: Best = 2.786792e+04 = [1, 1, 1, 1]	FES = 483100	Diversity = 8.5064e-01	k_t
Generation 4840: Best = 2.763380e+04 = [4, 4, 2, 2]	FES = 484100	Diversity = 9.1782e-01	k_t
Generation 4850: Best = 2.752325e+04 = [1, 1, 1, 1]	FES = 485100	Diversity = 8.7034e-01	k_t
Generation 4860: Best = 2.743261e+04 = [4, 2, 4, 4]	FES = 486100	Diversity = 8.9373e-01	k_t
Generation 4870: Best = 2.715610e+04 = [1, 1, 1, 1]	FES = 487100	Diversity = 8.3438e-01	k_t
Generation 4880: Best = 2.711118e+04 = [4, 3, 4, 4]	FES = 488100	Diversity = 8.1868e-01	k_t
Generation 4890: Best = 2.694822e+04 = [1, 1, 1, 1]	FES = 489100	Diversity = 7.5769e-01	k_t
Generation 4900: Best = 2.673907e+04 = [2, 4, 4, 2]	FES = 490100	Diversity = 7.0331e-01	k_t
Generation 4910: Best = 2.671965e+04 = [1, 1, 1, 1]	FES = 491100	Diversity = 7.3910e-01	k_t
Generation 4920: Best = 2.666422e+04 = [4, 4, 4, 3]	FES = 492100	Diversity = 7.0779e-01	k_t
Generation 4930: Best = 2.657153e+04 = [1, 1, 1, 1]	FES = 493100	Diversity = 6.8177e-01	k_t
Generation 4940: Best = 2.624160e+04 = [4, 3, 3, 2]	FES = 494100	Diversity = 6.8328e-01	k_t
Generation 4950: Best = 2.606220e+04 = [1, 1, 1, 1]	FES = 495100	Diversity = 7.8070e-01	k_t
Generation 4960: Best = 2.575436e+04 = [2, 2, 3, 2]	FES = 496100	Diversity = 9.0150e-01	k_t
Generation 4970: Best = 2.558269e+04 = [1, 1, 1, 1]	FES = 497100	Diversity = 9.5207e-01	k_t
Generation 4980: Best = 2.547643e+04 = [4, 4, 3, 3]	FES = 498100	Diversity = 9.3257e-01	k_t
Generation 4990: Best = 2.546514e+04 = [1, 1, 1, 1]	FES = 499100	Diversity = 9.1860e-01	k_t
Generation 5000: Best = 2.531318e+04 = [2, 2, 2, 4]	FES = 500100	Diversity = 8.7509e-01	k_t
Generation 5010: Best = 2.521688e+04 = [1, 1, 1, 1]	FES = 501100	Diversity = 7.8693e-01	k_t
Generation 5020: Best = 2.512416e+04 = [4, 3, 3, 2]	FES = 502100	Diversity = 7.3231e-01	k_t
Generation 5030: Best = 2.502263e+04 = [1, 1, 1, 1]	FES = 503100	Diversity = 7.4330e-01	k_t
Generation 5040: Best = 2.494791e+04 = [4, 4, 3, 4]	FES = 504100	Diversity = 7.2028e-01	k_t
Generation 5050: Best = 2.488139e+04 = [1, 1, 1, 1]	FES = 505100	Diversity = 7.1787e-01	k_t
Generation 5060: Best = 2.474523e+04 = [2, 3, 4, 2]	FES = 506100	Diversity = 7.1951e-01	k_t
Generation 5070: Best = 2.444474e+04 = [1, 1, 1, 1]	FES = 507100	Diversity = 8.0057e-01	k_t
Generation 5080: Best = 2.424192e+04 = [3, 2, 2, 4]	FES = 508100	Diversity = 9.8173e-01	k_t

Generation 5090: Best = 2.406128e+04 = [1, 1, 1, 1]	FES = 509100	Diversity = 9.6444e-01	k_t
Generation 5100: Best = 2.399005e+04 = [4, 2, 4, 3]	FES = 510100	Diversity = 8.9770e-01	k_t
Generation 5110: Best = 2.390585e+04 = [1, 1, 1, 1]	FES = 511100	Diversity = 8.1041e-01	k_t
Generation 5120: Best = 2.384917e+04 = [4, 4, 2, 2]	FES = 512100	Diversity = 7.2585e-01	k_t
Generation 5130: Best = 2.372529e+04 = [1, 1, 1, 1]	FES = 513100	Diversity = 7.2590e-01	k_t
Generation 5140: Best = 2.355528e+04 = [4, 4, 2, 3]	FES = 514100	Diversity = 7.4094e-01	k_t
Generation 5150: Best = 2.344678e+04 = [1, 1, 1, 1]	FES = 515100	Diversity = 7.7627e-01	k_t
Generation 5160: Best = 2.328424e+04 = [4, 3, 2, 2]	FES = 516100	Diversity = 7.8780e-01	k_t
Generation 5170: Best = 2.316305e+04 = [1, 1, 1, 1]	FES = 517100	Diversity = 8.4595e-01	k_t
Generation 5180: Best = 2.297448e+04 = [3, 3, 4, 3]	FES = 518100	Diversity = 8.9285e-01	k_t
Generation 5190: Best = 2.288459e+04 = [1, 1, 1, 1]	FES = 519100	Diversity = 9.0784e-01	k_t
Generation 5200: Best = 2.281998e+04 = [3, 2, 3, 2]	FES = 520100	Diversity = 8.1785e-01	k_t
Generation 5210: Best = 2.273879e+04 = [1, 1, 1, 1]	FES = 521100	Diversity = 7.7745e-01	k_t
Generation 5220: Best = 2.263432e+04 = [2, 2, 4, 3]	FES = 522100	Diversity = 7.5483e-01	k_t
Generation 5230: Best = 2.255500e+04 = [1, 1, 1, 1]	FES = 523100	Diversity = 7.0990e-01	k_t
Generation 5240: Best = 2.246933e+04 = [3, 4, 2, 4]	FES = 524100	Diversity = 6.6814e-01	k_t
Generation 5250: Best = 2.245387e+04 = [1, 1, 1, 1]	FES = 525100	Diversity = 6.4978e-01	k_t
Generation 5260: Best = 2.213888e+04 = [4, 2, 2, 4]	FES = 526100	Diversity = 6.3862e-01	k_t
Generation 5270: Best = 2.213888e+04 = [1, 1, 1, 1]	FES = 527100	Diversity = 7.2849e-01	k_t
Generation 5280: Best = 2.197129e+04 = [4, 3, 2, 2]	FES = 528100	Diversity = 7.1554e-01	k_t
Generation 5290: Best = 2.192157e+04 = [1, 1, 1, 1]	FES = 529100	Diversity = 7.1282e-01	k_t
Generation 5300: Best = 2.187464e+04 = [3, 3, 2, 4]	FES = 530100	Diversity = 6.8615e-01	k_t
Generation 5310: Best = 2.182116e+04 = [1, 1, 1, 1]	FES = 531100	Diversity = 6.2564e-01	k_t
Generation 5320: Best = 2.177545e+04 = [3, 2, 4, 2]	FES = 532100	Diversity = 5.5786e-01	k_t
Generation 5330: Best = 2.162579e+04 = [1, 1, 1, 1]	FES = 533100	Diversity = 5.0850e-01	k_t
Generation 5340: Best = 2.152800e+04 = [3, 4, 2, 4]	FES = 534100	Diversity = 5.4764e-01	k_t
Generation 5350: Best = 2.145654e+04 = [1, 1, 1, 1]	FES = 535100	Diversity = 6.1569e-01	k_t

Generation 5360: Best = 2.133948e+04 = [4, 3, 4, 3]	FES = 536100	Diversity = 6.2359e-01	k_t
Generation 5370: Best = 2.126043e+04 = [1, 1, 1, 1]	FES = 537100	Diversity = 6.7835e-01	k_t
Generation 5380: Best = 2.109814e+04 = [4, 2, 3, 4]	FES = 538100	Diversity = 7.7911e-01	k_t
Generation 5390: Best = 2.099721e+04 = [1, 1, 1, 1]	FES = 539100	Diversity = 8.7792e-01	k_t
Generation 5400: Best = 2.081891e+04 = [4, 4, 2, 2]	FES = 540100	Diversity = 8.8658e-01	k_t
Generation 5410: Best = 2.073571e+04 = [1, 1, 1, 1]	FES = 541100	Diversity = 7.9745e-01	k_t
Generation 5420: Best = 2.060601e+04 = [2, 4, 2, 3]	FES = 542100	Diversity = 7.8156e-01	k_t
Generation 5430: Best = 2.052002e+04 = [1, 1, 1, 1]	FES = 543100	Diversity = 7.5307e-01	k_t
Generation 5440: Best = 2.041497e+04 = [4, 4, 3, 2]	FES = 544100	Diversity = 8.1893e-01	k_t
Generation 5450: Best = 2.036522e+04 = [1, 1, 1, 1]	FES = 545100	Diversity = 8.0415e-01	k_t
Generation 5460: Best = 2.017351e+04 = [3, 3, 2, 4]	FES = 546100	Diversity = 7.6736e-01	k_t
Generation 5470: Best = 2.007765e+04 = [1, 1, 1, 1]	FES = 547100	Diversity = 7.4676e-01	k_t
Generation 5480: Best = 2.003681e+04 = [3, 4, 4, 2]	FES = 548100	Diversity = 6.9948e-01	k_t
Generation 5490: Best = 1.990216e+04 = [1, 1, 1, 1]	FES = 549100	Diversity = 6.3660e-01	k_t
Generation 5500: Best = 1.990216e+04 = [2, 3, 4, 3]	FES = 550100	Diversity = 6.2383e-01	k_t
Generation 5510: Best = 1.986936e+04 = [1, 1, 1, 1]	FES = 551100	Diversity = 5.9746e-01	k_t
Generation 5520: Best = 1.973194e+04 = [3, 4, 2, 4]	FES = 552100	Diversity = 5.5672e-01	k_t
Generation 5530: Best = 1.959083e+04 = [1, 1, 1, 1]	FES = 553100	Diversity = 6.6755e-01	k_t
Generation 5540: Best = 1.943850e+04 = [4, 2, 3, 3]	FES = 554100	Diversity = 8.0580e-01	k_t
Generation 5550: Best = 1.931737e+04 = [1, 1, 1, 1]	FES = 555100	Diversity = 7.8487e-01	k_t
Generation 5560: Best = 1.922310e+04 = [4, 2, 4, 4]	FES = 556100	Diversity = 7.1464e-01	k_t
Generation 5570: Best = 1.913706e+04 = [1, 1, 1, 1]	FES = 557100	Diversity = 7.0105e-01	k_t
Generation 5580: Best = 1.900622e+04 = [2, 2, 4, 3]	FES = 558100	Diversity = 6.7206e-01	k_t
Generation 5590: Best = 1.897306e+04 = [1, 1, 1, 1]	FES = 559100	Diversity = 7.0225e-01	k_t
Generation 5600: Best = 1.890549e+04 = [2, 3, 4, 4]	FES = 560100	Diversity = 6.4414e-01	k_t
Generation 5610: Best = 1.881440e+04 = [1, 1, 1, 1]	FES = 561100	Diversity = 6.5755e-01	k_t
Generation 5620: Best = 1.873006e+04 = [4, 2, 4, 2]	FES = 562100	Diversity = 5.8666e-01	k_t

Generation 5630: Best = 1.871230e+04 = [1, 1, 1, 1]	FES = 563100	Diversity = 5.4449e-01	k_t
Generation 5640: Best = 1.859674e+04 = [2, 3, 4, 3]	FES = 564100	Diversity = 5.2639e-01	k_t
Generation 5650: Best = 1.848920e+04 = [1, 1, 1, 1]	FES = 565100	Diversity = 5.8157e-01	k_t
Generation 5660: Best = 1.833795e+04 = [2, 4, 4, 3]	FES = 566100	Diversity = 6.2978e-01	k_t
Generation 5670: Best = 1.821431e+04 = [1, 1, 1, 1]	FES = 567100	Diversity = 6.4879e-01	k_t
Generation 5680: Best = 1.818035e+04 = [2, 4, 3, 4]	FES = 568100	Diversity = 6.3902e-01	k_t
Generation 5690: Best = 1.798754e+04 = [1, 1, 1, 1]	FES = 569100	Diversity = 6.6200e-01	k_t
Generation 5700: Best = 1.798754e+04 = [4, 2, 3, 4]	FES = 570100	Diversity = 6.8716e-01	k_t
Generation 5710: Best = 1.785671e+04 = [1, 1, 1, 1]	FES = 571100	Diversity = 6.5259e-01	k_t
Generation 5720: Best = 1.776376e+04 = [3, 4, 3, 3]	FES = 572100	Diversity = 6.0701e-01	k_t
Generation 5730: Best = 1.769482e+04 = [1, 1, 1, 1]	FES = 573100	Diversity = 6.2512e-01	k_t
Generation 5740: Best = 1.761372e+04 = [2, 4, 3, 3]	FES = 574100	Diversity = 6.1966e-01	k_t
Generation 5750: Best = 1.758241e+04 = [1, 1, 1, 1]	FES = 575100	Diversity = 5.8903e-01	k_t
Generation 5760: Best = 1.750943e+04 = [2, 2, 4, 2]	FES = 576100	Diversity = 5.3160e-01	k_t
Generation 5770: Best = 1.747836e+04 = [1, 1, 1, 1]	FES = 577100	Diversity = 4.9189e-01	k_t
Generation 5780: Best = 1.742492e+04 = [3, 2, 2, 3]	FES = 578100	Diversity = 4.7404e-01	k_t
Generation 5790: Best = 1.723130e+04 = [1, 1, 1, 1]	FES = 579100	Diversity = 4.3515e-01	k_t
Generation 5800: Best = 1.723130e+04 = [3, 3, 3, 3]	FES = 580100	Diversity = 4.9325e-01	k_t
Generation 5810: Best = 1.719684e+04 = [1, 1, 1, 1]	FES = 581100	Diversity = 5.0535e-01	k_t
Generation 5820: Best = 1.708323e+04 = [4, 4, 3, 2]	FES = 582100	Diversity = 5.4880e-01	k_t
Generation 5830: Best = 1.700310e+04 = [1, 1, 1, 1]	FES = 583100	Diversity = 6.3658e-01	k_t
Generation 5840: Best = 1.688981e+04 = [3, 4, 2, 4]	FES = 584100	Diversity = 5.9531e-01	k_t
Generation 5850: Best = 1.687294e+04 = [1, 1, 1, 1]	FES = 585100	Diversity = 6.0743e-01	k_t
Generation 5860: Best = 1.677429e+04 = [2, 2, 4, 3]	FES = 586100	Diversity = 6.3717e-01	k_t
Generation 5870: Best = 1.675874e+04 = [1, 1, 1, 1]	FES = 587100	Diversity = 6.0439e-01	k_t
Generation 5880: Best = 1.664234e+04 = [2, 3, 3, 4]	FES = 588100	Diversity = 5.1813e-01	k_t
Generation 5890: Best = 1.662252e+04 = [1, 1, 1, 1]	FES = 589100	Diversity = 5.2611e-01	k_t

Generation 5900: Best = 1.657133e+04 = [2, 2, 3, 3]	FES = 590100	Diversity = 5.2321e-01	k_t
Generation 5910: Best = 1.649557e+04 = [1, 1, 1, 1]	FES = 591100	Diversity = 5.3448e-01	k_t
Generation 5920: Best = 1.633520e+04 = [3, 3, 3, 4]	FES = 592100	Diversity = 5.7846e-01	k_t
Generation 5930: Best = 1.629715e+04 = [1, 1, 1, 1]	FES = 593100	Diversity = 6.0663e-01	k_t
Generation 5940: Best = 1.623806e+04 = [3, 2, 3, 4]	FES = 594100	Diversity = 5.6385e-01	k_t
Generation 5950: Best = 1.611856e+04 = [1, 1, 1, 1]	FES = 595100	Diversity = 6.4718e-01	k_t
Generation 5960: Best = 1.606762e+04 = [4, 4, 4, 3]	FES = 596100	Diversity = 6.3899e-01	k_t
Generation 5970: Best = 1.592943e+04 = [1, 1, 1, 1]	FES = 597100	Diversity = 6.5094e-01	k_t
Generation 5980: Best = 1.590068e+04 = [4, 3, 3, 2]	FES = 598100	Diversity = 5.8668e-01	k_t
Generation 5990: Best = 1.581319e+04 = [1, 1, 1, 1]	FES = 599100	Diversity = 5.7113e-01	k_t
Generation 6000: Best = 1.577720e+04 = [2, 3, 4, 3]	FES = 600100	Diversity = 5.5832e-01	k_t
Generation 6010: Best = 1.570067e+04 = [1, 1, 1, 1]	FES = 601100	Diversity = 5.8559e-01	k_t
Generation 6020: Best = 1.563499e+04 = [2, 3, 3, 3]	FES = 602100	Diversity = 5.5918e-01	k_t
Generation 6030: Best = 1.558303e+04 = [1, 1, 1, 1]	FES = 603100	Diversity = 5.6022e-01	k_t
Generation 6040: Best = 1.554055e+04 = [2, 4, 2, 3]	FES = 604100	Diversity = 5.3813e-01	k_t
Generation 6050: Best = 1.552621e+04 = [1, 1, 1, 1]	FES = 605100	Diversity = 4.8688e-01	k_t
Generation 6060: Best = 1.549369e+04 = [2, 4, 4, 2]	FES = 606100	Diversity = 4.1783e-01	k_t
Generation 6070: Best = 1.547124e+04 = [1, 1, 1, 1]	FES = 607100	Diversity = 4.1053e-01	k_t
Generation 6080: Best = 1.537806e+04 = [4, 4, 3, 3]	FES = 608100	Diversity = 4.1703e-01	k_t
Generation 6090: Best = 1.533390e+04 = [1, 1, 1, 1]	FES = 609100	Diversity = 4.7953e-01	k_t
Generation 6100: Best = 1.530675e+04 = [4, 3, 3, 3]	FES = 610100	Diversity = 4.6938e-01	k_t
Generation 6110: Best = 1.525427e+04 = [1, 1, 1, 1]	FES = 611100	Diversity = 4.5518e-01	k_t
Generation 6120: Best = 1.523953e+04 = [3, 2, 2, 4]	FES = 612100	Diversity = 4.8545e-01	k_t
Generation 6130: Best = 1.515780e+04 = [1, 1, 1, 1]	FES = 613100	Diversity = 4.8689e-01	k_t
Generation 6140: Best = 1.504378e+04 = [3, 4, 2, 4]	FES = 614100	Diversity = 5.4590e-01	k_t
Generation 6150: Best = 1.497535e+04 = [1, 1, 1, 1]	FES = 615100	Diversity = 5.5618e-01	k_t
Generation 6160: Best = 1.495379e+04 = [2, 4, 4, 2]	FES = 616100	Diversity = 5.7370e-01	k_t

Generation 6170: Best = 1.488975e+04 = [1, 1, 1, 1]	FES = 617100	Diversity = 6.0670e-01	k_t
Generation 6180: Best = 1.481045e+04 = [2, 4, 2, 4]	FES = 618100	Diversity = 6.2201e-01	k_t
Generation 6190: Best = 1.474580e+04 = [1, 1, 1, 1]	FES = 619100	Diversity = 6.4318e-01	k_t
Generation 6200: Best = 1.468255e+04 = [4, 2, 4, 3]	FES = 620100	Diversity = 5.9502e-01	k_t
Generation 6210: Best = 1.450752e+04 = [1, 1, 1, 1]	FES = 621100	Diversity = 6.0224e-01	k_t
Generation 6220: Best = 1.450752e+04 = [2, 2, 3, 4]	FES = 622100	Diversity = 6.5372e-01	k_t
Generation 6230: Best = 1.449766e+04 = [1, 1, 1, 1]	FES = 623100	Diversity = 6.7299e-01	k_t
Generation 6240: Best = 1.441852e+04 = [3, 4, 4, 2]	FES = 624100	Diversity = 6.3577e-01	k_t
Generation 6250: Best = 1.435911e+04 = [1, 1, 1, 1]	FES = 625100	Diversity = 5.8852e-01	k_t
Generation 6260: Best = 1.434183e+04 = [2, 4, 3, 4]	FES = 626100	Diversity = 5.4347e-01	k_t
Generation 6270: Best = 1.427849e+04 = [1, 1, 1, 1]	FES = 627100	Diversity = 5.5758e-01	k_t
Generation 6280: Best = 1.425944e+04 = [3, 4, 3, 2]	FES = 628100	Diversity = 5.0115e-01	k_t
Generation 6290: Best = 1.417755e+04 = [1, 1, 1, 1]	FES = 629100	Diversity = 4.4833e-01	k_t
Generation 6300: Best = 1.412412e+04 = [4, 4, 3, 2]	FES = 630100	Diversity = 4.8056e-01	k_t
Generation 6310: Best = 1.407598e+04 = [1, 1, 1, 1]	FES = 631100	Diversity = 5.1625e-01	k_t
Generation 6320: Best = 1.386515e+04 = [2, 2, 4, 2]	FES = 632100	Diversity = 5.6727e-01	k_t
Generation 6330: Best = 1.376653e+04 = [1, 1, 1, 1]	FES = 633100	Diversity = 6.8222e-01	k_t
Generation 6340: Best = 1.365255e+04 = [2, 4, 4, 2]	FES = 634100	Diversity = 6.8536e-01	k_t
Generation 6350: Best = 1.359546e+04 = [1, 1, 1, 1]	FES = 635100	Diversity = 7.1022e-01	k_t
Generation 6360: Best = 1.354804e+04 = [3, 3, 4, 4]	FES = 636100	Diversity = 7.3489e-01	k_t
Generation 6370: Best = 1.341861e+04 = [1, 1, 1, 1]	FES = 637100	Diversity = 6.8720e-01	k_t
Generation 6380: Best = 1.338796e+04 = [2, 4, 3, 2]	FES = 638100	Diversity = 6.6511e-01	k_t
Generation 6390: Best = 1.331878e+04 = [1, 1, 1, 1]	FES = 639100	Diversity = 7.0620e-01	k_t
Generation 6400: Best = 1.324101e+04 = [2, 2, 4, 4]	FES = 640100	Diversity = 6.2559e-01	k_t
Generation 6410: Best = 1.310438e+04 = [1, 1, 1, 1]	FES = 641100	Diversity = 5.8816e-01	k_t
Generation 6420: Best = 1.307257e+04 = [3, 2, 3, 2]	FES = 642100	Diversity = 5.5737e-01	k_t
Generation 6430: Best = 1.298862e+04 = [1, 1, 1, 1]	FES = 643100	Diversity = 5.8936e-01	k_t

Generation 6440: Best = 1.295050e+04 = [4, 4, 3, 2]	FES = 644100	Diversity = 5.6339e-01	k_t
Generation 6450: Best = 1.288584e+04 = [1, 1, 1, 1]	FES = 645100	Diversity = 5.2613e-01	k_t
Generation 6460: Best = 1.284662e+04 = [2, 3, 2, 3]	FES = 646100	Diversity = 5.1375e-01	k_t
Generation 6470: Best = 1.278728e+04 = [1, 1, 1, 1]	FES = 647100	Diversity = 5.0366e-01	k_t
Generation 6480: Best = 1.278104e+04 = [3, 2, 2, 3]	FES = 648100	Diversity = 4.8504e-01	k_t
Generation 6490: Best = 1.270378e+04 = [1, 1, 1, 1]	FES = 649100	Diversity = 4.5917e-01	k_t
Generation 6500: Best = 1.260241e+04 = [2, 3, 2, 4]	FES = 650100	Diversity = 5.3564e-01	k_t
Generation 6510: Best = 1.258136e+04 = [1, 1, 1, 1]	FES = 651100	Diversity = 5.4191e-01	k_t
Generation 6520: Best = 1.255610e+04 = [2, 3, 2, 3]	FES = 652100	Diversity = 5.1565e-01	k_t
Generation 6530: Best = 1.252059e+04 = [1, 1, 1, 1]	FES = 653100	Diversity = 4.9012e-01	k_t
Generation 6540: Best = 1.247655e+04 = [2, 3, 2, 2]	FES = 654100	Diversity = 4.5159e-01	k_t
Generation 6550: Best = 1.244101e+04 = [1, 1, 1, 1]	FES = 655100	Diversity = 4.7359e-01	k_t
Generation 6560: Best = 1.237301e+04 = [4, 3, 3, 2]	FES = 656100	Diversity = 4.9271e-01	k_t
Generation 6570: Best = 1.235570e+04 = [1, 1, 1, 1]	FES = 657100	Diversity = 4.6785e-01	k_t
Generation 6580: Best = 1.229783e+04 = [3, 3, 4, 4]	FES = 658100	Diversity = 4.8722e-01	k_t
Generation 6590: Best = 1.213167e+04 = [1, 1, 1, 1]	FES = 659100	Diversity = 5.6503e-01	k_t
Generation 6600: Best = 1.213167e+04 = [2, 4, 3, 4]	FES = 660100	Diversity = 5.3957e-01	k_t
Generation 6610: Best = 1.206522e+04 = [1, 1, 1, 1]	FES = 661100	Diversity = 5.3539e-01	k_t
Generation 6620: Best = 1.199711e+04 = [4, 2, 4, 3]	FES = 662100	Diversity = 5.6811e-01	k_t
Generation 6630: Best = 1.186735e+04 = [1, 1, 1, 1]	FES = 663100	Diversity = 6.1288e-01	k_t
Generation 6640: Best = 1.173242e+04 = [3, 4, 3, 3]	FES = 664100	Diversity = 6.4073e-01	k_t
Generation 6650: Best = 1.170747e+04 = [1, 1, 1, 1]	FES = 665100	Diversity = 6.7859e-01	k_t
Generation 6660: Best = 1.149806e+04 = [3, 2, 4, 4]	FES = 666100	Diversity = 6.4765e-01	k_t
Generation 6670: Best = 1.146585e+04 = [1, 1, 1, 1]	FES = 667100	Diversity = 6.4283e-01	k_t
Generation 6680: Best = 1.140592e+04 = [2, 4, 4, 4]	FES = 668100	Diversity = 6.2162e-01	k_t
Generation 6690: Best = 1.140592e+04 = [1, 1, 1, 1]	FES = 669100	Diversity = 5.6726e-01	k_t
Generation 6700: Best = 1.133780e+04 = [4, 4, 3, 2]	FES = 670100	Diversity = 5.2797e-01	k_t

Generation 6710: Best = 1.127963e+04 = [1, 1, 1, 1]	FES = 671100	Diversity = 4.5131e-01	k_t
Generation 6720: Best = 1.127394e+04 = [3, 3, 4, 3]	FES = 672100	Diversity = 4.5765e-01	k_t
Generation 6730: Best = 1.124908e+04 = [1, 1, 1, 1]	FES = 673100	Diversity = 4.0462e-01	k_t
Generation 6740: Best = 1.109861e+04 = [4, 3, 2, 3]	FES = 674100	Diversity = 3.5622e-01	k_t
Generation 6750: Best = 1.102215e+04 = [1, 1, 1, 1]	FES = 675100	Diversity = 4.4627e-01	k_t
Generation 6760: Best = 1.094800e+04 = [3, 2, 2, 3]	FES = 676100	Diversity = 5.3285e-01	k_t
Generation 6770: Best = 1.084559e+04 = [1, 1, 1, 1]	FES = 677100	Diversity = 6.0903e-01	k_t
Generation 6780: Best = 1.077734e+04 = [2, 4, 3, 3]	FES = 678100	Diversity = 6.9929e-01	k_t
Generation 6790: Best = 1.068651e+04 = [1, 1, 1, 1]	FES = 679100	Diversity = 6.9733e-01	k_t
Generation 6800: Best = 1.064885e+04 = [4, 4, 2, 2]	FES = 680100	Diversity = 6.3584e-01	k_t
Generation 6810: Best = 1.055899e+04 = [1, 1, 1, 1]	FES = 681100	Diversity = 6.3386e-01	k_t
Generation 6820: Best = 1.053233e+04 = [3, 3, 3, 4]	FES = 682100	Diversity = 6.0330e-01	k_t
Generation 6830: Best = 1.049009e+04 = [1, 1, 1, 1]	FES = 683100	Diversity = 5.7982e-01	k_t
Generation 6840: Best = 1.043253e+04 = [2, 2, 2, 3]	FES = 684100	Diversity = 5.5106e-01	k_t
Generation 6850: Best = 1.041127e+04 = [1, 1, 1, 1]	FES = 685100	Diversity = 5.5363e-01	k_t
Generation 6860: Best = 1.037686e+04 = [4, 2, 3, 4]	FES = 686100	Diversity = 5.0619e-01	k_t
Generation 6870: Best = 1.030520e+04 = [1, 1, 1, 1]	FES = 687100	Diversity = 4.7285e-01	k_t
Generation 6880: Best = 1.028747e+04 = [2, 3, 3, 3]	FES = 688100	Diversity = 4.9238e-01	k_t
Generation 6890: Best = 1.023000e+04 = [1, 1, 1, 1]	FES = 689100	Diversity = 4.6649e-01	k_t
Generation 6900: Best = 1.009466e+04 = [4, 3, 3, 2]	FES = 690100	Diversity = 5.1516e-01	k_t
Generation 6910: Best = 1.006545e+04 = [1, 1, 1, 1]	FES = 691100	Diversity = 6.1882e-01	k_t
Generation 6920: Best = 9.909400e+03 = [4, 3, 2, 2]	FES = 692100	Diversity = 6.0916e-01	k_t
Generation 6930: Best = 9.904418e+03 = [1, 1, 1, 1]	FES = 693100	Diversity = 6.1187e-01	k_t
Generation 6940: Best = 9.807772e+03 = [3, 4, 3, 4]	FES = 694100	Diversity = 5.5532e-01	k_t
Generation 6950: Best = 9.747895e+03 = [1, 1, 1, 1]	FES = 695100	Diversity = 5.5864e-01	k_t
Generation 6960: Best = 9.651428e+03 = [4, 2, 4, 2]	FES = 696100	Diversity = 5.9478e-01	k_t
Generation 6970: Best = 9.632389e+03 = [1, 1, 1, 1]	FES = 697100	Diversity = 6.0182e-01	k_t

Generation 6980: Best = 9.590329e+03 = [2, 2, 3, 4]	FES = 698100	Diversity = 5.9232e-01	k_t
Generation 6990: Best = 9.516269e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 5.6905e-01	k_t
Generation 7000: Best = 9.494274e+03 = [3, 3, 3, 3]	FES = 700100	Diversity = 5.6959e-01	k_t
Generation 7010: Best = 9.423450e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 5.3305e-01	k_t
Generation 7020: Best = 9.354930e+03 = [4, 3, 3, 2]	FES = 702100	Diversity = 5.2162e-01	k_t
Generation 7030: Best = 9.297888e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 5.1752e-01	k_t
Generation 7040: Best = 9.297888e+03 = [2, 2, 2, 3]	FES = 704100	Diversity = 5.0620e-01	k_t
Generation 7050: Best = 9.254649e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 4.8158e-01	k_t
Generation 7060: Best = 9.218588e+03 = [3, 4, 3, 2]	FES = 706100	Diversity = 4.6457e-01	k_t
Generation 7070: Best = 9.137246e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 4.8992e-01	k_t
Generation 7080: Best = 9.115545e+03 = [4, 4, 2, 2]	FES = 708100	Diversity = 5.1975e-01	k_t
Generation 7090: Best = 9.085437e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 4.7415e-01	k_t
Generation 7100: Best = 9.059430e+03 = [3, 4, 2, 4]	FES = 710100	Diversity = 4.7817e-01	k_t
Generation 7110: Best = 8.983225e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 4.5029e-01	k_t
Generation 7120: Best = 8.957045e+03 = [4, 2, 3, 2]	FES = 712100	Diversity = 4.4606e-01	k_t
Generation 7130: Best = 8.930250e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 4.5411e-01	k_t
Generation 7140: Best = 8.911375e+03 = [3, 3, 3, 3]	FES = 714100	Diversity = 4.3486e-01	k_t
Generation 7150: Best = 8.882700e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 4.4746e-01	k_t
Generation 7160: Best = 8.815943e+03 = [3, 2, 2, 2]	FES = 716100	Diversity = 4.6731e-01	k_t
Generation 7170: Best = 8.774969e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 4.5805e-01	k_t
Generation 7180: Best = 8.677322e+03 = [2, 3, 2, 4]	FES = 718100	Diversity = 4.7089e-01	k_t
Generation 7190: Best = 8.598363e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 5.2093e-01	k_t
Generation 7200: Best = 8.557973e+03 = [4, 2, 2, 4]	FES = 720100	Diversity = 5.5244e-01	k_t
Generation 7210: Best = 8.456512e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 5.4216e-01	k_t
Generation 7220: Best = 8.454843e+03 = [3, 2, 2, 4]	FES = 722100	Diversity = 5.5293e-01	k_t
Generation 7230: Best = 8.430924e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 5.5887e-01	k_t
Generation 7240: Best = 8.341400e+03 = [4, 4, 2, 4]	FES = 724100	Diversity = 5.5126e-01	k_t

Generation 7250: Best = 8.302047e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 5.7715e-01	k_t
Generation 7260: Best = 8.228330e+03 = [4, 4, 3, 2]	FES = 726100	Diversity = 5.7096e-01	k_t
Generation 7270: Best = 8.174570e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 5.7545e-01	k_t
Generation 7280: Best = 8.127933e+03 = [2, 2, 2, 2]	FES = 728100	Diversity = 5.4748e-01	k_t
Generation 7290: Best = 7.992509e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 5.7094e-01	k_t
Generation 7300: Best = 7.977895e+03 = [3, 3, 2, 2]	FES = 730100	Diversity = 5.8134e-01	k_t
Generation 7310: Best = 7.925363e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 5.4737e-01	k_t
Generation 7320: Best = 7.855982e+03 = [2, 4, 2, 4]	FES = 732100	Diversity = 5.1217e-01	k_t
Generation 7330: Best = 7.755877e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 5.4837e-01	k_t
Generation 7340: Best = 7.732413e+03 = [4, 3, 3, 4]	FES = 734100	Diversity = 5.9519e-01	k_t
Generation 7350: Best = 7.629807e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 6.2508e-01	k_t
Generation 7360: Best = 7.629807e+03 = [3, 2, 3, 3]	FES = 736100	Diversity = 6.3251e-01	k_t
Generation 7370: Best = 7.581433e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 6.0756e-01	k_t
Generation 7380: Best = 7.479799e+03 = [2, 3, 4, 4]	FES = 738100	Diversity = 5.5207e-01	k_t
Generation 7390: Best = 7.419023e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 6.2022e-01	k_t
Generation 7400: Best = 7.414409e+03 = [2, 4, 3, 4]	FES = 740100	Diversity = 6.0500e-01	k_t
Generation 7410: Best = 7.390649e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 5.4307e-01	k_t
Generation 7420: Best = 7.325644e+03 = [3, 3, 3, 3]	FES = 742100	Diversity = 5.2959e-01	k_t
Generation 7430: Best = 7.314087e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 4.5735e-01	k_t
Generation 7440: Best = 7.258841e+03 = [4, 4, 4, 2]	FES = 744100	Diversity = 4.1462e-01	k_t
Generation 7450: Best = 7.132807e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 3.9782e-01	k_t
Generation 7460: Best = 7.132779e+03 = [3, 3, 2, 2]	FES = 746100	Diversity = 4.9045e-01	k_t
Generation 7470: Best = 7.119834e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 4.6456e-01	k_t
Generation 7480: Best = 7.079554e+03 = [2, 2, 4, 3]	FES = 748100	Diversity = 4.4421e-01	k_t
Generation 7490: Best = 7.011548e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 4.5309e-01	k_t
Generation 7500: Best = 6.970442e+03 = [4, 3, 4, 4]	FES = 750100	Diversity = 4.7962e-01	k_t
Generation 7510: Best = 6.895456e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 4.8445e-01	k_t

Generation 7520: Best = 6.895456e+03 = [3, 3, 3, 3]	FES = 752100	Diversity = 5.0580e-01	k_t
Generation 7530: Best = 6.895456e+03 = [1, 1, 1, 1]	FES = 753100	Diversity = 4.5597e-01	k_t
Generation 7540: Best = 6.879507e+03 = [4, 2, 2, 4]	FES = 754100	Diversity = 3.9017e-01	k_t
Generation 7550: Best = 6.859201e+03 = [1, 1, 1, 1]	FES = 755100	Diversity = 3.3602e-01	k_t
Generation 7560: Best = 6.837371e+03 = [2, 4, 3, 4]	FES = 756100	Diversity = 3.3700e-01	k_t
Generation 7570: Best = 6.765211e+03 = [1, 1, 1, 1]	FES = 757100	Diversity = 3.0246e-01	k_t
Generation 7580: Best = 6.728177e+03 = [3, 4, 3, 4]	FES = 758100	Diversity = 3.2995e-01	k_t
Generation 7590: Best = 6.713821e+03 = [1, 1, 1, 1]	FES = 759100	Diversity = 4.0600e-01	k_t
Generation 7600: Best = 6.688050e+03 = [3, 4, 2, 4]	FES = 760100	Diversity = 4.3970e-01	k_t
Generation 7610: Best = 6.671743e+03 = [1, 1, 1, 1]	FES = 761100	Diversity = 4.2199e-01	k_t
Generation 7620: Best = 6.617594e+03 = [3, 3, 4, 4]	FES = 762100	Diversity = 4.0680e-01	k_t
Generation 7630: Best = 6.533281e+03 = [1, 1, 1, 1]	FES = 763100	Diversity = 4.6118e-01	k_t
Generation 7640: Best = 6.449534e+03 = [2, 3, 4, 4]	FES = 764100	Diversity = 4.9989e-01	k_t
Generation 7650: Best = 6.417272e+03 = [1, 1, 1, 1]	FES = 765100	Diversity = 4.7458e-01	k_t
Generation 7660: Best = 6.406866e+03 = [4, 2, 3, 2]	FES = 766100	Diversity = 4.4982e-01	k_t
Generation 7670: Best = 6.330168e+03 = [1, 1, 1, 1]	FES = 767100	Diversity = 4.0027e-01	k_t
Generation 7680: Best = 6.320137e+03 = [4, 4, 4, 4]	FES = 768100	Diversity = 4.3198e-01	k_t
Generation 7690: Best = 6.291942e+03 = [1, 1, 1, 1]	FES = 769100	Diversity = 4.0788e-01	k_t
Generation 7700: Best = 6.240658e+03 = [4, 2, 4, 3]	FES = 770100	Diversity = 3.9020e-01	k_t
Generation 7710: Best = 6.232731e+03 = [1, 1, 1, 1]	FES = 771100	Diversity = 3.5722e-01	k_t
Generation 7720: Best = 6.203709e+03 = [3, 2, 3, 4]	FES = 772100	Diversity = 3.6196e-01	k_t
Generation 7730: Best = 6.175800e+03 = [1, 1, 1, 1]	FES = 773100	Diversity = 3.7993e-01	k_t
Generation 7740: Best = 6.139776e+03 = [2, 2, 4, 2]	FES = 774100	Diversity = 3.6886e-01	k_t
Generation 7750: Best = 6.112115e+03 = [1, 1, 1, 1]	FES = 775100	Diversity = 3.4937e-01	k_t
Generation 7760: Best = 6.094989e+03 = [3, 2, 4, 4]	FES = 776100	Diversity = 3.6798e-01	k_t
Generation 7770: Best = 6.071463e+03 = [1, 1, 1, 1]	FES = 777100	Diversity = 3.6848e-01	k_t
Generation 7780: Best = 6.017770e+03 = [4, 4, 2, 2]	FES = 778100	Diversity = 3.6848e-01	k_t

Generation 7790: Best = 5.983227e+03 = [1, 1, 1, 1]	FES = 779100	Diversity = 3.8924e-01	k_t
Generation 7800: Best = 5.965912e+03 = [3, 2, 3, 3]	FES = 780100	Diversity = 3.8853e-01	k_t
Generation 7810: Best = 5.948753e+03 = [1, 1, 1, 1]	FES = 781100	Diversity = 4.0248e-01	k_t
Generation 7820: Best = 5.900122e+03 = [4, 2, 4, 3]	FES = 782100	Diversity = 3.9366e-01	k_t
Generation 7830: Best = 5.883063e+03 = [1, 1, 1, 1]	FES = 783100	Diversity = 3.8178e-01	k_t
Generation 7840: Best = 5.865844e+03 = [3, 2, 3, 3]	FES = 784100	Diversity = 3.7297e-01	k_t
Generation 7850: Best = 5.847781e+03 = [1, 1, 1, 1]	FES = 785100	Diversity = 3.5624e-01	k_t
Generation 7860: Best = 5.807117e+03 = [4, 2, 4, 3]	FES = 786100	Diversity = 3.3353e-01	k_t
Generation 7870: Best = 5.766924e+03 = [1, 1, 1, 1]	FES = 787100	Diversity = 3.4877e-01	k_t
Generation 7880: Best = 5.732115e+03 = [2, 4, 2, 2]	FES = 788100	Diversity = 3.9796e-01	k_t
Generation 7890: Best = 5.654487e+03 = [1, 1, 1, 1]	FES = 789100	Diversity = 4.5027e-01	k_t
Generation 7900: Best = 5.642647e+03 = [3, 4, 2, 2]	FES = 790100	Diversity = 4.7218e-01	k_t
Generation 7910: Best = 5.627892e+03 = [1, 1, 1, 1]	FES = 791100	Diversity = 4.5148e-01	k_t
Generation 7920: Best = 5.585956e+03 = [4, 2, 2, 2]	FES = 792100	Diversity = 4.2164e-01	k_t
Generation 7930: Best = 5.577270e+03 = [1, 1, 1, 1]	FES = 793100	Diversity = 3.8305e-01	k_t
Generation 7940: Best = 5.540719e+03 = [4, 2, 4, 2]	FES = 794100	Diversity = 3.6172e-01	k_t
Generation 7950: Best = 5.521444e+03 = [1, 1, 1, 1]	FES = 795100	Diversity = 3.3452e-01	k_t
Generation 7960: Best = 5.511201e+03 = [3, 4, 2, 4]	FES = 796100	Diversity = 3.2522e-01	k_t
Generation 7970: Best = 5.467493e+03 = [1, 1, 1, 1]	FES = 797100	Diversity = 3.5311e-01	k_t
Generation 7980: Best = 5.444524e+03 = [4, 3, 4, 3]	FES = 798100	Diversity = 3.7295e-01	k_t
Generation 7990: Best = 5.412032e+03 = [1, 1, 1, 1]	FES = 799100	Diversity = 3.5064e-01	k_t
Generation 8000: Best = 5.402331e+03 = [2, 4, 2, 3]	FES = 800100	Diversity = 3.4856e-01	k_t
Generation 8010: Best = 5.372246e+03 = [1, 1, 1, 1]	FES = 801100	Diversity = 3.5458e-01	k_t
Generation 8020: Best = 5.364966e+03 = [2, 4, 2, 3]	FES = 802100	Diversity = 3.3192e-01	k_t
Generation 8030: Best = 5.331812e+03 = [1, 1, 1, 1]	FES = 803100	Diversity = 3.1453e-01	k_t
Generation 8040: Best = 5.319516e+03 = [2, 4, 3, 3]	FES = 804100	Diversity = 3.1696e-01	k_t
Generation 8050: Best = 5.319250e+03 = [1, 1, 1, 1]	FES = 805100	Diversity = 3.2754e-01	k_t

Generation 8060: Best = 5.298912e+03 = [2, 2, 3, 2]	FES = 806100	Diversity = 2.9310e-01	k_t
Generation 8070: Best = 5.292183e+03 = [1, 1, 1, 1]	FES = 807100	Diversity = 2.6227e-01	k_t
Generation 8080: Best = 5.258265e+03 = [2, 4, 3, 3]	FES = 808100	Diversity = 2.5242e-01	k_t
Generation 8090: Best = 5.258265e+03 = [1, 1, 1, 1]	FES = 809100	Diversity = 2.6376e-01	k_t
Generation 8100: Best = 5.247614e+03 = [4, 2, 3, 2]	FES = 810100	Diversity = 2.6183e-01	k_t
Generation 8110: Best = 5.228882e+03 = [1, 1, 1, 1]	FES = 811100	Diversity = 2.7301e-01	k_t
Generation 8120: Best = 5.187982e+03 = [2, 2, 2, 4]	FES = 812100	Diversity = 3.1920e-01	k_t
Generation 8130: Best = 5.160931e+03 = [1, 1, 1, 1]	FES = 813100	Diversity = 3.1520e-01	k_t
Generation 8140: Best = 5.139445e+03 = [2, 4, 3, 4]	FES = 814100	Diversity = 3.5838e-01	k_t
Generation 8150: Best = 5.099793e+03 = [1, 1, 1, 1]	FES = 815100	Diversity = 3.8594e-01	k_t
Generation 8160: Best = 5.086513e+03 = [4, 4, 4, 2]	FES = 816100	Diversity = 3.8107e-01	k_t
Generation 8170: Best = 5.076609e+03 = [1, 1, 1, 1]	FES = 817100	Diversity = 3.7172e-01	k_t
Generation 8180: Best = 5.053865e+03 = [3, 3, 4, 3]	FES = 818100	Diversity = 3.5591e-01	k_t
Generation 8190: Best = 5.013531e+03 = [1, 1, 1, 1]	FES = 819100	Diversity = 3.3424e-01	k_t
Generation 8200: Best = 5.006134e+03 = [4, 3, 2, 2]	FES = 820100	Diversity = 3.0617e-01	k_t
Generation 8210: Best = 4.983612e+03 = [1, 1, 1, 1]	FES = 821100	Diversity = 3.0766e-01	k_t
Generation 8220: Best = 4.924654e+03 = [2, 3, 3, 3]	FES = 822100	Diversity = 3.2889e-01	k_t
Generation 8230: Best = 4.915170e+03 = [1, 1, 1, 1]	FES = 823100	Diversity = 3.9942e-01	k_t
Generation 8240: Best = 4.878904e+03 = [4, 4, 3, 4]	FES = 824100	Diversity = 4.0656e-01	k_t
Generation 8250: Best = 4.798416e+03 = [1, 1, 1, 1]	FES = 825100	Diversity = 4.0930e-01	k_t
Generation 8260: Best = 4.779077e+03 = [4, 4, 4, 3]	FES = 826100	Diversity = 3.9300e-01	k_t
Generation 8270: Best = 4.766205e+03 = [1, 1, 1, 1]	FES = 827100	Diversity = 4.1758e-01	k_t
Generation 8280: Best = 4.740856e+03 = [2, 3, 2, 3]	FES = 828100	Diversity = 3.7650e-01	k_t
Generation 8290: Best = 4.721581e+03 = [1, 1, 1, 1]	FES = 829100	Diversity = 3.6258e-01	k_t
Generation 8300: Best = 4.707742e+03 = [4, 4, 2, 3]	FES = 830100	Diversity = 3.4425e-01	k_t
Generation 8310: Best = 4.667000e+03 = [1, 1, 1, 1]	FES = 831100	Diversity = 3.3163e-01	k_t
Generation 8320: Best = 4.614844e+03 = [2, 3, 4, 3]	FES = 832100	Diversity = 3.8016e-01	k_t

Generation 8330: Best = 4.613560e+03 = [1, 1, 1, 1]	FES = 833100	Diversity = 3.6308e-01	k_t
Generation 8340: Best = 4.592583e+03 = [4, 4, 2, 2]	FES = 834100	Diversity = 3.7968e-01	k_t
Generation 8350: Best = 4.552687e+03 = [1, 1, 1, 1]	FES = 835100	Diversity = 3.6558e-01	k_t
Generation 8360: Best = 4.539533e+03 = [4, 4, 3, 4]	FES = 836100	Diversity = 3.5948e-01	k_t
Generation 8370: Best = 4.525313e+03 = [1, 1, 1, 1]	FES = 837100	Diversity = 3.4797e-01	k_t
Generation 8380: Best = 4.505324e+03 = [3, 2, 2, 3]	FES = 838100	Diversity = 3.4713e-01	k_t
Generation 8390: Best = 4.495482e+03 = [1, 1, 1, 1]	FES = 839100	Diversity = 3.1456e-01	k_t
Generation 8400: Best = 4.480191e+03 = [3, 3, 2, 4]	FES = 840100	Diversity = 2.7096e-01	k_t
Generation 8410: Best = 4.443081e+03 = [1, 1, 1, 1]	FES = 841100	Diversity = 2.9680e-01	k_t
Generation 8420: Best = 4.387322e+03 = [4, 3, 2, 4]	FES = 842100	Diversity = 3.4734e-01	k_t
Generation 8430: Best = 4.374357e+03 = [1, 1, 1, 1]	FES = 843100	Diversity = 3.8450e-01	k_t
Generation 8440: Best = 4.353019e+03 = [3, 4, 3, 4]	FES = 844100	Diversity = 3.5239e-01	k_t
Generation 8450: Best = 4.308189e+03 = [1, 1, 1, 1]	FES = 845100	Diversity = 3.5344e-01	k_t
Generation 8460: Best = 4.285717e+03 = [2, 3, 4, 3]	FES = 846100	Diversity = 3.6433e-01	k_t
Generation 8470: Best = 4.276730e+03 = [1, 1, 1, 1]	FES = 847100	Diversity = 3.6987e-01	k_t
Generation 8480: Best = 4.271495e+03 = [2, 2, 4, 4]	FES = 848100	Diversity = 3.6973e-01	k_t
Generation 8490: Best = 4.257081e+03 = [1, 1, 1, 1]	FES = 849100	Diversity = 3.1790e-01	k_t
Generation 8500: Best = 4.225987e+03 = [2, 4, 4, 4]	FES = 850100	Diversity = 3.1412e-01	k_t
Generation 8510: Best = 4.214784e+03 = [1, 1, 1, 1]	FES = 851100	Diversity = 3.3171e-01	k_t
Generation 8520: Best = 4.200221e+03 = [4, 3, 3, 2]	FES = 852100	Diversity = 3.3759e-01	k_t
Generation 8530: Best = 4.178164e+03 = [1, 1, 1, 1]	FES = 853100	Diversity = 3.2892e-01	k_t
Generation 8540: Best = 4.135208e+03 = [2, 2, 2, 2]	FES = 854100	Diversity = 3.0214e-01	k_t
Generation 8550: Best = 4.135208e+03 = [1, 1, 1, 1]	FES = 855100	Diversity = 3.1123e-01	k_t
Generation 8560: Best = 4.083470e+03 = [2, 4, 4, 3]	FES = 856100	Diversity = 3.2721e-01	k_t
Generation 8570: Best = 4.083470e+03 = [1, 1, 1, 1]	FES = 857100	Diversity = 3.5936e-01	k_t
Generation 8580: Best = 4.070672e+03 = [2, 2, 2, 3]	FES = 858100	Diversity = 3.2230e-01	k_t
Generation 8590: Best = 4.035574e+03 = [1, 1, 1, 1]	FES = 859100	Diversity = 2.8837e-01	k_t

Generation 8600: Best = 4.006201e+03 = [3, 4, 4, 4]	FES = 860100	Diversity = 3.1620e-01	k_t
Generation 8610: Best = 3.977244e+03 = [1, 1, 1, 1]	FES = 861100	Diversity = 3.3042e-01	k_t
Generation 8620: Best = 3.965106e+03 = [3, 3, 4, 3]	FES = 862100	Diversity = 3.2851e-01	k_t
Generation 8630: Best = 3.948356e+03 = [1, 1, 1, 1]	FES = 863100	Diversity = 3.0853e-01	k_t
Generation 8640: Best = 3.937013e+03 = [3, 3, 2, 2]	FES = 864100	Diversity = 3.0570e-01	k_t
Generation 8650: Best = 3.911412e+03 = [1, 1, 1, 1]	FES = 865100	Diversity = 3.0844e-01	k_t
Generation 8660: Best = 3.897898e+03 = [2, 2, 4, 3]	FES = 866100	Diversity = 3.1823e-01	k_t
Generation 8670: Best = 3.885382e+03 = [1, 1, 1, 1]	FES = 867100	Diversity = 3.2419e-01	k_t
Generation 8680: Best = 3.860375e+03 = [2, 3, 4, 4]	FES = 868100	Diversity = 3.2602e-01	k_t
Generation 8690: Best = 3.858296e+03 = [1, 1, 1, 1]	FES = 869100	Diversity = 3.1871e-01	k_t
Generation 8700: Best = 3.844511e+03 = [4, 2, 3, 4]	FES = 870100	Diversity = 3.0499e-01	k_t
Generation 8710: Best = 3.843363e+03 = [1, 1, 1, 1]	FES = 871100	Diversity = 2.9504e-01	k_t
Generation 8720: Best = 3.826916e+03 = [2, 4, 3, 4]	FES = 872100	Diversity = 2.9054e-01	k_t
Generation 8730: Best = 3.812588e+03 = [1, 1, 1, 1]	FES = 873100	Diversity = 2.9739e-01	k_t
Generation 8740: Best = 3.800296e+03 = [4, 3, 4, 2]	FES = 874100	Diversity = 3.1486e-01	k_t
Generation 8750: Best = 3.778697e+03 = [1, 1, 1, 1]	FES = 875100	Diversity = 2.9153e-01	k_t
Generation 8760: Best = 3.773804e+03 = [4, 4, 4, 3]	FES = 876100	Diversity = 2.6505e-01	k_t
Generation 8770: Best = 3.750260e+03 = [1, 1, 1, 1]	FES = 877100	Diversity = 2.6191e-01	k_t
Generation 8780: Best = 3.734235e+03 = [4, 2, 2, 4]	FES = 878100	Diversity = 2.7953e-01	k_t
Generation 8790: Best = 3.720130e+03 = [1, 1, 1, 1]	FES = 879100	Diversity = 3.0140e-01	k_t
Generation 8800: Best = 3.691895e+03 = [2, 4, 2, 3]	FES = 880100	Diversity = 3.2511e-01	k_t
Generation 8810: Best = 3.681591e+03 = [1, 1, 1, 1]	FES = 881100	Diversity = 3.2288e-01	k_t
Generation 8820: Best = 3.649842e+03 = [2, 4, 2, 4]	FES = 882100	Diversity = 3.3634e-01	k_t
Generation 8830: Best = 3.637740e+03 = [1, 1, 1, 1]	FES = 883100	Diversity = 3.7347e-01	k_t
Generation 8840: Best = 3.615826e+03 = [4, 4, 2, 4]	FES = 884100	Diversity = 3.7018e-01	k_t
Generation 8850: Best = 3.591558e+03 = [1, 1, 1, 1]	FES = 885100	Diversity = 3.8494e-01	k_t
Generation 8860: Best = 3.566612e+03 = [3, 4, 4, 3]	FES = 886100	Diversity = 3.8479e-01	k_t

Generation 8870: Best = 3.554882e+03 = [1, 1, 1, 1]	FES = 887100	Diversity = 3.7963e-01	k_t
Generation 8880: Best = 3.524224e+03 = [4, 3, 3, 4]	FES = 888100	Diversity = 3.6727e-01	k_t
Generation 8890: Best = 3.487986e+03 = [1, 1, 1, 1]	FES = 889100	Diversity = 3.6371e-01	k_t
Generation 8900: Best = 3.465572e+03 = [2, 4, 4, 3]	FES = 890100	Diversity = 3.6021e-01	k_t
Generation 8910: Best = 3.450951e+03 = [1, 1, 1, 1]	FES = 891100	Diversity = 4.0102e-01	k_t
Generation 8920: Best = 3.437542e+03 = [2, 4, 3, 2]	FES = 892100	Diversity = 3.7386e-01	k_t
Generation 8930: Best = 3.415813e+03 = [1, 1, 1, 1]	FES = 893100	Diversity = 3.3656e-01	k_t
Generation 8940: Best = 3.409740e+03 = [3, 3, 3, 4]	FES = 894100	Diversity = 3.1628e-01	k_t
Generation 8950: Best = 3.392258e+03 = [1, 1, 1, 1]	FES = 895100	Diversity = 2.6590e-01	k_t
Generation 8960: Best = 3.367530e+03 = [2, 2, 3, 3]	FES = 896100	Diversity = 2.6165e-01	k_t
Generation 8970: Best = 3.363046e+03 = [1, 1, 1, 1]	FES = 897100	Diversity = 2.6149e-01	k_t
Generation 8980: Best = 3.321205e+03 = [3, 2, 2, 3]	FES = 898100	Diversity = 2.6449e-01	k_t
Generation 8990: Best = 3.321205e+03 = [1, 1, 1, 1]	FES = 899100	Diversity = 2.6380e-01	k_t
Generation 9000: Best = 3.311583e+03 = [4, 3, 2, 2]	FES = 900100	Diversity = 2.5914e-01	k_t
Generation 9010: Best = 3.294644e+03 = [1, 1, 1, 1]	FES = 901100	Diversity = 2.5546e-01	k_t
Generation 9020: Best = 3.281698e+03 = [4, 2, 2, 2]	FES = 902100	Diversity = 2.6476e-01	k_t
Generation 9030: Best = 3.273203e+03 = [1, 1, 1, 1]	FES = 903100	Diversity = 2.6917e-01	k_t
Generation 9040: Best = 3.263162e+03 = [3, 4, 3, 3]	FES = 904100	Diversity = 2.5533e-01	k_t
Generation 9050: Best = 3.238847e+03 = [1, 1, 1, 1]	FES = 905100	Diversity = 2.6957e-01	k_t
Generation 9060: Best = 3.224224e+03 = [4, 2, 3, 3]	FES = 906100	Diversity = 2.6368e-01	k_t
Generation 9070: Best = 3.216648e+03 = [1, 1, 1, 1]	FES = 907100	Diversity = 2.5530e-01	k_t
Generation 9080: Best = 3.202321e+03 = [3, 3, 2, 3]	FES = 908100	Diversity = 2.5263e-01	k_t
Generation 9090: Best = 3.190386e+03 = [1, 1, 1, 1]	FES = 909100	Diversity = 2.5105e-01	k_t
Generation 9100: Best = 3.170739e+03 = [3, 3, 4, 4]	FES = 910100	Diversity = 2.3643e-01	k_t
Generation 9110: Best = 3.151747e+03 = [1, 1, 1, 1]	FES = 911100	Diversity = 2.5606e-01	k_t
Generation 9120: Best = 3.121790e+03 = [2, 4, 2, 2]	FES = 912100	Diversity = 2.9956e-01	k_t
Generation 9130: Best = 3.092496e+03 = [1, 1, 1, 1]	FES = 913100	Diversity = 2.9328e-01	k_t

Generation 9140: Best = 3.092496e+03 = [3, 4, 3, 2]	FES = 914100	Diversity = 2.9008e-01	k_t
Generation 9150: Best = 3.082192e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 3.0623e-01	k_t
Generation 9160: Best = 3.062195e+03 = [4, 3, 2, 4]	FES = 916100	Diversity = 3.3063e-01	k_t
Generation 9170: Best = 3.050822e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 3.2459e-01	k_t
Generation 9180: Best = 3.032581e+03 = [3, 3, 3, 4]	FES = 918100	Diversity = 3.3372e-01	k_t
Generation 9190: Best = 2.986191e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 3.3324e-01	k_t
Generation 9200: Best = 2.954658e+03 = [2, 4, 4, 3]	FES = 920100	Diversity = 3.4528e-01	k_t
Generation 9210: Best = 2.912676e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 3.6113e-01	k_t
Generation 9220: Best = 2.882959e+03 = [3, 4, 4, 2]	FES = 922100	Diversity = 4.1586e-01	k_t
Generation 9230: Best = 2.867600e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 4.3729e-01	k_t
Generation 9240: Best = 2.812832e+03 = [3, 4, 4, 3]	FES = 924100	Diversity = 4.2656e-01	k_t
Generation 9250: Best = 2.794239e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 4.0446e-01	k_t
Generation 9260: Best = 2.756779e+03 = [2, 3, 2, 2]	FES = 926100	Diversity = 4.2632e-01	k_t
Generation 9270: Best = 2.746970e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 4.3371e-01	k_t
Generation 9280: Best = 2.729529e+03 = [3, 2, 2, 4]	FES = 928100	Diversity = 4.3777e-01	k_t
Generation 9290: Best = 2.712926e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 3.8199e-01	k_t
Generation 9300: Best = 2.687071e+03 = [4, 3, 4, 2]	FES = 930100	Diversity = 3.6187e-01	k_t
Generation 9310: Best = 2.687071e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 3.4709e-01	k_t
Generation 9320: Best = 2.664380e+03 = [2, 2, 2, 3]	FES = 932100	Diversity = 3.0114e-01	k_t
Generation 9330: Best = 2.657432e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 2.7786e-01	k_t
Generation 9340: Best = 2.632902e+03 = [2, 3, 2, 4]	FES = 934100	Diversity = 2.8655e-01	k_t
Generation 9350: Best = 2.622358e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 2.9475e-01	k_t
Generation 9360: Best = 2.618453e+03 = [2, 4, 3, 3]	FES = 936100	Diversity = 2.9194e-01	k_t
Generation 9370: Best = 2.615540e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 2.6611e-01	k_t
Generation 9380: Best = 2.607711e+03 = [2, 2, 3, 2]	FES = 938100	Diversity = 2.4827e-01	k_t
Generation 9390: Best = 2.589292e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 2.2826e-01	k_t
Generation 9400: Best = 2.586313e+03 = [3, 3, 2, 2]	FES = 940100	Diversity = 2.3063e-01	k_t

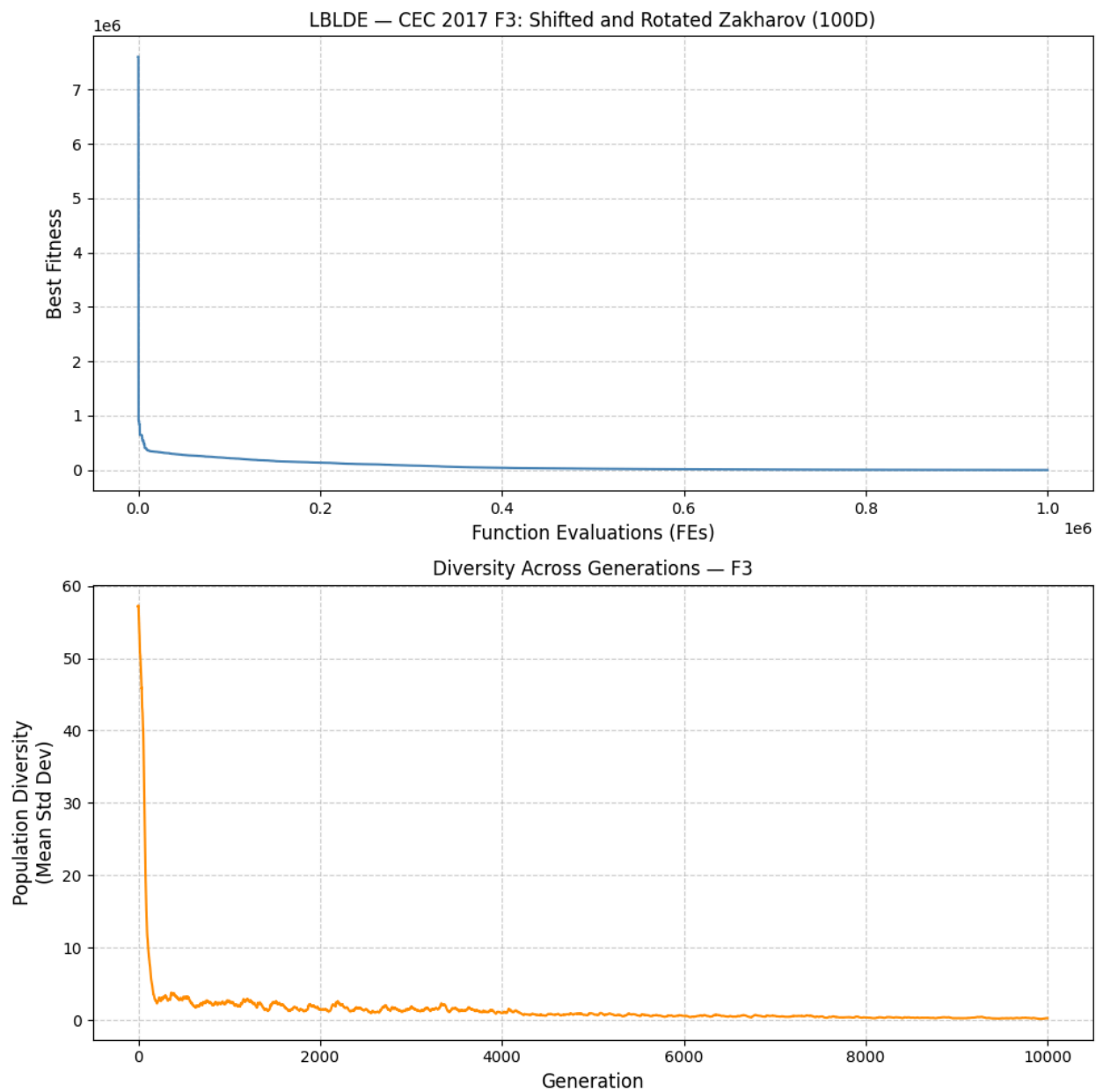
Generation 9410: Best = 2.567581e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 2.1004e-01	k_t
Generation 9420: Best = 2.541232e+03 = [4, 2, 2, 3]	FES = 942100	Diversity = 2.1759e-01	k_t
Generation 9430: Best = 2.541232e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 2.2092e-01	k_t
Generation 9440: Best = 2.535643e+03 = [3, 2, 2, 4]	FES = 944100	Diversity = 2.0636e-01	k_t
Generation 9450: Best = 2.526380e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 1.9584e-01	k_t
Generation 9460: Best = 2.514405e+03 = [4, 2, 4, 3]	FES = 946100	Diversity = 2.0454e-01	k_t
Generation 9470: Best = 2.510279e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 2.1129e-01	k_t
Generation 9480: Best = 2.480123e+03 = [4, 4, 4, 3]	FES = 948100	Diversity = 2.1706e-01	k_t
Generation 9490: Best = 2.469459e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 2.5556e-01	k_t
Generation 9500: Best = 2.461685e+03 = [4, 2, 3, 4]	FES = 950100	Diversity = 2.6402e-01	k_t
Generation 9510: Best = 2.453095e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 2.4316e-01	k_t
Generation 9520: Best = 2.437391e+03 = [4, 3, 4, 4]	FES = 952100	Diversity = 2.4024e-01	k_t
Generation 9530: Best = 2.423261e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 2.5289e-01	k_t
Generation 9540: Best = 2.416032e+03 = [2, 3, 2, 4]	FES = 954100	Diversity = 2.5918e-01	k_t
Generation 9550: Best = 2.391058e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 2.5248e-01	k_t
Generation 9560: Best = 2.376511e+03 = [4, 4, 4, 3]	FES = 956100	Diversity = 2.5517e-01	k_t
Generation 9570: Best = 2.376511e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 2.3995e-01	k_t
Generation 9580: Best = 2.352648e+03 = [3, 2, 4, 4]	FES = 958100	Diversity = 2.4355e-01	k_t
Generation 9590: Best = 2.350345e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 2.6782e-01	k_t
Generation 9600: Best = 2.339088e+03 = [3, 2, 3, 2]	FES = 960100	Diversity = 2.6865e-01	k_t
Generation 9610: Best = 2.328311e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 2.4059e-01	k_t
Generation 9620: Best = 2.313558e+03 = [4, 4, 4, 4]	FES = 962100	Diversity = 2.1925e-01	k_t
Generation 9630: Best = 2.291028e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 2.4600e-01	k_t
Generation 9640: Best = 2.276555e+03 = [3, 3, 4, 2]	FES = 964100	Diversity = 2.6693e-01	k_t
Generation 9650: Best = 2.270245e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 2.7924e-01	k_t
Generation 9660: Best = 2.252686e+03 = [3, 4, 3, 4]	FES = 966100	Diversity = 2.9725e-01	k_t
Generation 9670: Best = 2.238867e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 2.8682e-01	k_t

Generation 9680: Best = 2.224087e+03 = [4, 2, 4, 4]	FES = 968100	Diversity = 2.6408e-01	k_t
Generation 9690: Best = 2.217595e+03 = [1, 1, 1, 1]	FES = 969100	Diversity = 2.6368e-01	k_t
Generation 9700: Best = 2.204947e+03 = [3, 4, 2, 2]	FES = 970100	Diversity = 2.4990e-01	k_t
Generation 9710: Best = 2.163486e+03 = [1, 1, 1, 1]	FES = 971100	Diversity = 2.7190e-01	k_t
Generation 9720: Best = 2.162120e+03 = [4, 4, 3, 3]	FES = 972100	Diversity = 3.1565e-01	k_t
Generation 9730: Best = 2.138284e+03 = [1, 1, 1, 1]	FES = 973100	Diversity = 3.0125e-01	k_t
Generation 9740: Best = 2.126097e+03 = [3, 4, 3, 4]	FES = 974100	Diversity = 2.9485e-01	k_t
Generation 9750: Best = 2.112163e+03 = [1, 1, 1, 1]	FES = 975100	Diversity = 3.1157e-01	k_t
Generation 9760: Best = 2.096354e+03 = [4, 4, 3, 3]	FES = 976100	Diversity = 3.1592e-01	k_t
Generation 9770: Best = 2.081137e+03 = [1, 1, 1, 1]	FES = 977100	Diversity = 2.9328e-01	k_t
Generation 9780: Best = 2.056869e+03 = [3, 3, 3, 2]	FES = 978100	Diversity = 2.9336e-01	k_t
Generation 9790: Best = 2.046566e+03 = [1, 1, 1, 1]	FES = 979100	Diversity = 2.9215e-01	k_t
Generation 9800: Best = 2.033490e+03 = [4, 3, 4, 4]	FES = 980100	Diversity = 2.9173e-01	k_t
Generation 9810: Best = 2.021952e+03 = [1, 1, 1, 1]	FES = 981100	Diversity = 2.6731e-01	k_t
Generation 9820: Best = 2.001240e+03 = [2, 2, 2, 2]	FES = 982100	Diversity = 2.3902e-01	k_t
Generation 9830: Best = 1.973015e+03 = [1, 1, 1, 1]	FES = 983100	Diversity = 2.5865e-01	k_t
Generation 9840: Best = 1.963591e+03 = [4, 4, 4, 3]	FES = 984100	Diversity = 2.9434e-01	k_t
Generation 9850: Best = 1.962638e+03 = [1, 1, 1, 1]	FES = 985100	Diversity = 2.8550e-01	k_t
Generation 9860: Best = 1.949310e+03 = [3, 4, 3, 4]	FES = 986100	Diversity = 2.6569e-01	k_t
Generation 9870: Best = 1.942116e+03 = [1, 1, 1, 1]	FES = 987100	Diversity = 2.3108e-01	k_t
Generation 9880: Best = 1.938242e+03 = [2, 4, 4, 4]	FES = 988100	Diversity = 2.2502e-01	k_t
Generation 9890: Best = 1.933396e+03 = [1, 1, 1, 1]	FES = 989100	Diversity = 1.9962e-01	k_t
Generation 9900: Best = 1.930194e+03 = [3, 2, 3, 2]	FES = 990100	Diversity = 1.8136e-01	k_t
Generation 9910: Best = 1.923422e+03 = [1, 1, 1, 1]	FES = 991100	Diversity = 1.6755e-01	k_t
Generation 9920: Best = 1.919693e+03 = [2, 3, 3, 3]	FES = 992100	Diversity = 1.5987e-01	k_t
Generation 9930: Best = 1.905359e+03 = [1, 1, 1, 1]	FES = 993100	Diversity = 1.6201e-01	k_t
Generation 9940: Best = 1.900115e+03 = [3, 2, 2, 2]	FES = 994100	Diversity = 1.7045e-01	k_t

Generation 9950: Best = 1.889283e+03 FES = 995100 Diversity = 1.8811e-01 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 1.880633e+03 FES = 996100 Diversity = 2.0777e-01 k_t
= [4, 3, 4, 2]
Generation 9970: Best = 1.861464e+03 FES = 997100 Diversity = 2.2469e-01 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 1.852037e+03 FES = 998100 Diversity = 2.4644e-01 k_t
= [2, 3, 4, 4]
Generation 9990: Best = 1.842840e+03 FES = 999100 Diversity = 2.5239e-01 k_t
= [1, 1, 1, 1]

Optimization completed!
Final Best Fitness = 1.840295e+03
Total Function Evaluations = 1000000

=====
=
Results:
Best Fitness : 1.840295e+03
Global Optimum : 3.000000e+02
Error (|best-optimum|) : 1.540295e+03
Best Solution (5 dim) : [-51.22848182 4.81721518 33.85449755 8.54642978
-44.54422465]
D^(0) (init. diversity): 5.7220e+01
=====



Graph saved as convergence_F3.png

6.2 Multimodal Functions (F4-F10)

```
In [28]: # =====
# F4 – Shifted and Rotated Rosenbrock
# =====
run_and_plot('F4', make_wrapper(simple.f4), 400.0, "Shifted and Rotated Rosenb
```

```

=====
=
LBLDE_s on CEC 2017 F4 – Single Run
=====
=
Function      : Shifted and Rotated Rosenbrock's (100D)
Bounds        : [-100, 100]
Global optimum : 400.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 2.335375e+05 | Diversity = 5.7220e+01
Generation 10: Best = 8.179659e+04 FES = 1100 Diversity = 4.0735e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.801931e+04 FES = 2100 Diversity = 2.9132e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 1.922202e+04 FES = 3100 Diversity = 2.0666e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.066977e+04 FES = 4100 Diversity = 1.4678e+01 k_t
= [2, 2, 2, 4]
Generation 50: Best = 7.662542e+03 FES = 5100 Diversity = 1.0126e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 5.942143e+03 FES = 6100 Diversity = 7.8899e+00 k_t
= [2, 4, 3, 4]
Generation 70: Best = 4.628232e+03 FES = 7100 Diversity = 6.3465e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.737759e+03 FES = 8100 Diversity = 5.6175e+00 k_t
= [4, 3, 2, 3]
Generation 90: Best = 3.256210e+03 FES = 9100 Diversity = 4.7999e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.784697e+03 FES = 10100 Diversity = 4.3309e+00 k_t
= [3, 3, 2, 3]
Generation 110: Best = 2.481865e+03 FES = 11100 Diversity = 3.6359e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.231462e+03 FES = 12100 Diversity = 3.2155e+00 k_t
= [4, 3, 2, 2]
Generation 130: Best = 2.075216e+03 FES = 13100 Diversity = 2.8603e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.937803e+03 FES = 14100 Diversity = 2.4996e+00 k_t
= [2, 2, 3, 3]
Generation 150: Best = 1.782405e+03 FES = 15100 Diversity = 2.2862e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.673275e+03 FES = 16100 Diversity = 2.2311e+00 k_t
= [3, 4, 3, 4]
Generation 170: Best = 1.559846e+03 FES = 17100 Diversity = 2.1894e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.468372e+03 FES = 18100 Diversity = 1.9862e+00 k_t
= [4, 4, 2, 4]
Generation 190: Best = 1.375697e+03 FES = 19100 Diversity = 1.8704e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.303436e+03 FES = 20100 Diversity = 1.8351e+00 k_t
= [3, 4, 3, 4]
Generation 210: Best = 1.243370e+03 FES = 21100 Diversity = 1.7670e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.174989e+03 FES = 22100 Diversity = 1.6793e+00 k_t
= [3, 4, 2, 4]

```

Generation 230:	Best = 1.115718e+03	FES = 23100	Diversity = 1.5790e+00	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 1.082816e+03	FES = 24100	Diversity = 1.3960e+00	k_t
= [4, 2, 4, 3]				
Generation 250:	Best = 1.060444e+03	FES = 25100	Diversity = 1.1368e+00	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 1.031951e+03	FES = 26100	Diversity = 1.0207e+00	k_t
= [4, 2, 2, 2]				
Generation 270:	Best = 1.013041e+03	FES = 27100	Diversity = 8.9609e-01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 9.954363e+02	FES = 28100	Diversity = 8.5770e-01	k_t
= [2, 2, 2, 4]				
Generation 290:	Best = 9.801501e+02	FES = 29100	Diversity = 7.9564e-01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 9.651490e+02	FES = 30100	Diversity = 7.1982e-01	k_t
= [4, 4, 2, 4]				
Generation 310:	Best = 9.539127e+02	FES = 31100	Diversity = 7.1502e-01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 9.411313e+02	FES = 32100	Diversity = 6.7561e-01	k_t
= [3, 4, 4, 3]				
Generation 330:	Best = 9.298819e+02	FES = 33100	Diversity = 6.4789e-01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 9.160686e+02	FES = 34100	Diversity = 6.5321e-01	k_t
= [4, 4, 4, 4]				
Generation 350:	Best = 9.060062e+02	FES = 35100	Diversity = 6.2675e-01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 8.955156e+02	FES = 36100	Diversity = 6.1730e-01	k_t
= [2, 4, 2, 3]				
Generation 370:	Best = 8.835629e+02	FES = 37100	Diversity = 6.5778e-01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 8.710865e+02	FES = 38100	Diversity = 6.2072e-01	k_t
= [4, 2, 3, 4]				
Generation 390:	Best = 8.628451e+02	FES = 39100	Diversity = 5.6261e-01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 8.542789e+02	FES = 40100	Diversity = 5.6792e-01	k_t
= [2, 3, 3, 2]				
Generation 410:	Best = 8.462528e+02	FES = 41100	Diversity = 5.7147e-01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 8.388336e+02	FES = 42100	Diversity = 5.2842e-01	k_t
= [3, 4, 3, 3]				
Generation 430:	Best = 8.304818e+02	FES = 43100	Diversity = 5.5067e-01	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 8.235838e+02	FES = 44100	Diversity = 5.3261e-01	k_t
= [2, 3, 3, 2]				
Generation 450:	Best = 8.153209e+02	FES = 45100	Diversity = 5.6767e-01	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 8.105056e+02	FES = 46100	Diversity = 5.4209e-01	k_t
= [4, 4, 4, 4]				
Generation 470:	Best = 8.018330e+02	FES = 47100	Diversity = 5.6822e-01	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 7.940131e+02	FES = 48100	Diversity = 5.2262e-01	k_t
= [4, 4, 2, 4]				
Generation 490:	Best = 7.872467e+02	FES = 49100	Diversity = 5.0408e-01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 7.806565e+02	FES = 50100	Diversity = 4.8775e-01	k_t = [2, 2, 3, 3]
Generation 510:	Best = 7.730854e+02	FES = 51100	Diversity = 4.6390e-01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 7.685160e+02	FES = 52100	Diversity = 4.6369e-01	k_t = [3, 4, 2, 2]
Generation 530:	Best = 7.623140e+02	FES = 53100	Diversity = 4.6013e-01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 7.562146e+02	FES = 54100	Diversity = 4.4313e-01	k_t = [4, 3, 2, 3]
Generation 550:	Best = 7.516245e+02	FES = 55100	Diversity = 4.2882e-01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 7.479402e+02	FES = 56100	Diversity = 3.8210e-01	k_t = [2, 4, 3, 3]
Generation 570:	Best = 7.430850e+02	FES = 57100	Diversity = 3.7113e-01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 7.374565e+02	FES = 58100	Diversity = 4.1030e-01	k_t = [2, 3, 2, 2]
Generation 590:	Best = 7.323799e+02	FES = 59100	Diversity = 4.1711e-01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 7.283630e+02	FES = 60100	Diversity = 3.9372e-01	k_t = [4, 3, 4, 2]
Generation 610:	Best = 7.237273e+02	FES = 61100	Diversity = 3.9852e-01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 7.202569e+02	FES = 62100	Diversity = 3.9331e-01	k_t = [4, 4, 3, 3]
Generation 630:	Best = 7.150696e+02	FES = 63100	Diversity = 3.5285e-01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 7.123624e+02	FES = 64100	Diversity = 3.4185e-01	k_t = [2, 3, 3, 2]
Generation 650:	Best = 7.085284e+02	FES = 65100	Diversity = 3.3179e-01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 7.054410e+02	FES = 66100	Diversity = 3.3705e-01	k_t = [3, 3, 4, 3]
Generation 670:	Best = 7.028735e+02	FES = 67100	Diversity = 3.0105e-01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 6.993489e+02	FES = 68100	Diversity = 3.0731e-01	k_t = [4, 3, 3, 4]
Generation 690:	Best = 6.969934e+02	FES = 69100	Diversity = 3.2556e-01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 6.936854e+02	FES = 70100	Diversity = 3.1741e-01	k_t = [2, 3, 3, 4]
Generation 710:	Best = 6.909960e+02	FES = 71100	Diversity = 2.9574e-01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 6.878627e+02	FES = 72100	Diversity = 3.0778e-01	k_t = [4, 2, 4, 3]
Generation 730:	Best = 6.848955e+02	FES = 73100	Diversity = 2.9787e-01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 6.817038e+02	FES = 74100	Diversity = 2.9171e-01	k_t = [2, 4, 3, 3]
Generation 750:	Best = 6.792294e+02	FES = 75100	Diversity = 3.1141e-01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 6.761598e+02	FES = 76100	Diversity = 3.2271e-01	k_t = [3, 3, 2, 2]

Generation 770:	Best = 6.722825e+02	FES = 77100	Diversity = 3.2716e-01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 6.689793e+02	FES = 78100	Diversity = 2.9566e-01	k_t
= [3, 2, 4, 4]				
Generation 790:	Best = 6.661605e+02	FES = 79100	Diversity = 2.5307e-01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 6.638351e+02	FES = 80100	Diversity = 2.3839e-01	k_t
= [3, 3, 2, 3]				
Generation 810:	Best = 6.618474e+02	FES = 81100	Diversity = 2.2537e-01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 6.593999e+02	FES = 82100	Diversity = 2.2967e-01	k_t
= [4, 3, 2, 4]				
Generation 830:	Best = 6.574932e+02	FES = 83100	Diversity = 2.3165e-01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 6.555110e+02	FES = 84100	Diversity = 2.3912e-01	k_t
= [2, 4, 2, 2]				
Generation 850:	Best = 6.534306e+02	FES = 85100	Diversity = 2.4384e-01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 6.512458e+02	FES = 86100	Diversity = 2.3168e-01	k_t
= [4, 3, 2, 4]				
Generation 870:	Best = 6.494187e+02	FES = 87100	Diversity = 2.3560e-01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 6.475121e+02	FES = 88100	Diversity = 2.3714e-01	k_t
= [4, 4, 2, 2]				
Generation 890:	Best = 6.454512e+02	FES = 89100	Diversity = 2.5840e-01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 6.430156e+02	FES = 90100	Diversity = 2.6287e-01	k_t
= [3, 4, 3, 4]				
Generation 910:	Best = 6.405122e+02	FES = 91100	Diversity = 2.3928e-01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 6.384013e+02	FES = 92100	Diversity = 2.2901e-01	k_t
= [3, 3, 2, 2]				
Generation 930:	Best = 6.367938e+02	FES = 93100	Diversity = 2.3319e-01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 6.348440e+02	FES = 94100	Diversity = 2.2353e-01	k_t
= [4, 2, 3, 2]				
Generation 950:	Best = 6.334822e+02	FES = 95100	Diversity = 2.0525e-01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 6.320592e+02	FES = 96100	Diversity = 1.9546e-01	k_t
= [2, 3, 3, 2]				
Generation 970:	Best = 6.308454e+02	FES = 97100	Diversity = 1.8531e-01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 6.294592e+02	FES = 98100	Diversity = 1.7030e-01	k_t
= [3, 3, 4, 2]				
Generation 990:	Best = 6.282852e+02	FES = 99100	Diversity = 1.6344e-01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 6.273030e+02	FES = 100100	Diversity = 1.6611e-01	k_t
= [4, 4, 3, 4]				
Generation 1010:	Best = 6.259235e+02	FES = 101100	Diversity = 1.8910e-01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 6.247834e+02	FES = 102100	Diversity = 1.8467e-01	k_t
= [2, 2, 3, 4]				
Generation 1030:	Best = 6.234874e+02	FES = 103100	Diversity = 1.9666e-01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 6.225430e+02	FES = 104100	Diversity = 1.7622e-01	k_t
= [4, 2, 3, 4]			
Generation 1050: Best = 6.217016e+02	FES = 105100	Diversity = 1.6414e-01	k_t
= [1, 1, 1, 1]			
Generation 1060: Best = 6.207153e+02	FES = 106100	Diversity = 1.5310e-01	k_t
= [2, 4, 4, 3]			
Generation 1070: Best = 6.198853e+02	FES = 107100	Diversity = 1.4442e-01	k_t
= [1, 1, 1, 1]			
Generation 1080: Best = 6.190999e+02	FES = 108100	Diversity = 1.5979e-01	k_t
= [3, 4, 3, 2]			
Generation 1090: Best = 6.184198e+02	FES = 109100	Diversity = 1.4484e-01	k_t
= [1, 1, 1, 1]			
Generation 1100: Best = 6.177334e+02	FES = 110100	Diversity = 1.3253e-01	k_t
= [3, 4, 3, 3]			
Generation 1110: Best = 6.169935e+02	FES = 111100	Diversity = 1.2894e-01	k_t
= [1, 1, 1, 1]			
Generation 1120: Best = 6.161493e+02	FES = 112100	Diversity = 1.3388e-01	k_t
= [4, 4, 2, 2]			
Generation 1130: Best = 6.156804e+02	FES = 113100	Diversity = 1.2478e-01	k_t
= [1, 1, 1, 1]			
Generation 1140: Best = 6.150468e+02	FES = 114100	Diversity = 1.1372e-01	k_t
= [2, 3, 2, 2]			
Generation 1150: Best = 6.145398e+02	FES = 115100	Diversity = 1.0530e-01	k_t
= [1, 1, 1, 1]			
Generation 1160: Best = 6.141318e+02	FES = 116100	Diversity = 9.8789e-02	k_t
= [4, 3, 3, 3]			
Generation 1170: Best = 6.137000e+02	FES = 117100	Diversity = 9.7987e-02	k_t
= [1, 1, 1, 1]			
Generation 1180: Best = 6.130961e+02	FES = 118100	Diversity = 1.0624e-01	k_t
= [4, 2, 4, 3]			
Generation 1190: Best = 6.125302e+02	FES = 119100	Diversity = 1.1638e-01	k_t
= [1, 1, 1, 1]			
Generation 1200: Best = 6.118764e+02	FES = 120100	Diversity = 1.2604e-01	k_t
= [2, 2, 2, 4]			
Generation 1210: Best = 6.113423e+02	FES = 121100	Diversity = 1.1609e-01	k_t
= [1, 1, 1, 1]			
Generation 1220: Best = 6.107201e+02	FES = 122100	Diversity = 1.1420e-01	k_t
= [2, 4, 3, 2]			
Generation 1230: Best = 6.102650e+02	FES = 123100	Diversity = 1.0672e-01	k_t
= [1, 1, 1, 1]			
Generation 1240: Best = 6.096231e+02	FES = 124100	Diversity = 1.2045e-01	k_t
= [2, 4, 2, 3]			
Generation 1250: Best = 6.088716e+02	FES = 125100	Diversity = 1.2333e-01	k_t
= [1, 1, 1, 1]			
Generation 1260: Best = 6.083620e+02	FES = 126100	Diversity = 1.2096e-01	k_t
= [3, 4, 3, 4]			
Generation 1270: Best = 6.077947e+02	FES = 127100	Diversity = 1.1207e-01	k_t
= [1, 1, 1, 1]			
Generation 1280: Best = 6.073229e+02	FES = 128100	Diversity = 1.0727e-01	k_t
= [2, 2, 4, 2]			
Generation 1290: Best = 6.067143e+02	FES = 129100	Diversity = 1.0794e-01	k_t
= [1, 1, 1, 1]			
Generation 1300: Best = 6.062339e+02	FES = 130100	Diversity = 1.0877e-01	k_t
= [4, 3, 3, 4]			

Generation 1310: Best = 6.057355e+02	FES = 131100	Diversity = 9.9729e-02	k_t
= [1, 1, 1, 1]			
Generation 1320: Best = 6.053098e+02	FES = 132100	Diversity = 9.2822e-02	k_t
= [4, 2, 3, 4]			
Generation 1330: Best = 6.047643e+02	FES = 133100	Diversity = 9.9884e-02	k_t
= [1, 1, 1, 1]			
Generation 1340: Best = 6.043801e+02	FES = 134100	Diversity = 1.0199e-01	k_t
= [2, 4, 3, 3]			
Generation 1350: Best = 6.038774e+02	FES = 135100	Diversity = 1.0181e-01	k_t
= [1, 1, 1, 1]			
Generation 1360: Best = 6.032609e+02	FES = 136100	Diversity = 1.1214e-01	k_t
= [3, 2, 4, 2]			
Generation 1370: Best = 6.028059e+02	FES = 137100	Diversity = 1.1332e-01	k_t
= [1, 1, 1, 1]			
Generation 1380: Best = 6.022116e+02	FES = 138100	Diversity = 1.1341e-01	k_t
= [3, 4, 2, 3]			
Generation 1390: Best = 6.016939e+02	FES = 139100	Diversity = 1.0812e-01	k_t
= [1, 1, 1, 1]			
Generation 1400: Best = 6.011130e+02	FES = 140100	Diversity = 1.0581e-01	k_t
= [2, 3, 3, 3]			
Generation 1410: Best = 6.005553e+02	FES = 141100	Diversity = 1.0566e-01	k_t
= [1, 1, 1, 1]			
Generation 1420: Best = 6.000803e+02	FES = 142100	Diversity = 1.1462e-01	k_t
= [3, 4, 2, 2]			
Generation 1430: Best = 5.995356e+02	FES = 143100	Diversity = 1.2171e-01	k_t
= [1, 1, 1, 1]			
Generation 1440: Best = 5.989315e+02	FES = 144100	Diversity = 1.1336e-01	k_t
= [2, 4, 4, 4]			
Generation 1450: Best = 5.984212e+02	FES = 145100	Diversity = 1.1010e-01	k_t
= [1, 1, 1, 1]			
Generation 1460: Best = 5.978454e+02	FES = 146100	Diversity = 1.0337e-01	k_t
= [3, 4, 4, 3]			
Generation 1470: Best = 5.972940e+02	FES = 147100	Diversity = 1.0006e-01	k_t
= [1, 1, 1, 1]			
Generation 1480: Best = 5.967224e+02	FES = 148100	Diversity = 9.8427e-02	k_t
= [4, 3, 2, 3]			
Generation 1490: Best = 5.961982e+02	FES = 149100	Diversity = 1.0440e-01	k_t
= [1, 1, 1, 1]			
Generation 1500: Best = 5.956958e+02	FES = 150100	Diversity = 1.0426e-01	k_t
= [3, 2, 4, 2]			
Generation 1510: Best = 5.951832e+02	FES = 151100	Diversity = 9.6044e-02	k_t
= [1, 1, 1, 1]			
Generation 1520: Best = 5.946399e+02	FES = 152100	Diversity = 9.4617e-02	k_t
= [3, 3, 2, 4]			
Generation 1530: Best = 5.942723e+02	FES = 153100	Diversity = 9.6381e-02	k_t
= [1, 1, 1, 1]			
Generation 1540: Best = 5.935389e+02	FES = 154100	Diversity = 1.0930e-01	k_t
= [4, 2, 2, 4]			
Generation 1550: Best = 5.928545e+02	FES = 155100	Diversity = 1.1809e-01	k_t
= [1, 1, 1, 1]			
Generation 1560: Best = 5.922367e+02	FES = 156100	Diversity = 1.1801e-01	k_t
= [3, 4, 4, 4]			
Generation 1570: Best = 5.915495e+02	FES = 157100	Diversity = 1.2179e-01	k_t
= [1, 1, 1, 1]			

Generation 1580: Best = 5.908722e+02	FES = 158100	Diversity = 1.1548e-01	k_t
= [4, 3, 3, 4]			
Generation 1590: Best = 5.899871e+02	FES = 159100	Diversity = 1.2415e-01	k_t
= [1, 1, 1, 1]			
Generation 1600: Best = 5.894569e+02	FES = 160100	Diversity = 1.2163e-01	k_t
= [3, 2, 3, 2]			
Generation 1610: Best = 5.889326e+02	FES = 161100	Diversity = 1.0763e-01	k_t
= [1, 1, 1, 1]			
Generation 1620: Best = 5.883637e+02	FES = 162100	Diversity = 1.0756e-01	k_t
= [4, 4, 3, 4]			
Generation 1630: Best = 5.879190e+02	FES = 163100	Diversity = 1.1384e-01	k_t
= [1, 1, 1, 1]			
Generation 1640: Best = 5.875107e+02	FES = 164100	Diversity = 9.6584e-02	k_t
= [4, 2, 2, 4]			
Generation 1650: Best = 5.870490e+02	FES = 165100	Diversity = 9.9853e-02	k_t
= [1, 1, 1, 1]			
Generation 1660: Best = 5.865792e+02	FES = 166100	Diversity = 9.2093e-02	k_t
= [3, 2, 4, 2]			
Generation 1670: Best = 5.861921e+02	FES = 167100	Diversity = 9.0397e-02	k_t
= [1, 1, 1, 1]			
Generation 1680: Best = 5.857878e+02	FES = 168100	Diversity = 9.4852e-02	k_t
= [3, 2, 4, 2]			
Generation 1690: Best = 5.854277e+02	FES = 169100	Diversity = 1.0202e-01	k_t
= [1, 1, 1, 1]			
Generation 1700: Best = 5.849545e+02	FES = 170100	Diversity = 1.0318e-01	k_t
= [2, 3, 2, 4]			
Generation 1710: Best = 5.846658e+02	FES = 171100	Diversity = 8.8907e-02	k_t
= [1, 1, 1, 1]			
Generation 1720: Best = 5.843902e+02	FES = 172100	Diversity = 7.8560e-02	k_t
= [3, 3, 2, 4]			
Generation 1730: Best = 5.840301e+02	FES = 173100	Diversity = 8.2994e-02	k_t
= [1, 1, 1, 1]			
Generation 1740: Best = 5.837704e+02	FES = 174100	Diversity = 8.0563e-02	k_t
= [3, 4, 4, 4]			
Generation 1750: Best = 5.834634e+02	FES = 175100	Diversity = 8.1561e-02	k_t
= [1, 1, 1, 1]			
Generation 1760: Best = 5.831376e+02	FES = 176100	Diversity = 7.9798e-02	k_t
= [2, 3, 3, 4]			
Generation 1770: Best = 5.827842e+02	FES = 177100	Diversity = 7.4210e-02	k_t
= [1, 1, 1, 1]			
Generation 1780: Best = 5.825199e+02	FES = 178100	Diversity = 7.4890e-02	k_t
= [2, 3, 2, 3]			
Generation 1790: Best = 5.822370e+02	FES = 179100	Diversity = 7.0930e-02	k_t
= [1, 1, 1, 1]			
Generation 1800: Best = 5.819362e+02	FES = 180100	Diversity = 7.2123e-02	k_t
= [4, 4, 2, 2]			
Generation 1810: Best = 5.815837e+02	FES = 181100	Diversity = 7.5112e-02	k_t
= [1, 1, 1, 1]			
Generation 1820: Best = 5.811810e+02	FES = 182100	Diversity = 8.1124e-02	k_t
= [3, 4, 3, 3]			
Generation 1830: Best = 5.808061e+02	FES = 183100	Diversity = 9.4611e-02	k_t
= [1, 1, 1, 1]			
Generation 1840: Best = 5.803050e+02	FES = 184100	Diversity = 9.4824e-02	k_t
= [2, 3, 2, 4]			

Generation 1850: Best = 5.800206e+02 = [1, 1, 1, 1]	FES = 185100	Diversity = 8.8900e-02	k_t
Generation 1860: Best = 5.797365e+02 = [2, 3, 4, 2]	FES = 186100	Diversity = 8.1545e-02	k_t
Generation 1870: Best = 5.794859e+02 = [1, 1, 1, 1]	FES = 187100	Diversity = 8.0670e-02	k_t
Generation 1880: Best = 5.792453e+02 = [3, 2, 2, 3]	FES = 188100	Diversity = 7.6162e-02	k_t
Generation 1890: Best = 5.789643e+02 = [1, 1, 1, 1]	FES = 189100	Diversity = 7.6048e-02	k_t
Generation 1900: Best = 5.788121e+02 = [4, 2, 4, 2]	FES = 190100	Diversity = 6.8913e-02	k_t
Generation 1910: Best = 5.786092e+02 = [1, 1, 1, 1]	FES = 191100	Diversity = 6.4365e-02	k_t
Generation 1920: Best = 5.784885e+02 = [3, 4, 2, 3]	FES = 192100	Diversity = 5.9211e-02	k_t
Generation 1930: Best = 5.783894e+02 = [1, 1, 1, 1]	FES = 193100	Diversity = 5.3000e-02	k_t
Generation 1940: Best = 5.782429e+02 = [2, 4, 4, 3]	FES = 194100	Diversity = 5.3837e-02	k_t
Generation 1950: Best = 5.780833e+02 = [1, 1, 1, 1]	FES = 195100	Diversity = 5.4493e-02	k_t
Generation 1960: Best = 5.779666e+02 = [2, 2, 4, 4]	FES = 196100	Diversity = 5.0182e-02	k_t
Generation 1970: Best = 5.778504e+02 = [1, 1, 1, 1]	FES = 197100	Diversity = 4.5554e-02	k_t
Generation 1980: Best = 5.777494e+02 = [3, 3, 3, 3]	FES = 198100	Diversity = 4.1714e-02	k_t
Generation 1990: Best = 5.776617e+02 = [1, 1, 1, 1]	FES = 199100	Diversity = 4.6362e-02	k_t
Generation 2000: Best = 5.775834e+02 = [2, 2, 4, 4]	FES = 200100	Diversity = 4.5901e-02	k_t
Generation 2010: Best = 5.774546e+02 = [1, 1, 1, 1]	FES = 201100	Diversity = 4.8022e-02	k_t
Generation 2020: Best = 5.773526e+02 = [2, 3, 3, 3]	FES = 202100	Diversity = 4.8393e-02	k_t
Generation 2030: Best = 5.772541e+02 = [1, 1, 1, 1]	FES = 203100	Diversity = 4.0587e-02	k_t
Generation 2040: Best = 5.771700e+02 = [3, 4, 2, 2]	FES = 204100	Diversity = 4.5540e-02	k_t
Generation 2050: Best = 5.770832e+02 = [1, 1, 1, 1]	FES = 205100	Diversity = 5.0725e-02	k_t
Generation 2060: Best = 5.769687e+02 = [2, 4, 4, 4]	FES = 206100	Diversity = 5.0638e-02	k_t
Generation 2070: Best = 5.768701e+02 = [1, 1, 1, 1]	FES = 207100	Diversity = 4.8318e-02	k_t
Generation 2080: Best = 5.767724e+02 = [3, 4, 3, 2]	FES = 208100	Diversity = 4.3507e-02	k_t
Generation 2090: Best = 5.766939e+02 = [1, 1, 1, 1]	FES = 209100	Diversity = 3.9848e-02	k_t
Generation 2100: Best = 5.766194e+02 = [4, 3, 4, 4]	FES = 210100	Diversity = 3.7267e-02	k_t
Generation 2110: Best = 5.765539e+02 = [1, 1, 1, 1]	FES = 211100	Diversity = 3.3849e-02	k_t

Generation 2120: Best = 5.764791e+02 = [4, 2, 2, 4]	FES = 212100	Diversity = 3.6768e-02	k_t
Generation 2130: Best = 5.764399e+02 = [1, 1, 1, 1]	FES = 213100	Diversity = 3.8917e-02	k_t
Generation 2140: Best = 5.763803e+02 = [3, 2, 2, 3]	FES = 214100	Diversity = 3.3837e-02	k_t
Generation 2150: Best = 5.763205e+02 = [1, 1, 1, 1]	FES = 215100	Diversity = 3.5729e-02	k_t
Generation 2160: Best = 5.762747e+02 = [3, 2, 3, 2]	FES = 216100	Diversity = 3.3446e-02	k_t
Generation 2170: Best = 5.762367e+02 = [1, 1, 1, 1]	FES = 217100	Diversity = 2.7990e-02	k_t
Generation 2180: Best = 5.762017e+02 = [2, 3, 4, 4]	FES = 218100	Diversity = 2.3775e-02	k_t
Generation 2190: Best = 5.761748e+02 = [1, 1, 1, 1]	FES = 219100	Diversity = 2.1635e-02	k_t
Generation 2200: Best = 5.761491e+02 = [2, 3, 2, 2]	FES = 220100	Diversity = 2.1940e-02	k_t
Generation 2210: Best = 5.761179e+02 = [1, 1, 1, 1]	FES = 221100	Diversity = 2.3798e-02	k_t
Generation 2220: Best = 5.760863e+02 = [2, 4, 2, 4]	FES = 222100	Diversity = 2.3838e-02	k_t
Generation 2230: Best = 5.760684e+02 = [1, 1, 1, 1]	FES = 223100	Diversity = 2.0109e-02	k_t
Generation 2240: Best = 5.760478e+02 = [2, 3, 2, 3]	FES = 224100	Diversity = 1.9715e-02	k_t
Generation 2250: Best = 5.760292e+02 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.9565e-02	k_t
Generation 2260: Best = 5.760091e+02 = [4, 4, 2, 3]	FES = 226100	Diversity = 1.9699e-02	k_t
Generation 2270: Best = 5.759881e+02 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.8721e-02	k_t
Generation 2280: Best = 5.759706e+02 = [2, 2, 4, 4]	FES = 228100	Diversity = 1.8900e-02	k_t
Generation 2290: Best = 5.759444e+02 = [1, 1, 1, 1]	FES = 229100	Diversity = 2.1723e-02	k_t
Generation 2300: Best = 5.759267e+02 = [3, 4, 2, 4]	FES = 230100	Diversity = 2.0098e-02	k_t
Generation 2310: Best = 5.759098e+02 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.9511e-02	k_t
Generation 2320: Best = 5.758894e+02 = [4, 2, 2, 4]	FES = 232100	Diversity = 2.0613e-02	k_t
Generation 2330: Best = 5.758704e+02 = [1, 1, 1, 1]	FES = 233100	Diversity = 2.3622e-02	k_t
Generation 2340: Best = 5.758457e+02 = [3, 4, 2, 4]	FES = 234100	Diversity = 2.3545e-02	k_t
Generation 2350: Best = 5.758312e+02 = [1, 1, 1, 1]	FES = 235100	Diversity = 1.9093e-02	k_t
Generation 2360: Best = 5.758132e+02 = [3, 3, 4, 4]	FES = 236100	Diversity = 1.7311e-02	k_t
Generation 2370: Best = 5.757985e+02 = [1, 1, 1, 1]	FES = 237100	Diversity = 1.7618e-02	k_t
Generation 2380: Best = 5.757853e+02 = [4, 2, 4, 3]	FES = 238100	Diversity = 1.5602e-02	k_t

Generation 2390: Best = 5.757697e+02 = [1, 1, 1, 1]	FES = 239100	Diversity = 1.5985e-02	k_t
Generation 2400: Best = 5.757548e+02 = [3, 4, 3, 2]	FES = 240100	Diversity = 1.5699e-02	k_t
Generation 2410: Best = 5.757421e+02 = [1, 1, 1, 1]	FES = 241100	Diversity = 1.4469e-02	k_t
Generation 2420: Best = 5.757256e+02 = [3, 3, 4, 2]	FES = 242100	Diversity = 1.5290e-02	k_t
Generation 2430: Best = 5.757062e+02 = [1, 1, 1, 1]	FES = 243100	Diversity = 1.5538e-02	k_t
Generation 2440: Best = 5.756899e+02 = [2, 3, 4, 2]	FES = 244100	Diversity = 1.7389e-02	k_t
Generation 2450: Best = 5.756674e+02 = [1, 1, 1, 1]	FES = 245100	Diversity = 2.0102e-02	k_t
Generation 2460: Best = 5.756532e+02 = [2, 3, 4, 4]	FES = 246100	Diversity = 2.0100e-02	k_t
Generation 2470: Best = 5.756305e+02 = [1, 1, 1, 1]	FES = 247100	Diversity = 2.0801e-02	k_t
Generation 2480: Best = 5.756160e+02 = [2, 2, 4, 2]	FES = 248100	Diversity = 1.6662e-02	k_t
Generation 2490: Best = 5.755967e+02 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.7044e-02	k_t
Generation 2500: Best = 5.755836e+02 = [2, 3, 3, 2]	FES = 250100	Diversity = 1.7479e-02	k_t
Generation 2510: Best = 5.755667e+02 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.4616e-02	k_t
Generation 2520: Best = 5.755565e+02 = [4, 4, 3, 2]	FES = 252100	Diversity = 1.5580e-02	k_t
Generation 2530: Best = 5.755436e+02 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.6581e-02	k_t
Generation 2540: Best = 5.755331e+02 = [2, 2, 3, 3]	FES = 254100	Diversity = 1.5326e-02	k_t
Generation 2550: Best = 5.755235e+02 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.3933e-02	k_t
Generation 2560: Best = 5.755072e+02 = [3, 3, 3, 3]	FES = 256100	Diversity = 1.6278e-02	k_t
Generation 2570: Best = 5.754941e+02 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.4626e-02	k_t
Generation 2580: Best = 5.754853e+02 = [3, 2, 3, 4]	FES = 258100	Diversity = 1.2982e-02	k_t
Generation 2590: Best = 5.754711e+02 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.5291e-02	k_t
Generation 2600: Best = 5.754576e+02 = [4, 2, 4, 2]	FES = 260100	Diversity = 1.6470e-02	k_t
Generation 2610: Best = 5.754414e+02 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.6294e-02	k_t
Generation 2620: Best = 5.754253e+02 = [3, 4, 4, 4]	FES = 262100	Diversity = 1.6732e-02	k_t
Generation 2630: Best = 5.754112e+02 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.4946e-02	k_t
Generation 2640: Best = 5.753949e+02 = [3, 2, 2, 2]	FES = 264100	Diversity = 1.5031e-02	k_t
Generation 2650: Best = 5.753846e+02 = [1, 1, 1, 1]	FES = 265100	Diversity = 1.1992e-02	k_t

Generation 2660: Best = 5.753744e+02 = [3, 2, 3, 4]	FES = 266100	Diversity = 1.1569e-02	k_t
Generation 2670: Best = 5.753650e+02 = [1, 1, 1, 1]	FES = 267100	Diversity = 1.2917e-02	k_t
Generation 2680: Best = 5.753564e+02 = [3, 2, 2, 2]	FES = 268100	Diversity = 1.1966e-02	k_t
Generation 2690: Best = 5.753445e+02 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.3099e-02	k_t
Generation 2700: Best = 5.753347e+02 = [2, 2, 3, 3]	FES = 270100	Diversity = 1.1919e-02	k_t
Generation 2710: Best = 5.753252e+02 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.1392e-02	k_t
Generation 2720: Best = 5.753150e+02 = [4, 2, 3, 3]	FES = 272100	Diversity = 1.1723e-02	k_t
Generation 2730: Best = 5.753052e+02 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.3352e-02	k_t
Generation 2740: Best = 5.752975e+02 = [2, 3, 3, 3]	FES = 274100	Diversity = 1.0858e-02	k_t
Generation 2750: Best = 5.752875e+02 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.2675e-02	k_t
Generation 2760: Best = 5.752761e+02 = [4, 2, 2, 4]	FES = 276100	Diversity = 1.3688e-02	k_t
Generation 2770: Best = 5.752651e+02 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.3532e-02	k_t
Generation 2780: Best = 5.752503e+02 = [3, 4, 2, 4]	FES = 278100	Diversity = 1.5898e-02	k_t
Generation 2790: Best = 5.752376e+02 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.2508e-02	k_t
Generation 2800: Best = 5.752267e+02 = [3, 4, 4, 3]	FES = 280100	Diversity = 1.3931e-02	k_t
Generation 2810: Best = 5.752105e+02 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.4675e-02	k_t
Generation 2820: Best = 5.752023e+02 = [3, 3, 2, 4]	FES = 282100	Diversity = 1.4109e-02	k_t
Generation 2830: Best = 5.751890e+02 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.5657e-02	k_t
Generation 2840: Best = 5.751765e+02 = [4, 2, 4, 3]	FES = 284100	Diversity = 1.4228e-02	k_t
Generation 2850: Best = 5.751601e+02 = [1, 1, 1, 1]	FES = 285100	Diversity = 1.4632e-02	k_t
Generation 2860: Best = 5.751519e+02 = [3, 2, 2, 3]	FES = 286100	Diversity = 1.3132e-02	k_t
Generation 2870: Best = 5.751393e+02 = [1, 1, 1, 1]	FES = 287100	Diversity = 1.3424e-02	k_t
Generation 2880: Best = 5.751291e+02 = [4, 4, 4, 3]	FES = 288100	Diversity = 1.3838e-02	k_t
Generation 2890: Best = 5.751170e+02 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.2742e-02	k_t
Generation 2900: Best = 5.751042e+02 = [3, 3, 2, 3]	FES = 290100	Diversity = 1.2866e-02	k_t
Generation 2910: Best = 5.750977e+02 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.1136e-02	k_t
Generation 2920: Best = 5.750907e+02 = [3, 4, 4, 3]	FES = 292100	Diversity = 1.0483e-02	k_t

Generation 2930: Best = 5.750853e+02	FES = 293100	Diversity = 9.7115e-03	k_t
= [1, 1, 1, 1]			
Generation 2940: Best = 5.750796e+02	FES = 294100	Diversity = 9.3377e-03	k_t
= [2, 3, 2, 2]			
Generation 2950: Best = 5.750718e+02	FES = 295100	Diversity = 9.8751e-03	k_t
= [1, 1, 1, 1]			
Generation 2960: Best = 5.750630e+02	FES = 296100	Diversity = 1.1715e-02	k_t
= [3, 3, 2, 3]			
Generation 2970: Best = 5.750504e+02	FES = 297100	Diversity = 1.2532e-02	k_t
= [1, 1, 1, 1]			
Generation 2980: Best = 5.750426e+02	FES = 298100	Diversity = 1.2922e-02	k_t
= [2, 2, 4, 4]			
Generation 2990: Best = 5.750345e+02	FES = 299100	Diversity = 1.3611e-02	k_t
= [1, 1, 1, 1]			
Generation 3000: Best = 5.750221e+02	FES = 300100	Diversity = 1.4676e-02	k_t
= [2, 3, 2, 2]			
Generation 3010: Best = 5.750133e+02	FES = 301100	Diversity = 1.1707e-02	k_t
= [1, 1, 1, 1]			
Generation 3020: Best = 5.750049e+02	FES = 302100	Diversity = 1.2554e-02	k_t
= [3, 4, 3, 3]			
Generation 3030: Best = 5.749982e+02	FES = 303100	Diversity = 1.0931e-02	k_t
= [1, 1, 1, 1]			
Generation 3040: Best = 5.749889e+02	FES = 304100	Diversity = 1.1484e-02	k_t
= [3, 3, 4, 4]			
Generation 3050: Best = 5.749807e+02	FES = 305100	Diversity = 1.2967e-02	k_t
= [1, 1, 1, 1]			
Generation 3060: Best = 5.749727e+02	FES = 306100	Diversity = 1.0962e-02	k_t
= [4, 2, 4, 2]			
Generation 3070: Best = 5.749627e+02	FES = 307100	Diversity = 1.1942e-02	k_t
= [1, 1, 1, 1]			
Generation 3080: Best = 5.749541e+02	FES = 308100	Diversity = 1.2604e-02	k_t
= [3, 4, 2, 4]			
Generation 3090: Best = 5.749404e+02	FES = 309100	Diversity = 1.3568e-02	k_t
= [1, 1, 1, 1]			
Generation 3100: Best = 5.749265e+02	FES = 310100	Diversity = 1.2736e-02	k_t
= [2, 3, 4, 2]			
Generation 3110: Best = 5.749162e+02	FES = 311100	Diversity = 1.3092e-02	k_t
= [1, 1, 1, 1]			
Generation 3120: Best = 5.749047e+02	FES = 312100	Diversity = 1.4876e-02	k_t
= [4, 4, 2, 3]			
Generation 3130: Best = 5.748959e+02	FES = 313100	Diversity = 1.3440e-02	k_t
= [1, 1, 1, 1]			
Generation 3140: Best = 5.748852e+02	FES = 314100	Diversity = 1.3862e-02	k_t
= [3, 3, 4, 4]			
Generation 3150: Best = 5.748773e+02	FES = 315100	Diversity = 1.2259e-02	k_t
= [1, 1, 1, 1]			
Generation 3160: Best = 5.748677e+02	FES = 316100	Diversity = 1.2923e-02	k_t
= [2, 4, 4, 3]			
Generation 3170: Best = 5.748624e+02	FES = 317100	Diversity = 1.0184e-02	k_t
= [1, 1, 1, 1]			
Generation 3180: Best = 5.748549e+02	FES = 318100	Diversity = 1.0558e-02	k_t
= [4, 3, 3, 2]			
Generation 3190: Best = 5.748475e+02	FES = 319100	Diversity = 1.0062e-02	k_t
= [1, 1, 1, 1]			

Generation 3200: Best = 5.748400e+02 = [4, 2, 2, 4]	FES = 320100	Diversity = 9.5617e-03	k_t
Generation 3210: Best = 5.748310e+02 = [1, 1, 1, 1]	FES = 321100	Diversity = 1.1747e-02	k_t
Generation 3220: Best = 5.748152e+02 = [4, 4, 4, 3]	FES = 322100	Diversity = 1.4425e-02	k_t
Generation 3230: Best = 5.748055e+02 = [1, 1, 1, 1]	FES = 323100	Diversity = 1.2844e-02	k_t
Generation 3240: Best = 5.747918e+02 = [3, 3, 2, 2]	FES = 324100	Diversity = 1.4488e-02	k_t
Generation 3250: Best = 5.747801e+02 = [1, 1, 1, 1]	FES = 325100	Diversity = 1.2965e-02	k_t
Generation 3260: Best = 5.747713e+02 = [4, 4, 2, 4]	FES = 326100	Diversity = 1.0655e-02	k_t
Generation 3270: Best = 5.747630e+02 = [1, 1, 1, 1]	FES = 327100	Diversity = 1.1612e-02	k_t
Generation 3280: Best = 5.747497e+02 = [3, 2, 4, 2]	FES = 328100	Diversity = 1.2918e-02	k_t
Generation 3290: Best = 5.747375e+02 = [1, 1, 1, 1]	FES = 329100	Diversity = 1.3287e-02	k_t
Generation 3300: Best = 5.747288e+02 = [2, 2, 2, 4]	FES = 330100	Diversity = 1.1290e-02	k_t
Generation 3310: Best = 5.747192e+02 = [1, 1, 1, 1]	FES = 331100	Diversity = 1.1309e-02	k_t
Generation 3320: Best = 5.747121e+02 = [2, 2, 2, 4]	FES = 332100	Diversity = 1.0758e-02	k_t
Generation 3330: Best = 5.746965e+02 = [1, 1, 1, 1]	FES = 333100	Diversity = 1.5000e-02	k_t
Generation 3340: Best = 5.746820e+02 = [2, 3, 4, 3]	FES = 334100	Diversity = 1.5836e-02	k_t
Generation 3350: Best = 5.746731e+02 = [1, 1, 1, 1]	FES = 335100	Diversity = 1.3334e-02	k_t
Generation 3360: Best = 5.746637e+02 = [3, 2, 3, 2]	FES = 336100	Diversity = 1.1795e-02	k_t
Generation 3370: Best = 5.746502e+02 = [1, 1, 1, 1]	FES = 337100	Diversity = 1.3361e-02	k_t
Generation 3380: Best = 5.746414e+02 = [4, 3, 2, 4]	FES = 338100	Diversity = 1.3906e-02	k_t
Generation 3390: Best = 5.746291e+02 = [1, 1, 1, 1]	FES = 339100	Diversity = 1.3316e-02	k_t
Generation 3400: Best = 5.746195e+02 = [2, 2, 2, 4]	FES = 340100	Diversity = 1.1759e-02	k_t
Generation 3410: Best = 5.746089e+02 = [1, 1, 1, 1]	FES = 341100	Diversity = 1.3412e-02	k_t
Generation 3420: Best = 5.745935e+02 = [3, 3, 4, 2]	FES = 342100	Diversity = 1.4736e-02	k_t
Generation 3430: Best = 5.745813e+02 = [1, 1, 1, 1]	FES = 343100	Diversity = 1.2972e-02	k_t
Generation 3440: Best = 5.745713e+02 = [3, 2, 2, 2]	FES = 344100	Diversity = 1.1058e-02	k_t
Generation 3450: Best = 5.745576e+02 = [1, 1, 1, 1]	FES = 345100	Diversity = 1.2619e-02	k_t
Generation 3460: Best = 5.745430e+02 = [3, 4, 3, 2]	FES = 346100	Diversity = 1.1929e-02	k_t

Generation 3470: Best = 5.745315e+02 = [1, 1, 1, 1]	FES = 347100	Diversity = 1.3115e-02	k_t
Generation 3480: Best = 5.745199e+02 = [2, 2, 4, 3]	FES = 348100	Diversity = 1.3244e-02	k_t
Generation 3490: Best = 5.745072e+02 = [1, 1, 1, 1]	FES = 349100	Diversity = 1.2057e-02	k_t
Generation 3500: Best = 5.744987e+02 = [4, 3, 4, 4]	FES = 350100	Diversity = 1.2073e-02	k_t
Generation 3510: Best = 5.744832e+02 = [1, 1, 1, 1]	FES = 351100	Diversity = 1.4462e-02	k_t
Generation 3520: Best = 5.744707e+02 = [3, 2, 3, 3]	FES = 352100	Diversity = 1.2906e-02	k_t
Generation 3530: Best = 5.744583e+02 = [1, 1, 1, 1]	FES = 353100	Diversity = 1.3241e-02	k_t
Generation 3540: Best = 5.744469e+02 = [2, 2, 4, 3]	FES = 354100	Diversity = 1.3865e-02	k_t
Generation 3550: Best = 5.744306e+02 = [1, 1, 1, 1]	FES = 355100	Diversity = 1.5132e-02	k_t
Generation 3560: Best = 5.744195e+02 = [2, 3, 4, 2]	FES = 356100	Diversity = 1.2571e-02	k_t
Generation 3570: Best = 5.744076e+02 = [1, 1, 1, 1]	FES = 357100	Diversity = 1.2425e-02	k_t
Generation 3580: Best = 5.743948e+02 = [4, 2, 4, 4]	FES = 358100	Diversity = 1.3660e-02	k_t
Generation 3590: Best = 5.743778e+02 = [1, 1, 1, 1]	FES = 359100	Diversity = 1.5154e-02	k_t
Generation 3600: Best = 5.743645e+02 = [3, 3, 3, 2]	FES = 360100	Diversity = 1.5422e-02	k_t
Generation 3610: Best = 5.743513e+02 = [1, 1, 1, 1]	FES = 361100	Diversity = 1.5578e-02	k_t
Generation 3620: Best = 5.743355e+02 = [3, 3, 3, 3]	FES = 362100	Diversity = 1.4929e-02	k_t
Generation 3630: Best = 5.743228e+02 = [1, 1, 1, 1]	FES = 363100	Diversity = 1.4854e-02	k_t
Generation 3640: Best = 5.743060e+02 = [2, 4, 3, 2]	FES = 364100	Diversity = 1.5840e-02	k_t
Generation 3650: Best = 5.742906e+02 = [1, 1, 1, 1]	FES = 365100	Diversity = 1.5593e-02	k_t
Generation 3660: Best = 5.742794e+02 = [3, 2, 2, 4]	FES = 366100	Diversity = 1.3432e-02	k_t
Generation 3670: Best = 5.742670e+02 = [1, 1, 1, 1]	FES = 367100	Diversity = 1.4236e-02	k_t
Generation 3680: Best = 5.742568e+02 = [4, 3, 3, 3]	FES = 368100	Diversity = 1.2566e-02	k_t
Generation 3690: Best = 5.742433e+02 = [1, 1, 1, 1]	FES = 369100	Diversity = 1.2731e-02	k_t
Generation 3700: Best = 5.742285e+02 = [3, 2, 4, 2]	FES = 370100	Diversity = 1.3913e-02	k_t
Generation 3710: Best = 5.742156e+02 = [1, 1, 1, 1]	FES = 371100	Diversity = 1.5672e-02	k_t
Generation 3720: Best = 5.741944e+02 = [3, 3, 2, 4]	FES = 372100	Diversity = 1.7488e-02	k_t
Generation 3730: Best = 5.741774e+02 = [1, 1, 1, 1]	FES = 373100	Diversity = 1.5898e-02	k_t

Generation 3740: Best = 5.741683e+02 = [3, 3, 3, 3]	FES = 374100	Diversity = 1.3723e-02	k_t
Generation 3750: Best = 5.741547e+02 = [1, 1, 1, 1]	FES = 375100	Diversity = 1.6082e-02	k_t
Generation 3760: Best = 5.741395e+02 = [3, 4, 3, 4]	FES = 376100	Diversity = 1.5530e-02	k_t
Generation 3770: Best = 5.741285e+02 = [1, 1, 1, 1]	FES = 377100	Diversity = 1.1753e-02	k_t
Generation 3780: Best = 5.741184e+02 = [2, 3, 2, 3]	FES = 378100	Diversity = 1.1336e-02	k_t
Generation 3790: Best = 5.741068e+02 = [1, 1, 1, 1]	FES = 379100	Diversity = 1.3964e-02	k_t
Generation 3800: Best = 5.740907e+02 = [2, 2, 4, 4]	FES = 380100	Diversity = 1.5588e-02	k_t
Generation 3810: Best = 5.740773e+02 = [1, 1, 1, 1]	FES = 381100	Diversity = 1.6993e-02	k_t
Generation 3820: Best = 5.740613e+02 = [2, 4, 3, 3]	FES = 382100	Diversity = 1.5060e-02	k_t
Generation 3830: Best = 5.740489e+02 = [1, 1, 1, 1]	FES = 383100	Diversity = 1.3962e-02	k_t
Generation 3840: Best = 5.740398e+02 = [3, 4, 3, 2]	FES = 384100	Diversity = 1.2607e-02	k_t
Generation 3850: Best = 5.740319e+02 = [1, 1, 1, 1]	FES = 385100	Diversity = 1.1327e-02	k_t
Generation 3860: Best = 5.740207e+02 = [4, 2, 4, 2]	FES = 386100	Diversity = 1.1579e-02	k_t
Generation 3870: Best = 5.740102e+02 = [1, 1, 1, 1]	FES = 387100	Diversity = 1.2477e-02	k_t
Generation 3880: Best = 5.739972e+02 = [2, 3, 2, 3]	FES = 388100	Diversity = 1.2707e-02	k_t
Generation 3890: Best = 5.739882e+02 = [1, 1, 1, 1]	FES = 389100	Diversity = 1.1304e-02	k_t
Generation 3900: Best = 5.739757e+02 = [3, 2, 2, 4]	FES = 390100	Diversity = 1.1601e-02	k_t
Generation 3910: Best = 5.739666e+02 = [1, 1, 1, 1]	FES = 391100	Diversity = 1.1547e-02	k_t
Generation 3920: Best = 5.739543e+02 = [2, 2, 3, 3]	FES = 392100	Diversity = 1.2128e-02	k_t
Generation 3930: Best = 5.739472e+02 = [1, 1, 1, 1]	FES = 393100	Diversity = 1.0346e-02	k_t
Generation 3940: Best = 5.739354e+02 = [4, 2, 2, 4]	FES = 394100	Diversity = 1.1661e-02	k_t
Generation 3950: Best = 5.739248e+02 = [1, 1, 1, 1]	FES = 395100	Diversity = 1.1907e-02	k_t
Generation 3960: Best = 5.739107e+02 = [4, 2, 4, 3]	FES = 396100	Diversity = 1.4891e-02	k_t
Generation 3970: Best = 5.738959e+02 = [1, 1, 1, 1]	FES = 397100	Diversity = 1.4997e-02	k_t
Generation 3980: Best = 5.738835e+02 = [3, 3, 2, 2]	FES = 398100	Diversity = 1.3381e-02	k_t
Generation 3990: Best = 5.738737e+02 = [1, 1, 1, 1]	FES = 399100	Diversity = 1.1183e-02	k_t
Generation 4000: Best = 5.738630e+02 = [3, 2, 2, 3]	FES = 400100	Diversity = 1.0753e-02	k_t

Generation 4010: Best = 5.738523e+02	FES = 401100	Diversity = 1.1368e-02	k_t
= [1, 1, 1, 1]			
Generation 4020: Best = 5.738439e+02	FES = 402100	Diversity = 1.1120e-02	k_t
= [3, 4, 3, 3]			
Generation 4030: Best = 5.738349e+02	FES = 403100	Diversity = 1.0175e-02	k_t
= [1, 1, 1, 1]			
Generation 4040: Best = 5.738256e+02	FES = 404100	Diversity = 9.6698e-03	k_t
= [3, 2, 3, 2]			
Generation 4050: Best = 5.738109e+02	FES = 405100	Diversity = 1.2985e-02	k_t
= [1, 1, 1, 1]			
Generation 4060: Best = 5.737986e+02	FES = 406100	Diversity = 1.3273e-02	k_t
= [2, 3, 3, 2]			
Generation 4070: Best = 5.737876e+02	FES = 407100	Diversity = 1.1943e-02	k_t
= [1, 1, 1, 1]			
Generation 4080: Best = 5.737800e+02	FES = 408100	Diversity = 1.1004e-02	k_t
= [3, 2, 4, 2]			
Generation 4090: Best = 5.737691e+02	FES = 409100	Diversity = 1.2635e-02	k_t
= [1, 1, 1, 1]			
Generation 4100: Best = 5.737577e+02	FES = 410100	Diversity = 1.3259e-02	k_t
= [4, 2, 2, 4]			
Generation 4110: Best = 5.737449e+02	FES = 411100	Diversity = 1.3900e-02	k_t
= [1, 1, 1, 1]			
Generation 4120: Best = 5.737354e+02	FES = 412100	Diversity = 1.1229e-02	k_t
= [2, 4, 3, 2]			
Generation 4130: Best = 5.737282e+02	FES = 413100	Diversity = 9.9934e-03	k_t
= [1, 1, 1, 1]			
Generation 4140: Best = 5.737177e+02	FES = 414100	Diversity = 1.0783e-02	k_t
= [4, 4, 2, 4]			
Generation 4150: Best = 5.737098e+02	FES = 415100	Diversity = 1.0026e-02	k_t
= [1, 1, 1, 1]			
Generation 4160: Best = 5.737012e+02	FES = 416100	Diversity = 1.0225e-02	k_t
= [4, 4, 3, 4]			
Generation 4170: Best = 5.736924e+02	FES = 417100	Diversity = 1.0589e-02	k_t
= [1, 1, 1, 1]			
Generation 4180: Best = 5.736849e+02	FES = 418100	Diversity = 1.0842e-02	k_t
= [3, 2, 2, 4]			
Generation 4190: Best = 5.736737e+02	FES = 419100	Diversity = 1.1804e-02	k_t
= [1, 1, 1, 1]			
Generation 4200: Best = 5.736634e+02	FES = 420100	Diversity = 1.1548e-02	k_t
= [3, 3, 3, 4]			
Generation 4210: Best = 5.736546e+02	FES = 421100	Diversity = 9.5732e-03	k_t
= [1, 1, 1, 1]			
Generation 4220: Best = 5.736479e+02	FES = 422100	Diversity = 9.5344e-03	k_t
= [4, 4, 2, 4]			
Generation 4230: Best = 5.736379e+02	FES = 423100	Diversity = 1.1100e-02	k_t
= [1, 1, 1, 1]			
Generation 4240: Best = 5.736289e+02	FES = 424100	Diversity = 1.0888e-02	k_t
= [3, 2, 3, 4]			
Generation 4250: Best = 5.736165e+02	FES = 425100	Diversity = 1.2758e-02	k_t
= [1, 1, 1, 1]			
Generation 4260: Best = 5.736032e+02	FES = 426100	Diversity = 1.3338e-02	k_t
= [2, 2, 2, 4]			
Generation 4270: Best = 5.735880e+02	FES = 427100	Diversity = 1.3021e-02	k_t
= [1, 1, 1, 1]			

Generation 4280: Best = 5.735744e+02 = [2, 2, 3, 2]	FES = 428100	Diversity = 1.3200e-02	k_t
Generation 4290: Best = 5.735644e+02 = [1, 1, 1, 1]	FES = 429100	Diversity = 1.1985e-02	k_t
Generation 4300: Best = 5.735421e+02 = [3, 3, 2, 3]	FES = 430100	Diversity = 1.6785e-02	k_t
Generation 4310: Best = 5.735272e+02 = [1, 1, 1, 1]	FES = 431100	Diversity = 1.6407e-02	k_t
Generation 4320: Best = 5.735089e+02 = [4, 4, 3, 4]	FES = 432100	Diversity = 1.6139e-02	k_t
Generation 4330: Best = 5.734955e+02 = [1, 1, 1, 1]	FES = 433100	Diversity = 1.5221e-02	k_t
Generation 4340: Best = 5.734778e+02 = [4, 4, 2, 2]	FES = 434100	Diversity = 1.6097e-02	k_t
Generation 4350: Best = 5.734684e+02 = [1, 1, 1, 1]	FES = 435100	Diversity = 1.1690e-02	k_t
Generation 4360: Best = 5.734609e+02 = [3, 2, 4, 2]	FES = 436100	Diversity = 9.8547e-03	k_t
Generation 4370: Best = 5.734524e+02 = [1, 1, 1, 1]	FES = 437100	Diversity = 1.0752e-02	k_t
Generation 4380: Best = 5.734448e+02 = [2, 3, 2, 2]	FES = 438100	Diversity = 1.1464e-02	k_t
Generation 4390: Best = 5.734370e+02 = [1, 1, 1, 1]	FES = 439100	Diversity = 1.1395e-02	k_t
Generation 4400: Best = 5.734302e+02 = [2, 2, 4, 3]	FES = 440100	Diversity = 8.8426e-03	k_t
Generation 4410: Best = 5.734216e+02 = [1, 1, 1, 1]	FES = 441100	Diversity = 1.0303e-02	k_t
Generation 4420: Best = 5.734149e+02 = [3, 3, 4, 4]	FES = 442100	Diversity = 9.4906e-03	k_t
Generation 4430: Best = 5.734066e+02 = [1, 1, 1, 1]	FES = 443100	Diversity = 9.4336e-03	k_t
Generation 4440: Best = 5.733951e+02 = [4, 3, 2, 3]	FES = 444100	Diversity = 1.1841e-02	k_t
Generation 4450: Best = 5.733861e+02 = [1, 1, 1, 1]	FES = 445100	Diversity = 1.1544e-02	k_t
Generation 4460: Best = 5.733740e+02 = [2, 2, 4, 2]	FES = 446100	Diversity = 1.2038e-02	k_t
Generation 4470: Best = 5.733638e+02 = [1, 1, 1, 1]	FES = 447100	Diversity = 1.0937e-02	k_t
Generation 4480: Best = 5.733537e+02 = [2, 2, 2, 3]	FES = 448100	Diversity = 1.1816e-02	k_t
Generation 4490: Best = 5.733440e+02 = [1, 1, 1, 1]	FES = 449100	Diversity = 1.2079e-02	k_t
Generation 4500: Best = 5.733332e+02 = [2, 4, 3, 3]	FES = 450100	Diversity = 1.1493e-02	k_t
Generation 4510: Best = 5.733228e+02 = [1, 1, 1, 1]	FES = 451100	Diversity = 1.1809e-02	k_t
Generation 4520: Best = 5.733145e+02 = [2, 4, 2, 2]	FES = 452100	Diversity = 1.1919e-02	k_t
Generation 4530: Best = 5.733051e+02 = [1, 1, 1, 1]	FES = 453100	Diversity = 1.1198e-02	k_t
Generation 4540: Best = 5.732954e+02 = [3, 2, 4, 3]	FES = 454100	Diversity = 1.2773e-02	k_t

Generation 4550: Best = 5.732813e+02	FES = 455100	Diversity = 1.2990e-02	k_t
= [1, 1, 1, 1]			
Generation 4560: Best = 5.732709e+02	FES = 456100	Diversity = 1.2812e-02	k_t
= [3, 4, 4, 2]			
Generation 4570: Best = 5.732562e+02	FES = 457100	Diversity = 1.4196e-02	k_t
= [1, 1, 1, 1]			
Generation 4580: Best = 5.732435e+02	FES = 458100	Diversity = 1.3985e-02	k_t
= [4, 2, 2, 2]			
Generation 4590: Best = 5.732281e+02	FES = 459100	Diversity = 1.6438e-02	k_t
= [1, 1, 1, 1]			
Generation 4600: Best = 5.732124e+02	FES = 460100	Diversity = 1.5960e-02	k_t
= [3, 2, 2, 2]			
Generation 4610: Best = 5.731986e+02	FES = 461100	Diversity = 1.3604e-02	k_t
= [1, 1, 1, 1]			
Generation 4620: Best = 5.731836e+02	FES = 462100	Diversity = 1.5003e-02	k_t
= [3, 3, 2, 3]			
Generation 4630: Best = 5.731698e+02	FES = 463100	Diversity = 1.3004e-02	k_t
= [1, 1, 1, 1]			
Generation 4640: Best = 5.731582e+02	FES = 464100	Diversity = 1.3593e-02	k_t
= [4, 4, 4, 4]			
Generation 4650: Best = 5.731454e+02	FES = 465100	Diversity = 1.1323e-02	k_t
= [1, 1, 1, 1]			
Generation 4660: Best = 5.731358e+02	FES = 466100	Diversity = 1.2580e-02	k_t
= [4, 4, 4, 4]			
Generation 4670: Best = 5.731213e+02	FES = 467100	Diversity = 1.3730e-02	k_t
= [1, 1, 1, 1]			
Generation 4680: Best = 5.731075e+02	FES = 468100	Diversity = 1.3837e-02	k_t
= [4, 2, 4, 2]			
Generation 4690: Best = 5.730941e+02	FES = 469100	Diversity = 1.5275e-02	k_t
= [1, 1, 1, 1]			
Generation 4700: Best = 5.730785e+02	FES = 470100	Diversity = 1.4582e-02	k_t
= [2, 3, 2, 3]			
Generation 4710: Best = 5.730657e+02	FES = 471100	Diversity = 1.5045e-02	k_t
= [1, 1, 1, 1]			
Generation 4720: Best = 5.730512e+02	FES = 472100	Diversity = 1.5291e-02	k_t
= [3, 4, 2, 4]			
Generation 4730: Best = 5.730400e+02	FES = 473100	Diversity = 1.2590e-02	k_t
= [1, 1, 1, 1]			
Generation 4740: Best = 5.730294e+02	FES = 474100	Diversity = 1.0938e-02	k_t
= [2, 2, 3, 3]			
Generation 4750: Best = 5.730192e+02	FES = 475100	Diversity = 1.1829e-02	k_t
= [1, 1, 1, 1]			
Generation 4760: Best = 5.730043e+02	FES = 476100	Diversity = 1.3732e-02	k_t
= [4, 4, 4, 2]			
Generation 4770: Best = 5.729936e+02	FES = 477100	Diversity = 1.3060e-02	k_t
= [1, 1, 1, 1]			
Generation 4780: Best = 5.729826e+02	FES = 478100	Diversity = 1.1395e-02	k_t
= [2, 2, 4, 2]			
Generation 4790: Best = 5.729737e+02	FES = 479100	Diversity = 1.1710e-02	k_t
= [1, 1, 1, 1]			
Generation 4800: Best = 5.729642e+02	FES = 480100	Diversity = 1.1826e-02	k_t
= [3, 2, 2, 3]			
Generation 4810: Best = 5.729538e+02	FES = 481100	Diversity = 1.0708e-02	k_t
= [1, 1, 1, 1]			

Generation 4820: Best = 5.729437e+02 = [3, 4, 4, 3]	FES = 482100	Diversity = 1.0657e-02	k_t
Generation 4830: Best = 5.729342e+02 = [1, 1, 1, 1]	FES = 483100	Diversity = 1.1617e-02	k_t
Generation 4840: Best = 5.729243e+02 = [4, 4, 3, 4]	FES = 484100	Diversity = 1.0600e-02	k_t
Generation 4850: Best = 5.729133e+02 = [1, 1, 1, 1]	FES = 485100	Diversity = 1.2279e-02	k_t
Generation 4860: Best = 5.729041e+02 = [2, 3, 2, 4]	FES = 486100	Diversity = 1.2818e-02	k_t
Generation 4870: Best = 5.728907e+02 = [1, 1, 1, 1]	FES = 487100	Diversity = 1.4072e-02	k_t
Generation 4880: Best = 5.728818e+02 = [4, 2, 2, 4]	FES = 488100	Diversity = 1.1891e-02	k_t
Generation 4890: Best = 5.728695e+02 = [1, 1, 1, 1]	FES = 489100	Diversity = 1.3465e-02	k_t
Generation 4900: Best = 5.728559e+02 = [3, 4, 4, 4]	FES = 490100	Diversity = 1.3839e-02	k_t
Generation 4910: Best = 5.728440e+02 = [1, 1, 1, 1]	FES = 491100	Diversity = 1.3530e-02	k_t
Generation 4920: Best = 5.728366e+02 = [2, 2, 4, 3]	FES = 492100	Diversity = 1.0671e-02	k_t
Generation 4930: Best = 5.728278e+02 = [1, 1, 1, 1]	FES = 493100	Diversity = 1.0759e-02	k_t
Generation 4940: Best = 5.728202e+02 = [2, 4, 4, 3]	FES = 494100	Diversity = 1.0203e-02	k_t
Generation 4950: Best = 5.728113e+02 = [1, 1, 1, 1]	FES = 495100	Diversity = 9.7956e-03	k_t
Generation 4960: Best = 5.728044e+02 = [3, 4, 2, 2]	FES = 496100	Diversity = 9.2959e-03	k_t
Generation 4970: Best = 5.727956e+02 = [1, 1, 1, 1]	FES = 497100	Diversity = 8.4493e-03	k_t
Generation 4980: Best = 5.727879e+02 = [3, 4, 4, 3]	FES = 498100	Diversity = 1.0355e-02	k_t
Generation 4990: Best = 5.727792e+02 = [1, 1, 1, 1]	FES = 499100	Diversity = 1.0719e-02	k_t
Generation 5000: Best = 5.727697e+02 = [3, 4, 3, 2]	FES = 500100	Diversity = 1.1965e-02	k_t
Generation 5010: Best = 5.727581e+02 = [1, 1, 1, 1]	FES = 501100	Diversity = 1.3513e-02	k_t
Generation 5020: Best = 5.727415e+02 = [3, 4, 3, 4]	FES = 502100	Diversity = 1.6256e-02	k_t
Generation 5030: Best = 5.727221e+02 = [1, 1, 1, 1]	FES = 503100	Diversity = 1.4486e-02	k_t
Generation 5040: Best = 5.727114e+02 = [4, 4, 4, 4]	FES = 504100	Diversity = 1.4595e-02	k_t
Generation 5050: Best = 5.726968e+02 = [1, 1, 1, 1]	FES = 505100	Diversity = 1.6548e-02	k_t
Generation 5060: Best = 5.726786e+02 = [2, 4, 3, 4]	FES = 506100	Diversity = 1.4851e-02	k_t
Generation 5070: Best = 5.726649e+02 = [1, 1, 1, 1]	FES = 507100	Diversity = 1.6216e-02	k_t
Generation 5080: Best = 5.726509e+02 = [2, 4, 3, 3]	FES = 508100	Diversity = 1.4696e-02	k_t

Generation 5090: Best = 5.726340e+02	FES = 509100	Diversity = 1.5658e-02	k_t
= [1, 1, 1, 1]			
Generation 5100: Best = 5.726228e+02	FES = 510100	Diversity = 1.3729e-02	k_t
= [2, 3, 3, 2]			
Generation 5110: Best = 5.726131e+02	FES = 511100	Diversity = 1.2554e-02	k_t
= [1, 1, 1, 1]			
Generation 5120: Best = 5.726001e+02	FES = 512100	Diversity = 1.4867e-02	k_t
= [2, 3, 2, 3]			
Generation 5130: Best = 5.725885e+02	FES = 513100	Diversity = 1.3110e-02	k_t
= [1, 1, 1, 1]			
Generation 5140: Best = 5.725763e+02	FES = 514100	Diversity = 1.2335e-02	k_t
= [3, 2, 4, 3]			
Generation 5150: Best = 5.725678e+02	FES = 515100	Diversity = 1.1195e-02	k_t
= [1, 1, 1, 1]			
Generation 5160: Best = 5.725590e+02	FES = 516100	Diversity = 1.1635e-02	k_t
= [3, 2, 3, 2]			
Generation 5170: Best = 5.725504e+02	FES = 517100	Diversity = 1.2160e-02	k_t
= [1, 1, 1, 1]			
Generation 5180: Best = 5.725391e+02	FES = 518100	Diversity = 1.1916e-02	k_t
= [2, 3, 2, 3]			
Generation 5190: Best = 5.725305e+02	FES = 519100	Diversity = 1.2665e-02	k_t
= [1, 1, 1, 1]			
Generation 5200: Best = 5.725204e+02	FES = 520100	Diversity = 1.2072e-02	k_t
= [2, 4, 4, 4]			
Generation 5210: Best = 5.725127e+02	FES = 521100	Diversity = 1.0972e-02	k_t
= [1, 1, 1, 1]			
Generation 5220: Best = 5.725053e+02	FES = 522100	Diversity = 9.9228e-03	k_t
= [2, 2, 3, 4]			
Generation 5230: Best = 5.724985e+02	FES = 523100	Diversity = 9.6888e-03	k_t
= [1, 1, 1, 1]			
Generation 5240: Best = 5.724867e+02	FES = 524100	Diversity = 1.3275e-02	k_t
= [3, 2, 4, 2]			
Generation 5250: Best = 5.724760e+02	FES = 525100	Diversity = 1.4598e-02	k_t
= [1, 1, 1, 1]			
Generation 5260: Best = 5.724576e+02	FES = 526100	Diversity = 1.6409e-02	k_t
= [3, 2, 3, 3]			
Generation 5270: Best = 5.724428e+02	FES = 527100	Diversity = 1.5864e-02	k_t
= [1, 1, 1, 1]			
Generation 5280: Best = 5.724331e+02	FES = 528100	Diversity = 1.3642e-02	k_t
= [2, 4, 2, 3]			
Generation 5290: Best = 5.724226e+02	FES = 529100	Diversity = 1.2272e-02	k_t
= [1, 1, 1, 1]			
Generation 5300: Best = 5.724137e+02	FES = 530100	Diversity = 1.1849e-02	k_t
= [3, 2, 2, 4]			
Generation 5310: Best = 5.723992e+02	FES = 531100	Diversity = 1.3759e-02	k_t
= [1, 1, 1, 1]			
Generation 5320: Best = 5.723878e+02	FES = 532100	Diversity = 1.3347e-02	k_t
= [2, 2, 2, 3]			
Generation 5330: Best = 5.723780e+02	FES = 533100	Diversity = 1.2903e-02	k_t
= [1, 1, 1, 1]			
Generation 5340: Best = 5.723647e+02	FES = 534100	Diversity = 1.3835e-02	k_t
= [2, 2, 2, 3]			
Generation 5350: Best = 5.723577e+02	FES = 535100	Diversity = 1.1173e-02	k_t
= [1, 1, 1, 1]			

Generation 5360: Best = 5.723465e+02 = [4, 3, 4, 4]	FES = 536100	Diversity = 1.1681e-02	k_t
Generation 5370: Best = 5.723393e+02 = [1, 1, 1, 1]	FES = 537100	Diversity = 9.8095e-03	k_t
Generation 5380: Best = 5.723324e+02 = [2, 4, 4, 3]	FES = 538100	Diversity = 9.9575e-03	k_t
Generation 5390: Best = 5.723253e+02 = [1, 1, 1, 1]	FES = 539100	Diversity = 1.0186e-02	k_t
Generation 5400: Best = 5.723180e+02 = [4, 4, 2, 2]	FES = 540100	Diversity = 1.0354e-02	k_t
Generation 5410: Best = 5.723102e+02 = [1, 1, 1, 1]	FES = 541100	Diversity = 9.5321e-03	k_t
Generation 5420: Best = 5.723027e+02 = [3, 4, 4, 3]	FES = 542100	Diversity = 1.0353e-02	k_t
Generation 5430: Best = 5.722971e+02 = [1, 1, 1, 1]	FES = 543100	Diversity = 8.6684e-03	k_t
Generation 5440: Best = 5.722913e+02 = [4, 4, 4, 3]	FES = 544100	Diversity = 9.8134e-03	k_t
Generation 5450: Best = 5.722855e+02 = [1, 1, 1, 1]	FES = 545100	Diversity = 9.0664e-03	k_t
Generation 5460: Best = 5.722769e+02 = [2, 2, 3, 4]	FES = 546100	Diversity = 1.1603e-02	k_t
Generation 5470: Best = 5.722690e+02 = [1, 1, 1, 1]	FES = 547100	Diversity = 1.1070e-02	k_t
Generation 5480: Best = 5.722610e+02 = [2, 2, 2, 2]	FES = 548100	Diversity = 1.1251e-02	k_t
Generation 5490: Best = 5.722518e+02 = [1, 1, 1, 1]	FES = 549100	Diversity = 1.1934e-02	k_t
Generation 5500: Best = 5.722446e+02 = [4, 2, 2, 4]	FES = 550100	Diversity = 1.0131e-02	k_t
Generation 5510: Best = 5.722387e+02 = [1, 1, 1, 1]	FES = 551100	Diversity = 8.5770e-03	k_t
Generation 5520: Best = 5.722326e+02 = [4, 2, 2, 2]	FES = 552100	Diversity = 9.1228e-03	k_t
Generation 5530: Best = 5.722259e+02 = [1, 1, 1, 1]	FES = 553100	Diversity = 1.0009e-02	k_t
Generation 5540: Best = 5.722192e+02 = [2, 2, 2, 3]	FES = 554100	Diversity = 1.0956e-02	k_t
Generation 5550: Best = 5.722138e+02 = [1, 1, 1, 1]	FES = 555100	Diversity = 1.0162e-02	k_t
Generation 5560: Best = 5.722073e+02 = [3, 3, 3, 3]	FES = 556100	Diversity = 9.5858e-03	k_t
Generation 5570: Best = 5.722002e+02 = [1, 1, 1, 1]	FES = 557100	Diversity = 9.3349e-03	k_t
Generation 5580: Best = 5.721925e+02 = [2, 3, 4, 2]	FES = 558100	Diversity = 1.0116e-02	k_t
Generation 5590: Best = 5.721816e+02 = [1, 1, 1, 1]	FES = 559100	Diversity = 1.2534e-02	k_t
Generation 5600: Best = 5.721706e+02 = [2, 2, 3, 4]	FES = 560100	Diversity = 1.3880e-02	k_t
Generation 5610: Best = 5.721593e+02 = [1, 1, 1, 1]	FES = 561100	Diversity = 1.3633e-02	k_t
Generation 5620: Best = 5.721472e+02 = [3, 2, 4, 4]	FES = 562100	Diversity = 1.3156e-02	k_t

Generation 5630: Best = 5.721367e+02	FES = 563100	Diversity = 1.2888e-02	k_t
= [1, 1, 1, 1]			
Generation 5640: Best = 5.721232e+02	FES = 564100	Diversity = 1.3444e-02	k_t
= [3, 3, 3, 2]			
Generation 5650: Best = 5.721106e+02	FES = 565100	Diversity = 1.3424e-02	k_t
= [1, 1, 1, 1]			
Generation 5660: Best = 5.720988e+02	FES = 566100	Diversity = 1.3684e-02	k_t
= [3, 3, 3, 4]			
Generation 5670: Best = 5.720839e+02	FES = 567100	Diversity = 1.5405e-02	k_t
= [1, 1, 1, 1]			
Generation 5680: Best = 5.720722e+02	FES = 568100	Diversity = 1.2950e-02	k_t
= [4, 3, 4, 3]			
Generation 5690: Best = 5.720592e+02	FES = 569100	Diversity = 1.3029e-02	k_t
= [1, 1, 1, 1]			
Generation 5700: Best = 5.720479e+02	FES = 570100	Diversity = 1.2095e-02	k_t
= [4, 4, 4, 2]			
Generation 5710: Best = 5.720347e+02	FES = 571100	Diversity = 1.2858e-02	k_t
= [1, 1, 1, 1]			
Generation 5720: Best = 5.720255e+02	FES = 572100	Diversity = 1.1872e-02	k_t
= [2, 3, 2, 3]			
Generation 5730: Best = 5.720159e+02	FES = 573100	Diversity = 1.0362e-02	k_t
= [1, 1, 1, 1]			
Generation 5740: Best = 5.720098e+02	FES = 574100	Diversity = 9.3554e-03	k_t
= [4, 2, 2, 2]			
Generation 5750: Best = 5.720012e+02	FES = 575100	Diversity = 1.0695e-02	k_t
= [1, 1, 1, 1]			
Generation 5760: Best = 5.719918e+02	FES = 576100	Diversity = 9.9559e-03	k_t
= [3, 2, 4, 2]			
Generation 5770: Best = 5.719863e+02	FES = 577100	Diversity = 9.0647e-03	k_t
= [1, 1, 1, 1]			
Generation 5780: Best = 5.719774e+02	FES = 578100	Diversity = 1.0326e-02	k_t
= [2, 2, 4, 3]			
Generation 5790: Best = 5.719689e+02	FES = 579100	Diversity = 8.6885e-03	k_t
= [1, 1, 1, 1]			
Generation 5800: Best = 5.719580e+02	FES = 580100	Diversity = 1.0978e-02	k_t
= [2, 2, 4, 2]			
Generation 5810: Best = 5.719471e+02	FES = 581100	Diversity = 1.2981e-02	k_t
= [1, 1, 1, 1]			
Generation 5820: Best = 5.719353e+02	FES = 582100	Diversity = 1.3430e-02	k_t
= [2, 3, 4, 4]			
Generation 5830: Best = 5.719216e+02	FES = 583100	Diversity = 1.3980e-02	k_t
= [1, 1, 1, 1]			
Generation 5840: Best = 5.719115e+02	FES = 584100	Diversity = 1.2300e-02	k_t
= [4, 2, 3, 2]			
Generation 5850: Best = 5.718946e+02	FES = 585100	Diversity = 1.3303e-02	k_t
= [1, 1, 1, 1]			
Generation 5860: Best = 5.718828e+02	FES = 586100	Diversity = 1.3736e-02	k_t
= [4, 3, 2, 4]			
Generation 5870: Best = 5.718707e+02	FES = 587100	Diversity = 1.2080e-02	k_t
= [1, 1, 1, 1]			
Generation 5880: Best = 5.718566e+02	FES = 588100	Diversity = 1.4302e-02	k_t
= [3, 2, 3, 2]			
Generation 5890: Best = 5.718428e+02	FES = 589100	Diversity = 1.4291e-02	k_t
= [1, 1, 1, 1]			

Generation 5900: Best = 5.718324e+02 = [2, 2, 3, 2]	FES = 590100	Diversity = 1.3624e-02	k_t
Generation 5910: Best = 5.718214e+02 = [1, 1, 1, 1]	FES = 591100	Diversity = 1.2757e-02	k_t
Generation 5920: Best = 5.718080e+02 = [2, 3, 4, 3]	FES = 592100	Diversity = 1.2732e-02	k_t
Generation 5930: Best = 5.717983e+02 = [1, 1, 1, 1]	FES = 593100	Diversity = 1.2943e-02	k_t
Generation 5940: Best = 5.717893e+02 = [3, 4, 4, 3]	FES = 594100	Diversity = 1.1908e-02	k_t
Generation 5950: Best = 5.717795e+02 = [1, 1, 1, 1]	FES = 595100	Diversity = 1.0710e-02	k_t
Generation 5960: Best = 5.717717e+02 = [2, 3, 2, 4]	FES = 596100	Diversity = 1.0652e-02	k_t
Generation 5970: Best = 5.717596e+02 = [1, 1, 1, 1]	FES = 597100	Diversity = 1.1713e-02	k_t
Generation 5980: Best = 5.717513e+02 = [3, 4, 4, 2]	FES = 598100	Diversity = 1.0682e-02	k_t
Generation 5990: Best = 5.717418e+02 = [1, 1, 1, 1]	FES = 599100	Diversity = 1.0583e-02	k_t
Generation 6000: Best = 5.717324e+02 = [3, 4, 4, 3]	FES = 600100	Diversity = 1.2451e-02	k_t
Generation 6010: Best = 5.717194e+02 = [1, 1, 1, 1]	FES = 601100	Diversity = 1.3755e-02	k_t
Generation 6020: Best = 5.717081e+02 = [4, 2, 4, 3]	FES = 602100	Diversity = 1.4684e-02	k_t
Generation 6030: Best = 5.716972e+02 = [1, 1, 1, 1]	FES = 603100	Diversity = 1.1937e-02	k_t
Generation 6040: Best = 5.716887e+02 = [3, 3, 2, 2]	FES = 604100	Diversity = 1.0996e-02	k_t
Generation 6050: Best = 5.716771e+02 = [1, 1, 1, 1]	FES = 605100	Diversity = 1.3464e-02	k_t
Generation 6060: Best = 5.716679e+02 = [3, 3, 2, 3]	FES = 606100	Diversity = 1.2369e-02	k_t
Generation 6070: Best = 5.716576e+02 = [1, 1, 1, 1]	FES = 607100	Diversity = 1.2484e-02	k_t
Generation 6080: Best = 5.716487e+02 = [4, 2, 4, 4]	FES = 608100	Diversity = 1.1865e-02	k_t
Generation 6090: Best = 5.716395e+02 = [1, 1, 1, 1]	FES = 609100	Diversity = 1.1641e-02	k_t
Generation 6100: Best = 5.716302e+02 = [4, 4, 2, 2]	FES = 610100	Diversity = 1.0500e-02	k_t
Generation 6110: Best = 5.716181e+02 = [1, 1, 1, 1]	FES = 611100	Diversity = 1.2704e-02	k_t
Generation 6120: Best = 5.716031e+02 = [3, 2, 4, 3]	FES = 612100	Diversity = 1.5537e-02	k_t
Generation 6130: Best = 5.715869e+02 = [1, 1, 1, 1]	FES = 613100	Diversity = 1.3988e-02	k_t
Generation 6140: Best = 5.715726e+02 = [2, 4, 2, 3]	FES = 614100	Diversity = 1.3824e-02	k_t
Generation 6150: Best = 5.715600e+02 = [1, 1, 1, 1]	FES = 615100	Diversity = 1.2901e-02	k_t
Generation 6160: Best = 5.715490e+02 = [4, 2, 4, 4]	FES = 616100	Diversity = 1.2083e-02	k_t

Generation 6170: Best = 5.715376e+02 = [1, 1, 1, 1]	FES = 617100	Diversity = 1.2356e-02	k_t
Generation 6180: Best = 5.715254e+02 = [4, 2, 2, 4]	FES = 618100	Diversity = 1.2253e-02	k_t
Generation 6190: Best = 5.715159e+02 = [1, 1, 1, 1]	FES = 619100	Diversity = 1.2588e-02	k_t
Generation 6200: Best = 5.715018e+02 = [2, 2, 2, 4]	FES = 620100	Diversity = 1.3808e-02	k_t
Generation 6210: Best = 5.714925e+02 = [1, 1, 1, 1]	FES = 621100	Diversity = 1.1240e-02	k_t
Generation 6220: Best = 5.714796e+02 = [2, 4, 3, 4]	FES = 622100	Diversity = 1.1959e-02	k_t
Generation 6230: Best = 5.714660e+02 = [1, 1, 1, 1]	FES = 623100	Diversity = 1.4962e-02	k_t
Generation 6240: Best = 5.714529e+02 = [2, 3, 3, 2]	FES = 624100	Diversity = 1.2713e-02	k_t
Generation 6250: Best = 5.714428e+02 = [1, 1, 1, 1]	FES = 625100	Diversity = 1.2229e-02	k_t
Generation 6260: Best = 5.714335e+02 = [3, 4, 3, 4]	FES = 626100	Diversity = 1.2489e-02	k_t
Generation 6270: Best = 5.714217e+02 = [1, 1, 1, 1]	FES = 627100	Diversity = 1.2124e-02	k_t
Generation 6280: Best = 5.714122e+02 = [3, 4, 2, 2]	FES = 628100	Diversity = 1.2441e-02	k_t
Generation 6290: Best = 5.714003e+02 = [1, 1, 1, 1]	FES = 629100	Diversity = 1.2752e-02	k_t
Generation 6300: Best = 5.713901e+02 = [2, 3, 4, 4]	FES = 630100	Diversity = 1.2770e-02	k_t
Generation 6310: Best = 5.713763e+02 = [1, 1, 1, 1]	FES = 631100	Diversity = 1.5030e-02	k_t
Generation 6320: Best = 5.713617e+02 = [2, 4, 3, 4]	FES = 632100	Diversity = 1.5448e-02	k_t
Generation 6330: Best = 5.713491e+02 = [1, 1, 1, 1]	FES = 633100	Diversity = 1.4765e-02	k_t
Generation 6340: Best = 5.713374e+02 = [3, 4, 4, 2]	FES = 634100	Diversity = 1.5111e-02	k_t
Generation 6350: Best = 5.713242e+02 = [1, 1, 1, 1]	FES = 635100	Diversity = 1.3167e-02	k_t
Generation 6360: Best = 5.713131e+02 = [3, 3, 2, 2]	FES = 636100	Diversity = 1.3500e-02	k_t
Generation 6370: Best = 5.713038e+02 = [1, 1, 1, 1]	FES = 637100	Diversity = 1.1773e-02	k_t
Generation 6380: Best = 5.712929e+02 = [2, 2, 4, 2]	FES = 638100	Diversity = 1.3109e-02	k_t
Generation 6390: Best = 5.712813e+02 = [1, 1, 1, 1]	FES = 639100	Diversity = 1.2517e-02	k_t
Generation 6400: Best = 5.712717e+02 = [3, 2, 3, 4]	FES = 640100	Diversity = 1.1977e-02	k_t
Generation 6410: Best = 5.712618e+02 = [1, 1, 1, 1]	FES = 641100	Diversity = 1.1229e-02	k_t
Generation 6420: Best = 5.712505e+02 = [4, 3, 3, 3]	FES = 642100	Diversity = 1.2820e-02	k_t
Generation 6430: Best = 5.712371e+02 = [1, 1, 1, 1]	FES = 643100	Diversity = 1.2653e-02	k_t

Generation 6440: Best = 5.712252e+02 = [4, 2, 3, 3]	FES = 644100	Diversity = 1.3019e-02	k_t
Generation 6450: Best = 5.712160e+02 = [1, 1, 1, 1]	FES = 645100	Diversity = 1.2790e-02	k_t
Generation 6460: Best = 5.712039e+02 = [4, 4, 2, 4]	FES = 646100	Diversity = 1.2409e-02	k_t
Generation 6470: Best = 5.711949e+02 = [1, 1, 1, 1]	FES = 647100	Diversity = 1.1742e-02	k_t
Generation 6480: Best = 5.711861e+02 = [2, 4, 2, 4]	FES = 648100	Diversity = 1.1983e-02	k_t
Generation 6490: Best = 5.711777e+02 = [1, 1, 1, 1]	FES = 649100	Diversity = 9.9941e-03	k_t
Generation 6500: Best = 5.711704e+02 = [2, 2, 3, 2]	FES = 650100	Diversity = 1.0583e-02	k_t
Generation 6510: Best = 5.711622e+02 = [1, 1, 1, 1]	FES = 651100	Diversity = 1.0781e-02	k_t
Generation 6520: Best = 5.711536e+02 = [2, 2, 4, 4]	FES = 652100	Diversity = 1.1357e-02	k_t
Generation 6530: Best = 5.711442e+02 = [1, 1, 1, 1]	FES = 653100	Diversity = 1.2645e-02	k_t
Generation 6540: Best = 5.711288e+02 = [3, 2, 4, 2]	FES = 654100	Diversity = 1.4515e-02	k_t
Generation 6550: Best = 5.711156e+02 = [1, 1, 1, 1]	FES = 655100	Diversity = 1.4262e-02	k_t
Generation 6560: Best = 5.711054e+02 = [3, 2, 4, 4]	FES = 656100	Diversity = 1.5935e-02	k_t
Generation 6570: Best = 5.710961e+02 = [1, 1, 1, 1]	FES = 657100	Diversity = 1.1418e-02	k_t
Generation 6580: Best = 5.710895e+02 = [4, 4, 2, 2]	FES = 658100	Diversity = 9.7011e-03	k_t
Generation 6590: Best = 5.710799e+02 = [1, 1, 1, 1]	FES = 659100	Diversity = 1.1243e-02	k_t
Generation 6600: Best = 5.710704e+02 = [2, 3, 4, 4]	FES = 660100	Diversity = 1.0542e-02	k_t
Generation 6610: Best = 5.710593e+02 = [1, 1, 1, 1]	FES = 661100	Diversity = 1.1523e-02	k_t
Generation 6620: Best = 5.710543e+02 = [2, 4, 4, 2]	FES = 662100	Diversity = 1.0966e-02	k_t
Generation 6630: Best = 5.710467e+02 = [1, 1, 1, 1]	FES = 663100	Diversity = 1.0958e-02	k_t
Generation 6640: Best = 5.710376e+02 = [2, 4, 2, 4]	FES = 664100	Diversity = 1.1224e-02	k_t
Generation 6650: Best = 5.710281e+02 = [1, 1, 1, 1]	FES = 665100	Diversity = 1.0551e-02	k_t
Generation 6660: Best = 5.710202e+02 = [4, 3, 2, 2]	FES = 666100	Diversity = 9.8014e-03	k_t
Generation 6670: Best = 5.710115e+02 = [1, 1, 1, 1]	FES = 667100	Diversity = 1.0210e-02	k_t
Generation 6680: Best = 5.710039e+02 = [2, 3, 3, 2]	FES = 668100	Diversity = 1.0157e-02	k_t
Generation 6690: Best = 5.709960e+02 = [1, 1, 1, 1]	FES = 669100	Diversity = 1.1124e-02	k_t
Generation 6700: Best = 5.709883e+02 = [3, 2, 4, 4]	FES = 670100	Diversity = 1.0370e-02	k_t

Generation 6710: Best = 5.709778e+02	FES = 671100	Diversity = 1.2105e-02	k_t
= [1, 1, 1, 1]			
Generation 6720: Best = 5.709640e+02	FES = 672100	Diversity = 1.4046e-02	k_t
= [3, 4, 3, 2]			
Generation 6730: Best = 5.709547e+02	FES = 673100	Diversity = 1.4183e-02	k_t
= [1, 1, 1, 1]			
Generation 6740: Best = 5.709454e+02	FES = 674100	Diversity = 1.4643e-02	k_t
= [3, 4, 3, 4]			
Generation 6750: Best = 5.709341e+02	FES = 675100	Diversity = 1.3574e-02	k_t
= [1, 1, 1, 1]			
Generation 6760: Best = 5.709170e+02	FES = 676100	Diversity = 1.5537e-02	k_t
= [3, 3, 2, 2]			
Generation 6770: Best = 5.709051e+02	FES = 677100	Diversity = 1.3260e-02	k_t
= [1, 1, 1, 1]			
Generation 6780: Best = 5.708930e+02	FES = 678100	Diversity = 1.2714e-02	k_t
= [2, 4, 2, 4]			
Generation 6790: Best = 5.708820e+02	FES = 679100	Diversity = 1.3395e-02	k_t
= [1, 1, 1, 1]			
Generation 6800: Best = 5.708715e+02	FES = 680100	Diversity = 1.2245e-02	k_t
= [2, 3, 3, 2]			
Generation 6810: Best = 5.708653e+02	FES = 681100	Diversity = 9.5749e-03	k_t
= [1, 1, 1, 1]			
Generation 6820: Best = 5.708564e+02	FES = 682100	Diversity = 1.1389e-02	k_t
= [4, 4, 4, 4]			
Generation 6830: Best = 5.708453e+02	FES = 683100	Diversity = 1.4715e-02	k_t
= [1, 1, 1, 1]			
Generation 6840: Best = 5.708371e+02	FES = 684100	Diversity = 1.2578e-02	k_t
= [3, 4, 2, 4]			
Generation 6850: Best = 5.708253e+02	FES = 685100	Diversity = 1.3044e-02	k_t
= [1, 1, 1, 1]			
Generation 6860: Best = 5.708157e+02	FES = 686100	Diversity = 1.2386e-02	k_t
= [2, 4, 3, 4]			
Generation 6870: Best = 5.708034e+02	FES = 687100	Diversity = 1.3147e-02	k_t
= [1, 1, 1, 1]			
Generation 6880: Best = 5.707893e+02	FES = 688100	Diversity = 1.3945e-02	k_t
= [2, 4, 2, 2]			
Generation 6890: Best = 5.707796e+02	FES = 689100	Diversity = 1.2620e-02	k_t
= [1, 1, 1, 1]			
Generation 6900: Best = 5.707695e+02	FES = 690100	Diversity = 1.3028e-02	k_t
= [3, 2, 3, 3]			
Generation 6910: Best = 5.707609e+02	FES = 691100	Diversity = 1.2734e-02	k_t
= [1, 1, 1, 1]			
Generation 6920: Best = 5.707509e+02	FES = 692100	Diversity = 1.1384e-02	k_t
= [3, 4, 4, 3]			
Generation 6930: Best = 5.707404e+02	FES = 693100	Diversity = 1.3927e-02	k_t
= [1, 1, 1, 1]			
Generation 6940: Best = 5.707305e+02	FES = 694100	Diversity = 1.3741e-02	k_t
= [2, 2, 4, 2]			
Generation 6950: Best = 5.707189e+02	FES = 695100	Diversity = 1.2995e-02	k_t
= [1, 1, 1, 1]			
Generation 6960: Best = 5.707094e+02	FES = 696100	Diversity = 1.0912e-02	k_t
= [4, 3, 4, 3]			
Generation 6970: Best = 5.707017e+02	FES = 697100	Diversity = 1.0624e-02	k_t
= [1, 1, 1, 1]			

Generation 6980: Best = 5.706935e+02	FES = 698100	Diversity = 1.1899e-02	k_t
= [3, 3, 2, 2]			
Generation 6990: Best = 5.706857e+02	FES = 699100	Diversity = 1.1865e-02	k_t
= [1, 1, 1, 1]			
Generation 7000: Best = 5.706787e+02	FES = 700100	Diversity = 9.1484e-03	k_t
= [4, 2, 3, 4]			
Generation 7010: Best = 5.706742e+02	FES = 701100	Diversity = 8.6104e-03	k_t
= [1, 1, 1, 1]			
Generation 7020: Best = 5.706676e+02	FES = 702100	Diversity = 9.4756e-03	k_t
= [2, 4, 2, 3]			
Generation 7030: Best = 5.706618e+02	FES = 703100	Diversity = 9.5688e-03	k_t
= [1, 1, 1, 1]			
Generation 7040: Best = 5.706543e+02	FES = 704100	Diversity = 1.0406e-02	k_t
= [2, 2, 2, 2]			
Generation 7050: Best = 5.706472e+02	FES = 705100	Diversity = 1.2643e-02	k_t
= [1, 1, 1, 1]			
Generation 7060: Best = 5.706368e+02	FES = 706100	Diversity = 1.1989e-02	k_t
= [4, 3, 2, 4]			
Generation 7070: Best = 5.706270e+02	FES = 707100	Diversity = 1.2100e-02	k_t
= [1, 1, 1, 1]			
Generation 7080: Best = 5.706183e+02	FES = 708100	Diversity = 1.2698e-02	k_t
= [2, 3, 4, 2]			
Generation 7090: Best = 5.706065e+02	FES = 709100	Diversity = 1.3418e-02	k_t
= [1, 1, 1, 1]			
Generation 7100: Best = 5.705936e+02	FES = 710100	Diversity = 1.2688e-02	k_t
= [2, 4, 4, 3]			
Generation 7110: Best = 5.705845e+02	FES = 711100	Diversity = 1.2587e-02	k_t
= [1, 1, 1, 1]			
Generation 7120: Best = 5.705753e+02	FES = 712100	Diversity = 1.3525e-02	k_t
= [2, 4, 3, 4]			
Generation 7130: Best = 5.705596e+02	FES = 713100	Diversity = 1.5301e-02	k_t
= [1, 1, 1, 1]			
Generation 7140: Best = 5.705488e+02	FES = 714100	Diversity = 1.3278e-02	k_t
= [4, 4, 4, 4]			
Generation 7150: Best = 5.705357e+02	FES = 715100	Diversity = 1.2287e-02	k_t
= [1, 1, 1, 1]			
Generation 7160: Best = 5.705264e+02	FES = 716100	Diversity = 1.1339e-02	k_t
= [4, 2, 2, 4]			
Generation 7170: Best = 5.705153e+02	FES = 717100	Diversity = 1.1889e-02	k_t
= [1, 1, 1, 1]			
Generation 7180: Best = 5.705066e+02	FES = 718100	Diversity = 1.2817e-02	k_t
= [2, 3, 4, 2]			
Generation 7190: Best = 5.704918e+02	FES = 719100	Diversity = 1.4979e-02	k_t
= [1, 1, 1, 1]			
Generation 7200: Best = 5.704805e+02	FES = 720100	Diversity = 1.3936e-02	k_t
= [3, 4, 4, 4]			
Generation 7210: Best = 5.704705e+02	FES = 721100	Diversity = 1.3567e-02	k_t
= [1, 1, 1, 1]			
Generation 7220: Best = 5.704600e+02	FES = 722100	Diversity = 1.1609e-02	k_t
= [4, 4, 4, 4]			
Generation 7230: Best = 5.704516e+02	FES = 723100	Diversity = 1.1064e-02	k_t
= [1, 1, 1, 1]			
Generation 7240: Best = 5.704422e+02	FES = 724100	Diversity = 1.2480e-02	k_t
= [4, 2, 3, 3]			

Generation 7250: Best = 5.704300e+02 = [1, 1, 1, 1]	FES = 725100	Diversity = 1.3982e-02	k_t
Generation 7260: Best = 5.704188e+02 = [2, 2, 4, 4]	FES = 726100	Diversity = 1.4902e-02	k_t
Generation 7270: Best = 5.704065e+02 = [1, 1, 1, 1]	FES = 727100	Diversity = 1.2806e-02	k_t
Generation 7280: Best = 5.703927e+02 = [2, 3, 3, 3]	FES = 728100	Diversity = 1.4054e-02	k_t
Generation 7290: Best = 5.703781e+02 = [1, 1, 1, 1]	FES = 729100	Diversity = 1.4035e-02	k_t
Generation 7300: Best = 5.703679e+02 = [2, 4, 2, 4]	FES = 730100	Diversity = 1.3845e-02	k_t
Generation 7310: Best = 5.703539e+02 = [1, 1, 1, 1]	FES = 731100	Diversity = 1.4544e-02	k_t
Generation 7320: Best = 5.703419e+02 = [3, 2, 3, 2]	FES = 732100	Diversity = 1.3444e-02	k_t
Generation 7330: Best = 5.703306e+02 = [1, 1, 1, 1]	FES = 733100	Diversity = 1.2326e-02	k_t
Generation 7340: Best = 5.703228e+02 = [4, 2, 4, 2]	FES = 734100	Diversity = 1.1594e-02	k_t
Generation 7350: Best = 5.703127e+02 = [1, 1, 1, 1]	FES = 735100	Diversity = 1.2276e-02	k_t
Generation 7360: Best = 5.703033e+02 = [3, 4, 2, 4]	FES = 736100	Diversity = 1.0949e-02	k_t
Generation 7370: Best = 5.702923e+02 = [1, 1, 1, 1]	FES = 737100	Diversity = 1.1911e-02	k_t
Generation 7380: Best = 5.702796e+02 = [4, 3, 3, 2]	FES = 738100	Diversity = 1.2952e-02	k_t
Generation 7390: Best = 5.702688e+02 = [1, 1, 1, 1]	FES = 739100	Diversity = 1.3849e-02	k_t
Generation 7400: Best = 5.702592e+02 = [2, 3, 4, 4]	FES = 740100	Diversity = 1.3731e-02	k_t
Generation 7410: Best = 5.702505e+02 = [1, 1, 1, 1]	FES = 741100	Diversity = 1.0504e-02	k_t
Generation 7420: Best = 5.702437e+02 = [4, 2, 3, 3]	FES = 742100	Diversity = 9.9054e-03	k_t
Generation 7430: Best = 5.702346e+02 = [1, 1, 1, 1]	FES = 743100	Diversity = 1.0778e-02	k_t
Generation 7440: Best = 5.702270e+02 = [2, 2, 3, 4]	FES = 744100	Diversity = 1.1277e-02	k_t
Generation 7450: Best = 5.702212e+02 = [1, 1, 1, 1]	FES = 745100	Diversity = 8.8394e-03	k_t
Generation 7460: Best = 5.702123e+02 = [4, 3, 2, 2]	FES = 746100	Diversity = 1.0563e-02	k_t
Generation 7470: Best = 5.702021e+02 = [1, 1, 1, 1]	FES = 747100	Diversity = 1.1272e-02	k_t
Generation 7480: Best = 5.701928e+02 = [3, 4, 2, 4]	FES = 748100	Diversity = 1.2626e-02	k_t
Generation 7490: Best = 5.701829e+02 = [1, 1, 1, 1]	FES = 749100	Diversity = 1.1550e-02	k_t
Generation 7500: Best = 5.701707e+02 = [4, 3, 2, 2]	FES = 750100	Diversity = 1.3969e-02	k_t
Generation 7510: Best = 5.701604e+02 = [1, 1, 1, 1]	FES = 751100	Diversity = 1.3005e-02	k_t

Generation 7520: Best = 5.701499e+02 = [4, 3, 3, 4]	FES = 752100	Diversity = 1.3382e-02	k_t
Generation 7530: Best = 5.701391e+02 = [1, 1, 1, 1]	FES = 753100	Diversity = 1.3611e-02	k_t
Generation 7540: Best = 5.701287e+02 = [3, 3, 3, 4]	FES = 754100	Diversity = 1.3198e-02	k_t
Generation 7550: Best = 5.701183e+02 = [1, 1, 1, 1]	FES = 755100	Diversity = 1.3341e-02	k_t
Generation 7560: Best = 5.701083e+02 = [4, 4, 4, 4]	FES = 756100	Diversity = 1.3342e-02	k_t
Generation 7570: Best = 5.700996e+02 = [1, 1, 1, 1]	FES = 757100	Diversity = 1.1464e-02	k_t
Generation 7580: Best = 5.700898e+02 = [3, 4, 2, 4]	FES = 758100	Diversity = 1.2951e-02	k_t
Generation 7590: Best = 5.700802e+02 = [1, 1, 1, 1]	FES = 759100	Diversity = 1.1905e-02	k_t
Generation 7600: Best = 5.700745e+02 = [2, 4, 4, 3]	FES = 760100	Diversity = 1.1778e-02	k_t
Generation 7610: Best = 5.700685e+02 = [1, 1, 1, 1]	FES = 761100	Diversity = 9.7657e-03	k_t
Generation 7620: Best = 5.700622e+02 = [4, 4, 4, 3]	FES = 762100	Diversity = 8.9636e-03	k_t
Generation 7630: Best = 5.700535e+02 = [1, 1, 1, 1]	FES = 763100	Diversity = 1.0582e-02	k_t
Generation 7640: Best = 5.700477e+02 = [3, 2, 3, 2]	FES = 764100	Diversity = 9.7246e-03	k_t
Generation 7650: Best = 5.700410e+02 = [1, 1, 1, 1]	FES = 765100	Diversity = 9.8833e-03	k_t
Generation 7660: Best = 5.700360e+02 = [3, 4, 3, 4]	FES = 766100	Diversity = 9.3971e-03	k_t
Generation 7670: Best = 5.700307e+02 = [1, 1, 1, 1]	FES = 767100	Diversity = 8.7231e-03	k_t
Generation 7680: Best = 5.700267e+02 = [4, 3, 2, 4]	FES = 768100	Diversity = 7.3378e-03	k_t
Generation 7690: Best = 5.700231e+02 = [1, 1, 1, 1]	FES = 769100	Diversity = 7.5476e-03	k_t
Generation 7700: Best = 5.700170e+02 = [4, 4, 4, 3]	FES = 770100	Diversity = 8.8531e-03	k_t
Generation 7710: Best = 5.700110e+02 = [1, 1, 1, 1]	FES = 771100	Diversity = 9.2554e-03	k_t
Generation 7720: Best = 5.700047e+02 = [4, 3, 3, 3]	FES = 772100	Diversity = 1.0419e-02	k_t
Generation 7730: Best = 5.699987e+02 = [1, 1, 1, 1]	FES = 773100	Diversity = 9.4845e-03	k_t
Generation 7740: Best = 5.699914e+02 = [4, 3, 2, 3]	FES = 774100	Diversity = 1.0060e-02	k_t
Generation 7750: Best = 5.699848e+02 = [1, 1, 1, 1]	FES = 775100	Diversity = 9.2361e-03	k_t
Generation 7760: Best = 5.699784e+02 = [2, 3, 2, 3]	FES = 776100	Diversity = 8.3018e-03	k_t
Generation 7770: Best = 5.699714e+02 = [1, 1, 1, 1]	FES = 777100	Diversity = 1.0139e-02	k_t
Generation 7780: Best = 5.699645e+02 = [3, 4, 2, 2]	FES = 778100	Diversity = 9.3399e-03	k_t

Generation 7790: Best = 5.699572e+02	FES = 779100	Diversity = 9.8095e-03	k_t
= [1, 1, 1, 1]			
Generation 7800: Best = 5.699498e+02	FES = 780100	Diversity = 1.0575e-02	k_t
= [3, 2, 4, 2]			
Generation 7810: Best = 5.699427e+02	FES = 781100	Diversity = 1.1068e-02	k_t
= [1, 1, 1, 1]			
Generation 7820: Best = 5.699366e+02	FES = 782100	Diversity = 8.7618e-03	k_t
= [3, 4, 3, 2]			
Generation 7830: Best = 5.699301e+02	FES = 783100	Diversity = 9.7561e-03	k_t
= [1, 1, 1, 1]			
Generation 7840: Best = 5.699211e+02	FES = 784100	Diversity = 1.1180e-02	k_t
= [2, 3, 3, 2]			
Generation 7850: Best = 5.699078e+02	FES = 785100	Diversity = 1.3157e-02	k_t
= [1, 1, 1, 1]			
Generation 7860: Best = 5.698968e+02	FES = 786100	Diversity = 1.2950e-02	k_t
= [2, 3, 2, 3]			
Generation 7870: Best = 5.698832e+02	FES = 787100	Diversity = 1.3386e-02	k_t
= [1, 1, 1, 1]			
Generation 7880: Best = 5.698699e+02	FES = 788100	Diversity = 1.3022e-02	k_t
= [2, 4, 2, 4]			
Generation 7890: Best = 5.698589e+02	FES = 789100	Diversity = 1.3554e-02	k_t
= [1, 1, 1, 1]			
Generation 7900: Best = 5.698444e+02	FES = 790100	Diversity = 1.5645e-02	k_t
= [2, 2, 3, 2]			
Generation 7910: Best = 5.698315e+02	FES = 791100	Diversity = 1.5006e-02	k_t
= [1, 1, 1, 1]			
Generation 7920: Best = 5.698191e+02	FES = 792100	Diversity = 1.4606e-02	k_t
= [2, 2, 4, 3]			
Generation 7930: Best = 5.698078e+02	FES = 793100	Diversity = 1.2968e-02	k_t
= [1, 1, 1, 1]			
Generation 7940: Best = 5.697973e+02	FES = 794100	Diversity = 1.3446e-02	k_t
= [3, 4, 4, 2]			
Generation 7950: Best = 5.697859e+02	FES = 795100	Diversity = 1.2414e-02	k_t
= [1, 1, 1, 1]			
Generation 7960: Best = 5.697764e+02	FES = 796100	Diversity = 1.2983e-02	k_t
= [3, 3, 2, 2]			
Generation 7970: Best = 5.697639e+02	FES = 797100	Diversity = 1.2636e-02	k_t
= [1, 1, 1, 1]			
Generation 7980: Best = 5.697549e+02	FES = 798100	Diversity = 1.3028e-02	k_t
= [3, 4, 2, 4]			
Generation 7990: Best = 5.697442e+02	FES = 799100	Diversity = 1.2273e-02	k_t
= [1, 1, 1, 1]			
Generation 8000: Best = 5.697350e+02	FES = 800100	Diversity = 1.2234e-02	k_t
= [3, 3, 2, 3]			
Generation 8010: Best = 5.697232e+02	FES = 801100	Diversity = 1.3756e-02	k_t
= [1, 1, 1, 1]			
Generation 8020: Best = 5.697129e+02	FES = 802100	Diversity = 1.2994e-02	k_t
= [2, 4, 4, 2]			
Generation 8030: Best = 5.697020e+02	FES = 803100	Diversity = 1.2599e-02	k_t
= [1, 1, 1, 1]			
Generation 8040: Best = 5.696892e+02	FES = 804100	Diversity = 1.1539e-02	k_t
= [3, 4, 2, 3]			
Generation 8050: Best = 5.696833e+02	FES = 805100	Diversity = 1.0733e-02	k_t
= [1, 1, 1, 1]			

Generation 8060: Best = 5.696749e+02 = [3, 2, 2, 4]	FES = 806100	Diversity = 1.0683e-02	k_t
Generation 8070: Best = 5.696657e+02 = [1, 1, 1, 1]	FES = 807100	Diversity = 1.0615e-02	k_t
Generation 8080: Best = 5.696587e+02 = [4, 4, 4, 2]	FES = 808100	Diversity = 1.0666e-02	k_t
Generation 8090: Best = 5.696497e+02 = [1, 1, 1, 1]	FES = 809100	Diversity = 1.0907e-02	k_t
Generation 8100: Best = 5.696432e+02 = [2, 3, 4, 2]	FES = 810100	Diversity = 9.7872e-03	k_t
Generation 8110: Best = 5.696369e+02 = [1, 1, 1, 1]	FES = 811100	Diversity = 9.3483e-03	k_t
Generation 8120: Best = 5.696297e+02 = [4, 2, 3, 3]	FES = 812100	Diversity = 1.1006e-02	k_t
Generation 8130: Best = 5.696226e+02 = [1, 1, 1, 1]	FES = 813100	Diversity = 1.0002e-02	k_t
Generation 8140: Best = 5.696144e+02 = [2, 4, 3, 2]	FES = 814100	Diversity = 9.9823e-03	k_t
Generation 8150: Best = 5.696075e+02 = [1, 1, 1, 1]	FES = 815100	Diversity = 8.8210e-03	k_t
Generation 8160: Best = 5.696003e+02 = [3, 4, 2, 2]	FES = 816100	Diversity = 9.5659e-03	k_t
Generation 8170: Best = 5.695921e+02 = [1, 1, 1, 1]	FES = 817100	Diversity = 9.8397e-03	k_t
Generation 8180: Best = 5.695842e+02 = [2, 4, 4, 2]	FES = 818100	Diversity = 1.0575e-02	k_t
Generation 8190: Best = 5.695745e+02 = [1, 1, 1, 1]	FES = 819100	Diversity = 1.3158e-02	k_t
Generation 8200: Best = 5.695666e+02 = [4, 2, 4, 3]	FES = 820100	Diversity = 1.0415e-02	k_t
Generation 8210: Best = 5.695581e+02 = [1, 1, 1, 1]	FES = 821100	Diversity = 1.0634e-02	k_t
Generation 8220: Best = 5.695493e+02 = [4, 4, 2, 4]	FES = 822100	Diversity = 1.1362e-02	k_t
Generation 8230: Best = 5.695430e+02 = [1, 1, 1, 1]	FES = 823100	Diversity = 1.0496e-02	k_t
Generation 8240: Best = 5.695355e+02 = [4, 3, 4, 4]	FES = 824100	Diversity = 1.0391e-02	k_t
Generation 8250: Best = 5.695286e+02 = [1, 1, 1, 1]	FES = 825100	Diversity = 9.4401e-03	k_t
Generation 8260: Best = 5.695195e+02 = [4, 2, 2, 4]	FES = 826100	Diversity = 1.1968e-02	k_t
Generation 8270: Best = 5.695044e+02 = [1, 1, 1, 1]	FES = 827100	Diversity = 1.4619e-02	k_t
Generation 8280: Best = 5.694943e+02 = [4, 4, 4, 3]	FES = 828100	Diversity = 1.4359e-02	k_t
Generation 8290: Best = 5.694841e+02 = [1, 1, 1, 1]	FES = 829100	Diversity = 1.3271e-02	k_t
Generation 8300: Best = 5.694716e+02 = [4, 2, 2, 2]	FES = 830100	Diversity = 1.3545e-02	k_t
Generation 8310: Best = 5.694596e+02 = [1, 1, 1, 1]	FES = 831100	Diversity = 1.2790e-02	k_t
Generation 8320: Best = 5.694472e+02 = [2, 2, 3, 4]	FES = 832100	Diversity = 1.4829e-02	k_t

Generation 8330: Best = 5.694352e+02 = [1, 1, 1, 1]	FES = 833100	Diversity = 1.1810e-02	k_t
Generation 8340: Best = 5.694224e+02 = [3, 3, 3, 4]	FES = 834100	Diversity = 1.3867e-02	k_t
Generation 8350: Best = 5.694100e+02 = [1, 1, 1, 1]	FES = 835100	Diversity = 1.2409e-02	k_t
Generation 8360: Best = 5.693999e+02 = [2, 3, 4, 3]	FES = 836100	Diversity = 1.2770e-02	k_t
Generation 8370: Best = 5.693907e+02 = [1, 1, 1, 1]	FES = 837100	Diversity = 1.2685e-02	k_t
Generation 8380: Best = 5.693806e+02 = [4, 3, 3, 2]	FES = 838100	Diversity = 1.1787e-02	k_t
Generation 8390: Best = 5.693736e+02 = [1, 1, 1, 1]	FES = 839100	Diversity = 1.0644e-02	k_t
Generation 8400: Best = 5.693657e+02 = [4, 3, 4, 2]	FES = 840100	Diversity = 9.9976e-03	k_t
Generation 8410: Best = 5.693568e+02 = [1, 1, 1, 1]	FES = 841100	Diversity = 1.0317e-02	k_t
Generation 8420: Best = 5.693502e+02 = [2, 3, 4, 4]	FES = 842100	Diversity = 9.7537e-03	k_t
Generation 8430: Best = 5.693439e+02 = [1, 1, 1, 1]	FES = 843100	Diversity = 8.6440e-03	k_t
Generation 8440: Best = 5.693373e+02 = [3, 4, 4, 4]	FES = 844100	Diversity = 1.0284e-02	k_t
Generation 8450: Best = 5.693302e+02 = [1, 1, 1, 1]	FES = 845100	Diversity = 9.9311e-03	k_t
Generation 8460: Best = 5.693233e+02 = [3, 4, 4, 3]	FES = 846100	Diversity = 9.1044e-03	k_t
Generation 8470: Best = 5.693172e+02 = [1, 1, 1, 1]	FES = 847100	Diversity = 9.7883e-03	k_t
Generation 8480: Best = 5.693090e+02 = [2, 4, 3, 2]	FES = 848100	Diversity = 1.0787e-02	k_t
Generation 8490: Best = 5.693007e+02 = [1, 1, 1, 1]	FES = 849100	Diversity = 1.1170e-02	k_t
Generation 8500: Best = 5.692950e+02 = [3, 4, 3, 2]	FES = 850100	Diversity = 9.1557e-03	k_t
Generation 8510: Best = 5.692901e+02 = [1, 1, 1, 1]	FES = 851100	Diversity = 8.6677e-03	k_t
Generation 8520: Best = 5.692837e+02 = [4, 4, 3, 2]	FES = 852100	Diversity = 9.0516e-03	k_t
Generation 8530: Best = 5.692794e+02 = [1, 1, 1, 1]	FES = 853100	Diversity = 8.4060e-03	k_t
Generation 8540: Best = 5.692740e+02 = [4, 3, 4, 3]	FES = 854100	Diversity = 8.5038e-03	k_t
Generation 8550: Best = 5.692691e+02 = [1, 1, 1, 1]	FES = 855100	Diversity = 8.5011e-03	k_t
Generation 8560: Best = 5.692630e+02 = [4, 3, 3, 2]	FES = 856100	Diversity = 9.1927e-03	k_t
Generation 8570: Best = 5.692577e+02 = [1, 1, 1, 1]	FES = 857100	Diversity = 8.5009e-03	k_t
Generation 8580: Best = 5.692529e+02 = [2, 4, 4, 4]	FES = 858100	Diversity = 7.9452e-03	k_t
Generation 8590: Best = 5.692471e+02 = [1, 1, 1, 1]	FES = 859100	Diversity = 9.2715e-03	k_t

Generation 8600: Best = 5.692403e+02	FES = 860100	Diversity = 1.0148e-02	k_t
= [2, 3, 3, 4]			
Generation 8610: Best = 5.692347e+02	FES = 861100	Diversity = 9.5195e-03	k_t
= [1, 1, 1, 1]			
Generation 8620: Best = 5.692276e+02	FES = 862100	Diversity = 1.0448e-02	k_t
= [2, 4, 3, 3]			
Generation 8630: Best = 5.692212e+02	FES = 863100	Diversity = 9.7960e-03	k_t
= [1, 1, 1, 1]			
Generation 8640: Best = 5.692153e+02	FES = 864100	Diversity = 9.4080e-03	k_t
= [2, 4, 3, 4]			
Generation 8650: Best = 5.692098e+02	FES = 865100	Diversity = 8.8655e-03	k_t
= [1, 1, 1, 1]			
Generation 8660: Best = 5.692037e+02	FES = 866100	Diversity = 9.2158e-03	k_t
= [3, 2, 3, 4]			
Generation 8670: Best = 5.691990e+02	FES = 867100	Diversity = 8.1376e-03	k_t
= [1, 1, 1, 1]			
Generation 8680: Best = 5.691939e+02	FES = 868100	Diversity = 8.0818e-03	k_t
= [4, 2, 2, 3]			
Generation 8690: Best = 5.691892e+02	FES = 869100	Diversity = 7.5378e-03	k_t
= [1, 1, 1, 1]			
Generation 8700: Best = 5.691837e+02	FES = 870100	Diversity = 9.0630e-03	k_t
= [4, 2, 4, 2]			
Generation 8710: Best = 5.691789e+02	FES = 871100	Diversity = 8.7911e-03	k_t
= [1, 1, 1, 1]			
Generation 8720: Best = 5.691735e+02	FES = 872100	Diversity = 8.6987e-03	k_t
= [4, 2, 4, 4]			
Generation 8730: Best = 5.691659e+02	FES = 873100	Diversity = 1.0468e-02	k_t
= [1, 1, 1, 1]			
Generation 8740: Best = 5.691584e+02	FES = 874100	Diversity = 1.1268e-02	k_t
= [2, 4, 2, 2]			
Generation 8750: Best = 5.691514e+02	FES = 875100	Diversity = 1.0249e-02	k_t
= [1, 1, 1, 1]			
Generation 8760: Best = 5.691430e+02	FES = 876100	Diversity = 1.0391e-02	k_t
= [2, 2, 2, 2]			
Generation 8770: Best = 5.691363e+02	FES = 877100	Diversity = 1.0458e-02	k_t
= [1, 1, 1, 1]			
Generation 8780: Best = 5.691287e+02	FES = 878100	Diversity = 1.1190e-02	k_t
= [4, 3, 3, 2]			
Generation 8790: Best = 5.691195e+02	FES = 879100	Diversity = 1.1290e-02	k_t
= [1, 1, 1, 1]			
Generation 8800: Best = 5.691098e+02	FES = 880100	Diversity = 1.1284e-02	k_t
= [3, 2, 3, 4]			
Generation 8810: Best = 5.690977e+02	FES = 881100	Diversity = 1.2964e-02	k_t
= [1, 1, 1, 1]			
Generation 8820: Best = 5.690883e+02	FES = 882100	Diversity = 1.2763e-02	k_t
= [3, 3, 2, 3]			
Generation 8830: Best = 5.690768e+02	FES = 883100	Diversity = 1.2230e-02	k_t
= [1, 1, 1, 1]			
Generation 8840: Best = 5.690625e+02	FES = 884100	Diversity = 1.3860e-02	k_t
= [2, 3, 3, 2]			
Generation 8850: Best = 5.690513e+02	FES = 885100	Diversity = 1.4688e-02	k_t
= [1, 1, 1, 1]			
Generation 8860: Best = 5.690399e+02	FES = 886100	Diversity = 1.2828e-02	k_t
= [3, 3, 2, 3]			

Generation 8870: Best = 5.690319e+02 = [1, 1, 1, 1]	FES = 887100	Diversity = 1.2553e-02	k_t
Generation 8880: Best = 5.690225e+02 = [2, 3, 4, 2]	FES = 888100	Diversity = 1.1535e-02	k_t
Generation 8890: Best = 5.690150e+02 = [1, 1, 1, 1]	FES = 889100	Diversity = 1.0919e-02	k_t
Generation 8900: Best = 5.690038e+02 = [4, 2, 3, 3]	FES = 890100	Diversity = 1.2259e-02	k_t
Generation 8910: Best = 5.689974e+02 = [1, 1, 1, 1]	FES = 891100	Diversity = 1.0222e-02	k_t
Generation 8920: Best = 5.689892e+02 = [4, 3, 4, 4]	FES = 892100	Diversity = 1.0009e-02	k_t
Generation 8930: Best = 5.689820e+02 = [1, 1, 1, 1]	FES = 893100	Diversity = 1.1197e-02	k_t
Generation 8940: Best = 5.689710e+02 = [3, 3, 2, 3]	FES = 894100	Diversity = 1.2509e-02	k_t
Generation 8950: Best = 5.689611e+02 = [1, 1, 1, 1]	FES = 895100	Diversity = 1.2110e-02	k_t
Generation 8960: Best = 5.689521e+02 = [2, 3, 3, 4]	FES = 896100	Diversity = 1.1167e-02	k_t
Generation 8970: Best = 5.689374e+02 = [1, 1, 1, 1]	FES = 897100	Diversity = 1.2970e-02	k_t
Generation 8980: Best = 5.689232e+02 = [2, 4, 3, 4]	FES = 898100	Diversity = 1.4660e-02	k_t
Generation 8990: Best = 5.689114e+02 = [1, 1, 1, 1]	FES = 899100	Diversity = 1.5979e-02	k_t
Generation 9000: Best = 5.688998e+02 = [2, 4, 3, 4]	FES = 900100	Diversity = 1.2801e-02	k_t
Generation 9010: Best = 5.688894e+02 = [1, 1, 1, 1]	FES = 901100	Diversity = 1.1951e-02	k_t
Generation 9020: Best = 5.688816e+02 = [2, 4, 2, 4]	FES = 902100	Diversity = 9.9808e-03	k_t
Generation 9030: Best = 5.688725e+02 = [1, 1, 1, 1]	FES = 903100	Diversity = 1.0826e-02	k_t
Generation 9040: Best = 5.688660e+02 = [3, 3, 3, 3]	FES = 904100	Diversity = 9.1772e-03	k_t
Generation 9050: Best = 5.688604e+02 = [1, 1, 1, 1]	FES = 905100	Diversity = 9.0707e-03	k_t
Generation 9060: Best = 5.688519e+02 = [3, 3, 2, 3]	FES = 906100	Diversity = 1.1147e-02	k_t
Generation 9070: Best = 5.688420e+02 = [1, 1, 1, 1]	FES = 907100	Diversity = 1.1750e-02	k_t
Generation 9080: Best = 5.688342e+02 = [2, 2, 2, 2]	FES = 908100	Diversity = 1.0750e-02	k_t
Generation 9090: Best = 5.688250e+02 = [1, 1, 1, 1]	FES = 909100	Diversity = 1.1358e-02	k_t
Generation 9100: Best = 5.688181e+02 = [3, 2, 3, 2]	FES = 910100	Diversity = 9.3107e-03	k_t
Generation 9110: Best = 5.688107e+02 = [1, 1, 1, 1]	FES = 911100	Diversity = 9.9585e-03	k_t
Generation 9120: Best = 5.688038e+02 = [3, 4, 4, 4]	FES = 912100	Diversity = 1.0826e-02	k_t
Generation 9130: Best = 5.687962e+02 = [1, 1, 1, 1]	FES = 913100	Diversity = 9.5356e-03	k_t

Generation 9140: Best = 5.687885e+02 = [2, 4, 2, 4]	FES = 914100	Diversity = 9.3363e-03	k_t
Generation 9150: Best = 5.687786e+02 = [1, 1, 1, 1]	FES = 915100	Diversity = 1.1026e-02	k_t
Generation 9160: Best = 5.687650e+02 = [3, 4, 3, 2]	FES = 916100	Diversity = 1.4537e-02	k_t
Generation 9170: Best = 5.687518e+02 = [1, 1, 1, 1]	FES = 917100	Diversity = 1.3643e-02	k_t
Generation 9180: Best = 5.687350e+02 = [2, 4, 2, 3]	FES = 918100	Diversity = 1.5704e-02	k_t
Generation 9190: Best = 5.687198e+02 = [1, 1, 1, 1]	FES = 919100	Diversity = 1.5364e-02	k_t
Generation 9200: Best = 5.687080e+02 = [4, 2, 4, 4]	FES = 920100	Diversity = 1.5097e-02	k_t
Generation 9210: Best = 5.686974e+02 = [1, 1, 1, 1]	FES = 921100	Diversity = 1.1999e-02	k_t
Generation 9220: Best = 5.686884e+02 = [4, 2, 3, 3]	FES = 922100	Diversity = 1.1618e-02	k_t
Generation 9230: Best = 5.686802e+02 = [1, 1, 1, 1]	FES = 923100	Diversity = 1.1774e-02	k_t
Generation 9240: Best = 5.686671e+02 = [2, 3, 3, 3]	FES = 924100	Diversity = 1.3172e-02	k_t
Generation 9250: Best = 5.686560e+02 = [1, 1, 1, 1]	FES = 925100	Diversity = 1.2063e-02	k_t
Generation 9260: Best = 5.686464e+02 = [2, 4, 3, 3]	FES = 926100	Diversity = 1.0036e-02	k_t
Generation 9270: Best = 5.686387e+02 = [1, 1, 1, 1]	FES = 927100	Diversity = 9.6377e-03	k_t
Generation 9280: Best = 5.686316e+02 = [2, 2, 2, 3]	FES = 928100	Diversity = 1.0313e-02	k_t
Generation 9290: Best = 5.686242e+02 = [1, 1, 1, 1]	FES = 929100	Diversity = 9.3881e-03	k_t
Generation 9300: Best = 5.686187e+02 = [3, 2, 3, 3]	FES = 930100	Diversity = 7.9407e-03	k_t
Generation 9310: Best = 5.686126e+02 = [1, 1, 1, 1]	FES = 931100	Diversity = 7.4781e-03	k_t
Generation 9320: Best = 5.686082e+02 = [4, 3, 3, 3]	FES = 932100	Diversity = 7.1063e-03	k_t
Generation 9330: Best = 5.686030e+02 = [1, 1, 1, 1]	FES = 933100	Diversity = 8.7534e-03	k_t
Generation 9340: Best = 5.685967e+02 = [3, 2, 3, 3]	FES = 934100	Diversity = 9.6124e-03	k_t
Generation 9350: Best = 5.685896e+02 = [1, 1, 1, 1]	FES = 935100	Diversity = 1.0345e-02	k_t
Generation 9360: Best = 5.685840e+02 = [2, 2, 2, 3]	FES = 936100	Diversity = 9.1058e-03	k_t
Generation 9370: Best = 5.685789e+02 = [1, 1, 1, 1]	FES = 937100	Diversity = 8.2352e-03	k_t
Generation 9380: Best = 5.685746e+02 = [4, 3, 3, 3]	FES = 938100	Diversity = 9.3535e-03	k_t
Generation 9390: Best = 5.685696e+02 = [1, 1, 1, 1]	FES = 939100	Diversity = 8.5053e-03	k_t
Generation 9400: Best = 5.685645e+02 = [4, 2, 2, 2]	FES = 940100	Diversity = 8.5138e-03	k_t

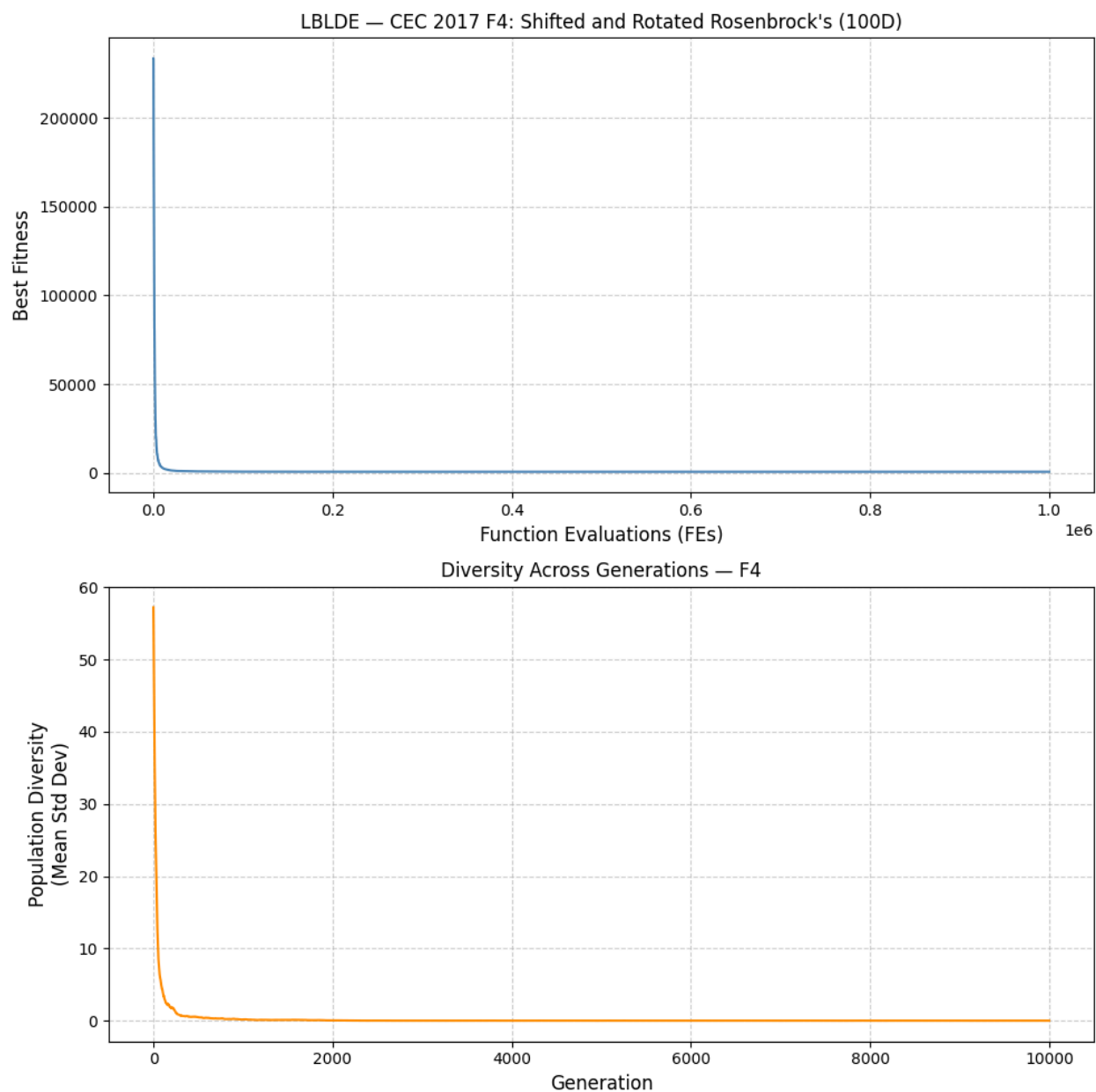
Generation 9410: Best = 5.685578e+02 = [1, 1, 1, 1]	FES = 941100	Diversity = 8.5574e-03	k_t
Generation 9420: Best = 5.685500e+02 = [4, 3, 3, 4]	FES = 942100	Diversity = 1.0401e-02	k_t
Generation 9430: Best = 5.685450e+02 = [1, 1, 1, 1]	FES = 943100	Diversity = 9.1456e-03	k_t
Generation 9440: Best = 5.685372e+02 = [4, 3, 3, 4]	FES = 944100	Diversity = 1.0412e-02	k_t
Generation 9450: Best = 5.685293e+02 = [1, 1, 1, 1]	FES = 945100	Diversity = 1.1075e-02	k_t
Generation 9460: Best = 5.685224e+02 = [3, 3, 3, 2]	FES = 946100	Diversity = 1.0036e-02	k_t
Generation 9470: Best = 5.685160e+02 = [1, 1, 1, 1]	FES = 947100	Diversity = 1.1269e-02	k_t
Generation 9480: Best = 5.685068e+02 = [4, 4, 4, 4]	FES = 948100	Diversity = 1.1715e-02	k_t
Generation 9490: Best = 5.684991e+02 = [1, 1, 1, 1]	FES = 949100	Diversity = 1.1605e-02	k_t
Generation 9500: Best = 5.684876e+02 = [2, 3, 3, 2]	FES = 950100	Diversity = 1.3362e-02	k_t
Generation 9510: Best = 5.684815e+02 = [1, 1, 1, 1]	FES = 951100	Diversity = 1.1307e-02	k_t
Generation 9520: Best = 5.684725e+02 = [4, 4, 4, 2]	FES = 952100	Diversity = 1.2321e-02	k_t
Generation 9530: Best = 5.684613e+02 = [1, 1, 1, 1]	FES = 953100	Diversity = 1.3320e-02	k_t
Generation 9540: Best = 5.684531e+02 = [4, 4, 4, 2]	FES = 954100	Diversity = 1.1334e-02	k_t
Generation 9550: Best = 5.684444e+02 = [1, 1, 1, 1]	FES = 955100	Diversity = 1.0006e-02	k_t
Generation 9560: Best = 5.684342e+02 = [3, 4, 4, 4]	FES = 956100	Diversity = 1.3556e-02	k_t
Generation 9570: Best = 5.684183e+02 = [1, 1, 1, 1]	FES = 957100	Diversity = 1.6650e-02	k_t
Generation 9580: Best = 5.684016e+02 = [4, 3, 3, 3]	FES = 958100	Diversity = 1.7043e-02	k_t
Generation 9590: Best = 5.683901e+02 = [1, 1, 1, 1]	FES = 959100	Diversity = 1.4877e-02	k_t
Generation 9600: Best = 5.683811e+02 = [2, 2, 4, 2]	FES = 960100	Diversity = 1.3052e-02	k_t
Generation 9610: Best = 5.683675e+02 = [1, 1, 1, 1]	FES = 961100	Diversity = 1.4710e-02	k_t
Generation 9620: Best = 5.683569e+02 = [2, 4, 3, 2]	FES = 962100	Diversity = 1.3148e-02	k_t
Generation 9630: Best = 5.683440e+02 = [1, 1, 1, 1]	FES = 963100	Diversity = 1.3643e-02	k_t
Generation 9640: Best = 5.683352e+02 = [4, 3, 4, 2]	FES = 964100	Diversity = 1.2530e-02	k_t
Generation 9650: Best = 5.683268e+02 = [1, 1, 1, 1]	FES = 965100	Diversity = 1.1487e-02	k_t
Generation 9660: Best = 5.683214e+02 = [2, 3, 4, 2]	FES = 966100	Diversity = 9.0672e-03	k_t
Generation 9670: Best = 5.683145e+02 = [1, 1, 1, 1]	FES = 967100	Diversity = 8.9966e-03	k_t

Generation 9680: Best = 5.683092e+02 = [2, 4, 3, 2]	FES = 968100	Diversity = 9.9709e-03	k_t
Generation 9690: Best = 5.683020e+02 = [1, 1, 1, 1]	FES = 969100	Diversity = 1.0769e-02	k_t
Generation 9700: Best = 5.682948e+02 = [4, 3, 2, 2]	FES = 970100	Diversity = 1.0232e-02	k_t
Generation 9710: Best = 5.682829e+02 = [1, 1, 1, 1]	FES = 971100	Diversity = 1.1790e-02	k_t
Generation 9720: Best = 5.682716e+02 = [4, 4, 3, 2]	FES = 972100	Diversity = 1.2729e-02	k_t
Generation 9730: Best = 5.682620e+02 = [1, 1, 1, 1]	FES = 973100	Diversity = 1.1540e-02	k_t
Generation 9740: Best = 5.682516e+02 = [2, 2, 3, 2]	FES = 974100	Diversity = 1.2635e-02	k_t
Generation 9750: Best = 5.682425e+02 = [1, 1, 1, 1]	FES = 975100	Diversity = 1.1715e-02	k_t
Generation 9760: Best = 5.682361e+02 = [2, 3, 2, 4]	FES = 976100	Diversity = 1.0160e-02	k_t
Generation 9770: Best = 5.682302e+02 = [1, 1, 1, 1]	FES = 977100	Diversity = 9.9115e-03	k_t
Generation 9780: Best = 5.682252e+02 = [3, 4, 2, 3]	FES = 978100	Diversity = 9.4700e-03	k_t
Generation 9790: Best = 5.682197e+02 = [1, 1, 1, 1]	FES = 979100	Diversity = 8.4713e-03	k_t
Generation 9800: Best = 5.682147e+02 = [3, 4, 3, 4]	FES = 980100	Diversity = 7.8723e-03	k_t
Generation 9810: Best = 5.682093e+02 = [1, 1, 1, 1]	FES = 981100	Diversity = 8.8749e-03	k_t
Generation 9820: Best = 5.682026e+02 = [3, 3, 2, 3]	FES = 982100	Diversity = 9.2000e-03	k_t
Generation 9830: Best = 5.681943e+02 = [1, 1, 1, 1]	FES = 983100	Diversity = 1.0335e-02	k_t
Generation 9840: Best = 5.681853e+02 = [4, 2, 3, 4]	FES = 984100	Diversity = 1.0257e-02	k_t
Generation 9850: Best = 5.681770e+02 = [1, 1, 1, 1]	FES = 985100	Diversity = 1.1180e-02	k_t
Generation 9860: Best = 5.681688e+02 = [2, 4, 2, 4]	FES = 986100	Diversity = 1.0564e-02	k_t
Generation 9870: Best = 5.681550e+02 = [1, 1, 1, 1]	FES = 987100	Diversity = 1.2206e-02	k_t
Generation 9880: Best = 5.681413e+02 = [2, 2, 4, 4]	FES = 988100	Diversity = 1.4315e-02	k_t
Generation 9890: Best = 5.681294e+02 = [1, 1, 1, 1]	FES = 989100	Diversity = 1.2649e-02	k_t
Generation 9900: Best = 5.681175e+02 = [3, 4, 3, 4]	FES = 990100	Diversity = 1.3104e-02	k_t
Generation 9910: Best = 5.681046e+02 = [1, 1, 1, 1]	FES = 991100	Diversity = 1.4012e-02	k_t
Generation 9920: Best = 5.680954e+02 = [4, 3, 2, 3]	FES = 992100	Diversity = 1.2413e-02	k_t
Generation 9930: Best = 5.680854e+02 = [1, 1, 1, 1]	FES = 993100	Diversity = 1.1340e-02	k_t
Generation 9940: Best = 5.680763e+02 = [4, 2, 2, 2]	FES = 994100	Diversity = 1.0314e-02	k_t

Generation 9950: Best = 5.680686e+02 FES = 995100 Diversity = 1.1195e-02 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 5.680593e+02 FES = 996100 Diversity = 1.2131e-02 k_t
= [3, 2, 4, 4]
Generation 9970: Best = 5.680517e+02 FES = 997100 Diversity = 1.0281e-02 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 5.680443e+02 FES = 998100 Diversity = 1.2207e-02 k_t
= [4, 2, 4, 3]
Generation 9990: Best = 5.680356e+02 FES = 999100 Diversity = 1.1330e-02 k_t
= [1, 1, 1, 1]

Optimization completed!
Final Best Fitness = 5.680283e+02
Total Function Evaluations = 1000000

=====
=
Results:
Best Fitness : 5.680283e+02
Global Optimum : 4.000000e+02
Error (|best-optimum|) : 1.680283e+02
Best Solution (5 dim) : [39.46591217 34.29248972 -56.62020118 74.32831848
39.15001888]
D^(0) (init. diversity): 5.7220e+01
=====



Graph saved as convergence_F4.png

```
In [29]: # =====
# F5 — Shifted and Rotated Rastrigin
# =====
run_and_plot('F5', make_wrapper(simple.f5), 500.0, "Shifted and Rotated Rastrigin")
```

```

=====
=
LBLDE_s on CEC 2017 F5 – Single Run
=====
=
Function      : Shifted and Rotated Rastrigin's (100D)
Bounds       : [-100, 100]
Global optimum : 500.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 3.063799e+03 | Diversity = 5.7220e+01
Generation 10: Best = 2.233173e+03 FES = 1100 Diversity = 3.8370e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.923551e+03 FES = 2100 Diversity = 2.9003e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 1.818317e+03 FES = 3100 Diversity = 2.3193e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.721203e+03 FES = 4100 Diversity = 2.0395e+01 k_t
= [2, 2, 2, 4]
Generation 50: Best = 1.686319e+03 FES = 5100 Diversity = 1.8595e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.637197e+03 FES = 6100 Diversity = 1.7128e+01 k_t
= [3, 4, 2, 3]
Generation 70: Best = 1.607942e+03 FES = 7100 Diversity = 1.6441e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.607942e+03 FES = 8100 Diversity = 1.6050e+01 k_t
= [3, 3, 3, 3]
Generation 90: Best = 1.606952e+03 FES = 9100 Diversity = 1.5655e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.528584e+03 FES = 10100 Diversity = 1.5425e+01 k_t
= [3, 3, 3, 2]
Generation 110: Best = 1.528584e+03 FES = 11100 Diversity = 1.5203e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.528584e+03 FES = 12100 Diversity = 1.5064e+01 k_t
= [4, 4, 3, 3]
Generation 130: Best = 1.526114e+03 FES = 13100 Diversity = 1.4946e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.515839e+03 FES = 14100 Diversity = 1.4889e+01 k_t
= [4, 4, 2, 3]
Generation 150: Best = 1.499047e+03 FES = 15100 Diversity = 1.4820e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.472086e+03 FES = 16100 Diversity = 1.4685e+01 k_t
= [4, 3, 2, 4]
Generation 170: Best = 1.472086e+03 FES = 17100 Diversity = 1.4728e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.472086e+03 FES = 18100 Diversity = 1.4686e+01 k_t
= [4, 3, 3, 3]
Generation 190: Best = 1.472086e+03 FES = 19100 Diversity = 1.4672e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.455595e+03 FES = 20100 Diversity = 1.4622e+01 k_t
= [4, 3, 2, 3]
Generation 210: Best = 1.455595e+03 FES = 21100 Diversity = 1.4606e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.446442e+03 FES = 22100 Diversity = 1.4694e+01 k_t
= [4, 2, 2, 2]

```

Generation 230:	Best = 1.446442e+03	FES = 23100	Diversity = 1.4677e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 1.432595e+03	FES = 24100	Diversity = 1.4585e+01	k_t
= [4, 3, 2, 3]				
Generation 250:	Best = 1.432595e+03	FES = 25100	Diversity = 1.4586e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 1.432595e+03	FES = 26100	Diversity = 1.4584e+01	k_t
= [3, 3, 4, 4]				
Generation 270:	Best = 1.414026e+03	FES = 27100	Diversity = 1.4590e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 1.414026e+03	FES = 28100	Diversity = 1.4587e+01	k_t
= [3, 4, 4, 3]				
Generation 290:	Best = 1.414026e+03	FES = 29100	Diversity = 1.4493e+01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 1.401980e+03	FES = 30100	Diversity = 1.4501e+01	k_t
= [3, 2, 2, 2]				
Generation 310:	Best = 1.401980e+03	FES = 31100	Diversity = 1.4503e+01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 1.384673e+03	FES = 32100	Diversity = 1.4507e+01	k_t
= [2, 4, 3, 2]				
Generation 330:	Best = 1.384673e+03	FES = 33100	Diversity = 1.4431e+01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 1.384673e+03	FES = 34100	Diversity = 1.4360e+01	k_t
= [4, 2, 2, 3]				
Generation 350:	Best = 1.384673e+03	FES = 35100	Diversity = 1.4422e+01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 1.384673e+03	FES = 36100	Diversity = 1.4405e+01	k_t
= [4, 2, 3, 2]				
Generation 370:	Best = 1.380903e+03	FES = 37100	Diversity = 1.4275e+01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 1.329544e+03	FES = 38100	Diversity = 1.4241e+01	k_t
= [2, 3, 2, 2]				
Generation 390:	Best = 1.328149e+03	FES = 39100	Diversity = 1.4306e+01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 1.328149e+03	FES = 40100	Diversity = 1.4330e+01	k_t
= [3, 4, 3, 2]				
Generation 410:	Best = 1.327577e+03	FES = 41100	Diversity = 1.4315e+01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 1.327577e+03	FES = 42100	Diversity = 1.4337e+01	k_t
= [4, 4, 4, 4]				
Generation 430:	Best = 1.327577e+03	FES = 43100	Diversity = 1.4359e+01	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 1.324077e+03	FES = 44100	Diversity = 1.4346e+01	k_t
= [3, 3, 2, 2]				
Generation 450:	Best = 1.324077e+03	FES = 45100	Diversity = 1.4336e+01	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 1.308934e+03	FES = 46100	Diversity = 1.4274e+01	k_t
= [2, 4, 2, 2]				
Generation 470:	Best = 1.304749e+03	FES = 47100	Diversity = 1.4225e+01	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 1.304749e+03	FES = 48100	Diversity = 1.4196e+01	k_t
= [2, 3, 3, 3]				
Generation 490:	Best = 1.304749e+03	FES = 49100	Diversity = 1.4174e+01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 1.304749e+03	FES = 50100	Diversity = 1.4178e+01	k_t = [2, 2, 4, 4]
Generation 510:	Best = 1.304749e+03	FES = 51100	Diversity = 1.4168e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.300461e+03	FES = 52100	Diversity = 1.4167e+01	k_t = [4, 3, 4, 2]
Generation 530:	Best = 1.296308e+03	FES = 53100	Diversity = 1.4164e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 1.296308e+03	FES = 54100	Diversity = 1.4151e+01	k_t = [4, 3, 4, 2]
Generation 550:	Best = 1.296308e+03	FES = 55100	Diversity = 1.4134e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 1.266610e+03	FES = 56100	Diversity = 1.4115e+01	k_t = [4, 3, 4, 3]
Generation 570:	Best = 1.266610e+03	FES = 57100	Diversity = 1.4086e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 1.266610e+03	FES = 58100	Diversity = 1.4047e+01	k_t = [4, 3, 3, 4]
Generation 590:	Best = 1.266610e+03	FES = 59100	Diversity = 1.4023e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 1.266610e+03	FES = 60100	Diversity = 1.4020e+01	k_t = [2, 4, 3, 2]
Generation 610:	Best = 1.266610e+03	FES = 61100	Diversity = 1.3997e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.248375e+03	FES = 62100	Diversity = 1.3939e+01	k_t = [4, 2, 2, 4]
Generation 630:	Best = 1.248375e+03	FES = 63100	Diversity = 1.3920e+01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.246875e+03	FES = 64100	Diversity = 1.3909e+01	k_t = [4, 4, 3, 4]
Generation 650:	Best = 1.242937e+03	FES = 65100	Diversity = 1.3901e+01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 1.241699e+03	FES = 66100	Diversity = 1.3883e+01	k_t = [4, 3, 4, 3]
Generation 670:	Best = 1.241699e+03	FES = 67100	Diversity = 1.3776e+01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 1.224202e+03	FES = 68100	Diversity = 1.3802e+01	k_t = [2, 4, 2, 3]
Generation 690:	Best = 1.222436e+03	FES = 69100	Diversity = 1.3787e+01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 1.213313e+03	FES = 70100	Diversity = 1.3783e+01	k_t = [4, 3, 3, 4]
Generation 710:	Best = 1.213313e+03	FES = 71100	Diversity = 1.3760e+01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 1.204835e+03	FES = 72100	Diversity = 1.3741e+01	k_t = [4, 3, 3, 3]
Generation 730:	Best = 1.204835e+03	FES = 73100	Diversity = 1.3667e+01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 1.203890e+03	FES = 74100	Diversity = 1.3664e+01	k_t = [3, 3, 3, 2]
Generation 750:	Best = 1.203890e+03	FES = 75100	Diversity = 1.3636e+01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 1.203890e+03	FES = 76100	Diversity = 1.3604e+01	k_t = [2, 2, 3, 2]

Generation 770:	Best = 1.203541e+03	FES = 77100	Diversity = 1.3485e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.203541e+03	FES = 78100	Diversity = 1.3434e+01	k_t
= [3, 3, 3, 4]				
Generation 790:	Best = 1.203541e+03	FES = 79100	Diversity = 1.3426e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.199546e+03	FES = 80100	Diversity = 1.3396e+01	k_t
= [2, 4, 2, 3]				
Generation 810:	Best = 1.193630e+03	FES = 81100	Diversity = 1.3411e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.193630e+03	FES = 82100	Diversity = 1.3505e+01	k_t
= [4, 3, 4, 3]				
Generation 830:	Best = 1.193630e+03	FES = 83100	Diversity = 1.3492e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.193630e+03	FES = 84100	Diversity = 1.3520e+01	k_t
= [3, 3, 4, 2]				
Generation 850:	Best = 1.193593e+03	FES = 85100	Diversity = 1.3515e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.193593e+03	FES = 86100	Diversity = 1.3479e+01	k_t
= [4, 3, 2, 4]				
Generation 870:	Best = 1.193593e+03	FES = 87100	Diversity = 1.3442e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.185507e+03	FES = 88100	Diversity = 1.3440e+01	k_t
= [4, 3, 4, 4]				
Generation 890:	Best = 1.185507e+03	FES = 89100	Diversity = 1.3439e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.185507e+03	FES = 90100	Diversity = 1.3424e+01	k_t
= [2, 2, 2, 2]				
Generation 910:	Best = 1.176453e+03	FES = 91100	Diversity = 1.3341e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.176176e+03	FES = 92100	Diversity = 1.3273e+01	k_t
= [2, 2, 2, 2]				
Generation 930:	Best = 1.172860e+03	FES = 93100	Diversity = 1.3281e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.170613e+03	FES = 94100	Diversity = 1.3255e+01	k_t
= [2, 2, 2, 3]				
Generation 950:	Best = 1.165327e+03	FES = 95100	Diversity = 1.3253e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.165327e+03	FES = 96100	Diversity = 1.3246e+01	k_t
= [2, 4, 3, 4]				
Generation 970:	Best = 1.159502e+03	FES = 97100	Diversity = 1.3245e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.159502e+03	FES = 98100	Diversity = 1.3216e+01	k_t
= [3, 3, 2, 3]				
Generation 990:	Best = 1.157657e+03	FES = 99100	Diversity = 1.3169e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 1.157011e+03	FES = 100100	Diversity = 1.3162e+01	k_t
= [3, 3, 2, 4]				
Generation 1010:	Best = 1.157011e+03	FES = 101100	Diversity = 1.3152e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.157011e+03	FES = 102100	Diversity = 1.3096e+01	k_t
= [3, 2, 2, 2]				
Generation 1030:	Best = 1.157011e+03	FES = 103100	Diversity = 1.3028e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.157011e+03 = [3, 2, 2, 2]	FES = 104100	Diversity = 1.2994e+01	k_t
Generation 1050: Best = 1.157011e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.2992e+01	k_t
Generation 1060: Best = 1.157011e+03 = [4, 2, 3, 2]	FES = 106100	Diversity = 1.2987e+01	k_t
Generation 1070: Best = 1.157011e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 1.2963e+01	k_t
Generation 1080: Best = 1.157011e+03 = [2, 4, 4, 4]	FES = 108100	Diversity = 1.2964e+01	k_t
Generation 1090: Best = 1.157011e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.2940e+01	k_t
Generation 1100: Best = 1.156422e+03 = [2, 3, 3, 3]	FES = 110100	Diversity = 1.2855e+01	k_t
Generation 1110: Best = 1.156422e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.2838e+01	k_t
Generation 1120: Best = 1.139981e+03 = [3, 3, 3, 4]	FES = 112100	Diversity = 1.2837e+01	k_t
Generation 1130: Best = 1.139981e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.2832e+01	k_t
Generation 1140: Best = 1.139981e+03 = [4, 4, 2, 3]	FES = 114100	Diversity = 1.2756e+01	k_t
Generation 1150: Best = 1.134339e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.2578e+01	k_t
Generation 1160: Best = 1.134339e+03 = [4, 4, 2, 3]	FES = 116100	Diversity = 1.2573e+01	k_t
Generation 1170: Best = 1.128865e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.2519e+01	k_t
Generation 1180: Best = 1.128865e+03 = [4, 3, 3, 4]	FES = 118100	Diversity = 1.2473e+01	k_t
Generation 1190: Best = 1.128865e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.2488e+01	k_t
Generation 1200: Best = 1.128865e+03 = [2, 3, 4, 2]	FES = 120100	Diversity = 1.2460e+01	k_t
Generation 1210: Best = 1.128865e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.2412e+01	k_t
Generation 1220: Best = 1.122085e+03 = [4, 3, 4, 3]	FES = 122100	Diversity = 1.2439e+01	k_t
Generation 1230: Best = 1.122085e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.2444e+01	k_t
Generation 1240: Best = 1.122085e+03 = [2, 2, 3, 4]	FES = 124100	Diversity = 1.2440e+01	k_t
Generation 1250: Best = 1.119216e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 1.2400e+01	k_t
Generation 1260: Best = 1.119216e+03 = [2, 3, 2, 4]	FES = 126100	Diversity = 1.2320e+01	k_t
Generation 1270: Best = 1.115730e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.2317e+01	k_t
Generation 1280: Best = 1.115730e+03 = [3, 4, 2, 4]	FES = 128100	Diversity = 1.2320e+01	k_t
Generation 1290: Best = 1.115730e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.2321e+01	k_t
Generation 1300: Best = 1.115730e+03 = [2, 4, 2, 4]	FES = 130100	Diversity = 1.2266e+01	k_t

Generation 1310: Best = 1.115730e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.2274e+01	k_t
Generation 1320: Best = 1.115730e+03 = [3, 3, 3, 2]	FES = 132100	Diversity = 1.2271e+01	k_t
Generation 1330: Best = 1.115730e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 1.2268e+01	k_t
Generation 1340: Best = 1.115730e+03 = [4, 2, 2, 2]	FES = 134100	Diversity = 1.2291e+01	k_t
Generation 1350: Best = 1.115730e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.2290e+01	k_t
Generation 1360: Best = 1.112470e+03 = [3, 3, 2, 4]	FES = 136100	Diversity = 1.2291e+01	k_t
Generation 1370: Best = 1.112470e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 1.2289e+01	k_t
Generation 1380: Best = 1.083839e+03 = [3, 4, 2, 4]	FES = 138100	Diversity = 1.2282e+01	k_t
Generation 1390: Best = 1.083839e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 1.2232e+01	k_t
Generation 1400: Best = 1.075635e+03 = [2, 4, 3, 4]	FES = 140100	Diversity = 1.2225e+01	k_t
Generation 1410: Best = 1.075635e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 1.2191e+01	k_t
Generation 1420: Best = 1.075635e+03 = [2, 2, 4, 2]	FES = 142100	Diversity = 1.2151e+01	k_t
Generation 1430: Best = 1.075635e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 1.2147e+01	k_t
Generation 1440: Best = 1.075635e+03 = [2, 2, 3, 3]	FES = 144100	Diversity = 1.2102e+01	k_t
Generation 1450: Best = 1.075635e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 1.2099e+01	k_t
Generation 1460: Best = 1.075635e+03 = [4, 4, 3, 4]	FES = 146100	Diversity = 1.2089e+01	k_t
Generation 1470: Best = 1.075635e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 1.2045e+01	k_t
Generation 1480: Best = 1.075635e+03 = [3, 4, 4, 3]	FES = 148100	Diversity = 1.2055e+01	k_t
Generation 1490: Best = 1.075635e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 1.2050e+01	k_t
Generation 1500: Best = 1.075635e+03 = [4, 3, 4, 3]	FES = 150100	Diversity = 1.2045e+01	k_t
Generation 1510: Best = 1.073164e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 1.2040e+01	k_t
Generation 1520: Best = 1.073164e+03 = [2, 2, 3, 3]	FES = 152100	Diversity = 1.2005e+01	k_t
Generation 1530: Best = 1.073164e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 1.1999e+01	k_t
Generation 1540: Best = 1.073164e+03 = [3, 3, 2, 4]	FES = 154100	Diversity = 1.1971e+01	k_t
Generation 1550: Best = 1.073164e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 1.1967e+01	k_t
Generation 1560: Best = 1.073164e+03 = [3, 2, 4, 2]	FES = 156100	Diversity = 1.1951e+01	k_t
Generation 1570: Best = 1.073164e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 1.1949e+01	k_t

Generation 1580: Best = 1.073164e+03 = [3, 2, 2, 3]	FES = 158100	Diversity = 1.1949e+01	k_t
Generation 1590: Best = 1.073164e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 1.1946e+01	k_t
Generation 1600: Best = 1.071495e+03 = [4, 3, 3, 2]	FES = 160100	Diversity = 1.1945e+01	k_t
Generation 1610: Best = 1.071495e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 1.1945e+01	k_t
Generation 1620: Best = 1.071495e+03 = [4, 2, 2, 2]	FES = 162100	Diversity = 1.1920e+01	k_t
Generation 1630: Best = 1.071495e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.1883e+01	k_t
Generation 1640: Best = 1.071495e+03 = [2, 3, 2, 2]	FES = 164100	Diversity = 1.1884e+01	k_t
Generation 1650: Best = 1.071371e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.1848e+01	k_t
Generation 1660: Best = 1.071371e+03 = [2, 4, 3, 2]	FES = 166100	Diversity = 1.1823e+01	k_t
Generation 1670: Best = 1.067561e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.1831e+01	k_t
Generation 1680: Best = 1.067561e+03 = [3, 2, 4, 2]	FES = 168100	Diversity = 1.1836e+01	k_t
Generation 1690: Best = 1.067561e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 1.1831e+01	k_t
Generation 1700: Best = 1.067561e+03 = [4, 3, 2, 4]	FES = 170100	Diversity = 1.1830e+01	k_t
Generation 1710: Best = 1.066886e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.1803e+01	k_t
Generation 1720: Best = 1.060472e+03 = [3, 4, 2, 2]	FES = 172100	Diversity = 1.1801e+01	k_t
Generation 1730: Best = 1.057069e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 1.1800e+01	k_t
Generation 1740: Best = 1.053326e+03 = [3, 4, 2, 3]	FES = 174100	Diversity = 1.1759e+01	k_t
Generation 1750: Best = 1.036207e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 1.1757e+01	k_t
Generation 1760: Best = 1.032302e+03 = [2, 2, 4, 2]	FES = 176100	Diversity = 1.1747e+01	k_t
Generation 1770: Best = 1.032302e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 1.1742e+01	k_t
Generation 1780: Best = 1.032302e+03 = [2, 4, 3, 2]	FES = 178100	Diversity = 1.1729e+01	k_t
Generation 1790: Best = 1.030238e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.1712e+01	k_t
Generation 1800: Best = 1.025398e+03 = [2, 2, 2, 3]	FES = 180100	Diversity = 1.1715e+01	k_t
Generation 1810: Best = 1.025131e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 1.1706e+01	k_t
Generation 1820: Best = 1.023921e+03 = [2, 2, 3, 2]	FES = 182100	Diversity = 1.1695e+01	k_t
Generation 1830: Best = 1.020190e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.1693e+01	k_t
Generation 1840: Best = 1.019885e+03 = [4, 2, 3, 4]	FES = 184100	Diversity = 1.1691e+01	k_t

Generation 1850: Best = 1.019885e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.1683e+01	k_t
Generation 1860: Best = 1.014638e+03 = [4, 4, 3, 4]	FES = 186100	Diversity = 1.1653e+01	k_t
Generation 1870: Best = 1.014638e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.1636e+01	k_t
Generation 1880: Best = 1.014380e+03 = [2, 4, 3, 3]	FES = 188100	Diversity = 1.1638e+01	k_t
Generation 1890: Best = 1.011191e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 1.1624e+01	k_t
Generation 1900: Best = 1.011191e+03 = [2, 3, 2, 4]	FES = 190100	Diversity = 1.1624e+01	k_t
Generation 1910: Best = 1.011191e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 1.1613e+01	k_t
Generation 1920: Best = 1.002548e+03 = [4, 2, 3, 2]	FES = 192100	Diversity = 1.1602e+01	k_t
Generation 1930: Best = 1.002548e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 1.1564e+01	k_t
Generation 1940: Best = 1.002548e+03 = [4, 4, 2, 2]	FES = 194100	Diversity = 1.1539e+01	k_t
Generation 1950: Best = 9.991404e+02 = [1, 1, 1, 1]	FES = 195100	Diversity = 1.1572e+01	k_t
Generation 1960: Best = 9.991404e+02 = [4, 3, 2, 2]	FES = 196100	Diversity = 1.1559e+01	k_t
Generation 1970: Best = 9.962582e+02 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.1549e+01	k_t
Generation 1980: Best = 9.947462e+02 = [3, 3, 2, 2]	FES = 198100	Diversity = 1.1510e+01	k_t
Generation 1990: Best = 9.947462e+02 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.1463e+01	k_t
Generation 2000: Best = 9.926752e+02 = [2, 2, 4, 2]	FES = 200100	Diversity = 1.1485e+01	k_t
Generation 2010: Best = 9.926752e+02 = [1, 1, 1, 1]	FES = 201100	Diversity = 1.1466e+01	k_t
Generation 2020: Best = 9.791222e+02 = [2, 3, 2, 3]	FES = 202100	Diversity = 1.1452e+01	k_t
Generation 2030: Best = 9.791222e+02 = [1, 1, 1, 1]	FES = 203100	Diversity = 1.1378e+01	k_t
Generation 2040: Best = 9.668846e+02 = [3, 4, 3, 4]	FES = 204100	Diversity = 1.1304e+01	k_t
Generation 2050: Best = 9.668846e+02 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.1310e+01	k_t
Generation 2060: Best = 9.642303e+02 = [2, 4, 3, 3]	FES = 206100	Diversity = 1.1278e+01	k_t
Generation 2070: Best = 9.642303e+02 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.1255e+01	k_t
Generation 2080: Best = 9.569426e+02 = [3, 4, 3, 2]	FES = 208100	Diversity = 1.1253e+01	k_t
Generation 2090: Best = 9.567406e+02 = [1, 1, 1, 1]	FES = 209100	Diversity = 1.1232e+01	k_t
Generation 2100: Best = 9.455330e+02 = [4, 2, 3, 2]	FES = 210100	Diversity = 1.1161e+01	k_t
Generation 2110: Best = 9.391888e+02 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.1150e+01	k_t

Generation 2120: Best = 9.385722e+02	FES = 212100	Diversity = 1.1031e+01	k_t
= [4, 4, 4, 3]			
Generation 2130: Best = 9.385722e+02	FES = 213100	Diversity = 1.1016e+01	k_t
= [1, 1, 1, 1]			
Generation 2140: Best = 9.329105e+02	FES = 214100	Diversity = 1.0910e+01	k_t
= [2, 2, 3, 4]			
Generation 2150: Best = 9.225883e+02	FES = 215100	Diversity = 1.0830e+01	k_t
= [1, 1, 1, 1]			
Generation 2160: Best = 9.158756e+02	FES = 216100	Diversity = 1.0730e+01	k_t
= [4, 3, 4, 2]			
Generation 2170: Best = 9.133696e+02	FES = 217100	Diversity = 1.0668e+01	k_t
= [1, 1, 1, 1]			
Generation 2180: Best = 9.133696e+02	FES = 218100	Diversity = 1.0609e+01	k_t
= [4, 4, 2, 2]			
Generation 2190: Best = 9.133696e+02	FES = 219100	Diversity = 1.0443e+01	k_t
= [1, 1, 1, 1]			
Generation 2200: Best = 9.133696e+02	FES = 220100	Diversity = 1.0086e+01	k_t
= [4, 2, 3, 3]			
Generation 2210: Best = 9.133696e+02	FES = 221100	Diversity = 1.0030e+01	k_t
= [1, 1, 1, 1]			
Generation 2220: Best = 9.133696e+02	FES = 222100	Diversity = 9.8253e+00	k_t
= [4, 2, 4, 4]			
Generation 2230: Best = 9.067676e+02	FES = 223100	Diversity = 9.6584e+00	k_t
= [1, 1, 1, 1]			
Generation 2240: Best = 9.020909e+02	FES = 224100	Diversity = 9.2292e+00	k_t
= [4, 3, 3, 2]			
Generation 2250: Best = 9.020909e+02	FES = 225100	Diversity = 8.6913e+00	k_t
= [1, 1, 1, 1]			
Generation 2260: Best = 8.996930e+02	FES = 226100	Diversity = 7.6677e+00	k_t
= [3, 3, 3, 4]			
Generation 2270: Best = 8.928393e+02	FES = 227100	Diversity = 6.9652e+00	k_t
= [1, 1, 1, 1]			
Generation 2280: Best = 8.733724e+02	FES = 228100	Diversity = 6.6306e+00	k_t
= [4, 4, 3, 3]			
Generation 2290: Best = 8.643214e+02	FES = 229100	Diversity = 6.2929e+00	k_t
= [1, 1, 1, 1]			
Generation 2300: Best = 8.550023e+02	FES = 230100	Diversity = 5.9585e+00	k_t
= [4, 2, 3, 2]			
Generation 2310: Best = 8.462553e+02	FES = 231100	Diversity = 5.6248e+00	k_t
= [1, 1, 1, 1]			
Generation 2320: Best = 8.324136e+02	FES = 232100	Diversity = 5.1665e+00	k_t
= [3, 3, 2, 4]			
Generation 2330: Best = 8.263974e+02	FES = 233100	Diversity = 4.2086e+00	k_t
= [1, 1, 1, 1]			
Generation 2340: Best = 8.185847e+02	FES = 234100	Diversity = 3.0642e+00	k_t
= [3, 3, 4, 4]			
Generation 2350: Best = 8.095481e+02	FES = 235100	Diversity = 2.4023e+00	k_t
= [1, 1, 1, 1]			
Generation 2360: Best = 8.014029e+02	FES = 236100	Diversity = 1.7166e+00	k_t
= [4, 2, 2, 2]			
Generation 2370: Best = 7.952222e+02	FES = 237100	Diversity = 3.5713e-01	k_t
= [1, 1, 1, 1]			
Generation 2380: Best = 7.918975e+02	FES = 238100	Diversity = 2.7465e-01	k_t
= [2, 2, 3, 4]			

Generation 2390: Best = 7.888055e+02	FES = 239100	Diversity = 2.2035e-01	k_t
= [1, 1, 1, 1]			
Generation 2400: Best = 7.868067e+02	FES = 240100	Diversity = 1.7686e-01	k_t
= [2, 4, 4, 4]			
Generation 2410: Best = 7.854160e+02	FES = 241100	Diversity = 1.4360e-01	k_t
= [1, 1, 1, 1]			
Generation 2420: Best = 7.843507e+02	FES = 242100	Diversity = 1.1143e-01	k_t
= [3, 4, 2, 4]			
Generation 2430: Best = 7.838380e+02	FES = 243100	Diversity = 9.1101e-02	k_t
= [1, 1, 1, 1]			
Generation 2440: Best = 7.832622e+02	FES = 244100	Diversity = 8.1488e-02	k_t
= [3, 4, 2, 3]			
Generation 2450: Best = 7.827544e+02	FES = 245100	Diversity = 7.1564e-02	k_t
= [1, 1, 1, 1]			
Generation 2460: Best = 7.822371e+02	FES = 246100	Diversity = 7.0995e-02	k_t
= [3, 2, 4, 2]			
Generation 2470: Best = 7.819846e+02	FES = 247100	Diversity = 6.5484e-02	k_t
= [1, 1, 1, 1]			
Generation 2480: Best = 7.817208e+02	FES = 248100	Diversity = 5.6921e-02	k_t
= [4, 4, 3, 3]			
Generation 2490: Best = 7.815528e+02	FES = 249100	Diversity = 4.7862e-02	k_t
= [1, 1, 1, 1]			
Generation 2500: Best = 7.813776e+02	FES = 250100	Diversity = 4.3088e-02	k_t
= [2, 2, 4, 2]			
Generation 2510: Best = 7.812173e+02	FES = 251100	Diversity = 3.8512e-02	k_t
= [1, 1, 1, 1]			
Generation 2520: Best = 7.811356e+02	FES = 252100	Diversity = 3.4275e-02	k_t
= [3, 3, 3, 2]			
Generation 2530: Best = 7.810409e+02	FES = 253100	Diversity = 3.0276e-02	k_t
= [1, 1, 1, 1]			
Generation 2540: Best = 7.809887e+02	FES = 254100	Diversity = 2.7776e-02	k_t
= [4, 4, 2, 2]			
Generation 2550: Best = 7.809121e+02	FES = 255100	Diversity = 2.6451e-02	k_t
= [1, 1, 1, 1]			
Generation 2560: Best = 7.808454e+02	FES = 256100	Diversity = 2.3297e-02	k_t
= [2, 2, 2, 2]			
Generation 2570: Best = 7.808052e+02	FES = 257100	Diversity = 2.1458e-02	k_t
= [1, 1, 1, 1]			
Generation 2580: Best = 7.807661e+02	FES = 258100	Diversity = 2.0500e-02	k_t
= [2, 4, 3, 3]			
Generation 2590: Best = 7.807377e+02	FES = 259100	Diversity = 1.9917e-02	k_t
= [1, 1, 1, 1]			
Generation 2600: Best = 7.806899e+02	FES = 260100	Diversity = 1.8410e-02	k_t
= [3, 3, 3, 3]			
Generation 2610: Best = 7.806671e+02	FES = 261100	Diversity = 1.7450e-02	k_t
= [1, 1, 1, 1]			
Generation 2620: Best = 7.806481e+02	FES = 262100	Diversity = 1.6835e-02	k_t
= [4, 3, 4, 2]			
Generation 2630: Best = 7.806329e+02	FES = 263100	Diversity = 1.3865e-02	k_t
= [1, 1, 1, 1]			
Generation 2640: Best = 7.806209e+02	FES = 264100	Diversity = 1.1213e-02	k_t
= [3, 4, 2, 4]			
Generation 2650: Best = 7.806162e+02	FES = 265100	Diversity = 9.9116e-03	k_t
= [1, 1, 1, 1]			

Generation 2660: Best = 7.806085e+02 = [2, 2, 3, 4]	FES = 266100	Diversity = 8.1434e-03	k_t
Generation 2670: Best = 7.806031e+02 = [1, 1, 1, 1]	FES = 267100	Diversity = 7.3921e-03	k_t
Generation 2680: Best = 7.805976e+02 = [4, 4, 4, 2]	FES = 268100	Diversity = 6.9149e-03	k_t
Generation 2690: Best = 7.805945e+02 = [1, 1, 1, 1]	FES = 269100	Diversity = 6.1731e-03	k_t
Generation 2700: Best = 7.805923e+02 = [4, 3, 2, 3]	FES = 270100	Diversity = 5.2418e-03	k_t
Generation 2710: Best = 7.805899e+02 = [1, 1, 1, 1]	FES = 271100	Diversity = 4.4565e-03	k_t
Generation 2720: Best = 7.805879e+02 = [2, 4, 2, 4]	FES = 272100	Diversity = 4.1297e-03	k_t
Generation 2730: Best = 7.805859e+02 = [1, 1, 1, 1]	FES = 273100	Diversity = 3.9563e-03	k_t
Generation 2740: Best = 7.805842e+02 = [4, 4, 3, 2]	FES = 274100	Diversity = 3.7915e-03	k_t
Generation 2750: Best = 7.805829e+02 = [1, 1, 1, 1]	FES = 275100	Diversity = 3.4827e-03	k_t
Generation 2760: Best = 7.805820e+02 = [4, 4, 4, 4]	FES = 276100	Diversity = 3.1634e-03	k_t
Generation 2770: Best = 7.805811e+02 = [1, 1, 1, 1]	FES = 277100	Diversity = 2.7502e-03	k_t
Generation 2780: Best = 7.805805e+02 = [4, 4, 3, 3]	FES = 278100	Diversity = 2.5633e-03	k_t
Generation 2790: Best = 7.805799e+02 = [1, 1, 1, 1]	FES = 279100	Diversity = 2.2375e-03	k_t
Generation 2800: Best = 7.805794e+02 = [4, 3, 3, 2]	FES = 280100	Diversity = 2.0512e-03	k_t
Generation 2810: Best = 7.805790e+02 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.7961e-03	k_t
Generation 2820: Best = 7.805787e+02 = [3, 2, 3, 2]	FES = 282100	Diversity = 1.6732e-03	k_t
Generation 2830: Best = 7.805785e+02 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.4716e-03	k_t
Generation 2840: Best = 7.805782e+02 = [2, 3, 4, 3]	FES = 284100	Diversity = 1.3566e-03	k_t
Generation 2850: Best = 7.805781e+02 = [1, 1, 1, 1]	FES = 285100	Diversity = 1.1894e-03	k_t
Generation 2860: Best = 7.805780e+02 = [3, 3, 4, 4]	FES = 286100	Diversity = 1.0673e-03	k_t
Generation 2870: Best = 7.805779e+02 = [1, 1, 1, 1]	FES = 287100	Diversity = 9.8044e-04	k_t
Generation 2880: Best = 7.805778e+02 = [4, 2, 4, 3]	FES = 288100	Diversity = 8.4299e-04	k_t
Generation 2890: Best = 7.805778e+02 = [1, 1, 1, 1]	FES = 289100	Diversity = 7.0628e-04	k_t
Generation 2900: Best = 7.805777e+02 = [2, 2, 2, 3]	FES = 290100	Diversity = 6.1483e-04	k_t
Generation 2910: Best = 7.805777e+02 = [1, 1, 1, 1]	FES = 291100	Diversity = 5.9613e-04	k_t
Generation 2920: Best = 7.805776e+02 = [4, 4, 3, 4]	FES = 292100	Diversity = 6.1534e-04	k_t

Generation 2930: Best = 7.805776e+02 = [1, 1, 1, 1]	FES = 293100	Diversity = 5.6802e-04	k_t
Generation 2940: Best = 7.805775e+02 = [3, 2, 3, 3]	FES = 294100	Diversity = 5.3817e-04	k_t
Generation 2950: Best = 7.805775e+02 = [1, 1, 1, 1]	FES = 295100	Diversity = 5.0196e-04	k_t
Generation 2960: Best = 7.805775e+02 = [3, 2, 2, 4]	FES = 296100	Diversity = 4.6293e-04	k_t
Generation 2970: Best = 7.805775e+02 = [1, 1, 1, 1]	FES = 297100	Diversity = 3.9655e-04	k_t
Generation 2980: Best = 7.805775e+02 = [4, 4, 4, 4]	FES = 298100	Diversity = 3.6025e-04	k_t
Generation 2990: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 299100	Diversity = 3.2217e-04	k_t
Generation 3000: Best = 7.805774e+02 = [2, 4, 3, 3]	FES = 300100	Diversity = 2.8218e-04	k_t
Generation 3010: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 301100	Diversity = 2.3807e-04	k_t
Generation 3020: Best = 7.805774e+02 = [2, 2, 3, 3]	FES = 302100	Diversity = 1.9895e-04	k_t
Generation 3030: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 303100	Diversity = 1.6812e-04	k_t
Generation 3040: Best = 7.805774e+02 = [4, 3, 4, 4]	FES = 304100	Diversity = 1.4409e-04	k_t
Generation 3050: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 305100	Diversity = 1.3026e-04	k_t
Generation 3060: Best = 7.805774e+02 = [3, 3, 2, 2]	FES = 306100	Diversity = 1.2333e-04	k_t
Generation 3070: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 307100	Diversity = 1.1547e-04	k_t
Generation 3080: Best = 7.805774e+02 = [2, 4, 2, 4]	FES = 308100	Diversity = 1.0954e-04	k_t
Generation 3090: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 309100	Diversity = 1.0316e-04	k_t
Generation 3100: Best = 7.805774e+02 = [4, 4, 2, 4]	FES = 310100	Diversity = 9.6179e-05	k_t
Generation 3110: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 311100	Diversity = 8.3870e-05	k_t
Generation 3120: Best = 7.805774e+02 = [4, 4, 3, 2]	FES = 312100	Diversity = 7.8382e-05	k_t
Generation 3130: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 313100	Diversity = 7.4895e-05	k_t
Generation 3140: Best = 7.805774e+02 = [4, 3, 4, 4]	FES = 314100	Diversity = 6.8169e-05	k_t
Generation 3150: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 315100	Diversity = 6.5078e-05	k_t
Generation 3160: Best = 7.805774e+02 = [4, 3, 3, 2]	FES = 316100	Diversity = 5.8855e-05	k_t
Generation 3170: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 317100	Diversity = 5.2564e-05	k_t
Generation 3180: Best = 7.805774e+02 = [4, 2, 4, 2]	FES = 318100	Diversity = 4.7691e-05	k_t
Generation 3190: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 319100	Diversity = 4.2130e-05	k_t

Generation 3200: Best = 7.805774e+02 = [4, 2, 2, 4]	FES = 320100	Diversity = 3.8093e-05	k_t
Generation 3210: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 321100	Diversity = 3.5666e-05	k_t
Generation 3220: Best = 7.805774e+02 = [2, 4, 2, 3]	FES = 322100	Diversity = 3.2661e-05	k_t
Generation 3230: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 323100	Diversity = 2.8206e-05	k_t
Generation 3240: Best = 7.805774e+02 = [4, 2, 4, 3]	FES = 324100	Diversity = 2.5615e-05	k_t
Generation 3250: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 325100	Diversity = 2.5196e-05	k_t
Generation 3260: Best = 7.805774e+02 = [3, 4, 2, 4]	FES = 326100	Diversity = 2.1447e-05	k_t
Generation 3270: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 327100	Diversity = 1.7996e-05	k_t
Generation 3280: Best = 7.805774e+02 = [3, 3, 2, 4]	FES = 328100	Diversity = 1.7627e-05	k_t
Generation 3290: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 329100	Diversity = 1.5622e-05	k_t
Generation 3300: Best = 7.805774e+02 = [4, 3, 3, 2]	FES = 330100	Diversity = 1.3477e-05	k_t
Generation 3310: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 331100	Diversity = 1.1296e-05	k_t
Generation 3320: Best = 7.805774e+02 = [2, 3, 3, 3]	FES = 332100	Diversity = 1.0264e-05	k_t
Generation 3330: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 333100	Diversity = 9.2045e-06	k_t
Generation 3340: Best = 7.805774e+02 = [4, 3, 2, 3]	FES = 334100	Diversity = 8.3682e-06	k_t
Generation 3350: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 335100	Diversity = 7.0920e-06	k_t
Generation 3360: Best = 7.805774e+02 = [4, 3, 2, 3]	FES = 336100	Diversity = 6.2986e-06	k_t
Generation 3370: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 337100	Diversity = 5.6718e-06	k_t
Generation 3380: Best = 7.805774e+02 = [4, 4, 3, 2]	FES = 338100	Diversity = 5.5104e-06	k_t
Generation 3390: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 339100	Diversity = 5.2804e-06	k_t
Generation 3400: Best = 7.805774e+02 = [3, 2, 2, 4]	FES = 340100	Diversity = 4.6067e-06	k_t
Generation 3410: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 341100	Diversity = 4.0425e-06	k_t
Generation 3420: Best = 7.805774e+02 = [4, 3, 4, 4]	FES = 342100	Diversity = 3.6446e-06	k_t
Generation 3430: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 343100	Diversity = 3.4473e-06	k_t
Generation 3440: Best = 7.805774e+02 = [2, 4, 4, 4]	FES = 344100	Diversity = 3.1193e-06	k_t
Generation 3450: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 345100	Diversity = 2.8560e-06	k_t
Generation 3460: Best = 7.805774e+02 = [4, 3, 4, 2]	FES = 346100	Diversity = 2.6579e-06	k_t

Generation 3470: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 347100	Diversity = 2.4335e-06	k_t
Generation 3480: Best = 7.805774e+02 = [4, 4, 2, 2]	FES = 348100	Diversity = 2.3670e-06	k_t
Generation 3490: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 349100	Diversity = 2.2232e-06	k_t
Generation 3500: Best = 7.805774e+02 = [3, 4, 2, 3]	FES = 350100	Diversity = 2.0228e-06	k_t
Generation 3510: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 351100	Diversity = 1.8694e-06	k_t
Generation 3520: Best = 7.805774e+02 = [3, 2, 4, 4]	FES = 352100	Diversity = 1.6231e-06	k_t
Generation 3530: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 353100	Diversity = 1.3539e-06	k_t
Generation 3540: Best = 7.805774e+02 = [2, 3, 2, 2]	FES = 354100	Diversity = 1.1664e-06	k_t
Generation 3550: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 355100	Diversity = 1.0428e-06	k_t
Generation 3560: Best = 7.805774e+02 = [2, 3, 3, 4]	FES = 356100	Diversity = 9.6147e-07	k_t
Generation 3570: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 357100	Diversity = 8.4678e-07	k_t
Generation 3580: Best = 7.805774e+02 = [3, 4, 3, 3]	FES = 358100	Diversity = 7.6617e-07	k_t
Generation 3590: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 359100	Diversity = 7.2888e-07	k_t
Generation 3600: Best = 7.805774e+02 = [4, 2, 4, 4]	FES = 360100	Diversity = 6.9999e-07	k_t
Generation 3610: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 361100	Diversity = 6.0469e-07	k_t
Generation 3620: Best = 7.805774e+02 = [2, 2, 2, 3]	FES = 362100	Diversity = 5.6533e-07	k_t
Generation 3630: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 363100	Diversity = 5.1296e-07	k_t
Generation 3640: Best = 7.805774e+02 = [4, 3, 4, 4]	FES = 364100	Diversity = 4.7333e-07	k_t
Generation 3650: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 365100	Diversity = 3.9689e-07	k_t
Generation 3660: Best = 7.805774e+02 = [4, 2, 4, 3]	FES = 366100	Diversity = 3.4732e-07	k_t
Generation 3670: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 367100	Diversity = 2.9956e-07	k_t
Generation 3680: Best = 7.805774e+02 = [2, 3, 2, 2]	FES = 368100	Diversity = 2.4976e-07	k_t
Generation 3690: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 369100	Diversity = 2.3976e-07	k_t
Generation 3700: Best = 7.805774e+02 = [2, 3, 3, 4]	FES = 370100	Diversity = 2.5707e-07	k_t
Generation 3710: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 371100	Diversity = 2.6212e-07	k_t
Generation 3720: Best = 7.805774e+02 = [2, 2, 3, 4]	FES = 372100	Diversity = 2.6208e-07	k_t
Generation 3730: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 373100	Diversity = 2.4582e-07	k_t

Generation 3740: Best = 7.805774e+02 = [4, 3, 2, 3]	FES = 374100	Diversity = 2.2252e-07	k_t
Generation 3750: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 375100	Diversity = 2.0463e-07	k_t
Generation 3760: Best = 7.805774e+02 = [3, 4, 4, 2]	FES = 376100	Diversity = 1.7435e-07	k_t
Generation 3770: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 377100	Diversity = 1.5287e-07	k_t
Generation 3780: Best = 7.805774e+02 = [3, 3, 2, 3]	FES = 378100	Diversity = 1.2298e-07	k_t
Generation 3790: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 379100	Diversity = 1.0709e-07	k_t
Generation 3800: Best = 7.805774e+02 = [2, 2, 3, 4]	FES = 380100	Diversity = 9.5872e-08	k_t
Generation 3810: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 381100	Diversity = 8.6159e-08	k_t
Generation 3820: Best = 7.805774e+02 = [3, 2, 3, 3]	FES = 382100	Diversity = 8.3121e-08	k_t
Generation 3830: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 383100	Diversity = 7.7665e-08	k_t
Generation 3840: Best = 7.805774e+02 = [4, 4, 2, 4]	FES = 384100	Diversity = 7.7111e-08	k_t
Generation 3850: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 385100	Diversity = 7.2859e-08	k_t
Generation 3860: Best = 7.805774e+02 = [2, 3, 4, 3]	FES = 386100	Diversity = 6.8356e-08	k_t
Generation 3870: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 387100	Diversity = 6.2186e-08	k_t
Generation 3880: Best = 7.805774e+02 = [4, 2, 2, 3]	FES = 388100	Diversity = 6.3433e-08	k_t
Generation 3890: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 389100	Diversity = 6.1761e-08	k_t
Generation 3900: Best = 7.805774e+02 = [2, 4, 3, 3]	FES = 390100	Diversity = 5.6084e-08	k_t
Generation 3910: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 391100	Diversity = 5.7252e-08	k_t
Generation 3920: Best = 7.805774e+02 = [4, 3, 2, 4]	FES = 392100	Diversity = 5.4295e-08	k_t
Generation 3930: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 393100	Diversity = 5.2115e-08	k_t
Generation 3940: Best = 7.805774e+02 = [3, 3, 2, 2]	FES = 394100	Diversity = 5.3701e-08	k_t
Generation 3950: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 395100	Diversity = 5.4077e-08	k_t
Generation 3960: Best = 7.805774e+02 = [2, 2, 2, 2]	FES = 396100	Diversity = 5.5806e-08	k_t
Generation 3970: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 397100	Diversity = 5.0735e-08	k_t
Generation 3980: Best = 7.805774e+02 = [4, 4, 3, 4]	FES = 398100	Diversity = 4.9489e-08	k_t
Generation 3990: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 399100	Diversity = 5.0450e-08	k_t
Generation 4000: Best = 7.805774e+02 = [3, 3, 2, 4]	FES = 400100	Diversity = 4.9671e-08	k_t

Generation 4010: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 401100	Diversity = 5.1266e-08	k_t
Generation 4020: Best = 7.805774e+02 = [2, 4, 4, 4]	FES = 402100	Diversity = 5.2114e-08	k_t
Generation 4030: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 403100	Diversity = 5.3266e-08	k_t
Generation 4040: Best = 7.805774e+02 = [2, 4, 4, 2]	FES = 404100	Diversity = 5.1726e-08	k_t
Generation 4050: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 405100	Diversity = 4.9353e-08	k_t
Generation 4060: Best = 7.805774e+02 = [3, 2, 4, 3]	FES = 406100	Diversity = 4.8497e-08	k_t
Generation 4070: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 407100	Diversity = 4.4815e-08	k_t
Generation 4080: Best = 7.805774e+02 = [2, 2, 2, 3]	FES = 408100	Diversity = 4.1663e-08	k_t
Generation 4090: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 409100	Diversity = 4.0996e-08	k_t
Generation 4100: Best = 7.805774e+02 = [3, 4, 3, 4]	FES = 410100	Diversity = 4.0295e-08	k_t
Generation 4110: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 411100	Diversity = 4.0283e-08	k_t
Generation 4120: Best = 7.805774e+02 = [3, 2, 2, 4]	FES = 412100	Diversity = 4.0014e-08	k_t
Generation 4130: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 413100	Diversity = 4.0660e-08	k_t
Generation 4140: Best = 7.805774e+02 = [2, 4, 4, 3]	FES = 414100	Diversity = 4.1517e-08	k_t
Generation 4150: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 415100	Diversity = 4.1512e-08	k_t
Generation 4160: Best = 7.805774e+02 = [4, 2, 2, 4]	FES = 416100	Diversity = 4.1871e-08	k_t
Generation 4170: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 417100	Diversity = 4.1222e-08	k_t
Generation 4180: Best = 7.805774e+02 = [2, 2, 3, 3]	FES = 418100	Diversity = 4.1701e-08	k_t
Generation 4190: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 419100	Diversity = 4.2117e-08	k_t
Generation 4200: Best = 7.805774e+02 = [2, 3, 4, 2]	FES = 420100	Diversity = 4.2359e-08	k_t
Generation 4210: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 421100	Diversity = 4.2494e-08	k_t
Generation 4220: Best = 7.805774e+02 = [4, 3, 3, 4]	FES = 422100	Diversity = 4.2349e-08	k_t
Generation 4230: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 423100	Diversity = 4.3813e-08	k_t
Generation 4240: Best = 7.805774e+02 = [3, 3, 3, 2]	FES = 424100	Diversity = 4.3738e-08	k_t
Generation 4250: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 425100	Diversity = 4.4003e-08	k_t
Generation 4260: Best = 7.805774e+02 = [4, 2, 2, 2]	FES = 426100	Diversity = 4.5476e-08	k_t
Generation 4270: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 427100	Diversity = 4.6118e-08	k_t

Generation 4280: Best = 7.805774e+02	FES = 428100	Diversity = 4.7628e-08	k_t
= [3, 2, 3, 2]			
Generation 4290: Best = 7.805774e+02	FES = 429100	Diversity = 4.8927e-08	k_t
= [1, 1, 1, 1]			
Generation 4300: Best = 7.805774e+02	FES = 430100	Diversity = 5.0123e-08	k_t
= [2, 4, 3, 3]			
Generation 4310: Best = 7.805774e+02	FES = 431100	Diversity = 5.0459e-08	k_t
= [1, 1, 1, 1]			
Generation 4320: Best = 7.805774e+02	FES = 432100	Diversity = 5.1940e-08	k_t
= [2, 2, 4, 2]			
Generation 4330: Best = 7.805774e+02	FES = 433100	Diversity = 5.1651e-08	k_t
= [1, 1, 1, 1]			
Generation 4340: Best = 7.805774e+02	FES = 434100	Diversity = 5.1049e-08	k_t
= [2, 3, 2, 3]			
Generation 4350: Best = 7.805774e+02	FES = 435100	Diversity = 5.0527e-08	k_t
= [1, 1, 1, 1]			
Generation 4360: Best = 7.805774e+02	FES = 436100	Diversity = 4.9489e-08	k_t
= [3, 4, 2, 3]			
Generation 4370: Best = 7.805774e+02	FES = 437100	Diversity = 4.8581e-08	k_t
= [1, 1, 1, 1]			
Generation 4380: Best = 7.805774e+02	FES = 438100	Diversity = 4.8439e-08	k_t
= [3, 2, 4, 4]			
Generation 4390: Best = 7.805774e+02	FES = 439100	Diversity = 4.8876e-08	k_t
= [1, 1, 1, 1]			
Generation 4400: Best = 7.805774e+02	FES = 440100	Diversity = 5.0202e-08	k_t
= [4, 4, 4, 4]			
Generation 4410: Best = 7.805774e+02	FES = 441100	Diversity = 4.9781e-08	k_t
= [1, 1, 1, 1]			
Generation 4420: Best = 7.805774e+02	FES = 442100	Diversity = 5.0275e-08	k_t
= [2, 3, 2, 3]			
Generation 4430: Best = 7.805774e+02	FES = 443100	Diversity = 5.1413e-08	k_t
= [1, 1, 1, 1]			
Generation 4440: Best = 7.805774e+02	FES = 444100	Diversity = 5.1645e-08	k_t
= [3, 2, 3, 4]			
Generation 4450: Best = 7.805774e+02	FES = 445100	Diversity = 5.1122e-08	k_t
= [1, 1, 1, 1]			
Generation 4460: Best = 7.805774e+02	FES = 446100	Diversity = 5.1584e-08	k_t
= [4, 2, 3, 2]			
Generation 4470: Best = 7.805774e+02	FES = 447100	Diversity = 5.0822e-08	k_t
= [1, 1, 1, 1]			
Generation 4480: Best = 7.805774e+02	FES = 448100	Diversity = 4.9697e-08	k_t
= [3, 3, 3, 4]			
Generation 4490: Best = 7.805774e+02	FES = 449100	Diversity = 5.0224e-08	k_t
= [1, 1, 1, 1]			
Generation 4500: Best = 7.805774e+02	FES = 450100	Diversity = 5.0682e-08	k_t
= [2, 4, 2, 4]			
Generation 4510: Best = 7.805774e+02	FES = 451100	Diversity = 5.0463e-08	k_t
= [1, 1, 1, 1]			
Generation 4520: Best = 7.805774e+02	FES = 452100	Diversity = 4.9567e-08	k_t
= [2, 2, 2, 4]			
Generation 4530: Best = 7.805774e+02	FES = 453100	Diversity = 5.0543e-08	k_t
= [1, 1, 1, 1]			
Generation 4540: Best = 7.805774e+02	FES = 454100	Diversity = 5.0480e-08	k_t
= [2, 3, 2, 4]			

Generation 4550: Best = 7.805774e+02	FES = 455100	Diversity = 5.1019e-08	k_t
= [1, 1, 1, 1]			
Generation 4560: Best = 7.805774e+02	FES = 456100	Diversity = 5.1014e-08	k_t
= [3, 2, 3, 4]			
Generation 4570: Best = 7.805774e+02	FES = 457100	Diversity = 5.1067e-08	k_t
= [1, 1, 1, 1]			
Generation 4580: Best = 7.805774e+02	FES = 458100	Diversity = 5.0870e-08	k_t
= [3, 3, 2, 2]			
Generation 4590: Best = 7.805774e+02	FES = 459100	Diversity = 5.1738e-08	k_t
= [1, 1, 1, 1]			
Generation 4600: Best = 7.805774e+02	FES = 460100	Diversity = 5.2639e-08	k_t
= [2, 4, 4, 2]			
Generation 4610: Best = 7.805774e+02	FES = 461100	Diversity = 5.1317e-08	k_t
= [1, 1, 1, 1]			
Generation 4620: Best = 7.805774e+02	FES = 462100	Diversity = 5.0966e-08	k_t
= [4, 2, 2, 3]			
Generation 4630: Best = 7.805774e+02	FES = 463100	Diversity = 5.1957e-08	k_t
= [1, 1, 1, 1]			
Generation 4640: Best = 7.805774e+02	FES = 464100	Diversity = 5.2292e-08	k_t
= [2, 2, 3, 3]			
Generation 4650: Best = 7.805774e+02	FES = 465100	Diversity = 5.2572e-08	k_t
= [1, 1, 1, 1]			
Generation 4660: Best = 7.805774e+02	FES = 466100	Diversity = 5.1659e-08	k_t
= [2, 4, 4, 3]			
Generation 4670: Best = 7.805774e+02	FES = 467100	Diversity = 5.1034e-08	k_t
= [1, 1, 1, 1]			
Generation 4680: Best = 7.805774e+02	FES = 468100	Diversity = 5.0692e-08	k_t
= [4, 4, 4, 2]			
Generation 4690: Best = 7.805774e+02	FES = 469100	Diversity = 4.9034e-08	k_t
= [1, 1, 1, 1]			
Generation 4700: Best = 7.805774e+02	FES = 470100	Diversity = 4.5345e-08	k_t
= [3, 2, 2, 2]			
Generation 4710: Best = 7.805774e+02	FES = 471100	Diversity = 4.2696e-08	k_t
= [1, 1, 1, 1]			
Generation 4720: Best = 7.805774e+02	FES = 472100	Diversity = 4.2070e-08	k_t
= [4, 4, 2, 2]			
Generation 4730: Best = 7.805774e+02	FES = 473100	Diversity = 3.8979e-08	k_t
= [1, 1, 1, 1]			
Generation 4740: Best = 7.805774e+02	FES = 474100	Diversity = 3.6591e-08	k_t
= [3, 2, 4, 3]			
Generation 4750: Best = 7.805774e+02	FES = 475100	Diversity = 3.5093e-08	k_t
= [1, 1, 1, 1]			
Generation 4760: Best = 7.805774e+02	FES = 476100	Diversity = 3.4070e-08	k_t
= [3, 3, 4, 4]			
Generation 4770: Best = 7.805774e+02	FES = 477100	Diversity = 3.3513e-08	k_t
= [1, 1, 1, 1]			
Generation 4780: Best = 7.805774e+02	FES = 478100	Diversity = 3.3391e-08	k_t
= [3, 3, 3, 4]			
Generation 4790: Best = 7.805774e+02	FES = 479100	Diversity = 3.3542e-08	k_t
= [1, 1, 1, 1]			
Generation 4800: Best = 7.805774e+02	FES = 480100	Diversity = 3.3389e-08	k_t
= [2, 4, 4, 3]			
Generation 4810: Best = 7.805774e+02	FES = 481100	Diversity = 3.3431e-08	k_t
= [1, 1, 1, 1]			

Generation 4820: Best = 7.805774e+02 = [3, 3, 3, 4]	FES = 482100	Diversity = 3.3172e-08	k_t
Generation 4830: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 483100	Diversity = 3.3851e-08	k_t
Generation 4840: Best = 7.805774e+02 = [4, 4, 4, 3]	FES = 484100	Diversity = 3.3457e-08	k_t
Generation 4850: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 485100	Diversity = 3.3491e-08	k_t
Generation 4860: Best = 7.805774e+02 = [4, 4, 4, 3]	FES = 486100	Diversity = 3.3435e-08	k_t
Generation 4870: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 487100	Diversity = 3.3636e-08	k_t
Generation 4880: Best = 7.805774e+02 = [3, 2, 2, 3]	FES = 488100	Diversity = 3.3965e-08	k_t
Generation 4890: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 489100	Diversity = 3.3403e-08	k_t
Generation 4900: Best = 7.805774e+02 = [3, 3, 3, 2]	FES = 490100	Diversity = 3.3061e-08	k_t
Generation 4910: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 491100	Diversity = 3.3649e-08	k_t
Generation 4920: Best = 7.805774e+02 = [3, 3, 3, 3]	FES = 492100	Diversity = 3.3602e-08	k_t
Generation 4930: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 493100	Diversity = 3.4358e-08	k_t
Generation 4940: Best = 7.805774e+02 = [3, 3, 2, 4]	FES = 494100	Diversity = 3.4095e-08	k_t
Generation 4950: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 495100	Diversity = 3.3406e-08	k_t
Generation 4960: Best = 7.805774e+02 = [3, 2, 3, 4]	FES = 496100	Diversity = 3.3654e-08	k_t
Generation 4970: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 497100	Diversity = 3.3701e-08	k_t
Generation 4980: Best = 7.805774e+02 = [4, 3, 3, 3]	FES = 498100	Diversity = 3.3669e-08	k_t
Generation 4990: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 499100	Diversity = 3.3696e-08	k_t
Generation 5000: Best = 7.805774e+02 = [3, 3, 3, 4]	FES = 500100	Diversity = 3.3610e-08	k_t
Generation 5010: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 501100	Diversity = 3.3628e-08	k_t
Generation 5020: Best = 7.805774e+02 = [4, 2, 3, 4]	FES = 502100	Diversity = 3.4134e-08	k_t
Generation 5030: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 503100	Diversity = 3.4424e-08	k_t
Generation 5040: Best = 7.805774e+02 = [4, 2, 3, 2]	FES = 504100	Diversity = 3.4619e-08	k_t
Generation 5050: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 505100	Diversity = 3.4947e-08	k_t
Generation 5060: Best = 7.805774e+02 = [2, 3, 2, 4]	FES = 506100	Diversity = 3.4804e-08	k_t
Generation 5070: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 507100	Diversity = 3.4776e-08	k_t
Generation 5080: Best = 7.805774e+02 = [4, 3, 3, 3]	FES = 508100	Diversity = 3.4735e-08	k_t

Generation 5090: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 509100	Diversity = 3.4580e-08	k_t
Generation 5100: Best = 7.805774e+02 = [3, 4, 2, 2]	FES = 510100	Diversity = 3.4204e-08	k_t
Generation 5110: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 511100	Diversity = 3.4808e-08	k_t
Generation 5120: Best = 7.805774e+02 = [3, 4, 2, 3]	FES = 512100	Diversity = 3.4391e-08	k_t
Generation 5130: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 513100	Diversity = 3.3991e-08	k_t
Generation 5140: Best = 7.805774e+02 = [4, 2, 2, 4]	FES = 514100	Diversity = 3.4299e-08	k_t
Generation 5150: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 515100	Diversity = 3.3884e-08	k_t
Generation 5160: Best = 7.805774e+02 = [4, 4, 2, 3]	FES = 516100	Diversity = 3.3357e-08	k_t
Generation 5170: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 517100	Diversity = 3.3992e-08	k_t
Generation 5180: Best = 7.805774e+02 = [2, 2, 4, 2]	FES = 518100	Diversity = 3.3664e-08	k_t
Generation 5190: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 519100	Diversity = 3.4845e-08	k_t
Generation 5200: Best = 7.805774e+02 = [3, 2, 2, 4]	FES = 520100	Diversity = 3.4729e-08	k_t
Generation 5210: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 521100	Diversity = 3.5768e-08	k_t
Generation 5220: Best = 7.805774e+02 = [4, 3, 2, 2]	FES = 522100	Diversity = 3.5992e-08	k_t
Generation 5230: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 523100	Diversity = 3.6177e-08	k_t
Generation 5240: Best = 7.805774e+02 = [3, 4, 3, 2]	FES = 524100	Diversity = 3.5721e-08	k_t
Generation 5250: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 525100	Diversity = 3.5726e-08	k_t
Generation 5260: Best = 7.805774e+02 = [4, 4, 4, 2]	FES = 526100	Diversity = 3.6002e-08	k_t
Generation 5270: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 527100	Diversity = 3.5292e-08	k_t
Generation 5280: Best = 7.805774e+02 = [2, 4, 2, 3]	FES = 528100	Diversity = 3.4994e-08	k_t
Generation 5290: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 529100	Diversity = 3.4629e-08	k_t
Generation 5300: Best = 7.805774e+02 = [4, 3, 3, 4]	FES = 530100	Diversity = 3.4872e-08	k_t
Generation 5310: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 531100	Diversity = 3.4986e-08	k_t
Generation 5320: Best = 7.805774e+02 = [2, 3, 4, 3]	FES = 532100	Diversity = 3.4825e-08	k_t
Generation 5330: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 533100	Diversity = 3.4352e-08	k_t
Generation 5340: Best = 7.805774e+02 = [4, 4, 4, 3]	FES = 534100	Diversity = 3.5093e-08	k_t
Generation 5350: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 535100	Diversity = 3.5161e-08	k_t

Generation 5360: Best = 7.805774e+02 = [2, 4, 4, 3]	FES = 536100	Diversity = 3.5776e-08	k_t
Generation 5370: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 537100	Diversity = 3.5977e-08	k_t
Generation 5380: Best = 7.805774e+02 = [3, 4, 4, 2]	FES = 538100	Diversity = 3.5509e-08	k_t
Generation 5390: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 539100	Diversity = 3.4854e-08	k_t
Generation 5400: Best = 7.805774e+02 = [3, 4, 4, 3]	FES = 540100	Diversity = 3.4581e-08	k_t
Generation 5410: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 541100	Diversity = 3.4087e-08	k_t
Generation 5420: Best = 7.805774e+02 = [4, 2, 2, 4]	FES = 542100	Diversity = 3.3895e-08	k_t
Generation 5430: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 543100	Diversity = 3.3502e-08	k_t
Generation 5440: Best = 7.805774e+02 = [4, 3, 4, 3]	FES = 544100	Diversity = 3.3269e-08	k_t
Generation 5450: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 545100	Diversity = 3.3062e-08	k_t
Generation 5460: Best = 7.805774e+02 = [4, 3, 3, 4]	FES = 546100	Diversity = 3.3091e-08	k_t
Generation 5470: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 547100	Diversity = 3.3195e-08	k_t
Generation 5480: Best = 7.805774e+02 = [4, 2, 4, 4]	FES = 548100	Diversity = 3.2662e-08	k_t
Generation 5490: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 549100	Diversity = 3.2388e-08	k_t
Generation 5500: Best = 7.805774e+02 = [3, 4, 2, 4]	FES = 550100	Diversity = 3.2371e-08	k_t
Generation 5510: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 551100	Diversity = 3.1815e-08	k_t
Generation 5520: Best = 7.805774e+02 = [3, 4, 4, 4]	FES = 552100	Diversity = 3.2521e-08	k_t
Generation 5530: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 553100	Diversity = 3.2666e-08	k_t
Generation 5540: Best = 7.805774e+02 = [4, 3, 3, 2]	FES = 554100	Diversity = 3.2348e-08	k_t
Generation 5550: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 555100	Diversity = 3.2116e-08	k_t
Generation 5560: Best = 7.805774e+02 = [3, 3, 4, 4]	FES = 556100	Diversity = 3.2154e-08	k_t
Generation 5570: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 557100	Diversity = 3.2071e-08	k_t
Generation 5580: Best = 7.805774e+02 = [2, 3, 2, 2]	FES = 558100	Diversity = 3.2106e-08	k_t
Generation 5590: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 559100	Diversity = 3.2605e-08	k_t
Generation 5600: Best = 7.805774e+02 = [3, 4, 2, 3]	FES = 560100	Diversity = 3.2660e-08	k_t
Generation 5610: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 561100	Diversity = 3.2333e-08	k_t
Generation 5620: Best = 7.805774e+02 = [2, 3, 3, 4]	FES = 562100	Diversity = 3.2859e-08	k_t

Generation 5630: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 563100	Diversity = 3.3017e-08	k_t
Generation 5640: Best = 7.805774e+02 = [2, 2, 3, 3]	FES = 564100	Diversity = 3.2988e-08	k_t
Generation 5650: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 565100	Diversity = 3.3136e-08	k_t
Generation 5660: Best = 7.805774e+02 = [3, 2, 3, 2]	FES = 566100	Diversity = 3.3512e-08	k_t
Generation 5670: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 567100	Diversity = 3.3656e-08	k_t
Generation 5680: Best = 7.805774e+02 = [4, 2, 4, 4]	FES = 568100	Diversity = 3.3821e-08	k_t
Generation 5690: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 569100	Diversity = 3.3747e-08	k_t
Generation 5700: Best = 7.805774e+02 = [3, 4, 3, 2]	FES = 570100	Diversity = 3.4017e-08	k_t
Generation 5710: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 571100	Diversity = 3.3649e-08	k_t
Generation 5720: Best = 7.805774e+02 = [4, 4, 4, 4]	FES = 572100	Diversity = 3.4240e-08	k_t
Generation 5730: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 573100	Diversity = 3.4572e-08	k_t
Generation 5740: Best = 7.805774e+02 = [2, 2, 2, 3]	FES = 574100	Diversity = 3.4501e-08	k_t
Generation 5750: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 575100	Diversity = 3.4836e-08	k_t
Generation 5760: Best = 7.805774e+02 = [4, 2, 3, 3]	FES = 576100	Diversity = 3.4270e-08	k_t
Generation 5770: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 577100	Diversity = 3.3769e-08	k_t
Generation 5780: Best = 7.805774e+02 = [4, 4, 3, 4]	FES = 578100	Diversity = 3.3840e-08	k_t
Generation 5790: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 579100	Diversity = 3.4294e-08	k_t
Generation 5800: Best = 7.805774e+02 = [3, 2, 3, 3]	FES = 580100	Diversity = 3.4265e-08	k_t
Generation 5810: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 581100	Diversity = 3.3933e-08	k_t
Generation 5820: Best = 7.805774e+02 = [4, 3, 2, 3]	FES = 582100	Diversity = 3.4033e-08	k_t
Generation 5830: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 583100	Diversity = 3.4480e-08	k_t
Generation 5840: Best = 7.805774e+02 = [2, 3, 4, 3]	FES = 584100	Diversity = 3.4207e-08	k_t
Generation 5850: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 585100	Diversity = 3.4246e-08	k_t
Generation 5860: Best = 7.805774e+02 = [2, 2, 4, 3]	FES = 586100	Diversity = 3.4298e-08	k_t
Generation 5870: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 587100	Diversity = 3.3936e-08	k_t
Generation 5880: Best = 7.805774e+02 = [4, 2, 2, 2]	FES = 588100	Diversity = 3.4073e-08	k_t
Generation 5890: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 589100	Diversity = 3.3938e-08	k_t

Generation 5900: Best = 7.805774e+02 = [3, 2, 4, 3]	FES = 590100	Diversity = 3.3631e-08	k_t
Generation 5910: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 591100	Diversity = 3.4000e-08	k_t
Generation 5920: Best = 7.805774e+02 = [4, 2, 3, 3]	FES = 592100	Diversity = 3.3995e-08	k_t
Generation 5930: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 593100	Diversity = 3.3673e-08	k_t
Generation 5940: Best = 7.805774e+02 = [4, 3, 4, 2]	FES = 594100	Diversity = 3.4236e-08	k_t
Generation 5950: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 595100	Diversity = 3.4268e-08	k_t
Generation 5960: Best = 7.805774e+02 = [2, 4, 4, 4]	FES = 596100	Diversity = 3.3999e-08	k_t
Generation 5970: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 597100	Diversity = 3.4275e-08	k_t
Generation 5980: Best = 7.805774e+02 = [2, 2, 2, 2]	FES = 598100	Diversity = 3.4416e-08	k_t
Generation 5990: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 599100	Diversity = 3.4328e-08	k_t
Generation 6000: Best = 7.805774e+02 = [4, 3, 2, 2]	FES = 600100	Diversity = 3.3934e-08	k_t
Generation 6010: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 601100	Diversity = 3.3755e-08	k_t
Generation 6020: Best = 7.805774e+02 = [3, 2, 2, 4]	FES = 602100	Diversity = 3.3800e-08	k_t
Generation 6030: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 603100	Diversity = 3.3767e-08	k_t
Generation 6040: Best = 7.805774e+02 = [4, 3, 2, 4]	FES = 604100	Diversity = 3.3615e-08	k_t
Generation 6050: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 605100	Diversity = 3.3977e-08	k_t
Generation 6060: Best = 7.805774e+02 = [3, 4, 4, 2]	FES = 606100	Diversity = 3.4123e-08	k_t
Generation 6070: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 607100	Diversity = 3.4612e-08	k_t
Generation 6080: Best = 7.805774e+02 = [2, 2, 4, 2]	FES = 608100	Diversity = 3.4686e-08	k_t
Generation 6090: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 609100	Diversity = 3.4690e-08	k_t
Generation 6100: Best = 7.805774e+02 = [4, 2, 2, 2]	FES = 610100	Diversity = 3.5077e-08	k_t
Generation 6110: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 611100	Diversity = 3.6215e-08	k_t
Generation 6120: Best = 7.805774e+02 = [4, 3, 4, 4]	FES = 612100	Diversity = 3.5780e-08	k_t
Generation 6130: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 613100	Diversity = 3.5542e-08	k_t
Generation 6140: Best = 7.805774e+02 = [3, 2, 2, 4]	FES = 614100	Diversity = 3.5523e-08	k_t
Generation 6150: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 615100	Diversity = 3.6274e-08	k_t
Generation 6160: Best = 7.805774e+02 = [3, 3, 4, 3]	FES = 616100	Diversity = 3.6409e-08	k_t

Generation 6170: Best = 7.805774e+02	FES = 617100	Diversity = 3.6954e-08	k_t
= [1, 1, 1, 1]			
Generation 6180: Best = 7.805774e+02	FES = 618100	Diversity = 3.7428e-08	k_t
= [2, 4, 4, 2]			
Generation 6190: Best = 7.805774e+02	FES = 619100	Diversity = 3.7157e-08	k_t
= [1, 1, 1, 1]			
Generation 6200: Best = 7.805774e+02	FES = 620100	Diversity = 3.7998e-08	k_t
= [4, 2, 4, 2]			
Generation 6210: Best = 7.805774e+02	FES = 621100	Diversity = 3.7855e-08	k_t
= [1, 1, 1, 1]			
Generation 6220: Best = 7.805774e+02	FES = 622100	Diversity = 3.7810e-08	k_t
= [2, 3, 4, 2]			
Generation 6230: Best = 7.805774e+02	FES = 623100	Diversity = 3.7676e-08	k_t
= [1, 1, 1, 1]			
Generation 6240: Best = 7.805774e+02	FES = 624100	Diversity = 3.7626e-08	k_t
= [3, 2, 4, 3]			
Generation 6250: Best = 7.805774e+02	FES = 625100	Diversity = 3.6930e-08	k_t
= [1, 1, 1, 1]			
Generation 6260: Best = 7.805774e+02	FES = 626100	Diversity = 3.6936e-08	k_t
= [2, 4, 3, 3]			
Generation 6270: Best = 7.805774e+02	FES = 627100	Diversity = 3.7001e-08	k_t
= [1, 1, 1, 1]			
Generation 6280: Best = 7.805774e+02	FES = 628100	Diversity = 3.7272e-08	k_t
= [3, 3, 4, 3]			
Generation 6290: Best = 7.805774e+02	FES = 629100	Diversity = 3.7317e-08	k_t
= [1, 1, 1, 1]			
Generation 6300: Best = 7.805774e+02	FES = 630100	Diversity = 3.7805e-08	k_t
= [3, 2, 2, 3]			
Generation 6310: Best = 7.805774e+02	FES = 631100	Diversity = 3.7140e-08	k_t
= [1, 1, 1, 1]			
Generation 6320: Best = 7.805774e+02	FES = 632100	Diversity = 3.7196e-08	k_t
= [3, 3, 2, 4]			
Generation 6330: Best = 7.805774e+02	FES = 633100	Diversity = 3.6940e-08	k_t
= [1, 1, 1, 1]			
Generation 6340: Best = 7.805774e+02	FES = 634100	Diversity = 3.7459e-08	k_t
= [2, 2, 2, 2]			
Generation 6350: Best = 7.805774e+02	FES = 635100	Diversity = 3.6827e-08	k_t
= [1, 1, 1, 1]			
Generation 6360: Best = 7.805774e+02	FES = 636100	Diversity = 3.6958e-08	k_t
= [3, 4, 4, 4]			
Generation 6370: Best = 7.805774e+02	FES = 637100	Diversity = 3.6693e-08	k_t
= [1, 1, 1, 1]			
Generation 6380: Best = 7.805774e+02	FES = 638100	Diversity = 3.6681e-08	k_t
= [3, 2, 2, 4]			
Generation 6390: Best = 7.805774e+02	FES = 639100	Diversity = 3.5965e-08	k_t
= [1, 1, 1, 1]			
Generation 6400: Best = 7.805774e+02	FES = 640100	Diversity = 3.5891e-08	k_t
= [3, 4, 2, 2]			
Generation 6410: Best = 7.805774e+02	FES = 641100	Diversity = 3.5413e-08	k_t
= [1, 1, 1, 1]			
Generation 6420: Best = 7.805774e+02	FES = 642100	Diversity = 3.5439e-08	k_t
= [2, 2, 3, 3]			
Generation 6430: Best = 7.805774e+02	FES = 643100	Diversity = 3.5117e-08	k_t
= [1, 1, 1, 1]			

Generation 6440: Best = 7.805774e+02	FES = 644100	Diversity = 3.5250e-08	k_t
= [3, 3, 2, 4]			
Generation 6450: Best = 7.805774e+02	FES = 645100	Diversity = 3.5298e-08	k_t
= [1, 1, 1, 1]			
Generation 6460: Best = 7.805774e+02	FES = 646100	Diversity = 3.5026e-08	k_t
= [4, 2, 2, 2]			
Generation 6470: Best = 7.805774e+02	FES = 647100	Diversity = 3.5171e-08	k_t
= [1, 1, 1, 1]			
Generation 6480: Best = 7.805774e+02	FES = 648100	Diversity = 3.5862e-08	k_t
= [2, 4, 2, 4]			
Generation 6490: Best = 7.805774e+02	FES = 649100	Diversity = 3.6324e-08	k_t
= [1, 1, 1, 1]			
Generation 6500: Best = 7.805774e+02	FES = 650100	Diversity = 3.6156e-08	k_t
= [3, 3, 3, 2]			
Generation 6510: Best = 7.805774e+02	FES = 651100	Diversity = 3.5844e-08	k_t
= [1, 1, 1, 1]			
Generation 6520: Best = 7.805774e+02	FES = 652100	Diversity = 3.5481e-08	k_t
= [2, 2, 2, 4]			
Generation 6530: Best = 7.805774e+02	FES = 653100	Diversity = 3.5888e-08	k_t
= [1, 1, 1, 1]			
Generation 6540: Best = 7.805774e+02	FES = 654100	Diversity = 3.6118e-08	k_t
= [3, 2, 2, 4]			
Generation 6550: Best = 7.805774e+02	FES = 655100	Diversity = 3.5326e-08	k_t
= [1, 1, 1, 1]			
Generation 6560: Best = 7.805774e+02	FES = 656100	Diversity = 3.4969e-08	k_t
= [4, 4, 2, 4]			
Generation 6570: Best = 7.805774e+02	FES = 657100	Diversity = 3.5050e-08	k_t
= [1, 1, 1, 1]			
Generation 6580: Best = 7.805774e+02	FES = 658100	Diversity = 3.4954e-08	k_t
= [4, 3, 4, 2]			
Generation 6590: Best = 7.805774e+02	FES = 659100	Diversity = 3.3916e-08	k_t
= [1, 1, 1, 1]			
Generation 6600: Best = 7.805774e+02	FES = 660100	Diversity = 3.4093e-08	k_t
= [4, 4, 3, 2]			
Generation 6610: Best = 7.805774e+02	FES = 661100	Diversity = 3.4594e-08	k_t
= [1, 1, 1, 1]			
Generation 6620: Best = 7.805774e+02	FES = 662100	Diversity = 3.4049e-08	k_t
= [3, 3, 2, 2]			
Generation 6630: Best = 7.805774e+02	FES = 663100	Diversity = 3.4015e-08	k_t
= [1, 1, 1, 1]			
Generation 6640: Best = 7.805774e+02	FES = 664100	Diversity = 3.4403e-08	k_t
= [3, 2, 3, 4]			
Generation 6650: Best = 7.805774e+02	FES = 665100	Diversity = 3.4485e-08	k_t
= [1, 1, 1, 1]			
Generation 6660: Best = 7.805774e+02	FES = 666100	Diversity = 3.5075e-08	k_t
= [4, 4, 4, 4]			
Generation 6670: Best = 7.805774e+02	FES = 667100	Diversity = 3.4619e-08	k_t
= [1, 1, 1, 1]			
Generation 6680: Best = 7.805774e+02	FES = 668100	Diversity = 3.4587e-08	k_t
= [4, 3, 2, 4]			
Generation 6690: Best = 7.805774e+02	FES = 669100	Diversity = 3.5186e-08	k_t
= [1, 1, 1, 1]			
Generation 6700: Best = 7.805774e+02	FES = 670100	Diversity = 3.4767e-08	k_t
= [3, 2, 3, 4]			

Generation 6710: Best = 7.805774e+02	FES = 671100	Diversity = 3.5081e-08	k_t
= [1, 1, 1, 1]			
Generation 6720: Best = 7.805774e+02	FES = 672100	Diversity = 3.4943e-08	k_t
= [4, 2, 2, 3]			
Generation 6730: Best = 7.805774e+02	FES = 673100	Diversity = 3.5237e-08	k_t
= [1, 1, 1, 1]			
Generation 6740: Best = 7.805774e+02	FES = 674100	Diversity = 3.4937e-08	k_t
= [4, 3, 4, 4]			
Generation 6750: Best = 7.805774e+02	FES = 675100	Diversity = 3.5352e-08	k_t
= [1, 1, 1, 1]			
Generation 6760: Best = 7.805774e+02	FES = 676100	Diversity = 3.5568e-08	k_t
= [3, 3, 3, 3]			
Generation 6770: Best = 7.805774e+02	FES = 677100	Diversity = 3.5268e-08	k_t
= [1, 1, 1, 1]			
Generation 6780: Best = 7.805774e+02	FES = 678100	Diversity = 3.5763e-08	k_t
= [3, 4, 4, 2]			
Generation 6790: Best = 7.805774e+02	FES = 679100	Diversity = 3.6107e-08	k_t
= [1, 1, 1, 1]			
Generation 6800: Best = 7.805774e+02	FES = 680100	Diversity = 3.6391e-08	k_t
= [4, 2, 3, 2]			
Generation 6810: Best = 7.805774e+02	FES = 681100	Diversity = 3.6049e-08	k_t
= [1, 1, 1, 1]			
Generation 6820: Best = 7.805774e+02	FES = 682100	Diversity = 3.6155e-08	k_t
= [4, 2, 4, 3]			
Generation 6830: Best = 7.805774e+02	FES = 683100	Diversity = 3.6328e-08	k_t
= [1, 1, 1, 1]			
Generation 6840: Best = 7.805774e+02	FES = 684100	Diversity = 3.7016e-08	k_t
= [4, 2, 2, 4]			
Generation 6850: Best = 7.805774e+02	FES = 685100	Diversity = 3.7563e-08	k_t
= [1, 1, 1, 1]			
Generation 6860: Best = 7.805774e+02	FES = 686100	Diversity = 3.7547e-08	k_t
= [4, 2, 4, 4]			
Generation 6870: Best = 7.805774e+02	FES = 687100	Diversity = 3.7933e-08	k_t
= [1, 1, 1, 1]			
Generation 6880: Best = 7.805774e+02	FES = 688100	Diversity = 3.7917e-08	k_t
= [3, 2, 2, 4]			
Generation 6890: Best = 7.805774e+02	FES = 689100	Diversity = 3.7764e-08	k_t
= [1, 1, 1, 1]			
Generation 6900: Best = 7.805774e+02	FES = 690100	Diversity = 3.7583e-08	k_t
= [2, 2, 2, 4]			
Generation 6910: Best = 7.805774e+02	FES = 691100	Diversity = 3.7516e-08	k_t
= [1, 1, 1, 1]			
Generation 6920: Best = 7.805774e+02	FES = 692100	Diversity = 3.7154e-08	k_t
= [4, 3, 4, 2]			
Generation 6930: Best = 7.805774e+02	FES = 693100	Diversity = 3.7346e-08	k_t
= [1, 1, 1, 1]			
Generation 6940: Best = 7.805774e+02	FES = 694100	Diversity = 3.7185e-08	k_t
= [4, 2, 4, 2]			
Generation 6950: Best = 7.805774e+02	FES = 695100	Diversity = 3.7348e-08	k_t
= [1, 1, 1, 1]			
Generation 6960: Best = 7.805774e+02	FES = 696100	Diversity = 3.6963e-08	k_t
= [3, 3, 3, 3]			
Generation 6970: Best = 7.805774e+02	FES = 697100	Diversity = 3.7022e-08	k_t
= [1, 1, 1, 1]			

Generation 6980: Best = 7.805774e+02	FES = 698100	Diversity = 3.6974e-08	k_t
= [3, 4, 4, 2]			
Generation 6990: Best = 7.805774e+02	FES = 699100	Diversity = 3.7276e-08	k_t
= [1, 1, 1, 1]			
Generation 7000: Best = 7.805774e+02	FES = 700100	Diversity = 3.6886e-08	k_t
= [4, 4, 3, 3]			
Generation 7010: Best = 7.805774e+02	FES = 701100	Diversity = 3.7209e-08	k_t
= [1, 1, 1, 1]			
Generation 7020: Best = 7.805774e+02	FES = 702100	Diversity = 3.7544e-08	k_t
= [2, 3, 2, 3]			
Generation 7030: Best = 7.805774e+02	FES = 703100	Diversity = 3.7510e-08	k_t
= [1, 1, 1, 1]			
Generation 7040: Best = 7.805774e+02	FES = 704100	Diversity = 3.7787e-08	k_t
= [2, 2, 3, 2]			
Generation 7050: Best = 7.805774e+02	FES = 705100	Diversity = 3.7833e-08	k_t
= [1, 1, 1, 1]			
Generation 7060: Best = 7.805774e+02	FES = 706100	Diversity = 3.7260e-08	k_t
= [4, 2, 2, 4]			
Generation 7070: Best = 7.805774e+02	FES = 707100	Diversity = 3.6899e-08	k_t
= [1, 1, 1, 1]			
Generation 7080: Best = 7.805774e+02	FES = 708100	Diversity = 3.6860e-08	k_t
= [3, 4, 2, 3]			
Generation 7090: Best = 7.805774e+02	FES = 709100	Diversity = 3.6591e-08	k_t
= [1, 1, 1, 1]			
Generation 7100: Best = 7.805774e+02	FES = 710100	Diversity = 3.6588e-08	k_t
= [2, 4, 2, 4]			
Generation 7110: Best = 7.805774e+02	FES = 711100	Diversity = 3.6056e-08	k_t
= [1, 1, 1, 1]			
Generation 7120: Best = 7.805774e+02	FES = 712100	Diversity = 3.6313e-08	k_t
= [2, 3, 3, 3]			
Generation 7130: Best = 7.805774e+02	FES = 713100	Diversity = 3.6785e-08	k_t
= [1, 1, 1, 1]			
Generation 7140: Best = 7.805774e+02	FES = 714100	Diversity = 3.7405e-08	k_t
= [2, 3, 4, 2]			
Generation 7150: Best = 7.805774e+02	FES = 715100	Diversity = 3.7145e-08	k_t
= [1, 1, 1, 1]			
Generation 7160: Best = 7.805774e+02	FES = 716100	Diversity = 3.7975e-08	k_t
= [4, 4, 2, 2]			
Generation 7170: Best = 7.805774e+02	FES = 717100	Diversity = 3.7605e-08	k_t
= [1, 1, 1, 1]			
Generation 7180: Best = 7.805774e+02	FES = 718100	Diversity = 3.7401e-08	k_t
= [2, 2, 4, 2]			
Generation 7190: Best = 7.805774e+02	FES = 719100	Diversity = 3.6865e-08	k_t
= [1, 1, 1, 1]			
Generation 7200: Best = 7.805774e+02	FES = 720100	Diversity = 3.6488e-08	k_t
= [2, 3, 3, 3]			
Generation 7210: Best = 7.805774e+02	FES = 721100	Diversity = 3.6175e-08	k_t
= [1, 1, 1, 1]			
Generation 7220: Best = 7.805774e+02	FES = 722100	Diversity = 3.5916e-08	k_t
= [3, 2, 2, 2]			
Generation 7230: Best = 7.805774e+02	FES = 723100	Diversity = 3.6180e-08	k_t
= [1, 1, 1, 1]			
Generation 7240: Best = 7.805774e+02	FES = 724100	Diversity = 3.6450e-08	k_t
= [2, 2, 3, 2]			

Generation 7250: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 725100	Diversity = 3.6577e-08	k_t
Generation 7260: Best = 7.805774e+02 = [2, 2, 4, 4]	FES = 726100	Diversity = 3.6058e-08	k_t
Generation 7270: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 727100	Diversity = 3.6196e-08	k_t
Generation 7280: Best = 7.805774e+02 = [2, 2, 3, 3]	FES = 728100	Diversity = 3.6699e-08	k_t
Generation 7290: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 729100	Diversity = 3.6175e-08	k_t
Generation 7300: Best = 7.805774e+02 = [4, 4, 3, 3]	FES = 730100	Diversity = 3.6306e-08	k_t
Generation 7310: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 731100	Diversity = 3.5636e-08	k_t
Generation 7320: Best = 7.805774e+02 = [4, 2, 2, 3]	FES = 732100	Diversity = 3.5483e-08	k_t
Generation 7330: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 733100	Diversity = 3.5186e-08	k_t
Generation 7340: Best = 7.805774e+02 = [2, 4, 3, 3]	FES = 734100	Diversity = 3.5205e-08	k_t
Generation 7350: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 735100	Diversity = 3.5406e-08	k_t
Generation 7360: Best = 7.805774e+02 = [3, 2, 2, 3]	FES = 736100	Diversity = 3.4966e-08	k_t
Generation 7370: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 737100	Diversity = 3.4628e-08	k_t
Generation 7380: Best = 7.805774e+02 = [2, 3, 4, 3]	FES = 738100	Diversity = 3.4477e-08	k_t
Generation 7390: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 739100	Diversity = 3.4133e-08	k_t
Generation 7400: Best = 7.805774e+02 = [3, 3, 3, 4]	FES = 740100	Diversity = 3.4189e-08	k_t
Generation 7410: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 741100	Diversity = 3.4632e-08	k_t
Generation 7420: Best = 7.805774e+02 = [3, 3, 4, 4]	FES = 742100	Diversity = 3.4537e-08	k_t
Generation 7430: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 743100	Diversity = 3.5131e-08	k_t
Generation 7440: Best = 7.805774e+02 = [4, 3, 2, 2]	FES = 744100	Diversity = 3.5394e-08	k_t
Generation 7450: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 745100	Diversity = 3.5350e-08	k_t
Generation 7460: Best = 7.805774e+02 = [4, 3, 3, 3]	FES = 746100	Diversity = 3.5672e-08	k_t
Generation 7470: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 747100	Diversity = 3.5652e-08	k_t
Generation 7480: Best = 7.805774e+02 = [4, 4, 4, 4]	FES = 748100	Diversity = 3.5993e-08	k_t
Generation 7490: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 749100	Diversity = 3.5987e-08	k_t
Generation 7500: Best = 7.805774e+02 = [2, 4, 3, 3]	FES = 750100	Diversity = 3.6516e-08	k_t
Generation 7510: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 751100	Diversity = 3.6175e-08	k_t

Generation 7520: Best = 7.805774e+02 = [4, 2, 2, 4]	FES = 752100	Diversity = 3.6847e-08	k_t
Generation 7530: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 753100	Diversity = 3.7620e-08	k_t
Generation 7540: Best = 7.805774e+02 = [2, 3, 2, 4]	FES = 754100	Diversity = 3.7386e-08	k_t
Generation 7550: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 755100	Diversity = 3.7548e-08	k_t
Generation 7560: Best = 7.805774e+02 = [2, 3, 2, 2]	FES = 756100	Diversity = 3.7722e-08	k_t
Generation 7570: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 757100	Diversity = 3.8007e-08	k_t
Generation 7580: Best = 7.805774e+02 = [2, 4, 4, 3]	FES = 758100	Diversity = 3.7841e-08	k_t
Generation 7590: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 759100	Diversity = 3.7734e-08	k_t
Generation 7600: Best = 7.805774e+02 = [3, 3, 2, 4]	FES = 760100	Diversity = 3.7501e-08	k_t
Generation 7610: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 761100	Diversity = 3.7982e-08	k_t
Generation 7620: Best = 7.805774e+02 = [4, 3, 2, 4]	FES = 762100	Diversity = 3.7433e-08	k_t
Generation 7630: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 763100	Diversity = 3.7612e-08	k_t
Generation 7640: Best = 7.805774e+02 = [2, 3, 2, 3]	FES = 764100	Diversity = 3.7320e-08	k_t
Generation 7650: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 765100	Diversity = 3.6161e-08	k_t
Generation 7660: Best = 7.805774e+02 = [3, 4, 3, 2]	FES = 766100	Diversity = 3.6382e-08	k_t
Generation 7670: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 767100	Diversity = 3.6853e-08	k_t
Generation 7680: Best = 7.805774e+02 = [2, 3, 2, 3]	FES = 768100	Diversity = 3.5686e-08	k_t
Generation 7690: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 769100	Diversity = 3.5093e-08	k_t
Generation 7700: Best = 7.805774e+02 = [3, 2, 4, 4]	FES = 770100	Diversity = 3.5556e-08	k_t
Generation 7710: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 771100	Diversity = 3.5925e-08	k_t
Generation 7720: Best = 7.805774e+02 = [2, 2, 4, 3]	FES = 772100	Diversity = 3.5790e-08	k_t
Generation 7730: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 773100	Diversity = 3.5820e-08	k_t
Generation 7740: Best = 7.805774e+02 = [2, 2, 4, 2]	FES = 774100	Diversity = 3.5505e-08	k_t
Generation 7750: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 775100	Diversity = 3.6083e-08	k_t
Generation 7760: Best = 7.805774e+02 = [4, 3, 4, 3]	FES = 776100	Diversity = 3.6172e-08	k_t
Generation 7770: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 777100	Diversity = 3.5426e-08	k_t
Generation 7780: Best = 7.805774e+02 = [4, 2, 4, 2]	FES = 778100	Diversity = 3.5848e-08	k_t

Generation 7790: Best = 7.805774e+02	FES = 779100	Diversity = 3.6041e-08	k_t
= [1, 1, 1, 1]			
Generation 7800: Best = 7.805774e+02	FES = 780100	Diversity = 3.6311e-08	k_t
= [4, 4, 4, 2]			
Generation 7810: Best = 7.805774e+02	FES = 781100	Diversity = 3.5802e-08	k_t
= [1, 1, 1, 1]			
Generation 7820: Best = 7.805774e+02	FES = 782100	Diversity = 3.5456e-08	k_t
= [4, 4, 4, 4]			
Generation 7830: Best = 7.805774e+02	FES = 783100	Diversity = 3.5209e-08	k_t
= [1, 1, 1, 1]			
Generation 7840: Best = 7.805774e+02	FES = 784100	Diversity = 3.5865e-08	k_t
= [2, 3, 2, 2]			
Generation 7850: Best = 7.805774e+02	FES = 785100	Diversity = 3.6663e-08	k_t
= [1, 1, 1, 1]			
Generation 7860: Best = 7.805774e+02	FES = 786100	Diversity = 3.7642e-08	k_t
= [4, 2, 4, 3]			
Generation 7870: Best = 7.805774e+02	FES = 787100	Diversity = 3.8031e-08	k_t
= [1, 1, 1, 1]			
Generation 7880: Best = 7.805774e+02	FES = 788100	Diversity = 3.8128e-08	k_t
= [2, 4, 2, 4]			
Generation 7890: Best = 7.805774e+02	FES = 789100	Diversity = 3.8440e-08	k_t
= [1, 1, 1, 1]			
Generation 7900: Best = 7.805774e+02	FES = 790100	Diversity = 3.8687e-08	k_t
= [4, 4, 4, 2]			
Generation 7910: Best = 7.805774e+02	FES = 791100	Diversity = 3.8536e-08	k_t
= [1, 1, 1, 1]			
Generation 7920: Best = 7.805774e+02	FES = 792100	Diversity = 3.8194e-08	k_t
= [2, 4, 4, 2]			
Generation 7930: Best = 7.805774e+02	FES = 793100	Diversity = 3.8667e-08	k_t
= [1, 1, 1, 1]			
Generation 7940: Best = 7.805774e+02	FES = 794100	Diversity = 3.8340e-08	k_t
= [4, 4, 2, 2]			
Generation 7950: Best = 7.805774e+02	FES = 795100	Diversity = 3.8094e-08	k_t
= [1, 1, 1, 1]			
Generation 7960: Best = 7.805774e+02	FES = 796100	Diversity = 3.7670e-08	k_t
= [4, 4, 2, 3]			
Generation 7970: Best = 7.805774e+02	FES = 797100	Diversity = 3.8225e-08	k_t
= [1, 1, 1, 1]			
Generation 7980: Best = 7.805774e+02	FES = 798100	Diversity = 3.8106e-08	k_t
= [4, 4, 4, 3]			
Generation 7990: Best = 7.805774e+02	FES = 799100	Diversity = 3.8443e-08	k_t
= [1, 1, 1, 1]			
Generation 8000: Best = 7.805774e+02	FES = 800100	Diversity = 3.8387e-08	k_t
= [4, 3, 2, 3]			
Generation 8010: Best = 7.805774e+02	FES = 801100	Diversity = 3.7198e-08	k_t
= [1, 1, 1, 1]			
Generation 8020: Best = 7.805774e+02	FES = 802100	Diversity = 3.7613e-08	k_t
= [4, 4, 2, 4]			
Generation 8030: Best = 7.805774e+02	FES = 803100	Diversity = 3.6878e-08	k_t
= [1, 1, 1, 1]			
Generation 8040: Best = 7.805774e+02	FES = 804100	Diversity = 3.6678e-08	k_t
= [2, 3, 3, 3]			
Generation 8050: Best = 7.805774e+02	FES = 805100	Diversity = 3.7092e-08	k_t
= [1, 1, 1, 1]			

Generation 8060: Best = 7.805774e+02 = [3, 4, 3, 2]	FES = 806100	Diversity = 3.7277e-08	k_t
Generation 8070: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 807100	Diversity = 3.7010e-08	k_t
Generation 8080: Best = 7.805774e+02 = [3, 4, 4, 4]	FES = 808100	Diversity = 3.7023e-08	k_t
Generation 8090: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 809100	Diversity = 3.6964e-08	k_t
Generation 8100: Best = 7.805774e+02 = [3, 3, 3, 2]	FES = 810100	Diversity = 3.6637e-08	k_t
Generation 8110: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 811100	Diversity = 3.6752e-08	k_t
Generation 8120: Best = 7.805774e+02 = [2, 3, 2, 2]	FES = 812100	Diversity = 3.6627e-08	k_t
Generation 8130: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 813100	Diversity = 3.6184e-08	k_t
Generation 8140: Best = 7.805774e+02 = [4, 4, 2, 2]	FES = 814100	Diversity = 3.7033e-08	k_t
Generation 8150: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 815100	Diversity = 3.7430e-08	k_t
Generation 8160: Best = 7.805774e+02 = [4, 4, 4, 4]	FES = 816100	Diversity = 3.7351e-08	k_t
Generation 8170: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 817100	Diversity = 3.7884e-08	k_t
Generation 8180: Best = 7.805774e+02 = [2, 4, 3, 3]	FES = 818100	Diversity = 3.7671e-08	k_t
Generation 8190: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 819100	Diversity = 3.7269e-08	k_t
Generation 8200: Best = 7.805774e+02 = [2, 3, 4, 3]	FES = 820100	Diversity = 3.7306e-08	k_t
Generation 8210: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 821100	Diversity = 3.7267e-08	k_t
Generation 8220: Best = 7.805774e+02 = [4, 3, 2, 2]	FES = 822100	Diversity = 3.7194e-08	k_t
Generation 8230: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 823100	Diversity = 3.6859e-08	k_t
Generation 8240: Best = 7.805774e+02 = [3, 2, 2, 4]	FES = 824100	Diversity = 3.6713e-08	k_t
Generation 8250: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 825100	Diversity = 3.5937e-08	k_t
Generation 8260: Best = 7.805774e+02 = [2, 4, 3, 4]	FES = 826100	Diversity = 3.5424e-08	k_t
Generation 8270: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 827100	Diversity = 3.5572e-08	k_t
Generation 8280: Best = 7.805774e+02 = [3, 2, 4, 2]	FES = 828100	Diversity = 3.5628e-08	k_t
Generation 8290: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 829100	Diversity = 3.5873e-08	k_t
Generation 8300: Best = 7.805774e+02 = [4, 3, 2, 3]	FES = 830100	Diversity = 3.6467e-08	k_t
Generation 8310: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 831100	Diversity = 3.6839e-08	k_t
Generation 8320: Best = 7.805774e+02 = [4, 2, 2, 2]	FES = 832100	Diversity = 3.7024e-08	k_t

Generation 8330: Best = 7.805774e+02	FES = 833100	Diversity = 3.6552e-08	k_t
= [1, 1, 1, 1]			
Generation 8340: Best = 7.805774e+02	FES = 834100	Diversity = 3.6873e-08	k_t
= [3, 4, 4, 3]			
Generation 8350: Best = 7.805774e+02	FES = 835100	Diversity = 3.7450e-08	k_t
= [1, 1, 1, 1]			
Generation 8360: Best = 7.805774e+02	FES = 836100	Diversity = 3.7324e-08	k_t
= [4, 3, 2, 2]			
Generation 8370: Best = 7.805774e+02	FES = 837100	Diversity = 3.7073e-08	k_t
= [1, 1, 1, 1]			
Generation 8380: Best = 7.805774e+02	FES = 838100	Diversity = 3.6724e-08	k_t
= [2, 4, 3, 3]			
Generation 8390: Best = 7.805774e+02	FES = 839100	Diversity = 3.6801e-08	k_t
= [1, 1, 1, 1]			
Generation 8400: Best = 7.805774e+02	FES = 840100	Diversity = 3.6305e-08	k_t
= [3, 4, 3, 2]			
Generation 8410: Best = 7.805774e+02	FES = 841100	Diversity = 3.6579e-08	k_t
= [1, 1, 1, 1]			
Generation 8420: Best = 7.805774e+02	FES = 842100	Diversity = 3.6465e-08	k_t
= [2, 3, 2, 2]			
Generation 8430: Best = 7.805774e+02	FES = 843100	Diversity = 3.6335e-08	k_t
= [1, 1, 1, 1]			
Generation 8440: Best = 7.805774e+02	FES = 844100	Diversity = 3.6450e-08	k_t
= [2, 2, 4, 2]			
Generation 8450: Best = 7.805774e+02	FES = 845100	Diversity = 3.5607e-08	k_t
= [1, 1, 1, 1]			
Generation 8460: Best = 7.805774e+02	FES = 846100	Diversity = 3.6295e-08	k_t
= [4, 3, 2, 2]			
Generation 8470: Best = 7.805774e+02	FES = 847100	Diversity = 3.6454e-08	k_t
= [1, 1, 1, 1]			
Generation 8480: Best = 7.805774e+02	FES = 848100	Diversity = 3.6189e-08	k_t
= [3, 2, 4, 4]			
Generation 8490: Best = 7.805774e+02	FES = 849100	Diversity = 3.6344e-08	k_t
= [1, 1, 1, 1]			
Generation 8500: Best = 7.805774e+02	FES = 850100	Diversity = 3.6398e-08	k_t
= [3, 4, 4, 2]			
Generation 8510: Best = 7.805774e+02	FES = 851100	Diversity = 3.6417e-08	k_t
= [1, 1, 1, 1]			
Generation 8520: Best = 7.805774e+02	FES = 852100	Diversity = 3.5663e-08	k_t
= [2, 3, 4, 3]			
Generation 8530: Best = 7.805774e+02	FES = 853100	Diversity = 3.5420e-08	k_t
= [1, 1, 1, 1]			
Generation 8540: Best = 7.805774e+02	FES = 854100	Diversity = 3.5686e-08	k_t
= [4, 2, 4, 2]			
Generation 8550: Best = 7.805774e+02	FES = 855100	Diversity = 3.5457e-08	k_t
= [1, 1, 1, 1]			
Generation 8560: Best = 7.805774e+02	FES = 856100	Diversity = 3.4737e-08	k_t
= [2, 2, 3, 4]			
Generation 8570: Best = 7.805774e+02	FES = 857100	Diversity = 3.4323e-08	k_t
= [1, 1, 1, 1]			
Generation 8580: Best = 7.805774e+02	FES = 858100	Diversity = 3.4989e-08	k_t
= [3, 3, 2, 3]			
Generation 8590: Best = 7.805774e+02	FES = 859100	Diversity = 3.5061e-08	k_t
= [1, 1, 1, 1]			

Generation 8600: Best = 7.805774e+02	FES = 860100	Diversity = 3.4741e-08	k_t
= [3, 2, 2, 4]			
Generation 8610: Best = 7.805774e+02	FES = 861100	Diversity = 3.5169e-08	k_t
= [1, 1, 1, 1]			
Generation 8620: Best = 7.805774e+02	FES = 862100	Diversity = 3.4687e-08	k_t
= [4, 2, 2, 4]			
Generation 8630: Best = 7.805774e+02	FES = 863100	Diversity = 3.4498e-08	k_t
= [1, 1, 1, 1]			
Generation 8640: Best = 7.805774e+02	FES = 864100	Diversity = 3.4630e-08	k_t
= [4, 3, 2, 4]			
Generation 8650: Best = 7.805774e+02	FES = 865100	Diversity = 3.5283e-08	k_t
= [1, 1, 1, 1]			
Generation 8660: Best = 7.805774e+02	FES = 866100	Diversity = 3.5663e-08	k_t
= [2, 2, 3, 4]			
Generation 8670: Best = 7.805774e+02	FES = 867100	Diversity = 3.5080e-08	k_t
= [1, 1, 1, 1]			
Generation 8680: Best = 7.805774e+02	FES = 868100	Diversity = 3.5411e-08	k_t
= [4, 3, 4, 2]			
Generation 8690: Best = 7.805774e+02	FES = 869100	Diversity = 3.5976e-08	k_t
= [1, 1, 1, 1]			
Generation 8700: Best = 7.805774e+02	FES = 870100	Diversity = 3.6163e-08	k_t
= [2, 4, 2, 3]			
Generation 8710: Best = 7.805774e+02	FES = 871100	Diversity = 3.6124e-08	k_t
= [1, 1, 1, 1]			
Generation 8720: Best = 7.805774e+02	FES = 872100	Diversity = 3.6544e-08	k_t
= [3, 2, 2, 4]			
Generation 8730: Best = 7.805774e+02	FES = 873100	Diversity = 3.6859e-08	k_t
= [1, 1, 1, 1]			
Generation 8740: Best = 7.805774e+02	FES = 874100	Diversity = 3.6941e-08	k_t
= [4, 2, 2, 2]			
Generation 8750: Best = 7.805774e+02	FES = 875100	Diversity = 3.7875e-08	k_t
= [1, 1, 1, 1]			
Generation 8760: Best = 7.805774e+02	FES = 876100	Diversity = 3.7816e-08	k_t
= [3, 2, 2, 4]			
Generation 8770: Best = 7.805774e+02	FES = 877100	Diversity = 3.7161e-08	k_t
= [1, 1, 1, 1]			
Generation 8780: Best = 7.805774e+02	FES = 878100	Diversity = 3.6669e-08	k_t
= [2, 4, 4, 3]			
Generation 8790: Best = 7.805774e+02	FES = 879100	Diversity = 3.6699e-08	k_t
= [1, 1, 1, 1]			
Generation 8800: Best = 7.805774e+02	FES = 880100	Diversity = 3.6456e-08	k_t
= [3, 3, 4, 4]			
Generation 8810: Best = 7.805774e+02	FES = 881100	Diversity = 3.5664e-08	k_t
= [1, 1, 1, 1]			
Generation 8820: Best = 7.805774e+02	FES = 882100	Diversity = 3.6122e-08	k_t
= [4, 3, 3, 2]			
Generation 8830: Best = 7.805774e+02	FES = 883100	Diversity = 3.5665e-08	k_t
= [1, 1, 1, 1]			
Generation 8840: Best = 7.805774e+02	FES = 884100	Diversity = 3.6074e-08	k_t
= [4, 4, 4, 2]			
Generation 8850: Best = 7.805774e+02	FES = 885100	Diversity = 3.5659e-08	k_t
= [1, 1, 1, 1]			
Generation 8860: Best = 7.805774e+02	FES = 886100	Diversity = 3.5603e-08	k_t
= [4, 2, 3, 2]			

Generation 8870: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 887100	Diversity = 3.5644e-08	k_t
Generation 8880: Best = 7.805774e+02 = [2, 3, 3, 4]	FES = 888100	Diversity = 3.6168e-08	k_t
Generation 8890: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 889100	Diversity = 3.6478e-08	k_t
Generation 8900: Best = 7.805774e+02 = [2, 2, 4, 3]	FES = 890100	Diversity = 3.7014e-08	k_t
Generation 8910: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 891100	Diversity = 3.7406e-08	k_t
Generation 8920: Best = 7.805774e+02 = [4, 3, 4, 3]	FES = 892100	Diversity = 3.7349e-08	k_t
Generation 8930: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 893100	Diversity = 3.7727e-08	k_t
Generation 8940: Best = 7.805774e+02 = [4, 4, 3, 4]	FES = 894100	Diversity = 3.7908e-08	k_t
Generation 8950: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 895100	Diversity = 3.7343e-08	k_t
Generation 8960: Best = 7.805774e+02 = [4, 2, 3, 3]	FES = 896100	Diversity = 3.7704e-08	k_t
Generation 8970: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 897100	Diversity = 3.7622e-08	k_t
Generation 8980: Best = 7.805774e+02 = [3, 4, 3, 4]	FES = 898100	Diversity = 3.6951e-08	k_t
Generation 8990: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 899100	Diversity = 3.6832e-08	k_t
Generation 9000: Best = 7.805774e+02 = [3, 4, 4, 4]	FES = 900100	Diversity = 3.7260e-08	k_t
Generation 9010: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 901100	Diversity = 3.7215e-08	k_t
Generation 9020: Best = 7.805774e+02 = [4, 3, 2, 4]	FES = 902100	Diversity = 3.7259e-08	k_t
Generation 9030: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 903100	Diversity = 3.7703e-08	k_t
Generation 9040: Best = 7.805774e+02 = [3, 2, 4, 3]	FES = 904100	Diversity = 3.7250e-08	k_t
Generation 9050: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 905100	Diversity = 3.7014e-08	k_t
Generation 9060: Best = 7.805774e+02 = [4, 2, 3, 4]	FES = 906100	Diversity = 3.6762e-08	k_t
Generation 9070: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 907100	Diversity = 3.6571e-08	k_t
Generation 9080: Best = 7.805774e+02 = [3, 2, 4, 4]	FES = 908100	Diversity = 3.6429e-08	k_t
Generation 9090: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 909100	Diversity = 3.5944e-08	k_t
Generation 9100: Best = 7.805774e+02 = [2, 2, 4, 4]	FES = 910100	Diversity = 3.5903e-08	k_t
Generation 9110: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 911100	Diversity = 3.5454e-08	k_t
Generation 9120: Best = 7.805774e+02 = [2, 2, 3, 2]	FES = 912100	Diversity = 3.5890e-08	k_t
Generation 9130: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 913100	Diversity = 3.6181e-08	k_t

Generation 9140: Best = 7.805774e+02	FES = 914100	Diversity = 3.5756e-08	k_t
= [3, 2, 4, 4]			
Generation 9150: Best = 7.805774e+02	FES = 915100	Diversity = 3.5794e-08	k_t
= [1, 1, 1, 1]			
Generation 9160: Best = 7.805774e+02	FES = 916100	Diversity = 3.5843e-08	k_t
= [2, 2, 2, 4]			
Generation 9170: Best = 7.805774e+02	FES = 917100	Diversity = 3.6110e-08	k_t
= [1, 1, 1, 1]			
Generation 9180: Best = 7.805774e+02	FES = 918100	Diversity = 3.5596e-08	k_t
= [3, 2, 4, 3]			
Generation 9190: Best = 7.805774e+02	FES = 919100	Diversity = 3.5768e-08	k_t
= [1, 1, 1, 1]			
Generation 9200: Best = 7.805774e+02	FES = 920100	Diversity = 3.5695e-08	k_t
= [2, 4, 2, 4]			
Generation 9210: Best = 7.805774e+02	FES = 921100	Diversity = 3.5451e-08	k_t
= [1, 1, 1, 1]			
Generation 9220: Best = 7.805774e+02	FES = 922100	Diversity = 3.5478e-08	k_t
= [4, 3, 3, 4]			
Generation 9230: Best = 7.805774e+02	FES = 923100	Diversity = 3.5719e-08	k_t
= [1, 1, 1, 1]			
Generation 9240: Best = 7.805774e+02	FES = 924100	Diversity = 3.5868e-08	k_t
= [3, 4, 2, 3]			
Generation 9250: Best = 7.805774e+02	FES = 925100	Diversity = 3.6410e-08	k_t
= [1, 1, 1, 1]			
Generation 9260: Best = 7.805774e+02	FES = 926100	Diversity = 3.6153e-08	k_t
= [4, 4, 4, 4]			
Generation 9270: Best = 7.805774e+02	FES = 927100	Diversity = 3.6587e-08	k_t
= [1, 1, 1, 1]			
Generation 9280: Best = 7.805774e+02	FES = 928100	Diversity = 3.7372e-08	k_t
= [4, 4, 3, 4]			
Generation 9290: Best = 7.805774e+02	FES = 929100	Diversity = 3.7014e-08	k_t
= [1, 1, 1, 1]			
Generation 9300: Best = 7.805774e+02	FES = 930100	Diversity = 3.6489e-08	k_t
= [4, 2, 3, 3]			
Generation 9310: Best = 7.805774e+02	FES = 931100	Diversity = 3.6135e-08	k_t
= [1, 1, 1, 1]			
Generation 9320: Best = 7.805774e+02	FES = 932100	Diversity = 3.5657e-08	k_t
= [3, 3, 4, 4]			
Generation 9330: Best = 7.805774e+02	FES = 933100	Diversity = 3.6214e-08	k_t
= [1, 1, 1, 1]			
Generation 9340: Best = 7.805774e+02	FES = 934100	Diversity = 3.6393e-08	k_t
= [3, 2, 2, 4]			
Generation 9350: Best = 7.805774e+02	FES = 935100	Diversity = 3.5806e-08	k_t
= [1, 1, 1, 1]			
Generation 9360: Best = 7.805774e+02	FES = 936100	Diversity = 3.6243e-08	k_t
= [3, 2, 2, 3]			
Generation 9370: Best = 7.805774e+02	FES = 937100	Diversity = 3.5799e-08	k_t
= [1, 1, 1, 1]			
Generation 9380: Best = 7.805774e+02	FES = 938100	Diversity = 3.5760e-08	k_t
= [2, 4, 2, 4]			
Generation 9390: Best = 7.805774e+02	FES = 939100	Diversity = 3.5867e-08	k_t
= [1, 1, 1, 1]			
Generation 9400: Best = 7.805774e+02	FES = 940100	Diversity = 3.5570e-08	k_t
= [3, 2, 2, 4]			

Generation 9410: Best = 7.805774e+02	FES = 941100	Diversity = 3.5587e-08	k_t
= [1, 1, 1, 1]			
Generation 9420: Best = 7.805774e+02	FES = 942100	Diversity = 3.5227e-08	k_t
= [2, 3, 3, 2]			
Generation 9430: Best = 7.805774e+02	FES = 943100	Diversity = 3.5500e-08	k_t
= [1, 1, 1, 1]			
Generation 9440: Best = 7.805774e+02	FES = 944100	Diversity = 3.6059e-08	k_t
= [3, 2, 2, 4]			
Generation 9450: Best = 7.805774e+02	FES = 945100	Diversity = 3.6273e-08	k_t
= [1, 1, 1, 1]			
Generation 9460: Best = 7.805774e+02	FES = 946100	Diversity = 3.6350e-08	k_t
= [3, 2, 3, 3]			
Generation 9470: Best = 7.805774e+02	FES = 947100	Diversity = 3.7010e-08	k_t
= [1, 1, 1, 1]			
Generation 9480: Best = 7.805774e+02	FES = 948100	Diversity = 3.6619e-08	k_t
= [3, 3, 4, 4]			
Generation 9490: Best = 7.805774e+02	FES = 949100	Diversity = 3.6039e-08	k_t
= [1, 1, 1, 1]			
Generation 9500: Best = 7.805774e+02	FES = 950100	Diversity = 3.6194e-08	k_t
= [4, 4, 2, 3]			
Generation 9510: Best = 7.805774e+02	FES = 951100	Diversity = 3.6800e-08	k_t
= [1, 1, 1, 1]			
Generation 9520: Best = 7.805774e+02	FES = 952100	Diversity = 3.6091e-08	k_t
= [4, 3, 4, 3]			
Generation 9530: Best = 7.805774e+02	FES = 953100	Diversity = 3.6286e-08	k_t
= [1, 1, 1, 1]			
Generation 9540: Best = 7.805774e+02	FES = 954100	Diversity = 3.6361e-08	k_t
= [4, 4, 4, 3]			
Generation 9550: Best = 7.805774e+02	FES = 955100	Diversity = 3.6971e-08	k_t
= [1, 1, 1, 1]			
Generation 9560: Best = 7.805774e+02	FES = 956100	Diversity = 3.7014e-08	k_t
= [3, 2, 3, 4]			
Generation 9570: Best = 7.805774e+02	FES = 957100	Diversity = 3.6858e-08	k_t
= [1, 1, 1, 1]			
Generation 9580: Best = 7.805774e+02	FES = 958100	Diversity = 3.6575e-08	k_t
= [3, 2, 4, 3]			
Generation 9590: Best = 7.805774e+02	FES = 959100	Diversity = 3.6864e-08	k_t
= [1, 1, 1, 1]			
Generation 9600: Best = 7.805774e+02	FES = 960100	Diversity = 3.7210e-08	k_t
= [2, 4, 2, 2]			
Generation 9610: Best = 7.805774e+02	FES = 961100	Diversity = 3.6620e-08	k_t
= [1, 1, 1, 1]			
Generation 9620: Best = 7.805774e+02	FES = 962100	Diversity = 3.6117e-08	k_t
= [2, 4, 3, 2]			
Generation 9630: Best = 7.805774e+02	FES = 963100	Diversity = 3.6159e-08	k_t
= [1, 1, 1, 1]			
Generation 9640: Best = 7.805774e+02	FES = 964100	Diversity = 3.6418e-08	k_t
= [4, 3, 3, 3]			
Generation 9650: Best = 7.805774e+02	FES = 965100	Diversity = 3.5386e-08	k_t
= [1, 1, 1, 1]			
Generation 9660: Best = 7.805774e+02	FES = 966100	Diversity = 3.5597e-08	k_t
= [2, 4, 2, 2]			
Generation 9670: Best = 7.805774e+02	FES = 967100	Diversity = 3.4635e-08	k_t
= [1, 1, 1, 1]			

Generation 9680: Best = 7.805774e+02 = [3, 4, 4, 4]	FES = 968100	Diversity = 3.5371e-08	k_t
Generation 9690: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 969100	Diversity = 3.5619e-08	k_t
Generation 9700: Best = 7.805774e+02 = [4, 4, 4, 2]	FES = 970100	Diversity = 3.6391e-08	k_t
Generation 9710: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 971100	Diversity = 3.6854e-08	k_t
Generation 9720: Best = 7.805774e+02 = [2, 2, 4, 2]	FES = 972100	Diversity = 3.7123e-08	k_t
Generation 9730: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 973100	Diversity = 3.7331e-08	k_t
Generation 9740: Best = 7.805774e+02 = [4, 4, 2, 2]	FES = 974100	Diversity = 3.7906e-08	k_t
Generation 9750: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 975100	Diversity = 3.7546e-08	k_t
Generation 9760: Best = 7.805774e+02 = [3, 3, 4, 3]	FES = 976100	Diversity = 3.7449e-08	k_t
Generation 9770: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 977100	Diversity = 3.7265e-08	k_t
Generation 9780: Best = 7.805774e+02 = [2, 4, 4, 4]	FES = 978100	Diversity = 3.7470e-08	k_t
Generation 9790: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 979100	Diversity = 3.6942e-08	k_t
Generation 9800: Best = 7.805774e+02 = [4, 3, 3, 2]	FES = 980100	Diversity = 3.6513e-08	k_t
Generation 9810: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 981100	Diversity = 3.6093e-08	k_t
Generation 9820: Best = 7.805774e+02 = [2, 2, 2, 2]	FES = 982100	Diversity = 3.5922e-08	k_t
Generation 9830: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 983100	Diversity = 3.5451e-08	k_t
Generation 9840: Best = 7.805774e+02 = [4, 2, 2, 4]	FES = 984100	Diversity = 3.5418e-08	k_t
Generation 9850: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 985100	Diversity = 3.5676e-08	k_t
Generation 9860: Best = 7.805774e+02 = [3, 4, 3, 4]	FES = 986100	Diversity = 3.5611e-08	k_t
Generation 9870: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 987100	Diversity = 3.5543e-08	k_t
Generation 9880: Best = 7.805774e+02 = [4, 4, 3, 4]	FES = 988100	Diversity = 3.5808e-08	k_t
Generation 9890: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 989100	Diversity = 3.5313e-08	k_t
Generation 9900: Best = 7.805774e+02 = [4, 2, 3, 3]	FES = 990100	Diversity = 3.5291e-08	k_t
Generation 9910: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 991100	Diversity = 3.5096e-08	k_t
Generation 9920: Best = 7.805774e+02 = [4, 4, 4, 2]	FES = 992100	Diversity = 3.4800e-08	k_t
Generation 9930: Best = 7.805774e+02 = [1, 1, 1, 1]	FES = 993100	Diversity = 3.4524e-08	k_t
Generation 9940: Best = 7.805774e+02 = [2, 4, 2, 3]	FES = 994100	Diversity = 3.5220e-08	k_t

Generation 9950: Best = 7.805774e+02 FES = 995100 Diversity = 3.5579e-08 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 7.805774e+02 FES = 996100 Diversity = 3.5076e-08 k_t
= [3, 2, 3, 2]
Generation 9970: Best = 7.805774e+02 FES = 997100 Diversity = 3.5023e-08 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 7.805774e+02 FES = 998100 Diversity = 3.4862e-08 k_t
= [4, 3, 2, 3]
Generation 9990: Best = 7.805774e+02 FES = 999100 Diversity = 3.5307e-08 k_t
= [1, 1, 1, 1]

Optimization completed!

Final Best Fitness = 7.805774e+02

Total Function Evaluations = 1000000

=====

=

Results:

Best Fitness : 7.805774e+02

Global Optimum : 5.000000e+02

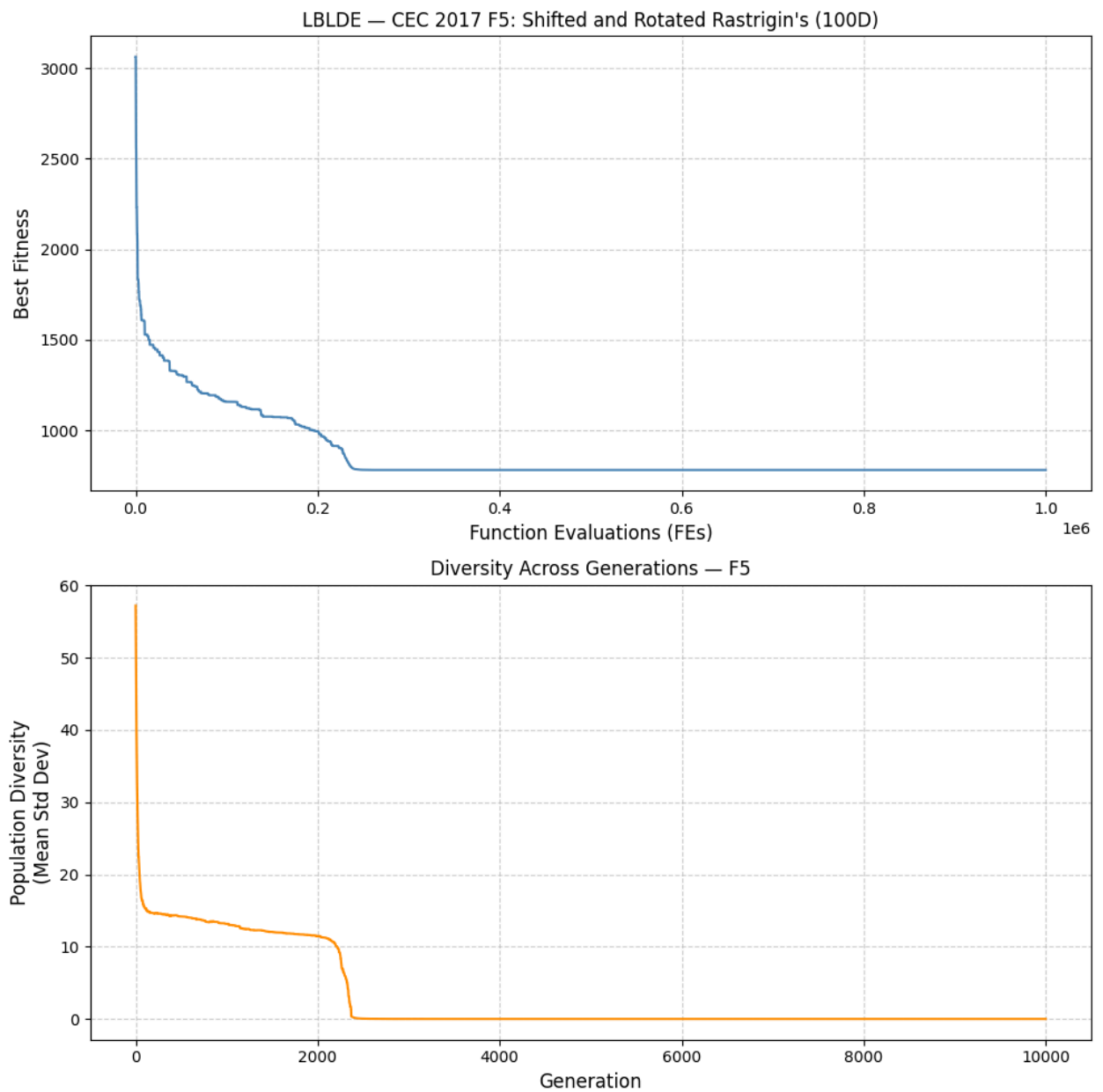
Error (|best-optimum|) : 2.805774e+02

Best Solution (5 dim) : [-29.13959033 73.1607261 -25.41103249 5.9734855
-15.79275569]

D^(0) (init. diversity): 5.7220e+01

=====

=



Graph saved as convergence_F5.png

```
In [30]: # =====
# F6 – Shifted and Rotated Schaffer's F7
# =====
run_and_plot('F6', make_wrapper(simple.f6), 600.0, "Shifted and Rotated Schaff
```

```

=====
=
LBLDE_s on CEC 2017 F6 – Single Run
=====
=
Function      : Shifted and Rotated Schaffer's F7 (100D)
Bounds       : [-100, 100]
Global optimum : 600.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 7.936410e+02 | Diversity = 5.7220e+01
Generation 10: Best = 7.526629e+02 FES = 1100 Diversity = 4.3520e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 7.174827e+02 FES = 2100 Diversity = 3.4514e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 7.030206e+02 FES = 3100 Diversity = 2.7000e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 6.931588e+02 FES = 4100 Diversity = 2.2554e+01 k_t
= [4, 2, 2, 4]
Generation 50: Best = 6.834784e+02 FES = 5100 Diversity = 1.8771e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 6.759668e+02 FES = 6100 Diversity = 1.6690e+01 k_t
= [4, 4, 2, 3]
Generation 70: Best = 6.691880e+02 FES = 7100 Diversity = 1.5593e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 6.671483e+02 FES = 8100 Diversity = 1.4463e+01 k_t
= [4, 3, 2, 3]
Generation 90: Best = 6.607705e+02 FES = 9100 Diversity = 1.3578e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 6.588117e+02 FES = 10100 Diversity = 1.3025e+01 k_t
= [3, 3, 2, 3]
Generation 110: Best = 6.559106e+02 FES = 11100 Diversity = 1.1870e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 6.538023e+02 FES = 12100 Diversity = 1.1201e+01 k_t
= [4, 2, 2, 3]
Generation 130: Best = 6.525239e+02 FES = 13100 Diversity = 1.0579e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 6.510343e+02 FES = 14100 Diversity = 9.8400e+00 k_t
= [4, 4, 2, 2]
Generation 150: Best = 6.488773e+02 FES = 15100 Diversity = 9.1593e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 6.460294e+02 FES = 16100 Diversity = 8.7304e+00 k_t
= [3, 3, 4, 4]
Generation 170: Best = 6.447922e+02 FES = 17100 Diversity = 8.4796e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 6.438647e+02 FES = 18100 Diversity = 8.1701e+00 k_t
= [2, 4, 3, 4]
Generation 190: Best = 6.427042e+02 FES = 19100 Diversity = 7.9262e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 6.427042e+02 FES = 20100 Diversity = 7.8488e+00 k_t
= [3, 3, 3, 4]
Generation 210: Best = 6.420587e+02 FES = 21100 Diversity = 7.7653e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 6.417615e+02 FES = 22100 Diversity = 7.5610e+00 k_t
= [4, 2, 3, 3]

```

Generation 230:	Best = 6.405584e+02	FES = 23100	Diversity = 7.4647e+00	k_t = [1, 1, 1, 1]
Generation 240:	Best = 6.399446e+02	FES = 24100	Diversity = 7.3908e+00	k_t = [2, 2, 3, 2]
Generation 250:	Best = 6.394119e+02	FES = 25100	Diversity = 7.3261e+00	k_t = [1, 1, 1, 1]
Generation 260:	Best = 6.376511e+02	FES = 26100	Diversity = 7.2359e+00	k_t = [4, 4, 2, 2]
Generation 270:	Best = 6.376511e+02	FES = 27100	Diversity = 7.2235e+00	k_t = [1, 1, 1, 1]
Generation 280:	Best = 6.374205e+02	FES = 28100	Diversity = 7.0614e+00	k_t = [4, 4, 4, 3]
Generation 290:	Best = 6.369167e+02	FES = 29100	Diversity = 6.9353e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 6.362322e+02	FES = 30100	Diversity = 6.8683e+00	k_t = [2, 2, 2, 4]
Generation 310:	Best = 6.359334e+02	FES = 31100	Diversity = 6.8560e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 6.359206e+02	FES = 32100	Diversity = 6.7748e+00	k_t = [2, 4, 3, 2]
Generation 330:	Best = 6.342083e+02	FES = 33100	Diversity = 6.6853e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 6.342083e+02	FES = 34100	Diversity = 6.5968e+00	k_t = [4, 4, 3, 2]
Generation 350:	Best = 6.335301e+02	FES = 35100	Diversity = 6.5038e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 6.335301e+02	FES = 36100	Diversity = 6.5231e+00	k_t = [3, 4, 2, 2]
Generation 370:	Best = 6.330650e+02	FES = 37100	Diversity = 6.4569e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 6.330650e+02	FES = 38100	Diversity = 6.4698e+00	k_t = [3, 3, 2, 4]
Generation 390:	Best = 6.327709e+02	FES = 39100	Diversity = 6.4346e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 6.327709e+02	FES = 40100	Diversity = 6.4788e+00	k_t = [3, 2, 2, 4]
Generation 410:	Best = 6.327134e+02	FES = 41100	Diversity = 6.5050e+00	k_t = [1, 1, 1, 1]
Generation 420:	Best = 6.326362e+02	FES = 42100	Diversity = 6.4462e+00	k_t = [4, 3, 4, 2]
Generation 430:	Best = 6.326362e+02	FES = 43100	Diversity = 6.3953e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 6.322329e+02	FES = 44100	Diversity = 6.4354e+00	k_t = [3, 4, 2, 3]
Generation 450:	Best = 6.322329e+02	FES = 45100	Diversity = 6.4585e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 6.322329e+02	FES = 46100	Diversity = 6.4763e+00	k_t = [2, 3, 3, 4]
Generation 470:	Best = 6.322329e+02	FES = 47100	Diversity = 6.4966e+00	k_t = [1, 1, 1, 1]
Generation 480:	Best = 6.322329e+02	FES = 48100	Diversity = 6.4816e+00	k_t = [2, 4, 2, 2]
Generation 490:	Best = 6.311556e+02	FES = 49100	Diversity = 6.4608e+00	k_t = [1, 1, 1, 1]

Generation 500:	Best = 6.303130e+02	FES = 50100	Diversity = 6.4111e+00	k_t = [4, 2, 4, 2]
Generation 510:	Best = 6.303130e+02	FES = 51100	Diversity = 6.3285e+00	k_t = [1, 1, 1, 1]
Generation 520:	Best = 6.302421e+02	FES = 52100	Diversity = 6.3110e+00	k_t = [3, 2, 2, 4]
Generation 530:	Best = 6.302421e+02	FES = 53100	Diversity = 6.3232e+00	k_t = [1, 1, 1, 1]
Generation 540:	Best = 6.302421e+02	FES = 54100	Diversity = 6.3356e+00	k_t = [4, 4, 3, 4]
Generation 550:	Best = 6.302421e+02	FES = 55100	Diversity = 6.2281e+00	k_t = [1, 1, 1, 1]
Generation 560:	Best = 6.302421e+02	FES = 56100	Diversity = 6.1897e+00	k_t = [2, 2, 3, 2]
Generation 570:	Best = 6.296223e+02	FES = 57100	Diversity = 6.1642e+00	k_t = [1, 1, 1, 1]
Generation 580:	Best = 6.296223e+02	FES = 58100	Diversity = 6.1447e+00	k_t = [4, 2, 4, 4]
Generation 590:	Best = 6.296223e+02	FES = 59100	Diversity = 6.1227e+00	k_t = [1, 1, 1, 1]
Generation 600:	Best = 6.296223e+02	FES = 60100	Diversity = 6.0858e+00	k_t = [2, 3, 4, 4]
Generation 610:	Best = 6.296223e+02	FES = 61100	Diversity = 5.9978e+00	k_t = [1, 1, 1, 1]
Generation 620:	Best = 6.296223e+02	FES = 62100	Diversity = 5.9722e+00	k_t = [2, 2, 4, 4]
Generation 630:	Best = 6.296223e+02	FES = 63100	Diversity = 5.9641e+00	k_t = [1, 1, 1, 1]
Generation 640:	Best = 6.295420e+02	FES = 64100	Diversity = 5.9674e+00	k_t = [4, 4, 3, 4]
Generation 650:	Best = 6.289689e+02	FES = 65100	Diversity = 5.9727e+00	k_t = [1, 1, 1, 1]
Generation 660:	Best = 6.288575e+02	FES = 66100	Diversity = 5.9597e+00	k_t = [2, 3, 2, 3]
Generation 670:	Best = 6.288575e+02	FES = 67100	Diversity = 5.9546e+00	k_t = [1, 1, 1, 1]
Generation 680:	Best = 6.287567e+02	FES = 68100	Diversity = 5.8804e+00	k_t = [4, 4, 4, 4]
Generation 690:	Best = 6.283954e+02	FES = 69100	Diversity = 5.8848e+00	k_t = [1, 1, 1, 1]
Generation 700:	Best = 6.283954e+02	FES = 70100	Diversity = 5.8887e+00	k_t = [3, 4, 4, 2]
Generation 710:	Best = 6.283954e+02	FES = 71100	Diversity = 5.8585e+00	k_t = [1, 1, 1, 1]
Generation 720:	Best = 6.282927e+02	FES = 72100	Diversity = 5.7705e+00	k_t = [4, 3, 3, 4]
Generation 730:	Best = 6.266755e+02	FES = 73100	Diversity = 5.7658e+00	k_t = [1, 1, 1, 1]
Generation 740:	Best = 6.266755e+02	FES = 74100	Diversity = 5.7463e+00	k_t = [3, 2, 2, 3]
Generation 750:	Best = 6.266755e+02	FES = 75100	Diversity = 5.6913e+00	k_t = [1, 1, 1, 1]
Generation 760:	Best = 6.266755e+02	FES = 76100	Diversity = 5.6865e+00	k_t = [3, 2, 4, 3]

Generation 770:	Best = 6.266755e+02	FES = 77100	Diversity = 5.6644e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 6.266755e+02	FES = 78100	Diversity = 5.6226e+00	k_t
= [2, 3, 4, 3]				
Generation 790:	Best = 6.266755e+02	FES = 79100	Diversity = 5.6095e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 6.266755e+02	FES = 80100	Diversity = 5.6055e+00	k_t
= [4, 4, 2, 3]				
Generation 810:	Best = 6.266755e+02	FES = 81100	Diversity = 5.5640e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 6.266755e+02	FES = 82100	Diversity = 5.5249e+00	k_t
= [3, 4, 3, 3]				
Generation 830:	Best = 6.266755e+02	FES = 83100	Diversity = 5.4657e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 6.266755e+02	FES = 84100	Diversity = 5.4068e+00	k_t
= [3, 3, 3, 4]				
Generation 850:	Best = 6.265889e+02	FES = 85100	Diversity = 5.3814e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 6.265889e+02	FES = 86100	Diversity = 5.3743e+00	k_t
= [4, 3, 4, 2]				
Generation 870:	Best = 6.265889e+02	FES = 87100	Diversity = 5.3560e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 6.265889e+02	FES = 88100	Diversity = 5.3344e+00	k_t
= [3, 3, 2, 2]				
Generation 890:	Best = 6.265889e+02	FES = 89100	Diversity = 5.3201e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 6.265889e+02	FES = 90100	Diversity = 5.2925e+00	k_t
= [3, 4, 4, 3]				
Generation 910:	Best = 6.265889e+02	FES = 91100	Diversity = 5.2800e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 6.265889e+02	FES = 92100	Diversity = 5.2623e+00	k_t
= [4, 4, 4, 2]				
Generation 930:	Best = 6.265889e+02	FES = 93100	Diversity = 5.2771e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 6.265889e+02	FES = 94100	Diversity = 5.2669e+00	k_t
= [4, 3, 2, 4]				
Generation 950:	Best = 6.265889e+02	FES = 95100	Diversity = 5.2031e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 6.265889e+02	FES = 96100	Diversity = 5.1971e+00	k_t
= [3, 3, 4, 4]				
Generation 970:	Best = 6.261878e+02	FES = 97100	Diversity = 5.1727e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 6.261878e+02	FES = 98100	Diversity = 5.1562e+00	k_t
= [2, 4, 3, 2]				
Generation 990:	Best = 6.261878e+02	FES = 99100	Diversity = 5.1108e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 6.261878e+02	FES = 100100	Diversity = 5.0665e+00	k_t
= [2, 2, 2, 2]				
Generation 1010:	Best = 6.261878e+02	FES = 101100	Diversity = 5.0789e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 6.261878e+02	FES = 102100	Diversity = 5.0741e+00	k_t
= [4, 2, 4, 2]				
Generation 1030:	Best = 6.261878e+02	FES = 103100	Diversity = 5.0704e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 6.260950e+02	FES = 104100	Diversity = 5.0720e+00	k_t
= [3, 3, 4, 4]			
Generation 1050: Best = 6.256896e+02	FES = 105100	Diversity = 5.0722e+00	k_t
= [1, 1, 1, 1]			
Generation 1060: Best = 6.256896e+02	FES = 106100	Diversity = 5.0615e+00	k_t
= [4, 2, 4, 4]			
Generation 1070: Best = 6.256896e+02	FES = 107100	Diversity = 5.0497e+00	k_t
= [1, 1, 1, 1]			
Generation 1080: Best = 6.256896e+02	FES = 108100	Diversity = 5.0226e+00	k_t
= [3, 3, 3, 2]			
Generation 1090: Best = 6.252267e+02	FES = 109100	Diversity = 5.0073e+00	k_t
= [1, 1, 1, 1]			
Generation 1100: Best = 6.248513e+02	FES = 110100	Diversity = 5.0043e+00	k_t
= [2, 2, 2, 4]			
Generation 1110: Best = 6.248513e+02	FES = 111100	Diversity = 5.0060e+00	k_t
= [1, 1, 1, 1]			
Generation 1120: Best = 6.248513e+02	FES = 112100	Diversity = 4.9850e+00	k_t
= [3, 3, 2, 2]			
Generation 1130: Best = 6.248177e+02	FES = 113100	Diversity = 4.9605e+00	k_t
= [1, 1, 1, 1]			
Generation 1140: Best = 6.248177e+02	FES = 114100	Diversity = 4.9391e+00	k_t
= [4, 3, 2, 3]			
Generation 1150: Best = 6.248177e+02	FES = 115100	Diversity = 4.9445e+00	k_t
= [1, 1, 1, 1]			
Generation 1160: Best = 6.248177e+02	FES = 116100	Diversity = 4.9481e+00	k_t
= [2, 2, 3, 2]			
Generation 1170: Best = 6.248177e+02	FES = 117100	Diversity = 4.9458e+00	k_t
= [1, 1, 1, 1]			
Generation 1180: Best = 6.242438e+02	FES = 118100	Diversity = 4.9434e+00	k_t
= [2, 2, 2, 3]			
Generation 1190: Best = 6.229029e+02	FES = 119100	Diversity = 4.9407e+00	k_t
= [1, 1, 1, 1]			
Generation 1200: Best = 6.229029e+02	FES = 120100	Diversity = 4.9450e+00	k_t
= [4, 3, 3, 4]			
Generation 1210: Best = 6.229029e+02	FES = 121100	Diversity = 4.9502e+00	k_t
= [1, 1, 1, 1]			
Generation 1220: Best = 6.229029e+02	FES = 122100	Diversity = 4.9460e+00	k_t
= [4, 2, 3, 3]			
Generation 1230: Best = 6.227386e+02	FES = 123100	Diversity = 4.9047e+00	k_t
= [1, 1, 1, 1]			
Generation 1240: Best = 6.227386e+02	FES = 124100	Diversity = 4.9149e+00	k_t
= [3, 4, 4, 4]			
Generation 1250: Best = 6.227386e+02	FES = 125100	Diversity = 4.8453e+00	k_t
= [1, 1, 1, 1]			
Generation 1260: Best = 6.227386e+02	FES = 126100	Diversity = 4.7738e+00	k_t
= [2, 2, 2, 2]			
Generation 1270: Best = 6.227386e+02	FES = 127100	Diversity = 4.7702e+00	k_t
= [1, 1, 1, 1]			
Generation 1280: Best = 6.227386e+02	FES = 128100	Diversity = 4.7420e+00	k_t
= [4, 4, 4, 3]			
Generation 1290: Best = 6.227386e+02	FES = 129100	Diversity = 4.7095e+00	k_t
= [1, 1, 1, 1]			
Generation 1300: Best = 6.227386e+02	FES = 130100	Diversity = 4.6616e+00	k_t
= [4, 4, 2, 4]			

Generation 1310: Best = 6.227386e+02	FES = 131100	Diversity = 4.6606e+00	k_t
= [1, 1, 1, 1]			
Generation 1320: Best = 6.227137e+02	FES = 132100	Diversity = 4.5758e+00	k_t
= [4, 4, 3, 2]			
Generation 1330: Best = 6.225999e+02	FES = 133100	Diversity = 4.5695e+00	k_t
= [1, 1, 1, 1]			
Generation 1340: Best = 6.225999e+02	FES = 134100	Diversity = 4.5688e+00	k_t
= [2, 2, 3, 3]			
Generation 1350: Best = 6.225999e+02	FES = 135100	Diversity = 4.5793e+00	k_t
= [1, 1, 1, 1]			
Generation 1360: Best = 6.225999e+02	FES = 136100	Diversity = 4.5363e+00	k_t
= [3, 2, 4, 2]			
Generation 1370: Best = 6.225999e+02	FES = 137100	Diversity = 4.5356e+00	k_t
= [1, 1, 1, 1]			
Generation 1380: Best = 6.225999e+02	FES = 138100	Diversity = 4.5368e+00	k_t
= [4, 2, 3, 2]			
Generation 1390: Best = 6.225999e+02	FES = 139100	Diversity = 4.5351e+00	k_t
= [1, 1, 1, 1]			
Generation 1400: Best = 6.225877e+02	FES = 140100	Diversity = 4.5568e+00	k_t
= [2, 3, 3, 4]			
Generation 1410: Best = 6.225877e+02	FES = 141100	Diversity = 4.5552e+00	k_t
= [1, 1, 1, 1]			
Generation 1420: Best = 6.225151e+02	FES = 142100	Diversity = 4.5578e+00	k_t
= [3, 2, 2, 2]			
Generation 1430: Best = 6.225151e+02	FES = 143100	Diversity = 4.5565e+00	k_t
= [1, 1, 1, 1]			
Generation 1440: Best = 6.225151e+02	FES = 144100	Diversity = 4.5556e+00	k_t
= [2, 3, 3, 3]			
Generation 1450: Best = 6.225151e+02	FES = 145100	Diversity = 4.5507e+00	k_t
= [1, 1, 1, 1]			
Generation 1460: Best = 6.225151e+02	FES = 146100	Diversity = 4.5493e+00	k_t
= [4, 3, 2, 2]			
Generation 1470: Best = 6.224588e+02	FES = 147100	Diversity = 4.5385e+00	k_t
= [1, 1, 1, 1]			
Generation 1480: Best = 6.224588e+02	FES = 148100	Diversity = 4.5234e+00	k_t
= [4, 3, 3, 4]			
Generation 1490: Best = 6.223452e+02	FES = 149100	Diversity = 4.4950e+00	k_t
= [1, 1, 1, 1]			
Generation 1500: Best = 6.223452e+02	FES = 150100	Diversity = 4.4938e+00	k_t
= [2, 4, 3, 2]			
Generation 1510: Best = 6.223452e+02	FES = 151100	Diversity = 4.4930e+00	k_t
= [1, 1, 1, 1]			
Generation 1520: Best = 6.223452e+02	FES = 152100	Diversity = 4.4838e+00	k_t
= [4, 3, 3, 3]			
Generation 1530: Best = 6.223452e+02	FES = 153100	Diversity = 4.4843e+00	k_t
= [1, 1, 1, 1]			
Generation 1540: Best = 6.223452e+02	FES = 154100	Diversity = 4.4774e+00	k_t
= [4, 3, 2, 4]			
Generation 1550: Best = 6.223452e+02	FES = 155100	Diversity = 4.4757e+00	k_t
= [1, 1, 1, 1]			
Generation 1560: Best = 6.223452e+02	FES = 156100	Diversity = 4.4434e+00	k_t
= [4, 4, 4, 2]			
Generation 1570: Best = 6.223013e+02	FES = 157100	Diversity = 4.4080e+00	k_t
= [1, 1, 1, 1]			

Generation 1580: Best = 6.223013e+02 = [3, 4, 4, 4]	FES = 158100	Diversity = 4.4085e+00	k_t
Generation 1590: Best = 6.223013e+02 = [1, 1, 1, 1]	FES = 159100	Diversity = 4.4139e+00	k_t
Generation 1600: Best = 6.223013e+02 = [4, 2, 3, 3]	FES = 160100	Diversity = 4.3801e+00	k_t
Generation 1610: Best = 6.223013e+02 = [1, 1, 1, 1]	FES = 161100	Diversity = 4.3793e+00	k_t
Generation 1620: Best = 6.223013e+02 = [3, 4, 2, 2]	FES = 162100	Diversity = 4.3797e+00	k_t
Generation 1630: Best = 6.222821e+02 = [1, 1, 1, 1]	FES = 163100	Diversity = 4.3880e+00	k_t
Generation 1640: Best = 6.222821e+02 = [2, 3, 3, 3]	FES = 164100	Diversity = 4.3876e+00	k_t
Generation 1650: Best = 6.222821e+02 = [1, 1, 1, 1]	FES = 165100	Diversity = 4.3725e+00	k_t
Generation 1660: Best = 6.222821e+02 = [2, 2, 4, 4]	FES = 166100	Diversity = 4.3719e+00	k_t
Generation 1670: Best = 6.222821e+02 = [1, 1, 1, 1]	FES = 167100	Diversity = 4.3484e+00	k_t
Generation 1680: Best = 6.222821e+02 = [4, 4, 2, 3]	FES = 168100	Diversity = 4.3407e+00	k_t
Generation 1690: Best = 6.222518e+02 = [1, 1, 1, 1]	FES = 169100	Diversity = 4.3384e+00	k_t
Generation 1700: Best = 6.222068e+02 = [3, 3, 3, 2]	FES = 170100	Diversity = 4.3010e+00	k_t
Generation 1710: Best = 6.221551e+02 = [1, 1, 1, 1]	FES = 171100	Diversity = 4.2294e+00	k_t
Generation 1720: Best = 6.221551e+02 = [4, 4, 4, 3]	FES = 172100	Diversity = 4.2282e+00	k_t
Generation 1730: Best = 6.220390e+02 = [1, 1, 1, 1]	FES = 173100	Diversity = 4.2249e+00	k_t
Generation 1740: Best = 6.220390e+02 = [2, 2, 3, 4]	FES = 174100	Diversity = 4.2249e+00	k_t
Generation 1750: Best = 6.220036e+02 = [1, 1, 1, 1]	FES = 175100	Diversity = 4.2264e+00	k_t
Generation 1760: Best = 6.219927e+02 = [4, 2, 4, 4]	FES = 176100	Diversity = 4.2241e+00	k_t
Generation 1770: Best = 6.217925e+02 = [1, 1, 1, 1]	FES = 177100	Diversity = 4.2233e+00	k_t
Generation 1780: Best = 6.217925e+02 = [2, 2, 4, 2]	FES = 178100	Diversity = 4.1833e+00	k_t
Generation 1790: Best = 6.217925e+02 = [1, 1, 1, 1]	FES = 179100	Diversity = 4.1893e+00	k_t
Generation 1800: Best = 6.217925e+02 = [3, 4, 2, 4]	FES = 180100	Diversity = 4.1907e+00	k_t
Generation 1810: Best = 6.214452e+02 = [1, 1, 1, 1]	FES = 181100	Diversity = 4.1769e+00	k_t
Generation 1820: Best = 6.214452e+02 = [2, 4, 4, 2]	FES = 182100	Diversity = 4.1766e+00	k_t
Generation 1830: Best = 6.214452e+02 = [1, 1, 1, 1]	FES = 183100	Diversity = 4.1758e+00	k_t
Generation 1840: Best = 6.214452e+02 = [2, 2, 4, 2]	FES = 184100	Diversity = 4.1324e+00	k_t

Generation 1850: Best = 6.214452e+02 = [1, 1, 1, 1]	FES = 185100	Diversity = 4.1315e+00	k_t
Generation 1860: Best = 6.212991e+02 = [3, 3, 2, 2]	FES = 186100	Diversity = 4.1307e+00	k_t
Generation 1870: Best = 6.212991e+02 = [1, 1, 1, 1]	FES = 187100	Diversity = 4.1296e+00	k_t
Generation 1880: Best = 6.212991e+02 = [3, 4, 2, 2]	FES = 188100	Diversity = 4.1279e+00	k_t
Generation 1890: Best = 6.212669e+02 = [1, 1, 1, 1]	FES = 189100	Diversity = 4.1089e+00	k_t
Generation 1900: Best = 6.212669e+02 = [4, 2, 3, 2]	FES = 190100	Diversity = 4.1410e+00	k_t
Generation 1910: Best = 6.212669e+02 = [1, 1, 1, 1]	FES = 191100	Diversity = 4.0501e+00	k_t
Generation 1920: Best = 6.211737e+02 = [4, 2, 4, 4]	FES = 192100	Diversity = 3.9918e+00	k_t
Generation 1930: Best = 6.211737e+02 = [1, 1, 1, 1]	FES = 193100	Diversity = 3.9954e+00	k_t
Generation 1940: Best = 6.211737e+02 = [2, 3, 4, 3]	FES = 194100	Diversity = 3.9944e+00	k_t
Generation 1950: Best = 6.211737e+02 = [1, 1, 1, 1]	FES = 195100	Diversity = 3.9926e+00	k_t
Generation 1960: Best = 6.210854e+02 = [4, 2, 4, 4]	FES = 196100	Diversity = 3.9465e+00	k_t
Generation 1970: Best = 6.210854e+02 = [1, 1, 1, 1]	FES = 197100	Diversity = 3.9290e+00	k_t
Generation 1980: Best = 6.210854e+02 = [4, 2, 2, 2]	FES = 198100	Diversity = 3.9041e+00	k_t
Generation 1990: Best = 6.210854e+02 = [1, 1, 1, 1]	FES = 199100	Diversity = 3.9031e+00	k_t
Generation 2000: Best = 6.210854e+02 = [2, 2, 3, 3]	FES = 200100	Diversity = 3.8797e+00	k_t
Generation 2010: Best = 6.210854e+02 = [1, 1, 1, 1]	FES = 201100	Diversity = 3.8787e+00	k_t
Generation 2020: Best = 6.210854e+02 = [4, 3, 3, 2]	FES = 202100	Diversity = 3.8225e+00	k_t
Generation 2030: Best = 6.210854e+02 = [1, 1, 1, 1]	FES = 203100	Diversity = 3.8223e+00	k_t
Generation 2040: Best = 6.210854e+02 = [4, 3, 3, 4]	FES = 204100	Diversity = 3.7835e+00	k_t
Generation 2050: Best = 6.210854e+02 = [1, 1, 1, 1]	FES = 205100	Diversity = 3.7221e+00	k_t
Generation 2060: Best = 6.210854e+02 = [2, 2, 2, 3]	FES = 206100	Diversity = 3.6900e+00	k_t
Generation 2070: Best = 6.210854e+02 = [1, 1, 1, 1]	FES = 207100	Diversity = 3.6896e+00	k_t
Generation 2080: Best = 6.210459e+02 = [2, 4, 2, 4]	FES = 208100	Diversity = 3.6900e+00	k_t
Generation 2090: Best = 6.210459e+02 = [1, 1, 1, 1]	FES = 209100	Diversity = 3.6277e+00	k_t
Generation 2100: Best = 6.210459e+02 = [2, 4, 3, 2]	FES = 210100	Diversity = 3.6277e+00	k_t
Generation 2110: Best = 6.210336e+02 = [1, 1, 1, 1]	FES = 211100	Diversity = 3.6275e+00	k_t

Generation 2120: Best = 6.209171e+02 = [4, 3, 2, 3]	FES = 212100	Diversity = 3.6267e+00	k_t
Generation 2130: Best = 6.209171e+02 = [1, 1, 1, 1]	FES = 213100	Diversity = 3.6265e+00	k_t
Generation 2140: Best = 6.209171e+02 = [2, 2, 2, 2]	FES = 214100	Diversity = 3.6271e+00	k_t
Generation 2150: Best = 6.209171e+02 = [1, 1, 1, 1]	FES = 215100	Diversity = 3.6276e+00	k_t
Generation 2160: Best = 6.209171e+02 = [3, 2, 4, 3]	FES = 216100	Diversity = 3.6222e+00	k_t
Generation 2170: Best = 6.209171e+02 = [1, 1, 1, 1]	FES = 217100	Diversity = 3.6218e+00	k_t
Generation 2180: Best = 6.208376e+02 = [3, 3, 4, 4]	FES = 218100	Diversity = 3.6219e+00	k_t
Generation 2190: Best = 6.208376e+02 = [1, 1, 1, 1]	FES = 219100	Diversity = 3.6205e+00	k_t
Generation 2200: Best = 6.207662e+02 = [3, 4, 2, 3]	FES = 220100	Diversity = 3.6378e+00	k_t
Generation 2210: Best = 6.207662e+02 = [1, 1, 1, 1]	FES = 221100	Diversity = 3.6214e+00	k_t
Generation 2220: Best = 6.207662e+02 = [4, 3, 2, 4]	FES = 222100	Diversity = 3.6209e+00	k_t
Generation 2230: Best = 6.207662e+02 = [1, 1, 1, 1]	FES = 223100	Diversity = 3.6196e+00	k_t
Generation 2240: Best = 6.207662e+02 = [2, 2, 2, 3]	FES = 224100	Diversity = 3.6205e+00	k_t
Generation 2250: Best = 6.207662e+02 = [1, 1, 1, 1]	FES = 225100	Diversity = 3.6203e+00	k_t
Generation 2260: Best = 6.207662e+02 = [4, 3, 4, 4]	FES = 226100	Diversity = 3.6160e+00	k_t
Generation 2270: Best = 6.207662e+02 = [1, 1, 1, 1]	FES = 227100	Diversity = 3.6184e+00	k_t
Generation 2280: Best = 6.207662e+02 = [2, 2, 3, 3]	FES = 228100	Diversity = 3.6185e+00	k_t
Generation 2290: Best = 6.207662e+02 = [1, 1, 1, 1]	FES = 229100	Diversity = 3.6182e+00	k_t
Generation 2300: Best = 6.207662e+02 = [4, 2, 2, 2]	FES = 230100	Diversity = 3.6144e+00	k_t
Generation 2310: Best = 6.207662e+02 = [1, 1, 1, 1]	FES = 231100	Diversity = 3.6145e+00	k_t
Generation 2320: Best = 6.207662e+02 = [3, 2, 3, 4]	FES = 232100	Diversity = 3.6139e+00	k_t
Generation 2330: Best = 6.207654e+02 = [1, 1, 1, 1]	FES = 233100	Diversity = 3.6139e+00	k_t
Generation 2340: Best = 6.207654e+02 = [4, 2, 3, 3]	FES = 234100	Diversity = 3.6126e+00	k_t
Generation 2350: Best = 6.207152e+02 = [1, 1, 1, 1]	FES = 235100	Diversity = 3.6119e+00	k_t
Generation 2360: Best = 6.206350e+02 = [2, 4, 4, 4]	FES = 236100	Diversity = 3.5897e+00	k_t
Generation 2370: Best = 6.205459e+02 = [1, 1, 1, 1]	FES = 237100	Diversity = 3.5887e+00	k_t
Generation 2380: Best = 6.205459e+02 = [3, 3, 2, 3]	FES = 238100	Diversity = 3.5879e+00	k_t

Generation 2390: Best = 6.205154e+02	FES = 239100	Diversity = 3.5829e+00	k_t = [1, 1, 1, 1]
Generation 2400: Best = 6.205154e+02	FES = 240100	Diversity = 3.5828e+00	k_t = [2, 2, 2, 4]
Generation 2410: Best = 6.204429e+02	FES = 241100	Diversity = 3.5665e+00	k_t = [1, 1, 1, 1]
Generation 2420: Best = 6.204429e+02	FES = 242100	Diversity = 3.5657e+00	k_t = [2, 2, 2, 3]
Generation 2430: Best = 6.204429e+02	FES = 243100	Diversity = 3.5654e+00	k_t = [1, 1, 1, 1]
Generation 2440: Best = 6.204429e+02	FES = 244100	Diversity = 3.5655e+00	k_t = [2, 3, 4, 4]
Generation 2450: Best = 6.204429e+02	FES = 245100	Diversity = 3.5393e+00	k_t = [1, 1, 1, 1]
Generation 2460: Best = 6.203146e+02	FES = 246100	Diversity = 3.5112e+00	k_t = [3, 2, 3, 2]
Generation 2470: Best = 6.203146e+02	FES = 247100	Diversity = 3.4805e+00	k_t = [1, 1, 1, 1]
Generation 2480: Best = 6.203146e+02	FES = 248100	Diversity = 3.4083e+00	k_t = [2, 3, 4, 4]
Generation 2490: Best = 6.203135e+02	FES = 249100	Diversity = 3.3739e+00	k_t = [1, 1, 1, 1]
Generation 2500: Best = 6.202071e+02	FES = 250100	Diversity = 3.3740e+00	k_t = [4, 3, 2, 4]
Generation 2510: Best = 6.202071e+02	FES = 251100	Diversity = 3.3338e+00	k_t = [1, 1, 1, 1]
Generation 2520: Best = 6.200419e+02	FES = 252100	Diversity = 3.3337e+00	k_t = [3, 3, 2, 3]
Generation 2530: Best = 6.200419e+02	FES = 253100	Diversity = 3.3266e+00	k_t = [1, 1, 1, 1]
Generation 2540: Best = 6.200419e+02	FES = 254100	Diversity = 3.2410e+00	k_t = [2, 3, 3, 4]
Generation 2550: Best = 6.199934e+02	FES = 255100	Diversity = 3.1588e+00	k_t = [1, 1, 1, 1]
Generation 2560: Best = 6.199934e+02	FES = 256100	Diversity = 3.0259e+00	k_t = [4, 2, 4, 3]
Generation 2570: Best = 6.199934e+02	FES = 257100	Diversity = 3.0165e+00	k_t = [1, 1, 1, 1]
Generation 2580: Best = 6.199932e+02	FES = 258100	Diversity = 2.9865e+00	k_t = [3, 2, 4, 4]
Generation 2590: Best = 6.199932e+02	FES = 259100	Diversity = 2.9351e+00	k_t = [1, 1, 1, 1]
Generation 2600: Best = 6.199932e+02	FES = 260100	Diversity = 2.8364e+00	k_t = [2, 3, 2, 3]
Generation 2610: Best = 6.199283e+02	FES = 261100	Diversity = 2.7983e+00	k_t = [1, 1, 1, 1]
Generation 2620: Best = 6.199283e+02	FES = 262100	Diversity = 2.7848e+00	k_t = [4, 4, 2, 2]
Generation 2630: Best = 6.199205e+02	FES = 263100	Diversity = 2.7733e+00	k_t = [1, 1, 1, 1]
Generation 2640: Best = 6.198538e+02	FES = 264100	Diversity = 2.7487e+00	k_t = [4, 4, 3, 3]
Generation 2650: Best = 6.198538e+02	FES = 265100	Diversity = 2.7487e+00	k_t = [1, 1, 1, 1]

Generation 2660: Best = 6.198538e+02	FES = 266100	Diversity = 2.7482e+00	k_t
= [3, 4, 4, 4]			
Generation 2670: Best = 6.197923e+02	FES = 267100	Diversity = 2.7478e+00	k_t
= [1, 1, 1, 1]			
Generation 2680: Best = 6.197923e+02	FES = 268100	Diversity = 2.7414e+00	k_t
= [2, 2, 3, 3]			
Generation 2690: Best = 6.197796e+02	FES = 269100	Diversity = 2.6996e+00	k_t
= [1, 1, 1, 1]			
Generation 2700: Best = 6.197796e+02	FES = 270100	Diversity = 2.6795e+00	k_t
= [4, 3, 4, 4]			
Generation 2710: Best = 6.197796e+02	FES = 271100	Diversity = 2.5474e+00	k_t
= [1, 1, 1, 1]			
Generation 2720: Best = 6.197796e+02	FES = 272100	Diversity = 2.4952e+00	k_t
= [2, 4, 4, 3]			
Generation 2730: Best = 6.197495e+02	FES = 273100	Diversity = 2.4496e+00	k_t
= [1, 1, 1, 1]			
Generation 2740: Best = 6.196649e+02	FES = 274100	Diversity = 2.4282e+00	k_t
= [2, 2, 4, 4]			
Generation 2750: Best = 6.193837e+02	FES = 275100	Diversity = 2.4068e+00	k_t
= [1, 1, 1, 1]			
Generation 2760: Best = 6.193837e+02	FES = 276100	Diversity = 2.3933e+00	k_t
= [3, 4, 3, 4]			
Generation 2770: Best = 6.193777e+02	FES = 277100	Diversity = 2.3673e+00	k_t
= [1, 1, 1, 1]			
Generation 2780: Best = 6.193323e+02	FES = 278100	Diversity = 2.2387e+00	k_t
= [3, 2, 2, 3]			
Generation 2790: Best = 6.193310e+02	FES = 279100	Diversity = 2.1698e+00	k_t
= [1, 1, 1, 1]			
Generation 2800: Best = 6.193310e+02	FES = 280100	Diversity = 2.0994e+00	k_t
= [2, 3, 4, 2]			
Generation 2810: Best = 6.193310e+02	FES = 281100	Diversity = 1.9153e+00	k_t
= [1, 1, 1, 1]			
Generation 2820: Best = 6.193310e+02	FES = 282100	Diversity = 1.8953e+00	k_t
= [4, 4, 4, 2]			
Generation 2830: Best = 6.193310e+02	FES = 283100	Diversity = 1.8220e+00	k_t
= [1, 1, 1, 1]			
Generation 2840: Best = 6.193310e+02	FES = 284100	Diversity = 1.8205e+00	k_t
= [4, 4, 4, 3]			
Generation 2850: Best = 6.192206e+02	FES = 285100	Diversity = 1.7849e+00	k_t
= [1, 1, 1, 1]			
Generation 2860: Best = 6.191384e+02	FES = 286100	Diversity = 1.7362e+00	k_t
= [3, 2, 4, 3]			
Generation 2870: Best = 6.191384e+02	FES = 287100	Diversity = 1.5847e+00	k_t
= [1, 1, 1, 1]			
Generation 2880: Best = 6.191384e+02	FES = 288100	Diversity = 1.5562e+00	k_t
= [2, 3, 4, 3]			
Generation 2890: Best = 6.191127e+02	FES = 289100	Diversity = 1.4527e+00	k_t
= [1, 1, 1, 1]			
Generation 2900: Best = 6.190198e+02	FES = 290100	Diversity = 1.2503e+00	k_t
= [4, 4, 3, 2]			
Generation 2910: Best = 6.190198e+02	FES = 291100	Diversity = 1.2509e+00	k_t
= [1, 1, 1, 1]			
Generation 2920: Best = 6.190198e+02	FES = 292100	Diversity = 1.2413e+00	k_t
= [3, 3, 2, 4]			

Generation 2930: Best = 6.188955e+02 = [1, 1, 1, 1]	FES = 293100	Diversity = 1.1009e+00	k_t
Generation 2940: Best = 6.187784e+02 = [2, 3, 2, 4]	FES = 294100	Diversity = 1.0508e+00	k_t
Generation 2950: Best = 6.187784e+02 = [1, 1, 1, 1]	FES = 295100	Diversity = 9.8193e-01	k_t
Generation 2960: Best = 6.187784e+02 = [3, 4, 3, 3]	FES = 296100	Diversity = 9.8431e-01	k_t
Generation 2970: Best = 6.187784e+02 = [1, 1, 1, 1]	FES = 297100	Diversity = 9.6926e-01	k_t
Generation 2980: Best = 6.187784e+02 = [4, 4, 3, 4]	FES = 298100	Diversity = 9.0921e-01	k_t
Generation 2990: Best = 6.187248e+02 = [1, 1, 1, 1]	FES = 299100	Diversity = 9.1047e-01	k_t
Generation 3000: Best = 6.187248e+02 = [2, 4, 2, 3]	FES = 300100	Diversity = 9.0968e-01	k_t
Generation 3010: Best = 6.186986e+02 = [1, 1, 1, 1]	FES = 301100	Diversity = 9.0899e-01	k_t
Generation 3020: Best = 6.186986e+02 = [2, 2, 2, 3]	FES = 302100	Diversity = 8.3043e-01	k_t
Generation 3030: Best = 6.186597e+02 = [1, 1, 1, 1]	FES = 303100	Diversity = 8.3031e-01	k_t
Generation 3040: Best = 6.186061e+02 = [3, 2, 2, 3]	FES = 304100	Diversity = 8.2867e-01	k_t
Generation 3050: Best = 6.185520e+02 = [1, 1, 1, 1]	FES = 305100	Diversity = 8.2718e-01	k_t
Generation 3060: Best = 6.184561e+02 = [2, 3, 4, 2]	FES = 306100	Diversity = 8.0529e-01	k_t
Generation 3070: Best = 6.184561e+02 = [1, 1, 1, 1]	FES = 307100	Diversity = 8.0594e-01	k_t
Generation 3080: Best = 6.184561e+02 = [2, 3, 2, 4]	FES = 308100	Diversity = 8.0563e-01	k_t
Generation 3090: Best = 6.183995e+02 = [1, 1, 1, 1]	FES = 309100	Diversity = 7.1869e-01	k_t
Generation 3100: Best = 6.183391e+02 = [3, 3, 3, 3]	FES = 310100	Diversity = 7.0561e-01	k_t
Generation 3110: Best = 6.183391e+02 = [1, 1, 1, 1]	FES = 311100	Diversity = 7.0653e-01	k_t
Generation 3120: Best = 6.182842e+02 = [4, 2, 3, 3]	FES = 312100	Diversity = 7.0661e-01	k_t
Generation 3130: Best = 6.182232e+02 = [1, 1, 1, 1]	FES = 313100	Diversity = 7.0677e-01	k_t
Generation 3140: Best = 6.182141e+02 = [3, 2, 2, 4]	FES = 314100	Diversity = 7.0640e-01	k_t
Generation 3150: Best = 6.180757e+02 = [1, 1, 1, 1]	FES = 315100	Diversity = 7.0670e-01	k_t
Generation 3160: Best = 6.180757e+02 = [2, 3, 2, 3]	FES = 316100	Diversity = 7.0545e-01	k_t
Generation 3170: Best = 6.180364e+02 = [1, 1, 1, 1]	FES = 317100	Diversity = 7.0476e-01	k_t
Generation 3180: Best = 6.180364e+02 = [2, 4, 3, 4]	FES = 318100	Diversity = 5.6963e-01	k_t
Generation 3190: Best = 6.180355e+02 = [1, 1, 1, 1]	FES = 319100	Diversity = 5.6547e-01	k_t

Generation 3200: Best = 6.179507e+02	FES = 320100	Diversity = 5.6100e-01	k_t
= [2, 2, 2, 3]			
Generation 3210: Best = 6.179507e+02	FES = 321100	Diversity = 5.5984e-01	k_t
= [1, 1, 1, 1]			
Generation 3220: Best = 6.179025e+02	FES = 322100	Diversity = 5.5867e-01	k_t
= [3, 2, 3, 2]			
Generation 3230: Best = 6.179025e+02	FES = 323100	Diversity = 5.5249e-01	k_t
= [1, 1, 1, 1]			
Generation 3240: Best = 6.178712e+02	FES = 324100	Diversity = 5.5176e-01	k_t
= [4, 4, 2, 3]			
Generation 3250: Best = 6.178402e+02	FES = 325100	Diversity = 5.4751e-01	k_t
= [1, 1, 1, 1]			
Generation 3260: Best = 6.178402e+02	FES = 326100	Diversity = 5.4242e-01	k_t
= [2, 4, 3, 4]			
Generation 3270: Best = 6.178326e+02	FES = 327100	Diversity = 5.3314e-01	k_t
= [1, 1, 1, 1]			
Generation 3280: Best = 6.178298e+02	FES = 328100	Diversity = 5.2452e-01	k_t
= [2, 4, 3, 2]			
Generation 3290: Best = 6.178150e+02	FES = 329100	Diversity = 5.0668e-01	k_t
= [1, 1, 1, 1]			
Generation 3300: Best = 6.178150e+02	FES = 330100	Diversity = 4.9751e-01	k_t
= [4, 3, 2, 4]			
Generation 3310: Best = 6.177821e+02	FES = 331100	Diversity = 4.9293e-01	k_t
= [1, 1, 1, 1]			
Generation 3320: Best = 6.177761e+02	FES = 332100	Diversity = 4.8929e-01	k_t
= [2, 3, 3, 4]			
Generation 3330: Best = 6.177528e+02	FES = 333100	Diversity = 4.5977e-01	k_t
= [1, 1, 1, 1]			
Generation 3340: Best = 6.177085e+02	FES = 334100	Diversity = 4.5913e-01	k_t
= [3, 4, 2, 3]			
Generation 3350: Best = 6.176946e+02	FES = 335100	Diversity = 4.0877e-01	k_t
= [1, 1, 1, 1]			
Generation 3360: Best = 6.176272e+02	FES = 336100	Diversity = 4.0056e-01	k_t
= [4, 4, 2, 4]			
Generation 3370: Best = 6.176262e+02	FES = 337100	Diversity = 3.8094e-01	k_t
= [1, 1, 1, 1]			
Generation 3380: Best = 6.176118e+02	FES = 338100	Diversity = 3.7085e-01	k_t
= [3, 3, 2, 3]			
Generation 3390: Best = 6.176118e+02	FES = 339100	Diversity = 3.5814e-01	k_t
= [1, 1, 1, 1]			
Generation 3400: Best = 6.175661e+02	FES = 340100	Diversity = 2.9458e-01	k_t
= [3, 2, 4, 2]			
Generation 3410: Best = 6.175661e+02	FES = 341100	Diversity = 2.5520e-01	k_t
= [1, 1, 1, 1]			
Generation 3420: Best = 6.175481e+02	FES = 342100	Diversity = 2.2576e-01	k_t
= [2, 3, 3, 3]			
Generation 3430: Best = 6.175325e+02	FES = 343100	Diversity = 2.2513e-01	k_t
= [1, 1, 1, 1]			
Generation 3440: Best = 6.174956e+02	FES = 344100	Diversity = 2.2437e-01	k_t
= [2, 2, 3, 2]			
Generation 3450: Best = 6.174693e+02	FES = 345100	Diversity = 2.2550e-01	k_t
= [1, 1, 1, 1]			
Generation 3460: Best = 6.174450e+02	FES = 346100	Diversity = 1.8214e-01	k_t
= [3, 2, 2, 2]			

Generation 3470: Best = 6.174429e+02 = [1, 1, 1, 1]	FES = 347100	Diversity = 1.8013e-01	k_t
Generation 3480: Best = 6.174010e+02 = [2, 3, 4, 2]	FES = 348100	Diversity = 1.7968e-01	k_t
Generation 3490: Best = 6.173211e+02 = [1, 1, 1, 1]	FES = 349100	Diversity = 1.8026e-01	k_t
Generation 3500: Best = 6.173061e+02 = [4, 4, 3, 3]	FES = 350100	Diversity = 1.7889e-01	k_t
Generation 3510: Best = 6.172959e+02 = [1, 1, 1, 1]	FES = 351100	Diversity = 1.7791e-01	k_t
Generation 3520: Best = 6.172909e+02 = [2, 3, 4, 2]	FES = 352100	Diversity = 1.7516e-01	k_t
Generation 3530: Best = 6.172909e+02 = [1, 1, 1, 1]	FES = 353100	Diversity = 1.7192e-01	k_t
Generation 3540: Best = 6.172367e+02 = [3, 3, 4, 3]	FES = 354100	Diversity = 1.7048e-01	k_t
Generation 3550: Best = 6.172190e+02 = [1, 1, 1, 1]	FES = 355100	Diversity = 1.5464e-01	k_t
Generation 3560: Best = 6.171300e+02 = [3, 4, 3, 3]	FES = 356100	Diversity = 1.5435e-01	k_t
Generation 3570: Best = 6.170879e+02 = [1, 1, 1, 1]	FES = 357100	Diversity = 1.5354e-01	k_t
Generation 3580: Best = 6.170723e+02 = [4, 2, 3, 3]	FES = 358100	Diversity = 1.4765e-01	k_t
Generation 3590: Best = 6.170237e+02 = [1, 1, 1, 1]	FES = 359100	Diversity = 1.4713e-01	k_t
Generation 3600: Best = 6.170193e+02 = [4, 3, 3, 3]	FES = 360100	Diversity = 1.4684e-01	k_t
Generation 3610: Best = 6.170185e+02 = [1, 1, 1, 1]	FES = 361100	Diversity = 1.3852e-01	k_t
Generation 3620: Best = 6.170173e+02 = [4, 3, 3, 3]	FES = 362100	Diversity = 1.3005e-01	k_t
Generation 3630: Best = 6.170044e+02 = [1, 1, 1, 1]	FES = 363100	Diversity = 1.2507e-01	k_t
Generation 3640: Best = 6.169986e+02 = [2, 2, 2, 2]	FES = 364100	Diversity = 1.2115e-01	k_t
Generation 3650: Best = 6.169333e+02 = [1, 1, 1, 1]	FES = 365100	Diversity = 1.1713e-01	k_t
Generation 3660: Best = 6.169333e+02 = [2, 3, 4, 2]	FES = 366100	Diversity = 1.1325e-01	k_t
Generation 3670: Best = 6.168973e+02 = [1, 1, 1, 1]	FES = 367100	Diversity = 1.0776e-01	k_t
Generation 3680: Best = 6.168883e+02 = [3, 2, 4, 4]	FES = 368100	Diversity = 1.0569e-01	k_t
Generation 3690: Best = 6.168714e+02 = [1, 1, 1, 1]	FES = 369100	Diversity = 1.0315e-01	k_t
Generation 3700: Best = 6.168714e+02 = [3, 4, 2, 2]	FES = 370100	Diversity = 9.1025e-02	k_t
Generation 3710: Best = 6.168656e+02 = [1, 1, 1, 1]	FES = 371100	Diversity = 8.3692e-02	k_t
Generation 3720: Best = 6.168167e+02 = [4, 2, 3, 2]	FES = 372100	Diversity = 7.5308e-02	k_t
Generation 3730: Best = 6.168125e+02 = [1, 1, 1, 1]	FES = 373100	Diversity = 7.0380e-02	k_t

Generation 3740: Best = 6.167556e+02	FES = 374100	Diversity = 6.3039e-02	k_t
= [2, 2, 4, 4]			
Generation 3750: Best = 6.167484e+02	FES = 375100	Diversity = 5.5644e-02	k_t
= [1, 1, 1, 1]			
Generation 3760: Best = 6.167319e+02	FES = 376100	Diversity = 5.1349e-02	k_t
= [2, 2, 3, 4]			
Generation 3770: Best = 6.167188e+02	FES = 377100	Diversity = 4.3222e-02	k_t
= [1, 1, 1, 1]			
Generation 3780: Best = 6.166924e+02	FES = 378100	Diversity = 4.1166e-02	k_t
= [3, 3, 3, 3]			
Generation 3790: Best = 6.166805e+02	FES = 379100	Diversity = 3.8168e-02	k_t
= [1, 1, 1, 1]			
Generation 3800: Best = 6.166787e+02	FES = 380100	Diversity = 3.5056e-02	k_t
= [3, 3, 2, 3]			
Generation 3810: Best = 6.166587e+02	FES = 381100	Diversity = 3.2977e-02	k_t
= [1, 1, 1, 1]			
Generation 3820: Best = 6.166519e+02	FES = 382100	Diversity = 3.2888e-02	k_t
= [3, 2, 3, 3]			
Generation 3830: Best = 6.166394e+02	FES = 383100	Diversity = 3.0577e-02	k_t
= [1, 1, 1, 1]			
Generation 3840: Best = 6.166215e+02	FES = 384100	Diversity = 2.8630e-02	k_t
= [3, 3, 3, 4]			
Generation 3850: Best = 6.166055e+02	FES = 385100	Diversity = 2.6877e-02	k_t
= [1, 1, 1, 1]			
Generation 3860: Best = 6.165901e+02	FES = 386100	Diversity = 2.3851e-02	k_t
= [3, 2, 3, 2]			
Generation 3870: Best = 6.165849e+02	FES = 387100	Diversity = 2.1918e-02	k_t
= [1, 1, 1, 1]			
Generation 3880: Best = 6.165737e+02	FES = 388100	Diversity = 2.0453e-02	k_t
= [4, 2, 2, 3]			
Generation 3890: Best = 6.165610e+02	FES = 389100	Diversity = 1.7465e-02	k_t
= [1, 1, 1, 1]			
Generation 3900: Best = 6.165552e+02	FES = 390100	Diversity = 1.6075e-02	k_t
= [4, 4, 2, 2]			
Generation 3910: Best = 6.165462e+02	FES = 391100	Diversity = 1.4415e-02	k_t
= [1, 1, 1, 1]			
Generation 3920: Best = 6.165343e+02	FES = 392100	Diversity = 1.2894e-02	k_t
= [4, 3, 2, 2]			
Generation 3930: Best = 6.165289e+02	FES = 393100	Diversity = 1.1494e-02	k_t
= [1, 1, 1, 1]			
Generation 3940: Best = 6.165222e+02	FES = 394100	Diversity = 1.0275e-02	k_t
= [2, 3, 4, 2]			
Generation 3950: Best = 6.165187e+02	FES = 395100	Diversity = 9.6371e-03	k_t
= [1, 1, 1, 1]			
Generation 3960: Best = 6.165145e+02	FES = 396100	Diversity = 8.1134e-03	k_t
= [4, 2, 3, 2]			
Generation 3970: Best = 6.165128e+02	FES = 397100	Diversity = 7.2242e-03	k_t
= [1, 1, 1, 1]			
Generation 3980: Best = 6.165090e+02	FES = 398100	Diversity = 6.6636e-03	k_t
= [2, 4, 3, 2]			
Generation 3990: Best = 6.165057e+02	FES = 399100	Diversity = 6.2164e-03	k_t
= [1, 1, 1, 1]			
Generation 4000: Best = 6.165032e+02	FES = 400100	Diversity = 5.9083e-03	k_t
= [4, 2, 2, 3]			

Generation 4010: Best = 6.165009e+02 = [1, 1, 1, 1]	FES = 401100	Diversity = 5.7832e-03	k_t
Generation 4020: Best = 6.164975e+02 = [2, 4, 2, 4]	FES = 402100	Diversity = 5.6167e-03	k_t
Generation 4030: Best = 6.164925e+02 = [1, 1, 1, 1]	FES = 403100	Diversity = 5.6918e-03	k_t
Generation 4040: Best = 6.164874e+02 = [4, 3, 2, 4]	FES = 404100	Diversity = 6.1550e-03	k_t
Generation 4050: Best = 6.164822e+02 = [1, 1, 1, 1]	FES = 405100	Diversity = 6.6907e-03	k_t
Generation 4060: Best = 6.164775e+02 = [3, 3, 3, 2]	FES = 406100	Diversity = 7.0931e-03	k_t
Generation 4070: Best = 6.164727e+02 = [1, 1, 1, 1]	FES = 407100	Diversity = 7.4044e-03	k_t
Generation 4080: Best = 6.164630e+02 = [4, 4, 3, 2]	FES = 408100	Diversity = 7.5908e-03	k_t
Generation 4090: Best = 6.164630e+02 = [1, 1, 1, 1]	FES = 409100	Diversity = 7.6698e-03	k_t
Generation 4100: Best = 6.164573e+02 = [2, 4, 4, 3]	FES = 410100	Diversity = 7.5227e-03	k_t
Generation 4110: Best = 6.164552e+02 = [1, 1, 1, 1]	FES = 411100	Diversity = 7.3328e-03	k_t
Generation 4120: Best = 6.164503e+02 = [4, 3, 3, 4]	FES = 412100	Diversity = 7.4000e-03	k_t
Generation 4130: Best = 6.164477e+02 = [1, 1, 1, 1]	FES = 413100	Diversity = 7.0568e-03	k_t
Generation 4140: Best = 6.164407e+02 = [3, 2, 3, 3]	FES = 414100	Diversity = 7.0832e-03	k_t
Generation 4150: Best = 6.164404e+02 = [1, 1, 1, 1]	FES = 415100	Diversity = 6.5023e-03	k_t
Generation 4160: Best = 6.164345e+02 = [4, 4, 2, 2]	FES = 416100	Diversity = 6.6647e-03	k_t
Generation 4170: Best = 6.164342e+02 = [1, 1, 1, 1]	FES = 417100	Diversity = 6.5685e-03	k_t
Generation 4180: Best = 6.164318e+02 = [2, 2, 3, 2]	FES = 418100	Diversity = 5.9861e-03	k_t
Generation 4190: Best = 6.164292e+02 = [1, 1, 1, 1]	FES = 419100	Diversity = 5.8098e-03	k_t
Generation 4200: Best = 6.164270e+02 = [3, 2, 4, 4]	FES = 420100	Diversity = 5.2737e-03	k_t
Generation 4210: Best = 6.164235e+02 = [1, 1, 1, 1]	FES = 421100	Diversity = 4.9865e-03	k_t
Generation 4220: Best = 6.164232e+02 = [4, 2, 4, 2]	FES = 422100	Diversity = 5.0023e-03	k_t
Generation 4230: Best = 6.164214e+02 = [1, 1, 1, 1]	FES = 423100	Diversity = 4.6430e-03	k_t
Generation 4240: Best = 6.164203e+02 = [4, 4, 2, 3]	FES = 424100	Diversity = 4.2873e-03	k_t
Generation 4250: Best = 6.164191e+02 = [1, 1, 1, 1]	FES = 425100	Diversity = 3.8476e-03	k_t
Generation 4260: Best = 6.164176e+02 = [3, 4, 2, 4]	FES = 426100	Diversity = 3.5110e-03	k_t
Generation 4270: Best = 6.164167e+02 = [1, 1, 1, 1]	FES = 427100	Diversity = 3.4241e-03	k_t

Generation 4280: Best = 6.164154e+02	FES = 428100	Diversity = 3.2409e-03	k_t
= [2, 3, 2, 4]			
Generation 4290: Best = 6.164145e+02	FES = 429100	Diversity = 3.2607e-03	k_t
= [1, 1, 1, 1]			
Generation 4300: Best = 6.164130e+02	FES = 430100	Diversity = 3.3867e-03	k_t
= [3, 2, 4, 4]			
Generation 4310: Best = 6.164118e+02	FES = 431100	Diversity = 3.4014e-03	k_t
= [1, 1, 1, 1]			
Generation 4320: Best = 6.164104e+02	FES = 432100	Diversity = 3.7403e-03	k_t
= [2, 4, 4, 2]			
Generation 4330: Best = 6.164083e+02	FES = 433100	Diversity = 3.7473e-03	k_t
= [1, 1, 1, 1]			
Generation 4340: Best = 6.164071e+02	FES = 434100	Diversity = 4.0990e-03	k_t
= [4, 2, 4, 2]			
Generation 4350: Best = 6.164047e+02	FES = 435100	Diversity = 4.1131e-03	k_t
= [1, 1, 1, 1]			
Generation 4360: Best = 6.164031e+02	FES = 436100	Diversity = 4.2948e-03	k_t
= [3, 4, 3, 2]			
Generation 4370: Best = 6.163999e+02	FES = 437100	Diversity = 4.4144e-03	k_t
= [1, 1, 1, 1]			
Generation 4380: Best = 6.163985e+02	FES = 438100	Diversity = 4.8515e-03	k_t
= [2, 3, 3, 2]			
Generation 4390: Best = 6.163966e+02	FES = 439100	Diversity = 5.1653e-03	k_t
= [1, 1, 1, 1]			
Generation 4400: Best = 6.163944e+02	FES = 440100	Diversity = 5.1781e-03	k_t
= [4, 4, 2, 2]			
Generation 4410: Best = 6.163924e+02	FES = 441100	Diversity = 5.5369e-03	k_t
= [1, 1, 1, 1]			
Generation 4420: Best = 6.163898e+02	FES = 442100	Diversity = 5.6093e-03	k_t
= [3, 2, 3, 2]			
Generation 4430: Best = 6.163861e+02	FES = 443100	Diversity = 5.5471e-03	k_t
= [1, 1, 1, 1]			
Generation 4440: Best = 6.163852e+02	FES = 444100	Diversity = 5.4291e-03	k_t
= [3, 4, 3, 3]			
Generation 4450: Best = 6.163815e+02	FES = 445100	Diversity = 5.8096e-03	k_t
= [1, 1, 1, 1]			
Generation 4460: Best = 6.163764e+02	FES = 446100	Diversity = 5.9300e-03	k_t
= [4, 4, 2, 4]			
Generation 4470: Best = 6.163709e+02	FES = 447100	Diversity = 6.0369e-03	k_t
= [1, 1, 1, 1]			
Generation 4480: Best = 6.163708e+02	FES = 448100	Diversity = 6.3566e-03	k_t
= [2, 4, 4, 4]			
Generation 4490: Best = 6.163672e+02	FES = 449100	Diversity = 6.2566e-03	k_t
= [1, 1, 1, 1]			
Generation 4500: Best = 6.163643e+02	FES = 450100	Diversity = 6.1217e-03	k_t
= [4, 3, 2, 2]			
Generation 4510: Best = 6.163582e+02	FES = 451100	Diversity = 6.4054e-03	k_t
= [1, 1, 1, 1]			
Generation 4520: Best = 6.163560e+02	FES = 452100	Diversity = 6.4368e-03	k_t
= [4, 3, 3, 4]			
Generation 4530: Best = 6.163541e+02	FES = 453100	Diversity = 6.3861e-03	k_t
= [1, 1, 1, 1]			
Generation 4540: Best = 6.163503e+02	FES = 454100	Diversity = 6.4184e-03	k_t
= [4, 4, 4, 3]			

Generation 4550: Best = 6.163469e+02	FES = 455100	Diversity = 6.2243e-03	k_t
= [1, 1, 1, 1]			
Generation 4560: Best = 6.163418e+02	FES = 456100	Diversity = 5.9141e-03	k_t
= [3, 4, 2, 4]			
Generation 4570: Best = 6.163414e+02	FES = 457100	Diversity = 5.7844e-03	k_t
= [1, 1, 1, 1]			
Generation 4580: Best = 6.163379e+02	FES = 458100	Diversity = 5.7012e-03	k_t
= [2, 2, 2, 3]			
Generation 4590: Best = 6.163375e+02	FES = 459100	Diversity = 5.3381e-03	k_t
= [1, 1, 1, 1]			
Generation 4600: Best = 6.163343e+02	FES = 460100	Diversity = 5.5676e-03	k_t
= [4, 2, 3, 3]			
Generation 4610: Best = 6.163289e+02	FES = 461100	Diversity = 5.5949e-03	k_t
= [1, 1, 1, 1]			
Generation 4620: Best = 6.163269e+02	FES = 462100	Diversity = 5.4275e-03	k_t
= [3, 4, 3, 4]			
Generation 4630: Best = 6.163248e+02	FES = 463100	Diversity = 4.9514e-03	k_t
= [1, 1, 1, 1]			
Generation 4640: Best = 6.163222e+02	FES = 464100	Diversity = 4.6615e-03	k_t
= [3, 2, 2, 3]			
Generation 4650: Best = 6.163179e+02	FES = 465100	Diversity = 4.8644e-03	k_t
= [1, 1, 1, 1]			
Generation 4660: Best = 6.163161e+02	FES = 466100	Diversity = 5.2035e-03	k_t
= [4, 4, 2, 4]			
Generation 4670: Best = 6.163126e+02	FES = 467100	Diversity = 5.0678e-03	k_t
= [1, 1, 1, 1]			
Generation 4680: Best = 6.163099e+02	FES = 468100	Diversity = 5.2621e-03	k_t
= [2, 4, 4, 3]			
Generation 4690: Best = 6.163084e+02	FES = 469100	Diversity = 5.4261e-03	k_t
= [1, 1, 1, 1]			
Generation 4700: Best = 6.163063e+02	FES = 470100	Diversity = 5.6253e-03	k_t
= [3, 4, 2, 2]			
Generation 4710: Best = 6.163038e+02	FES = 471100	Diversity = 5.4288e-03	k_t
= [1, 1, 1, 1]			
Generation 4720: Best = 6.163034e+02	FES = 472100	Diversity = 5.2485e-03	k_t
= [3, 2, 2, 2]			
Generation 4730: Best = 6.163006e+02	FES = 473100	Diversity = 5.1262e-03	k_t
= [1, 1, 1, 1]			
Generation 4740: Best = 6.162985e+02	FES = 474100	Diversity = 5.2858e-03	k_t
= [3, 2, 2, 3]			
Generation 4750: Best = 6.162956e+02	FES = 475100	Diversity = 5.3746e-03	k_t
= [1, 1, 1, 1]			
Generation 4760: Best = 6.162932e+02	FES = 476100	Diversity = 5.0235e-03	k_t
= [3, 2, 2, 4]			
Generation 4770: Best = 6.162923e+02	FES = 477100	Diversity = 5.0813e-03	k_t
= [1, 1, 1, 1]			
Generation 4780: Best = 6.162894e+02	FES = 478100	Diversity = 5.2356e-03	k_t
= [2, 3, 4, 3]			
Generation 4790: Best = 6.162879e+02	FES = 479100	Diversity = 5.6811e-03	k_t
= [1, 1, 1, 1]			
Generation 4800: Best = 6.162860e+02	FES = 480100	Diversity = 6.0130e-03	k_t
= [3, 2, 4, 3]			
Generation 4810: Best = 6.162831e+02	FES = 481100	Diversity = 5.9573e-03	k_t
= [1, 1, 1, 1]			

Generation 4820: Best = 6.162801e+02	FES = 482100	Diversity = 6.1316e-03	k_t
= [4, 2, 2, 2]			
Generation 4830: Best = 6.162772e+02	FES = 483100	Diversity = 6.1716e-03	k_t
= [1, 1, 1, 1]			
Generation 4840: Best = 6.162743e+02	FES = 484100	Diversity = 6.2298e-03	k_t
= [2, 4, 3, 4]			
Generation 4850: Best = 6.162721e+02	FES = 485100	Diversity = 6.0143e-03	k_t
= [1, 1, 1, 1]			
Generation 4860: Best = 6.162699e+02	FES = 486100	Diversity = 5.8092e-03	k_t
= [3, 2, 2, 3]			
Generation 4870: Best = 6.162687e+02	FES = 487100	Diversity = 5.7826e-03	k_t
= [1, 1, 1, 1]			
Generation 4880: Best = 6.162655e+02	FES = 488100	Diversity = 5.3840e-03	k_t
= [4, 2, 3, 2]			
Generation 4890: Best = 6.162638e+02	FES = 489100	Diversity = 5.3283e-03	k_t
= [1, 1, 1, 1]			
Generation 4900: Best = 6.162608e+02	FES = 490100	Diversity = 5.0982e-03	k_t
= [2, 3, 4, 4]			
Generation 4910: Best = 6.162605e+02	FES = 491100	Diversity = 4.6949e-03	k_t
= [1, 1, 1, 1]			
Generation 4920: Best = 6.162571e+02	FES = 492100	Diversity = 4.6736e-03	k_t
= [4, 4, 2, 2]			
Generation 4930: Best = 6.162556e+02	FES = 493100	Diversity = 4.9941e-03	k_t
= [1, 1, 1, 1]			
Generation 4940: Best = 6.162525e+02	FES = 494100	Diversity = 5.4779e-03	k_t
= [2, 2, 3, 4]			
Generation 4950: Best = 6.162478e+02	FES = 495100	Diversity = 5.5802e-03	k_t
= [1, 1, 1, 1]			
Generation 4960: Best = 6.162463e+02	FES = 496100	Diversity = 5.3566e-03	k_t
= [3, 3, 3, 3]			
Generation 4970: Best = 6.162446e+02	FES = 497100	Diversity = 5.3938e-03	k_t
= [1, 1, 1, 1]			
Generation 4980: Best = 6.162438e+02	FES = 498100	Diversity = 5.3265e-03	k_t
= [2, 2, 2, 4]			
Generation 4990: Best = 6.162422e+02	FES = 499100	Diversity = 5.3521e-03	k_t
= [1, 1, 1, 1]			
Generation 5000: Best = 6.162400e+02	FES = 500100	Diversity = 5.4080e-03	k_t
= [2, 3, 4, 2]			
Generation 5010: Best = 6.162373e+02	FES = 501100	Diversity = 5.4895e-03	k_t
= [1, 1, 1, 1]			
Generation 5020: Best = 6.162363e+02	FES = 502100	Diversity = 5.2541e-03	k_t
= [3, 2, 3, 4]			
Generation 5030: Best = 6.162353e+02	FES = 503100	Diversity = 4.8224e-03	k_t
= [1, 1, 1, 1]			
Generation 5040: Best = 6.162323e+02	FES = 504100	Diversity = 4.5291e-03	k_t
= [3, 3, 3, 3]			
Generation 5050: Best = 6.162313e+02	FES = 505100	Diversity = 4.5409e-03	k_t
= [1, 1, 1, 1]			
Generation 5060: Best = 6.162305e+02	FES = 506100	Diversity = 4.4341e-03	k_t
= [2, 2, 3, 4]			
Generation 5070: Best = 6.162294e+02	FES = 507100	Diversity = 4.2333e-03	k_t
= [1, 1, 1, 1]			
Generation 5080: Best = 6.162276e+02	FES = 508100	Diversity = 3.9965e-03	k_t
= [3, 2, 3, 3]			

Generation 5090: Best = 6.162260e+02 = [1, 1, 1, 1]	FES = 509100	Diversity = 3.8654e-03	k_t
Generation 5100: Best = 6.162250e+02 = [4, 2, 2, 3]	FES = 510100	Diversity = 3.5829e-03	k_t
Generation 5110: Best = 6.162238e+02 = [1, 1, 1, 1]	FES = 511100	Diversity = 3.2708e-03	k_t
Generation 5120: Best = 6.162236e+02 = [2, 4, 3, 2]	FES = 512100	Diversity = 3.1441e-03	k_t
Generation 5130: Best = 6.162227e+02 = [1, 1, 1, 1]	FES = 513100	Diversity = 3.0936e-03	k_t
Generation 5140: Best = 6.162215e+02 = [3, 2, 4, 4]	FES = 514100	Diversity = 3.1599e-03	k_t
Generation 5150: Best = 6.162198e+02 = [1, 1, 1, 1]	FES = 515100	Diversity = 3.4664e-03	k_t
Generation 5160: Best = 6.162186e+02 = [3, 2, 4, 2]	FES = 516100	Diversity = 3.7166e-03	k_t
Generation 5170: Best = 6.162170e+02 = [1, 1, 1, 1]	FES = 517100	Diversity = 4.0249e-03	k_t
Generation 5180: Best = 6.162159e+02 = [4, 2, 3, 2]	FES = 518100	Diversity = 3.9306e-03	k_t
Generation 5190: Best = 6.162143e+02 = [1, 1, 1, 1]	FES = 519100	Diversity = 4.3895e-03	k_t
Generation 5200: Best = 6.162119e+02 = [3, 2, 3, 3]	FES = 520100	Diversity = 4.3965e-03	k_t
Generation 5210: Best = 6.162101e+02 = [1, 1, 1, 1]	FES = 521100	Diversity = 4.4184e-03	k_t
Generation 5220: Best = 6.162090e+02 = [2, 2, 2, 2]	FES = 522100	Diversity = 4.4570e-03	k_t
Generation 5230: Best = 6.162084e+02 = [1, 1, 1, 1]	FES = 523100	Diversity = 4.0422e-03	k_t
Generation 5240: Best = 6.162071e+02 = [4, 2, 3, 4]	FES = 524100	Diversity = 3.7759e-03	k_t
Generation 5250: Best = 6.162055e+02 = [1, 1, 1, 1]	FES = 525100	Diversity = 3.7942e-03	k_t
Generation 5260: Best = 6.162050e+02 = [4, 3, 2, 2]	FES = 526100	Diversity = 3.6039e-03	k_t
Generation 5270: Best = 6.162048e+02 = [1, 1, 1, 1]	FES = 527100	Diversity = 3.4975e-03	k_t
Generation 5280: Best = 6.162032e+02 = [2, 3, 3, 3]	FES = 528100	Diversity = 3.2552e-03	k_t
Generation 5290: Best = 6.162030e+02 = [1, 1, 1, 1]	FES = 529100	Diversity = 3.2513e-03	k_t
Generation 5300: Best = 6.162017e+02 = [4, 3, 3, 3]	FES = 530100	Diversity = 3.4718e-03	k_t
Generation 5310: Best = 6.162009e+02 = [1, 1, 1, 1]	FES = 531100	Diversity = 3.6305e-03	k_t
Generation 5320: Best = 6.162003e+02 = [4, 3, 3, 4]	FES = 532100	Diversity = 3.7396e-03	k_t
Generation 5330: Best = 6.161989e+02 = [1, 1, 1, 1]	FES = 533100	Diversity = 3.7444e-03	k_t
Generation 5340: Best = 6.161979e+02 = [3, 3, 3, 4]	FES = 534100	Diversity = 3.7213e-03	k_t
Generation 5350: Best = 6.161972e+02 = [1, 1, 1, 1]	FES = 535100	Diversity = 3.5560e-03	k_t

Generation 5360: Best = 6.161962e+02 = [3, 4, 4, 3]	FES = 536100	Diversity = 3.5432e-03	k_t
Generation 5370: Best = 6.161950e+02 = [1, 1, 1, 1]	FES = 537100	Diversity = 3.4495e-03	k_t
Generation 5380: Best = 6.161944e+02 = [2, 4, 4, 3]	FES = 538100	Diversity = 3.5828e-03	k_t
Generation 5390: Best = 6.161933e+02 = [1, 1, 1, 1]	FES = 539100	Diversity = 3.5166e-03	k_t
Generation 5400: Best = 6.161928e+02 = [2, 2, 3, 4]	FES = 540100	Diversity = 3.4816e-03	k_t
Generation 5410: Best = 6.161914e+02 = [1, 1, 1, 1]	FES = 541100	Diversity = 3.6094e-03	k_t
Generation 5420: Best = 6.161909e+02 = [2, 2, 4, 4]	FES = 542100	Diversity = 3.4895e-03	k_t
Generation 5430: Best = 6.161903e+02 = [1, 1, 1, 1]	FES = 543100	Diversity = 3.2714e-03	k_t
Generation 5440: Best = 6.161887e+02 = [4, 2, 4, 2]	FES = 544100	Diversity = 3.2773e-03	k_t
Generation 5450: Best = 6.161882e+02 = [1, 1, 1, 1]	FES = 545100	Diversity = 3.4581e-03	k_t
Generation 5460: Best = 6.161880e+02 = [4, 4, 3, 4]	FES = 546100	Diversity = 3.6524e-03	k_t
Generation 5470: Best = 6.161869e+02 = [1, 1, 1, 1]	FES = 547100	Diversity = 3.6068e-03	k_t
Generation 5480: Best = 6.161863e+02 = [2, 4, 2, 3]	FES = 548100	Diversity = 3.6870e-03	k_t
Generation 5490: Best = 6.161857e+02 = [1, 1, 1, 1]	FES = 549100	Diversity = 3.6697e-03	k_t
Generation 5500: Best = 6.161846e+02 = [4, 2, 3, 2]	FES = 550100	Diversity = 3.7127e-03	k_t
Generation 5510: Best = 6.161833e+02 = [1, 1, 1, 1]	FES = 551100	Diversity = 3.7036e-03	k_t
Generation 5520: Best = 6.161821e+02 = [3, 2, 2, 3]	FES = 552100	Diversity = 3.9606e-03	k_t
Generation 5530: Best = 6.161805e+02 = [1, 1, 1, 1]	FES = 553100	Diversity = 4.0050e-03	k_t
Generation 5540: Best = 6.161802e+02 = [4, 2, 3, 3]	FES = 554100	Diversity = 3.5446e-03	k_t
Generation 5550: Best = 6.161790e+02 = [1, 1, 1, 1]	FES = 555100	Diversity = 3.3008e-03	k_t
Generation 5560: Best = 6.161781e+02 = [4, 3, 4, 2]	FES = 556100	Diversity = 3.2385e-03	k_t
Generation 5570: Best = 6.161773e+02 = [1, 1, 1, 1]	FES = 557100	Diversity = 3.2966e-03	k_t
Generation 5580: Best = 6.161760e+02 = [3, 4, 4, 4]	FES = 558100	Diversity = 3.5528e-03	k_t
Generation 5590: Best = 6.161748e+02 = [1, 1, 1, 1]	FES = 559100	Diversity = 3.5955e-03	k_t
Generation 5600: Best = 6.161738e+02 = [2, 3, 4, 4]	FES = 560100	Diversity = 3.6050e-03	k_t
Generation 5610: Best = 6.161727e+02 = [1, 1, 1, 1]	FES = 561100	Diversity = 3.6569e-03	k_t
Generation 5620: Best = 6.161720e+02 = [4, 2, 4, 2]	FES = 562100	Diversity = 3.5543e-03	k_t

Generation 5630: Best = 6.161708e+02	FES = 563100	Diversity = 4.0293e-03	k_t
= [1, 1, 1, 1]			
Generation 5640: Best = 6.161691e+02	FES = 564100	Diversity = 3.9577e-03	k_t
= [3, 4, 2, 3]			
Generation 5650: Best = 6.161682e+02	FES = 565100	Diversity = 4.0947e-03	k_t
= [1, 1, 1, 1]			
Generation 5660: Best = 6.161673e+02	FES = 566100	Diversity = 4.2618e-03	k_t
= [4, 2, 3, 4]			
Generation 5670: Best = 6.161635e+02	FES = 567100	Diversity = 4.4388e-03	k_t
= [1, 1, 1, 1]			
Generation 5680: Best = 6.161635e+02	FES = 568100	Diversity = 4.3898e-03	k_t
= [3, 3, 4, 4]			
Generation 5690: Best = 6.161613e+02	FES = 569100	Diversity = 4.2926e-03	k_t
= [1, 1, 1, 1]			
Generation 5700: Best = 6.161609e+02	FES = 570100	Diversity = 4.1435e-03	k_t
= [3, 4, 3, 4]			
Generation 5710: Best = 6.161602e+02	FES = 571100	Diversity = 4.1016e-03	k_t
= [1, 1, 1, 1]			
Generation 5720: Best = 6.161588e+02	FES = 572100	Diversity = 4.0897e-03	k_t
= [2, 3, 3, 2]			
Generation 5730: Best = 6.161575e+02	FES = 573100	Diversity = 4.4290e-03	k_t
= [1, 1, 1, 1]			
Generation 5740: Best = 6.161566e+02	FES = 574100	Diversity = 4.4042e-03	k_t
= [4, 3, 4, 2]			
Generation 5750: Best = 6.161558e+02	FES = 575100	Diversity = 4.2255e-03	k_t
= [1, 1, 1, 1]			
Generation 5760: Best = 6.161540e+02	FES = 576100	Diversity = 4.2946e-03	k_t
= [2, 4, 4, 2]			
Generation 5770: Best = 6.161516e+02	FES = 577100	Diversity = 4.4205e-03	k_t
= [1, 1, 1, 1]			
Generation 5780: Best = 6.161506e+02	FES = 578100	Diversity = 4.0302e-03	k_t
= [3, 3, 4, 3]			
Generation 5790: Best = 6.161495e+02	FES = 579100	Diversity = 4.1245e-03	k_t
= [1, 1, 1, 1]			
Generation 5800: Best = 6.161483e+02	FES = 580100	Diversity = 3.9672e-03	k_t
= [2, 3, 3, 4]			
Generation 5810: Best = 6.161473e+02	FES = 581100	Diversity = 4.0518e-03	k_t
= [1, 1, 1, 1]			
Generation 5820: Best = 6.161450e+02	FES = 582100	Diversity = 4.1458e-03	k_t
= [2, 4, 2, 3]			
Generation 5830: Best = 6.161442e+02	FES = 583100	Diversity = 4.0016e-03	k_t
= [1, 1, 1, 1]			
Generation 5840: Best = 6.161431e+02	FES = 584100	Diversity = 3.6751e-03	k_t
= [2, 4, 4, 2]			
Generation 5850: Best = 6.161425e+02	FES = 585100	Diversity = 3.5737e-03	k_t
= [1, 1, 1, 1]			
Generation 5860: Best = 6.161410e+02	FES = 586100	Diversity = 3.3886e-03	k_t
= [2, 4, 4, 2]			
Generation 5870: Best = 6.161387e+02	FES = 587100	Diversity = 3.4448e-03	k_t
= [1, 1, 1, 1]			
Generation 5880: Best = 6.161378e+02	FES = 588100	Diversity = 3.8780e-03	k_t
= [3, 4, 4, 3]			
Generation 5890: Best = 6.161370e+02	FES = 589100	Diversity = 3.7215e-03	k_t
= [1, 1, 1, 1]			

Generation 5900: Best = 6.161352e+02 = [3, 4, 2, 4]	FES = 590100	Diversity = 3.7182e-03	k_t
Generation 5910: Best = 6.161337e+02 = [1, 1, 1, 1]	FES = 591100	Diversity = 4.0515e-03	k_t
Generation 5920: Best = 6.161328e+02 = [3, 2, 3, 3]	FES = 592100	Diversity = 4.2364e-03	k_t
Generation 5930: Best = 6.161315e+02 = [1, 1, 1, 1]	FES = 593100	Diversity = 4.0698e-03	k_t
Generation 5940: Best = 6.161306e+02 = [2, 3, 3, 4]	FES = 594100	Diversity = 3.8227e-03	k_t
Generation 5950: Best = 6.161291e+02 = [1, 1, 1, 1]	FES = 595100	Diversity = 3.7828e-03	k_t
Generation 5960: Best = 6.161264e+02 = [2, 2, 2, 4]	FES = 596100	Diversity = 4.0258e-03	k_t
Generation 5970: Best = 6.161245e+02 = [1, 1, 1, 1]	FES = 597100	Diversity = 4.4517e-03	k_t
Generation 5980: Best = 6.161224e+02 = [4, 3, 4, 2]	FES = 598100	Diversity = 4.1691e-03	k_t
Generation 5990: Best = 6.161214e+02 = [1, 1, 1, 1]	FES = 599100	Diversity = 4.2129e-03	k_t
Generation 6000: Best = 6.161207e+02 = [3, 4, 2, 2]	FES = 600100	Diversity = 3.9081e-03	k_t
Generation 6010: Best = 6.161187e+02 = [1, 1, 1, 1]	FES = 601100	Diversity = 3.8689e-03	k_t
Generation 6020: Best = 6.161158e+02 = [3, 4, 2, 4]	FES = 602100	Diversity = 3.8882e-03	k_t
Generation 6030: Best = 6.161157e+02 = [1, 1, 1, 1]	FES = 603100	Diversity = 3.7196e-03	k_t
Generation 6040: Best = 6.161153e+02 = [3, 2, 4, 3]	FES = 604100	Diversity = 3.7492e-03	k_t
Generation 6050: Best = 6.161139e+02 = [1, 1, 1, 1]	FES = 605100	Diversity = 3.7402e-03	k_t
Generation 6060: Best = 6.161132e+02 = [4, 2, 4, 4]	FES = 606100	Diversity = 4.0623e-03	k_t
Generation 6070: Best = 6.161115e+02 = [1, 1, 1, 1]	FES = 607100	Diversity = 3.9818e-03	k_t
Generation 6080: Best = 6.161107e+02 = [4, 4, 2, 2]	FES = 608100	Diversity = 4.0200e-03	k_t
Generation 6090: Best = 6.161094e+02 = [1, 1, 1, 1]	FES = 609100	Diversity = 3.9101e-03	k_t
Generation 6100: Best = 6.161078e+02 = [3, 2, 3, 4]	FES = 610100	Diversity = 3.9388e-03	k_t
Generation 6110: Best = 6.161064e+02 = [1, 1, 1, 1]	FES = 611100	Diversity = 4.1130e-03	k_t
Generation 6120: Best = 6.161045e+02 = [3, 3, 3, 4]	FES = 612100	Diversity = 4.1251e-03	k_t
Generation 6130: Best = 6.161019e+02 = [1, 1, 1, 1]	FES = 613100	Diversity = 4.3920e-03	k_t
Generation 6140: Best = 6.161009e+02 = [2, 2, 2, 2]	FES = 614100	Diversity = 4.4948e-03	k_t
Generation 6150: Best = 6.161001e+02 = [1, 1, 1, 1]	FES = 615100	Diversity = 4.3405e-03	k_t
Generation 6160: Best = 6.160981e+02 = [2, 4, 2, 2]	FES = 616100	Diversity = 4.2502e-03	k_t

Generation 6170: Best = 6.160966e+02 = [1, 1, 1, 1]	FES = 617100	Diversity = 4.1428e-03	k_t
Generation 6180: Best = 6.160951e+02 = [4, 4, 2, 3]	FES = 618100	Diversity = 4.2102e-03	k_t
Generation 6190: Best = 6.160943e+02 = [1, 1, 1, 1]	FES = 619100	Diversity = 3.9418e-03	k_t
Generation 6200: Best = 6.160930e+02 = [4, 3, 2, 3]	FES = 620100	Diversity = 3.7117e-03	k_t
Generation 6210: Best = 6.160919e+02 = [1, 1, 1, 1]	FES = 621100	Diversity = 3.7313e-03	k_t
Generation 6220: Best = 6.160908e+02 = [4, 2, 2, 2]	FES = 622100	Diversity = 3.6898e-03	k_t
Generation 6230: Best = 6.160891e+02 = [1, 1, 1, 1]	FES = 623100	Diversity = 3.6834e-03	k_t
Generation 6240: Best = 6.160887e+02 = [4, 4, 4, 2]	FES = 624100	Diversity = 3.5305e-03	k_t
Generation 6250: Best = 6.160882e+02 = [1, 1, 1, 1]	FES = 625100	Diversity = 3.4708e-03	k_t
Generation 6260: Best = 6.160874e+02 = [2, 2, 4, 2]	FES = 626100	Diversity = 3.3474e-03	k_t
Generation 6270: Best = 6.160866e+02 = [1, 1, 1, 1]	FES = 627100	Diversity = 3.3118e-03	k_t
Generation 6280: Best = 6.160860e+02 = [2, 3, 4, 2]	FES = 628100	Diversity = 3.1401e-03	k_t
Generation 6290: Best = 6.160850e+02 = [1, 1, 1, 1]	FES = 629100	Diversity = 2.9705e-03	k_t
Generation 6300: Best = 6.160848e+02 = [2, 2, 4, 4]	FES = 630100	Diversity = 2.6464e-03	k_t
Generation 6310: Best = 6.160840e+02 = [1, 1, 1, 1]	FES = 631100	Diversity = 2.5356e-03	k_t
Generation 6320: Best = 6.160832e+02 = [4, 3, 4, 2]	FES = 632100	Diversity = 2.5766e-03	k_t
Generation 6330: Best = 6.160828e+02 = [1, 1, 1, 1]	FES = 633100	Diversity = 2.6544e-03	k_t
Generation 6340: Best = 6.160822e+02 = [2, 4, 4, 3]	FES = 634100	Diversity = 2.5837e-03	k_t
Generation 6350: Best = 6.160817e+02 = [1, 1, 1, 1]	FES = 635100	Diversity = 2.7156e-03	k_t
Generation 6360: Best = 6.160807e+02 = [2, 2, 3, 4]	FES = 636100	Diversity = 2.6164e-03	k_t
Generation 6370: Best = 6.160800e+02 = [1, 1, 1, 1]	FES = 637100	Diversity = 3.0832e-03	k_t
Generation 6380: Best = 6.160790e+02 = [3, 3, 3, 3]	FES = 638100	Diversity = 3.3946e-03	k_t
Generation 6390: Best = 6.160784e+02 = [1, 1, 1, 1]	FES = 639100	Diversity = 3.5506e-03	k_t
Generation 6400: Best = 6.160775e+02 = [3, 3, 2, 3]	FES = 640100	Diversity = 3.6910e-03	k_t
Generation 6410: Best = 6.160767e+02 = [1, 1, 1, 1]	FES = 641100	Diversity = 3.6532e-03	k_t
Generation 6420: Best = 6.160761e+02 = [4, 2, 3, 4]	FES = 642100	Diversity = 3.5956e-03	k_t
Generation 6430: Best = 6.160745e+02 = [1, 1, 1, 1]	FES = 643100	Diversity = 3.4448e-03	k_t

Generation 6440: Best = 6.160716e+02	FES = 644100	Diversity = 3.4953e-03	k_t
= [3, 4, 4, 4]			
Generation 6450: Best = 6.160709e+02	FES = 645100	Diversity = 4.0788e-03	k_t
= [1, 1, 1, 1]			
Generation 6460: Best = 6.160697e+02	FES = 646100	Diversity = 4.3745e-03	k_t
= [2, 4, 2, 3]			
Generation 6470: Best = 6.160682e+02	FES = 647100	Diversity = 4.1258e-03	k_t
= [1, 1, 1, 1]			
Generation 6480: Best = 6.160668e+02	FES = 648100	Diversity = 3.7859e-03	k_t
= [4, 3, 3, 4]			
Generation 6490: Best = 6.160652e+02	FES = 649100	Diversity = 3.9628e-03	k_t
= [1, 1, 1, 1]			
Generation 6500: Best = 6.160635e+02	FES = 650100	Diversity = 4.2124e-03	k_t
= [4, 3, 3, 2]			
Generation 6510: Best = 6.160629e+02	FES = 651100	Diversity = 4.3474e-03	k_t
= [1, 1, 1, 1]			
Generation 6520: Best = 6.160621e+02	FES = 652100	Diversity = 4.2008e-03	k_t
= [2, 2, 4, 2]			
Generation 6530: Best = 6.160613e+02	FES = 653100	Diversity = 3.8185e-03	k_t
= [1, 1, 1, 1]			
Generation 6540: Best = 6.160601e+02	FES = 654100	Diversity = 3.7746e-03	k_t
= [4, 3, 2, 2]			
Generation 6550: Best = 6.160587e+02	FES = 655100	Diversity = 3.5515e-03	k_t
= [1, 1, 1, 1]			
Generation 6560: Best = 6.160586e+02	FES = 656100	Diversity = 3.3853e-03	k_t
= [3, 4, 2, 3]			
Generation 6570: Best = 6.160569e+02	FES = 657100	Diversity = 3.2718e-03	k_t
= [1, 1, 1, 1]			
Generation 6580: Best = 6.160566e+02	FES = 658100	Diversity = 3.3840e-03	k_t
= [3, 4, 3, 4]			
Generation 6590: Best = 6.160555e+02	FES = 659100	Diversity = 3.2015e-03	k_t
= [1, 1, 1, 1]			
Generation 6600: Best = 6.160549e+02	FES = 660100	Diversity = 2.9855e-03	k_t
= [4, 2, 3, 2]			
Generation 6610: Best = 6.160536e+02	FES = 661100	Diversity = 3.0970e-03	k_t
= [1, 1, 1, 1]			
Generation 6620: Best = 6.160527e+02	FES = 662100	Diversity = 3.1541e-03	k_t
= [3, 3, 2, 2]			
Generation 6630: Best = 6.160510e+02	FES = 663100	Diversity = 3.4595e-03	k_t
= [1, 1, 1, 1]			
Generation 6640: Best = 6.160497e+02	FES = 664100	Diversity = 3.6875e-03	k_t
= [4, 3, 4, 2]			
Generation 6650: Best = 6.160490e+02	FES = 665100	Diversity = 3.7530e-03	k_t
= [1, 1, 1, 1]			
Generation 6660: Best = 6.160478e+02	FES = 666100	Diversity = 3.8283e-03	k_t
= [2, 2, 2, 3]			
Generation 6670: Best = 6.160464e+02	FES = 667100	Diversity = 3.8495e-03	k_t
= [1, 1, 1, 1]			
Generation 6680: Best = 6.160462e+02	FES = 668100	Diversity = 3.7510e-03	k_t
= [4, 3, 2, 2]			
Generation 6690: Best = 6.160448e+02	FES = 669100	Diversity = 3.7934e-03	k_t
= [1, 1, 1, 1]			
Generation 6700: Best = 6.160433e+02	FES = 670100	Diversity = 3.5897e-03	k_t
= [2, 2, 3, 3]			

Generation 6710: Best = 6.160426e+02 = [1, 1, 1, 1]	FES = 671100	Diversity = 3.4285e-03	k_t
Generation 6720: Best = 6.160411e+02 = [3, 4, 2, 2]	FES = 672100	Diversity = 3.3856e-03	k_t
Generation 6730: Best = 6.160393e+02 = [1, 1, 1, 1]	FES = 673100	Diversity = 3.4260e-03	k_t
Generation 6740: Best = 6.160372e+02 = [2, 3, 4, 3]	FES = 674100	Diversity = 3.5046e-03	k_t
Generation 6750: Best = 6.160372e+02 = [1, 1, 1, 1]	FES = 675100	Diversity = 3.5618e-03	k_t
Generation 6760: Best = 6.160366e+02 = [3, 2, 2, 3]	FES = 676100	Diversity = 3.6909e-03	k_t
Generation 6770: Best = 6.160349e+02 = [1, 1, 1, 1]	FES = 677100	Diversity = 3.7409e-03	k_t
Generation 6780: Best = 6.160334e+02 = [4, 3, 4, 3]	FES = 678100	Diversity = 3.7256e-03	k_t
Generation 6790: Best = 6.160330e+02 = [1, 1, 1, 1]	FES = 679100	Diversity = 3.5563e-03	k_t
Generation 6800: Best = 6.160323e+02 = [4, 4, 2, 4]	FES = 680100	Diversity = 3.4457e-03	k_t
Generation 6810: Best = 6.160314e+02 = [1, 1, 1, 1]	FES = 681100	Diversity = 3.3181e-03	k_t
Generation 6820: Best = 6.160303e+02 = [2, 3, 4, 4]	FES = 682100	Diversity = 3.2316e-03	k_t
Generation 6830: Best = 6.160296e+02 = [1, 1, 1, 1]	FES = 683100	Diversity = 3.3341e-03	k_t
Generation 6840: Best = 6.160287e+02 = [4, 3, 4, 4]	FES = 684100	Diversity = 3.4097e-03	k_t
Generation 6850: Best = 6.160276e+02 = [1, 1, 1, 1]	FES = 685100	Diversity = 3.2902e-03	k_t
Generation 6860: Best = 6.160269e+02 = [3, 2, 4, 3]	FES = 686100	Diversity = 3.3336e-03	k_t
Generation 6870: Best = 6.160263e+02 = [1, 1, 1, 1]	FES = 687100	Diversity = 3.2497e-03	k_t
Generation 6880: Best = 6.160254e+02 = [2, 3, 4, 3]	FES = 688100	Diversity = 3.3386e-03	k_t
Generation 6890: Best = 6.160242e+02 = [1, 1, 1, 1]	FES = 689100	Diversity = 3.3984e-03	k_t
Generation 6900: Best = 6.160224e+02 = [2, 4, 2, 3]	FES = 690100	Diversity = 3.4417e-03	k_t
Generation 6910: Best = 6.160218e+02 = [1, 1, 1, 1]	FES = 691100	Diversity = 3.4591e-03	k_t
Generation 6920: Best = 6.160201e+02 = [4, 3, 4, 3]	FES = 692100	Diversity = 3.3614e-03	k_t
Generation 6930: Best = 6.160182e+02 = [1, 1, 1, 1]	FES = 693100	Diversity = 3.6784e-03	k_t
Generation 6940: Best = 6.160179e+02 = [4, 3, 2, 3]	FES = 694100	Diversity = 4.0985e-03	k_t
Generation 6950: Best = 6.160173e+02 = [1, 1, 1, 1]	FES = 695100	Diversity = 4.1194e-03	k_t
Generation 6960: Best = 6.160168e+02 = [4, 2, 4, 4]	FES = 696100	Diversity = 3.9215e-03	k_t
Generation 6970: Best = 6.160153e+02 = [1, 1, 1, 1]	FES = 697100	Diversity = 3.8843e-03	k_t

Generation 6980: Best = 6.160134e+02	FES = 698100	Diversity = 3.9302e-03	k_t
= [3, 2, 4, 4]			
Generation 6990: Best = 6.160133e+02	FES = 699100	Diversity = 3.7707e-03	k_t
= [1, 1, 1, 1]			
Generation 7000: Best = 6.160128e+02	FES = 700100	Diversity = 3.7778e-03	k_t
= [3, 2, 3, 4]			
Generation 7010: Best = 6.160114e+02	FES = 701100	Diversity = 3.8432e-03	k_t
= [1, 1, 1, 1]			
Generation 7020: Best = 6.160105e+02	FES = 702100	Diversity = 3.7225e-03	k_t
= [2, 3, 4, 4]			
Generation 7030: Best = 6.160098e+02	FES = 703100	Diversity = 3.6446e-03	k_t
= [1, 1, 1, 1]			
Generation 7040: Best = 6.160088e+02	FES = 704100	Diversity = 3.5722e-03	k_t
= [4, 2, 4, 3]			
Generation 7050: Best = 6.160085e+02	FES = 705100	Diversity = 3.3099e-03	k_t
= [1, 1, 1, 1]			
Generation 7060: Best = 6.160076e+02	FES = 706100	Diversity = 3.2115e-03	k_t
= [4, 4, 4, 4]			
Generation 7070: Best = 6.160058e+02	FES = 707100	Diversity = 3.3326e-03	k_t
= [1, 1, 1, 1]			
Generation 7080: Best = 6.160052e+02	FES = 708100	Diversity = 3.2943e-03	k_t
= [4, 4, 4, 3]			
Generation 7090: Best = 6.160040e+02	FES = 709100	Diversity = 3.2078e-03	k_t
= [1, 1, 1, 1]			
Generation 7100: Best = 6.160035e+02	FES = 710100	Diversity = 3.0759e-03	k_t
= [4, 2, 4, 4]			
Generation 7110: Best = 6.160026e+02	FES = 711100	Diversity = 2.8869e-03	k_t
= [1, 1, 1, 1]			
Generation 7120: Best = 6.160019e+02	FES = 712100	Diversity = 2.9457e-03	k_t
= [3, 2, 3, 2]			
Generation 7130: Best = 6.160001e+02	FES = 713100	Diversity = 3.3870e-03	k_t
= [1, 1, 1, 1]			
Generation 7140: Best = 6.159996e+02	FES = 714100	Diversity = 3.5365e-03	k_t
= [3, 4, 2, 3]			
Generation 7150: Best = 6.159984e+02	FES = 715100	Diversity = 3.5508e-03	k_t
= [1, 1, 1, 1]			
Generation 7160: Best = 6.159976e+02	FES = 716100	Diversity = 3.4520e-03	k_t
= [4, 2, 2, 3]			
Generation 7170: Best = 6.159962e+02	FES = 717100	Diversity = 3.4358e-03	k_t
= [1, 1, 1, 1]			
Generation 7180: Best = 6.159945e+02	FES = 718100	Diversity = 3.5566e-03	k_t
= [3, 2, 3, 4]			
Generation 7190: Best = 6.159943e+02	FES = 719100	Diversity = 3.5805e-03	k_t
= [1, 1, 1, 1]			
Generation 7200: Best = 6.159937e+02	FES = 720100	Diversity = 3.7178e-03	k_t
= [2, 4, 2, 2]			
Generation 7210: Best = 6.159923e+02	FES = 721100	Diversity = 3.7824e-03	k_t
= [1, 1, 1, 1]			
Generation 7220: Best = 6.159913e+02	FES = 722100	Diversity = 3.6636e-03	k_t
= [3, 2, 3, 2]			
Generation 7230: Best = 6.159895e+02	FES = 723100	Diversity = 3.7599e-03	k_t
= [1, 1, 1, 1]			
Generation 7240: Best = 6.159888e+02	FES = 724100	Diversity = 3.8413e-03	k_t
= [2, 4, 4, 4]			

Generation 7250: Best = 6.159871e+02	FES = 725100	Diversity = 3.7975e-03	k_t
= [1, 1, 1, 1]			
Generation 7260: Best = 6.159864e+02	FES = 726100	Diversity = 3.6440e-03	k_t
= [3, 4, 2, 2]			
Generation 7270: Best = 6.159857e+02	FES = 727100	Diversity = 3.7313e-03	k_t
= [1, 1, 1, 1]			
Generation 7280: Best = 6.159847e+02	FES = 728100	Diversity = 3.6926e-03	k_t
= [4, 2, 3, 2]			
Generation 7290: Best = 6.159838e+02	FES = 729100	Diversity = 3.5889e-03	k_t
= [1, 1, 1, 1]			
Generation 7300: Best = 6.159829e+02	FES = 730100	Diversity = 3.3273e-03	k_t
= [4, 4, 3, 2]			
Generation 7310: Best = 6.159816e+02	FES = 731100	Diversity = 3.2087e-03	k_t
= [1, 1, 1, 1]			
Generation 7320: Best = 6.159813e+02	FES = 732100	Diversity = 3.1275e-03	k_t
= [3, 3, 4, 3]			
Generation 7330: Best = 6.159807e+02	FES = 733100	Diversity = 3.0462e-03	k_t
= [1, 1, 1, 1]			
Generation 7340: Best = 6.159795e+02	FES = 734100	Diversity = 3.1760e-03	k_t
= [2, 4, 4, 2]			
Generation 7350: Best = 6.159783e+02	FES = 735100	Diversity = 3.6937e-03	k_t
= [1, 1, 1, 1]			
Generation 7360: Best = 6.159772e+02	FES = 736100	Diversity = 4.0886e-03	k_t
= [3, 3, 3, 2]			
Generation 7370: Best = 6.159767e+02	FES = 737100	Diversity = 4.1964e-03	k_t
= [1, 1, 1, 1]			
Generation 7380: Best = 6.159754e+02	FES = 738100	Diversity = 3.8637e-03	k_t
= [4, 2, 3, 4]			
Generation 7390: Best = 6.159746e+02	FES = 739100	Diversity = 3.7152e-03	k_t
= [1, 1, 1, 1]			
Generation 7400: Best = 6.159729e+02	FES = 740100	Diversity = 3.6008e-03	k_t
= [4, 3, 3, 2]			
Generation 7410: Best = 6.159724e+02	FES = 741100	Diversity = 3.5838e-03	k_t
= [1, 1, 1, 1]			
Generation 7420: Best = 6.159715e+02	FES = 742100	Diversity = 3.6083e-03	k_t
= [4, 4, 2, 2]			
Generation 7430: Best = 6.159709e+02	FES = 743100	Diversity = 3.7517e-03	k_t
= [1, 1, 1, 1]			
Generation 7440: Best = 6.159690e+02	FES = 744100	Diversity = 3.8116e-03	k_t
= [2, 3, 4, 2]			
Generation 7450: Best = 6.159678e+02	FES = 745100	Diversity = 3.9134e-03	k_t
= [1, 1, 1, 1]			
Generation 7460: Best = 6.159667e+02	FES = 746100	Diversity = 3.6450e-03	k_t
= [3, 3, 3, 3]			
Generation 7470: Best = 6.159659e+02	FES = 747100	Diversity = 3.4635e-03	k_t
= [1, 1, 1, 1]			
Generation 7480: Best = 6.159655e+02	FES = 748100	Diversity = 3.4105e-03	k_t
= [3, 4, 4, 3]			
Generation 7490: Best = 6.159644e+02	FES = 749100	Diversity = 3.3650e-03	k_t
= [1, 1, 1, 1]			
Generation 7500: Best = 6.159632e+02	FES = 750100	Diversity = 3.3898e-03	k_t
= [4, 3, 3, 4]			
Generation 7510: Best = 6.159621e+02	FES = 751100	Diversity = 3.5230e-03	k_t
= [1, 1, 1, 1]			

Generation 7520: Best = 6.159616e+02 = [2, 3, 2, 4]	FES = 752100	Diversity = 3.4206e-03	k_t
Generation 7530: Best = 6.159610e+02 = [1, 1, 1, 1]	FES = 753100	Diversity = 3.3713e-03	k_t
Generation 7540: Best = 6.159595e+02 = [2, 4, 3, 4]	FES = 754100	Diversity = 3.3578e-03	k_t
Generation 7550: Best = 6.159591e+02 = [1, 1, 1, 1]	FES = 755100	Diversity = 3.5329e-03	k_t
Generation 7560: Best = 6.159583e+02 = [3, 4, 3, 2]	FES = 756100	Diversity = 3.5996e-03	k_t
Generation 7570: Best = 6.159574e+02 = [1, 1, 1, 1]	FES = 757100	Diversity = 3.4260e-03	k_t
Generation 7580: Best = 6.159561e+02 = [3, 2, 3, 4]	FES = 758100	Diversity = 3.4663e-03	k_t
Generation 7590: Best = 6.159550e+02 = [1, 1, 1, 1]	FES = 759100	Diversity = 3.5200e-03	k_t
Generation 7600: Best = 6.159542e+02 = [3, 3, 2, 3]	FES = 760100	Diversity = 3.5937e-03	k_t
Generation 7610: Best = 6.159529e+02 = [1, 1, 1, 1]	FES = 761100	Diversity = 3.5371e-03	k_t
Generation 7620: Best = 6.159528e+02 = [2, 2, 4, 4]	FES = 762100	Diversity = 3.3416e-03	k_t
Generation 7630: Best = 6.159518e+02 = [1, 1, 1, 1]	FES = 763100	Diversity = 3.2294e-03	k_t
Generation 7640: Best = 6.159512e+02 = [2, 4, 4, 4]	FES = 764100	Diversity = 3.0346e-03	k_t
Generation 7650: Best = 6.159507e+02 = [1, 1, 1, 1]	FES = 765100	Diversity = 3.0227e-03	k_t
Generation 7660: Best = 6.159498e+02 = [2, 2, 2, 4]	FES = 766100	Diversity = 3.1815e-03	k_t
Generation 7670: Best = 6.159491e+02 = [1, 1, 1, 1]	FES = 767100	Diversity = 3.2379e-03	k_t
Generation 7680: Best = 6.159483e+02 = [2, 4, 3, 3]	FES = 768100	Diversity = 3.2511e-03	k_t
Generation 7690: Best = 6.159478e+02 = [1, 1, 1, 1]	FES = 769100	Diversity = 3.1624e-03	k_t
Generation 7700: Best = 6.159472e+02 = [3, 3, 3, 2]	FES = 770100	Diversity = 3.0146e-03	k_t
Generation 7710: Best = 6.159468e+02 = [1, 1, 1, 1]	FES = 771100	Diversity = 2.9898e-03	k_t
Generation 7720: Best = 6.159463e+02 = [4, 3, 3, 4]	FES = 772100	Diversity = 2.9825e-03	k_t
Generation 7730: Best = 6.159455e+02 = [1, 1, 1, 1]	FES = 773100	Diversity = 3.0569e-03	k_t
Generation 7740: Best = 6.159447e+02 = [4, 4, 4, 4]	FES = 774100	Diversity = 3.2730e-03	k_t
Generation 7750: Best = 6.159439e+02 = [1, 1, 1, 1]	FES = 775100	Diversity = 3.4476e-03	k_t
Generation 7760: Best = 6.159428e+02 = [2, 2, 2, 3]	FES = 776100	Diversity = 3.6311e-03	k_t
Generation 7770: Best = 6.159425e+02 = [1, 1, 1, 1]	FES = 777100	Diversity = 3.7509e-03	k_t
Generation 7780: Best = 6.159414e+02 = [4, 4, 3, 3]	FES = 778100	Diversity = 3.5210e-03	k_t

Generation 7790: Best = 6.159399e+02 = [1, 1, 1, 1]	FES = 779100	Diversity = 3.5311e-03	k_t
Generation 7800: Best = 6.159393e+02 = [4, 2, 3, 3]	FES = 780100	Diversity = 3.5143e-03	k_t
Generation 7810: Best = 6.159387e+02 = [1, 1, 1, 1]	FES = 781100	Diversity = 3.4081e-03	k_t
Generation 7820: Best = 6.159378e+02 = [4, 2, 4, 4]	FES = 782100	Diversity = 3.5782e-03	k_t
Generation 7830: Best = 6.159364e+02 = [1, 1, 1, 1]	FES = 783100	Diversity = 3.4236e-03	k_t
Generation 7840: Best = 6.159358e+02 = [2, 4, 3, 3]	FES = 784100	Diversity = 3.4709e-03	k_t
Generation 7850: Best = 6.159350e+02 = [1, 1, 1, 1]	FES = 785100	Diversity = 3.4411e-03	k_t
Generation 7860: Best = 6.159336e+02 = [3, 3, 2, 4]	FES = 786100	Diversity = 3.7062e-03	k_t
Generation 7870: Best = 6.159329e+02 = [1, 1, 1, 1]	FES = 787100	Diversity = 3.7003e-03	k_t
Generation 7880: Best = 6.159323e+02 = [4, 3, 4, 4]	FES = 788100	Diversity = 3.6355e-03	k_t
Generation 7890: Best = 6.159315e+02 = [1, 1, 1, 1]	FES = 789100	Diversity = 3.4158e-03	k_t
Generation 7900: Best = 6.159304e+02 = [2, 3, 2, 2]	FES = 790100	Diversity = 3.4202e-03	k_t
Generation 7910: Best = 6.159299e+02 = [1, 1, 1, 1]	FES = 791100	Diversity = 3.5800e-03	k_t
Generation 7920: Best = 6.159291e+02 = [3, 2, 4, 3]	FES = 792100	Diversity = 3.4498e-03	k_t
Generation 7930: Best = 6.159279e+02 = [1, 1, 1, 1]	FES = 793100	Diversity = 3.2916e-03	k_t
Generation 7940: Best = 6.159269e+02 = [3, 2, 3, 3]	FES = 794100	Diversity = 3.1399e-03	k_t
Generation 7950: Best = 6.159267e+02 = [1, 1, 1, 1]	FES = 795100	Diversity = 2.9678e-03	k_t
Generation 7960: Best = 6.159255e+02 = [3, 3, 4, 4]	FES = 796100	Diversity = 3.0005e-03	k_t
Generation 7970: Best = 6.159254e+02 = [1, 1, 1, 1]	FES = 797100	Diversity = 2.9278e-03	k_t
Generation 7980: Best = 6.159240e+02 = [4, 2, 4, 4]	FES = 798100	Diversity = 2.9363e-03	k_t
Generation 7990: Best = 6.159239e+02 = [1, 1, 1, 1]	FES = 799100	Diversity = 2.8049e-03	k_t
Generation 8000: Best = 6.159234e+02 = [2, 2, 4, 2]	FES = 800100	Diversity = 2.8297e-03	k_t
Generation 8010: Best = 6.159220e+02 = [1, 1, 1, 1]	FES = 801100	Diversity = 3.0675e-03	k_t
Generation 8020: Best = 6.159217e+02 = [2, 4, 3, 3]	FES = 802100	Diversity = 3.3538e-03	k_t
Generation 8030: Best = 6.159208e+02 = [1, 1, 1, 1]	FES = 803100	Diversity = 3.4859e-03	k_t
Generation 8040: Best = 6.159201e+02 = [2, 4, 3, 3]	FES = 804100	Diversity = 3.4921e-03	k_t
Generation 8050: Best = 6.159185e+02 = [1, 1, 1, 1]	FES = 805100	Diversity = 3.5606e-03	k_t

Generation 8060: Best = 6.159180e+02	FES = 806100	Diversity = 3.5883e-03	k_t
= [4, 2, 3, 3]			
Generation 8070: Best = 6.159179e+02	FES = 807100	Diversity = 3.8381e-03	k_t
= [1, 1, 1, 1]			
Generation 8080: Best = 6.159170e+02	FES = 808100	Diversity = 3.5129e-03	k_t
= [3, 4, 2, 2]			
Generation 8090: Best = 6.159162e+02	FES = 809100	Diversity = 3.5034e-03	k_t
= [1, 1, 1, 1]			
Generation 8100: Best = 6.159148e+02	FES = 810100	Diversity = 3.1811e-03	k_t
= [2, 3, 3, 2]			
Generation 8110: Best = 6.159146e+02	FES = 811100	Diversity = 3.0494e-03	k_t
= [1, 1, 1, 1]			
Generation 8120: Best = 6.159143e+02	FES = 812100	Diversity = 2.8953e-03	k_t
= [3, 2, 4, 3]			
Generation 8130: Best = 6.159133e+02	FES = 813100	Diversity = 3.1645e-03	k_t
= [1, 1, 1, 1]			
Generation 8140: Best = 6.159127e+02	FES = 814100	Diversity = 3.1887e-03	k_t
= [4, 2, 3, 4]			
Generation 8150: Best = 6.159109e+02	FES = 815100	Diversity = 3.1547e-03	k_t
= [1, 1, 1, 1]			
Generation 8160: Best = 6.159102e+02	FES = 816100	Diversity = 3.2679e-03	k_t
= [4, 4, 4, 3]			
Generation 8170: Best = 6.159095e+02	FES = 817100	Diversity = 3.2835e-03	k_t
= [1, 1, 1, 1]			
Generation 8180: Best = 6.159087e+02	FES = 818100	Diversity = 3.4412e-03	k_t
= [4, 4, 3, 3]			
Generation 8190: Best = 6.159078e+02	FES = 819100	Diversity = 3.5040e-03	k_t
= [1, 1, 1, 1]			
Generation 8200: Best = 6.159061e+02	FES = 820100	Diversity = 3.6578e-03	k_t
= [4, 4, 3, 2]			
Generation 8210: Best = 6.159058e+02	FES = 821100	Diversity = 3.9012e-03	k_t
= [1, 1, 1, 1]			
Generation 8220: Best = 6.159049e+02	FES = 822100	Diversity = 3.8042e-03	k_t
= [2, 4, 3, 2]			
Generation 8230: Best = 6.159042e+02	FES = 823100	Diversity = 3.5371e-03	k_t
= [1, 1, 1, 1]			
Generation 8240: Best = 6.159033e+02	FES = 824100	Diversity = 3.5016e-03	k_t
= [2, 2, 2, 2]			
Generation 8250: Best = 6.159028e+02	FES = 825100	Diversity = 3.4213e-03	k_t
= [1, 1, 1, 1]			
Generation 8260: Best = 6.159014e+02	FES = 826100	Diversity = 3.2583e-03	k_t
= [2, 3, 2, 3]			
Generation 8270: Best = 6.159009e+02	FES = 827100	Diversity = 2.9797e-03	k_t
= [1, 1, 1, 1]			
Generation 8280: Best = 6.159005e+02	FES = 828100	Diversity = 2.9208e-03	k_t
= [3, 3, 4, 4]			
Generation 8290: Best = 6.158998e+02	FES = 829100	Diversity = 2.9373e-03	k_t
= [1, 1, 1, 1]			
Generation 8300: Best = 6.158987e+02	FES = 830100	Diversity = 2.9690e-03	k_t
= [3, 2, 4, 3]			
Generation 8310: Best = 6.158975e+02	FES = 831100	Diversity = 3.4836e-03	k_t
= [1, 1, 1, 1]			
Generation 8320: Best = 6.158963e+02	FES = 832100	Diversity = 3.5796e-03	k_t
= [2, 4, 2, 3]			

Generation 8330: Best = 6.158961e+02	FES = 833100	Diversity = 3.2945e-03	k_t
= [1, 1, 1, 1]			
Generation 8340: Best = 6.158956e+02	FES = 834100	Diversity = 3.2942e-03	k_t
= [4, 2, 3, 4]			
Generation 8350: Best = 6.158943e+02	FES = 835100	Diversity = 3.2538e-03	k_t
= [1, 1, 1, 1]			
Generation 8360: Best = 6.158938e+02	FES = 836100	Diversity = 3.2556e-03	k_t
= [4, 2, 4, 2]			
Generation 8370: Best = 6.158929e+02	FES = 837100	Diversity = 3.2018e-03	k_t
= [1, 1, 1, 1]			
Generation 8380: Best = 6.158925e+02	FES = 838100	Diversity = 3.0015e-03	k_t
= [3, 4, 3, 4]			
Generation 8390: Best = 6.158916e+02	FES = 839100	Diversity = 3.0419e-03	k_t
= [1, 1, 1, 1]			
Generation 8400: Best = 6.158915e+02	FES = 840100	Diversity = 3.1045e-03	k_t
= [3, 4, 3, 2]			
Generation 8410: Best = 6.158903e+02	FES = 841100	Diversity = 2.9695e-03	k_t
= [1, 1, 1, 1]			
Generation 8420: Best = 6.158899e+02	FES = 842100	Diversity = 2.9543e-03	k_t
= [2, 4, 4, 4]			
Generation 8430: Best = 6.158893e+02	FES = 843100	Diversity = 2.9235e-03	k_t
= [1, 1, 1, 1]			
Generation 8440: Best = 6.158888e+02	FES = 844100	Diversity = 2.8643e-03	k_t
= [3, 4, 3, 3]			
Generation 8450: Best = 6.158875e+02	FES = 845100	Diversity = 2.8557e-03	k_t
= [1, 1, 1, 1]			
Generation 8460: Best = 6.158875e+02	FES = 846100	Diversity = 2.8962e-03	k_t
= [3, 2, 4, 3]			
Generation 8470: Best = 6.158871e+02	FES = 847100	Diversity = 2.6696e-03	k_t
= [1, 1, 1, 1]			
Generation 8480: Best = 6.158864e+02	FES = 848100	Diversity = 2.4440e-03	k_t
= [3, 4, 3, 3]			
Generation 8490: Best = 6.158861e+02	FES = 849100	Diversity = 2.3419e-03	k_t
= [1, 1, 1, 1]			
Generation 8500: Best = 6.158859e+02	FES = 850100	Diversity = 2.1903e-03	k_t
= [2, 3, 2, 3]			
Generation 8510: Best = 6.158852e+02	FES = 851100	Diversity = 2.0730e-03	k_t
= [1, 1, 1, 1]			
Generation 8520: Best = 6.158850e+02	FES = 852100	Diversity = 1.9939e-03	k_t
= [3, 4, 2, 4]			
Generation 8530: Best = 6.158848e+02	FES = 853100	Diversity = 2.2394e-03	k_t
= [1, 1, 1, 1]			
Generation 8540: Best = 6.158844e+02	FES = 854100	Diversity = 2.0891e-03	k_t
= [3, 3, 2, 4]			
Generation 8550: Best = 6.158841e+02	FES = 855100	Diversity = 2.0392e-03	k_t
= [1, 1, 1, 1]			
Generation 8560: Best = 6.158838e+02	FES = 856100	Diversity = 2.0747e-03	k_t
= [3, 2, 2, 4]			
Generation 8570: Best = 6.158837e+02	FES = 857100	Diversity = 2.0266e-03	k_t
= [1, 1, 1, 1]			
Generation 8580: Best = 6.158835e+02	FES = 858100	Diversity = 1.8991e-03	k_t
= [3, 3, 4, 2]			
Generation 8590: Best = 6.158830e+02	FES = 859100	Diversity = 1.8121e-03	k_t
= [1, 1, 1, 1]			

Generation 8600: Best = 6.158829e+02	FES = 860100	Diversity = 1.7199e-03	k_t
= [2, 3, 3, 2]			
Generation 8610: Best = 6.158826e+02	FES = 861100	Diversity = 1.7128e-03	k_t
= [1, 1, 1, 1]			
Generation 8620: Best = 6.158822e+02	FES = 862100	Diversity = 1.6581e-03	k_t
= [3, 4, 3, 2]			
Generation 8630: Best = 6.158821e+02	FES = 863100	Diversity = 1.6517e-03	k_t
= [1, 1, 1, 1]			
Generation 8640: Best = 6.158818e+02	FES = 864100	Diversity = 1.7214e-03	k_t
= [2, 4, 2, 2]			
Generation 8650: Best = 6.158813e+02	FES = 865100	Diversity = 1.7555e-03	k_t
= [1, 1, 1, 1]			
Generation 8660: Best = 6.158808e+02	FES = 866100	Diversity = 1.8401e-03	k_t
= [2, 3, 4, 3]			
Generation 8670: Best = 6.158805e+02	FES = 867100	Diversity = 2.0588e-03	k_t
= [1, 1, 1, 1]			
Generation 8680: Best = 6.158801e+02	FES = 868100	Diversity = 1.9293e-03	k_t
= [3, 3, 3, 3]			
Generation 8690: Best = 6.158799e+02	FES = 869100	Diversity = 1.9855e-03	k_t
= [1, 1, 1, 1]			
Generation 8700: Best = 6.158796e+02	FES = 870100	Diversity = 1.9976e-03	k_t
= [2, 3, 4, 3]			
Generation 8710: Best = 6.158791e+02	FES = 871100	Diversity = 1.8731e-03	k_t
= [1, 1, 1, 1]			
Generation 8720: Best = 6.158790e+02	FES = 872100	Diversity = 1.8338e-03	k_t
= [2, 3, 3, 2]			
Generation 8730: Best = 6.158785e+02	FES = 873100	Diversity = 1.8063e-03	k_t
= [1, 1, 1, 1]			
Generation 8740: Best = 6.158781e+02	FES = 874100	Diversity = 1.8733e-03	k_t
= [4, 3, 4, 3]			
Generation 8750: Best = 6.158778e+02	FES = 875100	Diversity = 1.9736e-03	k_t
= [1, 1, 1, 1]			
Generation 8760: Best = 6.158774e+02	FES = 876100	Diversity = 1.9001e-03	k_t
= [4, 3, 2, 4]			
Generation 8770: Best = 6.158772e+02	FES = 877100	Diversity = 1.8972e-03	k_t
= [1, 1, 1, 1]			
Generation 8780: Best = 6.158768e+02	FES = 878100	Diversity = 1.9532e-03	k_t
= [2, 2, 4, 2]			
Generation 8790: Best = 6.158764e+02	FES = 879100	Diversity = 1.9462e-03	k_t
= [1, 1, 1, 1]			
Generation 8800: Best = 6.158761e+02	FES = 880100	Diversity = 1.9523e-03	k_t
= [4, 2, 2, 3]			
Generation 8810: Best = 6.158759e+02	FES = 881100	Diversity = 1.9416e-03	k_t
= [1, 1, 1, 1]			
Generation 8820: Best = 6.158755e+02	FES = 882100	Diversity = 1.9478e-03	k_t
= [2, 2, 3, 2]			
Generation 8830: Best = 6.158751e+02	FES = 883100	Diversity = 1.9773e-03	k_t
= [1, 1, 1, 1]			
Generation 8840: Best = 6.158749e+02	FES = 884100	Diversity = 1.9090e-03	k_t
= [3, 2, 4, 2]			
Generation 8850: Best = 6.158747e+02	FES = 885100	Diversity = 1.8765e-03	k_t
= [1, 1, 1, 1]			
Generation 8860: Best = 6.158745e+02	FES = 886100	Diversity = 1.7472e-03	k_t
= [4, 3, 4, 4]			

Generation 8870: Best = 6.158742e+02 = [1, 1, 1, 1]	FES = 887100	Diversity = 1.7980e-03	k_t
Generation 8880: Best = 6.158739e+02 = [4, 2, 2, 2]	FES = 888100	Diversity = 1.8236e-03	k_t
Generation 8890: Best = 6.158736e+02 = [1, 1, 1, 1]	FES = 889100	Diversity = 1.7717e-03	k_t
Generation 8900: Best = 6.158734e+02 = [2, 4, 4, 4]	FES = 890100	Diversity = 1.7499e-03	k_t
Generation 8910: Best = 6.158730e+02 = [1, 1, 1, 1]	FES = 891100	Diversity = 1.7904e-03	k_t
Generation 8920: Best = 6.158728e+02 = [4, 2, 2, 2]	FES = 892100	Diversity = 1.8397e-03	k_t
Generation 8930: Best = 6.158725e+02 = [1, 1, 1, 1]	FES = 893100	Diversity = 1.8865e-03	k_t
Generation 8940: Best = 6.158721e+02 = [4, 2, 4, 4]	FES = 894100	Diversity = 1.8961e-03	k_t
Generation 8950: Best = 6.158718e+02 = [1, 1, 1, 1]	FES = 895100	Diversity = 1.9465e-03	k_t
Generation 8960: Best = 6.158714e+02 = [2, 2, 4, 3]	FES = 896100	Diversity = 2.0595e-03	k_t
Generation 8970: Best = 6.158710e+02 = [1, 1, 1, 1]	FES = 897100	Diversity = 1.8732e-03	k_t
Generation 8980: Best = 6.158706e+02 = [2, 2, 3, 4]	FES = 898100	Diversity = 1.8323e-03	k_t
Generation 8990: Best = 6.158702e+02 = [1, 1, 1, 1]	FES = 899100	Diversity = 1.8247e-03	k_t
Generation 9000: Best = 6.158696e+02 = [4, 3, 3, 2]	FES = 900100	Diversity = 2.0294e-03	k_t
Generation 9010: Best = 6.158695e+02 = [1, 1, 1, 1]	FES = 901100	Diversity = 2.0368e-03	k_t
Generation 9020: Best = 6.158693e+02 = [3, 2, 3, 3]	FES = 902100	Diversity = 2.0934e-03	k_t
Generation 9030: Best = 6.158686e+02 = [1, 1, 1, 1]	FES = 903100	Diversity = 1.9056e-03	k_t
Generation 9040: Best = 6.158683e+02 = [3, 3, 2, 4]	FES = 904100	Diversity = 1.8773e-03	k_t
Generation 9050: Best = 6.158677e+02 = [1, 1, 1, 1]	FES = 905100	Diversity = 2.0346e-03	k_t
Generation 9060: Best = 6.158675e+02 = [3, 4, 4, 4]	FES = 906100	Diversity = 2.0757e-03	k_t
Generation 9070: Best = 6.158670e+02 = [1, 1, 1, 1]	FES = 907100	Diversity = 2.0786e-03	k_t
Generation 9080: Best = 6.158669e+02 = [4, 4, 4, 3]	FES = 908100	Diversity = 2.0011e-03	k_t
Generation 9090: Best = 6.158665e+02 = [1, 1, 1, 1]	FES = 909100	Diversity = 1.9442e-03	k_t
Generation 9100: Best = 6.158662e+02 = [3, 2, 3, 4]	FES = 910100	Diversity = 1.8367e-03	k_t
Generation 9110: Best = 6.158660e+02 = [1, 1, 1, 1]	FES = 911100	Diversity = 1.8500e-03	k_t
Generation 9120: Best = 6.158657e+02 = [4, 3, 2, 2]	FES = 912100	Diversity = 1.7654e-03	k_t
Generation 9130: Best = 6.158654e+02 = [1, 1, 1, 1]	FES = 913100	Diversity = 1.8587e-03	k_t

Generation 9140: Best = 6.158649e+02 = [2, 4, 4, 4]	FES = 914100	Diversity = 1.8007e-03	k_t
Generation 9150: Best = 6.158647e+02 = [1, 1, 1, 1]	FES = 915100	Diversity = 1.8702e-03	k_t
Generation 9160: Best = 6.158645e+02 = [2, 4, 2, 2]	FES = 916100	Diversity = 1.8215e-03	k_t
Generation 9170: Best = 6.158642e+02 = [1, 1, 1, 1]	FES = 917100	Diversity = 1.7674e-03	k_t
Generation 9180: Best = 6.158640e+02 = [4, 2, 2, 3]	FES = 918100	Diversity = 1.6510e-03	k_t
Generation 9190: Best = 6.158637e+02 = [1, 1, 1, 1]	FES = 919100	Diversity = 1.5053e-03	k_t
Generation 9200: Best = 6.158636e+02 = [2, 2, 4, 4]	FES = 920100	Diversity = 1.5069e-03	k_t
Generation 9210: Best = 6.158633e+02 = [1, 1, 1, 1]	FES = 921100	Diversity = 1.5840e-03	k_t
Generation 9220: Best = 6.158630e+02 = [2, 2, 3, 2]	FES = 922100	Diversity = 1.5561e-03	k_t
Generation 9230: Best = 6.158627e+02 = [1, 1, 1, 1]	FES = 923100	Diversity = 1.4738e-03	k_t
Generation 9240: Best = 6.158625e+02 = [4, 2, 3, 3]	FES = 924100	Diversity = 1.4428e-03	k_t
Generation 9250: Best = 6.158624e+02 = [1, 1, 1, 1]	FES = 925100	Diversity = 1.4110e-03	k_t
Generation 9260: Best = 6.158622e+02 = [2, 4, 2, 4]	FES = 926100	Diversity = 1.4219e-03	k_t
Generation 9270: Best = 6.158620e+02 = [1, 1, 1, 1]	FES = 927100	Diversity = 1.3879e-03	k_t
Generation 9280: Best = 6.158620e+02 = [3, 4, 4, 2]	FES = 928100	Diversity = 1.4164e-03	k_t
Generation 9290: Best = 6.158618e+02 = [1, 1, 1, 1]	FES = 929100	Diversity = 1.5010e-03	k_t
Generation 9300: Best = 6.158614e+02 = [2, 2, 2, 4]	FES = 930100	Diversity = 1.6874e-03	k_t
Generation 9310: Best = 6.158609e+02 = [1, 1, 1, 1]	FES = 931100	Diversity = 1.8804e-03	k_t
Generation 9320: Best = 6.158607e+02 = [2, 3, 4, 2]	FES = 932100	Diversity = 1.9504e-03	k_t
Generation 9330: Best = 6.158601e+02 = [1, 1, 1, 1]	FES = 933100	Diversity = 2.0645e-03	k_t
Generation 9340: Best = 6.158598e+02 = [2, 4, 3, 2]	FES = 934100	Diversity = 2.1636e-03	k_t
Generation 9350: Best = 6.158592e+02 = [1, 1, 1, 1]	FES = 935100	Diversity = 2.0447e-03	k_t
Generation 9360: Best = 6.158592e+02 = [3, 4, 2, 4]	FES = 936100	Diversity = 1.9644e-03	k_t
Generation 9370: Best = 6.158588e+02 = [1, 1, 1, 1]	FES = 937100	Diversity = 2.0891e-03	k_t
Generation 9380: Best = 6.158585e+02 = [3, 4, 2, 3]	FES = 938100	Diversity = 2.1248e-03	k_t
Generation 9390: Best = 6.158581e+02 = [1, 1, 1, 1]	FES = 939100	Diversity = 2.1939e-03	k_t
Generation 9400: Best = 6.158579e+02 = [2, 3, 4, 3]	FES = 940100	Diversity = 2.1781e-03	k_t

Generation 9410: Best = 6.158575e+02	FES = 941100	Diversity = 2.1557e-03	k_t
= [1, 1, 1, 1]			
Generation 9420: Best = 6.158571e+02	FES = 942100	Diversity = 1.9916e-03	k_t
= [4, 2, 2, 4]			
Generation 9430: Best = 6.158570e+02	FES = 943100	Diversity = 1.9391e-03	k_t
= [1, 1, 1, 1]			
Generation 9440: Best = 6.158564e+02	FES = 944100	Diversity = 1.8632e-03	k_t
= [2, 3, 4, 3]			
Generation 9450: Best = 6.158564e+02	FES = 945100	Diversity = 1.7594e-03	k_t
= [1, 1, 1, 1]			
Generation 9460: Best = 6.158562e+02	FES = 946100	Diversity = 1.7403e-03	k_t
= [2, 3, 3, 4]			
Generation 9470: Best = 6.158560e+02	FES = 947100	Diversity = 1.6886e-03	k_t
= [1, 1, 1, 1]			
Generation 9480: Best = 6.158557e+02	FES = 948100	Diversity = 1.6178e-03	k_t
= [4, 4, 2, 2]			
Generation 9490: Best = 6.158555e+02	FES = 949100	Diversity = 1.6248e-03	k_t
= [1, 1, 1, 1]			
Generation 9500: Best = 6.158552e+02	FES = 950100	Diversity = 1.6277e-03	k_t
= [4, 4, 3, 4]			
Generation 9510: Best = 6.158551e+02	FES = 951100	Diversity = 1.5161e-03	k_t
= [1, 1, 1, 1]			
Generation 9520: Best = 6.158550e+02	FES = 952100	Diversity = 1.4273e-03	k_t
= [2, 2, 2, 3]			
Generation 9530: Best = 6.158549e+02	FES = 953100	Diversity = 1.4443e-03	k_t
= [1, 1, 1, 1]			
Generation 9540: Best = 6.158546e+02	FES = 954100	Diversity = 1.4024e-03	k_t
= [2, 2, 2, 3]			
Generation 9550: Best = 6.158544e+02	FES = 955100	Diversity = 1.5326e-03	k_t
= [1, 1, 1, 1]			
Generation 9560: Best = 6.158541e+02	FES = 956100	Diversity = 1.5288e-03	k_t
= [2, 4, 2, 2]			
Generation 9570: Best = 6.158538e+02	FES = 957100	Diversity = 1.6651e-03	k_t
= [1, 1, 1, 1]			
Generation 9580: Best = 6.158536e+02	FES = 958100	Diversity = 1.6533e-03	k_t
= [3, 3, 3, 2]			
Generation 9590: Best = 6.158533e+02	FES = 959100	Diversity = 1.6778e-03	k_t
= [1, 1, 1, 1]			
Generation 9600: Best = 6.158529e+02	FES = 960100	Diversity = 1.8445e-03	k_t
= [3, 2, 3, 4]			
Generation 9610: Best = 6.158527e+02	FES = 961100	Diversity = 2.0758e-03	k_t
= [1, 1, 1, 1]			
Generation 9620: Best = 6.158523e+02	FES = 962100	Diversity = 2.0948e-03	k_t
= [2, 2, 3, 2]			
Generation 9630: Best = 6.158521e+02	FES = 963100	Diversity = 2.1062e-03	k_t
= [1, 1, 1, 1]			
Generation 9640: Best = 6.158517e+02	FES = 964100	Diversity = 2.0686e-03	k_t
= [2, 3, 4, 4]			
Generation 9650: Best = 6.158516e+02	FES = 965100	Diversity = 1.9785e-03	k_t
= [1, 1, 1, 1]			
Generation 9660: Best = 6.158512e+02	FES = 966100	Diversity = 1.9632e-03	k_t
= [4, 2, 3, 2]			
Generation 9670: Best = 6.158510e+02	FES = 967100	Diversity = 1.9381e-03	k_t
= [1, 1, 1, 1]			

Generation 9680: Best = 6.158508e+02	FES = 968100	Diversity = 1.9067e-03	k_t = [4, 2, 4, 4]
Generation 9690: Best = 6.158506e+02	FES = 969100	Diversity = 1.9034e-03	k_t = [1, 1, 1, 1]
Generation 9700: Best = 6.158502e+02	FES = 970100	Diversity = 1.8609e-03	k_t = [3, 2, 2, 3]
Generation 9710: Best = 6.158501e+02	FES = 971100	Diversity = 1.8514e-03	k_t = [1, 1, 1, 1]
Generation 9720: Best = 6.158499e+02	FES = 972100	Diversity = 1.7825e-03	k_t = [2, 3, 3, 2]
Generation 9730: Best = 6.158496e+02	FES = 973100	Diversity = 1.9533e-03	k_t = [1, 1, 1, 1]
Generation 9740: Best = 6.158493e+02	FES = 974100	Diversity = 1.9242e-03	k_t = [3, 2, 3, 3]
Generation 9750: Best = 6.158489e+02	FES = 975100	Diversity = 1.8129e-03	k_t = [1, 1, 1, 1]
Generation 9760: Best = 6.158488e+02	FES = 976100	Diversity = 1.7747e-03	k_t = [3, 2, 3, 4]
Generation 9770: Best = 6.158487e+02	FES = 977100	Diversity = 1.7787e-03	k_t = [1, 1, 1, 1]
Generation 9780: Best = 6.158485e+02	FES = 978100	Diversity = 1.7595e-03	k_t = [4, 4, 2, 4]
Generation 9790: Best = 6.158484e+02	FES = 979100	Diversity = 1.8242e-03	k_t = [1, 1, 1, 1]
Generation 9800: Best = 6.158479e+02	FES = 980100	Diversity = 1.7987e-03	k_t = [4, 3, 2, 2]
Generation 9810: Best = 6.158477e+02	FES = 981100	Diversity = 1.6164e-03	k_t = [1, 1, 1, 1]
Generation 9820: Best = 6.158474e+02	FES = 982100	Diversity = 1.7100e-03	k_t = [2, 3, 2, 2]
Generation 9830: Best = 6.158473e+02	FES = 983100	Diversity = 1.6839e-03	k_t = [1, 1, 1, 1]
Generation 9840: Best = 6.158470e+02	FES = 984100	Diversity = 1.6391e-03	k_t = [3, 4, 2, 4]
Generation 9850: Best = 6.158467e+02	FES = 985100	Diversity = 1.6418e-03	k_t = [1, 1, 1, 1]
Generation 9860: Best = 6.158465e+02	FES = 986100	Diversity = 1.7353e-03	k_t = [2, 2, 3, 3]
Generation 9870: Best = 6.158461e+02	FES = 987100	Diversity = 1.8031e-03	k_t = [1, 1, 1, 1]
Generation 9880: Best = 6.158460e+02	FES = 988100	Diversity = 1.7437e-03	k_t = [2, 3, 4, 2]
Generation 9890: Best = 6.158457e+02	FES = 989100	Diversity = 1.6687e-03	k_t = [1, 1, 1, 1]
Generation 9900: Best = 6.158453e+02	FES = 990100	Diversity = 1.6378e-03	k_t = [2, 3, 2, 2]
Generation 9910: Best = 6.158452e+02	FES = 991100	Diversity = 1.6570e-03	k_t = [1, 1, 1, 1]
Generation 9920: Best = 6.158450e+02	FES = 992100	Diversity = 1.6757e-03	k_t = [4, 3, 3, 2]
Generation 9930: Best = 6.158447e+02	FES = 993100	Diversity = 1.6879e-03	k_t = [1, 1, 1, 1]
Generation 9940: Best = 6.158446e+02	FES = 994100	Diversity = 1.7642e-03	k_t = [2, 4, 2, 3]

Generation 9950: Best = 6.158443e+02 FES = 995100 Diversity = 1.8588e-03 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 6.158440e+02 FES = 996100 Diversity = 1.7528e-03 k_t
= [3, 2, 3, 4]
Generation 9970: Best = 6.158438e+02 FES = 997100 Diversity = 1.6268e-03 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 6.158438e+02 FES = 998100 Diversity = 1.4852e-03 k_t
= [4, 4, 4, 4]
Generation 9990: Best = 6.158436e+02 FES = 999100 Diversity = 1.4654e-03 k_t
= [1, 1, 1, 1]

Optimization completed!

Final Best Fitness = 6.158432e+02

Total Function Evaluations = 1000000

=====
=

Results:

Best Fitness : 6.158432e+02

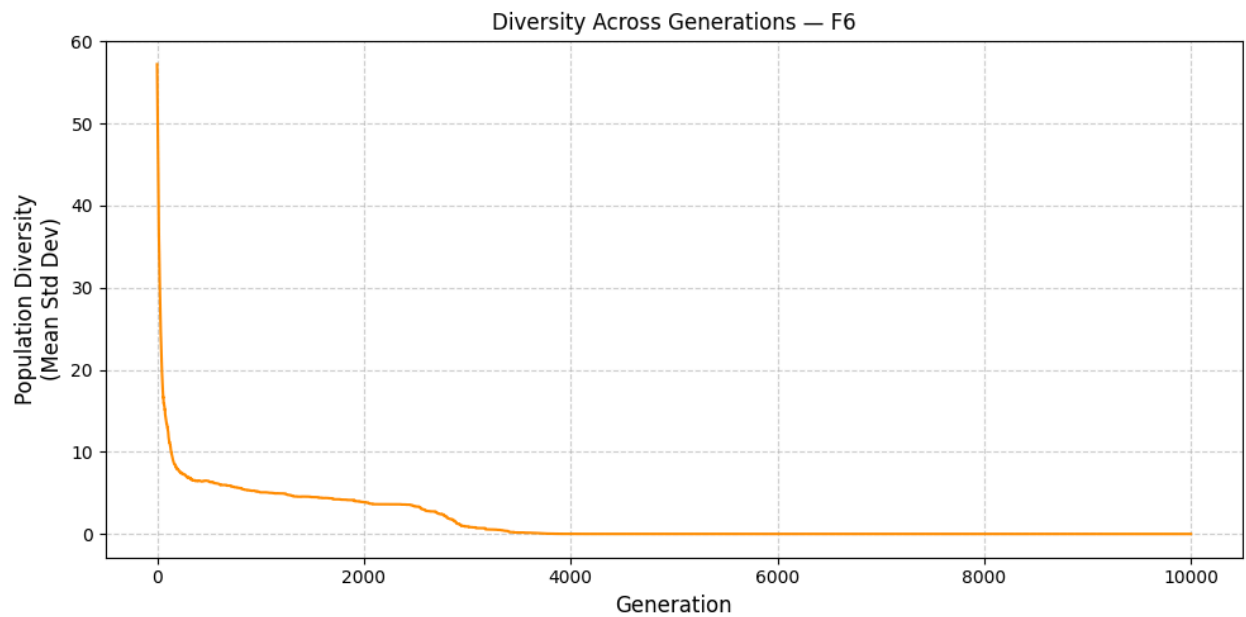
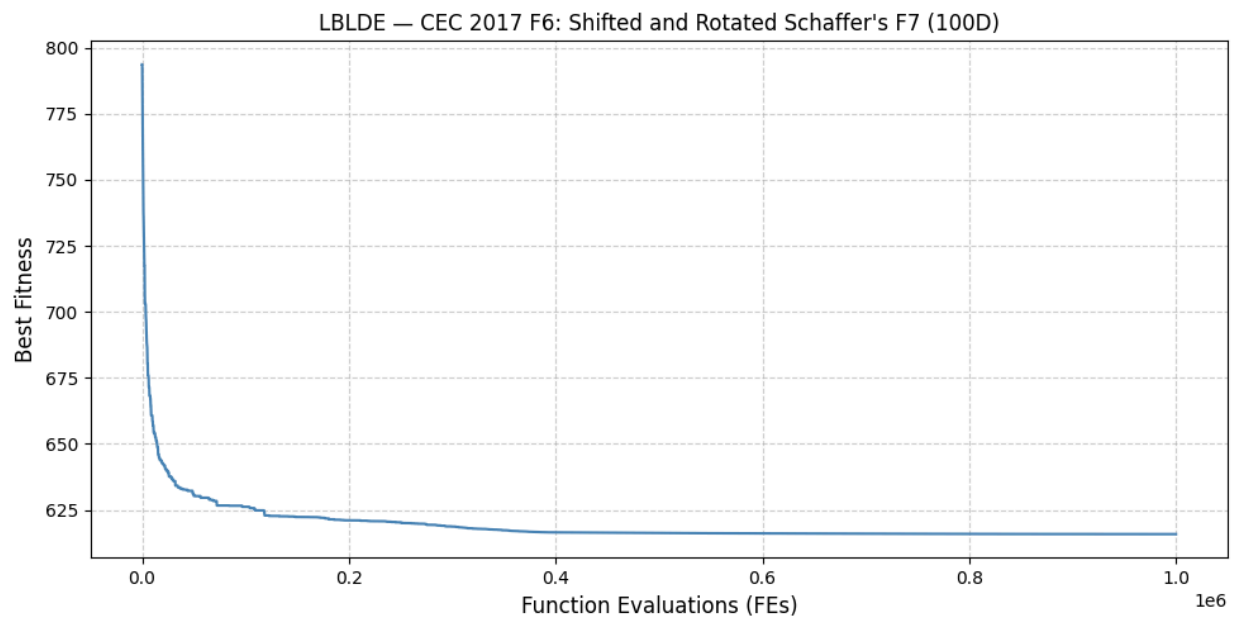
Global Optimum : 6.000000e+02

Error (|best-optimum|) : 1.584323e+01

Best Solution (5 dim) : [69.05559572 -16.85195308 6.69825428 -33.46299485
54.4121201]

D^(0) (init. diversity): 5.7220e+01

=====
=



Graph saved as convergence_F6.png

```
In [31]: # =====
# F7 — Shifted and Rotated Lunacek Bi-Rastrigin
# =====
run_and_plot('F7', make_wrapper(simple.f7), 700.0, "Shifted and Rotated Lunacek Bi-Rastrigin")
```

```

=====
=
LBLDE_s on CEC 2017 F7 – Single Run
=====
=
Function      : Shifted and Rotated Lunacek Bi-Rastrigin's (100D)
Bounds       : [-100, 100]
Global optimum : 700.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 1.400178e+04 | Diversity = 5.7220e+01
Generation 10: Best = 6.260204e+03 FES = 1100 Diversity = 3.3296e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 4.102519e+03 FES = 2100 Diversity = 2.1009e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 3.187306e+03 FES = 3100 Diversity = 1.4059e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.668470e+03 FES = 4100 Diversity = 9.8856e+00 k_t
= [2, 2, 2, 4]
Generation 50: Best = 2.473260e+03 FES = 5100 Diversity = 7.8792e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.291593e+03 FES = 6100 Diversity = 6.7769e+00 k_t
= [2, 4, 3, 4]
Generation 70: Best = 2.190108e+03 FES = 7100 Diversity = 6.0032e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.116652e+03 FES = 8100 Diversity = 5.5301e+00 k_t
= [4, 3, 2, 3]
Generation 90: Best = 2.064707e+03 FES = 9100 Diversity = 5.1971e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.024630e+03 FES = 10100 Diversity = 5.0035e+00 k_t
= [3, 3, 2, 3]
Generation 110: Best = 1.990640e+03 FES = 11100 Diversity = 4.9480e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.990640e+03 FES = 12100 Diversity = 4.8882e+00 k_t
= [4, 2, 2, 3]
Generation 130: Best = 1.990640e+03 FES = 13100 Diversity = 4.7652e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.952185e+03 FES = 14100 Diversity = 4.6935e+00 k_t
= [4, 4, 4, 2]
Generation 150: Best = 1.937860e+03 FES = 15100 Diversity = 4.5994e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.918799e+03 FES = 16100 Diversity = 4.5948e+00 k_t
= [2, 3, 4, 4]
Generation 170: Best = 1.906847e+03 FES = 17100 Diversity = 4.6194e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.898103e+03 FES = 18100 Diversity = 4.6563e+00 k_t
= [2, 4, 3, 4]
Generation 190: Best = 1.878643e+03 FES = 19100 Diversity = 4.6353e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.869899e+03 FES = 20100 Diversity = 4.6160e+00 k_t
= [4, 3, 2, 2]
Generation 210: Best = 1.858994e+03 FES = 21100 Diversity = 4.5813e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.839800e+03 FES = 22100 Diversity = 4.5332e+00 k_t
= [4, 3, 4, 2]

```

Generation 230:	Best = 1.839800e+03	FES = 23100	Diversity = 4.5238e+00	k_t = [1, 1, 1, 1]
Generation 240:	Best = 1.839800e+03	FES = 24100	Diversity = 4.5336e+00	k_t = [2, 2, 3, 4]
Generation 250:	Best = 1.820145e+03	FES = 25100	Diversity = 4.5282e+00	k_t = [1, 1, 1, 1]
Generation 260:	Best = 1.820145e+03	FES = 26100	Diversity = 4.5256e+00	k_t = [4, 2, 4, 2]
Generation 270:	Best = 1.771947e+03	FES = 27100	Diversity = 4.5094e+00	k_t = [1, 1, 1, 1]
Generation 280:	Best = 1.747794e+03	FES = 28100	Diversity = 4.5047e+00	k_t = [4, 4, 4, 3]
Generation 290:	Best = 1.747794e+03	FES = 29100	Diversity = 4.4309e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 1.747794e+03	FES = 30100	Diversity = 4.4429e+00	k_t = [4, 4, 4, 4]
Generation 310:	Best = 1.747794e+03	FES = 31100	Diversity = 4.4386e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 1.747794e+03	FES = 32100	Diversity = 4.4317e+00	k_t = [4, 2, 2, 4]
Generation 330:	Best = 1.747794e+03	FES = 33100	Diversity = 4.4075e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 1.747794e+03	FES = 34100	Diversity = 4.3898e+00	k_t = [4, 2, 3, 3]
Generation 350:	Best = 1.743917e+03	FES = 35100	Diversity = 4.4126e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 1.743917e+03	FES = 36100	Diversity = 4.4064e+00	k_t = [3, 2, 4, 4]
Generation 370:	Best = 1.743917e+03	FES = 37100	Diversity = 4.3986e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.740323e+03	FES = 38100	Diversity = 4.3734e+00	k_t = [3, 3, 4, 2]
Generation 390:	Best = 1.736327e+03	FES = 39100	Diversity = 4.3705e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.736327e+03	FES = 40100	Diversity = 4.3695e+00	k_t = [4, 3, 3, 2]
Generation 410:	Best = 1.736327e+03	FES = 41100	Diversity = 4.3487e+00	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.736008e+03	FES = 42100	Diversity = 4.3297e+00	k_t = [3, 4, 2, 3]
Generation 430:	Best = 1.736008e+03	FES = 43100	Diversity = 4.3357e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.720314e+03	FES = 44100	Diversity = 4.3468e+00	k_t = [2, 4, 4, 4]
Generation 450:	Best = 1.720314e+03	FES = 45100	Diversity = 4.3493e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.704650e+03	FES = 46100	Diversity = 4.3402e+00	k_t = [3, 3, 4, 2]
Generation 470:	Best = 1.704650e+03	FES = 47100	Diversity = 4.3354e+00	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.704650e+03	FES = 48100	Diversity = 4.3312e+00	k_t = [4, 2, 2, 4]
Generation 490:	Best = 1.703743e+03	FES = 49100	Diversity = 4.3217e+00	k_t = [1, 1, 1, 1]

Generation 500: Best = 1.693716e+03	FES = 50100	Diversity = 4.2870e+00	k_t
= [2, 4, 3, 4]			
Generation 510: Best = 1.693716e+03	FES = 51100	Diversity = 4.2495e+00	k_t
= [1, 1, 1, 1]			
Generation 520: Best = 1.693716e+03	FES = 52100	Diversity = 4.2497e+00	k_t
= [2, 3, 4, 4]			
Generation 530: Best = 1.693716e+03	FES = 53100	Diversity = 4.2497e+00	k_t
= [1, 1, 1, 1]			
Generation 540: Best = 1.693716e+03	FES = 54100	Diversity = 4.2138e+00	k_t
= [3, 2, 3, 2]			
Generation 550: Best = 1.687184e+03	FES = 55100	Diversity = 4.2125e+00	k_t
= [1, 1, 1, 1]			
Generation 560: Best = 1.687184e+03	FES = 56100	Diversity = 4.1751e+00	k_t
= [2, 2, 2, 4]			
Generation 570: Best = 1.682933e+03	FES = 57100	Diversity = 4.1812e+00	k_t
= [1, 1, 1, 1]			
Generation 580: Best = 1.682933e+03	FES = 58100	Diversity = 4.1807e+00	k_t
= [3, 3, 4, 2]			
Generation 590: Best = 1.676541e+03	FES = 59100	Diversity = 4.1620e+00	k_t
= [1, 1, 1, 1]			
Generation 600: Best = 1.671030e+03	FES = 60100	Diversity = 4.1632e+00	k_t
= [4, 2, 4, 4]			
Generation 610: Best = 1.664950e+03	FES = 61100	Diversity = 4.1533e+00	k_t
= [1, 1, 1, 1]			
Generation 620: Best = 1.664950e+03	FES = 62100	Diversity = 4.1463e+00	k_t
= [3, 3, 4, 3]			
Generation 630: Best = 1.664950e+03	FES = 63100	Diversity = 4.1457e+00	k_t
= [1, 1, 1, 1]			
Generation 640: Best = 1.638302e+03	FES = 64100	Diversity = 4.1346e+00	k_t
= [3, 2, 3, 2]			
Generation 650: Best = 1.629499e+03	FES = 65100	Diversity = 4.1254e+00	k_t
= [1, 1, 1, 1]			
Generation 660: Best = 1.629499e+03	FES = 66100	Diversity = 4.1228e+00	k_t
= [2, 2, 4, 2]			
Generation 670: Best = 1.587048e+03	FES = 67100	Diversity = 4.1205e+00	k_t
= [1, 1, 1, 1]			
Generation 680: Best = 1.577934e+03	FES = 68100	Diversity = 4.1185e+00	k_t
= [3, 4, 4, 3]			
Generation 690: Best = 1.577934e+03	FES = 69100	Diversity = 4.1126e+00	k_t
= [1, 1, 1, 1]			
Generation 700: Best = 1.577934e+03	FES = 70100	Diversity = 4.1045e+00	k_t
= [3, 2, 4, 4]			
Generation 710: Best = 1.576145e+03	FES = 71100	Diversity = 4.1058e+00	k_t
= [1, 1, 1, 1]			
Generation 720: Best = 1.576145e+03	FES = 72100	Diversity = 4.1069e+00	k_t
= [4, 2, 4, 4]			
Generation 730: Best = 1.576145e+03	FES = 73100	Diversity = 4.0872e+00	k_t
= [1, 1, 1, 1]			
Generation 740: Best = 1.576145e+03	FES = 74100	Diversity = 4.0870e+00	k_t
= [4, 3, 2, 2]			
Generation 750: Best = 1.576145e+03	FES = 75100	Diversity = 4.0845e+00	k_t
= [1, 1, 1, 1]			
Generation 760: Best = 1.575960e+03	FES = 76100	Diversity = 4.0805e+00	k_t
= [3, 4, 3, 4]			

Generation 770:	Best = 1.575960e+03	FES = 77100	Diversity = 4.0777e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.575960e+03	FES = 78100	Diversity = 4.0758e+00	k_t
= [4, 2, 2, 2]				
Generation 790:	Best = 1.573282e+03	FES = 79100	Diversity = 4.0733e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.573282e+03	FES = 80100	Diversity = 4.0654e+00	k_t
= [3, 4, 3, 3]				
Generation 810:	Best = 1.564331e+03	FES = 81100	Diversity = 4.0559e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.564331e+03	FES = 82100	Diversity = 4.0525e+00	k_t
= [3, 3, 2, 3]				
Generation 830:	Best = 1.546672e+03	FES = 83100	Diversity = 4.0330e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.542616e+03	FES = 84100	Diversity = 4.0283e+00	k_t
= [4, 4, 4, 4]				
Generation 850:	Best = 1.542616e+03	FES = 85100	Diversity = 4.0270e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.528485e+03	FES = 86100	Diversity = 4.0151e+00	k_t
= [3, 4, 4, 2]				
Generation 870:	Best = 1.528485e+03	FES = 87100	Diversity = 4.0124e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.528485e+03	FES = 88100	Diversity = 4.0122e+00	k_t
= [3, 4, 2, 2]				
Generation 890:	Best = 1.528485e+03	FES = 89100	Diversity = 4.0114e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.528485e+03	FES = 90100	Diversity = 4.0043e+00	k_t
= [2, 4, 2, 4]				
Generation 910:	Best = 1.528485e+03	FES = 91100	Diversity = 4.0017e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.528485e+03	FES = 92100	Diversity = 4.0081e+00	k_t
= [4, 2, 2, 3]				
Generation 930:	Best = 1.528485e+03	FES = 93100	Diversity = 4.0183e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.528485e+03	FES = 94100	Diversity = 4.0167e+00	k_t
= [3, 2, 3, 4]				
Generation 950:	Best = 1.516519e+03	FES = 95100	Diversity = 4.0207e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.516519e+03	FES = 96100	Diversity = 4.0311e+00	k_t
= [2, 2, 2, 3]				
Generation 970:	Best = 1.516519e+03	FES = 97100	Diversity = 4.0296e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.495654e+03	FES = 98100	Diversity = 4.0392e+00	k_t
= [3, 4, 3, 4]				
Generation 990:	Best = 1.494456e+03	FES = 99100	Diversity = 4.0375e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 1.494456e+03	FES = 100100	Diversity = 4.0299e+00	k_t
= [3, 2, 4, 2]				
Generation 1010:	Best = 1.487270e+03	FES = 101100	Diversity = 4.0280e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.487270e+03	FES = 102100	Diversity = 4.0264e+00	k_t
= [4, 3, 2, 4]				
Generation 1030:	Best = 1.487270e+03	FES = 103100	Diversity = 4.0232e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.487270e+03 = [2, 4, 4, 2]	FES = 104100	Diversity = 4.0122e+00	k_t
Generation 1050: Best = 1.487270e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 3.9938e+00	k_t
Generation 1060: Best = 1.485350e+03 = [2, 3, 3, 2]	FES = 106100	Diversity = 3.9948e+00	k_t
Generation 1070: Best = 1.485350e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 3.9984e+00	k_t
Generation 1080: Best = 1.474467e+03 = [4, 2, 4, 3]	FES = 108100	Diversity = 3.9976e+00	k_t
Generation 1090: Best = 1.474467e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 3.9717e+00	k_t
Generation 1100: Best = 1.474467e+03 = [4, 2, 4, 3]	FES = 110100	Diversity = 3.9734e+00	k_t
Generation 1110: Best = 1.473750e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 3.9712e+00	k_t
Generation 1120: Best = 1.473750e+03 = [4, 4, 4, 3]	FES = 112100	Diversity = 3.9408e+00	k_t
Generation 1130: Best = 1.473750e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 3.9326e+00	k_t
Generation 1140: Best = 1.473750e+03 = [2, 3, 3, 3]	FES = 114100	Diversity = 3.9211e+00	k_t
Generation 1150: Best = 1.473750e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 3.9210e+00	k_t
Generation 1160: Best = 1.473750e+03 = [3, 3, 4, 2]	FES = 116100	Diversity = 3.9273e+00	k_t
Generation 1170: Best = 1.473750e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 3.9260e+00	k_t
Generation 1180: Best = 1.473307e+03 = [3, 4, 3, 4]	FES = 118100	Diversity = 3.9207e+00	k_t
Generation 1190: Best = 1.469606e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 3.9322e+00	k_t
Generation 1200: Best = 1.467937e+03 = [4, 4, 2, 3]	FES = 120100	Diversity = 3.9317e+00	k_t
Generation 1210: Best = 1.467937e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 3.9243e+00	k_t
Generation 1220: Best = 1.466240e+03 = [2, 3, 4, 4]	FES = 122100	Diversity = 3.9235e+00	k_t
Generation 1230: Best = 1.465660e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 3.9072e+00	k_t
Generation 1240: Best = 1.465660e+03 = [2, 4, 4, 2]	FES = 124100	Diversity = 3.9062e+00	k_t
Generation 1250: Best = 1.464061e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 3.9060e+00	k_t
Generation 1260: Best = 1.464061e+03 = [4, 2, 3, 4]	FES = 126100	Diversity = 3.9024e+00	k_t
Generation 1270: Best = 1.461926e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 3.8934e+00	k_t
Generation 1280: Best = 1.461926e+03 = [2, 4, 2, 2]	FES = 128100	Diversity = 3.8942e+00	k_t
Generation 1290: Best = 1.461926e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 3.8870e+00	k_t
Generation 1300: Best = 1.461926e+03 = [3, 4, 2, 2]	FES = 130100	Diversity = 3.8731e+00	k_t

Generation 1310: Best = 1.461926e+03	FES = 131100	Diversity = 3.8729e+00	k_t
= [1, 1, 1, 1]			
Generation 1320: Best = 1.461926e+03	FES = 132100	Diversity = 3.8720e+00	k_t
= [3, 4, 2, 2]			
Generation 1330: Best = 1.452979e+03	FES = 133100	Diversity = 3.8718e+00	k_t
= [1, 1, 1, 1]			
Generation 1340: Best = 1.452979e+03	FES = 134100	Diversity = 3.8705e+00	k_t
= [4, 2, 4, 4]			
Generation 1350: Best = 1.452979e+03	FES = 135100	Diversity = 3.8708e+00	k_t
= [1, 1, 1, 1]			
Generation 1360: Best = 1.452979e+03	FES = 136100	Diversity = 3.8694e+00	k_t
= [2, 2, 4, 3]			
Generation 1370: Best = 1.426314e+03	FES = 137100	Diversity = 3.8683e+00	k_t
= [1, 1, 1, 1]			
Generation 1380: Best = 1.426314e+03	FES = 138100	Diversity = 3.8711e+00	k_t
= [4, 4, 4, 4]			
Generation 1390: Best = 1.426314e+03	FES = 139100	Diversity = 3.8797e+00	k_t
= [1, 1, 1, 1]			
Generation 1400: Best = 1.426314e+03	FES = 140100	Diversity = 3.8800e+00	k_t
= [4, 4, 4, 2]			
Generation 1410: Best = 1.426314e+03	FES = 141100	Diversity = 3.8487e+00	k_t
= [1, 1, 1, 1]			
Generation 1420: Best = 1.426314e+03	FES = 142100	Diversity = 3.8195e+00	k_t
= [4, 2, 4, 3]			
Generation 1430: Best = 1.426314e+03	FES = 143100	Diversity = 3.8002e+00	k_t
= [1, 1, 1, 1]			
Generation 1440: Best = 1.426314e+03	FES = 144100	Diversity = 3.7826e+00	k_t
= [3, 3, 4, 2]			
Generation 1450: Best = 1.426314e+03	FES = 145100	Diversity = 3.7824e+00	k_t
= [1, 1, 1, 1]			
Generation 1460: Best = 1.426314e+03	FES = 146100	Diversity = 3.7656e+00	k_t
= [2, 2, 4, 4]			
Generation 1470: Best = 1.426314e+03	FES = 147100	Diversity = 3.7651e+00	k_t
= [1, 1, 1, 1]			
Generation 1480: Best = 1.426314e+03	FES = 148100	Diversity = 3.7647e+00	k_t
= [2, 2, 4, 4]			
Generation 1490: Best = 1.426314e+03	FES = 149100	Diversity = 3.7561e+00	k_t
= [1, 1, 1, 1]			
Generation 1500: Best = 1.426314e+03	FES = 150100	Diversity = 3.7280e+00	k_t
= [2, 4, 3, 3]			
Generation 1510: Best = 1.423998e+03	FES = 151100	Diversity = 3.7179e+00	k_t
= [1, 1, 1, 1]			
Generation 1520: Best = 1.423998e+03	FES = 152100	Diversity = 3.6927e+00	k_t
= [3, 3, 4, 2]			
Generation 1530: Best = 1.423998e+03	FES = 153100	Diversity = 3.6672e+00	k_t
= [1, 1, 1, 1]			
Generation 1540: Best = 1.419943e+03	FES = 154100	Diversity = 3.6505e+00	k_t
= [4, 3, 2, 2]			
Generation 1550: Best = 1.419943e+03	FES = 155100	Diversity = 3.6435e+00	k_t
= [1, 1, 1, 1]			
Generation 1560: Best = 1.419943e+03	FES = 156100	Diversity = 3.6412e+00	k_t
= [2, 3, 4, 3]			
Generation 1570: Best = 1.418854e+03	FES = 157100	Diversity = 3.6410e+00	k_t
= [1, 1, 1, 1]			

Generation 1580: Best = 1.418854e+03 = [4, 4, 3, 2]	FES = 158100	Diversity = 3.6168e+00	k_t
Generation 1590: Best = 1.418621e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 3.5997e+00	k_t
Generation 1600: Best = 1.416080e+03 = [3, 4, 2, 2]	FES = 160100	Diversity = 3.6008e+00	k_t
Generation 1610: Best = 1.416080e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 3.6026e+00	k_t
Generation 1620: Best = 1.416080e+03 = [4, 3, 3, 3]	FES = 162100	Diversity = 3.5992e+00	k_t
Generation 1630: Best = 1.416080e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 3.5994e+00	k_t
Generation 1640: Best = 1.408150e+03 = [4, 3, 3, 2]	FES = 164100	Diversity = 3.6010e+00	k_t
Generation 1650: Best = 1.408150e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 3.6004e+00	k_t
Generation 1660: Best = 1.408150e+03 = [2, 2, 4, 2]	FES = 166100	Diversity = 3.5998e+00	k_t
Generation 1670: Best = 1.408150e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 3.6008e+00	k_t
Generation 1680: Best = 1.406581e+03 = [3, 4, 3, 2]	FES = 168100	Diversity = 3.5994e+00	k_t
Generation 1690: Best = 1.405670e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 3.5938e+00	k_t
Generation 1700: Best = 1.405670e+03 = [3, 3, 3, 3]	FES = 170100	Diversity = 3.5951e+00	k_t
Generation 1710: Best = 1.393678e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 3.5889e+00	k_t
Generation 1720: Best = 1.380624e+03 = [4, 3, 2, 3]	FES = 172100	Diversity = 3.5805e+00	k_t
Generation 1730: Best = 1.380624e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 3.5786e+00	k_t
Generation 1740: Best = 1.380624e+03 = [2, 3, 2, 4]	FES = 174100	Diversity = 3.5120e+00	k_t
Generation 1750: Best = 1.380624e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 3.5121e+00	k_t
Generation 1760: Best = 1.379399e+03 = [2, 4, 4, 4]	FES = 176100	Diversity = 3.4772e+00	k_t
Generation 1770: Best = 1.377362e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 3.4697e+00	k_t
Generation 1780: Best = 1.375369e+03 = [2, 4, 4, 3]	FES = 178100	Diversity = 3.4596e+00	k_t
Generation 1790: Best = 1.375369e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 3.4221e+00	k_t
Generation 1800: Best = 1.370927e+03 = [2, 2, 4, 2]	FES = 180100	Diversity = 3.4141e+00	k_t
Generation 1810: Best = 1.370927e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 3.3493e+00	k_t
Generation 1820: Best = 1.370927e+03 = [3, 2, 3, 4]	FES = 182100	Diversity = 3.3470e+00	k_t
Generation 1830: Best = 1.370927e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 3.2988e+00	k_t
Generation 1840: Best = 1.370927e+03 = [2, 4, 4, 3]	FES = 184100	Diversity = 3.2650e+00	k_t

Generation 1850: Best = 1.370927e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 3.2501e+00	k_t
Generation 1860: Best = 1.370927e+03 = [3, 4, 3, 2]	FES = 186100	Diversity = 3.1869e+00	k_t
Generation 1870: Best = 1.368768e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 3.1810e+00	k_t
Generation 1880: Best = 1.366333e+03 = [3, 4, 2, 4]	FES = 188100	Diversity = 3.1606e+00	k_t
Generation 1890: Best = 1.366333e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 3.1423e+00	k_t
Generation 1900: Best = 1.364971e+03 = [2, 3, 2, 4]	FES = 190100	Diversity = 3.1276e+00	k_t
Generation 1910: Best = 1.346875e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 3.0858e+00	k_t
Generation 1920: Best = 1.345788e+03 = [3, 4, 3, 4]	FES = 192100	Diversity = 3.0759e+00	k_t
Generation 1930: Best = 1.344169e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 3.0746e+00	k_t
Generation 1940: Best = 1.344169e+03 = [2, 4, 4, 2]	FES = 194100	Diversity = 3.0433e+00	k_t
Generation 1950: Best = 1.342831e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 3.0308e+00	k_t
Generation 1960: Best = 1.340036e+03 = [3, 3, 2, 2]	FES = 196100	Diversity = 3.0017e+00	k_t
Generation 1970: Best = 1.327767e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 2.9748e+00	k_t
Generation 1980: Best = 1.317777e+03 = [2, 3, 4, 2]	FES = 198100	Diversity = 2.9357e+00	k_t
Generation 1990: Best = 1.317729e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 2.8588e+00	k_t
Generation 2000: Best = 1.317574e+03 = [3, 4, 3, 4]	FES = 200100	Diversity = 2.8000e+00	k_t
Generation 2010: Best = 1.317574e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 2.7526e+00	k_t
Generation 2020: Best = 1.317574e+03 = [2, 3, 4, 3]	FES = 202100	Diversity = 2.6820e+00	k_t
Generation 2030: Best = 1.317574e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 2.6835e+00	k_t
Generation 2040: Best = 1.316024e+03 = [2, 3, 2, 4]	FES = 204100	Diversity = 2.6333e+00	k_t
Generation 2050: Best = 1.316024e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 2.5157e+00	k_t
Generation 2060: Best = 1.316024e+03 = [2, 2, 2, 2]	FES = 206100	Diversity = 2.3691e+00	k_t
Generation 2070: Best = 1.316007e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 2.3264e+00	k_t
Generation 2080: Best = 1.312914e+03 = [2, 2, 3, 3]	FES = 208100	Diversity = 2.2343e+00	k_t
Generation 2090: Best = 1.312914e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 2.1189e+00	k_t
Generation 2100: Best = 1.312914e+03 = [2, 2, 4, 3]	FES = 210100	Diversity = 1.9028e+00	k_t
Generation 2110: Best = 1.312294e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.7858e+00	k_t

Generation 2120: Best = 1.306957e+03 = [3, 4, 4, 3]	FES = 212100	Diversity = 1.7668e+00	k_t
Generation 2130: Best = 1.306957e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 1.6987e+00	k_t
Generation 2140: Best = 1.306768e+03 = [2, 2, 2, 3]	FES = 214100	Diversity = 1.6152e+00	k_t
Generation 2150: Best = 1.293433e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 1.5214e+00	k_t
Generation 2160: Best = 1.290942e+03 = [2, 3, 4, 4]	FES = 216100	Diversity = 1.5231e+00	k_t
Generation 2170: Best = 1.276730e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 1.5135e+00	k_t
Generation 2180: Best = 1.270164e+03 = [2, 2, 3, 3]	FES = 218100	Diversity = 1.5038e+00	k_t
Generation 2190: Best = 1.250004e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 1.4789e+00	k_t
Generation 2200: Best = 1.227714e+03 = [4, 3, 4, 2]	FES = 220100	Diversity = 1.4105e+00	k_t
Generation 2210: Best = 1.218973e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.2674e+00	k_t
Generation 2220: Best = 1.200941e+03 = [3, 3, 2, 3]	FES = 222100	Diversity = 8.0982e-01	k_t
Generation 2230: Best = 1.188475e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 5.0574e-01	k_t
Generation 2240: Best = 1.183531e+03 = [4, 2, 2, 4]	FES = 224100	Diversity = 3.2744e-01	k_t
Generation 2250: Best = 1.170966e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.3485e-01	k_t
Generation 2260: Best = 1.166236e+03 = [3, 2, 3, 3]	FES = 226100	Diversity = 1.0929e-01	k_t
Generation 2270: Best = 1.160080e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 8.7312e-02	k_t
Generation 2280: Best = 1.155269e+03 = [3, 4, 4, 2]	FES = 228100	Diversity = 6.8120e-02	k_t
Generation 2290: Best = 1.153465e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 5.3861e-02	k_t
Generation 2300: Best = 1.151967e+03 = [3, 2, 3, 4]	FES = 230100	Diversity = 4.4118e-02	k_t
Generation 2310: Best = 1.150580e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 3.4172e-02	k_t
Generation 2320: Best = 1.149628e+03 = [3, 3, 4, 2]	FES = 232100	Diversity = 2.7256e-02	k_t
Generation 2330: Best = 1.149186e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 2.2694e-02	k_t
Generation 2340: Best = 1.148831e+03 = [3, 3, 2, 2]	FES = 234100	Diversity = 1.8097e-02	k_t
Generation 2350: Best = 1.148482e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 1.5095e-02	k_t
Generation 2360: Best = 1.148278e+03 = [4, 3, 2, 2]	FES = 236100	Diversity = 1.3092e-02	k_t
Generation 2370: Best = 1.148123e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 1.0909e-02	k_t
Generation 2380: Best = 1.147971e+03 = [2, 2, 3, 4]	FES = 238100	Diversity = 1.0021e-02	k_t

Generation 2390: Best = 1.147850e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 9.1480e-03	k_t
Generation 2400: Best = 1.147751e+03 = [2, 2, 3, 3]	FES = 240100	Diversity = 8.2624e-03	k_t
Generation 2410: Best = 1.147695e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 7.4156e-03	k_t
Generation 2420: Best = 1.147638e+03 = [2, 2, 3, 4]	FES = 242100	Diversity = 6.3792e-03	k_t
Generation 2430: Best = 1.147600e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 5.4807e-03	k_t
Generation 2440: Best = 1.147575e+03 = [3, 4, 4, 4]	FES = 244100	Diversity = 4.5533e-03	k_t
Generation 2450: Best = 1.147557e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 3.8784e-03	k_t
Generation 2460: Best = 1.147541e+03 = [4, 4, 3, 4]	FES = 246100	Diversity = 3.2346e-03	k_t
Generation 2470: Best = 1.147528e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 2.9660e-03	k_t
Generation 2480: Best = 1.147513e+03 = [3, 4, 2, 4]	FES = 248100	Diversity = 2.7453e-03	k_t
Generation 2490: Best = 1.147505e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 2.4587e-03	k_t
Generation 2500: Best = 1.147500e+03 = [2, 4, 2, 2]	FES = 250100	Diversity = 2.1432e-03	k_t
Generation 2510: Best = 1.147494e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.7813e-03	k_t
Generation 2520: Best = 1.147491e+03 = [4, 4, 4, 4]	FES = 252100	Diversity = 1.6144e-03	k_t
Generation 2530: Best = 1.147488e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.3658e-03	k_t
Generation 2540: Best = 1.147486e+03 = [3, 3, 4, 3]	FES = 254100	Diversity = 1.2337e-03	k_t
Generation 2550: Best = 1.147484e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.0852e-03	k_t
Generation 2560: Best = 1.147483e+03 = [4, 4, 2, 4]	FES = 256100	Diversity = 9.3738e-04	k_t
Generation 2570: Best = 1.147482e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 8.1951e-04	k_t
Generation 2580: Best = 1.147481e+03 = [3, 3, 4, 4]	FES = 258100	Diversity = 6.8728e-04	k_t
Generation 2590: Best = 1.147481e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 6.0134e-04	k_t
Generation 2600: Best = 1.147480e+03 = [2, 2, 4, 4]	FES = 260100	Diversity = 5.7156e-04	k_t
Generation 2610: Best = 1.147480e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 5.3657e-04	k_t
Generation 2620: Best = 1.147480e+03 = [3, 4, 4, 3]	FES = 262100	Diversity = 4.6911e-04	k_t
Generation 2630: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 4.0825e-04	k_t
Generation 2640: Best = 1.147479e+03 = [3, 4, 4, 3]	FES = 264100	Diversity = 3.4429e-04	k_t
Generation 2650: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 3.0763e-04	k_t

Generation 2660: Best = 1.147479e+03 = [4, 4, 3, 4]	FES = 266100	Diversity = 2.8018e-04	k_t
Generation 2670: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 2.4388e-04	k_t
Generation 2680: Best = 1.147479e+03 = [2, 4, 4, 4]	FES = 268100	Diversity = 2.1198e-04	k_t
Generation 2690: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.9786e-04	k_t
Generation 2700: Best = 1.147479e+03 = [4, 3, 4, 2]	FES = 270100	Diversity = 1.7422e-04	k_t
Generation 2710: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.4543e-04	k_t
Generation 2720: Best = 1.147479e+03 = [2, 3, 3, 2]	FES = 272100	Diversity = 1.2905e-04	k_t
Generation 2730: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.0983e-04	k_t
Generation 2740: Best = 1.147479e+03 = [2, 4, 3, 3]	FES = 274100	Diversity = 9.3828e-05	k_t
Generation 2750: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 8.0335e-05	k_t
Generation 2760: Best = 1.147479e+03 = [3, 3, 3, 3]	FES = 276100	Diversity = 7.6169e-05	k_t
Generation 2770: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 6.8675e-05	k_t
Generation 2780: Best = 1.147479e+03 = [2, 4, 2, 4]	FES = 278100	Diversity = 6.1436e-05	k_t
Generation 2790: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 5.4934e-05	k_t
Generation 2800: Best = 1.147479e+03 = [2, 3, 2, 2]	FES = 280100	Diversity = 5.0556e-05	k_t
Generation 2810: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 4.6115e-05	k_t
Generation 2820: Best = 1.147479e+03 = [3, 4, 3, 4]	FES = 282100	Diversity = 4.2504e-05	k_t
Generation 2830: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 3.9526e-05	k_t
Generation 2840: Best = 1.147479e+03 = [2, 4, 4, 4]	FES = 284100	Diversity = 3.5369e-05	k_t
Generation 2850: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 3.3466e-05	k_t
Generation 2860: Best = 1.147479e+03 = [3, 3, 4, 4]	FES = 286100	Diversity = 3.3593e-05	k_t
Generation 2870: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 3.0268e-05	k_t
Generation 2880: Best = 1.147479e+03 = [4, 2, 2, 3]	FES = 288100	Diversity = 2.6084e-05	k_t
Generation 2890: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 2.2638e-05	k_t
Generation 2900: Best = 1.147479e+03 = [2, 2, 4, 4]	FES = 290100	Diversity = 1.9203e-05	k_t
Generation 2910: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.8979e-05	k_t
Generation 2920: Best = 1.147479e+03 = [4, 3, 3, 3]	FES = 292100	Diversity = 1.9132e-05	k_t

Generation 2930: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 293100	Diversity = 1.8332e-05	k_t
Generation 2940: Best = 1.147479e+03 = [3, 4, 4, 3]	FES = 294100	Diversity = 1.7650e-05	k_t
Generation 2950: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 295100	Diversity = 1.7946e-05	k_t
Generation 2960: Best = 1.147479e+03 = [2, 4, 4, 4]	FES = 296100	Diversity = 1.7125e-05	k_t
Generation 2970: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 297100	Diversity = 1.4084e-05	k_t
Generation 2980: Best = 1.147479e+03 = [3, 2, 2, 3]	FES = 298100	Diversity = 1.1744e-05	k_t
Generation 2990: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 299100	Diversity = 1.0258e-05	k_t
Generation 3000: Best = 1.147479e+03 = [3, 3, 3, 3]	FES = 300100	Diversity = 8.7753e-06	k_t
Generation 3010: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 301100	Diversity = 8.3277e-06	k_t
Generation 3020: Best = 1.147479e+03 = [3, 2, 4, 3]	FES = 302100	Diversity = 8.1888e-06	k_t
Generation 3030: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 303100	Diversity = 8.0099e-06	k_t
Generation 3040: Best = 1.147479e+03 = [4, 2, 2, 2]	FES = 304100	Diversity = 7.9009e-06	k_t
Generation 3050: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 305100	Diversity = 7.3549e-06	k_t
Generation 3060: Best = 1.147479e+03 = [3, 2, 2, 4]	FES = 306100	Diversity = 6.4338e-06	k_t
Generation 3070: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 307100	Diversity = 5.6323e-06	k_t
Generation 3080: Best = 1.147479e+03 = [3, 4, 4, 4]	FES = 308100	Diversity = 5.7121e-06	k_t
Generation 3090: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 309100	Diversity = 5.0431e-06	k_t
Generation 3100: Best = 1.147479e+03 = [2, 4, 4, 2]	FES = 310100	Diversity = 4.6674e-06	k_t
Generation 3110: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 311100	Diversity = 3.8278e-06	k_t
Generation 3120: Best = 1.147479e+03 = [3, 4, 4, 4]	FES = 312100	Diversity = 3.2802e-06	k_t
Generation 3130: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 313100	Diversity = 2.7819e-06	k_t
Generation 3140: Best = 1.147479e+03 = [4, 2, 2, 4]	FES = 314100	Diversity = 2.3983e-06	k_t
Generation 3150: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 315100	Diversity = 2.2440e-06	k_t
Generation 3160: Best = 1.147479e+03 = [3, 4, 4, 2]	FES = 316100	Diversity = 1.9437e-06	k_t
Generation 3170: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 317100	Diversity = 1.7919e-06	k_t
Generation 3180: Best = 1.147479e+03 = [2, 4, 4, 3]	FES = 318100	Diversity = 1.5194e-06	k_t
Generation 3190: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 319100	Diversity = 1.4638e-06	k_t

Generation 3200: Best = 1.147479e+03 = [2, 2, 3, 2]	FES = 320100	Diversity = 1.3910e-06	k_t
Generation 3210: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 321100	Diversity = 1.1822e-06	k_t
Generation 3220: Best = 1.147479e+03 = [4, 4, 4, 4]	FES = 322100	Diversity = 1.2556e-06	k_t
Generation 3230: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 323100	Diversity = 1.1552e-06	k_t
Generation 3240: Best = 1.147479e+03 = [2, 4, 2, 4]	FES = 324100	Diversity = 1.0291e-06	k_t
Generation 3250: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 325100	Diversity = 9.5749e-07	k_t
Generation 3260: Best = 1.147479e+03 = [2, 2, 2, 3]	FES = 326100	Diversity = 8.8771e-07	k_t
Generation 3270: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 327100	Diversity = 8.0431e-07	k_t
Generation 3280: Best = 1.147479e+03 = [3, 2, 4, 3]	FES = 328100	Diversity = 7.0297e-07	k_t
Generation 3290: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 329100	Diversity = 6.0250e-07	k_t
Generation 3300: Best = 1.147479e+03 = [4, 4, 3, 3]	FES = 330100	Diversity = 5.1856e-07	k_t
Generation 3310: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 331100	Diversity = 4.8608e-07	k_t
Generation 3320: Best = 1.147479e+03 = [2, 4, 2, 2]	FES = 332100	Diversity = 4.0878e-07	k_t
Generation 3330: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 333100	Diversity = 3.8598e-07	k_t
Generation 3340: Best = 1.147479e+03 = [3, 2, 4, 3]	FES = 334100	Diversity = 3.8082e-07	k_t
Generation 3350: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 335100	Diversity = 3.6200e-07	k_t
Generation 3360: Best = 1.147479e+03 = [3, 2, 4, 3]	FES = 336100	Diversity = 3.1909e-07	k_t
Generation 3370: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 337100	Diversity = 2.7097e-07	k_t
Generation 3380: Best = 1.147479e+03 = [2, 2, 2, 4]	FES = 338100	Diversity = 2.2185e-07	k_t
Generation 3390: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 339100	Diversity = 2.0835e-07	k_t
Generation 3400: Best = 1.147479e+03 = [4, 4, 3, 3]	FES = 340100	Diversity = 2.1578e-07	k_t
Generation 3410: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 341100	Diversity = 1.9812e-07	k_t
Generation 3420: Best = 1.147479e+03 = [4, 4, 4, 3]	FES = 342100	Diversity = 1.7253e-07	k_t
Generation 3430: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 343100	Diversity = 1.5512e-07	k_t
Generation 3440: Best = 1.147479e+03 = [2, 3, 4, 3]	FES = 344100	Diversity = 1.5205e-07	k_t
Generation 3450: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 345100	Diversity = 1.3736e-07	k_t
Generation 3460: Best = 1.147479e+03 = [4, 3, 3, 3]	FES = 346100	Diversity = 1.1160e-07	k_t

Generation 3470: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 347100	Diversity = 9.2144e-08	k_t
Generation 3480: Best = 1.147479e+03 = [3, 4, 3, 3]	FES = 348100	Diversity = 9.7094e-08	k_t
Generation 3490: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 349100	Diversity = 8.7449e-08	k_t
Generation 3500: Best = 1.147479e+03 = [4, 4, 4, 4]	FES = 350100	Diversity = 8.2877e-08	k_t
Generation 3510: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 351100	Diversity = 7.6102e-08	k_t
Generation 3520: Best = 1.147479e+03 = [3, 4, 4, 3]	FES = 352100	Diversity = 7.0228e-08	k_t
Generation 3530: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 353100	Diversity = 5.9695e-08	k_t
Generation 3540: Best = 1.147479e+03 = [3, 2, 2, 3]	FES = 354100	Diversity = 5.5555e-08	k_t
Generation 3550: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 355100	Diversity = 4.8796e-08	k_t
Generation 3560: Best = 1.147479e+03 = [2, 4, 4, 4]	FES = 356100	Diversity = 4.4418e-08	k_t
Generation 3570: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 357100	Diversity = 3.9334e-08	k_t
Generation 3580: Best = 1.147479e+03 = [2, 3, 3, 3]	FES = 358100	Diversity = 3.6001e-08	k_t
Generation 3590: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 359100	Diversity = 3.2433e-08	k_t
Generation 3600: Best = 1.147479e+03 = [3, 3, 4, 3]	FES = 360100	Diversity = 2.7871e-08	k_t
Generation 3610: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 361100	Diversity = 2.7165e-08	k_t
Generation 3620: Best = 1.147479e+03 = [4, 4, 3, 4]	FES = 362100	Diversity = 2.3954e-08	k_t
Generation 3630: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 363100	Diversity = 2.2825e-08	k_t
Generation 3640: Best = 1.147479e+03 = [4, 2, 3, 3]	FES = 364100	Diversity = 2.1972e-08	k_t
Generation 3650: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 365100	Diversity = 1.9555e-08	k_t
Generation 3660: Best = 1.147479e+03 = [4, 4, 2, 4]	FES = 366100	Diversity = 1.9137e-08	k_t
Generation 3670: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 367100	Diversity = 1.8184e-08	k_t
Generation 3680: Best = 1.147479e+03 = [3, 3, 3, 2]	FES = 368100	Diversity = 1.9013e-08	k_t
Generation 3690: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 369100	Diversity = 1.8608e-08	k_t
Generation 3700: Best = 1.147479e+03 = [2, 4, 2, 3]	FES = 370100	Diversity = 1.8339e-08	k_t
Generation 3710: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 371100	Diversity = 1.8123e-08	k_t
Generation 3720: Best = 1.147479e+03 = [2, 3, 3, 2]	FES = 372100	Diversity = 1.7304e-08	k_t
Generation 3730: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 373100	Diversity = 1.7052e-08	k_t

Generation 3740: Best = 1.147479e+03 = [3, 2, 3, 4]	FES = 374100	Diversity = 1.6723e-08	k_t
Generation 3750: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 375100	Diversity = 1.6690e-08	k_t
Generation 3760: Best = 1.147479e+03 = [3, 4, 4, 4]	FES = 376100	Diversity = 1.5084e-08	k_t
Generation 3770: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 377100	Diversity = 1.4625e-08	k_t
Generation 3780: Best = 1.147479e+03 = [3, 4, 4, 2]	FES = 378100	Diversity = 1.4143e-08	k_t
Generation 3790: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 379100	Diversity = 1.3958e-08	k_t
Generation 3800: Best = 1.147479e+03 = [2, 4, 2, 3]	FES = 380100	Diversity = 1.3556e-08	k_t
Generation 3810: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 381100	Diversity = 1.3735e-08	k_t
Generation 3820: Best = 1.147479e+03 = [2, 2, 2, 4]	FES = 382100	Diversity = 1.3711e-08	k_t
Generation 3830: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 383100	Diversity = 1.3887e-08	k_t
Generation 3840: Best = 1.147479e+03 = [4, 4, 4, 3]	FES = 384100	Diversity = 1.3374e-08	k_t
Generation 3850: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 385100	Diversity = 1.3398e-08	k_t
Generation 3860: Best = 1.147479e+03 = [4, 3, 3, 3]	FES = 386100	Diversity = 1.3226e-08	k_t
Generation 3870: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 387100	Diversity = 1.2735e-08	k_t
Generation 3880: Best = 1.147479e+03 = [3, 3, 2, 3]	FES = 388100	Diversity = 1.2877e-08	k_t
Generation 3890: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 389100	Diversity = 1.2305e-08	k_t
Generation 3900: Best = 1.147479e+03 = [2, 2, 4, 4]	FES = 390100	Diversity = 1.2540e-08	k_t
Generation 3910: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 391100	Diversity = 1.2323e-08	k_t
Generation 3920: Best = 1.147479e+03 = [3, 2, 3, 3]	FES = 392100	Diversity = 1.2314e-08	k_t
Generation 3930: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 393100	Diversity = 1.2338e-08	k_t
Generation 3940: Best = 1.147479e+03 = [2, 3, 2, 4]	FES = 394100	Diversity = 1.2367e-08	k_t
Generation 3950: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 395100	Diversity = 1.2328e-08	k_t
Generation 3960: Best = 1.147479e+03 = [2, 4, 3, 3]	FES = 396100	Diversity = 1.2193e-08	k_t
Generation 3970: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 397100	Diversity = 1.2312e-08	k_t
Generation 3980: Best = 1.147479e+03 = [2, 3, 4, 2]	FES = 398100	Diversity = 1.2474e-08	k_t
Generation 3990: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 399100	Diversity = 1.2381e-08	k_t
Generation 4000: Best = 1.147479e+03 = [2, 2, 3, 2]	FES = 400100	Diversity = 1.2595e-08	k_t

Generation 4010: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 401100	Diversity = 1.2724e-08	k_t
Generation 4020: Best = 1.147479e+03 = [2, 3, 3, 3]	FES = 402100	Diversity = 1.2883e-08	k_t
Generation 4030: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 403100	Diversity = 1.2585e-08	k_t
Generation 4040: Best = 1.147479e+03 = [4, 4, 2, 4]	FES = 404100	Diversity = 1.2493e-08	k_t
Generation 4050: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 405100	Diversity = 1.2553e-08	k_t
Generation 4060: Best = 1.147479e+03 = [2, 2, 3, 2]	FES = 406100	Diversity = 1.2801e-08	k_t
Generation 4070: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 407100	Diversity = 1.2670e-08	k_t
Generation 4080: Best = 1.147479e+03 = [3, 4, 2, 2]	FES = 408100	Diversity = 1.2699e-08	k_t
Generation 4090: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 409100	Diversity = 1.2701e-08	k_t
Generation 4100: Best = 1.147479e+03 = [4, 4, 2, 3]	FES = 410100	Diversity = 1.2875e-08	k_t
Generation 4110: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 411100	Diversity = 1.2924e-08	k_t
Generation 4120: Best = 1.147479e+03 = [3, 2, 4, 2]	FES = 412100	Diversity = 1.2987e-08	k_t
Generation 4130: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 413100	Diversity = 1.2954e-08	k_t
Generation 4140: Best = 1.147479e+03 = [2, 3, 3, 3]	FES = 414100	Diversity = 1.2974e-08	k_t
Generation 4150: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 415100	Diversity = 1.3007e-08	k_t
Generation 4160: Best = 1.147479e+03 = [4, 4, 4, 3]	FES = 416100	Diversity = 1.3074e-08	k_t
Generation 4170: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 417100	Diversity = 1.3356e-08	k_t
Generation 4180: Best = 1.147479e+03 = [2, 3, 2, 2]	FES = 418100	Diversity = 1.3344e-08	k_t
Generation 4190: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 419100	Diversity = 1.3481e-08	k_t
Generation 4200: Best = 1.147479e+03 = [4, 2, 2, 4]	FES = 420100	Diversity = 1.3478e-08	k_t
Generation 4210: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 421100	Diversity = 1.3487e-08	k_t
Generation 4220: Best = 1.147479e+03 = [3, 4, 3, 4]	FES = 422100	Diversity = 1.3555e-08	k_t
Generation 4230: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 423100	Diversity = 1.3642e-08	k_t
Generation 4240: Best = 1.147479e+03 = [4, 3, 3, 2]	FES = 424100	Diversity = 1.3694e-08	k_t
Generation 4250: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 425100	Diversity = 1.3771e-08	k_t
Generation 4260: Best = 1.147479e+03 = [4, 4, 4, 3]	FES = 426100	Diversity = 1.3835e-08	k_t
Generation 4270: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 427100	Diversity = 1.3865e-08	k_t

Generation 4280: Best = 1.147479e+03 = [3, 2, 4, 4]	FES = 428100	Diversity = 1.3944e-08	k_t
Generation 4290: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 429100	Diversity = 1.4008e-08	k_t
Generation 4300: Best = 1.147479e+03 = [4, 3, 4, 4]	FES = 430100	Diversity = 1.4105e-08	k_t
Generation 4310: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 431100	Diversity = 1.4154e-08	k_t
Generation 4320: Best = 1.147479e+03 = [3, 2, 2, 2]	FES = 432100	Diversity = 1.4159e-08	k_t
Generation 4330: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 433100	Diversity = 1.4292e-08	k_t
Generation 4340: Best = 1.147479e+03 = [2, 3, 4, 3]	FES = 434100	Diversity = 1.4369e-08	k_t
Generation 4350: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 435100	Diversity = 1.4494e-08	k_t
Generation 4360: Best = 1.147479e+03 = [4, 3, 3, 2]	FES = 436100	Diversity = 1.4564e-08	k_t
Generation 4370: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 437100	Diversity = 1.4567e-08	k_t
Generation 4380: Best = 1.147479e+03 = [4, 4, 4, 3]	FES = 438100	Diversity = 1.4643e-08	k_t
Generation 4390: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 439100	Diversity = 1.4712e-08	k_t
Generation 4400: Best = 1.147479e+03 = [2, 4, 4, 4]	FES = 440100	Diversity = 1.4830e-08	k_t
Generation 4410: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 441100	Diversity = 1.4955e-08	k_t
Generation 4420: Best = 1.147479e+03 = [4, 3, 2, 2]	FES = 442100	Diversity = 1.4974e-08	k_t
Generation 4430: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 443100	Diversity = 1.5052e-08	k_t
Generation 4440: Best = 1.147479e+03 = [3, 4, 3, 4]	FES = 444100	Diversity = 1.5097e-08	k_t
Generation 4450: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 445100	Diversity = 1.5156e-08	k_t
Generation 4460: Best = 1.147479e+03 = [2, 2, 2, 3]	FES = 446100	Diversity = 1.5192e-08	k_t
Generation 4470: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 447100	Diversity = 1.5170e-08	k_t
Generation 4480: Best = 1.147479e+03 = [3, 2, 2, 4]	FES = 448100	Diversity = 1.5100e-08	k_t
Generation 4490: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 449100	Diversity = 1.5178e-08	k_t
Generation 4500: Best = 1.147479e+03 = [4, 3, 4, 3]	FES = 450100	Diversity = 1.5148e-08	k_t
Generation 4510: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 451100	Diversity = 1.5144e-08	k_t
Generation 4520: Best = 1.147479e+03 = [2, 3, 4, 2]	FES = 452100	Diversity = 1.5257e-08	k_t
Generation 4530: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 453100	Diversity = 1.5429e-08	k_t
Generation 4540: Best = 1.147479e+03 = [2, 3, 2, 2]	FES = 454100	Diversity = 1.5526e-08	k_t

Generation 4550: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 455100	Diversity = 1.5452e-08	k_t
Generation 4560: Best = 1.147479e+03 = [4, 3, 4, 4]	FES = 456100	Diversity = 1.5414e-08	k_t
Generation 4570: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 457100	Diversity = 1.5492e-08	k_t
Generation 4580: Best = 1.147479e+03 = [2, 2, 4, 3]	FES = 458100	Diversity = 1.5416e-08	k_t
Generation 4590: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 459100	Diversity = 1.5442e-08	k_t
Generation 4600: Best = 1.147479e+03 = [2, 4, 3, 2]	FES = 460100	Diversity = 1.5507e-08	k_t
Generation 4610: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 461100	Diversity = 1.5714e-08	k_t
Generation 4620: Best = 1.147479e+03 = [3, 4, 4, 4]	FES = 462100	Diversity = 1.5790e-08	k_t
Generation 4630: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 463100	Diversity = 1.5908e-08	k_t
Generation 4640: Best = 1.147479e+03 = [3, 2, 4, 2]	FES = 464100	Diversity = 1.6053e-08	k_t
Generation 4650: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 465100	Diversity = 1.6260e-08	k_t
Generation 4660: Best = 1.147479e+03 = [4, 3, 4, 3]	FES = 466100	Diversity = 1.6160e-08	k_t
Generation 4670: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 467100	Diversity = 1.6220e-08	k_t
Generation 4680: Best = 1.147479e+03 = [4, 4, 3, 2]	FES = 468100	Diversity = 1.6115e-08	k_t
Generation 4690: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 469100	Diversity = 1.6095e-08	k_t
Generation 4700: Best = 1.147479e+03 = [2, 3, 4, 2]	FES = 470100	Diversity = 1.6189e-08	k_t
Generation 4710: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 471100	Diversity = 1.6259e-08	k_t
Generation 4720: Best = 1.147479e+03 = [4, 3, 4, 3]	FES = 472100	Diversity = 1.6295e-08	k_t
Generation 4730: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 473100	Diversity = 1.6309e-08	k_t
Generation 4740: Best = 1.147479e+03 = [4, 2, 2, 4]	FES = 474100	Diversity = 1.6499e-08	k_t
Generation 4750: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 475100	Diversity = 1.6476e-08	k_t
Generation 4760: Best = 1.147479e+03 = [4, 2, 3, 4]	FES = 476100	Diversity = 1.6659e-08	k_t
Generation 4770: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 477100	Diversity = 1.6862e-08	k_t
Generation 4780: Best = 1.147479e+03 = [2, 2, 4, 4]	FES = 478100	Diversity = 1.6934e-08	k_t
Generation 4790: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 479100	Diversity = 1.6897e-08	k_t
Generation 4800: Best = 1.147479e+03 = [2, 3, 3, 3]	FES = 480100	Diversity = 1.7103e-08	k_t
Generation 4810: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 481100	Diversity = 1.7157e-08	k_t

Generation 4820: Best = 1.147479e+03 = [2, 2, 3, 4]	FES = 482100	Diversity = 1.7169e-08	k_t
Generation 4830: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 483100	Diversity = 1.7199e-08	k_t
Generation 4840: Best = 1.147479e+03 = [3, 3, 4, 2]	FES = 484100	Diversity = 1.7264e-08	k_t
Generation 4850: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 485100	Diversity = 1.7250e-08	k_t
Generation 4860: Best = 1.147479e+03 = [2, 4, 2, 2]	FES = 486100	Diversity = 1.7234e-08	k_t
Generation 4870: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 487100	Diversity = 1.7140e-08	k_t
Generation 4880: Best = 1.147479e+03 = [4, 4, 2, 2]	FES = 488100	Diversity = 1.7234e-08	k_t
Generation 4890: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 489100	Diversity = 1.7095e-08	k_t
Generation 4900: Best = 1.147479e+03 = [2, 3, 2, 4]	FES = 490100	Diversity = 1.6994e-08	k_t
Generation 4910: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 491100	Diversity = 1.6956e-08	k_t
Generation 4920: Best = 1.147479e+03 = [2, 4, 3, 4]	FES = 492100	Diversity = 1.7071e-08	k_t
Generation 4930: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 493100	Diversity = 1.7166e-08	k_t
Generation 4940: Best = 1.147479e+03 = [3, 2, 4, 2]	FES = 494100	Diversity = 1.7139e-08	k_t
Generation 4950: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 495100	Diversity = 1.7392e-08	k_t
Generation 4960: Best = 1.147479e+03 = [4, 4, 2, 4]	FES = 496100	Diversity = 1.7580e-08	k_t
Generation 4970: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 497100	Diversity = 1.7851e-08	k_t
Generation 4980: Best = 1.147479e+03 = [2, 2, 2, 2]	FES = 498100	Diversity = 1.8058e-08	k_t
Generation 4990: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 499100	Diversity = 1.7891e-08	k_t
Generation 5000: Best = 1.147479e+03 = [4, 2, 3, 3]	FES = 500100	Diversity = 1.7950e-08	k_t
Generation 5010: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 501100	Diversity = 1.7932e-08	k_t
Generation 5020: Best = 1.147479e+03 = [2, 2, 3, 4]	FES = 502100	Diversity = 1.8107e-08	k_t
Generation 5030: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 503100	Diversity = 1.8158e-08	k_t
Generation 5040: Best = 1.147479e+03 = [3, 3, 4, 2]	FES = 504100	Diversity = 1.7875e-08	k_t
Generation 5050: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 505100	Diversity = 1.7813e-08	k_t
Generation 5060: Best = 1.147479e+03 = [4, 4, 3, 4]	FES = 506100	Diversity = 1.8286e-08	k_t
Generation 5070: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 507100	Diversity = 1.8118e-08	k_t
Generation 5080: Best = 1.147479e+03 = [3, 2, 2, 3]	FES = 508100	Diversity = 1.8529e-08	k_t

Generation 5090: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 509100	Diversity = 1.8702e-08	k_t
Generation 5100: Best = 1.147479e+03 = [3, 3, 3, 3]	FES = 510100	Diversity = 1.8939e-08	k_t
Generation 5110: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 511100	Diversity = 1.9217e-08	k_t
Generation 5120: Best = 1.147479e+03 = [2, 4, 3, 4]	FES = 512100	Diversity = 1.9270e-08	k_t
Generation 5130: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 513100	Diversity = 1.8578e-08	k_t
Generation 5140: Best = 1.147479e+03 = [2, 2, 3, 4]	FES = 514100	Diversity = 1.9040e-08	k_t
Generation 5150: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 515100	Diversity = 1.9133e-08	k_t
Generation 5160: Best = 1.147479e+03 = [4, 3, 2, 3]	FES = 516100	Diversity = 1.9014e-08	k_t
Generation 5170: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 517100	Diversity = 1.9118e-08	k_t
Generation 5180: Best = 1.147479e+03 = [2, 2, 4, 2]	FES = 518100	Diversity = 1.9316e-08	k_t
Generation 5190: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 519100	Diversity = 1.9109e-08	k_t
Generation 5200: Best = 1.147479e+03 = [4, 3, 4, 2]	FES = 520100	Diversity = 1.8742e-08	k_t
Generation 5210: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 521100	Diversity = 1.8064e-08	k_t
Generation 5220: Best = 1.147479e+03 = [4, 2, 3, 3]	FES = 522100	Diversity = 1.7686e-08	k_t
Generation 5230: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 523100	Diversity = 1.7260e-08	k_t
Generation 5240: Best = 1.147479e+03 = [2, 2, 3, 2]	FES = 524100	Diversity = 1.6726e-08	k_t
Generation 5250: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 525100	Diversity = 1.6221e-08	k_t
Generation 5260: Best = 1.147479e+03 = [2, 4, 2, 2]	FES = 526100	Diversity = 1.6071e-08	k_t
Generation 5270: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 527100	Diversity = 1.5913e-08	k_t
Generation 5280: Best = 1.147479e+03 = [3, 4, 2, 2]	FES = 528100	Diversity = 1.6077e-08	k_t
Generation 5290: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 529100	Diversity = 1.6087e-08	k_t
Generation 5300: Best = 1.147479e+03 = [4, 2, 2, 2]	FES = 530100	Diversity = 1.6000e-08	k_t
Generation 5310: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 531100	Diversity = 1.5831e-08	k_t
Generation 5320: Best = 1.147479e+03 = [3, 2, 4, 4]	FES = 532100	Diversity = 1.5673e-08	k_t
Generation 5330: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 533100	Diversity = 1.5774e-08	k_t
Generation 5340: Best = 1.147479e+03 = [2, 4, 4, 4]	FES = 534100	Diversity = 1.6051e-08	k_t
Generation 5350: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 535100	Diversity = 1.6100e-08	k_t

Generation 5360: Best = 1.147479e+03	FES = 536100	Diversity = 1.6203e-08	k_t
= [4, 3, 2, 4]			
Generation 5370: Best = 1.147479e+03	FES = 537100	Diversity = 1.6299e-08	k_t
= [1, 1, 1, 1]			
Generation 5380: Best = 1.147479e+03	FES = 538100	Diversity = 1.6410e-08	k_t
= [4, 4, 3, 3]			
Generation 5390: Best = 1.147479e+03	FES = 539100	Diversity = 1.6503e-08	k_t
= [1, 1, 1, 1]			
Generation 5400: Best = 1.147479e+03	FES = 540100	Diversity = 1.6637e-08	k_t
= [4, 4, 4, 3]			
Generation 5410: Best = 1.147479e+03	FES = 541100	Diversity = 1.6753e-08	k_t
= [1, 1, 1, 1]			
Generation 5420: Best = 1.147479e+03	FES = 542100	Diversity = 1.6886e-08	k_t
= [4, 3, 3, 3]			
Generation 5430: Best = 1.147479e+03	FES = 543100	Diversity = 1.6642e-08	k_t
= [1, 1, 1, 1]			
Generation 5440: Best = 1.147479e+03	FES = 544100	Diversity = 1.6662e-08	k_t
= [4, 2, 3, 4]			
Generation 5450: Best = 1.147479e+03	FES = 545100	Diversity = 1.6627e-08	k_t
= [1, 1, 1, 1]			
Generation 5460: Best = 1.147479e+03	FES = 546100	Diversity = 1.6731e-08	k_t
= [2, 4, 3, 2]			
Generation 5470: Best = 1.147479e+03	FES = 547100	Diversity = 1.6814e-08	k_t
= [1, 1, 1, 1]			
Generation 5480: Best = 1.147479e+03	FES = 548100	Diversity = 1.6889e-08	k_t
= [4, 3, 3, 3]			
Generation 5490: Best = 1.147479e+03	FES = 549100	Diversity = 1.6972e-08	k_t
= [1, 1, 1, 1]			
Generation 5500: Best = 1.147479e+03	FES = 550100	Diversity = 1.7012e-08	k_t
= [3, 4, 4, 4]			
Generation 5510: Best = 1.147479e+03	FES = 551100	Diversity = 1.7094e-08	k_t
= [1, 1, 1, 1]			
Generation 5520: Best = 1.147479e+03	FES = 552100	Diversity = 1.7135e-08	k_t
= [4, 2, 3, 4]			
Generation 5530: Best = 1.147479e+03	FES = 553100	Diversity = 1.6973e-08	k_t
= [1, 1, 1, 1]			
Generation 5540: Best = 1.147479e+03	FES = 554100	Diversity = 1.6938e-08	k_t
= [2, 4, 2, 4]			
Generation 5550: Best = 1.147479e+03	FES = 555100	Diversity = 1.6967e-08	k_t
= [1, 1, 1, 1]			
Generation 5560: Best = 1.147479e+03	FES = 556100	Diversity = 1.6859e-08	k_t
= [2, 4, 3, 3]			
Generation 5570: Best = 1.147479e+03	FES = 557100	Diversity = 1.6827e-08	k_t
= [1, 1, 1, 1]			
Generation 5580: Best = 1.147479e+03	FES = 558100	Diversity = 1.6815e-08	k_t
= [2, 3, 3, 3]			
Generation 5590: Best = 1.147479e+03	FES = 559100	Diversity = 1.6814e-08	k_t
= [1, 1, 1, 1]			
Generation 5600: Best = 1.147479e+03	FES = 560100	Diversity = 1.6783e-08	k_t
= [3, 4, 3, 3]			
Generation 5610: Best = 1.147479e+03	FES = 561100	Diversity = 1.6918e-08	k_t
= [1, 1, 1, 1]			
Generation 5620: Best = 1.147479e+03	FES = 562100	Diversity = 1.6929e-08	k_t
= [2, 3, 4, 2]			

Generation 5630: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 563100	Diversity = 1.6946e-08	k_t
Generation 5640: Best = 1.147479e+03 = [3, 3, 4, 4]	FES = 564100	Diversity = 1.6950e-08	k_t
Generation 5650: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 565100	Diversity = 1.6950e-08	k_t
Generation 5660: Best = 1.147479e+03 = [4, 3, 3, 4]	FES = 566100	Diversity = 1.6958e-08	k_t
Generation 5670: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 567100	Diversity = 1.7041e-08	k_t
Generation 5680: Best = 1.147479e+03 = [3, 2, 3, 3]	FES = 568100	Diversity = 1.7081e-08	k_t
Generation 5690: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 569100	Diversity = 1.7100e-08	k_t
Generation 5700: Best = 1.147479e+03 = [2, 4, 4, 2]	FES = 570100	Diversity = 1.7033e-08	k_t
Generation 5710: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 571100	Diversity = 1.7053e-08	k_t
Generation 5720: Best = 1.147479e+03 = [3, 2, 3, 3]	FES = 572100	Diversity = 1.7019e-08	k_t
Generation 5730: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 573100	Diversity = 1.7049e-08	k_t
Generation 5740: Best = 1.147479e+03 = [2, 3, 4, 2]	FES = 574100	Diversity = 1.7052e-08	k_t
Generation 5750: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 575100	Diversity = 1.6914e-08	k_t
Generation 5760: Best = 1.147479e+03 = [4, 2, 3, 3]	FES = 576100	Diversity = 1.7004e-08	k_t
Generation 5770: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 577100	Diversity = 1.7175e-08	k_t
Generation 5780: Best = 1.147479e+03 = [4, 2, 3, 4]	FES = 578100	Diversity = 1.6993e-08	k_t
Generation 5790: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 579100	Diversity = 1.7082e-08	k_t
Generation 5800: Best = 1.147479e+03 = [2, 4, 2, 2]	FES = 580100	Diversity = 1.7077e-08	k_t
Generation 5810: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 581100	Diversity = 1.6838e-08	k_t
Generation 5820: Best = 1.147479e+03 = [2, 3, 4, 4]	FES = 582100	Diversity = 1.6617e-08	k_t
Generation 5830: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 583100	Diversity = 1.6571e-08	k_t
Generation 5840: Best = 1.147479e+03 = [3, 2, 2, 2]	FES = 584100	Diversity = 1.6472e-08	k_t
Generation 5850: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 585100	Diversity = 1.6497e-08	k_t
Generation 5860: Best = 1.147479e+03 = [3, 2, 3, 2]	FES = 586100	Diversity = 1.6593e-08	k_t
Generation 5870: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 587100	Diversity = 1.6458e-08	k_t
Generation 5880: Best = 1.147479e+03 = [2, 3, 4, 3]	FES = 588100	Diversity = 1.6467e-08	k_t
Generation 5890: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 589100	Diversity = 1.6376e-08	k_t

Generation 5900: Best = 1.147479e+03 = [2, 3, 4, 2]	FES = 590100	Diversity = 1.6144e-08	k_t
Generation 5910: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 591100	Diversity = 1.5995e-08	k_t
Generation 5920: Best = 1.147479e+03 = [3, 4, 2, 3]	FES = 592100	Diversity = 1.5951e-08	k_t
Generation 5930: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 593100	Diversity = 1.5790e-08	k_t
Generation 5940: Best = 1.147479e+03 = [4, 3, 3, 2]	FES = 594100	Diversity = 1.5807e-08	k_t
Generation 5950: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 595100	Diversity = 1.5797e-08	k_t
Generation 5960: Best = 1.147479e+03 = [4, 3, 3, 3]	FES = 596100	Diversity = 1.5829e-08	k_t
Generation 5970: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 597100	Diversity = 1.5756e-08	k_t
Generation 5980: Best = 1.147479e+03 = [3, 2, 2, 2]	FES = 598100	Diversity = 1.5798e-08	k_t
Generation 5990: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 599100	Diversity = 1.5654e-08	k_t
Generation 6000: Best = 1.147479e+03 = [3, 4, 4, 3]	FES = 600100	Diversity = 1.5566e-08	k_t
Generation 6010: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 601100	Diversity = 1.5602e-08	k_t
Generation 6020: Best = 1.147479e+03 = [4, 3, 4, 4]	FES = 602100	Diversity = 1.5283e-08	k_t
Generation 6030: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 603100	Diversity = 1.5347e-08	k_t
Generation 6040: Best = 1.147479e+03 = [2, 2, 4, 4]	FES = 604100	Diversity = 1.4979e-08	k_t
Generation 6050: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 605100	Diversity = 1.4806e-08	k_t
Generation 6060: Best = 1.147479e+03 = [2, 3, 2, 2]	FES = 606100	Diversity = 1.4556e-08	k_t
Generation 6070: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 607100	Diversity = 1.4631e-08	k_t
Generation 6080: Best = 1.147479e+03 = [2, 2, 3, 3]	FES = 608100	Diversity = 1.4556e-08	k_t
Generation 6090: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 609100	Diversity = 1.4484e-08	k_t
Generation 6100: Best = 1.147479e+03 = [4, 4, 2, 2]	FES = 610100	Diversity = 1.4294e-08	k_t
Generation 6110: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 611100	Diversity = 1.4390e-08	k_t
Generation 6120: Best = 1.147479e+03 = [4, 2, 3, 2]	FES = 612100	Diversity = 1.4486e-08	k_t
Generation 6130: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 613100	Diversity = 1.4519e-08	k_t
Generation 6140: Best = 1.147479e+03 = [4, 3, 3, 3]	FES = 614100	Diversity = 1.4436e-08	k_t
Generation 6150: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 615100	Diversity = 1.4326e-08	k_t
Generation 6160: Best = 1.147479e+03 = [2, 4, 2, 4]	FES = 616100	Diversity = 1.4287e-08	k_t

Generation 6170: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 617100	Diversity = 1.4070e-08	k_t
Generation 6180: Best = 1.147479e+03 = [4, 4, 2, 2]	FES = 618100	Diversity = 1.4086e-08	k_t
Generation 6190: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 619100	Diversity = 1.4094e-08	k_t
Generation 6200: Best = 1.147479e+03 = [3, 4, 4, 2]	FES = 620100	Diversity = 1.3903e-08	k_t
Generation 6210: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 621100	Diversity = 1.3747e-08	k_t
Generation 6220: Best = 1.147479e+03 = [2, 3, 2, 3]	FES = 622100	Diversity = 1.3559e-08	k_t
Generation 6230: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 623100	Diversity = 1.3297e-08	k_t
Generation 6240: Best = 1.147479e+03 = [4, 4, 3, 4]	FES = 624100	Diversity = 1.3296e-08	k_t
Generation 6250: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 625100	Diversity = 1.3280e-08	k_t
Generation 6260: Best = 1.147479e+03 = [2, 4, 3, 3]	FES = 626100	Diversity = 1.3112e-08	k_t
Generation 6270: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 627100	Diversity = 1.2960e-08	k_t
Generation 6280: Best = 1.147479e+03 = [4, 4, 4, 2]	FES = 628100	Diversity = 1.2910e-08	k_t
Generation 6290: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 629100	Diversity = 1.2428e-08	k_t
Generation 6300: Best = 1.147479e+03 = [4, 3, 4, 2]	FES = 630100	Diversity = 1.2693e-08	k_t
Generation 6310: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 631100	Diversity = 1.2762e-08	k_t
Generation 6320: Best = 1.147479e+03 = [3, 4, 3, 4]	FES = 632100	Diversity = 1.2648e-08	k_t
Generation 6330: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 633100	Diversity = 1.2452e-08	k_t
Generation 6340: Best = 1.147479e+03 = [3, 2, 3, 3]	FES = 634100	Diversity = 1.2466e-08	k_t
Generation 6350: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 635100	Diversity = 1.2599e-08	k_t
Generation 6360: Best = 1.147479e+03 = [2, 4, 2, 2]	FES = 636100	Diversity = 1.2349e-08	k_t
Generation 6370: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 637100	Diversity = 1.2594e-08	k_t
Generation 6380: Best = 1.147479e+03 = [4, 3, 2, 4]	FES = 638100	Diversity = 1.2216e-08	k_t
Generation 6390: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 639100	Diversity = 1.2173e-08	k_t
Generation 6400: Best = 1.147479e+03 = [2, 3, 4, 2]	FES = 640100	Diversity = 1.2352e-08	k_t
Generation 6410: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 641100	Diversity = 1.2320e-08	k_t
Generation 6420: Best = 1.147479e+03 = [2, 2, 3, 4]	FES = 642100	Diversity = 1.2259e-08	k_t
Generation 6430: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 643100	Diversity = 1.2140e-08	k_t

Generation 6440: Best = 1.147479e+03 = [3, 2, 4, 3]	FES = 644100	Diversity = 1.2176e-08	k_t
Generation 6450: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 645100	Diversity = 1.2223e-08	k_t
Generation 6460: Best = 1.147479e+03 = [4, 4, 4, 4]	FES = 646100	Diversity = 1.2281e-08	k_t
Generation 6470: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 647100	Diversity = 1.2038e-08	k_t
Generation 6480: Best = 1.147479e+03 = [4, 3, 4, 2]	FES = 648100	Diversity = 1.1916e-08	k_t
Generation 6490: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 649100	Diversity = 1.1819e-08	k_t
Generation 6500: Best = 1.147479e+03 = [4, 3, 2, 2]	FES = 650100	Diversity = 1.1631e-08	k_t
Generation 6510: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 651100	Diversity = 1.1713e-08	k_t
Generation 6520: Best = 1.147479e+03 = [2, 4, 3, 2]	FES = 652100	Diversity = 1.1438e-08	k_t
Generation 6530: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 653100	Diversity = 1.1507e-08	k_t
Generation 6540: Best = 1.147479e+03 = [4, 2, 3, 2]	FES = 654100	Diversity = 1.1596e-08	k_t
Generation 6550: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 655100	Diversity = 1.1433e-08	k_t
Generation 6560: Best = 1.147479e+03 = [2, 3, 3, 2]	FES = 656100	Diversity = 1.1324e-08	k_t
Generation 6570: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 657100	Diversity = 1.1303e-08	k_t
Generation 6580: Best = 1.147479e+03 = [3, 4, 3, 2]	FES = 658100	Diversity = 1.1213e-08	k_t
Generation 6590: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 659100	Diversity = 1.1254e-08	k_t
Generation 6600: Best = 1.147479e+03 = [2, 2, 2, 3]	FES = 660100	Diversity = 1.1138e-08	k_t
Generation 6610: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 661100	Diversity = 1.0885e-08	k_t
Generation 6620: Best = 1.147479e+03 = [2, 3, 4, 2]	FES = 662100	Diversity = 1.1067e-08	k_t
Generation 6630: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 663100	Diversity = 1.0814e-08	k_t
Generation 6640: Best = 1.147479e+03 = [4, 3, 4, 2]	FES = 664100	Diversity = 1.0891e-08	k_t
Generation 6650: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 665100	Diversity = 1.1190e-08	k_t
Generation 6660: Best = 1.147479e+03 = [4, 3, 3, 3]	FES = 666100	Diversity = 1.0773e-08	k_t
Generation 6670: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 667100	Diversity = 1.0532e-08	k_t
Generation 6680: Best = 1.147479e+03 = [3, 4, 3, 4]	FES = 668100	Diversity = 1.0789e-08	k_t
Generation 6690: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 669100	Diversity = 1.0517e-08	k_t
Generation 6700: Best = 1.147479e+03 = [4, 4, 3, 3]	FES = 670100	Diversity = 1.0516e-08	k_t

Generation 6710: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 671100	Diversity = 1.0602e-08	k_t
Generation 6720: Best = 1.147479e+03 = [3, 3, 2, 2]	FES = 672100	Diversity = 1.0499e-08	k_t
Generation 6730: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 673100	Diversity = 1.0360e-08	k_t
Generation 6740: Best = 1.147479e+03 = [2, 3, 4, 3]	FES = 674100	Diversity = 1.0545e-08	k_t
Generation 6750: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 675100	Diversity = 1.0413e-08	k_t
Generation 6760: Best = 1.147479e+03 = [4, 4, 3, 4]	FES = 676100	Diversity = 1.0467e-08	k_t
Generation 6770: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 677100	Diversity = 1.0521e-08	k_t
Generation 6780: Best = 1.147479e+03 = [4, 2, 3, 3]	FES = 678100	Diversity = 1.0424e-08	k_t
Generation 6790: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 679100	Diversity = 1.0362e-08	k_t
Generation 6800: Best = 1.147479e+03 = [2, 2, 4, 3]	FES = 680100	Diversity = 1.0368e-08	k_t
Generation 6810: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 681100	Diversity = 1.0432e-08	k_t
Generation 6820: Best = 1.147479e+03 = [3, 4, 4, 4]	FES = 682100	Diversity = 1.0395e-08	k_t
Generation 6830: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 683100	Diversity = 1.0373e-08	k_t
Generation 6840: Best = 1.147479e+03 = [4, 2, 4, 3]	FES = 684100	Diversity = 1.0406e-08	k_t
Generation 6850: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 685100	Diversity = 1.0250e-08	k_t
Generation 6860: Best = 1.147479e+03 = [2, 4, 2, 3]	FES = 686100	Diversity = 1.0389e-08	k_t
Generation 6870: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 687100	Diversity = 1.0442e-08	k_t
Generation 6880: Best = 1.147479e+03 = [4, 3, 2, 2]	FES = 688100	Diversity = 1.0460e-08	k_t
Generation 6890: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 689100	Diversity = 1.0277e-08	k_t
Generation 6900: Best = 1.147479e+03 = [2, 4, 4, 2]	FES = 690100	Diversity = 1.0339e-08	k_t
Generation 6910: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 691100	Diversity = 1.0314e-08	k_t
Generation 6920: Best = 1.147479e+03 = [4, 2, 4, 4]	FES = 692100	Diversity = 1.0397e-08	k_t
Generation 6930: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 693100	Diversity = 1.0313e-08	k_t
Generation 6940: Best = 1.147479e+03 = [2, 2, 4, 3]	FES = 694100	Diversity = 1.0321e-08	k_t
Generation 6950: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 695100	Diversity = 1.0342e-08	k_t
Generation 6960: Best = 1.147479e+03 = [4, 4, 2, 4]	FES = 696100	Diversity = 1.0308e-08	k_t
Generation 6970: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 697100	Diversity = 1.0164e-08	k_t

Generation 6980: Best = 1.147479e+03 = [3, 2, 2, 4]	FES = 698100	Diversity = 1.0261e-08	k_t
Generation 6990: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 1.0358e-08	k_t
Generation 7000: Best = 1.147479e+03 = [2, 3, 3, 2]	FES = 700100	Diversity = 1.0255e-08	k_t
Generation 7010: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 1.0313e-08	k_t
Generation 7020: Best = 1.147479e+03 = [4, 2, 4, 3]	FES = 702100	Diversity = 1.0282e-08	k_t
Generation 7030: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 1.0305e-08	k_t
Generation 7040: Best = 1.147479e+03 = [2, 2, 4, 3]	FES = 704100	Diversity = 1.0396e-08	k_t
Generation 7050: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 1.0402e-08	k_t
Generation 7060: Best = 1.147479e+03 = [2, 2, 2, 3]	FES = 706100	Diversity = 1.0417e-08	k_t
Generation 7070: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 1.0434e-08	k_t
Generation 7080: Best = 1.147479e+03 = [3, 4, 2, 2]	FES = 708100	Diversity = 1.0286e-08	k_t
Generation 7090: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 1.0291e-08	k_t
Generation 7100: Best = 1.147479e+03 = [2, 2, 3, 4]	FES = 710100	Diversity = 1.0349e-08	k_t
Generation 7110: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 1.0461e-08	k_t
Generation 7120: Best = 1.147479e+03 = [3, 2, 4, 3]	FES = 712100	Diversity = 1.0347e-08	k_t
Generation 7130: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 1.0411e-08	k_t
Generation 7140: Best = 1.147479e+03 = [4, 4, 3, 4]	FES = 714100	Diversity = 1.0376e-08	k_t
Generation 7150: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 1.0381e-08	k_t
Generation 7160: Best = 1.147479e+03 = [3, 3, 3, 3]	FES = 716100	Diversity = 1.0341e-08	k_t
Generation 7170: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 1.0361e-08	k_t
Generation 7180: Best = 1.147479e+03 = [3, 4, 3, 4]	FES = 718100	Diversity = 1.0286e-08	k_t
Generation 7190: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 1.0245e-08	k_t
Generation 7200: Best = 1.147479e+03 = [4, 4, 4, 2]	FES = 720100	Diversity = 1.0157e-08	k_t
Generation 7210: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 1.0241e-08	k_t
Generation 7220: Best = 1.147479e+03 = [4, 4, 3, 4]	FES = 722100	Diversity = 1.0181e-08	k_t
Generation 7230: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 1.0274e-08	k_t
Generation 7240: Best = 1.147479e+03 = [3, 4, 3, 2]	FES = 724100	Diversity = 1.0293e-08	k_t

Generation 7250: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 1.0218e-08	k_t
Generation 7260: Best = 1.147479e+03 = [3, 2, 4, 4]	FES = 726100	Diversity = 1.0214e-08	k_t
Generation 7270: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 1.0202e-08	k_t
Generation 7280: Best = 1.147479e+03 = [2, 2, 4, 4]	FES = 728100	Diversity = 1.0222e-08	k_t
Generation 7290: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 1.0199e-08	k_t
Generation 7300: Best = 1.147479e+03 = [2, 3, 3, 2]	FES = 730100	Diversity = 1.0118e-08	k_t
Generation 7310: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 1.0030e-08	k_t
Generation 7320: Best = 1.147479e+03 = [4, 3, 2, 2]	FES = 732100	Diversity = 9.9964e-09	k_t
Generation 7330: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 9.9034e-09	k_t
Generation 7340: Best = 1.147479e+03 = [2, 3, 4, 3]	FES = 734100	Diversity = 9.8972e-09	k_t
Generation 7350: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 9.8087e-09	k_t
Generation 7360: Best = 1.147479e+03 = [3, 3, 4, 2]	FES = 736100	Diversity = 9.6962e-09	k_t
Generation 7370: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 9.5378e-09	k_t
Generation 7380: Best = 1.147479e+03 = [2, 4, 2, 3]	FES = 738100	Diversity = 9.4656e-09	k_t
Generation 7390: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 9.3133e-09	k_t
Generation 7400: Best = 1.147479e+03 = [3, 4, 4, 4]	FES = 740100	Diversity = 9.2823e-09	k_t
Generation 7410: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 9.1804e-09	k_t
Generation 7420: Best = 1.147479e+03 = [4, 4, 2, 2]	FES = 742100	Diversity = 9.0969e-09	k_t
Generation 7430: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 9.0435e-09	k_t
Generation 7440: Best = 1.147479e+03 = [3, 2, 4, 4]	FES = 744100	Diversity = 9.1349e-09	k_t
Generation 7450: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 9.0938e-09	k_t
Generation 7460: Best = 1.147479e+03 = [3, 2, 3, 2]	FES = 746100	Diversity = 9.0569e-09	k_t
Generation 7470: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 9.1009e-09	k_t
Generation 7480: Best = 1.147479e+03 = [2, 4, 2, 3]	FES = 748100	Diversity = 9.0916e-09	k_t
Generation 7490: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 9.0329e-09	k_t
Generation 7500: Best = 1.147479e+03 = [4, 3, 3, 3]	FES = 750100	Diversity = 9.0717e-09	k_t
Generation 7510: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 9.0664e-09	k_t

Generation 7520: Best = 1.147479e+03 = [4, 2, 4, 4]	FES = 752100	Diversity = 9.1199e-09	k_t
Generation 7530: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 753100	Diversity = 9.1570e-09	k_t
Generation 7540: Best = 1.147479e+03 = [3, 2, 4, 3]	FES = 754100	Diversity = 9.1968e-09	k_t
Generation 7550: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 755100	Diversity = 9.1596e-09	k_t
Generation 7560: Best = 1.147479e+03 = [2, 4, 2, 4]	FES = 756100	Diversity = 9.1302e-09	k_t
Generation 7570: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 757100	Diversity = 9.1678e-09	k_t
Generation 7580: Best = 1.147479e+03 = [4, 2, 4, 3]	FES = 758100	Diversity = 9.2129e-09	k_t
Generation 7590: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 759100	Diversity = 9.1904e-09	k_t
Generation 7600: Best = 1.147479e+03 = [3, 4, 2, 2]	FES = 760100	Diversity = 9.1819e-09	k_t
Generation 7610: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 761100	Diversity = 9.1662e-09	k_t
Generation 7620: Best = 1.147479e+03 = [2, 3, 3, 2]	FES = 762100	Diversity = 9.1249e-09	k_t
Generation 7630: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 763100	Diversity = 9.1242e-09	k_t
Generation 7640: Best = 1.147479e+03 = [2, 4, 2, 3]	FES = 764100	Diversity = 9.1405e-09	k_t
Generation 7650: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 765100	Diversity = 9.1148e-09	k_t
Generation 7660: Best = 1.147479e+03 = [3, 4, 2, 4]	FES = 766100	Diversity = 9.1206e-09	k_t
Generation 7670: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 767100	Diversity = 9.0628e-09	k_t
Generation 7680: Best = 1.147479e+03 = [3, 3, 3, 2]	FES = 768100	Diversity = 9.0801e-09	k_t
Generation 7690: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 769100	Diversity = 9.0716e-09	k_t
Generation 7700: Best = 1.147479e+03 = [3, 4, 2, 4]	FES = 770100	Diversity = 9.0780e-09	k_t
Generation 7710: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 771100	Diversity = 9.0344e-09	k_t
Generation 7720: Best = 1.147479e+03 = [4, 3, 3, 2]	FES = 772100	Diversity = 9.0432e-09	k_t
Generation 7730: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 773100	Diversity = 9.1335e-09	k_t
Generation 7740: Best = 1.147479e+03 = [3, 4, 3, 3]	FES = 774100	Diversity = 9.0673e-09	k_t
Generation 7750: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 775100	Diversity = 9.2545e-09	k_t
Generation 7760: Best = 1.147479e+03 = [3, 2, 3, 4]	FES = 776100	Diversity = 9.2610e-09	k_t
Generation 7770: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 777100	Diversity = 9.2218e-09	k_t
Generation 7780: Best = 1.147479e+03 = [2, 2, 2, 4]	FES = 778100	Diversity = 9.2541e-09	k_t

Generation 7790: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 779100	Diversity = 9.2593e-09	k_t
Generation 7800: Best = 1.147479e+03 = [3, 4, 3, 3]	FES = 780100	Diversity = 9.1966e-09	k_t
Generation 7810: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 781100	Diversity = 9.2477e-09	k_t
Generation 7820: Best = 1.147479e+03 = [3, 3, 3, 3]	FES = 782100	Diversity = 9.2760e-09	k_t
Generation 7830: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 783100	Diversity = 9.2487e-09	k_t
Generation 7840: Best = 1.147479e+03 = [3, 2, 2, 4]	FES = 784100	Diversity = 9.2562e-09	k_t
Generation 7850: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 785100	Diversity = 9.3429e-09	k_t
Generation 7860: Best = 1.147479e+03 = [2, 4, 3, 3]	FES = 786100	Diversity = 9.2889e-09	k_t
Generation 7870: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 787100	Diversity = 9.2770e-09	k_t
Generation 7880: Best = 1.147479e+03 = [2, 2, 4, 2]	FES = 788100	Diversity = 9.2286e-09	k_t
Generation 7890: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 789100	Diversity = 9.1936e-09	k_t
Generation 7900: Best = 1.147479e+03 = [2, 2, 2, 2]	FES = 790100	Diversity = 9.2011e-09	k_t
Generation 7910: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 791100	Diversity = 9.2631e-09	k_t
Generation 7920: Best = 1.147479e+03 = [4, 2, 2, 2]	FES = 792100	Diversity = 9.2460e-09	k_t
Generation 7930: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 793100	Diversity = 9.2689e-09	k_t
Generation 7940: Best = 1.147479e+03 = [3, 3, 4, 2]	FES = 794100	Diversity = 9.3091e-09	k_t
Generation 7950: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 795100	Diversity = 9.3332e-09	k_t
Generation 7960: Best = 1.147479e+03 = [4, 2, 2, 2]	FES = 796100	Diversity = 9.3982e-09	k_t
Generation 7970: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 797100	Diversity = 9.3116e-09	k_t
Generation 7980: Best = 1.147479e+03 = [4, 2, 3, 3]	FES = 798100	Diversity = 9.3056e-09	k_t
Generation 7990: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 799100	Diversity = 9.3123e-09	k_t
Generation 8000: Best = 1.147479e+03 = [4, 3, 4, 2]	FES = 800100	Diversity = 9.3245e-09	k_t
Generation 8010: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 801100	Diversity = 9.3360e-09	k_t
Generation 8020: Best = 1.147479e+03 = [4, 3, 3, 3]	FES = 802100	Diversity = 9.3310e-09	k_t
Generation 8030: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 803100	Diversity = 9.3355e-09	k_t
Generation 8040: Best = 1.147479e+03 = [3, 3, 3, 2]	FES = 804100	Diversity = 9.3140e-09	k_t
Generation 8050: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 805100	Diversity = 9.3256e-09	k_t

Generation 8060: Best = 1.147479e+03 = [2, 2, 4, 3]	FES = 806100	Diversity = 9.3285e-09	k_t
Generation 8070: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 807100	Diversity = 9.3021e-09	k_t
Generation 8080: Best = 1.147479e+03 = [3, 3, 2, 3]	FES = 808100	Diversity = 9.2565e-09	k_t
Generation 8090: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 809100	Diversity = 9.2542e-09	k_t
Generation 8100: Best = 1.147479e+03 = [2, 2, 2, 3]	FES = 810100	Diversity = 9.2579e-09	k_t
Generation 8110: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 811100	Diversity = 9.2489e-09	k_t
Generation 8120: Best = 1.147479e+03 = [3, 3, 3, 3]	FES = 812100	Diversity = 9.2958e-09	k_t
Generation 8130: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 813100	Diversity = 9.2903e-09	k_t
Generation 8140: Best = 1.147479e+03 = [2, 3, 3, 4]	FES = 814100	Diversity = 9.2899e-09	k_t
Generation 8150: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 815100	Diversity = 9.2846e-09	k_t
Generation 8160: Best = 1.147479e+03 = [4, 3, 4, 4]	FES = 816100	Diversity = 9.3404e-09	k_t
Generation 8170: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 817100	Diversity = 9.4109e-09	k_t
Generation 8180: Best = 1.147479e+03 = [4, 4, 4, 4]	FES = 818100	Diversity = 9.4105e-09	k_t
Generation 8190: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 819100	Diversity = 9.4165e-09	k_t
Generation 8200: Best = 1.147479e+03 = [2, 2, 4, 4]	FES = 820100	Diversity = 9.4263e-09	k_t
Generation 8210: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 821100	Diversity = 9.4275e-09	k_t
Generation 8220: Best = 1.147479e+03 = [3, 2, 2, 2]	FES = 822100	Diversity = 9.4531e-09	k_t
Generation 8230: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 823100	Diversity = 9.4971e-09	k_t
Generation 8240: Best = 1.147479e+03 = [4, 3, 4, 4]	FES = 824100	Diversity = 9.4967e-09	k_t
Generation 8250: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 825100	Diversity = 9.5631e-09	k_t
Generation 8260: Best = 1.147479e+03 = [2, 3, 4, 2]	FES = 826100	Diversity = 9.5733e-09	k_t
Generation 8270: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 827100	Diversity = 9.5875e-09	k_t
Generation 8280: Best = 1.147479e+03 = [2, 3, 4, 2]	FES = 828100	Diversity = 9.5906e-09	k_t
Generation 8290: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 829100	Diversity = 9.5773e-09	k_t
Generation 8300: Best = 1.147479e+03 = [3, 2, 3, 2]	FES = 830100	Diversity = 9.5981e-09	k_t
Generation 8310: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 831100	Diversity = 9.5488e-09	k_t
Generation 8320: Best = 1.147479e+03 = [2, 3, 4, 2]	FES = 832100	Diversity = 9.5923e-09	k_t

Generation 8330: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 833100	Diversity = 9.6073e-09	k_t
Generation 8340: Best = 1.147479e+03 = [3, 4, 4, 3]	FES = 834100	Diversity = 9.6323e-09	k_t
Generation 8350: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 835100	Diversity = 9.5527e-09	k_t
Generation 8360: Best = 1.147479e+03 = [2, 3, 4, 2]	FES = 836100	Diversity = 9.5393e-09	k_t
Generation 8370: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 837100	Diversity = 9.5767e-09	k_t
Generation 8380: Best = 1.147479e+03 = [2, 3, 2, 2]	FES = 838100	Diversity = 9.5825e-09	k_t
Generation 8390: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 839100	Diversity = 9.5707e-09	k_t
Generation 8400: Best = 1.147479e+03 = [4, 3, 4, 4]	FES = 840100	Diversity = 9.5487e-09	k_t
Generation 8410: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 841100	Diversity = 9.5662e-09	k_t
Generation 8420: Best = 1.147479e+03 = [4, 2, 2, 3]	FES = 842100	Diversity = 9.5681e-09	k_t
Generation 8430: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 843100	Diversity = 9.5378e-09	k_t
Generation 8440: Best = 1.147479e+03 = [3, 2, 3, 3]	FES = 844100	Diversity = 9.5703e-09	k_t
Generation 8450: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 845100	Diversity = 9.5946e-09	k_t
Generation 8460: Best = 1.147479e+03 = [4, 2, 3, 4]	FES = 846100	Diversity = 9.6012e-09	k_t
Generation 8470: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 847100	Diversity = 9.6055e-09	k_t
Generation 8480: Best = 1.147479e+03 = [3, 4, 2, 3]	FES = 848100	Diversity = 9.6037e-09	k_t
Generation 8490: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 849100	Diversity = 9.5639e-09	k_t
Generation 8500: Best = 1.147479e+03 = [4, 3, 2, 3]	FES = 850100	Diversity = 9.5735e-09	k_t
Generation 8510: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 851100	Diversity = 9.5908e-09	k_t
Generation 8520: Best = 1.147479e+03 = [2, 2, 4, 4]	FES = 852100	Diversity = 9.5680e-09	k_t
Generation 8530: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 853100	Diversity = 9.5987e-09	k_t
Generation 8540: Best = 1.147479e+03 = [3, 4, 4, 4]	FES = 854100	Diversity = 9.5783e-09	k_t
Generation 8550: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 855100	Diversity = 9.5991e-09	k_t
Generation 8560: Best = 1.147479e+03 = [3, 3, 2, 4]	FES = 856100	Diversity = 9.6060e-09	k_t
Generation 8570: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 857100	Diversity = 9.6058e-09	k_t
Generation 8580: Best = 1.147479e+03 = [3, 2, 4, 4]	FES = 858100	Diversity = 9.7055e-09	k_t
Generation 8590: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 859100	Diversity = 9.7236e-09	k_t

Generation 8600: Best = 1.147479e+03 = [4, 2, 4, 4]	FES = 860100	Diversity = 9.6644e-09	k_t
Generation 8610: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 861100	Diversity = 9.6723e-09	k_t
Generation 8620: Best = 1.147479e+03 = [4, 4, 4, 3]	FES = 862100	Diversity = 9.6552e-09	k_t
Generation 8630: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 863100	Diversity = 9.6377e-09	k_t
Generation 8640: Best = 1.147479e+03 = [4, 3, 3, 2]	FES = 864100	Diversity = 9.5834e-09	k_t
Generation 8650: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 865100	Diversity = 9.5500e-09	k_t
Generation 8660: Best = 1.147479e+03 = [4, 2, 4, 3]	FES = 866100	Diversity = 9.5475e-09	k_t
Generation 8670: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 867100	Diversity = 9.5520e-09	k_t
Generation 8680: Best = 1.147479e+03 = [4, 4, 4, 3]	FES = 868100	Diversity = 9.5546e-09	k_t
Generation 8690: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 869100	Diversity = 9.6019e-09	k_t
Generation 8700: Best = 1.147479e+03 = [2, 3, 2, 4]	FES = 870100	Diversity = 9.6143e-09	k_t
Generation 8710: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 871100	Diversity = 9.5802e-09	k_t
Generation 8720: Best = 1.147479e+03 = [4, 3, 3, 2]	FES = 872100	Diversity = 9.5709e-09	k_t
Generation 8730: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 873100	Diversity = 9.5819e-09	k_t
Generation 8740: Best = 1.147479e+03 = [3, 4, 2, 2]	FES = 874100	Diversity = 9.5903e-09	k_t
Generation 8750: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 875100	Diversity = 9.6373e-09	k_t
Generation 8760: Best = 1.147479e+03 = [4, 2, 2, 4]	FES = 876100	Diversity = 9.6268e-09	k_t
Generation 8770: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 877100	Diversity = 9.6602e-09	k_t
Generation 8780: Best = 1.147479e+03 = [2, 2, 2, 2]	FES = 878100	Diversity = 9.6652e-09	k_t
Generation 8790: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 879100	Diversity = 9.6896e-09	k_t
Generation 8800: Best = 1.147479e+03 = [3, 2, 3, 4]	FES = 880100	Diversity = 9.6859e-09	k_t
Generation 8810: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 881100	Diversity = 9.6755e-09	k_t
Generation 8820: Best = 1.147479e+03 = [3, 3, 4, 4]	FES = 882100	Diversity = 9.6678e-09	k_t
Generation 8830: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 883100	Diversity = 9.6822e-09	k_t
Generation 8840: Best = 1.147479e+03 = [3, 3, 3, 3]	FES = 884100	Diversity = 9.6791e-09	k_t
Generation 8850: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 885100	Diversity = 9.6558e-09	k_t
Generation 8860: Best = 1.147479e+03 = [3, 4, 2, 3]	FES = 886100	Diversity = 9.6728e-09	k_t

Generation 8870: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 887100	Diversity = 9.6785e-09	k_t
Generation 8880: Best = 1.147479e+03 = [3, 3, 4, 2]	FES = 888100	Diversity = 9.6892e-09	k_t
Generation 8890: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 889100	Diversity = 9.6896e-09	k_t
Generation 8900: Best = 1.147479e+03 = [4, 3, 4, 4]	FES = 890100	Diversity = 9.6673e-09	k_t
Generation 8910: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 891100	Diversity = 9.6610e-09	k_t
Generation 8920: Best = 1.147479e+03 = [4, 2, 2, 2]	FES = 892100	Diversity = 9.6570e-09	k_t
Generation 8930: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 893100	Diversity = 9.7027e-09	k_t
Generation 8940: Best = 1.147479e+03 = [4, 4, 2, 4]	FES = 894100	Diversity = 9.7035e-09	k_t
Generation 8950: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 895100	Diversity = 9.7250e-09	k_t
Generation 8960: Best = 1.147479e+03 = [3, 3, 3, 4]	FES = 896100	Diversity = 9.7046e-09	k_t
Generation 8970: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 897100	Diversity = 9.7115e-09	k_t
Generation 8980: Best = 1.147479e+03 = [2, 3, 4, 4]	FES = 898100	Diversity = 9.7864e-09	k_t
Generation 8990: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 899100	Diversity = 9.7903e-09	k_t
Generation 9000: Best = 1.147479e+03 = [2, 2, 2, 2]	FES = 900100	Diversity = 9.7612e-09	k_t
Generation 9010: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 901100	Diversity = 9.7819e-09	k_t
Generation 9020: Best = 1.147479e+03 = [2, 2, 4, 3]	FES = 902100	Diversity = 9.7826e-09	k_t
Generation 9030: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 903100	Diversity = 9.8104e-09	k_t
Generation 9040: Best = 1.147479e+03 = [3, 4, 4, 3]	FES = 904100	Diversity = 9.8223e-09	k_t
Generation 9050: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 905100	Diversity = 9.8336e-09	k_t
Generation 9060: Best = 1.147479e+03 = [4, 2, 2, 3]	FES = 906100	Diversity = 9.8451e-09	k_t
Generation 9070: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 907100	Diversity = 9.8662e-09	k_t
Generation 9080: Best = 1.147479e+03 = [3, 3, 3, 2]	FES = 908100	Diversity = 9.8584e-09	k_t
Generation 9090: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 909100	Diversity = 9.8855e-09	k_t
Generation 9100: Best = 1.147479e+03 = [4, 2, 4, 4]	FES = 910100	Diversity = 9.9392e-09	k_t
Generation 9110: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 911100	Diversity = 9.9198e-09	k_t
Generation 9120: Best = 1.147479e+03 = [4, 4, 4, 2]	FES = 912100	Diversity = 9.9215e-09	k_t
Generation 9130: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 913100	Diversity = 1.0002e-08	k_t

Generation 9140: Best = 1.147479e+03 = [3, 3, 2, 3]	FES = 914100	Diversity = 9.9939e-09	k_t
Generation 9150: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 9.9903e-09	k_t
Generation 9160: Best = 1.147479e+03 = [3, 3, 3, 2]	FES = 916100	Diversity = 9.9619e-09	k_t
Generation 9170: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 9.9825e-09	k_t
Generation 9180: Best = 1.147479e+03 = [4, 4, 2, 3]	FES = 918100	Diversity = 1.0014e-08	k_t
Generation 9190: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 9.9986e-09	k_t
Generation 9200: Best = 1.147479e+03 = [3, 4, 2, 2]	FES = 920100	Diversity = 9.9544e-09	k_t
Generation 9210: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 9.9608e-09	k_t
Generation 9220: Best = 1.147479e+03 = [2, 4, 2, 4]	FES = 922100	Diversity = 9.9400e-09	k_t
Generation 9230: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 9.9078e-09	k_t
Generation 9240: Best = 1.147479e+03 = [4, 4, 3, 2]	FES = 924100	Diversity = 9.9508e-09	k_t
Generation 9250: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 9.9645e-09	k_t
Generation 9260: Best = 1.147479e+03 = [2, 2, 3, 4]	FES = 926100	Diversity = 9.9620e-09	k_t
Generation 9270: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 9.9749e-09	k_t
Generation 9280: Best = 1.147479e+03 = [3, 3, 3, 4]	FES = 928100	Diversity = 9.9847e-09	k_t
Generation 9290: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 9.9740e-09	k_t
Generation 9300: Best = 1.147479e+03 = [3, 3, 4, 4]	FES = 930100	Diversity = 9.9412e-09	k_t
Generation 9310: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 9.9712e-09	k_t
Generation 9320: Best = 1.147479e+03 = [2, 4, 3, 3]	FES = 932100	Diversity = 9.9619e-09	k_t
Generation 9330: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 9.9998e-09	k_t
Generation 9340: Best = 1.147479e+03 = [3, 4, 4, 3]	FES = 934100	Diversity = 9.9995e-09	k_t
Generation 9350: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 1.0026e-08	k_t
Generation 9360: Best = 1.147479e+03 = [2, 4, 3, 4]	FES = 936100	Diversity = 1.0030e-08	k_t
Generation 9370: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 1.0045e-08	k_t
Generation 9380: Best = 1.147479e+03 = [4, 4, 4, 4]	FES = 938100	Diversity = 1.0060e-08	k_t
Generation 9390: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 1.0057e-08	k_t
Generation 9400: Best = 1.147479e+03 = [2, 4, 2, 2]	FES = 940100	Diversity = 1.0069e-08	k_t

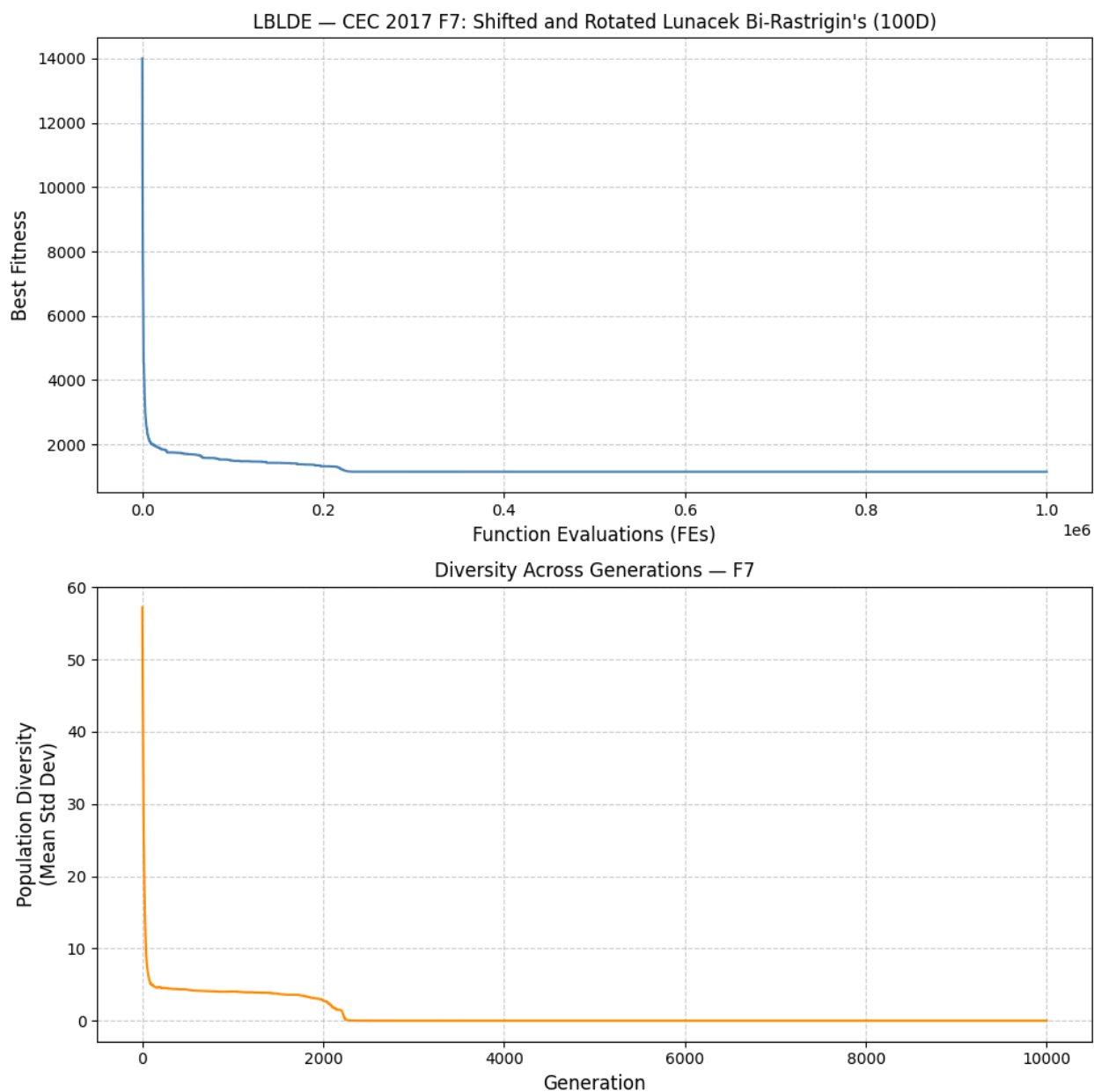
Generation 9410: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 1.0105e-08	k_t
Generation 9420: Best = 1.147479e+03 = [2, 4, 3, 4]	FES = 942100	Diversity = 1.0139e-08	k_t
Generation 9430: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 1.0150e-08	k_t
Generation 9440: Best = 1.147479e+03 = [2, 4, 4, 3]	FES = 944100	Diversity = 1.0146e-08	k_t
Generation 9450: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 1.0147e-08	k_t
Generation 9460: Best = 1.147479e+03 = [4, 3, 4, 4]	FES = 946100	Diversity = 1.0155e-08	k_t
Generation 9470: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 1.0165e-08	k_t
Generation 9480: Best = 1.147479e+03 = [2, 2, 2, 3]	FES = 948100	Diversity = 1.0200e-08	k_t
Generation 9490: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 1.0231e-08	k_t
Generation 9500: Best = 1.147479e+03 = [3, 4, 4, 4]	FES = 950100	Diversity = 1.0235e-08	k_t
Generation 9510: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 1.0264e-08	k_t
Generation 9520: Best = 1.147479e+03 = [4, 2, 4, 4]	FES = 952100	Diversity = 1.0259e-08	k_t
Generation 9530: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 1.0251e-08	k_t
Generation 9540: Best = 1.147479e+03 = [4, 3, 3, 4]	FES = 954100	Diversity = 1.0196e-08	k_t
Generation 9550: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 1.0223e-08	k_t
Generation 9560: Best = 1.147479e+03 = [2, 3, 2, 2]	FES = 956100	Diversity = 1.0198e-08	k_t
Generation 9570: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 1.0225e-08	k_t
Generation 9580: Best = 1.147479e+03 = [2, 2, 3, 2]	FES = 958100	Diversity = 1.0200e-08	k_t
Generation 9590: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 1.0199e-08	k_t
Generation 9600: Best = 1.147479e+03 = [3, 2, 2, 2]	FES = 960100	Diversity = 1.0211e-08	k_t
Generation 9610: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 1.0202e-08	k_t
Generation 9620: Best = 1.147479e+03 = [4, 4, 3, 2]	FES = 962100	Diversity = 1.0214e-08	k_t
Generation 9630: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 1.0224e-08	k_t
Generation 9640: Best = 1.147479e+03 = [4, 4, 2, 4]	FES = 964100	Diversity = 1.0207e-08	k_t
Generation 9650: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 1.0259e-08	k_t
Generation 9660: Best = 1.147479e+03 = [3, 2, 2, 3]	FES = 966100	Diversity = 1.0235e-08	k_t
Generation 9670: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 1.0255e-08	k_t

Generation 9680: Best = 1.147479e+03 = [3, 2, 2, 4]	FES = 968100	Diversity = 1.0244e-08	k_t
Generation 9690: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 969100	Diversity = 1.0202e-08	k_t
Generation 9700: Best = 1.147479e+03 = [4, 2, 3, 3]	FES = 970100	Diversity = 1.0196e-08	k_t
Generation 9710: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 971100	Diversity = 1.0177e-08	k_t
Generation 9720: Best = 1.147479e+03 = [4, 4, 3, 4]	FES = 972100	Diversity = 1.0210e-08	k_t
Generation 9730: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 973100	Diversity = 1.0196e-08	k_t
Generation 9740: Best = 1.147479e+03 = [3, 2, 4, 2]	FES = 974100	Diversity = 1.0231e-08	k_t
Generation 9750: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 975100	Diversity = 1.0265e-08	k_t
Generation 9760: Best = 1.147479e+03 = [3, 3, 2, 2]	FES = 976100	Diversity = 1.0266e-08	k_t
Generation 9770: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 977100	Diversity = 1.0269e-08	k_t
Generation 9780: Best = 1.147479e+03 = [4, 2, 2, 3]	FES = 978100	Diversity = 1.0274e-08	k_t
Generation 9790: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 979100	Diversity = 1.0265e-08	k_t
Generation 9800: Best = 1.147479e+03 = [2, 3, 4, 3]	FES = 980100	Diversity = 1.0270e-08	k_t
Generation 9810: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 981100	Diversity = 1.0260e-08	k_t
Generation 9820: Best = 1.147479e+03 = [3, 3, 3, 3]	FES = 982100	Diversity = 1.0276e-08	k_t
Generation 9830: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 983100	Diversity = 1.0298e-08	k_t
Generation 9840: Best = 1.147479e+03 = [4, 3, 4, 2]	FES = 984100	Diversity = 1.0328e-08	k_t
Generation 9850: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 985100	Diversity = 1.0342e-08	k_t
Generation 9860: Best = 1.147479e+03 = [3, 2, 4, 3]	FES = 986100	Diversity = 1.0415e-08	k_t
Generation 9870: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 987100	Diversity = 1.0408e-08	k_t
Generation 9880: Best = 1.147479e+03 = [2, 4, 3, 4]	FES = 988100	Diversity = 1.0394e-08	k_t
Generation 9890: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 989100	Diversity = 1.0443e-08	k_t
Generation 9900: Best = 1.147479e+03 = [3, 2, 4, 4]	FES = 990100	Diversity = 1.0457e-08	k_t
Generation 9910: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 991100	Diversity = 1.0469e-08	k_t
Generation 9920: Best = 1.147479e+03 = [3, 4, 2, 2]	FES = 992100	Diversity = 1.0498e-08	k_t
Generation 9930: Best = 1.147479e+03 = [1, 1, 1, 1]	FES = 993100	Diversity = 1.0510e-08	k_t
Generation 9940: Best = 1.147479e+03 = [3, 4, 3, 3]	FES = 994100	Diversity = 1.0520e-08	k_t

Generation 9950: Best = 1.147479e+03 FES = 995100 Diversity = 1.0521e-08 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 1.147479e+03 FES = 996100 Diversity = 1.0561e-08 k_t
= [3, 3, 3, 2]
Generation 9970: Best = 1.147479e+03 FES = 997100 Diversity = 1.0572e-08 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 1.147479e+03 FES = 998100 Diversity = 1.0550e-08 k_t
= [3, 4, 4, 2]
Generation 9990: Best = 1.147479e+03 FES = 999100 Diversity = 1.0554e-08 k_t
= [1, 1, 1, 1]

Optimization completed!
Final Best Fitness = 1.147479e+03
Total Function Evaluations = 1000000

=====
=
Results:
Best Fitness : 1.147479e+03
Global Optimum : 7.000000e+02
Error (|best-optimum|) : 4.474785e+02
Best Solution (5 dim) : [-24.45690087 1.78242748 -16.42719739 -1.79768511
12.16594462]
D^(0) (init. diversity): 5.7220e+01
=====



Graph saved as convergence_F7.png

```
In [32]: # =====
# F8 — Shifted and Rotated Non-Continuous Rastrigin
# =====
run_and_plot('F8', make_wrapper(simple.f8), 800.0, "Shifted and Rotated Non-Co
```

```

=====
=
LBLDE_s on CEC 2017 F8 – Single Run
=====
=
Function      : Shifted and Rotated Non-Continuous Rastrigin's (100D)
Bounds       : [-100, 100]
Global optimum : 800.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 3.562373e+03 | Diversity = 5.7220e+01
Generation 10: Best = 2.646132e+03 FES = 1100 Diversity = 4.0148e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.306524e+03 FES = 2100 Diversity = 2.9686e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 2.189246e+03 FES = 3100 Diversity = 2.3882e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.081743e+03 FES = 4100 Diversity = 2.0923e+01 k_t
= [2, 3, 4, 3]
Generation 50: Best = 2.033944e+03 FES = 5100 Diversity = 1.8958e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.963596e+03 FES = 6100 Diversity = 1.7775e+01 k_t
= [3, 3, 4, 4]
Generation 70: Best = 1.942888e+03 FES = 7100 Diversity = 1.6997e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.861938e+03 FES = 8100 Diversity = 1.6226e+01 k_t
= [2, 4, 3, 4]
Generation 90: Best = 1.861938e+03 FES = 9100 Diversity = 1.5684e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.825278e+03 FES = 10100 Diversity = 1.5187e+01 k_t
= [3, 3, 2, 3]
Generation 110: Best = 1.824503e+03 FES = 11100 Diversity = 1.4934e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.796289e+03 FES = 12100 Diversity = 1.4908e+01 k_t
= [4, 4, 4, 2]
Generation 130: Best = 1.769419e+03 FES = 13100 Diversity = 1.4973e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.769419e+03 FES = 14100 Diversity = 1.4696e+01 k_t
= [2, 2, 3, 3]
Generation 150: Best = 1.759876e+03 FES = 15100 Diversity = 1.4629e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.759876e+03 FES = 16100 Diversity = 1.4559e+01 k_t
= [4, 3, 4, 4]
Generation 170: Best = 1.759876e+03 FES = 17100 Diversity = 1.4548e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.756437e+03 FES = 18100 Diversity = 1.4623e+01 k_t
= [4, 3, 3, 4]
Generation 190: Best = 1.748130e+03 FES = 19100 Diversity = 1.4588e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.690187e+03 FES = 20100 Diversity = 1.4549e+01 k_t
= [2, 2, 4, 4]
Generation 210: Best = 1.690187e+03 FES = 21100 Diversity = 1.4624e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.682945e+03 FES = 22100 Diversity = 1.4574e+01 k_t
= [4, 2, 3, 3]

```

Generation 230:	Best = 1.682945e+03	FES = 23100	Diversity = 1.4529e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 1.682945e+03	FES = 24100	Diversity = 1.4505e+01	k_t
= [3, 3, 2, 3]				
Generation 250:	Best = 1.679223e+03	FES = 25100	Diversity = 1.4522e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 1.679223e+03	FES = 26100	Diversity = 1.4522e+01	k_t
= [4, 3, 2, 3]				
Generation 270:	Best = 1.679223e+03	FES = 27100	Diversity = 1.4385e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 1.679223e+03	FES = 28100	Diversity = 1.4327e+01	k_t
= [2, 4, 2, 2]				
Generation 290:	Best = 1.679223e+03	FES = 29100	Diversity = 1.4347e+01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 1.679223e+03	FES = 30100	Diversity = 1.4364e+01	k_t
= [2, 3, 3, 2]				
Generation 310:	Best = 1.676296e+03	FES = 31100	Diversity = 1.4349e+01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 1.660147e+03	FES = 32100	Diversity = 1.4361e+01	k_t
= [3, 3, 3, 4]				
Generation 330:	Best = 1.631826e+03	FES = 33100	Diversity = 1.4257e+01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 1.631826e+03	FES = 34100	Diversity = 1.4198e+01	k_t
= [3, 4, 2, 3]				
Generation 350:	Best = 1.631826e+03	FES = 35100	Diversity = 1.4189e+01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 1.631826e+03	FES = 36100	Diversity = 1.4160e+01	k_t
= [3, 4, 2, 2]				
Generation 370:	Best = 1.631826e+03	FES = 37100	Diversity = 1.4229e+01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 1.631826e+03	FES = 38100	Diversity = 1.4068e+01	k_t
= [2, 4, 2, 2]				
Generation 390:	Best = 1.631826e+03	FES = 39100	Diversity = 1.3992e+01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 1.631826e+03	FES = 40100	Diversity = 1.3819e+01	k_t
= [4, 4, 4, 2]				
Generation 410:	Best = 1.631826e+03	FES = 41100	Diversity = 1.3668e+01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 1.631392e+03	FES = 42100	Diversity = 1.3636e+01	k_t
= [3, 4, 2, 2]				
Generation 430:	Best = 1.627443e+03	FES = 43100	Diversity = 1.3636e+01	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 1.627443e+03	FES = 44100	Diversity = 1.3648e+01	k_t
= [4, 2, 4, 4]				
Generation 450:	Best = 1.615586e+03	FES = 45100	Diversity = 1.3642e+01	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 1.612038e+03	FES = 46100	Diversity = 1.3627e+01	k_t
= [2, 4, 3, 3]				
Generation 470:	Best = 1.612038e+03	FES = 47100	Diversity = 1.3613e+01	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 1.612038e+03	FES = 48100	Diversity = 1.3602e+01	k_t
= [3, 3, 2, 2]				
Generation 490:	Best = 1.589953e+03	FES = 49100	Diversity = 1.3541e+01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 1.582374e+03	FES = 50100	Diversity = 1.3548e+01	k_t = [2, 2, 2, 4]
Generation 510:	Best = 1.580156e+03	FES = 51100	Diversity = 1.3551e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.577267e+03	FES = 52100	Diversity = 1.3547e+01	k_t = [4, 3, 2, 4]
Generation 530:	Best = 1.577267e+03	FES = 53100	Diversity = 1.3526e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 1.571673e+03	FES = 54100	Diversity = 1.3529e+01	k_t = [3, 2, 4, 4]
Generation 550:	Best = 1.559893e+03	FES = 55100	Diversity = 1.3481e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 1.559893e+03	FES = 56100	Diversity = 1.3415e+01	k_t = [4, 3, 2, 2]
Generation 570:	Best = 1.543771e+03	FES = 57100	Diversity = 1.3425e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 1.543771e+03	FES = 58100	Diversity = 1.3423e+01	k_t = [4, 2, 2, 4]
Generation 590:	Best = 1.543771e+03	FES = 59100	Diversity = 1.3381e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 1.543771e+03	FES = 60100	Diversity = 1.3369e+01	k_t = [4, 3, 3, 4]
Generation 610:	Best = 1.543771e+03	FES = 61100	Diversity = 1.3313e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.543771e+03	FES = 62100	Diversity = 1.3290e+01	k_t = [3, 4, 4, 4]
Generation 630:	Best = 1.540028e+03	FES = 63100	Diversity = 1.3283e+01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.540028e+03	FES = 64100	Diversity = 1.3252e+01	k_t = [4, 3, 2, 3]
Generation 650:	Best = 1.533020e+03	FES = 65100	Diversity = 1.3238e+01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 1.533020e+03	FES = 66100	Diversity = 1.3198e+01	k_t = [4, 3, 2, 4]
Generation 670:	Best = 1.533020e+03	FES = 67100	Diversity = 1.3200e+01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 1.533020e+03	FES = 68100	Diversity = 1.3199e+01	k_t = [4, 3, 4, 2]
Generation 690:	Best = 1.518131e+03	FES = 69100	Diversity = 1.3186e+01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 1.518131e+03	FES = 70100	Diversity = 1.3175e+01	k_t = [3, 2, 3, 3]
Generation 710:	Best = 1.516601e+03	FES = 71100	Diversity = 1.3139e+01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 1.510986e+03	FES = 72100	Diversity = 1.3181e+01	k_t = [3, 3, 3, 4]
Generation 730:	Best = 1.507774e+03	FES = 73100	Diversity = 1.3153e+01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 1.507774e+03	FES = 74100	Diversity = 1.3146e+01	k_t = [4, 4, 2, 3]
Generation 750:	Best = 1.498437e+03	FES = 75100	Diversity = 1.3132e+01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 1.493994e+03	FES = 76100	Diversity = 1.3106e+01	k_t = [4, 4, 4, 3]

Generation 770:	Best = 1.493994e+03	FES = 77100	Diversity = 1.3086e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.493994e+03	FES = 78100	Diversity = 1.3072e+01	k_t
= [4, 4, 4, 3]				
Generation 790:	Best = 1.485375e+03	FES = 79100	Diversity = 1.3061e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.485375e+03	FES = 80100	Diversity = 1.3049e+01	k_t
= [3, 3, 4, 3]				
Generation 810:	Best = 1.455509e+03	FES = 81100	Diversity = 1.2987e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.454415e+03	FES = 82100	Diversity = 1.2970e+01	k_t
= [3, 4, 2, 4]				
Generation 830:	Best = 1.454415e+03	FES = 83100	Diversity = 1.2965e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.454415e+03	FES = 84100	Diversity = 1.2938e+01	k_t
= [3, 4, 4, 4]				
Generation 850:	Best = 1.454415e+03	FES = 85100	Diversity = 1.2892e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.454415e+03	FES = 86100	Diversity = 1.2877e+01	k_t
= [2, 2, 4, 4]				
Generation 870:	Best = 1.441619e+03	FES = 87100	Diversity = 1.2876e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.441322e+03	FES = 88100	Diversity = 1.2967e+01	k_t
= [2, 3, 4, 3]				
Generation 890:	Best = 1.438518e+03	FES = 89100	Diversity = 1.2883e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.438518e+03	FES = 90100	Diversity = 1.2802e+01	k_t
= [4, 2, 2, 3]				
Generation 910:	Best = 1.438518e+03	FES = 91100	Diversity = 1.2820e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.438518e+03	FES = 92100	Diversity = 1.2806e+01	k_t
= [3, 4, 3, 4]				
Generation 930:	Best = 1.425169e+03	FES = 93100	Diversity = 1.2801e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.420036e+03	FES = 94100	Diversity = 1.2810e+01	k_t
= [3, 4, 4, 2]				
Generation 950:	Best = 1.420036e+03	FES = 95100	Diversity = 1.2780e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.420036e+03	FES = 96100	Diversity = 1.2767e+01	k_t
= [2, 3, 4, 2]				
Generation 970:	Best = 1.420036e+03	FES = 97100	Diversity = 1.2757e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.420036e+03	FES = 98100	Diversity = 1.2643e+01	k_t
= [3, 4, 2, 3]				
Generation 990:	Best = 1.420036e+03	FES = 99100	Diversity = 1.2586e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 1.420036e+03	FES = 100100	Diversity = 1.2574e+01	k_t
= [3, 3, 3, 3]				
Generation 1010:	Best = 1.420036e+03	FES = 101100	Diversity = 1.2572e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.420036e+03	FES = 102100	Diversity = 1.2567e+01	k_t
= [3, 2, 4, 3]				
Generation 1030:	Best = 1.420036e+03	FES = 103100	Diversity = 1.2564e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.420036e+03 = [3, 2, 2, 2]	FES = 104100	Diversity = 1.2555e+01	k_t
Generation 1050: Best = 1.406356e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.2529e+01	k_t
Generation 1060: Best = 1.403069e+03 = [4, 4, 2, 3]	FES = 106100	Diversity = 1.2505e+01	k_t
Generation 1070: Best = 1.403069e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 1.2503e+01	k_t
Generation 1080: Best = 1.391686e+03 = [3, 3, 4, 2]	FES = 108100	Diversity = 1.2487e+01	k_t
Generation 1090: Best = 1.391686e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.2482e+01	k_t
Generation 1100: Best = 1.391686e+03 = [3, 2, 2, 3]	FES = 110100	Diversity = 1.2480e+01	k_t
Generation 1110: Best = 1.388241e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.2420e+01	k_t
Generation 1120: Best = 1.388241e+03 = [2, 2, 3, 3]	FES = 112100	Diversity = 1.2418e+01	k_t
Generation 1130: Best = 1.376230e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.2410e+01	k_t
Generation 1140: Best = 1.376230e+03 = [4, 2, 2, 4]	FES = 114100	Diversity = 1.2404e+01	k_t
Generation 1150: Best = 1.374568e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.2391e+01	k_t
Generation 1160: Best = 1.371076e+03 = [2, 2, 4, 4]	FES = 116100	Diversity = 1.2372e+01	k_t
Generation 1170: Best = 1.356079e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.2358e+01	k_t
Generation 1180: Best = 1.352429e+03 = [2, 3, 4, 4]	FES = 118100	Diversity = 1.2346e+01	k_t
Generation 1190: Best = 1.352429e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.2339e+01	k_t
Generation 1200: Best = 1.340581e+03 = [4, 4, 2, 2]	FES = 120100	Diversity = 1.2333e+01	k_t
Generation 1210: Best = 1.340581e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.2329e+01	k_t
Generation 1220: Best = 1.340581e+03 = [2, 2, 3, 2]	FES = 122100	Diversity = 1.2325e+01	k_t
Generation 1230: Best = 1.340581e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.2321e+01	k_t
Generation 1240: Best = 1.340473e+03 = [3, 2, 4, 4]	FES = 124100	Diversity = 1.2321e+01	k_t
Generation 1250: Best = 1.339251e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 1.2316e+01	k_t
Generation 1260: Best = 1.339251e+03 = [3, 3, 2, 4]	FES = 126100	Diversity = 1.2307e+01	k_t
Generation 1270: Best = 1.330354e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.2304e+01	k_t
Generation 1280: Best = 1.330354e+03 = [2, 2, 4, 4]	FES = 128100	Diversity = 1.2300e+01	k_t
Generation 1290: Best = 1.328998e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.2292e+01	k_t
Generation 1300: Best = 1.328998e+03 = [2, 2, 3, 2]	FES = 130100	Diversity = 1.2243e+01	k_t

Generation 1310: Best = 1.323517e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.2263e+01	k_t
Generation 1320: Best = 1.323517e+03 = [2, 3, 3, 4]	FES = 132100	Diversity = 1.2254e+01	k_t
Generation 1330: Best = 1.323517e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 1.2218e+01	k_t
Generation 1340: Best = 1.315690e+03 = [3, 4, 4, 2]	FES = 134100	Diversity = 1.2215e+01	k_t
Generation 1350: Best = 1.315690e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.2203e+01	k_t
Generation 1360: Best = 1.315690e+03 = [2, 2, 2, 2]	FES = 136100	Diversity = 1.2203e+01	k_t
Generation 1370: Best = 1.307330e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 1.2201e+01	k_t
Generation 1380: Best = 1.307330e+03 = [4, 2, 4, 3]	FES = 138100	Diversity = 1.2200e+01	k_t
Generation 1390: Best = 1.307330e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 1.2198e+01	k_t
Generation 1400: Best = 1.307330e+03 = [4, 2, 2, 4]	FES = 140100	Diversity = 1.2200e+01	k_t
Generation 1410: Best = 1.306506e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 1.2189e+01	k_t
Generation 1420: Best = 1.303780e+03 = [2, 4, 2, 4]	FES = 142100	Diversity = 1.2187e+01	k_t
Generation 1430: Best = 1.303301e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 1.2193e+01	k_t
Generation 1440: Best = 1.303301e+03 = [2, 3, 4, 3]	FES = 144100	Diversity = 1.2158e+01	k_t
Generation 1450: Best = 1.303301e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 1.2154e+01	k_t
Generation 1460: Best = 1.303301e+03 = [2, 4, 2, 4]	FES = 146100	Diversity = 1.2149e+01	k_t
Generation 1470: Best = 1.301943e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 1.2126e+01	k_t
Generation 1480: Best = 1.301943e+03 = [2, 3, 3, 2]	FES = 148100	Diversity = 1.2123e+01	k_t
Generation 1490: Best = 1.299808e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 1.2119e+01	k_t
Generation 1500: Best = 1.299808e+03 = [2, 3, 3, 3]	FES = 150100	Diversity = 1.2116e+01	k_t
Generation 1510: Best = 1.299808e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 1.2102e+01	k_t
Generation 1520: Best = 1.299808e+03 = [2, 3, 4, 2]	FES = 152100	Diversity = 1.2093e+01	k_t
Generation 1530: Best = 1.286603e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 1.2091e+01	k_t
Generation 1540: Best = 1.286603e+03 = [3, 2, 2, 2]	FES = 154100	Diversity = 1.2011e+01	k_t
Generation 1550: Best = 1.282636e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 1.2006e+01	k_t
Generation 1560: Best = 1.281518e+03 = [2, 4, 4, 2]	FES = 156100	Diversity = 1.1967e+01	k_t
Generation 1570: Best = 1.281518e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 1.1970e+01	k_t

Generation 1580: Best = 1.281518e+03 = [4, 4, 2, 2]	FES = 158100	Diversity = 1.1968e+01	k_t
Generation 1590: Best = 1.281518e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 1.1937e+01	k_t
Generation 1600: Best = 1.281518e+03 = [4, 3, 3, 4]	FES = 160100	Diversity = 1.1936e+01	k_t
Generation 1610: Best = 1.281082e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 1.1931e+01	k_t
Generation 1620: Best = 1.281082e+03 = [3, 4, 4, 2]	FES = 162100	Diversity = 1.1932e+01	k_t
Generation 1630: Best = 1.281082e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.1933e+01	k_t
Generation 1640: Best = 1.280018e+03 = [2, 2, 4, 3]	FES = 164100	Diversity = 1.1931e+01	k_t
Generation 1650: Best = 1.280018e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.1930e+01	k_t
Generation 1660: Best = 1.280018e+03 = [2, 3, 3, 3]	FES = 166100	Diversity = 1.1932e+01	k_t
Generation 1670: Best = 1.280018e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.1929e+01	k_t
Generation 1680: Best = 1.280018e+03 = [4, 4, 3, 3]	FES = 168100	Diversity = 1.1903e+01	k_t
Generation 1690: Best = 1.280018e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 1.1858e+01	k_t
Generation 1700: Best = 1.280018e+03 = [3, 3, 4, 4]	FES = 170100	Diversity = 1.1860e+01	k_t
Generation 1710: Best = 1.280018e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.1861e+01	k_t
Generation 1720: Best = 1.280018e+03 = [3, 2, 3, 2]	FES = 172100	Diversity = 1.1858e+01	k_t
Generation 1730: Best = 1.279975e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 1.1850e+01	k_t
Generation 1740: Best = 1.279975e+03 = [4, 3, 2, 2]	FES = 174100	Diversity = 1.1847e+01	k_t
Generation 1750: Best = 1.279975e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 1.1847e+01	k_t
Generation 1760: Best = 1.279975e+03 = [3, 2, 4, 2]	FES = 176100	Diversity = 1.1827e+01	k_t
Generation 1770: Best = 1.279230e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 1.1828e+01	k_t
Generation 1780: Best = 1.279230e+03 = [4, 4, 3, 3]	FES = 178100	Diversity = 1.1805e+01	k_t
Generation 1790: Best = 1.271509e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.1773e+01	k_t
Generation 1800: Best = 1.270195e+03 = [3, 4, 4, 2]	FES = 180100	Diversity = 1.1773e+01	k_t
Generation 1810: Best = 1.270195e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 1.1809e+01	k_t
Generation 1820: Best = 1.270195e+03 = [3, 2, 3, 2]	FES = 182100	Diversity = 1.1802e+01	k_t
Generation 1830: Best = 1.270195e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.1684e+01	k_t
Generation 1840: Best = 1.270195e+03 = [2, 3, 4, 2]	FES = 184100	Diversity = 1.1590e+01	k_t

Generation 1850: Best = 1.264971e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.1586e+01	k_t
Generation 1860: Best = 1.264971e+03 = [3, 2, 2, 4]	FES = 186100	Diversity = 1.1587e+01	k_t
Generation 1870: Best = 1.264971e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.1585e+01	k_t
Generation 1880: Best = 1.264519e+03 = [2, 2, 2, 2]	FES = 188100	Diversity = 1.1583e+01	k_t
Generation 1890: Best = 1.263317e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 1.1581e+01	k_t
Generation 1900: Best = 1.263317e+03 = [4, 2, 2, 3]	FES = 190100	Diversity = 1.1576e+01	k_t
Generation 1910: Best = 1.263072e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 1.1541e+01	k_t
Generation 1920: Best = 1.262189e+03 = [4, 4, 2, 3]	FES = 192100	Diversity = 1.1499e+01	k_t
Generation 1930: Best = 1.250145e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 1.1494e+01	k_t
Generation 1940: Best = 1.250145e+03 = [3, 4, 3, 2]	FES = 194100	Diversity = 1.1491e+01	k_t
Generation 1950: Best = 1.248734e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 1.1464e+01	k_t
Generation 1960: Best = 1.232274e+03 = [2, 3, 3, 4]	FES = 196100	Diversity = 1.1463e+01	k_t
Generation 1970: Best = 1.232274e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.1444e+01	k_t
Generation 1980: Best = 1.232274e+03 = [3, 3, 4, 3]	FES = 198100	Diversity = 1.1443e+01	k_t
Generation 1990: Best = 1.228475e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.1417e+01	k_t
Generation 2000: Best = 1.225817e+03 = [4, 3, 2, 3]	FES = 200100	Diversity = 1.1416e+01	k_t
Generation 2010: Best = 1.225817e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 1.1414e+01	k_t
Generation 2020: Best = 1.225817e+03 = [4, 4, 3, 4]	FES = 202100	Diversity = 1.1308e+01	k_t
Generation 2030: Best = 1.222230e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 1.1295e+01	k_t
Generation 2040: Best = 1.222230e+03 = [2, 2, 2, 2]	FES = 204100	Diversity = 1.1257e+01	k_t
Generation 2050: Best = 1.219637e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.1261e+01	k_t
Generation 2060: Best = 1.210810e+03 = [3, 3, 2, 2]	FES = 206100	Diversity = 1.1206e+01	k_t
Generation 2070: Best = 1.208963e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.1180e+01	k_t
Generation 2080: Best = 1.208963e+03 = [3, 3, 2, 2]	FES = 208100	Diversity = 1.1177e+01	k_t
Generation 2090: Best = 1.208963e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 1.1187e+01	k_t
Generation 2100: Best = 1.202788e+03 = [3, 2, 2, 4]	FES = 210100	Diversity = 1.1032e+01	k_t
Generation 2110: Best = 1.200788e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.1015e+01	k_t

Generation 2120: Best = 1.195310e+03 = [3, 4, 2, 2]	FES = 212100	Diversity = 1.0973e+01	k_t
Generation 2130: Best = 1.195310e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 1.0972e+01	k_t
Generation 2140: Best = 1.195123e+03 = [2, 3, 2, 3]	FES = 214100	Diversity = 1.0958e+01	k_t
Generation 2150: Best = 1.183707e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 1.0914e+01	k_t
Generation 2160: Best = 1.181725e+03 = [4, 3, 2, 4]	FES = 216100	Diversity = 1.0894e+01	k_t
Generation 2170: Best = 1.170052e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 1.0660e+01	k_t
Generation 2180: Best = 1.165434e+03 = [2, 3, 3, 3]	FES = 218100	Diversity = 1.0561e+01	k_t
Generation 2190: Best = 1.154261e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 1.0465e+01	k_t
Generation 2200: Best = 1.145544e+03 = [3, 2, 2, 4]	FES = 220100	Diversity = 1.0296e+01	k_t
Generation 2210: Best = 1.143831e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.0085e+01	k_t
Generation 2220: Best = 1.133606e+03 = [3, 4, 2, 4]	FES = 222100	Diversity = 9.9916e+00	k_t
Generation 2230: Best = 1.120756e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 9.4341e+00	k_t
Generation 2240: Best = 1.109980e+03 = [4, 4, 2, 3]	FES = 224100	Diversity = 7.9002e+00	k_t
Generation 2250: Best = 1.101276e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 4.0695e+00	k_t
Generation 2260: Best = 1.098257e+03 = [2, 4, 4, 3]	FES = 226100	Diversity = 2.6174e+00	k_t
Generation 2270: Best = 1.088915e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.4605e+00	k_t
Generation 2280: Best = 1.080804e+03 = [4, 3, 4, 4]	FES = 228100	Diversity = 4.7701e-01	k_t
Generation 2290: Best = 1.073389e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 3.9664e-01	k_t
Generation 2300: Best = 1.069257e+03 = [2, 2, 4, 3]	FES = 230100	Diversity = 3.2806e-01	k_t
Generation 2310: Best = 1.064828e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 2.6957e-01	k_t
Generation 2320: Best = 1.062233e+03 = [2, 2, 2, 3]	FES = 232100	Diversity = 2.1116e-01	k_t
Generation 2330: Best = 1.060365e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 1.6840e-01	k_t
Generation 2340: Best = 1.059240e+03 = [4, 2, 2, 4]	FES = 234100	Diversity = 1.4369e-01	k_t
Generation 2350: Best = 1.058277e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 1.2322e-01	k_t
Generation 2360: Best = 1.057077e+03 = [2, 2, 3, 4]	FES = 236100	Diversity = 1.0670e-01	k_t
Generation 2370: Best = 1.056776e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 9.6064e-02	k_t
Generation 2380: Best = 1.056353e+03 = [2, 2, 2, 4]	FES = 238100	Diversity = 8.1721e-02	k_t

Generation 2390: Best = 1.056047e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 6.8485e-02	k_t
Generation 2400: Best = 1.055782e+03 = [4, 4, 3, 3]	FES = 240100	Diversity = 5.8266e-02	k_t
Generation 2410: Best = 1.055539e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 4.9631e-02	k_t
Generation 2420: Best = 1.055379e+03 = [3, 4, 4, 2]	FES = 242100	Diversity = 4.4172e-02	k_t
Generation 2430: Best = 1.055267e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 3.9673e-02	k_t
Generation 2440: Best = 1.055201e+03 = [4, 4, 2, 3]	FES = 244100	Diversity = 3.4408e-02	k_t
Generation 2450: Best = 1.055137e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 2.9634e-02	k_t
Generation 2460: Best = 1.055090e+03 = [3, 4, 3, 2]	FES = 246100	Diversity = 2.6325e-02	k_t
Generation 2470: Best = 1.055036e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 2.4013e-02	k_t
Generation 2480: Best = 1.054994e+03 = [2, 2, 4, 2]	FES = 248100	Diversity = 2.2804e-02	k_t
Generation 2490: Best = 1.054963e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.9909e-02	k_t
Generation 2500: Best = 1.054932e+03 = [3, 2, 4, 4]	FES = 250100	Diversity = 1.7786e-02	k_t
Generation 2510: Best = 1.054903e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.6346e-02	k_t
Generation 2520: Best = 1.054891e+03 = [4, 3, 4, 4]	FES = 252100	Diversity = 1.4337e-02	k_t
Generation 2530: Best = 1.054870e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.3484e-02	k_t
Generation 2540: Best = 1.054855e+03 = [4, 2, 4, 2]	FES = 254100	Diversity = 1.2996e-02	k_t
Generation 2550: Best = 1.054844e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.1250e-02	k_t
Generation 2560: Best = 1.054838e+03 = [3, 4, 2, 2]	FES = 256100	Diversity = 9.8115e-03	k_t
Generation 2570: Best = 1.054832e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 8.1217e-03	k_t
Generation 2580: Best = 1.054828e+03 = [3, 2, 2, 2]	FES = 258100	Diversity = 7.1066e-03	k_t
Generation 2590: Best = 1.054823e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 6.4940e-03	k_t
Generation 2600: Best = 1.054820e+03 = [4, 4, 3, 3]	FES = 260100	Diversity = 5.8506e-03	k_t
Generation 2610: Best = 1.054818e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 4.8853e-03	k_t
Generation 2620: Best = 1.054817e+03 = [2, 2, 4, 2]	FES = 262100	Diversity = 4.5328e-03	k_t
Generation 2630: Best = 1.054815e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 4.0211e-03	k_t
Generation 2640: Best = 1.054814e+03 = [3, 4, 3, 4]	FES = 264100	Diversity = 3.7205e-03	k_t
Generation 2650: Best = 1.054813e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 3.6587e-03	k_t

Generation 2660: Best = 1.054812e+03 = [3, 4, 3, 2]	FES = 266100	Diversity = 3.3423e-03	k_t
Generation 2670: Best = 1.054811e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 3.0275e-03	k_t
Generation 2680: Best = 1.054810e+03 = [3, 2, 4, 2]	FES = 268100	Diversity = 2.6603e-03	k_t
Generation 2690: Best = 1.054810e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 2.5314e-03	k_t
Generation 2700: Best = 1.054809e+03 = [2, 2, 3, 4]	FES = 270100	Diversity = 2.2930e-03	k_t
Generation 2710: Best = 1.054809e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 2.0602e-03	k_t
Generation 2720: Best = 1.054809e+03 = [4, 2, 2, 3]	FES = 272100	Diversity = 1.7761e-03	k_t
Generation 2730: Best = 1.054808e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.6075e-03	k_t
Generation 2740: Best = 1.054808e+03 = [2, 2, 2, 2]	FES = 274100	Diversity = 1.5089e-03	k_t
Generation 2750: Best = 1.054808e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.3522e-03	k_t
Generation 2760: Best = 1.054808e+03 = [2, 3, 2, 4]	FES = 276100	Diversity = 1.2187e-03	k_t
Generation 2770: Best = 1.054808e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.0809e-03	k_t
Generation 2780: Best = 1.054808e+03 = [2, 4, 3, 3]	FES = 278100	Diversity = 9.4611e-04	k_t
Generation 2790: Best = 1.054808e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 7.7907e-04	k_t
Generation 2800: Best = 1.054808e+03 = [4, 3, 2, 4]	FES = 280100	Diversity = 6.9357e-04	k_t
Generation 2810: Best = 1.054808e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 6.3011e-04	k_t
Generation 2820: Best = 1.054807e+03 = [2, 3, 4, 4]	FES = 282100	Diversity = 6.0957e-04	k_t
Generation 2830: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 5.8713e-04	k_t
Generation 2840: Best = 1.054807e+03 = [4, 2, 2, 2]	FES = 284100	Diversity = 5.5108e-04	k_t
Generation 2850: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 4.9665e-04	k_t
Generation 2860: Best = 1.054807e+03 = [4, 2, 4, 4]	FES = 286100	Diversity = 4.4204e-04	k_t
Generation 2870: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 4.3746e-04	k_t
Generation 2880: Best = 1.054807e+03 = [4, 2, 2, 3]	FES = 288100	Diversity = 3.9710e-04	k_t
Generation 2890: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 3.5844e-04	k_t
Generation 2900: Best = 1.054807e+03 = [3, 3, 2, 4]	FES = 290100	Diversity = 3.1531e-04	k_t
Generation 2910: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 2.8927e-04	k_t
Generation 2920: Best = 1.054807e+03 = [4, 4, 3, 3]	FES = 292100	Diversity = 2.3902e-04	k_t

Generation 2930: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 293100	Diversity = 2.1606e-04	k_t
Generation 2940: Best = 1.054807e+03 = [4, 2, 2, 2]	FES = 294100	Diversity = 1.9869e-04	k_t
Generation 2950: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 295100	Diversity = 1.8127e-04	k_t
Generation 2960: Best = 1.054807e+03 = [2, 4, 3, 3]	FES = 296100	Diversity = 1.6302e-04	k_t
Generation 2970: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 297100	Diversity = 1.4142e-04	k_t
Generation 2980: Best = 1.054807e+03 = [4, 2, 3, 2]	FES = 298100	Diversity = 1.2503e-04	k_t
Generation 2990: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 299100	Diversity = 1.1127e-04	k_t
Generation 3000: Best = 1.054807e+03 = [4, 3, 4, 2]	FES = 300100	Diversity = 9.7739e-05	k_t
Generation 3010: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 301100	Diversity = 9.1375e-05	k_t
Generation 3020: Best = 1.054807e+03 = [4, 2, 2, 3]	FES = 302100	Diversity = 8.2966e-05	k_t
Generation 3030: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 303100	Diversity = 7.4921e-05	k_t
Generation 3040: Best = 1.054807e+03 = [2, 2, 4, 2]	FES = 304100	Diversity = 7.1157e-05	k_t
Generation 3050: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 305100	Diversity = 6.3448e-05	k_t
Generation 3060: Best = 1.054807e+03 = [2, 2, 2, 2]	FES = 306100	Diversity = 5.8078e-05	k_t
Generation 3070: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 307100	Diversity = 5.0400e-05	k_t
Generation 3080: Best = 1.054807e+03 = [3, 4, 4, 2]	FES = 308100	Diversity = 4.7454e-05	k_t
Generation 3090: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 309100	Diversity = 3.9947e-05	k_t
Generation 3100: Best = 1.054807e+03 = [3, 2, 3, 4]	FES = 310100	Diversity = 3.3161e-05	k_t
Generation 3110: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 311100	Diversity = 2.8014e-05	k_t
Generation 3120: Best = 1.054807e+03 = [3, 4, 3, 4]	FES = 312100	Diversity = 2.5135e-05	k_t
Generation 3130: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 313100	Diversity = 2.2126e-05	k_t
Generation 3140: Best = 1.054807e+03 = [4, 4, 2, 2]	FES = 314100	Diversity = 2.0114e-05	k_t
Generation 3150: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 315100	Diversity = 1.8436e-05	k_t
Generation 3160: Best = 1.054807e+03 = [2, 4, 4, 2]	FES = 316100	Diversity = 1.7004e-05	k_t
Generation 3170: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 317100	Diversity = 1.5207e-05	k_t
Generation 3180: Best = 1.054807e+03 = [3, 3, 3, 2]	FES = 318100	Diversity = 1.3427e-05	k_t
Generation 3190: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 319100	Diversity = 1.1829e-05	k_t

Generation 3200: Best = 1.054807e+03 = [4, 4, 4, 2]	FES = 320100	Diversity = 1.0675e-05	k_t
Generation 3210: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 321100	Diversity = 9.5230e-06	k_t
Generation 3220: Best = 1.054807e+03 = [4, 3, 4, 3]	FES = 322100	Diversity = 8.8205e-06	k_t
Generation 3230: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 323100	Diversity = 8.6335e-06	k_t
Generation 3240: Best = 1.054807e+03 = [4, 4, 3, 3]	FES = 324100	Diversity = 7.9391e-06	k_t
Generation 3250: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 325100	Diversity = 7.5942e-06	k_t
Generation 3260: Best = 1.054807e+03 = [4, 3, 2, 2]	FES = 326100	Diversity = 7.6507e-06	k_t
Generation 3270: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 327100	Diversity = 7.3064e-06	k_t
Generation 3280: Best = 1.054807e+03 = [2, 4, 2, 2]	FES = 328100	Diversity = 6.8516e-06	k_t
Generation 3290: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 329100	Diversity = 5.7690e-06	k_t
Generation 3300: Best = 1.054807e+03 = [2, 3, 2, 3]	FES = 330100	Diversity = 4.9606e-06	k_t
Generation 3310: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 331100	Diversity = 4.2345e-06	k_t
Generation 3320: Best = 1.054807e+03 = [3, 2, 4, 4]	FES = 332100	Diversity = 3.8934e-06	k_t
Generation 3330: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 333100	Diversity = 3.6670e-06	k_t
Generation 3340: Best = 1.054807e+03 = [2, 2, 3, 2]	FES = 334100	Diversity = 3.2871e-06	k_t
Generation 3350: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 335100	Diversity = 3.0433e-06	k_t
Generation 3360: Best = 1.054807e+03 = [2, 2, 4, 3]	FES = 336100	Diversity = 2.7973e-06	k_t
Generation 3370: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 337100	Diversity = 2.6872e-06	k_t
Generation 3380: Best = 1.054807e+03 = [2, 4, 2, 2]	FES = 338100	Diversity = 2.4342e-06	k_t
Generation 3390: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 339100	Diversity = 2.1678e-06	k_t
Generation 3400: Best = 1.054807e+03 = [4, 4, 4, 4]	FES = 340100	Diversity = 1.9725e-06	k_t
Generation 3410: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 341100	Diversity = 1.7579e-06	k_t
Generation 3420: Best = 1.054807e+03 = [4, 3, 3, 4]	FES = 342100	Diversity = 1.4641e-06	k_t
Generation 3430: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 343100	Diversity = 1.2124e-06	k_t
Generation 3440: Best = 1.054807e+03 = [2, 3, 2, 2]	FES = 344100	Diversity = 1.0196e-06	k_t
Generation 3450: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 345100	Diversity = 8.5387e-07	k_t
Generation 3460: Best = 1.054807e+03 = [3, 3, 4, 3]	FES = 346100	Diversity = 7.6982e-07	k_t

Generation 3470: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 347100	Diversity = 7.6533e-07	k_t
Generation 3480: Best = 1.054807e+03 = [2, 3, 4, 3]	FES = 348100	Diversity = 6.8636e-07	k_t
Generation 3490: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 349100	Diversity = 6.3498e-07	k_t
Generation 3500: Best = 1.054807e+03 = [3, 2, 2, 3]	FES = 350100	Diversity = 5.8277e-07	k_t
Generation 3510: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 351100	Diversity = 5.6895e-07	k_t
Generation 3520: Best = 1.054807e+03 = [2, 2, 4, 3]	FES = 352100	Diversity = 5.4216e-07	k_t
Generation 3530: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 353100	Diversity = 4.9466e-07	k_t
Generation 3540: Best = 1.054807e+03 = [3, 4, 3, 3]	FES = 354100	Diversity = 4.3540e-07	k_t
Generation 3550: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 355100	Diversity = 4.0678e-07	k_t
Generation 3560: Best = 1.054807e+03 = [2, 3, 3, 2]	FES = 356100	Diversity = 3.6848e-07	k_t
Generation 3570: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 357100	Diversity = 3.4494e-07	k_t
Generation 3580: Best = 1.054807e+03 = [2, 2, 2, 2]	FES = 358100	Diversity = 2.9573e-07	k_t
Generation 3590: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 359100	Diversity = 2.5756e-07	k_t
Generation 3600: Best = 1.054807e+03 = [2, 2, 4, 4]	FES = 360100	Diversity = 2.3431e-07	k_t
Generation 3610: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 361100	Diversity = 2.0779e-07	k_t
Generation 3620: Best = 1.054807e+03 = [3, 4, 4, 2]	FES = 362100	Diversity = 2.1400e-07	k_t
Generation 3630: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 363100	Diversity = 1.8881e-07	k_t
Generation 3640: Best = 1.054807e+03 = [2, 3, 4, 2]	FES = 364100	Diversity = 1.7625e-07	k_t
Generation 3650: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 365100	Diversity = 1.6301e-07	k_t
Generation 3660: Best = 1.054807e+03 = [3, 4, 4, 2]	FES = 366100	Diversity = 1.5185e-07	k_t
Generation 3670: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 367100	Diversity = 1.3792e-07	k_t
Generation 3680: Best = 1.054807e+03 = [3, 3, 3, 3]	FES = 368100	Diversity = 1.2584e-07	k_t
Generation 3690: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 369100	Diversity = 1.0832e-07	k_t
Generation 3700: Best = 1.054807e+03 = [2, 2, 3, 3]	FES = 370100	Diversity = 1.0110e-07	k_t
Generation 3710: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 371100	Diversity = 9.7556e-08	k_t
Generation 3720: Best = 1.054807e+03 = [3, 3, 3, 2]	FES = 372100	Diversity = 8.5051e-08	k_t
Generation 3730: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 373100	Diversity = 8.2117e-08	k_t

Generation 3740: Best = 1.054807e+03 = [2, 2, 2, 4]	FES = 374100	Diversity = 8.0420e-08	k_t
Generation 3750: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 375100	Diversity = 7.9843e-08	k_t
Generation 3760: Best = 1.054807e+03 = [3, 3, 4, 2]	FES = 376100	Diversity = 8.3100e-08	k_t
Generation 3770: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 377100	Diversity = 7.7505e-08	k_t
Generation 3780: Best = 1.054807e+03 = [3, 2, 2, 2]	FES = 378100	Diversity = 6.9746e-08	k_t
Generation 3790: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 379100	Diversity = 7.1456e-08	k_t
Generation 3800: Best = 1.054807e+03 = [4, 2, 3, 4]	FES = 380100	Diversity = 6.7947e-08	k_t
Generation 3810: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 381100	Diversity = 6.8260e-08	k_t
Generation 3820: Best = 1.054807e+03 = [4, 3, 4, 2]	FES = 382100	Diversity = 6.5890e-08	k_t
Generation 3830: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 383100	Diversity = 6.5892e-08	k_t
Generation 3840: Best = 1.054807e+03 = [3, 4, 4, 3]	FES = 384100	Diversity = 6.3997e-08	k_t
Generation 3850: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 385100	Diversity = 6.1096e-08	k_t
Generation 3860: Best = 1.054807e+03 = [3, 2, 4, 4]	FES = 386100	Diversity = 5.9928e-08	k_t
Generation 3870: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 387100	Diversity = 5.8418e-08	k_t
Generation 3880: Best = 1.054807e+03 = [3, 2, 2, 2]	FES = 388100	Diversity = 5.8181e-08	k_t
Generation 3890: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 389100	Diversity = 5.6813e-08	k_t
Generation 3900: Best = 1.054807e+03 = [4, 4, 2, 2]	FES = 390100	Diversity = 5.4873e-08	k_t
Generation 3910: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 391100	Diversity = 5.2939e-08	k_t
Generation 3920: Best = 1.054807e+03 = [4, 3, 2, 3]	FES = 392100	Diversity = 4.9593e-08	k_t
Generation 3930: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 393100	Diversity = 4.7743e-08	k_t
Generation 3940: Best = 1.054807e+03 = [3, 3, 3, 2]	FES = 394100	Diversity = 4.8747e-08	k_t
Generation 3950: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 395100	Diversity = 4.7576e-08	k_t
Generation 3960: Best = 1.054807e+03 = [3, 3, 3, 4]	FES = 396100	Diversity = 4.6877e-08	k_t
Generation 3970: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 397100	Diversity = 4.7099e-08	k_t
Generation 3980: Best = 1.054807e+03 = [4, 4, 3, 2]	FES = 398100	Diversity = 4.6366e-08	k_t
Generation 3990: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 399100	Diversity = 4.5766e-08	k_t
Generation 4000: Best = 1.054807e+03 = [2, 2, 2, 3]	FES = 400100	Diversity = 4.6351e-08	k_t

Generation 4010: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 401100	Diversity = 4.7456e-08	k_t
Generation 4020: Best = 1.054807e+03 = [2, 3, 4, 3]	FES = 402100	Diversity = 4.6593e-08	k_t
Generation 4030: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 403100	Diversity = 4.6766e-08	k_t
Generation 4040: Best = 1.054807e+03 = [2, 2, 3, 4]	FES = 404100	Diversity = 4.7064e-08	k_t
Generation 4050: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 405100	Diversity = 4.7680e-08	k_t
Generation 4060: Best = 1.054807e+03 = [2, 4, 3, 2]	FES = 406100	Diversity = 4.8332e-08	k_t
Generation 4070: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 407100	Diversity = 4.8555e-08	k_t
Generation 4080: Best = 1.054807e+03 = [3, 3, 2, 3]	FES = 408100	Diversity = 4.9409e-08	k_t
Generation 4090: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 409100	Diversity = 5.0428e-08	k_t
Generation 4100: Best = 1.054807e+03 = [3, 2, 3, 2]	FES = 410100	Diversity = 5.0792e-08	k_t
Generation 4110: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 411100	Diversity = 5.0591e-08	k_t
Generation 4120: Best = 1.054807e+03 = [4, 2, 3, 3]	FES = 412100	Diversity = 5.0095e-08	k_t
Generation 4130: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 413100	Diversity = 5.0680e-08	k_t
Generation 4140: Best = 1.054807e+03 = [3, 4, 4, 3]	FES = 414100	Diversity = 5.0178e-08	k_t
Generation 4150: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 415100	Diversity = 5.1508e-08	k_t
Generation 4160: Best = 1.054807e+03 = [3, 3, 2, 4]	FES = 416100	Diversity = 5.1407e-08	k_t
Generation 4170: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 417100	Diversity = 5.2314e-08	k_t
Generation 4180: Best = 1.054807e+03 = [4, 2, 4, 4]	FES = 418100	Diversity = 5.1961e-08	k_t
Generation 4190: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 419100	Diversity = 5.2598e-08	k_t
Generation 4200: Best = 1.054807e+03 = [4, 3, 2, 3]	FES = 420100	Diversity = 5.2356e-08	k_t
Generation 4210: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 421100	Diversity = 5.1887e-08	k_t
Generation 4220: Best = 1.054807e+03 = [2, 4, 4, 3]	FES = 422100	Diversity = 5.2037e-08	k_t
Generation 4230: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 423100	Diversity = 5.1521e-08	k_t
Generation 4240: Best = 1.054807e+03 = [4, 2, 3, 4]	FES = 424100	Diversity = 5.2722e-08	k_t
Generation 4250: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 425100	Diversity = 5.2842e-08	k_t
Generation 4260: Best = 1.054807e+03 = [4, 4, 3, 4]	FES = 426100	Diversity = 5.2314e-08	k_t
Generation 4270: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 427100	Diversity = 5.2436e-08	k_t

Generation 4280: Best = 1.054807e+03 = [3, 4, 4, 4]	FES = 428100	Diversity = 5.1408e-08	k_t
Generation 4290: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 429100	Diversity = 5.1210e-08	k_t
Generation 4300: Best = 1.054807e+03 = [2, 2, 3, 4]	FES = 430100	Diversity = 5.1314e-08	k_t
Generation 4310: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 431100	Diversity = 5.2090e-08	k_t
Generation 4320: Best = 1.054807e+03 = [3, 2, 2, 4]	FES = 432100	Diversity = 5.2780e-08	k_t
Generation 4330: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 433100	Diversity = 5.1814e-08	k_t
Generation 4340: Best = 1.054807e+03 = [4, 3, 3, 3]	FES = 434100	Diversity = 5.3695e-08	k_t
Generation 4350: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 435100	Diversity = 5.5206e-08	k_t
Generation 4360: Best = 1.054807e+03 = [3, 4, 4, 2]	FES = 436100	Diversity = 5.7631e-08	k_t
Generation 4370: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 437100	Diversity = 5.7540e-08	k_t
Generation 4380: Best = 1.054807e+03 = [4, 2, 3, 2]	FES = 438100	Diversity = 5.7924e-08	k_t
Generation 4390: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 439100	Diversity = 5.8307e-08	k_t
Generation 4400: Best = 1.054807e+03 = [4, 3, 3, 2]	FES = 440100	Diversity = 5.7672e-08	k_t
Generation 4410: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 441100	Diversity = 5.9776e-08	k_t
Generation 4420: Best = 1.054807e+03 = [4, 3, 2, 2]	FES = 442100	Diversity = 5.9694e-08	k_t
Generation 4430: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 443100	Diversity = 6.0051e-08	k_t
Generation 4440: Best = 1.054807e+03 = [3, 2, 3, 2]	FES = 444100	Diversity = 5.9906e-08	k_t
Generation 4450: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 445100	Diversity = 5.9107e-08	k_t
Generation 4460: Best = 1.054807e+03 = [3, 3, 3, 3]	FES = 446100	Diversity = 5.8288e-08	k_t
Generation 4470: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 447100	Diversity = 5.7584e-08	k_t
Generation 4480: Best = 1.054807e+03 = [4, 3, 4, 2]	FES = 448100	Diversity = 5.6376e-08	k_t
Generation 4490: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 449100	Diversity = 5.7211e-08	k_t
Generation 4500: Best = 1.054807e+03 = [4, 3, 4, 3]	FES = 450100	Diversity = 5.7367e-08	k_t
Generation 4510: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 451100	Diversity = 5.7779e-08	k_t
Generation 4520: Best = 1.054807e+03 = [2, 2, 3, 4]	FES = 452100	Diversity = 5.8072e-08	k_t
Generation 4530: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 453100	Diversity = 5.8470e-08	k_t
Generation 4540: Best = 1.054807e+03 = [4, 4, 3, 3]	FES = 454100	Diversity = 5.7951e-08	k_t

Generation 4550: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 455100	Diversity = 5.8818e-08	k_t
Generation 4560: Best = 1.054807e+03 = [4, 4, 4, 3]	FES = 456100	Diversity = 5.7321e-08	k_t
Generation 4570: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 457100	Diversity = 5.7726e-08	k_t
Generation 4580: Best = 1.054807e+03 = [2, 2, 4, 2]	FES = 458100	Diversity = 5.7665e-08	k_t
Generation 4590: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 459100	Diversity = 5.6986e-08	k_t
Generation 4600: Best = 1.054807e+03 = [2, 2, 4, 3]	FES = 460100	Diversity = 5.6006e-08	k_t
Generation 4610: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 461100	Diversity = 5.6069e-08	k_t
Generation 4620: Best = 1.054807e+03 = [2, 4, 3, 3]	FES = 462100	Diversity = 5.4496e-08	k_t
Generation 4630: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 463100	Diversity = 5.5466e-08	k_t
Generation 4640: Best = 1.054807e+03 = [3, 3, 2, 2]	FES = 464100	Diversity = 5.6110e-08	k_t
Generation 4650: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 465100	Diversity = 5.5880e-08	k_t
Generation 4660: Best = 1.054807e+03 = [2, 4, 2, 3]	FES = 466100	Diversity = 5.6301e-08	k_t
Generation 4670: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 467100	Diversity = 5.6003e-08	k_t
Generation 4680: Best = 1.054807e+03 = [2, 4, 2, 2]	FES = 468100	Diversity = 5.5209e-08	k_t
Generation 4690: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 469100	Diversity = 5.4851e-08	k_t
Generation 4700: Best = 1.054807e+03 = [2, 2, 2, 2]	FES = 470100	Diversity = 5.4967e-08	k_t
Generation 4710: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 471100	Diversity = 5.5444e-08	k_t
Generation 4720: Best = 1.054807e+03 = [2, 2, 4, 2]	FES = 472100	Diversity = 5.5884e-08	k_t
Generation 4730: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 473100	Diversity = 5.5904e-08	k_t
Generation 4740: Best = 1.054807e+03 = [3, 4, 3, 2]	FES = 474100	Diversity = 5.5932e-08	k_t
Generation 4750: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 475100	Diversity = 5.5762e-08	k_t
Generation 4760: Best = 1.054807e+03 = [2, 4, 4, 4]	FES = 476100	Diversity = 5.5834e-08	k_t
Generation 4770: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 477100	Diversity = 5.5349e-08	k_t
Generation 4780: Best = 1.054807e+03 = [4, 3, 2, 4]	FES = 478100	Diversity = 5.4871e-08	k_t
Generation 4790: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 479100	Diversity = 5.5528e-08	k_t
Generation 4800: Best = 1.054807e+03 = [4, 2, 4, 2]	FES = 480100	Diversity = 5.5380e-08	k_t
Generation 4810: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 481100	Diversity = 5.4845e-08	k_t

Generation 4820: Best = 1.054807e+03 = [3, 4, 3, 4]	FES = 482100	Diversity = 5.6065e-08	k_t
Generation 4830: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 483100	Diversity = 5.7084e-08	k_t
Generation 4840: Best = 1.054807e+03 = [4, 2, 3, 3]	FES = 484100	Diversity = 5.7993e-08	k_t
Generation 4850: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 485100	Diversity = 5.8606e-08	k_t
Generation 4860: Best = 1.054807e+03 = [2, 2, 3, 4]	FES = 486100	Diversity = 5.9188e-08	k_t
Generation 4870: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 487100	Diversity = 5.8507e-08	k_t
Generation 4880: Best = 1.054807e+03 = [4, 2, 2, 3]	FES = 488100	Diversity = 5.9004e-08	k_t
Generation 4890: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 489100	Diversity = 5.7844e-08	k_t
Generation 4900: Best = 1.054807e+03 = [3, 4, 2, 2]	FES = 490100	Diversity = 5.7534e-08	k_t
Generation 4910: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 491100	Diversity = 5.6383e-08	k_t
Generation 4920: Best = 1.054807e+03 = [4, 2, 3, 2]	FES = 492100	Diversity = 5.6619e-08	k_t
Generation 4930: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 493100	Diversity = 5.7976e-08	k_t
Generation 4940: Best = 1.054807e+03 = [3, 3, 2, 3]	FES = 494100	Diversity = 5.8225e-08	k_t
Generation 4950: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 495100	Diversity = 5.9228e-08	k_t
Generation 4960: Best = 1.054807e+03 = [3, 3, 3, 3]	FES = 496100	Diversity = 5.9784e-08	k_t
Generation 4970: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 497100	Diversity = 5.9832e-08	k_t
Generation 4980: Best = 1.054807e+03 = [4, 2, 4, 4]	FES = 498100	Diversity = 5.8978e-08	k_t
Generation 4990: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 499100	Diversity = 5.9146e-08	k_t
Generation 5000: Best = 1.054807e+03 = [3, 3, 4, 4]	FES = 500100	Diversity = 5.8595e-08	k_t
Generation 5010: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 501100	Diversity = 5.9887e-08	k_t
Generation 5020: Best = 1.054807e+03 = [3, 3, 2, 2]	FES = 502100	Diversity = 5.9412e-08	k_t
Generation 5030: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 503100	Diversity = 5.9312e-08	k_t
Generation 5040: Best = 1.054807e+03 = [2, 4, 4, 3]	FES = 504100	Diversity = 5.8907e-08	k_t
Generation 5050: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 505100	Diversity = 5.7936e-08	k_t
Generation 5060: Best = 1.054807e+03 = [2, 4, 2, 2]	FES = 506100	Diversity = 5.8427e-08	k_t
Generation 5070: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 507100	Diversity = 5.6839e-08	k_t
Generation 5080: Best = 1.054807e+03 = [4, 2, 2, 4]	FES = 508100	Diversity = 5.7974e-08	k_t

Generation 5090: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 509100	Diversity = 5.9181e-08	k_t
Generation 5100: Best = 1.054807e+03 = [4, 3, 3, 4]	FES = 510100	Diversity = 6.0366e-08	k_t
Generation 5110: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 511100	Diversity = 5.9482e-08	k_t
Generation 5120: Best = 1.054807e+03 = [4, 3, 3, 4]	FES = 512100	Diversity = 6.0433e-08	k_t
Generation 5130: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 513100	Diversity = 5.8789e-08	k_t
Generation 5140: Best = 1.054807e+03 = [2, 3, 3, 3]	FES = 514100	Diversity = 5.9298e-08	k_t
Generation 5150: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 515100	Diversity = 5.7518e-08	k_t
Generation 5160: Best = 1.054807e+03 = [4, 3, 2, 2]	FES = 516100	Diversity = 5.7749e-08	k_t
Generation 5170: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 517100	Diversity = 5.7418e-08	k_t
Generation 5180: Best = 1.054807e+03 = [3, 4, 4, 2]	FES = 518100	Diversity = 5.7995e-08	k_t
Generation 5190: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 519100	Diversity = 5.6720e-08	k_t
Generation 5200: Best = 1.054807e+03 = [3, 3, 4, 2]	FES = 520100	Diversity = 5.7285e-08	k_t
Generation 5210: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 521100	Diversity = 5.7852e-08	k_t
Generation 5220: Best = 1.054807e+03 = [2, 3, 3, 4]	FES = 522100	Diversity = 5.7379e-08	k_t
Generation 5230: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 523100	Diversity = 5.5897e-08	k_t
Generation 5240: Best = 1.054807e+03 = [3, 3, 3, 4]	FES = 524100	Diversity = 5.4844e-08	k_t
Generation 5250: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 525100	Diversity = 5.6222e-08	k_t
Generation 5260: Best = 1.054807e+03 = [2, 4, 4, 3]	FES = 526100	Diversity = 5.5693e-08	k_t
Generation 5270: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 527100	Diversity = 5.4050e-08	k_t
Generation 5280: Best = 1.054807e+03 = [2, 2, 3, 4]	FES = 528100	Diversity = 5.5522e-08	k_t
Generation 5290: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 529100	Diversity = 5.5639e-08	k_t
Generation 5300: Best = 1.054807e+03 = [2, 4, 3, 4]	FES = 530100	Diversity = 5.4418e-08	k_t
Generation 5310: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 531100	Diversity = 5.2535e-08	k_t
Generation 5320: Best = 1.054807e+03 = [3, 3, 4, 3]	FES = 532100	Diversity = 5.1826e-08	k_t
Generation 5330: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 533100	Diversity = 4.9522e-08	k_t
Generation 5340: Best = 1.054807e+03 = [3, 2, 4, 4]	FES = 534100	Diversity = 5.0761e-08	k_t
Generation 5350: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 535100	Diversity = 4.9438e-08	k_t

Generation 5360: Best = 1.054807e+03	FES = 536100	Diversity = 4.7679e-08	k_t
= [4, 3, 2, 3]			
Generation 5370: Best = 1.054807e+03	FES = 537100	Diversity = 4.5645e-08	k_t
= [1, 1, 1, 1]			
Generation 5380: Best = 1.054807e+03	FES = 538100	Diversity = 4.3588e-08	k_t
= [4, 4, 4, 4]			
Generation 5390: Best = 1.054807e+03	FES = 539100	Diversity = 4.4028e-08	k_t
= [1, 1, 1, 1]			
Generation 5400: Best = 1.054807e+03	FES = 540100	Diversity = 4.2985e-08	k_t
= [2, 3, 4, 2]			
Generation 5410: Best = 1.054807e+03	FES = 541100	Diversity = 4.3093e-08	k_t
= [1, 1, 1, 1]			
Generation 5420: Best = 1.054807e+03	FES = 542100	Diversity = 4.2170e-08	k_t
= [4, 4, 3, 3]			
Generation 5430: Best = 1.054807e+03	FES = 543100	Diversity = 4.2286e-08	k_t
= [1, 1, 1, 1]			
Generation 5440: Best = 1.054807e+03	FES = 544100	Diversity = 4.2337e-08	k_t
= [2, 4, 2, 3]			
Generation 5450: Best = 1.054807e+03	FES = 545100	Diversity = 4.2622e-08	k_t
= [1, 1, 1, 1]			
Generation 5460: Best = 1.054807e+03	FES = 546100	Diversity = 4.2886e-08	k_t
= [3, 4, 4, 3]			
Generation 5470: Best = 1.054807e+03	FES = 547100	Diversity = 4.2428e-08	k_t
= [1, 1, 1, 1]			
Generation 5480: Best = 1.054807e+03	FES = 548100	Diversity = 4.3420e-08	k_t
= [3, 4, 2, 2]			
Generation 5490: Best = 1.054807e+03	FES = 549100	Diversity = 4.2748e-08	k_t
= [1, 1, 1, 1]			
Generation 5500: Best = 1.054807e+03	FES = 550100	Diversity = 4.2824e-08	k_t
= [4, 2, 4, 3]			
Generation 5510: Best = 1.054807e+03	FES = 551100	Diversity = 4.3070e-08	k_t
= [1, 1, 1, 1]			
Generation 5520: Best = 1.054807e+03	FES = 552100	Diversity = 4.2635e-08	k_t
= [3, 3, 3, 4]			
Generation 5530: Best = 1.054807e+03	FES = 553100	Diversity = 4.1497e-08	k_t
= [1, 1, 1, 1]			
Generation 5540: Best = 1.054807e+03	FES = 554100	Diversity = 4.0929e-08	k_t
= [4, 2, 2, 3]			
Generation 5550: Best = 1.054807e+03	FES = 555100	Diversity = 4.0691e-08	k_t
= [1, 1, 1, 1]			
Generation 5560: Best = 1.054807e+03	FES = 556100	Diversity = 4.0704e-08	k_t
= [3, 3, 3, 3]			
Generation 5570: Best = 1.054807e+03	FES = 557100	Diversity = 4.0205e-08	k_t
= [1, 1, 1, 1]			
Generation 5580: Best = 1.054807e+03	FES = 558100	Diversity = 3.9736e-08	k_t
= [4, 2, 3, 4]			
Generation 5590: Best = 1.054807e+03	FES = 559100	Diversity = 4.1446e-08	k_t
= [1, 1, 1, 1]			
Generation 5600: Best = 1.054807e+03	FES = 560100	Diversity = 4.1165e-08	k_t
= [4, 3, 2, 2]			
Generation 5610: Best = 1.054807e+03	FES = 561100	Diversity = 4.0736e-08	k_t
= [1, 1, 1, 1]			
Generation 5620: Best = 1.054807e+03	FES = 562100	Diversity = 4.0815e-08	k_t
= [4, 2, 4, 2]			

Generation 5630: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 563100	Diversity = 4.0650e-08	k_t
Generation 5640: Best = 1.054807e+03 = [3, 4, 4, 4]	FES = 564100	Diversity = 4.0443e-08	k_t
Generation 5650: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 565100	Diversity = 3.9981e-08	k_t
Generation 5660: Best = 1.054807e+03 = [2, 3, 3, 4]	FES = 566100	Diversity = 4.0167e-08	k_t
Generation 5670: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 567100	Diversity = 4.0496e-08	k_t
Generation 5680: Best = 1.054807e+03 = [2, 2, 4, 4]	FES = 568100	Diversity = 3.9863e-08	k_t
Generation 5690: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 569100	Diversity = 3.9971e-08	k_t
Generation 5700: Best = 1.054807e+03 = [3, 3, 3, 3]	FES = 570100	Diversity = 3.9752e-08	k_t
Generation 5710: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 571100	Diversity = 3.8833e-08	k_t
Generation 5720: Best = 1.054807e+03 = [2, 2, 2, 3]	FES = 572100	Diversity = 3.9578e-08	k_t
Generation 5730: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 573100	Diversity = 3.9805e-08	k_t
Generation 5740: Best = 1.054807e+03 = [2, 2, 4, 4]	FES = 574100	Diversity = 3.9476e-08	k_t
Generation 5750: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 575100	Diversity = 3.9697e-08	k_t
Generation 5760: Best = 1.054807e+03 = [4, 3, 3, 2]	FES = 576100	Diversity = 4.0756e-08	k_t
Generation 5770: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 577100	Diversity = 4.0129e-08	k_t
Generation 5780: Best = 1.054807e+03 = [3, 2, 4, 3]	FES = 578100	Diversity = 4.0323e-08	k_t
Generation 5790: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 579100	Diversity = 4.0281e-08	k_t
Generation 5800: Best = 1.054807e+03 = [4, 4, 2, 4]	FES = 580100	Diversity = 4.0335e-08	k_t
Generation 5810: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 581100	Diversity = 3.9778e-08	k_t
Generation 5820: Best = 1.054807e+03 = [3, 3, 2, 4]	FES = 582100	Diversity = 3.9887e-08	k_t
Generation 5830: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 583100	Diversity = 4.0400e-08	k_t
Generation 5840: Best = 1.054807e+03 = [3, 2, 4, 4]	FES = 584100	Diversity = 4.0669e-08	k_t
Generation 5850: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 585100	Diversity = 4.1202e-08	k_t
Generation 5860: Best = 1.054807e+03 = [3, 3, 2, 4]	FES = 586100	Diversity = 4.2452e-08	k_t
Generation 5870: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 587100	Diversity = 4.2525e-08	k_t
Generation 5880: Best = 1.054807e+03 = [2, 4, 4, 3]	FES = 588100	Diversity = 4.1960e-08	k_t
Generation 5890: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 589100	Diversity = 4.2739e-08	k_t

Generation 5900: Best = 1.054807e+03 = [3, 3, 2, 3]	FES = 590100	Diversity = 4.2364e-08	k_t
Generation 5910: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 591100	Diversity = 4.2168e-08	k_t
Generation 5920: Best = 1.054807e+03 = [3, 2, 4, 4]	FES = 592100	Diversity = 4.2575e-08	k_t
Generation 5930: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 593100	Diversity = 4.2033e-08	k_t
Generation 5940: Best = 1.054807e+03 = [2, 2, 2, 3]	FES = 594100	Diversity = 4.1577e-08	k_t
Generation 5950: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 595100	Diversity = 4.2069e-08	k_t
Generation 5960: Best = 1.054807e+03 = [2, 4, 4, 2]	FES = 596100	Diversity = 4.1512e-08	k_t
Generation 5970: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 597100	Diversity = 4.1293e-08	k_t
Generation 5980: Best = 1.054807e+03 = [2, 4, 2, 3]	FES = 598100	Diversity = 4.0379e-08	k_t
Generation 5990: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 599100	Diversity = 4.0038e-08	k_t
Generation 6000: Best = 1.054807e+03 = [2, 3, 2, 4]	FES = 600100	Diversity = 4.0288e-08	k_t
Generation 6010: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 601100	Diversity = 3.9916e-08	k_t
Generation 6020: Best = 1.054807e+03 = [4, 2, 4, 3]	FES = 602100	Diversity = 4.0366e-08	k_t
Generation 6030: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 603100	Diversity = 4.1039e-08	k_t
Generation 6040: Best = 1.054807e+03 = [4, 4, 3, 3]	FES = 604100	Diversity = 3.9921e-08	k_t
Generation 6050: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 605100	Diversity = 4.0570e-08	k_t
Generation 6060: Best = 1.054807e+03 = [2, 4, 3, 4]	FES = 606100	Diversity = 3.9285e-08	k_t
Generation 6070: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 607100	Diversity = 4.0114e-08	k_t
Generation 6080: Best = 1.054807e+03 = [3, 4, 2, 2]	FES = 608100	Diversity = 3.9737e-08	k_t
Generation 6090: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 609100	Diversity = 4.0300e-08	k_t
Generation 6100: Best = 1.054807e+03 = [4, 4, 4, 3]	FES = 610100	Diversity = 3.9870e-08	k_t
Generation 6110: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 611100	Diversity = 4.0561e-08	k_t
Generation 6120: Best = 1.054807e+03 = [4, 3, 4, 2]	FES = 612100	Diversity = 4.0119e-08	k_t
Generation 6130: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 613100	Diversity = 3.9076e-08	k_t
Generation 6140: Best = 1.054807e+03 = [3, 3, 3, 4]	FES = 614100	Diversity = 3.8417e-08	k_t
Generation 6150: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 615100	Diversity = 3.8347e-08	k_t
Generation 6160: Best = 1.054807e+03 = [4, 2, 3, 4]	FES = 616100	Diversity = 3.8289e-08	k_t

Generation 6170: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 617100	Diversity = 3.8565e-08	k_t
Generation 6180: Best = 1.054807e+03 = [2, 3, 3, 3]	FES = 618100	Diversity = 3.9173e-08	k_t
Generation 6190: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 619100	Diversity = 3.9401e-08	k_t
Generation 6200: Best = 1.054807e+03 = [3, 4, 4, 4]	FES = 620100	Diversity = 3.9839e-08	k_t
Generation 6210: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 621100	Diversity = 4.0719e-08	k_t
Generation 6220: Best = 1.054807e+03 = [2, 4, 4, 4]	FES = 622100	Diversity = 4.0746e-08	k_t
Generation 6230: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 623100	Diversity = 4.0554e-08	k_t
Generation 6240: Best = 1.054807e+03 = [2, 2, 3, 4]	FES = 624100	Diversity = 4.1290e-08	k_t
Generation 6250: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 625100	Diversity = 4.1205e-08	k_t
Generation 6260: Best = 1.054807e+03 = [2, 4, 2, 2]	FES = 626100	Diversity = 4.1246e-08	k_t
Generation 6270: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 627100	Diversity = 4.0596e-08	k_t
Generation 6280: Best = 1.054807e+03 = [3, 2, 2, 2]	FES = 628100	Diversity = 4.0678e-08	k_t
Generation 6290: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 629100	Diversity = 4.0182e-08	k_t
Generation 6300: Best = 1.054807e+03 = [4, 3, 2, 4]	FES = 630100	Diversity = 3.9677e-08	k_t
Generation 6310: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 631100	Diversity = 4.0086e-08	k_t
Generation 6320: Best = 1.054807e+03 = [2, 3, 3, 3]	FES = 632100	Diversity = 3.9142e-08	k_t
Generation 6330: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 633100	Diversity = 3.9063e-08	k_t
Generation 6340: Best = 1.054807e+03 = [2, 4, 4, 4]	FES = 634100	Diversity = 3.9232e-08	k_t
Generation 6350: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 635100	Diversity = 3.9277e-08	k_t
Generation 6360: Best = 1.054807e+03 = [4, 3, 2, 3]	FES = 636100	Diversity = 3.9324e-08	k_t
Generation 6370: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 637100	Diversity = 3.9931e-08	k_t
Generation 6380: Best = 1.054807e+03 = [2, 4, 2, 3]	FES = 638100	Diversity = 4.0437e-08	k_t
Generation 6390: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 639100	Diversity = 4.0774e-08	k_t
Generation 6400: Best = 1.054807e+03 = [2, 3, 3, 2]	FES = 640100	Diversity = 4.0949e-08	k_t
Generation 6410: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 641100	Diversity = 4.0848e-08	k_t
Generation 6420: Best = 1.054807e+03 = [4, 4, 4, 2]	FES = 642100	Diversity = 4.0470e-08	k_t
Generation 6430: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 643100	Diversity = 4.0135e-08	k_t

Generation 6440: Best = 1.054807e+03 = [2, 3, 2, 2]	FES = 644100	Diversity = 4.0549e-08	k_t
Generation 6450: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 645100	Diversity = 4.0382e-08	k_t
Generation 6460: Best = 1.054807e+03 = [4, 3, 4, 3]	FES = 646100	Diversity = 3.9663e-08	k_t
Generation 6470: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 647100	Diversity = 3.9646e-08	k_t
Generation 6480: Best = 1.054807e+03 = [4, 2, 3, 2]	FES = 648100	Diversity = 3.9526e-08	k_t
Generation 6490: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 649100	Diversity = 3.9265e-08	k_t
Generation 6500: Best = 1.054807e+03 = [4, 2, 2, 2]	FES = 650100	Diversity = 3.8941e-08	k_t
Generation 6510: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 651100	Diversity = 3.8684e-08	k_t
Generation 6520: Best = 1.054807e+03 = [2, 2, 3, 3]	FES = 652100	Diversity = 3.9007e-08	k_t
Generation 6530: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 653100	Diversity = 3.9977e-08	k_t
Generation 6540: Best = 1.054807e+03 = [3, 2, 3, 4]	FES = 654100	Diversity = 3.9599e-08	k_t
Generation 6550: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 655100	Diversity = 3.9482e-08	k_t
Generation 6560: Best = 1.054807e+03 = [4, 3, 3, 3]	FES = 656100	Diversity = 3.9297e-08	k_t
Generation 6570: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 657100	Diversity = 3.9685e-08	k_t
Generation 6580: Best = 1.054807e+03 = [4, 2, 4, 4]	FES = 658100	Diversity = 4.0350e-08	k_t
Generation 6590: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 659100	Diversity = 4.1480e-08	k_t
Generation 6600: Best = 1.054807e+03 = [4, 3, 3, 4]	FES = 660100	Diversity = 4.1637e-08	k_t
Generation 6610: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 661100	Diversity = 4.1775e-08	k_t
Generation 6620: Best = 1.054807e+03 = [3, 4, 4, 2]	FES = 662100	Diversity = 4.1807e-08	k_t
Generation 6630: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 663100	Diversity = 4.1385e-08	k_t
Generation 6640: Best = 1.054807e+03 = [4, 4, 3, 2]	FES = 664100	Diversity = 4.1914e-08	k_t
Generation 6650: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 665100	Diversity = 4.2128e-08	k_t
Generation 6660: Best = 1.054807e+03 = [2, 2, 3, 2]	FES = 666100	Diversity = 4.1469e-08	k_t
Generation 6670: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 667100	Diversity = 4.1456e-08	k_t
Generation 6680: Best = 1.054807e+03 = [4, 3, 4, 3]	FES = 668100	Diversity = 4.1160e-08	k_t
Generation 6690: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 669100	Diversity = 4.1037e-08	k_t
Generation 6700: Best = 1.054807e+03 = [4, 4, 2, 4]	FES = 670100	Diversity = 4.1046e-08	k_t

Generation 6710: Best = 1.054807e+03	FES = 671100	Diversity = 4.0897e-08	k_t
= [1, 1, 1, 1]			
Generation 6720: Best = 1.054807e+03	FES = 672100	Diversity = 4.0566e-08	k_t
= [2, 2, 2, 3]			
Generation 6730: Best = 1.054807e+03	FES = 673100	Diversity = 4.1326e-08	k_t
= [1, 1, 1, 1]			
Generation 6740: Best = 1.054807e+03	FES = 674100	Diversity = 4.1447e-08	k_t
= [3, 3, 2, 2]			
Generation 6750: Best = 1.054807e+03	FES = 675100	Diversity = 4.1483e-08	k_t
= [1, 1, 1, 1]			
Generation 6760: Best = 1.054807e+03	FES = 676100	Diversity = 4.1113e-08	k_t
= [2, 4, 2, 2]			
Generation 6770: Best = 1.054807e+03	FES = 677100	Diversity = 4.0688e-08	k_t
= [1, 1, 1, 1]			
Generation 6780: Best = 1.054807e+03	FES = 678100	Diversity = 4.0614e-08	k_t
= [3, 3, 3, 2]			
Generation 6790: Best = 1.054807e+03	FES = 679100	Diversity = 4.0280e-08	k_t
= [1, 1, 1, 1]			
Generation 6800: Best = 1.054807e+03	FES = 680100	Diversity = 4.0015e-08	k_t
= [2, 4, 3, 2]			
Generation 6810: Best = 1.054807e+03	FES = 681100	Diversity = 3.9821e-08	k_t
= [1, 1, 1, 1]			
Generation 6820: Best = 1.054807e+03	FES = 682100	Diversity = 3.9058e-08	k_t
= [3, 2, 2, 3]			
Generation 6830: Best = 1.054807e+03	FES = 683100	Diversity = 3.8975e-08	k_t
= [1, 1, 1, 1]			
Generation 6840: Best = 1.054807e+03	FES = 684100	Diversity = 3.8562e-08	k_t
= [2, 4, 4, 4]			
Generation 6850: Best = 1.054807e+03	FES = 685100	Diversity = 3.8621e-08	k_t
= [1, 1, 1, 1]			
Generation 6860: Best = 1.054807e+03	FES = 686100	Diversity = 3.8911e-08	k_t
= [4, 4, 3, 3]			
Generation 6870: Best = 1.054807e+03	FES = 687100	Diversity = 3.9080e-08	k_t
= [1, 1, 1, 1]			
Generation 6880: Best = 1.054807e+03	FES = 688100	Diversity = 3.8625e-08	k_t
= [3, 2, 2, 2]			
Generation 6890: Best = 1.054807e+03	FES = 689100	Diversity = 3.7474e-08	k_t
= [1, 1, 1, 1]			
Generation 6900: Best = 1.054807e+03	FES = 690100	Diversity = 3.7394e-08	k_t
= [3, 4, 3, 4]			
Generation 6910: Best = 1.054807e+03	FES = 691100	Diversity = 3.7194e-08	k_t
= [1, 1, 1, 1]			
Generation 6920: Best = 1.054807e+03	FES = 692100	Diversity = 3.6986e-08	k_t
= [4, 3, 3, 3]			
Generation 6930: Best = 1.054807e+03	FES = 693100	Diversity = 3.6396e-08	k_t
= [1, 1, 1, 1]			
Generation 6940: Best = 1.054807e+03	FES = 694100	Diversity = 3.6935e-08	k_t
= [2, 2, 4, 2]			
Generation 6950: Best = 1.054807e+03	FES = 695100	Diversity = 3.6492e-08	k_t
= [1, 1, 1, 1]			
Generation 6960: Best = 1.054807e+03	FES = 696100	Diversity = 3.5870e-08	k_t
= [3, 4, 2, 3]			
Generation 6970: Best = 1.054807e+03	FES = 697100	Diversity = 3.5989e-08	k_t
= [1, 1, 1, 1]			

Generation 6980: Best = 1.054807e+03 = [2, 4, 4, 3]	FES = 698100	Diversity = 3.6028e-08	k_t
Generation 6990: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 3.5956e-08	k_t
Generation 7000: Best = 1.054807e+03 = [4, 3, 2, 4]	FES = 700100	Diversity = 3.6071e-08	k_t
Generation 7010: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 3.6123e-08	k_t
Generation 7020: Best = 1.054807e+03 = [2, 4, 2, 4]	FES = 702100	Diversity = 3.5782e-08	k_t
Generation 7030: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 3.5595e-08	k_t
Generation 7040: Best = 1.054807e+03 = [2, 2, 4, 3]	FES = 704100	Diversity = 3.6286e-08	k_t
Generation 7050: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 3.7527e-08	k_t
Generation 7060: Best = 1.054807e+03 = [2, 3, 2, 2]	FES = 706100	Diversity = 3.8383e-08	k_t
Generation 7070: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 3.7831e-08	k_t
Generation 7080: Best = 1.054807e+03 = [4, 3, 3, 2]	FES = 708100	Diversity = 3.8752e-08	k_t
Generation 7090: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 3.9317e-08	k_t
Generation 7100: Best = 1.054807e+03 = [3, 3, 2, 3]	FES = 710100	Diversity = 3.9486e-08	k_t
Generation 7110: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 3.9479e-08	k_t
Generation 7120: Best = 1.054807e+03 = [4, 3, 3, 4]	FES = 712100	Diversity = 3.9372e-08	k_t
Generation 7130: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 3.9459e-08	k_t
Generation 7140: Best = 1.054807e+03 = [2, 4, 2, 3]	FES = 714100	Diversity = 3.9564e-08	k_t
Generation 7150: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 4.0077e-08	k_t
Generation 7160: Best = 1.054807e+03 = [4, 2, 2, 2]	FES = 716100	Diversity = 4.0223e-08	k_t
Generation 7170: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 3.8936e-08	k_t
Generation 7180: Best = 1.054807e+03 = [3, 4, 4, 2]	FES = 718100	Diversity = 3.9183e-08	k_t
Generation 7190: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 3.8617e-08	k_t
Generation 7200: Best = 1.054807e+03 = [3, 3, 3, 3]	FES = 720100	Diversity = 3.7815e-08	k_t
Generation 7210: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 3.7320e-08	k_t
Generation 7220: Best = 1.054807e+03 = [4, 4, 4, 4]	FES = 722100	Diversity = 3.7165e-08	k_t
Generation 7230: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 3.7086e-08	k_t
Generation 7240: Best = 1.054807e+03 = [4, 4, 3, 4]	FES = 724100	Diversity = 3.6187e-08	k_t

Generation 7250: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 3.6426e-08	k_t
Generation 7260: Best = 1.054807e+03 = [4, 4, 3, 2]	FES = 726100	Diversity = 3.7227e-08	k_t
Generation 7270: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 3.7775e-08	k_t
Generation 7280: Best = 1.054807e+03 = [2, 3, 2, 2]	FES = 728100	Diversity = 3.7703e-08	k_t
Generation 7290: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 3.8400e-08	k_t
Generation 7300: Best = 1.054807e+03 = [2, 4, 2, 3]	FES = 730100	Diversity = 3.8087e-08	k_t
Generation 7310: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 3.7187e-08	k_t
Generation 7320: Best = 1.054807e+03 = [4, 3, 3, 2]	FES = 732100	Diversity = 3.7099e-08	k_t
Generation 7330: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 3.7732e-08	k_t
Generation 7340: Best = 1.054807e+03 = [3, 4, 3, 4]	FES = 734100	Diversity = 3.7622e-08	k_t
Generation 7350: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 3.7452e-08	k_t
Generation 7360: Best = 1.054807e+03 = [3, 2, 3, 4]	FES = 736100	Diversity = 3.8046e-08	k_t
Generation 7370: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 3.7997e-08	k_t
Generation 7380: Best = 1.054807e+03 = [3, 4, 2, 3]	FES = 738100	Diversity = 3.8847e-08	k_t
Generation 7390: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 3.9115e-08	k_t
Generation 7400: Best = 1.054807e+03 = [4, 2, 4, 3]	FES = 740100	Diversity = 3.8870e-08	k_t
Generation 7410: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 3.8805e-08	k_t
Generation 7420: Best = 1.054807e+03 = [4, 2, 2, 4]	FES = 742100	Diversity = 3.8183e-08	k_t
Generation 7430: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 3.8219e-08	k_t
Generation 7440: Best = 1.054807e+03 = [3, 4, 3, 2]	FES = 744100	Diversity = 3.8103e-08	k_t
Generation 7450: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 3.8072e-08	k_t
Generation 7460: Best = 1.054807e+03 = [3, 2, 2, 3]	FES = 746100	Diversity = 3.7995e-08	k_t
Generation 7470: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 3.7369e-08	k_t
Generation 7480: Best = 1.054807e+03 = [3, 4, 3, 3]	FES = 748100	Diversity = 3.8305e-08	k_t
Generation 7490: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 3.9332e-08	k_t
Generation 7500: Best = 1.054807e+03 = [3, 4, 2, 3]	FES = 750100	Diversity = 3.9974e-08	k_t
Generation 7510: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 4.0374e-08	k_t

Generation 7520: Best = 1.054807e+03	FES = 752100	Diversity = 4.0218e-08	k_t
= [4, 2, 2, 4]			
Generation 7530: Best = 1.054807e+03	FES = 753100	Diversity = 3.9934e-08	k_t
= [1, 1, 1, 1]			
Generation 7540: Best = 1.054807e+03	FES = 754100	Diversity = 3.9445e-08	k_t
= [2, 2, 2, 2]			
Generation 7550: Best = 1.054807e+03	FES = 755100	Diversity = 3.9517e-08	k_t
= [1, 1, 1, 1]			
Generation 7560: Best = 1.054807e+03	FES = 756100	Diversity = 3.8570e-08	k_t
= [3, 3, 2, 4]			
Generation 7570: Best = 1.054807e+03	FES = 757100	Diversity = 3.7999e-08	k_t
= [1, 1, 1, 1]			
Generation 7580: Best = 1.054807e+03	FES = 758100	Diversity = 3.8177e-08	k_t
= [4, 2, 4, 4]			
Generation 7590: Best = 1.054807e+03	FES = 759100	Diversity = 3.8775e-08	k_t
= [1, 1, 1, 1]			
Generation 7600: Best = 1.054807e+03	FES = 760100	Diversity = 3.8563e-08	k_t
= [3, 2, 2, 4]			
Generation 7610: Best = 1.054807e+03	FES = 761100	Diversity = 3.9258e-08	k_t
= [1, 1, 1, 1]			
Generation 7620: Best = 1.054807e+03	FES = 762100	Diversity = 3.9368e-08	k_t
= [4, 3, 2, 3]			
Generation 7630: Best = 1.054807e+03	FES = 763100	Diversity = 3.8885e-08	k_t
= [1, 1, 1, 1]			
Generation 7640: Best = 1.054807e+03	FES = 764100	Diversity = 3.9527e-08	k_t
= [4, 2, 3, 2]			
Generation 7650: Best = 1.054807e+03	FES = 765100	Diversity = 3.9733e-08	k_t
= [1, 1, 1, 1]			
Generation 7660: Best = 1.054807e+03	FES = 766100	Diversity = 4.0358e-08	k_t
= [2, 3, 2, 4]			
Generation 7670: Best = 1.054807e+03	FES = 767100	Diversity = 4.1018e-08	k_t
= [1, 1, 1, 1]			
Generation 7680: Best = 1.054807e+03	FES = 768100	Diversity = 4.0869e-08	k_t
= [2, 2, 3, 2]			
Generation 7690: Best = 1.054807e+03	FES = 769100	Diversity = 4.0829e-08	k_t
= [1, 1, 1, 1]			
Generation 7700: Best = 1.054807e+03	FES = 770100	Diversity = 4.0795e-08	k_t
= [4, 2, 4, 2]			
Generation 7710: Best = 1.054807e+03	FES = 771100	Diversity = 3.9841e-08	k_t
= [1, 1, 1, 1]			
Generation 7720: Best = 1.054807e+03	FES = 772100	Diversity = 4.0102e-08	k_t
= [3, 4, 4, 3]			
Generation 7730: Best = 1.054807e+03	FES = 773100	Diversity = 3.8437e-08	k_t
= [1, 1, 1, 1]			
Generation 7740: Best = 1.054807e+03	FES = 774100	Diversity = 3.8446e-08	k_t
= [2, 4, 2, 3]			
Generation 7750: Best = 1.054807e+03	FES = 775100	Diversity = 3.8152e-08	k_t
= [1, 1, 1, 1]			
Generation 7760: Best = 1.054807e+03	FES = 776100	Diversity = 3.8698e-08	k_t
= [3, 2, 4, 4]			
Generation 7770: Best = 1.054807e+03	FES = 777100	Diversity = 3.8235e-08	k_t
= [1, 1, 1, 1]			
Generation 7780: Best = 1.054807e+03	FES = 778100	Diversity = 3.8380e-08	k_t
= [2, 2, 2, 4]			

Generation 7790: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 779100	Diversity = 3.8168e-08	k_t
Generation 7800: Best = 1.054807e+03 = [2, 2, 3, 2]	FES = 780100	Diversity = 3.7814e-08	k_t
Generation 7810: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 781100	Diversity = 3.8029e-08	k_t
Generation 7820: Best = 1.054807e+03 = [2, 3, 4, 2]	FES = 782100	Diversity = 3.6888e-08	k_t
Generation 7830: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 783100	Diversity = 3.6877e-08	k_t
Generation 7840: Best = 1.054807e+03 = [4, 2, 2, 2]	FES = 784100	Diversity = 3.7632e-08	k_t
Generation 7850: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 785100	Diversity = 3.7608e-08	k_t
Generation 7860: Best = 1.054807e+03 = [3, 2, 3, 2]	FES = 786100	Diversity = 3.7740e-08	k_t
Generation 7870: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 787100	Diversity = 3.8772e-08	k_t
Generation 7880: Best = 1.054807e+03 = [3, 4, 4, 3]	FES = 788100	Diversity = 3.8357e-08	k_t
Generation 7890: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 789100	Diversity = 3.8490e-08	k_t
Generation 7900: Best = 1.054807e+03 = [2, 4, 2, 3]	FES = 790100	Diversity = 3.7911e-08	k_t
Generation 7910: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 791100	Diversity = 3.7016e-08	k_t
Generation 7920: Best = 1.054807e+03 = [4, 4, 4, 2]	FES = 792100	Diversity = 3.8282e-08	k_t
Generation 7930: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 793100	Diversity = 3.7939e-08	k_t
Generation 7940: Best = 1.054807e+03 = [4, 2, 3, 3]	FES = 794100	Diversity = 3.8334e-08	k_t
Generation 7950: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 795100	Diversity = 3.7649e-08	k_t
Generation 7960: Best = 1.054807e+03 = [2, 3, 3, 4]	FES = 796100	Diversity = 3.7697e-08	k_t
Generation 7970: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 797100	Diversity = 3.7840e-08	k_t
Generation 7980: Best = 1.054807e+03 = [2, 4, 3, 3]	FES = 798100	Diversity = 3.8027e-08	k_t
Generation 7990: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 799100	Diversity = 3.8019e-08	k_t
Generation 8000: Best = 1.054807e+03 = [4, 4, 2, 2]	FES = 800100	Diversity = 3.7291e-08	k_t
Generation 8010: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 801100	Diversity = 3.8124e-08	k_t
Generation 8020: Best = 1.054807e+03 = [3, 2, 2, 3]	FES = 802100	Diversity = 3.8768e-08	k_t
Generation 8030: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 803100	Diversity = 3.8186e-08	k_t
Generation 8040: Best = 1.054807e+03 = [2, 3, 2, 3]	FES = 804100	Diversity = 3.8676e-08	k_t
Generation 8050: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 805100	Diversity = 3.9206e-08	k_t

Generation 8060: Best = 1.054807e+03 = [3, 4, 4, 3]	FES = 806100	Diversity = 3.9455e-08	k_t
Generation 8070: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 807100	Diversity = 4.0137e-08	k_t
Generation 8080: Best = 1.054807e+03 = [3, 2, 4, 3]	FES = 808100	Diversity = 3.9570e-08	k_t
Generation 8090: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 809100	Diversity = 3.9235e-08	k_t
Generation 8100: Best = 1.054807e+03 = [2, 4, 4, 2]	FES = 810100	Diversity = 3.9515e-08	k_t
Generation 8110: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 811100	Diversity = 3.9269e-08	k_t
Generation 8120: Best = 1.054807e+03 = [2, 4, 3, 4]	FES = 812100	Diversity = 3.8831e-08	k_t
Generation 8130: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 813100	Diversity = 3.8925e-08	k_t
Generation 8140: Best = 1.054807e+03 = [3, 4, 2, 4]	FES = 814100	Diversity = 3.8761e-08	k_t
Generation 8150: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 815100	Diversity = 3.9496e-08	k_t
Generation 8160: Best = 1.054807e+03 = [2, 3, 4, 2]	FES = 816100	Diversity = 3.9300e-08	k_t
Generation 8170: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 817100	Diversity = 3.8268e-08	k_t
Generation 8180: Best = 1.054807e+03 = [4, 4, 3, 4]	FES = 818100	Diversity = 3.8293e-08	k_t
Generation 8190: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 819100	Diversity = 3.8114e-08	k_t
Generation 8200: Best = 1.054807e+03 = [4, 4, 3, 2]	FES = 820100	Diversity = 3.7090e-08	k_t
Generation 8210: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 821100	Diversity = 3.7720e-08	k_t
Generation 8220: Best = 1.054807e+03 = [4, 3, 3, 4]	FES = 822100	Diversity = 3.8073e-08	k_t
Generation 8230: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 823100	Diversity = 3.8491e-08	k_t
Generation 8240: Best = 1.054807e+03 = [3, 3, 4, 3]	FES = 824100	Diversity = 3.8926e-08	k_t
Generation 8250: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 825100	Diversity = 3.8013e-08	k_t
Generation 8260: Best = 1.054807e+03 = [4, 2, 3, 2]	FES = 826100	Diversity = 3.8414e-08	k_t
Generation 8270: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 827100	Diversity = 3.7581e-08	k_t
Generation 8280: Best = 1.054807e+03 = [3, 2, 3, 3]	FES = 828100	Diversity = 3.8152e-08	k_t
Generation 8290: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 829100	Diversity = 3.7568e-08	k_t
Generation 8300: Best = 1.054807e+03 = [3, 3, 2, 3]	FES = 830100	Diversity = 3.7735e-08	k_t
Generation 8310: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 831100	Diversity = 3.8463e-08	k_t
Generation 8320: Best = 1.054807e+03 = [4, 3, 2, 4]	FES = 832100	Diversity = 3.8769e-08	k_t

Generation 8330: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 833100	Diversity = 3.8804e-08	k_t
Generation 8340: Best = 1.054807e+03 = [2, 2, 4, 2]	FES = 834100	Diversity = 3.9165e-08	k_t
Generation 8350: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 835100	Diversity = 3.9582e-08	k_t
Generation 8360: Best = 1.054807e+03 = [2, 3, 2, 2]	FES = 836100	Diversity = 3.9033e-08	k_t
Generation 8370: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 837100	Diversity = 3.8781e-08	k_t
Generation 8380: Best = 1.054807e+03 = [3, 4, 3, 4]	FES = 838100	Diversity = 3.9365e-08	k_t
Generation 8390: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 839100	Diversity = 4.0256e-08	k_t
Generation 8400: Best = 1.054807e+03 = [4, 2, 3, 3]	FES = 840100	Diversity = 3.9670e-08	k_t
Generation 8410: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 841100	Diversity = 4.0282e-08	k_t
Generation 8420: Best = 1.054807e+03 = [4, 3, 3, 2]	FES = 842100	Diversity = 3.9962e-08	k_t
Generation 8430: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 843100	Diversity = 4.0354e-08	k_t
Generation 8440: Best = 1.054807e+03 = [3, 3, 3, 3]	FES = 844100	Diversity = 4.0342e-08	k_t
Generation 8450: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 845100	Diversity = 3.9940e-08	k_t
Generation 8460: Best = 1.054807e+03 = [2, 4, 4, 4]	FES = 846100	Diversity = 3.9095e-08	k_t
Generation 8470: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 847100	Diversity = 3.8585e-08	k_t
Generation 8480: Best = 1.054807e+03 = [3, 4, 3, 3]	FES = 848100	Diversity = 3.9765e-08	k_t
Generation 8490: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 849100	Diversity = 3.9506e-08	k_t
Generation 8500: Best = 1.054807e+03 = [3, 4, 2, 3]	FES = 850100	Diversity = 3.8917e-08	k_t
Generation 8510: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 851100	Diversity = 3.9806e-08	k_t
Generation 8520: Best = 1.054807e+03 = [2, 4, 3, 4]	FES = 852100	Diversity = 3.9654e-08	k_t
Generation 8530: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 853100	Diversity = 3.9790e-08	k_t
Generation 8540: Best = 1.054807e+03 = [3, 4, 2, 2]	FES = 854100	Diversity = 4.0069e-08	k_t
Generation 8550: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 855100	Diversity = 3.9870e-08	k_t
Generation 8560: Best = 1.054807e+03 = [4, 4, 3, 2]	FES = 856100	Diversity = 4.0384e-08	k_t
Generation 8570: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 857100	Diversity = 4.0585e-08	k_t
Generation 8580: Best = 1.054807e+03 = [4, 4, 2, 2]	FES = 858100	Diversity = 3.9982e-08	k_t
Generation 8590: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 859100	Diversity = 4.0303e-08	k_t

Generation 8600: Best = 1.054807e+03 = [2, 4, 3, 2]	FES = 860100	Diversity = 3.9788e-08	k_t
Generation 8610: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 861100	Diversity = 4.0174e-08	k_t
Generation 8620: Best = 1.054807e+03 = [4, 2, 4, 2]	FES = 862100	Diversity = 3.9353e-08	k_t
Generation 8630: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 863100	Diversity = 3.9746e-08	k_t
Generation 8640: Best = 1.054807e+03 = [2, 3, 4, 3]	FES = 864100	Diversity = 4.0592e-08	k_t
Generation 8650: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 865100	Diversity = 4.0990e-08	k_t
Generation 8660: Best = 1.054807e+03 = [2, 2, 2, 3]	FES = 866100	Diversity = 4.0660e-08	k_t
Generation 8670: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 867100	Diversity = 4.0758e-08	k_t
Generation 8680: Best = 1.054807e+03 = [4, 2, 2, 2]	FES = 868100	Diversity = 4.0990e-08	k_t
Generation 8690: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 869100	Diversity = 4.1479e-08	k_t
Generation 8700: Best = 1.054807e+03 = [3, 4, 2, 3]	FES = 870100	Diversity = 4.1393e-08	k_t
Generation 8710: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 871100	Diversity = 4.0525e-08	k_t
Generation 8720: Best = 1.054807e+03 = [3, 2, 2, 3]	FES = 872100	Diversity = 4.0252e-08	k_t
Generation 8730: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 873100	Diversity = 4.0342e-08	k_t
Generation 8740: Best = 1.054807e+03 = [4, 4, 3, 3]	FES = 874100	Diversity = 4.0358e-08	k_t
Generation 8750: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 875100	Diversity = 3.9976e-08	k_t
Generation 8760: Best = 1.054807e+03 = [2, 2, 3, 3]	FES = 876100	Diversity = 4.0120e-08	k_t
Generation 8770: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 877100	Diversity = 3.9303e-08	k_t
Generation 8780: Best = 1.054807e+03 = [2, 3, 2, 2]	FES = 878100	Diversity = 3.8285e-08	k_t
Generation 8790: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 879100	Diversity = 3.8397e-08	k_t
Generation 8800: Best = 1.054807e+03 = [3, 3, 2, 4]	FES = 880100	Diversity = 3.8388e-08	k_t
Generation 8810: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 881100	Diversity = 3.9309e-08	k_t
Generation 8820: Best = 1.054807e+03 = [2, 2, 2, 4]	FES = 882100	Diversity = 3.9126e-08	k_t
Generation 8830: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 883100	Diversity = 3.9006e-08	k_t
Generation 8840: Best = 1.054807e+03 = [4, 3, 2, 2]	FES = 884100	Diversity = 3.9536e-08	k_t
Generation 8850: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 885100	Diversity = 3.8611e-08	k_t
Generation 8860: Best = 1.054807e+03 = [3, 3, 3, 2]	FES = 886100	Diversity = 3.9060e-08	k_t

Generation 8870: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 887100	Diversity = 3.9235e-08	k_t
Generation 8880: Best = 1.054807e+03 = [3, 3, 4, 3]	FES = 888100	Diversity = 3.8785e-08	k_t
Generation 8890: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 889100	Diversity = 3.8747e-08	k_t
Generation 8900: Best = 1.054807e+03 = [3, 2, 3, 3]	FES = 890100	Diversity = 3.8420e-08	k_t
Generation 8910: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 891100	Diversity = 3.8094e-08	k_t
Generation 8920: Best = 1.054807e+03 = [3, 4, 2, 2]	FES = 892100	Diversity = 3.8343e-08	k_t
Generation 8930: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 893100	Diversity = 3.8701e-08	k_t
Generation 8940: Best = 1.054807e+03 = [3, 2, 3, 3]	FES = 894100	Diversity = 3.9291e-08	k_t
Generation 8950: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 895100	Diversity = 3.9509e-08	k_t
Generation 8960: Best = 1.054807e+03 = [2, 4, 3, 2]	FES = 896100	Diversity = 3.9165e-08	k_t
Generation 8970: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 897100	Diversity = 3.9816e-08	k_t
Generation 8980: Best = 1.054807e+03 = [2, 4, 2, 3]	FES = 898100	Diversity = 4.0128e-08	k_t
Generation 8990: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 899100	Diversity = 4.1175e-08	k_t
Generation 9000: Best = 1.054807e+03 = [2, 4, 2, 4]	FES = 900100	Diversity = 4.1745e-08	k_t
Generation 9010: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 901100	Diversity = 4.1246e-08	k_t
Generation 9020: Best = 1.054807e+03 = [2, 3, 4, 4]	FES = 902100	Diversity = 4.1561e-08	k_t
Generation 9030: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 903100	Diversity = 4.1612e-08	k_t
Generation 9040: Best = 1.054807e+03 = [4, 4, 4, 3]	FES = 904100	Diversity = 4.1108e-08	k_t
Generation 9050: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 905100	Diversity = 4.0840e-08	k_t
Generation 9060: Best = 1.054807e+03 = [2, 3, 3, 2]	FES = 906100	Diversity = 4.1307e-08	k_t
Generation 9070: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 907100	Diversity = 4.0572e-08	k_t
Generation 9080: Best = 1.054807e+03 = [3, 4, 2, 3]	FES = 908100	Diversity = 4.0582e-08	k_t
Generation 9090: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 909100	Diversity = 4.0640e-08	k_t
Generation 9100: Best = 1.054807e+03 = [3, 3, 4, 2]	FES = 910100	Diversity = 4.0987e-08	k_t
Generation 9110: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 911100	Diversity = 4.1128e-08	k_t
Generation 9120: Best = 1.054807e+03 = [4, 2, 4, 3]	FES = 912100	Diversity = 4.1405e-08	k_t
Generation 9130: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 913100	Diversity = 4.1043e-08	k_t

Generation 9140: Best = 1.054807e+03 = [2, 3, 4, 4]	FES = 914100	Diversity = 4.0265e-08	k_t
Generation 9150: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 3.9801e-08	k_t
Generation 9160: Best = 1.054807e+03 = [2, 4, 2, 2]	FES = 916100	Diversity = 4.0309e-08	k_t
Generation 9170: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 4.0645e-08	k_t
Generation 9180: Best = 1.054807e+03 = [3, 2, 2, 3]	FES = 918100	Diversity = 3.9495e-08	k_t
Generation 9190: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 3.9779e-08	k_t
Generation 9200: Best = 1.054807e+03 = [4, 3, 4, 2]	FES = 920100	Diversity = 3.9244e-08	k_t
Generation 9210: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 3.9500e-08	k_t
Generation 9220: Best = 1.054807e+03 = [4, 3, 3, 4]	FES = 922100	Diversity = 4.0255e-08	k_t
Generation 9230: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 3.9231e-08	k_t
Generation 9240: Best = 1.054807e+03 = [3, 4, 4, 4]	FES = 924100	Diversity = 3.9324e-08	k_t
Generation 9250: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 3.9444e-08	k_t
Generation 9260: Best = 1.054807e+03 = [3, 2, 3, 3]	FES = 926100	Diversity = 3.9919e-08	k_t
Generation 9270: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 4.0089e-08	k_t
Generation 9280: Best = 1.054807e+03 = [3, 3, 4, 4]	FES = 928100	Diversity = 3.9764e-08	k_t
Generation 9290: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 3.9973e-08	k_t
Generation 9300: Best = 1.054807e+03 = [3, 2, 2, 4]	FES = 930100	Diversity = 4.0007e-08	k_t
Generation 9310: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 4.0888e-08	k_t
Generation 9320: Best = 1.054807e+03 = [3, 3, 2, 2]	FES = 932100	Diversity = 4.0695e-08	k_t
Generation 9330: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 4.0483e-08	k_t
Generation 9340: Best = 1.054807e+03 = [3, 4, 2, 4]	FES = 934100	Diversity = 4.0615e-08	k_t
Generation 9350: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 4.1452e-08	k_t
Generation 9360: Best = 1.054807e+03 = [3, 3, 3, 3]	FES = 936100	Diversity = 4.2035e-08	k_t
Generation 9370: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 4.2005e-08	k_t
Generation 9380: Best = 1.054807e+03 = [4, 4, 2, 2]	FES = 938100	Diversity = 4.2017e-08	k_t
Generation 9390: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 4.0782e-08	k_t
Generation 9400: Best = 1.054807e+03 = [3, 3, 2, 4]	FES = 940100	Diversity = 4.0598e-08	k_t

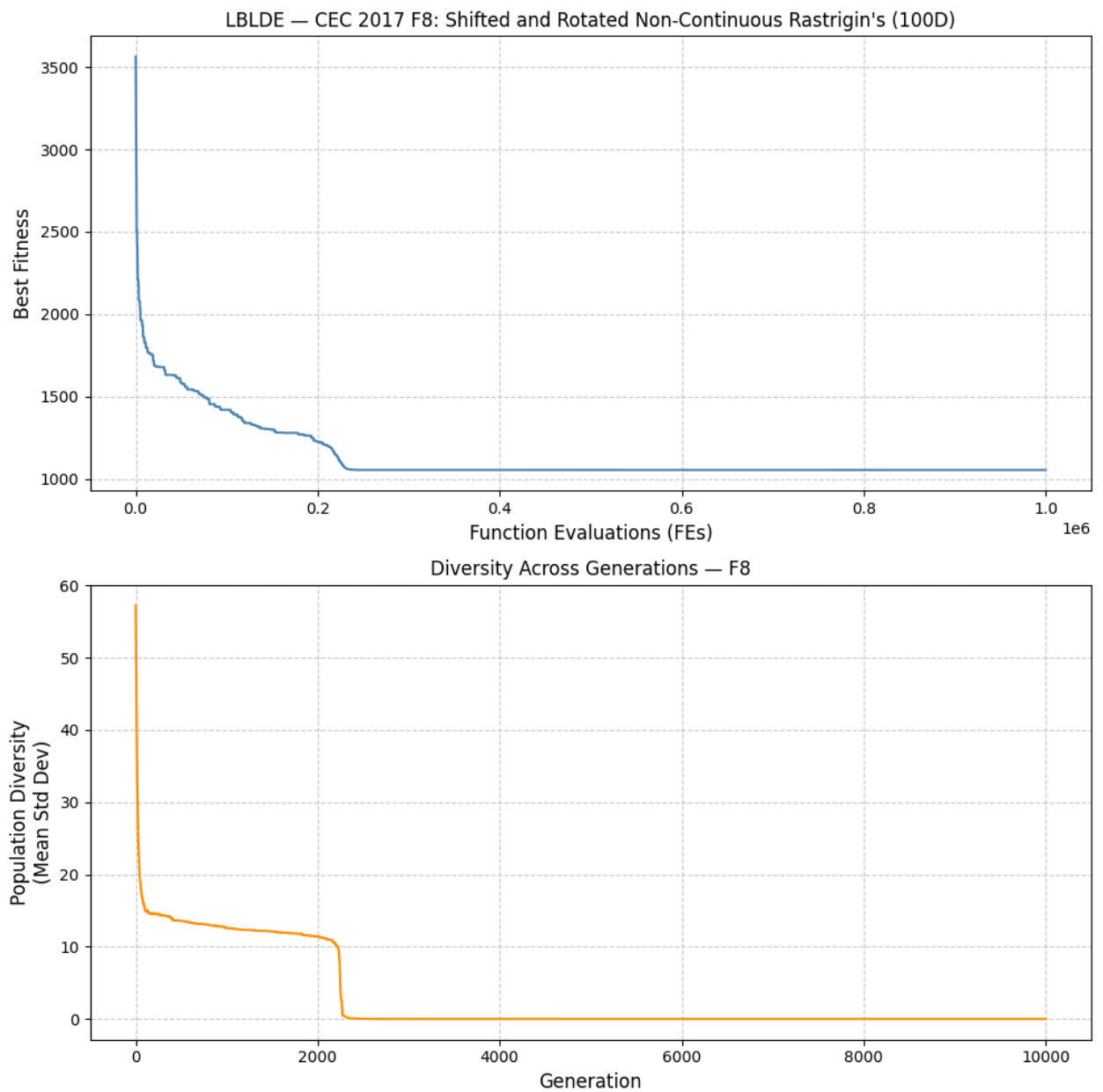
Generation 9410: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 3.9939e-08	k_t
Generation 9420: Best = 1.054807e+03 = [4, 3, 3, 3]	FES = 942100	Diversity = 3.9022e-08	k_t
Generation 9430: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 3.9506e-08	k_t
Generation 9440: Best = 1.054807e+03 = [2, 2, 3, 4]	FES = 944100	Diversity = 3.9206e-08	k_t
Generation 9450: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 3.8875e-08	k_t
Generation 9460: Best = 1.054807e+03 = [2, 2, 3, 2]	FES = 946100	Diversity = 3.8281e-08	k_t
Generation 9470: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 3.8934e-08	k_t
Generation 9480: Best = 1.054807e+03 = [3, 2, 4, 4]	FES = 948100	Diversity = 3.8918e-08	k_t
Generation 9490: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 3.9635e-08	k_t
Generation 9500: Best = 1.054807e+03 = [4, 3, 2, 2]	FES = 950100	Diversity = 3.9451e-08	k_t
Generation 9510: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 4.0550e-08	k_t
Generation 9520: Best = 1.054807e+03 = [4, 4, 2, 3]	FES = 952100	Diversity = 4.1279e-08	k_t
Generation 9530: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 4.0312e-08	k_t
Generation 9540: Best = 1.054807e+03 = [4, 2, 2, 2]	FES = 954100	Diversity = 4.0047e-08	k_t
Generation 9550: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 4.0449e-08	k_t
Generation 9560: Best = 1.054807e+03 = [2, 3, 2, 3]	FES = 956100	Diversity = 3.9858e-08	k_t
Generation 9570: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 3.9250e-08	k_t
Generation 9580: Best = 1.054807e+03 = [3, 3, 2, 3]	FES = 958100	Diversity = 3.9038e-08	k_t
Generation 9590: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 3.8847e-08	k_t
Generation 9600: Best = 1.054807e+03 = [3, 4, 2, 3]	FES = 960100	Diversity = 3.9380e-08	k_t
Generation 9610: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 3.9276e-08	k_t
Generation 9620: Best = 1.054807e+03 = [2, 4, 3, 4]	FES = 962100	Diversity = 3.9952e-08	k_t
Generation 9630: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 3.9442e-08	k_t
Generation 9640: Best = 1.054807e+03 = [4, 4, 2, 2]	FES = 964100	Diversity = 3.8509e-08	k_t
Generation 9650: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 3.8687e-08	k_t
Generation 9660: Best = 1.054807e+03 = [3, 2, 2, 3]	FES = 966100	Diversity = 3.8055e-08	k_t
Generation 9670: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 3.8695e-08	k_t

Generation 9680: Best = 1.054807e+03 = [2, 2, 4, 2]	FES = 968100	Diversity = 3.9226e-08	k_t
Generation 9690: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 969100	Diversity = 3.9216e-08	k_t
Generation 9700: Best = 1.054807e+03 = [2, 2, 3, 4]	FES = 970100	Diversity = 3.9707e-08	k_t
Generation 9710: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 971100	Diversity = 3.9126e-08	k_t
Generation 9720: Best = 1.054807e+03 = [4, 3, 3, 4]	FES = 972100	Diversity = 3.8981e-08	k_t
Generation 9730: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 973100	Diversity = 3.9882e-08	k_t
Generation 9740: Best = 1.054807e+03 = [2, 3, 4, 4]	FES = 974100	Diversity = 4.0348e-08	k_t
Generation 9750: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 975100	Diversity = 4.0668e-08	k_t
Generation 9760: Best = 1.054807e+03 = [4, 4, 3, 4]	FES = 976100	Diversity = 4.1168e-08	k_t
Generation 9770: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 977100	Diversity = 4.0974e-08	k_t
Generation 9780: Best = 1.054807e+03 = [4, 2, 4, 2]	FES = 978100	Diversity = 4.1583e-08	k_t
Generation 9790: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 979100	Diversity = 4.1806e-08	k_t
Generation 9800: Best = 1.054807e+03 = [4, 3, 3, 4]	FES = 980100	Diversity = 4.2120e-08	k_t
Generation 9810: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 981100	Diversity = 4.1944e-08	k_t
Generation 9820: Best = 1.054807e+03 = [4, 3, 2, 3]	FES = 982100	Diversity = 4.1812e-08	k_t
Generation 9830: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 983100	Diversity = 4.2957e-08	k_t
Generation 9840: Best = 1.054807e+03 = [4, 2, 2, 3]	FES = 984100	Diversity = 4.3731e-08	k_t
Generation 9850: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 985100	Diversity = 4.2824e-08	k_t
Generation 9860: Best = 1.054807e+03 = [4, 2, 2, 2]	FES = 986100	Diversity = 4.2150e-08	k_t
Generation 9870: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 987100	Diversity = 4.1589e-08	k_t
Generation 9880: Best = 1.054807e+03 = [2, 4, 2, 2]	FES = 988100	Diversity = 4.1676e-08	k_t
Generation 9890: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 989100	Diversity = 4.1311e-08	k_t
Generation 9900: Best = 1.054807e+03 = [4, 4, 3, 3]	FES = 990100	Diversity = 4.0538e-08	k_t
Generation 9910: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 991100	Diversity = 4.0810e-08	k_t
Generation 9920: Best = 1.054807e+03 = [3, 2, 4, 3]	FES = 992100	Diversity = 4.1573e-08	k_t
Generation 9930: Best = 1.054807e+03 = [1, 1, 1, 1]	FES = 993100	Diversity = 4.1814e-08	k_t
Generation 9940: Best = 1.054807e+03 = [3, 2, 2, 3]	FES = 994100	Diversity = 4.1197e-08	k_t

Generation 9950: Best = 1.054807e+03 FES = 995100 Diversity = 4.1620e-08 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 1.054807e+03 FES = 996100 Diversity = 4.1119e-08 k_t
= [4, 4, 3, 2]
Generation 9970: Best = 1.054807e+03 FES = 997100 Diversity = 4.1832e-08 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 1.054807e+03 FES = 998100 Diversity = 4.1077e-08 k_t
= [2, 4, 3, 4]
Generation 9990: Best = 1.054807e+03 FES = 999100 Diversity = 4.0411e-08 k_t
= [1, 1, 1, 1]

Optimization completed!
Final Best Fitness = 1.054807e+03
Total Function Evaluations = 1000000

=====
=
Results:
Best Fitness : 1.054807e+03
Global Optimum : 8.000000e+02
Error (|best-optimum|) : 2.548073e+02
Best Solution (5 dim) : [2.18456894 -48.73881062 13.94992048 12.46366191
-11.5676619]
D^(0) (init. diversity): 5.7220e+01
=====



Graph saved as convergence_F8.png

```
In [33]: # =====  
# F9 – Shifted and Rotated Levy  
# =====  
run_and_plot('F9', make_wrapper(simple.f9), 900.0, 'Shifted and Rotated Levy')
```

```

=====
=
LBLDE_s on CEC 2017 F9 – Single Run
=====
=
Function      : Shifted and Rotated Levy (100D)
Bounds       : [-100, 100]
Global optimum : 900.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 2.313486e+05 | Diversity = 5.7220e+01
Generation 10: Best = 1.152850e+05 FES = 1100 Diversity = 4.3654e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 6.388299e+04 FES = 2100 Diversity = 3.3946e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 6.298371e+04 FES = 3100 Diversity = 2.6903e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 5.044079e+04 FES = 4100 Diversity = 2.3110e+01 k_t
= [2, 2, 4, 4]
Generation 50: Best = 3.850386e+04 FES = 5100 Diversity = 1.9878e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.850386e+04 FES = 6100 Diversity = 1.7114e+01 k_t
= [4, 2, 4, 2]
Generation 70: Best = 3.142334e+04 FES = 7100 Diversity = 1.6048e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.805690e+04 FES = 8100 Diversity = 1.4832e+01 k_t
= [4, 4, 3, 3]
Generation 90: Best = 2.573185e+04 FES = 9100 Diversity = 1.3746e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.408825e+04 FES = 10100 Diversity = 1.2887e+01 k_t
= [4, 2, 2, 2]
Generation 110: Best = 2.229316e+04 FES = 11100 Diversity = 1.2532e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.189193e+04 FES = 12100 Diversity = 1.1921e+01 k_t
= [2, 2, 2, 2]
Generation 130: Best = 1.934805e+04 FES = 13100 Diversity = 1.1287e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.726000e+04 FES = 14100 Diversity = 1.0470e+01 k_t
= [3, 2, 4, 2]
Generation 150: Best = 1.637862e+04 FES = 15100 Diversity = 1.0274e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.334058e+04 FES = 16100 Diversity = 9.8058e+00 k_t
= [2, 2, 4, 4]
Generation 170: Best = 1.334058e+04 FES = 17100 Diversity = 9.5545e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.334058e+04 FES = 18100 Diversity = 9.1081e+00 k_t
= [3, 2, 3, 3]
Generation 190: Best = 1.304473e+04 FES = 19100 Diversity = 8.7744e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.135685e+04 FES = 20100 Diversity = 8.5109e+00 k_t
= [4, 4, 4, 3]
Generation 210: Best = 1.135685e+04 FES = 21100 Diversity = 8.4961e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.135685e+04 FES = 22100 Diversity = 8.2458e+00 k_t
= [3, 3, 3, 4]

```

Generation 230:	Best = 1.087320e+04	FES = 23100	Diversity = 8.2529e+00	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 9.667466e+03	FES = 24100	Diversity = 8.2617e+00	k_t
= [3, 2, 4, 2]				
Generation 250:	Best = 9.457901e+03	FES = 25100	Diversity = 8.1887e+00	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 9.457901e+03	FES = 26100	Diversity = 8.0901e+00	k_t
= [3, 2, 4, 2]				
Generation 270:	Best = 9.420348e+03	FES = 27100	Diversity = 8.0775e+00	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 7.985416e+03	FES = 28100	Diversity = 8.0854e+00	k_t
= [3, 3, 4, 2]				
Generation 290:	Best = 7.985416e+03	FES = 29100	Diversity = 8.0319e+00	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 7.985416e+03	FES = 30100	Diversity = 8.0773e+00	k_t
= [4, 3, 4, 3]				
Generation 310:	Best = 7.985416e+03	FES = 31100	Diversity = 8.0875e+00	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 7.985416e+03	FES = 32100	Diversity = 8.1651e+00	k_t
= [3, 4, 4, 2]				
Generation 330:	Best = 7.480122e+03	FES = 33100	Diversity = 8.1904e+00	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 7.480122e+03	FES = 34100	Diversity = 8.2392e+00	k_t
= [3, 4, 2, 2]				
Generation 350:	Best = 7.480122e+03	FES = 35100	Diversity = 8.2017e+00	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 7.480122e+03	FES = 36100	Diversity = 8.1455e+00	k_t
= [2, 2, 3, 3]				
Generation 370:	Best = 7.480122e+03	FES = 37100	Diversity = 8.1282e+00	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 7.480122e+03	FES = 38100	Diversity = 8.0175e+00	k_t
= [2, 3, 4, 2]				
Generation 390:	Best = 7.293071e+03	FES = 39100	Diversity = 7.9791e+00	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 6.702352e+03	FES = 40100	Diversity = 7.7951e+00	k_t
= [4, 3, 2, 4]				
Generation 410:	Best = 6.614077e+03	FES = 41100	Diversity = 7.8168e+00	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 6.311824e+03	FES = 42100	Diversity = 7.8056e+00	k_t
= [3, 2, 3, 3]				
Generation 430:	Best = 6.311824e+03	FES = 43100	Diversity = 7.8129e+00	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 6.311824e+03	FES = 44100	Diversity = 7.7036e+00	k_t
= [4, 2, 3, 3]				
Generation 450:	Best = 6.311824e+03	FES = 45100	Diversity = 7.6474e+00	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 5.706098e+03	FES = 46100	Diversity = 7.5930e+00	k_t
= [2, 3, 4, 3]				
Generation 470:	Best = 5.456813e+03	FES = 47100	Diversity = 7.4099e+00	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 5.456813e+03	FES = 48100	Diversity = 7.2838e+00	k_t
= [2, 3, 2, 4]				
Generation 490:	Best = 5.456813e+03	FES = 49100	Diversity = 7.1107e+00	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 5.456813e+03	FES = 50100	Diversity = 7.0729e+00	k_t
= [3, 2, 3, 3]				
Generation 510:	Best = 5.456813e+03	FES = 51100	Diversity = 7.0036e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 4.934541e+03	FES = 52100	Diversity = 6.9755e+00	k_t
= [2, 4, 2, 2]				
Generation 530:	Best = 4.934541e+03	FES = 53100	Diversity = 7.0076e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 4.934541e+03	FES = 54100	Diversity = 7.0279e+00	k_t
= [4, 3, 4, 4]				
Generation 550:	Best = 4.934541e+03	FES = 55100	Diversity = 6.9987e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 4.934541e+03	FES = 56100	Diversity = 6.9789e+00	k_t
= [3, 4, 2, 4]				
Generation 570:	Best = 4.898954e+03	FES = 57100	Diversity = 6.9401e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 4.886729e+03	FES = 58100	Diversity = 6.8646e+00	k_t
= [3, 4, 3, 4]				
Generation 590:	Best = 4.559136e+03	FES = 59100	Diversity = 6.8405e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 4.559136e+03	FES = 60100	Diversity = 6.8543e+00	k_t
= [4, 2, 4, 2]				
Generation 610:	Best = 4.559136e+03	FES = 61100	Diversity = 6.7543e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 4.559136e+03	FES = 62100	Diversity = 6.7064e+00	k_t
= [4, 3, 4, 2]				
Generation 630:	Best = 4.474878e+03	FES = 63100	Diversity = 6.5745e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 4.394560e+03	FES = 64100	Diversity = 6.5008e+00	k_t
= [4, 3, 2, 3]				
Generation 650:	Best = 4.392654e+03	FES = 65100	Diversity = 6.4658e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 4.207268e+03	FES = 66100	Diversity = 6.3996e+00	k_t
= [4, 4, 4, 3]				
Generation 670:	Best = 4.065088e+03	FES = 67100	Diversity = 6.3318e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 4.065088e+03	FES = 68100	Diversity = 6.3376e+00	k_t
= [2, 4, 2, 2]				
Generation 690:	Best = 4.045109e+03	FES = 69100	Diversity = 6.2780e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 4.045109e+03	FES = 70100	Diversity = 6.3456e+00	k_t
= [4, 2, 4, 4]				
Generation 710:	Best = 4.045109e+03	FES = 71100	Diversity = 6.2240e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 3.708718e+03	FES = 72100	Diversity = 6.1502e+00	k_t
= [4, 4, 4, 3]				
Generation 730:	Best = 3.700554e+03	FES = 73100	Diversity = 6.0658e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 3.633942e+03	FES = 74100	Diversity = 6.0401e+00	k_t
= [4, 3, 4, 2]				
Generation 750:	Best = 3.633942e+03	FES = 75100	Diversity = 5.9329e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 3.633942e+03	FES = 76100	Diversity = 5.8871e+00	k_t
= [4, 4, 2, 4]				

Generation 770:	Best = 3.603623e+03	FES = 77100	Diversity = 5.7452e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 3.524961e+03	FES = 78100	Diversity = 5.7355e+00	k_t
= [3, 4, 3, 4]				
Generation 790:	Best = 3.515608e+03	FES = 79100	Diversity = 5.7206e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 3.387464e+03	FES = 80100	Diversity = 5.6307e+00	k_t
= [3, 4, 2, 4]				
Generation 810:	Best = 3.346091e+03	FES = 81100	Diversity = 5.5601e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 3.196671e+03	FES = 82100	Diversity = 5.5100e+00	k_t
= [4, 2, 3, 2]				
Generation 830:	Best = 3.104474e+03	FES = 83100	Diversity = 5.4536e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 3.104474e+03	FES = 84100	Diversity = 5.4120e+00	k_t
= [4, 4, 4, 4]				
Generation 850:	Best = 3.104474e+03	FES = 85100	Diversity = 5.3793e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 3.104474e+03	FES = 86100	Diversity = 5.3397e+00	k_t
= [3, 4, 3, 4]				
Generation 870:	Best = 3.104474e+03	FES = 87100	Diversity = 5.1941e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 3.096971e+03	FES = 88100	Diversity = 5.1204e+00	k_t
= [2, 3, 2, 4]				
Generation 890:	Best = 3.047241e+03	FES = 89100	Diversity = 5.1401e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 3.037996e+03	FES = 90100	Diversity = 5.1055e+00	k_t
= [2, 4, 4, 3]				
Generation 910:	Best = 3.037996e+03	FES = 91100	Diversity = 5.0703e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 3.037996e+03	FES = 92100	Diversity = 4.9643e+00	k_t
= [4, 2, 4, 2]				
Generation 930:	Best = 3.037996e+03	FES = 93100	Diversity = 4.8809e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 3.037996e+03	FES = 94100	Diversity = 4.8246e+00	k_t
= [4, 4, 4, 3]				
Generation 950:	Best = 3.004267e+03	FES = 95100	Diversity = 4.7973e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 3.004267e+03	FES = 96100	Diversity = 4.7868e+00	k_t
= [4, 2, 3, 2]				
Generation 970:	Best = 2.998353e+03	FES = 97100	Diversity = 4.7803e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.982089e+03	FES = 98100	Diversity = 4.7235e+00	k_t
= [2, 4, 4, 2]				
Generation 990:	Best = 2.967016e+03	FES = 99100	Diversity = 4.6769e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 2.862208e+03	FES = 100100	Diversity = 4.6584e+00	k_t
= [3, 4, 2, 4]				
Generation 1010:	Best = 2.707535e+03	FES = 101100	Diversity = 4.6540e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 2.707535e+03	FES = 102100	Diversity = 4.6359e+00	k_t
= [4, 3, 4, 4]				
Generation 1030:	Best = 2.650244e+03	FES = 103100	Diversity = 4.5679e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 2.650244e+03 = [2, 2, 4, 4]	FES = 104100	Diversity = 4.5444e+00	k_t
Generation 1050: Best = 2.561727e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 4.5411e+00	k_t
Generation 1060: Best = 2.516405e+03 = [4, 2, 3, 3]	FES = 106100	Diversity = 4.5225e+00	k_t
Generation 1070: Best = 2.509984e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 4.4881e+00	k_t
Generation 1080: Best = 2.509984e+03 = [4, 4, 3, 3]	FES = 108100	Diversity = 4.4543e+00	k_t
Generation 1090: Best = 2.472613e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 4.4225e+00	k_t
Generation 1100: Best = 2.472613e+03 = [3, 4, 3, 4]	FES = 110100	Diversity = 4.4318e+00	k_t
Generation 1110: Best = 2.451227e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 4.4165e+00	k_t
Generation 1120: Best = 2.451227e+03 = [2, 2, 2, 3]	FES = 112100	Diversity = 4.4103e+00	k_t
Generation 1130: Best = 2.373199e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 4.3602e+00	k_t
Generation 1140: Best = 2.286222e+03 = [2, 3, 4, 3]	FES = 114100	Diversity = 4.3106e+00	k_t
Generation 1150: Best = 2.286222e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 4.3321e+00	k_t
Generation 1160: Best = 2.286222e+03 = [4, 3, 2, 3]	FES = 116100	Diversity = 4.2455e+00	k_t
Generation 1170: Best = 2.286222e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 4.1690e+00	k_t
Generation 1180: Best = 2.196344e+03 = [2, 4, 4, 3]	FES = 118100	Diversity = 4.1576e+00	k_t
Generation 1190: Best = 2.196344e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 4.1645e+00	k_t
Generation 1200: Best = 2.195212e+03 = [4, 2, 2, 2]	FES = 120100	Diversity = 4.0911e+00	k_t
Generation 1210: Best = 2.195212e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 4.0479e+00	k_t
Generation 1220: Best = 2.163757e+03 = [3, 3, 2, 4]	FES = 122100	Diversity = 4.0274e+00	k_t
Generation 1230: Best = 2.163757e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 4.0137e+00	k_t
Generation 1240: Best = 2.163757e+03 = [4, 2, 2, 4]	FES = 124100	Diversity = 4.0209e+00	k_t
Generation 1250: Best = 2.160169e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 3.9754e+00	k_t
Generation 1260: Best = 2.160169e+03 = [2, 4, 3, 2]	FES = 126100	Diversity = 3.9418e+00	k_t
Generation 1270: Best = 2.133244e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 3.8971e+00	k_t
Generation 1280: Best = 2.133244e+03 = [2, 2, 3, 3]	FES = 128100	Diversity = 3.8904e+00	k_t
Generation 1290: Best = 2.101028e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 3.8703e+00	k_t
Generation 1300: Best = 2.096679e+03 = [4, 3, 3, 2]	FES = 130100	Diversity = 3.8720e+00	k_t

Generation 1310: Best = 2.096679e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 3.8649e+00	k_t
Generation 1320: Best = 2.091713e+03 = [4, 3, 4, 3]	FES = 132100	Diversity = 3.8369e+00	k_t
Generation 1330: Best = 2.091713e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 3.8424e+00	k_t
Generation 1340: Best = 2.091713e+03 = [2, 4, 2, 3]	FES = 134100	Diversity = 3.8432e+00	k_t
Generation 1350: Best = 2.091713e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 3.8404e+00	k_t
Generation 1360: Best = 2.091713e+03 = [4, 2, 3, 3]	FES = 136100	Diversity = 3.8142e+00	k_t
Generation 1370: Best = 2.091713e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 3.7992e+00	k_t
Generation 1380: Best = 2.091713e+03 = [3, 3, 2, 4]	FES = 138100	Diversity = 3.7829e+00	k_t
Generation 1390: Best = 2.091713e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 3.7602e+00	k_t
Generation 1400: Best = 2.070527e+03 = [2, 2, 3, 2]	FES = 140100	Diversity = 3.7693e+00	k_t
Generation 1410: Best = 2.070527e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 3.7376e+00	k_t
Generation 1420: Best = 2.070527e+03 = [2, 4, 2, 2]	FES = 142100	Diversity = 3.7152e+00	k_t
Generation 1430: Best = 2.055571e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 3.6914e+00	k_t
Generation 1440: Best = 2.055571e+03 = [3, 4, 4, 2]	FES = 144100	Diversity = 3.6550e+00	k_t
Generation 1450: Best = 2.055571e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 3.6561e+00	k_t
Generation 1460: Best = 2.025436e+03 = [4, 3, 3, 2]	FES = 146100	Diversity = 3.6795e+00	k_t
Generation 1470: Best = 2.004208e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 3.6320e+00	k_t
Generation 1480: Best = 2.004208e+03 = [4, 3, 3, 3]	FES = 148100	Diversity = 3.5968e+00	k_t
Generation 1490: Best = 2.004208e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 3.5812e+00	k_t
Generation 1500: Best = 2.004208e+03 = [4, 2, 3, 3]	FES = 150100	Diversity = 3.5049e+00	k_t
Generation 1510: Best = 2.004208e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 3.4805e+00	k_t
Generation 1520: Best = 1.999179e+03 = [4, 4, 4, 2]	FES = 152100	Diversity = 3.4070e+00	k_t
Generation 1530: Best = 1.999179e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 3.4034e+00	k_t
Generation 1540: Best = 1.999179e+03 = [3, 2, 2, 4]	FES = 154100	Diversity = 3.4128e+00	k_t
Generation 1550: Best = 1.999179e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 3.4111e+00	k_t
Generation 1560: Best = 1.994265e+03 = [2, 2, 2, 4]	FES = 156100	Diversity = 3.3659e+00	k_t
Generation 1570: Best = 1.900173e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 3.3084e+00	k_t

Generation 1580: Best = 1.900173e+03 = [3, 4, 2, 3]	FES = 158100	Diversity = 3.2529e+00	k_t
Generation 1590: Best = 1.897179e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 3.1975e+00	k_t
Generation 1600: Best = 1.887932e+03 = [2, 4, 2, 3]	FES = 160100	Diversity = 3.1951e+00	k_t
Generation 1610: Best = 1.876170e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 3.1664e+00	k_t
Generation 1620: Best = 1.876170e+03 = [4, 2, 4, 3]	FES = 162100	Diversity = 3.1466e+00	k_t
Generation 1630: Best = 1.863407e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 3.1270e+00	k_t
Generation 1640: Best = 1.854286e+03 = [2, 4, 2, 4]	FES = 164100	Diversity = 3.0773e+00	k_t
Generation 1650: Best = 1.854172e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 3.0436e+00	k_t
Generation 1660: Best = 1.827878e+03 = [2, 2, 3, 2]	FES = 166100	Diversity = 3.0195e+00	k_t
Generation 1670: Best = 1.815722e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 2.8614e+00	k_t
Generation 1680: Best = 1.789611e+03 = [2, 3, 4, 2]	FES = 168100	Diversity = 2.8206e+00	k_t
Generation 1690: Best = 1.787703e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 2.8164e+00	k_t
Generation 1700: Best = 1.740948e+03 = [3, 3, 2, 2]	FES = 170100	Diversity = 2.8053e+00	k_t
Generation 1710: Best = 1.740948e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 2.7732e+00	k_t
Generation 1720: Best = 1.730282e+03 = [3, 3, 2, 3]	FES = 172100	Diversity = 2.7295e+00	k_t
Generation 1730: Best = 1.730025e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 2.6200e+00	k_t
Generation 1740: Best = 1.728818e+03 = [3, 4, 4, 2]	FES = 174100	Diversity = 2.5958e+00	k_t
Generation 1750: Best = 1.728818e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 2.5223e+00	k_t
Generation 1760: Best = 1.728818e+03 = [4, 4, 4, 2]	FES = 176100	Diversity = 2.4403e+00	k_t
Generation 1770: Best = 1.708945e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 2.2295e+00	k_t
Generation 1780: Best = 1.707728e+03 = [2, 3, 3, 3]	FES = 178100	Diversity = 2.0583e+00	k_t
Generation 1790: Best = 1.707728e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.8005e+00	k_t
Generation 1800: Best = 1.707728e+03 = [3, 4, 2, 3]	FES = 180100	Diversity = 1.5005e+00	k_t
Generation 1810: Best = 1.687914e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 1.3604e+00	k_t
Generation 1820: Best = 1.687914e+03 = [3, 3, 2, 4]	FES = 182100	Diversity = 1.2945e+00	k_t
Generation 1830: Best = 1.687914e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.1349e+00	k_t
Generation 1840: Best = 1.684590e+03 = [2, 4, 3, 3]	FES = 184100	Diversity = 1.0444e+00	k_t

Generation 1850: Best = 1.672073e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 9.3853e-01	k_t
Generation 1860: Best = 1.666344e+03 = [2, 4, 2, 3]	FES = 186100	Diversity = 9.0541e-01	k_t
Generation 1870: Best = 1.663743e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 8.6657e-01	k_t
Generation 1880: Best = 1.661736e+03 = [2, 4, 2, 2]	FES = 188100	Diversity = 8.5135e-01	k_t
Generation 1890: Best = 1.643093e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 8.4482e-01	k_t
Generation 1900: Best = 1.636743e+03 = [2, 3, 3, 2]	FES = 190100	Diversity = 8.2867e-01	k_t
Generation 1910: Best = 1.632585e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 8.0898e-01	k_t
Generation 1920: Best = 1.632585e+03 = [2, 4, 3, 2]	FES = 192100	Diversity = 7.9243e-01	k_t
Generation 1930: Best = 1.619788e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 7.7574e-01	k_t
Generation 1940: Best = 1.606321e+03 = [2, 2, 4, 4]	FES = 194100	Diversity = 7.3213e-01	k_t
Generation 1950: Best = 1.593038e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 6.8382e-01	k_t
Generation 1960: Best = 1.582555e+03 = [2, 3, 4, 2]	FES = 196100	Diversity = 6.7339e-01	k_t
Generation 1970: Best = 1.582555e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 6.4317e-01	k_t
Generation 1980: Best = 1.567317e+03 = [3, 2, 3, 4]	FES = 198100	Diversity = 6.3823e-01	k_t
Generation 1990: Best = 1.561527e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 5.9794e-01	k_t
Generation 2000: Best = 1.560039e+03 = [2, 4, 2, 3]	FES = 200100	Diversity = 5.8032e-01	k_t
Generation 2010: Best = 1.528328e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 5.5968e-01	k_t
Generation 2020: Best = 1.525137e+03 = [3, 3, 2, 2]	FES = 202100	Diversity = 5.4353e-01	k_t
Generation 2030: Best = 1.520586e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 5.0870e-01	k_t
Generation 2040: Best = 1.509923e+03 = [3, 2, 4, 3]	FES = 204100	Diversity = 4.6390e-01	k_t
Generation 2050: Best = 1.507477e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 4.4030e-01	k_t
Generation 2060: Best = 1.504992e+03 = [2, 2, 4, 4]	FES = 206100	Diversity = 4.1532e-01	k_t
Generation 2070: Best = 1.502407e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 3.9456e-01	k_t
Generation 2080: Best = 1.484651e+03 = [2, 4, 4, 4]	FES = 208100	Diversity = 3.8012e-01	k_t
Generation 2090: Best = 1.484651e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 3.6533e-01	k_t
Generation 2100: Best = 1.480594e+03 = [3, 2, 4, 2]	FES = 210100	Diversity = 3.5821e-01	k_t
Generation 2110: Best = 1.478108e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 3.4407e-01	k_t

Generation 2120: Best = 1.472544e+03 = [4, 4, 3, 3]	FES = 212100	Diversity = 3.3499e-01	k_t
Generation 2130: Best = 1.464531e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 3.0610e-01	k_t
Generation 2140: Best = 1.458455e+03 = [2, 2, 3, 4]	FES = 214100	Diversity = 2.8246e-01	k_t
Generation 2150: Best = 1.458238e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 2.4555e-01	k_t
Generation 2160: Best = 1.453374e+03 = [4, 2, 2, 4]	FES = 216100	Diversity = 2.2047e-01	k_t
Generation 2170: Best = 1.449670e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 1.7093e-01	k_t
Generation 2180: Best = 1.445348e+03 = [2, 4, 2, 4]	FES = 218100	Diversity = 1.4289e-01	k_t
Generation 2190: Best = 1.440812e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 1.1916e-01	k_t
Generation 2200: Best = 1.438453e+03 = [3, 4, 3, 4]	FES = 220100	Diversity = 9.5249e-02	k_t
Generation 2210: Best = 1.436810e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 7.9733e-02	k_t
Generation 2220: Best = 1.435163e+03 = [4, 2, 4, 4]	FES = 222100	Diversity = 6.4669e-02	k_t
Generation 2230: Best = 1.433187e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 5.6845e-02	k_t
Generation 2240: Best = 1.431231e+03 = [4, 4, 3, 2]	FES = 224100	Diversity = 4.9606e-02	k_t
Generation 2250: Best = 1.429713e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 4.2679e-02	k_t
Generation 2260: Best = 1.428652e+03 = [2, 2, 3, 2]	FES = 226100	Diversity = 4.0966e-02	k_t
Generation 2270: Best = 1.427110e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 4.2601e-02	k_t
Generation 2280: Best = 1.426048e+03 = [2, 4, 4, 4]	FES = 228100	Diversity = 3.9063e-02	k_t
Generation 2290: Best = 1.425052e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 3.5089e-02	k_t
Generation 2300: Best = 1.424173e+03 = [3, 4, 3, 2]	FES = 230100	Diversity = 3.4968e-02	k_t
Generation 2310: Best = 1.423176e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 3.2491e-02	k_t
Generation 2320: Best = 1.422253e+03 = [2, 3, 4, 4]	FES = 232100	Diversity = 3.0659e-02	k_t
Generation 2330: Best = 1.421723e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 2.9667e-02	k_t
Generation 2340: Best = 1.420704e+03 = [2, 3, 3, 2]	FES = 234100	Diversity = 3.1177e-02	k_t
Generation 2350: Best = 1.419805e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 3.1795e-02	k_t
Generation 2360: Best = 1.418931e+03 = [3, 4, 4, 2]	FES = 236100	Diversity = 3.2051e-02	k_t
Generation 2370: Best = 1.418141e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 3.3342e-02	k_t
Generation 2380: Best = 1.417195e+03 = [4, 2, 3, 4]	FES = 238100	Diversity = 3.2397e-02	k_t

Generation 2390: Best = 1.416556e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 3.1875e-02	k_t
Generation 2400: Best = 1.415730e+03 = [4, 3, 2, 2]	FES = 240100	Diversity = 3.1097e-02	k_t
Generation 2410: Best = 1.415185e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 2.9346e-02	k_t
Generation 2420: Best = 1.414624e+03 = [2, 4, 2, 3]	FES = 242100	Diversity = 2.6903e-02	k_t
Generation 2430: Best = 1.414184e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 2.4075e-02	k_t
Generation 2440: Best = 1.413572e+03 = [4, 4, 3, 4]	FES = 244100	Diversity = 2.2416e-02	k_t
Generation 2450: Best = 1.413230e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 2.1717e-02	k_t
Generation 2460: Best = 1.412787e+03 = [3, 4, 2, 4]	FES = 246100	Diversity = 2.0778e-02	k_t
Generation 2470: Best = 1.412337e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 2.1362e-02	k_t
Generation 2480: Best = 1.411601e+03 = [2, 4, 2, 3]	FES = 248100	Diversity = 2.3751e-02	k_t
Generation 2490: Best = 1.410823e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 2.6186e-02	k_t
Generation 2500: Best = 1.410508e+03 = [2, 4, 3, 4]	FES = 250100	Diversity = 2.8668e-02	k_t
Generation 2510: Best = 1.409813e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 2.7299e-02	k_t
Generation 2520: Best = 1.408927e+03 = [4, 2, 3, 2]	FES = 252100	Diversity = 2.6846e-02	k_t
Generation 2530: Best = 1.408322e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 2.7944e-02	k_t
Generation 2540: Best = 1.407520e+03 = [2, 4, 4, 2]	FES = 254100	Diversity = 2.7990e-02	k_t
Generation 2550: Best = 1.406578e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 2.9095e-02	k_t
Generation 2560: Best = 1.405956e+03 = [3, 4, 4, 3]	FES = 256100	Diversity = 2.9159e-02	k_t
Generation 2570: Best = 1.405075e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 3.2365e-02	k_t
Generation 2580: Best = 1.403866e+03 = [2, 2, 2, 4]	FES = 258100	Diversity = 3.1318e-02	k_t
Generation 2590: Best = 1.402672e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 3.1890e-02	k_t
Generation 2600: Best = 1.401712e+03 = [4, 4, 4, 2]	FES = 260100	Diversity = 3.1546e-02	k_t
Generation 2610: Best = 1.400714e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 3.1840e-02	k_t
Generation 2620: Best = 1.400024e+03 = [4, 2, 2, 3]	FES = 262100	Diversity = 2.8212e-02	k_t
Generation 2630: Best = 1.399473e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 2.7316e-02	k_t
Generation 2640: Best = 1.398925e+03 = [2, 3, 3, 4]	FES = 264100	Diversity = 2.3502e-02	k_t
Generation 2650: Best = 1.398528e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 2.3429e-02	k_t

Generation 2660: Best = 1.397904e+03 = [4, 4, 4, 3]	FES = 266100	Diversity = 2.2268e-02	k_t
Generation 2670: Best = 1.397330e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 2.2307e-02	k_t
Generation 2680: Best = 1.396790e+03 = [4, 2, 2, 4]	FES = 268100	Diversity = 2.3016e-02	k_t
Generation 2690: Best = 1.396142e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 2.3798e-02	k_t
Generation 2700: Best = 1.395669e+03 = [3, 4, 2, 4]	FES = 270100	Diversity = 2.4327e-02	k_t
Generation 2710: Best = 1.394860e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 2.4303e-02	k_t
Generation 2720: Best = 1.394052e+03 = [2, 4, 2, 3]	FES = 272100	Diversity = 2.6194e-02	k_t
Generation 2730: Best = 1.393186e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 2.9026e-02	k_t
Generation 2740: Best = 1.392660e+03 = [4, 2, 2, 3]	FES = 274100	Diversity = 2.9032e-02	k_t
Generation 2750: Best = 1.392103e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 2.8861e-02	k_t
Generation 2760: Best = 1.391173e+03 = [4, 2, 4, 4]	FES = 276100	Diversity = 2.7320e-02	k_t
Generation 2770: Best = 1.390653e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 2.6667e-02	k_t
Generation 2780: Best = 1.389922e+03 = [4, 2, 4, 3]	FES = 278100	Diversity = 2.7177e-02	k_t
Generation 2790: Best = 1.388751e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 2.6913e-02	k_t
Generation 2800: Best = 1.388292e+03 = [3, 4, 4, 2]	FES = 280100	Diversity = 2.7929e-02	k_t
Generation 2810: Best = 1.387891e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 2.7530e-02	k_t
Generation 2820: Best = 1.387299e+03 = [3, 2, 4, 4]	FES = 282100	Diversity = 2.5576e-02	k_t
Generation 2830: Best = 1.386827e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 2.4365e-02	k_t
Generation 2840: Best = 1.386492e+03 = [3, 4, 2, 2]	FES = 284100	Diversity = 2.2574e-02	k_t
Generation 2850: Best = 1.386149e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 2.1662e-02	k_t
Generation 2860: Best = 1.385919e+03 = [2, 3, 4, 3]	FES = 286100	Diversity = 1.9443e-02	k_t
Generation 2870: Best = 1.385589e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 1.8172e-02	k_t
Generation 2880: Best = 1.385400e+03 = [2, 4, 3, 3]	FES = 288100	Diversity = 1.7132e-02	k_t
Generation 2890: Best = 1.385239e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.5090e-02	k_t
Generation 2900: Best = 1.385062e+03 = [3, 3, 3, 4]	FES = 290100	Diversity = 1.3418e-02	k_t
Generation 2910: Best = 1.384961e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.2162e-02	k_t
Generation 2920: Best = 1.384828e+03 = [4, 2, 2, 4]	FES = 292100	Diversity = 1.0473e-02	k_t

Generation 2930: Best = 1.384753e+03 = [1, 1, 1, 1]	FES = 293100	Diversity = 9.3967e-03	k_t
Generation 2940: Best = 1.384693e+03 = [3, 2, 4, 3]	FES = 294100	Diversity = 8.7991e-03	k_t
Generation 2950: Best = 1.384586e+03 = [1, 1, 1, 1]	FES = 295100	Diversity = 8.7854e-03	k_t
Generation 2960: Best = 1.384538e+03 = [4, 4, 2, 3]	FES = 296100	Diversity = 8.5905e-03	k_t
Generation 2970: Best = 1.384470e+03 = [1, 1, 1, 1]	FES = 297100	Diversity = 8.2038e-03	k_t
Generation 2980: Best = 1.384421e+03 = [4, 2, 2, 4]	FES = 298100	Diversity = 7.6536e-03	k_t
Generation 2990: Best = 1.384364e+03 = [1, 1, 1, 1]	FES = 299100	Diversity = 7.0393e-03	k_t
Generation 3000: Best = 1.384311e+03 = [2, 2, 4, 4]	FES = 300100	Diversity = 7.2447e-03	k_t
Generation 3010: Best = 1.384254e+03 = [1, 1, 1, 1]	FES = 301100	Diversity = 7.2607e-03	k_t
Generation 3020: Best = 1.384185e+03 = [4, 4, 3, 4]	FES = 302100	Diversity = 7.6731e-03	k_t
Generation 3030: Best = 1.384109e+03 = [1, 1, 1, 1]	FES = 303100	Diversity = 7.4938e-03	k_t
Generation 3040: Best = 1.384035e+03 = [2, 3, 2, 3]	FES = 304100	Diversity = 7.7813e-03	k_t
Generation 3050: Best = 1.383977e+03 = [1, 1, 1, 1]	FES = 305100	Diversity = 7.9705e-03	k_t
Generation 3060: Best = 1.383939e+03 = [3, 4, 4, 4]	FES = 306100	Diversity = 7.9478e-03	k_t
Generation 3070: Best = 1.383867e+03 = [1, 1, 1, 1]	FES = 307100	Diversity = 7.2404e-03	k_t
Generation 3080: Best = 1.383842e+03 = [3, 3, 4, 4]	FES = 308100	Diversity = 7.0911e-03	k_t
Generation 3090: Best = 1.383787e+03 = [1, 1, 1, 1]	FES = 309100	Diversity = 7.0811e-03	k_t
Generation 3100: Best = 1.383756e+03 = [2, 4, 2, 2]	FES = 310100	Diversity = 6.6316e-03	k_t
Generation 3110: Best = 1.383719e+03 = [1, 1, 1, 1]	FES = 311100	Diversity = 6.2945e-03	k_t
Generation 3120: Best = 1.383672e+03 = [4, 4, 4, 3]	FES = 312100	Diversity = 6.1961e-03	k_t
Generation 3130: Best = 1.383633e+03 = [1, 1, 1, 1]	FES = 313100	Diversity = 5.9807e-03	k_t
Generation 3140: Best = 1.383601e+03 = [3, 2, 3, 4]	FES = 314100	Diversity = 5.7453e-03	k_t
Generation 3150: Best = 1.383545e+03 = [1, 1, 1, 1]	FES = 315100	Diversity = 5.5313e-03	k_t
Generation 3160: Best = 1.383503e+03 = [4, 4, 4, 2]	FES = 316100	Diversity = 6.1063e-03	k_t
Generation 3170: Best = 1.383475e+03 = [1, 1, 1, 1]	FES = 317100	Diversity = 6.1676e-03	k_t
Generation 3180: Best = 1.383446e+03 = [3, 4, 3, 4]	FES = 318100	Diversity = 5.7644e-03	k_t
Generation 3190: Best = 1.383408e+03 = [1, 1, 1, 1]	FES = 319100	Diversity = 5.7201e-03	k_t

Generation 3200: Best = 1.383374e+03 = [2, 2, 2, 2]	FES = 320100	Diversity = 5.5051e-03	k_t
Generation 3210: Best = 1.383345e+03 = [1, 1, 1, 1]	FES = 321100	Diversity = 5.3236e-03	k_t
Generation 3220: Best = 1.383318e+03 = [4, 3, 3, 4]	FES = 322100	Diversity = 5.1122e-03	k_t
Generation 3230: Best = 1.383295e+03 = [1, 1, 1, 1]	FES = 323100	Diversity = 4.7987e-03	k_t
Generation 3240: Best = 1.383265e+03 = [3, 3, 4, 4]	FES = 324100	Diversity = 4.6585e-03	k_t
Generation 3250: Best = 1.383245e+03 = [1, 1, 1, 1]	FES = 325100	Diversity = 4.9934e-03	k_t
Generation 3260: Best = 1.383222e+03 = [4, 3, 4, 2]	FES = 326100	Diversity = 4.7329e-03	k_t
Generation 3270: Best = 1.383207e+03 = [1, 1, 1, 1]	FES = 327100	Diversity = 4.5385e-03	k_t
Generation 3280: Best = 1.383181e+03 = [3, 4, 4, 3]	FES = 328100	Diversity = 4.4948e-03	k_t
Generation 3290: Best = 1.383164e+03 = [1, 1, 1, 1]	FES = 329100	Diversity = 4.6070e-03	k_t
Generation 3300: Best = 1.383142e+03 = [2, 3, 4, 4]	FES = 330100	Diversity = 4.7685e-03	k_t
Generation 3310: Best = 1.383121e+03 = [1, 1, 1, 1]	FES = 331100	Diversity = 4.9241e-03	k_t
Generation 3320: Best = 1.383097e+03 = [2, 2, 3, 3]	FES = 332100	Diversity = 5.1180e-03	k_t
Generation 3330: Best = 1.383081e+03 = [1, 1, 1, 1]	FES = 333100	Diversity = 5.0854e-03	k_t
Generation 3340: Best = 1.383062e+03 = [4, 4, 4, 2]	FES = 334100	Diversity = 4.7894e-03	k_t
Generation 3350: Best = 1.383039e+03 = [1, 1, 1, 1]	FES = 335100	Diversity = 4.7097e-03	k_t
Generation 3360: Best = 1.383021e+03 = [4, 2, 3, 3]	FES = 336100	Diversity = 4.4530e-03	k_t
Generation 3370: Best = 1.383005e+03 = [1, 1, 1, 1]	FES = 337100	Diversity = 4.1140e-03	k_t
Generation 3380: Best = 1.382981e+03 = [2, 4, 2, 2]	FES = 338100	Diversity = 4.1205e-03	k_t
Generation 3390: Best = 1.382971e+03 = [1, 1, 1, 1]	FES = 339100	Diversity = 4.3116e-03	k_t
Generation 3400: Best = 1.382951e+03 = [2, 2, 3, 3]	FES = 340100	Diversity = 4.2866e-03	k_t
Generation 3410: Best = 1.382937e+03 = [1, 1, 1, 1]	FES = 341100	Diversity = 4.1070e-03	k_t
Generation 3420: Best = 1.382922e+03 = [2, 3, 4, 3]	FES = 342100	Diversity = 4.1288e-03	k_t
Generation 3430: Best = 1.382904e+03 = [1, 1, 1, 1]	FES = 343100	Diversity = 4.2740e-03	k_t
Generation 3440: Best = 1.382889e+03 = [3, 2, 3, 4]	FES = 344100	Diversity = 3.9991e-03	k_t
Generation 3450: Best = 1.382877e+03 = [1, 1, 1, 1]	FES = 345100	Diversity = 4.0048e-03	k_t
Generation 3460: Best = 1.382864e+03 = [3, 3, 2, 2]	FES = 346100	Diversity = 4.0102e-03	k_t

Generation 3470: Best = 1.382851e+03 = [1, 1, 1, 1]	FES = 347100	Diversity = 3.9605e-03	k_t
Generation 3480: Best = 1.382839e+03 = [2, 2, 4, 4]	FES = 348100	Diversity = 3.8901e-03	k_t
Generation 3490: Best = 1.382820e+03 = [1, 1, 1, 1]	FES = 349100	Diversity = 4.0194e-03	k_t
Generation 3500: Best = 1.382801e+03 = [3, 2, 2, 4]	FES = 350100	Diversity = 4.5479e-03	k_t
Generation 3510: Best = 1.382786e+03 = [1, 1, 1, 1]	FES = 351100	Diversity = 4.3755e-03	k_t
Generation 3520: Best = 1.382775e+03 = [4, 4, 3, 4]	FES = 352100	Diversity = 4.0041e-03	k_t
Generation 3530: Best = 1.382765e+03 = [1, 1, 1, 1]	FES = 353100	Diversity = 3.7815e-03	k_t
Generation 3540: Best = 1.382747e+03 = [3, 4, 3, 3]	FES = 354100	Diversity = 3.8526e-03	k_t
Generation 3550: Best = 1.382738e+03 = [1, 1, 1, 1]	FES = 355100	Diversity = 3.8393e-03	k_t
Generation 3560: Best = 1.382723e+03 = [2, 2, 2, 2]	FES = 356100	Diversity = 3.7431e-03	k_t
Generation 3570: Best = 1.382711e+03 = [1, 1, 1, 1]	FES = 357100	Diversity = 3.5228e-03	k_t
Generation 3580: Best = 1.382697e+03 = [2, 2, 4, 3]	FES = 358100	Diversity = 3.4815e-03	k_t
Generation 3590: Best = 1.382687e+03 = [1, 1, 1, 1]	FES = 359100	Diversity = 3.5291e-03	k_t
Generation 3600: Best = 1.382675e+03 = [3, 4, 4, 4]	FES = 360100	Diversity = 3.2222e-03	k_t
Generation 3610: Best = 1.382666e+03 = [1, 1, 1, 1]	FES = 361100	Diversity = 2.9364e-03	k_t
Generation 3620: Best = 1.382656e+03 = [3, 3, 2, 2]	FES = 362100	Diversity = 3.0647e-03	k_t
Generation 3630: Best = 1.382644e+03 = [1, 1, 1, 1]	FES = 363100	Diversity = 3.2865e-03	k_t
Generation 3640: Best = 1.382636e+03 = [4, 3, 4, 3]	FES = 364100	Diversity = 3.1869e-03	k_t
Generation 3650: Best = 1.382622e+03 = [1, 1, 1, 1]	FES = 365100	Diversity = 3.2206e-03	k_t
Generation 3660: Best = 1.382614e+03 = [4, 3, 3, 2]	FES = 366100	Diversity = 3.2880e-03	k_t
Generation 3670: Best = 1.382605e+03 = [1, 1, 1, 1]	FES = 367100	Diversity = 3.2368e-03	k_t
Generation 3680: Best = 1.382595e+03 = [2, 2, 2, 4]	FES = 368100	Diversity = 3.0217e-03	k_t
Generation 3690: Best = 1.382581e+03 = [1, 1, 1, 1]	FES = 369100	Diversity = 3.0751e-03	k_t
Generation 3700: Best = 1.382577e+03 = [2, 3, 3, 2]	FES = 370100	Diversity = 2.9594e-03	k_t
Generation 3710: Best = 1.382567e+03 = [1, 1, 1, 1]	FES = 371100	Diversity = 3.0440e-03	k_t
Generation 3720: Best = 1.382557e+03 = [4, 2, 4, 3]	FES = 372100	Diversity = 3.0633e-03	k_t
Generation 3730: Best = 1.382549e+03 = [1, 1, 1, 1]	FES = 373100	Diversity = 3.0158e-03	k_t

Generation 3740: Best = 1.382540e+03 = [2, 3, 3, 3]	FES = 374100	Diversity = 3.0878e-03	k_t
Generation 3750: Best = 1.382533e+03 = [1, 1, 1, 1]	FES = 375100	Diversity = 3.2187e-03	k_t
Generation 3760: Best = 1.382523e+03 = [3, 3, 2, 2]	FES = 376100	Diversity = 2.9833e-03	k_t
Generation 3770: Best = 1.382517e+03 = [1, 1, 1, 1]	FES = 377100	Diversity = 2.9369e-03	k_t
Generation 3780: Best = 1.382509e+03 = [4, 2, 4, 2]	FES = 378100	Diversity = 2.8460e-03	k_t
Generation 3790: Best = 1.382500e+03 = [1, 1, 1, 1]	FES = 379100	Diversity = 2.6885e-03	k_t
Generation 3800: Best = 1.382491e+03 = [4, 4, 2, 4]	FES = 380100	Diversity = 2.6948e-03	k_t
Generation 3810: Best = 1.382483e+03 = [1, 1, 1, 1]	FES = 381100	Diversity = 2.9138e-03	k_t
Generation 3820: Best = 1.382477e+03 = [3, 2, 4, 2]	FES = 382100	Diversity = 2.9370e-03	k_t
Generation 3830: Best = 1.382469e+03 = [1, 1, 1, 1]	FES = 383100	Diversity = 2.8552e-03	k_t
Generation 3840: Best = 1.382460e+03 = [2, 2, 2, 3]	FES = 384100	Diversity = 2.8655e-03	k_t
Generation 3850: Best = 1.382454e+03 = [1, 1, 1, 1]	FES = 385100	Diversity = 2.8863e-03	k_t
Generation 3860: Best = 1.382448e+03 = [2, 4, 3, 3]	FES = 386100	Diversity = 2.9063e-03	k_t
Generation 3870: Best = 1.382441e+03 = [1, 1, 1, 1]	FES = 387100	Diversity = 2.6711e-03	k_t
Generation 3880: Best = 1.382434e+03 = [3, 2, 2, 2]	FES = 388100	Diversity = 2.7086e-03	k_t
Generation 3890: Best = 1.382428e+03 = [1, 1, 1, 1]	FES = 389100	Diversity = 2.8478e-03	k_t
Generation 3900: Best = 1.382420e+03 = [3, 4, 4, 3]	FES = 390100	Diversity = 2.8062e-03	k_t
Generation 3910: Best = 1.382414e+03 = [1, 1, 1, 1]	FES = 391100	Diversity = 2.9049e-03	k_t
Generation 3920: Best = 1.382406e+03 = [4, 3, 4, 3]	FES = 392100	Diversity = 2.8281e-03	k_t
Generation 3930: Best = 1.382399e+03 = [1, 1, 1, 1]	FES = 393100	Diversity = 2.6457e-03	k_t
Generation 3940: Best = 1.382393e+03 = [4, 3, 4, 4]	FES = 394100	Diversity = 2.6652e-03	k_t
Generation 3950: Best = 1.382387e+03 = [1, 1, 1, 1]	FES = 395100	Diversity = 2.6692e-03	k_t
Generation 3960: Best = 1.382380e+03 = [2, 2, 3, 2]	FES = 396100	Diversity = 2.7007e-03	k_t
Generation 3970: Best = 1.382373e+03 = [1, 1, 1, 1]	FES = 397100	Diversity = 2.6146e-03	k_t
Generation 3980: Best = 1.382369e+03 = [2, 4, 4, 2]	FES = 398100	Diversity = 2.5695e-03	k_t
Generation 3990: Best = 1.382361e+03 = [1, 1, 1, 1]	FES = 399100	Diversity = 2.6247e-03	k_t
Generation 4000: Best = 1.382355e+03 = [4, 3, 3, 4]	FES = 400100	Diversity = 2.4759e-03	k_t

Generation 4010: Best = 1.382350e+03 = [1, 1, 1, 1]	FES = 401100	Diversity = 2.5230e-03	k_t
Generation 4020: Best = 1.382344e+03 = [3, 4, 4, 4]	FES = 402100	Diversity = 2.5519e-03	k_t
Generation 4030: Best = 1.382336e+03 = [1, 1, 1, 1]	FES = 403100	Diversity = 2.4318e-03	k_t
Generation 4040: Best = 1.382330e+03 = [4, 3, 3, 2]	FES = 404100	Diversity = 2.4300e-03	k_t
Generation 4050: Best = 1.382325e+03 = [1, 1, 1, 1]	FES = 405100	Diversity = 2.4362e-03	k_t
Generation 4060: Best = 1.382317e+03 = [2, 2, 4, 3]	FES = 406100	Diversity = 2.6087e-03	k_t
Generation 4070: Best = 1.382310e+03 = [1, 1, 1, 1]	FES = 407100	Diversity = 2.6949e-03	k_t
Generation 4080: Best = 1.382302e+03 = [4, 4, 3, 3]	FES = 408100	Diversity = 2.7466e-03	k_t
Generation 4090: Best = 1.382297e+03 = [1, 1, 1, 1]	FES = 409100	Diversity = 2.6419e-03	k_t
Generation 4100: Best = 1.382291e+03 = [3, 2, 4, 4]	FES = 410100	Diversity = 2.5823e-03	k_t
Generation 4110: Best = 1.382286e+03 = [1, 1, 1, 1]	FES = 411100	Diversity = 2.4923e-03	k_t
Generation 4120: Best = 1.382280e+03 = [2, 2, 2, 4]	FES = 412100	Diversity = 2.4960e-03	k_t
Generation 4130: Best = 1.382274e+03 = [1, 1, 1, 1]	FES = 413100	Diversity = 2.3046e-03	k_t
Generation 4140: Best = 1.382267e+03 = [2, 3, 4, 3]	FES = 414100	Diversity = 2.2909e-03	k_t
Generation 4150: Best = 1.382263e+03 = [1, 1, 1, 1]	FES = 415100	Diversity = 2.2074e-03	k_t
Generation 4160: Best = 1.382258e+03 = [2, 4, 3, 3]	FES = 416100	Diversity = 2.1565e-03	k_t
Generation 4170: Best = 1.382250e+03 = [1, 1, 1, 1]	FES = 417100	Diversity = 2.1445e-03	k_t
Generation 4180: Best = 1.382246e+03 = [3, 2, 2, 3]	FES = 418100	Diversity = 2.1737e-03	k_t
Generation 4190: Best = 1.382239e+03 = [1, 1, 1, 1]	FES = 419100	Diversity = 2.3392e-03	k_t
Generation 4200: Best = 1.382235e+03 = [2, 3, 4, 2]	FES = 420100	Diversity = 2.3393e-03	k_t
Generation 4210: Best = 1.382227e+03 = [1, 1, 1, 1]	FES = 421100	Diversity = 2.3344e-03	k_t
Generation 4220: Best = 1.382224e+03 = [2, 3, 4, 2]	FES = 422100	Diversity = 2.3074e-03	k_t
Generation 4230: Best = 1.382218e+03 = [1, 1, 1, 1]	FES = 423100	Diversity = 2.3670e-03	k_t
Generation 4240: Best = 1.382213e+03 = [4, 2, 3, 2]	FES = 424100	Diversity = 2.4951e-03	k_t
Generation 4250: Best = 1.382209e+03 = [1, 1, 1, 1]	FES = 425100	Diversity = 2.4168e-03	k_t
Generation 4260: Best = 1.382204e+03 = [2, 4, 4, 3]	FES = 426100	Diversity = 2.4015e-03	k_t
Generation 4270: Best = 1.382199e+03 = [1, 1, 1, 1]	FES = 427100	Diversity = 2.2519e-03	k_t

Generation 4280: Best = 1.382196e+03 = [2, 4, 2, 4]	FES = 428100	Diversity = 2.0603e-03	k_t
Generation 4290: Best = 1.382191e+03 = [1, 1, 1, 1]	FES = 429100	Diversity = 2.0644e-03	k_t
Generation 4300: Best = 1.382188e+03 = [2, 2, 3, 4]	FES = 430100	Diversity = 1.9868e-03	k_t
Generation 4310: Best = 1.382183e+03 = [1, 1, 1, 1]	FES = 431100	Diversity = 1.9953e-03	k_t
Generation 4320: Best = 1.382181e+03 = [4, 2, 3, 3]	FES = 432100	Diversity = 1.9021e-03	k_t
Generation 4330: Best = 1.382176e+03 = [1, 1, 1, 1]	FES = 433100	Diversity = 1.9767e-03	k_t
Generation 4340: Best = 1.382171e+03 = [3, 4, 4, 4]	FES = 434100	Diversity = 2.0664e-03	k_t
Generation 4350: Best = 1.382168e+03 = [1, 1, 1, 1]	FES = 435100	Diversity = 2.0472e-03	k_t
Generation 4360: Best = 1.382163e+03 = [4, 4, 2, 3]	FES = 436100	Diversity = 2.1172e-03	k_t
Generation 4370: Best = 1.382159e+03 = [1, 1, 1, 1]	FES = 437100	Diversity = 2.0999e-03	k_t
Generation 4380: Best = 1.382155e+03 = [3, 2, 2, 2]	FES = 438100	Diversity = 2.0584e-03	k_t
Generation 4390: Best = 1.382150e+03 = [1, 1, 1, 1]	FES = 439100	Diversity = 1.8854e-03	k_t
Generation 4400: Best = 1.382146e+03 = [3, 3, 3, 4]	FES = 440100	Diversity = 1.8013e-03	k_t
Generation 4410: Best = 1.382142e+03 = [1, 1, 1, 1]	FES = 441100	Diversity = 1.7391e-03	k_t
Generation 4420: Best = 1.382139e+03 = [2, 3, 4, 2]	FES = 442100	Diversity = 1.6943e-03	k_t
Generation 4430: Best = 1.382136e+03 = [1, 1, 1, 1]	FES = 443100	Diversity = 1.7261e-03	k_t
Generation 4440: Best = 1.382131e+03 = [4, 4, 2, 2]	FES = 444100	Diversity = 1.8409e-03	k_t
Generation 4450: Best = 1.382129e+03 = [1, 1, 1, 1]	FES = 445100	Diversity = 1.7950e-03	k_t
Generation 4460: Best = 1.382126e+03 = [2, 4, 4, 2]	FES = 446100	Diversity = 1.9172e-03	k_t
Generation 4470: Best = 1.382122e+03 = [1, 1, 1, 1]	FES = 447100	Diversity = 1.8849e-03	k_t
Generation 4480: Best = 1.382119e+03 = [2, 4, 2, 4]	FES = 448100	Diversity = 1.8495e-03	k_t
Generation 4490: Best = 1.382115e+03 = [1, 1, 1, 1]	FES = 449100	Diversity = 1.7115e-03	k_t
Generation 4500: Best = 1.382110e+03 = [3, 4, 4, 3]	FES = 450100	Diversity = 1.7513e-03	k_t
Generation 4510: Best = 1.382107e+03 = [1, 1, 1, 1]	FES = 451100	Diversity = 1.8826e-03	k_t
Generation 4520: Best = 1.382102e+03 = [4, 2, 4, 3]	FES = 452100	Diversity = 1.9748e-03	k_t
Generation 4530: Best = 1.382098e+03 = [1, 1, 1, 1]	FES = 453100	Diversity = 1.9761e-03	k_t
Generation 4540: Best = 1.382095e+03 = [4, 2, 3, 2]	FES = 454100	Diversity = 1.9485e-03	k_t

Generation 4550: Best = 1.382091e+03 = [1, 1, 1, 1]	FES = 455100	Diversity = 1.9699e-03	k_t
Generation 4560: Best = 1.382088e+03 = [4, 3, 4, 4]	FES = 456100	Diversity = 1.8945e-03	k_t
Generation 4570: Best = 1.382085e+03 = [1, 1, 1, 1]	FES = 457100	Diversity = 1.8259e-03	k_t
Generation 4580: Best = 1.382081e+03 = [2, 4, 3, 2]	FES = 458100	Diversity = 1.7628e-03	k_t
Generation 4590: Best = 1.382077e+03 = [1, 1, 1, 1]	FES = 459100	Diversity = 1.8580e-03	k_t
Generation 4600: Best = 1.382073e+03 = [4, 4, 4, 3]	FES = 460100	Diversity = 1.9500e-03	k_t
Generation 4610: Best = 1.382068e+03 = [1, 1, 1, 1]	FES = 461100	Diversity = 1.8630e-03	k_t
Generation 4620: Best = 1.382065e+03 = [4, 2, 2, 3]	FES = 462100	Diversity = 1.9322e-03	k_t
Generation 4630: Best = 1.382061e+03 = [1, 1, 1, 1]	FES = 463100	Diversity = 1.8145e-03	k_t
Generation 4640: Best = 1.382058e+03 = [3, 4, 4, 3]	FES = 464100	Diversity = 1.8225e-03	k_t
Generation 4650: Best = 1.382055e+03 = [1, 1, 1, 1]	FES = 465100	Diversity = 1.8402e-03	k_t
Generation 4660: Best = 1.382052e+03 = [3, 2, 4, 4]	FES = 466100	Diversity = 1.8020e-03	k_t
Generation 4670: Best = 1.382050e+03 = [1, 1, 1, 1]	FES = 467100	Diversity = 1.7538e-03	k_t
Generation 4680: Best = 1.382046e+03 = [2, 4, 3, 4]	FES = 468100	Diversity = 1.8800e-03	k_t
Generation 4690: Best = 1.382042e+03 = [1, 1, 1, 1]	FES = 469100	Diversity = 1.8340e-03	k_t
Generation 4700: Best = 1.382039e+03 = [4, 3, 4, 4]	FES = 470100	Diversity = 1.8673e-03	k_t
Generation 4710: Best = 1.382035e+03 = [1, 1, 1, 1]	FES = 471100	Diversity = 1.7284e-03	k_t
Generation 4720: Best = 1.382032e+03 = [4, 4, 2, 3]	FES = 472100	Diversity = 1.7132e-03	k_t
Generation 4730: Best = 1.382029e+03 = [1, 1, 1, 1]	FES = 473100	Diversity = 1.6369e-03	k_t
Generation 4740: Best = 1.382027e+03 = [4, 4, 2, 4]	FES = 474100	Diversity = 1.5341e-03	k_t
Generation 4750: Best = 1.382024e+03 = [1, 1, 1, 1]	FES = 475100	Diversity = 1.5172e-03	k_t
Generation 4760: Best = 1.382022e+03 = [4, 4, 3, 3]	FES = 476100	Diversity = 1.5971e-03	k_t
Generation 4770: Best = 1.382019e+03 = [1, 1, 1, 1]	FES = 477100	Diversity = 1.6079e-03	k_t
Generation 4780: Best = 1.382015e+03 = [4, 2, 4, 2]	FES = 478100	Diversity = 1.6045e-03	k_t
Generation 4790: Best = 1.382014e+03 = [1, 1, 1, 1]	FES = 479100	Diversity = 1.6073e-03	k_t
Generation 4800: Best = 1.382011e+03 = [2, 2, 3, 4]	FES = 480100	Diversity = 1.5518e-03	k_t
Generation 4810: Best = 1.382008e+03 = [1, 1, 1, 1]	FES = 481100	Diversity = 1.5722e-03	k_t

Generation 4820: Best = 1.382006e+03 = [2, 2, 3, 3]	FES = 482100	Diversity = 1.4481e-03	k_t
Generation 4830: Best = 1.382003e+03 = [1, 1, 1, 1]	FES = 483100	Diversity = 1.4850e-03	k_t
Generation 4840: Best = 1.382000e+03 = [2, 3, 4, 2]	FES = 484100	Diversity = 1.5297e-03	k_t
Generation 4850: Best = 1.381997e+03 = [1, 1, 1, 1]	FES = 485100	Diversity = 1.5219e-03	k_t
Generation 4860: Best = 1.381994e+03 = [2, 3, 2, 2]	FES = 486100	Diversity = 1.6628e-03	k_t
Generation 4870: Best = 1.381990e+03 = [1, 1, 1, 1]	FES = 487100	Diversity = 1.7013e-03	k_t
Generation 4880: Best = 1.381988e+03 = [4, 4, 2, 3]	FES = 488100	Diversity = 1.7045e-03	k_t
Generation 4890: Best = 1.381985e+03 = [1, 1, 1, 1]	FES = 489100	Diversity = 1.6261e-03	k_t
Generation 4900: Best = 1.381983e+03 = [4, 3, 4, 4]	FES = 490100	Diversity = 1.5597e-03	k_t
Generation 4910: Best = 1.381980e+03 = [1, 1, 1, 1]	FES = 491100	Diversity = 1.4476e-03	k_t
Generation 4920: Best = 1.381978e+03 = [2, 2, 3, 3]	FES = 492100	Diversity = 1.4021e-03	k_t
Generation 4930: Best = 1.381976e+03 = [1, 1, 1, 1]	FES = 493100	Diversity = 1.4350e-03	k_t
Generation 4940: Best = 1.381974e+03 = [2, 2, 4, 2]	FES = 494100	Diversity = 1.4706e-03	k_t
Generation 4950: Best = 1.381971e+03 = [1, 1, 1, 1]	FES = 495100	Diversity = 1.3784e-03	k_t
Generation 4960: Best = 1.381970e+03 = [4, 2, 4, 2]	FES = 496100	Diversity = 1.3757e-03	k_t
Generation 4970: Best = 1.381968e+03 = [1, 1, 1, 1]	FES = 497100	Diversity = 1.2542e-03	k_t
Generation 4980: Best = 1.381967e+03 = [4, 4, 4, 2]	FES = 498100	Diversity = 1.2179e-03	k_t
Generation 4990: Best = 1.381965e+03 = [1, 1, 1, 1]	FES = 499100	Diversity = 1.1433e-03	k_t
Generation 5000: Best = 1.381963e+03 = [4, 4, 2, 3]	FES = 500100	Diversity = 1.1659e-03	k_t
Generation 5010: Best = 1.381962e+03 = [1, 1, 1, 1]	FES = 501100	Diversity = 1.1827e-03	k_t
Generation 5020: Best = 1.381959e+03 = [2, 2, 2, 4]	FES = 502100	Diversity = 1.2513e-03	k_t
Generation 5030: Best = 1.381957e+03 = [1, 1, 1, 1]	FES = 503100	Diversity = 1.2909e-03	k_t
Generation 5040: Best = 1.381955e+03 = [2, 3, 4, 3]	FES = 504100	Diversity = 1.2509e-03	k_t
Generation 5050: Best = 1.381953e+03 = [1, 1, 1, 1]	FES = 505100	Diversity = 1.3181e-03	k_t
Generation 5060: Best = 1.381951e+03 = [3, 4, 4, 3]	FES = 506100	Diversity = 1.4079e-03	k_t
Generation 5070: Best = 1.381949e+03 = [1, 1, 1, 1]	FES = 507100	Diversity = 1.2870e-03	k_t
Generation 5080: Best = 1.381947e+03 = [2, 3, 2, 4]	FES = 508100	Diversity = 1.2545e-03	k_t

Generation 5090: Best = 1.381945e+03 = [1, 1, 1, 1]	FES = 509100	Diversity = 1.3165e-03	k_t
Generation 5100: Best = 1.381944e+03 = [2, 3, 4, 4]	FES = 510100	Diversity = 1.2253e-03	k_t
Generation 5110: Best = 1.381942e+03 = [1, 1, 1, 1]	FES = 511100	Diversity = 1.1747e-03	k_t
Generation 5120: Best = 1.381941e+03 = [4, 2, 4, 4]	FES = 512100	Diversity = 1.1949e-03	k_t
Generation 5130: Best = 1.381939e+03 = [1, 1, 1, 1]	FES = 513100	Diversity = 1.2164e-03	k_t
Generation 5140: Best = 1.381937e+03 = [2, 2, 2, 2]	FES = 514100	Diversity = 1.2455e-03	k_t
Generation 5150: Best = 1.381936e+03 = [1, 1, 1, 1]	FES = 515100	Diversity = 1.2147e-03	k_t
Generation 5160: Best = 1.381934e+03 = [3, 2, 2, 2]	FES = 516100	Diversity = 1.2081e-03	k_t
Generation 5170: Best = 1.381933e+03 = [1, 1, 1, 1]	FES = 517100	Diversity = 1.1146e-03	k_t
Generation 5180: Best = 1.381931e+03 = [3, 2, 4, 2]	FES = 518100	Diversity = 1.0871e-03	k_t
Generation 5190: Best = 1.381929e+03 = [1, 1, 1, 1]	FES = 519100	Diversity = 1.0833e-03	k_t
Generation 5200: Best = 1.381927e+03 = [2, 4, 2, 4]	FES = 520100	Diversity = 1.1326e-03	k_t
Generation 5210: Best = 1.381925e+03 = [1, 1, 1, 1]	FES = 521100	Diversity = 1.1900e-03	k_t
Generation 5220: Best = 1.381923e+03 = [2, 2, 3, 3]	FES = 522100	Diversity = 1.2423e-03	k_t
Generation 5230: Best = 1.381920e+03 = [1, 1, 1, 1]	FES = 523100	Diversity = 1.3012e-03	k_t
Generation 5240: Best = 1.381919e+03 = [4, 4, 4, 4]	FES = 524100	Diversity = 1.2827e-03	k_t
Generation 5250: Best = 1.381917e+03 = [1, 1, 1, 1]	FES = 525100	Diversity = 1.1776e-03	k_t
Generation 5260: Best = 1.381913e+03 = [3, 4, 3, 3]	FES = 526100	Diversity = 1.2254e-03	k_t
Generation 5270: Best = 1.381911e+03 = [1, 1, 1, 1]	FES = 527100	Diversity = 1.1283e-03	k_t
Generation 5280: Best = 1.381909e+03 = [3, 4, 2, 4]	FES = 528100	Diversity = 1.1861e-03	k_t
Generation 5290: Best = 1.381907e+03 = [1, 1, 1, 1]	FES = 529100	Diversity = 1.2317e-03	k_t
Generation 5300: Best = 1.381906e+03 = [2, 4, 3, 2]	FES = 530100	Diversity = 1.1965e-03	k_t
Generation 5310: Best = 1.381904e+03 = [1, 1, 1, 1]	FES = 531100	Diversity = 1.2703e-03	k_t
Generation 5320: Best = 1.381901e+03 = [2, 3, 4, 4]	FES = 532100	Diversity = 1.3094e-03	k_t
Generation 5330: Best = 1.381900e+03 = [1, 1, 1, 1]	FES = 533100	Diversity = 1.3481e-03	k_t
Generation 5340: Best = 1.381897e+03 = [2, 2, 3, 2]	FES = 534100	Diversity = 1.2981e-03	k_t
Generation 5350: Best = 1.381895e+03 = [1, 1, 1, 1]	FES = 535100	Diversity = 1.3053e-03	k_t

Generation 5360: Best = 1.381893e+03 = [2, 3, 2, 2]	FES = 536100	Diversity = 1.2530e-03	k_t
Generation 5370: Best = 1.381891e+03 = [1, 1, 1, 1]	FES = 537100	Diversity = 1.1614e-03	k_t
Generation 5380: Best = 1.381889e+03 = [4, 4, 4, 2]	FES = 538100	Diversity = 1.1795e-03	k_t
Generation 5390: Best = 1.381887e+03 = [1, 1, 1, 1]	FES = 539100	Diversity = 1.1489e-03	k_t
Generation 5400: Best = 1.381885e+03 = [3, 2, 4, 3]	FES = 540100	Diversity = 1.1172e-03	k_t
Generation 5410: Best = 1.381883e+03 = [1, 1, 1, 1]	FES = 541100	Diversity = 1.1656e-03	k_t
Generation 5420: Best = 1.381882e+03 = [3, 2, 3, 4]	FES = 542100	Diversity = 1.2243e-03	k_t
Generation 5430: Best = 1.381880e+03 = [1, 1, 1, 1]	FES = 543100	Diversity = 1.2498e-03	k_t
Generation 5440: Best = 1.381877e+03 = [3, 3, 4, 4]	FES = 544100	Diversity = 1.2451e-03	k_t
Generation 5450: Best = 1.381876e+03 = [1, 1, 1, 1]	FES = 545100	Diversity = 1.2009e-03	k_t
Generation 5460: Best = 1.381874e+03 = [4, 3, 4, 3]	FES = 546100	Diversity = 1.1307e-03	k_t
Generation 5470: Best = 1.381873e+03 = [1, 1, 1, 1]	FES = 547100	Diversity = 1.0357e-03	k_t
Generation 5480: Best = 1.381872e+03 = [2, 4, 3, 3]	FES = 548100	Diversity = 9.5003e-04	k_t
Generation 5490: Best = 1.381871e+03 = [1, 1, 1, 1]	FES = 549100	Diversity = 9.1515e-04	k_t
Generation 5500: Best = 1.381870e+03 = [2, 3, 3, 3]	FES = 550100	Diversity = 7.9119e-04	k_t
Generation 5510: Best = 1.381868e+03 = [1, 1, 1, 1]	FES = 551100	Diversity = 7.4431e-04	k_t
Generation 5520: Best = 1.381868e+03 = [2, 4, 3, 4]	FES = 552100	Diversity = 7.3710e-04	k_t
Generation 5530: Best = 1.381866e+03 = [1, 1, 1, 1]	FES = 553100	Diversity = 7.7428e-04	k_t
Generation 5540: Best = 1.381865e+03 = [4, 3, 4, 2]	FES = 554100	Diversity = 8.1543e-04	k_t
Generation 5550: Best = 1.381864e+03 = [1, 1, 1, 1]	FES = 555100	Diversity = 9.9538e-04	k_t
Generation 5560: Best = 1.381862e+03 = [3, 2, 2, 4]	FES = 556100	Diversity = 1.0187e-03	k_t
Generation 5570: Best = 1.381861e+03 = [1, 1, 1, 1]	FES = 557100	Diversity = 8.9877e-04	k_t
Generation 5580: Best = 1.381860e+03 = [4, 3, 4, 2]	FES = 558100	Diversity = 9.1378e-04	k_t
Generation 5590: Best = 1.381859e+03 = [1, 1, 1, 1]	FES = 559100	Diversity = 8.0949e-04	k_t
Generation 5600: Best = 1.381857e+03 = [3, 4, 4, 4]	FES = 560100	Diversity = 9.1248e-04	k_t
Generation 5610: Best = 1.381856e+03 = [1, 1, 1, 1]	FES = 561100	Diversity = 9.6755e-04	k_t
Generation 5620: Best = 1.381854e+03 = [4, 2, 3, 2]	FES = 562100	Diversity = 9.1292e-04	k_t

Generation 5630: Best = 1.381853e+03 = [1, 1, 1, 1]	FES = 563100	Diversity = 9.8378e-04	k_t
Generation 5640: Best = 1.381851e+03 = [4, 3, 2, 2]	FES = 564100	Diversity = 8.3885e-04	k_t
Generation 5650: Best = 1.381850e+03 = [1, 1, 1, 1]	FES = 565100	Diversity = 8.0332e-04	k_t
Generation 5660: Best = 1.381849e+03 = [2, 3, 2, 2]	FES = 566100	Diversity = 7.5001e-04	k_t
Generation 5670: Best = 1.381848e+03 = [1, 1, 1, 1]	FES = 567100	Diversity = 7.5168e-04	k_t
Generation 5680: Best = 1.381847e+03 = [2, 2, 3, 3]	FES = 568100	Diversity = 7.1397e-04	k_t
Generation 5690: Best = 1.381846e+03 = [1, 1, 1, 1]	FES = 569100	Diversity = 8.0597e-04	k_t
Generation 5700: Best = 1.381844e+03 = [3, 2, 4, 2]	FES = 570100	Diversity = 8.4496e-04	k_t
Generation 5710: Best = 1.381843e+03 = [1, 1, 1, 1]	FES = 571100	Diversity = 9.2026e-04	k_t
Generation 5720: Best = 1.381841e+03 = [3, 3, 4, 3]	FES = 572100	Diversity = 9.1483e-04	k_t
Generation 5730: Best = 1.381840e+03 = [1, 1, 1, 1]	FES = 573100	Diversity = 9.7290e-04	k_t
Generation 5740: Best = 1.381839e+03 = [4, 2, 4, 4]	FES = 574100	Diversity = 9.9992e-04	k_t
Generation 5750: Best = 1.381837e+03 = [1, 1, 1, 1]	FES = 575100	Diversity = 9.3375e-04	k_t
Generation 5760: Best = 1.381836e+03 = [4, 4, 2, 2]	FES = 576100	Diversity = 8.8361e-04	k_t
Generation 5770: Best = 1.381835e+03 = [1, 1, 1, 1]	FES = 577100	Diversity = 8.2233e-04	k_t
Generation 5780: Best = 1.381833e+03 = [3, 3, 2, 2]	FES = 578100	Diversity = 8.1904e-04	k_t
Generation 5790: Best = 1.381832e+03 = [1, 1, 1, 1]	FES = 579100	Diversity = 8.2392e-04	k_t
Generation 5800: Best = 1.381831e+03 = [4, 4, 3, 3]	FES = 580100	Diversity = 7.2494e-04	k_t
Generation 5810: Best = 1.381830e+03 = [1, 1, 1, 1]	FES = 581100	Diversity = 6.4713e-04	k_t
Generation 5820: Best = 1.381829e+03 = [2, 4, 3, 4]	FES = 582100	Diversity = 6.2628e-04	k_t
Generation 5830: Best = 1.381828e+03 = [1, 1, 1, 1]	FES = 583100	Diversity = 7.6600e-04	k_t
Generation 5840: Best = 1.381826e+03 = [2, 4, 2, 4]	FES = 584100	Diversity = 7.7928e-04	k_t
Generation 5850: Best = 1.381825e+03 = [1, 1, 1, 1]	FES = 585100	Diversity = 8.8974e-04	k_t
Generation 5860: Best = 1.381824e+03 = [3, 3, 2, 2]	FES = 586100	Diversity = 9.0076e-04	k_t
Generation 5870: Best = 1.381822e+03 = [1, 1, 1, 1]	FES = 587100	Diversity = 7.9746e-04	k_t
Generation 5880: Best = 1.381821e+03 = [3, 2, 2, 4]	FES = 588100	Diversity = 8.7220e-04	k_t
Generation 5890: Best = 1.381820e+03 = [1, 1, 1, 1]	FES = 589100	Diversity = 7.9523e-04	k_t

Generation 5900: Best = 1.381819e+03 = [3, 2, 3, 4]	FES = 590100	Diversity = 7.8122e-04	k_t
Generation 5910: Best = 1.381818e+03 = [1, 1, 1, 1]	FES = 591100	Diversity = 7.4911e-04	k_t
Generation 5920: Best = 1.381817e+03 = [2, 4, 3, 2]	FES = 592100	Diversity = 7.5764e-04	k_t
Generation 5930: Best = 1.381816e+03 = [1, 1, 1, 1]	FES = 593100	Diversity = 6.7510e-04	k_t
Generation 5940: Best = 1.381816e+03 = [2, 2, 4, 2]	FES = 594100	Diversity = 5.8519e-04	k_t
Generation 5950: Best = 1.381815e+03 = [1, 1, 1, 1]	FES = 595100	Diversity = 5.7791e-04	k_t
Generation 5960: Best = 1.381814e+03 = [3, 4, 3, 3]	FES = 596100	Diversity = 6.1721e-04	k_t
Generation 5970: Best = 1.381813e+03 = [1, 1, 1, 1]	FES = 597100	Diversity = 5.9401e-04	k_t
Generation 5980: Best = 1.381812e+03 = [4, 2, 3, 2]	FES = 598100	Diversity = 6.1894e-04	k_t
Generation 5990: Best = 1.381811e+03 = [1, 1, 1, 1]	FES = 599100	Diversity = 6.0505e-04	k_t
Generation 6000: Best = 1.381810e+03 = [2, 2, 3, 2]	FES = 600100	Diversity = 5.3417e-04	k_t
Generation 6010: Best = 1.381810e+03 = [1, 1, 1, 1]	FES = 601100	Diversity = 5.8174e-04	k_t
Generation 6020: Best = 1.381809e+03 = [4, 4, 3, 4]	FES = 602100	Diversity = 5.9292e-04	k_t
Generation 6030: Best = 1.381808e+03 = [1, 1, 1, 1]	FES = 603100	Diversity = 6.0871e-04	k_t
Generation 6040: Best = 1.381807e+03 = [4, 2, 4, 4]	FES = 604100	Diversity = 5.8772e-04	k_t
Generation 6050: Best = 1.381807e+03 = [1, 1, 1, 1]	FES = 605100	Diversity = 5.7204e-04	k_t
Generation 6060: Best = 1.381806e+03 = [2, 2, 4, 2]	FES = 606100	Diversity = 5.9661e-04	k_t
Generation 6070: Best = 1.381805e+03 = [1, 1, 1, 1]	FES = 607100	Diversity = 5.7002e-04	k_t
Generation 6080: Best = 1.381805e+03 = [4, 4, 2, 3]	FES = 608100	Diversity = 5.4379e-04	k_t
Generation 6090: Best = 1.381804e+03 = [1, 1, 1, 1]	FES = 609100	Diversity = 5.7497e-04	k_t
Generation 6100: Best = 1.381803e+03 = [4, 2, 4, 3]	FES = 610100	Diversity = 6.0211e-04	k_t
Generation 6110: Best = 1.381803e+03 = [1, 1, 1, 1]	FES = 611100	Diversity = 6.1810e-04	k_t
Generation 6120: Best = 1.381802e+03 = [3, 3, 3, 4]	FES = 612100	Diversity = 6.2968e-04	k_t
Generation 6130: Best = 1.381801e+03 = [1, 1, 1, 1]	FES = 613100	Diversity = 5.0192e-04	k_t
Generation 6140: Best = 1.381801e+03 = [4, 2, 2, 3]	FES = 614100	Diversity = 4.7939e-04	k_t
Generation 6150: Best = 1.381800e+03 = [1, 1, 1, 1]	FES = 615100	Diversity = 5.0003e-04	k_t
Generation 6160: Best = 1.381800e+03 = [3, 2, 3, 2]	FES = 616100	Diversity = 5.2661e-04	k_t

Generation 6170: Best = 1.381799e+03 = [1, 1, 1, 1]	FES = 617100	Diversity = 4.6081e-04	k_t
Generation 6180: Best = 1.381799e+03 = [2, 2, 4, 2]	FES = 618100	Diversity = 4.7074e-04	k_t
Generation 6190: Best = 1.381798e+03 = [1, 1, 1, 1]	FES = 619100	Diversity = 4.4606e-04	k_t
Generation 6200: Best = 1.381798e+03 = [2, 3, 4, 3]	FES = 620100	Diversity = 4.2348e-04	k_t
Generation 6210: Best = 1.381797e+03 = [1, 1, 1, 1]	FES = 621100	Diversity = 4.7650e-04	k_t
Generation 6220: Best = 1.381796e+03 = [4, 3, 3, 2]	FES = 622100	Diversity = 5.6224e-04	k_t
Generation 6230: Best = 1.381796e+03 = [1, 1, 1, 1]	FES = 623100	Diversity = 5.7162e-04	k_t
Generation 6240: Best = 1.381795e+03 = [4, 2, 2, 4]	FES = 624100	Diversity = 6.0140e-04	k_t
Generation 6250: Best = 1.381794e+03 = [1, 1, 1, 1]	FES = 625100	Diversity = 5.8551e-04	k_t
Generation 6260: Best = 1.381793e+03 = [3, 2, 2, 3]	FES = 626100	Diversity = 5.4829e-04	k_t
Generation 6270: Best = 1.381793e+03 = [1, 1, 1, 1]	FES = 627100	Diversity = 5.2028e-04	k_t
Generation 6280: Best = 1.381792e+03 = [2, 3, 3, 2]	FES = 628100	Diversity = 4.5383e-04	k_t
Generation 6290: Best = 1.381791e+03 = [1, 1, 1, 1]	FES = 629100	Diversity = 4.5748e-04	k_t
Generation 6300: Best = 1.381791e+03 = [4, 4, 4, 2]	FES = 630100	Diversity = 4.5430e-04	k_t
Generation 6310: Best = 1.381790e+03 = [1, 1, 1, 1]	FES = 631100	Diversity = 4.0773e-04	k_t
Generation 6320: Best = 1.381790e+03 = [4, 4, 4, 3]	FES = 632100	Diversity = 4.3023e-04	k_t
Generation 6330: Best = 1.381789e+03 = [1, 1, 1, 1]	FES = 633100	Diversity = 5.0026e-04	k_t
Generation 6340: Best = 1.381788e+03 = [2, 3, 4, 2]	FES = 634100	Diversity = 4.8428e-04	k_t
Generation 6350: Best = 1.381788e+03 = [1, 1, 1, 1]	FES = 635100	Diversity = 4.9477e-04	k_t
Generation 6360: Best = 1.381787e+03 = [3, 3, 4, 3]	FES = 636100	Diversity = 4.3841e-04	k_t
Generation 6370: Best = 1.381786e+03 = [1, 1, 1, 1]	FES = 637100	Diversity = 4.2974e-04	k_t
Generation 6380: Best = 1.381786e+03 = [2, 2, 2, 4]	FES = 638100	Diversity = 4.0489e-04	k_t
Generation 6390: Best = 1.381786e+03 = [1, 1, 1, 1]	FES = 639100	Diversity = 3.7680e-04	k_t
Generation 6400: Best = 1.381785e+03 = [4, 3, 4, 2]	FES = 640100	Diversity = 3.4477e-04	k_t
Generation 6410: Best = 1.381785e+03 = [1, 1, 1, 1]	FES = 641100	Diversity = 3.9615e-04	k_t
Generation 6420: Best = 1.381784e+03 = [3, 4, 4, 4]	FES = 642100	Diversity = 3.6800e-04	k_t
Generation 6430: Best = 1.381784e+03 = [1, 1, 1, 1]	FES = 643100	Diversity = 4.0760e-04	k_t

Generation 6440: Best = 1.381783e+03 = [4, 4, 2, 2]	FES = 644100	Diversity = 4.6328e-04	k_t
Generation 6450: Best = 1.381783e+03 = [1, 1, 1, 1]	FES = 645100	Diversity = 4.5541e-04	k_t
Generation 6460: Best = 1.381782e+03 = [3, 2, 4, 3]	FES = 646100	Diversity = 3.9713e-04	k_t
Generation 6470: Best = 1.381782e+03 = [1, 1, 1, 1]	FES = 647100	Diversity = 3.7489e-04	k_t
Generation 6480: Best = 1.381781e+03 = [3, 4, 4, 3]	FES = 648100	Diversity = 4.4862e-04	k_t
Generation 6490: Best = 1.381781e+03 = [1, 1, 1, 1]	FES = 649100	Diversity = 4.7409e-04	k_t
Generation 6500: Best = 1.381780e+03 = [3, 3, 4, 4]	FES = 650100	Diversity = 4.1600e-04	k_t
Generation 6510: Best = 1.381780e+03 = [1, 1, 1, 1]	FES = 651100	Diversity = 3.9553e-04	k_t
Generation 6520: Best = 1.381779e+03 = [2, 2, 2, 3]	FES = 652100	Diversity = 4.0908e-04	k_t
Generation 6530: Best = 1.381779e+03 = [1, 1, 1, 1]	FES = 653100	Diversity = 4.8997e-04	k_t
Generation 6540: Best = 1.381778e+03 = [2, 2, 3, 2]	FES = 654100	Diversity = 4.8701e-04	k_t
Generation 6550: Best = 1.381778e+03 = [1, 1, 1, 1]	FES = 655100	Diversity = 4.8973e-04	k_t
Generation 6560: Best = 1.381777e+03 = [3, 2, 2, 3]	FES = 656100	Diversity = 4.7946e-04	k_t
Generation 6570: Best = 1.381776e+03 = [1, 1, 1, 1]	FES = 657100	Diversity = 4.2908e-04	k_t
Generation 6580: Best = 1.381776e+03 = [2, 4, 3, 4]	FES = 658100	Diversity = 3.0603e-04	k_t
Generation 6590: Best = 1.381775e+03 = [1, 1, 1, 1]	FES = 659100	Diversity = 3.9146e-04	k_t
Generation 6600: Best = 1.381775e+03 = [3, 4, 2, 2]	FES = 660100	Diversity = 4.5687e-04	k_t
Generation 6610: Best = 1.381774e+03 = [1, 1, 1, 1]	FES = 661100	Diversity = 3.9876e-04	k_t
Generation 6620: Best = 1.381774e+03 = [3, 4, 3, 4]	FES = 662100	Diversity = 4.3676e-04	k_t
Generation 6630: Best = 1.381773e+03 = [1, 1, 1, 1]	FES = 663100	Diversity = 4.2088e-04	k_t
Generation 6640: Best = 1.381772e+03 = [4, 4, 2, 2]	FES = 664100	Diversity = 4.5808e-04	k_t
Generation 6650: Best = 1.381771e+03 = [1, 1, 1, 1]	FES = 665100	Diversity = 5.6548e-04	k_t
Generation 6660: Best = 1.381770e+03 = [4, 3, 3, 2]	FES = 666100	Diversity = 5.3665e-04	k_t
Generation 6670: Best = 1.381770e+03 = [1, 1, 1, 1]	FES = 667100	Diversity = 4.8868e-04	k_t
Generation 6680: Best = 1.381769e+03 = [3, 2, 2, 2]	FES = 668100	Diversity = 4.9737e-04	k_t
Generation 6690: Best = 1.381768e+03 = [1, 1, 1, 1]	FES = 669100	Diversity = 5.3453e-04	k_t
Generation 6700: Best = 1.381767e+03 = [2, 4, 4, 4]	FES = 670100	Diversity = 5.7865e-04	k_t

Generation 6710: Best = 1.381766e+03 = [1, 1, 1, 1]	FES = 671100	Diversity = 5.2929e-04	k_t
Generation 6720: Best = 1.381765e+03 = [3, 3, 2, 4]	FES = 672100	Diversity = 5.3506e-04	k_t
Generation 6730: Best = 1.381765e+03 = [1, 1, 1, 1]	FES = 673100	Diversity = 4.7702e-04	k_t
Generation 6740: Best = 1.381764e+03 = [2, 3, 3, 2]	FES = 674100	Diversity = 5.7098e-04	k_t
Generation 6750: Best = 1.381763e+03 = [1, 1, 1, 1]	FES = 675100	Diversity = 5.3729e-04	k_t
Generation 6760: Best = 1.381763e+03 = [3, 4, 3, 3]	FES = 676100	Diversity = 4.8605e-04	k_t
Generation 6770: Best = 1.381762e+03 = [1, 1, 1, 1]	FES = 677100	Diversity = 3.9830e-04	k_t
Generation 6780: Best = 1.381761e+03 = [3, 2, 4, 3]	FES = 678100	Diversity = 4.7361e-04	k_t
Generation 6790: Best = 1.381761e+03 = [1, 1, 1, 1]	FES = 679100	Diversity = 4.1266e-04	k_t
Generation 6800: Best = 1.381760e+03 = [3, 3, 2, 2]	FES = 680100	Diversity = 4.2796e-04	k_t
Generation 6810: Best = 1.381760e+03 = [1, 1, 1, 1]	FES = 681100	Diversity = 3.7835e-04	k_t
Generation 6820: Best = 1.381759e+03 = [4, 4, 4, 4]	FES = 682100	Diversity = 3.9284e-04	k_t
Generation 6830: Best = 1.381759e+03 = [1, 1, 1, 1]	FES = 683100	Diversity = 3.4491e-04	k_t
Generation 6840: Best = 1.381758e+03 = [2, 3, 3, 2]	FES = 684100	Diversity = 4.0398e-04	k_t
Generation 6850: Best = 1.381758e+03 = [1, 1, 1, 1]	FES = 685100	Diversity = 4.0341e-04	k_t
Generation 6860: Best = 1.381757e+03 = [4, 3, 4, 4]	FES = 686100	Diversity = 5.2222e-04	k_t
Generation 6870: Best = 1.381756e+03 = [1, 1, 1, 1]	FES = 687100	Diversity = 4.2353e-04	k_t
Generation 6880: Best = 1.381756e+03 = [3, 3, 4, 4]	FES = 688100	Diversity = 4.4952e-04	k_t
Generation 6890: Best = 1.381755e+03 = [1, 1, 1, 1]	FES = 689100	Diversity = 4.2940e-04	k_t
Generation 6900: Best = 1.381755e+03 = [2, 4, 4, 3]	FES = 690100	Diversity = 3.6205e-04	k_t
Generation 6910: Best = 1.381754e+03 = [1, 1, 1, 1]	FES = 691100	Diversity = 4.4483e-04	k_t
Generation 6920: Best = 1.381754e+03 = [4, 2, 3, 2]	FES = 692100	Diversity = 4.5274e-04	k_t
Generation 6930: Best = 1.381753e+03 = [1, 1, 1, 1]	FES = 693100	Diversity = 5.0053e-04	k_t
Generation 6940: Best = 1.381753e+03 = [4, 4, 3, 3]	FES = 694100	Diversity = 3.8427e-04	k_t
Generation 6950: Best = 1.381752e+03 = [1, 1, 1, 1]	FES = 695100	Diversity = 4.5816e-04	k_t
Generation 6960: Best = 1.381752e+03 = [3, 4, 2, 3]	FES = 696100	Diversity = 4.5089e-04	k_t
Generation 6970: Best = 1.381751e+03 = [1, 1, 1, 1]	FES = 697100	Diversity = 4.2348e-04	k_t

Generation 6980: Best = 1.381751e+03 = [4, 3, 3, 3]	FES = 698100	Diversity = 4.2307e-04	k_t
Generation 6990: Best = 1.381750e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 4.9751e-04	k_t
Generation 7000: Best = 1.381750e+03 = [3, 3, 3, 3]	FES = 700100	Diversity = 5.0832e-04	k_t
Generation 7010: Best = 1.381749e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 4.2967e-04	k_t
Generation 7020: Best = 1.381749e+03 = [4, 3, 4, 3]	FES = 702100	Diversity = 4.6457e-04	k_t
Generation 7030: Best = 1.381748e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 4.0431e-04	k_t
Generation 7040: Best = 1.381748e+03 = [2, 2, 3, 4]	FES = 704100	Diversity = 3.8704e-04	k_t
Generation 7050: Best = 1.381747e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 3.8941e-04	k_t
Generation 7060: Best = 1.381747e+03 = [4, 2, 3, 4]	FES = 706100	Diversity = 3.7323e-04	k_t
Generation 7070: Best = 1.381747e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 3.6457e-04	k_t
Generation 7080: Best = 1.381746e+03 = [2, 2, 2, 4]	FES = 708100	Diversity = 3.5205e-04	k_t
Generation 7090: Best = 1.381746e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 3.2425e-04	k_t
Generation 7100: Best = 1.381745e+03 = [2, 3, 4, 2]	FES = 710100	Diversity = 3.1606e-04	k_t
Generation 7110: Best = 1.381745e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 3.6183e-04	k_t
Generation 7120: Best = 1.381745e+03 = [4, 3, 3, 4]	FES = 712100	Diversity = 3.5114e-04	k_t
Generation 7130: Best = 1.381744e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 3.3484e-04	k_t
Generation 7140: Best = 1.381744e+03 = [4, 3, 4, 4]	FES = 714100	Diversity = 3.9910e-04	k_t
Generation 7150: Best = 1.381744e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 4.0548e-04	k_t
Generation 7160: Best = 1.381743e+03 = [4, 3, 4, 4]	FES = 716100	Diversity = 3.9921e-04	k_t
Generation 7170: Best = 1.381743e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 3.4631e-04	k_t
Generation 7180: Best = 1.381742e+03 = [3, 2, 3, 4]	FES = 718100	Diversity = 3.2947e-04	k_t
Generation 7190: Best = 1.381742e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 3.6700e-04	k_t
Generation 7200: Best = 1.381742e+03 = [3, 3, 3, 3]	FES = 720100	Diversity = 3.7785e-04	k_t
Generation 7210: Best = 1.381741e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 3.3427e-04	k_t
Generation 7220: Best = 1.381741e+03 = [2, 4, 3, 3]	FES = 722100	Diversity = 3.3928e-04	k_t
Generation 7230: Best = 1.381741e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 3.4698e-04	k_t
Generation 7240: Best = 1.381740e+03 = [2, 4, 2, 3]	FES = 724100	Diversity = 3.2879e-04	k_t

Generation 7250: Best = 1.381740e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 3.5630e-04	k_t
Generation 7260: Best = 1.381739e+03 = [2, 3, 3, 4]	FES = 726100	Diversity = 3.1112e-04	k_t
Generation 7270: Best = 1.381739e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 3.3678e-04	k_t
Generation 7280: Best = 1.381739e+03 = [3, 2, 2, 2]	FES = 728100	Diversity = 3.4278e-04	k_t
Generation 7290: Best = 1.381738e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 4.1570e-04	k_t
Generation 7300: Best = 1.381738e+03 = [4, 2, 3, 3]	FES = 730100	Diversity = 4.1638e-04	k_t
Generation 7310: Best = 1.381738e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 3.8152e-04	k_t
Generation 7320: Best = 1.381737e+03 = [3, 2, 4, 3]	FES = 732100	Diversity = 3.4822e-04	k_t
Generation 7330: Best = 1.381737e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 2.7262e-04	k_t
Generation 7340: Best = 1.381736e+03 = [4, 4, 3, 4]	FES = 734100	Diversity = 3.0947e-04	k_t
Generation 7350: Best = 1.381736e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 3.0009e-04	k_t
Generation 7360: Best = 1.381736e+03 = [4, 4, 3, 4]	FES = 736100	Diversity = 3.8502e-04	k_t
Generation 7370: Best = 1.381735e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 4.2203e-04	k_t
Generation 7380: Best = 1.381734e+03 = [3, 4, 3, 4]	FES = 738100	Diversity = 4.0500e-04	k_t
Generation 7390: Best = 1.381734e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 4.0762e-04	k_t
Generation 7400: Best = 1.381733e+03 = [2, 2, 3, 3]	FES = 740100	Diversity = 3.9771e-04	k_t
Generation 7410: Best = 1.381733e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 4.0587e-04	k_t
Generation 7420: Best = 1.381732e+03 = [3, 3, 4, 2]	FES = 742100	Diversity = 3.9400e-04	k_t
Generation 7430: Best = 1.381732e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 3.9765e-04	k_t
Generation 7440: Best = 1.381731e+03 = [2, 2, 3, 3]	FES = 744100	Diversity = 3.2017e-04	k_t
Generation 7450: Best = 1.381731e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 3.8354e-04	k_t
Generation 7460: Best = 1.381730e+03 = [2, 3, 4, 3]	FES = 746100	Diversity = 3.9305e-04	k_t
Generation 7470: Best = 1.381730e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 4.0023e-04	k_t
Generation 7480: Best = 1.381729e+03 = [2, 3, 3, 4]	FES = 748100	Diversity = 2.6634e-04	k_t
Generation 7490: Best = 1.381729e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 2.9748e-04	k_t
Generation 7500: Best = 1.381729e+03 = [2, 2, 3, 4]	FES = 750100	Diversity = 3.5367e-04	k_t
Generation 7510: Best = 1.381728e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 4.3586e-04	k_t

Generation 7520: Best = 1.381728e+03 = [4, 3, 2, 3]	FES = 752100	Diversity = 3.6836e-04	k_t
Generation 7530: Best = 1.381728e+03 = [1, 1, 1, 1]	FES = 753100	Diversity = 3.6799e-04	k_t
Generation 7540: Best = 1.381727e+03 = [3, 4, 2, 4]	FES = 754100	Diversity = 3.0716e-04	k_t
Generation 7550: Best = 1.381727e+03 = [1, 1, 1, 1]	FES = 755100	Diversity = 2.9723e-04	k_t
Generation 7560: Best = 1.381727e+03 = [2, 3, 3, 3]	FES = 756100	Diversity = 2.9845e-04	k_t
Generation 7570: Best = 1.381726e+03 = [1, 1, 1, 1]	FES = 757100	Diversity = 3.3167e-04	k_t
Generation 7580: Best = 1.381726e+03 = [3, 2, 2, 3]	FES = 758100	Diversity = 3.0146e-04	k_t
Generation 7590: Best = 1.381726e+03 = [1, 1, 1, 1]	FES = 759100	Diversity = 3.6317e-04	k_t
Generation 7600: Best = 1.381725e+03 = [2, 4, 3, 3]	FES = 760100	Diversity = 3.3618e-04	k_t
Generation 7610: Best = 1.381725e+03 = [1, 1, 1, 1]	FES = 761100	Diversity = 3.0970e-04	k_t
Generation 7620: Best = 1.381725e+03 = [2, 4, 2, 2]	FES = 762100	Diversity = 3.3521e-04	k_t
Generation 7630: Best = 1.381724e+03 = [1, 1, 1, 1]	FES = 763100	Diversity = 3.1307e-04	k_t
Generation 7640: Best = 1.381724e+03 = [4, 3, 4, 4]	FES = 764100	Diversity = 3.1182e-04	k_t
Generation 7650: Best = 1.381724e+03 = [1, 1, 1, 1]	FES = 765100	Diversity = 3.0424e-04	k_t
Generation 7660: Best = 1.381723e+03 = [2, 3, 2, 3]	FES = 766100	Diversity = 2.9122e-04	k_t
Generation 7670: Best = 1.381723e+03 = [1, 1, 1, 1]	FES = 767100	Diversity = 3.1326e-04	k_t
Generation 7680: Best = 1.381723e+03 = [3, 4, 4, 4]	FES = 768100	Diversity = 3.3220e-04	k_t
Generation 7690: Best = 1.381722e+03 = [1, 1, 1, 1]	FES = 769100	Diversity = 3.3243e-04	k_t
Generation 7700: Best = 1.381722e+03 = [3, 2, 2, 2]	FES = 770100	Diversity = 3.1240e-04	k_t
Generation 7710: Best = 1.381722e+03 = [1, 1, 1, 1]	FES = 771100	Diversity = 3.2691e-04	k_t
Generation 7720: Best = 1.381721e+03 = [2, 3, 2, 3]	FES = 772100	Diversity = 3.8963e-04	k_t
Generation 7730: Best = 1.381721e+03 = [1, 1, 1, 1]	FES = 773100	Diversity = 3.7777e-04	k_t
Generation 7740: Best = 1.381720e+03 = [2, 3, 4, 4]	FES = 774100	Diversity = 4.1354e-04	k_t
Generation 7750: Best = 1.381720e+03 = [1, 1, 1, 1]	FES = 775100	Diversity = 3.9795e-04	k_t
Generation 7760: Best = 1.381719e+03 = [4, 3, 3, 2]	FES = 776100	Diversity = 3.8407e-04	k_t
Generation 7770: Best = 1.381719e+03 = [1, 1, 1, 1]	FES = 777100	Diversity = 5.3160e-04	k_t
Generation 7780: Best = 1.381718e+03 = [4, 3, 3, 4]	FES = 778100	Diversity = 4.9849e-04	k_t

Generation 7790: Best = 1.381717e+03 = [1, 1, 1, 1]	FES = 779100	Diversity = 5.2866e-04	k_t
Generation 7800: Best = 1.381716e+03 = [3, 2, 2, 2]	FES = 780100	Diversity = 5.8348e-04	k_t
Generation 7810: Best = 1.381716e+03 = [1, 1, 1, 1]	FES = 781100	Diversity = 4.4245e-04	k_t
Generation 7820: Best = 1.381715e+03 = [2, 4, 2, 4]	FES = 782100	Diversity = 4.3829e-04	k_t
Generation 7830: Best = 1.381715e+03 = [1, 1, 1, 1]	FES = 783100	Diversity = 4.5628e-04	k_t
Generation 7840: Best = 1.381714e+03 = [3, 2, 2, 4]	FES = 784100	Diversity = 4.4741e-04	k_t
Generation 7850: Best = 1.381714e+03 = [1, 1, 1, 1]	FES = 785100	Diversity = 4.0158e-04	k_t
Generation 7860: Best = 1.381713e+03 = [4, 3, 2, 4]	FES = 786100	Diversity = 3.6962e-04	k_t
Generation 7870: Best = 1.381713e+03 = [1, 1, 1, 1]	FES = 787100	Diversity = 3.3619e-04	k_t
Generation 7880: Best = 1.381713e+03 = [2, 4, 2, 3]	FES = 788100	Diversity = 3.1077e-04	k_t
Generation 7890: Best = 1.381713e+03 = [1, 1, 1, 1]	FES = 789100	Diversity = 3.1312e-04	k_t
Generation 7900: Best = 1.381712e+03 = [2, 3, 2, 3]	FES = 790100	Diversity = 3.2220e-04	k_t
Generation 7910: Best = 1.381712e+03 = [1, 1, 1, 1]	FES = 791100	Diversity = 3.5157e-04	k_t
Generation 7920: Best = 1.381712e+03 = [3, 4, 3, 3]	FES = 792100	Diversity = 3.0220e-04	k_t
Generation 7930: Best = 1.381711e+03 = [1, 1, 1, 1]	FES = 793100	Diversity = 3.1386e-04	k_t
Generation 7940: Best = 1.381711e+03 = [3, 2, 2, 3]	FES = 794100	Diversity = 2.7965e-04	k_t
Generation 7950: Best = 1.381711e+03 = [1, 1, 1, 1]	FES = 795100	Diversity = 2.9041e-04	k_t
Generation 7960: Best = 1.381711e+03 = [3, 4, 4, 3]	FES = 796100	Diversity = 3.1954e-04	k_t
Generation 7970: Best = 1.381710e+03 = [1, 1, 1, 1]	FES = 797100	Diversity = 2.8069e-04	k_t
Generation 7980: Best = 1.381710e+03 = [4, 2, 4, 4]	FES = 798100	Diversity = 3.4715e-04	k_t
Generation 7990: Best = 1.381710e+03 = [1, 1, 1, 1]	FES = 799100	Diversity = 3.4227e-04	k_t
Generation 8000: Best = 1.381709e+03 = [3, 3, 3, 2]	FES = 800100	Diversity = 3.9363e-04	k_t
Generation 8010: Best = 1.381709e+03 = [1, 1, 1, 1]	FES = 801100	Diversity = 4.1371e-04	k_t
Generation 8020: Best = 1.381708e+03 = [4, 3, 4, 3]	FES = 802100	Diversity = 2.9898e-04	k_t
Generation 8030: Best = 1.381708e+03 = [1, 1, 1, 1]	FES = 803100	Diversity = 2.8872e-04	k_t
Generation 8040: Best = 1.381708e+03 = [4, 4, 4, 2]	FES = 804100	Diversity = 2.8239e-04	k_t
Generation 8050: Best = 1.381707e+03 = [1, 1, 1, 1]	FES = 805100	Diversity = 3.0585e-04	k_t

Generation 8060: Best = 1.381707e+03	FES = 806100	Diversity = 3.1686e-04	k_t
= [3, 3, 4, 3]			
Generation 8070: Best = 1.381707e+03	FES = 807100	Diversity = 3.2056e-04	k_t
= [1, 1, 1, 1]			
Generation 8080: Best = 1.381706e+03	FES = 808100	Diversity = 2.9930e-04	k_t
= [2, 2, 3, 2]			
Generation 8090: Best = 1.381706e+03	FES = 809100	Diversity = 3.0474e-04	k_t
= [1, 1, 1, 1]			
Generation 8100: Best = 1.381706e+03	FES = 810100	Diversity = 3.0975e-04	k_t
= [2, 2, 3, 3]			
Generation 8110: Best = 1.381706e+03	FES = 811100	Diversity = 2.5384e-04	k_t
= [1, 1, 1, 1]			
Generation 8120: Best = 1.381705e+03	FES = 812100	Diversity = 2.5513e-04	k_t
= [3, 4, 2, 2]			
Generation 8130: Best = 1.381705e+03	FES = 813100	Diversity = 2.7523e-04	k_t
= [1, 1, 1, 1]			
Generation 8140: Best = 1.381705e+03	FES = 814100	Diversity = 2.7004e-04	k_t
= [3, 2, 4, 2]			
Generation 8150: Best = 1.381705e+03	FES = 815100	Diversity = 3.0641e-04	k_t
= [1, 1, 1, 1]			
Generation 8160: Best = 1.381704e+03	FES = 816100	Diversity = 3.0164e-04	k_t
= [2, 3, 3, 4]			
Generation 8170: Best = 1.381704e+03	FES = 817100	Diversity = 3.2027e-04	k_t
= [1, 1, 1, 1]			
Generation 8180: Best = 1.381704e+03	FES = 818100	Diversity = 3.2037e-04	k_t
= [2, 4, 4, 2]			
Generation 8190: Best = 1.381703e+03	FES = 819100	Diversity = 3.0871e-04	k_t
= [1, 1, 1, 1]			
Generation 8200: Best = 1.381703e+03	FES = 820100	Diversity = 2.8030e-04	k_t
= [2, 3, 3, 2]			
Generation 8210: Best = 1.381703e+03	FES = 821100	Diversity = 2.6526e-04	k_t
= [1, 1, 1, 1]			
Generation 8220: Best = 1.381703e+03	FES = 822100	Diversity = 2.7241e-04	k_t
= [2, 4, 2, 2]			
Generation 8230: Best = 1.381702e+03	FES = 823100	Diversity = 2.5006e-04	k_t
= [1, 1, 1, 1]			
Generation 8240: Best = 1.381702e+03	FES = 824100	Diversity = 2.4043e-04	k_t
= [4, 3, 2, 4]			
Generation 8250: Best = 1.381702e+03	FES = 825100	Diversity = 2.1916e-04	k_t
= [1, 1, 1, 1]			
Generation 8260: Best = 1.381702e+03	FES = 826100	Diversity = 2.1004e-04	k_t
= [2, 3, 4, 2]			
Generation 8270: Best = 1.381702e+03	FES = 827100	Diversity = 2.3160e-04	k_t
= [1, 1, 1, 1]			
Generation 8280: Best = 1.381702e+03	FES = 828100	Diversity = 2.5482e-04	k_t
= [3, 4, 4, 4]			
Generation 8290: Best = 1.381701e+03	FES = 829100	Diversity = 2.3066e-04	k_t
= [1, 1, 1, 1]			
Generation 8300: Best = 1.381701e+03	FES = 830100	Diversity = 2.6634e-04	k_t
= [3, 4, 4, 4]			
Generation 8310: Best = 1.381701e+03	FES = 831100	Diversity = 2.6508e-04	k_t
= [1, 1, 1, 1]			
Generation 8320: Best = 1.381701e+03	FES = 832100	Diversity = 2.8966e-04	k_t
= [2, 4, 4, 4]			

Generation 8330: Best = 1.381700e+03	FES = 833100	Diversity = 2.8439e-04	k_t
= [1, 1, 1, 1]			
Generation 8340: Best = 1.381700e+03	FES = 834100	Diversity = 3.0371e-04	k_t
= [2, 3, 3, 4]			
Generation 8350: Best = 1.381700e+03	FES = 835100	Diversity = 3.4338e-04	k_t
= [1, 1, 1, 1]			
Generation 8360: Best = 1.381700e+03	FES = 836100	Diversity = 3.3350e-04	k_t
= [2, 2, 4, 4]			
Generation 8370: Best = 1.381699e+03	FES = 837100	Diversity = 2.9153e-04	k_t
= [1, 1, 1, 1]			
Generation 8380: Best = 1.381699e+03	FES = 838100	Diversity = 3.4804e-04	k_t
= [4, 4, 3, 4]			
Generation 8390: Best = 1.381698e+03	FES = 839100	Diversity = 3.3478e-04	k_t
= [1, 1, 1, 1]			
Generation 8400: Best = 1.381698e+03	FES = 840100	Diversity = 2.8388e-04	k_t
= [3, 2, 3, 3]			
Generation 8410: Best = 1.381698e+03	FES = 841100	Diversity = 3.3599e-04	k_t
= [1, 1, 1, 1]			
Generation 8420: Best = 1.381697e+03	FES = 842100	Diversity = 2.8178e-04	k_t
= [3, 4, 2, 3]			
Generation 8430: Best = 1.381697e+03	FES = 843100	Diversity = 2.2400e-04	k_t
= [1, 1, 1, 1]			
Generation 8440: Best = 1.381697e+03	FES = 844100	Diversity = 2.2105e-04	k_t
= [2, 3, 2, 3]			
Generation 8450: Best = 1.381697e+03	FES = 845100	Diversity = 1.9792e-04	k_t
= [1, 1, 1, 1]			
Generation 8460: Best = 1.381697e+03	FES = 846100	Diversity = 2.4160e-04	k_t
= [4, 4, 4, 3]			
Generation 8470: Best = 1.381696e+03	FES = 847100	Diversity = 2.6092e-04	k_t
= [1, 1, 1, 1]			
Generation 8480: Best = 1.381696e+03	FES = 848100	Diversity = 2.8374e-04	k_t
= [2, 2, 3, 2]			
Generation 8490: Best = 1.381696e+03	FES = 849100	Diversity = 2.6779e-04	k_t
= [1, 1, 1, 1]			
Generation 8500: Best = 1.381696e+03	FES = 850100	Diversity = 2.6031e-04	k_t
= [2, 3, 3, 3]			
Generation 8510: Best = 1.381696e+03	FES = 851100	Diversity = 2.2861e-04	k_t
= [1, 1, 1, 1]			
Generation 8520: Best = 1.381695e+03	FES = 852100	Diversity = 2.2051e-04	k_t
= [3, 4, 4, 3]			
Generation 8530: Best = 1.381695e+03	FES = 853100	Diversity = 1.9215e-04	k_t
= [1, 1, 1, 1]			
Generation 8540: Best = 1.381695e+03	FES = 854100	Diversity = 2.4982e-04	k_t
= [2, 2, 2, 4]			
Generation 8550: Best = 1.381695e+03	FES = 855100	Diversity = 2.4468e-04	k_t
= [1, 1, 1, 1]			
Generation 8560: Best = 1.381695e+03	FES = 856100	Diversity = 2.4618e-04	k_t
= [4, 4, 4, 4]			
Generation 8570: Best = 1.381695e+03	FES = 857100	Diversity = 2.1616e-04	k_t
= [1, 1, 1, 1]			
Generation 8580: Best = 1.381694e+03	FES = 858100	Diversity = 1.8694e-04	k_t
= [2, 3, 4, 4]			
Generation 8590: Best = 1.381694e+03	FES = 859100	Diversity = 2.0160e-04	k_t
= [1, 1, 1, 1]			

Generation 8600: Best = 1.381694e+03 = [2, 3, 4, 4]	FES = 860100	Diversity = 2.5709e-04	k_t
Generation 8610: Best = 1.381694e+03 = [1, 1, 1, 1]	FES = 861100	Diversity = 2.5112e-04	k_t
Generation 8620: Best = 1.381694e+03 = [4, 3, 2, 2]	FES = 862100	Diversity = 2.5933e-04	k_t
Generation 8630: Best = 1.381693e+03 = [1, 1, 1, 1]	FES = 863100	Diversity = 2.2172e-04	k_t
Generation 8640: Best = 1.381693e+03 = [4, 3, 2, 2]	FES = 864100	Diversity = 2.1627e-04	k_t
Generation 8650: Best = 1.381693e+03 = [1, 1, 1, 1]	FES = 865100	Diversity = 2.6124e-04	k_t
Generation 8660: Best = 1.381693e+03 = [2, 2, 3, 2]	FES = 866100	Diversity = 3.2356e-04	k_t
Generation 8670: Best = 1.381693e+03 = [1, 1, 1, 1]	FES = 867100	Diversity = 2.3851e-04	k_t
Generation 8680: Best = 1.381692e+03 = [3, 3, 3, 2]	FES = 868100	Diversity = 2.4844e-04	k_t
Generation 8690: Best = 1.381692e+03 = [1, 1, 1, 1]	FES = 869100	Diversity = 2.4890e-04	k_t
Generation 8700: Best = 1.381692e+03 = [4, 2, 4, 4]	FES = 870100	Diversity = 2.7651e-04	k_t
Generation 8710: Best = 1.381692e+03 = [1, 1, 1, 1]	FES = 871100	Diversity = 2.7190e-04	k_t
Generation 8720: Best = 1.381691e+03 = [4, 3, 3, 4]	FES = 872100	Diversity = 2.7357e-04	k_t
Generation 8730: Best = 1.381691e+03 = [1, 1, 1, 1]	FES = 873100	Diversity = 2.3533e-04	k_t
Generation 8740: Best = 1.381691e+03 = [2, 2, 4, 3]	FES = 874100	Diversity = 2.2199e-04	k_t
Generation 8750: Best = 1.381691e+03 = [1, 1, 1, 1]	FES = 875100	Diversity = 2.3633e-04	k_t
Generation 8760: Best = 1.381690e+03 = [2, 4, 3, 4]	FES = 876100	Diversity = 2.7671e-04	k_t
Generation 8770: Best = 1.381690e+03 = [1, 1, 1, 1]	FES = 877100	Diversity = 2.5805e-04	k_t
Generation 8780: Best = 1.381690e+03 = [2, 4, 2, 2]	FES = 878100	Diversity = 3.3887e-04	k_t
Generation 8790: Best = 1.381690e+03 = [1, 1, 1, 1]	FES = 879100	Diversity = 2.9462e-04	k_t
Generation 8800: Best = 1.381689e+03 = [3, 2, 4, 4]	FES = 880100	Diversity = 2.5873e-04	k_t
Generation 8810: Best = 1.381689e+03 = [1, 1, 1, 1]	FES = 881100	Diversity = 2.5605e-04	k_t
Generation 8820: Best = 1.381689e+03 = [4, 3, 2, 2]	FES = 882100	Diversity = 2.0379e-04	k_t
Generation 8830: Best = 1.381689e+03 = [1, 1, 1, 1]	FES = 883100	Diversity = 2.3783e-04	k_t
Generation 8840: Best = 1.381689e+03 = [4, 3, 3, 2]	FES = 884100	Diversity = 2.3086e-04	k_t
Generation 8850: Best = 1.381688e+03 = [1, 1, 1, 1]	FES = 885100	Diversity = 2.3691e-04	k_t
Generation 8860: Best = 1.381688e+03 = [3, 4, 4, 2]	FES = 886100	Diversity = 2.3022e-04	k_t

Generation 8870: Best = 1.381688e+03	FES = 887100	Diversity = 2.4136e-04	k_t
= [1, 1, 1, 1]			
Generation 8880: Best = 1.381688e+03	FES = 888100	Diversity = 2.7784e-04	k_t
= [4, 2, 2, 3]			
Generation 8890: Best = 1.381688e+03	FES = 889100	Diversity = 2.9403e-04	k_t
= [1, 1, 1, 1]			
Generation 8900: Best = 1.381687e+03	FES = 890100	Diversity = 2.9253e-04	k_t
= [3, 3, 3, 3]			
Generation 8910: Best = 1.381687e+03	FES = 891100	Diversity = 2.5495e-04	k_t
= [1, 1, 1, 1]			
Generation 8920: Best = 1.381687e+03	FES = 892100	Diversity = 2.9823e-04	k_t
= [3, 2, 3, 4]			
Generation 8930: Best = 1.381687e+03	FES = 893100	Diversity = 3.3841e-04	k_t
= [1, 1, 1, 1]			
Generation 8940: Best = 1.381686e+03	FES = 894100	Diversity = 4.1669e-04	k_t
= [2, 3, 2, 4]			
Generation 8950: Best = 1.381686e+03	FES = 895100	Diversity = 3.7406e-04	k_t
= [1, 1, 1, 1]			
Generation 8960: Best = 1.381686e+03	FES = 896100	Diversity = 3.9403e-04	k_t
= [4, 2, 2, 4]			
Generation 8970: Best = 1.381685e+03	FES = 897100	Diversity = 3.9409e-04	k_t
= [1, 1, 1, 1]			
Generation 8980: Best = 1.381685e+03	FES = 898100	Diversity = 4.2669e-04	k_t
= [3, 3, 3, 4]			
Generation 8990: Best = 1.381684e+03	FES = 899100	Diversity = 3.8235e-04	k_t
= [1, 1, 1, 1]			
Generation 9000: Best = 1.381684e+03	FES = 900100	Diversity = 3.5736e-04	k_t
= [4, 3, 2, 4]			
Generation 9010: Best = 1.381684e+03	FES = 901100	Diversity = 3.1414e-04	k_t
= [1, 1, 1, 1]			
Generation 9020: Best = 1.381683e+03	FES = 902100	Diversity = 3.1496e-04	k_t
= [3, 2, 2, 2]			
Generation 9030: Best = 1.381683e+03	FES = 903100	Diversity = 3.1656e-04	k_t
= [1, 1, 1, 1]			
Generation 9040: Best = 1.381682e+03	FES = 904100	Diversity = 3.1018e-04	k_t
= [3, 3, 3, 4]			
Generation 9050: Best = 1.381682e+03	FES = 905100	Diversity = 2.9525e-04	k_t
= [1, 1, 1, 1]			
Generation 9060: Best = 1.381682e+03	FES = 906100	Diversity = 3.3950e-04	k_t
= [3, 4, 4, 2]			
Generation 9070: Best = 1.381681e+03	FES = 907100	Diversity = 3.4930e-04	k_t
= [1, 1, 1, 1]			
Generation 9080: Best = 1.381681e+03	FES = 908100	Diversity = 3.5228e-04	k_t
= [4, 2, 3, 4]			
Generation 9090: Best = 1.381680e+03	FES = 909100	Diversity = 3.3787e-04	k_t
= [1, 1, 1, 1]			
Generation 9100: Best = 1.381680e+03	FES = 910100	Diversity = 4.0282e-04	k_t
= [4, 4, 4, 3]			
Generation 9110: Best = 1.381680e+03	FES = 911100	Diversity = 3.4281e-04	k_t
= [1, 1, 1, 1]			
Generation 9120: Best = 1.381679e+03	FES = 912100	Diversity = 3.0635e-04	k_t
= [3, 4, 2, 4]			
Generation 9130: Best = 1.381679e+03	FES = 913100	Diversity = 3.6710e-04	k_t
= [1, 1, 1, 1]			

Generation 9140: Best = 1.381678e+03 = [2, 3, 4, 3]	FES = 914100	Diversity = 3.9628e-04	k_t
Generation 9150: Best = 1.381678e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 3.4492e-04	k_t
Generation 9160: Best = 1.381677e+03 = [2, 4, 2, 2]	FES = 916100	Diversity = 3.3409e-04	k_t
Generation 9170: Best = 1.381677e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 2.7410e-04	k_t
Generation 9180: Best = 1.381677e+03 = [3, 4, 4, 3]	FES = 918100	Diversity = 2.4901e-04	k_t
Generation 9190: Best = 1.381677e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 2.7908e-04	k_t
Generation 9200: Best = 1.381676e+03 = [3, 3, 2, 4]	FES = 920100	Diversity = 3.1430e-04	k_t
Generation 9210: Best = 1.381676e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 3.0678e-04	k_t
Generation 9220: Best = 1.381676e+03 = [4, 3, 2, 3]	FES = 922100	Diversity = 2.4295e-04	k_t
Generation 9230: Best = 1.381676e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 2.4312e-04	k_t
Generation 9240: Best = 1.381675e+03 = [3, 3, 2, 3]	FES = 924100	Diversity = 2.5395e-04	k_t
Generation 9250: Best = 1.381675e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 2.9912e-04	k_t
Generation 9260: Best = 1.381675e+03 = [4, 4, 4, 2]	FES = 926100	Diversity = 2.8273e-04	k_t
Generation 9270: Best = 1.381674e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 3.2136e-04	k_t
Generation 9280: Best = 1.381674e+03 = [4, 2, 4, 4]	FES = 928100	Diversity = 3.1129e-04	k_t
Generation 9290: Best = 1.381674e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 2.9860e-04	k_t
Generation 9300: Best = 1.381673e+03 = [4, 3, 4, 3]	FES = 930100	Diversity = 2.6845e-04	k_t
Generation 9310: Best = 1.381673e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 2.4791e-04	k_t
Generation 9320: Best = 1.381673e+03 = [4, 3, 2, 2]	FES = 932100	Diversity = 2.1489e-04	k_t
Generation 9330: Best = 1.381673e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 2.9139e-04	k_t
Generation 9340: Best = 1.381672e+03 = [2, 2, 2, 2]	FES = 934100	Diversity = 2.8912e-04	k_t
Generation 9350: Best = 1.381672e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 2.4522e-04	k_t
Generation 9360: Best = 1.381672e+03 = [3, 2, 2, 3]	FES = 936100	Diversity = 2.8424e-04	k_t
Generation 9370: Best = 1.381672e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 2.5249e-04	k_t
Generation 9380: Best = 1.381672e+03 = [2, 3, 4, 2]	FES = 938100	Diversity = 2.6465e-04	k_t
Generation 9390: Best = 1.381671e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 3.0227e-04	k_t
Generation 9400: Best = 1.381671e+03 = [4, 4, 2, 3]	FES = 940100	Diversity = 3.0561e-04	k_t

Generation 9410: Best = 1.381671e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 3.0167e-04	k_t
Generation 9420: Best = 1.381671e+03 = [4, 2, 3, 3]	FES = 942100	Diversity = 3.1033e-04	k_t
Generation 9430: Best = 1.381670e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 2.7822e-04	k_t
Generation 9440: Best = 1.381670e+03 = [4, 3, 2, 3]	FES = 944100	Diversity = 2.2058e-04	k_t
Generation 9450: Best = 1.381670e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 2.2860e-04	k_t
Generation 9460: Best = 1.381670e+03 = [2, 2, 2, 3]	FES = 946100	Diversity = 2.6675e-04	k_t
Generation 9470: Best = 1.381670e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 2.9005e-04	k_t
Generation 9480: Best = 1.381669e+03 = [2, 4, 2, 3]	FES = 948100	Diversity = 2.8475e-04	k_t
Generation 9490: Best = 1.381669e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 3.1007e-04	k_t
Generation 9500: Best = 1.381669e+03 = [4, 3, 3, 2]	FES = 950100	Diversity = 3.8451e-04	k_t
Generation 9510: Best = 1.381668e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 4.0527e-04	k_t
Generation 9520: Best = 1.381668e+03 = [2, 4, 2, 3]	FES = 952100	Diversity = 4.0486e-04	k_t
Generation 9530: Best = 1.381668e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 3.6780e-04	k_t
Generation 9540: Best = 1.381667e+03 = [2, 4, 2, 3]	FES = 954100	Diversity = 2.6130e-04	k_t
Generation 9550: Best = 1.381667e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 2.4303e-04	k_t
Generation 9560: Best = 1.381667e+03 = [3, 2, 2, 2]	FES = 956100	Diversity = 2.5205e-04	k_t
Generation 9570: Best = 1.381666e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 3.0114e-04	k_t
Generation 9580: Best = 1.381666e+03 = [2, 2, 4, 2]	FES = 958100	Diversity = 2.4889e-04	k_t
Generation 9590: Best = 1.381666e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 2.6190e-04	k_t
Generation 9600: Best = 1.381666e+03 = [3, 3, 3, 3]	FES = 960100	Diversity = 2.2441e-04	k_t
Generation 9610: Best = 1.381665e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 2.5369e-04	k_t
Generation 9620: Best = 1.381665e+03 = [3, 3, 3, 2]	FES = 962100	Diversity = 2.6771e-04	k_t
Generation 9630: Best = 1.381665e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 2.8812e-04	k_t
Generation 9640: Best = 1.381665e+03 = [4, 3, 3, 2]	FES = 964100	Diversity = 2.5409e-04	k_t
Generation 9650: Best = 1.381665e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 2.7487e-04	k_t
Generation 9660: Best = 1.381664e+03 = [2, 3, 3, 2]	FES = 966100	Diversity = 2.9147e-04	k_t
Generation 9670: Best = 1.381664e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 2.3844e-04	k_t

Generation 9680: Best = 1.381664e+03 = [4, 3, 2, 4]	FES = 968100	Diversity = 2.6263e-04	k_t
Generation 9690: Best = 1.381664e+03 = [1, 1, 1, 1]	FES = 969100	Diversity = 2.6662e-04	k_t
Generation 9700: Best = 1.381663e+03 = [4, 2, 4, 2]	FES = 970100	Diversity = 2.0581e-04	k_t
Generation 9710: Best = 1.381663e+03 = [1, 1, 1, 1]	FES = 971100	Diversity = 2.3485e-04	k_t
Generation 9720: Best = 1.381663e+03 = [3, 4, 4, 2]	FES = 972100	Diversity = 2.3063e-04	k_t
Generation 9730: Best = 1.381663e+03 = [1, 1, 1, 1]	FES = 973100	Diversity = 2.3458e-04	k_t
Generation 9740: Best = 1.381663e+03 = [3, 2, 2, 2]	FES = 974100	Diversity = 2.4137e-04	k_t
Generation 9750: Best = 1.381662e+03 = [1, 1, 1, 1]	FES = 975100	Diversity = 3.1166e-04	k_t
Generation 9760: Best = 1.381662e+03 = [4, 3, 2, 3]	FES = 976100	Diversity = 2.7818e-04	k_t
Generation 9770: Best = 1.381662e+03 = [1, 1, 1, 1]	FES = 977100	Diversity = 2.6790e-04	k_t
Generation 9780: Best = 1.381662e+03 = [4, 3, 2, 3]	FES = 978100	Diversity = 2.9385e-04	k_t
Generation 9790: Best = 1.381661e+03 = [1, 1, 1, 1]	FES = 979100	Diversity = 2.7438e-04	k_t
Generation 9800: Best = 1.381661e+03 = [4, 4, 3, 2]	FES = 980100	Diversity = 3.8695e-04	k_t
Generation 9810: Best = 1.381661e+03 = [1, 1, 1, 1]	FES = 981100	Diversity = 3.9798e-04	k_t
Generation 9820: Best = 1.381660e+03 = [2, 3, 4, 3]	FES = 982100	Diversity = 3.0958e-04	k_t
Generation 9830: Best = 1.381660e+03 = [1, 1, 1, 1]	FES = 983100	Diversity = 2.7155e-04	k_t
Generation 9840: Best = 1.381660e+03 = [2, 2, 4, 4]	FES = 984100	Diversity = 3.0918e-04	k_t
Generation 9850: Best = 1.381659e+03 = [1, 1, 1, 1]	FES = 985100	Diversity = 3.1907e-04	k_t
Generation 9860: Best = 1.381659e+03 = [3, 3, 4, 4]	FES = 986100	Diversity = 2.9494e-04	k_t
Generation 9870: Best = 1.381659e+03 = [1, 1, 1, 1]	FES = 987100	Diversity = 3.0998e-04	k_t
Generation 9880: Best = 1.381658e+03 = [2, 3, 4, 3]	FES = 988100	Diversity = 3.2597e-04	k_t
Generation 9890: Best = 1.381658e+03 = [1, 1, 1, 1]	FES = 989100	Diversity = 3.2788e-04	k_t
Generation 9900: Best = 1.381658e+03 = [2, 4, 4, 2]	FES = 990100	Diversity = 3.2203e-04	k_t
Generation 9910: Best = 1.381657e+03 = [1, 1, 1, 1]	FES = 991100	Diversity = 3.4566e-04	k_t
Generation 9920: Best = 1.381657e+03 = [3, 4, 2, 4]	FES = 992100	Diversity = 3.5443e-04	k_t
Generation 9930: Best = 1.381657e+03 = [1, 1, 1, 1]	FES = 993100	Diversity = 3.9247e-04	k_t
Generation 9940: Best = 1.381656e+03 = [3, 4, 2, 3]	FES = 994100	Diversity = 3.5266e-04	k_t

Generation 9950: Best = 1.381656e+03 FES = 995100 Diversity = 2.9859e-04 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 1.381656e+03 FES = 996100 Diversity = 2.7779e-04 k_t
= [3, 3, 2, 2]
Generation 9970: Best = 1.381656e+03 FES = 997100 Diversity = 2.5020e-04 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 1.381655e+03 FES = 998100 Diversity = 2.6222e-04 k_t
= [2, 2, 2, 3]
Generation 9990: Best = 1.381655e+03 FES = 999100 Diversity = 2.2192e-04 k_t
= [1, 1, 1, 1]

Optimization completed!

Final Best Fitness = 1.381655e+03

Total Function Evaluations = 1000000

=====
=

Results:

Best Fitness : 1.381655e+03

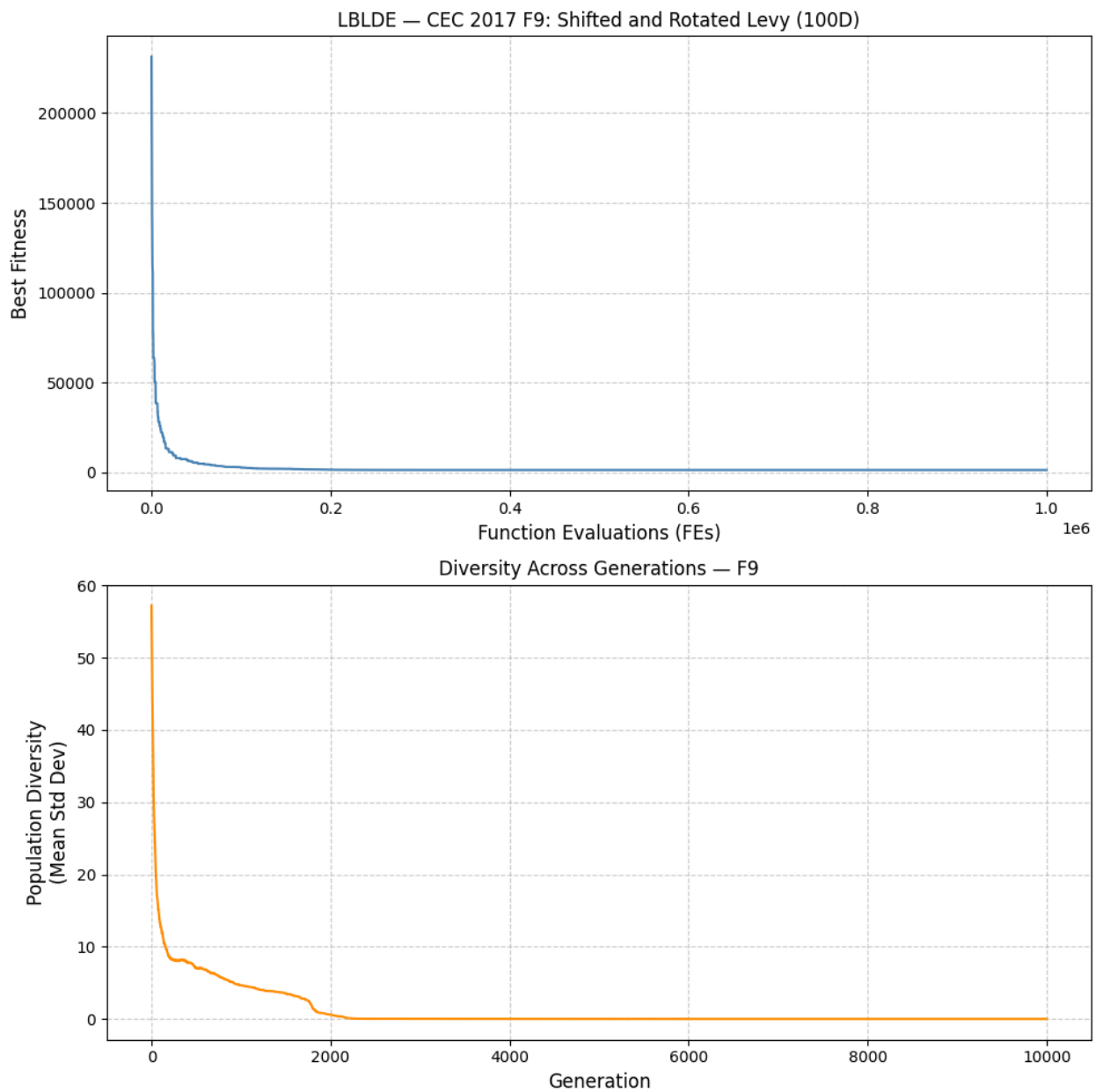
Global Optimum : 9.000000e+02

Error (|best-optimum|) : 4.816550e+02

Best Solution (5 dim) : [-28.64986292 6.1129623 39.75410227 58.32136598
-23.02229486]

D^(0) (init. diversity): 5.7220e+01

=====
=



Graph saved as convergence_F9.png

```
In [34]: # =====
# F10 – Shifted and Rotated Schwefel
# =====
run_and_plot('F10', make_wrapper(simple.f10), 1000.0, "Shifted and Rotated Sch
```



```

=====
=
LBLDE_s on CEC 2017 F10 – Single Run
=====
=
Function      : Shifted and Rotated Schwefel's (100D)
Bounds       : [-100, 100]
Global optimum : 1000.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 3.685326e+04 | Diversity = 5.7220e+01
Generation 10: Best = 3.402748e+04 FES = 1100 Diversity = 5.5455e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.307363e+04 FES = 2100 Diversity = 5.5617e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 3.307363e+04 FES = 3100 Diversity = 5.6039e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 3.304676e+04 FES = 4100 Diversity = 5.6054e+01 k_t
= [4, 3, 4, 2]
Generation 50: Best = 3.276292e+04 FES = 5100 Diversity = 5.6627e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.230560e+04 FES = 6100 Diversity = 5.6517e+01 k_t
= [4, 2, 3, 4]
Generation 70: Best = 3.179768e+04 FES = 7100 Diversity = 5.6717e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.178864e+04 FES = 8100 Diversity = 5.7044e+01 k_t
= [2, 4, 4, 2]
Generation 90: Best = 3.178864e+04 FES = 9100 Diversity = 5.6910e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.107707e+04 FES = 10100 Diversity = 5.6601e+01 k_t
= [2, 4, 4, 2]
Generation 110: Best = 3.107707e+04 FES = 11100 Diversity = 5.6432e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.107707e+04 FES = 12100 Diversity = 5.6365e+01 k_t
= [3, 3, 4, 3]
Generation 130: Best = 3.096682e+04 FES = 13100 Diversity = 5.6558e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.019933e+04 FES = 14100 Diversity = 5.6668e+01 k_t
= [2, 4, 2, 3]
Generation 150: Best = 2.959856e+04 FES = 15100 Diversity = 5.6602e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 2.959856e+04 FES = 16100 Diversity = 5.6912e+01 k_t
= [3, 4, 3, 2]
Generation 170: Best = 2.921417e+04 FES = 17100 Diversity = 5.6996e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 2.921417e+04 FES = 18100 Diversity = 5.7006e+01 k_t
= [4, 4, 2, 2]
Generation 190: Best = 2.921417e+04 FES = 19100 Diversity = 5.6849e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 2.921417e+04 FES = 20100 Diversity = 5.6941e+01 k_t
= [3, 2, 3, 2]
Generation 210: Best = 2.921417e+04 FES = 21100 Diversity = 5.6905e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.830509e+04 FES = 22100 Diversity = 5.6697e+01 k_t
= [3, 2, 2, 4]

```

Generation 230:	Best = 2.830509e+04	FES = 23100	Diversity = 5.6722e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 2.830509e+04	FES = 24100	Diversity = 5.6850e+01	k_t
= [2, 4, 4, 3]				
Generation 250:	Best = 2.830509e+04	FES = 25100	Diversity = 5.6704e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 2.830509e+04	FES = 26100	Diversity = 5.6588e+01	k_t
= [3, 3, 4, 4]				
Generation 270:	Best = 2.829474e+04	FES = 27100	Diversity = 5.6571e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 2.829474e+04	FES = 28100	Diversity = 5.6730e+01	k_t
= [4, 3, 4, 3]				
Generation 290:	Best = 2.829474e+04	FES = 29100	Diversity = 5.6740e+01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 2.775469e+04	FES = 30100	Diversity = 5.6600e+01	k_t
= [2, 2, 4, 3]				
Generation 310:	Best = 2.775469e+04	FES = 31100	Diversity = 5.6602e+01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 2.775469e+04	FES = 32100	Diversity = 5.6639e+01	k_t
= [2, 2, 4, 3]				
Generation 330:	Best = 2.775469e+04	FES = 33100	Diversity = 5.6706e+01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 2.772826e+04	FES = 34100	Diversity = 5.6274e+01	k_t
= [3, 2, 3, 3]				
Generation 350:	Best = 2.772826e+04	FES = 35100	Diversity = 5.6145e+01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 2.772826e+04	FES = 36100	Diversity = 5.6184e+01	k_t
= [4, 4, 2, 3]				
Generation 370:	Best = 2.772826e+04	FES = 37100	Diversity = 5.6183e+01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 2.772826e+04	FES = 38100	Diversity = 5.6122e+01	k_t
= [3, 4, 3, 3]				
Generation 390:	Best = 2.772826e+04	FES = 39100	Diversity = 5.6269e+01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 2.772826e+04	FES = 40100	Diversity = 5.6070e+01	k_t
= [3, 2, 3, 2]				
Generation 410:	Best = 2.772826e+04	FES = 41100	Diversity = 5.6045e+01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 2.759513e+04	FES = 42100	Diversity = 5.6053e+01	k_t
= [2, 4, 2, 2]				
Generation 430:	Best = 2.759513e+04	FES = 43100	Diversity = 5.6074e+01	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 2.759513e+04	FES = 44100	Diversity = 5.5830e+01	k_t
= [3, 3, 2, 2]				
Generation 450:	Best = 2.735782e+04	FES = 45100	Diversity = 5.5710e+01	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 2.705821e+04	FES = 46100	Diversity = 5.5738e+01	k_t
= [2, 4, 2, 3]				
Generation 470:	Best = 2.705821e+04	FES = 47100	Diversity = 5.5438e+01	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 2.705821e+04	FES = 48100	Diversity = 5.5473e+01	k_t
= [4, 4, 2, 2]				
Generation 490:	Best = 2.705821e+04	FES = 49100	Diversity = 5.5486e+01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 2.705821e+04	FES = 50100	Diversity = 5.5563e+01	k_t
= [3, 3, 4, 4]				
Generation 510:	Best = 2.612161e+04	FES = 51100	Diversity = 5.5504e+01	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 2.612161e+04	FES = 52100	Diversity = 5.5407e+01	k_t
= [2, 2, 3, 4]				
Generation 530:	Best = 2.612161e+04	FES = 53100	Diversity = 5.5391e+01	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 2.612161e+04	FES = 54100	Diversity = 5.5533e+01	k_t
= [2, 4, 2, 2]				
Generation 550:	Best = 2.612161e+04	FES = 55100	Diversity = 5.5392e+01	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 2.612161e+04	FES = 56100	Diversity = 5.5375e+01	k_t
= [4, 4, 2, 2]				
Generation 570:	Best = 2.612161e+04	FES = 57100	Diversity = 5.5169e+01	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 2.612161e+04	FES = 58100	Diversity = 5.5004e+01	k_t
= [3, 2, 2, 2]				
Generation 590:	Best = 2.602797e+04	FES = 59100	Diversity = 5.4926e+01	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 2.602797e+04	FES = 60100	Diversity = 5.4880e+01	k_t
= [4, 4, 4, 4]				
Generation 610:	Best = 2.602797e+04	FES = 61100	Diversity = 5.5030e+01	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 2.602797e+04	FES = 62100	Diversity = 5.5044e+01	k_t
= [2, 4, 3, 4]				
Generation 630:	Best = 2.602797e+04	FES = 63100	Diversity = 5.5039e+01	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 2.602797e+04	FES = 64100	Diversity = 5.4981e+01	k_t
= [2, 4, 3, 4]				
Generation 650:	Best = 2.602797e+04	FES = 65100	Diversity = 5.4967e+01	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 2.602797e+04	FES = 66100	Diversity = 5.4510e+01	k_t
= [2, 2, 4, 2]				
Generation 670:	Best = 2.602797e+04	FES = 67100	Diversity = 5.4366e+01	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 2.602797e+04	FES = 68100	Diversity = 5.4365e+01	k_t
= [2, 3, 2, 4]				
Generation 690:	Best = 2.602797e+04	FES = 69100	Diversity = 5.4090e+01	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 2.602797e+04	FES = 70100	Diversity = 5.4068e+01	k_t
= [3, 3, 4, 2]				
Generation 710:	Best = 2.602797e+04	FES = 71100	Diversity = 5.4108e+01	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 2.602797e+04	FES = 72100	Diversity = 5.3933e+01	k_t
= [4, 2, 4, 2]				
Generation 730:	Best = 2.602797e+04	FES = 73100	Diversity = 5.3484e+01	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 2.602797e+04	FES = 74100	Diversity = 5.3244e+01	k_t
= [2, 2, 4, 2]				
Generation 750:	Best = 2.601230e+04	FES = 75100	Diversity = 5.3387e+01	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 2.601230e+04	FES = 76100	Diversity = 5.3371e+01	k_t
= [3, 4, 2, 3]				

Generation 770:	Best = 2.589415e+04	FES = 77100	Diversity = 5.2784e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.589415e+04	FES = 78100	Diversity = 5.2160e+01	k_t
= [4, 2, 3, 4]				
Generation 790:	Best = 2.589415e+04	FES = 79100	Diversity = 5.1874e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.589415e+04	FES = 80100	Diversity = 5.1525e+01	k_t
= [4, 4, 2, 2]				
Generation 810:	Best = 2.501960e+04	FES = 81100	Diversity = 5.1001e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.501960e+04	FES = 82100	Diversity = 5.0564e+01	k_t
= [3, 2, 2, 3]				
Generation 830:	Best = 2.501960e+04	FES = 83100	Diversity = 5.0459e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.478979e+04	FES = 84100	Diversity = 4.9746e+01	k_t
= [2, 4, 3, 2]				
Generation 850:	Best = 2.455952e+04	FES = 85100	Diversity = 4.8740e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.455952e+04	FES = 86100	Diversity = 4.7537e+01	k_t
= [3, 4, 3, 2]				
Generation 870:	Best = 2.423797e+04	FES = 87100	Diversity = 4.6822e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.423797e+04	FES = 88100	Diversity = 4.6224e+01	k_t
= [3, 2, 2, 3]				
Generation 890:	Best = 2.408788e+04	FES = 89100	Diversity = 4.4317e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.408766e+04	FES = 90100	Diversity = 4.4671e+01	k_t
= [4, 4, 2, 3]				
Generation 910:	Best = 2.408766e+04	FES = 91100	Diversity = 4.3011e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.387979e+04	FES = 92100	Diversity = 4.1351e+01	k_t
= [3, 2, 4, 2]				
Generation 930:	Best = 2.385234e+04	FES = 93100	Diversity = 4.1698e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.349686e+04	FES = 94100	Diversity = 4.0336e+01	k_t
= [2, 4, 2, 2]				
Generation 950:	Best = 2.334171e+04	FES = 95100	Diversity = 3.8210e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.307833e+04	FES = 96100	Diversity = 3.6409e+01	k_t
= [3, 3, 3, 3]				
Generation 970:	Best = 2.184318e+04	FES = 97100	Diversity = 3.5028e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.176058e+04	FES = 98100	Diversity = 3.1673e+01	k_t
= [3, 2, 4, 2]				
Generation 990:	Best = 2.100357e+04	FES = 99100	Diversity = 3.0916e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 2.034170e+04	FES = 100100	Diversity = 2.8868e+01	k_t
= [3, 3, 2, 3]				
Generation 1010:	Best = 2.022515e+04	FES = 101100	Diversity = 2.6854e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.992769e+04	FES = 102100	Diversity = 2.3443e+01	k_t
= [3, 2, 2, 4]				
Generation 1030:	Best = 1.943199e+04	FES = 103100	Diversity = 2.0384e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.918442e+04 = [2, 2, 2, 3]	FES = 104100	Diversity = 1.7749e+01	k_t
Generation 1050: Best = 1.880992e+04 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.3740e+01	k_t
Generation 1060: Best = 1.808243e+04 = [2, 2, 2, 2]	FES = 106100	Diversity = 8.6900e+00	k_t
Generation 1070: Best = 1.798531e+04 = [1, 1, 1, 1]	FES = 107100	Diversity = 8.6621e+00	k_t
Generation 1080: Best = 1.754569e+04 = [3, 2, 4, 3]	FES = 108100	Diversity = 3.0215e+00	k_t
Generation 1090: Best = 1.733965e+04 = [1, 1, 1, 1]	FES = 109100	Diversity = 2.8536e+00	k_t
Generation 1100: Best = 1.714868e+04 = [4, 2, 3, 4]	FES = 110100	Diversity = 2.8062e+00	k_t
Generation 1110: Best = 1.670664e+04 = [1, 1, 1, 1]	FES = 111100	Diversity = 2.7867e+00	k_t
Generation 1120: Best = 1.670664e+04 = [2, 4, 4, 2]	FES = 112100	Diversity = 2.6982e+00	k_t
Generation 1130: Best = 1.633105e+04 = [1, 1, 1, 1]	FES = 113100	Diversity = 2.6138e+00	k_t
Generation 1140: Best = 1.621571e+04 = [2, 4, 3, 3]	FES = 114100	Diversity = 2.3922e+00	k_t
Generation 1150: Best = 1.615771e+04 = [1, 1, 1, 1]	FES = 115100	Diversity = 2.1794e+00	k_t
Generation 1160: Best = 1.593363e+04 = [4, 2, 3, 4]	FES = 116100	Diversity = 1.8872e+00	k_t
Generation 1170: Best = 1.563281e+04 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.8511e+00	k_t
Generation 1180: Best = 1.561853e+04 = [3, 4, 4, 4]	FES = 118100	Diversity = 1.7908e+00	k_t
Generation 1190: Best = 1.550882e+04 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.7894e+00	k_t
Generation 1200: Best = 1.535462e+04 = [4, 3, 2, 3]	FES = 120100	Diversity = 1.7172e+00	k_t
Generation 1210: Best = 1.528983e+04 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.7188e+00	k_t
Generation 1220: Best = 1.517692e+04 = [4, 2, 4, 3]	FES = 122100	Diversity = 1.6191e+00	k_t
Generation 1230: Best = 1.509446e+04 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.6026e+00	k_t
Generation 1240: Best = 1.495746e+04 = [4, 2, 3, 3]	FES = 124100	Diversity = 1.5972e+00	k_t
Generation 1250: Best = 1.487126e+04 = [1, 1, 1, 1]	FES = 125100	Diversity = 1.5533e+00	k_t
Generation 1260: Best = 1.484978e+04 = [3, 3, 4, 4]	FES = 126100	Diversity = 1.5126e+00	k_t
Generation 1270: Best = 1.477917e+04 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.4930e+00	k_t
Generation 1280: Best = 1.463600e+04 = [4, 3, 2, 3]	FES = 128100	Diversity = 1.4860e+00	k_t
Generation 1290: Best = 1.460927e+04 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.4497e+00	k_t
Generation 1300: Best = 1.457076e+04 = [2, 3, 3, 3]	FES = 130100	Diversity = 1.4108e+00	k_t

Generation 1310: Best = 1.455222e+04 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.3935e+00	k_t
Generation 1320: Best = 1.449498e+04 = [2, 4, 3, 4]	FES = 132100	Diversity = 1.2843e+00	k_t
Generation 1330: Best = 1.441991e+04 = [1, 1, 1, 1]	FES = 133100	Diversity = 1.2515e+00	k_t
Generation 1340: Best = 1.437802e+04 = [3, 2, 2, 4]	FES = 134100	Diversity = 1.2219e+00	k_t
Generation 1350: Best = 1.433485e+04 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.1664e+00	k_t
Generation 1360: Best = 1.423077e+04 = [3, 3, 2, 3]	FES = 136100	Diversity = 1.1130e+00	k_t
Generation 1370: Best = 1.419038e+04 = [1, 1, 1, 1]	FES = 137100	Diversity = 9.8807e-01	k_t
Generation 1380: Best = 1.412985e+04 = [4, 2, 3, 4]	FES = 138100	Diversity = 8.8075e-01	k_t
Generation 1390: Best = 1.409636e+04 = [1, 1, 1, 1]	FES = 139100	Diversity = 6.9211e-01	k_t
Generation 1400: Best = 1.405564e+04 = [2, 3, 4, 2]	FES = 140100	Diversity = 4.8919e-01	k_t
Generation 1410: Best = 1.400267e+04 = [1, 1, 1, 1]	FES = 141100	Diversity = 2.5118e-01	k_t
Generation 1420: Best = 1.395709e+04 = [4, 2, 2, 3]	FES = 142100	Diversity = 2.0733e-01	k_t
Generation 1430: Best = 1.392741e+04 = [1, 1, 1, 1]	FES = 143100	Diversity = 1.7100e-01	k_t
Generation 1440: Best = 1.390250e+04 = [2, 3, 2, 2]	FES = 144100	Diversity = 1.3253e-01	k_t
Generation 1450: Best = 1.389177e+04 = [1, 1, 1, 1]	FES = 145100	Diversity = 1.0596e-01	k_t
Generation 1460: Best = 1.387928e+04 = [4, 3, 3, 3]	FES = 146100	Diversity = 9.4134e-02	k_t
Generation 1470: Best = 1.386902e+04 = [1, 1, 1, 1]	FES = 147100	Diversity = 8.2355e-02	k_t
Generation 1480: Best = 1.386165e+04 = [3, 3, 2, 3]	FES = 148100	Diversity = 6.8732e-02	k_t
Generation 1490: Best = 1.385618e+04 = [1, 1, 1, 1]	FES = 149100	Diversity = 6.0923e-02	k_t
Generation 1500: Best = 1.385073e+04 = [3, 4, 2, 2]	FES = 150100	Diversity = 5.3018e-02	k_t
Generation 1510: Best = 1.384617e+04 = [1, 1, 1, 1]	FES = 151100	Diversity = 4.7875e-02	k_t
Generation 1520: Best = 1.384360e+04 = [2, 3, 3, 2]	FES = 152100	Diversity = 4.0605e-02	k_t
Generation 1530: Best = 1.384098e+04 = [1, 1, 1, 1]	FES = 153100	Diversity = 3.7923e-02	k_t
Generation 1540: Best = 1.383800e+04 = [3, 3, 4, 3]	FES = 154100	Diversity = 3.7074e-02	k_t
Generation 1550: Best = 1.383611e+04 = [1, 1, 1, 1]	FES = 155100	Diversity = 3.5039e-02	k_t
Generation 1560: Best = 1.383357e+04 = [3, 3, 4, 4]	FES = 156100	Diversity = 3.3037e-02	k_t
Generation 1570: Best = 1.382905e+04 = [1, 1, 1, 1]	FES = 157100	Diversity = 3.5085e-02	k_t

Generation 1580: Best = 1.382610e+04 = [4, 2, 2, 2]	FES = 158100	Diversity = 3.5780e-02	k_t
Generation 1590: Best = 1.382487e+04 = [1, 1, 1, 1]	FES = 159100	Diversity = 3.3217e-02	k_t
Generation 1600: Best = 1.382356e+04 = [4, 2, 4, 3]	FES = 160100	Diversity = 2.9747e-02	k_t
Generation 1610: Best = 1.382249e+04 = [1, 1, 1, 1]	FES = 161100	Diversity = 2.6864e-02	k_t
Generation 1620: Best = 1.382194e+04 = [3, 4, 4, 2]	FES = 162100	Diversity = 2.4182e-02	k_t
Generation 1630: Best = 1.382117e+04 = [1, 1, 1, 1]	FES = 163100	Diversity = 2.0809e-02	k_t
Generation 1640: Best = 1.382082e+04 = [4, 3, 2, 2]	FES = 164100	Diversity = 1.8119e-02	k_t
Generation 1650: Best = 1.382049e+04 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.6247e-02	k_t
Generation 1660: Best = 1.382008e+04 = [3, 4, 3, 3]	FES = 166100	Diversity = 1.4900e-02	k_t
Generation 1670: Best = 1.381987e+04 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.3473e-02	k_t
Generation 1680: Best = 1.381956e+04 = [2, 4, 4, 4]	FES = 168100	Diversity = 1.2546e-02	k_t
Generation 1690: Best = 1.381942e+04 = [1, 1, 1, 1]	FES = 169100	Diversity = 1.1367e-02	k_t
Generation 1700: Best = 1.381929e+04 = [3, 3, 4, 3]	FES = 170100	Diversity = 1.0415e-02	k_t
Generation 1710: Best = 1.381912e+04 = [1, 1, 1, 1]	FES = 171100	Diversity = 9.7158e-03	k_t
Generation 1720: Best = 1.381899e+04 = [4, 2, 3, 2]	FES = 172100	Diversity = 9.3960e-03	k_t
Generation 1730: Best = 1.381882e+04 = [1, 1, 1, 1]	FES = 173100	Diversity = 8.4974e-03	k_t
Generation 1740: Best = 1.381874e+04 = [3, 4, 2, 2]	FES = 174100	Diversity = 7.9591e-03	k_t
Generation 1750: Best = 1.381861e+04 = [1, 1, 1, 1]	FES = 175100	Diversity = 8.0409e-03	k_t
Generation 1760: Best = 1.381849e+04 = [3, 2, 3, 3]	FES = 176100	Diversity = 7.6489e-03	k_t
Generation 1770: Best = 1.381841e+04 = [1, 1, 1, 1]	FES = 177100	Diversity = 7.5118e-03	k_t
Generation 1780: Best = 1.381831e+04 = [3, 3, 2, 3]	FES = 178100	Diversity = 7.0606e-03	k_t
Generation 1790: Best = 1.381821e+04 = [1, 1, 1, 1]	FES = 179100	Diversity = 6.9171e-03	k_t
Generation 1800: Best = 1.381816e+04 = [4, 2, 3, 3]	FES = 180100	Diversity = 7.1352e-03	k_t
Generation 1810: Best = 1.381805e+04 = [1, 1, 1, 1]	FES = 181100	Diversity = 7.2395e-03	k_t
Generation 1820: Best = 1.381799e+04 = [2, 4, 4, 2]	FES = 182100	Diversity = 7.4794e-03	k_t
Generation 1830: Best = 1.381789e+04 = [1, 1, 1, 1]	FES = 183100	Diversity = 7.6027e-03	k_t
Generation 1840: Best = 1.381783e+04 = [2, 2, 4, 4]	FES = 184100	Diversity = 7.3376e-03	k_t

Generation 1850: Best = 1.381779e+04 = [1, 1, 1, 1]	FES = 185100	Diversity = 6.4386e-03	k_t
Generation 1860: Best = 1.381776e+04 = [3, 3, 2, 3]	FES = 186100	Diversity = 5.8715e-03	k_t
Generation 1870: Best = 1.381773e+04 = [1, 1, 1, 1]	FES = 187100	Diversity = 5.3394e-03	k_t
Generation 1880: Best = 1.381769e+04 = [4, 2, 3, 3]	FES = 188100	Diversity = 4.8520e-03	k_t
Generation 1890: Best = 1.381765e+04 = [1, 1, 1, 1]	FES = 189100	Diversity = 4.7498e-03	k_t
Generation 1900: Best = 1.381763e+04 = [3, 3, 3, 3]	FES = 190100	Diversity = 4.4899e-03	k_t
Generation 1910: Best = 1.381761e+04 = [1, 1, 1, 1]	FES = 191100	Diversity = 4.2219e-03	k_t
Generation 1920: Best = 1.381758e+04 = [4, 3, 4, 3]	FES = 192100	Diversity = 3.8249e-03	k_t
Generation 1930: Best = 1.381756e+04 = [1, 1, 1, 1]	FES = 193100	Diversity = 3.5067e-03	k_t
Generation 1940: Best = 1.381754e+04 = [4, 3, 2, 4]	FES = 194100	Diversity = 3.2850e-03	k_t
Generation 1950: Best = 1.381753e+04 = [1, 1, 1, 1]	FES = 195100	Diversity = 3.0574e-03	k_t
Generation 1960: Best = 1.381751e+04 = [4, 3, 4, 3]	FES = 196100	Diversity = 3.0032e-03	k_t
Generation 1970: Best = 1.381750e+04 = [1, 1, 1, 1]	FES = 197100	Diversity = 2.8731e-03	k_t
Generation 1980: Best = 1.381748e+04 = [4, 4, 4, 3]	FES = 198100	Diversity = 2.7991e-03	k_t
Generation 1990: Best = 1.381748e+04 = [1, 1, 1, 1]	FES = 199100	Diversity = 2.7841e-03	k_t
Generation 2000: Best = 1.381747e+04 = [3, 2, 4, 2]	FES = 200100	Diversity = 2.8117e-03	k_t
Generation 2010: Best = 1.381745e+04 = [1, 1, 1, 1]	FES = 201100	Diversity = 2.8619e-03	k_t
Generation 2020: Best = 1.381744e+04 = [3, 4, 3, 3]	FES = 202100	Diversity = 2.7570e-03	k_t
Generation 2030: Best = 1.381742e+04 = [1, 1, 1, 1]	FES = 203100	Diversity = 2.6745e-03	k_t
Generation 2040: Best = 1.381741e+04 = [3, 2, 3, 3]	FES = 204100	Diversity = 2.6831e-03	k_t
Generation 2050: Best = 1.381741e+04 = [1, 1, 1, 1]	FES = 205100	Diversity = 2.5710e-03	k_t
Generation 2060: Best = 1.381740e+04 = [2, 4, 4, 4]	FES = 206100	Diversity = 2.3637e-03	k_t
Generation 2070: Best = 1.381739e+04 = [1, 1, 1, 1]	FES = 207100	Diversity = 2.1570e-03	k_t
Generation 2080: Best = 1.381739e+04 = [2, 4, 4, 4]	FES = 208100	Diversity = 2.0155e-03	k_t
Generation 2090: Best = 1.381738e+04 = [1, 1, 1, 1]	FES = 209100	Diversity = 2.0391e-03	k_t
Generation 2100: Best = 1.381738e+04 = [4, 3, 4, 2]	FES = 210100	Diversity = 2.0041e-03	k_t
Generation 2110: Best = 1.381737e+04 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.8874e-03	k_t

Generation 2120: Best = 1.381737e+04	FES = 212100	Diversity = 1.7798e-03	k_t
= [2, 3, 3, 2]			
Generation 2130: Best = 1.381736e+04	FES = 213100	Diversity = 1.7956e-03	k_t
= [1, 1, 1, 1]			
Generation 2140: Best = 1.381736e+04	FES = 214100	Diversity = 1.8643e-03	k_t
= [4, 4, 3, 2]			
Generation 2150: Best = 1.381736e+04	FES = 215100	Diversity = 1.8218e-03	k_t
= [1, 1, 1, 1]			
Generation 2160: Best = 1.381735e+04	FES = 216100	Diversity = 1.6459e-03	k_t
= [4, 3, 3, 4]			
Generation 2170: Best = 1.381735e+04	FES = 217100	Diversity = 1.5611e-03	k_t
= [1, 1, 1, 1]			
Generation 2180: Best = 1.381735e+04	FES = 218100	Diversity = 1.5699e-03	k_t
= [3, 3, 4, 2]			
Generation 2190: Best = 1.381734e+04	FES = 219100	Diversity = 1.4970e-03	k_t
= [1, 1, 1, 1]			
Generation 2200: Best = 1.381734e+04	FES = 220100	Diversity = 1.4696e-03	k_t
= [2, 2, 2, 4]			
Generation 2210: Best = 1.381734e+04	FES = 221100	Diversity = 1.4281e-03	k_t
= [1, 1, 1, 1]			
Generation 2220: Best = 1.381734e+04	FES = 222100	Diversity = 1.3891e-03	k_t
= [2, 3, 4, 2]			
Generation 2230: Best = 1.381734e+04	FES = 223100	Diversity = 1.3904e-03	k_t
= [1, 1, 1, 1]			
Generation 2240: Best = 1.381733e+04	FES = 224100	Diversity = 1.3758e-03	k_t
= [4, 3, 2, 4]			
Generation 2250: Best = 1.381733e+04	FES = 225100	Diversity = 1.3970e-03	k_t
= [1, 1, 1, 1]			
Generation 2260: Best = 1.381733e+04	FES = 226100	Diversity = 1.3591e-03	k_t
= [3, 2, 4, 3]			
Generation 2270: Best = 1.381732e+04	FES = 227100	Diversity = 1.2968e-03	k_t
= [1, 1, 1, 1]			
Generation 2280: Best = 1.381732e+04	FES = 228100	Diversity = 1.2710e-03	k_t
= [3, 4, 3, 4]			
Generation 2290: Best = 1.381732e+04	FES = 229100	Diversity = 1.2187e-03	k_t
= [1, 1, 1, 1]			
Generation 2300: Best = 1.381732e+04	FES = 230100	Diversity = 1.2190e-03	k_t
= [3, 3, 3, 4]			
Generation 2310: Best = 1.381731e+04	FES = 231100	Diversity = 1.2053e-03	k_t
= [1, 1, 1, 1]			
Generation 2320: Best = 1.381731e+04	FES = 232100	Diversity = 1.2020e-03	k_t
= [2, 3, 3, 2]			
Generation 2330: Best = 1.381731e+04	FES = 233100	Diversity = 1.1608e-03	k_t
= [1, 1, 1, 1]			
Generation 2340: Best = 1.381731e+04	FES = 234100	Diversity = 1.1304e-03	k_t
= [3, 2, 2, 3]			
Generation 2350: Best = 1.381731e+04	FES = 235100	Diversity = 1.0856e-03	k_t
= [1, 1, 1, 1]			
Generation 2360: Best = 1.381731e+04	FES = 236100	Diversity = 1.0278e-03	k_t
= [3, 3, 2, 4]			
Generation 2370: Best = 1.381730e+04	FES = 237100	Diversity = 9.8234e-04	k_t
= [1, 1, 1, 1]			
Generation 2380: Best = 1.381730e+04	FES = 238100	Diversity = 9.7249e-04	k_t
= [4, 2, 3, 2]			

Generation 2390: Best = 1.381730e+04 = [1, 1, 1, 1]	FES = 239100	Diversity = 9.2756e-04	k_t
Generation 2400: Best = 1.381730e+04 = [2, 2, 2, 4]	FES = 240100	Diversity = 9.1184e-04	k_t
Generation 2410: Best = 1.381730e+04 = [1, 1, 1, 1]	FES = 241100	Diversity = 8.6865e-04	k_t
Generation 2420: Best = 1.381730e+04 = [4, 3, 4, 3]	FES = 242100	Diversity = 8.0552e-04	k_t
Generation 2430: Best = 1.381730e+04 = [1, 1, 1, 1]	FES = 243100	Diversity = 7.9674e-04	k_t
Generation 2440: Best = 1.381730e+04 = [2, 4, 2, 3]	FES = 244100	Diversity = 7.6949e-04	k_t
Generation 2450: Best = 1.381730e+04 = [1, 1, 1, 1]	FES = 245100	Diversity = 7.8172e-04	k_t
Generation 2460: Best = 1.381730e+04 = [2, 4, 4, 4]	FES = 246100	Diversity = 7.7308e-04	k_t
Generation 2470: Best = 1.381730e+04 = [1, 1, 1, 1]	FES = 247100	Diversity = 8.3459e-04	k_t
Generation 2480: Best = 1.381729e+04 = [3, 2, 4, 4]	FES = 248100	Diversity = 8.0841e-04	k_t
Generation 2490: Best = 1.381729e+04 = [1, 1, 1, 1]	FES = 249100	Diversity = 9.0477e-04	k_t
Generation 2500: Best = 1.381729e+04 = [3, 4, 3, 2]	FES = 250100	Diversity = 9.3288e-04	k_t
Generation 2510: Best = 1.381729e+04 = [1, 1, 1, 1]	FES = 251100	Diversity = 9.7426e-04	k_t
Generation 2520: Best = 1.381729e+04 = [3, 3, 2, 2]	FES = 252100	Diversity = 9.8290e-04	k_t
Generation 2530: Best = 1.381729e+04 = [1, 1, 1, 1]	FES = 253100	Diversity = 9.2166e-04	k_t
Generation 2540: Best = 1.381729e+04 = [2, 3, 2, 4]	FES = 254100	Diversity = 8.6110e-04	k_t
Generation 2550: Best = 1.381729e+04 = [1, 1, 1, 1]	FES = 255100	Diversity = 8.0596e-04	k_t
Generation 2560: Best = 1.381729e+04 = [3, 4, 2, 3]	FES = 256100	Diversity = 7.5747e-04	k_t
Generation 2570: Best = 1.381728e+04 = [1, 1, 1, 1]	FES = 257100	Diversity = 7.5880e-04	k_t
Generation 2580: Best = 1.381728e+04 = [2, 2, 4, 2]	FES = 258100	Diversity = 7.4989e-04	k_t
Generation 2590: Best = 1.381728e+04 = [1, 1, 1, 1]	FES = 259100	Diversity = 7.3338e-04	k_t
Generation 2600: Best = 1.381728e+04 = [4, 2, 4, 4]	FES = 260100	Diversity = 7.7173e-04	k_t
Generation 2610: Best = 1.381728e+04 = [1, 1, 1, 1]	FES = 261100	Diversity = 8.6405e-04	k_t
Generation 2620: Best = 1.381728e+04 = [2, 3, 2, 4]	FES = 262100	Diversity = 9.9955e-04	k_t
Generation 2630: Best = 1.381728e+04 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.0426e-03	k_t
Generation 2640: Best = 1.381727e+04 = [4, 2, 2, 4]	FES = 264100	Diversity = 9.9352e-04	k_t
Generation 2650: Best = 1.381727e+04 = [1, 1, 1, 1]	FES = 265100	Diversity = 9.5335e-04	k_t

Generation 2660: Best = 1.381727e+04 = [2, 2, 3, 2]	FES = 266100	Diversity = 9.5902e-04	k_t
Generation 2670: Best = 1.381727e+04 = [1, 1, 1, 1]	FES = 267100	Diversity = 9.5773e-04	k_t
Generation 2680: Best = 1.381727e+04 = [2, 2, 4, 3]	FES = 268100	Diversity = 9.5187e-04	k_t
Generation 2690: Best = 1.381726e+04 = [1, 1, 1, 1]	FES = 269100	Diversity = 9.5861e-04	k_t
Generation 2700: Best = 1.381726e+04 = [2, 3, 2, 2]	FES = 270100	Diversity = 1.0012e-03	k_t
Generation 2710: Best = 1.381726e+04 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.0091e-03	k_t
Generation 2720: Best = 1.381726e+04 = [4, 4, 2, 2]	FES = 272100	Diversity = 9.9841e-04	k_t
Generation 2730: Best = 1.381726e+04 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.0088e-03	k_t
Generation 2740: Best = 1.381726e+04 = [3, 4, 3, 2]	FES = 274100	Diversity = 9.3817e-04	k_t
Generation 2750: Best = 1.381726e+04 = [1, 1, 1, 1]	FES = 275100	Diversity = 8.5258e-04	k_t
Generation 2760: Best = 1.381726e+04 = [4, 4, 2, 3]	FES = 276100	Diversity = 7.9755e-04	k_t
Generation 2770: Best = 1.381726e+04 = [1, 1, 1, 1]	FES = 277100	Diversity = 7.6202e-04	k_t
Generation 2780: Best = 1.381726e+04 = [3, 3, 3, 3]	FES = 278100	Diversity = 7.6702e-04	k_t
Generation 2790: Best = 1.381726e+04 = [1, 1, 1, 1]	FES = 279100	Diversity = 7.4770e-04	k_t
Generation 2800: Best = 1.381725e+04 = [3, 4, 3, 2]	FES = 280100	Diversity = 8.4992e-04	k_t
Generation 2810: Best = 1.381725e+04 = [1, 1, 1, 1]	FES = 281100	Diversity = 8.6940e-04	k_t
Generation 2820: Best = 1.381725e+04 = [3, 2, 4, 2]	FES = 282100	Diversity = 8.4091e-04	k_t
Generation 2830: Best = 1.381725e+04 = [1, 1, 1, 1]	FES = 283100	Diversity = 8.2633e-04	k_t
Generation 2840: Best = 1.381725e+04 = [2, 4, 4, 4]	FES = 284100	Diversity = 8.3876e-04	k_t
Generation 2850: Best = 1.381725e+04 = [1, 1, 1, 1]	FES = 285100	Diversity = 8.7416e-04	k_t
Generation 2860: Best = 1.381725e+04 = [3, 2, 3, 3]	FES = 286100	Diversity = 8.6591e-04	k_t
Generation 2870: Best = 1.381725e+04 = [1, 1, 1, 1]	FES = 287100	Diversity = 8.3877e-04	k_t
Generation 2880: Best = 1.381725e+04 = [4, 3, 3, 4]	FES = 288100	Diversity = 8.3682e-04	k_t
Generation 2890: Best = 1.381725e+04 = [1, 1, 1, 1]	FES = 289100	Diversity = 8.2837e-04	k_t
Generation 2900: Best = 1.381724e+04 = [3, 2, 3, 4]	FES = 290100	Diversity = 7.7396e-04	k_t
Generation 2910: Best = 1.381724e+04 = [1, 1, 1, 1]	FES = 291100	Diversity = 7.4344e-04	k_t
Generation 2920: Best = 1.381724e+04 = [4, 4, 3, 2]	FES = 292100	Diversity = 7.1066e-04	k_t

Generation 2930: Best = 1.381724e+04 = [1, 1, 1, 1]	FES = 293100	Diversity = 6.9867e-04	k_t
Generation 2940: Best = 1.381724e+04 = [4, 2, 2, 3]	FES = 294100	Diversity = 6.6510e-04	k_t
Generation 2950: Best = 1.381724e+04 = [1, 1, 1, 1]	FES = 295100	Diversity = 6.5346e-04	k_t
Generation 2960: Best = 1.381724e+04 = [3, 4, 3, 2]	FES = 296100	Diversity = 6.4152e-04	k_t
Generation 2970: Best = 1.381724e+04 = [1, 1, 1, 1]	FES = 297100	Diversity = 6.2310e-04	k_t
Generation 2980: Best = 1.381724e+04 = [3, 2, 3, 4]	FES = 298100	Diversity = 6.3428e-04	k_t
Generation 2990: Best = 1.381724e+04 = [1, 1, 1, 1]	FES = 299100	Diversity = 6.5163e-04	k_t
Generation 3000: Best = 1.381724e+04 = [4, 2, 2, 2]	FES = 300100	Diversity = 6.2448e-04	k_t
Generation 3010: Best = 1.381724e+04 = [1, 1, 1, 1]	FES = 301100	Diversity = 6.1680e-04	k_t
Generation 3020: Best = 1.381724e+04 = [3, 2, 4, 2]	FES = 302100	Diversity = 6.1056e-04	k_t
Generation 3030: Best = 1.381724e+04 = [1, 1, 1, 1]	FES = 303100	Diversity = 6.1865e-04	k_t
Generation 3040: Best = 1.381724e+04 = [3, 3, 2, 4]	FES = 304100	Diversity = 6.3513e-04	k_t
Generation 3050: Best = 1.381723e+04 = [1, 1, 1, 1]	FES = 305100	Diversity = 6.2629e-04	k_t
Generation 3060: Best = 1.381723e+04 = [2, 2, 3, 3]	FES = 306100	Diversity = 6.4642e-04	k_t
Generation 3070: Best = 1.381723e+04 = [1, 1, 1, 1]	FES = 307100	Diversity = 6.3764e-04	k_t
Generation 3080: Best = 1.381723e+04 = [3, 4, 2, 3]	FES = 308100	Diversity = 6.4945e-04	k_t
Generation 3090: Best = 1.381723e+04 = [1, 1, 1, 1]	FES = 309100	Diversity = 6.6620e-04	k_t
Generation 3100: Best = 1.381723e+04 = [2, 3, 3, 2]	FES = 310100	Diversity = 6.5360e-04	k_t
Generation 3110: Best = 1.381723e+04 = [1, 1, 1, 1]	FES = 311100	Diversity = 6.2651e-04	k_t
Generation 3120: Best = 1.381723e+04 = [4, 4, 2, 2]	FES = 312100	Diversity = 5.8409e-04	k_t
Generation 3130: Best = 1.381723e+04 = [1, 1, 1, 1]	FES = 313100	Diversity = 5.8886e-04	k_t
Generation 3140: Best = 1.381723e+04 = [2, 2, 2, 3]	FES = 314100	Diversity = 6.3984e-04	k_t
Generation 3150: Best = 1.381723e+04 = [1, 1, 1, 1]	FES = 315100	Diversity = 6.4134e-04	k_t
Generation 3160: Best = 1.381723e+04 = [2, 4, 3, 4]	FES = 316100	Diversity = 6.3985e-04	k_t
Generation 3170: Best = 1.381723e+04 = [1, 1, 1, 1]	FES = 317100	Diversity = 6.4057e-04	k_t
Generation 3180: Best = 1.381723e+04 = [2, 3, 2, 2]	FES = 318100	Diversity = 6.3847e-04	k_t
Generation 3190: Best = 1.381723e+04 = [1, 1, 1, 1]	FES = 319100	Diversity = 6.4225e-04	k_t

Generation 3200: Best = 1.381723e+04 = [3, 3, 3, 3]	FES = 320100	Diversity = 6.0043e-04	k_t
Generation 3210: Best = 1.381723e+04 = [1, 1, 1, 1]	FES = 321100	Diversity = 5.8002e-04	k_t
Generation 3220: Best = 1.381722e+04 = [3, 4, 2, 3]	FES = 322100	Diversity = 5.5117e-04	k_t
Generation 3230: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 323100	Diversity = 5.3463e-04	k_t
Generation 3240: Best = 1.381722e+04 = [3, 3, 3, 4]	FES = 324100	Diversity = 5.6868e-04	k_t
Generation 3250: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 325100	Diversity = 5.3141e-04	k_t
Generation 3260: Best = 1.381722e+04 = [4, 3, 4, 3]	FES = 326100	Diversity = 5.4438e-04	k_t
Generation 3270: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 327100	Diversity = 5.7314e-04	k_t
Generation 3280: Best = 1.381722e+04 = [3, 2, 3, 2]	FES = 328100	Diversity = 5.5771e-04	k_t
Generation 3290: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 329100	Diversity = 5.1632e-04	k_t
Generation 3300: Best = 1.381722e+04 = [4, 4, 3, 4]	FES = 330100	Diversity = 4.9446e-04	k_t
Generation 3310: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 331100	Diversity = 4.8017e-04	k_t
Generation 3320: Best = 1.381722e+04 = [4, 2, 3, 4]	FES = 332100	Diversity = 4.3897e-04	k_t
Generation 3330: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 333100	Diversity = 4.2154e-04	k_t
Generation 3340: Best = 1.381722e+04 = [2, 4, 2, 3]	FES = 334100	Diversity = 4.3544e-04	k_t
Generation 3350: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 335100	Diversity = 4.4611e-04	k_t
Generation 3360: Best = 1.381722e+04 = [4, 4, 4, 2]	FES = 336100	Diversity = 4.6051e-04	k_t
Generation 3370: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 337100	Diversity = 4.5483e-04	k_t
Generation 3380: Best = 1.381722e+04 = [4, 2, 3, 3]	FES = 338100	Diversity = 4.6629e-04	k_t
Generation 3390: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 339100	Diversity = 4.3829e-04	k_t
Generation 3400: Best = 1.381722e+04 = [2, 4, 3, 4]	FES = 340100	Diversity = 4.0757e-04	k_t
Generation 3410: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 341100	Diversity = 4.1791e-04	k_t
Generation 3420: Best = 1.381722e+04 = [2, 2, 4, 2]	FES = 342100	Diversity = 4.1326e-04	k_t
Generation 3430: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 343100	Diversity = 4.1574e-04	k_t
Generation 3440: Best = 1.381722e+04 = [2, 4, 2, 2]	FES = 344100	Diversity = 3.9689e-04	k_t
Generation 3450: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 345100	Diversity = 3.6826e-04	k_t
Generation 3460: Best = 1.381722e+04 = [2, 2, 4, 3]	FES = 346100	Diversity = 3.6917e-04	k_t

Generation 3470: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 347100	Diversity = 3.5476e-04	k_t
Generation 3480: Best = 1.381722e+04 = [3, 4, 2, 2]	FES = 348100	Diversity = 3.4122e-04	k_t
Generation 3490: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 349100	Diversity = 3.2179e-04	k_t
Generation 3500: Best = 1.381722e+04 = [4, 4, 2, 3]	FES = 350100	Diversity = 2.9042e-04	k_t
Generation 3510: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 351100	Diversity = 2.8717e-04	k_t
Generation 3520: Best = 1.381722e+04 = [4, 4, 3, 4]	FES = 352100	Diversity = 2.9307e-04	k_t
Generation 3530: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 353100	Diversity = 3.0392e-04	k_t
Generation 3540: Best = 1.381722e+04 = [3, 2, 2, 2]	FES = 354100	Diversity = 3.1484e-04	k_t
Generation 3550: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 355100	Diversity = 3.0233e-04	k_t
Generation 3560: Best = 1.381722e+04 = [3, 4, 2, 2]	FES = 356100	Diversity = 2.8430e-04	k_t
Generation 3570: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 357100	Diversity = 3.0225e-04	k_t
Generation 3580: Best = 1.381722e+04 = [3, 4, 2, 2]	FES = 358100	Diversity = 2.8785e-04	k_t
Generation 3590: Best = 1.381722e+04 = [1, 1, 1, 1]	FES = 359100	Diversity = 2.9381e-04	k_t
Generation 3600: Best = 1.381722e+04 = [3, 3, 2, 4]	FES = 360100	Diversity = 3.0213e-04	k_t
Generation 3610: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 361100	Diversity = 3.0557e-04	k_t
Generation 3620: Best = 1.381721e+04 = [2, 4, 4, 3]	FES = 362100	Diversity = 3.1130e-04	k_t
Generation 3630: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 363100	Diversity = 3.0512e-04	k_t
Generation 3640: Best = 1.381721e+04 = [3, 3, 3, 2]	FES = 364100	Diversity = 2.8610e-04	k_t
Generation 3650: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 365100	Diversity = 2.7116e-04	k_t
Generation 3660: Best = 1.381721e+04 = [3, 4, 2, 3]	FES = 366100	Diversity = 2.6103e-04	k_t
Generation 3670: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 367100	Diversity = 2.5773e-04	k_t
Generation 3680: Best = 1.381721e+04 = [3, 4, 3, 4]	FES = 368100	Diversity = 2.5062e-04	k_t
Generation 3690: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 369100	Diversity = 2.3948e-04	k_t
Generation 3700: Best = 1.381721e+04 = [2, 2, 3, 2]	FES = 370100	Diversity = 2.3337e-04	k_t
Generation 3710: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 371100	Diversity = 2.2524e-04	k_t
Generation 3720: Best = 1.381721e+04 = [4, 2, 4, 4]	FES = 372100	Diversity = 2.3076e-04	k_t
Generation 3730: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 373100	Diversity = 2.2414e-04	k_t

Generation 3740: Best = 1.381721e+04 = [2, 3, 4, 2]	FES = 374100	Diversity = 2.1846e-04	k_t
Generation 3750: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 375100	Diversity = 2.1729e-04	k_t
Generation 3760: Best = 1.381721e+04 = [4, 2, 2, 3]	FES = 376100	Diversity = 2.2243e-04	k_t
Generation 3770: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 377100	Diversity = 2.1201e-04	k_t
Generation 3780: Best = 1.381721e+04 = [2, 4, 4, 4]	FES = 378100	Diversity = 2.0625e-04	k_t
Generation 3790: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 379100	Diversity = 1.9506e-04	k_t
Generation 3800: Best = 1.381721e+04 = [4, 2, 4, 2]	FES = 380100	Diversity = 1.9401e-04	k_t
Generation 3810: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 381100	Diversity = 1.9280e-04	k_t
Generation 3820: Best = 1.381721e+04 = [2, 4, 2, 3]	FES = 382100	Diversity = 1.8857e-04	k_t
Generation 3830: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 383100	Diversity = 1.7395e-04	k_t
Generation 3840: Best = 1.381721e+04 = [4, 4, 2, 2]	FES = 384100	Diversity = 1.8225e-04	k_t
Generation 3850: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 385100	Diversity = 1.9377e-04	k_t
Generation 3860: Best = 1.381721e+04 = [4, 3, 2, 4]	FES = 386100	Diversity = 2.0273e-04	k_t
Generation 3870: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 387100	Diversity = 2.2288e-04	k_t
Generation 3880: Best = 1.381721e+04 = [3, 2, 4, 3]	FES = 388100	Diversity = 2.2405e-04	k_t
Generation 3890: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 389100	Diversity = 2.3858e-04	k_t
Generation 3900: Best = 1.381721e+04 = [3, 2, 4, 4]	FES = 390100	Diversity = 2.4679e-04	k_t
Generation 3910: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 391100	Diversity = 2.5395e-04	k_t
Generation 3920: Best = 1.381721e+04 = [4, 3, 2, 3]	FES = 392100	Diversity = 2.3847e-04	k_t
Generation 3930: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 393100	Diversity = 2.4196e-04	k_t
Generation 3940: Best = 1.381721e+04 = [2, 4, 4, 2]	FES = 394100	Diversity = 2.3816e-04	k_t
Generation 3950: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 395100	Diversity = 2.3287e-04	k_t
Generation 3960: Best = 1.381721e+04 = [3, 4, 4, 4]	FES = 396100	Diversity = 2.2653e-04	k_t
Generation 3970: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 397100	Diversity = 2.1207e-04	k_t
Generation 3980: Best = 1.381721e+04 = [2, 2, 2, 2]	FES = 398100	Diversity = 2.0131e-04	k_t
Generation 3990: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 399100	Diversity = 1.9583e-04	k_t
Generation 4000: Best = 1.381721e+04 = [4, 2, 2, 4]	FES = 400100	Diversity = 2.0031e-04	k_t

Generation 4010: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 401100	Diversity = 1.9273e-04	k_t
Generation 4020: Best = 1.381721e+04 = [3, 4, 2, 4]	FES = 402100	Diversity = 1.8491e-04	k_t
Generation 4030: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 403100	Diversity = 1.7142e-04	k_t
Generation 4040: Best = 1.381721e+04 = [2, 4, 3, 4]	FES = 404100	Diversity = 1.6104e-04	k_t
Generation 4050: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 405100	Diversity = 1.6203e-04	k_t
Generation 4060: Best = 1.381721e+04 = [4, 2, 4, 3]	FES = 406100	Diversity = 1.6841e-04	k_t
Generation 4070: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 407100	Diversity = 1.6539e-04	k_t
Generation 4080: Best = 1.381721e+04 = [3, 4, 4, 3]	FES = 408100	Diversity = 1.5836e-04	k_t
Generation 4090: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 409100	Diversity = 1.5632e-04	k_t
Generation 4100: Best = 1.381721e+04 = [2, 2, 3, 4]	FES = 410100	Diversity = 1.5206e-04	k_t
Generation 4110: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 411100	Diversity = 1.4331e-04	k_t
Generation 4120: Best = 1.381721e+04 = [2, 3, 4, 2]	FES = 412100	Diversity = 1.3477e-04	k_t
Generation 4130: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 413100	Diversity = 1.2492e-04	k_t
Generation 4140: Best = 1.381721e+04 = [2, 2, 4, 3]	FES = 414100	Diversity = 1.2309e-04	k_t
Generation 4150: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 415100	Diversity = 1.1707e-04	k_t
Generation 4160: Best = 1.381721e+04 = [4, 3, 3, 3]	FES = 416100	Diversity = 1.0787e-04	k_t
Generation 4170: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 417100	Diversity = 1.0551e-04	k_t
Generation 4180: Best = 1.381721e+04 = [3, 2, 2, 2]	FES = 418100	Diversity = 1.0311e-04	k_t
Generation 4190: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 419100	Diversity = 9.9520e-05	k_t
Generation 4200: Best = 1.381721e+04 = [4, 3, 3, 3]	FES = 420100	Diversity = 1.0117e-04	k_t
Generation 4210: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 421100	Diversity = 9.0632e-05	k_t
Generation 4220: Best = 1.381721e+04 = [2, 3, 2, 4]	FES = 422100	Diversity = 8.5373e-05	k_t
Generation 4230: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 423100	Diversity = 8.1098e-05	k_t
Generation 4240: Best = 1.381721e+04 = [4, 4, 4, 3]	FES = 424100	Diversity = 7.9673e-05	k_t
Generation 4250: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 425100	Diversity = 8.0340e-05	k_t
Generation 4260: Best = 1.381721e+04 = [3, 4, 2, 3]	FES = 426100	Diversity = 7.7991e-05	k_t
Generation 4270: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 427100	Diversity = 7.3400e-05	k_t

Generation 4280: Best = 1.381721e+04 = [2, 4, 2, 3]	FES = 428100	Diversity = 6.9825e-05	k_t
Generation 4290: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 429100	Diversity = 6.3430e-05	k_t
Generation 4300: Best = 1.381721e+04 = [3, 3, 4, 2]	FES = 430100	Diversity = 6.0467e-05	k_t
Generation 4310: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 431100	Diversity = 5.8204e-05	k_t
Generation 4320: Best = 1.381721e+04 = [2, 4, 2, 3]	FES = 432100	Diversity = 5.4137e-05	k_t
Generation 4330: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 433100	Diversity = 5.3092e-05	k_t
Generation 4340: Best = 1.381721e+04 = [2, 3, 4, 3]	FES = 434100	Diversity = 5.2674e-05	k_t
Generation 4350: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 435100	Diversity = 5.3487e-05	k_t
Generation 4360: Best = 1.381721e+04 = [4, 2, 4, 3]	FES = 436100	Diversity = 5.3868e-05	k_t
Generation 4370: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 437100	Diversity = 5.1929e-05	k_t
Generation 4380: Best = 1.381721e+04 = [3, 2, 2, 4]	FES = 438100	Diversity = 5.1466e-05	k_t
Generation 4390: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 439100	Diversity = 5.1295e-05	k_t
Generation 4400: Best = 1.381721e+04 = [4, 4, 3, 3]	FES = 440100	Diversity = 4.9695e-05	k_t
Generation 4410: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 441100	Diversity = 4.8506e-05	k_t
Generation 4420: Best = 1.381721e+04 = [2, 4, 4, 3]	FES = 442100	Diversity = 4.6743e-05	k_t
Generation 4430: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 443100	Diversity = 4.8591e-05	k_t
Generation 4440: Best = 1.381721e+04 = [4, 3, 3, 3]	FES = 444100	Diversity = 4.8815e-05	k_t
Generation 4450: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 445100	Diversity = 5.0852e-05	k_t
Generation 4460: Best = 1.381721e+04 = [3, 3, 3, 4]	FES = 446100	Diversity = 5.2354e-05	k_t
Generation 4470: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 447100	Diversity = 5.1074e-05	k_t
Generation 4480: Best = 1.381721e+04 = [3, 3, 4, 4]	FES = 448100	Diversity = 5.0815e-05	k_t
Generation 4490: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 449100	Diversity = 4.9277e-05	k_t
Generation 4500: Best = 1.381721e+04 = [4, 3, 4, 2]	FES = 450100	Diversity = 4.6081e-05	k_t
Generation 4510: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 451100	Diversity = 4.4603e-05	k_t
Generation 4520: Best = 1.381721e+04 = [2, 4, 3, 4]	FES = 452100	Diversity = 4.3465e-05	k_t
Generation 4530: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 453100	Diversity = 4.3922e-05	k_t
Generation 4540: Best = 1.381721e+04 = [4, 2, 2, 2]	FES = 454100	Diversity = 4.1023e-05	k_t

Generation 4550: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 455100	Diversity = 4.0311e-05	k_t
Generation 4560: Best = 1.381721e+04 = [2, 2, 4, 2]	FES = 456100	Diversity = 4.2061e-05	k_t
Generation 4570: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 457100	Diversity = 4.0509e-05	k_t
Generation 4580: Best = 1.381721e+04 = [3, 3, 3, 2]	FES = 458100	Diversity = 3.8967e-05	k_t
Generation 4590: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 459100	Diversity = 3.7625e-05	k_t
Generation 4600: Best = 1.381721e+04 = [2, 2, 3, 2]	FES = 460100	Diversity = 3.5465e-05	k_t
Generation 4610: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 461100	Diversity = 3.3776e-05	k_t
Generation 4620: Best = 1.381721e+04 = [4, 3, 3, 4]	FES = 462100	Diversity = 3.2808e-05	k_t
Generation 4630: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 463100	Diversity = 3.1270e-05	k_t
Generation 4640: Best = 1.381721e+04 = [4, 4, 2, 2]	FES = 464100	Diversity = 3.0824e-05	k_t
Generation 4650: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 465100	Diversity = 3.1657e-05	k_t
Generation 4660: Best = 1.381721e+04 = [4, 4, 3, 3]	FES = 466100	Diversity = 3.0278e-05	k_t
Generation 4670: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 467100	Diversity = 3.0682e-05	k_t
Generation 4680: Best = 1.381721e+04 = [2, 3, 3, 4]	FES = 468100	Diversity = 3.0595e-05	k_t
Generation 4690: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 469100	Diversity = 2.9656e-05	k_t
Generation 4700: Best = 1.381721e+04 = [3, 3, 2, 3]	FES = 470100	Diversity = 2.9165e-05	k_t
Generation 4710: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 471100	Diversity = 2.8626e-05	k_t
Generation 4720: Best = 1.381721e+04 = [3, 2, 3, 4]	FES = 472100	Diversity = 2.8042e-05	k_t
Generation 4730: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 473100	Diversity = 2.8388e-05	k_t
Generation 4740: Best = 1.381721e+04 = [4, 2, 2, 3]	FES = 474100	Diversity = 2.8095e-05	k_t
Generation 4750: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 475100	Diversity = 2.7188e-05	k_t
Generation 4760: Best = 1.381721e+04 = [4, 2, 3, 4]	FES = 476100	Diversity = 2.6872e-05	k_t
Generation 4770: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 477100	Diversity = 2.7013e-05	k_t
Generation 4780: Best = 1.381721e+04 = [3, 4, 2, 2]	FES = 478100	Diversity = 2.6797e-05	k_t
Generation 4790: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 479100	Diversity = 2.6208e-05	k_t
Generation 4800: Best = 1.381721e+04 = [2, 3, 4, 4]	FES = 480100	Diversity = 2.6760e-05	k_t
Generation 4810: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 481100	Diversity = 2.7763e-05	k_t

Generation 4820: Best = 1.381721e+04	FES = 482100	Diversity = 2.8689e-05	k_t
= [4, 3, 2, 2]			
Generation 4830: Best = 1.381721e+04	FES = 483100	Diversity = 2.8591e-05	k_t
= [1, 1, 1, 1]			
Generation 4840: Best = 1.381721e+04	FES = 484100	Diversity = 2.9662e-05	k_t
= [2, 3, 3, 2]			
Generation 4850: Best = 1.381721e+04	FES = 485100	Diversity = 2.8999e-05	k_t
= [1, 1, 1, 1]			
Generation 4860: Best = 1.381721e+04	FES = 486100	Diversity = 2.8341e-05	k_t
= [4, 4, 2, 3]			
Generation 4870: Best = 1.381721e+04	FES = 487100	Diversity = 2.7636e-05	k_t
= [1, 1, 1, 1]			
Generation 4880: Best = 1.381721e+04	FES = 488100	Diversity = 2.8147e-05	k_t
= [2, 4, 3, 2]			
Generation 4890: Best = 1.381721e+04	FES = 489100	Diversity = 2.7919e-05	k_t
= [1, 1, 1, 1]			
Generation 4900: Best = 1.381721e+04	FES = 490100	Diversity = 2.6846e-05	k_t
= [2, 4, 3, 3]			
Generation 4910: Best = 1.381721e+04	FES = 491100	Diversity = 2.7689e-05	k_t
= [1, 1, 1, 1]			
Generation 4920: Best = 1.381721e+04	FES = 492100	Diversity = 2.9343e-05	k_t
= [4, 2, 3, 2]			
Generation 4930: Best = 1.381721e+04	FES = 493100	Diversity = 2.8297e-05	k_t
= [1, 1, 1, 1]			
Generation 4940: Best = 1.381721e+04	FES = 494100	Diversity = 2.8407e-05	k_t
= [2, 4, 4, 3]			
Generation 4950: Best = 1.381721e+04	FES = 495100	Diversity = 2.8833e-05	k_t
= [1, 1, 1, 1]			
Generation 4960: Best = 1.381721e+04	FES = 496100	Diversity = 2.7986e-05	k_t
= [2, 3, 4, 3]			
Generation 4970: Best = 1.381721e+04	FES = 497100	Diversity = 2.8209e-05	k_t
= [1, 1, 1, 1]			
Generation 4980: Best = 1.381721e+04	FES = 498100	Diversity = 2.7586e-05	k_t
= [3, 2, 2, 3]			
Generation 4990: Best = 1.381721e+04	FES = 499100	Diversity = 2.6637e-05	k_t
= [1, 1, 1, 1]			
Generation 5000: Best = 1.381721e+04	FES = 500100	Diversity = 2.6644e-05	k_t
= [2, 3, 2, 4]			
Generation 5010: Best = 1.381721e+04	FES = 501100	Diversity = 2.5902e-05	k_t
= [1, 1, 1, 1]			
Generation 5020: Best = 1.381721e+04	FES = 502100	Diversity = 2.5388e-05	k_t
= [2, 2, 3, 2]			
Generation 5030: Best = 1.381721e+04	FES = 503100	Diversity = 2.4687e-05	k_t
= [1, 1, 1, 1]			
Generation 5040: Best = 1.381721e+04	FES = 504100	Diversity = 2.4523e-05	k_t
= [4, 4, 2, 3]			
Generation 5050: Best = 1.381721e+04	FES = 505100	Diversity = 2.3099e-05	k_t
= [1, 1, 1, 1]			
Generation 5060: Best = 1.381721e+04	FES = 506100	Diversity = 2.3291e-05	k_t
= [2, 2, 2, 3]			
Generation 5070: Best = 1.381721e+04	FES = 507100	Diversity = 2.3760e-05	k_t
= [1, 1, 1, 1]			
Generation 5080: Best = 1.381721e+04	FES = 508100	Diversity = 2.4543e-05	k_t
= [3, 4, 2, 4]			

Generation 5090: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 509100	Diversity = 2.4444e-05	k_t
Generation 5100: Best = 1.381721e+04 = [4, 3, 3, 2]	FES = 510100	Diversity = 2.6182e-05	k_t
Generation 5110: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 511100	Diversity = 2.8294e-05	k_t
Generation 5120: Best = 1.381721e+04 = [3, 3, 3, 3]	FES = 512100	Diversity = 2.6191e-05	k_t
Generation 5130: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 513100	Diversity = 2.5547e-05	k_t
Generation 5140: Best = 1.381721e+04 = [4, 3, 4, 3]	FES = 514100	Diversity = 2.5145e-05	k_t
Generation 5150: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 515100	Diversity = 2.3710e-05	k_t
Generation 5160: Best = 1.381721e+04 = [4, 4, 4, 3]	FES = 516100	Diversity = 2.4066e-05	k_t
Generation 5170: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 517100	Diversity = 2.3528e-05	k_t
Generation 5180: Best = 1.381721e+04 = [3, 3, 2, 2]	FES = 518100	Diversity = 2.1819e-05	k_t
Generation 5190: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 519100	Diversity = 2.1121e-05	k_t
Generation 5200: Best = 1.381721e+04 = [2, 2, 4, 3]	FES = 520100	Diversity = 2.1993e-05	k_t
Generation 5210: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 521100	Diversity = 2.2999e-05	k_t
Generation 5220: Best = 1.381721e+04 = [2, 3, 2, 2]	FES = 522100	Diversity = 2.5308e-05	k_t
Generation 5230: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 523100	Diversity = 2.4289e-05	k_t
Generation 5240: Best = 1.381721e+04 = [4, 2, 3, 3]	FES = 524100	Diversity = 2.4038e-05	k_t
Generation 5250: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 525100	Diversity = 2.3412e-05	k_t
Generation 5260: Best = 1.381721e+04 = [3, 4, 3, 2]	FES = 526100	Diversity = 2.4122e-05	k_t
Generation 5270: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 527100	Diversity = 2.3370e-05	k_t
Generation 5280: Best = 1.381721e+04 = [4, 3, 2, 4]	FES = 528100	Diversity = 2.2417e-05	k_t
Generation 5290: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 529100	Diversity = 2.0866e-05	k_t
Generation 5300: Best = 1.381721e+04 = [3, 3, 3, 4]	FES = 530100	Diversity = 2.0146e-05	k_t
Generation 5310: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 531100	Diversity = 1.9795e-05	k_t
Generation 5320: Best = 1.381721e+04 = [2, 3, 2, 2]	FES = 532100	Diversity = 1.8239e-05	k_t
Generation 5330: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 533100	Diversity = 1.6913e-05	k_t
Generation 5340: Best = 1.381721e+04 = [4, 4, 4, 3]	FES = 534100	Diversity = 1.6541e-05	k_t
Generation 5350: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 535100	Diversity = 1.6958e-05	k_t

Generation 5360: Best = 1.381721e+04 = [4, 2, 2, 3]	FES = 536100	Diversity = 1.7176e-05	k_t
Generation 5370: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 537100	Diversity = 1.6944e-05	k_t
Generation 5380: Best = 1.381721e+04 = [4, 4, 3, 3]	FES = 538100	Diversity = 1.6562e-05	k_t
Generation 5390: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 539100	Diversity = 1.7488e-05	k_t
Generation 5400: Best = 1.381721e+04 = [4, 3, 2, 3]	FES = 540100	Diversity = 1.7663e-05	k_t
Generation 5410: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 541100	Diversity = 1.6695e-05	k_t
Generation 5420: Best = 1.381721e+04 = [4, 3, 2, 2]	FES = 542100	Diversity = 1.6158e-05	k_t
Generation 5430: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 543100	Diversity = 1.6429e-05	k_t
Generation 5440: Best = 1.381721e+04 = [4, 3, 4, 2]	FES = 544100	Diversity = 1.6775e-05	k_t
Generation 5450: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 545100	Diversity = 1.6611e-05	k_t
Generation 5460: Best = 1.381721e+04 = [3, 2, 4, 3]	FES = 546100	Diversity = 1.6088e-05	k_t
Generation 5470: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 547100	Diversity = 1.4785e-05	k_t
Generation 5480: Best = 1.381721e+04 = [3, 3, 3, 4]	FES = 548100	Diversity = 1.4358e-05	k_t
Generation 5490: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 549100	Diversity = 1.3248e-05	k_t
Generation 5500: Best = 1.381721e+04 = [3, 2, 4, 4]	FES = 550100	Diversity = 1.2410e-05	k_t
Generation 5510: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 551100	Diversity = 1.2535e-05	k_t
Generation 5520: Best = 1.381721e+04 = [2, 2, 4, 3]	FES = 552100	Diversity = 1.1663e-05	k_t
Generation 5530: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 553100	Diversity = 1.0859e-05	k_t
Generation 5540: Best = 1.381721e+04 = [3, 4, 2, 2]	FES = 554100	Diversity = 1.0336e-05	k_t
Generation 5550: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 555100	Diversity = 1.0022e-05	k_t
Generation 5560: Best = 1.381721e+04 = [3, 3, 2, 3]	FES = 556100	Diversity = 1.0767e-05	k_t
Generation 5570: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 557100	Diversity = 1.1191e-05	k_t
Generation 5580: Best = 1.381721e+04 = [2, 3, 2, 4]	FES = 558100	Diversity = 1.1207e-05	k_t
Generation 5590: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 559100	Diversity = 1.2465e-05	k_t
Generation 5600: Best = 1.381721e+04 = [4, 2, 2, 3]	FES = 560100	Diversity = 1.2052e-05	k_t
Generation 5610: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 561100	Diversity = 1.1398e-05	k_t
Generation 5620: Best = 1.381721e+04 = [4, 4, 4, 2]	FES = 562100	Diversity = 1.0923e-05	k_t

Generation 5630: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 563100	Diversity = 1.2001e-05	k_t
Generation 5640: Best = 1.381721e+04 = [3, 3, 4, 3]	FES = 564100	Diversity = 1.2279e-05	k_t
Generation 5650: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 565100	Diversity = 1.2544e-05	k_t
Generation 5660: Best = 1.381721e+04 = [3, 2, 2, 4]	FES = 566100	Diversity = 1.4086e-05	k_t
Generation 5670: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 567100	Diversity = 1.3745e-05	k_t
Generation 5680: Best = 1.381721e+04 = [2, 4, 4, 4]	FES = 568100	Diversity = 1.3192e-05	k_t
Generation 5690: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 569100	Diversity = 1.3180e-05	k_t
Generation 5700: Best = 1.381721e+04 = [4, 2, 4, 3]	FES = 570100	Diversity = 1.2802e-05	k_t
Generation 5710: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 571100	Diversity = 1.2790e-05	k_t
Generation 5720: Best = 1.381721e+04 = [4, 4, 4, 2]	FES = 572100	Diversity = 1.1994e-05	k_t
Generation 5730: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 573100	Diversity = 1.1849e-05	k_t
Generation 5740: Best = 1.381721e+04 = [4, 2, 3, 4]	FES = 574100	Diversity = 1.1953e-05	k_t
Generation 5750: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 575100	Diversity = 1.1936e-05	k_t
Generation 5760: Best = 1.381721e+04 = [3, 3, 3, 3]	FES = 576100	Diversity = 1.1592e-05	k_t
Generation 5770: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 577100	Diversity = 1.1300e-05	k_t
Generation 5780: Best = 1.381721e+04 = [3, 2, 2, 4]	FES = 578100	Diversity = 1.0931e-05	k_t
Generation 5790: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 579100	Diversity = 1.0130e-05	k_t
Generation 5800: Best = 1.381721e+04 = [2, 4, 4, 4]	FES = 580100	Diversity = 1.0239e-05	k_t
Generation 5810: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 581100	Diversity = 1.1668e-05	k_t
Generation 5820: Best = 1.381721e+04 = [2, 4, 2, 4]	FES = 582100	Diversity = 1.2896e-05	k_t
Generation 5830: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 583100	Diversity = 1.3226e-05	k_t
Generation 5840: Best = 1.381721e+04 = [3, 2, 4, 4]	FES = 584100	Diversity = 1.3982e-05	k_t
Generation 5850: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 585100	Diversity = 1.3409e-05	k_t
Generation 5860: Best = 1.381721e+04 = [4, 4, 3, 2]	FES = 586100	Diversity = 1.3330e-05	k_t
Generation 5870: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 587100	Diversity = 1.3198e-05	k_t
Generation 5880: Best = 1.381721e+04 = [4, 4, 2, 3]	FES = 588100	Diversity = 1.3069e-05	k_t
Generation 5890: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 589100	Diversity = 1.2067e-05	k_t

Generation 5900: Best = 1.381721e+04 = [2, 3, 3, 3]	FES = 590100	Diversity = 1.1758e-05	k_t
Generation 5910: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 591100	Diversity = 1.1750e-05	k_t
Generation 5920: Best = 1.381721e+04 = [2, 2, 2, 3]	FES = 592100	Diversity = 1.1875e-05	k_t
Generation 5930: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 593100	Diversity = 1.2544e-05	k_t
Generation 5940: Best = 1.381721e+04 = [4, 2, 2, 2]	FES = 594100	Diversity = 1.2177e-05	k_t
Generation 5950: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 595100	Diversity = 1.2313e-05	k_t
Generation 5960: Best = 1.381721e+04 = [2, 4, 4, 4]	FES = 596100	Diversity = 1.1880e-05	k_t
Generation 5970: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 597100	Diversity = 1.1394e-05	k_t
Generation 5980: Best = 1.381721e+04 = [2, 3, 2, 3]	FES = 598100	Diversity = 1.1490e-05	k_t
Generation 5990: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 599100	Diversity = 1.0951e-05	k_t
Generation 6000: Best = 1.381721e+04 = [2, 4, 2, 3]	FES = 600100	Diversity = 1.0596e-05	k_t
Generation 6010: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 601100	Diversity = 1.0145e-05	k_t
Generation 6020: Best = 1.381721e+04 = [4, 3, 3, 2]	FES = 602100	Diversity = 9.7971e-06	k_t
Generation 6030: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 603100	Diversity = 9.4050e-06	k_t
Generation 6040: Best = 1.381721e+04 = [3, 3, 4, 3]	FES = 604100	Diversity = 9.0075e-06	k_t
Generation 6050: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 605100	Diversity = 8.3320e-06	k_t
Generation 6060: Best = 1.381721e+04 = [4, 2, 4, 4]	FES = 606100	Diversity = 8.2668e-06	k_t
Generation 6070: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 607100	Diversity = 7.9134e-06	k_t
Generation 6080: Best = 1.381721e+04 = [2, 3, 3, 3]	FES = 608100	Diversity = 7.6900e-06	k_t
Generation 6090: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 609100	Diversity = 8.1009e-06	k_t
Generation 6100: Best = 1.381721e+04 = [2, 2, 3, 3]	FES = 610100	Diversity = 7.9470e-06	k_t
Generation 6110: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 611100	Diversity = 8.0142e-06	k_t
Generation 6120: Best = 1.381721e+04 = [3, 4, 3, 3]	FES = 612100	Diversity = 8.1357e-06	k_t
Generation 6130: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 613100	Diversity = 7.8496e-06	k_t
Generation 6140: Best = 1.381721e+04 = [4, 3, 3, 3]	FES = 614100	Diversity = 7.8173e-06	k_t
Generation 6150: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 615100	Diversity = 7.6595e-06	k_t
Generation 6160: Best = 1.381721e+04 = [3, 4, 4, 3]	FES = 616100	Diversity = 7.0335e-06	k_t

Generation 6170: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 617100	Diversity = 6.4785e-06	k_t
Generation 6180: Best = 1.381721e+04 = [4, 2, 4, 3]	FES = 618100	Diversity = 6.5381e-06	k_t
Generation 6190: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 619100	Diversity = 6.7827e-06	k_t
Generation 6200: Best = 1.381721e+04 = [4, 3, 4, 3]	FES = 620100	Diversity = 6.6913e-06	k_t
Generation 6210: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 621100	Diversity = 6.6445e-06	k_t
Generation 6220: Best = 1.381721e+04 = [4, 2, 4, 3]	FES = 622100	Diversity = 6.6415e-06	k_t
Generation 6230: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 623100	Diversity = 6.1874e-06	k_t
Generation 6240: Best = 1.381721e+04 = [2, 3, 3, 4]	FES = 624100	Diversity = 6.2141e-06	k_t
Generation 6250: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 625100	Diversity = 5.9185e-06	k_t
Generation 6260: Best = 1.381721e+04 = [2, 3, 4, 2]	FES = 626100	Diversity = 5.6624e-06	k_t
Generation 6270: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 627100	Diversity = 5.3515e-06	k_t
Generation 6280: Best = 1.381721e+04 = [2, 4, 4, 3]	FES = 628100	Diversity = 5.2439e-06	k_t
Generation 6290: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 629100	Diversity = 4.8914e-06	k_t
Generation 6300: Best = 1.381721e+04 = [4, 2, 2, 2]	FES = 630100	Diversity = 4.8621e-06	k_t
Generation 6310: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 631100	Diversity = 4.9921e-06	k_t
Generation 6320: Best = 1.381721e+04 = [4, 4, 4, 3]	FES = 632100	Diversity = 4.8700e-06	k_t
Generation 6330: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 633100	Diversity = 4.8556e-06	k_t
Generation 6340: Best = 1.381721e+04 = [4, 4, 4, 4]	FES = 634100	Diversity = 4.9749e-06	k_t
Generation 6350: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 635100	Diversity = 4.8211e-06	k_t
Generation 6360: Best = 1.381721e+04 = [2, 2, 3, 2]	FES = 636100	Diversity = 4.6731e-06	k_t
Generation 6370: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 637100	Diversity = 4.5046e-06	k_t
Generation 6380: Best = 1.381721e+04 = [3, 2, 2, 2]	FES = 638100	Diversity = 4.5117e-06	k_t
Generation 6390: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 639100	Diversity = 4.7422e-06	k_t
Generation 6400: Best = 1.381721e+04 = [3, 3, 2, 3]	FES = 640100	Diversity = 5.1276e-06	k_t
Generation 6410: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 641100	Diversity = 5.1802e-06	k_t
Generation 6420: Best = 1.381721e+04 = [3, 3, 3, 4]	FES = 642100	Diversity = 5.1371e-06	k_t
Generation 6430: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 643100	Diversity = 4.9333e-06	k_t

Generation 6440: Best = 1.381721e+04 = [3, 2, 3, 4]	FES = 644100	Diversity = 4.5697e-06	k_t
Generation 6450: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 645100	Diversity = 4.0470e-06	k_t
Generation 6460: Best = 1.381721e+04 = [3, 4, 3, 2]	FES = 646100	Diversity = 3.8768e-06	k_t
Generation 6470: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 647100	Diversity = 3.8165e-06	k_t
Generation 6480: Best = 1.381721e+04 = [4, 4, 3, 3]	FES = 648100	Diversity = 3.7160e-06	k_t
Generation 6490: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 649100	Diversity = 3.8289e-06	k_t
Generation 6500: Best = 1.381721e+04 = [4, 3, 3, 3]	FES = 650100	Diversity = 4.0032e-06	k_t
Generation 6510: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 651100	Diversity = 4.3729e-06	k_t
Generation 6520: Best = 1.381721e+04 = [4, 3, 3, 3]	FES = 652100	Diversity = 4.5102e-06	k_t
Generation 6530: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 653100	Diversity = 4.4508e-06	k_t
Generation 6540: Best = 1.381721e+04 = [3, 3, 4, 2]	FES = 654100	Diversity = 4.5110e-06	k_t
Generation 6550: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 655100	Diversity = 4.4691e-06	k_t
Generation 6560: Best = 1.381721e+04 = [4, 4, 4, 2]	FES = 656100	Diversity = 4.7654e-06	k_t
Generation 6570: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 657100	Diversity = 5.1010e-06	k_t
Generation 6580: Best = 1.381721e+04 = [4, 2, 4, 3]	FES = 658100	Diversity = 5.4754e-06	k_t
Generation 6590: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 659100	Diversity = 5.9087e-06	k_t
Generation 6600: Best = 1.381721e+04 = [2, 3, 2, 2]	FES = 660100	Diversity = 5.5988e-06	k_t
Generation 6610: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 661100	Diversity = 5.4625e-06	k_t
Generation 6620: Best = 1.381721e+04 = [3, 3, 3, 4]	FES = 662100	Diversity = 5.7345e-06	k_t
Generation 6630: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 663100	Diversity = 5.7085e-06	k_t
Generation 6640: Best = 1.381721e+04 = [4, 2, 2, 4]	FES = 664100	Diversity = 5.7060e-06	k_t
Generation 6650: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 665100	Diversity = 5.7656e-06	k_t
Generation 6660: Best = 1.381721e+04 = [3, 3, 2, 4]	FES = 666100	Diversity = 5.5432e-06	k_t
Generation 6670: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 667100	Diversity = 5.1615e-06	k_t
Generation 6680: Best = 1.381721e+04 = [2, 3, 4, 4]	FES = 668100	Diversity = 4.8564e-06	k_t
Generation 6690: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 669100	Diversity = 4.2881e-06	k_t
Generation 6700: Best = 1.381721e+04 = [2, 4, 2, 2]	FES = 670100	Diversity = 3.9893e-06	k_t

Generation 6710: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 671100	Diversity = 3.6648e-06	k_t
Generation 6720: Best = 1.381721e+04 = [2, 3, 2, 2]	FES = 672100	Diversity = 3.5030e-06	k_t
Generation 6730: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 673100	Diversity = 3.5204e-06	k_t
Generation 6740: Best = 1.381721e+04 = [2, 4, 4, 3]	FES = 674100	Diversity = 3.3945e-06	k_t
Generation 6750: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 675100	Diversity = 3.4632e-06	k_t
Generation 6760: Best = 1.381721e+04 = [2, 3, 4, 4]	FES = 676100	Diversity = 3.8994e-06	k_t
Generation 6770: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 677100	Diversity = 4.6357e-06	k_t
Generation 6780: Best = 1.381721e+04 = [2, 3, 2, 4]	FES = 678100	Diversity = 4.9515e-06	k_t
Generation 6790: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 679100	Diversity = 4.7052e-06	k_t
Generation 6800: Best = 1.381721e+04 = [2, 3, 3, 3]	FES = 680100	Diversity = 5.0671e-06	k_t
Generation 6810: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 681100	Diversity = 5.2235e-06	k_t
Generation 6820: Best = 1.381721e+04 = [3, 3, 3, 3]	FES = 682100	Diversity = 5.1991e-06	k_t
Generation 6830: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 683100	Diversity = 5.3546e-06	k_t
Generation 6840: Best = 1.381721e+04 = [3, 4, 4, 3]	FES = 684100	Diversity = 5.1586e-06	k_t
Generation 6850: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 685100	Diversity = 5.0515e-06	k_t
Generation 6860: Best = 1.381721e+04 = [3, 3, 3, 3]	FES = 686100	Diversity = 5.1015e-06	k_t
Generation 6870: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 687100	Diversity = 5.1068e-06	k_t
Generation 6880: Best = 1.381721e+04 = [2, 3, 4, 2]	FES = 688100	Diversity = 5.1007e-06	k_t
Generation 6890: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 689100	Diversity = 5.8472e-06	k_t
Generation 6900: Best = 1.381721e+04 = [3, 4, 3, 3]	FES = 690100	Diversity = 6.9230e-06	k_t
Generation 6910: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 691100	Diversity = 8.1063e-06	k_t
Generation 6920: Best = 1.381721e+04 = [3, 2, 3, 3]	FES = 692100	Diversity = 8.4288e-06	k_t
Generation 6930: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 693100	Diversity = 9.0476e-06	k_t
Generation 6940: Best = 1.381721e+04 = [2, 4, 3, 2]	FES = 694100	Diversity = 9.3806e-06	k_t
Generation 6950: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 695100	Diversity = 9.3261e-06	k_t
Generation 6960: Best = 1.381721e+04 = [2, 2, 2, 4]	FES = 696100	Diversity = 9.7273e-06	k_t
Generation 6970: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 697100	Diversity = 9.7148e-06	k_t

Generation 6980: Best = 1.381721e+04 = [3, 4, 4, 3]	FES = 698100	Diversity = 9.4044e-06	k_t
Generation 6990: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 699100	Diversity = 9.7617e-06	k_t
Generation 7000: Best = 1.381721e+04 = [3, 3, 3, 2]	FES = 700100	Diversity = 1.0267e-05	k_t
Generation 7010: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 701100	Diversity = 9.4798e-06	k_t
Generation 7020: Best = 1.381721e+04 = [4, 4, 4, 4]	FES = 702100	Diversity = 8.9593e-06	k_t
Generation 7030: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 703100	Diversity = 8.6739e-06	k_t
Generation 7040: Best = 1.381721e+04 = [3, 2, 4, 4]	FES = 704100	Diversity = 8.5237e-06	k_t
Generation 7050: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 705100	Diversity = 8.6075e-06	k_t
Generation 7060: Best = 1.381721e+04 = [3, 2, 4, 3]	FES = 706100	Diversity = 1.0026e-05	k_t
Generation 7070: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 707100	Diversity = 1.0485e-05	k_t
Generation 7080: Best = 1.381721e+04 = [4, 4, 3, 4]	FES = 708100	Diversity = 1.0168e-05	k_t
Generation 7090: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 709100	Diversity = 1.0741e-05	k_t
Generation 7100: Best = 1.381721e+04 = [4, 2, 3, 4]	FES = 710100	Diversity = 1.0839e-05	k_t
Generation 7110: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 711100	Diversity = 1.1570e-05	k_t
Generation 7120: Best = 1.381721e+04 = [2, 3, 3, 2]	FES = 712100	Diversity = 1.1905e-05	k_t
Generation 7130: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 713100	Diversity = 1.1458e-05	k_t
Generation 7140: Best = 1.381721e+04 = [2, 2, 4, 4]	FES = 714100	Diversity = 1.0669e-05	k_t
Generation 7150: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 715100	Diversity = 9.8949e-06	k_t
Generation 7160: Best = 1.381721e+04 = [3, 2, 2, 4]	FES = 716100	Diversity = 9.0058e-06	k_t
Generation 7170: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 717100	Diversity = 8.2193e-06	k_t
Generation 7180: Best = 1.381721e+04 = [4, 2, 3, 2]	FES = 718100	Diversity = 7.8672e-06	k_t
Generation 7190: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 719100	Diversity = 7.7844e-06	k_t
Generation 7200: Best = 1.381721e+04 = [4, 3, 4, 3]	FES = 720100	Diversity = 7.2309e-06	k_t
Generation 7210: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 721100	Diversity = 7.4390e-06	k_t
Generation 7220: Best = 1.381721e+04 = [2, 3, 2, 2]	FES = 722100	Diversity = 7.3774e-06	k_t
Generation 7230: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 723100	Diversity = 7.0129e-06	k_t
Generation 7240: Best = 1.381721e+04 = [3, 4, 3, 2]	FES = 724100	Diversity = 7.8112e-06	k_t

Generation 7250: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 725100	Diversity = 8.1102e-06	k_t
Generation 7260: Best = 1.381721e+04 = [3, 4, 3, 4]	FES = 726100	Diversity = 8.2901e-06	k_t
Generation 7270: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 727100	Diversity = 8.7886e-06	k_t
Generation 7280: Best = 1.381721e+04 = [3, 4, 4, 3]	FES = 728100	Diversity = 9.7341e-06	k_t
Generation 7290: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 729100	Diversity = 9.7946e-06	k_t
Generation 7300: Best = 1.381721e+04 = [4, 4, 4, 4]	FES = 730100	Diversity = 1.1089e-05	k_t
Generation 7310: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 731100	Diversity = 1.1068e-05	k_t
Generation 7320: Best = 1.381721e+04 = [4, 2, 4, 4]	FES = 732100	Diversity = 1.1272e-05	k_t
Generation 7330: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 733100	Diversity = 1.1131e-05	k_t
Generation 7340: Best = 1.381721e+04 = [4, 3, 2, 4]	FES = 734100	Diversity = 1.1407e-05	k_t
Generation 7350: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 735100	Diversity = 1.1790e-05	k_t
Generation 7360: Best = 1.381721e+04 = [3, 2, 3, 4]	FES = 736100	Diversity = 1.2005e-05	k_t
Generation 7370: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 737100	Diversity = 1.1727e-05	k_t
Generation 7380: Best = 1.381721e+04 = [2, 3, 4, 3]	FES = 738100	Diversity = 1.1316e-05	k_t
Generation 7390: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 739100	Diversity = 1.2023e-05	k_t
Generation 7400: Best = 1.381721e+04 = [4, 2, 3, 4]	FES = 740100	Diversity = 1.1786e-05	k_t
Generation 7410: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 741100	Diversity = 1.2539e-05	k_t
Generation 7420: Best = 1.381721e+04 = [4, 3, 2, 3]	FES = 742100	Diversity = 1.4306e-05	k_t
Generation 7430: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 743100	Diversity = 1.3618e-05	k_t
Generation 7440: Best = 1.381721e+04 = [2, 3, 3, 3]	FES = 744100	Diversity = 1.3745e-05	k_t
Generation 7450: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 745100	Diversity = 1.4459e-05	k_t
Generation 7460: Best = 1.381721e+04 = [2, 3, 4, 2]	FES = 746100	Diversity = 1.4432e-05	k_t
Generation 7470: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 747100	Diversity = 1.5782e-05	k_t
Generation 7480: Best = 1.381721e+04 = [4, 2, 4, 3]	FES = 748100	Diversity = 1.6841e-05	k_t
Generation 7490: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 749100	Diversity = 1.7538e-05	k_t
Generation 7500: Best = 1.381721e+04 = [2, 2, 4, 3]	FES = 750100	Diversity = 1.6674e-05	k_t
Generation 7510: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 751100	Diversity = 1.7598e-05	k_t

Generation 7520: Best = 1.381721e+04 = [3, 4, 3, 2]	FES = 752100	Diversity = 1.7746e-05	k_t
Generation 7530: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 753100	Diversity = 1.8574e-05	k_t
Generation 7540: Best = 1.381721e+04 = [3, 4, 2, 4]	FES = 754100	Diversity = 1.9164e-05	k_t
Generation 7550: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 755100	Diversity = 1.8964e-05	k_t
Generation 7560: Best = 1.381721e+04 = [4, 4, 3, 4]	FES = 756100	Diversity = 1.8097e-05	k_t
Generation 7570: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 757100	Diversity = 1.6892e-05	k_t
Generation 7580: Best = 1.381721e+04 = [2, 4, 2, 2]	FES = 758100	Diversity = 1.5182e-05	k_t
Generation 7590: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 759100	Diversity = 1.4503e-05	k_t
Generation 7600: Best = 1.381721e+04 = [2, 4, 3, 4]	FES = 760100	Diversity = 1.4294e-05	k_t
Generation 7610: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 761100	Diversity = 1.4333e-05	k_t
Generation 7620: Best = 1.381721e+04 = [3, 2, 2, 3]	FES = 762100	Diversity = 1.4171e-05	k_t
Generation 7630: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 763100	Diversity = 1.4215e-05	k_t
Generation 7640: Best = 1.381721e+04 = [3, 3, 4, 4]	FES = 764100	Diversity = 1.3987e-05	k_t
Generation 7650: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 765100	Diversity = 1.4006e-05	k_t
Generation 7660: Best = 1.381721e+04 = [4, 4, 3, 2]	FES = 766100	Diversity = 1.4282e-05	k_t
Generation 7670: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 767100	Diversity = 1.4053e-05	k_t
Generation 7680: Best = 1.381721e+04 = [2, 4, 2, 2]	FES = 768100	Diversity = 1.3532e-05	k_t
Generation 7690: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 769100	Diversity = 1.3175e-05	k_t
Generation 7700: Best = 1.381721e+04 = [4, 2, 3, 2]	FES = 770100	Diversity = 1.2412e-05	k_t
Generation 7710: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 771100	Diversity = 1.1199e-05	k_t
Generation 7720: Best = 1.381721e+04 = [2, 4, 4, 4]	FES = 772100	Diversity = 1.0227e-05	k_t
Generation 7730: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 773100	Diversity = 9.5959e-06	k_t
Generation 7740: Best = 1.381721e+04 = [2, 2, 2, 4]	FES = 774100	Diversity = 8.8783e-06	k_t
Generation 7750: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 775100	Diversity = 8.4591e-06	k_t
Generation 7760: Best = 1.381721e+04 = [2, 2, 2, 4]	FES = 776100	Diversity = 8.1455e-06	k_t
Generation 7770: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 777100	Diversity = 8.2219e-06	k_t
Generation 7780: Best = 1.381721e+04 = [4, 3, 2, 4]	FES = 778100	Diversity = 7.9534e-06	k_t

Generation 7790: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 779100	Diversity = 8.2532e-06	k_t
Generation 7800: Best = 1.381721e+04 = [2, 4, 2, 4]	FES = 780100	Diversity = 8.3393e-06	k_t
Generation 7810: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 781100	Diversity = 8.3196e-06	k_t
Generation 7820: Best = 1.381721e+04 = [3, 3, 3, 4]	FES = 782100	Diversity = 8.0147e-06	k_t
Generation 7830: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 783100	Diversity = 7.9797e-06	k_t
Generation 7840: Best = 1.381721e+04 = [3, 4, 2, 4]	FES = 784100	Diversity = 7.8193e-06	k_t
Generation 7850: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 785100	Diversity = 7.6911e-06	k_t
Generation 7860: Best = 1.381721e+04 = [4, 2, 4, 2]	FES = 786100	Diversity = 7.4453e-06	k_t
Generation 7870: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 787100	Diversity = 6.8533e-06	k_t
Generation 7880: Best = 1.381721e+04 = [4, 4, 4, 4]	FES = 788100	Diversity = 6.9650e-06	k_t
Generation 7890: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 789100	Diversity = 6.8969e-06	k_t
Generation 7900: Best = 1.381721e+04 = [3, 3, 2, 3]	FES = 790100	Diversity = 6.8238e-06	k_t
Generation 7910: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 791100	Diversity = 6.3885e-06	k_t
Generation 7920: Best = 1.381721e+04 = [2, 4, 4, 3]	FES = 792100	Diversity = 6.3035e-06	k_t
Generation 7930: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 793100	Diversity = 6.0491e-06	k_t
Generation 7940: Best = 1.381721e+04 = [3, 2, 3, 3]	FES = 794100	Diversity = 5.9857e-06	k_t
Generation 7950: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 795100	Diversity = 5.7051e-06	k_t
Generation 7960: Best = 1.381721e+04 = [4, 4, 3, 2]	FES = 796100	Diversity = 5.9879e-06	k_t
Generation 7970: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 797100	Diversity = 6.0005e-06	k_t
Generation 7980: Best = 1.381721e+04 = [2, 3, 2, 2]	FES = 798100	Diversity = 6.0064e-06	k_t
Generation 7990: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 799100	Diversity = 6.0830e-06	k_t
Generation 8000: Best = 1.381721e+04 = [3, 4, 4, 4]	FES = 800100	Diversity = 5.9963e-06	k_t
Generation 8010: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 801100	Diversity = 6.1134e-06	k_t
Generation 8020: Best = 1.381721e+04 = [2, 3, 4, 2]	FES = 802100	Diversity = 5.9630e-06	k_t
Generation 8030: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 803100	Diversity = 5.9365e-06	k_t
Generation 8040: Best = 1.381721e+04 = [2, 2, 2, 3]	FES = 804100	Diversity = 5.9501e-06	k_t
Generation 8050: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 805100	Diversity = 5.5903e-06	k_t

Generation 8060: Best = 1.381721e+04 = [3, 3, 3, 3]	FES = 806100	Diversity = 5.4909e-06	k_t
Generation 8070: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 807100	Diversity = 5.2452e-06	k_t
Generation 8080: Best = 1.381721e+04 = [3, 3, 3, 2]	FES = 808100	Diversity = 4.8773e-06	k_t
Generation 8090: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 809100	Diversity = 4.7748e-06	k_t
Generation 8100: Best = 1.381721e+04 = [2, 4, 3, 2]	FES = 810100	Diversity = 4.5932e-06	k_t
Generation 8110: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 811100	Diversity = 4.3257e-06	k_t
Generation 8120: Best = 1.381721e+04 = [2, 2, 2, 4]	FES = 812100	Diversity = 4.1095e-06	k_t
Generation 8130: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 813100	Diversity = 4.1752e-06	k_t
Generation 8140: Best = 1.381721e+04 = [3, 4, 2, 4]	FES = 814100	Diversity = 4.0873e-06	k_t
Generation 8150: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 815100	Diversity = 3.9115e-06	k_t
Generation 8160: Best = 1.381721e+04 = [2, 4, 3, 2]	FES = 816100	Diversity = 3.7315e-06	k_t
Generation 8170: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 817100	Diversity = 3.6240e-06	k_t
Generation 8180: Best = 1.381721e+04 = [4, 4, 4, 3]	FES = 818100	Diversity = 3.7739e-06	k_t
Generation 8190: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 819100	Diversity = 3.7564e-06	k_t
Generation 8200: Best = 1.381721e+04 = [4, 4, 4, 4]	FES = 820100	Diversity = 3.9161e-06	k_t
Generation 8210: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 821100	Diversity = 4.2722e-06	k_t
Generation 8220: Best = 1.381721e+04 = [4, 4, 4, 2]	FES = 822100	Diversity = 4.1156e-06	k_t
Generation 8230: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 823100	Diversity = 4.1179e-06	k_t
Generation 8240: Best = 1.381721e+04 = [2, 3, 2, 2]	FES = 824100	Diversity = 4.5913e-06	k_t
Generation 8250: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 825100	Diversity = 4.9387e-06	k_t
Generation 8260: Best = 1.381721e+04 = [3, 3, 3, 4]	FES = 826100	Diversity = 5.2201e-06	k_t
Generation 8270: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 827100	Diversity = 5.5657e-06	k_t
Generation 8280: Best = 1.381721e+04 = [3, 2, 3, 3]	FES = 828100	Diversity = 5.2647e-06	k_t
Generation 8290: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 829100	Diversity = 4.8934e-06	k_t
Generation 8300: Best = 1.381721e+04 = [3, 3, 4, 2]	FES = 830100	Diversity = 4.7372e-06	k_t
Generation 8310: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 831100	Diversity = 4.6980e-06	k_t
Generation 8320: Best = 1.381721e+04 = [3, 4, 4, 2]	FES = 832100	Diversity = 4.5291e-06	k_t

Generation 8330: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 833100	Diversity = 4.3412e-06	k_t
Generation 8340: Best = 1.381721e+04 = [3, 4, 3, 4]	FES = 834100	Diversity = 4.3900e-06	k_t
Generation 8350: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 835100	Diversity = 4.4093e-06	k_t
Generation 8360: Best = 1.381721e+04 = [3, 4, 4, 4]	FES = 836100	Diversity = 4.4694e-06	k_t
Generation 8370: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 837100	Diversity = 4.3777e-06	k_t
Generation 8380: Best = 1.381721e+04 = [2, 3, 3, 3]	FES = 838100	Diversity = 4.3807e-06	k_t
Generation 8390: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 839100	Diversity = 4.1490e-06	k_t
Generation 8400: Best = 1.381721e+04 = [4, 3, 3, 3]	FES = 840100	Diversity = 4.0079e-06	k_t
Generation 8410: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 841100	Diversity = 3.8313e-06	k_t
Generation 8420: Best = 1.381721e+04 = [3, 3, 3, 4]	FES = 842100	Diversity = 3.9107e-06	k_t
Generation 8430: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 843100	Diversity = 3.8169e-06	k_t
Generation 8440: Best = 1.381721e+04 = [4, 4, 2, 2]	FES = 844100	Diversity = 3.6751e-06	k_t
Generation 8450: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 845100	Diversity = 3.5688e-06	k_t
Generation 8460: Best = 1.381721e+04 = [2, 2, 2, 4]	FES = 846100	Diversity = 3.4706e-06	k_t
Generation 8470: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 847100	Diversity = 3.2524e-06	k_t
Generation 8480: Best = 1.381721e+04 = [4, 2, 3, 2]	FES = 848100	Diversity = 3.1730e-06	k_t
Generation 8490: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 849100	Diversity = 3.1937e-06	k_t
Generation 8500: Best = 1.381721e+04 = [2, 4, 4, 2]	FES = 850100	Diversity = 3.1694e-06	k_t
Generation 8510: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 851100	Diversity = 2.8499e-06	k_t
Generation 8520: Best = 1.381721e+04 = [4, 3, 2, 4]	FES = 852100	Diversity = 2.6580e-06	k_t
Generation 8530: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 853100	Diversity = 2.4447e-06	k_t
Generation 8540: Best = 1.381721e+04 = [4, 2, 2, 4]	FES = 854100	Diversity = 2.3594e-06	k_t
Generation 8550: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 855100	Diversity = 2.3455e-06	k_t
Generation 8560: Best = 1.381721e+04 = [2, 3, 3, 3]	FES = 856100	Diversity = 2.2597e-06	k_t
Generation 8570: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 857100	Diversity = 2.0926e-06	k_t
Generation 8580: Best = 1.381721e+04 = [2, 2, 2, 2]	FES = 858100	Diversity = 1.8644e-06	k_t
Generation 8590: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 859100	Diversity = 1.8044e-06	k_t

Generation 8600: Best = 1.381721e+04 = [4, 4, 2, 2]	FES = 860100	Diversity = 1.7938e-06	k_t
Generation 8610: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 861100	Diversity = 1.7683e-06	k_t
Generation 8620: Best = 1.381721e+04 = [2, 3, 4, 4]	FES = 862100	Diversity = 1.8036e-06	k_t
Generation 8630: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 863100	Diversity = 1.7976e-06	k_t
Generation 8640: Best = 1.381721e+04 = [2, 4, 4, 4]	FES = 864100	Diversity = 1.8766e-06	k_t
Generation 8650: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 865100	Diversity = 1.9853e-06	k_t
Generation 8660: Best = 1.381721e+04 = [2, 4, 4, 2]	FES = 866100	Diversity = 2.0255e-06	k_t
Generation 8670: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 867100	Diversity = 1.9962e-06	k_t
Generation 8680: Best = 1.381721e+04 = [4, 2, 2, 3]	FES = 868100	Diversity = 1.9286e-06	k_t
Generation 8690: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 869100	Diversity = 1.8456e-06	k_t
Generation 8700: Best = 1.381721e+04 = [3, 2, 2, 3]	FES = 870100	Diversity = 1.8096e-06	k_t
Generation 8710: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 871100	Diversity = 1.9234e-06	k_t
Generation 8720: Best = 1.381721e+04 = [4, 2, 2, 4]	FES = 872100	Diversity = 1.9201e-06	k_t
Generation 8730: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 873100	Diversity = 1.8852e-06	k_t
Generation 8740: Best = 1.381721e+04 = [2, 4, 3, 2]	FES = 874100	Diversity = 1.8396e-06	k_t
Generation 8750: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 875100	Diversity = 1.8334e-06	k_t
Generation 8760: Best = 1.381721e+04 = [4, 4, 4, 3]	FES = 876100	Diversity = 1.7535e-06	k_t
Generation 8770: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 877100	Diversity = 1.6414e-06	k_t
Generation 8780: Best = 1.381721e+04 = [3, 3, 2, 2]	FES = 878100	Diversity = 1.5939e-06	k_t
Generation 8790: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 879100	Diversity = 1.5614e-06	k_t
Generation 8800: Best = 1.381721e+04 = [3, 4, 3, 3]	FES = 880100	Diversity = 1.4960e-06	k_t
Generation 8810: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 881100	Diversity = 1.3943e-06	k_t
Generation 8820: Best = 1.381721e+04 = [4, 3, 4, 3]	FES = 882100	Diversity = 1.3437e-06	k_t
Generation 8830: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 883100	Diversity = 1.3294e-06	k_t
Generation 8840: Best = 1.381721e+04 = [3, 4, 3, 3]	FES = 884100	Diversity = 1.2937e-06	k_t
Generation 8850: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 885100	Diversity = 1.2954e-06	k_t
Generation 8860: Best = 1.381721e+04 = [4, 3, 4, 3]	FES = 886100	Diversity = 1.3116e-06	k_t

Generation 8870: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 887100	Diversity = 1.3110e-06	k_t
Generation 8880: Best = 1.381721e+04 = [3, 3, 4, 2]	FES = 888100	Diversity = 1.2838e-06	k_t
Generation 8890: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 889100	Diversity = 1.2867e-06	k_t
Generation 8900: Best = 1.381721e+04 = [2, 4, 2, 4]	FES = 890100	Diversity = 1.2342e-06	k_t
Generation 8910: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 891100	Diversity = 1.2460e-06	k_t
Generation 8920: Best = 1.381721e+04 = [2, 2, 2, 4]	FES = 892100	Diversity = 1.2266e-06	k_t
Generation 8930: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 893100	Diversity = 1.2164e-06	k_t
Generation 8940: Best = 1.381721e+04 = [2, 4, 4, 2]	FES = 894100	Diversity = 1.1914e-06	k_t
Generation 8950: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 895100	Diversity = 1.1118e-06	k_t
Generation 8960: Best = 1.381721e+04 = [3, 2, 2, 3]	FES = 896100	Diversity = 1.0744e-06	k_t
Generation 8970: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 897100	Diversity = 1.0605e-06	k_t
Generation 8980: Best = 1.381721e+04 = [3, 3, 3, 4]	FES = 898100	Diversity = 1.0589e-06	k_t
Generation 8990: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 899100	Diversity = 1.0934e-06	k_t
Generation 9000: Best = 1.381721e+04 = [4, 2, 4, 4]	FES = 900100	Diversity = 1.0582e-06	k_t
Generation 9010: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 901100	Diversity = 1.0304e-06	k_t
Generation 9020: Best = 1.381721e+04 = [2, 3, 4, 4]	FES = 902100	Diversity = 1.0065e-06	k_t
Generation 9030: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 903100	Diversity = 1.0413e-06	k_t
Generation 9040: Best = 1.381721e+04 = [4, 3, 4, 4]	FES = 904100	Diversity = 9.4320e-07	k_t
Generation 9050: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 905100	Diversity = 8.6180e-07	k_t
Generation 9060: Best = 1.381721e+04 = [4, 3, 4, 4]	FES = 906100	Diversity = 8.2489e-07	k_t
Generation 9070: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 907100	Diversity = 7.8562e-07	k_t
Generation 9080: Best = 1.381721e+04 = [3, 3, 4, 3]	FES = 908100	Diversity = 7.6306e-07	k_t
Generation 9090: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 909100	Diversity = 7.7913e-07	k_t
Generation 9100: Best = 1.381721e+04 = [3, 2, 3, 2]	FES = 910100	Diversity = 7.3310e-07	k_t
Generation 9110: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 911100	Diversity = 7.0469e-07	k_t
Generation 9120: Best = 1.381721e+04 = [2, 4, 2, 2]	FES = 912100	Diversity = 6.7320e-07	k_t
Generation 9130: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 913100	Diversity = 6.7175e-07	k_t

Generation 9140: Best = 1.381721e+04 = [3, 4, 3, 2]	FES = 914100	Diversity = 6.1886e-07	k_t
Generation 9150: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 915100	Diversity = 6.1365e-07	k_t
Generation 9160: Best = 1.381721e+04 = [2, 2, 3, 2]	FES = 916100	Diversity = 6.2006e-07	k_t
Generation 9170: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 917100	Diversity = 6.2356e-07	k_t
Generation 9180: Best = 1.381721e+04 = [3, 4, 3, 3]	FES = 918100	Diversity = 6.4982e-07	k_t
Generation 9190: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 919100	Diversity = 6.7502e-07	k_t
Generation 9200: Best = 1.381721e+04 = [3, 2, 2, 2]	FES = 920100	Diversity = 7.2349e-07	k_t
Generation 9210: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 921100	Diversity = 7.3766e-07	k_t
Generation 9220: Best = 1.381721e+04 = [2, 3, 2, 4]	FES = 922100	Diversity = 7.8621e-07	k_t
Generation 9230: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 923100	Diversity = 7.6809e-07	k_t
Generation 9240: Best = 1.381721e+04 = [2, 4, 3, 3]	FES = 924100	Diversity = 7.5630e-07	k_t
Generation 9250: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 925100	Diversity = 7.4737e-07	k_t
Generation 9260: Best = 1.381721e+04 = [2, 2, 2, 3]	FES = 926100	Diversity = 7.3740e-07	k_t
Generation 9270: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 927100	Diversity = 7.1084e-07	k_t
Generation 9280: Best = 1.381721e+04 = [3, 2, 4, 3]	FES = 928100	Diversity = 6.9984e-07	k_t
Generation 9290: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 929100	Diversity = 6.9022e-07	k_t
Generation 9300: Best = 1.381721e+04 = [2, 2, 2, 4]	FES = 930100	Diversity = 6.8040e-07	k_t
Generation 9310: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 931100	Diversity = 7.4592e-07	k_t
Generation 9320: Best = 1.381721e+04 = [4, 4, 2, 4]	FES = 932100	Diversity = 7.5936e-07	k_t
Generation 9330: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 933100	Diversity = 7.5061e-07	k_t
Generation 9340: Best = 1.381721e+04 = [4, 3, 4, 3]	FES = 934100	Diversity = 7.2508e-07	k_t
Generation 9350: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 935100	Diversity = 6.8789e-07	k_t
Generation 9360: Best = 1.381721e+04 = [2, 4, 3, 4]	FES = 936100	Diversity = 6.7454e-07	k_t
Generation 9370: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 937100	Diversity = 6.3558e-07	k_t
Generation 9380: Best = 1.381721e+04 = [3, 4, 4, 4]	FES = 938100	Diversity = 6.2206e-07	k_t
Generation 9390: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 939100	Diversity = 6.0364e-07	k_t
Generation 9400: Best = 1.381721e+04 = [4, 4, 3, 2]	FES = 940100	Diversity = 5.8150e-07	k_t

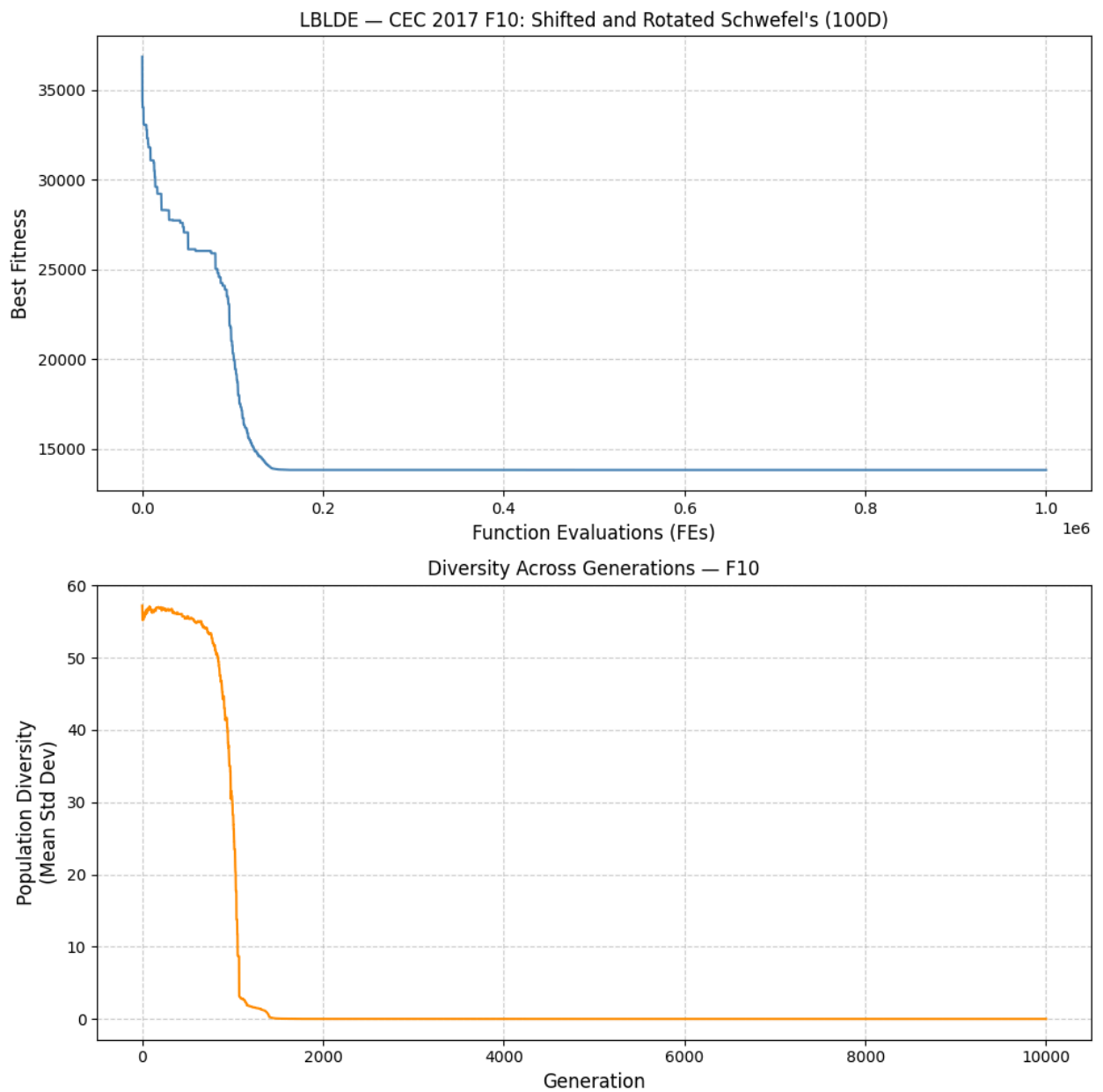
Generation 9410: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 941100	Diversity = 5.6840e-07	k_t
Generation 9420: Best = 1.381721e+04 = [2, 2, 4, 3]	FES = 942100	Diversity = 5.5447e-07	k_t
Generation 9430: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 943100	Diversity = 5.4208e-07	k_t
Generation 9440: Best = 1.381721e+04 = [2, 3, 4, 3]	FES = 944100	Diversity = 5.2768e-07	k_t
Generation 9450: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 945100	Diversity = 5.1170e-07	k_t
Generation 9460: Best = 1.381721e+04 = [2, 4, 3, 3]	FES = 946100	Diversity = 5.5493e-07	k_t
Generation 9470: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 947100	Diversity = 6.1737e-07	k_t
Generation 9480: Best = 1.381721e+04 = [3, 2, 3, 2]	FES = 948100	Diversity = 6.2124e-07	k_t
Generation 9490: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 949100	Diversity = 6.3670e-07	k_t
Generation 9500: Best = 1.381721e+04 = [2, 2, 2, 3]	FES = 950100	Diversity = 6.2968e-07	k_t
Generation 9510: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 951100	Diversity = 6.4584e-07	k_t
Generation 9520: Best = 1.381721e+04 = [4, 2, 3, 2]	FES = 952100	Diversity = 6.4956e-07	k_t
Generation 9530: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 953100	Diversity = 6.5618e-07	k_t
Generation 9540: Best = 1.381721e+04 = [2, 4, 3, 4]	FES = 954100	Diversity = 6.6571e-07	k_t
Generation 9550: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 955100	Diversity = 6.4370e-07	k_t
Generation 9560: Best = 1.381721e+04 = [2, 4, 2, 2]	FES = 956100	Diversity = 6.2434e-07	k_t
Generation 9570: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 957100	Diversity = 6.2461e-07	k_t
Generation 9580: Best = 1.381721e+04 = [4, 4, 2, 4]	FES = 958100	Diversity = 6.4409e-07	k_t
Generation 9590: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 959100	Diversity = 6.3818e-07	k_t
Generation 9600: Best = 1.381721e+04 = [3, 4, 4, 4]	FES = 960100	Diversity = 6.1849e-07	k_t
Generation 9610: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 961100	Diversity = 5.9606e-07	k_t
Generation 9620: Best = 1.381721e+04 = [3, 3, 2, 3]	FES = 962100	Diversity = 5.8386e-07	k_t
Generation 9630: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 963100	Diversity = 5.8456e-07	k_t
Generation 9640: Best = 1.381721e+04 = [2, 3, 3, 3]	FES = 964100	Diversity = 5.5661e-07	k_t
Generation 9650: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 965100	Diversity = 5.0393e-07	k_t
Generation 9660: Best = 1.381721e+04 = [2, 4, 2, 3]	FES = 966100	Diversity = 4.7591e-07	k_t
Generation 9670: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 967100	Diversity = 4.7675e-07	k_t

Generation 9680: Best = 1.381721e+04 = [2, 2, 2, 2]	FES = 968100	Diversity = 4.7884e-07	k_t
Generation 9690: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 969100	Diversity = 4.5988e-07	k_t
Generation 9700: Best = 1.381721e+04 = [3, 3, 4, 2]	FES = 970100	Diversity = 4.2682e-07	k_t
Generation 9710: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 971100	Diversity = 4.2729e-07	k_t
Generation 9720: Best = 1.381721e+04 = [3, 2, 2, 3]	FES = 972100	Diversity = 4.1559e-07	k_t
Generation 9730: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 973100	Diversity = 4.1248e-07	k_t
Generation 9740: Best = 1.381721e+04 = [4, 2, 3, 3]	FES = 974100	Diversity = 3.9295e-07	k_t
Generation 9750: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 975100	Diversity = 3.8263e-07	k_t
Generation 9760: Best = 1.381721e+04 = [2, 2, 2, 3]	FES = 976100	Diversity = 3.6381e-07	k_t
Generation 9770: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 977100	Diversity = 3.6668e-07	k_t
Generation 9780: Best = 1.381721e+04 = [3, 2, 4, 4]	FES = 978100	Diversity = 3.8740e-07	k_t
Generation 9790: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 979100	Diversity = 3.7281e-07	k_t
Generation 9800: Best = 1.381721e+04 = [4, 4, 2, 3]	FES = 980100	Diversity = 3.4414e-07	k_t
Generation 9810: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 981100	Diversity = 3.2845e-07	k_t
Generation 9820: Best = 1.381721e+04 = [3, 2, 3, 2]	FES = 982100	Diversity = 3.2684e-07	k_t
Generation 9830: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 983100	Diversity = 3.2720e-07	k_t
Generation 9840: Best = 1.381721e+04 = [2, 2, 4, 2]	FES = 984100	Diversity = 3.2730e-07	k_t
Generation 9850: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 985100	Diversity = 3.2465e-07	k_t
Generation 9860: Best = 1.381721e+04 = [4, 3, 3, 2]	FES = 986100	Diversity = 3.1931e-07	k_t
Generation 9870: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 987100	Diversity = 3.2724e-07	k_t
Generation 9880: Best = 1.381721e+04 = [2, 4, 4, 4]	FES = 988100	Diversity = 3.1938e-07	k_t
Generation 9890: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 989100	Diversity = 3.2188e-07	k_t
Generation 9900: Best = 1.381721e+04 = [2, 2, 4, 4]	FES = 990100	Diversity = 3.1818e-07	k_t
Generation 9910: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 991100	Diversity = 3.4852e-07	k_t
Generation 9920: Best = 1.381721e+04 = [4, 4, 4, 4]	FES = 992100	Diversity = 3.8483e-07	k_t
Generation 9930: Best = 1.381721e+04 = [1, 1, 1, 1]	FES = 993100	Diversity = 4.0569e-07	k_t
Generation 9940: Best = 1.381721e+04 = [3, 2, 3, 3]	FES = 994100	Diversity = 4.2473e-07	k_t

Generation 9950: Best = 1.381721e+04 FES = 995100 Diversity = 4.6029e-07 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 1.381721e+04 FES = 996100 Diversity = 4.8114e-07 k_t
= [3, 2, 2, 4]
Generation 9970: Best = 1.381721e+04 FES = 997100 Diversity = 4.7537e-07 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 1.381721e+04 FES = 998100 Diversity = 4.6930e-07 k_t
= [2, 4, 4, 3]
Generation 9990: Best = 1.381721e+04 FES = 999100 Diversity = 4.7833e-07 k_t
= [1, 1, 1, 1]

Optimization completed!
Final Best Fitness = 1.381721e+04
Total Function Evaluations = 1000000

=====
=
Results:
Best Fitness : 1.381721e+04
Global Optimum : 1.000000e+03
Error (|best-optimum|) : 1.281721e+04
Best Solution (5 dim) : [24.53954141 -88.70397249 13.33704218 -27.5266173
-78.63424069]
D^(0) (init. diversity): 5.7220e+01
=====



Graph saved as convergence_F10.png

6.3 Hybrid Functions (F11-F20)

```
In [35]: # =====
# F11 — Hybrid Function 1 (Zakharov; Rosenbrock; Rastrigin)
# =====
run_and_plot('F11', make_wrapper(hybrid.f11), 1100.0, 'Hybrid: Zakharov; Roser
```

```

=====
=
LBLDE_s on CEC 2017 F11 – Single Run
=====
=
Function      : Hybrid: Zakharov; Rosenbrock; Rastrigin (100D)
Bounds        : [-100, 100]
Global optimum : 1100.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 7.262506e+05 | Diversity = 5.7220e+01
Generation 10: Best = 3.657557e+05 FES = 1100 Diversity = 5.5052e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.662910e+05 FES = 2100 Diversity = 4.9896e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 2.280606e+05 FES = 3100 Diversity = 4.3093e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.907895e+05 FES = 4100 Diversity = 3.5114e+01 k_t
= [2, 3, 3, 3]
Generation 50: Best = 1.478524e+05 FES = 5100 Diversity = 2.9213e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.138892e+05 FES = 6100 Diversity = 2.3469e+01 k_t
= [4, 3, 3, 2]
Generation 70: Best = 9.320363e+04 FES = 7100 Diversity = 1.8631e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 8.685890e+04 FES = 8100 Diversity = 1.6452e+01 k_t
= [2, 4, 2, 3]
Generation 90: Best = 6.855721e+04 FES = 9100 Diversity = 1.4399e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 6.684414e+04 FES = 10100 Diversity = 1.3882e+01 k_t
= [2, 3, 4, 3]
Generation 110: Best = 6.088636e+04 FES = 11100 Diversity = 1.2843e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 5.425148e+04 FES = 12100 Diversity = 1.2665e+01 k_t
= [4, 3, 2, 2]
Generation 130: Best = 5.130294e+04 FES = 13100 Diversity = 1.1973e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 4.861154e+04 FES = 14100 Diversity = 1.1332e+01 k_t
= [2, 3, 4, 4]
Generation 150: Best = 4.561489e+04 FES = 15100 Diversity = 1.1329e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 4.284888e+04 FES = 16100 Diversity = 1.0838e+01 k_t
= [2, 4, 4, 4]
Generation 170: Best = 4.151946e+04 FES = 17100 Diversity = 1.0287e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.875967e+04 FES = 18100 Diversity = 1.0611e+01 k_t
= [3, 4, 4, 3]
Generation 190: Best = 3.584175e+04 FES = 19100 Diversity = 1.0465e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.471799e+04 FES = 20100 Diversity = 1.0386e+01 k_t
= [3, 4, 2, 2]
Generation 210: Best = 3.192379e+04 FES = 21100 Diversity = 1.0479e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.035925e+04 FES = 22100 Diversity = 1.0507e+01 k_t
= [4, 4, 2, 3]

```


Generation 230:	Best = 2.921476e+04	FES = 23100	Diversity = 1.1274e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 2.658398e+04	FES = 24100	Diversity = 1.1625e+01	k_t = [4, 4, 2, 4]
Generation 250:	Best = 2.633953e+04	FES = 25100	Diversity = 1.1537e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 2.511565e+04	FES = 26100	Diversity = 1.1749e+01	k_t = [3, 3, 4, 3]
Generation 270:	Best = 2.403812e+04	FES = 27100	Diversity = 1.1424e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 2.349415e+04	FES = 28100	Diversity = 1.1277e+01	k_t = [3, 3, 2, 2]
Generation 290:	Best = 2.253486e+04	FES = 29100	Diversity = 1.0975e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 2.179141e+04	FES = 30100	Diversity = 1.0939e+01	k_t = [2, 4, 3, 2]
Generation 310:	Best = 2.130466e+04	FES = 31100	Diversity = 1.1228e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 2.049419e+04	FES = 32100	Diversity = 1.0975e+01	k_t = [3, 4, 2, 4]
Generation 330:	Best = 2.010497e+04	FES = 33100	Diversity = 1.0567e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 1.911263e+04	FES = 34100	Diversity = 1.0117e+01	k_t = [2, 2, 2, 2]
Generation 350:	Best = 1.871778e+04	FES = 35100	Diversity = 9.8303e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 1.814740e+04	FES = 36100	Diversity = 9.9375e+00	k_t = [4, 2, 2, 4]
Generation 370:	Best = 1.775597e+04	FES = 37100	Diversity = 1.0197e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.726552e+04	FES = 38100	Diversity = 9.9474e+00	k_t = [4, 2, 4, 2]
Generation 390:	Best = 1.682742e+04	FES = 39100	Diversity = 9.8704e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.622447e+04	FES = 40100	Diversity = 9.7758e+00	k_t = [2, 4, 2, 2]
Generation 410:	Best = 1.612947e+04	FES = 41100	Diversity = 1.0303e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.553889e+04	FES = 42100	Diversity = 9.8671e+00	k_t = [2, 3, 2, 3]
Generation 430:	Best = 1.539086e+04	FES = 43100	Diversity = 9.8151e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.480121e+04	FES = 44100	Diversity = 9.5204e+00	k_t = [3, 4, 4, 4]
Generation 450:	Best = 1.457224e+04	FES = 45100	Diversity = 9.7877e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.379301e+04	FES = 46100	Diversity = 9.5755e+00	k_t = [2, 3, 4, 2]
Generation 470:	Best = 1.349779e+04	FES = 47100	Diversity = 1.0047e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.343249e+04	FES = 48100	Diversity = 1.0359e+01	k_t = [3, 4, 4, 4]
Generation 490:	Best = 1.317776e+04	FES = 49100	Diversity = 1.0010e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.301624e+04	FES = 50100	Diversity = 9.4689e+00	k_t
= [3, 3, 4, 2]				
Generation 510:	Best = 1.271175e+04	FES = 51100	Diversity = 9.2455e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 1.241762e+04	FES = 52100	Diversity = 9.1619e+00	k_t
= [4, 3, 3, 4]				
Generation 530:	Best = 1.207643e+04	FES = 53100	Diversity = 9.3262e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 1.183609e+04	FES = 54100	Diversity = 9.5110e+00	k_t
= [4, 2, 3, 3]				
Generation 550:	Best = 1.159253e+04	FES = 55100	Diversity = 9.4415e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 1.149319e+04	FES = 56100	Diversity = 8.4603e+00	k_t
= [3, 2, 3, 3]				
Generation 570:	Best = 1.116449e+04	FES = 57100	Diversity = 8.0282e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 1.114568e+04	FES = 58100	Diversity = 8.1145e+00	k_t
= [4, 4, 3, 3]				
Generation 590:	Best = 1.082615e+04	FES = 59100	Diversity = 7.6686e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 1.063869e+04	FES = 60100	Diversity = 7.7922e+00	k_t
= [4, 2, 4, 2]				
Generation 610:	Best = 1.042547e+04	FES = 61100	Diversity = 7.4649e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 1.041707e+04	FES = 62100	Diversity = 7.2174e+00	k_t
= [3, 4, 4, 2]				
Generation 630:	Best = 1.002748e+04	FES = 63100	Diversity = 6.9234e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 1.002589e+04	FES = 64100	Diversity = 6.8807e+00	k_t
= [4, 3, 2, 4]				
Generation 650:	Best = 9.939303e+03	FES = 65100	Diversity = 6.5524e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 9.813047e+03	FES = 66100	Diversity = 6.3440e+00	k_t
= [2, 2, 2, 2]				
Generation 670:	Best = 9.617080e+03	FES = 67100	Diversity = 6.4641e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 9.567029e+03	FES = 68100	Diversity = 6.5094e+00	k_t
= [3, 2, 3, 3]				
Generation 690:	Best = 9.368313e+03	FES = 69100	Diversity = 6.5666e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 9.207005e+03	FES = 70100	Diversity = 6.7222e+00	k_t
= [4, 3, 3, 4]				
Generation 710:	Best = 9.076911e+03	FES = 71100	Diversity = 6.7304e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 8.889475e+03	FES = 72100	Diversity = 6.3333e+00	k_t
= [2, 3, 2, 4]				
Generation 730:	Best = 8.800777e+03	FES = 73100	Diversity = 6.7787e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 8.665303e+03	FES = 74100	Diversity = 6.7760e+00	k_t
= [3, 4, 2, 2]				
Generation 750:	Best = 8.619390e+03	FES = 75100	Diversity = 6.6323e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 8.486437e+03	FES = 76100	Diversity = 6.5846e+00	k_t
= [3, 4, 2, 3]				

Generation 770:	Best = 8.367471e+03	FES = 77100	Diversity = 6.5842e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 8.201228e+03	FES = 78100	Diversity = 6.8291e+00	k_t
= [3, 3, 3, 3]				
Generation 790:	Best = 8.142260e+03	FES = 79100	Diversity = 6.6315e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 7.987799e+03	FES = 80100	Diversity = 6.9909e+00	k_t
= [2, 2, 4, 2]				
Generation 810:	Best = 7.747232e+03	FES = 81100	Diversity = 7.3901e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 7.581216e+03	FES = 82100	Diversity = 7.7066e+00	k_t
= [4, 4, 3, 4]				
Generation 830:	Best = 7.543487e+03	FES = 83100	Diversity = 7.9492e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 7.469336e+03	FES = 84100	Diversity = 7.3944e+00	k_t
= [4, 4, 4, 4]				
Generation 850:	Best = 7.364939e+03	FES = 85100	Diversity = 7.0818e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 7.208012e+03	FES = 86100	Diversity = 6.6540e+00	k_t
= [2, 2, 3, 3]				
Generation 870:	Best = 7.171708e+03	FES = 87100	Diversity = 6.4055e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 7.073239e+03	FES = 88100	Diversity = 5.9904e+00	k_t
= [4, 3, 2, 4]				
Generation 890:	Best = 6.977285e+03	FES = 89100	Diversity = 5.7398e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 6.900076e+03	FES = 90100	Diversity = 5.7195e+00	k_t
= [4, 2, 4, 2]				
Generation 910:	Best = 6.809051e+03	FES = 91100	Diversity = 5.7375e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 6.746548e+03	FES = 92100	Diversity = 5.8945e+00	k_t
= [4, 3, 2, 2]				
Generation 930:	Best = 6.668923e+03	FES = 93100	Diversity = 5.9148e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 6.552858e+03	FES = 94100	Diversity = 5.8087e+00	k_t
= [2, 2, 4, 3]				
Generation 950:	Best = 6.516748e+03	FES = 95100	Diversity = 5.8430e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 6.516748e+03	FES = 96100	Diversity = 6.1378e+00	k_t
= [2, 4, 3, 3]				
Generation 970:	Best = 6.355101e+03	FES = 97100	Diversity = 6.6025e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 6.247759e+03	FES = 98100	Diversity = 6.3430e+00	k_t
= [2, 2, 2, 2]				
Generation 990:	Best = 6.247759e+03	FES = 99100	Diversity = 6.0636e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 6.135062e+03	FES = 100100	Diversity = 5.9909e+00	k_t
= [3, 3, 4, 2]				
Generation 1010:	Best = 6.114391e+03	FES = 101100	Diversity = 6.1450e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 6.099878e+03	FES = 102100	Diversity = 6.0500e+00	k_t
= [3, 3, 2, 4]				
Generation 1030:	Best = 6.057541e+03	FES = 103100	Diversity = 5.8982e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 6.006035e+03 = [2, 2, 4, 4]	FES = 104100	Diversity = 5.6432e+00	k_t
Generation 1050: Best = 5.967644e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 5.4369e+00	k_t
Generation 1060: Best = 5.923058e+03 = [3, 3, 2, 4]	FES = 106100	Diversity = 5.5650e+00	k_t
Generation 1070: Best = 5.908352e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 5.5623e+00	k_t
Generation 1080: Best = 5.877677e+03 = [4, 3, 4, 4]	FES = 108100	Diversity = 5.5687e+00	k_t
Generation 1090: Best = 5.835015e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 5.3844e+00	k_t
Generation 1100: Best = 5.768605e+03 = [2, 3, 2, 4]	FES = 110100	Diversity = 5.4048e+00	k_t
Generation 1110: Best = 5.711429e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 6.1353e+00	k_t
Generation 1120: Best = 5.627141e+03 = [3, 3, 2, 3]	FES = 112100	Diversity = 6.5668e+00	k_t
Generation 1130: Best = 5.536133e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 6.8374e+00	k_t
Generation 1140: Best = 5.388635e+03 = [3, 2, 3, 2]	FES = 114100	Diversity = 6.7827e+00	k_t
Generation 1150: Best = 5.350601e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 6.9672e+00	k_t
Generation 1160: Best = 5.297985e+03 = [4, 2, 4, 4]	FES = 116100	Diversity = 6.8171e+00	k_t
Generation 1170: Best = 5.297985e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 7.0193e+00	k_t
Generation 1180: Best = 5.170268e+03 = [2, 3, 2, 4]	FES = 118100	Diversity = 7.2337e+00	k_t
Generation 1190: Best = 5.166992e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 6.8614e+00	k_t
Generation 1200: Best = 5.091532e+03 = [4, 3, 3, 2]	FES = 120100	Diversity = 6.7044e+00	k_t
Generation 1210: Best = 5.077725e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 6.9256e+00	k_t
Generation 1220: Best = 4.973205e+03 = [3, 4, 2, 2]	FES = 122100	Diversity = 6.8627e+00	k_t
Generation 1230: Best = 4.887783e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 6.7006e+00	k_t
Generation 1240: Best = 4.802782e+03 = [2, 2, 3, 3]	FES = 124100	Diversity = 6.8052e+00	k_t
Generation 1250: Best = 4.727699e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 7.2073e+00	k_t
Generation 1260: Best = 4.638720e+03 = [4, 4, 3, 3]	FES = 126100	Diversity = 7.4947e+00	k_t
Generation 1270: Best = 4.574700e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 7.2378e+00	k_t
Generation 1280: Best = 4.545886e+03 = [2, 2, 2, 4]	FES = 128100	Diversity = 6.9893e+00	k_t
Generation 1290: Best = 4.460557e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 6.8393e+00	k_t
Generation 1300: Best = 4.375207e+03 = [4, 2, 2, 2]	FES = 130100	Diversity = 6.4148e+00	k_t

Generation 1310: Best = 4.342083e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 6.0885e+00	k_t
Generation 1320: Best = 4.307526e+03 = [4, 4, 4, 4]	FES = 132100	Diversity = 5.6391e+00	k_t
Generation 1330: Best = 4.250645e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 5.4810e+00	k_t
Generation 1340: Best = 4.180818e+03 = [2, 4, 3, 3]	FES = 134100	Diversity = 5.4543e+00	k_t
Generation 1350: Best = 4.175535e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 5.4630e+00	k_t
Generation 1360: Best = 4.139081e+03 = [3, 2, 3, 4]	FES = 136100	Diversity = 5.4683e+00	k_t
Generation 1370: Best = 4.047980e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 5.6189e+00	k_t
Generation 1380: Best = 3.984133e+03 = [3, 2, 3, 2]	FES = 138100	Diversity = 6.3192e+00	k_t
Generation 1390: Best = 3.947772e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 6.6691e+00	k_t
Generation 1400: Best = 3.922671e+03 = [3, 2, 4, 3]	FES = 140100	Diversity = 6.5689e+00	k_t
Generation 1410: Best = 3.850010e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 6.3483e+00	k_t
Generation 1420: Best = 3.850010e+03 = [2, 4, 2, 2]	FES = 142100	Diversity = 6.0563e+00	k_t
Generation 1430: Best = 3.728878e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 5.8844e+00	k_t
Generation 1440: Best = 3.710963e+03 = [2, 3, 2, 2]	FES = 144100	Diversity = 5.9875e+00	k_t
Generation 1450: Best = 3.710963e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 5.7932e+00	k_t
Generation 1460: Best = 3.671945e+03 = [4, 2, 3, 3]	FES = 146100	Diversity = 5.7512e+00	k_t
Generation 1470: Best = 3.597354e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 5.8635e+00	k_t
Generation 1480: Best = 3.562628e+03 = [4, 2, 2, 3]	FES = 148100	Diversity = 5.7733e+00	k_t
Generation 1490: Best = 3.533286e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 5.7785e+00	k_t
Generation 1500: Best = 3.487054e+03 = [3, 2, 3, 4]	FES = 150100	Diversity = 5.8585e+00	k_t
Generation 1510: Best = 3.467829e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 5.8899e+00	k_t
Generation 1520: Best = 3.408709e+03 = [2, 2, 4, 4]	FES = 152100	Diversity = 5.9519e+00	k_t
Generation 1530: Best = 3.352144e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 5.8338e+00	k_t
Generation 1540: Best = 3.352144e+03 = [3, 2, 4, 3]	FES = 154100	Diversity = 5.8750e+00	k_t
Generation 1550: Best = 3.295657e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 5.6654e+00	k_t
Generation 1560: Best = 3.259821e+03 = [4, 2, 2, 4]	FES = 156100	Diversity = 5.4738e+00	k_t
Generation 1570: Best = 3.249598e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 5.3776e+00	k_t

Generation 1580: Best = 3.212969e+03 = [2, 2, 3, 2]	FES = 158100	Diversity = 5.6551e+00	k_t
Generation 1590: Best = 3.150025e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 5.7540e+00	k_t
Generation 1600: Best = 3.137651e+03 = [2, 2, 3, 2]	FES = 160100	Diversity = 5.9413e+00	k_t
Generation 1610: Best = 3.105568e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 5.7738e+00	k_t
Generation 1620: Best = 3.085207e+03 = [4, 2, 4, 3]	FES = 162100	Diversity = 5.5577e+00	k_t
Generation 1630: Best = 3.080452e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 5.5853e+00	k_t
Generation 1640: Best = 3.080452e+03 = [4, 2, 4, 2]	FES = 164100	Diversity = 5.5311e+00	k_t
Generation 1650: Best = 3.077493e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 5.5237e+00	k_t
Generation 1660: Best = 3.025651e+03 = [4, 3, 3, 3]	FES = 166100	Diversity = 5.4635e+00	k_t
Generation 1670: Best = 2.962938e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 5.3526e+00	k_t
Generation 1680: Best = 2.960198e+03 = [4, 3, 2, 2]	FES = 168100	Diversity = 5.3795e+00	k_t
Generation 1690: Best = 2.960034e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 5.2918e+00	k_t
Generation 1700: Best = 2.946184e+03 = [2, 4, 2, 4]	FES = 170100	Diversity = 5.0955e+00	k_t
Generation 1710: Best = 2.935281e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 4.8359e+00	k_t
Generation 1720: Best = 2.887589e+03 = [3, 2, 2, 3]	FES = 172100	Diversity = 4.9138e+00	k_t
Generation 1730: Best = 2.884048e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 5.0737e+00	k_t
Generation 1740: Best = 2.880940e+03 = [2, 4, 3, 4]	FES = 174100	Diversity = 5.1060e+00	k_t
Generation 1750: Best = 2.878562e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 5.1899e+00	k_t
Generation 1760: Best = 2.856784e+03 = [4, 3, 3, 4]	FES = 176100	Diversity = 5.1050e+00	k_t
Generation 1770: Best = 2.819267e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 5.0084e+00	k_t
Generation 1780: Best = 2.806726e+03 = [4, 2, 4, 2]	FES = 178100	Diversity = 5.0612e+00	k_t
Generation 1790: Best = 2.780723e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 4.9637e+00	k_t
Generation 1800: Best = 2.780723e+03 = [4, 2, 2, 2]	FES = 180100	Diversity = 4.8961e+00	k_t
Generation 1810: Best = 2.752401e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 4.8422e+00	k_t
Generation 1820: Best = 2.724704e+03 = [2, 4, 3, 4]	FES = 182100	Diversity = 4.7676e+00	k_t
Generation 1830: Best = 2.724704e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 4.7443e+00	k_t
Generation 1840: Best = 2.720168e+03 = [4, 4, 3, 2]	FES = 184100	Diversity = 4.7158e+00	k_t

Generation 1850: Best = 2.720168e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 4.7302e+00	k_t
Generation 1860: Best = 2.718955e+03 = [2, 4, 3, 3]	FES = 186100	Diversity = 4.7702e+00	k_t
Generation 1870: Best = 2.717356e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 4.7527e+00	k_t
Generation 1880: Best = 2.714084e+03 = [2, 4, 4, 3]	FES = 188100	Diversity = 4.6855e+00	k_t
Generation 1890: Best = 2.694990e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 4.6894e+00	k_t
Generation 1900: Best = 2.686497e+03 = [3, 3, 4, 2]	FES = 190100	Diversity = 4.6367e+00	k_t
Generation 1910: Best = 2.673612e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 4.6310e+00	k_t
Generation 1920: Best = 2.667854e+03 = [3, 3, 2, 3]	FES = 192100	Diversity = 4.6027e+00	k_t
Generation 1930: Best = 2.665516e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 4.6308e+00	k_t
Generation 1940: Best = 2.665516e+03 = [3, 2, 3, 4]	FES = 194100	Diversity = 4.5826e+00	k_t
Generation 1950: Best = 2.657212e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 4.5340e+00	k_t
Generation 1960: Best = 2.640474e+03 = [4, 4, 2, 3]	FES = 196100	Diversity = 4.5647e+00	k_t
Generation 1970: Best = 2.640474e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 4.6338e+00	k_t
Generation 1980: Best = 2.612717e+03 = [2, 3, 2, 3]	FES = 198100	Diversity = 4.6410e+00	k_t
Generation 1990: Best = 2.597387e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 4.7037e+00	k_t
Generation 2000: Best = 2.596336e+03 = [2, 2, 4, 3]	FES = 200100	Diversity = 4.7170e+00	k_t
Generation 2010: Best = 2.589352e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 4.7721e+00	k_t
Generation 2020: Best = 2.589352e+03 = [4, 4, 3, 2]	FES = 202100	Diversity = 4.8625e+00	k_t
Generation 2030: Best = 2.555827e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 4.8935e+00	k_t
Generation 2040: Best = 2.555827e+03 = [3, 3, 3, 2]	FES = 204100	Diversity = 5.0059e+00	k_t
Generation 2050: Best = 2.548602e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 4.9665e+00	k_t
Generation 2060: Best = 2.548602e+03 = [3, 2, 2, 2]	FES = 206100	Diversity = 4.9288e+00	k_t
Generation 2070: Best = 2.546766e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 4.9721e+00	k_t
Generation 2080: Best = 2.543419e+03 = [3, 4, 3, 2]	FES = 208100	Diversity = 4.9245e+00	k_t
Generation 2090: Best = 2.543419e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 4.8405e+00	k_t
Generation 2100: Best = 2.543419e+03 = [2, 2, 2, 3]	FES = 210100	Diversity = 4.8755e+00	k_t
Generation 2110: Best = 2.540129e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 4.8298e+00	k_t

Generation 2120: Best = 2.526308e+03 = [2, 4, 3, 4]	FES = 212100	Diversity = 4.7824e+00	k_t
Generation 2130: Best = 2.507125e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 4.8135e+00	k_t
Generation 2140: Best = 2.498587e+03 = [2, 4, 2, 3]	FES = 214100	Diversity = 4.8287e+00	k_t
Generation 2150: Best = 2.498587e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 4.8795e+00	k_t
Generation 2160: Best = 2.466471e+03 = [3, 3, 2, 4]	FES = 216100	Diversity = 4.8074e+00	k_t
Generation 2170: Best = 2.458169e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 4.8324e+00	k_t
Generation 2180: Best = 2.452526e+03 = [3, 4, 4, 4]	FES = 218100	Diversity = 4.7243e+00	k_t
Generation 2190: Best = 2.442202e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 4.7475e+00	k_t
Generation 2200: Best = 2.442202e+03 = [2, 3, 4, 2]	FES = 220100	Diversity = 4.7867e+00	k_t
Generation 2210: Best = 2.442202e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 4.6912e+00	k_t
Generation 2220: Best = 2.441899e+03 = [3, 2, 2, 4]	FES = 222100	Diversity = 4.6998e+00	k_t
Generation 2230: Best = 2.441899e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 4.7168e+00	k_t
Generation 2240: Best = 2.439242e+03 = [3, 4, 4, 3]	FES = 224100	Diversity = 4.7189e+00	k_t
Generation 2250: Best = 2.439242e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 4.5937e+00	k_t
Generation 2260: Best = 2.435646e+03 = [3, 3, 2, 3]	FES = 226100	Diversity = 4.5567e+00	k_t
Generation 2270: Best = 2.415156e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 4.5847e+00	k_t
Generation 2280: Best = 2.415156e+03 = [3, 4, 4, 3]	FES = 228100	Diversity = 4.5737e+00	k_t
Generation 2290: Best = 2.371777e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 4.5466e+00	k_t
Generation 2300: Best = 2.371777e+03 = [3, 2, 4, 4]	FES = 230100	Diversity = 4.4899e+00	k_t
Generation 2310: Best = 2.368517e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 4.4330e+00	k_t
Generation 2320: Best = 2.368517e+03 = [4, 3, 2, 3]	FES = 232100	Diversity = 4.3795e+00	k_t
Generation 2330: Best = 2.367792e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 4.5000e+00	k_t
Generation 2340: Best = 2.367792e+03 = [3, 4, 4, 3]	FES = 234100	Diversity = 4.4836e+00	k_t
Generation 2350: Best = 2.367792e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 4.4786e+00	k_t
Generation 2360: Best = 2.351727e+03 = [2, 3, 4, 2]	FES = 236100	Diversity = 4.4844e+00	k_t
Generation 2370: Best = 2.351727e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 4.4452e+00	k_t
Generation 2380: Best = 2.342780e+03 = [2, 3, 4, 4]	FES = 238100	Diversity = 4.4752e+00	k_t

Generation 2390: Best = 2.320227e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 4.5165e+00	k_t
Generation 2400: Best = 2.305134e+03 = [2, 3, 3, 4]	FES = 240100	Diversity = 4.5331e+00	k_t
Generation 2410: Best = 2.293295e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 4.5721e+00	k_t
Generation 2420: Best = 2.267661e+03 = [2, 4, 3, 4]	FES = 242100	Diversity = 4.6545e+00	k_t
Generation 2430: Best = 2.267661e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 5.0687e+00	k_t
Generation 2440: Best = 2.246464e+03 = [2, 2, 3, 3]	FES = 244100	Diversity = 5.4832e+00	k_t
Generation 2450: Best = 2.246464e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 5.6858e+00	k_t
Generation 2460: Best = 2.216136e+03 = [4, 4, 3, 2]	FES = 246100	Diversity = 5.5214e+00	k_t
Generation 2470: Best = 2.181638e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 5.4487e+00	k_t
Generation 2480: Best = 2.172173e+03 = [4, 4, 3, 2]	FES = 248100	Diversity = 5.5874e+00	k_t
Generation 2490: Best = 2.172173e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 5.3454e+00	k_t
Generation 2500: Best = 2.134743e+03 = [4, 2, 3, 2]	FES = 250100	Diversity = 5.2130e+00	k_t
Generation 2510: Best = 2.111698e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 5.1557e+00	k_t
Generation 2520: Best = 2.085841e+03 = [3, 2, 4, 3]	FES = 252100	Diversity = 5.1062e+00	k_t
Generation 2530: Best = 2.074854e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 5.0089e+00	k_t
Generation 2540: Best = 2.049458e+03 = [2, 2, 2, 2]	FES = 254100	Diversity = 5.0241e+00	k_t
Generation 2550: Best = 2.033665e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 4.9941e+00	k_t
Generation 2560: Best = 2.031440e+03 = [4, 4, 3, 2]	FES = 256100	Diversity = 5.0232e+00	k_t
Generation 2570: Best = 2.026702e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 4.8511e+00	k_t
Generation 2580: Best = 2.003794e+03 = [2, 2, 4, 4]	FES = 258100	Diversity = 4.8337e+00	k_t
Generation 2590: Best = 2.001327e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 4.8814e+00	k_t
Generation 2600: Best = 2.001114e+03 = [3, 4, 4, 4]	FES = 260100	Diversity = 4.8680e+00	k_t
Generation 2610: Best = 1.998690e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 4.7827e+00	k_t
Generation 2620: Best = 1.998690e+03 = [4, 4, 2, 2]	FES = 262100	Diversity = 4.7926e+00	k_t
Generation 2630: Best = 1.985605e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 4.7066e+00	k_t
Generation 2640: Best = 1.985605e+03 = [2, 4, 3, 4]	FES = 264100	Diversity = 4.6674e+00	k_t
Generation 2650: Best = 1.985605e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 4.6640e+00	k_t

Generation 2660: Best = 1.967733e+03 = [4, 4, 4, 2]	FES = 266100	Diversity = 4.6684e+00	k_t
Generation 2670: Best = 1.967155e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 4.6123e+00	k_t
Generation 2680: Best = 1.967155e+03 = [2, 3, 2, 3]	FES = 268100	Diversity = 4.5763e+00	k_t
Generation 2690: Best = 1.963073e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 4.5577e+00	k_t
Generation 2700: Best = 1.963073e+03 = [2, 4, 2, 4]	FES = 270100	Diversity = 4.4725e+00	k_t
Generation 2710: Best = 1.963073e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 4.4463e+00	k_t
Generation 2720: Best = 1.960477e+03 = [4, 3, 3, 4]	FES = 272100	Diversity = 4.4061e+00	k_t
Generation 2730: Best = 1.944207e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 4.3268e+00	k_t
Generation 2740: Best = 1.908357e+03 = [2, 3, 2, 2]	FES = 274100	Diversity = 4.2710e+00	k_t
Generation 2750: Best = 1.908357e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 4.2240e+00	k_t
Generation 2760: Best = 1.908357e+03 = [2, 4, 3, 3]	FES = 276100	Diversity = 4.2105e+00	k_t
Generation 2770: Best = 1.907919e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 4.1819e+00	k_t
Generation 2780: Best = 1.906967e+03 = [3, 3, 4, 2]	FES = 278100	Diversity = 4.1094e+00	k_t
Generation 2790: Best = 1.906967e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 4.0685e+00	k_t
Generation 2800: Best = 1.906967e+03 = [2, 3, 4, 3]	FES = 280100	Diversity = 4.0373e+00	k_t
Generation 2810: Best = 1.906967e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 4.0091e+00	k_t
Generation 2820: Best = 1.906967e+03 = [4, 4, 2, 3]	FES = 282100	Diversity = 4.0970e+00	k_t
Generation 2830: Best = 1.906967e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 4.0673e+00	k_t
Generation 2840: Best = 1.906967e+03 = [3, 4, 3, 4]	FES = 284100	Diversity = 4.0449e+00	k_t
Generation 2850: Best = 1.906154e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 4.0656e+00	k_t
Generation 2860: Best = 1.887278e+03 = [2, 4, 3, 2]	FES = 286100	Diversity = 4.0519e+00	k_t
Generation 2870: Best = 1.887278e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 4.0210e+00	k_t
Generation 2880: Best = 1.887278e+03 = [3, 3, 2, 3]	FES = 288100	Diversity = 3.9674e+00	k_t
Generation 2890: Best = 1.887278e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 3.9137e+00	k_t
Generation 2900: Best = 1.887278e+03 = [4, 4, 4, 4]	FES = 290100	Diversity = 3.8829e+00	k_t
Generation 2910: Best = 1.885396e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 3.8951e+00	k_t
Generation 2920: Best = 1.874488e+03 = [4, 2, 4, 4]	FES = 292100	Diversity = 3.8788e+00	k_t

Generation 2930: Best = 1.871294e+03 = [1, 1, 1, 1]	FES = 293100	Diversity = 3.8009e+00	k_t
Generation 2940: Best = 1.871294e+03 = [4, 3, 3, 2]	FES = 294100	Diversity = 3.8017e+00	k_t
Generation 2950: Best = 1.868160e+03 = [1, 1, 1, 1]	FES = 295100	Diversity = 3.7998e+00	k_t
Generation 2960: Best = 1.861032e+03 = [3, 2, 2, 4]	FES = 296100	Diversity = 3.7691e+00	k_t
Generation 2970: Best = 1.855120e+03 = [1, 1, 1, 1]	FES = 297100	Diversity = 3.7029e+00	k_t
Generation 2980: Best = 1.855120e+03 = [4, 3, 2, 3]	FES = 298100	Diversity = 3.6684e+00	k_t
Generation 2990: Best = 1.855120e+03 = [1, 1, 1, 1]	FES = 299100	Diversity = 3.6261e+00	k_t
Generation 3000: Best = 1.854047e+03 = [2, 4, 3, 2]	FES = 300100	Diversity = 3.6138e+00	k_t
Generation 3010: Best = 1.852305e+03 = [1, 1, 1, 1]	FES = 301100	Diversity = 3.6181e+00	k_t
Generation 3020: Best = 1.852305e+03 = [3, 2, 4, 4]	FES = 302100	Diversity = 3.5786e+00	k_t
Generation 3030: Best = 1.852305e+03 = [1, 1, 1, 1]	FES = 303100	Diversity = 3.5210e+00	k_t
Generation 3040: Best = 1.852305e+03 = [4, 3, 2, 2]	FES = 304100	Diversity = 3.5398e+00	k_t
Generation 3050: Best = 1.849363e+03 = [1, 1, 1, 1]	FES = 305100	Diversity = 3.4875e+00	k_t
Generation 3060: Best = 1.849062e+03 = [3, 3, 2, 2]	FES = 306100	Diversity = 3.4862e+00	k_t
Generation 3070: Best = 1.845225e+03 = [1, 1, 1, 1]	FES = 307100	Diversity = 3.4675e+00	k_t
Generation 3080: Best = 1.836122e+03 = [2, 3, 4, 4]	FES = 308100	Diversity = 3.4737e+00	k_t
Generation 3090: Best = 1.833051e+03 = [1, 1, 1, 1]	FES = 309100	Diversity = 3.4739e+00	k_t
Generation 3100: Best = 1.827223e+03 = [2, 2, 3, 2]	FES = 310100	Diversity = 3.4515e+00	k_t
Generation 3110: Best = 1.819664e+03 = [1, 1, 1, 1]	FES = 311100	Diversity = 3.4587e+00	k_t
Generation 3120: Best = 1.812349e+03 = [4, 4, 3, 4]	FES = 312100	Diversity = 3.4378e+00	k_t
Generation 3130: Best = 1.806644e+03 = [1, 1, 1, 1]	FES = 313100	Diversity = 3.3861e+00	k_t
Generation 3140: Best = 1.803127e+03 = [4, 3, 4, 2]	FES = 314100	Diversity = 3.3944e+00	k_t
Generation 3150: Best = 1.803127e+03 = [1, 1, 1, 1]	FES = 315100	Diversity = 3.3888e+00	k_t
Generation 3160: Best = 1.803127e+03 = [3, 4, 4, 4]	FES = 316100	Diversity = 3.3219e+00	k_t
Generation 3170: Best = 1.803127e+03 = [1, 1, 1, 1]	FES = 317100	Diversity = 3.3185e+00	k_t
Generation 3180: Best = 1.793774e+03 = [4, 3, 2, 2]	FES = 318100	Diversity = 3.3129e+00	k_t
Generation 3190: Best = 1.779152e+03 = [1, 1, 1, 1]	FES = 319100	Diversity = 3.3080e+00	k_t

Generation 3200: Best = 1.779121e+03 = [3, 2, 2, 3]	FES = 320100	Diversity = 3.2718e+00	k_t
Generation 3210: Best = 1.779041e+03 = [1, 1, 1, 1]	FES = 321100	Diversity = 3.2586e+00	k_t
Generation 3220: Best = 1.778841e+03 = [3, 4, 4, 4]	FES = 322100	Diversity = 3.2664e+00	k_t
Generation 3230: Best = 1.778025e+03 = [1, 1, 1, 1]	FES = 323100	Diversity = 3.2063e+00	k_t
Generation 3240: Best = 1.777124e+03 = [2, 3, 4, 3]	FES = 324100	Diversity = 3.2279e+00	k_t
Generation 3250: Best = 1.764916e+03 = [1, 1, 1, 1]	FES = 325100	Diversity = 3.2259e+00	k_t
Generation 3260: Best = 1.760562e+03 = [2, 2, 4, 2]	FES = 326100	Diversity = 3.1638e+00	k_t
Generation 3270: Best = 1.754411e+03 = [1, 1, 1, 1]	FES = 327100	Diversity = 3.1453e+00	k_t
Generation 3280: Best = 1.747761e+03 = [3, 4, 4, 3]	FES = 328100	Diversity = 3.0418e+00	k_t
Generation 3290: Best = 1.743260e+03 = [1, 1, 1, 1]	FES = 329100	Diversity = 2.9341e+00	k_t
Generation 3300: Best = 1.741726e+03 = [2, 3, 4, 3]	FES = 330100	Diversity = 2.7322e+00	k_t
Generation 3310: Best = 1.736783e+03 = [1, 1, 1, 1]	FES = 331100	Diversity = 2.4780e+00	k_t
Generation 3320: Best = 1.728914e+03 = [3, 4, 4, 2]	FES = 332100	Diversity = 2.0558e+00	k_t
Generation 3330: Best = 1.716299e+03 = [1, 1, 1, 1]	FES = 333100	Diversity = 1.6239e+00	k_t
Generation 3340: Best = 1.713294e+03 = [4, 2, 2, 3]	FES = 334100	Diversity = 1.4314e+00	k_t
Generation 3350: Best = 1.699453e+03 = [1, 1, 1, 1]	FES = 335100	Diversity = 1.1952e+00	k_t
Generation 3360: Best = 1.690893e+03 = [2, 4, 4, 4]	FES = 336100	Diversity = 1.1481e+00	k_t
Generation 3370: Best = 1.681240e+03 = [1, 1, 1, 1]	FES = 337100	Diversity = 9.1154e-01	k_t
Generation 3380: Best = 1.666038e+03 = [3, 4, 2, 4]	FES = 338100	Diversity = 6.1831e-01	k_t
Generation 3390: Best = 1.662195e+03 = [1, 1, 1, 1]	FES = 339100	Diversity = 5.5499e-01	k_t
Generation 3400: Best = 1.651591e+03 = [2, 3, 4, 3]	FES = 340100	Diversity = 4.8091e-01	k_t
Generation 3410: Best = 1.642006e+03 = [1, 1, 1, 1]	FES = 341100	Diversity = 4.4433e-01	k_t
Generation 3420: Best = 1.632852e+03 = [3, 3, 3, 4]	FES = 342100	Diversity = 4.2247e-01	k_t
Generation 3430: Best = 1.627670e+03 = [1, 1, 1, 1]	FES = 343100	Diversity = 3.9275e-01	k_t
Generation 3440: Best = 1.620665e+03 = [2, 2, 2, 4]	FES = 344100	Diversity = 3.6151e-01	k_t
Generation 3450: Best = 1.617742e+03 = [1, 1, 1, 1]	FES = 345100	Diversity = 3.3963e-01	k_t
Generation 3460: Best = 1.611076e+03 = [2, 2, 3, 2]	FES = 346100	Diversity = 3.3109e-01	k_t

Generation 3470: Best = 1.608295e+03 = [1, 1, 1, 1]	FES = 347100	Diversity = 3.2522e-01	k_t
Generation 3480: Best = 1.602533e+03 = [4, 4, 2, 3]	FES = 348100	Diversity = 3.2255e-01	k_t
Generation 3490: Best = 1.598551e+03 = [1, 1, 1, 1]	FES = 349100	Diversity = 3.1077e-01	k_t
Generation 3500: Best = 1.594227e+03 = [3, 2, 2, 4]	FES = 350100	Diversity = 3.2054e-01	k_t
Generation 3510: Best = 1.586386e+03 = [1, 1, 1, 1]	FES = 351100	Diversity = 3.1681e-01	k_t
Generation 3520: Best = 1.578989e+03 = [2, 2, 3, 2]	FES = 352100	Diversity = 3.1441e-01	k_t
Generation 3530: Best = 1.574458e+03 = [1, 1, 1, 1]	FES = 353100	Diversity = 3.3202e-01	k_t
Generation 3540: Best = 1.571983e+03 = [2, 4, 2, 3]	FES = 354100	Diversity = 3.1201e-01	k_t
Generation 3550: Best = 1.567795e+03 = [1, 1, 1, 1]	FES = 355100	Diversity = 3.1650e-01	k_t
Generation 3560: Best = 1.562605e+03 = [4, 2, 2, 2]	FES = 356100	Diversity = 2.9918e-01	k_t
Generation 3570: Best = 1.557598e+03 = [1, 1, 1, 1]	FES = 357100	Diversity = 2.8376e-01	k_t
Generation 3580: Best = 1.554131e+03 = [2, 3, 4, 4]	FES = 358100	Diversity = 2.9070e-01	k_t
Generation 3590: Best = 1.551469e+03 = [1, 1, 1, 1]	FES = 359100	Diversity = 2.9000e-01	k_t
Generation 3600: Best = 1.547696e+03 = [3, 3, 3, 4]	FES = 360100	Diversity = 2.8197e-01	k_t
Generation 3610: Best = 1.542570e+03 = [1, 1, 1, 1]	FES = 361100	Diversity = 3.0101e-01	k_t
Generation 3620: Best = 1.537009e+03 = [3, 4, 2, 3]	FES = 362100	Diversity = 2.9461e-01	k_t
Generation 3630: Best = 1.529584e+03 = [1, 1, 1, 1]	FES = 363100	Diversity = 3.1057e-01	k_t
Generation 3640: Best = 1.529077e+03 = [4, 3, 4, 4]	FES = 364100	Diversity = 3.2168e-01	k_t
Generation 3650: Best = 1.526240e+03 = [1, 1, 1, 1]	FES = 365100	Diversity = 3.2721e-01	k_t
Generation 3660: Best = 1.519656e+03 = [3, 2, 2, 2]	FES = 366100	Diversity = 3.3408e-01	k_t
Generation 3670: Best = 1.514584e+03 = [1, 1, 1, 1]	FES = 367100	Diversity = 3.3956e-01	k_t
Generation 3680: Best = 1.509989e+03 = [4, 2, 2, 3]	FES = 368100	Diversity = 3.3160e-01	k_t
Generation 3690: Best = 1.503622e+03 = [1, 1, 1, 1]	FES = 369100	Diversity = 3.3949e-01	k_t
Generation 3700: Best = 1.497579e+03 = [3, 4, 3, 4]	FES = 370100	Diversity = 3.4972e-01	k_t
Generation 3710: Best = 1.493009e+03 = [1, 1, 1, 1]	FES = 371100	Diversity = 3.3879e-01	k_t
Generation 3720: Best = 1.486025e+03 = [3, 4, 3, 3]	FES = 372100	Diversity = 3.5480e-01	k_t
Generation 3730: Best = 1.482385e+03 = [1, 1, 1, 1]	FES = 373100	Diversity = 3.4008e-01	k_t

Generation 3740: Best = 1.477903e+03 = [3, 4, 3, 3]	FES = 374100	Diversity = 3.1191e-01	k_t
Generation 3750: Best = 1.475067e+03 = [1, 1, 1, 1]	FES = 375100	Diversity = 2.9605e-01	k_t
Generation 3760: Best = 1.467287e+03 = [3, 2, 4, 4]	FES = 376100	Diversity = 2.8447e-01	k_t
Generation 3770: Best = 1.464834e+03 = [1, 1, 1, 1]	FES = 377100	Diversity = 2.8149e-01	k_t
Generation 3780: Best = 1.461899e+03 = [4, 2, 2, 2]	FES = 378100	Diversity = 2.7486e-01	k_t
Generation 3790: Best = 1.457590e+03 = [1, 1, 1, 1]	FES = 379100	Diversity = 2.7325e-01	k_t
Generation 3800: Best = 1.454278e+03 = [4, 3, 4, 3]	FES = 380100	Diversity = 2.5570e-01	k_t
Generation 3810: Best = 1.450630e+03 = [1, 1, 1, 1]	FES = 381100	Diversity = 2.4738e-01	k_t
Generation 3820: Best = 1.447795e+03 = [4, 4, 4, 2]	FES = 382100	Diversity = 2.4746e-01	k_t
Generation 3830: Best = 1.444765e+03 = [1, 1, 1, 1]	FES = 383100	Diversity = 2.4731e-01	k_t
Generation 3840: Best = 1.438853e+03 = [3, 4, 3, 4]	FES = 384100	Diversity = 2.5532e-01	k_t
Generation 3850: Best = 1.437003e+03 = [1, 1, 1, 1]	FES = 385100	Diversity = 2.5586e-01	k_t
Generation 3860: Best = 1.434178e+03 = [3, 2, 2, 2]	FES = 386100	Diversity = 2.6239e-01	k_t
Generation 3870: Best = 1.430832e+03 = [1, 1, 1, 1]	FES = 387100	Diversity = 2.6034e-01	k_t
Generation 3880: Best = 1.428504e+03 = [2, 4, 4, 2]	FES = 388100	Diversity = 2.6456e-01	k_t
Generation 3890: Best = 1.426364e+03 = [1, 1, 1, 1]	FES = 389100	Diversity = 2.7146e-01	k_t
Generation 3900: Best = 1.423643e+03 = [3, 3, 2, 4]	FES = 390100	Diversity = 2.7699e-01	k_t
Generation 3910: Best = 1.418827e+03 = [1, 1, 1, 1]	FES = 391100	Diversity = 2.7740e-01	k_t
Generation 3920: Best = 1.417490e+03 = [2, 3, 3, 2]	FES = 392100	Diversity = 2.6268e-01	k_t
Generation 3930: Best = 1.415461e+03 = [1, 1, 1, 1]	FES = 393100	Diversity = 2.5424e-01	k_t
Generation 3940: Best = 1.412368e+03 = [3, 2, 4, 4]	FES = 394100	Diversity = 2.5434e-01	k_t
Generation 3950: Best = 1.409653e+03 = [1, 1, 1, 1]	FES = 395100	Diversity = 2.6643e-01	k_t
Generation 3960: Best = 1.407411e+03 = [3, 3, 3, 4]	FES = 396100	Diversity = 2.5058e-01	k_t
Generation 3970: Best = 1.405779e+03 = [1, 1, 1, 1]	FES = 397100	Diversity = 2.4809e-01	k_t
Generation 3980: Best = 1.401980e+03 = [4, 2, 2, 4]	FES = 398100	Diversity = 2.3974e-01	k_t
Generation 3990: Best = 1.399153e+03 = [1, 1, 1, 1]	FES = 399100	Diversity = 2.4989e-01	k_t
Generation 4000: Best = 1.397169e+03 = [3, 4, 2, 2]	FES = 400100	Diversity = 2.4725e-01	k_t

Generation 4010: Best = 1.393842e+03 = [1, 1, 1, 1]	FES = 401100	Diversity = 2.5048e-01	k_t
Generation 4020: Best = 1.393581e+03 = [4, 2, 2, 3]	FES = 402100	Diversity = 2.3783e-01	k_t
Generation 4030: Best = 1.390866e+03 = [1, 1, 1, 1]	FES = 403100	Diversity = 2.2614e-01	k_t
Generation 4040: Best = 1.389249e+03 = [4, 4, 2, 4]	FES = 404100	Diversity = 2.1881e-01	k_t
Generation 4050: Best = 1.388080e+03 = [1, 1, 1, 1]	FES = 405100	Diversity = 2.0990e-01	k_t
Generation 4060: Best = 1.386471e+03 = [4, 4, 3, 3]	FES = 406100	Diversity = 2.1258e-01	k_t
Generation 4070: Best = 1.383643e+03 = [1, 1, 1, 1]	FES = 407100	Diversity = 2.1339e-01	k_t
Generation 4080: Best = 1.382172e+03 = [4, 4, 2, 4]	FES = 408100	Diversity = 1.9797e-01	k_t
Generation 4090: Best = 1.380157e+03 = [1, 1, 1, 1]	FES = 409100	Diversity = 1.8720e-01	k_t
Generation 4100: Best = 1.378917e+03 = [2, 2, 2, 3]	FES = 410100	Diversity = 1.8751e-01	k_t
Generation 4110: Best = 1.377448e+03 = [1, 1, 1, 1]	FES = 411100	Diversity = 1.7789e-01	k_t
Generation 4120: Best = 1.376076e+03 = [2, 4, 4, 2]	FES = 412100	Diversity = 1.7175e-01	k_t
Generation 4130: Best = 1.373277e+03 = [1, 1, 1, 1]	FES = 413100	Diversity = 1.7682e-01	k_t
Generation 4140: Best = 1.372422e+03 = [3, 3, 2, 4]	FES = 414100	Diversity = 1.8702e-01	k_t
Generation 4150: Best = 1.371069e+03 = [1, 1, 1, 1]	FES = 415100	Diversity = 1.9090e-01	k_t
Generation 4160: Best = 1.370011e+03 = [2, 2, 2, 4]	FES = 416100	Diversity = 1.8140e-01	k_t
Generation 4170: Best = 1.369095e+03 = [1, 1, 1, 1]	FES = 417100	Diversity = 1.7073e-01	k_t
Generation 4180: Best = 1.367648e+03 = [4, 2, 2, 2]	FES = 418100	Diversity = 1.6943e-01	k_t
Generation 4190: Best = 1.365748e+03 = [1, 1, 1, 1]	FES = 419100	Diversity = 1.6661e-01	k_t
Generation 4200: Best = 1.364328e+03 = [4, 4, 4, 2]	FES = 420100	Diversity = 1.8009e-01	k_t
Generation 4210: Best = 1.361996e+03 = [1, 1, 1, 1]	FES = 421100	Diversity = 1.7915e-01	k_t
Generation 4220: Best = 1.360841e+03 = [3, 4, 2, 4]	FES = 422100	Diversity = 1.8428e-01	k_t
Generation 4230: Best = 1.359402e+03 = [1, 1, 1, 1]	FES = 423100	Diversity = 1.8400e-01	k_t
Generation 4240: Best = 1.357878e+03 = [3, 2, 3, 4]	FES = 424100	Diversity = 1.7169e-01	k_t
Generation 4250: Best = 1.357164e+03 = [1, 1, 1, 1]	FES = 425100	Diversity = 1.6171e-01	k_t
Generation 4260: Best = 1.355934e+03 = [3, 4, 3, 3]	FES = 426100	Diversity = 1.5260e-01	k_t
Generation 4270: Best = 1.354804e+03 = [1, 1, 1, 1]	FES = 427100	Diversity = 1.5402e-01	k_t

Generation 4280: Best = 1.353969e+03 = [3, 4, 2, 2]	FES = 428100	Diversity = 1.4382e-01	k_t
Generation 4290: Best = 1.352572e+03 = [1, 1, 1, 1]	FES = 429100	Diversity = 1.4058e-01	k_t
Generation 4300: Best = 1.351580e+03 = [3, 4, 3, 3]	FES = 430100	Diversity = 1.4469e-01	k_t
Generation 4310: Best = 1.350886e+03 = [1, 1, 1, 1]	FES = 431100	Diversity = 1.4521e-01	k_t
Generation 4320: Best = 1.350413e+03 = [2, 3, 3, 2]	FES = 432100	Diversity = 1.4515e-01	k_t
Generation 4330: Best = 1.349190e+03 = [1, 1, 1, 1]	FES = 433100	Diversity = 1.4000e-01	k_t
Generation 4340: Best = 1.348798e+03 = [4, 3, 4, 4]	FES = 434100	Diversity = 1.3427e-01	k_t
Generation 4350: Best = 1.347717e+03 = [1, 1, 1, 1]	FES = 435100	Diversity = 1.3494e-01	k_t
Generation 4360: Best = 1.346970e+03 = [4, 4, 4, 2]	FES = 436100	Diversity = 1.2943e-01	k_t
Generation 4370: Best = 1.346074e+03 = [1, 1, 1, 1]	FES = 437100	Diversity = 1.2143e-01	k_t
Generation 4380: Best = 1.345667e+03 = [3, 3, 2, 4]	FES = 438100	Diversity = 1.1948e-01	k_t
Generation 4390: Best = 1.344777e+03 = [1, 1, 1, 1]	FES = 439100	Diversity = 1.1336e-01	k_t
Generation 4400: Best = 1.344260e+03 = [3, 3, 2, 4]	FES = 440100	Diversity = 1.0975e-01	k_t
Generation 4410: Best = 1.343757e+03 = [1, 1, 1, 1]	FES = 441100	Diversity = 1.0007e-01	k_t
Generation 4420: Best = 1.343069e+03 = [4, 4, 3, 3]	FES = 442100	Diversity = 9.5288e-02	k_t
Generation 4430: Best = 1.342440e+03 = [1, 1, 1, 1]	FES = 443100	Diversity = 9.5726e-02	k_t
Generation 4440: Best = 1.342101e+03 = [3, 3, 3, 2]	FES = 444100	Diversity = 8.8532e-02	k_t
Generation 4450: Best = 1.341558e+03 = [1, 1, 1, 1]	FES = 445100	Diversity = 8.8726e-02	k_t
Generation 4460: Best = 1.340865e+03 = [4, 4, 4, 2]	FES = 446100	Diversity = 9.7456e-02	k_t
Generation 4470: Best = 1.340059e+03 = [1, 1, 1, 1]	FES = 447100	Diversity = 1.0332e-01	k_t
Generation 4480: Best = 1.339520e+03 = [4, 4, 4, 2]	FES = 448100	Diversity = 1.0016e-01	k_t
Generation 4490: Best = 1.339149e+03 = [1, 1, 1, 1]	FES = 449100	Diversity = 9.9437e-02	k_t
Generation 4500: Best = 1.338689e+03 = [3, 3, 2, 4]	FES = 450100	Diversity = 9.6551e-02	k_t
Generation 4510: Best = 1.337990e+03 = [1, 1, 1, 1]	FES = 451100	Diversity = 9.5708e-02	k_t
Generation 4520: Best = 1.337909e+03 = [2, 3, 3, 3]	FES = 452100	Diversity = 9.5831e-02	k_t
Generation 4530: Best = 1.337303e+03 = [1, 1, 1, 1]	FES = 453100	Diversity = 9.2048e-02	k_t
Generation 4540: Best = 1.336957e+03 = [2, 3, 4, 4]	FES = 454100	Diversity = 9.1569e-02	k_t

Generation 4550: Best = 1.336552e+03 = [1, 1, 1, 1]	FES = 455100	Diversity = 8.5797e-02	k_t
Generation 4560: Best = 1.336131e+03 = [2, 2, 2, 3]	FES = 456100	Diversity = 8.5703e-02	k_t
Generation 4570: Best = 1.335813e+03 = [1, 1, 1, 1]	FES = 457100	Diversity = 8.3317e-02	k_t
Generation 4580: Best = 1.335478e+03 = [3, 3, 4, 4]	FES = 458100	Diversity = 8.1739e-02	k_t
Generation 4590: Best = 1.334914e+03 = [1, 1, 1, 1]	FES = 459100	Diversity = 8.2462e-02	k_t
Generation 4600: Best = 1.334774e+03 = [3, 3, 3, 4]	FES = 460100	Diversity = 7.7404e-02	k_t
Generation 4610: Best = 1.334369e+03 = [1, 1, 1, 1]	FES = 461100	Diversity = 7.7307e-02	k_t
Generation 4620: Best = 1.334061e+03 = [4, 3, 2, 3]	FES = 462100	Diversity = 7.0452e-02	k_t
Generation 4630: Best = 1.333563e+03 = [1, 1, 1, 1]	FES = 463100	Diversity = 7.3045e-02	k_t
Generation 4640: Best = 1.333213e+03 = [4, 2, 2, 4]	FES = 464100	Diversity = 8.2398e-02	k_t
Generation 4650: Best = 1.332870e+03 = [1, 1, 1, 1]	FES = 465100	Diversity = 7.9463e-02	k_t
Generation 4660: Best = 1.332229e+03 = [3, 4, 3, 3]	FES = 466100	Diversity = 8.0627e-02	k_t
Generation 4670: Best = 1.331822e+03 = [1, 1, 1, 1]	FES = 467100	Diversity = 8.8488e-02	k_t
Generation 4680: Best = 1.331343e+03 = [3, 3, 2, 4]	FES = 468100	Diversity = 8.6497e-02	k_t
Generation 4690: Best = 1.330826e+03 = [1, 1, 1, 1]	FES = 469100	Diversity = 8.7283e-02	k_t
Generation 4700: Best = 1.330589e+03 = [4, 4, 2, 4]	FES = 470100	Diversity = 8.2588e-02	k_t
Generation 4710: Best = 1.330147e+03 = [1, 1, 1, 1]	FES = 471100	Diversity = 8.6569e-02	k_t
Generation 4720: Best = 1.329923e+03 = [3, 3, 2, 3]	FES = 472100	Diversity = 8.4691e-02	k_t
Generation 4730: Best = 1.329701e+03 = [1, 1, 1, 1]	FES = 473100	Diversity = 7.9026e-02	k_t
Generation 4740: Best = 1.329383e+03 = [4, 2, 4, 4]	FES = 474100	Diversity = 7.0847e-02	k_t
Generation 4750: Best = 1.329065e+03 = [1, 1, 1, 1]	FES = 475100	Diversity = 6.9382e-02	k_t
Generation 4760: Best = 1.328756e+03 = [4, 3, 3, 4]	FES = 476100	Diversity = 6.8068e-02	k_t
Generation 4770: Best = 1.328508e+03 = [1, 1, 1, 1]	FES = 477100	Diversity = 6.9286e-02	k_t
Generation 4780: Best = 1.328154e+03 = [3, 2, 2, 4]	FES = 478100	Diversity = 6.9214e-02	k_t
Generation 4790: Best = 1.327928e+03 = [1, 1, 1, 1]	FES = 479100	Diversity = 6.5568e-02	k_t
Generation 4800: Best = 1.327785e+03 = [2, 3, 3, 3]	FES = 480100	Diversity = 6.5093e-02	k_t
Generation 4810: Best = 1.327566e+03 = [1, 1, 1, 1]	FES = 481100	Diversity = 6.3930e-02	k_t

Generation 4820: Best = 1.327344e+03 = [4, 3, 4, 3]	FES = 482100	Diversity = 6.5265e-02	k_t
Generation 4830: Best = 1.327182e+03 = [1, 1, 1, 1]	FES = 483100	Diversity = 6.5945e-02	k_t
Generation 4840: Best = 1.327007e+03 = [2, 3, 3, 3]	FES = 484100	Diversity = 6.2961e-02	k_t
Generation 4850: Best = 1.326530e+03 = [1, 1, 1, 1]	FES = 485100	Diversity = 6.6635e-02	k_t
Generation 4860: Best = 1.326423e+03 = [3, 4, 3, 3]	FES = 486100	Diversity = 6.4749e-02	k_t
Generation 4870: Best = 1.326191e+03 = [1, 1, 1, 1]	FES = 487100	Diversity = 6.6498e-02	k_t
Generation 4880: Best = 1.326006e+03 = [4, 2, 2, 4]	FES = 488100	Diversity = 6.3043e-02	k_t
Generation 4890: Best = 1.325846e+03 = [1, 1, 1, 1]	FES = 489100	Diversity = 5.8028e-02	k_t
Generation 4900: Best = 1.325667e+03 = [2, 3, 3, 4]	FES = 490100	Diversity = 5.6601e-02	k_t
Generation 4910: Best = 1.325391e+03 = [1, 1, 1, 1]	FES = 491100	Diversity = 5.4439e-02	k_t
Generation 4920: Best = 1.325344e+03 = [3, 3, 3, 2]	FES = 492100	Diversity = 5.2527e-02	k_t
Generation 4930: Best = 1.325131e+03 = [1, 1, 1, 1]	FES = 493100	Diversity = 5.1944e-02	k_t
Generation 4940: Best = 1.324890e+03 = [3, 4, 2, 4]	FES = 494100	Diversity = 5.1954e-02	k_t
Generation 4950: Best = 1.324789e+03 = [1, 1, 1, 1]	FES = 495100	Diversity = 5.4254e-02	k_t
Generation 4960: Best = 1.324636e+03 = [2, 3, 4, 4]	FES = 496100	Diversity = 5.5099e-02	k_t
Generation 4970: Best = 1.324469e+03 = [1, 1, 1, 1]	FES = 497100	Diversity = 5.6414e-02	k_t
Generation 4980: Best = 1.324282e+03 = [4, 3, 3, 2]	FES = 498100	Diversity = 5.6680e-02	k_t
Generation 4990: Best = 1.324173e+03 = [1, 1, 1, 1]	FES = 499100	Diversity = 5.4981e-02	k_t
Generation 5000: Best = 1.324031e+03 = [3, 2, 3, 3]	FES = 500100	Diversity = 5.5470e-02	k_t
Generation 5010: Best = 1.323816e+03 = [1, 1, 1, 1]	FES = 501100	Diversity = 5.6232e-02	k_t
Generation 5020: Best = 1.323720e+03 = [4, 4, 2, 3]	FES = 502100	Diversity = 5.3217e-02	k_t
Generation 5030: Best = 1.323598e+03 = [1, 1, 1, 1]	FES = 503100	Diversity = 5.1513e-02	k_t
Generation 5040: Best = 1.323502e+03 = [3, 4, 4, 3]	FES = 504100	Diversity = 5.2165e-02	k_t
Generation 5050: Best = 1.323369e+03 = [1, 1, 1, 1]	FES = 505100	Diversity = 5.2617e-02	k_t
Generation 5060: Best = 1.323151e+03 = [3, 3, 2, 3]	FES = 506100	Diversity = 5.1612e-02	k_t
Generation 5070: Best = 1.323112e+03 = [1, 1, 1, 1]	FES = 507100	Diversity = 4.9590e-02	k_t
Generation 5080: Best = 1.322947e+03 = [2, 3, 3, 2]	FES = 508100	Diversity = 4.9048e-02	k_t

Generation 5090: Best = 1.322850e+03 = [1, 1, 1, 1]	FES = 509100	Diversity = 4.7661e-02	k_t
Generation 5100: Best = 1.322723e+03 = [3, 2, 4, 2]	FES = 510100	Diversity = 4.4654e-02	k_t
Generation 5110: Best = 1.322607e+03 = [1, 1, 1, 1]	FES = 511100	Diversity = 4.2126e-02	k_t
Generation 5120: Best = 1.322510e+03 = [4, 2, 2, 2]	FES = 512100	Diversity = 4.2388e-02	k_t
Generation 5130: Best = 1.322416e+03 = [1, 1, 1, 1]	FES = 513100	Diversity = 4.3136e-02	k_t
Generation 5140: Best = 1.322327e+03 = [2, 4, 2, 2]	FES = 514100	Diversity = 4.3164e-02	k_t
Generation 5150: Best = 1.322268e+03 = [1, 1, 1, 1]	FES = 515100	Diversity = 4.2238e-02	k_t
Generation 5160: Best = 1.322169e+03 = [2, 4, 4, 2]	FES = 516100	Diversity = 4.1827e-02	k_t
Generation 5170: Best = 1.322100e+03 = [1, 1, 1, 1]	FES = 517100	Diversity = 4.1879e-02	k_t
Generation 5180: Best = 1.322047e+03 = [2, 4, 3, 3]	FES = 518100	Diversity = 3.9200e-02	k_t
Generation 5190: Best = 1.321955e+03 = [1, 1, 1, 1]	FES = 519100	Diversity = 3.6960e-02	k_t
Generation 5200: Best = 1.321884e+03 = [2, 2, 2, 4]	FES = 520100	Diversity = 3.8014e-02	k_t
Generation 5210: Best = 1.321790e+03 = [1, 1, 1, 1]	FES = 521100	Diversity = 3.5264e-02	k_t
Generation 5220: Best = 1.321733e+03 = [3, 2, 2, 4]	FES = 522100	Diversity = 3.6131e-02	k_t
Generation 5230: Best = 1.321642e+03 = [1, 1, 1, 1]	FES = 523100	Diversity = 3.5935e-02	k_t
Generation 5240: Best = 1.321577e+03 = [4, 4, 3, 2]	FES = 524100	Diversity = 3.5142e-02	k_t
Generation 5250: Best = 1.321565e+03 = [1, 1, 1, 1]	FES = 525100	Diversity = 3.5214e-02	k_t
Generation 5260: Best = 1.321494e+03 = [2, 2, 4, 3]	FES = 526100	Diversity = 3.5933e-02	k_t
Generation 5270: Best = 1.321433e+03 = [1, 1, 1, 1]	FES = 527100	Diversity = 3.5141e-02	k_t
Generation 5280: Best = 1.321367e+03 = [3, 3, 3, 4]	FES = 528100	Diversity = 3.5776e-02	k_t
Generation 5290: Best = 1.321296e+03 = [1, 1, 1, 1]	FES = 529100	Diversity = 3.4166e-02	k_t
Generation 5300: Best = 1.321227e+03 = [2, 2, 3, 3]	FES = 530100	Diversity = 3.3187e-02	k_t
Generation 5310: Best = 1.321180e+03 = [1, 1, 1, 1]	FES = 531100	Diversity = 3.4125e-02	k_t
Generation 5320: Best = 1.321093e+03 = [2, 3, 3, 3]	FES = 532100	Diversity = 3.3687e-02	k_t
Generation 5330: Best = 1.321048e+03 = [1, 1, 1, 1]	FES = 533100	Diversity = 3.6499e-02	k_t
Generation 5340: Best = 1.320981e+03 = [3, 4, 3, 2]	FES = 534100	Diversity = 3.8644e-02	k_t
Generation 5350: Best = 1.320927e+03 = [1, 1, 1, 1]	FES = 535100	Diversity = 3.5185e-02	k_t

Generation 5360: Best = 1.320866e+03 = [3, 2, 3, 3]	FES = 536100	Diversity = 3.3925e-02	k_t
Generation 5370: Best = 1.320793e+03 = [1, 1, 1, 1]	FES = 537100	Diversity = 3.4150e-02	k_t
Generation 5380: Best = 1.320729e+03 = [2, 3, 4, 2]	FES = 538100	Diversity = 3.2563e-02	k_t
Generation 5390: Best = 1.320673e+03 = [1, 1, 1, 1]	FES = 539100	Diversity = 3.1184e-02	k_t
Generation 5400: Best = 1.320640e+03 = [3, 2, 2, 3]	FES = 540100	Diversity = 2.9722e-02	k_t
Generation 5410: Best = 1.320575e+03 = [1, 1, 1, 1]	FES = 541100	Diversity = 2.8913e-02	k_t
Generation 5420: Best = 1.320522e+03 = [4, 4, 4, 2]	FES = 542100	Diversity = 2.7780e-02	k_t
Generation 5430: Best = 1.320495e+03 = [1, 1, 1, 1]	FES = 543100	Diversity = 2.7011e-02	k_t
Generation 5440: Best = 1.320443e+03 = [3, 3, 3, 2]	FES = 544100	Diversity = 2.7598e-02	k_t
Generation 5450: Best = 1.320375e+03 = [1, 1, 1, 1]	FES = 545100	Diversity = 2.8755e-02	k_t
Generation 5460: Best = 1.320309e+03 = [2, 3, 3, 2]	FES = 546100	Diversity = 2.9825e-02	k_t
Generation 5470: Best = 1.320267e+03 = [1, 1, 1, 1]	FES = 547100	Diversity = 3.0572e-02	k_t
Generation 5480: Best = 1.320200e+03 = [2, 3, 3, 4]	FES = 548100	Diversity = 2.9858e-02	k_t
Generation 5490: Best = 1.320164e+03 = [1, 1, 1, 1]	FES = 549100	Diversity = 3.0870e-02	k_t
Generation 5500: Best = 1.320113e+03 = [2, 2, 3, 2]	FES = 550100	Diversity = 3.0188e-02	k_t
Generation 5510: Best = 1.320042e+03 = [1, 1, 1, 1]	FES = 551100	Diversity = 3.1179e-02	k_t
Generation 5520: Best = 1.319994e+03 = [3, 2, 2, 4]	FES = 552100	Diversity = 3.1132e-02	k_t
Generation 5530: Best = 1.319933e+03 = [1, 1, 1, 1]	FES = 553100	Diversity = 3.0062e-02	k_t
Generation 5540: Best = 1.319875e+03 = [4, 2, 3, 4]	FES = 554100	Diversity = 3.0415e-02	k_t
Generation 5550: Best = 1.319834e+03 = [1, 1, 1, 1]	FES = 555100	Diversity = 3.0500e-02	k_t
Generation 5560: Best = 1.319785e+03 = [3, 3, 4, 3]	FES = 556100	Diversity = 2.9398e-02	k_t
Generation 5570: Best = 1.319753e+03 = [1, 1, 1, 1]	FES = 557100	Diversity = 2.8171e-02	k_t
Generation 5580: Best = 1.319697e+03 = [2, 2, 3, 4]	FES = 558100	Diversity = 2.6667e-02	k_t
Generation 5590: Best = 1.319659e+03 = [1, 1, 1, 1]	FES = 559100	Diversity = 2.7475e-02	k_t
Generation 5600: Best = 1.319625e+03 = [3, 3, 3, 2]	FES = 560100	Diversity = 2.7623e-02	k_t
Generation 5610: Best = 1.319578e+03 = [1, 1, 1, 1]	FES = 561100	Diversity = 2.5720e-02	k_t
Generation 5620: Best = 1.319544e+03 = [3, 3, 4, 2]	FES = 562100	Diversity = 2.5148e-02	k_t

Generation 5630: Best = 1.319488e+03 = [1, 1, 1, 1]	FES = 563100	Diversity = 2.5062e-02	k_t
Generation 5640: Best = 1.319455e+03 = [3, 3, 4, 4]	FES = 564100	Diversity = 2.4290e-02	k_t
Generation 5650: Best = 1.319423e+03 = [1, 1, 1, 1]	FES = 565100	Diversity = 2.7106e-02	k_t
Generation 5660: Best = 1.319354e+03 = [3, 3, 4, 3]	FES = 566100	Diversity = 2.7252e-02	k_t
Generation 5670: Best = 1.319308e+03 = [1, 1, 1, 1]	FES = 567100	Diversity = 2.6684e-02	k_t
Generation 5680: Best = 1.319282e+03 = [3, 3, 3, 3]	FES = 568100	Diversity = 2.4995e-02	k_t
Generation 5690: Best = 1.319242e+03 = [1, 1, 1, 1]	FES = 569100	Diversity = 2.2183e-02	k_t
Generation 5700: Best = 1.319202e+03 = [4, 2, 2, 4]	FES = 570100	Diversity = 2.3096e-02	k_t
Generation 5710: Best = 1.319168e+03 = [1, 1, 1, 1]	FES = 571100	Diversity = 2.2180e-02	k_t
Generation 5720: Best = 1.319136e+03 = [2, 4, 2, 2]	FES = 572100	Diversity = 2.4088e-02	k_t
Generation 5730: Best = 1.319111e+03 = [1, 1, 1, 1]	FES = 573100	Diversity = 2.4253e-02	k_t
Generation 5740: Best = 1.319061e+03 = [2, 3, 3, 2]	FES = 574100	Diversity = 2.5141e-02	k_t
Generation 5750: Best = 1.318996e+03 = [1, 1, 1, 1]	FES = 575100	Diversity = 2.7561e-02	k_t
Generation 5760: Best = 1.318950e+03 = [3, 3, 3, 3]	FES = 576100	Diversity = 2.8067e-02	k_t
Generation 5770: Best = 1.318913e+03 = [1, 1, 1, 1]	FES = 577100	Diversity = 2.8968e-02	k_t
Generation 5780: Best = 1.318871e+03 = [4, 2, 4, 4]	FES = 578100	Diversity = 2.8676e-02	k_t
Generation 5790: Best = 1.318824e+03 = [1, 1, 1, 1]	FES = 579100	Diversity = 2.9068e-02	k_t
Generation 5800: Best = 1.318776e+03 = [4, 2, 4, 4]	FES = 580100	Diversity = 2.7646e-02	k_t
Generation 5810: Best = 1.318758e+03 = [1, 1, 1, 1]	FES = 581100	Diversity = 2.6038e-02	k_t
Generation 5820: Best = 1.318719e+03 = [3, 4, 4, 4]	FES = 582100	Diversity = 2.5701e-02	k_t
Generation 5830: Best = 1.318674e+03 = [1, 1, 1, 1]	FES = 583100	Diversity = 2.5109e-02	k_t
Generation 5840: Best = 1.318646e+03 = [3, 4, 2, 2]	FES = 584100	Diversity = 2.4218e-02	k_t
Generation 5850: Best = 1.318614e+03 = [1, 1, 1, 1]	FES = 585100	Diversity = 2.3962e-02	k_t
Generation 5860: Best = 1.318560e+03 = [4, 2, 3, 3]	FES = 586100	Diversity = 2.4371e-02	k_t
Generation 5870: Best = 1.318537e+03 = [1, 1, 1, 1]	FES = 587100	Diversity = 2.3969e-02	k_t
Generation 5880: Best = 1.318510e+03 = [4, 4, 4, 4]	FES = 588100	Diversity = 2.2494e-02	k_t
Generation 5890: Best = 1.318477e+03 = [1, 1, 1, 1]	FES = 589100	Diversity = 2.2654e-02	k_t

Generation 5900: Best = 1.318436e+03	FES = 590100	Diversity = 2.3662e-02	k_t
= [2, 3, 4, 2]			
Generation 5910: Best = 1.318407e+03	FES = 591100	Diversity = 2.3586e-02	k_t
= [1, 1, 1, 1]			
Generation 5920: Best = 1.318389e+03	FES = 592100	Diversity = 2.2449e-02	k_t
= [4, 2, 3, 4]			
Generation 5930: Best = 1.318362e+03	FES = 593100	Diversity = 2.2307e-02	k_t
= [1, 1, 1, 1]			
Generation 5940: Best = 1.318328e+03	FES = 594100	Diversity = 2.1400e-02	k_t
= [2, 2, 4, 3]			
Generation 5950: Best = 1.318300e+03	FES = 595100	Diversity = 2.1740e-02	k_t
= [1, 1, 1, 1]			
Generation 5960: Best = 1.318259e+03	FES = 596100	Diversity = 2.3436e-02	k_t
= [3, 3, 3, 3]			
Generation 5970: Best = 1.318238e+03	FES = 597100	Diversity = 2.2102e-02	k_t
= [1, 1, 1, 1]			
Generation 5980: Best = 1.318212e+03	FES = 598100	Diversity = 2.0525e-02	k_t
= [4, 4, 3, 3]			
Generation 5990: Best = 1.318198e+03	FES = 599100	Diversity = 2.0787e-02	k_t
= [1, 1, 1, 1]			
Generation 6000: Best = 1.318173e+03	FES = 600100	Diversity = 1.9912e-02	k_t
= [2, 4, 3, 3]			
Generation 6010: Best = 1.318147e+03	FES = 601100	Diversity = 2.0085e-02	k_t
= [1, 1, 1, 1]			
Generation 6020: Best = 1.318127e+03	FES = 602100	Diversity = 2.0670e-02	k_t
= [4, 3, 3, 4]			
Generation 6030: Best = 1.318099e+03	FES = 603100	Diversity = 2.1240e-02	k_t
= [1, 1, 1, 1]			
Generation 6040: Best = 1.318064e+03	FES = 604100	Diversity = 2.2033e-02	k_t
= [3, 3, 2, 4]			
Generation 6050: Best = 1.318025e+03	FES = 605100	Diversity = 2.4341e-02	k_t
= [1, 1, 1, 1]			
Generation 6060: Best = 1.318007e+03	FES = 606100	Diversity = 2.3774e-02	k_t
= [4, 2, 4, 3]			
Generation 6070: Best = 1.317970e+03	FES = 607100	Diversity = 2.4051e-02	k_t
= [1, 1, 1, 1]			
Generation 6080: Best = 1.317941e+03	FES = 608100	Diversity = 2.3720e-02	k_t
= [3, 2, 3, 3]			
Generation 6090: Best = 1.317918e+03	FES = 609100	Diversity = 2.3170e-02	k_t
= [1, 1, 1, 1]			
Generation 6100: Best = 1.317895e+03	FES = 610100	Diversity = 2.2729e-02	k_t
= [2, 3, 3, 4]			
Generation 6110: Best = 1.317866e+03	FES = 611100	Diversity = 2.2675e-02	k_t
= [1, 1, 1, 1]			
Generation 6120: Best = 1.317836e+03	FES = 612100	Diversity = 2.3138e-02	k_t
= [4, 4, 3, 2]			
Generation 6130: Best = 1.317796e+03	FES = 613100	Diversity = 2.3888e-02	k_t
= [1, 1, 1, 1]			
Generation 6140: Best = 1.317788e+03	FES = 614100	Diversity = 2.3985e-02	k_t
= [3, 4, 4, 4]			
Generation 6150: Best = 1.317757e+03	FES = 615100	Diversity = 2.5346e-02	k_t
= [1, 1, 1, 1]			
Generation 6160: Best = 1.317716e+03	FES = 616100	Diversity = 2.6138e-02	k_t
= [2, 3, 4, 2]			

Generation 6170: Best = 1.317699e+03 = [1, 1, 1, 1]	FES = 617100	Diversity = 2.5370e-02	k_t
Generation 6180: Best = 1.317674e+03 = [4, 3, 2, 3]	FES = 618100	Diversity = 2.3783e-02	k_t
Generation 6190: Best = 1.317643e+03 = [1, 1, 1, 1]	FES = 619100	Diversity = 2.4183e-02	k_t
Generation 6200: Best = 1.317629e+03 = [2, 4, 2, 3]	FES = 620100	Diversity = 2.3984e-02	k_t
Generation 6210: Best = 1.317582e+03 = [1, 1, 1, 1]	FES = 621100	Diversity = 2.4439e-02	k_t
Generation 6220: Best = 1.317564e+03 = [3, 3, 2, 3]	FES = 622100	Diversity = 2.3809e-02	k_t
Generation 6230: Best = 1.317541e+03 = [1, 1, 1, 1]	FES = 623100	Diversity = 2.4064e-02	k_t
Generation 6240: Best = 1.317517e+03 = [4, 2, 4, 4]	FES = 624100	Diversity = 2.3532e-02	k_t
Generation 6250: Best = 1.317481e+03 = [1, 1, 1, 1]	FES = 625100	Diversity = 2.3326e-02	k_t
Generation 6260: Best = 1.317474e+03 = [2, 2, 2, 3]	FES = 626100	Diversity = 2.3470e-02	k_t
Generation 6270: Best = 1.317443e+03 = [1, 1, 1, 1]	FES = 627100	Diversity = 2.2651e-02	k_t
Generation 6280: Best = 1.317423e+03 = [2, 3, 3, 2]	FES = 628100	Diversity = 2.1300e-02	k_t
Generation 6290: Best = 1.317404e+03 = [1, 1, 1, 1]	FES = 629100	Diversity = 1.9420e-02	k_t
Generation 6300: Best = 1.317384e+03 = [3, 3, 4, 2]	FES = 630100	Diversity = 1.8539e-02	k_t
Generation 6310: Best = 1.317367e+03 = [1, 1, 1, 1]	FES = 631100	Diversity = 1.7407e-02	k_t
Generation 6320: Best = 1.317349e+03 = [4, 2, 2, 4]	FES = 632100	Diversity = 1.7138e-02	k_t
Generation 6330: Best = 1.317333e+03 = [1, 1, 1, 1]	FES = 633100	Diversity = 1.5798e-02	k_t
Generation 6340: Best = 1.317319e+03 = [2, 3, 3, 3]	FES = 634100	Diversity = 1.6072e-02	k_t
Generation 6350: Best = 1.317305e+03 = [1, 1, 1, 1]	FES = 635100	Diversity = 1.5691e-02	k_t
Generation 6360: Best = 1.317283e+03 = [4, 4, 2, 3]	FES = 636100	Diversity = 1.5727e-02	k_t
Generation 6370: Best = 1.317267e+03 = [1, 1, 1, 1]	FES = 637100	Diversity = 1.6164e-02	k_t
Generation 6380: Best = 1.317257e+03 = [2, 2, 2, 2]	FES = 638100	Diversity = 1.5757e-02	k_t
Generation 6390: Best = 1.317241e+03 = [1, 1, 1, 1]	FES = 639100	Diversity = 1.6443e-02	k_t
Generation 6400: Best = 1.317225e+03 = [2, 2, 3, 3]	FES = 640100	Diversity = 1.7535e-02	k_t
Generation 6410: Best = 1.317208e+03 = [1, 1, 1, 1]	FES = 641100	Diversity = 1.8291e-02	k_t
Generation 6420: Best = 1.317195e+03 = [3, 2, 4, 2]	FES = 642100	Diversity = 1.8412e-02	k_t
Generation 6430: Best = 1.317180e+03 = [1, 1, 1, 1]	FES = 643100	Diversity = 1.8434e-02	k_t

Generation 6440: Best = 1.317165e+03 = [3, 4, 3, 2]	FES = 644100	Diversity = 1.9159e-02	k_t
Generation 6450: Best = 1.317150e+03 = [1, 1, 1, 1]	FES = 645100	Diversity = 1.8381e-02	k_t
Generation 6460: Best = 1.317136e+03 = [2, 4, 3, 4]	FES = 646100	Diversity = 1.7753e-02	k_t
Generation 6470: Best = 1.317126e+03 = [1, 1, 1, 1]	FES = 647100	Diversity = 1.6282e-02	k_t
Generation 6480: Best = 1.317098e+03 = [4, 3, 3, 4]	FES = 648100	Diversity = 1.5713e-02	k_t
Generation 6490: Best = 1.317091e+03 = [1, 1, 1, 1]	FES = 649100	Diversity = 1.6487e-02	k_t
Generation 6500: Best = 1.317079e+03 = [3, 2, 3, 3]	FES = 650100	Diversity = 1.6316e-02	k_t
Generation 6510: Best = 1.317060e+03 = [1, 1, 1, 1]	FES = 651100	Diversity = 1.5960e-02	k_t
Generation 6520: Best = 1.317055e+03 = [2, 4, 2, 4]	FES = 652100	Diversity = 1.5326e-02	k_t
Generation 6530: Best = 1.317033e+03 = [1, 1, 1, 1]	FES = 653100	Diversity = 1.5651e-02	k_t
Generation 6540: Best = 1.317024e+03 = [4, 3, 2, 4]	FES = 654100	Diversity = 1.5923e-02	k_t
Generation 6550: Best = 1.317012e+03 = [1, 1, 1, 1]	FES = 655100	Diversity = 1.5724e-02	k_t
Generation 6560: Best = 1.317003e+03 = [3, 3, 2, 2]	FES = 656100	Diversity = 1.5313e-02	k_t
Generation 6570: Best = 1.316984e+03 = [1, 1, 1, 1]	FES = 657100	Diversity = 1.5617e-02	k_t
Generation 6580: Best = 1.316974e+03 = [3, 2, 4, 4]	FES = 658100	Diversity = 1.5279e-02	k_t
Generation 6590: Best = 1.316958e+03 = [1, 1, 1, 1]	FES = 659100	Diversity = 1.5804e-02	k_t
Generation 6600: Best = 1.316947e+03 = [3, 3, 2, 4]	FES = 660100	Diversity = 1.5362e-02	k_t
Generation 6610: Best = 1.316934e+03 = [1, 1, 1, 1]	FES = 661100	Diversity = 1.4752e-02	k_t
Generation 6620: Best = 1.316922e+03 = [2, 4, 2, 3]	FES = 662100	Diversity = 1.5266e-02	k_t
Generation 6630: Best = 1.316906e+03 = [1, 1, 1, 1]	FES = 663100	Diversity = 1.5348e-02	k_t
Generation 6640: Best = 1.316892e+03 = [2, 4, 3, 3]	FES = 664100	Diversity = 1.5324e-02	k_t
Generation 6650: Best = 1.316879e+03 = [1, 1, 1, 1]	FES = 665100	Diversity = 1.5544e-02	k_t
Generation 6660: Best = 1.316865e+03 = [4, 3, 2, 4]	FES = 666100	Diversity = 1.6354e-02	k_t
Generation 6670: Best = 1.316855e+03 = [1, 1, 1, 1]	FES = 667100	Diversity = 1.5823e-02	k_t
Generation 6680: Best = 1.316842e+03 = [4, 4, 4, 2]	FES = 668100	Diversity = 1.5771e-02	k_t
Generation 6690: Best = 1.316828e+03 = [1, 1, 1, 1]	FES = 669100	Diversity = 1.5344e-02	k_t
Generation 6700: Best = 1.316812e+03 = [2, 3, 4, 3]	FES = 670100	Diversity = 1.5110e-02	k_t

Generation 6710: Best = 1.316802e+03 = [1, 1, 1, 1]	FES = 671100	Diversity = 1.4019e-02	k_t
Generation 6720: Best = 1.316788e+03 = [4, 3, 3, 3]	FES = 672100	Diversity = 1.3248e-02	k_t
Generation 6730: Best = 1.316780e+03 = [1, 1, 1, 1]	FES = 673100	Diversity = 1.2781e-02	k_t
Generation 6740: Best = 1.316771e+03 = [2, 2, 4, 4]	FES = 674100	Diversity = 1.2221e-02	k_t
Generation 6750: Best = 1.316760e+03 = [1, 1, 1, 1]	FES = 675100	Diversity = 1.2007e-02	k_t
Generation 6760: Best = 1.316752e+03 = [3, 4, 2, 3]	FES = 676100	Diversity = 1.2316e-02	k_t
Generation 6770: Best = 1.316736e+03 = [1, 1, 1, 1]	FES = 677100	Diversity = 1.2308e-02	k_t
Generation 6780: Best = 1.316730e+03 = [2, 4, 4, 2]	FES = 678100	Diversity = 1.3552e-02	k_t
Generation 6790: Best = 1.316720e+03 = [1, 1, 1, 1]	FES = 679100	Diversity = 1.3346e-02	k_t
Generation 6800: Best = 1.316712e+03 = [4, 3, 3, 2]	FES = 680100	Diversity = 1.3314e-02	k_t
Generation 6810: Best = 1.316703e+03 = [1, 1, 1, 1]	FES = 681100	Diversity = 1.2833e-02	k_t
Generation 6820: Best = 1.316692e+03 = [3, 3, 2, 3]	FES = 682100	Diversity = 1.3151e-02	k_t
Generation 6830: Best = 1.316679e+03 = [1, 1, 1, 1]	FES = 683100	Diversity = 1.3420e-02	k_t
Generation 6840: Best = 1.316668e+03 = [2, 4, 3, 4]	FES = 684100	Diversity = 1.3664e-02	k_t
Generation 6850: Best = 1.316651e+03 = [1, 1, 1, 1]	FES = 685100	Diversity = 1.3903e-02	k_t
Generation 6860: Best = 1.316642e+03 = [2, 2, 3, 4]	FES = 686100	Diversity = 1.4426e-02	k_t
Generation 6870: Best = 1.316627e+03 = [1, 1, 1, 1]	FES = 687100	Diversity = 1.4998e-02	k_t
Generation 6880: Best = 1.316618e+03 = [4, 4, 3, 4]	FES = 688100	Diversity = 1.5868e-02	k_t
Generation 6890: Best = 1.316602e+03 = [1, 1, 1, 1]	FES = 689100	Diversity = 1.5633e-02	k_t
Generation 6900: Best = 1.316591e+03 = [4, 4, 3, 3]	FES = 690100	Diversity = 1.6273e-02	k_t
Generation 6910: Best = 1.316577e+03 = [1, 1, 1, 1]	FES = 691100	Diversity = 1.4922e-02	k_t
Generation 6920: Best = 1.316572e+03 = [4, 2, 3, 3]	FES = 692100	Diversity = 1.4054e-02	k_t
Generation 6930: Best = 1.316552e+03 = [1, 1, 1, 1]	FES = 693100	Diversity = 1.4674e-02	k_t
Generation 6940: Best = 1.316540e+03 = [2, 3, 4, 2]	FES = 694100	Diversity = 1.4483e-02	k_t
Generation 6950: Best = 1.316532e+03 = [1, 1, 1, 1]	FES = 695100	Diversity = 1.4755e-02	k_t
Generation 6960: Best = 1.316521e+03 = [4, 4, 3, 3]	FES = 696100	Diversity = 1.4442e-02	k_t
Generation 6970: Best = 1.316510e+03 = [1, 1, 1, 1]	FES = 697100	Diversity = 1.3901e-02	k_t

Generation 6980: Best = 1.316499e+03 = [4, 2, 2, 3]	FES = 698100	Diversity = 1.3746e-02	k_t
Generation 6990: Best = 1.316487e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 1.3770e-02	k_t
Generation 7000: Best = 1.316478e+03 = [2, 4, 3, 2]	FES = 700100	Diversity = 1.3635e-02	k_t
Generation 7010: Best = 1.316466e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 1.3734e-02	k_t
Generation 7020: Best = 1.316455e+03 = [3, 4, 2, 3]	FES = 702100	Diversity = 1.3403e-02	k_t
Generation 7030: Best = 1.316442e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 1.4099e-02	k_t
Generation 7040: Best = 1.316435e+03 = [3, 3, 4, 3]	FES = 704100	Diversity = 1.3531e-02	k_t
Generation 7050: Best = 1.316424e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 1.3907e-02	k_t
Generation 7060: Best = 1.316416e+03 = [2, 2, 3, 3]	FES = 706100	Diversity = 1.3872e-02	k_t
Generation 7070: Best = 1.316399e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 1.4210e-02	k_t
Generation 7080: Best = 1.316393e+03 = [3, 4, 4, 4]	FES = 708100	Diversity = 1.4182e-02	k_t
Generation 7090: Best = 1.316383e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 1.3832e-02	k_t
Generation 7100: Best = 1.316369e+03 = [4, 4, 2, 4]	FES = 710100	Diversity = 1.3471e-02	k_t
Generation 7110: Best = 1.316358e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 1.3189e-02	k_t
Generation 7120: Best = 1.316347e+03 = [2, 2, 3, 4]	FES = 712100	Diversity = 1.2825e-02	k_t
Generation 7130: Best = 1.316341e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 1.2058e-02	k_t
Generation 7140: Best = 1.316332e+03 = [3, 2, 3, 2]	FES = 714100	Diversity = 1.1054e-02	k_t
Generation 7150: Best = 1.316320e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 1.1158e-02	k_t
Generation 7160: Best = 1.316311e+03 = [3, 3, 3, 2]	FES = 716100	Diversity = 1.2189e-02	k_t
Generation 7170: Best = 1.316302e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 1.2264e-02	k_t
Generation 7180: Best = 1.316289e+03 = [3, 2, 4, 2]	FES = 718100	Diversity = 1.3138e-02	k_t
Generation 7190: Best = 1.316280e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 1.2503e-02	k_t
Generation 7200: Best = 1.316272e+03 = [4, 3, 2, 2]	FES = 720100	Diversity = 1.2033e-02	k_t
Generation 7210: Best = 1.316265e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 1.2514e-02	k_t
Generation 7220: Best = 1.316253e+03 = [3, 3, 4, 4]	FES = 722100	Diversity = 1.1725e-02	k_t
Generation 7230: Best = 1.316243e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 1.1177e-02	k_t
Generation 7240: Best = 1.316237e+03 = [3, 2, 4, 3]	FES = 724100	Diversity = 1.1260e-02	k_t

Generation 7250: Best = 1.316231e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 1.0697e-02	k_t
Generation 7260: Best = 1.316222e+03 = [3, 3, 2, 4]	FES = 726100	Diversity = 1.0669e-02	k_t
Generation 7270: Best = 1.316215e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 1.0273e-02	k_t
Generation 7280: Best = 1.316210e+03 = [2, 2, 2, 4]	FES = 728100	Diversity = 1.0216e-02	k_t
Generation 7290: Best = 1.316200e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 1.0244e-02	k_t
Generation 7300: Best = 1.316190e+03 = [4, 3, 2, 3]	FES = 730100	Diversity = 1.1689e-02	k_t
Generation 7310: Best = 1.316180e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 1.2435e-02	k_t
Generation 7320: Best = 1.316168e+03 = [3, 3, 2, 3]	FES = 732100	Diversity = 1.2797e-02	k_t
Generation 7330: Best = 1.316152e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 1.3519e-02	k_t
Generation 7340: Best = 1.316148e+03 = [2, 4, 2, 2]	FES = 734100	Diversity = 1.3379e-02	k_t
Generation 7350: Best = 1.316141e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 1.2730e-02	k_t
Generation 7360: Best = 1.316131e+03 = [2, 4, 4, 2]	FES = 736100	Diversity = 1.2326e-02	k_t
Generation 7370: Best = 1.316124e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 1.1289e-02	k_t
Generation 7380: Best = 1.316117e+03 = [2, 4, 4, 3]	FES = 738100	Diversity = 1.1608e-02	k_t
Generation 7390: Best = 1.316109e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 1.1352e-02	k_t
Generation 7400: Best = 1.316103e+03 = [4, 4, 4, 2]	FES = 740100	Diversity = 1.0648e-02	k_t
Generation 7410: Best = 1.316095e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 1.0573e-02	k_t
Generation 7420: Best = 1.316090e+03 = [2, 3, 3, 4]	FES = 742100	Diversity = 1.0721e-02	k_t
Generation 7430: Best = 1.316082e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 1.0485e-02	k_t
Generation 7440: Best = 1.316070e+03 = [2, 2, 4, 4]	FES = 744100	Diversity = 1.0594e-02	k_t
Generation 7450: Best = 1.316065e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 1.0599e-02	k_t
Generation 7460: Best = 1.316056e+03 = [3, 2, 2, 2]	FES = 746100	Diversity = 1.1013e-02	k_t
Generation 7470: Best = 1.316051e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 1.1273e-02	k_t
Generation 7480: Best = 1.316044e+03 = [2, 2, 4, 3]	FES = 748100	Diversity = 1.0743e-02	k_t
Generation 7490: Best = 1.316034e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 1.1468e-02	k_t
Generation 7500: Best = 1.316028e+03 = [3, 4, 3, 4]	FES = 750100	Diversity = 1.1801e-02	k_t
Generation 7510: Best = 1.316017e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 1.2454e-02	k_t

Generation 7520: Best = 1.316007e+03 = [3, 3, 2, 3]	FES = 752100	Diversity = 1.3272e-02	k_t
Generation 7530: Best = 1.315996e+03 = [1, 1, 1, 1]	FES = 753100	Diversity = 1.2706e-02	k_t
Generation 7540: Best = 1.315987e+03 = [3, 3, 3, 3]	FES = 754100	Diversity = 1.2393e-02	k_t
Generation 7550: Best = 1.315973e+03 = [1, 1, 1, 1]	FES = 755100	Diversity = 1.1938e-02	k_t
Generation 7560: Best = 1.315967e+03 = [4, 2, 2, 3]	FES = 756100	Diversity = 1.1635e-02	k_t
Generation 7570: Best = 1.315957e+03 = [1, 1, 1, 1]	FES = 757100	Diversity = 1.2207e-02	k_t
Generation 7580: Best = 1.315947e+03 = [2, 4, 4, 3]	FES = 758100	Diversity = 1.2574e-02	k_t
Generation 7590: Best = 1.315935e+03 = [1, 1, 1, 1]	FES = 759100	Diversity = 1.3258e-02	k_t
Generation 7600: Best = 1.315923e+03 = [4, 4, 2, 4]	FES = 760100	Diversity = 1.3860e-02	k_t
Generation 7610: Best = 1.315908e+03 = [1, 1, 1, 1]	FES = 761100	Diversity = 1.4305e-02	k_t
Generation 7620: Best = 1.315898e+03 = [2, 2, 2, 2]	FES = 762100	Diversity = 1.4939e-02	k_t
Generation 7630: Best = 1.315885e+03 = [1, 1, 1, 1]	FES = 763100	Diversity = 1.5617e-02	k_t
Generation 7640: Best = 1.315877e+03 = [3, 2, 2, 4]	FES = 764100	Diversity = 1.4902e-02	k_t
Generation 7650: Best = 1.315868e+03 = [1, 1, 1, 1]	FES = 765100	Diversity = 1.3537e-02	k_t
Generation 7660: Best = 1.315864e+03 = [3, 4, 3, 2]	FES = 766100	Diversity = 1.2966e-02	k_t
Generation 7670: Best = 1.315852e+03 = [1, 1, 1, 1]	FES = 767100	Diversity = 1.2617e-02	k_t
Generation 7680: Best = 1.315848e+03 = [2, 3, 2, 3]	FES = 768100	Diversity = 1.2466e-02	k_t
Generation 7690: Best = 1.315839e+03 = [1, 1, 1, 1]	FES = 769100	Diversity = 1.2319e-02	k_t
Generation 7700: Best = 1.315833e+03 = [2, 3, 2, 2]	FES = 770100	Diversity = 1.2104e-02	k_t
Generation 7710: Best = 1.315826e+03 = [1, 1, 1, 1]	FES = 771100	Diversity = 1.1281e-02	k_t
Generation 7720: Best = 1.315818e+03 = [2, 2, 4, 3]	FES = 772100	Diversity = 1.0968e-02	k_t
Generation 7730: Best = 1.315812e+03 = [1, 1, 1, 1]	FES = 773100	Diversity = 1.1389e-02	k_t
Generation 7740: Best = 1.315804e+03 = [2, 3, 4, 3]	FES = 774100	Diversity = 1.1232e-02	k_t
Generation 7750: Best = 1.315797e+03 = [1, 1, 1, 1]	FES = 775100	Diversity = 1.0937e-02	k_t
Generation 7760: Best = 1.315786e+03 = [2, 4, 3, 4]	FES = 776100	Diversity = 1.0996e-02	k_t
Generation 7770: Best = 1.315778e+03 = [1, 1, 1, 1]	FES = 777100	Diversity = 1.0844e-02	k_t
Generation 7780: Best = 1.315774e+03 = [3, 3, 3, 4]	FES = 778100	Diversity = 1.0789e-02	k_t

Generation 7790: Best = 1.315763e+03 = [1, 1, 1, 1]	FES = 779100	Diversity = 1.2038e-02	k_t
Generation 7800: Best = 1.315755e+03 = [4, 2, 4, 3]	FES = 780100	Diversity = 1.3130e-02	k_t
Generation 7810: Best = 1.315741e+03 = [1, 1, 1, 1]	FES = 781100	Diversity = 1.4964e-02	k_t
Generation 7820: Best = 1.315731e+03 = [2, 3, 2, 3]	FES = 782100	Diversity = 1.4821e-02	k_t
Generation 7830: Best = 1.315718e+03 = [1, 1, 1, 1]	FES = 783100	Diversity = 1.5056e-02	k_t
Generation 7840: Best = 1.315707e+03 = [2, 3, 3, 3]	FES = 784100	Diversity = 1.5759e-02	k_t
Generation 7850: Best = 1.315694e+03 = [1, 1, 1, 1]	FES = 785100	Diversity = 1.5122e-02	k_t
Generation 7860: Best = 1.315683e+03 = [3, 4, 4, 4]	FES = 786100	Diversity = 1.5723e-02	k_t
Generation 7870: Best = 1.315672e+03 = [1, 1, 1, 1]	FES = 787100	Diversity = 1.4870e-02	k_t
Generation 7880: Best = 1.315663e+03 = [3, 4, 4, 3]	FES = 788100	Diversity = 1.4524e-02	k_t
Generation 7890: Best = 1.315653e+03 = [1, 1, 1, 1]	FES = 789100	Diversity = 1.3896e-02	k_t
Generation 7900: Best = 1.315645e+03 = [4, 4, 2, 3]	FES = 790100	Diversity = 1.3409e-02	k_t
Generation 7910: Best = 1.315636e+03 = [1, 1, 1, 1]	FES = 791100	Diversity = 1.3633e-02	k_t
Generation 7920: Best = 1.315626e+03 = [2, 2, 4, 3]	FES = 792100	Diversity = 1.2701e-02	k_t
Generation 7930: Best = 1.315615e+03 = [1, 1, 1, 1]	FES = 793100	Diversity = 1.2924e-02	k_t
Generation 7940: Best = 1.315605e+03 = [3, 2, 2, 2]	FES = 794100	Diversity = 1.3302e-02	k_t
Generation 7950: Best = 1.315598e+03 = [1, 1, 1, 1]	FES = 795100	Diversity = 1.2708e-02	k_t
Generation 7960: Best = 1.315590e+03 = [4, 3, 3, 3]	FES = 796100	Diversity = 1.2661e-02	k_t
Generation 7970: Best = 1.315583e+03 = [1, 1, 1, 1]	FES = 797100	Diversity = 1.1971e-02	k_t
Generation 7980: Best = 1.315570e+03 = [3, 3, 4, 3]	FES = 798100	Diversity = 1.1772e-02	k_t
Generation 7990: Best = 1.315564e+03 = [1, 1, 1, 1]	FES = 799100	Diversity = 1.2237e-02	k_t
Generation 8000: Best = 1.315553e+03 = [4, 3, 3, 3]	FES = 800100	Diversity = 1.2096e-02	k_t
Generation 8010: Best = 1.315545e+03 = [1, 1, 1, 1]	FES = 801100	Diversity = 1.2561e-02	k_t
Generation 8020: Best = 1.315534e+03 = [4, 3, 3, 4]	FES = 802100	Diversity = 1.1932e-02	k_t
Generation 8030: Best = 1.315524e+03 = [1, 1, 1, 1]	FES = 803100	Diversity = 1.1091e-02	k_t
Generation 8040: Best = 1.315520e+03 = [3, 2, 3, 3]	FES = 804100	Diversity = 1.1111e-02	k_t
Generation 8050: Best = 1.315513e+03 = [1, 1, 1, 1]	FES = 805100	Diversity = 1.0253e-02	k_t

Generation 8060: Best = 1.315507e+03 = [4, 3, 3, 4]	FES = 806100	Diversity = 9.7486e-03	k_t
Generation 8070: Best = 1.315497e+03 = [1, 1, 1, 1]	FES = 807100	Diversity = 1.0432e-02	k_t
Generation 8080: Best = 1.315491e+03 = [3, 4, 3, 4]	FES = 808100	Diversity = 1.0846e-02	k_t
Generation 8090: Best = 1.315485e+03 = [1, 1, 1, 1]	FES = 809100	Diversity = 1.0414e-02	k_t
Generation 8100: Best = 1.315475e+03 = [2, 2, 3, 3]	FES = 810100	Diversity = 1.0630e-02	k_t
Generation 8110: Best = 1.315463e+03 = [1, 1, 1, 1]	FES = 811100	Diversity = 1.1386e-02	k_t
Generation 8120: Best = 1.315455e+03 = [2, 2, 3, 2]	FES = 812100	Diversity = 1.1205e-02	k_t
Generation 8130: Best = 1.315449e+03 = [1, 1, 1, 1]	FES = 813100	Diversity = 1.1261e-02	k_t
Generation 8140: Best = 1.315437e+03 = [2, 2, 4, 4]	FES = 814100	Diversity = 1.1906e-02	k_t
Generation 8150: Best = 1.315430e+03 = [1, 1, 1, 1]	FES = 815100	Diversity = 1.2366e-02	k_t
Generation 8160: Best = 1.315423e+03 = [4, 3, 2, 4]	FES = 816100	Diversity = 1.1909e-02	k_t
Generation 8170: Best = 1.315412e+03 = [1, 1, 1, 1]	FES = 817100	Diversity = 1.1337e-02	k_t
Generation 8180: Best = 1.315402e+03 = [2, 4, 3, 2]	FES = 818100	Diversity = 1.1621e-02	k_t
Generation 8190: Best = 1.315397e+03 = [1, 1, 1, 1]	FES = 819100	Diversity = 1.1703e-02	k_t
Generation 8200: Best = 1.315390e+03 = [3, 3, 3, 4]	FES = 820100	Diversity = 1.1644e-02	k_t
Generation 8210: Best = 1.315379e+03 = [1, 1, 1, 1]	FES = 821100	Diversity = 1.2741e-02	k_t
Generation 8220: Best = 1.315368e+03 = [2, 4, 4, 2]	FES = 822100	Diversity = 1.3683e-02	k_t
Generation 8230: Best = 1.315356e+03 = [1, 1, 1, 1]	FES = 823100	Diversity = 1.3837e-02	k_t
Generation 8240: Best = 1.315341e+03 = [3, 3, 4, 2]	FES = 824100	Diversity = 1.4828e-02	k_t
Generation 8250: Best = 1.315329e+03 = [1, 1, 1, 1]	FES = 825100	Diversity = 1.5561e-02	k_t
Generation 8260: Best = 1.315319e+03 = [3, 4, 2, 2]	FES = 826100	Diversity = 1.5248e-02	k_t
Generation 8270: Best = 1.315306e+03 = [1, 1, 1, 1]	FES = 827100	Diversity = 1.4567e-02	k_t
Generation 8280: Best = 1.315289e+03 = [3, 3, 3, 3]	FES = 828100	Diversity = 1.4874e-02	k_t
Generation 8290: Best = 1.315278e+03 = [1, 1, 1, 1]	FES = 829100	Diversity = 1.4799e-02	k_t
Generation 8300: Best = 1.315262e+03 = [3, 4, 4, 3]	FES = 830100	Diversity = 1.5288e-02	k_t
Generation 8310: Best = 1.315248e+03 = [1, 1, 1, 1]	FES = 831100	Diversity = 1.5433e-02	k_t
Generation 8320: Best = 1.315239e+03 = [3, 2, 2, 2]	FES = 832100	Diversity = 1.6119e-02	k_t

Generation 8330: Best = 1.315225e+03 = [1, 1, 1, 1]	FES = 833100	Diversity = 1.5977e-02	k_t
Generation 8340: Best = 1.315216e+03 = [2, 2, 4, 2]	FES = 834100	Diversity = 1.6603e-02	k_t
Generation 8350: Best = 1.315200e+03 = [1, 1, 1, 1]	FES = 835100	Diversity = 1.6559e-02	k_t
Generation 8360: Best = 1.315185e+03 = [3, 2, 4, 4]	FES = 836100	Diversity = 1.7247e-02	k_t
Generation 8370: Best = 1.315164e+03 = [1, 1, 1, 1]	FES = 837100	Diversity = 1.6408e-02	k_t
Generation 8380: Best = 1.315156e+03 = [2, 4, 4, 2]	FES = 838100	Diversity = 1.5941e-02	k_t
Generation 8390: Best = 1.315144e+03 = [1, 1, 1, 1]	FES = 839100	Diversity = 1.6116e-02	k_t
Generation 8400: Best = 1.315128e+03 = [4, 3, 2, 3]	FES = 840100	Diversity = 1.7257e-02	k_t
Generation 8410: Best = 1.315110e+03 = [1, 1, 1, 1]	FES = 841100	Diversity = 1.7308e-02	k_t
Generation 8420: Best = 1.315107e+03 = [3, 3, 2, 2]	FES = 842100	Diversity = 1.6338e-02	k_t
Generation 8430: Best = 1.315094e+03 = [1, 1, 1, 1]	FES = 843100	Diversity = 1.5209e-02	k_t
Generation 8440: Best = 1.315082e+03 = [2, 3, 3, 4]	FES = 844100	Diversity = 1.4533e-02	k_t
Generation 8450: Best = 1.315073e+03 = [1, 1, 1, 1]	FES = 845100	Diversity = 1.4779e-02	k_t
Generation 8460: Best = 1.315059e+03 = [2, 4, 3, 4]	FES = 846100	Diversity = 1.4690e-02	k_t
Generation 8470: Best = 1.315050e+03 = [1, 1, 1, 1]	FES = 847100	Diversity = 1.4158e-02	k_t
Generation 8480: Best = 1.315039e+03 = [3, 3, 2, 4]	FES = 848100	Diversity = 1.4001e-02	k_t
Generation 8490: Best = 1.315030e+03 = [1, 1, 1, 1]	FES = 849100	Diversity = 1.3311e-02	k_t
Generation 8500: Best = 1.315018e+03 = [2, 4, 4, 2]	FES = 850100	Diversity = 1.2351e-02	k_t
Generation 8510: Best = 1.315007e+03 = [1, 1, 1, 1]	FES = 851100	Diversity = 1.2194e-02	k_t
Generation 8520: Best = 1.314997e+03 = [2, 3, 2, 4]	FES = 852100	Diversity = 1.1980e-02	k_t
Generation 8530: Best = 1.314984e+03 = [1, 1, 1, 1]	FES = 853100	Diversity = 1.3257e-02	k_t
Generation 8540: Best = 1.314971e+03 = [3, 2, 2, 3]	FES = 854100	Diversity = 1.3530e-02	k_t
Generation 8550: Best = 1.314957e+03 = [1, 1, 1, 1]	FES = 855100	Diversity = 1.4762e-02	k_t
Generation 8560: Best = 1.314931e+03 = [4, 2, 2, 2]	FES = 856100	Diversity = 1.7313e-02	k_t
Generation 8570: Best = 1.314919e+03 = [1, 1, 1, 1]	FES = 857100	Diversity = 1.7586e-02	k_t
Generation 8580: Best = 1.314902e+03 = [3, 2, 2, 4]	FES = 858100	Diversity = 1.8132e-02	k_t
Generation 8590: Best = 1.314885e+03 = [1, 1, 1, 1]	FES = 859100	Diversity = 1.8442e-02	k_t

Generation 8600: Best = 1.314854e+03 = [2, 3, 2, 2]	FES = 860100	Diversity = 1.7614e-02	k_t
Generation 8610: Best = 1.314842e+03 = [1, 1, 1, 1]	FES = 861100	Diversity = 1.8275e-02	k_t
Generation 8620: Best = 1.314828e+03 = [2, 3, 2, 2]	FES = 862100	Diversity = 1.9335e-02	k_t
Generation 8630: Best = 1.314803e+03 = [1, 1, 1, 1]	FES = 863100	Diversity = 1.9801e-02	k_t
Generation 8640: Best = 1.314790e+03 = [4, 2, 2, 3]	FES = 864100	Diversity = 1.9217e-02	k_t
Generation 8650: Best = 1.314768e+03 = [1, 1, 1, 1]	FES = 865100	Diversity = 1.9018e-02	k_t
Generation 8660: Best = 1.314753e+03 = [3, 3, 4, 4]	FES = 866100	Diversity = 1.8094e-02	k_t
Generation 8670: Best = 1.314733e+03 = [1, 1, 1, 1]	FES = 867100	Diversity = 1.5843e-02	k_t
Generation 8680: Best = 1.314723e+03 = [3, 3, 4, 3]	FES = 868100	Diversity = 1.4288e-02	k_t
Generation 8690: Best = 1.314714e+03 = [1, 1, 1, 1]	FES = 869100	Diversity = 1.3715e-02	k_t
Generation 8700: Best = 1.314698e+03 = [2, 2, 2, 4]	FES = 870100	Diversity = 1.4400e-02	k_t
Generation 8710: Best = 1.314687e+03 = [1, 1, 1, 1]	FES = 871100	Diversity = 1.3659e-02	k_t
Generation 8720: Best = 1.314672e+03 = [4, 2, 2, 3]	FES = 872100	Diversity = 1.4183e-02	k_t
Generation 8730: Best = 1.314652e+03 = [1, 1, 1, 1]	FES = 873100	Diversity = 1.5395e-02	k_t
Generation 8740: Best = 1.314641e+03 = [2, 4, 3, 4]	FES = 874100	Diversity = 1.5519e-02	k_t
Generation 8750: Best = 1.314626e+03 = [1, 1, 1, 1]	FES = 875100	Diversity = 1.6340e-02	k_t
Generation 8760: Best = 1.314602e+03 = [4, 3, 4, 3]	FES = 876100	Diversity = 1.5909e-02	k_t
Generation 8770: Best = 1.314592e+03 = [1, 1, 1, 1]	FES = 877100	Diversity = 1.5636e-02	k_t
Generation 8780: Best = 1.314583e+03 = [3, 2, 3, 4]	FES = 878100	Diversity = 1.5204e-02	k_t
Generation 8790: Best = 1.314571e+03 = [1, 1, 1, 1]	FES = 879100	Diversity = 1.5840e-02	k_t
Generation 8800: Best = 1.314560e+03 = [4, 2, 2, 2]	FES = 880100	Diversity = 1.4245e-02	k_t
Generation 8810: Best = 1.314552e+03 = [1, 1, 1, 1]	FES = 881100	Diversity = 1.2681e-02	k_t
Generation 8820: Best = 1.314543e+03 = [3, 2, 2, 3]	FES = 882100	Diversity = 1.2034e-02	k_t
Generation 8830: Best = 1.314536e+03 = [1, 1, 1, 1]	FES = 883100	Diversity = 1.1217e-02	k_t
Generation 8840: Best = 1.314527e+03 = [3, 2, 4, 2]	FES = 884100	Diversity = 1.0550e-02	k_t
Generation 8850: Best = 1.314513e+03 = [1, 1, 1, 1]	FES = 885100	Diversity = 1.1030e-02	k_t
Generation 8860: Best = 1.314503e+03 = [4, 4, 4, 3]	FES = 886100	Diversity = 1.1790e-02	k_t

Generation 8870: Best = 1.314489e+03 = [1, 1, 1, 1]	FES = 887100	Diversity = 1.2677e-02	k_t
Generation 8880: Best = 1.314480e+03 = [2, 2, 2, 4]	FES = 888100	Diversity = 1.2929e-02	k_t
Generation 8890: Best = 1.314466e+03 = [1, 1, 1, 1]	FES = 889100	Diversity = 1.3817e-02	k_t
Generation 8900: Best = 1.314454e+03 = [2, 2, 4, 2]	FES = 890100	Diversity = 1.4640e-02	k_t
Generation 8910: Best = 1.314427e+03 = [1, 1, 1, 1]	FES = 891100	Diversity = 1.6665e-02	k_t
Generation 8920: Best = 1.314415e+03 = [3, 4, 4, 2]	FES = 892100	Diversity = 1.9391e-02	k_t
Generation 8930: Best = 1.314396e+03 = [1, 1, 1, 1]	FES = 893100	Diversity = 1.9961e-02	k_t
Generation 8940: Best = 1.314382e+03 = [3, 4, 2, 4]	FES = 894100	Diversity = 1.8222e-02	k_t
Generation 8950: Best = 1.314365e+03 = [1, 1, 1, 1]	FES = 895100	Diversity = 1.6894e-02	k_t
Generation 8960: Best = 1.314354e+03 = [2, 4, 2, 3]	FES = 896100	Diversity = 1.5483e-02	k_t
Generation 8970: Best = 1.314342e+03 = [1, 1, 1, 1]	FES = 897100	Diversity = 1.4147e-02	k_t
Generation 8980: Best = 1.314330e+03 = [3, 2, 3, 4]	FES = 898100	Diversity = 1.3526e-02	k_t
Generation 8990: Best = 1.314318e+03 = [1, 1, 1, 1]	FES = 899100	Diversity = 1.3779e-02	k_t
Generation 9000: Best = 1.314308e+03 = [3, 3, 4, 2]	FES = 900100	Diversity = 1.4143e-02	k_t
Generation 9010: Best = 1.314289e+03 = [1, 1, 1, 1]	FES = 901100	Diversity = 1.4467e-02	k_t
Generation 9020: Best = 1.314281e+03 = [2, 4, 2, 3]	FES = 902100	Diversity = 1.4268e-02	k_t
Generation 9030: Best = 1.314270e+03 = [1, 1, 1, 1]	FES = 903100	Diversity = 1.3564e-02	k_t
Generation 9040: Best = 1.314248e+03 = [3, 2, 3, 4]	FES = 904100	Diversity = 1.4123e-02	k_t
Generation 9050: Best = 1.314239e+03 = [1, 1, 1, 1]	FES = 905100	Diversity = 1.4647e-02	k_t
Generation 9060: Best = 1.314227e+03 = [4, 3, 4, 2]	FES = 906100	Diversity = 1.3941e-02	k_t
Generation 9070: Best = 1.314209e+03 = [1, 1, 1, 1]	FES = 907100	Diversity = 1.5157e-02	k_t
Generation 9080: Best = 1.314196e+03 = [4, 4, 4, 2]	FES = 908100	Diversity = 1.4545e-02	k_t
Generation 9090: Best = 1.314180e+03 = [1, 1, 1, 1]	FES = 909100	Diversity = 1.4837e-02	k_t
Generation 9100: Best = 1.314163e+03 = [2, 3, 3, 2]	FES = 910100	Diversity = 1.5698e-02	k_t
Generation 9110: Best = 1.314155e+03 = [1, 1, 1, 1]	FES = 911100	Diversity = 1.6027e-02	k_t
Generation 9120: Best = 1.314129e+03 = [3, 3, 4, 4]	FES = 912100	Diversity = 1.6077e-02	k_t
Generation 9130: Best = 1.314128e+03 = [1, 1, 1, 1]	FES = 913100	Diversity = 1.6043e-02	k_t

Generation 9140: Best = 1.314110e+03 = [2, 2, 3, 3]	FES = 914100	Diversity = 1.7200e-02	k_t
Generation 9150: Best = 1.314083e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 1.7511e-02	k_t
Generation 9160: Best = 1.314074e+03 = [4, 4, 3, 2]	FES = 916100	Diversity = 1.7435e-02	k_t
Generation 9170: Best = 1.314060e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 1.6709e-02	k_t
Generation 9180: Best = 1.314049e+03 = [3, 4, 2, 3]	FES = 918100	Diversity = 1.5839e-02	k_t
Generation 9190: Best = 1.314030e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 1.5375e-02	k_t
Generation 9200: Best = 1.314013e+03 = [2, 4, 2, 3]	FES = 920100	Diversity = 1.5955e-02	k_t
Generation 9210: Best = 1.313998e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 1.5924e-02	k_t
Generation 9220: Best = 1.313976e+03 = [3, 4, 4, 2]	FES = 922100	Diversity = 1.6124e-02	k_t
Generation 9230: Best = 1.313967e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 1.6201e-02	k_t
Generation 9240: Best = 1.313947e+03 = [4, 3, 3, 3]	FES = 924100	Diversity = 1.7930e-02	k_t
Generation 9250: Best = 1.313930e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 1.8651e-02	k_t
Generation 9260: Best = 1.313910e+03 = [4, 4, 2, 4]	FES = 926100	Diversity = 1.9371e-02	k_t
Generation 9270: Best = 1.313886e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 1.9428e-02	k_t
Generation 9280: Best = 1.313870e+03 = [4, 3, 4, 3]	FES = 928100	Diversity = 1.9232e-02	k_t
Generation 9290: Best = 1.313849e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 2.0347e-02	k_t
Generation 9300: Best = 1.313833e+03 = [4, 3, 4, 2]	FES = 930100	Diversity = 1.8906e-02	k_t
Generation 9310: Best = 1.313811e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 1.8893e-02	k_t
Generation 9320: Best = 1.313792e+03 = [2, 2, 4, 3]	FES = 932100	Diversity = 1.8716e-02	k_t
Generation 9330: Best = 1.313778e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 1.6593e-02	k_t
Generation 9340: Best = 1.313764e+03 = [2, 3, 2, 4]	FES = 934100	Diversity = 1.6243e-02	k_t
Generation 9350: Best = 1.313744e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 1.6438e-02	k_t
Generation 9360: Best = 1.313736e+03 = [4, 3, 3, 3]	FES = 936100	Diversity = 1.6221e-02	k_t
Generation 9370: Best = 1.313721e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 1.5998e-02	k_t
Generation 9380: Best = 1.313702e+03 = [3, 3, 4, 3]	FES = 938100	Diversity = 1.6246e-02	k_t
Generation 9390: Best = 1.313689e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 1.5920e-02	k_t
Generation 9400: Best = 1.313675e+03 = [3, 4, 4, 3]	FES = 940100	Diversity = 1.6174e-02	k_t

Generation 9410: Best = 1.313657e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 1.5491e-02	k_t
Generation 9420: Best = 1.313645e+03 = [2, 4, 4, 3]	FES = 942100	Diversity = 1.4975e-02	k_t
Generation 9430: Best = 1.313632e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 1.5380e-02	k_t
Generation 9440: Best = 1.313615e+03 = [2, 3, 4, 2]	FES = 944100	Diversity = 1.5805e-02	k_t
Generation 9450: Best = 1.313603e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 1.5564e-02	k_t
Generation 9460: Best = 1.313585e+03 = [2, 2, 2, 2]	FES = 946100	Diversity = 1.6040e-02	k_t
Generation 9470: Best = 1.313571e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 1.6023e-02	k_t
Generation 9480: Best = 1.313558e+03 = [4, 3, 2, 4]	FES = 948100	Diversity = 1.5355e-02	k_t
Generation 9490: Best = 1.313541e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 1.5710e-02	k_t
Generation 9500: Best = 1.313518e+03 = [3, 3, 3, 4]	FES = 950100	Diversity = 1.6138e-02	k_t
Generation 9510: Best = 1.313500e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 1.6066e-02	k_t
Generation 9520: Best = 1.313485e+03 = [2, 3, 2, 4]	FES = 952100	Diversity = 1.6626e-02	k_t
Generation 9530: Best = 1.313468e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 1.6910e-02	k_t
Generation 9540: Best = 1.313455e+03 = [3, 3, 3, 2]	FES = 954100	Diversity = 1.7497e-02	k_t
Generation 9550: Best = 1.313439e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 1.8044e-02	k_t
Generation 9560: Best = 1.313418e+03 = [2, 3, 4, 3]	FES = 956100	Diversity = 1.8098e-02	k_t
Generation 9570: Best = 1.313400e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 1.7768e-02	k_t
Generation 9580: Best = 1.313391e+03 = [2, 4, 4, 2]	FES = 958100	Diversity = 1.6823e-02	k_t
Generation 9590: Best = 1.313375e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 1.6050e-02	k_t
Generation 9600: Best = 1.313362e+03 = [3, 2, 3, 2]	FES = 960100	Diversity = 1.5897e-02	k_t
Generation 9610: Best = 1.313348e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 1.5291e-02	k_t
Generation 9620: Best = 1.313333e+03 = [4, 3, 3, 4]	FES = 962100	Diversity = 1.4380e-02	k_t
Generation 9630: Best = 1.313323e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 1.3564e-02	k_t
Generation 9640: Best = 1.313314e+03 = [4, 2, 4, 2]	FES = 964100	Diversity = 1.3328e-02	k_t
Generation 9650: Best = 1.313301e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 1.3469e-02	k_t
Generation 9660: Best = 1.313290e+03 = [4, 3, 4, 4]	FES = 966100	Diversity = 1.3622e-02	k_t
Generation 9670: Best = 1.313273e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 1.3458e-02	k_t

Generation 9680: Best = 1.313259e+03 = [2, 4, 2, 3]	FES = 968100	Diversity = 1.4105e-02	k_t
Generation 9690: Best = 1.313252e+03 = [1, 1, 1, 1]	FES = 969100	Diversity = 1.4397e-02	k_t
Generation 9700: Best = 1.313229e+03 = [4, 2, 3, 4]	FES = 970100	Diversity = 1.5943e-02	k_t
Generation 9710: Best = 1.313208e+03 = [1, 1, 1, 1]	FES = 971100	Diversity = 1.6988e-02	k_t
Generation 9720: Best = 1.313192e+03 = [4, 4, 2, 2]	FES = 972100	Diversity = 1.6808e-02	k_t
Generation 9730: Best = 1.313178e+03 = [1, 1, 1, 1]	FES = 973100	Diversity = 1.5553e-02	k_t
Generation 9740: Best = 1.313164e+03 = [2, 4, 2, 4]	FES = 974100	Diversity = 1.5824e-02	k_t
Generation 9750: Best = 1.313154e+03 = [1, 1, 1, 1]	FES = 975100	Diversity = 1.5404e-02	k_t
Generation 9760: Best = 1.313138e+03 = [4, 2, 4, 4]	FES = 976100	Diversity = 1.5890e-02	k_t
Generation 9770: Best = 1.313123e+03 = [1, 1, 1, 1]	FES = 977100	Diversity = 1.4732e-02	k_t
Generation 9780: Best = 1.313108e+03 = [3, 4, 3, 4]	FES = 978100	Diversity = 1.3885e-02	k_t
Generation 9790: Best = 1.313093e+03 = [1, 1, 1, 1]	FES = 979100	Diversity = 1.4534e-02	k_t
Generation 9800: Best = 1.313073e+03 = [3, 2, 2, 4]	FES = 980100	Diversity = 1.3934e-02	k_t
Generation 9810: Best = 1.313059e+03 = [1, 1, 1, 1]	FES = 981100	Diversity = 1.4472e-02	k_t
Generation 9820: Best = 1.313043e+03 = [3, 3, 4, 4]	FES = 982100	Diversity = 1.5188e-02	k_t
Generation 9830: Best = 1.313031e+03 = [1, 1, 1, 1]	FES = 983100	Diversity = 1.4357e-02	k_t
Generation 9840: Best = 1.313020e+03 = [3, 2, 3, 2]	FES = 984100	Diversity = 1.4555e-02	k_t
Generation 9850: Best = 1.313002e+03 = [1, 1, 1, 1]	FES = 985100	Diversity = 1.5443e-02	k_t
Generation 9860: Best = 1.312990e+03 = [3, 4, 3, 3]	FES = 986100	Diversity = 1.5968e-02	k_t
Generation 9870: Best = 1.312975e+03 = [1, 1, 1, 1]	FES = 987100	Diversity = 1.6254e-02	k_t
Generation 9880: Best = 1.312960e+03 = [4, 4, 4, 4]	FES = 988100	Diversity = 1.7376e-02	k_t
Generation 9890: Best = 1.312948e+03 = [1, 1, 1, 1]	FES = 989100	Diversity = 1.7197e-02	k_t
Generation 9900: Best = 1.312933e+03 = [2, 3, 4, 4]	FES = 990100	Diversity = 1.6891e-02	k_t
Generation 9910: Best = 1.312920e+03 = [1, 1, 1, 1]	FES = 991100	Diversity = 1.6867e-02	k_t
Generation 9920: Best = 1.312909e+03 = [2, 3, 4, 3]	FES = 992100	Diversity = 1.7199e-02	k_t
Generation 9930: Best = 1.312886e+03 = [1, 1, 1, 1]	FES = 993100	Diversity = 1.7631e-02	k_t
Generation 9940: Best = 1.312868e+03 = [4, 2, 3, 3]	FES = 994100	Diversity = 1.7736e-02	k_t

Generation 9950: Best = 1.312858e+03 FES = 995100 Diversity = 1.7142e-02 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 1.312844e+03 FES = 996100 Diversity = 1.6070e-02 k_t
= [3, 3, 3, 2]
Generation 9970: Best = 1.312827e+03 FES = 997100 Diversity = 1.6091e-02 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 1.312810e+03 FES = 998100 Diversity = 1.7212e-02 k_t
= [2, 3, 2, 3]
Generation 9990: Best = 1.312786e+03 FES = 999100 Diversity = 1.7639e-02 k_t
= [1, 1, 1, 1]

Optimization completed!

Final Best Fitness = 1.312781e+03

Total Function Evaluations = 1000000

=====

=

Results:

Best Fitness : 1.312781e+03

Global Optimum : 1.100000e+03

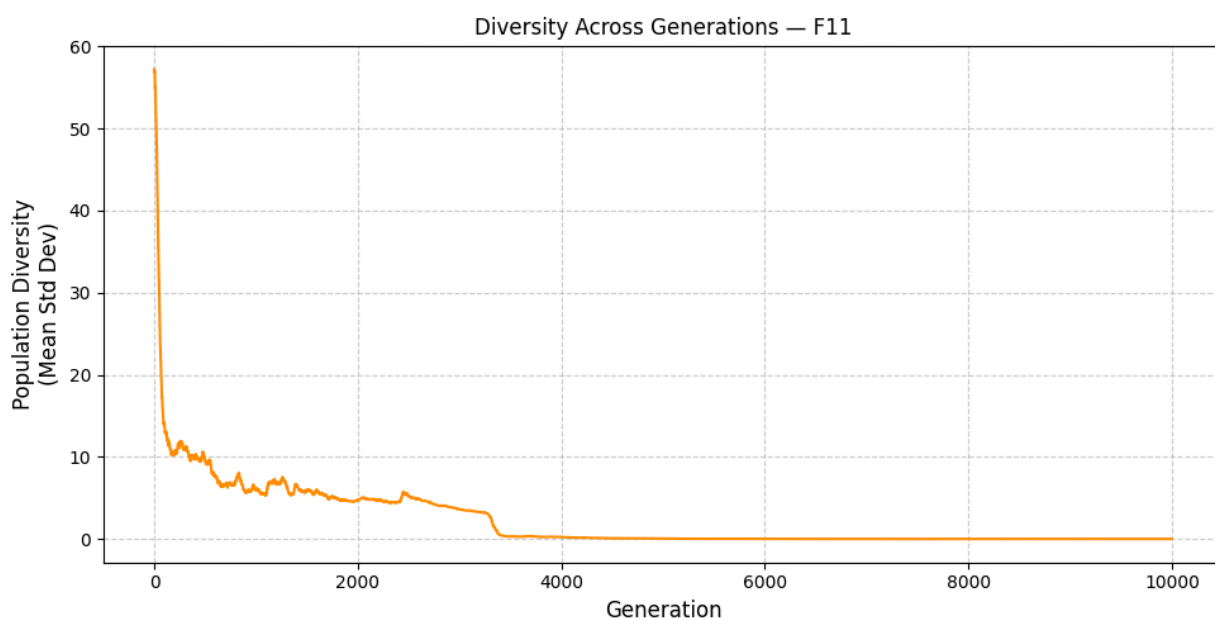
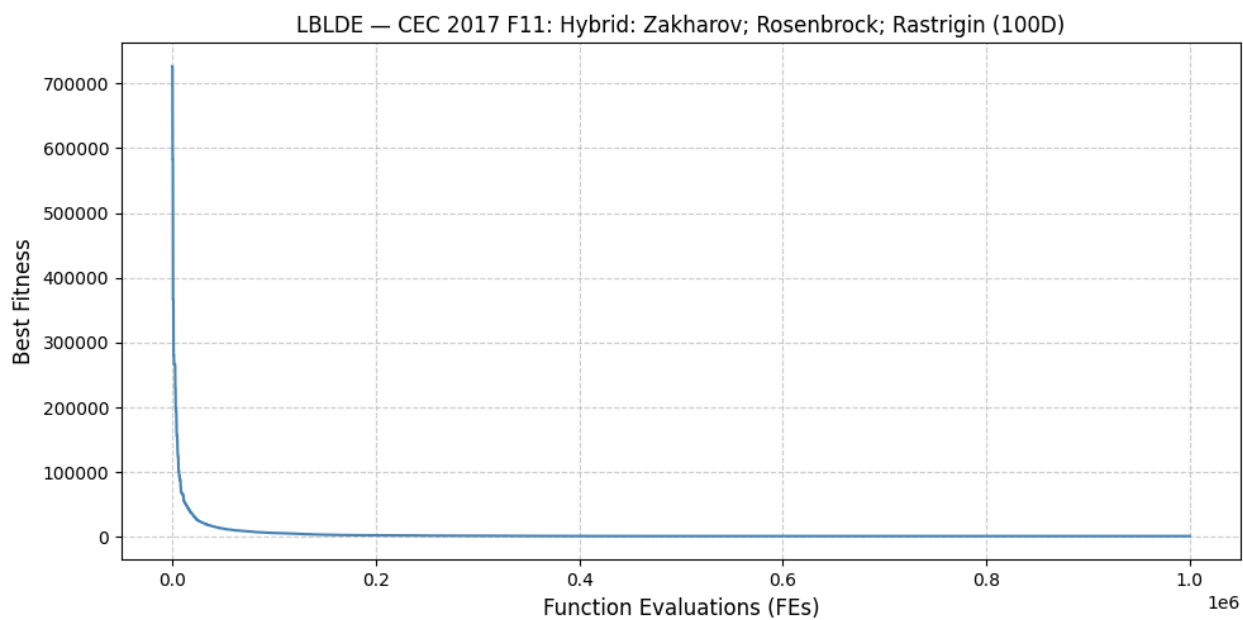
Error (|best-optimum|) : 2.127808e+02

Best Solution (5 dim) : [-24.76011887 -16.49503909 -26.25135991 9.08819915
19.53123924]

D^(0) (init. diversity): 5.7220e+01

=====

=



Graph saved as convergence_F11.png

```
In [36]: # =====
# F12 – Hybrid Function 2 (Elliptic; Schwefel; Bent Cigar)
# =====
run_and_plot('F12', make_wrapper(hybrid.f12), 1200.0, 'Hybrid: Elliptic; Schwe
```

```

=====
=
LBLDE_s on CEC 2017 F12 – Single Run
=====
=
Function      : Hybrid: Elliptic; Schwefel; Bent Cigar (100D)
Bounds       : [-100, 100]
Global optimum : 1200.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 3.383138e+12 | Diversity = 5.7220e+01
Generation 10: Best = 1.018195e+12 FES = 1100 Diversity = 3.9894e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.617675e+11 FES = 2100 Diversity = 3.0850e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 2.107147e+11 FES = 3100 Diversity = 2.3054e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.126512e+11 FES = 4100 Diversity = 1.9465e+01 k_t
= [2, 2, 4, 4]
Generation 50: Best = 5.825033e+10 FES = 5100 Diversity = 1.8047e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.212060e+10 FES = 6100 Diversity = 1.5192e+01 k_t
= [4, 2, 4, 2]
Generation 70: Best = 1.779292e+10 FES = 7100 Diversity = 1.3824e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 9.100842e+09 FES = 8100 Diversity = 1.2242e+01 k_t
= [4, 4, 3, 3]
Generation 90: Best = 6.632459e+09 FES = 9100 Diversity = 1.1030e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 4.369860e+09 FES = 10100 Diversity = 1.0359e+01 k_t
= [3, 2, 2, 4]
Generation 110: Best = 3.272828e+09 FES = 11100 Diversity = 9.0411e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.153057e+09 FES = 12100 Diversity = 8.3859e+00 k_t
= [4, 2, 2, 3]
Generation 130: Best = 1.718547e+09 FES = 13100 Diversity = 6.8679e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.342199e+09 FES = 14100 Diversity = 6.5318e+00 k_t
= [4, 3, 3, 2]
Generation 150: Best = 1.019034e+09 FES = 15100 Diversity = 5.9267e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 7.987484e+08 FES = 16100 Diversity = 5.7462e+00 k_t
= [3, 4, 3, 4]
Generation 170: Best = 5.737323e+08 FES = 17100 Diversity = 5.6237e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 4.469239e+08 FES = 18100 Diversity = 5.0585e+00 k_t
= [4, 2, 2, 4]
Generation 190: Best = 3.720635e+08 FES = 19100 Diversity = 5.0023e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.157920e+08 FES = 20100 Diversity = 4.7011e+00 k_t
= [4, 3, 4, 2]
Generation 210: Best = 2.655358e+08 FES = 21100 Diversity = 4.6000e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.186227e+08 FES = 22100 Diversity = 4.1635e+00 k_t
= [3, 4, 3, 3]

```

Generation 230:	Best = 1.990960e+08	FES = 23100	Diversity = 4.1527e+00	k_t = [1, 1, 1, 1]
Generation 240:	Best = 1.682689e+08	FES = 24100	Diversity = 4.1142e+00	k_t = [2, 3, 2, 2]
Generation 250:	Best = 1.522266e+08	FES = 25100	Diversity = 4.0109e+00	k_t = [1, 1, 1, 1]
Generation 260:	Best = 1.366416e+08	FES = 26100	Diversity = 3.7835e+00	k_t = [3, 2, 4, 3]
Generation 270:	Best = 1.166519e+08	FES = 27100	Diversity = 3.6947e+00	k_t = [1, 1, 1, 1]
Generation 280:	Best = 1.064620e+08	FES = 28100	Diversity = 3.7409e+00	k_t = [4, 4, 4, 3]
Generation 290:	Best = 9.670130e+07	FES = 29100	Diversity = 3.7760e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 8.621242e+07	FES = 30100	Diversity = 3.9311e+00	k_t = [2, 3, 4, 4]
Generation 310:	Best = 7.766234e+07	FES = 31100	Diversity = 4.2425e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 7.030898e+07	FES = 32100	Diversity = 4.0842e+00	k_t = [3, 4, 2, 2]
Generation 330:	Best = 6.372461e+07	FES = 33100	Diversity = 3.9563e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 6.000800e+07	FES = 34100	Diversity = 3.9056e+00	k_t = [4, 4, 2, 2]
Generation 350:	Best = 5.554120e+07	FES = 35100	Diversity = 4.2430e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 4.934635e+07	FES = 36100	Diversity = 4.2513e+00	k_t = [2, 4, 2, 4]
Generation 370:	Best = 4.768873e+07	FES = 37100	Diversity = 4.3341e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 4.435609e+07	FES = 38100	Diversity = 4.2295e+00	k_t = [2, 3, 2, 3]
Generation 390:	Best = 4.145530e+07	FES = 39100	Diversity = 4.3192e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 3.890626e+07	FES = 40100	Diversity = 4.6082e+00	k_t = [2, 2, 2, 4]
Generation 410:	Best = 3.746031e+07	FES = 41100	Diversity = 4.6803e+00	k_t = [1, 1, 1, 1]
Generation 420:	Best = 3.519931e+07	FES = 42100	Diversity = 4.6371e+00	k_t = [2, 3, 3, 4]
Generation 430:	Best = 3.338851e+07	FES = 43100	Diversity = 4.2650e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 3.213734e+07	FES = 44100	Diversity = 4.2568e+00	k_t = [4, 3, 2, 2]
Generation 450:	Best = 3.102349e+07	FES = 45100	Diversity = 4.4248e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 2.971875e+07	FES = 46100	Diversity = 4.4646e+00	k_t = [2, 2, 2, 4]
Generation 470:	Best = 2.914319e+07	FES = 47100	Diversity = 4.7813e+00	k_t = [1, 1, 1, 1]
Generation 480:	Best = 2.814480e+07	FES = 48100	Diversity = 4.9509e+00	k_t = [2, 4, 3, 2]
Generation 490:	Best = 2.753256e+07	FES = 49100	Diversity = 5.0383e+00	k_t = [1, 1, 1, 1]

Generation 500:	Best = 2.677936e+07	FES = 50100	Diversity = 5.3339e+00	k_t
= [2, 2, 4, 2]				
Generation 510:	Best = 2.628748e+07	FES = 51100	Diversity = 6.0215e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 2.560309e+07	FES = 52100	Diversity = 6.2961e+00	k_t
= [3, 4, 3, 3]				
Generation 530:	Best = 2.480957e+07	FES = 53100	Diversity = 6.4256e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 2.410393e+07	FES = 54100	Diversity = 6.8415e+00	k_t
= [4, 2, 3, 2]				
Generation 550:	Best = 2.367941e+07	FES = 55100	Diversity = 7.4086e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 2.323584e+07	FES = 56100	Diversity = 7.0600e+00	k_t
= [2, 2, 2, 3]				
Generation 570:	Best = 2.289631e+07	FES = 57100	Diversity = 7.1869e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 2.247008e+07	FES = 58100	Diversity = 7.7991e+00	k_t
= [4, 4, 3, 2]				
Generation 590:	Best = 2.214036e+07	FES = 59100	Diversity = 7.8132e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 2.170301e+07	FES = 60100	Diversity = 7.4930e+00	k_t
= [3, 2, 4, 3]				
Generation 610:	Best = 2.139385e+07	FES = 61100	Diversity = 7.9794e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 2.097467e+07	FES = 62100	Diversity = 8.0427e+00	k_t
= [4, 3, 3, 4]				
Generation 630:	Best = 2.064611e+07	FES = 63100	Diversity = 8.5803e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 2.038419e+07	FES = 64100	Diversity = 7.9924e+00	k_t
= [4, 4, 4, 3]				
Generation 650:	Best = 2.007158e+07	FES = 65100	Diversity = 7.6446e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 1.957809e+07	FES = 66100	Diversity = 7.6079e+00	k_t
= [2, 2, 3, 3]				
Generation 670:	Best = 1.935728e+07	FES = 67100	Diversity = 7.8587e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 1.902209e+07	FES = 68100	Diversity = 7.6187e+00	k_t
= [2, 3, 2, 3]				
Generation 690:	Best = 1.877227e+07	FES = 69100	Diversity = 7.8465e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 1.852664e+07	FES = 70100	Diversity = 7.3242e+00	k_t
= [3, 2, 3, 4]				
Generation 710:	Best = 1.822334e+07	FES = 71100	Diversity = 6.9308e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 1.807034e+07	FES = 72100	Diversity = 6.8541e+00	k_t
= [4, 3, 3, 3]				
Generation 730:	Best = 1.790485e+07	FES = 73100	Diversity = 6.6189e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 1.766609e+07	FES = 74100	Diversity = 6.9943e+00	k_t
= [2, 2, 3, 3]				
Generation 750:	Best = 1.752115e+07	FES = 75100	Diversity = 7.1294e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 1.735142e+07	FES = 76100	Diversity = 7.2675e+00	k_t
= [2, 2, 4, 3]				

Generation 770:	Best = 1.696973e+07	FES = 77100	Diversity = 7.5298e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.687269e+07	FES = 78100	Diversity = 7.9015e+00	k_t
= [2, 4, 2, 2]				
Generation 790:	Best = 1.666032e+07	FES = 79100	Diversity = 7.4741e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.658267e+07	FES = 80100	Diversity = 7.1244e+00	k_t
= [3, 2, 2, 4]				
Generation 810:	Best = 1.633937e+07	FES = 81100	Diversity = 6.5832e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.605409e+07	FES = 82100	Diversity = 6.7545e+00	k_t
= [3, 4, 2, 4]				
Generation 830:	Best = 1.599296e+07	FES = 83100	Diversity = 7.3691e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.572070e+07	FES = 84100	Diversity = 7.2854e+00	k_t
= [4, 4, 3, 3]				
Generation 850:	Best = 1.563418e+07	FES = 85100	Diversity = 7.3557e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.551298e+07	FES = 86100	Diversity = 7.4301e+00	k_t
= [2, 4, 3, 3]				
Generation 870:	Best = 1.540785e+07	FES = 87100	Diversity = 7.3049e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.525332e+07	FES = 88100	Diversity = 6.6466e+00	k_t
= [4, 4, 3, 2]				
Generation 890:	Best = 1.507499e+07	FES = 89100	Diversity = 7.2098e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.493586e+07	FES = 90100	Diversity = 7.5572e+00	k_t
= [4, 2, 2, 4]				
Generation 910:	Best = 1.483155e+07	FES = 91100	Diversity = 7.2028e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.471086e+07	FES = 92100	Diversity = 6.7215e+00	k_t
= [4, 4, 4, 3]				
Generation 930:	Best = 1.457719e+07	FES = 93100	Diversity = 6.6209e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.446698e+07	FES = 94100	Diversity = 6.4822e+00	k_t
= [2, 2, 4, 3]				
Generation 950:	Best = 1.433944e+07	FES = 95100	Diversity = 6.4785e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.426822e+07	FES = 96100	Diversity = 6.4573e+00	k_t
= [3, 3, 4, 4]				
Generation 970:	Best = 1.415771e+07	FES = 97100	Diversity = 6.6146e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.406906e+07	FES = 98100	Diversity = 6.7649e+00	k_t
= [4, 4, 2, 2]				
Generation 990:	Best = 1.396425e+07	FES = 99100	Diversity = 6.8913e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 1.384014e+07	FES = 100100	Diversity = 7.1010e+00	k_t
= [3, 4, 3, 3]				
Generation 1010:	Best = 1.375672e+07	FES = 101100	Diversity = 7.2622e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.364406e+07	FES = 102100	Diversity = 7.5125e+00	k_t
= [4, 4, 4, 2]				
Generation 1030:	Best = 1.347065e+07	FES = 103100	Diversity = 7.8431e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.328763e+07 = [3, 4, 3, 3]	FES = 104100	Diversity = 8.1899e+00	k_t
Generation 1050: Best = 1.317673e+07 = [1, 1, 1, 1]	FES = 105100	Diversity = 8.2992e+00	k_t
Generation 1060: Best = 1.306364e+07 = [4, 4, 4, 3]	FES = 106100	Diversity = 8.1832e+00	k_t
Generation 1070: Best = 1.296708e+07 = [1, 1, 1, 1]	FES = 107100	Diversity = 8.0054e+00	k_t
Generation 1080: Best = 1.287102e+07 = [2, 2, 3, 3]	FES = 108100	Diversity = 7.7310e+00	k_t
Generation 1090: Best = 1.277856e+07 = [1, 1, 1, 1]	FES = 109100	Diversity = 7.8662e+00	k_t
Generation 1100: Best = 1.267296e+07 = [4, 3, 2, 2]	FES = 110100	Diversity = 7.5380e+00	k_t
Generation 1110: Best = 1.258855e+07 = [1, 1, 1, 1]	FES = 111100	Diversity = 7.4607e+00	k_t
Generation 1120: Best = 1.243199e+07 = [3, 2, 3, 2]	FES = 112100	Diversity = 7.3456e+00	k_t
Generation 1130: Best = 1.232734e+07 = [1, 1, 1, 1]	FES = 113100	Diversity = 7.3494e+00	k_t
Generation 1140: Best = 1.226602e+07 = [3, 3, 4, 2]	FES = 114100	Diversity = 7.0979e+00	k_t
Generation 1150: Best = 1.216484e+07 = [1, 1, 1, 1]	FES = 115100	Diversity = 7.5276e+00	k_t
Generation 1160: Best = 1.208434e+07 = [2, 3, 2, 3]	FES = 116100	Diversity = 7.7897e+00	k_t
Generation 1170: Best = 1.196289e+07 = [1, 1, 1, 1]	FES = 117100	Diversity = 7.8358e+00	k_t
Generation 1180: Best = 1.186798e+07 = [2, 4, 4, 2]	FES = 118100	Diversity = 7.8561e+00	k_t
Generation 1190: Best = 1.171479e+07 = [1, 1, 1, 1]	FES = 119100	Diversity = 8.2718e+00	k_t
Generation 1200: Best = 1.157816e+07 = [3, 3, 4, 3]	FES = 120100	Diversity = 8.7918e+00	k_t
Generation 1210: Best = 1.152639e+07 = [1, 1, 1, 1]	FES = 121100	Diversity = 8.6549e+00	k_t
Generation 1220: Best = 1.130633e+07 = [4, 2, 4, 3]	FES = 122100	Diversity = 9.2299e+00	k_t
Generation 1230: Best = 1.124016e+07 = [1, 1, 1, 1]	FES = 123100	Diversity = 9.4112e+00	k_t
Generation 1240: Best = 1.120163e+07 = [2, 4, 4, 3]	FES = 124100	Diversity = 9.1453e+00	k_t
Generation 1250: Best = 1.107004e+07 = [1, 1, 1, 1]	FES = 125100	Diversity = 8.5587e+00	k_t
Generation 1260: Best = 1.099348e+07 = [4, 4, 4, 3]	FES = 126100	Diversity = 8.6805e+00	k_t
Generation 1270: Best = 1.082579e+07 = [1, 1, 1, 1]	FES = 127100	Diversity = 8.0685e+00	k_t
Generation 1280: Best = 1.075696e+07 = [3, 2, 4, 4]	FES = 128100	Diversity = 7.8219e+00	k_t
Generation 1290: Best = 1.063336e+07 = [1, 1, 1, 1]	FES = 129100	Diversity = 7.6322e+00	k_t
Generation 1300: Best = 1.055632e+07 = [4, 2, 2, 4]	FES = 130100	Diversity = 6.9757e+00	k_t

Generation 1310: Best = 1.049577e+07 = [1, 1, 1, 1]	FES = 131100	Diversity = 6.5061e+00	k_t
Generation 1320: Best = 1.034762e+07 = [4, 4, 3, 4]	FES = 132100	Diversity = 6.5080e+00	k_t
Generation 1330: Best = 1.028533e+07 = [1, 1, 1, 1]	FES = 133100	Diversity = 6.7937e+00	k_t
Generation 1340: Best = 1.020136e+07 = [3, 2, 4, 2]	FES = 134100	Diversity = 6.8616e+00	k_t
Generation 1350: Best = 1.015215e+07 = [1, 1, 1, 1]	FES = 135100	Diversity = 6.3836e+00	k_t
Generation 1360: Best = 1.006715e+07 = [2, 3, 3, 3]	FES = 136100	Diversity = 6.4610e+00	k_t
Generation 1370: Best = 9.968022e+06 = [1, 1, 1, 1]	FES = 137100	Diversity = 6.9714e+00	k_t
Generation 1380: Best = 9.865539e+06 = [4, 3, 4, 3]	FES = 138100	Diversity = 6.9740e+00	k_t
Generation 1390: Best = 9.778553e+06 = [1, 1, 1, 1]	FES = 139100	Diversity = 7.3940e+00	k_t
Generation 1400: Best = 9.702014e+06 = [3, 3, 4, 3]	FES = 140100	Diversity = 7.3645e+00	k_t
Generation 1410: Best = 9.622830e+06 = [1, 1, 1, 1]	FES = 141100	Diversity = 7.7352e+00	k_t
Generation 1420: Best = 9.441556e+06 = [4, 4, 4, 3]	FES = 142100	Diversity = 7.6779e+00	k_t
Generation 1430: Best = 9.441556e+06 = [1, 1, 1, 1]	FES = 143100	Diversity = 6.9796e+00	k_t
Generation 1440: Best = 9.374857e+06 = [3, 4, 2, 2]	FES = 144100	Diversity = 7.0720e+00	k_t
Generation 1450: Best = 9.278890e+06 = [1, 1, 1, 1]	FES = 145100	Diversity = 7.2674e+00	k_t
Generation 1460: Best = 9.235092e+06 = [3, 3, 2, 3]	FES = 146100	Diversity = 6.5963e+00	k_t
Generation 1470: Best = 9.151581e+06 = [1, 1, 1, 1]	FES = 147100	Diversity = 6.0767e+00	k_t
Generation 1480: Best = 9.106251e+06 = [2, 2, 4, 4]	FES = 148100	Diversity = 5.9393e+00	k_t
Generation 1490: Best = 9.067138e+06 = [1, 1, 1, 1]	FES = 149100	Diversity = 6.4282e+00	k_t
Generation 1500: Best = 8.976483e+06 = [3, 4, 4, 2]	FES = 150100	Diversity = 6.5572e+00	k_t
Generation 1510: Best = 8.936203e+06 = [1, 1, 1, 1]	FES = 151100	Diversity = 6.3116e+00	k_t
Generation 1520: Best = 8.854491e+06 = [3, 3, 2, 4]	FES = 152100	Diversity = 6.8641e+00	k_t
Generation 1530: Best = 8.772532e+06 = [1, 1, 1, 1]	FES = 153100	Diversity = 6.9447e+00	k_t
Generation 1540: Best = 8.743772e+06 = [3, 4, 4, 4]	FES = 154100	Diversity = 6.4611e+00	k_t
Generation 1550: Best = 8.657831e+06 = [1, 1, 1, 1]	FES = 155100	Diversity = 6.6342e+00	k_t
Generation 1560: Best = 8.587847e+06 = [4, 2, 4, 2]	FES = 156100	Diversity = 6.6163e+00	k_t
Generation 1570: Best = 8.528808e+06 = [1, 1, 1, 1]	FES = 157100	Diversity = 6.4483e+00	k_t

Generation 1580: Best = 8.432423e+06 = [4, 3, 2, 3]	FES = 158100	Diversity = 6.8085e+00	k_t
Generation 1590: Best = 8.395978e+06 = [1, 1, 1, 1]	FES = 159100	Diversity = 6.3615e+00	k_t
Generation 1600: Best = 8.346825e+06 = [3, 3, 3, 4]	FES = 160100	Diversity = 6.3297e+00	k_t
Generation 1610: Best = 8.233625e+06 = [1, 1, 1, 1]	FES = 161100	Diversity = 6.4454e+00	k_t
Generation 1620: Best = 8.180362e+06 = [2, 3, 2, 3]	FES = 162100	Diversity = 6.6163e+00	k_t
Generation 1630: Best = 8.122556e+06 = [1, 1, 1, 1]	FES = 163100	Diversity = 6.7695e+00	k_t
Generation 1640: Best = 7.995104e+06 = [4, 4, 3, 4]	FES = 164100	Diversity = 6.4333e+00	k_t
Generation 1650: Best = 7.954995e+06 = [1, 1, 1, 1]	FES = 165100	Diversity = 6.9967e+00	k_t
Generation 1660: Best = 7.897094e+06 = [2, 3, 3, 4]	FES = 166100	Diversity = 6.8874e+00	k_t
Generation 1670: Best = 7.828496e+06 = [1, 1, 1, 1]	FES = 167100	Diversity = 7.0324e+00	k_t
Generation 1680: Best = 7.751990e+06 = [3, 3, 2, 2]	FES = 168100	Diversity = 7.3428e+00	k_t
Generation 1690: Best = 7.640891e+06 = [1, 1, 1, 1]	FES = 169100	Diversity = 7.4730e+00	k_t
Generation 1700: Best = 7.595476e+06 = [4, 2, 4, 2]	FES = 170100	Diversity = 7.5202e+00	k_t
Generation 1710: Best = 7.541629e+06 = [1, 1, 1, 1]	FES = 171100	Diversity = 7.0771e+00	k_t
Generation 1720: Best = 7.485483e+06 = [4, 4, 2, 3]	FES = 172100	Diversity = 6.7907e+00	k_t
Generation 1730: Best = 7.425682e+06 = [1, 1, 1, 1]	FES = 173100	Diversity = 6.7375e+00	k_t
Generation 1740: Best = 7.364550e+06 = [4, 3, 3, 2]	FES = 174100	Diversity = 6.2820e+00	k_t
Generation 1750: Best = 7.336513e+06 = [1, 1, 1, 1]	FES = 175100	Diversity = 6.4551e+00	k_t
Generation 1760: Best = 7.243964e+06 = [3, 4, 4, 4]	FES = 176100	Diversity = 6.5513e+00	k_t
Generation 1770: Best = 7.206421e+06 = [1, 1, 1, 1]	FES = 177100	Diversity = 6.7385e+00	k_t
Generation 1780: Best = 7.156187e+06 = [3, 2, 2, 4]	FES = 178100	Diversity = 6.7738e+00	k_t
Generation 1790: Best = 7.111755e+06 = [1, 1, 1, 1]	FES = 179100	Diversity = 6.1207e+00	k_t
Generation 1800: Best = 7.050559e+06 = [4, 2, 2, 4]	FES = 180100	Diversity = 6.1984e+00	k_t
Generation 1810: Best = 7.013384e+06 = [1, 1, 1, 1]	FES = 181100	Diversity = 5.8425e+00	k_t
Generation 1820: Best = 6.941031e+06 = [3, 3, 4, 4]	FES = 182100	Diversity = 5.4425e+00	k_t
Generation 1830: Best = 6.841280e+06 = [1, 1, 1, 1]	FES = 183100	Diversity = 5.2624e+00	k_t
Generation 1840: Best = 6.820009e+06 = [4, 4, 4, 3]	FES = 184100	Diversity = 5.0795e+00	k_t

Generation 1850: Best = 6.784494e+06 = [1, 1, 1, 1]	FES = 185100	Diversity = 4.9133e+00	k_t
Generation 1860: Best = 6.728281e+06 = [3, 3, 4, 3]	FES = 186100	Diversity = 5.0965e+00	k_t
Generation 1870: Best = 6.693413e+06 = [1, 1, 1, 1]	FES = 187100	Diversity = 5.1815e+00	k_t
Generation 1880: Best = 6.638584e+06 = [2, 2, 4, 4]	FES = 188100	Diversity = 5.4024e+00	k_t
Generation 1890: Best = 6.612015e+06 = [1, 1, 1, 1]	FES = 189100	Diversity = 5.8377e+00	k_t
Generation 1900: Best = 6.576345e+06 = [2, 2, 3, 3]	FES = 190100	Diversity = 5.6219e+00	k_t
Generation 1910: Best = 6.541905e+06 = [1, 1, 1, 1]	FES = 191100	Diversity = 5.4909e+00	k_t
Generation 1920: Best = 6.487882e+06 = [4, 2, 4, 2]	FES = 192100	Diversity = 5.5492e+00	k_t
Generation 1930: Best = 6.451186e+06 = [1, 1, 1, 1]	FES = 193100	Diversity = 5.9053e+00	k_t
Generation 1940: Best = 6.418490e+06 = [4, 3, 2, 3]	FES = 194100	Diversity = 5.8656e+00	k_t
Generation 1950: Best = 6.351994e+06 = [1, 1, 1, 1]	FES = 195100	Diversity = 5.5629e+00	k_t
Generation 1960: Best = 6.329050e+06 = [2, 2, 4, 4]	FES = 196100	Diversity = 5.6766e+00	k_t
Generation 1970: Best = 6.287529e+06 = [1, 1, 1, 1]	FES = 197100	Diversity = 5.3955e+00	k_t
Generation 1980: Best = 6.246173e+06 = [4, 2, 2, 2]	FES = 198100	Diversity = 5.2641e+00	k_t
Generation 1990: Best = 6.202710e+06 = [1, 1, 1, 1]	FES = 199100	Diversity = 5.0751e+00	k_t
Generation 2000: Best = 6.170560e+06 = [3, 2, 3, 4]	FES = 200100	Diversity = 5.3358e+00	k_t
Generation 2010: Best = 6.126710e+06 = [1, 1, 1, 1]	FES = 201100	Diversity = 5.4749e+00	k_t
Generation 2020: Best = 6.106288e+06 = [4, 2, 2, 2]	FES = 202100	Diversity = 5.0558e+00	k_t
Generation 2030: Best = 6.077429e+06 = [1, 1, 1, 1]	FES = 203100	Diversity = 5.0996e+00	k_t
Generation 2040: Best = 6.038453e+06 = [3, 3, 3, 3]	FES = 204100	Diversity = 5.2190e+00	k_t
Generation 2050: Best = 6.010293e+06 = [1, 1, 1, 1]	FES = 205100	Diversity = 4.9755e+00	k_t
Generation 2060: Best = 5.969232e+06 = [2, 4, 3, 4]	FES = 206100	Diversity = 4.7302e+00	k_t
Generation 2070: Best = 5.923964e+06 = [1, 1, 1, 1]	FES = 207100	Diversity = 4.6847e+00	k_t
Generation 2080: Best = 5.873114e+06 = [2, 3, 2, 4]	FES = 208100	Diversity = 5.0126e+00	k_t
Generation 2090: Best = 5.819384e+06 = [1, 1, 1, 1]	FES = 209100	Diversity = 5.5141e+00	k_t
Generation 2100: Best = 5.773651e+06 = [3, 4, 2, 2]	FES = 210100	Diversity = 6.0760e+00	k_t
Generation 2110: Best = 5.746852e+06 = [1, 1, 1, 1]	FES = 211100	Diversity = 6.4911e+00	k_t

Generation 2120: Best = 5.709444e+06 = [3, 4, 4, 2]	FES = 212100	Diversity = 6.3782e+00	k_t
Generation 2130: Best = 5.681285e+06 = [1, 1, 1, 1]	FES = 213100	Diversity = 5.6453e+00	k_t
Generation 2140: Best = 5.638475e+06 = [2, 2, 2, 4]	FES = 214100	Diversity = 5.3031e+00	k_t
Generation 2150: Best = 5.607352e+06 = [1, 1, 1, 1]	FES = 215100	Diversity = 5.3336e+00	k_t
Generation 2160: Best = 5.570524e+06 = [3, 4, 4, 3]	FES = 216100	Diversity = 5.7106e+00	k_t
Generation 2170: Best = 5.542129e+06 = [1, 1, 1, 1]	FES = 217100	Diversity = 5.7117e+00	k_t
Generation 2180: Best = 5.523939e+06 = [3, 4, 3, 3]	FES = 218100	Diversity = 6.2494e+00	k_t
Generation 2190: Best = 5.480889e+06 = [1, 1, 1, 1]	FES = 219100	Diversity = 6.2879e+00	k_t
Generation 2200: Best = 5.451834e+06 = [3, 4, 2, 2]	FES = 220100	Diversity = 6.7315e+00	k_t
Generation 2210: Best = 5.433186e+06 = [1, 1, 1, 1]	FES = 221100	Diversity = 6.6057e+00	k_t
Generation 2220: Best = 5.411586e+06 = [2, 4, 3, 4]	FES = 222100	Diversity = 6.2845e+00	k_t
Generation 2230: Best = 5.384278e+06 = [1, 1, 1, 1]	FES = 223100	Diversity = 5.8554e+00	k_t
Generation 2240: Best = 5.355577e+06 = [3, 2, 3, 2]	FES = 224100	Diversity = 5.8929e+00	k_t
Generation 2250: Best = 5.331507e+06 = [1, 1, 1, 1]	FES = 225100	Diversity = 5.7804e+00	k_t
Generation 2260: Best = 5.291997e+06 = [3, 2, 2, 2]	FES = 226100	Diversity = 6.1903e+00	k_t
Generation 2270: Best = 5.261628e+06 = [1, 1, 1, 1]	FES = 227100	Diversity = 6.0573e+00	k_t
Generation 2280: Best = 5.230721e+06 = [2, 3, 4, 3]	FES = 228100	Diversity = 6.2990e+00	k_t
Generation 2290: Best = 5.207291e+06 = [1, 1, 1, 1]	FES = 229100	Diversity = 6.3517e+00	k_t
Generation 2300: Best = 5.163076e+06 = [2, 2, 2, 2]	FES = 230100	Diversity = 6.2342e+00	k_t
Generation 2310: Best = 5.121463e+06 = [1, 1, 1, 1]	FES = 231100	Diversity = 6.5293e+00	k_t
Generation 2320: Best = 5.097985e+06 = [4, 2, 3, 4]	FES = 232100	Diversity = 6.6175e+00	k_t
Generation 2330: Best = 5.050699e+06 = [1, 1, 1, 1]	FES = 233100	Diversity = 6.7351e+00	k_t
Generation 2340: Best = 5.010908e+06 = [3, 2, 3, 3]	FES = 234100	Diversity = 6.2557e+00	k_t
Generation 2350: Best = 4.985624e+06 = [1, 1, 1, 1]	FES = 235100	Diversity = 6.4903e+00	k_t
Generation 2360: Best = 4.945574e+06 = [3, 4, 3, 3]	FES = 236100	Diversity = 5.8997e+00	k_t
Generation 2370: Best = 4.903226e+06 = [1, 1, 1, 1]	FES = 237100	Diversity = 5.8501e+00	k_t
Generation 2380: Best = 4.884219e+06 = [2, 3, 4, 2]	FES = 238100	Diversity = 6.0072e+00	k_t

Generation 2390: Best = 4.844183e+06 = [1, 1, 1, 1]	FES = 239100	Diversity = 6.1663e+00	k_t
Generation 2400: Best = 4.819867e+06 = [2, 3, 3, 2]	FES = 240100	Diversity = 6.3591e+00	k_t
Generation 2410: Best = 4.794008e+06 = [1, 1, 1, 1]	FES = 241100	Diversity = 6.1400e+00	k_t
Generation 2420: Best = 4.744627e+06 = [4, 4, 2, 2]	FES = 242100	Diversity = 6.5425e+00	k_t
Generation 2430: Best = 4.724362e+06 = [1, 1, 1, 1]	FES = 243100	Diversity = 6.0969e+00	k_t
Generation 2440: Best = 4.690872e+06 = [3, 4, 2, 4]	FES = 244100	Diversity = 5.7756e+00	k_t
Generation 2450: Best = 4.658118e+06 = [1, 1, 1, 1]	FES = 245100	Diversity = 5.4660e+00	k_t
Generation 2460: Best = 4.626703e+06 = [3, 2, 3, 4]	FES = 246100	Diversity = 5.5849e+00	k_t
Generation 2470: Best = 4.597633e+06 = [1, 1, 1, 1]	FES = 247100	Diversity = 5.8615e+00	k_t
Generation 2480: Best = 4.573448e+06 = [3, 3, 2, 4]	FES = 248100	Diversity = 5.8666e+00	k_t
Generation 2490: Best = 4.548859e+06 = [1, 1, 1, 1]	FES = 249100	Diversity = 5.5950e+00	k_t
Generation 2500: Best = 4.524880e+06 = [2, 4, 2, 4]	FES = 250100	Diversity = 5.5073e+00	k_t
Generation 2510: Best = 4.499492e+06 = [1, 1, 1, 1]	FES = 251100	Diversity = 6.1148e+00	k_t
Generation 2520: Best = 4.481837e+06 = [2, 2, 3, 2]	FES = 252100	Diversity = 5.9653e+00	k_t
Generation 2530: Best = 4.462131e+06 = [1, 1, 1, 1]	FES = 253100	Diversity = 5.6707e+00	k_t
Generation 2540: Best = 4.435825e+06 = [4, 4, 4, 4]	FES = 254100	Diversity = 5.6584e+00	k_t
Generation 2550: Best = 4.421814e+06 = [1, 1, 1, 1]	FES = 255100	Diversity = 5.5949e+00	k_t
Generation 2560: Best = 4.403772e+06 = [3, 4, 2, 4]	FES = 256100	Diversity = 6.1053e+00	k_t
Generation 2570: Best = 4.382800e+06 = [1, 1, 1, 1]	FES = 257100	Diversity = 5.9971e+00	k_t
Generation 2580: Best = 4.353796e+06 = [3, 4, 3, 4]	FES = 258100	Diversity = 5.7543e+00	k_t
Generation 2590: Best = 4.327908e+06 = [1, 1, 1, 1]	FES = 259100	Diversity = 5.4960e+00	k_t
Generation 2600: Best = 4.313710e+06 = [4, 4, 3, 4]	FES = 260100	Diversity = 5.5907e+00	k_t
Generation 2610: Best = 4.295054e+06 = [1, 1, 1, 1]	FES = 261100	Diversity = 5.8213e+00	k_t
Generation 2620: Best = 4.268425e+06 = [3, 3, 2, 4]	FES = 262100	Diversity = 5.7944e+00	k_t
Generation 2630: Best = 4.251096e+06 = [1, 1, 1, 1]	FES = 263100	Diversity = 6.0039e+00	k_t
Generation 2640: Best = 4.230337e+06 = [4, 4, 4, 3]	FES = 264100	Diversity = 6.5410e+00	k_t
Generation 2650: Best = 4.214007e+06 = [1, 1, 1, 1]	FES = 265100	Diversity = 6.5840e+00	k_t

Generation 2660: Best = 4.174640e+06 = [2, 3, 4, 3]	FES = 266100	Diversity = 7.3112e+00	k_t
Generation 2670: Best = 4.150728e+06 = [1, 1, 1, 1]	FES = 267100	Diversity = 7.2459e+00	k_t
Generation 2680: Best = 4.117304e+06 = [2, 2, 4, 4]	FES = 268100	Diversity = 7.0833e+00	k_t
Generation 2690: Best = 4.094936e+06 = [1, 1, 1, 1]	FES = 269100	Diversity = 7.4095e+00	k_t
Generation 2700: Best = 4.066114e+06 = [2, 3, 4, 2]	FES = 270100	Diversity = 7.2158e+00	k_t
Generation 2710: Best = 4.047379e+06 = [1, 1, 1, 1]	FES = 271100	Diversity = 7.3339e+00	k_t
Generation 2720: Best = 4.007699e+06 = [3, 3, 2, 2]	FES = 272100	Diversity = 6.9748e+00	k_t
Generation 2730: Best = 3.982320e+06 = [1, 1, 1, 1]	FES = 273100	Diversity = 6.8052e+00	k_t
Generation 2740: Best = 3.961822e+06 = [3, 4, 2, 4]	FES = 274100	Diversity = 6.2632e+00	k_t
Generation 2750: Best = 3.941166e+06 = [1, 1, 1, 1]	FES = 275100	Diversity = 5.6660e+00	k_t
Generation 2760: Best = 3.934440e+06 = [3, 3, 2, 3]	FES = 276100	Diversity = 5.5062e+00	k_t
Generation 2770: Best = 3.917450e+06 = [1, 1, 1, 1]	FES = 277100	Diversity = 5.6139e+00	k_t
Generation 2780: Best = 3.898277e+06 = [4, 4, 2, 2]	FES = 278100	Diversity = 5.2469e+00	k_t
Generation 2790: Best = 3.880356e+06 = [1, 1, 1, 1]	FES = 279100	Diversity = 4.8463e+00	k_t
Generation 2800: Best = 3.857836e+06 = [4, 3, 4, 2]	FES = 280100	Diversity = 5.2764e+00	k_t
Generation 2810: Best = 3.838234e+06 = [1, 1, 1, 1]	FES = 281100	Diversity = 5.4785e+00	k_t
Generation 2820: Best = 3.821261e+06 = [2, 4, 2, 3]	FES = 282100	Diversity = 5.3457e+00	k_t
Generation 2830: Best = 3.813988e+06 = [1, 1, 1, 1]	FES = 283100	Diversity = 5.2760e+00	k_t
Generation 2840: Best = 3.797490e+06 = [3, 2, 2, 3]	FES = 284100	Diversity = 5.6600e+00	k_t
Generation 2850: Best = 3.772330e+06 = [1, 1, 1, 1]	FES = 285100	Diversity = 5.6318e+00	k_t
Generation 2860: Best = 3.761622e+06 = [3, 3, 3, 3]	FES = 286100	Diversity = 5.7810e+00	k_t
Generation 2870: Best = 3.748851e+06 = [1, 1, 1, 1]	FES = 287100	Diversity = 5.7521e+00	k_t
Generation 2880: Best = 3.744764e+06 = [4, 4, 3, 3]	FES = 288100	Diversity = 5.3319e+00	k_t
Generation 2890: Best = 3.727702e+06 = [1, 1, 1, 1]	FES = 289100	Diversity = 5.1607e+00	k_t
Generation 2900: Best = 3.713840e+06 = [2, 4, 4, 3]	FES = 290100	Diversity = 4.9816e+00	k_t
Generation 2910: Best = 3.693376e+06 = [1, 1, 1, 1]	FES = 291100	Diversity = 5.0477e+00	k_t
Generation 2920: Best = 3.680301e+06 = [4, 4, 4, 3]	FES = 292100	Diversity = 5.2967e+00	k_t

Generation 2930: Best = 3.665019e+06 = [1, 1, 1, 1]	FES = 293100	Diversity = 5.3794e+00	k_t
Generation 2940: Best = 3.641164e+06 = [2, 3, 2, 2]	FES = 294100	Diversity = 5.2937e+00	k_t
Generation 2950: Best = 3.630167e+06 = [1, 1, 1, 1]	FES = 295100	Diversity = 5.2550e+00	k_t
Generation 2960: Best = 3.608521e+06 = [4, 4, 4, 3]	FES = 296100	Diversity = 5.0551e+00	k_t
Generation 2970: Best = 3.598187e+06 = [1, 1, 1, 1]	FES = 297100	Diversity = 5.2942e+00	k_t
Generation 2980: Best = 3.571962e+06 = [2, 2, 2, 4]	FES = 298100	Diversity = 5.3673e+00	k_t
Generation 2990: Best = 3.559318e+06 = [1, 1, 1, 1]	FES = 299100	Diversity = 5.3487e+00	k_t
Generation 3000: Best = 3.541065e+06 = [2, 3, 3, 3]	FES = 300100	Diversity = 5.2122e+00	k_t
Generation 3010: Best = 3.511881e+06 = [1, 1, 1, 1]	FES = 301100	Diversity = 4.7783e+00	k_t
Generation 3020: Best = 3.509768e+06 = [3, 2, 3, 2]	FES = 302100	Diversity = 4.6124e+00	k_t
Generation 3030: Best = 3.491342e+06 = [1, 1, 1, 1]	FES = 303100	Diversity = 4.5431e+00	k_t
Generation 3040: Best = 3.469205e+06 = [2, 3, 3, 4]	FES = 304100	Diversity = 4.6575e+00	k_t
Generation 3050: Best = 3.460779e+06 = [1, 1, 1, 1]	FES = 305100	Diversity = 4.7904e+00	k_t
Generation 3060: Best = 3.429644e+06 = [2, 3, 4, 3]	FES = 306100	Diversity = 4.8674e+00	k_t
Generation 3070: Best = 3.413011e+06 = [1, 1, 1, 1]	FES = 307100	Diversity = 5.4804e+00	k_t
Generation 3080: Best = 3.395875e+06 = [4, 4, 2, 2]	FES = 308100	Diversity = 5.4888e+00	k_t
Generation 3090: Best = 3.383261e+06 = [1, 1, 1, 1]	FES = 309100	Diversity = 5.3349e+00	k_t
Generation 3100: Best = 3.350223e+06 = [3, 2, 4, 4]	FES = 310100	Diversity = 5.1366e+00	k_t
Generation 3110: Best = 3.335051e+06 = [1, 1, 1, 1]	FES = 311100	Diversity = 4.8687e+00	k_t
Generation 3120: Best = 3.323554e+06 = [2, 2, 3, 2]	FES = 312100	Diversity = 4.4135e+00	k_t
Generation 3130: Best = 3.303451e+06 = [1, 1, 1, 1]	FES = 313100	Diversity = 4.1926e+00	k_t
Generation 3140: Best = 3.285409e+06 = [4, 3, 2, 4]	FES = 314100	Diversity = 4.1941e+00	k_t
Generation 3150: Best = 3.275347e+06 = [1, 1, 1, 1]	FES = 315100	Diversity = 3.9507e+00	k_t
Generation 3160: Best = 3.263776e+06 = [2, 2, 3, 4]	FES = 316100	Diversity = 3.7063e+00	k_t
Generation 3170: Best = 3.234053e+06 = [1, 1, 1, 1]	FES = 317100	Diversity = 3.8786e+00	k_t
Generation 3180: Best = 3.224473e+06 = [2, 4, 3, 3]	FES = 318100	Diversity = 3.9447e+00	k_t
Generation 3190: Best = 3.211884e+06 = [1, 1, 1, 1]	FES = 319100	Diversity = 3.9768e+00	k_t

Generation 3200: Best = 3.192763e+06 = [3, 2, 3, 4]	FES = 320100	Diversity = 3.9721e+00	k_t
Generation 3210: Best = 3.186389e+06 = [1, 1, 1, 1]	FES = 321100	Diversity = 4.0019e+00	k_t
Generation 3220: Best = 3.165791e+06 = [3, 2, 4, 4]	FES = 322100	Diversity = 4.2195e+00	k_t
Generation 3230: Best = 3.143342e+06 = [1, 1, 1, 1]	FES = 323100	Diversity = 4.5329e+00	k_t
Generation 3240: Best = 3.136651e+06 = [2, 2, 4, 2]	FES = 324100	Diversity = 4.0949e+00	k_t
Generation 3250: Best = 3.116515e+06 = [1, 1, 1, 1]	FES = 325100	Diversity = 4.1448e+00	k_t
Generation 3260: Best = 3.095868e+06 = [2, 3, 2, 2]	FES = 326100	Diversity = 4.3523e+00	k_t
Generation 3270: Best = 3.070882e+06 = [1, 1, 1, 1]	FES = 327100	Diversity = 4.4046e+00	k_t
Generation 3280: Best = 3.066346e+06 = [4, 2, 3, 3]	FES = 328100	Diversity = 4.2940e+00	k_t
Generation 3290: Best = 3.058628e+06 = [1, 1, 1, 1]	FES = 329100	Diversity = 4.1182e+00	k_t
Generation 3300: Best = 3.042305e+06 = [3, 3, 4, 3]	FES = 330100	Diversity = 3.9316e+00	k_t
Generation 3310: Best = 3.028801e+06 = [1, 1, 1, 1]	FES = 331100	Diversity = 3.9571e+00	k_t
Generation 3320: Best = 3.017943e+06 = [4, 2, 2, 4]	FES = 332100	Diversity = 3.8583e+00	k_t
Generation 3330: Best = 3.000149e+06 = [1, 1, 1, 1]	FES = 333100	Diversity = 3.6270e+00	k_t
Generation 3340: Best = 2.984196e+06 = [3, 4, 2, 3]	FES = 334100	Diversity = 3.9996e+00	k_t
Generation 3350: Best = 2.975632e+06 = [1, 1, 1, 1]	FES = 335100	Diversity = 3.7835e+00	k_t
Generation 3360: Best = 2.969892e+06 = [2, 3, 2, 2]	FES = 336100	Diversity = 3.7898e+00	k_t
Generation 3370: Best = 2.955943e+06 = [1, 1, 1, 1]	FES = 337100	Diversity = 3.6890e+00	k_t
Generation 3380: Best = 2.949505e+06 = [4, 4, 2, 3]	FES = 338100	Diversity = 3.6251e+00	k_t
Generation 3390: Best = 2.932385e+06 = [1, 1, 1, 1]	FES = 339100	Diversity = 3.5091e+00	k_t
Generation 3400: Best = 2.915821e+06 = [4, 2, 2, 2]	FES = 340100	Diversity = 3.6672e+00	k_t
Generation 3410: Best = 2.908305e+06 = [1, 1, 1, 1]	FES = 341100	Diversity = 3.6572e+00	k_t
Generation 3420: Best = 2.895995e+06 = [2, 3, 2, 3]	FES = 342100	Diversity = 3.8371e+00	k_t
Generation 3430: Best = 2.884486e+06 = [1, 1, 1, 1]	FES = 343100	Diversity = 3.8962e+00	k_t
Generation 3440: Best = 2.881127e+06 = [2, 4, 3, 2]	FES = 344100	Diversity = 3.9807e+00	k_t
Generation 3450: Best = 2.874989e+06 = [1, 1, 1, 1]	FES = 345100	Diversity = 3.7005e+00	k_t
Generation 3460: Best = 2.861097e+06 = [4, 4, 4, 3]	FES = 346100	Diversity = 3.8296e+00	k_t

Generation 3470: Best = 2.856522e+06 = [1, 1, 1, 1]	FES = 347100	Diversity = 3.9667e+00	k_t
Generation 3480: Best = 2.837692e+06 = [4, 4, 3, 2]	FES = 348100	Diversity = 3.8768e+00	k_t
Generation 3490: Best = 2.831595e+06 = [1, 1, 1, 1]	FES = 349100	Diversity = 3.7053e+00	k_t
Generation 3500: Best = 2.827800e+06 = [4, 2, 2, 2]	FES = 350100	Diversity = 3.8509e+00	k_t
Generation 3510: Best = 2.823415e+06 = [1, 1, 1, 1]	FES = 351100	Diversity = 3.9255e+00	k_t
Generation 3520: Best = 2.816276e+06 = [4, 2, 3, 2]	FES = 352100	Diversity = 3.6852e+00	k_t
Generation 3530: Best = 2.809747e+06 = [1, 1, 1, 1]	FES = 353100	Diversity = 3.6499e+00	k_t
Generation 3540: Best = 2.798686e+06 = [3, 4, 2, 2]	FES = 354100	Diversity = 3.5568e+00	k_t
Generation 3550: Best = 2.788626e+06 = [1, 1, 1, 1]	FES = 355100	Diversity = 4.0341e+00	k_t
Generation 3560: Best = 2.781291e+06 = [3, 2, 4, 4]	FES = 356100	Diversity = 4.0870e+00	k_t
Generation 3570: Best = 2.770887e+06 = [1, 1, 1, 1]	FES = 357100	Diversity = 4.3154e+00	k_t
Generation 3580: Best = 2.767493e+06 = [3, 3, 2, 4]	FES = 358100	Diversity = 4.5506e+00	k_t
Generation 3590: Best = 2.758836e+06 = [1, 1, 1, 1]	FES = 359100	Diversity = 4.5098e+00	k_t
Generation 3600: Best = 2.753887e+06 = [3, 4, 4, 4]	FES = 360100	Diversity = 4.6257e+00	k_t
Generation 3610: Best = 2.747994e+06 = [1, 1, 1, 1]	FES = 361100	Diversity = 4.5296e+00	k_t
Generation 3620: Best = 2.742029e+06 = [3, 4, 2, 2]	FES = 362100	Diversity = 4.5212e+00	k_t
Generation 3630: Best = 2.736784e+06 = [1, 1, 1, 1]	FES = 363100	Diversity = 4.2640e+00	k_t
Generation 3640: Best = 2.724044e+06 = [3, 4, 2, 3]	FES = 364100	Diversity = 3.8247e+00	k_t
Generation 3650: Best = 2.717980e+06 = [1, 1, 1, 1]	FES = 365100	Diversity = 3.7286e+00	k_t
Generation 3660: Best = 2.716358e+06 = [2, 2, 3, 4]	FES = 366100	Diversity = 3.7545e+00	k_t
Generation 3670: Best = 2.710129e+06 = [1, 1, 1, 1]	FES = 367100	Diversity = 3.9485e+00	k_t
Generation 3680: Best = 2.704044e+06 = [3, 4, 3, 3]	FES = 368100	Diversity = 3.5793e+00	k_t
Generation 3690: Best = 2.696799e+06 = [1, 1, 1, 1]	FES = 369100	Diversity = 3.3910e+00	k_t
Generation 3700: Best = 2.693512e+06 = [4, 3, 4, 4]	FES = 370100	Diversity = 3.1092e+00	k_t
Generation 3710: Best = 2.685344e+06 = [1, 1, 1, 1]	FES = 371100	Diversity = 3.1241e+00	k_t
Generation 3720: Best = 2.679150e+06 = [4, 4, 4, 4]	FES = 372100	Diversity = 3.0863e+00	k_t
Generation 3730: Best = 2.673522e+06 = [1, 1, 1, 1]	FES = 373100	Diversity = 3.0483e+00	k_t

Generation 3740: Best = 2.666356e+06 = [4, 4, 4, 3]	FES = 374100	Diversity = 3.0135e+00	k_t
Generation 3750: Best = 2.661188e+06 = [1, 1, 1, 1]	FES = 375100	Diversity = 3.3935e+00	k_t
Generation 3760: Best = 2.655302e+06 = [4, 2, 2, 3]	FES = 376100	Diversity = 3.4466e+00	k_t
Generation 3770: Best = 2.651565e+06 = [1, 1, 1, 1]	FES = 377100	Diversity = 3.5675e+00	k_t
Generation 3780: Best = 2.643594e+06 = [2, 3, 2, 3]	FES = 378100	Diversity = 3.8432e+00	k_t
Generation 3790: Best = 2.636384e+06 = [1, 1, 1, 1]	FES = 379100	Diversity = 4.0641e+00	k_t
Generation 3800: Best = 2.627833e+06 = [4, 2, 4, 3]	FES = 380100	Diversity = 3.9660e+00	k_t
Generation 3810: Best = 2.620251e+06 = [1, 1, 1, 1]	FES = 381100	Diversity = 3.8352e+00	k_t
Generation 3820: Best = 2.614856e+06 = [4, 3, 3, 4]	FES = 382100	Diversity = 3.7899e+00	k_t
Generation 3830: Best = 2.606169e+06 = [1, 1, 1, 1]	FES = 383100	Diversity = 3.7094e+00	k_t
Generation 3840: Best = 2.596533e+06 = [4, 3, 2, 4]	FES = 384100	Diversity = 3.9794e+00	k_t
Generation 3850: Best = 2.592675e+06 = [1, 1, 1, 1]	FES = 385100	Diversity = 3.9147e+00	k_t
Generation 3860: Best = 2.580410e+06 = [4, 2, 2, 2]	FES = 386100	Diversity = 4.0719e+00	k_t
Generation 3870: Best = 2.575061e+06 = [1, 1, 1, 1]	FES = 387100	Diversity = 4.1007e+00	k_t
Generation 3880: Best = 2.567361e+06 = [4, 4, 2, 2]	FES = 388100	Diversity = 3.8992e+00	k_t
Generation 3890: Best = 2.558591e+06 = [1, 1, 1, 1]	FES = 389100	Diversity = 4.0416e+00	k_t
Generation 3900: Best = 2.553686e+06 = [2, 2, 4, 2]	FES = 390100	Diversity = 3.9738e+00	k_t
Generation 3910: Best = 2.541878e+06 = [1, 1, 1, 1]	FES = 391100	Diversity = 3.9695e+00	k_t
Generation 3920: Best = 2.537404e+06 = [3, 2, 2, 4]	FES = 392100	Diversity = 3.9614e+00	k_t
Generation 3930: Best = 2.526902e+06 = [1, 1, 1, 1]	FES = 393100	Diversity = 3.7975e+00	k_t
Generation 3940: Best = 2.523928e+06 = [2, 4, 4, 2]	FES = 394100	Diversity = 3.7408e+00	k_t
Generation 3950: Best = 2.515664e+06 = [1, 1, 1, 1]	FES = 395100	Diversity = 3.5645e+00	k_t
Generation 3960: Best = 2.503857e+06 = [3, 3, 2, 4]	FES = 396100	Diversity = 3.4244e+00	k_t
Generation 3970: Best = 2.499557e+06 = [1, 1, 1, 1]	FES = 397100	Diversity = 3.3790e+00	k_t
Generation 3980: Best = 2.495372e+06 = [4, 4, 2, 2]	FES = 398100	Diversity = 3.3288e+00	k_t
Generation 3990: Best = 2.490099e+06 = [1, 1, 1, 1]	FES = 399100	Diversity = 3.5225e+00	k_t
Generation 4000: Best = 2.482011e+06 = [3, 3, 3, 2]	FES = 400100	Diversity = 3.6476e+00	k_t

Generation 4010: Best = 2.476018e+06 = [1, 1, 1, 1]	FES = 401100	Diversity = 3.7287e+00	k_t
Generation 4020: Best = 2.473520e+06 = [2, 3, 4, 3]	FES = 402100	Diversity = 3.6256e+00	k_t
Generation 4030: Best = 2.467315e+06 = [1, 1, 1, 1]	FES = 403100	Diversity = 3.7955e+00	k_t
Generation 4040: Best = 2.463933e+06 = [3, 2, 4, 4]	FES = 404100	Diversity = 3.7592e+00	k_t
Generation 4050: Best = 2.451970e+06 = [1, 1, 1, 1]	FES = 405100	Diversity = 3.8441e+00	k_t
Generation 4060: Best = 2.451970e+06 = [3, 2, 3, 2]	FES = 406100	Diversity = 4.4592e+00	k_t
Generation 4070: Best = 2.445896e+06 = [1, 1, 1, 1]	FES = 407100	Diversity = 4.3916e+00	k_t
Generation 4080: Best = 2.434928e+06 = [3, 3, 3, 2]	FES = 408100	Diversity = 4.1918e+00	k_t
Generation 4090: Best = 2.430586e+06 = [1, 1, 1, 1]	FES = 409100	Diversity = 4.0507e+00	k_t
Generation 4100: Best = 2.422192e+06 = [3, 2, 3, 2]	FES = 410100	Diversity = 4.0093e+00	k_t
Generation 4110: Best = 2.413846e+06 = [1, 1, 1, 1]	FES = 411100	Diversity = 4.0799e+00	k_t
Generation 4120: Best = 2.409471e+06 = [4, 4, 2, 3]	FES = 412100	Diversity = 4.1475e+00	k_t
Generation 4130: Best = 2.401225e+06 = [1, 1, 1, 1]	FES = 413100	Diversity = 4.4124e+00	k_t
Generation 4140: Best = 2.399821e+06 = [4, 3, 3, 3]	FES = 414100	Diversity = 4.1447e+00	k_t
Generation 4150: Best = 2.389437e+06 = [1, 1, 1, 1]	FES = 415100	Diversity = 3.8041e+00	k_t
Generation 4160: Best = 2.383228e+06 = [3, 4, 3, 2]	FES = 416100	Diversity = 3.7366e+00	k_t
Generation 4170: Best = 2.377727e+06 = [1, 1, 1, 1]	FES = 417100	Diversity = 3.6283e+00	k_t
Generation 4180: Best = 2.376043e+06 = [2, 4, 3, 3]	FES = 418100	Diversity = 3.6503e+00	k_t
Generation 4190: Best = 2.367055e+06 = [1, 1, 1, 1]	FES = 419100	Diversity = 3.6287e+00	k_t
Generation 4200: Best = 2.363321e+06 = [4, 4, 2, 3]	FES = 420100	Diversity = 3.8052e+00	k_t
Generation 4210: Best = 2.357949e+06 = [1, 1, 1, 1]	FES = 421100	Diversity = 3.8416e+00	k_t
Generation 4220: Best = 2.350559e+06 = [2, 4, 4, 4]	FES = 422100	Diversity = 3.9645e+00	k_t
Generation 4230: Best = 2.345801e+06 = [1, 1, 1, 1]	FES = 423100	Diversity = 4.0639e+00	k_t
Generation 4240: Best = 2.343258e+06 = [4, 4, 3, 3]	FES = 424100	Diversity = 3.8585e+00	k_t
Generation 4250: Best = 2.338189e+06 = [1, 1, 1, 1]	FES = 425100	Diversity = 3.6655e+00	k_t
Generation 4260: Best = 2.332047e+06 = [2, 4, 2, 3]	FES = 426100	Diversity = 3.7988e+00	k_t
Generation 4270: Best = 2.328164e+06 = [1, 1, 1, 1]	FES = 427100	Diversity = 3.7223e+00	k_t

Generation 4280: Best = 2.321163e+06 = [4, 2, 4, 4]	FES = 428100	Diversity = 3.9663e+00	k_t
Generation 4290: Best = 2.319937e+06 = [1, 1, 1, 1]	FES = 429100	Diversity = 3.7429e+00	k_t
Generation 4300: Best = 2.314509e+06 = [3, 2, 4, 3]	FES = 430100	Diversity = 3.9205e+00	k_t
Generation 4310: Best = 2.308254e+06 = [1, 1, 1, 1]	FES = 431100	Diversity = 3.8524e+00	k_t
Generation 4320: Best = 2.303250e+06 = [2, 4, 4, 2]	FES = 432100	Diversity = 4.0058e+00	k_t
Generation 4330: Best = 2.300116e+06 = [1, 1, 1, 1]	FES = 433100	Diversity = 3.9761e+00	k_t
Generation 4340: Best = 2.299027e+06 = [2, 2, 2, 2]	FES = 434100	Diversity = 3.6470e+00	k_t
Generation 4350: Best = 2.293532e+06 = [1, 1, 1, 1]	FES = 435100	Diversity = 3.6167e+00	k_t
Generation 4360: Best = 2.291402e+06 = [3, 4, 4, 3]	FES = 436100	Diversity = 3.4906e+00	k_t
Generation 4370: Best = 2.286696e+06 = [1, 1, 1, 1]	FES = 437100	Diversity = 3.7297e+00	k_t
Generation 4380: Best = 2.282886e+06 = [4, 2, 3, 4]	FES = 438100	Diversity = 3.6853e+00	k_t
Generation 4390: Best = 2.281128e+06 = [1, 1, 1, 1]	FES = 439100	Diversity = 3.5627e+00	k_t
Generation 4400: Best = 2.277521e+06 = [2, 2, 4, 4]	FES = 440100	Diversity = 3.6555e+00	k_t
Generation 4410: Best = 2.274543e+06 = [1, 1, 1, 1]	FES = 441100	Diversity = 3.7012e+00	k_t
Generation 4420: Best = 2.270928e+06 = [2, 4, 4, 3]	FES = 442100	Diversity = 3.9085e+00	k_t
Generation 4430: Best = 2.267906e+06 = [1, 1, 1, 1]	FES = 443100	Diversity = 3.8500e+00	k_t
Generation 4440: Best = 2.260680e+06 = [2, 2, 4, 3]	FES = 444100	Diversity = 3.7634e+00	k_t
Generation 4450: Best = 2.254896e+06 = [1, 1, 1, 1]	FES = 445100	Diversity = 3.9691e+00	k_t
Generation 4460: Best = 2.254224e+06 = [4, 3, 2, 3]	FES = 446100	Diversity = 4.1773e+00	k_t
Generation 4470: Best = 2.248731e+06 = [1, 1, 1, 1]	FES = 447100	Diversity = 4.1766e+00	k_t
Generation 4480: Best = 2.243809e+06 = [4, 2, 4, 3]	FES = 448100	Diversity = 3.8361e+00	k_t
Generation 4490: Best = 2.241761e+06 = [1, 1, 1, 1]	FES = 449100	Diversity = 3.9442e+00	k_t
Generation 4500: Best = 2.239678e+06 = [4, 4, 2, 2]	FES = 450100	Diversity = 4.2399e+00	k_t
Generation 4510: Best = 2.234705e+06 = [1, 1, 1, 1]	FES = 451100	Diversity = 4.1077e+00	k_t
Generation 4520: Best = 2.232369e+06 = [3, 4, 2, 4]	FES = 452100	Diversity = 3.8397e+00	k_t
Generation 4530: Best = 2.227629e+06 = [1, 1, 1, 1]	FES = 453100	Diversity = 3.9174e+00	k_t
Generation 4540: Best = 2.224751e+06 = [4, 2, 3, 4]	FES = 454100	Diversity = 3.7711e+00	k_t

Generation 4550: Best = 2.223715e+06	FES = 455100	Diversity = 3.8362e+00	k_t
= [1, 1, 1, 1]			
Generation 4560: Best = 2.219983e+06	FES = 456100	Diversity = 3.7797e+00	k_t
= [3, 3, 4, 2]			
Generation 4570: Best = 2.216804e+06	FES = 457100	Diversity = 3.9374e+00	k_t
= [1, 1, 1, 1]			
Generation 4580: Best = 2.211731e+06	FES = 458100	Diversity = 4.1822e+00	k_t
= [4, 4, 3, 2]			
Generation 4590: Best = 2.209379e+06	FES = 459100	Diversity = 4.2171e+00	k_t
= [1, 1, 1, 1]			
Generation 4600: Best = 2.205368e+06	FES = 460100	Diversity = 4.2944e+00	k_t
= [3, 4, 2, 2]			
Generation 4610: Best = 2.203877e+06	FES = 461100	Diversity = 4.3643e+00	k_t
= [1, 1, 1, 1]			
Generation 4620: Best = 2.198846e+06	FES = 462100	Diversity = 4.3890e+00	k_t
= [4, 2, 4, 4]			
Generation 4630: Best = 2.195376e+06	FES = 463100	Diversity = 4.4191e+00	k_t
= [1, 1, 1, 1]			
Generation 4640: Best = 2.189348e+06	FES = 464100	Diversity = 4.2648e+00	k_t
= [2, 3, 4, 4]			
Generation 4650: Best = 2.186213e+06	FES = 465100	Diversity = 4.3612e+00	k_t
= [1, 1, 1, 1]			
Generation 4660: Best = 2.181170e+06	FES = 466100	Diversity = 4.6377e+00	k_t
= [3, 2, 4, 2]			
Generation 4670: Best = 2.178605e+06	FES = 467100	Diversity = 4.6915e+00	k_t
= [1, 1, 1, 1]			
Generation 4680: Best = 2.171533e+06	FES = 468100	Diversity = 4.5509e+00	k_t
= [4, 3, 3, 3]			
Generation 4690: Best = 2.168812e+06	FES = 469100	Diversity = 4.2926e+00	k_t
= [1, 1, 1, 1]			
Generation 4700: Best = 2.162826e+06	FES = 470100	Diversity = 4.5269e+00	k_t
= [3, 4, 2, 3]			
Generation 4710: Best = 2.158938e+06	FES = 471100	Diversity = 4.5916e+00	k_t
= [1, 1, 1, 1]			
Generation 4720: Best = 2.157537e+06	FES = 472100	Diversity = 4.3308e+00	k_t
= [4, 2, 3, 4]			
Generation 4730: Best = 2.153934e+06	FES = 473100	Diversity = 4.4579e+00	k_t
= [1, 1, 1, 1]			
Generation 4740: Best = 2.151046e+06	FES = 474100	Diversity = 4.2079e+00	k_t
= [2, 4, 3, 2]			
Generation 4750: Best = 2.147765e+06	FES = 475100	Diversity = 4.1802e+00	k_t
= [1, 1, 1, 1]			
Generation 4760: Best = 2.144193e+06	FES = 476100	Diversity = 4.0772e+00	k_t
= [2, 3, 2, 2]			
Generation 4770: Best = 2.139863e+06	FES = 477100	Diversity = 3.8921e+00	k_t
= [1, 1, 1, 1]			
Generation 4780: Best = 2.132757e+06	FES = 478100	Diversity = 3.6610e+00	k_t
= [3, 4, 4, 3]			
Generation 4790: Best = 2.130560e+06	FES = 479100	Diversity = 3.6264e+00	k_t
= [1, 1, 1, 1]			
Generation 4800: Best = 2.127241e+06	FES = 480100	Diversity = 3.6703e+00	k_t
= [2, 2, 4, 4]			
Generation 4810: Best = 2.124670e+06	FES = 481100	Diversity = 3.5488e+00	k_t
= [1, 1, 1, 1]			

Generation 4820: Best = 2.122181e+06 = [3, 4, 2, 4]	FES = 482100	Diversity = 3.4406e+00	k_t
Generation 4830: Best = 2.118496e+06 = [1, 1, 1, 1]	FES = 483100	Diversity = 3.3919e+00	k_t
Generation 4840: Best = 2.116855e+06 = [4, 3, 4, 4]	FES = 484100	Diversity = 3.2644e+00	k_t
Generation 4850: Best = 2.114730e+06 = [1, 1, 1, 1]	FES = 485100	Diversity = 3.2726e+00	k_t
Generation 4860: Best = 2.111810e+06 = [4, 3, 3, 2]	FES = 486100	Diversity = 3.1754e+00	k_t
Generation 4870: Best = 2.109986e+06 = [1, 1, 1, 1]	FES = 487100	Diversity = 3.0842e+00	k_t
Generation 4880: Best = 2.106793e+06 = [4, 2, 4, 4]	FES = 488100	Diversity = 3.1809e+00	k_t
Generation 4890: Best = 2.104114e+06 = [1, 1, 1, 1]	FES = 489100	Diversity = 3.1744e+00	k_t
Generation 4900: Best = 2.101590e+06 = [4, 4, 3, 4]	FES = 490100	Diversity = 3.0978e+00	k_t
Generation 4910: Best = 2.100867e+06 = [1, 1, 1, 1]	FES = 491100	Diversity = 3.2541e+00	k_t
Generation 4920: Best = 2.095976e+06 = [2, 3, 2, 4]	FES = 492100	Diversity = 3.4350e+00	k_t
Generation 4930: Best = 2.095976e+06 = [1, 1, 1, 1]	FES = 493100	Diversity = 3.5461e+00	k_t
Generation 4940: Best = 2.093166e+06 = [4, 4, 2, 3]	FES = 494100	Diversity = 3.2674e+00	k_t
Generation 4950: Best = 2.088414e+06 = [1, 1, 1, 1]	FES = 495100	Diversity = 3.2605e+00	k_t
Generation 4960: Best = 2.087242e+06 = [2, 3, 3, 2]	FES = 496100	Diversity = 3.3355e+00	k_t
Generation 4970: Best = 2.085601e+06 = [1, 1, 1, 1]	FES = 497100	Diversity = 3.4127e+00	k_t
Generation 4980: Best = 2.083201e+06 = [2, 3, 3, 4]	FES = 498100	Diversity = 3.6460e+00	k_t
Generation 4990: Best = 2.078505e+06 = [1, 1, 1, 1]	FES = 499100	Diversity = 4.0422e+00	k_t
Generation 5000: Best = 2.075754e+06 = [2, 2, 3, 4]	FES = 500100	Diversity = 4.2008e+00	k_t
Generation 5010: Best = 2.071582e+06 = [1, 1, 1, 1]	FES = 501100	Diversity = 4.0932e+00	k_t
Generation 5020: Best = 2.067644e+06 = [2, 2, 4, 4]	FES = 502100	Diversity = 3.9937e+00	k_t
Generation 5030: Best = 2.064071e+06 = [1, 1, 1, 1]	FES = 503100	Diversity = 3.8009e+00	k_t
Generation 5040: Best = 2.061568e+06 = [4, 4, 2, 3]	FES = 504100	Diversity = 3.6440e+00	k_t
Generation 5050: Best = 2.059785e+06 = [1, 1, 1, 1]	FES = 505100	Diversity = 3.6370e+00	k_t
Generation 5060: Best = 2.057124e+06 = [4, 4, 2, 4]	FES = 506100	Diversity = 3.8650e+00	k_t
Generation 5070: Best = 2.055106e+06 = [1, 1, 1, 1]	FES = 507100	Diversity = 3.8922e+00	k_t
Generation 5080: Best = 2.051281e+06 = [2, 2, 4, 2]	FES = 508100	Diversity = 3.6777e+00	k_t

Generation 5090: Best = 2.049346e+06 = [1, 1, 1, 1]	FES = 509100	Diversity = 3.9008e+00	k_t
Generation 5100: Best = 2.045739e+06 = [3, 2, 4, 3]	FES = 510100	Diversity = 3.8786e+00	k_t
Generation 5110: Best = 2.042614e+06 = [1, 1, 1, 1]	FES = 511100	Diversity = 4.1085e+00	k_t
Generation 5120: Best = 2.037540e+06 = [2, 3, 2, 2]	FES = 512100	Diversity = 3.9775e+00	k_t
Generation 5130: Best = 2.035094e+06 = [1, 1, 1, 1]	FES = 513100	Diversity = 4.0966e+00	k_t
Generation 5140: Best = 2.029562e+06 = [2, 2, 3, 4]	FES = 514100	Diversity = 4.3379e+00	k_t
Generation 5150: Best = 2.027151e+06 = [1, 1, 1, 1]	FES = 515100	Diversity = 4.2744e+00	k_t
Generation 5160: Best = 2.024171e+06 = [4, 4, 3, 4]	FES = 516100	Diversity = 4.1561e+00	k_t
Generation 5170: Best = 2.020267e+06 = [1, 1, 1, 1]	FES = 517100	Diversity = 4.0912e+00	k_t
Generation 5180: Best = 2.018505e+06 = [3, 2, 2, 4]	FES = 518100	Diversity = 3.9269e+00	k_t
Generation 5190: Best = 2.014335e+06 = [1, 1, 1, 1]	FES = 519100	Diversity = 3.7445e+00	k_t
Generation 5200: Best = 2.010068e+06 = [4, 4, 3, 3]	FES = 520100	Diversity = 3.6357e+00	k_t
Generation 5210: Best = 2.007085e+06 = [1, 1, 1, 1]	FES = 521100	Diversity = 3.7241e+00	k_t
Generation 5220: Best = 2.003744e+06 = [4, 2, 4, 2]	FES = 522100	Diversity = 3.8480e+00	k_t
Generation 5230: Best = 2.000858e+06 = [1, 1, 1, 1]	FES = 523100	Diversity = 4.0354e+00	k_t
Generation 5240: Best = 1.996866e+06 = [2, 4, 4, 4]	FES = 524100	Diversity = 4.0554e+00	k_t
Generation 5250: Best = 1.992972e+06 = [1, 1, 1, 1]	FES = 525100	Diversity = 4.4429e+00	k_t
Generation 5260: Best = 1.991155e+06 = [2, 2, 4, 2]	FES = 526100	Diversity = 4.3801e+00	k_t
Generation 5270: Best = 1.989278e+06 = [1, 1, 1, 1]	FES = 527100	Diversity = 4.4865e+00	k_t
Generation 5280: Best = 1.985467e+06 = [3, 2, 3, 2]	FES = 528100	Diversity = 4.1592e+00	k_t
Generation 5290: Best = 1.981603e+06 = [1, 1, 1, 1]	FES = 529100	Diversity = 4.0611e+00	k_t
Generation 5300: Best = 1.979253e+06 = [2, 3, 4, 3]	FES = 530100	Diversity = 4.1141e+00	k_t
Generation 5310: Best = 1.975122e+06 = [1, 1, 1, 1]	FES = 531100	Diversity = 4.3248e+00	k_t
Generation 5320: Best = 1.971614e+06 = [4, 2, 4, 3]	FES = 532100	Diversity = 4.2261e+00	k_t
Generation 5330: Best = 1.965269e+06 = [1, 1, 1, 1]	FES = 533100	Diversity = 4.2173e+00	k_t
Generation 5340: Best = 1.958979e+06 = [3, 2, 4, 2]	FES = 534100	Diversity = 4.6825e+00	k_t
Generation 5350: Best = 1.955584e+06 = [1, 1, 1, 1]	FES = 535100	Diversity = 4.8353e+00	k_t

Generation 5360: Best = 1.950058e+06 = [4, 3, 3, 2]	FES = 536100	Diversity = 4.9110e+00	k_t
Generation 5370: Best = 1.947216e+06 = [1, 1, 1, 1]	FES = 537100	Diversity = 4.9769e+00	k_t
Generation 5380: Best = 1.943095e+06 = [2, 3, 2, 2]	FES = 538100	Diversity = 4.5876e+00	k_t
Generation 5390: Best = 1.941070e+06 = [1, 1, 1, 1]	FES = 539100	Diversity = 4.3730e+00	k_t
Generation 5400: Best = 1.935626e+06 = [4, 4, 2, 3]	FES = 540100	Diversity = 4.0433e+00	k_t
Generation 5410: Best = 1.934440e+06 = [1, 1, 1, 1]	FES = 541100	Diversity = 4.2110e+00	k_t
Generation 5420: Best = 1.928268e+06 = [3, 3, 3, 3]	FES = 542100	Diversity = 4.6640e+00	k_t
Generation 5430: Best = 1.922742e+06 = [1, 1, 1, 1]	FES = 543100	Diversity = 4.3031e+00	k_t
Generation 5440: Best = 1.919166e+06 = [4, 3, 3, 3]	FES = 544100	Diversity = 4.3039e+00	k_t
Generation 5450: Best = 1.916917e+06 = [1, 1, 1, 1]	FES = 545100	Diversity = 4.1643e+00	k_t
Generation 5460: Best = 1.913531e+06 = [2, 4, 3, 2]	FES = 546100	Diversity = 4.0732e+00	k_t
Generation 5470: Best = 1.911108e+06 = [1, 1, 1, 1]	FES = 547100	Diversity = 4.0548e+00	k_t
Generation 5480: Best = 1.906960e+06 = [2, 2, 3, 2]	FES = 548100	Diversity = 3.9029e+00	k_t
Generation 5490: Best = 1.903814e+06 = [1, 1, 1, 1]	FES = 549100	Diversity = 3.9830e+00	k_t
Generation 5500: Best = 1.900848e+06 = [4, 3, 4, 3]	FES = 550100	Diversity = 3.9809e+00	k_t
Generation 5510: Best = 1.899972e+06 = [1, 1, 1, 1]	FES = 551100	Diversity = 3.7753e+00	k_t
Generation 5520: Best = 1.898012e+06 = [3, 3, 4, 4]	FES = 552100	Diversity = 3.7526e+00	k_t
Generation 5530: Best = 1.895502e+06 = [1, 1, 1, 1]	FES = 553100	Diversity = 3.6909e+00	k_t
Generation 5540: Best = 1.893628e+06 = [4, 3, 2, 3]	FES = 554100	Diversity = 3.6892e+00	k_t
Generation 5550: Best = 1.891069e+06 = [1, 1, 1, 1]	FES = 555100	Diversity = 3.7344e+00	k_t
Generation 5560: Best = 1.888224e+06 = [4, 3, 2, 4]	FES = 556100	Diversity = 3.6349e+00	k_t
Generation 5570: Best = 1.885625e+06 = [1, 1, 1, 1]	FES = 557100	Diversity = 3.5480e+00	k_t
Generation 5580: Best = 1.884083e+06 = [4, 2, 2, 2]	FES = 558100	Diversity = 3.4867e+00	k_t
Generation 5590: Best = 1.880991e+06 = [1, 1, 1, 1]	FES = 559100	Diversity = 3.3678e+00	k_t
Generation 5600: Best = 1.878484e+06 = [4, 4, 4, 4]	FES = 560100	Diversity = 3.3435e+00	k_t
Generation 5610: Best = 1.876581e+06 = [1, 1, 1, 1]	FES = 561100	Diversity = 3.5930e+00	k_t
Generation 5620: Best = 1.873386e+06 = [3, 2, 3, 2]	FES = 562100	Diversity = 3.7203e+00	k_t

Generation 5630: Best = 1.870221e+06 = [1, 1, 1, 1]	FES = 563100	Diversity = 3.7697e+00	k_t
Generation 5640: Best = 1.869422e+06 = [3, 2, 3, 2]	FES = 564100	Diversity = 3.6800e+00	k_t
Generation 5650: Best = 1.866590e+06 = [1, 1, 1, 1]	FES = 565100	Diversity = 3.6126e+00	k_t
Generation 5660: Best = 1.866337e+06 = [3, 3, 2, 4]	FES = 566100	Diversity = 3.7324e+00	k_t
Generation 5670: Best = 1.863722e+06 = [1, 1, 1, 1]	FES = 567100	Diversity = 3.7172e+00	k_t
Generation 5680: Best = 1.862541e+06 = [4, 3, 2, 2]	FES = 568100	Diversity = 3.7169e+00	k_t
Generation 5690: Best = 1.861430e+06 = [1, 1, 1, 1]	FES = 569100	Diversity = 3.5830e+00	k_t
Generation 5700: Best = 1.859301e+06 = [3, 3, 2, 4]	FES = 570100	Diversity = 3.7242e+00	k_t
Generation 5710: Best = 1.856114e+06 = [1, 1, 1, 1]	FES = 571100	Diversity = 4.0397e+00	k_t
Generation 5720: Best = 1.853084e+06 = [3, 4, 3, 2]	FES = 572100	Diversity = 4.4294e+00	k_t
Generation 5730: Best = 1.852074e+06 = [1, 1, 1, 1]	FES = 573100	Diversity = 4.6500e+00	k_t
Generation 5740: Best = 1.850065e+06 = [2, 2, 4, 2]	FES = 574100	Diversity = 4.9312e+00	k_t
Generation 5750: Best = 1.846784e+06 = [1, 1, 1, 1]	FES = 575100	Diversity = 4.6603e+00	k_t
Generation 5760: Best = 1.842724e+06 = [2, 2, 3, 3]	FES = 576100	Diversity = 4.3024e+00	k_t
Generation 5770: Best = 1.841987e+06 = [1, 1, 1, 1]	FES = 577100	Diversity = 4.2697e+00	k_t
Generation 5780: Best = 1.840976e+06 = [3, 3, 4, 4]	FES = 578100	Diversity = 4.4695e+00	k_t
Generation 5790: Best = 1.838167e+06 = [1, 1, 1, 1]	FES = 579100	Diversity = 4.1063e+00	k_t
Generation 5800: Best = 1.837496e+06 = [2, 3, 3, 3]	FES = 580100	Diversity = 4.1800e+00	k_t
Generation 5810: Best = 1.833820e+06 = [1, 1, 1, 1]	FES = 581100	Diversity = 4.0648e+00	k_t
Generation 5820: Best = 1.831410e+06 = [2, 3, 2, 4]	FES = 582100	Diversity = 4.2905e+00	k_t
Generation 5830: Best = 1.829590e+06 = [1, 1, 1, 1]	FES = 583100	Diversity = 4.1487e+00	k_t
Generation 5840: Best = 1.826595e+06 = [2, 3, 3, 4]	FES = 584100	Diversity = 3.9686e+00	k_t
Generation 5850: Best = 1.822858e+06 = [1, 1, 1, 1]	FES = 585100	Diversity = 3.5773e+00	k_t
Generation 5860: Best = 1.822518e+06 = [2, 3, 3, 4]	FES = 586100	Diversity = 3.4718e+00	k_t
Generation 5870: Best = 1.820505e+06 = [1, 1, 1, 1]	FES = 587100	Diversity = 3.4144e+00	k_t
Generation 5880: Best = 1.818643e+06 = [3, 3, 2, 4]	FES = 588100	Diversity = 3.3351e+00	k_t
Generation 5890: Best = 1.816896e+06 = [1, 1, 1, 1]	FES = 589100	Diversity = 3.5246e+00	k_t

Generation 5900: Best = 1.816301e+06 = [2, 3, 3, 4]	FES = 590100	Diversity = 3.5816e+00	k_t
Generation 5910: Best = 1.812468e+06 = [1, 1, 1, 1]	FES = 591100	Diversity = 3.3825e+00	k_t
Generation 5920: Best = 1.811841e+06 = [2, 4, 3, 2]	FES = 592100	Diversity = 3.3166e+00	k_t
Generation 5930: Best = 1.809671e+06 = [1, 1, 1, 1]	FES = 593100	Diversity = 3.4113e+00	k_t
Generation 5940: Best = 1.805959e+06 = [2, 2, 2, 4]	FES = 594100	Diversity = 3.2435e+00	k_t
Generation 5950: Best = 1.804908e+06 = [1, 1, 1, 1]	FES = 595100	Diversity = 3.2552e+00	k_t
Generation 5960: Best = 1.802572e+06 = [3, 3, 4, 4]	FES = 596100	Diversity = 3.3110e+00	k_t
Generation 5970: Best = 1.798816e+06 = [1, 1, 1, 1]	FES = 597100	Diversity = 3.1963e+00	k_t
Generation 5980: Best = 1.797580e+06 = [2, 2, 2, 2]	FES = 598100	Diversity = 3.2218e+00	k_t
Generation 5990: Best = 1.796634e+06 = [1, 1, 1, 1]	FES = 599100	Diversity = 3.1212e+00	k_t
Generation 6000: Best = 1.795354e+06 = [3, 3, 4, 2]	FES = 600100	Diversity = 3.1368e+00	k_t
Generation 6010: Best = 1.793135e+06 = [1, 1, 1, 1]	FES = 601100	Diversity = 3.1636e+00	k_t
Generation 6020: Best = 1.792451e+06 = [3, 4, 4, 3]	FES = 602100	Diversity = 3.1295e+00	k_t
Generation 6030: Best = 1.791229e+06 = [1, 1, 1, 1]	FES = 603100	Diversity = 3.0479e+00	k_t
Generation 6040: Best = 1.789674e+06 = [3, 4, 2, 2]	FES = 604100	Diversity = 3.0838e+00	k_t
Generation 6050: Best = 1.788678e+06 = [1, 1, 1, 1]	FES = 605100	Diversity = 3.0004e+00	k_t
Generation 6060: Best = 1.784280e+06 = [4, 4, 2, 2]	FES = 606100	Diversity = 3.2371e+00	k_t
Generation 6070: Best = 1.783812e+06 = [1, 1, 1, 1]	FES = 607100	Diversity = 3.2197e+00	k_t
Generation 6080: Best = 1.782303e+06 = [4, 3, 4, 4]	FES = 608100	Diversity = 3.3028e+00	k_t
Generation 6090: Best = 1.779634e+06 = [1, 1, 1, 1]	FES = 609100	Diversity = 3.2822e+00	k_t
Generation 6100: Best = 1.775997e+06 = [2, 4, 3, 2]	FES = 610100	Diversity = 3.4619e+00	k_t
Generation 6110: Best = 1.775622e+06 = [1, 1, 1, 1]	FES = 611100	Diversity = 3.2917e+00	k_t
Generation 6120: Best = 1.774059e+06 = [4, 4, 2, 4]	FES = 612100	Diversity = 3.2690e+00	k_t
Generation 6130: Best = 1.771884e+06 = [1, 1, 1, 1]	FES = 613100	Diversity = 3.4848e+00	k_t
Generation 6140: Best = 1.770094e+06 = [4, 2, 4, 2]	FES = 614100	Diversity = 3.4787e+00	k_t
Generation 6150: Best = 1.768865e+06 = [1, 1, 1, 1]	FES = 615100	Diversity = 3.4712e+00	k_t
Generation 6160: Best = 1.767713e+06 = [3, 2, 2, 3]	FES = 616100	Diversity = 3.3162e+00	k_t

Generation 6170: Best = 1.765833e+06 = [1, 1, 1, 1]	FES = 617100	Diversity = 3.2796e+00	k_t
Generation 6180: Best = 1.764045e+06 = [4, 3, 4, 3]	FES = 618100	Diversity = 3.2452e+00	k_t
Generation 6190: Best = 1.763671e+06 = [1, 1, 1, 1]	FES = 619100	Diversity = 3.1363e+00	k_t
Generation 6200: Best = 1.762800e+06 = [3, 4, 2, 2]	FES = 620100	Diversity = 3.1488e+00	k_t
Generation 6210: Best = 1.760735e+06 = [1, 1, 1, 1]	FES = 621100	Diversity = 3.0787e+00	k_t
Generation 6220: Best = 1.759305e+06 = [2, 3, 3, 3]	FES = 622100	Diversity = 3.0537e+00	k_t
Generation 6230: Best = 1.759038e+06 = [1, 1, 1, 1]	FES = 623100	Diversity = 3.1276e+00	k_t
Generation 6240: Best = 1.757768e+06 = [4, 4, 3, 3]	FES = 624100	Diversity = 3.1631e+00	k_t
Generation 6250: Best = 1.757076e+06 = [1, 1, 1, 1]	FES = 625100	Diversity = 3.1774e+00	k_t
Generation 6260: Best = 1.755723e+06 = [3, 2, 2, 3]	FES = 626100	Diversity = 3.3029e+00	k_t
Generation 6270: Best = 1.755169e+06 = [1, 1, 1, 1]	FES = 627100	Diversity = 3.4191e+00	k_t
Generation 6280: Best = 1.753261e+06 = [3, 4, 2, 3]	FES = 628100	Diversity = 3.6164e+00	k_t
Generation 6290: Best = 1.751128e+06 = [1, 1, 1, 1]	FES = 629100	Diversity = 3.9333e+00	k_t
Generation 6300: Best = 1.748196e+06 = [2, 2, 2, 4]	FES = 630100	Diversity = 3.9949e+00	k_t
Generation 6310: Best = 1.745978e+06 = [1, 1, 1, 1]	FES = 631100	Diversity = 4.2063e+00	k_t
Generation 6320: Best = 1.743340e+06 = [3, 4, 2, 4]	FES = 632100	Diversity = 4.6324e+00	k_t
Generation 6330: Best = 1.740127e+06 = [1, 1, 1, 1]	FES = 633100	Diversity = 4.8148e+00	k_t
Generation 6340: Best = 1.736449e+06 = [4, 2, 3, 3]	FES = 634100	Diversity = 4.6895e+00	k_t
Generation 6350: Best = 1.736387e+06 = [1, 1, 1, 1]	FES = 635100	Diversity = 4.5999e+00	k_t
Generation 6360: Best = 1.734320e+06 = [3, 2, 2, 2]	FES = 636100	Diversity = 4.4821e+00	k_t
Generation 6370: Best = 1.731428e+06 = [1, 1, 1, 1]	FES = 637100	Diversity = 4.3438e+00	k_t
Generation 6380: Best = 1.730879e+06 = [4, 3, 4, 4]	FES = 638100	Diversity = 4.5768e+00	k_t
Generation 6390: Best = 1.727674e+06 = [1, 1, 1, 1]	FES = 639100	Diversity = 4.7560e+00	k_t
Generation 6400: Best = 1.727245e+06 = [3, 2, 2, 3]	FES = 640100	Diversity = 4.9707e+00	k_t
Generation 6410: Best = 1.725374e+06 = [1, 1, 1, 1]	FES = 641100	Diversity = 4.8709e+00	k_t
Generation 6420: Best = 1.723737e+06 = [3, 2, 3, 4]	FES = 642100	Diversity = 4.6399e+00	k_t
Generation 6430: Best = 1.721489e+06 = [1, 1, 1, 1]	FES = 643100	Diversity = 4.5008e+00	k_t

Generation 6440: Best = 1.720203e+06 = [4, 2, 4, 4]	FES = 644100	Diversity = 4.4959e+00	k_t
Generation 6450: Best = 1.718567e+06 = [1, 1, 1, 1]	FES = 645100	Diversity = 4.3175e+00	k_t
Generation 6460: Best = 1.716302e+06 = [3, 3, 2, 2]	FES = 646100	Diversity = 4.4578e+00	k_t
Generation 6470: Best = 1.714001e+06 = [1, 1, 1, 1]	FES = 647100	Diversity = 4.2961e+00	k_t
Generation 6480: Best = 1.712283e+06 = [4, 2, 2, 3]	FES = 648100	Diversity = 4.3132e+00	k_t
Generation 6490: Best = 1.710057e+06 = [1, 1, 1, 1]	FES = 649100	Diversity = 4.3037e+00	k_t
Generation 6500: Best = 1.709183e+06 = [3, 2, 3, 4]	FES = 650100	Diversity = 4.0281e+00	k_t
Generation 6510: Best = 1.704520e+06 = [1, 1, 1, 1]	FES = 651100	Diversity = 4.2220e+00	k_t
Generation 6520: Best = 1.702770e+06 = [3, 2, 4, 2]	FES = 652100	Diversity = 4.4003e+00	k_t
Generation 6530: Best = 1.700999e+06 = [1, 1, 1, 1]	FES = 653100	Diversity = 4.5381e+00	k_t
Generation 6540: Best = 1.699634e+06 = [4, 3, 2, 2]	FES = 654100	Diversity = 4.5044e+00	k_t
Generation 6550: Best = 1.697063e+06 = [1, 1, 1, 1]	FES = 655100	Diversity = 4.3314e+00	k_t
Generation 6560: Best = 1.695622e+06 = [2, 3, 4, 4]	FES = 656100	Diversity = 4.0005e+00	k_t
Generation 6570: Best = 1.692954e+06 = [1, 1, 1, 1]	FES = 657100	Diversity = 3.9563e+00	k_t
Generation 6580: Best = 1.691394e+06 = [3, 4, 2, 4]	FES = 658100	Diversity = 3.9037e+00	k_t
Generation 6590: Best = 1.690177e+06 = [1, 1, 1, 1]	FES = 659100	Diversity = 3.9491e+00	k_t
Generation 6600: Best = 1.687965e+06 = [2, 4, 3, 3]	FES = 660100	Diversity = 3.9912e+00	k_t
Generation 6610: Best = 1.687139e+06 = [1, 1, 1, 1]	FES = 661100	Diversity = 3.9989e+00	k_t
Generation 6620: Best = 1.684329e+06 = [4, 2, 3, 4]	FES = 662100	Diversity = 4.0167e+00	k_t
Generation 6630: Best = 1.682120e+06 = [1, 1, 1, 1]	FES = 663100	Diversity = 4.0341e+00	k_t
Generation 6640: Best = 1.681342e+06 = [4, 3, 4, 2]	FES = 664100	Diversity = 3.9912e+00	k_t
Generation 6650: Best = 1.678894e+06 = [1, 1, 1, 1]	FES = 665100	Diversity = 3.9383e+00	k_t
Generation 6660: Best = 1.677970e+06 = [3, 3, 2, 2]	FES = 666100	Diversity = 3.8407e+00	k_t
Generation 6670: Best = 1.674825e+06 = [1, 1, 1, 1]	FES = 667100	Diversity = 3.8175e+00	k_t
Generation 6680: Best = 1.674521e+06 = [4, 2, 4, 2]	FES = 668100	Diversity = 3.8967e+00	k_t
Generation 6690: Best = 1.674005e+06 = [1, 1, 1, 1]	FES = 669100	Diversity = 3.8751e+00	k_t
Generation 6700: Best = 1.671212e+06 = [4, 4, 2, 4]	FES = 670100	Diversity = 4.0255e+00	k_t

Generation 6710: Best = 1.670024e+06 = [1, 1, 1, 1]	FES = 671100	Diversity = 4.0172e+00	k_t
Generation 6720: Best = 1.668072e+06 = [4, 2, 2, 2]	FES = 672100	Diversity = 4.1432e+00	k_t
Generation 6730: Best = 1.667257e+06 = [1, 1, 1, 1]	FES = 673100	Diversity = 4.1201e+00	k_t
Generation 6740: Best = 1.664793e+06 = [4, 4, 3, 4]	FES = 674100	Diversity = 4.5419e+00	k_t
Generation 6750: Best = 1.662117e+06 = [1, 1, 1, 1]	FES = 675100	Diversity = 4.7637e+00	k_t
Generation 6760: Best = 1.661154e+06 = [3, 4, 4, 3]	FES = 676100	Diversity = 4.3363e+00	k_t
Generation 6770: Best = 1.658965e+06 = [1, 1, 1, 1]	FES = 677100	Diversity = 4.1736e+00	k_t
Generation 6780: Best = 1.658184e+06 = [2, 2, 2, 4]	FES = 678100	Diversity = 3.9548e+00	k_t
Generation 6790: Best = 1.657114e+06 = [1, 1, 1, 1]	FES = 679100	Diversity = 3.8431e+00	k_t
Generation 6800: Best = 1.654694e+06 = [3, 4, 4, 3]	FES = 680100	Diversity = 4.0034e+00	k_t
Generation 6810: Best = 1.652832e+06 = [1, 1, 1, 1]	FES = 681100	Diversity = 4.1899e+00	k_t
Generation 6820: Best = 1.650836e+06 = [2, 2, 4, 2]	FES = 682100	Diversity = 4.2857e+00	k_t
Generation 6830: Best = 1.649572e+06 = [1, 1, 1, 1]	FES = 683100	Diversity = 4.2675e+00	k_t
Generation 6840: Best = 1.646062e+06 = [3, 2, 4, 2]	FES = 684100	Diversity = 4.3637e+00	k_t
Generation 6850: Best = 1.646008e+06 = [1, 1, 1, 1]	FES = 685100	Diversity = 4.0819e+00	k_t
Generation 6860: Best = 1.644844e+06 = [4, 4, 3, 3]	FES = 686100	Diversity = 4.0333e+00	k_t
Generation 6870: Best = 1.642191e+06 = [1, 1, 1, 1]	FES = 687100	Diversity = 4.1632e+00	k_t
Generation 6880: Best = 1.640499e+06 = [3, 4, 4, 3]	FES = 688100	Diversity = 4.3422e+00	k_t
Generation 6890: Best = 1.638792e+06 = [1, 1, 1, 1]	FES = 689100	Diversity = 4.2938e+00	k_t
Generation 6900: Best = 1.633730e+06 = [4, 3, 2, 2]	FES = 690100	Diversity = 4.2626e+00	k_t
Generation 6910: Best = 1.633522e+06 = [1, 1, 1, 1]	FES = 691100	Diversity = 4.1081e+00	k_t
Generation 6920: Best = 1.628919e+06 = [2, 2, 2, 4]	FES = 692100	Diversity = 4.1048e+00	k_t
Generation 6930: Best = 1.627167e+06 = [1, 1, 1, 1]	FES = 693100	Diversity = 4.1656e+00	k_t
Generation 6940: Best = 1.624869e+06 = [3, 3, 3, 2]	FES = 694100	Diversity = 3.8408e+00	k_t
Generation 6950: Best = 1.622592e+06 = [1, 1, 1, 1]	FES = 695100	Diversity = 3.9049e+00	k_t
Generation 6960: Best = 1.620300e+06 = [3, 2, 4, 2]	FES = 696100	Diversity = 3.7806e+00	k_t
Generation 6970: Best = 1.618023e+06 = [1, 1, 1, 1]	FES = 697100	Diversity = 3.7326e+00	k_t

Generation 6980: Best = 1.615442e+06 = [4, 4, 3, 4]	FES = 698100	Diversity = 3.8583e+00	k_t
Generation 6990: Best = 1.613349e+06 = [1, 1, 1, 1]	FES = 699100	Diversity = 3.7646e+00	k_t
Generation 7000: Best = 1.609472e+06 = [2, 4, 2, 2]	FES = 700100	Diversity = 3.7730e+00	k_t
Generation 7010: Best = 1.607973e+06 = [1, 1, 1, 1]	FES = 701100	Diversity = 3.7230e+00	k_t
Generation 7020: Best = 1.606463e+06 = [2, 3, 4, 2]	FES = 702100	Diversity = 3.8123e+00	k_t
Generation 7030: Best = 1.602060e+06 = [1, 1, 1, 1]	FES = 703100	Diversity = 3.4815e+00	k_t
Generation 7040: Best = 1.600843e+06 = [2, 2, 3, 4]	FES = 704100	Diversity = 3.2634e+00	k_t
Generation 7050: Best = 1.599314e+06 = [1, 1, 1, 1]	FES = 705100	Diversity = 3.3351e+00	k_t
Generation 7060: Best = 1.596847e+06 = [3, 2, 4, 2]	FES = 706100	Diversity = 3.2329e+00	k_t
Generation 7070: Best = 1.594286e+06 = [1, 1, 1, 1]	FES = 707100	Diversity = 3.3463e+00	k_t
Generation 7080: Best = 1.591794e+06 = [3, 4, 2, 3]	FES = 708100	Diversity = 3.3544e+00	k_t
Generation 7090: Best = 1.590594e+06 = [1, 1, 1, 1]	FES = 709100	Diversity = 3.3267e+00	k_t
Generation 7100: Best = 1.587097e+06 = [2, 4, 2, 4]	FES = 710100	Diversity = 3.4444e+00	k_t
Generation 7110: Best = 1.586774e+06 = [1, 1, 1, 1]	FES = 711100	Diversity = 3.1619e+00	k_t
Generation 7120: Best = 1.585255e+06 = [4, 4, 4, 2]	FES = 712100	Diversity = 3.7306e+00	k_t
Generation 7130: Best = 1.581852e+06 = [1, 1, 1, 1]	FES = 713100	Diversity = 3.5914e+00	k_t
Generation 7140: Best = 1.580976e+06 = [2, 3, 3, 3]	FES = 714100	Diversity = 3.7350e+00	k_t
Generation 7150: Best = 1.578588e+06 = [1, 1, 1, 1]	FES = 715100	Diversity = 3.5608e+00	k_t
Generation 7160: Best = 1.577870e+06 = [4, 4, 3, 4]	FES = 716100	Diversity = 3.6001e+00	k_t
Generation 7170: Best = 1.576103e+06 = [1, 1, 1, 1]	FES = 717100	Diversity = 3.6187e+00	k_t
Generation 7180: Best = 1.575137e+06 = [2, 3, 4, 4]	FES = 718100	Diversity = 3.5576e+00	k_t
Generation 7190: Best = 1.573531e+06 = [1, 1, 1, 1]	FES = 719100	Diversity = 3.4043e+00	k_t
Generation 7200: Best = 1.572922e+06 = [4, 3, 4, 3]	FES = 720100	Diversity = 3.4674e+00	k_t
Generation 7210: Best = 1.570214e+06 = [1, 1, 1, 1]	FES = 721100	Diversity = 3.2458e+00	k_t
Generation 7220: Best = 1.569879e+06 = [2, 4, 2, 2]	FES = 722100	Diversity = 3.2794e+00	k_t
Generation 7230: Best = 1.568000e+06 = [1, 1, 1, 1]	FES = 723100	Diversity = 3.2886e+00	k_t
Generation 7240: Best = 1.567635e+06 = [4, 4, 3, 3]	FES = 724100	Diversity = 3.3001e+00	k_t

Generation 7250: Best = 1.566287e+06 = [1, 1, 1, 1]	FES = 725100	Diversity = 3.1470e+00	k_t
Generation 7260: Best = 1.565941e+06 = [3, 3, 3, 4]	FES = 726100	Diversity = 3.0704e+00	k_t
Generation 7270: Best = 1.565597e+06 = [1, 1, 1, 1]	FES = 727100	Diversity = 3.0106e+00	k_t
Generation 7280: Best = 1.563878e+06 = [3, 4, 3, 4]	FES = 728100	Diversity = 2.9939e+00	k_t
Generation 7290: Best = 1.562639e+06 = [1, 1, 1, 1]	FES = 729100	Diversity = 2.8733e+00	k_t
Generation 7300: Best = 1.560733e+06 = [4, 3, 4, 3]	FES = 730100	Diversity = 3.0257e+00	k_t
Generation 7310: Best = 1.560213e+06 = [1, 1, 1, 1]	FES = 731100	Diversity = 3.0427e+00	k_t
Generation 7320: Best = 1.558917e+06 = [4, 4, 2, 4]	FES = 732100	Diversity = 3.1448e+00	k_t
Generation 7330: Best = 1.557885e+06 = [1, 1, 1, 1]	FES = 733100	Diversity = 3.4165e+00	k_t
Generation 7340: Best = 1.556176e+06 = [2, 4, 4, 2]	FES = 734100	Diversity = 3.4418e+00	k_t
Generation 7350: Best = 1.554602e+06 = [1, 1, 1, 1]	FES = 735100	Diversity = 3.3486e+00	k_t
Generation 7360: Best = 1.553274e+06 = [4, 4, 2, 4]	FES = 736100	Diversity = 3.2771e+00	k_t
Generation 7370: Best = 1.552036e+06 = [1, 1, 1, 1]	FES = 737100	Diversity = 3.0987e+00	k_t
Generation 7380: Best = 1.550646e+06 = [2, 4, 2, 4]	FES = 738100	Diversity = 3.1398e+00	k_t
Generation 7390: Best = 1.548784e+06 = [1, 1, 1, 1]	FES = 739100	Diversity = 3.2213e+00	k_t
Generation 7400: Best = 1.547531e+06 = [3, 4, 2, 4]	FES = 740100	Diversity = 3.0586e+00	k_t
Generation 7410: Best = 1.545966e+06 = [1, 1, 1, 1]	FES = 741100	Diversity = 3.0843e+00	k_t
Generation 7420: Best = 1.545215e+06 = [2, 4, 2, 3]	FES = 742100	Diversity = 2.9907e+00	k_t
Generation 7430: Best = 1.542813e+06 = [1, 1, 1, 1]	FES = 743100	Diversity = 2.9486e+00	k_t
Generation 7440: Best = 1.542251e+06 = [3, 3, 2, 3]	FES = 744100	Diversity = 3.0030e+00	k_t
Generation 7450: Best = 1.541540e+06 = [1, 1, 1, 1]	FES = 745100	Diversity = 2.9637e+00	k_t
Generation 7460: Best = 1.538579e+06 = [2, 2, 3, 3]	FES = 746100	Diversity = 2.9572e+00	k_t
Generation 7470: Best = 1.536190e+06 = [1, 1, 1, 1]	FES = 747100	Diversity = 2.9750e+00	k_t
Generation 7480: Best = 1.535122e+06 = [2, 2, 3, 4]	FES = 748100	Diversity = 2.9022e+00	k_t
Generation 7490: Best = 1.532212e+06 = [1, 1, 1, 1]	FES = 749100	Diversity = 2.8218e+00	k_t
Generation 7500: Best = 1.529257e+06 = [3, 4, 2, 3]	FES = 750100	Diversity = 2.7315e+00	k_t
Generation 7510: Best = 1.528776e+06 = [1, 1, 1, 1]	FES = 751100	Diversity = 2.9068e+00	k_t

Generation 7520: Best = 1.527821e+06 = [3, 2, 3, 4]	FES = 752100	Diversity = 2.7822e+00	k_t
Generation 7530: Best = 1.527393e+06 = [1, 1, 1, 1]	FES = 753100	Diversity = 2.8762e+00	k_t
Generation 7540: Best = 1.526714e+06 = [2, 4, 4, 2]	FES = 754100	Diversity = 2.9217e+00	k_t
Generation 7550: Best = 1.526178e+06 = [1, 1, 1, 1]	FES = 755100	Diversity = 2.9039e+00	k_t
Generation 7560: Best = 1.524414e+06 = [2, 3, 2, 4]	FES = 756100	Diversity = 2.8451e+00	k_t
Generation 7570: Best = 1.523720e+06 = [1, 1, 1, 1]	FES = 757100	Diversity = 2.8813e+00	k_t
Generation 7580: Best = 1.523262e+06 = [2, 3, 3, 3]	FES = 758100	Diversity = 2.9864e+00	k_t
Generation 7590: Best = 1.521842e+06 = [1, 1, 1, 1]	FES = 759100	Diversity = 2.9998e+00	k_t
Generation 7600: Best = 1.521381e+06 = [3, 2, 4, 4]	FES = 760100	Diversity = 2.9100e+00	k_t
Generation 7610: Best = 1.521246e+06 = [1, 1, 1, 1]	FES = 761100	Diversity = 2.8183e+00	k_t
Generation 7620: Best = 1.519221e+06 = [4, 3, 2, 3]	FES = 762100	Diversity = 2.9966e+00	k_t
Generation 7630: Best = 1.518411e+06 = [1, 1, 1, 1]	FES = 763100	Diversity = 3.0020e+00	k_t
Generation 7640: Best = 1.516909e+06 = [2, 3, 3, 4]	FES = 764100	Diversity = 3.1793e+00	k_t
Generation 7650: Best = 1.516366e+06 = [1, 1, 1, 1]	FES = 765100	Diversity = 3.0308e+00	k_t
Generation 7660: Best = 1.515555e+06 = [4, 2, 4, 4]	FES = 766100	Diversity = 3.0525e+00	k_t
Generation 7670: Best = 1.514845e+06 = [1, 1, 1, 1]	FES = 767100	Diversity = 3.2041e+00	k_t
Generation 7680: Best = 1.513400e+06 = [4, 2, 3, 3]	FES = 768100	Diversity = 3.2592e+00	k_t
Generation 7690: Best = 1.511705e+06 = [1, 1, 1, 1]	FES = 769100	Diversity = 3.2769e+00	k_t
Generation 7700: Best = 1.510554e+06 = [3, 4, 3, 4]	FES = 770100	Diversity = 3.1766e+00	k_t
Generation 7710: Best = 1.509422e+06 = [1, 1, 1, 1]	FES = 771100	Diversity = 3.2067e+00	k_t
Generation 7720: Best = 1.507518e+06 = [4, 4, 4, 3]	FES = 772100	Diversity = 3.3625e+00	k_t
Generation 7730: Best = 1.506940e+06 = [1, 1, 1, 1]	FES = 773100	Diversity = 3.1949e+00	k_t
Generation 7740: Best = 1.506232e+06 = [2, 4, 4, 3]	FES = 774100	Diversity = 3.2012e+00	k_t
Generation 7750: Best = 1.504817e+06 = [1, 1, 1, 1]	FES = 775100	Diversity = 3.2147e+00	k_t
Generation 7760: Best = 1.504174e+06 = [2, 3, 2, 4]	FES = 776100	Diversity = 3.2390e+00	k_t
Generation 7770: Best = 1.499032e+06 = [1, 1, 1, 1]	FES = 777100	Diversity = 3.1085e+00	k_t
Generation 7780: Best = 1.499032e+06 = [3, 4, 2, 3]	FES = 778100	Diversity = 3.2271e+00	k_t

Generation 7790: Best = 1.497700e+06 = [1, 1, 1, 1]	FES = 779100	Diversity = 3.3445e+00	k_t
Generation 7800: Best = 1.495504e+06 = [3, 3, 2, 4]	FES = 780100	Diversity = 3.4080e+00	k_t
Generation 7810: Best = 1.493064e+06 = [1, 1, 1, 1]	FES = 781100	Diversity = 3.5560e+00	k_t
Generation 7820: Best = 1.490706e+06 = [4, 2, 4, 2]	FES = 782100	Diversity = 3.3276e+00	k_t
Generation 7830: Best = 1.487753e+06 = [1, 1, 1, 1]	FES = 783100	Diversity = 3.4466e+00	k_t
Generation 7840: Best = 1.486518e+06 = [4, 2, 3, 2]	FES = 784100	Diversity = 3.4600e+00	k_t
Generation 7850: Best = 1.483443e+06 = [1, 1, 1, 1]	FES = 785100	Diversity = 3.6533e+00	k_t
Generation 7860: Best = 1.483291e+06 = [4, 4, 3, 3]	FES = 786100	Diversity = 3.6297e+00	k_t
Generation 7870: Best = 1.481838e+06 = [1, 1, 1, 1]	FES = 787100	Diversity = 3.6332e+00	k_t
Generation 7880: Best = 1.479845e+06 = [2, 4, 3, 4]	FES = 788100	Diversity = 3.5619e+00	k_t
Generation 7890: Best = 1.475816e+06 = [1, 1, 1, 1]	FES = 789100	Diversity = 3.5538e+00	k_t
Generation 7900: Best = 1.475500e+06 = [4, 3, 3, 2]	FES = 790100	Diversity = 3.4223e+00	k_t
Generation 7910: Best = 1.473174e+06 = [1, 1, 1, 1]	FES = 791100	Diversity = 3.4957e+00	k_t
Generation 7920: Best = 1.472806e+06 = [4, 3, 4, 2]	FES = 792100	Diversity = 3.3601e+00	k_t
Generation 7930: Best = 1.470669e+06 = [1, 1, 1, 1]	FES = 793100	Diversity = 3.5116e+00	k_t
Generation 7940: Best = 1.469317e+06 = [3, 4, 4, 4]	FES = 794100	Diversity = 3.5625e+00	k_t
Generation 7950: Best = 1.468590e+06 = [1, 1, 1, 1]	FES = 795100	Diversity = 3.4996e+00	k_t
Generation 7960: Best = 1.467322e+06 = [3, 4, 2, 2]	FES = 796100	Diversity = 3.4953e+00	k_t
Generation 7970: Best = 1.465592e+06 = [1, 1, 1, 1]	FES = 797100	Diversity = 3.4587e+00	k_t
Generation 7980: Best = 1.464616e+06 = [4, 3, 4, 3]	FES = 798100	Diversity = 3.4381e+00	k_t
Generation 7990: Best = 1.462718e+06 = [1, 1, 1, 1]	FES = 799100	Diversity = 3.5775e+00	k_t
Generation 8000: Best = 1.462710e+06 = [4, 3, 2, 3]	FES = 800100	Diversity = 3.5528e+00	k_t
Generation 8010: Best = 1.460113e+06 = [1, 1, 1, 1]	FES = 801100	Diversity = 3.6126e+00	k_t
Generation 8020: Best = 1.459501e+06 = [2, 4, 4, 3]	FES = 802100	Diversity = 3.7205e+00	k_t
Generation 8030: Best = 1.456954e+06 = [1, 1, 1, 1]	FES = 803100	Diversity = 3.8530e+00	k_t
Generation 8040: Best = 1.454975e+06 = [3, 4, 2, 4]	FES = 804100	Diversity = 3.8007e+00	k_t
Generation 8050: Best = 1.453057e+06 = [1, 1, 1, 1]	FES = 805100	Diversity = 3.7949e+00	k_t

Generation 8060: Best = 1.452071e+06 = [3, 2, 2, 3]	FES = 806100	Diversity = 3.7036e+00	k_t
Generation 8070: Best = 1.450518e+06 = [1, 1, 1, 1]	FES = 807100	Diversity = 3.7228e+00	k_t
Generation 8080: Best = 1.449198e+06 = [2, 2, 3, 3]	FES = 808100	Diversity = 3.6398e+00	k_t
Generation 8090: Best = 1.448148e+06 = [1, 1, 1, 1]	FES = 809100	Diversity = 3.5752e+00	k_t
Generation 8100: Best = 1.447717e+06 = [3, 4, 3, 3]	FES = 810100	Diversity = 3.6200e+00	k_t
Generation 8110: Best = 1.444221e+06 = [1, 1, 1, 1]	FES = 811100	Diversity = 3.6360e+00	k_t
Generation 8120: Best = 1.444105e+06 = [4, 4, 3, 2]	FES = 812100	Diversity = 3.4498e+00	k_t
Generation 8130: Best = 1.442586e+06 = [1, 1, 1, 1]	FES = 813100	Diversity = 3.5676e+00	k_t
Generation 8140: Best = 1.441736e+06 = [2, 3, 2, 2]	FES = 814100	Diversity = 3.5778e+00	k_t
Generation 8150: Best = 1.438450e+06 = [1, 1, 1, 1]	FES = 815100	Diversity = 3.5629e+00	k_t
Generation 8160: Best = 1.438058e+06 = [2, 3, 2, 3]	FES = 816100	Diversity = 3.5104e+00	k_t
Generation 8170: Best = 1.436457e+06 = [1, 1, 1, 1]	FES = 817100	Diversity = 3.4364e+00	k_t
Generation 8180: Best = 1.434448e+06 = [2, 2, 3, 3]	FES = 818100	Diversity = 3.6799e+00	k_t
Generation 8190: Best = 1.431653e+06 = [1, 1, 1, 1]	FES = 819100	Diversity = 3.7980e+00	k_t
Generation 8200: Best = 1.429522e+06 = [4, 3, 3, 4]	FES = 820100	Diversity = 3.6440e+00	k_t
Generation 8210: Best = 1.428751e+06 = [1, 1, 1, 1]	FES = 821100	Diversity = 3.7649e+00	k_t
Generation 8220: Best = 1.426373e+06 = [4, 3, 4, 4]	FES = 822100	Diversity = 3.6341e+00	k_t
Generation 8230: Best = 1.425540e+06 = [1, 1, 1, 1]	FES = 823100	Diversity = 3.5629e+00	k_t
Generation 8240: Best = 1.422793e+06 = [3, 2, 2, 4]	FES = 824100	Diversity = 3.6684e+00	k_t
Generation 8250: Best = 1.420722e+06 = [1, 1, 1, 1]	FES = 825100	Diversity = 3.5460e+00	k_t
Generation 8260: Best = 1.418335e+06 = [3, 4, 2, 3]	FES = 826100	Diversity = 3.5078e+00	k_t
Generation 8270: Best = 1.416594e+06 = [1, 1, 1, 1]	FES = 827100	Diversity = 3.3462e+00	k_t
Generation 8280: Best = 1.415852e+06 = [4, 2, 2, 4]	FES = 828100	Diversity = 3.3502e+00	k_t
Generation 8290: Best = 1.413314e+06 = [1, 1, 1, 1]	FES = 829100	Diversity = 3.3791e+00	k_t
Generation 8300: Best = 1.411802e+06 = [2, 4, 4, 3]	FES = 830100	Diversity = 3.3906e+00	k_t
Generation 8310: Best = 1.409983e+06 = [1, 1, 1, 1]	FES = 831100	Diversity = 3.2316e+00	k_t
Generation 8320: Best = 1.408728e+06 = [3, 4, 2, 2]	FES = 832100	Diversity = 2.9902e+00	k_t

Generation 8330: Best = 1.406597e+06 = [1, 1, 1, 1]	FES = 833100	Diversity = 2.9495e+00	k_t
Generation 8340: Best = 1.405413e+06 = [3, 4, 3, 3]	FES = 834100	Diversity = 3.0112e+00	k_t
Generation 8350: Best = 1.402930e+06 = [1, 1, 1, 1]	FES = 835100	Diversity = 3.1342e+00	k_t
Generation 8360: Best = 1.400591e+06 = [2, 4, 2, 3]	FES = 836100	Diversity = 3.4013e+00	k_t
Generation 8370: Best = 1.398634e+06 = [1, 1, 1, 1]	FES = 837100	Diversity = 3.5911e+00	k_t
Generation 8380: Best = 1.395610e+06 = [3, 4, 2, 2]	FES = 838100	Diversity = 3.8133e+00	k_t
Generation 8390: Best = 1.392336e+06 = [1, 1, 1, 1]	FES = 839100	Diversity = 3.7863e+00	k_t
Generation 8400: Best = 1.391211e+06 = [3, 2, 4, 4]	FES = 840100	Diversity = 3.7748e+00	k_t
Generation 8410: Best = 1.387256e+06 = [1, 1, 1, 1]	FES = 841100	Diversity = 3.6854e+00	k_t
Generation 8420: Best = 1.385524e+06 = [4, 3, 3, 3]	FES = 842100	Diversity = 3.7537e+00	k_t
Generation 8430: Best = 1.383092e+06 = [1, 1, 1, 1]	FES = 843100	Diversity = 3.8213e+00	k_t
Generation 8440: Best = 1.379072e+06 = [4, 2, 2, 3]	FES = 844100	Diversity = 3.9477e+00	k_t
Generation 8450: Best = 1.376758e+06 = [1, 1, 1, 1]	FES = 845100	Diversity = 4.0442e+00	k_t
Generation 8460: Best = 1.375018e+06 = [2, 3, 4, 4]	FES = 846100	Diversity = 4.0192e+00	k_t
Generation 8470: Best = 1.373846e+06 = [1, 1, 1, 1]	FES = 847100	Diversity = 3.7988e+00	k_t
Generation 8480: Best = 1.372244e+06 = [3, 2, 3, 2]	FES = 848100	Diversity = 3.5357e+00	k_t
Generation 8490: Best = 1.369372e+06 = [1, 1, 1, 1]	FES = 849100	Diversity = 3.4506e+00	k_t
Generation 8500: Best = 1.368356e+06 = [4, 3, 4, 2]	FES = 850100	Diversity = 3.5880e+00	k_t
Generation 8510: Best = 1.367001e+06 = [1, 1, 1, 1]	FES = 851100	Diversity = 3.8022e+00	k_t
Generation 8520: Best = 1.366266e+06 = [3, 4, 4, 2]	FES = 852100	Diversity = 3.8285e+00	k_t
Generation 8530: Best = 1.364354e+06 = [1, 1, 1, 1]	FES = 853100	Diversity = 3.8165e+00	k_t
Generation 8540: Best = 1.362579e+06 = [4, 3, 3, 3]	FES = 854100	Diversity = 3.7852e+00	k_t
Generation 8550: Best = 1.361493e+06 = [1, 1, 1, 1]	FES = 855100	Diversity = 3.6051e+00	k_t
Generation 8560: Best = 1.360545e+06 = [2, 3, 4, 2]	FES = 856100	Diversity = 3.8557e+00	k_t
Generation 8570: Best = 1.360541e+06 = [1, 1, 1, 1]	FES = 857100	Diversity = 3.8071e+00	k_t
Generation 8580: Best = 1.358297e+06 = [3, 2, 2, 4]	FES = 858100	Diversity = 3.7853e+00	k_t
Generation 8590: Best = 1.354348e+06 = [1, 1, 1, 1]	FES = 859100	Diversity = 3.8997e+00	k_t

Generation 8600: Best = 1.354348e+06 = [4, 2, 2, 3]	FES = 860100	Diversity = 3.7726e+00	k_t
Generation 8610: Best = 1.353827e+06 = [1, 1, 1, 1]	FES = 861100	Diversity = 3.6496e+00	k_t
Generation 8620: Best = 1.351643e+06 = [4, 3, 2, 2]	FES = 862100	Diversity = 3.5043e+00	k_t
Generation 8630: Best = 1.350030e+06 = [1, 1, 1, 1]	FES = 863100	Diversity = 3.6980e+00	k_t
Generation 8640: Best = 1.349331e+06 = [2, 4, 4, 4]	FES = 864100	Diversity = 3.6922e+00	k_t
Generation 8650: Best = 1.348151e+06 = [1, 1, 1, 1]	FES = 865100	Diversity = 3.5271e+00	k_t
Generation 8660: Best = 1.346537e+06 = [2, 2, 2, 4]	FES = 866100	Diversity = 3.6746e+00	k_t
Generation 8670: Best = 1.345553e+06 = [1, 1, 1, 1]	FES = 867100	Diversity = 3.6950e+00	k_t
Generation 8680: Best = 1.344615e+06 = [3, 3, 2, 4]	FES = 868100	Diversity = 3.7564e+00	k_t
Generation 8690: Best = 1.343730e+06 = [1, 1, 1, 1]	FES = 869100	Diversity = 3.6366e+00	k_t
Generation 8700: Best = 1.342609e+06 = [2, 3, 3, 2]	FES = 870100	Diversity = 3.6308e+00	k_t
Generation 8710: Best = 1.340991e+06 = [1, 1, 1, 1]	FES = 871100	Diversity = 3.5556e+00	k_t
Generation 8720: Best = 1.340570e+06 = [2, 4, 2, 2]	FES = 872100	Diversity = 3.5667e+00	k_t
Generation 8730: Best = 1.339000e+06 = [1, 1, 1, 1]	FES = 873100	Diversity = 3.8394e+00	k_t
Generation 8740: Best = 1.334834e+06 = [4, 4, 4, 4]	FES = 874100	Diversity = 4.0881e+00	k_t
Generation 8750: Best = 1.333893e+06 = [1, 1, 1, 1]	FES = 875100	Diversity = 4.4361e+00	k_t
Generation 8760: Best = 1.329693e+06 = [4, 3, 4, 2]	FES = 876100	Diversity = 4.6884e+00	k_t
Generation 8770: Best = 1.327880e+06 = [1, 1, 1, 1]	FES = 877100	Diversity = 4.7830e+00	k_t
Generation 8780: Best = 1.325118e+06 = [3, 3, 4, 3]	FES = 878100	Diversity = 4.3730e+00	k_t
Generation 8790: Best = 1.324971e+06 = [1, 1, 1, 1]	FES = 879100	Diversity = 4.1725e+00	k_t
Generation 8800: Best = 1.322967e+06 = [2, 3, 4, 3]	FES = 880100	Diversity = 3.9500e+00	k_t
Generation 8810: Best = 1.321096e+06 = [1, 1, 1, 1]	FES = 881100	Diversity = 3.9174e+00	k_t
Generation 8820: Best = 1.320290e+06 = [2, 3, 4, 4]	FES = 882100	Diversity = 3.8825e+00	k_t
Generation 8830: Best = 1.319072e+06 = [1, 1, 1, 1]	FES = 883100	Diversity = 3.9337e+00	k_t
Generation 8840: Best = 1.317132e+06 = [2, 4, 4, 3]	FES = 884100	Diversity = 3.8408e+00	k_t
Generation 8850: Best = 1.315791e+06 = [1, 1, 1, 1]	FES = 885100	Diversity = 3.8757e+00	k_t
Generation 8860: Best = 1.313993e+06 = [2, 2, 2, 4]	FES = 886100	Diversity = 3.8883e+00	k_t

Generation 8870: Best = 1.313975e+06 = [1, 1, 1, 1]	FES = 887100	Diversity = 3.9059e+00	k_t
Generation 8880: Best = 1.311537e+06 = [3, 4, 3, 3]	FES = 888100	Diversity = 4.0585e+00	k_t
Generation 8890: Best = 1.311080e+06 = [1, 1, 1, 1]	FES = 889100	Diversity = 4.3852e+00	k_t
Generation 8900: Best = 1.307191e+06 = [4, 3, 4, 2]	FES = 890100	Diversity = 4.4445e+00	k_t
Generation 8910: Best = 1.305419e+06 = [1, 1, 1, 1]	FES = 891100	Diversity = 4.5570e+00	k_t
Generation 8920: Best = 1.305149e+06 = [4, 2, 2, 4]	FES = 892100	Diversity = 4.2706e+00	k_t
Generation 8930: Best = 1.302980e+06 = [1, 1, 1, 1]	FES = 893100	Diversity = 4.2989e+00	k_t
Generation 8940: Best = 1.301770e+06 = [2, 2, 4, 2]	FES = 894100	Diversity = 4.2702e+00	k_t
Generation 8950: Best = 1.301453e+06 = [1, 1, 1, 1]	FES = 895100	Diversity = 4.4606e+00	k_t
Generation 8960: Best = 1.298748e+06 = [3, 4, 2, 4]	FES = 896100	Diversity = 4.6498e+00	k_t
Generation 8970: Best = 1.297338e+06 = [1, 1, 1, 1]	FES = 897100	Diversity = 4.6667e+00	k_t
Generation 8980: Best = 1.296406e+06 = [2, 2, 3, 3]	FES = 898100	Diversity = 4.4336e+00	k_t
Generation 8990: Best = 1.294086e+06 = [1, 1, 1, 1]	FES = 899100	Diversity = 4.4517e+00	k_t
Generation 9000: Best = 1.293873e+06 = [4, 2, 3, 3]	FES = 900100	Diversity = 4.4846e+00	k_t
Generation 9010: Best = 1.292214e+06 = [1, 1, 1, 1]	FES = 901100	Diversity = 4.3338e+00	k_t
Generation 9020: Best = 1.291653e+06 = [4, 3, 2, 2]	FES = 902100	Diversity = 4.0183e+00	k_t
Generation 9030: Best = 1.289991e+06 = [1, 1, 1, 1]	FES = 903100	Diversity = 3.9528e+00	k_t
Generation 9040: Best = 1.286769e+06 = [2, 2, 3, 3]	FES = 904100	Diversity = 4.0160e+00	k_t
Generation 9050: Best = 1.285466e+06 = [1, 1, 1, 1]	FES = 905100	Diversity = 4.1875e+00	k_t
Generation 9060: Best = 1.283845e+06 = [4, 4, 2, 2]	FES = 906100	Diversity = 4.0380e+00	k_t
Generation 9070: Best = 1.282532e+06 = [1, 1, 1, 1]	FES = 907100	Diversity = 3.7121e+00	k_t
Generation 9080: Best = 1.280154e+06 = [3, 4, 2, 3]	FES = 908100	Diversity = 3.5994e+00	k_t
Generation 9090: Best = 1.279992e+06 = [1, 1, 1, 1]	FES = 909100	Diversity = 3.5308e+00	k_t
Generation 9100: Best = 1.278239e+06 = [4, 4, 2, 3]	FES = 910100	Diversity = 3.3459e+00	k_t
Generation 9110: Best = 1.277474e+06 = [1, 1, 1, 1]	FES = 911100	Diversity = 3.2286e+00	k_t
Generation 9120: Best = 1.275724e+06 = [3, 4, 2, 2]	FES = 912100	Diversity = 3.1282e+00	k_t
Generation 9130: Best = 1.274946e+06 = [1, 1, 1, 1]	FES = 913100	Diversity = 2.9969e+00	k_t

Generation 9140: Best = 1.274265e+06 = [3, 2, 2, 4]	FES = 914100	Diversity = 2.7694e+00	k_t
Generation 9150: Best = 1.273483e+06 = [1, 1, 1, 1]	FES = 915100	Diversity = 2.7089e+00	k_t
Generation 9160: Best = 1.272990e+06 = [3, 2, 3, 2]	FES = 916100	Diversity = 2.6181e+00	k_t
Generation 9170: Best = 1.271365e+06 = [1, 1, 1, 1]	FES = 917100	Diversity = 2.4920e+00	k_t
Generation 9180: Best = 1.269598e+06 = [4, 2, 3, 2]	FES = 918100	Diversity = 2.3936e+00	k_t
Generation 9190: Best = 1.269598e+06 = [1, 1, 1, 1]	FES = 919100	Diversity = 2.3349e+00	k_t
Generation 9200: Best = 1.268924e+06 = [4, 2, 3, 3]	FES = 920100	Diversity = 2.3361e+00	k_t
Generation 9210: Best = 1.268324e+06 = [1, 1, 1, 1]	FES = 921100	Diversity = 2.2552e+00	k_t
Generation 9220: Best = 1.267257e+06 = [3, 2, 2, 3]	FES = 922100	Diversity = 2.2535e+00	k_t
Generation 9230: Best = 1.265946e+06 = [1, 1, 1, 1]	FES = 923100	Diversity = 2.3392e+00	k_t
Generation 9240: Best = 1.264980e+06 = [2, 2, 3, 2]	FES = 924100	Diversity = 2.3737e+00	k_t
Generation 9250: Best = 1.263877e+06 = [1, 1, 1, 1]	FES = 925100	Diversity = 2.4550e+00	k_t
Generation 9260: Best = 1.263000e+06 = [2, 3, 4, 3]	FES = 926100	Diversity = 2.5195e+00	k_t
Generation 9270: Best = 1.261601e+06 = [1, 1, 1, 1]	FES = 927100	Diversity = 2.6208e+00	k_t
Generation 9280: Best = 1.260713e+06 = [2, 2, 4, 3]	FES = 928100	Diversity = 2.7562e+00	k_t
Generation 9290: Best = 1.257823e+06 = [1, 1, 1, 1]	FES = 929100	Diversity = 2.7286e+00	k_t
Generation 9300: Best = 1.257701e+06 = [4, 4, 4, 3]	FES = 930100	Diversity = 2.7540e+00	k_t
Generation 9310: Best = 1.256621e+06 = [1, 1, 1, 1]	FES = 931100	Diversity = 2.8986e+00	k_t
Generation 9320: Best = 1.255482e+06 = [4, 4, 2, 4]	FES = 932100	Diversity = 2.8963e+00	k_t
Generation 9330: Best = 1.253566e+06 = [1, 1, 1, 1]	FES = 933100	Diversity = 2.9012e+00	k_t
Generation 9340: Best = 1.251474e+06 = [4, 4, 2, 2]	FES = 934100	Diversity = 2.8872e+00	k_t
Generation 9350: Best = 1.249243e+06 = [1, 1, 1, 1]	FES = 935100	Diversity = 3.0011e+00	k_t
Generation 9360: Best = 1.247869e+06 = [2, 3, 2, 2]	FES = 936100	Diversity = 3.1784e+00	k_t
Generation 9370: Best = 1.246253e+06 = [1, 1, 1, 1]	FES = 937100	Diversity = 3.2644e+00	k_t
Generation 9380: Best = 1.244136e+06 = [2, 2, 2, 3]	FES = 938100	Diversity = 3.4802e+00	k_t
Generation 9390: Best = 1.243691e+06 = [1, 1, 1, 1]	FES = 939100	Diversity = 3.4281e+00	k_t
Generation 9400: Best = 1.241977e+06 = [4, 4, 4, 2]	FES = 940100	Diversity = 3.3832e+00	k_t

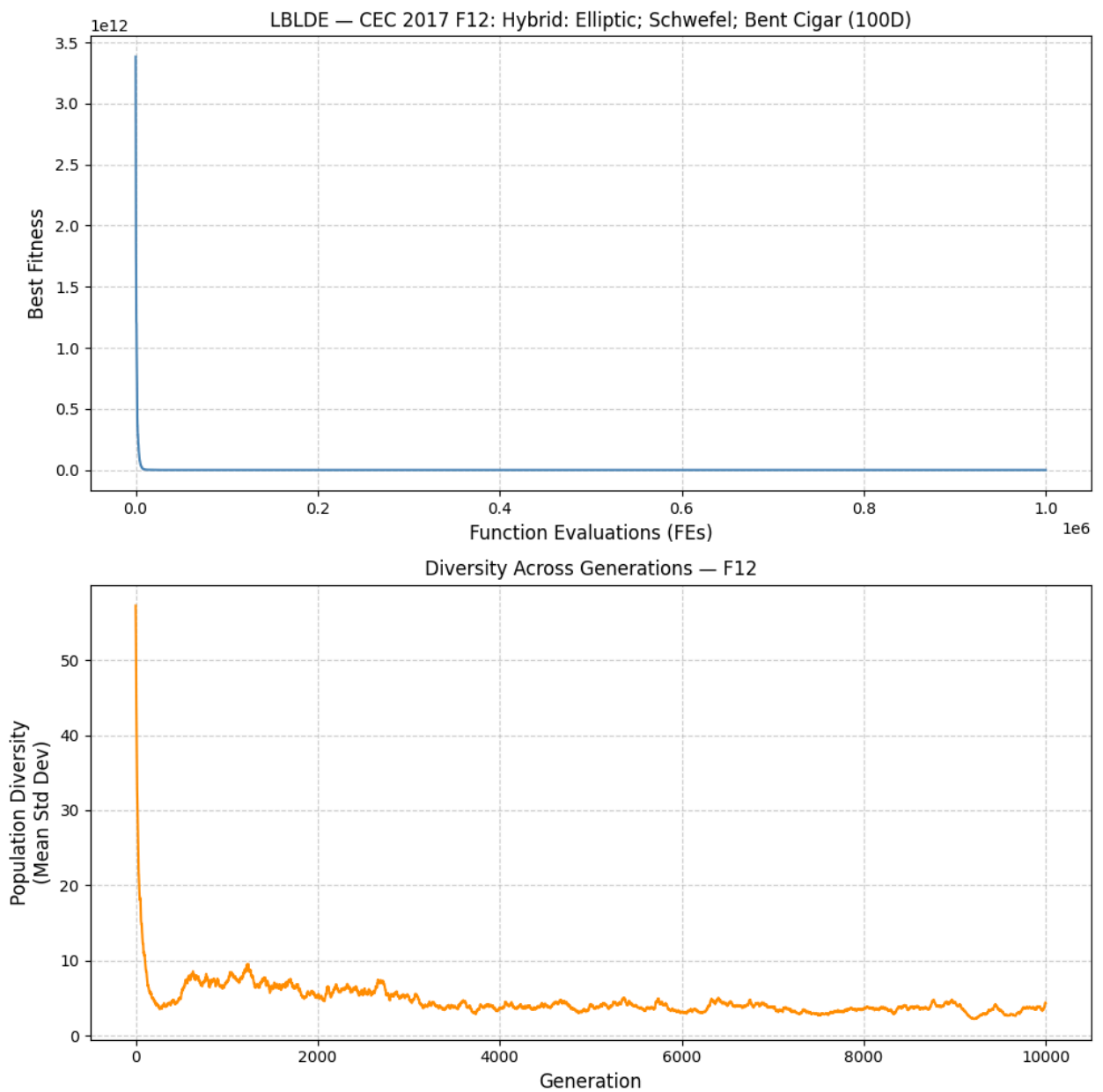
Generation 9410: Best = 1.240165e+06 = [1, 1, 1, 1]	FES = 941100	Diversity = 3.5416e+00	k_t
Generation 9420: Best = 1.238965e+06 = [3, 3, 2, 4]	FES = 942100	Diversity = 3.4997e+00	k_t
Generation 9430: Best = 1.235675e+06 = [1, 1, 1, 1]	FES = 943100	Diversity = 3.8377e+00	k_t
Generation 9440: Best = 1.233314e+06 = [3, 3, 3, 3]	FES = 944100	Diversity = 4.0351e+00	k_t
Generation 9450: Best = 1.232605e+06 = [1, 1, 1, 1]	FES = 945100	Diversity = 4.0907e+00	k_t
Generation 9460: Best = 1.231651e+06 = [4, 3, 3, 2]	FES = 946100	Diversity = 3.9088e+00	k_t
Generation 9470: Best = 1.230145e+06 = [1, 1, 1, 1]	FES = 947100	Diversity = 3.7023e+00	k_t
Generation 9480: Best = 1.228280e+06 = [2, 3, 4, 3]	FES = 948100	Diversity = 3.4926e+00	k_t
Generation 9490: Best = 1.226644e+06 = [1, 1, 1, 1]	FES = 949100	Diversity = 3.5381e+00	k_t
Generation 9500: Best = 1.226536e+06 = [4, 2, 4, 2]	FES = 950100	Diversity = 3.3763e+00	k_t
Generation 9510: Best = 1.224050e+06 = [1, 1, 1, 1]	FES = 951100	Diversity = 3.3012e+00	k_t
Generation 9520: Best = 1.223687e+06 = [3, 3, 2, 3]	FES = 952100	Diversity = 3.1799e+00	k_t
Generation 9530: Best = 1.221740e+06 = [1, 1, 1, 1]	FES = 953100	Diversity = 3.0759e+00	k_t
Generation 9540: Best = 1.221740e+06 = [4, 3, 2, 2]	FES = 954100	Diversity = 2.8702e+00	k_t
Generation 9550: Best = 1.221316e+06 = [1, 1, 1, 1]	FES = 955100	Diversity = 2.7706e+00	k_t
Generation 9560: Best = 1.220962e+06 = [2, 2, 4, 4]	FES = 956100	Diversity = 2.7387e+00	k_t
Generation 9570: Best = 1.220194e+06 = [1, 1, 1, 1]	FES = 957100	Diversity = 2.7440e+00	k_t
Generation 9580: Best = 1.220194e+06 = [4, 2, 2, 3]	FES = 958100	Diversity = 2.6637e+00	k_t
Generation 9590: Best = 1.218987e+06 = [1, 1, 1, 1]	FES = 959100	Diversity = 2.7057e+00	k_t
Generation 9600: Best = 1.218584e+06 = [2, 3, 4, 4]	FES = 960100	Diversity = 2.7038e+00	k_t
Generation 9610: Best = 1.217747e+06 = [1, 1, 1, 1]	FES = 961100	Diversity = 2.8167e+00	k_t
Generation 9620: Best = 1.216903e+06 = [4, 4, 3, 2]	FES = 962100	Diversity = 2.8517e+00	k_t
Generation 9630: Best = 1.216853e+06 = [1, 1, 1, 1]	FES = 963100	Diversity = 2.8586e+00	k_t
Generation 9640: Best = 1.215638e+06 = [3, 3, 4, 2]	FES = 964100	Diversity = 2.7943e+00	k_t
Generation 9650: Best = 1.214640e+06 = [1, 1, 1, 1]	FES = 965100	Diversity = 2.7305e+00	k_t
Generation 9660: Best = 1.214370e+06 = [2, 2, 3, 4]	FES = 966100	Diversity = 2.6719e+00	k_t
Generation 9670: Best = 1.213185e+06 = [1, 1, 1, 1]	FES = 967100	Diversity = 2.6765e+00	k_t

Generation 9680: Best = 1.212790e+06 = [2, 2, 4, 2]	FES = 968100	Diversity = 2.9678e+00	k_t
Generation 9690: Best = 1.211741e+06 = [1, 1, 1, 1]	FES = 969100	Diversity = 3.0343e+00	k_t
Generation 9700: Best = 1.210574e+06 = [3, 3, 4, 2]	FES = 970100	Diversity = 3.0762e+00	k_t
Generation 9710: Best = 1.210346e+06 = [1, 1, 1, 1]	FES = 971100	Diversity = 2.9355e+00	k_t
Generation 9720: Best = 1.208825e+06 = [3, 2, 2, 3]	FES = 972100	Diversity = 3.1078e+00	k_t
Generation 9730: Best = 1.207489e+06 = [1, 1, 1, 1]	FES = 973100	Diversity = 3.2555e+00	k_t
Generation 9740: Best = 1.206340e+06 = [3, 2, 2, 3]	FES = 974100	Diversity = 3.4703e+00	k_t
Generation 9750: Best = 1.204382e+06 = [1, 1, 1, 1]	FES = 975100	Diversity = 3.7291e+00	k_t
Generation 9760: Best = 1.202881e+06 = [4, 3, 4, 4]	FES = 976100	Diversity = 3.7090e+00	k_t
Generation 9770: Best = 1.202104e+06 = [1, 1, 1, 1]	FES = 977100	Diversity = 3.7374e+00	k_t
Generation 9780: Best = 1.200240e+06 = [3, 2, 2, 2]	FES = 978100	Diversity = 3.7099e+00	k_t
Generation 9790: Best = 1.198628e+06 = [1, 1, 1, 1]	FES = 979100	Diversity = 3.8064e+00	k_t
Generation 9800: Best = 1.197194e+06 = [2, 2, 2, 4]	FES = 980100	Diversity = 3.7613e+00	k_t
Generation 9810: Best = 1.195636e+06 = [1, 1, 1, 1]	FES = 981100	Diversity = 3.6315e+00	k_t
Generation 9820: Best = 1.194131e+06 = [3, 4, 2, 4]	FES = 982100	Diversity = 3.8119e+00	k_t
Generation 9830: Best = 1.192043e+06 = [1, 1, 1, 1]	FES = 983100	Diversity = 3.7305e+00	k_t
Generation 9840: Best = 1.191577e+06 = [2, 2, 2, 4]	FES = 984100	Diversity = 3.6407e+00	k_t
Generation 9850: Best = 1.190347e+06 = [1, 1, 1, 1]	FES = 985100	Diversity = 3.7086e+00	k_t
Generation 9860: Best = 1.187607e+06 = [4, 4, 4, 3]	FES = 986100	Diversity = 3.8112e+00	k_t
Generation 9870: Best = 1.184943e+06 = [1, 1, 1, 1]	FES = 987100	Diversity = 3.6925e+00	k_t
Generation 9880: Best = 1.183398e+06 = [4, 4, 4, 2]	FES = 988100	Diversity = 3.6458e+00	k_t
Generation 9890: Best = 1.182671e+06 = [1, 1, 1, 1]	FES = 989100	Diversity = 3.5204e+00	k_t
Generation 9900: Best = 1.179754e+06 = [3, 2, 4, 2]	FES = 990100	Diversity = 3.6791e+00	k_t
Generation 9910: Best = 1.178189e+06 = [1, 1, 1, 1]	FES = 991100	Diversity = 3.8463e+00	k_t
Generation 9920: Best = 1.176289e+06 = [4, 3, 4, 2]	FES = 992100	Diversity = 3.8678e+00	k_t
Generation 9930: Best = 1.174644e+06 = [1, 1, 1, 1]	FES = 993100	Diversity = 3.8815e+00	k_t
Generation 9940: Best = 1.173583e+06 = [2, 4, 4, 4]	FES = 994100	Diversity = 3.7303e+00	k_t

Generation 9950: Best = 1.171779e+06 FES = 995100 Diversity = 3.5942e+00 k_t
 = [1, 1, 1, 1]
 Generation 9960: Best = 1.170798e+06 FES = 996100 Diversity = 3.4258e+00 k_t
 = [3, 4, 4, 3]
 Generation 9970: Best = 1.169501e+06 FES = 997100 Diversity = 3.4906e+00 k_t
 = [1, 1, 1, 1]
 Generation 9980: Best = 1.168349e+06 FES = 998100 Diversity = 3.7011e+00 k_t
 = [3, 3, 2, 3]
 Generation 9990: Best = 1.165774e+06 FES = 999100 Diversity = 4.1613e+00 k_t
 = [1, 1, 1, 1]

Optimization completed!
 Final Best Fitness = 1.165080e+06
 Total Function Evaluations = 1000000

=====
 =
 Results:
 Best Fitness : 1.165080e+06
 Global Optimum : 1.200000e+03
 Error (|best-optimum|) : 1.163880e+06
 Best Solution (5 dim) : [58.08713921 -55.0104836 -13.23807331 29.4106
 93 -100.]
 D^(0) (init. diversity): 5.7220e+01
 =====
 =



Graph saved as convergence_F12.png

```
In [37]: # =====
# F13 – Hybrid Function 3 (Bent Cigar; Rosenbrock; Bi-Rastrigin)
# =====
run_and_plot('F13', make_wrapper(hybrid.f13), 1300.0, 'Hybrid: Bent Cigar; Ros
```

```

=====
=
LBLDE_s on CEC 2017 F13 – Single Run
=====
=
Function      : Hybrid: Bent Cigar; Rosenbrock; Bi-Rastrigin (100D)
Bounds       : [-100, 100]
Global optimum : 1300.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 9.446380e+11 | Diversity = 5.7220e+01
Generation 10: Best = 3.668865e+11 FES = 1100 Diversity = 4.2590e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.212631e+11 FES = 2100 Diversity = 3.4898e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 4.381050e+10 FES = 3100 Diversity = 2.9656e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.502031e+10 FES = 4100 Diversity = 2.5586e+01 k_t
= [4, 3, 4, 2]
Generation 50: Best = 4.789221e+09 FES = 5100 Diversity = 2.4384e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.064891e+09 FES = 6100 Diversity = 2.1107e+01 k_t
= [4, 3, 4, 3]
Generation 70: Best = 6.303341e+08 FES = 7100 Diversity = 2.0684e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.470219e+08 FES = 8100 Diversity = 1.9926e+01 k_t
= [4, 3, 2, 3]
Generation 90: Best = 4.916460e+07 FES = 9100 Diversity = 1.8889e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.410361e+07 FES = 10100 Diversity = 1.8866e+01 k_t
= [4, 2, 4, 3]
Generation 110: Best = 8.235719e+06 FES = 11100 Diversity = 1.7381e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.089391e+06 FES = 12100 Diversity = 1.5961e+01 k_t
= [4, 2, 2, 3]
Generation 130: Best = 1.034510e+06 FES = 13100 Diversity = 1.3997e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 4.736661e+05 FES = 14100 Diversity = 1.4030e+01 k_t
= [3, 2, 4, 2]
Generation 150: Best = 1.935384e+05 FES = 15100 Diversity = 1.4354e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.058975e+05 FES = 16100 Diversity = 1.2598e+01 k_t
= [4, 3, 2, 3]
Generation 170: Best = 5.384429e+04 FES = 17100 Diversity = 1.0633e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.371986e+04 FES = 18100 Diversity = 9.4541e+00 k_t
= [4, 4, 2, 4]
Generation 190: Best = 2.515944e+04 FES = 19100 Diversity = 7.7405e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.944538e+04 FES = 20100 Diversity = 6.5757e+00 k_t
= [3, 4, 3, 4]
Generation 210: Best = 1.610139e+04 FES = 21100 Diversity = 6.2147e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.354743e+04 FES = 22100 Diversity = 5.8232e+00 k_t
= [4, 4, 4, 2]

```

Generation 230:	Best = 1.202768e+04	FES = 23100	Diversity = 5.5944e+00	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 1.075230e+04	FES = 24100	Diversity = 5.4814e+00	k_t
= [2, 3, 2, 3]				
Generation 250:	Best = 9.800968e+03	FES = 25100	Diversity = 5.2163e+00	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 9.003881e+03	FES = 26100	Diversity = 4.9983e+00	k_t
= [3, 3, 4, 3]				
Generation 270:	Best = 8.352113e+03	FES = 27100	Diversity = 4.7084e+00	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 7.636513e+03	FES = 28100	Diversity = 4.4711e+00	k_t
= [3, 3, 4, 4]				
Generation 290:	Best = 7.249653e+03	FES = 29100	Diversity = 4.3441e+00	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 6.621449e+03	FES = 30100	Diversity = 4.5775e+00	k_t
= [4, 2, 4, 4]				
Generation 310:	Best = 6.392339e+03	FES = 31100	Diversity = 4.2726e+00	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 5.997667e+03	FES = 32100	Diversity = 3.8649e+00	k_t
= [3, 3, 2, 4]				
Generation 330:	Best = 5.839602e+03	FES = 33100	Diversity = 3.5647e+00	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 5.589568e+03	FES = 34100	Diversity = 3.2879e+00	k_t
= [3, 2, 4, 4]				
Generation 350:	Best = 5.439472e+03	FES = 35100	Diversity = 3.3334e+00	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 5.262961e+03	FES = 36100	Diversity = 3.3103e+00	k_t
= [4, 4, 4, 2]				
Generation 370:	Best = 5.060425e+03	FES = 37100	Diversity = 3.2745e+00	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 4.998888e+03	FES = 38100	Diversity = 3.1489e+00	k_t
= [4, 2, 2, 4]				
Generation 390:	Best = 4.912583e+03	FES = 39100	Diversity = 2.7964e+00	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 4.875510e+03	FES = 40100	Diversity = 2.7052e+00	k_t
= [3, 4, 3, 3]				
Generation 410:	Best = 4.820836e+03	FES = 41100	Diversity = 2.5881e+00	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 4.750619e+03	FES = 42100	Diversity = 2.4072e+00	k_t
= [3, 4, 2, 2]				
Generation 430:	Best = 4.750619e+03	FES = 43100	Diversity = 2.4005e+00	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 4.750619e+03	FES = 44100	Diversity = 2.3250e+00	k_t
= [4, 2, 4, 4]				
Generation 450:	Best = 4.701178e+03	FES = 45100	Diversity = 2.2268e+00	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 4.701178e+03	FES = 46100	Diversity = 2.2105e+00	k_t
= [3, 3, 2, 3]				
Generation 470:	Best = 4.682490e+03	FES = 47100	Diversity = 2.1627e+00	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 4.682490e+03	FES = 48100	Diversity = 2.1026e+00	k_t
= [3, 4, 2, 3]				
Generation 490:	Best = 4.676617e+03	FES = 49100	Diversity = 2.0655e+00	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 4.640217e+03	FES = 50100	Diversity = 2.0792e+00	k_t
= [4, 4, 4, 2]				
Generation 510:	Best = 4.640217e+03	FES = 51100	Diversity = 2.0302e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 4.634403e+03	FES = 52100	Diversity = 1.9746e+00	k_t
= [4, 2, 2, 3]				
Generation 530:	Best = 4.609348e+03	FES = 53100	Diversity = 1.9382e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 4.608844e+03	FES = 54100	Diversity = 1.8967e+00	k_t
= [4, 3, 2, 3]				
Generation 550:	Best = 4.595856e+03	FES = 55100	Diversity = 1.8757e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 4.578631e+03	FES = 56100	Diversity = 1.8244e+00	k_t
= [2, 4, 2, 3]				
Generation 570:	Best = 4.570578e+03	FES = 57100	Diversity = 1.7679e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 4.556374e+03	FES = 58100	Diversity = 1.7456e+00	k_t
= [3, 4, 3, 3]				
Generation 590:	Best = 4.551292e+03	FES = 59100	Diversity = 1.7521e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 4.532993e+03	FES = 60100	Diversity = 1.7284e+00	k_t
= [2, 4, 2, 4]				
Generation 610:	Best = 4.532993e+03	FES = 61100	Diversity = 1.7298e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 4.532605e+03	FES = 62100	Diversity = 1.7135e+00	k_t
= [3, 4, 2, 4]				
Generation 630:	Best = 4.531183e+03	FES = 63100	Diversity = 1.7251e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 4.499734e+03	FES = 64100	Diversity = 1.7018e+00	k_t
= [3, 3, 2, 2]				
Generation 650:	Best = 4.499734e+03	FES = 65100	Diversity = 1.7097e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 4.499734e+03	FES = 66100	Diversity = 1.6911e+00	k_t
= [3, 2, 2, 3]				
Generation 670:	Best = 4.499630e+03	FES = 67100	Diversity = 1.6853e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 4.497812e+03	FES = 68100	Diversity = 1.7210e+00	k_t
= [2, 2, 4, 4]				
Generation 690:	Best = 4.493842e+03	FES = 69100	Diversity = 1.6998e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 4.482971e+03	FES = 70100	Diversity = 1.6906e+00	k_t
= [2, 3, 3, 4]				
Generation 710:	Best = 4.482971e+03	FES = 71100	Diversity = 1.6807e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 4.482971e+03	FES = 72100	Diversity = 1.6657e+00	k_t
= [4, 4, 2, 4]				
Generation 730:	Best = 4.480423e+03	FES = 73100	Diversity = 1.6780e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 4.464958e+03	FES = 74100	Diversity = 1.6822e+00	k_t
= [3, 4, 2, 2]				
Generation 750:	Best = 4.464958e+03	FES = 75100	Diversity = 1.6741e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 4.461059e+03	FES = 76100	Diversity = 1.6640e+00	k_t
= [2, 3, 2, 3]				

Generation 770:	Best = 4.452657e+03	FES = 77100	Diversity = 1.6635e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 4.452657e+03	FES = 78100	Diversity = 1.6581e+00	k_t
= [2, 4, 2, 3]				
Generation 790:	Best = 4.446581e+03	FES = 79100	Diversity = 1.6563e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 4.446272e+03	FES = 80100	Diversity = 1.6717e+00	k_t
= [4, 4, 2, 2]				
Generation 810:	Best = 4.446272e+03	FES = 81100	Diversity = 1.6544e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 4.445878e+03	FES = 82100	Diversity = 1.6429e+00	k_t
= [3, 3, 2, 2]				
Generation 830:	Best = 4.438951e+03	FES = 83100	Diversity = 1.6474e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 4.435129e+03	FES = 84100	Diversity = 1.6443e+00	k_t
= [3, 4, 2, 3]				
Generation 850:	Best = 4.435129e+03	FES = 85100	Diversity = 1.6450e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 4.435129e+03	FES = 86100	Diversity = 1.6589e+00	k_t
= [3, 2, 2, 2]				
Generation 870:	Best = 4.434218e+03	FES = 87100	Diversity = 1.6504e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 4.432161e+03	FES = 88100	Diversity = 1.6380e+00	k_t
= [4, 2, 4, 2]				
Generation 890:	Best = 4.429606e+03	FES = 89100	Diversity = 1.6453e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 4.407355e+03	FES = 90100	Diversity = 1.6450e+00	k_t
= [4, 4, 4, 3]				
Generation 910:	Best = 4.397053e+03	FES = 91100	Diversity = 1.6409e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 4.397053e+03	FES = 92100	Diversity = 1.6429e+00	k_t
= [3, 4, 4, 2]				
Generation 930:	Best = 4.397053e+03	FES = 93100	Diversity = 1.6465e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 4.397053e+03	FES = 94100	Diversity = 1.6462e+00	k_t
= [2, 2, 2, 2]				
Generation 950:	Best = 4.393074e+03	FES = 95100	Diversity = 1.6274e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 4.392649e+03	FES = 96100	Diversity = 1.6340e+00	k_t
= [2, 4, 3, 2]				
Generation 970:	Best = 4.392649e+03	FES = 97100	Diversity = 1.6326e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 4.392040e+03	FES = 98100	Diversity = 1.6265e+00	k_t
= [4, 4, 3, 2]				
Generation 990:	Best = 4.390491e+03	FES = 99100	Diversity = 1.6192e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 4.390491e+03	FES = 100100	Diversity = 1.6152e+00	k_t
= [3, 2, 4, 4]				
Generation 1010:	Best = 4.387224e+03	FES = 101100	Diversity = 1.6076e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 4.385957e+03	FES = 102100	Diversity = 1.6045e+00	k_t
= [2, 3, 4, 3]				
Generation 1030:	Best = 4.384362e+03	FES = 103100	Diversity = 1.5754e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 4.382155e+03 = [3, 2, 3, 2]	FES = 104100	Diversity = 1.5694e+00	k_t
Generation 1050: Best = 4.380409e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.5659e+00	k_t
Generation 1060: Best = 4.371161e+03 = [3, 2, 4, 3]	FES = 106100	Diversity = 1.5767e+00	k_t
Generation 1070: Best = 4.369589e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 1.5680e+00	k_t
Generation 1080: Best = 4.357418e+03 = [2, 3, 2, 2]	FES = 108100	Diversity = 1.5638e+00	k_t
Generation 1090: Best = 4.350382e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.5059e+00	k_t
Generation 1100: Best = 4.341603e+03 = [3, 2, 4, 2]	FES = 110100	Diversity = 1.4209e+00	k_t
Generation 1110: Best = 4.326919e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.2285e+00	k_t
Generation 1120: Best = 4.310235e+03 = [4, 3, 3, 4]	FES = 112100	Diversity = 8.5657e-01	k_t
Generation 1130: Best = 4.304823e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 6.6104e-01	k_t
Generation 1140: Best = 4.295320e+03 = [3, 2, 2, 2]	FES = 114100	Diversity = 5.5693e-01	k_t
Generation 1150: Best = 4.283134e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 5.0654e-01	k_t
Generation 1160: Best = 4.274827e+03 = [4, 4, 2, 2]	FES = 116100	Diversity = 4.3787e-01	k_t
Generation 1170: Best = 4.264300e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 4.1239e-01	k_t
Generation 1180: Best = 4.260007e+03 = [2, 4, 3, 3]	FES = 118100	Diversity = 3.3685e-01	k_t
Generation 1190: Best = 4.254697e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 2.9062e-01	k_t
Generation 1200: Best = 4.250968e+03 = [3, 3, 2, 3]	FES = 120100	Diversity = 2.8084e-01	k_t
Generation 1210: Best = 4.244332e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 2.7083e-01	k_t
Generation 1220: Best = 4.241710e+03 = [3, 3, 2, 4]	FES = 122100	Diversity = 2.5472e-01	k_t
Generation 1230: Best = 4.237687e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 2.3801e-01	k_t
Generation 1240: Best = 4.234239e+03 = [4, 2, 3, 3]	FES = 124100	Diversity = 2.3108e-01	k_t
Generation 1250: Best = 4.230259e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 2.2824e-01	k_t
Generation 1260: Best = 4.228537e+03 = [4, 2, 4, 4]	FES = 126100	Diversity = 2.0022e-01	k_t
Generation 1270: Best = 4.225969e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.9335e-01	k_t
Generation 1280: Best = 4.223111e+03 = [2, 4, 3, 2]	FES = 128100	Diversity = 1.8370e-01	k_t
Generation 1290: Best = 4.220666e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.7998e-01	k_t
Generation 1300: Best = 4.219450e+03 = [2, 2, 3, 3]	FES = 130100	Diversity = 1.6078e-01	k_t

Generation 1310: Best = 4.218316e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.4383e-01	k_t
Generation 1320: Best = 4.216798e+03 = [4, 2, 3, 4]	FES = 132100	Diversity = 1.3340e-01	k_t
Generation 1330: Best = 4.216050e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 1.2157e-01	k_t
Generation 1340: Best = 4.215133e+03 = [2, 4, 3, 3]	FES = 134100	Diversity = 1.0800e-01	k_t
Generation 1350: Best = 4.214487e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.0047e-01	k_t
Generation 1360: Best = 4.213705e+03 = [2, 3, 4, 2]	FES = 136100	Diversity = 9.2897e-02	k_t
Generation 1370: Best = 4.213007e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 8.8403e-02	k_t
Generation 1380: Best = 4.212608e+03 = [2, 2, 3, 4]	FES = 138100	Diversity = 8.5308e-02	k_t
Generation 1390: Best = 4.211831e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 8.0847e-02	k_t
Generation 1400: Best = 4.211225e+03 = [4, 2, 2, 3]	FES = 140100	Diversity = 8.0337e-02	k_t
Generation 1410: Best = 4.210772e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 7.2761e-02	k_t
Generation 1420: Best = 4.210234e+03 = [4, 4, 4, 2]	FES = 142100	Diversity = 7.4867e-02	k_t
Generation 1430: Best = 4.209725e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 7.4992e-02	k_t
Generation 1440: Best = 4.209211e+03 = [2, 4, 2, 2]	FES = 144100	Diversity = 7.7092e-02	k_t
Generation 1450: Best = 4.208702e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 7.6811e-02	k_t
Generation 1460: Best = 4.208105e+03 = [4, 2, 4, 3]	FES = 146100	Diversity = 7.6605e-02	k_t
Generation 1470: Best = 4.207572e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 7.5196e-02	k_t
Generation 1480: Best = 4.207150e+03 = [2, 3, 3, 3]	FES = 148100	Diversity = 7.5828e-02	k_t
Generation 1490: Best = 4.206440e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 7.8773e-02	k_t
Generation 1500: Best = 4.205883e+03 = [3, 4, 3, 3]	FES = 150100	Diversity = 7.9487e-02	k_t
Generation 1510: Best = 4.205425e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 8.3442e-02	k_t
Generation 1520: Best = 4.204737e+03 = [3, 4, 2, 4]	FES = 152100	Diversity = 7.7511e-02	k_t
Generation 1530: Best = 4.204103e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 7.8576e-02	k_t
Generation 1540: Best = 4.203278e+03 = [2, 3, 2, 2]	FES = 154100	Diversity = 8.8681e-02	k_t
Generation 1550: Best = 4.202322e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 8.8421e-02	k_t
Generation 1560: Best = 4.201830e+03 = [3, 4, 4, 2]	FES = 156100	Diversity = 9.1843e-02	k_t
Generation 1570: Best = 4.201357e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 8.1687e-02	k_t

Generation 1580: Best = 4.200548e+03 = [4, 2, 2, 3]	FES = 158100	Diversity = 8.1351e-02	k_t
Generation 1590: Best = 4.199990e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 8.3989e-02	k_t
Generation 1600: Best = 4.199181e+03 = [4, 4, 3, 2]	FES = 160100	Diversity = 8.3926e-02	k_t
Generation 1610: Best = 4.198448e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 9.5862e-02	k_t
Generation 1620: Best = 4.197783e+03 = [2, 3, 2, 4]	FES = 162100	Diversity = 8.8987e-02	k_t
Generation 1630: Best = 4.196976e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 8.2343e-02	k_t
Generation 1640: Best = 4.196364e+03 = [2, 4, 4, 4]	FES = 164100	Diversity = 7.7873e-02	k_t
Generation 1650: Best = 4.195418e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 8.2180e-02	k_t
Generation 1660: Best = 4.194608e+03 = [3, 3, 2, 2]	FES = 166100	Diversity = 8.6783e-02	k_t
Generation 1670: Best = 4.193359e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 9.6018e-02	k_t
Generation 1680: Best = 4.192614e+03 = [3, 3, 4, 4]	FES = 168100	Diversity = 9.1713e-02	k_t
Generation 1690: Best = 4.191594e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 9.7771e-02	k_t
Generation 1700: Best = 4.189910e+03 = [2, 3, 2, 2]	FES = 170100	Diversity = 1.1003e-01	k_t
Generation 1710: Best = 4.188635e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.1950e-01	k_t
Generation 1720: Best = 4.186861e+03 = [3, 4, 4, 4]	FES = 172100	Diversity = 1.1790e-01	k_t
Generation 1730: Best = 4.185685e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 1.1193e-01	k_t
Generation 1740: Best = 4.184550e+03 = [3, 2, 2, 2]	FES = 174100	Diversity = 1.0871e-01	k_t
Generation 1750: Best = 4.183154e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 1.1042e-01	k_t
Generation 1760: Best = 4.182008e+03 = [2, 4, 2, 2]	FES = 176100	Diversity = 1.0890e-01	k_t
Generation 1770: Best = 4.180844e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 1.1584e-01	k_t
Generation 1780: Best = 4.179241e+03 = [2, 4, 3, 4]	FES = 178100	Diversity = 1.2169e-01	k_t
Generation 1790: Best = 4.177936e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.2596e-01	k_t
Generation 1800: Best = 4.176399e+03 = [4, 2, 4, 2]	FES = 180100	Diversity = 1.2115e-01	k_t
Generation 1810: Best = 4.175450e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 1.1488e-01	k_t
Generation 1820: Best = 4.174227e+03 = [4, 3, 4, 2]	FES = 182100	Diversity = 1.1535e-01	k_t
Generation 1830: Best = 4.173141e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.1163e-01	k_t
Generation 1840: Best = 4.172382e+03 = [4, 3, 4, 2]	FES = 184100	Diversity = 9.5538e-02	k_t

Generation 1850: Best = 4.171686e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 9.2363e-02	k_t
Generation 1860: Best = 4.171111e+03 = [4, 2, 3, 3]	FES = 186100	Diversity = 8.2589e-02	k_t
Generation 1870: Best = 4.170450e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 7.9878e-02	k_t
Generation 1880: Best = 4.169958e+03 = [4, 3, 2, 3]	FES = 188100	Diversity = 7.1244e-02	k_t
Generation 1890: Best = 4.169554e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 6.4684e-02	k_t
Generation 1900: Best = 4.169119e+03 = [2, 2, 3, 3]	FES = 190100	Diversity = 6.3861e-02	k_t
Generation 1910: Best = 4.168693e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 6.1314e-02	k_t
Generation 1920: Best = 4.168274e+03 = [4, 4, 2, 2]	FES = 192100	Diversity = 6.0717e-02	k_t
Generation 1930: Best = 4.167991e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 6.0340e-02	k_t
Generation 1940: Best = 4.167733e+03 = [2, 4, 4, 4]	FES = 194100	Diversity = 5.4654e-02	k_t
Generation 1950: Best = 4.167500e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 4.6099e-02	k_t
Generation 1960: Best = 4.167322e+03 = [2, 3, 3, 3]	FES = 196100	Diversity = 4.4789e-02	k_t
Generation 1970: Best = 4.167157e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 4.1118e-02	k_t
Generation 1980: Best = 4.166973e+03 = [2, 3, 3, 2]	FES = 198100	Diversity = 4.0180e-02	k_t
Generation 1990: Best = 4.166833e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 3.6111e-02	k_t
Generation 2000: Best = 4.166656e+03 = [2, 2, 2, 3]	FES = 200100	Diversity = 3.8221e-02	k_t
Generation 2010: Best = 4.166534e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 3.4658e-02	k_t
Generation 2020: Best = 4.166453e+03 = [4, 2, 3, 3]	FES = 202100	Diversity = 3.1162e-02	k_t
Generation 2030: Best = 4.166368e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 2.9277e-02	k_t
Generation 2040: Best = 4.166286e+03 = [2, 3, 2, 3]	FES = 204100	Diversity = 2.7493e-02	k_t
Generation 2050: Best = 4.166209e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 2.6380e-02	k_t
Generation 2060: Best = 4.166146e+03 = [3, 2, 2, 4]	FES = 206100	Diversity = 2.5772e-02	k_t
Generation 2070: Best = 4.166075e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 2.6082e-02	k_t
Generation 2080: Best = 4.166023e+03 = [2, 4, 3, 4]	FES = 208100	Diversity = 2.3235e-02	k_t
Generation 2090: Best = 4.165975e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 2.1578e-02	k_t
Generation 2100: Best = 4.165951e+03 = [4, 2, 3, 4]	FES = 210100	Diversity = 2.0567e-02	k_t
Generation 2110: Best = 4.165906e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.9722e-02	k_t

Generation 2120: Best = 4.165886e+03 = [4, 2, 3, 4]	FES = 212100	Diversity = 1.5753e-02	k_t
Generation 2130: Best = 4.165871e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 1.4151e-02	k_t
Generation 2140: Best = 4.165846e+03 = [3, 4, 4, 3]	FES = 214100	Diversity = 1.2804e-02	k_t
Generation 2150: Best = 4.165825e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 1.3132e-02	k_t
Generation 2160: Best = 4.165804e+03 = [2, 3, 4, 3]	FES = 216100	Diversity = 1.3844e-02	k_t
Generation 2170: Best = 4.165779e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 1.3467e-02	k_t
Generation 2180: Best = 4.165765e+03 = [2, 4, 4, 4]	FES = 218100	Diversity = 1.3185e-02	k_t
Generation 2190: Best = 4.165745e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 1.2407e-02	k_t
Generation 2200: Best = 4.165732e+03 = [2, 2, 4, 3]	FES = 220100	Diversity = 1.1694e-02	k_t
Generation 2210: Best = 4.165709e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.2430e-02	k_t
Generation 2220: Best = 4.165698e+03 = [2, 4, 2, 3]	FES = 222100	Diversity = 1.2189e-02	k_t
Generation 2230: Best = 4.165677e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 1.1981e-02	k_t
Generation 2240: Best = 4.165660e+03 = [4, 2, 3, 4]	FES = 224100	Diversity = 1.2384e-02	k_t
Generation 2250: Best = 4.165643e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.2332e-02	k_t
Generation 2260: Best = 4.165628e+03 = [3, 4, 4, 3]	FES = 226100	Diversity = 1.2120e-02	k_t
Generation 2270: Best = 4.165613e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.2803e-02	k_t
Generation 2280: Best = 4.165599e+03 = [3, 2, 3, 3]	FES = 228100	Diversity = 1.2134e-02	k_t
Generation 2290: Best = 4.165579e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.3009e-02	k_t
Generation 2300: Best = 4.165560e+03 = [2, 2, 2, 2]	FES = 230100	Diversity = 1.3668e-02	k_t
Generation 2310: Best = 4.165544e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.2589e-02	k_t
Generation 2320: Best = 4.165529e+03 = [4, 2, 4, 2]	FES = 232100	Diversity = 1.2435e-02	k_t
Generation 2330: Best = 4.165519e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 1.1416e-02	k_t
Generation 2340: Best = 4.165503e+03 = [3, 2, 4, 2]	FES = 234100	Diversity = 1.1230e-02	k_t
Generation 2350: Best = 4.165488e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 1.0929e-02	k_t
Generation 2360: Best = 4.165474e+03 = [4, 2, 3, 2]	FES = 236100	Diversity = 1.0340e-02	k_t
Generation 2370: Best = 4.165464e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 9.4314e-03	k_t
Generation 2380: Best = 4.165450e+03 = [3, 4, 2, 2]	FES = 238100	Diversity = 9.8630e-03	k_t

Generation 2390: Best = 4.165438e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 9.9960e-03	k_t
Generation 2400: Best = 4.165423e+03 = [3, 4, 4, 3]	FES = 240100	Diversity = 1.0242e-02	k_t
Generation 2410: Best = 4.165411e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 1.1095e-02	k_t
Generation 2420: Best = 4.165394e+03 = [3, 2, 3, 4]	FES = 242100	Diversity = 1.0743e-02	k_t
Generation 2430: Best = 4.165376e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 1.0334e-02	k_t
Generation 2440: Best = 4.165360e+03 = [2, 4, 3, 2]	FES = 244100	Diversity = 1.0232e-02	k_t
Generation 2450: Best = 4.165346e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 1.0670e-02	k_t
Generation 2460: Best = 4.165334e+03 = [3, 3, 4, 3]	FES = 246100	Diversity = 1.1209e-02	k_t
Generation 2470: Best = 4.165320e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 1.1366e-02	k_t
Generation 2480: Best = 4.165307e+03 = [3, 4, 3, 3]	FES = 248100	Diversity = 1.1013e-02	k_t
Generation 2490: Best = 4.165298e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.0341e-02	k_t
Generation 2500: Best = 4.165286e+03 = [4, 3, 3, 3]	FES = 250100	Diversity = 8.8773e-03	k_t
Generation 2510: Best = 4.165276e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 8.0981e-03	k_t
Generation 2520: Best = 4.165268e+03 = [3, 3, 4, 2]	FES = 252100	Diversity = 8.2768e-03	k_t
Generation 2530: Best = 4.165257e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 8.7187e-03	k_t
Generation 2540: Best = 4.165248e+03 = [3, 2, 4, 3]	FES = 254100	Diversity = 8.1601e-03	k_t
Generation 2550: Best = 4.165239e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 8.5934e-03	k_t
Generation 2560: Best = 4.165226e+03 = [3, 3, 4, 3]	FES = 256100	Diversity = 9.2490e-03	k_t
Generation 2570: Best = 4.165221e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 7.4655e-03	k_t
Generation 2580: Best = 4.165211e+03 = [3, 4, 4, 3]	FES = 258100	Diversity = 7.6306e-03	k_t
Generation 2590: Best = 4.165200e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 8.9957e-03	k_t
Generation 2600: Best = 4.165184e+03 = [3, 4, 2, 2]	FES = 260100	Diversity = 8.6221e-03	k_t
Generation 2610: Best = 4.165176e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 8.5431e-03	k_t
Generation 2620: Best = 4.165164e+03 = [4, 4, 4, 4]	FES = 262100	Diversity = 9.0761e-03	k_t
Generation 2630: Best = 4.165152e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 8.7001e-03	k_t
Generation 2640: Best = 4.165139e+03 = [3, 4, 3, 2]	FES = 264100	Diversity = 9.5044e-03	k_t
Generation 2650: Best = 4.165126e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 9.2387e-03	k_t

Generation 2660: Best = 4.165115e+03 = [2, 2, 3, 3]	FES = 266100	Diversity = 8.7908e-03	k_t
Generation 2670: Best = 4.165102e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 9.7183e-03	k_t
Generation 2680: Best = 4.165090e+03 = [2, 3, 3, 2]	FES = 268100	Diversity = 9.6959e-03	k_t
Generation 2690: Best = 4.165079e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 8.4674e-03	k_t
Generation 2700: Best = 4.165069e+03 = [2, 4, 3, 4]	FES = 270100	Diversity = 7.2453e-03	k_t
Generation 2710: Best = 4.165063e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 6.7848e-03	k_t
Generation 2720: Best = 4.165054e+03 = [3, 3, 3, 4]	FES = 272100	Diversity = 6.3949e-03	k_t
Generation 2730: Best = 4.165046e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 6.6406e-03	k_t
Generation 2740: Best = 4.165039e+03 = [4, 4, 4, 3]	FES = 274100	Diversity = 6.0894e-03	k_t
Generation 2750: Best = 4.165033e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 5.5248e-03	k_t
Generation 2760: Best = 4.165027e+03 = [3, 2, 4, 2]	FES = 276100	Diversity = 5.8300e-03	k_t
Generation 2770: Best = 4.165018e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 6.9375e-03	k_t
Generation 2780: Best = 4.165009e+03 = [3, 4, 4, 2]	FES = 278100	Diversity = 7.7186e-03	k_t
Generation 2790: Best = 4.164997e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 8.4502e-03	k_t
Generation 2800: Best = 4.164989e+03 = [2, 2, 3, 4]	FES = 280100	Diversity = 7.3229e-03	k_t
Generation 2810: Best = 4.164980e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 7.7535e-03	k_t
Generation 2820: Best = 4.164972e+03 = [4, 4, 3, 3]	FES = 282100	Diversity = 8.3191e-03	k_t
Generation 2830: Best = 4.164964e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 7.7883e-03	k_t
Generation 2840: Best = 4.164956e+03 = [3, 3, 2, 2]	FES = 284100	Diversity = 7.4197e-03	k_t
Generation 2850: Best = 4.164948e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 6.5614e-03	k_t
Generation 2860: Best = 4.164943e+03 = [2, 2, 3, 4]	FES = 286100	Diversity = 5.6214e-03	k_t
Generation 2870: Best = 4.164935e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 6.1588e-03	k_t
Generation 2880: Best = 4.164927e+03 = [4, 4, 3, 3]	FES = 288100	Diversity = 5.9660e-03	k_t
Generation 2890: Best = 4.164922e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 6.0864e-03	k_t
Generation 2900: Best = 4.164910e+03 = [3, 3, 2, 4]	FES = 290100	Diversity = 6.9803e-03	k_t
Generation 2910: Best = 4.164903e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 6.9045e-03	k_t
Generation 2920: Best = 4.164897e+03 = [2, 4, 4, 4]	FES = 292100	Diversity = 5.9239e-03	k_t

Generation 2930: Best = 4.164890e+03 = [1, 1, 1, 1]	FES = 293100	Diversity = 5.4222e-03	k_t
Generation 2940: Best = 4.164885e+03 = [2, 3, 2, 4]	FES = 294100	Diversity = 4.9571e-03	k_t
Generation 2950: Best = 4.164879e+03 = [1, 1, 1, 1]	FES = 295100	Diversity = 4.9860e-03	k_t
Generation 2960: Best = 4.164874e+03 = [4, 2, 2, 4]	FES = 296100	Diversity = 4.5124e-03	k_t
Generation 2970: Best = 4.164870e+03 = [1, 1, 1, 1]	FES = 297100	Diversity = 4.4748e-03	k_t
Generation 2980: Best = 4.164865e+03 = [3, 3, 4, 3]	FES = 298100	Diversity = 4.4956e-03	k_t
Generation 2990: Best = 4.164860e+03 = [1, 1, 1, 1]	FES = 299100	Diversity = 4.6145e-03	k_t
Generation 3000: Best = 4.164855e+03 = [2, 4, 3, 4]	FES = 300100	Diversity = 4.8375e-03	k_t
Generation 3010: Best = 4.164850e+03 = [1, 1, 1, 1]	FES = 301100	Diversity = 4.8690e-03	k_t
Generation 3020: Best = 4.164844e+03 = [4, 4, 4, 2]	FES = 302100	Diversity = 5.6742e-03	k_t
Generation 3030: Best = 4.164838e+03 = [1, 1, 1, 1]	FES = 303100	Diversity = 6.1101e-03	k_t
Generation 3040: Best = 4.164830e+03 = [4, 2, 3, 4]	FES = 304100	Diversity = 6.4218e-03	k_t
Generation 3050: Best = 4.164825e+03 = [1, 1, 1, 1]	FES = 305100	Diversity = 6.2254e-03	k_t
Generation 3060: Best = 4.164818e+03 = [3, 3, 2, 4]	FES = 306100	Diversity = 6.2798e-03	k_t
Generation 3070: Best = 4.164812e+03 = [1, 1, 1, 1]	FES = 307100	Diversity = 6.1765e-03	k_t
Generation 3080: Best = 4.164807e+03 = [2, 4, 3, 4]	FES = 308100	Diversity = 5.6837e-03	k_t
Generation 3090: Best = 4.164801e+03 = [1, 1, 1, 1]	FES = 309100	Diversity = 5.3276e-03	k_t
Generation 3100: Best = 4.164798e+03 = [3, 2, 3, 2]	FES = 310100	Diversity = 4.5287e-03	k_t
Generation 3110: Best = 4.164794e+03 = [1, 1, 1, 1]	FES = 311100	Diversity = 4.4591e-03	k_t
Generation 3120: Best = 4.164791e+03 = [4, 2, 3, 4]	FES = 312100	Diversity = 3.7446e-03	k_t
Generation 3130: Best = 4.164787e+03 = [1, 1, 1, 1]	FES = 313100	Diversity = 3.5037e-03	k_t
Generation 3140: Best = 4.164785e+03 = [4, 3, 4, 3]	FES = 314100	Diversity = 3.6092e-03	k_t
Generation 3150: Best = 4.164782e+03 = [1, 1, 1, 1]	FES = 315100	Diversity = 3.8435e-03	k_t
Generation 3160: Best = 4.164778e+03 = [2, 2, 4, 3]	FES = 316100	Diversity = 3.7485e-03	k_t
Generation 3170: Best = 4.164775e+03 = [1, 1, 1, 1]	FES = 317100	Diversity = 3.8751e-03	k_t
Generation 3180: Best = 4.164770e+03 = [2, 4, 2, 2]	FES = 318100	Diversity = 3.8357e-03	k_t
Generation 3190: Best = 4.164766e+03 = [1, 1, 1, 1]	FES = 319100	Diversity = 4.6843e-03	k_t

Generation 3200: Best = 4.164760e+03 = [2, 4, 2, 2]	FES = 320100	Diversity = 6.0130e-03	k_t
Generation 3210: Best = 4.164753e+03 = [1, 1, 1, 1]	FES = 321100	Diversity = 7.5869e-03	k_t
Generation 3220: Best = 4.164746e+03 = [2, 3, 4, 2]	FES = 322100	Diversity = 6.3951e-03	k_t
Generation 3230: Best = 4.164739e+03 = [1, 1, 1, 1]	FES = 323100	Diversity = 6.0034e-03	k_t
Generation 3240: Best = 4.164732e+03 = [3, 3, 2, 3]	FES = 324100	Diversity = 5.8754e-03	k_t
Generation 3250: Best = 4.164726e+03 = [1, 1, 1, 1]	FES = 325100	Diversity = 6.0444e-03	k_t
Generation 3260: Best = 4.164720e+03 = [2, 2, 4, 4]	FES = 326100	Diversity = 6.0463e-03	k_t
Generation 3270: Best = 4.164713e+03 = [1, 1, 1, 1]	FES = 327100	Diversity = 5.5686e-03	k_t
Generation 3280: Best = 4.164703e+03 = [2, 4, 3, 2]	FES = 328100	Diversity = 6.4876e-03	k_t
Generation 3290: Best = 4.164697e+03 = [1, 1, 1, 1]	FES = 329100	Diversity = 6.1878e-03	k_t
Generation 3300: Best = 4.164689e+03 = [3, 4, 2, 2]	FES = 330100	Diversity = 6.1129e-03	k_t
Generation 3310: Best = 4.164682e+03 = [1, 1, 1, 1]	FES = 331100	Diversity = 6.7106e-03	k_t
Generation 3320: Best = 4.164675e+03 = [2, 4, 2, 4]	FES = 332100	Diversity = 6.1294e-03	k_t
Generation 3330: Best = 4.164668e+03 = [1, 1, 1, 1]	FES = 333100	Diversity = 6.4189e-03	k_t
Generation 3340: Best = 4.164661e+03 = [4, 3, 2, 3]	FES = 334100	Diversity = 5.9344e-03	k_t
Generation 3350: Best = 4.164655e+03 = [1, 1, 1, 1]	FES = 335100	Diversity = 5.0133e-03	k_t
Generation 3360: Best = 4.164651e+03 = [2, 4, 4, 2]	FES = 336100	Diversity = 4.4024e-03	k_t
Generation 3370: Best = 4.164645e+03 = [1, 1, 1, 1]	FES = 337100	Diversity = 4.1951e-03	k_t
Generation 3380: Best = 4.164639e+03 = [2, 3, 2, 4]	FES = 338100	Diversity = 4.2118e-03	k_t
Generation 3390: Best = 4.164634e+03 = [1, 1, 1, 1]	FES = 339100	Diversity = 4.4451e-03	k_t
Generation 3400: Best = 4.164628e+03 = [3, 3, 3, 3]	FES = 340100	Diversity = 4.0354e-03	k_t
Generation 3410: Best = 4.164623e+03 = [1, 1, 1, 1]	FES = 341100	Diversity = 4.4185e-03	k_t
Generation 3420: Best = 4.164617e+03 = [4, 3, 2, 3]	FES = 342100	Diversity = 5.0539e-03	k_t
Generation 3430: Best = 4.164610e+03 = [1, 1, 1, 1]	FES = 343100	Diversity = 5.5316e-03	k_t
Generation 3440: Best = 4.164603e+03 = [4, 3, 4, 4]	FES = 344100	Diversity = 5.6189e-03	k_t
Generation 3450: Best = 4.164598e+03 = [1, 1, 1, 1]	FES = 345100	Diversity = 5.4533e-03	k_t
Generation 3460: Best = 4.164592e+03 = [4, 4, 2, 4]	FES = 346100	Diversity = 5.2215e-03	k_t

Generation 3470: Best = 4.164586e+03 = [1, 1, 1, 1]	FES = 347100	Diversity = 4.9533e-03	k_t
Generation 3480: Best = 4.164583e+03 = [3, 2, 3, 3]	FES = 348100	Diversity = 3.8588e-03	k_t
Generation 3490: Best = 4.164580e+03 = [1, 1, 1, 1]	FES = 349100	Diversity = 3.3915e-03	k_t
Generation 3500: Best = 4.164577e+03 = [2, 3, 3, 2]	FES = 350100	Diversity = 3.1827e-03	k_t
Generation 3510: Best = 4.164574e+03 = [1, 1, 1, 1]	FES = 351100	Diversity = 3.3687e-03	k_t
Generation 3520: Best = 4.164572e+03 = [3, 2, 2, 4]	FES = 352100	Diversity = 2.9279e-03	k_t
Generation 3530: Best = 4.164569e+03 = [1, 1, 1, 1]	FES = 353100	Diversity = 3.8609e-03	k_t
Generation 3540: Best = 4.164565e+03 = [3, 4, 2, 4]	FES = 354100	Diversity = 4.2051e-03	k_t
Generation 3550: Best = 4.164560e+03 = [1, 1, 1, 1]	FES = 355100	Diversity = 4.6854e-03	k_t
Generation 3560: Best = 4.164556e+03 = [4, 4, 3, 4]	FES = 356100	Diversity = 4.3399e-03	k_t
Generation 3570: Best = 4.164553e+03 = [1, 1, 1, 1]	FES = 357100	Diversity = 3.9351e-03	k_t
Generation 3580: Best = 4.164549e+03 = [2, 2, 4, 4]	FES = 358100	Diversity = 3.6213e-03	k_t
Generation 3590: Best = 4.164547e+03 = [1, 1, 1, 1]	FES = 359100	Diversity = 3.4703e-03	k_t
Generation 3600: Best = 4.164544e+03 = [2, 3, 4, 3]	FES = 360100	Diversity = 3.1467e-03	k_t
Generation 3610: Best = 4.164541e+03 = [1, 1, 1, 1]	FES = 361100	Diversity = 3.2039e-03	k_t
Generation 3620: Best = 4.164539e+03 = [4, 3, 4, 2]	FES = 362100	Diversity = 3.4142e-03	k_t
Generation 3630: Best = 4.164532e+03 = [1, 1, 1, 1]	FES = 363100	Diversity = 4.2171e-03	k_t
Generation 3640: Best = 4.164529e+03 = [4, 3, 3, 2]	FES = 364100	Diversity = 4.3500e-03	k_t
Generation 3650: Best = 4.164523e+03 = [1, 1, 1, 1]	FES = 365100	Diversity = 4.3861e-03	k_t
Generation 3660: Best = 4.164519e+03 = [4, 4, 2, 4]	FES = 366100	Diversity = 4.8096e-03	k_t
Generation 3670: Best = 4.164513e+03 = [1, 1, 1, 1]	FES = 367100	Diversity = 5.1699e-03	k_t
Generation 3680: Best = 4.164509e+03 = [3, 4, 3, 2]	FES = 368100	Diversity = 4.9501e-03	k_t
Generation 3690: Best = 4.164504e+03 = [1, 1, 1, 1]	FES = 369100	Diversity = 4.6038e-03	k_t
Generation 3700: Best = 4.164500e+03 = [4, 3, 2, 4]	FES = 370100	Diversity = 4.7782e-03	k_t
Generation 3710: Best = 4.164497e+03 = [1, 1, 1, 1]	FES = 371100	Diversity = 3.8054e-03	k_t
Generation 3720: Best = 4.164493e+03 = [3, 2, 4, 3]	FES = 372100	Diversity = 3.8065e-03	k_t
Generation 3730: Best = 4.164489e+03 = [1, 1, 1, 1]	FES = 373100	Diversity = 3.4110e-03	k_t

Generation 3740: Best = 4.164486e+03 = [4, 3, 4, 3]	FES = 374100	Diversity = 3.4449e-03	k_t
Generation 3750: Best = 4.164483e+03 = [1, 1, 1, 1]	FES = 375100	Diversity = 3.3412e-03	k_t
Generation 3760: Best = 4.164480e+03 = [2, 4, 2, 3]	FES = 376100	Diversity = 3.5340e-03	k_t
Generation 3770: Best = 4.164477e+03 = [1, 1, 1, 1]	FES = 377100	Diversity = 3.8116e-03	k_t
Generation 3780: Best = 4.164475e+03 = [2, 3, 4, 3]	FES = 378100	Diversity = 3.1363e-03	k_t
Generation 3790: Best = 4.164472e+03 = [1, 1, 1, 1]	FES = 379100	Diversity = 3.6504e-03	k_t
Generation 3800: Best = 4.164468e+03 = [2, 4, 4, 3]	FES = 380100	Diversity = 3.7456e-03	k_t
Generation 3810: Best = 4.164462e+03 = [1, 1, 1, 1]	FES = 381100	Diversity = 4.0174e-03	k_t
Generation 3820: Best = 4.164457e+03 = [3, 2, 2, 4]	FES = 382100	Diversity = 3.9878e-03	k_t
Generation 3830: Best = 4.164452e+03 = [1, 1, 1, 1]	FES = 383100	Diversity = 4.9336e-03	k_t
Generation 3840: Best = 4.164447e+03 = [4, 4, 4, 4]	FES = 384100	Diversity = 4.3640e-03	k_t
Generation 3850: Best = 4.164444e+03 = [1, 1, 1, 1]	FES = 385100	Diversity = 4.5740e-03	k_t
Generation 3860: Best = 4.164439e+03 = [2, 3, 4, 3]	FES = 386100	Diversity = 4.1540e-03	k_t
Generation 3870: Best = 4.164436e+03 = [1, 1, 1, 1]	FES = 387100	Diversity = 3.6969e-03	k_t
Generation 3880: Best = 4.164432e+03 = [2, 3, 3, 3]	FES = 388100	Diversity = 3.9475e-03	k_t
Generation 3890: Best = 4.164426e+03 = [1, 1, 1, 1]	FES = 389100	Diversity = 4.7873e-03	k_t
Generation 3900: Best = 4.164421e+03 = [4, 3, 3, 3]	FES = 390100	Diversity = 4.8434e-03	k_t
Generation 3910: Best = 4.164416e+03 = [1, 1, 1, 1]	FES = 391100	Diversity = 4.7595e-03	k_t
Generation 3920: Best = 4.164411e+03 = [4, 2, 4, 3]	FES = 392100	Diversity = 4.6249e-03	k_t
Generation 3930: Best = 4.164406e+03 = [1, 1, 1, 1]	FES = 393100	Diversity = 4.5331e-03	k_t
Generation 3940: Best = 4.164401e+03 = [2, 3, 3, 4]	FES = 394100	Diversity = 5.1814e-03	k_t
Generation 3950: Best = 4.164396e+03 = [1, 1, 1, 1]	FES = 395100	Diversity = 4.5826e-03	k_t
Generation 3960: Best = 4.164393e+03 = [3, 4, 3, 2]	FES = 396100	Diversity = 3.4967e-03	k_t
Generation 3970: Best = 4.164390e+03 = [1, 1, 1, 1]	FES = 397100	Diversity = 3.7036e-03	k_t
Generation 3980: Best = 4.164386e+03 = [4, 3, 4, 3]	FES = 398100	Diversity = 4.0214e-03	k_t
Generation 3990: Best = 4.164381e+03 = [1, 1, 1, 1]	FES = 399100	Diversity = 4.3626e-03	k_t
Generation 4000: Best = 4.164378e+03 = [4, 2, 4, 3]	FES = 400100	Diversity = 4.0177e-03	k_t

Generation 4010: Best = 4.164375e+03 = [1, 1, 1, 1]	FES = 401100	Diversity = 3.6517e-03	k_t
Generation 4020: Best = 4.164373e+03 = [2, 3, 2, 4]	FES = 402100	Diversity = 3.7050e-03	k_t
Generation 4030: Best = 4.164369e+03 = [1, 1, 1, 1]	FES = 403100	Diversity = 4.1678e-03	k_t
Generation 4040: Best = 4.164365e+03 = [3, 4, 3, 2]	FES = 404100	Diversity = 4.3806e-03	k_t
Generation 4050: Best = 4.164361e+03 = [1, 1, 1, 1]	FES = 405100	Diversity = 4.0930e-03	k_t
Generation 4060: Best = 4.164358e+03 = [3, 4, 4, 4]	FES = 406100	Diversity = 3.9224e-03	k_t
Generation 4070: Best = 4.164355e+03 = [1, 1, 1, 1]	FES = 407100	Diversity = 4.0343e-03	k_t
Generation 4080: Best = 4.164351e+03 = [4, 4, 3, 3]	FES = 408100	Diversity = 4.4547e-03	k_t
Generation 4090: Best = 4.164346e+03 = [1, 1, 1, 1]	FES = 409100	Diversity = 4.4023e-03	k_t
Generation 4100: Best = 4.164342e+03 = [4, 3, 4, 3]	FES = 410100	Diversity = 4.1715e-03	k_t
Generation 4110: Best = 4.164339e+03 = [1, 1, 1, 1]	FES = 411100	Diversity = 3.8045e-03	k_t
Generation 4120: Best = 4.164335e+03 = [2, 4, 4, 3]	FES = 412100	Diversity = 3.8175e-03	k_t
Generation 4130: Best = 4.164332e+03 = [1, 1, 1, 1]	FES = 413100	Diversity = 3.5190e-03	k_t
Generation 4140: Best = 4.164329e+03 = [3, 3, 2, 2]	FES = 414100	Diversity = 3.6919e-03	k_t
Generation 4150: Best = 4.164327e+03 = [1, 1, 1, 1]	FES = 415100	Diversity = 3.5348e-03	k_t
Generation 4160: Best = 4.164323e+03 = [2, 4, 4, 3]	FES = 416100	Diversity = 3.4237e-03	k_t
Generation 4170: Best = 4.164321e+03 = [1, 1, 1, 1]	FES = 417100	Diversity = 3.0799e-03	k_t
Generation 4180: Best = 4.164318e+03 = [2, 4, 2, 3]	FES = 418100	Diversity = 3.8901e-03	k_t
Generation 4190: Best = 4.164314e+03 = [1, 1, 1, 1]	FES = 419100	Diversity = 3.8443e-03	k_t
Generation 4200: Best = 4.164312e+03 = [4, 2, 4, 4]	FES = 420100	Diversity = 3.5840e-03	k_t
Generation 4210: Best = 4.164308e+03 = [1, 1, 1, 1]	FES = 421100	Diversity = 3.8309e-03	k_t
Generation 4220: Best = 4.164303e+03 = [4, 3, 3, 3]	FES = 422100	Diversity = 4.1860e-03	k_t
Generation 4230: Best = 4.164300e+03 = [1, 1, 1, 1]	FES = 423100	Diversity = 3.9134e-03	k_t
Generation 4240: Best = 4.164298e+03 = [3, 2, 4, 3]	FES = 424100	Diversity = 3.7948e-03	k_t
Generation 4250: Best = 4.164295e+03 = [1, 1, 1, 1]	FES = 425100	Diversity = 3.1996e-03	k_t
Generation 4260: Best = 4.164293e+03 = [2, 2, 3, 2]	FES = 426100	Diversity = 3.2625e-03	k_t
Generation 4270: Best = 4.164289e+03 = [1, 1, 1, 1]	FES = 427100	Diversity = 3.4618e-03	k_t

Generation 4280: Best = 4.164288e+03 = [4, 2, 4, 4]	FES = 428100	Diversity = 3.2602e-03	k_t
Generation 4290: Best = 4.164286e+03 = [1, 1, 1, 1]	FES = 429100	Diversity = 2.5348e-03	k_t
Generation 4300: Best = 4.164285e+03 = [3, 2, 3, 4]	FES = 430100	Diversity = 2.0059e-03	k_t
Generation 4310: Best = 4.164284e+03 = [1, 1, 1, 1]	FES = 431100	Diversity = 1.7463e-03	k_t
Generation 4320: Best = 4.164283e+03 = [2, 4, 4, 4]	FES = 432100	Diversity = 1.9321e-03	k_t
Generation 4330: Best = 4.164282e+03 = [1, 1, 1, 1]	FES = 433100	Diversity = 1.8075e-03	k_t
Generation 4340: Best = 4.164281e+03 = [3, 2, 3, 4]	FES = 434100	Diversity = 1.6226e-03	k_t
Generation 4350: Best = 4.164280e+03 = [1, 1, 1, 1]	FES = 435100	Diversity = 1.7047e-03	k_t
Generation 4360: Best = 4.164280e+03 = [3, 3, 4, 3]	FES = 436100	Diversity = 1.7328e-03	k_t
Generation 4370: Best = 4.164279e+03 = [1, 1, 1, 1]	FES = 437100	Diversity = 1.6571e-03	k_t
Generation 4380: Best = 4.164278e+03 = [2, 4, 3, 2]	FES = 438100	Diversity = 1.7646e-03	k_t
Generation 4390: Best = 4.164276e+03 = [1, 1, 1, 1]	FES = 439100	Diversity = 2.4929e-03	k_t
Generation 4400: Best = 4.164275e+03 = [2, 2, 4, 3]	FES = 440100	Diversity = 2.5736e-03	k_t
Generation 4410: Best = 4.164273e+03 = [1, 1, 1, 1]	FES = 441100	Diversity = 2.4347e-03	k_t
Generation 4420: Best = 4.164272e+03 = [4, 3, 2, 4]	FES = 442100	Diversity = 2.5082e-03	k_t
Generation 4430: Best = 4.164270e+03 = [1, 1, 1, 1]	FES = 443100	Diversity = 2.6232e-03	k_t
Generation 4440: Best = 4.164268e+03 = [4, 3, 2, 2]	FES = 444100	Diversity = 2.9011e-03	k_t
Generation 4450: Best = 4.164264e+03 = [1, 1, 1, 1]	FES = 445100	Diversity = 4.0940e-03	k_t
Generation 4460: Best = 4.164262e+03 = [4, 3, 2, 2]	FES = 446100	Diversity = 3.3791e-03	k_t
Generation 4470: Best = 4.164259e+03 = [1, 1, 1, 1]	FES = 447100	Diversity = 3.1422e-03	k_t
Generation 4480: Best = 4.164257e+03 = [4, 3, 3, 2]	FES = 448100	Diversity = 3.1300e-03	k_t
Generation 4490: Best = 4.164255e+03 = [1, 1, 1, 1]	FES = 449100	Diversity = 2.9390e-03	k_t
Generation 4500: Best = 4.164253e+03 = [4, 2, 2, 4]	FES = 450100	Diversity = 2.4284e-03	k_t
Generation 4510: Best = 4.164251e+03 = [1, 1, 1, 1]	FES = 451100	Diversity = 2.5231e-03	k_t
Generation 4520: Best = 4.164249e+03 = [4, 2, 4, 4]	FES = 452100	Diversity = 2.3289e-03	k_t
Generation 4530: Best = 4.164247e+03 = [1, 1, 1, 1]	FES = 453100	Diversity = 2.5009e-03	k_t
Generation 4540: Best = 4.164245e+03 = [4, 4, 2, 2]	FES = 454100	Diversity = 2.4753e-03	k_t

Generation 4550: Best = 4.164243e+03 = [1, 1, 1, 1]	FES = 455100	Diversity = 2.8308e-03	k_t
Generation 4560: Best = 4.164240e+03 = [3, 2, 2, 3]	FES = 456100	Diversity = 2.9546e-03	k_t
Generation 4570: Best = 4.164237e+03 = [1, 1, 1, 1]	FES = 457100	Diversity = 2.8414e-03	k_t
Generation 4580: Best = 4.164234e+03 = [2, 3, 3, 2]	FES = 458100	Diversity = 3.5391e-03	k_t
Generation 4590: Best = 4.164230e+03 = [1, 1, 1, 1]	FES = 459100	Diversity = 3.8207e-03	k_t
Generation 4600: Best = 4.164227e+03 = [3, 4, 4, 4]	FES = 460100	Diversity = 3.6363e-03	k_t
Generation 4610: Best = 4.164222e+03 = [1, 1, 1, 1]	FES = 461100	Diversity = 4.1547e-03	k_t
Generation 4620: Best = 4.164217e+03 = [4, 4, 4, 4]	FES = 462100	Diversity = 4.8982e-03	k_t
Generation 4630: Best = 4.164212e+03 = [1, 1, 1, 1]	FES = 463100	Diversity = 4.7437e-03	k_t
Generation 4640: Best = 4.164206e+03 = [3, 3, 2, 2]	FES = 464100	Diversity = 5.3197e-03	k_t
Generation 4650: Best = 4.164199e+03 = [1, 1, 1, 1]	FES = 465100	Diversity = 5.5014e-03	k_t
Generation 4660: Best = 4.164193e+03 = [3, 2, 4, 2]	FES = 466100	Diversity = 4.3948e-03	k_t
Generation 4670: Best = 4.164188e+03 = [1, 1, 1, 1]	FES = 467100	Diversity = 4.3257e-03	k_t
Generation 4680: Best = 4.164183e+03 = [3, 3, 3, 2]	FES = 468100	Diversity = 4.3096e-03	k_t
Generation 4690: Best = 4.164179e+03 = [1, 1, 1, 1]	FES = 469100	Diversity = 4.1440e-03	k_t
Generation 4700: Best = 4.164175e+03 = [2, 4, 2, 3]	FES = 470100	Diversity = 3.8861e-03	k_t
Generation 4710: Best = 4.164172e+03 = [1, 1, 1, 1]	FES = 471100	Diversity = 3.7709e-03	k_t
Generation 4720: Best = 4.164169e+03 = [2, 4, 2, 4]	FES = 472100	Diversity = 3.7024e-03	k_t
Generation 4730: Best = 4.164167e+03 = [1, 1, 1, 1]	FES = 473100	Diversity = 2.9112e-03	k_t
Generation 4740: Best = 4.164164e+03 = [3, 3, 4, 3]	FES = 474100	Diversity = 3.0024e-03	k_t
Generation 4750: Best = 4.164162e+03 = [1, 1, 1, 1]	FES = 475100	Diversity = 2.8233e-03	k_t
Generation 4760: Best = 4.164159e+03 = [2, 2, 3, 3]	FES = 476100	Diversity = 2.9649e-03	k_t
Generation 4770: Best = 4.164157e+03 = [1, 1, 1, 1]	FES = 477100	Diversity = 2.6947e-03	k_t
Generation 4780: Best = 4.164155e+03 = [3, 3, 2, 4]	FES = 478100	Diversity = 3.0694e-03	k_t
Generation 4790: Best = 4.164152e+03 = [1, 1, 1, 1]	FES = 479100	Diversity = 2.8768e-03	k_t
Generation 4800: Best = 4.164149e+03 = [2, 2, 4, 3]	FES = 480100	Diversity = 2.7388e-03	k_t
Generation 4810: Best = 4.164148e+03 = [1, 1, 1, 1]	FES = 481100	Diversity = 2.4874e-03	k_t

Generation 4820: Best = 4.164146e+03 = [4, 4, 4, 2]	FES = 482100	Diversity = 2.1608e-03	k_t
Generation 4830: Best = 4.164142e+03 = [1, 1, 1, 1]	FES = 483100	Diversity = 3.0943e-03	k_t
Generation 4840: Best = 4.164140e+03 = [4, 2, 3, 2]	FES = 484100	Diversity = 3.3033e-03	k_t
Generation 4850: Best = 4.164138e+03 = [1, 1, 1, 1]	FES = 485100	Diversity = 2.8408e-03	k_t
Generation 4860: Best = 4.164135e+03 = [4, 2, 3, 4]	FES = 486100	Diversity = 3.0210e-03	k_t
Generation 4870: Best = 4.164133e+03 = [1, 1, 1, 1]	FES = 487100	Diversity = 2.5905e-03	k_t
Generation 4880: Best = 4.164131e+03 = [3, 3, 3, 2]	FES = 488100	Diversity = 2.7453e-03	k_t
Generation 4890: Best = 4.164129e+03 = [1, 1, 1, 1]	FES = 489100	Diversity = 2.7062e-03	k_t
Generation 4900: Best = 4.164127e+03 = [4, 2, 4, 4]	FES = 490100	Diversity = 2.7104e-03	k_t
Generation 4910: Best = 4.164124e+03 = [1, 1, 1, 1]	FES = 491100	Diversity = 3.0951e-03	k_t
Generation 4920: Best = 4.164122e+03 = [2, 4, 3, 2]	FES = 492100	Diversity = 2.4607e-03	k_t
Generation 4930: Best = 4.164119e+03 = [1, 1, 1, 1]	FES = 493100	Diversity = 2.5625e-03	k_t
Generation 4940: Best = 4.164117e+03 = [4, 4, 2, 3]	FES = 494100	Diversity = 2.4216e-03	k_t
Generation 4950: Best = 4.164115e+03 = [1, 1, 1, 1]	FES = 495100	Diversity = 2.6611e-03	k_t
Generation 4960: Best = 4.164113e+03 = [4, 4, 2, 4]	FES = 496100	Diversity = 2.8200e-03	k_t
Generation 4970: Best = 4.164110e+03 = [1, 1, 1, 1]	FES = 497100	Diversity = 3.0037e-03	k_t
Generation 4980: Best = 4.164107e+03 = [3, 4, 4, 4]	FES = 498100	Diversity = 2.9327e-03	k_t
Generation 4990: Best = 4.164105e+03 = [1, 1, 1, 1]	FES = 499100	Diversity = 3.1019e-03	k_t
Generation 5000: Best = 4.164103e+03 = [2, 4, 2, 3]	FES = 500100	Diversity = 2.8298e-03	k_t
Generation 5010: Best = 4.164101e+03 = [1, 1, 1, 1]	FES = 501100	Diversity = 2.4390e-03	k_t
Generation 5020: Best = 4.164098e+03 = [3, 4, 4, 4]	FES = 502100	Diversity = 3.0783e-03	k_t
Generation 5030: Best = 4.164096e+03 = [1, 1, 1, 1]	FES = 503100	Diversity = 2.8923e-03	k_t
Generation 5040: Best = 4.164092e+03 = [2, 2, 2, 4]	FES = 504100	Diversity = 3.1821e-03	k_t
Generation 5050: Best = 4.164089e+03 = [1, 1, 1, 1]	FES = 505100	Diversity = 2.9420e-03	k_t
Generation 5060: Best = 4.164088e+03 = [3, 2, 4, 2]	FES = 506100	Diversity = 2.5248e-03	k_t
Generation 5070: Best = 4.164086e+03 = [1, 1, 1, 1]	FES = 507100	Diversity = 2.3110e-03	k_t
Generation 5080: Best = 4.164085e+03 = [3, 4, 2, 4]	FES = 508100	Diversity = 2.4636e-03	k_t

Generation 5090: Best = 4.164082e+03 = [1, 1, 1, 1]	FES = 509100	Diversity = 2.5114e-03	k_t
Generation 5100: Best = 4.164080e+03 = [3, 3, 2, 4]	FES = 510100	Diversity = 2.1864e-03	k_t
Generation 5110: Best = 4.164078e+03 = [1, 1, 1, 1]	FES = 511100	Diversity = 2.2964e-03	k_t
Generation 5120: Best = 4.164077e+03 = [4, 2, 3, 3]	FES = 512100	Diversity = 2.4486e-03	k_t
Generation 5130: Best = 4.164075e+03 = [1, 1, 1, 1]	FES = 513100	Diversity = 2.5879e-03	k_t
Generation 5140: Best = 4.164072e+03 = [2, 3, 4, 3]	FES = 514100	Diversity = 2.9163e-03	k_t
Generation 5150: Best = 4.164070e+03 = [1, 1, 1, 1]	FES = 515100	Diversity = 2.9667e-03	k_t
Generation 5160: Best = 4.164069e+03 = [4, 2, 4, 3]	FES = 516100	Diversity = 2.2422e-03	k_t
Generation 5170: Best = 4.164067e+03 = [1, 1, 1, 1]	FES = 517100	Diversity = 2.1720e-03	k_t
Generation 5180: Best = 4.164066e+03 = [4, 4, 2, 4]	FES = 518100	Diversity = 1.8934e-03	k_t
Generation 5190: Best = 4.164065e+03 = [1, 1, 1, 1]	FES = 519100	Diversity = 1.7813e-03	k_t
Generation 5200: Best = 4.164063e+03 = [2, 2, 4, 4]	FES = 520100	Diversity = 1.8739e-03	k_t
Generation 5210: Best = 4.164063e+03 = [1, 1, 1, 1]	FES = 521100	Diversity = 1.6054e-03	k_t
Generation 5220: Best = 4.164062e+03 = [3, 4, 3, 3]	FES = 522100	Diversity = 1.7102e-03	k_t
Generation 5230: Best = 4.164061e+03 = [1, 1, 1, 1]	FES = 523100	Diversity = 1.5489e-03	k_t
Generation 5240: Best = 4.164060e+03 = [3, 3, 2, 3]	FES = 524100	Diversity = 1.6042e-03	k_t
Generation 5250: Best = 4.164059e+03 = [1, 1, 1, 1]	FES = 525100	Diversity = 1.4237e-03	k_t
Generation 5260: Best = 4.164058e+03 = [2, 2, 4, 2]	FES = 526100	Diversity = 1.4443e-03	k_t
Generation 5270: Best = 4.164058e+03 = [1, 1, 1, 1]	FES = 527100	Diversity = 1.4140e-03	k_t
Generation 5280: Best = 4.164057e+03 = [2, 4, 3, 4]	FES = 528100	Diversity = 1.3089e-03	k_t
Generation 5290: Best = 4.164056e+03 = [1, 1, 1, 1]	FES = 529100	Diversity = 1.3878e-03	k_t
Generation 5300: Best = 4.164056e+03 = [4, 2, 4, 2]	FES = 530100	Diversity = 1.3655e-03	k_t
Generation 5310: Best = 4.164055e+03 = [1, 1, 1, 1]	FES = 531100	Diversity = 1.4645e-03	k_t
Generation 5320: Best = 4.164053e+03 = [4, 3, 2, 3]	FES = 532100	Diversity = 1.6373e-03	k_t
Generation 5330: Best = 4.164053e+03 = [1, 1, 1, 1]	FES = 533100	Diversity = 1.5290e-03	k_t
Generation 5340: Best = 4.164052e+03 = [4, 4, 4, 4]	FES = 534100	Diversity = 1.6369e-03	k_t
Generation 5350: Best = 4.164050e+03 = [1, 1, 1, 1]	FES = 535100	Diversity = 1.7863e-03	k_t

Generation 5360: Best = 4.164050e+03 = [2, 2, 2, 3]	FES = 536100	Diversity = 1.7205e-03	k_t
Generation 5370: Best = 4.164048e+03 = [1, 1, 1, 1]	FES = 537100	Diversity = 1.8138e-03	k_t
Generation 5380: Best = 4.164047e+03 = [4, 3, 4, 3]	FES = 538100	Diversity = 1.7315e-03	k_t
Generation 5390: Best = 4.164045e+03 = [1, 1, 1, 1]	FES = 539100	Diversity = 2.0485e-03	k_t
Generation 5400: Best = 4.164044e+03 = [4, 2, 3, 2]	FES = 540100	Diversity = 1.9479e-03	k_t
Generation 5410: Best = 4.164042e+03 = [1, 1, 1, 1]	FES = 541100	Diversity = 2.1674e-03	k_t
Generation 5420: Best = 4.164040e+03 = [2, 2, 3, 4]	FES = 542100	Diversity = 2.0064e-03	k_t
Generation 5430: Best = 4.164037e+03 = [1, 1, 1, 1]	FES = 543100	Diversity = 2.7381e-03	k_t
Generation 5440: Best = 4.164033e+03 = [2, 4, 4, 2]	FES = 544100	Diversity = 3.0777e-03	k_t
Generation 5450: Best = 4.164031e+03 = [1, 1, 1, 1]	FES = 545100	Diversity = 2.4380e-03	k_t
Generation 5460: Best = 4.164029e+03 = [3, 2, 4, 3]	FES = 546100	Diversity = 2.6263e-03	k_t
Generation 5470: Best = 4.164027e+03 = [1, 1, 1, 1]	FES = 547100	Diversity = 2.2550e-03	k_t
Generation 5480: Best = 4.164026e+03 = [4, 2, 4, 2]	FES = 548100	Diversity = 1.7363e-03	k_t
Generation 5490: Best = 4.164025e+03 = [1, 1, 1, 1]	FES = 549100	Diversity = 1.4281e-03	k_t
Generation 5500: Best = 4.164024e+03 = [2, 2, 2, 4]	FES = 550100	Diversity = 1.4106e-03	k_t
Generation 5510: Best = 4.164023e+03 = [1, 1, 1, 1]	FES = 551100	Diversity = 1.5236e-03	k_t
Generation 5520: Best = 4.164021e+03 = [2, 4, 3, 2]	FES = 552100	Diversity = 1.9560e-03	k_t
Generation 5530: Best = 4.164019e+03 = [1, 1, 1, 1]	FES = 553100	Diversity = 2.2138e-03	k_t
Generation 5540: Best = 4.164017e+03 = [4, 3, 2, 4]	FES = 554100	Diversity = 1.9718e-03	k_t
Generation 5550: Best = 4.164016e+03 = [1, 1, 1, 1]	FES = 555100	Diversity = 1.9639e-03	k_t
Generation 5560: Best = 4.164014e+03 = [3, 3, 4, 2]	FES = 556100	Diversity = 2.1503e-03	k_t
Generation 5570: Best = 4.164012e+03 = [1, 1, 1, 1]	FES = 557100	Diversity = 2.1599e-03	k_t
Generation 5580: Best = 4.164010e+03 = [3, 4, 3, 2]	FES = 558100	Diversity = 1.9449e-03	k_t
Generation 5590: Best = 4.164009e+03 = [1, 1, 1, 1]	FES = 559100	Diversity = 1.7982e-03	k_t
Generation 5600: Best = 4.164007e+03 = [4, 4, 2, 2]	FES = 560100	Diversity = 1.8046e-03	k_t
Generation 5610: Best = 4.164005e+03 = [1, 1, 1, 1]	FES = 561100	Diversity = 2.4355e-03	k_t
Generation 5620: Best = 4.164003e+03 = [4, 3, 3, 4]	FES = 562100	Diversity = 2.3803e-03	k_t

Generation 5630: Best = 4.164001e+03 = [1, 1, 1, 1]	FES = 563100	Diversity = 2.1519e-03	k_t
Generation 5640: Best = 4.163999e+03 = [3, 3, 2, 4]	FES = 564100	Diversity = 2.1583e-03	k_t
Generation 5650: Best = 4.163998e+03 = [1, 1, 1, 1]	FES = 565100	Diversity = 1.9799e-03	k_t
Generation 5660: Best = 4.163997e+03 = [4, 4, 2, 2]	FES = 566100	Diversity = 2.0795e-03	k_t
Generation 5670: Best = 4.163995e+03 = [1, 1, 1, 1]	FES = 567100	Diversity = 2.1068e-03	k_t
Generation 5680: Best = 4.163993e+03 = [3, 3, 4, 2]	FES = 568100	Diversity = 2.0163e-03	k_t
Generation 5690: Best = 4.163992e+03 = [1, 1, 1, 1]	FES = 569100	Diversity = 1.6204e-03	k_t
Generation 5700: Best = 4.163991e+03 = [3, 3, 3, 4]	FES = 570100	Diversity = 1.4658e-03	k_t
Generation 5710: Best = 4.163989e+03 = [1, 1, 1, 1]	FES = 571100	Diversity = 1.5908e-03	k_t
Generation 5720: Best = 4.163988e+03 = [4, 3, 3, 3]	FES = 572100	Diversity = 1.7497e-03	k_t
Generation 5730: Best = 4.163987e+03 = [1, 1, 1, 1]	FES = 573100	Diversity = 1.7217e-03	k_t
Generation 5740: Best = 4.163986e+03 = [4, 2, 4, 3]	FES = 574100	Diversity = 1.7775e-03	k_t
Generation 5750: Best = 4.163985e+03 = [1, 1, 1, 1]	FES = 575100	Diversity = 1.9096e-03	k_t
Generation 5760: Best = 4.163983e+03 = [3, 3, 2, 2]	FES = 576100	Diversity = 2.0763e-03	k_t
Generation 5770: Best = 4.163982e+03 = [1, 1, 1, 1]	FES = 577100	Diversity = 1.8188e-03	k_t
Generation 5780: Best = 4.163980e+03 = [4, 3, 2, 4]	FES = 578100	Diversity = 1.6644e-03	k_t
Generation 5790: Best = 4.163979e+03 = [1, 1, 1, 1]	FES = 579100	Diversity = 1.4883e-03	k_t
Generation 5800: Best = 4.163978e+03 = [2, 3, 2, 4]	FES = 580100	Diversity = 1.7590e-03	k_t
Generation 5810: Best = 4.163977e+03 = [1, 1, 1, 1]	FES = 581100	Diversity = 1.6541e-03	k_t
Generation 5820: Best = 4.163975e+03 = [2, 4, 3, 2]	FES = 582100	Diversity = 1.6675e-03	k_t
Generation 5830: Best = 4.163974e+03 = [1, 1, 1, 1]	FES = 583100	Diversity = 1.7828e-03	k_t
Generation 5840: Best = 4.163973e+03 = [2, 2, 3, 4]	FES = 584100	Diversity = 1.8259e-03	k_t
Generation 5850: Best = 4.163971e+03 = [1, 1, 1, 1]	FES = 585100	Diversity = 1.9730e-03	k_t
Generation 5860: Best = 4.163970e+03 = [3, 2, 3, 2]	FES = 586100	Diversity = 1.8143e-03	k_t
Generation 5870: Best = 4.163969e+03 = [1, 1, 1, 1]	FES = 587100	Diversity = 1.3601e-03	k_t
Generation 5880: Best = 4.163968e+03 = [3, 3, 4, 2]	FES = 588100	Diversity = 1.8361e-03	k_t
Generation 5890: Best = 4.163966e+03 = [1, 1, 1, 1]	FES = 589100	Diversity = 1.7801e-03	k_t

Generation 5900: Best = 4.163964e+03 = [3, 2, 2, 3]	FES = 590100	Diversity = 1.7019e-03	k_t
Generation 5910: Best = 4.163963e+03 = [1, 1, 1, 1]	FES = 591100	Diversity = 1.6842e-03	k_t
Generation 5920: Best = 4.163962e+03 = [2, 3, 3, 4]	FES = 592100	Diversity = 2.0333e-03	k_t
Generation 5930: Best = 4.163960e+03 = [1, 1, 1, 1]	FES = 593100	Diversity = 2.0463e-03	k_t
Generation 5940: Best = 4.163959e+03 = [2, 2, 2, 4]	FES = 594100	Diversity = 2.2817e-03	k_t
Generation 5950: Best = 4.163957e+03 = [1, 1, 1, 1]	FES = 595100	Diversity = 1.7680e-03	k_t
Generation 5960: Best = 4.163956e+03 = [4, 3, 4, 3]	FES = 596100	Diversity = 1.5559e-03	k_t
Generation 5970: Best = 4.163956e+03 = [1, 1, 1, 1]	FES = 597100	Diversity = 1.4432e-03	k_t
Generation 5980: Best = 4.163955e+03 = [2, 3, 4, 2]	FES = 598100	Diversity = 1.5762e-03	k_t
Generation 5990: Best = 4.163954e+03 = [1, 1, 1, 1]	FES = 599100	Diversity = 1.3596e-03	k_t
Generation 6000: Best = 4.163953e+03 = [3, 4, 2, 3]	FES = 600100	Diversity = 1.6073e-03	k_t
Generation 6010: Best = 4.163953e+03 = [1, 1, 1, 1]	FES = 601100	Diversity = 1.3294e-03	k_t
Generation 6020: Best = 4.163952e+03 = [3, 4, 2, 3]	FES = 602100	Diversity = 1.4119e-03	k_t
Generation 6030: Best = 4.163951e+03 = [1, 1, 1, 1]	FES = 603100	Diversity = 1.5895e-03	k_t
Generation 6040: Best = 4.163950e+03 = [4, 2, 3, 4]	FES = 604100	Diversity = 1.5399e-03	k_t
Generation 6050: Best = 4.163949e+03 = [1, 1, 1, 1]	FES = 605100	Diversity = 1.6159e-03	k_t
Generation 6060: Best = 4.163947e+03 = [3, 3, 3, 2]	FES = 606100	Diversity = 1.8433e-03	k_t
Generation 6070: Best = 4.163946e+03 = [1, 1, 1, 1]	FES = 607100	Diversity = 1.9970e-03	k_t
Generation 6080: Best = 4.163945e+03 = [2, 2, 4, 3]	FES = 608100	Diversity = 1.8680e-03	k_t
Generation 6090: Best = 4.163944e+03 = [1, 1, 1, 1]	FES = 609100	Diversity = 1.6699e-03	k_t
Generation 6100: Best = 4.163942e+03 = [4, 2, 4, 4]	FES = 610100	Diversity = 1.6530e-03	k_t
Generation 6110: Best = 4.163941e+03 = [1, 1, 1, 1]	FES = 611100	Diversity = 1.3147e-03	k_t
Generation 6120: Best = 4.163940e+03 = [3, 4, 2, 3]	FES = 612100	Diversity = 1.4583e-03	k_t
Generation 6130: Best = 4.163939e+03 = [1, 1, 1, 1]	FES = 613100	Diversity = 1.6763e-03	k_t
Generation 6140: Best = 4.163938e+03 = [3, 4, 2, 2]	FES = 614100	Diversity = 1.6756e-03	k_t
Generation 6150: Best = 4.163937e+03 = [1, 1, 1, 1]	FES = 615100	Diversity = 1.8378e-03	k_t
Generation 6160: Best = 4.163936e+03 = [2, 2, 2, 4]	FES = 616100	Diversity = 1.8293e-03	k_t

Generation 6170: Best = 4.163934e+03 = [1, 1, 1, 1]	FES = 617100	Diversity = 1.5715e-03	k_t
Generation 6180: Best = 4.163933e+03 = [2, 3, 4, 3]	FES = 618100	Diversity = 1.5316e-03	k_t
Generation 6190: Best = 4.163933e+03 = [1, 1, 1, 1]	FES = 619100	Diversity = 1.2548e-03	k_t
Generation 6200: Best = 4.163932e+03 = [3, 4, 3, 3]	FES = 620100	Diversity = 1.1931e-03	k_t
Generation 6210: Best = 4.163931e+03 = [1, 1, 1, 1]	FES = 621100	Diversity = 1.2676e-03	k_t
Generation 6220: Best = 4.163931e+03 = [3, 4, 3, 2]	FES = 622100	Diversity = 1.2530e-03	k_t
Generation 6230: Best = 4.163930e+03 = [1, 1, 1, 1]	FES = 623100	Diversity = 1.4236e-03	k_t
Generation 6240: Best = 4.163929e+03 = [2, 4, 3, 2]	FES = 624100	Diversity = 1.3350e-03	k_t
Generation 6250: Best = 4.163928e+03 = [1, 1, 1, 1]	FES = 625100	Diversity = 1.8657e-03	k_t
Generation 6260: Best = 4.163927e+03 = [4, 2, 2, 4]	FES = 626100	Diversity = 1.7444e-03	k_t
Generation 6270: Best = 4.163926e+03 = [1, 1, 1, 1]	FES = 627100	Diversity = 1.5935e-03	k_t
Generation 6280: Best = 4.163925e+03 = [4, 4, 3, 3]	FES = 628100	Diversity = 1.7844e-03	k_t
Generation 6290: Best = 4.163924e+03 = [1, 1, 1, 1]	FES = 629100	Diversity = 1.7831e-03	k_t
Generation 6300: Best = 4.163922e+03 = [3, 4, 2, 2]	FES = 630100	Diversity = 1.8007e-03	k_t
Generation 6310: Best = 4.163921e+03 = [1, 1, 1, 1]	FES = 631100	Diversity = 1.9245e-03	k_t
Generation 6320: Best = 4.163919e+03 = [4, 4, 4, 3]	FES = 632100	Diversity = 2.2148e-03	k_t
Generation 6330: Best = 4.163918e+03 = [1, 1, 1, 1]	FES = 633100	Diversity = 2.0370e-03	k_t
Generation 6340: Best = 4.163917e+03 = [4, 3, 2, 4]	FES = 634100	Diversity = 1.8215e-03	k_t
Generation 6350: Best = 4.163916e+03 = [1, 1, 1, 1]	FES = 635100	Diversity = 1.8501e-03	k_t
Generation 6360: Best = 4.163915e+03 = [4, 3, 2, 3]	FES = 636100	Diversity = 1.7060e-03	k_t
Generation 6370: Best = 4.163914e+03 = [1, 1, 1, 1]	FES = 637100	Diversity = 1.5840e-03	k_t
Generation 6380: Best = 4.163912e+03 = [4, 3, 2, 4]	FES = 638100	Diversity = 2.2876e-03	k_t
Generation 6390: Best = 4.163911e+03 = [1, 1, 1, 1]	FES = 639100	Diversity = 2.7301e-03	k_t
Generation 6400: Best = 4.163909e+03 = [2, 2, 4, 4]	FES = 640100	Diversity = 2.3939e-03	k_t
Generation 6410: Best = 4.163908e+03 = [1, 1, 1, 1]	FES = 641100	Diversity = 2.2217e-03	k_t
Generation 6420: Best = 4.163906e+03 = [3, 4, 4, 3]	FES = 642100	Diversity = 2.2694e-03	k_t
Generation 6430: Best = 4.163904e+03 = [1, 1, 1, 1]	FES = 643100	Diversity = 2.4833e-03	k_t

Generation 6440: Best = 4.163902e+03 = [4, 2, 3, 3]	FES = 644100	Diversity = 2.4177e-03	k_t
Generation 6450: Best = 4.163899e+03 = [1, 1, 1, 1]	FES = 645100	Diversity = 2.9319e-03	k_t
Generation 6460: Best = 4.163896e+03 = [2, 3, 3, 3]	FES = 646100	Diversity = 3.2057e-03	k_t
Generation 6470: Best = 4.163894e+03 = [1, 1, 1, 1]	FES = 647100	Diversity = 3.4962e-03	k_t
Generation 6480: Best = 4.163892e+03 = [4, 4, 3, 4]	FES = 648100	Diversity = 2.7802e-03	k_t
Generation 6490: Best = 4.163889e+03 = [1, 1, 1, 1]	FES = 649100	Diversity = 2.6637e-03	k_t
Generation 6500: Best = 4.163887e+03 = [2, 2, 2, 2]	FES = 650100	Diversity = 2.4992e-03	k_t
Generation 6510: Best = 4.163885e+03 = [1, 1, 1, 1]	FES = 651100	Diversity = 2.2474e-03	k_t
Generation 6520: Best = 4.163884e+03 = [2, 3, 3, 3]	FES = 652100	Diversity = 2.0199e-03	k_t
Generation 6530: Best = 4.163883e+03 = [1, 1, 1, 1]	FES = 653100	Diversity = 1.8445e-03	k_t
Generation 6540: Best = 4.163882e+03 = [4, 3, 3, 3]	FES = 654100	Diversity = 1.7108e-03	k_t
Generation 6550: Best = 4.163881e+03 = [1, 1, 1, 1]	FES = 655100	Diversity = 1.5760e-03	k_t
Generation 6560: Best = 4.163880e+03 = [3, 3, 3, 4]	FES = 656100	Diversity = 1.6080e-03	k_t
Generation 6570: Best = 4.163879e+03 = [1, 1, 1, 1]	FES = 657100	Diversity = 1.6727e-03	k_t
Generation 6580: Best = 4.163878e+03 = [4, 4, 4, 4]	FES = 658100	Diversity = 1.9344e-03	k_t
Generation 6590: Best = 4.163877e+03 = [1, 1, 1, 1]	FES = 659100	Diversity = 1.6893e-03	k_t
Generation 6600: Best = 4.163876e+03 = [3, 2, 3, 3]	FES = 660100	Diversity = 1.5338e-03	k_t
Generation 6610: Best = 4.163875e+03 = [1, 1, 1, 1]	FES = 661100	Diversity = 1.6759e-03	k_t
Generation 6620: Best = 4.163874e+03 = [4, 2, 3, 3]	FES = 662100	Diversity = 1.6321e-03	k_t
Generation 6630: Best = 4.163873e+03 = [1, 1, 1, 1]	FES = 663100	Diversity = 1.5774e-03	k_t
Generation 6640: Best = 4.163872e+03 = [3, 4, 3, 2]	FES = 664100	Diversity = 1.5458e-03	k_t
Generation 6650: Best = 4.163871e+03 = [1, 1, 1, 1]	FES = 665100	Diversity = 1.4639e-03	k_t
Generation 6660: Best = 4.163870e+03 = [3, 3, 3, 2]	FES = 666100	Diversity = 1.9211e-03	k_t
Generation 6670: Best = 4.163869e+03 = [1, 1, 1, 1]	FES = 667100	Diversity = 2.0649e-03	k_t
Generation 6680: Best = 4.163868e+03 = [3, 2, 2, 2]	FES = 668100	Diversity = 2.0698e-03	k_t
Generation 6690: Best = 4.163867e+03 = [1, 1, 1, 1]	FES = 669100	Diversity = 2.1661e-03	k_t
Generation 6700: Best = 4.163865e+03 = [3, 3, 2, 2]	FES = 670100	Diversity = 1.9729e-03	k_t

Generation 6710: Best = 4.163864e+03 = [1, 1, 1, 1]	FES = 671100	Diversity = 1.6392e-03	k_t
Generation 6720: Best = 4.163863e+03 = [4, 3, 2, 4]	FES = 672100	Diversity = 1.2708e-03	k_t
Generation 6730: Best = 4.163863e+03 = [1, 1, 1, 1]	FES = 673100	Diversity = 9.9581e-04	k_t
Generation 6740: Best = 4.163862e+03 = [4, 4, 3, 2]	FES = 674100	Diversity = 1.0049e-03	k_t
Generation 6750: Best = 4.163862e+03 = [1, 1, 1, 1]	FES = 675100	Diversity = 1.0885e-03	k_t
Generation 6760: Best = 4.163861e+03 = [3, 3, 3, 4]	FES = 676100	Diversity = 1.0814e-03	k_t
Generation 6770: Best = 4.163861e+03 = [1, 1, 1, 1]	FES = 677100	Diversity = 1.1397e-03	k_t
Generation 6780: Best = 4.163860e+03 = [2, 2, 4, 2]	FES = 678100	Diversity = 9.1307e-04	k_t
Generation 6790: Best = 4.163860e+03 = [1, 1, 1, 1]	FES = 679100	Diversity = 1.0490e-03	k_t
Generation 6800: Best = 4.163859e+03 = [4, 4, 2, 4]	FES = 680100	Diversity = 1.1471e-03	k_t
Generation 6810: Best = 4.163859e+03 = [1, 1, 1, 1]	FES = 681100	Diversity = 1.2795e-03	k_t
Generation 6820: Best = 4.163858e+03 = [4, 3, 2, 2]	FES = 682100	Diversity = 1.3555e-03	k_t
Generation 6830: Best = 4.163857e+03 = [1, 1, 1, 1]	FES = 683100	Diversity = 1.4232e-03	k_t
Generation 6840: Best = 4.163857e+03 = [4, 3, 3, 4]	FES = 684100	Diversity = 1.2778e-03	k_t
Generation 6850: Best = 4.163856e+03 = [1, 1, 1, 1]	FES = 685100	Diversity = 1.1549e-03	k_t
Generation 6860: Best = 4.163856e+03 = [2, 4, 4, 2]	FES = 686100	Diversity = 1.1541e-03	k_t
Generation 6870: Best = 4.163855e+03 = [1, 1, 1, 1]	FES = 687100	Diversity = 1.3594e-03	k_t
Generation 6880: Best = 4.163855e+03 = [2, 3, 3, 2]	FES = 688100	Diversity = 1.3432e-03	k_t
Generation 6890: Best = 4.163853e+03 = [1, 1, 1, 1]	FES = 689100	Diversity = 1.8608e-03	k_t
Generation 6900: Best = 4.163852e+03 = [4, 3, 2, 3]	FES = 690100	Diversity = 1.6510e-03	k_t
Generation 6910: Best = 4.163851e+03 = [1, 1, 1, 1]	FES = 691100	Diversity = 1.2958e-03	k_t
Generation 6920: Best = 4.163851e+03 = [2, 2, 3, 4]	FES = 692100	Diversity = 1.3675e-03	k_t
Generation 6930: Best = 4.163850e+03 = [1, 1, 1, 1]	FES = 693100	Diversity = 1.4150e-03	k_t
Generation 6940: Best = 4.163849e+03 = [4, 4, 3, 2]	FES = 694100	Diversity = 1.7114e-03	k_t
Generation 6950: Best = 4.163848e+03 = [1, 1, 1, 1]	FES = 695100	Diversity = 1.8690e-03	k_t
Generation 6960: Best = 4.163847e+03 = [3, 2, 3, 4]	FES = 696100	Diversity = 1.7607e-03	k_t
Generation 6970: Best = 4.163846e+03 = [1, 1, 1, 1]	FES = 697100	Diversity = 1.7662e-03	k_t

Generation 6980: Best = 4.163844e+03 = [2, 4, 4, 2]	FES = 698100	Diversity = 2.6650e-03	k_t
Generation 6990: Best = 4.163842e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 2.2177e-03	k_t
Generation 7000: Best = 4.163838e+03 = [3, 2, 3, 2]	FES = 700100	Diversity = 2.7224e-03	k_t
Generation 7010: Best = 4.163836e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 2.7775e-03	k_t
Generation 7020: Best = 4.163832e+03 = [4, 4, 2, 4]	FES = 702100	Diversity = 3.0224e-03	k_t
Generation 7030: Best = 4.163828e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 3.3766e-03	k_t
Generation 7040: Best = 4.163825e+03 = [3, 3, 4, 2]	FES = 704100	Diversity = 3.1846e-03	k_t
Generation 7050: Best = 4.163823e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 2.2540e-03	k_t
Generation 7060: Best = 4.163822e+03 = [4, 4, 3, 3]	FES = 706100	Diversity = 1.8695e-03	k_t
Generation 7070: Best = 4.163820e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 1.6920e-03	k_t
Generation 7080: Best = 4.163819e+03 = [3, 2, 4, 3]	FES = 708100	Diversity = 1.7941e-03	k_t
Generation 7090: Best = 4.163818e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 2.0663e-03	k_t
Generation 7100: Best = 4.163816e+03 = [2, 3, 4, 2]	FES = 710100	Diversity = 2.1027e-03	k_t
Generation 7110: Best = 4.163815e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 1.8177e-03	k_t
Generation 7120: Best = 4.163813e+03 = [3, 2, 3, 4]	FES = 712100	Diversity = 1.7646e-03	k_t
Generation 7130: Best = 4.163812e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 2.0996e-03	k_t
Generation 7140: Best = 4.163810e+03 = [3, 2, 3, 3]	FES = 714100	Diversity = 1.9214e-03	k_t
Generation 7150: Best = 4.163808e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 2.0513e-03	k_t
Generation 7160: Best = 4.163807e+03 = [4, 4, 2, 3]	FES = 716100	Diversity = 1.9795e-03	k_t
Generation 7170: Best = 4.163805e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 1.8971e-03	k_t
Generation 7180: Best = 4.163804e+03 = [3, 2, 4, 3]	FES = 718100	Diversity = 1.6743e-03	k_t
Generation 7190: Best = 4.163803e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 1.7473e-03	k_t
Generation 7200: Best = 4.163801e+03 = [3, 2, 2, 2]	FES = 720100	Diversity = 1.8949e-03	k_t
Generation 7210: Best = 4.163799e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 1.6763e-03	k_t
Generation 7220: Best = 4.163798e+03 = [4, 2, 3, 3]	FES = 722100	Diversity = 1.6955e-03	k_t
Generation 7230: Best = 4.163797e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 1.8057e-03	k_t
Generation 7240: Best = 4.163796e+03 = [3, 3, 2, 3]	FES = 724100	Diversity = 2.0068e-03	k_t

Generation 7250: Best = 4.163794e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 1.8311e-03	k_t
Generation 7260: Best = 4.163793e+03 = [2, 2, 2, 3]	FES = 726100	Diversity = 2.0342e-03	k_t
Generation 7270: Best = 4.163791e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 2.1867e-03	k_t
Generation 7280: Best = 4.163789e+03 = [2, 3, 4, 3]	FES = 728100	Diversity = 2.0086e-03	k_t
Generation 7290: Best = 4.163788e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 2.1006e-03	k_t
Generation 7300: Best = 4.163786e+03 = [3, 4, 3, 2]	FES = 730100	Diversity = 1.8488e-03	k_t
Generation 7310: Best = 4.163785e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 1.7546e-03	k_t
Generation 7320: Best = 4.163783e+03 = [3, 4, 4, 2]	FES = 732100	Diversity = 1.8141e-03	k_t
Generation 7330: Best = 4.163782e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 1.7858e-03	k_t
Generation 7340: Best = 4.163780e+03 = [3, 4, 3, 4]	FES = 734100	Diversity = 1.8218e-03	k_t
Generation 7350: Best = 4.163779e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 1.9357e-03	k_t
Generation 7360: Best = 4.163777e+03 = [4, 3, 3, 3]	FES = 736100	Diversity = 1.9975e-03	k_t
Generation 7370: Best = 4.163775e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 1.7580e-03	k_t
Generation 7380: Best = 4.163774e+03 = [4, 4, 3, 4]	FES = 738100	Diversity = 1.5328e-03	k_t
Generation 7390: Best = 4.163773e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 1.6081e-03	k_t
Generation 7400: Best = 4.163771e+03 = [2, 3, 4, 4]	FES = 740100	Diversity = 1.4922e-03	k_t
Generation 7410: Best = 4.163770e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 1.4735e-03	k_t
Generation 7420: Best = 4.163770e+03 = [2, 4, 3, 3]	FES = 742100	Diversity = 1.4453e-03	k_t
Generation 7430: Best = 4.163768e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 1.6169e-03	k_t
Generation 7440: Best = 4.163767e+03 = [4, 3, 2, 3]	FES = 744100	Diversity = 1.6204e-03	k_t
Generation 7450: Best = 4.163766e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 1.2952e-03	k_t
Generation 7460: Best = 4.163765e+03 = [4, 2, 4, 4]	FES = 746100	Diversity = 1.2269e-03	k_t
Generation 7470: Best = 4.163765e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 1.2278e-03	k_t
Generation 7480: Best = 4.163764e+03 = [2, 2, 4, 4]	FES = 748100	Diversity = 1.2672e-03	k_t
Generation 7490: Best = 4.163763e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 1.4728e-03	k_t
Generation 7500: Best = 4.163762e+03 = [4, 4, 4, 4]	FES = 750100	Diversity = 1.5045e-03	k_t
Generation 7510: Best = 4.163761e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 1.7467e-03	k_t

Generation 7520: Best = 4.163760e+03 = [3, 4, 4, 3]	FES = 752100	Diversity = 1.5917e-03	k_t
Generation 7530: Best = 4.163758e+03 = [1, 1, 1, 1]	FES = 753100	Diversity = 1.5628e-03	k_t
Generation 7540: Best = 4.163756e+03 = [4, 3, 4, 3]	FES = 754100	Diversity = 2.2440e-03	k_t
Generation 7550: Best = 4.163753e+03 = [1, 1, 1, 1]	FES = 755100	Diversity = 2.7274e-03	k_t
Generation 7560: Best = 4.163750e+03 = [2, 4, 2, 3]	FES = 756100	Diversity = 2.8205e-03	k_t
Generation 7570: Best = 4.163748e+03 = [1, 1, 1, 1]	FES = 757100	Diversity = 2.3507e-03	k_t
Generation 7580: Best = 4.163746e+03 = [2, 2, 4, 4]	FES = 758100	Diversity = 2.2142e-03	k_t
Generation 7590: Best = 4.163744e+03 = [1, 1, 1, 1]	FES = 759100	Diversity = 2.1017e-03	k_t
Generation 7600: Best = 4.163743e+03 = [3, 4, 4, 4]	FES = 760100	Diversity = 1.6630e-03	k_t
Generation 7610: Best = 4.163742e+03 = [1, 1, 1, 1]	FES = 761100	Diversity = 1.5651e-03	k_t
Generation 7620: Best = 4.163740e+03 = [4, 3, 4, 3]	FES = 762100	Diversity = 2.1444e-03	k_t
Generation 7630: Best = 4.163739e+03 = [1, 1, 1, 1]	FES = 763100	Diversity = 2.2780e-03	k_t
Generation 7640: Best = 4.163737e+03 = [4, 2, 3, 3]	FES = 764100	Diversity = 1.9466e-03	k_t
Generation 7650: Best = 4.163736e+03 = [1, 1, 1, 1]	FES = 765100	Diversity = 2.2871e-03	k_t
Generation 7660: Best = 4.163734e+03 = [4, 2, 3, 2]	FES = 766100	Diversity = 2.4537e-03	k_t
Generation 7670: Best = 4.163733e+03 = [1, 1, 1, 1]	FES = 767100	Diversity = 1.8180e-03	k_t
Generation 7680: Best = 4.163732e+03 = [3, 4, 4, 3]	FES = 768100	Diversity = 1.6715e-03	k_t
Generation 7690: Best = 4.163731e+03 = [1, 1, 1, 1]	FES = 769100	Diversity = 1.7572e-03	k_t
Generation 7700: Best = 4.163729e+03 = [4, 4, 2, 4]	FES = 770100	Diversity = 2.0490e-03	k_t
Generation 7710: Best = 4.163727e+03 = [1, 1, 1, 1]	FES = 771100	Diversity = 2.5135e-03	k_t
Generation 7720: Best = 4.163725e+03 = [4, 2, 3, 2]	FES = 772100	Diversity = 2.6846e-03	k_t
Generation 7730: Best = 4.163724e+03 = [1, 1, 1, 1]	FES = 773100	Diversity = 2.0942e-03	k_t
Generation 7740: Best = 4.163723e+03 = [3, 2, 3, 2]	FES = 774100	Diversity = 2.0042e-03	k_t
Generation 7750: Best = 4.163721e+03 = [1, 1, 1, 1]	FES = 775100	Diversity = 2.3135e-03	k_t
Generation 7760: Best = 4.163720e+03 = [3, 2, 3, 4]	FES = 776100	Diversity = 2.0085e-03	k_t
Generation 7770: Best = 4.163719e+03 = [1, 1, 1, 1]	FES = 777100	Diversity = 1.5412e-03	k_t
Generation 7780: Best = 4.163718e+03 = [2, 2, 4, 3]	FES = 778100	Diversity = 1.4945e-03	k_t

Generation 7790: Best = 4.163717e+03 = [1, 1, 1, 1]	FES = 779100	Diversity = 1.6401e-03	k_t
Generation 7800: Best = 4.163717e+03 = [4, 3, 4, 4]	FES = 780100	Diversity = 1.2643e-03	k_t
Generation 7810: Best = 4.163716e+03 = [1, 1, 1, 1]	FES = 781100	Diversity = 1.3875e-03	k_t
Generation 7820: Best = 4.163715e+03 = [3, 3, 2, 4]	FES = 782100	Diversity = 1.3935e-03	k_t
Generation 7830: Best = 4.163715e+03 = [1, 1, 1, 1]	FES = 783100	Diversity = 1.2944e-03	k_t
Generation 7840: Best = 4.163714e+03 = [2, 3, 3, 4]	FES = 784100	Diversity = 1.2479e-03	k_t
Generation 7850: Best = 4.163714e+03 = [1, 1, 1, 1]	FES = 785100	Diversity = 1.1867e-03	k_t
Generation 7860: Best = 4.163713e+03 = [4, 2, 3, 4]	FES = 786100	Diversity = 1.1718e-03	k_t
Generation 7870: Best = 4.163712e+03 = [1, 1, 1, 1]	FES = 787100	Diversity = 1.2717e-03	k_t
Generation 7880: Best = 4.163712e+03 = [2, 3, 4, 4]	FES = 788100	Diversity = 1.3272e-03	k_t
Generation 7890: Best = 4.163711e+03 = [1, 1, 1, 1]	FES = 789100	Diversity = 1.2028e-03	k_t
Generation 7900: Best = 4.163711e+03 = [2, 4, 2, 3]	FES = 790100	Diversity = 1.2409e-03	k_t
Generation 7910: Best = 4.163710e+03 = [1, 1, 1, 1]	FES = 791100	Diversity = 1.3471e-03	k_t
Generation 7920: Best = 4.163710e+03 = [4, 4, 2, 4]	FES = 792100	Diversity = 1.3500e-03	k_t
Generation 7930: Best = 4.163709e+03 = [1, 1, 1, 1]	FES = 793100	Diversity = 1.5344e-03	k_t
Generation 7940: Best = 4.163708e+03 = [4, 4, 4, 4]	FES = 794100	Diversity = 1.4302e-03	k_t
Generation 7950: Best = 4.163708e+03 = [1, 1, 1, 1]	FES = 795100	Diversity = 1.3500e-03	k_t
Generation 7960: Best = 4.163707e+03 = [4, 2, 4, 2]	FES = 796100	Diversity = 1.3057e-03	k_t
Generation 7970: Best = 4.163707e+03 = [1, 1, 1, 1]	FES = 797100	Diversity = 1.2524e-03	k_t
Generation 7980: Best = 4.163706e+03 = [2, 4, 4, 2]	FES = 798100	Diversity = 1.3200e-03	k_t
Generation 7990: Best = 4.163705e+03 = [1, 1, 1, 1]	FES = 799100	Diversity = 1.3357e-03	k_t
Generation 8000: Best = 4.163705e+03 = [4, 2, 3, 3]	FES = 800100	Diversity = 1.2905e-03	k_t
Generation 8010: Best = 4.163704e+03 = [1, 1, 1, 1]	FES = 801100	Diversity = 1.3428e-03	k_t
Generation 8020: Best = 4.163704e+03 = [2, 2, 4, 4]	FES = 802100	Diversity = 1.6238e-03	k_t
Generation 8030: Best = 4.163703e+03 = [1, 1, 1, 1]	FES = 803100	Diversity = 1.6423e-03	k_t
Generation 8040: Best = 4.163702e+03 = [4, 4, 2, 3]	FES = 804100	Diversity = 1.5173e-03	k_t
Generation 8050: Best = 4.163702e+03 = [1, 1, 1, 1]	FES = 805100	Diversity = 1.1571e-03	k_t

Generation 8060: Best = 4.163701e+03 = [2, 2, 4, 2]	FES = 806100	Diversity = 1.1231e-03	k_t
Generation 8070: Best = 4.163701e+03 = [1, 1, 1, 1]	FES = 807100	Diversity = 1.0057e-03	k_t
Generation 8080: Best = 4.163701e+03 = [3, 2, 3, 4]	FES = 808100	Diversity = 8.5242e-04	k_t
Generation 8090: Best = 4.163701e+03 = [1, 1, 1, 1]	FES = 809100	Diversity = 8.8293e-04	k_t
Generation 8100: Best = 4.163700e+03 = [2, 4, 2, 3]	FES = 810100	Diversity = 9.2734e-04	k_t
Generation 8110: Best = 4.163700e+03 = [1, 1, 1, 1]	FES = 811100	Diversity = 1.1468e-03	k_t
Generation 8120: Best = 4.163700e+03 = [4, 3, 4, 4]	FES = 812100	Diversity = 1.0005e-03	k_t
Generation 8130: Best = 4.163699e+03 = [1, 1, 1, 1]	FES = 813100	Diversity = 1.0966e-03	k_t
Generation 8140: Best = 4.163699e+03 = [2, 3, 3, 4]	FES = 814100	Diversity = 9.6985e-04	k_t
Generation 8150: Best = 4.163699e+03 = [1, 1, 1, 1]	FES = 815100	Diversity = 8.9293e-04	k_t
Generation 8160: Best = 4.163698e+03 = [3, 3, 2, 3]	FES = 816100	Diversity = 8.4327e-04	k_t
Generation 8170: Best = 4.163698e+03 = [1, 1, 1, 1]	FES = 817100	Diversity = 8.5736e-04	k_t
Generation 8180: Best = 4.163698e+03 = [2, 4, 4, 4]	FES = 818100	Diversity = 1.0189e-03	k_t
Generation 8190: Best = 4.163697e+03 = [1, 1, 1, 1]	FES = 819100	Diversity = 1.0624e-03	k_t
Generation 8200: Best = 4.163697e+03 = [4, 4, 4, 4]	FES = 820100	Diversity = 8.0923e-04	k_t
Generation 8210: Best = 4.163697e+03 = [1, 1, 1, 1]	FES = 821100	Diversity = 8.6790e-04	k_t
Generation 8220: Best = 4.163697e+03 = [2, 2, 4, 2]	FES = 822100	Diversity = 9.6470e-04	k_t
Generation 8230: Best = 4.163696e+03 = [1, 1, 1, 1]	FES = 823100	Diversity = 8.9332e-04	k_t
Generation 8240: Best = 4.163696e+03 = [2, 2, 2, 2]	FES = 824100	Diversity = 8.0909e-04	k_t
Generation 8250: Best = 4.163696e+03 = [1, 1, 1, 1]	FES = 825100	Diversity = 6.7382e-04	k_t
Generation 8260: Best = 4.163696e+03 = [3, 3, 2, 2]	FES = 826100	Diversity = 9.4459e-04	k_t
Generation 8270: Best = 4.163695e+03 = [1, 1, 1, 1]	FES = 827100	Diversity = 1.0301e-03	k_t
Generation 8280: Best = 4.163695e+03 = [4, 2, 4, 2]	FES = 828100	Diversity = 1.1771e-03	k_t
Generation 8290: Best = 4.163694e+03 = [1, 1, 1, 1]	FES = 829100	Diversity = 1.0519e-03	k_t
Generation 8300: Best = 4.163694e+03 = [4, 2, 4, 3]	FES = 830100	Diversity = 9.1142e-04	k_t
Generation 8310: Best = 4.163693e+03 = [1, 1, 1, 1]	FES = 831100	Diversity = 1.1377e-03	k_t
Generation 8320: Best = 4.163692e+03 = [4, 4, 4, 2]	FES = 832100	Diversity = 1.4583e-03	k_t

Generation 8330: Best = 4.163692e+03 = [1, 1, 1, 1]	FES = 833100	Diversity = 1.6644e-03	k_t
Generation 8340: Best = 4.163691e+03 = [2, 3, 3, 4]	FES = 834100	Diversity = 1.7352e-03	k_t
Generation 8350: Best = 4.163690e+03 = [1, 1, 1, 1]	FES = 835100	Diversity = 1.7470e-03	k_t
Generation 8360: Best = 4.163689e+03 = [3, 4, 3, 3]	FES = 836100	Diversity = 2.0174e-03	k_t
Generation 8370: Best = 4.163688e+03 = [1, 1, 1, 1]	FES = 837100	Diversity = 2.0489e-03	k_t
Generation 8380: Best = 4.163687e+03 = [2, 4, 4, 3]	FES = 838100	Diversity = 1.9232e-03	k_t
Generation 8390: Best = 4.163686e+03 = [1, 1, 1, 1]	FES = 839100	Diversity = 1.6904e-03	k_t
Generation 8400: Best = 4.163685e+03 = [2, 3, 4, 2]	FES = 840100	Diversity = 1.7715e-03	k_t
Generation 8410: Best = 4.163684e+03 = [1, 1, 1, 1]	FES = 841100	Diversity = 1.9962e-03	k_t
Generation 8420: Best = 4.163682e+03 = [2, 2, 4, 4]	FES = 842100	Diversity = 2.0711e-03	k_t
Generation 8430: Best = 4.163681e+03 = [1, 1, 1, 1]	FES = 843100	Diversity = 1.7866e-03	k_t
Generation 8440: Best = 4.163681e+03 = [2, 4, 2, 2]	FES = 844100	Diversity = 1.6834e-03	k_t
Generation 8450: Best = 4.163680e+03 = [1, 1, 1, 1]	FES = 845100	Diversity = 1.4499e-03	k_t
Generation 8460: Best = 4.163679e+03 = [3, 2, 4, 3]	FES = 846100	Diversity = 1.2804e-03	k_t
Generation 8470: Best = 4.163679e+03 = [1, 1, 1, 1]	FES = 847100	Diversity = 1.0526e-03	k_t
Generation 8480: Best = 4.163679e+03 = [3, 3, 2, 4]	FES = 848100	Diversity = 8.4870e-04	k_t
Generation 8490: Best = 4.163679e+03 = [1, 1, 1, 1]	FES = 849100	Diversity = 1.0170e-03	k_t
Generation 8500: Best = 4.163678e+03 = [3, 2, 4, 2]	FES = 850100	Diversity = 9.2388e-04	k_t
Generation 8510: Best = 4.163678e+03 = [1, 1, 1, 1]	FES = 851100	Diversity = 9.4605e-04	k_t
Generation 8520: Best = 4.163677e+03 = [3, 2, 4, 3]	FES = 852100	Diversity = 1.3153e-03	k_t
Generation 8530: Best = 4.163677e+03 = [1, 1, 1, 1]	FES = 853100	Diversity = 1.3692e-03	k_t
Generation 8540: Best = 4.163676e+03 = [3, 4, 3, 4]	FES = 854100	Diversity = 1.2984e-03	k_t
Generation 8550: Best = 4.163676e+03 = [1, 1, 1, 1]	FES = 855100	Diversity = 1.2862e-03	k_t
Generation 8560: Best = 4.163675e+03 = [2, 3, 2, 4]	FES = 856100	Diversity = 1.0620e-03	k_t
Generation 8570: Best = 4.163675e+03 = [1, 1, 1, 1]	FES = 857100	Diversity = 1.0935e-03	k_t
Generation 8580: Best = 4.163674e+03 = [4, 4, 4, 4]	FES = 858100	Diversity = 9.4110e-04	k_t
Generation 8590: Best = 4.163674e+03 = [1, 1, 1, 1]	FES = 859100	Diversity = 7.9544e-04	k_t

Generation 8600: Best = 4.163674e+03 = [4, 2, 3, 2]	FES = 860100	Diversity = 8.9781e-04	k_t
Generation 8610: Best = 4.163673e+03 = [1, 1, 1, 1]	FES = 861100	Diversity = 9.2232e-04	k_t
Generation 8620: Best = 4.163673e+03 = [3, 4, 3, 2]	FES = 862100	Diversity = 7.3919e-04	k_t
Generation 8630: Best = 4.163673e+03 = [1, 1, 1, 1]	FES = 863100	Diversity = 6.0395e-04	k_t
Generation 8640: Best = 4.163672e+03 = [2, 2, 4, 2]	FES = 864100	Diversity = 6.9075e-04	k_t
Generation 8650: Best = 4.163672e+03 = [1, 1, 1, 1]	FES = 865100	Diversity = 8.3870e-04	k_t
Generation 8660: Best = 4.163672e+03 = [3, 4, 3, 3]	FES = 866100	Diversity = 1.0056e-03	k_t
Generation 8670: Best = 4.163671e+03 = [1, 1, 1, 1]	FES = 867100	Diversity = 1.0017e-03	k_t
Generation 8680: Best = 4.163671e+03 = [3, 4, 4, 4]	FES = 868100	Diversity = 1.0114e-03	k_t
Generation 8690: Best = 4.163670e+03 = [1, 1, 1, 1]	FES = 869100	Diversity = 1.1484e-03	k_t
Generation 8700: Best = 4.163670e+03 = [4, 4, 2, 2]	FES = 870100	Diversity = 1.1586e-03	k_t
Generation 8710: Best = 4.163669e+03 = [1, 1, 1, 1]	FES = 871100	Diversity = 1.1511e-03	k_t
Generation 8720: Best = 4.163669e+03 = [3, 2, 4, 2]	FES = 872100	Diversity = 1.0151e-03	k_t
Generation 8730: Best = 4.163669e+03 = [1, 1, 1, 1]	FES = 873100	Diversity = 8.2869e-04	k_t
Generation 8740: Best = 4.163669e+03 = [2, 3, 3, 4]	FES = 874100	Diversity = 7.0342e-04	k_t
Generation 8750: Best = 4.163668e+03 = [1, 1, 1, 1]	FES = 875100	Diversity = 6.9881e-04	k_t
Generation 8760: Best = 4.163668e+03 = [2, 3, 4, 2]	FES = 876100	Diversity = 7.4820e-04	k_t
Generation 8770: Best = 4.163668e+03 = [1, 1, 1, 1]	FES = 877100	Diversity = 6.8147e-04	k_t
Generation 8780: Best = 4.163668e+03 = [3, 4, 2, 3]	FES = 878100	Diversity = 6.3502e-04	k_t
Generation 8790: Best = 4.163668e+03 = [1, 1, 1, 1]	FES = 879100	Diversity = 7.2036e-04	k_t
Generation 8800: Best = 4.163667e+03 = [4, 4, 4, 3]	FES = 880100	Diversity = 7.5408e-04	k_t
Generation 8810: Best = 4.163667e+03 = [1, 1, 1, 1]	FES = 881100	Diversity = 7.1805e-04	k_t
Generation 8820: Best = 4.163667e+03 = [2, 4, 4, 4]	FES = 882100	Diversity = 7.2730e-04	k_t
Generation 8830: Best = 4.163667e+03 = [1, 1, 1, 1]	FES = 883100	Diversity = 7.4385e-04	k_t
Generation 8840: Best = 4.163667e+03 = [3, 4, 3, 4]	FES = 884100	Diversity = 6.4423e-04	k_t
Generation 8850: Best = 4.163667e+03 = [1, 1, 1, 1]	FES = 885100	Diversity = 6.3847e-04	k_t
Generation 8860: Best = 4.163666e+03 = [4, 3, 3, 3]	FES = 886100	Diversity = 7.1400e-04	k_t

Generation 8870: Best = 4.163666e+03 = [1, 1, 1, 1]	FES = 887100	Diversity = 7.8241e-04	k_t
Generation 8880: Best = 4.163666e+03 = [4, 3, 3, 4]	FES = 888100	Diversity = 8.6809e-04	k_t
Generation 8890: Best = 4.163666e+03 = [1, 1, 1, 1]	FES = 889100	Diversity = 9.1434e-04	k_t
Generation 8900: Best = 4.163665e+03 = [2, 3, 3, 2]	FES = 890100	Diversity = 7.8537e-04	k_t
Generation 8910: Best = 4.163665e+03 = [1, 1, 1, 1]	FES = 891100	Diversity = 8.4874e-04	k_t
Generation 8920: Best = 4.163665e+03 = [2, 3, 3, 2]	FES = 892100	Diversity = 1.0630e-03	k_t
Generation 8930: Best = 4.163664e+03 = [1, 1, 1, 1]	FES = 893100	Diversity = 1.0785e-03	k_t
Generation 8940: Best = 4.163664e+03 = [2, 4, 4, 2]	FES = 894100	Diversity = 1.0302e-03	k_t
Generation 8950: Best = 4.163663e+03 = [1, 1, 1, 1]	FES = 895100	Diversity = 1.1223e-03	k_t
Generation 8960: Best = 4.163663e+03 = [3, 2, 3, 4]	FES = 896100	Diversity = 1.0987e-03	k_t
Generation 8970: Best = 4.163662e+03 = [1, 1, 1, 1]	FES = 897100	Diversity = 1.1463e-03	k_t
Generation 8980: Best = 4.163662e+03 = [4, 4, 3, 4]	FES = 898100	Diversity = 9.6045e-04	k_t
Generation 8990: Best = 4.163662e+03 = [1, 1, 1, 1]	FES = 899100	Diversity = 9.0204e-04	k_t
Generation 9000: Best = 4.163662e+03 = [3, 4, 2, 4]	FES = 900100	Diversity = 9.8259e-04	k_t
Generation 9010: Best = 4.163661e+03 = [1, 1, 1, 1]	FES = 901100	Diversity = 1.0048e-03	k_t
Generation 9020: Best = 4.163661e+03 = [2, 2, 4, 4]	FES = 902100	Diversity = 8.3523e-04	k_t
Generation 9030: Best = 4.163661e+03 = [1, 1, 1, 1]	FES = 903100	Diversity = 7.5307e-04	k_t
Generation 9040: Best = 4.163661e+03 = [3, 2, 2, 4]	FES = 904100	Diversity = 6.5842e-04	k_t
Generation 9050: Best = 4.163661e+03 = [1, 1, 1, 1]	FES = 905100	Diversity = 6.5194e-04	k_t
Generation 9060: Best = 4.163660e+03 = [4, 3, 4, 2]	FES = 906100	Diversity = 7.5110e-04	k_t
Generation 9070: Best = 4.163660e+03 = [1, 1, 1, 1]	FES = 907100	Diversity = 7.2241e-04	k_t
Generation 9080: Best = 4.163660e+03 = [2, 2, 4, 4]	FES = 908100	Diversity = 7.8379e-04	k_t
Generation 9090: Best = 4.163660e+03 = [1, 1, 1, 1]	FES = 909100	Diversity = 8.9551e-04	k_t
Generation 9100: Best = 4.163660e+03 = [3, 2, 3, 3]	FES = 910100	Diversity = 8.5071e-04	k_t
Generation 9110: Best = 4.163659e+03 = [1, 1, 1, 1]	FES = 911100	Diversity = 8.0833e-04	k_t
Generation 9120: Best = 4.163659e+03 = [3, 2, 3, 3]	FES = 912100	Diversity = 8.6549e-04	k_t
Generation 9130: Best = 4.163659e+03 = [1, 1, 1, 1]	FES = 913100	Diversity = 1.0644e-03	k_t

Generation 9140: Best = 4.163658e+03 = [4, 3, 3, 4]	FES = 914100	Diversity = 1.1845e-03	k_t
Generation 9150: Best = 4.163658e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 1.2529e-03	k_t
Generation 9160: Best = 4.163657e+03 = [3, 4, 2, 3]	FES = 916100	Diversity = 1.1416e-03	k_t
Generation 9170: Best = 4.163657e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 1.3287e-03	k_t
Generation 9180: Best = 4.163656e+03 = [2, 4, 3, 4]	FES = 918100	Diversity = 1.4953e-03	k_t
Generation 9190: Best = 4.163655e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 1.7666e-03	k_t
Generation 9200: Best = 4.163654e+03 = [4, 2, 2, 4]	FES = 920100	Diversity = 1.7658e-03	k_t
Generation 9210: Best = 4.163653e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 2.0976e-03	k_t
Generation 9220: Best = 4.163650e+03 = [2, 2, 2, 3]	FES = 922100	Diversity = 2.8018e-03	k_t
Generation 9230: Best = 4.163648e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 2.8349e-03	k_t
Generation 9240: Best = 4.163646e+03 = [2, 3, 2, 2]	FES = 924100	Diversity = 2.5683e-03	k_t
Generation 9250: Best = 4.163642e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 2.9499e-03	k_t
Generation 9260: Best = 4.163640e+03 = [2, 2, 2, 2]	FES = 926100	Diversity = 2.4550e-03	k_t
Generation 9270: Best = 4.163638e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 2.4321e-03	k_t
Generation 9280: Best = 4.163636e+03 = [2, 4, 3, 2]	FES = 928100	Diversity = 2.6165e-03	k_t
Generation 9290: Best = 4.163635e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 1.9218e-03	k_t
Generation 9300: Best = 4.163634e+03 = [3, 3, 3, 3]	FES = 930100	Diversity = 1.5954e-03	k_t
Generation 9310: Best = 4.163632e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 1.8270e-03	k_t
Generation 9320: Best = 4.163631e+03 = [4, 3, 2, 3]	FES = 932100	Diversity = 1.7799e-03	k_t
Generation 9330: Best = 4.163630e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 2.0470e-03	k_t
Generation 9340: Best = 4.163628e+03 = [4, 2, 4, 2]	FES = 934100	Diversity = 2.3120e-03	k_t
Generation 9350: Best = 4.163626e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 2.2769e-03	k_t
Generation 9360: Best = 4.163624e+03 = [2, 4, 3, 2]	FES = 936100	Diversity = 2.2447e-03	k_t
Generation 9370: Best = 4.163621e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 2.6750e-03	k_t
Generation 9380: Best = 4.163619e+03 = [2, 2, 3, 2]	FES = 938100	Diversity = 2.9393e-03	k_t
Generation 9390: Best = 4.163616e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 2.6756e-03	k_t
Generation 9400: Best = 4.163615e+03 = [3, 4, 3, 4]	FES = 940100	Diversity = 2.2382e-03	k_t

Generation 9410: Best = 4.163614e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 1.8238e-03	k_t
Generation 9420: Best = 4.163613e+03 = [4, 4, 3, 2]	FES = 942100	Diversity = 1.5133e-03	k_t
Generation 9430: Best = 4.163612e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 1.1978e-03	k_t
Generation 9440: Best = 4.163612e+03 = [2, 3, 2, 2]	FES = 944100	Diversity = 1.3313e-03	k_t
Generation 9450: Best = 4.163611e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 1.1619e-03	k_t
Generation 9460: Best = 4.163610e+03 = [3, 3, 4, 2]	FES = 946100	Diversity = 1.3326e-03	k_t
Generation 9470: Best = 4.163609e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 1.4380e-03	k_t
Generation 9480: Best = 4.163609e+03 = [4, 3, 4, 2]	FES = 948100	Diversity = 1.4278e-03	k_t
Generation 9490: Best = 4.163608e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 1.4443e-03	k_t
Generation 9500: Best = 4.163607e+03 = [4, 4, 2, 4]	FES = 950100	Diversity = 1.4056e-03	k_t
Generation 9510: Best = 4.163607e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 1.5323e-03	k_t
Generation 9520: Best = 4.163606e+03 = [3, 3, 3, 3]	FES = 952100	Diversity = 1.2352e-03	k_t
Generation 9530: Best = 4.163605e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 1.2158e-03	k_t
Generation 9540: Best = 4.163605e+03 = [4, 4, 2, 4]	FES = 954100	Diversity = 1.1531e-03	k_t
Generation 9550: Best = 4.163604e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 1.3318e-03	k_t
Generation 9560: Best = 4.163604e+03 = [2, 2, 3, 4]	FES = 956100	Diversity = 1.3100e-03	k_t
Generation 9570: Best = 4.163603e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 1.1796e-03	k_t
Generation 9580: Best = 4.163602e+03 = [2, 2, 2, 3]	FES = 958100	Diversity = 1.1978e-03	k_t
Generation 9590: Best = 4.163601e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 1.2618e-03	k_t
Generation 9600: Best = 4.163601e+03 = [4, 4, 3, 3]	FES = 960100	Diversity = 1.2844e-03	k_t
Generation 9610: Best = 4.163600e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 1.2154e-03	k_t
Generation 9620: Best = 4.163600e+03 = [3, 2, 2, 4]	FES = 962100	Diversity = 1.3973e-03	k_t
Generation 9630: Best = 4.163599e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 1.1724e-03	k_t
Generation 9640: Best = 4.163598e+03 = [2, 4, 4, 2]	FES = 964100	Diversity = 1.5092e-03	k_t
Generation 9650: Best = 4.163598e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 1.3223e-03	k_t
Generation 9660: Best = 4.163597e+03 = [2, 4, 3, 2]	FES = 966100	Diversity = 1.2042e-03	k_t
Generation 9670: Best = 4.163597e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 1.3135e-03	k_t

Generation 9680: Best = 4.163596e+03 = [2, 2, 3, 3]	FES = 968100	Diversity = 1.0488e-03	k_t
Generation 9690: Best = 4.163596e+03 = [1, 1, 1, 1]	FES = 969100	Diversity = 9.4535e-04	k_t
Generation 9700: Best = 4.163596e+03 = [4, 4, 2, 4]	FES = 970100	Diversity = 1.1746e-03	k_t
Generation 9710: Best = 4.163595e+03 = [1, 1, 1, 1]	FES = 971100	Diversity = 1.1965e-03	k_t
Generation 9720: Best = 4.163595e+03 = [2, 3, 2, 3]	FES = 972100	Diversity = 1.2937e-03	k_t
Generation 9730: Best = 4.163594e+03 = [1, 1, 1, 1]	FES = 973100	Diversity = 1.5280e-03	k_t
Generation 9740: Best = 4.163593e+03 = [3, 3, 2, 2]	FES = 974100	Diversity = 1.5892e-03	k_t
Generation 9750: Best = 4.163592e+03 = [1, 1, 1, 1]	FES = 975100	Diversity = 1.7657e-03	k_t
Generation 9760: Best = 4.163591e+03 = [2, 4, 3, 4]	FES = 976100	Diversity = 1.9570e-03	k_t
Generation 9770: Best = 4.163590e+03 = [1, 1, 1, 1]	FES = 977100	Diversity = 1.7385e-03	k_t
Generation 9780: Best = 4.163589e+03 = [4, 2, 2, 2]	FES = 978100	Diversity = 1.6021e-03	k_t
Generation 9790: Best = 4.163588e+03 = [1, 1, 1, 1]	FES = 979100	Diversity = 1.4801e-03	k_t
Generation 9800: Best = 4.163587e+03 = [3, 4, 3, 3]	FES = 980100	Diversity = 1.4863e-03	k_t
Generation 9810: Best = 4.163586e+03 = [1, 1, 1, 1]	FES = 981100	Diversity = 1.4572e-03	k_t
Generation 9820: Best = 4.163586e+03 = [2, 2, 2, 3]	FES = 982100	Diversity = 1.5990e-03	k_t
Generation 9830: Best = 4.163584e+03 = [1, 1, 1, 1]	FES = 983100	Diversity = 1.8365e-03	k_t
Generation 9840: Best = 4.163583e+03 = [2, 4, 4, 3]	FES = 984100	Diversity = 1.7372e-03	k_t
Generation 9850: Best = 4.163582e+03 = [1, 1, 1, 1]	FES = 985100	Diversity = 2.0823e-03	k_t
Generation 9860: Best = 4.163580e+03 = [2, 4, 4, 3]	FES = 986100	Diversity = 1.9925e-03	k_t
Generation 9870: Best = 4.163579e+03 = [1, 1, 1, 1]	FES = 987100	Diversity = 2.0297e-03	k_t
Generation 9880: Best = 4.163578e+03 = [2, 4, 2, 2]	FES = 988100	Diversity = 1.8782e-03	k_t
Generation 9890: Best = 4.163577e+03 = [1, 1, 1, 1]	FES = 989100	Diversity = 1.7442e-03	k_t
Generation 9900: Best = 4.163576e+03 = [2, 2, 2, 3]	FES = 990100	Diversity = 1.7332e-03	k_t
Generation 9910: Best = 4.163575e+03 = [1, 1, 1, 1]	FES = 991100	Diversity = 1.5602e-03	k_t
Generation 9920: Best = 4.163574e+03 = [2, 4, 3, 2]	FES = 992100	Diversity = 1.3498e-03	k_t
Generation 9930: Best = 4.163573e+03 = [1, 1, 1, 1]	FES = 993100	Diversity = 1.4895e-03	k_t
Generation 9940: Best = 4.163572e+03 = [2, 2, 3, 4]	FES = 994100	Diversity = 1.8737e-03	k_t

Generation 9950: Best = 4.163570e+03 FES = 995100 Diversity = 1.7676e-03 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 4.163569e+03 FES = 996100 Diversity = 1.7831e-03 k_t
= [2, 4, 3, 2]
Generation 9970: Best = 4.163568e+03 FES = 997100 Diversity = 1.7589e-03 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 4.163567e+03 FES = 998100 Diversity = 1.6240e-03 k_t
= [4, 4, 4, 4]
Generation 9990: Best = 4.163567e+03 FES = 999100 Diversity = 1.4032e-03 k_t
= [1, 1, 1, 1]

Optimization completed!

Final Best Fitness = 4.163566e+03

Total Function Evaluations = 1000000

=====

=

Results:

Best Fitness : 4.163566e+03

Global Optimum : 1.300000e+03

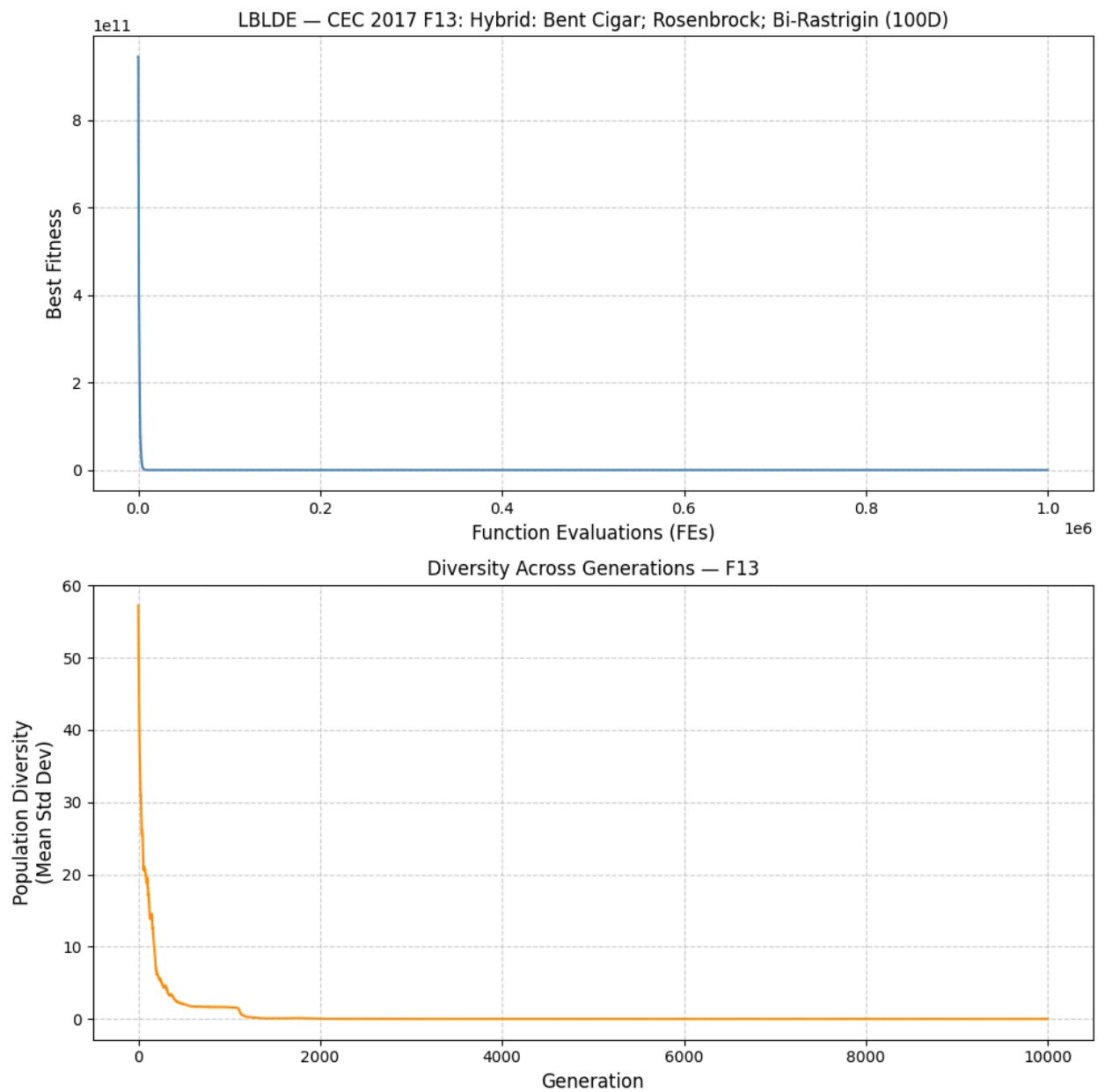
Error (|best-optimum|) : 2.863566e+03

Best Solution (5 dim) : [7.00735609 -70.4218114 -8.7978873 -12.84655019
25.93506638]

D^(0) (init. diversity): 5.7220e+01

=====

=



Graph saved as convergence_F13.png

```
In [38]: # =====
# F14 – Hybrid Function 4 (Elliptic; Ackley; Schaffer; Rastrigin)
# =====
run_and_plot('F14', make_wrapper(hybrid.f14), 1400.0, 'Hybrid: Elliptic; Ackle
```

```

=====
=
LBLDE_s on CEC 2017 F14 – Single Run
=====
=
Function      : Hybrid: Elliptic; Ackley; Schaffer; Rastrigin (100D)
Bounds       : [-100, 100]
Global optimum : 1400.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 7.040180e+08 | Diversity = 5.7220e+01
Generation 10: Best = 1.216529e+08 FES = 1100 Diversity = 4.9633e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 4.991195e+07 FES = 2100 Diversity = 4.4460e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 1.950754e+07 FES = 3100 Diversity = 4.0131e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.219308e+07 FES = 4100 Diversity = 3.6359e+01 k_t
= [2, 3, 4, 3]
Generation 50: Best = 8.492419e+06 FES = 5100 Diversity = 3.0905e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 4.975315e+06 FES = 6100 Diversity = 2.6425e+01 k_t
= [3, 4, 4, 2]
Generation 70: Best = 3.690946e+06 FES = 7100 Diversity = 2.5368e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.601939e+06 FES = 8100 Diversity = 2.2504e+01 k_t
= [4, 2, 2, 3]
Generation 90: Best = 1.832585e+06 FES = 9100 Diversity = 2.2151e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.160878e+06 FES = 10100 Diversity = 2.1286e+01 k_t
= [4, 3, 3, 4]
Generation 110: Best = 9.384946e+05 FES = 11100 Diversity = 2.0045e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 7.425233e+05 FES = 12100 Diversity = 1.9266e+01 k_t
= [3, 3, 3, 4]
Generation 130: Best = 6.130466e+05 FES = 13100 Diversity = 1.7512e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 5.597559e+05 FES = 14100 Diversity = 1.6234e+01 k_t
= [3, 3, 3, 2]
Generation 150: Best = 5.030619e+05 FES = 15100 Diversity = 1.7010e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 4.679370e+05 FES = 16100 Diversity = 1.7089e+01 k_t
= [3, 2, 3, 2]
Generation 170: Best = 4.380345e+05 FES = 17100 Diversity = 1.5484e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 4.072049e+05 FES = 18100 Diversity = 1.3968e+01 k_t
= [3, 4, 3, 4]
Generation 190: Best = 3.723942e+05 FES = 19100 Diversity = 1.5180e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.263000e+05 FES = 20100 Diversity = 1.8553e+01 k_t
= [4, 2, 4, 4]
Generation 210: Best = 3.262688e+05 FES = 21100 Diversity = 1.8703e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.045840e+05 FES = 22100 Diversity = 1.9748e+01 k_t
= [3, 2, 4, 4]

```

Generation 230:	Best = 2.829521e+05	FES = 23100	Diversity = 2.0828e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 2.699883e+05	FES = 24100	Diversity = 2.3081e+01	k_t = [4, 3, 3, 4]
Generation 250:	Best = 2.505575e+05	FES = 25100	Diversity = 2.3841e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 2.359812e+05	FES = 26100	Diversity = 2.4865e+01	k_t = [2, 2, 2, 4]
Generation 270:	Best = 2.336985e+05	FES = 27100	Diversity = 2.3293e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 2.285808e+05	FES = 28100	Diversity = 2.3077e+01	k_t = [4, 3, 2, 2]
Generation 290:	Best = 2.182322e+05	FES = 29100	Diversity = 2.2552e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 2.182002e+05	FES = 30100	Diversity = 2.1238e+01	k_t = [4, 3, 4, 3]
Generation 310:	Best = 2.164789e+05	FES = 31100	Diversity = 2.2465e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 2.145242e+05	FES = 32100	Diversity = 2.3207e+01	k_t = [3, 4, 4, 4]
Generation 330:	Best = 2.125530e+05	FES = 33100	Diversity = 2.3778e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 2.057183e+05	FES = 34100	Diversity = 2.4267e+01	k_t = [4, 4, 3, 4]
Generation 350:	Best = 2.046028e+05	FES = 35100	Diversity = 2.4911e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 2.027405e+05	FES = 36100	Diversity = 2.3736e+01	k_t = [2, 2, 3, 4]
Generation 370:	Best = 1.986077e+05	FES = 37100	Diversity = 2.4405e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 1.950733e+05	FES = 38100	Diversity = 2.5010e+01	k_t = [3, 4, 3, 2]
Generation 390:	Best = 1.930312e+05	FES = 39100	Diversity = 2.4894e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 1.867571e+05	FES = 40100	Diversity = 2.5640e+01	k_t = [3, 4, 2, 2]
Generation 410:	Best = 1.866648e+05	FES = 41100	Diversity = 2.6298e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.860006e+05	FES = 42100	Diversity = 2.4155e+01	k_t = [4, 2, 2, 4]
Generation 430:	Best = 1.851866e+05	FES = 43100	Diversity = 2.3191e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.851108e+05	FES = 44100	Diversity = 2.2763e+01	k_t = [4, 3, 2, 2]
Generation 450:	Best = 1.844884e+05	FES = 45100	Diversity = 2.2765e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.837977e+05	FES = 46100	Diversity = 2.3316e+01	k_t = [3, 3, 2, 3]
Generation 470:	Best = 1.828995e+05	FES = 47100	Diversity = 2.3206e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.813812e+05	FES = 48100	Diversity = 2.2818e+01	k_t = [4, 2, 2, 4]
Generation 490:	Best = 1.805552e+05	FES = 49100	Diversity = 2.3156e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.791162e+05	FES = 50100	Diversity = 2.3861e+01	k_t = [2, 4, 2, 3]
Generation 510:	Best = 1.777277e+05	FES = 51100	Diversity = 2.4815e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 1.774277e+05	FES = 52100	Diversity = 2.5247e+01	k_t = [3, 2, 2, 4]
Generation 530:	Best = 1.762749e+05	FES = 53100	Diversity = 2.5071e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 1.738634e+05	FES = 54100	Diversity = 2.4749e+01	k_t = [2, 2, 3, 4]
Generation 550:	Best = 1.706589e+05	FES = 55100	Diversity = 2.8508e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 1.694858e+05	FES = 56100	Diversity = 2.9554e+01	k_t = [2, 4, 2, 4]
Generation 570:	Best = 1.670900e+05	FES = 57100	Diversity = 3.2302e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 1.665753e+05	FES = 58100	Diversity = 3.2745e+01	k_t = [4, 2, 4, 4]
Generation 590:	Best = 1.658276e+05	FES = 59100	Diversity = 3.1486e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 1.643824e+05	FES = 60100	Diversity = 3.0541e+01	k_t = [4, 3, 3, 2]
Generation 610:	Best = 1.643390e+05	FES = 61100	Diversity = 3.0413e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 1.630810e+05	FES = 62100	Diversity = 2.7989e+01	k_t = [3, 4, 3, 2]
Generation 630:	Best = 1.630317e+05	FES = 63100	Diversity = 2.5882e+01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 1.623341e+05	FES = 64100	Diversity = 2.4354e+01	k_t = [3, 4, 2, 2]
Generation 650:	Best = 1.618650e+05	FES = 65100	Diversity = 2.3801e+01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 1.615168e+05	FES = 66100	Diversity = 2.4094e+01	k_t = [3, 3, 2, 3]
Generation 670:	Best = 1.611001e+05	FES = 67100	Diversity = 2.4952e+01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 1.602435e+05	FES = 68100	Diversity = 2.6552e+01	k_t = [3, 2, 2, 3]
Generation 690:	Best = 1.597230e+05	FES = 69100	Diversity = 2.8885e+01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 1.589764e+05	FES = 70100	Diversity = 2.8856e+01	k_t = [2, 4, 4, 2]
Generation 710:	Best = 1.579481e+05	FES = 71100	Diversity = 2.9274e+01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 1.577361e+05	FES = 72100	Diversity = 2.8849e+01	k_t = [4, 3, 2, 3]
Generation 730:	Best = 1.570691e+05	FES = 73100	Diversity = 2.7976e+01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 1.568119e+05	FES = 74100	Diversity = 2.7748e+01	k_t = [4, 3, 4, 2]
Generation 750:	Best = 1.563868e+05	FES = 75100	Diversity = 2.6566e+01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 1.561169e+05	FES = 76100	Diversity = 2.6552e+01	k_t = [3, 4, 4, 2]

Generation 770:	Best = 1.556042e+05	FES = 77100	Diversity = 2.7472e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 1.544894e+05	FES = 78100	Diversity = 2.9949e+01	k_t
= [3, 2, 2, 4]				
Generation 790:	Best = 1.535088e+05	FES = 79100	Diversity = 3.0075e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 1.535088e+05	FES = 80100	Diversity = 3.1441e+01	k_t
= [2, 2, 4, 2]				
Generation 810:	Best = 1.529432e+05	FES = 81100	Diversity = 2.9894e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 1.523321e+05	FES = 82100	Diversity = 3.0080e+01	k_t
= [2, 2, 3, 3]				
Generation 830:	Best = 1.520331e+05	FES = 83100	Diversity = 3.0888e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 1.514491e+05	FES = 84100	Diversity = 3.0318e+01	k_t
= [2, 4, 4, 4]				
Generation 850:	Best = 1.512394e+05	FES = 85100	Diversity = 3.0615e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 1.505745e+05	FES = 86100	Diversity = 3.1104e+01	k_t
= [2, 4, 3, 4]				
Generation 870:	Best = 1.501138e+05	FES = 87100	Diversity = 2.9121e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 1.499006e+05	FES = 88100	Diversity = 2.8135e+01	k_t
= [4, 3, 2, 3]				
Generation 890:	Best = 1.495620e+05	FES = 89100	Diversity = 2.8078e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 1.494966e+05	FES = 90100	Diversity = 2.8967e+01	k_t
= [2, 4, 4, 4]				
Generation 910:	Best = 1.490817e+05	FES = 91100	Diversity = 2.8514e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 1.484965e+05	FES = 92100	Diversity = 2.9153e+01	k_t
= [3, 3, 3, 4]				
Generation 930:	Best = 1.479457e+05	FES = 93100	Diversity = 3.0097e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 1.478784e+05	FES = 94100	Diversity = 3.0199e+01	k_t
= [4, 4, 3, 2]				
Generation 950:	Best = 1.476385e+05	FES = 95100	Diversity = 2.9709e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.468421e+05	FES = 96100	Diversity = 2.8718e+01	k_t
= [4, 4, 3, 4]				
Generation 970:	Best = 1.465040e+05	FES = 97100	Diversity = 2.8846e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.463318e+05	FES = 98100	Diversity = 2.8902e+01	k_t
= [2, 4, 4, 4]				
Generation 990:	Best = 1.454871e+05	FES = 99100	Diversity = 2.9141e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 1.449037e+05	FES = 100100	Diversity = 2.9818e+01	k_t
= [2, 3, 4, 4]				
Generation 1010:	Best = 1.447216e+05	FES = 101100	Diversity = 3.1394e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.441766e+05	FES = 102100	Diversity = 3.1602e+01	k_t
= [3, 2, 2, 4]				
Generation 1030:	Best = 1.437339e+05	FES = 103100	Diversity = 3.0843e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.433462e+05 = [4, 3, 4, 3]	FES = 104100	Diversity = 3.1621e+01	k_t
Generation 1050: Best = 1.429251e+05 = [1, 1, 1, 1]	FES = 105100	Diversity = 3.3043e+01	k_t
Generation 1060: Best = 1.419159e+05 = [2, 4, 2, 3]	FES = 106100	Diversity = 3.4090e+01	k_t
Generation 1070: Best = 1.406425e+05 = [1, 1, 1, 1]	FES = 107100	Diversity = 3.3496e+01	k_t
Generation 1080: Best = 1.399933e+05 = [4, 2, 2, 3]	FES = 108100	Diversity = 3.3490e+01	k_t
Generation 1090: Best = 1.399009e+05 = [1, 1, 1, 1]	FES = 109100	Diversity = 3.2333e+01	k_t
Generation 1100: Best = 1.390746e+05 = [2, 4, 2, 2]	FES = 110100	Diversity = 3.2087e+01	k_t
Generation 1110: Best = 1.380532e+05 = [1, 1, 1, 1]	FES = 111100	Diversity = 3.2100e+01	k_t
Generation 1120: Best = 1.378763e+05 = [4, 4, 2, 3]	FES = 112100	Diversity = 3.3346e+01	k_t
Generation 1130: Best = 1.372870e+05 = [1, 1, 1, 1]	FES = 113100	Diversity = 3.2770e+01	k_t
Generation 1140: Best = 1.369395e+05 = [4, 4, 4, 4]	FES = 114100	Diversity = 3.1015e+01	k_t
Generation 1150: Best = 1.365181e+05 = [1, 1, 1, 1]	FES = 115100	Diversity = 2.8995e+01	k_t
Generation 1160: Best = 1.360364e+05 = [2, 4, 2, 2]	FES = 116100	Diversity = 2.8981e+01	k_t
Generation 1170: Best = 1.360077e+05 = [1, 1, 1, 1]	FES = 117100	Diversity = 2.8357e+01	k_t
Generation 1180: Best = 1.353959e+05 = [2, 2, 3, 3]	FES = 118100	Diversity = 2.9937e+01	k_t
Generation 1190: Best = 1.343502e+05 = [1, 1, 1, 1]	FES = 119100	Diversity = 3.2981e+01	k_t
Generation 1200: Best = 1.335041e+05 = [3, 4, 3, 3]	FES = 120100	Diversity = 3.3185e+01	k_t
Generation 1210: Best = 1.325990e+05 = [1, 1, 1, 1]	FES = 121100	Diversity = 3.4140e+01	k_t
Generation 1220: Best = 1.318463e+05 = [4, 4, 3, 4]	FES = 122100	Diversity = 3.3535e+01	k_t
Generation 1230: Best = 1.311192e+05 = [1, 1, 1, 1]	FES = 123100	Diversity = 3.1868e+01	k_t
Generation 1240: Best = 1.309750e+05 = [3, 3, 3, 2]	FES = 124100	Diversity = 3.0780e+01	k_t
Generation 1250: Best = 1.305881e+05 = [1, 1, 1, 1]	FES = 125100	Diversity = 2.8983e+01	k_t
Generation 1260: Best = 1.297499e+05 = [3, 3, 4, 3]	FES = 126100	Diversity = 2.7705e+01	k_t
Generation 1270: Best = 1.289851e+05 = [1, 1, 1, 1]	FES = 127100	Diversity = 2.8380e+01	k_t
Generation 1280: Best = 1.288565e+05 = [4, 4, 4, 3]	FES = 128100	Diversity = 2.8733e+01	k_t
Generation 1290: Best = 1.286382e+05 = [1, 1, 1, 1]	FES = 129100	Diversity = 2.6371e+01	k_t
Generation 1300: Best = 1.281726e+05 = [3, 3, 2, 3]	FES = 130100	Diversity = 2.7078e+01	k_t

Generation 1310: Best = 1.279058e+05 = [1, 1, 1, 1]	FES = 131100	Diversity = 2.6100e+01	k_t
Generation 1320: Best = 1.272742e+05 = [4, 2, 4, 3]	FES = 132100	Diversity = 2.4647e+01	k_t
Generation 1330: Best = 1.271033e+05 = [1, 1, 1, 1]	FES = 133100	Diversity = 2.4829e+01	k_t
Generation 1340: Best = 1.269900e+05 = [2, 3, 3, 3]	FES = 134100	Diversity = 2.4650e+01	k_t
Generation 1350: Best = 1.263464e+05 = [1, 1, 1, 1]	FES = 135100	Diversity = 2.5672e+01	k_t
Generation 1360: Best = 1.258807e+05 = [4, 2, 2, 2]	FES = 136100	Diversity = 2.6374e+01	k_t
Generation 1370: Best = 1.254442e+05 = [1, 1, 1, 1]	FES = 137100	Diversity = 2.7351e+01	k_t
Generation 1380: Best = 1.253238e+05 = [4, 2, 4, 2]	FES = 138100	Diversity = 2.7046e+01	k_t
Generation 1390: Best = 1.248022e+05 = [1, 1, 1, 1]	FES = 139100	Diversity = 2.6812e+01	k_t
Generation 1400: Best = 1.246223e+05 = [3, 2, 3, 3]	FES = 140100	Diversity = 2.6693e+01	k_t
Generation 1410: Best = 1.243392e+05 = [1, 1, 1, 1]	FES = 141100	Diversity = 2.6024e+01	k_t
Generation 1420: Best = 1.241532e+05 = [2, 4, 3, 3]	FES = 142100	Diversity = 2.5342e+01	k_t
Generation 1430: Best = 1.240355e+05 = [1, 1, 1, 1]	FES = 143100	Diversity = 2.4086e+01	k_t
Generation 1440: Best = 1.236408e+05 = [2, 2, 2, 2]	FES = 144100	Diversity = 2.5085e+01	k_t
Generation 1450: Best = 1.233059e+05 = [1, 1, 1, 1]	FES = 145100	Diversity = 2.3756e+01	k_t
Generation 1460: Best = 1.229152e+05 = [4, 2, 4, 2]	FES = 146100	Diversity = 2.4670e+01	k_t
Generation 1470: Best = 1.223926e+05 = [1, 1, 1, 1]	FES = 147100	Diversity = 2.7162e+01	k_t
Generation 1480: Best = 1.219673e+05 = [4, 2, 3, 3]	FES = 148100	Diversity = 2.6017e+01	k_t
Generation 1490: Best = 1.216609e+05 = [1, 1, 1, 1]	FES = 149100	Diversity = 2.6926e+01	k_t
Generation 1500: Best = 1.214517e+05 = [4, 3, 3, 2]	FES = 150100	Diversity = 2.7509e+01	k_t
Generation 1510: Best = 1.210185e+05 = [1, 1, 1, 1]	FES = 151100	Diversity = 2.8147e+01	k_t
Generation 1520: Best = 1.207951e+05 = [3, 4, 4, 2]	FES = 152100	Diversity = 2.6150e+01	k_t
Generation 1530: Best = 1.204080e+05 = [1, 1, 1, 1]	FES = 153100	Diversity = 2.6200e+01	k_t
Generation 1540: Best = 1.192783e+05 = [2, 3, 2, 3]	FES = 154100	Diversity = 2.8431e+01	k_t
Generation 1550: Best = 1.189546e+05 = [1, 1, 1, 1]	FES = 155100	Diversity = 2.9093e+01	k_t
Generation 1560: Best = 1.187347e+05 = [3, 4, 2, 4]	FES = 156100	Diversity = 2.7815e+01	k_t
Generation 1570: Best = 1.180263e+05 = [1, 1, 1, 1]	FES = 157100	Diversity = 2.7611e+01	k_t

Generation 1580: Best = 1.176082e+05 = [2, 2, 3, 4]	FES = 158100	Diversity = 2.5761e+01	k_t
Generation 1590: Best = 1.171597e+05 = [1, 1, 1, 1]	FES = 159100	Diversity = 2.6274e+01	k_t
Generation 1600: Best = 1.166996e+05 = [3, 3, 2, 3]	FES = 160100	Diversity = 2.7188e+01	k_t
Generation 1610: Best = 1.164662e+05 = [1, 1, 1, 1]	FES = 161100	Diversity = 2.8261e+01	k_t
Generation 1620: Best = 1.160558e+05 = [4, 3, 3, 4]	FES = 162100	Diversity = 2.7249e+01	k_t
Generation 1630: Best = 1.156941e+05 = [1, 1, 1, 1]	FES = 163100	Diversity = 2.4990e+01	k_t
Generation 1640: Best = 1.154089e+05 = [3, 3, 2, 2]	FES = 164100	Diversity = 2.3304e+01	k_t
Generation 1650: Best = 1.150991e+05 = [1, 1, 1, 1]	FES = 165100	Diversity = 2.2865e+01	k_t
Generation 1660: Best = 1.149169e+05 = [2, 2, 3, 2]	FES = 166100	Diversity = 2.3409e+01	k_t
Generation 1670: Best = 1.147498e+05 = [1, 1, 1, 1]	FES = 167100	Diversity = 2.4177e+01	k_t
Generation 1680: Best = 1.145641e+05 = [3, 4, 3, 3]	FES = 168100	Diversity = 2.4882e+01	k_t
Generation 1690: Best = 1.142381e+05 = [1, 1, 1, 1]	FES = 169100	Diversity = 2.6629e+01	k_t
Generation 1700: Best = 1.140274e+05 = [3, 2, 2, 2]	FES = 170100	Diversity = 2.8076e+01	k_t
Generation 1710: Best = 1.139593e+05 = [1, 1, 1, 1]	FES = 171100	Diversity = 2.8130e+01	k_t
Generation 1720: Best = 1.135907e+05 = [2, 4, 2, 3]	FES = 172100	Diversity = 2.7155e+01	k_t
Generation 1730: Best = 1.133233e+05 = [1, 1, 1, 1]	FES = 173100	Diversity = 2.8128e+01	k_t
Generation 1740: Best = 1.128823e+05 = [2, 4, 3, 2]	FES = 174100	Diversity = 2.9160e+01	k_t
Generation 1750: Best = 1.117989e+05 = [1, 1, 1, 1]	FES = 175100	Diversity = 3.0022e+01	k_t
Generation 1760: Best = 1.110428e+05 = [3, 4, 2, 4]	FES = 176100	Diversity = 3.2237e+01	k_t
Generation 1770: Best = 1.107188e+05 = [1, 1, 1, 1]	FES = 177100	Diversity = 3.1719e+01	k_t
Generation 1780: Best = 1.101525e+05 = [3, 4, 2, 2]	FES = 178100	Diversity = 3.1775e+01	k_t
Generation 1790: Best = 1.099537e+05 = [1, 1, 1, 1]	FES = 179100	Diversity = 3.0515e+01	k_t
Generation 1800: Best = 1.094937e+05 = [2, 3, 3, 2]	FES = 180100	Diversity = 2.8903e+01	k_t
Generation 1810: Best = 1.092492e+05 = [1, 1, 1, 1]	FES = 181100	Diversity = 2.9538e+01	k_t
Generation 1820: Best = 1.086576e+05 = [3, 2, 4, 4]	FES = 182100	Diversity = 3.0689e+01	k_t
Generation 1830: Best = 1.083704e+05 = [1, 1, 1, 1]	FES = 183100	Diversity = 3.1110e+01	k_t
Generation 1840: Best = 1.075499e+05 = [3, 2, 3, 4]	FES = 184100	Diversity = 3.2361e+01	k_t

Generation 1850: Best = 1.069151e+05 = [1, 1, 1, 1]	FES = 185100	Diversity = 3.0928e+01	k_t
Generation 1860: Best = 1.067006e+05 = [2, 2, 4, 3]	FES = 186100	Diversity = 2.9617e+01	k_t
Generation 1870: Best = 1.062121e+05 = [1, 1, 1, 1]	FES = 187100	Diversity = 2.8903e+01	k_t
Generation 1880: Best = 1.060034e+05 = [4, 4, 2, 4]	FES = 188100	Diversity = 2.7507e+01	k_t
Generation 1890: Best = 1.055724e+05 = [1, 1, 1, 1]	FES = 189100	Diversity = 2.5625e+01	k_t
Generation 1900: Best = 1.050411e+05 = [4, 3, 2, 3]	FES = 190100	Diversity = 2.4792e+01	k_t
Generation 1910: Best = 1.045698e+05 = [1, 1, 1, 1]	FES = 191100	Diversity = 2.6570e+01	k_t
Generation 1920: Best = 1.041368e+05 = [3, 4, 4, 3]	FES = 192100	Diversity = 2.5240e+01	k_t
Generation 1930: Best = 1.038525e+05 = [1, 1, 1, 1]	FES = 193100	Diversity = 2.4093e+01	k_t
Generation 1940: Best = 1.029521e+05 = [2, 4, 4, 3]	FES = 194100	Diversity = 2.6111e+01	k_t
Generation 1950: Best = 1.023192e+05 = [1, 1, 1, 1]	FES = 195100	Diversity = 2.9585e+01	k_t
Generation 1960: Best = 1.017210e+05 = [3, 2, 2, 2]	FES = 196100	Diversity = 3.3200e+01	k_t
Generation 1970: Best = 1.010033e+05 = [1, 1, 1, 1]	FES = 197100	Diversity = 3.3167e+01	k_t
Generation 1980: Best = 1.006938e+05 = [3, 3, 4, 3]	FES = 198100	Diversity = 3.2685e+01	k_t
Generation 1990: Best = 9.977639e+04 = [1, 1, 1, 1]	FES = 199100	Diversity = 3.3632e+01	k_t
Generation 2000: Best = 9.842355e+04 = [4, 4, 3, 4]	FES = 200100	Diversity = 3.5241e+01	k_t
Generation 2010: Best = 9.817344e+04 = [1, 1, 1, 1]	FES = 201100	Diversity = 3.1757e+01	k_t
Generation 2020: Best = 9.752535e+04 = [4, 2, 4, 2]	FES = 202100	Diversity = 3.0870e+01	k_t
Generation 2030: Best = 9.735854e+04 = [1, 1, 1, 1]	FES = 203100	Diversity = 3.0368e+01	k_t
Generation 2040: Best = 9.717987e+04 = [2, 3, 3, 2]	FES = 204100	Diversity = 2.9288e+01	k_t
Generation 2050: Best = 9.685852e+04 = [1, 1, 1, 1]	FES = 205100	Diversity = 2.8150e+01	k_t
Generation 2060: Best = 9.647012e+04 = [3, 3, 3, 3]	FES = 206100	Diversity = 2.6757e+01	k_t
Generation 2070: Best = 9.631238e+04 = [1, 1, 1, 1]	FES = 207100	Diversity = 2.7427e+01	k_t
Generation 2080: Best = 9.612655e+04 = [2, 2, 3, 2]	FES = 208100	Diversity = 2.7256e+01	k_t
Generation 2090: Best = 9.599376e+04 = [1, 1, 1, 1]	FES = 209100	Diversity = 2.6027e+01	k_t
Generation 2100: Best = 9.590205e+04 = [4, 3, 2, 3]	FES = 210100	Diversity = 2.4312e+01	k_t
Generation 2110: Best = 9.572127e+04 = [1, 1, 1, 1]	FES = 211100	Diversity = 2.3149e+01	k_t

Generation 2120: Best = 9.568489e+04 = [2, 2, 2, 4]	FES = 212100	Diversity = 2.2639e+01	k_t
Generation 2130: Best = 9.537269e+04 = [1, 1, 1, 1]	FES = 213100	Diversity = 2.3505e+01	k_t
Generation 2140: Best = 9.520366e+04 = [3, 3, 2, 3]	FES = 214100	Diversity = 2.5609e+01	k_t
Generation 2150: Best = 9.504854e+04 = [1, 1, 1, 1]	FES = 215100	Diversity = 2.6108e+01	k_t
Generation 2160: Best = 9.489383e+04 = [2, 3, 2, 4]	FES = 216100	Diversity = 2.5453e+01	k_t
Generation 2170: Best = 9.478107e+04 = [1, 1, 1, 1]	FES = 217100	Diversity = 2.5417e+01	k_t
Generation 2180: Best = 9.470967e+04 = [3, 3, 2, 2]	FES = 218100	Diversity = 2.5764e+01	k_t
Generation 2190: Best = 9.448426e+04 = [1, 1, 1, 1]	FES = 219100	Diversity = 2.4652e+01	k_t
Generation 2200: Best = 9.434739e+04 = [3, 4, 3, 3]	FES = 220100	Diversity = 2.3334e+01	k_t
Generation 2210: Best = 9.423643e+04 = [1, 1, 1, 1]	FES = 221100	Diversity = 2.2647e+01	k_t
Generation 2220: Best = 9.410274e+04 = [4, 4, 4, 2]	FES = 222100	Diversity = 2.3144e+01	k_t
Generation 2230: Best = 9.384118e+04 = [1, 1, 1, 1]	FES = 223100	Diversity = 2.3681e+01	k_t
Generation 2240: Best = 9.344124e+04 = [4, 4, 2, 4]	FES = 224100	Diversity = 2.4470e+01	k_t
Generation 2250: Best = 9.318936e+04 = [1, 1, 1, 1]	FES = 225100	Diversity = 2.5211e+01	k_t
Generation 2260: Best = 9.289872e+04 = [3, 2, 4, 4]	FES = 226100	Diversity = 2.5507e+01	k_t
Generation 2270: Best = 9.258634e+04 = [1, 1, 1, 1]	FES = 227100	Diversity = 2.4488e+01	k_t
Generation 2280: Best = 9.247340e+04 = [3, 3, 4, 2]	FES = 228100	Diversity = 2.5237e+01	k_t
Generation 2290: Best = 9.225241e+04 = [1, 1, 1, 1]	FES = 229100	Diversity = 2.5993e+01	k_t
Generation 2300: Best = 9.187573e+04 = [3, 4, 2, 2]	FES = 230100	Diversity = 2.5803e+01	k_t
Generation 2310: Best = 9.175870e+04 = [1, 1, 1, 1]	FES = 231100	Diversity = 2.6961e+01	k_t
Generation 2320: Best = 9.139357e+04 = [4, 2, 3, 3]	FES = 232100	Diversity = 2.7175e+01	k_t
Generation 2330: Best = 9.111728e+04 = [1, 1, 1, 1]	FES = 233100	Diversity = 2.8486e+01	k_t
Generation 2340: Best = 9.072288e+04 = [3, 2, 3, 2]	FES = 234100	Diversity = 2.9050e+01	k_t
Generation 2350: Best = 9.040232e+04 = [1, 1, 1, 1]	FES = 235100	Diversity = 2.8300e+01	k_t
Generation 2360: Best = 8.981695e+04 = [3, 3, 3, 4]	FES = 236100	Diversity = 2.9225e+01	k_t
Generation 2370: Best = 8.949011e+04 = [1, 1, 1, 1]	FES = 237100	Diversity = 3.1836e+01	k_t
Generation 2380: Best = 8.905601e+04 = [2, 2, 3, 2]	FES = 238100	Diversity = 3.0805e+01	k_t

Generation 2390: Best = 8.878243e+04 = [1, 1, 1, 1]	FES = 239100	Diversity = 2.8855e+01	k_t
Generation 2400: Best = 8.854465e+04 = [4, 2, 3, 2]	FES = 240100	Diversity = 2.6662e+01	k_t
Generation 2410: Best = 8.827978e+04 = [1, 1, 1, 1]	FES = 241100	Diversity = 2.6259e+01	k_t
Generation 2420: Best = 8.751312e+04 = [2, 4, 2, 2]	FES = 242100	Diversity = 2.6491e+01	k_t
Generation 2430: Best = 8.663064e+04 = [1, 1, 1, 1]	FES = 243100	Diversity = 3.1938e+01	k_t
Generation 2440: Best = 8.630613e+04 = [4, 4, 4, 2]	FES = 244100	Diversity = 3.1234e+01	k_t
Generation 2450: Best = 8.507753e+04 = [1, 1, 1, 1]	FES = 245100	Diversity = 3.1407e+01	k_t
Generation 2460: Best = 8.425937e+04 = [3, 2, 2, 2]	FES = 246100	Diversity = 3.1742e+01	k_t
Generation 2470: Best = 8.313140e+04 = [1, 1, 1, 1]	FES = 247100	Diversity = 3.1177e+01	k_t
Generation 2480: Best = 8.263759e+04 = [2, 2, 3, 2]	FES = 248100	Diversity = 3.2133e+01	k_t
Generation 2490: Best = 8.182687e+04 = [1, 1, 1, 1]	FES = 249100	Diversity = 3.1920e+01	k_t
Generation 2500: Best = 8.119269e+04 = [4, 2, 4, 4]	FES = 250100	Diversity = 3.2675e+01	k_t
Generation 2510: Best = 8.075657e+04 = [1, 1, 1, 1]	FES = 251100	Diversity = 3.3536e+01	k_t
Generation 2520: Best = 8.032762e+04 = [4, 4, 3, 2]	FES = 252100	Diversity = 3.3153e+01	k_t
Generation 2530: Best = 7.955085e+04 = [1, 1, 1, 1]	FES = 253100	Diversity = 3.2885e+01	k_t
Generation 2540: Best = 7.946097e+04 = [3, 2, 3, 4]	FES = 254100	Diversity = 3.0811e+01	k_t
Generation 2550: Best = 7.883216e+04 = [1, 1, 1, 1]	FES = 255100	Diversity = 2.7925e+01	k_t
Generation 2560: Best = 7.834712e+04 = [3, 2, 3, 3]	FES = 256100	Diversity = 2.9273e+01	k_t
Generation 2570: Best = 7.764056e+04 = [1, 1, 1, 1]	FES = 257100	Diversity = 2.9139e+01	k_t
Generation 2580: Best = 7.757571e+04 = [4, 2, 3, 4]	FES = 258100	Diversity = 2.7269e+01	k_t
Generation 2590: Best = 7.650165e+04 = [1, 1, 1, 1]	FES = 259100	Diversity = 2.7334e+01	k_t
Generation 2600: Best = 7.637456e+04 = [4, 2, 3, 3]	FES = 260100	Diversity = 2.7980e+01	k_t
Generation 2610: Best = 7.592343e+04 = [1, 1, 1, 1]	FES = 261100	Diversity = 2.5804e+01	k_t
Generation 2620: Best = 7.577033e+04 = [2, 4, 2, 4]	FES = 262100	Diversity = 2.5888e+01	k_t
Generation 2630: Best = 7.541126e+04 = [1, 1, 1, 1]	FES = 263100	Diversity = 2.7317e+01	k_t
Generation 2640: Best = 7.532063e+04 = [4, 4, 3, 4]	FES = 264100	Diversity = 2.7626e+01	k_t
Generation 2650: Best = 7.504355e+04 = [1, 1, 1, 1]	FES = 265100	Diversity = 2.7454e+01	k_t

Generation 2660: Best = 7.477218e+04 = [3, 2, 2, 3]	FES = 266100	Diversity = 2.6896e+01	k_t
Generation 2670: Best = 7.429510e+04 = [1, 1, 1, 1]	FES = 267100	Diversity = 2.5947e+01	k_t
Generation 2680: Best = 7.358680e+04 = [2, 3, 4, 3]	FES = 268100	Diversity = 2.8916e+01	k_t
Generation 2690: Best = 7.358680e+04 = [1, 1, 1, 1]	FES = 269100	Diversity = 2.9937e+01	k_t
Generation 2700: Best = 7.315664e+04 = [3, 2, 3, 2]	FES = 270100	Diversity = 2.9947e+01	k_t
Generation 2710: Best = 7.296313e+04 = [1, 1, 1, 1]	FES = 271100	Diversity = 3.0093e+01	k_t
Generation 2720: Best = 7.285389e+04 = [4, 2, 3, 3]	FES = 272100	Diversity = 2.9185e+01	k_t
Generation 2730: Best = 7.230458e+04 = [1, 1, 1, 1]	FES = 273100	Diversity = 2.8959e+01	k_t
Generation 2740: Best = 7.134929e+04 = [4, 2, 3, 4]	FES = 274100	Diversity = 3.0417e+01	k_t
Generation 2750: Best = 7.095925e+04 = [1, 1, 1, 1]	FES = 275100	Diversity = 3.1575e+01	k_t
Generation 2760: Best = 7.068560e+04 = [2, 3, 4, 3]	FES = 276100	Diversity = 3.1647e+01	k_t
Generation 2770: Best = 6.977750e+04 = [1, 1, 1, 1]	FES = 277100	Diversity = 3.2562e+01	k_t
Generation 2780: Best = 6.923941e+04 = [4, 3, 2, 4]	FES = 278100	Diversity = 3.2189e+01	k_t
Generation 2790: Best = 6.886450e+04 = [1, 1, 1, 1]	FES = 279100	Diversity = 3.2117e+01	k_t
Generation 2800: Best = 6.840360e+04 = [4, 2, 4, 3]	FES = 280100	Diversity = 3.0882e+01	k_t
Generation 2810: Best = 6.806395e+04 = [1, 1, 1, 1]	FES = 281100	Diversity = 3.0330e+01	k_t
Generation 2820: Best = 6.693318e+04 = [2, 2, 3, 3]	FES = 282100	Diversity = 3.1351e+01	k_t
Generation 2830: Best = 6.636442e+04 = [1, 1, 1, 1]	FES = 283100	Diversity = 3.0188e+01	k_t
Generation 2840: Best = 6.619569e+04 = [2, 3, 2, 2]	FES = 284100	Diversity = 3.1774e+01	k_t
Generation 2850: Best = 6.573789e+04 = [1, 1, 1, 1]	FES = 285100	Diversity = 3.1791e+01	k_t
Generation 2860: Best = 6.530382e+04 = [4, 2, 3, 2]	FES = 286100	Diversity = 3.2251e+01	k_t
Generation 2870: Best = 6.508409e+04 = [1, 1, 1, 1]	FES = 287100	Diversity = 3.0960e+01	k_t
Generation 2880: Best = 6.435253e+04 = [2, 3, 2, 3]	FES = 288100	Diversity = 3.1573e+01	k_t
Generation 2890: Best = 6.402836e+04 = [1, 1, 1, 1]	FES = 289100	Diversity = 3.0660e+01	k_t
Generation 2900: Best = 6.346877e+04 = [2, 2, 3, 4]	FES = 290100	Diversity = 2.8973e+01	k_t
Generation 2910: Best = 6.268871e+04 = [1, 1, 1, 1]	FES = 291100	Diversity = 3.1045e+01	k_t
Generation 2920: Best = 6.219904e+04 = [2, 2, 3, 3]	FES = 292100	Diversity = 3.2959e+01	k_t

Generation 2930: Best = 6.205099e+04 = [1, 1, 1, 1]	FES = 293100	Diversity = 3.1546e+01	k_t
Generation 2940: Best = 6.135830e+04 = [3, 3, 2, 2]	FES = 294100	Diversity = 3.1321e+01	k_t
Generation 2950: Best = 6.112535e+04 = [1, 1, 1, 1]	FES = 295100	Diversity = 3.2141e+01	k_t
Generation 2960: Best = 6.091359e+04 = [4, 4, 3, 2]	FES = 296100	Diversity = 3.1452e+01	k_t
Generation 2970: Best = 6.069212e+04 = [1, 1, 1, 1]	FES = 297100	Diversity = 3.0540e+01	k_t
Generation 2980: Best = 6.017602e+04 = [4, 2, 2, 3]	FES = 298100	Diversity = 2.9834e+01	k_t
Generation 2990: Best = 5.972182e+04 = [1, 1, 1, 1]	FES = 299100	Diversity = 2.9228e+01	k_t
Generation 3000: Best = 5.947766e+04 = [4, 2, 2, 2]	FES = 300100	Diversity = 3.0447e+01	k_t
Generation 3010: Best = 5.900240e+04 = [1, 1, 1, 1]	FES = 301100	Diversity = 3.1266e+01	k_t
Generation 3020: Best = 5.897946e+04 = [4, 2, 3, 2]	FES = 302100	Diversity = 2.9765e+01	k_t
Generation 3030: Best = 5.885502e+04 = [1, 1, 1, 1]	FES = 303100	Diversity = 2.9967e+01	k_t
Generation 3040: Best = 5.859943e+04 = [3, 4, 4, 2]	FES = 304100	Diversity = 2.7642e+01	k_t
Generation 3050: Best = 5.857186e+04 = [1, 1, 1, 1]	FES = 305100	Diversity = 2.6470e+01	k_t
Generation 3060: Best = 5.835406e+04 = [4, 4, 4, 4]	FES = 306100	Diversity = 2.5344e+01	k_t
Generation 3070: Best = 5.831400e+04 = [1, 1, 1, 1]	FES = 307100	Diversity = 2.4929e+01	k_t
Generation 3080: Best = 5.824009e+04 = [3, 3, 4, 3]	FES = 308100	Diversity = 2.4494e+01	k_t
Generation 3090: Best = 5.821333e+04 = [1, 1, 1, 1]	FES = 309100	Diversity = 2.3007e+01	k_t
Generation 3100: Best = 5.808928e+04 = [4, 3, 4, 4]	FES = 310100	Diversity = 2.3348e+01	k_t
Generation 3110: Best = 5.799719e+04 = [1, 1, 1, 1]	FES = 311100	Diversity = 2.2180e+01	k_t
Generation 3120: Best = 5.791248e+04 = [2, 2, 2, 4]	FES = 312100	Diversity = 2.1598e+01	k_t
Generation 3130: Best = 5.784708e+04 = [1, 1, 1, 1]	FES = 313100	Diversity = 2.1404e+01	k_t
Generation 3140: Best = 5.771774e+04 = [4, 3, 4, 3]	FES = 314100	Diversity = 2.0728e+01	k_t
Generation 3150: Best = 5.766438e+04 = [1, 1, 1, 1]	FES = 315100	Diversity = 1.9846e+01	k_t
Generation 3160: Best = 5.760910e+04 = [2, 3, 3, 4]	FES = 316100	Diversity = 2.0032e+01	k_t
Generation 3170: Best = 5.751759e+04 = [1, 1, 1, 1]	FES = 317100	Diversity = 1.9634e+01	k_t
Generation 3180: Best = 5.737287e+04 = [3, 2, 3, 4]	FES = 318100	Diversity = 2.0188e+01	k_t
Generation 3190: Best = 5.727409e+04 = [1, 1, 1, 1]	FES = 319100	Diversity = 1.9607e+01	k_t

Generation 3200: Best = 5.714205e+04 = [2, 4, 3, 2]	FES = 320100	Diversity = 2.0496e+01	k_t
Generation 3210: Best = 5.697038e+04 = [1, 1, 1, 1]	FES = 321100	Diversity = 2.2021e+01	k_t
Generation 3220: Best = 5.668085e+04 = [4, 2, 2, 3]	FES = 322100	Diversity = 2.5151e+01	k_t
Generation 3230: Best = 5.632831e+04 = [1, 1, 1, 1]	FES = 323100	Diversity = 2.6693e+01	k_t
Generation 3240: Best = 5.608787e+04 = [4, 4, 3, 2]	FES = 324100	Diversity = 2.6940e+01	k_t
Generation 3250: Best = 5.588912e+04 = [1, 1, 1, 1]	FES = 325100	Diversity = 2.8215e+01	k_t
Generation 3260: Best = 5.540572e+04 = [3, 4, 4, 4]	FES = 326100	Diversity = 3.0125e+01	k_t
Generation 3270: Best = 5.517558e+04 = [1, 1, 1, 1]	FES = 327100	Diversity = 3.1675e+01	k_t
Generation 3280: Best = 5.468938e+04 = [2, 4, 2, 3]	FES = 328100	Diversity = 3.1894e+01	k_t
Generation 3290: Best = 5.442643e+04 = [1, 1, 1, 1]	FES = 329100	Diversity = 3.0502e+01	k_t
Generation 3300: Best = 5.376429e+04 = [4, 2, 3, 2]	FES = 330100	Diversity = 3.0829e+01	k_t
Generation 3310: Best = 5.313985e+04 = [1, 1, 1, 1]	FES = 331100	Diversity = 2.9399e+01	k_t
Generation 3320: Best = 5.256318e+04 = [4, 4, 3, 4]	FES = 332100	Diversity = 3.0916e+01	k_t
Generation 3330: Best = 5.227150e+04 = [1, 1, 1, 1]	FES = 333100	Diversity = 3.0718e+01	k_t
Generation 3340: Best = 5.190808e+04 = [4, 3, 2, 3]	FES = 334100	Diversity = 2.9648e+01	k_t
Generation 3350: Best = 5.158354e+04 = [1, 1, 1, 1]	FES = 335100	Diversity = 2.9958e+01	k_t
Generation 3360: Best = 5.120005e+04 = [2, 4, 2, 2]	FES = 336100	Diversity = 2.9827e+01	k_t
Generation 3370: Best = 5.105958e+04 = [1, 1, 1, 1]	FES = 337100	Diversity = 2.7974e+01	k_t
Generation 3380: Best = 5.095577e+04 = [3, 2, 3, 4]	FES = 338100	Diversity = 2.8463e+01	k_t
Generation 3390: Best = 5.050827e+04 = [1, 1, 1, 1]	FES = 339100	Diversity = 2.9285e+01	k_t
Generation 3400: Best = 5.027700e+04 = [4, 3, 3, 3]	FES = 340100	Diversity = 2.9016e+01	k_t
Generation 3410: Best = 4.969129e+04 = [1, 1, 1, 1]	FES = 341100	Diversity = 3.0690e+01	k_t
Generation 3420: Best = 4.952288e+04 = [3, 4, 3, 3]	FES = 342100	Diversity = 3.3706e+01	k_t
Generation 3430: Best = 4.858784e+04 = [1, 1, 1, 1]	FES = 343100	Diversity = 3.2735e+01	k_t
Generation 3440: Best = 4.790900e+04 = [4, 3, 4, 4]	FES = 344100	Diversity = 3.1259e+01	k_t
Generation 3450: Best = 4.756760e+04 = [1, 1, 1, 1]	FES = 345100	Diversity = 3.0591e+01	k_t
Generation 3460: Best = 4.731710e+04 = [3, 2, 2, 3]	FES = 346100	Diversity = 2.8998e+01	k_t

Generation 3470: Best = 4.670879e+04	FES = 347100	Diversity = 2.7910e+01	k_t
= [1, 1, 1, 1]			
Generation 3480: Best = 4.648504e+04	FES = 348100	Diversity = 2.6531e+01	k_t
= [2, 4, 3, 2]			
Generation 3490: Best = 4.596146e+04	FES = 349100	Diversity = 2.5403e+01	k_t
= [1, 1, 1, 1]			
Generation 3500: Best = 4.536539e+04	FES = 350100	Diversity = 2.5656e+01	k_t
= [4, 3, 3, 3]			
Generation 3510: Best = 4.491996e+04	FES = 351100	Diversity = 2.6870e+01	k_t
= [1, 1, 1, 1]			
Generation 3520: Best = 4.439707e+04	FES = 352100	Diversity = 2.6308e+01	k_t
= [3, 2, 3, 4]			
Generation 3530: Best = 4.380727e+04	FES = 353100	Diversity = 2.7405e+01	k_t
= [1, 1, 1, 1]			
Generation 3540: Best = 4.372725e+04	FES = 354100	Diversity = 2.6882e+01	k_t
= [2, 3, 2, 4]			
Generation 3550: Best = 4.354883e+04	FES = 355100	Diversity = 2.7675e+01	k_t
= [1, 1, 1, 1]			
Generation 3560: Best = 4.330101e+04	FES = 356100	Diversity = 2.6794e+01	k_t
= [2, 3, 3, 3]			
Generation 3570: Best = 4.305957e+04	FES = 357100	Diversity = 2.6912e+01	k_t
= [1, 1, 1, 1]			
Generation 3580: Best = 4.251315e+04	FES = 358100	Diversity = 2.8907e+01	k_t
= [4, 4, 4, 4]			
Generation 3590: Best = 4.223461e+04	FES = 359100	Diversity = 2.9611e+01	k_t
= [1, 1, 1, 1]			
Generation 3600: Best = 4.212101e+04	FES = 360100	Diversity = 2.9279e+01	k_t
= [2, 3, 3, 3]			
Generation 3610: Best = 4.166824e+04	FES = 361100	Diversity = 2.7668e+01	k_t
= [1, 1, 1, 1]			
Generation 3620: Best = 4.140199e+04	FES = 362100	Diversity = 2.9513e+01	k_t
= [2, 4, 4, 4]			
Generation 3630: Best = 4.115506e+04	FES = 363100	Diversity = 3.0459e+01	k_t
= [1, 1, 1, 1]			
Generation 3640: Best = 4.098488e+04	FES = 364100	Diversity = 2.9570e+01	k_t
= [2, 3, 3, 4]			
Generation 3650: Best = 4.034912e+04	FES = 365100	Diversity = 3.1542e+01	k_t
= [1, 1, 1, 1]			
Generation 3660: Best = 4.030335e+04	FES = 366100	Diversity = 3.1420e+01	k_t
= [2, 3, 3, 2]			
Generation 3670: Best = 4.019458e+04	FES = 367100	Diversity = 2.8453e+01	k_t
= [1, 1, 1, 1]			
Generation 3680: Best = 3.972258e+04	FES = 368100	Diversity = 2.7647e+01	k_t
= [3, 3, 2, 4]			
Generation 3690: Best = 3.919511e+04	FES = 369100	Diversity = 2.8048e+01	k_t
= [1, 1, 1, 1]			
Generation 3700: Best = 3.915607e+04	FES = 370100	Diversity = 2.5496e+01	k_t
= [2, 2, 2, 2]			
Generation 3710: Best = 3.898061e+04	FES = 371100	Diversity = 2.4240e+01	k_t
= [1, 1, 1, 1]			
Generation 3720: Best = 3.832408e+04	FES = 372100	Diversity = 2.6528e+01	k_t
= [2, 2, 2, 4]			
Generation 3730: Best = 3.814617e+04	FES = 373100	Diversity = 2.7867e+01	k_t
= [1, 1, 1, 1]			

Generation 3740: Best = 3.788741e+04 = [4, 4, 3, 2]	FES = 374100	Diversity = 2.7772e+01	k_t
Generation 3750: Best = 3.752257e+04 = [1, 1, 1, 1]	FES = 375100	Diversity = 2.6829e+01	k_t
Generation 3760: Best = 3.751003e+04 = [3, 2, 3, 3]	FES = 376100	Diversity = 2.6269e+01	k_t
Generation 3770: Best = 3.708608e+04 = [1, 1, 1, 1]	FES = 377100	Diversity = 2.6281e+01	k_t
Generation 3780: Best = 3.701744e+04 = [4, 4, 3, 2]	FES = 378100	Diversity = 2.6288e+01	k_t
Generation 3790: Best = 3.671128e+04 = [1, 1, 1, 1]	FES = 379100	Diversity = 2.6406e+01	k_t
Generation 3800: Best = 3.650279e+04 = [4, 3, 2, 2]	FES = 380100	Diversity = 2.7797e+01	k_t
Generation 3810: Best = 3.624000e+04 = [1, 1, 1, 1]	FES = 381100	Diversity = 2.7879e+01	k_t
Generation 3820: Best = 3.604645e+04 = [4, 3, 4, 3]	FES = 382100	Diversity = 2.6431e+01	k_t
Generation 3830: Best = 3.579904e+04 = [1, 1, 1, 1]	FES = 383100	Diversity = 2.5887e+01	k_t
Generation 3840: Best = 3.556606e+04 = [2, 4, 2, 4]	FES = 384100	Diversity = 2.5357e+01	k_t
Generation 3850: Best = 3.539502e+04 = [1, 1, 1, 1]	FES = 385100	Diversity = 2.6235e+01	k_t
Generation 3860: Best = 3.523268e+04 = [4, 2, 3, 2]	FES = 386100	Diversity = 2.6011e+01	k_t
Generation 3870: Best = 3.474894e+04 = [1, 1, 1, 1]	FES = 387100	Diversity = 2.7725e+01	k_t
Generation 3880: Best = 3.440503e+04 = [2, 2, 3, 3]	FES = 388100	Diversity = 3.0280e+01	k_t
Generation 3890: Best = 3.419132e+04 = [1, 1, 1, 1]	FES = 389100	Diversity = 2.7921e+01	k_t
Generation 3900: Best = 3.390708e+04 = [2, 3, 2, 3]	FES = 390100	Diversity = 2.8467e+01	k_t
Generation 3910: Best = 3.381713e+04 = [1, 1, 1, 1]	FES = 391100	Diversity = 2.9397e+01	k_t
Generation 3920: Best = 3.359554e+04 = [3, 2, 4, 2]	FES = 392100	Diversity = 2.8713e+01	k_t
Generation 3930: Best = 3.343661e+04 = [1, 1, 1, 1]	FES = 393100	Diversity = 2.6655e+01	k_t
Generation 3940: Best = 3.324485e+04 = [2, 3, 2, 2]	FES = 394100	Diversity = 2.5898e+01	k_t
Generation 3950: Best = 3.300625e+04 = [1, 1, 1, 1]	FES = 395100	Diversity = 2.4631e+01	k_t
Generation 3960: Best = 3.299720e+04 = [4, 4, 4, 3]	FES = 396100	Diversity = 2.5802e+01	k_t
Generation 3970: Best = 3.294503e+04 = [1, 1, 1, 1]	FES = 397100	Diversity = 2.4995e+01	k_t
Generation 3980: Best = 3.281054e+04 = [4, 4, 2, 2]	FES = 398100	Diversity = 2.4436e+01	k_t
Generation 3990: Best = 3.265909e+04 = [1, 1, 1, 1]	FES = 399100	Diversity = 2.3132e+01	k_t
Generation 4000: Best = 3.256244e+04 = [4, 2, 2, 2]	FES = 400100	Diversity = 2.4147e+01	k_t

Generation 4010: Best = 3.249091e+04 = [1, 1, 1, 1]	FES = 401100	Diversity = 2.4424e+01	k_t
Generation 4020: Best = 3.236542e+04 = [2, 4, 3, 4]	FES = 402100	Diversity = 2.4426e+01	k_t
Generation 4030: Best = 3.229172e+04 = [1, 1, 1, 1]	FES = 403100	Diversity = 2.4351e+01	k_t
Generation 4040: Best = 3.224640e+04 = [2, 3, 2, 4]	FES = 404100	Diversity = 2.3317e+01	k_t
Generation 4050: Best = 3.205871e+04 = [1, 1, 1, 1]	FES = 405100	Diversity = 2.2808e+01	k_t
Generation 4060: Best = 3.205871e+04 = [2, 4, 4, 2]	FES = 406100	Diversity = 2.2521e+01	k_t
Generation 4070: Best = 3.193758e+04 = [1, 1, 1, 1]	FES = 407100	Diversity = 2.1576e+01	k_t
Generation 4080: Best = 3.182189e+04 = [2, 2, 3, 2]	FES = 408100	Diversity = 2.0894e+01	k_t
Generation 4090: Best = 3.181648e+04 = [1, 1, 1, 1]	FES = 409100	Diversity = 1.9603e+01	k_t
Generation 4100: Best = 3.179296e+04 = [2, 2, 4, 3]	FES = 410100	Diversity = 2.0286e+01	k_t
Generation 4110: Best = 3.170421e+04 = [1, 1, 1, 1]	FES = 411100	Diversity = 2.0606e+01	k_t
Generation 4120: Best = 3.151875e+04 = [3, 3, 4, 2]	FES = 412100	Diversity = 2.1022e+01	k_t
Generation 4130: Best = 3.151828e+04 = [1, 1, 1, 1]	FES = 413100	Diversity = 2.1342e+01	k_t
Generation 4140: Best = 3.128030e+04 = [4, 4, 2, 2]	FES = 414100	Diversity = 2.2657e+01	k_t
Generation 4150: Best = 3.126426e+04 = [1, 1, 1, 1]	FES = 415100	Diversity = 2.2654e+01	k_t
Generation 4160: Best = 3.119465e+04 = [3, 2, 2, 2]	FES = 416100	Diversity = 2.3297e+01	k_t
Generation 4170: Best = 3.112382e+04 = [1, 1, 1, 1]	FES = 417100	Diversity = 2.3785e+01	k_t
Generation 4180: Best = 3.090299e+04 = [3, 2, 2, 4]	FES = 418100	Diversity = 2.2846e+01	k_t
Generation 4190: Best = 3.079339e+04 = [1, 1, 1, 1]	FES = 419100	Diversity = 2.3875e+01	k_t
Generation 4200: Best = 3.073856e+04 = [4, 4, 3, 3]	FES = 420100	Diversity = 2.5663e+01	k_t
Generation 4210: Best = 3.044492e+04 = [1, 1, 1, 1]	FES = 421100	Diversity = 2.5065e+01	k_t
Generation 4220: Best = 3.042132e+04 = [3, 4, 2, 2]	FES = 422100	Diversity = 2.3802e+01	k_t
Generation 4230: Best = 3.034700e+04 = [1, 1, 1, 1]	FES = 423100	Diversity = 2.3252e+01	k_t
Generation 4240: Best = 3.027355e+04 = [3, 2, 3, 2]	FES = 424100	Diversity = 2.2250e+01	k_t
Generation 4250: Best = 3.013867e+04 = [1, 1, 1, 1]	FES = 425100	Diversity = 2.0983e+01	k_t
Generation 4260: Best = 3.007695e+04 = [2, 2, 2, 4]	FES = 426100	Diversity = 2.0696e+01	k_t
Generation 4270: Best = 3.005655e+04 = [1, 1, 1, 1]	FES = 427100	Diversity = 1.9466e+01	k_t

Generation 4280: Best = 3.003948e+04 = [3, 3, 4, 3]	FES = 428100	Diversity = 1.8688e+01	k_t
Generation 4290: Best = 2.999669e+04 = [1, 1, 1, 1]	FES = 429100	Diversity = 1.7874e+01	k_t
Generation 4300: Best = 2.998277e+04 = [3, 2, 2, 4]	FES = 430100	Diversity = 1.7255e+01	k_t
Generation 4310: Best = 2.994323e+04 = [1, 1, 1, 1]	FES = 431100	Diversity = 1.6525e+01	k_t
Generation 4320: Best = 2.992828e+04 = [3, 4, 3, 3]	FES = 432100	Diversity = 1.6501e+01	k_t
Generation 4330: Best = 2.987142e+04 = [1, 1, 1, 1]	FES = 433100	Diversity = 1.6781e+01	k_t
Generation 4340: Best = 2.983519e+04 = [3, 2, 3, 4]	FES = 434100	Diversity = 1.6332e+01	k_t
Generation 4350: Best = 2.980404e+04 = [1, 1, 1, 1]	FES = 435100	Diversity = 1.6508e+01	k_t
Generation 4360: Best = 2.972972e+04 = [3, 4, 2, 4]	FES = 436100	Diversity = 1.6166e+01	k_t
Generation 4370: Best = 2.972972e+04 = [1, 1, 1, 1]	FES = 437100	Diversity = 1.5984e+01	k_t
Generation 4380: Best = 2.970226e+04 = [3, 2, 2, 4]	FES = 438100	Diversity = 1.6023e+01	k_t
Generation 4390: Best = 2.970054e+04 = [1, 1, 1, 1]	FES = 439100	Diversity = 1.5620e+01	k_t
Generation 4400: Best = 2.962019e+04 = [3, 3, 4, 4]	FES = 440100	Diversity = 1.5625e+01	k_t
Generation 4410: Best = 2.949055e+04 = [1, 1, 1, 1]	FES = 441100	Diversity = 1.5900e+01	k_t
Generation 4420: Best = 2.948425e+04 = [3, 2, 4, 2]	FES = 442100	Diversity = 1.5828e+01	k_t
Generation 4430: Best = 2.940732e+04 = [1, 1, 1, 1]	FES = 443100	Diversity = 1.6554e+01	k_t
Generation 4440: Best = 2.928836e+04 = [4, 4, 4, 4]	FES = 444100	Diversity = 1.8309e+01	k_t
Generation 4450: Best = 2.915955e+04 = [1, 1, 1, 1]	FES = 445100	Diversity = 2.0356e+01	k_t
Generation 4460: Best = 2.902511e+04 = [3, 4, 2, 4]	FES = 446100	Diversity = 2.0219e+01	k_t
Generation 4470: Best = 2.900267e+04 = [1, 1, 1, 1]	FES = 447100	Diversity = 2.0719e+01	k_t
Generation 4480: Best = 2.893171e+04 = [3, 3, 4, 3]	FES = 448100	Diversity = 2.0487e+01	k_t
Generation 4490: Best = 2.882550e+04 = [1, 1, 1, 1]	FES = 449100	Diversity = 2.0083e+01	k_t
Generation 4500: Best = 2.873488e+04 = [3, 3, 4, 4]	FES = 450100	Diversity = 2.0289e+01	k_t
Generation 4510: Best = 2.869500e+04 = [1, 1, 1, 1]	FES = 451100	Diversity = 2.0100e+01	k_t
Generation 4520: Best = 2.869427e+04 = [3, 4, 2, 4]	FES = 452100	Diversity = 1.9564e+01	k_t
Generation 4530: Best = 2.864693e+04 = [1, 1, 1, 1]	FES = 453100	Diversity = 1.9228e+01	k_t
Generation 4540: Best = 2.860561e+04 = [2, 2, 4, 4]	FES = 454100	Diversity = 1.8683e+01	k_t

Generation 4550: Best = 2.856535e+04 = [1, 1, 1, 1]	FES = 455100	Diversity = 1.8963e+01	k_t
Generation 4560: Best = 2.855877e+04 = [3, 3, 2, 4]	FES = 456100	Diversity = 1.8083e+01	k_t
Generation 4570: Best = 2.851761e+04 = [1, 1, 1, 1]	FES = 457100	Diversity = 1.7714e+01	k_t
Generation 4580: Best = 2.851761e+04 = [2, 4, 3, 4]	FES = 458100	Diversity = 1.7557e+01	k_t
Generation 4590: Best = 2.849951e+04 = [1, 1, 1, 1]	FES = 459100	Diversity = 1.7016e+01	k_t
Generation 4600: Best = 2.844543e+04 = [4, 4, 2, 4]	FES = 460100	Diversity = 1.6668e+01	k_t
Generation 4610: Best = 2.844543e+04 = [1, 1, 1, 1]	FES = 461100	Diversity = 1.6367e+01	k_t
Generation 4620: Best = 2.844240e+04 = [4, 2, 2, 4]	FES = 462100	Diversity = 1.6690e+01	k_t
Generation 4630: Best = 2.842658e+04 = [1, 1, 1, 1]	FES = 463100	Diversity = 1.6838e+01	k_t
Generation 4640: Best = 2.838517e+04 = [4, 4, 2, 3]	FES = 464100	Diversity = 1.6475e+01	k_t
Generation 4650: Best = 2.838517e+04 = [1, 1, 1, 1]	FES = 465100	Diversity = 1.6682e+01	k_t
Generation 4660: Best = 2.838517e+04 = [4, 3, 4, 3]	FES = 466100	Diversity = 1.5985e+01	k_t
Generation 4670: Best = 2.838517e+04 = [1, 1, 1, 1]	FES = 467100	Diversity = 1.5609e+01	k_t
Generation 4680: Best = 2.836093e+04 = [4, 2, 3, 2]	FES = 468100	Diversity = 1.5828e+01	k_t
Generation 4690: Best = 2.836093e+04 = [1, 1, 1, 1]	FES = 469100	Diversity = 1.5671e+01	k_t
Generation 4700: Best = 2.836084e+04 = [3, 4, 2, 2]	FES = 470100	Diversity = 1.5439e+01	k_t
Generation 4710: Best = 2.836084e+04 = [1, 1, 1, 1]	FES = 471100	Diversity = 1.5458e+01	k_t
Generation 4720: Best = 2.835771e+04 = [3, 4, 4, 4]	FES = 472100	Diversity = 1.5196e+01	k_t
Generation 4730: Best = 2.835771e+04 = [1, 1, 1, 1]	FES = 473100	Diversity = 1.4935e+01	k_t
Generation 4740: Best = 2.835385e+04 = [4, 4, 2, 2]	FES = 474100	Diversity = 1.4854e+01	k_t
Generation 4750: Best = 2.835385e+04 = [1, 1, 1, 1]	FES = 475100	Diversity = 1.4807e+01	k_t
Generation 4760: Best = 2.835385e+04 = [2, 3, 3, 4]	FES = 476100	Diversity = 1.4814e+01	k_t
Generation 4770: Best = 2.834755e+04 = [1, 1, 1, 1]	FES = 477100	Diversity = 1.4658e+01	k_t
Generation 4780: Best = 2.834734e+04 = [4, 4, 3, 2]	FES = 478100	Diversity = 1.4690e+01	k_t
Generation 4790: Best = 2.834214e+04 = [1, 1, 1, 1]	FES = 479100	Diversity = 1.4546e+01	k_t
Generation 4800: Best = 2.834214e+04 = [3, 4, 4, 2]	FES = 480100	Diversity = 1.4746e+01	k_t
Generation 4810: Best = 2.833946e+04 = [1, 1, 1, 1]	FES = 481100	Diversity = 1.4659e+01	k_t

Generation 4820: Best = 2.832681e+04 = [2, 2, 2, 4]	FES = 482100	Diversity = 1.4762e+01	k_t
Generation 4830: Best = 2.832681e+04 = [1, 1, 1, 1]	FES = 483100	Diversity = 1.4822e+01	k_t
Generation 4840: Best = 2.832681e+04 = [2, 2, 3, 4]	FES = 484100	Diversity = 1.4713e+01	k_t
Generation 4850: Best = 2.830899e+04 = [1, 1, 1, 1]	FES = 485100	Diversity = 1.4528e+01	k_t
Generation 4860: Best = 2.830899e+04 = [3, 2, 4, 4]	FES = 486100	Diversity = 1.4539e+01	k_t
Generation 4870: Best = 2.830838e+04 = [1, 1, 1, 1]	FES = 487100	Diversity = 1.4540e+01	k_t
Generation 4880: Best = 2.829847e+04 = [3, 2, 4, 3]	FES = 488100	Diversity = 1.4388e+01	k_t
Generation 4890: Best = 2.829847e+04 = [1, 1, 1, 1]	FES = 489100	Diversity = 1.4507e+01	k_t
Generation 4900: Best = 2.829847e+04 = [2, 4, 2, 2]	FES = 490100	Diversity = 1.4517e+01	k_t
Generation 4910: Best = 2.829847e+04 = [1, 1, 1, 1]	FES = 491100	Diversity = 1.4242e+01	k_t
Generation 4920: Best = 2.827244e+04 = [2, 2, 3, 2]	FES = 492100	Diversity = 1.4216e+01	k_t
Generation 4930: Best = 2.827244e+04 = [1, 1, 1, 1]	FES = 493100	Diversity = 1.4193e+01	k_t
Generation 4940: Best = 2.825974e+04 = [3, 3, 4, 2]	FES = 494100	Diversity = 1.4133e+01	k_t
Generation 4950: Best = 2.825203e+04 = [1, 1, 1, 1]	FES = 495100	Diversity = 1.3950e+01	k_t
Generation 4960: Best = 2.825061e+04 = [4, 3, 2, 3]	FES = 496100	Diversity = 1.4048e+01	k_t
Generation 4970: Best = 2.825061e+04 = [1, 1, 1, 1]	FES = 497100	Diversity = 1.4159e+01	k_t
Generation 4980: Best = 2.823995e+04 = [3, 2, 4, 2]	FES = 498100	Diversity = 1.4081e+01	k_t
Generation 4990: Best = 2.823995e+04 = [1, 1, 1, 1]	FES = 499100	Diversity = 1.3992e+01	k_t
Generation 5000: Best = 2.823995e+04 = [3, 3, 4, 2]	FES = 500100	Diversity = 1.3926e+01	k_t
Generation 5010: Best = 2.823995e+04 = [1, 1, 1, 1]	FES = 501100	Diversity = 1.3908e+01	k_t
Generation 5020: Best = 2.823995e+04 = [4, 3, 3, 3]	FES = 502100	Diversity = 1.3805e+01	k_t
Generation 5030: Best = 2.823944e+04 = [1, 1, 1, 1]	FES = 503100	Diversity = 1.3658e+01	k_t
Generation 5040: Best = 2.823944e+04 = [3, 4, 4, 3]	FES = 504100	Diversity = 1.3569e+01	k_t
Generation 5050: Best = 2.823944e+04 = [1, 1, 1, 1]	FES = 505100	Diversity = 1.3561e+01	k_t
Generation 5060: Best = 2.823186e+04 = [4, 4, 2, 2]	FES = 506100	Diversity = 1.3450e+01	k_t
Generation 5070: Best = 2.823186e+04 = [1, 1, 1, 1]	FES = 507100	Diversity = 1.3435e+01	k_t
Generation 5080: Best = 2.822161e+04 = [3, 2, 2, 3]	FES = 508100	Diversity = 1.3505e+01	k_t

Generation 5090: Best = 2.822161e+04 = [1, 1, 1, 1]	FES = 509100	Diversity = 1.3495e+01	k_t
Generation 5100: Best = 2.822161e+04 = [3, 3, 3, 3]	FES = 510100	Diversity = 1.3478e+01	k_t
Generation 5110: Best = 2.822161e+04 = [1, 1, 1, 1]	FES = 511100	Diversity = 1.3376e+01	k_t
Generation 5120: Best = 2.822161e+04 = [2, 4, 2, 4]	FES = 512100	Diversity = 1.3386e+01	k_t
Generation 5130: Best = 2.822161e+04 = [1, 1, 1, 1]	FES = 513100	Diversity = 1.3413e+01	k_t
Generation 5140: Best = 2.822161e+04 = [2, 2, 4, 2]	FES = 514100	Diversity = 1.3377e+01	k_t
Generation 5150: Best = 2.822161e+04 = [1, 1, 1, 1]	FES = 515100	Diversity = 1.3327e+01	k_t
Generation 5160: Best = 2.822161e+04 = [3, 4, 3, 4]	FES = 516100	Diversity = 1.3233e+01	k_t
Generation 5170: Best = 2.822161e+04 = [1, 1, 1, 1]	FES = 517100	Diversity = 1.3210e+01	k_t
Generation 5180: Best = 2.822161e+04 = [2, 3, 2, 3]	FES = 518100	Diversity = 1.3122e+01	k_t
Generation 5190: Best = 2.822161e+04 = [1, 1, 1, 1]	FES = 519100	Diversity = 1.3110e+01	k_t
Generation 5200: Best = 2.822138e+04 = [2, 2, 2, 4]	FES = 520100	Diversity = 1.3134e+01	k_t
Generation 5210: Best = 2.821035e+04 = [1, 1, 1, 1]	FES = 521100	Diversity = 1.3187e+01	k_t
Generation 5220: Best = 2.821035e+04 = [4, 2, 2, 4]	FES = 522100	Diversity = 1.3129e+01	k_t
Generation 5230: Best = 2.821035e+04 = [1, 1, 1, 1]	FES = 523100	Diversity = 1.3215e+01	k_t
Generation 5240: Best = 2.821035e+04 = [2, 2, 4, 4]	FES = 524100	Diversity = 1.3188e+01	k_t
Generation 5250: Best = 2.820968e+04 = [1, 1, 1, 1]	FES = 525100	Diversity = 1.3183e+01	k_t
Generation 5260: Best = 2.820968e+04 = [4, 3, 3, 3]	FES = 526100	Diversity = 1.3100e+01	k_t
Generation 5270: Best = 2.820614e+04 = [1, 1, 1, 1]	FES = 527100	Diversity = 1.3202e+01	k_t
Generation 5280: Best = 2.820614e+04 = [4, 4, 4, 3]	FES = 528100	Diversity = 1.3116e+01	k_t
Generation 5290: Best = 2.820614e+04 = [1, 1, 1, 1]	FES = 529100	Diversity = 1.3134e+01	k_t
Generation 5300: Best = 2.819783e+04 = [2, 2, 4, 4]	FES = 530100	Diversity = 1.3010e+01	k_t
Generation 5310: Best = 2.819783e+04 = [1, 1, 1, 1]	FES = 531100	Diversity = 1.2966e+01	k_t
Generation 5320: Best = 2.819783e+04 = [3, 3, 2, 2]	FES = 532100	Diversity = 1.3025e+01	k_t
Generation 5330: Best = 2.817661e+04 = [1, 1, 1, 1]	FES = 533100	Diversity = 1.3026e+01	k_t
Generation 5340: Best = 2.816831e+04 = [3, 4, 3, 3]	FES = 534100	Diversity = 1.3025e+01	k_t
Generation 5350: Best = 2.816245e+04 = [1, 1, 1, 1]	FES = 535100	Diversity = 1.3052e+01	k_t

Generation 5360: Best = 2.815608e+04 = [4, 3, 3, 3]	FES = 536100	Diversity = 1.3007e+01	k_t
Generation 5370: Best = 2.815608e+04 = [1, 1, 1, 1]	FES = 537100	Diversity = 1.3107e+01	k_t
Generation 5380: Best = 2.814366e+04 = [4, 3, 4, 4]	FES = 538100	Diversity = 1.3073e+01	k_t
Generation 5390: Best = 2.814366e+04 = [1, 1, 1, 1]	FES = 539100	Diversity = 1.3141e+01	k_t
Generation 5400: Best = 2.814366e+04 = [4, 4, 2, 2]	FES = 540100	Diversity = 1.3149e+01	k_t
Generation 5410: Best = 2.814366e+04 = [1, 1, 1, 1]	FES = 541100	Diversity = 1.3127e+01	k_t
Generation 5420: Best = 2.814352e+04 = [2, 2, 4, 2]	FES = 542100	Diversity = 1.3061e+01	k_t
Generation 5430: Best = 2.813805e+04 = [1, 1, 1, 1]	FES = 543100	Diversity = 1.3166e+01	k_t
Generation 5440: Best = 2.813805e+04 = [4, 4, 2, 4]	FES = 544100	Diversity = 1.3073e+01	k_t
Generation 5450: Best = 2.813805e+04 = [1, 1, 1, 1]	FES = 545100	Diversity = 1.2611e+01	k_t
Generation 5460: Best = 2.811388e+04 = [4, 3, 4, 4]	FES = 546100	Diversity = 1.2389e+01	k_t
Generation 5470: Best = 2.810143e+04 = [1, 1, 1, 1]	FES = 547100	Diversity = 1.2360e+01	k_t
Generation 5480: Best = 2.808124e+04 = [3, 4, 4, 3]	FES = 548100	Diversity = 1.2250e+01	k_t
Generation 5490: Best = 2.808124e+04 = [1, 1, 1, 1]	FES = 549100	Diversity = 1.2114e+01	k_t
Generation 5500: Best = 2.807693e+04 = [3, 4, 3, 4]	FES = 550100	Diversity = 1.1322e+01	k_t
Generation 5510: Best = 2.806293e+04 = [1, 1, 1, 1]	FES = 551100	Diversity = 1.0714e+01	k_t
Generation 5520: Best = 2.804776e+04 = [3, 3, 2, 2]	FES = 552100	Diversity = 9.7708e+00	k_t
Generation 5530: Best = 2.804136e+04 = [1, 1, 1, 1]	FES = 553100	Diversity = 9.0266e+00	k_t
Generation 5540: Best = 2.800187e+04 = [2, 2, 3, 3]	FES = 554100	Diversity = 8.2923e+00	k_t
Generation 5550: Best = 2.796492e+04 = [1, 1, 1, 1]	FES = 555100	Diversity = 7.2695e+00	k_t
Generation 5560: Best = 2.792325e+04 = [3, 4, 4, 3]	FES = 556100	Diversity = 6.7851e+00	k_t
Generation 5570: Best = 2.789642e+04 = [1, 1, 1, 1]	FES = 557100	Diversity = 6.3926e+00	k_t
Generation 5580: Best = 2.788497e+04 = [3, 4, 2, 2]	FES = 558100	Diversity = 6.3504e+00	k_t
Generation 5590: Best = 2.786501e+04 = [1, 1, 1, 1]	FES = 559100	Diversity = 6.1853e+00	k_t
Generation 5600: Best = 2.783308e+04 = [2, 2, 4, 4]	FES = 560100	Diversity = 5.6365e+00	k_t
Generation 5610: Best = 2.779177e+04 = [1, 1, 1, 1]	FES = 561100	Diversity = 5.0704e+00	k_t
Generation 5620: Best = 2.777143e+04 = [3, 2, 3, 3]	FES = 562100	Diversity = 5.0603e+00	k_t

Generation 5630: Best = 2.772331e+04 = [1, 1, 1, 1]	FES = 563100	Diversity = 4.7837e+00	k_t
Generation 5640: Best = 2.770230e+04 = [2, 2, 3, 2]	FES = 564100	Diversity = 4.8887e+00	k_t
Generation 5650: Best = 2.767631e+04 = [1, 1, 1, 1]	FES = 565100	Diversity = 4.8374e+00	k_t
Generation 5660: Best = 2.764837e+04 = [3, 3, 3, 2]	FES = 566100	Diversity = 4.6783e+00	k_t
Generation 5670: Best = 2.764523e+04 = [1, 1, 1, 1]	FES = 567100	Diversity = 4.5366e+00	k_t
Generation 5680: Best = 2.763092e+04 = [3, 2, 2, 2]	FES = 568100	Diversity = 4.1897e+00	k_t
Generation 5690: Best = 2.760815e+04 = [1, 1, 1, 1]	FES = 569100	Diversity = 3.8443e+00	k_t
Generation 5700: Best = 2.759763e+04 = [3, 2, 3, 2]	FES = 570100	Diversity = 3.7012e+00	k_t
Generation 5710: Best = 2.758851e+04 = [1, 1, 1, 1]	FES = 571100	Diversity = 3.5128e+00	k_t
Generation 5720: Best = 2.757904e+04 = [3, 3, 2, 3]	FES = 572100	Diversity = 3.5283e+00	k_t
Generation 5730: Best = 2.757398e+04 = [1, 1, 1, 1]	FES = 573100	Diversity = 3.3849e+00	k_t
Generation 5740: Best = 2.755793e+04 = [4, 4, 3, 3]	FES = 574100	Diversity = 3.4455e+00	k_t
Generation 5750: Best = 2.754685e+04 = [1, 1, 1, 1]	FES = 575100	Diversity = 3.4138e+00	k_t
Generation 5760: Best = 2.754087e+04 = [3, 3, 3, 2]	FES = 576100	Diversity = 3.4714e+00	k_t
Generation 5770: Best = 2.751743e+04 = [1, 1, 1, 1]	FES = 577100	Diversity = 3.4844e+00	k_t
Generation 5780: Best = 2.750660e+04 = [3, 4, 2, 2]	FES = 578100	Diversity = 3.3100e+00	k_t
Generation 5790: Best = 2.749815e+04 = [1, 1, 1, 1]	FES = 579100	Diversity = 3.3098e+00	k_t
Generation 5800: Best = 2.748626e+04 = [4, 3, 4, 2]	FES = 580100	Diversity = 3.4294e+00	k_t
Generation 5810: Best = 2.747606e+04 = [1, 1, 1, 1]	FES = 581100	Diversity = 3.1846e+00	k_t
Generation 5820: Best = 2.747083e+04 = [4, 3, 3, 2]	FES = 582100	Diversity = 3.1718e+00	k_t
Generation 5830: Best = 2.745196e+04 = [1, 1, 1, 1]	FES = 583100	Diversity = 3.2166e+00	k_t
Generation 5840: Best = 2.744791e+04 = [3, 2, 3, 4]	FES = 584100	Diversity = 3.3052e+00	k_t
Generation 5850: Best = 2.743565e+04 = [1, 1, 1, 1]	FES = 585100	Diversity = 3.3535e+00	k_t
Generation 5860: Best = 2.742849e+04 = [4, 4, 2, 4]	FES = 586100	Diversity = 3.3490e+00	k_t
Generation 5870: Best = 2.741306e+04 = [1, 1, 1, 1]	FES = 587100	Diversity = 3.3922e+00	k_t
Generation 5880: Best = 2.740731e+04 = [2, 3, 4, 2]	FES = 588100	Diversity = 3.4419e+00	k_t
Generation 5890: Best = 2.739632e+04 = [1, 1, 1, 1]	FES = 589100	Diversity = 3.3880e+00	k_t

Generation 5900: Best = 2.739135e+04 = [3, 3, 2, 2]	FES = 590100	Diversity = 3.2640e+00	k_t
Generation 5910: Best = 2.737850e+04 = [1, 1, 1, 1]	FES = 591100	Diversity = 3.1396e+00	k_t
Generation 5920: Best = 2.737198e+04 = [4, 3, 4, 2]	FES = 592100	Diversity = 3.3089e+00	k_t
Generation 5930: Best = 2.736591e+04 = [1, 1, 1, 1]	FES = 593100	Diversity = 3.3072e+00	k_t
Generation 5940: Best = 2.735940e+04 = [4, 2, 2, 2]	FES = 594100	Diversity = 3.4226e+00	k_t
Generation 5950: Best = 2.734715e+04 = [1, 1, 1, 1]	FES = 595100	Diversity = 3.4948e+00	k_t
Generation 5960: Best = 2.734562e+04 = [4, 3, 4, 4]	FES = 596100	Diversity = 3.6002e+00	k_t
Generation 5970: Best = 2.733913e+04 = [1, 1, 1, 1]	FES = 597100	Diversity = 3.5887e+00	k_t
Generation 5980: Best = 2.733091e+04 = [4, 3, 4, 2]	FES = 598100	Diversity = 3.6863e+00	k_t
Generation 5990: Best = 2.732079e+04 = [1, 1, 1, 1]	FES = 599100	Diversity = 4.0197e+00	k_t
Generation 6000: Best = 2.730605e+04 = [3, 4, 4, 3]	FES = 600100	Diversity = 4.0887e+00	k_t
Generation 6010: Best = 2.729108e+04 = [1, 1, 1, 1]	FES = 601100	Diversity = 4.0842e+00	k_t
Generation 6020: Best = 2.728025e+04 = [2, 3, 2, 3]	FES = 602100	Diversity = 4.3471e+00	k_t
Generation 6030: Best = 2.724829e+04 = [1, 1, 1, 1]	FES = 603100	Diversity = 4.6923e+00	k_t
Generation 6040: Best = 2.723400e+04 = [3, 4, 4, 2]	FES = 604100	Diversity = 5.1155e+00	k_t
Generation 6050: Best = 2.720706e+04 = [1, 1, 1, 1]	FES = 605100	Diversity = 5.4055e+00	k_t
Generation 6060: Best = 2.717464e+04 = [4, 2, 2, 4]	FES = 606100	Diversity = 5.7663e+00	k_t
Generation 6070: Best = 2.715252e+04 = [1, 1, 1, 1]	FES = 607100	Diversity = 6.0562e+00	k_t
Generation 6080: Best = 2.709884e+04 = [2, 2, 2, 2]	FES = 608100	Diversity = 6.6924e+00	k_t
Generation 6090: Best = 2.707815e+04 = [1, 1, 1, 1]	FES = 609100	Diversity = 7.2873e+00	k_t
Generation 6100: Best = 2.704653e+04 = [2, 3, 2, 3]	FES = 610100	Diversity = 7.1302e+00	k_t
Generation 6110: Best = 2.700195e+04 = [1, 1, 1, 1]	FES = 611100	Diversity = 7.4256e+00	k_t
Generation 6120: Best = 2.693777e+04 = [3, 4, 4, 2]	FES = 612100	Diversity = 7.0730e+00	k_t
Generation 6130: Best = 2.685706e+04 = [1, 1, 1, 1]	FES = 613100	Diversity = 7.9141e+00	k_t
Generation 6140: Best = 2.684669e+04 = [3, 4, 2, 2]	FES = 614100	Diversity = 7.7970e+00	k_t
Generation 6150: Best = 2.677231e+04 = [1, 1, 1, 1]	FES = 615100	Diversity = 7.6117e+00	k_t
Generation 6160: Best = 2.675663e+04 = [3, 2, 4, 4]	FES = 616100	Diversity = 7.6465e+00	k_t

Generation 6170: Best = 2.672819e+04 = [1, 1, 1, 1]	FES = 617100	Diversity = 8.0265e+00	k_t
Generation 6180: Best = 2.665893e+04 = [4, 2, 4, 2]	FES = 618100	Diversity = 8.2308e+00	k_t
Generation 6190: Best = 2.658566e+04 = [1, 1, 1, 1]	FES = 619100	Diversity = 8.8652e+00	k_t
Generation 6200: Best = 2.652947e+04 = [4, 2, 4, 2]	FES = 620100	Diversity = 9.1892e+00	k_t
Generation 6210: Best = 2.643603e+04 = [1, 1, 1, 1]	FES = 621100	Diversity = 1.0087e+01	k_t
Generation 6220: Best = 2.637733e+04 = [4, 3, 4, 2]	FES = 622100	Diversity = 1.0175e+01	k_t
Generation 6230: Best = 2.627900e+04 = [1, 1, 1, 1]	FES = 623100	Diversity = 1.1345e+01	k_t
Generation 6240: Best = 2.621467e+04 = [3, 3, 3, 3]	FES = 624100	Diversity = 1.1187e+01	k_t
Generation 6250: Best = 2.596643e+04 = [1, 1, 1, 1]	FES = 625100	Diversity = 1.2737e+01	k_t
Generation 6260: Best = 2.576080e+04 = [4, 4, 3, 2]	FES = 626100	Diversity = 1.3892e+01	k_t
Generation 6270: Best = 2.556407e+04 = [1, 1, 1, 1]	FES = 627100	Diversity = 1.3422e+01	k_t
Generation 6280: Best = 2.542348e+04 = [4, 4, 2, 2]	FES = 628100	Diversity = 1.6648e+01	k_t
Generation 6290: Best = 2.516849e+04 = [1, 1, 1, 1]	FES = 629100	Diversity = 1.7145e+01	k_t
Generation 6300: Best = 2.489609e+04 = [4, 4, 2, 3]	FES = 630100	Diversity = 1.9483e+01	k_t
Generation 6310: Best = 2.483054e+04 = [1, 1, 1, 1]	FES = 631100	Diversity = 2.0687e+01	k_t
Generation 6320: Best = 2.473258e+04 = [4, 3, 4, 3]	FES = 632100	Diversity = 2.0238e+01	k_t
Generation 6330: Best = 2.451606e+04 = [1, 1, 1, 1]	FES = 633100	Diversity = 2.1458e+01	k_t
Generation 6340: Best = 2.443371e+04 = [3, 3, 4, 2]	FES = 634100	Diversity = 2.1966e+01	k_t
Generation 6350: Best = 2.427774e+04 = [1, 1, 1, 1]	FES = 635100	Diversity = 2.2137e+01	k_t
Generation 6360: Best = 2.419326e+04 = [3, 3, 4, 2]	FES = 636100	Diversity = 2.3365e+01	k_t
Generation 6370: Best = 2.405813e+04 = [1, 1, 1, 1]	FES = 637100	Diversity = 2.3325e+01	k_t
Generation 6380: Best = 2.398264e+04 = [3, 4, 4, 3]	FES = 638100	Diversity = 2.4034e+01	k_t
Generation 6390: Best = 2.383155e+04 = [1, 1, 1, 1]	FES = 639100	Diversity = 2.4778e+01	k_t
Generation 6400: Best = 2.376519e+04 = [4, 2, 4, 2]	FES = 640100	Diversity = 2.5939e+01	k_t
Generation 6410: Best = 2.360672e+04 = [1, 1, 1, 1]	FES = 641100	Diversity = 2.5450e+01	k_t
Generation 6420: Best = 2.347632e+04 = [3, 2, 2, 3]	FES = 642100	Diversity = 2.4871e+01	k_t
Generation 6430: Best = 2.316226e+04 = [1, 1, 1, 1]	FES = 643100	Diversity = 2.6599e+01	k_t

Generation 6440: Best = 2.295434e+04 = [3, 4, 4, 3]	FES = 644100	Diversity = 2.7708e+01	k_t
Generation 6450: Best = 2.267543e+04 = [1, 1, 1, 1]	FES = 645100	Diversity = 2.7718e+01	k_t
Generation 6460: Best = 2.251520e+04 = [3, 2, 2, 3]	FES = 646100	Diversity = 2.9610e+01	k_t
Generation 6470: Best = 2.206295e+04 = [1, 1, 1, 1]	FES = 647100	Diversity = 3.1431e+01	k_t
Generation 6480: Best = 2.190732e+04 = [2, 2, 2, 3]	FES = 648100	Diversity = 3.2626e+01	k_t
Generation 6490: Best = 2.170674e+04 = [1, 1, 1, 1]	FES = 649100	Diversity = 3.1999e+01	k_t
Generation 6500: Best = 2.163527e+04 = [4, 2, 2, 3]	FES = 650100	Diversity = 3.0101e+01	k_t
Generation 6510: Best = 2.146548e+04 = [1, 1, 1, 1]	FES = 651100	Diversity = 2.7654e+01	k_t
Generation 6520: Best = 2.142790e+04 = [3, 3, 2, 4]	FES = 652100	Diversity = 2.6640e+01	k_t
Generation 6530: Best = 2.135853e+04 = [1, 1, 1, 1]	FES = 653100	Diversity = 2.4527e+01	k_t
Generation 6540: Best = 2.127952e+04 = [3, 4, 2, 3]	FES = 654100	Diversity = 2.3213e+01	k_t
Generation 6550: Best = 2.120129e+04 = [1, 1, 1, 1]	FES = 655100	Diversity = 2.3007e+01	k_t
Generation 6560: Best = 2.110583e+04 = [3, 2, 4, 2]	FES = 656100	Diversity = 2.2596e+01	k_t
Generation 6570: Best = 2.105262e+04 = [1, 1, 1, 1]	FES = 657100	Diversity = 2.2548e+01	k_t
Generation 6580: Best = 2.090613e+04 = [2, 4, 4, 3]	FES = 658100	Diversity = 2.2365e+01	k_t
Generation 6590: Best = 2.080182e+04 = [1, 1, 1, 1]	FES = 659100	Diversity = 2.2393e+01	k_t
Generation 6600: Best = 2.080182e+04 = [3, 3, 2, 3]	FES = 660100	Diversity = 2.1541e+01	k_t
Generation 6610: Best = 2.074948e+04 = [1, 1, 1, 1]	FES = 661100	Diversity = 2.0620e+01	k_t
Generation 6620: Best = 2.070315e+04 = [4, 2, 2, 3]	FES = 662100	Diversity = 2.0527e+01	k_t
Generation 6630: Best = 2.064378e+04 = [1, 1, 1, 1]	FES = 663100	Diversity = 1.9777e+01	k_t
Generation 6640: Best = 2.059337e+04 = [2, 4, 4, 2]	FES = 664100	Diversity = 1.8723e+01	k_t
Generation 6650: Best = 2.053164e+04 = [1, 1, 1, 1]	FES = 665100	Diversity = 1.8157e+01	k_t
Generation 6660: Best = 2.050654e+04 = [3, 2, 3, 4]	FES = 666100	Diversity = 1.7647e+01	k_t
Generation 6670: Best = 2.049131e+04 = [1, 1, 1, 1]	FES = 667100	Diversity = 1.7678e+01	k_t
Generation 6680: Best = 2.048817e+04 = [2, 4, 3, 3]	FES = 668100	Diversity = 1.7326e+01	k_t
Generation 6690: Best = 2.046814e+04 = [1, 1, 1, 1]	FES = 669100	Diversity = 1.6859e+01	k_t
Generation 6700: Best = 2.044661e+04 = [4, 2, 2, 3]	FES = 670100	Diversity = 1.6841e+01	k_t

Generation 6710: Best = 2.043018e+04 = [1, 1, 1, 1]	FES = 671100	Diversity = 1.6785e+01	k_t
Generation 6720: Best = 2.042297e+04 = [2, 4, 2, 2]	FES = 672100	Diversity = 1.6806e+01	k_t
Generation 6730: Best = 2.041335e+04 = [1, 1, 1, 1]	FES = 673100	Diversity = 1.6181e+01	k_t
Generation 6740: Best = 2.040766e+04 = [2, 3, 4, 2]	FES = 674100	Diversity = 1.6031e+01	k_t
Generation 6750: Best = 2.039283e+04 = [1, 1, 1, 1]	FES = 675100	Diversity = 1.5985e+01	k_t
Generation 6760: Best = 2.039283e+04 = [2, 4, 4, 2]	FES = 676100	Diversity = 1.5642e+01	k_t
Generation 6770: Best = 2.033695e+04 = [1, 1, 1, 1]	FES = 677100	Diversity = 1.5335e+01	k_t
Generation 6780: Best = 2.033695e+04 = [3, 4, 2, 2]	FES = 678100	Diversity = 1.5195e+01	k_t
Generation 6790: Best = 2.033695e+04 = [1, 1, 1, 1]	FES = 679100	Diversity = 1.5445e+01	k_t
Generation 6800: Best = 2.033695e+04 = [2, 3, 4, 3]	FES = 680100	Diversity = 1.5331e+01	k_t
Generation 6810: Best = 2.031651e+04 = [1, 1, 1, 1]	FES = 681100	Diversity = 1.5258e+01	k_t
Generation 6820: Best = 2.030489e+04 = [4, 2, 2, 4]	FES = 682100	Diversity = 1.4805e+01	k_t
Generation 6830: Best = 2.030483e+04 = [1, 1, 1, 1]	FES = 683100	Diversity = 1.4550e+01	k_t
Generation 6840: Best = 2.029587e+04 = [2, 2, 3, 2]	FES = 684100	Diversity = 1.4437e+01	k_t
Generation 6850: Best = 2.029265e+04 = [1, 1, 1, 1]	FES = 685100	Diversity = 1.4406e+01	k_t
Generation 6860: Best = 2.028627e+04 = [2, 4, 3, 3]	FES = 686100	Diversity = 1.4451e+01	k_t
Generation 6870: Best = 2.027807e+04 = [1, 1, 1, 1]	FES = 687100	Diversity = 1.4609e+01	k_t
Generation 6880: Best = 2.023108e+04 = [4, 2, 4, 4]	FES = 688100	Diversity = 1.4473e+01	k_t
Generation 6890: Best = 2.023007e+04 = [1, 1, 1, 1]	FES = 689100	Diversity = 1.4476e+01	k_t
Generation 6900: Best = 2.022920e+04 = [3, 4, 4, 3]	FES = 690100	Diversity = 1.4573e+01	k_t
Generation 6910: Best = 2.021025e+04 = [1, 1, 1, 1]	FES = 691100	Diversity = 1.4083e+01	k_t
Generation 6920: Best = 2.021025e+04 = [3, 4, 2, 4]	FES = 692100	Diversity = 1.3837e+01	k_t
Generation 6930: Best = 2.021025e+04 = [1, 1, 1, 1]	FES = 693100	Diversity = 1.3882e+01	k_t
Generation 6940: Best = 2.019548e+04 = [4, 2, 3, 4]	FES = 694100	Diversity = 1.3764e+01	k_t
Generation 6950: Best = 2.018958e+04 = [1, 1, 1, 1]	FES = 695100	Diversity = 1.3619e+01	k_t
Generation 6960: Best = 2.018958e+04 = [4, 3, 3, 2]	FES = 696100	Diversity = 1.3641e+01	k_t
Generation 6970: Best = 2.018958e+04 = [1, 1, 1, 1]	FES = 697100	Diversity = 1.3788e+01	k_t

Generation 6980: Best = 2.018308e+04 = [3, 2, 4, 4]	FES = 698100	Diversity = 1.3677e+01	k_t
Generation 6990: Best = 2.018308e+04 = [1, 1, 1, 1]	FES = 699100	Diversity = 1.3545e+01	k_t
Generation 7000: Best = 2.018308e+04 = [3, 3, 2, 3]	FES = 700100	Diversity = 1.3616e+01	k_t
Generation 7010: Best = 2.018308e+04 = [1, 1, 1, 1]	FES = 701100	Diversity = 1.3535e+01	k_t
Generation 7020: Best = 2.018308e+04 = [4, 4, 2, 4]	FES = 702100	Diversity = 1.3416e+01	k_t
Generation 7030: Best = 2.018308e+04 = [1, 1, 1, 1]	FES = 703100	Diversity = 1.3488e+01	k_t
Generation 7040: Best = 2.018308e+04 = [3, 2, 4, 4]	FES = 704100	Diversity = 1.3526e+01	k_t
Generation 7050: Best = 2.017175e+04 = [1, 1, 1, 1]	FES = 705100	Diversity = 1.3338e+01	k_t
Generation 7060: Best = 2.017175e+04 = [3, 2, 2, 2]	FES = 706100	Diversity = 1.3242e+01	k_t
Generation 7070: Best = 2.015467e+04 = [1, 1, 1, 1]	FES = 707100	Diversity = 1.3242e+01	k_t
Generation 7080: Best = 2.015445e+04 = [2, 3, 4, 3]	FES = 708100	Diversity = 1.3208e+01	k_t
Generation 7090: Best = 2.015445e+04 = [1, 1, 1, 1]	FES = 709100	Diversity = 1.3272e+01	k_t
Generation 7100: Best = 2.014256e+04 = [2, 2, 3, 2]	FES = 710100	Diversity = 1.3300e+01	k_t
Generation 7110: Best = 2.014256e+04 = [1, 1, 1, 1]	FES = 711100	Diversity = 1.3236e+01	k_t
Generation 7120: Best = 2.014256e+04 = [3, 4, 3, 2]	FES = 712100	Diversity = 1.3177e+01	k_t
Generation 7130: Best = 2.014256e+04 = [1, 1, 1, 1]	FES = 713100	Diversity = 1.3238e+01	k_t
Generation 7140: Best = 2.014063e+04 = [4, 4, 2, 3]	FES = 714100	Diversity = 1.3281e+01	k_t
Generation 7150: Best = 2.012755e+04 = [1, 1, 1, 1]	FES = 715100	Diversity = 1.3141e+01	k_t
Generation 7160: Best = 2.011209e+04 = [4, 2, 3, 2]	FES = 716100	Diversity = 1.2885e+01	k_t
Generation 7170: Best = 2.011209e+04 = [1, 1, 1, 1]	FES = 717100	Diversity = 1.2882e+01	k_t
Generation 7180: Best = 2.011185e+04 = [3, 4, 2, 4]	FES = 718100	Diversity = 1.2878e+01	k_t
Generation 7190: Best = 2.011163e+04 = [1, 1, 1, 1]	FES = 719100	Diversity = 1.2806e+01	k_t
Generation 7200: Best = 2.011163e+04 = [3, 4, 3, 4]	FES = 720100	Diversity = 1.2895e+01	k_t
Generation 7210: Best = 2.011163e+04 = [1, 1, 1, 1]	FES = 721100	Diversity = 1.2712e+01	k_t
Generation 7220: Best = 2.011163e+04 = [3, 4, 3, 3]	FES = 722100	Diversity = 1.2690e+01	k_t
Generation 7230: Best = 2.011163e+04 = [1, 1, 1, 1]	FES = 723100	Diversity = 1.2759e+01	k_t
Generation 7240: Best = 2.010990e+04 = [2, 3, 3, 3]	FES = 724100	Diversity = 1.2559e+01	k_t

Generation 7250: Best = 2.010990e+04 = [1, 1, 1, 1]	FES = 725100	Diversity = 1.2552e+01	k_t
Generation 7260: Best = 2.010990e+04 = [3, 2, 4, 4]	FES = 726100	Diversity = 1.2549e+01	k_t
Generation 7270: Best = 2.010983e+04 = [1, 1, 1, 1]	FES = 727100	Diversity = 1.2555e+01	k_t
Generation 7280: Best = 2.010908e+04 = [2, 3, 4, 2]	FES = 728100	Diversity = 1.2451e+01	k_t
Generation 7290: Best = 2.010898e+04 = [1, 1, 1, 1]	FES = 729100	Diversity = 1.2423e+01	k_t
Generation 7300: Best = 2.009715e+04 = [2, 2, 3, 3]	FES = 730100	Diversity = 1.2341e+01	k_t
Generation 7310: Best = 2.009282e+04 = [1, 1, 1, 1]	FES = 731100	Diversity = 1.2347e+01	k_t
Generation 7320: Best = 2.009264e+04 = [3, 4, 4, 3]	FES = 732100	Diversity = 1.2510e+01	k_t
Generation 7330: Best = 2.008830e+04 = [1, 1, 1, 1]	FES = 733100	Diversity = 1.2457e+01	k_t
Generation 7340: Best = 2.008830e+04 = [4, 4, 3, 4]	FES = 734100	Diversity = 1.2298e+01	k_t
Generation 7350: Best = 2.007162e+04 = [1, 1, 1, 1]	FES = 735100	Diversity = 1.2315e+01	k_t
Generation 7360: Best = 2.005866e+04 = [4, 4, 3, 2]	FES = 736100	Diversity = 1.2323e+01	k_t
Generation 7370: Best = 2.005866e+04 = [1, 1, 1, 1]	FES = 737100	Diversity = 1.2316e+01	k_t
Generation 7380: Best = 2.005866e+04 = [3, 2, 4, 3]	FES = 738100	Diversity = 1.2218e+01	k_t
Generation 7390: Best = 2.005861e+04 = [1, 1, 1, 1]	FES = 739100	Diversity = 1.2207e+01	k_t
Generation 7400: Best = 2.005861e+04 = [2, 2, 2, 4]	FES = 740100	Diversity = 1.2089e+01	k_t
Generation 7410: Best = 2.005861e+04 = [1, 1, 1, 1]	FES = 741100	Diversity = 1.2157e+01	k_t
Generation 7420: Best = 2.005861e+04 = [4, 4, 3, 3]	FES = 742100	Diversity = 1.2098e+01	k_t
Generation 7430: Best = 2.005366e+04 = [1, 1, 1, 1]	FES = 743100	Diversity = 1.1979e+01	k_t
Generation 7440: Best = 2.005366e+04 = [2, 4, 2, 2]	FES = 744100	Diversity = 1.1972e+01	k_t
Generation 7450: Best = 2.005346e+04 = [1, 1, 1, 1]	FES = 745100	Diversity = 1.1982e+01	k_t
Generation 7460: Best = 2.002246e+04 = [3, 2, 4, 4]	FES = 746100	Diversity = 1.1855e+01	k_t
Generation 7470: Best = 2.002228e+04 = [1, 1, 1, 1]	FES = 747100	Diversity = 1.1796e+01	k_t
Generation 7480: Best = 2.002228e+04 = [3, 2, 2, 3]	FES = 748100	Diversity = 1.1846e+01	k_t
Generation 7490: Best = 2.001822e+04 = [1, 1, 1, 1]	FES = 749100	Diversity = 1.1811e+01	k_t
Generation 7500: Best = 2.001477e+04 = [2, 2, 4, 4]	FES = 750100	Diversity = 1.1712e+01	k_t
Generation 7510: Best = 2.001477e+04 = [1, 1, 1, 1]	FES = 751100	Diversity = 1.1737e+01	k_t

Generation 7520: Best = 2.001477e+04 = [3, 4, 2, 3]	FES = 752100	Diversity = 1.1906e+01	k_t
Generation 7530: Best = 2.001477e+04 = [1, 1, 1, 1]	FES = 753100	Diversity = 1.1808e+01	k_t
Generation 7540: Best = 2.000437e+04 = [2, 3, 3, 4]	FES = 754100	Diversity = 1.1755e+01	k_t
Generation 7550: Best = 2.000332e+04 = [1, 1, 1, 1]	FES = 755100	Diversity = 1.1777e+01	k_t
Generation 7560: Best = 2.000332e+04 = [2, 4, 2, 4]	FES = 756100	Diversity = 1.1776e+01	k_t
Generation 7570: Best = 2.000332e+04 = [1, 1, 1, 1]	FES = 757100	Diversity = 1.1800e+01	k_t
Generation 7580: Best = 1.999140e+04 = [3, 3, 4, 3]	FES = 758100	Diversity = 1.1713e+01	k_t
Generation 7590: Best = 1.998803e+04 = [1, 1, 1, 1]	FES = 759100	Diversity = 1.1644e+01	k_t
Generation 7600: Best = 1.998319e+04 = [4, 3, 3, 2]	FES = 760100	Diversity = 1.1665e+01	k_t
Generation 7610: Best = 1.998319e+04 = [1, 1, 1, 1]	FES = 761100	Diversity = 1.1563e+01	k_t
Generation 7620: Best = 1.998319e+04 = [2, 3, 2, 3]	FES = 762100	Diversity = 1.1535e+01	k_t
Generation 7630: Best = 1.997739e+04 = [1, 1, 1, 1]	FES = 763100	Diversity = 1.1416e+01	k_t
Generation 7640: Best = 1.997739e+04 = [2, 2, 3, 4]	FES = 764100	Diversity = 1.1439e+01	k_t
Generation 7650: Best = 1.996481e+04 = [1, 1, 1, 1]	FES = 765100	Diversity = 1.1427e+01	k_t
Generation 7660: Best = 1.996481e+04 = [3, 2, 2, 2]	FES = 766100	Diversity = 1.1361e+01	k_t
Generation 7670: Best = 1.996481e+04 = [1, 1, 1, 1]	FES = 767100	Diversity = 1.1330e+01	k_t
Generation 7680: Best = 1.996335e+04 = [4, 2, 2, 3]	FES = 768100	Diversity = 1.1359e+01	k_t
Generation 7690: Best = 1.996335e+04 = [1, 1, 1, 1]	FES = 769100	Diversity = 1.1407e+01	k_t
Generation 7700: Best = 1.996327e+04 = [3, 2, 3, 4]	FES = 770100	Diversity = 1.1428e+01	k_t
Generation 7710: Best = 1.996290e+04 = [1, 1, 1, 1]	FES = 771100	Diversity = 1.1412e+01	k_t
Generation 7720: Best = 1.995887e+04 = [2, 4, 2, 2]	FES = 772100	Diversity = 1.1382e+01	k_t
Generation 7730: Best = 1.995887e+04 = [1, 1, 1, 1]	FES = 773100	Diversity = 1.1453e+01	k_t
Generation 7740: Best = 1.995887e+04 = [4, 4, 2, 2]	FES = 774100	Diversity = 1.1348e+01	k_t
Generation 7750: Best = 1.995872e+04 = [1, 1, 1, 1]	FES = 775100	Diversity = 1.1260e+01	k_t
Generation 7760: Best = 1.995756e+04 = [4, 2, 2, 2]	FES = 776100	Diversity = 1.1176e+01	k_t
Generation 7770: Best = 1.995224e+04 = [1, 1, 1, 1]	FES = 777100	Diversity = 1.1074e+01	k_t
Generation 7780: Best = 1.995224e+04 = [3, 2, 4, 4]	FES = 778100	Diversity = 1.1011e+01	k_t

Generation 7790: Best = 1.995224e+04 = [1, 1, 1, 1]	FES = 779100	Diversity = 1.0876e+01	k_t
Generation 7800: Best = 1.994698e+04 = [2, 3, 3, 2]	FES = 780100	Diversity = 1.0805e+01	k_t
Generation 7810: Best = 1.994447e+04 = [1, 1, 1, 1]	FES = 781100	Diversity = 1.0716e+01	k_t
Generation 7820: Best = 1.994090e+04 = [2, 4, 4, 4]	FES = 782100	Diversity = 1.0578e+01	k_t
Generation 7830: Best = 1.991588e+04 = [1, 1, 1, 1]	FES = 783100	Diversity = 1.0427e+01	k_t
Generation 7840: Best = 1.991588e+04 = [3, 2, 4, 3]	FES = 784100	Diversity = 1.0259e+01	k_t
Generation 7850: Best = 1.991340e+04 = [1, 1, 1, 1]	FES = 785100	Diversity = 1.0011e+01	k_t
Generation 7860: Best = 1.991340e+04 = [3, 3, 4, 2]	FES = 786100	Diversity = 1.0016e+01	k_t
Generation 7870: Best = 1.991284e+04 = [1, 1, 1, 1]	FES = 787100	Diversity = 9.8989e+00	k_t
Generation 7880: Best = 1.991284e+04 = [3, 2, 2, 4]	FES = 788100	Diversity = 9.7302e+00	k_t
Generation 7890: Best = 1.990875e+04 = [1, 1, 1, 1]	FES = 789100	Diversity = 9.4536e+00	k_t
Generation 7900: Best = 1.990875e+04 = [2, 3, 4, 3]	FES = 790100	Diversity = 9.0741e+00	k_t
Generation 7910: Best = 1.990469e+04 = [1, 1, 1, 1]	FES = 791100	Diversity = 8.8310e+00	k_t
Generation 7920: Best = 1.990192e+04 = [2, 2, 2, 2]	FES = 792100	Diversity = 8.4513e+00	k_t
Generation 7930: Best = 1.989221e+04 = [1, 1, 1, 1]	FES = 793100	Diversity = 7.7702e+00	k_t
Generation 7940: Best = 1.987694e+04 = [3, 3, 3, 3]	FES = 794100	Diversity = 6.9491e+00	k_t
Generation 7950: Best = 1.987664e+04 = [1, 1, 1, 1]	FES = 795100	Diversity = 5.8453e+00	k_t
Generation 7960: Best = 1.987664e+04 = [4, 4, 3, 4]	FES = 796100	Diversity = 4.7254e+00	k_t
Generation 7970: Best = 1.986089e+04 = [1, 1, 1, 1]	FES = 797100	Diversity = 4.6944e+00	k_t
Generation 7980: Best = 1.985236e+04 = [4, 4, 4, 4]	FES = 798100	Diversity = 4.1258e+00	k_t
Generation 7990: Best = 1.983846e+04 = [1, 1, 1, 1]	FES = 799100	Diversity = 4.3741e+00	k_t
Generation 8000: Best = 1.983586e+04 = [2, 2, 2, 3]	FES = 800100	Diversity = 4.9498e+00	k_t
Generation 8010: Best = 1.982543e+04 = [1, 1, 1, 1]	FES = 801100	Diversity = 5.2751e+00	k_t
Generation 8020: Best = 1.981500e+04 = [2, 2, 2, 3]	FES = 802100	Diversity = 5.1142e+00	k_t
Generation 8030: Best = 1.980148e+04 = [1, 1, 1, 1]	FES = 803100	Diversity = 4.9088e+00	k_t
Generation 8040: Best = 1.978873e+04 = [3, 2, 4, 4]	FES = 804100	Diversity = 4.9386e+00	k_t
Generation 8050: Best = 1.978305e+04 = [1, 1, 1, 1]	FES = 805100	Diversity = 4.7617e+00	k_t

Generation 8060: Best = 1.976314e+04 = [3, 4, 3, 4]	FES = 806100	Diversity = 4.7970e+00	k_t
Generation 8070: Best = 1.976083e+04 = [1, 1, 1, 1]	FES = 807100	Diversity = 4.7132e+00	k_t
Generation 8080: Best = 1.974971e+04 = [4, 4, 3, 2]	FES = 808100	Diversity = 4.7439e+00	k_t
Generation 8090: Best = 1.973811e+04 = [1, 1, 1, 1]	FES = 809100	Diversity = 4.7730e+00	k_t
Generation 8100: Best = 1.971474e+04 = [3, 4, 2, 3]	FES = 810100	Diversity = 4.5042e+00	k_t
Generation 8110: Best = 1.970554e+04 = [1, 1, 1, 1]	FES = 811100	Diversity = 4.6598e+00	k_t
Generation 8120: Best = 1.969740e+04 = [4, 3, 2, 3]	FES = 812100	Diversity = 4.6623e+00	k_t
Generation 8130: Best = 1.969018e+04 = [1, 1, 1, 1]	FES = 813100	Diversity = 4.5428e+00	k_t
Generation 8140: Best = 1.967705e+04 = [2, 2, 2, 2]	FES = 814100	Diversity = 4.4777e+00	k_t
Generation 8150: Best = 1.967373e+04 = [1, 1, 1, 1]	FES = 815100	Diversity = 4.4099e+00	k_t
Generation 8160: Best = 1.966256e+04 = [3, 3, 4, 3]	FES = 816100	Diversity = 4.3396e+00	k_t
Generation 8170: Best = 1.965026e+04 = [1, 1, 1, 1]	FES = 817100	Diversity = 4.2261e+00	k_t
Generation 8180: Best = 1.964303e+04 = [4, 4, 4, 2]	FES = 818100	Diversity = 4.1900e+00	k_t
Generation 8190: Best = 1.963992e+04 = [1, 1, 1, 1]	FES = 819100	Diversity = 4.1649e+00	k_t
Generation 8200: Best = 1.963883e+04 = [3, 4, 4, 3]	FES = 820100	Diversity = 4.0520e+00	k_t
Generation 8210: Best = 1.963111e+04 = [1, 1, 1, 1]	FES = 821100	Diversity = 4.2374e+00	k_t
Generation 8220: Best = 1.962695e+04 = [4, 4, 2, 4]	FES = 822100	Diversity = 4.1511e+00	k_t
Generation 8230: Best = 1.962473e+04 = [1, 1, 1, 1]	FES = 823100	Diversity = 4.0760e+00	k_t
Generation 8240: Best = 1.962208e+04 = [4, 3, 4, 2]	FES = 824100	Diversity = 4.0832e+00	k_t
Generation 8250: Best = 1.961919e+04 = [1, 1, 1, 1]	FES = 825100	Diversity = 3.9832e+00	k_t
Generation 8260: Best = 1.961689e+04 = [3, 4, 4, 3]	FES = 826100	Diversity = 3.9222e+00	k_t
Generation 8270: Best = 1.961277e+04 = [1, 1, 1, 1]	FES = 827100	Diversity = 3.8087e+00	k_t
Generation 8280: Best = 1.961138e+04 = [2, 3, 4, 2]	FES = 828100	Diversity = 3.7125e+00	k_t
Generation 8290: Best = 1.961053e+04 = [1, 1, 1, 1]	FES = 829100	Diversity = 3.6108e+00	k_t
Generation 8300: Best = 1.960759e+04 = [3, 4, 2, 2]	FES = 830100	Diversity = 3.6369e+00	k_t
Generation 8310: Best = 1.960305e+04 = [1, 1, 1, 1]	FES = 831100	Diversity = 3.5590e+00	k_t
Generation 8320: Best = 1.960216e+04 = [2, 4, 3, 3]	FES = 832100	Diversity = 3.4125e+00	k_t

Generation 8330: Best = 1.960187e+04 = [1, 1, 1, 1]	FES = 833100	Diversity = 3.4345e+00	k_t
Generation 8340: Best = 1.960107e+04 = [2, 3, 2, 4]	FES = 834100	Diversity = 3.5048e+00	k_t
Generation 8350: Best = 1.959792e+04 = [1, 1, 1, 1]	FES = 835100	Diversity = 3.4314e+00	k_t
Generation 8360: Best = 1.959792e+04 = [4, 3, 4, 2]	FES = 836100	Diversity = 3.4866e+00	k_t
Generation 8370: Best = 1.959634e+04 = [1, 1, 1, 1]	FES = 837100	Diversity = 3.5308e+00	k_t
Generation 8380: Best = 1.959322e+04 = [2, 3, 3, 4]	FES = 838100	Diversity = 3.4481e+00	k_t
Generation 8390: Best = 1.959322e+04 = [1, 1, 1, 1]	FES = 839100	Diversity = 3.3872e+00	k_t
Generation 8400: Best = 1.959195e+04 = [3, 3, 2, 2]	FES = 840100	Diversity = 3.3425e+00	k_t
Generation 8410: Best = 1.958639e+04 = [1, 1, 1, 1]	FES = 841100	Diversity = 3.4049e+00	k_t
Generation 8420: Best = 1.958103e+04 = [4, 3, 4, 3]	FES = 842100	Diversity = 3.5237e+00	k_t
Generation 8430: Best = 1.957523e+04 = [1, 1, 1, 1]	FES = 843100	Diversity = 3.5783e+00	k_t
Generation 8440: Best = 1.957472e+04 = [2, 2, 2, 3]	FES = 844100	Diversity = 3.6337e+00	k_t
Generation 8450: Best = 1.957190e+04 = [1, 1, 1, 1]	FES = 845100	Diversity = 3.5438e+00	k_t
Generation 8460: Best = 1.956920e+04 = [4, 4, 4, 4]	FES = 846100	Diversity = 3.5275e+00	k_t
Generation 8470: Best = 1.956888e+04 = [1, 1, 1, 1]	FES = 847100	Diversity = 3.4368e+00	k_t
Generation 8480: Best = 1.956447e+04 = [2, 2, 3, 4]	FES = 848100	Diversity = 3.4734e+00	k_t
Generation 8490: Best = 1.956153e+04 = [1, 1, 1, 1]	FES = 849100	Diversity = 3.5728e+00	k_t
Generation 8500: Best = 1.955973e+04 = [4, 2, 4, 4]	FES = 850100	Diversity = 3.4651e+00	k_t
Generation 8510: Best = 1.955965e+04 = [1, 1, 1, 1]	FES = 851100	Diversity = 3.5654e+00	k_t
Generation 8520: Best = 1.955632e+04 = [3, 2, 2, 4]	FES = 852100	Diversity = 3.4433e+00	k_t
Generation 8530: Best = 1.955157e+04 = [1, 1, 1, 1]	FES = 853100	Diversity = 3.4864e+00	k_t
Generation 8540: Best = 1.955157e+04 = [4, 3, 3, 2]	FES = 854100	Diversity = 3.6203e+00	k_t
Generation 8550: Best = 1.954404e+04 = [1, 1, 1, 1]	FES = 855100	Diversity = 3.8166e+00	k_t
Generation 8560: Best = 1.954340e+04 = [2, 4, 2, 3]	FES = 856100	Diversity = 3.8550e+00	k_t
Generation 8570: Best = 1.953728e+04 = [1, 1, 1, 1]	FES = 857100	Diversity = 3.9949e+00	k_t
Generation 8580: Best = 1.953400e+04 = [4, 4, 3, 3]	FES = 858100	Diversity = 4.1842e+00	k_t
Generation 8590: Best = 1.953323e+04 = [1, 1, 1, 1]	FES = 859100	Diversity = 3.9971e+00	k_t

Generation 8600: Best = 1.952847e+04 = [3, 4, 2, 3]	FES = 860100	Diversity = 4.1365e+00	k_t
Generation 8610: Best = 1.952513e+04 = [1, 1, 1, 1]	FES = 861100	Diversity = 3.8793e+00	k_t
Generation 8620: Best = 1.951960e+04 = [2, 2, 2, 3]	FES = 862100	Diversity = 3.6967e+00	k_t
Generation 8630: Best = 1.951894e+04 = [1, 1, 1, 1]	FES = 863100	Diversity = 3.7313e+00	k_t
Generation 8640: Best = 1.951810e+04 = [2, 2, 2, 4]	FES = 864100	Diversity = 3.6804e+00	k_t
Generation 8650: Best = 1.951604e+04 = [1, 1, 1, 1]	FES = 865100	Diversity = 3.7273e+00	k_t
Generation 8660: Best = 1.951256e+04 = [2, 3, 2, 3]	FES = 866100	Diversity = 3.7221e+00	k_t
Generation 8670: Best = 1.951108e+04 = [1, 1, 1, 1]	FES = 867100	Diversity = 3.6470e+00	k_t
Generation 8680: Best = 1.950808e+04 = [3, 3, 3, 3]	FES = 868100	Diversity = 3.6809e+00	k_t
Generation 8690: Best = 1.950614e+04 = [1, 1, 1, 1]	FES = 869100	Diversity = 3.7896e+00	k_t
Generation 8700: Best = 1.950322e+04 = [3, 2, 2, 4]	FES = 870100	Diversity = 3.6047e+00	k_t
Generation 8710: Best = 1.950091e+04 = [1, 1, 1, 1]	FES = 871100	Diversity = 3.4842e+00	k_t
Generation 8720: Best = 1.949799e+04 = [3, 2, 3, 3]	FES = 872100	Diversity = 3.5074e+00	k_t
Generation 8730: Best = 1.949538e+04 = [1, 1, 1, 1]	FES = 873100	Diversity = 3.4745e+00	k_t
Generation 8740: Best = 1.949287e+04 = [4, 2, 4, 3]	FES = 874100	Diversity = 3.5339e+00	k_t
Generation 8750: Best = 1.949019e+04 = [1, 1, 1, 1]	FES = 875100	Diversity = 3.4669e+00	k_t
Generation 8760: Best = 1.949019e+04 = [4, 4, 4, 4]	FES = 876100	Diversity = 3.5761e+00	k_t
Generation 8770: Best = 1.948787e+04 = [1, 1, 1, 1]	FES = 877100	Diversity = 3.8968e+00	k_t
Generation 8780: Best = 1.948482e+04 = [3, 3, 3, 2]	FES = 878100	Diversity = 3.7650e+00	k_t
Generation 8790: Best = 1.948424e+04 = [1, 1, 1, 1]	FES = 879100	Diversity = 3.8958e+00	k_t
Generation 8800: Best = 1.948140e+04 = [3, 2, 2, 3]	FES = 880100	Diversity = 3.9662e+00	k_t
Generation 8810: Best = 1.948053e+04 = [1, 1, 1, 1]	FES = 881100	Diversity = 3.8748e+00	k_t
Generation 8820: Best = 1.947853e+04 = [2, 2, 3, 4]	FES = 882100	Diversity = 3.7482e+00	k_t
Generation 8830: Best = 1.947532e+04 = [1, 1, 1, 1]	FES = 883100	Diversity = 3.8326e+00	k_t
Generation 8840: Best = 1.947119e+04 = [2, 3, 4, 2]	FES = 884100	Diversity = 4.0597e+00	k_t
Generation 8850: Best = 1.946810e+04 = [1, 1, 1, 1]	FES = 885100	Diversity = 4.1863e+00	k_t
Generation 8860: Best = 1.946318e+04 = [3, 3, 3, 2]	FES = 886100	Diversity = 4.0974e+00	k_t

Generation 8870: Best = 1.946151e+04 = [1, 1, 1, 1]	FES = 887100	Diversity = 4.0256e+00	k_t
Generation 8880: Best = 1.945824e+04 = [4, 2, 4, 4]	FES = 888100	Diversity = 3.9015e+00	k_t
Generation 8890: Best = 1.945689e+04 = [1, 1, 1, 1]	FES = 889100	Diversity = 3.8064e+00	k_t
Generation 8900: Best = 1.945043e+04 = [4, 4, 4, 3]	FES = 890100	Diversity = 3.6891e+00	k_t
Generation 8910: Best = 1.944880e+04 = [1, 1, 1, 1]	FES = 891100	Diversity = 3.7426e+00	k_t
Generation 8920: Best = 1.944473e+04 = [4, 3, 2, 3]	FES = 892100	Diversity = 3.7614e+00	k_t
Generation 8930: Best = 1.943957e+04 = [1, 1, 1, 1]	FES = 893100	Diversity = 3.8175e+00	k_t
Generation 8940: Best = 1.943746e+04 = [2, 4, 2, 4]	FES = 894100	Diversity = 4.0085e+00	k_t
Generation 8950: Best = 1.942912e+04 = [1, 1, 1, 1]	FES = 895100	Diversity = 4.2467e+00	k_t
Generation 8960: Best = 1.942584e+04 = [3, 3, 3, 4]	FES = 896100	Diversity = 4.4090e+00	k_t
Generation 8970: Best = 1.942054e+04 = [1, 1, 1, 1]	FES = 897100	Diversity = 4.2136e+00	k_t
Generation 8980: Best = 1.941742e+04 = [4, 3, 3, 2]	FES = 898100	Diversity = 4.3032e+00	k_t
Generation 8990: Best = 1.941412e+04 = [1, 1, 1, 1]	FES = 899100	Diversity = 4.2867e+00	k_t
Generation 9000: Best = 1.941067e+04 = [4, 2, 2, 3]	FES = 900100	Diversity = 4.2745e+00	k_t
Generation 9010: Best = 1.940521e+04 = [1, 1, 1, 1]	FES = 901100	Diversity = 4.1800e+00	k_t
Generation 9020: Best = 1.940373e+04 = [3, 3, 2, 4]	FES = 902100	Diversity = 3.7781e+00	k_t
Generation 9030: Best = 1.940321e+04 = [1, 1, 1, 1]	FES = 903100	Diversity = 3.5385e+00	k_t
Generation 9040: Best = 1.940197e+04 = [2, 3, 2, 2]	FES = 904100	Diversity = 3.3557e+00	k_t
Generation 9050: Best = 1.940024e+04 = [1, 1, 1, 1]	FES = 905100	Diversity = 3.2611e+00	k_t
Generation 9060: Best = 1.939719e+04 = [2, 4, 2, 4]	FES = 906100	Diversity = 3.4399e+00	k_t
Generation 9070: Best = 1.939474e+04 = [1, 1, 1, 1]	FES = 907100	Diversity = 3.1433e+00	k_t
Generation 9080: Best = 1.939236e+04 = [2, 3, 4, 2]	FES = 908100	Diversity = 2.9479e+00	k_t
Generation 9090: Best = 1.939180e+04 = [1, 1, 1, 1]	FES = 909100	Diversity = 2.7762e+00	k_t
Generation 9100: Best = 1.939112e+04 = [2, 2, 3, 2]	FES = 910100	Diversity = 2.7225e+00	k_t
Generation 9110: Best = 1.938947e+04 = [1, 1, 1, 1]	FES = 911100	Diversity = 2.6298e+00	k_t
Generation 9120: Best = 1.938774e+04 = [4, 4, 4, 2]	FES = 912100	Diversity = 2.5697e+00	k_t
Generation 9130: Best = 1.938587e+04 = [1, 1, 1, 1]	FES = 913100	Diversity = 2.8008e+00	k_t

Generation 9140: Best = 1.938392e+04 = [4, 2, 3, 3]	FES = 914100	Diversity = 2.9516e+00	k_t
Generation 9150: Best = 1.938059e+04 = [1, 1, 1, 1]	FES = 915100	Diversity = 3.2557e+00	k_t
Generation 9160: Best = 1.938011e+04 = [3, 4, 3, 2]	FES = 916100	Diversity = 3.1453e+00	k_t
Generation 9170: Best = 1.937844e+04 = [1, 1, 1, 1]	FES = 917100	Diversity = 3.0507e+00	k_t
Generation 9180: Best = 1.937560e+04 = [4, 2, 4, 3]	FES = 918100	Diversity = 3.2444e+00	k_t
Generation 9190: Best = 1.937470e+04 = [1, 1, 1, 1]	FES = 919100	Diversity = 3.4203e+00	k_t
Generation 9200: Best = 1.937323e+04 = [4, 4, 4, 2]	FES = 920100	Diversity = 3.6952e+00	k_t
Generation 9210: Best = 1.937112e+04 = [1, 1, 1, 1]	FES = 921100	Diversity = 3.6373e+00	k_t
Generation 9220: Best = 1.936866e+04 = [4, 4, 2, 3]	FES = 922100	Diversity = 3.7099e+00	k_t
Generation 9230: Best = 1.936608e+04 = [1, 1, 1, 1]	FES = 923100	Diversity = 3.9147e+00	k_t
Generation 9240: Best = 1.936519e+04 = [3, 3, 4, 3]	FES = 924100	Diversity = 3.8256e+00	k_t
Generation 9250: Best = 1.936405e+04 = [1, 1, 1, 1]	FES = 925100	Diversity = 3.8496e+00	k_t
Generation 9260: Best = 1.936207e+04 = [3, 4, 4, 3]	FES = 926100	Diversity = 3.6848e+00	k_t
Generation 9270: Best = 1.935964e+04 = [1, 1, 1, 1]	FES = 927100	Diversity = 3.6564e+00	k_t
Generation 9280: Best = 1.935707e+04 = [4, 3, 2, 3]	FES = 928100	Diversity = 3.6273e+00	k_t
Generation 9290: Best = 1.935638e+04 = [1, 1, 1, 1]	FES = 929100	Diversity = 3.7505e+00	k_t
Generation 9300: Best = 1.935425e+04 = [4, 3, 4, 4]	FES = 930100	Diversity = 3.6423e+00	k_t
Generation 9310: Best = 1.935217e+04 = [1, 1, 1, 1]	FES = 931100	Diversity = 3.6592e+00	k_t
Generation 9320: Best = 1.935025e+04 = [4, 2, 2, 3]	FES = 932100	Diversity = 3.7624e+00	k_t
Generation 9330: Best = 1.934751e+04 = [1, 1, 1, 1]	FES = 933100	Diversity = 3.6680e+00	k_t
Generation 9340: Best = 1.934613e+04 = [2, 2, 3, 3]	FES = 934100	Diversity = 3.5761e+00	k_t
Generation 9350: Best = 1.934439e+04 = [1, 1, 1, 1]	FES = 935100	Diversity = 3.5524e+00	k_t
Generation 9360: Best = 1.934247e+04 = [2, 4, 2, 3]	FES = 936100	Diversity = 3.6486e+00	k_t
Generation 9370: Best = 1.934201e+04 = [1, 1, 1, 1]	FES = 937100	Diversity = 3.6565e+00	k_t
Generation 9380: Best = 1.934097e+04 = [2, 4, 3, 3]	FES = 938100	Diversity = 3.6104e+00	k_t
Generation 9390: Best = 1.933904e+04 = [1, 1, 1, 1]	FES = 939100	Diversity = 3.3547e+00	k_t
Generation 9400: Best = 1.933655e+04 = [3, 4, 2, 4]	FES = 940100	Diversity = 3.5071e+00	k_t

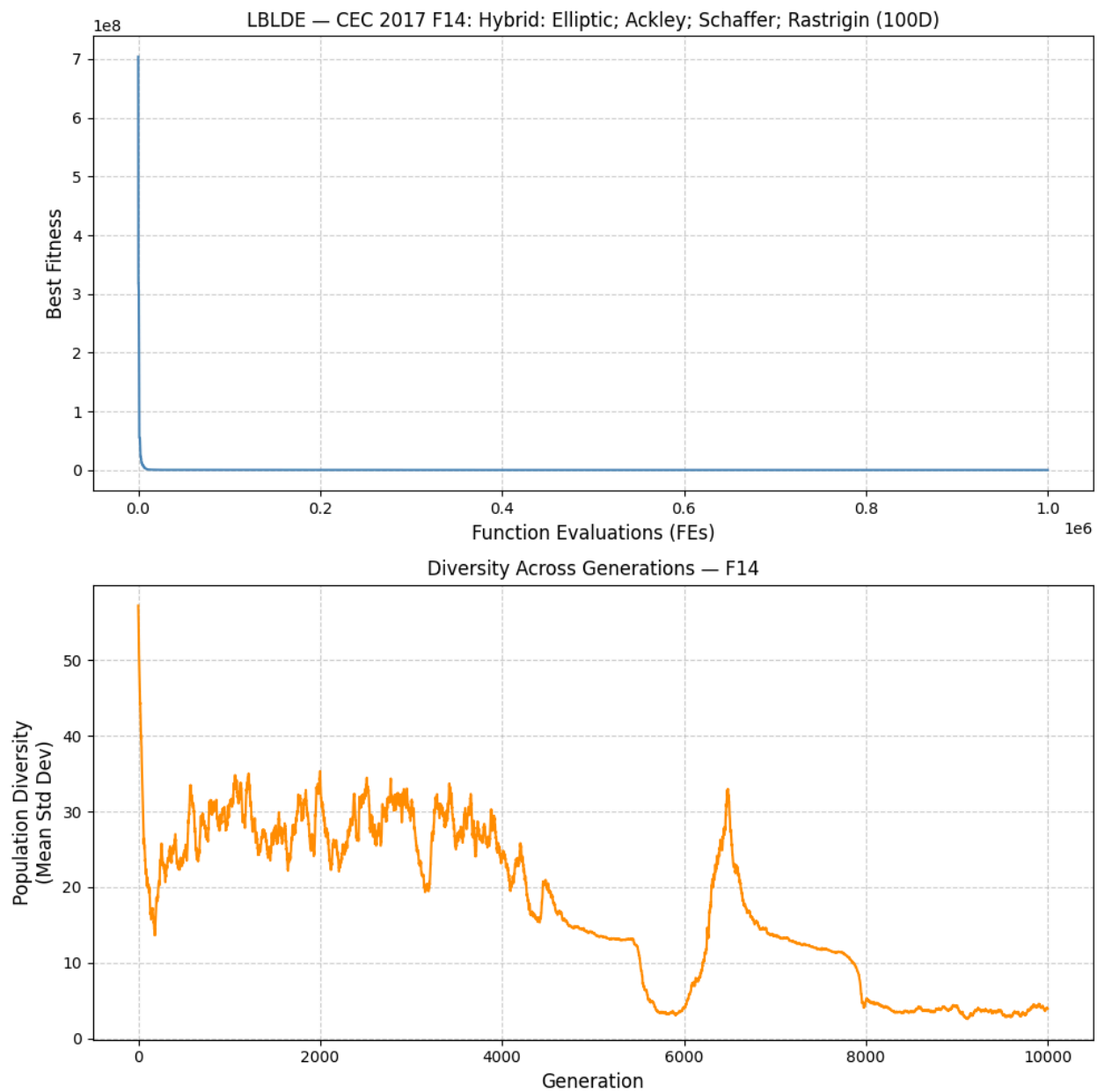
Generation 9410: Best = 1.933563e+04 = [1, 1, 1, 1]	FES = 941100	Diversity = 3.5445e+00	k_t
Generation 9420: Best = 1.933389e+04 = [4, 2, 4, 3]	FES = 942100	Diversity = 3.6241e+00	k_t
Generation 9430: Best = 1.933184e+04 = [1, 1, 1, 1]	FES = 943100	Diversity = 3.3765e+00	k_t
Generation 9440: Best = 1.933049e+04 = [2, 4, 3, 4]	FES = 944100	Diversity = 3.5821e+00	k_t
Generation 9450: Best = 1.932838e+04 = [1, 1, 1, 1]	FES = 945100	Diversity = 3.5990e+00	k_t
Generation 9460: Best = 1.932677e+04 = [2, 4, 2, 3]	FES = 946100	Diversity = 3.5273e+00	k_t
Generation 9470: Best = 1.932426e+04 = [1, 1, 1, 1]	FES = 947100	Diversity = 3.6652e+00	k_t
Generation 9480: Best = 1.932247e+04 = [2, 4, 3, 4]	FES = 948100	Diversity = 3.6054e+00	k_t
Generation 9490: Best = 1.931991e+04 = [1, 1, 1, 1]	FES = 949100	Diversity = 3.5389e+00	k_t
Generation 9500: Best = 1.931608e+04 = [3, 4, 3, 3]	FES = 950100	Diversity = 3.2176e+00	k_t
Generation 9510: Best = 1.931361e+04 = [1, 1, 1, 1]	FES = 951100	Diversity = 3.0737e+00	k_t
Generation 9520: Best = 1.931219e+04 = [4, 3, 3, 3]	FES = 952100	Diversity = 3.1681e+00	k_t
Generation 9530: Best = 1.931067e+04 = [1, 1, 1, 1]	FES = 953100	Diversity = 2.8934e+00	k_t
Generation 9540: Best = 1.930946e+04 = [2, 3, 2, 4]	FES = 954100	Diversity = 2.8776e+00	k_t
Generation 9550: Best = 1.930870e+04 = [1, 1, 1, 1]	FES = 955100	Diversity = 2.8666e+00	k_t
Generation 9560: Best = 1.930614e+04 = [4, 2, 3, 4]	FES = 956100	Diversity = 2.9694e+00	k_t
Generation 9570: Best = 1.930340e+04 = [1, 1, 1, 1]	FES = 957100	Diversity = 2.9022e+00	k_t
Generation 9580: Best = 1.930290e+04 = [4, 2, 4, 4]	FES = 958100	Diversity = 3.0602e+00	k_t
Generation 9590: Best = 1.930146e+04 = [1, 1, 1, 1]	FES = 959100	Diversity = 2.8802e+00	k_t
Generation 9600: Best = 1.929940e+04 = [3, 2, 4, 3]	FES = 960100	Diversity = 2.8657e+00	k_t
Generation 9610: Best = 1.929677e+04 = [1, 1, 1, 1]	FES = 961100	Diversity = 3.0255e+00	k_t
Generation 9620: Best = 1.929662e+04 = [3, 4, 4, 2]	FES = 962100	Diversity = 3.2508e+00	k_t
Generation 9630: Best = 1.929486e+04 = [1, 1, 1, 1]	FES = 963100	Diversity = 3.4600e+00	k_t
Generation 9640: Best = 1.929295e+04 = [4, 3, 4, 4]	FES = 964100	Diversity = 3.4124e+00	k_t
Generation 9650: Best = 1.929080e+04 = [1, 1, 1, 1]	FES = 965100	Diversity = 3.3785e+00	k_t
Generation 9660: Best = 1.928920e+04 = [2, 3, 3, 2]	FES = 966100	Diversity = 3.3286e+00	k_t
Generation 9670: Best = 1.928684e+04 = [1, 1, 1, 1]	FES = 967100	Diversity = 3.4847e+00	k_t

Generation 9680: Best = 1.928533e+04	FES = 968100	Diversity = 3.2855e+00	k_t
= [4, 3, 3, 2]			
Generation 9690: Best = 1.928391e+04	FES = 969100	Diversity = 3.1585e+00	k_t
= [1, 1, 1, 1]			
Generation 9700: Best = 1.928208e+04	FES = 970100	Diversity = 2.9282e+00	k_t
= [3, 3, 4, 3]			
Generation 9710: Best = 1.927982e+04	FES = 971100	Diversity = 3.1124e+00	k_t
= [1, 1, 1, 1]			
Generation 9720: Best = 1.927888e+04	FES = 972100	Diversity = 3.3606e+00	k_t
= [3, 3, 4, 4]			
Generation 9730: Best = 1.927532e+04	FES = 973100	Diversity = 3.3193e+00	k_t
= [1, 1, 1, 1]			
Generation 9740: Best = 1.927461e+04	FES = 974100	Diversity = 3.1205e+00	k_t
= [4, 3, 3, 2]			
Generation 9750: Best = 1.927143e+04	FES = 975100	Diversity = 3.1028e+00	k_t
= [1, 1, 1, 1]			
Generation 9760: Best = 1.927073e+04	FES = 976100	Diversity = 3.2127e+00	k_t
= [4, 2, 2, 4]			
Generation 9770: Best = 1.926698e+04	FES = 977100	Diversity = 3.4324e+00	k_t
= [1, 1, 1, 1]			
Generation 9780: Best = 1.926650e+04	FES = 978100	Diversity = 3.7411e+00	k_t
= [4, 2, 4, 3]			
Generation 9790: Best = 1.926562e+04	FES = 979100	Diversity = 3.7900e+00	k_t
= [1, 1, 1, 1]			
Generation 9800: Best = 1.926377e+04	FES = 980100	Diversity = 3.7511e+00	k_t
= [3, 3, 3, 4]			
Generation 9810: Best = 1.926135e+04	FES = 981100	Diversity = 4.0007e+00	k_t
= [1, 1, 1, 1]			
Generation 9820: Best = 1.925882e+04	FES = 982100	Diversity = 4.0619e+00	k_t
= [4, 4, 3, 2]			
Generation 9830: Best = 1.925723e+04	FES = 983100	Diversity = 4.1148e+00	k_t
= [1, 1, 1, 1]			
Generation 9840: Best = 1.925598e+04	FES = 984100	Diversity = 4.2587e+00	k_t
= [2, 2, 4, 2]			
Generation 9850: Best = 1.925383e+04	FES = 985100	Diversity = 4.3286e+00	k_t
= [1, 1, 1, 1]			
Generation 9860: Best = 1.925262e+04	FES = 986100	Diversity = 4.2872e+00	k_t
= [2, 3, 2, 4]			
Generation 9870: Best = 1.925072e+04	FES = 987100	Diversity = 4.1241e+00	k_t
= [1, 1, 1, 1]			
Generation 9880: Best = 1.924957e+04	FES = 988100	Diversity = 4.0650e+00	k_t
= [3, 3, 3, 4]			
Generation 9890: Best = 1.924734e+04	FES = 989100	Diversity = 4.2397e+00	k_t
= [1, 1, 1, 1]			
Generation 9900: Best = 1.924617e+04	FES = 990100	Diversity = 4.5749e+00	k_t
= [3, 3, 2, 2]			
Generation 9910: Best = 1.924435e+04	FES = 991100	Diversity = 4.4292e+00	k_t
= [1, 1, 1, 1]			
Generation 9920: Best = 1.924283e+04	FES = 992100	Diversity = 4.1400e+00	k_t
= [2, 2, 4, 3]			
Generation 9930: Best = 1.924152e+04	FES = 993100	Diversity = 4.0827e+00	k_t
= [1, 1, 1, 1]			
Generation 9940: Best = 1.924011e+04	FES = 994100	Diversity = 4.2828e+00	k_t
= [2, 2, 2, 4]			

Generation 9950: Best = 1.923851e+04 FES = 995100 Diversity = 4.0164e+00 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 1.923777e+04 FES = 996100 Diversity = 3.8561e+00 k_t
= [2, 4, 4, 3]
Generation 9970: Best = 1.923653e+04 FES = 997100 Diversity = 3.7044e+00 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 1.923350e+04 FES = 998100 Diversity = 3.9271e+00 k_t
= [3, 2, 2, 2]
Generation 9990: Best = 1.923252e+04 FES = 999100 Diversity = 3.9915e+00 k_t
= [1, 1, 1, 1]

Optimization completed!
Final Best Fitness = 1.923153e+04
Total Function Evaluations = 1000000

=====
=
Results:
Best Fitness : 1.923153e+04
Global Optimum : 1.400000e+03
Error (|best-optimum|) : 1.783153e+04
Best Solution (5 dim) : [-54.67179208 -64.0317636 -93.55060304 78.61606976
-16.4941301]
D^(0) (init. diversity): 5.7220e+01
=====



Graph saved as convergence_F14.png

```
In [39]: # =====
# F15 – Hybrid Function 5 (Bent Cigar; HGBat; Rastrigin; Rosenbrock)
# =====
run_and_plot('F15', make_wrapper(hybrid.f15), 1500.0, 'Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock')
```

```

=====
=
LBLDE_s on CEC 2017 F15 – Single Run
=====
=
Function      : Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock (100D)
Bounds       : [-100, 100]
Global optimum : 1500.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 3.781522e+11 | Diversity = 5.7220e+01
Generation 10: Best = 1.135764e+11 FES = 1100 Diversity = 4.4635e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.978048e+10 FES = 2100 Diversity = 3.6244e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 7.492971e+09 FES = 3100 Diversity = 2.9968e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.668483e+09 FES = 4100 Diversity = 2.8494e+01 k_t
= [4, 3, 4, 2]
Generation 50: Best = 4.816285e+08 FES = 5100 Diversity = 2.6625e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.397028e+08 FES = 6100 Diversity = 2.6086e+01 k_t
= [4, 4, 2, 3]
Generation 70: Best = 4.103336e+07 FES = 7100 Diversity = 2.4881e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.516390e+07 FES = 8100 Diversity = 2.3704e+01 k_t
= [4, 2, 4, 4]
Generation 90: Best = 5.356727e+06 FES = 9100 Diversity = 2.2043e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.364740e+06 FES = 10100 Diversity = 1.9778e+01 k_t
= [4, 2, 4, 3]
Generation 110: Best = 1.191147e+06 FES = 11100 Diversity = 1.8163e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 5.429196e+05 FES = 12100 Diversity = 1.7176e+01 k_t
= [4, 2, 2, 3]
Generation 130: Best = 2.795720e+05 FES = 13100 Diversity = 1.6016e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.551329e+05 FES = 14100 Diversity = 1.3995e+01 k_t
= [4, 4, 4, 2]
Generation 150: Best = 1.102276e+05 FES = 15100 Diversity = 1.2706e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 8.936469e+04 FES = 16100 Diversity = 1.0837e+01 k_t
= [3, 3, 4, 4]
Generation 170: Best = 7.096264e+04 FES = 17100 Diversity = 9.4388e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 5.422058e+04 FES = 18100 Diversity = 9.0305e+00 k_t
= [3, 2, 4, 2]
Generation 190: Best = 4.558153e+04 FES = 19100 Diversity = 9.0021e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 4.113379e+04 FES = 20100 Diversity = 8.4015e+00 k_t
= [2, 2, 2, 2]
Generation 210: Best = 3.129428e+04 FES = 21100 Diversity = 7.8127e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 2.931019e+04 FES = 22100 Diversity = 7.2356e+00 k_t
= [3, 4, 4, 4]

```

Generation 230:	Best = 2.444763e+04	FES = 23100	Diversity = 7.1042e+00	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 2.263389e+04	FES = 24100	Diversity = 7.1892e+00	k_t
= [3, 3, 3, 4]				
Generation 250:	Best = 2.074222e+04	FES = 25100	Diversity = 7.2914e+00	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 1.737993e+04	FES = 26100	Diversity = 7.3581e+00	k_t
= [4, 2, 3, 4]				
Generation 270:	Best = 1.545986e+04	FES = 27100	Diversity = 7.9173e+00	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 1.424602e+04	FES = 28100	Diversity = 8.5947e+00	k_t
= [3, 4, 3, 2]				
Generation 290:	Best = 1.142235e+04	FES = 29100	Diversity = 8.7846e+00	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 1.092007e+04	FES = 30100	Diversity = 8.6269e+00	k_t
= [3, 4, 4, 4]				
Generation 310:	Best = 9.726758e+03	FES = 31100	Diversity = 8.7240e+00	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 8.849507e+03	FES = 32100	Diversity = 8.8320e+00	k_t
= [4, 3, 4, 2]				
Generation 330:	Best = 8.002338e+03	FES = 33100	Diversity = 8.4596e+00	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 7.375294e+03	FES = 34100	Diversity = 9.1505e+00	k_t
= [4, 3, 3, 3]				
Generation 350:	Best = 6.699908e+03	FES = 35100	Diversity = 9.7067e+00	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 6.203036e+03	FES = 36100	Diversity = 9.7857e+00	k_t
= [3, 2, 2, 3]				
Generation 370:	Best = 5.323014e+03	FES = 37100	Diversity = 9.7165e+00	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 4.947520e+03	FES = 38100	Diversity = 1.0320e+01	k_t
= [3, 4, 4, 2]				
Generation 390:	Best = 4.767547e+03	FES = 39100	Diversity = 1.0218e+01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 4.313153e+03	FES = 40100	Diversity = 9.1670e+00	k_t
= [3, 2, 3, 4]				
Generation 410:	Best = 4.178801e+03	FES = 41100	Diversity = 8.9789e+00	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 4.048530e+03	FES = 42100	Diversity = 8.7165e+00	k_t
= [2, 2, 2, 4]				
Generation 430:	Best = 3.831441e+03	FES = 43100	Diversity = 7.9880e+00	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 3.732831e+03	FES = 44100	Diversity = 7.6083e+00	k_t
= [4, 4, 3, 3]				
Generation 450:	Best = 3.654756e+03	FES = 45100	Diversity = 6.7623e+00	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 3.544847e+03	FES = 46100	Diversity = 6.2779e+00	k_t
= [2, 2, 4, 4]				
Generation 470:	Best = 3.474992e+03	FES = 47100	Diversity = 6.0602e+00	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 3.474992e+03	FES = 48100	Diversity = 5.8442e+00	k_t
= [2, 4, 3, 4]				
Generation 490:	Best = 3.441518e+03	FES = 49100	Diversity = 5.4907e+00	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 3.418158e+03	FES = 50100	Diversity = 5.2818e+00	k_t
= [3, 2, 4, 4]				
Generation 510:	Best = 3.381554e+03	FES = 51100	Diversity = 5.2225e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 3.353315e+03	FES = 52100	Diversity = 5.1794e+00	k_t
= [2, 4, 4, 2]				
Generation 530:	Best = 3.328487e+03	FES = 53100	Diversity = 5.1415e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 3.301857e+03	FES = 54100	Diversity = 5.0769e+00	k_t
= [4, 4, 2, 2]				
Generation 550:	Best = 3.297914e+03	FES = 55100	Diversity = 5.0228e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 3.272299e+03	FES = 56100	Diversity = 5.0291e+00	k_t
= [4, 2, 4, 2]				
Generation 570:	Best = 3.241857e+03	FES = 57100	Diversity = 4.9443e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 3.229888e+03	FES = 58100	Diversity = 4.9331e+00	k_t
= [3, 2, 3, 3]				
Generation 590:	Best = 3.219893e+03	FES = 59100	Diversity = 4.8899e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 3.203945e+03	FES = 60100	Diversity = 4.9361e+00	k_t
= [3, 2, 3, 3]				
Generation 610:	Best = 3.193701e+03	FES = 61100	Diversity = 4.8677e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 3.188767e+03	FES = 62100	Diversity = 4.8415e+00	k_t
= [4, 2, 3, 3]				
Generation 630:	Best = 3.153424e+03	FES = 63100	Diversity = 4.8357e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 3.152266e+03	FES = 64100	Diversity = 4.8436e+00	k_t
= [4, 4, 3, 2]				
Generation 650:	Best = 3.128937e+03	FES = 65100	Diversity = 4.7990e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 3.113148e+03	FES = 66100	Diversity = 4.7938e+00	k_t
= [4, 3, 2, 4]				
Generation 670:	Best = 3.107025e+03	FES = 67100	Diversity = 4.7600e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 3.105035e+03	FES = 68100	Diversity = 4.7612e+00	k_t
= [2, 4, 3, 4]				
Generation 690:	Best = 3.098143e+03	FES = 69100	Diversity = 4.8115e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 3.091323e+03	FES = 70100	Diversity = 4.8110e+00	k_t
= [2, 3, 2, 2]				
Generation 710:	Best = 3.083436e+03	FES = 71100	Diversity = 4.7378e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 3.081563e+03	FES = 72100	Diversity = 4.7022e+00	k_t
= [2, 3, 3, 4]				
Generation 730:	Best = 3.077311e+03	FES = 73100	Diversity = 4.6370e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 3.071399e+03	FES = 74100	Diversity = 4.6367e+00	k_t
= [2, 3, 2, 2]				
Generation 750:	Best = 3.064584e+03	FES = 75100	Diversity = 4.5404e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 3.056187e+03	FES = 76100	Diversity = 4.5720e+00	k_t
= [3, 3, 2, 3]				

Generation 770:	Best = 3.045151e+03	FES = 77100	Diversity = 4.5481e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 3.034204e+03	FES = 78100	Diversity = 4.5284e+00	k_t
= [3, 4, 4, 2]				
Generation 790:	Best = 3.024545e+03	FES = 79100	Diversity = 4.4879e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 3.021185e+03	FES = 80100	Diversity = 4.5103e+00	k_t
= [2, 3, 3, 2]				
Generation 810:	Best = 3.010141e+03	FES = 81100	Diversity = 4.4333e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.998230e+03	FES = 82100	Diversity = 4.3561e+00	k_t
= [4, 4, 3, 2]				
Generation 830:	Best = 2.996834e+03	FES = 83100	Diversity = 4.3841e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.995689e+03	FES = 84100	Diversity = 4.3603e+00	k_t
= [2, 3, 4, 4]				
Generation 850:	Best = 2.989113e+03	FES = 85100	Diversity = 4.3532e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.989113e+03	FES = 86100	Diversity = 4.3250e+00	k_t
= [2, 3, 2, 3]				
Generation 870:	Best = 2.983687e+03	FES = 87100	Diversity = 4.2937e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.983687e+03	FES = 88100	Diversity = 4.3003e+00	k_t
= [4, 4, 3, 2]				
Generation 890:	Best = 2.976276e+03	FES = 89100	Diversity = 4.2777e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.972082e+03	FES = 90100	Diversity = 4.2303e+00	k_t
= [3, 4, 4, 3]				
Generation 910:	Best = 2.969876e+03	FES = 91100	Diversity = 4.1443e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.968529e+03	FES = 92100	Diversity = 4.1075e+00	k_t
= [2, 4, 2, 3]				
Generation 930:	Best = 2.968529e+03	FES = 93100	Diversity = 4.0870e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.968529e+03	FES = 94100	Diversity = 4.0303e+00	k_t
= [2, 4, 4, 4]				
Generation 950:	Best = 2.967948e+03	FES = 95100	Diversity = 4.0390e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.967948e+03	FES = 96100	Diversity = 4.0091e+00	k_t
= [3, 3, 4, 4]				
Generation 970:	Best = 2.964216e+03	FES = 97100	Diversity = 3.9564e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.964032e+03	FES = 98100	Diversity = 3.8335e+00	k_t
= [2, 4, 3, 2]				
Generation 990:	Best = 2.962069e+03	FES = 99100	Diversity = 3.8329e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 2.951962e+03	FES = 100100	Diversity = 3.8145e+00	k_t
= [4, 4, 2, 3]				
Generation 1010:	Best = 2.951776e+03	FES = 101100	Diversity = 3.8252e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 2.934450e+03	FES = 102100	Diversity = 3.7745e+00	k_t
= [4, 2, 3, 4]				
Generation 1030:	Best = 2.933983e+03	FES = 103100	Diversity = 3.7820e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 2.925316e+03 = [2, 2, 2, 2]	FES = 104100	Diversity = 3.7606e+00	k_t
Generation 1050: Best = 2.920369e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 3.7394e+00	k_t
Generation 1060: Best = 2.905682e+03 = [4, 3, 3, 2]	FES = 106100	Diversity = 3.8754e+00	k_t
Generation 1070: Best = 2.889560e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 3.7367e+00	k_t
Generation 1080: Best = 2.875011e+03 = [2, 2, 3, 3]	FES = 108100	Diversity = 3.3132e+00	k_t
Generation 1090: Best = 2.864227e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 2.8454e+00	k_t
Generation 1100: Best = 2.849129e+03 = [3, 3, 3, 4]	FES = 110100	Diversity = 2.2246e+00	k_t
Generation 1110: Best = 2.839729e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.8107e+00	k_t
Generation 1120: Best = 2.829643e+03 = [3, 3, 2, 2]	FES = 112100	Diversity = 1.6205e+00	k_t
Generation 1130: Best = 2.824083e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.4258e+00	k_t
Generation 1140: Best = 2.817930e+03 = [2, 2, 3, 2]	FES = 114100	Diversity = 1.2284e+00	k_t
Generation 1150: Best = 2.810959e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.1838e+00	k_t
Generation 1160: Best = 2.805229e+03 = [4, 2, 3, 4]	FES = 116100	Diversity = 1.0787e+00	k_t
Generation 1170: Best = 2.800813e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.0461e+00	k_t
Generation 1180: Best = 2.798236e+03 = [2, 2, 4, 3]	FES = 118100	Diversity = 1.0032e+00	k_t
Generation 1190: Best = 2.795062e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 9.4495e-01	k_t
Generation 1200: Best = 2.792230e+03 = [3, 3, 4, 3]	FES = 120100	Diversity = 9.0393e-01	k_t
Generation 1210: Best = 2.790609e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 8.8236e-01	k_t
Generation 1220: Best = 2.789017e+03 = [2, 4, 4, 2]	FES = 122100	Diversity = 8.9127e-01	k_t
Generation 1230: Best = 2.787866e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 8.3704e-01	k_t
Generation 1240: Best = 2.786164e+03 = [4, 3, 4, 4]	FES = 124100	Diversity = 8.0869e-01	k_t
Generation 1250: Best = 2.785244e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 7.8837e-01	k_t
Generation 1260: Best = 2.784497e+03 = [3, 3, 3, 3]	FES = 126100	Diversity = 7.6041e-01	k_t
Generation 1270: Best = 2.783472e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 7.5861e-01	k_t
Generation 1280: Best = 2.782457e+03 = [4, 2, 2, 4]	FES = 128100	Diversity = 7.2987e-01	k_t
Generation 1290: Best = 2.782123e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 7.1544e-01	k_t
Generation 1300: Best = 2.781331e+03 = [4, 4, 4, 3]	FES = 130100	Diversity = 6.8836e-01	k_t

Generation 1310: Best = 2.781122e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 6.7290e-01	k_t
Generation 1320: Best = 2.780477e+03 = [3, 3, 4, 4]	FES = 132100	Diversity = 6.4818e-01	k_t
Generation 1330: Best = 2.780066e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 6.3640e-01	k_t
Generation 1340: Best = 2.779249e+03 = [4, 2, 4, 4]	FES = 134100	Diversity = 6.2627e-01	k_t
Generation 1350: Best = 2.779249e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 6.1206e-01	k_t
Generation 1360: Best = 2.778669e+03 = [4, 4, 3, 3]	FES = 136100	Diversity = 5.9647e-01	k_t
Generation 1370: Best = 2.778239e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 5.9266e-01	k_t
Generation 1380: Best = 2.777533e+03 = [4, 4, 4, 3]	FES = 138100	Diversity = 6.0937e-01	k_t
Generation 1390: Best = 2.777371e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 6.2765e-01	k_t
Generation 1400: Best = 2.777119e+03 = [3, 4, 2, 2]	FES = 140100	Diversity = 6.3168e-01	k_t
Generation 1410: Best = 2.776654e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 6.1992e-01	k_t
Generation 1420: Best = 2.776447e+03 = [4, 3, 2, 2]	FES = 142100	Diversity = 6.2601e-01	k_t
Generation 1430: Best = 2.775904e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 6.2841e-01	k_t
Generation 1440: Best = 2.775832e+03 = [2, 2, 4, 4]	FES = 144100	Diversity = 6.3239e-01	k_t
Generation 1450: Best = 2.775613e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 6.3856e-01	k_t
Generation 1460: Best = 2.775047e+03 = [4, 2, 2, 3]	FES = 146100	Diversity = 6.4103e-01	k_t
Generation 1470: Best = 2.774891e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 6.4163e-01	k_t
Generation 1480: Best = 2.774756e+03 = [2, 3, 4, 2]	FES = 148100	Diversity = 6.4261e-01	k_t
Generation 1490: Best = 2.774670e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 6.3429e-01	k_t
Generation 1500: Best = 2.774520e+03 = [4, 4, 4, 3]	FES = 150100	Diversity = 6.3841e-01	k_t
Generation 1510: Best = 2.774229e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 6.3455e-01	k_t
Generation 1520: Best = 2.774196e+03 = [2, 2, 3, 4]	FES = 152100	Diversity = 6.1877e-01	k_t
Generation 1530: Best = 2.773934e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 6.0139e-01	k_t
Generation 1540: Best = 2.773781e+03 = [2, 2, 2, 4]	FES = 154100	Diversity = 6.1678e-01	k_t
Generation 1550: Best = 2.773550e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 6.0838e-01	k_t
Generation 1560: Best = 2.773455e+03 = [4, 3, 4, 2]	FES = 156100	Diversity = 6.0690e-01	k_t
Generation 1570: Best = 2.773054e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 6.0643e-01	k_t

Generation 1580: Best = 2.772887e+03 = [4, 3, 4, 2]	FES = 158100	Diversity = 6.1884e-01	k_t
Generation 1590: Best = 2.772703e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 6.3567e-01	k_t
Generation 1600: Best = 2.772655e+03 = [2, 4, 2, 3]	FES = 160100	Diversity = 6.4574e-01	k_t
Generation 1610: Best = 2.772530e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 6.4677e-01	k_t
Generation 1620: Best = 2.772484e+03 = [2, 4, 3, 2]	FES = 162100	Diversity = 6.4184e-01	k_t
Generation 1630: Best = 2.772266e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 6.4800e-01	k_t
Generation 1640: Best = 2.772266e+03 = [3, 4, 3, 2]	FES = 164100	Diversity = 6.4049e-01	k_t
Generation 1650: Best = 2.772063e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 6.4895e-01	k_t
Generation 1660: Best = 2.771991e+03 = [2, 3, 4, 4]	FES = 166100	Diversity = 6.4770e-01	k_t
Generation 1670: Best = 2.771776e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 6.4017e-01	k_t
Generation 1680: Best = 2.771686e+03 = [4, 3, 3, 4]	FES = 168100	Diversity = 6.4365e-01	k_t
Generation 1690: Best = 2.771671e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 6.4290e-01	k_t
Generation 1700: Best = 2.771469e+03 = [4, 3, 4, 2]	FES = 170100	Diversity = 6.4358e-01	k_t
Generation 1710: Best = 2.771010e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 6.4491e-01	k_t
Generation 1720: Best = 2.770988e+03 = [4, 4, 3, 3]	FES = 172100	Diversity = 6.4060e-01	k_t
Generation 1730: Best = 2.770988e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 6.3606e-01	k_t
Generation 1740: Best = 2.770871e+03 = [3, 4, 4, 4]	FES = 174100	Diversity = 6.3008e-01	k_t
Generation 1750: Best = 2.770869e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 6.1300e-01	k_t
Generation 1760: Best = 2.770649e+03 = [2, 2, 3, 3]	FES = 176100	Diversity = 6.0929e-01	k_t
Generation 1770: Best = 2.770516e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 5.9740e-01	k_t
Generation 1780: Best = 2.770466e+03 = [2, 4, 3, 2]	FES = 178100	Diversity = 6.1056e-01	k_t
Generation 1790: Best = 2.770279e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 6.1173e-01	k_t
Generation 1800: Best = 2.770259e+03 = [3, 2, 2, 2]	FES = 180100	Diversity = 6.0791e-01	k_t
Generation 1810: Best = 2.770230e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 6.1207e-01	k_t
Generation 1820: Best = 2.770192e+03 = [4, 4, 2, 2]	FES = 182100	Diversity = 6.1722e-01	k_t
Generation 1830: Best = 2.770086e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 6.0902e-01	k_t
Generation 1840: Best = 2.769826e+03 = [4, 3, 3, 2]	FES = 184100	Diversity = 6.1366e-01	k_t

Generation 1850: Best = 2.769826e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 6.1327e-01	k_t
Generation 1860: Best = 2.769826e+03 = [3, 2, 3, 3]	FES = 186100	Diversity = 6.1143e-01	k_t
Generation 1870: Best = 2.769750e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 6.1389e-01	k_t
Generation 1880: Best = 2.769665e+03 = [2, 3, 4, 3]	FES = 188100	Diversity = 6.1452e-01	k_t
Generation 1890: Best = 2.769665e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 6.2123e-01	k_t
Generation 1900: Best = 2.769665e+03 = [2, 2, 4, 3]	FES = 190100	Diversity = 6.2056e-01	k_t
Generation 1910: Best = 2.769653e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 6.2236e-01	k_t
Generation 1920: Best = 2.769505e+03 = [3, 2, 2, 2]	FES = 192100	Diversity = 6.2031e-01	k_t
Generation 1930: Best = 2.769488e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 6.2521e-01	k_t
Generation 1940: Best = 2.769365e+03 = [2, 4, 2, 3]	FES = 194100	Diversity = 6.2118e-01	k_t
Generation 1950: Best = 2.769358e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 6.2602e-01	k_t
Generation 1960: Best = 2.769358e+03 = [4, 4, 3, 3]	FES = 196100	Diversity = 6.3189e-01	k_t
Generation 1970: Best = 2.769358e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 6.3541e-01	k_t
Generation 1980: Best = 2.769341e+03 = [3, 4, 4, 4]	FES = 198100	Diversity = 6.3821e-01	k_t
Generation 1990: Best = 2.769279e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 6.3427e-01	k_t
Generation 2000: Best = 2.769209e+03 = [2, 4, 3, 2]	FES = 200100	Diversity = 6.3067e-01	k_t
Generation 2010: Best = 2.769208e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 6.3193e-01	k_t
Generation 2020: Best = 2.769197e+03 = [4, 4, 2, 2]	FES = 202100	Diversity = 6.3718e-01	k_t
Generation 2030: Best = 2.769071e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 6.3994e-01	k_t
Generation 2040: Best = 2.769062e+03 = [2, 3, 4, 4]	FES = 204100	Diversity = 6.3918e-01	k_t
Generation 2050: Best = 2.768982e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 6.4548e-01	k_t
Generation 2060: Best = 2.768982e+03 = [3, 3, 4, 2]	FES = 206100	Diversity = 6.4664e-01	k_t
Generation 2070: Best = 2.768982e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 6.4061e-01	k_t
Generation 2080: Best = 2.768982e+03 = [4, 4, 4, 3]	FES = 208100	Diversity = 6.4075e-01	k_t
Generation 2090: Best = 2.768913e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 6.4187e-01	k_t
Generation 2100: Best = 2.768883e+03 = [4, 4, 4, 2]	FES = 210100	Diversity = 6.4289e-01	k_t
Generation 2110: Best = 2.768838e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 6.4475e-01	k_t

Generation 2120: Best = 2.768811e+03 = [4, 4, 4, 2]	FES = 212100	Diversity = 6.4179e-01	k_t
Generation 2130: Best = 2.768742e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 6.4138e-01	k_t
Generation 2140: Best = 2.768729e+03 = [3, 2, 3, 3]	FES = 214100	Diversity = 6.3862e-01	k_t
Generation 2150: Best = 2.768630e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 6.4102e-01	k_t
Generation 2160: Best = 2.768629e+03 = [3, 3, 3, 4]	FES = 216100	Diversity = 6.4312e-01	k_t
Generation 2170: Best = 2.768589e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 6.4521e-01	k_t
Generation 2180: Best = 2.768482e+03 = [3, 4, 4, 4]	FES = 218100	Diversity = 6.4492e-01	k_t
Generation 2190: Best = 2.768386e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 6.4409e-01	k_t
Generation 2200: Best = 2.768386e+03 = [2, 2, 4, 2]	FES = 220100	Diversity = 6.4040e-01	k_t
Generation 2210: Best = 2.768368e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 6.4096e-01	k_t
Generation 2220: Best = 2.768364e+03 = [3, 2, 4, 4]	FES = 222100	Diversity = 6.4358e-01	k_t
Generation 2230: Best = 2.768364e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 6.4533e-01	k_t
Generation 2240: Best = 2.768360e+03 = [4, 3, 2, 4]	FES = 224100	Diversity = 6.4562e-01	k_t
Generation 2250: Best = 2.768357e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 6.4665e-01	k_t
Generation 2260: Best = 2.768335e+03 = [2, 4, 3, 2]	FES = 226100	Diversity = 6.5393e-01	k_t
Generation 2270: Best = 2.768272e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 6.5426e-01	k_t
Generation 2280: Best = 2.768234e+03 = [3, 2, 2, 2]	FES = 228100	Diversity = 6.5309e-01	k_t
Generation 2290: Best = 2.768205e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 6.5262e-01	k_t
Generation 2300: Best = 2.768168e+03 = [4, 4, 2, 4]	FES = 230100	Diversity = 6.5438e-01	k_t
Generation 2310: Best = 2.768168e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 6.5021e-01	k_t
Generation 2320: Best = 2.768158e+03 = [3, 4, 2, 2]	FES = 232100	Diversity = 6.5369e-01	k_t
Generation 2330: Best = 2.768151e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 6.5586e-01	k_t
Generation 2340: Best = 2.768104e+03 = [2, 2, 2, 4]	FES = 234100	Diversity = 6.5784e-01	k_t
Generation 2350: Best = 2.768104e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 6.6055e-01	k_t
Generation 2360: Best = 2.768094e+03 = [2, 2, 4, 4]	FES = 236100	Diversity = 6.6243e-01	k_t
Generation 2370: Best = 2.767924e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 6.6452e-01	k_t
Generation 2380: Best = 2.767879e+03 = [3, 2, 4, 3]	FES = 238100	Diversity = 6.6389e-01	k_t

Generation 2390: Best = 2.767879e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 6.6760e-01	k_t
Generation 2400: Best = 2.767852e+03 = [2, 3, 3, 2]	FES = 240100	Diversity = 6.6918e-01	k_t
Generation 2410: Best = 2.767852e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 6.7206e-01	k_t
Generation 2420: Best = 2.767844e+03 = [3, 3, 4, 3]	FES = 242100	Diversity = 6.7222e-01	k_t
Generation 2430: Best = 2.767830e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 6.7645e-01	k_t
Generation 2440: Best = 2.767830e+03 = [4, 2, 3, 4]	FES = 244100	Diversity = 6.7957e-01	k_t
Generation 2450: Best = 2.767824e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 6.8179e-01	k_t
Generation 2460: Best = 2.767777e+03 = [4, 3, 2, 2]	FES = 246100	Diversity = 6.7981e-01	k_t
Generation 2470: Best = 2.767768e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 6.7683e-01	k_t
Generation 2480: Best = 2.767716e+03 = [4, 4, 2, 3]	FES = 248100	Diversity = 6.7747e-01	k_t
Generation 2490: Best = 2.767711e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 6.7500e-01	k_t
Generation 2500: Best = 2.767640e+03 = [3, 4, 3, 3]	FES = 250100	Diversity = 6.7540e-01	k_t
Generation 2510: Best = 2.767637e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 6.8184e-01	k_t
Generation 2520: Best = 2.767637e+03 = [2, 4, 4, 3]	FES = 252100	Diversity = 6.8329e-01	k_t
Generation 2530: Best = 2.767600e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 6.8654e-01	k_t
Generation 2540: Best = 2.767490e+03 = [2, 4, 4, 4]	FES = 254100	Diversity = 6.8477e-01	k_t
Generation 2550: Best = 2.767490e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 6.8657e-01	k_t
Generation 2560: Best = 2.767486e+03 = [2, 2, 3, 2]	FES = 256100	Diversity = 6.8101e-01	k_t
Generation 2570: Best = 2.767474e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 6.7913e-01	k_t
Generation 2580: Best = 2.767474e+03 = [4, 2, 2, 2]	FES = 258100	Diversity = 6.8314e-01	k_t
Generation 2590: Best = 2.767474e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 6.8743e-01	k_t
Generation 2600: Best = 2.767474e+03 = [4, 2, 3, 2]	FES = 260100	Diversity = 6.8681e-01	k_t
Generation 2610: Best = 2.767470e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 6.8641e-01	k_t
Generation 2620: Best = 2.767470e+03 = [4, 2, 3, 3]	FES = 262100	Diversity = 6.8485e-01	k_t
Generation 2630: Best = 2.767429e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 6.8601e-01	k_t
Generation 2640: Best = 2.767429e+03 = [4, 3, 3, 4]	FES = 264100	Diversity = 6.8459e-01	k_t
Generation 2650: Best = 2.767426e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 6.8535e-01	k_t

Generation 2660: Best = 2.767423e+03 = [3, 4, 4, 3]	FES = 266100	Diversity = 6.8723e-01	k_t
Generation 2670: Best = 2.767410e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 6.8817e-01	k_t
Generation 2680: Best = 2.767410e+03 = [3, 3, 3, 4]	FES = 268100	Diversity = 6.8652e-01	k_t
Generation 2690: Best = 2.767410e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 6.8544e-01	k_t
Generation 2700: Best = 2.767408e+03 = [3, 2, 2, 4]	FES = 270100	Diversity = 6.8385e-01	k_t
Generation 2710: Best = 2.767391e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 6.8461e-01	k_t
Generation 2720: Best = 2.767383e+03 = [4, 3, 4, 2]	FES = 272100	Diversity = 6.8149e-01	k_t
Generation 2730: Best = 2.767371e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 6.8240e-01	k_t
Generation 2740: Best = 2.767371e+03 = [3, 3, 3, 2]	FES = 274100	Diversity = 6.8141e-01	k_t
Generation 2750: Best = 2.767368e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 6.8236e-01	k_t
Generation 2760: Best = 2.767333e+03 = [3, 4, 2, 2]	FES = 276100	Diversity = 6.8130e-01	k_t
Generation 2770: Best = 2.767281e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 6.7803e-01	k_t
Generation 2780: Best = 2.767270e+03 = [3, 4, 2, 3]	FES = 278100	Diversity = 6.7896e-01	k_t
Generation 2790: Best = 2.767261e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 6.7897e-01	k_t
Generation 2800: Best = 2.767259e+03 = [2, 3, 2, 2]	FES = 280100	Diversity = 6.7929e-01	k_t
Generation 2810: Best = 2.767259e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 6.7777e-01	k_t
Generation 2820: Best = 2.767259e+03 = [2, 2, 3, 2]	FES = 282100	Diversity = 6.7556e-01	k_t
Generation 2830: Best = 2.767259e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 6.7427e-01	k_t
Generation 2840: Best = 2.767257e+03 = [4, 2, 2, 2]	FES = 284100	Diversity = 6.7361e-01	k_t
Generation 2850: Best = 2.767209e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 6.7530e-01	k_t
Generation 2860: Best = 2.767198e+03 = [3, 4, 4, 2]	FES = 286100	Diversity = 6.7611e-01	k_t
Generation 2870: Best = 2.767198e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 6.7840e-01	k_t
Generation 2880: Best = 2.767146e+03 = [4, 4, 4, 2]	FES = 288100	Diversity = 6.7836e-01	k_t
Generation 2890: Best = 2.767146e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 6.7869e-01	k_t
Generation 2900: Best = 2.767145e+03 = [2, 2, 2, 4]	FES = 290100	Diversity = 6.7903e-01	k_t
Generation 2910: Best = 2.767097e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 6.7677e-01	k_t
Generation 2920: Best = 2.767090e+03 = [2, 4, 3, 4]	FES = 292100	Diversity = 6.7657e-01	k_t

Generation 2930: Best = 2.767086e+03 = [1, 1, 1, 1]	FES = 293100	Diversity = 6.7549e-01	k_t
Generation 2940: Best = 2.767069e+03 = [3, 2, 2, 4]	FES = 294100	Diversity = 6.7503e-01	k_t
Generation 2950: Best = 2.767069e+03 = [1, 1, 1, 1]	FES = 295100	Diversity = 6.7032e-01	k_t
Generation 2960: Best = 2.767062e+03 = [2, 2, 2, 2]	FES = 296100	Diversity = 6.6886e-01	k_t
Generation 2970: Best = 2.766997e+03 = [1, 1, 1, 1]	FES = 297100	Diversity = 6.6745e-01	k_t
Generation 2980: Best = 2.766997e+03 = [3, 3, 4, 4]	FES = 298100	Diversity = 6.6613e-01	k_t
Generation 2990: Best = 2.766997e+03 = [1, 1, 1, 1]	FES = 299100	Diversity = 6.6845e-01	k_t
Generation 3000: Best = 2.766994e+03 = [2, 2, 4, 4]	FES = 300100	Diversity = 6.6685e-01	k_t
Generation 3010: Best = 2.766994e+03 = [1, 1, 1, 1]	FES = 301100	Diversity = 6.6794e-01	k_t
Generation 3020: Best = 2.766988e+03 = [4, 3, 4, 2]	FES = 302100	Diversity = 6.6657e-01	k_t
Generation 3030: Best = 2.766988e+03 = [1, 1, 1, 1]	FES = 303100	Diversity = 6.6286e-01	k_t
Generation 3040: Best = 2.766966e+03 = [4, 4, 3, 4]	FES = 304100	Diversity = 6.6144e-01	k_t
Generation 3050: Best = 2.766950e+03 = [1, 1, 1, 1]	FES = 305100	Diversity = 6.6067e-01	k_t
Generation 3060: Best = 2.766950e+03 = [4, 3, 4, 2]	FES = 306100	Diversity = 6.6124e-01	k_t
Generation 3070: Best = 2.766921e+03 = [1, 1, 1, 1]	FES = 307100	Diversity = 6.6314e-01	k_t
Generation 3080: Best = 2.766921e+03 = [3, 2, 4, 2]	FES = 308100	Diversity = 6.6473e-01	k_t
Generation 3090: Best = 2.766917e+03 = [1, 1, 1, 1]	FES = 309100	Diversity = 6.6222e-01	k_t
Generation 3100: Best = 2.766906e+03 = [4, 2, 3, 3]	FES = 310100	Diversity = 6.6261e-01	k_t
Generation 3110: Best = 2.766906e+03 = [1, 1, 1, 1]	FES = 311100	Diversity = 6.5983e-01	k_t
Generation 3120: Best = 2.766896e+03 = [3, 4, 3, 4]	FES = 312100	Diversity = 6.5589e-01	k_t
Generation 3130: Best = 2.766896e+03 = [1, 1, 1, 1]	FES = 313100	Diversity = 6.5552e-01	k_t
Generation 3140: Best = 2.766893e+03 = [3, 4, 2, 4]	FES = 314100	Diversity = 6.5558e-01	k_t
Generation 3150: Best = 2.766846e+03 = [1, 1, 1, 1]	FES = 315100	Diversity = 6.5199e-01	k_t
Generation 3160: Best = 2.766845e+03 = [3, 2, 4, 3]	FES = 316100	Diversity = 6.5130e-01	k_t
Generation 3170: Best = 2.766845e+03 = [1, 1, 1, 1]	FES = 317100	Diversity = 6.5179e-01	k_t
Generation 3180: Best = 2.766812e+03 = [3, 4, 3, 2]	FES = 318100	Diversity = 6.5059e-01	k_t
Generation 3190: Best = 2.766796e+03 = [1, 1, 1, 1]	FES = 319100	Diversity = 6.5169e-01	k_t

Generation 3200: Best = 2.766796e+03 = [4, 2, 4, 4]	FES = 320100	Diversity = 6.4862e-01	k_t
Generation 3210: Best = 2.766790e+03 = [1, 1, 1, 1]	FES = 321100	Diversity = 6.4689e-01	k_t
Generation 3220: Best = 2.766755e+03 = [4, 4, 4, 4]	FES = 322100	Diversity = 6.4612e-01	k_t
Generation 3230: Best = 2.766750e+03 = [1, 1, 1, 1]	FES = 323100	Diversity = 6.4591e-01	k_t
Generation 3240: Best = 2.766750e+03 = [2, 2, 3, 3]	FES = 324100	Diversity = 6.4617e-01	k_t
Generation 3250: Best = 2.766750e+03 = [1, 1, 1, 1]	FES = 325100	Diversity = 6.4721e-01	k_t
Generation 3260: Best = 2.766750e+03 = [3, 2, 3, 4]	FES = 326100	Diversity = 6.4666e-01	k_t
Generation 3270: Best = 2.766749e+03 = [1, 1, 1, 1]	FES = 327100	Diversity = 6.4740e-01	k_t
Generation 3280: Best = 2.766719e+03 = [2, 2, 4, 2]	FES = 328100	Diversity = 6.4866e-01	k_t
Generation 3290: Best = 2.766696e+03 = [1, 1, 1, 1]	FES = 329100	Diversity = 6.4432e-01	k_t
Generation 3300: Best = 2.766658e+03 = [4, 3, 4, 2]	FES = 330100	Diversity = 6.4484e-01	k_t
Generation 3310: Best = 2.766651e+03 = [1, 1, 1, 1]	FES = 331100	Diversity = 6.4388e-01	k_t
Generation 3320: Best = 2.766651e+03 = [2, 4, 4, 4]	FES = 332100	Diversity = 6.4081e-01	k_t
Generation 3330: Best = 2.766603e+03 = [1, 1, 1, 1]	FES = 333100	Diversity = 6.4216e-01	k_t
Generation 3340: Best = 2.766602e+03 = [2, 4, 3, 3]	FES = 334100	Diversity = 6.3852e-01	k_t
Generation 3350: Best = 2.766589e+03 = [1, 1, 1, 1]	FES = 335100	Diversity = 6.3922e-01	k_t
Generation 3360: Best = 2.766547e+03 = [2, 4, 2, 3]	FES = 336100	Diversity = 6.3926e-01	k_t
Generation 3370: Best = 2.766544e+03 = [1, 1, 1, 1]	FES = 337100	Diversity = 6.3618e-01	k_t
Generation 3380: Best = 2.766543e+03 = [4, 2, 2, 4]	FES = 338100	Diversity = 6.3268e-01	k_t
Generation 3390: Best = 2.766543e+03 = [1, 1, 1, 1]	FES = 339100	Diversity = 6.3565e-01	k_t
Generation 3400: Best = 2.766541e+03 = [3, 3, 4, 4]	FES = 340100	Diversity = 6.3131e-01	k_t
Generation 3410: Best = 2.766510e+03 = [1, 1, 1, 1]	FES = 341100	Diversity = 6.2801e-01	k_t
Generation 3420: Best = 2.766496e+03 = [2, 3, 2, 4]	FES = 342100	Diversity = 6.2495e-01	k_t
Generation 3430: Best = 2.766496e+03 = [1, 1, 1, 1]	FES = 343100	Diversity = 6.2294e-01	k_t
Generation 3440: Best = 2.766496e+03 = [4, 4, 2, 2]	FES = 344100	Diversity = 6.2160e-01	k_t
Generation 3450: Best = 2.766496e+03 = [1, 1, 1, 1]	FES = 345100	Diversity = 6.2050e-01	k_t
Generation 3460: Best = 2.766495e+03 = [3, 4, 3, 4]	FES = 346100	Diversity = 6.1946e-01	k_t

Generation 3470: Best = 2.766494e+03 = [1, 1, 1, 1]	FES = 347100	Diversity = 6.2585e-01	k_t
Generation 3480: Best = 2.766494e+03 = [4, 2, 4, 4]	FES = 348100	Diversity = 6.2602e-01	k_t
Generation 3490: Best = 2.766491e+03 = [1, 1, 1, 1]	FES = 349100	Diversity = 6.2434e-01	k_t
Generation 3500: Best = 2.766490e+03 = [4, 3, 4, 2]	FES = 350100	Diversity = 6.2803e-01	k_t
Generation 3510: Best = 2.766474e+03 = [1, 1, 1, 1]	FES = 351100	Diversity = 6.3227e-01	k_t
Generation 3520: Best = 2.766456e+03 = [3, 4, 3, 3]	FES = 352100	Diversity = 6.3745e-01	k_t
Generation 3530: Best = 2.766439e+03 = [1, 1, 1, 1]	FES = 353100	Diversity = 6.3865e-01	k_t
Generation 3540: Best = 2.766413e+03 = [2, 3, 2, 2]	FES = 354100	Diversity = 6.3399e-01	k_t
Generation 3550: Best = 2.766407e+03 = [1, 1, 1, 1]	FES = 355100	Diversity = 6.3570e-01	k_t
Generation 3560: Best = 2.766372e+03 = [2, 4, 2, 4]	FES = 356100	Diversity = 6.3829e-01	k_t
Generation 3570: Best = 2.766356e+03 = [1, 1, 1, 1]	FES = 357100	Diversity = 6.3611e-01	k_t
Generation 3580: Best = 2.766356e+03 = [4, 4, 3, 4]	FES = 358100	Diversity = 6.3762e-01	k_t
Generation 3590: Best = 2.766354e+03 = [1, 1, 1, 1]	FES = 359100	Diversity = 6.3620e-01	k_t
Generation 3600: Best = 2.766354e+03 = [4, 4, 3, 3]	FES = 360100	Diversity = 6.3450e-01	k_t
Generation 3610: Best = 2.766342e+03 = [1, 1, 1, 1]	FES = 361100	Diversity = 6.3583e-01	k_t
Generation 3620: Best = 2.766342e+03 = [2, 4, 3, 4]	FES = 362100	Diversity = 6.3138e-01	k_t
Generation 3630: Best = 2.766336e+03 = [1, 1, 1, 1]	FES = 363100	Diversity = 6.2395e-01	k_t
Generation 3640: Best = 2.766319e+03 = [4, 2, 3, 2]	FES = 364100	Diversity = 6.2241e-01	k_t
Generation 3650: Best = 2.766319e+03 = [1, 1, 1, 1]	FES = 365100	Diversity = 6.2327e-01	k_t
Generation 3660: Best = 2.766305e+03 = [4, 2, 4, 4]	FES = 366100	Diversity = 6.1781e-01	k_t
Generation 3670: Best = 2.766289e+03 = [1, 1, 1, 1]	FES = 367100	Diversity = 6.1437e-01	k_t
Generation 3680: Best = 2.766289e+03 = [2, 3, 3, 2]	FES = 368100	Diversity = 6.1866e-01	k_t
Generation 3690: Best = 2.766267e+03 = [1, 1, 1, 1]	FES = 369100	Diversity = 6.1559e-01	k_t
Generation 3700: Best = 2.766255e+03 = [2, 4, 3, 2]	FES = 370100	Diversity = 6.1810e-01	k_t
Generation 3710: Best = 2.766250e+03 = [1, 1, 1, 1]	FES = 371100	Diversity = 6.1820e-01	k_t
Generation 3720: Best = 2.766234e+03 = [3, 2, 3, 3]	FES = 372100	Diversity = 6.1066e-01	k_t
Generation 3730: Best = 2.766225e+03 = [1, 1, 1, 1]	FES = 373100	Diversity = 6.0881e-01	k_t

Generation 3740: Best = 2.766190e+03 = [2, 4, 3, 2]	FES = 374100	Diversity = 6.0563e-01	k_t
Generation 3750: Best = 2.766189e+03 = [1, 1, 1, 1]	FES = 375100	Diversity = 6.0712e-01	k_t
Generation 3760: Best = 2.766183e+03 = [4, 4, 4, 4]	FES = 376100	Diversity = 6.0977e-01	k_t
Generation 3770: Best = 2.766175e+03 = [1, 1, 1, 1]	FES = 377100	Diversity = 6.1001e-01	k_t
Generation 3780: Best = 2.766150e+03 = [4, 3, 3, 4]	FES = 378100	Diversity = 6.0960e-01	k_t
Generation 3790: Best = 2.766141e+03 = [1, 1, 1, 1]	FES = 379100	Diversity = 6.0966e-01	k_t
Generation 3800: Best = 2.766128e+03 = [4, 2, 3, 4]	FES = 380100	Diversity = 6.0093e-01	k_t
Generation 3810: Best = 2.766082e+03 = [1, 1, 1, 1]	FES = 381100	Diversity = 5.9587e-01	k_t
Generation 3820: Best = 2.766080e+03 = [2, 3, 4, 3]	FES = 382100	Diversity = 5.8942e-01	k_t
Generation 3830: Best = 2.766046e+03 = [1, 1, 1, 1]	FES = 383100	Diversity = 5.7699e-01	k_t
Generation 3840: Best = 2.766042e+03 = [3, 2, 4, 4]	FES = 384100	Diversity = 5.7052e-01	k_t
Generation 3850: Best = 2.765982e+03 = [1, 1, 1, 1]	FES = 385100	Diversity = 5.5540e-01	k_t
Generation 3860: Best = 2.765931e+03 = [3, 4, 4, 3]	FES = 386100	Diversity = 5.5304e-01	k_t
Generation 3870: Best = 2.765931e+03 = [1, 1, 1, 1]	FES = 387100	Diversity = 5.4841e-01	k_t
Generation 3880: Best = 2.765899e+03 = [3, 4, 3, 3]	FES = 388100	Diversity = 5.3338e-01	k_t
Generation 3890: Best = 2.765879e+03 = [1, 1, 1, 1]	FES = 389100	Diversity = 5.1721e-01	k_t
Generation 3900: Best = 2.765839e+03 = [3, 4, 3, 2]	FES = 390100	Diversity = 4.5252e-01	k_t
Generation 3910: Best = 2.765794e+03 = [1, 1, 1, 1]	FES = 391100	Diversity = 3.9784e-01	k_t
Generation 3920: Best = 2.765721e+03 = [3, 3, 4, 2]	FES = 392100	Diversity = 3.3071e-01	k_t
Generation 3930: Best = 2.765666e+03 = [1, 1, 1, 1]	FES = 393100	Diversity = 2.3049e-01	k_t
Generation 3940: Best = 2.765585e+03 = [4, 2, 3, 3]	FES = 394100	Diversity = 1.6659e-01	k_t
Generation 3950: Best = 2.765521e+03 = [1, 1, 1, 1]	FES = 395100	Diversity = 1.3593e-01	k_t
Generation 3960: Best = 2.765438e+03 = [4, 2, 2, 3]	FES = 396100	Diversity = 6.4541e-02	k_t
Generation 3970: Best = 2.765348e+03 = [1, 1, 1, 1]	FES = 397100	Diversity = 5.8134e-02	k_t
Generation 3980: Best = 2.765300e+03 = [3, 2, 2, 3]	FES = 398100	Diversity = 5.3571e-02	k_t
Generation 3990: Best = 2.765212e+03 = [1, 1, 1, 1]	FES = 399100	Diversity = 5.0559e-02	k_t
Generation 4000: Best = 2.765149e+03 = [3, 3, 2, 4]	FES = 400100	Diversity = 4.4163e-02	k_t

Generation 4010: Best = 2.765094e+03 = [1, 1, 1, 1]	FES = 401100	Diversity = 3.9243e-02	k_t
Generation 4020: Best = 2.765049e+03 = [4, 4, 3, 4]	FES = 402100	Diversity = 3.5146e-02	k_t
Generation 4030: Best = 2.765013e+03 = [1, 1, 1, 1]	FES = 403100	Diversity = 3.3751e-02	k_t
Generation 4040: Best = 2.764969e+03 = [3, 2, 3, 2]	FES = 404100	Diversity = 3.3005e-02	k_t
Generation 4050: Best = 2.764926e+03 = [1, 1, 1, 1]	FES = 405100	Diversity = 3.0382e-02	k_t
Generation 4060: Best = 2.764891e+03 = [4, 4, 4, 4]	FES = 406100	Diversity = 2.9507e-02	k_t
Generation 4070: Best = 2.764861e+03 = [1, 1, 1, 1]	FES = 407100	Diversity = 2.7261e-02	k_t
Generation 4080: Best = 2.764842e+03 = [4, 3, 3, 3]	FES = 408100	Diversity = 2.5257e-02	k_t
Generation 4090: Best = 2.764826e+03 = [1, 1, 1, 1]	FES = 409100	Diversity = 2.3118e-02	k_t
Generation 4100: Best = 2.764807e+03 = [3, 3, 2, 3]	FES = 410100	Diversity = 2.2456e-02	k_t
Generation 4110: Best = 2.764785e+03 = [1, 1, 1, 1]	FES = 411100	Diversity = 2.2503e-02	k_t
Generation 4120: Best = 2.764775e+03 = [2, 4, 4, 4]	FES = 412100	Diversity = 2.1694e-02	k_t
Generation 4130: Best = 2.764757e+03 = [1, 1, 1, 1]	FES = 413100	Diversity = 2.1131e-02	k_t
Generation 4140: Best = 2.764755e+03 = [3, 4, 4, 4]	FES = 414100	Diversity = 1.9191e-02	k_t
Generation 4150: Best = 2.764724e+03 = [1, 1, 1, 1]	FES = 415100	Diversity = 2.0406e-02	k_t
Generation 4160: Best = 2.764711e+03 = [2, 4, 4, 2]	FES = 416100	Diversity = 2.1956e-02	k_t
Generation 4170: Best = 2.764680e+03 = [1, 1, 1, 1]	FES = 417100	Diversity = 2.2837e-02	k_t
Generation 4180: Best = 2.764668e+03 = [3, 3, 4, 2]	FES = 418100	Diversity = 2.2719e-02	k_t
Generation 4190: Best = 2.764646e+03 = [1, 1, 1, 1]	FES = 419100	Diversity = 2.4917e-02	k_t
Generation 4200: Best = 2.764629e+03 = [4, 4, 3, 4]	FES = 420100	Diversity = 2.5030e-02	k_t
Generation 4210: Best = 2.764605e+03 = [1, 1, 1, 1]	FES = 421100	Diversity = 2.5413e-02	k_t
Generation 4220: Best = 2.764581e+03 = [3, 2, 4, 3]	FES = 422100	Diversity = 2.5130e-02	k_t
Generation 4230: Best = 2.764563e+03 = [1, 1, 1, 1]	FES = 423100	Diversity = 2.6779e-02	k_t
Generation 4240: Best = 2.764538e+03 = [2, 2, 4, 3]	FES = 424100	Diversity = 2.8356e-02	k_t
Generation 4250: Best = 2.764511e+03 = [1, 1, 1, 1]	FES = 425100	Diversity = 2.8739e-02	k_t
Generation 4260: Best = 2.764488e+03 = [2, 3, 3, 2]	FES = 426100	Diversity = 2.9505e-02	k_t
Generation 4270: Best = 2.764452e+03 = [1, 1, 1, 1]	FES = 427100	Diversity = 3.1243e-02	k_t

Generation 4280: Best = 2.764431e+03 = [3, 4, 3, 2]	FES = 428100	Diversity = 3.2917e-02	k_t
Generation 4290: Best = 2.764394e+03 = [1, 1, 1, 1]	FES = 429100	Diversity = 3.3708e-02	k_t
Generation 4300: Best = 2.764375e+03 = [3, 4, 4, 4]	FES = 430100	Diversity = 3.2556e-02	k_t
Generation 4310: Best = 2.764344e+03 = [1, 1, 1, 1]	FES = 431100	Diversity = 2.8448e-02	k_t
Generation 4320: Best = 2.764334e+03 = [2, 2, 4, 4]	FES = 432100	Diversity = 2.7059e-02	k_t
Generation 4330: Best = 2.764317e+03 = [1, 1, 1, 1]	FES = 433100	Diversity = 2.6025e-02	k_t
Generation 4340: Best = 2.764300e+03 = [3, 3, 3, 2]	FES = 434100	Diversity = 2.4889e-02	k_t
Generation 4350: Best = 2.764277e+03 = [1, 1, 1, 1]	FES = 435100	Diversity = 2.4666e-02	k_t
Generation 4360: Best = 2.764258e+03 = [2, 4, 3, 3]	FES = 436100	Diversity = 2.4161e-02	k_t
Generation 4370: Best = 2.764244e+03 = [1, 1, 1, 1]	FES = 437100	Diversity = 2.3420e-02	k_t
Generation 4380: Best = 2.764218e+03 = [2, 2, 2, 4]	FES = 438100	Diversity = 2.3457e-02	k_t
Generation 4390: Best = 2.764210e+03 = [1, 1, 1, 1]	FES = 439100	Diversity = 2.3351e-02	k_t
Generation 4400: Best = 2.764192e+03 = [3, 2, 2, 4]	FES = 440100	Diversity = 2.2986e-02	k_t
Generation 4410: Best = 2.764183e+03 = [1, 1, 1, 1]	FES = 441100	Diversity = 2.1978e-02	k_t
Generation 4420: Best = 2.764165e+03 = [3, 3, 3, 4]	FES = 442100	Diversity = 2.2710e-02	k_t
Generation 4430: Best = 2.764150e+03 = [1, 1, 1, 1]	FES = 443100	Diversity = 2.3136e-02	k_t
Generation 4440: Best = 2.764138e+03 = [2, 2, 2, 3]	FES = 444100	Diversity = 2.2445e-02	k_t
Generation 4450: Best = 2.764132e+03 = [1, 1, 1, 1]	FES = 445100	Diversity = 2.2805e-02	k_t
Generation 4460: Best = 2.764113e+03 = [3, 4, 2, 4]	FES = 446100	Diversity = 2.2320e-02	k_t
Generation 4470: Best = 2.764101e+03 = [1, 1, 1, 1]	FES = 447100	Diversity = 2.1343e-02	k_t
Generation 4480: Best = 2.764095e+03 = [4, 2, 4, 2]	FES = 448100	Diversity = 2.2173e-02	k_t
Generation 4490: Best = 2.764071e+03 = [1, 1, 1, 1]	FES = 449100	Diversity = 2.3507e-02	k_t
Generation 4500: Best = 2.764059e+03 = [2, 2, 2, 2]	FES = 450100	Diversity = 2.3385e-02	k_t
Generation 4510: Best = 2.764045e+03 = [1, 1, 1, 1]	FES = 451100	Diversity = 2.6240e-02	k_t
Generation 4520: Best = 2.764008e+03 = [3, 2, 3, 3]	FES = 452100	Diversity = 2.7811e-02	k_t
Generation 4530: Best = 2.763978e+03 = [1, 1, 1, 1]	FES = 453100	Diversity = 2.8752e-02	k_t
Generation 4540: Best = 2.763957e+03 = [4, 3, 4, 2]	FES = 454100	Diversity = 2.8277e-02	k_t

Generation 4550: Best = 2.763945e+03 = [1, 1, 1, 1]	FES = 455100	Diversity = 2.8017e-02	k_t
Generation 4560: Best = 2.763933e+03 = [2, 3, 3, 3]	FES = 456100	Diversity = 2.6118e-02	k_t
Generation 4570: Best = 2.763908e+03 = [1, 1, 1, 1]	FES = 457100	Diversity = 2.5812e-02	k_t
Generation 4580: Best = 2.763891e+03 = [2, 4, 2, 2]	FES = 458100	Diversity = 2.6192e-02	k_t
Generation 4590: Best = 2.763875e+03 = [1, 1, 1, 1]	FES = 459100	Diversity = 2.6171e-02	k_t
Generation 4600: Best = 2.763861e+03 = [2, 3, 3, 3]	FES = 460100	Diversity = 2.4827e-02	k_t
Generation 4610: Best = 2.763844e+03 = [1, 1, 1, 1]	FES = 461100	Diversity = 2.4763e-02	k_t
Generation 4620: Best = 2.763826e+03 = [4, 3, 4, 4]	FES = 462100	Diversity = 2.3670e-02	k_t
Generation 4630: Best = 2.763807e+03 = [1, 1, 1, 1]	FES = 463100	Diversity = 2.2841e-02	k_t
Generation 4640: Best = 2.763797e+03 = [3, 2, 3, 2]	FES = 464100	Diversity = 2.2874e-02	k_t
Generation 4650: Best = 2.763788e+03 = [1, 1, 1, 1]	FES = 465100	Diversity = 2.3044e-02	k_t
Generation 4660: Best = 2.763770e+03 = [2, 3, 4, 2]	FES = 466100	Diversity = 2.3262e-02	k_t
Generation 4670: Best = 2.763751e+03 = [1, 1, 1, 1]	FES = 467100	Diversity = 2.3906e-02	k_t
Generation 4680: Best = 2.763734e+03 = [3, 4, 3, 3]	FES = 468100	Diversity = 2.4164e-02	k_t
Generation 4690: Best = 2.763710e+03 = [1, 1, 1, 1]	FES = 469100	Diversity = 2.4334e-02	k_t
Generation 4700: Best = 2.763689e+03 = [2, 2, 4, 3]	FES = 470100	Diversity = 2.5507e-02	k_t
Generation 4710: Best = 2.763674e+03 = [1, 1, 1, 1]	FES = 471100	Diversity = 2.5345e-02	k_t
Generation 4720: Best = 2.763658e+03 = [4, 2, 4, 4]	FES = 472100	Diversity = 2.5169e-02	k_t
Generation 4730: Best = 2.763636e+03 = [1, 1, 1, 1]	FES = 473100	Diversity = 2.4604e-02	k_t
Generation 4740: Best = 2.763623e+03 = [4, 3, 3, 2]	FES = 474100	Diversity = 2.4336e-02	k_t
Generation 4750: Best = 2.763604e+03 = [1, 1, 1, 1]	FES = 475100	Diversity = 2.5686e-02	k_t
Generation 4760: Best = 2.763579e+03 = [3, 3, 2, 3]	FES = 476100	Diversity = 2.7708e-02	k_t
Generation 4770: Best = 2.763546e+03 = [1, 1, 1, 1]	FES = 477100	Diversity = 2.9891e-02	k_t
Generation 4780: Best = 2.763535e+03 = [2, 2, 2, 2]	FES = 478100	Diversity = 3.0493e-02	k_t
Generation 4790: Best = 2.763508e+03 = [1, 1, 1, 1]	FES = 479100	Diversity = 2.8818e-02	k_t
Generation 4800: Best = 2.763479e+03 = [4, 2, 2, 2]	FES = 480100	Diversity = 2.6616e-02	k_t
Generation 4810: Best = 2.763467e+03 = [1, 1, 1, 1]	FES = 481100	Diversity = 2.7162e-02	k_t

Generation 4820: Best = 2.763450e+03 = [4, 3, 2, 4]	FES = 482100	Diversity = 2.6313e-02	k_t
Generation 4830: Best = 2.763435e+03 = [1, 1, 1, 1]	FES = 483100	Diversity = 2.6465e-02	k_t
Generation 4840: Best = 2.763405e+03 = [2, 4, 4, 3]	FES = 484100	Diversity = 2.5360e-02	k_t
Generation 4850: Best = 2.763383e+03 = [1, 1, 1, 1]	FES = 485100	Diversity = 2.5629e-02	k_t
Generation 4860: Best = 2.763355e+03 = [4, 2, 4, 2]	FES = 486100	Diversity = 2.6315e-02	k_t
Generation 4870: Best = 2.763346e+03 = [1, 1, 1, 1]	FES = 487100	Diversity = 2.7054e-02	k_t
Generation 4880: Best = 2.763316e+03 = [4, 3, 4, 2]	FES = 488100	Diversity = 2.7439e-02	k_t
Generation 4890: Best = 2.763295e+03 = [1, 1, 1, 1]	FES = 489100	Diversity = 3.0934e-02	k_t
Generation 4900: Best = 2.763275e+03 = [2, 4, 2, 2]	FES = 490100	Diversity = 2.9041e-02	k_t
Generation 4910: Best = 2.763259e+03 = [1, 1, 1, 1]	FES = 491100	Diversity = 2.7991e-02	k_t
Generation 4920: Best = 2.763227e+03 = [2, 2, 4, 3]	FES = 492100	Diversity = 2.6933e-02	k_t
Generation 4930: Best = 2.763191e+03 = [1, 1, 1, 1]	FES = 493100	Diversity = 2.8231e-02	k_t
Generation 4940: Best = 2.763178e+03 = [2, 4, 4, 3]	FES = 494100	Diversity = 2.9116e-02	k_t
Generation 4950: Best = 2.763146e+03 = [1, 1, 1, 1]	FES = 495100	Diversity = 2.5652e-02	k_t
Generation 4960: Best = 2.763132e+03 = [4, 4, 4, 3]	FES = 496100	Diversity = 2.4990e-02	k_t
Generation 4970: Best = 2.763120e+03 = [1, 1, 1, 1]	FES = 497100	Diversity = 2.5221e-02	k_t
Generation 4980: Best = 2.763104e+03 = [3, 2, 3, 4]	FES = 498100	Diversity = 2.4276e-02	k_t
Generation 4990: Best = 2.763079e+03 = [1, 1, 1, 1]	FES = 499100	Diversity = 2.2995e-02	k_t
Generation 5000: Best = 2.763076e+03 = [3, 4, 3, 3]	FES = 500100	Diversity = 2.1298e-02	k_t
Generation 5010: Best = 2.763058e+03 = [1, 1, 1, 1]	FES = 501100	Diversity = 2.1831e-02	k_t
Generation 5020: Best = 2.763052e+03 = [4, 2, 2, 2]	FES = 502100	Diversity = 2.1208e-02	k_t
Generation 5030: Best = 2.763034e+03 = [1, 1, 1, 1]	FES = 503100	Diversity = 2.1174e-02	k_t
Generation 5040: Best = 2.763020e+03 = [2, 4, 3, 4]	FES = 504100	Diversity = 2.1321e-02	k_t
Generation 5050: Best = 2.763011e+03 = [1, 1, 1, 1]	FES = 505100	Diversity = 2.0687e-02	k_t
Generation 5060: Best = 2.763004e+03 = [4, 3, 3, 2]	FES = 506100	Diversity = 1.9951e-02	k_t
Generation 5070: Best = 2.762997e+03 = [1, 1, 1, 1]	FES = 507100	Diversity = 2.0388e-02	k_t
Generation 5080: Best = 2.762991e+03 = [3, 2, 2, 4]	FES = 508100	Diversity = 2.0157e-02	k_t

Generation 5090: Best = 2.762976e+03 = [1, 1, 1, 1]	FES = 509100	Diversity = 1.8971e-02	k_t
Generation 5100: Best = 2.762964e+03 = [4, 2, 4, 2]	FES = 510100	Diversity = 1.8078e-02	k_t
Generation 5110: Best = 2.762953e+03 = [1, 1, 1, 1]	FES = 511100	Diversity = 1.7634e-02	k_t
Generation 5120: Best = 2.762950e+03 = [3, 3, 2, 3]	FES = 512100	Diversity = 1.7828e-02	k_t
Generation 5130: Best = 2.762942e+03 = [1, 1, 1, 1]	FES = 513100	Diversity = 1.6200e-02	k_t
Generation 5140: Best = 2.762931e+03 = [4, 4, 3, 2]	FES = 514100	Diversity = 1.4968e-02	k_t
Generation 5150: Best = 2.762921e+03 = [1, 1, 1, 1]	FES = 515100	Diversity = 1.5959e-02	k_t
Generation 5160: Best = 2.762910e+03 = [3, 2, 2, 4]	FES = 516100	Diversity = 1.6357e-02	k_t
Generation 5170: Best = 2.762898e+03 = [1, 1, 1, 1]	FES = 517100	Diversity = 1.7364e-02	k_t
Generation 5180: Best = 2.762889e+03 = [2, 3, 2, 4]	FES = 518100	Diversity = 1.6551e-02	k_t
Generation 5190: Best = 2.762869e+03 = [1, 1, 1, 1]	FES = 519100	Diversity = 1.7199e-02	k_t
Generation 5200: Best = 2.762865e+03 = [2, 2, 2, 4]	FES = 520100	Diversity = 1.7687e-02	k_t
Generation 5210: Best = 2.762852e+03 = [1, 1, 1, 1]	FES = 521100	Diversity = 1.9168e-02	k_t
Generation 5220: Best = 2.762842e+03 = [4, 4, 4, 4]	FES = 522100	Diversity = 1.8660e-02	k_t
Generation 5230: Best = 2.762828e+03 = [1, 1, 1, 1]	FES = 523100	Diversity = 1.9289e-02	k_t
Generation 5240: Best = 2.762822e+03 = [4, 2, 2, 4]	FES = 524100	Diversity = 1.8636e-02	k_t
Generation 5250: Best = 2.762810e+03 = [1, 1, 1, 1]	FES = 525100	Diversity = 1.9563e-02	k_t
Generation 5260: Best = 2.762802e+03 = [3, 4, 2, 4]	FES = 526100	Diversity = 2.0761e-02	k_t
Generation 5270: Best = 2.762792e+03 = [1, 1, 1, 1]	FES = 527100	Diversity = 2.2104e-02	k_t
Generation 5280: Best = 2.762776e+03 = [2, 4, 4, 4]	FES = 528100	Diversity = 2.2661e-02	k_t
Generation 5290: Best = 2.762764e+03 = [1, 1, 1, 1]	FES = 529100	Diversity = 2.3064e-02	k_t
Generation 5300: Best = 2.762760e+03 = [4, 3, 2, 4]	FES = 530100	Diversity = 2.2651e-02	k_t
Generation 5310: Best = 2.762748e+03 = [1, 1, 1, 1]	FES = 531100	Diversity = 2.1891e-02	k_t
Generation 5320: Best = 2.762732e+03 = [2, 4, 3, 2]	FES = 532100	Diversity = 2.1916e-02	k_t
Generation 5330: Best = 2.762717e+03 = [1, 1, 1, 1]	FES = 533100	Diversity = 2.3183e-02	k_t
Generation 5340: Best = 2.762700e+03 = [2, 2, 4, 4]	FES = 534100	Diversity = 2.3634e-02	k_t
Generation 5350: Best = 2.762682e+03 = [1, 1, 1, 1]	FES = 535100	Diversity = 2.3713e-02	k_t

Generation 5360: Best = 2.762667e+03 = [4, 3, 2, 4]	FES = 536100	Diversity = 2.3714e-02	k_t
Generation 5370: Best = 2.762658e+03 = [1, 1, 1, 1]	FES = 537100	Diversity = 2.5464e-02	k_t
Generation 5380: Best = 2.762642e+03 = [4, 4, 3, 4]	FES = 538100	Diversity = 2.5876e-02	k_t
Generation 5390: Best = 2.762629e+03 = [1, 1, 1, 1]	FES = 539100	Diversity = 2.6156e-02	k_t
Generation 5400: Best = 2.762615e+03 = [3, 4, 2, 4]	FES = 540100	Diversity = 2.4590e-02	k_t
Generation 5410: Best = 2.762603e+03 = [1, 1, 1, 1]	FES = 541100	Diversity = 2.5604e-02	k_t
Generation 5420: Best = 2.762585e+03 = [2, 2, 4, 2]	FES = 542100	Diversity = 2.3240e-02	k_t
Generation 5430: Best = 2.762576e+03 = [1, 1, 1, 1]	FES = 543100	Diversity = 2.2834e-02	k_t
Generation 5440: Best = 2.762557e+03 = [2, 2, 3, 3]	FES = 544100	Diversity = 2.1988e-02	k_t
Generation 5450: Best = 2.762549e+03 = [1, 1, 1, 1]	FES = 545100	Diversity = 2.1892e-02	k_t
Generation 5460: Best = 2.762541e+03 = [4, 4, 2, 2]	FES = 546100	Diversity = 2.0816e-02	k_t
Generation 5470: Best = 2.762533e+03 = [1, 1, 1, 1]	FES = 547100	Diversity = 2.0095e-02	k_t
Generation 5480: Best = 2.762521e+03 = [4, 2, 4, 4]	FES = 548100	Diversity = 1.9492e-02	k_t
Generation 5490: Best = 2.762513e+03 = [1, 1, 1, 1]	FES = 549100	Diversity = 1.8623e-02	k_t
Generation 5500: Best = 2.762502e+03 = [4, 2, 4, 2]	FES = 550100	Diversity = 1.8319e-02	k_t
Generation 5510: Best = 2.762484e+03 = [1, 1, 1, 1]	FES = 551100	Diversity = 1.9632e-02	k_t
Generation 5520: Best = 2.762475e+03 = [2, 2, 3, 4]	FES = 552100	Diversity = 2.0380e-02	k_t
Generation 5530: Best = 2.762463e+03 = [1, 1, 1, 1]	FES = 553100	Diversity = 1.9685e-02	k_t
Generation 5540: Best = 2.762452e+03 = [3, 4, 2, 2]	FES = 554100	Diversity = 2.1329e-02	k_t
Generation 5550: Best = 2.762438e+03 = [1, 1, 1, 1]	FES = 555100	Diversity = 2.0290e-02	k_t
Generation 5560: Best = 2.762428e+03 = [3, 2, 2, 3]	FES = 556100	Diversity = 1.9059e-02	k_t
Generation 5570: Best = 2.762419e+03 = [1, 1, 1, 1]	FES = 557100	Diversity = 1.7982e-02	k_t
Generation 5580: Best = 2.762416e+03 = [4, 3, 4, 4]	FES = 558100	Diversity = 1.7238e-02	k_t
Generation 5590: Best = 2.762405e+03 = [1, 1, 1, 1]	FES = 559100	Diversity = 1.6996e-02	k_t
Generation 5600: Best = 2.762398e+03 = [3, 3, 3, 2]	FES = 560100	Diversity = 1.6610e-02	k_t
Generation 5610: Best = 2.762389e+03 = [1, 1, 1, 1]	FES = 561100	Diversity = 1.6287e-02	k_t
Generation 5620: Best = 2.762383e+03 = [3, 3, 3, 2]	FES = 562100	Diversity = 1.6928e-02	k_t

Generation 5630: Best = 2.762378e+03 = [1, 1, 1, 1]	FES = 563100	Diversity = 1.7058e-02	k_t
Generation 5640: Best = 2.762362e+03 = [4, 3, 4, 2]	FES = 564100	Diversity = 1.7596e-02	k_t
Generation 5650: Best = 2.762351e+03 = [1, 1, 1, 1]	FES = 565100	Diversity = 1.7702e-02	k_t
Generation 5660: Best = 2.762344e+03 = [2, 2, 2, 3]	FES = 566100	Diversity = 1.7114e-02	k_t
Generation 5670: Best = 2.762336e+03 = [1, 1, 1, 1]	FES = 567100	Diversity = 1.8558e-02	k_t
Generation 5680: Best = 2.762323e+03 = [2, 3, 3, 4]	FES = 568100	Diversity = 1.9276e-02	k_t
Generation 5690: Best = 2.762312e+03 = [1, 1, 1, 1]	FES = 569100	Diversity = 1.9528e-02	k_t
Generation 5700: Best = 2.762304e+03 = [2, 3, 3, 3]	FES = 570100	Diversity = 1.9239e-02	k_t
Generation 5710: Best = 2.762292e+03 = [1, 1, 1, 1]	FES = 571100	Diversity = 1.8965e-02	k_t
Generation 5720: Best = 2.762276e+03 = [4, 2, 2, 4]	FES = 572100	Diversity = 2.1174e-02	k_t
Generation 5730: Best = 2.762263e+03 = [1, 1, 1, 1]	FES = 573100	Diversity = 2.3548e-02	k_t
Generation 5740: Best = 2.762250e+03 = [3, 4, 2, 3]	FES = 574100	Diversity = 2.3017e-02	k_t
Generation 5750: Best = 2.762238e+03 = [1, 1, 1, 1]	FES = 575100	Diversity = 2.3795e-02	k_t
Generation 5760: Best = 2.762219e+03 = [2, 4, 4, 4]	FES = 576100	Diversity = 2.5472e-02	k_t
Generation 5770: Best = 2.762198e+03 = [1, 1, 1, 1]	FES = 577100	Diversity = 2.6348e-02	k_t
Generation 5780: Best = 2.762185e+03 = [2, 4, 3, 3]	FES = 578100	Diversity = 2.6638e-02	k_t
Generation 5790: Best = 2.762173e+03 = [1, 1, 1, 1]	FES = 579100	Diversity = 2.5817e-02	k_t
Generation 5800: Best = 2.762152e+03 = [4, 3, 2, 4]	FES = 580100	Diversity = 2.6205e-02	k_t
Generation 5810: Best = 2.762138e+03 = [1, 1, 1, 1]	FES = 581100	Diversity = 2.7169e-02	k_t
Generation 5820: Best = 2.762114e+03 = [3, 4, 3, 4]	FES = 582100	Diversity = 2.6127e-02	k_t
Generation 5830: Best = 2.762096e+03 = [1, 1, 1, 1]	FES = 583100	Diversity = 2.6958e-02	k_t
Generation 5840: Best = 2.762065e+03 = [3, 2, 3, 4]	FES = 584100	Diversity = 2.9889e-02	k_t
Generation 5850: Best = 2.762039e+03 = [1, 1, 1, 1]	FES = 585100	Diversity = 3.3263e-02	k_t
Generation 5860: Best = 2.761992e+03 = [3, 2, 3, 2]	FES = 586100	Diversity = 3.4360e-02	k_t
Generation 5870: Best = 2.761951e+03 = [1, 1, 1, 1]	FES = 587100	Diversity = 3.5835e-02	k_t
Generation 5880: Best = 2.761944e+03 = [2, 2, 3, 4]	FES = 588100	Diversity = 3.6696e-02	k_t
Generation 5890: Best = 2.761915e+03 = [1, 1, 1, 1]	FES = 589100	Diversity = 3.7711e-02	k_t

Generation 5900: Best = 2.761883e+03 = [3, 3, 3, 3]	FES = 590100	Diversity = 3.5529e-02	k_t
Generation 5910: Best = 2.761860e+03 = [1, 1, 1, 1]	FES = 591100	Diversity = 3.3445e-02	k_t
Generation 5920: Best = 2.761832e+03 = [3, 3, 3, 4]	FES = 592100	Diversity = 3.2087e-02	k_t
Generation 5930: Best = 2.761802e+03 = [1, 1, 1, 1]	FES = 593100	Diversity = 3.2606e-02	k_t
Generation 5940: Best = 2.761783e+03 = [4, 2, 4, 3]	FES = 594100	Diversity = 3.1253e-02	k_t
Generation 5950: Best = 2.761744e+03 = [1, 1, 1, 1]	FES = 595100	Diversity = 3.1335e-02	k_t
Generation 5960: Best = 2.761731e+03 = [3, 3, 4, 4]	FES = 596100	Diversity = 3.2542e-02	k_t
Generation 5970: Best = 2.761709e+03 = [1, 1, 1, 1]	FES = 597100	Diversity = 3.2155e-02	k_t
Generation 5980: Best = 2.761694e+03 = [4, 4, 3, 4]	FES = 598100	Diversity = 3.1136e-02	k_t
Generation 5990: Best = 2.761672e+03 = [1, 1, 1, 1]	FES = 599100	Diversity = 2.8665e-02	k_t
Generation 6000: Best = 2.761663e+03 = [3, 4, 3, 4]	FES = 600100	Diversity = 2.6062e-02	k_t
Generation 6010: Best = 2.761647e+03 = [1, 1, 1, 1]	FES = 601100	Diversity = 2.5402e-02	k_t
Generation 6020: Best = 2.761631e+03 = [3, 2, 4, 2]	FES = 602100	Diversity = 2.2871e-02	k_t
Generation 6030: Best = 2.761623e+03 = [1, 1, 1, 1]	FES = 603100	Diversity = 2.1484e-02	k_t
Generation 6040: Best = 2.761619e+03 = [4, 3, 4, 2]	FES = 604100	Diversity = 2.0176e-02	k_t
Generation 6050: Best = 2.761606e+03 = [1, 1, 1, 1]	FES = 605100	Diversity = 1.9403e-02	k_t
Generation 6060: Best = 2.761591e+03 = [2, 2, 2, 2]	FES = 606100	Diversity = 1.8324e-02	k_t
Generation 6070: Best = 2.761570e+03 = [1, 1, 1, 1]	FES = 607100	Diversity = 1.9381e-02	k_t
Generation 6080: Best = 2.761566e+03 = [2, 2, 4, 4]	FES = 608100	Diversity = 1.9603e-02	k_t
Generation 6090: Best = 2.761553e+03 = [1, 1, 1, 1]	FES = 609100	Diversity = 1.9683e-02	k_t
Generation 6100: Best = 2.761543e+03 = [3, 2, 4, 4]	FES = 610100	Diversity = 1.8886e-02	k_t
Generation 6110: Best = 2.761532e+03 = [1, 1, 1, 1]	FES = 611100	Diversity = 1.9086e-02	k_t
Generation 6120: Best = 2.761521e+03 = [4, 2, 4, 3]	FES = 612100	Diversity = 1.8355e-02	k_t
Generation 6130: Best = 2.761504e+03 = [1, 1, 1, 1]	FES = 613100	Diversity = 1.9226e-02	k_t
Generation 6140: Best = 2.761492e+03 = [4, 3, 2, 3]	FES = 614100	Diversity = 2.0385e-02	k_t
Generation 6150: Best = 2.761475e+03 = [1, 1, 1, 1]	FES = 615100	Diversity = 2.1980e-02	k_t
Generation 6160: Best = 2.761457e+03 = [4, 4, 4, 4]	FES = 616100	Diversity = 2.2924e-02	k_t

Generation 6170: Best = 2.761445e+03 = [1, 1, 1, 1]	FES = 617100	Diversity = 2.6427e-02	k_t
Generation 6180: Best = 2.761426e+03 = [4, 4, 2, 3]	FES = 618100	Diversity = 2.5064e-02	k_t
Generation 6190: Best = 2.761410e+03 = [1, 1, 1, 1]	FES = 619100	Diversity = 2.5721e-02	k_t
Generation 6200: Best = 2.761398e+03 = [2, 2, 3, 4]	FES = 620100	Diversity = 2.5054e-02	k_t
Generation 6210: Best = 2.761381e+03 = [1, 1, 1, 1]	FES = 621100	Diversity = 2.4277e-02	k_t
Generation 6220: Best = 2.761360e+03 = [2, 4, 4, 2]	FES = 622100	Diversity = 2.5413e-02	k_t
Generation 6230: Best = 2.761354e+03 = [1, 1, 1, 1]	FES = 623100	Diversity = 2.4517e-02	k_t
Generation 6240: Best = 2.761345e+03 = [3, 3, 4, 2]	FES = 624100	Diversity = 2.3224e-02	k_t
Generation 6250: Best = 2.761329e+03 = [1, 1, 1, 1]	FES = 625100	Diversity = 2.2155e-02	k_t
Generation 6260: Best = 2.761306e+03 = [4, 2, 3, 3]	FES = 626100	Diversity = 2.4048e-02	k_t
Generation 6270: Best = 2.761286e+03 = [1, 1, 1, 1]	FES = 627100	Diversity = 2.4516e-02	k_t
Generation 6280: Best = 2.761259e+03 = [4, 3, 4, 2]	FES = 628100	Diversity = 2.5453e-02	k_t
Generation 6290: Best = 2.761240e+03 = [1, 1, 1, 1]	FES = 629100	Diversity = 2.4959e-02	k_t
Generation 6300: Best = 2.761230e+03 = [2, 3, 3, 3]	FES = 630100	Diversity = 2.5760e-02	k_t
Generation 6310: Best = 2.761210e+03 = [1, 1, 1, 1]	FES = 631100	Diversity = 2.4673e-02	k_t
Generation 6320: Best = 2.761209e+03 = [4, 3, 4, 4]	FES = 632100	Diversity = 2.3520e-02	k_t
Generation 6330: Best = 2.761192e+03 = [1, 1, 1, 1]	FES = 633100	Diversity = 2.2352e-02	k_t
Generation 6340: Best = 2.761186e+03 = [4, 2, 4, 3]	FES = 634100	Diversity = 2.1169e-02	k_t
Generation 6350: Best = 2.761171e+03 = [1, 1, 1, 1]	FES = 635100	Diversity = 2.1803e-02	k_t
Generation 6360: Best = 2.761148e+03 = [3, 4, 3, 3]	FES = 636100	Diversity = 2.2120e-02	k_t
Generation 6370: Best = 2.761141e+03 = [1, 1, 1, 1]	FES = 637100	Diversity = 2.1419e-02	k_t
Generation 6380: Best = 2.761131e+03 = [3, 4, 3, 4]	FES = 638100	Diversity = 2.1186e-02	k_t
Generation 6390: Best = 2.761112e+03 = [1, 1, 1, 1]	FES = 639100	Diversity = 2.2273e-02	k_t
Generation 6400: Best = 2.761101e+03 = [2, 4, 4, 3]	FES = 640100	Diversity = 2.2200e-02	k_t
Generation 6410: Best = 2.761090e+03 = [1, 1, 1, 1]	FES = 641100	Diversity = 2.1204e-02	k_t
Generation 6420: Best = 2.761070e+03 = [4, 2, 3, 3]	FES = 642100	Diversity = 2.3897e-02	k_t
Generation 6430: Best = 2.761060e+03 = [1, 1, 1, 1]	FES = 643100	Diversity = 2.3520e-02	k_t

Generation 6440: Best = 2.761048e+03 = [3, 3, 4, 2]	FES = 644100	Diversity = 2.1763e-02	k_t
Generation 6450: Best = 2.761036e+03 = [1, 1, 1, 1]	FES = 645100	Diversity = 2.2816e-02	k_t
Generation 6460: Best = 2.761020e+03 = [3, 3, 4, 2]	FES = 646100	Diversity = 2.0663e-02	k_t
Generation 6470: Best = 2.761006e+03 = [1, 1, 1, 1]	FES = 647100	Diversity = 2.0941e-02	k_t
Generation 6480: Best = 2.760998e+03 = [3, 3, 4, 3]	FES = 648100	Diversity = 1.9545e-02	k_t
Generation 6490: Best = 2.760991e+03 = [1, 1, 1, 1]	FES = 649100	Diversity = 1.8144e-02	k_t
Generation 6500: Best = 2.760981e+03 = [4, 3, 3, 4]	FES = 650100	Diversity = 1.8141e-02	k_t
Generation 6510: Best = 2.760971e+03 = [1, 1, 1, 1]	FES = 651100	Diversity = 1.8652e-02	k_t
Generation 6520: Best = 2.760948e+03 = [4, 3, 3, 4]	FES = 652100	Diversity = 2.0236e-02	k_t
Generation 6530: Best = 2.760921e+03 = [1, 1, 1, 1]	FES = 653100	Diversity = 2.3662e-02	k_t
Generation 6540: Best = 2.760921e+03 = [4, 4, 2, 2]	FES = 654100	Diversity = 2.4567e-02	k_t
Generation 6550: Best = 2.760905e+03 = [1, 1, 1, 1]	FES = 655100	Diversity = 2.3388e-02	k_t
Generation 6560: Best = 2.760890e+03 = [4, 3, 2, 3]	FES = 656100	Diversity = 2.3352e-02	k_t
Generation 6570: Best = 2.760880e+03 = [1, 1, 1, 1]	FES = 657100	Diversity = 2.3636e-02	k_t
Generation 6580: Best = 2.760867e+03 = [4, 2, 4, 4]	FES = 658100	Diversity = 2.4636e-02	k_t
Generation 6590: Best = 2.760847e+03 = [1, 1, 1, 1]	FES = 659100	Diversity = 2.4001e-02	k_t
Generation 6600: Best = 2.760836e+03 = [2, 3, 2, 3]	FES = 660100	Diversity = 2.3967e-02	k_t
Generation 6610: Best = 2.760820e+03 = [1, 1, 1, 1]	FES = 661100	Diversity = 2.3587e-02	k_t
Generation 6620: Best = 2.760806e+03 = [4, 3, 3, 4]	FES = 662100	Diversity = 2.3553e-02	k_t
Generation 6630: Best = 2.760792e+03 = [1, 1, 1, 1]	FES = 663100	Diversity = 2.4155e-02	k_t
Generation 6640: Best = 2.760778e+03 = [3, 3, 3, 4]	FES = 664100	Diversity = 2.2749e-02	k_t
Generation 6650: Best = 2.760766e+03 = [1, 1, 1, 1]	FES = 665100	Diversity = 2.1127e-02	k_t
Generation 6660: Best = 2.760754e+03 = [2, 2, 2, 2]	FES = 666100	Diversity = 2.1480e-02	k_t
Generation 6670: Best = 2.760741e+03 = [1, 1, 1, 1]	FES = 667100	Diversity = 2.2230e-02	k_t
Generation 6680: Best = 2.760726e+03 = [2, 4, 3, 3]	FES = 668100	Diversity = 2.2971e-02	k_t
Generation 6690: Best = 2.760712e+03 = [1, 1, 1, 1]	FES = 669100	Diversity = 2.2505e-02	k_t
Generation 6700: Best = 2.760695e+03 = [4, 4, 4, 4]	FES = 670100	Diversity = 2.2868e-02	k_t

Generation 6710: Best = 2.760674e+03 = [1, 1, 1, 1]	FES = 671100	Diversity = 2.4227e-02	k_t
Generation 6720: Best = 2.760663e+03 = [4, 4, 3, 4]	FES = 672100	Diversity = 2.3700e-02	k_t
Generation 6730: Best = 2.760651e+03 = [1, 1, 1, 1]	FES = 673100	Diversity = 2.4316e-02	k_t
Generation 6740: Best = 2.760640e+03 = [4, 2, 3, 2]	FES = 674100	Diversity = 2.4579e-02	k_t
Generation 6750: Best = 2.760633e+03 = [1, 1, 1, 1]	FES = 675100	Diversity = 2.3424e-02	k_t
Generation 6760: Best = 2.760609e+03 = [3, 2, 4, 4]	FES = 676100	Diversity = 2.2822e-02	k_t
Generation 6770: Best = 2.760587e+03 = [1, 1, 1, 1]	FES = 677100	Diversity = 2.4695e-02	k_t
Generation 6780: Best = 2.760577e+03 = [3, 3, 2, 4]	FES = 678100	Diversity = 2.4841e-02	k_t
Generation 6790: Best = 2.760556e+03 = [1, 1, 1, 1]	FES = 679100	Diversity = 2.3297e-02	k_t
Generation 6800: Best = 2.760537e+03 = [3, 3, 3, 3]	FES = 680100	Diversity = 2.3902e-02	k_t
Generation 6810: Best = 2.760524e+03 = [1, 1, 1, 1]	FES = 681100	Diversity = 2.3718e-02	k_t
Generation 6820: Best = 2.760509e+03 = [3, 3, 2, 2]	FES = 682100	Diversity = 2.2872e-02	k_t
Generation 6830: Best = 2.760498e+03 = [1, 1, 1, 1]	FES = 683100	Diversity = 2.4699e-02	k_t
Generation 6840: Best = 2.760481e+03 = [2, 2, 2, 4]	FES = 684100	Diversity = 2.4131e-02	k_t
Generation 6850: Best = 2.760462e+03 = [1, 1, 1, 1]	FES = 685100	Diversity = 2.4007e-02	k_t
Generation 6860: Best = 2.760456e+03 = [3, 3, 4, 2]	FES = 686100	Diversity = 2.3957e-02	k_t
Generation 6870: Best = 2.760431e+03 = [1, 1, 1, 1]	FES = 687100	Diversity = 2.3849e-02	k_t
Generation 6880: Best = 2.760419e+03 = [2, 3, 4, 4]	FES = 688100	Diversity = 2.3051e-02	k_t
Generation 6890: Best = 2.760398e+03 = [1, 1, 1, 1]	FES = 689100	Diversity = 2.2196e-02	k_t
Generation 6900: Best = 2.760383e+03 = [4, 4, 3, 4]	FES = 690100	Diversity = 2.4866e-02	k_t
Generation 6910: Best = 2.760358e+03 = [1, 1, 1, 1]	FES = 691100	Diversity = 2.5761e-02	k_t
Generation 6920: Best = 2.760349e+03 = [4, 3, 3, 3]	FES = 692100	Diversity = 2.7329e-02	k_t
Generation 6930: Best = 2.760331e+03 = [1, 1, 1, 1]	FES = 693100	Diversity = 2.6837e-02	k_t
Generation 6940: Best = 2.760300e+03 = [2, 2, 2, 4]	FES = 694100	Diversity = 2.7371e-02	k_t
Generation 6950: Best = 2.760273e+03 = [1, 1, 1, 1]	FES = 695100	Diversity = 2.9234e-02	k_t
Generation 6960: Best = 2.760252e+03 = [2, 4, 2, 2]	FES = 696100	Diversity = 3.0605e-02	k_t
Generation 6970: Best = 2.760237e+03 = [1, 1, 1, 1]	FES = 697100	Diversity = 3.1246e-02	k_t

Generation 6980: Best = 2.760208e+03 = [2, 4, 2, 4]	FES = 698100	Diversity = 2.9940e-02	k_t
Generation 6990: Best = 2.760189e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 2.9621e-02	k_t
Generation 7000: Best = 2.760173e+03 = [4, 3, 2, 3]	FES = 700100	Diversity = 2.8540e-02	k_t
Generation 7010: Best = 2.760157e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 2.6761e-02	k_t
Generation 7020: Best = 2.760141e+03 = [4, 2, 3, 2]	FES = 702100	Diversity = 2.5775e-02	k_t
Generation 7030: Best = 2.760103e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 2.4930e-02	k_t
Generation 7040: Best = 2.760101e+03 = [2, 3, 4, 2]	FES = 704100	Diversity = 2.4567e-02	k_t
Generation 7050: Best = 2.760075e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 2.4508e-02	k_t
Generation 7060: Best = 2.760058e+03 = [4, 4, 4, 2]	FES = 706100	Diversity = 2.4245e-02	k_t
Generation 7070: Best = 2.760037e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 2.4244e-02	k_t
Generation 7080: Best = 2.760026e+03 = [4, 3, 4, 4]	FES = 708100	Diversity = 2.2578e-02	k_t
Generation 7090: Best = 2.760015e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 2.1553e-02	k_t
Generation 7100: Best = 2.760001e+03 = [2, 4, 4, 3]	FES = 710100	Diversity = 2.0874e-02	k_t
Generation 7110: Best = 2.759984e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 2.0903e-02	k_t
Generation 7120: Best = 2.759967e+03 = [3, 4, 3, 4]	FES = 712100	Diversity = 2.2347e-02	k_t
Generation 7130: Best = 2.759953e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 2.2433e-02	k_t
Generation 7140: Best = 2.759933e+03 = [2, 2, 3, 4]	FES = 714100	Diversity = 2.3079e-02	k_t
Generation 7150: Best = 2.759909e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 2.3562e-02	k_t
Generation 7160: Best = 2.759885e+03 = [2, 3, 4, 3]	FES = 716100	Diversity = 2.4067e-02	k_t
Generation 7170: Best = 2.759863e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 2.6324e-02	k_t
Generation 7180: Best = 2.759841e+03 = [4, 3, 2, 3]	FES = 718100	Diversity = 2.7669e-02	k_t
Generation 7190: Best = 2.759831e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 2.6289e-02	k_t
Generation 7200: Best = 2.759813e+03 = [3, 3, 4, 4]	FES = 720100	Diversity = 2.6656e-02	k_t
Generation 7210: Best = 2.759794e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 2.5608e-02	k_t
Generation 7220: Best = 2.759776e+03 = [2, 4, 3, 3]	FES = 722100	Diversity = 2.4449e-02	k_t
Generation 7230: Best = 2.759757e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 2.4604e-02	k_t
Generation 7240: Best = 2.759733e+03 = [2, 2, 3, 2]	FES = 724100	Diversity = 2.5380e-02	k_t

Generation 7250: Best = 2.759702e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 2.5570e-02	k_t
Generation 7260: Best = 2.759694e+03 = [4, 4, 2, 2]	FES = 726100	Diversity = 2.4981e-02	k_t
Generation 7270: Best = 2.759667e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 2.4613e-02	k_t
Generation 7280: Best = 2.759642e+03 = [4, 3, 2, 3]	FES = 728100	Diversity = 2.5872e-02	k_t
Generation 7290: Best = 2.759631e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 2.5777e-02	k_t
Generation 7300: Best = 2.759598e+03 = [2, 2, 4, 3]	FES = 730100	Diversity = 2.6544e-02	k_t
Generation 7310: Best = 2.759587e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 2.6330e-02	k_t
Generation 7320: Best = 2.759563e+03 = [3, 2, 4, 2]	FES = 732100	Diversity = 2.3905e-02	k_t
Generation 7330: Best = 2.759530e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 2.4683e-02	k_t
Generation 7340: Best = 2.759524e+03 = [4, 3, 4, 3]	FES = 734100	Diversity = 2.5081e-02	k_t
Generation 7350: Best = 2.759499e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 2.3618e-02	k_t
Generation 7360: Best = 2.759470e+03 = [4, 2, 3, 2]	FES = 736100	Diversity = 2.4620e-02	k_t
Generation 7370: Best = 2.759453e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 2.4895e-02	k_t
Generation 7380: Best = 2.759420e+03 = [4, 2, 2, 3]	FES = 738100	Diversity = 2.5192e-02	k_t
Generation 7390: Best = 2.759405e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 2.6794e-02	k_t
Generation 7400: Best = 2.759368e+03 = [4, 4, 4, 2]	FES = 740100	Diversity = 2.7590e-02	k_t
Generation 7410: Best = 2.759348e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 2.6859e-02	k_t
Generation 7420: Best = 2.759321e+03 = [2, 3, 4, 3]	FES = 742100	Diversity = 2.5823e-02	k_t
Generation 7430: Best = 2.759296e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 2.4247e-02	k_t
Generation 7440: Best = 2.759274e+03 = [3, 2, 2, 3]	FES = 744100	Diversity = 2.4829e-02	k_t
Generation 7450: Best = 2.759262e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 2.5822e-02	k_t
Generation 7460: Best = 2.759229e+03 = [4, 2, 3, 4]	FES = 746100	Diversity = 2.7660e-02	k_t
Generation 7470: Best = 2.759197e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 2.9618e-02	k_t
Generation 7480: Best = 2.759176e+03 = [4, 2, 4, 2]	FES = 748100	Diversity = 3.0346e-02	k_t
Generation 7490: Best = 2.759133e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 3.2300e-02	k_t
Generation 7500: Best = 2.759102e+03 = [4, 3, 3, 2]	FES = 750100	Diversity = 3.0895e-02	k_t
Generation 7510: Best = 2.759092e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 3.0202e-02	k_t

Generation 7520: Best = 2.759079e+03 = [3, 4, 2, 4]	FES = 752100	Diversity = 2.8679e-02	k_t
Generation 7530: Best = 2.759034e+03 = [1, 1, 1, 1]	FES = 753100	Diversity = 2.9395e-02	k_t
Generation 7540: Best = 2.759011e+03 = [2, 2, 2, 3]	FES = 754100	Diversity = 2.9899e-02	k_t
Generation 7550: Best = 2.758984e+03 = [1, 1, 1, 1]	FES = 755100	Diversity = 3.1355e-02	k_t
Generation 7560: Best = 2.758951e+03 = [2, 2, 4, 2]	FES = 756100	Diversity = 3.1531e-02	k_t
Generation 7570: Best = 2.758918e+03 = [1, 1, 1, 1]	FES = 757100	Diversity = 3.2334e-02	k_t
Generation 7580: Best = 2.758900e+03 = [4, 2, 4, 3]	FES = 758100	Diversity = 3.0220e-02	k_t
Generation 7590: Best = 2.758848e+03 = [1, 1, 1, 1]	FES = 759100	Diversity = 3.1353e-02	k_t
Generation 7600: Best = 2.758826e+03 = [4, 3, 4, 4]	FES = 760100	Diversity = 3.2260e-02	k_t
Generation 7610: Best = 2.758805e+03 = [1, 1, 1, 1]	FES = 761100	Diversity = 3.1686e-02	k_t
Generation 7620: Best = 2.758778e+03 = [3, 2, 4, 2]	FES = 762100	Diversity = 3.1028e-02	k_t
Generation 7630: Best = 2.758753e+03 = [1, 1, 1, 1]	FES = 763100	Diversity = 3.1198e-02	k_t
Generation 7640: Best = 2.758726e+03 = [2, 3, 3, 2]	FES = 764100	Diversity = 2.9425e-02	k_t
Generation 7650: Best = 2.758704e+03 = [1, 1, 1, 1]	FES = 765100	Diversity = 2.9640e-02	k_t
Generation 7660: Best = 2.758670e+03 = [2, 2, 3, 2]	FES = 766100	Diversity = 2.9911e-02	k_t
Generation 7670: Best = 2.758646e+03 = [1, 1, 1, 1]	FES = 767100	Diversity = 2.9147e-02	k_t
Generation 7680: Best = 2.758625e+03 = [2, 3, 3, 4]	FES = 768100	Diversity = 2.8499e-02	k_t
Generation 7690: Best = 2.758608e+03 = [1, 1, 1, 1]	FES = 769100	Diversity = 2.9384e-02	k_t
Generation 7700: Best = 2.758575e+03 = [3, 4, 3, 3]	FES = 770100	Diversity = 2.9138e-02	k_t
Generation 7710: Best = 2.758545e+03 = [1, 1, 1, 1]	FES = 771100	Diversity = 2.8550e-02	k_t
Generation 7720: Best = 2.758537e+03 = [2, 3, 4, 3]	FES = 772100	Diversity = 2.7699e-02	k_t
Generation 7730: Best = 2.758484e+03 = [1, 1, 1, 1]	FES = 773100	Diversity = 2.8884e-02	k_t
Generation 7740: Best = 2.758458e+03 = [3, 4, 3, 4]	FES = 774100	Diversity = 2.8704e-02	k_t
Generation 7750: Best = 2.758428e+03 = [1, 1, 1, 1]	FES = 775100	Diversity = 3.2628e-02	k_t
Generation 7760: Best = 2.758403e+03 = [3, 4, 3, 3]	FES = 776100	Diversity = 3.4112e-02	k_t
Generation 7770: Best = 2.758371e+03 = [1, 1, 1, 1]	FES = 777100	Diversity = 3.3856e-02	k_t
Generation 7780: Best = 2.758355e+03 = [4, 4, 2, 3]	FES = 778100	Diversity = 3.3255e-02	k_t

Generation 7790: Best = 2.758267e+03 = [1, 1, 1, 1]	FES = 779100	Diversity = 3.4892e-02	k_t
Generation 7800: Best = 2.758267e+03 = [2, 4, 2, 4]	FES = 780100	Diversity = 3.8421e-02	k_t
Generation 7810: Best = 2.758217e+03 = [1, 1, 1, 1]	FES = 781100	Diversity = 3.8547e-02	k_t
Generation 7820: Best = 2.758161e+03 = [3, 3, 2, 4]	FES = 782100	Diversity = 4.0143e-02	k_t
Generation 7830: Best = 2.758115e+03 = [1, 1, 1, 1]	FES = 783100	Diversity = 4.2949e-02	k_t
Generation 7840: Best = 2.758060e+03 = [4, 4, 3, 4]	FES = 784100	Diversity = 4.4450e-02	k_t
Generation 7850: Best = 2.758003e+03 = [1, 1, 1, 1]	FES = 785100	Diversity = 4.5454e-02	k_t
Generation 7860: Best = 2.757944e+03 = [3, 4, 4, 2]	FES = 786100	Diversity = 4.5374e-02	k_t
Generation 7870: Best = 2.757899e+03 = [1, 1, 1, 1]	FES = 787100	Diversity = 4.3999e-02	k_t
Generation 7880: Best = 2.757838e+03 = [3, 3, 4, 2]	FES = 788100	Diversity = 4.2977e-02	k_t
Generation 7890: Best = 2.757802e+03 = [1, 1, 1, 1]	FES = 789100	Diversity = 4.1675e-02	k_t
Generation 7900: Best = 2.757766e+03 = [3, 2, 3, 3]	FES = 790100	Diversity = 3.8996e-02	k_t
Generation 7910: Best = 2.757743e+03 = [1, 1, 1, 1]	FES = 791100	Diversity = 3.3411e-02	k_t
Generation 7920: Best = 2.757683e+03 = [3, 2, 2, 2]	FES = 792100	Diversity = 3.1926e-02	k_t
Generation 7930: Best = 2.757678e+03 = [1, 1, 1, 1]	FES = 793100	Diversity = 2.9736e-02	k_t
Generation 7940: Best = 2.757647e+03 = [4, 2, 4, 3]	FES = 794100	Diversity = 2.8124e-02	k_t
Generation 7950: Best = 2.757602e+03 = [1, 1, 1, 1]	FES = 795100	Diversity = 2.8489e-02	k_t
Generation 7960: Best = 2.757573e+03 = [4, 2, 3, 3]	FES = 796100	Diversity = 2.7508e-02	k_t
Generation 7970: Best = 2.757520e+03 = [1, 1, 1, 1]	FES = 797100	Diversity = 3.1205e-02	k_t
Generation 7980: Best = 2.757477e+03 = [2, 2, 2, 2]	FES = 798100	Diversity = 3.4495e-02	k_t
Generation 7990: Best = 2.757427e+03 = [1, 1, 1, 1]	FES = 799100	Diversity = 3.8768e-02	k_t
Generation 8000: Best = 2.757354e+03 = [2, 3, 4, 2]	FES = 800100	Diversity = 3.9226e-02	k_t
Generation 8010: Best = 2.757283e+03 = [1, 1, 1, 1]	FES = 801100	Diversity = 4.1626e-02	k_t
Generation 8020: Best = 2.757265e+03 = [3, 2, 3, 2]	FES = 802100	Diversity = 4.2303e-02	k_t
Generation 8030: Best = 2.757164e+03 = [1, 1, 1, 1]	FES = 803100	Diversity = 4.3008e-02	k_t
Generation 8040: Best = 2.757155e+03 = [2, 3, 3, 2]	FES = 804100	Diversity = 4.3932e-02	k_t
Generation 8050: Best = 2.757082e+03 = [1, 1, 1, 1]	FES = 805100	Diversity = 4.5056e-02	k_t

Generation 8060: Best = 2.757010e+03 = [2, 3, 3, 4]	FES = 806100	Diversity = 4.7012e-02	k_t
Generation 8070: Best = 2.756926e+03 = [1, 1, 1, 1]	FES = 807100	Diversity = 4.5134e-02	k_t
Generation 8080: Best = 2.756869e+03 = [4, 4, 2, 4]	FES = 808100	Diversity = 4.9570e-02	k_t
Generation 8090: Best = 2.756810e+03 = [1, 1, 1, 1]	FES = 809100	Diversity = 4.9653e-02	k_t
Generation 8100: Best = 2.756751e+03 = [3, 4, 4, 2]	FES = 810100	Diversity = 4.9780e-02	k_t
Generation 8110: Best = 2.756585e+03 = [1, 1, 1, 1]	FES = 811100	Diversity = 5.2221e-02	k_t
Generation 8120: Best = 2.756566e+03 = [2, 4, 2, 4]	FES = 812100	Diversity = 5.5772e-02	k_t
Generation 8130: Best = 2.756476e+03 = [1, 1, 1, 1]	FES = 813100	Diversity = 6.2223e-02	k_t
Generation 8140: Best = 2.756346e+03 = [3, 4, 3, 2]	FES = 814100	Diversity = 6.6248e-02	k_t
Generation 8150: Best = 2.756195e+03 = [1, 1, 1, 1]	FES = 815100	Diversity = 7.1483e-02	k_t
Generation 8160: Best = 2.755994e+03 = [3, 4, 4, 3]	FES = 816100	Diversity = 7.7785e-02	k_t
Generation 8170: Best = 2.755805e+03 = [1, 1, 1, 1]	FES = 817100	Diversity = 7.8867e-02	k_t
Generation 8180: Best = 2.755680e+03 = [4, 4, 2, 2]	FES = 818100	Diversity = 7.7454e-02	k_t
Generation 8190: Best = 2.755526e+03 = [1, 1, 1, 1]	FES = 819100	Diversity = 7.5499e-02	k_t
Generation 8200: Best = 2.755320e+03 = [3, 4, 3, 2]	FES = 820100	Diversity = 7.7574e-02	k_t
Generation 8210: Best = 2.755195e+03 = [1, 1, 1, 1]	FES = 821100	Diversity = 7.6546e-02	k_t
Generation 8220: Best = 2.755041e+03 = [3, 4, 4, 3]	FES = 822100	Diversity = 7.3274e-02	k_t
Generation 8230: Best = 2.754991e+03 = [1, 1, 1, 1]	FES = 823100	Diversity = 6.7116e-02	k_t
Generation 8240: Best = 2.754885e+03 = [2, 4, 2, 3]	FES = 824100	Diversity = 5.9380e-02	k_t
Generation 8250: Best = 2.754731e+03 = [1, 1, 1, 1]	FES = 825100	Diversity = 5.8111e-02	k_t
Generation 8260: Best = 2.754647e+03 = [3, 2, 3, 3]	FES = 826100	Diversity = 5.8794e-02	k_t
Generation 8270: Best = 2.754527e+03 = [1, 1, 1, 1]	FES = 827100	Diversity = 5.7666e-02	k_t
Generation 8280: Best = 2.754453e+03 = [2, 3, 2, 3]	FES = 828100	Diversity = 6.1102e-02	k_t
Generation 8290: Best = 2.754370e+03 = [1, 1, 1, 1]	FES = 829100	Diversity = 6.3066e-02	k_t
Generation 8300: Best = 2.754290e+03 = [2, 4, 4, 2]	FES = 830100	Diversity = 5.9480e-02	k_t
Generation 8310: Best = 2.754202e+03 = [1, 1, 1, 1]	FES = 831100	Diversity = 5.9710e-02	k_t
Generation 8320: Best = 2.754105e+03 = [4, 2, 4, 3]	FES = 832100	Diversity = 5.7420e-02	k_t

Generation 8330: Best = 2.754011e+03 = [1, 1, 1, 1]	FES = 833100	Diversity = 5.4375e-02	k_t
Generation 8340: Best = 2.753966e+03 = [4, 2, 3, 4]	FES = 834100	Diversity = 5.4980e-02	k_t
Generation 8350: Best = 2.753770e+03 = [1, 1, 1, 1]	FES = 835100	Diversity = 5.8847e-02	k_t
Generation 8360: Best = 2.753732e+03 = [3, 4, 3, 3]	FES = 836100	Diversity = 6.1360e-02	k_t
Generation 8370: Best = 2.753691e+03 = [1, 1, 1, 1]	FES = 837100	Diversity = 5.8746e-02	k_t
Generation 8380: Best = 2.753641e+03 = [2, 4, 2, 3]	FES = 838100	Diversity = 5.9979e-02	k_t
Generation 8390: Best = 2.753475e+03 = [1, 1, 1, 1]	FES = 839100	Diversity = 6.4824e-02	k_t
Generation 8400: Best = 2.753408e+03 = [4, 4, 2, 2]	FES = 840100	Diversity = 6.6800e-02	k_t
Generation 8410: Best = 2.753243e+03 = [1, 1, 1, 1]	FES = 841100	Diversity = 6.9473e-02	k_t
Generation 8420: Best = 2.753128e+03 = [2, 4, 2, 4]	FES = 842100	Diversity = 7.4265e-02	k_t
Generation 8430: Best = 2.753087e+03 = [1, 1, 1, 1]	FES = 843100	Diversity = 7.4321e-02	k_t
Generation 8440: Best = 2.752940e+03 = [4, 3, 4, 2]	FES = 844100	Diversity = 7.6429e-02	k_t
Generation 8450: Best = 2.752812e+03 = [1, 1, 1, 1]	FES = 845100	Diversity = 8.0374e-02	k_t
Generation 8460: Best = 2.752691e+03 = [3, 3, 3, 3]	FES = 846100	Diversity = 8.2919e-02	k_t
Generation 8470: Best = 2.752578e+03 = [1, 1, 1, 1]	FES = 847100	Diversity = 8.3581e-02	k_t
Generation 8480: Best = 2.752408e+03 = [2, 4, 2, 2]	FES = 848100	Diversity = 8.7877e-02	k_t
Generation 8490: Best = 2.752224e+03 = [1, 1, 1, 1]	FES = 849100	Diversity = 8.6481e-02	k_t
Generation 8500: Best = 2.752176e+03 = [2, 2, 2, 2]	FES = 850100	Diversity = 8.5293e-02	k_t
Generation 8510: Best = 2.752093e+03 = [1, 1, 1, 1]	FES = 851100	Diversity = 8.2925e-02	k_t
Generation 8520: Best = 2.751984e+03 = [4, 2, 3, 4]	FES = 852100	Diversity = 8.7131e-02	k_t
Generation 8530: Best = 2.751822e+03 = [1, 1, 1, 1]	FES = 853100	Diversity = 8.9643e-02	k_t
Generation 8540: Best = 2.751693e+03 = [4, 4, 2, 2]	FES = 854100	Diversity = 8.7204e-02	k_t
Generation 8550: Best = 2.751587e+03 = [1, 1, 1, 1]	FES = 855100	Diversity = 8.3119e-02	k_t
Generation 8560: Best = 2.751516e+03 = [2, 2, 3, 3]	FES = 856100	Diversity = 8.4439e-02	k_t
Generation 8570: Best = 2.751388e+03 = [1, 1, 1, 1]	FES = 857100	Diversity = 8.5030e-02	k_t
Generation 8580: Best = 2.751201e+03 = [4, 2, 2, 4]	FES = 858100	Diversity = 8.7073e-02	k_t
Generation 8590: Best = 2.751015e+03 = [1, 1, 1, 1]	FES = 859100	Diversity = 9.4873e-02	k_t

Generation 8600: Best = 2.750766e+03 = [3, 3, 3, 2]	FES = 860100	Diversity = 1.1289e-01	k_t
Generation 8610: Best = 2.750465e+03 = [1, 1, 1, 1]	FES = 861100	Diversity = 1.1914e-01	k_t
Generation 8620: Best = 2.750048e+03 = [3, 2, 2, 4]	FES = 862100	Diversity = 1.3339e-01	k_t
Generation 8630: Best = 2.749699e+03 = [1, 1, 1, 1]	FES = 863100	Diversity = 1.3500e-01	k_t
Generation 8640: Best = 2.749339e+03 = [2, 2, 2, 4]	FES = 864100	Diversity = 1.3461e-01	k_t
Generation 8650: Best = 2.749051e+03 = [1, 1, 1, 1]	FES = 865100	Diversity = 1.3464e-01	k_t
Generation 8660: Best = 2.748916e+03 = [3, 3, 2, 2]	FES = 866100	Diversity = 1.3817e-01	k_t
Generation 8670: Best = 2.748635e+03 = [1, 1, 1, 1]	FES = 867100	Diversity = 1.3632e-01	k_t
Generation 8680: Best = 2.748484e+03 = [2, 3, 4, 3]	FES = 868100	Diversity = 1.3294e-01	k_t
Generation 8690: Best = 2.748208e+03 = [1, 1, 1, 1]	FES = 869100	Diversity = 1.2606e-01	k_t
Generation 8700: Best = 2.748061e+03 = [4, 3, 3, 4]	FES = 870100	Diversity = 1.2890e-01	k_t
Generation 8710: Best = 2.747840e+03 = [1, 1, 1, 1]	FES = 871100	Diversity = 1.3142e-01	k_t
Generation 8720: Best = 2.747286e+03 = [2, 4, 4, 2]	FES = 872100	Diversity = 1.5345e-01	k_t
Generation 8730: Best = 2.747015e+03 = [1, 1, 1, 1]	FES = 873100	Diversity = 1.6087e-01	k_t
Generation 8740: Best = 2.746425e+03 = [4, 4, 2, 3]	FES = 874100	Diversity = 1.7417e-01	k_t
Generation 8750: Best = 2.745797e+03 = [1, 1, 1, 1]	FES = 875100	Diversity = 1.9561e-01	k_t
Generation 8760: Best = 2.745076e+03 = [2, 4, 2, 4]	FES = 876100	Diversity = 2.1710e-01	k_t
Generation 8770: Best = 2.744177e+03 = [1, 1, 1, 1]	FES = 877100	Diversity = 2.2952e-01	k_t
Generation 8780: Best = 2.743950e+03 = [2, 3, 3, 4]	FES = 878100	Diversity = 2.2504e-01	k_t
Generation 8790: Best = 2.743392e+03 = [1, 1, 1, 1]	FES = 879100	Diversity = 2.2238e-01	k_t
Generation 8800: Best = 2.742898e+03 = [2, 3, 3, 3]	FES = 880100	Diversity = 2.1106e-01	k_t
Generation 8810: Best = 2.742501e+03 = [1, 1, 1, 1]	FES = 881100	Diversity = 2.0192e-01	k_t
Generation 8820: Best = 2.741868e+03 = [4, 4, 4, 2]	FES = 882100	Diversity = 1.9928e-01	k_t
Generation 8830: Best = 2.741386e+03 = [1, 1, 1, 1]	FES = 883100	Diversity = 2.0159e-01	k_t
Generation 8840: Best = 2.741129e+03 = [2, 3, 2, 4]	FES = 884100	Diversity = 2.0062e-01	k_t
Generation 8850: Best = 2.740464e+03 = [1, 1, 1, 1]	FES = 885100	Diversity = 2.0149e-01	k_t
Generation 8860: Best = 2.740292e+03 = [3, 2, 3, 4]	FES = 886100	Diversity = 1.9593e-01	k_t

Generation 8870: Best = 2.739976e+03 = [1, 1, 1, 1]	FES = 887100	Diversity = 1.9763e-01	k_t
Generation 8880: Best = 2.739310e+03 = [4, 2, 3, 4]	FES = 888100	Diversity = 2.0739e-01	k_t
Generation 8890: Best = 2.738773e+03 = [1, 1, 1, 1]	FES = 889100	Diversity = 2.0886e-01	k_t
Generation 8900: Best = 2.738337e+03 = [4, 3, 4, 2]	FES = 890100	Diversity = 2.1235e-01	k_t
Generation 8910: Best = 2.737913e+03 = [1, 1, 1, 1]	FES = 891100	Diversity = 2.1886e-01	k_t
Generation 8920: Best = 2.737209e+03 = [4, 4, 2, 4]	FES = 892100	Diversity = 2.2784e-01	k_t
Generation 8930: Best = 2.736916e+03 = [1, 1, 1, 1]	FES = 893100	Diversity = 2.3518e-01	k_t
Generation 8940: Best = 2.736437e+03 = [2, 2, 3, 4]	FES = 894100	Diversity = 2.4331e-01	k_t
Generation 8950: Best = 2.736122e+03 = [1, 1, 1, 1]	FES = 895100	Diversity = 2.5050e-01	k_t
Generation 8960: Best = 2.735098e+03 = [3, 2, 4, 3]	FES = 896100	Diversity = 2.5946e-01	k_t
Generation 8970: Best = 2.733809e+03 = [1, 1, 1, 1]	FES = 897100	Diversity = 2.8331e-01	k_t
Generation 8980: Best = 2.733597e+03 = [3, 4, 4, 2]	FES = 898100	Diversity = 3.0162e-01	k_t
Generation 8990: Best = 2.732961e+03 = [1, 1, 1, 1]	FES = 899100	Diversity = 3.1023e-01	k_t
Generation 9000: Best = 2.731939e+03 = [3, 4, 2, 2]	FES = 900100	Diversity = 3.1445e-01	k_t
Generation 9010: Best = 2.731033e+03 = [1, 1, 1, 1]	FES = 901100	Diversity = 3.4289e-01	k_t
Generation 9020: Best = 2.729123e+03 = [4, 3, 3, 3]	FES = 902100	Diversity = 3.7250e-01	k_t
Generation 9030: Best = 2.728367e+03 = [1, 1, 1, 1]	FES = 903100	Diversity = 4.2803e-01	k_t
Generation 9040: Best = 2.726285e+03 = [3, 4, 2, 3]	FES = 904100	Diversity = 4.5481e-01	k_t
Generation 9050: Best = 2.725329e+03 = [1, 1, 1, 1]	FES = 905100	Diversity = 4.5227e-01	k_t
Generation 9060: Best = 2.724272e+03 = [3, 2, 3, 2]	FES = 906100	Diversity = 4.7249e-01	k_t
Generation 9070: Best = 2.723118e+03 = [1, 1, 1, 1]	FES = 907100	Diversity = 4.9218e-01	k_t
Generation 9080: Best = 2.721840e+03 = [4, 3, 4, 2]	FES = 908100	Diversity = 5.3156e-01	k_t
Generation 9090: Best = 2.720801e+03 = [1, 1, 1, 1]	FES = 909100	Diversity = 5.1578e-01	k_t
Generation 9100: Best = 2.720084e+03 = [4, 3, 2, 4]	FES = 910100	Diversity = 5.0034e-01	k_t
Generation 9110: Best = 2.719248e+03 = [1, 1, 1, 1]	FES = 911100	Diversity = 5.1030e-01	k_t
Generation 9120: Best = 2.717441e+03 = [4, 2, 4, 4]	FES = 912100	Diversity = 5.2783e-01	k_t
Generation 9130: Best = 2.716543e+03 = [1, 1, 1, 1]	FES = 913100	Diversity = 5.7025e-01	k_t

Generation 9140: Best = 2.714925e+03 = [3, 4, 2, 2]	FES = 914100	Diversity = 5.7230e-01	k_t
Generation 9150: Best = 2.713980e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 5.6068e-01	k_t
Generation 9160: Best = 2.712700e+03 = [3, 2, 3, 2]	FES = 916100	Diversity = 5.6351e-01	k_t
Generation 9170: Best = 2.711725e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 5.6245e-01	k_t
Generation 9180: Best = 2.711167e+03 = [2, 4, 3, 2]	FES = 918100	Diversity = 5.6363e-01	k_t
Generation 9190: Best = 2.709966e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 5.6585e-01	k_t
Generation 9200: Best = 2.709806e+03 = [4, 3, 2, 2]	FES = 920100	Diversity = 5.4210e-01	k_t
Generation 9210: Best = 2.708525e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 5.3319e-01	k_t
Generation 9220: Best = 2.707719e+03 = [4, 2, 4, 2]	FES = 922100	Diversity = 5.2967e-01	k_t
Generation 9230: Best = 2.707385e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 5.3356e-01	k_t
Generation 9240: Best = 2.707134e+03 = [2, 3, 4, 2]	FES = 924100	Diversity = 5.1023e-01	k_t
Generation 9250: Best = 2.706722e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 5.0284e-01	k_t
Generation 9260: Best = 2.706426e+03 = [2, 2, 4, 3]	FES = 926100	Diversity = 4.9483e-01	k_t
Generation 9270: Best = 2.705877e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 4.9337e-01	k_t
Generation 9280: Best = 2.705814e+03 = [4, 4, 4, 3]	FES = 928100	Diversity = 4.7714e-01	k_t
Generation 9290: Best = 2.705614e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 4.6532e-01	k_t
Generation 9300: Best = 2.705128e+03 = [4, 3, 3, 2]	FES = 930100	Diversity = 4.6183e-01	k_t
Generation 9310: Best = 2.704990e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 4.6074e-01	k_t
Generation 9320: Best = 2.704576e+03 = [4, 2, 2, 2]	FES = 932100	Diversity = 4.5461e-01	k_t
Generation 9330: Best = 2.704287e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 4.5797e-01	k_t
Generation 9340: Best = 2.704249e+03 = [2, 2, 4, 2]	FES = 934100	Diversity = 4.6257e-01	k_t
Generation 9350: Best = 2.703780e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 4.7035e-01	k_t
Generation 9360: Best = 2.703441e+03 = [3, 3, 3, 3]	FES = 936100	Diversity = 4.7538e-01	k_t
Generation 9370: Best = 2.703015e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 4.8396e-01	k_t
Generation 9380: Best = 2.702955e+03 = [4, 4, 2, 4]	FES = 938100	Diversity = 4.8224e-01	k_t
Generation 9390: Best = 2.702646e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 4.9010e-01	k_t
Generation 9400: Best = 2.702029e+03 = [3, 2, 2, 3]	FES = 940100	Diversity = 4.9167e-01	k_t

Generation 9410: Best = 2.701619e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 4.9123e-01	k_t
Generation 9420: Best = 2.701431e+03 = [4, 2, 3, 3]	FES = 942100	Diversity = 5.0618e-01	k_t
Generation 9430: Best = 2.700825e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 5.2168e-01	k_t
Generation 9440: Best = 2.700516e+03 = [3, 3, 3, 2]	FES = 944100	Diversity = 5.3493e-01	k_t
Generation 9450: Best = 2.700204e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 5.4031e-01	k_t
Generation 9460: Best = 2.699589e+03 = [2, 3, 4, 2]	FES = 946100	Diversity = 5.3758e-01	k_t
Generation 9470: Best = 2.699290e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 5.4066e-01	k_t
Generation 9480: Best = 2.698818e+03 = [3, 4, 4, 2]	FES = 948100	Diversity = 5.3714e-01	k_t
Generation 9490: Best = 2.698404e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 5.4820e-01	k_t
Generation 9500: Best = 2.698221e+03 = [2, 4, 3, 3]	FES = 950100	Diversity = 5.4669e-01	k_t
Generation 9510: Best = 2.697809e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 5.2929e-01	k_t
Generation 9520: Best = 2.697301e+03 = [2, 3, 2, 4]	FES = 952100	Diversity = 5.1847e-01	k_t
Generation 9530: Best = 2.697097e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 5.1551e-01	k_t
Generation 9540: Best = 2.696425e+03 = [3, 3, 4, 2]	FES = 954100	Diversity = 5.2888e-01	k_t
Generation 9550: Best = 2.696153e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 5.2159e-01	k_t
Generation 9560: Best = 2.695214e+03 = [3, 2, 2, 2]	FES = 956100	Diversity = 5.1185e-01	k_t
Generation 9570: Best = 2.694633e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 4.9112e-01	k_t
Generation 9580: Best = 2.694148e+03 = [4, 2, 4, 2]	FES = 958100	Diversity = 4.6929e-01	k_t
Generation 9590: Best = 2.693600e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 4.7544e-01	k_t
Generation 9600: Best = 2.692993e+03 = [4, 2, 3, 3]	FES = 960100	Diversity = 4.8568e-01	k_t
Generation 9610: Best = 2.692364e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 5.0541e-01	k_t
Generation 9620: Best = 2.691725e+03 = [2, 2, 2, 4]	FES = 962100	Diversity = 5.1615e-01	k_t
Generation 9630: Best = 2.691251e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 5.0890e-01	k_t
Generation 9640: Best = 2.690346e+03 = [2, 2, 3, 4]	FES = 964100	Diversity = 5.5058e-01	k_t
Generation 9650: Best = 2.689333e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 5.9649e-01	k_t
Generation 9660: Best = 2.688793e+03 = [3, 4, 4, 3]	FES = 966100	Diversity = 6.0179e-01	k_t
Generation 9670: Best = 2.687702e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 6.0448e-01	k_t

Generation 9680: Best = 2.687145e+03 = [4, 3, 3, 2]	FES = 968100	Diversity = 6.1538e-01	k_t
Generation 9690: Best = 2.686776e+03 = [1, 1, 1, 1]	FES = 969100	Diversity = 6.6233e-01	k_t
Generation 9700: Best = 2.686086e+03 = [3, 3, 3, 4]	FES = 970100	Diversity = 6.9249e-01	k_t
Generation 9710: Best = 2.685001e+03 = [1, 1, 1, 1]	FES = 971100	Diversity = 7.1141e-01	k_t
Generation 9720: Best = 2.683928e+03 = [2, 3, 2, 4]	FES = 972100	Diversity = 7.4720e-01	k_t
Generation 9730: Best = 2.683454e+03 = [1, 1, 1, 1]	FES = 973100	Diversity = 7.8470e-01	k_t
Generation 9740: Best = 2.681797e+03 = [2, 3, 3, 4]	FES = 974100	Diversity = 8.3525e-01	k_t
Generation 9750: Best = 2.680874e+03 = [1, 1, 1, 1]	FES = 975100	Diversity = 8.6798e-01	k_t
Generation 9760: Best = 2.679536e+03 = [2, 4, 3, 3]	FES = 976100	Diversity = 8.7003e-01	k_t
Generation 9770: Best = 2.679026e+03 = [1, 1, 1, 1]	FES = 977100	Diversity = 8.7988e-01	k_t
Generation 9780: Best = 2.677143e+03 = [4, 3, 4, 2]	FES = 978100	Diversity = 8.7702e-01	k_t
Generation 9790: Best = 2.677100e+03 = [1, 1, 1, 1]	FES = 979100	Diversity = 8.5326e-01	k_t
Generation 9800: Best = 2.676193e+03 = [3, 3, 4, 4]	FES = 980100	Diversity = 8.2916e-01	k_t
Generation 9810: Best = 2.675050e+03 = [1, 1, 1, 1]	FES = 981100	Diversity = 8.2342e-01	k_t
Generation 9820: Best = 2.674576e+03 = [3, 2, 3, 3]	FES = 982100	Diversity = 8.3388e-01	k_t
Generation 9830: Best = 2.673490e+03 = [1, 1, 1, 1]	FES = 983100	Diversity = 8.3407e-01	k_t
Generation 9840: Best = 2.672466e+03 = [4, 2, 3, 2]	FES = 984100	Diversity = 8.2577e-01	k_t
Generation 9850: Best = 2.671435e+03 = [1, 1, 1, 1]	FES = 985100	Diversity = 8.1200e-01	k_t
Generation 9860: Best = 2.670292e+03 = [2, 4, 2, 4]	FES = 986100	Diversity = 8.3905e-01	k_t
Generation 9870: Best = 2.669491e+03 = [1, 1, 1, 1]	FES = 987100	Diversity = 8.3604e-01	k_t
Generation 9880: Best = 2.669118e+03 = [4, 4, 4, 4]	FES = 988100	Diversity = 8.1020e-01	k_t
Generation 9890: Best = 2.667859e+03 = [1, 1, 1, 1]	FES = 989100	Diversity = 8.0398e-01	k_t
Generation 9900: Best = 2.667546e+03 = [3, 3, 4, 3]	FES = 990100	Diversity = 7.7049e-01	k_t
Generation 9910: Best = 2.667350e+03 = [1, 1, 1, 1]	FES = 991100	Diversity = 7.4999e-01	k_t
Generation 9920: Best = 2.667350e+03 = [3, 4, 4, 4]	FES = 992100	Diversity = 7.5487e-01	k_t
Generation 9930: Best = 2.666812e+03 = [1, 1, 1, 1]	FES = 993100	Diversity = 7.8687e-01	k_t
Generation 9940: Best = 2.666502e+03 = [3, 4, 4, 2]	FES = 994100	Diversity = 7.7917e-01	k_t

Generation 9950: Best = 2.666059e+03 FES = 995100 Diversity = 7.7803e-01 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 2.666036e+03 FES = 996100 Diversity = 7.7042e-01 k_t
= [4, 3, 4, 3]
Generation 9970: Best = 2.665698e+03 FES = 997100 Diversity = 7.7643e-01 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 2.665377e+03 FES = 998100 Diversity = 7.8496e-01 k_t
= [3, 4, 4, 3]
Generation 9990: Best = 2.664941e+03 FES = 999100 Diversity = 7.7290e-01 k_t
= [1, 1, 1, 1]

Optimization completed!

Final Best Fitness = 2.664671e+03

Total Function Evaluations = 1000000

=====

=

Results:

Best Fitness : 2.664671e+03

Global Optimum : 1.500000e+03

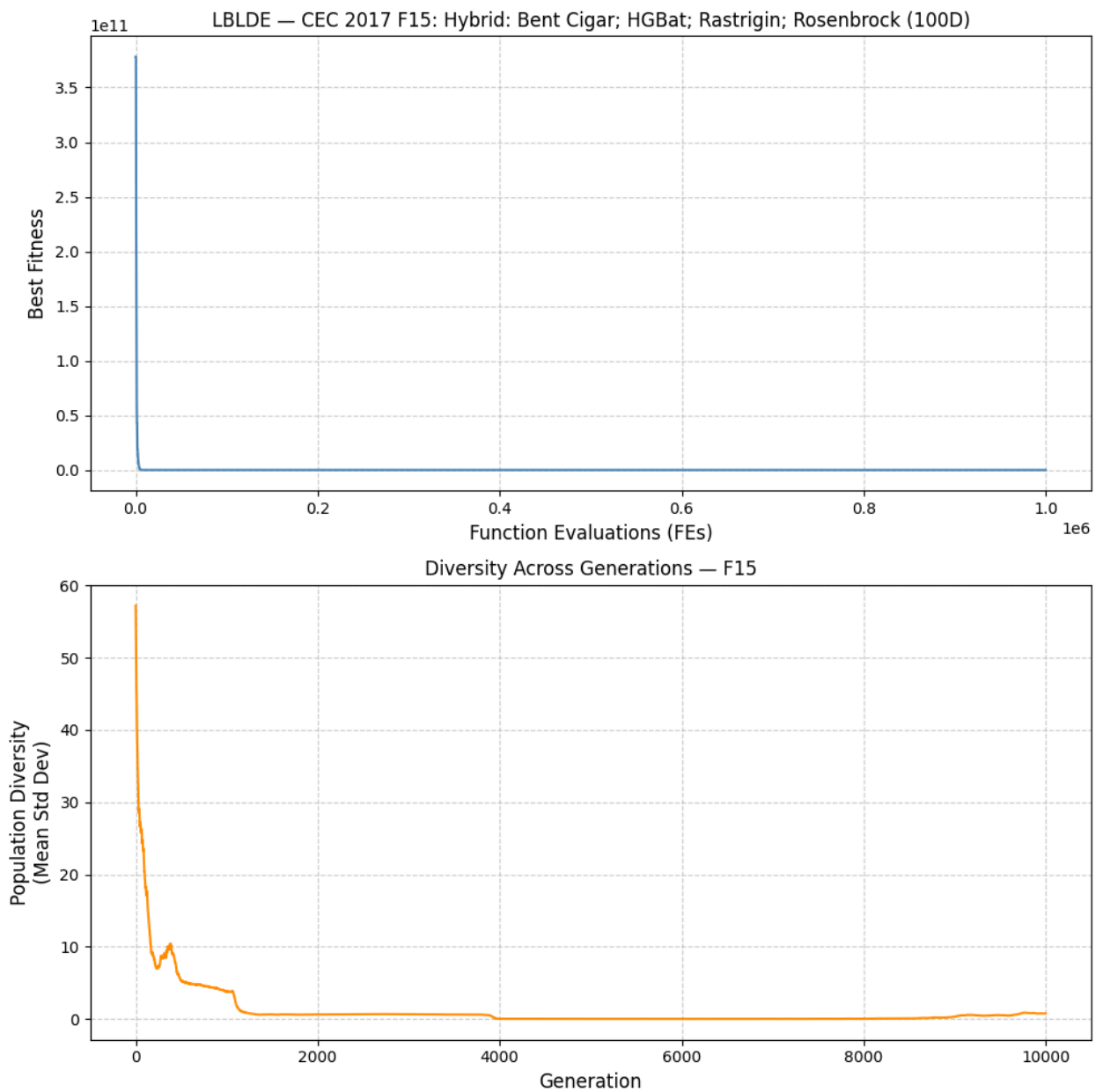
Error (|best-optimum|) : 1.164671e+03

Best Solution (5 dim) : [6.03286017 -11.14182555 50.2319618 15.48369193
-14.09266919]

D^(0) (init. diversity): 5.7220e+01

=====

=



Graph saved as convergence_F15.png

```
In [40]: # =====
# F16 – Hybrid Function 6 (Schaffer; HGBat; Rosenbrock; Schwefel)
# =====
run_and_plot('F16', make_wrapper(hybrid.f16), 1600.0, 'Hybrid: Schaffer; HGBat
```

```

=====
=
LBLDE_s on CEC 2017 F16 – Single Run
=====
=
Function      : Hybrid: Schaffer; HGBat; Rosenbrock; Schwefel (100D)
Bounds       : [-100, 100]
Global optimum : 1600.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 3.124977e+04 | Diversity = 5.7220e+01
Generation 10: Best = 1.704994e+04 FES = 1100 Diversity = 4.4042e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.486056e+04 FES = 2100 Diversity = 3.7596e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 1.263194e+04 FES = 3100 Diversity = 3.4332e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.188023e+04 FES = 4100 Diversity = 3.2857e+01 k_t
= [2, 3, 4, 3]
Generation 50: Best = 1.188023e+04 FES = 5100 Diversity = 3.1185e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.188023e+04 FES = 6100 Diversity = 3.0592e+01 k_t
= [4, 3, 4, 3]
Generation 70: Best = 1.174373e+04 FES = 7100 Diversity = 2.9164e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.078038e+04 FES = 8100 Diversity = 2.8441e+01 k_t
= [4, 3, 4, 4]
Generation 90: Best = 1.078038e+04 FES = 9100 Diversity = 2.7634e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 9.947563e+03 FES = 10100 Diversity = 2.7053e+01 k_t
= [2, 2, 4, 4]
Generation 110: Best = 9.947563e+03 FES = 11100 Diversity = 2.6571e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 9.947562e+03 FES = 12100 Diversity = 2.6087e+01 k_t
= [3, 2, 2, 3]
Generation 130: Best = 9.947562e+03 FES = 13100 Diversity = 2.5915e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 9.404565e+03 FES = 14100 Diversity = 2.5731e+01 k_t
= [2, 2, 3, 4]
Generation 150: Best = 9.404565e+03 FES = 15100 Diversity = 2.5878e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 9.404565e+03 FES = 16100 Diversity = 2.5653e+01 k_t
= [4, 2, 4, 4]
Generation 170: Best = 9.404565e+03 FES = 17100 Diversity = 2.5654e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 9.404565e+03 FES = 18100 Diversity = 2.5714e+01 k_t
= [3, 3, 3, 3]
Generation 190: Best = 9.385133e+03 FES = 19100 Diversity = 2.5752e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 9.385133e+03 FES = 20100 Diversity = 2.5914e+01 k_t
= [3, 4, 3, 3]
Generation 210: Best = 9.017924e+03 FES = 21100 Diversity = 2.5850e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 9.017924e+03 FES = 22100 Diversity = 2.5780e+01 k_t
= [2, 2, 4, 3]

```

Generation 230:	Best = 9.017924e+03	FES = 23100	Diversity = 2.5855e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 9.017924e+03	FES = 24100	Diversity = 2.5774e+01	k_t
= [3, 3, 4, 4]				
Generation 250:	Best = 9.017924e+03	FES = 25100	Diversity = 2.5898e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 8.624613e+03	FES = 26100	Diversity = 2.5787e+01	k_t
= [2, 4, 4, 3]				
Generation 270:	Best = 8.399356e+03	FES = 27100	Diversity = 2.5690e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 8.399356e+03	FES = 28100	Diversity = 2.5695e+01	k_t
= [4, 2, 3, 3]				
Generation 290:	Best = 8.399356e+03	FES = 29100	Diversity = 2.5716e+01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 8.399356e+03	FES = 30100	Diversity = 2.5560e+01	k_t
= [2, 4, 3, 2]				
Generation 310:	Best = 8.399356e+03	FES = 31100	Diversity = 2.5692e+01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 8.399356e+03	FES = 32100	Diversity = 2.5603e+01	k_t
= [2, 2, 2, 4]				
Generation 330:	Best = 8.396935e+03	FES = 33100	Diversity = 2.5494e+01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 8.396935e+03	FES = 34100	Diversity = 2.5534e+01	k_t
= [2, 2, 2, 3]				
Generation 350:	Best = 8.386896e+03	FES = 35100	Diversity = 2.5514e+01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 8.386896e+03	FES = 36100	Diversity = 2.5391e+01	k_t
= [4, 3, 4, 4]				
Generation 370:	Best = 8.386896e+03	FES = 37100	Diversity = 2.5353e+01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 8.386896e+03	FES = 38100	Diversity = 2.5419e+01	k_t
= [4, 2, 3, 4]				
Generation 390:	Best = 8.386896e+03	FES = 39100	Diversity = 2.5439e+01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 8.385965e+03	FES = 40100	Diversity = 2.5535e+01	k_t
= [3, 3, 2, 4]				
Generation 410:	Best = 8.385965e+03	FES = 41100	Diversity = 2.5460e+01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 8.320632e+03	FES = 42100	Diversity = 2.5477e+01	k_t
= [2, 2, 3, 2]				
Generation 430:	Best = 8.320632e+03	FES = 43100	Diversity = 2.5408e+01	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 8.320327e+03	FES = 44100	Diversity = 2.5378e+01	k_t
= [4, 2, 4, 3]				
Generation 450:	Best = 8.287079e+03	FES = 45100	Diversity = 2.5340e+01	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 8.035569e+03	FES = 46100	Diversity = 2.5630e+01	k_t
= [2, 4, 3, 2]				
Generation 470:	Best = 8.028824e+03	FES = 47100	Diversity = 2.5679e+01	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 8.005168e+03	FES = 48100	Diversity = 2.5540e+01	k_t
= [4, 4, 4, 2]				
Generation 490:	Best = 8.001120e+03	FES = 49100	Diversity = 2.5269e+01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 8.001120e+03	FES = 50100	Diversity = 2.5143e+01	k_t
= [4, 4, 2, 3]				
Generation 510:	Best = 7.971360e+03	FES = 51100	Diversity = 2.5316e+01	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 7.927525e+03	FES = 52100	Diversity = 2.5187e+01	k_t
= [4, 3, 3, 4]				
Generation 530:	Best = 7.867771e+03	FES = 53100	Diversity = 2.5117e+01	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 7.720112e+03	FES = 54100	Diversity = 2.5343e+01	k_t
= [3, 2, 2, 2]				
Generation 550:	Best = 7.538007e+03	FES = 55100	Diversity = 2.5360e+01	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 7.538007e+03	FES = 56100	Diversity = 2.5322e+01	k_t
= [2, 3, 4, 3]				
Generation 570:	Best = 7.339593e+03	FES = 57100	Diversity = 2.5237e+01	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 7.198306e+03	FES = 58100	Diversity = 2.5225e+01	k_t
= [2, 3, 2, 3]				
Generation 590:	Best = 6.791493e+03	FES = 59100	Diversity = 2.5330e+01	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 6.791115e+03	FES = 60100	Diversity = 2.5141e+01	k_t
= [3, 4, 4, 3]				
Generation 610:	Best = 6.591547e+03	FES = 61100	Diversity = 2.5041e+01	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 6.573389e+03	FES = 62100	Diversity = 2.4968e+01	k_t
= [2, 2, 4, 4]				
Generation 630:	Best = 6.573389e+03	FES = 63100	Diversity = 2.4922e+01	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 6.569335e+03	FES = 64100	Diversity = 2.4940e+01	k_t
= [2, 3, 4, 4]				
Generation 650:	Best = 6.526692e+03	FES = 65100	Diversity = 2.4918e+01	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 6.500270e+03	FES = 66100	Diversity = 2.4919e+01	k_t
= [3, 2, 4, 4]				
Generation 670:	Best = 6.461604e+03	FES = 67100	Diversity = 2.4935e+01	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 6.451753e+03	FES = 68100	Diversity = 2.4881e+01	k_t
= [4, 3, 2, 4]				
Generation 690:	Best = 6.415368e+03	FES = 69100	Diversity = 2.4833e+01	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 6.415368e+03	FES = 70100	Diversity = 2.4380e+01	k_t
= [3, 2, 3, 3]				
Generation 710:	Best = 6.388774e+03	FES = 71100	Diversity = 2.3330e+01	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 6.206959e+03	FES = 72100	Diversity = 2.3084e+01	k_t
= [4, 4, 4, 3]				
Generation 730:	Best = 6.197640e+03	FES = 73100	Diversity = 2.2031e+01	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 6.068626e+03	FES = 74100	Diversity = 1.9970e+01	k_t
= [3, 2, 4, 4]				
Generation 750:	Best = 6.026083e+03	FES = 75100	Diversity = 1.7436e+01	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 5.933764e+03	FES = 76100	Diversity = 1.6247e+01	k_t
= [2, 4, 3, 4]				

Generation 770:	Best = 5.774365e+03	FES = 77100	Diversity = 1.5182e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 5.693819e+03	FES = 78100	Diversity = 1.4524e+01	k_t
= [3, 4, 2, 4]				
Generation 790:	Best = 5.559585e+03	FES = 79100	Diversity = 1.3270e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 5.493085e+03	FES = 80100	Diversity = 1.2461e+01	k_t
= [3, 3, 2, 4]				
Generation 810:	Best = 5.480282e+03	FES = 81100	Diversity = 1.1707e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 5.440954e+03	FES = 82100	Diversity = 1.0164e+01	k_t
= [4, 2, 4, 3]				
Generation 830:	Best = 5.410831e+03	FES = 83100	Diversity = 8.8189e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 5.379739e+03	FES = 84100	Diversity = 7.0026e+00	k_t
= [3, 2, 4, 3]				
Generation 850:	Best = 5.364446e+03	FES = 85100	Diversity = 6.0355e+00	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 5.351240e+03	FES = 86100	Diversity = 5.1422e+00	k_t
= [4, 2, 3, 4]				
Generation 870:	Best = 5.338824e+03	FES = 87100	Diversity = 4.7162e+00	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 5.334314e+03	FES = 88100	Diversity = 4.1762e+00	k_t
= [3, 4, 4, 2]				
Generation 890:	Best = 5.327967e+03	FES = 89100	Diversity = 3.8104e+00	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 5.324341e+03	FES = 90100	Diversity = 3.6664e+00	k_t
= [2, 2, 3, 2]				
Generation 910:	Best = 5.319982e+03	FES = 91100	Diversity = 3.3429e+00	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 5.315509e+03	FES = 92100	Diversity = 3.2146e+00	k_t
= [4, 4, 4, 3]				
Generation 930:	Best = 5.311641e+03	FES = 93100	Diversity = 3.2441e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 5.306470e+03	FES = 94100	Diversity = 3.5950e+00	k_t
= [3, 4, 2, 3]				
Generation 950:	Best = 5.299364e+03	FES = 95100	Diversity = 4.0765e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 5.297371e+03	FES = 96100	Diversity = 4.4877e+00	k_t
= [4, 2, 2, 2]				
Generation 970:	Best = 5.294869e+03	FES = 97100	Diversity = 4.3243e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 5.292681e+03	FES = 98100	Diversity = 4.4123e+00	k_t
= [4, 3, 3, 2]				
Generation 990:	Best = 5.291221e+03	FES = 99100	Diversity = 4.4597e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 5.289503e+03	FES = 100100	Diversity = 4.5910e+00	k_t
= [3, 3, 4, 2]				
Generation 1010:	Best = 5.287732e+03	FES = 101100	Diversity = 4.3834e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 5.285248e+03	FES = 102100	Diversity = 4.2845e+00	k_t
= [3, 3, 3, 4]				
Generation 1030:	Best = 5.285223e+03	FES = 103100	Diversity = 4.2136e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 5.283678e+03 = [2, 4, 4, 2]	FES = 104100	Diversity = 4.3113e+00	k_t
Generation 1050: Best = 5.282284e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 4.2465e+00	k_t
Generation 1060: Best = 5.280819e+03 = [4, 2, 2, 3]	FES = 106100	Diversity = 4.3599e+00	k_t
Generation 1070: Best = 5.278299e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 4.4168e+00	k_t
Generation 1080: Best = 5.278136e+03 = [4, 4, 4, 4]	FES = 108100	Diversity = 4.3628e+00	k_t
Generation 1090: Best = 5.276966e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 4.0164e+00	k_t
Generation 1100: Best = 5.276132e+03 = [3, 3, 2, 3]	FES = 110100	Diversity = 4.1309e+00	k_t
Generation 1110: Best = 5.273631e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 4.3389e+00	k_t
Generation 1120: Best = 5.273631e+03 = [3, 4, 2, 3]	FES = 112100	Diversity = 4.4476e+00	k_t
Generation 1130: Best = 5.272608e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 4.3879e+00	k_t
Generation 1140: Best = 5.272421e+03 = [2, 3, 3, 3]	FES = 114100	Diversity = 4.5418e+00	k_t
Generation 1150: Best = 5.271053e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 4.5695e+00	k_t
Generation 1160: Best = 5.270770e+03 = [4, 2, 3, 2]	FES = 116100	Diversity = 4.5226e+00	k_t
Generation 1170: Best = 5.269477e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 4.3759e+00	k_t
Generation 1180: Best = 5.269030e+03 = [3, 4, 2, 3]	FES = 118100	Diversity = 4.2389e+00	k_t
Generation 1190: Best = 5.267760e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 4.2403e+00	k_t
Generation 1200: Best = 5.266646e+03 = [2, 4, 4, 2]	FES = 120100	Diversity = 4.0367e+00	k_t
Generation 1210: Best = 5.266646e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 3.9878e+00	k_t
Generation 1220: Best = 5.266142e+03 = [3, 3, 2, 3]	FES = 122100	Diversity = 3.8880e+00	k_t
Generation 1230: Best = 5.265279e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 4.1177e+00	k_t
Generation 1240: Best = 5.265099e+03 = [2, 3, 2, 3]	FES = 124100	Diversity = 4.0741e+00	k_t
Generation 1250: Best = 5.264223e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 4.1237e+00	k_t
Generation 1260: Best = 5.264015e+03 = [4, 4, 3, 3]	FES = 126100	Diversity = 4.2398e+00	k_t
Generation 1270: Best = 5.263268e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 4.1545e+00	k_t
Generation 1280: Best = 5.263005e+03 = [3, 4, 4, 2]	FES = 128100	Diversity = 4.1377e+00	k_t
Generation 1290: Best = 5.262577e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 4.0828e+00	k_t
Generation 1300: Best = 5.262577e+03 = [2, 3, 4, 3]	FES = 130100	Diversity = 4.1058e+00	k_t

Generation 1310: Best = 5.261764e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 4.1043e+00	k_t
Generation 1320: Best = 5.261679e+03 = [3, 4, 3, 2]	FES = 132100	Diversity = 4.0765e+00	k_t
Generation 1330: Best = 5.261609e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 4.0614e+00	k_t
Generation 1340: Best = 5.260565e+03 = [2, 3, 4, 2]	FES = 134100	Diversity = 4.0733e+00	k_t
Generation 1350: Best = 5.260565e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 4.1233e+00	k_t
Generation 1360: Best = 5.260565e+03 = [2, 2, 4, 4]	FES = 136100	Diversity = 4.0739e+00	k_t
Generation 1370: Best = 5.260213e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 4.1476e+00	k_t
Generation 1380: Best = 5.260213e+03 = [4, 4, 2, 3]	FES = 138100	Diversity = 4.1441e+00	k_t
Generation 1390: Best = 5.259917e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 4.1536e+00	k_t
Generation 1400: Best = 5.259819e+03 = [3, 3, 2, 2]	FES = 140100	Diversity = 4.2078e+00	k_t
Generation 1410: Best = 5.259819e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 4.3185e+00	k_t
Generation 1420: Best = 5.259610e+03 = [4, 4, 2, 2]	FES = 142100	Diversity = 4.3440e+00	k_t
Generation 1430: Best = 5.259282e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 4.3451e+00	k_t
Generation 1440: Best = 5.258895e+03 = [4, 3, 2, 4]	FES = 144100	Diversity = 4.4144e+00	k_t
Generation 1450: Best = 5.257746e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 4.4925e+00	k_t
Generation 1460: Best = 5.257694e+03 = [3, 4, 4, 3]	FES = 146100	Diversity = 4.4669e+00	k_t
Generation 1470: Best = 5.257441e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 4.5018e+00	k_t
Generation 1480: Best = 5.257310e+03 = [4, 3, 3, 3]	FES = 148100	Diversity = 4.5870e+00	k_t
Generation 1490: Best = 5.256525e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 4.6452e+00	k_t
Generation 1500: Best = 5.256388e+03 = [3, 3, 3, 3]	FES = 150100	Diversity = 4.6266e+00	k_t
Generation 1510: Best = 5.255935e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 4.7588e+00	k_t
Generation 1520: Best = 5.255697e+03 = [3, 4, 2, 3]	FES = 152100	Diversity = 4.7327e+00	k_t
Generation 1530: Best = 5.255162e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 4.7852e+00	k_t
Generation 1540: Best = 5.254005e+03 = [3, 2, 4, 3]	FES = 154100	Diversity = 4.7303e+00	k_t
Generation 1550: Best = 5.252643e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 4.8111e+00	k_t
Generation 1560: Best = 5.252578e+03 = [4, 2, 2, 2]	FES = 156100	Diversity = 5.0523e+00	k_t
Generation 1570: Best = 5.252577e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 5.0583e+00	k_t

Generation 1580: Best = 5.252249e+03 = [2, 3, 4, 2]	FES = 158100	Diversity = 5.2155e+00	k_t
Generation 1590: Best = 5.251561e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 5.2573e+00	k_t
Generation 1600: Best = 5.250849e+03 = [3, 3, 3, 3]	FES = 160100	Diversity = 5.3494e+00	k_t
Generation 1610: Best = 5.249894e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 5.3318e+00	k_t
Generation 1620: Best = 5.249666e+03 = [2, 4, 4, 2]	FES = 162100	Diversity = 5.4247e+00	k_t
Generation 1630: Best = 5.248891e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 5.4799e+00	k_t
Generation 1640: Best = 5.247374e+03 = [4, 2, 2, 4]	FES = 164100	Diversity = 5.2820e+00	k_t
Generation 1650: Best = 5.247246e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 5.2171e+00	k_t
Generation 1660: Best = 5.246389e+03 = [2, 4, 2, 4]	FES = 166100	Diversity = 5.0511e+00	k_t
Generation 1670: Best = 5.245126e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 4.8849e+00	k_t
Generation 1680: Best = 5.244766e+03 = [2, 4, 4, 3]	FES = 168100	Diversity = 4.8173e+00	k_t
Generation 1690: Best = 5.243711e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 4.8657e+00	k_t
Generation 1700: Best = 5.243711e+03 = [4, 3, 2, 2]	FES = 170100	Diversity = 4.6649e+00	k_t
Generation 1710: Best = 5.243558e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 4.7170e+00	k_t
Generation 1720: Best = 5.243183e+03 = [2, 2, 4, 4]	FES = 172100	Diversity = 4.5963e+00	k_t
Generation 1730: Best = 5.242210e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 4.5405e+00	k_t
Generation 1740: Best = 5.242089e+03 = [2, 3, 2, 2]	FES = 174100	Diversity = 4.4927e+00	k_t
Generation 1750: Best = 5.241633e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 4.6183e+00	k_t
Generation 1760: Best = 5.240504e+03 = [3, 3, 3, 2]	FES = 176100	Diversity = 4.7150e+00	k_t
Generation 1770: Best = 5.240083e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 4.9744e+00	k_t
Generation 1780: Best = 5.239728e+03 = [3, 2, 3, 3]	FES = 178100	Diversity = 4.9376e+00	k_t
Generation 1790: Best = 5.239344e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 4.9317e+00	k_t
Generation 1800: Best = 5.237805e+03 = [4, 4, 3, 3]	FES = 180100	Diversity = 5.0276e+00	k_t
Generation 1810: Best = 5.237796e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 5.0551e+00	k_t
Generation 1820: Best = 5.236796e+03 = [4, 3, 3, 2]	FES = 182100	Diversity = 4.8454e+00	k_t
Generation 1830: Best = 5.235768e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 4.9356e+00	k_t
Generation 1840: Best = 5.234046e+03 = [4, 2, 3, 4]	FES = 184100	Diversity = 4.9598e+00	k_t

Generation 1850: Best = 5.234046e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 5.0745e+00	k_t
Generation 1860: Best = 5.233697e+03 = [2, 3, 3, 2]	FES = 186100	Diversity = 5.1857e+00	k_t
Generation 1870: Best = 5.232442e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 5.0506e+00	k_t
Generation 1880: Best = 5.231174e+03 = [3, 4, 2, 4]	FES = 188100	Diversity = 5.0777e+00	k_t
Generation 1890: Best = 5.230132e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 4.9534e+00	k_t
Generation 1900: Best = 5.229019e+03 = [2, 4, 2, 3]	FES = 190100	Diversity = 5.0346e+00	k_t
Generation 1910: Best = 5.229019e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 5.0100e+00	k_t
Generation 1920: Best = 5.228365e+03 = [4, 2, 4, 3]	FES = 192100	Diversity = 5.0843e+00	k_t
Generation 1930: Best = 5.227451e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 5.0130e+00	k_t
Generation 1940: Best = 5.226671e+03 = [3, 4, 4, 3]	FES = 194100	Diversity = 5.2672e+00	k_t
Generation 1950: Best = 5.225381e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 5.1408e+00	k_t
Generation 1960: Best = 5.225042e+03 = [2, 4, 3, 3]	FES = 196100	Diversity = 5.0978e+00	k_t
Generation 1970: Best = 5.222754e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 5.2699e+00	k_t
Generation 1980: Best = 5.222754e+03 = [3, 4, 3, 3]	FES = 198100	Diversity = 5.1513e+00	k_t
Generation 1990: Best = 5.221666e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 5.0282e+00	k_t
Generation 2000: Best = 5.221627e+03 = [4, 2, 4, 2]	FES = 200100	Diversity = 5.0080e+00	k_t
Generation 2010: Best = 5.221111e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 4.9804e+00	k_t
Generation 2020: Best = 5.220379e+03 = [2, 4, 2, 2]	FES = 202100	Diversity = 4.8415e+00	k_t
Generation 2030: Best = 5.219858e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 4.7392e+00	k_t
Generation 2040: Best = 5.219353e+03 = [2, 2, 4, 2]	FES = 204100	Diversity = 4.5645e+00	k_t
Generation 2050: Best = 5.218879e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 4.4366e+00	k_t
Generation 2060: Best = 5.218879e+03 = [2, 4, 3, 3]	FES = 206100	Diversity = 4.3479e+00	k_t
Generation 2070: Best = 5.218126e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 4.2081e+00	k_t
Generation 2080: Best = 5.217545e+03 = [4, 2, 3, 4]	FES = 208100	Diversity = 4.2551e+00	k_t
Generation 2090: Best = 5.217410e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 4.1297e+00	k_t
Generation 2100: Best = 5.216499e+03 = [4, 2, 2, 2]	FES = 210100	Diversity = 3.9715e+00	k_t
Generation 2110: Best = 5.216499e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 3.9958e+00	k_t

Generation 2120: Best = 5.216499e+03 = [4, 3, 2, 3]	FES = 212100	Diversity = 3.9255e+00	k_t
Generation 2130: Best = 5.216224e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 3.9028e+00	k_t
Generation 2140: Best = 5.216016e+03 = [4, 3, 3, 2]	FES = 214100	Diversity = 3.8526e+00	k_t
Generation 2150: Best = 5.216016e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 3.8439e+00	k_t
Generation 2160: Best = 5.215470e+03 = [2, 2, 4, 3]	FES = 216100	Diversity = 3.7966e+00	k_t
Generation 2170: Best = 5.215251e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 3.7820e+00	k_t
Generation 2180: Best = 5.214998e+03 = [4, 3, 2, 4]	FES = 218100	Diversity = 3.7176e+00	k_t
Generation 2190: Best = 5.214787e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 3.6563e+00	k_t
Generation 2200: Best = 5.214124e+03 = [2, 2, 4, 3]	FES = 220100	Diversity = 3.6870e+00	k_t
Generation 2210: Best = 5.214097e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 3.7251e+00	k_t
Generation 2220: Best = 5.214031e+03 = [2, 3, 2, 2]	FES = 222100	Diversity = 3.7025e+00	k_t
Generation 2230: Best = 5.213958e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 3.7445e+00	k_t
Generation 2240: Best = 5.213644e+03 = [3, 3, 2, 2]	FES = 224100	Diversity = 3.7872e+00	k_t
Generation 2250: Best = 5.213552e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 3.7593e+00	k_t
Generation 2260: Best = 5.213266e+03 = [2, 2, 2, 2]	FES = 226100	Diversity = 3.7440e+00	k_t
Generation 2270: Best = 5.212963e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 3.7047e+00	k_t
Generation 2280: Best = 5.212963e+03 = [3, 2, 4, 2]	FES = 228100	Diversity = 3.6703e+00	k_t
Generation 2290: Best = 5.212963e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 3.7109e+00	k_t
Generation 2300: Best = 5.212796e+03 = [2, 3, 2, 4]	FES = 230100	Diversity = 3.7393e+00	k_t
Generation 2310: Best = 5.212771e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 3.6975e+00	k_t
Generation 2320: Best = 5.212634e+03 = [2, 3, 2, 2]	FES = 232100	Diversity = 3.6899e+00	k_t
Generation 2330: Best = 5.212409e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 3.7398e+00	k_t
Generation 2340: Best = 5.212409e+03 = [3, 3, 2, 3]	FES = 234100	Diversity = 3.7268e+00	k_t
Generation 2350: Best = 5.212409e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 3.7529e+00	k_t
Generation 2360: Best = 5.212183e+03 = [2, 3, 4, 3]	FES = 236100	Diversity = 3.7257e+00	k_t
Generation 2370: Best = 5.212080e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 3.7186e+00	k_t
Generation 2380: Best = 5.212080e+03 = [4, 3, 3, 2]	FES = 238100	Diversity = 3.7814e+00	k_t

Generation 2390: Best = 5.212052e+03	FES = 239100	Diversity = 3.7883e+00	k_t
= [1, 1, 1, 1]			
Generation 2400: Best = 5.211999e+03	FES = 240100	Diversity = 3.7410e+00	k_t
= [4, 4, 3, 4]			
Generation 2410: Best = 5.211999e+03	FES = 241100	Diversity = 3.8269e+00	k_t
= [1, 1, 1, 1]			
Generation 2420: Best = 5.211999e+03	FES = 242100	Diversity = 3.8064e+00	k_t
= [4, 4, 3, 3]			
Generation 2430: Best = 5.211956e+03	FES = 243100	Diversity = 3.7755e+00	k_t
= [1, 1, 1, 1]			
Generation 2440: Best = 5.211956e+03	FES = 244100	Diversity = 3.7700e+00	k_t
= [2, 4, 2, 3]			
Generation 2450: Best = 5.211747e+03	FES = 245100	Diversity = 3.7394e+00	k_t
= [1, 1, 1, 1]			
Generation 2460: Best = 5.211694e+03	FES = 246100	Diversity = 3.7017e+00	k_t
= [3, 4, 3, 3]			
Generation 2470: Best = 5.211694e+03	FES = 247100	Diversity = 3.6652e+00	k_t
= [1, 1, 1, 1]			
Generation 2480: Best = 5.211582e+03	FES = 248100	Diversity = 3.6473e+00	k_t
= [4, 3, 3, 3]			
Generation 2490: Best = 5.211582e+03	FES = 249100	Diversity = 3.6752e+00	k_t
= [1, 1, 1, 1]			
Generation 2500: Best = 5.211582e+03	FES = 250100	Diversity = 3.6212e+00	k_t
= [3, 2, 2, 4]			
Generation 2510: Best = 5.211582e+03	FES = 251100	Diversity = 3.5957e+00	k_t
= [1, 1, 1, 1]			
Generation 2520: Best = 5.211582e+03	FES = 252100	Diversity = 3.5812e+00	k_t
= [4, 4, 3, 2]			
Generation 2530: Best = 5.211582e+03	FES = 253100	Diversity = 3.6260e+00	k_t
= [1, 1, 1, 1]			
Generation 2540: Best = 5.211495e+03	FES = 254100	Diversity = 3.5825e+00	k_t
= [2, 2, 3, 3]			
Generation 2550: Best = 5.211495e+03	FES = 255100	Diversity = 3.5685e+00	k_t
= [1, 1, 1, 1]			
Generation 2560: Best = 5.211495e+03	FES = 256100	Diversity = 3.6019e+00	k_t
= [2, 4, 2, 2]			
Generation 2570: Best = 5.211495e+03	FES = 257100	Diversity = 3.6329e+00	k_t
= [1, 1, 1, 1]			
Generation 2580: Best = 5.211495e+03	FES = 258100	Diversity = 3.5970e+00	k_t
= [4, 2, 3, 2]			
Generation 2590: Best = 5.211495e+03	FES = 259100	Diversity = 3.5952e+00	k_t
= [1, 1, 1, 1]			
Generation 2600: Best = 5.211443e+03	FES = 260100	Diversity = 3.6470e+00	k_t
= [2, 3, 2, 2]			
Generation 2610: Best = 5.211368e+03	FES = 261100	Diversity = 3.6278e+00	k_t
= [1, 1, 1, 1]			
Generation 2620: Best = 5.211313e+03	FES = 262100	Diversity = 3.6029e+00	k_t
= [3, 4, 3, 3]			
Generation 2630: Best = 5.211313e+03	FES = 263100	Diversity = 3.6275e+00	k_t
= [1, 1, 1, 1]			
Generation 2640: Best = 5.211313e+03	FES = 264100	Diversity = 3.6566e+00	k_t
= [2, 4, 4, 2]			
Generation 2650: Best = 5.211313e+03	FES = 265100	Diversity = 3.6860e+00	k_t
= [1, 1, 1, 1]			

Generation 2660: Best = 5.211313e+03 = [3, 2, 3, 3]	FES = 266100	Diversity = 3.6980e+00	k_t
Generation 2670: Best = 5.211313e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 3.7188e+00	k_t
Generation 2680: Best = 5.211313e+03 = [4, 4, 4, 3]	FES = 268100	Diversity = 3.7042e+00	k_t
Generation 2690: Best = 5.211313e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 3.6853e+00	k_t
Generation 2700: Best = 5.211188e+03 = [4, 3, 4, 3]	FES = 270100	Diversity = 3.7082e+00	k_t
Generation 2710: Best = 5.211129e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 3.7101e+00	k_t
Generation 2720: Best = 5.211129e+03 = [2, 3, 4, 4]	FES = 272100	Diversity = 3.6829e+00	k_t
Generation 2730: Best = 5.211129e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 3.7017e+00	k_t
Generation 2740: Best = 5.211002e+03 = [3, 2, 4, 4]	FES = 274100	Diversity = 3.7048e+00	k_t
Generation 2750: Best = 5.210876e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 3.7701e+00	k_t
Generation 2760: Best = 5.210876e+03 = [2, 3, 2, 3]	FES = 276100	Diversity = 3.7693e+00	k_t
Generation 2770: Best = 5.210876e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 3.7654e+00	k_t
Generation 2780: Best = 5.210876e+03 = [3, 3, 4, 4]	FES = 278100	Diversity = 3.7806e+00	k_t
Generation 2790: Best = 5.210834e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 3.7442e+00	k_t
Generation 2800: Best = 5.210834e+03 = [3, 3, 3, 4]	FES = 280100	Diversity = 3.6902e+00	k_t
Generation 2810: Best = 5.210400e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 3.7231e+00	k_t
Generation 2820: Best = 5.210400e+03 = [2, 2, 4, 3]	FES = 282100	Diversity = 3.8185e+00	k_t
Generation 2830: Best = 5.210400e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 3.8151e+00	k_t
Generation 2840: Best = 5.210400e+03 = [2, 3, 2, 2]	FES = 284100	Diversity = 3.8540e+00	k_t
Generation 2850: Best = 5.210354e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 3.8146e+00	k_t
Generation 2860: Best = 5.210354e+03 = [3, 2, 4, 4]	FES = 286100	Diversity = 3.8243e+00	k_t
Generation 2870: Best = 5.210173e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 3.8796e+00	k_t
Generation 2880: Best = 5.210173e+03 = [4, 4, 2, 2]	FES = 288100	Diversity = 3.8885e+00	k_t
Generation 2890: Best = 5.210173e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 3.9663e+00	k_t
Generation 2900: Best = 5.210173e+03 = [4, 3, 4, 2]	FES = 290100	Diversity = 4.0121e+00	k_t
Generation 2910: Best = 5.210132e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 4.0341e+00	k_t
Generation 2920: Best = 5.210132e+03 = [3, 2, 3, 4]	FES = 292100	Diversity = 4.0924e+00	k_t

Generation 2930: Best = 5.209987e+03 = [1, 1, 1, 1]	FES = 293100	Diversity = 4.1816e+00	k_t
Generation 2940: Best = 5.209753e+03 = [2, 2, 2, 2]	FES = 294100	Diversity = 4.1643e+00	k_t
Generation 2950: Best = 5.209666e+03 = [1, 1, 1, 1]	FES = 295100	Diversity = 4.1291e+00	k_t
Generation 2960: Best = 5.209666e+03 = [2, 2, 3, 4]	FES = 296100	Diversity = 4.1990e+00	k_t
Generation 2970: Best = 5.209568e+03 = [1, 1, 1, 1]	FES = 297100	Diversity = 4.3116e+00	k_t
Generation 2980: Best = 5.209300e+03 = [3, 3, 4, 2]	FES = 298100	Diversity = 4.3346e+00	k_t
Generation 2990: Best = 5.209300e+03 = [1, 1, 1, 1]	FES = 299100	Diversity = 4.3693e+00	k_t
Generation 3000: Best = 5.209300e+03 = [4, 4, 3, 2]	FES = 300100	Diversity = 4.3373e+00	k_t
Generation 3010: Best = 5.209300e+03 = [1, 1, 1, 1]	FES = 301100	Diversity = 4.2741e+00	k_t
Generation 3020: Best = 5.209300e+03 = [2, 4, 2, 2]	FES = 302100	Diversity = 4.2278e+00	k_t
Generation 3030: Best = 5.208725e+03 = [1, 1, 1, 1]	FES = 303100	Diversity = 4.2395e+00	k_t
Generation 3040: Best = 5.208680e+03 = [3, 3, 4, 3]	FES = 304100	Diversity = 4.1279e+00	k_t
Generation 3050: Best = 5.208673e+03 = [1, 1, 1, 1]	FES = 305100	Diversity = 4.0560e+00	k_t
Generation 3060: Best = 5.208673e+03 = [4, 2, 3, 2]	FES = 306100	Diversity = 4.0254e+00	k_t
Generation 3070: Best = 5.208673e+03 = [1, 1, 1, 1]	FES = 307100	Diversity = 3.8908e+00	k_t
Generation 3080: Best = 5.208673e+03 = [4, 4, 4, 2]	FES = 308100	Diversity = 3.8323e+00	k_t
Generation 3090: Best = 5.208673e+03 = [1, 1, 1, 1]	FES = 309100	Diversity = 3.8125e+00	k_t
Generation 3100: Best = 5.208663e+03 = [2, 3, 2, 4]	FES = 310100	Diversity = 3.7172e+00	k_t
Generation 3110: Best = 5.208663e+03 = [1, 1, 1, 1]	FES = 311100	Diversity = 3.6860e+00	k_t
Generation 3120: Best = 5.208663e+03 = [2, 4, 3, 2]	FES = 312100	Diversity = 3.7050e+00	k_t
Generation 3130: Best = 5.208660e+03 = [1, 1, 1, 1]	FES = 313100	Diversity = 3.7394e+00	k_t
Generation 3140: Best = 5.208611e+03 = [3, 2, 3, 3]	FES = 314100	Diversity = 3.7550e+00	k_t
Generation 3150: Best = 5.208523e+03 = [1, 1, 1, 1]	FES = 315100	Diversity = 3.7305e+00	k_t
Generation 3160: Best = 5.208396e+03 = [2, 2, 4, 3]	FES = 316100	Diversity = 3.7397e+00	k_t
Generation 3170: Best = 5.208085e+03 = [1, 1, 1, 1]	FES = 317100	Diversity = 3.8050e+00	k_t
Generation 3180: Best = 5.207994e+03 = [4, 2, 4, 3]	FES = 318100	Diversity = 3.7772e+00	k_t
Generation 3190: Best = 5.207885e+03 = [1, 1, 1, 1]	FES = 319100	Diversity = 3.7872e+00	k_t

Generation 3200: Best = 5.207885e+03 = [2, 2, 4, 2]	FES = 320100	Diversity = 3.7426e+00	k_t
Generation 3210: Best = 5.207885e+03 = [1, 1, 1, 1]	FES = 321100	Diversity = 3.7134e+00	k_t
Generation 3220: Best = 5.207885e+03 = [2, 2, 3, 2]	FES = 322100	Diversity = 3.6529e+00	k_t
Generation 3230: Best = 5.207885e+03 = [1, 1, 1, 1]	FES = 323100	Diversity = 3.6483e+00	k_t
Generation 3240: Best = 5.207885e+03 = [4, 2, 2, 3]	FES = 324100	Diversity = 3.6794e+00	k_t
Generation 3250: Best = 5.207885e+03 = [1, 1, 1, 1]	FES = 325100	Diversity = 3.6489e+00	k_t
Generation 3260: Best = 5.207491e+03 = [2, 3, 3, 3]	FES = 326100	Diversity = 3.6446e+00	k_t
Generation 3270: Best = 5.207491e+03 = [1, 1, 1, 1]	FES = 327100	Diversity = 3.6306e+00	k_t
Generation 3280: Best = 5.207491e+03 = [2, 2, 3, 3]	FES = 328100	Diversity = 3.6447e+00	k_t
Generation 3290: Best = 5.207357e+03 = [1, 1, 1, 1]	FES = 329100	Diversity = 3.7255e+00	k_t
Generation 3300: Best = 5.207357e+03 = [3, 2, 3, 4]	FES = 330100	Diversity = 3.7434e+00	k_t
Generation 3310: Best = 5.207357e+03 = [1, 1, 1, 1]	FES = 331100	Diversity = 3.7675e+00	k_t
Generation 3320: Best = 5.207348e+03 = [3, 2, 3, 4]	FES = 332100	Diversity = 3.6826e+00	k_t
Generation 3330: Best = 5.207348e+03 = [1, 1, 1, 1]	FES = 333100	Diversity = 3.7358e+00	k_t
Generation 3340: Best = 5.207218e+03 = [3, 2, 4, 2]	FES = 334100	Diversity = 3.7434e+00	k_t
Generation 3350: Best = 5.207218e+03 = [1, 1, 1, 1]	FES = 335100	Diversity = 3.7483e+00	k_t
Generation 3360: Best = 5.207218e+03 = [4, 3, 4, 3]	FES = 336100	Diversity = 3.7565e+00	k_t
Generation 3370: Best = 5.206823e+03 = [1, 1, 1, 1]	FES = 337100	Diversity = 3.7358e+00	k_t
Generation 3380: Best = 5.206823e+03 = [2, 3, 2, 2]	FES = 338100	Diversity = 3.7151e+00	k_t
Generation 3390: Best = 5.206823e+03 = [1, 1, 1, 1]	FES = 339100	Diversity = 3.7440e+00	k_t
Generation 3400: Best = 5.206823e+03 = [4, 4, 4, 4]	FES = 340100	Diversity = 3.7705e+00	k_t
Generation 3410: Best = 5.206823e+03 = [1, 1, 1, 1]	FES = 341100	Diversity = 3.8119e+00	k_t
Generation 3420: Best = 5.206776e+03 = [2, 4, 4, 2]	FES = 342100	Diversity = 3.9263e+00	k_t
Generation 3430: Best = 5.206763e+03 = [1, 1, 1, 1]	FES = 343100	Diversity = 3.9853e+00	k_t
Generation 3440: Best = 5.206763e+03 = [3, 2, 3, 4]	FES = 344100	Diversity = 3.9816e+00	k_t
Generation 3450: Best = 5.206763e+03 = [1, 1, 1, 1]	FES = 345100	Diversity = 3.9566e+00	k_t
Generation 3460: Best = 5.206763e+03 = [2, 2, 2, 4]	FES = 346100	Diversity = 3.9937e+00	k_t

Generation 3470: Best = 5.206758e+03 = [1, 1, 1, 1]	FES = 347100	Diversity = 3.9795e+00	k_t
Generation 3480: Best = 5.206758e+03 = [4, 3, 4, 2]	FES = 348100	Diversity = 3.9602e+00	k_t
Generation 3490: Best = 5.206613e+03 = [1, 1, 1, 1]	FES = 349100	Diversity = 3.9823e+00	k_t
Generation 3500: Best = 5.206606e+03 = [2, 2, 2, 2]	FES = 350100	Diversity = 3.9801e+00	k_t
Generation 3510: Best = 5.206342e+03 = [1, 1, 1, 1]	FES = 351100	Diversity = 4.0183e+00	k_t
Generation 3520: Best = 5.206342e+03 = [3, 2, 2, 4]	FES = 352100	Diversity = 4.0318e+00	k_t
Generation 3530: Best = 5.206334e+03 = [1, 1, 1, 1]	FES = 353100	Diversity = 3.9775e+00	k_t
Generation 3540: Best = 5.206189e+03 = [4, 3, 3, 2]	FES = 354100	Diversity = 3.9560e+00	k_t
Generation 3550: Best = 5.206189e+03 = [1, 1, 1, 1]	FES = 355100	Diversity = 3.9282e+00	k_t
Generation 3560: Best = 5.206189e+03 = [3, 4, 2, 4]	FES = 356100	Diversity = 3.9228e+00	k_t
Generation 3570: Best = 5.206189e+03 = [1, 1, 1, 1]	FES = 357100	Diversity = 3.9671e+00	k_t
Generation 3580: Best = 5.206119e+03 = [2, 3, 4, 2]	FES = 358100	Diversity = 4.0209e+00	k_t
Generation 3590: Best = 5.206070e+03 = [1, 1, 1, 1]	FES = 359100	Diversity = 4.0218e+00	k_t
Generation 3600: Best = 5.206070e+03 = [4, 3, 2, 2]	FES = 360100	Diversity = 4.0275e+00	k_t
Generation 3610: Best = 5.206070e+03 = [1, 1, 1, 1]	FES = 361100	Diversity = 4.0207e+00	k_t
Generation 3620: Best = 5.206070e+03 = [3, 4, 2, 3]	FES = 362100	Diversity = 3.9725e+00	k_t
Generation 3630: Best = 5.206048e+03 = [1, 1, 1, 1]	FES = 363100	Diversity = 3.9437e+00	k_t
Generation 3640: Best = 5.206048e+03 = [4, 2, 2, 4]	FES = 364100	Diversity = 3.9431e+00	k_t
Generation 3650: Best = 5.205988e+03 = [1, 1, 1, 1]	FES = 365100	Diversity = 3.9621e+00	k_t
Generation 3660: Best = 5.205988e+03 = [2, 3, 2, 2]	FES = 366100	Diversity = 3.9212e+00	k_t
Generation 3670: Best = 5.205839e+03 = [1, 1, 1, 1]	FES = 367100	Diversity = 3.9397e+00	k_t
Generation 3680: Best = 5.205839e+03 = [2, 4, 2, 4]	FES = 368100	Diversity = 3.9148e+00	k_t
Generation 3690: Best = 5.205839e+03 = [1, 1, 1, 1]	FES = 369100	Diversity = 3.9039e+00	k_t
Generation 3700: Best = 5.205732e+03 = [3, 2, 2, 2]	FES = 370100	Diversity = 3.9209e+00	k_t
Generation 3710: Best = 5.205732e+03 = [1, 1, 1, 1]	FES = 371100	Diversity = 3.9116e+00	k_t
Generation 3720: Best = 5.205732e+03 = [3, 2, 2, 4]	FES = 372100	Diversity = 3.8341e+00	k_t
Generation 3730: Best = 5.205732e+03 = [1, 1, 1, 1]	FES = 373100	Diversity = 3.8311e+00	k_t

Generation 3740: Best = 5.205683e+03 = [4, 3, 3, 3]	FES = 374100	Diversity = 3.7996e+00	k_t
Generation 3750: Best = 5.205654e+03 = [1, 1, 1, 1]	FES = 375100	Diversity = 3.7890e+00	k_t
Generation 3760: Best = 5.205617e+03 = [2, 4, 4, 4]	FES = 376100	Diversity = 3.7988e+00	k_t
Generation 3770: Best = 5.205568e+03 = [1, 1, 1, 1]	FES = 377100	Diversity = 3.7511e+00	k_t
Generation 3780: Best = 5.205568e+03 = [4, 2, 2, 3]	FES = 378100	Diversity = 3.7411e+00	k_t
Generation 3790: Best = 5.205541e+03 = [1, 1, 1, 1]	FES = 379100	Diversity = 3.7657e+00	k_t
Generation 3800: Best = 5.205540e+03 = [4, 3, 2, 4]	FES = 380100	Diversity = 3.8198e+00	k_t
Generation 3810: Best = 5.205536e+03 = [1, 1, 1, 1]	FES = 381100	Diversity = 3.7767e+00	k_t
Generation 3820: Best = 5.205465e+03 = [3, 3, 3, 3]	FES = 382100	Diversity = 3.7352e+00	k_t
Generation 3830: Best = 5.205465e+03 = [1, 1, 1, 1]	FES = 383100	Diversity = 3.7131e+00	k_t
Generation 3840: Best = 5.205465e+03 = [2, 3, 3, 3]	FES = 384100	Diversity = 3.7359e+00	k_t
Generation 3850: Best = 5.205465e+03 = [1, 1, 1, 1]	FES = 385100	Diversity = 3.7059e+00	k_t
Generation 3860: Best = 5.205465e+03 = [2, 3, 3, 2]	FES = 386100	Diversity = 3.7009e+00	k_t
Generation 3870: Best = 5.205465e+03 = [1, 1, 1, 1]	FES = 387100	Diversity = 3.6168e+00	k_t
Generation 3880: Best = 5.205434e+03 = [4, 3, 4, 3]	FES = 388100	Diversity = 3.5845e+00	k_t
Generation 3890: Best = 5.205434e+03 = [1, 1, 1, 1]	FES = 389100	Diversity = 3.5879e+00	k_t
Generation 3900: Best = 5.205434e+03 = [4, 2, 3, 4]	FES = 390100	Diversity = 3.6219e+00	k_t
Generation 3910: Best = 5.205434e+03 = [1, 1, 1, 1]	FES = 391100	Diversity = 3.6079e+00	k_t
Generation 3920: Best = 5.205434e+03 = [2, 4, 4, 4]	FES = 392100	Diversity = 3.6169e+00	k_t
Generation 3930: Best = 5.205434e+03 = [1, 1, 1, 1]	FES = 393100	Diversity = 3.5585e+00	k_t
Generation 3940: Best = 5.205434e+03 = [3, 4, 2, 3]	FES = 394100	Diversity = 3.5582e+00	k_t
Generation 3950: Best = 5.205282e+03 = [1, 1, 1, 1]	FES = 395100	Diversity = 3.5687e+00	k_t
Generation 3960: Best = 5.205282e+03 = [3, 2, 3, 4]	FES = 396100	Diversity = 3.5584e+00	k_t
Generation 3970: Best = 5.205282e+03 = [1, 1, 1, 1]	FES = 397100	Diversity = 3.5343e+00	k_t
Generation 3980: Best = 5.205282e+03 = [2, 4, 4, 2]	FES = 398100	Diversity = 3.5118e+00	k_t
Generation 3990: Best = 5.205003e+03 = [1, 1, 1, 1]	FES = 399100	Diversity = 3.4730e+00	k_t
Generation 4000: Best = 5.204641e+03 = [2, 4, 2, 2]	FES = 400100	Diversity = 3.4744e+00	k_t

Generation 4010: Best = 5.204641e+03 = [1, 1, 1, 1]	FES = 401100	Diversity = 3.4681e+00	k_t
Generation 4020: Best = 5.204641e+03 = [4, 2, 2, 4]	FES = 402100	Diversity = 3.4688e+00	k_t
Generation 4030: Best = 5.204641e+03 = [1, 1, 1, 1]	FES = 403100	Diversity = 3.4721e+00	k_t
Generation 4040: Best = 5.204629e+03 = [3, 3, 4, 3]	FES = 404100	Diversity = 3.4605e+00	k_t
Generation 4050: Best = 5.204629e+03 = [1, 1, 1, 1]	FES = 405100	Diversity = 3.4465e+00	k_t
Generation 4060: Best = 5.204629e+03 = [4, 3, 4, 3]	FES = 406100	Diversity = 3.4545e+00	k_t
Generation 4070: Best = 5.204629e+03 = [1, 1, 1, 1]	FES = 407100	Diversity = 3.4572e+00	k_t
Generation 4080: Best = 5.204629e+03 = [3, 4, 4, 3]	FES = 408100	Diversity = 3.4596e+00	k_t
Generation 4090: Best = 5.204502e+03 = [1, 1, 1, 1]	FES = 409100	Diversity = 3.4390e+00	k_t
Generation 4100: Best = 5.204502e+03 = [4, 4, 2, 4]	FES = 410100	Diversity = 3.3809e+00	k_t
Generation 4110: Best = 5.204502e+03 = [1, 1, 1, 1]	FES = 411100	Diversity = 3.3653e+00	k_t
Generation 4120: Best = 5.204502e+03 = [3, 2, 2, 2]	FES = 412100	Diversity = 3.3602e+00	k_t
Generation 4130: Best = 5.204491e+03 = [1, 1, 1, 1]	FES = 413100	Diversity = 3.3602e+00	k_t
Generation 4140: Best = 5.204483e+03 = [3, 2, 2, 4]	FES = 414100	Diversity = 3.3480e+00	k_t
Generation 4150: Best = 5.204483e+03 = [1, 1, 1, 1]	FES = 415100	Diversity = 3.3719e+00	k_t
Generation 4160: Best = 5.204483e+03 = [2, 4, 2, 3]	FES = 416100	Diversity = 3.3528e+00	k_t
Generation 4170: Best = 5.204483e+03 = [1, 1, 1, 1]	FES = 417100	Diversity = 3.3547e+00	k_t
Generation 4180: Best = 5.204449e+03 = [4, 4, 4, 3]	FES = 418100	Diversity = 3.3397e+00	k_t
Generation 4190: Best = 5.204449e+03 = [1, 1, 1, 1]	FES = 419100	Diversity = 3.3110e+00	k_t
Generation 4200: Best = 5.204445e+03 = [2, 2, 4, 3]	FES = 420100	Diversity = 3.2757e+00	k_t
Generation 4210: Best = 5.204445e+03 = [1, 1, 1, 1]	FES = 421100	Diversity = 3.2845e+00	k_t
Generation 4220: Best = 5.204445e+03 = [2, 4, 2, 4]	FES = 422100	Diversity = 3.2933e+00	k_t
Generation 4230: Best = 5.204445e+03 = [1, 1, 1, 1]	FES = 423100	Diversity = 3.2949e+00	k_t
Generation 4240: Best = 5.204439e+03 = [3, 2, 2, 2]	FES = 424100	Diversity = 3.2817e+00	k_t
Generation 4250: Best = 5.204430e+03 = [1, 1, 1, 1]	FES = 425100	Diversity = 3.2810e+00	k_t
Generation 4260: Best = 5.204430e+03 = [3, 2, 4, 4]	FES = 426100	Diversity = 3.2808e+00	k_t
Generation 4270: Best = 5.204430e+03 = [1, 1, 1, 1]	FES = 427100	Diversity = 3.2673e+00	k_t

Generation 4280: Best = 5.204424e+03	FES = 428100	Diversity = 3.2697e+00	k_t
= [2, 3, 2, 3]			
Generation 4290: Best = 5.204424e+03	FES = 429100	Diversity = 3.2695e+00	k_t
= [1, 1, 1, 1]			
Generation 4300: Best = 5.204418e+03	FES = 430100	Diversity = 3.2752e+00	k_t
= [2, 2, 3, 4]			
Generation 4310: Best = 5.204418e+03	FES = 431100	Diversity = 3.2747e+00	k_t
= [1, 1, 1, 1]			
Generation 4320: Best = 5.204418e+03	FES = 432100	Diversity = 3.3148e+00	k_t
= [3, 3, 2, 2]			
Generation 4330: Best = 5.204418e+03	FES = 433100	Diversity = 3.3148e+00	k_t
= [1, 1, 1, 1]			
Generation 4340: Best = 5.204418e+03	FES = 434100	Diversity = 3.3161e+00	k_t
= [2, 2, 3, 4]			
Generation 4350: Best = 5.204418e+03	FES = 435100	Diversity = 3.3288e+00	k_t
= [1, 1, 1, 1]			
Generation 4360: Best = 5.204396e+03	FES = 436100	Diversity = 3.3180e+00	k_t
= [2, 2, 2, 3]			
Generation 4370: Best = 5.204396e+03	FES = 437100	Diversity = 3.3088e+00	k_t
= [1, 1, 1, 1]			
Generation 4380: Best = 5.204396e+03	FES = 438100	Diversity = 3.3109e+00	k_t
= [2, 2, 4, 3]			
Generation 4390: Best = 5.204396e+03	FES = 439100	Diversity = 3.3049e+00	k_t
= [1, 1, 1, 1]			
Generation 4400: Best = 5.204396e+03	FES = 440100	Diversity = 3.2941e+00	k_t
= [2, 4, 2, 3]			
Generation 4410: Best = 5.204385e+03	FES = 441100	Diversity = 3.3102e+00	k_t
= [1, 1, 1, 1]			
Generation 4420: Best = 5.204385e+03	FES = 442100	Diversity = 3.3005e+00	k_t
= [3, 2, 2, 4]			
Generation 4430: Best = 5.204382e+03	FES = 443100	Diversity = 3.3122e+00	k_t
= [1, 1, 1, 1]			
Generation 4440: Best = 5.204382e+03	FES = 444100	Diversity = 3.3657e+00	k_t
= [4, 4, 3, 3]			
Generation 4450: Best = 5.204382e+03	FES = 445100	Diversity = 3.3488e+00	k_t
= [1, 1, 1, 1]			
Generation 4460: Best = 5.204382e+03	FES = 446100	Diversity = 3.3430e+00	k_t
= [4, 2, 3, 2]			
Generation 4470: Best = 5.204382e+03	FES = 447100	Diversity = 3.2992e+00	k_t
= [1, 1, 1, 1]			
Generation 4480: Best = 5.204382e+03	FES = 448100	Diversity = 3.3038e+00	k_t
= [2, 2, 2, 3]			
Generation 4490: Best = 5.204363e+03	FES = 449100	Diversity = 3.2365e+00	k_t
= [1, 1, 1, 1]			
Generation 4500: Best = 5.204363e+03	FES = 450100	Diversity = 3.2394e+00	k_t
= [2, 3, 3, 4]			
Generation 4510: Best = 5.204363e+03	FES = 451100	Diversity = 3.2622e+00	k_t
= [1, 1, 1, 1]			
Generation 4520: Best = 5.204339e+03	FES = 452100	Diversity = 3.2153e+00	k_t
= [4, 2, 3, 4]			
Generation 4530: Best = 5.204339e+03	FES = 453100	Diversity = 3.2126e+00	k_t
= [1, 1, 1, 1]			
Generation 4540: Best = 5.204339e+03	FES = 454100	Diversity = 3.1994e+00	k_t
= [2, 3, 4, 4]			

Generation 4550: Best = 5.204339e+03 = [1, 1, 1, 1]	FES = 455100	Diversity = 3.2185e+00	k_t
Generation 4560: Best = 5.204339e+03 = [2, 4, 2, 2]	FES = 456100	Diversity = 3.1828e+00	k_t
Generation 4570: Best = 5.204316e+03 = [1, 1, 1, 1]	FES = 457100	Diversity = 3.1345e+00	k_t
Generation 4580: Best = 5.204316e+03 = [3, 4, 2, 4]	FES = 458100	Diversity = 3.1347e+00	k_t
Generation 4590: Best = 5.204316e+03 = [1, 1, 1, 1]	FES = 459100	Diversity = 3.1130e+00	k_t
Generation 4600: Best = 5.204316e+03 = [2, 2, 3, 4]	FES = 460100	Diversity = 3.0793e+00	k_t
Generation 4610: Best = 5.204316e+03 = [1, 1, 1, 1]	FES = 461100	Diversity = 3.0677e+00	k_t
Generation 4620: Best = 5.204310e+03 = [2, 3, 2, 3]	FES = 462100	Diversity = 3.0514e+00	k_t
Generation 4630: Best = 5.204291e+03 = [1, 1, 1, 1]	FES = 463100	Diversity = 3.0516e+00	k_t
Generation 4640: Best = 5.204291e+03 = [4, 3, 4, 4]	FES = 464100	Diversity = 3.0399e+00	k_t
Generation 4650: Best = 5.204162e+03 = [1, 1, 1, 1]	FES = 465100	Diversity = 3.0400e+00	k_t
Generation 4660: Best = 5.204162e+03 = [4, 4, 2, 4]	FES = 466100	Diversity = 3.0354e+00	k_t
Generation 4670: Best = 5.204162e+03 = [1, 1, 1, 1]	FES = 467100	Diversity = 2.9788e+00	k_t
Generation 4680: Best = 5.204162e+03 = [3, 2, 3, 2]	FES = 468100	Diversity = 2.9209e+00	k_t
Generation 4690: Best = 5.204162e+03 = [1, 1, 1, 1]	FES = 469100	Diversity = 2.9150e+00	k_t
Generation 4700: Best = 5.204162e+03 = [3, 2, 3, 3]	FES = 470100	Diversity = 2.7899e+00	k_t
Generation 4710: Best = 5.204149e+03 = [1, 1, 1, 1]	FES = 471100	Diversity = 2.7477e+00	k_t
Generation 4720: Best = 5.204149e+03 = [4, 3, 4, 2]	FES = 472100	Diversity = 2.7392e+00	k_t
Generation 4730: Best = 5.204047e+03 = [1, 1, 1, 1]	FES = 473100	Diversity = 2.6982e+00	k_t
Generation 4740: Best = 5.203928e+03 = [2, 4, 2, 4]	FES = 474100	Diversity = 2.6494e+00	k_t
Generation 4750: Best = 5.203928e+03 = [1, 1, 1, 1]	FES = 475100	Diversity = 2.6177e+00	k_t
Generation 4760: Best = 5.203919e+03 = [4, 4, 2, 4]	FES = 476100	Diversity = 2.5894e+00	k_t
Generation 4770: Best = 5.203889e+03 = [1, 1, 1, 1]	FES = 477100	Diversity = 2.5372e+00	k_t
Generation 4780: Best = 5.203869e+03 = [4, 2, 2, 3]	FES = 478100	Diversity = 2.4355e+00	k_t
Generation 4790: Best = 5.203799e+03 = [1, 1, 1, 1]	FES = 479100	Diversity = 2.3819e+00	k_t
Generation 4800: Best = 5.203797e+03 = [4, 3, 2, 3]	FES = 480100	Diversity = 2.2457e+00	k_t
Generation 4810: Best = 5.203781e+03 = [1, 1, 1, 1]	FES = 481100	Diversity = 2.0192e+00	k_t

Generation 4820: Best = 5.203595e+03 = [3, 3, 4, 3]	FES = 482100	Diversity = 1.7862e+00	k_t
Generation 4830: Best = 5.203595e+03 = [1, 1, 1, 1]	FES = 483100	Diversity = 1.7133e+00	k_t
Generation 4840: Best = 5.203595e+03 = [3, 4, 4, 4]	FES = 484100	Diversity = 1.4392e+00	k_t
Generation 4850: Best = 5.203595e+03 = [1, 1, 1, 1]	FES = 485100	Diversity = 1.1706e+00	k_t
Generation 4860: Best = 5.203557e+03 = [4, 3, 4, 2]	FES = 486100	Diversity = 9.1590e-01	k_t
Generation 4870: Best = 5.203557e+03 = [1, 1, 1, 1]	FES = 487100	Diversity = 5.3890e-01	k_t
Generation 4880: Best = 5.203545e+03 = [4, 3, 4, 3]	FES = 488100	Diversity = 5.3481e-01	k_t
Generation 4890: Best = 5.203486e+03 = [1, 1, 1, 1]	FES = 489100	Diversity = 5.2726e-01	k_t
Generation 4900: Best = 5.203474e+03 = [2, 4, 4, 4]	FES = 490100	Diversity = 5.2222e-01	k_t
Generation 4910: Best = 5.203440e+03 = [1, 1, 1, 1]	FES = 491100	Diversity = 1.2736e-01	k_t
Generation 4920: Best = 5.203413e+03 = [4, 3, 4, 2]	FES = 492100	Diversity = 1.1886e-01	k_t
Generation 4930: Best = 5.203402e+03 = [1, 1, 1, 1]	FES = 493100	Diversity = 1.0824e-01	k_t
Generation 4940: Best = 5.203362e+03 = [3, 2, 2, 2]	FES = 494100	Diversity = 1.0155e-01	k_t
Generation 4950: Best = 5.203359e+03 = [1, 1, 1, 1]	FES = 495100	Diversity = 9.2876e-02	k_t
Generation 4960: Best = 5.203351e+03 = [3, 4, 2, 2]	FES = 496100	Diversity = 9.0306e-02	k_t
Generation 4970: Best = 5.203347e+03 = [1, 1, 1, 1]	FES = 497100	Diversity = 8.4307e-02	k_t
Generation 4980: Best = 5.203317e+03 = [3, 2, 4, 4]	FES = 498100	Diversity = 7.4465e-02	k_t
Generation 4990: Best = 5.203314e+03 = [1, 1, 1, 1]	FES = 499100	Diversity = 7.2257e-02	k_t
Generation 5000: Best = 5.203304e+03 = [4, 2, 2, 4]	FES = 500100	Diversity = 6.9801e-02	k_t
Generation 5010: Best = 5.203280e+03 = [1, 1, 1, 1]	FES = 501100	Diversity = 6.8448e-02	k_t
Generation 5020: Best = 5.203260e+03 = [3, 3, 4, 4]	FES = 502100	Diversity = 6.4136e-02	k_t
Generation 5030: Best = 5.203230e+03 = [1, 1, 1, 1]	FES = 503100	Diversity = 6.1706e-02	k_t
Generation 5040: Best = 5.203230e+03 = [2, 3, 3, 3]	FES = 504100	Diversity = 5.9411e-02	k_t
Generation 5050: Best = 5.203230e+03 = [1, 1, 1, 1]	FES = 505100	Diversity = 5.8867e-02	k_t
Generation 5060: Best = 5.203230e+03 = [4, 4, 2, 3]	FES = 506100	Diversity = 5.7496e-02	k_t
Generation 5070: Best = 5.203226e+03 = [1, 1, 1, 1]	FES = 507100	Diversity = 5.4229e-02	k_t
Generation 5080: Best = 5.203214e+03 = [2, 4, 4, 4]	FES = 508100	Diversity = 5.3302e-02	k_t

Generation 5090: Best = 5.203205e+03 = [1, 1, 1, 1]	FES = 509100	Diversity = 5.0800e-02	k_t
Generation 5100: Best = 5.203203e+03 = [2, 2, 2, 2]	FES = 510100	Diversity = 5.0318e-02	k_t
Generation 5110: Best = 5.203203e+03 = [1, 1, 1, 1]	FES = 511100	Diversity = 5.0121e-02	k_t
Generation 5120: Best = 5.203191e+03 = [4, 4, 2, 4]	FES = 512100	Diversity = 4.8588e-02	k_t
Generation 5130: Best = 5.203185e+03 = [1, 1, 1, 1]	FES = 513100	Diversity = 4.7181e-02	k_t
Generation 5140: Best = 5.203176e+03 = [3, 2, 3, 4]	FES = 514100	Diversity = 4.6036e-02	k_t
Generation 5150: Best = 5.203175e+03 = [1, 1, 1, 1]	FES = 515100	Diversity = 4.5322e-02	k_t
Generation 5160: Best = 5.203173e+03 = [3, 3, 3, 4]	FES = 516100	Diversity = 4.3126e-02	k_t
Generation 5170: Best = 5.203168e+03 = [1, 1, 1, 1]	FES = 517100	Diversity = 4.1906e-02	k_t
Generation 5180: Best = 5.203151e+03 = [4, 4, 2, 3]	FES = 518100	Diversity = 3.9738e-02	k_t
Generation 5190: Best = 5.203151e+03 = [1, 1, 1, 1]	FES = 519100	Diversity = 3.9008e-02	k_t
Generation 5200: Best = 5.203147e+03 = [2, 3, 3, 3]	FES = 520100	Diversity = 3.8375e-02	k_t
Generation 5210: Best = 5.203147e+03 = [1, 1, 1, 1]	FES = 521100	Diversity = 3.7748e-02	k_t
Generation 5220: Best = 5.203142e+03 = [3, 4, 2, 4]	FES = 522100	Diversity = 3.6674e-02	k_t
Generation 5230: Best = 5.203136e+03 = [1, 1, 1, 1]	FES = 523100	Diversity = 3.5805e-02	k_t
Generation 5240: Best = 5.203136e+03 = [3, 3, 2, 3]	FES = 524100	Diversity = 3.5174e-02	k_t
Generation 5250: Best = 5.203136e+03 = [1, 1, 1, 1]	FES = 525100	Diversity = 3.4144e-02	k_t
Generation 5260: Best = 5.203129e+03 = [2, 2, 3, 3]	FES = 526100	Diversity = 3.2748e-02	k_t
Generation 5270: Best = 5.203123e+03 = [1, 1, 1, 1]	FES = 527100	Diversity = 3.2247e-02	k_t
Generation 5280: Best = 5.203116e+03 = [4, 4, 2, 3]	FES = 528100	Diversity = 3.0958e-02	k_t
Generation 5290: Best = 5.203106e+03 = [1, 1, 1, 1]	FES = 529100	Diversity = 2.8087e-02	k_t
Generation 5300: Best = 5.203106e+03 = [4, 4, 3, 4]	FES = 530100	Diversity = 2.7979e-02	k_t
Generation 5310: Best = 5.203103e+03 = [1, 1, 1, 1]	FES = 531100	Diversity = 2.7730e-02	k_t
Generation 5320: Best = 5.203100e+03 = [2, 3, 2, 4]	FES = 532100	Diversity = 2.6258e-02	k_t
Generation 5330: Best = 5.203094e+03 = [1, 1, 1, 1]	FES = 533100	Diversity = 2.5473e-02	k_t
Generation 5340: Best = 5.203092e+03 = [2, 3, 3, 2]	FES = 534100	Diversity = 2.5492e-02	k_t
Generation 5350: Best = 5.203092e+03 = [1, 1, 1, 1]	FES = 535100	Diversity = 2.5977e-02	k_t

Generation 5360: Best = 5.203091e+03 = [2, 2, 2, 4]	FES = 536100	Diversity = 2.5156e-02	k_t
Generation 5370: Best = 5.203086e+03 = [1, 1, 1, 1]	FES = 537100	Diversity = 2.4626e-02	k_t
Generation 5380: Best = 5.203086e+03 = [2, 2, 4, 3]	FES = 538100	Diversity = 2.4187e-02	k_t
Generation 5390: Best = 5.203081e+03 = [1, 1, 1, 1]	FES = 539100	Diversity = 2.4296e-02	k_t
Generation 5400: Best = 5.203075e+03 = [3, 2, 3, 3]	FES = 540100	Diversity = 2.4519e-02	k_t
Generation 5410: Best = 5.203075e+03 = [1, 1, 1, 1]	FES = 541100	Diversity = 2.4099e-02	k_t
Generation 5420: Best = 5.203072e+03 = [4, 2, 2, 3]	FES = 542100	Diversity = 2.4423e-02	k_t
Generation 5430: Best = 5.203063e+03 = [1, 1, 1, 1]	FES = 543100	Diversity = 2.2823e-02	k_t
Generation 5440: Best = 5.203063e+03 = [4, 4, 3, 3]	FES = 544100	Diversity = 2.2820e-02	k_t
Generation 5450: Best = 5.203063e+03 = [1, 1, 1, 1]	FES = 545100	Diversity = 2.2584e-02	k_t
Generation 5460: Best = 5.203053e+03 = [4, 3, 2, 3]	FES = 546100	Diversity = 2.2232e-02	k_t
Generation 5470: Best = 5.203053e+03 = [1, 1, 1, 1]	FES = 547100	Diversity = 2.1878e-02	k_t
Generation 5480: Best = 5.203048e+03 = [3, 3, 4, 3]	FES = 548100	Diversity = 2.1430e-02	k_t
Generation 5490: Best = 5.203048e+03 = [1, 1, 1, 1]	FES = 549100	Diversity = 2.2078e-02	k_t
Generation 5500: Best = 5.203045e+03 = [2, 2, 2, 2]	FES = 550100	Diversity = 2.1584e-02	k_t
Generation 5510: Best = 5.203041e+03 = [1, 1, 1, 1]	FES = 551100	Diversity = 2.0478e-02	k_t
Generation 5520: Best = 5.203041e+03 = [3, 3, 3, 2]	FES = 552100	Diversity = 1.9071e-02	k_t
Generation 5530: Best = 5.203041e+03 = [1, 1, 1, 1]	FES = 553100	Diversity = 1.8425e-02	k_t
Generation 5540: Best = 5.203040e+03 = [2, 2, 4, 3]	FES = 554100	Diversity = 1.8355e-02	k_t
Generation 5550: Best = 5.203036e+03 = [1, 1, 1, 1]	FES = 555100	Diversity = 1.8336e-02	k_t
Generation 5560: Best = 5.203033e+03 = [3, 3, 2, 3]	FES = 556100	Diversity = 1.8555e-02	k_t
Generation 5570: Best = 5.203032e+03 = [1, 1, 1, 1]	FES = 557100	Diversity = 1.8908e-02	k_t
Generation 5580: Best = 5.203030e+03 = [3, 3, 4, 2]	FES = 558100	Diversity = 1.8709e-02	k_t
Generation 5590: Best = 5.203029e+03 = [1, 1, 1, 1]	FES = 559100	Diversity = 1.8759e-02	k_t
Generation 5600: Best = 5.203026e+03 = [4, 2, 2, 4]	FES = 560100	Diversity = 1.8367e-02	k_t
Generation 5610: Best = 5.203023e+03 = [1, 1, 1, 1]	FES = 561100	Diversity = 1.8612e-02	k_t
Generation 5620: Best = 5.203014e+03 = [3, 2, 3, 4]	FES = 562100	Diversity = 1.9495e-02	k_t

Generation 5630: Best = 5.203013e+03 = [1, 1, 1, 1]	FES = 563100	Diversity = 1.9001e-02	k_t
Generation 5640: Best = 5.203013e+03 = [2, 3, 2, 2]	FES = 564100	Diversity = 1.8454e-02	k_t
Generation 5650: Best = 5.203004e+03 = [1, 1, 1, 1]	FES = 565100	Diversity = 1.8080e-02	k_t
Generation 5660: Best = 5.203001e+03 = [4, 2, 2, 4]	FES = 566100	Diversity = 1.8425e-02	k_t
Generation 5670: Best = 5.202998e+03 = [1, 1, 1, 1]	FES = 567100	Diversity = 1.9078e-02	k_t
Generation 5680: Best = 5.202991e+03 = [4, 3, 4, 4]	FES = 568100	Diversity = 2.0630e-02	k_t
Generation 5690: Best = 5.202987e+03 = [1, 1, 1, 1]	FES = 569100	Diversity = 2.0002e-02	k_t
Generation 5700: Best = 5.202984e+03 = [2, 3, 3, 3]	FES = 570100	Diversity = 1.9887e-02	k_t
Generation 5710: Best = 5.202981e+03 = [1, 1, 1, 1]	FES = 571100	Diversity = 1.9682e-02	k_t
Generation 5720: Best = 5.202978e+03 = [2, 2, 3, 3]	FES = 572100	Diversity = 1.9041e-02	k_t
Generation 5730: Best = 5.202976e+03 = [1, 1, 1, 1]	FES = 573100	Diversity = 1.8186e-02	k_t
Generation 5740: Best = 5.202970e+03 = [2, 4, 2, 3]	FES = 574100	Diversity = 1.8306e-02	k_t
Generation 5750: Best = 5.202963e+03 = [1, 1, 1, 1]	FES = 575100	Diversity = 1.8017e-02	k_t
Generation 5760: Best = 5.202960e+03 = [4, 2, 4, 2]	FES = 576100	Diversity = 1.8035e-02	k_t
Generation 5770: Best = 5.202952e+03 = [1, 1, 1, 1]	FES = 577100	Diversity = 1.8243e-02	k_t
Generation 5780: Best = 5.202951e+03 = [3, 4, 3, 2]	FES = 578100	Diversity = 1.8099e-02	k_t
Generation 5790: Best = 5.202951e+03 = [1, 1, 1, 1]	FES = 579100	Diversity = 1.8665e-02	k_t
Generation 5800: Best = 5.202946e+03 = [2, 2, 3, 2]	FES = 580100	Diversity = 1.7974e-02	k_t
Generation 5810: Best = 5.202946e+03 = [1, 1, 1, 1]	FES = 581100	Diversity = 1.7619e-02	k_t
Generation 5820: Best = 5.202943e+03 = [3, 3, 3, 3]	FES = 582100	Diversity = 1.7416e-02	k_t
Generation 5830: Best = 5.202939e+03 = [1, 1, 1, 1]	FES = 583100	Diversity = 1.8196e-02	k_t
Generation 5840: Best = 5.202931e+03 = [4, 4, 2, 3]	FES = 584100	Diversity = 1.9941e-02	k_t
Generation 5850: Best = 5.202931e+03 = [1, 1, 1, 1]	FES = 585100	Diversity = 1.8765e-02	k_t
Generation 5860: Best = 5.202931e+03 = [4, 3, 2, 3]	FES = 586100	Diversity = 1.8370e-02	k_t
Generation 5870: Best = 5.202928e+03 = [1, 1, 1, 1]	FES = 587100	Diversity = 1.7226e-02	k_t
Generation 5880: Best = 5.202924e+03 = [2, 2, 3, 3]	FES = 588100	Diversity = 1.6341e-02	k_t
Generation 5890: Best = 5.202922e+03 = [1, 1, 1, 1]	FES = 589100	Diversity = 1.5200e-02	k_t

Generation 5900: Best = 5.202921e+03 = [4, 4, 4, 2]	FES = 590100	Diversity = 1.4240e-02	k_t
Generation 5910: Best = 5.202919e+03 = [1, 1, 1, 1]	FES = 591100	Diversity = 1.4251e-02	k_t
Generation 5920: Best = 5.202915e+03 = [3, 2, 4, 3]	FES = 592100	Diversity = 1.4007e-02	k_t
Generation 5930: Best = 5.202913e+03 = [1, 1, 1, 1]	FES = 593100	Diversity = 1.4542e-02	k_t
Generation 5940: Best = 5.202911e+03 = [3, 3, 2, 2]	FES = 594100	Diversity = 1.5241e-02	k_t
Generation 5950: Best = 5.202908e+03 = [1, 1, 1, 1]	FES = 595100	Diversity = 1.4978e-02	k_t
Generation 5960: Best = 5.202905e+03 = [4, 4, 2, 3]	FES = 596100	Diversity = 1.5228e-02	k_t
Generation 5970: Best = 5.202902e+03 = [1, 1, 1, 1]	FES = 597100	Diversity = 1.5547e-02	k_t
Generation 5980: Best = 5.202897e+03 = [4, 4, 2, 2]	FES = 598100	Diversity = 1.4974e-02	k_t
Generation 5990: Best = 5.202893e+03 = [1, 1, 1, 1]	FES = 599100	Diversity = 1.5977e-02	k_t
Generation 6000: Best = 5.202890e+03 = [4, 3, 4, 2]	FES = 600100	Diversity = 1.5788e-02	k_t
Generation 6010: Best = 5.202887e+03 = [1, 1, 1, 1]	FES = 601100	Diversity = 1.5737e-02	k_t
Generation 6020: Best = 5.202885e+03 = [4, 2, 4, 4]	FES = 602100	Diversity = 1.6494e-02	k_t
Generation 6030: Best = 5.202882e+03 = [1, 1, 1, 1]	FES = 603100	Diversity = 1.7008e-02	k_t
Generation 6040: Best = 5.202879e+03 = [3, 2, 3, 4]	FES = 604100	Diversity = 1.6803e-02	k_t
Generation 6050: Best = 5.202877e+03 = [1, 1, 1, 1]	FES = 605100	Diversity = 1.6702e-02	k_t
Generation 6060: Best = 5.202873e+03 = [2, 3, 2, 4]	FES = 606100	Diversity = 1.6534e-02	k_t
Generation 6070: Best = 5.202872e+03 = [1, 1, 1, 1]	FES = 607100	Diversity = 1.6915e-02	k_t
Generation 6080: Best = 5.202870e+03 = [2, 4, 4, 2]	FES = 608100	Diversity = 1.6214e-02	k_t
Generation 6090: Best = 5.202868e+03 = [1, 1, 1, 1]	FES = 609100	Diversity = 1.6566e-02	k_t
Generation 6100: Best = 5.202865e+03 = [3, 3, 2, 2]	FES = 610100	Diversity = 1.6432e-02	k_t
Generation 6110: Best = 5.202864e+03 = [1, 1, 1, 1]	FES = 611100	Diversity = 1.5605e-02	k_t
Generation 6120: Best = 5.202860e+03 = [4, 3, 3, 3]	FES = 612100	Diversity = 1.3898e-02	k_t
Generation 6130: Best = 5.202860e+03 = [1, 1, 1, 1]	FES = 613100	Diversity = 1.3319e-02	k_t
Generation 6140: Best = 5.202859e+03 = [2, 2, 2, 2]	FES = 614100	Diversity = 1.2760e-02	k_t
Generation 6150: Best = 5.202859e+03 = [1, 1, 1, 1]	FES = 615100	Diversity = 1.1938e-02	k_t
Generation 6160: Best = 5.202857e+03 = [2, 4, 2, 3]	FES = 616100	Diversity = 1.1695e-02	k_t

Generation 6170: Best = 5.202856e+03 = [1, 1, 1, 1]	FES = 617100	Diversity = 1.1551e-02	k_t
Generation 6180: Best = 5.202855e+03 = [3, 4, 2, 2]	FES = 618100	Diversity = 1.1124e-02	k_t
Generation 6190: Best = 5.202853e+03 = [1, 1, 1, 1]	FES = 619100	Diversity = 1.1858e-02	k_t
Generation 6200: Best = 5.202851e+03 = [4, 2, 2, 3]	FES = 620100	Diversity = 1.1553e-02	k_t
Generation 6210: Best = 5.202850e+03 = [1, 1, 1, 1]	FES = 621100	Diversity = 1.1428e-02	k_t
Generation 6220: Best = 5.202848e+03 = [4, 3, 3, 3]	FES = 622100	Diversity = 1.1779e-02	k_t
Generation 6230: Best = 5.202847e+03 = [1, 1, 1, 1]	FES = 623100	Diversity = 1.1306e-02	k_t
Generation 6240: Best = 5.202844e+03 = [2, 3, 3, 2]	FES = 624100	Diversity = 1.1365e-02	k_t
Generation 6250: Best = 5.202842e+03 = [1, 1, 1, 1]	FES = 625100	Diversity = 1.2716e-02	k_t
Generation 6260: Best = 5.202841e+03 = [4, 4, 4, 3]	FES = 626100	Diversity = 1.2793e-02	k_t
Generation 6270: Best = 5.202839e+03 = [1, 1, 1, 1]	FES = 627100	Diversity = 1.1993e-02	k_t
Generation 6280: Best = 5.202839e+03 = [3, 4, 2, 2]	FES = 628100	Diversity = 1.0562e-02	k_t
Generation 6290: Best = 5.202837e+03 = [1, 1, 1, 1]	FES = 629100	Diversity = 1.1242e-02	k_t
Generation 6300: Best = 5.202837e+03 = [3, 4, 2, 3]	FES = 630100	Diversity = 1.0721e-02	k_t
Generation 6310: Best = 5.202836e+03 = [1, 1, 1, 1]	FES = 631100	Diversity = 1.0547e-02	k_t
Generation 6320: Best = 5.202835e+03 = [4, 4, 4, 4]	FES = 632100	Diversity = 9.9186e-03	k_t
Generation 6330: Best = 5.202834e+03 = [1, 1, 1, 1]	FES = 633100	Diversity = 9.7358e-03	k_t
Generation 6340: Best = 5.202833e+03 = [2, 3, 4, 4]	FES = 634100	Diversity = 9.2110e-03	k_t
Generation 6350: Best = 5.202832e+03 = [1, 1, 1, 1]	FES = 635100	Diversity = 8.2189e-03	k_t
Generation 6360: Best = 5.202832e+03 = [3, 3, 3, 2]	FES = 636100	Diversity = 8.3184e-03	k_t
Generation 6370: Best = 5.202830e+03 = [1, 1, 1, 1]	FES = 637100	Diversity = 7.8415e-03	k_t
Generation 6380: Best = 5.202829e+03 = [2, 4, 4, 2]	FES = 638100	Diversity = 8.9161e-03	k_t
Generation 6390: Best = 5.202828e+03 = [1, 1, 1, 1]	FES = 639100	Diversity = 9.1045e-03	k_t
Generation 6400: Best = 5.202826e+03 = [2, 4, 4, 2]	FES = 640100	Diversity = 8.9582e-03	k_t
Generation 6410: Best = 5.202824e+03 = [1, 1, 1, 1]	FES = 641100	Diversity = 9.1258e-03	k_t
Generation 6420: Best = 5.202820e+03 = [4, 3, 3, 4]	FES = 642100	Diversity = 9.8284e-03	k_t
Generation 6430: Best = 5.202820e+03 = [1, 1, 1, 1]	FES = 643100	Diversity = 1.0440e-02	k_t

Generation 6440: Best = 5.202818e+03 = [4, 2, 3, 3]	FES = 644100	Diversity = 9.9259e-03	k_t
Generation 6450: Best = 5.202817e+03 = [1, 1, 1, 1]	FES = 645100	Diversity = 9.4459e-03	k_t
Generation 6460: Best = 5.202814e+03 = [4, 4, 4, 3]	FES = 646100	Diversity = 8.8747e-03	k_t
Generation 6470: Best = 5.202813e+03 = [1, 1, 1, 1]	FES = 647100	Diversity = 9.2864e-03	k_t
Generation 6480: Best = 5.202812e+03 = [2, 2, 2, 4]	FES = 648100	Diversity = 9.3527e-03	k_t
Generation 6490: Best = 5.202811e+03 = [1, 1, 1, 1]	FES = 649100	Diversity = 1.0096e-02	k_t
Generation 6500: Best = 5.202809e+03 = [3, 3, 3, 4]	FES = 650100	Diversity = 1.0494e-02	k_t
Generation 6510: Best = 5.202806e+03 = [1, 1, 1, 1]	FES = 651100	Diversity = 1.0744e-02	k_t
Generation 6520: Best = 5.202805e+03 = [4, 4, 3, 4]	FES = 652100	Diversity = 1.1307e-02	k_t
Generation 6530: Best = 5.202799e+03 = [1, 1, 1, 1]	FES = 653100	Diversity = 1.2354e-02	k_t
Generation 6540: Best = 5.202797e+03 = [3, 2, 3, 4]	FES = 654100	Diversity = 1.3045e-02	k_t
Generation 6550: Best = 5.202794e+03 = [1, 1, 1, 1]	FES = 655100	Diversity = 1.2869e-02	k_t
Generation 6560: Best = 5.202794e+03 = [2, 3, 3, 2]	FES = 656100	Diversity = 1.3716e-02	k_t
Generation 6570: Best = 5.202790e+03 = [1, 1, 1, 1]	FES = 657100	Diversity = 1.3106e-02	k_t
Generation 6580: Best = 5.202788e+03 = [2, 3, 2, 3]	FES = 658100	Diversity = 1.3347e-02	k_t
Generation 6590: Best = 5.202784e+03 = [1, 1, 1, 1]	FES = 659100	Diversity = 1.3485e-02	k_t
Generation 6600: Best = 5.202784e+03 = [3, 4, 3, 4]	FES = 660100	Diversity = 1.3959e-02	k_t
Generation 6610: Best = 5.202781e+03 = [1, 1, 1, 1]	FES = 661100	Diversity = 1.3586e-02	k_t
Generation 6620: Best = 5.202779e+03 = [4, 2, 3, 4]	FES = 662100	Diversity = 1.2831e-02	k_t
Generation 6630: Best = 5.202776e+03 = [1, 1, 1, 1]	FES = 663100	Diversity = 1.3248e-02	k_t
Generation 6640: Best = 5.202775e+03 = [4, 2, 2, 3]	FES = 664100	Diversity = 1.3256e-02	k_t
Generation 6650: Best = 5.202774e+03 = [1, 1, 1, 1]	FES = 665100	Diversity = 1.3406e-02	k_t
Generation 6660: Best = 5.202772e+03 = [3, 2, 2, 2]	FES = 666100	Diversity = 1.2594e-02	k_t
Generation 6670: Best = 5.202768e+03 = [1, 1, 1, 1]	FES = 667100	Diversity = 1.1908e-02	k_t
Generation 6680: Best = 5.202767e+03 = [3, 3, 3, 2]	FES = 668100	Diversity = 1.1298e-02	k_t
Generation 6690: Best = 5.202766e+03 = [1, 1, 1, 1]	FES = 669100	Diversity = 1.0475e-02	k_t
Generation 6700: Best = 5.202766e+03 = [4, 2, 2, 2]	FES = 670100	Diversity = 9.6957e-03	k_t

Generation 6710: Best = 5.202764e+03 = [1, 1, 1, 1]	FES = 671100	Diversity = 9.6419e-03	k_t
Generation 6720: Best = 5.202763e+03 = [3, 4, 2, 2]	FES = 672100	Diversity = 9.0204e-03	k_t
Generation 6730: Best = 5.202761e+03 = [1, 1, 1, 1]	FES = 673100	Diversity = 9.3297e-03	k_t
Generation 6740: Best = 5.202761e+03 = [3, 2, 2, 2]	FES = 674100	Diversity = 9.5909e-03	k_t
Generation 6750: Best = 5.202759e+03 = [1, 1, 1, 1]	FES = 675100	Diversity = 1.0132e-02	k_t
Generation 6760: Best = 5.202756e+03 = [4, 4, 3, 2]	FES = 676100	Diversity = 9.7887e-03	k_t
Generation 6770: Best = 5.202755e+03 = [1, 1, 1, 1]	FES = 677100	Diversity = 1.0622e-02	k_t
Generation 6780: Best = 5.202754e+03 = [4, 2, 4, 4]	FES = 678100	Diversity = 1.0327e-02	k_t
Generation 6790: Best = 5.202752e+03 = [1, 1, 1, 1]	FES = 679100	Diversity = 1.0188e-02	k_t
Generation 6800: Best = 5.202751e+03 = [2, 2, 4, 2]	FES = 680100	Diversity = 1.0403e-02	k_t
Generation 6810: Best = 5.202750e+03 = [1, 1, 1, 1]	FES = 681100	Diversity = 1.0843e-02	k_t
Generation 6820: Best = 5.202749e+03 = [2, 2, 3, 3]	FES = 682100	Diversity = 1.0947e-02	k_t
Generation 6830: Best = 5.202747e+03 = [1, 1, 1, 1]	FES = 683100	Diversity = 1.1097e-02	k_t
Generation 6840: Best = 5.202743e+03 = [2, 3, 4, 2]	FES = 684100	Diversity = 1.2222e-02	k_t
Generation 6850: Best = 5.202741e+03 = [1, 1, 1, 1]	FES = 685100	Diversity = 1.2110e-02	k_t
Generation 6860: Best = 5.202740e+03 = [2, 3, 4, 2]	FES = 686100	Diversity = 1.1532e-02	k_t
Generation 6870: Best = 5.202738e+03 = [1, 1, 1, 1]	FES = 687100	Diversity = 1.1232e-02	k_t
Generation 6880: Best = 5.202737e+03 = [4, 2, 2, 4]	FES = 688100	Diversity = 1.0561e-02	k_t
Generation 6890: Best = 5.202736e+03 = [1, 1, 1, 1]	FES = 689100	Diversity = 1.0529e-02	k_t
Generation 6900: Best = 5.202735e+03 = [4, 3, 4, 4]	FES = 690100	Diversity = 1.0208e-02	k_t
Generation 6910: Best = 5.202733e+03 = [1, 1, 1, 1]	FES = 691100	Diversity = 8.7397e-03	k_t
Generation 6920: Best = 5.202733e+03 = [4, 4, 3, 4]	FES = 692100	Diversity = 8.9054e-03	k_t
Generation 6930: Best = 5.202730e+03 = [1, 1, 1, 1]	FES = 693100	Diversity = 8.8584e-03	k_t
Generation 6940: Best = 5.202730e+03 = [3, 4, 3, 2]	FES = 694100	Diversity = 8.5983e-03	k_t
Generation 6950: Best = 5.202728e+03 = [1, 1, 1, 1]	FES = 695100	Diversity = 8.8497e-03	k_t
Generation 6960: Best = 5.202726e+03 = [2, 4, 2, 4]	FES = 696100	Diversity = 8.9743e-03	k_t
Generation 6970: Best = 5.202725e+03 = [1, 1, 1, 1]	FES = 697100	Diversity = 8.6268e-03	k_t

Generation 6980: Best = 5.202724e+03 = [4, 2, 2, 2]	FES = 698100	Diversity = 8.5046e-03	k_t
Generation 6990: Best = 5.202723e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 7.8807e-03	k_t
Generation 7000: Best = 5.202722e+03 = [2, 4, 3, 3]	FES = 700100	Diversity = 7.0276e-03	k_t
Generation 7010: Best = 5.202722e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 6.7287e-03	k_t
Generation 7020: Best = 5.202721e+03 = [4, 3, 4, 4]	FES = 702100	Diversity = 6.5661e-03	k_t
Generation 7030: Best = 5.202720e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 6.4874e-03	k_t
Generation 7040: Best = 5.202719e+03 = [4, 2, 4, 3]	FES = 704100	Diversity = 6.5174e-03	k_t
Generation 7050: Best = 5.202718e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 6.6823e-03	k_t
Generation 7060: Best = 5.202718e+03 = [3, 4, 2, 3]	FES = 706100	Diversity = 7.3334e-03	k_t
Generation 7070: Best = 5.202716e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 7.8623e-03	k_t
Generation 7080: Best = 5.202716e+03 = [2, 4, 2, 2]	FES = 708100	Diversity = 7.9744e-03	k_t
Generation 7090: Best = 5.202714e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 8.6738e-03	k_t
Generation 7100: Best = 5.202714e+03 = [4, 4, 3, 2]	FES = 710100	Diversity = 9.0014e-03	k_t
Generation 7110: Best = 5.202713e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 9.1644e-03	k_t
Generation 7120: Best = 5.202712e+03 = [3, 4, 3, 4]	FES = 712100	Diversity = 8.5321e-03	k_t
Generation 7130: Best = 5.202712e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 7.1798e-03	k_t
Generation 7140: Best = 5.202711e+03 = [3, 2, 4, 3]	FES = 714100	Diversity = 6.6761e-03	k_t
Generation 7150: Best = 5.202709e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 6.4833e-03	k_t
Generation 7160: Best = 5.202708e+03 = [4, 4, 4, 3]	FES = 716100	Diversity = 7.7411e-03	k_t
Generation 7170: Best = 5.202707e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 8.1020e-03	k_t
Generation 7180: Best = 5.202706e+03 = [3, 3, 2, 2]	FES = 718100	Diversity = 8.6517e-03	k_t
Generation 7190: Best = 5.202705e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 8.4498e-03	k_t
Generation 7200: Best = 5.202703e+03 = [4, 2, 4, 3]	FES = 720100	Diversity = 8.2210e-03	k_t
Generation 7210: Best = 5.202702e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 9.6192e-03	k_t
Generation 7220: Best = 5.202697e+03 = [2, 4, 4, 4]	FES = 722100	Diversity = 1.1374e-02	k_t
Generation 7230: Best = 5.202694e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 1.4028e-02	k_t
Generation 7240: Best = 5.202693e+03 = [3, 2, 3, 4]	FES = 724100	Diversity = 1.2565e-02	k_t

Generation 7250: Best = 5.202691e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 1.1979e-02	k_t
Generation 7260: Best = 5.202690e+03 = [3, 4, 3, 3]	FES = 726100	Diversity = 1.1580e-02	k_t
Generation 7270: Best = 5.202687e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 1.0948e-02	k_t
Generation 7280: Best = 5.202686e+03 = [4, 2, 2, 4]	FES = 728100	Diversity = 1.0410e-02	k_t
Generation 7290: Best = 5.202684e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 1.0202e-02	k_t
Generation 7300: Best = 5.202684e+03 = [4, 2, 3, 2]	FES = 730100	Diversity = 9.6787e-03	k_t
Generation 7310: Best = 5.202683e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 9.7898e-03	k_t
Generation 7320: Best = 5.202681e+03 = [2, 2, 3, 2]	FES = 732100	Diversity = 9.8453e-03	k_t
Generation 7330: Best = 5.202680e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 9.3217e-03	k_t
Generation 7340: Best = 5.202679e+03 = [4, 2, 3, 3]	FES = 734100	Diversity = 9.3035e-03	k_t
Generation 7350: Best = 5.202678e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 9.5993e-03	k_t
Generation 7360: Best = 5.202677e+03 = [4, 2, 4, 3]	FES = 736100	Diversity = 9.7975e-03	k_t
Generation 7370: Best = 5.202675e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 9.5277e-03	k_t
Generation 7380: Best = 5.202672e+03 = [4, 2, 4, 2]	FES = 738100	Diversity = 9.1619e-03	k_t
Generation 7390: Best = 5.202672e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 9.1458e-03	k_t
Generation 7400: Best = 5.202671e+03 = [3, 3, 3, 3]	FES = 740100	Diversity = 8.6030e-03	k_t
Generation 7410: Best = 5.202669e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 8.3354e-03	k_t
Generation 7420: Best = 5.202668e+03 = [2, 2, 4, 2]	FES = 742100	Diversity = 8.6355e-03	k_t
Generation 7430: Best = 5.202667e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 7.7204e-03	k_t
Generation 7440: Best = 5.202666e+03 = [4, 3, 4, 4]	FES = 744100	Diversity = 7.8864e-03	k_t
Generation 7450: Best = 5.202665e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 7.7452e-03	k_t
Generation 7460: Best = 5.202664e+03 = [4, 4, 4, 2]	FES = 746100	Diversity = 8.1374e-03	k_t
Generation 7470: Best = 5.202663e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 7.6956e-03	k_t
Generation 7480: Best = 5.202662e+03 = [2, 3, 4, 4]	FES = 748100	Diversity = 7.3433e-03	k_t
Generation 7490: Best = 5.202662e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 7.4086e-03	k_t
Generation 7500: Best = 5.202661e+03 = [3, 3, 2, 2]	FES = 750100	Diversity = 7.3717e-03	k_t
Generation 7510: Best = 5.202660e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 6.7978e-03	k_t

Generation 7520: Best = 5.202660e+03 = [4, 2, 4, 3]	FES = 752100	Diversity = 6.4310e-03	k_t
Generation 7530: Best = 5.202659e+03 = [1, 1, 1, 1]	FES = 753100	Diversity = 6.4671e-03	k_t
Generation 7540: Best = 5.202658e+03 = [4, 2, 2, 3]	FES = 754100	Diversity = 6.7922e-03	k_t
Generation 7550: Best = 5.202656e+03 = [1, 1, 1, 1]	FES = 755100	Diversity = 6.9723e-03	k_t
Generation 7560: Best = 5.202656e+03 = [2, 2, 4, 4]	FES = 756100	Diversity = 8.0575e-03	k_t
Generation 7570: Best = 5.202655e+03 = [1, 1, 1, 1]	FES = 757100	Diversity = 7.8236e-03	k_t
Generation 7580: Best = 5.202654e+03 = [2, 2, 2, 4]	FES = 758100	Diversity = 7.2918e-03	k_t
Generation 7590: Best = 5.202653e+03 = [1, 1, 1, 1]	FES = 759100	Diversity = 6.6728e-03	k_t
Generation 7600: Best = 5.202652e+03 = [2, 3, 4, 4]	FES = 760100	Diversity = 6.4377e-03	k_t
Generation 7610: Best = 5.202651e+03 = [1, 1, 1, 1]	FES = 761100	Diversity = 6.6761e-03	k_t
Generation 7620: Best = 5.202650e+03 = [3, 3, 2, 4]	FES = 762100	Diversity = 7.1222e-03	k_t
Generation 7630: Best = 5.202650e+03 = [1, 1, 1, 1]	FES = 763100	Diversity = 6.7454e-03	k_t
Generation 7640: Best = 5.202648e+03 = [3, 3, 3, 2]	FES = 764100	Diversity = 6.6642e-03	k_t
Generation 7650: Best = 5.202648e+03 = [1, 1, 1, 1]	FES = 765100	Diversity = 7.1871e-03	k_t
Generation 7660: Best = 5.202647e+03 = [4, 4, 4, 4]	FES = 766100	Diversity = 6.6646e-03	k_t
Generation 7670: Best = 5.202646e+03 = [1, 1, 1, 1]	FES = 767100	Diversity = 6.7748e-03	k_t
Generation 7680: Best = 5.202646e+03 = [2, 3, 2, 4]	FES = 768100	Diversity = 6.1193e-03	k_t
Generation 7690: Best = 5.202645e+03 = [1, 1, 1, 1]	FES = 769100	Diversity = 5.8695e-03	k_t
Generation 7700: Best = 5.202645e+03 = [2, 3, 4, 2]	FES = 770100	Diversity = 5.6715e-03	k_t
Generation 7710: Best = 5.202644e+03 = [1, 1, 1, 1]	FES = 771100	Diversity = 5.9620e-03	k_t
Generation 7720: Best = 5.202644e+03 = [2, 3, 2, 2]	FES = 772100	Diversity = 6.2352e-03	k_t
Generation 7730: Best = 5.202643e+03 = [1, 1, 1, 1]	FES = 773100	Diversity = 6.1141e-03	k_t
Generation 7740: Best = 5.202643e+03 = [3, 2, 3, 4]	FES = 774100	Diversity = 5.7804e-03	k_t
Generation 7750: Best = 5.202642e+03 = [1, 1, 1, 1]	FES = 775100	Diversity = 5.4338e-03	k_t
Generation 7760: Best = 5.202642e+03 = [3, 4, 4, 2]	FES = 776100	Diversity = 5.5373e-03	k_t
Generation 7770: Best = 5.202641e+03 = [1, 1, 1, 1]	FES = 777100	Diversity = 5.3833e-03	k_t
Generation 7780: Best = 5.202641e+03 = [3, 2, 4, 2]	FES = 778100	Diversity = 5.4441e-03	k_t

Generation 7790: Best = 5.202640e+03 = [1, 1, 1, 1]	FES = 779100	Diversity = 5.1163e-03	k_t
Generation 7800: Best = 5.202640e+03 = [4, 3, 3, 2]	FES = 780100	Diversity = 4.4789e-03	k_t
Generation 7810: Best = 5.202640e+03 = [1, 1, 1, 1]	FES = 781100	Diversity = 4.3104e-03	k_t
Generation 7820: Best = 5.202639e+03 = [3, 3, 3, 4]	FES = 782100	Diversity = 4.4971e-03	k_t
Generation 7830: Best = 5.202639e+03 = [1, 1, 1, 1]	FES = 783100	Diversity = 4.4152e-03	k_t
Generation 7840: Best = 5.202639e+03 = [3, 3, 3, 2]	FES = 784100	Diversity = 4.4003e-03	k_t
Generation 7850: Best = 5.202638e+03 = [1, 1, 1, 1]	FES = 785100	Diversity = 4.1729e-03	k_t
Generation 7860: Best = 5.202637e+03 = [3, 2, 2, 4]	FES = 786100	Diversity = 4.3995e-03	k_t
Generation 7870: Best = 5.202637e+03 = [1, 1, 1, 1]	FES = 787100	Diversity = 5.0517e-03	k_t
Generation 7880: Best = 5.202637e+03 = [3, 3, 4, 4]	FES = 788100	Diversity = 4.8184e-03	k_t
Generation 7890: Best = 5.202636e+03 = [1, 1, 1, 1]	FES = 789100	Diversity = 4.5858e-03	k_t
Generation 7900: Best = 5.202636e+03 = [3, 3, 3, 2]	FES = 790100	Diversity = 4.6078e-03	k_t
Generation 7910: Best = 5.202636e+03 = [1, 1, 1, 1]	FES = 791100	Diversity = 4.9598e-03	k_t
Generation 7920: Best = 5.202635e+03 = [2, 2, 2, 4]	FES = 792100	Diversity = 4.9234e-03	k_t
Generation 7930: Best = 5.202635e+03 = [1, 1, 1, 1]	FES = 793100	Diversity = 4.7816e-03	k_t
Generation 7940: Best = 5.202634e+03 = [3, 3, 3, 4]	FES = 794100	Diversity = 4.4949e-03	k_t
Generation 7950: Best = 5.202634e+03 = [1, 1, 1, 1]	FES = 795100	Diversity = 4.5395e-03	k_t
Generation 7960: Best = 5.202634e+03 = [4, 2, 2, 4]	FES = 796100	Diversity = 4.7581e-03	k_t
Generation 7970: Best = 5.202633e+03 = [1, 1, 1, 1]	FES = 797100	Diversity = 4.9662e-03	k_t
Generation 7980: Best = 5.202633e+03 = [3, 4, 2, 3]	FES = 798100	Diversity = 5.3443e-03	k_t
Generation 7990: Best = 5.202632e+03 = [1, 1, 1, 1]	FES = 799100	Diversity = 5.5278e-03	k_t
Generation 8000: Best = 5.202632e+03 = [3, 4, 3, 4]	FES = 800100	Diversity = 5.2548e-03	k_t
Generation 8010: Best = 5.202631e+03 = [1, 1, 1, 1]	FES = 801100	Diversity = 5.3702e-03	k_t
Generation 8020: Best = 5.202630e+03 = [4, 4, 2, 2]	FES = 802100	Diversity = 5.7535e-03	k_t
Generation 8030: Best = 5.202630e+03 = [1, 1, 1, 1]	FES = 803100	Diversity = 5.4852e-03	k_t
Generation 8040: Best = 5.202629e+03 = [2, 2, 3, 4]	FES = 804100	Diversity = 5.5407e-03	k_t
Generation 8050: Best = 5.202629e+03 = [1, 1, 1, 1]	FES = 805100	Diversity = 5.7405e-03	k_t

Generation 8060: Best = 5.202628e+03 = [2, 4, 3, 4]	FES = 806100	Diversity = 5.6809e-03	k_t
Generation 8070: Best = 5.202628e+03 = [1, 1, 1, 1]	FES = 807100	Diversity = 5.8281e-03	k_t
Generation 8080: Best = 5.202627e+03 = [2, 3, 3, 3]	FES = 808100	Diversity = 6.3958e-03	k_t
Generation 8090: Best = 5.202626e+03 = [1, 1, 1, 1]	FES = 809100	Diversity = 6.3820e-03	k_t
Generation 8100: Best = 5.202626e+03 = [4, 3, 4, 2]	FES = 810100	Diversity = 5.9071e-03	k_t
Generation 8110: Best = 5.202626e+03 = [1, 1, 1, 1]	FES = 811100	Diversity = 5.3946e-03	k_t
Generation 8120: Best = 5.202625e+03 = [4, 4, 3, 3]	FES = 812100	Diversity = 5.1489e-03	k_t
Generation 8130: Best = 5.202625e+03 = [1, 1, 1, 1]	FES = 813100	Diversity = 4.9656e-03	k_t
Generation 8140: Best = 5.202625e+03 = [3, 4, 3, 3]	FES = 814100	Diversity = 5.0449e-03	k_t
Generation 8150: Best = 5.202623e+03 = [1, 1, 1, 1]	FES = 815100	Diversity = 5.1813e-03	k_t
Generation 8160: Best = 5.202623e+03 = [2, 3, 3, 4]	FES = 816100	Diversity = 5.1734e-03	k_t
Generation 8170: Best = 5.202623e+03 = [1, 1, 1, 1]	FES = 817100	Diversity = 5.2213e-03	k_t
Generation 8180: Best = 5.202622e+03 = [2, 3, 4, 4]	FES = 818100	Diversity = 5.3960e-03	k_t
Generation 8190: Best = 5.202622e+03 = [1, 1, 1, 1]	FES = 819100	Diversity = 5.3461e-03	k_t
Generation 8200: Best = 5.202621e+03 = [4, 4, 3, 2]	FES = 820100	Diversity = 5.3796e-03	k_t
Generation 8210: Best = 5.202621e+03 = [1, 1, 1, 1]	FES = 821100	Diversity = 5.0366e-03	k_t
Generation 8220: Best = 5.202621e+03 = [2, 4, 2, 3]	FES = 822100	Diversity = 5.0046e-03	k_t
Generation 8230: Best = 5.202620e+03 = [1, 1, 1, 1]	FES = 823100	Diversity = 4.6562e-03	k_t
Generation 8240: Best = 5.202620e+03 = [4, 2, 4, 4]	FES = 824100	Diversity = 4.3020e-03	k_t
Generation 8250: Best = 5.202620e+03 = [1, 1, 1, 1]	FES = 825100	Diversity = 3.6960e-03	k_t
Generation 8260: Best = 5.202619e+03 = [4, 4, 2, 4]	FES = 826100	Diversity = 3.8163e-03	k_t
Generation 8270: Best = 5.202619e+03 = [1, 1, 1, 1]	FES = 827100	Diversity = 3.9886e-03	k_t
Generation 8280: Best = 5.202619e+03 = [4, 2, 2, 2]	FES = 828100	Diversity = 4.4300e-03	k_t
Generation 8290: Best = 5.202618e+03 = [1, 1, 1, 1]	FES = 829100	Diversity = 4.3839e-03	k_t
Generation 8300: Best = 5.202618e+03 = [2, 3, 2, 2]	FES = 830100	Diversity = 4.5311e-03	k_t
Generation 8310: Best = 5.202617e+03 = [1, 1, 1, 1]	FES = 831100	Diversity = 4.6844e-03	k_t
Generation 8320: Best = 5.202617e+03 = [4, 2, 4, 3]	FES = 832100	Diversity = 5.2231e-03	k_t

Generation 8330: Best = 5.202616e+03	FES = 833100	Diversity = 5.1159e-03	k_t
= [1, 1, 1, 1]			
Generation 8340: Best = 5.202616e+03	FES = 834100	Diversity = 4.7232e-03	k_t
= [3, 4, 3, 4]			
Generation 8350: Best = 5.202615e+03	FES = 835100	Diversity = 4.7147e-03	k_t
= [1, 1, 1, 1]			
Generation 8360: Best = 5.202615e+03	FES = 836100	Diversity = 4.5730e-03	k_t
= [4, 3, 2, 3]			
Generation 8370: Best = 5.202614e+03	FES = 837100	Diversity = 4.8982e-03	k_t
= [1, 1, 1, 1]			
Generation 8380: Best = 5.202614e+03	FES = 838100	Diversity = 4.8877e-03	k_t
= [3, 3, 4, 3]			
Generation 8390: Best = 5.202614e+03	FES = 839100	Diversity = 4.4959e-03	k_t
= [1, 1, 1, 1]			
Generation 8400: Best = 5.202613e+03	FES = 840100	Diversity = 5.0097e-03	k_t
= [3, 3, 3, 4]			
Generation 8410: Best = 5.202613e+03	FES = 841100	Diversity = 5.0413e-03	k_t
= [1, 1, 1, 1]			
Generation 8420: Best = 5.202612e+03	FES = 842100	Diversity = 5.1737e-03	k_t
= [2, 4, 2, 4]			
Generation 8430: Best = 5.202612e+03	FES = 843100	Diversity = 5.5864e-03	k_t
= [1, 1, 1, 1]			
Generation 8440: Best = 5.202611e+03	FES = 844100	Diversity = 5.4203e-03	k_t
= [4, 2, 2, 4]			
Generation 8450: Best = 5.202611e+03	FES = 845100	Diversity = 5.6800e-03	k_t
= [1, 1, 1, 1]			
Generation 8460: Best = 5.202610e+03	FES = 846100	Diversity = 5.4851e-03	k_t
= [2, 3, 3, 4]			
Generation 8470: Best = 5.202610e+03	FES = 847100	Diversity = 5.7712e-03	k_t
= [1, 1, 1, 1]			
Generation 8480: Best = 5.202609e+03	FES = 848100	Diversity = 5.1962e-03	k_t
= [3, 4, 4, 4]			
Generation 8490: Best = 5.202609e+03	FES = 849100	Diversity = 5.1922e-03	k_t
= [1, 1, 1, 1]			
Generation 8500: Best = 5.202608e+03	FES = 850100	Diversity = 5.1539e-03	k_t
= [4, 4, 4, 3]			
Generation 8510: Best = 5.202608e+03	FES = 851100	Diversity = 5.6847e-03	k_t
= [1, 1, 1, 1]			
Generation 8520: Best = 5.202607e+03	FES = 852100	Diversity = 6.0765e-03	k_t
= [2, 2, 4, 2]			
Generation 8530: Best = 5.202607e+03	FES = 853100	Diversity = 5.7782e-03	k_t
= [1, 1, 1, 1]			
Generation 8540: Best = 5.202606e+03	FES = 854100	Diversity = 5.4921e-03	k_t
= [3, 2, 4, 2]			
Generation 8550: Best = 5.202605e+03	FES = 855100	Diversity = 6.1012e-03	k_t
= [1, 1, 1, 1]			
Generation 8560: Best = 5.202605e+03	FES = 856100	Diversity = 6.1327e-03	k_t
= [2, 2, 2, 2]			
Generation 8570: Best = 5.202604e+03	FES = 857100	Diversity = 6.7626e-03	k_t
= [1, 1, 1, 1]			
Generation 8580: Best = 5.202604e+03	FES = 858100	Diversity = 6.5419e-03	k_t
= [4, 4, 3, 4]			
Generation 8590: Best = 5.202603e+03	FES = 859100	Diversity = 6.1028e-03	k_t
= [1, 1, 1, 1]			

Generation 8600: Best = 5.202603e+03 = [3, 3, 2, 2]	FES = 860100	Diversity = 5.4691e-03	k_t
Generation 8610: Best = 5.202602e+03 = [1, 1, 1, 1]	FES = 861100	Diversity = 5.1014e-03	k_t
Generation 8620: Best = 5.202601e+03 = [4, 3, 3, 3]	FES = 862100	Diversity = 4.9406e-03	k_t
Generation 8630: Best = 5.202601e+03 = [1, 1, 1, 1]	FES = 863100	Diversity = 5.2002e-03	k_t
Generation 8640: Best = 5.202601e+03 = [3, 3, 3, 3]	FES = 864100	Diversity = 5.3413e-03	k_t
Generation 8650: Best = 5.202600e+03 = [1, 1, 1, 1]	FES = 865100	Diversity = 5.0590e-03	k_t
Generation 8660: Best = 5.202600e+03 = [4, 3, 3, 3]	FES = 866100	Diversity = 5.2094e-03	k_t
Generation 8670: Best = 5.202599e+03 = [1, 1, 1, 1]	FES = 867100	Diversity = 5.1357e-03	k_t
Generation 8680: Best = 5.202599e+03 = [3, 4, 2, 4]	FES = 868100	Diversity = 4.4371e-03	k_t
Generation 8690: Best = 5.202599e+03 = [1, 1, 1, 1]	FES = 869100	Diversity = 4.5058e-03	k_t
Generation 8700: Best = 5.202598e+03 = [2, 3, 4, 4]	FES = 870100	Diversity = 4.4710e-03	k_t
Generation 8710: Best = 5.202598e+03 = [1, 1, 1, 1]	FES = 871100	Diversity = 4.3232e-03	k_t
Generation 8720: Best = 5.202598e+03 = [3, 2, 3, 3]	FES = 872100	Diversity = 4.6671e-03	k_t
Generation 8730: Best = 5.202597e+03 = [1, 1, 1, 1]	FES = 873100	Diversity = 4.7083e-03	k_t
Generation 8740: Best = 5.202597e+03 = [4, 3, 2, 4]	FES = 874100	Diversity = 4.7862e-03	k_t
Generation 8750: Best = 5.202596e+03 = [1, 1, 1, 1]	FES = 875100	Diversity = 4.8159e-03	k_t
Generation 8760: Best = 5.202596e+03 = [2, 4, 4, 3]	FES = 876100	Diversity = 4.9358e-03	k_t
Generation 8770: Best = 5.202595e+03 = [1, 1, 1, 1]	FES = 877100	Diversity = 5.0032e-03	k_t
Generation 8780: Best = 5.202595e+03 = [2, 2, 3, 2]	FES = 878100	Diversity = 4.6328e-03	k_t
Generation 8790: Best = 5.202594e+03 = [1, 1, 1, 1]	FES = 879100	Diversity = 4.8545e-03	k_t
Generation 8800: Best = 5.202594e+03 = [4, 3, 2, 4]	FES = 880100	Diversity = 6.2547e-03	k_t
Generation 8810: Best = 5.202593e+03 = [1, 1, 1, 1]	FES = 881100	Diversity = 6.1464e-03	k_t
Generation 8820: Best = 5.202592e+03 = [4, 4, 3, 4]	FES = 882100	Diversity = 6.7095e-03	k_t
Generation 8830: Best = 5.202592e+03 = [1, 1, 1, 1]	FES = 883100	Diversity = 6.2272e-03	k_t
Generation 8840: Best = 5.202591e+03 = [3, 4, 2, 2]	FES = 884100	Diversity = 6.6616e-03	k_t
Generation 8850: Best = 5.202590e+03 = [1, 1, 1, 1]	FES = 885100	Diversity = 6.8542e-03	k_t
Generation 8860: Best = 5.202589e+03 = [4, 3, 3, 2]	FES = 886100	Diversity = 7.1411e-03	k_t

Generation 8870: Best = 5.202589e+03 = [1, 1, 1, 1]	FES = 887100	Diversity = 7.3323e-03	k_t
Generation 8880: Best = 5.202588e+03 = [3, 2, 3, 4]	FES = 888100	Diversity = 7.7618e-03	k_t
Generation 8890: Best = 5.202587e+03 = [1, 1, 1, 1]	FES = 889100	Diversity = 7.3030e-03	k_t
Generation 8900: Best = 5.202585e+03 = [2, 4, 4, 4]	FES = 890100	Diversity = 7.4350e-03	k_t
Generation 8910: Best = 5.202584e+03 = [1, 1, 1, 1]	FES = 891100	Diversity = 7.3064e-03	k_t
Generation 8920: Best = 5.202583e+03 = [3, 2, 4, 4]	FES = 892100	Diversity = 7.4906e-03	k_t
Generation 8930: Best = 5.202583e+03 = [1, 1, 1, 1]	FES = 893100	Diversity = 7.0033e-03	k_t
Generation 8940: Best = 5.202582e+03 = [2, 3, 2, 2]	FES = 894100	Diversity = 6.2505e-03	k_t
Generation 8950: Best = 5.202581e+03 = [1, 1, 1, 1]	FES = 895100	Diversity = 5.5571e-03	k_t
Generation 8960: Best = 5.202581e+03 = [3, 2, 2, 2]	FES = 896100	Diversity = 4.9694e-03	k_t
Generation 8970: Best = 5.202580e+03 = [1, 1, 1, 1]	FES = 897100	Diversity = 4.8309e-03	k_t
Generation 8980: Best = 5.202580e+03 = [4, 2, 4, 3]	FES = 898100	Diversity = 5.0272e-03	k_t
Generation 8990: Best = 5.202579e+03 = [1, 1, 1, 1]	FES = 899100	Diversity = 4.7646e-03	k_t
Generation 9000: Best = 5.202578e+03 = [3, 3, 4, 4]	FES = 900100	Diversity = 5.0540e-03	k_t
Generation 9010: Best = 5.202578e+03 = [1, 1, 1, 1]	FES = 901100	Diversity = 5.5453e-03	k_t
Generation 9020: Best = 5.202577e+03 = [2, 2, 4, 4]	FES = 902100	Diversity = 5.8416e-03	k_t
Generation 9030: Best = 5.202576e+03 = [1, 1, 1, 1]	FES = 903100	Diversity = 5.4180e-03	k_t
Generation 9040: Best = 5.202575e+03 = [3, 4, 4, 2]	FES = 904100	Diversity = 6.4671e-03	k_t
Generation 9050: Best = 5.202574e+03 = [1, 1, 1, 1]	FES = 905100	Diversity = 6.9371e-03	k_t
Generation 9060: Best = 5.202573e+03 = [2, 3, 2, 3]	FES = 906100	Diversity = 7.2548e-03	k_t
Generation 9070: Best = 5.202572e+03 = [1, 1, 1, 1]	FES = 907100	Diversity = 6.9447e-03	k_t
Generation 9080: Best = 5.202571e+03 = [3, 4, 3, 4]	FES = 908100	Diversity = 7.1802e-03	k_t
Generation 9090: Best = 5.202569e+03 = [1, 1, 1, 1]	FES = 909100	Diversity = 7.1072e-03	k_t
Generation 9100: Best = 5.202569e+03 = [2, 4, 3, 3]	FES = 910100	Diversity = 7.6444e-03	k_t
Generation 9110: Best = 5.202567e+03 = [1, 1, 1, 1]	FES = 911100	Diversity = 7.9470e-03	k_t
Generation 9120: Best = 5.202567e+03 = [3, 4, 3, 2]	FES = 912100	Diversity = 8.4952e-03	k_t
Generation 9130: Best = 5.202566e+03 = [1, 1, 1, 1]	FES = 913100	Diversity = 8.4263e-03	k_t

Generation 9140: Best = 5.202565e+03 = [4, 3, 4, 3]	FES = 914100	Diversity = 8.3646e-03	k_t
Generation 9150: Best = 5.202563e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 8.3205e-03	k_t
Generation 9160: Best = 5.202562e+03 = [3, 3, 4, 4]	FES = 916100	Diversity = 7.8768e-03	k_t
Generation 9170: Best = 5.202561e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 7.4029e-03	k_t
Generation 9180: Best = 5.202561e+03 = [2, 3, 4, 3]	FES = 918100	Diversity = 7.0553e-03	k_t
Generation 9190: Best = 5.202561e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 6.5499e-03	k_t
Generation 9200: Best = 5.202560e+03 = [3, 2, 3, 2]	FES = 920100	Diversity = 6.4880e-03	k_t
Generation 9210: Best = 5.202560e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 6.5366e-03	k_t
Generation 9220: Best = 5.202559e+03 = [3, 3, 3, 3]	FES = 922100	Diversity = 6.1961e-03	k_t
Generation 9230: Best = 5.202558e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 5.8553e-03	k_t
Generation 9240: Best = 5.202558e+03 = [3, 2, 3, 2]	FES = 924100	Diversity = 5.6696e-03	k_t
Generation 9250: Best = 5.202558e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 5.3602e-03	k_t
Generation 9260: Best = 5.202557e+03 = [3, 4, 2, 3]	FES = 926100	Diversity = 4.8764e-03	k_t
Generation 9270: Best = 5.202556e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 4.6581e-03	k_t
Generation 9280: Best = 5.202556e+03 = [2, 2, 2, 2]	FES = 928100	Diversity = 4.6356e-03	k_t
Generation 9290: Best = 5.202555e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 5.5098e-03	k_t
Generation 9300: Best = 5.202554e+03 = [4, 2, 4, 2]	FES = 930100	Diversity = 6.2126e-03	k_t
Generation 9310: Best = 5.202553e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 6.1505e-03	k_t
Generation 9320: Best = 5.202553e+03 = [4, 3, 3, 4]	FES = 932100	Diversity = 5.8240e-03	k_t
Generation 9330: Best = 5.202552e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 5.7210e-03	k_t
Generation 9340: Best = 5.202551e+03 = [4, 4, 2, 2]	FES = 934100	Diversity = 5.4319e-03	k_t
Generation 9350: Best = 5.202551e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 5.8612e-03	k_t
Generation 9360: Best = 5.202549e+03 = [4, 2, 3, 2]	FES = 936100	Diversity = 6.3663e-03	k_t
Generation 9370: Best = 5.202548e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 7.1039e-03	k_t
Generation 9380: Best = 5.202548e+03 = [3, 2, 3, 3]	FES = 938100	Diversity = 7.5419e-03	k_t
Generation 9390: Best = 5.202547e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 7.0635e-03	k_t
Generation 9400: Best = 5.202546e+03 = [4, 3, 3, 3]	FES = 940100	Diversity = 6.7091e-03	k_t

Generation 9410: Best = 5.202546e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 6.6488e-03	k_t
Generation 9420: Best = 5.202545e+03 = [3, 3, 4, 2]	FES = 942100	Diversity = 6.0157e-03	k_t
Generation 9430: Best = 5.202545e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 6.2480e-03	k_t
Generation 9440: Best = 5.202544e+03 = [2, 4, 3, 4]	FES = 944100	Diversity = 6.0045e-03	k_t
Generation 9450: Best = 5.202543e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 5.6732e-03	k_t
Generation 9460: Best = 5.202543e+03 = [2, 4, 3, 2]	FES = 946100	Diversity = 5.5119e-03	k_t
Generation 9470: Best = 5.202542e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 5.6727e-03	k_t
Generation 9480: Best = 5.202542e+03 = [2, 3, 3, 3]	FES = 948100	Diversity = 5.6589e-03	k_t
Generation 9490: Best = 5.202541e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 5.6715e-03	k_t
Generation 9500: Best = 5.202540e+03 = [3, 4, 2, 2]	FES = 950100	Diversity = 5.9680e-03	k_t
Generation 9510: Best = 5.202539e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 6.7474e-03	k_t
Generation 9520: Best = 5.202538e+03 = [3, 3, 4, 2]	FES = 952100	Diversity = 6.8526e-03	k_t
Generation 9530: Best = 5.202538e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 6.6318e-03	k_t
Generation 9540: Best = 5.202538e+03 = [2, 2, 2, 4]	FES = 954100	Diversity = 6.2338e-03	k_t
Generation 9550: Best = 5.202537e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 6.1909e-03	k_t
Generation 9560: Best = 5.202535e+03 = [3, 3, 2, 2]	FES = 956100	Diversity = 6.1356e-03	k_t
Generation 9570: Best = 5.202535e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 5.9301e-03	k_t
Generation 9580: Best = 5.202535e+03 = [2, 4, 2, 4]	FES = 958100	Diversity = 5.6792e-03	k_t
Generation 9590: Best = 5.202534e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 5.7631e-03	k_t
Generation 9600: Best = 5.202533e+03 = [2, 2, 4, 2]	FES = 960100	Diversity = 5.7462e-03	k_t
Generation 9610: Best = 5.202533e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 6.0144e-03	k_t
Generation 9620: Best = 5.202532e+03 = [4, 3, 3, 2]	FES = 962100	Diversity = 5.8340e-03	k_t
Generation 9630: Best = 5.202532e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 5.8275e-03	k_t
Generation 9640: Best = 5.202531e+03 = [4, 3, 2, 3]	FES = 964100	Diversity = 5.2390e-03	k_t
Generation 9650: Best = 5.202531e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 5.0992e-03	k_t
Generation 9660: Best = 5.202531e+03 = [4, 2, 3, 2]	FES = 966100	Diversity = 5.5698e-03	k_t
Generation 9670: Best = 5.202530e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 5.1669e-03	k_t

Generation 9680: Best = 5.202530e+03 = [2, 4, 4, 2]	FES = 968100	Diversity = 5.1567e-03	k_t
Generation 9690: Best = 5.202529e+03 = [1, 1, 1, 1]	FES = 969100	Diversity = 5.6207e-03	k_t
Generation 9700: Best = 5.202529e+03 = [4, 3, 3, 4]	FES = 970100	Diversity = 5.7515e-03	k_t
Generation 9710: Best = 5.202528e+03 = [1, 1, 1, 1]	FES = 971100	Diversity = 5.6708e-03	k_t
Generation 9720: Best = 5.202527e+03 = [2, 4, 3, 2]	FES = 972100	Diversity = 6.1338e-03	k_t
Generation 9730: Best = 5.202527e+03 = [1, 1, 1, 1]	FES = 973100	Diversity = 6.1332e-03	k_t
Generation 9740: Best = 5.202526e+03 = [3, 3, 3, 2]	FES = 974100	Diversity = 5.8092e-03	k_t
Generation 9750: Best = 5.202526e+03 = [1, 1, 1, 1]	FES = 975100	Diversity = 5.6527e-03	k_t
Generation 9760: Best = 5.202525e+03 = [3, 4, 2, 3]	FES = 976100	Diversity = 5.4461e-03	k_t
Generation 9770: Best = 5.202525e+03 = [1, 1, 1, 1]	FES = 977100	Diversity = 5.1931e-03	k_t
Generation 9780: Best = 5.202524e+03 = [3, 3, 2, 4]	FES = 978100	Diversity = 5.5165e-03	k_t
Generation 9790: Best = 5.202523e+03 = [1, 1, 1, 1]	FES = 979100	Diversity = 5.5820e-03	k_t
Generation 9800: Best = 5.202523e+03 = [2, 2, 3, 4]	FES = 980100	Diversity = 5.1640e-03	k_t
Generation 9810: Best = 5.202522e+03 = [1, 1, 1, 1]	FES = 981100	Diversity = 5.2307e-03	k_t
Generation 9820: Best = 5.202522e+03 = [2, 3, 4, 3]	FES = 982100	Diversity = 4.8874e-03	k_t
Generation 9830: Best = 5.202521e+03 = [1, 1, 1, 1]	FES = 983100	Diversity = 4.8170e-03	k_t
Generation 9840: Best = 5.202521e+03 = [4, 2, 3, 4]	FES = 984100	Diversity = 4.6972e-03	k_t
Generation 9850: Best = 5.202521e+03 = [1, 1, 1, 1]	FES = 985100	Diversity = 4.3054e-03	k_t
Generation 9860: Best = 5.202520e+03 = [2, 2, 2, 2]	FES = 986100	Diversity = 4.3857e-03	k_t
Generation 9870: Best = 5.202520e+03 = [1, 1, 1, 1]	FES = 987100	Diversity = 4.2945e-03	k_t
Generation 9880: Best = 5.202520e+03 = [4, 2, 4, 4]	FES = 988100	Diversity = 3.7266e-03	k_t
Generation 9890: Best = 5.202519e+03 = [1, 1, 1, 1]	FES = 989100	Diversity = 3.9363e-03	k_t
Generation 9900: Best = 5.202519e+03 = [4, 3, 3, 3]	FES = 990100	Diversity = 3.9691e-03	k_t
Generation 9910: Best = 5.202519e+03 = [1, 1, 1, 1]	FES = 991100	Diversity = 4.0234e-03	k_t
Generation 9920: Best = 5.202519e+03 = [4, 2, 3, 2]	FES = 992100	Diversity = 3.8298e-03	k_t
Generation 9930: Best = 5.202518e+03 = [1, 1, 1, 1]	FES = 993100	Diversity = 3.9263e-03	k_t
Generation 9940: Best = 5.202518e+03 = [3, 3, 3, 4]	FES = 994100	Diversity = 3.8812e-03	k_t

Generation 9950: Best = 5.202518e+03 FES = 995100 Diversity = 3.6884e-03 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 5.202517e+03 FES = 996100 Diversity = 4.0496e-03 k_t
= [3, 2, 2, 2]
Generation 9970: Best = 5.202517e+03 FES = 997100 Diversity = 4.4024e-03 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 5.202517e+03 FES = 998100 Diversity = 4.8744e-03 k_t
= [4, 2, 4, 2]
Generation 9990: Best = 5.202516e+03 FES = 999100 Diversity = 4.8174e-03 k_t
= [1, 1, 1, 1]

Optimization completed!

Final Best Fitness = 5.202516e+03

Total Function Evaluations = 1000000

=====

=

Results:

Best Fitness : 5.202516e+03

Global Optimum : 1.600000e+03

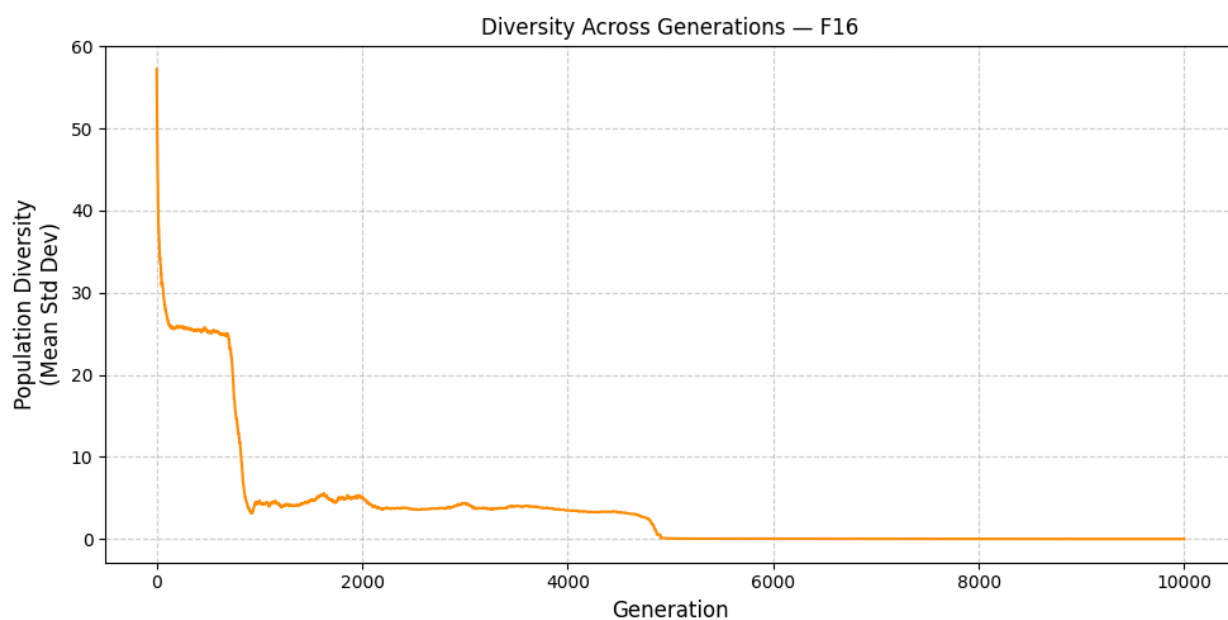
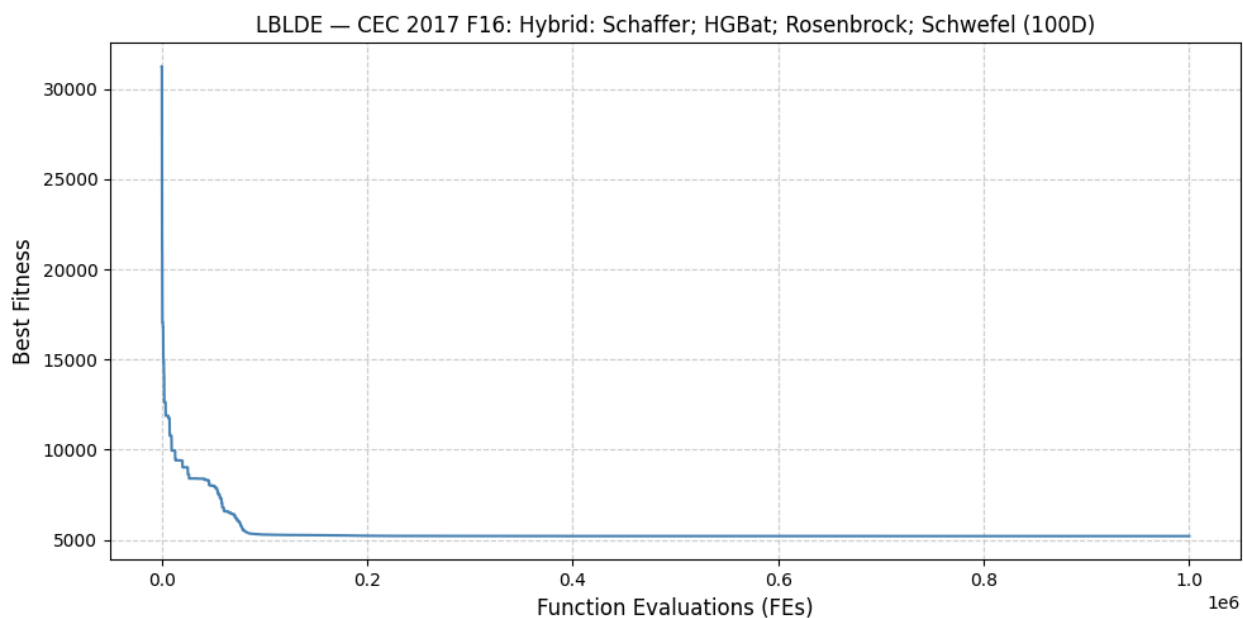
Error (|best-optimum|) : 3.602516e+03

Best Solution (5 dim) : [36.00008468 49.81935728 52.25242385 -30.41849259
-79.03677182]

D^(0) (init. diversity): 5.7220e+01

=====

=



Graph saved as convergence_F16.png

```
In [41]: # =====
# F17 – Hybrid Function 7 (Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Ra
# =====
run_and_plot('F17', make_wrapper(hybrid.f17), 1700.0, 'Hybrid: Katsuura; Ackle
```

```

=====
=
LBLDE_s on CEC 2017 F17 – Single Run
=====
=
Function      : Hybrid: Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastr
igin (100D)
Bounds       : [-100, 100]
Global optimum : 1700.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 3.179716e+07 | Diversity = 5.7220e+01
Generation 10: Best = 2.804091e+05 FES = 1100 Diversity = 4.5832e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.335256e+04 FES = 2100 Diversity = 3.9453e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 9.425536e+03 FES = 3100 Diversity = 3.3112e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 8.597837e+03 FES = 4100 Diversity = 3.1321e+01 k_t
= [4, 3, 4, 2]
Generation 50: Best = 8.123152e+03 FES = 5100 Diversity = 3.0794e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 8.008423e+03 FES = 6100 Diversity = 3.0307e+01 k_t
= [4, 4, 2, 3]
Generation 70: Best = 8.008423e+03 FES = 7100 Diversity = 3.0222e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 7.825244e+03 FES = 8100 Diversity = 2.9774e+01 k_t
= [2, 4, 3, 4]
Generation 90: Best = 7.065359e+03 FES = 9100 Diversity = 2.9274e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 6.927641e+03 FES = 10100 Diversity = 2.9191e+01 k_t
= [4, 3, 2, 3]
Generation 110: Best = 6.861627e+03 FES = 11100 Diversity = 2.8954e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 6.767802e+03 FES = 12100 Diversity = 2.8967e+01 k_t
= [3, 3, 3, 4]
Generation 130: Best = 6.767802e+03 FES = 13100 Diversity = 2.9115e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 6.688288e+03 FES = 14100 Diversity = 2.9172e+01 k_t
= [3, 4, 3, 4]
Generation 150: Best = 6.688288e+03 FES = 15100 Diversity = 2.9143e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 6.688288e+03 FES = 16100 Diversity = 2.9099e+01 k_t
= [2, 2, 4, 4]
Generation 170: Best = 6.688288e+03 FES = 17100 Diversity = 2.9189e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 6.681583e+03 FES = 18100 Diversity = 2.9618e+01 k_t
= [4, 4, 4, 3]
Generation 190: Best = 6.681583e+03 FES = 19100 Diversity = 2.9628e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 6.266111e+03 FES = 20100 Diversity = 2.9487e+01 k_t
= [4, 4, 3, 3]
Generation 210: Best = 6.157636e+03 FES = 21100 Diversity = 2.9358e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 6.157636e+03 FES = 22100 Diversity = 2.9159e+01 k_t

```

```

= [2, 2, 2, 4]
Generation 230: Best = 6.157636e+03 FES = 23100 Diversity = 2.9155e+01 k_t
= [1, 1, 1, 1]
Generation 240: Best = 6.157636e+03 FES = 24100 Diversity = 2.9234e+01 k_t
= [3, 3, 3, 2]
Generation 250: Best = 6.157636e+03 FES = 25100 Diversity = 2.9193e+01 k_t
= [1, 1, 1, 1]
Generation 260: Best = 6.157352e+03 FES = 26100 Diversity = 2.9075e+01 k_t
= [2, 2, 2, 4]
Generation 270: Best = 6.100311e+03 FES = 27100 Diversity = 2.9007e+01 k_t
= [1, 1, 1, 1]
Generation 280: Best = 6.100311e+03 FES = 28100 Diversity = 2.8979e+01 k_t
= [4, 2, 3, 2]
Generation 290: Best = 6.100311e+03 FES = 29100 Diversity = 2.8908e+01 k_t
= [1, 1, 1, 1]
Generation 300: Best = 6.100311e+03 FES = 30100 Diversity = 2.9073e+01 k_t
= [4, 2, 4, 4]
Generation 310: Best = 6.100311e+03 FES = 31100 Diversity = 2.8937e+01 k_t
= [1, 1, 1, 1]
Generation 320: Best = 5.990251e+03 FES = 32100 Diversity = 2.8728e+01 k_t
= [4, 4, 2, 3]
Generation 330: Best = 5.990251e+03 FES = 33100 Diversity = 2.8678e+01 k_t
= [1, 1, 1, 1]
Generation 340: Best = 5.990251e+03 FES = 34100 Diversity = 2.8675e+01 k_t
= [4, 4, 3, 4]
Generation 350: Best = 5.981208e+03 FES = 35100 Diversity = 2.8555e+01 k_t
= [1, 1, 1, 1]
Generation 360: Best = 5.937357e+03 FES = 36100 Diversity = 2.8368e+01 k_t
= [4, 2, 2, 4]
Generation 370: Best = 5.915474e+03 FES = 37100 Diversity = 2.8338e+01 k_t
= [1, 1, 1, 1]
Generation 380: Best = 5.914828e+03 FES = 38100 Diversity = 2.8396e+01 k_t
= [4, 4, 3, 2]
Generation 390: Best = 5.914755e+03 FES = 39100 Diversity = 2.8399e+01 k_t
= [1, 1, 1, 1]
Generation 400: Best = 5.910961e+03 FES = 40100 Diversity = 2.8332e+01 k_t
= [3, 3, 3, 3]
Generation 410: Best = 5.910961e+03 FES = 41100 Diversity = 2.8307e+01 k_t
= [1, 1, 1, 1]
Generation 420: Best = 5.826725e+03 FES = 42100 Diversity = 2.8459e+01 k_t
= [4, 3, 4, 4]
Generation 430: Best = 5.780252e+03 FES = 43100 Diversity = 2.8384e+01 k_t
= [1, 1, 1, 1]
Generation 440: Best = 5.780252e+03 FES = 44100 Diversity = 2.8523e+01 k_t
= [3, 3, 4, 4]
Generation 450: Best = 5.780252e+03 FES = 45100 Diversity = 2.8676e+01 k_t
= [1, 1, 1, 1]
Generation 460: Best = 5.759238e+03 FES = 46100 Diversity = 2.8808e+01 k_t
= [2, 3, 2, 3]
Generation 470: Best = 5.759238e+03 FES = 47100 Diversity = 2.8543e+01 k_t
= [1, 1, 1, 1]
Generation 480: Best = 5.759238e+03 FES = 48100 Diversity = 2.8568e+01 k_t
= [2, 3, 4, 2]
Generation 490: Best = 5.759238e+03 FES = 49100 Diversity = 2.8755e+01 k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 5.736309e+03 FES = 50100 Diversity = 2.8747e+01 k_t
= [2, 3, 3, 2]
Generation 510: Best = 5.736309e+03 FES = 51100 Diversity = 2.8721e+01 k_t
= [1, 1, 1, 1]
Generation 520: Best = 5.643967e+03 FES = 52100 Diversity = 2.8591e+01 k_t
= [2, 4, 4, 4]
Generation 530: Best = 5.643967e+03 FES = 53100 Diversity = 2.8977e+01 k_t
= [1, 1, 1, 1]
Generation 540: Best = 5.604005e+03 FES = 54100 Diversity = 2.9235e+01 k_t
= [2, 4, 3, 3]
Generation 550: Best = 5.570650e+03 FES = 55100 Diversity = 2.9256e+01 k_t
= [1, 1, 1, 1]
Generation 560: Best = 5.463698e+03 FES = 56100 Diversity = 2.9196e+01 k_t
= [4, 2, 2, 2]
Generation 570: Best = 5.446687e+03 FES = 57100 Diversity = 2.9076e+01 k_t
= [1, 1, 1, 1]
Generation 580: Best = 5.441992e+03 FES = 58100 Diversity = 2.9033e+01 k_t
= [2, 2, 3, 4]
Generation 590: Best = 5.284709e+03 FES = 59100 Diversity = 2.9005e+01 k_t
= [1, 1, 1, 1]
Generation 600: Best = 5.227045e+03 FES = 60100 Diversity = 2.9196e+01 k_t
= [4, 2, 3, 4]
Generation 610: Best = 5.184315e+03 FES = 61100 Diversity = 2.8937e+01 k_t
= [1, 1, 1, 1]
Generation 620: Best = 5.073096e+03 FES = 62100 Diversity = 2.8400e+01 k_t
= [3, 3, 3, 3]
Generation 630: Best = 5.009293e+03 FES = 63100 Diversity = 2.7550e+01 k_t
= [1, 1, 1, 1]
Generation 640: Best = 4.914323e+03 FES = 64100 Diversity = 2.6694e+01 k_t
= [2, 2, 2, 4]
Generation 650: Best = 4.798856e+03 FES = 65100 Diversity = 2.6212e+01 k_t
= [1, 1, 1, 1]
Generation 660: Best = 4.788113e+03 FES = 66100 Diversity = 2.5928e+01 k_t
= [3, 2, 4, 4]
Generation 670: Best = 4.688679e+03 FES = 67100 Diversity = 2.5511e+01 k_t
= [1, 1, 1, 1]
Generation 680: Best = 4.639928e+03 FES = 68100 Diversity = 2.5387e+01 k_t
= [2, 2, 4, 2]
Generation 690: Best = 4.553365e+03 FES = 69100 Diversity = 2.5182e+01 k_t
= [1, 1, 1, 1]
Generation 700: Best = 4.496326e+03 FES = 70100 Diversity = 2.5223e+01 k_t
= [4, 3, 4, 4]
Generation 710: Best = 4.474370e+03 FES = 71100 Diversity = 2.4846e+01 k_t
= [1, 1, 1, 1]
Generation 720: Best = 4.417108e+03 FES = 72100 Diversity = 2.5033e+01 k_t
= [4, 4, 2, 4]
Generation 730: Best = 4.417108e+03 FES = 73100 Diversity = 2.4811e+01 k_t
= [1, 1, 1, 1]
Generation 740: Best = 4.365344e+03 FES = 74100 Diversity = 2.4616e+01 k_t
= [2, 2, 2, 2]
Generation 750: Best = 4.353085e+03 FES = 75100 Diversity = 2.4238e+01 k_t
= [1, 1, 1, 1]
Generation 760: Best = 4.345782e+03 FES = 76100 Diversity = 2.3334e+01 k_t

```

```

= [2, 3, 3, 4]
Generation 770: Best = 4.289720e+03 FES = 77100 Diversity = 2.2009e+01 k_t
= [1, 1, 1, 1]
Generation 780: Best = 4.289280e+03 FES = 78100 Diversity = 2.0392e+01 k_t
= [3, 2, 4, 3]
Generation 790: Best = 4.278507e+03 FES = 79100 Diversity = 1.8809e+01 k_t
= [1, 1, 1, 1]
Generation 800: Best = 4.258899e+03 FES = 80100 Diversity = 1.6892e+01 k_t
= [4, 2, 4, 2]
Generation 810: Best = 4.195467e+03 FES = 81100 Diversity = 1.6206e+01 k_t
= [1, 1, 1, 1]
Generation 820: Best = 4.195467e+03 FES = 82100 Diversity = 1.5542e+01 k_t
= [2, 4, 4, 3]
Generation 830: Best = 4.170555e+03 FES = 83100 Diversity = 1.5177e+01 k_t
= [1, 1, 1, 1]
Generation 840: Best = 4.165299e+03 FES = 84100 Diversity = 1.4015e+01 k_t
= [4, 4, 2, 3]
Generation 850: Best = 4.165299e+03 FES = 85100 Diversity = 1.3858e+01 k_t
= [1, 1, 1, 1]
Generation 860: Best = 4.165299e+03 FES = 86100 Diversity = 1.3586e+01 k_t
= [3, 4, 2, 2]
Generation 870: Best = 4.150445e+03 FES = 87100 Diversity = 1.3369e+01 k_t
= [1, 1, 1, 1]
Generation 880: Best = 4.115318e+03 FES = 88100 Diversity = 1.3434e+01 k_t
= [3, 4, 2, 2]
Generation 890: Best = 4.105110e+03 FES = 89100 Diversity = 1.3325e+01 k_t
= [1, 1, 1, 1]
Generation 900: Best = 4.105110e+03 FES = 90100 Diversity = 1.3169e+01 k_t
= [3, 3, 2, 3]
Generation 910: Best = 4.105110e+03 FES = 91100 Diversity = 1.3138e+01 k_t
= [1, 1, 1, 1]
Generation 920: Best = 4.105110e+03 FES = 92100 Diversity = 1.3036e+01 k_t
= [4, 3, 2, 4]
Generation 930: Best = 4.096676e+03 FES = 93100 Diversity = 1.2955e+01 k_t
= [1, 1, 1, 1]
Generation 940: Best = 4.095385e+03 FES = 94100 Diversity = 1.2804e+01 k_t
= [2, 2, 4, 3]
Generation 950: Best = 4.093883e+03 FES = 95100 Diversity = 1.2711e+01 k_t
= [1, 1, 1, 1]
Generation 960: Best = 4.093883e+03 FES = 96100 Diversity = 1.2643e+01 k_t
= [4, 2, 3, 3]
Generation 970: Best = 4.093883e+03 FES = 97100 Diversity = 1.2595e+01 k_t
= [1, 1, 1, 1]
Generation 980: Best = 4.083646e+03 FES = 98100 Diversity = 1.2503e+01 k_t
= [4, 4, 4, 4]
Generation 990: Best = 4.083646e+03 FES = 99100 Diversity = 1.2439e+01 k_t
= [1, 1, 1, 1]
Generation 1000: Best = 4.083646e+03 FES = 100100 Diversity = 1.2475e+01 k_t
= [3, 4, 2, 2]
Generation 1010: Best = 4.078376e+03 FES = 101100 Diversity = 1.2399e+01 k_t
= [1, 1, 1, 1]
Generation 1020: Best = 4.076488e+03 FES = 102100 Diversity = 1.2350e+01 k_t
= [4, 3, 4, 4]
Generation 1030: Best = 4.076488e+03 FES = 103100 Diversity = 1.2438e+01 k_t

```

```

= [1, 1, 1, 1]
Generation 1040: Best = 4.070513e+03  FES = 104100  Diversity = 1.2435e+01  k_t
= [2, 2, 2, 2]
Generation 1050: Best = 4.066214e+03  FES = 105100  Diversity = 1.2503e+01  k_t
= [1, 1, 1, 1]
Generation 1060: Best = 4.064755e+03  FES = 106100  Diversity = 1.2502e+01  k_t
= [2, 3, 3, 4]
Generation 1070: Best = 4.062921e+03  FES = 107100  Diversity = 1.2590e+01  k_t
= [1, 1, 1, 1]
Generation 1080: Best = 4.062921e+03  FES = 108100  Diversity = 1.2616e+01  k_t
= [2, 2, 2, 3]
Generation 1090: Best = 4.062921e+03  FES = 109100  Diversity = 1.2528e+01  k_t
= [1, 1, 1, 1]
Generation 1100: Best = 4.062921e+03  FES = 110100  Diversity = 1.2558e+01  k_t
= [4, 3, 4, 2]
Generation 1110: Best = 4.062921e+03  FES = 111100  Diversity = 1.2553e+01  k_t
= [1, 1, 1, 1]
Generation 1120: Best = 4.062921e+03  FES = 112100  Diversity = 1.2578e+01  k_t
= [3, 4, 4, 3]
Generation 1130: Best = 4.053934e+03  FES = 113100  Diversity = 1.2579e+01  k_t
= [1, 1, 1, 1]
Generation 1140: Best = 4.053934e+03  FES = 114100  Diversity = 1.2544e+01  k_t
= [3, 2, 4, 3]
Generation 1150: Best = 4.053934e+03  FES = 115100  Diversity = 1.2481e+01  k_t
= [1, 1, 1, 1]
Generation 1160: Best = 4.053786e+03  FES = 116100  Diversity = 1.2502e+01  k_t
= [3, 2, 3, 2]
Generation 1170: Best = 4.053252e+03  FES = 117100  Diversity = 1.2527e+01  k_t
= [1, 1, 1, 1]
Generation 1180: Best = 4.053252e+03  FES = 118100  Diversity = 1.2570e+01  k_t
= [2, 3, 2, 2]
Generation 1190: Best = 4.053252e+03  FES = 119100  Diversity = 1.2575e+01  k_t
= [1, 1, 1, 1]
Generation 1200: Best = 4.053252e+03  FES = 120100  Diversity = 1.2598e+01  k_t
= [2, 3, 4, 3]
Generation 1210: Best = 4.049821e+03  FES = 121100  Diversity = 1.2624e+01  k_t
= [1, 1, 1, 1]
Generation 1220: Best = 4.036482e+03  FES = 122100  Diversity = 1.2638e+01  k_t
= [3, 3, 2, 4]
Generation 1230: Best = 4.036151e+03  FES = 123100  Diversity = 1.2559e+01  k_t
= [1, 1, 1, 1]
Generation 1240: Best = 4.036151e+03  FES = 124100  Diversity = 1.2574e+01  k_t
= [2, 3, 2, 3]
Generation 1250: Best = 4.035823e+03  FES = 125100  Diversity = 1.2580e+01  k_t
= [1, 1, 1, 1]
Generation 1260: Best = 4.035823e+03  FES = 126100  Diversity = 1.2638e+01  k_t
= [4, 4, 4, 2]
Generation 1270: Best = 4.035823e+03  FES = 127100  Diversity = 1.2634e+01  k_t
= [1, 1, 1, 1]
Generation 1280: Best = 4.035823e+03  FES = 128100  Diversity = 1.2626e+01  k_t
= [4, 3, 3, 2]
Generation 1290: Best = 4.035790e+03  FES = 129100  Diversity = 1.2608e+01  k_t
= [1, 1, 1, 1]
Generation 1300: Best = 4.035790e+03  FES = 130100  Diversity = 1.2640e+01  k_t

```

```

= [2, 4, 4, 4]
Generation 1310: Best = 4.035790e+03 FES = 131100 Diversity = 1.2557e+01 k_t
= [1, 1, 1, 1]
Generation 1320: Best = 4.035790e+03 FES = 132100 Diversity = 1.2582e+01 k_t
= [2, 4, 2, 4]
Generation 1330: Best = 4.033315e+03 FES = 133100 Diversity = 1.2579e+01 k_t
= [1, 1, 1, 1]
Generation 1340: Best = 4.031116e+03 FES = 134100 Diversity = 1.2560e+01 k_t
= [3, 4, 3, 4]
Generation 1350: Best = 4.031116e+03 FES = 135100 Diversity = 1.2528e+01 k_t
= [1, 1, 1, 1]
Generation 1360: Best = 4.031116e+03 FES = 136100 Diversity = 1.2592e+01 k_t
= [4, 3, 2, 3]
Generation 1370: Best = 4.031116e+03 FES = 137100 Diversity = 1.2664e+01 k_t
= [1, 1, 1, 1]
Generation 1380: Best = 4.031116e+03 FES = 138100 Diversity = 1.2624e+01 k_t
= [2, 3, 4, 2]
Generation 1390: Best = 4.027480e+03 FES = 139100 Diversity = 1.2580e+01 k_t
= [1, 1, 1, 1]
Generation 1400: Best = 4.022258e+03 FES = 140100 Diversity = 1.2590e+01 k_t
= [3, 4, 4, 4]
Generation 1410: Best = 4.022258e+03 FES = 141100 Diversity = 1.2597e+01 k_t
= [1, 1, 1, 1]
Generation 1420: Best = 4.022258e+03 FES = 142100 Diversity = 1.2581e+01 k_t
= [2, 4, 4, 4]
Generation 1430: Best = 4.022258e+03 FES = 143100 Diversity = 1.2592e+01 k_t
= [1, 1, 1, 1]
Generation 1440: Best = 4.022258e+03 FES = 144100 Diversity = 1.2580e+01 k_t
= [2, 3, 3, 2]
Generation 1450: Best = 4.004330e+03 FES = 145100 Diversity = 1.2585e+01 k_t
= [1, 1, 1, 1]
Generation 1460: Best = 4.004330e+03 FES = 146100 Diversity = 1.2549e+01 k_t
= [2, 4, 3, 2]
Generation 1470: Best = 4.004330e+03 FES = 147100 Diversity = 1.2559e+01 k_t
= [1, 1, 1, 1]
Generation 1480: Best = 3.998608e+03 FES = 148100 Diversity = 1.2586e+01 k_t
= [2, 2, 4, 4]
Generation 1490: Best = 3.998608e+03 FES = 149100 Diversity = 1.2587e+01 k_t
= [1, 1, 1, 1]
Generation 1500: Best = 3.998561e+03 FES = 150100 Diversity = 1.2611e+01 k_t
= [3, 2, 4, 3]
Generation 1510: Best = 3.998561e+03 FES = 151100 Diversity = 1.2571e+01 k_t
= [1, 1, 1, 1]
Generation 1520: Best = 3.998561e+03 FES = 152100 Diversity = 1.2570e+01 k_t
= [3, 3, 2, 3]
Generation 1530: Best = 3.998561e+03 FES = 153100 Diversity = 1.2605e+01 k_t
= [1, 1, 1, 1]
Generation 1540: Best = 3.998519e+03 FES = 154100 Diversity = 1.2646e+01 k_t
= [3, 4, 4, 2]
Generation 1550: Best = 3.998519e+03 FES = 155100 Diversity = 1.2653e+01 k_t
= [1, 1, 1, 1]
Generation 1560: Best = 3.997225e+03 FES = 156100 Diversity = 1.2643e+01 k_t
= [4, 4, 3, 2]
Generation 1570: Best = 3.997225e+03 FES = 157100 Diversity = 1.2626e+01 k_t

```

```

= [1, 1, 1, 1]
Generation 1580: Best = 3.997225e+03  FES = 158100  Diversity = 1.2731e+01  k_t
= [2, 4, 2, 3]
Generation 1590: Best = 3.997225e+03  FES = 159100  Diversity = 1.2637e+01  k_t
= [1, 1, 1, 1]
Generation 1600: Best = 3.997225e+03  FES = 160100  Diversity = 1.2599e+01  k_t
= [4, 2, 3, 2]
Generation 1610: Best = 3.995301e+03  FES = 161100  Diversity = 1.2605e+01  k_t
= [1, 1, 1, 1]
Generation 1620: Best = 3.993020e+03  FES = 162100  Diversity = 1.2545e+01  k_t
= [3, 2, 2, 2]
Generation 1630: Best = 3.993020e+03  FES = 163100  Diversity = 1.2554e+01  k_t
= [1, 1, 1, 1]
Generation 1640: Best = 3.992125e+03  FES = 164100  Diversity = 1.2543e+01  k_t
= [4, 3, 4, 2]
Generation 1650: Best = 3.992125e+03  FES = 165100  Diversity = 1.2543e+01  k_t
= [1, 1, 1, 1]
Generation 1660: Best = 3.992125e+03  FES = 166100  Diversity = 1.2515e+01  k_t
= [3, 2, 4, 4]
Generation 1670: Best = 3.991822e+03  FES = 167100  Diversity = 1.2535e+01  k_t
= [1, 1, 1, 1]
Generation 1680: Best = 3.975490e+03  FES = 168100  Diversity = 1.2502e+01  k_t
= [2, 4, 4, 4]
Generation 1690: Best = 3.975490e+03  FES = 169100  Diversity = 1.2521e+01  k_t
= [1, 1, 1, 1]
Generation 1700: Best = 3.972281e+03  FES = 170100  Diversity = 1.2548e+01  k_t
= [2, 4, 3, 2]
Generation 1710: Best = 3.967507e+03  FES = 171100  Diversity = 1.2504e+01  k_t
= [1, 1, 1, 1]
Generation 1720: Best = 3.967507e+03  FES = 172100  Diversity = 1.2426e+01  k_t
= [3, 3, 4, 4]
Generation 1730: Best = 3.963968e+03  FES = 173100  Diversity = 1.2418e+01  k_t
= [1, 1, 1, 1]
Generation 1740: Best = 3.955670e+03  FES = 174100  Diversity = 1.2420e+01  k_t
= [4, 3, 2, 4]
Generation 1750: Best = 3.955670e+03  FES = 175100  Diversity = 1.2440e+01  k_t
= [1, 1, 1, 1]
Generation 1760: Best = 3.945253e+03  FES = 176100  Diversity = 1.2358e+01  k_t
= [4, 2, 3, 4]
Generation 1770: Best = 3.941487e+03  FES = 177100  Diversity = 1.2179e+01  k_t
= [1, 1, 1, 1]
Generation 1780: Best = 3.938035e+03  FES = 178100  Diversity = 1.2011e+01  k_t
= [2, 4, 3, 4]
Generation 1790: Best = 3.936191e+03  FES = 179100  Diversity = 1.1830e+01  k_t
= [1, 1, 1, 1]
Generation 1800: Best = 3.930246e+03  FES = 180100  Diversity = 1.1708e+01  k_t
= [4, 2, 3, 2]
Generation 1810: Best = 3.930246e+03  FES = 181100  Diversity = 1.1472e+01  k_t
= [1, 1, 1, 1]
Generation 1820: Best = 3.930246e+03  FES = 182100  Diversity = 1.1193e+01  k_t
= [3, 3, 3, 3]
Generation 1830: Best = 3.930006e+03  FES = 183100  Diversity = 1.1049e+01  k_t
= [1, 1, 1, 1]
Generation 1840: Best = 3.929321e+03  FES = 184100  Diversity = 1.0807e+01  k_t

```



```

= [3, 3, 3, 2]
Generation 1850: Best = 3.926224e+03 FES = 185100 Diversity = 1.0119e+01 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 3.919124e+03 FES = 186100 Diversity = 9.8191e+00 k_t
= [4, 4, 2, 2]
Generation 1870: Best = 3.917327e+03 FES = 187100 Diversity = 9.2220e+00 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 3.914363e+03 FES = 188100 Diversity = 8.7792e+00 k_t
= [3, 4, 3, 3]
Generation 1890: Best = 3.907953e+03 FES = 189100 Diversity = 8.4361e+00 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 3.907953e+03 FES = 190100 Diversity = 7.8193e+00 k_t
= [2, 3, 2, 4]
Generation 1910: Best = 3.906019e+03 FES = 191100 Diversity = 7.5556e+00 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 3.904160e+03 FES = 192100 Diversity = 7.2527e+00 k_t
= [3, 2, 4, 3]
Generation 1930: Best = 3.904021e+03 FES = 193100 Diversity = 6.8770e+00 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 3.900589e+03 FES = 194100 Diversity = 6.8397e+00 k_t
= [2, 4, 3, 4]
Generation 1950: Best = 3.898560e+03 FES = 195100 Diversity = 6.6127e+00 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 3.893390e+03 FES = 196100 Diversity = 6.3653e+00 k_t
= [2, 2, 4, 3]
Generation 1970: Best = 3.893390e+03 FES = 197100 Diversity = 6.2368e+00 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 3.891542e+03 FES = 198100 Diversity = 6.1517e+00 k_t
= [3, 2, 4, 4]
Generation 1990: Best = 3.891542e+03 FES = 199100 Diversity = 6.0595e+00 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 3.890402e+03 FES = 200100 Diversity = 5.9466e+00 k_t
= [4, 3, 3, 3]
Generation 2010: Best = 3.890402e+03 FES = 201100 Diversity = 5.9399e+00 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 3.890402e+03 FES = 202100 Diversity = 5.7522e+00 k_t
= [4, 4, 4, 4]
Generation 2030: Best = 3.888977e+03 FES = 203100 Diversity = 5.5945e+00 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 3.888829e+03 FES = 204100 Diversity = 5.5243e+00 k_t
= [3, 3, 2, 4]
Generation 2050: Best = 3.886694e+03 FES = 205100 Diversity = 5.4711e+00 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 3.886539e+03 FES = 206100 Diversity = 5.3718e+00 k_t
= [4, 2, 4, 2]
Generation 2070: Best = 3.886539e+03 FES = 207100 Diversity = 5.4077e+00 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 3.885672e+03 FES = 208100 Diversity = 5.3417e+00 k_t
= [2, 2, 4, 2]
Generation 2090: Best = 3.885672e+03 FES = 209100 Diversity = 5.3310e+00 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 3.885672e+03 FES = 210100 Diversity = 5.2813e+00 k_t
= [3, 2, 3, 3]
Generation 2110: Best = 3.885290e+03 FES = 211100 Diversity = 5.3313e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 2120: Best = 3.883930e+03 FES = 212100 Diversity = 5.3036e+00 k_t
= [3, 2, 2, 3]
Generation 2130: Best = 3.883644e+03 FES = 213100 Diversity = 5.3066e+00 k_t
= [1, 1, 1, 1]
Generation 2140: Best = 3.883644e+03 FES = 214100 Diversity = 5.3372e+00 k_t
= [2, 4, 3, 2]
Generation 2150: Best = 3.882082e+03 FES = 215100 Diversity = 5.2963e+00 k_t
= [1, 1, 1, 1]
Generation 2160: Best = 3.882082e+03 FES = 216100 Diversity = 5.2102e+00 k_t
= [4, 2, 4, 3]
Generation 2170: Best = 3.882082e+03 FES = 217100 Diversity = 5.1770e+00 k_t
= [1, 1, 1, 1]
Generation 2180: Best = 3.881166e+03 FES = 218100 Diversity = 5.1952e+00 k_t
= [3, 4, 3, 3]
Generation 2190: Best = 3.881166e+03 FES = 219100 Diversity = 5.1684e+00 k_t
= [1, 1, 1, 1]
Generation 2200: Best = 3.881065e+03 FES = 220100 Diversity = 5.1977e+00 k_t
= [3, 4, 2, 3]
Generation 2210: Best = 3.881043e+03 FES = 221100 Diversity = 5.2214e+00 k_t
= [1, 1, 1, 1]
Generation 2220: Best = 3.880756e+03 FES = 222100 Diversity = 5.2531e+00 k_t
= [4, 3, 3, 2]
Generation 2230: Best = 3.880544e+03 FES = 223100 Diversity = 5.2285e+00 k_t
= [1, 1, 1, 1]
Generation 2240: Best = 3.880544e+03 FES = 224100 Diversity = 5.2151e+00 k_t
= [4, 2, 2, 3]
Generation 2250: Best = 3.879566e+03 FES = 225100 Diversity = 5.2168e+00 k_t
= [1, 1, 1, 1]
Generation 2260: Best = 3.879566e+03 FES = 226100 Diversity = 5.2641e+00 k_t
= [4, 3, 2, 3]
Generation 2270: Best = 3.879566e+03 FES = 227100 Diversity = 5.2820e+00 k_t
= [1, 1, 1, 1]
Generation 2280: Best = 3.879566e+03 FES = 228100 Diversity = 5.2643e+00 k_t
= [2, 2, 2, 3]
Generation 2290: Best = 3.879566e+03 FES = 229100 Diversity = 5.2806e+00 k_t
= [1, 1, 1, 1]
Generation 2300: Best = 3.879157e+03 FES = 230100 Diversity = 5.2279e+00 k_t
= [3, 3, 3, 4]
Generation 2310: Best = 3.879157e+03 FES = 231100 Diversity = 5.1804e+00 k_t
= [1, 1, 1, 1]
Generation 2320: Best = 3.879157e+03 FES = 232100 Diversity = 5.1614e+00 k_t
= [4, 3, 3, 2]
Generation 2330: Best = 3.879062e+03 FES = 233100 Diversity = 5.1767e+00 k_t
= [1, 1, 1, 1]
Generation 2340: Best = 3.879062e+03 FES = 234100 Diversity = 5.1795e+00 k_t
= [2, 2, 2, 2]
Generation 2350: Best = 3.878975e+03 FES = 235100 Diversity = 5.2211e+00 k_t
= [1, 1, 1, 1]
Generation 2360: Best = 3.878869e+03 FES = 236100 Diversity = 5.2385e+00 k_t
= [4, 3, 2, 3]
Generation 2370: Best = 3.878869e+03 FES = 237100 Diversity = 5.2041e+00 k_t
= [1, 1, 1, 1]
Generation 2380: Best = 3.878826e+03 FES = 238100 Diversity = 5.2621e+00 k_t

```

```

= [4, 2, 4, 2]
Generation 2390: Best = 3.878826e+03 FES = 239100 Diversity = 5.3088e+00 k_t
= [1, 1, 1, 1]
Generation 2400: Best = 3.877058e+03 FES = 240100 Diversity = 5.3225e+00 k_t
= [3, 4, 3, 4]
Generation 2410: Best = 3.877058e+03 FES = 241100 Diversity = 5.3434e+00 k_t
= [1, 1, 1, 1]
Generation 2420: Best = 3.877058e+03 FES = 242100 Diversity = 5.3525e+00 k_t
= [3, 2, 3, 4]
Generation 2430: Best = 3.877058e+03 FES = 243100 Diversity = 5.3542e+00 k_t
= [1, 1, 1, 1]
Generation 2440: Best = 3.877058e+03 FES = 244100 Diversity = 5.3795e+00 k_t
= [4, 4, 4, 3]
Generation 2450: Best = 3.877058e+03 FES = 245100 Diversity = 5.4567e+00 k_t
= [1, 1, 1, 1]
Generation 2460: Best = 3.877058e+03 FES = 246100 Diversity = 5.4515e+00 k_t
= [2, 3, 4, 4]
Generation 2470: Best = 3.877044e+03 FES = 247100 Diversity = 5.5185e+00 k_t
= [1, 1, 1, 1]
Generation 2480: Best = 3.876291e+03 FES = 248100 Diversity = 5.5080e+00 k_t
= [3, 4, 4, 4]
Generation 2490: Best = 3.876291e+03 FES = 249100 Diversity = 5.6028e+00 k_t
= [1, 1, 1, 1]
Generation 2500: Best = 3.876291e+03 FES = 250100 Diversity = 5.6200e+00 k_t
= [4, 3, 4, 4]
Generation 2510: Best = 3.876287e+03 FES = 251100 Diversity = 5.6194e+00 k_t
= [1, 1, 1, 1]
Generation 2520: Best = 3.876264e+03 FES = 252100 Diversity = 5.6287e+00 k_t
= [3, 2, 4, 3]
Generation 2530: Best = 3.876264e+03 FES = 253100 Diversity = 5.5547e+00 k_t
= [1, 1, 1, 1]
Generation 2540: Best = 3.876264e+03 FES = 254100 Diversity = 5.5710e+00 k_t
= [2, 2, 4, 3]
Generation 2550: Best = 3.876253e+03 FES = 255100 Diversity = 5.5783e+00 k_t
= [1, 1, 1, 1]
Generation 2560: Best = 3.876253e+03 FES = 256100 Diversity = 5.6275e+00 k_t
= [3, 2, 4, 2]
Generation 2570: Best = 3.876253e+03 FES = 257100 Diversity = 5.6517e+00 k_t
= [1, 1, 1, 1]
Generation 2580: Best = 3.876253e+03 FES = 258100 Diversity = 5.6737e+00 k_t
= [4, 4, 3, 2]
Generation 2590: Best = 3.876253e+03 FES = 259100 Diversity = 5.6583e+00 k_t
= [1, 1, 1, 1]
Generation 2600: Best = 3.876253e+03 FES = 260100 Diversity = 5.6033e+00 k_t
= [4, 3, 4, 4]
Generation 2610: Best = 3.875990e+03 FES = 261100 Diversity = 5.5284e+00 k_t
= [1, 1, 1, 1]
Generation 2620: Best = 3.875990e+03 FES = 262100 Diversity = 5.5440e+00 k_t
= [4, 2, 3, 3]
Generation 2630: Best = 3.875990e+03 FES = 263100 Diversity = 5.5453e+00 k_t
= [1, 1, 1, 1]
Generation 2640: Best = 3.875990e+03 FES = 264100 Diversity = 5.5476e+00 k_t
= [2, 4, 2, 2]
Generation 2650: Best = 3.875612e+03 FES = 265100 Diversity = 5.5361e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 2660: Best = 3.875612e+03 FES = 266100 Diversity = 5.5860e+00 k_t
= [4, 3, 4, 2]
Generation 2670: Best = 3.875106e+03 FES = 267100 Diversity = 5.6075e+00 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 3.874990e+03 FES = 268100 Diversity = 5.6319e+00 k_t
= [3, 3, 4, 4]
Generation 2690: Best = 3.874533e+03 FES = 269100 Diversity = 5.6553e+00 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 3.874462e+03 FES = 270100 Diversity = 5.6957e+00 k_t
= [3, 4, 2, 2]
Generation 2710: Best = 3.874462e+03 FES = 271100 Diversity = 5.5795e+00 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 3.874462e+03 FES = 272100 Diversity = 5.6047e+00 k_t
= [3, 2, 3, 3]
Generation 2730: Best = 3.873699e+03 FES = 273100 Diversity = 5.5864e+00 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 3.873699e+03 FES = 274100 Diversity = 5.5303e+00 k_t
= [2, 3, 2, 2]
Generation 2750: Best = 3.873699e+03 FES = 275100 Diversity = 5.5061e+00 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 3.873699e+03 FES = 276100 Diversity = 5.4820e+00 k_t
= [4, 4, 3, 2]
Generation 2770: Best = 3.873699e+03 FES = 277100 Diversity = 5.4260e+00 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 3.873625e+03 FES = 278100 Diversity = 5.4104e+00 k_t
= [2, 2, 3, 2]
Generation 2790: Best = 3.873625e+03 FES = 279100 Diversity = 5.4197e+00 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 3.873625e+03 FES = 280100 Diversity = 5.5286e+00 k_t
= [4, 4, 3, 3]
Generation 2810: Best = 3.873625e+03 FES = 281100 Diversity = 5.4954e+00 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 3.873458e+03 FES = 282100 Diversity = 5.4232e+00 k_t
= [2, 4, 3, 4]
Generation 2830: Best = 3.873423e+03 FES = 283100 Diversity = 5.4224e+00 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 3.873423e+03 FES = 284100 Diversity = 5.3792e+00 k_t
= [4, 2, 3, 2]
Generation 2850: Best = 3.873423e+03 FES = 285100 Diversity = 5.3844e+00 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 3.873418e+03 FES = 286100 Diversity = 5.3200e+00 k_t
= [4, 4, 2, 2]
Generation 2870: Best = 3.873386e+03 FES = 287100 Diversity = 5.2508e+00 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 3.873386e+03 FES = 288100 Diversity = 5.2184e+00 k_t
= [2, 4, 3, 3]
Generation 2890: Best = 3.873234e+03 FES = 289100 Diversity = 5.2504e+00 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 3.873234e+03 FES = 290100 Diversity = 5.1898e+00 k_t
= [4, 4, 2, 2]
Generation 2910: Best = 3.873234e+03 FES = 291100 Diversity = 5.2675e+00 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 3.872933e+03 FES = 292100 Diversity = 5.1915e+00 k_t

```

```

= [3, 3, 4, 3]
Generation 2930: Best = 3.872916e+03  FES = 293100  Diversity = 5.1452e+00  k_t
= [1, 1, 1, 1]
Generation 2940: Best = 3.872702e+03  FES = 294100  Diversity = 5.1263e+00  k_t
= [4, 4, 3, 4]
Generation 2950: Best = 3.872702e+03  FES = 295100  Diversity = 5.1564e+00  k_t
= [1, 1, 1, 1]
Generation 2960: Best = 3.871601e+03  FES = 296100  Diversity = 5.2330e+00  k_t
= [2, 3, 2, 2]
Generation 2970: Best = 3.871474e+03  FES = 297100  Diversity = 5.1677e+00  k_t
= [1, 1, 1, 1]
Generation 2980: Best = 3.871474e+03  FES = 298100  Diversity = 5.2094e+00  k_t
= [4, 4, 4, 4]
Generation 2990: Best = 3.871474e+03  FES = 299100  Diversity = 5.2375e+00  k_t
= [1, 1, 1, 1]
Generation 3000: Best = 3.871474e+03  FES = 300100  Diversity = 5.2221e+00  k_t
= [4, 3, 3, 4]
Generation 3010: Best = 3.871474e+03  FES = 301100  Diversity = 5.2032e+00  k_t
= [1, 1, 1, 1]
Generation 3020: Best = 3.871465e+03  FES = 302100  Diversity = 5.1793e+00  k_t
= [2, 4, 3, 4]
Generation 3030: Best = 3.871210e+03  FES = 303100  Diversity = 5.1478e+00  k_t
= [1, 1, 1, 1]
Generation 3040: Best = 3.871067e+03  FES = 304100  Diversity = 5.1417e+00  k_t
= [3, 2, 2, 2]
Generation 3050: Best = 3.871067e+03  FES = 305100  Diversity = 5.1394e+00  k_t
= [1, 1, 1, 1]
Generation 3060: Best = 3.871067e+03  FES = 306100  Diversity = 5.1411e+00  k_t
= [4, 3, 4, 4]
Generation 3070: Best = 3.871067e+03  FES = 307100  Diversity = 5.1289e+00  k_t
= [1, 1, 1, 1]
Generation 3080: Best = 3.871067e+03  FES = 308100  Diversity = 5.1500e+00  k_t
= [4, 3, 2, 4]
Generation 3090: Best = 3.871042e+03  FES = 309100  Diversity = 5.1593e+00  k_t
= [1, 1, 1, 1]
Generation 3100: Best = 3.871042e+03  FES = 310100  Diversity = 5.1789e+00  k_t
= [4, 4, 3, 4]
Generation 3110: Best = 3.870656e+03  FES = 311100  Diversity = 5.1527e+00  k_t
= [1, 1, 1, 1]
Generation 3120: Best = 3.870656e+03  FES = 312100  Diversity = 5.2088e+00  k_t
= [3, 3, 4, 2]
Generation 3130: Best = 3.870491e+03  FES = 313100  Diversity = 5.2131e+00  k_t
= [1, 1, 1, 1]
Generation 3140: Best = 3.869710e+03  FES = 314100  Diversity = 5.2103e+00  k_t
= [3, 2, 2, 4]
Generation 3150: Best = 3.868694e+03  FES = 315100  Diversity = 5.2347e+00  k_t
= [1, 1, 1, 1]
Generation 3160: Best = 3.868694e+03  FES = 316100  Diversity = 5.2050e+00  k_t
= [3, 4, 2, 3]
Generation 3170: Best = 3.868694e+03  FES = 317100  Diversity = 5.1474e+00  k_t
= [1, 1, 1, 1]
Generation 3180: Best = 3.868694e+03  FES = 318100  Diversity = 5.1834e+00  k_t
= [3, 4, 2, 2]
Generation 3190: Best = 3.868534e+03  FES = 319100  Diversity = 5.1940e+00  k_t

```

```

= [1, 1, 1, 1]
Generation 3200: Best = 3.868534e+03 FES = 320100 Diversity = 5.2549e+00 k_t
= [2, 3, 4, 4]
Generation 3210: Best = 3.868534e+03 FES = 321100 Diversity = 5.2825e+00 k_t
= [1, 1, 1, 1]
Generation 3220: Best = 3.868477e+03 FES = 322100 Diversity = 5.2208e+00 k_t
= [3, 2, 3, 2]
Generation 3230: Best = 3.867996e+03 FES = 323100 Diversity = 5.1567e+00 k_t
= [1, 1, 1, 1]
Generation 3240: Best = 3.867567e+03 FES = 324100 Diversity = 5.1434e+00 k_t
= [2, 3, 4, 2]
Generation 3250: Best = 3.867258e+03 FES = 325100 Diversity = 5.0684e+00 k_t
= [1, 1, 1, 1]
Generation 3260: Best = 3.867217e+03 FES = 326100 Diversity = 5.0477e+00 k_t
= [2, 4, 4, 4]
Generation 3270: Best = 3.866882e+03 FES = 327100 Diversity = 5.0164e+00 k_t
= [1, 1, 1, 1]
Generation 3280: Best = 3.866572e+03 FES = 328100 Diversity = 4.9582e+00 k_t
= [2, 2, 2, 3]
Generation 3290: Best = 3.866480e+03 FES = 329100 Diversity = 4.8675e+00 k_t
= [1, 1, 1, 1]
Generation 3300: Best = 3.866431e+03 FES = 330100 Diversity = 4.8549e+00 k_t
= [3, 4, 3, 4]
Generation 3310: Best = 3.866431e+03 FES = 331100 Diversity = 4.8313e+00 k_t
= [1, 1, 1, 1]
Generation 3320: Best = 3.866431e+03 FES = 332100 Diversity = 4.8260e+00 k_t
= [2, 4, 3, 3]
Generation 3330: Best = 3.866209e+03 FES = 333100 Diversity = 4.8134e+00 k_t
= [1, 1, 1, 1]
Generation 3340: Best = 3.865896e+03 FES = 334100 Diversity = 4.6067e+00 k_t
= [3, 3, 3, 4]
Generation 3350: Best = 3.865684e+03 FES = 335100 Diversity = 4.5142e+00 k_t
= [1, 1, 1, 1]
Generation 3360: Best = 3.865133e+03 FES = 336100 Diversity = 4.0450e+00 k_t
= [4, 4, 3, 3]
Generation 3370: Best = 3.864995e+03 FES = 337100 Diversity = 3.6469e+00 k_t
= [1, 1, 1, 1]
Generation 3380: Best = 3.864525e+03 FES = 338100 Diversity = 3.3950e+00 k_t
= [4, 3, 2, 3]
Generation 3390: Best = 3.864258e+03 FES = 339100 Diversity = 3.3282e+00 k_t
= [1, 1, 1, 1]
Generation 3400: Best = 3.863907e+03 FES = 340100 Diversity = 2.9932e+00 k_t
= [3, 2, 2, 2]
Generation 3410: Best = 3.863727e+03 FES = 341100 Diversity = 2.6744e+00 k_t
= [1, 1, 1, 1]
Generation 3420: Best = 3.863586e+03 FES = 342100 Diversity = 2.4645e+00 k_t
= [4, 4, 2, 4]
Generation 3430: Best = 3.863451e+03 FES = 343100 Diversity = 2.3375e+00 k_t
= [1, 1, 1, 1]
Generation 3440: Best = 3.863451e+03 FES = 344100 Diversity = 2.2035e+00 k_t
= [2, 4, 3, 3]
Generation 3450: Best = 3.863390e+03 FES = 345100 Diversity = 2.0861e+00 k_t
= [1, 1, 1, 1]
Generation 3460: Best = 3.863390e+03 FES = 346100 Diversity = 2.0282e+00 k_t

```

```

= [2, 2, 4, 2]
Generation 3470: Best = 3.863313e+03 FES = 347100 Diversity = 2.0275e+00 k_t
= [1, 1, 1, 1]
Generation 3480: Best = 3.863303e+03 FES = 348100 Diversity = 2.0198e+00 k_t
= [2, 3, 3, 3]
Generation 3490: Best = 3.863175e+03 FES = 349100 Diversity = 2.0382e+00 k_t
= [1, 1, 1, 1]
Generation 3500: Best = 3.863172e+03 FES = 350100 Diversity = 1.9377e+00 k_t
= [2, 4, 2, 3]
Generation 3510: Best = 3.863042e+03 FES = 351100 Diversity = 1.9068e+00 k_t
= [1, 1, 1, 1]
Generation 3520: Best = 3.863040e+03 FES = 352100 Diversity = 1.9141e+00 k_t
= [3, 3, 4, 2]
Generation 3530: Best = 3.863014e+03 FES = 353100 Diversity = 1.8776e+00 k_t
= [1, 1, 1, 1]
Generation 3540: Best = 3.862995e+03 FES = 354100 Diversity = 1.8686e+00 k_t
= [4, 3, 2, 4]
Generation 3550: Best = 3.862852e+03 FES = 355100 Diversity = 1.8618e+00 k_t
= [1, 1, 1, 1]
Generation 3560: Best = 3.862556e+03 FES = 356100 Diversity = 1.8927e+00 k_t
= [2, 4, 3, 4]
Generation 3570: Best = 3.862556e+03 FES = 357100 Diversity = 1.8851e+00 k_t
= [1, 1, 1, 1]
Generation 3580: Best = 3.862556e+03 FES = 358100 Diversity = 1.9142e+00 k_t
= [3, 2, 2, 2]
Generation 3590: Best = 3.862556e+03 FES = 359100 Diversity = 1.9060e+00 k_t
= [1, 1, 1, 1]
Generation 3600: Best = 3.862556e+03 FES = 360100 Diversity = 1.9285e+00 k_t
= [2, 3, 2, 3]
Generation 3610: Best = 3.862556e+03 FES = 361100 Diversity = 1.9105e+00 k_t
= [1, 1, 1, 1]
Generation 3620: Best = 3.862556e+03 FES = 362100 Diversity = 1.9000e+00 k_t
= [3, 4, 3, 4]
Generation 3630: Best = 3.862532e+03 FES = 363100 Diversity = 1.9314e+00 k_t
= [1, 1, 1, 1]
Generation 3640: Best = 3.862517e+03 FES = 364100 Diversity = 1.9191e+00 k_t
= [2, 4, 4, 3]
Generation 3650: Best = 3.862508e+03 FES = 365100 Diversity = 1.9224e+00 k_t
= [1, 1, 1, 1]
Generation 3660: Best = 3.862508e+03 FES = 366100 Diversity = 1.9051e+00 k_t
= [3, 3, 3, 4]
Generation 3670: Best = 3.862508e+03 FES = 367100 Diversity = 1.9131e+00 k_t
= [1, 1, 1, 1]
Generation 3680: Best = 3.862508e+03 FES = 368100 Diversity = 1.9331e+00 k_t
= [4, 3, 3, 4]
Generation 3690: Best = 3.862480e+03 FES = 369100 Diversity = 1.9774e+00 k_t
= [1, 1, 1, 1]
Generation 3700: Best = 3.862480e+03 FES = 370100 Diversity = 1.9794e+00 k_t
= [4, 2, 2, 2]
Generation 3710: Best = 3.862480e+03 FES = 371100 Diversity = 1.9515e+00 k_t
= [1, 1, 1, 1]
Generation 3720: Best = 3.862480e+03 FES = 372100 Diversity = 1.8696e+00 k_t
= [3, 2, 2, 4]
Generation 3730: Best = 3.862480e+03 FES = 373100 Diversity = 1.8977e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 3740: Best = 3.862362e+03 FES = 374100 Diversity = 1.9184e+00 k_t
= [3, 4, 2, 2]
Generation 3750: Best = 3.862347e+03 FES = 375100 Diversity = 1.9312e+00 k_t
= [1, 1, 1, 1]
Generation 3760: Best = 3.862323e+03 FES = 376100 Diversity = 1.9108e+00 k_t
= [4, 2, 3, 3]
Generation 3770: Best = 3.862313e+03 FES = 377100 Diversity = 1.9072e+00 k_t
= [1, 1, 1, 1]
Generation 3780: Best = 3.862250e+03 FES = 378100 Diversity = 1.9269e+00 k_t
= [3, 2, 4, 3]
Generation 3790: Best = 3.862250e+03 FES = 379100 Diversity = 1.9525e+00 k_t
= [1, 1, 1, 1]
Generation 3800: Best = 3.862250e+03 FES = 380100 Diversity = 1.9215e+00 k_t
= [2, 2, 4, 3]
Generation 3810: Best = 3.862250e+03 FES = 381100 Diversity = 1.9589e+00 k_t
= [1, 1, 1, 1]
Generation 3820: Best = 3.862250e+03 FES = 382100 Diversity = 1.9990e+00 k_t
= [3, 4, 3, 3]
Generation 3830: Best = 3.862250e+03 FES = 383100 Diversity = 2.0405e+00 k_t
= [1, 1, 1, 1]
Generation 3840: Best = 3.862235e+03 FES = 384100 Diversity = 2.0740e+00 k_t
= [3, 2, 4, 2]
Generation 3850: Best = 3.862112e+03 FES = 385100 Diversity = 2.1047e+00 k_t
= [1, 1, 1, 1]
Generation 3860: Best = 3.862112e+03 FES = 386100 Diversity = 2.1092e+00 k_t
= [2, 2, 4, 2]
Generation 3870: Best = 3.862112e+03 FES = 387100 Diversity = 2.0186e+00 k_t
= [1, 1, 1, 1]
Generation 3880: Best = 3.862112e+03 FES = 388100 Diversity = 2.0157e+00 k_t
= [2, 2, 3, 4]
Generation 3890: Best = 3.862067e+03 FES = 389100 Diversity = 2.0162e+00 k_t
= [1, 1, 1, 1]
Generation 3900: Best = 3.862047e+03 FES = 390100 Diversity = 2.0275e+00 k_t
= [2, 3, 2, 4]
Generation 3910: Best = 3.862001e+03 FES = 391100 Diversity = 2.0129e+00 k_t
= [1, 1, 1, 1]
Generation 3920: Best = 3.861913e+03 FES = 392100 Diversity = 1.9824e+00 k_t
= [4, 2, 2, 2]
Generation 3930: Best = 3.861896e+03 FES = 393100 Diversity = 1.9251e+00 k_t
= [1, 1, 1, 1]
Generation 3940: Best = 3.861895e+03 FES = 394100 Diversity = 1.9150e+00 k_t
= [4, 3, 3, 3]
Generation 3950: Best = 3.861895e+03 FES = 395100 Diversity = 1.8896e+00 k_t
= [1, 1, 1, 1]
Generation 3960: Best = 3.861895e+03 FES = 396100 Diversity = 1.9133e+00 k_t
= [3, 2, 4, 2]
Generation 3970: Best = 3.861893e+03 FES = 397100 Diversity = 1.8683e+00 k_t
= [1, 1, 1, 1]
Generation 3980: Best = 3.861833e+03 FES = 398100 Diversity = 1.8538e+00 k_t
= [2, 4, 4, 3]
Generation 3990: Best = 3.861833e+03 FES = 399100 Diversity = 1.8473e+00 k_t
= [1, 1, 1, 1]
Generation 4000: Best = 3.861830e+03 FES = 400100 Diversity = 1.8107e+00 k_t

```



```

= [4, 3, 3, 4]
Generation 4010: Best = 3.861830e+03 FES = 401100 Diversity = 1.8312e+00 k_t
= [1, 1, 1, 1]
Generation 4020: Best = 3.861830e+03 FES = 402100 Diversity = 1.8579e+00 k_t
= [4, 3, 2, 4]
Generation 4030: Best = 3.861584e+03 FES = 403100 Diversity = 1.8500e+00 k_t
= [1, 1, 1, 1]
Generation 4040: Best = 3.861584e+03 FES = 404100 Diversity = 1.8370e+00 k_t
= [2, 2, 4, 3]
Generation 4050: Best = 3.861584e+03 FES = 405100 Diversity = 1.8142e+00 k_t
= [1, 1, 1, 1]
Generation 4060: Best = 3.861541e+03 FES = 406100 Diversity = 1.8340e+00 k_t
= [3, 3, 4, 3]
Generation 4070: Best = 3.861485e+03 FES = 407100 Diversity = 1.8467e+00 k_t
= [1, 1, 1, 1]
Generation 4080: Best = 3.861485e+03 FES = 408100 Diversity = 1.8255e+00 k_t
= [3, 3, 2, 2]
Generation 4090: Best = 3.861485e+03 FES = 409100 Diversity = 1.8243e+00 k_t
= [1, 1, 1, 1]
Generation 4100: Best = 3.861485e+03 FES = 410100 Diversity = 1.8367e+00 k_t
= [2, 2, 4, 2]
Generation 4110: Best = 3.861485e+03 FES = 411100 Diversity = 1.8208e+00 k_t
= [1, 1, 1, 1]
Generation 4120: Best = 3.861485e+03 FES = 412100 Diversity = 1.8076e+00 k_t
= [4, 3, 3, 4]
Generation 4130: Best = 3.861485e+03 FES = 413100 Diversity = 1.8144e+00 k_t
= [1, 1, 1, 1]
Generation 4140: Best = 3.861474e+03 FES = 414100 Diversity = 1.8228e+00 k_t
= [4, 2, 4, 4]
Generation 4150: Best = 3.861474e+03 FES = 415100 Diversity = 1.8289e+00 k_t
= [1, 1, 1, 1]
Generation 4160: Best = 3.861442e+03 FES = 416100 Diversity = 1.8153e+00 k_t
= [2, 4, 4, 4]
Generation 4170: Best = 3.861442e+03 FES = 417100 Diversity = 1.8186e+00 k_t
= [1, 1, 1, 1]
Generation 4180: Best = 3.861442e+03 FES = 418100 Diversity = 1.8216e+00 k_t
= [3, 4, 4, 4]
Generation 4190: Best = 3.861442e+03 FES = 419100 Diversity = 1.8083e+00 k_t
= [1, 1, 1, 1]
Generation 4200: Best = 3.861442e+03 FES = 420100 Diversity = 1.7937e+00 k_t
= [4, 4, 3, 4]
Generation 4210: Best = 3.861442e+03 FES = 421100 Diversity = 1.7912e+00 k_t
= [1, 1, 1, 1]
Generation 4220: Best = 3.861442e+03 FES = 422100 Diversity = 1.7932e+00 k_t
= [4, 4, 3, 4]
Generation 4230: Best = 3.861422e+03 FES = 423100 Diversity = 1.8138e+00 k_t
= [1, 1, 1, 1]
Generation 4240: Best = 3.861422e+03 FES = 424100 Diversity = 1.8085e+00 k_t
= [3, 4, 3, 4]
Generation 4250: Best = 3.861359e+03 FES = 425100 Diversity = 1.7941e+00 k_t
= [1, 1, 1, 1]
Generation 4260: Best = 3.861359e+03 FES = 426100 Diversity = 1.7939e+00 k_t
= [4, 4, 4, 4]
Generation 4270: Best = 3.861359e+03 FES = 427100 Diversity = 1.7944e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 4280: Best = 3.861359e+03 FES = 428100 Diversity = 1.7740e+00 k_t
= [3, 3, 3, 3]
Generation 4290: Best = 3.861359e+03 FES = 429100 Diversity = 1.7720e+00 k_t
= [1, 1, 1, 1]
Generation 4300: Best = 3.861354e+03 FES = 430100 Diversity = 1.8099e+00 k_t
= [3, 3, 2, 2]
Generation 4310: Best = 3.861333e+03 FES = 431100 Diversity = 1.8194e+00 k_t
= [1, 1, 1, 1]
Generation 4320: Best = 3.861325e+03 FES = 432100 Diversity = 1.8289e+00 k_t
= [2, 4, 3, 4]
Generation 4330: Best = 3.861325e+03 FES = 433100 Diversity = 1.8127e+00 k_t
= [1, 1, 1, 1]
Generation 4340: Best = 3.861317e+03 FES = 434100 Diversity = 1.8150e+00 k_t
= [2, 4, 3, 4]
Generation 4350: Best = 3.861317e+03 FES = 435100 Diversity = 1.8208e+00 k_t
= [1, 1, 1, 1]
Generation 4360: Best = 3.861317e+03 FES = 436100 Diversity = 1.8136e+00 k_t
= [2, 4, 4, 4]
Generation 4370: Best = 3.861317e+03 FES = 437100 Diversity = 1.7852e+00 k_t
= [1, 1, 1, 1]
Generation 4380: Best = 3.861317e+03 FES = 438100 Diversity = 1.7643e+00 k_t
= [4, 3, 2, 3]
Generation 4390: Best = 3.861317e+03 FES = 439100 Diversity = 1.7691e+00 k_t
= [1, 1, 1, 1]
Generation 4400: Best = 3.861317e+03 FES = 440100 Diversity = 1.7725e+00 k_t
= [4, 2, 4, 4]
Generation 4410: Best = 3.861317e+03 FES = 441100 Diversity = 1.7557e+00 k_t
= [1, 1, 1, 1]
Generation 4420: Best = 3.861317e+03 FES = 442100 Diversity = 1.7531e+00 k_t
= [3, 4, 4, 4]
Generation 4430: Best = 3.861304e+03 FES = 443100 Diversity = 1.7440e+00 k_t
= [1, 1, 1, 1]
Generation 4440: Best = 3.861265e+03 FES = 444100 Diversity = 1.7766e+00 k_t
= [2, 3, 2, 3]
Generation 4450: Best = 3.861265e+03 FES = 445100 Diversity = 1.7775e+00 k_t
= [1, 1, 1, 1]
Generation 4460: Best = 3.861265e+03 FES = 446100 Diversity = 1.7707e+00 k_t
= [2, 2, 3, 3]
Generation 4470: Best = 3.861265e+03 FES = 447100 Diversity = 1.7512e+00 k_t
= [1, 1, 1, 1]
Generation 4480: Best = 3.861265e+03 FES = 448100 Diversity = 1.7603e+00 k_t
= [4, 2, 2, 2]
Generation 4490: Best = 3.861265e+03 FES = 449100 Diversity = 1.7601e+00 k_t
= [1, 1, 1, 1]
Generation 4500: Best = 3.861265e+03 FES = 450100 Diversity = 1.7732e+00 k_t
= [3, 2, 3, 3]
Generation 4510: Best = 3.861265e+03 FES = 451100 Diversity = 1.7735e+00 k_t
= [1, 1, 1, 1]
Generation 4520: Best = 3.861264e+03 FES = 452100 Diversity = 1.7868e+00 k_t
= [4, 2, 4, 2]
Generation 4530: Best = 3.861264e+03 FES = 453100 Diversity = 1.7776e+00 k_t
= [1, 1, 1, 1]
Generation 4540: Best = 3.861264e+03 FES = 454100 Diversity = 1.7681e+00 k_t

```

```

= [2, 4, 4, 4]
Generation 4550: Best = 3.861264e+03 FES = 455100 Diversity = 1.7451e+00 k_t
= [1, 1, 1, 1]
Generation 4560: Best = 3.861264e+03 FES = 456100 Diversity = 1.7382e+00 k_t
= [4, 4, 2, 3]
Generation 4570: Best = 3.861264e+03 FES = 457100 Diversity = 1.7459e+00 k_t
= [1, 1, 1, 1]
Generation 4580: Best = 3.861234e+03 FES = 458100 Diversity = 1.7249e+00 k_t
= [4, 3, 2, 4]
Generation 4590: Best = 3.861234e+03 FES = 459100 Diversity = 1.6988e+00 k_t
= [1, 1, 1, 1]
Generation 4600: Best = 3.861234e+03 FES = 460100 Diversity = 1.6913e+00 k_t
= [4, 4, 2, 4]
Generation 4610: Best = 3.861227e+03 FES = 461100 Diversity = 1.6900e+00 k_t
= [1, 1, 1, 1]
Generation 4620: Best = 3.861218e+03 FES = 462100 Diversity = 1.6899e+00 k_t
= [3, 4, 3, 4]
Generation 4630: Best = 3.861218e+03 FES = 463100 Diversity = 1.6899e+00 k_t
= [1, 1, 1, 1]
Generation 4640: Best = 3.861190e+03 FES = 464100 Diversity = 1.7176e+00 k_t
= [3, 2, 2, 3]
Generation 4650: Best = 3.861178e+03 FES = 465100 Diversity = 1.7195e+00 k_t
= [1, 1, 1, 1]
Generation 4660: Best = 3.861178e+03 FES = 466100 Diversity = 1.7154e+00 k_t
= [4, 2, 4, 4]
Generation 4670: Best = 3.861178e+03 FES = 467100 Diversity = 1.7105e+00 k_t
= [1, 1, 1, 1]
Generation 4680: Best = 3.861178e+03 FES = 468100 Diversity = 1.6976e+00 k_t
= [2, 2, 2, 2]
Generation 4690: Best = 3.861171e+03 FES = 469100 Diversity = 1.7081e+00 k_t
= [1, 1, 1, 1]
Generation 4700: Best = 3.861171e+03 FES = 470100 Diversity = 1.7189e+00 k_t
= [3, 2, 3, 4]
Generation 4710: Best = 3.861171e+03 FES = 471100 Diversity = 1.7189e+00 k_t
= [1, 1, 1, 1]
Generation 4720: Best = 3.861086e+03 FES = 472100 Diversity = 1.7129e+00 k_t
= [4, 4, 2, 2]
Generation 4730: Best = 3.861086e+03 FES = 473100 Diversity = 1.7226e+00 k_t
= [1, 1, 1, 1]
Generation 4740: Best = 3.861086e+03 FES = 474100 Diversity = 1.7316e+00 k_t
= [4, 4, 2, 2]
Generation 4750: Best = 3.861086e+03 FES = 475100 Diversity = 1.7289e+00 k_t
= [1, 1, 1, 1]
Generation 4760: Best = 3.861086e+03 FES = 476100 Diversity = 1.7347e+00 k_t
= [2, 3, 4, 2]
Generation 4770: Best = 3.861086e+03 FES = 477100 Diversity = 1.7537e+00 k_t
= [1, 1, 1, 1]
Generation 4780: Best = 3.861085e+03 FES = 478100 Diversity = 1.7465e+00 k_t
= [2, 4, 2, 3]
Generation 4790: Best = 3.861085e+03 FES = 479100 Diversity = 1.7294e+00 k_t
= [1, 1, 1, 1]
Generation 4800: Best = 3.861085e+03 FES = 480100 Diversity = 1.7083e+00 k_t
= [4, 2, 3, 4]
Generation 4810: Best = 3.861083e+03 FES = 481100 Diversity = 1.6858e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 4820: Best = 3.861042e+03 FES = 482100 Diversity = 1.6744e+00 k_t
= [3, 4, 2, 4]
Generation 4830: Best = 3.861042e+03 FES = 483100 Diversity = 1.6670e+00 k_t
= [1, 1, 1, 1]
Generation 4840: Best = 3.861042e+03 FES = 484100 Diversity = 1.6252e+00 k_t
= [4, 2, 4, 4]
Generation 4850: Best = 3.861042e+03 FES = 485100 Diversity = 1.6250e+00 k_t
= [1, 1, 1, 1]
Generation 4860: Best = 3.861041e+03 FES = 486100 Diversity = 1.6214e+00 k_t
= [2, 2, 2, 2]
Generation 4870: Best = 3.861041e+03 FES = 487100 Diversity = 1.6300e+00 k_t
= [1, 1, 1, 1]
Generation 4880: Best = 3.861041e+03 FES = 488100 Diversity = 1.6303e+00 k_t
= [4, 2, 2, 2]
Generation 4890: Best = 3.861041e+03 FES = 489100 Diversity = 1.6323e+00 k_t
= [1, 1, 1, 1]
Generation 4900: Best = 3.861041e+03 FES = 490100 Diversity = 1.6338e+00 k_t
= [4, 3, 4, 4]
Generation 4910: Best = 3.861028e+03 FES = 491100 Diversity = 1.6201e+00 k_t
= [1, 1, 1, 1]
Generation 4920: Best = 3.861007e+03 FES = 492100 Diversity = 1.6206e+00 k_t
= [2, 3, 4, 3]
Generation 4930: Best = 3.860926e+03 FES = 493100 Diversity = 1.6285e+00 k_t
= [1, 1, 1, 1]
Generation 4940: Best = 3.860924e+03 FES = 494100 Diversity = 1.6321e+00 k_t
= [4, 3, 3, 4]
Generation 4950: Best = 3.860924e+03 FES = 495100 Diversity = 1.5931e+00 k_t
= [1, 1, 1, 1]
Generation 4960: Best = 3.860924e+03 FES = 496100 Diversity = 1.5621e+00 k_t
= [2, 4, 2, 3]
Generation 4970: Best = 3.860924e+03 FES = 497100 Diversity = 1.5708e+00 k_t
= [1, 1, 1, 1]
Generation 4980: Best = 3.860924e+03 FES = 498100 Diversity = 1.5636e+00 k_t
= [2, 2, 3, 3]
Generation 4990: Best = 3.860894e+03 FES = 499100 Diversity = 1.5574e+00 k_t
= [1, 1, 1, 1]
Generation 5000: Best = 3.860894e+03 FES = 500100 Diversity = 1.5557e+00 k_t
= [4, 3, 4, 3]
Generation 5010: Best = 3.860831e+03 FES = 501100 Diversity = 1.5682e+00 k_t
= [1, 1, 1, 1]
Generation 5020: Best = 3.860831e+03 FES = 502100 Diversity = 1.5758e+00 k_t
= [2, 4, 2, 2]
Generation 5030: Best = 3.860831e+03 FES = 503100 Diversity = 1.5716e+00 k_t
= [1, 1, 1, 1]
Generation 5040: Best = 3.860831e+03 FES = 504100 Diversity = 1.5673e+00 k_t
= [2, 4, 2, 2]
Generation 5050: Best = 3.860724e+03 FES = 505100 Diversity = 1.5672e+00 k_t
= [1, 1, 1, 1]
Generation 5060: Best = 3.860724e+03 FES = 506100 Diversity = 1.5674e+00 k_t
= [3, 4, 4, 2]
Generation 5070: Best = 3.860724e+03 FES = 507100 Diversity = 1.5558e+00 k_t
= [1, 1, 1, 1]
Generation 5080: Best = 3.860724e+03 FES = 508100 Diversity = 1.5401e+00 k_t

```

```

= [4, 2, 2, 3]
Generation 5090: Best = 3.860724e+03 FES = 509100 Diversity = 1.5463e+00 k_t
= [1, 1, 1, 1]
Generation 5100: Best = 3.860716e+03 FES = 510100 Diversity = 1.5462e+00 k_t
= [3, 3, 2, 3]
Generation 5110: Best = 3.860606e+03 FES = 511100 Diversity = 1.4914e+00 k_t
= [1, 1, 1, 1]
Generation 5120: Best = 3.860598e+03 FES = 512100 Diversity = 1.4810e+00 k_t
= [4, 2, 4, 2]
Generation 5130: Best = 3.860598e+03 FES = 513100 Diversity = 1.4466e+00 k_t
= [1, 1, 1, 1]
Generation 5140: Best = 3.860572e+03 FES = 514100 Diversity = 1.4164e+00 k_t
= [2, 2, 4, 2]
Generation 5150: Best = 3.860480e+03 FES = 515100 Diversity = 1.2948e+00 k_t
= [1, 1, 1, 1]
Generation 5160: Best = 3.860480e+03 FES = 516100 Diversity = 1.2140e+00 k_t
= [3, 2, 3, 3]
Generation 5170: Best = 3.860459e+03 FES = 517100 Diversity = 1.1891e+00 k_t
= [1, 1, 1, 1]
Generation 5180: Best = 3.860459e+03 FES = 518100 Diversity = 1.1636e+00 k_t
= [4, 3, 4, 4]
Generation 5190: Best = 3.860459e+03 FES = 519100 Diversity = 1.1158e+00 k_t
= [1, 1, 1, 1]
Generation 5200: Best = 3.860456e+03 FES = 520100 Diversity = 1.0569e+00 k_t
= [4, 2, 4, 4]
Generation 5210: Best = 3.860436e+03 FES = 521100 Diversity = 1.0203e+00 k_t
= [1, 1, 1, 1]
Generation 5220: Best = 3.860421e+03 FES = 522100 Diversity = 9.5104e-01 k_t
= [3, 3, 3, 4]
Generation 5230: Best = 3.860321e+03 FES = 523100 Diversity = 9.3253e-01 k_t
= [1, 1, 1, 1]
Generation 5240: Best = 3.860321e+03 FES = 524100 Diversity = 8.3949e-01 k_t
= [2, 2, 3, 4]
Generation 5250: Best = 3.860321e+03 FES = 525100 Diversity = 7.2697e-01 k_t
= [1, 1, 1, 1]
Generation 5260: Best = 3.860167e+03 FES = 526100 Diversity = 5.8966e-01 k_t
= [3, 2, 2, 2]
Generation 5270: Best = 3.860165e+03 FES = 527100 Diversity = 5.0336e-01 k_t
= [1, 1, 1, 1]
Generation 5280: Best = 3.860114e+03 FES = 528100 Diversity = 4.3206e-01 k_t
= [2, 4, 2, 3]
Generation 5290: Best = 3.860009e+03 FES = 529100 Diversity = 2.8311e-01 k_t
= [1, 1, 1, 1]
Generation 5300: Best = 3.859976e+03 FES = 530100 Diversity = 1.9404e-02 k_t
= [4, 4, 2, 4]
Generation 5310: Best = 3.859925e+03 FES = 531100 Diversity = 1.7248e-02 k_t
= [1, 1, 1, 1]
Generation 5320: Best = 3.859833e+03 FES = 532100 Diversity = 1.4190e-02 k_t
= [2, 2, 3, 4]
Generation 5330: Best = 3.859815e+03 FES = 533100 Diversity = 1.2958e-02 k_t
= [1, 1, 1, 1]
Generation 5340: Best = 3.859790e+03 FES = 534100 Diversity = 1.1490e-02 k_t
= [4, 3, 4, 3]
Generation 5350: Best = 3.859772e+03 FES = 535100 Diversity = 1.0812e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 5360: Best = 3.859743e+03 FES = 536100 Diversity = 9.6165e-03 k_t
= [3, 3, 4, 4]
Generation 5370: Best = 3.859724e+03 FES = 537100 Diversity = 9.2191e-03 k_t
= [1, 1, 1, 1]
Generation 5380: Best = 3.859690e+03 FES = 538100 Diversity = 8.7962e-03 k_t
= [3, 3, 4, 4]
Generation 5390: Best = 3.859679e+03 FES = 539100 Diversity = 8.5374e-03 k_t
= [1, 1, 1, 1]
Generation 5400: Best = 3.859666e+03 FES = 540100 Diversity = 8.3755e-03 k_t
= [3, 4, 4, 3]
Generation 5410: Best = 3.859638e+03 FES = 541100 Diversity = 8.0616e-03 k_t
= [1, 1, 1, 1]
Generation 5420: Best = 3.859612e+03 FES = 542100 Diversity = 7.7669e-03 k_t
= [4, 2, 4, 3]
Generation 5430: Best = 3.859599e+03 FES = 543100 Diversity = 7.3765e-03 k_t
= [1, 1, 1, 1]
Generation 5440: Best = 3.859577e+03 FES = 544100 Diversity = 6.9356e-03 k_t
= [2, 3, 4, 3]
Generation 5450: Best = 3.859562e+03 FES = 545100 Diversity = 6.9149e-03 k_t
= [1, 1, 1, 1]
Generation 5460: Best = 3.859554e+03 FES = 546100 Diversity = 6.6787e-03 k_t
= [3, 2, 3, 3]
Generation 5470: Best = 3.859539e+03 FES = 547100 Diversity = 6.1565e-03 k_t
= [1, 1, 1, 1]
Generation 5480: Best = 3.859533e+03 FES = 548100 Diversity = 6.0224e-03 k_t
= [3, 3, 2, 2]
Generation 5490: Best = 3.859511e+03 FES = 549100 Diversity = 5.8836e-03 k_t
= [1, 1, 1, 1]
Generation 5500: Best = 3.859502e+03 FES = 550100 Diversity = 6.1667e-03 k_t
= [2, 2, 2, 2]
Generation 5510: Best = 3.859481e+03 FES = 551100 Diversity = 5.7811e-03 k_t
= [1, 1, 1, 1]
Generation 5520: Best = 3.859470e+03 FES = 552100 Diversity = 5.8888e-03 k_t
= [3, 4, 4, 4]
Generation 5530: Best = 3.859461e+03 FES = 553100 Diversity = 6.4346e-03 k_t
= [1, 1, 1, 1]
Generation 5540: Best = 3.859428e+03 FES = 554100 Diversity = 6.5726e-03 k_t
= [4, 2, 3, 4]
Generation 5550: Best = 3.859412e+03 FES = 555100 Diversity = 6.9967e-03 k_t
= [1, 1, 1, 1]
Generation 5560: Best = 3.859393e+03 FES = 556100 Diversity = 6.8027e-03 k_t
= [3, 2, 4, 4]
Generation 5570: Best = 3.859361e+03 FES = 557100 Diversity = 6.9709e-03 k_t
= [1, 1, 1, 1]
Generation 5580: Best = 3.859340e+03 FES = 558100 Diversity = 6.9682e-03 k_t
= [2, 3, 3, 2]
Generation 5590: Best = 3.859313e+03 FES = 559100 Diversity = 7.4201e-03 k_t
= [1, 1, 1, 1]
Generation 5600: Best = 3.859288e+03 FES = 560100 Diversity = 7.7265e-03 k_t
= [3, 3, 4, 3]
Generation 5610: Best = 3.859284e+03 FES = 561100 Diversity = 7.8231e-03 k_t
= [1, 1, 1, 1]
Generation 5620: Best = 3.859263e+03 FES = 562100 Diversity = 7.6980e-03 k_t

```

```

= [4, 4, 2, 3]
Generation 5630: Best = 3.859232e+03 FES = 563100 Diversity = 7.5521e-03 k_t
= [1, 1, 1, 1]
Generation 5640: Best = 3.859200e+03 FES = 564100 Diversity = 7.8822e-03 k_t
= [4, 3, 4, 2]
Generation 5650: Best = 3.859196e+03 FES = 565100 Diversity = 7.7234e-03 k_t
= [1, 1, 1, 1]
Generation 5660: Best = 3.859130e+03 FES = 566100 Diversity = 8.2366e-03 k_t
= [4, 3, 3, 3]
Generation 5670: Best = 3.859115e+03 FES = 567100 Diversity = 8.8623e-03 k_t
= [1, 1, 1, 1]
Generation 5680: Best = 3.859075e+03 FES = 568100 Diversity = 9.4670e-03 k_t
= [3, 2, 3, 2]
Generation 5690: Best = 3.859002e+03 FES = 569100 Diversity = 1.1223e-02 k_t
= [1, 1, 1, 1]
Generation 5700: Best = 3.858908e+03 FES = 570100 Diversity = 1.2921e-02 k_t
= [3, 4, 4, 2]
Generation 5710: Best = 3.858834e+03 FES = 571100 Diversity = 1.3957e-02 k_t
= [1, 1, 1, 1]
Generation 5720: Best = 3.858771e+03 FES = 572100 Diversity = 1.3337e-02 k_t
= [3, 2, 3, 2]
Generation 5730: Best = 3.858713e+03 FES = 573100 Diversity = 1.2894e-02 k_t
= [1, 1, 1, 1]
Generation 5740: Best = 3.858681e+03 FES = 574100 Diversity = 1.2256e-02 k_t
= [3, 4, 4, 4]
Generation 5750: Best = 3.858647e+03 FES = 575100 Diversity = 1.1384e-02 k_t
= [1, 1, 1, 1]
Generation 5760: Best = 3.858605e+03 FES = 576100 Diversity = 1.0168e-02 k_t
= [2, 2, 4, 4]
Generation 5770: Best = 3.858592e+03 FES = 577100 Diversity = 9.0437e-03 k_t
= [1, 1, 1, 1]
Generation 5780: Best = 3.858579e+03 FES = 578100 Diversity = 8.3531e-03 k_t
= [3, 3, 4, 4]
Generation 5790: Best = 3.858558e+03 FES = 579100 Diversity = 7.3909e-03 k_t
= [1, 1, 1, 1]
Generation 5800: Best = 3.858539e+03 FES = 580100 Diversity = 6.7265e-03 k_t
= [2, 4, 4, 2]
Generation 5810: Best = 3.858522e+03 FES = 581100 Diversity = 6.4220e-03 k_t
= [1, 1, 1, 1]
Generation 5820: Best = 3.858515e+03 FES = 582100 Diversity = 6.3599e-03 k_t
= [2, 3, 3, 3]
Generation 5830: Best = 3.858502e+03 FES = 583100 Diversity = 5.6169e-03 k_t
= [1, 1, 1, 1]
Generation 5840: Best = 3.858493e+03 FES = 584100 Diversity = 5.3051e-03 k_t
= [4, 4, 4, 4]
Generation 5850: Best = 3.858486e+03 FES = 585100 Diversity = 4.8519e-03 k_t
= [1, 1, 1, 1]
Generation 5860: Best = 3.858478e+03 FES = 586100 Diversity = 4.6215e-03 k_t
= [3, 3, 4, 4]
Generation 5870: Best = 3.858468e+03 FES = 587100 Diversity = 4.7280e-03 k_t
= [1, 1, 1, 1]
Generation 5880: Best = 3.858452e+03 FES = 588100 Diversity = 5.3284e-03 k_t
= [2, 2, 3, 3]
Generation 5890: Best = 3.858438e+03 FES = 589100 Diversity = 5.7616e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 5900: Best = 3.858434e+03 FES = 590100 Diversity = 5.6792e-03 k_t
= [2, 4, 2, 2]
Generation 5910: Best = 3.858421e+03 FES = 591100 Diversity = 5.3468e-03 k_t
= [1, 1, 1, 1]
Generation 5920: Best = 3.858409e+03 FES = 592100 Diversity = 5.4896e-03 k_t
= [4, 4, 3, 2]
Generation 5930: Best = 3.858403e+03 FES = 593100 Diversity = 4.9560e-03 k_t
= [1, 1, 1, 1]
Generation 5940: Best = 3.858393e+03 FES = 594100 Diversity = 4.6261e-03 k_t
= [2, 2, 4, 4]
Generation 5950: Best = 3.858391e+03 FES = 595100 Diversity = 4.4252e-03 k_t
= [1, 1, 1, 1]
Generation 5960: Best = 3.858384e+03 FES = 596100 Diversity = 3.9774e-03 k_t
= [2, 2, 3, 4]
Generation 5970: Best = 3.858377e+03 FES = 597100 Diversity = 3.4557e-03 k_t
= [1, 1, 1, 1]
Generation 5980: Best = 3.858370e+03 FES = 598100 Diversity = 3.6932e-03 k_t
= [2, 3, 2, 2]
Generation 5990: Best = 3.858364e+03 FES = 599100 Diversity = 3.9354e-03 k_t
= [1, 1, 1, 1]
Generation 6000: Best = 3.858358e+03 FES = 600100 Diversity = 4.0038e-03 k_t
= [2, 3, 3, 2]
Generation 6010: Best = 3.858350e+03 FES = 601100 Diversity = 3.9019e-03 k_t
= [1, 1, 1, 1]
Generation 6020: Best = 3.858345e+03 FES = 602100 Diversity = 3.8672e-03 k_t
= [3, 3, 3, 2]
Generation 6030: Best = 3.858336e+03 FES = 603100 Diversity = 3.9258e-03 k_t
= [1, 1, 1, 1]
Generation 6040: Best = 3.858327e+03 FES = 604100 Diversity = 4.4401e-03 k_t
= [3, 3, 4, 2]
Generation 6050: Best = 3.858319e+03 FES = 605100 Diversity = 4.5438e-03 k_t
= [1, 1, 1, 1]
Generation 6060: Best = 3.858311e+03 FES = 606100 Diversity = 4.5632e-03 k_t
= [3, 2, 2, 3]
Generation 6070: Best = 3.858308e+03 FES = 607100 Diversity = 4.7297e-03 k_t
= [1, 1, 1, 1]
Generation 6080: Best = 3.858291e+03 FES = 608100 Diversity = 4.9369e-03 k_t
= [2, 4, 3, 3]
Generation 6090: Best = 3.858263e+03 FES = 609100 Diversity = 5.7608e-03 k_t
= [1, 1, 1, 1]
Generation 6100: Best = 3.858247e+03 FES = 610100 Diversity = 6.5345e-03 k_t
= [2, 2, 4, 4]
Generation 6110: Best = 3.858233e+03 FES = 611100 Diversity = 6.8110e-03 k_t
= [1, 1, 1, 1]
Generation 6120: Best = 3.858225e+03 FES = 612100 Diversity = 6.4146e-03 k_t
= [3, 3, 3, 2]
Generation 6130: Best = 3.858216e+03 FES = 613100 Diversity = 6.5474e-03 k_t
= [1, 1, 1, 1]
Generation 6140: Best = 3.858199e+03 FES = 614100 Diversity = 6.3337e-03 k_t
= [4, 3, 4, 4]
Generation 6150: Best = 3.858181e+03 FES = 615100 Diversity = 6.2827e-03 k_t
= [1, 1, 1, 1]
Generation 6160: Best = 3.858174e+03 FES = 616100 Diversity = 6.2501e-03 k_t

```



```

= [4, 4, 2, 3]
Generation 6170: Best = 3.858168e+03 FES = 617100 Diversity = 5.6561e-03 k_t
= [1, 1, 1, 1]
Generation 6180: Best = 3.858157e+03 FES = 618100 Diversity = 5.6138e-03 k_t
= [2, 2, 4, 4]
Generation 6190: Best = 3.858144e+03 FES = 619100 Diversity = 5.0303e-03 k_t
= [1, 1, 1, 1]
Generation 6200: Best = 3.858143e+03 FES = 620100 Diversity = 4.8654e-03 k_t
= [2, 4, 2, 4]
Generation 6210: Best = 3.858134e+03 FES = 621100 Diversity = 4.7676e-03 k_t
= [1, 1, 1, 1]
Generation 6220: Best = 3.858128e+03 FES = 622100 Diversity = 4.5712e-03 k_t
= [2, 4, 3, 2]
Generation 6230: Best = 3.858122e+03 FES = 623100 Diversity = 4.4177e-03 k_t
= [1, 1, 1, 1]
Generation 6240: Best = 3.858113e+03 FES = 624100 Diversity = 4.1076e-03 k_t
= [4, 4, 4, 2]
Generation 6250: Best = 3.858112e+03 FES = 625100 Diversity = 4.0041e-03 k_t
= [1, 1, 1, 1]
Generation 6260: Best = 3.858108e+03 FES = 626100 Diversity = 3.6299e-03 k_t
= [2, 3, 4, 2]
Generation 6270: Best = 3.858101e+03 FES = 627100 Diversity = 3.4211e-03 k_t
= [1, 1, 1, 1]
Generation 6280: Best = 3.858096e+03 FES = 628100 Diversity = 3.2289e-03 k_t
= [3, 3, 4, 2]
Generation 6290: Best = 3.858092e+03 FES = 629100 Diversity = 3.3959e-03 k_t
= [1, 1, 1, 1]
Generation 6300: Best = 3.858088e+03 FES = 630100 Diversity = 3.2485e-03 k_t
= [2, 4, 2, 4]
Generation 6310: Best = 3.858083e+03 FES = 631100 Diversity = 3.2902e-03 k_t
= [1, 1, 1, 1]
Generation 6320: Best = 3.858078e+03 FES = 632100 Diversity = 3.4125e-03 k_t
= [4, 3, 3, 3]
Generation 6330: Best = 3.858074e+03 FES = 633100 Diversity = 3.2413e-03 k_t
= [1, 1, 1, 1]
Generation 6340: Best = 3.858066e+03 FES = 634100 Diversity = 3.5452e-03 k_t
= [4, 3, 4, 2]
Generation 6350: Best = 3.858064e+03 FES = 635100 Diversity = 3.3885e-03 k_t
= [1, 1, 1, 1]
Generation 6360: Best = 3.858060e+03 FES = 636100 Diversity = 3.4710e-03 k_t
= [3, 4, 3, 4]
Generation 6370: Best = 3.858059e+03 FES = 637100 Diversity = 3.4474e-03 k_t
= [1, 1, 1, 1]
Generation 6380: Best = 3.858056e+03 FES = 638100 Diversity = 3.2013e-03 k_t
= [3, 3, 2, 3]
Generation 6390: Best = 3.858054e+03 FES = 639100 Diversity = 3.0600e-03 k_t
= [1, 1, 1, 1]
Generation 6400: Best = 3.858050e+03 FES = 640100 Diversity = 3.0566e-03 k_t
= [2, 4, 3, 4]
Generation 6410: Best = 3.858049e+03 FES = 641100 Diversity = 3.0092e-03 k_t
= [1, 1, 1, 1]
Generation 6420: Best = 3.858046e+03 FES = 642100 Diversity = 2.8666e-03 k_t
= [3, 4, 4, 4]
Generation 6430: Best = 3.858042e+03 FES = 643100 Diversity = 2.9044e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 6440: Best = 3.858039e+03 FES = 644100 Diversity = 2.8796e-03 k_t
= [3, 4, 3, 3]
Generation 6450: Best = 3.858036e+03 FES = 645100 Diversity = 2.8110e-03 k_t
= [1, 1, 1, 1]
Generation 6460: Best = 3.858032e+03 FES = 646100 Diversity = 3.0170e-03 k_t
= [2, 2, 2, 2]
Generation 6470: Best = 3.858027e+03 FES = 647100 Diversity = 3.1707e-03 k_t
= [1, 1, 1, 1]
Generation 6480: Best = 3.858025e+03 FES = 648100 Diversity = 3.0312e-03 k_t
= [3, 4, 4, 3]
Generation 6490: Best = 3.858020e+03 FES = 649100 Diversity = 2.9939e-03 k_t
= [1, 1, 1, 1]
Generation 6500: Best = 3.858020e+03 FES = 650100 Diversity = 2.9643e-03 k_t
= [2, 3, 3, 3]
Generation 6510: Best = 3.858016e+03 FES = 651100 Diversity = 2.7967e-03 k_t
= [1, 1, 1, 1]
Generation 6520: Best = 3.858015e+03 FES = 652100 Diversity = 2.6983e-03 k_t
= [4, 4, 2, 4]
Generation 6530: Best = 3.858012e+03 FES = 653100 Diversity = 2.6199e-03 k_t
= [1, 1, 1, 1]
Generation 6540: Best = 3.858010e+03 FES = 654100 Diversity = 2.5336e-03 k_t
= [2, 3, 2, 2]
Generation 6550: Best = 3.858008e+03 FES = 655100 Diversity = 2.3746e-03 k_t
= [1, 1, 1, 1]
Generation 6560: Best = 3.858005e+03 FES = 656100 Diversity = 2.2644e-03 k_t
= [3, 2, 2, 4]
Generation 6570: Best = 3.858004e+03 FES = 657100 Diversity = 2.3195e-03 k_t
= [1, 1, 1, 1]
Generation 6580: Best = 3.858002e+03 FES = 658100 Diversity = 2.1319e-03 k_t
= [3, 2, 2, 3]
Generation 6590: Best = 3.857999e+03 FES = 659100 Diversity = 2.2167e-03 k_t
= [1, 1, 1, 1]
Generation 6600: Best = 3.857998e+03 FES = 660100 Diversity = 2.2013e-03 k_t
= [3, 4, 4, 2]
Generation 6610: Best = 3.857994e+03 FES = 661100 Diversity = 2.2255e-03 k_t
= [1, 1, 1, 1]
Generation 6620: Best = 3.857994e+03 FES = 662100 Diversity = 2.1298e-03 k_t
= [3, 3, 2, 4]
Generation 6630: Best = 3.857993e+03 FES = 663100 Diversity = 2.0229e-03 k_t
= [1, 1, 1, 1]
Generation 6640: Best = 3.857991e+03 FES = 664100 Diversity = 2.1070e-03 k_t
= [4, 2, 2, 2]
Generation 6650: Best = 3.857988e+03 FES = 665100 Diversity = 2.0175e-03 k_t
= [1, 1, 1, 1]
Generation 6660: Best = 3.857988e+03 FES = 666100 Diversity = 1.9719e-03 k_t
= [3, 4, 2, 3]
Generation 6670: Best = 3.857986e+03 FES = 667100 Diversity = 1.9302e-03 k_t
= [1, 1, 1, 1]
Generation 6680: Best = 3.857984e+03 FES = 668100 Diversity = 1.9263e-03 k_t
= [4, 2, 3, 4]
Generation 6690: Best = 3.857981e+03 FES = 669100 Diversity = 2.0093e-03 k_t
= [1, 1, 1, 1]
Generation 6700: Best = 3.857980e+03 FES = 670100 Diversity = 1.9173e-03 k_t

```

```

= [3, 2, 2, 3]
Generation 6710: Best = 3.857978e+03 FES = 671100 Diversity = 1.9888e-03 k_t
= [1, 1, 1, 1]
Generation 6720: Best = 3.857977e+03 FES = 672100 Diversity = 2.1473e-03 k_t
= [3, 3, 4, 2]
Generation 6730: Best = 3.857973e+03 FES = 673100 Diversity = 2.2695e-03 k_t
= [1, 1, 1, 1]
Generation 6740: Best = 3.857971e+03 FES = 674100 Diversity = 2.3835e-03 k_t
= [3, 2, 2, 4]
Generation 6750: Best = 3.857968e+03 FES = 675100 Diversity = 2.6041e-03 k_t
= [1, 1, 1, 1]
Generation 6760: Best = 3.857967e+03 FES = 676100 Diversity = 2.6059e-03 k_t
= [3, 4, 4, 3]
Generation 6770: Best = 3.857965e+03 FES = 677100 Diversity = 2.5605e-03 k_t
= [1, 1, 1, 1]
Generation 6780: Best = 3.857962e+03 FES = 678100 Diversity = 2.4575e-03 k_t
= [3, 3, 4, 3]
Generation 6790: Best = 3.857962e+03 FES = 679100 Diversity = 2.4170e-03 k_t
= [1, 1, 1, 1]
Generation 6800: Best = 3.857959e+03 FES = 680100 Diversity = 2.2092e-03 k_t
= [3, 4, 3, 4]
Generation 6810: Best = 3.857959e+03 FES = 681100 Diversity = 2.0195e-03 k_t
= [1, 1, 1, 1]
Generation 6820: Best = 3.857957e+03 FES = 682100 Diversity = 1.8832e-03 k_t
= [2, 4, 4, 3]
Generation 6830: Best = 3.857955e+03 FES = 683100 Diversity = 1.8948e-03 k_t
= [1, 1, 1, 1]
Generation 6840: Best = 3.857954e+03 FES = 684100 Diversity = 1.7585e-03 k_t
= [3, 3, 3, 2]
Generation 6850: Best = 3.857953e+03 FES = 685100 Diversity = 1.6792e-03 k_t
= [1, 1, 1, 1]
Generation 6860: Best = 3.857952e+03 FES = 686100 Diversity = 1.6423e-03 k_t
= [2, 3, 2, 4]
Generation 6870: Best = 3.857951e+03 FES = 687100 Diversity = 1.6046e-03 k_t
= [1, 1, 1, 1]
Generation 6880: Best = 3.857950e+03 FES = 688100 Diversity = 1.5691e-03 k_t
= [2, 4, 4, 2]
Generation 6890: Best = 3.857948e+03 FES = 689100 Diversity = 1.5087e-03 k_t
= [1, 1, 1, 1]
Generation 6900: Best = 3.857947e+03 FES = 690100 Diversity = 1.5396e-03 k_t
= [2, 4, 2, 2]
Generation 6910: Best = 3.857945e+03 FES = 691100 Diversity = 1.5683e-03 k_t
= [1, 1, 1, 1]
Generation 6920: Best = 3.857944e+03 FES = 692100 Diversity = 1.7423e-03 k_t
= [3, 4, 3, 2]
Generation 6930: Best = 3.857942e+03 FES = 693100 Diversity = 1.8005e-03 k_t
= [1, 1, 1, 1]
Generation 6940: Best = 3.857941e+03 FES = 694100 Diversity = 1.8469e-03 k_t
= [4, 4, 4, 3]
Generation 6950: Best = 3.857939e+03 FES = 695100 Diversity = 2.0542e-03 k_t
= [1, 1, 1, 1]
Generation 6960: Best = 3.857937e+03 FES = 696100 Diversity = 2.2851e-03 k_t
= [4, 3, 2, 3]
Generation 6970: Best = 3.857936e+03 FES = 697100 Diversity = 2.3545e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 6980: Best = 3.857935e+03 FES = 698100 Diversity = 2.3177e-03 k_t
= [4, 4, 4, 3]
Generation 6990: Best = 3.857931e+03 FES = 699100 Diversity = 2.3995e-03 k_t
= [1, 1, 1, 1]
Generation 7000: Best = 3.857930e+03 FES = 700100 Diversity = 2.2649e-03 k_t
= [2, 3, 3, 4]
Generation 7010: Best = 3.857928e+03 FES = 701100 Diversity = 2.1374e-03 k_t
= [1, 1, 1, 1]
Generation 7020: Best = 3.857927e+03 FES = 702100 Diversity = 2.0421e-03 k_t
= [2, 4, 2, 4]
Generation 7030: Best = 3.857926e+03 FES = 703100 Diversity = 2.0519e-03 k_t
= [1, 1, 1, 1]
Generation 7040: Best = 3.857924e+03 FES = 704100 Diversity = 2.0047e-03 k_t
= [2, 2, 4, 4]
Generation 7050: Best = 3.857923e+03 FES = 705100 Diversity = 1.9615e-03 k_t
= [1, 1, 1, 1]
Generation 7060: Best = 3.857922e+03 FES = 706100 Diversity = 1.7959e-03 k_t
= [4, 4, 4, 3]
Generation 7070: Best = 3.857921e+03 FES = 707100 Diversity = 1.6902e-03 k_t
= [1, 1, 1, 1]
Generation 7080: Best = 3.857919e+03 FES = 708100 Diversity = 1.7235e-03 k_t
= [2, 2, 3, 3]
Generation 7090: Best = 3.857918e+03 FES = 709100 Diversity = 1.7352e-03 k_t
= [1, 1, 1, 1]
Generation 7100: Best = 3.857916e+03 FES = 710100 Diversity = 1.7873e-03 k_t
= [2, 2, 4, 4]
Generation 7110: Best = 3.857916e+03 FES = 711100 Diversity = 1.7103e-03 k_t
= [1, 1, 1, 1]
Generation 7120: Best = 3.857915e+03 FES = 712100 Diversity = 1.6542e-03 k_t
= [3, 3, 2, 4]
Generation 7130: Best = 3.857914e+03 FES = 713100 Diversity = 1.5762e-03 k_t
= [1, 1, 1, 1]
Generation 7140: Best = 3.857913e+03 FES = 714100 Diversity = 1.6044e-03 k_t
= [3, 3, 4, 3]
Generation 7150: Best = 3.857912e+03 FES = 715100 Diversity = 1.5938e-03 k_t
= [1, 1, 1, 1]
Generation 7160: Best = 3.857911e+03 FES = 716100 Diversity = 1.5064e-03 k_t
= [4, 2, 2, 3]
Generation 7170: Best = 3.857910e+03 FES = 717100 Diversity = 1.4434e-03 k_t
= [1, 1, 1, 1]
Generation 7180: Best = 3.857909e+03 FES = 718100 Diversity = 1.3664e-03 k_t
= [4, 3, 3, 3]
Generation 7190: Best = 3.857908e+03 FES = 719100 Diversity = 1.2952e-03 k_t
= [1, 1, 1, 1]
Generation 7200: Best = 3.857908e+03 FES = 720100 Diversity = 1.3115e-03 k_t
= [3, 4, 3, 2]
Generation 7210: Best = 3.857906e+03 FES = 721100 Diversity = 1.3383e-03 k_t
= [1, 1, 1, 1]
Generation 7220: Best = 3.857906e+03 FES = 722100 Diversity = 1.4414e-03 k_t
= [2, 3, 2, 3]
Generation 7230: Best = 3.857905e+03 FES = 723100 Diversity = 1.4464e-03 k_t
= [1, 1, 1, 1]
Generation 7240: Best = 3.857904e+03 FES = 724100 Diversity = 1.4080e-03 k_t

```

```

= [3, 3, 3, 2]
Generation 7250: Best = 3.857903e+03 FES = 725100 Diversity = 1.3755e-03 k_t
= [1, 1, 1, 1]
Generation 7260: Best = 3.857902e+03 FES = 726100 Diversity = 1.4406e-03 k_t
= [4, 2, 2, 2]
Generation 7270: Best = 3.857902e+03 FES = 727100 Diversity = 1.4623e-03 k_t
= [1, 1, 1, 1]
Generation 7280: Best = 3.857900e+03 FES = 728100 Diversity = 1.3816e-03 k_t
= [2, 3, 3, 4]
Generation 7290: Best = 3.857899e+03 FES = 729100 Diversity = 1.5727e-03 k_t
= [1, 1, 1, 1]
Generation 7300: Best = 3.857897e+03 FES = 730100 Diversity = 1.6581e-03 k_t
= [4, 3, 2, 4]
Generation 7310: Best = 3.857896e+03 FES = 731100 Diversity = 1.8365e-03 k_t
= [1, 1, 1, 1]
Generation 7320: Best = 3.857895e+03 FES = 732100 Diversity = 2.0175e-03 k_t
= [3, 4, 4, 2]
Generation 7330: Best = 3.857893e+03 FES = 733100 Diversity = 1.9881e-03 k_t
= [1, 1, 1, 1]
Generation 7340: Best = 3.857893e+03 FES = 734100 Diversity = 1.8407e-03 k_t
= [3, 2, 4, 4]
Generation 7350: Best = 3.857892e+03 FES = 735100 Diversity = 1.7746e-03 k_t
= [1, 1, 1, 1]
Generation 7360: Best = 3.857891e+03 FES = 736100 Diversity = 1.6340e-03 k_t
= [3, 2, 2, 2]
Generation 7370: Best = 3.857889e+03 FES = 737100 Diversity = 1.4903e-03 k_t
= [1, 1, 1, 1]
Generation 7380: Best = 3.857888e+03 FES = 738100 Diversity = 1.5923e-03 k_t
= [2, 2, 2, 3]
Generation 7390: Best = 3.857888e+03 FES = 739100 Diversity = 1.6431e-03 k_t
= [1, 1, 1, 1]
Generation 7400: Best = 3.857887e+03 FES = 740100 Diversity = 1.5738e-03 k_t
= [4, 3, 4, 3]
Generation 7410: Best = 3.857886e+03 FES = 741100 Diversity = 1.5261e-03 k_t
= [1, 1, 1, 1]
Generation 7420: Best = 3.857885e+03 FES = 742100 Diversity = 1.5214e-03 k_t
= [3, 2, 2, 3]
Generation 7430: Best = 3.857884e+03 FES = 743100 Diversity = 1.5578e-03 k_t
= [1, 1, 1, 1]
Generation 7440: Best = 3.857883e+03 FES = 744100 Diversity = 1.4927e-03 k_t
= [3, 2, 2, 2]
Generation 7450: Best = 3.857883e+03 FES = 745100 Diversity = 1.3812e-03 k_t
= [1, 1, 1, 1]
Generation 7460: Best = 3.857882e+03 FES = 746100 Diversity = 1.3313e-03 k_t
= [2, 3, 4, 2]
Generation 7470: Best = 3.857880e+03 FES = 747100 Diversity = 1.4799e-03 k_t
= [1, 1, 1, 1]
Generation 7480: Best = 3.857879e+03 FES = 748100 Diversity = 1.6040e-03 k_t
= [3, 3, 2, 2]
Generation 7490: Best = 3.857878e+03 FES = 749100 Diversity = 1.5929e-03 k_t
= [1, 1, 1, 1]
Generation 7500: Best = 3.857877e+03 FES = 750100 Diversity = 1.6594e-03 k_t
= [2, 3, 3, 2]
Generation 7510: Best = 3.857876e+03 FES = 751100 Diversity = 1.6569e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 7520: Best = 3.857875e+03 FES = 752100 Diversity = 1.6563e-03 k_t
= [4, 4, 4, 4]
Generation 7530: Best = 3.857874e+03 FES = 753100 Diversity = 1.7359e-03 k_t
= [1, 1, 1, 1]
Generation 7540: Best = 3.857872e+03 FES = 754100 Diversity = 1.6668e-03 k_t
= [4, 3, 2, 2]
Generation 7550: Best = 3.857872e+03 FES = 755100 Diversity = 1.5219e-03 k_t
= [1, 1, 1, 1]
Generation 7560: Best = 3.857871e+03 FES = 756100 Diversity = 1.4435e-03 k_t
= [2, 2, 3, 2]
Generation 7570: Best = 3.857871e+03 FES = 757100 Diversity = 1.3210e-03 k_t
= [1, 1, 1, 1]
Generation 7580: Best = 3.857870e+03 FES = 758100 Diversity = 1.3061e-03 k_t
= [2, 2, 4, 4]
Generation 7590: Best = 3.857869e+03 FES = 759100 Diversity = 1.3122e-03 k_t
= [1, 1, 1, 1]
Generation 7600: Best = 3.857869e+03 FES = 760100 Diversity = 1.2803e-03 k_t
= [2, 2, 2, 2]
Generation 7610: Best = 3.857868e+03 FES = 761100 Diversity = 1.2721e-03 k_t
= [1, 1, 1, 1]
Generation 7620: Best = 3.857868e+03 FES = 762100 Diversity = 1.2830e-03 k_t
= [3, 2, 4, 2]
Generation 7630: Best = 3.857867e+03 FES = 763100 Diversity = 1.1926e-03 k_t
= [1, 1, 1, 1]
Generation 7640: Best = 3.857866e+03 FES = 764100 Diversity = 1.1394e-03 k_t
= [3, 3, 3, 2]
Generation 7650: Best = 3.857865e+03 FES = 765100 Diversity = 1.1540e-03 k_t
= [1, 1, 1, 1]
Generation 7660: Best = 3.857864e+03 FES = 766100 Diversity = 1.2497e-03 k_t
= [4, 2, 3, 4]
Generation 7670: Best = 3.857864e+03 FES = 767100 Diversity = 1.3708e-03 k_t
= [1, 1, 1, 1]
Generation 7680: Best = 3.857863e+03 FES = 768100 Diversity = 1.5176e-03 k_t
= [2, 3, 3, 2]
Generation 7690: Best = 3.857861e+03 FES = 769100 Diversity = 1.6071e-03 k_t
= [1, 1, 1, 1]
Generation 7700: Best = 3.857860e+03 FES = 770100 Diversity = 1.7464e-03 k_t
= [2, 4, 2, 3]
Generation 7710: Best = 3.857859e+03 FES = 771100 Diversity = 1.9053e-03 k_t
= [1, 1, 1, 1]
Generation 7720: Best = 3.857858e+03 FES = 772100 Diversity = 1.9266e-03 k_t
= [3, 4, 2, 4]
Generation 7730: Best = 3.857857e+03 FES = 773100 Diversity = 1.8131e-03 k_t
= [1, 1, 1, 1]
Generation 7740: Best = 3.857854e+03 FES = 774100 Diversity = 1.7275e-03 k_t
= [4, 4, 3, 4]
Generation 7750: Best = 3.857854e+03 FES = 775100 Diversity = 1.7584e-03 k_t
= [1, 1, 1, 1]
Generation 7760: Best = 3.857852e+03 FES = 776100 Diversity = 1.7602e-03 k_t
= [2, 3, 2, 2]
Generation 7770: Best = 3.857851e+03 FES = 777100 Diversity = 1.7833e-03 k_t
= [1, 1, 1, 1]
Generation 7780: Best = 3.857849e+03 FES = 778100 Diversity = 1.9654e-03 k_t

```

```

= [2, 2, 4, 2]
Generation 7790: Best = 3.857848e+03 FES = 779100 Diversity = 1.9891e-03 k_t
= [1, 1, 1, 1]
Generation 7800: Best = 3.857847e+03 FES = 780100 Diversity = 1.8906e-03 k_t
= [4, 3, 4, 4]
Generation 7810: Best = 3.857846e+03 FES = 781100 Diversity = 1.8480e-03 k_t
= [1, 1, 1, 1]
Generation 7820: Best = 3.857845e+03 FES = 782100 Diversity = 1.7896e-03 k_t
= [3, 4, 2, 4]
Generation 7830: Best = 3.857844e+03 FES = 783100 Diversity = 1.6116e-03 k_t
= [1, 1, 1, 1]
Generation 7840: Best = 3.857843e+03 FES = 784100 Diversity = 1.5298e-03 k_t
= [3, 2, 2, 3]
Generation 7850: Best = 3.857843e+03 FES = 785100 Diversity = 1.5122e-03 k_t
= [1, 1, 1, 1]
Generation 7860: Best = 3.857842e+03 FES = 786100 Diversity = 1.4871e-03 k_t
= [4, 2, 2, 3]
Generation 7870: Best = 3.857841e+03 FES = 787100 Diversity = 1.4348e-03 k_t
= [1, 1, 1, 1]
Generation 7880: Best = 3.857841e+03 FES = 788100 Diversity = 1.4164e-03 k_t
= [4, 4, 2, 4]
Generation 7890: Best = 3.857840e+03 FES = 789100 Diversity = 1.3278e-03 k_t
= [1, 1, 1, 1]
Generation 7900: Best = 3.857840e+03 FES = 790100 Diversity = 1.2357e-03 k_t
= [3, 4, 2, 4]
Generation 7910: Best = 3.857839e+03 FES = 791100 Diversity = 1.1770e-03 k_t
= [1, 1, 1, 1]
Generation 7920: Best = 3.857838e+03 FES = 792100 Diversity = 1.1854e-03 k_t
= [2, 2, 2, 4]
Generation 7930: Best = 3.857838e+03 FES = 793100 Diversity = 1.1980e-03 k_t
= [1, 1, 1, 1]
Generation 7940: Best = 3.857837e+03 FES = 794100 Diversity = 1.1781e-03 k_t
= [2, 2, 3, 3]
Generation 7950: Best = 3.857837e+03 FES = 795100 Diversity = 1.0489e-03 k_t
= [1, 1, 1, 1]
Generation 7960: Best = 3.857836e+03 FES = 796100 Diversity = 1.0462e-03 k_t
= [3, 3, 2, 4]
Generation 7970: Best = 3.857836e+03 FES = 797100 Diversity = 1.0718e-03 k_t
= [1, 1, 1, 1]
Generation 7980: Best = 3.857835e+03 FES = 798100 Diversity = 1.0561e-03 k_t
= [2, 2, 4, 4]
Generation 7990: Best = 3.857835e+03 FES = 799100 Diversity = 1.0854e-03 k_t
= [1, 1, 1, 1]
Generation 8000: Best = 3.857834e+03 FES = 800100 Diversity = 1.0979e-03 k_t
= [4, 2, 2, 4]
Generation 8010: Best = 3.857834e+03 FES = 801100 Diversity = 1.0818e-03 k_t
= [1, 1, 1, 1]
Generation 8020: Best = 3.857834e+03 FES = 802100 Diversity = 1.0619e-03 k_t
= [2, 2, 2, 2]
Generation 8030: Best = 3.857833e+03 FES = 803100 Diversity = 1.0164e-03 k_t
= [1, 1, 1, 1]
Generation 8040: Best = 3.857833e+03 FES = 804100 Diversity = 9.9072e-04 k_t
= [4, 2, 4, 4]
Generation 8050: Best = 3.857832e+03 FES = 805100 Diversity = 9.2364e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 8060: Best = 3.857832e+03 FES = 806100 Diversity = 9.1307e-04 k_t
= [2, 2, 2, 4]
Generation 8070: Best = 3.857831e+03 FES = 807100 Diversity = 9.5828e-04 k_t
= [1, 1, 1, 1]
Generation 8080: Best = 3.857831e+03 FES = 808100 Diversity = 9.0056e-04 k_t
= [2, 4, 4, 2]
Generation 8090: Best = 3.857831e+03 FES = 809100 Diversity = 9.3238e-04 k_t
= [1, 1, 1, 1]
Generation 8100: Best = 3.857830e+03 FES = 810100 Diversity = 9.5280e-04 k_t
= [2, 4, 2, 2]
Generation 8110: Best = 3.857830e+03 FES = 811100 Diversity = 9.4419e-04 k_t
= [1, 1, 1, 1]
Generation 8120: Best = 3.857830e+03 FES = 812100 Diversity = 9.7193e-04 k_t
= [4, 3, 4, 4]
Generation 8130: Best = 3.857830e+03 FES = 813100 Diversity = 9.5981e-04 k_t
= [1, 1, 1, 1]
Generation 8140: Best = 3.857829e+03 FES = 814100 Diversity = 1.0461e-03 k_t
= [4, 4, 4, 2]
Generation 8150: Best = 3.857829e+03 FES = 815100 Diversity = 1.0804e-03 k_t
= [1, 1, 1, 1]
Generation 8160: Best = 3.857828e+03 FES = 816100 Diversity = 1.0111e-03 k_t
= [4, 2, 4, 3]
Generation 8170: Best = 3.857828e+03 FES = 817100 Diversity = 1.0026e-03 k_t
= [1, 1, 1, 1]
Generation 8180: Best = 3.857828e+03 FES = 818100 Diversity = 9.7116e-04 k_t
= [2, 2, 2, 3]
Generation 8190: Best = 3.857827e+03 FES = 819100 Diversity = 9.4564e-04 k_t
= [1, 1, 1, 1]
Generation 8200: Best = 3.857827e+03 FES = 820100 Diversity = 8.4576e-04 k_t
= [2, 3, 4, 3]
Generation 8210: Best = 3.857827e+03 FES = 821100 Diversity = 8.8981e-04 k_t
= [1, 1, 1, 1]
Generation 8220: Best = 3.857826e+03 FES = 822100 Diversity = 9.1058e-04 k_t
= [2, 4, 4, 4]
Generation 8230: Best = 3.857825e+03 FES = 823100 Diversity = 1.0387e-03 k_t
= [1, 1, 1, 1]
Generation 8240: Best = 3.857824e+03 FES = 824100 Diversity = 1.1594e-03 k_t
= [2, 3, 2, 3]
Generation 8250: Best = 3.857824e+03 FES = 825100 Diversity = 1.1655e-03 k_t
= [1, 1, 1, 1]
Generation 8260: Best = 3.857823e+03 FES = 826100 Diversity = 1.1681e-03 k_t
= [4, 3, 3, 3]
Generation 8270: Best = 3.857822e+03 FES = 827100 Diversity = 1.1409e-03 k_t
= [1, 1, 1, 1]
Generation 8280: Best = 3.857822e+03 FES = 828100 Diversity = 1.2020e-03 k_t
= [4, 4, 2, 4]
Generation 8290: Best = 3.857821e+03 FES = 829100 Diversity = 1.2508e-03 k_t
= [1, 1, 1, 1]
Generation 8300: Best = 3.857820e+03 FES = 830100 Diversity = 1.2967e-03 k_t
= [4, 2, 2, 4]
Generation 8310: Best = 3.857820e+03 FES = 831100 Diversity = 1.3374e-03 k_t
= [1, 1, 1, 1]
Generation 8320: Best = 3.857819e+03 FES = 832100 Diversity = 1.3105e-03 k_t

```



```

= [4, 3, 4, 4]
Generation 8330: Best = 3.857819e+03 FES = 833100 Diversity = 1.2655e-03 k_t
= [1, 1, 1, 1]
Generation 8340: Best = 3.857818e+03 FES = 834100 Diversity = 1.1293e-03 k_t
= [2, 4, 3, 2]
Generation 8350: Best = 3.857817e+03 FES = 835100 Diversity = 1.0460e-03 k_t
= [1, 1, 1, 1]
Generation 8360: Best = 3.857817e+03 FES = 836100 Diversity = 1.0646e-03 k_t
= [2, 2, 3, 2]
Generation 8370: Best = 3.857816e+03 FES = 837100 Diversity = 1.0504e-03 k_t
= [1, 1, 1, 1]
Generation 8380: Best = 3.857816e+03 FES = 838100 Diversity = 1.1672e-03 k_t
= [4, 2, 4, 3]
Generation 8390: Best = 3.857816e+03 FES = 839100 Diversity = 1.0903e-03 k_t
= [1, 1, 1, 1]
Generation 8400: Best = 3.857815e+03 FES = 840100 Diversity = 1.0712e-03 k_t
= [2, 4, 2, 4]
Generation 8410: Best = 3.857815e+03 FES = 841100 Diversity = 1.1023e-03 k_t
= [1, 1, 1, 1]
Generation 8420: Best = 3.857814e+03 FES = 842100 Diversity = 1.1004e-03 k_t
= [2, 2, 3, 4]
Generation 8430: Best = 3.857814e+03 FES = 843100 Diversity = 1.0356e-03 k_t
= [1, 1, 1, 1]
Generation 8440: Best = 3.857813e+03 FES = 844100 Diversity = 1.0177e-03 k_t
= [3, 2, 4, 2]
Generation 8450: Best = 3.857813e+03 FES = 845100 Diversity = 9.2168e-04 k_t
= [1, 1, 1, 1]
Generation 8460: Best = 3.857812e+03 FES = 846100 Diversity = 8.6937e-04 k_t
= [3, 3, 2, 2]
Generation 8470: Best = 3.857812e+03 FES = 847100 Diversity = 9.5709e-04 k_t
= [1, 1, 1, 1]
Generation 8480: Best = 3.857812e+03 FES = 848100 Diversity = 9.4048e-04 k_t
= [3, 2, 2, 4]
Generation 8490: Best = 3.857811e+03 FES = 849100 Diversity = 9.2122e-04 k_t
= [1, 1, 1, 1]
Generation 8500: Best = 3.857811e+03 FES = 850100 Diversity = 8.8740e-04 k_t
= [3, 4, 3, 2]
Generation 8510: Best = 3.857810e+03 FES = 851100 Diversity = 8.9534e-04 k_t
= [1, 1, 1, 1]
Generation 8520: Best = 3.857810e+03 FES = 852100 Diversity = 9.6651e-04 k_t
= [4, 3, 3, 4]
Generation 8530: Best = 3.857809e+03 FES = 853100 Diversity = 9.5529e-04 k_t
= [1, 1, 1, 1]
Generation 8540: Best = 3.857809e+03 FES = 854100 Diversity = 9.5349e-04 k_t
= [3, 3, 2, 4]
Generation 8550: Best = 3.857809e+03 FES = 855100 Diversity = 9.3691e-04 k_t
= [1, 1, 1, 1]
Generation 8560: Best = 3.857808e+03 FES = 856100 Diversity = 8.8147e-04 k_t
= [2, 4, 4, 4]
Generation 8570: Best = 3.857808e+03 FES = 857100 Diversity = 9.1173e-04 k_t
= [1, 1, 1, 1]
Generation 8580: Best = 3.857807e+03 FES = 858100 Diversity = 9.7738e-04 k_t
= [3, 2, 2, 3]
Generation 8590: Best = 3.857807e+03 FES = 859100 Diversity = 9.5688e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 8600: Best = 3.857807e+03 FES = 860100 Diversity = 9.3776e-04 k_t
= [3, 3, 3, 3]
Generation 8610: Best = 3.857806e+03 FES = 861100 Diversity = 9.2108e-04 k_t
= [1, 1, 1, 1]
Generation 8620: Best = 3.857806e+03 FES = 862100 Diversity = 8.7862e-04 k_t
= [2, 4, 4, 2]
Generation 8630: Best = 3.857806e+03 FES = 863100 Diversity = 8.9109e-04 k_t
= [1, 1, 1, 1]
Generation 8640: Best = 3.857805e+03 FES = 864100 Diversity = 8.8724e-04 k_t
= [4, 3, 4, 2]
Generation 8650: Best = 3.857805e+03 FES = 865100 Diversity = 8.8056e-04 k_t
= [1, 1, 1, 1]
Generation 8660: Best = 3.857804e+03 FES = 866100 Diversity = 8.8257e-04 k_t
= [4, 3, 3, 2]
Generation 8670: Best = 3.857804e+03 FES = 867100 Diversity = 8.5817e-04 k_t
= [1, 1, 1, 1]
Generation 8680: Best = 3.857804e+03 FES = 868100 Diversity = 8.4071e-04 k_t
= [2, 4, 4, 3]
Generation 8690: Best = 3.857803e+03 FES = 869100 Diversity = 7.7216e-04 k_t
= [1, 1, 1, 1]
Generation 8700: Best = 3.857803e+03 FES = 870100 Diversity = 7.3812e-04 k_t
= [3, 3, 2, 3]
Generation 8710: Best = 3.857803e+03 FES = 871100 Diversity = 7.7228e-04 k_t
= [1, 1, 1, 1]
Generation 8720: Best = 3.857802e+03 FES = 872100 Diversity = 7.8009e-04 k_t
= [2, 2, 4, 4]
Generation 8730: Best = 3.857802e+03 FES = 873100 Diversity = 8.1159e-04 k_t
= [1, 1, 1, 1]
Generation 8740: Best = 3.857802e+03 FES = 874100 Diversity = 8.2578e-04 k_t
= [3, 3, 2, 2]
Generation 8750: Best = 3.857801e+03 FES = 875100 Diversity = 8.1282e-04 k_t
= [1, 1, 1, 1]
Generation 8760: Best = 3.857801e+03 FES = 876100 Diversity = 8.4545e-04 k_t
= [3, 4, 4, 4]
Generation 8770: Best = 3.857801e+03 FES = 877100 Diversity = 8.3722e-04 k_t
= [1, 1, 1, 1]
Generation 8780: Best = 3.857800e+03 FES = 878100 Diversity = 8.4080e-04 k_t
= [3, 2, 4, 3]
Generation 8790: Best = 3.857800e+03 FES = 879100 Diversity = 8.0808e-04 k_t
= [1, 1, 1, 1]
Generation 8800: Best = 3.857800e+03 FES = 880100 Diversity = 8.4546e-04 k_t
= [3, 4, 2, 2]
Generation 8810: Best = 3.857800e+03 FES = 881100 Diversity = 8.7184e-04 k_t
= [1, 1, 1, 1]
Generation 8820: Best = 3.857799e+03 FES = 882100 Diversity = 8.8188e-04 k_t
= [2, 4, 4, 3]
Generation 8830: Best = 3.857799e+03 FES = 883100 Diversity = 9.1264e-04 k_t
= [1, 1, 1, 1]
Generation 8840: Best = 3.857799e+03 FES = 884100 Diversity = 9.1200e-04 k_t
= [2, 2, 3, 3]
Generation 8850: Best = 3.857798e+03 FES = 885100 Diversity = 9.1079e-04 k_t
= [1, 1, 1, 1]
Generation 8860: Best = 3.857798e+03 FES = 886100 Diversity = 9.0035e-04 k_t

```

```

= [4, 4, 2, 4]
Generation 8870: Best = 3.857798e+03  FES = 887100  Diversity = 9.2473e-04  k_t
= [1, 1, 1, 1]
Generation 8880: Best = 3.857797e+03  FES = 888100  Diversity = 9.2484e-04  k_t
= [2, 3, 3, 2]
Generation 8890: Best = 3.857797e+03  FES = 889100  Diversity = 9.0073e-04  k_t
= [1, 1, 1, 1]
Generation 8900: Best = 3.857797e+03  FES = 890100  Diversity = 8.7770e-04  k_t
= [4, 2, 4, 4]
Generation 8910: Best = 3.857796e+03  FES = 891100  Diversity = 8.9112e-04  k_t
= [1, 1, 1, 1]
Generation 8920: Best = 3.857796e+03  FES = 892100  Diversity = 8.7253e-04  k_t
= [4, 3, 4, 2]
Generation 8930: Best = 3.857795e+03  FES = 893100  Diversity = 8.5304e-04  k_t
= [1, 1, 1, 1]
Generation 8940: Best = 3.857795e+03  FES = 894100  Diversity = 8.9110e-04  k_t
= [3, 2, 2, 2]
Generation 8950: Best = 3.857795e+03  FES = 895100  Diversity = 9.0039e-04  k_t
= [1, 1, 1, 1]
Generation 8960: Best = 3.857795e+03  FES = 896100  Diversity = 8.7765e-04  k_t
= [4, 4, 3, 3]
Generation 8970: Best = 3.857794e+03  FES = 897100  Diversity = 8.4635e-04  k_t
= [1, 1, 1, 1]
Generation 8980: Best = 3.857794e+03  FES = 898100  Diversity = 8.3892e-04  k_t
= [3, 4, 3, 2]
Generation 8990: Best = 3.857794e+03  FES = 899100  Diversity = 8.3860e-04  k_t
= [1, 1, 1, 1]
Generation 9000: Best = 3.857793e+03  FES = 900100  Diversity = 8.9028e-04  k_t
= [2, 4, 3, 3]
Generation 9010: Best = 3.857793e+03  FES = 901100  Diversity = 8.9063e-04  k_t
= [1, 1, 1, 1]
Generation 9020: Best = 3.857793e+03  FES = 902100  Diversity = 9.3861e-04  k_t
= [4, 4, 2, 2]
Generation 9030: Best = 3.857792e+03  FES = 903100  Diversity = 9.3067e-04  k_t
= [1, 1, 1, 1]
Generation 9040: Best = 3.857792e+03  FES = 904100  Diversity = 9.5127e-04  k_t
= [3, 3, 2, 3]
Generation 9050: Best = 3.857792e+03  FES = 905100  Diversity = 8.8502e-04  k_t
= [1, 1, 1, 1]
Generation 9060: Best = 3.857791e+03  FES = 906100  Diversity = 8.3487e-04  k_t
= [4, 4, 4, 4]
Generation 9070: Best = 3.857791e+03  FES = 907100  Diversity = 8.3593e-04  k_t
= [1, 1, 1, 1]
Generation 9080: Best = 3.857791e+03  FES = 908100  Diversity = 7.8489e-04  k_t
= [3, 4, 2, 2]
Generation 9090: Best = 3.857791e+03  FES = 909100  Diversity = 8.1020e-04  k_t
= [1, 1, 1, 1]
Generation 9100: Best = 3.857790e+03  FES = 910100  Diversity = 7.3168e-04  k_t
= [3, 2, 2, 2]
Generation 9110: Best = 3.857790e+03  FES = 911100  Diversity = 6.7579e-04  k_t
= [1, 1, 1, 1]
Generation 9120: Best = 3.857790e+03  FES = 912100  Diversity = 7.1094e-04  k_t
= [2, 3, 4, 3]
Generation 9130: Best = 3.857790e+03  FES = 913100  Diversity = 7.3948e-04  k_t

```

```

= [1, 1, 1, 1]
Generation 9140: Best = 3.857789e+03 FES = 914100 Diversity = 7.9684e-04 k_t
= [3, 2, 2, 2]
Generation 9150: Best = 3.857789e+03 FES = 915100 Diversity = 7.7697e-04 k_t
= [1, 1, 1, 1]
Generation 9160: Best = 3.857789e+03 FES = 916100 Diversity = 7.4329e-04 k_t
= [3, 4, 3, 2]
Generation 9170: Best = 3.857789e+03 FES = 917100 Diversity = 7.8166e-04 k_t
= [1, 1, 1, 1]
Generation 9180: Best = 3.857788e+03 FES = 918100 Diversity = 8.5169e-04 k_t
= [2, 4, 2, 3]
Generation 9190: Best = 3.857788e+03 FES = 919100 Diversity = 8.5248e-04 k_t
= [1, 1, 1, 1]
Generation 9200: Best = 3.857788e+03 FES = 920100 Diversity = 8.4927e-04 k_t
= [3, 4, 4, 2]
Generation 9210: Best = 3.857787e+03 FES = 921100 Diversity = 7.9552e-04 k_t
= [1, 1, 1, 1]
Generation 9220: Best = 3.857787e+03 FES = 922100 Diversity = 8.1129e-04 k_t
= [3, 4, 4, 2]
Generation 9230: Best = 3.857787e+03 FES = 923100 Diversity = 7.8269e-04 k_t
= [1, 1, 1, 1]
Generation 9240: Best = 3.857787e+03 FES = 924100 Diversity = 8.1254e-04 k_t
= [4, 2, 3, 4]
Generation 9250: Best = 3.857786e+03 FES = 925100 Diversity = 8.2219e-04 k_t
= [1, 1, 1, 1]
Generation 9260: Best = 3.857786e+03 FES = 926100 Diversity = 8.2105e-04 k_t
= [4, 4, 2, 4]
Generation 9270: Best = 3.857786e+03 FES = 927100 Diversity = 8.3525e-04 k_t
= [1, 1, 1, 1]
Generation 9280: Best = 3.857786e+03 FES = 928100 Diversity = 7.9883e-04 k_t
= [2, 2, 2, 4]
Generation 9290: Best = 3.857786e+03 FES = 929100 Diversity = 8.1155e-04 k_t
= [1, 1, 1, 1]
Generation 9300: Best = 3.857785e+03 FES = 930100 Diversity = 7.8191e-04 k_t
= [4, 2, 4, 2]
Generation 9310: Best = 3.857785e+03 FES = 931100 Diversity = 8.5029e-04 k_t
= [1, 1, 1, 1]
Generation 9320: Best = 3.857785e+03 FES = 932100 Diversity = 8.5471e-04 k_t
= [2, 3, 3, 3]
Generation 9330: Best = 3.857784e+03 FES = 933100 Diversity = 9.2248e-04 k_t
= [1, 1, 1, 1]
Generation 9340: Best = 3.857784e+03 FES = 934100 Diversity = 9.7763e-04 k_t
= [3, 3, 4, 3]
Generation 9350: Best = 3.857784e+03 FES = 935100 Diversity = 9.8170e-04 k_t
= [1, 1, 1, 1]
Generation 9360: Best = 3.857783e+03 FES = 936100 Diversity = 9.8663e-04 k_t
= [4, 2, 3, 4]
Generation 9370: Best = 3.857782e+03 FES = 937100 Diversity = 9.8497e-04 k_t
= [1, 1, 1, 1]
Generation 9380: Best = 3.857782e+03 FES = 938100 Diversity = 9.8176e-04 k_t
= [3, 4, 4, 4]
Generation 9390: Best = 3.857782e+03 FES = 939100 Diversity = 9.5213e-04 k_t
= [1, 1, 1, 1]
Generation 9400: Best = 3.857782e+03 FES = 940100 Diversity = 9.3423e-04 k_t

```

```

= [3, 4, 3, 2]
Generation 9410: Best = 3.857781e+03 FES = 941100 Diversity = 9.0130e-04 k_t
= [1, 1, 1, 1]
Generation 9420: Best = 3.857781e+03 FES = 942100 Diversity = 8.8767e-04 k_t
= [2, 3, 4, 4]
Generation 9430: Best = 3.857781e+03 FES = 943100 Diversity = 8.4442e-04 k_t
= [1, 1, 1, 1]
Generation 9440: Best = 3.857780e+03 FES = 944100 Diversity = 8.0666e-04 k_t
= [4, 4, 4, 4]
Generation 9450: Best = 3.857780e+03 FES = 945100 Diversity = 7.6567e-04 k_t
= [1, 1, 1, 1]
Generation 9460: Best = 3.857780e+03 FES = 946100 Diversity = 7.4741e-04 k_t
= [3, 4, 3, 4]
Generation 9470: Best = 3.857780e+03 FES = 947100 Diversity = 7.7432e-04 k_t
= [1, 1, 1, 1]
Generation 9480: Best = 3.857780e+03 FES = 948100 Diversity = 7.6849e-04 k_t
= [3, 4, 3, 3]
Generation 9490: Best = 3.857779e+03 FES = 949100 Diversity = 7.6439e-04 k_t
= [1, 1, 1, 1]
Generation 9500: Best = 3.857779e+03 FES = 950100 Diversity = 8.3041e-04 k_t
= [3, 2, 2, 2]
Generation 9510: Best = 3.857778e+03 FES = 951100 Diversity = 8.6044e-04 k_t
= [1, 1, 1, 1]
Generation 9520: Best = 3.857778e+03 FES = 952100 Diversity = 8.3797e-04 k_t
= [3, 4, 4, 4]
Generation 9530: Best = 3.857778e+03 FES = 953100 Diversity = 8.2127e-04 k_t
= [1, 1, 1, 1]
Generation 9540: Best = 3.857778e+03 FES = 954100 Diversity = 7.8360e-04 k_t
= [2, 3, 3, 2]
Generation 9550: Best = 3.857778e+03 FES = 955100 Diversity = 7.4018e-04 k_t
= [1, 1, 1, 1]
Generation 9560: Best = 3.857777e+03 FES = 956100 Diversity = 7.7208e-04 k_t
= [4, 4, 4, 2]
Generation 9570: Best = 3.857777e+03 FES = 957100 Diversity = 7.7689e-04 k_t
= [1, 1, 1, 1]
Generation 9580: Best = 3.857776e+03 FES = 958100 Diversity = 8.7855e-04 k_t
= [3, 3, 3, 3]
Generation 9590: Best = 3.857776e+03 FES = 959100 Diversity = 8.2670e-04 k_t
= [1, 1, 1, 1]
Generation 9600: Best = 3.857776e+03 FES = 960100 Diversity = 7.8996e-04 k_t
= [4, 2, 3, 4]
Generation 9610: Best = 3.857776e+03 FES = 961100 Diversity = 7.3574e-04 k_t
= [1, 1, 1, 1]
Generation 9620: Best = 3.857776e+03 FES = 962100 Diversity = 7.0902e-04 k_t
= [3, 3, 4, 4]
Generation 9630: Best = 3.857776e+03 FES = 963100 Diversity = 7.0872e-04 k_t
= [1, 1, 1, 1]
Generation 9640: Best = 3.857775e+03 FES = 964100 Diversity = 6.8040e-04 k_t
= [3, 3, 3, 2]
Generation 9650: Best = 3.857775e+03 FES = 965100 Diversity = 7.1649e-04 k_t
= [1, 1, 1, 1]
Generation 9660: Best = 3.857775e+03 FES = 966100 Diversity = 7.4833e-04 k_t
= [2, 2, 2, 4]
Generation 9670: Best = 3.857775e+03 FES = 967100 Diversity = 7.9747e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 9680: Best = 3.857774e+03 FES = 968100 Diversity = 8.0717e-04 k_t
= [3, 4, 3, 2]
Generation 9690: Best = 3.857774e+03 FES = 969100 Diversity = 8.1068e-04 k_t
= [1, 1, 1, 1]
Generation 9700: Best = 3.857774e+03 FES = 970100 Diversity = 7.9854e-04 k_t
= [3, 4, 3, 2]
Generation 9710: Best = 3.857774e+03 FES = 971100 Diversity = 7.6824e-04 k_t
= [1, 1, 1, 1]
Generation 9720: Best = 3.857773e+03 FES = 972100 Diversity = 7.4429e-04 k_t
= [3, 4, 4, 4]
Generation 9730: Best = 3.857773e+03 FES = 973100 Diversity = 7.4676e-04 k_t
= [1, 1, 1, 1]
Generation 9740: Best = 3.857773e+03 FES = 974100 Diversity = 7.0926e-04 k_t
= [4, 2, 2, 2]
Generation 9750: Best = 3.857773e+03 FES = 975100 Diversity = 7.2751e-04 k_t
= [1, 1, 1, 1]
Generation 9760: Best = 3.857773e+03 FES = 976100 Diversity = 7.1957e-04 k_t
= [4, 2, 2, 4]
Generation 9770: Best = 3.857772e+03 FES = 977100 Diversity = 7.2304e-04 k_t
= [1, 1, 1, 1]
Generation 9780: Best = 3.857772e+03 FES = 978100 Diversity = 7.4242e-04 k_t
= [3, 2, 4, 3]
Generation 9790: Best = 3.857772e+03 FES = 979100 Diversity = 7.6693e-04 k_t
= [1, 1, 1, 1]
Generation 9800: Best = 3.857772e+03 FES = 980100 Diversity = 7.5448e-04 k_t
= [3, 4, 3, 2]
Generation 9810: Best = 3.857772e+03 FES = 981100 Diversity = 7.1542e-04 k_t
= [1, 1, 1, 1]
Generation 9820: Best = 3.857772e+03 FES = 982100 Diversity = 6.9052e-04 k_t
= [4, 4, 3, 2]
Generation 9830: Best = 3.857771e+03 FES = 983100 Diversity = 7.1312e-04 k_t
= [1, 1, 1, 1]
Generation 9840: Best = 3.857771e+03 FES = 984100 Diversity = 6.3927e-04 k_t
= [4, 2, 3, 2]
Generation 9850: Best = 3.857771e+03 FES = 985100 Diversity = 6.4796e-04 k_t
= [1, 1, 1, 1]
Generation 9860: Best = 3.857771e+03 FES = 986100 Diversity = 6.3802e-04 k_t
= [4, 4, 2, 3]
Generation 9870: Best = 3.857771e+03 FES = 987100 Diversity = 5.9948e-04 k_t
= [1, 1, 1, 1]
Generation 9880: Best = 3.857770e+03 FES = 988100 Diversity = 6.2191e-04 k_t
= [3, 2, 4, 2]
Generation 9890: Best = 3.857770e+03 FES = 989100 Diversity = 6.4384e-04 k_t
= [1, 1, 1, 1]
Generation 9900: Best = 3.857770e+03 FES = 990100 Diversity = 6.5154e-04 k_t
= [3, 4, 2, 3]
Generation 9910: Best = 3.857770e+03 FES = 991100 Diversity = 6.3008e-04 k_t
= [1, 1, 1, 1]
Generation 9920: Best = 3.857770e+03 FES = 992100 Diversity = 6.0330e-04 k_t
= [4, 3, 3, 2]
Generation 9930: Best = 3.857770e+03 FES = 993100 Diversity = 5.8096e-04 k_t
= [1, 1, 1, 1]
Generation 9940: Best = 3.857770e+03 FES = 994100 Diversity = 5.6617e-04 k_t

```

```

= [4, 4, 4, 2]
Generation 9950: Best = 3.857770e+03  FES = 995100  Diversity = 5.6095e-04  k_t
= [1, 1, 1, 1]
Generation 9960: Best = 3.857769e+03  FES = 996100  Diversity = 5.5563e-04  k_t
= [3, 4, 3, 2]
Generation 9970: Best = 3.857769e+03  FES = 997100  Diversity = 5.6665e-04  k_t
= [1, 1, 1, 1]
Generation 9980: Best = 3.857769e+03  FES = 998100  Diversity = 5.6989e-04  k_t
= [2, 3, 3, 3]
Generation 9990: Best = 3.857769e+03  FES = 999100  Diversity = 5.9754e-04  k_t
= [1, 1, 1, 1]

```

```

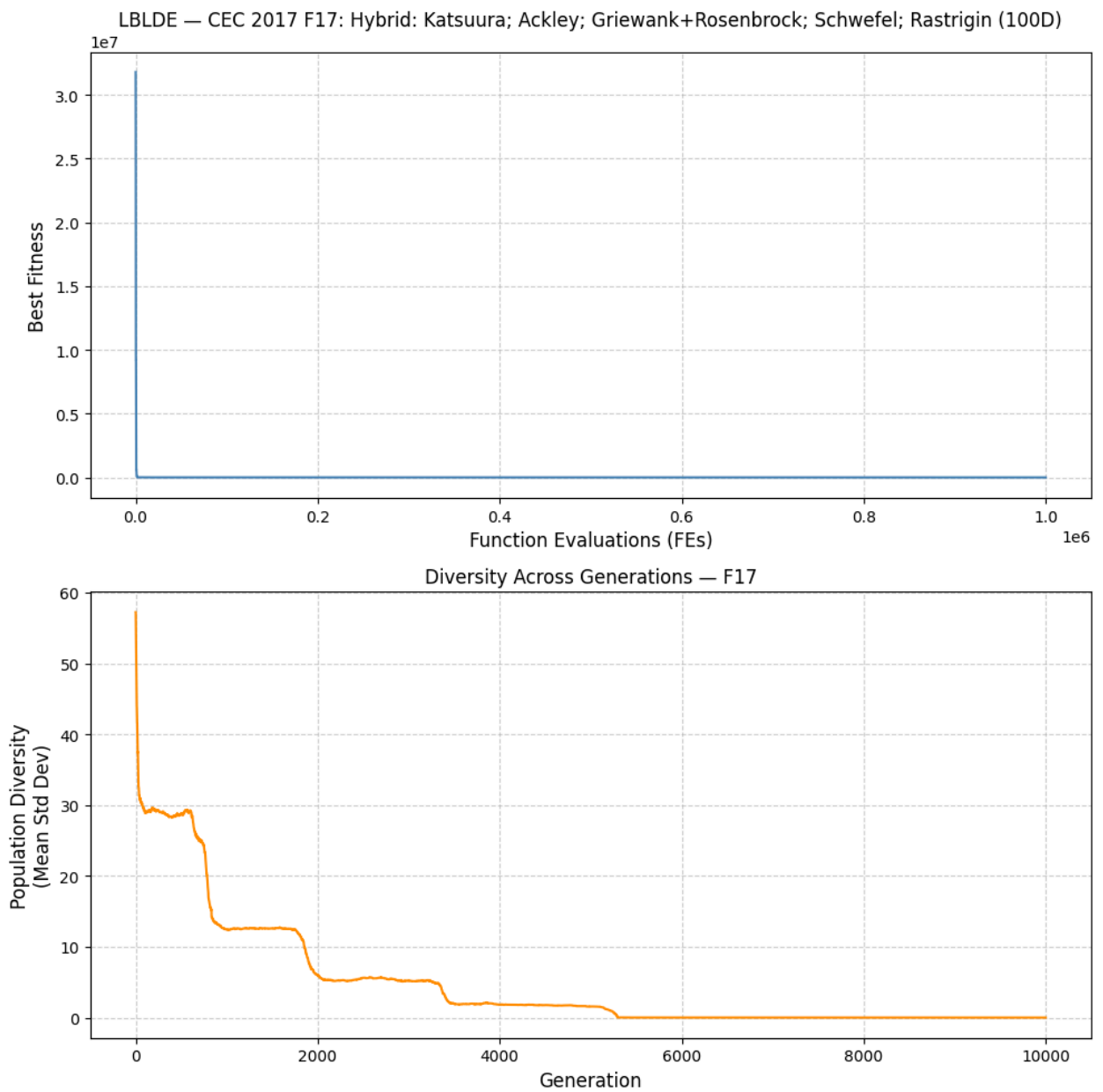
Optimization completed!
Final Best Fitness = 3.857769e+03
Total Function Evaluations = 1000000

```

```

=====
=
Results:
  Best Fitness           : 3.857769e+03
  Global Optimum         : 1.700000e+03
  Error (|best-optimum|) : 2.157769e+03
  Best Solution (5 dim)  : [-54.6331353    0.93328262    3.41952005   53.79173872
-33.21887127]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```



Graph saved as convergence_F17.png

```
In [42]: # =====
# F18 — Hybrid Function 8 (Elliptic; Ackley; Rastrigin; HGBat; Discus)
# =====
run_and_plot('F18', make_wrapper(hybrid.f18), 1800.0, 'Hybrid: Elliptic; Ackle
```



```

=====
=
LBLDE_s on CEC 2017 F18 – Single Run
=====
=
Function      : Hybrid: Elliptic; Ackley; Rastrigin; HGBat; Discus (100D)
Bounds       : [-100, 100]
Global optimum : 1800.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 1.395496e+09 | Diversity = 5.7220e+01
Generation 10: Best = 2.051486e+08 FES = 1100 Diversity = 5.2335e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.153103e+08 FES = 2100 Diversity = 4.7499e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 4.199349e+07 FES = 3100 Diversity = 3.9607e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.332839e+07 FES = 4100 Diversity = 3.6567e+01 k_t
= [3, 4, 4, 4]
Generation 50: Best = 1.392914e+07 FES = 5100 Diversity = 3.1440e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 5.575996e+06 FES = 6100 Diversity = 2.8303e+01 k_t
= [3, 4, 4, 2]
Generation 70: Best = 3.771820e+06 FES = 7100 Diversity = 2.4329e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 2.582841e+06 FES = 8100 Diversity = 2.1494e+01 k_t
= [3, 2, 4, 3]
Generation 90: Best = 1.882804e+06 FES = 9100 Diversity = 1.9399e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.210064e+06 FES = 10100 Diversity = 1.7607e+01 k_t
= [3, 4, 4, 3]
Generation 110: Best = 9.562028e+05 FES = 11100 Diversity = 1.6213e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 8.278326e+05 FES = 12100 Diversity = 1.4699e+01 k_t
= [2, 2, 2, 3]
Generation 130: Best = 6.404464e+05 FES = 13100 Diversity = 1.2906e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 4.922429e+05 FES = 14100 Diversity = 1.1810e+01 k_t
= [4, 3, 4, 4]
Generation 150: Best = 4.922428e+05 FES = 15100 Diversity = 1.0833e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 4.441622e+05 FES = 16100 Diversity = 9.2454e+00 k_t
= [4, 3, 4, 2]
Generation 170: Best = 3.931920e+05 FES = 17100 Diversity = 9.3993e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.842596e+05 FES = 18100 Diversity = 8.2695e+00 k_t
= [2, 3, 2, 2]
Generation 190: Best = 3.561784e+05 FES = 19100 Diversity = 7.6534e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.378459e+05 FES = 20100 Diversity = 8.0342e+00 k_t
= [3, 3, 3, 4]
Generation 210: Best = 3.326643e+05 FES = 21100 Diversity = 8.1883e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.271235e+05 FES = 22100 Diversity = 7.5519e+00 k_t
= [3, 2, 3, 3]

```

Generation 230:	Best = 3.149059e+05	FES = 23100	Diversity = 6.8946e+00	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 3.116037e+05	FES = 24100	Diversity = 7.2064e+00	k_t
= [2, 2, 2, 2]				
Generation 250:	Best = 3.060366e+05	FES = 25100	Diversity = 7.3061e+00	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 3.008605e+05	FES = 26100	Diversity = 7.0299e+00	k_t
= [4, 2, 2, 2]				
Generation 270:	Best = 2.960067e+05	FES = 27100	Diversity = 6.6640e+00	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 2.898922e+05	FES = 28100	Diversity = 6.2715e+00	k_t
= [2, 3, 4, 3]				
Generation 290:	Best = 2.886263e+05	FES = 29100	Diversity = 6.5264e+00	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 2.858566e+05	FES = 30100	Diversity = 6.1155e+00	k_t
= [4, 2, 4, 4]				
Generation 310:	Best = 2.832696e+05	FES = 31100	Diversity = 6.0837e+00	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 2.821998e+05	FES = 32100	Diversity = 5.9443e+00	k_t
= [4, 3, 2, 3]				
Generation 330:	Best = 2.803636e+05	FES = 33100	Diversity = 5.4239e+00	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 2.798316e+05	FES = 34100	Diversity = 5.1870e+00	k_t
= [4, 2, 3, 2]				
Generation 350:	Best = 2.772375e+05	FES = 35100	Diversity = 5.5523e+00	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 2.753499e+05	FES = 36100	Diversity = 5.9493e+00	k_t
= [4, 4, 2, 3]				
Generation 370:	Best = 2.740066e+05	FES = 37100	Diversity = 7.1653e+00	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 2.713617e+05	FES = 38100	Diversity = 8.0667e+00	k_t
= [3, 4, 2, 2]				
Generation 390:	Best = 2.703416e+05	FES = 39100	Diversity = 7.9937e+00	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 2.675385e+05	FES = 40100	Diversity = 9.2771e+00	k_t
= [3, 2, 3, 2]				
Generation 410:	Best = 2.669289e+05	FES = 41100	Diversity = 9.0228e+00	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 2.647790e+05	FES = 42100	Diversity = 9.8545e+00	k_t
= [3, 3, 4, 4]				
Generation 430:	Best = 2.637835e+05	FES = 43100	Diversity = 1.0556e+01	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 2.623454e+05	FES = 44100	Diversity = 1.0869e+01	k_t
= [3, 2, 3, 4]				
Generation 450:	Best = 2.604967e+05	FES = 45100	Diversity = 1.1474e+01	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 2.597962e+05	FES = 46100	Diversity = 1.1648e+01	k_t
= [3, 3, 3, 2]				
Generation 470:	Best = 2.577178e+05	FES = 47100	Diversity = 1.2455e+01	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 2.547516e+05	FES = 48100	Diversity = 1.3308e+01	k_t
= [4, 3, 2, 3]				
Generation 490:	Best = 2.543012e+05	FES = 49100	Diversity = 1.3044e+01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 2.537597e+05	FES = 50100	Diversity = 1.2590e+01	k_t
= [3, 3, 4, 2]				
Generation 510:	Best = 2.527803e+05	FES = 51100	Diversity = 1.2059e+01	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 2.517059e+05	FES = 52100	Diversity = 1.1070e+01	k_t
= [4, 2, 4, 2]				
Generation 530:	Best = 2.516973e+05	FES = 53100	Diversity = 1.0840e+01	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 2.509288e+05	FES = 54100	Diversity = 1.1376e+01	k_t
= [3, 4, 2, 3]				
Generation 550:	Best = 2.503110e+05	FES = 55100	Diversity = 1.0997e+01	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 2.496088e+05	FES = 56100	Diversity = 1.0672e+01	k_t
= [2, 4, 2, 2]				
Generation 570:	Best = 2.488932e+05	FES = 57100	Diversity = 1.0819e+01	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 2.486124e+05	FES = 58100	Diversity = 1.0496e+01	k_t
= [2, 3, 4, 3]				
Generation 590:	Best = 2.480007e+05	FES = 59100	Diversity = 1.1171e+01	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 2.463223e+05	FES = 60100	Diversity = 1.2144e+01	k_t
= [4, 4, 4, 4]				
Generation 610:	Best = 2.453100e+05	FES = 61100	Diversity = 1.3264e+01	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 2.453060e+05	FES = 62100	Diversity = 1.1725e+01	k_t
= [2, 2, 3, 4]				
Generation 630:	Best = 2.450082e+05	FES = 63100	Diversity = 1.0576e+01	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 2.435893e+05	FES = 64100	Diversity = 1.1020e+01	k_t
= [4, 3, 3, 4]				
Generation 650:	Best = 2.418012e+05	FES = 65100	Diversity = 1.2932e+01	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 2.407681e+05	FES = 66100	Diversity = 1.3698e+01	k_t
= [3, 2, 3, 3]				
Generation 670:	Best = 2.398730e+05	FES = 67100	Diversity = 1.4638e+01	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 2.383024e+05	FES = 68100	Diversity = 1.6005e+01	k_t
= [4, 3, 4, 3]				
Generation 690:	Best = 2.373545e+05	FES = 69100	Diversity = 1.5402e+01	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 2.369633e+05	FES = 70100	Diversity = 1.5755e+01	k_t
= [3, 3, 4, 2]				
Generation 710:	Best = 2.351587e+05	FES = 71100	Diversity = 1.7420e+01	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 2.343839e+05	FES = 72100	Diversity = 1.6907e+01	k_t
= [2, 2, 3, 2]				
Generation 730:	Best = 2.338972e+05	FES = 73100	Diversity = 1.5549e+01	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 2.328951e+05	FES = 74100	Diversity = 1.6887e+01	k_t
= [4, 4, 2, 2]				
Generation 750:	Best = 2.324421e+05	FES = 75100	Diversity = 1.6644e+01	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 2.318644e+05	FES = 76100	Diversity = 1.7891e+01	k_t
= [2, 2, 3, 2]				

Generation 770:	Best = 2.307748e+05	FES = 77100	Diversity = 1.8518e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.296772e+05	FES = 78100	Diversity = 1.8247e+01	k_t
= [3, 2, 4, 3]				
Generation 790:	Best = 2.291274e+05	FES = 79100	Diversity = 1.7006e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.290498e+05	FES = 80100	Diversity = 1.6039e+01	k_t
= [4, 3, 3, 2]				
Generation 810:	Best = 2.283010e+05	FES = 81100	Diversity = 1.5503e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.274961e+05	FES = 82100	Diversity = 1.4659e+01	k_t
= [2, 2, 2, 2]				
Generation 830:	Best = 2.274866e+05	FES = 83100	Diversity = 1.4044e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.270909e+05	FES = 84100	Diversity = 1.4666e+01	k_t
= [3, 2, 3, 2]				
Generation 850:	Best = 2.267885e+05	FES = 85100	Diversity = 1.4779e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.262461e+05	FES = 86100	Diversity = 1.3694e+01	k_t
= [3, 2, 3, 2]				
Generation 870:	Best = 2.258571e+05	FES = 87100	Diversity = 1.3751e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.252549e+05	FES = 88100	Diversity = 1.3074e+01	k_t
= [4, 4, 3, 2]				
Generation 890:	Best = 2.252292e+05	FES = 89100	Diversity = 1.4210e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.236201e+05	FES = 90100	Diversity = 1.4662e+01	k_t
= [4, 3, 3, 4]				
Generation 910:	Best = 2.234265e+05	FES = 91100	Diversity = 1.5877e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.227497e+05	FES = 92100	Diversity = 1.6276e+01	k_t
= [2, 3, 2, 2]				
Generation 930:	Best = 2.225262e+05	FES = 93100	Diversity = 1.6896e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.219070e+05	FES = 94100	Diversity = 1.6567e+01	k_t
= [2, 4, 2, 4]				
Generation 950:	Best = 2.214332e+05	FES = 95100	Diversity = 1.6047e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.199639e+05	FES = 96100	Diversity = 1.7505e+01	k_t
= [3, 4, 2, 3]				
Generation 970:	Best = 2.196724e+05	FES = 97100	Diversity = 1.9398e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.191805e+05	FES = 98100	Diversity = 1.9072e+01	k_t
= [3, 3, 2, 3]				
Generation 990:	Best = 2.185810e+05	FES = 99100	Diversity = 1.9044e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 2.180936e+05	FES = 100100	Diversity = 1.8924e+01	k_t
= [4, 3, 4, 4]				
Generation 1010:	Best = 2.174669e+05	FES = 101100	Diversity = 1.9675e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 2.165589e+05	FES = 102100	Diversity = 1.9693e+01	k_t
= [4, 3, 4, 4]				
Generation 1030:	Best = 2.158186e+05	FES = 103100	Diversity = 1.9093e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 2.150625e+05	FES = 104100	Diversity = 1.7485e+01	k_t
= [2, 4, 3, 2]			
Generation 1050: Best = 2.145253e+05	FES = 105100	Diversity = 1.7825e+01	k_t
= [1, 1, 1, 1]			
Generation 1060: Best = 2.143078e+05	FES = 106100	Diversity = 1.7022e+01	k_t
= [4, 3, 2, 3]			
Generation 1070: Best = 2.135859e+05	FES = 107100	Diversity = 1.6494e+01	k_t
= [1, 1, 1, 1]			
Generation 1080: Best = 2.129461e+05	FES = 108100	Diversity = 1.7145e+01	k_t
= [3, 2, 4, 2]			
Generation 1090: Best = 2.124143e+05	FES = 109100	Diversity = 1.8276e+01	k_t
= [1, 1, 1, 1]			
Generation 1100: Best = 2.121838e+05	FES = 110100	Diversity = 1.7617e+01	k_t
= [4, 3, 4, 4]			
Generation 1110: Best = 2.115265e+05	FES = 111100	Diversity = 1.6265e+01	k_t
= [1, 1, 1, 1]			
Generation 1120: Best = 2.109703e+05	FES = 112100	Diversity = 1.6557e+01	k_t
= [3, 4, 2, 3]			
Generation 1130: Best = 2.098829e+05	FES = 113100	Diversity = 1.6739e+01	k_t
= [1, 1, 1, 1]			
Generation 1140: Best = 2.092207e+05	FES = 114100	Diversity = 1.8507e+01	k_t
= [2, 3, 2, 2]			
Generation 1150: Best = 2.091469e+05	FES = 115100	Diversity = 1.8680e+01	k_t
= [1, 1, 1, 1]			
Generation 1160: Best = 2.085283e+05	FES = 116100	Diversity = 1.8364e+01	k_t
= [4, 2, 3, 3]			
Generation 1170: Best = 2.082364e+05	FES = 117100	Diversity = 1.6375e+01	k_t
= [1, 1, 1, 1]			
Generation 1180: Best = 2.075237e+05	FES = 118100	Diversity = 1.7373e+01	k_t
= [2, 3, 3, 3]			
Generation 1190: Best = 2.068831e+05	FES = 119100	Diversity = 1.7931e+01	k_t
= [1, 1, 1, 1]			
Generation 1200: Best = 2.064815e+05	FES = 120100	Diversity = 1.7941e+01	k_t
= [3, 2, 3, 3]			
Generation 1210: Best = 2.057544e+05	FES = 121100	Diversity = 1.8442e+01	k_t
= [1, 1, 1, 1]			
Generation 1220: Best = 2.051933e+05	FES = 122100	Diversity = 1.8970e+01	k_t
= [4, 2, 2, 3]			
Generation 1230: Best = 2.044989e+05	FES = 123100	Diversity = 1.8891e+01	k_t
= [1, 1, 1, 1]			
Generation 1240: Best = 2.041639e+05	FES = 124100	Diversity = 1.9104e+01	k_t
= [2, 2, 2, 4]			
Generation 1250: Best = 2.036964e+05	FES = 125100	Diversity = 1.8249e+01	k_t
= [1, 1, 1, 1]			
Generation 1260: Best = 2.035049e+05	FES = 126100	Diversity = 1.7689e+01	k_t
= [4, 3, 4, 3]			
Generation 1270: Best = 2.033849e+05	FES = 127100	Diversity = 1.7025e+01	k_t
= [1, 1, 1, 1]			
Generation 1280: Best = 2.030678e+05	FES = 128100	Diversity = 1.7051e+01	k_t
= [4, 3, 4, 2]			
Generation 1290: Best = 2.027762e+05	FES = 129100	Diversity = 1.6507e+01	k_t
= [1, 1, 1, 1]			
Generation 1300: Best = 2.024752e+05	FES = 130100	Diversity = 1.6215e+01	k_t
= [3, 2, 3, 3]			

Generation 1310: Best = 2.022919e+05 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.6349e+01	k_t
Generation 1320: Best = 2.019900e+05 = [3, 4, 3, 4]	FES = 132100	Diversity = 1.6383e+01	k_t
Generation 1330: Best = 2.018301e+05 = [1, 1, 1, 1]	FES = 133100	Diversity = 1.6788e+01	k_t
Generation 1340: Best = 2.015437e+05 = [3, 4, 4, 3]	FES = 134100	Diversity = 1.6625e+01	k_t
Generation 1350: Best = 2.012937e+05 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.7664e+01	k_t
Generation 1360: Best = 2.010291e+05 = [4, 4, 4, 3]	FES = 136100	Diversity = 1.7094e+01	k_t
Generation 1370: Best = 2.009804e+05 = [1, 1, 1, 1]	FES = 137100	Diversity = 1.7968e+01	k_t
Generation 1380: Best = 2.005444e+05 = [3, 4, 3, 2]	FES = 138100	Diversity = 1.7498e+01	k_t
Generation 1390: Best = 2.001064e+05 = [1, 1, 1, 1]	FES = 139100	Diversity = 1.8552e+01	k_t
Generation 1400: Best = 2.000632e+05 = [3, 2, 2, 2]	FES = 140100	Diversity = 1.8900e+01	k_t
Generation 1410: Best = 1.998571e+05 = [1, 1, 1, 1]	FES = 141100	Diversity = 1.8721e+01	k_t
Generation 1420: Best = 1.987486e+05 = [2, 2, 2, 4]	FES = 142100	Diversity = 2.0110e+01	k_t
Generation 1430: Best = 1.978505e+05 = [1, 1, 1, 1]	FES = 143100	Diversity = 2.3231e+01	k_t
Generation 1440: Best = 1.976905e+05 = [4, 2, 3, 3]	FES = 144100	Diversity = 2.4533e+01	k_t
Generation 1450: Best = 1.966453e+05 = [1, 1, 1, 1]	FES = 145100	Diversity = 2.6091e+01	k_t
Generation 1460: Best = 1.959110e+05 = [4, 3, 2, 2]	FES = 146100	Diversity = 2.4618e+01	k_t
Generation 1470: Best = 1.955737e+05 = [1, 1, 1, 1]	FES = 147100	Diversity = 2.3074e+01	k_t
Generation 1480: Best = 1.944470e+05 = [4, 3, 3, 3]	FES = 148100	Diversity = 2.2422e+01	k_t
Generation 1490: Best = 1.935855e+05 = [1, 1, 1, 1]	FES = 149100	Diversity = 2.1874e+01	k_t
Generation 1500: Best = 1.927301e+05 = [4, 3, 3, 3]	FES = 150100	Diversity = 2.1636e+01	k_t
Generation 1510: Best = 1.920799e+05 = [1, 1, 1, 1]	FES = 151100	Diversity = 2.2356e+01	k_t
Generation 1520: Best = 1.911376e+05 = [3, 4, 3, 3]	FES = 152100	Diversity = 2.2018e+01	k_t
Generation 1530: Best = 1.910465e+05 = [1, 1, 1, 1]	FES = 153100	Diversity = 2.0691e+01	k_t
Generation 1540: Best = 1.893949e+05 = [3, 3, 2, 3]	FES = 154100	Diversity = 2.0994e+01	k_t
Generation 1550: Best = 1.893949e+05 = [1, 1, 1, 1]	FES = 155100	Diversity = 2.2214e+01	k_t
Generation 1560: Best = 1.890818e+05 = [2, 3, 4, 2]	FES = 156100	Diversity = 2.1821e+01	k_t
Generation 1570: Best = 1.890488e+05 = [1, 1, 1, 1]	FES = 157100	Diversity = 2.0453e+01	k_t

Generation 1580: Best = 1.885459e+05 = [2, 3, 3, 2]	FES = 158100	Diversity = 1.9764e+01	k_t
Generation 1590: Best = 1.877488e+05 = [1, 1, 1, 1]	FES = 159100	Diversity = 2.0785e+01	k_t
Generation 1600: Best = 1.870563e+05 = [4, 3, 4, 2]	FES = 160100	Diversity = 2.1521e+01	k_t
Generation 1610: Best = 1.866276e+05 = [1, 1, 1, 1]	FES = 161100	Diversity = 2.0603e+01	k_t
Generation 1620: Best = 1.862061e+05 = [3, 2, 4, 4]	FES = 162100	Diversity = 1.7625e+01	k_t
Generation 1630: Best = 1.857752e+05 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.7546e+01	k_t
Generation 1640: Best = 1.854756e+05 = [3, 2, 2, 3]	FES = 164100	Diversity = 1.6597e+01	k_t
Generation 1650: Best = 1.849521e+05 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.5823e+01	k_t
Generation 1660: Best = 1.846732e+05 = [4, 2, 2, 3]	FES = 166100	Diversity = 1.6358e+01	k_t
Generation 1670: Best = 1.842340e+05 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.4589e+01	k_t
Generation 1680: Best = 1.837784e+05 = [3, 3, 2, 2]	FES = 168100	Diversity = 1.4716e+01	k_t
Generation 1690: Best = 1.835380e+05 = [1, 1, 1, 1]	FES = 169100	Diversity = 1.3889e+01	k_t
Generation 1700: Best = 1.824661e+05 = [4, 3, 4, 2]	FES = 170100	Diversity = 1.4018e+01	k_t
Generation 1710: Best = 1.824661e+05 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.4037e+01	k_t
Generation 1720: Best = 1.823569e+05 = [2, 4, 3, 4]	FES = 172100	Diversity = 1.3714e+01	k_t
Generation 1730: Best = 1.821017e+05 = [1, 1, 1, 1]	FES = 173100	Diversity = 1.3100e+01	k_t
Generation 1740: Best = 1.817853e+05 = [3, 4, 3, 2]	FES = 174100	Diversity = 1.3387e+01	k_t
Generation 1750: Best = 1.816885e+05 = [1, 1, 1, 1]	FES = 175100	Diversity = 1.1958e+01	k_t
Generation 1760: Best = 1.810717e+05 = [2, 3, 3, 2]	FES = 176100	Diversity = 1.1992e+01	k_t
Generation 1770: Best = 1.810608e+05 = [1, 1, 1, 1]	FES = 177100	Diversity = 1.2878e+01	k_t
Generation 1780: Best = 1.803384e+05 = [3, 3, 3, 3]	FES = 178100	Diversity = 1.2867e+01	k_t
Generation 1790: Best = 1.803384e+05 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.3353e+01	k_t
Generation 1800: Best = 1.801463e+05 = [4, 2, 4, 2]	FES = 180100	Diversity = 1.3213e+01	k_t
Generation 1810: Best = 1.798722e+05 = [1, 1, 1, 1]	FES = 181100	Diversity = 1.2756e+01	k_t
Generation 1820: Best = 1.797501e+05 = [2, 4, 3, 4]	FES = 182100	Diversity = 1.2236e+01	k_t
Generation 1830: Best = 1.794737e+05 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.1608e+01	k_t
Generation 1840: Best = 1.794330e+05 = [3, 2, 4, 4]	FES = 184100	Diversity = 1.1337e+01	k_t

Generation 1850: Best = 1.792433e+05 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.1098e+01	k_t
Generation 1860: Best = 1.790485e+05 = [2, 3, 3, 2]	FES = 186100	Diversity = 1.0926e+01	k_t
Generation 1870: Best = 1.789846e+05 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.0766e+01	k_t
Generation 1880: Best = 1.785839e+05 = [3, 4, 2, 2]	FES = 188100	Diversity = 1.0953e+01	k_t
Generation 1890: Best = 1.782868e+05 = [1, 1, 1, 1]	FES = 189100	Diversity = 1.2496e+01	k_t
Generation 1900: Best = 1.778238e+05 = [4, 2, 3, 3]	FES = 190100	Diversity = 1.3556e+01	k_t
Generation 1910: Best = 1.777346e+05 = [1, 1, 1, 1]	FES = 191100	Diversity = 1.3870e+01	k_t
Generation 1920: Best = 1.772582e+05 = [3, 4, 4, 3]	FES = 192100	Diversity = 1.3201e+01	k_t
Generation 1930: Best = 1.760968e+05 = [1, 1, 1, 1]	FES = 193100	Diversity = 1.4393e+01	k_t
Generation 1940: Best = 1.751676e+05 = [4, 4, 2, 3]	FES = 194100	Diversity = 1.5045e+01	k_t
Generation 1950: Best = 1.749157e+05 = [1, 1, 1, 1]	FES = 195100	Diversity = 1.7692e+01	k_t
Generation 1960: Best = 1.741260e+05 = [2, 3, 2, 4]	FES = 196100	Diversity = 1.8364e+01	k_t
Generation 1970: Best = 1.735043e+05 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.8939e+01	k_t
Generation 1980: Best = 1.730475e+05 = [4, 2, 4, 2]	FES = 198100	Diversity = 1.7913e+01	k_t
Generation 1990: Best = 1.714321e+05 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.7671e+01	k_t
Generation 2000: Best = 1.714208e+05 = [3, 4, 3, 3]	FES = 200100	Diversity = 1.8095e+01	k_t
Generation 2010: Best = 1.706187e+05 = [1, 1, 1, 1]	FES = 201100	Diversity = 1.8336e+01	k_t
Generation 2020: Best = 1.695195e+05 = [4, 4, 3, 4]	FES = 202100	Diversity = 1.8294e+01	k_t
Generation 2030: Best = 1.692952e+05 = [1, 1, 1, 1]	FES = 203100	Diversity = 1.7011e+01	k_t
Generation 2040: Best = 1.687704e+05 = [4, 3, 4, 2]	FES = 204100	Diversity = 1.7243e+01	k_t
Generation 2050: Best = 1.683827e+05 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.6299e+01	k_t
Generation 2060: Best = 1.679897e+05 = [2, 2, 3, 4]	FES = 206100	Diversity = 1.5729e+01	k_t
Generation 2070: Best = 1.678112e+05 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.4510e+01	k_t
Generation 2080: Best = 1.676337e+05 = [4, 2, 3, 3]	FES = 208100	Diversity = 1.3554e+01	k_t
Generation 2090: Best = 1.674867e+05 = [1, 1, 1, 1]	FES = 209100	Diversity = 1.2845e+01	k_t
Generation 2100: Best = 1.673796e+05 = [2, 4, 2, 3]	FES = 210100	Diversity = 1.1959e+01	k_t
Generation 2110: Best = 1.672437e+05 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.1157e+01	k_t

Generation 2120: Best = 1.669138e+05 = [3, 2, 3, 3]	FES = 212100	Diversity = 1.1158e+01	k_t
Generation 2130: Best = 1.667860e+05 = [1, 1, 1, 1]	FES = 213100	Diversity = 1.1965e+01	k_t
Generation 2140: Best = 1.666728e+05 = [2, 2, 2, 4]	FES = 214100	Diversity = 1.1657e+01	k_t
Generation 2150: Best = 1.665715e+05 = [1, 1, 1, 1]	FES = 215100	Diversity = 1.0612e+01	k_t
Generation 2160: Best = 1.664138e+05 = [2, 3, 4, 3]	FES = 216100	Diversity = 9.8546e+00	k_t
Generation 2170: Best = 1.664135e+05 = [1, 1, 1, 1]	FES = 217100	Diversity = 1.0368e+01	k_t
Generation 2180: Best = 1.662096e+05 = [2, 2, 2, 3]	FES = 218100	Diversity = 8.9849e+00	k_t
Generation 2190: Best = 1.661747e+05 = [1, 1, 1, 1]	FES = 219100	Diversity = 9.6954e+00	k_t
Generation 2200: Best = 1.661379e+05 = [4, 3, 3, 3]	FES = 220100	Diversity = 1.0811e+01	k_t
Generation 2210: Best = 1.660326e+05 = [1, 1, 1, 1]	FES = 221100	Diversity = 1.0733e+01	k_t
Generation 2220: Best = 1.656818e+05 = [2, 4, 4, 4]	FES = 222100	Diversity = 1.1620e+01	k_t
Generation 2230: Best = 1.654691e+05 = [1, 1, 1, 1]	FES = 223100	Diversity = 1.3695e+01	k_t
Generation 2240: Best = 1.651685e+05 = [3, 4, 4, 2]	FES = 224100	Diversity = 1.3732e+01	k_t
Generation 2250: Best = 1.646770e+05 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.5992e+01	k_t
Generation 2260: Best = 1.643205e+05 = [3, 2, 2, 4]	FES = 226100	Diversity = 1.6323e+01	k_t
Generation 2270: Best = 1.642187e+05 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.7492e+01	k_t
Generation 2280: Best = 1.637321e+05 = [4, 2, 4, 3]	FES = 228100	Diversity = 1.7632e+01	k_t
Generation 2290: Best = 1.633959e+05 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.8961e+01	k_t
Generation 2300: Best = 1.631574e+05 = [3, 3, 3, 3]	FES = 230100	Diversity = 1.9509e+01	k_t
Generation 2310: Best = 1.627827e+05 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.8843e+01	k_t
Generation 2320: Best = 1.626343e+05 = [2, 2, 4, 3]	FES = 232100	Diversity = 1.8823e+01	k_t
Generation 2330: Best = 1.623864e+05 = [1, 1, 1, 1]	FES = 233100	Diversity = 2.0349e+01	k_t
Generation 2340: Best = 1.621704e+05 = [3, 2, 2, 3]	FES = 234100	Diversity = 1.8487e+01	k_t
Generation 2350: Best = 1.621111e+05 = [1, 1, 1, 1]	FES = 235100	Diversity = 1.7410e+01	k_t
Generation 2360: Best = 1.618261e+05 = [3, 2, 3, 3]	FES = 236100	Diversity = 1.7409e+01	k_t
Generation 2370: Best = 1.615023e+05 = [1, 1, 1, 1]	FES = 237100	Diversity = 1.8422e+01	k_t
Generation 2380: Best = 1.613883e+05 = [3, 4, 3, 4]	FES = 238100	Diversity = 1.6785e+01	k_t

Generation 2390: Best = 1.613345e+05 = [1, 1, 1, 1]	FES = 239100	Diversity = 1.5171e+01	k_t
Generation 2400: Best = 1.612264e+05 = [3, 3, 4, 3]	FES = 240100	Diversity = 1.4613e+01	k_t
Generation 2410: Best = 1.611201e+05 = [1, 1, 1, 1]	FES = 241100	Diversity = 1.3808e+01	k_t
Generation 2420: Best = 1.611201e+05 = [2, 2, 4, 2]	FES = 242100	Diversity = 1.3093e+01	k_t
Generation 2430: Best = 1.609377e+05 = [1, 1, 1, 1]	FES = 243100	Diversity = 1.4016e+01	k_t
Generation 2440: Best = 1.608141e+05 = [3, 4, 2, 3]	FES = 244100	Diversity = 1.4579e+01	k_t
Generation 2450: Best = 1.607258e+05 = [1, 1, 1, 1]	FES = 245100	Diversity = 1.3955e+01	k_t
Generation 2460: Best = 1.603557e+05 = [2, 2, 4, 2]	FES = 246100	Diversity = 1.6345e+01	k_t
Generation 2470: Best = 1.602687e+05 = [1, 1, 1, 1]	FES = 247100	Diversity = 1.7731e+01	k_t
Generation 2480: Best = 1.600876e+05 = [3, 4, 3, 2]	FES = 248100	Diversity = 1.7724e+01	k_t
Generation 2490: Best = 1.599595e+05 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.8791e+01	k_t
Generation 2500: Best = 1.593497e+05 = [4, 4, 4, 3]	FES = 250100	Diversity = 1.9891e+01	k_t
Generation 2510: Best = 1.589949e+05 = [1, 1, 1, 1]	FES = 251100	Diversity = 2.0209e+01	k_t
Generation 2520: Best = 1.587079e+05 = [2, 3, 3, 4]	FES = 252100	Diversity = 1.9568e+01	k_t
Generation 2530: Best = 1.580540e+05 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.9559e+01	k_t
Generation 2540: Best = 1.577282e+05 = [4, 4, 2, 3]	FES = 254100	Diversity = 2.0503e+01	k_t
Generation 2550: Best = 1.575588e+05 = [1, 1, 1, 1]	FES = 255100	Diversity = 2.0107e+01	k_t
Generation 2560: Best = 1.574174e+05 = [3, 3, 3, 3]	FES = 256100	Diversity = 1.8613e+01	k_t
Generation 2570: Best = 1.566513e+05 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.7380e+01	k_t
Generation 2580: Best = 1.566513e+05 = [4, 2, 3, 2]	FES = 258100	Diversity = 1.7369e+01	k_t
Generation 2590: Best = 1.561036e+05 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.7146e+01	k_t
Generation 2600: Best = 1.560205e+05 = [3, 4, 3, 4]	FES = 260100	Diversity = 1.7434e+01	k_t
Generation 2610: Best = 1.557557e+05 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.7344e+01	k_t
Generation 2620: Best = 1.552837e+05 = [4, 4, 3, 3]	FES = 262100	Diversity = 1.5904e+01	k_t
Generation 2630: Best = 1.552837e+05 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.4473e+01	k_t
Generation 2640: Best = 1.551549e+05 = [4, 3, 2, 3]	FES = 264100	Diversity = 1.4336e+01	k_t
Generation 2650: Best = 1.547934e+05 = [1, 1, 1, 1]	FES = 265100	Diversity = 1.5259e+01	k_t

Generation 2660: Best = 1.547362e+05 = [2, 3, 2, 3]	FES = 266100	Diversity = 1.5778e+01	k_t
Generation 2670: Best = 1.544466e+05 = [1, 1, 1, 1]	FES = 267100	Diversity = 1.5999e+01	k_t
Generation 2680: Best = 1.539901e+05 = [4, 4, 4, 2]	FES = 268100	Diversity = 1.6386e+01	k_t
Generation 2690: Best = 1.538863e+05 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.8284e+01	k_t
Generation 2700: Best = 1.531749e+05 = [2, 2, 2, 2]	FES = 270100	Diversity = 1.8302e+01	k_t
Generation 2710: Best = 1.525441e+05 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.8844e+01	k_t
Generation 2720: Best = 1.522524e+05 = [3, 3, 3, 3]	FES = 272100	Diversity = 1.9631e+01	k_t
Generation 2730: Best = 1.519815e+05 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.7790e+01	k_t
Generation 2740: Best = 1.516140e+05 = [2, 4, 2, 3]	FES = 274100	Diversity = 1.7113e+01	k_t
Generation 2750: Best = 1.513562e+05 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.7795e+01	k_t
Generation 2760: Best = 1.509699e+05 = [3, 4, 3, 4]	FES = 276100	Diversity = 1.9611e+01	k_t
Generation 2770: Best = 1.506164e+05 = [1, 1, 1, 1]	FES = 277100	Diversity = 2.0247e+01	k_t
Generation 2780: Best = 1.494587e+05 = [2, 2, 3, 3]	FES = 278100	Diversity = 2.0358e+01	k_t
Generation 2790: Best = 1.494248e+05 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.9059e+01	k_t
Generation 2800: Best = 1.491807e+05 = [4, 2, 2, 2]	FES = 280100	Diversity = 1.8417e+01	k_t
Generation 2810: Best = 1.489766e+05 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.8084e+01	k_t
Generation 2820: Best = 1.487007e+05 = [4, 3, 4, 4]	FES = 282100	Diversity = 1.6367e+01	k_t
Generation 2830: Best = 1.485730e+05 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.6198e+01	k_t
Generation 2840: Best = 1.481722e+05 = [2, 4, 2, 2]	FES = 284100	Diversity = 1.5812e+01	k_t
Generation 2850: Best = 1.476666e+05 = [1, 1, 1, 1]	FES = 285100	Diversity = 1.8057e+01	k_t
Generation 2860: Best = 1.472487e+05 = [2, 3, 2, 2]	FES = 286100	Diversity = 1.9258e+01	k_t
Generation 2870: Best = 1.471403e+05 = [1, 1, 1, 1]	FES = 287100	Diversity = 1.8146e+01	k_t
Generation 2880: Best = 1.466587e+05 = [3, 4, 4, 4]	FES = 288100	Diversity = 1.7817e+01	k_t
Generation 2890: Best = 1.465388e+05 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.7553e+01	k_t
Generation 2900: Best = 1.462915e+05 = [3, 4, 4, 4]	FES = 290100	Diversity = 1.7881e+01	k_t
Generation 2910: Best = 1.460397e+05 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.7294e+01	k_t
Generation 2920: Best = 1.458052e+05 = [2, 4, 2, 4]	FES = 292100	Diversity = 1.6172e+01	k_t

Generation 2930: Best = 1.456019e+05 = [1, 1, 1, 1]	FES = 293100	Diversity = 1.5099e+01	k_t
Generation 2940: Best = 1.453341e+05 = [2, 3, 3, 3]	FES = 294100	Diversity = 1.5006e+01	k_t
Generation 2950: Best = 1.452217e+05 = [1, 1, 1, 1]	FES = 295100	Diversity = 1.4027e+01	k_t
Generation 2960: Best = 1.449908e+05 = [3, 3, 2, 2]	FES = 296100	Diversity = 1.3206e+01	k_t
Generation 2970: Best = 1.447430e+05 = [1, 1, 1, 1]	FES = 297100	Diversity = 1.2422e+01	k_t
Generation 2980: Best = 1.446082e+05 = [4, 3, 3, 3]	FES = 298100	Diversity = 1.3014e+01	k_t
Generation 2990: Best = 1.443567e+05 = [1, 1, 1, 1]	FES = 299100	Diversity = 1.2664e+01	k_t
Generation 3000: Best = 1.443203e+05 = [3, 2, 4, 3]	FES = 300100	Diversity = 1.2235e+01	k_t
Generation 3010: Best = 1.439704e+05 = [1, 1, 1, 1]	FES = 301100	Diversity = 1.1746e+01	k_t
Generation 3020: Best = 1.439346e+05 = [4, 2, 3, 4]	FES = 302100	Diversity = 1.2094e+01	k_t
Generation 3030: Best = 1.435600e+05 = [1, 1, 1, 1]	FES = 303100	Diversity = 1.3272e+01	k_t
Generation 3040: Best = 1.433224e+05 = [2, 3, 4, 2]	FES = 304100	Diversity = 1.3777e+01	k_t
Generation 3050: Best = 1.431187e+05 = [1, 1, 1, 1]	FES = 305100	Diversity = 1.4744e+01	k_t
Generation 3060: Best = 1.430170e+05 = [4, 4, 3, 3]	FES = 306100	Diversity = 1.5480e+01	k_t
Generation 3070: Best = 1.427928e+05 = [1, 1, 1, 1]	FES = 307100	Diversity = 1.4709e+01	k_t
Generation 3080: Best = 1.426224e+05 = [3, 4, 3, 3]	FES = 308100	Diversity = 1.4104e+01	k_t
Generation 3090: Best = 1.425138e+05 = [1, 1, 1, 1]	FES = 309100	Diversity = 1.3666e+01	k_t
Generation 3100: Best = 1.424240e+05 = [2, 2, 3, 3]	FES = 310100	Diversity = 1.3716e+01	k_t
Generation 3110: Best = 1.423128e+05 = [1, 1, 1, 1]	FES = 311100	Diversity = 1.3994e+01	k_t
Generation 3120: Best = 1.421743e+05 = [2, 4, 4, 3]	FES = 312100	Diversity = 1.5905e+01	k_t
Generation 3130: Best = 1.420784e+05 = [1, 1, 1, 1]	FES = 313100	Diversity = 1.5827e+01	k_t
Generation 3140: Best = 1.418408e+05 = [2, 3, 3, 3]	FES = 314100	Diversity = 1.4582e+01	k_t
Generation 3150: Best = 1.416371e+05 = [1, 1, 1, 1]	FES = 315100	Diversity = 1.4207e+01	k_t
Generation 3160: Best = 1.416226e+05 = [3, 4, 3, 3]	FES = 316100	Diversity = 1.5113e+01	k_t
Generation 3170: Best = 1.413278e+05 = [1, 1, 1, 1]	FES = 317100	Diversity = 1.6339e+01	k_t
Generation 3180: Best = 1.410402e+05 = [4, 4, 4, 2]	FES = 318100	Diversity = 1.6555e+01	k_t
Generation 3190: Best = 1.409210e+05 = [1, 1, 1, 1]	FES = 319100	Diversity = 1.6593e+01	k_t

Generation 3200: Best = 1.403552e+05 = [3, 4, 3, 2]	FES = 320100	Diversity = 1.6979e+01	k_t
Generation 3210: Best = 1.399597e+05 = [1, 1, 1, 1]	FES = 321100	Diversity = 1.9512e+01	k_t
Generation 3220: Best = 1.398086e+05 = [4, 4, 3, 2]	FES = 322100	Diversity = 1.8680e+01	k_t
Generation 3230: Best = 1.393168e+05 = [1, 1, 1, 1]	FES = 323100	Diversity = 2.0076e+01	k_t
Generation 3240: Best = 1.388972e+05 = [4, 2, 3, 3]	FES = 324100	Diversity = 2.0936e+01	k_t
Generation 3250: Best = 1.387608e+05 = [1, 1, 1, 1]	FES = 325100	Diversity = 2.1941e+01	k_t
Generation 3260: Best = 1.382035e+05 = [2, 2, 4, 3]	FES = 326100	Diversity = 2.2756e+01	k_t
Generation 3270: Best = 1.372879e+05 = [1, 1, 1, 1]	FES = 327100	Diversity = 2.3123e+01	k_t
Generation 3280: Best = 1.368353e+05 = [4, 2, 2, 4]	FES = 328100	Diversity = 2.2958e+01	k_t
Generation 3290: Best = 1.360565e+05 = [1, 1, 1, 1]	FES = 329100	Diversity = 2.3502e+01	k_t
Generation 3300: Best = 1.360409e+05 = [2, 3, 3, 4]	FES = 330100	Diversity = 2.3362e+01	k_t
Generation 3310: Best = 1.353130e+05 = [1, 1, 1, 1]	FES = 331100	Diversity = 2.4269e+01	k_t
Generation 3320: Best = 1.348131e+05 = [3, 3, 2, 3]	FES = 332100	Diversity = 2.2972e+01	k_t
Generation 3330: Best = 1.342419e+05 = [1, 1, 1, 1]	FES = 333100	Diversity = 2.2927e+01	k_t
Generation 3340: Best = 1.342015e+05 = [4, 3, 2, 4]	FES = 334100	Diversity = 2.1867e+01	k_t
Generation 3350: Best = 1.334832e+05 = [1, 1, 1, 1]	FES = 335100	Diversity = 2.0814e+01	k_t
Generation 3360: Best = 1.333577e+05 = [2, 4, 3, 2]	FES = 336100	Diversity = 2.0940e+01	k_t
Generation 3370: Best = 1.325507e+05 = [1, 1, 1, 1]	FES = 337100	Diversity = 2.0067e+01	k_t
Generation 3380: Best = 1.323088e+05 = [2, 2, 3, 2]	FES = 338100	Diversity = 1.8780e+01	k_t
Generation 3390: Best = 1.320611e+05 = [1, 1, 1, 1]	FES = 339100	Diversity = 1.7225e+01	k_t
Generation 3400: Best = 1.317035e+05 = [3, 2, 2, 4]	FES = 340100	Diversity = 1.5486e+01	k_t
Generation 3410: Best = 1.314941e+05 = [1, 1, 1, 1]	FES = 341100	Diversity = 1.5619e+01	k_t
Generation 3420: Best = 1.313039e+05 = [3, 4, 3, 2]	FES = 342100	Diversity = 1.5304e+01	k_t
Generation 3430: Best = 1.312384e+05 = [1, 1, 1, 1]	FES = 343100	Diversity = 1.3794e+01	k_t
Generation 3440: Best = 1.309525e+05 = [2, 4, 3, 4]	FES = 344100	Diversity = 1.3734e+01	k_t
Generation 3450: Best = 1.306435e+05 = [1, 1, 1, 1]	FES = 345100	Diversity = 1.4116e+01	k_t
Generation 3460: Best = 1.305413e+05 = [2, 4, 3, 3]	FES = 346100	Diversity = 1.3684e+01	k_t

Generation 3470: Best = 1.302931e+05 = [1, 1, 1, 1]	FES = 347100	Diversity = 1.4014e+01	k_t
Generation 3480: Best = 1.301954e+05 = [4, 4, 2, 4]	FES = 348100	Diversity = 1.3592e+01	k_t
Generation 3490: Best = 1.301059e+05 = [1, 1, 1, 1]	FES = 349100	Diversity = 1.3047e+01	k_t
Generation 3500: Best = 1.299674e+05 = [3, 4, 2, 4]	FES = 350100	Diversity = 1.3258e+01	k_t
Generation 3510: Best = 1.298310e+05 = [1, 1, 1, 1]	FES = 351100	Diversity = 1.3726e+01	k_t
Generation 3520: Best = 1.296398e+05 = [3, 2, 4, 3]	FES = 352100	Diversity = 1.5025e+01	k_t
Generation 3530: Best = 1.293851e+05 = [1, 1, 1, 1]	FES = 353100	Diversity = 1.4623e+01	k_t
Generation 3540: Best = 1.292139e+05 = [4, 2, 2, 2]	FES = 354100	Diversity = 1.4833e+01	k_t
Generation 3550: Best = 1.291841e+05 = [1, 1, 1, 1]	FES = 355100	Diversity = 1.3812e+01	k_t
Generation 3560: Best = 1.290426e+05 = [3, 3, 3, 3]	FES = 356100	Diversity = 1.3844e+01	k_t
Generation 3570: Best = 1.289555e+05 = [1, 1, 1, 1]	FES = 357100	Diversity = 1.2061e+01	k_t
Generation 3580: Best = 1.288674e+05 = [4, 4, 3, 4]	FES = 358100	Diversity = 1.1004e+01	k_t
Generation 3590: Best = 1.287691e+05 = [1, 1, 1, 1]	FES = 359100	Diversity = 1.0937e+01	k_t
Generation 3600: Best = 1.286037e+05 = [4, 4, 2, 4]	FES = 360100	Diversity = 9.9319e+00	k_t
Generation 3610: Best = 1.284862e+05 = [1, 1, 1, 1]	FES = 361100	Diversity = 1.1052e+01	k_t
Generation 3620: Best = 1.284566e+05 = [3, 4, 2, 4]	FES = 362100	Diversity = 1.1135e+01	k_t
Generation 3630: Best = 1.283841e+05 = [1, 1, 1, 1]	FES = 363100	Diversity = 1.0146e+01	k_t
Generation 3640: Best = 1.283193e+05 = [3, 4, 3, 3]	FES = 364100	Diversity = 1.0097e+01	k_t
Generation 3650: Best = 1.282159e+05 = [1, 1, 1, 1]	FES = 365100	Diversity = 1.1225e+01	k_t
Generation 3660: Best = 1.281640e+05 = [2, 2, 4, 4]	FES = 366100	Diversity = 1.1808e+01	k_t
Generation 3670: Best = 1.280373e+05 = [1, 1, 1, 1]	FES = 367100	Diversity = 1.1718e+01	k_t
Generation 3680: Best = 1.279889e+05 = [4, 3, 4, 2]	FES = 368100	Diversity = 1.1496e+01	k_t
Generation 3690: Best = 1.279687e+05 = [1, 1, 1, 1]	FES = 369100	Diversity = 1.0959e+01	k_t
Generation 3700: Best = 1.279384e+05 = [2, 4, 4, 2]	FES = 370100	Diversity = 1.0321e+01	k_t
Generation 3710: Best = 1.278789e+05 = [1, 1, 1, 1]	FES = 371100	Diversity = 1.0975e+01	k_t
Generation 3720: Best = 1.277907e+05 = [4, 3, 4, 3]	FES = 372100	Diversity = 1.1273e+01	k_t
Generation 3730: Best = 1.277574e+05 = [1, 1, 1, 1]	FES = 373100	Diversity = 1.1315e+01	k_t

Generation 3740: Best = 1.276871e+05 = [3, 4, 4, 3]	FES = 374100	Diversity = 1.1707e+01	k_t
Generation 3750: Best = 1.276518e+05 = [1, 1, 1, 1]	FES = 375100	Diversity = 1.1685e+01	k_t
Generation 3760: Best = 1.275364e+05 = [3, 2, 3, 2]	FES = 376100	Diversity = 1.1650e+01	k_t
Generation 3770: Best = 1.275009e+05 = [1, 1, 1, 1]	FES = 377100	Diversity = 1.1897e+01	k_t
Generation 3780: Best = 1.274422e+05 = [4, 3, 3, 4]	FES = 378100	Diversity = 1.2728e+01	k_t
Generation 3790: Best = 1.273646e+05 = [1, 1, 1, 1]	FES = 379100	Diversity = 1.3015e+01	k_t
Generation 3800: Best = 1.272564e+05 = [2, 3, 2, 2]	FES = 380100	Diversity = 1.2809e+01	k_t
Generation 3810: Best = 1.271778e+05 = [1, 1, 1, 1]	FES = 381100	Diversity = 1.3509e+01	k_t
Generation 3820: Best = 1.270664e+05 = [2, 3, 3, 3]	FES = 382100	Diversity = 1.4418e+01	k_t
Generation 3830: Best = 1.270003e+05 = [1, 1, 1, 1]	FES = 383100	Diversity = 1.4291e+01	k_t
Generation 3840: Best = 1.269240e+05 = [3, 2, 2, 4]	FES = 384100	Diversity = 1.2989e+01	k_t
Generation 3850: Best = 1.268506e+05 = [1, 1, 1, 1]	FES = 385100	Diversity = 1.3343e+01	k_t
Generation 3860: Best = 1.267396e+05 = [4, 4, 2, 3]	FES = 386100	Diversity = 1.2454e+01	k_t
Generation 3870: Best = 1.266778e+05 = [1, 1, 1, 1]	FES = 387100	Diversity = 1.1741e+01	k_t
Generation 3880: Best = 1.266345e+05 = [2, 4, 3, 3]	FES = 388100	Diversity = 1.2695e+01	k_t
Generation 3890: Best = 1.265449e+05 = [1, 1, 1, 1]	FES = 389100	Diversity = 1.2420e+01	k_t
Generation 3900: Best = 1.264473e+05 = [2, 2, 4, 2]	FES = 390100	Diversity = 1.2424e+01	k_t
Generation 3910: Best = 1.264108e+05 = [1, 1, 1, 1]	FES = 391100	Diversity = 1.3454e+01	k_t
Generation 3920: Best = 1.262852e+05 = [4, 4, 4, 2]	FES = 392100	Diversity = 1.3631e+01	k_t
Generation 3930: Best = 1.262728e+05 = [1, 1, 1, 1]	FES = 393100	Diversity = 1.2485e+01	k_t
Generation 3940: Best = 1.262416e+05 = [2, 2, 3, 4]	FES = 394100	Diversity = 1.2038e+01	k_t
Generation 3950: Best = 1.261380e+05 = [1, 1, 1, 1]	FES = 395100	Diversity = 1.1430e+01	k_t
Generation 3960: Best = 1.261297e+05 = [4, 2, 3, 4]	FES = 396100	Diversity = 1.0590e+01	k_t
Generation 3970: Best = 1.260951e+05 = [1, 1, 1, 1]	FES = 397100	Diversity = 1.1201e+01	k_t
Generation 3980: Best = 1.260234e+05 = [3, 4, 3, 2]	FES = 398100	Diversity = 1.1428e+01	k_t
Generation 3990: Best = 1.259703e+05 = [1, 1, 1, 1]	FES = 399100	Diversity = 1.2119e+01	k_t
Generation 4000: Best = 1.259368e+05 = [4, 3, 2, 3]	FES = 400100	Diversity = 1.2292e+01	k_t

Generation 4010: Best = 1.259181e+05 = [1, 1, 1, 1]	FES = 401100	Diversity = 1.2825e+01	k_t
Generation 4020: Best = 1.257683e+05 = [3, 3, 4, 3]	FES = 402100	Diversity = 1.3196e+01	k_t
Generation 4030: Best = 1.256557e+05 = [1, 1, 1, 1]	FES = 403100	Diversity = 1.4748e+01	k_t
Generation 4040: Best = 1.255538e+05 = [2, 4, 3, 2]	FES = 404100	Diversity = 1.5278e+01	k_t
Generation 4050: Best = 1.254468e+05 = [1, 1, 1, 1]	FES = 405100	Diversity = 1.5701e+01	k_t
Generation 4060: Best = 1.253436e+05 = [4, 2, 4, 2]	FES = 406100	Diversity = 1.5151e+01	k_t
Generation 4070: Best = 1.252054e+05 = [1, 1, 1, 1]	FES = 407100	Diversity = 1.4093e+01	k_t
Generation 4080: Best = 1.251024e+05 = [4, 2, 4, 4]	FES = 408100	Diversity = 1.3717e+01	k_t
Generation 4090: Best = 1.250274e+05 = [1, 1, 1, 1]	FES = 409100	Diversity = 1.2876e+01	k_t
Generation 4100: Best = 1.249576e+05 = [2, 2, 4, 4]	FES = 410100	Diversity = 1.2345e+01	k_t
Generation 4110: Best = 1.248711e+05 = [1, 1, 1, 1]	FES = 411100	Diversity = 1.2114e+01	k_t
Generation 4120: Best = 1.247104e+05 = [4, 3, 4, 3]	FES = 412100	Diversity = 1.2668e+01	k_t
Generation 4130: Best = 1.246729e+05 = [1, 1, 1, 1]	FES = 413100	Diversity = 1.3265e+01	k_t
Generation 4140: Best = 1.244745e+05 = [2, 2, 3, 3]	FES = 414100	Diversity = 1.3289e+01	k_t
Generation 4150: Best = 1.243789e+05 = [1, 1, 1, 1]	FES = 415100	Diversity = 1.3799e+01	k_t
Generation 4160: Best = 1.242621e+05 = [2, 4, 3, 3]	FES = 416100	Diversity = 1.3371e+01	k_t
Generation 4170: Best = 1.242169e+05 = [1, 1, 1, 1]	FES = 417100	Diversity = 1.2189e+01	k_t
Generation 4180: Best = 1.241285e+05 = [4, 3, 2, 2]	FES = 418100	Diversity = 1.2362e+01	k_t
Generation 4190: Best = 1.239769e+05 = [1, 1, 1, 1]	FES = 419100	Diversity = 1.2777e+01	k_t
Generation 4200: Best = 1.239469e+05 = [4, 2, 3, 2]	FES = 420100	Diversity = 1.2928e+01	k_t
Generation 4210: Best = 1.238595e+05 = [1, 1, 1, 1]	FES = 421100	Diversity = 1.3925e+01	k_t
Generation 4220: Best = 1.237164e+05 = [4, 3, 3, 3]	FES = 422100	Diversity = 1.3774e+01	k_t
Generation 4230: Best = 1.235615e+05 = [1, 1, 1, 1]	FES = 423100	Diversity = 1.3070e+01	k_t
Generation 4240: Best = 1.235615e+05 = [4, 2, 2, 3]	FES = 424100	Diversity = 1.2066e+01	k_t
Generation 4250: Best = 1.235081e+05 = [1, 1, 1, 1]	FES = 425100	Diversity = 1.1633e+01	k_t
Generation 4260: Best = 1.234098e+05 = [3, 3, 3, 4]	FES = 426100	Diversity = 1.1657e+01	k_t
Generation 4270: Best = 1.234065e+05 = [1, 1, 1, 1]	FES = 427100	Diversity = 1.2285e+01	k_t

Generation 4280: Best = 1.233536e+05 = [4, 2, 4, 2]	FES = 428100	Diversity = 1.2195e+01	k_t
Generation 4290: Best = 1.232762e+05 = [1, 1, 1, 1]	FES = 429100	Diversity = 1.3034e+01	k_t
Generation 4300: Best = 1.232068e+05 = [2, 2, 2, 2]	FES = 430100	Diversity = 1.3093e+01	k_t
Generation 4310: Best = 1.231085e+05 = [1, 1, 1, 1]	FES = 431100	Diversity = 1.3181e+01	k_t
Generation 4320: Best = 1.229676e+05 = [4, 4, 4, 4]	FES = 432100	Diversity = 1.2218e+01	k_t
Generation 4330: Best = 1.228689e+05 = [1, 1, 1, 1]	FES = 433100	Diversity = 1.3630e+01	k_t
Generation 4340: Best = 1.227201e+05 = [3, 4, 2, 2]	FES = 434100	Diversity = 1.5122e+01	k_t
Generation 4350: Best = 1.226221e+05 = [1, 1, 1, 1]	FES = 435100	Diversity = 1.4767e+01	k_t
Generation 4360: Best = 1.225573e+05 = [2, 4, 3, 4]	FES = 436100	Diversity = 1.5296e+01	k_t
Generation 4370: Best = 1.223875e+05 = [1, 1, 1, 1]	FES = 437100	Diversity = 1.4335e+01	k_t
Generation 4380: Best = 1.222211e+05 = [2, 3, 3, 3]	FES = 438100	Diversity = 1.4328e+01	k_t
Generation 4390: Best = 1.221561e+05 = [1, 1, 1, 1]	FES = 439100	Diversity = 1.3258e+01	k_t
Generation 4400: Best = 1.221374e+05 = [2, 4, 4, 2]	FES = 440100	Diversity = 1.2655e+01	k_t
Generation 4410: Best = 1.220240e+05 = [1, 1, 1, 1]	FES = 441100	Diversity = 1.1732e+01	k_t
Generation 4420: Best = 1.219485e+05 = [4, 2, 4, 3]	FES = 442100	Diversity = 1.2345e+01	k_t
Generation 4430: Best = 1.218910e+05 = [1, 1, 1, 1]	FES = 443100	Diversity = 1.2178e+01	k_t
Generation 4440: Best = 1.217914e+05 = [3, 2, 2, 3]	FES = 444100	Diversity = 1.2232e+01	k_t
Generation 4450: Best = 1.216800e+05 = [1, 1, 1, 1]	FES = 445100	Diversity = 1.1999e+01	k_t
Generation 4460: Best = 1.215834e+05 = [2, 2, 3, 3]	FES = 446100	Diversity = 1.1773e+01	k_t
Generation 4470: Best = 1.214999e+05 = [1, 1, 1, 1]	FES = 447100	Diversity = 1.1306e+01	k_t
Generation 4480: Best = 1.213581e+05 = [4, 3, 4, 4]	FES = 448100	Diversity = 1.1283e+01	k_t
Generation 4490: Best = 1.213444e+05 = [1, 1, 1, 1]	FES = 449100	Diversity = 1.0942e+01	k_t
Generation 4500: Best = 1.212921e+05 = [3, 2, 2, 3]	FES = 450100	Diversity = 1.0886e+01	k_t
Generation 4510: Best = 1.212237e+05 = [1, 1, 1, 1]	FES = 451100	Diversity = 1.0466e+01	k_t
Generation 4520: Best = 1.211525e+05 = [4, 4, 3, 3]	FES = 452100	Diversity = 1.0430e+01	k_t
Generation 4530: Best = 1.211294e+05 = [1, 1, 1, 1]	FES = 453100	Diversity = 1.1252e+01	k_t
Generation 4540: Best = 1.210274e+05 = [2, 4, 3, 4]	FES = 454100	Diversity = 1.0903e+01	k_t

Generation 4550: Best = 1.209250e+05 = [1, 1, 1, 1]	FES = 455100	Diversity = 1.0535e+01	k_t
Generation 4560: Best = 1.208785e+05 = [3, 3, 4, 2]	FES = 456100	Diversity = 1.1226e+01	k_t
Generation 4570: Best = 1.207913e+05 = [1, 1, 1, 1]	FES = 457100	Diversity = 1.1237e+01	k_t
Generation 4580: Best = 1.207456e+05 = [3, 2, 3, 2]	FES = 458100	Diversity = 1.2066e+01	k_t
Generation 4590: Best = 1.206848e+05 = [1, 1, 1, 1]	FES = 459100	Diversity = 1.2053e+01	k_t
Generation 4600: Best = 1.206848e+05 = [4, 2, 3, 3]	FES = 460100	Diversity = 1.2269e+01	k_t
Generation 4610: Best = 1.206032e+05 = [1, 1, 1, 1]	FES = 461100	Diversity = 1.1706e+01	k_t
Generation 4620: Best = 1.205241e+05 = [4, 2, 2, 2]	FES = 462100	Diversity = 1.1240e+01	k_t
Generation 4630: Best = 1.204499e+05 = [1, 1, 1, 1]	FES = 463100	Diversity = 1.1230e+01	k_t
Generation 4640: Best = 1.204293e+05 = [2, 4, 4, 4]	FES = 464100	Diversity = 1.1316e+01	k_t
Generation 4650: Best = 1.203292e+05 = [1, 1, 1, 1]	FES = 465100	Diversity = 1.0869e+01	k_t
Generation 4660: Best = 1.202366e+05 = [4, 3, 2, 4]	FES = 466100	Diversity = 1.0320e+01	k_t
Generation 4670: Best = 1.201856e+05 = [1, 1, 1, 1]	FES = 467100	Diversity = 1.0248e+01	k_t
Generation 4680: Best = 1.201597e+05 = [3, 4, 2, 3]	FES = 468100	Diversity = 1.0830e+01	k_t
Generation 4690: Best = 1.200814e+05 = [1, 1, 1, 1]	FES = 469100	Diversity = 1.0925e+01	k_t
Generation 4700: Best = 1.200214e+05 = [4, 2, 2, 4]	FES = 470100	Diversity = 1.1721e+01	k_t
Generation 4710: Best = 1.198514e+05 = [1, 1, 1, 1]	FES = 471100	Diversity = 1.2288e+01	k_t
Generation 4720: Best = 1.197885e+05 = [3, 2, 2, 3]	FES = 472100	Diversity = 1.2274e+01	k_t
Generation 4730: Best = 1.195910e+05 = [1, 1, 1, 1]	FES = 473100	Diversity = 1.2809e+01	k_t
Generation 4740: Best = 1.194225e+05 = [4, 2, 3, 3]	FES = 474100	Diversity = 1.3642e+01	k_t
Generation 4750: Best = 1.192521e+05 = [1, 1, 1, 1]	FES = 475100	Diversity = 1.5241e+01	k_t
Generation 4760: Best = 1.191150e+05 = [2, 2, 3, 2]	FES = 476100	Diversity = 1.5643e+01	k_t
Generation 4770: Best = 1.190721e+05 = [1, 1, 1, 1]	FES = 477100	Diversity = 1.6016e+01	k_t
Generation 4780: Best = 1.189289e+05 = [3, 4, 4, 3]	FES = 478100	Diversity = 1.5450e+01	k_t
Generation 4790: Best = 1.188653e+05 = [1, 1, 1, 1]	FES = 479100	Diversity = 1.4468e+01	k_t
Generation 4800: Best = 1.187518e+05 = [3, 3, 4, 4]	FES = 480100	Diversity = 1.5045e+01	k_t
Generation 4810: Best = 1.186603e+05 = [1, 1, 1, 1]	FES = 481100	Diversity = 1.5136e+01	k_t

Generation 4820: Best = 1.184760e+05 = [3, 4, 4, 4]	FES = 482100	Diversity = 1.4394e+01	k_t
Generation 4830: Best = 1.184489e+05 = [1, 1, 1, 1]	FES = 483100	Diversity = 1.3366e+01	k_t
Generation 4840: Best = 1.183702e+05 = [4, 3, 2, 2]	FES = 484100	Diversity = 1.3408e+01	k_t
Generation 4850: Best = 1.182713e+05 = [1, 1, 1, 1]	FES = 485100	Diversity = 1.2999e+01	k_t
Generation 4860: Best = 1.181310e+05 = [3, 4, 2, 3]	FES = 486100	Diversity = 1.4842e+01	k_t
Generation 4870: Best = 1.180070e+05 = [1, 1, 1, 1]	FES = 487100	Diversity = 1.4485e+01	k_t
Generation 4880: Best = 1.179461e+05 = [4, 3, 4, 3]	FES = 488100	Diversity = 1.4304e+01	k_t
Generation 4890: Best = 1.178517e+05 = [1, 1, 1, 1]	FES = 489100	Diversity = 1.3925e+01	k_t
Generation 4900: Best = 1.176494e+05 = [4, 4, 4, 3]	FES = 490100	Diversity = 1.3804e+01	k_t
Generation 4910: Best = 1.176445e+05 = [1, 1, 1, 1]	FES = 491100	Diversity = 1.2716e+01	k_t
Generation 4920: Best = 1.174498e+05 = [2, 4, 2, 2]	FES = 492100	Diversity = 1.2784e+01	k_t
Generation 4930: Best = 1.173809e+05 = [1, 1, 1, 1]	FES = 493100	Diversity = 1.3129e+01	k_t
Generation 4940: Best = 1.173053e+05 = [2, 3, 4, 3]	FES = 494100	Diversity = 1.3879e+01	k_t
Generation 4950: Best = 1.170525e+05 = [1, 1, 1, 1]	FES = 495100	Diversity = 1.3911e+01	k_t
Generation 4960: Best = 1.168758e+05 = [4, 4, 3, 2]	FES = 496100	Diversity = 1.5251e+01	k_t
Generation 4970: Best = 1.167132e+05 = [1, 1, 1, 1]	FES = 497100	Diversity = 1.6598e+01	k_t
Generation 4980: Best = 1.165048e+05 = [2, 4, 2, 4]	FES = 498100	Diversity = 1.6943e+01	k_t
Generation 4990: Best = 1.164553e+05 = [1, 1, 1, 1]	FES = 499100	Diversity = 1.5213e+01	k_t
Generation 5000: Best = 1.161693e+05 = [3, 2, 3, 3]	FES = 500100	Diversity = 1.5752e+01	k_t
Generation 5010: Best = 1.161323e+05 = [1, 1, 1, 1]	FES = 501100	Diversity = 1.5168e+01	k_t
Generation 5020: Best = 1.157859e+05 = [3, 2, 3, 4]	FES = 502100	Diversity = 1.5575e+01	k_t
Generation 5030: Best = 1.156146e+05 = [1, 1, 1, 1]	FES = 503100	Diversity = 1.6439e+01	k_t
Generation 5040: Best = 1.153819e+05 = [4, 2, 3, 3]	FES = 504100	Diversity = 1.6412e+01	k_t
Generation 5050: Best = 1.150242e+05 = [1, 1, 1, 1]	FES = 505100	Diversity = 1.6885e+01	k_t
Generation 5060: Best = 1.149009e+05 = [2, 3, 2, 2]	FES = 506100	Diversity = 1.7365e+01	k_t
Generation 5070: Best = 1.147053e+05 = [1, 1, 1, 1]	FES = 507100	Diversity = 1.6816e+01	k_t
Generation 5080: Best = 1.145771e+05 = [2, 3, 3, 4]	FES = 508100	Diversity = 1.5589e+01	k_t

Generation 5090: Best = 1.145767e+05 = [1, 1, 1, 1]	FES = 509100	Diversity = 1.4668e+01	k_t
Generation 5100: Best = 1.143810e+05 = [4, 3, 4, 2]	FES = 510100	Diversity = 1.3857e+01	k_t
Generation 5110: Best = 1.143578e+05 = [1, 1, 1, 1]	FES = 511100	Diversity = 1.3366e+01	k_t
Generation 5120: Best = 1.143509e+05 = [2, 4, 2, 4]	FES = 512100	Diversity = 1.3107e+01	k_t
Generation 5130: Best = 1.142488e+05 = [1, 1, 1, 1]	FES = 513100	Diversity = 1.2360e+01	k_t
Generation 5140: Best = 1.141835e+05 = [2, 3, 3, 2]	FES = 514100	Diversity = 1.2610e+01	k_t
Generation 5150: Best = 1.141092e+05 = [1, 1, 1, 1]	FES = 515100	Diversity = 1.2071e+01	k_t
Generation 5160: Best = 1.140154e+05 = [4, 3, 2, 3]	FES = 516100	Diversity = 1.1406e+01	k_t
Generation 5170: Best = 1.138885e+05 = [1, 1, 1, 1]	FES = 517100	Diversity = 1.1257e+01	k_t
Generation 5180: Best = 1.138375e+05 = [2, 2, 4, 3]	FES = 518100	Diversity = 1.1834e+01	k_t
Generation 5190: Best = 1.137379e+05 = [1, 1, 1, 1]	FES = 519100	Diversity = 1.1836e+01	k_t
Generation 5200: Best = 1.136801e+05 = [2, 4, 3, 4]	FES = 520100	Diversity = 1.1433e+01	k_t
Generation 5210: Best = 1.136357e+05 = [1, 1, 1, 1]	FES = 521100	Diversity = 1.0808e+01	k_t
Generation 5220: Best = 1.134842e+05 = [3, 2, 2, 2]	FES = 522100	Diversity = 1.0977e+01	k_t
Generation 5230: Best = 1.134602e+05 = [1, 1, 1, 1]	FES = 523100	Diversity = 1.1534e+01	k_t
Generation 5240: Best = 1.133997e+05 = [4, 2, 4, 4]	FES = 524100	Diversity = 1.1734e+01	k_t
Generation 5250: Best = 1.133020e+05 = [1, 1, 1, 1]	FES = 525100	Diversity = 1.1189e+01	k_t
Generation 5260: Best = 1.131949e+05 = [3, 4, 3, 2]	FES = 526100	Diversity = 1.1481e+01	k_t
Generation 5270: Best = 1.131949e+05 = [1, 1, 1, 1]	FES = 527100	Diversity = 1.1304e+01	k_t
Generation 5280: Best = 1.130985e+05 = [3, 2, 2, 2]	FES = 528100	Diversity = 1.0426e+01	k_t
Generation 5290: Best = 1.130961e+05 = [1, 1, 1, 1]	FES = 529100	Diversity = 1.0546e+01	k_t
Generation 5300: Best = 1.129595e+05 = [2, 2, 2, 2]	FES = 530100	Diversity = 1.1801e+01	k_t
Generation 5310: Best = 1.127612e+05 = [1, 1, 1, 1]	FES = 531100	Diversity = 1.2906e+01	k_t
Generation 5320: Best = 1.126808e+05 = [4, 3, 3, 3]	FES = 532100	Diversity = 1.2213e+01	k_t
Generation 5330: Best = 1.125341e+05 = [1, 1, 1, 1]	FES = 533100	Diversity = 1.2644e+01	k_t
Generation 5340: Best = 1.124554e+05 = [2, 4, 2, 3]	FES = 534100	Diversity = 1.2855e+01	k_t
Generation 5350: Best = 1.123176e+05 = [1, 1, 1, 1]	FES = 535100	Diversity = 1.2772e+01	k_t

Generation 5360: Best = 1.121514e+05 = [4, 4, 4, 4]	FES = 536100	Diversity = 1.2803e+01	k_t
Generation 5370: Best = 1.121277e+05 = [1, 1, 1, 1]	FES = 537100	Diversity = 1.1846e+01	k_t
Generation 5380: Best = 1.119954e+05 = [3, 4, 3, 2]	FES = 538100	Diversity = 1.2303e+01	k_t
Generation 5390: Best = 1.118831e+05 = [1, 1, 1, 1]	FES = 539100	Diversity = 1.2698e+01	k_t
Generation 5400: Best = 1.117112e+05 = [2, 2, 2, 2]	FES = 540100	Diversity = 1.2948e+01	k_t
Generation 5410: Best = 1.116292e+05 = [1, 1, 1, 1]	FES = 541100	Diversity = 1.3719e+01	k_t
Generation 5420: Best = 1.114641e+05 = [2, 2, 3, 2]	FES = 542100	Diversity = 1.2994e+01	k_t
Generation 5430: Best = 1.113857e+05 = [1, 1, 1, 1]	FES = 543100	Diversity = 1.1542e+01	k_t
Generation 5440: Best = 1.112829e+05 = [3, 2, 2, 2]	FES = 544100	Diversity = 1.1959e+01	k_t
Generation 5450: Best = 1.112328e+05 = [1, 1, 1, 1]	FES = 545100	Diversity = 1.2712e+01	k_t
Generation 5460: Best = 1.111529e+05 = [3, 2, 3, 2]	FES = 546100	Diversity = 1.2401e+01	k_t
Generation 5470: Best = 1.110765e+05 = [1, 1, 1, 1]	FES = 547100	Diversity = 1.2503e+01	k_t
Generation 5480: Best = 1.109793e+05 = [3, 2, 2, 2]	FES = 548100	Diversity = 1.2133e+01	k_t
Generation 5490: Best = 1.107904e+05 = [1, 1, 1, 1]	FES = 549100	Diversity = 1.1407e+01	k_t
Generation 5500: Best = 1.107443e+05 = [3, 3, 4, 4]	FES = 550100	Diversity = 1.1783e+01	k_t
Generation 5510: Best = 1.106918e+05 = [1, 1, 1, 1]	FES = 551100	Diversity = 1.1686e+01	k_t
Generation 5520: Best = 1.106036e+05 = [4, 2, 2, 4]	FES = 552100	Diversity = 1.1140e+01	k_t
Generation 5530: Best = 1.105590e+05 = [1, 1, 1, 1]	FES = 553100	Diversity = 1.1345e+01	k_t
Generation 5540: Best = 1.104166e+05 = [3, 3, 2, 4]	FES = 554100	Diversity = 1.1517e+01	k_t
Generation 5550: Best = 1.103669e+05 = [1, 1, 1, 1]	FES = 555100	Diversity = 1.1811e+01	k_t
Generation 5560: Best = 1.103162e+05 = [2, 4, 2, 3]	FES = 556100	Diversity = 1.1305e+01	k_t
Generation 5570: Best = 1.102078e+05 = [1, 1, 1, 1]	FES = 557100	Diversity = 1.1186e+01	k_t
Generation 5580: Best = 1.101649e+05 = [2, 4, 2, 2]	FES = 558100	Diversity = 1.1058e+01	k_t
Generation 5590: Best = 1.101368e+05 = [1, 1, 1, 1]	FES = 559100	Diversity = 1.0722e+01	k_t
Generation 5600: Best = 1.100513e+05 = [4, 4, 2, 3]	FES = 560100	Diversity = 1.1195e+01	k_t
Generation 5610: Best = 1.099481e+05 = [1, 1, 1, 1]	FES = 561100	Diversity = 1.2524e+01	k_t
Generation 5620: Best = 1.099177e+05 = [4, 2, 3, 3]	FES = 562100	Diversity = 1.2606e+01	k_t

Generation 5630: Best = 1.098992e+05 = [1, 1, 1, 1]	FES = 563100	Diversity = 1.2238e+01	k_t
Generation 5640: Best = 1.098061e+05 = [2, 2, 4, 2]	FES = 564100	Diversity = 1.1979e+01	k_t
Generation 5650: Best = 1.097928e+05 = [1, 1, 1, 1]	FES = 565100	Diversity = 1.1698e+01	k_t
Generation 5660: Best = 1.097428e+05 = [3, 2, 2, 2]	FES = 566100	Diversity = 1.1219e+01	k_t
Generation 5670: Best = 1.096687e+05 = [1, 1, 1, 1]	FES = 567100	Diversity = 1.1040e+01	k_t
Generation 5680: Best = 1.096687e+05 = [2, 3, 4, 2]	FES = 568100	Diversity = 1.0460e+01	k_t
Generation 5690: Best = 1.095793e+05 = [1, 1, 1, 1]	FES = 569100	Diversity = 1.0121e+01	k_t
Generation 5700: Best = 1.095359e+05 = [2, 4, 2, 3]	FES = 570100	Diversity = 9.6702e+00	k_t
Generation 5710: Best = 1.094906e+05 = [1, 1, 1, 1]	FES = 571100	Diversity = 1.0093e+01	k_t
Generation 5720: Best = 1.094906e+05 = [3, 4, 2, 2]	FES = 572100	Diversity = 1.0113e+01	k_t
Generation 5730: Best = 1.094220e+05 = [1, 1, 1, 1]	FES = 573100	Diversity = 1.0338e+01	k_t
Generation 5740: Best = 1.093854e+05 = [2, 2, 3, 3]	FES = 574100	Diversity = 1.0265e+01	k_t
Generation 5750: Best = 1.092840e+05 = [1, 1, 1, 1]	FES = 575100	Diversity = 1.0344e+01	k_t
Generation 5760: Best = 1.092840e+05 = [4, 4, 4, 2]	FES = 576100	Diversity = 1.0728e+01	k_t
Generation 5770: Best = 1.092377e+05 = [1, 1, 1, 1]	FES = 577100	Diversity = 1.0387e+01	k_t
Generation 5780: Best = 1.091593e+05 = [2, 4, 2, 2]	FES = 578100	Diversity = 9.9804e+00	k_t
Generation 5790: Best = 1.091580e+05 = [1, 1, 1, 1]	FES = 579100	Diversity = 1.0264e+01	k_t
Generation 5800: Best = 1.090825e+05 = [2, 4, 2, 3]	FES = 580100	Diversity = 1.0498e+01	k_t
Generation 5810: Best = 1.089634e+05 = [1, 1, 1, 1]	FES = 581100	Diversity = 1.0910e+01	k_t
Generation 5820: Best = 1.087812e+05 = [4, 2, 2, 4]	FES = 582100	Diversity = 1.2387e+01	k_t
Generation 5830: Best = 1.086269e+05 = [1, 1, 1, 1]	FES = 583100	Diversity = 1.4982e+01	k_t
Generation 5840: Best = 1.083472e+05 = [3, 4, 2, 3]	FES = 584100	Diversity = 1.4304e+01	k_t
Generation 5850: Best = 1.082667e+05 = [1, 1, 1, 1]	FES = 585100	Diversity = 1.4793e+01	k_t
Generation 5860: Best = 1.079563e+05 = [2, 2, 4, 3]	FES = 586100	Diversity = 1.6453e+01	k_t
Generation 5870: Best = 1.077902e+05 = [1, 1, 1, 1]	FES = 587100	Diversity = 1.6344e+01	k_t
Generation 5880: Best = 1.076635e+05 = [4, 2, 2, 4]	FES = 588100	Diversity = 1.5876e+01	k_t
Generation 5890: Best = 1.074648e+05 = [1, 1, 1, 1]	FES = 589100	Diversity = 1.6189e+01	k_t

Generation 5900: Best = 1.073719e+05	FES = 590100	Diversity = 1.6197e+01	k_t
= [4, 2, 3, 4]			
Generation 5910: Best = 1.073016e+05	FES = 591100	Diversity = 1.6901e+01	k_t
= [1, 1, 1, 1]			
Generation 5920: Best = 1.070002e+05	FES = 592100	Diversity = 1.7698e+01	k_t
= [4, 4, 3, 3]			
Generation 5930: Best = 1.067548e+05	FES = 593100	Diversity = 1.6581e+01	k_t
= [1, 1, 1, 1]			
Generation 5940: Best = 1.066864e+05	FES = 594100	Diversity = 1.6324e+01	k_t
= [2, 4, 3, 2]			
Generation 5950: Best = 1.065259e+05	FES = 595100	Diversity = 1.5292e+01	k_t
= [1, 1, 1, 1]			
Generation 5960: Best = 1.064485e+05	FES = 596100	Diversity = 1.4632e+01	k_t
= [4, 2, 4, 2]			
Generation 5970: Best = 1.063721e+05	FES = 597100	Diversity = 1.3573e+01	k_t
= [1, 1, 1, 1]			
Generation 5980: Best = 1.063083e+05	FES = 598100	Diversity = 1.3108e+01	k_t
= [4, 4, 3, 4]			
Generation 5990: Best = 1.062495e+05	FES = 599100	Diversity = 1.3215e+01	k_t
= [1, 1, 1, 1]			
Generation 6000: Best = 1.061625e+05	FES = 600100	Diversity = 1.2749e+01	k_t
= [4, 2, 3, 2]			
Generation 6010: Best = 1.060343e+05	FES = 601100	Diversity = 1.2305e+01	k_t
= [1, 1, 1, 1]			
Generation 6020: Best = 1.058529e+05	FES = 602100	Diversity = 1.2843e+01	k_t
= [2, 4, 4, 3]			
Generation 6030: Best = 1.057523e+05	FES = 603100	Diversity = 1.4242e+01	k_t
= [1, 1, 1, 1]			
Generation 6040: Best = 1.056335e+05	FES = 604100	Diversity = 1.4013e+01	k_t
= [4, 3, 3, 2]			
Generation 6050: Best = 1.055463e+05	FES = 605100	Diversity = 1.3345e+01	k_t
= [1, 1, 1, 1]			
Generation 6060: Best = 1.054115e+05	FES = 606100	Diversity = 1.3444e+01	k_t
= [2, 3, 2, 3]			
Generation 6070: Best = 1.051836e+05	FES = 607100	Diversity = 1.3727e+01	k_t
= [1, 1, 1, 1]			
Generation 6080: Best = 1.051664e+05	FES = 608100	Diversity = 1.3816e+01	k_t
= [4, 2, 4, 2]			
Generation 6090: Best = 1.050394e+05	FES = 609100	Diversity = 1.3340e+01	k_t
= [1, 1, 1, 1]			
Generation 6100: Best = 1.048749e+05	FES = 610100	Diversity = 1.3490e+01	k_t
= [4, 4, 4, 4]			
Generation 6110: Best = 1.048167e+05	FES = 611100	Diversity = 1.3559e+01	k_t
= [1, 1, 1, 1]			
Generation 6120: Best = 1.047614e+05	FES = 612100	Diversity = 1.2937e+01	k_t
= [3, 4, 3, 3]			
Generation 6130: Best = 1.046713e+05	FES = 613100	Diversity = 1.2397e+01	k_t
= [1, 1, 1, 1]			
Generation 6140: Best = 1.046132e+05	FES = 614100	Diversity = 1.2204e+01	k_t
= [4, 2, 4, 3]			
Generation 6150: Best = 1.045071e+05	FES = 615100	Diversity = 1.2088e+01	k_t
= [1, 1, 1, 1]			
Generation 6160: Best = 1.044726e+05	FES = 616100	Diversity = 1.1960e+01	k_t
= [4, 2, 4, 3]			

Generation 6170: Best = 1.044074e+05 = [1, 1, 1, 1]	FES = 617100	Diversity = 1.1508e+01	k_t
Generation 6180: Best = 1.043338e+05 = [3, 2, 3, 4]	FES = 618100	Diversity = 1.1138e+01	k_t
Generation 6190: Best = 1.042496e+05 = [1, 1, 1, 1]	FES = 619100	Diversity = 1.0641e+01	k_t
Generation 6200: Best = 1.042489e+05 = [3, 4, 2, 2]	FES = 620100	Diversity = 1.0142e+01	k_t
Generation 6210: Best = 1.041620e+05 = [1, 1, 1, 1]	FES = 621100	Diversity = 1.0345e+01	k_t
Generation 6220: Best = 1.040896e+05 = [4, 3, 3, 3]	FES = 622100	Diversity = 1.0836e+01	k_t
Generation 6230: Best = 1.040450e+05 = [1, 1, 1, 1]	FES = 623100	Diversity = 1.1514e+01	k_t
Generation 6240: Best = 1.040244e+05 = [4, 3, 2, 4]	FES = 624100	Diversity = 1.0892e+01	k_t
Generation 6250: Best = 1.039952e+05 = [1, 1, 1, 1]	FES = 625100	Diversity = 1.0944e+01	k_t
Generation 6260: Best = 1.039563e+05 = [2, 2, 2, 4]	FES = 626100	Diversity = 1.0901e+01	k_t
Generation 6270: Best = 1.038568e+05 = [1, 1, 1, 1]	FES = 627100	Diversity = 1.1773e+01	k_t
Generation 6280: Best = 1.037336e+05 = [2, 4, 3, 4]	FES = 628100	Diversity = 1.2896e+01	k_t
Generation 6290: Best = 1.036231e+05 = [1, 1, 1, 1]	FES = 629100	Diversity = 1.3450e+01	k_t
Generation 6300: Best = 1.034964e+05 = [4, 2, 2, 4]	FES = 630100	Diversity = 1.3890e+01	k_t
Generation 6310: Best = 1.034321e+05 = [1, 1, 1, 1]	FES = 631100	Diversity = 1.4081e+01	k_t
Generation 6320: Best = 1.032919e+05 = [4, 2, 2, 3]	FES = 632100	Diversity = 1.4053e+01	k_t
Generation 6330: Best = 1.032022e+05 = [1, 1, 1, 1]	FES = 633100	Diversity = 1.4665e+01	k_t
Generation 6340: Best = 1.030220e+05 = [4, 4, 2, 4]	FES = 634100	Diversity = 1.4376e+01	k_t
Generation 6350: Best = 1.030069e+05 = [1, 1, 1, 1]	FES = 635100	Diversity = 1.5160e+01	k_t
Generation 6360: Best = 1.029355e+05 = [4, 4, 4, 3]	FES = 636100	Diversity = 1.5388e+01	k_t
Generation 6370: Best = 1.028725e+05 = [1, 1, 1, 1]	FES = 637100	Diversity = 1.4378e+01	k_t
Generation 6380: Best = 1.028033e+05 = [4, 3, 2, 4]	FES = 638100	Diversity = 1.3606e+01	k_t
Generation 6390: Best = 1.026810e+05 = [1, 1, 1, 1]	FES = 639100	Diversity = 1.2619e+01	k_t
Generation 6400: Best = 1.025621e+05 = [2, 2, 3, 2]	FES = 640100	Diversity = 1.2561e+01	k_t
Generation 6410: Best = 1.025155e+05 = [1, 1, 1, 1]	FES = 641100	Diversity = 1.2131e+01	k_t
Generation 6420: Best = 1.024178e+05 = [4, 4, 4, 2]	FES = 642100	Diversity = 1.2619e+01	k_t
Generation 6430: Best = 1.024018e+05 = [1, 1, 1, 1]	FES = 643100	Diversity = 1.2003e+01	k_t

Generation 6440: Best = 1.023129e+05 = [4, 4, 2, 4]	FES = 644100	Diversity = 1.0689e+01	k_t
Generation 6450: Best = 1.023129e+05 = [1, 1, 1, 1]	FES = 645100	Diversity = 1.0200e+01	k_t
Generation 6460: Best = 1.022766e+05 = [3, 3, 4, 3]	FES = 646100	Diversity = 9.8600e+00	k_t
Generation 6470: Best = 1.022232e+05 = [1, 1, 1, 1]	FES = 647100	Diversity = 9.8564e+00	k_t
Generation 6480: Best = 1.021944e+05 = [2, 4, 2, 4]	FES = 648100	Diversity = 9.7374e+00	k_t
Generation 6490: Best = 1.021095e+05 = [1, 1, 1, 1]	FES = 649100	Diversity = 1.0015e+01	k_t
Generation 6500: Best = 1.020837e+05 = [3, 4, 2, 4]	FES = 650100	Diversity = 1.0363e+01	k_t
Generation 6510: Best = 1.019895e+05 = [1, 1, 1, 1]	FES = 651100	Diversity = 1.0674e+01	k_t
Generation 6520: Best = 1.019188e+05 = [2, 4, 3, 4]	FES = 652100	Diversity = 1.1063e+01	k_t
Generation 6530: Best = 1.018960e+05 = [1, 1, 1, 1]	FES = 653100	Diversity = 1.1651e+01	k_t
Generation 6540: Best = 1.018239e+05 = [2, 4, 3, 4]	FES = 654100	Diversity = 1.1283e+01	k_t
Generation 6550: Best = 1.017274e+05 = [1, 1, 1, 1]	FES = 655100	Diversity = 1.1397e+01	k_t
Generation 6560: Best = 1.016357e+05 = [3, 4, 4, 2]	FES = 656100	Diversity = 1.2234e+01	k_t
Generation 6570: Best = 1.015472e+05 = [1, 1, 1, 1]	FES = 657100	Diversity = 1.2721e+01	k_t
Generation 6580: Best = 1.014495e+05 = [2, 4, 4, 4]	FES = 658100	Diversity = 1.1640e+01	k_t
Generation 6590: Best = 1.014063e+05 = [1, 1, 1, 1]	FES = 659100	Diversity = 1.2031e+01	k_t
Generation 6600: Best = 1.013808e+05 = [4, 2, 2, 4]	FES = 660100	Diversity = 1.2109e+01	k_t
Generation 6610: Best = 1.013382e+05 = [1, 1, 1, 1]	FES = 661100	Diversity = 1.1510e+01	k_t
Generation 6620: Best = 1.013075e+05 = [2, 3, 2, 4]	FES = 662100	Diversity = 1.1769e+01	k_t
Generation 6630: Best = 1.011971e+05 = [1, 1, 1, 1]	FES = 663100	Diversity = 1.2005e+01	k_t
Generation 6640: Best = 1.011971e+05 = [3, 4, 3, 2]	FES = 664100	Diversity = 1.1514e+01	k_t
Generation 6650: Best = 1.011791e+05 = [1, 1, 1, 1]	FES = 665100	Diversity = 1.0977e+01	k_t
Generation 6660: Best = 1.011556e+05 = [3, 4, 4, 4]	FES = 666100	Diversity = 1.0768e+01	k_t
Generation 6670: Best = 1.011345e+05 = [1, 1, 1, 1]	FES = 667100	Diversity = 1.0796e+01	k_t
Generation 6680: Best = 1.011128e+05 = [2, 4, 3, 2]	FES = 668100	Diversity = 1.0332e+01	k_t
Generation 6690: Best = 1.010667e+05 = [1, 1, 1, 1]	FES = 669100	Diversity = 9.8415e+00	k_t
Generation 6700: Best = 1.010388e+05 = [4, 4, 2, 4]	FES = 670100	Diversity = 9.7279e+00	k_t

Generation 6710: Best = 1.010171e+05 = [1, 1, 1, 1]	FES = 671100	Diversity = 9.5065e+00	k_t
Generation 6720: Best = 1.009296e+05 = [2, 2, 2, 3]	FES = 672100	Diversity = 1.0008e+01	k_t
Generation 6730: Best = 1.009084e+05 = [1, 1, 1, 1]	FES = 673100	Diversity = 1.0097e+01	k_t
Generation 6740: Best = 1.008677e+05 = [4, 2, 2, 4]	FES = 674100	Diversity = 1.0590e+01	k_t
Generation 6750: Best = 1.007944e+05 = [1, 1, 1, 1]	FES = 675100	Diversity = 1.0254e+01	k_t
Generation 6760: Best = 1.007455e+05 = [4, 4, 3, 2]	FES = 676100	Diversity = 1.0393e+01	k_t
Generation 6770: Best = 1.006682e+05 = [1, 1, 1, 1]	FES = 677100	Diversity = 1.0758e+01	k_t
Generation 6780: Best = 1.006349e+05 = [3, 3, 2, 2]	FES = 678100	Diversity = 1.0590e+01	k_t
Generation 6790: Best = 1.005991e+05 = [1, 1, 1, 1]	FES = 679100	Diversity = 1.0347e+01	k_t
Generation 6800: Best = 1.005639e+05 = [3, 4, 2, 2]	FES = 680100	Diversity = 1.0059e+01	k_t
Generation 6810: Best = 1.005111e+05 = [1, 1, 1, 1]	FES = 681100	Diversity = 9.4243e+00	k_t
Generation 6820: Best = 1.004534e+05 = [2, 4, 3, 2]	FES = 682100	Diversity = 9.2428e+00	k_t
Generation 6830: Best = 1.004534e+05 = [1, 1, 1, 1]	FES = 683100	Diversity = 9.6547e+00	k_t
Generation 6840: Best = 1.003931e+05 = [3, 3, 2, 2]	FES = 684100	Diversity = 1.0249e+01	k_t
Generation 6850: Best = 1.003098e+05 = [1, 1, 1, 1]	FES = 685100	Diversity = 1.0997e+01	k_t
Generation 6860: Best = 1.002675e+05 = [2, 4, 2, 3]	FES = 686100	Diversity = 1.1121e+01	k_t
Generation 6870: Best = 1.002554e+05 = [1, 1, 1, 1]	FES = 687100	Diversity = 1.1623e+01	k_t
Generation 6880: Best = 1.001642e+05 = [4, 2, 3, 3]	FES = 688100	Diversity = 1.1745e+01	k_t
Generation 6890: Best = 1.000946e+05 = [1, 1, 1, 1]	FES = 689100	Diversity = 1.3575e+01	k_t
Generation 6900: Best = 9.991110e+04 = [3, 2, 4, 2]	FES = 690100	Diversity = 1.4367e+01	k_t
Generation 6910: Best = 9.975892e+04 = [1, 1, 1, 1]	FES = 691100	Diversity = 1.5437e+01	k_t
Generation 6920: Best = 9.965796e+04 = [2, 4, 3, 4]	FES = 692100	Diversity = 1.5046e+01	k_t
Generation 6930: Best = 9.962817e+04 = [1, 1, 1, 1]	FES = 693100	Diversity = 1.5001e+01	k_t
Generation 6940: Best = 9.955146e+04 = [4, 3, 4, 2]	FES = 694100	Diversity = 1.3553e+01	k_t
Generation 6950: Best = 9.947451e+04 = [1, 1, 1, 1]	FES = 695100	Diversity = 1.3358e+01	k_t
Generation 6960: Best = 9.938026e+04 = [4, 4, 2, 3]	FES = 696100	Diversity = 1.3146e+01	k_t
Generation 6970: Best = 9.930163e+04 = [1, 1, 1, 1]	FES = 697100	Diversity = 1.2900e+01	k_t

Generation 6980: Best = 9.924206e+04 = [4, 3, 4, 2]	FES = 698100	Diversity = 1.2492e+01	k_t
Generation 6990: Best = 9.916170e+04 = [1, 1, 1, 1]	FES = 699100	Diversity = 1.2676e+01	k_t
Generation 7000: Best = 9.914319e+04 = [3, 2, 2, 4]	FES = 700100	Diversity = 1.2221e+01	k_t
Generation 7010: Best = 9.902598e+04 = [1, 1, 1, 1]	FES = 701100	Diversity = 1.1511e+01	k_t
Generation 7020: Best = 9.901599e+04 = [4, 3, 3, 4]	FES = 702100	Diversity = 1.0982e+01	k_t
Generation 7030: Best = 9.895918e+04 = [1, 1, 1, 1]	FES = 703100	Diversity = 1.1596e+01	k_t
Generation 7040: Best = 9.884913e+04 = [4, 4, 3, 4]	FES = 704100	Diversity = 1.0828e+01	k_t
Generation 7050: Best = 9.867466e+04 = [1, 1, 1, 1]	FES = 705100	Diversity = 1.2082e+01	k_t
Generation 7060: Best = 9.852100e+04 = [4, 3, 3, 3]	FES = 706100	Diversity = 1.2855e+01	k_t
Generation 7070: Best = 9.835073e+04 = [1, 1, 1, 1]	FES = 707100	Diversity = 1.4004e+01	k_t
Generation 7080: Best = 9.816878e+04 = [3, 3, 3, 2]	FES = 708100	Diversity = 1.5157e+01	k_t
Generation 7090: Best = 9.815197e+04 = [1, 1, 1, 1]	FES = 709100	Diversity = 1.5005e+01	k_t
Generation 7100: Best = 9.780901e+04 = [3, 2, 4, 3]	FES = 710100	Diversity = 1.5145e+01	k_t
Generation 7110: Best = 9.769711e+04 = [1, 1, 1, 1]	FES = 711100	Diversity = 1.6134e+01	k_t
Generation 7120: Best = 9.749056e+04 = [3, 3, 4, 2]	FES = 712100	Diversity = 1.4729e+01	k_t
Generation 7130: Best = 9.730549e+04 = [1, 1, 1, 1]	FES = 713100	Diversity = 1.5064e+01	k_t
Generation 7140: Best = 9.714855e+04 = [2, 4, 3, 4]	FES = 714100	Diversity = 1.4113e+01	k_t
Generation 7150: Best = 9.714418e+04 = [1, 1, 1, 1]	FES = 715100	Diversity = 1.2987e+01	k_t
Generation 7160: Best = 9.704393e+04 = [3, 2, 3, 3]	FES = 716100	Diversity = 1.1871e+01	k_t
Generation 7170: Best = 9.700964e+04 = [1, 1, 1, 1]	FES = 717100	Diversity = 1.1658e+01	k_t
Generation 7180: Best = 9.692608e+04 = [4, 3, 3, 4]	FES = 718100	Diversity = 1.0384e+01	k_t
Generation 7190: Best = 9.676496e+04 = [1, 1, 1, 1]	FES = 719100	Diversity = 9.9839e+00	k_t
Generation 7200: Best = 9.676496e+04 = [4, 2, 2, 3]	FES = 720100	Diversity = 1.0479e+01	k_t
Generation 7210: Best = 9.665073e+04 = [1, 1, 1, 1]	FES = 721100	Diversity = 1.1036e+01	k_t
Generation 7220: Best = 9.650567e+04 = [4, 4, 2, 3]	FES = 722100	Diversity = 1.2241e+01	k_t
Generation 7230: Best = 9.630020e+04 = [1, 1, 1, 1]	FES = 723100	Diversity = 1.3874e+01	k_t
Generation 7240: Best = 9.613967e+04 = [2, 4, 2, 4]	FES = 724100	Diversity = 1.4202e+01	k_t

Generation 7250: Best = 9.594534e+04 = [1, 1, 1, 1]	FES = 725100	Diversity = 1.4658e+01	k_t
Generation 7260: Best = 9.588940e+04 = [4, 2, 4, 2]	FES = 726100	Diversity = 1.4720e+01	k_t
Generation 7270: Best = 9.574451e+04 = [1, 1, 1, 1]	FES = 727100	Diversity = 1.4485e+01	k_t
Generation 7280: Best = 9.556618e+04 = [3, 3, 2, 3]	FES = 728100	Diversity = 1.4355e+01	k_t
Generation 7290: Best = 9.536590e+04 = [1, 1, 1, 1]	FES = 729100	Diversity = 1.4113e+01	k_t
Generation 7300: Best = 9.529390e+04 = [2, 2, 3, 2]	FES = 730100	Diversity = 1.3567e+01	k_t
Generation 7310: Best = 9.513858e+04 = [1, 1, 1, 1]	FES = 731100	Diversity = 1.2996e+01	k_t
Generation 7320: Best = 9.503264e+04 = [4, 2, 4, 2]	FES = 732100	Diversity = 1.1782e+01	k_t
Generation 7330: Best = 9.497892e+04 = [1, 1, 1, 1]	FES = 733100	Diversity = 1.0762e+01	k_t
Generation 7340: Best = 9.493345e+04 = [3, 3, 3, 3]	FES = 734100	Diversity = 1.0053e+01	k_t
Generation 7350: Best = 9.481540e+04 = [1, 1, 1, 1]	FES = 735100	Diversity = 1.0136e+01	k_t
Generation 7360: Best = 9.477306e+04 = [2, 2, 2, 3]	FES = 736100	Diversity = 1.0439e+01	k_t
Generation 7370: Best = 9.469267e+04 = [1, 1, 1, 1]	FES = 737100	Diversity = 1.0736e+01	k_t
Generation 7380: Best = 9.468465e+04 = [4, 2, 3, 3]	FES = 738100	Diversity = 1.1192e+01	k_t
Generation 7390: Best = 9.459173e+04 = [1, 1, 1, 1]	FES = 739100	Diversity = 1.0717e+01	k_t
Generation 7400: Best = 9.448561e+04 = [4, 2, 3, 3]	FES = 740100	Diversity = 1.0670e+01	k_t
Generation 7410: Best = 9.448474e+04 = [1, 1, 1, 1]	FES = 741100	Diversity = 1.0714e+01	k_t
Generation 7420: Best = 9.435990e+04 = [4, 4, 4, 2]	FES = 742100	Diversity = 1.0775e+01	k_t
Generation 7430: Best = 9.428637e+04 = [1, 1, 1, 1]	FES = 743100	Diversity = 1.1482e+01	k_t
Generation 7440: Best = 9.419804e+04 = [4, 4, 2, 4]	FES = 744100	Diversity = 1.1937e+01	k_t
Generation 7450: Best = 9.407585e+04 = [1, 1, 1, 1]	FES = 745100	Diversity = 1.2264e+01	k_t
Generation 7460: Best = 9.401334e+04 = [2, 4, 3, 2]	FES = 746100	Diversity = 1.2887e+01	k_t
Generation 7470: Best = 9.390086e+04 = [1, 1, 1, 1]	FES = 747100	Diversity = 1.3037e+01	k_t
Generation 7480: Best = 9.381052e+04 = [2, 2, 2, 2]	FES = 748100	Diversity = 1.2668e+01	k_t
Generation 7490: Best = 9.372237e+04 = [1, 1, 1, 1]	FES = 749100	Diversity = 1.2587e+01	k_t
Generation 7500: Best = 9.360630e+04 = [3, 2, 2, 3]	FES = 750100	Diversity = 1.2770e+01	k_t
Generation 7510: Best = 9.354526e+04 = [1, 1, 1, 1]	FES = 751100	Diversity = 1.2866e+01	k_t

Generation 7520: Best = 9.350814e+04 = [2, 4, 3, 3]	FES = 752100	Diversity = 1.2856e+01	k_t
Generation 7530: Best = 9.340074e+04 = [1, 1, 1, 1]	FES = 753100	Diversity = 1.2499e+01	k_t
Generation 7540: Best = 9.330006e+04 = [4, 3, 3, 4]	FES = 754100	Diversity = 1.2347e+01	k_t
Generation 7550: Best = 9.324636e+04 = [1, 1, 1, 1]	FES = 755100	Diversity = 1.2422e+01	k_t
Generation 7560: Best = 9.317502e+04 = [3, 2, 4, 3]	FES = 756100	Diversity = 1.2745e+01	k_t
Generation 7570: Best = 9.310908e+04 = [1, 1, 1, 1]	FES = 757100	Diversity = 1.3040e+01	k_t
Generation 7580: Best = 9.302635e+04 = [4, 2, 4, 4]	FES = 758100	Diversity = 1.2999e+01	k_t
Generation 7590: Best = 9.295610e+04 = [1, 1, 1, 1]	FES = 759100	Diversity = 1.3598e+01	k_t
Generation 7600: Best = 9.287540e+04 = [3, 2, 4, 2]	FES = 760100	Diversity = 1.4269e+01	k_t
Generation 7610: Best = 9.278403e+04 = [1, 1, 1, 1]	FES = 761100	Diversity = 1.4257e+01	k_t
Generation 7620: Best = 9.266833e+04 = [4, 2, 2, 4]	FES = 762100	Diversity = 1.3090e+01	k_t
Generation 7630: Best = 9.262006e+04 = [1, 1, 1, 1]	FES = 763100	Diversity = 1.2705e+01	k_t
Generation 7640: Best = 9.260493e+04 = [2, 4, 4, 3]	FES = 764100	Diversity = 1.2315e+01	k_t
Generation 7650: Best = 9.257203e+04 = [1, 1, 1, 1]	FES = 765100	Diversity = 1.1414e+01	k_t
Generation 7660: Best = 9.253805e+04 = [4, 4, 3, 4]	FES = 766100	Diversity = 1.1505e+01	k_t
Generation 7670: Best = 9.248568e+04 = [1, 1, 1, 1]	FES = 767100	Diversity = 1.1845e+01	k_t
Generation 7680: Best = 9.235202e+04 = [3, 4, 2, 2]	FES = 768100	Diversity = 1.2462e+01	k_t
Generation 7690: Best = 9.235202e+04 = [1, 1, 1, 1]	FES = 769100	Diversity = 1.3174e+01	k_t
Generation 7700: Best = 9.229383e+04 = [2, 3, 4, 4]	FES = 770100	Diversity = 1.3138e+01	k_t
Generation 7710: Best = 9.225590e+04 = [1, 1, 1, 1]	FES = 771100	Diversity = 1.2645e+01	k_t
Generation 7720: Best = 9.209863e+04 = [2, 3, 2, 4]	FES = 772100	Diversity = 1.2188e+01	k_t
Generation 7730: Best = 9.209863e+04 = [1, 1, 1, 1]	FES = 773100	Diversity = 1.2854e+01	k_t
Generation 7740: Best = 9.206602e+04 = [4, 4, 3, 3]	FES = 774100	Diversity = 1.2268e+01	k_t
Generation 7750: Best = 9.203744e+04 = [1, 1, 1, 1]	FES = 775100	Diversity = 1.1514e+01	k_t
Generation 7760: Best = 9.198968e+04 = [4, 4, 3, 4]	FES = 776100	Diversity = 1.0553e+01	k_t
Generation 7770: Best = 9.193439e+04 = [1, 1, 1, 1]	FES = 777100	Diversity = 1.0557e+01	k_t
Generation 7780: Best = 9.193439e+04 = [3, 2, 3, 4]	FES = 778100	Diversity = 1.0126e+01	k_t

Generation 7790: Best = 9.190714e+04 = [1, 1, 1, 1]	FES = 779100	Diversity = 9.5143e+00	k_t
Generation 7800: Best = 9.190714e+04 = [2, 4, 2, 2]	FES = 780100	Diversity = 9.2022e+00	k_t
Generation 7810: Best = 9.185983e+04 = [1, 1, 1, 1]	FES = 781100	Diversity = 9.1397e+00	k_t
Generation 7820: Best = 9.183752e+04 = [4, 3, 4, 2]	FES = 782100	Diversity = 8.8105e+00	k_t
Generation 7830: Best = 9.173739e+04 = [1, 1, 1, 1]	FES = 783100	Diversity = 9.1930e+00	k_t
Generation 7840: Best = 9.173713e+04 = [4, 4, 3, 4]	FES = 784100	Diversity = 9.4493e+00	k_t
Generation 7850: Best = 9.169960e+04 = [1, 1, 1, 1]	FES = 785100	Diversity = 9.8860e+00	k_t
Generation 7860: Best = 9.163153e+04 = [2, 2, 2, 2]	FES = 786100	Diversity = 1.0495e+01	k_t
Generation 7870: Best = 9.155731e+04 = [1, 1, 1, 1]	FES = 787100	Diversity = 1.0658e+01	k_t
Generation 7880: Best = 9.154379e+04 = [3, 3, 4, 3]	FES = 788100	Diversity = 1.1334e+01	k_t
Generation 7890: Best = 9.152116e+04 = [1, 1, 1, 1]	FES = 789100	Diversity = 1.1891e+01	k_t
Generation 7900: Best = 9.136932e+04 = [2, 3, 4, 4]	FES = 790100	Diversity = 1.2866e+01	k_t
Generation 7910: Best = 9.130959e+04 = [1, 1, 1, 1]	FES = 791100	Diversity = 1.3001e+01	k_t
Generation 7920: Best = 9.125849e+04 = [2, 4, 2, 4]	FES = 792100	Diversity = 1.2684e+01	k_t
Generation 7930: Best = 9.106494e+04 = [1, 1, 1, 1]	FES = 793100	Diversity = 1.3320e+01	k_t
Generation 7940: Best = 9.088773e+04 = [3, 4, 3, 2]	FES = 794100	Diversity = 1.2946e+01	k_t
Generation 7950: Best = 9.077745e+04 = [1, 1, 1, 1]	FES = 795100	Diversity = 1.2911e+01	k_t
Generation 7960: Best = 9.067034e+04 = [4, 2, 4, 3]	FES = 796100	Diversity = 1.3493e+01	k_t
Generation 7970: Best = 9.058443e+04 = [1, 1, 1, 1]	FES = 797100	Diversity = 1.3449e+01	k_t
Generation 7980: Best = 9.033579e+04 = [2, 4, 4, 3]	FES = 798100	Diversity = 1.4983e+01	k_t
Generation 7990: Best = 9.028684e+04 = [1, 1, 1, 1]	FES = 799100	Diversity = 1.5301e+01	k_t
Generation 8000: Best = 9.015892e+04 = [2, 4, 3, 3]	FES = 800100	Diversity = 1.4147e+01	k_t
Generation 8010: Best = 8.998143e+04 = [1, 1, 1, 1]	FES = 801100	Diversity = 1.3157e+01	k_t
Generation 8020: Best = 8.988302e+04 = [4, 4, 4, 3]	FES = 802100	Diversity = 1.2243e+01	k_t
Generation 8030: Best = 8.978703e+04 = [1, 1, 1, 1]	FES = 803100	Diversity = 1.2571e+01	k_t
Generation 8040: Best = 8.970913e+04 = [2, 4, 3, 3]	FES = 804100	Diversity = 1.3712e+01	k_t
Generation 8050: Best = 8.967469e+04 = [1, 1, 1, 1]	FES = 805100	Diversity = 1.2829e+01	k_t

Generation 8060: Best = 8.950027e+04 = [4, 3, 2, 2]	FES = 806100	Diversity = 1.2769e+01	k_t
Generation 8070: Best = 8.945224e+04 = [1, 1, 1, 1]	FES = 807100	Diversity = 1.4035e+01	k_t
Generation 8080: Best = 8.930209e+04 = [4, 4, 3, 2]	FES = 808100	Diversity = 1.3857e+01	k_t
Generation 8090: Best = 8.930209e+04 = [1, 1, 1, 1]	FES = 809100	Diversity = 1.3138e+01	k_t
Generation 8100: Best = 8.919938e+04 = [2, 3, 4, 2]	FES = 810100	Diversity = 1.2663e+01	k_t
Generation 8110: Best = 8.914948e+04 = [1, 1, 1, 1]	FES = 811100	Diversity = 1.1867e+01	k_t
Generation 8120: Best = 8.908297e+04 = [2, 3, 3, 2]	FES = 812100	Diversity = 1.1549e+01	k_t
Generation 8130: Best = 8.899542e+04 = [1, 1, 1, 1]	FES = 813100	Diversity = 1.1441e+01	k_t
Generation 8140: Best = 8.895998e+04 = [2, 2, 2, 3]	FES = 814100	Diversity = 1.2117e+01	k_t
Generation 8150: Best = 8.889680e+04 = [1, 1, 1, 1]	FES = 815100	Diversity = 1.1984e+01	k_t
Generation 8160: Best = 8.889480e+04 = [2, 2, 4, 3]	FES = 816100	Diversity = 1.1597e+01	k_t
Generation 8170: Best = 8.884468e+04 = [1, 1, 1, 1]	FES = 817100	Diversity = 1.0953e+01	k_t
Generation 8180: Best = 8.883782e+04 = [2, 4, 3, 4]	FES = 818100	Diversity = 1.0534e+01	k_t
Generation 8190: Best = 8.879702e+04 = [1, 1, 1, 1]	FES = 819100	Diversity = 1.0354e+01	k_t
Generation 8200: Best = 8.875255e+04 = [2, 3, 2, 4]	FES = 820100	Diversity = 1.0046e+01	k_t
Generation 8210: Best = 8.873624e+04 = [1, 1, 1, 1]	FES = 821100	Diversity = 1.0142e+01	k_t
Generation 8220: Best = 8.870573e+04 = [2, 4, 3, 3]	FES = 822100	Diversity = 9.8599e+00	k_t
Generation 8230: Best = 8.867229e+04 = [1, 1, 1, 1]	FES = 823100	Diversity = 9.5305e+00	k_t
Generation 8240: Best = 8.863930e+04 = [3, 2, 3, 2]	FES = 824100	Diversity = 9.6936e+00	k_t
Generation 8250: Best = 8.859135e+04 = [1, 1, 1, 1]	FES = 825100	Diversity = 1.0290e+01	k_t
Generation 8260: Best = 8.853196e+04 = [2, 2, 3, 2]	FES = 826100	Diversity = 1.0993e+01	k_t
Generation 8270: Best = 8.851031e+04 = [1, 1, 1, 1]	FES = 827100	Diversity = 1.0932e+01	k_t
Generation 8280: Best = 8.849507e+04 = [3, 2, 3, 3]	FES = 828100	Diversity = 1.1274e+01	k_t
Generation 8290: Best = 8.846742e+04 = [1, 1, 1, 1]	FES = 829100	Diversity = 1.0723e+01	k_t
Generation 8300: Best = 8.839298e+04 = [4, 2, 2, 3]	FES = 830100	Diversity = 1.0790e+01	k_t
Generation 8310: Best = 8.839298e+04 = [1, 1, 1, 1]	FES = 831100	Diversity = 1.0040e+01	k_t
Generation 8320: Best = 8.838129e+04 = [3, 3, 4, 2]	FES = 832100	Diversity = 1.0269e+01	k_t

Generation 8330: Best = 8.835327e+04 = [1, 1, 1, 1]	FES = 833100	Diversity = 1.0285e+01	k_t
Generation 8340: Best = 8.835327e+04 = [4, 2, 3, 4]	FES = 834100	Diversity = 1.0161e+01	k_t
Generation 8350: Best = 8.834846e+04 = [1, 1, 1, 1]	FES = 835100	Diversity = 1.0386e+01	k_t
Generation 8360: Best = 8.833251e+04 = [4, 4, 3, 4]	FES = 836100	Diversity = 1.0166e+01	k_t
Generation 8370: Best = 8.826603e+04 = [1, 1, 1, 1]	FES = 837100	Diversity = 1.0050e+01	k_t
Generation 8380: Best = 8.823940e+04 = [2, 2, 4, 2]	FES = 838100	Diversity = 9.9719e+00	k_t
Generation 8390: Best = 8.823494e+04 = [1, 1, 1, 1]	FES = 839100	Diversity = 1.0318e+01	k_t
Generation 8400: Best = 8.823494e+04 = [3, 2, 4, 3]	FES = 840100	Diversity = 1.0099e+01	k_t
Generation 8410: Best = 8.819808e+04 = [1, 1, 1, 1]	FES = 841100	Diversity = 1.0867e+01	k_t
Generation 8420: Best = 8.816465e+04 = [3, 3, 2, 3]	FES = 842100	Diversity = 1.1127e+01	k_t
Generation 8430: Best = 8.812161e+04 = [1, 1, 1, 1]	FES = 843100	Diversity = 1.1158e+01	k_t
Generation 8440: Best = 8.809828e+04 = [2, 2, 2, 2]	FES = 844100	Diversity = 1.1171e+01	k_t
Generation 8450: Best = 8.808467e+04 = [1, 1, 1, 1]	FES = 845100	Diversity = 1.1273e+01	k_t
Generation 8460: Best = 8.804303e+04 = [4, 3, 2, 2]	FES = 846100	Diversity = 1.1386e+01	k_t
Generation 8470: Best = 8.802433e+04 = [1, 1, 1, 1]	FES = 847100	Diversity = 1.1673e+01	k_t
Generation 8480: Best = 8.795674e+04 = [4, 3, 4, 4]	FES = 848100	Diversity = 1.1728e+01	k_t
Generation 8490: Best = 8.792044e+04 = [1, 1, 1, 1]	FES = 849100	Diversity = 1.1859e+01	k_t
Generation 8500: Best = 8.788330e+04 = [2, 3, 4, 2]	FES = 850100	Diversity = 1.2102e+01	k_t
Generation 8510: Best = 8.785469e+04 = [1, 1, 1, 1]	FES = 851100	Diversity = 1.2245e+01	k_t
Generation 8520: Best = 8.781161e+04 = [3, 2, 3, 4]	FES = 852100	Diversity = 1.2745e+01	k_t
Generation 8530: Best = 8.772920e+04 = [1, 1, 1, 1]	FES = 853100	Diversity = 1.3500e+01	k_t
Generation 8540: Best = 8.766612e+04 = [2, 3, 3, 2]	FES = 854100	Diversity = 1.3910e+01	k_t
Generation 8550: Best = 8.750661e+04 = [1, 1, 1, 1]	FES = 855100	Diversity = 1.6852e+01	k_t
Generation 8560: Best = 8.699352e+04 = [3, 2, 2, 2]	FES = 856100	Diversity = 1.9861e+01	k_t
Generation 8570: Best = 8.676885e+04 = [1, 1, 1, 1]	FES = 857100	Diversity = 2.1432e+01	k_t
Generation 8580: Best = 8.635158e+04 = [2, 4, 3, 3]	FES = 858100	Diversity = 2.1010e+01	k_t
Generation 8590: Best = 8.593839e+04 = [1, 1, 1, 1]	FES = 859100	Diversity = 2.2577e+01	k_t

Generation 8600: Best = 8.589319e+04 = [2, 3, 4, 3]	FES = 860100	Diversity = 2.2491e+01	k_t
Generation 8610: Best = 8.552082e+04 = [1, 1, 1, 1]	FES = 861100	Diversity = 2.2610e+01	k_t
Generation 8620: Best = 8.530021e+04 = [2, 2, 4, 4]	FES = 862100	Diversity = 2.1189e+01	k_t
Generation 8630: Best = 8.522320e+04 = [1, 1, 1, 1]	FES = 863100	Diversity = 2.0323e+01	k_t
Generation 8640: Best = 8.492831e+04 = [3, 4, 3, 2]	FES = 864100	Diversity = 1.8425e+01	k_t
Generation 8650: Best = 8.492831e+04 = [1, 1, 1, 1]	FES = 865100	Diversity = 1.6608e+01	k_t
Generation 8660: Best = 8.487260e+04 = [4, 4, 2, 3]	FES = 866100	Diversity = 1.5272e+01	k_t
Generation 8670: Best = 8.487260e+04 = [1, 1, 1, 1]	FES = 867100	Diversity = 1.4532e+01	k_t
Generation 8680: Best = 8.475215e+04 = [4, 3, 2, 2]	FES = 868100	Diversity = 1.3053e+01	k_t
Generation 8690: Best = 8.470239e+04 = [1, 1, 1, 1]	FES = 869100	Diversity = 1.3169e+01	k_t
Generation 8700: Best = 8.466307e+04 = [4, 4, 3, 3]	FES = 870100	Diversity = 1.2833e+01	k_t
Generation 8710: Best = 8.462487e+04 = [1, 1, 1, 1]	FES = 871100	Diversity = 1.2255e+01	k_t
Generation 8720: Best = 8.455485e+04 = [4, 4, 2, 2]	FES = 872100	Diversity = 1.1959e+01	k_t
Generation 8730: Best = 8.452502e+04 = [1, 1, 1, 1]	FES = 873100	Diversity = 1.2080e+01	k_t
Generation 8740: Best = 8.446960e+04 = [4, 2, 4, 3]	FES = 874100	Diversity = 1.1249e+01	k_t
Generation 8750: Best = 8.442769e+04 = [1, 1, 1, 1]	FES = 875100	Diversity = 1.0630e+01	k_t
Generation 8760: Best = 8.437966e+04 = [4, 2, 3, 4]	FES = 876100	Diversity = 1.0075e+01	k_t
Generation 8770: Best = 8.433626e+04 = [1, 1, 1, 1]	FES = 877100	Diversity = 9.8989e+00	k_t
Generation 8780: Best = 8.431758e+04 = [4, 3, 2, 4]	FES = 878100	Diversity = 1.0744e+01	k_t
Generation 8790: Best = 8.428160e+04 = [1, 1, 1, 1]	FES = 879100	Diversity = 1.0927e+01	k_t
Generation 8800: Best = 8.421418e+04 = [2, 3, 3, 4]	FES = 880100	Diversity = 1.0772e+01	k_t
Generation 8810: Best = 8.416729e+04 = [1, 1, 1, 1]	FES = 881100	Diversity = 1.0858e+01	k_t
Generation 8820: Best = 8.410295e+04 = [4, 2, 3, 3]	FES = 882100	Diversity = 1.1346e+01	k_t
Generation 8830: Best = 8.397803e+04 = [1, 1, 1, 1]	FES = 883100	Diversity = 1.1971e+01	k_t
Generation 8840: Best = 8.394145e+04 = [4, 2, 3, 3]	FES = 884100	Diversity = 1.2884e+01	k_t
Generation 8850: Best = 8.386324e+04 = [1, 1, 1, 1]	FES = 885100	Diversity = 1.2892e+01	k_t
Generation 8860: Best = 8.376031e+04 = [3, 4, 4, 4]	FES = 886100	Diversity = 1.2172e+01	k_t

Generation 8870: Best = 8.372904e+04 = [1, 1, 1, 1]	FES = 887100	Diversity = 1.1735e+01	k_t
Generation 8880: Best = 8.364060e+04 = [3, 4, 3, 2]	FES = 888100	Diversity = 1.2229e+01	k_t
Generation 8890: Best = 8.357257e+04 = [1, 1, 1, 1]	FES = 889100	Diversity = 1.1665e+01	k_t
Generation 8900: Best = 8.357257e+04 = [2, 2, 2, 4]	FES = 890100	Diversity = 1.0955e+01	k_t
Generation 8910: Best = 8.351941e+04 = [1, 1, 1, 1]	FES = 891100	Diversity = 9.9663e+00	k_t
Generation 8920: Best = 8.349938e+04 = [2, 4, 2, 4]	FES = 892100	Diversity = 9.3570e+00	k_t
Generation 8930: Best = 8.344272e+04 = [1, 1, 1, 1]	FES = 893100	Diversity = 9.0164e+00	k_t
Generation 8940: Best = 8.344272e+04 = [4, 2, 2, 3]	FES = 894100	Diversity = 9.1446e+00	k_t
Generation 8950: Best = 8.341655e+04 = [1, 1, 1, 1]	FES = 895100	Diversity = 9.1292e+00	k_t
Generation 8960: Best = 8.339901e+04 = [4, 3, 4, 2]	FES = 896100	Diversity = 8.9559e+00	k_t
Generation 8970: Best = 8.335466e+04 = [1, 1, 1, 1]	FES = 897100	Diversity = 9.4569e+00	k_t
Generation 8980: Best = 8.329495e+04 = [2, 4, 2, 3]	FES = 898100	Diversity = 1.0294e+01	k_t
Generation 8990: Best = 8.326561e+04 = [1, 1, 1, 1]	FES = 899100	Diversity = 1.0604e+01	k_t
Generation 9000: Best = 8.321085e+04 = [2, 3, 3, 4]	FES = 900100	Diversity = 1.0941e+01	k_t
Generation 9010: Best = 8.316358e+04 = [1, 1, 1, 1]	FES = 901100	Diversity = 1.0506e+01	k_t
Generation 9020: Best = 8.307496e+04 = [3, 4, 3, 4]	FES = 902100	Diversity = 1.0357e+01	k_t
Generation 9030: Best = 8.301623e+04 = [1, 1, 1, 1]	FES = 903100	Diversity = 1.1133e+01	k_t
Generation 9040: Best = 8.293955e+04 = [3, 2, 4, 2]	FES = 904100	Diversity = 1.1008e+01	k_t
Generation 9050: Best = 8.291140e+04 = [1, 1, 1, 1]	FES = 905100	Diversity = 1.1505e+01	k_t
Generation 9060: Best = 8.280668e+04 = [3, 4, 4, 2]	FES = 906100	Diversity = 1.1655e+01	k_t
Generation 9070: Best = 8.269048e+04 = [1, 1, 1, 1]	FES = 907100	Diversity = 1.1824e+01	k_t
Generation 9080: Best = 8.265662e+04 = [4, 4, 2, 2]	FES = 908100	Diversity = 1.2263e+01	k_t
Generation 9090: Best = 8.256313e+04 = [1, 1, 1, 1]	FES = 909100	Diversity = 1.1868e+01	k_t
Generation 9100: Best = 8.241792e+04 = [3, 4, 4, 4]	FES = 910100	Diversity = 1.1311e+01	k_t
Generation 9110: Best = 8.236462e+04 = [1, 1, 1, 1]	FES = 911100	Diversity = 1.1348e+01	k_t
Generation 9120: Best = 8.232090e+04 = [4, 2, 2, 4]	FES = 912100	Diversity = 1.2065e+01	k_t
Generation 9130: Best = 8.228880e+04 = [1, 1, 1, 1]	FES = 913100	Diversity = 1.1042e+01	k_t

Generation 9140: Best = 8.222411e+04 = [2, 2, 3, 2]	FES = 914100	Diversity = 1.0748e+01	k_t
Generation 9150: Best = 8.212869e+04 = [1, 1, 1, 1]	FES = 915100	Diversity = 1.0463e+01	k_t
Generation 9160: Best = 8.210902e+04 = [2, 2, 4, 3]	FES = 916100	Diversity = 1.0484e+01	k_t
Generation 9170: Best = 8.198619e+04 = [1, 1, 1, 1]	FES = 917100	Diversity = 1.0074e+01	k_t
Generation 9180: Best = 8.192435e+04 = [3, 3, 3, 2]	FES = 918100	Diversity = 1.0270e+01	k_t
Generation 9190: Best = 8.183523e+04 = [1, 1, 1, 1]	FES = 919100	Diversity = 1.0871e+01	k_t
Generation 9200: Best = 8.173206e+04 = [4, 4, 4, 2]	FES = 920100	Diversity = 1.2905e+01	k_t
Generation 9210: Best = 8.162847e+04 = [1, 1, 1, 1]	FES = 921100	Diversity = 1.2095e+01	k_t
Generation 9220: Best = 8.159228e+04 = [4, 4, 4, 4]	FES = 922100	Diversity = 1.2061e+01	k_t
Generation 9230: Best = 8.152418e+04 = [1, 1, 1, 1]	FES = 923100	Diversity = 1.1734e+01	k_t
Generation 9240: Best = 8.146435e+04 = [2, 2, 4, 2]	FES = 924100	Diversity = 1.0918e+01	k_t
Generation 9250: Best = 8.146435e+04 = [1, 1, 1, 1]	FES = 925100	Diversity = 1.0282e+01	k_t
Generation 9260: Best = 8.139668e+04 = [2, 4, 2, 4]	FES = 926100	Diversity = 1.0539e+01	k_t
Generation 9270: Best = 8.134455e+04 = [1, 1, 1, 1]	FES = 927100	Diversity = 1.0777e+01	k_t
Generation 9280: Best = 8.127584e+04 = [2, 3, 3, 3]	FES = 928100	Diversity = 1.0629e+01	k_t
Generation 9290: Best = 8.119894e+04 = [1, 1, 1, 1]	FES = 929100	Diversity = 1.1012e+01	k_t
Generation 9300: Best = 8.102195e+04 = [4, 3, 4, 3]	FES = 930100	Diversity = 1.2825e+01	k_t
Generation 9310: Best = 8.098169e+04 = [1, 1, 1, 1]	FES = 931100	Diversity = 1.3509e+01	k_t
Generation 9320: Best = 8.076088e+04 = [4, 2, 4, 2]	FES = 932100	Diversity = 1.2457e+01	k_t
Generation 9330: Best = 8.066299e+04 = [1, 1, 1, 1]	FES = 933100	Diversity = 1.2900e+01	k_t
Generation 9340: Best = 8.063887e+04 = [4, 4, 4, 2]	FES = 934100	Diversity = 1.2248e+01	k_t
Generation 9350: Best = 8.055840e+04 = [1, 1, 1, 1]	FES = 935100	Diversity = 1.1757e+01	k_t
Generation 9360: Best = 8.049456e+04 = [4, 3, 4, 3]	FES = 936100	Diversity = 1.0921e+01	k_t
Generation 9370: Best = 8.049456e+04 = [1, 1, 1, 1]	FES = 937100	Diversity = 9.9440e+00	k_t
Generation 9380: Best = 8.039910e+04 = [4, 3, 2, 2]	FES = 938100	Diversity = 9.8908e+00	k_t
Generation 9390: Best = 8.039031e+04 = [1, 1, 1, 1]	FES = 939100	Diversity = 9.8247e+00	k_t
Generation 9400: Best = 8.025627e+04 = [2, 4, 4, 4]	FES = 940100	Diversity = 1.0448e+01	k_t

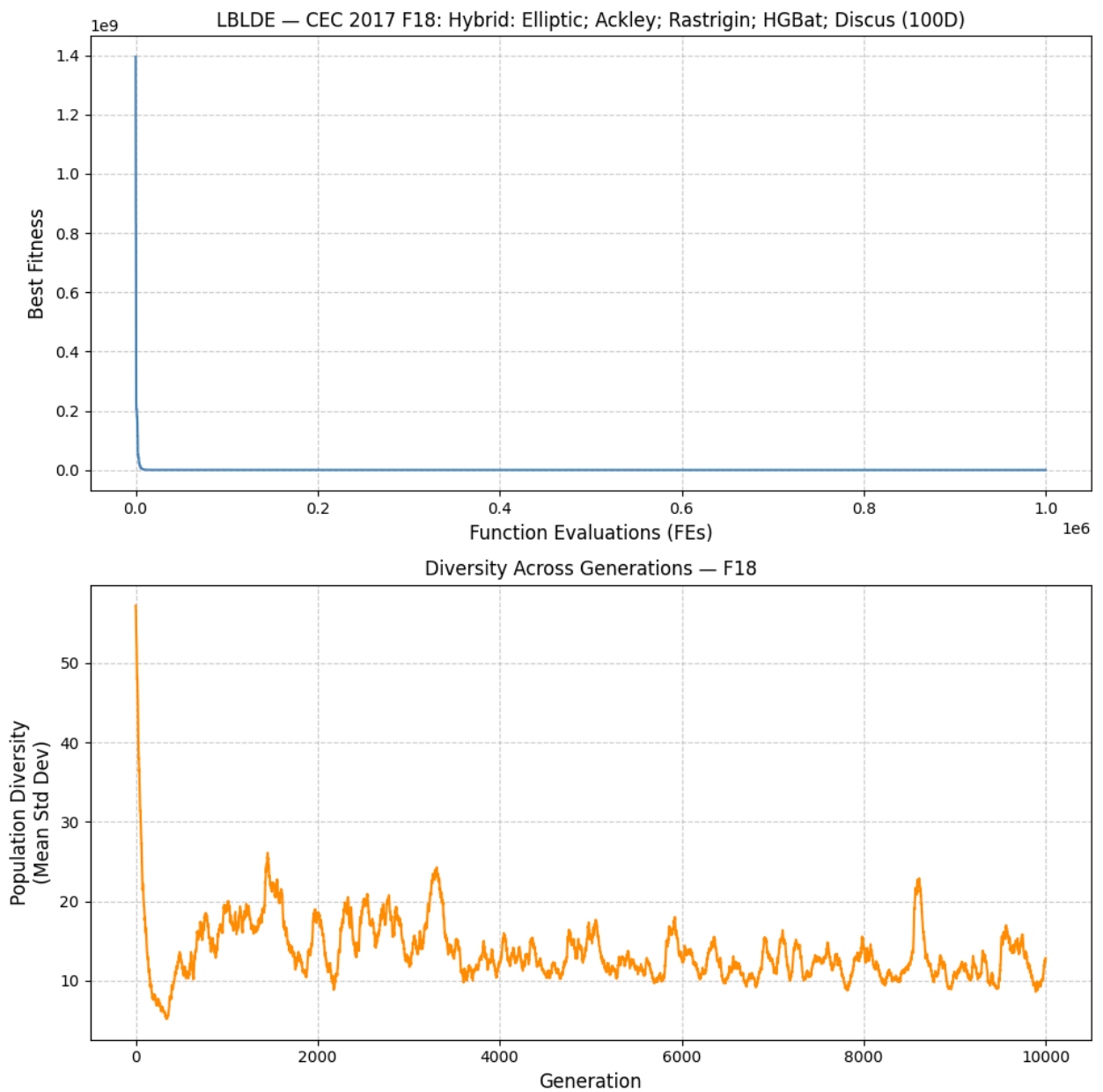
Generation 9410: Best = 8.025627e+04 = [1, 1, 1, 1]	FES = 941100	Diversity = 1.0988e+01	k_t
Generation 9420: Best = 8.024527e+04 = [4, 3, 4, 2]	FES = 942100	Diversity = 1.0256e+01	k_t
Generation 9430: Best = 8.019733e+04 = [1, 1, 1, 1]	FES = 943100	Diversity = 9.6768e+00	k_t
Generation 9440: Best = 8.016554e+04 = [2, 4, 2, 4]	FES = 944100	Diversity = 9.3657e+00	k_t
Generation 9450: Best = 8.016458e+04 = [1, 1, 1, 1]	FES = 945100	Diversity = 9.2513e+00	k_t
Generation 9460: Best = 8.012817e+04 = [3, 2, 3, 2]	FES = 946100	Diversity = 9.1580e+00	k_t
Generation 9470: Best = 8.010440e+04 = [1, 1, 1, 1]	FES = 947100	Diversity = 9.0032e+00	k_t
Generation 9480: Best = 8.003592e+04 = [4, 4, 2, 4]	FES = 948100	Diversity = 9.3980e+00	k_t
Generation 9490: Best = 7.997290e+04 = [1, 1, 1, 1]	FES = 949100	Diversity = 1.0370e+01	k_t
Generation 9500: Best = 7.983659e+04 = [4, 4, 3, 2]	FES = 950100	Diversity = 1.1962e+01	k_t
Generation 9510: Best = 7.968642e+04 = [1, 1, 1, 1]	FES = 951100	Diversity = 1.4550e+01	k_t
Generation 9520: Best = 7.955827e+04 = [3, 4, 4, 4]	FES = 952100	Diversity = 1.4914e+01	k_t
Generation 9530: Best = 7.947334e+04 = [1, 1, 1, 1]	FES = 953100	Diversity = 1.4748e+01	k_t
Generation 9540: Best = 7.929230e+04 = [2, 4, 2, 3]	FES = 954100	Diversity = 1.5973e+01	k_t
Generation 9550: Best = 7.914026e+04 = [1, 1, 1, 1]	FES = 955100	Diversity = 1.5343e+01	k_t
Generation 9560: Best = 7.881072e+04 = [3, 4, 3, 2]	FES = 956100	Diversity = 1.6515e+01	k_t
Generation 9570: Best = 7.867347e+04 = [1, 1, 1, 1]	FES = 957100	Diversity = 1.6512e+01	k_t
Generation 9580: Best = 7.861839e+04 = [2, 4, 4, 4]	FES = 958100	Diversity = 1.5950e+01	k_t
Generation 9590: Best = 7.833472e+04 = [1, 1, 1, 1]	FES = 959100	Diversity = 1.5182e+01	k_t
Generation 9600: Best = 7.818322e+04 = [4, 4, 3, 4]	FES = 960100	Diversity = 1.4755e+01	k_t
Generation 9610: Best = 7.801828e+04 = [1, 1, 1, 1]	FES = 961100	Diversity = 1.4051e+01	k_t
Generation 9620: Best = 7.780850e+04 = [2, 3, 3, 4]	FES = 962100	Diversity = 1.4222e+01	k_t
Generation 9630: Best = 7.773592e+04 = [1, 1, 1, 1]	FES = 963100	Diversity = 1.3758e+01	k_t
Generation 9640: Best = 7.729469e+04 = [2, 2, 3, 3]	FES = 964100	Diversity = 1.3648e+01	k_t
Generation 9650: Best = 7.716376e+04 = [1, 1, 1, 1]	FES = 965100	Diversity = 1.3516e+01	k_t
Generation 9660: Best = 7.685213e+04 = [3, 4, 4, 2]	FES = 966100	Diversity = 1.4331e+01	k_t
Generation 9670: Best = 7.662950e+04 = [1, 1, 1, 1]	FES = 967100	Diversity = 1.4954e+01	k_t

Generation 9680: Best = 7.640187e+04 = [4, 2, 2, 4]	FES = 968100	Diversity = 1.4615e+01	k_t
Generation 9690: Best = 7.628672e+04 = [1, 1, 1, 1]	FES = 969100	Diversity = 1.5186e+01	k_t
Generation 9700: Best = 7.608451e+04 = [3, 3, 4, 2]	FES = 970100	Diversity = 1.3496e+01	k_t
Generation 9710: Best = 7.583273e+04 = [1, 1, 1, 1]	FES = 971100	Diversity = 1.3937e+01	k_t
Generation 9720: Best = 7.575858e+04 = [3, 4, 3, 4]	FES = 972100	Diversity = 1.4320e+01	k_t
Generation 9730: Best = 7.555095e+04 = [1, 1, 1, 1]	FES = 973100	Diversity = 1.4949e+01	k_t
Generation 9740: Best = 7.509950e+04 = [3, 2, 4, 2]	FES = 974100	Diversity = 1.5726e+01	k_t
Generation 9750: Best = 7.505811e+04 = [1, 1, 1, 1]	FES = 975100	Diversity = 1.4693e+01	k_t
Generation 9760: Best = 7.491996e+04 = [3, 4, 2, 4]	FES = 976100	Diversity = 1.3322e+01	k_t
Generation 9770: Best = 7.462203e+04 = [1, 1, 1, 1]	FES = 977100	Diversity = 1.2938e+01	k_t
Generation 9780: Best = 7.431658e+04 = [4, 3, 2, 3]	FES = 978100	Diversity = 1.3542e+01	k_t
Generation 9790: Best = 7.412142e+04 = [1, 1, 1, 1]	FES = 979100	Diversity = 1.3092e+01	k_t
Generation 9800: Best = 7.399963e+04 = [3, 3, 2, 3]	FES = 980100	Diversity = 1.2237e+01	k_t
Generation 9810: Best = 7.378726e+04 = [1, 1, 1, 1]	FES = 981100	Diversity = 1.1642e+01	k_t
Generation 9820: Best = 7.364900e+04 = [3, 2, 3, 3]	FES = 982100	Diversity = 1.1713e+01	k_t
Generation 9830: Best = 7.349413e+04 = [1, 1, 1, 1]	FES = 983100	Diversity = 1.1099e+01	k_t
Generation 9840: Best = 7.344644e+04 = [3, 2, 3, 2]	FES = 984100	Diversity = 1.0521e+01	k_t
Generation 9850: Best = 7.337833e+04 = [1, 1, 1, 1]	FES = 985100	Diversity = 1.0651e+01	k_t
Generation 9860: Best = 7.327453e+04 = [3, 4, 3, 4]	FES = 986100	Diversity = 9.5137e+00	k_t
Generation 9870: Best = 7.320275e+04 = [1, 1, 1, 1]	FES = 987100	Diversity = 9.5807e+00	k_t
Generation 9880: Best = 7.315928e+04 = [3, 3, 2, 2]	FES = 988100	Diversity = 9.4226e+00	k_t
Generation 9890: Best = 7.304836e+04 = [1, 1, 1, 1]	FES = 989100	Diversity = 8.5891e+00	k_t
Generation 9900: Best = 7.300769e+04 = [3, 2, 2, 2]	FES = 990100	Diversity = 9.8916e+00	k_t
Generation 9910: Best = 7.293048e+04 = [1, 1, 1, 1]	FES = 991100	Diversity = 8.8101e+00	k_t
Generation 9920: Best = 7.288737e+04 = [2, 2, 2, 2]	FES = 992100	Diversity = 9.3491e+00	k_t
Generation 9930: Best = 7.275077e+04 = [1, 1, 1, 1]	FES = 993100	Diversity = 9.7703e+00	k_t
Generation 9940: Best = 7.274707e+04 = [3, 3, 4, 4]	FES = 994100	Diversity = 9.3083e+00	k_t

Generation 9950: Best = 7.269514e+04 FES = 995100 Diversity = 9.8336e+00 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 7.259137e+04 FES = 996100 Diversity = 9.9178e+00 k_t
= [4, 3, 4, 4]
Generation 9970: Best = 7.242787e+04 FES = 997100 Diversity = 1.0504e+01 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 7.229769e+04 FES = 998100 Diversity = 1.1682e+01 k_t
= [4, 4, 2, 3]
Generation 9990: Best = 7.218659e+04 FES = 999100 Diversity = 1.2208e+01 k_t
= [1, 1, 1, 1]

Optimization completed!
Final Best Fitness = 7.212491e+04
Total Function Evaluations = 1000000

=====
=
Results:
Best Fitness : 7.212491e+04
Global Optimum : 1.800000e+03
Error (|best-optimum|) : 7.032491e+04
Best Solution (5 dim) : [-2.78707411 -17.18676095 29.66315523 -43.51792169
6.55145982]
D^(0) (init. diversity): 5.7220e+01
=====



Graph saved as convergence_F18.png

```
In [43]: # =====
# F19 – Hybrid Function 9 (Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierst
# =====
run_and_plot('F19', make_wrapper(hybrid.f19), 1900.0, 'Hybrid: Bent Cigar; Ras
```

```

=====
=
LBLDE_s on CEC 2017 F19 – Single Run
=====
=
Function      : Hybrid: Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstras
s; Schaffer (100D)
Bounds       : [-100, 100]
Global optimum : 1900.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 4.192174e+11 | Diversity = 5.7220e+01
Generation 10: Best = 1.615010e+11 FES = 1100 Diversity = 4.2684e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.613015e+10 FES = 2100 Diversity = 3.6652e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 8.015113e+09 FES = 3100 Diversity = 3.2276e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.552453e+09 FES = 4100 Diversity = 2.6712e+01 k_t
= [4, 3, 4, 2]
Generation 50: Best = 4.367560e+08 FES = 5100 Diversity = 2.3897e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 2.252686e+08 FES = 6100 Diversity = 2.2370e+01 k_t
= [4, 3, 4, 3]
Generation 70: Best = 6.489205e+07 FES = 7100 Diversity = 2.1428e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.863053e+07 FES = 8100 Diversity = 1.9837e+01 k_t
= [4, 3, 2, 3]
Generation 90: Best = 6.241380e+06 FES = 9100 Diversity = 1.8040e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 2.591865e+06 FES = 10100 Diversity = 1.5584e+01 k_t
= [4, 2, 4, 3]
Generation 110: Best = 1.000434e+06 FES = 11100 Diversity = 1.3433e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 5.213089e+05 FES = 12100 Diversity = 1.1417e+01 k_t
= [2, 2, 3, 2]
Generation 130: Best = 3.098141e+05 FES = 13100 Diversity = 9.4855e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.376753e+05 FES = 14100 Diversity = 8.0199e+00 k_t
= [2, 3, 2, 3]
Generation 150: Best = 6.140824e+04 FES = 15100 Diversity = 7.2239e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.497764e+04 FES = 16100 Diversity = 6.3279e+00 k_t
= [3, 4, 2, 3]
Generation 170: Best = 2.449471e+04 FES = 17100 Diversity = 5.0459e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.616181e+04 FES = 18100 Diversity = 4.8957e+00 k_t
= [3, 3, 3, 2]
Generation 190: Best = 1.156284e+04 FES = 19100 Diversity = 5.5861e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 8.258972e+03 FES = 20100 Diversity = 6.3692e+00 k_t
= [4, 4, 4, 2]
Generation 210: Best = 6.448689e+03 FES = 21100 Diversity = 6.6912e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 5.749892e+03 FES = 22100 Diversity = 7.4382e+00 k_t

```



```

= [3, 4, 3, 3]
Generation 230: Best = 5.323451e+03 FES = 23100 Diversity = 8.5886e+00 k_t
= [1, 1, 1, 1]
Generation 240: Best = 5.107179e+03 FES = 24100 Diversity = 8.6315e+00 k_t
= [3, 4, 4, 3]
Generation 250: Best = 5.011255e+03 FES = 25100 Diversity = 8.8003e+00 k_t
= [1, 1, 1, 1]
Generation 260: Best = 4.954710e+03 FES = 26100 Diversity = 8.8146e+00 k_t
= [2, 2, 4, 3]
Generation 270: Best = 4.935797e+03 FES = 27100 Diversity = 8.5170e+00 k_t
= [1, 1, 1, 1]
Generation 280: Best = 4.905100e+03 FES = 28100 Diversity = 8.3386e+00 k_t
= [3, 2, 4, 4]
Generation 290: Best = 4.882669e+03 FES = 29100 Diversity = 8.1892e+00 k_t
= [1, 1, 1, 1]
Generation 300: Best = 4.876150e+03 FES = 30100 Diversity = 7.9161e+00 k_t
= [2, 3, 4, 4]
Generation 310: Best = 4.849266e+03 FES = 31100 Diversity = 7.7827e+00 k_t
= [1, 1, 1, 1]
Generation 320: Best = 4.846948e+03 FES = 32100 Diversity = 7.5870e+00 k_t
= [3, 3, 3, 2]
Generation 330: Best = 4.841365e+03 FES = 33100 Diversity = 7.2126e+00 k_t
= [1, 1, 1, 1]
Generation 340: Best = 4.841275e+03 FES = 34100 Diversity = 7.0968e+00 k_t
= [4, 2, 3, 4]
Generation 350: Best = 4.812736e+03 FES = 35100 Diversity = 6.9002e+00 k_t
= [1, 1, 1, 1]
Generation 360: Best = 4.808480e+03 FES = 36100 Diversity = 6.7618e+00 k_t
= [2, 3, 3, 4]
Generation 370: Best = 4.807549e+03 FES = 37100 Diversity = 6.7165e+00 k_t
= [1, 1, 1, 1]
Generation 380: Best = 4.807549e+03 FES = 38100 Diversity = 6.6732e+00 k_t
= [3, 3, 4, 2]
Generation 390: Best = 4.793570e+03 FES = 39100 Diversity = 6.6894e+00 k_t
= [1, 1, 1, 1]
Generation 400: Best = 4.780005e+03 FES = 40100 Diversity = 6.6314e+00 k_t
= [2, 2, 2, 4]
Generation 410: Best = 4.780005e+03 FES = 41100 Diversity = 6.5187e+00 k_t
= [1, 1, 1, 1]
Generation 420: Best = 4.776623e+03 FES = 42100 Diversity = 6.5096e+00 k_t
= [3, 4, 2, 3]
Generation 430: Best = 4.776444e+03 FES = 43100 Diversity = 6.5239e+00 k_t
= [1, 1, 1, 1]
Generation 440: Best = 4.776444e+03 FES = 44100 Diversity = 6.4765e+00 k_t
= [3, 2, 3, 3]
Generation 450: Best = 4.769531e+03 FES = 45100 Diversity = 6.4480e+00 k_t
= [1, 1, 1, 1]
Generation 460: Best = 4.766907e+03 FES = 46100 Diversity = 6.3886e+00 k_t
= [4, 2, 3, 4]
Generation 470: Best = 4.766907e+03 FES = 47100 Diversity = 6.3436e+00 k_t
= [1, 1, 1, 1]
Generation 480: Best = 4.764506e+03 FES = 48100 Diversity = 6.3904e+00 k_t
= [2, 2, 4, 2]
Generation 490: Best = 4.761027e+03 FES = 49100 Diversity = 6.2744e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 4.761027e+03 FES = 50100 Diversity = 6.3838e+00 k_t
= [2, 2, 4, 2]
Generation 510: Best = 4.758913e+03 FES = 51100 Diversity = 6.2714e+00 k_t
= [1, 1, 1, 1]
Generation 520: Best = 4.748087e+03 FES = 52100 Diversity = 6.3809e+00 k_t
= [3, 3, 2, 3]
Generation 530: Best = 4.747566e+03 FES = 53100 Diversity = 6.4744e+00 k_t
= [1, 1, 1, 1]
Generation 540: Best = 4.747268e+03 FES = 54100 Diversity = 6.4509e+00 k_t
= [4, 4, 2, 2]
Generation 550: Best = 4.747268e+03 FES = 55100 Diversity = 6.5106e+00 k_t
= [1, 1, 1, 1]
Generation 560: Best = 4.741209e+03 FES = 56100 Diversity = 6.5286e+00 k_t
= [2, 4, 4, 4]
Generation 570: Best = 4.740399e+03 FES = 57100 Diversity = 6.6140e+00 k_t
= [1, 1, 1, 1]
Generation 580: Best = 4.738809e+03 FES = 58100 Diversity = 6.7501e+00 k_t
= [4, 3, 3, 3]
Generation 590: Best = 4.737429e+03 FES = 59100 Diversity = 6.8781e+00 k_t
= [1, 1, 1, 1]
Generation 600: Best = 4.731285e+03 FES = 60100 Diversity = 6.8608e+00 k_t
= [4, 4, 4, 4]
Generation 610: Best = 4.728579e+03 FES = 61100 Diversity = 6.9105e+00 k_t
= [1, 1, 1, 1]
Generation 620: Best = 4.724884e+03 FES = 62100 Diversity = 6.8982e+00 k_t
= [2, 3, 4, 4]
Generation 630: Best = 4.721930e+03 FES = 63100 Diversity = 6.8787e+00 k_t
= [1, 1, 1, 1]
Generation 640: Best = 4.721930e+03 FES = 64100 Diversity = 6.8817e+00 k_t
= [4, 3, 4, 2]
Generation 650: Best = 4.721930e+03 FES = 65100 Diversity = 6.9896e+00 k_t
= [1, 1, 1, 1]
Generation 660: Best = 4.721930e+03 FES = 66100 Diversity = 6.8380e+00 k_t
= [3, 3, 2, 2]
Generation 670: Best = 4.721930e+03 FES = 67100 Diversity = 6.8413e+00 k_t
= [1, 1, 1, 1]
Generation 680: Best = 4.721563e+03 FES = 68100 Diversity = 6.8427e+00 k_t
= [2, 3, 4, 4]
Generation 690: Best = 4.719275e+03 FES = 69100 Diversity = 6.8332e+00 k_t
= [1, 1, 1, 1]
Generation 700: Best = 4.719275e+03 FES = 70100 Diversity = 6.9169e+00 k_t
= [2, 4, 4, 3]
Generation 710: Best = 4.718094e+03 FES = 71100 Diversity = 6.9991e+00 k_t
= [1, 1, 1, 1]
Generation 720: Best = 4.717895e+03 FES = 72100 Diversity = 6.9778e+00 k_t
= [2, 4, 3, 2]
Generation 730: Best = 4.714809e+03 FES = 73100 Diversity = 7.0406e+00 k_t
= [1, 1, 1, 1]
Generation 740: Best = 4.714809e+03 FES = 74100 Diversity = 7.0743e+00 k_t
= [2, 2, 2, 3]
Generation 750: Best = 4.714505e+03 FES = 75100 Diversity = 7.0592e+00 k_t
= [1, 1, 1, 1]
Generation 760: Best = 4.714505e+03 FES = 76100 Diversity = 7.0920e+00 k_t

```

```

= [4, 2, 4, 4]
Generation 770: Best = 4.708519e+03 FES = 77100 Diversity = 7.0987e+00 k_t
= [1, 1, 1, 1]
Generation 780: Best = 4.706502e+03 FES = 78100 Diversity = 7.1770e+00 k_t
= [2, 3, 2, 3]
Generation 790: Best = 4.705705e+03 FES = 79100 Diversity = 7.1572e+00 k_t
= [1, 1, 1, 1]
Generation 800: Best = 4.705705e+03 FES = 80100 Diversity = 7.1997e+00 k_t
= [3, 4, 3, 4]
Generation 810: Best = 4.705705e+03 FES = 81100 Diversity = 7.2598e+00 k_t
= [1, 1, 1, 1]
Generation 820: Best = 4.703775e+03 FES = 82100 Diversity = 7.2614e+00 k_t
= [3, 3, 3, 2]
Generation 830: Best = 4.701004e+03 FES = 83100 Diversity = 7.2889e+00 k_t
= [1, 1, 1, 1]
Generation 840: Best = 4.701004e+03 FES = 84100 Diversity = 7.2589e+00 k_t
= [2, 4, 4, 4]
Generation 850: Best = 4.701004e+03 FES = 85100 Diversity = 7.2115e+00 k_t
= [1, 1, 1, 1]
Generation 860: Best = 4.695026e+03 FES = 86100 Diversity = 7.1348e+00 k_t
= [4, 3, 3, 2]
Generation 870: Best = 4.694857e+03 FES = 87100 Diversity = 7.1413e+00 k_t
= [1, 1, 1, 1]
Generation 880: Best = 4.694669e+03 FES = 88100 Diversity = 7.2606e+00 k_t
= [2, 4, 3, 2]
Generation 890: Best = 4.694669e+03 FES = 89100 Diversity = 7.1930e+00 k_t
= [1, 1, 1, 1]
Generation 900: Best = 4.691795e+03 FES = 90100 Diversity = 7.1625e+00 k_t
= [2, 2, 2, 2]
Generation 910: Best = 4.691795e+03 FES = 91100 Diversity = 7.1975e+00 k_t
= [1, 1, 1, 1]
Generation 920: Best = 4.691436e+03 FES = 92100 Diversity = 7.3180e+00 k_t
= [2, 2, 4, 3]
Generation 930: Best = 4.690171e+03 FES = 93100 Diversity = 7.3514e+00 k_t
= [1, 1, 1, 1]
Generation 940: Best = 4.690171e+03 FES = 94100 Diversity = 7.4834e+00 k_t
= [4, 2, 4, 3]
Generation 950: Best = 4.689081e+03 FES = 95100 Diversity = 7.5153e+00 k_t
= [1, 1, 1, 1]
Generation 960: Best = 4.689081e+03 FES = 96100 Diversity = 7.5606e+00 k_t
= [3, 3, 2, 4]
Generation 970: Best = 4.689081e+03 FES = 97100 Diversity = 7.6390e+00 k_t
= [1, 1, 1, 1]
Generation 980: Best = 4.687545e+03 FES = 98100 Diversity = 7.5249e+00 k_t
= [2, 3, 2, 4]
Generation 990: Best = 4.686040e+03 FES = 99100 Diversity = 7.5945e+00 k_t
= [1, 1, 1, 1]
Generation 1000: Best = 4.686040e+03 FES = 100100 Diversity = 7.5270e+00 k_t
= [3, 3, 4, 2]
Generation 1010: Best = 4.684912e+03 FES = 101100 Diversity = 7.5579e+00 k_t
= [1, 1, 1, 1]
Generation 1020: Best = 4.684612e+03 FES = 102100 Diversity = 7.6357e+00 k_t
= [2, 3, 3, 3]
Generation 1030: Best = 4.684188e+03 FES = 103100 Diversity = 7.6771e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 1040: Best = 4.684188e+03 FES = 104100 Diversity = 7.7341e+00 k_t
= [4, 3, 2, 2]
Generation 1050: Best = 4.683460e+03 FES = 105100 Diversity = 7.5791e+00 k_t
= [1, 1, 1, 1]
Generation 1060: Best = 4.682947e+03 FES = 106100 Diversity = 7.5771e+00 k_t
= [4, 3, 2, 2]
Generation 1070: Best = 4.681676e+03 FES = 107100 Diversity = 7.5847e+00 k_t
= [1, 1, 1, 1]
Generation 1080: Best = 4.681644e+03 FES = 108100 Diversity = 7.5992e+00 k_t
= [2, 4, 2, 2]
Generation 1090: Best = 4.681302e+03 FES = 109100 Diversity = 7.6071e+00 k_t
= [1, 1, 1, 1]
Generation 1100: Best = 4.678394e+03 FES = 110100 Diversity = 7.5035e+00 k_t
= [3, 4, 3, 2]
Generation 1110: Best = 4.678394e+03 FES = 111100 Diversity = 7.5215e+00 k_t
= [1, 1, 1, 1]
Generation 1120: Best = 4.676521e+03 FES = 112100 Diversity = 7.5881e+00 k_t
= [2, 3, 2, 2]
Generation 1130: Best = 4.675786e+03 FES = 113100 Diversity = 7.6055e+00 k_t
= [1, 1, 1, 1]
Generation 1140: Best = 4.674368e+03 FES = 114100 Diversity = 7.6118e+00 k_t
= [4, 3, 3, 4]
Generation 1150: Best = 4.674368e+03 FES = 115100 Diversity = 7.4657e+00 k_t
= [1, 1, 1, 1]
Generation 1160: Best = 4.673058e+03 FES = 116100 Diversity = 7.4233e+00 k_t
= [4, 2, 2, 4]
Generation 1170: Best = 4.672345e+03 FES = 117100 Diversity = 7.2339e+00 k_t
= [1, 1, 1, 1]
Generation 1180: Best = 4.672303e+03 FES = 118100 Diversity = 7.0965e+00 k_t
= [2, 2, 3, 4]
Generation 1190: Best = 4.672303e+03 FES = 119100 Diversity = 6.9794e+00 k_t
= [1, 1, 1, 1]
Generation 1200: Best = 4.671548e+03 FES = 120100 Diversity = 6.7668e+00 k_t
= [3, 4, 4, 4]
Generation 1210: Best = 4.671492e+03 FES = 121100 Diversity = 6.4478e+00 k_t
= [1, 1, 1, 1]
Generation 1220: Best = 4.670474e+03 FES = 122100 Diversity = 6.3852e+00 k_t
= [3, 2, 2, 2]
Generation 1230: Best = 4.670242e+03 FES = 123100 Diversity = 6.3294e+00 k_t
= [1, 1, 1, 1]
Generation 1240: Best = 4.670065e+03 FES = 124100 Diversity = 6.2771e+00 k_t
= [3, 4, 4, 4]
Generation 1250: Best = 4.669379e+03 FES = 125100 Diversity = 6.2902e+00 k_t
= [1, 1, 1, 1]
Generation 1260: Best = 4.667613e+03 FES = 126100 Diversity = 6.3221e+00 k_t
= [3, 3, 3, 4]
Generation 1270: Best = 4.667613e+03 FES = 127100 Diversity = 6.1928e+00 k_t
= [1, 1, 1, 1]
Generation 1280: Best = 4.667613e+03 FES = 128100 Diversity = 6.2597e+00 k_t
= [4, 3, 2, 4]
Generation 1290: Best = 4.667613e+03 FES = 129100 Diversity = 6.2963e+00 k_t
= [1, 1, 1, 1]
Generation 1300: Best = 4.667483e+03 FES = 130100 Diversity = 6.3216e+00 k_t

```

```

= [3, 3, 3, 2]
Generation 1310: Best = 4.667483e+03  FES = 131100  Diversity = 6.2703e+00  k_t
= [1, 1, 1, 1]
Generation 1320: Best = 4.665815e+03  FES = 132100  Diversity = 6.1315e+00  k_t
= [3, 4, 4, 4]
Generation 1330: Best = 4.665815e+03  FES = 133100  Diversity = 6.0622e+00  k_t
= [1, 1, 1, 1]
Generation 1340: Best = 4.665815e+03  FES = 134100  Diversity = 5.9903e+00  k_t
= [2, 3, 2, 4]
Generation 1350: Best = 4.665200e+03  FES = 135100  Diversity = 5.9562e+00  k_t
= [1, 1, 1, 1]
Generation 1360: Best = 4.664720e+03  FES = 136100  Diversity = 5.9520e+00  k_t
= [2, 3, 4, 2]
Generation 1370: Best = 4.664705e+03  FES = 137100  Diversity = 5.9270e+00  k_t
= [1, 1, 1, 1]
Generation 1380: Best = 4.664670e+03  FES = 138100  Diversity = 5.9242e+00  k_t
= [2, 2, 3, 4]
Generation 1390: Best = 4.664356e+03  FES = 139100  Diversity = 5.9737e+00  k_t
= [1, 1, 1, 1]
Generation 1400: Best = 4.663969e+03  FES = 140100  Diversity = 6.0452e+00  k_t
= [2, 3, 4, 2]
Generation 1410: Best = 4.662590e+03  FES = 141100  Diversity = 6.0291e+00  k_t
= [1, 1, 1, 1]
Generation 1420: Best = 4.662590e+03  FES = 142100  Diversity = 5.9858e+00  k_t
= [3, 2, 3, 3]
Generation 1430: Best = 4.662590e+03  FES = 143100  Diversity = 5.9433e+00  k_t
= [1, 1, 1, 1]
Generation 1440: Best = 4.662590e+03  FES = 144100  Diversity = 5.9269e+00  k_t
= [2, 3, 4, 2]
Generation 1450: Best = 4.662428e+03  FES = 145100  Diversity = 5.9730e+00  k_t
= [1, 1, 1, 1]
Generation 1460: Best = 4.662428e+03  FES = 146100  Diversity = 5.9788e+00  k_t
= [2, 4, 4, 2]
Generation 1470: Best = 4.662414e+03  FES = 147100  Diversity = 6.0117e+00  k_t
= [1, 1, 1, 1]
Generation 1480: Best = 4.662056e+03  FES = 148100  Diversity = 6.0127e+00  k_t
= [3, 3, 2, 4]
Generation 1490: Best = 4.662048e+03  FES = 149100  Diversity = 5.9641e+00  k_t
= [1, 1, 1, 1]
Generation 1500: Best = 4.662048e+03  FES = 150100  Diversity = 5.9597e+00  k_t
= [3, 3, 2, 3]
Generation 1510: Best = 4.662007e+03  FES = 151100  Diversity = 5.9302e+00  k_t
= [1, 1, 1, 1]
Generation 1520: Best = 4.661910e+03  FES = 152100  Diversity = 5.9051e+00  k_t
= [4, 2, 3, 3]
Generation 1530: Best = 4.661202e+03  FES = 153100  Diversity = 5.8916e+00  k_t
= [1, 1, 1, 1]
Generation 1540: Best = 4.661185e+03  FES = 154100  Diversity = 5.8452e+00  k_t
= [2, 2, 3, 4]
Generation 1550: Best = 4.661112e+03  FES = 155100  Diversity = 5.7583e+00  k_t
= [1, 1, 1, 1]
Generation 1560: Best = 4.660859e+03  FES = 156100  Diversity = 5.6771e+00  k_t
= [2, 3, 2, 4]
Generation 1570: Best = 4.660680e+03  FES = 157100  Diversity = 5.6723e+00  k_t

```

```

= [1, 1, 1, 1]
Generation 1580: Best = 4.660680e+03 FES = 158100 Diversity = 5.6944e+00 k_t
= [2, 3, 2, 3]
Generation 1590: Best = 4.660680e+03 FES = 159100 Diversity = 5.6694e+00 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 4.660680e+03 FES = 160100 Diversity = 5.6952e+00 k_t
= [4, 4, 2, 4]
Generation 1610: Best = 4.660490e+03 FES = 161100 Diversity = 5.7274e+00 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 4.660490e+03 FES = 162100 Diversity = 5.7249e+00 k_t
= [4, 3, 3, 2]
Generation 1630: Best = 4.659666e+03 FES = 163100 Diversity = 5.7571e+00 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 4.659666e+03 FES = 164100 Diversity = 5.7389e+00 k_t
= [4, 3, 3, 2]
Generation 1650: Best = 4.658689e+03 FES = 165100 Diversity = 5.7470e+00 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 4.658689e+03 FES = 166100 Diversity = 5.7783e+00 k_t
= [2, 3, 4, 2]
Generation 1670: Best = 4.658689e+03 FES = 167100 Diversity = 5.7191e+00 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 4.658673e+03 FES = 168100 Diversity = 5.6940e+00 k_t
= [3, 3, 4, 3]
Generation 1690: Best = 4.658426e+03 FES = 169100 Diversity = 5.7074e+00 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 4.658426e+03 FES = 170100 Diversity = 5.6753e+00 k_t
= [3, 4, 4, 2]
Generation 1710: Best = 4.658307e+03 FES = 171100 Diversity = 5.6510e+00 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 4.658307e+03 FES = 172100 Diversity = 5.6827e+00 k_t
= [3, 2, 2, 3]
Generation 1730: Best = 4.658248e+03 FES = 173100 Diversity = 5.6774e+00 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 4.658119e+03 FES = 174100 Diversity = 5.6803e+00 k_t
= [2, 3, 3, 3]
Generation 1750: Best = 4.657470e+03 FES = 175100 Diversity = 5.7371e+00 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 4.657470e+03 FES = 176100 Diversity = 5.7457e+00 k_t
= [3, 4, 2, 4]
Generation 1770: Best = 4.657468e+03 FES = 177100 Diversity = 5.7275e+00 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 4.657461e+03 FES = 178100 Diversity = 5.7850e+00 k_t
= [2, 4, 2, 4]
Generation 1790: Best = 4.657114e+03 FES = 179100 Diversity = 5.7717e+00 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 4.656635e+03 FES = 180100 Diversity = 5.7887e+00 k_t
= [2, 2, 2, 2]
Generation 1810: Best = 4.656062e+03 FES = 181100 Diversity = 5.7867e+00 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 4.656062e+03 FES = 182100 Diversity = 5.8168e+00 k_t
= [3, 3, 3, 2]
Generation 1830: Best = 4.656062e+03 FES = 183100 Diversity = 5.7445e+00 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 4.655888e+03 FES = 184100 Diversity = 5.7960e+00 k_t

```

```

= [3, 3, 4, 4]
Generation 1850: Best = 4.655888e+03  FES = 185100  Diversity = 5.7727e+00  k_t
= [1, 1, 1, 1]
Generation 1860: Best = 4.655888e+03  FES = 186100  Diversity = 5.7890e+00  k_t
= [2, 2, 2, 2]
Generation 1870: Best = 4.655435e+03  FES = 187100  Diversity = 5.7713e+00  k_t
= [1, 1, 1, 1]
Generation 1880: Best = 4.655311e+03  FES = 188100  Diversity = 5.7859e+00  k_t
= [2, 4, 2, 2]
Generation 1890: Best = 4.655311e+03  FES = 189100  Diversity = 5.8263e+00  k_t
= [1, 1, 1, 1]
Generation 1900: Best = 4.654915e+03  FES = 190100  Diversity = 5.8665e+00  k_t
= [4, 4, 4, 3]
Generation 1910: Best = 4.654915e+03  FES = 191100  Diversity = 5.8321e+00  k_t
= [1, 1, 1, 1]
Generation 1920: Best = 4.654915e+03  FES = 192100  Diversity = 5.8266e+00  k_t
= [4, 4, 2, 3]
Generation 1930: Best = 4.654809e+03  FES = 193100  Diversity = 5.7789e+00  k_t
= [1, 1, 1, 1]
Generation 1940: Best = 4.654808e+03  FES = 194100  Diversity = 5.8229e+00  k_t
= [2, 3, 3, 4]
Generation 1950: Best = 4.654808e+03  FES = 195100  Diversity = 5.7680e+00  k_t
= [1, 1, 1, 1]
Generation 1960: Best = 4.654745e+03  FES = 196100  Diversity = 5.7549e+00  k_t
= [3, 3, 4, 4]
Generation 1970: Best = 4.653950e+03  FES = 197100  Diversity = 5.7164e+00  k_t
= [1, 1, 1, 1]
Generation 1980: Best = 4.653950e+03  FES = 198100  Diversity = 5.7140e+00  k_t
= [3, 4, 3, 2]
Generation 1990: Best = 4.653950e+03  FES = 199100  Diversity = 5.7025e+00  k_t
= [1, 1, 1, 1]
Generation 2000: Best = 4.653813e+03  FES = 200100  Diversity = 5.7343e+00  k_t
= [4, 2, 2, 4]
Generation 2010: Best = 4.653632e+03  FES = 201100  Diversity = 5.7545e+00  k_t
= [1, 1, 1, 1]
Generation 2020: Best = 4.653632e+03  FES = 202100  Diversity = 5.7709e+00  k_t
= [2, 3, 3, 2]
Generation 2030: Best = 4.652462e+03  FES = 203100  Diversity = 5.6816e+00  k_t
= [1, 1, 1, 1]
Generation 2040: Best = 4.652462e+03  FES = 204100  Diversity = 5.6351e+00  k_t
= [3, 4, 3, 4]
Generation 2050: Best = 4.652435e+03  FES = 205100  Diversity = 5.5981e+00  k_t
= [1, 1, 1, 1]
Generation 2060: Best = 4.651921e+03  FES = 206100  Diversity = 5.5025e+00  k_t
= [4, 3, 3, 4]
Generation 2070: Best = 4.651799e+03  FES = 207100  Diversity = 5.3753e+00  k_t
= [1, 1, 1, 1]
Generation 2080: Best = 4.651542e+03  FES = 208100  Diversity = 5.3228e+00  k_t
= [4, 3, 4, 2]
Generation 2090: Best = 4.651531e+03  FES = 209100  Diversity = 5.0241e+00  k_t
= [1, 1, 1, 1]
Generation 2100: Best = 4.651521e+03  FES = 210100  Diversity = 4.8699e+00  k_t
= [4, 3, 3, 2]
Generation 2110: Best = 4.650535e+03  FES = 211100  Diversity = 4.4633e+00  k_t

```

```

= [1, 1, 1, 1]
Generation 2120: Best = 4.650535e+03  FES = 212100  Diversity = 4.1547e+00  k_t
= [2, 2, 2, 4]
Generation 2130: Best = 4.650459e+03  FES = 213100  Diversity = 3.8910e+00  k_t
= [1, 1, 1, 1]
Generation 2140: Best = 4.650153e+03  FES = 214100  Diversity = 3.6007e+00  k_t
= [3, 3, 3, 4]
Generation 2150: Best = 4.649879e+03  FES = 215100  Diversity = 3.5110e+00  k_t
= [1, 1, 1, 1]
Generation 2160: Best = 4.649382e+03  FES = 216100  Diversity = 3.4870e+00  k_t
= [4, 4, 3, 2]
Generation 2170: Best = 4.648899e+03  FES = 217100  Diversity = 3.2616e+00  k_t
= [1, 1, 1, 1]
Generation 2180: Best = 4.648899e+03  FES = 218100  Diversity = 3.1080e+00  k_t
= [3, 4, 3, 4]
Generation 2190: Best = 4.648591e+03  FES = 219100  Diversity = 2.9758e+00  k_t
= [1, 1, 1, 1]
Generation 2200: Best = 4.648331e+03  FES = 220100  Diversity = 2.9614e+00  k_t
= [2, 4, 3, 3]
Generation 2210: Best = 4.647930e+03  FES = 221100  Diversity = 2.9095e+00  k_t
= [1, 1, 1, 1]
Generation 2220: Best = 4.647930e+03  FES = 222100  Diversity = 2.7983e+00  k_t
= [3, 4, 3, 3]
Generation 2230: Best = 4.647787e+03  FES = 223100  Diversity = 2.6466e+00  k_t
= [1, 1, 1, 1]
Generation 2240: Best = 4.647786e+03  FES = 224100  Diversity = 2.6125e+00  k_t
= [3, 4, 4, 3]
Generation 2250: Best = 4.647661e+03  FES = 225100  Diversity = 2.6003e+00  k_t
= [1, 1, 1, 1]
Generation 2260: Best = 4.647596e+03  FES = 226100  Diversity = 2.5921e+00  k_t
= [3, 3, 2, 3]
Generation 2270: Best = 4.647503e+03  FES = 227100  Diversity = 2.5543e+00  k_t
= [1, 1, 1, 1]
Generation 2280: Best = 4.647320e+03  FES = 228100  Diversity = 2.5107e+00  k_t
= [4, 3, 4, 4]
Generation 2290: Best = 4.647201e+03  FES = 229100  Diversity = 2.5049e+00  k_t
= [1, 1, 1, 1]
Generation 2300: Best = 4.647182e+03  FES = 230100  Diversity = 2.5421e+00  k_t
= [3, 4, 3, 4]
Generation 2310: Best = 4.647169e+03  FES = 231100  Diversity = 2.5278e+00  k_t
= [1, 1, 1, 1]
Generation 2320: Best = 4.647040e+03  FES = 232100  Diversity = 2.5181e+00  k_t
= [4, 3, 3, 4]
Generation 2330: Best = 4.646969e+03  FES = 233100  Diversity = 2.5209e+00  k_t
= [1, 1, 1, 1]
Generation 2340: Best = 4.646962e+03  FES = 234100  Diversity = 2.5418e+00  k_t
= [2, 3, 2, 3]
Generation 2350: Best = 4.646961e+03  FES = 235100  Diversity = 2.5172e+00  k_t
= [1, 1, 1, 1]
Generation 2360: Best = 4.646886e+03  FES = 236100  Diversity = 2.5579e+00  k_t
= [3, 4, 4, 4]
Generation 2370: Best = 4.646825e+03  FES = 237100  Diversity = 2.5919e+00  k_t
= [1, 1, 1, 1]
Generation 2380: Best = 4.646789e+03  FES = 238100  Diversity = 2.5851e+00  k_t

```



```

= [4, 4, 4, 4]
Generation 2390: Best = 4.646789e+03  FES = 239100  Diversity = 2.5570e+00  k_t
= [1, 1, 1, 1]
Generation 2400: Best = 4.646773e+03  FES = 240100  Diversity = 2.5481e+00  k_t
= [3, 4, 2, 2]
Generation 2410: Best = 4.646773e+03  FES = 241100  Diversity = 2.5659e+00  k_t
= [1, 1, 1, 1]
Generation 2420: Best = 4.646741e+03  FES = 242100  Diversity = 2.5581e+00  k_t
= [2, 3, 4, 4]
Generation 2430: Best = 4.646741e+03  FES = 243100  Diversity = 2.5100e+00  k_t
= [1, 1, 1, 1]
Generation 2440: Best = 4.646741e+03  FES = 244100  Diversity = 2.5016e+00  k_t
= [2, 3, 4, 3]
Generation 2450: Best = 4.646729e+03  FES = 245100  Diversity = 2.4710e+00  k_t
= [1, 1, 1, 1]
Generation 2460: Best = 4.646573e+03  FES = 246100  Diversity = 2.4434e+00  k_t
= [3, 3, 3, 4]
Generation 2470: Best = 4.646536e+03  FES = 247100  Diversity = 2.4492e+00  k_t
= [1, 1, 1, 1]
Generation 2480: Best = 4.646456e+03  FES = 248100  Diversity = 2.4457e+00  k_t
= [3, 3, 4, 3]
Generation 2490: Best = 4.646442e+03  FES = 249100  Diversity = 2.4339e+00  k_t
= [1, 1, 1, 1]
Generation 2500: Best = 4.646442e+03  FES = 250100  Diversity = 2.4239e+00  k_t
= [3, 4, 4, 3]
Generation 2510: Best = 4.646417e+03  FES = 251100  Diversity = 2.3964e+00  k_t
= [1, 1, 1, 1]
Generation 2520: Best = 4.646417e+03  FES = 252100  Diversity = 2.4362e+00  k_t
= [2, 2, 4, 2]
Generation 2530: Best = 4.646416e+03  FES = 253100  Diversity = 2.3955e+00  k_t
= [1, 1, 1, 1]
Generation 2540: Best = 4.646406e+03  FES = 254100  Diversity = 2.3668e+00  k_t
= [3, 4, 4, 2]
Generation 2550: Best = 4.646406e+03  FES = 255100  Diversity = 2.3440e+00  k_t
= [1, 1, 1, 1]
Generation 2560: Best = 4.646406e+03  FES = 256100  Diversity = 2.3047e+00  k_t
= [3, 4, 3, 2]
Generation 2570: Best = 4.646399e+03  FES = 257100  Diversity = 2.3017e+00  k_t
= [1, 1, 1, 1]
Generation 2580: Best = 4.646399e+03  FES = 258100  Diversity = 2.3151e+00  k_t
= [2, 2, 3, 4]
Generation 2590: Best = 4.646399e+03  FES = 259100  Diversity = 2.2883e+00  k_t
= [1, 1, 1, 1]
Generation 2600: Best = 4.646323e+03  FES = 260100  Diversity = 2.2495e+00  k_t
= [2, 4, 3, 2]
Generation 2610: Best = 4.646240e+03  FES = 261100  Diversity = 2.2612e+00  k_t
= [1, 1, 1, 1]
Generation 2620: Best = 4.646222e+03  FES = 262100  Diversity = 2.2692e+00  k_t
= [3, 2, 2, 4]
Generation 2630: Best = 4.646221e+03  FES = 263100  Diversity = 2.2334e+00  k_t
= [1, 1, 1, 1]
Generation 2640: Best = 4.646170e+03  FES = 264100  Diversity = 2.2384e+00  k_t
= [3, 3, 3, 3]
Generation 2650: Best = 4.646072e+03  FES = 265100  Diversity = 2.2366e+00  k_t

```

```

= [1, 1, 1, 1]
Generation 2660: Best = 4.646072e+03 FES = 266100 Diversity = 2.2298e+00 k_t
= [4, 4, 4, 2]
Generation 2670: Best = 4.646009e+03 FES = 267100 Diversity = 2.2471e+00 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 4.646009e+03 FES = 268100 Diversity = 2.2438e+00 k_t
= [4, 2, 2, 3]
Generation 2690: Best = 4.646006e+03 FES = 269100 Diversity = 2.2799e+00 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 4.646006e+03 FES = 270100 Diversity = 2.2953e+00 k_t
= [2, 3, 3, 2]
Generation 2710: Best = 4.645985e+03 FES = 271100 Diversity = 2.2796e+00 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 4.645981e+03 FES = 272100 Diversity = 2.2734e+00 k_t
= [2, 4, 2, 4]
Generation 2730: Best = 4.645981e+03 FES = 273100 Diversity = 2.2479e+00 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 4.645980e+03 FES = 274100 Diversity = 2.2384e+00 k_t
= [4, 2, 2, 3]
Generation 2750: Best = 4.645974e+03 FES = 275100 Diversity = 2.2428e+00 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 4.645951e+03 FES = 276100 Diversity = 2.2305e+00 k_t
= [4, 2, 4, 2]
Generation 2770: Best = 4.645888e+03 FES = 277100 Diversity = 2.2364e+00 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 4.645835e+03 FES = 278100 Diversity = 2.2445e+00 k_t
= [3, 3, 4, 4]
Generation 2790: Best = 4.645831e+03 FES = 279100 Diversity = 2.2502e+00 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 4.645777e+03 FES = 280100 Diversity = 2.2361e+00 k_t
= [2, 4, 4, 2]
Generation 2810: Best = 4.645777e+03 FES = 281100 Diversity = 2.2308e+00 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 4.645777e+03 FES = 282100 Diversity = 2.2498e+00 k_t
= [3, 3, 2, 3]
Generation 2830: Best = 4.645777e+03 FES = 283100 Diversity = 2.2578e+00 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 4.645772e+03 FES = 284100 Diversity = 2.2420e+00 k_t
= [3, 3, 4, 4]
Generation 2850: Best = 4.645730e+03 FES = 285100 Diversity = 2.2659e+00 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 4.645730e+03 FES = 286100 Diversity = 2.2683e+00 k_t
= [4, 2, 4, 2]
Generation 2870: Best = 4.645730e+03 FES = 287100 Diversity = 2.2604e+00 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 4.645730e+03 FES = 288100 Diversity = 2.2709e+00 k_t
= [4, 4, 3, 2]
Generation 2890: Best = 4.645730e+03 FES = 289100 Diversity = 2.2742e+00 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 4.645673e+03 FES = 290100 Diversity = 2.2828e+00 k_t
= [3, 3, 2, 4]
Generation 2910: Best = 4.645668e+03 FES = 291100 Diversity = 2.2588e+00 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 4.645668e+03 FES = 292100 Diversity = 2.2710e+00 k_t

```

```

= [2, 2, 4, 4]
Generation 2930: Best = 4.645649e+03 FES = 293100 Diversity = 2.2709e+00 k_t
= [1, 1, 1, 1]
Generation 2940: Best = 4.645649e+03 FES = 294100 Diversity = 2.2866e+00 k_t
= [3, 2, 2, 4]
Generation 2950: Best = 4.645622e+03 FES = 295100 Diversity = 2.2932e+00 k_t
= [1, 1, 1, 1]
Generation 2960: Best = 4.645611e+03 FES = 296100 Diversity = 2.3028e+00 k_t
= [4, 2, 4, 2]
Generation 2970: Best = 4.645583e+03 FES = 297100 Diversity = 2.2720e+00 k_t
= [1, 1, 1, 1]
Generation 2980: Best = 4.645537e+03 FES = 298100 Diversity = 2.2770e+00 k_t
= [2, 2, 2, 4]
Generation 2990: Best = 4.645537e+03 FES = 299100 Diversity = 2.2821e+00 k_t
= [1, 1, 1, 1]
Generation 3000: Best = 4.645537e+03 FES = 300100 Diversity = 2.3123e+00 k_t
= [3, 4, 2, 3]
Generation 3010: Best = 4.645537e+03 FES = 301100 Diversity = 2.3061e+00 k_t
= [1, 1, 1, 1]
Generation 3020: Best = 4.645530e+03 FES = 302100 Diversity = 2.3125e+00 k_t
= [3, 2, 4, 2]
Generation 3030: Best = 4.645530e+03 FES = 303100 Diversity = 2.3421e+00 k_t
= [1, 1, 1, 1]
Generation 3040: Best = 4.645530e+03 FES = 304100 Diversity = 2.3082e+00 k_t
= [2, 4, 3, 2]
Generation 3050: Best = 4.645507e+03 FES = 305100 Diversity = 2.2882e+00 k_t
= [1, 1, 1, 1]
Generation 3060: Best = 4.645507e+03 FES = 306100 Diversity = 2.2907e+00 k_t
= [2, 4, 4, 4]
Generation 3070: Best = 4.645507e+03 FES = 307100 Diversity = 2.3050e+00 k_t
= [1, 1, 1, 1]
Generation 3080: Best = 4.645507e+03 FES = 308100 Diversity = 2.3049e+00 k_t
= [3, 2, 3, 2]
Generation 3090: Best = 4.645488e+03 FES = 309100 Diversity = 2.3035e+00 k_t
= [1, 1, 1, 1]
Generation 3100: Best = 4.645478e+03 FES = 310100 Diversity = 2.3027e+00 k_t
= [4, 4, 2, 2]
Generation 3110: Best = 4.645454e+03 FES = 311100 Diversity = 2.2828e+00 k_t
= [1, 1, 1, 1]
Generation 3120: Best = 4.645437e+03 FES = 312100 Diversity = 2.2411e+00 k_t
= [4, 2, 2, 4]
Generation 3130: Best = 4.645358e+03 FES = 313100 Diversity = 2.2606e+00 k_t
= [1, 1, 1, 1]
Generation 3140: Best = 4.645358e+03 FES = 314100 Diversity = 2.2658e+00 k_t
= [2, 3, 3, 3]
Generation 3150: Best = 4.645350e+03 FES = 315100 Diversity = 2.2584e+00 k_t
= [1, 1, 1, 1]
Generation 3160: Best = 4.645350e+03 FES = 316100 Diversity = 2.2563e+00 k_t
= [4, 2, 2, 3]
Generation 3170: Best = 4.645350e+03 FES = 317100 Diversity = 2.2886e+00 k_t
= [1, 1, 1, 1]
Generation 3180: Best = 4.645330e+03 FES = 318100 Diversity = 2.2857e+00 k_t
= [2, 2, 3, 3]
Generation 3190: Best = 4.645330e+03 FES = 319100 Diversity = 2.2811e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 3200: Best = 4.645330e+03 FES = 320100 Diversity = 2.2780e+00 k_t
= [4, 4, 3, 2]
Generation 3210: Best = 4.645330e+03 FES = 321100 Diversity = 2.3112e+00 k_t
= [1, 1, 1, 1]
Generation 3220: Best = 4.645324e+03 FES = 322100 Diversity = 2.3065e+00 k_t
= [4, 4, 4, 4]
Generation 3230: Best = 4.645282e+03 FES = 323100 Diversity = 2.3107e+00 k_t
= [1, 1, 1, 1]
Generation 3240: Best = 4.645282e+03 FES = 324100 Diversity = 2.2930e+00 k_t
= [2, 3, 4, 2]
Generation 3250: Best = 4.645252e+03 FES = 325100 Diversity = 2.2744e+00 k_t
= [1, 1, 1, 1]
Generation 3260: Best = 4.645244e+03 FES = 326100 Diversity = 2.2714e+00 k_t
= [2, 2, 2, 3]
Generation 3270: Best = 4.645218e+03 FES = 327100 Diversity = 2.2847e+00 k_t
= [1, 1, 1, 1]
Generation 3280: Best = 4.645174e+03 FES = 328100 Diversity = 2.2711e+00 k_t
= [2, 2, 3, 2]
Generation 3290: Best = 4.645161e+03 FES = 329100 Diversity = 2.2606e+00 k_t
= [1, 1, 1, 1]
Generation 3300: Best = 4.645161e+03 FES = 330100 Diversity = 2.2613e+00 k_t
= [2, 2, 4, 4]
Generation 3310: Best = 4.645161e+03 FES = 331100 Diversity = 2.3011e+00 k_t
= [1, 1, 1, 1]
Generation 3320: Best = 4.645161e+03 FES = 332100 Diversity = 2.2982e+00 k_t
= [3, 3, 4, 2]
Generation 3330: Best = 4.645161e+03 FES = 333100 Diversity = 2.2855e+00 k_t
= [1, 1, 1, 1]
Generation 3340: Best = 4.645161e+03 FES = 334100 Diversity = 2.2941e+00 k_t
= [4, 4, 4, 2]
Generation 3350: Best = 4.645096e+03 FES = 335100 Diversity = 2.2972e+00 k_t
= [1, 1, 1, 1]
Generation 3360: Best = 4.645088e+03 FES = 336100 Diversity = 2.3028e+00 k_t
= [3, 4, 4, 3]
Generation 3370: Best = 4.645088e+03 FES = 337100 Diversity = 2.3042e+00 k_t
= [1, 1, 1, 1]
Generation 3380: Best = 4.645026e+03 FES = 338100 Diversity = 2.2948e+00 k_t
= [4, 2, 4, 4]
Generation 3390: Best = 4.645025e+03 FES = 339100 Diversity = 2.3020e+00 k_t
= [1, 1, 1, 1]
Generation 3400: Best = 4.645023e+03 FES = 340100 Diversity = 2.3113e+00 k_t
= [2, 3, 4, 3]
Generation 3410: Best = 4.645023e+03 FES = 341100 Diversity = 2.2853e+00 k_t
= [1, 1, 1, 1]
Generation 3420: Best = 4.645023e+03 FES = 342100 Diversity = 2.2898e+00 k_t
= [4, 2, 3, 4]
Generation 3430: Best = 4.645023e+03 FES = 343100 Diversity = 2.2900e+00 k_t
= [1, 1, 1, 1]
Generation 3440: Best = 4.644985e+03 FES = 344100 Diversity = 2.3042e+00 k_t
= [2, 3, 3, 2]
Generation 3450: Best = 4.644906e+03 FES = 345100 Diversity = 2.3028e+00 k_t
= [1, 1, 1, 1]
Generation 3460: Best = 4.644906e+03 FES = 346100 Diversity = 2.2730e+00 k_t

```

```

= [4, 2, 4, 2]
Generation 3470: Best = 4.644840e+03 FES = 347100 Diversity = 2.2750e+00 k_t
= [1, 1, 1, 1]
Generation 3480: Best = 4.644839e+03 FES = 348100 Diversity = 2.2717e+00 k_t
= [3, 4, 2, 2]
Generation 3490: Best = 4.644806e+03 FES = 349100 Diversity = 2.2579e+00 k_t
= [1, 1, 1, 1]
Generation 3500: Best = 4.644776e+03 FES = 350100 Diversity = 2.2675e+00 k_t
= [4, 2, 3, 2]
Generation 3510: Best = 4.644775e+03 FES = 351100 Diversity = 2.2681e+00 k_t
= [1, 1, 1, 1]
Generation 3520: Best = 4.644737e+03 FES = 352100 Diversity = 2.2646e+00 k_t
= [4, 4, 3, 4]
Generation 3530: Best = 4.644737e+03 FES = 353100 Diversity = 2.2726e+00 k_t
= [1, 1, 1, 1]
Generation 3540: Best = 4.644705e+03 FES = 354100 Diversity = 2.2727e+00 k_t
= [4, 3, 4, 4]
Generation 3550: Best = 4.644662e+03 FES = 355100 Diversity = 2.2627e+00 k_t
= [1, 1, 1, 1]
Generation 3560: Best = 4.644649e+03 FES = 356100 Diversity = 2.2422e+00 k_t
= [4, 2, 3, 2]
Generation 3570: Best = 4.644649e+03 FES = 357100 Diversity = 2.2330e+00 k_t
= [1, 1, 1, 1]
Generation 3580: Best = 4.644649e+03 FES = 358100 Diversity = 2.2399e+00 k_t
= [4, 3, 2, 4]
Generation 3590: Best = 4.644641e+03 FES = 359100 Diversity = 2.2365e+00 k_t
= [1, 1, 1, 1]
Generation 3600: Best = 4.644641e+03 FES = 360100 Diversity = 2.2240e+00 k_t
= [3, 3, 2, 3]
Generation 3610: Best = 4.644629e+03 FES = 361100 Diversity = 2.2101e+00 k_t
= [1, 1, 1, 1]
Generation 3620: Best = 4.644619e+03 FES = 362100 Diversity = 2.1783e+00 k_t
= [4, 3, 2, 4]
Generation 3630: Best = 4.644577e+03 FES = 363100 Diversity = 2.1597e+00 k_t
= [1, 1, 1, 1]
Generation 3640: Best = 4.644577e+03 FES = 364100 Diversity = 2.1561e+00 k_t
= [3, 4, 2, 2]
Generation 3650: Best = 4.644566e+03 FES = 365100 Diversity = 2.1432e+00 k_t
= [1, 1, 1, 1]
Generation 3660: Best = 4.644560e+03 FES = 366100 Diversity = 2.1146e+00 k_t
= [2, 4, 4, 4]
Generation 3670: Best = 4.644560e+03 FES = 367100 Diversity = 2.0916e+00 k_t
= [1, 1, 1, 1]
Generation 3680: Best = 4.644492e+03 FES = 368100 Diversity = 1.9004e+00 k_t
= [2, 3, 4, 4]
Generation 3690: Best = 4.644491e+03 FES = 369100 Diversity = 1.7884e+00 k_t
= [1, 1, 1, 1]
Generation 3700: Best = 4.644483e+03 FES = 370100 Diversity = 1.7144e+00 k_t
= [3, 3, 3, 4]
Generation 3710: Best = 4.644483e+03 FES = 371100 Diversity = 1.6545e+00 k_t
= [1, 1, 1, 1]
Generation 3720: Best = 4.644470e+03 FES = 372100 Diversity = 1.6063e+00 k_t
= [4, 4, 3, 3]
Generation 3730: Best = 4.644452e+03 FES = 373100 Diversity = 1.5349e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 3740: Best = 4.644421e+03 FES = 374100 Diversity = 1.4077e+00 k_t
= [2, 2, 2, 2]
Generation 3750: Best = 4.644333e+03 FES = 375100 Diversity = 1.2931e+00 k_t
= [1, 1, 1, 1]
Generation 3760: Best = 4.644308e+03 FES = 376100 Diversity = 1.2916e+00 k_t
= [3, 4, 2, 4]
Generation 3770: Best = 4.644288e+03 FES = 377100 Diversity = 1.2909e+00 k_t
= [1, 1, 1, 1]
Generation 3780: Best = 4.644238e+03 FES = 378100 Diversity = 1.2979e+00 k_t
= [4, 3, 4, 4]
Generation 3790: Best = 4.644227e+03 FES = 379100 Diversity = 1.3091e+00 k_t
= [1, 1, 1, 1]
Generation 3800: Best = 4.644216e+03 FES = 380100 Diversity = 1.3087e+00 k_t
= [3, 2, 4, 3]
Generation 3810: Best = 4.644134e+03 FES = 381100 Diversity = 1.3088e+00 k_t
= [1, 1, 1, 1]
Generation 3820: Best = 4.644131e+03 FES = 382100 Diversity = 1.3126e+00 k_t
= [4, 2, 3, 3]
Generation 3830: Best = 4.644123e+03 FES = 383100 Diversity = 1.3168e+00 k_t
= [1, 1, 1, 1]
Generation 3840: Best = 4.644090e+03 FES = 384100 Diversity = 1.3206e+00 k_t
= [3, 2, 4, 2]
Generation 3850: Best = 4.644090e+03 FES = 385100 Diversity = 1.3264e+00 k_t
= [1, 1, 1, 1]
Generation 3860: Best = 4.644073e+03 FES = 386100 Diversity = 1.3311e+00 k_t
= [3, 2, 4, 4]
Generation 3870: Best = 4.644055e+03 FES = 387100 Diversity = 1.3319e+00 k_t
= [1, 1, 1, 1]
Generation 3880: Best = 4.644038e+03 FES = 388100 Diversity = 1.3311e+00 k_t
= [4, 2, 2, 4]
Generation 3890: Best = 4.644030e+03 FES = 389100 Diversity = 1.3302e+00 k_t
= [1, 1, 1, 1]
Generation 3900: Best = 4.644013e+03 FES = 390100 Diversity = 1.3234e+00 k_t
= [3, 2, 3, 4]
Generation 3910: Best = 4.643965e+03 FES = 391100 Diversity = 1.3118e+00 k_t
= [1, 1, 1, 1]
Generation 3920: Best = 4.643931e+03 FES = 392100 Diversity = 1.2952e+00 k_t
= [2, 4, 4, 2]
Generation 3930: Best = 4.643922e+03 FES = 393100 Diversity = 1.2947e+00 k_t
= [1, 1, 1, 1]
Generation 3940: Best = 4.643908e+03 FES = 394100 Diversity = 1.2939e+00 k_t
= [3, 3, 2, 4]
Generation 3950: Best = 4.643869e+03 FES = 395100 Diversity = 1.2806e+00 k_t
= [1, 1, 1, 1]
Generation 3960: Best = 4.643868e+03 FES = 396100 Diversity = 1.2569e+00 k_t
= [4, 3, 3, 2]
Generation 3970: Best = 4.643853e+03 FES = 397100 Diversity = 1.1791e+00 k_t
= [1, 1, 1, 1]
Generation 3980: Best = 4.643846e+03 FES = 398100 Diversity = 1.1346e+00 k_t
= [4, 3, 2, 2]
Generation 3990: Best = 4.643834e+03 FES = 399100 Diversity = 9.7504e-01 k_t
= [1, 1, 1, 1]
Generation 4000: Best = 4.643832e+03 FES = 400100 Diversity = 8.9479e-01 k_t

```

```

= [2, 2, 2, 4]
Generation 4010: Best = 4.643810e+03 FES = 401100 Diversity = 6.8195e-01 k_t
= [1, 1, 1, 1]
Generation 4020: Best = 4.643780e+03 FES = 402100 Diversity = 4.6068e-01 k_t
= [2, 3, 4, 3]
Generation 4030: Best = 4.643771e+03 FES = 403100 Diversity = 2.9726e-02 k_t
= [1, 1, 1, 1]
Generation 4040: Best = 4.643737e+03 FES = 404100 Diversity = 2.7463e-02 k_t
= [3, 2, 2, 4]
Generation 4050: Best = 4.643725e+03 FES = 405100 Diversity = 2.5459e-02 k_t
= [1, 1, 1, 1]
Generation 4060: Best = 4.643707e+03 FES = 406100 Diversity = 2.5358e-02 k_t
= [3, 3, 4, 3]
Generation 4070: Best = 4.643683e+03 FES = 407100 Diversity = 2.2593e-02 k_t
= [1, 1, 1, 1]
Generation 4080: Best = 4.643668e+03 FES = 408100 Diversity = 1.9484e-02 k_t
= [2, 2, 4, 2]
Generation 4090: Best = 4.643645e+03 FES = 409100 Diversity = 1.7810e-02 k_t
= [1, 1, 1, 1]
Generation 4100: Best = 4.643635e+03 FES = 410100 Diversity = 1.6447e-02 k_t
= [4, 4, 4, 2]
Generation 4110: Best = 4.643619e+03 FES = 411100 Diversity = 1.5144e-02 k_t
= [1, 1, 1, 1]
Generation 4120: Best = 4.643608e+03 FES = 412100 Diversity = 1.4814e-02 k_t
= [3, 3, 2, 3]
Generation 4130: Best = 4.643598e+03 FES = 413100 Diversity = 1.3743e-02 k_t
= [1, 1, 1, 1]
Generation 4140: Best = 4.643590e+03 FES = 414100 Diversity = 1.3961e-02 k_t
= [4, 3, 4, 3]
Generation 4150: Best = 4.643575e+03 FES = 415100 Diversity = 1.3681e-02 k_t
= [1, 1, 1, 1]
Generation 4160: Best = 4.643563e+03 FES = 416100 Diversity = 1.3186e-02 k_t
= [4, 3, 4, 4]
Generation 4170: Best = 4.643544e+03 FES = 417100 Diversity = 1.1828e-02 k_t
= [1, 1, 1, 1]
Generation 4180: Best = 4.643534e+03 FES = 418100 Diversity = 1.1845e-02 k_t
= [2, 4, 2, 4]
Generation 4190: Best = 4.643531e+03 FES = 419100 Diversity = 1.1612e-02 k_t
= [1, 1, 1, 1]
Generation 4200: Best = 4.643512e+03 FES = 420100 Diversity = 1.1284e-02 k_t
= [4, 2, 4, 3]
Generation 4210: Best = 4.643501e+03 FES = 421100 Diversity = 1.1041e-02 k_t
= [1, 1, 1, 1]
Generation 4220: Best = 4.643492e+03 FES = 422100 Diversity = 1.0901e-02 k_t
= [3, 3, 2, 3]
Generation 4230: Best = 4.643485e+03 FES = 423100 Diversity = 1.0322e-02 k_t
= [1, 1, 1, 1]
Generation 4240: Best = 4.643473e+03 FES = 424100 Diversity = 9.5643e-03 k_t
= [4, 2, 2, 2]
Generation 4250: Best = 4.643467e+03 FES = 425100 Diversity = 9.9034e-03 k_t
= [1, 1, 1, 1]
Generation 4260: Best = 4.643455e+03 FES = 426100 Diversity = 9.2834e-03 k_t
= [4, 2, 3, 2]
Generation 4270: Best = 4.643451e+03 FES = 427100 Diversity = 8.7426e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 4280: Best = 4.643445e+03 FES = 428100 Diversity = 8.5710e-03 k_t
= [3, 3, 2, 3]
Generation 4290: Best = 4.643438e+03 FES = 429100 Diversity = 8.3322e-03 k_t
= [1, 1, 1, 1]
Generation 4300: Best = 4.643431e+03 FES = 430100 Diversity = 7.5825e-03 k_t
= [4, 3, 3, 4]
Generation 4310: Best = 4.643423e+03 FES = 431100 Diversity = 7.3430e-03 k_t
= [1, 1, 1, 1]
Generation 4320: Best = 4.643419e+03 FES = 432100 Diversity = 7.2401e-03 k_t
= [4, 3, 2, 2]
Generation 4330: Best = 4.643410e+03 FES = 433100 Diversity = 7.2232e-03 k_t
= [1, 1, 1, 1]
Generation 4340: Best = 4.643405e+03 FES = 434100 Diversity = 7.7644e-03 k_t
= [2, 4, 3, 3]
Generation 4350: Best = 4.643396e+03 FES = 435100 Diversity = 8.0601e-03 k_t
= [1, 1, 1, 1]
Generation 4360: Best = 4.643386e+03 FES = 436100 Diversity = 8.6894e-03 k_t
= [4, 4, 2, 4]
Generation 4370: Best = 4.643382e+03 FES = 437100 Diversity = 8.1781e-03 k_t
= [1, 1, 1, 1]
Generation 4380: Best = 4.643375e+03 FES = 438100 Diversity = 7.8573e-03 k_t
= [3, 2, 4, 2]
Generation 4390: Best = 4.643370e+03 FES = 439100 Diversity = 7.6529e-03 k_t
= [1, 1, 1, 1]
Generation 4400: Best = 4.643366e+03 FES = 440100 Diversity = 7.1736e-03 k_t
= [2, 4, 3, 4]
Generation 4410: Best = 4.643362e+03 FES = 441100 Diversity = 6.5974e-03 k_t
= [1, 1, 1, 1]
Generation 4420: Best = 4.643359e+03 FES = 442100 Diversity = 6.2573e-03 k_t
= [4, 4, 4, 3]
Generation 4430: Best = 4.643351e+03 FES = 443100 Diversity = 6.0970e-03 k_t
= [1, 1, 1, 1]
Generation 4440: Best = 4.643346e+03 FES = 444100 Diversity = 6.6094e-03 k_t
= [3, 3, 2, 4]
Generation 4450: Best = 4.643342e+03 FES = 445100 Diversity = 7.0127e-03 k_t
= [1, 1, 1, 1]
Generation 4460: Best = 4.643338e+03 FES = 446100 Diversity = 6.9570e-03 k_t
= [4, 2, 3, 2]
Generation 4470: Best = 4.643333e+03 FES = 447100 Diversity = 6.8612e-03 k_t
= [1, 1, 1, 1]
Generation 4480: Best = 4.643325e+03 FES = 448100 Diversity = 6.8708e-03 k_t
= [3, 4, 3, 3]
Generation 4490: Best = 4.643321e+03 FES = 449100 Diversity = 6.9730e-03 k_t
= [1, 1, 1, 1]
Generation 4500: Best = 4.643316e+03 FES = 450100 Diversity = 7.2512e-03 k_t
= [4, 4, 2, 4]
Generation 4510: Best = 4.643313e+03 FES = 451100 Diversity = 7.1546e-03 k_t
= [1, 1, 1, 1]
Generation 4520: Best = 4.643306e+03 FES = 452100 Diversity = 7.2632e-03 k_t
= [4, 4, 2, 2]
Generation 4530: Best = 4.643304e+03 FES = 453100 Diversity = 7.2821e-03 k_t
= [1, 1, 1, 1]
Generation 4540: Best = 4.643298e+03 FES = 454100 Diversity = 6.7418e-03 k_t

```



```

= [2, 3, 2, 4]
Generation 4550: Best = 4.643295e+03 FES = 455100 Diversity = 6.3498e-03 k_t
= [1, 1, 1, 1]
Generation 4560: Best = 4.643291e+03 FES = 456100 Diversity = 6.1273e-03 k_t
= [2, 4, 4, 4]
Generation 4570: Best = 4.643286e+03 FES = 457100 Diversity = 6.0698e-03 k_t
= [1, 1, 1, 1]
Generation 4580: Best = 4.643281e+03 FES = 458100 Diversity = 5.8145e-03 k_t
= [3, 2, 2, 2]
Generation 4590: Best = 4.643277e+03 FES = 459100 Diversity = 5.9203e-03 k_t
= [1, 1, 1, 1]
Generation 4600: Best = 4.643270e+03 FES = 460100 Diversity = 5.8369e-03 k_t
= [2, 4, 2, 2]
Generation 4610: Best = 4.643268e+03 FES = 461100 Diversity = 5.6243e-03 k_t
= [1, 1, 1, 1]
Generation 4620: Best = 4.643262e+03 FES = 462100 Diversity = 5.7941e-03 k_t
= [4, 3, 3, 4]
Generation 4630: Best = 4.643258e+03 FES = 463100 Diversity = 6.1472e-03 k_t
= [1, 1, 1, 1]
Generation 4640: Best = 4.643253e+03 FES = 464100 Diversity = 6.7292e-03 k_t
= [3, 4, 4, 3]
Generation 4650: Best = 4.643248e+03 FES = 465100 Diversity = 6.5671e-03 k_t
= [1, 1, 1, 1]
Generation 4660: Best = 4.643242e+03 FES = 466100 Diversity = 6.5176e-03 k_t
= [3, 3, 2, 4]
Generation 4670: Best = 4.643235e+03 FES = 467100 Diversity = 6.7094e-03 k_t
= [1, 1, 1, 1]
Generation 4680: Best = 4.643230e+03 FES = 468100 Diversity = 6.7308e-03 k_t
= [3, 2, 4, 3]
Generation 4690: Best = 4.643222e+03 FES = 469100 Diversity = 6.9994e-03 k_t
= [1, 1, 1, 1]
Generation 4700: Best = 4.643217e+03 FES = 470100 Diversity = 7.0961e-03 k_t
= [4, 3, 4, 2]
Generation 4710: Best = 4.643213e+03 FES = 471100 Diversity = 7.0582e-03 k_t
= [1, 1, 1, 1]
Generation 4720: Best = 4.643203e+03 FES = 472100 Diversity = 7.3271e-03 k_t
= [4, 4, 3, 3]
Generation 4730: Best = 4.643198e+03 FES = 473100 Diversity = 7.3976e-03 k_t
= [1, 1, 1, 1]
Generation 4740: Best = 4.643188e+03 FES = 474100 Diversity = 7.8746e-03 k_t
= [2, 2, 2, 3]
Generation 4750: Best = 4.643180e+03 FES = 475100 Diversity = 8.2580e-03 k_t
= [1, 1, 1, 1]
Generation 4760: Best = 4.643175e+03 FES = 476100 Diversity = 8.4484e-03 k_t
= [2, 4, 4, 2]
Generation 4770: Best = 4.643168e+03 FES = 477100 Diversity = 8.8068e-03 k_t
= [1, 1, 1, 1]
Generation 4780: Best = 4.643162e+03 FES = 478100 Diversity = 8.1766e-03 k_t
= [3, 4, 4, 2]
Generation 4790: Best = 4.643157e+03 FES = 479100 Diversity = 7.3107e-03 k_t
= [1, 1, 1, 1]
Generation 4800: Best = 4.643153e+03 FES = 480100 Diversity = 6.4393e-03 k_t
= [3, 4, 2, 2]
Generation 4810: Best = 4.643147e+03 FES = 481100 Diversity = 6.1450e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 4820: Best = 4.643141e+03 FES = 482100 Diversity = 6.2713e-03 k_t
= [2, 4, 3, 3]
Generation 4830: Best = 4.643133e+03 FES = 483100 Diversity = 6.3878e-03 k_t
= [1, 1, 1, 1]
Generation 4840: Best = 4.643128e+03 FES = 484100 Diversity = 7.2039e-03 k_t
= [3, 2, 2, 4]
Generation 4850: Best = 4.643123e+03 FES = 485100 Diversity = 7.7679e-03 k_t
= [1, 1, 1, 1]
Generation 4860: Best = 4.643119e+03 FES = 486100 Diversity = 7.2734e-03 k_t
= [2, 3, 4, 2]
Generation 4870: Best = 4.643113e+03 FES = 487100 Diversity = 7.1881e-03 k_t
= [1, 1, 1, 1]
Generation 4880: Best = 4.643107e+03 FES = 488100 Diversity = 7.7355e-03 k_t
= [4, 3, 3, 4]
Generation 4890: Best = 4.643103e+03 FES = 489100 Diversity = 8.0029e-03 k_t
= [1, 1, 1, 1]
Generation 4900: Best = 4.643098e+03 FES = 490100 Diversity = 7.2903e-03 k_t
= [4, 3, 4, 4]
Generation 4910: Best = 4.643093e+03 FES = 491100 Diversity = 7.0284e-03 k_t
= [1, 1, 1, 1]
Generation 4920: Best = 4.643088e+03 FES = 492100 Diversity = 6.9401e-03 k_t
= [2, 4, 3, 2]
Generation 4930: Best = 4.643081e+03 FES = 493100 Diversity = 7.1488e-03 k_t
= [1, 1, 1, 1]
Generation 4940: Best = 4.643078e+03 FES = 494100 Diversity = 7.2210e-03 k_t
= [2, 2, 2, 3]
Generation 4950: Best = 4.643073e+03 FES = 495100 Diversity = 6.9192e-03 k_t
= [1, 1, 1, 1]
Generation 4960: Best = 4.643067e+03 FES = 496100 Diversity = 7.1113e-03 k_t
= [4, 3, 2, 2]
Generation 4970: Best = 4.643062e+03 FES = 497100 Diversity = 6.7680e-03 k_t
= [1, 1, 1, 1]
Generation 4980: Best = 4.643058e+03 FES = 498100 Diversity = 6.7162e-03 k_t
= [3, 2, 2, 3]
Generation 4990: Best = 4.643054e+03 FES = 499100 Diversity = 6.3912e-03 k_t
= [1, 1, 1, 1]
Generation 5000: Best = 4.643051e+03 FES = 500100 Diversity = 5.9844e-03 k_t
= [2, 4, 2, 4]
Generation 5010: Best = 4.643046e+03 FES = 501100 Diversity = 5.9657e-03 k_t
= [1, 1, 1, 1]
Generation 5020: Best = 4.643041e+03 FES = 502100 Diversity = 5.6959e-03 k_t
= [4, 3, 4, 3]
Generation 5030: Best = 4.643037e+03 FES = 503100 Diversity = 6.2669e-03 k_t
= [1, 1, 1, 1]
Generation 5040: Best = 4.643034e+03 FES = 504100 Diversity = 5.9057e-03 k_t
= [4, 3, 2, 4]
Generation 5050: Best = 4.643028e+03 FES = 505100 Diversity = 5.6640e-03 k_t
= [1, 1, 1, 1]
Generation 5060: Best = 4.643025e+03 FES = 506100 Diversity = 5.9786e-03 k_t
= [2, 2, 2, 4]
Generation 5070: Best = 4.643021e+03 FES = 507100 Diversity = 5.7714e-03 k_t
= [1, 1, 1, 1]
Generation 5080: Best = 4.643015e+03 FES = 508100 Diversity = 6.2431e-03 k_t

```

```

= [2, 2, 2, 3]
Generation 5090: Best = 4.643012e+03 FES = 509100 Diversity = 6.3517e-03 k_t
= [1, 1, 1, 1]
Generation 5100: Best = 4.643009e+03 FES = 510100 Diversity = 5.7273e-03 k_t
= [3, 3, 3, 3]
Generation 5110: Best = 4.643003e+03 FES = 511100 Diversity = 5.8206e-03 k_t
= [1, 1, 1, 1]
Generation 5120: Best = 4.642997e+03 FES = 512100 Diversity = 5.9465e-03 k_t
= [2, 2, 2, 4]
Generation 5130: Best = 4.642989e+03 FES = 513100 Diversity = 7.2137e-03 k_t
= [1, 1, 1, 1]
Generation 5140: Best = 4.642983e+03 FES = 514100 Diversity = 7.4693e-03 k_t
= [4, 4, 4, 3]
Generation 5150: Best = 4.642976e+03 FES = 515100 Diversity = 7.5009e-03 k_t
= [1, 1, 1, 1]
Generation 5160: Best = 4.642968e+03 FES = 516100 Diversity = 7.4522e-03 k_t
= [3, 2, 4, 2]
Generation 5170: Best = 4.642962e+03 FES = 517100 Diversity = 7.0611e-03 k_t
= [1, 1, 1, 1]
Generation 5180: Best = 4.642960e+03 FES = 518100 Diversity = 6.3416e-03 k_t
= [3, 2, 3, 4]
Generation 5190: Best = 4.642955e+03 FES = 519100 Diversity = 6.4948e-03 k_t
= [1, 1, 1, 1]
Generation 5200: Best = 4.642951e+03 FES = 520100 Diversity = 6.5717e-03 k_t
= [4, 4, 4, 4]
Generation 5210: Best = 4.642949e+03 FES = 521100 Diversity = 6.3599e-03 k_t
= [1, 1, 1, 1]
Generation 5220: Best = 4.642945e+03 FES = 522100 Diversity = 5.7447e-03 k_t
= [4, 4, 3, 2]
Generation 5230: Best = 4.642943e+03 FES = 523100 Diversity = 5.4297e-03 k_t
= [1, 1, 1, 1]
Generation 5240: Best = 4.642939e+03 FES = 524100 Diversity = 4.8285e-03 k_t
= [4, 2, 4, 4]
Generation 5250: Best = 4.642936e+03 FES = 525100 Diversity = 4.7443e-03 k_t
= [1, 1, 1, 1]
Generation 5260: Best = 4.642933e+03 FES = 526100 Diversity = 4.6577e-03 k_t
= [2, 2, 4, 3]
Generation 5270: Best = 4.642929e+03 FES = 527100 Diversity = 4.9092e-03 k_t
= [1, 1, 1, 1]
Generation 5280: Best = 4.642926e+03 FES = 528100 Diversity = 4.7815e-03 k_t
= [4, 3, 2, 3]
Generation 5290: Best = 4.642923e+03 FES = 529100 Diversity = 4.6248e-03 k_t
= [1, 1, 1, 1]
Generation 5300: Best = 4.642920e+03 FES = 530100 Diversity = 4.5057e-03 k_t
= [4, 4, 3, 3]
Generation 5310: Best = 4.642915e+03 FES = 531100 Diversity = 4.3257e-03 k_t
= [1, 1, 1, 1]
Generation 5320: Best = 4.642911e+03 FES = 532100 Diversity = 5.2087e-03 k_t
= [3, 2, 3, 3]
Generation 5330: Best = 4.642905e+03 FES = 533100 Diversity = 5.1887e-03 k_t
= [1, 1, 1, 1]
Generation 5340: Best = 4.642899e+03 FES = 534100 Diversity = 5.2237e-03 k_t
= [4, 2, 2, 4]
Generation 5350: Best = 4.642895e+03 FES = 535100 Diversity = 5.1189e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 5360: Best = 4.642889e+03 FES = 536100 Diversity = 5.6586e-03 k_t
= [4, 3, 3, 2]
Generation 5370: Best = 4.642884e+03 FES = 537100 Diversity = 5.0152e-03 k_t
= [1, 1, 1, 1]
Generation 5380: Best = 4.642878e+03 FES = 538100 Diversity = 4.7462e-03 k_t
= [4, 2, 3, 3]
Generation 5390: Best = 4.642874e+03 FES = 539100 Diversity = 4.7353e-03 k_t
= [1, 1, 1, 1]
Generation 5400: Best = 4.642869e+03 FES = 540100 Diversity = 4.5551e-03 k_t
= [2, 2, 2, 2]
Generation 5410: Best = 4.642865e+03 FES = 541100 Diversity = 4.5585e-03 k_t
= [1, 1, 1, 1]
Generation 5420: Best = 4.642861e+03 FES = 542100 Diversity = 4.2821e-03 k_t
= [3, 2, 3, 3]
Generation 5430: Best = 4.642857e+03 FES = 543100 Diversity = 4.1398e-03 k_t
= [1, 1, 1, 1]
Generation 5440: Best = 4.642854e+03 FES = 544100 Diversity = 3.9628e-03 k_t
= [2, 4, 4, 4]
Generation 5450: Best = 4.642850e+03 FES = 545100 Diversity = 3.7607e-03 k_t
= [1, 1, 1, 1]
Generation 5460: Best = 4.642847e+03 FES = 546100 Diversity = 3.3938e-03 k_t
= [3, 3, 3, 3]
Generation 5470: Best = 4.642843e+03 FES = 547100 Diversity = 3.5327e-03 k_t
= [1, 1, 1, 1]
Generation 5480: Best = 4.642839e+03 FES = 548100 Diversity = 3.7587e-03 k_t
= [3, 4, 3, 3]
Generation 5490: Best = 4.642835e+03 FES = 549100 Diversity = 3.9378e-03 k_t
= [1, 1, 1, 1]
Generation 5500: Best = 4.642831e+03 FES = 550100 Diversity = 4.1910e-03 k_t
= [4, 2, 2, 3]
Generation 5510: Best = 4.642826e+03 FES = 551100 Diversity = 4.6488e-03 k_t
= [1, 1, 1, 1]
Generation 5520: Best = 4.642823e+03 FES = 552100 Diversity = 4.9225e-03 k_t
= [2, 4, 4, 4]
Generation 5530: Best = 4.642818e+03 FES = 553100 Diversity = 4.6762e-03 k_t
= [1, 1, 1, 1]
Generation 5540: Best = 4.642815e+03 FES = 554100 Diversity = 4.8264e-03 k_t
= [4, 4, 2, 3]
Generation 5550: Best = 4.642810e+03 FES = 555100 Diversity = 4.9082e-03 k_t
= [1, 1, 1, 1]
Generation 5560: Best = 4.642804e+03 FES = 556100 Diversity = 5.0190e-03 k_t
= [3, 4, 2, 2]
Generation 5570: Best = 4.642799e+03 FES = 557100 Diversity = 4.9301e-03 k_t
= [1, 1, 1, 1]
Generation 5580: Best = 4.642794e+03 FES = 558100 Diversity = 4.7191e-03 k_t
= [2, 2, 2, 4]
Generation 5590: Best = 4.642789e+03 FES = 559100 Diversity = 5.0111e-03 k_t
= [1, 1, 1, 1]
Generation 5600: Best = 4.642784e+03 FES = 560100 Diversity = 5.5205e-03 k_t
= [2, 2, 4, 2]
Generation 5610: Best = 4.642779e+03 FES = 561100 Diversity = 5.6926e-03 k_t
= [1, 1, 1, 1]
Generation 5620: Best = 4.642775e+03 FES = 562100 Diversity = 5.5187e-03 k_t

```

```

= [2, 2, 4, 2]
Generation 5630: Best = 4.642771e+03 FES = 563100 Diversity = 4.8617e-03 k_t
= [1, 1, 1, 1]
Generation 5640: Best = 4.642768e+03 FES = 564100 Diversity = 4.8084e-03 k_t
= [4, 2, 4, 4]
Generation 5650: Best = 4.642764e+03 FES = 565100 Diversity = 5.1460e-03 k_t
= [1, 1, 1, 1]
Generation 5660: Best = 4.642760e+03 FES = 566100 Diversity = 4.5296e-03 k_t
= [4, 2, 4, 3]
Generation 5670: Best = 4.642756e+03 FES = 567100 Diversity = 4.0815e-03 k_t
= [1, 1, 1, 1]
Generation 5680: Best = 4.642753e+03 FES = 568100 Diversity = 4.0214e-03 k_t
= [2, 2, 2, 3]
Generation 5690: Best = 4.642749e+03 FES = 569100 Diversity = 3.8030e-03 k_t
= [1, 1, 1, 1]
Generation 5700: Best = 4.642747e+03 FES = 570100 Diversity = 3.6579e-03 k_t
= [4, 2, 2, 2]
Generation 5710: Best = 4.642742e+03 FES = 571100 Diversity = 3.8141e-03 k_t
= [1, 1, 1, 1]
Generation 5720: Best = 4.642739e+03 FES = 572100 Diversity = 3.6700e-03 k_t
= [4, 2, 2, 3]
Generation 5730: Best = 4.642736e+03 FES = 573100 Diversity = 3.7860e-03 k_t
= [1, 1, 1, 1]
Generation 5740: Best = 4.642732e+03 FES = 574100 Diversity = 4.2841e-03 k_t
= [2, 3, 3, 2]
Generation 5750: Best = 4.642728e+03 FES = 575100 Diversity = 4.6383e-03 k_t
= [1, 1, 1, 1]
Generation 5760: Best = 4.642725e+03 FES = 576100 Diversity = 4.9376e-03 k_t
= [2, 3, 3, 3]
Generation 5770: Best = 4.642723e+03 FES = 577100 Diversity = 4.6936e-03 k_t
= [1, 1, 1, 1]
Generation 5780: Best = 4.642720e+03 FES = 578100 Diversity = 4.6875e-03 k_t
= [3, 2, 2, 2]
Generation 5790: Best = 4.642717e+03 FES = 579100 Diversity = 4.4334e-03 k_t
= [1, 1, 1, 1]
Generation 5800: Best = 4.642715e+03 FES = 580100 Diversity = 4.1102e-03 k_t
= [3, 3, 3, 3]
Generation 5810: Best = 4.642712e+03 FES = 581100 Diversity = 4.1861e-03 k_t
= [1, 1, 1, 1]
Generation 5820: Best = 4.642709e+03 FES = 582100 Diversity = 4.1280e-03 k_t
= [3, 2, 2, 4]
Generation 5830: Best = 4.642706e+03 FES = 583100 Diversity = 4.3967e-03 k_t
= [1, 1, 1, 1]
Generation 5840: Best = 4.642703e+03 FES = 584100 Diversity = 4.0566e-03 k_t
= [2, 4, 2, 4]
Generation 5850: Best = 4.642699e+03 FES = 585100 Diversity = 4.1130e-03 k_t
= [1, 1, 1, 1]
Generation 5860: Best = 4.642697e+03 FES = 586100 Diversity = 3.9039e-03 k_t
= [4, 4, 2, 4]
Generation 5870: Best = 4.642694e+03 FES = 587100 Diversity = 4.0415e-03 k_t
= [1, 1, 1, 1]
Generation 5880: Best = 4.642689e+03 FES = 588100 Diversity = 4.2922e-03 k_t
= [2, 2, 4, 2]
Generation 5890: Best = 4.642685e+03 FES = 589100 Diversity = 4.5071e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 5900: Best = 4.642681e+03 FES = 590100 Diversity = 4.3499e-03 k_t
= [4, 2, 2, 4]
Generation 5910: Best = 4.642677e+03 FES = 591100 Diversity = 4.4891e-03 k_t
= [1, 1, 1, 1]
Generation 5920: Best = 4.642674e+03 FES = 592100 Diversity = 4.3074e-03 k_t
= [2, 4, 2, 2]
Generation 5930: Best = 4.642671e+03 FES = 593100 Diversity = 3.9780e-03 k_t
= [1, 1, 1, 1]
Generation 5940: Best = 4.642668e+03 FES = 594100 Diversity = 3.8667e-03 k_t
= [3, 4, 3, 3]
Generation 5950: Best = 4.642666e+03 FES = 595100 Diversity = 3.9057e-03 k_t
= [1, 1, 1, 1]
Generation 5960: Best = 4.642664e+03 FES = 596100 Diversity = 3.5490e-03 k_t
= [3, 4, 2, 3]
Generation 5970: Best = 4.642662e+03 FES = 597100 Diversity = 3.4906e-03 k_t
= [1, 1, 1, 1]
Generation 5980: Best = 4.642661e+03 FES = 598100 Diversity = 3.4160e-03 k_t
= [2, 2, 3, 2]
Generation 5990: Best = 4.642659e+03 FES = 599100 Diversity = 3.2609e-03 k_t
= [1, 1, 1, 1]
Generation 6000: Best = 4.642657e+03 FES = 600100 Diversity = 3.0273e-03 k_t
= [2, 2, 2, 2]
Generation 6010: Best = 4.642655e+03 FES = 601100 Diversity = 3.0227e-03 k_t
= [1, 1, 1, 1]
Generation 6020: Best = 4.642654e+03 FES = 602100 Diversity = 2.8088e-03 k_t
= [4, 2, 4, 2]
Generation 6030: Best = 4.642653e+03 FES = 603100 Diversity = 2.5583e-03 k_t
= [1, 1, 1, 1]
Generation 6040: Best = 4.642650e+03 FES = 604100 Diversity = 2.6641e-03 k_t
= [3, 2, 2, 3]
Generation 6050: Best = 4.642648e+03 FES = 605100 Diversity = 3.4791e-03 k_t
= [1, 1, 1, 1]
Generation 6060: Best = 4.642646e+03 FES = 606100 Diversity = 3.1617e-03 k_t
= [4, 3, 2, 3]
Generation 6070: Best = 4.642643e+03 FES = 607100 Diversity = 3.0964e-03 k_t
= [1, 1, 1, 1]
Generation 6080: Best = 4.642638e+03 FES = 608100 Diversity = 4.0214e-03 k_t
= [3, 4, 3, 2]
Generation 6090: Best = 4.642635e+03 FES = 609100 Diversity = 4.0117e-03 k_t
= [1, 1, 1, 1]
Generation 6100: Best = 4.642632e+03 FES = 610100 Diversity = 3.9996e-03 k_t
= [3, 3, 3, 4]
Generation 6110: Best = 4.642629e+03 FES = 611100 Diversity = 4.2235e-03 k_t
= [1, 1, 1, 1]
Generation 6120: Best = 4.642626e+03 FES = 612100 Diversity = 4.3197e-03 k_t
= [2, 2, 4, 4]
Generation 6130: Best = 4.642623e+03 FES = 613100 Diversity = 4.2420e-03 k_t
= [1, 1, 1, 1]
Generation 6140: Best = 4.642620e+03 FES = 614100 Diversity = 3.9083e-03 k_t
= [3, 2, 4, 2]
Generation 6150: Best = 4.642617e+03 FES = 615100 Diversity = 3.7082e-03 k_t
= [1, 1, 1, 1]
Generation 6160: Best = 4.642614e+03 FES = 616100 Diversity = 3.2756e-03 k_t

```

```

= [2, 4, 4, 3]
Generation 6170: Best = 4.642612e+03 FES = 617100 Diversity = 3.1025e-03 k_t
= [1, 1, 1, 1]
Generation 6180: Best = 4.642609e+03 FES = 618100 Diversity = 2.6213e-03 k_t
= [2, 3, 2, 2]
Generation 6190: Best = 4.642608e+03 FES = 619100 Diversity = 2.2738e-03 k_t
= [1, 1, 1, 1]
Generation 6200: Best = 4.642605e+03 FES = 620100 Diversity = 2.4819e-03 k_t
= [4, 4, 4, 3]
Generation 6210: Best = 4.642603e+03 FES = 621100 Diversity = 2.4694e-03 k_t
= [1, 1, 1, 1]
Generation 6220: Best = 4.642602e+03 FES = 622100 Diversity = 2.4624e-03 k_t
= [3, 3, 4, 2]
Generation 6230: Best = 4.642600e+03 FES = 623100 Diversity = 2.2123e-03 k_t
= [1, 1, 1, 1]
Generation 6240: Best = 4.642598e+03 FES = 624100 Diversity = 2.1982e-03 k_t
= [4, 2, 3, 3]
Generation 6250: Best = 4.642597e+03 FES = 625100 Diversity = 2.4414e-03 k_t
= [1, 1, 1, 1]
Generation 6260: Best = 4.642595e+03 FES = 626100 Diversity = 2.6085e-03 k_t
= [2, 4, 4, 3]
Generation 6270: Best = 4.642593e+03 FES = 627100 Diversity = 2.9105e-03 k_t
= [1, 1, 1, 1]
Generation 6280: Best = 4.642590e+03 FES = 628100 Diversity = 2.9206e-03 k_t
= [2, 3, 3, 4]
Generation 6290: Best = 4.642588e+03 FES = 629100 Diversity = 3.0857e-03 k_t
= [1, 1, 1, 1]
Generation 6300: Best = 4.642586e+03 FES = 630100 Diversity = 2.7315e-03 k_t
= [4, 3, 3, 4]
Generation 6310: Best = 4.642584e+03 FES = 631100 Diversity = 2.9986e-03 k_t
= [1, 1, 1, 1]
Generation 6320: Best = 4.642582e+03 FES = 632100 Diversity = 2.7827e-03 k_t
= [4, 4, 3, 2]
Generation 6330: Best = 4.642581e+03 FES = 633100 Diversity = 2.8361e-03 k_t
= [1, 1, 1, 1]
Generation 6340: Best = 4.642579e+03 FES = 634100 Diversity = 2.6213e-03 k_t
= [2, 2, 4, 4]
Generation 6350: Best = 4.642577e+03 FES = 635100 Diversity = 2.5725e-03 k_t
= [1, 1, 1, 1]
Generation 6360: Best = 4.642576e+03 FES = 636100 Diversity = 2.6667e-03 k_t
= [2, 4, 2, 3]
Generation 6370: Best = 4.642575e+03 FES = 637100 Diversity = 2.3762e-03 k_t
= [1, 1, 1, 1]
Generation 6380: Best = 4.642573e+03 FES = 638100 Diversity = 2.4819e-03 k_t
= [2, 4, 2, 2]
Generation 6390: Best = 4.642572e+03 FES = 639100 Diversity = 2.3495e-03 k_t
= [1, 1, 1, 1]
Generation 6400: Best = 4.642571e+03 FES = 640100 Diversity = 1.9294e-03 k_t
= [2, 2, 3, 3]
Generation 6410: Best = 4.642570e+03 FES = 641100 Diversity = 1.8005e-03 k_t
= [1, 1, 1, 1]
Generation 6420: Best = 4.642568e+03 FES = 642100 Diversity = 1.7105e-03 k_t
= [3, 3, 4, 4]
Generation 6430: Best = 4.642567e+03 FES = 643100 Diversity = 1.8686e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 6440: Best = 4.642566e+03 FES = 644100 Diversity = 1.9242e-03 k_t
= [4, 4, 2, 4]
Generation 6450: Best = 4.642564e+03 FES = 645100 Diversity = 1.9311e-03 k_t
= [1, 1, 1, 1]
Generation 6460: Best = 4.642563e+03 FES = 646100 Diversity = 1.7722e-03 k_t
= [4, 4, 2, 2]
Generation 6470: Best = 4.642561e+03 FES = 647100 Diversity = 1.6743e-03 k_t
= [1, 1, 1, 1]
Generation 6480: Best = 4.642560e+03 FES = 648100 Diversity = 1.7060e-03 k_t
= [2, 4, 2, 3]
Generation 6490: Best = 4.642559e+03 FES = 649100 Diversity = 1.7064e-03 k_t
= [1, 1, 1, 1]
Generation 6500: Best = 4.642557e+03 FES = 650100 Diversity = 1.8574e-03 k_t
= [4, 4, 3, 4]
Generation 6510: Best = 4.642556e+03 FES = 651100 Diversity = 1.8522e-03 k_t
= [1, 1, 1, 1]
Generation 6520: Best = 4.642554e+03 FES = 652100 Diversity = 2.0313e-03 k_t
= [2, 2, 3, 2]
Generation 6530: Best = 4.642553e+03 FES = 653100 Diversity = 1.9899e-03 k_t
= [1, 1, 1, 1]
Generation 6540: Best = 4.642551e+03 FES = 654100 Diversity = 2.0664e-03 k_t
= [3, 2, 2, 4]
Generation 6550: Best = 4.642551e+03 FES = 655100 Diversity = 1.9858e-03 k_t
= [1, 1, 1, 1]
Generation 6560: Best = 4.642550e+03 FES = 656100 Diversity = 1.9375e-03 k_t
= [2, 2, 3, 4]
Generation 6570: Best = 4.642549e+03 FES = 657100 Diversity = 1.9690e-03 k_t
= [1, 1, 1, 1]
Generation 6580: Best = 4.642548e+03 FES = 658100 Diversity = 2.1318e-03 k_t
= [2, 2, 3, 4]
Generation 6590: Best = 4.642547e+03 FES = 659100 Diversity = 2.0785e-03 k_t
= [1, 1, 1, 1]
Generation 6600: Best = 4.642546e+03 FES = 660100 Diversity = 1.9880e-03 k_t
= [4, 4, 4, 4]
Generation 6610: Best = 4.642545e+03 FES = 661100 Diversity = 1.8699e-03 k_t
= [1, 1, 1, 1]
Generation 6620: Best = 4.642544e+03 FES = 662100 Diversity = 2.0342e-03 k_t
= [4, 4, 2, 2]
Generation 6630: Best = 4.642543e+03 FES = 663100 Diversity = 2.1834e-03 k_t
= [1, 1, 1, 1]
Generation 6640: Best = 4.642542e+03 FES = 664100 Diversity = 2.0978e-03 k_t
= [4, 4, 4, 4]
Generation 6650: Best = 4.642541e+03 FES = 665100 Diversity = 2.1300e-03 k_t
= [1, 1, 1, 1]
Generation 6660: Best = 4.642539e+03 FES = 666100 Diversity = 2.2067e-03 k_t
= [2, 4, 2, 4]
Generation 6670: Best = 4.642538e+03 FES = 667100 Diversity = 2.1167e-03 k_t
= [1, 1, 1, 1]
Generation 6680: Best = 4.642537e+03 FES = 668100 Diversity = 2.0666e-03 k_t
= [3, 4, 3, 4]
Generation 6690: Best = 4.642536e+03 FES = 669100 Diversity = 2.1091e-03 k_t
= [1, 1, 1, 1]
Generation 6700: Best = 4.642534e+03 FES = 670100 Diversity = 2.2572e-03 k_t

```



```

= [4, 2, 2, 4]
Generation 6710: Best = 4.642533e+03 FES = 671100 Diversity = 2.1746e-03 k_t
= [1, 1, 1, 1]
Generation 6720: Best = 4.642532e+03 FES = 672100 Diversity = 2.2981e-03 k_t
= [2, 3, 4, 3]
Generation 6730: Best = 4.642531e+03 FES = 673100 Diversity = 1.9223e-03 k_t
= [1, 1, 1, 1]
Generation 6740: Best = 4.642531e+03 FES = 674100 Diversity = 2.1579e-03 k_t
= [2, 3, 2, 4]
Generation 6750: Best = 4.642530e+03 FES = 675100 Diversity = 1.9116e-03 k_t
= [1, 1, 1, 1]
Generation 6760: Best = 4.642529e+03 FES = 676100 Diversity = 2.1113e-03 k_t
= [2, 2, 3, 2]
Generation 6770: Best = 4.642528e+03 FES = 677100 Diversity = 2.1078e-03 k_t
= [1, 1, 1, 1]
Generation 6780: Best = 4.642527e+03 FES = 678100 Diversity = 1.8433e-03 k_t
= [4, 2, 4, 4]
Generation 6790: Best = 4.642526e+03 FES = 679100 Diversity = 1.7863e-03 k_t
= [1, 1, 1, 1]
Generation 6800: Best = 4.642525e+03 FES = 680100 Diversity = 1.7617e-03 k_t
= [2, 2, 3, 2]
Generation 6810: Best = 4.642525e+03 FES = 681100 Diversity = 1.7829e-03 k_t
= [1, 1, 1, 1]
Generation 6820: Best = 4.642524e+03 FES = 682100 Diversity = 1.6337e-03 k_t
= [3, 4, 3, 4]
Generation 6830: Best = 4.642523e+03 FES = 683100 Diversity = 1.6947e-03 k_t
= [1, 1, 1, 1]
Generation 6840: Best = 4.642522e+03 FES = 684100 Diversity = 1.7391e-03 k_t
= [3, 2, 2, 2]
Generation 6850: Best = 4.642522e+03 FES = 685100 Diversity = 1.7874e-03 k_t
= [1, 1, 1, 1]
Generation 6860: Best = 4.642521e+03 FES = 686100 Diversity = 1.8241e-03 k_t
= [4, 2, 3, 4]
Generation 6870: Best = 4.642520e+03 FES = 687100 Diversity = 1.7722e-03 k_t
= [1, 1, 1, 1]
Generation 6880: Best = 4.642520e+03 FES = 688100 Diversity = 1.6926e-03 k_t
= [2, 2, 2, 2]
Generation 6890: Best = 4.642519e+03 FES = 689100 Diversity = 1.7082e-03 k_t
= [1, 1, 1, 1]
Generation 6900: Best = 4.642518e+03 FES = 690100 Diversity = 1.8968e-03 k_t
= [3, 4, 4, 2]
Generation 6910: Best = 4.642518e+03 FES = 691100 Diversity = 2.0947e-03 k_t
= [1, 1, 1, 1]
Generation 6920: Best = 4.642517e+03 FES = 692100 Diversity = 2.0650e-03 k_t
= [2, 4, 4, 2]
Generation 6930: Best = 4.642516e+03 FES = 693100 Diversity = 2.2090e-03 k_t
= [1, 1, 1, 1]
Generation 6940: Best = 4.642515e+03 FES = 694100 Diversity = 2.2113e-03 k_t
= [3, 3, 3, 4]
Generation 6950: Best = 4.642514e+03 FES = 695100 Diversity = 2.3055e-03 k_t
= [1, 1, 1, 1]
Generation 6960: Best = 4.642513e+03 FES = 696100 Diversity = 2.0091e-03 k_t
= [4, 3, 3, 3]
Generation 6970: Best = 4.642512e+03 FES = 697100 Diversity = 2.0383e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 6980: Best = 4.642512e+03 FES = 698100 Diversity = 2.2105e-03 k_t
= [3, 3, 2, 4]
Generation 6990: Best = 4.642511e+03 FES = 699100 Diversity = 2.0539e-03 k_t
= [1, 1, 1, 1]
Generation 7000: Best = 4.642510e+03 FES = 700100 Diversity = 1.6958e-03 k_t
= [4, 4, 2, 2]
Generation 7010: Best = 4.642510e+03 FES = 701100 Diversity = 1.4635e-03 k_t
= [1, 1, 1, 1]
Generation 7020: Best = 4.642510e+03 FES = 702100 Diversity = 1.2419e-03 k_t
= [2, 4, 4, 2]
Generation 7030: Best = 4.642509e+03 FES = 703100 Diversity = 1.1479e-03 k_t
= [1, 1, 1, 1]
Generation 7040: Best = 4.642509e+03 FES = 704100 Diversity = 1.0799e-03 k_t
= [3, 3, 2, 4]
Generation 7050: Best = 4.642509e+03 FES = 705100 Diversity = 1.2521e-03 k_t
= [1, 1, 1, 1]
Generation 7060: Best = 4.642508e+03 FES = 706100 Diversity = 1.3492e-03 k_t
= [4, 3, 4, 4]
Generation 7070: Best = 4.642508e+03 FES = 707100 Diversity = 1.2041e-03 k_t
= [1, 1, 1, 1]
Generation 7080: Best = 4.642507e+03 FES = 708100 Diversity = 1.2549e-03 k_t
= [4, 4, 3, 2]
Generation 7090: Best = 4.642507e+03 FES = 709100 Diversity = 1.2310e-03 k_t
= [1, 1, 1, 1]
Generation 7100: Best = 4.642506e+03 FES = 710100 Diversity = 1.3204e-03 k_t
= [2, 2, 3, 4]
Generation 7110: Best = 4.642506e+03 FES = 711100 Diversity = 1.3963e-03 k_t
= [1, 1, 1, 1]
Generation 7120: Best = 4.642505e+03 FES = 712100 Diversity = 1.4690e-03 k_t
= [3, 4, 2, 4]
Generation 7130: Best = 4.642505e+03 FES = 713100 Diversity = 1.2335e-03 k_t
= [1, 1, 1, 1]
Generation 7140: Best = 4.642505e+03 FES = 714100 Diversity = 9.9411e-04 k_t
= [4, 3, 3, 4]
Generation 7150: Best = 4.642504e+03 FES = 715100 Diversity = 9.9951e-04 k_t
= [1, 1, 1, 1]
Generation 7160: Best = 4.642504e+03 FES = 716100 Diversity = 1.1260e-03 k_t
= [4, 4, 4, 4]
Generation 7170: Best = 4.642504e+03 FES = 717100 Diversity = 1.0155e-03 k_t
= [1, 1, 1, 1]
Generation 7180: Best = 4.642503e+03 FES = 718100 Diversity = 1.1279e-03 k_t
= [3, 4, 4, 2]
Generation 7190: Best = 4.642503e+03 FES = 719100 Diversity = 9.4608e-04 k_t
= [1, 1, 1, 1]
Generation 7200: Best = 4.642503e+03 FES = 720100 Diversity = 1.0210e-03 k_t
= [3, 4, 2, 3]
Generation 7210: Best = 4.642502e+03 FES = 721100 Diversity = 1.0633e-03 k_t
= [1, 1, 1, 1]
Generation 7220: Best = 4.642502e+03 FES = 722100 Diversity = 1.1231e-03 k_t
= [2, 3, 2, 3]
Generation 7230: Best = 4.642502e+03 FES = 723100 Diversity = 1.0003e-03 k_t
= [1, 1, 1, 1]
Generation 7240: Best = 4.642502e+03 FES = 724100 Diversity = 9.3813e-04 k_t

```

```

= [4, 4, 3, 4]
Generation 7250: Best = 4.642501e+03 FES = 725100 Diversity = 8.6379e-04 k_t
= [1, 1, 1, 1]
Generation 7260: Best = 4.642501e+03 FES = 726100 Diversity = 1.0717e-03 k_t
= [3, 2, 3, 3]
Generation 7270: Best = 4.642500e+03 FES = 727100 Diversity = 1.3260e-03 k_t
= [1, 1, 1, 1]
Generation 7280: Best = 4.642500e+03 FES = 728100 Diversity = 1.3383e-03 k_t
= [3, 4, 2, 4]
Generation 7290: Best = 4.642499e+03 FES = 729100 Diversity = 1.3382e-03 k_t
= [1, 1, 1, 1]
Generation 7300: Best = 4.642499e+03 FES = 730100 Diversity = 1.2249e-03 k_t
= [3, 2, 4, 3]
Generation 7310: Best = 4.642499e+03 FES = 731100 Diversity = 1.1413e-03 k_t
= [1, 1, 1, 1]
Generation 7320: Best = 4.642498e+03 FES = 732100 Diversity = 1.0185e-03 k_t
= [2, 4, 2, 2]
Generation 7330: Best = 4.642498e+03 FES = 733100 Diversity = 9.6530e-04 k_t
= [1, 1, 1, 1]
Generation 7340: Best = 4.642498e+03 FES = 734100 Diversity = 1.1791e-03 k_t
= [4, 3, 2, 4]
Generation 7350: Best = 4.642497e+03 FES = 735100 Diversity = 1.0627e-03 k_t
= [1, 1, 1, 1]
Generation 7360: Best = 4.642497e+03 FES = 736100 Diversity = 1.0500e-03 k_t
= [4, 2, 4, 4]
Generation 7370: Best = 4.642497e+03 FES = 737100 Diversity = 1.2489e-03 k_t
= [1, 1, 1, 1]
Generation 7380: Best = 4.642496e+03 FES = 738100 Diversity = 1.2013e-03 k_t
= [2, 2, 3, 3]
Generation 7390: Best = 4.642496e+03 FES = 739100 Diversity = 1.1359e-03 k_t
= [1, 1, 1, 1]
Generation 7400: Best = 4.642495e+03 FES = 740100 Diversity = 1.1735e-03 k_t
= [3, 3, 4, 4]
Generation 7410: Best = 4.642495e+03 FES = 741100 Diversity = 1.2144e-03 k_t
= [1, 1, 1, 1]
Generation 7420: Best = 4.642494e+03 FES = 742100 Diversity = 1.2207e-03 k_t
= [4, 4, 2, 4]
Generation 7430: Best = 4.642494e+03 FES = 743100 Diversity = 1.1588e-03 k_t
= [1, 1, 1, 1]
Generation 7440: Best = 4.642494e+03 FES = 744100 Diversity = 1.1215e-03 k_t
= [2, 3, 4, 4]
Generation 7450: Best = 4.642493e+03 FES = 745100 Diversity = 9.9587e-04 k_t
= [1, 1, 1, 1]
Generation 7460: Best = 4.642493e+03 FES = 746100 Diversity = 1.0252e-03 k_t
= [4, 2, 2, 2]
Generation 7470: Best = 4.642493e+03 FES = 747100 Diversity = 8.1029e-04 k_t
= [1, 1, 1, 1]
Generation 7480: Best = 4.642492e+03 FES = 748100 Diversity = 9.6668e-04 k_t
= [2, 4, 4, 4]
Generation 7490: Best = 4.642492e+03 FES = 749100 Diversity = 8.5157e-04 k_t
= [1, 1, 1, 1]
Generation 7500: Best = 4.642492e+03 FES = 750100 Diversity = 7.3697e-04 k_t
= [2, 2, 3, 4]
Generation 7510: Best = 4.642491e+03 FES = 751100 Diversity = 8.0474e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 7520: Best = 4.642491e+03 FES = 752100 Diversity = 8.6746e-04 k_t
= [2, 3, 2, 3]
Generation 7530: Best = 4.642491e+03 FES = 753100 Diversity = 8.5151e-04 k_t
= [1, 1, 1, 1]
Generation 7540: Best = 4.642490e+03 FES = 754100 Diversity = 7.0946e-04 k_t
= [3, 2, 4, 3]
Generation 7550: Best = 4.642490e+03 FES = 755100 Diversity = 6.1073e-04 k_t
= [1, 1, 1, 1]
Generation 7560: Best = 4.642490e+03 FES = 756100 Diversity = 9.0171e-04 k_t
= [4, 3, 2, 3]
Generation 7570: Best = 4.642490e+03 FES = 757100 Diversity = 8.6457e-04 k_t
= [1, 1, 1, 1]
Generation 7580: Best = 4.642489e+03 FES = 758100 Diversity = 8.0960e-04 k_t
= [2, 2, 2, 2]
Generation 7590: Best = 4.642489e+03 FES = 759100 Diversity = 8.5436e-04 k_t
= [1, 1, 1, 1]
Generation 7600: Best = 4.642489e+03 FES = 760100 Diversity = 9.2002e-04 k_t
= [2, 2, 4, 2]
Generation 7610: Best = 4.642489e+03 FES = 761100 Diversity = 8.7654e-04 k_t
= [1, 1, 1, 1]
Generation 7620: Best = 4.642489e+03 FES = 762100 Diversity = 1.0635e-03 k_t
= [3, 3, 4, 4]
Generation 7630: Best = 4.642488e+03 FES = 763100 Diversity = 8.7848e-04 k_t
= [1, 1, 1, 1]
Generation 7640: Best = 4.642488e+03 FES = 764100 Diversity = 8.2391e-04 k_t
= [2, 4, 3, 2]
Generation 7650: Best = 4.642488e+03 FES = 765100 Diversity = 7.5509e-04 k_t
= [1, 1, 1, 1]
Generation 7660: Best = 4.642488e+03 FES = 766100 Diversity = 6.8762e-04 k_t
= [4, 4, 3, 3]
Generation 7670: Best = 4.642488e+03 FES = 767100 Diversity = 8.0172e-04 k_t
= [1, 1, 1, 1]
Generation 7680: Best = 4.642487e+03 FES = 768100 Diversity = 8.7140e-04 k_t
= [4, 2, 4, 4]
Generation 7690: Best = 4.642487e+03 FES = 769100 Diversity = 9.2875e-04 k_t
= [1, 1, 1, 1]
Generation 7700: Best = 4.642487e+03 FES = 770100 Diversity = 9.8742e-04 k_t
= [3, 4, 3, 4]
Generation 7710: Best = 4.642487e+03 FES = 771100 Diversity = 7.9662e-04 k_t
= [1, 1, 1, 1]
Generation 7720: Best = 4.642487e+03 FES = 772100 Diversity = 6.9151e-04 k_t
= [3, 4, 4, 2]
Generation 7730: Best = 4.642487e+03 FES = 773100 Diversity = 5.5312e-04 k_t
= [1, 1, 1, 1]
Generation 7740: Best = 4.642486e+03 FES = 774100 Diversity = 5.8115e-04 k_t
= [3, 3, 2, 4]
Generation 7750: Best = 4.642486e+03 FES = 775100 Diversity = 5.1021e-04 k_t
= [1, 1, 1, 1]
Generation 7760: Best = 4.642486e+03 FES = 776100 Diversity = 4.9262e-04 k_t
= [3, 3, 4, 4]
Generation 7770: Best = 4.642486e+03 FES = 777100 Diversity = 4.6219e-04 k_t
= [1, 1, 1, 1]
Generation 7780: Best = 4.642486e+03 FES = 778100 Diversity = 6.7100e-04 k_t

```

```

= [4, 4, 4, 2]
Generation 7790: Best = 4.642486e+03 FES = 779100 Diversity = 6.7896e-04 k_t
= [1, 1, 1, 1]
Generation 7800: Best = 4.642486e+03 FES = 780100 Diversity = 5.2116e-04 k_t
= [2, 2, 4, 2]
Generation 7810: Best = 4.642486e+03 FES = 781100 Diversity = 5.3014e-04 k_t
= [1, 1, 1, 1]
Generation 7820: Best = 4.642486e+03 FES = 782100 Diversity = 4.5713e-04 k_t
= [3, 3, 4, 3]
Generation 7830: Best = 4.642485e+03 FES = 783100 Diversity = 4.5640e-04 k_t
= [1, 1, 1, 1]
Generation 7840: Best = 4.642485e+03 FES = 784100 Diversity = 4.6170e-04 k_t
= [4, 4, 3, 4]
Generation 7850: Best = 4.642485e+03 FES = 785100 Diversity = 4.9189e-04 k_t
= [1, 1, 1, 1]
Generation 7860: Best = 4.642485e+03 FES = 786100 Diversity = 4.1792e-04 k_t
= [2, 4, 2, 4]
Generation 7870: Best = 4.642485e+03 FES = 787100 Diversity = 3.7457e-04 k_t
= [1, 1, 1, 1]
Generation 7880: Best = 4.642485e+03 FES = 788100 Diversity = 3.9567e-04 k_t
= [2, 2, 4, 2]
Generation 7890: Best = 4.642485e+03 FES = 789100 Diversity = 3.5197e-04 k_t
= [1, 1, 1, 1]
Generation 7900: Best = 4.642485e+03 FES = 790100 Diversity = 3.9866e-04 k_t
= [2, 3, 4, 4]
Generation 7910: Best = 4.642485e+03 FES = 791100 Diversity = 4.1641e-04 k_t
= [1, 1, 1, 1]
Generation 7920: Best = 4.642485e+03 FES = 792100 Diversity = 5.3248e-04 k_t
= [3, 4, 3, 3]
Generation 7930: Best = 4.642485e+03 FES = 793100 Diversity = 6.7125e-04 k_t
= [1, 1, 1, 1]
Generation 7940: Best = 4.642484e+03 FES = 794100 Diversity = 6.4005e-04 k_t
= [3, 3, 3, 3]
Generation 7950: Best = 4.642484e+03 FES = 795100 Diversity = 5.8517e-04 k_t
= [1, 1, 1, 1]
Generation 7960: Best = 4.642484e+03 FES = 796100 Diversity = 5.9752e-04 k_t
= [2, 4, 4, 4]
Generation 7970: Best = 4.642484e+03 FES = 797100 Diversity = 7.0013e-04 k_t
= [1, 1, 1, 1]
Generation 7980: Best = 4.642484e+03 FES = 798100 Diversity = 7.0920e-04 k_t
= [3, 2, 3, 4]
Generation 7990: Best = 4.642483e+03 FES = 799100 Diversity = 8.4950e-04 k_t
= [1, 1, 1, 1]
Generation 8000: Best = 4.642483e+03 FES = 800100 Diversity = 9.0990e-04 k_t
= [4, 4, 2, 2]
Generation 8010: Best = 4.642483e+03 FES = 801100 Diversity = 8.2755e-04 k_t
= [1, 1, 1, 1]
Generation 8020: Best = 4.642483e+03 FES = 802100 Diversity = 8.0766e-04 k_t
= [3, 2, 4, 3]
Generation 8030: Best = 4.642482e+03 FES = 803100 Diversity = 8.0424e-04 k_t
= [1, 1, 1, 1]
Generation 8040: Best = 4.642482e+03 FES = 804100 Diversity = 7.4870e-04 k_t
= [2, 3, 3, 3]
Generation 8050: Best = 4.642482e+03 FES = 805100 Diversity = 8.1153e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 8060: Best = 4.642482e+03 FES = 806100 Diversity = 8.3710e-04 k_t
= [3, 3, 2, 4]
Generation 8070: Best = 4.642481e+03 FES = 807100 Diversity = 8.2480e-04 k_t
= [1, 1, 1, 1]
Generation 8080: Best = 4.642481e+03 FES = 808100 Diversity = 6.8911e-04 k_t
= [3, 3, 2, 4]
Generation 8090: Best = 4.642481e+03 FES = 809100 Diversity = 6.1541e-04 k_t
= [1, 1, 1, 1]
Generation 8100: Best = 4.642481e+03 FES = 810100 Diversity = 6.7260e-04 k_t
= [4, 2, 2, 3]
Generation 8110: Best = 4.642481e+03 FES = 811100 Diversity = 5.6814e-04 k_t
= [1, 1, 1, 1]
Generation 8120: Best = 4.642481e+03 FES = 812100 Diversity = 5.7660e-04 k_t
= [2, 4, 3, 2]
Generation 8130: Best = 4.642481e+03 FES = 813100 Diversity = 5.4033e-04 k_t
= [1, 1, 1, 1]
Generation 8140: Best = 4.642480e+03 FES = 814100 Diversity = 5.5128e-04 k_t
= [2, 4, 4, 3]
Generation 8150: Best = 4.642480e+03 FES = 815100 Diversity = 4.9784e-04 k_t
= [1, 1, 1, 1]
Generation 8160: Best = 4.642480e+03 FES = 816100 Diversity = 5.1916e-04 k_t
= [2, 4, 4, 4]
Generation 8170: Best = 4.642480e+03 FES = 817100 Diversity = 5.2022e-04 k_t
= [1, 1, 1, 1]
Generation 8180: Best = 4.642480e+03 FES = 818100 Diversity = 5.9396e-04 k_t
= [4, 3, 4, 3]
Generation 8190: Best = 4.642480e+03 FES = 819100 Diversity = 5.6152e-04 k_t
= [1, 1, 1, 1]
Generation 8200: Best = 4.642479e+03 FES = 820100 Diversity = 5.7222e-04 k_t
= [2, 2, 4, 2]
Generation 8210: Best = 4.642479e+03 FES = 821100 Diversity = 6.5005e-04 k_t
= [1, 1, 1, 1]
Generation 8220: Best = 4.642479e+03 FES = 822100 Diversity = 6.2124e-04 k_t
= [3, 2, 4, 2]
Generation 8230: Best = 4.642479e+03 FES = 823100 Diversity = 5.9089e-04 k_t
= [1, 1, 1, 1]
Generation 8240: Best = 4.642479e+03 FES = 824100 Diversity = 7.6896e-04 k_t
= [4, 4, 2, 4]
Generation 8250: Best = 4.642479e+03 FES = 825100 Diversity = 6.2520e-04 k_t
= [1, 1, 1, 1]
Generation 8260: Best = 4.642478e+03 FES = 826100 Diversity = 4.9999e-04 k_t
= [3, 3, 3, 3]
Generation 8270: Best = 4.642478e+03 FES = 827100 Diversity = 4.1939e-04 k_t
= [1, 1, 1, 1]
Generation 8280: Best = 4.642478e+03 FES = 828100 Diversity = 4.3323e-04 k_t
= [2, 2, 4, 3]
Generation 8290: Best = 4.642478e+03 FES = 829100 Diversity = 4.8438e-04 k_t
= [1, 1, 1, 1]
Generation 8300: Best = 4.642478e+03 FES = 830100 Diversity = 3.8393e-04 k_t
= [3, 2, 4, 2]
Generation 8310: Best = 4.642478e+03 FES = 831100 Diversity = 4.6753e-04 k_t
= [1, 1, 1, 1]
Generation 8320: Best = 4.642478e+03 FES = 832100 Diversity = 4.4063e-04 k_t

```

```

= [3, 2, 2, 3]
Generation 8330: Best = 4.642478e+03 FES = 833100 Diversity = 5.4786e-04 k_t
= [1, 1, 1, 1]
Generation 8340: Best = 4.642478e+03 FES = 834100 Diversity = 5.9243e-04 k_t
= [4, 4, 3, 3]
Generation 8350: Best = 4.642477e+03 FES = 835100 Diversity = 6.2686e-04 k_t
= [1, 1, 1, 1]
Generation 8360: Best = 4.642477e+03 FES = 836100 Diversity = 4.6738e-04 k_t
= [3, 4, 3, 3]
Generation 8370: Best = 4.642477e+03 FES = 837100 Diversity = 5.3350e-04 k_t
= [1, 1, 1, 1]
Generation 8380: Best = 4.642477e+03 FES = 838100 Diversity = 5.9963e-04 k_t
= [4, 2, 4, 2]
Generation 8390: Best = 4.642477e+03 FES = 839100 Diversity = 5.5565e-04 k_t
= [1, 1, 1, 1]
Generation 8400: Best = 4.642477e+03 FES = 840100 Diversity = 5.4418e-04 k_t
= [2, 4, 4, 2]
Generation 8410: Best = 4.642477e+03 FES = 841100 Diversity = 6.1538e-04 k_t
= [1, 1, 1, 1]
Generation 8420: Best = 4.642476e+03 FES = 842100 Diversity = 6.6438e-04 k_t
= [2, 2, 3, 2]
Generation 8430: Best = 4.642476e+03 FES = 843100 Diversity = 6.2782e-04 k_t
= [1, 1, 1, 1]
Generation 8440: Best = 4.642476e+03 FES = 844100 Diversity = 4.8764e-04 k_t
= [2, 4, 4, 4]
Generation 8450: Best = 4.642476e+03 FES = 845100 Diversity = 4.9267e-04 k_t
= [1, 1, 1, 1]
Generation 8460: Best = 4.642476e+03 FES = 846100 Diversity = 3.9310e-04 k_t
= [3, 2, 4, 3]
Generation 8470: Best = 4.642476e+03 FES = 847100 Diversity = 4.8225e-04 k_t
= [1, 1, 1, 1]
Generation 8480: Best = 4.642476e+03 FES = 848100 Diversity = 4.9633e-04 k_t
= [3, 4, 4, 4]
Generation 8490: Best = 4.642476e+03 FES = 849100 Diversity = 4.2217e-04 k_t
= [1, 1, 1, 1]
Generation 8500: Best = 4.642476e+03 FES = 850100 Diversity = 4.9620e-04 k_t
= [3, 2, 4, 2]
Generation 8510: Best = 4.642475e+03 FES = 851100 Diversity = 4.8085e-04 k_t
= [1, 1, 1, 1]
Generation 8520: Best = 4.642475e+03 FES = 852100 Diversity = 4.4608e-04 k_t
= [2, 3, 4, 3]
Generation 8530: Best = 4.642475e+03 FES = 853100 Diversity = 5.1525e-04 k_t
= [1, 1, 1, 1]
Generation 8540: Best = 4.642475e+03 FES = 854100 Diversity = 5.3588e-04 k_t
= [4, 3, 3, 3]
Generation 8550: Best = 4.642475e+03 FES = 855100 Diversity = 5.1240e-04 k_t
= [1, 1, 1, 1]
Generation 8560: Best = 4.642475e+03 FES = 856100 Diversity = 4.7731e-04 k_t
= [2, 2, 4, 2]
Generation 8570: Best = 4.642475e+03 FES = 857100 Diversity = 4.6822e-04 k_t
= [1, 1, 1, 1]
Generation 8580: Best = 4.642475e+03 FES = 858100 Diversity = 5.0564e-04 k_t
= [4, 3, 4, 3]
Generation 8590: Best = 4.642475e+03 FES = 859100 Diversity = 5.5549e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 8600: Best = 4.642474e+03 FES = 860100 Diversity = 5.1850e-04 k_t
= [2, 4, 2, 4]
Generation 8610: Best = 4.642474e+03 FES = 861100 Diversity = 4.5423e-04 k_t
= [1, 1, 1, 1]
Generation 8620: Best = 4.642474e+03 FES = 862100 Diversity = 3.6655e-04 k_t
= [2, 3, 2, 4]
Generation 8630: Best = 4.642474e+03 FES = 863100 Diversity = 3.1750e-04 k_t
= [1, 1, 1, 1]
Generation 8640: Best = 4.642474e+03 FES = 864100 Diversity = 2.9969e-04 k_t
= [4, 2, 4, 4]
Generation 8650: Best = 4.642474e+03 FES = 865100 Diversity = 2.8224e-04 k_t
= [1, 1, 1, 1]
Generation 8660: Best = 4.642474e+03 FES = 866100 Diversity = 3.5136e-04 k_t
= [2, 4, 4, 4]
Generation 8670: Best = 4.642474e+03 FES = 867100 Diversity = 5.6296e-04 k_t
= [1, 1, 1, 1]
Generation 8680: Best = 4.642474e+03 FES = 868100 Diversity = 4.7223e-04 k_t
= [2, 2, 3, 3]
Generation 8690: Best = 4.642474e+03 FES = 869100 Diversity = 4.2289e-04 k_t
= [1, 1, 1, 1]
Generation 8700: Best = 4.642474e+03 FES = 870100 Diversity = 5.0877e-04 k_t
= [2, 3, 4, 2]
Generation 8710: Best = 4.642473e+03 FES = 871100 Diversity = 5.1478e-04 k_t
= [1, 1, 1, 1]
Generation 8720: Best = 4.642473e+03 FES = 872100 Diversity = 5.8947e-04 k_t
= [4, 3, 2, 2]
Generation 8730: Best = 4.642473e+03 FES = 873100 Diversity = 5.1710e-04 k_t
= [1, 1, 1, 1]
Generation 8740: Best = 4.642473e+03 FES = 874100 Diversity = 4.7767e-04 k_t
= [3, 2, 2, 2]
Generation 8750: Best = 4.642473e+03 FES = 875100 Diversity = 4.4319e-04 k_t
= [1, 1, 1, 1]
Generation 8760: Best = 4.642473e+03 FES = 876100 Diversity = 4.0759e-04 k_t
= [4, 3, 4, 4]
Generation 8770: Best = 4.642473e+03 FES = 877100 Diversity = 4.1925e-04 k_t
= [1, 1, 1, 1]
Generation 8780: Best = 4.642473e+03 FES = 878100 Diversity = 4.3669e-04 k_t
= [2, 2, 2, 3]
Generation 8790: Best = 4.642473e+03 FES = 879100 Diversity = 4.3737e-04 k_t
= [1, 1, 1, 1]
Generation 8800: Best = 4.642473e+03 FES = 880100 Diversity = 3.5898e-04 k_t
= [3, 2, 3, 3]
Generation 8810: Best = 4.642473e+03 FES = 881100 Diversity = 3.4414e-04 k_t
= [1, 1, 1, 1]
Generation 8820: Best = 4.642473e+03 FES = 882100 Diversity = 3.3380e-04 k_t
= [3, 2, 4, 2]
Generation 8830: Best = 4.642473e+03 FES = 883100 Diversity = 3.5194e-04 k_t
= [1, 1, 1, 1]
Generation 8840: Best = 4.642472e+03 FES = 884100 Diversity = 3.3844e-04 k_t
= [2, 3, 2, 2]
Generation 8850: Best = 4.642472e+03 FES = 885100 Diversity = 3.2294e-04 k_t
= [1, 1, 1, 1]
Generation 8860: Best = 4.642472e+03 FES = 886100 Diversity = 2.9260e-04 k_t

```



```

= [4, 4, 3, 2]
Generation 8870: Best = 4.642472e+03 FES = 887100 Diversity = 3.6352e-04 k_t
= [1, 1, 1, 1]
Generation 8880: Best = 4.642472e+03 FES = 888100 Diversity = 4.3398e-04 k_t
= [2, 2, 3, 2]
Generation 8890: Best = 4.642472e+03 FES = 889100 Diversity = 4.1463e-04 k_t
= [1, 1, 1, 1]
Generation 8900: Best = 4.642472e+03 FES = 890100 Diversity = 4.9809e-04 k_t
= [2, 2, 2, 4]
Generation 8910: Best = 4.642472e+03 FES = 891100 Diversity = 4.6101e-04 k_t
= [1, 1, 1, 1]
Generation 8920: Best = 4.642472e+03 FES = 892100 Diversity = 4.1073e-04 k_t
= [2, 2, 2, 2]
Generation 8930: Best = 4.642472e+03 FES = 893100 Diversity = 3.4790e-04 k_t
= [1, 1, 1, 1]
Generation 8940: Best = 4.642472e+03 FES = 894100 Diversity = 3.5760e-04 k_t
= [4, 3, 3, 3]
Generation 8950: Best = 4.642472e+03 FES = 895100 Diversity = 3.7055e-04 k_t
= [1, 1, 1, 1]
Generation 8960: Best = 4.642471e+03 FES = 896100 Diversity = 5.4263e-04 k_t
= [3, 2, 4, 2]
Generation 8970: Best = 4.642471e+03 FES = 897100 Diversity = 6.1239e-04 k_t
= [1, 1, 1, 1]
Generation 8980: Best = 4.642471e+03 FES = 898100 Diversity = 6.8093e-04 k_t
= [4, 4, 2, 2]
Generation 8990: Best = 4.642471e+03 FES = 899100 Diversity = 6.7019e-04 k_t
= [1, 1, 1, 1]
Generation 9000: Best = 4.642470e+03 FES = 900100 Diversity = 7.4172e-04 k_t
= [2, 4, 2, 3]
Generation 9010: Best = 4.642470e+03 FES = 901100 Diversity = 9.8156e-04 k_t
= [1, 1, 1, 1]
Generation 9020: Best = 4.642470e+03 FES = 902100 Diversity = 7.4106e-04 k_t
= [3, 2, 3, 4]
Generation 9030: Best = 4.642469e+03 FES = 903100 Diversity = 7.5458e-04 k_t
= [1, 1, 1, 1]
Generation 9040: Best = 4.642469e+03 FES = 904100 Diversity = 7.3295e-04 k_t
= [3, 3, 2, 2]
Generation 9050: Best = 4.642469e+03 FES = 905100 Diversity = 7.4917e-04 k_t
= [1, 1, 1, 1]
Generation 9060: Best = 4.642469e+03 FES = 906100 Diversity = 6.7322e-04 k_t
= [3, 4, 4, 2]
Generation 9070: Best = 4.642469e+03 FES = 907100 Diversity = 5.3960e-04 k_t
= [1, 1, 1, 1]
Generation 9080: Best = 4.642469e+03 FES = 908100 Diversity = 5.7324e-04 k_t
= [4, 3, 4, 2]
Generation 9090: Best = 4.642469e+03 FES = 909100 Diversity = 5.5209e-04 k_t
= [1, 1, 1, 1]
Generation 9100: Best = 4.642468e+03 FES = 910100 Diversity = 5.9541e-04 k_t
= [3, 3, 4, 3]
Generation 9110: Best = 4.642468e+03 FES = 911100 Diversity = 5.8528e-04 k_t
= [1, 1, 1, 1]
Generation 9120: Best = 4.642468e+03 FES = 912100 Diversity = 6.2463e-04 k_t
= [3, 3, 2, 3]
Generation 9130: Best = 4.642468e+03 FES = 913100 Diversity = 6.2956e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 9140: Best = 4.642468e+03 FES = 914100 Diversity = 6.0467e-04 k_t
= [3, 4, 4, 3]
Generation 9150: Best = 4.642468e+03 FES = 915100 Diversity = 5.4401e-04 k_t
= [1, 1, 1, 1]
Generation 9160: Best = 4.642467e+03 FES = 916100 Diversity = 5.9985e-04 k_t
= [2, 3, 4, 4]
Generation 9170: Best = 4.642467e+03 FES = 917100 Diversity = 5.3539e-04 k_t
= [1, 1, 1, 1]
Generation 9180: Best = 4.642467e+03 FES = 918100 Diversity = 4.9070e-04 k_t
= [4, 4, 4, 4]
Generation 9190: Best = 4.642467e+03 FES = 919100 Diversity = 5.2180e-04 k_t
= [1, 1, 1, 1]
Generation 9200: Best = 4.642467e+03 FES = 920100 Diversity = 5.6161e-04 k_t
= [4, 2, 2, 3]
Generation 9210: Best = 4.642467e+03 FES = 921100 Diversity = 5.0239e-04 k_t
= [1, 1, 1, 1]
Generation 9220: Best = 4.642467e+03 FES = 922100 Diversity = 4.4592e-04 k_t
= [4, 4, 2, 2]
Generation 9230: Best = 4.642467e+03 FES = 923100 Diversity = 5.3477e-04 k_t
= [1, 1, 1, 1]
Generation 9240: Best = 4.642466e+03 FES = 924100 Diversity = 6.4048e-04 k_t
= [2, 2, 2, 2]
Generation 9250: Best = 4.642466e+03 FES = 925100 Diversity = 6.2906e-04 k_t
= [1, 1, 1, 1]
Generation 9260: Best = 4.642466e+03 FES = 926100 Diversity = 6.3181e-04 k_t
= [2, 3, 3, 4]
Generation 9270: Best = 4.642466e+03 FES = 927100 Diversity = 6.0400e-04 k_t
= [1, 1, 1, 1]
Generation 9280: Best = 4.642466e+03 FES = 928100 Diversity = 6.1739e-04 k_t
= [4, 4, 3, 2]
Generation 9290: Best = 4.642466e+03 FES = 929100 Diversity = 5.5134e-04 k_t
= [1, 1, 1, 1]
Generation 9300: Best = 4.642465e+03 FES = 930100 Diversity = 5.4174e-04 k_t
= [2, 3, 2, 3]
Generation 9310: Best = 4.642465e+03 FES = 931100 Diversity = 4.5018e-04 k_t
= [1, 1, 1, 1]
Generation 9320: Best = 4.642465e+03 FES = 932100 Diversity = 4.9628e-04 k_t
= [3, 3, 3, 3]
Generation 9330: Best = 4.642465e+03 FES = 933100 Diversity = 4.3163e-04 k_t
= [1, 1, 1, 1]
Generation 9340: Best = 4.642465e+03 FES = 934100 Diversity = 4.1978e-04 k_t
= [4, 2, 4, 4]
Generation 9350: Best = 4.642465e+03 FES = 935100 Diversity = 4.3492e-04 k_t
= [1, 1, 1, 1]
Generation 9360: Best = 4.642465e+03 FES = 936100 Diversity = 4.0324e-04 k_t
= [4, 4, 2, 3]
Generation 9370: Best = 4.642464e+03 FES = 937100 Diversity = 3.9297e-04 k_t
= [1, 1, 1, 1]
Generation 9380: Best = 4.642464e+03 FES = 938100 Diversity = 4.1397e-04 k_t
= [2, 4, 4, 2]
Generation 9390: Best = 4.642464e+03 FES = 939100 Diversity = 3.9075e-04 k_t
= [1, 1, 1, 1]
Generation 9400: Best = 4.642464e+03 FES = 940100 Diversity = 3.6152e-04 k_t

```

```

= [2, 2, 2, 4]
Generation 9410: Best = 4.642464e+03 FES = 941100 Diversity = 4.1609e-04 k_t
= [1, 1, 1, 1]
Generation 9420: Best = 4.642464e+03 FES = 942100 Diversity = 5.1060e-04 k_t
= [2, 3, 3, 3]
Generation 9430: Best = 4.642464e+03 FES = 943100 Diversity = 4.8999e-04 k_t
= [1, 1, 1, 1]
Generation 9440: Best = 4.642464e+03 FES = 944100 Diversity = 4.4662e-04 k_t
= [2, 4, 2, 2]
Generation 9450: Best = 4.642464e+03 FES = 945100 Diversity = 4.4634e-04 k_t
= [1, 1, 1, 1]
Generation 9460: Best = 4.642463e+03 FES = 946100 Diversity = 4.4758e-04 k_t
= [3, 4, 3, 3]
Generation 9470: Best = 4.642463e+03 FES = 947100 Diversity = 4.6350e-04 k_t
= [1, 1, 1, 1]
Generation 9480: Best = 4.642463e+03 FES = 948100 Diversity = 3.9520e-04 k_t
= [3, 4, 4, 4]
Generation 9490: Best = 4.642463e+03 FES = 949100 Diversity = 3.4731e-04 k_t
= [1, 1, 1, 1]
Generation 9500: Best = 4.642463e+03 FES = 950100 Diversity = 3.5636e-04 k_t
= [4, 4, 2, 3]
Generation 9510: Best = 4.642463e+03 FES = 951100 Diversity = 3.9592e-04 k_t
= [1, 1, 1, 1]
Generation 9520: Best = 4.642463e+03 FES = 952100 Diversity = 3.9182e-04 k_t
= [4, 2, 3, 3]
Generation 9530: Best = 4.642463e+03 FES = 953100 Diversity = 4.7321e-04 k_t
= [1, 1, 1, 1]
Generation 9540: Best = 4.642463e+03 FES = 954100 Diversity = 4.5164e-04 k_t
= [3, 4, 2, 2]
Generation 9550: Best = 4.642463e+03 FES = 955100 Diversity = 5.0842e-04 k_t
= [1, 1, 1, 1]
Generation 9560: Best = 4.642462e+03 FES = 956100 Diversity = 4.9293e-04 k_t
= [3, 4, 2, 2]
Generation 9570: Best = 4.642462e+03 FES = 957100 Diversity = 4.8830e-04 k_t
= [1, 1, 1, 1]
Generation 9580: Best = 4.642462e+03 FES = 958100 Diversity = 4.9964e-04 k_t
= [4, 4, 4, 4]
Generation 9590: Best = 4.642462e+03 FES = 959100 Diversity = 4.4494e-04 k_t
= [1, 1, 1, 1]
Generation 9600: Best = 4.642462e+03 FES = 960100 Diversity = 4.3123e-04 k_t
= [2, 2, 4, 4]
Generation 9610: Best = 4.642462e+03 FES = 961100 Diversity = 4.3779e-04 k_t
= [1, 1, 1, 1]
Generation 9620: Best = 4.642462e+03 FES = 962100 Diversity = 3.6441e-04 k_t
= [2, 3, 4, 4]
Generation 9630: Best = 4.642462e+03 FES = 963100 Diversity = 3.5758e-04 k_t
= [1, 1, 1, 1]
Generation 9640: Best = 4.642462e+03 FES = 964100 Diversity = 4.1508e-04 k_t
= [3, 4, 2, 2]
Generation 9650: Best = 4.642462e+03 FES = 965100 Diversity = 4.1426e-04 k_t
= [1, 1, 1, 1]
Generation 9660: Best = 4.642461e+03 FES = 966100 Diversity = 3.9681e-04 k_t
= [2, 3, 2, 3]
Generation 9670: Best = 4.642461e+03 FES = 967100 Diversity = 3.9463e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 9680: Best = 4.642461e+03 FES = 968100 Diversity = 3.9755e-04 k_t
= [2, 2, 2, 2]
Generation 9690: Best = 4.642461e+03 FES = 969100 Diversity = 3.9172e-04 k_t
= [1, 1, 1, 1]
Generation 9700: Best = 4.642461e+03 FES = 970100 Diversity = 3.3119e-04 k_t
= [4, 3, 4, 2]
Generation 9710: Best = 4.642461e+03 FES = 971100 Diversity = 3.0715e-04 k_t
= [1, 1, 1, 1]
Generation 9720: Best = 4.642461e+03 FES = 972100 Diversity = 2.9827e-04 k_t
= [2, 3, 3, 3]
Generation 9730: Best = 4.642461e+03 FES = 973100 Diversity = 3.1226e-04 k_t
= [1, 1, 1, 1]
Generation 9740: Best = 4.642461e+03 FES = 974100 Diversity = 2.8808e-04 k_t
= [4, 2, 3, 4]
Generation 9750: Best = 4.642461e+03 FES = 975100 Diversity = 2.5464e-04 k_t
= [1, 1, 1, 1]
Generation 9760: Best = 4.642461e+03 FES = 976100 Diversity = 2.3824e-04 k_t
= [3, 2, 2, 4]
Generation 9770: Best = 4.642461e+03 FES = 977100 Diversity = 2.8845e-04 k_t
= [1, 1, 1, 1]
Generation 9780: Best = 4.642461e+03 FES = 978100 Diversity = 2.5783e-04 k_t
= [2, 3, 4, 3]
Generation 9790: Best = 4.642461e+03 FES = 979100 Diversity = 3.0103e-04 k_t
= [1, 1, 1, 1]
Generation 9800: Best = 4.642461e+03 FES = 980100 Diversity = 2.9834e-04 k_t
= [4, 4, 4, 3]
Generation 9810: Best = 4.642461e+03 FES = 981100 Diversity = 3.8227e-04 k_t
= [1, 1, 1, 1]
Generation 9820: Best = 4.642460e+03 FES = 982100 Diversity = 5.0115e-04 k_t
= [4, 3, 3, 3]
Generation 9830: Best = 4.642460e+03 FES = 983100 Diversity = 4.7686e-04 k_t
= [1, 1, 1, 1]
Generation 9840: Best = 4.642460e+03 FES = 984100 Diversity = 4.8921e-04 k_t
= [4, 4, 3, 2]
Generation 9850: Best = 4.642460e+03 FES = 985100 Diversity = 6.3357e-04 k_t
= [1, 1, 1, 1]
Generation 9860: Best = 4.642460e+03 FES = 986100 Diversity = 5.4137e-04 k_t
= [2, 3, 3, 3]
Generation 9870: Best = 4.642460e+03 FES = 987100 Diversity = 5.1039e-04 k_t
= [1, 1, 1, 1]
Generation 9880: Best = 4.642460e+03 FES = 988100 Diversity = 4.1897e-04 k_t
= [3, 3, 4, 2]
Generation 9890: Best = 4.642460e+03 FES = 989100 Diversity = 4.1087e-04 k_t
= [1, 1, 1, 1]
Generation 9900: Best = 4.642460e+03 FES = 990100 Diversity = 3.5696e-04 k_t
= [4, 3, 3, 4]
Generation 9910: Best = 4.642460e+03 FES = 991100 Diversity = 3.8798e-04 k_t
= [1, 1, 1, 1]
Generation 9920: Best = 4.642460e+03 FES = 992100 Diversity = 4.7258e-04 k_t
= [3, 3, 4, 4]
Generation 9930: Best = 4.642460e+03 FES = 993100 Diversity = 4.8084e-04 k_t
= [1, 1, 1, 1]
Generation 9940: Best = 4.642460e+03 FES = 994100 Diversity = 5.0752e-04 k_t

```

```

= [2, 3, 4, 2]
Generation 9950: Best = 4.642460e+03  FES = 995100  Diversity = 3.6979e-04  k_t
= [1, 1, 1, 1]
Generation 9960: Best = 4.642459e+03  FES = 996100  Diversity = 4.6498e-04  k_t
= [4, 2, 4, 2]
Generation 9970: Best = 4.642459e+03  FES = 997100  Diversity = 5.7884e-04  k_t
= [1, 1, 1, 1]
Generation 9980: Best = 4.642459e+03  FES = 998100  Diversity = 5.8901e-04  k_t
= [2, 2, 3, 4]
Generation 9990: Best = 4.642459e+03  FES = 999100  Diversity = 5.9256e-04  k_t
= [1, 1, 1, 1]

```

```

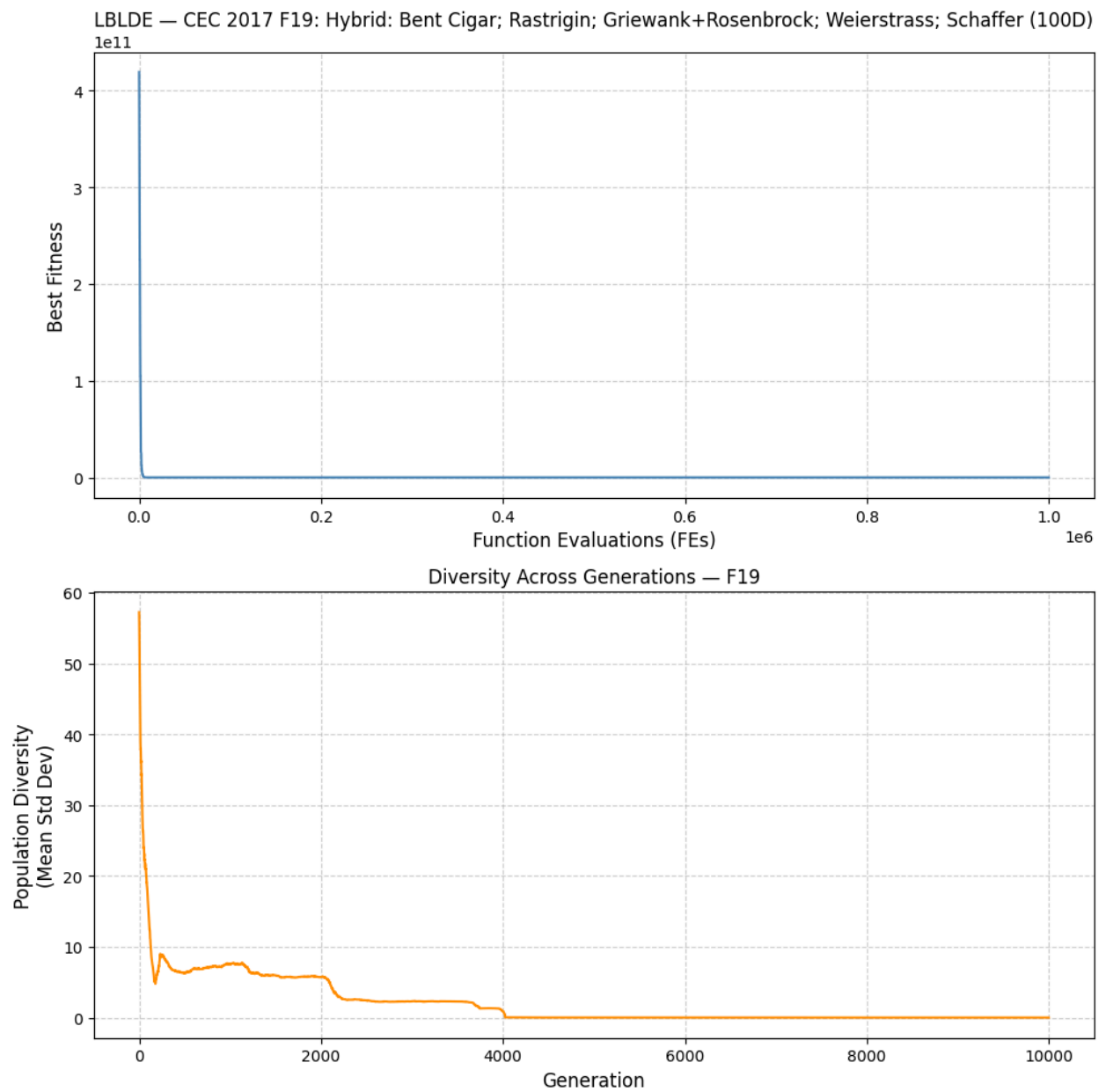
Optimization completed!
Final Best Fitness = 4.642459e+03
Total Function Evaluations = 1000000

```

```

=====
=
Results:
  Best Fitness      : 4.642459e+03
  Global Optimum    : 1.900000e+03
  Error (|best-optimum|) : 2.742459e+03
  Best Solution (5 dim) : [ 18.49661584  -2.82199298 -30.84462971  61.33594857
45.13459553]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```



Graph saved as convergence_F19.png

```
In [44]: # =====
# F20 — Hybrid Function 10 (HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; S
# =====
run_and_plot('F20', make_wrapper(hybrid.f20), 2000.0, 'Hybrid: HappyCat; Katsu
```

```

=====
=
LBLDE_s on CEC 2017 F20 – Single Run
=====
=
Function      : Hybrid: HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaf
fer (100D)
Bounds       : [-100, 100]
Global optimum : 2000.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 8.995901e+03 | Diversity = 5.7220e+01
Generation 10: Best = 8.776137e+03 FES = 1100 Diversity = 5.4676e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 8.442835e+03 FES = 2100 Diversity = 5.4112e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 8.274444e+03 FES = 3100 Diversity = 5.3860e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 8.274444e+03 FES = 4100 Diversity = 5.2721e+01 k_t
= [4, 3, 4, 2]
Generation 50: Best = 8.213404e+03 FES = 5100 Diversity = 5.2867e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 8.213404e+03 FES = 6100 Diversity = 5.3205e+01 k_t
= [4, 4, 4, 3]
Generation 70: Best = 7.353439e+03 FES = 7100 Diversity = 5.3590e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 7.168834e+03 FES = 8100 Diversity = 5.3477e+01 k_t
= [3, 3, 2, 4]
Generation 90: Best = 7.145821e+03 FES = 9100 Diversity = 5.3009e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 7.145821e+03 FES = 10100 Diversity = 5.3179e+01 k_t
= [3, 3, 2, 2]
Generation 110: Best = 7.101931e+03 FES = 11100 Diversity = 5.3609e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 7.060108e+03 FES = 12100 Diversity = 5.3409e+01 k_t
= [4, 3, 2, 2]
Generation 130: Best = 7.060108e+03 FES = 13100 Diversity = 5.3277e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 7.060108e+03 FES = 14100 Diversity = 5.3394e+01 k_t
= [4, 4, 2, 2]
Generation 150: Best = 6.743938e+03 FES = 15100 Diversity = 5.3224e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 6.743938e+03 FES = 16100 Diversity = 5.3518e+01 k_t
= [2, 2, 2, 3]
Generation 170: Best = 6.743938e+03 FES = 17100 Diversity = 5.3796e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 6.509814e+03 FES = 18100 Diversity = 5.3718e+01 k_t
= [3, 3, 3, 4]
Generation 190: Best = 6.473939e+03 FES = 19100 Diversity = 5.3450e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 6.473819e+03 FES = 20100 Diversity = 5.3072e+01 k_t
= [3, 4, 4, 3]
Generation 210: Best = 6.408366e+03 FES = 21100 Diversity = 5.2987e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 6.408366e+03 FES = 22100 Diversity = 5.2984e+01 k_t

```

```

= [4, 4, 2, 3]
Generation 230: Best = 6.408366e+03  FES = 23100  Diversity = 5.3344e+01  k_t
= [1, 1, 1, 1]
Generation 240: Best = 6.371258e+03  FES = 24100  Diversity = 5.3282e+01  k_t
= [4, 2, 3, 2]
Generation 250: Best = 6.371258e+03  FES = 25100  Diversity = 5.3304e+01  k_t
= [1, 1, 1, 1]
Generation 260: Best = 6.371258e+03  FES = 26100  Diversity = 5.3154e+01  k_t
= [3, 2, 3, 2]
Generation 270: Best = 6.364185e+03  FES = 27100  Diversity = 5.3173e+01  k_t
= [1, 1, 1, 1]
Generation 280: Best = 6.364185e+03  FES = 28100  Diversity = 5.3264e+01  k_t
= [4, 2, 4, 3]
Generation 290: Best = 6.359647e+03  FES = 29100  Diversity = 5.2855e+01  k_t
= [1, 1, 1, 1]
Generation 300: Best = 6.359647e+03  FES = 30100  Diversity = 5.2648e+01  k_t
= [4, 4, 2, 4]
Generation 310: Best = 6.359647e+03  FES = 31100  Diversity = 5.2512e+01  k_t
= [1, 1, 1, 1]
Generation 320: Best = 6.359647e+03  FES = 32100  Diversity = 5.2300e+01  k_t
= [2, 2, 4, 4]
Generation 330: Best = 6.359568e+03  FES = 33100  Diversity = 5.2047e+01  k_t
= [1, 1, 1, 1]
Generation 340: Best = 6.359568e+03  FES = 34100  Diversity = 5.1636e+01  k_t
= [2, 2, 4, 3]
Generation 350: Best = 6.359568e+03  FES = 35100  Diversity = 5.1662e+01  k_t
= [1, 1, 1, 1]
Generation 360: Best = 6.359568e+03  FES = 36100  Diversity = 5.1626e+01  k_t
= [2, 4, 2, 3]
Generation 370: Best = 6.349563e+03  FES = 37100  Diversity = 5.1637e+01  k_t
= [1, 1, 1, 1]
Generation 380: Best = 6.349563e+03  FES = 38100  Diversity = 5.1426e+01  k_t
= [4, 2, 3, 3]
Generation 390: Best = 6.349563e+03  FES = 39100  Diversity = 5.1362e+01  k_t
= [1, 1, 1, 1]
Generation 400: Best = 6.349563e+03  FES = 40100  Diversity = 5.1011e+01  k_t
= [4, 2, 3, 3]
Generation 410: Best = 6.349563e+03  FES = 41100  Diversity = 5.0856e+01  k_t
= [1, 1, 1, 1]
Generation 420: Best = 6.332848e+03  FES = 42100  Diversity = 5.0589e+01  k_t
= [4, 2, 4, 3]
Generation 430: Best = 6.332848e+03  FES = 43100  Diversity = 5.0808e+01  k_t
= [1, 1, 1, 1]
Generation 440: Best = 6.332848e+03  FES = 44100  Diversity = 5.0486e+01  k_t
= [3, 4, 4, 4]
Generation 450: Best = 5.979065e+03  FES = 45100  Diversity = 5.0303e+01  k_t
= [1, 1, 1, 1]
Generation 460: Best = 5.979065e+03  FES = 46100  Diversity = 4.9941e+01  k_t
= [4, 3, 2, 4]
Generation 470: Best = 5.964458e+03  FES = 47100  Diversity = 4.9475e+01  k_t
= [1, 1, 1, 1]
Generation 480: Best = 5.964458e+03  FES = 48100  Diversity = 4.9241e+01  k_t
= [2, 3, 2, 4]
Generation 490: Best = 5.856622e+03  FES = 49100  Diversity = 4.8926e+01  k_t

```



```

= [1, 1, 1, 1]
Generation 500: Best = 5.812813e+03 FES = 50100 Diversity = 4.8968e+01 k_t
= [3, 3, 4, 4]
Generation 510: Best = 5.557877e+03 FES = 51100 Diversity = 4.9306e+01 k_t
= [1, 1, 1, 1]
Generation 520: Best = 5.501582e+03 FES = 52100 Diversity = 4.9178e+01 k_t
= [2, 4, 3, 2]
Generation 530: Best = 5.220336e+03 FES = 53100 Diversity = 4.9450e+01 k_t
= [1, 1, 1, 1]
Generation 540: Best = 5.178466e+03 FES = 54100 Diversity = 4.9548e+01 k_t
= [2, 4, 4, 4]
Generation 550: Best = 5.124732e+03 FES = 55100 Diversity = 4.9338e+01 k_t
= [1, 1, 1, 1]
Generation 560: Best = 4.858508e+03 FES = 56100 Diversity = 4.8688e+01 k_t
= [2, 3, 3, 3]
Generation 570: Best = 4.757044e+03 FES = 57100 Diversity = 4.8876e+01 k_t
= [1, 1, 1, 1]
Generation 580: Best = 4.596097e+03 FES = 58100 Diversity = 4.7529e+01 k_t
= [3, 2, 2, 3]
Generation 590: Best = 4.519418e+03 FES = 59100 Diversity = 4.4906e+01 k_t
= [1, 1, 1, 1]
Generation 600: Best = 4.297048e+03 FES = 60100 Diversity = 4.1432e+01 k_t
= [2, 2, 2, 4]
Generation 610: Best = 4.248918e+03 FES = 61100 Diversity = 3.9117e+01 k_t
= [1, 1, 1, 1]
Generation 620: Best = 4.122820e+03 FES = 62100 Diversity = 3.5882e+01 k_t
= [4, 2, 3, 2]
Generation 630: Best = 4.054684e+03 FES = 63100 Diversity = 3.3604e+01 k_t
= [1, 1, 1, 1]
Generation 640: Best = 3.998411e+03 FES = 64100 Diversity = 3.0742e+01 k_t
= [4, 3, 2, 3]
Generation 650: Best = 3.971840e+03 FES = 65100 Diversity = 2.8254e+01 k_t
= [1, 1, 1, 1]
Generation 660: Best = 3.959026e+03 FES = 66100 Diversity = 2.5816e+01 k_t
= [3, 3, 4, 3]
Generation 670: Best = 3.934743e+03 FES = 67100 Diversity = 2.3670e+01 k_t
= [1, 1, 1, 1]
Generation 680: Best = 3.928319e+03 FES = 68100 Diversity = 2.3224e+01 k_t
= [4, 4, 4, 4]
Generation 690: Best = 3.884668e+03 FES = 69100 Diversity = 2.2222e+01 k_t
= [1, 1, 1, 1]
Generation 700: Best = 3.879835e+03 FES = 70100 Diversity = 2.1833e+01 k_t
= [4, 2, 4, 4]
Generation 710: Best = 3.879835e+03 FES = 71100 Diversity = 2.0829e+01 k_t
= [1, 1, 1, 1]
Generation 720: Best = 3.832432e+03 FES = 72100 Diversity = 2.0581e+01 k_t
= [3, 3, 4, 4]
Generation 730: Best = 3.808733e+03 FES = 73100 Diversity = 2.0369e+01 k_t
= [1, 1, 1, 1]
Generation 740: Best = 3.808733e+03 FES = 74100 Diversity = 1.9828e+01 k_t
= [4, 4, 4, 3]
Generation 750: Best = 3.808733e+03 FES = 75100 Diversity = 1.9484e+01 k_t
= [1, 1, 1, 1]
Generation 760: Best = 3.805577e+03 FES = 76100 Diversity = 1.9255e+01 k_t

```

```

= [3, 2, 3, 4]
Generation 770: Best = 3.804480e+03  FES = 77100  Diversity = 1.8897e+01  k_t
= [1, 1, 1, 1]
Generation 780: Best = 3.802187e+03  FES = 78100  Diversity = 1.8804e+01  k_t
= [3, 3, 4, 3]
Generation 790: Best = 3.802187e+03  FES = 79100  Diversity = 1.8627e+01  k_t
= [1, 1, 1, 1]
Generation 800: Best = 3.802187e+03  FES = 80100  Diversity = 1.8420e+01  k_t
= [3, 3, 2, 3]
Generation 810: Best = 3.802187e+03  FES = 81100  Diversity = 1.8381e+01  k_t
= [1, 1, 1, 1]
Generation 820: Best = 3.802187e+03  FES = 82100  Diversity = 1.8373e+01  k_t
= [3, 4, 3, 4]
Generation 830: Best = 3.802187e+03  FES = 83100  Diversity = 1.8402e+01  k_t
= [1, 1, 1, 1]
Generation 840: Best = 3.801522e+03  FES = 84100  Diversity = 1.8328e+01  k_t
= [4, 2, 4, 3]
Generation 850: Best = 3.801522e+03  FES = 85100  Diversity = 1.8282e+01  k_t
= [1, 1, 1, 1]
Generation 860: Best = 3.801522e+03  FES = 86100  Diversity = 1.8160e+01  k_t
= [4, 2, 4, 3]
Generation 870: Best = 3.801522e+03  FES = 87100  Diversity = 1.8165e+01  k_t
= [1, 1, 1, 1]
Generation 880: Best = 3.799375e+03  FES = 88100  Diversity = 1.8022e+01  k_t
= [2, 3, 3, 4]
Generation 890: Best = 3.793006e+03  FES = 89100  Diversity = 1.7718e+01  k_t
= [1, 1, 1, 1]
Generation 900: Best = 3.791399e+03  FES = 90100  Diversity = 1.7720e+01  k_t
= [3, 3, 3, 3]
Generation 910: Best = 3.790396e+03  FES = 91100  Diversity = 1.7603e+01  k_t
= [1, 1, 1, 1]
Generation 920: Best = 3.789736e+03  FES = 92100  Diversity = 1.7674e+01  k_t
= [2, 3, 2, 2]
Generation 930: Best = 3.789736e+03  FES = 93100  Diversity = 1.7613e+01  k_t
= [1, 1, 1, 1]
Generation 940: Best = 3.789736e+03  FES = 94100  Diversity = 1.7585e+01  k_t
= [3, 3, 2, 2]
Generation 950: Best = 3.789736e+03  FES = 95100  Diversity = 1.7620e+01  k_t
= [1, 1, 1, 1]
Generation 960: Best = 3.789736e+03  FES = 96100  Diversity = 1.7640e+01  k_t
= [2, 4, 3, 2]
Generation 970: Best = 3.789736e+03  FES = 97100  Diversity = 1.7533e+01  k_t
= [1, 1, 1, 1]
Generation 980: Best = 3.789673e+03  FES = 98100  Diversity = 1.7517e+01  k_t
= [4, 2, 3, 3]
Generation 990: Best = 3.782210e+03  FES = 99100  Diversity = 1.7525e+01  k_t
= [1, 1, 1, 1]
Generation 1000: Best = 3.782210e+03  FES = 100100  Diversity = 1.7479e+01  k_t
= [4, 3, 4, 2]
Generation 1010: Best = 3.781584e+03  FES = 101100  Diversity = 1.7481e+01  k_t
= [1, 1, 1, 1]
Generation 1020: Best = 3.781584e+03  FES = 102100  Diversity = 1.7440e+01  k_t
= [3, 2, 4, 3]
Generation 1030: Best = 3.780782e+03  FES = 103100  Diversity = 1.7442e+01  k_t

```

```

= [1, 1, 1, 1]
Generation 1040: Best = 3.778170e+03  FES = 104100  Diversity = 1.7406e+01  k_t
= [3, 4, 2, 3]
Generation 1050: Best = 3.777089e+03  FES = 105100  Diversity = 1.7423e+01  k_t
= [1, 1, 1, 1]
Generation 1060: Best = 3.768285e+03  FES = 106100  Diversity = 1.7517e+01  k_t
= [2, 3, 4, 4]
Generation 1070: Best = 3.768285e+03  FES = 107100  Diversity = 1.7479e+01  k_t
= [1, 1, 1, 1]
Generation 1080: Best = 3.768285e+03  FES = 108100  Diversity = 1.7474e+01  k_t
= [3, 2, 3, 3]
Generation 1090: Best = 3.766805e+03  FES = 109100  Diversity = 1.7516e+01  k_t
= [1, 1, 1, 1]
Generation 1100: Best = 3.766805e+03  FES = 110100  Diversity = 1.7498e+01  k_t
= [2, 3, 4, 4]
Generation 1110: Best = 3.766805e+03  FES = 111100  Diversity = 1.7560e+01  k_t
= [1, 1, 1, 1]
Generation 1120: Best = 3.759556e+03  FES = 112100  Diversity = 1.7680e+01  k_t
= [2, 4, 4, 2]
Generation 1130: Best = 3.759556e+03  FES = 113100  Diversity = 1.7665e+01  k_t
= [1, 1, 1, 1]
Generation 1140: Best = 3.759556e+03  FES = 114100  Diversity = 1.7654e+01  k_t
= [3, 2, 3, 2]
Generation 1150: Best = 3.759556e+03  FES = 115100  Diversity = 1.7823e+01  k_t
= [1, 1, 1, 1]
Generation 1160: Best = 3.759197e+03  FES = 116100  Diversity = 1.7889e+01  k_t
= [3, 4, 4, 3]
Generation 1170: Best = 3.756368e+03  FES = 117100  Diversity = 1.7868e+01  k_t
= [1, 1, 1, 1]
Generation 1180: Best = 3.756368e+03  FES = 118100  Diversity = 1.7981e+01  k_t
= [4, 2, 2, 2]
Generation 1190: Best = 3.756368e+03  FES = 119100  Diversity = 1.7947e+01  k_t
= [1, 1, 1, 1]
Generation 1200: Best = 3.751034e+03  FES = 120100  Diversity = 1.7939e+01  k_t
= [3, 2, 3, 3]
Generation 1210: Best = 3.744509e+03  FES = 121100  Diversity = 1.8025e+01  k_t
= [1, 1, 1, 1]
Generation 1220: Best = 3.744509e+03  FES = 122100  Diversity = 1.8092e+01  k_t
= [4, 4, 3, 2]
Generation 1230: Best = 3.739418e+03  FES = 123100  Diversity = 1.8082e+01  k_t
= [1, 1, 1, 1]
Generation 1240: Best = 3.739418e+03  FES = 124100  Diversity = 1.8160e+01  k_t
= [3, 3, 4, 3]
Generation 1250: Best = 3.738565e+03  FES = 125100  Diversity = 1.8092e+01  k_t
= [1, 1, 1, 1]
Generation 1260: Best = 3.738565e+03  FES = 126100  Diversity = 1.8065e+01  k_t
= [3, 4, 2, 2]
Generation 1270: Best = 3.738565e+03  FES = 127100  Diversity = 1.8150e+01  k_t
= [1, 1, 1, 1]
Generation 1280: Best = 3.738565e+03  FES = 128100  Diversity = 1.8243e+01  k_t
= [2, 3, 4, 4]
Generation 1290: Best = 3.738565e+03  FES = 129100  Diversity = 1.8193e+01  k_t
= [1, 1, 1, 1]
Generation 1300: Best = 3.738565e+03  FES = 130100  Diversity = 1.8117e+01  k_t

```

```

= [3, 4, 3, 4]
Generation 1310: Best = 3.738565e+03  FES = 131100  Diversity = 1.8114e+01  k_t
= [1, 1, 1, 1]
Generation 1320: Best = 3.738565e+03  FES = 132100  Diversity = 1.8155e+01  k_t
= [3, 3, 3, 3]
Generation 1330: Best = 3.738565e+03  FES = 133100  Diversity = 1.8175e+01  k_t
= [1, 1, 1, 1]
Generation 1340: Best = 3.738565e+03  FES = 134100  Diversity = 1.8155e+01  k_t
= [4, 2, 3, 4]
Generation 1350: Best = 3.731735e+03  FES = 135100  Diversity = 1.8113e+01  k_t
= [1, 1, 1, 1]
Generation 1360: Best = 3.731735e+03  FES = 136100  Diversity = 1.8226e+01  k_t
= [3, 4, 4, 3]
Generation 1370: Best = 3.728516e+03  FES = 137100  Diversity = 1.8359e+01  k_t
= [1, 1, 1, 1]
Generation 1380: Best = 3.728348e+03  FES = 138100  Diversity = 1.8333e+01  k_t
= [2, 2, 3, 4]
Generation 1390: Best = 3.728316e+03  FES = 139100  Diversity = 1.8237e+01  k_t
= [1, 1, 1, 1]
Generation 1400: Best = 3.728316e+03  FES = 140100  Diversity = 1.8287e+01  k_t
= [3, 2, 3, 3]
Generation 1410: Best = 3.728161e+03  FES = 141100  Diversity = 1.8168e+01  k_t
= [1, 1, 1, 1]
Generation 1420: Best = 3.728099e+03  FES = 142100  Diversity = 1.8098e+01  k_t
= [4, 4, 4, 2]
Generation 1430: Best = 3.728099e+03  FES = 143100  Diversity = 1.8039e+01  k_t
= [1, 1, 1, 1]
Generation 1440: Best = 3.722661e+03  FES = 144100  Diversity = 1.8086e+01  k_t
= [3, 3, 4, 4]
Generation 1450: Best = 3.722661e+03  FES = 145100  Diversity = 1.7981e+01  k_t
= [1, 1, 1, 1]
Generation 1460: Best = 3.720126e+03  FES = 146100  Diversity = 1.7881e+01  k_t
= [2, 4, 4, 4]
Generation 1470: Best = 3.720110e+03  FES = 147100  Diversity = 1.7863e+01  k_t
= [1, 1, 1, 1]
Generation 1480: Best = 3.718013e+03  FES = 148100  Diversity = 1.7748e+01  k_t
= [4, 2, 3, 3]
Generation 1490: Best = 3.714738e+03  FES = 149100  Diversity = 1.7650e+01  k_t
= [1, 1, 1, 1]
Generation 1500: Best = 3.714738e+03  FES = 150100  Diversity = 1.7694e+01  k_t
= [2, 2, 4, 4]
Generation 1510: Best = 3.708703e+03  FES = 151100  Diversity = 1.7596e+01  k_t
= [1, 1, 1, 1]
Generation 1520: Best = 3.696421e+03  FES = 152100  Diversity = 1.7442e+01  k_t
= [4, 2, 3, 4]
Generation 1530: Best = 3.695951e+03  FES = 153100  Diversity = 1.7505e+01  k_t
= [1, 1, 1, 1]
Generation 1540: Best = 3.695951e+03  FES = 154100  Diversity = 1.7411e+01  k_t
= [2, 2, 4, 4]
Generation 1550: Best = 3.695336e+03  FES = 155100  Diversity = 1.7384e+01  k_t
= [1, 1, 1, 1]
Generation 1560: Best = 3.691330e+03  FES = 156100  Diversity = 1.7294e+01  k_t
= [4, 3, 4, 3]
Generation 1570: Best = 3.691330e+03  FES = 157100  Diversity = 1.7198e+01  k_t

```

```

= [1, 1, 1, 1]
Generation 1580: Best = 3.690166e+03 FES = 158100 Diversity = 1.7044e+01 k_t
= [3, 3, 2, 2]
Generation 1590: Best = 3.690166e+03 FES = 159100 Diversity = 1.6991e+01 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 3.690166e+03 FES = 160100 Diversity = 1.6801e+01 k_t
= [3, 3, 2, 4]
Generation 1610: Best = 3.689302e+03 FES = 161100 Diversity = 1.6701e+01 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 3.678818e+03 FES = 162100 Diversity = 1.6245e+01 k_t
= [2, 4, 4, 2]
Generation 1630: Best = 3.677012e+03 FES = 163100 Diversity = 1.5722e+01 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 3.677012e+03 FES = 164100 Diversity = 1.4869e+01 k_t
= [3, 3, 4, 4]
Generation 1650: Best = 3.677012e+03 FES = 165100 Diversity = 1.4488e+01 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 3.677012e+03 FES = 166100 Diversity = 1.4417e+01 k_t
= [3, 3, 2, 3]
Generation 1670: Best = 3.675937e+03 FES = 167100 Diversity = 1.4042e+01 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 3.673622e+03 FES = 168100 Diversity = 1.3736e+01 k_t
= [3, 3, 4, 3]
Generation 1690: Best = 3.668831e+03 FES = 169100 Diversity = 1.3357e+01 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 3.665766e+03 FES = 170100 Diversity = 1.3189e+01 k_t
= [4, 3, 2, 3]
Generation 1710: Best = 3.652388e+03 FES = 171100 Diversity = 1.3247e+01 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 3.652388e+03 FES = 172100 Diversity = 1.3216e+01 k_t
= [2, 4, 3, 2]
Generation 1730: Best = 3.651753e+03 FES = 173100 Diversity = 1.3064e+01 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 3.651753e+03 FES = 174100 Diversity = 1.2935e+01 k_t
= [3, 2, 2, 3]
Generation 1750: Best = 3.649505e+03 FES = 175100 Diversity = 1.2963e+01 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 3.648593e+03 FES = 176100 Diversity = 1.2411e+01 k_t
= [3, 4, 2, 3]
Generation 1770: Best = 3.648456e+03 FES = 177100 Diversity = 1.2183e+01 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 3.648343e+03 FES = 178100 Diversity = 1.2249e+01 k_t
= [4, 2, 2, 4]
Generation 1790: Best = 3.648343e+03 FES = 179100 Diversity = 1.2182e+01 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 3.648232e+03 FES = 180100 Diversity = 1.2047e+01 k_t
= [4, 4, 3, 4]
Generation 1810: Best = 3.647063e+03 FES = 181100 Diversity = 1.1886e+01 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 3.647016e+03 FES = 182100 Diversity = 1.1914e+01 k_t
= [2, 3, 4, 4]
Generation 1830: Best = 3.647016e+03 FES = 183100 Diversity = 1.1833e+01 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 3.644012e+03 FES = 184100 Diversity = 1.1787e+01 k_t

```

```

= [2, 2, 4, 4]
Generation 1850: Best = 3.643337e+03 FES = 185100 Diversity = 1.1799e+01 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 3.643164e+03 FES = 186100 Diversity = 1.1789e+01 k_t
= [3, 3, 4, 3]
Generation 1870: Best = 3.642972e+03 FES = 187100 Diversity = 1.1835e+01 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 3.640792e+03 FES = 188100 Diversity = 1.1562e+01 k_t
= [2, 4, 2, 4]
Generation 1890: Best = 3.640792e+03 FES = 189100 Diversity = 1.1653e+01 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 3.640792e+03 FES = 190100 Diversity = 1.1577e+01 k_t
= [4, 4, 2, 2]
Generation 1910: Best = 3.640169e+03 FES = 191100 Diversity = 1.1525e+01 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 3.639568e+03 FES = 192100 Diversity = 1.1511e+01 k_t
= [2, 3, 2, 4]
Generation 1930: Best = 3.637318e+03 FES = 193100 Diversity = 1.1568e+01 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 3.637318e+03 FES = 194100 Diversity = 1.1474e+01 k_t
= [4, 2, 2, 3]
Generation 1950: Best = 3.637318e+03 FES = 195100 Diversity = 1.1321e+01 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 3.637318e+03 FES = 196100 Diversity = 1.1258e+01 k_t
= [4, 4, 4, 4]
Generation 1970: Best = 3.636870e+03 FES = 197100 Diversity = 1.1287e+01 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 3.632574e+03 FES = 198100 Diversity = 1.1228e+01 k_t
= [2, 4, 3, 3]
Generation 1990: Best = 3.632574e+03 FES = 199100 Diversity = 1.1248e+01 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 3.631820e+03 FES = 200100 Diversity = 1.1116e+01 k_t
= [2, 2, 4, 2]
Generation 2010: Best = 3.629352e+03 FES = 201100 Diversity = 1.1081e+01 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 3.628463e+03 FES = 202100 Diversity = 1.1057e+01 k_t
= [4, 4, 3, 3]
Generation 2030: Best = 3.628463e+03 FES = 203100 Diversity = 1.0878e+01 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 3.628463e+03 FES = 204100 Diversity = 1.0876e+01 k_t
= [3, 4, 3, 3]
Generation 2050: Best = 3.628167e+03 FES = 205100 Diversity = 1.0771e+01 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 3.628167e+03 FES = 206100 Diversity = 1.0770e+01 k_t
= [4, 3, 3, 3]
Generation 2070: Best = 3.628167e+03 FES = 207100 Diversity = 1.0871e+01 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 3.627885e+03 FES = 208100 Diversity = 1.0883e+01 k_t
= [4, 3, 2, 2]
Generation 2090: Best = 3.627885e+03 FES = 209100 Diversity = 1.0864e+01 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 3.627885e+03 FES = 210100 Diversity = 1.0879e+01 k_t
= [3, 3, 2, 3]
Generation 2110: Best = 3.627885e+03 FES = 211100 Diversity = 1.0900e+01 k_t

```

```

= [1, 1, 1, 1]
Generation 2120: Best = 3.627833e+03  FES = 212100  Diversity = 1.0957e+01  k_t
= [4, 2, 2, 3]
Generation 2130: Best = 3.627833e+03  FES = 213100  Diversity = 1.0960e+01  k_t
= [1, 1, 1, 1]
Generation 2140: Best = 3.627833e+03  FES = 214100  Diversity = 1.0947e+01  k_t
= [3, 2, 2, 3]
Generation 2150: Best = 3.627833e+03  FES = 215100  Diversity = 1.0962e+01  k_t
= [1, 1, 1, 1]
Generation 2160: Best = 3.625500e+03  FES = 216100  Diversity = 1.0946e+01  k_t
= [2, 4, 3, 3]
Generation 2170: Best = 3.625500e+03  FES = 217100  Diversity = 1.0922e+01  k_t
= [1, 1, 1, 1]
Generation 2180: Best = 3.625399e+03  FES = 218100  Diversity = 1.0897e+01  k_t
= [4, 3, 4, 4]
Generation 2190: Best = 3.624671e+03  FES = 219100  Diversity = 1.0964e+01  k_t
= [1, 1, 1, 1]
Generation 2200: Best = 3.624671e+03  FES = 220100  Diversity = 1.1001e+01  k_t
= [4, 4, 3, 4]
Generation 2210: Best = 3.624671e+03  FES = 221100  Diversity = 1.1003e+01  k_t
= [1, 1, 1, 1]
Generation 2220: Best = 3.624346e+03  FES = 222100  Diversity = 1.0920e+01  k_t
= [4, 4, 4, 2]
Generation 2230: Best = 3.624346e+03  FES = 223100  Diversity = 1.0996e+01  k_t
= [1, 1, 1, 1]
Generation 2240: Best = 3.624346e+03  FES = 224100  Diversity = 1.0964e+01  k_t
= [4, 3, 3, 3]
Generation 2250: Best = 3.624346e+03  FES = 225100  Diversity = 1.1000e+01  k_t
= [1, 1, 1, 1]
Generation 2260: Best = 3.623928e+03  FES = 226100  Diversity = 1.1044e+01  k_t
= [2, 2, 4, 4]
Generation 2270: Best = 3.623928e+03  FES = 227100  Diversity = 1.1084e+01  k_t
= [1, 1, 1, 1]
Generation 2280: Best = 3.623928e+03  FES = 228100  Diversity = 1.1115e+01  k_t
= [2, 4, 2, 2]
Generation 2290: Best = 3.623928e+03  FES = 229100  Diversity = 1.1189e+01  k_t
= [1, 1, 1, 1]
Generation 2300: Best = 3.623834e+03  FES = 230100  Diversity = 1.1113e+01  k_t
= [3, 2, 4, 4]
Generation 2310: Best = 3.623663e+03  FES = 231100  Diversity = 1.0996e+01  k_t
= [1, 1, 1, 1]
Generation 2320: Best = 3.623603e+03  FES = 232100  Diversity = 1.0970e+01  k_t
= [2, 3, 3, 2]
Generation 2330: Best = 3.623338e+03  FES = 233100  Diversity = 1.0984e+01  k_t
= [1, 1, 1, 1]
Generation 2340: Best = 3.622471e+03  FES = 234100  Diversity = 1.0972e+01  k_t
= [2, 3, 4, 3]
Generation 2350: Best = 3.622471e+03  FES = 235100  Diversity = 1.0826e+01  k_t
= [1, 1, 1, 1]
Generation 2360: Best = 3.620353e+03  FES = 236100  Diversity = 1.0819e+01  k_t
= [2, 3, 2, 2]
Generation 2370: Best = 3.620353e+03  FES = 237100  Diversity = 1.0778e+01  k_t
= [1, 1, 1, 1]
Generation 2380: Best = 3.620353e+03  FES = 238100  Diversity = 1.0719e+01  k_t

```

```

= [4, 3, 4, 3]
Generation 2390: Best = 3.620353e+03  FES = 239100  Diversity = 1.0766e+01  k_t
= [1, 1, 1, 1]
Generation 2400: Best = 3.620353e+03  FES = 240100  Diversity = 1.0815e+01  k_t
= [3, 3, 3, 2]
Generation 2410: Best = 3.620353e+03  FES = 241100  Diversity = 1.0668e+01  k_t
= [1, 1, 1, 1]
Generation 2420: Best = 3.620353e+03  FES = 242100  Diversity = 1.0638e+01  k_t
= [3, 3, 3, 3]
Generation 2430: Best = 3.620353e+03  FES = 243100  Diversity = 1.0643e+01  k_t
= [1, 1, 1, 1]
Generation 2440: Best = 3.620353e+03  FES = 244100  Diversity = 1.0580e+01  k_t
= [4, 2, 4, 3]
Generation 2450: Best = 3.620255e+03  FES = 245100  Diversity = 1.0458e+01  k_t
= [1, 1, 1, 1]
Generation 2460: Best = 3.620253e+03  FES = 246100  Diversity = 1.0499e+01  k_t
= [2, 2, 2, 2]
Generation 2470: Best = 3.620059e+03  FES = 247100  Diversity = 1.0491e+01  k_t
= [1, 1, 1, 1]
Generation 2480: Best = 3.617647e+03  FES = 248100  Diversity = 1.0503e+01  k_t
= [3, 2, 4, 4]
Generation 2490: Best = 3.617647e+03  FES = 249100  Diversity = 1.0489e+01  k_t
= [1, 1, 1, 1]
Generation 2500: Best = 3.617544e+03  FES = 250100  Diversity = 1.0438e+01  k_t
= [2, 3, 4, 2]
Generation 2510: Best = 3.617544e+03  FES = 251100  Diversity = 1.0408e+01  k_t
= [1, 1, 1, 1]
Generation 2520: Best = 3.617525e+03  FES = 252100  Diversity = 1.0386e+01  k_t
= [3, 4, 2, 4]
Generation 2530: Best = 3.617525e+03  FES = 253100  Diversity = 1.0417e+01  k_t
= [1, 1, 1, 1]
Generation 2540: Best = 3.617525e+03  FES = 254100  Diversity = 1.0427e+01  k_t
= [4, 3, 4, 2]
Generation 2550: Best = 3.617525e+03  FES = 255100  Diversity = 1.0355e+01  k_t
= [1, 1, 1, 1]
Generation 2560: Best = 3.617464e+03  FES = 256100  Diversity = 1.0321e+01  k_t
= [2, 4, 3, 4]
Generation 2570: Best = 3.617414e+03  FES = 257100  Diversity = 1.0189e+01  k_t
= [1, 1, 1, 1]
Generation 2580: Best = 3.617414e+03  FES = 258100  Diversity = 1.0204e+01  k_t
= [4, 2, 3, 3]
Generation 2590: Best = 3.617414e+03  FES = 259100  Diversity = 1.0198e+01  k_t
= [1, 1, 1, 1]
Generation 2600: Best = 3.617414e+03  FES = 260100  Diversity = 1.0203e+01  k_t
= [4, 4, 4, 2]
Generation 2610: Best = 3.617346e+03  FES = 261100  Diversity = 1.0106e+01  k_t
= [1, 1, 1, 1]
Generation 2620: Best = 3.617346e+03  FES = 262100  Diversity = 1.0039e+01  k_t
= [4, 3, 2, 4]
Generation 2630: Best = 3.617346e+03  FES = 263100  Diversity = 9.9145e+00  k_t
= [1, 1, 1, 1]
Generation 2640: Best = 3.617093e+03  FES = 264100  Diversity = 9.8817e+00  k_t
= [4, 4, 2, 2]
Generation 2650: Best = 3.617093e+03  FES = 265100  Diversity = 9.7875e+00  k_t

```



```

= [1, 1, 1, 1]
Generation 2660: Best = 3.617093e+03 FES = 266100 Diversity = 9.6891e+00 k_t
= [2, 4, 4, 4]
Generation 2670: Best = 3.616794e+03 FES = 267100 Diversity = 9.7148e+00 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 3.616480e+03 FES = 268100 Diversity = 9.7385e+00 k_t
= [3, 3, 4, 2]
Generation 2690: Best = 3.616480e+03 FES = 269100 Diversity = 9.7803e+00 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 3.616467e+03 FES = 270100 Diversity = 9.7515e+00 k_t
= [3, 4, 4, 2]
Generation 2710: Best = 3.616464e+03 FES = 271100 Diversity = 9.7259e+00 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 3.616197e+03 FES = 272100 Diversity = 9.8565e+00 k_t
= [4, 3, 4, 2]
Generation 2730: Best = 3.616140e+03 FES = 273100 Diversity = 9.8355e+00 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 3.616140e+03 FES = 274100 Diversity = 9.7772e+00 k_t
= [4, 4, 4, 3]
Generation 2750: Best = 3.615805e+03 FES = 275100 Diversity = 9.8556e+00 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 3.615718e+03 FES = 276100 Diversity = 9.8792e+00 k_t
= [4, 4, 3, 4]
Generation 2770: Best = 3.615718e+03 FES = 277100 Diversity = 9.8435e+00 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 3.615718e+03 FES = 278100 Diversity = 9.8150e+00 k_t
= [3, 3, 3, 3]
Generation 2790: Best = 3.615580e+03 FES = 279100 Diversity = 9.7487e+00 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 3.615538e+03 FES = 280100 Diversity = 9.7588e+00 k_t
= [2, 4, 3, 2]
Generation 2810: Best = 3.615538e+03 FES = 281100 Diversity = 9.5935e+00 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 3.615538e+03 FES = 282100 Diversity = 9.5535e+00 k_t
= [4, 3, 4, 2]
Generation 2830: Best = 3.615535e+03 FES = 283100 Diversity = 9.4552e+00 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 3.615535e+03 FES = 284100 Diversity = 9.2738e+00 k_t
= [2, 4, 4, 4]
Generation 2850: Best = 3.615453e+03 FES = 285100 Diversity = 9.2052e+00 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 3.614624e+03 FES = 286100 Diversity = 9.1664e+00 k_t
= [2, 4, 4, 2]
Generation 2870: Best = 3.613955e+03 FES = 287100 Diversity = 9.1918e+00 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 3.613955e+03 FES = 288100 Diversity = 9.1335e+00 k_t
= [2, 3, 4, 3]
Generation 2890: Best = 3.613615e+03 FES = 289100 Diversity = 9.1264e+00 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 3.613615e+03 FES = 290100 Diversity = 9.1654e+00 k_t
= [4, 2, 3, 3]
Generation 2910: Best = 3.613615e+03 FES = 291100 Diversity = 9.1275e+00 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 3.613615e+03 FES = 292100 Diversity = 8.9262e+00 k_t

```

```

= [3, 4, 2, 3]
Generation 2930: Best = 3.613615e+03 FES = 293100 Diversity = 8.8641e+00 k_t
= [1, 1, 1, 1]
Generation 2940: Best = 3.613615e+03 FES = 294100 Diversity = 8.7987e+00 k_t
= [4, 2, 4, 4]
Generation 2950: Best = 3.613615e+03 FES = 295100 Diversity = 8.7796e+00 k_t
= [1, 1, 1, 1]
Generation 2960: Best = 3.613615e+03 FES = 296100 Diversity = 8.7695e+00 k_t
= [4, 4, 2, 4]
Generation 2970: Best = 3.613422e+03 FES = 297100 Diversity = 8.6900e+00 k_t
= [1, 1, 1, 1]
Generation 2980: Best = 3.613422e+03 FES = 298100 Diversity = 8.6458e+00 k_t
= [4, 2, 4, 2]
Generation 2990: Best = 3.613422e+03 FES = 299100 Diversity = 8.5594e+00 k_t
= [1, 1, 1, 1]
Generation 3000: Best = 3.613422e+03 FES = 300100 Diversity = 8.5155e+00 k_t
= [2, 3, 4, 4]
Generation 3010: Best = 3.613305e+03 FES = 301100 Diversity = 8.3171e+00 k_t
= [1, 1, 1, 1]
Generation 3020: Best = 3.612782e+03 FES = 302100 Diversity = 8.2637e+00 k_t
= [4, 4, 2, 2]
Generation 3030: Best = 3.612782e+03 FES = 303100 Diversity = 8.2950e+00 k_t
= [1, 1, 1, 1]
Generation 3040: Best = 3.612741e+03 FES = 304100 Diversity = 8.2889e+00 k_t
= [2, 4, 3, 4]
Generation 3050: Best = 3.612741e+03 FES = 305100 Diversity = 8.2511e+00 k_t
= [1, 1, 1, 1]
Generation 3060: Best = 3.612741e+03 FES = 306100 Diversity = 8.1924e+00 k_t
= [3, 2, 3, 2]
Generation 3070: Best = 3.612741e+03 FES = 307100 Diversity = 8.1469e+00 k_t
= [1, 1, 1, 1]
Generation 3080: Best = 3.612693e+03 FES = 308100 Diversity = 8.1060e+00 k_t
= [4, 3, 3, 2]
Generation 3090: Best = 3.612687e+03 FES = 309100 Diversity = 8.1212e+00 k_t
= [1, 1, 1, 1]
Generation 3100: Best = 3.612572e+03 FES = 310100 Diversity = 8.0877e+00 k_t
= [4, 2, 4, 4]
Generation 3110: Best = 3.612572e+03 FES = 311100 Diversity = 8.1454e+00 k_t
= [1, 1, 1, 1]
Generation 3120: Best = 3.612413e+03 FES = 312100 Diversity = 8.1239e+00 k_t
= [2, 3, 2, 3]
Generation 3130: Best = 3.612413e+03 FES = 313100 Diversity = 8.1403e+00 k_t
= [1, 1, 1, 1]
Generation 3140: Best = 3.612413e+03 FES = 314100 Diversity = 8.1115e+00 k_t
= [3, 2, 3, 2]
Generation 3150: Best = 3.612413e+03 FES = 315100 Diversity = 8.0484e+00 k_t
= [1, 1, 1, 1]
Generation 3160: Best = 3.612292e+03 FES = 316100 Diversity = 7.9129e+00 k_t
= [3, 3, 3, 3]
Generation 3170: Best = 3.612155e+03 FES = 317100 Diversity = 7.8957e+00 k_t
= [1, 1, 1, 1]
Generation 3180: Best = 3.611939e+03 FES = 318100 Diversity = 7.9368e+00 k_t
= [3, 2, 3, 3]
Generation 3190: Best = 3.611909e+03 FES = 319100 Diversity = 7.9516e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 3200: Best = 3.611909e+03 FES = 320100 Diversity = 7.9475e+00 k_t
= [3, 3, 2, 2]
Generation 3210: Best = 3.611909e+03 FES = 321100 Diversity = 7.9688e+00 k_t
= [1, 1, 1, 1]
Generation 3220: Best = 3.611909e+03 FES = 322100 Diversity = 7.9881e+00 k_t
= [3, 3, 2, 4]
Generation 3230: Best = 3.611668e+03 FES = 323100 Diversity = 7.9492e+00 k_t
= [1, 1, 1, 1]
Generation 3240: Best = 3.611484e+03 FES = 324100 Diversity = 7.9837e+00 k_t
= [3, 2, 2, 4]
Generation 3250: Best = 3.611457e+03 FES = 325100 Diversity = 7.9901e+00 k_t
= [1, 1, 1, 1]
Generation 3260: Best = 3.611359e+03 FES = 326100 Diversity = 7.9766e+00 k_t
= [3, 3, 2, 2]
Generation 3270: Best = 3.610990e+03 FES = 327100 Diversity = 7.9933e+00 k_t
= [1, 1, 1, 1]
Generation 3280: Best = 3.610612e+03 FES = 328100 Diversity = 7.9791e+00 k_t
= [3, 4, 2, 3]
Generation 3290: Best = 3.610612e+03 FES = 329100 Diversity = 7.9437e+00 k_t
= [1, 1, 1, 1]
Generation 3300: Best = 3.609807e+03 FES = 330100 Diversity = 7.9288e+00 k_t
= [4, 2, 3, 3]
Generation 3310: Best = 3.609807e+03 FES = 331100 Diversity = 7.9576e+00 k_t
= [1, 1, 1, 1]
Generation 3320: Best = 3.609807e+03 FES = 332100 Diversity = 7.9531e+00 k_t
= [2, 4, 4, 3]
Generation 3330: Best = 3.609807e+03 FES = 333100 Diversity = 7.9268e+00 k_t
= [1, 1, 1, 1]
Generation 3340: Best = 3.609807e+03 FES = 334100 Diversity = 7.9079e+00 k_t
= [3, 3, 4, 3]
Generation 3350: Best = 3.609805e+03 FES = 335100 Diversity = 7.8614e+00 k_t
= [1, 1, 1, 1]
Generation 3360: Best = 3.609805e+03 FES = 336100 Diversity = 7.8413e+00 k_t
= [3, 3, 4, 2]
Generation 3370: Best = 3.609805e+03 FES = 337100 Diversity = 7.8491e+00 k_t
= [1, 1, 1, 1]
Generation 3380: Best = 3.609805e+03 FES = 338100 Diversity = 7.8690e+00 k_t
= [3, 2, 3, 3]
Generation 3390: Best = 3.609336e+03 FES = 339100 Diversity = 7.8211e+00 k_t
= [1, 1, 1, 1]
Generation 3400: Best = 3.609251e+03 FES = 340100 Diversity = 7.8768e+00 k_t
= [3, 4, 2, 3]
Generation 3410: Best = 3.609244e+03 FES = 341100 Diversity = 7.8115e+00 k_t
= [1, 1, 1, 1]
Generation 3420: Best = 3.608527e+03 FES = 342100 Diversity = 7.7411e+00 k_t
= [3, 3, 4, 4]
Generation 3430: Best = 3.607999e+03 FES = 343100 Diversity = 7.7388e+00 k_t
= [1, 1, 1, 1]
Generation 3440: Best = 3.607999e+03 FES = 344100 Diversity = 7.7728e+00 k_t
= [3, 2, 3, 2]
Generation 3450: Best = 3.607999e+03 FES = 345100 Diversity = 7.7880e+00 k_t
= [1, 1, 1, 1]
Generation 3460: Best = 3.607999e+03 FES = 346100 Diversity = 7.8232e+00 k_t

```

```

= [4, 2, 4, 2]
Generation 3470: Best = 3.607032e+03 FES = 347100 Diversity = 7.6601e+00 k_t
= [1, 1, 1, 1]
Generation 3480: Best = 3.606469e+03 FES = 348100 Diversity = 7.6700e+00 k_t
= [4, 2, 3, 3]
Generation 3490: Best = 3.606469e+03 FES = 349100 Diversity = 7.8269e+00 k_t
= [1, 1, 1, 1]
Generation 3500: Best = 3.606412e+03 FES = 350100 Diversity = 7.8853e+00 k_t
= [4, 4, 2, 3]
Generation 3510: Best = 3.606412e+03 FES = 351100 Diversity = 7.8343e+00 k_t
= [1, 1, 1, 1]
Generation 3520: Best = 3.606390e+03 FES = 352100 Diversity = 7.7592e+00 k_t
= [2, 2, 2, 4]
Generation 3530: Best = 3.606228e+03 FES = 353100 Diversity = 7.6568e+00 k_t
= [1, 1, 1, 1]
Generation 3540: Best = 3.606122e+03 FES = 354100 Diversity = 7.4952e+00 k_t
= [3, 2, 3, 3]
Generation 3550: Best = 3.606122e+03 FES = 355100 Diversity = 7.4454e+00 k_t
= [1, 1, 1, 1]
Generation 3560: Best = 3.606050e+03 FES = 356100 Diversity = 7.1915e+00 k_t
= [2, 3, 4, 4]
Generation 3570: Best = 3.605816e+03 FES = 357100 Diversity = 7.1350e+00 k_t
= [1, 1, 1, 1]
Generation 3580: Best = 3.605626e+03 FES = 358100 Diversity = 7.1139e+00 k_t
= [3, 3, 4, 3]
Generation 3590: Best = 3.605233e+03 FES = 359100 Diversity = 7.0907e+00 k_t
= [1, 1, 1, 1]
Generation 3600: Best = 3.604424e+03 FES = 360100 Diversity = 6.8416e+00 k_t
= [4, 2, 2, 3]
Generation 3610: Best = 3.604424e+03 FES = 361100 Diversity = 6.4066e+00 k_t
= [1, 1, 1, 1]
Generation 3620: Best = 3.604394e+03 FES = 362100 Diversity = 6.3489e+00 k_t
= [3, 3, 3, 3]
Generation 3630: Best = 3.604266e+03 FES = 363100 Diversity = 6.1326e+00 k_t
= [1, 1, 1, 1]
Generation 3640: Best = 3.603622e+03 FES = 364100 Diversity = 5.9140e+00 k_t
= [3, 4, 3, 2]
Generation 3650: Best = 3.603230e+03 FES = 365100 Diversity = 5.7174e+00 k_t
= [1, 1, 1, 1]
Generation 3660: Best = 3.603089e+03 FES = 366100 Diversity = 5.5668e+00 k_t
= [3, 2, 2, 3]
Generation 3670: Best = 3.602288e+03 FES = 367100 Diversity = 5.4177e+00 k_t
= [1, 1, 1, 1]
Generation 3680: Best = 3.602288e+03 FES = 368100 Diversity = 5.3107e+00 k_t
= [2, 4, 2, 2]
Generation 3690: Best = 3.601985e+03 FES = 369100 Diversity = 5.0847e+00 k_t
= [1, 1, 1, 1]
Generation 3700: Best = 3.601985e+03 FES = 370100 Diversity = 4.9724e+00 k_t
= [4, 2, 2, 2]
Generation 3710: Best = 3.601985e+03 FES = 371100 Diversity = 4.8666e+00 k_t
= [1, 1, 1, 1]
Generation 3720: Best = 3.601985e+03 FES = 372100 Diversity = 4.7095e+00 k_t
= [3, 3, 2, 3]
Generation 3730: Best = 3.601985e+03 FES = 373100 Diversity = 4.5772e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 3740: Best = 3.601944e+03 FES = 374100 Diversity = 4.4735e+00 k_t
= [3, 2, 4, 2]
Generation 3750: Best = 3.601313e+03 FES = 375100 Diversity = 4.3130e+00 k_t
= [1, 1, 1, 1]
Generation 3760: Best = 3.601313e+03 FES = 376100 Diversity = 4.3645e+00 k_t
= [2, 2, 4, 4]
Generation 3770: Best = 3.601313e+03 FES = 377100 Diversity = 4.2981e+00 k_t
= [1, 1, 1, 1]
Generation 3780: Best = 3.601257e+03 FES = 378100 Diversity = 4.2700e+00 k_t
= [4, 2, 4, 2]
Generation 3790: Best = 3.601257e+03 FES = 379100 Diversity = 4.2300e+00 k_t
= [1, 1, 1, 1]
Generation 3800: Best = 3.601257e+03 FES = 380100 Diversity = 4.0941e+00 k_t
= [4, 3, 4, 3]
Generation 3810: Best = 3.601148e+03 FES = 381100 Diversity = 3.9381e+00 k_t
= [1, 1, 1, 1]
Generation 3820: Best = 3.601060e+03 FES = 382100 Diversity = 3.9002e+00 k_t
= [3, 2, 4, 4]
Generation 3830: Best = 3.601060e+03 FES = 383100 Diversity = 3.8519e+00 k_t
= [1, 1, 1, 1]
Generation 3840: Best = 3.601009e+03 FES = 384100 Diversity = 3.7866e+00 k_t
= [2, 4, 2, 3]
Generation 3850: Best = 3.600991e+03 FES = 385100 Diversity = 3.7465e+00 k_t
= [1, 1, 1, 1]
Generation 3860: Best = 3.600991e+03 FES = 386100 Diversity = 3.7009e+00 k_t
= [2, 3, 2, 2]
Generation 3870: Best = 3.600921e+03 FES = 387100 Diversity = 3.5836e+00 k_t
= [1, 1, 1, 1]
Generation 3880: Best = 3.600886e+03 FES = 388100 Diversity = 3.5991e+00 k_t
= [2, 4, 3, 2]
Generation 3890: Best = 3.600869e+03 FES = 389100 Diversity = 3.5781e+00 k_t
= [1, 1, 1, 1]
Generation 3900: Best = 3.600707e+03 FES = 390100 Diversity = 3.5232e+00 k_t
= [4, 4, 3, 3]
Generation 3910: Best = 3.600707e+03 FES = 391100 Diversity = 3.5167e+00 k_t
= [1, 1, 1, 1]
Generation 3920: Best = 3.600637e+03 FES = 392100 Diversity = 3.5249e+00 k_t
= [3, 3, 3, 4]
Generation 3930: Best = 3.600637e+03 FES = 393100 Diversity = 3.4895e+00 k_t
= [1, 1, 1, 1]
Generation 3940: Best = 3.600637e+03 FES = 394100 Diversity = 3.4545e+00 k_t
= [2, 4, 3, 3]
Generation 3950: Best = 3.600566e+03 FES = 395100 Diversity = 3.3774e+00 k_t
= [1, 1, 1, 1]
Generation 3960: Best = 3.600566e+03 FES = 396100 Diversity = 3.3429e+00 k_t
= [3, 3, 3, 2]
Generation 3970: Best = 3.600433e+03 FES = 397100 Diversity = 3.3470e+00 k_t
= [1, 1, 1, 1]
Generation 3980: Best = 3.600429e+03 FES = 398100 Diversity = 3.3320e+00 k_t
= [3, 3, 3, 4]
Generation 3990: Best = 3.600429e+03 FES = 399100 Diversity = 3.2870e+00 k_t
= [1, 1, 1, 1]
Generation 4000: Best = 3.600429e+03 FES = 400100 Diversity = 3.2776e+00 k_t

```

```

= [2, 4, 3, 3]
Generation 4010: Best = 3.600429e+03 FES = 401100 Diversity = 3.2669e+00 k_t
= [1, 1, 1, 1]
Generation 4020: Best = 3.600413e+03 FES = 402100 Diversity = 3.2791e+00 k_t
= [3, 3, 3, 2]
Generation 4030: Best = 3.600413e+03 FES = 403100 Diversity = 3.2602e+00 k_t
= [1, 1, 1, 1]
Generation 4040: Best = 3.600413e+03 FES = 404100 Diversity = 3.2320e+00 k_t
= [4, 4, 4, 3]
Generation 4050: Best = 3.600389e+03 FES = 405100 Diversity = 3.2139e+00 k_t
= [1, 1, 1, 1]
Generation 4060: Best = 3.600389e+03 FES = 406100 Diversity = 3.1968e+00 k_t
= [4, 2, 3, 4]
Generation 4070: Best = 3.600389e+03 FES = 407100 Diversity = 3.1651e+00 k_t
= [1, 1, 1, 1]
Generation 4080: Best = 3.600389e+03 FES = 408100 Diversity = 3.1562e+00 k_t
= [2, 2, 3, 3]
Generation 4090: Best = 3.600389e+03 FES = 409100 Diversity = 3.1372e+00 k_t
= [1, 1, 1, 1]
Generation 4100: Best = 3.600389e+03 FES = 410100 Diversity = 3.1533e+00 k_t
= [4, 2, 3, 4]
Generation 4110: Best = 3.600389e+03 FES = 411100 Diversity = 3.1080e+00 k_t
= [1, 1, 1, 1]
Generation 4120: Best = 3.600379e+03 FES = 412100 Diversity = 3.1205e+00 k_t
= [3, 3, 3, 3]
Generation 4130: Best = 3.600358e+03 FES = 413100 Diversity = 3.1367e+00 k_t
= [1, 1, 1, 1]
Generation 4140: Best = 3.600358e+03 FES = 414100 Diversity = 3.1248e+00 k_t
= [3, 3, 4, 2]
Generation 4150: Best = 3.600358e+03 FES = 415100 Diversity = 3.1328e+00 k_t
= [1, 1, 1, 1]
Generation 4160: Best = 3.600216e+03 FES = 416100 Diversity = 3.0961e+00 k_t
= [3, 3, 3, 4]
Generation 4170: Best = 3.600168e+03 FES = 417100 Diversity = 3.0872e+00 k_t
= [1, 1, 1, 1]
Generation 4180: Best = 3.600164e+03 FES = 418100 Diversity = 3.0421e+00 k_t
= [4, 4, 2, 3]
Generation 4190: Best = 3.600140e+03 FES = 419100 Diversity = 3.0202e+00 k_t
= [1, 1, 1, 1]
Generation 4200: Best = 3.600140e+03 FES = 420100 Diversity = 2.9961e+00 k_t
= [4, 3, 3, 3]
Generation 4210: Best = 3.600137e+03 FES = 421100 Diversity = 2.9650e+00 k_t
= [1, 1, 1, 1]
Generation 4220: Best = 3.600137e+03 FES = 422100 Diversity = 2.9400e+00 k_t
= [3, 4, 4, 2]
Generation 4230: Best = 3.600137e+03 FES = 423100 Diversity = 2.9124e+00 k_t
= [1, 1, 1, 1]
Generation 4240: Best = 3.599954e+03 FES = 424100 Diversity = 2.9156e+00 k_t
= [3, 4, 4, 2]
Generation 4250: Best = 3.599954e+03 FES = 425100 Diversity = 2.9013e+00 k_t
= [1, 1, 1, 1]
Generation 4260: Best = 3.599950e+03 FES = 426100 Diversity = 2.9106e+00 k_t
= [4, 4, 3, 2]
Generation 4270: Best = 3.599950e+03 FES = 427100 Diversity = 2.9143e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 4280: Best = 3.599950e+03 FES = 428100 Diversity = 2.8642e+00 k_t
= [4, 2, 4, 4]
Generation 4290: Best = 3.599950e+03 FES = 429100 Diversity = 2.8646e+00 k_t
= [1, 1, 1, 1]
Generation 4300: Best = 3.599950e+03 FES = 430100 Diversity = 2.8800e+00 k_t
= [4, 4, 4, 3]
Generation 4310: Best = 3.599950e+03 FES = 431100 Diversity = 2.8684e+00 k_t
= [1, 1, 1, 1]
Generation 4320: Best = 3.599860e+03 FES = 432100 Diversity = 2.8266e+00 k_t
= [3, 3, 2, 2]
Generation 4330: Best = 3.599860e+03 FES = 433100 Diversity = 2.8181e+00 k_t
= [1, 1, 1, 1]
Generation 4340: Best = 3.599860e+03 FES = 434100 Diversity = 2.8315e+00 k_t
= [3, 3, 2, 4]
Generation 4350: Best = 3.599860e+03 FES = 435100 Diversity = 2.8209e+00 k_t
= [1, 1, 1, 1]
Generation 4360: Best = 3.599860e+03 FES = 436100 Diversity = 2.8103e+00 k_t
= [2, 2, 2, 2]
Generation 4370: Best = 3.599860e+03 FES = 437100 Diversity = 2.7854e+00 k_t
= [1, 1, 1, 1]
Generation 4380: Best = 3.599860e+03 FES = 438100 Diversity = 2.7456e+00 k_t
= [4, 4, 2, 2]
Generation 4390: Best = 3.599860e+03 FES = 439100 Diversity = 2.7460e+00 k_t
= [1, 1, 1, 1]
Generation 4400: Best = 3.599848e+03 FES = 440100 Diversity = 2.7131e+00 k_t
= [3, 3, 2, 4]
Generation 4410: Best = 3.599848e+03 FES = 441100 Diversity = 2.7026e+00 k_t
= [1, 1, 1, 1]
Generation 4420: Best = 3.599848e+03 FES = 442100 Diversity = 2.6917e+00 k_t
= [2, 4, 2, 3]
Generation 4430: Best = 3.599848e+03 FES = 443100 Diversity = 2.6996e+00 k_t
= [1, 1, 1, 1]
Generation 4440: Best = 3.599848e+03 FES = 444100 Diversity = 2.6999e+00 k_t
= [4, 4, 4, 2]
Generation 4450: Best = 3.599750e+03 FES = 445100 Diversity = 2.6986e+00 k_t
= [1, 1, 1, 1]
Generation 4460: Best = 3.599750e+03 FES = 446100 Diversity = 2.6820e+00 k_t
= [4, 3, 3, 2]
Generation 4470: Best = 3.599750e+03 FES = 447100 Diversity = 2.6814e+00 k_t
= [1, 1, 1, 1]
Generation 4480: Best = 3.599748e+03 FES = 448100 Diversity = 2.6851e+00 k_t
= [2, 2, 3, 3]
Generation 4490: Best = 3.599748e+03 FES = 449100 Diversity = 2.6856e+00 k_t
= [1, 1, 1, 1]
Generation 4500: Best = 3.599748e+03 FES = 450100 Diversity = 2.6847e+00 k_t
= [3, 4, 3, 3]
Generation 4510: Best = 3.599748e+03 FES = 451100 Diversity = 2.6843e+00 k_t
= [1, 1, 1, 1]
Generation 4520: Best = 3.599748e+03 FES = 452100 Diversity = 2.6824e+00 k_t
= [4, 3, 3, 4]
Generation 4530: Best = 3.599748e+03 FES = 453100 Diversity = 2.6780e+00 k_t
= [1, 1, 1, 1]
Generation 4540: Best = 3.599748e+03 FES = 454100 Diversity = 2.6744e+00 k_t

```

```

= [4, 2, 2, 2]
Generation 4550: Best = 3.599748e+03 FES = 455100 Diversity = 2.6608e+00 k_t
= [1, 1, 1, 1]
Generation 4560: Best = 3.599748e+03 FES = 456100 Diversity = 2.6573e+00 k_t
= [3, 2, 2, 3]
Generation 4570: Best = 3.599748e+03 FES = 457100 Diversity = 2.6575e+00 k_t
= [1, 1, 1, 1]
Generation 4580: Best = 3.599748e+03 FES = 458100 Diversity = 2.6596e+00 k_t
= [4, 4, 4, 4]
Generation 4590: Best = 3.599748e+03 FES = 459100 Diversity = 2.6433e+00 k_t
= [1, 1, 1, 1]
Generation 4600: Best = 3.599748e+03 FES = 460100 Diversity = 2.6438e+00 k_t
= [2, 3, 4, 2]
Generation 4610: Best = 3.599701e+03 FES = 461100 Diversity = 2.6397e+00 k_t
= [1, 1, 1, 1]
Generation 4620: Best = 3.599617e+03 FES = 462100 Diversity = 2.6395e+00 k_t
= [4, 3, 4, 2]
Generation 4630: Best = 3.599617e+03 FES = 463100 Diversity = 2.6163e+00 k_t
= [1, 1, 1, 1]
Generation 4640: Best = 3.599617e+03 FES = 464100 Diversity = 2.6147e+00 k_t
= [2, 2, 2, 4]
Generation 4650: Best = 3.599617e+03 FES = 465100 Diversity = 2.6006e+00 k_t
= [1, 1, 1, 1]
Generation 4660: Best = 3.599584e+03 FES = 466100 Diversity = 2.6007e+00 k_t
= [4, 3, 4, 2]
Generation 4670: Best = 3.599584e+03 FES = 467100 Diversity = 2.5761e+00 k_t
= [1, 1, 1, 1]
Generation 4680: Best = 3.599584e+03 FES = 468100 Diversity = 2.5765e+00 k_t
= [2, 3, 2, 4]
Generation 4690: Best = 3.599579e+03 FES = 469100 Diversity = 2.5665e+00 k_t
= [1, 1, 1, 1]
Generation 4700: Best = 3.599579e+03 FES = 470100 Diversity = 2.5538e+00 k_t
= [3, 4, 2, 4]
Generation 4710: Best = 3.599579e+03 FES = 471100 Diversity = 2.5223e+00 k_t
= [1, 1, 1, 1]
Generation 4720: Best = 3.599579e+03 FES = 472100 Diversity = 2.5256e+00 k_t
= [3, 2, 3, 2]
Generation 4730: Best = 3.599579e+03 FES = 473100 Diversity = 2.5284e+00 k_t
= [1, 1, 1, 1]
Generation 4740: Best = 3.599579e+03 FES = 474100 Diversity = 2.5116e+00 k_t
= [3, 2, 3, 4]
Generation 4750: Best = 3.599579e+03 FES = 475100 Diversity = 2.5022e+00 k_t
= [1, 1, 1, 1]
Generation 4760: Best = 3.599579e+03 FES = 476100 Diversity = 2.4856e+00 k_t
= [2, 4, 4, 4]
Generation 4770: Best = 3.599579e+03 FES = 477100 Diversity = 2.4607e+00 k_t
= [1, 1, 1, 1]
Generation 4780: Best = 3.599579e+03 FES = 478100 Diversity = 2.4670e+00 k_t
= [2, 2, 3, 4]
Generation 4790: Best = 3.599560e+03 FES = 479100 Diversity = 2.4659e+00 k_t
= [1, 1, 1, 1]
Generation 4800: Best = 3.599546e+03 FES = 480100 Diversity = 2.4566e+00 k_t
= [2, 2, 2, 2]
Generation 4810: Best = 3.599546e+03 FES = 481100 Diversity = 2.4389e+00 k_t

```



```

= [1, 1, 1, 1]
Generation 4820: Best = 3.599501e+03 FES = 482100 Diversity = 2.4216e+00 k_t
= [3, 4, 3, 3]
Generation 4830: Best = 3.599501e+03 FES = 483100 Diversity = 2.4216e+00 k_t
= [1, 1, 1, 1]
Generation 4840: Best = 3.599444e+03 FES = 484100 Diversity = 2.4103e+00 k_t
= [4, 4, 4, 4]
Generation 4850: Best = 3.599444e+03 FES = 485100 Diversity = 2.3991e+00 k_t
= [1, 1, 1, 1]
Generation 4860: Best = 3.599444e+03 FES = 486100 Diversity = 2.3402e+00 k_t
= [4, 2, 4, 2]
Generation 4870: Best = 3.599444e+03 FES = 487100 Diversity = 2.3360e+00 k_t
= [1, 1, 1, 1]
Generation 4880: Best = 3.599444e+03 FES = 488100 Diversity = 2.3294e+00 k_t
= [3, 3, 3, 4]
Generation 4890: Best = 3.599444e+03 FES = 489100 Diversity = 2.3280e+00 k_t
= [1, 1, 1, 1]
Generation 4900: Best = 3.599292e+03 FES = 490100 Diversity = 2.3040e+00 k_t
= [2, 3, 4, 2]
Generation 4910: Best = 3.599292e+03 FES = 491100 Diversity = 2.2913e+00 k_t
= [1, 1, 1, 1]
Generation 4920: Best = 3.599292e+03 FES = 492100 Diversity = 2.2853e+00 k_t
= [4, 4, 3, 4]
Generation 4930: Best = 3.599292e+03 FES = 493100 Diversity = 2.2607e+00 k_t
= [1, 1, 1, 1]
Generation 4940: Best = 3.599292e+03 FES = 494100 Diversity = 2.2023e+00 k_t
= [3, 2, 2, 4]
Generation 4950: Best = 3.599290e+03 FES = 495100 Diversity = 2.1798e+00 k_t
= [1, 1, 1, 1]
Generation 4960: Best = 3.599290e+03 FES = 496100 Diversity = 2.1402e+00 k_t
= [2, 3, 3, 2]
Generation 4970: Best = 3.599196e+03 FES = 497100 Diversity = 2.0236e+00 k_t
= [1, 1, 1, 1]
Generation 4980: Best = 3.599196e+03 FES = 498100 Diversity = 1.9944e+00 k_t
= [3, 3, 4, 2]
Generation 4990: Best = 3.599175e+03 FES = 499100 Diversity = 1.9563e+00 k_t
= [1, 1, 1, 1]
Generation 5000: Best = 3.599174e+03 FES = 500100 Diversity = 1.8533e+00 k_t
= [2, 4, 4, 4]
Generation 5010: Best = 3.599135e+03 FES = 501100 Diversity = 1.8267e+00 k_t
= [1, 1, 1, 1]
Generation 5020: Best = 3.599077e+03 FES = 502100 Diversity = 1.8232e+00 k_t
= [4, 3, 2, 3]
Generation 5030: Best = 3.599001e+03 FES = 503100 Diversity = 1.7364e+00 k_t
= [1, 1, 1, 1]
Generation 5040: Best = 3.599001e+03 FES = 504100 Diversity = 1.5839e+00 k_t
= [3, 3, 4, 3]
Generation 5050: Best = 3.598915e+03 FES = 505100 Diversity = 1.0202e+00 k_t
= [1, 1, 1, 1]
Generation 5060: Best = 3.598884e+03 FES = 506100 Diversity = 3.8397e-01 k_t
= [4, 4, 3, 4]
Generation 5070: Best = 3.598773e+03 FES = 507100 Diversity = 1.7204e-01 k_t
= [1, 1, 1, 1]
Generation 5080: Best = 3.598755e+03 FES = 508100 Diversity = 1.5274e-02 k_t

```

```

= [3, 4, 4, 3]
Generation 5090: Best = 3.598673e+03 FES = 509100 Diversity = 1.5710e-02 k_t
= [1, 1, 1, 1]
Generation 5100: Best = 3.598648e+03 FES = 510100 Diversity = 1.6513e-02 k_t
= [2, 3, 4, 2]
Generation 5110: Best = 3.598450e+03 FES = 511100 Diversity = 1.6985e-02 k_t
= [1, 1, 1, 1]
Generation 5120: Best = 3.598450e+03 FES = 512100 Diversity = 1.6894e-02 k_t
= [4, 2, 4, 3]
Generation 5130: Best = 3.598450e+03 FES = 513100 Diversity = 1.5830e-02 k_t
= [1, 1, 1, 1]
Generation 5140: Best = 3.598426e+03 FES = 514100 Diversity = 1.4798e-02 k_t
= [4, 3, 4, 2]
Generation 5150: Best = 3.598354e+03 FES = 515100 Diversity = 1.3440e-02 k_t
= [1, 1, 1, 1]
Generation 5160: Best = 3.598347e+03 FES = 516100 Diversity = 1.2764e-02 k_t
= [4, 2, 4, 4]
Generation 5170: Best = 3.598313e+03 FES = 517100 Diversity = 1.2053e-02 k_t
= [1, 1, 1, 1]
Generation 5180: Best = 3.598261e+03 FES = 518100 Diversity = 1.1432e-02 k_t
= [3, 3, 2, 3]
Generation 5190: Best = 3.598261e+03 FES = 519100 Diversity = 1.0617e-02 k_t
= [1, 1, 1, 1]
Generation 5200: Best = 3.598242e+03 FES = 520100 Diversity = 1.0045e-02 k_t
= [4, 4, 4, 4]
Generation 5210: Best = 3.598214e+03 FES = 521100 Diversity = 8.8669e-03 k_t
= [1, 1, 1, 1]
Generation 5220: Best = 3.598208e+03 FES = 522100 Diversity = 8.0316e-03 k_t
= [3, 3, 4, 4]
Generation 5230: Best = 3.598185e+03 FES = 523100 Diversity = 7.5259e-03 k_t
= [1, 1, 1, 1]
Generation 5240: Best = 3.598164e+03 FES = 524100 Diversity = 7.1604e-03 k_t
= [3, 2, 2, 2]
Generation 5250: Best = 3.598151e+03 FES = 525100 Diversity = 6.9441e-03 k_t
= [1, 1, 1, 1]
Generation 5260: Best = 3.598135e+03 FES = 526100 Diversity = 6.9637e-03 k_t
= [4, 3, 2, 3]
Generation 5270: Best = 3.598135e+03 FES = 527100 Diversity = 6.9431e-03 k_t
= [1, 1, 1, 1]
Generation 5280: Best = 3.598130e+03 FES = 528100 Diversity = 6.7199e-03 k_t
= [2, 4, 3, 2]
Generation 5290: Best = 3.598092e+03 FES = 529100 Diversity = 6.4613e-03 k_t
= [1, 1, 1, 1]
Generation 5300: Best = 3.598089e+03 FES = 530100 Diversity = 6.3910e-03 k_t
= [4, 4, 2, 4]
Generation 5310: Best = 3.598076e+03 FES = 531100 Diversity = 6.2751e-03 k_t
= [1, 1, 1, 1]
Generation 5320: Best = 3.598057e+03 FES = 532100 Diversity = 5.8833e-03 k_t
= [2, 4, 2, 3]
Generation 5330: Best = 3.598049e+03 FES = 533100 Diversity = 5.6696e-03 k_t
= [1, 1, 1, 1]
Generation 5340: Best = 3.598041e+03 FES = 534100 Diversity = 5.3527e-03 k_t
= [2, 3, 4, 2]
Generation 5350: Best = 3.598040e+03 FES = 535100 Diversity = 5.1524e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 5360: Best = 3.598037e+03 FES = 536100 Diversity = 4.9131e-03 k_t
= [4, 2, 3, 3]
Generation 5370: Best = 3.598026e+03 FES = 537100 Diversity = 4.6287e-03 k_t
= [1, 1, 1, 1]
Generation 5380: Best = 3.598017e+03 FES = 538100 Diversity = 4.5259e-03 k_t
= [4, 2, 2, 4]
Generation 5390: Best = 3.598013e+03 FES = 539100 Diversity = 4.2808e-03 k_t
= [1, 1, 1, 1]
Generation 5400: Best = 3.598009e+03 FES = 540100 Diversity = 4.0966e-03 k_t
= [3, 3, 4, 3]
Generation 5410: Best = 3.598006e+03 FES = 541100 Diversity = 4.0812e-03 k_t
= [1, 1, 1, 1]
Generation 5420: Best = 3.598003e+03 FES = 542100 Diversity = 4.1148e-03 k_t
= [3, 4, 3, 4]
Generation 5430: Best = 3.597989e+03 FES = 543100 Diversity = 4.1345e-03 k_t
= [1, 1, 1, 1]
Generation 5440: Best = 3.597989e+03 FES = 544100 Diversity = 4.2629e-03 k_t
= [3, 3, 2, 2]
Generation 5450: Best = 3.597975e+03 FES = 545100 Diversity = 4.1310e-03 k_t
= [1, 1, 1, 1]
Generation 5460: Best = 3.597975e+03 FES = 546100 Diversity = 4.1628e-03 k_t
= [2, 4, 4, 4]
Generation 5470: Best = 3.597973e+03 FES = 547100 Diversity = 4.1205e-03 k_t
= [1, 1, 1, 1]
Generation 5480: Best = 3.597970e+03 FES = 548100 Diversity = 4.1416e-03 k_t
= [3, 4, 3, 3]
Generation 5490: Best = 3.597949e+03 FES = 549100 Diversity = 3.9960e-03 k_t
= [1, 1, 1, 1]
Generation 5500: Best = 3.597946e+03 FES = 550100 Diversity = 4.1664e-03 k_t
= [3, 2, 2, 4]
Generation 5510: Best = 3.597942e+03 FES = 551100 Diversity = 4.1167e-03 k_t
= [1, 1, 1, 1]
Generation 5520: Best = 3.597935e+03 FES = 552100 Diversity = 4.1122e-03 k_t
= [2, 2, 3, 4]
Generation 5530: Best = 3.597919e+03 FES = 553100 Diversity = 4.1105e-03 k_t
= [1, 1, 1, 1]
Generation 5540: Best = 3.597914e+03 FES = 554100 Diversity = 4.1184e-03 k_t
= [4, 4, 4, 4]
Generation 5550: Best = 3.597906e+03 FES = 555100 Diversity = 4.1408e-03 k_t
= [1, 1, 1, 1]
Generation 5560: Best = 3.597899e+03 FES = 556100 Diversity = 3.9604e-03 k_t
= [3, 3, 2, 4]
Generation 5570: Best = 3.597897e+03 FES = 557100 Diversity = 3.6231e-03 k_t
= [1, 1, 1, 1]
Generation 5580: Best = 3.597886e+03 FES = 558100 Diversity = 3.2918e-03 k_t
= [3, 3, 4, 2]
Generation 5590: Best = 3.597886e+03 FES = 559100 Diversity = 3.0875e-03 k_t
= [1, 1, 1, 1]
Generation 5600: Best = 3.597880e+03 FES = 560100 Diversity = 3.0480e-03 k_t
= [4, 3, 2, 2]
Generation 5610: Best = 3.597879e+03 FES = 561100 Diversity = 3.0473e-03 k_t
= [1, 1, 1, 1]
Generation 5620: Best = 3.597876e+03 FES = 562100 Diversity = 3.0454e-03 k_t

```

```

= [2, 2, 2, 3]
Generation 5630: Best = 3.597865e+03 FES = 563100 Diversity = 3.0855e-03 k_t
= [1, 1, 1, 1]
Generation 5640: Best = 3.597863e+03 FES = 564100 Diversity = 3.1067e-03 k_t
= [4, 3, 4, 4]
Generation 5650: Best = 3.597859e+03 FES = 565100 Diversity = 3.1523e-03 k_t
= [1, 1, 1, 1]
Generation 5660: Best = 3.597859e+03 FES = 566100 Diversity = 3.1109e-03 k_t
= [3, 4, 3, 4]
Generation 5670: Best = 3.597851e+03 FES = 567100 Diversity = 3.0246e-03 k_t
= [1, 1, 1, 1]
Generation 5680: Best = 3.597848e+03 FES = 568100 Diversity = 3.0213e-03 k_t
= [4, 2, 3, 3]
Generation 5690: Best = 3.597842e+03 FES = 569100 Diversity = 2.9909e-03 k_t
= [1, 1, 1, 1]
Generation 5700: Best = 3.597842e+03 FES = 570100 Diversity = 2.7438e-03 k_t
= [4, 2, 3, 4]
Generation 5710: Best = 3.597839e+03 FES = 571100 Diversity = 2.7455e-03 k_t
= [1, 1, 1, 1]
Generation 5720: Best = 3.597835e+03 FES = 572100 Diversity = 2.6119e-03 k_t
= [2, 4, 3, 2]
Generation 5730: Best = 3.597835e+03 FES = 573100 Diversity = 2.5033e-03 k_t
= [1, 1, 1, 1]
Generation 5740: Best = 3.597829e+03 FES = 574100 Diversity = 2.4535e-03 k_t
= [4, 4, 3, 2]
Generation 5750: Best = 3.597825e+03 FES = 575100 Diversity = 2.4181e-03 k_t
= [1, 1, 1, 1]
Generation 5760: Best = 3.597825e+03 FES = 576100 Diversity = 2.5426e-03 k_t
= [2, 2, 3, 4]
Generation 5770: Best = 3.597824e+03 FES = 577100 Diversity = 2.7078e-03 k_t
= [1, 1, 1, 1]
Generation 5780: Best = 3.597820e+03 FES = 578100 Diversity = 2.9080e-03 k_t
= [3, 2, 3, 4]
Generation 5790: Best = 3.597817e+03 FES = 579100 Diversity = 2.8790e-03 k_t
= [1, 1, 1, 1]
Generation 5800: Best = 3.597812e+03 FES = 580100 Diversity = 2.9369e-03 k_t
= [3, 2, 4, 3]
Generation 5810: Best = 3.597809e+03 FES = 581100 Diversity = 2.7915e-03 k_t
= [1, 1, 1, 1]
Generation 5820: Best = 3.597803e+03 FES = 582100 Diversity = 2.7821e-03 k_t
= [4, 2, 3, 4]
Generation 5830: Best = 3.597801e+03 FES = 583100 Diversity = 2.8605e-03 k_t
= [1, 1, 1, 1]
Generation 5840: Best = 3.597799e+03 FES = 584100 Diversity = 2.8516e-03 k_t
= [2, 2, 2, 3]
Generation 5850: Best = 3.597795e+03 FES = 585100 Diversity = 2.9234e-03 k_t
= [1, 1, 1, 1]
Generation 5860: Best = 3.597791e+03 FES = 586100 Diversity = 2.8841e-03 k_t
= [2, 4, 2, 2]
Generation 5870: Best = 3.597789e+03 FES = 587100 Diversity = 2.8519e-03 k_t
= [1, 1, 1, 1]
Generation 5880: Best = 3.597784e+03 FES = 588100 Diversity = 2.7588e-03 k_t
= [2, 4, 4, 2]
Generation 5890: Best = 3.597783e+03 FES = 589100 Diversity = 2.8276e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 5900: Best = 3.597781e+03 FES = 590100 Diversity = 2.5438e-03 k_t
= [4, 2, 4, 3]
Generation 5910: Best = 3.597776e+03 FES = 591100 Diversity = 2.4594e-03 k_t
= [1, 1, 1, 1]
Generation 5920: Best = 3.597773e+03 FES = 592100 Diversity = 2.4115e-03 k_t
= [4, 2, 4, 3]
Generation 5930: Best = 3.597769e+03 FES = 593100 Diversity = 2.3608e-03 k_t
= [1, 1, 1, 1]
Generation 5940: Best = 3.597765e+03 FES = 594100 Diversity = 2.5397e-03 k_t
= [2, 3, 3, 3]
Generation 5950: Best = 3.597761e+03 FES = 595100 Diversity = 2.6508e-03 k_t
= [1, 1, 1, 1]
Generation 5960: Best = 3.597754e+03 FES = 596100 Diversity = 2.7072e-03 k_t
= [4, 2, 3, 3]
Generation 5970: Best = 3.597752e+03 FES = 597100 Diversity = 2.7554e-03 k_t
= [1, 1, 1, 1]
Generation 5980: Best = 3.597752e+03 FES = 598100 Diversity = 2.7820e-03 k_t
= [4, 4, 2, 2]
Generation 5990: Best = 3.597748e+03 FES = 599100 Diversity = 2.7216e-03 k_t
= [1, 1, 1, 1]
Generation 6000: Best = 3.597745e+03 FES = 600100 Diversity = 2.7049e-03 k_t
= [4, 2, 2, 3]
Generation 6010: Best = 3.597745e+03 FES = 601100 Diversity = 2.8677e-03 k_t
= [1, 1, 1, 1]
Generation 6020: Best = 3.597739e+03 FES = 602100 Diversity = 2.7257e-03 k_t
= [2, 4, 3, 3]
Generation 6030: Best = 3.597735e+03 FES = 603100 Diversity = 2.8116e-03 k_t
= [1, 1, 1, 1]
Generation 6040: Best = 3.597732e+03 FES = 604100 Diversity = 2.8782e-03 k_t
= [2, 3, 3, 4]
Generation 6050: Best = 3.597728e+03 FES = 605100 Diversity = 2.7791e-03 k_t
= [1, 1, 1, 1]
Generation 6060: Best = 3.597726e+03 FES = 606100 Diversity = 2.8127e-03 k_t
= [3, 2, 2, 3]
Generation 6070: Best = 3.597720e+03 FES = 607100 Diversity = 2.8303e-03 k_t
= [1, 1, 1, 1]
Generation 6080: Best = 3.597715e+03 FES = 608100 Diversity = 2.9455e-03 k_t
= [4, 4, 4, 2]
Generation 6090: Best = 3.597710e+03 FES = 609100 Diversity = 2.9130e-03 k_t
= [1, 1, 1, 1]
Generation 6100: Best = 3.597708e+03 FES = 610100 Diversity = 2.6570e-03 k_t
= [2, 4, 4, 2]
Generation 6110: Best = 3.597701e+03 FES = 611100 Diversity = 2.5819e-03 k_t
= [1, 1, 1, 1]
Generation 6120: Best = 3.597692e+03 FES = 612100 Diversity = 2.5772e-03 k_t
= [3, 3, 4, 3]
Generation 6130: Best = 3.597688e+03 FES = 613100 Diversity = 2.7839e-03 k_t
= [1, 1, 1, 1]
Generation 6140: Best = 3.597685e+03 FES = 614100 Diversity = 2.8755e-03 k_t
= [3, 2, 4, 2]
Generation 6150: Best = 3.597683e+03 FES = 615100 Diversity = 2.5794e-03 k_t
= [1, 1, 1, 1]
Generation 6160: Best = 3.597679e+03 FES = 616100 Diversity = 2.4684e-03 k_t

```

```

= [3, 4, 4, 4]
Generation 6170: Best = 3.597675e+03 FES = 617100 Diversity = 2.4400e-03 k_t
= [1, 1, 1, 1]
Generation 6180: Best = 3.597673e+03 FES = 618100 Diversity = 2.5378e-03 k_t
= [2, 2, 3, 4]
Generation 6190: Best = 3.597670e+03 FES = 619100 Diversity = 2.4345e-03 k_t
= [1, 1, 1, 1]
Generation 6200: Best = 3.597663e+03 FES = 620100 Diversity = 2.4284e-03 k_t
= [4, 3, 2, 4]
Generation 6210: Best = 3.597661e+03 FES = 621100 Diversity = 2.3436e-03 k_t
= [1, 1, 1, 1]
Generation 6220: Best = 3.597659e+03 FES = 622100 Diversity = 2.5554e-03 k_t
= [4, 4, 2, 3]
Generation 6230: Best = 3.597653e+03 FES = 623100 Diversity = 2.6095e-03 k_t
= [1, 1, 1, 1]
Generation 6240: Best = 3.597651e+03 FES = 624100 Diversity = 2.5670e-03 k_t
= [3, 4, 4, 3]
Generation 6250: Best = 3.597641e+03 FES = 625100 Diversity = 2.5773e-03 k_t
= [1, 1, 1, 1]
Generation 6260: Best = 3.597641e+03 FES = 626100 Diversity = 2.5727e-03 k_t
= [2, 4, 2, 3]
Generation 6270: Best = 3.597641e+03 FES = 627100 Diversity = 2.5372e-03 k_t
= [1, 1, 1, 1]
Generation 6280: Best = 3.597641e+03 FES = 628100 Diversity = 2.5196e-03 k_t
= [2, 2, 4, 2]
Generation 6290: Best = 3.597635e+03 FES = 629100 Diversity = 2.4164e-03 k_t
= [1, 1, 1, 1]
Generation 6300: Best = 3.597635e+03 FES = 630100 Diversity = 2.2658e-03 k_t
= [4, 4, 2, 3]
Generation 6310: Best = 3.597632e+03 FES = 631100 Diversity = 2.2198e-03 k_t
= [1, 1, 1, 1]
Generation 6320: Best = 3.597632e+03 FES = 632100 Diversity = 2.0714e-03 k_t
= [2, 4, 4, 4]
Generation 6330: Best = 3.597630e+03 FES = 633100 Diversity = 2.0746e-03 k_t
= [1, 1, 1, 1]
Generation 6340: Best = 3.597628e+03 FES = 634100 Diversity = 1.8937e-03 k_t
= [4, 2, 4, 2]
Generation 6350: Best = 3.597627e+03 FES = 635100 Diversity = 1.8235e-03 k_t
= [1, 1, 1, 1]
Generation 6360: Best = 3.597626e+03 FES = 636100 Diversity = 1.7006e-03 k_t
= [2, 4, 4, 4]
Generation 6370: Best = 3.597624e+03 FES = 637100 Diversity = 1.6312e-03 k_t
= [1, 1, 1, 1]
Generation 6380: Best = 3.597623e+03 FES = 638100 Diversity = 1.5485e-03 k_t
= [4, 2, 3, 3]
Generation 6390: Best = 3.597623e+03 FES = 639100 Diversity = 1.5351e-03 k_t
= [1, 1, 1, 1]
Generation 6400: Best = 3.597622e+03 FES = 640100 Diversity = 1.5008e-03 k_t
= [4, 2, 2, 3]
Generation 6410: Best = 3.597620e+03 FES = 641100 Diversity = 1.4904e-03 k_t
= [1, 1, 1, 1]
Generation 6420: Best = 3.597619e+03 FES = 642100 Diversity = 1.4531e-03 k_t
= [2, 3, 2, 3]
Generation 6430: Best = 3.597619e+03 FES = 643100 Diversity = 1.3972e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 6440: Best = 3.597619e+03 FES = 644100 Diversity = 1.3764e-03 k_t
= [4, 4, 4, 4]
Generation 6450: Best = 3.597619e+03 FES = 645100 Diversity = 1.3615e-03 k_t
= [1, 1, 1, 1]
Generation 6460: Best = 3.597618e+03 FES = 646100 Diversity = 1.3409e-03 k_t
= [2, 3, 4, 4]
Generation 6470: Best = 3.597617e+03 FES = 647100 Diversity = 1.2947e-03 k_t
= [1, 1, 1, 1]
Generation 6480: Best = 3.597617e+03 FES = 648100 Diversity = 1.2439e-03 k_t
= [2, 4, 4, 2]
Generation 6490: Best = 3.597617e+03 FES = 649100 Diversity = 1.1948e-03 k_t
= [1, 1, 1, 1]
Generation 6500: Best = 3.597615e+03 FES = 650100 Diversity = 1.1461e-03 k_t
= [4, 2, 4, 4]
Generation 6510: Best = 3.597613e+03 FES = 651100 Diversity = 1.1674e-03 k_t
= [1, 1, 1, 1]
Generation 6520: Best = 3.597613e+03 FES = 652100 Diversity = 1.2194e-03 k_t
= [4, 3, 3, 4]
Generation 6530: Best = 3.597613e+03 FES = 653100 Diversity = 1.2059e-03 k_t
= [1, 1, 1, 1]
Generation 6540: Best = 3.597611e+03 FES = 654100 Diversity = 1.2139e-03 k_t
= [2, 3, 4, 3]
Generation 6550: Best = 3.597610e+03 FES = 655100 Diversity = 1.2455e-03 k_t
= [1, 1, 1, 1]
Generation 6560: Best = 3.597609e+03 FES = 656100 Diversity = 1.1688e-03 k_t
= [3, 2, 4, 3]
Generation 6570: Best = 3.597608e+03 FES = 657100 Diversity = 1.2237e-03 k_t
= [1, 1, 1, 1]
Generation 6580: Best = 3.597608e+03 FES = 658100 Diversity = 1.2243e-03 k_t
= [2, 2, 3, 4]
Generation 6590: Best = 3.597608e+03 FES = 659100 Diversity = 1.1815e-03 k_t
= [1, 1, 1, 1]
Generation 6600: Best = 3.597606e+03 FES = 660100 Diversity = 1.2469e-03 k_t
= [4, 2, 4, 4]
Generation 6610: Best = 3.597606e+03 FES = 661100 Diversity = 1.2657e-03 k_t
= [1, 1, 1, 1]
Generation 6620: Best = 3.597606e+03 FES = 662100 Diversity = 1.2433e-03 k_t
= [4, 4, 3, 3]
Generation 6630: Best = 3.597605e+03 FES = 663100 Diversity = 1.1630e-03 k_t
= [1, 1, 1, 1]
Generation 6640: Best = 3.597605e+03 FES = 664100 Diversity = 1.0308e-03 k_t
= [4, 2, 3, 4]
Generation 6650: Best = 3.597604e+03 FES = 665100 Diversity = 9.6762e-04 k_t
= [1, 1, 1, 1]
Generation 6660: Best = 3.597604e+03 FES = 666100 Diversity = 9.4770e-04 k_t
= [4, 4, 3, 3]
Generation 6670: Best = 3.597604e+03 FES = 667100 Diversity = 9.2898e-04 k_t
= [1, 1, 1, 1]
Generation 6680: Best = 3.597603e+03 FES = 668100 Diversity = 9.5603e-04 k_t
= [4, 4, 4, 3]
Generation 6690: Best = 3.597603e+03 FES = 669100 Diversity = 9.4503e-04 k_t
= [1, 1, 1, 1]
Generation 6700: Best = 3.597603e+03 FES = 670100 Diversity = 9.2777e-04 k_t

```

```

= [4, 4, 4, 4]
Generation 6710: Best = 3.597603e+03  FES = 671100  Diversity = 8.9632e-04  k_t
= [1, 1, 1, 1]
Generation 6720: Best = 3.597602e+03  FES = 672100  Diversity = 9.1156e-04  k_t
= [3, 3, 2, 3]
Generation 6730: Best = 3.597602e+03  FES = 673100  Diversity = 8.8466e-04  k_t
= [1, 1, 1, 1]
Generation 6740: Best = 3.597602e+03  FES = 674100  Diversity = 9.0599e-04  k_t
= [4, 2, 2, 2]
Generation 6750: Best = 3.597602e+03  FES = 675100  Diversity = 8.7747e-04  k_t
= [1, 1, 1, 1]
Generation 6760: Best = 3.597602e+03  FES = 676100  Diversity = 8.5544e-04  k_t
= [2, 3, 2, 2]
Generation 6770: Best = 3.597602e+03  FES = 677100  Diversity = 8.6247e-04  k_t
= [1, 1, 1, 1]
Generation 6780: Best = 3.597602e+03  FES = 678100  Diversity = 8.4574e-04  k_t
= [2, 3, 3, 4]
Generation 6790: Best = 3.597602e+03  FES = 679100  Diversity = 8.4995e-04  k_t
= [1, 1, 1, 1]
Generation 6800: Best = 3.597602e+03  FES = 680100  Diversity = 8.4664e-04  k_t
= [2, 2, 4, 3]
Generation 6810: Best = 3.597601e+03  FES = 681100  Diversity = 8.3555e-04  k_t
= [1, 1, 1, 1]
Generation 6820: Best = 3.597601e+03  FES = 682100  Diversity = 8.4338e-04  k_t
= [4, 3, 4, 4]
Generation 6830: Best = 3.597601e+03  FES = 683100  Diversity = 8.3658e-04  k_t
= [1, 1, 1, 1]
Generation 6840: Best = 3.597601e+03  FES = 684100  Diversity = 8.4818e-04  k_t
= [2, 4, 3, 4]
Generation 6850: Best = 3.597600e+03  FES = 685100  Diversity = 8.1972e-04  k_t
= [1, 1, 1, 1]
Generation 6860: Best = 3.597600e+03  FES = 686100  Diversity = 8.0369e-04  k_t
= [4, 2, 4, 2]
Generation 6870: Best = 3.597600e+03  FES = 687100  Diversity = 8.0437e-04  k_t
= [1, 1, 1, 1]
Generation 6880: Best = 3.597600e+03  FES = 688100  Diversity = 8.0172e-04  k_t
= [2, 4, 2, 3]
Generation 6890: Best = 3.597600e+03  FES = 689100  Diversity = 7.9475e-04  k_t
= [1, 1, 1, 1]
Generation 6900: Best = 3.597599e+03  FES = 690100  Diversity = 8.0121e-04  k_t
= [2, 4, 4, 3]
Generation 6910: Best = 3.597599e+03  FES = 691100  Diversity = 7.9095e-04  k_t
= [1, 1, 1, 1]
Generation 6920: Best = 3.597599e+03  FES = 692100  Diversity = 7.8887e-04  k_t
= [2, 3, 2, 4]
Generation 6930: Best = 3.597598e+03  FES = 693100  Diversity = 7.7610e-04  k_t
= [1, 1, 1, 1]
Generation 6940: Best = 3.597598e+03  FES = 694100  Diversity = 7.8358e-04  k_t
= [4, 2, 4, 4]
Generation 6950: Best = 3.597598e+03  FES = 695100  Diversity = 7.6789e-04  k_t
= [1, 1, 1, 1]
Generation 6960: Best = 3.597597e+03  FES = 696100  Diversity = 7.8878e-04  k_t
= [4, 3, 2, 4]
Generation 6970: Best = 3.597597e+03  FES = 697100  Diversity = 7.8251e-04  k_t

```



```

= [1, 1, 1, 1]
Generation 6980: Best = 3.597597e+03 FES = 698100 Diversity = 7.4062e-04 k_t
= [2, 3, 3, 3]
Generation 6990: Best = 3.597597e+03 FES = 699100 Diversity = 7.0731e-04 k_t
= [1, 1, 1, 1]
Generation 7000: Best = 3.597597e+03 FES = 700100 Diversity = 7.0433e-04 k_t
= [2, 4, 4, 4]
Generation 7010: Best = 3.597596e+03 FES = 701100 Diversity = 7.1179e-04 k_t
= [1, 1, 1, 1]
Generation 7020: Best = 3.597596e+03 FES = 702100 Diversity = 6.9626e-04 k_t
= [4, 4, 2, 4]
Generation 7030: Best = 3.597596e+03 FES = 703100 Diversity = 6.5682e-04 k_t
= [1, 1, 1, 1]
Generation 7040: Best = 3.597596e+03 FES = 704100 Diversity = 6.4875e-04 k_t
= [3, 4, 3, 4]
Generation 7050: Best = 3.597596e+03 FES = 705100 Diversity = 6.4216e-04 k_t
= [1, 1, 1, 1]
Generation 7060: Best = 3.597596e+03 FES = 706100 Diversity = 6.1723e-04 k_t
= [2, 4, 2, 2]
Generation 7070: Best = 3.597595e+03 FES = 707100 Diversity = 6.1046e-04 k_t
= [1, 1, 1, 1]
Generation 7080: Best = 3.597595e+03 FES = 708100 Diversity = 5.6858e-04 k_t
= [4, 2, 4, 2]
Generation 7090: Best = 3.597595e+03 FES = 709100 Diversity = 5.9273e-04 k_t
= [1, 1, 1, 1]
Generation 7100: Best = 3.597595e+03 FES = 710100 Diversity = 5.7205e-04 k_t
= [4, 4, 4, 2]
Generation 7110: Best = 3.597595e+03 FES = 711100 Diversity = 5.3942e-04 k_t
= [1, 1, 1, 1]
Generation 7120: Best = 3.597595e+03 FES = 712100 Diversity = 5.2043e-04 k_t
= [4, 4, 2, 4]
Generation 7130: Best = 3.597595e+03 FES = 713100 Diversity = 5.0650e-04 k_t
= [1, 1, 1, 1]
Generation 7140: Best = 3.597595e+03 FES = 714100 Diversity = 5.0471e-04 k_t
= [2, 3, 4, 4]
Generation 7150: Best = 3.597595e+03 FES = 715100 Diversity = 4.9569e-04 k_t
= [1, 1, 1, 1]
Generation 7160: Best = 3.597594e+03 FES = 716100 Diversity = 4.7225e-04 k_t
= [3, 2, 3, 4]
Generation 7170: Best = 3.597594e+03 FES = 717100 Diversity = 4.7482e-04 k_t
= [1, 1, 1, 1]
Generation 7180: Best = 3.597594e+03 FES = 718100 Diversity = 4.7398e-04 k_t
= [3, 2, 4, 3]
Generation 7190: Best = 3.597594e+03 FES = 719100 Diversity = 4.7452e-04 k_t
= [1, 1, 1, 1]
Generation 7200: Best = 3.597594e+03 FES = 720100 Diversity = 4.6617e-04 k_t
= [2, 4, 2, 3]
Generation 7210: Best = 3.597594e+03 FES = 721100 Diversity = 4.6576e-04 k_t
= [1, 1, 1, 1]
Generation 7220: Best = 3.597594e+03 FES = 722100 Diversity = 4.7934e-04 k_t
= [3, 3, 2, 4]
Generation 7230: Best = 3.597594e+03 FES = 723100 Diversity = 4.9521e-04 k_t
= [1, 1, 1, 1]
Generation 7240: Best = 3.597594e+03 FES = 724100 Diversity = 4.7815e-04 k_t

```

```

= [4, 2, 2, 4]
Generation 7250: Best = 3.597593e+03 FES = 725100 Diversity = 4.9813e-04 k_t
= [1, 1, 1, 1]
Generation 7260: Best = 3.597593e+03 FES = 726100 Diversity = 5.3363e-04 k_t
= [4, 3, 4, 2]
Generation 7270: Best = 3.597593e+03 FES = 727100 Diversity = 5.3955e-04 k_t
= [1, 1, 1, 1]
Generation 7280: Best = 3.597593e+03 FES = 728100 Diversity = 6.3390e-04 k_t
= [2, 3, 4, 3]
Generation 7290: Best = 3.597593e+03 FES = 729100 Diversity = 6.6016e-04 k_t
= [1, 1, 1, 1]
Generation 7300: Best = 3.597592e+03 FES = 730100 Diversity = 6.4857e-04 k_t
= [4, 3, 2, 3]
Generation 7310: Best = 3.597592e+03 FES = 731100 Diversity = 6.4472e-04 k_t
= [1, 1, 1, 1]
Generation 7320: Best = 3.597592e+03 FES = 732100 Diversity = 6.1604e-04 k_t
= [3, 2, 4, 2]
Generation 7330: Best = 3.597592e+03 FES = 733100 Diversity = 5.9635e-04 k_t
= [1, 1, 1, 1]
Generation 7340: Best = 3.597592e+03 FES = 734100 Diversity = 5.9690e-04 k_t
= [2, 2, 3, 4]
Generation 7350: Best = 3.597592e+03 FES = 735100 Diversity = 5.9910e-04 k_t
= [1, 1, 1, 1]
Generation 7360: Best = 3.597592e+03 FES = 736100 Diversity = 5.8862e-04 k_t
= [3, 4, 4, 4]
Generation 7370: Best = 3.597591e+03 FES = 737100 Diversity = 5.6625e-04 k_t
= [1, 1, 1, 1]
Generation 7380: Best = 3.597591e+03 FES = 738100 Diversity = 5.1327e-04 k_t
= [3, 4, 3, 3]
Generation 7390: Best = 3.597591e+03 FES = 739100 Diversity = 5.0142e-04 k_t
= [1, 1, 1, 1]
Generation 7400: Best = 3.597591e+03 FES = 740100 Diversity = 4.9715e-04 k_t
= [4, 2, 4, 2]
Generation 7410: Best = 3.597591e+03 FES = 741100 Diversity = 4.9431e-04 k_t
= [1, 1, 1, 1]
Generation 7420: Best = 3.597591e+03 FES = 742100 Diversity = 4.7577e-04 k_t
= [3, 3, 3, 3]
Generation 7430: Best = 3.597591e+03 FES = 743100 Diversity = 4.7510e-04 k_t
= [1, 1, 1, 1]
Generation 7440: Best = 3.597591e+03 FES = 744100 Diversity = 4.5945e-04 k_t
= [3, 3, 2, 4]
Generation 7450: Best = 3.597591e+03 FES = 745100 Diversity = 4.3556e-04 k_t
= [1, 1, 1, 1]
Generation 7460: Best = 3.597591e+03 FES = 746100 Diversity = 4.2488e-04 k_t
= [4, 3, 2, 2]
Generation 7470: Best = 3.597591e+03 FES = 747100 Diversity = 4.2966e-04 k_t
= [1, 1, 1, 1]
Generation 7480: Best = 3.597591e+03 FES = 748100 Diversity = 4.3210e-04 k_t
= [2, 2, 2, 4]
Generation 7490: Best = 3.597591e+03 FES = 749100 Diversity = 4.4019e-04 k_t
= [1, 1, 1, 1]
Generation 7500: Best = 3.597591e+03 FES = 750100 Diversity = 4.3489e-04 k_t
= [3, 4, 3, 3]
Generation 7510: Best = 3.597591e+03 FES = 751100 Diversity = 4.4387e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 7520: Best = 3.597590e+03 FES = 752100 Diversity = 4.3624e-04 k_t
= [2, 2, 3, 3]
Generation 7530: Best = 3.597590e+03 FES = 753100 Diversity = 4.4270e-04 k_t
= [1, 1, 1, 1]
Generation 7540: Best = 3.597590e+03 FES = 754100 Diversity = 4.5041e-04 k_t
= [4, 3, 2, 4]
Generation 7550: Best = 3.597590e+03 FES = 755100 Diversity = 4.6398e-04 k_t
= [1, 1, 1, 1]
Generation 7560: Best = 3.597590e+03 FES = 756100 Diversity = 4.6722e-04 k_t
= [3, 3, 4, 3]
Generation 7570: Best = 3.597590e+03 FES = 757100 Diversity = 4.6943e-04 k_t
= [1, 1, 1, 1]
Generation 7580: Best = 3.597590e+03 FES = 758100 Diversity = 4.6758e-04 k_t
= [3, 3, 3, 3]
Generation 7590: Best = 3.597590e+03 FES = 759100 Diversity = 4.5104e-04 k_t
= [1, 1, 1, 1]
Generation 7600: Best = 3.597590e+03 FES = 760100 Diversity = 4.4068e-04 k_t
= [3, 2, 3, 2]
Generation 7610: Best = 3.597590e+03 FES = 761100 Diversity = 4.2493e-04 k_t
= [1, 1, 1, 1]
Generation 7620: Best = 3.597590e+03 FES = 762100 Diversity = 4.0804e-04 k_t
= [4, 2, 2, 3]
Generation 7630: Best = 3.597590e+03 FES = 763100 Diversity = 3.8768e-04 k_t
= [1, 1, 1, 1]
Generation 7640: Best = 3.597590e+03 FES = 764100 Diversity = 3.8736e-04 k_t
= [4, 2, 3, 2]
Generation 7650: Best = 3.597590e+03 FES = 765100 Diversity = 3.9496e-04 k_t
= [1, 1, 1, 1]
Generation 7660: Best = 3.597590e+03 FES = 766100 Diversity = 3.9021e-04 k_t
= [3, 3, 3, 4]
Generation 7670: Best = 3.597590e+03 FES = 767100 Diversity = 3.7419e-04 k_t
= [1, 1, 1, 1]
Generation 7680: Best = 3.597590e+03 FES = 768100 Diversity = 3.7012e-04 k_t
= [2, 4, 4, 2]
Generation 7690: Best = 3.597589e+03 FES = 769100 Diversity = 3.6157e-04 k_t
= [1, 1, 1, 1]
Generation 7700: Best = 3.597589e+03 FES = 770100 Diversity = 3.5148e-04 k_t
= [3, 4, 2, 4]
Generation 7710: Best = 3.597589e+03 FES = 771100 Diversity = 3.4510e-04 k_t
= [1, 1, 1, 1]
Generation 7720: Best = 3.597589e+03 FES = 772100 Diversity = 3.4068e-04 k_t
= [2, 4, 3, 2]
Generation 7730: Best = 3.597589e+03 FES = 773100 Diversity = 3.3097e-04 k_t
= [1, 1, 1, 1]
Generation 7740: Best = 3.597589e+03 FES = 774100 Diversity = 3.3185e-04 k_t
= [2, 2, 2, 4]
Generation 7750: Best = 3.597589e+03 FES = 775100 Diversity = 3.2526e-04 k_t
= [1, 1, 1, 1]
Generation 7760: Best = 3.597589e+03 FES = 776100 Diversity = 3.1220e-04 k_t
= [4, 3, 3, 4]
Generation 7770: Best = 3.597589e+03 FES = 777100 Diversity = 3.1136e-04 k_t
= [1, 1, 1, 1]
Generation 7780: Best = 3.597589e+03 FES = 778100 Diversity = 3.0545e-04 k_t

```

```

= [3, 4, 3, 3]
Generation 7790: Best = 3.597589e+03 FES = 779100 Diversity = 3.0251e-04 k_t
= [1, 1, 1, 1]
Generation 7800: Best = 3.597589e+03 FES = 780100 Diversity = 2.9856e-04 k_t
= [2, 3, 3, 2]
Generation 7810: Best = 3.597589e+03 FES = 781100 Diversity = 2.9828e-04 k_t
= [1, 1, 1, 1]
Generation 7820: Best = 3.597589e+03 FES = 782100 Diversity = 2.9791e-04 k_t
= [4, 2, 3, 4]
Generation 7830: Best = 3.597589e+03 FES = 783100 Diversity = 3.0541e-04 k_t
= [1, 1, 1, 1]
Generation 7840: Best = 3.597589e+03 FES = 784100 Diversity = 3.0927e-04 k_t
= [4, 2, 3, 2]
Generation 7850: Best = 3.597589e+03 FES = 785100 Diversity = 3.1015e-04 k_t
= [1, 1, 1, 1]
Generation 7860: Best = 3.597589e+03 FES = 786100 Diversity = 3.1199e-04 k_t
= [2, 2, 4, 4]
Generation 7870: Best = 3.597589e+03 FES = 787100 Diversity = 3.1443e-04 k_t
= [1, 1, 1, 1]
Generation 7880: Best = 3.597589e+03 FES = 788100 Diversity = 3.2197e-04 k_t
= [4, 3, 4, 3]
Generation 7890: Best = 3.597589e+03 FES = 789100 Diversity = 3.2910e-04 k_t
= [1, 1, 1, 1]
Generation 7900: Best = 3.597589e+03 FES = 790100 Diversity = 3.2919e-04 k_t
= [4, 3, 3, 4]
Generation 7910: Best = 3.597589e+03 FES = 791100 Diversity = 3.3156e-04 k_t
= [1, 1, 1, 1]
Generation 7920: Best = 3.597589e+03 FES = 792100 Diversity = 3.2857e-04 k_t
= [3, 3, 3, 3]
Generation 7930: Best = 3.597589e+03 FES = 793100 Diversity = 3.3831e-04 k_t
= [1, 1, 1, 1]
Generation 7940: Best = 3.597589e+03 FES = 794100 Diversity = 3.4480e-04 k_t
= [4, 4, 2, 4]
Generation 7950: Best = 3.597589e+03 FES = 795100 Diversity = 3.4099e-04 k_t
= [1, 1, 1, 1]
Generation 7960: Best = 3.597589e+03 FES = 796100 Diversity = 3.4951e-04 k_t
= [4, 3, 4, 2]
Generation 7970: Best = 3.597589e+03 FES = 797100 Diversity = 3.5592e-04 k_t
= [1, 1, 1, 1]
Generation 7980: Best = 3.597589e+03 FES = 798100 Diversity = 3.5974e-04 k_t
= [4, 4, 2, 4]
Generation 7990: Best = 3.597589e+03 FES = 799100 Diversity = 3.6369e-04 k_t
= [1, 1, 1, 1]
Generation 8000: Best = 3.597589e+03 FES = 800100 Diversity = 3.6525e-04 k_t
= [4, 3, 4, 3]
Generation 8010: Best = 3.597589e+03 FES = 801100 Diversity = 3.6501e-04 k_t
= [1, 1, 1, 1]
Generation 8020: Best = 3.597589e+03 FES = 802100 Diversity = 3.6794e-04 k_t
= [4, 2, 4, 3]
Generation 8030: Best = 3.597589e+03 FES = 803100 Diversity = 3.7583e-04 k_t
= [1, 1, 1, 1]
Generation 8040: Best = 3.597589e+03 FES = 804100 Diversity = 3.7174e-04 k_t
= [2, 2, 2, 2]
Generation 8050: Best = 3.597589e+03 FES = 805100 Diversity = 3.6870e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 8060: Best = 3.597589e+03 FES = 806100 Diversity = 3.6374e-04 k_t
= [4, 2, 3, 4]
Generation 8070: Best = 3.597589e+03 FES = 807100 Diversity = 3.6003e-04 k_t
= [1, 1, 1, 1]
Generation 8080: Best = 3.597589e+03 FES = 808100 Diversity = 3.6396e-04 k_t
= [3, 4, 2, 3]
Generation 8090: Best = 3.597589e+03 FES = 809100 Diversity = 3.6333e-04 k_t
= [1, 1, 1, 1]
Generation 8100: Best = 3.597589e+03 FES = 810100 Diversity = 3.6657e-04 k_t
= [4, 4, 2, 3]
Generation 8110: Best = 3.597589e+03 FES = 811100 Diversity = 3.6399e-04 k_t
= [1, 1, 1, 1]
Generation 8120: Best = 3.597589e+03 FES = 812100 Diversity = 3.7373e-04 k_t
= [3, 2, 2, 4]
Generation 8130: Best = 3.597589e+03 FES = 813100 Diversity = 3.7550e-04 k_t
= [1, 1, 1, 1]
Generation 8140: Best = 3.597589e+03 FES = 814100 Diversity = 3.8192e-04 k_t
= [3, 3, 3, 2]
Generation 8150: Best = 3.597589e+03 FES = 815100 Diversity = 3.6845e-04 k_t
= [1, 1, 1, 1]
Generation 8160: Best = 3.597589e+03 FES = 816100 Diversity = 3.7341e-04 k_t
= [4, 2, 2, 4]
Generation 8170: Best = 3.597589e+03 FES = 817100 Diversity = 3.5819e-04 k_t
= [1, 1, 1, 1]
Generation 8180: Best = 3.597589e+03 FES = 818100 Diversity = 3.5586e-04 k_t
= [3, 2, 2, 4]
Generation 8190: Best = 3.597589e+03 FES = 819100 Diversity = 3.6221e-04 k_t
= [1, 1, 1, 1]
Generation 8200: Best = 3.597589e+03 FES = 820100 Diversity = 3.5954e-04 k_t
= [3, 2, 3, 2]
Generation 8210: Best = 3.597589e+03 FES = 821100 Diversity = 3.5142e-04 k_t
= [1, 1, 1, 1]
Generation 8220: Best = 3.597589e+03 FES = 822100 Diversity = 3.4942e-04 k_t
= [3, 3, 4, 3]
Generation 8230: Best = 3.597589e+03 FES = 823100 Diversity = 3.4375e-04 k_t
= [1, 1, 1, 1]
Generation 8240: Best = 3.597589e+03 FES = 824100 Diversity = 3.4195e-04 k_t
= [3, 4, 3, 4]
Generation 8250: Best = 3.597589e+03 FES = 825100 Diversity = 3.3452e-04 k_t
= [1, 1, 1, 1]
Generation 8260: Best = 3.597589e+03 FES = 826100 Diversity = 3.2464e-04 k_t
= [3, 3, 4, 4]
Generation 8270: Best = 3.597589e+03 FES = 827100 Diversity = 3.1306e-04 k_t
= [1, 1, 1, 1]
Generation 8280: Best = 3.597589e+03 FES = 828100 Diversity = 2.9381e-04 k_t
= [2, 2, 4, 2]
Generation 8290: Best = 3.597589e+03 FES = 829100 Diversity = 2.8401e-04 k_t
= [1, 1, 1, 1]
Generation 8300: Best = 3.597589e+03 FES = 830100 Diversity = 2.6687e-04 k_t
= [4, 4, 4, 2]
Generation 8310: Best = 3.597588e+03 FES = 831100 Diversity = 2.6498e-04 k_t
= [1, 1, 1, 1]
Generation 8320: Best = 3.597588e+03 FES = 832100 Diversity = 2.7008e-04 k_t

```

```

= [2, 3, 4, 3]
Generation 8330: Best = 3.597588e+03 FES = 833100 Diversity = 2.6993e-04 k_t
= [1, 1, 1, 1]
Generation 8340: Best = 3.597588e+03 FES = 834100 Diversity = 2.6020e-04 k_t
= [2, 3, 2, 3]
Generation 8350: Best = 3.597588e+03 FES = 835100 Diversity = 2.5497e-04 k_t
= [1, 1, 1, 1]
Generation 8360: Best = 3.597588e+03 FES = 836100 Diversity = 2.5351e-04 k_t
= [2, 3, 3, 4]
Generation 8370: Best = 3.597588e+03 FES = 837100 Diversity = 2.4546e-04 k_t
= [1, 1, 1, 1]
Generation 8380: Best = 3.597588e+03 FES = 838100 Diversity = 2.5243e-04 k_t
= [2, 3, 3, 4]
Generation 8390: Best = 3.597588e+03 FES = 839100 Diversity = 2.6073e-04 k_t
= [1, 1, 1, 1]
Generation 8400: Best = 3.597588e+03 FES = 840100 Diversity = 2.6191e-04 k_t
= [2, 2, 3, 2]
Generation 8410: Best = 3.597588e+03 FES = 841100 Diversity = 2.6208e-04 k_t
= [1, 1, 1, 1]
Generation 8420: Best = 3.597588e+03 FES = 842100 Diversity = 2.6698e-04 k_t
= [4, 4, 2, 3]
Generation 8430: Best = 3.597588e+03 FES = 843100 Diversity = 2.6448e-04 k_t
= [1, 1, 1, 1]
Generation 8440: Best = 3.597588e+03 FES = 844100 Diversity = 2.6529e-04 k_t
= [2, 2, 2, 4]
Generation 8450: Best = 3.597588e+03 FES = 845100 Diversity = 2.5431e-04 k_t
= [1, 1, 1, 1]
Generation 8460: Best = 3.597588e+03 FES = 846100 Diversity = 2.5616e-04 k_t
= [3, 4, 4, 2]
Generation 8470: Best = 3.597588e+03 FES = 847100 Diversity = 2.5111e-04 k_t
= [1, 1, 1, 1]
Generation 8480: Best = 3.597588e+03 FES = 848100 Diversity = 2.4795e-04 k_t
= [3, 4, 2, 3]
Generation 8490: Best = 3.597588e+03 FES = 849100 Diversity = 2.4500e-04 k_t
= [1, 1, 1, 1]
Generation 8500: Best = 3.597588e+03 FES = 850100 Diversity = 2.4011e-04 k_t
= [3, 3, 2, 3]
Generation 8510: Best = 3.597588e+03 FES = 851100 Diversity = 2.4325e-04 k_t
= [1, 1, 1, 1]
Generation 8520: Best = 3.597588e+03 FES = 852100 Diversity = 2.4681e-04 k_t
= [2, 2, 4, 4]
Generation 8530: Best = 3.597588e+03 FES = 853100 Diversity = 2.4847e-04 k_t
= [1, 1, 1, 1]
Generation 8540: Best = 3.597588e+03 FES = 854100 Diversity = 2.4909e-04 k_t
= [2, 2, 3, 4]
Generation 8550: Best = 3.597588e+03 FES = 855100 Diversity = 2.4410e-04 k_t
= [1, 1, 1, 1]
Generation 8560: Best = 3.597588e+03 FES = 856100 Diversity = 2.4662e-04 k_t
= [2, 2, 2, 3]
Generation 8570: Best = 3.597588e+03 FES = 857100 Diversity = 2.5220e-04 k_t
= [1, 1, 1, 1]
Generation 8580: Best = 3.597588e+03 FES = 858100 Diversity = 2.4917e-04 k_t
= [3, 2, 2, 2]
Generation 8590: Best = 3.597588e+03 FES = 859100 Diversity = 2.4257e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 8600: Best = 3.597588e+03 FES = 860100 Diversity = 2.3324e-04 k_t
= [2, 3, 4, 4]
Generation 8610: Best = 3.597588e+03 FES = 861100 Diversity = 2.3401e-04 k_t
= [1, 1, 1, 1]
Generation 8620: Best = 3.597588e+03 FES = 862100 Diversity = 2.3054e-04 k_t
= [4, 2, 4, 3]
Generation 8630: Best = 3.597588e+03 FES = 863100 Diversity = 2.3134e-04 k_t
= [1, 1, 1, 1]
Generation 8640: Best = 3.597588e+03 FES = 864100 Diversity = 2.3337e-04 k_t
= [2, 4, 4, 2]
Generation 8650: Best = 3.597588e+03 FES = 865100 Diversity = 2.3143e-04 k_t
= [1, 1, 1, 1]
Generation 8660: Best = 3.597588e+03 FES = 866100 Diversity = 2.2567e-04 k_t
= [2, 4, 2, 3]
Generation 8670: Best = 3.597588e+03 FES = 867100 Diversity = 2.1073e-04 k_t
= [1, 1, 1, 1]
Generation 8680: Best = 3.597588e+03 FES = 868100 Diversity = 1.9555e-04 k_t
= [4, 4, 2, 2]
Generation 8690: Best = 3.597588e+03 FES = 869100 Diversity = 1.8948e-04 k_t
= [1, 1, 1, 1]
Generation 8700: Best = 3.597588e+03 FES = 870100 Diversity = 1.8526e-04 k_t
= [3, 4, 3, 4]
Generation 8710: Best = 3.597588e+03 FES = 871100 Diversity = 1.8681e-04 k_t
= [1, 1, 1, 1]
Generation 8720: Best = 3.597588e+03 FES = 872100 Diversity = 1.8627e-04 k_t
= [2, 2, 3, 3]
Generation 8730: Best = 3.597588e+03 FES = 873100 Diversity = 1.8266e-04 k_t
= [1, 1, 1, 1]
Generation 8740: Best = 3.597588e+03 FES = 874100 Diversity = 1.7685e-04 k_t
= [2, 4, 4, 4]
Generation 8750: Best = 3.597588e+03 FES = 875100 Diversity = 1.7104e-04 k_t
= [1, 1, 1, 1]
Generation 8760: Best = 3.597588e+03 FES = 876100 Diversity = 1.6820e-04 k_t
= [3, 4, 3, 4]
Generation 8770: Best = 3.597588e+03 FES = 877100 Diversity = 1.6907e-04 k_t
= [1, 1, 1, 1]
Generation 8780: Best = 3.597588e+03 FES = 878100 Diversity = 1.6641e-04 k_t
= [3, 4, 3, 4]
Generation 8790: Best = 3.597588e+03 FES = 879100 Diversity = 1.5809e-04 k_t
= [1, 1, 1, 1]
Generation 8800: Best = 3.597588e+03 FES = 880100 Diversity = 1.6045e-04 k_t
= [4, 4, 2, 2]
Generation 8810: Best = 3.597588e+03 FES = 881100 Diversity = 1.5975e-04 k_t
= [1, 1, 1, 1]
Generation 8820: Best = 3.597588e+03 FES = 882100 Diversity = 1.5342e-04 k_t
= [3, 3, 4, 4]
Generation 8830: Best = 3.597588e+03 FES = 883100 Diversity = 1.4563e-04 k_t
= [1, 1, 1, 1]
Generation 8840: Best = 3.597588e+03 FES = 884100 Diversity = 1.4446e-04 k_t
= [4, 2, 2, 3]
Generation 8850: Best = 3.597588e+03 FES = 885100 Diversity = 1.4167e-04 k_t
= [1, 1, 1, 1]
Generation 8860: Best = 3.597588e+03 FES = 886100 Diversity = 1.3700e-04 k_t

```

```

= [2, 4, 2, 2]
Generation 8870: Best = 3.597588e+03 FES = 887100 Diversity = 1.3715e-04 k_t
= [1, 1, 1, 1]
Generation 8880: Best = 3.597588e+03 FES = 888100 Diversity = 1.3187e-04 k_t
= [3, 3, 2, 4]
Generation 8890: Best = 3.597588e+03 FES = 889100 Diversity = 1.2760e-04 k_t
= [1, 1, 1, 1]
Generation 8900: Best = 3.597588e+03 FES = 890100 Diversity = 1.2189e-04 k_t
= [2, 4, 4, 4]
Generation 8910: Best = 3.597588e+03 FES = 891100 Diversity = 1.2153e-04 k_t
= [1, 1, 1, 1]
Generation 8920: Best = 3.597588e+03 FES = 892100 Diversity = 1.2046e-04 k_t
= [3, 3, 2, 4]
Generation 8930: Best = 3.597588e+03 FES = 893100 Diversity = 1.1866e-04 k_t
= [1, 1, 1, 1]
Generation 8940: Best = 3.597588e+03 FES = 894100 Diversity = 1.1309e-04 k_t
= [3, 2, 3, 3]
Generation 8950: Best = 3.597588e+03 FES = 895100 Diversity = 1.1181e-04 k_t
= [1, 1, 1, 1]
Generation 8960: Best = 3.597588e+03 FES = 896100 Diversity = 1.1146e-04 k_t
= [4, 3, 2, 2]
Generation 8970: Best = 3.597588e+03 FES = 897100 Diversity = 1.1205e-04 k_t
= [1, 1, 1, 1]
Generation 8980: Best = 3.597588e+03 FES = 898100 Diversity = 1.1035e-04 k_t
= [3, 3, 2, 4]
Generation 8990: Best = 3.597588e+03 FES = 899100 Diversity = 1.0749e-04 k_t
= [1, 1, 1, 1]
Generation 9000: Best = 3.597588e+03 FES = 900100 Diversity = 1.0825e-04 k_t
= [3, 3, 2, 3]
Generation 9010: Best = 3.597588e+03 FES = 901100 Diversity = 1.0543e-04 k_t
= [1, 1, 1, 1]
Generation 9020: Best = 3.597588e+03 FES = 902100 Diversity = 1.0110e-04 k_t
= [4, 2, 4, 4]
Generation 9030: Best = 3.597588e+03 FES = 903100 Diversity = 9.9176e-05 k_t
= [1, 1, 1, 1]
Generation 9040: Best = 3.597588e+03 FES = 904100 Diversity = 9.3889e-05 k_t
= [2, 2, 4, 2]
Generation 9050: Best = 3.597588e+03 FES = 905100 Diversity = 9.6710e-05 k_t
= [1, 1, 1, 1]
Generation 9060: Best = 3.597588e+03 FES = 906100 Diversity = 9.3401e-05 k_t
= [3, 3, 4, 4]
Generation 9070: Best = 3.597588e+03 FES = 907100 Diversity = 8.9717e-05 k_t
= [1, 1, 1, 1]
Generation 9080: Best = 3.597588e+03 FES = 908100 Diversity = 8.7654e-05 k_t
= [2, 4, 4, 4]
Generation 9090: Best = 3.597588e+03 FES = 909100 Diversity = 8.8790e-05 k_t
= [1, 1, 1, 1]
Generation 9100: Best = 3.597588e+03 FES = 910100 Diversity = 8.7861e-05 k_t
= [3, 4, 4, 2]
Generation 9110: Best = 3.597588e+03 FES = 911100 Diversity = 8.6633e-05 k_t
= [1, 1, 1, 1]
Generation 9120: Best = 3.597588e+03 FES = 912100 Diversity = 8.6595e-05 k_t
= [2, 4, 3, 2]
Generation 9130: Best = 3.597588e+03 FES = 913100 Diversity = 8.5046e-05 k_t

```



```

= [1, 1, 1, 1]
Generation 9140: Best = 3.597588e+03 FES = 914100 Diversity = 8.3875e-05 k_t
= [3, 3, 2, 2]
Generation 9150: Best = 3.597588e+03 FES = 915100 Diversity = 8.4862e-05 k_t
= [1, 1, 1, 1]
Generation 9160: Best = 3.597588e+03 FES = 916100 Diversity = 8.8369e-05 k_t
= [3, 4, 4, 4]
Generation 9170: Best = 3.597588e+03 FES = 917100 Diversity = 8.9829e-05 k_t
= [1, 1, 1, 1]
Generation 9180: Best = 3.597588e+03 FES = 918100 Diversity = 8.8173e-05 k_t
= [3, 2, 4, 2]
Generation 9190: Best = 3.597588e+03 FES = 919100 Diversity = 8.9694e-05 k_t
= [1, 1, 1, 1]
Generation 9200: Best = 3.597588e+03 FES = 920100 Diversity = 9.0599e-05 k_t
= [3, 3, 2, 2]
Generation 9210: Best = 3.597588e+03 FES = 921100 Diversity = 9.3487e-05 k_t
= [1, 1, 1, 1]
Generation 9220: Best = 3.597588e+03 FES = 922100 Diversity = 9.5669e-05 k_t
= [3, 2, 4, 3]
Generation 9230: Best = 3.597588e+03 FES = 923100 Diversity = 9.7936e-05 k_t
= [1, 1, 1, 1]
Generation 9240: Best = 3.597588e+03 FES = 924100 Diversity = 1.0055e-04 k_t
= [3, 3, 3, 4]
Generation 9250: Best = 3.597588e+03 FES = 925100 Diversity = 1.0489e-04 k_t
= [1, 1, 1, 1]
Generation 9260: Best = 3.597588e+03 FES = 926100 Diversity = 1.0479e-04 k_t
= [3, 4, 4, 3]
Generation 9270: Best = 3.597588e+03 FES = 927100 Diversity = 1.0438e-04 k_t
= [1, 1, 1, 1]
Generation 9280: Best = 3.597588e+03 FES = 928100 Diversity = 1.0832e-04 k_t
= [3, 3, 4, 3]
Generation 9290: Best = 3.597588e+03 FES = 929100 Diversity = 1.1186e-04 k_t
= [1, 1, 1, 1]
Generation 9300: Best = 3.597588e+03 FES = 930100 Diversity = 1.1141e-04 k_t
= [4, 2, 4, 2]
Generation 9310: Best = 3.597588e+03 FES = 931100 Diversity = 1.1195e-04 k_t
= [1, 1, 1, 1]
Generation 9320: Best = 3.597588e+03 FES = 932100 Diversity = 1.0896e-04 k_t
= [2, 2, 2, 4]
Generation 9330: Best = 3.597588e+03 FES = 933100 Diversity = 1.0409e-04 k_t
= [1, 1, 1, 1]
Generation 9340: Best = 3.597588e+03 FES = 934100 Diversity = 9.8904e-05 k_t
= [2, 3, 3, 2]
Generation 9350: Best = 3.597588e+03 FES = 935100 Diversity = 9.7776e-05 k_t
= [1, 1, 1, 1]
Generation 9360: Best = 3.597588e+03 FES = 936100 Diversity = 9.8116e-05 k_t
= [2, 2, 4, 2]
Generation 9370: Best = 3.597588e+03 FES = 937100 Diversity = 9.6366e-05 k_t
= [1, 1, 1, 1]
Generation 9380: Best = 3.597588e+03 FES = 938100 Diversity = 9.4353e-05 k_t
= [2, 4, 3, 3]
Generation 9390: Best = 3.597588e+03 FES = 939100 Diversity = 8.8029e-05 k_t
= [1, 1, 1, 1]
Generation 9400: Best = 3.597588e+03 FES = 940100 Diversity = 8.5656e-05 k_t

```

```

= [3, 4, 4, 4]
Generation 9410: Best = 3.597588e+03 FES = 941100 Diversity = 8.6654e-05 k_t
= [1, 1, 1, 1]
Generation 9420: Best = 3.597588e+03 FES = 942100 Diversity = 8.9076e-05 k_t
= [2, 2, 2, 4]
Generation 9430: Best = 3.597588e+03 FES = 943100 Diversity = 8.9083e-05 k_t
= [1, 1, 1, 1]
Generation 9440: Best = 3.597588e+03 FES = 944100 Diversity = 8.6210e-05 k_t
= [4, 4, 3, 2]
Generation 9450: Best = 3.597588e+03 FES = 945100 Diversity = 7.9095e-05 k_t
= [1, 1, 1, 1]
Generation 9460: Best = 3.597588e+03 FES = 946100 Diversity = 7.7416e-05 k_t
= [4, 4, 3, 3]
Generation 9470: Best = 3.597588e+03 FES = 947100 Diversity = 7.4043e-05 k_t
= [1, 1, 1, 1]
Generation 9480: Best = 3.597588e+03 FES = 948100 Diversity = 7.3023e-05 k_t
= [3, 2, 2, 2]
Generation 9490: Best = 3.597588e+03 FES = 949100 Diversity = 7.4305e-05 k_t
= [1, 1, 1, 1]
Generation 9500: Best = 3.597588e+03 FES = 950100 Diversity = 7.3543e-05 k_t
= [3, 2, 2, 2]
Generation 9510: Best = 3.597588e+03 FES = 951100 Diversity = 7.1518e-05 k_t
= [1, 1, 1, 1]
Generation 9520: Best = 3.597588e+03 FES = 952100 Diversity = 7.3275e-05 k_t
= [4, 2, 4, 4]
Generation 9530: Best = 3.597588e+03 FES = 953100 Diversity = 7.5553e-05 k_t
= [1, 1, 1, 1]
Generation 9540: Best = 3.597588e+03 FES = 954100 Diversity = 7.6716e-05 k_t
= [3, 2, 4, 2]
Generation 9550: Best = 3.597588e+03 FES = 955100 Diversity = 7.9349e-05 k_t
= [1, 1, 1, 1]
Generation 9560: Best = 3.597588e+03 FES = 956100 Diversity = 7.7694e-05 k_t
= [4, 2, 2, 3]
Generation 9570: Best = 3.597588e+03 FES = 957100 Diversity = 8.4711e-05 k_t
= [1, 1, 1, 1]
Generation 9580: Best = 3.597588e+03 FES = 958100 Diversity = 9.2245e-05 k_t
= [4, 4, 3, 4]
Generation 9590: Best = 3.597588e+03 FES = 959100 Diversity = 9.5598e-05 k_t
= [1, 1, 1, 1]
Generation 9600: Best = 3.597588e+03 FES = 960100 Diversity = 1.0219e-04 k_t
= [3, 2, 2, 2]
Generation 9610: Best = 3.597588e+03 FES = 961100 Diversity = 1.1262e-04 k_t
= [1, 1, 1, 1]
Generation 9620: Best = 3.597588e+03 FES = 962100 Diversity = 1.1248e-04 k_t
= [4, 3, 3, 4]
Generation 9630: Best = 3.597588e+03 FES = 963100 Diversity = 1.1866e-04 k_t
= [1, 1, 1, 1]
Generation 9640: Best = 3.597588e+03 FES = 964100 Diversity = 1.2165e-04 k_t
= [2, 3, 3, 4]
Generation 9650: Best = 3.597588e+03 FES = 965100 Diversity = 1.2281e-04 k_t
= [1, 1, 1, 1]
Generation 9660: Best = 3.597588e+03 FES = 966100 Diversity = 1.2751e-04 k_t
= [4, 2, 2, 2]
Generation 9670: Best = 3.597588e+03 FES = 967100 Diversity = 1.3497e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 9680: Best = 3.597588e+03 FES = 968100 Diversity = 1.4104e-04 k_t
= [4, 3, 2, 4]
Generation 9690: Best = 3.597588e+03 FES = 969100 Diversity = 1.4141e-04 k_t
= [1, 1, 1, 1]
Generation 9700: Best = 3.597588e+03 FES = 970100 Diversity = 1.4076e-04 k_t
= [3, 2, 2, 4]
Generation 9710: Best = 3.597588e+03 FES = 971100 Diversity = 1.3807e-04 k_t
= [1, 1, 1, 1]
Generation 9720: Best = 3.597588e+03 FES = 972100 Diversity = 1.3670e-04 k_t
= [2, 2, 3, 4]
Generation 9730: Best = 3.597588e+03 FES = 973100 Diversity = 1.3581e-04 k_t
= [1, 1, 1, 1]
Generation 9740: Best = 3.597588e+03 FES = 974100 Diversity = 1.3515e-04 k_t
= [4, 4, 3, 2]
Generation 9750: Best = 3.597588e+03 FES = 975100 Diversity = 1.3515e-04 k_t
= [1, 1, 1, 1]
Generation 9760: Best = 3.597588e+03 FES = 976100 Diversity = 1.4300e-04 k_t
= [4, 3, 4, 4]
Generation 9770: Best = 3.597588e+03 FES = 977100 Diversity = 1.4651e-04 k_t
= [1, 1, 1, 1]
Generation 9780: Best = 3.597588e+03 FES = 978100 Diversity = 1.4325e-04 k_t
= [2, 3, 4, 2]
Generation 9790: Best = 3.597588e+03 FES = 979100 Diversity = 1.4118e-04 k_t
= [1, 1, 1, 1]
Generation 9800: Best = 3.597588e+03 FES = 980100 Diversity = 1.3969e-04 k_t
= [2, 4, 4, 4]
Generation 9810: Best = 3.597588e+03 FES = 981100 Diversity = 1.3883e-04 k_t
= [1, 1, 1, 1]
Generation 9820: Best = 3.597588e+03 FES = 982100 Diversity = 1.3797e-04 k_t
= [3, 4, 2, 4]
Generation 9830: Best = 3.597588e+03 FES = 983100 Diversity = 1.3395e-04 k_t
= [1, 1, 1, 1]
Generation 9840: Best = 3.597588e+03 FES = 984100 Diversity = 1.2796e-04 k_t
= [3, 2, 2, 2]
Generation 9850: Best = 3.597588e+03 FES = 985100 Diversity = 1.2665e-04 k_t
= [1, 1, 1, 1]
Generation 9860: Best = 3.597588e+03 FES = 986100 Diversity = 1.1544e-04 k_t
= [2, 3, 2, 2]
Generation 9870: Best = 3.597588e+03 FES = 987100 Diversity = 1.1200e-04 k_t
= [1, 1, 1, 1]
Generation 9880: Best = 3.597588e+03 FES = 988100 Diversity = 1.1597e-04 k_t
= [2, 3, 3, 3]
Generation 9890: Best = 3.597588e+03 FES = 989100 Diversity = 1.1529e-04 k_t
= [1, 1, 1, 1]
Generation 9900: Best = 3.597588e+03 FES = 990100 Diversity = 1.1520e-04 k_t
= [4, 3, 2, 3]
Generation 9910: Best = 3.597588e+03 FES = 991100 Diversity = 1.1828e-04 k_t
= [1, 1, 1, 1]
Generation 9920: Best = 3.597588e+03 FES = 992100 Diversity = 1.2809e-04 k_t
= [2, 4, 4, 2]
Generation 9930: Best = 3.597588e+03 FES = 993100 Diversity = 1.3121e-04 k_t
= [1, 1, 1, 1]
Generation 9940: Best = 3.597588e+03 FES = 994100 Diversity = 1.3778e-04 k_t

```

```

= [3, 4, 3, 3]
Generation 9950: Best = 3.597588e+03  FES = 995100  Diversity = 1.4435e-04  k_t
= [1, 1, 1, 1]
Generation 9960: Best = 3.597588e+03  FES = 996100  Diversity = 1.4014e-04  k_t
= [4, 4, 2, 3]
Generation 9970: Best = 3.597588e+03  FES = 997100  Diversity = 1.4194e-04  k_t
= [1, 1, 1, 1]
Generation 9980: Best = 3.597588e+03  FES = 998100  Diversity = 1.4885e-04  k_t
= [3, 2, 3, 3]
Generation 9990: Best = 3.597588e+03  FES = 999100  Diversity = 1.5148e-04  k_t
= [1, 1, 1, 1]

```

```

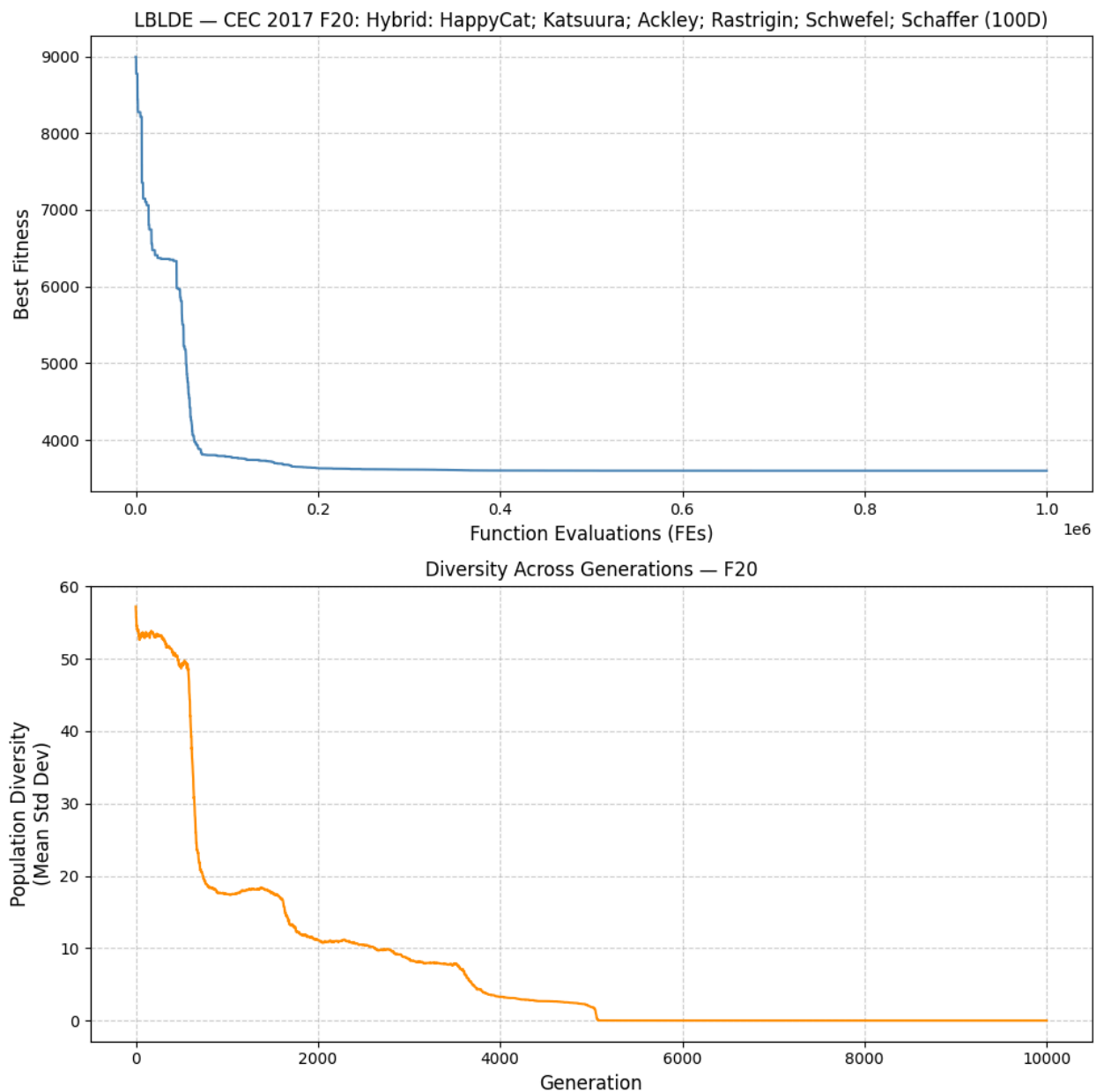
Optimization completed!
Final Best Fitness = 3.597588e+03
Total Function Evaluations = 1000000

```

```

=====
=
Results:
  Best Fitness           : 3.597588e+03
  Global Optimum         : 2.000000e+03
  Error (|best-optimum|) : 1.597588e+03
  Best Solution (5 dim)  : [-28.69068273   5.27523936 -86.5096224   45.59467027
97.2591724 ]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```



Graph saved as convergence_F20.png

6.4 Composite Functions (F21-F30)

```
In [45]: # =====
# F21 – Composition Function 1 (Rosenbrock; Elliptic; Rastrigin)
# =====
run_and_plot('F21', make_wrapper(composition.f21), 2100.0, 'Composition: Roser
```

```

=====
=
LBLDE_s on CEC 2017 F21 – Single Run
=====
=
Function      : Composition: Rosenbrock; Elliptic; Rastrigin (100D)
Bounds       : [-100, 100]
Global optimum : 2100.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 5.273291e+03 | Diversity = 5.7220e+01
Generation 10: Best = 4.490972e+03 FES = 1100 Diversity = 4.7328e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.960522e+03 FES = 2100 Diversity = 3.6898e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 3.638050e+03 FES = 3100 Diversity = 2.9142e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 3.569652e+03 FES = 4100 Diversity = 2.4112e+01 k_t
= [4, 3, 4, 2]
Generation 50: Best = 3.468986e+03 FES = 5100 Diversity = 2.2073e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.400809e+03 FES = 6100 Diversity = 1.9866e+01 k_t
= [4, 2, 4, 2]
Generation 70: Best = 3.400809e+03 FES = 7100 Diversity = 1.8719e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.341697e+03 FES = 8100 Diversity = 1.7907e+01 k_t
= [3, 2, 4, 3]
Generation 90: Best = 3.312410e+03 FES = 9100 Diversity = 1.7083e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.310966e+03 FES = 10100 Diversity = 1.6368e+01 k_t
= [4, 2, 4, 3]
Generation 110: Best = 3.298681e+03 FES = 11100 Diversity = 1.5942e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.290600e+03 FES = 12100 Diversity = 1.5537e+01 k_t
= [3, 3, 4, 3]
Generation 130: Best = 3.243433e+03 FES = 13100 Diversity = 1.5475e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.230163e+03 FES = 14100 Diversity = 1.5432e+01 k_t
= [3, 4, 3, 3]
Generation 150: Best = 3.230163e+03 FES = 15100 Diversity = 1.5208e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.220042e+03 FES = 16100 Diversity = 1.5321e+01 k_t
= [3, 2, 4, 3]
Generation 170: Best = 3.171982e+03 FES = 17100 Diversity = 1.5271e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.171982e+03 FES = 18100 Diversity = 1.5380e+01 k_t
= [2, 3, 3, 3]
Generation 190: Best = 3.171982e+03 FES = 19100 Diversity = 1.5220e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.171982e+03 FES = 20100 Diversity = 1.5328e+01 k_t
= [3, 4, 4, 2]
Generation 210: Best = 3.171982e+03 FES = 21100 Diversity = 1.5261e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.171982e+03 FES = 22100 Diversity = 1.5300e+01 k_t
= [2, 2, 4, 3]

```

Generation 230:	Best = 3.169681e+03	FES = 23100	Diversity = 1.5202e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 3.169681e+03	FES = 24100	Diversity = 1.5229e+01	k_t = [2, 4, 2, 2]
Generation 250:	Best = 3.158576e+03	FES = 25100	Diversity = 1.5186e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 3.155377e+03	FES = 26100	Diversity = 1.5114e+01	k_t = [3, 2, 4, 4]
Generation 270:	Best = 3.147954e+03	FES = 27100	Diversity = 1.5021e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 3.147954e+03	FES = 28100	Diversity = 1.4974e+01	k_t = [3, 2, 2, 4]
Generation 290:	Best = 3.142027e+03	FES = 29100	Diversity = 1.5011e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 3.142027e+03	FES = 30100	Diversity = 1.5024e+01	k_t = [3, 4, 4, 2]
Generation 310:	Best = 3.142027e+03	FES = 31100	Diversity = 1.5065e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 3.137510e+03	FES = 32100	Diversity = 1.5025e+01	k_t = [3, 3, 4, 4]
Generation 330:	Best = 3.137510e+03	FES = 33100	Diversity = 1.4996e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 3.127743e+03	FES = 34100	Diversity = 1.4953e+01	k_t = [3, 4, 2, 3]
Generation 350:	Best = 3.126977e+03	FES = 35100	Diversity = 1.4868e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 3.118826e+03	FES = 36100	Diversity = 1.4810e+01	k_t = [2, 4, 2, 2]
Generation 370:	Best = 3.118541e+03	FES = 37100	Diversity = 1.4790e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 3.091366e+03	FES = 38100	Diversity = 1.4742e+01	k_t = [2, 2, 2, 4]
Generation 390:	Best = 3.091366e+03	FES = 39100	Diversity = 1.4695e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 3.065675e+03	FES = 40100	Diversity = 1.4670e+01	k_t = [2, 3, 3, 2]
Generation 410:	Best = 3.065675e+03	FES = 41100	Diversity = 1.4709e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 3.059047e+03	FES = 42100	Diversity = 1.4703e+01	k_t = [3, 2, 3, 3]
Generation 430:	Best = 3.059047e+03	FES = 43100	Diversity = 1.4772e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 3.057095e+03	FES = 44100	Diversity = 1.4774e+01	k_t = [4, 2, 2, 4]
Generation 450:	Best = 3.057095e+03	FES = 45100	Diversity = 1.4766e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 3.057095e+03	FES = 46100	Diversity = 1.4730e+01	k_t = [3, 2, 2, 4]
Generation 470:	Best = 3.057095e+03	FES = 47100	Diversity = 1.4698e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 3.055162e+03	FES = 48100	Diversity = 1.4607e+01	k_t = [4, 4, 4, 3]
Generation 490:	Best = 3.042213e+03	FES = 49100	Diversity = 1.4560e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 3.028164e+03	FES = 50100	Diversity = 1.4607e+01	k_t = [4, 3, 3, 3]
Generation 510:	Best = 3.015580e+03	FES = 51100	Diversity = 1.4593e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 3.015150e+03	FES = 52100	Diversity = 1.4582e+01	k_t = [4, 2, 2, 3]
Generation 530:	Best = 3.015150e+03	FES = 53100	Diversity = 1.4562e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 3.015024e+03	FES = 54100	Diversity = 1.4539e+01	k_t = [4, 3, 3, 2]
Generation 550:	Best = 3.015024e+03	FES = 55100	Diversity = 1.4510e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 3.014859e+03	FES = 56100	Diversity = 1.4497e+01	k_t = [2, 4, 2, 4]
Generation 570:	Best = 2.993195e+03	FES = 57100	Diversity = 1.4484e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 2.990098e+03	FES = 58100	Diversity = 1.4462e+01	k_t = [2, 2, 2, 3]
Generation 590:	Best = 2.988859e+03	FES = 59100	Diversity = 1.4468e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 2.981321e+03	FES = 60100	Diversity = 1.4446e+01	k_t = [4, 4, 2, 3]
Generation 610:	Best = 2.981321e+03	FES = 61100	Diversity = 1.4359e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 2.981321e+03	FES = 62100	Diversity = 1.4344e+01	k_t = [4, 4, 3, 3]
Generation 630:	Best = 2.981321e+03	FES = 63100	Diversity = 1.4339e+01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 2.979306e+03	FES = 64100	Diversity = 1.4329e+01	k_t = [2, 2, 2, 2]
Generation 650:	Best = 2.968474e+03	FES = 65100	Diversity = 1.4310e+01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 2.968474e+03	FES = 66100	Diversity = 1.4299e+01	k_t = [4, 3, 2, 4]
Generation 670:	Best = 2.968474e+03	FES = 67100	Diversity = 1.4299e+01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 2.968474e+03	FES = 68100	Diversity = 1.4276e+01	k_t = [2, 4, 2, 4]
Generation 690:	Best = 2.968474e+03	FES = 69100	Diversity = 1.4231e+01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 2.968474e+03	FES = 70100	Diversity = 1.4219e+01	k_t = [3, 3, 3, 4]
Generation 710:	Best = 2.965063e+03	FES = 71100	Diversity = 1.4214e+01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 2.962911e+03	FES = 72100	Diversity = 1.4206e+01	k_t = [2, 4, 3, 3]
Generation 730:	Best = 2.960869e+03	FES = 73100	Diversity = 1.4193e+01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 2.960869e+03	FES = 74100	Diversity = 1.4176e+01	k_t = [4, 3, 4, 3]
Generation 750:	Best = 2.952496e+03	FES = 75100	Diversity = 1.4172e+01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 2.939288e+03	FES = 76100	Diversity = 1.4140e+01	k_t = [3, 3, 2, 4]

Generation 770:	Best = 2.939288e+03	FES = 77100	Diversity = 1.4130e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.939288e+03	FES = 78100	Diversity = 1.4093e+01	k_t
= [2, 3, 3, 2]				
Generation 790:	Best = 2.932610e+03	FES = 79100	Diversity = 1.4097e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.932610e+03	FES = 80100	Diversity = 1.4034e+01	k_t
= [3, 3, 4, 3]				
Generation 810:	Best = 2.915206e+03	FES = 81100	Diversity = 1.3860e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.914667e+03	FES = 82100	Diversity = 1.3786e+01	k_t
= [2, 3, 3, 4]				
Generation 830:	Best = 2.912018e+03	FES = 83100	Diversity = 1.3773e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.898584e+03	FES = 84100	Diversity = 1.3757e+01	k_t
= [3, 4, 2, 2]				
Generation 850:	Best = 2.898584e+03	FES = 85100	Diversity = 1.3755e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.893973e+03	FES = 86100	Diversity = 1.3743e+01	k_t
= [3, 4, 2, 3]				
Generation 870:	Best = 2.890016e+03	FES = 87100	Diversity = 1.3747e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.889572e+03	FES = 88100	Diversity = 1.3663e+01	k_t
= [4, 4, 4, 3]				
Generation 890:	Best = 2.889572e+03	FES = 89100	Diversity = 1.3652e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.889572e+03	FES = 90100	Diversity = 1.3637e+01	k_t
= [2, 2, 3, 2]				
Generation 910:	Best = 2.889572e+03	FES = 91100	Diversity = 1.3593e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.889572e+03	FES = 92100	Diversity = 1.3590e+01	k_t
= [3, 3, 4, 2]				
Generation 930:	Best = 2.889572e+03	FES = 93100	Diversity = 1.3497e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.889572e+03	FES = 94100	Diversity = 1.3485e+01	k_t
= [2, 2, 3, 4]				
Generation 950:	Best = 2.887641e+03	FES = 95100	Diversity = 1.3459e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 2.884818e+03	FES = 96100	Diversity = 1.3431e+01	k_t
= [3, 4, 4, 2]				
Generation 970:	Best = 2.884818e+03	FES = 97100	Diversity = 1.3356e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 2.870825e+03	FES = 98100	Diversity = 1.3354e+01	k_t
= [2, 3, 2, 4]				
Generation 990:	Best = 2.870825e+03	FES = 99100	Diversity = 1.3351e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 2.870825e+03	FES = 100100	Diversity = 1.3352e+01	k_t
= [3, 3, 3, 4]				
Generation 1010:	Best = 2.849706e+03	FES = 101100	Diversity = 1.3307e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 2.842548e+03	FES = 102100	Diversity = 1.3286e+01	k_t
= [3, 3, 2, 3]				
Generation 1030:	Best = 2.842548e+03	FES = 103100	Diversity = 1.3287e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 2.841020e+03 = [3, 2, 4, 2]	FES = 104100	Diversity = 1.3138e+01	k_t
Generation 1050: Best = 2.829251e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.3131e+01	k_t
Generation 1060: Best = 2.822331e+03 = [3, 4, 3, 4]	FES = 106100	Diversity = 1.3037e+01	k_t
Generation 1070: Best = 2.815201e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 1.2974e+01	k_t
Generation 1080: Best = 2.815201e+03 = [4, 4, 4, 3]	FES = 108100	Diversity = 1.2904e+01	k_t
Generation 1090: Best = 2.812767e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.2877e+01	k_t
Generation 1100: Best = 2.812296e+03 = [4, 4, 2, 4]	FES = 110100	Diversity = 1.2824e+01	k_t
Generation 1110: Best = 2.793623e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.2825e+01	k_t
Generation 1120: Best = 2.793623e+03 = [3, 3, 2, 4]	FES = 112100	Diversity = 1.2820e+01	k_t
Generation 1130: Best = 2.792187e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.2766e+01	k_t
Generation 1140: Best = 2.791610e+03 = [4, 3, 2, 3]	FES = 114100	Diversity = 1.2764e+01	k_t
Generation 1150: Best = 2.791610e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.2761e+01	k_t
Generation 1160: Best = 2.778473e+03 = [3, 3, 3, 3]	FES = 116100	Diversity = 1.2765e+01	k_t
Generation 1170: Best = 2.766961e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.2685e+01	k_t
Generation 1180: Best = 2.766961e+03 = [2, 3, 3, 3]	FES = 118100	Diversity = 1.2545e+01	k_t
Generation 1190: Best = 2.766961e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.2510e+01	k_t
Generation 1200: Best = 2.759034e+03 = [2, 2, 4, 2]	FES = 120100	Diversity = 1.2540e+01	k_t
Generation 1210: Best = 2.757602e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.2459e+01	k_t
Generation 1220: Best = 2.750812e+03 = [2, 2, 2, 2]	FES = 122100	Diversity = 1.2376e+01	k_t
Generation 1230: Best = 2.749528e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.2255e+01	k_t
Generation 1240: Best = 2.745652e+03 = [2, 2, 3, 3]	FES = 124100	Diversity = 1.2147e+01	k_t
Generation 1250: Best = 2.744635e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 1.2160e+01	k_t
Generation 1260: Best = 2.738857e+03 = [4, 2, 2, 2]	FES = 126100	Diversity = 1.2107e+01	k_t
Generation 1270: Best = 2.719722e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.1987e+01	k_t
Generation 1280: Best = 2.710668e+03 = [3, 4, 4, 4]	FES = 128100	Diversity = 1.1871e+01	k_t
Generation 1290: Best = 2.699408e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.1813e+01	k_t
Generation 1300: Best = 2.696161e+03 = [2, 3, 3, 3]	FES = 130100	Diversity = 1.1329e+01	k_t

Generation 1310: Best = 2.683464e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.0649e+01	k_t
Generation 1320: Best = 2.678149e+03 = [3, 2, 4, 3]	FES = 132100	Diversity = 1.0001e+01	k_t
Generation 1330: Best = 2.663503e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 8.9621e+00	k_t
Generation 1340: Best = 2.657923e+03 = [4, 4, 2, 4]	FES = 134100	Diversity = 7.1068e+00	k_t
Generation 1350: Best = 2.632797e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 4.6056e+00	k_t
Generation 1360: Best = 2.611894e+03 = [4, 4, 4, 4]	FES = 136100	Diversity = 3.2921e+00	k_t
Generation 1370: Best = 2.583246e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 1.7825e+00	k_t
Generation 1380: Best = 2.558362e+03 = [4, 3, 2, 2]	FES = 138100	Diversity = 7.4544e-01	k_t
Generation 1390: Best = 2.551757e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 6.3879e-01	k_t
Generation 1400: Best = 2.540301e+03 = [2, 3, 2, 3]	FES = 140100	Diversity = 5.2435e-01	k_t
Generation 1410: Best = 2.532997e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 4.2030e-01	k_t
Generation 1420: Best = 2.527115e+03 = [2, 2, 4, 4]	FES = 142100	Diversity = 3.2795e-01	k_t
Generation 1430: Best = 2.522277e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 2.5466e-01	k_t
Generation 1440: Best = 2.519293e+03 = [4, 2, 3, 2]	FES = 144100	Diversity = 2.2630e-01	k_t
Generation 1450: Best = 2.516155e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 1.9423e-01	k_t
Generation 1460: Best = 2.514325e+03 = [2, 3, 4, 3]	FES = 146100	Diversity = 1.6555e-01	k_t
Generation 1470: Best = 2.512943e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 1.3992e-01	k_t
Generation 1480: Best = 2.511685e+03 = [2, 3, 2, 3]	FES = 148100	Diversity = 1.2125e-01	k_t
Generation 1490: Best = 2.511008e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 1.0536e-01	k_t
Generation 1500: Best = 2.510583e+03 = [3, 4, 2, 2]	FES = 150100	Diversity = 9.0789e-02	k_t
Generation 1510: Best = 2.510076e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 7.2898e-02	k_t
Generation 1520: Best = 2.509875e+03 = [4, 3, 4, 4]	FES = 152100	Diversity = 5.9462e-02	k_t
Generation 1530: Best = 2.509574e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 5.2408e-02	k_t
Generation 1540: Best = 2.509305e+03 = [3, 4, 3, 4]	FES = 154100	Diversity = 4.7463e-02	k_t
Generation 1550: Best = 2.509179e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 4.1944e-02	k_t
Generation 1560: Best = 2.509055e+03 = [2, 3, 2, 4]	FES = 156100	Diversity = 3.7812e-02	k_t
Generation 1570: Best = 2.508932e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 3.5731e-02	k_t

Generation 1580: Best = 2.508864e+03 = [2, 2, 3, 4]	FES = 158100	Diversity = 3.2533e-02	k_t
Generation 1590: Best = 2.508795e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 2.9292e-02	k_t
Generation 1600: Best = 2.508748e+03 = [4, 4, 3, 2]	FES = 160100	Diversity = 2.5620e-02	k_t
Generation 1610: Best = 2.508697e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 2.2323e-02	k_t
Generation 1620: Best = 2.508676e+03 = [3, 4, 2, 4]	FES = 162100	Diversity = 1.9586e-02	k_t
Generation 1630: Best = 2.508655e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.6097e-02	k_t
Generation 1640: Best = 2.508634e+03 = [4, 2, 4, 3]	FES = 164100	Diversity = 1.4415e-02	k_t
Generation 1650: Best = 2.508616e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.3239e-02	k_t
Generation 1660: Best = 2.508601e+03 = [4, 4, 4, 3]	FES = 166100	Diversity = 1.2830e-02	k_t
Generation 1670: Best = 2.508587e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.1853e-02	k_t
Generation 1680: Best = 2.508579e+03 = [4, 3, 2, 3]	FES = 168100	Diversity = 1.0249e-02	k_t
Generation 1690: Best = 2.508572e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 9.0012e-03	k_t
Generation 1700: Best = 2.508567e+03 = [3, 4, 4, 4]	FES = 170100	Diversity = 7.4193e-03	k_t
Generation 1710: Best = 2.508563e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 6.3001e-03	k_t
Generation 1720: Best = 2.508560e+03 = [4, 2, 3, 4]	FES = 172100	Diversity = 5.3292e-03	k_t
Generation 1730: Best = 2.508557e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 4.8143e-03	k_t
Generation 1740: Best = 2.508555e+03 = [4, 3, 2, 2]	FES = 174100	Diversity = 4.3774e-03	k_t
Generation 1750: Best = 2.508554e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 4.0187e-03	k_t
Generation 1760: Best = 2.508552e+03 = [4, 4, 3, 2]	FES = 176100	Diversity = 3.6431e-03	k_t
Generation 1770: Best = 2.508551e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 3.1474e-03	k_t
Generation 1780: Best = 2.508550e+03 = [2, 4, 2, 4]	FES = 178100	Diversity = 2.8365e-03	k_t
Generation 1790: Best = 2.508550e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 2.6159e-03	k_t
Generation 1800: Best = 2.508549e+03 = [3, 2, 4, 3]	FES = 180100	Diversity = 2.4002e-03	k_t
Generation 1810: Best = 2.508549e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 2.1019e-03	k_t
Generation 1820: Best = 2.508548e+03 = [4, 2, 3, 2]	FES = 182100	Diversity = 1.9111e-03	k_t
Generation 1830: Best = 2.508548e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.7854e-03	k_t
Generation 1840: Best = 2.508548e+03 = [2, 4, 2, 4]	FES = 184100	Diversity = 1.6521e-03	k_t

Generation 1850: Best = 2.508547e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.7695e-03	k_t
Generation 1860: Best = 2.508547e+03 = [3, 3, 2, 4]	FES = 186100	Diversity = 1.7000e-03	k_t
Generation 1870: Best = 2.508547e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.4533e-03	k_t
Generation 1880: Best = 2.508547e+03 = [2, 2, 4, 2]	FES = 188100	Diversity = 1.2453e-03	k_t
Generation 1890: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 1.1302e-03	k_t
Generation 1900: Best = 2.508546e+03 = [3, 2, 3, 3]	FES = 190100	Diversity = 1.0313e-03	k_t
Generation 1910: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 8.7843e-04	k_t
Generation 1920: Best = 2.508546e+03 = [4, 4, 2, 3]	FES = 192100	Diversity = 7.5674e-04	k_t
Generation 1930: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 6.8220e-04	k_t
Generation 1940: Best = 2.508546e+03 = [2, 2, 2, 3]	FES = 194100	Diversity = 6.0221e-04	k_t
Generation 1950: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 5.5084e-04	k_t
Generation 1960: Best = 2.508546e+03 = [2, 2, 3, 2]	FES = 196100	Diversity = 4.8449e-04	k_t
Generation 1970: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 4.2786e-04	k_t
Generation 1980: Best = 2.508546e+03 = [3, 2, 4, 3]	FES = 198100	Diversity = 3.8727e-04	k_t
Generation 1990: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 3.5853e-04	k_t
Generation 2000: Best = 2.508546e+03 = [3, 2, 2, 4]	FES = 200100	Diversity = 3.2684e-04	k_t
Generation 2010: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 2.8795e-04	k_t
Generation 2020: Best = 2.508546e+03 = [3, 2, 4, 4]	FES = 202100	Diversity = 2.6548e-04	k_t
Generation 2030: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 2.4912e-04	k_t
Generation 2040: Best = 2.508546e+03 = [4, 2, 2, 4]	FES = 204100	Diversity = 2.1405e-04	k_t
Generation 2050: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.9633e-04	k_t
Generation 2060: Best = 2.508546e+03 = [3, 2, 4, 3]	FES = 206100	Diversity = 1.8526e-04	k_t
Generation 2070: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.7576e-04	k_t
Generation 2080: Best = 2.508546e+03 = [3, 3, 3, 4]	FES = 208100	Diversity = 1.5678e-04	k_t
Generation 2090: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 1.3312e-04	k_t
Generation 2100: Best = 2.508546e+03 = [2, 3, 3, 2]	FES = 210100	Diversity = 1.1491e-04	k_t
Generation 2110: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 1.0226e-04	k_t

Generation 2120: Best = 2.508546e+03 = [3, 2, 2, 3]	FES = 212100	Diversity = 9.1395e-05	k_t
Generation 2130: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 8.1820e-05	k_t
Generation 2140: Best = 2.508546e+03 = [2, 3, 3, 3]	FES = 214100	Diversity = 7.3265e-05	k_t
Generation 2150: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 6.2064e-05	k_t
Generation 2160: Best = 2.508546e+03 = [2, 4, 3, 3]	FES = 216100	Diversity = 5.4710e-05	k_t
Generation 2170: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 5.0192e-05	k_t
Generation 2180: Best = 2.508546e+03 = [2, 4, 2, 4]	FES = 218100	Diversity = 4.9739e-05	k_t
Generation 2190: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 4.8827e-05	k_t
Generation 2200: Best = 2.508546e+03 = [4, 4, 4, 4]	FES = 220100	Diversity = 4.4307e-05	k_t
Generation 2210: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 3.9821e-05	k_t
Generation 2220: Best = 2.508546e+03 = [4, 2, 3, 3]	FES = 222100	Diversity = 3.6722e-05	k_t
Generation 2230: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 3.3002e-05	k_t
Generation 2240: Best = 2.508546e+03 = [2, 3, 3, 3]	FES = 224100	Diversity = 3.0319e-05	k_t
Generation 2250: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 2.6757e-05	k_t
Generation 2260: Best = 2.508546e+03 = [4, 2, 4, 2]	FES = 226100	Diversity = 2.4109e-05	k_t
Generation 2270: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 2.1452e-05	k_t
Generation 2280: Best = 2.508546e+03 = [4, 2, 3, 2]	FES = 228100	Diversity = 1.6763e-05	k_t
Generation 2290: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.4971e-05	k_t
Generation 2300: Best = 2.508546e+03 = [3, 3, 2, 3]	FES = 230100	Diversity = 1.3630e-05	k_t
Generation 2310: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.3899e-05	k_t
Generation 2320: Best = 2.508546e+03 = [3, 4, 3, 4]	FES = 232100	Diversity = 1.2662e-05	k_t
Generation 2330: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 1.1553e-05	k_t
Generation 2340: Best = 2.508546e+03 = [2, 4, 3, 4]	FES = 234100	Diversity = 1.0477e-05	k_t
Generation 2350: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 9.7679e-06	k_t
Generation 2360: Best = 2.508546e+03 = [4, 4, 3, 2]	FES = 236100	Diversity = 9.0770e-06	k_t
Generation 2370: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 8.2806e-06	k_t
Generation 2380: Best = 2.508546e+03 = [3, 2, 4, 2]	FES = 238100	Diversity = 7.8694e-06	k_t

Generation 2390: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 6.5758e-06	k_t
Generation 2400: Best = 2.508546e+03 = [2, 4, 3, 2]	FES = 240100	Diversity = 6.0086e-06	k_t
Generation 2410: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 5.4260e-06	k_t
Generation 2420: Best = 2.508546e+03 = [2, 2, 4, 4]	FES = 242100	Diversity = 4.9915e-06	k_t
Generation 2430: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 4.4915e-06	k_t
Generation 2440: Best = 2.508546e+03 = [4, 2, 4, 4]	FES = 244100	Diversity = 4.1918e-06	k_t
Generation 2450: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 3.7377e-06	k_t
Generation 2460: Best = 2.508546e+03 = [2, 3, 2, 4]	FES = 246100	Diversity = 2.8753e-06	k_t
Generation 2470: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 2.5646e-06	k_t
Generation 2480: Best = 2.508546e+03 = [4, 3, 3, 4]	FES = 248100	Diversity = 2.1920e-06	k_t
Generation 2490: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 2.0317e-06	k_t
Generation 2500: Best = 2.508546e+03 = [3, 3, 4, 2]	FES = 250100	Diversity = 1.8490e-06	k_t
Generation 2510: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.6584e-06	k_t
Generation 2520: Best = 2.508546e+03 = [2, 2, 4, 2]	FES = 252100	Diversity = 1.5529e-06	k_t
Generation 2530: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.4439e-06	k_t
Generation 2540: Best = 2.508546e+03 = [2, 3, 3, 3]	FES = 254100	Diversity = 1.4351e-06	k_t
Generation 2550: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.3279e-06	k_t
Generation 2560: Best = 2.508546e+03 = [4, 4, 3, 4]	FES = 256100	Diversity = 1.1340e-06	k_t
Generation 2570: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.0004e-06	k_t
Generation 2580: Best = 2.508546e+03 = [2, 2, 4, 3]	FES = 258100	Diversity = 9.1096e-07	k_t
Generation 2590: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 8.0834e-07	k_t
Generation 2600: Best = 2.508546e+03 = [3, 4, 3, 2]	FES = 260100	Diversity = 7.6080e-07	k_t
Generation 2610: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 6.9817e-07	k_t
Generation 2620: Best = 2.508546e+03 = [2, 3, 3, 4]	FES = 262100	Diversity = 6.1708e-07	k_t
Generation 2630: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 5.5471e-07	k_t
Generation 2640: Best = 2.508546e+03 = [3, 2, 2, 3]	FES = 264100	Diversity = 5.3183e-07	k_t
Generation 2650: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 5.2432e-07	k_t

Generation 2660: Best = 2.508546e+03 = [4, 4, 2, 2]	FES = 266100	Diversity = 4.5478e-07	k_t
Generation 2670: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 4.5682e-07	k_t
Generation 2680: Best = 2.508546e+03 = [2, 4, 2, 2]	FES = 268100	Diversity = 3.9393e-07	k_t
Generation 2690: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 3.3398e-07	k_t
Generation 2700: Best = 2.508546e+03 = [3, 3, 3, 2]	FES = 270100	Diversity = 2.8849e-07	k_t
Generation 2710: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 2.5830e-07	k_t
Generation 2720: Best = 2.508546e+03 = [4, 4, 4, 2]	FES = 272100	Diversity = 2.2388e-07	k_t
Generation 2730: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.9433e-07	k_t
Generation 2740: Best = 2.508546e+03 = [3, 3, 2, 3]	FES = 274100	Diversity = 1.6963e-07	k_t
Generation 2750: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.4289e-07	k_t
Generation 2760: Best = 2.508546e+03 = [4, 3, 2, 4]	FES = 276100	Diversity = 1.4766e-07	k_t
Generation 2770: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.3651e-07	k_t
Generation 2780: Best = 2.508546e+03 = [4, 3, 4, 3]	FES = 278100	Diversity = 1.2484e-07	k_t
Generation 2790: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.1703e-07	k_t
Generation 2800: Best = 2.508546e+03 = [2, 4, 2, 4]	FES = 280100	Diversity = 1.0925e-07	k_t
Generation 2810: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.0131e-07	k_t
Generation 2820: Best = 2.508546e+03 = [3, 4, 4, 4]	FES = 282100	Diversity = 1.0203e-07	k_t
Generation 2830: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.0031e-07	k_t
Generation 2840: Best = 2.508546e+03 = [4, 2, 4, 3]	FES = 284100	Diversity = 1.0148e-07	k_t
Generation 2850: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 9.2569e-08	k_t
Generation 2860: Best = 2.508546e+03 = [3, 4, 3, 3]	FES = 286100	Diversity = 8.4603e-08	k_t
Generation 2870: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 7.9395e-08	k_t
Generation 2880: Best = 2.508546e+03 = [2, 2, 3, 2]	FES = 288100	Diversity = 7.7824e-08	k_t
Generation 2890: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 7.1646e-08	k_t
Generation 2900: Best = 2.508546e+03 = [4, 2, 2, 4]	FES = 290100	Diversity = 6.9919e-08	k_t
Generation 2910: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 7.5055e-08	k_t
Generation 2920: Best = 2.508546e+03 = [3, 3, 2, 3]	FES = 292100	Diversity = 7.0550e-08	k_t

Generation 2930: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 293100	Diversity = 6.7091e-08	k_t
Generation 2940: Best = 2.508546e+03 = [3, 2, 2, 4]	FES = 294100	Diversity = 6.4530e-08	k_t
Generation 2950: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 295100	Diversity = 6.4532e-08	k_t
Generation 2960: Best = 2.508546e+03 = [4, 2, 2, 2]	FES = 296100	Diversity = 6.4747e-08	k_t
Generation 2970: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 297100	Diversity = 6.4064e-08	k_t
Generation 2980: Best = 2.508546e+03 = [2, 2, 2, 2]	FES = 298100	Diversity = 6.3250e-08	k_t
Generation 2990: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 299100	Diversity = 6.4670e-08	k_t
Generation 3000: Best = 2.508546e+03 = [2, 3, 2, 2]	FES = 300100	Diversity = 6.5658e-08	k_t
Generation 3010: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 301100	Diversity = 6.7534e-08	k_t
Generation 3020: Best = 2.508546e+03 = [4, 4, 2, 2]	FES = 302100	Diversity = 6.6525e-08	k_t
Generation 3030: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 303100	Diversity = 7.0043e-08	k_t
Generation 3040: Best = 2.508546e+03 = [2, 2, 3, 2]	FES = 304100	Diversity = 7.2187e-08	k_t
Generation 3050: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 305100	Diversity = 7.4135e-08	k_t
Generation 3060: Best = 2.508546e+03 = [4, 2, 2, 3]	FES = 306100	Diversity = 7.5707e-08	k_t
Generation 3070: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 307100	Diversity = 7.6243e-08	k_t
Generation 3080: Best = 2.508546e+03 = [3, 3, 2, 4]	FES = 308100	Diversity = 7.9670e-08	k_t
Generation 3090: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 309100	Diversity = 8.1851e-08	k_t
Generation 3100: Best = 2.508546e+03 = [4, 3, 3, 4]	FES = 310100	Diversity = 8.4241e-08	k_t
Generation 3110: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 311100	Diversity = 8.5228e-08	k_t
Generation 3120: Best = 2.508546e+03 = [4, 3, 3, 2]	FES = 312100	Diversity = 8.7386e-08	k_t
Generation 3130: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 313100	Diversity = 8.8770e-08	k_t
Generation 3140: Best = 2.508546e+03 = [3, 2, 3, 4]	FES = 314100	Diversity = 8.8946e-08	k_t
Generation 3150: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 315100	Diversity = 8.9947e-08	k_t
Generation 3160: Best = 2.508546e+03 = [3, 3, 3, 3]	FES = 316100	Diversity = 8.7136e-08	k_t
Generation 3170: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 317100	Diversity = 8.1259e-08	k_t
Generation 3180: Best = 2.508546e+03 = [3, 2, 4, 3]	FES = 318100	Diversity = 7.9246e-08	k_t
Generation 3190: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 319100	Diversity = 7.6179e-08	k_t

Generation 3200: Best = 2.508546e+03 = [2, 2, 2, 4]	FES = 320100	Diversity = 7.4407e-08	k_t
Generation 3210: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 321100	Diversity = 7.4087e-08	k_t
Generation 3220: Best = 2.508546e+03 = [3, 2, 4, 2]	FES = 322100	Diversity = 7.4611e-08	k_t
Generation 3230: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 323100	Diversity = 7.6029e-08	k_t
Generation 3240: Best = 2.508546e+03 = [2, 4, 2, 4]	FES = 324100	Diversity = 7.4786e-08	k_t
Generation 3250: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 325100	Diversity = 7.5644e-08	k_t
Generation 3260: Best = 2.508546e+03 = [4, 2, 2, 4]	FES = 326100	Diversity = 7.5997e-08	k_t
Generation 3270: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 327100	Diversity = 7.4495e-08	k_t
Generation 3280: Best = 2.508546e+03 = [4, 2, 3, 4]	FES = 328100	Diversity = 7.2746e-08	k_t
Generation 3290: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 329100	Diversity = 7.1026e-08	k_t
Generation 3300: Best = 2.508546e+03 = [4, 2, 4, 4]	FES = 330100	Diversity = 6.9694e-08	k_t
Generation 3310: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 331100	Diversity = 7.0218e-08	k_t
Generation 3320: Best = 2.508546e+03 = [2, 2, 2, 2]	FES = 332100	Diversity = 6.9570e-08	k_t
Generation 3330: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 333100	Diversity = 7.0170e-08	k_t
Generation 3340: Best = 2.508546e+03 = [3, 2, 2, 3]	FES = 334100	Diversity = 6.7106e-08	k_t
Generation 3350: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 335100	Diversity = 6.8076e-08	k_t
Generation 3360: Best = 2.508546e+03 = [4, 2, 3, 4]	FES = 336100	Diversity = 6.8006e-08	k_t
Generation 3370: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 337100	Diversity = 6.8518e-08	k_t
Generation 3380: Best = 2.508546e+03 = [2, 2, 3, 2]	FES = 338100	Diversity = 6.6016e-08	k_t
Generation 3390: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 339100	Diversity = 6.2798e-08	k_t
Generation 3400: Best = 2.508546e+03 = [4, 3, 4, 4]	FES = 340100	Diversity = 6.1003e-08	k_t
Generation 3410: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 341100	Diversity = 5.8777e-08	k_t
Generation 3420: Best = 2.508546e+03 = [4, 3, 4, 4]	FES = 342100	Diversity = 5.7145e-08	k_t
Generation 3430: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 343100	Diversity = 5.8042e-08	k_t
Generation 3440: Best = 2.508546e+03 = [3, 2, 2, 2]	FES = 344100	Diversity = 5.7448e-08	k_t
Generation 3450: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 345100	Diversity = 5.7596e-08	k_t
Generation 3460: Best = 2.508546e+03 = [4, 2, 2, 2]	FES = 346100	Diversity = 5.8296e-08	k_t

Generation 3470: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 347100	Diversity = 5.8981e-08	k_t
Generation 3480: Best = 2.508546e+03 = [2, 3, 2, 4]	FES = 348100	Diversity = 5.8499e-08	k_t
Generation 3490: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 349100	Diversity = 5.8085e-08	k_t
Generation 3500: Best = 2.508546e+03 = [3, 4, 3, 4]	FES = 350100	Diversity = 5.8871e-08	k_t
Generation 3510: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 351100	Diversity = 5.7773e-08	k_t
Generation 3520: Best = 2.508546e+03 = [2, 4, 2, 2]	FES = 352100	Diversity = 5.7975e-08	k_t
Generation 3530: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 353100	Diversity = 5.7319e-08	k_t
Generation 3540: Best = 2.508546e+03 = [4, 3, 3, 3]	FES = 354100	Diversity = 5.6657e-08	k_t
Generation 3550: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 355100	Diversity = 5.7491e-08	k_t
Generation 3560: Best = 2.508546e+03 = [3, 2, 4, 4]	FES = 356100	Diversity = 5.7131e-08	k_t
Generation 3570: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 357100	Diversity = 5.6324e-08	k_t
Generation 3580: Best = 2.508546e+03 = [4, 2, 2, 3]	FES = 358100	Diversity = 5.6916e-08	k_t
Generation 3590: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 359100	Diversity = 5.7993e-08	k_t
Generation 3600: Best = 2.508546e+03 = [4, 3, 4, 3]	FES = 360100	Diversity = 5.6912e-08	k_t
Generation 3610: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 361100	Diversity = 5.6986e-08	k_t
Generation 3620: Best = 2.508546e+03 = [2, 3, 2, 3]	FES = 362100	Diversity = 5.7419e-08	k_t
Generation 3630: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 363100	Diversity = 5.8515e-08	k_t
Generation 3640: Best = 2.508546e+03 = [4, 2, 3, 2]	FES = 364100	Diversity = 6.0167e-08	k_t
Generation 3650: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 365100	Diversity = 6.1785e-08	k_t
Generation 3660: Best = 2.508546e+03 = [4, 2, 4, 2]	FES = 366100	Diversity = 6.2365e-08	k_t
Generation 3670: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 367100	Diversity = 6.2156e-08	k_t
Generation 3680: Best = 2.508546e+03 = [4, 3, 3, 4]	FES = 368100	Diversity = 6.2327e-08	k_t
Generation 3690: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 369100	Diversity = 6.4510e-08	k_t
Generation 3700: Best = 2.508546e+03 = [3, 2, 4, 4]	FES = 370100	Diversity = 6.3736e-08	k_t
Generation 3710: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 371100	Diversity = 6.5224e-08	k_t
Generation 3720: Best = 2.508546e+03 = [2, 4, 4, 4]	FES = 372100	Diversity = 6.6348e-08	k_t
Generation 3730: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 373100	Diversity = 6.5288e-08	k_t

Generation 3740: Best = 2.508546e+03 = [2, 4, 3, 4]	FES = 374100	Diversity = 6.5332e-08	k_t
Generation 3750: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 375100	Diversity = 6.6132e-08	k_t
Generation 3760: Best = 2.508546e+03 = [3, 4, 4, 4]	FES = 376100	Diversity = 6.6697e-08	k_t
Generation 3770: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 377100	Diversity = 6.7081e-08	k_t
Generation 3780: Best = 2.508546e+03 = [3, 4, 3, 4]	FES = 378100	Diversity = 6.7622e-08	k_t
Generation 3790: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 379100	Diversity = 6.7169e-08	k_t
Generation 3800: Best = 2.508546e+03 = [2, 3, 3, 3]	FES = 380100	Diversity = 6.6120e-08	k_t
Generation 3810: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 381100	Diversity = 6.5565e-08	k_t
Generation 3820: Best = 2.508546e+03 = [3, 2, 4, 4]	FES = 382100	Diversity = 6.5113e-08	k_t
Generation 3830: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 383100	Diversity = 6.5620e-08	k_t
Generation 3840: Best = 2.508546e+03 = [4, 3, 4, 2]	FES = 384100	Diversity = 6.5716e-08	k_t
Generation 3850: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 385100	Diversity = 6.5454e-08	k_t
Generation 3860: Best = 2.508546e+03 = [3, 3, 3, 2]	FES = 386100	Diversity = 6.5386e-08	k_t
Generation 3870: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 387100	Diversity = 6.4267e-08	k_t
Generation 3880: Best = 2.508546e+03 = [4, 4, 4, 3]	FES = 388100	Diversity = 6.4668e-08	k_t
Generation 3890: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 389100	Diversity = 6.5134e-08	k_t
Generation 3900: Best = 2.508546e+03 = [2, 2, 3, 2]	FES = 390100	Diversity = 6.5104e-08	k_t
Generation 3910: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 391100	Diversity = 6.5934e-08	k_t
Generation 3920: Best = 2.508546e+03 = [4, 4, 3, 4]	FES = 392100	Diversity = 6.6132e-08	k_t
Generation 3930: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 393100	Diversity = 6.6668e-08	k_t
Generation 3940: Best = 2.508546e+03 = [3, 2, 4, 2]	FES = 394100	Diversity = 6.6657e-08	k_t
Generation 3950: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 395100	Diversity = 6.7139e-08	k_t
Generation 3960: Best = 2.508546e+03 = [4, 3, 2, 3]	FES = 396100	Diversity = 6.6773e-08	k_t
Generation 3970: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 397100	Diversity = 6.8845e-08	k_t
Generation 3980: Best = 2.508546e+03 = [3, 2, 3, 4]	FES = 398100	Diversity = 6.7366e-08	k_t
Generation 3990: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 399100	Diversity = 6.6802e-08	k_t
Generation 4000: Best = 2.508546e+03 = [2, 2, 4, 3]	FES = 400100	Diversity = 6.7203e-08	k_t

Generation 4010: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 401100	Diversity = 6.4372e-08	k_t
Generation 4020: Best = 2.508546e+03 = [3, 2, 2, 4]	FES = 402100	Diversity = 6.4220e-08	k_t
Generation 4030: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 403100	Diversity = 6.4255e-08	k_t
Generation 4040: Best = 2.508546e+03 = [2, 2, 2, 2]	FES = 404100	Diversity = 6.3766e-08	k_t
Generation 4050: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 405100	Diversity = 6.2611e-08	k_t
Generation 4060: Best = 2.508546e+03 = [3, 4, 4, 3]	FES = 406100	Diversity = 6.1285e-08	k_t
Generation 4070: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 407100	Diversity = 6.1189e-08	k_t
Generation 4080: Best = 2.508546e+03 = [4, 2, 4, 3]	FES = 408100	Diversity = 6.2343e-08	k_t
Generation 4090: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 409100	Diversity = 6.2618e-08	k_t
Generation 4100: Best = 2.508546e+03 = [3, 3, 2, 4]	FES = 410100	Diversity = 6.3024e-08	k_t
Generation 4110: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 411100	Diversity = 6.2422e-08	k_t
Generation 4120: Best = 2.508546e+03 = [3, 4, 3, 4]	FES = 412100	Diversity = 6.1804e-08	k_t
Generation 4130: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 413100	Diversity = 6.2001e-08	k_t
Generation 4140: Best = 2.508546e+03 = [3, 3, 2, 2]	FES = 414100	Diversity = 6.3367e-08	k_t
Generation 4150: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 415100	Diversity = 6.3216e-08	k_t
Generation 4160: Best = 2.508546e+03 = [2, 3, 2, 3]	FES = 416100	Diversity = 6.3839e-08	k_t
Generation 4170: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 417100	Diversity = 6.3681e-08	k_t
Generation 4180: Best = 2.508546e+03 = [2, 3, 4, 3]	FES = 418100	Diversity = 6.3955e-08	k_t
Generation 4190: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 419100	Diversity = 6.4352e-08	k_t
Generation 4200: Best = 2.508546e+03 = [3, 2, 4, 3]	FES = 420100	Diversity = 6.5757e-08	k_t
Generation 4210: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 421100	Diversity = 6.4874e-08	k_t
Generation 4220: Best = 2.508546e+03 = [3, 2, 3, 4]	FES = 422100	Diversity = 6.3731e-08	k_t
Generation 4230: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 423100	Diversity = 6.4316e-08	k_t
Generation 4240: Best = 2.508546e+03 = [2, 2, 2, 2]	FES = 424100	Diversity = 6.4464e-08	k_t
Generation 4250: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 425100	Diversity = 6.2744e-08	k_t
Generation 4260: Best = 2.508546e+03 = [4, 4, 3, 3]	FES = 426100	Diversity = 6.3324e-08	k_t
Generation 4270: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 427100	Diversity = 6.3798e-08	k_t

Generation 4280: Best = 2.508546e+03 = [3, 2, 4, 2]	FES = 428100	Diversity = 6.5049e-08	k_t
Generation 4290: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 429100	Diversity = 6.5495e-08	k_t
Generation 4300: Best = 2.508546e+03 = [2, 2, 4, 3]	FES = 430100	Diversity = 6.5511e-08	k_t
Generation 4310: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 431100	Diversity = 6.5384e-08	k_t
Generation 4320: Best = 2.508546e+03 = [2, 3, 4, 3]	FES = 432100	Diversity = 6.7317e-08	k_t
Generation 4330: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 433100	Diversity = 6.9845e-08	k_t
Generation 4340: Best = 2.508546e+03 = [2, 2, 4, 2]	FES = 434100	Diversity = 7.0054e-08	k_t
Generation 4350: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 435100	Diversity = 6.9876e-08	k_t
Generation 4360: Best = 2.508546e+03 = [3, 2, 2, 4]	FES = 436100	Diversity = 6.9521e-08	k_t
Generation 4370: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 437100	Diversity = 6.8972e-08	k_t
Generation 4380: Best = 2.508546e+03 = [4, 4, 2, 4]	FES = 438100	Diversity = 6.9112e-08	k_t
Generation 4390: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 439100	Diversity = 6.8624e-08	k_t
Generation 4400: Best = 2.508546e+03 = [2, 4, 3, 4]	FES = 440100	Diversity = 6.9015e-08	k_t
Generation 4410: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 441100	Diversity = 6.9330e-08	k_t
Generation 4420: Best = 2.508546e+03 = [4, 3, 4, 3]	FES = 442100	Diversity = 6.8875e-08	k_t
Generation 4430: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 443100	Diversity = 6.6771e-08	k_t
Generation 4440: Best = 2.508546e+03 = [2, 2, 4, 3]	FES = 444100	Diversity = 6.6712e-08	k_t
Generation 4450: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 445100	Diversity = 6.6671e-08	k_t
Generation 4460: Best = 2.508546e+03 = [3, 4, 4, 2]	FES = 446100	Diversity = 6.6777e-08	k_t
Generation 4470: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 447100	Diversity = 6.7037e-08	k_t
Generation 4480: Best = 2.508546e+03 = [2, 3, 2, 4]	FES = 448100	Diversity = 6.6895e-08	k_t
Generation 4490: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 449100	Diversity = 6.6836e-08	k_t
Generation 4500: Best = 2.508546e+03 = [4, 4, 3, 3]	FES = 450100	Diversity = 6.6238e-08	k_t
Generation 4510: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 451100	Diversity = 6.5901e-08	k_t
Generation 4520: Best = 2.508546e+03 = [4, 4, 2, 2]	FES = 452100	Diversity = 6.6603e-08	k_t
Generation 4530: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 453100	Diversity = 6.7559e-08	k_t
Generation 4540: Best = 2.508546e+03 = [4, 4, 4, 4]	FES = 454100	Diversity = 6.8423e-08	k_t

Generation 4550: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 455100	Diversity = 6.8399e-08	k_t
Generation 4560: Best = 2.508546e+03 = [2, 3, 3, 2]	FES = 456100	Diversity = 6.8948e-08	k_t
Generation 4570: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 457100	Diversity = 6.7823e-08	k_t
Generation 4580: Best = 2.508546e+03 = [4, 4, 2, 4]	FES = 458100	Diversity = 6.8812e-08	k_t
Generation 4590: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 459100	Diversity = 6.7475e-08	k_t
Generation 4600: Best = 2.508546e+03 = [2, 4, 3, 3]	FES = 460100	Diversity = 6.7295e-08	k_t
Generation 4610: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 461100	Diversity = 6.6605e-08	k_t
Generation 4620: Best = 2.508546e+03 = [2, 2, 4, 4]	FES = 462100	Diversity = 6.6142e-08	k_t
Generation 4630: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 463100	Diversity = 6.5756e-08	k_t
Generation 4640: Best = 2.508546e+03 = [3, 3, 4, 3]	FES = 464100	Diversity = 6.5586e-08	k_t
Generation 4650: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 465100	Diversity = 6.5364e-08	k_t
Generation 4660: Best = 2.508546e+03 = [3, 4, 3, 3]	FES = 466100	Diversity = 6.6541e-08	k_t
Generation 4670: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 467100	Diversity = 6.7560e-08	k_t
Generation 4680: Best = 2.508546e+03 = [4, 2, 2, 2]	FES = 468100	Diversity = 6.5957e-08	k_t
Generation 4690: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 469100	Diversity = 6.7271e-08	k_t
Generation 4700: Best = 2.508546e+03 = [4, 2, 2, 3]	FES = 470100	Diversity = 6.7550e-08	k_t
Generation 4710: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 471100	Diversity = 6.8209e-08	k_t
Generation 4720: Best = 2.508546e+03 = [3, 4, 4, 3]	FES = 472100	Diversity = 6.8346e-08	k_t
Generation 4730: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 473100	Diversity = 6.9797e-08	k_t
Generation 4740: Best = 2.508546e+03 = [3, 4, 3, 2]	FES = 474100	Diversity = 6.9100e-08	k_t
Generation 4750: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 475100	Diversity = 6.9673e-08	k_t
Generation 4760: Best = 2.508546e+03 = [3, 2, 3, 3]	FES = 476100	Diversity = 7.0080e-08	k_t
Generation 4770: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 477100	Diversity = 7.0067e-08	k_t
Generation 4780: Best = 2.508546e+03 = [4, 4, 4, 3]	FES = 478100	Diversity = 6.9844e-08	k_t
Generation 4790: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 479100	Diversity = 6.9445e-08	k_t
Generation 4800: Best = 2.508546e+03 = [4, 4, 2, 3]	FES = 480100	Diversity = 6.9641e-08	k_t
Generation 4810: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 481100	Diversity = 6.8240e-08	k_t

Generation 4820: Best = 2.508546e+03 = [2, 3, 2, 3]	FES = 482100	Diversity = 6.7283e-08	k_t
Generation 4830: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 483100	Diversity = 6.6500e-08	k_t
Generation 4840: Best = 2.508546e+03 = [3, 2, 4, 4]	FES = 484100	Diversity = 6.7021e-08	k_t
Generation 4850: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 485100	Diversity = 6.5667e-08	k_t
Generation 4860: Best = 2.508546e+03 = [4, 2, 2, 2]	FES = 486100	Diversity = 6.4855e-08	k_t
Generation 4870: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 487100	Diversity = 6.4663e-08	k_t
Generation 4880: Best = 2.508546e+03 = [3, 2, 2, 3]	FES = 488100	Diversity = 6.4456e-08	k_t
Generation 4890: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 489100	Diversity = 6.4116e-08	k_t
Generation 4900: Best = 2.508546e+03 = [3, 4, 4, 2]	FES = 490100	Diversity = 6.4191e-08	k_t
Generation 4910: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 491100	Diversity = 6.4712e-08	k_t
Generation 4920: Best = 2.508546e+03 = [2, 3, 3, 4]	FES = 492100	Diversity = 6.5044e-08	k_t
Generation 4930: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 493100	Diversity = 6.6457e-08	k_t
Generation 4940: Best = 2.508546e+03 = [3, 3, 4, 2]	FES = 494100	Diversity = 6.7194e-08	k_t
Generation 4950: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 495100	Diversity = 6.7001e-08	k_t
Generation 4960: Best = 2.508546e+03 = [2, 4, 4, 3]	FES = 496100	Diversity = 6.5988e-08	k_t
Generation 4970: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 497100	Diversity = 6.6321e-08	k_t
Generation 4980: Best = 2.508546e+03 = [2, 2, 4, 2]	FES = 498100	Diversity = 6.6456e-08	k_t
Generation 4990: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 499100	Diversity = 6.7107e-08	k_t
Generation 5000: Best = 2.508546e+03 = [4, 3, 3, 2]	FES = 500100	Diversity = 6.6528e-08	k_t
Generation 5010: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 501100	Diversity = 6.4698e-08	k_t
Generation 5020: Best = 2.508546e+03 = [2, 2, 4, 3]	FES = 502100	Diversity = 6.4018e-08	k_t
Generation 5030: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 503100	Diversity = 6.3394e-08	k_t
Generation 5040: Best = 2.508546e+03 = [2, 2, 4, 3]	FES = 504100	Diversity = 6.3834e-08	k_t
Generation 5050: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 505100	Diversity = 6.3266e-08	k_t
Generation 5060: Best = 2.508546e+03 = [3, 3, 3, 4]	FES = 506100	Diversity = 6.2989e-08	k_t
Generation 5070: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 507100	Diversity = 6.2550e-08	k_t
Generation 5080: Best = 2.508546e+03 = [4, 2, 2, 2]	FES = 508100	Diversity = 6.3032e-08	k_t

Generation 5090: Best = 2.508546e+03	FES = 509100	Diversity = 6.2522e-08	k_t
= [1, 1, 1, 1]			
Generation 5100: Best = 2.508546e+03	FES = 510100	Diversity = 6.2722e-08	k_t
= [4, 4, 4, 4]			
Generation 5110: Best = 2.508546e+03	FES = 511100	Diversity = 6.1873e-08	k_t
= [1, 1, 1, 1]			
Generation 5120: Best = 2.508546e+03	FES = 512100	Diversity = 6.2556e-08	k_t
= [4, 2, 2, 4]			
Generation 5130: Best = 2.508546e+03	FES = 513100	Diversity = 6.1441e-08	k_t
= [1, 1, 1, 1]			
Generation 5140: Best = 2.508546e+03	FES = 514100	Diversity = 6.3946e-08	k_t
= [3, 4, 4, 3]			
Generation 5150: Best = 2.508546e+03	FES = 515100	Diversity = 6.3717e-08	k_t
= [1, 1, 1, 1]			
Generation 5160: Best = 2.508546e+03	FES = 516100	Diversity = 6.3797e-08	k_t
= [4, 4, 2, 2]			
Generation 5170: Best = 2.508546e+03	FES = 517100	Diversity = 6.2620e-08	k_t
= [1, 1, 1, 1]			
Generation 5180: Best = 2.508546e+03	FES = 518100	Diversity = 6.4006e-08	k_t
= [3, 3, 4, 3]			
Generation 5190: Best = 2.508546e+03	FES = 519100	Diversity = 6.3695e-08	k_t
= [1, 1, 1, 1]			
Generation 5200: Best = 2.508546e+03	FES = 520100	Diversity = 6.3741e-08	k_t
= [4, 2, 3, 2]			
Generation 5210: Best = 2.508546e+03	FES = 521100	Diversity = 6.3579e-08	k_t
= [1, 1, 1, 1]			
Generation 5220: Best = 2.508546e+03	FES = 522100	Diversity = 6.2909e-08	k_t
= [3, 4, 4, 2]			
Generation 5230: Best = 2.508546e+03	FES = 523100	Diversity = 6.3034e-08	k_t
= [1, 1, 1, 1]			
Generation 5240: Best = 2.508546e+03	FES = 524100	Diversity = 6.0785e-08	k_t
= [4, 2, 3, 4]			
Generation 5250: Best = 2.508546e+03	FES = 525100	Diversity = 6.2230e-08	k_t
= [1, 1, 1, 1]			
Generation 5260: Best = 2.508546e+03	FES = 526100	Diversity = 6.2734e-08	k_t
= [2, 4, 2, 4]			
Generation 5270: Best = 2.508546e+03	FES = 527100	Diversity = 6.3862e-08	k_t
= [1, 1, 1, 1]			
Generation 5280: Best = 2.508546e+03	FES = 528100	Diversity = 6.4750e-08	k_t
= [2, 3, 2, 4]			
Generation 5290: Best = 2.508546e+03	FES = 529100	Diversity = 6.5470e-08	k_t
= [1, 1, 1, 1]			
Generation 5300: Best = 2.508546e+03	FES = 530100	Diversity = 6.5974e-08	k_t
= [2, 4, 2, 2]			
Generation 5310: Best = 2.508546e+03	FES = 531100	Diversity = 6.4809e-08	k_t
= [1, 1, 1, 1]			
Generation 5320: Best = 2.508546e+03	FES = 532100	Diversity = 6.5085e-08	k_t
= [3, 4, 4, 4]			
Generation 5330: Best = 2.508546e+03	FES = 533100	Diversity = 6.5059e-08	k_t
= [1, 1, 1, 1]			
Generation 5340: Best = 2.508546e+03	FES = 534100	Diversity = 6.4734e-08	k_t
= [2, 4, 4, 2]			
Generation 5350: Best = 2.508546e+03	FES = 535100	Diversity = 6.4850e-08	k_t
= [1, 1, 1, 1]			

Generation 5360: Best = 2.508546e+03 = [2, 2, 3, 4]	FES = 536100	Diversity = 6.4852e-08	k_t
Generation 5370: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 537100	Diversity = 6.2929e-08	k_t
Generation 5380: Best = 2.508546e+03 = [2, 3, 3, 2]	FES = 538100	Diversity = 6.3746e-08	k_t
Generation 5390: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 539100	Diversity = 6.4640e-08	k_t
Generation 5400: Best = 2.508546e+03 = [2, 3, 2, 3]	FES = 540100	Diversity = 6.4985e-08	k_t
Generation 5410: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 541100	Diversity = 6.5260e-08	k_t
Generation 5420: Best = 2.508546e+03 = [4, 2, 3, 3]	FES = 542100	Diversity = 6.4900e-08	k_t
Generation 5430: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 543100	Diversity = 6.4588e-08	k_t
Generation 5440: Best = 2.508546e+03 = [2, 4, 4, 2]	FES = 544100	Diversity = 6.4153e-08	k_t
Generation 5450: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 545100	Diversity = 6.4518e-08	k_t
Generation 5460: Best = 2.508546e+03 = [3, 3, 3, 3]	FES = 546100	Diversity = 6.6035e-08	k_t
Generation 5470: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 547100	Diversity = 6.6445e-08	k_t
Generation 5480: Best = 2.508546e+03 = [3, 2, 2, 4]	FES = 548100	Diversity = 6.6075e-08	k_t
Generation 5490: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 549100	Diversity = 6.7179e-08	k_t
Generation 5500: Best = 2.508546e+03 = [3, 4, 3, 4]	FES = 550100	Diversity = 6.6318e-08	k_t
Generation 5510: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 551100	Diversity = 6.6071e-08	k_t
Generation 5520: Best = 2.508546e+03 = [2, 4, 3, 4]	FES = 552100	Diversity = 6.6940e-08	k_t
Generation 5530: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 553100	Diversity = 6.5820e-08	k_t
Generation 5540: Best = 2.508546e+03 = [3, 2, 4, 2]	FES = 554100	Diversity = 6.4740e-08	k_t
Generation 5550: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 555100	Diversity = 6.4770e-08	k_t
Generation 5560: Best = 2.508546e+03 = [3, 3, 2, 4]	FES = 556100	Diversity = 6.5824e-08	k_t
Generation 5570: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 557100	Diversity = 6.7409e-08	k_t
Generation 5580: Best = 2.508546e+03 = [3, 4, 4, 2]	FES = 558100	Diversity = 6.7858e-08	k_t
Generation 5590: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 559100	Diversity = 6.8895e-08	k_t
Generation 5600: Best = 2.508546e+03 = [2, 3, 3, 2]	FES = 560100	Diversity = 6.8141e-08	k_t
Generation 5610: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 561100	Diversity = 6.8044e-08	k_t
Generation 5620: Best = 2.508546e+03 = [3, 3, 4, 3]	FES = 562100	Diversity = 6.8487e-08	k_t

Generation 5630: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 563100	Diversity = 6.9168e-08	k_t
Generation 5640: Best = 2.508546e+03 = [2, 4, 2, 4]	FES = 564100	Diversity = 6.8188e-08	k_t
Generation 5650: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 565100	Diversity = 6.8188e-08	k_t
Generation 5660: Best = 2.508546e+03 = [4, 4, 3, 4]	FES = 566100	Diversity = 6.8560e-08	k_t
Generation 5670: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 567100	Diversity = 6.7348e-08	k_t
Generation 5680: Best = 2.508546e+03 = [2, 4, 4, 4]	FES = 568100	Diversity = 6.7859e-08	k_t
Generation 5690: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 569100	Diversity = 6.9071e-08	k_t
Generation 5700: Best = 2.508546e+03 = [3, 4, 2, 4]	FES = 570100	Diversity = 6.9543e-08	k_t
Generation 5710: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 571100	Diversity = 6.9491e-08	k_t
Generation 5720: Best = 2.508546e+03 = [2, 4, 2, 3]	FES = 572100	Diversity = 7.0417e-08	k_t
Generation 5730: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 573100	Diversity = 7.0089e-08	k_t
Generation 5740: Best = 2.508546e+03 = [4, 2, 3, 4]	FES = 574100	Diversity = 6.9770e-08	k_t
Generation 5750: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 575100	Diversity = 6.8989e-08	k_t
Generation 5760: Best = 2.508546e+03 = [2, 4, 3, 2]	FES = 576100	Diversity = 6.8604e-08	k_t
Generation 5770: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 577100	Diversity = 6.7320e-08	k_t
Generation 5780: Best = 2.508546e+03 = [2, 4, 2, 4]	FES = 578100	Diversity = 6.6349e-08	k_t
Generation 5790: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 579100	Diversity = 6.5750e-08	k_t
Generation 5800: Best = 2.508546e+03 = [3, 3, 4, 2]	FES = 580100	Diversity = 6.4704e-08	k_t
Generation 5810: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 581100	Diversity = 6.4514e-08	k_t
Generation 5820: Best = 2.508546e+03 = [2, 3, 3, 3]	FES = 582100	Diversity = 6.3237e-08	k_t
Generation 5830: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 583100	Diversity = 6.4167e-08	k_t
Generation 5840: Best = 2.508546e+03 = [3, 4, 4, 2]	FES = 584100	Diversity = 6.3938e-08	k_t
Generation 5850: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 585100	Diversity = 6.4596e-08	k_t
Generation 5860: Best = 2.508546e+03 = [3, 4, 2, 4]	FES = 586100	Diversity = 6.4311e-08	k_t
Generation 5870: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 587100	Diversity = 6.5400e-08	k_t
Generation 5880: Best = 2.508546e+03 = [2, 2, 4, 2]	FES = 588100	Diversity = 6.5740e-08	k_t
Generation 5890: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 589100	Diversity = 6.6422e-08	k_t

Generation 5900: Best = 2.508546e+03 = [3, 4, 3, 2]	FES = 590100	Diversity = 6.6229e-08	k_t
Generation 5910: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 591100	Diversity = 6.7313e-08	k_t
Generation 5920: Best = 2.508546e+03 = [2, 2, 4, 4]	FES = 592100	Diversity = 6.7288e-08	k_t
Generation 5930: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 593100	Diversity = 6.7761e-08	k_t
Generation 5940: Best = 2.508546e+03 = [3, 2, 4, 3]	FES = 594100	Diversity = 6.8373e-08	k_t
Generation 5950: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 595100	Diversity = 6.6748e-08	k_t
Generation 5960: Best = 2.508546e+03 = [3, 2, 2, 4]	FES = 596100	Diversity = 6.6344e-08	k_t
Generation 5970: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 597100	Diversity = 6.5435e-08	k_t
Generation 5980: Best = 2.508546e+03 = [4, 4, 4, 3]	FES = 598100	Diversity = 6.4633e-08	k_t
Generation 5990: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 599100	Diversity = 6.3517e-08	k_t
Generation 6000: Best = 2.508546e+03 = [3, 4, 2, 3]	FES = 600100	Diversity = 6.2773e-08	k_t
Generation 6010: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 601100	Diversity = 6.3121e-08	k_t
Generation 6020: Best = 2.508546e+03 = [4, 2, 2, 2]	FES = 602100	Diversity = 6.2311e-08	k_t
Generation 6030: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 603100	Diversity = 6.2155e-08	k_t
Generation 6040: Best = 2.508546e+03 = [4, 4, 2, 2]	FES = 604100	Diversity = 6.2389e-08	k_t
Generation 6050: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 605100	Diversity = 6.1554e-08	k_t
Generation 6060: Best = 2.508546e+03 = [4, 2, 3, 3]	FES = 606100	Diversity = 6.1833e-08	k_t
Generation 6070: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 607100	Diversity = 6.2188e-08	k_t
Generation 6080: Best = 2.508546e+03 = [2, 4, 2, 3]	FES = 608100	Diversity = 6.1740e-08	k_t
Generation 6090: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 609100	Diversity = 6.1866e-08	k_t
Generation 6100: Best = 2.508546e+03 = [4, 2, 2, 4]	FES = 610100	Diversity = 6.3816e-08	k_t
Generation 6110: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 611100	Diversity = 6.4239e-08	k_t
Generation 6120: Best = 2.508546e+03 = [2, 2, 3, 2]	FES = 612100	Diversity = 6.4426e-08	k_t
Generation 6130: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 613100	Diversity = 6.4813e-08	k_t
Generation 6140: Best = 2.508546e+03 = [3, 3, 3, 2]	FES = 614100	Diversity = 6.3247e-08	k_t
Generation 6150: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 615100	Diversity = 6.3499e-08	k_t
Generation 6160: Best = 2.508546e+03 = [2, 3, 2, 2]	FES = 616100	Diversity = 6.3570e-08	k_t

Generation 6170: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 617100	Diversity = 6.2733e-08	k_t
Generation 6180: Best = 2.508546e+03 = [4, 2, 3, 4]	FES = 618100	Diversity = 6.2091e-08	k_t
Generation 6190: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 619100	Diversity = 6.1965e-08	k_t
Generation 6200: Best = 2.508546e+03 = [4, 2, 3, 4]	FES = 620100	Diversity = 6.2304e-08	k_t
Generation 6210: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 621100	Diversity = 6.1152e-08	k_t
Generation 6220: Best = 2.508546e+03 = [4, 2, 3, 3]	FES = 622100	Diversity = 6.0873e-08	k_t
Generation 6230: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 623100	Diversity = 6.2066e-08	k_t
Generation 6240: Best = 2.508546e+03 = [3, 2, 4, 3]	FES = 624100	Diversity = 6.1197e-08	k_t
Generation 6250: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 625100	Diversity = 6.0991e-08	k_t
Generation 6260: Best = 2.508546e+03 = [2, 4, 3, 3]	FES = 626100	Diversity = 5.9686e-08	k_t
Generation 6270: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 627100	Diversity = 5.9964e-08	k_t
Generation 6280: Best = 2.508546e+03 = [3, 2, 4, 3]	FES = 628100	Diversity = 6.0100e-08	k_t
Generation 6290: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 629100	Diversity = 5.9688e-08	k_t
Generation 6300: Best = 2.508546e+03 = [4, 2, 4, 4]	FES = 630100	Diversity = 5.8655e-08	k_t
Generation 6310: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 631100	Diversity = 5.9158e-08	k_t
Generation 6320: Best = 2.508546e+03 = [2, 3, 4, 2]	FES = 632100	Diversity = 5.9453e-08	k_t
Generation 6330: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 633100	Diversity = 6.1719e-08	k_t
Generation 6340: Best = 2.508546e+03 = [2, 2, 3, 3]	FES = 634100	Diversity = 6.1571e-08	k_t
Generation 6350: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 635100	Diversity = 6.0744e-08	k_t
Generation 6360: Best = 2.508546e+03 = [3, 4, 4, 2]	FES = 636100	Diversity = 6.0935e-08	k_t
Generation 6370: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 637100	Diversity = 5.9989e-08	k_t
Generation 6380: Best = 2.508546e+03 = [4, 4, 4, 4]	FES = 638100	Diversity = 5.9243e-08	k_t
Generation 6390: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 639100	Diversity = 6.0375e-08	k_t
Generation 6400: Best = 2.508546e+03 = [2, 2, 3, 4]	FES = 640100	Diversity = 6.0738e-08	k_t
Generation 6410: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 641100	Diversity = 6.0061e-08	k_t
Generation 6420: Best = 2.508546e+03 = [2, 3, 4, 3]	FES = 642100	Diversity = 6.0348e-08	k_t
Generation 6430: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 643100	Diversity = 6.1457e-08	k_t

Generation 6440: Best = 2.508546e+03 = [4, 3, 3, 3]	FES = 644100	Diversity = 6.0952e-08	k_t
Generation 6450: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 645100	Diversity = 6.1770e-08	k_t
Generation 6460: Best = 2.508546e+03 = [2, 3, 4, 2]	FES = 646100	Diversity = 6.0930e-08	k_t
Generation 6470: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 647100	Diversity = 6.0281e-08	k_t
Generation 6480: Best = 2.508546e+03 = [2, 2, 4, 3]	FES = 648100	Diversity = 6.0033e-08	k_t
Generation 6490: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 649100	Diversity = 5.9590e-08	k_t
Generation 6500: Best = 2.508546e+03 = [4, 3, 2, 4]	FES = 650100	Diversity = 5.9874e-08	k_t
Generation 6510: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 651100	Diversity = 6.0561e-08	k_t
Generation 6520: Best = 2.508546e+03 = [3, 3, 2, 2]	FES = 652100	Diversity = 6.1292e-08	k_t
Generation 6530: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 653100	Diversity = 6.1139e-08	k_t
Generation 6540: Best = 2.508546e+03 = [4, 4, 2, 2]	FES = 654100	Diversity = 6.2435e-08	k_t
Generation 6550: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 655100	Diversity = 6.1791e-08	k_t
Generation 6560: Best = 2.508546e+03 = [4, 2, 4, 3]	FES = 656100	Diversity = 6.3145e-08	k_t
Generation 6570: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 657100	Diversity = 6.3679e-08	k_t
Generation 6580: Best = 2.508546e+03 = [3, 4, 3, 4]	FES = 658100	Diversity = 6.3499e-08	k_t
Generation 6590: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 659100	Diversity = 6.3840e-08	k_t
Generation 6600: Best = 2.508546e+03 = [4, 2, 2, 2]	FES = 660100	Diversity = 6.4097e-08	k_t
Generation 6610: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 661100	Diversity = 6.4792e-08	k_t
Generation 6620: Best = 2.508546e+03 = [2, 2, 3, 2]	FES = 662100	Diversity = 6.5372e-08	k_t
Generation 6630: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 663100	Diversity = 6.4498e-08	k_t
Generation 6640: Best = 2.508546e+03 = [2, 3, 2, 4]	FES = 664100	Diversity = 6.4679e-08	k_t
Generation 6650: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 665100	Diversity = 6.4361e-08	k_t
Generation 6660: Best = 2.508546e+03 = [2, 2, 4, 2]	FES = 666100	Diversity = 6.6396e-08	k_t
Generation 6670: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 667100	Diversity = 6.6869e-08	k_t
Generation 6680: Best = 2.508546e+03 = [3, 2, 3, 4]	FES = 668100	Diversity = 6.6375e-08	k_t
Generation 6690: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 669100	Diversity = 6.5309e-08	k_t
Generation 6700: Best = 2.508546e+03 = [3, 2, 2, 4]	FES = 670100	Diversity = 6.4493e-08	k_t

Generation 6710: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 671100	Diversity = 6.4382e-08	k_t
Generation 6720: Best = 2.508546e+03 = [2, 3, 4, 3]	FES = 672100	Diversity = 6.4810e-08	k_t
Generation 6730: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 673100	Diversity = 6.4504e-08	k_t
Generation 6740: Best = 2.508546e+03 = [2, 4, 3, 4]	FES = 674100	Diversity = 6.3875e-08	k_t
Generation 6750: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 675100	Diversity = 6.3656e-08	k_t
Generation 6760: Best = 2.508546e+03 = [3, 3, 2, 4]	FES = 676100	Diversity = 6.4403e-08	k_t
Generation 6770: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 677100	Diversity = 6.3015e-08	k_t
Generation 6780: Best = 2.508546e+03 = [3, 4, 4, 2]	FES = 678100	Diversity = 6.3463e-08	k_t
Generation 6790: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 679100	Diversity = 6.3954e-08	k_t
Generation 6800: Best = 2.508546e+03 = [3, 4, 2, 4]	FES = 680100	Diversity = 6.6071e-08	k_t
Generation 6810: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 681100	Diversity = 6.7044e-08	k_t
Generation 6820: Best = 2.508546e+03 = [3, 3, 3, 3]	FES = 682100	Diversity = 6.7116e-08	k_t
Generation 6830: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 683100	Diversity = 6.7877e-08	k_t
Generation 6840: Best = 2.508546e+03 = [4, 2, 4, 2]	FES = 684100	Diversity = 6.8142e-08	k_t
Generation 6850: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 685100	Diversity = 6.7312e-08	k_t
Generation 6860: Best = 2.508546e+03 = [2, 2, 4, 2]	FES = 686100	Diversity = 6.6428e-08	k_t
Generation 6870: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 687100	Diversity = 6.6078e-08	k_t
Generation 6880: Best = 2.508546e+03 = [3, 4, 4, 4]	FES = 688100	Diversity = 6.6249e-08	k_t
Generation 6890: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 689100	Diversity = 6.5584e-08	k_t
Generation 6900: Best = 2.508546e+03 = [2, 4, 3, 2]	FES = 690100	Diversity = 6.5386e-08	k_t
Generation 6910: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 691100	Diversity = 6.2900e-08	k_t
Generation 6920: Best = 2.508546e+03 = [2, 2, 2, 3]	FES = 692100	Diversity = 6.3305e-08	k_t
Generation 6930: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 693100	Diversity = 6.2448e-08	k_t
Generation 6940: Best = 2.508546e+03 = [2, 2, 4, 3]	FES = 694100	Diversity = 6.1847e-08	k_t
Generation 6950: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 695100	Diversity = 6.3099e-08	k_t
Generation 6960: Best = 2.508546e+03 = [3, 2, 2, 2]	FES = 696100	Diversity = 6.3630e-08	k_t
Generation 6970: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 697100	Diversity = 6.3169e-08	k_t

Generation 6980: Best = 2.508546e+03 = [4, 2, 4, 4]	FES = 698100	Diversity = 6.4684e-08	k_t
Generation 6990: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 6.3936e-08	k_t
Generation 7000: Best = 2.508546e+03 = [4, 4, 2, 2]	FES = 700100	Diversity = 6.3804e-08	k_t
Generation 7010: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 6.3273e-08	k_t
Generation 7020: Best = 2.508546e+03 = [2, 2, 2, 2]	FES = 702100	Diversity = 6.5179e-08	k_t
Generation 7030: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 6.2895e-08	k_t
Generation 7040: Best = 2.508546e+03 = [3, 3, 3, 3]	FES = 704100	Diversity = 6.3742e-08	k_t
Generation 7050: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 6.3484e-08	k_t
Generation 7060: Best = 2.508546e+03 = [2, 2, 4, 4]	FES = 706100	Diversity = 6.2916e-08	k_t
Generation 7070: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 6.2822e-08	k_t
Generation 7080: Best = 2.508546e+03 = [3, 3, 2, 2]	FES = 708100	Diversity = 6.2066e-08	k_t
Generation 7090: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 6.1933e-08	k_t
Generation 7100: Best = 2.508546e+03 = [3, 3, 2, 3]	FES = 710100	Diversity = 6.2027e-08	k_t
Generation 7110: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 6.2545e-08	k_t
Generation 7120: Best = 2.508546e+03 = [4, 4, 2, 3]	FES = 712100	Diversity = 6.2413e-08	k_t
Generation 7130: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 6.3231e-08	k_t
Generation 7140: Best = 2.508546e+03 = [3, 2, 3, 2]	FES = 714100	Diversity = 6.3284e-08	k_t
Generation 7150: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 6.3361e-08	k_t
Generation 7160: Best = 2.508546e+03 = [4, 2, 4, 4]	FES = 716100	Diversity = 6.4819e-08	k_t
Generation 7170: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 6.6548e-08	k_t
Generation 7180: Best = 2.508546e+03 = [2, 3, 4, 3]	FES = 718100	Diversity = 6.4600e-08	k_t
Generation 7190: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 6.4415e-08	k_t
Generation 7200: Best = 2.508546e+03 = [4, 3, 3, 4]	FES = 720100	Diversity = 6.4659e-08	k_t
Generation 7210: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 6.4195e-08	k_t
Generation 7220: Best = 2.508546e+03 = [4, 2, 4, 2]	FES = 722100	Diversity = 6.4211e-08	k_t
Generation 7230: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 6.4438e-08	k_t
Generation 7240: Best = 2.508546e+03 = [3, 3, 2, 2]	FES = 724100	Diversity = 6.6173e-08	k_t

Generation 7250: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 6.6508e-08	k_t
Generation 7260: Best = 2.508546e+03 = [3, 2, 3, 3]	FES = 726100	Diversity = 6.7061e-08	k_t
Generation 7270: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 6.6254e-08	k_t
Generation 7280: Best = 2.508546e+03 = [2, 3, 2, 2]	FES = 728100	Diversity = 6.6654e-08	k_t
Generation 7290: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 6.6961e-08	k_t
Generation 7300: Best = 2.508546e+03 = [2, 4, 4, 3]	FES = 730100	Diversity = 6.6591e-08	k_t
Generation 7310: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 6.7888e-08	k_t
Generation 7320: Best = 2.508546e+03 = [4, 4, 3, 2]	FES = 732100	Diversity = 6.9159e-08	k_t
Generation 7330: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 6.8580e-08	k_t
Generation 7340: Best = 2.508546e+03 = [3, 2, 3, 2]	FES = 734100	Diversity = 6.7755e-08	k_t
Generation 7350: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 6.7781e-08	k_t
Generation 7360: Best = 2.508546e+03 = [2, 4, 4, 3]	FES = 736100	Diversity = 6.7355e-08	k_t
Generation 7370: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 6.5733e-08	k_t
Generation 7380: Best = 2.508546e+03 = [3, 2, 2, 3]	FES = 738100	Diversity = 6.6022e-08	k_t
Generation 7390: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 6.4162e-08	k_t
Generation 7400: Best = 2.508546e+03 = [2, 3, 2, 4]	FES = 740100	Diversity = 6.2861e-08	k_t
Generation 7410: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 6.3099e-08	k_t
Generation 7420: Best = 2.508546e+03 = [2, 4, 4, 2]	FES = 742100	Diversity = 6.3169e-08	k_t
Generation 7430: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 6.2889e-08	k_t
Generation 7440: Best = 2.508546e+03 = [4, 3, 3, 3]	FES = 744100	Diversity = 6.2592e-08	k_t
Generation 7450: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 6.2090e-08	k_t
Generation 7460: Best = 2.508546e+03 = [2, 2, 2, 4]	FES = 746100	Diversity = 6.2676e-08	k_t
Generation 7470: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 6.1818e-08	k_t
Generation 7480: Best = 2.508546e+03 = [3, 4, 2, 4]	FES = 748100	Diversity = 6.1376e-08	k_t
Generation 7490: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 6.1531e-08	k_t
Generation 7500: Best = 2.508546e+03 = [4, 4, 2, 4]	FES = 750100	Diversity = 6.1962e-08	k_t
Generation 7510: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 6.1340e-08	k_t

Generation 7520: Best = 2.508546e+03 = [2, 2, 3, 2]	FES = 752100	Diversity = 6.0899e-08	k_t
Generation 7530: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 753100	Diversity = 6.2376e-08	k_t
Generation 7540: Best = 2.508546e+03 = [2, 2, 3, 4]	FES = 754100	Diversity = 6.3130e-08	k_t
Generation 7550: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 755100	Diversity = 6.2750e-08	k_t
Generation 7560: Best = 2.508546e+03 = [3, 4, 4, 3]	FES = 756100	Diversity = 6.3390e-08	k_t
Generation 7570: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 757100	Diversity = 6.4233e-08	k_t
Generation 7580: Best = 2.508546e+03 = [3, 3, 3, 3]	FES = 758100	Diversity = 6.3907e-08	k_t
Generation 7590: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 759100	Diversity = 6.3317e-08	k_t
Generation 7600: Best = 2.508546e+03 = [2, 2, 3, 3]	FES = 760100	Diversity = 6.5076e-08	k_t
Generation 7610: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 761100	Diversity = 6.3641e-08	k_t
Generation 7620: Best = 2.508546e+03 = [2, 2, 4, 3]	FES = 762100	Diversity = 6.4399e-08	k_t
Generation 7630: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 763100	Diversity = 6.4759e-08	k_t
Generation 7640: Best = 2.508546e+03 = [2, 2, 4, 2]	FES = 764100	Diversity = 6.5181e-08	k_t
Generation 7650: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 765100	Diversity = 6.5927e-08	k_t
Generation 7660: Best = 2.508546e+03 = [2, 2, 4, 4]	FES = 766100	Diversity = 6.6293e-08	k_t
Generation 7670: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 767100	Diversity = 6.6286e-08	k_t
Generation 7680: Best = 2.508546e+03 = [2, 4, 4, 4]	FES = 768100	Diversity = 6.6570e-08	k_t
Generation 7690: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 769100	Diversity = 6.6775e-08	k_t
Generation 7700: Best = 2.508546e+03 = [3, 4, 4, 3]	FES = 770100	Diversity = 6.7697e-08	k_t
Generation 7710: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 771100	Diversity = 6.7056e-08	k_t
Generation 7720: Best = 2.508546e+03 = [4, 4, 2, 3]	FES = 772100	Diversity = 6.7158e-08	k_t
Generation 7730: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 773100	Diversity = 6.7742e-08	k_t
Generation 7740: Best = 2.508546e+03 = [3, 3, 4, 3]	FES = 774100	Diversity = 6.7556e-08	k_t
Generation 7750: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 775100	Diversity = 6.7439e-08	k_t
Generation 7760: Best = 2.508546e+03 = [4, 3, 2, 3]	FES = 776100	Diversity = 6.7050e-08	k_t
Generation 7770: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 777100	Diversity = 6.6155e-08	k_t
Generation 7780: Best = 2.508546e+03 = [2, 3, 3, 3]	FES = 778100	Diversity = 6.7750e-08	k_t

Generation 7790: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 779100	Diversity = 6.6827e-08	k_t
Generation 7800: Best = 2.508546e+03 = [2, 4, 4, 4]	FES = 780100	Diversity = 6.7105e-08	k_t
Generation 7810: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 781100	Diversity = 6.6481e-08	k_t
Generation 7820: Best = 2.508546e+03 = [3, 2, 2, 3]	FES = 782100	Diversity = 6.6374e-08	k_t
Generation 7830: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 783100	Diversity = 6.6869e-08	k_t
Generation 7840: Best = 2.508546e+03 = [4, 2, 3, 2]	FES = 784100	Diversity = 6.7890e-08	k_t
Generation 7850: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 785100	Diversity = 6.5999e-08	k_t
Generation 7860: Best = 2.508546e+03 = [2, 4, 4, 2]	FES = 786100	Diversity = 6.3512e-08	k_t
Generation 7870: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 787100	Diversity = 6.4573e-08	k_t
Generation 7880: Best = 2.508546e+03 = [3, 4, 2, 3]	FES = 788100	Diversity = 6.3174e-08	k_t
Generation 7890: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 789100	Diversity = 6.4363e-08	k_t
Generation 7900: Best = 2.508546e+03 = [3, 3, 3, 2]	FES = 790100	Diversity = 6.5130e-08	k_t
Generation 7910: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 791100	Diversity = 6.3684e-08	k_t
Generation 7920: Best = 2.508546e+03 = [3, 4, 4, 3]	FES = 792100	Diversity = 6.2452e-08	k_t
Generation 7930: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 793100	Diversity = 6.2851e-08	k_t
Generation 7940: Best = 2.508546e+03 = [3, 3, 4, 3]	FES = 794100	Diversity = 6.2404e-08	k_t
Generation 7950: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 795100	Diversity = 6.2631e-08	k_t
Generation 7960: Best = 2.508546e+03 = [2, 3, 3, 4]	FES = 796100	Diversity = 6.2317e-08	k_t
Generation 7970: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 797100	Diversity = 6.2373e-08	k_t
Generation 7980: Best = 2.508546e+03 = [4, 2, 3, 4]	FES = 798100	Diversity = 6.3917e-08	k_t
Generation 7990: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 799100	Diversity = 6.2863e-08	k_t
Generation 8000: Best = 2.508546e+03 = [4, 4, 4, 2]	FES = 800100	Diversity = 6.2835e-08	k_t
Generation 8010: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 801100	Diversity = 6.3879e-08	k_t
Generation 8020: Best = 2.508546e+03 = [3, 3, 4, 3]	FES = 802100	Diversity = 6.2882e-08	k_t
Generation 8030: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 803100	Diversity = 6.2226e-08	k_t
Generation 8040: Best = 2.508546e+03 = [4, 4, 3, 4]	FES = 804100	Diversity = 6.2920e-08	k_t
Generation 8050: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 805100	Diversity = 6.3270e-08	k_t

Generation 8060: Best = 2.508546e+03 = [3, 2, 4, 4]	FES = 806100	Diversity = 6.3404e-08	k_t
Generation 8070: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 807100	Diversity = 6.3872e-08	k_t
Generation 8080: Best = 2.508546e+03 = [4, 4, 2, 2]	FES = 808100	Diversity = 6.3472e-08	k_t
Generation 8090: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 809100	Diversity = 6.3640e-08	k_t
Generation 8100: Best = 2.508546e+03 = [3, 3, 3, 2]	FES = 810100	Diversity = 6.4524e-08	k_t
Generation 8110: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 811100	Diversity = 6.4104e-08	k_t
Generation 8120: Best = 2.508546e+03 = [2, 4, 2, 4]	FES = 812100	Diversity = 6.4478e-08	k_t
Generation 8130: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 813100	Diversity = 6.4927e-08	k_t
Generation 8140: Best = 2.508546e+03 = [2, 2, 4, 2]	FES = 814100	Diversity = 6.4520e-08	k_t
Generation 8150: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 815100	Diversity = 6.3835e-08	k_t
Generation 8160: Best = 2.508546e+03 = [4, 2, 3, 2]	FES = 816100	Diversity = 6.2643e-08	k_t
Generation 8170: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 817100	Diversity = 6.2591e-08	k_t
Generation 8180: Best = 2.508546e+03 = [3, 4, 4, 2]	FES = 818100	Diversity = 6.1539e-08	k_t
Generation 8190: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 819100	Diversity = 6.1913e-08	k_t
Generation 8200: Best = 2.508546e+03 = [4, 2, 4, 4]	FES = 820100	Diversity = 6.1252e-08	k_t
Generation 8210: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 821100	Diversity = 6.1197e-08	k_t
Generation 8220: Best = 2.508546e+03 = [4, 3, 4, 3]	FES = 822100	Diversity = 6.1409e-08	k_t
Generation 8230: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 823100	Diversity = 6.2901e-08	k_t
Generation 8240: Best = 2.508546e+03 = [4, 2, 4, 3]	FES = 824100	Diversity = 6.4991e-08	k_t
Generation 8250: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 825100	Diversity = 6.5598e-08	k_t
Generation 8260: Best = 2.508546e+03 = [3, 4, 2, 3]	FES = 826100	Diversity = 6.5966e-08	k_t
Generation 8270: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 827100	Diversity = 6.5898e-08	k_t
Generation 8280: Best = 2.508546e+03 = [3, 3, 3, 2]	FES = 828100	Diversity = 6.7135e-08	k_t
Generation 8290: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 829100	Diversity = 6.7062e-08	k_t
Generation 8300: Best = 2.508546e+03 = [3, 2, 4, 2]	FES = 830100	Diversity = 6.7575e-08	k_t
Generation 8310: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 831100	Diversity = 6.7525e-08	k_t
Generation 8320: Best = 2.508546e+03 = [4, 2, 2, 2]	FES = 832100	Diversity = 6.6443e-08	k_t

Generation 8330: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 833100	Diversity = 6.7457e-08	k_t
Generation 8340: Best = 2.508546e+03 = [4, 3, 2, 4]	FES = 834100	Diversity = 6.7282e-08	k_t
Generation 8350: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 835100	Diversity = 6.6940e-08	k_t
Generation 8360: Best = 2.508546e+03 = [2, 4, 2, 4]	FES = 836100	Diversity = 6.7713e-08	k_t
Generation 8370: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 837100	Diversity = 6.8000e-08	k_t
Generation 8380: Best = 2.508546e+03 = [2, 2, 3, 3]	FES = 838100	Diversity = 6.7552e-08	k_t
Generation 8390: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 839100	Diversity = 6.7546e-08	k_t
Generation 8400: Best = 2.508546e+03 = [2, 2, 3, 3]	FES = 840100	Diversity = 6.8048e-08	k_t
Generation 8410: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 841100	Diversity = 6.8630e-08	k_t
Generation 8420: Best = 2.508546e+03 = [3, 3, 4, 4]	FES = 842100	Diversity = 6.7565e-08	k_t
Generation 8430: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 843100	Diversity = 6.6208e-08	k_t
Generation 8440: Best = 2.508546e+03 = [2, 4, 4, 3]	FES = 844100	Diversity = 6.5210e-08	k_t
Generation 8450: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 845100	Diversity = 6.4761e-08	k_t
Generation 8460: Best = 2.508546e+03 = [3, 3, 4, 4]	FES = 846100	Diversity = 6.6183e-08	k_t
Generation 8470: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 847100	Diversity = 6.6616e-08	k_t
Generation 8480: Best = 2.508546e+03 = [4, 4, 3, 3]	FES = 848100	Diversity = 6.6039e-08	k_t
Generation 8490: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 849100	Diversity = 6.5250e-08	k_t
Generation 8500: Best = 2.508546e+03 = [2, 4, 3, 4]	FES = 850100	Diversity = 6.4476e-08	k_t
Generation 8510: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 851100	Diversity = 6.5126e-08	k_t
Generation 8520: Best = 2.508546e+03 = [2, 4, 2, 3]	FES = 852100	Diversity = 6.5926e-08	k_t
Generation 8530: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 853100	Diversity = 6.5767e-08	k_t
Generation 8540: Best = 2.508546e+03 = [3, 3, 4, 3]	FES = 854100	Diversity = 6.4823e-08	k_t
Generation 8550: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 855100	Diversity = 6.3782e-08	k_t
Generation 8560: Best = 2.508546e+03 = [3, 3, 2, 4]	FES = 856100	Diversity = 6.3716e-08	k_t
Generation 8570: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 857100	Diversity = 6.5041e-08	k_t
Generation 8580: Best = 2.508546e+03 = [3, 2, 2, 2]	FES = 858100	Diversity = 6.5836e-08	k_t
Generation 8590: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 859100	Diversity = 6.5514e-08	k_t

Generation 8600: Best = 2.508546e+03 = [4, 2, 2, 4]	FES = 860100	Diversity = 6.3599e-08	k_t
Generation 8610: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 861100	Diversity = 6.4178e-08	k_t
Generation 8620: Best = 2.508546e+03 = [2, 2, 2, 4]	FES = 862100	Diversity = 6.4010e-08	k_t
Generation 8630: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 863100	Diversity = 6.3276e-08	k_t
Generation 8640: Best = 2.508546e+03 = [3, 2, 3, 3]	FES = 864100	Diversity = 6.3483e-08	k_t
Generation 8650: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 865100	Diversity = 6.3947e-08	k_t
Generation 8660: Best = 2.508546e+03 = [2, 4, 4, 2]	FES = 866100	Diversity = 6.3533e-08	k_t
Generation 8670: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 867100	Diversity = 6.4243e-08	k_t
Generation 8680: Best = 2.508546e+03 = [2, 2, 2, 2]	FES = 868100	Diversity = 6.4154e-08	k_t
Generation 8690: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 869100	Diversity = 6.3684e-08	k_t
Generation 8700: Best = 2.508546e+03 = [4, 3, 4, 4]	FES = 870100	Diversity = 6.4490e-08	k_t
Generation 8710: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 871100	Diversity = 6.3733e-08	k_t
Generation 8720: Best = 2.508546e+03 = [2, 4, 2, 3]	FES = 872100	Diversity = 6.3073e-08	k_t
Generation 8730: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 873100	Diversity = 6.3472e-08	k_t
Generation 8740: Best = 2.508546e+03 = [3, 2, 4, 2]	FES = 874100	Diversity = 6.4167e-08	k_t
Generation 8750: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 875100	Diversity = 6.3996e-08	k_t
Generation 8760: Best = 2.508546e+03 = [3, 2, 4, 2]	FES = 876100	Diversity = 6.4903e-08	k_t
Generation 8770: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 877100	Diversity = 6.4181e-08	k_t
Generation 8780: Best = 2.508546e+03 = [3, 2, 4, 4]	FES = 878100	Diversity = 6.4466e-08	k_t
Generation 8790: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 879100	Diversity = 6.4669e-08	k_t
Generation 8800: Best = 2.508546e+03 = [4, 3, 4, 2]	FES = 880100	Diversity = 6.3537e-08	k_t
Generation 8810: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 881100	Diversity = 6.4519e-08	k_t
Generation 8820: Best = 2.508546e+03 = [4, 2, 4, 4]	FES = 882100	Diversity = 6.3174e-08	k_t
Generation 8830: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 883100	Diversity = 6.4521e-08	k_t
Generation 8840: Best = 2.508546e+03 = [2, 4, 4, 2]	FES = 884100	Diversity = 6.6354e-08	k_t
Generation 8850: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 885100	Diversity = 6.6493e-08	k_t
Generation 8860: Best = 2.508546e+03 = [4, 4, 4, 3]	FES = 886100	Diversity = 6.7033e-08	k_t

Generation 8870: Best = 2.508546e+03	FES = 887100	Diversity = 6.7743e-08	k_t
= [1, 1, 1, 1]			
Generation 8880: Best = 2.508546e+03	FES = 888100	Diversity = 6.7330e-08	k_t
= [2, 3, 3, 3]			
Generation 8890: Best = 2.508546e+03	FES = 889100	Diversity = 6.7157e-08	k_t
= [1, 1, 1, 1]			
Generation 8900: Best = 2.508546e+03	FES = 890100	Diversity = 6.8113e-08	k_t
= [3, 2, 4, 3]			
Generation 8910: Best = 2.508546e+03	FES = 891100	Diversity = 6.8935e-08	k_t
= [1, 1, 1, 1]			
Generation 8920: Best = 2.508546e+03	FES = 892100	Diversity = 6.8779e-08	k_t
= [2, 4, 4, 2]			
Generation 8930: Best = 2.508546e+03	FES = 893100	Diversity = 6.8202e-08	k_t
= [1, 1, 1, 1]			
Generation 8940: Best = 2.508546e+03	FES = 894100	Diversity = 6.8105e-08	k_t
= [2, 4, 4, 3]			
Generation 8950: Best = 2.508546e+03	FES = 895100	Diversity = 6.8220e-08	k_t
= [1, 1, 1, 1]			
Generation 8960: Best = 2.508546e+03	FES = 896100	Diversity = 6.8199e-08	k_t
= [4, 3, 2, 3]			
Generation 8970: Best = 2.508546e+03	FES = 897100	Diversity = 6.8285e-08	k_t
= [1, 1, 1, 1]			
Generation 8980: Best = 2.508546e+03	FES = 898100	Diversity = 6.8637e-08	k_t
= [2, 3, 4, 2]			
Generation 8990: Best = 2.508546e+03	FES = 899100	Diversity = 6.7991e-08	k_t
= [1, 1, 1, 1]			
Generation 9000: Best = 2.508546e+03	FES = 900100	Diversity = 6.7570e-08	k_t
= [2, 4, 4, 2]			
Generation 9010: Best = 2.508546e+03	FES = 901100	Diversity = 6.6210e-08	k_t
= [1, 1, 1, 1]			
Generation 9020: Best = 2.508546e+03	FES = 902100	Diversity = 6.5995e-08	k_t
= [4, 2, 2, 2]			
Generation 9030: Best = 2.508546e+03	FES = 903100	Diversity = 6.6423e-08	k_t
= [1, 1, 1, 1]			
Generation 9040: Best = 2.508546e+03	FES = 904100	Diversity = 6.5551e-08	k_t
= [2, 4, 3, 2]			
Generation 9050: Best = 2.508546e+03	FES = 905100	Diversity = 6.6537e-08	k_t
= [1, 1, 1, 1]			
Generation 9060: Best = 2.508546e+03	FES = 906100	Diversity = 6.7043e-08	k_t
= [2, 4, 4, 3]			
Generation 9070: Best = 2.508546e+03	FES = 907100	Diversity = 6.6110e-08	k_t
= [1, 1, 1, 1]			
Generation 9080: Best = 2.508546e+03	FES = 908100	Diversity = 6.6172e-08	k_t
= [4, 4, 4, 4]			
Generation 9090: Best = 2.508546e+03	FES = 909100	Diversity = 6.5371e-08	k_t
= [1, 1, 1, 1]			
Generation 9100: Best = 2.508546e+03	FES = 910100	Diversity = 6.4390e-08	k_t
= [2, 4, 3, 4]			
Generation 9110: Best = 2.508546e+03	FES = 911100	Diversity = 6.5191e-08	k_t
= [1, 1, 1, 1]			
Generation 9120: Best = 2.508546e+03	FES = 912100	Diversity = 6.6168e-08	k_t
= [2, 2, 2, 3]			
Generation 9130: Best = 2.508546e+03	FES = 913100	Diversity = 6.7248e-08	k_t
= [1, 1, 1, 1]			

Generation 9140: Best = 2.508546e+03 = [4, 2, 3, 2]	FES = 914100	Diversity = 6.7725e-08	k_t
Generation 9150: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 6.8734e-08	k_t
Generation 9160: Best = 2.508546e+03 = [2, 4, 2, 2]	FES = 916100	Diversity = 6.6535e-08	k_t
Generation 9170: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 6.7420e-08	k_t
Generation 9180: Best = 2.508546e+03 = [2, 3, 3, 3]	FES = 918100	Diversity = 6.6181e-08	k_t
Generation 9190: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 6.7504e-08	k_t
Generation 9200: Best = 2.508546e+03 = [2, 4, 4, 4]	FES = 920100	Diversity = 6.7917e-08	k_t
Generation 9210: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 6.9077e-08	k_t
Generation 9220: Best = 2.508546e+03 = [3, 4, 2, 3]	FES = 922100	Diversity = 6.8258e-08	k_t
Generation 9230: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 6.8437e-08	k_t
Generation 9240: Best = 2.508546e+03 = [4, 3, 3, 4]	FES = 924100	Diversity = 6.8621e-08	k_t
Generation 9250: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 6.8663e-08	k_t
Generation 9260: Best = 2.508546e+03 = [2, 2, 2, 4]	FES = 926100	Diversity = 6.8765e-08	k_t
Generation 9270: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 6.8775e-08	k_t
Generation 9280: Best = 2.508546e+03 = [4, 2, 3, 3]	FES = 928100	Diversity = 6.8056e-08	k_t
Generation 9290: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 6.7704e-08	k_t
Generation 9300: Best = 2.508546e+03 = [2, 4, 2, 2]	FES = 930100	Diversity = 6.9316e-08	k_t
Generation 9310: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 6.8782e-08	k_t
Generation 9320: Best = 2.508546e+03 = [4, 2, 3, 4]	FES = 932100	Diversity = 6.7704e-08	k_t
Generation 9330: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 6.7825e-08	k_t
Generation 9340: Best = 2.508546e+03 = [3, 3, 3, 4]	FES = 934100	Diversity = 6.9155e-08	k_t
Generation 9350: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 6.9553e-08	k_t
Generation 9360: Best = 2.508546e+03 = [2, 4, 4, 4]	FES = 936100	Diversity = 6.9434e-08	k_t
Generation 9370: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 6.9802e-08	k_t
Generation 9380: Best = 2.508546e+03 = [2, 3, 3, 2]	FES = 938100	Diversity = 6.9681e-08	k_t
Generation 9390: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 6.8995e-08	k_t
Generation 9400: Best = 2.508546e+03 = [3, 3, 3, 3]	FES = 940100	Diversity = 6.8867e-08	k_t

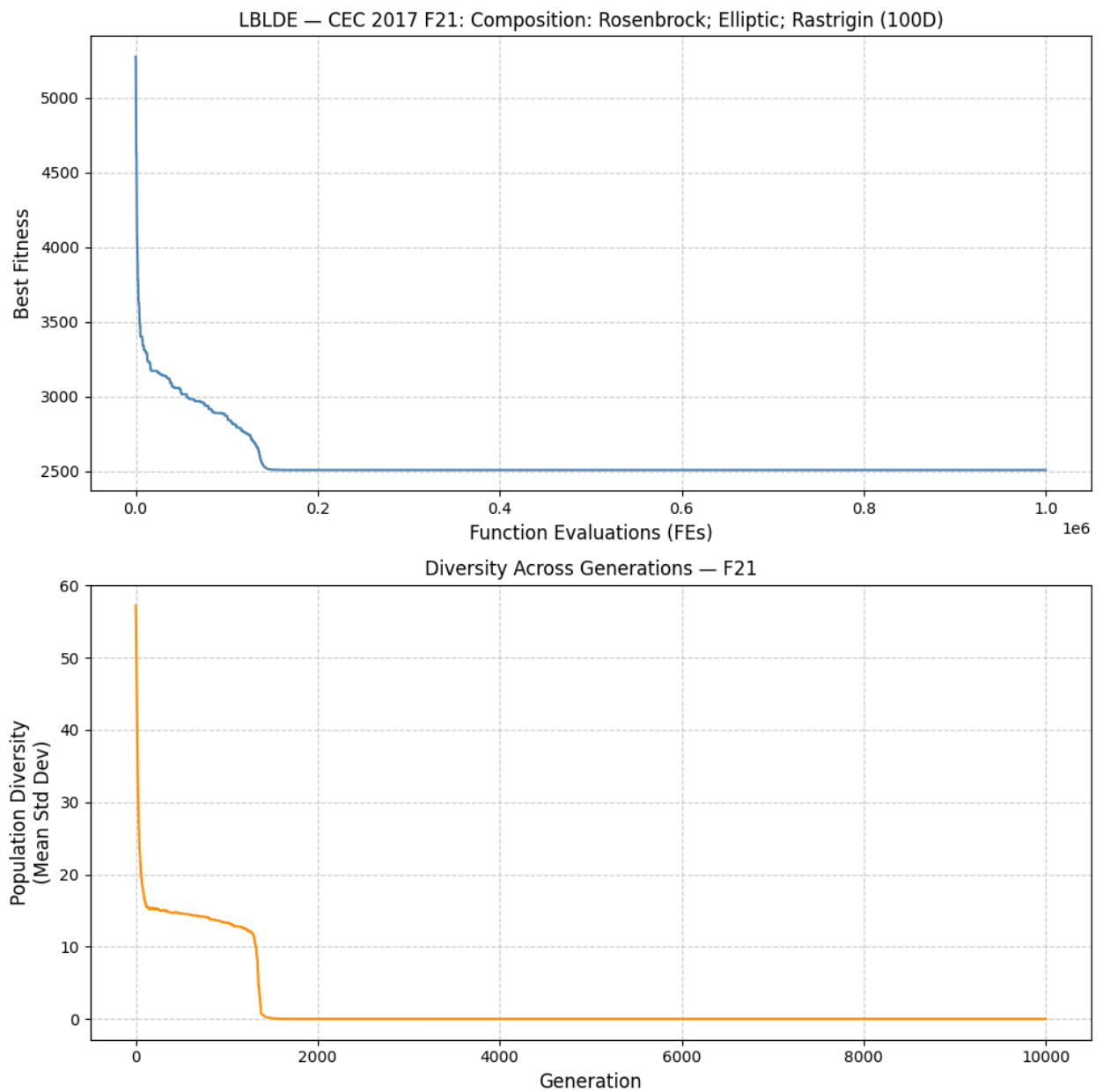
Generation 9410: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 6.8797e-08	k_t
Generation 9420: Best = 2.508546e+03 = [4, 3, 2, 4]	FES = 942100	Diversity = 6.8500e-08	k_t
Generation 9430: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 6.7940e-08	k_t
Generation 9440: Best = 2.508546e+03 = [4, 3, 3, 3]	FES = 944100	Diversity = 6.6399e-08	k_t
Generation 9450: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 6.5822e-08	k_t
Generation 9460: Best = 2.508546e+03 = [4, 4, 4, 2]	FES = 946100	Diversity = 6.4598e-08	k_t
Generation 9470: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 6.5089e-08	k_t
Generation 9480: Best = 2.508546e+03 = [3, 4, 3, 3]	FES = 948100	Diversity = 6.5047e-08	k_t
Generation 9490: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 6.6395e-08	k_t
Generation 9500: Best = 2.508546e+03 = [2, 3, 2, 3]	FES = 950100	Diversity = 6.5796e-08	k_t
Generation 9510: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 6.4789e-08	k_t
Generation 9520: Best = 2.508546e+03 = [4, 4, 3, 3]	FES = 952100	Diversity = 6.4362e-08	k_t
Generation 9530: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 6.4403e-08	k_t
Generation 9540: Best = 2.508546e+03 = [4, 3, 3, 2]	FES = 954100	Diversity = 6.4207e-08	k_t
Generation 9550: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 6.4367e-08	k_t
Generation 9560: Best = 2.508546e+03 = [4, 4, 2, 3]	FES = 956100	Diversity = 6.3677e-08	k_t
Generation 9570: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 6.4964e-08	k_t
Generation 9580: Best = 2.508546e+03 = [3, 3, 2, 3]	FES = 958100	Diversity = 6.4471e-08	k_t
Generation 9590: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 6.5057e-08	k_t
Generation 9600: Best = 2.508546e+03 = [4, 3, 2, 3]	FES = 960100	Diversity = 6.4597e-08	k_t
Generation 9610: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 6.3819e-08	k_t
Generation 9620: Best = 2.508546e+03 = [2, 4, 2, 3]	FES = 962100	Diversity = 6.3759e-08	k_t
Generation 9630: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 6.4497e-08	k_t
Generation 9640: Best = 2.508546e+03 = [4, 4, 4, 2]	FES = 964100	Diversity = 6.4854e-08	k_t
Generation 9650: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 6.5464e-08	k_t
Generation 9660: Best = 2.508546e+03 = [4, 2, 2, 2]	FES = 966100	Diversity = 6.6742e-08	k_t
Generation 9670: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 6.7160e-08	k_t

Generation 9680: Best = 2.508546e+03 = [3, 3, 2, 3]	FES = 968100	Diversity = 6.6779e-08	k_t
Generation 9690: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 969100	Diversity = 6.7385e-08	k_t
Generation 9700: Best = 2.508546e+03 = [4, 4, 4, 4]	FES = 970100	Diversity = 6.8228e-08	k_t
Generation 9710: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 971100	Diversity = 6.8526e-08	k_t
Generation 9720: Best = 2.508546e+03 = [3, 2, 2, 2]	FES = 972100	Diversity = 6.9106e-08	k_t
Generation 9730: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 973100	Diversity = 6.8821e-08	k_t
Generation 9740: Best = 2.508546e+03 = [2, 2, 4, 3]	FES = 974100	Diversity = 6.8805e-08	k_t
Generation 9750: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 975100	Diversity = 6.8219e-08	k_t
Generation 9760: Best = 2.508546e+03 = [2, 4, 4, 2]	FES = 976100	Diversity = 6.8563e-08	k_t
Generation 9770: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 977100	Diversity = 6.8826e-08	k_t
Generation 9780: Best = 2.508546e+03 = [4, 3, 3, 4]	FES = 978100	Diversity = 6.8633e-08	k_t
Generation 9790: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 979100	Diversity = 6.8597e-08	k_t
Generation 9800: Best = 2.508546e+03 = [2, 4, 3, 2]	FES = 980100	Diversity = 6.7975e-08	k_t
Generation 9810: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 981100	Diversity = 6.7616e-08	k_t
Generation 9820: Best = 2.508546e+03 = [3, 3, 3, 2]	FES = 982100	Diversity = 6.6346e-08	k_t
Generation 9830: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 983100	Diversity = 6.6230e-08	k_t
Generation 9840: Best = 2.508546e+03 = [2, 4, 2, 4]	FES = 984100	Diversity = 6.5978e-08	k_t
Generation 9850: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 985100	Diversity = 6.5738e-08	k_t
Generation 9860: Best = 2.508546e+03 = [3, 3, 3, 2]	FES = 986100	Diversity = 6.5090e-08	k_t
Generation 9870: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 987100	Diversity = 6.5182e-08	k_t
Generation 9880: Best = 2.508546e+03 = [3, 4, 3, 4]	FES = 988100	Diversity = 6.5671e-08	k_t
Generation 9890: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 989100	Diversity = 6.4930e-08	k_t
Generation 9900: Best = 2.508546e+03 = [4, 3, 4, 2]	FES = 990100	Diversity = 6.5205e-08	k_t
Generation 9910: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 991100	Diversity = 6.4703e-08	k_t
Generation 9920: Best = 2.508546e+03 = [3, 2, 4, 2]	FES = 992100	Diversity = 6.5316e-08	k_t
Generation 9930: Best = 2.508546e+03 = [1, 1, 1, 1]	FES = 993100	Diversity = 6.5183e-08	k_t
Generation 9940: Best = 2.508546e+03 = [2, 2, 3, 2]	FES = 994100	Diversity = 6.4388e-08	k_t

Generation 9950: Best = 2.508546e+03 FES = 995100 Diversity = 6.4956e-08 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 2.508546e+03 FES = 996100 Diversity = 6.6400e-08 k_t
= [2, 2, 3, 2]
Generation 9970: Best = 2.508546e+03 FES = 997100 Diversity = 6.6556e-08 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 2.508546e+03 FES = 998100 Diversity = 6.5628e-08 k_t
= [4, 4, 2, 2]
Generation 9990: Best = 2.508546e+03 FES = 999100 Diversity = 6.6486e-08 k_t
= [1, 1, 1, 1]

Optimization completed!
Final Best Fitness = 2.508546e+03
Total Function Evaluations = 1000000

=====
=
Results:
Best Fitness : 2.508546e+03
Global Optimum : 2.100000e+03
Error (|best-optimum|) : 4.085460e+02
Best Solution (5 dim) : [-14.76845081 13.61042736 33.43844352 37.63189077
67.07093032]
D^(0) (init. diversity): 5.7220e+01
=====



Graph saved as convergence_F21.png

```
In [46]: # =====
# F22 – Composition Function 2 (Rastrigin; Griewank; Schwefel)
# =====
run_and_plot('F22', make_wrapper(composition.f22), 2200.0, 'Composition: Rastr
```

```

=====
=
LBLDE_s on CEC 2017 F22 – Single Run
=====
=
Function      : Composition: Rastrigin; Griewank; Schwefel (100D)
Bounds       : [-100, 100]
Global optimum : 2200.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 4.002781e+04 | Diversity = 5.7220e+01
Generation 10: Best = 3.732482e+04 FES = 1100 Diversity = 5.6131e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.608708e+04 FES = 2100 Diversity = 5.6209e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 3.608708e+04 FES = 3100 Diversity = 5.6673e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 3.427968e+04 FES = 4100 Diversity = 5.7984e+01 k_t
= [2, 2, 2, 4]
Generation 50: Best = 3.427968e+04 FES = 5100 Diversity = 5.9451e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 3.427968e+04 FES = 6100 Diversity = 6.0051e+01 k_t
= [4, 4, 4, 3]
Generation 70: Best = 3.427968e+04 FES = 7100 Diversity = 5.9776e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.427968e+04 FES = 8100 Diversity = 5.9811e+01 k_t
= [4, 4, 3, 3]
Generation 90: Best = 3.427968e+04 FES = 9100 Diversity = 5.9901e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.427968e+04 FES = 10100 Diversity = 5.9623e+01 k_t
= [4, 2, 3, 3]
Generation 110: Best = 3.388922e+04 FES = 11100 Diversity = 5.9367e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.388922e+04 FES = 12100 Diversity = 5.9646e+01 k_t
= [4, 2, 2, 4]
Generation 130: Best = 3.388922e+04 FES = 13100 Diversity = 5.9205e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.343801e+04 FES = 14100 Diversity = 5.9070e+01 k_t
= [2, 4, 4, 2]
Generation 150: Best = 3.343801e+04 FES = 15100 Diversity = 5.9002e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.273557e+04 FES = 16100 Diversity = 5.8754e+01 k_t
= [3, 2, 4, 4]
Generation 170: Best = 3.179983e+04 FES = 17100 Diversity = 5.8493e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.179983e+04 FES = 18100 Diversity = 5.8607e+01 k_t
= [3, 4, 2, 4]
Generation 190: Best = 3.179983e+04 FES = 19100 Diversity = 5.8540e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.179983e+04 FES = 20100 Diversity = 5.8361e+01 k_t
= [2, 4, 4, 3]
Generation 210: Best = 3.179983e+04 FES = 21100 Diversity = 5.8306e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.179983e+04 FES = 22100 Diversity = 5.8153e+01 k_t
= [2, 4, 4, 3]

```

Generation 230:	Best = 3.179983e+04	FES = 23100	Diversity = 5.7798e+01	k_t
= [1, 1, 1, 1]				
Generation 240:	Best = 3.179983e+04	FES = 24100	Diversity = 5.7717e+01	k_t
= [4, 4, 2, 3]				
Generation 250:	Best = 3.147989e+04	FES = 25100	Diversity = 5.7693e+01	k_t
= [1, 1, 1, 1]				
Generation 260:	Best = 3.089795e+04	FES = 26100	Diversity = 5.7697e+01	k_t
= [4, 2, 4, 3]				
Generation 270:	Best = 3.089795e+04	FES = 27100	Diversity = 5.7690e+01	k_t
= [1, 1, 1, 1]				
Generation 280:	Best = 3.089795e+04	FES = 28100	Diversity = 5.7652e+01	k_t
= [3, 4, 4, 2]				
Generation 290:	Best = 3.089795e+04	FES = 29100	Diversity = 5.7536e+01	k_t
= [1, 1, 1, 1]				
Generation 300:	Best = 3.071119e+04	FES = 30100	Diversity = 5.7630e+01	k_t
= [4, 4, 3, 3]				
Generation 310:	Best = 3.071119e+04	FES = 31100	Diversity = 5.7651e+01	k_t
= [1, 1, 1, 1]				
Generation 320:	Best = 3.071119e+04	FES = 32100	Diversity = 5.7939e+01	k_t
= [3, 3, 2, 3]				
Generation 330:	Best = 3.071119e+04	FES = 33100	Diversity = 5.8044e+01	k_t
= [1, 1, 1, 1]				
Generation 340:	Best = 3.071119e+04	FES = 34100	Diversity = 5.8016e+01	k_t
= [3, 3, 4, 4]				
Generation 350:	Best = 3.071119e+04	FES = 35100	Diversity = 5.8024e+01	k_t
= [1, 1, 1, 1]				
Generation 360:	Best = 3.071119e+04	FES = 36100	Diversity = 5.7969e+01	k_t
= [4, 4, 3, 3]				
Generation 370:	Best = 3.071119e+04	FES = 37100	Diversity = 5.7964e+01	k_t
= [1, 1, 1, 1]				
Generation 380:	Best = 3.071119e+04	FES = 38100	Diversity = 5.7886e+01	k_t
= [4, 4, 2, 2]				
Generation 390:	Best = 3.071119e+04	FES = 39100	Diversity = 5.7653e+01	k_t
= [1, 1, 1, 1]				
Generation 400:	Best = 3.024568e+04	FES = 40100	Diversity = 5.7059e+01	k_t
= [2, 2, 2, 2]				
Generation 410:	Best = 2.996818e+04	FES = 41100	Diversity = 5.7210e+01	k_t
= [1, 1, 1, 1]				
Generation 420:	Best = 2.996818e+04	FES = 42100	Diversity = 5.7175e+01	k_t
= [4, 3, 2, 3]				
Generation 430:	Best = 2.996818e+04	FES = 43100	Diversity = 5.6684e+01	k_t
= [1, 1, 1, 1]				
Generation 440:	Best = 2.996818e+04	FES = 44100	Diversity = 5.6069e+01	k_t
= [4, 3, 2, 3]				
Generation 450:	Best = 2.996818e+04	FES = 45100	Diversity = 5.6109e+01	k_t
= [1, 1, 1, 1]				
Generation 460:	Best = 2.996818e+04	FES = 46100	Diversity = 5.5826e+01	k_t
= [2, 4, 2, 2]				
Generation 470:	Best = 2.917160e+04	FES = 47100	Diversity = 5.5636e+01	k_t
= [1, 1, 1, 1]				
Generation 480:	Best = 2.917160e+04	FES = 48100	Diversity = 5.5511e+01	k_t
= [3, 4, 4, 3]				
Generation 490:	Best = 2.850229e+04	FES = 49100	Diversity = 5.5637e+01	k_t
= [1, 1, 1, 1]				

Generation 500:	Best = 2.848518e+04	FES = 50100	Diversity = 5.5593e+01	k_t
= [3, 2, 4, 2]				
Generation 510:	Best = 2.848518e+04	FES = 51100	Diversity = 5.5532e+01	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 2.848518e+04	FES = 52100	Diversity = 5.5498e+01	k_t
= [3, 4, 2, 4]				
Generation 530:	Best = 2.848518e+04	FES = 53100	Diversity = 5.5426e+01	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 2.848518e+04	FES = 54100	Diversity = 5.5364e+01	k_t
= [4, 2, 3, 3]				
Generation 550:	Best = 2.847124e+04	FES = 55100	Diversity = 5.5082e+01	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 2.847124e+04	FES = 56100	Diversity = 5.4901e+01	k_t
= [3, 4, 4, 4]				
Generation 570:	Best = 2.825077e+04	FES = 57100	Diversity = 5.4396e+01	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 2.824721e+04	FES = 58100	Diversity = 5.4286e+01	k_t
= [3, 3, 2, 3]				
Generation 590:	Best = 2.811066e+04	FES = 59100	Diversity = 5.4258e+01	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 2.811066e+04	FES = 60100	Diversity = 5.4114e+01	k_t
= [4, 4, 4, 2]				
Generation 610:	Best = 2.804990e+04	FES = 61100	Diversity = 5.4019e+01	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 2.798133e+04	FES = 62100	Diversity = 5.3554e+01	k_t
= [3, 2, 4, 2]				
Generation 630:	Best = 2.761178e+04	FES = 63100	Diversity = 5.3290e+01	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 2.761178e+04	FES = 64100	Diversity = 5.2957e+01	k_t
= [3, 4, 3, 3]				
Generation 650:	Best = 2.761178e+04	FES = 65100	Diversity = 5.2694e+01	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 2.703388e+04	FES = 66100	Diversity = 5.2495e+01	k_t
= [3, 2, 4, 3]				
Generation 670:	Best = 2.695329e+04	FES = 67100	Diversity = 5.2297e+01	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 2.695329e+04	FES = 68100	Diversity = 5.2071e+01	k_t
= [3, 3, 2, 4]				
Generation 690:	Best = 2.681654e+04	FES = 69100	Diversity = 5.2062e+01	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 2.681654e+04	FES = 70100	Diversity = 5.1736e+01	k_t
= [4, 4, 3, 2]				
Generation 710:	Best = 2.681654e+04	FES = 71100	Diversity = 5.1477e+01	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 2.672798e+04	FES = 72100	Diversity = 5.0657e+01	k_t
= [4, 3, 3, 3]				
Generation 730:	Best = 2.670494e+04	FES = 73100	Diversity = 5.0220e+01	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 2.666273e+04	FES = 74100	Diversity = 4.9693e+01	k_t
= [3, 3, 3, 2]				
Generation 750:	Best = 2.605859e+04	FES = 75100	Diversity = 4.8837e+01	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 2.605859e+04	FES = 76100	Diversity = 4.7777e+01	k_t
= [3, 2, 3, 4]				

Generation 770:	Best = 2.561883e+04	FES = 77100	Diversity = 4.7229e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 2.559650e+04	FES = 78100	Diversity = 4.6493e+01	k_t
= [2, 4, 4, 4]				
Generation 790:	Best = 2.525996e+04	FES = 79100	Diversity = 4.5249e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 2.468850e+04	FES = 80100	Diversity = 4.1747e+01	k_t
= [4, 2, 4, 4]				
Generation 810:	Best = 2.336542e+04	FES = 81100	Diversity = 3.8365e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 2.336542e+04	FES = 82100	Diversity = 3.5081e+01	k_t
= [2, 2, 3, 4]				
Generation 830:	Best = 2.320082e+04	FES = 83100	Diversity = 2.8974e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 2.317425e+04	FES = 84100	Diversity = 2.5481e+01	k_t
= [4, 4, 4, 2]				
Generation 850:	Best = 2.280612e+04	FES = 85100	Diversity = 2.0210e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 2.280612e+04	FES = 86100	Diversity = 1.7543e+01	k_t
= [2, 4, 4, 4]				
Generation 870:	Best = 2.179534e+04	FES = 87100	Diversity = 1.5052e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 2.176518e+04	FES = 88100	Diversity = 1.3749e+01	k_t
= [2, 3, 4, 2]				
Generation 890:	Best = 2.147067e+04	FES = 89100	Diversity = 1.3621e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 2.089669e+04	FES = 90100	Diversity = 1.0520e+01	k_t
= [2, 2, 4, 4]				
Generation 910:	Best = 2.064140e+04	FES = 91100	Diversity = 1.0336e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 2.044415e+04	FES = 92100	Diversity = 8.0207e+00	k_t
= [4, 3, 3, 2]				
Generation 930:	Best = 2.044415e+04	FES = 93100	Diversity = 7.9700e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 2.013998e+04	FES = 94100	Diversity = 3.1443e+00	k_t
= [4, 2, 4, 2]				
Generation 950:	Best = 1.957720e+04	FES = 95100	Diversity = 3.0031e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 1.957720e+04	FES = 96100	Diversity = 2.8825e+00	k_t
= [3, 2, 3, 2]				
Generation 970:	Best = 1.948460e+04	FES = 97100	Diversity = 2.6558e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 1.938192e+04	FES = 98100	Diversity = 2.4815e+00	k_t
= [2, 4, 4, 4]				
Generation 990:	Best = 1.920947e+04	FES = 99100	Diversity = 2.3889e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 1.845403e+04	FES = 100100	Diversity = 2.3033e+00	k_t
= [2, 4, 4, 2]				
Generation 1010:	Best = 1.845403e+04	FES = 101100	Diversity = 2.1631e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 1.837162e+04	FES = 102100	Diversity = 1.7513e+00	k_t
= [2, 2, 2, 3]				
Generation 1030:	Best = 1.789389e+04	FES = 103100	Diversity = 1.7308e+00	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 1.766332e+04 = [4, 4, 2, 3]	FES = 104100	Diversity = 1.5072e+00	k_t
Generation 1050: Best = 1.757371e+04 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.3435e+00	k_t
Generation 1060: Best = 1.717484e+04 = [3, 3, 3, 2]	FES = 106100	Diversity = 1.0226e+00	k_t
Generation 1070: Best = 1.698139e+04 = [1, 1, 1, 1]	FES = 107100	Diversity = 9.3993e-01	k_t
Generation 1080: Best = 1.657481e+04 = [4, 4, 3, 3]	FES = 108100	Diversity = 7.7378e-01	k_t
Generation 1090: Best = 1.633735e+04 = [1, 1, 1, 1]	FES = 109100	Diversity = 6.4869e-01	k_t
Generation 1100: Best = 1.604019e+04 = [4, 2, 2, 4]	FES = 110100	Diversity = 4.8544e-01	k_t
Generation 1110: Best = 1.581844e+04 = [1, 1, 1, 1]	FES = 111100	Diversity = 4.0512e-01	k_t
Generation 1120: Best = 1.562715e+04 = [4, 4, 4, 3]	FES = 112100	Diversity = 3.5246e-01	k_t
Generation 1130: Best = 1.546159e+04 = [1, 1, 1, 1]	FES = 113100	Diversity = 3.0323e-01	k_t
Generation 1140: Best = 1.535321e+04 = [3, 3, 2, 4]	FES = 114100	Diversity = 2.6652e-01	k_t
Generation 1150: Best = 1.524288e+04 = [1, 1, 1, 1]	FES = 115100	Diversity = 2.3896e-01	k_t
Generation 1160: Best = 1.512221e+04 = [4, 2, 3, 2]	FES = 116100	Diversity = 2.0815e-01	k_t
Generation 1170: Best = 1.504120e+04 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.9547e-01	k_t
Generation 1180: Best = 1.492549e+04 = [3, 4, 4, 2]	FES = 118100	Diversity = 1.8800e-01	k_t
Generation 1190: Best = 1.480137e+04 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.9023e-01	k_t
Generation 1200: Best = 1.471631e+04 = [3, 2, 3, 4]	FES = 120100	Diversity = 1.8762e-01	k_t
Generation 1210: Best = 1.464394e+04 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.8150e-01	k_t
Generation 1220: Best = 1.457945e+04 = [4, 4, 2, 3]	FES = 122100	Diversity = 1.6623e-01	k_t
Generation 1230: Best = 1.452323e+04 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.5047e-01	k_t
Generation 1240: Best = 1.445889e+04 = [2, 2, 3, 4]	FES = 124100	Diversity = 1.4009e-01	k_t
Generation 1250: Best = 1.442379e+04 = [1, 1, 1, 1]	FES = 125100	Diversity = 1.2955e-01	k_t
Generation 1260: Best = 1.436238e+04 = [2, 4, 3, 3]	FES = 126100	Diversity = 1.2758e-01	k_t
Generation 1270: Best = 1.431926e+04 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.2265e-01	k_t
Generation 1280: Best = 1.424698e+04 = [4, 2, 3, 2]	FES = 128100	Diversity = 1.2134e-01	k_t
Generation 1290: Best = 1.417646e+04 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.3383e-01	k_t
Generation 1300: Best = 1.415276e+04 = [2, 4, 3, 2]	FES = 130100	Diversity = 1.2283e-01	k_t

Generation 1310: Best = 1.412353e+04 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.1784e-01	k_t
Generation 1320: Best = 1.406773e+04 = [2, 3, 4, 2]	FES = 132100	Diversity = 1.1237e-01	k_t
Generation 1330: Best = 1.405013e+04 = [1, 1, 1, 1]	FES = 133100	Diversity = 1.0787e-01	k_t
Generation 1340: Best = 1.402648e+04 = [3, 2, 3, 2]	FES = 134100	Diversity = 9.5659e-02	k_t
Generation 1350: Best = 1.402195e+04 = [1, 1, 1, 1]	FES = 135100	Diversity = 8.5443e-02	k_t
Generation 1360: Best = 1.401058e+04 = [4, 3, 2, 4]	FES = 136100	Diversity = 7.2044e-02	k_t
Generation 1370: Best = 1.400146e+04 = [1, 1, 1, 1]	FES = 137100	Diversity = 6.3413e-02	k_t
Generation 1380: Best = 1.398736e+04 = [3, 4, 4, 4]	FES = 138100	Diversity = 5.8768e-02	k_t
Generation 1390: Best = 1.398121e+04 = [1, 1, 1, 1]	FES = 139100	Diversity = 5.5487e-02	k_t
Generation 1400: Best = 1.398116e+04 = [4, 3, 4, 4]	FES = 140100	Diversity = 5.1540e-02	k_t
Generation 1410: Best = 1.397407e+04 = [1, 1, 1, 1]	FES = 141100	Diversity = 4.4911e-02	k_t
Generation 1420: Best = 1.397336e+04 = [4, 2, 2, 3]	FES = 142100	Diversity = 4.0696e-02	k_t
Generation 1430: Best = 1.397172e+04 = [1, 1, 1, 1]	FES = 143100	Diversity = 3.4401e-02	k_t
Generation 1440: Best = 1.397028e+04 = [3, 4, 3, 2]	FES = 144100	Diversity = 3.0594e-02	k_t
Generation 1450: Best = 1.396806e+04 = [1, 1, 1, 1]	FES = 145100	Diversity = 2.6864e-02	k_t
Generation 1460: Best = 1.396692e+04 = [3, 2, 2, 2]	FES = 146100	Diversity = 2.3387e-02	k_t
Generation 1470: Best = 1.396535e+04 = [1, 1, 1, 1]	FES = 147100	Diversity = 2.1423e-02	k_t
Generation 1480: Best = 1.396447e+04 = [3, 3, 2, 2]	FES = 148100	Diversity = 2.1014e-02	k_t
Generation 1490: Best = 1.396365e+04 = [1, 1, 1, 1]	FES = 149100	Diversity = 1.9396e-02	k_t
Generation 1500: Best = 1.396243e+04 = [2, 4, 4, 4]	FES = 150100	Diversity = 2.0447e-02	k_t
Generation 1510: Best = 1.396140e+04 = [1, 1, 1, 1]	FES = 151100	Diversity = 2.0345e-02	k_t
Generation 1520: Best = 1.395933e+04 = [2, 3, 3, 3]	FES = 152100	Diversity = 2.1723e-02	k_t
Generation 1530: Best = 1.395843e+04 = [1, 1, 1, 1]	FES = 153100	Diversity = 2.2078e-02	k_t
Generation 1540: Best = 1.395715e+04 = [2, 2, 2, 2]	FES = 154100	Diversity = 2.2460e-02	k_t
Generation 1550: Best = 1.395562e+04 = [1, 1, 1, 1]	FES = 155100	Diversity = 2.2194e-02	k_t
Generation 1560: Best = 1.395443e+04 = [3, 2, 4, 3]	FES = 156100	Diversity = 2.2488e-02	k_t
Generation 1570: Best = 1.395310e+04 = [1, 1, 1, 1]	FES = 157100	Diversity = 2.2447e-02	k_t

Generation 1580: Best = 1.395077e+04 = [4, 2, 2, 2]	FES = 158100	Diversity = 2.2694e-02	k_t
Generation 1590: Best = 1.395036e+04 = [1, 1, 1, 1]	FES = 159100	Diversity = 2.2560e-02	k_t
Generation 1600: Best = 1.394950e+04 = [3, 2, 4, 3]	FES = 160100	Diversity = 2.2387e-02	k_t
Generation 1610: Best = 1.394867e+04 = [1, 1, 1, 1]	FES = 161100	Diversity = 2.0583e-02	k_t
Generation 1620: Best = 1.394695e+04 = [2, 3, 2, 2]	FES = 162100	Diversity = 1.8738e-02	k_t
Generation 1630: Best = 1.394655e+04 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.7295e-02	k_t
Generation 1640: Best = 1.394612e+04 = [3, 2, 4, 3]	FES = 164100	Diversity = 1.7014e-02	k_t
Generation 1650: Best = 1.394529e+04 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.7149e-02	k_t
Generation 1660: Best = 1.394430e+04 = [2, 2, 3, 3]	FES = 166100	Diversity = 1.8379e-02	k_t
Generation 1670: Best = 1.394380e+04 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.6959e-02	k_t
Generation 1680: Best = 1.394299e+04 = [3, 2, 4, 2]	FES = 168100	Diversity = 1.6415e-02	k_t
Generation 1690: Best = 1.394272e+04 = [1, 1, 1, 1]	FES = 169100	Diversity = 1.4732e-02	k_t
Generation 1700: Best = 1.394213e+04 = [2, 4, 4, 2]	FES = 170100	Diversity = 1.2672e-02	k_t
Generation 1710: Best = 1.394176e+04 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.2198e-02	k_t
Generation 1720: Best = 1.394149e+04 = [2, 4, 3, 3]	FES = 172100	Diversity = 1.1857e-02	k_t
Generation 1730: Best = 1.394114e+04 = [1, 1, 1, 1]	FES = 173100	Diversity = 1.1715e-02	k_t
Generation 1740: Best = 1.394077e+04 = [3, 2, 2, 2]	FES = 174100	Diversity = 1.1653e-02	k_t
Generation 1750: Best = 1.394030e+04 = [1, 1, 1, 1]	FES = 175100	Diversity = 1.1639e-02	k_t
Generation 1760: Best = 1.393997e+04 = [3, 4, 2, 3]	FES = 176100	Diversity = 1.1007e-02	k_t
Generation 1770: Best = 1.393918e+04 = [1, 1, 1, 1]	FES = 177100	Diversity = 1.3466e-02	k_t
Generation 1780: Best = 1.393854e+04 = [2, 3, 2, 2]	FES = 178100	Diversity = 1.3881e-02	k_t
Generation 1790: Best = 1.393807e+04 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.4267e-02	k_t
Generation 1800: Best = 1.393755e+04 = [3, 3, 4, 4]	FES = 180100	Diversity = 1.4198e-02	k_t
Generation 1810: Best = 1.393711e+04 = [1, 1, 1, 1]	FES = 181100	Diversity = 1.3178e-02	k_t
Generation 1820: Best = 1.393652e+04 = [2, 3, 2, 3]	FES = 182100	Diversity = 1.3972e-02	k_t
Generation 1830: Best = 1.393625e+04 = [1, 1, 1, 1]	FES = 183100	Diversity = 1.3702e-02	k_t
Generation 1840: Best = 1.393555e+04 = [2, 3, 4, 4]	FES = 184100	Diversity = 1.2587e-02	k_t

Generation 1850: Best = 1.393540e+04 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.3018e-02	k_t
Generation 1860: Best = 1.393509e+04 = [3, 4, 4, 4]	FES = 186100	Diversity = 1.3263e-02	k_t
Generation 1870: Best = 1.393473e+04 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.2455e-02	k_t
Generation 1880: Best = 1.393443e+04 = [2, 3, 4, 2]	FES = 188100	Diversity = 1.2732e-02	k_t
Generation 1890: Best = 1.393410e+04 = [1, 1, 1, 1]	FES = 189100	Diversity = 1.2388e-02	k_t
Generation 1900: Best = 1.393394e+04 = [4, 2, 2, 4]	FES = 190100	Diversity = 1.0926e-02	k_t
Generation 1910: Best = 1.393372e+04 = [1, 1, 1, 1]	FES = 191100	Diversity = 1.0284e-02	k_t
Generation 1920: Best = 1.393350e+04 = [4, 4, 3, 2]	FES = 192100	Diversity = 1.0202e-02	k_t
Generation 1930: Best = 1.393325e+04 = [1, 1, 1, 1]	FES = 193100	Diversity = 1.1395e-02	k_t
Generation 1940: Best = 1.393296e+04 = [2, 2, 4, 4]	FES = 194100	Diversity = 1.0837e-02	k_t
Generation 1950: Best = 1.393258e+04 = [1, 1, 1, 1]	FES = 195100	Diversity = 1.0390e-02	k_t
Generation 1960: Best = 1.393245e+04 = [4, 3, 4, 2]	FES = 196100	Diversity = 1.0430e-02	k_t
Generation 1970: Best = 1.393229e+04 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.0164e-02	k_t
Generation 1980: Best = 1.393201e+04 = [2, 2, 2, 3]	FES = 198100	Diversity = 9.2551e-03	k_t
Generation 1990: Best = 1.393196e+04 = [1, 1, 1, 1]	FES = 199100	Diversity = 8.8919e-03	k_t
Generation 2000: Best = 1.393173e+04 = [4, 3, 4, 3]	FES = 200100	Diversity = 8.5558e-03	k_t
Generation 2010: Best = 1.393164e+04 = [1, 1, 1, 1]	FES = 201100	Diversity = 7.8630e-03	k_t
Generation 2020: Best = 1.393158e+04 = [4, 3, 2, 4]	FES = 202100	Diversity = 7.3952e-03	k_t
Generation 2030: Best = 1.393145e+04 = [1, 1, 1, 1]	FES = 203100	Diversity = 7.2217e-03	k_t
Generation 2040: Best = 1.393118e+04 = [2, 3, 2, 4]	FES = 204100	Diversity = 7.8252e-03	k_t
Generation 2050: Best = 1.393100e+04 = [1, 1, 1, 1]	FES = 205100	Diversity = 8.3359e-03	k_t
Generation 2060: Best = 1.393048e+04 = [2, 3, 4, 4]	FES = 206100	Diversity = 8.7699e-03	k_t
Generation 2070: Best = 1.393016e+04 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.0003e-02	k_t
Generation 2080: Best = 1.392935e+04 = [2, 4, 3, 3]	FES = 208100	Diversity = 1.0661e-02	k_t
Generation 2090: Best = 1.392861e+04 = [1, 1, 1, 1]	FES = 209100	Diversity = 1.2181e-02	k_t
Generation 2100: Best = 1.392744e+04 = [4, 4, 3, 2]	FES = 210100	Diversity = 1.5001e-02	k_t
Generation 2110: Best = 1.392057e+04 = [1, 1, 1, 1]	FES = 211100	Diversity = 2.0112e-02	k_t

Generation 2120: Best = 1.388732e+04 = [3, 2, 4, 3]	FES = 212100	Diversity = 3.9834e-02	k_t
Generation 2130: Best = 1.383636e+04 = [1, 1, 1, 1]	FES = 213100	Diversity = 7.2140e-02	k_t
Generation 2140: Best = 1.374650e+04 = [3, 4, 2, 4]	FES = 214100	Diversity = 1.1226e-01	k_t
Generation 2150: Best = 1.370867e+04 = [1, 1, 1, 1]	FES = 215100	Diversity = 1.3112e-01	k_t
Generation 2160: Best = 1.370357e+04 = [4, 4, 3, 3]	FES = 216100	Diversity = 1.1770e-01	k_t
Generation 2170: Best = 1.368811e+04 = [1, 1, 1, 1]	FES = 217100	Diversity = 1.0066e-01	k_t
Generation 2180: Best = 1.368301e+04 = [4, 3, 3, 3]	FES = 218100	Diversity = 8.7220e-02	k_t
Generation 2190: Best = 1.367802e+04 = [1, 1, 1, 1]	FES = 219100	Diversity = 7.3298e-02	k_t
Generation 2200: Best = 1.367452e+04 = [4, 3, 3, 2]	FES = 220100	Diversity = 6.0014e-02	k_t
Generation 2210: Best = 1.367185e+04 = [1, 1, 1, 1]	FES = 221100	Diversity = 4.9834e-02	k_t
Generation 2220: Best = 1.367041e+04 = [4, 3, 2, 3]	FES = 222100	Diversity = 3.8978e-02	k_t
Generation 2230: Best = 1.366881e+04 = [1, 1, 1, 1]	FES = 223100	Diversity = 3.1377e-02	k_t
Generation 2240: Best = 1.366795e+04 = [4, 2, 3, 2]	FES = 224100	Diversity = 2.4756e-02	k_t
Generation 2250: Best = 1.366722e+04 = [1, 1, 1, 1]	FES = 225100	Diversity = 2.0952e-02	k_t
Generation 2260: Best = 1.366641e+04 = [2, 3, 4, 4]	FES = 226100	Diversity = 1.7342e-02	k_t
Generation 2270: Best = 1.366618e+04 = [1, 1, 1, 1]	FES = 227100	Diversity = 1.4330e-02	k_t
Generation 2280: Best = 1.366608e+04 = [2, 4, 2, 4]	FES = 228100	Diversity = 1.2620e-02	k_t
Generation 2290: Best = 1.366597e+04 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.1001e-02	k_t
Generation 2300: Best = 1.366563e+04 = [2, 3, 2, 3]	FES = 230100	Diversity = 9.5146e-03	k_t
Generation 2310: Best = 1.366551e+04 = [1, 1, 1, 1]	FES = 231100	Diversity = 9.0778e-03	k_t
Generation 2320: Best = 1.366526e+04 = [4, 3, 4, 4]	FES = 232100	Diversity = 8.0458e-03	k_t
Generation 2330: Best = 1.366507e+04 = [1, 1, 1, 1]	FES = 233100	Diversity = 7.9112e-03	k_t
Generation 2340: Best = 1.366486e+04 = [3, 3, 3, 3]	FES = 234100	Diversity = 7.7939e-03	k_t
Generation 2350: Best = 1.366467e+04 = [1, 1, 1, 1]	FES = 235100	Diversity = 6.9475e-03	k_t
Generation 2360: Best = 1.366467e+04 = [3, 4, 3, 2]	FES = 236100	Diversity = 6.4119e-03	k_t
Generation 2370: Best = 1.366462e+04 = [1, 1, 1, 1]	FES = 237100	Diversity = 6.1278e-03	k_t
Generation 2380: Best = 1.366457e+04 = [3, 4, 3, 2]	FES = 238100	Diversity = 6.1134e-03	k_t

Generation 2390: Best = 1.366443e+04 = [1, 1, 1, 1]	FES = 239100	Diversity = 5.8185e-03	k_t
Generation 2400: Best = 1.366436e+04 = [3, 4, 4, 2]	FES = 240100	Diversity = 5.7906e-03	k_t
Generation 2410: Best = 1.366423e+04 = [1, 1, 1, 1]	FES = 241100	Diversity = 5.7168e-03	k_t
Generation 2420: Best = 1.366414e+04 = [3, 3, 4, 3]	FES = 242100	Diversity = 6.5224e-03	k_t
Generation 2430: Best = 1.366406e+04 = [1, 1, 1, 1]	FES = 243100	Diversity = 6.0719e-03	k_t
Generation 2440: Best = 1.366393e+04 = [3, 2, 4, 4]	FES = 244100	Diversity = 5.8412e-03	k_t
Generation 2450: Best = 1.366380e+04 = [1, 1, 1, 1]	FES = 245100	Diversity = 5.7550e-03	k_t
Generation 2460: Best = 1.366375e+04 = [3, 3, 3, 3]	FES = 246100	Diversity = 5.8690e-03	k_t
Generation 2470: Best = 1.366360e+04 = [1, 1, 1, 1]	FES = 247100	Diversity = 6.1122e-03	k_t
Generation 2480: Best = 1.366347e+04 = [3, 3, 2, 3]	FES = 248100	Diversity = 6.0174e-03	k_t
Generation 2490: Best = 1.366340e+04 = [1, 1, 1, 1]	FES = 249100	Diversity = 6.3863e-03	k_t
Generation 2500: Best = 1.366333e+04 = [2, 3, 2, 2]	FES = 250100	Diversity = 6.2354e-03	k_t
Generation 2510: Best = 1.366317e+04 = [1, 1, 1, 1]	FES = 251100	Diversity = 6.2234e-03	k_t
Generation 2520: Best = 1.366314e+04 = [4, 4, 4, 3]	FES = 252100	Diversity = 6.1565e-03	k_t
Generation 2530: Best = 1.366302e+04 = [1, 1, 1, 1]	FES = 253100	Diversity = 5.9194e-03	k_t
Generation 2540: Best = 1.366292e+04 = [3, 3, 4, 2]	FES = 254100	Diversity = 6.1925e-03	k_t
Generation 2550: Best = 1.366282e+04 = [1, 1, 1, 1]	FES = 255100	Diversity = 5.7407e-03	k_t
Generation 2560: Best = 1.366277e+04 = [4, 2, 3, 3]	FES = 256100	Diversity = 5.6768e-03	k_t
Generation 2570: Best = 1.366272e+04 = [1, 1, 1, 1]	FES = 257100	Diversity = 5.4059e-03	k_t
Generation 2580: Best = 1.366258e+04 = [3, 4, 2, 2]	FES = 258100	Diversity = 5.0619e-03	k_t
Generation 2590: Best = 1.366255e+04 = [1, 1, 1, 1]	FES = 259100	Diversity = 4.6436e-03	k_t
Generation 2600: Best = 1.366248e+04 = [3, 2, 4, 4]	FES = 260100	Diversity = 4.6457e-03	k_t
Generation 2610: Best = 1.366243e+04 = [1, 1, 1, 1]	FES = 261100	Diversity = 4.7740e-03	k_t
Generation 2620: Best = 1.366233e+04 = [4, 2, 2, 4]	FES = 262100	Diversity = 4.1808e-03	k_t
Generation 2630: Best = 1.366226e+04 = [1, 1, 1, 1]	FES = 263100	Diversity = 4.2089e-03	k_t
Generation 2640: Best = 1.366221e+04 = [4, 3, 4, 3]	FES = 264100	Diversity = 4.4496e-03	k_t
Generation 2650: Best = 1.366215e+04 = [1, 1, 1, 1]	FES = 265100	Diversity = 4.5172e-03	k_t

Generation 2660: Best = 1.366207e+04 = [4, 3, 2, 3]	FES = 266100	Diversity = 4.3939e-03	k_t
Generation 2670: Best = 1.366202e+04 = [1, 1, 1, 1]	FES = 267100	Diversity = 4.3978e-03	k_t
Generation 2680: Best = 1.366197e+04 = [3, 3, 2, 3]	FES = 268100	Diversity = 4.0643e-03	k_t
Generation 2690: Best = 1.366193e+04 = [1, 1, 1, 1]	FES = 269100	Diversity = 3.9173e-03	k_t
Generation 2700: Best = 1.366190e+04 = [3, 4, 2, 4]	FES = 270100	Diversity = 3.7071e-03	k_t
Generation 2710: Best = 1.366183e+04 = [1, 1, 1, 1]	FES = 271100	Diversity = 3.7628e-03	k_t
Generation 2720: Best = 1.366179e+04 = [3, 4, 2, 3]	FES = 272100	Diversity = 3.5982e-03	k_t
Generation 2730: Best = 1.366170e+04 = [1, 1, 1, 1]	FES = 273100	Diversity = 3.4307e-03	k_t
Generation 2740: Best = 1.366167e+04 = [3, 2, 2, 4]	FES = 274100	Diversity = 3.5459e-03	k_t
Generation 2750: Best = 1.366165e+04 = [1, 1, 1, 1]	FES = 275100	Diversity = 3.8084e-03	k_t
Generation 2760: Best = 1.366155e+04 = [4, 2, 3, 4]	FES = 276100	Diversity = 3.7097e-03	k_t
Generation 2770: Best = 1.366153e+04 = [1, 1, 1, 1]	FES = 277100	Diversity = 3.6933e-03	k_t
Generation 2780: Best = 1.366150e+04 = [4, 4, 3, 3]	FES = 278100	Diversity = 3.6515e-03	k_t
Generation 2790: Best = 1.366145e+04 = [1, 1, 1, 1]	FES = 279100	Diversity = 3.5838e-03	k_t
Generation 2800: Best = 1.366141e+04 = [2, 3, 3, 2]	FES = 280100	Diversity = 3.6406e-03	k_t
Generation 2810: Best = 1.366132e+04 = [1, 1, 1, 1]	FES = 281100	Diversity = 3.5745e-03	k_t
Generation 2820: Best = 1.366131e+04 = [3, 3, 2, 3]	FES = 282100	Diversity = 3.6939e-03	k_t
Generation 2830: Best = 1.366125e+04 = [1, 1, 1, 1]	FES = 283100	Diversity = 4.1968e-03	k_t
Generation 2840: Best = 1.366122e+04 = [3, 4, 2, 4]	FES = 284100	Diversity = 3.9875e-03	k_t
Generation 2850: Best = 1.366110e+04 = [1, 1, 1, 1]	FES = 285100	Diversity = 4.2826e-03	k_t
Generation 2860: Best = 1.366104e+04 = [2, 3, 2, 4]	FES = 286100	Diversity = 4.1374e-03	k_t
Generation 2870: Best = 1.366099e+04 = [1, 1, 1, 1]	FES = 287100	Diversity = 4.0010e-03	k_t
Generation 2880: Best = 1.366096e+04 = [2, 2, 4, 3]	FES = 288100	Diversity = 4.2526e-03	k_t
Generation 2890: Best = 1.366091e+04 = [1, 1, 1, 1]	FES = 289100	Diversity = 4.2200e-03	k_t
Generation 2900: Best = 1.366085e+04 = [3, 4, 3, 3]	FES = 290100	Diversity = 4.1893e-03	k_t
Generation 2910: Best = 1.366082e+04 = [1, 1, 1, 1]	FES = 291100	Diversity = 4.3483e-03	k_t
Generation 2920: Best = 1.366074e+04 = [3, 2, 4, 4]	FES = 292100	Diversity = 4.3077e-03	k_t

Generation 2930: Best = 1.366066e+04 = [1, 1, 1, 1]	FES = 293100	Diversity = 4.3085e-03	k_t
Generation 2940: Best = 1.366065e+04 = [4, 3, 4, 3]	FES = 294100	Diversity = 4.3139e-03	k_t
Generation 2950: Best = 1.366061e+04 = [1, 1, 1, 1]	FES = 295100	Diversity = 4.0827e-03	k_t
Generation 2960: Best = 1.366058e+04 = [2, 4, 3, 4]	FES = 296100	Diversity = 4.1945e-03	k_t
Generation 2970: Best = 1.366052e+04 = [1, 1, 1, 1]	FES = 297100	Diversity = 4.0403e-03	k_t
Generation 2980: Best = 1.366048e+04 = [3, 4, 3, 4]	FES = 298100	Diversity = 3.7704e-03	k_t
Generation 2990: Best = 1.366048e+04 = [1, 1, 1, 1]	FES = 299100	Diversity = 3.4577e-03	k_t
Generation 3000: Best = 1.366044e+04 = [2, 2, 3, 4]	FES = 300100	Diversity = 3.3077e-03	k_t
Generation 3010: Best = 1.366041e+04 = [1, 1, 1, 1]	FES = 301100	Diversity = 2.9267e-03	k_t
Generation 3020: Best = 1.366039e+04 = [2, 2, 4, 2]	FES = 302100	Diversity = 2.6345e-03	k_t
Generation 3030: Best = 1.366036e+04 = [1, 1, 1, 1]	FES = 303100	Diversity = 2.6056e-03	k_t
Generation 3040: Best = 1.366035e+04 = [2, 2, 4, 2]	FES = 304100	Diversity = 2.3811e-03	k_t
Generation 3050: Best = 1.366032e+04 = [1, 1, 1, 1]	FES = 305100	Diversity = 2.3831e-03	k_t
Generation 3060: Best = 1.366029e+04 = [2, 4, 4, 3]	FES = 306100	Diversity = 2.2608e-03	k_t
Generation 3070: Best = 1.366027e+04 = [1, 1, 1, 1]	FES = 307100	Diversity = 2.4427e-03	k_t
Generation 3080: Best = 1.366026e+04 = [2, 4, 4, 3]	FES = 308100	Diversity = 2.2996e-03	k_t
Generation 3090: Best = 1.366024e+04 = [1, 1, 1, 1]	FES = 309100	Diversity = 2.3512e-03	k_t
Generation 3100: Best = 1.366022e+04 = [2, 3, 2, 2]	FES = 310100	Diversity = 2.3928e-03	k_t
Generation 3110: Best = 1.366019e+04 = [1, 1, 1, 1]	FES = 311100	Diversity = 2.4767e-03	k_t
Generation 3120: Best = 1.366017e+04 = [4, 4, 3, 2]	FES = 312100	Diversity = 2.4074e-03	k_t
Generation 3130: Best = 1.366014e+04 = [1, 1, 1, 1]	FES = 313100	Diversity = 2.5781e-03	k_t
Generation 3140: Best = 1.366013e+04 = [3, 2, 4, 2]	FES = 314100	Diversity = 2.6222e-03	k_t
Generation 3150: Best = 1.366011e+04 = [1, 1, 1, 1]	FES = 315100	Diversity = 2.6358e-03	k_t
Generation 3160: Best = 1.366008e+04 = [4, 4, 4, 3]	FES = 316100	Diversity = 2.6887e-03	k_t
Generation 3170: Best = 1.366007e+04 = [1, 1, 1, 1]	FES = 317100	Diversity = 2.7144e-03	k_t
Generation 3180: Best = 1.366004e+04 = [4, 4, 4, 2]	FES = 318100	Diversity = 2.9609e-03	k_t
Generation 3190: Best = 1.366000e+04 = [1, 1, 1, 1]	FES = 319100	Diversity = 3.0084e-03	k_t

Generation 3200: Best = 1.365998e+04 = [2, 2, 3, 2]	FES = 320100	Diversity = 3.1949e-03	k_t
Generation 3210: Best = 1.365996e+04 = [1, 1, 1, 1]	FES = 321100	Diversity = 3.1760e-03	k_t
Generation 3220: Best = 1.365992e+04 = [2, 4, 3, 4]	FES = 322100	Diversity = 3.3823e-03	k_t
Generation 3230: Best = 1.365990e+04 = [1, 1, 1, 1]	FES = 323100	Diversity = 3.2163e-03	k_t
Generation 3240: Best = 1.365985e+04 = [3, 4, 4, 2]	FES = 324100	Diversity = 3.4689e-03	k_t
Generation 3250: Best = 1.365982e+04 = [1, 1, 1, 1]	FES = 325100	Diversity = 3.5458e-03	k_t
Generation 3260: Best = 1.365977e+04 = [3, 3, 4, 4]	FES = 326100	Diversity = 3.4490e-03	k_t
Generation 3270: Best = 1.365977e+04 = [1, 1, 1, 1]	FES = 327100	Diversity = 3.1817e-03	k_t
Generation 3280: Best = 1.365975e+04 = [3, 4, 3, 2]	FES = 328100	Diversity = 3.1000e-03	k_t
Generation 3290: Best = 1.365972e+04 = [1, 1, 1, 1]	FES = 329100	Diversity = 3.0087e-03	k_t
Generation 3300: Best = 1.365970e+04 = [4, 4, 4, 4]	FES = 330100	Diversity = 3.0696e-03	k_t
Generation 3310: Best = 1.365968e+04 = [1, 1, 1, 1]	FES = 331100	Diversity = 3.1108e-03	k_t
Generation 3320: Best = 1.365966e+04 = [4, 2, 2, 4]	FES = 332100	Diversity = 3.0741e-03	k_t
Generation 3330: Best = 1.365962e+04 = [1, 1, 1, 1]	FES = 333100	Diversity = 2.9824e-03	k_t
Generation 3340: Best = 1.365960e+04 = [2, 4, 4, 3]	FES = 334100	Diversity = 2.9644e-03	k_t
Generation 3350: Best = 1.365958e+04 = [1, 1, 1, 1]	FES = 335100	Diversity = 3.0333e-03	k_t
Generation 3360: Best = 1.365955e+04 = [4, 3, 3, 2]	FES = 336100	Diversity = 2.8195e-03	k_t
Generation 3370: Best = 1.365952e+04 = [1, 1, 1, 1]	FES = 337100	Diversity = 2.6649e-03	k_t
Generation 3380: Best = 1.365951e+04 = [3, 3, 2, 4]	FES = 338100	Diversity = 2.5283e-03	k_t
Generation 3390: Best = 1.365949e+04 = [1, 1, 1, 1]	FES = 339100	Diversity = 2.5515e-03	k_t
Generation 3400: Best = 1.365946e+04 = [4, 2, 3, 2]	FES = 340100	Diversity = 2.3431e-03	k_t
Generation 3410: Best = 1.365944e+04 = [1, 1, 1, 1]	FES = 341100	Diversity = 2.4381e-03	k_t
Generation 3420: Best = 1.365941e+04 = [3, 3, 3, 3]	FES = 342100	Diversity = 2.7355e-03	k_t
Generation 3430: Best = 1.365938e+04 = [1, 1, 1, 1]	FES = 343100	Diversity = 2.7670e-03	k_t
Generation 3440: Best = 1.365936e+04 = [4, 4, 3, 2]	FES = 344100	Diversity = 2.9374e-03	k_t
Generation 3450: Best = 1.365933e+04 = [1, 1, 1, 1]	FES = 345100	Diversity = 2.8566e-03	k_t
Generation 3460: Best = 1.365931e+04 = [2, 2, 2, 3]	FES = 346100	Diversity = 2.7106e-03	k_t

Generation 3470: Best = 1.365930e+04 = [1, 1, 1, 1]	FES = 347100	Diversity = 2.5920e-03	k_t
Generation 3480: Best = 1.365928e+04 = [4, 2, 3, 4]	FES = 348100	Diversity = 2.4754e-03	k_t
Generation 3490: Best = 1.365926e+04 = [1, 1, 1, 1]	FES = 349100	Diversity = 2.4354e-03	k_t
Generation 3500: Best = 1.365925e+04 = [3, 3, 3, 2]	FES = 350100	Diversity = 2.4165e-03	k_t
Generation 3510: Best = 1.365923e+04 = [1, 1, 1, 1]	FES = 351100	Diversity = 2.4162e-03	k_t
Generation 3520: Best = 1.365920e+04 = [2, 2, 4, 4]	FES = 352100	Diversity = 2.3614e-03	k_t
Generation 3530: Best = 1.365919e+04 = [1, 1, 1, 1]	FES = 353100	Diversity = 2.3164e-03	k_t
Generation 3540: Best = 1.365918e+04 = [4, 4, 3, 3]	FES = 354100	Diversity = 2.1395e-03	k_t
Generation 3550: Best = 1.365916e+04 = [1, 1, 1, 1]	FES = 355100	Diversity = 2.0148e-03	k_t
Generation 3560: Best = 1.365914e+04 = [2, 4, 4, 2]	FES = 356100	Diversity = 1.9101e-03	k_t
Generation 3570: Best = 1.365914e+04 = [1, 1, 1, 1]	FES = 357100	Diversity = 1.8943e-03	k_t
Generation 3580: Best = 1.365912e+04 = [4, 3, 3, 4]	FES = 358100	Diversity = 1.8860e-03	k_t
Generation 3590: Best = 1.365912e+04 = [1, 1, 1, 1]	FES = 359100	Diversity = 1.8436e-03	k_t
Generation 3600: Best = 1.365911e+04 = [3, 4, 3, 2]	FES = 360100	Diversity = 1.7003e-03	k_t
Generation 3610: Best = 1.365910e+04 = [1, 1, 1, 1]	FES = 361100	Diversity = 1.5306e-03	k_t
Generation 3620: Best = 1.365910e+04 = [3, 4, 2, 4]	FES = 362100	Diversity = 1.4472e-03	k_t
Generation 3630: Best = 1.365908e+04 = [1, 1, 1, 1]	FES = 363100	Diversity = 1.3991e-03	k_t
Generation 3640: Best = 1.365908e+04 = [3, 3, 3, 3]	FES = 364100	Diversity = 1.3385e-03	k_t
Generation 3650: Best = 1.365908e+04 = [1, 1, 1, 1]	FES = 365100	Diversity = 1.1656e-03	k_t
Generation 3660: Best = 1.365907e+04 = [2, 4, 4, 3]	FES = 366100	Diversity = 1.1828e-03	k_t
Generation 3670: Best = 1.365907e+04 = [1, 1, 1, 1]	FES = 367100	Diversity = 1.1087e-03	k_t
Generation 3680: Best = 1.365906e+04 = [4, 2, 4, 2]	FES = 368100	Diversity = 1.0021e-03	k_t
Generation 3690: Best = 1.365906e+04 = [1, 1, 1, 1]	FES = 369100	Diversity = 9.8847e-04	k_t
Generation 3700: Best = 1.365905e+04 = [4, 4, 4, 2]	FES = 370100	Diversity = 1.0645e-03	k_t
Generation 3710: Best = 1.365905e+04 = [1, 1, 1, 1]	FES = 371100	Diversity = 1.1110e-03	k_t
Generation 3720: Best = 1.365905e+04 = [4, 3, 3, 3]	FES = 372100	Diversity = 1.0439e-03	k_t
Generation 3730: Best = 1.365904e+04 = [1, 1, 1, 1]	FES = 373100	Diversity = 1.0647e-03	k_t

Generation 3740: Best = 1.365903e+04 = [2, 3, 2, 3]	FES = 374100	Diversity = 1.2434e-03	k_t
Generation 3750: Best = 1.365902e+04 = [1, 1, 1, 1]	FES = 375100	Diversity = 1.3682e-03	k_t
Generation 3760: Best = 1.365902e+04 = [2, 3, 3, 4]	FES = 376100	Diversity = 1.3440e-03	k_t
Generation 3770: Best = 1.365901e+04 = [1, 1, 1, 1]	FES = 377100	Diversity = 1.2555e-03	k_t
Generation 3780: Best = 1.365900e+04 = [4, 3, 3, 3]	FES = 378100	Diversity = 1.2265e-03	k_t
Generation 3790: Best = 1.365900e+04 = [1, 1, 1, 1]	FES = 379100	Diversity = 1.1394e-03	k_t
Generation 3800: Best = 1.365900e+04 = [4, 2, 3, 4]	FES = 380100	Diversity = 1.1274e-03	k_t
Generation 3810: Best = 1.365899e+04 = [1, 1, 1, 1]	FES = 381100	Diversity = 1.0413e-03	k_t
Generation 3820: Best = 1.365899e+04 = [3, 4, 4, 3]	FES = 382100	Diversity = 1.1143e-03	k_t
Generation 3830: Best = 1.365898e+04 = [1, 1, 1, 1]	FES = 383100	Diversity = 1.0768e-03	k_t
Generation 3840: Best = 1.365898e+04 = [3, 2, 2, 3]	FES = 384100	Diversity = 1.1532e-03	k_t
Generation 3850: Best = 1.365897e+04 = [1, 1, 1, 1]	FES = 385100	Diversity = 1.0897e-03	k_t
Generation 3860: Best = 1.365897e+04 = [3, 3, 3, 4]	FES = 386100	Diversity = 1.0378e-03	k_t
Generation 3870: Best = 1.365897e+04 = [1, 1, 1, 1]	FES = 387100	Diversity = 1.0294e-03	k_t
Generation 3880: Best = 1.365896e+04 = [2, 3, 4, 4]	FES = 388100	Diversity = 9.9737e-04	k_t
Generation 3890: Best = 1.365896e+04 = [1, 1, 1, 1]	FES = 389100	Diversity = 1.0376e-03	k_t
Generation 3900: Best = 1.365895e+04 = [3, 2, 2, 4]	FES = 390100	Diversity = 1.1383e-03	k_t
Generation 3910: Best = 1.365895e+04 = [1, 1, 1, 1]	FES = 391100	Diversity = 1.1597e-03	k_t
Generation 3920: Best = 1.365895e+04 = [2, 2, 3, 2]	FES = 392100	Diversity = 1.2220e-03	k_t
Generation 3930: Best = 1.365894e+04 = [1, 1, 1, 1]	FES = 393100	Diversity = 1.1043e-03	k_t
Generation 3940: Best = 1.365894e+04 = [4, 2, 3, 2]	FES = 394100	Diversity = 1.0475e-03	k_t
Generation 3950: Best = 1.365893e+04 = [1, 1, 1, 1]	FES = 395100	Diversity = 1.1343e-03	k_t
Generation 3960: Best = 1.365892e+04 = [3, 2, 4, 4]	FES = 396100	Diversity = 1.1333e-03	k_t
Generation 3970: Best = 1.365892e+04 = [1, 1, 1, 1]	FES = 397100	Diversity = 1.3131e-03	k_t
Generation 3980: Best = 1.365891e+04 = [3, 3, 3, 2]	FES = 398100	Diversity = 1.2587e-03	k_t
Generation 3990: Best = 1.365891e+04 = [1, 1, 1, 1]	FES = 399100	Diversity = 1.2143e-03	k_t
Generation 4000: Best = 1.365891e+04 = [2, 3, 4, 3]	FES = 400100	Diversity = 1.1202e-03	k_t

Generation 4010: Best = 1.365890e+04 = [1, 1, 1, 1]	FES = 401100	Diversity = 1.0890e-03	k_t
Generation 4020: Best = 1.365890e+04 = [2, 3, 3, 4]	FES = 402100	Diversity = 1.0034e-03	k_t
Generation 4030: Best = 1.365889e+04 = [1, 1, 1, 1]	FES = 403100	Diversity = 9.3869e-04	k_t
Generation 4040: Best = 1.365889e+04 = [4, 4, 4, 4]	FES = 404100	Diversity = 9.4520e-04	k_t
Generation 4050: Best = 1.365889e+04 = [1, 1, 1, 1]	FES = 405100	Diversity = 9.7554e-04	k_t
Generation 4060: Best = 1.365889e+04 = [4, 4, 2, 2]	FES = 406100	Diversity = 9.4030e-04	k_t
Generation 4070: Best = 1.365889e+04 = [1, 1, 1, 1]	FES = 407100	Diversity = 8.7327e-04	k_t
Generation 4080: Best = 1.365888e+04 = [4, 2, 4, 4]	FES = 408100	Diversity = 8.7462e-04	k_t
Generation 4090: Best = 1.365888e+04 = [1, 1, 1, 1]	FES = 409100	Diversity = 7.9999e-04	k_t
Generation 4100: Best = 1.365888e+04 = [2, 4, 2, 3]	FES = 410100	Diversity = 8.3724e-04	k_t
Generation 4110: Best = 1.365888e+04 = [1, 1, 1, 1]	FES = 411100	Diversity = 8.6342e-04	k_t
Generation 4120: Best = 1.365887e+04 = [3, 3, 3, 3]	FES = 412100	Diversity = 9.4030e-04	k_t
Generation 4130: Best = 1.365887e+04 = [1, 1, 1, 1]	FES = 413100	Diversity = 9.1126e-04	k_t
Generation 4140: Best = 1.365887e+04 = [2, 4, 4, 4]	FES = 414100	Diversity = 9.0614e-04	k_t
Generation 4150: Best = 1.365887e+04 = [1, 1, 1, 1]	FES = 415100	Diversity = 9.0081e-04	k_t
Generation 4160: Best = 1.365886e+04 = [2, 2, 3, 3]	FES = 416100	Diversity = 8.5931e-04	k_t
Generation 4170: Best = 1.365886e+04 = [1, 1, 1, 1]	FES = 417100	Diversity = 9.2974e-04	k_t
Generation 4180: Best = 1.365886e+04 = [3, 2, 4, 4]	FES = 418100	Diversity = 1.0100e-03	k_t
Generation 4190: Best = 1.365885e+04 = [1, 1, 1, 1]	FES = 419100	Diversity = 9.9813e-04	k_t
Generation 4200: Best = 1.365885e+04 = [2, 4, 3, 4]	FES = 420100	Diversity = 9.6865e-04	k_t
Generation 4210: Best = 1.365885e+04 = [1, 1, 1, 1]	FES = 421100	Diversity = 9.4103e-04	k_t
Generation 4220: Best = 1.365884e+04 = [4, 2, 2, 2]	FES = 422100	Diversity = 9.1629e-04	k_t
Generation 4230: Best = 1.365884e+04 = [1, 1, 1, 1]	FES = 423100	Diversity = 1.0009e-03	k_t
Generation 4240: Best = 1.365884e+04 = [4, 4, 2, 4]	FES = 424100	Diversity = 9.7696e-04	k_t
Generation 4250: Best = 1.365883e+04 = [1, 1, 1, 1]	FES = 425100	Diversity = 9.6050e-04	k_t
Generation 4260: Best = 1.365883e+04 = [3, 3, 4, 3]	FES = 426100	Diversity = 9.4916e-04	k_t
Generation 4270: Best = 1.365882e+04 = [1, 1, 1, 1]	FES = 427100	Diversity = 1.0168e-03	k_t

Generation 4280: Best = 1.365882e+04 = [2, 3, 2, 4]	FES = 428100	Diversity = 9.8009e-04	k_t
Generation 4290: Best = 1.365882e+04 = [1, 1, 1, 1]	FES = 429100	Diversity = 9.3282e-04	k_t
Generation 4300: Best = 1.365882e+04 = [2, 2, 3, 3]	FES = 430100	Diversity = 9.8098e-04	k_t
Generation 4310: Best = 1.365881e+04 = [1, 1, 1, 1]	FES = 431100	Diversity = 9.4580e-04	k_t
Generation 4320: Best = 1.365881e+04 = [4, 3, 3, 2]	FES = 432100	Diversity = 8.9748e-04	k_t
Generation 4330: Best = 1.365881e+04 = [1, 1, 1, 1]	FES = 433100	Diversity = 9.4659e-04	k_t
Generation 4340: Best = 1.365881e+04 = [4, 4, 4, 3]	FES = 434100	Diversity = 9.1425e-04	k_t
Generation 4350: Best = 1.365880e+04 = [1, 1, 1, 1]	FES = 435100	Diversity = 8.9098e-04	k_t
Generation 4360: Best = 1.365880e+04 = [3, 2, 4, 2]	FES = 436100	Diversity = 8.2782e-04	k_t
Generation 4370: Best = 1.365880e+04 = [1, 1, 1, 1]	FES = 437100	Diversity = 8.3045e-04	k_t
Generation 4380: Best = 1.365880e+04 = [2, 4, 2, 3]	FES = 438100	Diversity = 8.5880e-04	k_t
Generation 4390: Best = 1.365879e+04 = [1, 1, 1, 1]	FES = 439100	Diversity = 8.6744e-04	k_t
Generation 4400: Best = 1.365879e+04 = [4, 2, 4, 3]	FES = 440100	Diversity = 8.6549e-04	k_t
Generation 4410: Best = 1.365879e+04 = [1, 1, 1, 1]	FES = 441100	Diversity = 8.7743e-04	k_t
Generation 4420: Best = 1.365879e+04 = [2, 2, 2, 4]	FES = 442100	Diversity = 8.5219e-04	k_t
Generation 4430: Best = 1.365878e+04 = [1, 1, 1, 1]	FES = 443100	Diversity = 9.2724e-04	k_t
Generation 4440: Best = 1.365877e+04 = [3, 3, 4, 2]	FES = 444100	Diversity = 9.8467e-04	k_t
Generation 4450: Best = 1.365877e+04 = [1, 1, 1, 1]	FES = 445100	Diversity = 1.1654e-03	k_t
Generation 4460: Best = 1.365876e+04 = [4, 4, 3, 2]	FES = 446100	Diversity = 1.0838e-03	k_t
Generation 4470: Best = 1.365875e+04 = [1, 1, 1, 1]	FES = 447100	Diversity = 1.3267e-03	k_t
Generation 4480: Best = 1.365874e+04 = [2, 3, 3, 4]	FES = 448100	Diversity = 1.5034e-03	k_t
Generation 4490: Best = 1.365873e+04 = [1, 1, 1, 1]	FES = 449100	Diversity = 1.6125e-03	k_t
Generation 4500: Best = 1.365872e+04 = [2, 3, 2, 3]	FES = 450100	Diversity = 1.6875e-03	k_t
Generation 4510: Best = 1.365871e+04 = [1, 1, 1, 1]	FES = 451100	Diversity = 1.7182e-03	k_t
Generation 4520: Best = 1.365871e+04 = [3, 3, 4, 4]	FES = 452100	Diversity = 1.6453e-03	k_t
Generation 4530: Best = 1.365870e+04 = [1, 1, 1, 1]	FES = 453100	Diversity = 1.6548e-03	k_t
Generation 4540: Best = 1.365870e+04 = [4, 2, 2, 4]	FES = 454100	Diversity = 1.6382e-03	k_t

Generation 4550: Best = 1.365869e+04 = [1, 1, 1, 1]	FES = 455100	Diversity = 1.6438e-03	k_t
Generation 4560: Best = 1.365867e+04 = [3, 3, 3, 2]	FES = 456100	Diversity = 1.6514e-03	k_t
Generation 4570: Best = 1.365867e+04 = [1, 1, 1, 1]	FES = 457100	Diversity = 1.6607e-03	k_t
Generation 4580: Best = 1.365867e+04 = [2, 4, 3, 4]	FES = 458100	Diversity = 1.5236e-03	k_t
Generation 4590: Best = 1.365866e+04 = [1, 1, 1, 1]	FES = 459100	Diversity = 1.4336e-03	k_t
Generation 4600: Best = 1.365866e+04 = [4, 3, 2, 2]	FES = 460100	Diversity = 1.4204e-03	k_t
Generation 4610: Best = 1.365865e+04 = [1, 1, 1, 1]	FES = 461100	Diversity = 1.3260e-03	k_t
Generation 4620: Best = 1.365865e+04 = [3, 2, 4, 4]	FES = 462100	Diversity = 1.2767e-03	k_t
Generation 4630: Best = 1.365864e+04 = [1, 1, 1, 1]	FES = 463100	Diversity = 1.2206e-03	k_t
Generation 4640: Best = 1.365864e+04 = [3, 4, 4, 4]	FES = 464100	Diversity = 1.2242e-03	k_t
Generation 4650: Best = 1.365863e+04 = [1, 1, 1, 1]	FES = 465100	Diversity = 1.1977e-03	k_t
Generation 4660: Best = 1.365863e+04 = [2, 4, 2, 2]	FES = 466100	Diversity = 1.1285e-03	k_t
Generation 4670: Best = 1.365862e+04 = [1, 1, 1, 1]	FES = 467100	Diversity = 1.1468e-03	k_t
Generation 4680: Best = 1.365862e+04 = [2, 4, 2, 2]	FES = 468100	Diversity = 1.1165e-03	k_t
Generation 4690: Best = 1.365861e+04 = [1, 1, 1, 1]	FES = 469100	Diversity = 1.0557e-03	k_t
Generation 4700: Best = 1.365861e+04 = [4, 2, 3, 4]	FES = 470100	Diversity = 1.0533e-03	k_t
Generation 4710: Best = 1.365861e+04 = [1, 1, 1, 1]	FES = 471100	Diversity = 1.0586e-03	k_t
Generation 4720: Best = 1.365860e+04 = [4, 2, 2, 4]	FES = 472100	Diversity = 1.0692e-03	k_t
Generation 4730: Best = 1.365860e+04 = [1, 1, 1, 1]	FES = 473100	Diversity = 1.1298e-03	k_t
Generation 4740: Best = 1.365860e+04 = [2, 3, 3, 3]	FES = 474100	Diversity = 1.1582e-03	k_t
Generation 4750: Best = 1.365859e+04 = [1, 1, 1, 1]	FES = 475100	Diversity = 1.1034e-03	k_t
Generation 4760: Best = 1.365859e+04 = [2, 3, 4, 2]	FES = 476100	Diversity = 1.1854e-03	k_t
Generation 4770: Best = 1.365859e+04 = [1, 1, 1, 1]	FES = 477100	Diversity = 1.1526e-03	k_t
Generation 4780: Best = 1.365858e+04 = [4, 3, 3, 3]	FES = 478100	Diversity = 1.2200e-03	k_t
Generation 4790: Best = 1.365858e+04 = [1, 1, 1, 1]	FES = 479100	Diversity = 1.2279e-03	k_t
Generation 4800: Best = 1.365857e+04 = [4, 4, 4, 2]	FES = 480100	Diversity = 1.1508e-03	k_t
Generation 4810: Best = 1.365857e+04 = [1, 1, 1, 1]	FES = 481100	Diversity = 1.1277e-03	k_t

Generation 4820: Best = 1.365856e+04 = [4, 2, 4, 4]	FES = 482100	Diversity = 1.2132e-03	k_t
Generation 4830: Best = 1.365856e+04 = [1, 1, 1, 1]	FES = 483100	Diversity = 1.2770e-03	k_t
Generation 4840: Best = 1.365855e+04 = [2, 3, 3, 4]	FES = 484100	Diversity = 1.2833e-03	k_t
Generation 4850: Best = 1.365855e+04 = [1, 1, 1, 1]	FES = 485100	Diversity = 1.3717e-03	k_t
Generation 4860: Best = 1.365855e+04 = [3, 3, 2, 4]	FES = 486100	Diversity = 1.2868e-03	k_t
Generation 4870: Best = 1.365854e+04 = [1, 1, 1, 1]	FES = 487100	Diversity = 1.2518e-03	k_t
Generation 4880: Best = 1.365853e+04 = [4, 4, 3, 4]	FES = 488100	Diversity = 1.2935e-03	k_t
Generation 4890: Best = 1.365851e+04 = [1, 1, 1, 1]	FES = 489100	Diversity = 1.5122e-03	k_t
Generation 4900: Best = 1.365850e+04 = [3, 2, 2, 3]	FES = 490100	Diversity = 1.6911e-03	k_t
Generation 4910: Best = 1.365849e+04 = [1, 1, 1, 1]	FES = 491100	Diversity = 1.8293e-03	k_t
Generation 4920: Best = 1.365848e+04 = [3, 4, 2, 3]	FES = 492100	Diversity = 1.8701e-03	k_t
Generation 4930: Best = 1.365847e+04 = [1, 1, 1, 1]	FES = 493100	Diversity = 1.7043e-03	k_t
Generation 4940: Best = 1.365847e+04 = [3, 3, 3, 4]	FES = 494100	Diversity = 1.5335e-03	k_t
Generation 4950: Best = 1.365846e+04 = [1, 1, 1, 1]	FES = 495100	Diversity = 1.4304e-03	k_t
Generation 4960: Best = 1.365846e+04 = [4, 4, 4, 4]	FES = 496100	Diversity = 1.3607e-03	k_t
Generation 4970: Best = 1.365845e+04 = [1, 1, 1, 1]	FES = 497100	Diversity = 1.3818e-03	k_t
Generation 4980: Best = 1.365845e+04 = [2, 2, 2, 3]	FES = 498100	Diversity = 1.3221e-03	k_t
Generation 4990: Best = 1.365844e+04 = [1, 1, 1, 1]	FES = 499100	Diversity = 1.3136e-03	k_t
Generation 5000: Best = 1.365844e+04 = [4, 4, 2, 3]	FES = 500100	Diversity = 1.2424e-03	k_t
Generation 5010: Best = 1.365843e+04 = [1, 1, 1, 1]	FES = 501100	Diversity = 1.0917e-03	k_t
Generation 5020: Best = 1.365842e+04 = [4, 4, 4, 3]	FES = 502100	Diversity = 1.1463e-03	k_t
Generation 5030: Best = 1.365842e+04 = [1, 1, 1, 1]	FES = 503100	Diversity = 1.0446e-03	k_t
Generation 5040: Best = 1.365842e+04 = [3, 3, 3, 4]	FES = 504100	Diversity = 1.0057e-03	k_t
Generation 5050: Best = 1.365842e+04 = [1, 1, 1, 1]	FES = 505100	Diversity = 1.0132e-03	k_t
Generation 5060: Best = 1.365842e+04 = [4, 2, 2, 2]	FES = 506100	Diversity = 9.1678e-04	k_t
Generation 5070: Best = 1.365841e+04 = [1, 1, 1, 1]	FES = 507100	Diversity = 8.0083e-04	k_t
Generation 5080: Best = 1.365841e+04 = [3, 4, 2, 4]	FES = 508100	Diversity = 7.2888e-04	k_t

Generation 5090: Best = 1.365841e+04 = [1, 1, 1, 1]	FES = 509100	Diversity = 7.2123e-04	k_t
Generation 5100: Best = 1.365840e+04 = [2, 4, 4, 4]	FES = 510100	Diversity = 7.7413e-04	k_t
Generation 5110: Best = 1.365840e+04 = [1, 1, 1, 1]	FES = 511100	Diversity = 9.0763e-04	k_t
Generation 5120: Best = 1.365839e+04 = [3, 2, 2, 2]	FES = 512100	Diversity = 1.0022e-03	k_t
Generation 5130: Best = 1.365838e+04 = [1, 1, 1, 1]	FES = 513100	Diversity = 1.2383e-03	k_t
Generation 5140: Best = 1.365837e+04 = [3, 2, 4, 3]	FES = 514100	Diversity = 1.4234e-03	k_t
Generation 5150: Best = 1.365836e+04 = [1, 1, 1, 1]	FES = 515100	Diversity = 1.6862e-03	k_t
Generation 5160: Best = 1.365834e+04 = [2, 4, 4, 2]	FES = 516100	Diversity = 1.9454e-03	k_t
Generation 5170: Best = 1.365832e+04 = [1, 1, 1, 1]	FES = 517100	Diversity = 2.3418e-03	k_t
Generation 5180: Best = 1.365830e+04 = [2, 3, 2, 3]	FES = 518100	Diversity = 2.5810e-03	k_t
Generation 5190: Best = 1.365828e+04 = [1, 1, 1, 1]	FES = 519100	Diversity = 2.4968e-03	k_t
Generation 5200: Best = 1.365826e+04 = [3, 4, 4, 3]	FES = 520100	Diversity = 2.4944e-03	k_t
Generation 5210: Best = 1.365824e+04 = [1, 1, 1, 1]	FES = 521100	Diversity = 2.4763e-03	k_t
Generation 5220: Best = 1.365823e+04 = [3, 3, 2, 2]	FES = 522100	Diversity = 2.3569e-03	k_t
Generation 5230: Best = 1.365822e+04 = [1, 1, 1, 1]	FES = 523100	Diversity = 2.1780e-03	k_t
Generation 5240: Best = 1.365821e+04 = [3, 4, 2, 3]	FES = 524100	Diversity = 2.0067e-03	k_t
Generation 5250: Best = 1.365820e+04 = [1, 1, 1, 1]	FES = 525100	Diversity = 1.9980e-03	k_t
Generation 5260: Best = 1.365819e+04 = [2, 2, 3, 2]	FES = 526100	Diversity = 1.8654e-03	k_t
Generation 5270: Best = 1.365819e+04 = [1, 1, 1, 1]	FES = 527100	Diversity = 1.6202e-03	k_t
Generation 5280: Best = 1.365818e+04 = [4, 4, 4, 2]	FES = 528100	Diversity = 1.4660e-03	k_t
Generation 5290: Best = 1.365818e+04 = [1, 1, 1, 1]	FES = 529100	Diversity = 1.3180e-03	k_t
Generation 5300: Best = 1.365817e+04 = [4, 2, 3, 2]	FES = 530100	Diversity = 1.1921e-03	k_t
Generation 5310: Best = 1.365817e+04 = [1, 1, 1, 1]	FES = 531100	Diversity = 1.1594e-03	k_t
Generation 5320: Best = 1.365816e+04 = [2, 3, 3, 3]	FES = 532100	Diversity = 1.2309e-03	k_t
Generation 5330: Best = 1.365816e+04 = [1, 1, 1, 1]	FES = 533100	Diversity = 1.2185e-03	k_t
Generation 5340: Best = 1.365815e+04 = [3, 2, 4, 3]	FES = 534100	Diversity = 1.2085e-03	k_t
Generation 5350: Best = 1.365815e+04 = [1, 1, 1, 1]	FES = 535100	Diversity = 1.2032e-03	k_t

Generation 5360: Best = 1.365815e+04 = [4, 3, 3, 4]	FES = 536100	Diversity = 1.2814e-03	k_t
Generation 5370: Best = 1.365814e+04 = [1, 1, 1, 1]	FES = 537100	Diversity = 1.4735e-03	k_t
Generation 5380: Best = 1.365813e+04 = [2, 2, 3, 3]	FES = 538100	Diversity = 1.4402e-03	k_t
Generation 5390: Best = 1.365813e+04 = [1, 1, 1, 1]	FES = 539100	Diversity = 1.3987e-03	k_t
Generation 5400: Best = 1.365812e+04 = [3, 2, 4, 3]	FES = 540100	Diversity = 1.5061e-03	k_t
Generation 5410: Best = 1.365811e+04 = [1, 1, 1, 1]	FES = 541100	Diversity = 1.5020e-03	k_t
Generation 5420: Best = 1.365811e+04 = [4, 3, 4, 2]	FES = 542100	Diversity = 1.4021e-03	k_t
Generation 5430: Best = 1.365810e+04 = [1, 1, 1, 1]	FES = 543100	Diversity = 1.3145e-03	k_t
Generation 5440: Best = 1.365810e+04 = [2, 3, 3, 2]	FES = 544100	Diversity = 1.2938e-03	k_t
Generation 5450: Best = 1.365810e+04 = [1, 1, 1, 1]	FES = 545100	Diversity = 1.1781e-03	k_t
Generation 5460: Best = 1.365809e+04 = [4, 3, 2, 4]	FES = 546100	Diversity = 1.1196e-03	k_t
Generation 5470: Best = 1.365809e+04 = [1, 1, 1, 1]	FES = 547100	Diversity = 1.1023e-03	k_t
Generation 5480: Best = 1.365808e+04 = [3, 4, 3, 4]	FES = 548100	Diversity = 1.0968e-03	k_t
Generation 5490: Best = 1.365808e+04 = [1, 1, 1, 1]	FES = 549100	Diversity = 1.0352e-03	k_t
Generation 5500: Best = 1.365808e+04 = [3, 4, 4, 3]	FES = 550100	Diversity = 9.4740e-04	k_t
Generation 5510: Best = 1.365807e+04 = [1, 1, 1, 1]	FES = 551100	Diversity = 1.0746e-03	k_t
Generation 5520: Best = 1.365807e+04 = [4, 2, 2, 2]	FES = 552100	Diversity = 1.0673e-03	k_t
Generation 5530: Best = 1.365807e+04 = [1, 1, 1, 1]	FES = 553100	Diversity = 9.8410e-04	k_t
Generation 5540: Best = 1.365806e+04 = [2, 2, 4, 3]	FES = 554100	Diversity = 1.0033e-03	k_t
Generation 5550: Best = 1.365805e+04 = [1, 1, 1, 1]	FES = 555100	Diversity = 1.0437e-03	k_t
Generation 5560: Best = 1.365805e+04 = [2, 2, 4, 3]	FES = 556100	Diversity = 1.0868e-03	k_t
Generation 5570: Best = 1.365805e+04 = [1, 1, 1, 1]	FES = 557100	Diversity = 1.0848e-03	k_t
Generation 5580: Best = 1.365804e+04 = [2, 2, 2, 2]	FES = 558100	Diversity = 1.0631e-03	k_t
Generation 5590: Best = 1.365804e+04 = [1, 1, 1, 1]	FES = 559100	Diversity = 1.1031e-03	k_t
Generation 5600: Best = 1.365803e+04 = [4, 4, 2, 4]	FES = 560100	Diversity = 1.1526e-03	k_t
Generation 5610: Best = 1.365803e+04 = [1, 1, 1, 1]	FES = 561100	Diversity = 1.2839e-03	k_t
Generation 5620: Best = 1.365802e+04 = [4, 4, 3, 2]	FES = 562100	Diversity = 1.2370e-03	k_t

Generation 5630: Best = 1.365802e+04 = [1, 1, 1, 1]	FES = 563100	Diversity = 1.2453e-03	k_t
Generation 5640: Best = 1.365802e+04 = [3, 4, 3, 4]	FES = 564100	Diversity = 1.2030e-03	k_t
Generation 5650: Best = 1.365801e+04 = [1, 1, 1, 1]	FES = 565100	Diversity = 1.1175e-03	k_t
Generation 5660: Best = 1.365801e+04 = [2, 2, 4, 3]	FES = 566100	Diversity = 1.0164e-03	k_t
Generation 5670: Best = 1.365801e+04 = [1, 1, 1, 1]	FES = 567100	Diversity = 9.9169e-04	k_t
Generation 5680: Best = 1.365801e+04 = [2, 4, 2, 2]	FES = 568100	Diversity = 8.9961e-04	k_t
Generation 5690: Best = 1.365800e+04 = [1, 1, 1, 1]	FES = 569100	Diversity = 9.3098e-04	k_t
Generation 5700: Best = 1.365800e+04 = [2, 3, 3, 2]	FES = 570100	Diversity = 1.0377e-03	k_t
Generation 5710: Best = 1.365799e+04 = [1, 1, 1, 1]	FES = 571100	Diversity = 1.1295e-03	k_t
Generation 5720: Best = 1.365799e+04 = [3, 4, 2, 2]	FES = 572100	Diversity = 1.1459e-03	k_t
Generation 5730: Best = 1.365799e+04 = [1, 1, 1, 1]	FES = 573100	Diversity = 1.1156e-03	k_t
Generation 5740: Best = 1.365798e+04 = [4, 4, 2, 3]	FES = 574100	Diversity = 1.0886e-03	k_t
Generation 5750: Best = 1.365798e+04 = [1, 1, 1, 1]	FES = 575100	Diversity = 1.0852e-03	k_t
Generation 5760: Best = 1.365798e+04 = [2, 4, 4, 3]	FES = 576100	Diversity = 1.0350e-03	k_t
Generation 5770: Best = 1.365798e+04 = [1, 1, 1, 1]	FES = 577100	Diversity = 9.9024e-04	k_t
Generation 5780: Best = 1.365797e+04 = [3, 4, 2, 3]	FES = 578100	Diversity = 1.0206e-03	k_t
Generation 5790: Best = 1.365797e+04 = [1, 1, 1, 1]	FES = 579100	Diversity = 9.8926e-04	k_t
Generation 5800: Best = 1.365796e+04 = [3, 2, 3, 4]	FES = 580100	Diversity = 1.0605e-03	k_t
Generation 5810: Best = 1.365796e+04 = [1, 1, 1, 1]	FES = 581100	Diversity = 1.0710e-03	k_t
Generation 5820: Best = 1.365796e+04 = [4, 2, 2, 3]	FES = 582100	Diversity = 1.0743e-03	k_t
Generation 5830: Best = 1.365796e+04 = [1, 1, 1, 1]	FES = 583100	Diversity = 9.5605e-04	k_t
Generation 5840: Best = 1.365795e+04 = [2, 4, 2, 2]	FES = 584100	Diversity = 9.3914e-04	k_t
Generation 5850: Best = 1.365795e+04 = [1, 1, 1, 1]	FES = 585100	Diversity = 9.6878e-04	k_t
Generation 5860: Best = 1.365794e+04 = [3, 3, 2, 3]	FES = 586100	Diversity = 9.9102e-04	k_t
Generation 5870: Best = 1.365794e+04 = [1, 1, 1, 1]	FES = 587100	Diversity = 1.0390e-03	k_t
Generation 5880: Best = 1.365794e+04 = [3, 4, 2, 4]	FES = 588100	Diversity = 9.9897e-04	k_t
Generation 5890: Best = 1.365794e+04 = [1, 1, 1, 1]	FES = 589100	Diversity = 9.9175e-04	k_t

Generation 5900: Best = 1.365793e+04 = [4, 2, 3, 4]	FES = 590100	Diversity = 9.4194e-04	k_t
Generation 5910: Best = 1.365793e+04 = [1, 1, 1, 1]	FES = 591100	Diversity = 9.0073e-04	k_t
Generation 5920: Best = 1.365793e+04 = [4, 4, 4, 3]	FES = 592100	Diversity = 9.7312e-04	k_t
Generation 5930: Best = 1.365793e+04 = [1, 1, 1, 1]	FES = 593100	Diversity = 8.8290e-04	k_t
Generation 5940: Best = 1.365792e+04 = [3, 3, 2, 4]	FES = 594100	Diversity = 8.0611e-04	k_t
Generation 5950: Best = 1.365792e+04 = [1, 1, 1, 1]	FES = 595100	Diversity = 7.6493e-04	k_t
Generation 5960: Best = 1.365792e+04 = [4, 2, 4, 4]	FES = 596100	Diversity = 7.8360e-04	k_t
Generation 5970: Best = 1.365792e+04 = [1, 1, 1, 1]	FES = 597100	Diversity = 7.5696e-04	k_t
Generation 5980: Best = 1.365792e+04 = [3, 3, 2, 4]	FES = 598100	Diversity = 7.1418e-04	k_t
Generation 5990: Best = 1.365791e+04 = [1, 1, 1, 1]	FES = 599100	Diversity = 6.7248e-04	k_t
Generation 6000: Best = 1.365791e+04 = [2, 4, 2, 3]	FES = 600100	Diversity = 6.7840e-04	k_t
Generation 6010: Best = 1.365791e+04 = [1, 1, 1, 1]	FES = 601100	Diversity = 6.6591e-04	k_t
Generation 6020: Best = 1.365791e+04 = [3, 3, 2, 2]	FES = 602100	Diversity = 7.2001e-04	k_t
Generation 6030: Best = 1.365791e+04 = [1, 1, 1, 1]	FES = 603100	Diversity = 7.1727e-04	k_t
Generation 6040: Best = 1.365791e+04 = [3, 3, 2, 2]	FES = 604100	Diversity = 6.9911e-04	k_t
Generation 6050: Best = 1.365790e+04 = [1, 1, 1, 1]	FES = 605100	Diversity = 6.6999e-04	k_t
Generation 6060: Best = 1.365790e+04 = [4, 4, 3, 2]	FES = 606100	Diversity = 6.5719e-04	k_t
Generation 6070: Best = 1.365790e+04 = [1, 1, 1, 1]	FES = 607100	Diversity = 6.9941e-04	k_t
Generation 6080: Best = 1.365790e+04 = [3, 2, 2, 2]	FES = 608100	Diversity = 7.3435e-04	k_t
Generation 6090: Best = 1.365789e+04 = [1, 1, 1, 1]	FES = 609100	Diversity = 7.0325e-04	k_t
Generation 6100: Best = 1.365789e+04 = [3, 3, 2, 2]	FES = 610100	Diversity = 7.7721e-04	k_t
Generation 6110: Best = 1.365789e+04 = [1, 1, 1, 1]	FES = 611100	Diversity = 7.6173e-04	k_t
Generation 6120: Best = 1.365789e+04 = [4, 3, 2, 4]	FES = 612100	Diversity = 7.3840e-04	k_t
Generation 6130: Best = 1.365789e+04 = [1, 1, 1, 1]	FES = 613100	Diversity = 6.9943e-04	k_t
Generation 6140: Best = 1.365789e+04 = [4, 3, 3, 3]	FES = 614100	Diversity = 6.5780e-04	k_t
Generation 6150: Best = 1.365789e+04 = [1, 1, 1, 1]	FES = 615100	Diversity = 6.3141e-04	k_t
Generation 6160: Best = 1.365788e+04 = [3, 3, 2, 3]	FES = 616100	Diversity = 6.4965e-04	k_t

Generation 6170: Best = 1.365788e+04 = [1, 1, 1, 1]	FES = 617100	Diversity = 6.5088e-04	k_t
Generation 6180: Best = 1.365788e+04 = [4, 4, 4, 4]	FES = 618100	Diversity = 7.1916e-04	k_t
Generation 6190: Best = 1.365788e+04 = [1, 1, 1, 1]	FES = 619100	Diversity = 6.8620e-04	k_t
Generation 6200: Best = 1.365788e+04 = [2, 4, 4, 2]	FES = 620100	Diversity = 6.5961e-04	k_t
Generation 6210: Best = 1.365787e+04 = [1, 1, 1, 1]	FES = 621100	Diversity = 6.4239e-04	k_t
Generation 6220: Best = 1.365787e+04 = [3, 4, 4, 2]	FES = 622100	Diversity = 6.4386e-04	k_t
Generation 6230: Best = 1.365787e+04 = [1, 1, 1, 1]	FES = 623100	Diversity = 6.0818e-04	k_t
Generation 6240: Best = 1.365787e+04 = [4, 4, 4, 3]	FES = 624100	Diversity = 6.0676e-04	k_t
Generation 6250: Best = 1.365787e+04 = [1, 1, 1, 1]	FES = 625100	Diversity = 7.0014e-04	k_t
Generation 6260: Best = 1.365787e+04 = [4, 2, 2, 2]	FES = 626100	Diversity = 7.4014e-04	k_t
Generation 6270: Best = 1.365786e+04 = [1, 1, 1, 1]	FES = 627100	Diversity = 7.7149e-04	k_t
Generation 6280: Best = 1.365786e+04 = [3, 4, 2, 3]	FES = 628100	Diversity = 7.5196e-04	k_t
Generation 6290: Best = 1.365786e+04 = [1, 1, 1, 1]	FES = 629100	Diversity = 7.6001e-04	k_t
Generation 6300: Best = 1.365785e+04 = [4, 3, 2, 3]	FES = 630100	Diversity = 8.8901e-04	k_t
Generation 6310: Best = 1.365785e+04 = [1, 1, 1, 1]	FES = 631100	Diversity = 9.9700e-04	k_t
Generation 6320: Best = 1.365785e+04 = [4, 3, 3, 3]	FES = 632100	Diversity = 1.1072e-03	k_t
Generation 6330: Best = 1.365784e+04 = [1, 1, 1, 1]	FES = 633100	Diversity = 1.0636e-03	k_t
Generation 6340: Best = 1.365784e+04 = [2, 3, 4, 4]	FES = 634100	Diversity = 1.1469e-03	k_t
Generation 6350: Best = 1.365783e+04 = [1, 1, 1, 1]	FES = 635100	Diversity = 1.1025e-03	k_t
Generation 6360: Best = 1.365783e+04 = [4, 2, 2, 2]	FES = 636100	Diversity = 1.1138e-03	k_t
Generation 6370: Best = 1.365782e+04 = [1, 1, 1, 1]	FES = 637100	Diversity = 1.1363e-03	k_t
Generation 6380: Best = 1.365782e+04 = [3, 2, 3, 3]	FES = 638100	Diversity = 1.0748e-03	k_t
Generation 6390: Best = 1.365781e+04 = [1, 1, 1, 1]	FES = 639100	Diversity = 1.0984e-03	k_t
Generation 6400: Best = 1.365781e+04 = [2, 3, 4, 2]	FES = 640100	Diversity = 1.1435e-03	k_t
Generation 6410: Best = 1.365781e+04 = [1, 1, 1, 1]	FES = 641100	Diversity = 1.1740e-03	k_t
Generation 6420: Best = 1.365781e+04 = [3, 4, 3, 3]	FES = 642100	Diversity = 1.1530e-03	k_t
Generation 6430: Best = 1.365780e+04 = [1, 1, 1, 1]	FES = 643100	Diversity = 1.0472e-03	k_t

Generation 6440: Best = 1.365780e+04 = [3, 3, 2, 3]	FES = 644100	Diversity = 1.0366e-03	k_t
Generation 6450: Best = 1.365779e+04 = [1, 1, 1, 1]	FES = 645100	Diversity = 1.0731e-03	k_t
Generation 6460: Best = 1.365778e+04 = [3, 2, 3, 4]	FES = 646100	Diversity = 1.1860e-03	k_t
Generation 6470: Best = 1.365777e+04 = [1, 1, 1, 1]	FES = 647100	Diversity = 1.3544e-03	k_t
Generation 6480: Best = 1.365775e+04 = [3, 4, 4, 3]	FES = 648100	Diversity = 1.5250e-03	k_t
Generation 6490: Best = 1.365774e+04 = [1, 1, 1, 1]	FES = 649100	Diversity = 1.6902e-03	k_t
Generation 6500: Best = 1.365772e+04 = [2, 4, 3, 3]	FES = 650100	Diversity = 2.2419e-03	k_t
Generation 6510: Best = 1.365771e+04 = [1, 1, 1, 1]	FES = 651100	Diversity = 2.3680e-03	k_t
Generation 6520: Best = 1.365769e+04 = [2, 4, 2, 2]	FES = 652100	Diversity = 2.2490e-03	k_t
Generation 6530: Best = 1.365766e+04 = [1, 1, 1, 1]	FES = 653100	Diversity = 2.3224e-03	k_t
Generation 6540: Best = 1.365765e+04 = [4, 2, 2, 2]	FES = 654100	Diversity = 2.1180e-03	k_t
Generation 6550: Best = 1.365763e+04 = [1, 1, 1, 1]	FES = 655100	Diversity = 2.1279e-03	k_t
Generation 6560: Best = 1.365762e+04 = [4, 3, 3, 2]	FES = 656100	Diversity = 2.1244e-03	k_t
Generation 6570: Best = 1.365761e+04 = [1, 1, 1, 1]	FES = 657100	Diversity = 2.1708e-03	k_t
Generation 6580: Best = 1.365759e+04 = [3, 4, 3, 4]	FES = 658100	Diversity = 2.1734e-03	k_t
Generation 6590: Best = 1.365757e+04 = [1, 1, 1, 1]	FES = 659100	Diversity = 2.1995e-03	k_t
Generation 6600: Best = 1.365757e+04 = [2, 4, 4, 4]	FES = 660100	Diversity = 2.0636e-03	k_t
Generation 6610: Best = 1.365755e+04 = [1, 1, 1, 1]	FES = 661100	Diversity = 2.0660e-03	k_t
Generation 6620: Best = 1.365753e+04 = [3, 2, 4, 2]	FES = 662100	Diversity = 1.9995e-03	k_t
Generation 6630: Best = 1.365751e+04 = [1, 1, 1, 1]	FES = 663100	Diversity = 1.9114e-03	k_t
Generation 6640: Best = 1.365750e+04 = [3, 4, 4, 4]	FES = 664100	Diversity = 1.8219e-03	k_t
Generation 6650: Best = 1.365750e+04 = [1, 1, 1, 1]	FES = 665100	Diversity = 1.7226e-03	k_t
Generation 6660: Best = 1.365749e+04 = [2, 4, 4, 4]	FES = 666100	Diversity = 1.6762e-03	k_t
Generation 6670: Best = 1.365748e+04 = [1, 1, 1, 1]	FES = 667100	Diversity = 1.5198e-03	k_t
Generation 6680: Best = 1.365748e+04 = [2, 3, 4, 4]	FES = 668100	Diversity = 1.4094e-03	k_t
Generation 6690: Best = 1.365747e+04 = [1, 1, 1, 1]	FES = 669100	Diversity = 1.4106e-03	k_t
Generation 6700: Best = 1.365746e+04 = [2, 2, 2, 2]	FES = 670100	Diversity = 1.4128e-03	k_t

Generation 6710: Best = 1.365746e+04 = [1, 1, 1, 1]	FES = 671100	Diversity = 1.5076e-03	k_t
Generation 6720: Best = 1.365745e+04 = [2, 3, 4, 4]	FES = 672100	Diversity = 1.5143e-03	k_t
Generation 6730: Best = 1.365744e+04 = [1, 1, 1, 1]	FES = 673100	Diversity = 1.5517e-03	k_t
Generation 6740: Best = 1.365744e+04 = [3, 2, 3, 3]	FES = 674100	Diversity = 1.5145e-03	k_t
Generation 6750: Best = 1.365743e+04 = [1, 1, 1, 1]	FES = 675100	Diversity = 1.4958e-03	k_t
Generation 6760: Best = 1.365742e+04 = [4, 3, 4, 3]	FES = 676100	Diversity = 1.5227e-03	k_t
Generation 6770: Best = 1.365741e+04 = [1, 1, 1, 1]	FES = 677100	Diversity = 1.6663e-03	k_t
Generation 6780: Best = 1.365739e+04 = [3, 3, 4, 2]	FES = 678100	Diversity = 1.7922e-03	k_t
Generation 6790: Best = 1.365738e+04 = [1, 1, 1, 1]	FES = 679100	Diversity = 1.7936e-03	k_t
Generation 6800: Best = 1.365737e+04 = [2, 3, 2, 2]	FES = 680100	Diversity = 2.0231e-03	k_t
Generation 6810: Best = 1.365735e+04 = [1, 1, 1, 1]	FES = 681100	Diversity = 2.0256e-03	k_t
Generation 6820: Best = 1.365735e+04 = [4, 4, 3, 3]	FES = 682100	Diversity = 2.0106e-03	k_t
Generation 6830: Best = 1.365733e+04 = [1, 1, 1, 1]	FES = 683100	Diversity = 2.0348e-03	k_t
Generation 6840: Best = 1.365732e+04 = [4, 3, 2, 3]	FES = 684100	Diversity = 2.1882e-03	k_t
Generation 6850: Best = 1.365730e+04 = [1, 1, 1, 1]	FES = 685100	Diversity = 2.2452e-03	k_t
Generation 6860: Best = 1.365730e+04 = [3, 4, 3, 4]	FES = 686100	Diversity = 2.0832e-03	k_t
Generation 6870: Best = 1.365728e+04 = [1, 1, 1, 1]	FES = 687100	Diversity = 1.9114e-03	k_t
Generation 6880: Best = 1.365728e+04 = [4, 2, 3, 3]	FES = 688100	Diversity = 1.8269e-03	k_t
Generation 6890: Best = 1.365727e+04 = [1, 1, 1, 1]	FES = 689100	Diversity = 1.6824e-03	k_t
Generation 6900: Best = 1.365727e+04 = [4, 2, 4, 4]	FES = 690100	Diversity = 1.5891e-03	k_t
Generation 6910: Best = 1.365726e+04 = [1, 1, 1, 1]	FES = 691100	Diversity = 1.5273e-03	k_t
Generation 6920: Best = 1.365725e+04 = [3, 2, 4, 4]	FES = 692100	Diversity = 1.5138e-03	k_t
Generation 6930: Best = 1.365724e+04 = [1, 1, 1, 1]	FES = 693100	Diversity = 1.5882e-03	k_t
Generation 6940: Best = 1.365724e+04 = [4, 4, 4, 2]	FES = 694100	Diversity = 1.4910e-03	k_t
Generation 6950: Best = 1.365723e+04 = [1, 1, 1, 1]	FES = 695100	Diversity = 1.5275e-03	k_t
Generation 6960: Best = 1.365722e+04 = [4, 3, 4, 3]	FES = 696100	Diversity = 1.7816e-03	k_t
Generation 6970: Best = 1.365721e+04 = [1, 1, 1, 1]	FES = 697100	Diversity = 1.7414e-03	k_t

Generation 6980: Best = 1.365719e+04 = [4, 3, 4, 3]	FES = 698100	Diversity = 1.7314e-03	k_t
Generation 6990: Best = 1.365719e+04 = [1, 1, 1, 1]	FES = 699100	Diversity = 1.6185e-03	k_t
Generation 7000: Best = 1.365718e+04 = [3, 4, 3, 3]	FES = 700100	Diversity = 1.7514e-03	k_t
Generation 7010: Best = 1.365717e+04 = [1, 1, 1, 1]	FES = 701100	Diversity = 1.7062e-03	k_t
Generation 7020: Best = 1.365717e+04 = [3, 3, 2, 2]	FES = 702100	Diversity = 1.6876e-03	k_t
Generation 7030: Best = 1.365716e+04 = [1, 1, 1, 1]	FES = 703100	Diversity = 1.5361e-03	k_t
Generation 7040: Best = 1.365716e+04 = [2, 2, 2, 4]	FES = 704100	Diversity = 1.4894e-03	k_t
Generation 7050: Best = 1.365716e+04 = [1, 1, 1, 1]	FES = 705100	Diversity = 1.3828e-03	k_t
Generation 7060: Best = 1.365715e+04 = [3, 3, 4, 4]	FES = 706100	Diversity = 1.2461e-03	k_t
Generation 7070: Best = 1.365714e+04 = [1, 1, 1, 1]	FES = 707100	Diversity = 1.2023e-03	k_t
Generation 7080: Best = 1.365713e+04 = [4, 4, 4, 4]	FES = 708100	Diversity = 1.2498e-03	k_t
Generation 7090: Best = 1.365713e+04 = [1, 1, 1, 1]	FES = 709100	Diversity = 1.2926e-03	k_t
Generation 7100: Best = 1.365713e+04 = [3, 4, 3, 4]	FES = 710100	Diversity = 1.1683e-03	k_t
Generation 7110: Best = 1.365712e+04 = [1, 1, 1, 1]	FES = 711100	Diversity = 1.2053e-03	k_t
Generation 7120: Best = 1.365712e+04 = [3, 3, 4, 4]	FES = 712100	Diversity = 1.1661e-03	k_t
Generation 7130: Best = 1.365711e+04 = [1, 1, 1, 1]	FES = 713100	Diversity = 1.0682e-03	k_t
Generation 7140: Best = 1.365710e+04 = [3, 4, 3, 2]	FES = 714100	Diversity = 1.0706e-03	k_t
Generation 7150: Best = 1.365709e+04 = [1, 1, 1, 1]	FES = 715100	Diversity = 1.1358e-03	k_t
Generation 7160: Best = 1.365709e+04 = [3, 3, 3, 3]	FES = 716100	Diversity = 1.2071e-03	k_t
Generation 7170: Best = 1.365709e+04 = [1, 1, 1, 1]	FES = 717100	Diversity = 1.1874e-03	k_t
Generation 7180: Best = 1.365708e+04 = [2, 4, 2, 4]	FES = 718100	Diversity = 1.0875e-03	k_t
Generation 7190: Best = 1.365708e+04 = [1, 1, 1, 1]	FES = 719100	Diversity = 9.9246e-04	k_t
Generation 7200: Best = 1.365708e+04 = [2, 4, 3, 3]	FES = 720100	Diversity = 9.7295e-04	k_t
Generation 7210: Best = 1.365708e+04 = [1, 1, 1, 1]	FES = 721100	Diversity = 9.3795e-04	k_t
Generation 7220: Best = 1.365707e+04 = [3, 4, 3, 3]	FES = 722100	Diversity = 1.0212e-03	k_t
Generation 7230: Best = 1.365707e+04 = [1, 1, 1, 1]	FES = 723100	Diversity = 1.0441e-03	k_t
Generation 7240: Best = 1.365707e+04 = [4, 4, 3, 3]	FES = 724100	Diversity = 1.0400e-03	k_t

Generation 7250: Best = 1.365706e+04 = [1, 1, 1, 1]	FES = 725100	Diversity = 9.5641e-04	k_t
Generation 7260: Best = 1.365706e+04 = [4, 4, 2, 4]	FES = 726100	Diversity = 9.7500e-04	k_t
Generation 7270: Best = 1.365706e+04 = [1, 1, 1, 1]	FES = 727100	Diversity = 9.3827e-04	k_t
Generation 7280: Best = 1.365706e+04 = [3, 2, 4, 3]	FES = 728100	Diversity = 9.4624e-04	k_t
Generation 7290: Best = 1.365705e+04 = [1, 1, 1, 1]	FES = 729100	Diversity = 1.0270e-03	k_t
Generation 7300: Best = 1.365705e+04 = [3, 4, 2, 3]	FES = 730100	Diversity = 1.0011e-03	k_t
Generation 7310: Best = 1.365705e+04 = [1, 1, 1, 1]	FES = 731100	Diversity = 1.0352e-03	k_t
Generation 7320: Best = 1.365704e+04 = [3, 3, 2, 3]	FES = 732100	Diversity = 1.0544e-03	k_t
Generation 7330: Best = 1.365704e+04 = [1, 1, 1, 1]	FES = 733100	Diversity = 1.0583e-03	k_t
Generation 7340: Best = 1.365704e+04 = [3, 4, 3, 4]	FES = 734100	Diversity = 1.0229e-03	k_t
Generation 7350: Best = 1.365703e+04 = [1, 1, 1, 1]	FES = 735100	Diversity = 9.3656e-04	k_t
Generation 7360: Best = 1.365703e+04 = [2, 4, 3, 2]	FES = 736100	Diversity = 8.8539e-04	k_t
Generation 7370: Best = 1.365703e+04 = [1, 1, 1, 1]	FES = 737100	Diversity = 7.7482e-04	k_t
Generation 7380: Best = 1.365703e+04 = [3, 3, 4, 2]	FES = 738100	Diversity = 7.7815e-04	k_t
Generation 7390: Best = 1.365703e+04 = [1, 1, 1, 1]	FES = 739100	Diversity = 7.3569e-04	k_t
Generation 7400: Best = 1.365703e+04 = [2, 2, 3, 2]	FES = 740100	Diversity = 7.2533e-04	k_t
Generation 7410: Best = 1.365702e+04 = [1, 1, 1, 1]	FES = 741100	Diversity = 6.9058e-04	k_t
Generation 7420: Best = 1.365702e+04 = [3, 4, 3, 3]	FES = 742100	Diversity = 6.7183e-04	k_t
Generation 7430: Best = 1.365702e+04 = [1, 1, 1, 1]	FES = 743100	Diversity = 7.2309e-04	k_t
Generation 7440: Best = 1.365702e+04 = [4, 3, 3, 3]	FES = 744100	Diversity = 7.5148e-04	k_t
Generation 7450: Best = 1.365702e+04 = [1, 1, 1, 1]	FES = 745100	Diversity = 7.0482e-04	k_t
Generation 7460: Best = 1.365702e+04 = [4, 4, 4, 3]	FES = 746100	Diversity = 6.6738e-04	k_t
Generation 7470: Best = 1.365701e+04 = [1, 1, 1, 1]	FES = 747100	Diversity = 7.0147e-04	k_t
Generation 7480: Best = 1.365701e+04 = [4, 3, 2, 3]	FES = 748100	Diversity = 7.8601e-04	k_t
Generation 7490: Best = 1.365701e+04 = [1, 1, 1, 1]	FES = 749100	Diversity = 7.6125e-04	k_t
Generation 7500: Best = 1.365701e+04 = [4, 3, 3, 3]	FES = 750100	Diversity = 7.3665e-04	k_t
Generation 7510: Best = 1.365701e+04 = [1, 1, 1, 1]	FES = 751100	Diversity = 7.8321e-04	k_t

Generation 7520: Best = 1.365701e+04 = [2, 2, 4, 2]	FES = 752100	Diversity = 8.3088e-04	k_t
Generation 7530: Best = 1.365700e+04 = [1, 1, 1, 1]	FES = 753100	Diversity = 8.3295e-04	k_t
Generation 7540: Best = 1.365700e+04 = [2, 2, 2, 2]	FES = 754100	Diversity = 8.4127e-04	k_t
Generation 7550: Best = 1.365700e+04 = [1, 1, 1, 1]	FES = 755100	Diversity = 8.4783e-04	k_t
Generation 7560: Best = 1.365700e+04 = [4, 3, 2, 2]	FES = 756100	Diversity = 8.3819e-04	k_t
Generation 7570: Best = 1.365699e+04 = [1, 1, 1, 1]	FES = 757100	Diversity = 7.8994e-04	k_t
Generation 7580: Best = 1.365699e+04 = [4, 4, 3, 2]	FES = 758100	Diversity = 8.2988e-04	k_t
Generation 7590: Best = 1.365699e+04 = [1, 1, 1, 1]	FES = 759100	Diversity = 8.4389e-04	k_t
Generation 7600: Best = 1.365698e+04 = [4, 4, 2, 2]	FES = 760100	Diversity = 7.8280e-04	k_t
Generation 7610: Best = 1.365698e+04 = [1, 1, 1, 1]	FES = 761100	Diversity = 8.3419e-04	k_t
Generation 7620: Best = 1.365698e+04 = [4, 2, 4, 4]	FES = 762100	Diversity = 8.7560e-04	k_t
Generation 7630: Best = 1.365698e+04 = [1, 1, 1, 1]	FES = 763100	Diversity = 9.0365e-04	k_t
Generation 7640: Best = 1.365697e+04 = [3, 4, 2, 3]	FES = 764100	Diversity = 9.5875e-04	k_t
Generation 7650: Best = 1.365697e+04 = [1, 1, 1, 1]	FES = 765100	Diversity = 1.0137e-03	k_t
Generation 7660: Best = 1.365695e+04 = [4, 2, 2, 4]	FES = 766100	Diversity = 1.2818e-03	k_t
Generation 7670: Best = 1.365694e+04 = [1, 1, 1, 1]	FES = 767100	Diversity = 1.3964e-03	k_t
Generation 7680: Best = 1.365694e+04 = [3, 2, 4, 4]	FES = 768100	Diversity = 1.4306e-03	k_t
Generation 7690: Best = 1.365694e+04 = [1, 1, 1, 1]	FES = 769100	Diversity = 1.4810e-03	k_t
Generation 7700: Best = 1.365693e+04 = [4, 4, 2, 3]	FES = 770100	Diversity = 1.5224e-03	k_t
Generation 7710: Best = 1.365692e+04 = [1, 1, 1, 1]	FES = 771100	Diversity = 1.4514e-03	k_t
Generation 7720: Best = 1.365692e+04 = [2, 2, 3, 3]	FES = 772100	Diversity = 1.4448e-03	k_t
Generation 7730: Best = 1.365692e+04 = [1, 1, 1, 1]	FES = 773100	Diversity = 1.4789e-03	k_t
Generation 7740: Best = 1.365691e+04 = [4, 3, 2, 3]	FES = 774100	Diversity = 1.4414e-03	k_t
Generation 7750: Best = 1.365690e+04 = [1, 1, 1, 1]	FES = 775100	Diversity = 1.4135e-03	k_t
Generation 7760: Best = 1.365690e+04 = [2, 4, 2, 3]	FES = 776100	Diversity = 1.3084e-03	k_t
Generation 7770: Best = 1.365690e+04 = [1, 1, 1, 1]	FES = 777100	Diversity = 1.1953e-03	k_t
Generation 7780: Best = 1.365689e+04 = [2, 4, 4, 4]	FES = 778100	Diversity = 1.1528e-03	k_t

Generation 7790: Best = 1.365689e+04 = [1, 1, 1, 1]	FES = 779100	Diversity = 1.1504e-03	k_t
Generation 7800: Best = 1.365688e+04 = [2, 2, 3, 3]	FES = 780100	Diversity = 1.0957e-03	k_t
Generation 7810: Best = 1.365688e+04 = [1, 1, 1, 1]	FES = 781100	Diversity = 1.0356e-03	k_t
Generation 7820: Best = 1.365688e+04 = [4, 2, 4, 4]	FES = 782100	Diversity = 9.5124e-04	k_t
Generation 7830: Best = 1.365688e+04 = [1, 1, 1, 1]	FES = 783100	Diversity = 9.1775e-04	k_t
Generation 7840: Best = 1.365688e+04 = [2, 3, 3, 3]	FES = 784100	Diversity = 9.2142e-04	k_t
Generation 7850: Best = 1.365687e+04 = [1, 1, 1, 1]	FES = 785100	Diversity = 8.6262e-04	k_t
Generation 7860: Best = 1.365687e+04 = [3, 2, 4, 3]	FES = 786100	Diversity = 8.2563e-04	k_t
Generation 7870: Best = 1.365687e+04 = [1, 1, 1, 1]	FES = 787100	Diversity = 8.2357e-04	k_t
Generation 7880: Best = 1.365687e+04 = [2, 3, 4, 4]	FES = 788100	Diversity = 7.9193e-04	k_t
Generation 7890: Best = 1.365687e+04 = [1, 1, 1, 1]	FES = 789100	Diversity = 8.1157e-04	k_t
Generation 7900: Best = 1.365687e+04 = [2, 3, 4, 3]	FES = 790100	Diversity = 7.7195e-04	k_t
Generation 7910: Best = 1.365686e+04 = [1, 1, 1, 1]	FES = 791100	Diversity = 7.6497e-04	k_t
Generation 7920: Best = 1.365686e+04 = [3, 3, 4, 2]	FES = 792100	Diversity = 7.3987e-04	k_t
Generation 7930: Best = 1.365686e+04 = [1, 1, 1, 1]	FES = 793100	Diversity = 7.1834e-04	k_t
Generation 7940: Best = 1.365686e+04 = [3, 3, 4, 4]	FES = 794100	Diversity = 7.7221e-04	k_t
Generation 7950: Best = 1.365685e+04 = [1, 1, 1, 1]	FES = 795100	Diversity = 7.8488e-04	k_t
Generation 7960: Best = 1.365685e+04 = [2, 4, 2, 4]	FES = 796100	Diversity = 7.3955e-04	k_t
Generation 7970: Best = 1.365685e+04 = [1, 1, 1, 1]	FES = 797100	Diversity = 7.6221e-04	k_t
Generation 7980: Best = 1.365685e+04 = [2, 3, 2, 3]	FES = 798100	Diversity = 7.4359e-04	k_t
Generation 7990: Best = 1.365684e+04 = [1, 1, 1, 1]	FES = 799100	Diversity = 7.8614e-04	k_t
Generation 8000: Best = 1.365684e+04 = [3, 2, 4, 3]	FES = 800100	Diversity = 8.5361e-04	k_t
Generation 8010: Best = 1.365684e+04 = [1, 1, 1, 1]	FES = 801100	Diversity = 8.2428e-04	k_t
Generation 8020: Best = 1.365684e+04 = [2, 3, 4, 4]	FES = 802100	Diversity = 8.4503e-04	k_t
Generation 8030: Best = 1.365684e+04 = [1, 1, 1, 1]	FES = 803100	Diversity = 9.7792e-04	k_t
Generation 8040: Best = 1.365683e+04 = [3, 3, 2, 3]	FES = 804100	Diversity = 9.6213e-04	k_t
Generation 8050: Best = 1.365683e+04 = [1, 1, 1, 1]	FES = 805100	Diversity = 9.7725e-04	k_t

Generation 8060: Best = 1.365683e+04 = [4, 4, 4, 3]	FES = 806100	Diversity = 9.0909e-04	k_t
Generation 8070: Best = 1.365683e+04 = [1, 1, 1, 1]	FES = 807100	Diversity = 9.0126e-04	k_t
Generation 8080: Best = 1.365682e+04 = [4, 4, 4, 4]	FES = 808100	Diversity = 8.7158e-04	k_t
Generation 8090: Best = 1.365682e+04 = [1, 1, 1, 1]	FES = 809100	Diversity = 8.5566e-04	k_t
Generation 8100: Best = 1.365682e+04 = [3, 4, 3, 2]	FES = 810100	Diversity = 8.2887e-04	k_t
Generation 8110: Best = 1.365682e+04 = [1, 1, 1, 1]	FES = 811100	Diversity = 8.0936e-04	k_t
Generation 8120: Best = 1.365682e+04 = [3, 3, 3, 4]	FES = 812100	Diversity = 7.6596e-04	k_t
Generation 8130: Best = 1.365681e+04 = [1, 1, 1, 1]	FES = 813100	Diversity = 7.3955e-04	k_t
Generation 8140: Best = 1.365681e+04 = [4, 3, 3, 3]	FES = 814100	Diversity = 7.5614e-04	k_t
Generation 8150: Best = 1.365681e+04 = [1, 1, 1, 1]	FES = 815100	Diversity = 7.4807e-04	k_t
Generation 8160: Best = 1.365681e+04 = [4, 3, 4, 3]	FES = 816100	Diversity = 7.2241e-04	k_t
Generation 8170: Best = 1.365681e+04 = [1, 1, 1, 1]	FES = 817100	Diversity = 6.9887e-04	k_t
Generation 8180: Best = 1.365681e+04 = [3, 2, 3, 2]	FES = 818100	Diversity = 7.2446e-04	k_t
Generation 8190: Best = 1.365680e+04 = [1, 1, 1, 1]	FES = 819100	Diversity = 7.2129e-04	k_t
Generation 8200: Best = 1.365680e+04 = [4, 4, 2, 2]	FES = 820100	Diversity = 6.9190e-04	k_t
Generation 8210: Best = 1.365680e+04 = [1, 1, 1, 1]	FES = 821100	Diversity = 7.8643e-04	k_t
Generation 8220: Best = 1.365680e+04 = [2, 2, 4, 4]	FES = 822100	Diversity = 7.9830e-04	k_t
Generation 8230: Best = 1.365680e+04 = [1, 1, 1, 1]	FES = 823100	Diversity = 8.3978e-04	k_t
Generation 8240: Best = 1.365679e+04 = [3, 3, 3, 2]	FES = 824100	Diversity = 8.2672e-04	k_t
Generation 8250: Best = 1.365679e+04 = [1, 1, 1, 1]	FES = 825100	Diversity = 7.9900e-04	k_t
Generation 8260: Best = 1.365679e+04 = [2, 3, 3, 4]	FES = 826100	Diversity = 7.8179e-04	k_t
Generation 8270: Best = 1.365679e+04 = [1, 1, 1, 1]	FES = 827100	Diversity = 7.9743e-04	k_t
Generation 8280: Best = 1.365678e+04 = [2, 2, 4, 2]	FES = 828100	Diversity = 7.2863e-04	k_t
Generation 8290: Best = 1.365678e+04 = [1, 1, 1, 1]	FES = 829100	Diversity = 6.4898e-04	k_t
Generation 8300: Best = 1.365678e+04 = [4, 3, 3, 4]	FES = 830100	Diversity = 5.4537e-04	k_t
Generation 8310: Best = 1.365678e+04 = [1, 1, 1, 1]	FES = 831100	Diversity = 5.5210e-04	k_t
Generation 8320: Best = 1.365678e+04 = [2, 2, 2, 3]	FES = 832100	Diversity = 5.9816e-04	k_t

Generation 8330: Best = 1.365678e+04 = [1, 1, 1, 1]	FES = 833100	Diversity = 6.4425e-04	k_t
Generation 8340: Best = 1.365677e+04 = [3, 4, 2, 3]	FES = 834100	Diversity = 6.8887e-04	k_t
Generation 8350: Best = 1.365677e+04 = [1, 1, 1, 1]	FES = 835100	Diversity = 7.3731e-04	k_t
Generation 8360: Best = 1.365677e+04 = [3, 3, 2, 4]	FES = 836100	Diversity = 7.6303e-04	k_t
Generation 8370: Best = 1.365676e+04 = [1, 1, 1, 1]	FES = 837100	Diversity = 8.4470e-04	k_t
Generation 8380: Best = 1.365676e+04 = [4, 4, 2, 2]	FES = 838100	Diversity = 8.6307e-04	k_t
Generation 8390: Best = 1.365676e+04 = [1, 1, 1, 1]	FES = 839100	Diversity = 8.3210e-04	k_t
Generation 8400: Best = 1.365676e+04 = [2, 3, 2, 4]	FES = 840100	Diversity = 7.5757e-04	k_t
Generation 8410: Best = 1.365676e+04 = [1, 1, 1, 1]	FES = 841100	Diversity = 7.0479e-04	k_t
Generation 8420: Best = 1.365676e+04 = [4, 2, 2, 2]	FES = 842100	Diversity = 6.4769e-04	k_t
Generation 8430: Best = 1.365675e+04 = [1, 1, 1, 1]	FES = 843100	Diversity = 5.8738e-04	k_t
Generation 8440: Best = 1.365675e+04 = [2, 2, 4, 2]	FES = 844100	Diversity = 5.5672e-04	k_t
Generation 8450: Best = 1.365675e+04 = [1, 1, 1, 1]	FES = 845100	Diversity = 5.2922e-04	k_t
Generation 8460: Best = 1.365675e+04 = [4, 3, 2, 2]	FES = 846100	Diversity = 5.6544e-04	k_t
Generation 8470: Best = 1.365675e+04 = [1, 1, 1, 1]	FES = 847100	Diversity = 5.8630e-04	k_t
Generation 8480: Best = 1.365675e+04 = [4, 4, 2, 4]	FES = 848100	Diversity = 5.6196e-04	k_t
Generation 8490: Best = 1.365675e+04 = [1, 1, 1, 1]	FES = 849100	Diversity = 5.2900e-04	k_t
Generation 8500: Best = 1.365675e+04 = [2, 3, 3, 4]	FES = 850100	Diversity = 4.8614e-04	k_t
Generation 8510: Best = 1.365674e+04 = [1, 1, 1, 1]	FES = 851100	Diversity = 4.4154e-04	k_t
Generation 8520: Best = 1.365674e+04 = [3, 2, 3, 4]	FES = 852100	Diversity = 4.3095e-04	k_t
Generation 8530: Best = 1.365674e+04 = [1, 1, 1, 1]	FES = 853100	Diversity = 4.3919e-04	k_t
Generation 8540: Best = 1.365674e+04 = [2, 3, 3, 3]	FES = 854100	Diversity = 4.4308e-04	k_t
Generation 8550: Best = 1.365674e+04 = [1, 1, 1, 1]	FES = 855100	Diversity = 4.8988e-04	k_t
Generation 8560: Best = 1.365674e+04 = [2, 4, 2, 4]	FES = 856100	Diversity = 4.6700e-04	k_t
Generation 8570: Best = 1.365674e+04 = [1, 1, 1, 1]	FES = 857100	Diversity = 4.5808e-04	k_t
Generation 8580: Best = 1.365674e+04 = [2, 2, 3, 2]	FES = 858100	Diversity = 4.9771e-04	k_t
Generation 8590: Best = 1.365674e+04 = [1, 1, 1, 1]	FES = 859100	Diversity = 5.3681e-04	k_t

Generation 8600: Best = 1.365673e+04 = [2, 3, 2, 3]	FES = 860100	Diversity = 5.8325e-04	k_t
Generation 8610: Best = 1.365673e+04 = [1, 1, 1, 1]	FES = 861100	Diversity = 5.6842e-04	k_t
Generation 8620: Best = 1.365673e+04 = [3, 3, 2, 4]	FES = 862100	Diversity = 5.8339e-04	k_t
Generation 8630: Best = 1.365673e+04 = [1, 1, 1, 1]	FES = 863100	Diversity = 5.7817e-04	k_t
Generation 8640: Best = 1.365673e+04 = [2, 4, 4, 4]	FES = 864100	Diversity = 5.6391e-04	k_t
Generation 8650: Best = 1.365673e+04 = [1, 1, 1, 1]	FES = 865100	Diversity = 5.3029e-04	k_t
Generation 8660: Best = 1.365673e+04 = [2, 4, 4, 4]	FES = 866100	Diversity = 5.3651e-04	k_t
Generation 8670: Best = 1.365673e+04 = [1, 1, 1, 1]	FES = 867100	Diversity = 5.5037e-04	k_t
Generation 8680: Best = 1.365673e+04 = [4, 2, 2, 4]	FES = 868100	Diversity = 6.1616e-04	k_t
Generation 8690: Best = 1.365672e+04 = [1, 1, 1, 1]	FES = 869100	Diversity = 6.6326e-04	k_t
Generation 8700: Best = 1.365672e+04 = [4, 2, 3, 2]	FES = 870100	Diversity = 7.0176e-04	k_t
Generation 8710: Best = 1.365672e+04 = [1, 1, 1, 1]	FES = 871100	Diversity = 7.2704e-04	k_t
Generation 8720: Best = 1.365672e+04 = [3, 2, 2, 3]	FES = 872100	Diversity = 7.6887e-04	k_t
Generation 8730: Best = 1.365672e+04 = [1, 1, 1, 1]	FES = 873100	Diversity = 8.4129e-04	k_t
Generation 8740: Best = 1.365671e+04 = [2, 2, 4, 4]	FES = 874100	Diversity = 8.7001e-04	k_t
Generation 8750: Best = 1.365671e+04 = [1, 1, 1, 1]	FES = 875100	Diversity = 9.1441e-04	k_t
Generation 8760: Best = 1.365671e+04 = [2, 3, 2, 3]	FES = 876100	Diversity = 8.1980e-04	k_t
Generation 8770: Best = 1.365671e+04 = [1, 1, 1, 1]	FES = 877100	Diversity = 8.0217e-04	k_t
Generation 8780: Best = 1.365671e+04 = [2, 4, 3, 3]	FES = 878100	Diversity = 8.8473e-04	k_t
Generation 8790: Best = 1.365670e+04 = [1, 1, 1, 1]	FES = 879100	Diversity = 9.3760e-04	k_t
Generation 8800: Best = 1.365670e+04 = [3, 4, 3, 2]	FES = 880100	Diversity = 8.6443e-04	k_t
Generation 8810: Best = 1.365670e+04 = [1, 1, 1, 1]	FES = 881100	Diversity = 8.0815e-04	k_t
Generation 8820: Best = 1.365670e+04 = [3, 2, 2, 2]	FES = 882100	Diversity = 7.5231e-04	k_t
Generation 8830: Best = 1.365670e+04 = [1, 1, 1, 1]	FES = 883100	Diversity = 6.9327e-04	k_t
Generation 8840: Best = 1.365669e+04 = [4, 2, 3, 3]	FES = 884100	Diversity = 6.5139e-04	k_t
Generation 8850: Best = 1.365669e+04 = [1, 1, 1, 1]	FES = 885100	Diversity = 6.4125e-04	k_t
Generation 8860: Best = 1.365669e+04 = [4, 3, 4, 2]	FES = 886100	Diversity = 6.4091e-04	k_t

Generation 8870: Best = 1.365669e+04 = [1, 1, 1, 1]	FES = 887100	Diversity = 6.6164e-04	k_t
Generation 8880: Best = 1.365669e+04 = [3, 4, 3, 4]	FES = 888100	Diversity = 6.8644e-04	k_t
Generation 8890: Best = 1.365669e+04 = [1, 1, 1, 1]	FES = 889100	Diversity = 7.2593e-04	k_t
Generation 8900: Best = 1.365668e+04 = [3, 3, 4, 2]	FES = 890100	Diversity = 6.9037e-04	k_t
Generation 8910: Best = 1.365668e+04 = [1, 1, 1, 1]	FES = 891100	Diversity = 7.0410e-04	k_t
Generation 8920: Best = 1.365668e+04 = [4, 2, 2, 3]	FES = 892100	Diversity = 8.7319e-04	k_t
Generation 8930: Best = 1.365667e+04 = [1, 1, 1, 1]	FES = 893100	Diversity = 9.4033e-04	k_t
Generation 8940: Best = 1.365667e+04 = [2, 3, 3, 4]	FES = 894100	Diversity = 9.6813e-04	k_t
Generation 8950: Best = 1.365666e+04 = [1, 1, 1, 1]	FES = 895100	Diversity = 1.0843e-03	k_t
Generation 8960: Best = 1.365666e+04 = [3, 4, 4, 2]	FES = 896100	Diversity = 1.1853e-03	k_t
Generation 8970: Best = 1.365665e+04 = [1, 1, 1, 1]	FES = 897100	Diversity = 1.3580e-03	k_t
Generation 8980: Best = 1.365664e+04 = [4, 3, 3, 4]	FES = 898100	Diversity = 1.3690e-03	k_t
Generation 8990: Best = 1.365664e+04 = [1, 1, 1, 1]	FES = 899100	Diversity = 1.2995e-03	k_t
Generation 9000: Best = 1.365663e+04 = [3, 4, 3, 2]	FES = 900100	Diversity = 1.2818e-03	k_t
Generation 9010: Best = 1.365663e+04 = [1, 1, 1, 1]	FES = 901100	Diversity = 1.2960e-03	k_t
Generation 9020: Best = 1.365663e+04 = [4, 3, 4, 4]	FES = 902100	Diversity = 1.2219e-03	k_t
Generation 9030: Best = 1.365662e+04 = [1, 1, 1, 1]	FES = 903100	Diversity = 1.2717e-03	k_t
Generation 9040: Best = 1.365661e+04 = [2, 3, 3, 4]	FES = 904100	Diversity = 1.2163e-03	k_t
Generation 9050: Best = 1.365661e+04 = [1, 1, 1, 1]	FES = 905100	Diversity = 1.2425e-03	k_t
Generation 9060: Best = 1.365660e+04 = [3, 4, 2, 4]	FES = 906100	Diversity = 1.2831e-03	k_t
Generation 9070: Best = 1.365660e+04 = [1, 1, 1, 1]	FES = 907100	Diversity = 1.3146e-03	k_t
Generation 9080: Best = 1.365659e+04 = [4, 3, 4, 2]	FES = 908100	Diversity = 1.2773e-03	k_t
Generation 9090: Best = 1.365659e+04 = [1, 1, 1, 1]	FES = 909100	Diversity = 1.3083e-03	k_t
Generation 9100: Best = 1.365658e+04 = [2, 4, 2, 2]	FES = 910100	Diversity = 1.3198e-03	k_t
Generation 9110: Best = 1.365658e+04 = [1, 1, 1, 1]	FES = 911100	Diversity = 1.3507e-03	k_t
Generation 9120: Best = 1.365658e+04 = [4, 3, 4, 3]	FES = 912100	Diversity = 1.2599e-03	k_t
Generation 9130: Best = 1.365657e+04 = [1, 1, 1, 1]	FES = 913100	Diversity = 1.2526e-03	k_t

Generation 9140: Best = 1.365656e+04 = [2, 4, 4, 4]	FES = 914100	Diversity = 1.2641e-03	k_t
Generation 9150: Best = 1.365656e+04 = [1, 1, 1, 1]	FES = 915100	Diversity = 1.4569e-03	k_t
Generation 9160: Best = 1.365655e+04 = [3, 2, 2, 4]	FES = 916100	Diversity = 1.5248e-03	k_t
Generation 9170: Best = 1.365653e+04 = [1, 1, 1, 1]	FES = 917100	Diversity = 1.7527e-03	k_t
Generation 9180: Best = 1.365651e+04 = [3, 2, 3, 4]	FES = 918100	Diversity = 1.8561e-03	k_t
Generation 9190: Best = 1.365649e+04 = [1, 1, 1, 1]	FES = 919100	Diversity = 2.0995e-03	k_t
Generation 9200: Best = 1.365647e+04 = [4, 2, 4, 3]	FES = 920100	Diversity = 2.2644e-03	k_t
Generation 9210: Best = 1.365642e+04 = [1, 1, 1, 1]	FES = 921100	Diversity = 2.6969e-03	k_t
Generation 9220: Best = 1.365640e+04 = [2, 2, 4, 3]	FES = 922100	Diversity = 3.1620e-03	k_t
Generation 9230: Best = 1.365637e+04 = [1, 1, 1, 1]	FES = 923100	Diversity = 3.2717e-03	k_t
Generation 9240: Best = 1.365634e+04 = [2, 4, 2, 4]	FES = 924100	Diversity = 3.3669e-03	k_t
Generation 9250: Best = 1.365631e+04 = [1, 1, 1, 1]	FES = 925100	Diversity = 3.5408e-03	k_t
Generation 9260: Best = 1.365630e+04 = [2, 2, 3, 4]	FES = 926100	Diversity = 3.1070e-03	k_t
Generation 9270: Best = 1.365629e+04 = [1, 1, 1, 1]	FES = 927100	Diversity = 2.9626e-03	k_t
Generation 9280: Best = 1.365628e+04 = [4, 3, 2, 4]	FES = 928100	Diversity = 2.9376e-03	k_t
Generation 9290: Best = 1.365624e+04 = [1, 1, 1, 1]	FES = 929100	Diversity = 2.6932e-03	k_t
Generation 9300: Best = 1.365623e+04 = [2, 3, 4, 2]	FES = 930100	Diversity = 2.6560e-03	k_t
Generation 9310: Best = 1.365622e+04 = [1, 1, 1, 1]	FES = 931100	Diversity = 2.5254e-03	k_t
Generation 9320: Best = 1.365622e+04 = [4, 3, 2, 3]	FES = 932100	Diversity = 2.3145e-03	k_t
Generation 9330: Best = 1.365621e+04 = [1, 1, 1, 1]	FES = 933100	Diversity = 2.0991e-03	k_t
Generation 9340: Best = 1.365620e+04 = [2, 2, 2, 2]	FES = 934100	Diversity = 1.9828e-03	k_t
Generation 9350: Best = 1.365619e+04 = [1, 1, 1, 1]	FES = 935100	Diversity = 1.8409e-03	k_t
Generation 9360: Best = 1.365619e+04 = [4, 3, 3, 2]	FES = 936100	Diversity = 1.8560e-03	k_t
Generation 9370: Best = 1.365618e+04 = [1, 1, 1, 1]	FES = 937100	Diversity = 1.6521e-03	k_t
Generation 9380: Best = 1.365617e+04 = [3, 2, 4, 4]	FES = 938100	Diversity = 1.4854e-03	k_t
Generation 9390: Best = 1.365617e+04 = [1, 1, 1, 1]	FES = 939100	Diversity = 1.3814e-03	k_t
Generation 9400: Best = 1.365616e+04 = [2, 3, 3, 4]	FES = 940100	Diversity = 1.2447e-03	k_t

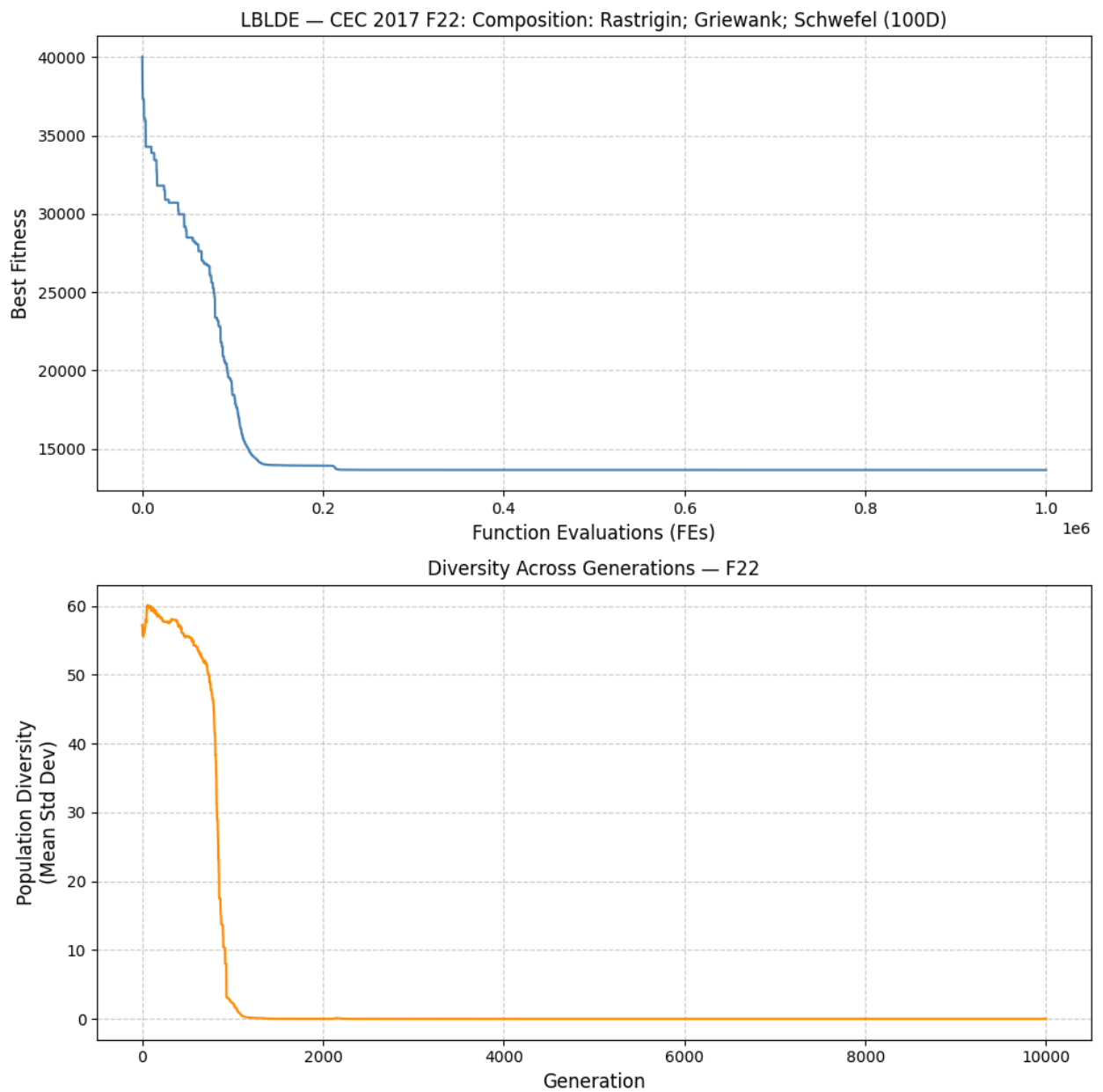
Generation 9410: Best = 1.365616e+04 = [1, 1, 1, 1]	FES = 941100	Diversity = 1.1463e-03	k_t
Generation 9420: Best = 1.365616e+04 = [2, 3, 3, 2]	FES = 942100	Diversity = 1.0913e-03	k_t
Generation 9430: Best = 1.365616e+04 = [1, 1, 1, 1]	FES = 943100	Diversity = 1.0249e-03	k_t
Generation 9440: Best = 1.365615e+04 = [4, 4, 2, 3]	FES = 944100	Diversity = 1.0204e-03	k_t
Generation 9450: Best = 1.365615e+04 = [1, 1, 1, 1]	FES = 945100	Diversity = 9.8598e-04	k_t
Generation 9460: Best = 1.365615e+04 = [2, 4, 3, 4]	FES = 946100	Diversity = 9.7191e-04	k_t
Generation 9470: Best = 1.365615e+04 = [1, 1, 1, 1]	FES = 947100	Diversity = 9.4913e-04	k_t
Generation 9480: Best = 1.365614e+04 = [2, 3, 4, 4]	FES = 948100	Diversity = 9.1450e-04	k_t
Generation 9490: Best = 1.365614e+04 = [1, 1, 1, 1]	FES = 949100	Diversity = 8.5478e-04	k_t
Generation 9500: Best = 1.365614e+04 = [3, 3, 3, 4]	FES = 950100	Diversity = 8.2707e-04	k_t
Generation 9510: Best = 1.365614e+04 = [1, 1, 1, 1]	FES = 951100	Diversity = 7.4104e-04	k_t
Generation 9520: Best = 1.365614e+04 = [3, 4, 2, 2]	FES = 952100	Diversity = 7.4091e-04	k_t
Generation 9530: Best = 1.365614e+04 = [1, 1, 1, 1]	FES = 953100	Diversity = 7.4488e-04	k_t
Generation 9540: Best = 1.365613e+04 = [4, 3, 3, 4]	FES = 954100	Diversity = 6.9634e-04	k_t
Generation 9550: Best = 1.365613e+04 = [1, 1, 1, 1]	FES = 955100	Diversity = 7.3970e-04	k_t
Generation 9560: Best = 1.365613e+04 = [2, 4, 4, 3]	FES = 956100	Diversity = 6.9977e-04	k_t
Generation 9570: Best = 1.365613e+04 = [1, 1, 1, 1]	FES = 957100	Diversity = 6.4900e-04	k_t
Generation 9580: Best = 1.365613e+04 = [4, 2, 3, 3]	FES = 958100	Diversity = 5.8425e-04	k_t
Generation 9590: Best = 1.365613e+04 = [1, 1, 1, 1]	FES = 959100	Diversity = 6.1239e-04	k_t
Generation 9600: Best = 1.365613e+04 = [2, 2, 4, 4]	FES = 960100	Diversity = 6.0829e-04	k_t
Generation 9610: Best = 1.365612e+04 = [1, 1, 1, 1]	FES = 961100	Diversity = 6.0447e-04	k_t
Generation 9620: Best = 1.365612e+04 = [4, 4, 4, 4]	FES = 962100	Diversity = 6.2215e-04	k_t
Generation 9630: Best = 1.365612e+04 = [1, 1, 1, 1]	FES = 963100	Diversity = 6.0807e-04	k_t
Generation 9640: Best = 1.365612e+04 = [4, 4, 3, 2]	FES = 964100	Diversity = 5.8141e-04	k_t
Generation 9650: Best = 1.365612e+04 = [1, 1, 1, 1]	FES = 965100	Diversity = 5.9476e-04	k_t
Generation 9660: Best = 1.365612e+04 = [4, 4, 2, 3]	FES = 966100	Diversity = 6.3041e-04	k_t
Generation 9670: Best = 1.365612e+04 = [1, 1, 1, 1]	FES = 967100	Diversity = 6.3168e-04	k_t

Generation 9680: Best = 1.365612e+04 = [4, 3, 4, 3]	FES = 968100	Diversity = 5.8964e-04	k_t
Generation 9690: Best = 1.365611e+04 = [1, 1, 1, 1]	FES = 969100	Diversity = 5.5078e-04	k_t
Generation 9700: Best = 1.365611e+04 = [4, 3, 4, 2]	FES = 970100	Diversity = 5.8382e-04	k_t
Generation 9710: Best = 1.365611e+04 = [1, 1, 1, 1]	FES = 971100	Diversity = 6.0187e-04	k_t
Generation 9720: Best = 1.365611e+04 = [2, 4, 4, 2]	FES = 972100	Diversity = 5.7564e-04	k_t
Generation 9730: Best = 1.365611e+04 = [1, 1, 1, 1]	FES = 973100	Diversity = 5.2458e-04	k_t
Generation 9740: Best = 1.365611e+04 = [4, 4, 4, 2]	FES = 974100	Diversity = 5.1268e-04	k_t
Generation 9750: Best = 1.365611e+04 = [1, 1, 1, 1]	FES = 975100	Diversity = 5.0551e-04	k_t
Generation 9760: Best = 1.365611e+04 = [3, 2, 3, 4]	FES = 976100	Diversity = 5.3598e-04	k_t
Generation 9770: Best = 1.365611e+04 = [1, 1, 1, 1]	FES = 977100	Diversity = 5.9530e-04	k_t
Generation 9780: Best = 1.365611e+04 = [2, 3, 2, 3]	FES = 978100	Diversity = 5.9707e-04	k_t
Generation 9790: Best = 1.365610e+04 = [1, 1, 1, 1]	FES = 979100	Diversity = 6.1014e-04	k_t
Generation 9800: Best = 1.365610e+04 = [4, 2, 3, 2]	FES = 980100	Diversity = 6.2649e-04	k_t
Generation 9810: Best = 1.365610e+04 = [1, 1, 1, 1]	FES = 981100	Diversity = 6.2031e-04	k_t
Generation 9820: Best = 1.365610e+04 = [3, 3, 3, 2]	FES = 982100	Diversity = 6.8039e-04	k_t
Generation 9830: Best = 1.365610e+04 = [1, 1, 1, 1]	FES = 983100	Diversity = 6.7590e-04	k_t
Generation 9840: Best = 1.365610e+04 = [2, 3, 4, 2]	FES = 984100	Diversity = 6.6674e-04	k_t
Generation 9850: Best = 1.365610e+04 = [1, 1, 1, 1]	FES = 985100	Diversity = 6.3538e-04	k_t
Generation 9860: Best = 1.365610e+04 = [4, 2, 4, 3]	FES = 986100	Diversity = 6.2264e-04	k_t
Generation 9870: Best = 1.365609e+04 = [1, 1, 1, 1]	FES = 987100	Diversity = 6.4260e-04	k_t
Generation 9880: Best = 1.365609e+04 = [3, 4, 2, 3]	FES = 988100	Diversity = 6.4328e-04	k_t
Generation 9890: Best = 1.365609e+04 = [1, 1, 1, 1]	FES = 989100	Diversity = 5.9761e-04	k_t
Generation 9900: Best = 1.365609e+04 = [2, 4, 3, 4]	FES = 990100	Diversity = 5.5004e-04	k_t
Generation 9910: Best = 1.365609e+04 = [1, 1, 1, 1]	FES = 991100	Diversity = 4.9232e-04	k_t
Generation 9920: Best = 1.365609e+04 = [3, 4, 3, 2]	FES = 992100	Diversity = 4.6710e-04	k_t
Generation 9930: Best = 1.365609e+04 = [1, 1, 1, 1]	FES = 993100	Diversity = 4.7289e-04	k_t
Generation 9940: Best = 1.365609e+04 = [3, 4, 4, 2]	FES = 994100	Diversity = 4.7890e-04	k_t

Generation 9950: Best = 1.365609e+04 FES = 995100 Diversity = 4.4516e-04 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 1.365609e+04 FES = 996100 Diversity = 4.4561e-04 k_t
= [4, 3, 2, 3]
Generation 9970: Best = 1.365609e+04 FES = 997100 Diversity = 4.9148e-04 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 1.365609e+04 FES = 998100 Diversity = 4.9638e-04 k_t
= [2, 3, 3, 3]
Generation 9990: Best = 1.365609e+04 FES = 999100 Diversity = 4.7606e-04 k_t
= [1, 1, 1, 1]

Optimization completed!
Final Best Fitness = 1.365609e+04
Total Function Evaluations = 1000000

=====
=
Results:
Best Fitness : 1.365609e+04
Global Optimum : 2.200000e+03
Error (|best-optimum|) : 1.145609e+04
Best Solution (5 dim) : [53.89566867 23.83138548 -7.30570557 -95.61502274
-21.70109578]
D^(0) (init. diversity): 5.7220e+01
=====



Graph saved as convergence_F22.png

```
In [47]: # =====
# F23 — Composition Function 3 (Rosenbrock; Ackley; Schwefel; Rastrigin)
# =====
run_and_plot('F23', make_wrapper(composition.f23), 2300.0, 'Composition: Rosen
```

```

=====
=
LBLDE_s on CEC 2017 F23 – Single Run
=====
=
Function      : Composition: Rosenbrock; Ackley; Schwefel; Rastrigin (100D)
Bounds       : [-100, 100]
Global optimum : 2300.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 8.120533e+03 | Diversity = 5.7220e+01
Generation 10: Best = 5.895303e+03 FES = 1100 Diversity = 5.5447e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 4.869165e+03 FES = 2100 Diversity = 4.7640e+01 k_t
= [4, 4, 2, 2]
Generation 30: Best = 4.407195e+03 FES = 3100 Diversity = 3.8928e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 4.176798e+03 FES = 4100 Diversity = 3.2892e+01 k_t
= [2, 2, 2, 4]
Generation 50: Best = 4.108730e+03 FES = 5100 Diversity = 2.8943e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 4.034725e+03 FES = 6100 Diversity = 2.6676e+01 k_t
= [4, 2, 2, 3]
Generation 70: Best = 3.933203e+03 FES = 7100 Diversity = 2.4672e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 3.919507e+03 FES = 8100 Diversity = 2.3133e+01 k_t
= [3, 3, 3, 3]
Generation 90: Best = 3.896298e+03 FES = 9100 Diversity = 2.1777e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 3.841344e+03 FES = 10100 Diversity = 2.1107e+01 k_t
= [2, 2, 4, 4]
Generation 110: Best = 3.834629e+03 FES = 11100 Diversity = 2.0088e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 3.819246e+03 FES = 12100 Diversity = 1.9816e+01 k_t
= [3, 3, 3, 3]
Generation 130: Best = 3.788052e+03 FES = 13100 Diversity = 1.9772e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.751895e+03 FES = 14100 Diversity = 1.9340e+01 k_t
= [4, 4, 3, 3]
Generation 150: Best = 3.731999e+03 FES = 15100 Diversity = 1.9008e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.731999e+03 FES = 16100 Diversity = 1.9050e+01 k_t
= [2, 2, 3, 2]
Generation 170: Best = 3.728271e+03 FES = 17100 Diversity = 1.8886e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.704478e+03 FES = 18100 Diversity = 1.8775e+01 k_t
= [2, 2, 3, 2]
Generation 190: Best = 3.693492e+03 FES = 19100 Diversity = 1.8719e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.684769e+03 FES = 20100 Diversity = 1.8652e+01 k_t
= [2, 3, 4, 4]
Generation 210: Best = 3.684769e+03 FES = 21100 Diversity = 1.8527e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.658523e+03 FES = 22100 Diversity = 1.8394e+01 k_t
= [2, 3, 3, 3]

```

Generation 230: Best = 3.643440e+03	FES = 23100	Diversity = 1.8418e+01	k_t
= [1, 1, 1, 1]			
Generation 240: Best = 3.638326e+03	FES = 24100	Diversity = 1.8336e+01	k_t
= [4, 3, 3, 4]			
Generation 250: Best = 3.628674e+03	FES = 25100	Diversity = 1.8216e+01	k_t
= [1, 1, 1, 1]			
Generation 260: Best = 3.610933e+03	FES = 26100	Diversity = 1.8135e+01	k_t
= [3, 2, 4, 2]			
Generation 270: Best = 3.610933e+03	FES = 27100	Diversity = 1.8126e+01	k_t
= [1, 1, 1, 1]			
Generation 280: Best = 3.601127e+03	FES = 28100	Diversity = 1.8129e+01	k_t
= [2, 3, 2, 4]			
Generation 290: Best = 3.592526e+03	FES = 29100	Diversity = 1.7984e+01	k_t
= [1, 1, 1, 1]			
Generation 300: Best = 3.592526e+03	FES = 30100	Diversity = 1.7828e+01	k_t
= [2, 3, 2, 2]			
Generation 310: Best = 3.592526e+03	FES = 31100	Diversity = 1.7786e+01	k_t
= [1, 1, 1, 1]			
Generation 320: Best = 3.592526e+03	FES = 32100	Diversity = 1.7787e+01	k_t
= [4, 3, 4, 2]			
Generation 330: Best = 3.574367e+03	FES = 33100	Diversity = 1.7731e+01	k_t
= [1, 1, 1, 1]			
Generation 340: Best = 3.574367e+03	FES = 34100	Diversity = 1.7725e+01	k_t
= [3, 3, 3, 3]			
Generation 350: Best = 3.574367e+03	FES = 35100	Diversity = 1.7670e+01	k_t
= [1, 1, 1, 1]			
Generation 360: Best = 3.574367e+03	FES = 36100	Diversity = 1.7664e+01	k_t
= [4, 3, 4, 3]			
Generation 370: Best = 3.574367e+03	FES = 37100	Diversity = 1.7516e+01	k_t
= [1, 1, 1, 1]			
Generation 380: Best = 3.574367e+03	FES = 38100	Diversity = 1.7493e+01	k_t
= [4, 3, 4, 3]			
Generation 390: Best = 3.552509e+03	FES = 39100	Diversity = 1.7548e+01	k_t
= [1, 1, 1, 1]			
Generation 400: Best = 3.552509e+03	FES = 40100	Diversity = 1.7555e+01	k_t
= [3, 3, 4, 2]			
Generation 410: Best = 3.545930e+03	FES = 41100	Diversity = 1.7548e+01	k_t
= [1, 1, 1, 1]			
Generation 420: Best = 3.545930e+03	FES = 42100	Diversity = 1.7510e+01	k_t
= [3, 3, 4, 4]			
Generation 430: Best = 3.545930e+03	FES = 43100	Diversity = 1.7537e+01	k_t
= [1, 1, 1, 1]			
Generation 440: Best = 3.538245e+03	FES = 44100	Diversity = 1.7519e+01	k_t
= [4, 3, 4, 4]			
Generation 450: Best = 3.500905e+03	FES = 45100	Diversity = 1.7404e+01	k_t
= [1, 1, 1, 1]			
Generation 460: Best = 3.497638e+03	FES = 46100	Diversity = 1.7540e+01	k_t
= [3, 4, 3, 2]			
Generation 470: Best = 3.497638e+03	FES = 47100	Diversity = 1.7582e+01	k_t
= [1, 1, 1, 1]			
Generation 480: Best = 3.497638e+03	FES = 48100	Diversity = 1.7492e+01	k_t
= [2, 2, 2, 4]			
Generation 490: Best = 3.497638e+03	FES = 49100	Diversity = 1.7490e+01	k_t
= [1, 1, 1, 1]			

Generation 500:	Best = 3.497638e+03	FES = 50100	Diversity = 1.7480e+01	k_t = [2, 2, 4, 4]
Generation 510:	Best = 3.497638e+03	FES = 51100	Diversity = 1.7452e+01	k_t = [1, 1, 1, 1]
Generation 520:	Best = 3.497638e+03	FES = 52100	Diversity = 1.7454e+01	k_t = [4, 2, 3, 3]
Generation 530:	Best = 3.497638e+03	FES = 53100	Diversity = 1.7477e+01	k_t = [1, 1, 1, 1]
Generation 540:	Best = 3.495120e+03	FES = 54100	Diversity = 1.7424e+01	k_t = [4, 2, 2, 3]
Generation 550:	Best = 3.495120e+03	FES = 55100	Diversity = 1.7397e+01	k_t = [1, 1, 1, 1]
Generation 560:	Best = 3.491054e+03	FES = 56100	Diversity = 1.7335e+01	k_t = [4, 3, 2, 4]
Generation 570:	Best = 3.446402e+03	FES = 57100	Diversity = 1.7330e+01	k_t = [1, 1, 1, 1]
Generation 580:	Best = 3.446402e+03	FES = 58100	Diversity = 1.7308e+01	k_t = [3, 3, 3, 2]
Generation 590:	Best = 3.446402e+03	FES = 59100	Diversity = 1.7327e+01	k_t = [1, 1, 1, 1]
Generation 600:	Best = 3.446402e+03	FES = 60100	Diversity = 1.7312e+01	k_t = [4, 3, 3, 2]
Generation 610:	Best = 3.446402e+03	FES = 61100	Diversity = 1.7317e+01	k_t = [1, 1, 1, 1]
Generation 620:	Best = 3.446402e+03	FES = 62100	Diversity = 1.7281e+01	k_t = [3, 3, 4, 2]
Generation 630:	Best = 3.446402e+03	FES = 63100	Diversity = 1.7279e+01	k_t = [1, 1, 1, 1]
Generation 640:	Best = 3.446402e+03	FES = 64100	Diversity = 1.7260e+01	k_t = [4, 4, 2, 4]
Generation 650:	Best = 3.446402e+03	FES = 65100	Diversity = 1.7199e+01	k_t = [1, 1, 1, 1]
Generation 660:	Best = 3.427164e+03	FES = 66100	Diversity = 1.7204e+01	k_t = [2, 4, 4, 3]
Generation 670:	Best = 3.427164e+03	FES = 67100	Diversity = 1.7174e+01	k_t = [1, 1, 1, 1]
Generation 680:	Best = 3.427164e+03	FES = 68100	Diversity = 1.7134e+01	k_t = [2, 3, 2, 2]
Generation 690:	Best = 3.427164e+03	FES = 69100	Diversity = 1.7122e+01	k_t = [1, 1, 1, 1]
Generation 700:	Best = 3.424708e+03	FES = 70100	Diversity = 1.7092e+01	k_t = [2, 3, 4, 2]
Generation 710:	Best = 3.422140e+03	FES = 71100	Diversity = 1.7081e+01	k_t = [1, 1, 1, 1]
Generation 720:	Best = 3.422140e+03	FES = 72100	Diversity = 1.7083e+01	k_t = [4, 3, 3, 4]
Generation 730:	Best = 3.418473e+03	FES = 73100	Diversity = 1.7102e+01	k_t = [1, 1, 1, 1]
Generation 740:	Best = 3.418473e+03	FES = 74100	Diversity = 1.7109e+01	k_t = [4, 3, 4, 2]
Generation 750:	Best = 3.418473e+03	FES = 75100	Diversity = 1.7109e+01	k_t = [1, 1, 1, 1]
Generation 760:	Best = 3.411842e+03	FES = 76100	Diversity = 1.7126e+01	k_t = [4, 2, 2, 3]

Generation 770:	Best = 3.410412e+03	FES = 77100	Diversity = 1.7118e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 3.410412e+03	FES = 78100	Diversity = 1.7114e+01	k_t
= [3, 2, 2, 4]				
Generation 790:	Best = 3.410412e+03	FES = 79100	Diversity = 1.7130e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 3.409083e+03	FES = 80100	Diversity = 1.7133e+01	k_t
= [4, 3, 4, 2]				
Generation 810:	Best = 3.398169e+03	FES = 81100	Diversity = 1.7121e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 3.398048e+03	FES = 82100	Diversity = 1.7143e+01	k_t
= [4, 3, 3, 2]				
Generation 830:	Best = 3.386937e+03	FES = 83100	Diversity = 1.7109e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 3.376350e+03	FES = 84100	Diversity = 1.7063e+01	k_t
= [3, 4, 2, 2]				
Generation 850:	Best = 3.375860e+03	FES = 85100	Diversity = 1.7051e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 3.375860e+03	FES = 86100	Diversity = 1.7001e+01	k_t
= [4, 3, 3, 3]				
Generation 870:	Best = 3.342719e+03	FES = 87100	Diversity = 1.6982e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 3.342719e+03	FES = 88100	Diversity = 1.6972e+01	k_t
= [3, 4, 3, 4]				
Generation 890:	Best = 3.342719e+03	FES = 89100	Diversity = 1.6901e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 3.342719e+03	FES = 90100	Diversity = 1.6873e+01	k_t
= [3, 3, 2, 4]				
Generation 910:	Best = 3.342719e+03	FES = 91100	Diversity = 1.6849e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 3.342719e+03	FES = 92100	Diversity = 1.6816e+01	k_t
= [3, 2, 4, 4]				
Generation 930:	Best = 3.342719e+03	FES = 93100	Diversity = 1.6804e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 3.342719e+03	FES = 94100	Diversity = 1.6789e+01	k_t
= [3, 3, 2, 3]				
Generation 950:	Best = 3.342719e+03	FES = 95100	Diversity = 1.6754e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 3.342233e+03	FES = 96100	Diversity = 1.6729e+01	k_t
= [2, 3, 3, 2]				
Generation 970:	Best = 3.333324e+03	FES = 97100	Diversity = 1.6666e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 3.333324e+03	FES = 98100	Diversity = 1.6639e+01	k_t
= [4, 4, 4, 2]				
Generation 990:	Best = 3.333324e+03	FES = 99100	Diversity = 1.6601e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 3.333324e+03	FES = 100100	Diversity = 1.6544e+01	k_t
= [2, 4, 3, 3]				
Generation 1010:	Best = 3.333324e+03	FES = 101100	Diversity = 1.6561e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 3.333324e+03	FES = 102100	Diversity = 1.6549e+01	k_t
= [4, 2, 2, 2]				
Generation 1030:	Best = 3.330672e+03	FES = 103100	Diversity = 1.6534e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 3.330672e+03 = [4, 4, 2, 2]	FES = 104100	Diversity = 1.6520e+01	k_t
Generation 1050: Best = 3.330672e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.6487e+01	k_t
Generation 1060: Best = 3.330672e+03 = [2, 3, 4, 4]	FES = 106100	Diversity = 1.6492e+01	k_t
Generation 1070: Best = 3.326025e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 1.6414e+01	k_t
Generation 1080: Best = 3.326025e+03 = [2, 2, 2, 2]	FES = 108100	Diversity = 1.6403e+01	k_t
Generation 1090: Best = 3.326025e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.6379e+01	k_t
Generation 1100: Best = 3.326025e+03 = [3, 3, 4, 4]	FES = 110100	Diversity = 1.6371e+01	k_t
Generation 1110: Best = 3.326025e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.6350e+01	k_t
Generation 1120: Best = 3.326025e+03 = [2, 2, 2, 3]	FES = 112100	Diversity = 1.6366e+01	k_t
Generation 1130: Best = 3.325754e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.6345e+01	k_t
Generation 1140: Best = 3.322417e+03 = [3, 2, 2, 3]	FES = 114100	Diversity = 1.6319e+01	k_t
Generation 1150: Best = 3.322417e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.6282e+01	k_t
Generation 1160: Best = 3.317724e+03 = [4, 4, 4, 2]	FES = 116100	Diversity = 1.6280e+01	k_t
Generation 1170: Best = 3.317724e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.6265e+01	k_t
Generation 1180: Best = 3.317724e+03 = [4, 3, 2, 4]	FES = 118100	Diversity = 1.6259e+01	k_t
Generation 1190: Best = 3.317724e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.6276e+01	k_t
Generation 1200: Best = 3.317724e+03 = [4, 4, 3, 2]	FES = 120100	Diversity = 1.6309e+01	k_t
Generation 1210: Best = 3.317724e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.6261e+01	k_t
Generation 1220: Best = 3.312738e+03 = [3, 4, 3, 4]	FES = 122100	Diversity = 1.6173e+01	k_t
Generation 1230: Best = 3.312738e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.6133e+01	k_t
Generation 1240: Best = 3.312738e+03 = [4, 3, 4, 4]	FES = 124100	Diversity = 1.6135e+01	k_t
Generation 1250: Best = 3.312738e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 1.6120e+01	k_t
Generation 1260: Best = 3.301650e+03 = [3, 4, 2, 4]	FES = 126100	Diversity = 1.6129e+01	k_t
Generation 1270: Best = 3.301650e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.6132e+01	k_t
Generation 1280: Best = 3.301650e+03 = [2, 2, 2, 3]	FES = 128100	Diversity = 1.6079e+01	k_t
Generation 1290: Best = 3.301650e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.6074e+01	k_t
Generation 1300: Best = 3.301650e+03 = [4, 2, 2, 3]	FES = 130100	Diversity = 1.6082e+01	k_t

Generation 1310: Best = 3.301650e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.5994e+01	k_t
Generation 1320: Best = 3.301650e+03 = [2, 4, 3, 4]	FES = 132100	Diversity = 1.5909e+01	k_t
Generation 1330: Best = 3.301650e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 1.5883e+01	k_t
Generation 1340: Best = 3.301650e+03 = [4, 3, 2, 2]	FES = 134100	Diversity = 1.5874e+01	k_t
Generation 1350: Best = 3.296571e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.5781e+01	k_t
Generation 1360: Best = 3.295043e+03 = [4, 2, 4, 3]	FES = 136100	Diversity = 1.5816e+01	k_t
Generation 1370: Best = 3.295043e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 1.5829e+01	k_t
Generation 1380: Best = 3.287945e+03 = [3, 3, 2, 2]	FES = 138100	Diversity = 1.5797e+01	k_t
Generation 1390: Best = 3.281379e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 1.5758e+01	k_t
Generation 1400: Best = 3.281119e+03 = [4, 3, 3, 4]	FES = 140100	Diversity = 1.5659e+01	k_t
Generation 1410: Best = 3.277043e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 1.5622e+01	k_t
Generation 1420: Best = 3.277043e+03 = [3, 2, 3, 4]	FES = 142100	Diversity = 1.5570e+01	k_t
Generation 1430: Best = 3.269075e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 1.5470e+01	k_t
Generation 1440: Best = 3.269075e+03 = [2, 4, 3, 2]	FES = 144100	Diversity = 1.5433e+01	k_t
Generation 1450: Best = 3.269075e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 1.5432e+01	k_t
Generation 1460: Best = 3.268186e+03 = [2, 2, 4, 4]	FES = 146100	Diversity = 1.5429e+01	k_t
Generation 1470: Best = 3.268186e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 1.5421e+01	k_t
Generation 1480: Best = 3.268186e+03 = [2, 2, 2, 4]	FES = 148100	Diversity = 1.5408e+01	k_t
Generation 1490: Best = 3.264193e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 1.5399e+01	k_t
Generation 1500: Best = 3.264193e+03 = [2, 4, 3, 2]	FES = 150100	Diversity = 1.5403e+01	k_t
Generation 1510: Best = 3.246333e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 1.5384e+01	k_t
Generation 1520: Best = 3.246333e+03 = [4, 2, 4, 4]	FES = 152100	Diversity = 1.5385e+01	k_t
Generation 1530: Best = 3.246333e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 1.5345e+01	k_t
Generation 1540: Best = 3.246333e+03 = [3, 2, 2, 2]	FES = 154100	Diversity = 1.5307e+01	k_t
Generation 1550: Best = 3.246333e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 1.5263e+01	k_t
Generation 1560: Best = 3.246333e+03 = [3, 2, 2, 3]	FES = 156100	Diversity = 1.5259e+01	k_t
Generation 1570: Best = 3.246225e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 1.5253e+01	k_t

Generation 1580: Best = 3.246225e+03	FES = 158100	Diversity = 1.5248e+01	k_t
= [3, 2, 3, 2]			
Generation 1590: Best = 3.246225e+03	FES = 159100	Diversity = 1.5243e+01	k_t
= [1, 1, 1, 1]			
Generation 1600: Best = 3.246225e+03	FES = 160100	Diversity = 1.5236e+01	k_t
= [2, 4, 2, 2]			
Generation 1610: Best = 3.229274e+03	FES = 161100	Diversity = 1.5217e+01	k_t
= [1, 1, 1, 1]			
Generation 1620: Best = 3.222739e+03	FES = 162100	Diversity = 1.5219e+01	k_t
= [4, 3, 4, 2]			
Generation 1630: Best = 3.216339e+03	FES = 163100	Diversity = 1.5222e+01	k_t
= [1, 1, 1, 1]			
Generation 1640: Best = 3.212316e+03	FES = 164100	Diversity = 1.5189e+01	k_t
= [4, 2, 2, 4]			
Generation 1650: Best = 3.209632e+03	FES = 165100	Diversity = 1.5185e+01	k_t
= [1, 1, 1, 1]			
Generation 1660: Best = 3.209632e+03	FES = 166100	Diversity = 1.5152e+01	k_t
= [3, 4, 2, 3]			
Generation 1670: Best = 3.209632e+03	FES = 167100	Diversity = 1.5143e+01	k_t
= [1, 1, 1, 1]			
Generation 1680: Best = 3.209632e+03	FES = 168100	Diversity = 1.5108e+01	k_t
= [3, 4, 4, 2]			
Generation 1690: Best = 3.209632e+03	FES = 169100	Diversity = 1.5056e+01	k_t
= [1, 1, 1, 1]			
Generation 1700: Best = 3.209632e+03	FES = 170100	Diversity = 1.5009e+01	k_t
= [2, 3, 4, 3]			
Generation 1710: Best = 3.208237e+03	FES = 171100	Diversity = 1.5020e+01	k_t
= [1, 1, 1, 1]			
Generation 1720: Best = 3.208237e+03	FES = 172100	Diversity = 1.5013e+01	k_t
= [3, 4, 3, 3]			
Generation 1730: Best = 3.208237e+03	FES = 173100	Diversity = 1.4898e+01	k_t
= [1, 1, 1, 1]			
Generation 1740: Best = 3.208237e+03	FES = 174100	Diversity = 1.4893e+01	k_t
= [3, 4, 4, 2]			
Generation 1750: Best = 3.208237e+03	FES = 175100	Diversity = 1.4886e+01	k_t
= [1, 1, 1, 1]			
Generation 1760: Best = 3.208237e+03	FES = 176100	Diversity = 1.4737e+01	k_t
= [4, 4, 4, 2]			
Generation 1770: Best = 3.206769e+03	FES = 177100	Diversity = 1.4745e+01	k_t
= [1, 1, 1, 1]			
Generation 1780: Best = 3.206769e+03	FES = 178100	Diversity = 1.4745e+01	k_t
= [2, 3, 3, 4]			
Generation 1790: Best = 3.191103e+03	FES = 179100	Diversity = 1.4709e+01	k_t
= [1, 1, 1, 1]			
Generation 1800: Best = 3.191103e+03	FES = 180100	Diversity = 1.4719e+01	k_t
= [4, 3, 4, 2]			
Generation 1810: Best = 3.191103e+03	FES = 181100	Diversity = 1.4676e+01	k_t
= [1, 1, 1, 1]			
Generation 1820: Best = 3.191103e+03	FES = 182100	Diversity = 1.4628e+01	k_t
= [3, 3, 4, 3]			
Generation 1830: Best = 3.191103e+03	FES = 183100	Diversity = 1.4591e+01	k_t
= [1, 1, 1, 1]			
Generation 1840: Best = 3.191103e+03	FES = 184100	Diversity = 1.4552e+01	k_t
= [4, 3, 4, 4]			

Generation 1850: Best = 3.191103e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 1.4551e+01	k_t
Generation 1860: Best = 3.191103e+03 = [3, 3, 3, 2]	FES = 186100	Diversity = 1.4502e+01	k_t
Generation 1870: Best = 3.191103e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 1.4478e+01	k_t
Generation 1880: Best = 3.191103e+03 = [3, 4, 3, 2]	FES = 188100	Diversity = 1.4378e+01	k_t
Generation 1890: Best = 3.191103e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 1.4261e+01	k_t
Generation 1900: Best = 3.188961e+03 = [4, 4, 4, 4]	FES = 190100	Diversity = 1.4177e+01	k_t
Generation 1910: Best = 3.187939e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 1.4153e+01	k_t
Generation 1920: Best = 3.187939e+03 = [2, 4, 2, 3]	FES = 192100	Diversity = 1.4086e+01	k_t
Generation 1930: Best = 3.187939e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 1.4071e+01	k_t
Generation 1940: Best = 3.187939e+03 = [2, 4, 3, 4]	FES = 194100	Diversity = 1.3923e+01	k_t
Generation 1950: Best = 3.185556e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 1.3689e+01	k_t
Generation 1960: Best = 3.185556e+03 = [3, 4, 3, 2]	FES = 196100	Diversity = 1.3557e+01	k_t
Generation 1970: Best = 3.179557e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 1.3110e+01	k_t
Generation 1980: Best = 3.163914e+03 = [3, 4, 4, 4]	FES = 198100	Diversity = 1.2969e+01	k_t
Generation 1990: Best = 3.163155e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 1.2872e+01	k_t
Generation 2000: Best = 3.162167e+03 = [3, 3, 2, 2]	FES = 200100	Diversity = 1.2684e+01	k_t
Generation 2010: Best = 3.161202e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 1.2425e+01	k_t
Generation 2020: Best = 3.138561e+03 = [2, 3, 3, 4]	FES = 202100	Diversity = 1.2347e+01	k_t
Generation 2030: Best = 3.133224e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 1.2236e+01	k_t
Generation 2040: Best = 3.133224e+03 = [3, 3, 4, 2]	FES = 204100	Diversity = 1.2143e+01	k_t
Generation 2050: Best = 3.127411e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 1.2017e+01	k_t
Generation 2060: Best = 3.124590e+03 = [4, 2, 4, 4]	FES = 206100	Diversity = 1.1828e+01	k_t
Generation 2070: Best = 3.108769e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 1.1582e+01	k_t
Generation 2080: Best = 3.100409e+03 = [2, 3, 2, 4]	FES = 208100	Diversity = 1.1208e+01	k_t
Generation 2090: Best = 3.094378e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 1.0923e+01	k_t
Generation 2100: Best = 3.087188e+03 = [3, 2, 4, 3]	FES = 210100	Diversity = 1.0145e+01	k_t
Generation 2110: Best = 3.080881e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 8.4357e+00	k_t

Generation 2120: Best = 3.076402e+03 = [3, 2, 2, 4]	FES = 212100	Diversity = 5.6345e+00	k_t
Generation 2130: Best = 3.065905e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 4.0135e+00	k_t
Generation 2140: Best = 3.061883e+03 = [2, 4, 2, 4]	FES = 214100	Diversity = 1.3193e+00	k_t
Generation 2150: Best = 3.053392e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 1.2201e+00	k_t
Generation 2160: Best = 3.047719e+03 = [3, 3, 2, 4]	FES = 216100	Diversity = 1.1635e+00	k_t
Generation 2170: Best = 3.041175e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 1.0350e+00	k_t
Generation 2180: Best = 3.035477e+03 = [3, 3, 2, 2]	FES = 218100	Diversity = 9.3768e-01	k_t
Generation 2190: Best = 3.030703e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 8.6370e-01	k_t
Generation 2200: Best = 3.026128e+03 = [3, 2, 4, 2]	FES = 220100	Diversity = 7.8697e-01	k_t
Generation 2210: Best = 3.024244e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 7.2449e-01	k_t
Generation 2220: Best = 3.020222e+03 = [2, 2, 2, 4]	FES = 222100	Diversity = 6.5476e-01	k_t
Generation 2230: Best = 3.016587e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 5.8001e-01	k_t
Generation 2240: Best = 3.014554e+03 = [2, 2, 2, 2]	FES = 224100	Diversity = 5.2242e-01	k_t
Generation 2250: Best = 3.011638e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 4.2288e-01	k_t
Generation 2260: Best = 3.010728e+03 = [3, 3, 2, 3]	FES = 226100	Diversity = 3.2319e-01	k_t
Generation 2270: Best = 3.009469e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 2.3925e-01	k_t
Generation 2280: Best = 3.008772e+03 = [4, 2, 4, 2]	FES = 228100	Diversity = 1.9405e-01	k_t
Generation 2290: Best = 3.007947e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 1.4730e-01	k_t
Generation 2300: Best = 3.007284e+03 = [2, 2, 4, 3]	FES = 230100	Diversity = 1.2423e-01	k_t
Generation 2310: Best = 3.006722e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 1.0390e-01	k_t
Generation 2320: Best = 3.006495e+03 = [4, 2, 2, 2]	FES = 232100	Diversity = 8.5905e-02	k_t
Generation 2330: Best = 3.006227e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 7.0809e-02	k_t
Generation 2340: Best = 3.006012e+03 = [3, 2, 3, 4]	FES = 234100	Diversity = 6.0257e-02	k_t
Generation 2350: Best = 3.005896e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 5.1436e-02	k_t
Generation 2360: Best = 3.005804e+03 = [3, 2, 2, 2]	FES = 236100	Diversity = 4.3000e-02	k_t
Generation 2370: Best = 3.005715e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 3.6396e-02	k_t
Generation 2380: Best = 3.005670e+03 = [4, 3, 3, 2]	FES = 238100	Diversity = 3.0885e-02	k_t

Generation 2390: Best = 3.005627e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 2.6303e-02	k_t
Generation 2400: Best = 3.005599e+03 = [2, 3, 3, 3]	FES = 240100	Diversity = 2.2529e-02	k_t
Generation 2410: Best = 3.005579e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 1.8653e-02	k_t
Generation 2420: Best = 3.005566e+03 = [4, 4, 4, 2]	FES = 242100	Diversity = 1.5263e-02	k_t
Generation 2430: Best = 3.005557e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 1.2647e-02	k_t
Generation 2440: Best = 3.005550e+03 = [4, 3, 3, 2]	FES = 244100	Diversity = 1.0589e-02	k_t
Generation 2450: Best = 3.005543e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 9.2293e-03	k_t
Generation 2460: Best = 3.005540e+03 = [4, 2, 4, 4]	FES = 246100	Diversity = 7.9828e-03	k_t
Generation 2470: Best = 3.005537e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 6.8219e-03	k_t
Generation 2480: Best = 3.005534e+03 = [4, 4, 4, 3]	FES = 248100	Diversity = 5.9958e-03	k_t
Generation 2490: Best = 3.005533e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 5.1839e-03	k_t
Generation 2500: Best = 3.005532e+03 = [4, 4, 4, 4]	FES = 250100	Diversity = 4.0632e-03	k_t
Generation 2510: Best = 3.005531e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 3.3519e-03	k_t
Generation 2520: Best = 3.005530e+03 = [4, 3, 4, 3]	FES = 252100	Diversity = 2.9558e-03	k_t
Generation 2530: Best = 3.005530e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 2.5385e-03	k_t
Generation 2540: Best = 3.005529e+03 = [3, 4, 4, 2]	FES = 254100	Diversity = 2.2088e-03	k_t
Generation 2550: Best = 3.005529e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.9392e-03	k_t
Generation 2560: Best = 3.005529e+03 = [3, 2, 2, 2]	FES = 256100	Diversity = 1.8366e-03	k_t
Generation 2570: Best = 3.005529e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.5445e-03	k_t
Generation 2580: Best = 3.005529e+03 = [4, 3, 4, 4]	FES = 258100	Diversity = 1.3524e-03	k_t
Generation 2590: Best = 3.005529e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.1547e-03	k_t
Generation 2600: Best = 3.005529e+03 = [2, 2, 3, 2]	FES = 260100	Diversity = 1.0123e-03	k_t
Generation 2610: Best = 3.005529e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 8.2182e-04	k_t
Generation 2620: Best = 3.005528e+03 = [4, 3, 3, 2]	FES = 262100	Diversity = 7.2086e-04	k_t
Generation 2630: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 6.2238e-04	k_t
Generation 2640: Best = 3.005528e+03 = [2, 2, 3, 2]	FES = 264100	Diversity = 5.4284e-04	k_t
Generation 2650: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 4.6940e-04	k_t

Generation 2660: Best = 3.005528e+03 = [2, 3, 3, 2]	FES = 266100	Diversity = 3.9176e-04	k_t
Generation 2670: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 3.5210e-04	k_t
Generation 2680: Best = 3.005528e+03 = [3, 4, 4, 3]	FES = 268100	Diversity = 3.1804e-04	k_t
Generation 2690: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 2.6651e-04	k_t
Generation 2700: Best = 3.005528e+03 = [4, 3, 2, 3]	FES = 270100	Diversity = 2.3197e-04	k_t
Generation 2710: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.9001e-04	k_t
Generation 2720: Best = 3.005528e+03 = [4, 4, 3, 4]	FES = 272100	Diversity = 1.5800e-04	k_t
Generation 2730: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.3470e-04	k_t
Generation 2740: Best = 3.005528e+03 = [4, 2, 3, 3]	FES = 274100	Diversity = 1.1310e-04	k_t
Generation 2750: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.0130e-04	k_t
Generation 2760: Best = 3.005528e+03 = [4, 2, 4, 2]	FES = 276100	Diversity = 8.9279e-05	k_t
Generation 2770: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 8.1437e-05	k_t
Generation 2780: Best = 3.005528e+03 = [2, 2, 3, 2]	FES = 278100	Diversity = 7.4340e-05	k_t
Generation 2790: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 6.4790e-05	k_t
Generation 2800: Best = 3.005528e+03 = [3, 2, 2, 2]	FES = 280100	Diversity = 5.4053e-05	k_t
Generation 2810: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 4.2352e-05	k_t
Generation 2820: Best = 3.005528e+03 = [4, 3, 3, 2]	FES = 282100	Diversity = 3.6597e-05	k_t
Generation 2830: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 3.1192e-05	k_t
Generation 2840: Best = 3.005528e+03 = [2, 4, 4, 2]	FES = 284100	Diversity = 2.7466e-05	k_t
Generation 2850: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 2.3199e-05	k_t
Generation 2860: Best = 3.005528e+03 = [3, 3, 3, 3]	FES = 286100	Diversity = 2.3408e-05	k_t
Generation 2870: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 2.2210e-05	k_t
Generation 2880: Best = 3.005528e+03 = [2, 3, 4, 3]	FES = 288100	Diversity = 2.0883e-05	k_t
Generation 2890: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.8610e-05	k_t
Generation 2900: Best = 3.005528e+03 = [2, 3, 4, 2]	FES = 290100	Diversity = 1.7789e-05	k_t
Generation 2910: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.4887e-05	k_t
Generation 2920: Best = 3.005528e+03 = [3, 3, 3, 4]	FES = 292100	Diversity = 1.2133e-05	k_t

Generation 2930: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 293100	Diversity = 9.8137e-06	k_t
Generation 2940: Best = 3.005528e+03 = [2, 3, 4, 4]	FES = 294100	Diversity = 7.8933e-06	k_t
Generation 2950: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 295100	Diversity = 6.2021e-06	k_t
Generation 2960: Best = 3.005528e+03 = [2, 4, 2, 3]	FES = 296100	Diversity = 5.0928e-06	k_t
Generation 2970: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 297100	Diversity = 4.0682e-06	k_t
Generation 2980: Best = 3.005528e+03 = [4, 2, 4, 4]	FES = 298100	Diversity = 3.1358e-06	k_t
Generation 2990: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 299100	Diversity = 2.7934e-06	k_t
Generation 3000: Best = 3.005528e+03 = [4, 4, 2, 3]	FES = 300100	Diversity = 2.5562e-06	k_t
Generation 3010: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 301100	Diversity = 2.3586e-06	k_t
Generation 3020: Best = 3.005528e+03 = [2, 4, 4, 3]	FES = 302100	Diversity = 2.1084e-06	k_t
Generation 3030: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 303100	Diversity = 1.8334e-06	k_t
Generation 3040: Best = 3.005528e+03 = [4, 2, 2, 2]	FES = 304100	Diversity = 1.6235e-06	k_t
Generation 3050: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 305100	Diversity = 1.4672e-06	k_t
Generation 3060: Best = 3.005528e+03 = [2, 3, 4, 2]	FES = 306100	Diversity = 1.4375e-06	k_t
Generation 3070: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 307100	Diversity = 1.5083e-06	k_t
Generation 3080: Best = 3.005528e+03 = [3, 2, 2, 4]	FES = 308100	Diversity = 1.4176e-06	k_t
Generation 3090: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 309100	Diversity = 1.1965e-06	k_t
Generation 3100: Best = 3.005528e+03 = [4, 2, 4, 4]	FES = 310100	Diversity = 1.0274e-06	k_t
Generation 3110: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 311100	Diversity = 8.0121e-07	k_t
Generation 3120: Best = 3.005528e+03 = [3, 4, 3, 2]	FES = 312100	Diversity = 6.8784e-07	k_t
Generation 3130: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 313100	Diversity = 5.8037e-07	k_t
Generation 3140: Best = 3.005528e+03 = [4, 2, 3, 3]	FES = 314100	Diversity = 4.6560e-07	k_t
Generation 3150: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 315100	Diversity = 3.8962e-07	k_t
Generation 3160: Best = 3.005528e+03 = [3, 4, 3, 4]	FES = 316100	Diversity = 3.1849e-07	k_t
Generation 3170: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 317100	Diversity = 2.7212e-07	k_t
Generation 3180: Best = 3.005528e+03 = [2, 4, 2, 2]	FES = 318100	Diversity = 2.4328e-07	k_t
Generation 3190: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 319100	Diversity = 2.2074e-07	k_t

Generation 3200: Best = 3.005528e+03 = [3, 4, 4, 4]	FES = 320100	Diversity = 1.9087e-07	k_t
Generation 3210: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 321100	Diversity = 1.6906e-07	k_t
Generation 3220: Best = 3.005528e+03 = [3, 3, 2, 3]	FES = 322100	Diversity = 1.5526e-07	k_t
Generation 3230: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 323100	Diversity = 1.3684e-07	k_t
Generation 3240: Best = 3.005528e+03 = [4, 4, 4, 2]	FES = 324100	Diversity = 1.4179e-07	k_t
Generation 3250: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 325100	Diversity = 1.4551e-07	k_t
Generation 3260: Best = 3.005528e+03 = [3, 2, 4, 2]	FES = 326100	Diversity = 1.3707e-07	k_t
Generation 3270: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 327100	Diversity = 1.3293e-07	k_t
Generation 3280: Best = 3.005528e+03 = [4, 4, 3, 3]	FES = 328100	Diversity = 1.1684e-07	k_t
Generation 3290: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 329100	Diversity = 1.1649e-07	k_t
Generation 3300: Best = 3.005528e+03 = [3, 4, 3, 4]	FES = 330100	Diversity = 1.1277e-07	k_t
Generation 3310: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 331100	Diversity = 1.0620e-07	k_t
Generation 3320: Best = 3.005528e+03 = [2, 3, 3, 3]	FES = 332100	Diversity = 1.0662e-07	k_t
Generation 3330: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 333100	Diversity = 1.0670e-07	k_t
Generation 3340: Best = 3.005528e+03 = [2, 4, 3, 2]	FES = 334100	Diversity = 1.0752e-07	k_t
Generation 3350: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 335100	Diversity = 1.1016e-07	k_t
Generation 3360: Best = 3.005528e+03 = [3, 4, 2, 2]	FES = 336100	Diversity = 1.0776e-07	k_t
Generation 3370: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 337100	Diversity = 1.0618e-07	k_t
Generation 3380: Best = 3.005528e+03 = [2, 3, 4, 2]	FES = 338100	Diversity = 1.0178e-07	k_t
Generation 3390: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 339100	Diversity = 1.0051e-07	k_t
Generation 3400: Best = 3.005528e+03 = [3, 2, 4, 4]	FES = 340100	Diversity = 9.5774e-08	k_t
Generation 3410: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 341100	Diversity = 9.3273e-08	k_t
Generation 3420: Best = 3.005528e+03 = [4, 4, 3, 2]	FES = 342100	Diversity = 8.9274e-08	k_t
Generation 3430: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 343100	Diversity = 8.8706e-08	k_t
Generation 3440: Best = 3.005528e+03 = [4, 4, 4, 2]	FES = 344100	Diversity = 8.5039e-08	k_t
Generation 3450: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 345100	Diversity = 8.4894e-08	k_t
Generation 3460: Best = 3.005528e+03 = [2, 4, 4, 2]	FES = 346100	Diversity = 8.4970e-08	k_t

Generation 3470: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 347100	Diversity = 8.4894e-08	k_t
Generation 3480: Best = 3.005528e+03 = [2, 3, 4, 3]	FES = 348100	Diversity = 8.6007e-08	k_t
Generation 3490: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 349100	Diversity = 8.7527e-08	k_t
Generation 3500: Best = 3.005528e+03 = [2, 2, 4, 3]	FES = 350100	Diversity = 8.7001e-08	k_t
Generation 3510: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 351100	Diversity = 8.8412e-08	k_t
Generation 3520: Best = 3.005528e+03 = [4, 4, 2, 4]	FES = 352100	Diversity = 8.7925e-08	k_t
Generation 3530: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 353100	Diversity = 8.8757e-08	k_t
Generation 3540: Best = 3.005528e+03 = [2, 2, 3, 4]	FES = 354100	Diversity = 8.8427e-08	k_t
Generation 3550: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 355100	Diversity = 8.8560e-08	k_t
Generation 3560: Best = 3.005528e+03 = [3, 4, 3, 4]	FES = 356100	Diversity = 8.8629e-08	k_t
Generation 3570: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 357100	Diversity = 8.8198e-08	k_t
Generation 3580: Best = 3.005528e+03 = [4, 4, 2, 3]	FES = 358100	Diversity = 8.9356e-08	k_t
Generation 3590: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 359100	Diversity = 9.0788e-08	k_t
Generation 3600: Best = 3.005528e+03 = [4, 4, 4, 2]	FES = 360100	Diversity = 9.1408e-08	k_t
Generation 3610: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 361100	Diversity = 9.0973e-08	k_t
Generation 3620: Best = 3.005528e+03 = [4, 4, 3, 2]	FES = 362100	Diversity = 9.1504e-08	k_t
Generation 3630: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 363100	Diversity = 9.1602e-08	k_t
Generation 3640: Best = 3.005528e+03 = [3, 2, 4, 4]	FES = 364100	Diversity = 9.1065e-08	k_t
Generation 3650: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 365100	Diversity = 9.2508e-08	k_t
Generation 3660: Best = 3.005528e+03 = [4, 3, 4, 4]	FES = 366100	Diversity = 9.2857e-08	k_t
Generation 3670: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 367100	Diversity = 9.2285e-08	k_t
Generation 3680: Best = 3.005528e+03 = [2, 3, 2, 4]	FES = 368100	Diversity = 9.2257e-08	k_t
Generation 3690: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 369100	Diversity = 9.3869e-08	k_t
Generation 3700: Best = 3.005528e+03 = [3, 4, 3, 2]	FES = 370100	Diversity = 9.3844e-08	k_t
Generation 3710: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 371100	Diversity = 9.3924e-08	k_t
Generation 3720: Best = 3.005528e+03 = [4, 2, 3, 2]	FES = 372100	Diversity = 9.3448e-08	k_t
Generation 3730: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 373100	Diversity = 9.4841e-08	k_t

Generation 3740: Best = 3.005528e+03 = [2, 3, 2, 3]	FES = 374100	Diversity = 9.4297e-08	k_t
Generation 3750: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 375100	Diversity = 9.5390e-08	k_t
Generation 3760: Best = 3.005528e+03 = [3, 2, 4, 2]	FES = 376100	Diversity = 9.5680e-08	k_t
Generation 3770: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 377100	Diversity = 9.6182e-08	k_t
Generation 3780: Best = 3.005528e+03 = [3, 2, 2, 3]	FES = 378100	Diversity = 9.6576e-08	k_t
Generation 3790: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 379100	Diversity = 9.5167e-08	k_t
Generation 3800: Best = 3.005528e+03 = [3, 2, 4, 3]	FES = 380100	Diversity = 9.3887e-08	k_t
Generation 3810: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 381100	Diversity = 9.6054e-08	k_t
Generation 3820: Best = 3.005528e+03 = [2, 3, 3, 3]	FES = 382100	Diversity = 9.6266e-08	k_t
Generation 3830: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 383100	Diversity = 9.7542e-08	k_t
Generation 3840: Best = 3.005528e+03 = [2, 3, 4, 2]	FES = 384100	Diversity = 9.7570e-08	k_t
Generation 3850: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 385100	Diversity = 9.7707e-08	k_t
Generation 3860: Best = 3.005528e+03 = [4, 4, 3, 3]	FES = 386100	Diversity = 9.7436e-08	k_t
Generation 3870: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 387100	Diversity = 9.7095e-08	k_t
Generation 3880: Best = 3.005528e+03 = [3, 2, 2, 3]	FES = 388100	Diversity = 9.6565e-08	k_t
Generation 3890: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 389100	Diversity = 9.5512e-08	k_t
Generation 3900: Best = 3.005528e+03 = [3, 4, 4, 4]	FES = 390100	Diversity = 9.5351e-08	k_t
Generation 3910: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 391100	Diversity = 9.4627e-08	k_t
Generation 3920: Best = 3.005528e+03 = [3, 3, 2, 2]	FES = 392100	Diversity = 9.4547e-08	k_t
Generation 3930: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 393100	Diversity = 9.5258e-08	k_t
Generation 3940: Best = 3.005528e+03 = [3, 3, 4, 3]	FES = 394100	Diversity = 9.5223e-08	k_t
Generation 3950: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 395100	Diversity = 9.4971e-08	k_t
Generation 3960: Best = 3.005528e+03 = [3, 4, 3, 2]	FES = 396100	Diversity = 9.5162e-08	k_t
Generation 3970: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 397100	Diversity = 9.5405e-08	k_t
Generation 3980: Best = 3.005528e+03 = [4, 4, 3, 3]	FES = 398100	Diversity = 9.5725e-08	k_t
Generation 3990: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 399100	Diversity = 9.6147e-08	k_t
Generation 4000: Best = 3.005528e+03 = [2, 2, 3, 4]	FES = 400100	Diversity = 9.5514e-08	k_t

Generation 4010: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 401100	Diversity = 9.4889e-08	k_t
Generation 4020: Best = 3.005528e+03 = [3, 3, 3, 3]	FES = 402100	Diversity = 9.6426e-08	k_t
Generation 4030: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 403100	Diversity = 9.5943e-08	k_t
Generation 4040: Best = 3.005528e+03 = [4, 3, 2, 4]	FES = 404100	Diversity = 9.6780e-08	k_t
Generation 4050: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 405100	Diversity = 9.5469e-08	k_t
Generation 4060: Best = 3.005528e+03 = [2, 2, 2, 2]	FES = 406100	Diversity = 9.6365e-08	k_t
Generation 4070: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 407100	Diversity = 9.6559e-08	k_t
Generation 4080: Best = 3.005528e+03 = [4, 2, 4, 4]	FES = 408100	Diversity = 9.6163e-08	k_t
Generation 4090: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 409100	Diversity = 9.5939e-08	k_t
Generation 4100: Best = 3.005528e+03 = [4, 2, 2, 3]	FES = 410100	Diversity = 9.4983e-08	k_t
Generation 4110: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 411100	Diversity = 9.4124e-08	k_t
Generation 4120: Best = 3.005528e+03 = [3, 2, 4, 3]	FES = 412100	Diversity = 9.3686e-08	k_t
Generation 4130: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 413100	Diversity = 9.3927e-08	k_t
Generation 4140: Best = 3.005528e+03 = [3, 2, 4, 2]	FES = 414100	Diversity = 9.4083e-08	k_t
Generation 4150: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 415100	Diversity = 9.4926e-08	k_t
Generation 4160: Best = 3.005528e+03 = [4, 4, 3, 4]	FES = 416100	Diversity = 9.4564e-08	k_t
Generation 4170: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 417100	Diversity = 9.4443e-08	k_t
Generation 4180: Best = 3.005528e+03 = [2, 4, 4, 2]	FES = 418100	Diversity = 9.5231e-08	k_t
Generation 4190: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 419100	Diversity = 9.5087e-08	k_t
Generation 4200: Best = 3.005528e+03 = [4, 2, 4, 4]	FES = 420100	Diversity = 9.6048e-08	k_t
Generation 4210: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 421100	Diversity = 9.6075e-08	k_t
Generation 4220: Best = 3.005528e+03 = [3, 4, 2, 2]	FES = 422100	Diversity = 9.5186e-08	k_t
Generation 4230: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 423100	Diversity = 9.5142e-08	k_t
Generation 4240: Best = 3.005528e+03 = [3, 4, 3, 4]	FES = 424100	Diversity = 9.4423e-08	k_t
Generation 4250: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 425100	Diversity = 9.4054e-08	k_t
Generation 4260: Best = 3.005528e+03 = [2, 4, 4, 3]	FES = 426100	Diversity = 9.5296e-08	k_t
Generation 4270: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 427100	Diversity = 9.6350e-08	k_t

Generation 4280: Best = 3.005528e+03 = [3, 2, 2, 2]	FES = 428100	Diversity = 9.7040e-08	k_t
Generation 4290: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 429100	Diversity = 9.6858e-08	k_t
Generation 4300: Best = 3.005528e+03 = [3, 3, 4, 4]	FES = 430100	Diversity = 9.5675e-08	k_t
Generation 4310: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 431100	Diversity = 9.6357e-08	k_t
Generation 4320: Best = 3.005528e+03 = [2, 3, 2, 2]	FES = 432100	Diversity = 9.5908e-08	k_t
Generation 4330: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 433100	Diversity = 9.5465e-08	k_t
Generation 4340: Best = 3.005528e+03 = [2, 4, 3, 3]	FES = 434100	Diversity = 9.5343e-08	k_t
Generation 4350: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 435100	Diversity = 9.4278e-08	k_t
Generation 4360: Best = 3.005528e+03 = [2, 4, 3, 4]	FES = 436100	Diversity = 9.4364e-08	k_t
Generation 4370: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 437100	Diversity = 9.6414e-08	k_t
Generation 4380: Best = 3.005528e+03 = [4, 3, 3, 3]	FES = 438100	Diversity = 9.6387e-08	k_t
Generation 4390: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 439100	Diversity = 9.6435e-08	k_t
Generation 4400: Best = 3.005528e+03 = [4, 2, 2, 2]	FES = 440100	Diversity = 9.6014e-08	k_t
Generation 4410: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 441100	Diversity = 9.7168e-08	k_t
Generation 4420: Best = 3.005528e+03 = [3, 2, 4, 4]	FES = 442100	Diversity = 9.5800e-08	k_t
Generation 4430: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 443100	Diversity = 9.6769e-08	k_t
Generation 4440: Best = 3.005528e+03 = [2, 4, 3, 4]	FES = 444100	Diversity = 9.6743e-08	k_t
Generation 4450: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 445100	Diversity = 9.5841e-08	k_t
Generation 4460: Best = 3.005528e+03 = [2, 2, 3, 3]	FES = 446100	Diversity = 9.6692e-08	k_t
Generation 4470: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 447100	Diversity = 9.6056e-08	k_t
Generation 4480: Best = 3.005528e+03 = [2, 4, 4, 3]	FES = 448100	Diversity = 9.4686e-08	k_t
Generation 4490: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 449100	Diversity = 9.4997e-08	k_t
Generation 4500: Best = 3.005528e+03 = [4, 3, 4, 4]	FES = 450100	Diversity = 9.2617e-08	k_t
Generation 4510: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 451100	Diversity = 9.1697e-08	k_t
Generation 4520: Best = 3.005528e+03 = [2, 2, 4, 3]	FES = 452100	Diversity = 9.2420e-08	k_t
Generation 4530: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 453100	Diversity = 9.2439e-08	k_t
Generation 4540: Best = 3.005528e+03 = [4, 4, 3, 4]	FES = 454100	Diversity = 9.2747e-08	k_t

Generation 4550: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 455100	Diversity = 9.3249e-08	k_t
Generation 4560: Best = 3.005528e+03 = [3, 3, 3, 3]	FES = 456100	Diversity = 9.3247e-08	k_t
Generation 4570: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 457100	Diversity = 9.3770e-08	k_t
Generation 4580: Best = 3.005528e+03 = [2, 4, 2, 3]	FES = 458100	Diversity = 9.4997e-08	k_t
Generation 4590: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 459100	Diversity = 9.6535e-08	k_t
Generation 4600: Best = 3.005528e+03 = [2, 2, 2, 3]	FES = 460100	Diversity = 9.5879e-08	k_t
Generation 4610: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 461100	Diversity = 9.6573e-08	k_t
Generation 4620: Best = 3.005528e+03 = [4, 4, 2, 2]	FES = 462100	Diversity = 9.6575e-08	k_t
Generation 4630: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 463100	Diversity = 9.6851e-08	k_t
Generation 4640: Best = 3.005528e+03 = [3, 3, 2, 3]	FES = 464100	Diversity = 9.5894e-08	k_t
Generation 4650: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 465100	Diversity = 9.4897e-08	k_t
Generation 4660: Best = 3.005528e+03 = [2, 4, 3, 3]	FES = 466100	Diversity = 9.4309e-08	k_t
Generation 4670: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 467100	Diversity = 9.4338e-08	k_t
Generation 4680: Best = 3.005528e+03 = [2, 4, 4, 2]	FES = 468100	Diversity = 9.4910e-08	k_t
Generation 4690: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 469100	Diversity = 9.3951e-08	k_t
Generation 4700: Best = 3.005528e+03 = [2, 3, 3, 3]	FES = 470100	Diversity = 9.3503e-08	k_t
Generation 4710: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 471100	Diversity = 9.3556e-08	k_t
Generation 4720: Best = 3.005528e+03 = [3, 3, 4, 4]	FES = 472100	Diversity = 9.1705e-08	k_t
Generation 4730: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 473100	Diversity = 9.2075e-08	k_t
Generation 4740: Best = 3.005528e+03 = [4, 3, 3, 4]	FES = 474100	Diversity = 9.3098e-08	k_t
Generation 4750: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 475100	Diversity = 9.2467e-08	k_t
Generation 4760: Best = 3.005528e+03 = [2, 4, 3, 3]	FES = 476100	Diversity = 9.3121e-08	k_t
Generation 4770: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 477100	Diversity = 9.3759e-08	k_t
Generation 4780: Best = 3.005528e+03 = [4, 4, 4, 2]	FES = 478100	Diversity = 9.5321e-08	k_t
Generation 4790: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 479100	Diversity = 9.4817e-08	k_t
Generation 4800: Best = 3.005528e+03 = [3, 4, 3, 3]	FES = 480100	Diversity = 9.6024e-08	k_t
Generation 4810: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 481100	Diversity = 9.6323e-08	k_t

Generation 4820: Best = 3.005528e+03 = [2, 2, 3, 2]	FES = 482100	Diversity = 9.6035e-08	k_t
Generation 4830: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 483100	Diversity = 9.6230e-08	k_t
Generation 4840: Best = 3.005528e+03 = [2, 4, 3, 4]	FES = 484100	Diversity = 9.7099e-08	k_t
Generation 4850: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 485100	Diversity = 9.7145e-08	k_t
Generation 4860: Best = 3.005528e+03 = [4, 2, 4, 4]	FES = 486100	Diversity = 9.7608e-08	k_t
Generation 4870: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 487100	Diversity = 9.7744e-08	k_t
Generation 4880: Best = 3.005528e+03 = [3, 2, 4, 2]	FES = 488100	Diversity = 9.7362e-08	k_t
Generation 4890: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 489100	Diversity = 9.7054e-08	k_t
Generation 4900: Best = 3.005528e+03 = [2, 3, 2, 2]	FES = 490100	Diversity = 9.7977e-08	k_t
Generation 4910: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 491100	Diversity = 9.6755e-08	k_t
Generation 4920: Best = 3.005528e+03 = [2, 4, 3, 2]	FES = 492100	Diversity = 9.6476e-08	k_t
Generation 4930: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 493100	Diversity = 9.7800e-08	k_t
Generation 4940: Best = 3.005528e+03 = [4, 2, 4, 3]	FES = 494100	Diversity = 9.6242e-08	k_t
Generation 4950: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 495100	Diversity = 9.6251e-08	k_t
Generation 4960: Best = 3.005528e+03 = [2, 2, 3, 3]	FES = 496100	Diversity = 9.6560e-08	k_t
Generation 4970: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 497100	Diversity = 9.6095e-08	k_t
Generation 4980: Best = 3.005528e+03 = [3, 2, 4, 3]	FES = 498100	Diversity = 9.6504e-08	k_t
Generation 4990: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 499100	Diversity = 9.5486e-08	k_t
Generation 5000: Best = 3.005528e+03 = [4, 3, 4, 2]	FES = 500100	Diversity = 9.4767e-08	k_t
Generation 5010: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 501100	Diversity = 9.6341e-08	k_t
Generation 5020: Best = 3.005528e+03 = [3, 3, 3, 4]	FES = 502100	Diversity = 9.7375e-08	k_t
Generation 5030: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 503100	Diversity = 9.7518e-08	k_t
Generation 5040: Best = 3.005528e+03 = [2, 4, 2, 2]	FES = 504100	Diversity = 9.7695e-08	k_t
Generation 5050: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 505100	Diversity = 9.7224e-08	k_t
Generation 5060: Best = 3.005528e+03 = [4, 4, 4, 3]	FES = 506100	Diversity = 9.7639e-08	k_t
Generation 5070: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 507100	Diversity = 9.6861e-08	k_t
Generation 5080: Best = 3.005528e+03 = [4, 2, 2, 3]	FES = 508100	Diversity = 9.9999e-08	k_t

Generation 5090: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 509100	Diversity = 9.9231e-08	k_t
Generation 5100: Best = 3.005528e+03 = [3, 2, 2, 4]	FES = 510100	Diversity = 9.7495e-08	k_t
Generation 5110: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 511100	Diversity = 9.7946e-08	k_t
Generation 5120: Best = 3.005528e+03 = [3, 2, 2, 2]	FES = 512100	Diversity = 9.8260e-08	k_t
Generation 5130: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 513100	Diversity = 9.6715e-08	k_t
Generation 5140: Best = 3.005528e+03 = [2, 2, 2, 3]	FES = 514100	Diversity = 9.7011e-08	k_t
Generation 5150: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 515100	Diversity = 9.7648e-08	k_t
Generation 5160: Best = 3.005528e+03 = [2, 4, 4, 4]	FES = 516100	Diversity = 9.5617e-08	k_t
Generation 5170: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 517100	Diversity = 9.5289e-08	k_t
Generation 5180: Best = 3.005528e+03 = [4, 4, 4, 3]	FES = 518100	Diversity = 9.3911e-08	k_t
Generation 5190: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 519100	Diversity = 9.4764e-08	k_t
Generation 5200: Best = 3.005528e+03 = [2, 2, 3, 4]	FES = 520100	Diversity = 9.4918e-08	k_t
Generation 5210: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 521100	Diversity = 9.4911e-08	k_t
Generation 5220: Best = 3.005528e+03 = [2, 4, 2, 2]	FES = 522100	Diversity = 9.5713e-08	k_t
Generation 5230: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 523100	Diversity = 9.3832e-08	k_t
Generation 5240: Best = 3.005528e+03 = [4, 4, 4, 4]	FES = 524100	Diversity = 9.4611e-08	k_t
Generation 5250: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 525100	Diversity = 9.5451e-08	k_t
Generation 5260: Best = 3.005528e+03 = [4, 2, 2, 4]	FES = 526100	Diversity = 9.5586e-08	k_t
Generation 5270: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 527100	Diversity = 9.6099e-08	k_t
Generation 5280: Best = 3.005528e+03 = [3, 3, 4, 3]	FES = 528100	Diversity = 9.5664e-08	k_t
Generation 5290: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 529100	Diversity = 9.4878e-08	k_t
Generation 5300: Best = 3.005528e+03 = [2, 3, 3, 3]	FES = 530100	Diversity = 9.3675e-08	k_t
Generation 5310: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 531100	Diversity = 9.4583e-08	k_t
Generation 5320: Best = 3.005528e+03 = [2, 3, 2, 4]	FES = 532100	Diversity = 9.4135e-08	k_t
Generation 5330: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 533100	Diversity = 9.2942e-08	k_t
Generation 5340: Best = 3.005528e+03 = [2, 2, 2, 2]	FES = 534100	Diversity = 9.0622e-08	k_t
Generation 5350: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 535100	Diversity = 9.1539e-08	k_t

Generation 5360: Best = 3.005528e+03 = [2, 2, 4, 4]	FES = 536100	Diversity = 9.2756e-08	k_t
Generation 5370: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 537100	Diversity = 9.2291e-08	k_t
Generation 5380: Best = 3.005528e+03 = [4, 3, 4, 3]	FES = 538100	Diversity = 9.3491e-08	k_t
Generation 5390: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 539100	Diversity = 9.3651e-08	k_t
Generation 5400: Best = 3.005528e+03 = [4, 3, 2, 3]	FES = 540100	Diversity = 9.1790e-08	k_t
Generation 5410: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 541100	Diversity = 9.2657e-08	k_t
Generation 5420: Best = 3.005528e+03 = [3, 3, 2, 2]	FES = 542100	Diversity = 9.2827e-08	k_t
Generation 5430: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 543100	Diversity = 9.0801e-08	k_t
Generation 5440: Best = 3.005528e+03 = [3, 3, 4, 3]	FES = 544100	Diversity = 8.9378e-08	k_t
Generation 5450: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 545100	Diversity = 8.7587e-08	k_t
Generation 5460: Best = 3.005528e+03 = [2, 3, 3, 3]	FES = 546100	Diversity = 8.6127e-08	k_t
Generation 5470: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 547100	Diversity = 8.6190e-08	k_t
Generation 5480: Best = 3.005528e+03 = [3, 2, 2, 3]	FES = 548100	Diversity = 8.3781e-08	k_t
Generation 5490: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 549100	Diversity = 8.2972e-08	k_t
Generation 5500: Best = 3.005528e+03 = [2, 2, 2, 4]	FES = 550100	Diversity = 8.2408e-08	k_t
Generation 5510: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 551100	Diversity = 8.4026e-08	k_t
Generation 5520: Best = 3.005528e+03 = [3, 3, 3, 4]	FES = 552100	Diversity = 8.3586e-08	k_t
Generation 5530: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 553100	Diversity = 8.3180e-08	k_t
Generation 5540: Best = 3.005528e+03 = [4, 4, 2, 4]	FES = 554100	Diversity = 8.1922e-08	k_t
Generation 5550: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 555100	Diversity = 8.1031e-08	k_t
Generation 5560: Best = 3.005528e+03 = [4, 4, 3, 2]	FES = 556100	Diversity = 8.1434e-08	k_t
Generation 5570: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 557100	Diversity = 8.1535e-08	k_t
Generation 5580: Best = 3.005528e+03 = [2, 3, 3, 2]	FES = 558100	Diversity = 8.1228e-08	k_t
Generation 5590: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 559100	Diversity = 8.1384e-08	k_t
Generation 5600: Best = 3.005528e+03 = [4, 3, 2, 4]	FES = 560100	Diversity = 8.0387e-08	k_t
Generation 5610: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 561100	Diversity = 8.0617e-08	k_t
Generation 5620: Best = 3.005528e+03 = [2, 4, 2, 4]	FES = 562100	Diversity = 8.1184e-08	k_t

Generation 5630: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 563100	Diversity = 7.8614e-08	k_t
Generation 5640: Best = 3.005528e+03 = [3, 4, 4, 3]	FES = 564100	Diversity = 7.8060e-08	k_t
Generation 5650: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 565100	Diversity = 7.8941e-08	k_t
Generation 5660: Best = 3.005528e+03 = [4, 4, 4, 4]	FES = 566100	Diversity = 7.8609e-08	k_t
Generation 5670: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 567100	Diversity = 7.7160e-08	k_t
Generation 5680: Best = 3.005528e+03 = [3, 4, 2, 4]	FES = 568100	Diversity = 7.9717e-08	k_t
Generation 5690: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 569100	Diversity = 8.1149e-08	k_t
Generation 5700: Best = 3.005528e+03 = [3, 4, 3, 2]	FES = 570100	Diversity = 7.9096e-08	k_t
Generation 5710: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 571100	Diversity = 7.9484e-08	k_t
Generation 5720: Best = 3.005528e+03 = [2, 2, 3, 4]	FES = 572100	Diversity = 8.0317e-08	k_t
Generation 5730: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 573100	Diversity = 8.1528e-08	k_t
Generation 5740: Best = 3.005528e+03 = [2, 3, 3, 3]	FES = 574100	Diversity = 8.0385e-08	k_t
Generation 5750: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 575100	Diversity = 8.0448e-08	k_t
Generation 5760: Best = 3.005528e+03 = [2, 4, 2, 2]	FES = 576100	Diversity = 7.9054e-08	k_t
Generation 5770: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 577100	Diversity = 7.9858e-08	k_t
Generation 5780: Best = 3.005528e+03 = [4, 2, 2, 4]	FES = 578100	Diversity = 7.8682e-08	k_t
Generation 5790: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 579100	Diversity = 8.0378e-08	k_t
Generation 5800: Best = 3.005528e+03 = [4, 3, 4, 3]	FES = 580100	Diversity = 8.1212e-08	k_t
Generation 5810: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 581100	Diversity = 8.0201e-08	k_t
Generation 5820: Best = 3.005528e+03 = [2, 4, 2, 3]	FES = 582100	Diversity = 7.9628e-08	k_t
Generation 5830: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 583100	Diversity = 7.9929e-08	k_t
Generation 5840: Best = 3.005528e+03 = [2, 2, 2, 3]	FES = 584100	Diversity = 7.9344e-08	k_t
Generation 5850: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 585100	Diversity = 7.9064e-08	k_t
Generation 5860: Best = 3.005528e+03 = [4, 2, 4, 3]	FES = 586100	Diversity = 7.9635e-08	k_t
Generation 5870: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 587100	Diversity = 7.8958e-08	k_t
Generation 5880: Best = 3.005528e+03 = [2, 3, 2, 4]	FES = 588100	Diversity = 7.8399e-08	k_t
Generation 5890: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 589100	Diversity = 7.8263e-08	k_t

Generation 5900: Best = 3.005528e+03 = [3, 2, 2, 3]	FES = 590100	Diversity = 7.7789e-08	k_t
Generation 5910: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 591100	Diversity = 7.7789e-08	k_t
Generation 5920: Best = 3.005528e+03 = [2, 3, 2, 4]	FES = 592100	Diversity = 7.7834e-08	k_t
Generation 5930: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 593100	Diversity = 7.7521e-08	k_t
Generation 5940: Best = 3.005528e+03 = [4, 3, 2, 4]	FES = 594100	Diversity = 7.6614e-08	k_t
Generation 5950: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 595100	Diversity = 7.6612e-08	k_t
Generation 5960: Best = 3.005528e+03 = [2, 4, 3, 3]	FES = 596100	Diversity = 7.8725e-08	k_t
Generation 5970: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 597100	Diversity = 7.8613e-08	k_t
Generation 5980: Best = 3.005528e+03 = [2, 3, 4, 2]	FES = 598100	Diversity = 7.9062e-08	k_t
Generation 5990: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 599100	Diversity = 7.8766e-08	k_t
Generation 6000: Best = 3.005528e+03 = [4, 4, 2, 3]	FES = 600100	Diversity = 7.8622e-08	k_t
Generation 6010: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 601100	Diversity = 7.7364e-08	k_t
Generation 6020: Best = 3.005528e+03 = [4, 2, 3, 3]	FES = 602100	Diversity = 7.7441e-08	k_t
Generation 6030: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 603100	Diversity = 7.7709e-08	k_t
Generation 6040: Best = 3.005528e+03 = [4, 3, 3, 4]	FES = 604100	Diversity = 7.7402e-08	k_t
Generation 6050: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 605100	Diversity = 7.6741e-08	k_t
Generation 6060: Best = 3.005528e+03 = [2, 4, 3, 3]	FES = 606100	Diversity = 7.6881e-08	k_t
Generation 6070: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 607100	Diversity = 7.6823e-08	k_t
Generation 6080: Best = 3.005528e+03 = [3, 3, 4, 4]	FES = 608100	Diversity = 7.5177e-08	k_t
Generation 6090: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 609100	Diversity = 7.3705e-08	k_t
Generation 6100: Best = 3.005528e+03 = [4, 3, 3, 3]	FES = 610100	Diversity = 7.2924e-08	k_t
Generation 6110: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 611100	Diversity = 7.5010e-08	k_t
Generation 6120: Best = 3.005528e+03 = [2, 3, 4, 4]	FES = 612100	Diversity = 7.5095e-08	k_t
Generation 6130: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 613100	Diversity = 7.5378e-08	k_t
Generation 6140: Best = 3.005528e+03 = [3, 4, 4, 4]	FES = 614100	Diversity = 7.5394e-08	k_t
Generation 6150: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 615100	Diversity = 7.4717e-08	k_t
Generation 6160: Best = 3.005528e+03 = [4, 3, 2, 3]	FES = 616100	Diversity = 7.7842e-08	k_t

Generation 6170: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 617100	Diversity = 7.7939e-08	k_t
Generation 6180: Best = 3.005528e+03 = [4, 3, 4, 3]	FES = 618100	Diversity = 8.0212e-08	k_t
Generation 6190: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 619100	Diversity = 7.8678e-08	k_t
Generation 6200: Best = 3.005528e+03 = [3, 2, 4, 3]	FES = 620100	Diversity = 7.6383e-08	k_t
Generation 6210: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 621100	Diversity = 7.7593e-08	k_t
Generation 6220: Best = 3.005528e+03 = [3, 4, 3, 2]	FES = 622100	Diversity = 7.6629e-08	k_t
Generation 6230: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 623100	Diversity = 7.7925e-08	k_t
Generation 6240: Best = 3.005528e+03 = [3, 2, 2, 2]	FES = 624100	Diversity = 7.6360e-08	k_t
Generation 6250: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 625100	Diversity = 7.6645e-08	k_t
Generation 6260: Best = 3.005528e+03 = [2, 4, 2, 4]	FES = 626100	Diversity = 7.5788e-08	k_t
Generation 6270: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 627100	Diversity = 7.2805e-08	k_t
Generation 6280: Best = 3.005528e+03 = [4, 2, 3, 3]	FES = 628100	Diversity = 7.3659e-08	k_t
Generation 6290: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 629100	Diversity = 7.3265e-08	k_t
Generation 6300: Best = 3.005528e+03 = [3, 2, 2, 4]	FES = 630100	Diversity = 7.4745e-08	k_t
Generation 6310: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 631100	Diversity = 7.7026e-08	k_t
Generation 6320: Best = 3.005528e+03 = [2, 3, 2, 2]	FES = 632100	Diversity = 7.6290e-08	k_t
Generation 6330: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 633100	Diversity = 7.5267e-08	k_t
Generation 6340: Best = 3.005528e+03 = [4, 2, 4, 4]	FES = 634100	Diversity = 7.5333e-08	k_t
Generation 6350: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 635100	Diversity = 7.6926e-08	k_t
Generation 6360: Best = 3.005528e+03 = [3, 2, 3, 3]	FES = 636100	Diversity = 7.7964e-08	k_t
Generation 6370: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 637100	Diversity = 7.7228e-08	k_t
Generation 6380: Best = 3.005528e+03 = [3, 2, 4, 3]	FES = 638100	Diversity = 7.6997e-08	k_t
Generation 6390: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 639100	Diversity = 7.8116e-08	k_t
Generation 6400: Best = 3.005528e+03 = [4, 4, 3, 3]	FES = 640100	Diversity = 7.6701e-08	k_t
Generation 6410: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 641100	Diversity = 7.6393e-08	k_t
Generation 6420: Best = 3.005528e+03 = [4, 3, 4, 4]	FES = 642100	Diversity = 7.6511e-08	k_t
Generation 6430: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 643100	Diversity = 7.4727e-08	k_t

Generation 6440: Best = 3.005528e+03 = [3, 2, 3, 4]	FES = 644100	Diversity = 7.4639e-08	k_t
Generation 6450: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 645100	Diversity = 7.3988e-08	k_t
Generation 6460: Best = 3.005528e+03 = [4, 2, 4, 4]	FES = 646100	Diversity = 7.4588e-08	k_t
Generation 6470: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 647100	Diversity = 7.3871e-08	k_t
Generation 6480: Best = 3.005528e+03 = [2, 4, 2, 2]	FES = 648100	Diversity = 7.3745e-08	k_t
Generation 6490: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 649100	Diversity = 7.4944e-08	k_t
Generation 6500: Best = 3.005528e+03 = [4, 2, 3, 3]	FES = 650100	Diversity = 7.5263e-08	k_t
Generation 6510: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 651100	Diversity = 7.5525e-08	k_t
Generation 6520: Best = 3.005528e+03 = [3, 3, 4, 4]	FES = 652100	Diversity = 7.5902e-08	k_t
Generation 6530: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 653100	Diversity = 7.6813e-08	k_t
Generation 6540: Best = 3.005528e+03 = [2, 4, 4, 4]	FES = 654100	Diversity = 7.7093e-08	k_t
Generation 6550: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 655100	Diversity = 7.6083e-08	k_t
Generation 6560: Best = 3.005528e+03 = [2, 4, 3, 4]	FES = 656100	Diversity = 7.5106e-08	k_t
Generation 6570: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 657100	Diversity = 7.5695e-08	k_t
Generation 6580: Best = 3.005528e+03 = [3, 3, 2, 2]	FES = 658100	Diversity = 7.5728e-08	k_t
Generation 6590: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 659100	Diversity = 7.5601e-08	k_t
Generation 6600: Best = 3.005528e+03 = [3, 4, 3, 3]	FES = 660100	Diversity = 7.4609e-08	k_t
Generation 6610: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 661100	Diversity = 7.2303e-08	k_t
Generation 6620: Best = 3.005528e+03 = [3, 4, 2, 4]	FES = 662100	Diversity = 7.4332e-08	k_t
Generation 6630: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 663100	Diversity = 7.1398e-08	k_t
Generation 6640: Best = 3.005528e+03 = [2, 2, 2, 2]	FES = 664100	Diversity = 7.1193e-08	k_t
Generation 6650: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 665100	Diversity = 6.9971e-08	k_t
Generation 6660: Best = 3.005528e+03 = [3, 4, 4, 4]	FES = 666100	Diversity = 7.0889e-08	k_t
Generation 6670: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 667100	Diversity = 7.1332e-08	k_t
Generation 6680: Best = 3.005528e+03 = [4, 2, 3, 4]	FES = 668100	Diversity = 6.9633e-08	k_t
Generation 6690: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 669100	Diversity = 6.9005e-08	k_t
Generation 6700: Best = 3.005528e+03 = [3, 3, 3, 3]	FES = 670100	Diversity = 7.0378e-08	k_t

Generation 6710: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 671100	Diversity = 6.9364e-08	k_t
Generation 6720: Best = 3.005528e+03 = [3, 4, 3, 3]	FES = 672100	Diversity = 7.0628e-08	k_t
Generation 6730: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 673100	Diversity = 7.0798e-08	k_t
Generation 6740: Best = 3.005528e+03 = [3, 2, 3, 2]	FES = 674100	Diversity = 7.0586e-08	k_t
Generation 6750: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 675100	Diversity = 7.0241e-08	k_t
Generation 6760: Best = 3.005528e+03 = [4, 4, 4, 2]	FES = 676100	Diversity = 6.9902e-08	k_t
Generation 6770: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 677100	Diversity = 7.0039e-08	k_t
Generation 6780: Best = 3.005528e+03 = [2, 2, 4, 4]	FES = 678100	Diversity = 7.2657e-08	k_t
Generation 6790: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 679100	Diversity = 6.9880e-08	k_t
Generation 6800: Best = 3.005528e+03 = [2, 3, 3, 3]	FES = 680100	Diversity = 6.8723e-08	k_t
Generation 6810: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 681100	Diversity = 7.0501e-08	k_t
Generation 6820: Best = 3.005528e+03 = [3, 4, 3, 4]	FES = 682100	Diversity = 6.9461e-08	k_t
Generation 6830: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 683100	Diversity = 6.8797e-08	k_t
Generation 6840: Best = 3.005528e+03 = [3, 3, 3, 2]	FES = 684100	Diversity = 7.0004e-08	k_t
Generation 6850: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 685100	Diversity = 6.9406e-08	k_t
Generation 6860: Best = 3.005528e+03 = [2, 2, 3, 4]	FES = 686100	Diversity = 6.9756e-08	k_t
Generation 6870: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 687100	Diversity = 6.8666e-08	k_t
Generation 6880: Best = 3.005528e+03 = [3, 4, 2, 2]	FES = 688100	Diversity = 6.9233e-08	k_t
Generation 6890: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 689100	Diversity = 6.8436e-08	k_t
Generation 6900: Best = 3.005528e+03 = [2, 2, 2, 2]	FES = 690100	Diversity = 6.8069e-08	k_t
Generation 6910: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 691100	Diversity = 6.7096e-08	k_t
Generation 6920: Best = 3.005528e+03 = [2, 4, 3, 4]	FES = 692100	Diversity = 6.5823e-08	k_t
Generation 6930: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 693100	Diversity = 6.5725e-08	k_t
Generation 6940: Best = 3.005528e+03 = [4, 4, 3, 4]	FES = 694100	Diversity = 6.6748e-08	k_t
Generation 6950: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 695100	Diversity = 6.4834e-08	k_t
Generation 6960: Best = 3.005528e+03 = [4, 4, 3, 3]	FES = 696100	Diversity = 6.7190e-08	k_t
Generation 6970: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 697100	Diversity = 6.7746e-08	k_t

Generation 6980: Best = 3.005528e+03 = [2, 4, 4, 4]	FES = 698100	Diversity = 6.6721e-08	k_t
Generation 6990: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 6.8373e-08	k_t
Generation 7000: Best = 3.005528e+03 = [4, 3, 2, 3]	FES = 700100	Diversity = 6.6541e-08	k_t
Generation 7010: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 6.5840e-08	k_t
Generation 7020: Best = 3.005528e+03 = [3, 4, 2, 4]	FES = 702100	Diversity = 6.5459e-08	k_t
Generation 7030: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 6.5042e-08	k_t
Generation 7040: Best = 3.005528e+03 = [4, 2, 2, 4]	FES = 704100	Diversity = 6.5796e-08	k_t
Generation 7050: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 6.5244e-08	k_t
Generation 7060: Best = 3.005528e+03 = [2, 2, 4, 2]	FES = 706100	Diversity = 6.4331e-08	k_t
Generation 7070: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 6.4003e-08	k_t
Generation 7080: Best = 3.005528e+03 = [3, 4, 2, 2]	FES = 708100	Diversity = 6.3587e-08	k_t
Generation 7090: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 6.4077e-08	k_t
Generation 7100: Best = 3.005528e+03 = [2, 2, 4, 3]	FES = 710100	Diversity = 6.3808e-08	k_t
Generation 7110: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 6.3616e-08	k_t
Generation 7120: Best = 3.005528e+03 = [4, 3, 3, 3]	FES = 712100	Diversity = 6.3413e-08	k_t
Generation 7130: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 6.2941e-08	k_t
Generation 7140: Best = 3.005528e+03 = [4, 3, 4, 3]	FES = 714100	Diversity = 6.2441e-08	k_t
Generation 7150: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 6.1581e-08	k_t
Generation 7160: Best = 3.005528e+03 = [4, 3, 3, 2]	FES = 716100	Diversity = 6.1577e-08	k_t
Generation 7170: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 6.3198e-08	k_t
Generation 7180: Best = 3.005528e+03 = [3, 2, 4, 2]	FES = 718100	Diversity = 6.3612e-08	k_t
Generation 7190: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 6.2592e-08	k_t
Generation 7200: Best = 3.005528e+03 = [3, 4, 3, 3]	FES = 720100	Diversity = 6.2289e-08	k_t
Generation 7210: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 6.2783e-08	k_t
Generation 7220: Best = 3.005528e+03 = [2, 3, 2, 2]	FES = 722100	Diversity = 6.3367e-08	k_t
Generation 7230: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 6.2510e-08	k_t
Generation 7240: Best = 3.005528e+03 = [2, 2, 3, 3]	FES = 724100	Diversity = 6.2935e-08	k_t

Generation 7250: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 6.3242e-08	k_t
Generation 7260: Best = 3.005528e+03 = [2, 2, 4, 2]	FES = 726100	Diversity = 6.3201e-08	k_t
Generation 7270: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 6.3650e-08	k_t
Generation 7280: Best = 3.005528e+03 = [3, 4, 2, 2]	FES = 728100	Diversity = 6.4041e-08	k_t
Generation 7290: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 6.3906e-08	k_t
Generation 7300: Best = 3.005528e+03 = [3, 4, 4, 3]	FES = 730100	Diversity = 6.4762e-08	k_t
Generation 7310: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 6.4162e-08	k_t
Generation 7320: Best = 3.005528e+03 = [4, 2, 4, 4]	FES = 732100	Diversity = 6.4167e-08	k_t
Generation 7330: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 6.5976e-08	k_t
Generation 7340: Best = 3.005528e+03 = [2, 3, 2, 4]	FES = 734100	Diversity = 6.5409e-08	k_t
Generation 7350: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 6.5476e-08	k_t
Generation 7360: Best = 3.005528e+03 = [2, 2, 2, 2]	FES = 736100	Diversity = 6.5400e-08	k_t
Generation 7370: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 6.4923e-08	k_t
Generation 7380: Best = 3.005528e+03 = [4, 2, 4, 3]	FES = 738100	Diversity = 6.5016e-08	k_t
Generation 7390: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 6.6184e-08	k_t
Generation 7400: Best = 3.005528e+03 = [3, 4, 3, 3]	FES = 740100	Diversity = 6.4973e-08	k_t
Generation 7410: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 6.5446e-08	k_t
Generation 7420: Best = 3.005528e+03 = [4, 2, 4, 2]	FES = 742100	Diversity = 6.5729e-08	k_t
Generation 7430: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 6.5118e-08	k_t
Generation 7440: Best = 3.005528e+03 = [2, 3, 3, 2]	FES = 744100	Diversity = 6.5875e-08	k_t
Generation 7450: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 6.5517e-08	k_t
Generation 7460: Best = 3.005528e+03 = [2, 2, 2, 4]	FES = 746100	Diversity = 6.5569e-08	k_t
Generation 7470: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 6.5528e-08	k_t
Generation 7480: Best = 3.005528e+03 = [4, 3, 2, 2]	FES = 748100	Diversity = 6.5248e-08	k_t
Generation 7490: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 6.5734e-08	k_t
Generation 7500: Best = 3.005528e+03 = [2, 2, 4, 3]	FES = 750100	Diversity = 6.4693e-08	k_t
Generation 7510: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 6.4547e-08	k_t

Generation 7520: Best = 3.005528e+03 = [4, 4, 2, 3]	FES = 752100	Diversity = 6.4350e-08	k_t
Generation 7530: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 753100	Diversity = 6.4426e-08	k_t
Generation 7540: Best = 3.005528e+03 = [3, 3, 4, 4]	FES = 754100	Diversity = 6.4227e-08	k_t
Generation 7550: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 755100	Diversity = 6.4056e-08	k_t
Generation 7560: Best = 3.005528e+03 = [4, 4, 3, 2]	FES = 756100	Diversity = 6.4147e-08	k_t
Generation 7570: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 757100	Diversity = 6.4068e-08	k_t
Generation 7580: Best = 3.005528e+03 = [3, 3, 4, 4]	FES = 758100	Diversity = 6.4223e-08	k_t
Generation 7590: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 759100	Diversity = 6.4236e-08	k_t
Generation 7600: Best = 3.005528e+03 = [4, 2, 2, 3]	FES = 760100	Diversity = 6.4462e-08	k_t
Generation 7610: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 761100	Diversity = 6.4442e-08	k_t
Generation 7620: Best = 3.005528e+03 = [3, 2, 2, 4]	FES = 762100	Diversity = 6.4200e-08	k_t
Generation 7630: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 763100	Diversity = 6.4454e-08	k_t
Generation 7640: Best = 3.005528e+03 = [2, 4, 3, 2]	FES = 764100	Diversity = 6.4267e-08	k_t
Generation 7650: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 765100	Diversity = 6.4089e-08	k_t
Generation 7660: Best = 3.005528e+03 = [4, 4, 3, 3]	FES = 766100	Diversity = 6.4017e-08	k_t
Generation 7670: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 767100	Diversity = 6.4217e-08	k_t
Generation 7680: Best = 3.005528e+03 = [2, 4, 2, 4]	FES = 768100	Diversity = 6.3780e-08	k_t
Generation 7690: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 769100	Diversity = 6.3221e-08	k_t
Generation 7700: Best = 3.005528e+03 = [2, 4, 2, 3]	FES = 770100	Diversity = 6.3167e-08	k_t
Generation 7710: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 771100	Diversity = 6.2847e-08	k_t
Generation 7720: Best = 3.005528e+03 = [2, 4, 3, 3]	FES = 772100	Diversity = 6.2432e-08	k_t
Generation 7730: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 773100	Diversity = 6.2951e-08	k_t
Generation 7740: Best = 3.005528e+03 = [4, 3, 4, 2]	FES = 774100	Diversity = 6.3165e-08	k_t
Generation 7750: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 775100	Diversity = 6.3564e-08	k_t
Generation 7760: Best = 3.005528e+03 = [2, 3, 3, 2]	FES = 776100	Diversity = 6.4192e-08	k_t
Generation 7770: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 777100	Diversity = 6.3670e-08	k_t
Generation 7780: Best = 3.005528e+03 = [3, 4, 4, 2]	FES = 778100	Diversity = 6.3220e-08	k_t

Generation 7790: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 779100	Diversity = 6.3712e-08	k_t
Generation 7800: Best = 3.005528e+03 = [2, 3, 2, 3]	FES = 780100	Diversity = 6.3930e-08	k_t
Generation 7810: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 781100	Diversity = 6.3871e-08	k_t
Generation 7820: Best = 3.005528e+03 = [3, 4, 3, 3]	FES = 782100	Diversity = 6.3679e-08	k_t
Generation 7830: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 783100	Diversity = 6.3635e-08	k_t
Generation 7840: Best = 3.005528e+03 = [3, 2, 2, 4]	FES = 784100	Diversity = 6.3621e-08	k_t
Generation 7850: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 785100	Diversity = 6.3444e-08	k_t
Generation 7860: Best = 3.005528e+03 = [4, 2, 2, 3]	FES = 786100	Diversity = 6.3124e-08	k_t
Generation 7870: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 787100	Diversity = 6.3012e-08	k_t
Generation 7880: Best = 3.005528e+03 = [4, 3, 3, 2]	FES = 788100	Diversity = 6.3171e-08	k_t
Generation 7890: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 789100	Diversity = 6.2700e-08	k_t
Generation 7900: Best = 3.005528e+03 = [3, 2, 3, 4]	FES = 790100	Diversity = 6.2502e-08	k_t
Generation 7910: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 791100	Diversity = 6.2587e-08	k_t
Generation 7920: Best = 3.005528e+03 = [3, 2, 3, 2]	FES = 792100	Diversity = 6.2266e-08	k_t
Generation 7930: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 793100	Diversity = 6.2186e-08	k_t
Generation 7940: Best = 3.005528e+03 = [4, 3, 4, 2]	FES = 794100	Diversity = 6.2595e-08	k_t
Generation 7950: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 795100	Diversity = 6.2549e-08	k_t
Generation 7960: Best = 3.005528e+03 = [3, 3, 3, 3]	FES = 796100	Diversity = 6.2133e-08	k_t
Generation 7970: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 797100	Diversity = 6.2046e-08	k_t
Generation 7980: Best = 3.005528e+03 = [4, 4, 4, 4]	FES = 798100	Diversity = 6.1963e-08	k_t
Generation 7990: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 799100	Diversity = 6.1837e-08	k_t
Generation 8000: Best = 3.005528e+03 = [4, 3, 3, 2]	FES = 800100	Diversity = 6.1762e-08	k_t
Generation 8010: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 801100	Diversity = 6.1402e-08	k_t
Generation 8020: Best = 3.005528e+03 = [3, 3, 3, 2]	FES = 802100	Diversity = 6.0917e-08	k_t
Generation 8030: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 803100	Diversity = 6.1100e-08	k_t
Generation 8040: Best = 3.005528e+03 = [4, 4, 2, 4]	FES = 804100	Diversity = 6.0830e-08	k_t
Generation 8050: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 805100	Diversity = 6.1010e-08	k_t

Generation 8060: Best = 3.005528e+03 = [4, 2, 3, 2]	FES = 806100	Diversity = 6.1125e-08	k_t
Generation 8070: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 807100	Diversity = 6.1318e-08	k_t
Generation 8080: Best = 3.005528e+03 = [3, 4, 3, 4]	FES = 808100	Diversity = 6.1323e-08	k_t
Generation 8090: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 809100	Diversity = 6.1198e-08	k_t
Generation 8100: Best = 3.005528e+03 = [3, 4, 2, 2]	FES = 810100	Diversity = 6.0427e-08	k_t
Generation 8110: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 811100	Diversity = 6.0162e-08	k_t
Generation 8120: Best = 3.005528e+03 = [4, 3, 3, 2]	FES = 812100	Diversity = 5.9596e-08	k_t
Generation 8130: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 813100	Diversity = 5.9295e-08	k_t
Generation 8140: Best = 3.005528e+03 = [2, 4, 3, 4]	FES = 814100	Diversity = 5.9499e-08	k_t
Generation 8150: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 815100	Diversity = 5.9836e-08	k_t
Generation 8160: Best = 3.005528e+03 = [2, 4, 3, 4]	FES = 816100	Diversity = 5.9642e-08	k_t
Generation 8170: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 817100	Diversity = 5.9318e-08	k_t
Generation 8180: Best = 3.005528e+03 = [2, 3, 3, 3]	FES = 818100	Diversity = 5.9486e-08	k_t
Generation 8190: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 819100	Diversity = 5.8931e-08	k_t
Generation 8200: Best = 3.005528e+03 = [4, 2, 2, 2]	FES = 820100	Diversity = 5.8597e-08	k_t
Generation 8210: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 821100	Diversity = 5.8392e-08	k_t
Generation 8220: Best = 3.005528e+03 = [3, 3, 4, 3]	FES = 822100	Diversity = 5.7602e-08	k_t
Generation 8230: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 823100	Diversity = 5.7206e-08	k_t
Generation 8240: Best = 3.005528e+03 = [4, 4, 2, 2]	FES = 824100	Diversity = 5.7052e-08	k_t
Generation 8250: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 825100	Diversity = 5.6951e-08	k_t
Generation 8260: Best = 3.005528e+03 = [2, 3, 2, 3]	FES = 826100	Diversity = 5.6805e-08	k_t
Generation 8270: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 827100	Diversity = 5.6895e-08	k_t
Generation 8280: Best = 3.005528e+03 = [3, 3, 3, 3]	FES = 828100	Diversity = 5.6896e-08	k_t
Generation 8290: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 829100	Diversity = 5.6750e-08	k_t
Generation 8300: Best = 3.005528e+03 = [3, 3, 3, 3]	FES = 830100	Diversity = 5.6506e-08	k_t
Generation 8310: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 831100	Diversity = 5.6541e-08	k_t
Generation 8320: Best = 3.005528e+03 = [3, 3, 4, 2]	FES = 832100	Diversity = 5.5794e-08	k_t

Generation 8330: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 833100	Diversity = 5.5501e-08	k_t
Generation 8340: Best = 3.005528e+03 = [4, 4, 4, 3]	FES = 834100	Diversity = 5.5211e-08	k_t
Generation 8350: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 835100	Diversity = 5.5218e-08	k_t
Generation 8360: Best = 3.005528e+03 = [2, 2, 3, 4]	FES = 836100	Diversity = 5.4985e-08	k_t
Generation 8370: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 837100	Diversity = 5.5041e-08	k_t
Generation 8380: Best = 3.005528e+03 = [3, 2, 3, 3]	FES = 838100	Diversity = 5.5034e-08	k_t
Generation 8390: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 839100	Diversity = 5.4516e-08	k_t
Generation 8400: Best = 3.005528e+03 = [2, 2, 2, 2]	FES = 840100	Diversity = 5.4100e-08	k_t
Generation 8410: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 841100	Diversity = 5.3742e-08	k_t
Generation 8420: Best = 3.005528e+03 = [3, 3, 3, 2]	FES = 842100	Diversity = 5.3666e-08	k_t
Generation 8430: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 843100	Diversity = 5.3297e-08	k_t
Generation 8440: Best = 3.005528e+03 = [3, 4, 3, 2]	FES = 844100	Diversity = 5.2951e-08	k_t
Generation 8450: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 845100	Diversity = 5.2288e-08	k_t
Generation 8460: Best = 3.005528e+03 = [2, 2, 3, 2]	FES = 846100	Diversity = 5.1910e-08	k_t
Generation 8470: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 847100	Diversity = 5.1137e-08	k_t
Generation 8480: Best = 3.005528e+03 = [4, 4, 3, 2]	FES = 848100	Diversity = 5.0423e-08	k_t
Generation 8490: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 849100	Diversity = 5.0035e-08	k_t
Generation 8500: Best = 3.005528e+03 = [3, 3, 4, 3]	FES = 850100	Diversity = 4.9795e-08	k_t
Generation 8510: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 851100	Diversity = 4.9727e-08	k_t
Generation 8520: Best = 3.005528e+03 = [4, 4, 2, 2]	FES = 852100	Diversity = 4.8944e-08	k_t
Generation 8530: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 853100	Diversity = 4.8611e-08	k_t
Generation 8540: Best = 3.005528e+03 = [2, 2, 2, 4]	FES = 854100	Diversity = 4.8202e-08	k_t
Generation 8550: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 855100	Diversity = 4.7019e-08	k_t
Generation 8560: Best = 3.005528e+03 = [4, 4, 4, 4]	FES = 856100	Diversity = 4.5904e-08	k_t
Generation 8570: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 857100	Diversity = 4.4434e-08	k_t
Generation 8580: Best = 3.005528e+03 = [2, 3, 2, 2]	FES = 858100	Diversity = 4.3352e-08	k_t
Generation 8590: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 859100	Diversity = 4.2748e-08	k_t

Generation 8600: Best = 3.005528e+03 = [2, 3, 3, 2]	FES = 860100	Diversity = 4.1594e-08	k_t
Generation 8610: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 861100	Diversity = 4.1423e-08	k_t
Generation 8620: Best = 3.005528e+03 = [2, 4, 3, 4]	FES = 862100	Diversity = 3.8672e-08	k_t
Generation 8630: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 863100	Diversity = 3.6133e-08	k_t
Generation 8640: Best = 3.005528e+03 = [2, 4, 3, 3]	FES = 864100	Diversity = 3.3560e-08	k_t
Generation 8650: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 865100	Diversity = 3.1518e-08	k_t
Generation 8660: Best = 3.005528e+03 = [3, 4, 4, 3]	FES = 866100	Diversity = 2.8414e-08	k_t
Generation 8670: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 867100	Diversity = 2.1617e-08	k_t
Generation 8680: Best = 3.005528e+03 = [4, 2, 2, 4]	FES = 868100	Diversity = 2.0560e-08	k_t
Generation 8690: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 869100	Diversity = 2.0084e-08	k_t
Generation 8700: Best = 3.005528e+03 = [2, 3, 4, 2]	FES = 870100	Diversity = 2.0363e-08	k_t
Generation 8710: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 871100	Diversity = 1.9938e-08	k_t
Generation 8720: Best = 3.005528e+03 = [2, 2, 3, 4]	FES = 872100	Diversity = 2.0282e-08	k_t
Generation 8730: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 873100	Diversity = 2.0722e-08	k_t
Generation 8740: Best = 3.005528e+03 = [3, 4, 4, 2]	FES = 874100	Diversity = 2.0916e-08	k_t
Generation 8750: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 875100	Diversity = 2.0667e-08	k_t
Generation 8760: Best = 3.005528e+03 = [4, 3, 2, 4]	FES = 876100	Diversity = 2.0645e-08	k_t
Generation 8770: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 877100	Diversity = 2.0735e-08	k_t
Generation 8780: Best = 3.005528e+03 = [3, 2, 2, 2]	FES = 878100	Diversity = 2.0703e-08	k_t
Generation 8790: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 879100	Diversity = 2.1081e-08	k_t
Generation 8800: Best = 3.005528e+03 = [3, 3, 3, 3]	FES = 880100	Diversity = 2.1042e-08	k_t
Generation 8810: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 881100	Diversity = 2.0951e-08	k_t
Generation 8820: Best = 3.005528e+03 = [4, 4, 2, 3]	FES = 882100	Diversity = 2.0989e-08	k_t
Generation 8830: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 883100	Diversity = 2.1023e-08	k_t
Generation 8840: Best = 3.005528e+03 = [2, 4, 3, 3]	FES = 884100	Diversity = 2.1103e-08	k_t
Generation 8850: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 885100	Diversity = 2.1134e-08	k_t
Generation 8860: Best = 3.005528e+03 = [3, 4, 2, 2]	FES = 886100	Diversity = 2.1124e-08	k_t

Generation 8870: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 887100	Diversity = 2.1133e-08	k_t
Generation 8880: Best = 3.005528e+03 = [2, 3, 2, 4]	FES = 888100	Diversity = 2.1285e-08	k_t
Generation 8890: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 889100	Diversity = 2.1202e-08	k_t
Generation 8900: Best = 3.005528e+03 = [4, 2, 3, 2]	FES = 890100	Diversity = 2.1346e-08	k_t
Generation 8910: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 891100	Diversity = 2.1280e-08	k_t
Generation 8920: Best = 3.005528e+03 = [2, 2, 2, 4]	FES = 892100	Diversity = 2.1314e-08	k_t
Generation 8930: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 893100	Diversity = 2.1300e-08	k_t
Generation 8940: Best = 3.005528e+03 = [4, 4, 4, 2]	FES = 894100	Diversity = 2.1181e-08	k_t
Generation 8950: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 895100	Diversity = 2.1220e-08	k_t
Generation 8960: Best = 3.005528e+03 = [2, 2, 2, 3]	FES = 896100	Diversity = 2.1365e-08	k_t
Generation 8970: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 897100	Diversity = 2.1273e-08	k_t
Generation 8980: Best = 3.005528e+03 = [4, 2, 2, 3]	FES = 898100	Diversity = 2.1351e-08	k_t
Generation 8990: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 899100	Diversity = 2.1421e-08	k_t
Generation 9000: Best = 3.005528e+03 = [3, 2, 3, 4]	FES = 900100	Diversity = 2.1520e-08	k_t
Generation 9010: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 901100	Diversity = 2.1347e-08	k_t
Generation 9020: Best = 3.005528e+03 = [4, 3, 4, 4]	FES = 902100	Diversity = 2.1358e-08	k_t
Generation 9030: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 903100	Diversity = 2.1472e-08	k_t
Generation 9040: Best = 3.005528e+03 = [3, 3, 3, 2]	FES = 904100	Diversity = 2.1529e-08	k_t
Generation 9050: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 905100	Diversity = 2.1477e-08	k_t
Generation 9060: Best = 3.005528e+03 = [2, 2, 3, 2]	FES = 906100	Diversity = 2.1344e-08	k_t
Generation 9070: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 907100	Diversity = 2.1449e-08	k_t
Generation 9080: Best = 3.005528e+03 = [2, 4, 3, 4]	FES = 908100	Diversity = 2.1478e-08	k_t
Generation 9090: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 909100	Diversity = 2.1571e-08	k_t
Generation 9100: Best = 3.005528e+03 = [3, 2, 4, 4]	FES = 910100	Diversity = 2.1735e-08	k_t
Generation 9110: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 911100	Diversity = 2.1719e-08	k_t
Generation 9120: Best = 3.005528e+03 = [2, 4, 2, 4]	FES = 912100	Diversity = 2.1714e-08	k_t
Generation 9130: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 913100	Diversity = 2.1832e-08	k_t

Generation 9140: Best = 3.005528e+03 = [2, 4, 2, 4]	FES = 914100	Diversity = 2.3356e-08	k_t
Generation 9150: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 2.3391e-08	k_t
Generation 9160: Best = 3.005528e+03 = [2, 4, 4, 4]	FES = 916100	Diversity = 2.3477e-08	k_t
Generation 9170: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 2.3565e-08	k_t
Generation 9180: Best = 3.005528e+03 = [2, 4, 3, 4]	FES = 918100	Diversity = 2.3853e-08	k_t
Generation 9190: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 2.3862e-08	k_t
Generation 9200: Best = 3.005528e+03 = [2, 4, 2, 4]	FES = 920100	Diversity = 2.3984e-08	k_t
Generation 9210: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 2.4189e-08	k_t
Generation 9220: Best = 3.005528e+03 = [4, 2, 2, 2]	FES = 922100	Diversity = 2.4210e-08	k_t
Generation 9230: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 2.4300e-08	k_t
Generation 9240: Best = 3.005528e+03 = [2, 2, 2, 4]	FES = 924100	Diversity = 2.4328e-08	k_t
Generation 9250: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 2.4307e-08	k_t
Generation 9260: Best = 3.005528e+03 = [2, 4, 3, 2]	FES = 926100	Diversity = 2.4227e-08	k_t
Generation 9270: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 2.4395e-08	k_t
Generation 9280: Best = 3.005528e+03 = [2, 4, 2, 2]	FES = 928100	Diversity = 2.3000e-08	k_t
Generation 9290: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 2.3196e-08	k_t
Generation 9300: Best = 3.005528e+03 = [2, 2, 4, 4]	FES = 930100	Diversity = 2.3239e-08	k_t
Generation 9310: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 2.3399e-08	k_t
Generation 9320: Best = 3.005528e+03 = [3, 4, 3, 3]	FES = 932100	Diversity = 2.3301e-08	k_t
Generation 9330: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 2.3310e-08	k_t
Generation 9340: Best = 3.005528e+03 = [3, 3, 2, 3]	FES = 934100	Diversity = 2.3364e-08	k_t
Generation 9350: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 2.3379e-08	k_t
Generation 9360: Best = 3.005528e+03 = [2, 3, 3, 3]	FES = 936100	Diversity = 2.3291e-08	k_t
Generation 9370: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 2.3475e-08	k_t
Generation 9380: Best = 3.005528e+03 = [4, 2, 4, 2]	FES = 938100	Diversity = 2.3287e-08	k_t
Generation 9390: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 2.3432e-08	k_t
Generation 9400: Best = 3.005528e+03 = [2, 4, 2, 2]	FES = 940100	Diversity = 2.3475e-08	k_t

Generation 9410: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 2.3397e-08	k_t
Generation 9420: Best = 3.005528e+03 = [4, 4, 2, 3]	FES = 942100	Diversity = 2.3708e-08	k_t
Generation 9430: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 2.3872e-08	k_t
Generation 9440: Best = 3.005528e+03 = [3, 3, 3, 2]	FES = 944100	Diversity = 2.3696e-08	k_t
Generation 9450: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 2.3609e-08	k_t
Generation 9460: Best = 3.005528e+03 = [3, 4, 2, 2]	FES = 946100	Diversity = 2.3651e-08	k_t
Generation 9470: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 2.3436e-08	k_t
Generation 9480: Best = 3.005528e+03 = [3, 4, 2, 4]	FES = 948100	Diversity = 2.3394e-08	k_t
Generation 9490: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 2.3477e-08	k_t
Generation 9500: Best = 3.005528e+03 = [3, 4, 4, 4]	FES = 950100	Diversity = 2.3418e-08	k_t
Generation 9510: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 2.3427e-08	k_t
Generation 9520: Best = 3.005528e+03 = [3, 2, 4, 2]	FES = 952100	Diversity = 2.3484e-08	k_t
Generation 9530: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 2.3256e-08	k_t
Generation 9540: Best = 3.005528e+03 = [2, 2, 4, 4]	FES = 954100	Diversity = 2.4086e-08	k_t
Generation 9550: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 2.3739e-08	k_t
Generation 9560: Best = 3.005528e+03 = [3, 3, 2, 4]	FES = 956100	Diversity = 2.3714e-08	k_t
Generation 9570: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 2.3822e-08	k_t
Generation 9580: Best = 3.005528e+03 = [3, 4, 3, 4]	FES = 958100	Diversity = 2.3793e-08	k_t
Generation 9590: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 2.3789e-08	k_t
Generation 9600: Best = 3.005528e+03 = [2, 3, 3, 4]	FES = 960100	Diversity = 2.3740e-08	k_t
Generation 9610: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 2.3226e-08	k_t
Generation 9620: Best = 3.005528e+03 = [4, 2, 2, 4]	FES = 962100	Diversity = 2.3242e-08	k_t
Generation 9630: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 2.3254e-08	k_t
Generation 9640: Best = 3.005528e+03 = [2, 4, 3, 3]	FES = 964100	Diversity = 2.3365e-08	k_t
Generation 9650: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 2.3581e-08	k_t
Generation 9660: Best = 3.005528e+03 = [2, 3, 4, 2]	FES = 966100	Diversity = 2.4848e-08	k_t
Generation 9670: Best = 3.005528e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 2.4859e-08	k_t

Generation 9680: Best = 3.005528e+03	FES = 968100	Diversity = 2.4896e-08	k_t
= [3, 2, 2, 2]			
Generation 9690: Best = 3.005528e+03	FES = 969100	Diversity = 2.4910e-08	k_t
= [1, 1, 1, 1]			
Generation 9700: Best = 3.005528e+03	FES = 970100	Diversity = 2.6026e-08	k_t
= [3, 2, 4, 2]			
Generation 9710: Best = 3.005528e+03	FES = 971100	Diversity = 2.5920e-08	k_t
= [1, 1, 1, 1]			
Generation 9720: Best = 3.005528e+03	FES = 972100	Diversity = 2.4975e-08	k_t
= [2, 4, 2, 2]			
Generation 9730: Best = 3.005528e+03	FES = 973100	Diversity = 2.5068e-08	k_t
= [1, 1, 1, 1]			
Generation 9740: Best = 3.005528e+03	FES = 974100	Diversity = 2.4965e-08	k_t
= [2, 4, 3, 2]			
Generation 9750: Best = 3.005528e+03	FES = 975100	Diversity = 2.5214e-08	k_t
= [1, 1, 1, 1]			
Generation 9760: Best = 3.005528e+03	FES = 976100	Diversity = 2.5206e-08	k_t
= [2, 4, 3, 2]			
Generation 9770: Best = 3.005528e+03	FES = 977100	Diversity = 2.5127e-08	k_t
= [1, 1, 1, 1]			
Generation 9780: Best = 3.005528e+03	FES = 978100	Diversity = 2.5151e-08	k_t
= [2, 3, 4, 3]			
Generation 9790: Best = 3.005528e+03	FES = 979100	Diversity = 2.5226e-08	k_t
= [1, 1, 1, 1]			
Generation 9800: Best = 3.005528e+03	FES = 980100	Diversity = 2.5150e-08	k_t
= [3, 3, 3, 3]			
Generation 9810: Best = 3.005528e+03	FES = 981100	Diversity = 2.5166e-08	k_t
= [1, 1, 1, 1]			
Generation 9820: Best = 3.005528e+03	FES = 982100	Diversity = 2.3877e-08	k_t
= [2, 4, 3, 4]			
Generation 9830: Best = 3.005528e+03	FES = 983100	Diversity = 2.4094e-08	k_t
= [1, 1, 1, 1]			
Generation 9840: Best = 3.005528e+03	FES = 984100	Diversity = 2.4102e-08	k_t
= [2, 4, 2, 2]			
Generation 9850: Best = 3.005528e+03	FES = 985100	Diversity = 2.6375e-08	k_t
= [1, 1, 1, 1]			
Generation 9860: Best = 3.005528e+03	FES = 986100	Diversity = 2.6384e-08	k_t
= [3, 4, 2, 3]			
Generation 9870: Best = 3.005528e+03	FES = 987100	Diversity = 2.6464e-08	k_t
= [1, 1, 1, 1]			
Generation 9880: Best = 3.005528e+03	FES = 988100	Diversity = 2.6396e-08	k_t
= [2, 2, 4, 3]			
Generation 9890: Best = 3.005528e+03	FES = 989100	Diversity = 2.6443e-08	k_t
= [1, 1, 1, 1]			
Generation 9900: Best = 3.005528e+03	FES = 990100	Diversity = 2.6446e-08	k_t
= [3, 4, 2, 2]			
Generation 9910: Best = 3.005528e+03	FES = 991100	Diversity = 2.5424e-08	k_t
= [1, 1, 1, 1]			
Generation 9920: Best = 3.005528e+03	FES = 992100	Diversity = 2.5327e-08	k_t
= [2, 2, 2, 3]			
Generation 9930: Best = 3.005528e+03	FES = 993100	Diversity = 2.5426e-08	k_t
= [1, 1, 1, 1]			
Generation 9940: Best = 3.005528e+03	FES = 994100	Diversity = 2.5427e-08	k_t
= [3, 2, 4, 2]			

Generation 9950: Best = 3.005528e+03 FES = 995100 Diversity = 2.5456e-08 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 3.005528e+03 FES = 996100 Diversity = 2.5530e-08 k_t
= [2, 2, 2, 2]
Generation 9970: Best = 3.005528e+03 FES = 997100 Diversity = 2.5583e-08 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 3.005528e+03 FES = 998100 Diversity = 2.5657e-08 k_t
= [4, 2, 2, 2]
Generation 9990: Best = 3.005528e+03 FES = 999100 Diversity = 2.5700e-08 k_t
= [1, 1, 1, 1]

Optimization completed!

Final Best Fitness = 3.005528e+03

Total Function Evaluations = 1000000

=====

=

Results:

Best Fitness : 3.005528e+03

Global Optimum : 2.300000e+03

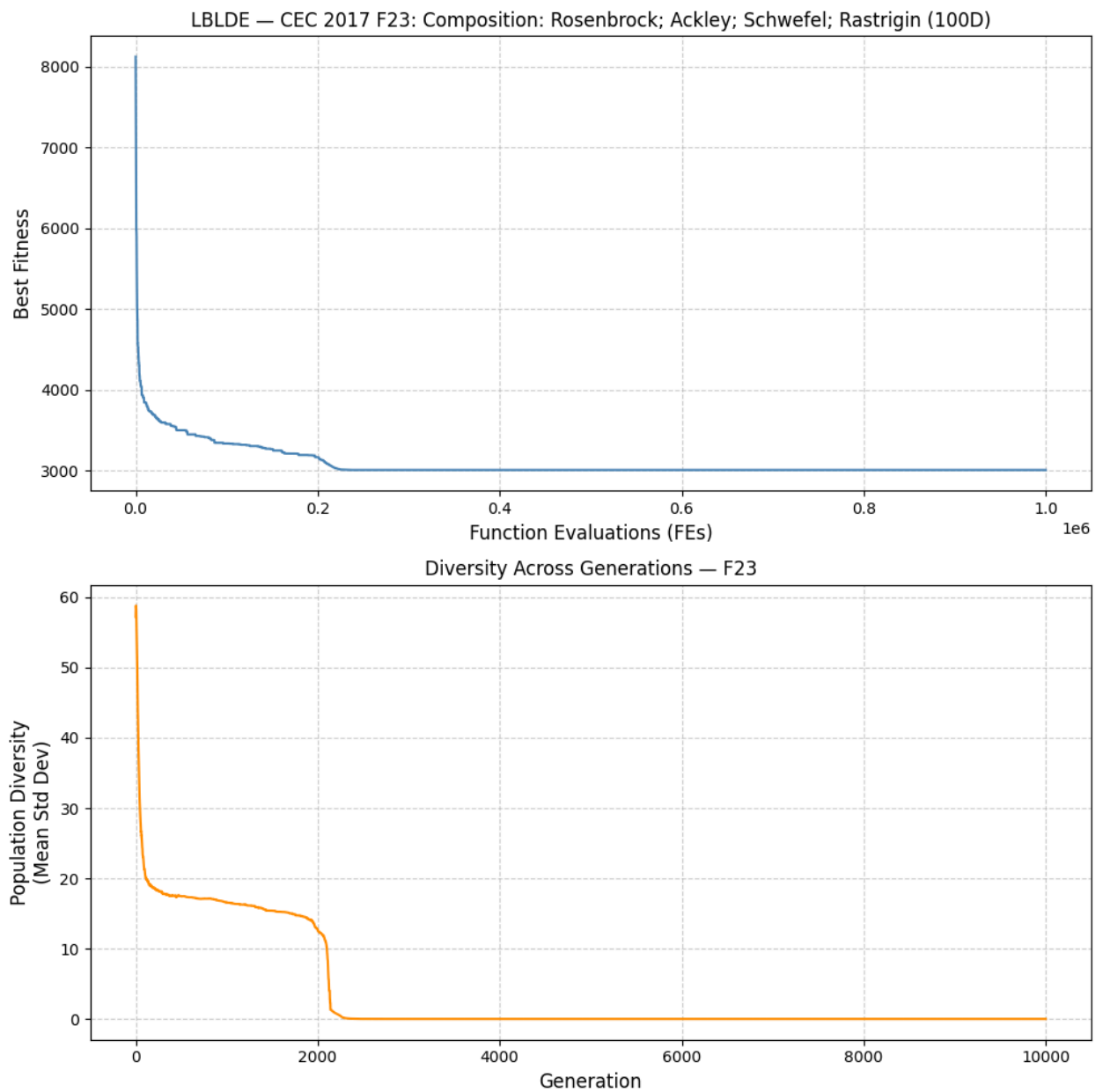
Error (|best-optimum|) : 7.055284e+02

Best Solution (5 dim) : [-88.64002064 -99.36829093 -95.99685991 78.94086985
-96.95168499]

D^(0) (init. diversity): 5.7220e+01

=====

=



Graph saved as convergence_F23.png

```
In [48]: # =====
# F24 — Composition Function 4 (Ackley; Elliptic; Griewank; Rastrigin)
# =====
run_and_plot('F24', make_wrapper(composition.f24), 2400.0, 'Composition: Ackle
```

```

=====
=
LBLDE_s on CEC 2017 F24 – Single Run
=====
=
Function      : Composition: Ackley; Elliptic; Griewank; Rastrigin (100D)
Bounds       : [-100, 100]
Global optimum : 2400.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 1.419682e+04 | Diversity = 5.7220e+01
Generation 10: Best = 8.860021e+03 FES = 1100 Diversity = 5.2910e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 6.770698e+03 FES = 2100 Diversity = 4.4496e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 5.458228e+03 FES = 3100 Diversity = 3.8364e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 5.093461e+03 FES = 4100 Diversity = 3.1328e+01 k_t
= [4, 3, 4, 2]
Generation 50: Best = 4.758332e+03 FES = 5100 Diversity = 2.7145e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 4.549504e+03 FES = 6100 Diversity = 2.2492e+01 k_t
= [2, 2, 3, 3]
Generation 70: Best = 4.430472e+03 FES = 7100 Diversity = 1.9222e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 4.430472e+03 FES = 8100 Diversity = 1.7662e+01 k_t
= [4, 2, 2, 2]
Generation 90: Best = 4.328067e+03 FES = 9100 Diversity = 1.6217e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 4.312849e+03 FES = 10100 Diversity = 1.5130e+01 k_t
= [3, 3, 4, 4]
Generation 110: Best = 4.296620e+03 FES = 11100 Diversity = 1.4681e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 4.232115e+03 FES = 12100 Diversity = 1.4382e+01 k_t
= [2, 4, 4, 4]
Generation 130: Best = 4.232115e+03 FES = 13100 Diversity = 1.4032e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 4.232115e+03 FES = 14100 Diversity = 1.3681e+01 k_t
= [4, 2, 2, 3]
Generation 150: Best = 4.164908e+03 FES = 15100 Diversity = 1.3446e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 4.161734e+03 FES = 16100 Diversity = 1.3251e+01 k_t
= [2, 3, 4, 4]
Generation 170: Best = 4.161734e+03 FES = 17100 Diversity = 1.3062e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 4.108675e+03 FES = 18100 Diversity = 1.2887e+01 k_t
= [2, 2, 4, 4]
Generation 190: Best = 4.108675e+03 FES = 19100 Diversity = 1.2916e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 4.108675e+03 FES = 20100 Diversity = 1.2971e+01 k_t
= [4, 3, 2, 3]
Generation 210: Best = 4.106940e+03 FES = 21100 Diversity = 1.2868e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 4.106940e+03 FES = 22100 Diversity = 1.2806e+01 k_t
= [3, 3, 3, 2]

```

Generation 230:	Best = 4.106940e+03	FES = 23100	Diversity = 1.2738e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 4.101483e+03	FES = 24100	Diversity = 1.2685e+01	k_t = [2, 2, 3, 4]
Generation 250:	Best = 4.084727e+03	FES = 25100	Diversity = 1.2684e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 4.084727e+03	FES = 26100	Diversity = 1.2549e+01	k_t = [4, 2, 4, 2]
Generation 270:	Best = 4.079606e+03	FES = 27100	Diversity = 1.2535e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 4.063474e+03	FES = 28100	Diversity = 1.2555e+01	k_t = [4, 4, 4, 3]
Generation 290:	Best = 4.063474e+03	FES = 29100	Diversity = 1.2404e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 4.063474e+03	FES = 30100	Diversity = 1.2402e+01	k_t = [4, 4, 4, 4]
Generation 310:	Best = 4.059431e+03	FES = 31100	Diversity = 1.2420e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 4.059431e+03	FES = 32100	Diversity = 1.2424e+01	k_t = [2, 2, 2, 2]
Generation 330:	Best = 4.048022e+03	FES = 33100	Diversity = 1.2432e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 4.048022e+03	FES = 34100	Diversity = 1.2351e+01	k_t = [2, 3, 4, 3]
Generation 350:	Best = 4.048022e+03	FES = 35100	Diversity = 1.2331e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 4.048022e+03	FES = 36100	Diversity = 1.2309e+01	k_t = [3, 2, 4, 2]
Generation 370:	Best = 4.028292e+03	FES = 37100	Diversity = 1.2288e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 4.027774e+03	FES = 38100	Diversity = 1.2246e+01	k_t = [4, 2, 4, 3]
Generation 390:	Best = 4.017453e+03	FES = 39100	Diversity = 1.2102e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 4.017453e+03	FES = 40100	Diversity = 1.2139e+01	k_t = [4, 4, 4, 4]
Generation 410:	Best = 4.017453e+03	FES = 41100	Diversity = 1.2066e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 4.015609e+03	FES = 42100	Diversity = 1.2026e+01	k_t = [4, 4, 4, 2]
Generation 430:	Best = 4.015609e+03	FES = 43100	Diversity = 1.2001e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 4.015609e+03	FES = 44100	Diversity = 1.1980e+01	k_t = [2, 2, 2, 2]
Generation 450:	Best = 4.005786e+03	FES = 45100	Diversity = 1.1980e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 4.005610e+03	FES = 46100	Diversity = 1.1970e+01	k_t = [3, 3, 2, 2]
Generation 470:	Best = 4.004963e+03	FES = 47100	Diversity = 1.1918e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 3.967520e+03	FES = 48100	Diversity = 1.1908e+01	k_t = [2, 4, 2, 4]
Generation 490:	Best = 3.950991e+03	FES = 49100	Diversity = 1.1901e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 3.950991e+03	FES = 50100	Diversity = 1.1881e+01	k_t
= [2, 3, 2, 2]				
Generation 510:	Best = 3.950991e+03	FES = 51100	Diversity = 1.1877e+01	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 3.950991e+03	FES = 52100	Diversity = 1.1857e+01	k_t
= [2, 4, 4, 2]				
Generation 530:	Best = 3.943080e+03	FES = 53100	Diversity = 1.1838e+01	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 3.943080e+03	FES = 54100	Diversity = 1.1817e+01	k_t
= [2, 2, 4, 2]				
Generation 550:	Best = 3.943080e+03	FES = 55100	Diversity = 1.1802e+01	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 3.943080e+03	FES = 56100	Diversity = 1.1776e+01	k_t
= [2, 4, 2, 3]				
Generation 570:	Best = 3.943080e+03	FES = 57100	Diversity = 1.1774e+01	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 3.943080e+03	FES = 58100	Diversity = 1.1724e+01	k_t
= [3, 3, 3, 4]				
Generation 590:	Best = 3.937744e+03	FES = 59100	Diversity = 1.1724e+01	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 3.937744e+03	FES = 60100	Diversity = 1.1691e+01	k_t
= [4, 3, 2, 2]				
Generation 610:	Best = 3.923950e+03	FES = 61100	Diversity = 1.1674e+01	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 3.909270e+03	FES = 62100	Diversity = 1.1680e+01	k_t
= [4, 3, 3, 4]				
Generation 630:	Best = 3.907427e+03	FES = 63100	Diversity = 1.1683e+01	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 3.907427e+03	FES = 64100	Diversity = 1.1651e+01	k_t
= [3, 4, 4, 4]				
Generation 650:	Best = 3.907427e+03	FES = 65100	Diversity = 1.1656e+01	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 3.907427e+03	FES = 66100	Diversity = 1.1648e+01	k_t
= [3, 3, 4, 4]				
Generation 670:	Best = 3.906541e+03	FES = 67100	Diversity = 1.1634e+01	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 3.882753e+03	FES = 68100	Diversity = 1.1664e+01	k_t
= [3, 3, 2, 4]				
Generation 690:	Best = 3.882753e+03	FES = 69100	Diversity = 1.1658e+01	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 3.882753e+03	FES = 70100	Diversity = 1.1635e+01	k_t
= [3, 4, 2, 3]				
Generation 710:	Best = 3.882753e+03	FES = 71100	Diversity = 1.1637e+01	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 3.877354e+03	FES = 72100	Diversity = 1.1561e+01	k_t
= [4, 4, 2, 4]				
Generation 730:	Best = 3.877354e+03	FES = 73100	Diversity = 1.1623e+01	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 3.877354e+03	FES = 74100	Diversity = 1.1658e+01	k_t
= [2, 4, 3, 3]				
Generation 750:	Best = 3.875416e+03	FES = 75100	Diversity = 1.1650e+01	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 3.875416e+03	FES = 76100	Diversity = 1.1576e+01	k_t
= [3, 2, 4, 2]				

Generation 770:	Best = 3.869153e+03	FES = 77100	Diversity = 1.1574e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 3.868401e+03	FES = 78100	Diversity = 1.1564e+01	k_t
= [2, 2, 3, 4]				
Generation 790:	Best = 3.864248e+03	FES = 79100	Diversity = 1.1555e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 3.864248e+03	FES = 80100	Diversity = 1.1432e+01	k_t
= [3, 2, 2, 4]				
Generation 810:	Best = 3.864054e+03	FES = 81100	Diversity = 1.1379e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 3.862252e+03	FES = 82100	Diversity = 1.1378e+01	k_t
= [2, 2, 4, 4]				
Generation 830:	Best = 3.862252e+03	FES = 83100	Diversity = 1.1307e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 3.861797e+03	FES = 84100	Diversity = 1.1316e+01	k_t
= [4, 4, 3, 2]				
Generation 850:	Best = 3.861797e+03	FES = 85100	Diversity = 1.1284e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 3.861697e+03	FES = 86100	Diversity = 1.1226e+01	k_t
= [2, 4, 3, 2]				
Generation 870:	Best = 3.861697e+03	FES = 87100	Diversity = 1.1229e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 3.856665e+03	FES = 88100	Diversity = 1.1206e+01	k_t
= [3, 3, 4, 4]				
Generation 890:	Best = 3.854839e+03	FES = 89100	Diversity = 1.1234e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 3.854839e+03	FES = 90100	Diversity = 1.1234e+01	k_t
= [3, 3, 4, 2]				
Generation 910:	Best = 3.850039e+03	FES = 91100	Diversity = 1.1210e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 3.850039e+03	FES = 92100	Diversity = 1.1209e+01	k_t
= [3, 4, 4, 4]				
Generation 930:	Best = 3.850039e+03	FES = 93100	Diversity = 1.1163e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 3.850039e+03	FES = 94100	Diversity = 1.1158e+01	k_t
= [2, 3, 2, 2]				
Generation 950:	Best = 3.836209e+03	FES = 95100	Diversity = 1.1149e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 3.830014e+03	FES = 96100	Diversity = 1.1108e+01	k_t
= [3, 3, 3, 4]				
Generation 970:	Best = 3.829043e+03	FES = 97100	Diversity = 1.1106e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 3.829043e+03	FES = 98100	Diversity = 1.1099e+01	k_t
= [4, 3, 2, 2]				
Generation 990:	Best = 3.822981e+03	FES = 99100	Diversity = 1.1073e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 3.822981e+03	FES = 100100	Diversity = 1.1069e+01	k_t
= [3, 2, 2, 2]				
Generation 1010:	Best = 3.817424e+03	FES = 101100	Diversity = 1.1068e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 3.816587e+03	FES = 102100	Diversity = 1.1061e+01	k_t
= [2, 2, 3, 3]				
Generation 1030:	Best = 3.806359e+03	FES = 103100	Diversity = 1.1059e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 3.804139e+03 = [2, 4, 2, 3]	FES = 104100	Diversity = 1.1056e+01	k_t
Generation 1050: Best = 3.804139e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 1.1045e+01	k_t
Generation 1060: Best = 3.783168e+03 = [4, 2, 2, 4]	FES = 106100	Diversity = 1.1019e+01	k_t
Generation 1070: Best = 3.783004e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 1.1008e+01	k_t
Generation 1080: Best = 3.779972e+03 = [4, 2, 3, 2]	FES = 108100	Diversity = 1.0957e+01	k_t
Generation 1090: Best = 3.779972e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 1.0957e+01	k_t
Generation 1100: Best = 3.779972e+03 = [4, 4, 3, 3]	FES = 110100	Diversity = 1.0934e+01	k_t
Generation 1110: Best = 3.769777e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.0926e+01	k_t
Generation 1120: Best = 3.769777e+03 = [2, 2, 2, 4]	FES = 112100	Diversity = 1.0916e+01	k_t
Generation 1130: Best = 3.766465e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.0899e+01	k_t
Generation 1140: Best = 3.766122e+03 = [3, 3, 2, 2]	FES = 114100	Diversity = 1.0871e+01	k_t
Generation 1150: Best = 3.766122e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.0828e+01	k_t
Generation 1160: Best = 3.766122e+03 = [4, 2, 2, 3]	FES = 116100	Diversity = 1.0823e+01	k_t
Generation 1170: Best = 3.759851e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.0800e+01	k_t
Generation 1180: Best = 3.759821e+03 = [3, 3, 3, 4]	FES = 118100	Diversity = 1.0789e+01	k_t
Generation 1190: Best = 3.758398e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.0785e+01	k_t
Generation 1200: Best = 3.755085e+03 = [4, 2, 3, 3]	FES = 120100	Diversity = 1.0782e+01	k_t
Generation 1210: Best = 3.754259e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.0780e+01	k_t
Generation 1220: Best = 3.754259e+03 = [4, 2, 3, 2]	FES = 122100	Diversity = 1.0778e+01	k_t
Generation 1230: Best = 3.748415e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 1.0739e+01	k_t
Generation 1240: Best = 3.740589e+03 = [4, 4, 2, 2]	FES = 124100	Diversity = 1.0732e+01	k_t
Generation 1250: Best = 3.737246e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 1.0733e+01	k_t
Generation 1260: Best = 3.723739e+03 = [4, 2, 2, 3]	FES = 126100	Diversity = 1.0727e+01	k_t
Generation 1270: Best = 3.722197e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 1.0724e+01	k_t
Generation 1280: Best = 3.722197e+03 = [3, 4, 3, 4]	FES = 128100	Diversity = 1.0719e+01	k_t
Generation 1290: Best = 3.721378e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 1.0680e+01	k_t
Generation 1300: Best = 3.719789e+03 = [4, 3, 4, 3]	FES = 130100	Diversity = 1.0680e+01	k_t

Generation 1310: Best = 3.719789e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 1.0640e+01	k_t
Generation 1320: Best = 3.718729e+03 = [2, 4, 4, 4]	FES = 132100	Diversity = 1.0637e+01	k_t
Generation 1330: Best = 3.709453e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 1.0611e+01	k_t
Generation 1340: Best = 3.709453e+03 = [2, 3, 4, 4]	FES = 134100	Diversity = 1.0613e+01	k_t
Generation 1350: Best = 3.709453e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 1.0614e+01	k_t
Generation 1360: Best = 3.709453e+03 = [2, 2, 3, 3]	FES = 136100	Diversity = 1.0612e+01	k_t
Generation 1370: Best = 3.709453e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 1.0614e+01	k_t
Generation 1380: Best = 3.702495e+03 = [4, 4, 2, 3]	FES = 138100	Diversity = 1.0612e+01	k_t
Generation 1390: Best = 3.697596e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 1.0603e+01	k_t
Generation 1400: Best = 3.697596e+03 = [4, 2, 2, 4]	FES = 140100	Diversity = 1.0585e+01	k_t
Generation 1410: Best = 3.694163e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 1.0584e+01	k_t
Generation 1420: Best = 3.694163e+03 = [3, 4, 3, 4]	FES = 142100	Diversity = 1.0575e+01	k_t
Generation 1430: Best = 3.678504e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 1.0588e+01	k_t
Generation 1440: Best = 3.678504e+03 = [4, 3, 2, 4]	FES = 144100	Diversity = 1.0543e+01	k_t
Generation 1450: Best = 3.677803e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 1.0542e+01	k_t
Generation 1460: Best = 3.677371e+03 = [4, 3, 3, 2]	FES = 146100	Diversity = 1.0536e+01	k_t
Generation 1470: Best = 3.677371e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 1.0504e+01	k_t
Generation 1480: Best = 3.673092e+03 = [3, 2, 3, 4]	FES = 148100	Diversity = 1.0504e+01	k_t
Generation 1490: Best = 3.673092e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 1.0473e+01	k_t
Generation 1500: Best = 3.672483e+03 = [2, 4, 3, 4]	FES = 150100	Diversity = 1.0469e+01	k_t
Generation 1510: Best = 3.656625e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 1.0468e+01	k_t
Generation 1520: Best = 3.656614e+03 = [2, 4, 3, 2]	FES = 152100	Diversity = 1.0462e+01	k_t
Generation 1530: Best = 3.654437e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 1.0461e+01	k_t
Generation 1540: Best = 3.653147e+03 = [3, 3, 2, 2]	FES = 154100	Diversity = 1.0465e+01	k_t
Generation 1550: Best = 3.649297e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 1.0464e+01	k_t
Generation 1560: Best = 3.642967e+03 = [3, 4, 2, 4]	FES = 156100	Diversity = 1.0428e+01	k_t
Generation 1570: Best = 3.638381e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 1.0421e+01	k_t

Generation 1580: Best = 3.638381e+03 = [3, 2, 2, 3]	FES = 158100	Diversity = 1.0441e+01	k_t
Generation 1590: Best = 3.638308e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 1.0416e+01	k_t
Generation 1600: Best = 3.631296e+03 = [2, 3, 3, 2]	FES = 160100	Diversity = 1.0402e+01	k_t
Generation 1610: Best = 3.630159e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 1.0416e+01	k_t
Generation 1620: Best = 3.630159e+03 = [2, 3, 4, 3]	FES = 162100	Diversity = 1.0409e+01	k_t
Generation 1630: Best = 3.630159e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 1.0383e+01	k_t
Generation 1640: Best = 3.629188e+03 = [3, 4, 4, 4]	FES = 164100	Diversity = 1.0376e+01	k_t
Generation 1650: Best = 3.626659e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 1.0379e+01	k_t
Generation 1660: Best = 3.626659e+03 = [2, 4, 4, 2]	FES = 166100	Diversity = 1.0383e+01	k_t
Generation 1670: Best = 3.625905e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 1.0389e+01	k_t
Generation 1680: Best = 3.625712e+03 = [3, 4, 3, 4]	FES = 168100	Diversity = 1.0328e+01	k_t
Generation 1690: Best = 3.625053e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 1.0324e+01	k_t
Generation 1700: Best = 3.625053e+03 = [2, 2, 3, 4]	FES = 170100	Diversity = 1.0329e+01	k_t
Generation 1710: Best = 3.624434e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 1.0337e+01	k_t
Generation 1720: Best = 3.623596e+03 = [4, 3, 2, 2]	FES = 172100	Diversity = 1.0337e+01	k_t
Generation 1730: Best = 3.623596e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 1.0335e+01	k_t
Generation 1740: Best = 3.612522e+03 = [2, 2, 4, 3]	FES = 174100	Diversity = 1.0337e+01	k_t
Generation 1750: Best = 3.607256e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 1.0322e+01	k_t
Generation 1760: Best = 3.607177e+03 = [2, 2, 3, 4]	FES = 176100	Diversity = 1.0292e+01	k_t
Generation 1770: Best = 3.606686e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 1.0250e+01	k_t
Generation 1780: Best = 3.606508e+03 = [4, 2, 2, 2]	FES = 178100	Diversity = 1.0163e+01	k_t
Generation 1790: Best = 3.606486e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 1.0161e+01	k_t
Generation 1800: Best = 3.596306e+03 = [2, 4, 3, 3]	FES = 180100	Diversity = 1.0106e+01	k_t
Generation 1810: Best = 3.596306e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 9.9845e+00	k_t
Generation 1820: Best = 3.596306e+03 = [2, 2, 4, 3]	FES = 182100	Diversity = 9.8697e+00	k_t
Generation 1830: Best = 3.584388e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 9.7112e+00	k_t
Generation 1840: Best = 3.580328e+03 = [4, 4, 4, 3]	FES = 184100	Diversity = 9.4964e+00	k_t

Generation 1850: Best = 3.580328e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 9.4525e+00	k_t
Generation 1860: Best = 3.577930e+03 = [4, 4, 3, 3]	FES = 186100	Diversity = 9.4159e+00	k_t
Generation 1870: Best = 3.574492e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 9.3147e+00	k_t
Generation 1880: Best = 3.566237e+03 = [2, 2, 4, 4]	FES = 188100	Diversity = 9.0100e+00	k_t
Generation 1890: Best = 3.555281e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 8.4670e+00	k_t
Generation 1900: Best = 3.542651e+03 = [3, 2, 4, 4]	FES = 190100	Diversity = 7.4033e+00	k_t
Generation 1910: Best = 3.530604e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 5.1942e+00	k_t
Generation 1920: Best = 3.521453e+03 = [3, 4, 3, 4]	FES = 192100	Diversity = 2.7923e+00	k_t
Generation 1930: Best = 3.511821e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 1.0563e+00	k_t
Generation 1940: Best = 3.505263e+03 = [4, 2, 4, 4]	FES = 194100	Diversity = 8.0377e-01	k_t
Generation 1950: Best = 3.495153e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 6.0109e-01	k_t
Generation 1960: Best = 3.487476e+03 = [3, 2, 3, 3]	FES = 196100	Diversity = 5.0135e-01	k_t
Generation 1970: Best = 3.481359e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 3.9349e-01	k_t
Generation 1980: Best = 3.474585e+03 = [2, 3, 4, 4]	FES = 198100	Diversity = 3.0598e-01	k_t
Generation 1990: Best = 3.473324e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 2.3458e-01	k_t
Generation 2000: Best = 3.471071e+03 = [3, 3, 4, 4]	FES = 200100	Diversity = 1.9588e-01	k_t
Generation 2010: Best = 3.469263e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 1.6026e-01	k_t
Generation 2020: Best = 3.468387e+03 = [2, 3, 3, 3]	FES = 202100	Diversity = 1.3586e-01	k_t
Generation 2030: Best = 3.467370e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 1.1910e-01	k_t
Generation 2040: Best = 3.466348e+03 = [2, 3, 2, 2]	FES = 204100	Diversity = 1.0151e-01	k_t
Generation 2050: Best = 3.465755e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 8.7790e-02	k_t
Generation 2060: Best = 3.465400e+03 = [2, 4, 4, 2]	FES = 206100	Diversity = 7.6968e-02	k_t
Generation 2070: Best = 3.465002e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 6.6874e-02	k_t
Generation 2080: Best = 3.464806e+03 = [2, 3, 2, 3]	FES = 208100	Diversity = 5.6979e-02	k_t
Generation 2090: Best = 3.464561e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 5.0033e-02	k_t
Generation 2100: Best = 3.464390e+03 = [2, 2, 3, 4]	FES = 210100	Diversity = 4.6020e-02	k_t
Generation 2110: Best = 3.464247e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 4.1764e-02	k_t

Generation 2120: Best = 3.464112e+03 = [4, 2, 4, 4]	FES = 212100	Diversity = 3.8474e-02	k_t
Generation 2130: Best = 3.463978e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 3.6796e-02	k_t
Generation 2140: Best = 3.463885e+03 = [3, 4, 3, 3]	FES = 214100	Diversity = 3.7173e-02	k_t
Generation 2150: Best = 3.463745e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 3.6304e-02	k_t
Generation 2160: Best = 3.463666e+03 = [2, 4, 4, 2]	FES = 216100	Diversity = 3.4538e-02	k_t
Generation 2170: Best = 3.463580e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 3.1907e-02	k_t
Generation 2180: Best = 3.463514e+03 = [3, 3, 4, 2]	FES = 218100	Diversity = 2.8133e-02	k_t
Generation 2190: Best = 3.463468e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 2.5186e-02	k_t
Generation 2200: Best = 3.463420e+03 = [3, 2, 4, 2]	FES = 220100	Diversity = 2.2507e-02	k_t
Generation 2210: Best = 3.463386e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 2.0428e-02	k_t
Generation 2220: Best = 3.463359e+03 = [2, 4, 4, 2]	FES = 222100	Diversity = 1.8381e-02	k_t
Generation 2230: Best = 3.463332e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 1.6148e-02	k_t
Generation 2240: Best = 3.463317e+03 = [3, 4, 4, 4]	FES = 224100	Diversity = 1.5014e-02	k_t
Generation 2250: Best = 3.463302e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 1.2860e-02	k_t
Generation 2260: Best = 3.463294e+03 = [4, 2, 4, 4]	FES = 226100	Diversity = 1.0653e-02	k_t
Generation 2270: Best = 3.463288e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 9.3554e-03	k_t
Generation 2280: Best = 3.463281e+03 = [3, 2, 2, 4]	FES = 228100	Diversity = 8.6857e-03	k_t
Generation 2290: Best = 3.463275e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 7.6930e-03	k_t
Generation 2300: Best = 3.463270e+03 = [3, 3, 2, 3]	FES = 230100	Diversity = 7.3798e-03	k_t
Generation 2310: Best = 3.463266e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 6.6622e-03	k_t
Generation 2320: Best = 3.463262e+03 = [3, 3, 2, 3]	FES = 232100	Diversity = 5.8172e-03	k_t
Generation 2330: Best = 3.463261e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 5.1714e-03	k_t
Generation 2340: Best = 3.463257e+03 = [2, 2, 3, 4]	FES = 234100	Diversity = 4.6775e-03	k_t
Generation 2350: Best = 3.463255e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 4.4179e-03	k_t
Generation 2360: Best = 3.463254e+03 = [3, 2, 3, 2]	FES = 236100	Diversity = 4.1456e-03	k_t
Generation 2370: Best = 3.463253e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 3.6266e-03	k_t
Generation 2380: Best = 3.463252e+03 = [4, 3, 4, 3]	FES = 238100	Diversity = 3.2544e-03	k_t

Generation 2390: Best = 3.463251e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 2.9707e-03	k_t
Generation 2400: Best = 3.463250e+03 = [3, 2, 3, 4]	FES = 240100	Diversity = 2.7626e-03	k_t
Generation 2410: Best = 3.463249e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 2.6363e-03	k_t
Generation 2420: Best = 3.463249e+03 = [2, 3, 2, 4]	FES = 242100	Diversity = 2.4544e-03	k_t
Generation 2430: Best = 3.463248e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 2.0418e-03	k_t
Generation 2440: Best = 3.463248e+03 = [3, 3, 4, 3]	FES = 244100	Diversity = 2.0073e-03	k_t
Generation 2450: Best = 3.463247e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 1.8387e-03	k_t
Generation 2460: Best = 3.463247e+03 = [4, 3, 4, 4]	FES = 246100	Diversity = 1.7228e-03	k_t
Generation 2470: Best = 3.463247e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 1.5384e-03	k_t
Generation 2480: Best = 3.463247e+03 = [4, 3, 4, 2]	FES = 248100	Diversity = 1.4434e-03	k_t
Generation 2490: Best = 3.463247e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.3310e-03	k_t
Generation 2500: Best = 3.463246e+03 = [4, 2, 2, 4]	FES = 250100	Diversity = 1.1949e-03	k_t
Generation 2510: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.0603e-03	k_t
Generation 2520: Best = 3.463246e+03 = [3, 2, 2, 2]	FES = 252100	Diversity = 9.0534e-04	k_t
Generation 2530: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 8.1165e-04	k_t
Generation 2540: Best = 3.463246e+03 = [2, 3, 2, 4]	FES = 254100	Diversity = 7.2537e-04	k_t
Generation 2550: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 6.4617e-04	k_t
Generation 2560: Best = 3.463246e+03 = [2, 2, 3, 4]	FES = 256100	Diversity = 6.1955e-04	k_t
Generation 2570: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 6.0017e-04	k_t
Generation 2580: Best = 3.463246e+03 = [4, 4, 4, 4]	FES = 258100	Diversity = 5.4435e-04	k_t
Generation 2590: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 4.9529e-04	k_t
Generation 2600: Best = 3.463246e+03 = [3, 3, 2, 3]	FES = 260100	Diversity = 4.3680e-04	k_t
Generation 2610: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 4.1088e-04	k_t
Generation 2620: Best = 3.463246e+03 = [3, 3, 2, 3]	FES = 262100	Diversity = 3.7948e-04	k_t
Generation 2630: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 3.2190e-04	k_t
Generation 2640: Best = 3.463246e+03 = [4, 3, 4, 2]	FES = 264100	Diversity = 2.7003e-04	k_t
Generation 2650: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 2.4514e-04	k_t

Generation 2660: Best = 3.463246e+03 = [3, 2, 2, 4]	FES = 266100	Diversity = 2.2498e-04	k_t
Generation 2670: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 2.2127e-04	k_t
Generation 2680: Best = 3.463246e+03 = [4, 2, 2, 4]	FES = 268100	Diversity = 2.0172e-04	k_t
Generation 2690: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.8531e-04	k_t
Generation 2700: Best = 3.463246e+03 = [3, 4, 2, 3]	FES = 270100	Diversity = 1.8447e-04	k_t
Generation 2710: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.7709e-04	k_t
Generation 2720: Best = 3.463246e+03 = [4, 3, 2, 3]	FES = 272100	Diversity = 1.6268e-04	k_t
Generation 2730: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.5700e-04	k_t
Generation 2740: Best = 3.463246e+03 = [4, 4, 2, 3]	FES = 274100	Diversity = 1.4514e-04	k_t
Generation 2750: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.3303e-04	k_t
Generation 2760: Best = 3.463246e+03 = [2, 3, 4, 3]	FES = 276100	Diversity = 1.1478e-04	k_t
Generation 2770: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.0766e-04	k_t
Generation 2780: Best = 3.463246e+03 = [4, 3, 2, 2]	FES = 278100	Diversity = 1.0188e-04	k_t
Generation 2790: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 8.5881e-05	k_t
Generation 2800: Best = 3.463246e+03 = [3, 4, 4, 4]	FES = 280100	Diversity = 7.4436e-05	k_t
Generation 2810: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 6.5681e-05	k_t
Generation 2820: Best = 3.463246e+03 = [3, 3, 4, 4]	FES = 282100	Diversity = 5.9371e-05	k_t
Generation 2830: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 5.1051e-05	k_t
Generation 2840: Best = 3.463246e+03 = [3, 2, 4, 4]	FES = 284100	Diversity = 4.8689e-05	k_t
Generation 2850: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 4.3476e-05	k_t
Generation 2860: Best = 3.463246e+03 = [2, 3, 2, 3]	FES = 286100	Diversity = 4.2320e-05	k_t
Generation 2870: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 3.7396e-05	k_t
Generation 2880: Best = 3.463246e+03 = [4, 2, 3, 3]	FES = 288100	Diversity = 3.3051e-05	k_t
Generation 2890: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 3.0323e-05	k_t
Generation 2900: Best = 3.463246e+03 = [2, 4, 3, 2]	FES = 290100	Diversity = 2.7344e-05	k_t
Generation 2910: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 2.6114e-05	k_t
Generation 2920: Best = 3.463246e+03 = [4, 2, 3, 2]	FES = 292100	Diversity = 2.6016e-05	k_t

Generation 2930: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 293100	Diversity = 2.3343e-05	k_t
Generation 2940: Best = 3.463246e+03 = [2, 2, 3, 2]	FES = 294100	Diversity = 2.0922e-05	k_t
Generation 2950: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 295100	Diversity = 2.0074e-05	k_t
Generation 2960: Best = 3.463246e+03 = [3, 3, 3, 3]	FES = 296100	Diversity = 1.7381e-05	k_t
Generation 2970: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 297100	Diversity = 1.5843e-05	k_t
Generation 2980: Best = 3.463246e+03 = [3, 3, 2, 2]	FES = 298100	Diversity = 1.4941e-05	k_t
Generation 2990: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 299100	Diversity = 1.3624e-05	k_t
Generation 3000: Best = 3.463246e+03 = [4, 2, 2, 3]	FES = 300100	Diversity = 1.1620e-05	k_t
Generation 3010: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 301100	Diversity = 1.0826e-05	k_t
Generation 3020: Best = 3.463246e+03 = [2, 2, 2, 2]	FES = 302100	Diversity = 1.0006e-05	k_t
Generation 3030: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 303100	Diversity = 9.6455e-06	k_t
Generation 3040: Best = 3.463246e+03 = [2, 2, 4, 2]	FES = 304100	Diversity = 8.6166e-06	k_t
Generation 3050: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 305100	Diversity = 7.4132e-06	k_t
Generation 3060: Best = 3.463246e+03 = [4, 2, 2, 4]	FES = 306100	Diversity = 6.5734e-06	k_t
Generation 3070: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 307100	Diversity = 5.7154e-06	k_t
Generation 3080: Best = 3.463246e+03 = [4, 4, 2, 3]	FES = 308100	Diversity = 5.5220e-06	k_t
Generation 3090: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 309100	Diversity = 4.8842e-06	k_t
Generation 3100: Best = 3.463246e+03 = [3, 2, 2, 2]	FES = 310100	Diversity = 4.3131e-06	k_t
Generation 3110: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 311100	Diversity = 4.2289e-06	k_t
Generation 3120: Best = 3.463246e+03 = [4, 4, 2, 3]	FES = 312100	Diversity = 3.7103e-06	k_t
Generation 3130: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 313100	Diversity = 3.4461e-06	k_t
Generation 3140: Best = 3.463246e+03 = [3, 3, 4, 3]	FES = 314100	Diversity = 3.2686e-06	k_t
Generation 3150: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 315100	Diversity = 3.0389e-06	k_t
Generation 3160: Best = 3.463246e+03 = [3, 3, 2, 4]	FES = 316100	Diversity = 2.8147e-06	k_t
Generation 3170: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 317100	Diversity = 2.4637e-06	k_t
Generation 3180: Best = 3.463246e+03 = [2, 4, 4, 3]	FES = 318100	Diversity = 2.2468e-06	k_t
Generation 3190: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 319100	Diversity = 2.0223e-06	k_t

Generation 3200: Best = 3.463246e+03 = [2, 3, 2, 2]	FES = 320100	Diversity = 1.7703e-06	k_t
Generation 3210: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 321100	Diversity = 1.4966e-06	k_t
Generation 3220: Best = 3.463246e+03 = [3, 2, 3, 2]	FES = 322100	Diversity = 1.3638e-06	k_t
Generation 3230: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 323100	Diversity = 1.1849e-06	k_t
Generation 3240: Best = 3.463246e+03 = [3, 3, 3, 3]	FES = 324100	Diversity = 1.0952e-06	k_t
Generation 3250: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 325100	Diversity = 9.8490e-07	k_t
Generation 3260: Best = 3.463246e+03 = [2, 3, 2, 3]	FES = 326100	Diversity = 9.4851e-07	k_t
Generation 3270: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 327100	Diversity = 9.7538e-07	k_t
Generation 3280: Best = 3.463246e+03 = [4, 3, 3, 4]	FES = 328100	Diversity = 9.3030e-07	k_t
Generation 3290: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 329100	Diversity = 8.0885e-07	k_t
Generation 3300: Best = 3.463246e+03 = [2, 3, 3, 3]	FES = 330100	Diversity = 7.3094e-07	k_t
Generation 3310: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 331100	Diversity = 6.4865e-07	k_t
Generation 3320: Best = 3.463246e+03 = [4, 3, 2, 4]	FES = 332100	Diversity = 5.7550e-07	k_t
Generation 3330: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 333100	Diversity = 5.1279e-07	k_t
Generation 3340: Best = 3.463246e+03 = [2, 3, 2, 3]	FES = 334100	Diversity = 4.2435e-07	k_t
Generation 3350: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 335100	Diversity = 3.8586e-07	k_t
Generation 3360: Best = 3.463246e+03 = [4, 2, 3, 2]	FES = 336100	Diversity = 3.4471e-07	k_t
Generation 3370: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 337100	Diversity = 3.2903e-07	k_t
Generation 3380: Best = 3.463246e+03 = [2, 3, 2, 3]	FES = 338100	Diversity = 2.9339e-07	k_t
Generation 3390: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 339100	Diversity = 2.8499e-07	k_t
Generation 3400: Best = 3.463246e+03 = [3, 2, 3, 3]	FES = 340100	Diversity = 2.5600e-07	k_t
Generation 3410: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 341100	Diversity = 2.2793e-07	k_t
Generation 3420: Best = 3.463246e+03 = [4, 2, 4, 4]	FES = 342100	Diversity = 2.1821e-07	k_t
Generation 3430: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 343100	Diversity = 2.0021e-07	k_t
Generation 3440: Best = 3.463246e+03 = [2, 2, 2, 2]	FES = 344100	Diversity = 1.8337e-07	k_t
Generation 3450: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 345100	Diversity = 1.6468e-07	k_t
Generation 3460: Best = 3.463246e+03 = [3, 2, 4, 2]	FES = 346100	Diversity = 1.3905e-07	k_t

Generation 3470: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 347100	Diversity = 1.2737e-07	k_t
Generation 3480: Best = 3.463246e+03 = [3, 4, 3, 3]	FES = 348100	Diversity = 1.1510e-07	k_t
Generation 3490: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 349100	Diversity = 1.0936e-07	k_t
Generation 3500: Best = 3.463246e+03 = [2, 2, 4, 4]	FES = 350100	Diversity = 1.0816e-07	k_t
Generation 3510: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 351100	Diversity = 1.0516e-07	k_t
Generation 3520: Best = 3.463246e+03 = [3, 3, 3, 2]	FES = 352100	Diversity = 1.1422e-07	k_t
Generation 3530: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 353100	Diversity = 1.0896e-07	k_t
Generation 3540: Best = 3.463246e+03 = [3, 4, 3, 3]	FES = 354100	Diversity = 1.1092e-07	k_t
Generation 3550: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 355100	Diversity = 1.1557e-07	k_t
Generation 3560: Best = 3.463246e+03 = [3, 4, 3, 4]	FES = 356100	Diversity = 1.1162e-07	k_t
Generation 3570: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 357100	Diversity = 1.1013e-07	k_t
Generation 3580: Best = 3.463246e+03 = [3, 4, 3, 4]	FES = 358100	Diversity = 1.0734e-07	k_t
Generation 3590: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 359100	Diversity = 1.0368e-07	k_t
Generation 3600: Best = 3.463246e+03 = [3, 2, 4, 2]	FES = 360100	Diversity = 1.0260e-07	k_t
Generation 3610: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 361100	Diversity = 1.0268e-07	k_t
Generation 3620: Best = 3.463246e+03 = [3, 4, 3, 4]	FES = 362100	Diversity = 9.9594e-08	k_t
Generation 3630: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 363100	Diversity = 1.0175e-07	k_t
Generation 3640: Best = 3.463246e+03 = [2, 4, 3, 2]	FES = 364100	Diversity = 1.1064e-07	k_t
Generation 3650: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 365100	Diversity = 1.0981e-07	k_t
Generation 3660: Best = 3.463246e+03 = [2, 4, 3, 4]	FES = 366100	Diversity = 1.0700e-07	k_t
Generation 3670: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 367100	Diversity = 1.0379e-07	k_t
Generation 3680: Best = 3.463246e+03 = [3, 4, 3, 3]	FES = 368100	Diversity = 9.9794e-08	k_t
Generation 3690: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 369100	Diversity = 9.9081e-08	k_t
Generation 3700: Best = 3.463246e+03 = [2, 2, 2, 4]	FES = 370100	Diversity = 9.5020e-08	k_t
Generation 3710: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 371100	Diversity = 9.3923e-08	k_t
Generation 3720: Best = 3.463246e+03 = [4, 2, 2, 4]	FES = 372100	Diversity = 9.2991e-08	k_t
Generation 3730: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 373100	Diversity = 9.2531e-08	k_t

Generation 3740: Best = 3.463246e+03 = [2, 3, 2, 4]	FES = 374100	Diversity = 9.2650e-08	k_t
Generation 3750: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 375100	Diversity = 9.3174e-08	k_t
Generation 3760: Best = 3.463246e+03 = [4, 4, 3, 4]	FES = 376100	Diversity = 9.3658e-08	k_t
Generation 3770: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 377100	Diversity = 9.4742e-08	k_t
Generation 3780: Best = 3.463246e+03 = [4, 2, 4, 2]	FES = 378100	Diversity = 9.6722e-08	k_t
Generation 3790: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 379100	Diversity = 9.8567e-08	k_t
Generation 3800: Best = 3.463246e+03 = [2, 2, 3, 4]	FES = 380100	Diversity = 9.7965e-08	k_t
Generation 3810: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 381100	Diversity = 9.8561e-08	k_t
Generation 3820: Best = 3.463246e+03 = [4, 4, 4, 4]	FES = 382100	Diversity = 9.9214e-08	k_t
Generation 3830: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 383100	Diversity = 9.9165e-08	k_t
Generation 3840: Best = 3.463246e+03 = [4, 3, 3, 4]	FES = 384100	Diversity = 9.8446e-08	k_t
Generation 3850: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 385100	Diversity = 9.6690e-08	k_t
Generation 3860: Best = 3.463246e+03 = [2, 3, 4, 2]	FES = 386100	Diversity = 9.6256e-08	k_t
Generation 3870: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 387100	Diversity = 9.7465e-08	k_t
Generation 3880: Best = 3.463246e+03 = [3, 4, 4, 3]	FES = 388100	Diversity = 9.6653e-08	k_t
Generation 3890: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 389100	Diversity = 9.5667e-08	k_t
Generation 3900: Best = 3.463246e+03 = [4, 4, 2, 2]	FES = 390100	Diversity = 9.6181e-08	k_t
Generation 3910: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 391100	Diversity = 9.5735e-08	k_t
Generation 3920: Best = 3.463246e+03 = [2, 4, 2, 2]	FES = 392100	Diversity = 9.5193e-08	k_t
Generation 3930: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 393100	Diversity = 9.2571e-08	k_t
Generation 3940: Best = 3.463246e+03 = [2, 3, 3, 4]	FES = 394100	Diversity = 9.0695e-08	k_t
Generation 3950: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 395100	Diversity = 8.9426e-08	k_t
Generation 3960: Best = 3.463246e+03 = [2, 3, 4, 3]	FES = 396100	Diversity = 8.9692e-08	k_t
Generation 3970: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 397100	Diversity = 8.8813e-08	k_t
Generation 3980: Best = 3.463246e+03 = [3, 2, 3, 3]	FES = 398100	Diversity = 8.7862e-08	k_t
Generation 3990: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 399100	Diversity = 8.7749e-08	k_t
Generation 4000: Best = 3.463246e+03 = [2, 3, 2, 4]	FES = 400100	Diversity = 8.7249e-08	k_t

Generation 4010: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 401100	Diversity = 8.6282e-08	k_t
Generation 4020: Best = 3.463246e+03 = [2, 2, 4, 2]	FES = 402100	Diversity = 8.6157e-08	k_t
Generation 4030: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 403100	Diversity = 8.5827e-08	k_t
Generation 4040: Best = 3.463246e+03 = [4, 4, 2, 3]	FES = 404100	Diversity = 8.5977e-08	k_t
Generation 4050: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 405100	Diversity = 8.5502e-08	k_t
Generation 4060: Best = 3.463246e+03 = [3, 3, 2, 3]	FES = 406100	Diversity = 8.5175e-08	k_t
Generation 4070: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 407100	Diversity = 8.5034e-08	k_t
Generation 4080: Best = 3.463246e+03 = [4, 3, 3, 3]	FES = 408100	Diversity = 8.5570e-08	k_t
Generation 4090: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 409100	Diversity = 8.5722e-08	k_t
Generation 4100: Best = 3.463246e+03 = [4, 2, 2, 3]	FES = 410100	Diversity = 8.5483e-08	k_t
Generation 4110: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 411100	Diversity = 8.6329e-08	k_t
Generation 4120: Best = 3.463246e+03 = [2, 2, 4, 4]	FES = 412100	Diversity = 8.6258e-08	k_t
Generation 4130: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 413100	Diversity = 8.6853e-08	k_t
Generation 4140: Best = 3.463246e+03 = [4, 4, 3, 4]	FES = 414100	Diversity = 8.7363e-08	k_t
Generation 4150: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 415100	Diversity = 8.7335e-08	k_t
Generation 4160: Best = 3.463246e+03 = [4, 2, 4, 4]	FES = 416100	Diversity = 8.8242e-08	k_t
Generation 4170: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 417100	Diversity = 8.8569e-08	k_t
Generation 4180: Best = 3.463246e+03 = [4, 4, 4, 2]	FES = 418100	Diversity = 8.9211e-08	k_t
Generation 4190: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 419100	Diversity = 8.9645e-08	k_t
Generation 4200: Best = 3.463246e+03 = [3, 2, 2, 3]	FES = 420100	Diversity = 8.8838e-08	k_t
Generation 4210: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 421100	Diversity = 8.8938e-08	k_t
Generation 4220: Best = 3.463246e+03 = [4, 4, 4, 2]	FES = 422100	Diversity = 8.8804e-08	k_t
Generation 4230: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 423100	Diversity = 8.8886e-08	k_t
Generation 4240: Best = 3.463246e+03 = [2, 4, 2, 3]	FES = 424100	Diversity = 8.8439e-08	k_t
Generation 4250: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 425100	Diversity = 8.8840e-08	k_t
Generation 4260: Best = 3.463246e+03 = [2, 2, 4, 3]	FES = 426100	Diversity = 8.9263e-08	k_t
Generation 4270: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 427100	Diversity = 8.9823e-08	k_t

Generation 4280: Best = 3.463246e+03 = [3, 4, 2, 2]	FES = 428100	Diversity = 9.0216e-08	k_t
Generation 4290: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 429100	Diversity = 9.0286e-08	k_t
Generation 4300: Best = 3.463246e+03 = [4, 3, 2, 4]	FES = 430100	Diversity = 9.1228e-08	k_t
Generation 4310: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 431100	Diversity = 9.1427e-08	k_t
Generation 4320: Best = 3.463246e+03 = [3, 3, 3, 3]	FES = 432100	Diversity = 9.2321e-08	k_t
Generation 4330: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 433100	Diversity = 9.2470e-08	k_t
Generation 4340: Best = 3.463246e+03 = [3, 4, 3, 3]	FES = 434100	Diversity = 9.2545e-08	k_t
Generation 4350: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 435100	Diversity = 9.2943e-08	k_t
Generation 4360: Best = 3.463246e+03 = [3, 3, 3, 4]	FES = 436100	Diversity = 9.3331e-08	k_t
Generation 4370: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 437100	Diversity = 9.3023e-08	k_t
Generation 4380: Best = 3.463246e+03 = [4, 3, 2, 2]	FES = 438100	Diversity = 9.2992e-08	k_t
Generation 4390: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 439100	Diversity = 9.3333e-08	k_t
Generation 4400: Best = 3.463246e+03 = [4, 4, 2, 2]	FES = 440100	Diversity = 9.3265e-08	k_t
Generation 4410: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 441100	Diversity = 9.3489e-08	k_t
Generation 4420: Best = 3.463246e+03 = [2, 4, 2, 3]	FES = 442100	Diversity = 9.3848e-08	k_t
Generation 4430: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 443100	Diversity = 9.4065e-08	k_t
Generation 4440: Best = 3.463246e+03 = [2, 3, 4, 3]	FES = 444100	Diversity = 9.5308e-08	k_t
Generation 4450: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 445100	Diversity = 9.5458e-08	k_t
Generation 4460: Best = 3.463246e+03 = [3, 2, 2, 3]	FES = 446100	Diversity = 9.5803e-08	k_t
Generation 4470: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 447100	Diversity = 9.5782e-08	k_t
Generation 4480: Best = 3.463246e+03 = [2, 4, 3, 2]	FES = 448100	Diversity = 9.6119e-08	k_t
Generation 4490: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 449100	Diversity = 9.6960e-08	k_t
Generation 4500: Best = 3.463246e+03 = [3, 2, 3, 4]	FES = 450100	Diversity = 9.7264e-08	k_t
Generation 4510: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 451100	Diversity = 9.7577e-08	k_t
Generation 4520: Best = 3.463246e+03 = [3, 4, 2, 3]	FES = 452100	Diversity = 9.8293e-08	k_t
Generation 4530: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 453100	Diversity = 9.8699e-08	k_t
Generation 4540: Best = 3.463246e+03 = [4, 4, 3, 4]	FES = 454100	Diversity = 9.9334e-08	k_t

Generation 4550: Best = 3.463246e+03	FES = 455100	Diversity = 9.9358e-08	k_t
= [1, 1, 1, 1]			
Generation 4560: Best = 3.463246e+03	FES = 456100	Diversity = 9.9537e-08	k_t
= [2, 4, 4, 2]			
Generation 4570: Best = 3.463246e+03	FES = 457100	Diversity = 9.9639e-08	k_t
= [1, 1, 1, 1]			
Generation 4580: Best = 3.463246e+03	FES = 458100	Diversity = 9.9740e-08	k_t
= [2, 4, 2, 2]			
Generation 4590: Best = 3.463246e+03	FES = 459100	Diversity = 1.0039e-07	k_t
= [1, 1, 1, 1]			
Generation 4600: Best = 3.463246e+03	FES = 460100	Diversity = 1.0025e-07	k_t
= [3, 4, 3, 4]			
Generation 4610: Best = 3.463246e+03	FES = 461100	Diversity = 9.9900e-08	k_t
= [1, 1, 1, 1]			
Generation 4620: Best = 3.463246e+03	FES = 462100	Diversity = 9.9722e-08	k_t
= [3, 4, 4, 3]			
Generation 4630: Best = 3.463246e+03	FES = 463100	Diversity = 9.9465e-08	k_t
= [1, 1, 1, 1]			
Generation 4640: Best = 3.463246e+03	FES = 464100	Diversity = 9.9060e-08	k_t
= [3, 2, 2, 2]			
Generation 4650: Best = 3.463246e+03	FES = 465100	Diversity = 9.8741e-08	k_t
= [1, 1, 1, 1]			
Generation 4660: Best = 3.463246e+03	FES = 466100	Diversity = 9.8513e-08	k_t
= [3, 4, 4, 4]			
Generation 4670: Best = 3.463246e+03	FES = 467100	Diversity = 9.8272e-08	k_t
= [1, 1, 1, 1]			
Generation 4680: Best = 3.463246e+03	FES = 468100	Diversity = 9.8890e-08	k_t
= [4, 2, 4, 4]			
Generation 4690: Best = 3.463246e+03	FES = 469100	Diversity = 9.8342e-08	k_t
= [1, 1, 1, 1]			
Generation 4700: Best = 3.463246e+03	FES = 470100	Diversity = 9.7526e-08	k_t
= [3, 4, 3, 2]			
Generation 4710: Best = 3.463246e+03	FES = 471100	Diversity = 9.7455e-08	k_t
= [1, 1, 1, 1]			
Generation 4720: Best = 3.463246e+03	FES = 472100	Diversity = 9.7107e-08	k_t
= [4, 2, 2, 2]			
Generation 4730: Best = 3.463246e+03	FES = 473100	Diversity = 9.7376e-08	k_t
= [1, 1, 1, 1]			
Generation 4740: Best = 3.463246e+03	FES = 474100	Diversity = 9.7092e-08	k_t
= [3, 2, 2, 4]			
Generation 4750: Best = 3.463246e+03	FES = 475100	Diversity = 9.7752e-08	k_t
= [1, 1, 1, 1]			
Generation 4760: Best = 3.463246e+03	FES = 476100	Diversity = 9.7721e-08	k_t
= [4, 3, 3, 2]			
Generation 4770: Best = 3.463246e+03	FES = 477100	Diversity = 9.8225e-08	k_t
= [1, 1, 1, 1]			
Generation 4780: Best = 3.463246e+03	FES = 478100	Diversity = 9.8232e-08	k_t
= [2, 4, 3, 3]			
Generation 4790: Best = 3.463246e+03	FES = 479100	Diversity = 9.8026e-08	k_t
= [1, 1, 1, 1]			
Generation 4800: Best = 3.463246e+03	FES = 480100	Diversity = 9.8336e-08	k_t
= [4, 4, 2, 3]			
Generation 4810: Best = 3.463246e+03	FES = 481100	Diversity = 9.7668e-08	k_t
= [1, 1, 1, 1]			

Generation 4820: Best = 3.463246e+03 = [3, 4, 2, 3]	FES = 482100	Diversity = 9.7755e-08	k_t
Generation 4830: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 483100	Diversity = 9.7841e-08	k_t
Generation 4840: Best = 3.463246e+03 = [3, 2, 2, 3]	FES = 484100	Diversity = 9.7925e-08	k_t
Generation 4850: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 485100	Diversity = 9.8267e-08	k_t
Generation 4860: Best = 3.463246e+03 = [2, 4, 4, 3]	FES = 486100	Diversity = 9.8373e-08	k_t
Generation 4870: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 487100	Diversity = 9.8599e-08	k_t
Generation 4880: Best = 3.463246e+03 = [3, 4, 3, 3]	FES = 488100	Diversity = 9.8607e-08	k_t
Generation 4890: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 489100	Diversity = 9.8347e-08	k_t
Generation 4900: Best = 3.463246e+03 = [2, 3, 2, 3]	FES = 490100	Diversity = 9.8545e-08	k_t
Generation 4910: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 491100	Diversity = 9.8598e-08	k_t
Generation 4920: Best = 3.463246e+03 = [4, 2, 3, 2]	FES = 492100	Diversity = 9.8122e-08	k_t
Generation 4930: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 493100	Diversity = 9.7594e-08	k_t
Generation 4940: Best = 3.463246e+03 = [4, 3, 3, 2]	FES = 494100	Diversity = 9.7937e-08	k_t
Generation 4950: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 495100	Diversity = 9.8354e-08	k_t
Generation 4960: Best = 3.463246e+03 = [2, 4, 4, 4]	FES = 496100	Diversity = 9.7943e-08	k_t
Generation 4970: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 497100	Diversity = 9.7941e-08	k_t
Generation 4980: Best = 3.463246e+03 = [4, 4, 4, 2]	FES = 498100	Diversity = 9.8229e-08	k_t
Generation 4990: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 499100	Diversity = 9.8538e-08	k_t
Generation 5000: Best = 3.463246e+03 = [4, 3, 3, 3]	FES = 500100	Diversity = 9.8547e-08	k_t
Generation 5010: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 501100	Diversity = 9.8968e-08	k_t
Generation 5020: Best = 3.463246e+03 = [3, 4, 3, 3]	FES = 502100	Diversity = 9.9083e-08	k_t
Generation 5030: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 503100	Diversity = 9.9288e-08	k_t
Generation 5040: Best = 3.463246e+03 = [3, 2, 2, 4]	FES = 504100	Diversity = 9.9845e-08	k_t
Generation 5050: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 505100	Diversity = 1.0056e-07	k_t
Generation 5060: Best = 3.463246e+03 = [3, 4, 4, 2]	FES = 506100	Diversity = 1.0090e-07	k_t
Generation 5070: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 507100	Diversity = 1.0105e-07	k_t
Generation 5080: Best = 3.463246e+03 = [2, 4, 4, 2]	FES = 508100	Diversity = 1.0101e-07	k_t

Generation 5090: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 509100	Diversity = 1.0096e-07	k_t
Generation 5100: Best = 3.463246e+03 = [4, 4, 4, 4]	FES = 510100	Diversity = 1.0172e-07	k_t
Generation 5110: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 511100	Diversity = 1.0131e-07	k_t
Generation 5120: Best = 3.463246e+03 = [3, 2, 2, 3]	FES = 512100	Diversity = 1.0118e-07	k_t
Generation 5130: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 513100	Diversity = 1.0042e-07	k_t
Generation 5140: Best = 3.463246e+03 = [3, 2, 3, 2]	FES = 514100	Diversity = 1.0041e-07	k_t
Generation 5150: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 515100	Diversity = 1.0102e-07	k_t
Generation 5160: Best = 3.463246e+03 = [4, 2, 3, 2]	FES = 516100	Diversity = 1.0100e-07	k_t
Generation 5170: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 517100	Diversity = 1.0084e-07	k_t
Generation 5180: Best = 3.463246e+03 = [2, 3, 4, 3]	FES = 518100	Diversity = 1.0055e-07	k_t
Generation 5190: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 519100	Diversity = 1.0050e-07	k_t
Generation 5200: Best = 3.463246e+03 = [2, 2, 2, 2]	FES = 520100	Diversity = 1.0023e-07	k_t
Generation 5210: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 521100	Diversity = 1.0027e-07	k_t
Generation 5220: Best = 3.463246e+03 = [3, 2, 3, 2]	FES = 522100	Diversity = 9.9839e-08	k_t
Generation 5230: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 523100	Diversity = 9.9292e-08	k_t
Generation 5240: Best = 3.463246e+03 = [3, 4, 3, 2]	FES = 524100	Diversity = 9.9211e-08	k_t
Generation 5250: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 525100	Diversity = 9.9099e-08	k_t
Generation 5260: Best = 3.463246e+03 = [3, 2, 2, 3]	FES = 526100	Diversity = 9.9069e-08	k_t
Generation 5270: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 527100	Diversity = 9.8914e-08	k_t
Generation 5280: Best = 3.463246e+03 = [4, 2, 2, 3]	FES = 528100	Diversity = 9.7604e-08	k_t
Generation 5290: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 529100	Diversity = 9.7033e-08	k_t
Generation 5300: Best = 3.463246e+03 = [4, 3, 4, 3]	FES = 530100	Diversity = 9.6610e-08	k_t
Generation 5310: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 531100	Diversity = 9.6729e-08	k_t
Generation 5320: Best = 3.463246e+03 = [2, 3, 4, 2]	FES = 532100	Diversity = 9.6381e-08	k_t
Generation 5330: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 533100	Diversity = 9.6128e-08	k_t
Generation 5340: Best = 3.463246e+03 = [2, 4, 4, 3]	FES = 534100	Diversity = 9.5918e-08	k_t
Generation 5350: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 535100	Diversity = 9.5248e-08	k_t

Generation 5360: Best = 3.463246e+03 = [4, 4, 4, 2]	FES = 536100	Diversity = 9.5213e-08	k_t
Generation 5370: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 537100	Diversity = 9.4235e-08	k_t
Generation 5380: Best = 3.463246e+03 = [4, 2, 4, 2]	FES = 538100	Diversity = 9.4151e-08	k_t
Generation 5390: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 539100	Diversity = 9.3380e-08	k_t
Generation 5400: Best = 3.463246e+03 = [2, 3, 3, 3]	FES = 540100	Diversity = 9.3319e-08	k_t
Generation 5410: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 541100	Diversity = 9.2961e-08	k_t
Generation 5420: Best = 3.463246e+03 = [4, 4, 4, 3]	FES = 542100	Diversity = 9.2228e-08	k_t
Generation 5430: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 543100	Diversity = 9.2423e-08	k_t
Generation 5440: Best = 3.463246e+03 = [3, 2, 3, 2]	FES = 544100	Diversity = 9.2496e-08	k_t
Generation 5450: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 545100	Diversity = 9.2341e-08	k_t
Generation 5460: Best = 3.463246e+03 = [4, 2, 2, 2]	FES = 546100	Diversity = 9.2976e-08	k_t
Generation 5470: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 547100	Diversity = 9.2236e-08	k_t
Generation 5480: Best = 3.463246e+03 = [4, 4, 2, 2]	FES = 548100	Diversity = 9.1725e-08	k_t
Generation 5490: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 549100	Diversity = 8.8915e-08	k_t
Generation 5500: Best = 3.463246e+03 = [4, 3, 2, 4]	FES = 550100	Diversity = 8.8733e-08	k_t
Generation 5510: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 551100	Diversity = 8.7311e-08	k_t
Generation 5520: Best = 3.463246e+03 = [3, 4, 4, 3]	FES = 552100	Diversity = 8.6984e-08	k_t
Generation 5530: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 553100	Diversity = 8.5538e-08	k_t
Generation 5540: Best = 3.463246e+03 = [2, 3, 3, 4]	FES = 554100	Diversity = 8.4075e-08	k_t
Generation 5550: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 555100	Diversity = 8.2822e-08	k_t
Generation 5560: Best = 3.463246e+03 = [4, 2, 4, 3]	FES = 556100	Diversity = 8.2738e-08	k_t
Generation 5570: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 557100	Diversity = 8.2637e-08	k_t
Generation 5580: Best = 3.463246e+03 = [4, 3, 4, 2]	FES = 558100	Diversity = 8.1372e-08	k_t
Generation 5590: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 559100	Diversity = 8.0163e-08	k_t
Generation 5600: Best = 3.463246e+03 = [4, 3, 4, 2]	FES = 560100	Diversity = 7.9714e-08	k_t
Generation 5610: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 561100	Diversity = 7.8339e-08	k_t
Generation 5620: Best = 3.463246e+03 = [3, 2, 2, 4]	FES = 562100	Diversity = 7.9036e-08	k_t

Generation 5630: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 563100	Diversity = 7.7678e-08	k_t
Generation 5640: Best = 3.463246e+03 = [4, 4, 2, 2]	FES = 564100	Diversity = 7.7619e-08	k_t
Generation 5650: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 565100	Diversity = 7.7603e-08	k_t
Generation 5660: Best = 3.463246e+03 = [3, 3, 2, 3]	FES = 566100	Diversity = 7.6831e-08	k_t
Generation 5670: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 567100	Diversity = 7.7934e-08	k_t
Generation 5680: Best = 3.463246e+03 = [4, 3, 3, 2]	FES = 568100	Diversity = 7.7408e-08	k_t
Generation 5690: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 569100	Diversity = 7.7352e-08	k_t
Generation 5700: Best = 3.463246e+03 = [3, 4, 3, 2]	FES = 570100	Diversity = 7.7291e-08	k_t
Generation 5710: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 571100	Diversity = 7.6060e-08	k_t
Generation 5720: Best = 3.463246e+03 = [3, 4, 2, 3]	FES = 572100	Diversity = 7.6387e-08	k_t
Generation 5730: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 573100	Diversity = 7.6717e-08	k_t
Generation 5740: Best = 3.463246e+03 = [2, 3, 2, 3]	FES = 574100	Diversity = 7.6188e-08	k_t
Generation 5750: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 575100	Diversity = 7.5318e-08	k_t
Generation 5760: Best = 3.463246e+03 = [2, 3, 3, 3]	FES = 576100	Diversity = 7.5684e-08	k_t
Generation 5770: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 577100	Diversity = 7.5628e-08	k_t
Generation 5780: Best = 3.463246e+03 = [3, 3, 4, 4]	FES = 578100	Diversity = 7.5130e-08	k_t
Generation 5790: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 579100	Diversity = 7.4086e-08	k_t
Generation 5800: Best = 3.463246e+03 = [2, 2, 4, 3]	FES = 580100	Diversity = 7.3263e-08	k_t
Generation 5810: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 581100	Diversity = 7.3329e-08	k_t
Generation 5820: Best = 3.463246e+03 = [2, 3, 3, 3]	FES = 582100	Diversity = 7.3950e-08	k_t
Generation 5830: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 583100	Diversity = 7.4570e-08	k_t
Generation 5840: Best = 3.463246e+03 = [3, 3, 2, 3]	FES = 584100	Diversity = 7.4025e-08	k_t
Generation 5850: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 585100	Diversity = 7.3099e-08	k_t
Generation 5860: Best = 3.463246e+03 = [3, 3, 2, 4]	FES = 586100	Diversity = 7.3802e-08	k_t
Generation 5870: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 587100	Diversity = 7.3664e-08	k_t
Generation 5880: Best = 3.463246e+03 = [4, 2, 2, 4]	FES = 588100	Diversity = 7.3795e-08	k_t
Generation 5890: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 589100	Diversity = 7.4110e-08	k_t

Generation 5900: Best = 3.463246e+03 = [2, 3, 4, 4]	FES = 590100	Diversity = 7.3944e-08	k_t
Generation 5910: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 591100	Diversity = 7.4013e-08	k_t
Generation 5920: Best = 3.463246e+03 = [3, 2, 3, 2]	FES = 592100	Diversity = 7.4362e-08	k_t
Generation 5930: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 593100	Diversity = 7.4296e-08	k_t
Generation 5940: Best = 3.463246e+03 = [2, 2, 4, 3]	FES = 594100	Diversity = 7.4603e-08	k_t
Generation 5950: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 595100	Diversity = 7.3453e-08	k_t
Generation 5960: Best = 3.463246e+03 = [4, 4, 4, 2]	FES = 596100	Diversity = 7.3366e-08	k_t
Generation 5970: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 597100	Diversity = 7.2485e-08	k_t
Generation 5980: Best = 3.463246e+03 = [3, 4, 2, 3]	FES = 598100	Diversity = 7.2842e-08	k_t
Generation 5990: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 599100	Diversity = 7.2624e-08	k_t
Generation 6000: Best = 3.463246e+03 = [4, 4, 2, 4]	FES = 600100	Diversity = 7.2495e-08	k_t
Generation 6010: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 601100	Diversity = 7.3178e-08	k_t
Generation 6020: Best = 3.463246e+03 = [3, 3, 2, 2]	FES = 602100	Diversity = 7.3220e-08	k_t
Generation 6030: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 603100	Diversity = 7.3205e-08	k_t
Generation 6040: Best = 3.463246e+03 = [3, 4, 4, 4]	FES = 604100	Diversity = 7.3267e-08	k_t
Generation 6050: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 605100	Diversity = 7.3166e-08	k_t
Generation 6060: Best = 3.463246e+03 = [3, 2, 3, 4]	FES = 606100	Diversity = 7.3074e-08	k_t
Generation 6070: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 607100	Diversity = 7.2226e-08	k_t
Generation 6080: Best = 3.463246e+03 = [3, 2, 4, 2]	FES = 608100	Diversity = 7.2170e-08	k_t
Generation 6090: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 609100	Diversity = 7.2224e-08	k_t
Generation 6100: Best = 3.463246e+03 = [2, 4, 4, 3]	FES = 610100	Diversity = 7.1817e-08	k_t
Generation 6110: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 611100	Diversity = 7.1270e-08	k_t
Generation 6120: Best = 3.463246e+03 = [4, 3, 4, 2]	FES = 612100	Diversity = 7.1289e-08	k_t
Generation 6130: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 613100	Diversity = 7.0684e-08	k_t
Generation 6140: Best = 3.463246e+03 = [2, 2, 3, 3]	FES = 614100	Diversity = 7.0440e-08	k_t
Generation 6150: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 615100	Diversity = 6.9854e-08	k_t
Generation 6160: Best = 3.463246e+03 = [4, 3, 3, 2]	FES = 616100	Diversity = 6.8155e-08	k_t

Generation 6170: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 617100	Diversity = 6.6046e-08	k_t
Generation 6180: Best = 3.463246e+03 = [3, 3, 4, 2]	FES = 618100	Diversity = 6.4784e-08	k_t
Generation 6190: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 619100	Diversity = 6.3377e-08	k_t
Generation 6200: Best = 3.463246e+03 = [3, 2, 4, 3]	FES = 620100	Diversity = 6.2388e-08	k_t
Generation 6210: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 621100	Diversity = 6.1365e-08	k_t
Generation 6220: Best = 3.463246e+03 = [2, 2, 3, 4]	FES = 622100	Diversity = 6.0119e-08	k_t
Generation 6230: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 623100	Diversity = 5.6944e-08	k_t
Generation 6240: Best = 3.463246e+03 = [2, 3, 2, 3]	FES = 624100	Diversity = 5.6461e-08	k_t
Generation 6250: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 625100	Diversity = 5.7545e-08	k_t
Generation 6260: Best = 3.463246e+03 = [2, 4, 3, 2]	FES = 626100	Diversity = 5.7755e-08	k_t
Generation 6270: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 627100	Diversity = 5.7219e-08	k_t
Generation 6280: Best = 3.463246e+03 = [4, 2, 4, 3]	FES = 628100	Diversity = 5.6807e-08	k_t
Generation 6290: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 629100	Diversity = 5.6986e-08	k_t
Generation 6300: Best = 3.463246e+03 = [2, 3, 3, 3]	FES = 630100	Diversity = 5.6835e-08	k_t
Generation 6310: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 631100	Diversity = 5.6433e-08	k_t
Generation 6320: Best = 3.463246e+03 = [4, 3, 2, 3]	FES = 632100	Diversity = 5.5848e-08	k_t
Generation 6330: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 633100	Diversity = 5.6372e-08	k_t
Generation 6340: Best = 3.463246e+03 = [4, 3, 2, 2]	FES = 634100	Diversity = 5.6344e-08	k_t
Generation 6350: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 635100	Diversity = 5.6067e-08	k_t
Generation 6360: Best = 3.463246e+03 = [4, 4, 3, 3]	FES = 636100	Diversity = 5.6526e-08	k_t
Generation 6370: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 637100	Diversity = 5.6720e-08	k_t
Generation 6380: Best = 3.463246e+03 = [3, 3, 3, 4]	FES = 638100	Diversity = 5.6949e-08	k_t
Generation 6390: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 639100	Diversity = 5.7282e-08	k_t
Generation 6400: Best = 3.463246e+03 = [2, 2, 4, 4]	FES = 640100	Diversity = 5.7383e-08	k_t
Generation 6410: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 641100	Diversity = 5.7166e-08	k_t
Generation 6420: Best = 3.463246e+03 = [3, 2, 2, 2]	FES = 642100	Diversity = 5.7557e-08	k_t
Generation 6430: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 643100	Diversity = 5.7730e-08	k_t

Generation 6440: Best = 3.463246e+03 = [2, 2, 4, 4]	FES = 644100	Diversity = 5.7929e-08	k_t
Generation 6450: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 645100	Diversity = 5.7478e-08	k_t
Generation 6460: Best = 3.463246e+03 = [4, 3, 3, 2]	FES = 646100	Diversity = 5.7767e-08	k_t
Generation 6470: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 647100	Diversity = 5.7446e-08	k_t
Generation 6480: Best = 3.463246e+03 = [4, 2, 3, 2]	FES = 648100	Diversity = 5.7577e-08	k_t
Generation 6490: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 649100	Diversity = 5.7597e-08	k_t
Generation 6500: Best = 3.463246e+03 = [4, 3, 4, 2]	FES = 650100	Diversity = 5.7606e-08	k_t
Generation 6510: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 651100	Diversity = 5.7473e-08	k_t
Generation 6520: Best = 3.463246e+03 = [4, 3, 4, 2]	FES = 652100	Diversity = 5.7778e-08	k_t
Generation 6530: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 653100	Diversity = 5.7551e-08	k_t
Generation 6540: Best = 3.463246e+03 = [4, 4, 3, 2]	FES = 654100	Diversity = 5.6915e-08	k_t
Generation 6550: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 655100	Diversity = 5.6203e-08	k_t
Generation 6560: Best = 3.463246e+03 = [4, 2, 3, 2]	FES = 656100	Diversity = 5.6153e-08	k_t
Generation 6570: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 657100	Diversity = 5.5915e-08	k_t
Generation 6580: Best = 3.463246e+03 = [2, 2, 3, 4]	FES = 658100	Diversity = 5.6011e-08	k_t
Generation 6590: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 659100	Diversity = 5.6127e-08	k_t
Generation 6600: Best = 3.463246e+03 = [4, 2, 3, 2]	FES = 660100	Diversity = 5.6896e-08	k_t
Generation 6610: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 661100	Diversity = 5.7015e-08	k_t
Generation 6620: Best = 3.463246e+03 = [3, 2, 4, 4]	FES = 662100	Diversity = 5.7067e-08	k_t
Generation 6630: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 663100	Diversity = 5.7459e-08	k_t
Generation 6640: Best = 3.463246e+03 = [4, 2, 2, 3]	FES = 664100	Diversity = 5.7614e-08	k_t
Generation 6650: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 665100	Diversity = 5.7627e-08	k_t
Generation 6660: Best = 3.463246e+03 = [3, 4, 4, 2]	FES = 666100	Diversity = 5.7367e-08	k_t
Generation 6670: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 667100	Diversity = 5.7020e-08	k_t
Generation 6680: Best = 3.463246e+03 = [2, 4, 4, 4]	FES = 668100	Diversity = 5.7287e-08	k_t
Generation 6690: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 669100	Diversity = 5.7300e-08	k_t
Generation 6700: Best = 3.463246e+03 = [2, 3, 4, 3]	FES = 670100	Diversity = 5.7271e-08	k_t

Generation 6710: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 671100	Diversity = 5.7781e-08	k_t
Generation 6720: Best = 3.463246e+03 = [2, 2, 3, 2]	FES = 672100	Diversity = 5.8021e-08	k_t
Generation 6730: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 673100	Diversity = 5.7581e-08	k_t
Generation 6740: Best = 3.463246e+03 = [2, 2, 4, 4]	FES = 674100	Diversity = 5.7583e-08	k_t
Generation 6750: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 675100	Diversity = 5.7575e-08	k_t
Generation 6760: Best = 3.463246e+03 = [4, 4, 4, 2]	FES = 676100	Diversity = 5.7637e-08	k_t
Generation 6770: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 677100	Diversity = 5.7361e-08	k_t
Generation 6780: Best = 3.463246e+03 = [3, 3, 2, 3]	FES = 678100	Diversity = 5.7727e-08	k_t
Generation 6790: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 679100	Diversity = 5.7437e-08	k_t
Generation 6800: Best = 3.463246e+03 = [3, 3, 3, 2]	FES = 680100	Diversity = 5.7575e-08	k_t
Generation 6810: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 681100	Diversity = 5.7653e-08	k_t
Generation 6820: Best = 3.463246e+03 = [3, 4, 3, 3]	FES = 682100	Diversity = 5.7595e-08	k_t
Generation 6830: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 683100	Diversity = 5.6930e-08	k_t
Generation 6840: Best = 3.463246e+03 = [2, 4, 3, 3]	FES = 684100	Diversity = 5.7238e-08	k_t
Generation 6850: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 685100	Diversity = 5.7185e-08	k_t
Generation 6860: Best = 3.463246e+03 = [3, 3, 3, 4]	FES = 686100	Diversity = 5.7264e-08	k_t
Generation 6870: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 687100	Diversity = 5.7182e-08	k_t
Generation 6880: Best = 3.463246e+03 = [4, 2, 2, 4]	FES = 688100	Diversity = 5.7156e-08	k_t
Generation 6890: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 689100	Diversity = 5.7224e-08	k_t
Generation 6900: Best = 3.463246e+03 = [3, 2, 4, 2]	FES = 690100	Diversity = 5.7462e-08	k_t
Generation 6910: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 691100	Diversity = 5.7537e-08	k_t
Generation 6920: Best = 3.463246e+03 = [4, 3, 2, 2]	FES = 692100	Diversity = 5.7735e-08	k_t
Generation 6930: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 693100	Diversity = 5.7617e-08	k_t
Generation 6940: Best = 3.463246e+03 = [2, 3, 2, 3]	FES = 694100	Diversity = 5.7697e-08	k_t
Generation 6950: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 695100	Diversity = 5.7835e-08	k_t
Generation 6960: Best = 3.463246e+03 = [4, 4, 2, 2]	FES = 696100	Diversity = 5.7849e-08	k_t
Generation 6970: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 697100	Diversity = 5.7920e-08	k_t

Generation 6980: Best = 3.463246e+03 = [3, 2, 3, 3]	FES = 698100	Diversity = 5.8206e-08	k_t
Generation 6990: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 5.7989e-08	k_t
Generation 7000: Best = 3.463246e+03 = [4, 2, 2, 3]	FES = 700100	Diversity = 5.7515e-08	k_t
Generation 7010: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 5.7415e-08	k_t
Generation 7020: Best = 3.463246e+03 = [3, 2, 2, 3]	FES = 702100	Diversity = 5.7728e-08	k_t
Generation 7030: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 5.7743e-08	k_t
Generation 7040: Best = 3.463246e+03 = [2, 3, 2, 2]	FES = 704100	Diversity = 5.7921e-08	k_t
Generation 7050: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 5.7755e-08	k_t
Generation 7060: Best = 3.463246e+03 = [2, 4, 2, 2]	FES = 706100	Diversity = 5.8167e-08	k_t
Generation 7070: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 5.8264e-08	k_t
Generation 7080: Best = 3.463246e+03 = [2, 2, 2, 3]	FES = 708100	Diversity = 5.8308e-08	k_t
Generation 7090: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 5.8040e-08	k_t
Generation 7100: Best = 3.463246e+03 = [2, 3, 3, 4]	FES = 710100	Diversity = 5.8199e-08	k_t
Generation 7110: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 5.8645e-08	k_t
Generation 7120: Best = 3.463246e+03 = [3, 4, 2, 4]	FES = 712100	Diversity = 5.8396e-08	k_t
Generation 7130: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 5.8626e-08	k_t
Generation 7140: Best = 3.463246e+03 = [4, 2, 3, 2]	FES = 714100	Diversity = 5.9274e-08	k_t
Generation 7150: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 5.9267e-08	k_t
Generation 7160: Best = 3.463246e+03 = [3, 3, 3, 3]	FES = 716100	Diversity = 5.9123e-08	k_t
Generation 7170: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 5.9289e-08	k_t
Generation 7180: Best = 3.463246e+03 = [3, 4, 3, 3]	FES = 718100	Diversity = 5.9316e-08	k_t
Generation 7190: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 5.9353e-08	k_t
Generation 7200: Best = 3.463246e+03 = [3, 2, 4, 4]	FES = 720100	Diversity = 5.8804e-08	k_t
Generation 7210: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 5.8864e-08	k_t
Generation 7220: Best = 3.463246e+03 = [3, 2, 4, 2]	FES = 722100	Diversity = 5.9028e-08	k_t
Generation 7230: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 5.8653e-08	k_t
Generation 7240: Best = 3.463246e+03 = [2, 2, 2, 2]	FES = 724100	Diversity = 5.8512e-08	k_t

Generation 7250: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 5.8726e-08	k_t
Generation 7260: Best = 3.463246e+03 = [4, 4, 3, 3]	FES = 726100	Diversity = 5.8862e-08	k_t
Generation 7270: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 5.9297e-08	k_t
Generation 7280: Best = 3.463246e+03 = [4, 2, 2, 3]	FES = 728100	Diversity = 5.9355e-08	k_t
Generation 7290: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 5.9322e-08	k_t
Generation 7300: Best = 3.463246e+03 = [4, 3, 3, 3]	FES = 730100	Diversity = 5.9591e-08	k_t
Generation 7310: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 5.9567e-08	k_t
Generation 7320: Best = 3.463246e+03 = [3, 3, 3, 4]	FES = 732100	Diversity = 5.9556e-08	k_t
Generation 7330: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 5.9592e-08	k_t
Generation 7340: Best = 3.463246e+03 = [3, 2, 2, 2]	FES = 734100	Diversity = 5.9386e-08	k_t
Generation 7350: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 5.9480e-08	k_t
Generation 7360: Best = 3.463246e+03 = [2, 4, 4, 4]	FES = 736100	Diversity = 5.9451e-08	k_t
Generation 7370: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 5.9874e-08	k_t
Generation 7380: Best = 3.463246e+03 = [2, 3, 4, 2]	FES = 738100	Diversity = 5.9669e-08	k_t
Generation 7390: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 5.9717e-08	k_t
Generation 7400: Best = 3.463246e+03 = [3, 2, 2, 4]	FES = 740100	Diversity = 5.9729e-08	k_t
Generation 7410: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 5.9796e-08	k_t
Generation 7420: Best = 3.463246e+03 = [3, 3, 4, 2]	FES = 742100	Diversity = 5.9790e-08	k_t
Generation 7430: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 5.9966e-08	k_t
Generation 7440: Best = 3.463246e+03 = [4, 2, 4, 4]	FES = 744100	Diversity = 6.0070e-08	k_t
Generation 7450: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 5.9936e-08	k_t
Generation 7460: Best = 3.463246e+03 = [4, 3, 2, 4]	FES = 746100	Diversity = 6.0088e-08	k_t
Generation 7470: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 6.0053e-08	k_t
Generation 7480: Best = 3.463246e+03 = [4, 2, 3, 2]	FES = 748100	Diversity = 5.9822e-08	k_t
Generation 7490: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 5.9853e-08	k_t
Generation 7500: Best = 3.463246e+03 = [4, 2, 3, 3]	FES = 750100	Diversity = 6.0171e-08	k_t
Generation 7510: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 5.9851e-08	k_t

Generation 7520: Best = 3.463246e+03 = [2, 3, 4, 4]	FES = 752100	Diversity = 5.9577e-08	k_t
Generation 7530: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 753100	Diversity = 5.9793e-08	k_t
Generation 7540: Best = 3.463246e+03 = [2, 4, 4, 4]	FES = 754100	Diversity = 5.9820e-08	k_t
Generation 7550: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 755100	Diversity = 5.9756e-08	k_t
Generation 7560: Best = 3.463246e+03 = [3, 4, 4, 2]	FES = 756100	Diversity = 5.9802e-08	k_t
Generation 7570: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 757100	Diversity = 5.9826e-08	k_t
Generation 7580: Best = 3.463246e+03 = [4, 3, 3, 2]	FES = 758100	Diversity = 6.0331e-08	k_t
Generation 7590: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 759100	Diversity = 6.0489e-08	k_t
Generation 7600: Best = 3.463246e+03 = [3, 4, 4, 3]	FES = 760100	Diversity = 6.0593e-08	k_t
Generation 7610: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 761100	Diversity = 6.0527e-08	k_t
Generation 7620: Best = 3.463246e+03 = [3, 4, 3, 3]	FES = 762100	Diversity = 6.0818e-08	k_t
Generation 7630: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 763100	Diversity = 6.0409e-08	k_t
Generation 7640: Best = 3.463246e+03 = [4, 3, 3, 3]	FES = 764100	Diversity = 6.0407e-08	k_t
Generation 7650: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 765100	Diversity = 6.0415e-08	k_t
Generation 7660: Best = 3.463246e+03 = [4, 3, 2, 4]	FES = 766100	Diversity = 6.0382e-08	k_t
Generation 7670: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 767100	Diversity = 6.0433e-08	k_t
Generation 7680: Best = 3.463246e+03 = [2, 4, 3, 2]	FES = 768100	Diversity = 6.0447e-08	k_t
Generation 7690: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 769100	Diversity = 6.0455e-08	k_t
Generation 7700: Best = 3.463246e+03 = [4, 4, 2, 2]	FES = 770100	Diversity = 6.0555e-08	k_t
Generation 7710: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 771100	Diversity = 6.0483e-08	k_t
Generation 7720: Best = 3.463246e+03 = [4, 3, 2, 4]	FES = 772100	Diversity = 6.0327e-08	k_t
Generation 7730: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 773100	Diversity = 6.0310e-08	k_t
Generation 7740: Best = 3.463246e+03 = [3, 2, 2, 2]	FES = 774100	Diversity = 6.0343e-08	k_t
Generation 7750: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 775100	Diversity = 6.0584e-08	k_t
Generation 7760: Best = 3.463246e+03 = [3, 2, 2, 3]	FES = 776100	Diversity = 6.0503e-08	k_t
Generation 7770: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 777100	Diversity = 6.0658e-08	k_t
Generation 7780: Best = 3.463246e+03 = [3, 2, 2, 4]	FES = 778100	Diversity = 6.0692e-08	k_t

Generation 7790: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 779100	Diversity = 6.0530e-08	k_t
Generation 7800: Best = 3.463246e+03 = [3, 3, 4, 4]	FES = 780100	Diversity = 6.0460e-08	k_t
Generation 7810: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 781100	Diversity = 6.0822e-08	k_t
Generation 7820: Best = 3.463246e+03 = [4, 2, 2, 3]	FES = 782100	Diversity = 6.0283e-08	k_t
Generation 7830: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 783100	Diversity = 6.0436e-08	k_t
Generation 7840: Best = 3.463246e+03 = [2, 2, 3, 4]	FES = 784100	Diversity = 6.0319e-08	k_t
Generation 7850: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 785100	Diversity = 6.0606e-08	k_t
Generation 7860: Best = 3.463246e+03 = [2, 2, 4, 3]	FES = 786100	Diversity = 6.0143e-08	k_t
Generation 7870: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 787100	Diversity = 6.0156e-08	k_t
Generation 7880: Best = 3.463246e+03 = [4, 4, 3, 3]	FES = 788100	Diversity = 6.0791e-08	k_t
Generation 7890: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 789100	Diversity = 6.0820e-08	k_t
Generation 7900: Best = 3.463246e+03 = [2, 3, 3, 4]	FES = 790100	Diversity = 6.0839e-08	k_t
Generation 7910: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 791100	Diversity = 6.0976e-08	k_t
Generation 7920: Best = 3.463246e+03 = [4, 2, 2, 4]	FES = 792100	Diversity = 6.1024e-08	k_t
Generation 7930: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 793100	Diversity = 6.1261e-08	k_t
Generation 7940: Best = 3.463246e+03 = [2, 3, 4, 2]	FES = 794100	Diversity = 6.1151e-08	k_t
Generation 7950: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 795100	Diversity = 6.1003e-08	k_t
Generation 7960: Best = 3.463246e+03 = [4, 4, 4, 2]	FES = 796100	Diversity = 6.1031e-08	k_t
Generation 7970: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 797100	Diversity = 6.0912e-08	k_t
Generation 7980: Best = 3.463246e+03 = [3, 3, 3, 3]	FES = 798100	Diversity = 6.0904e-08	k_t
Generation 7990: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 799100	Diversity = 6.1236e-08	k_t
Generation 8000: Best = 3.463246e+03 = [4, 3, 4, 2]	FES = 800100	Diversity = 6.1691e-08	k_t
Generation 8010: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 801100	Diversity = 6.1928e-08	k_t
Generation 8020: Best = 3.463246e+03 = [2, 2, 2, 2]	FES = 802100	Diversity = 6.1943e-08	k_t
Generation 8030: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 803100	Diversity = 6.2018e-08	k_t
Generation 8040: Best = 3.463246e+03 = [2, 4, 2, 4]	FES = 804100	Diversity = 6.2666e-08	k_t
Generation 8050: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 805100	Diversity = 6.2836e-08	k_t

Generation 8060: Best = 3.463246e+03 = [2, 4, 3, 3]	FES = 806100	Diversity = 6.2863e-08	k_t
Generation 8070: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 807100	Diversity = 6.2737e-08	k_t
Generation 8080: Best = 3.463246e+03 = [2, 2, 4, 4]	FES = 808100	Diversity = 6.2590e-08	k_t
Generation 8090: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 809100	Diversity = 6.2575e-08	k_t
Generation 8100: Best = 3.463246e+03 = [3, 2, 3, 2]	FES = 810100	Diversity = 6.2827e-08	k_t
Generation 8110: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 811100	Diversity = 6.2853e-08	k_t
Generation 8120: Best = 3.463246e+03 = [2, 3, 3, 3]	FES = 812100	Diversity = 6.2992e-08	k_t
Generation 8130: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 813100	Diversity = 6.3028e-08	k_t
Generation 8140: Best = 3.463246e+03 = [3, 3, 3, 4]	FES = 814100	Diversity = 6.2952e-08	k_t
Generation 8150: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 815100	Diversity = 6.3104e-08	k_t
Generation 8160: Best = 3.463246e+03 = [4, 4, 4, 3]	FES = 816100	Diversity = 6.3153e-08	k_t
Generation 8170: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 817100	Diversity = 6.3245e-08	k_t
Generation 8180: Best = 3.463246e+03 = [3, 2, 4, 4]	FES = 818100	Diversity = 6.2951e-08	k_t
Generation 8190: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 819100	Diversity = 6.2967e-08	k_t
Generation 8200: Best = 3.463246e+03 = [4, 4, 4, 4]	FES = 820100	Diversity = 6.2952e-08	k_t
Generation 8210: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 821100	Diversity = 6.2969e-08	k_t
Generation 8220: Best = 3.463246e+03 = [4, 3, 4, 2]	FES = 822100	Diversity = 6.2920e-08	k_t
Generation 8230: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 823100	Diversity = 6.2665e-08	k_t
Generation 8240: Best = 3.463246e+03 = [3, 4, 3, 3]	FES = 824100	Diversity = 6.2935e-08	k_t
Generation 8250: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 825100	Diversity = 6.2912e-08	k_t
Generation 8260: Best = 3.463246e+03 = [3, 3, 3, 3]	FES = 826100	Diversity = 6.2836e-08	k_t
Generation 8270: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 827100	Diversity = 6.2771e-08	k_t
Generation 8280: Best = 3.463246e+03 = [2, 3, 3, 3]	FES = 828100	Diversity = 6.2844e-08	k_t
Generation 8290: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 829100	Diversity = 6.2838e-08	k_t
Generation 8300: Best = 3.463246e+03 = [3, 4, 4, 2]	FES = 830100	Diversity = 6.3022e-08	k_t
Generation 8310: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 831100	Diversity = 6.2989e-08	k_t
Generation 8320: Best = 3.463246e+03 = [2, 2, 4, 2]	FES = 832100	Diversity = 6.2870e-08	k_t

Generation 8330: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 833100	Diversity = 6.3111e-08	k_t
Generation 8340: Best = 3.463246e+03 = [4, 3, 2, 3]	FES = 834100	Diversity = 6.3481e-08	k_t
Generation 8350: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 835100	Diversity = 6.3867e-08	k_t
Generation 8360: Best = 3.463246e+03 = [4, 3, 2, 2]	FES = 836100	Diversity = 6.3826e-08	k_t
Generation 8370: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 837100	Diversity = 6.3447e-08	k_t
Generation 8380: Best = 3.463246e+03 = [3, 4, 3, 3]	FES = 838100	Diversity = 6.3446e-08	k_t
Generation 8390: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 839100	Diversity = 6.3758e-08	k_t
Generation 8400: Best = 3.463246e+03 = [4, 4, 4, 2]	FES = 840100	Diversity = 6.3727e-08	k_t
Generation 8410: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 841100	Diversity = 6.3642e-08	k_t
Generation 8420: Best = 3.463246e+03 = [4, 4, 3, 3]	FES = 842100	Diversity = 6.3818e-08	k_t
Generation 8430: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 843100	Diversity = 6.3639e-08	k_t
Generation 8440: Best = 3.463246e+03 = [2, 4, 2, 3]	FES = 844100	Diversity = 6.3604e-08	k_t
Generation 8450: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 845100	Diversity = 6.3684e-08	k_t
Generation 8460: Best = 3.463246e+03 = [3, 2, 2, 2]	FES = 846100	Diversity = 6.3621e-08	k_t
Generation 8470: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 847100	Diversity = 6.3744e-08	k_t
Generation 8480: Best = 3.463246e+03 = [4, 4, 3, 2]	FES = 848100	Diversity = 6.3741e-08	k_t
Generation 8490: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 849100	Diversity = 6.3906e-08	k_t
Generation 8500: Best = 3.463246e+03 = [3, 2, 2, 2]	FES = 850100	Diversity = 6.4017e-08	k_t
Generation 8510: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 851100	Diversity = 6.4347e-08	k_t
Generation 8520: Best = 3.463246e+03 = [2, 3, 4, 4]	FES = 852100	Diversity = 6.4494e-08	k_t
Generation 8530: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 853100	Diversity = 6.4567e-08	k_t
Generation 8540: Best = 3.463246e+03 = [4, 3, 2, 2]	FES = 854100	Diversity = 6.4605e-08	k_t
Generation 8550: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 855100	Diversity = 6.4682e-08	k_t
Generation 8560: Best = 3.463246e+03 = [3, 4, 3, 2]	FES = 856100	Diversity = 6.4846e-08	k_t
Generation 8570: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 857100	Diversity = 6.4882e-08	k_t
Generation 8580: Best = 3.463246e+03 = [4, 2, 2, 3]	FES = 858100	Diversity = 6.4555e-08	k_t
Generation 8590: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 859100	Diversity = 6.4545e-08	k_t

Generation 8600: Best = 3.463246e+03 = [3, 2, 4, 4]	FES = 860100	Diversity = 6.4565e-08	k_t
Generation 8610: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 861100	Diversity = 6.4613e-08	k_t
Generation 8620: Best = 3.463246e+03 = [2, 2, 4, 3]	FES = 862100	Diversity = 6.4600e-08	k_t
Generation 8630: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 863100	Diversity = 6.4793e-08	k_t
Generation 8640: Best = 3.463246e+03 = [3, 4, 3, 4]	FES = 864100	Diversity = 6.4946e-08	k_t
Generation 8650: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 865100	Diversity = 6.5346e-08	k_t
Generation 8660: Best = 3.463246e+03 = [2, 3, 4, 3]	FES = 866100	Diversity = 6.5457e-08	k_t
Generation 8670: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 867100	Diversity = 6.5423e-08	k_t
Generation 8680: Best = 3.463246e+03 = [4, 4, 2, 4]	FES = 868100	Diversity = 6.5396e-08	k_t
Generation 8690: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 869100	Diversity = 6.5527e-08	k_t
Generation 8700: Best = 3.463246e+03 = [2, 3, 4, 4]	FES = 870100	Diversity = 6.5707e-08	k_t
Generation 8710: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 871100	Diversity = 6.6051e-08	k_t
Generation 8720: Best = 3.463246e+03 = [4, 4, 2, 2]	FES = 872100	Diversity = 6.6294e-08	k_t
Generation 8730: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 873100	Diversity = 6.6430e-08	k_t
Generation 8740: Best = 3.463246e+03 = [3, 4, 4, 4]	FES = 874100	Diversity = 6.6374e-08	k_t
Generation 8750: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 875100	Diversity = 6.6326e-08	k_t
Generation 8760: Best = 3.463246e+03 = [2, 3, 3, 2]	FES = 876100	Diversity = 6.6100e-08	k_t
Generation 8770: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 877100	Diversity = 6.6335e-08	k_t
Generation 8780: Best = 3.463246e+03 = [4, 3, 3, 4]	FES = 878100	Diversity = 6.6271e-08	k_t
Generation 8790: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 879100	Diversity = 6.6306e-08	k_t
Generation 8800: Best = 3.463246e+03 = [4, 2, 4, 3]	FES = 880100	Diversity = 6.6409e-08	k_t
Generation 8810: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 881100	Diversity = 6.6322e-08	k_t
Generation 8820: Best = 3.463246e+03 = [3, 3, 2, 2]	FES = 882100	Diversity = 6.6109e-08	k_t
Generation 8830: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 883100	Diversity = 6.6408e-08	k_t
Generation 8840: Best = 3.463246e+03 = [4, 4, 3, 2]	FES = 884100	Diversity = 6.6636e-08	k_t
Generation 8850: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 885100	Diversity = 6.6554e-08	k_t
Generation 8860: Best = 3.463246e+03 = [3, 4, 3, 2]	FES = 886100	Diversity = 6.6574e-08	k_t

Generation 8870: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 887100	Diversity = 6.6600e-08	k_t
Generation 8880: Best = 3.463246e+03 = [4, 2, 3, 2]	FES = 888100	Diversity = 6.6382e-08	k_t
Generation 8890: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 889100	Diversity = 6.6086e-08	k_t
Generation 8900: Best = 3.463246e+03 = [2, 2, 3, 2]	FES = 890100	Diversity = 6.6197e-08	k_t
Generation 8910: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 891100	Diversity = 6.5386e-08	k_t
Generation 8920: Best = 3.463246e+03 = [4, 3, 2, 3]	FES = 892100	Diversity = 6.5434e-08	k_t
Generation 8930: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 893100	Diversity = 6.5843e-08	k_t
Generation 8940: Best = 3.463246e+03 = [2, 3, 2, 3]	FES = 894100	Diversity = 6.6056e-08	k_t
Generation 8950: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 895100	Diversity = 6.5653e-08	k_t
Generation 8960: Best = 3.463246e+03 = [2, 2, 4, 3]	FES = 896100	Diversity = 6.5622e-08	k_t
Generation 8970: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 897100	Diversity = 6.5066e-08	k_t
Generation 8980: Best = 3.463246e+03 = [3, 2, 2, 4]	FES = 898100	Diversity = 6.4668e-08	k_t
Generation 8990: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 899100	Diversity = 6.4422e-08	k_t
Generation 9000: Best = 3.463246e+03 = [3, 2, 4, 4]	FES = 900100	Diversity = 6.4771e-08	k_t
Generation 9010: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 901100	Diversity = 6.4217e-08	k_t
Generation 9020: Best = 3.463246e+03 = [4, 3, 4, 3]	FES = 902100	Diversity = 6.3844e-08	k_t
Generation 9030: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 903100	Diversity = 6.2767e-08	k_t
Generation 9040: Best = 3.463246e+03 = [4, 4, 2, 3]	FES = 904100	Diversity = 6.2182e-08	k_t
Generation 9050: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 905100	Diversity = 5.9692e-08	k_t
Generation 9060: Best = 3.463246e+03 = [4, 2, 4, 3]	FES = 906100	Diversity = 5.7506e-08	k_t
Generation 9070: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 907100	Diversity = 5.4869e-08	k_t
Generation 9080: Best = 3.463246e+03 = [2, 2, 3, 3]	FES = 908100	Diversity = 4.4179e-08	k_t
Generation 9090: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 909100	Diversity = 1.7160e-08	k_t
Generation 9100: Best = 3.463246e+03 = [3, 2, 2, 3]	FES = 910100	Diversity = 2.4017e-09	k_t
Generation 9110: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 911100	Diversity = 2.3999e-09	k_t
Generation 9120: Best = 3.463246e+03 = [2, 4, 4, 3]	FES = 912100	Diversity = 2.5277e-09	k_t
Generation 9130: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 913100	Diversity = 2.4844e-09	k_t

Generation 9140: Best = 3.463246e+03 = [3, 2, 4, 4]	FES = 914100	Diversity = 2.5484e-09	k_t
Generation 9150: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 2.5326e-09	k_t
Generation 9160: Best = 3.463246e+03 = [2, 4, 4, 3]	FES = 916100	Diversity = 2.4600e-09	k_t
Generation 9170: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 2.3945e-09	k_t
Generation 9180: Best = 3.463246e+03 = [2, 4, 2, 3]	FES = 918100	Diversity = 2.4584e-09	k_t
Generation 9190: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 2.4562e-09	k_t
Generation 9200: Best = 3.463246e+03 = [4, 4, 2, 2]	FES = 920100	Diversity = 2.4978e-09	k_t
Generation 9210: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 2.3130e-09	k_t
Generation 9220: Best = 3.463246e+03 = [3, 3, 4, 2]	FES = 922100	Diversity = 2.2306e-09	k_t
Generation 9230: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 2.2812e-09	k_t
Generation 9240: Best = 3.463246e+03 = [2, 3, 2, 2]	FES = 924100	Diversity = 2.2634e-09	k_t
Generation 9250: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 2.2965e-09	k_t
Generation 9260: Best = 3.463246e+03 = [4, 4, 2, 2]	FES = 926100	Diversity = 2.2265e-09	k_t
Generation 9270: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 2.3357e-09	k_t
Generation 9280: Best = 3.463246e+03 = [3, 2, 2, 4]	FES = 928100	Diversity = 2.2265e-09	k_t
Generation 9290: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 2.3175e-09	k_t
Generation 9300: Best = 3.463246e+03 = [4, 4, 3, 2]	FES = 930100	Diversity = 2.3965e-09	k_t
Generation 9310: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 2.2779e-09	k_t
Generation 9320: Best = 3.463246e+03 = [2, 2, 3, 4]	FES = 932100	Diversity = 2.3870e-09	k_t
Generation 9330: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 2.3090e-09	k_t
Generation 9340: Best = 3.463246e+03 = [4, 4, 4, 4]	FES = 934100	Diversity = 2.1909e-09	k_t
Generation 9350: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 2.1559e-09	k_t
Generation 9360: Best = 3.463246e+03 = [3, 4, 4, 3]	FES = 936100	Diversity = 2.0894e-09	k_t
Generation 9370: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 2.1860e-09	k_t
Generation 9380: Best = 3.463246e+03 = [4, 4, 3, 2]	FES = 938100	Diversity = 2.1133e-09	k_t
Generation 9390: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 2.1212e-09	k_t
Generation 9400: Best = 3.463246e+03 = [4, 4, 4, 2]	FES = 940100	Diversity = 2.1944e-09	k_t

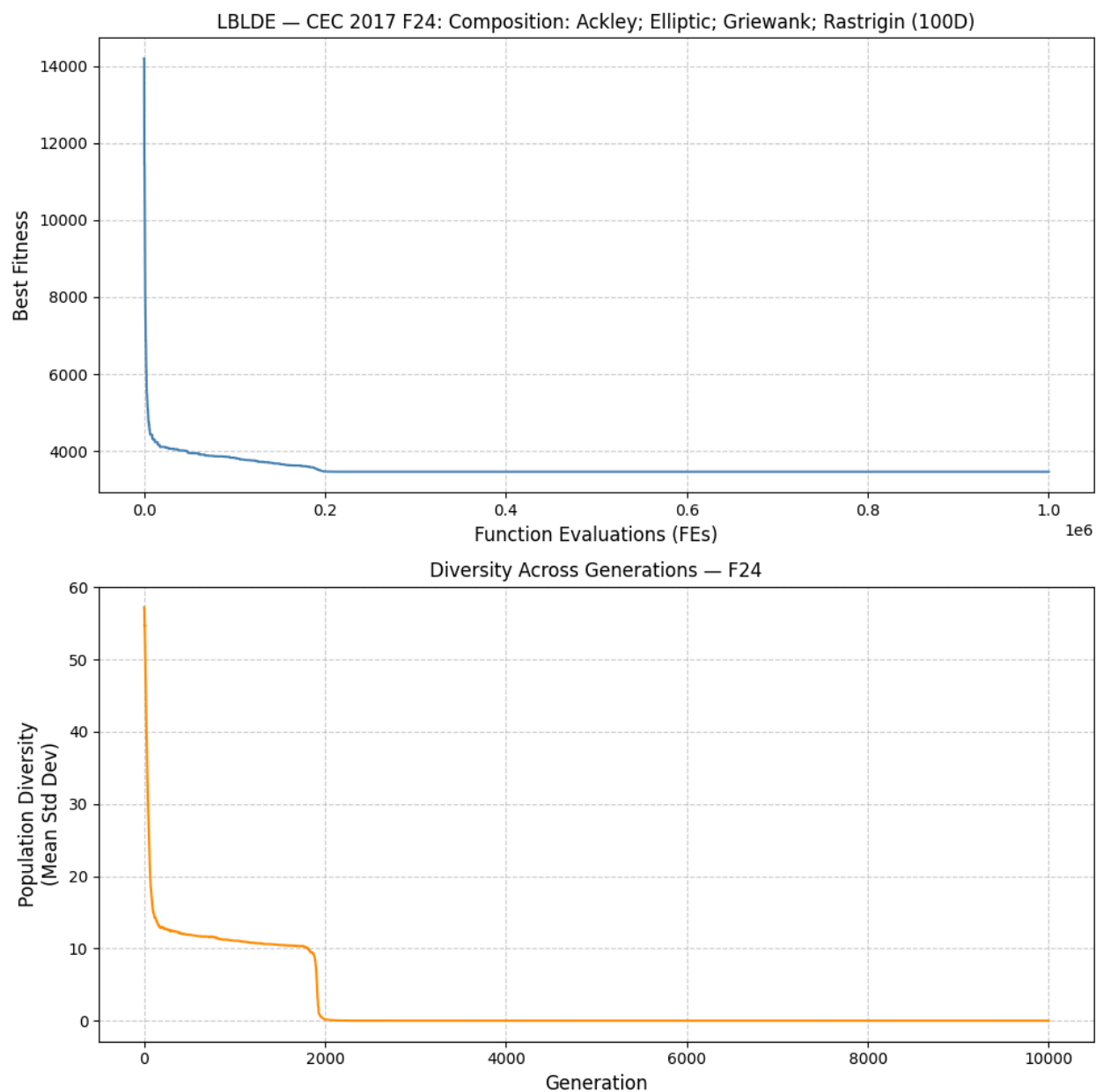
Generation 9410: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 2.3259e-09	k_t
Generation 9420: Best = 3.463246e+03 = [3, 2, 4, 3]	FES = 942100	Diversity = 2.2984e-09	k_t
Generation 9430: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 2.0910e-09	k_t
Generation 9440: Best = 3.463246e+03 = [4, 3, 3, 2]	FES = 944100	Diversity = 2.0527e-09	k_t
Generation 9450: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 2.2643e-09	k_t
Generation 9460: Best = 3.463246e+03 = [2, 4, 2, 3]	FES = 946100	Diversity = 2.2894e-09	k_t
Generation 9470: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 2.5604e-09	k_t
Generation 9480: Best = 3.463246e+03 = [2, 3, 3, 2]	FES = 948100	Diversity = 2.4456e-09	k_t
Generation 9490: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 2.3909e-09	k_t
Generation 9500: Best = 3.463246e+03 = [2, 4, 2, 4]	FES = 950100	Diversity = 2.3962e-09	k_t
Generation 9510: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 2.3688e-09	k_t
Generation 9520: Best = 3.463246e+03 = [3, 4, 3, 2]	FES = 952100	Diversity = 2.4828e-09	k_t
Generation 9530: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 2.5016e-09	k_t
Generation 9540: Best = 3.463246e+03 = [3, 4, 4, 4]	FES = 954100	Diversity = 2.4940e-09	k_t
Generation 9550: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 2.4543e-09	k_t
Generation 9560: Best = 3.463246e+03 = [4, 2, 2, 4]	FES = 956100	Diversity = 2.4046e-09	k_t
Generation 9570: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 2.4136e-09	k_t
Generation 9580: Best = 3.463246e+03 = [2, 2, 4, 4]	FES = 958100	Diversity = 2.4475e-09	k_t
Generation 9590: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 2.4186e-09	k_t
Generation 9600: Best = 3.463246e+03 = [4, 2, 4, 2]	FES = 960100	Diversity = 2.4496e-09	k_t
Generation 9610: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 2.4343e-09	k_t
Generation 9620: Best = 3.463246e+03 = [2, 2, 2, 2]	FES = 962100	Diversity = 2.5088e-09	k_t
Generation 9630: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 2.5050e-09	k_t
Generation 9640: Best = 3.463246e+03 = [4, 4, 2, 3]	FES = 964100	Diversity = 2.5387e-09	k_t
Generation 9650: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 2.5797e-09	k_t
Generation 9660: Best = 3.463246e+03 = [4, 3, 4, 3]	FES = 966100	Diversity = 2.5361e-09	k_t
Generation 9670: Best = 3.463246e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 2.4976e-09	k_t

Generation 9680: Best = 3.463246e+03	FES = 968100	Diversity = 2.4893e-09	k_t
= [2, 4, 2, 4]			
Generation 9690: Best = 3.463246e+03	FES = 969100	Diversity = 2.5049e-09	k_t
= [1, 1, 1, 1]			
Generation 9700: Best = 3.463246e+03	FES = 970100	Diversity = 2.4670e-09	k_t
= [3, 4, 3, 3]			
Generation 9710: Best = 3.463246e+03	FES = 971100	Diversity = 2.4116e-09	k_t
= [1, 1, 1, 1]			
Generation 9720: Best = 3.463246e+03	FES = 972100	Diversity = 2.4417e-09	k_t
= [3, 4, 2, 2]			
Generation 9730: Best = 3.463246e+03	FES = 973100	Diversity = 2.4117e-09	k_t
= [1, 1, 1, 1]			
Generation 9740: Best = 3.463246e+03	FES = 974100	Diversity = 2.4152e-09	k_t
= [3, 4, 3, 4]			
Generation 9750: Best = 3.463246e+03	FES = 975100	Diversity = 2.3873e-09	k_t
= [1, 1, 1, 1]			
Generation 9760: Best = 3.463246e+03	FES = 976100	Diversity = 2.3729e-09	k_t
= [2, 4, 3, 3]			
Generation 9770: Best = 3.463246e+03	FES = 977100	Diversity = 2.3566e-09	k_t
= [1, 1, 1, 1]			
Generation 9780: Best = 3.463246e+03	FES = 978100	Diversity = 2.4594e-09	k_t
= [3, 4, 3, 3]			
Generation 9790: Best = 3.463246e+03	FES = 979100	Diversity = 2.4880e-09	k_t
= [1, 1, 1, 1]			
Generation 9800: Best = 3.463246e+03	FES = 980100	Diversity = 2.4993e-09	k_t
= [2, 4, 3, 4]			
Generation 9810: Best = 3.463246e+03	FES = 981100	Diversity = 2.5908e-09	k_t
= [1, 1, 1, 1]			
Generation 9820: Best = 3.463246e+03	FES = 982100	Diversity = 2.6141e-09	k_t
= [3, 3, 2, 2]			
Generation 9830: Best = 3.463246e+03	FES = 983100	Diversity = 2.5896e-09	k_t
= [1, 1, 1, 1]			
Generation 9840: Best = 3.463246e+03	FES = 984100	Diversity = 2.5277e-09	k_t
= [3, 2, 2, 3]			
Generation 9850: Best = 3.463246e+03	FES = 985100	Diversity = 2.5355e-09	k_t
= [1, 1, 1, 1]			
Generation 9860: Best = 3.463246e+03	FES = 986100	Diversity = 2.5419e-09	k_t
= [4, 3, 3, 3]			
Generation 9870: Best = 3.463246e+03	FES = 987100	Diversity = 2.5334e-09	k_t
= [1, 1, 1, 1]			
Generation 9880: Best = 3.463246e+03	FES = 988100	Diversity = 2.4372e-09	k_t
= [3, 3, 4, 2]			
Generation 9890: Best = 3.463246e+03	FES = 989100	Diversity = 2.4414e-09	k_t
= [1, 1, 1, 1]			
Generation 9900: Best = 3.463246e+03	FES = 990100	Diversity = 2.5236e-09	k_t
= [4, 4, 2, 2]			
Generation 9910: Best = 3.463246e+03	FES = 991100	Diversity = 2.5725e-09	k_t
= [1, 1, 1, 1]			
Generation 9920: Best = 3.463246e+03	FES = 992100	Diversity = 2.5710e-09	k_t
= [2, 4, 4, 4]			
Generation 9930: Best = 3.463246e+03	FES = 993100	Diversity = 2.6152e-09	k_t
= [1, 1, 1, 1]			
Generation 9940: Best = 3.463246e+03	FES = 994100	Diversity = 2.6131e-09	k_t
= [2, 3, 3, 4]			

Generation 9950: Best = 3.463246e+03 FES = 995100 Diversity = 2.6171e-09 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 3.463246e+03 FES = 996100 Diversity = 2.6023e-09 k_t
= [4, 2, 2, 4]
Generation 9970: Best = 3.463246e+03 FES = 997100 Diversity = 2.5743e-09 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 3.463246e+03 FES = 998100 Diversity = 2.5889e-09 k_t
= [2, 2, 3, 3]
Generation 9990: Best = 3.463246e+03 FES = 999100 Diversity = 2.5694e-09 k_t
= [1, 1, 1, 1]

Optimization completed!
Final Best Fitness = 3.463246e+03
Total Function Evaluations = 1000000

=====
=
Results:
Best Fitness : 3.463246e+03
Global Optimum : 2.400000e+03
Error (|best-optimum|) : 1.063246e+03
Best Solution (5 dim) : [68.48847471 76.51347433 -84.81797203 -70.70534101
72.92938842]
D^(0) (init. diversity): 5.7220e+01
=====



Graph saved as convergence_F24.png

```
In [49]: # =====
# F25 — Composition Function 5 (Rastrigin; HappyCat; Ackley; Discus; Rosenbrock)
# =====
run_and_plot('F25', make_wrapper(composition.f25), 2500.0, 'Composition: Rastr
```

```

=====
=
LBLDE_s on CEC 2017 F25 – Single Run
=====
=
Function      : Composition: Rastrigin; HappyCat; Ackley; Discus; Rosenbrock
(100D)
Bounds       : [-100, 100]
Global optimum : 2500.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 1.233610e+05 | Diversity = 5.7220e+01
Generation 10: Best = 3.642115e+04 FES = 1100 Diversity = 3.7050e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.020277e+04 FES = 2100 Diversity = 2.4162e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 1.388809e+04 FES = 3100 Diversity = 1.6821e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.069678e+04 FES = 4100 Diversity = 1.1716e+01 k_t
= [2, 2, 2, 4]
Generation 50: Best = 8.878153e+03 FES = 5100 Diversity = 8.4606e+00 k_t
= [1, 1, 1, 1]
Generation 60: Best = 7.546511e+03 FES = 6100 Diversity = 6.7941e+00 k_t
= [2, 4, 3, 4]
Generation 70: Best = 6.845326e+03 FES = 7100 Diversity = 5.5567e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 6.314199e+03 FES = 8100 Diversity = 4.5739e+00 k_t
= [4, 3, 2, 3]
Generation 90: Best = 5.995519e+03 FES = 9100 Diversity = 3.9904e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 5.636556e+03 FES = 10100 Diversity = 3.4816e+00 k_t
= [3, 3, 2, 3]
Generation 110: Best = 5.428608e+03 FES = 11100 Diversity = 3.1677e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 5.242095e+03 FES = 12100 Diversity = 2.7563e+00 k_t
= [4, 3, 2, 2]
Generation 130: Best = 5.040748e+03 FES = 13100 Diversity = 2.6506e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 4.915299e+03 FES = 14100 Diversity = 2.5286e+00 k_t
= [2, 2, 3, 3]
Generation 150: Best = 4.812171e+03 FES = 15100 Diversity = 2.3352e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 4.676594e+03 FES = 16100 Diversity = 2.2264e+00 k_t
= [3, 4, 3, 4]
Generation 170: Best = 4.598043e+03 FES = 17100 Diversity = 2.1391e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 4.504832e+03 FES = 18100 Diversity = 2.1419e+00 k_t
= [4, 4, 2, 4]
Generation 190: Best = 4.399332e+03 FES = 19100 Diversity = 2.0983e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 4.275273e+03 FES = 20100 Diversity = 2.3938e+00 k_t
= [2, 2, 3, 4]
Generation 210: Best = 4.176020e+03 FES = 21100 Diversity = 2.3044e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 4.098282e+03 FES = 22100 Diversity = 2.2244e+00 k_t

```

```

= [3, 4, 2, 4]
Generation 230: Best = 4.036736e+03 FES = 23100 Diversity = 2.0632e+00 k_t
= [1, 1, 1, 1]
Generation 240: Best = 3.979074e+03 FES = 24100 Diversity = 1.9119e+00 k_t
= [4, 2, 4, 3]
Generation 250: Best = 3.921288e+03 FES = 25100 Diversity = 1.7320e+00 k_t
= [1, 1, 1, 1]
Generation 260: Best = 3.875760e+03 FES = 26100 Diversity = 1.5560e+00 k_t
= [2, 2, 3, 4]
Generation 270: Best = 3.841400e+03 FES = 27100 Diversity = 1.4026e+00 k_t
= [1, 1, 1, 1]
Generation 280: Best = 3.816459e+03 FES = 28100 Diversity = 1.1918e+00 k_t
= [2, 2, 2, 4]
Generation 290: Best = 3.795850e+03 FES = 29100 Diversity = 1.0346e+00 k_t
= [1, 1, 1, 1]
Generation 300: Best = 3.775023e+03 FES = 30100 Diversity = 1.0165e+00 k_t
= [4, 4, 2, 4]
Generation 310: Best = 3.747975e+03 FES = 31100 Diversity = 9.8030e-01 k_t
= [1, 1, 1, 1]
Generation 320: Best = 3.731865e+03 FES = 32100 Diversity = 9.1599e-01 k_t
= [3, 4, 4, 3]
Generation 330: Best = 3.711427e+03 FES = 33100 Diversity = 9.8104e-01 k_t
= [1, 1, 1, 1]
Generation 340: Best = 3.693768e+03 FES = 34100 Diversity = 1.0217e+00 k_t
= [4, 4, 4, 4]
Generation 350: Best = 3.674889e+03 FES = 35100 Diversity = 1.0384e+00 k_t
= [1, 1, 1, 1]
Generation 360: Best = 3.661569e+03 FES = 36100 Diversity = 1.0302e+00 k_t
= [2, 4, 2, 3]
Generation 370: Best = 3.645800e+03 FES = 37100 Diversity = 9.6275e-01 k_t
= [1, 1, 1, 1]
Generation 380: Best = 3.631166e+03 FES = 38100 Diversity = 9.3546e-01 k_t
= [4, 2, 3, 4]
Generation 390: Best = 3.621605e+03 FES = 39100 Diversity = 8.1418e-01 k_t
= [1, 1, 1, 1]
Generation 400: Best = 3.610787e+03 FES = 40100 Diversity = 7.4702e-01 k_t
= [2, 3, 3, 2]
Generation 410: Best = 3.602625e+03 FES = 41100 Diversity = 6.6113e-01 k_t
= [1, 1, 1, 1]
Generation 420: Best = 3.594976e+03 FES = 42100 Diversity = 6.1793e-01 k_t
= [3, 4, 3, 3]
Generation 430: Best = 3.587168e+03 FES = 43100 Diversity = 5.9746e-01 k_t
= [1, 1, 1, 1]
Generation 440: Best = 3.579608e+03 FES = 44100 Diversity = 5.9014e-01 k_t
= [2, 3, 3, 2]
Generation 450: Best = 3.570920e+03 FES = 45100 Diversity = 5.5690e-01 k_t
= [1, 1, 1, 1]
Generation 460: Best = 3.563850e+03 FES = 46100 Diversity = 5.6060e-01 k_t
= [4, 4, 4, 4]
Generation 470: Best = 3.557573e+03 FES = 47100 Diversity = 5.7429e-01 k_t
= [1, 1, 1, 1]
Generation 480: Best = 3.550954e+03 FES = 48100 Diversity = 5.5100e-01 k_t
= [4, 4, 2, 4]
Generation 490: Best = 3.546398e+03 FES = 49100 Diversity = 5.2782e-01 k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 3.538579e+03 FES = 50100 Diversity = 5.1573e-01 k_t
= [2, 2, 3, 3]
Generation 510: Best = 3.534464e+03 FES = 51100 Diversity = 4.7156e-01 k_t
= [1, 1, 1, 1]
Generation 520: Best = 3.531463e+03 FES = 52100 Diversity = 4.1763e-01 k_t
= [3, 4, 2, 2]
Generation 530: Best = 3.527451e+03 FES = 53100 Diversity = 4.1356e-01 k_t
= [1, 1, 1, 1]
Generation 540: Best = 3.522647e+03 FES = 54100 Diversity = 4.3119e-01 k_t
= [4, 3, 2, 3]
Generation 550: Best = 3.517284e+03 FES = 55100 Diversity = 4.5706e-01 k_t
= [1, 1, 1, 1]
Generation 560: Best = 3.511732e+03 FES = 56100 Diversity = 4.9380e-01 k_t
= [2, 4, 3, 3]
Generation 570: Best = 3.507871e+03 FES = 57100 Diversity = 4.5749e-01 k_t
= [1, 1, 1, 1]
Generation 580: Best = 3.501816e+03 FES = 58100 Diversity = 4.5468e-01 k_t
= [2, 3, 2, 2]
Generation 590: Best = 3.496857e+03 FES = 59100 Diversity = 4.5008e-01 k_t
= [1, 1, 1, 1]
Generation 600: Best = 3.491288e+03 FES = 60100 Diversity = 4.8530e-01 k_t
= [4, 3, 4, 2]
Generation 610: Best = 3.483304e+03 FES = 61100 Diversity = 4.8976e-01 k_t
= [1, 1, 1, 1]
Generation 620: Best = 3.479214e+03 FES = 62100 Diversity = 4.9959e-01 k_t
= [4, 4, 3, 3]
Generation 630: Best = 3.472602e+03 FES = 63100 Diversity = 4.9846e-01 k_t
= [1, 1, 1, 1]
Generation 640: Best = 3.465549e+03 FES = 64100 Diversity = 5.1450e-01 k_t
= [2, 3, 3, 2]
Generation 650: Best = 3.457653e+03 FES = 65100 Diversity = 5.0976e-01 k_t
= [1, 1, 1, 1]
Generation 660: Best = 3.451545e+03 FES = 66100 Diversity = 5.7197e-01 k_t
= [3, 3, 4, 3]
Generation 670: Best = 3.441625e+03 FES = 67100 Diversity = 5.7349e-01 k_t
= [1, 1, 1, 1]
Generation 680: Best = 3.434790e+03 FES = 68100 Diversity = 5.4978e-01 k_t
= [4, 2, 2, 2]
Generation 690: Best = 3.427972e+03 FES = 69100 Diversity = 5.4065e-01 k_t
= [1, 1, 1, 1]
Generation 700: Best = 3.423371e+03 FES = 70100 Diversity = 4.9849e-01 k_t
= [4, 3, 2, 4]
Generation 710: Best = 3.417023e+03 FES = 71100 Diversity = 4.6896e-01 k_t
= [1, 1, 1, 1]
Generation 720: Best = 3.413373e+03 FES = 72100 Diversity = 4.5943e-01 k_t
= [3, 4, 4, 2]
Generation 730: Best = 3.407703e+03 FES = 73100 Diversity = 4.1966e-01 k_t
= [1, 1, 1, 1]
Generation 740: Best = 3.403032e+03 FES = 74100 Diversity = 3.9999e-01 k_t
= [3, 3, 4, 3]
Generation 750: Best = 3.399737e+03 FES = 75100 Diversity = 3.7525e-01 k_t
= [1, 1, 1, 1]
Generation 760: Best = 3.396364e+03 FES = 76100 Diversity = 3.5099e-01 k_t

```

```

= [3, 3, 2, 2]
Generation 770: Best = 3.393093e+03  FES = 77100  Diversity = 3.2879e-01  k_t
= [1, 1, 1, 1]
Generation 780: Best = 3.389378e+03  FES = 78100  Diversity = 3.3615e-01  k_t
= [3, 2, 4, 4]
Generation 790: Best = 3.385750e+03  FES = 79100  Diversity = 3.3510e-01  k_t
= [1, 1, 1, 1]
Generation 800: Best = 3.382884e+03  FES = 80100  Diversity = 3.2658e-01  k_t
= [3, 3, 2, 3]
Generation 810: Best = 3.379729e+03  FES = 81100  Diversity = 3.1428e-01  k_t
= [1, 1, 1, 1]
Generation 820: Best = 3.376316e+03  FES = 82100  Diversity = 3.2503e-01  k_t
= [4, 3, 2, 4]
Generation 830: Best = 3.373293e+03  FES = 83100  Diversity = 3.2004e-01  k_t
= [1, 1, 1, 1]
Generation 840: Best = 3.370456e+03  FES = 84100  Diversity = 3.0904e-01  k_t
= [2, 4, 2, 2]
Generation 850: Best = 3.367680e+03  FES = 85100  Diversity = 3.0711e-01  k_t
= [1, 1, 1, 1]
Generation 860: Best = 3.364226e+03  FES = 86100  Diversity = 3.2540e-01  k_t
= [4, 3, 2, 4]
Generation 870: Best = 3.359794e+03  FES = 87100  Diversity = 3.2291e-01  k_t
= [1, 1, 1, 1]
Generation 880: Best = 3.356869e+03  FES = 88100  Diversity = 3.3492e-01  k_t
= [4, 4, 2, 2]
Generation 890: Best = 3.354334e+03  FES = 89100  Diversity = 3.0947e-01  k_t
= [1, 1, 1, 1]
Generation 900: Best = 3.351757e+03  FES = 90100  Diversity = 2.8877e-01  k_t
= [3, 4, 3, 4]
Generation 910: Best = 3.349933e+03  FES = 91100  Diversity = 2.5725e-01  k_t
= [1, 1, 1, 1]
Generation 920: Best = 3.348101e+03  FES = 92100  Diversity = 2.4610e-01  k_t
= [3, 3, 2, 2]
Generation 930: Best = 3.345595e+03  FES = 93100  Diversity = 2.6888e-01  k_t
= [1, 1, 1, 1]
Generation 940: Best = 3.342279e+03  FES = 94100  Diversity = 2.7617e-01  k_t
= [4, 2, 3, 2]
Generation 950: Best = 3.339961e+03  FES = 95100  Diversity = 3.1624e-01  k_t
= [1, 1, 1, 1]
Generation 960: Best = 3.337037e+03  FES = 96100  Diversity = 3.0984e-01  k_t
= [2, 3, 3, 2]
Generation 970: Best = 3.333900e+03  FES = 97100  Diversity = 3.1833e-01  k_t
= [1, 1, 1, 1]
Generation 980: Best = 3.330859e+03  FES = 98100  Diversity = 2.9581e-01  k_t
= [3, 3, 4, 2]
Generation 990: Best = 3.327841e+03  FES = 99100  Diversity = 3.0001e-01  k_t
= [1, 1, 1, 1]
Generation 1000: Best = 3.324848e+03  FES = 100100  Diversity = 3.0492e-01  k_t
= [4, 4, 3, 4]
Generation 1010: Best = 3.321536e+03  FES = 101100  Diversity = 3.3015e-01  k_t
= [1, 1, 1, 1]
Generation 1020: Best = 3.317779e+03  FES = 102100  Diversity = 3.6309e-01  k_t
= [2, 2, 3, 4]
Generation 1030: Best = 3.314064e+03  FES = 103100  Diversity = 3.2795e-01  k_t

```

```

= [1, 1, 1, 1]
Generation 1040: Best = 3.312468e+03 FES = 104100 Diversity = 2.8302e-01 k_t
= [4, 2, 3, 4]
Generation 1050: Best = 3.310310e+03 FES = 105100 Diversity = 2.6610e-01 k_t
= [1, 1, 1, 1]
Generation 1060: Best = 3.308111e+03 FES = 106100 Diversity = 2.5566e-01 k_t
= [2, 4, 4, 3]
Generation 1070: Best = 3.305528e+03 FES = 107100 Diversity = 2.6648e-01 k_t
= [1, 1, 1, 1]
Generation 1080: Best = 3.303953e+03 FES = 108100 Diversity = 2.3162e-01 k_t
= [3, 4, 3, 2]
Generation 1090: Best = 3.302160e+03 FES = 109100 Diversity = 2.1505e-01 k_t
= [1, 1, 1, 1]
Generation 1100: Best = 3.300770e+03 FES = 110100 Diversity = 2.1788e-01 k_t
= [3, 4, 3, 3]
Generation 1110: Best = 3.299521e+03 FES = 111100 Diversity = 1.9669e-01 k_t
= [1, 1, 1, 1]
Generation 1120: Best = 3.298653e+03 FES = 112100 Diversity = 1.6660e-01 k_t
= [4, 4, 2, 2]
Generation 1130: Best = 3.297726e+03 FES = 113100 Diversity = 1.6242e-01 k_t
= [1, 1, 1, 1]
Generation 1140: Best = 3.296900e+03 FES = 114100 Diversity = 1.4167e-01 k_t
= [2, 3, 2, 2]
Generation 1150: Best = 3.296285e+03 FES = 115100 Diversity = 1.4219e-01 k_t
= [1, 1, 1, 1]
Generation 1160: Best = 3.295463e+03 FES = 116100 Diversity = 1.4272e-01 k_t
= [4, 3, 3, 3]
Generation 1170: Best = 3.294890e+03 FES = 117100 Diversity = 1.3198e-01 k_t
= [1, 1, 1, 1]
Generation 1180: Best = 3.294333e+03 FES = 118100 Diversity = 1.1627e-01 k_t
= [4, 2, 4, 3]
Generation 1190: Best = 3.293837e+03 FES = 119100 Diversity = 1.0117e-01 k_t
= [1, 1, 1, 1]
Generation 1200: Best = 3.293575e+03 FES = 120100 Diversity = 8.5001e-02 k_t
= [2, 2, 2, 4]
Generation 1210: Best = 3.293246e+03 FES = 121100 Diversity = 7.6542e-02 k_t
= [1, 1, 1, 1]
Generation 1220: Best = 3.292941e+03 FES = 122100 Diversity = 7.2320e-02 k_t
= [2, 4, 2, 3]
Generation 1230: Best = 3.292637e+03 FES = 123100 Diversity = 6.7358e-02 k_t
= [1, 1, 1, 1]
Generation 1240: Best = 3.292264e+03 FES = 124100 Diversity = 6.7065e-02 k_t
= [2, 3, 3, 3]
Generation 1250: Best = 3.291493e+03 FES = 125100 Diversity = 8.9210e-02 k_t
= [1, 1, 1, 1]
Generation 1260: Best = 3.290137e+03 FES = 126100 Diversity = 1.3144e-01 k_t
= [4, 4, 2, 2]
Generation 1270: Best = 3.288399e+03 FES = 127100 Diversity = 1.7837e-01 k_t
= [1, 1, 1, 1]
Generation 1280: Best = 3.287391e+03 FES = 128100 Diversity = 1.6259e-01 k_t
= [2, 4, 3, 4]
Generation 1290: Best = 3.286355e+03 FES = 129100 Diversity = 1.5787e-01 k_t
= [1, 1, 1, 1]
Generation 1300: Best = 3.285658e+03 FES = 130100 Diversity = 1.6955e-01 k_t

```

```

= [3, 2, 4, 2]
Generation 1310: Best = 3.284525e+03 FES = 131100 Diversity = 1.6247e-01 k_t
= [1, 1, 1, 1]
Generation 1320: Best = 3.283670e+03 FES = 132100 Diversity = 1.6553e-01 k_t
= [4, 4, 4, 4]
Generation 1330: Best = 3.283172e+03 FES = 133100 Diversity = 1.3769e-01 k_t
= [1, 1, 1, 1]
Generation 1340: Best = 3.282413e+03 FES = 134100 Diversity = 1.3427e-01 k_t
= [3, 3, 3, 3]
Generation 1350: Best = 3.281683e+03 FES = 135100 Diversity = 1.3287e-01 k_t
= [1, 1, 1, 1]
Generation 1360: Best = 3.281042e+03 FES = 136100 Diversity = 1.1666e-01 k_t
= [2, 2, 2, 4]
Generation 1370: Best = 3.280628e+03 FES = 137100 Diversity = 1.0460e-01 k_t
= [1, 1, 1, 1]
Generation 1380: Best = 3.280190e+03 FES = 138100 Diversity = 9.7441e-02 k_t
= [2, 3, 3, 4]
Generation 1390: Best = 3.279774e+03 FES = 139100 Diversity = 9.3541e-02 k_t
= [1, 1, 1, 1]
Generation 1400: Best = 3.279358e+03 FES = 140100 Diversity = 8.8436e-02 k_t
= [2, 4, 2, 3]
Generation 1410: Best = 3.278959e+03 FES = 141100 Diversity = 8.4762e-02 k_t
= [1, 1, 1, 1]
Generation 1420: Best = 3.278616e+03 FES = 142100 Diversity = 7.9534e-02 k_t
= [2, 2, 3, 3]
Generation 1430: Best = 3.278235e+03 FES = 143100 Diversity = 7.4907e-02 k_t
= [1, 1, 1, 1]
Generation 1440: Best = 3.277853e+03 FES = 144100 Diversity = 7.5820e-02 k_t
= [3, 3, 4, 4]
Generation 1450: Best = 3.277548e+03 FES = 145100 Diversity = 8.1036e-02 k_t
= [1, 1, 1, 1]
Generation 1460: Best = 3.277286e+03 FES = 146100 Diversity = 7.7350e-02 k_t
= [3, 2, 2, 4]
Generation 1470: Best = 3.277000e+03 FES = 147100 Diversity = 7.7897e-02 k_t
= [1, 1, 1, 1]
Generation 1480: Best = 3.276752e+03 FES = 148100 Diversity = 7.1809e-02 k_t
= [4, 3, 3, 4]
Generation 1490: Best = 3.276450e+03 FES = 149100 Diversity = 7.3147e-02 k_t
= [1, 1, 1, 1]
Generation 1500: Best = 3.276054e+03 FES = 150100 Diversity = 8.8699e-02 k_t
= [3, 4, 2, 4]
Generation 1510: Best = 3.275837e+03 FES = 151100 Diversity = 7.0818e-02 k_t
= [1, 1, 1, 1]
Generation 1520: Best = 3.275639e+03 FES = 152100 Diversity = 5.5215e-02 k_t
= [2, 3, 4, 2]
Generation 1530: Best = 3.275449e+03 FES = 153100 Diversity = 5.7392e-02 k_t
= [1, 1, 1, 1]
Generation 1540: Best = 3.275209e+03 FES = 154100 Diversity = 6.2477e-02 k_t
= [4, 2, 3, 2]
Generation 1550: Best = 3.275050e+03 FES = 155100 Diversity = 5.2193e-02 k_t
= [1, 1, 1, 1]
Generation 1560: Best = 3.274846e+03 FES = 156100 Diversity = 4.9985e-02 k_t
= [3, 3, 2, 3]
Generation 1570: Best = 3.274729e+03 FES = 157100 Diversity = 4.5320e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 1580: Best = 3.274615e+03 FES = 158100 Diversity = 4.0843e-02 k_t
= [3, 3, 2, 2]
Generation 1590: Best = 3.274495e+03 FES = 159100 Diversity = 4.2966e-02 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 3.274361e+03 FES = 160100 Diversity = 4.1004e-02 k_t
= [2, 2, 4, 3]
Generation 1610: Best = 3.274236e+03 FES = 161100 Diversity = 4.5383e-02 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 3.274079e+03 FES = 162100 Diversity = 5.4300e-02 k_t
= [3, 4, 2, 4]
Generation 1630: Best = 3.273898e+03 FES = 163100 Diversity = 5.5855e-02 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 3.273736e+03 FES = 164100 Diversity = 5.5281e-02 k_t
= [4, 2, 2, 3]
Generation 1650: Best = 3.273596e+03 FES = 165100 Diversity = 5.0166e-02 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 3.273467e+03 FES = 166100 Diversity = 4.3852e-02 k_t
= [2, 2, 4, 2]
Generation 1670: Best = 3.273348e+03 FES = 167100 Diversity = 4.3439e-02 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 3.273226e+03 FES = 168100 Diversity = 4.1454e-02 k_t
= [4, 2, 3, 4]
Generation 1690: Best = 3.273124e+03 FES = 169100 Diversity = 3.8766e-02 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 3.273058e+03 FES = 170100 Diversity = 3.0758e-02 k_t
= [4, 3, 2, 4]
Generation 1710: Best = 3.272995e+03 FES = 171100 Diversity = 2.8485e-02 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 3.272940e+03 FES = 172100 Diversity = 2.3236e-02 k_t
= [4, 3, 3, 2]
Generation 1730: Best = 3.272891e+03 FES = 173100 Diversity = 2.1695e-02 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 3.272842e+03 FES = 174100 Diversity = 2.1279e-02 k_t
= [3, 3, 4, 3]
Generation 1750: Best = 3.272795e+03 FES = 175100 Diversity = 2.0473e-02 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 3.272749e+03 FES = 176100 Diversity = 1.9259e-02 k_t
= [4, 3, 4, 4]
Generation 1770: Best = 3.272707e+03 FES = 177100 Diversity = 1.9156e-02 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 3.272657e+03 FES = 178100 Diversity = 1.9552e-02 k_t
= [3, 2, 3, 2]
Generation 1790: Best = 3.272617e+03 FES = 179100 Diversity = 1.8536e-02 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 3.272576e+03 FES = 180100 Diversity = 1.7061e-02 k_t
= [4, 2, 3, 4]
Generation 1810: Best = 3.272553e+03 FES = 181100 Diversity = 1.6266e-02 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 3.272524e+03 FES = 182100 Diversity = 1.5893e-02 k_t
= [2, 2, 2, 3]
Generation 1830: Best = 3.272477e+03 FES = 183100 Diversity = 1.9410e-02 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 3.272442e+03 FES = 184100 Diversity = 2.0744e-02 k_t

```



```

= [2, 3, 3, 3]
Generation 1850: Best = 3.272385e+03 FES = 185100 Diversity = 2.0896e-02 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 3.272325e+03 FES = 186100 Diversity = 2.3148e-02 k_t
= [2, 2, 4, 3]
Generation 1870: Best = 3.272287e+03 FES = 187100 Diversity = 2.1036e-02 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 3.272253e+03 FES = 188100 Diversity = 1.8579e-02 k_t
= [4, 3, 4, 4]
Generation 1890: Best = 3.272232e+03 FES = 189100 Diversity = 1.5241e-02 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 3.272215e+03 FES = 190100 Diversity = 1.2976e-02 k_t
= [3, 3, 2, 3]
Generation 1910: Best = 3.272195e+03 FES = 191100 Diversity = 1.3688e-02 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 3.272180e+03 FES = 192100 Diversity = 1.1727e-02 k_t
= [2, 3, 4, 2]
Generation 1930: Best = 3.272169e+03 FES = 193100 Diversity = 1.0643e-02 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 3.272133e+03 FES = 194100 Diversity = 1.5700e-02 k_t
= [3, 2, 3, 3]
Generation 1950: Best = 3.272112e+03 FES = 195100 Diversity = 1.4349e-02 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 3.272090e+03 FES = 196100 Diversity = 1.3734e-02 k_t
= [2, 3, 4, 2]
Generation 1970: Best = 3.272066e+03 FES = 197100 Diversity = 1.5700e-02 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 3.272032e+03 FES = 198100 Diversity = 1.8175e-02 k_t
= [3, 2, 3, 4]
Generation 1990: Best = 3.272000e+03 FES = 199100 Diversity = 1.8507e-02 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 3.271969e+03 FES = 200100 Diversity = 1.8232e-02 k_t
= [3, 4, 3, 3]
Generation 2010: Best = 3.271927e+03 FES = 201100 Diversity = 1.8147e-02 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 3.271896e+03 FES = 202100 Diversity = 1.5652e-02 k_t
= [3, 4, 3, 3]
Generation 2030: Best = 3.271872e+03 FES = 203100 Diversity = 1.3511e-02 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 3.271853e+03 FES = 204100 Diversity = 1.3274e-02 k_t
= [3, 4, 2, 4]
Generation 2050: Best = 3.271828e+03 FES = 205100 Diversity = 1.4475e-02 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 3.271807e+03 FES = 206100 Diversity = 1.3358e-02 k_t
= [4, 3, 3, 2]
Generation 2070: Best = 3.271793e+03 FES = 207100 Diversity = 1.1593e-02 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 3.271778e+03 FES = 208100 Diversity = 1.1398e-02 k_t
= [3, 4, 4, 3]
Generation 2090: Best = 3.271766e+03 FES = 209100 Diversity = 1.0215e-02 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 3.271753e+03 FES = 210100 Diversity = 1.0466e-02 k_t
= [3, 3, 4, 2]
Generation 2110: Best = 3.271736e+03 FES = 211100 Diversity = 1.1821e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 2120: Best = 3.271715e+03 FES = 212100 Diversity = 1.2973e-02 k_t
= [2, 3, 3, 4]
Generation 2130: Best = 3.271693e+03 FES = 213100 Diversity = 1.3416e-02 k_t
= [1, 1, 1, 1]
Generation 2140: Best = 3.271672e+03 FES = 214100 Diversity = 1.1206e-02 k_t
= [4, 2, 4, 4]
Generation 2150: Best = 3.271655e+03 FES = 215100 Diversity = 1.1865e-02 k_t
= [1, 1, 1, 1]
Generation 2160: Best = 3.271641e+03 FES = 216100 Diversity = 1.1763e-02 k_t
= [2, 2, 4, 4]
Generation 2170: Best = 3.271627e+03 FES = 217100 Diversity = 1.1315e-02 k_t
= [1, 1, 1, 1]
Generation 2180: Best = 3.271612e+03 FES = 218100 Diversity = 1.1304e-02 k_t
= [3, 3, 4, 4]
Generation 2190: Best = 3.271595e+03 FES = 219100 Diversity = 1.0608e-02 k_t
= [1, 1, 1, 1]
Generation 2200: Best = 3.271582e+03 FES = 220100 Diversity = 1.1198e-02 k_t
= [3, 3, 3, 4]
Generation 2210: Best = 3.271568e+03 FES = 221100 Diversity = 1.0590e-02 k_t
= [1, 1, 1, 1]
Generation 2220: Best = 3.271553e+03 FES = 222100 Diversity = 1.2298e-02 k_t
= [2, 4, 3, 3]
Generation 2230: Best = 3.271537e+03 FES = 223100 Diversity = 1.2794e-02 k_t
= [1, 1, 1, 1]
Generation 2240: Best = 3.271504e+03 FES = 224100 Diversity = 1.7102e-02 k_t
= [2, 2, 3, 3]
Generation 2250: Best = 3.271466e+03 FES = 225100 Diversity = 1.9014e-02 k_t
= [1, 1, 1, 1]
Generation 2260: Best = 3.271443e+03 FES = 226100 Diversity = 1.6575e-02 k_t
= [3, 2, 2, 2]
Generation 2270: Best = 3.271417e+03 FES = 227100 Diversity = 1.6036e-02 k_t
= [1, 1, 1, 1]
Generation 2280: Best = 3.271388e+03 FES = 228100 Diversity = 1.8357e-02 k_t
= [2, 4, 3, 3]
Generation 2290: Best = 3.271345e+03 FES = 229100 Diversity = 2.1901e-02 k_t
= [1, 1, 1, 1]
Generation 2300: Best = 3.271291e+03 FES = 230100 Diversity = 2.2903e-02 k_t
= [2, 2, 4, 2]
Generation 2310: Best = 3.271221e+03 FES = 231100 Diversity = 2.7340e-02 k_t
= [1, 1, 1, 1]
Generation 2320: Best = 3.271160e+03 FES = 232100 Diversity = 2.5741e-02 k_t
= [2, 3, 4, 3]
Generation 2330: Best = 3.271127e+03 FES = 233100 Diversity = 1.9689e-02 k_t
= [1, 1, 1, 1]
Generation 2340: Best = 3.271085e+03 FES = 234100 Diversity = 2.3157e-02 k_t
= [3, 3, 2, 4]
Generation 2350: Best = 3.271026e+03 FES = 235100 Diversity = 2.4064e-02 k_t
= [1, 1, 1, 1]
Generation 2360: Best = 3.270971e+03 FES = 236100 Diversity = 2.3901e-02 k_t
= [3, 2, 2, 3]
Generation 2370: Best = 3.270931e+03 FES = 237100 Diversity = 2.0325e-02 k_t
= [1, 1, 1, 1]
Generation 2380: Best = 3.270893e+03 FES = 238100 Diversity = 1.8259e-02 k_t

```

```

= [2, 3, 3, 4]
Generation 2390: Best = 3.270869e+03 FES = 239100 Diversity = 1.6044e-02 k_t
= [1, 1, 1, 1]
Generation 2400: Best = 3.270847e+03 FES = 240100 Diversity = 1.4249e-02 k_t
= [4, 2, 2, 4]
Generation 2410: Best = 3.270829e+03 FES = 241100 Diversity = 1.1968e-02 k_t
= [1, 1, 1, 1]
Generation 2420: Best = 3.270820e+03 FES = 242100 Diversity = 9.1200e-03 k_t
= [4, 4, 4, 4]
Generation 2430: Best = 3.270811e+03 FES = 243100 Diversity = 8.4789e-03 k_t
= [1, 1, 1, 1]
Generation 2440: Best = 3.270804e+03 FES = 244100 Diversity = 7.5030e-03 k_t
= [2, 4, 4, 4]
Generation 2450: Best = 3.270799e+03 FES = 245100 Diversity = 6.5997e-03 k_t
= [1, 1, 1, 1]
Generation 2460: Best = 3.270793e+03 FES = 246100 Diversity = 5.5294e-03 k_t
= [4, 2, 3, 2]
Generation 2470: Best = 3.270790e+03 FES = 247100 Diversity = 5.3906e-03 k_t
= [1, 1, 1, 1]
Generation 2480: Best = 3.270785e+03 FES = 248100 Diversity = 5.1700e-03 k_t
= [3, 3, 2, 2]
Generation 2490: Best = 3.270782e+03 FES = 249100 Diversity = 4.9732e-03 k_t
= [1, 1, 1, 1]
Generation 2500: Best = 3.270778e+03 FES = 250100 Diversity = 5.5075e-03 k_t
= [4, 4, 3, 4]
Generation 2510: Best = 3.270774e+03 FES = 251100 Diversity = 5.3070e-03 k_t
= [1, 1, 1, 1]
Generation 2520: Best = 3.270771e+03 FES = 252100 Diversity = 5.1960e-03 k_t
= [2, 4, 2, 3]
Generation 2530: Best = 3.270767e+03 FES = 253100 Diversity = 6.1427e-03 k_t
= [1, 1, 1, 1]
Generation 2540: Best = 3.270756e+03 FES = 254100 Diversity = 8.1245e-03 k_t
= [2, 4, 4, 2]
Generation 2550: Best = 3.270745e+03 FES = 255100 Diversity = 9.2927e-03 k_t
= [1, 1, 1, 1]
Generation 2560: Best = 3.270734e+03 FES = 256100 Diversity = 9.4923e-03 k_t
= [3, 3, 3, 3]
Generation 2570: Best = 3.270726e+03 FES = 257100 Diversity = 8.9045e-03 k_t
= [1, 1, 1, 1]
Generation 2580: Best = 3.270721e+03 FES = 258100 Diversity = 6.3660e-03 k_t
= [2, 4, 2, 2]
Generation 2590: Best = 3.270717e+03 FES = 259100 Diversity = 5.7543e-03 k_t
= [1, 1, 1, 1]
Generation 2600: Best = 3.270713e+03 FES = 260100 Diversity = 5.4206e-03 k_t
= [2, 2, 3, 2]
Generation 2610: Best = 3.270709e+03 FES = 261100 Diversity = 4.4395e-03 k_t
= [1, 1, 1, 1]
Generation 2620: Best = 3.270707e+03 FES = 262100 Diversity = 4.1324e-03 k_t
= [2, 2, 3, 3]
Generation 2630: Best = 3.270704e+03 FES = 263100 Diversity = 4.0712e-03 k_t
= [1, 1, 1, 1]
Generation 2640: Best = 3.270703e+03 FES = 264100 Diversity = 3.9308e-03 k_t
= [4, 3, 3, 3]
Generation 2650: Best = 3.270701e+03 FES = 265100 Diversity = 3.0230e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 2660: Best = 3.270700e+03 FES = 266100 Diversity = 2.8032e-03 k_t
= [3, 2, 4, 4]
Generation 2670: Best = 3.270698e+03 FES = 267100 Diversity = 2.8902e-03 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 3.270697e+03 FES = 268100 Diversity = 3.0673e-03 k_t
= [4, 2, 2, 4]
Generation 2690: Best = 3.270696e+03 FES = 269100 Diversity = 2.9974e-03 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 3.270695e+03 FES = 270100 Diversity = 2.4561e-03 k_t
= [3, 3, 4, 2]
Generation 2710: Best = 3.270694e+03 FES = 271100 Diversity = 2.3929e-03 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 3.270693e+03 FES = 272100 Diversity = 2.5607e-03 k_t
= [4, 2, 2, 4]
Generation 2730: Best = 3.270693e+03 FES = 273100 Diversity = 2.1826e-03 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 3.270692e+03 FES = 274100 Diversity = 1.9253e-03 k_t
= [2, 2, 2, 3]
Generation 2750: Best = 3.270692e+03 FES = 275100 Diversity = 1.8917e-03 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 3.270691e+03 FES = 276100 Diversity = 2.1017e-03 k_t
= [4, 3, 2, 3]
Generation 2770: Best = 3.270691e+03 FES = 277100 Diversity = 1.7931e-03 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 3.270690e+03 FES = 278100 Diversity = 1.7049e-03 k_t
= [2, 2, 3, 4]
Generation 2790: Best = 3.270690e+03 FES = 279100 Diversity = 1.8054e-03 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 3.270689e+03 FES = 280100 Diversity = 1.7445e-03 k_t
= [2, 3, 3, 2]
Generation 2810: Best = 3.270689e+03 FES = 281100 Diversity = 1.5242e-03 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 3.270689e+03 FES = 282100 Diversity = 1.5360e-03 k_t
= [3, 2, 4, 2]
Generation 2830: Best = 3.270688e+03 FES = 283100 Diversity = 1.5679e-03 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 3.270688e+03 FES = 284100 Diversity = 1.6746e-03 k_t
= [3, 3, 4, 4]
Generation 2850: Best = 3.270688e+03 FES = 285100 Diversity = 1.6164e-03 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 3.270687e+03 FES = 286100 Diversity = 1.7242e-03 k_t
= [2, 3, 3, 4]
Generation 2870: Best = 3.270687e+03 FES = 287100 Diversity = 1.7690e-03 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 3.270686e+03 FES = 288100 Diversity = 1.5646e-03 k_t
= [4, 3, 3, 4]
Generation 2890: Best = 3.270686e+03 FES = 289100 Diversity = 1.6381e-03 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 3.270685e+03 FES = 290100 Diversity = 1.7424e-03 k_t
= [2, 2, 4, 2]
Generation 2910: Best = 3.270685e+03 FES = 291100 Diversity = 2.1396e-03 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 3.270684e+03 FES = 292100 Diversity = 2.4292e-03 k_t

```

```

= [2, 3, 4, 2]
Generation 2930: Best = 3.270683e+03 FES = 293100 Diversity = 2.3031e-03 k_t
= [1, 1, 1, 1]
Generation 2940: Best = 3.270682e+03 FES = 294100 Diversity = 2.2699e-03 k_t
= [2, 3, 2, 2]
Generation 2950: Best = 3.270682e+03 FES = 295100 Diversity = 1.8821e-03 k_t
= [1, 1, 1, 1]
Generation 2960: Best = 3.270681e+03 FES = 296100 Diversity = 1.8586e-03 k_t
= [3, 3, 2, 3]
Generation 2970: Best = 3.270681e+03 FES = 297100 Diversity = 1.7947e-03 k_t
= [1, 1, 1, 1]
Generation 2980: Best = 3.270680e+03 FES = 298100 Diversity = 1.9679e-03 k_t
= [2, 2, 4, 4]
Generation 2990: Best = 3.270680e+03 FES = 299100 Diversity = 1.8341e-03 k_t
= [1, 1, 1, 1]
Generation 3000: Best = 3.270679e+03 FES = 300100 Diversity = 1.7572e-03 k_t
= [2, 3, 2, 2]
Generation 3010: Best = 3.270679e+03 FES = 301100 Diversity = 2.0177e-03 k_t
= [1, 1, 1, 1]
Generation 3020: Best = 3.270678e+03 FES = 302100 Diversity = 1.9189e-03 k_t
= [3, 4, 3, 3]
Generation 3030: Best = 3.270678e+03 FES = 303100 Diversity = 2.1386e-03 k_t
= [1, 1, 1, 1]
Generation 3040: Best = 3.270677e+03 FES = 304100 Diversity = 2.1557e-03 k_t
= [3, 3, 4, 4]
Generation 3050: Best = 3.270676e+03 FES = 305100 Diversity = 2.2304e-03 k_t
= [1, 1, 1, 1]
Generation 3060: Best = 3.270675e+03 FES = 306100 Diversity = 2.2883e-03 k_t
= [4, 2, 4, 2]
Generation 3070: Best = 3.270675e+03 FES = 307100 Diversity = 2.2261e-03 k_t
= [1, 1, 1, 1]
Generation 3080: Best = 3.270674e+03 FES = 308100 Diversity = 2.6189e-03 k_t
= [3, 4, 2, 4]
Generation 3090: Best = 3.270673e+03 FES = 309100 Diversity = 2.4935e-03 k_t
= [1, 1, 1, 1]
Generation 3100: Best = 3.270672e+03 FES = 310100 Diversity = 2.4119e-03 k_t
= [2, 3, 4, 2]
Generation 3110: Best = 3.270671e+03 FES = 311100 Diversity = 2.3415e-03 k_t
= [1, 1, 1, 1]
Generation 3120: Best = 3.270670e+03 FES = 312100 Diversity = 2.9596e-03 k_t
= [4, 4, 2, 3]
Generation 3130: Best = 3.270668e+03 FES = 313100 Diversity = 2.9192e-03 k_t
= [1, 1, 1, 1]
Generation 3140: Best = 3.270667e+03 FES = 314100 Diversity = 3.6196e-03 k_t
= [3, 3, 4, 4]
Generation 3150: Best = 3.270665e+03 FES = 315100 Diversity = 3.8138e-03 k_t
= [1, 1, 1, 1]
Generation 3160: Best = 3.270663e+03 FES = 316100 Diversity = 4.0823e-03 k_t
= [2, 4, 4, 3]
Generation 3170: Best = 3.270660e+03 FES = 317100 Diversity = 4.6748e-03 k_t
= [1, 1, 1, 1]
Generation 3180: Best = 3.270657e+03 FES = 318100 Diversity = 4.0064e-03 k_t
= [4, 3, 3, 2]
Generation 3190: Best = 3.270655e+03 FES = 319100 Diversity = 3.9353e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 3200: Best = 3.270653e+03 FES = 320100 Diversity = 3.7957e-03 k_t
= [4, 2, 2, 4]
Generation 3210: Best = 3.270650e+03 FES = 321100 Diversity = 4.0369e-03 k_t
= [1, 1, 1, 1]
Generation 3220: Best = 3.270647e+03 FES = 322100 Diversity = 4.4663e-03 k_t
= [4, 4, 4, 3]
Generation 3230: Best = 3.270645e+03 FES = 323100 Diversity = 3.6204e-03 k_t
= [1, 1, 1, 1]
Generation 3240: Best = 3.270643e+03 FES = 324100 Diversity = 3.5045e-03 k_t
= [3, 3, 2, 2]
Generation 3250: Best = 3.270641e+03 FES = 325100 Diversity = 3.6909e-03 k_t
= [1, 1, 1, 1]
Generation 3260: Best = 3.270639e+03 FES = 326100 Diversity = 3.3571e-03 k_t
= [4, 4, 2, 4]
Generation 3270: Best = 3.270638e+03 FES = 327100 Diversity = 3.2313e-03 k_t
= [1, 1, 1, 1]
Generation 3280: Best = 3.270636e+03 FES = 328100 Diversity = 3.1923e-03 k_t
= [3, 2, 4, 2]
Generation 3290: Best = 3.270634e+03 FES = 329100 Diversity = 3.1123e-03 k_t
= [1, 1, 1, 1]
Generation 3300: Best = 3.270633e+03 FES = 330100 Diversity = 2.9241e-03 k_t
= [2, 2, 2, 4]
Generation 3310: Best = 3.270632e+03 FES = 331100 Diversity = 2.3574e-03 k_t
= [1, 1, 1, 1]
Generation 3320: Best = 3.270631e+03 FES = 332100 Diversity = 2.6515e-03 k_t
= [2, 2, 2, 4]
Generation 3330: Best = 3.270630e+03 FES = 333100 Diversity = 2.9587e-03 k_t
= [1, 1, 1, 1]
Generation 3340: Best = 3.270628e+03 FES = 334100 Diversity = 3.3630e-03 k_t
= [2, 3, 4, 3]
Generation 3350: Best = 3.270627e+03 FES = 335100 Diversity = 2.8045e-03 k_t
= [1, 1, 1, 1]
Generation 3360: Best = 3.270625e+03 FES = 336100 Diversity = 2.8694e-03 k_t
= [3, 2, 3, 2]
Generation 3370: Best = 3.270624e+03 FES = 337100 Diversity = 2.8033e-03 k_t
= [1, 1, 1, 1]
Generation 3380: Best = 3.270623e+03 FES = 338100 Diversity = 2.7286e-03 k_t
= [3, 2, 2, 4]
Generation 3390: Best = 3.270622e+03 FES = 339100 Diversity = 2.7665e-03 k_t
= [1, 1, 1, 1]
Generation 3400: Best = 3.270621e+03 FES = 340100 Diversity = 2.6505e-03 k_t
= [2, 4, 3, 3]
Generation 3410: Best = 3.270620e+03 FES = 341100 Diversity = 2.6664e-03 k_t
= [1, 1, 1, 1]
Generation 3420: Best = 3.270619e+03 FES = 342100 Diversity = 2.3592e-03 k_t
= [4, 3, 4, 4]
Generation 3430: Best = 3.270618e+03 FES = 343100 Diversity = 1.8393e-03 k_t
= [1, 1, 1, 1]
Generation 3440: Best = 3.270618e+03 FES = 344100 Diversity = 1.8108e-03 k_t
= [3, 2, 2, 2]
Generation 3450: Best = 3.270617e+03 FES = 345100 Diversity = 1.8641e-03 k_t
= [1, 1, 1, 1]
Generation 3460: Best = 3.270617e+03 FES = 346100 Diversity = 1.7143e-03 k_t

```

```

= [3, 3, 3, 3]
Generation 3470: Best = 3.270616e+03 FES = 347100 Diversity = 2.0545e-03 k_t
= [1, 1, 1, 1]
Generation 3480: Best = 3.270616e+03 FES = 348100 Diversity = 1.7485e-03 k_t
= [4, 4, 3, 4]
Generation 3490: Best = 3.270615e+03 FES = 349100 Diversity = 1.7121e-03 k_t
= [1, 1, 1, 1]
Generation 3500: Best = 3.270615e+03 FES = 350100 Diversity = 1.7602e-03 k_t
= [4, 2, 3, 4]
Generation 3510: Best = 3.270614e+03 FES = 351100 Diversity = 1.5556e-03 k_t
= [1, 1, 1, 1]
Generation 3520: Best = 3.270614e+03 FES = 352100 Diversity = 1.5094e-03 k_t
= [3, 3, 3, 3]
Generation 3530: Best = 3.270614e+03 FES = 353100 Diversity = 1.5402e-03 k_t
= [1, 1, 1, 1]
Generation 3540: Best = 3.270613e+03 FES = 354100 Diversity = 1.2783e-03 k_t
= [2, 2, 3, 4]
Generation 3550: Best = 3.270613e+03 FES = 355100 Diversity = 1.5416e-03 k_t
= [1, 1, 1, 1]
Generation 3560: Best = 3.270612e+03 FES = 356100 Diversity = 1.7575e-03 k_t
= [4, 4, 4, 3]
Generation 3570: Best = 3.270612e+03 FES = 357100 Diversity = 1.9907e-03 k_t
= [1, 1, 1, 1]
Generation 3580: Best = 3.270611e+03 FES = 358100 Diversity = 1.9275e-03 k_t
= [3, 3, 2, 3]
Generation 3590: Best = 3.270611e+03 FES = 359100 Diversity = 1.6136e-03 k_t
= [1, 1, 1, 1]
Generation 3600: Best = 3.270610e+03 FES = 360100 Diversity = 1.4669e-03 k_t
= [4, 4, 4, 4]
Generation 3610: Best = 3.270610e+03 FES = 361100 Diversity = 1.3454e-03 k_t
= [1, 1, 1, 1]
Generation 3620: Best = 3.270610e+03 FES = 362100 Diversity = 1.5341e-03 k_t
= [3, 3, 4, 3]
Generation 3630: Best = 3.270609e+03 FES = 363100 Diversity = 1.2990e-03 k_t
= [1, 1, 1, 1]
Generation 3640: Best = 3.270609e+03 FES = 364100 Diversity = 1.0743e-03 k_t
= [3, 4, 3, 3]
Generation 3650: Best = 3.270609e+03 FES = 365100 Diversity = 1.0087e-03 k_t
= [1, 1, 1, 1]
Generation 3660: Best = 3.270609e+03 FES = 366100 Diversity = 9.2523e-04 k_t
= [4, 3, 3, 2]
Generation 3670: Best = 3.270609e+03 FES = 367100 Diversity = 1.1587e-03 k_t
= [1, 1, 1, 1]
Generation 3680: Best = 3.270608e+03 FES = 368100 Diversity = 1.4343e-03 k_t
= [2, 3, 2, 2]
Generation 3690: Best = 3.270608e+03 FES = 369100 Diversity = 1.2657e-03 k_t
= [1, 1, 1, 1]
Generation 3700: Best = 3.270608e+03 FES = 370100 Diversity = 1.3687e-03 k_t
= [3, 4, 4, 2]
Generation 3710: Best = 3.270607e+03 FES = 371100 Diversity = 1.4404e-03 k_t
= [1, 1, 1, 1]
Generation 3720: Best = 3.270607e+03 FES = 372100 Diversity = 1.3697e-03 k_t
= [2, 4, 3, 2]
Generation 3730: Best = 3.270607e+03 FES = 373100 Diversity = 1.6006e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 3740: Best = 3.270606e+03 FES = 374100 Diversity = 2.0451e-03 k_t
= [2, 2, 3, 4]
Generation 3750: Best = 3.270605e+03 FES = 375100 Diversity = 2.1162e-03 k_t
= [1, 1, 1, 1]
Generation 3760: Best = 3.270604e+03 FES = 376100 Diversity = 2.2535e-03 k_t
= [3, 3, 4, 2]
Generation 3770: Best = 3.270602e+03 FES = 377100 Diversity = 2.6939e-03 k_t
= [1, 1, 1, 1]
Generation 3780: Best = 3.270600e+03 FES = 378100 Diversity = 3.1735e-03 k_t
= [3, 2, 3, 4]
Generation 3790: Best = 3.270599e+03 FES = 379100 Diversity = 3.3262e-03 k_t
= [1, 1, 1, 1]
Generation 3800: Best = 3.270596e+03 FES = 380100 Diversity = 3.6774e-03 k_t
= [4, 2, 3, 2]
Generation 3810: Best = 3.270594e+03 FES = 381100 Diversity = 4.4010e-03 k_t
= [1, 1, 1, 1]
Generation 3820: Best = 3.270588e+03 FES = 382100 Diversity = 5.1826e-03 k_t
= [3, 2, 2, 4]
Generation 3830: Best = 3.270585e+03 FES = 383100 Diversity = 4.8852e-03 k_t
= [1, 1, 1, 1]
Generation 3840: Best = 3.270578e+03 FES = 384100 Diversity = 5.9987e-03 k_t
= [4, 4, 2, 3]
Generation 3850: Best = 3.270572e+03 FES = 385100 Diversity = 7.3661e-03 k_t
= [1, 1, 1, 1]
Generation 3860: Best = 3.270566e+03 FES = 386100 Diversity = 7.6516e-03 k_t
= [4, 4, 3, 2]
Generation 3870: Best = 3.270559e+03 FES = 387100 Diversity = 6.7183e-03 k_t
= [1, 1, 1, 1]
Generation 3880: Best = 3.270551e+03 FES = 388100 Diversity = 5.6617e-03 k_t
= [2, 2, 2, 4]
Generation 3890: Best = 3.270546e+03 FES = 389100 Diversity = 5.6053e-03 k_t
= [1, 1, 1, 1]
Generation 3900: Best = 3.270541e+03 FES = 390100 Diversity = 5.7834e-03 k_t
= [2, 4, 3, 2]
Generation 3910: Best = 3.270534e+03 FES = 391100 Diversity = 5.9221e-03 k_t
= [1, 1, 1, 1]
Generation 3920: Best = 3.270530e+03 FES = 392100 Diversity = 5.2011e-03 k_t
= [2, 3, 3, 2]
Generation 3930: Best = 3.270525e+03 FES = 393100 Diversity = 5.0247e-03 k_t
= [1, 1, 1, 1]
Generation 3940: Best = 3.270522e+03 FES = 394100 Diversity = 4.1444e-03 k_t
= [4, 2, 2, 4]
Generation 3950: Best = 3.270518e+03 FES = 395100 Diversity = 4.5444e-03 k_t
= [1, 1, 1, 1]
Generation 3960: Best = 3.270514e+03 FES = 396100 Diversity = 5.0745e-03 k_t
= [2, 3, 4, 4]
Generation 3970: Best = 3.270510e+03 FES = 397100 Diversity = 4.5596e-03 k_t
= [1, 1, 1, 1]
Generation 3980: Best = 3.270504e+03 FES = 398100 Diversity = 5.9433e-03 k_t
= [4, 2, 2, 2]
Generation 3990: Best = 3.270498e+03 FES = 399100 Diversity = 6.4248e-03 k_t
= [1, 1, 1, 1]
Generation 4000: Best = 3.270492e+03 FES = 400100 Diversity = 6.4122e-03 k_t

```



```

= [2, 2, 2, 4]
Generation 4010: Best = 3.270486e+03 FES = 401100 Diversity = 5.7147e-03 k_t
= [1, 1, 1, 1]
Generation 4020: Best = 3.270482e+03 FES = 402100 Diversity = 4.9273e-03 k_t
= [2, 2, 3, 4]
Generation 4030: Best = 3.270479e+03 FES = 403100 Diversity = 4.3731e-03 k_t
= [1, 1, 1, 1]
Generation 4040: Best = 3.270476e+03 FES = 404100 Diversity = 3.6097e-03 k_t
= [3, 2, 2, 2]
Generation 4050: Best = 3.270473e+03 FES = 405100 Diversity = 3.9643e-03 k_t
= [1, 1, 1, 1]
Generation 4060: Best = 3.270470e+03 FES = 406100 Diversity = 4.0141e-03 k_t
= [4, 4, 2, 2]
Generation 4070: Best = 3.270467e+03 FES = 407100 Diversity = 4.2856e-03 k_t
= [1, 1, 1, 1]
Generation 4080: Best = 3.270465e+03 FES = 408100 Diversity = 3.9302e-03 k_t
= [4, 2, 2, 4]
Generation 4090: Best = 3.270463e+03 FES = 409100 Diversity = 3.4554e-03 k_t
= [1, 1, 1, 1]
Generation 4100: Best = 3.270461e+03 FES = 410100 Diversity = 3.2245e-03 k_t
= [4, 3, 2, 3]
Generation 4110: Best = 3.270459e+03 FES = 411100 Diversity = 3.3890e-03 k_t
= [1, 1, 1, 1]
Generation 4120: Best = 3.270458e+03 FES = 412100 Diversity = 3.3216e-03 k_t
= [4, 3, 4, 4]
Generation 4130: Best = 3.270456e+03 FES = 413100 Diversity = 3.5255e-03 k_t
= [1, 1, 1, 1]
Generation 4140: Best = 3.270454e+03 FES = 414100 Diversity = 3.7048e-03 k_t
= [4, 2, 4, 2]
Generation 4150: Best = 3.270451e+03 FES = 415100 Diversity = 3.9573e-03 k_t
= [1, 1, 1, 1]
Generation 4160: Best = 3.270449e+03 FES = 416100 Diversity = 4.1751e-03 k_t
= [2, 3, 4, 2]
Generation 4170: Best = 3.270446e+03 FES = 417100 Diversity = 4.1620e-03 k_t
= [1, 1, 1, 1]
Generation 4180: Best = 3.270443e+03 FES = 418100 Diversity = 4.3440e-03 k_t
= [3, 2, 2, 2]
Generation 4190: Best = 3.270442e+03 FES = 419100 Diversity = 3.5646e-03 k_t
= [1, 1, 1, 1]
Generation 4200: Best = 3.270440e+03 FES = 420100 Diversity = 3.4192e-03 k_t
= [4, 4, 4, 3]
Generation 4210: Best = 3.270438e+03 FES = 421100 Diversity = 3.1977e-03 k_t
= [1, 1, 1, 1]
Generation 4220: Best = 3.270437e+03 FES = 422100 Diversity = 2.9786e-03 k_t
= [2, 3, 2, 4]
Generation 4230: Best = 3.270436e+03 FES = 423100 Diversity = 2.6453e-03 k_t
= [1, 1, 1, 1]
Generation 4240: Best = 3.270434e+03 FES = 424100 Diversity = 2.8712e-03 k_t
= [3, 2, 3, 2]
Generation 4250: Best = 3.270433e+03 FES = 425100 Diversity = 2.8017e-03 k_t
= [1, 1, 1, 1]
Generation 4260: Best = 3.270431e+03 FES = 426100 Diversity = 3.5767e-03 k_t
= [3, 4, 3, 3]
Generation 4270: Best = 3.270428e+03 FES = 427100 Diversity = 3.6960e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 4280: Best = 3.270426e+03 FES = 428100 Diversity = 3.8209e-03 k_t
= [3, 2, 2, 3]
Generation 4290: Best = 3.270423e+03 FES = 429100 Diversity = 4.0878e-03 k_t
= [1, 1, 1, 1]
Generation 4300: Best = 3.270417e+03 FES = 430100 Diversity = 5.4375e-03 k_t
= [2, 4, 2, 2]
Generation 4310: Best = 3.270414e+03 FES = 431100 Diversity = 4.8825e-03 k_t
= [1, 1, 1, 1]
Generation 4320: Best = 3.270410e+03 FES = 432100 Diversity = 5.6883e-03 k_t
= [3, 4, 3, 3]
Generation 4330: Best = 3.270406e+03 FES = 433100 Diversity = 5.1195e-03 k_t
= [1, 1, 1, 1]
Generation 4340: Best = 3.270401e+03 FES = 434100 Diversity = 5.7226e-03 k_t
= [2, 2, 4, 2]
Generation 4350: Best = 3.270398e+03 FES = 435100 Diversity = 4.6810e-03 k_t
= [1, 1, 1, 1]
Generation 4360: Best = 3.270395e+03 FES = 436100 Diversity = 4.4791e-03 k_t
= [3, 2, 2, 3]
Generation 4370: Best = 3.270391e+03 FES = 437100 Diversity = 4.5149e-03 k_t
= [1, 1, 1, 1]
Generation 4380: Best = 3.270387e+03 FES = 438100 Diversity = 5.5163e-03 k_t
= [2, 4, 3, 2]
Generation 4390: Best = 3.270382e+03 FES = 439100 Diversity = 5.9563e-03 k_t
= [1, 1, 1, 1]
Generation 4400: Best = 3.270376e+03 FES = 440100 Diversity = 6.2975e-03 k_t
= [2, 3, 3, 2]
Generation 4410: Best = 3.270371e+03 FES = 441100 Diversity = 5.6314e-03 k_t
= [1, 1, 1, 1]
Generation 4420: Best = 3.270367e+03 FES = 442100 Diversity = 5.0476e-03 k_t
= [4, 2, 4, 4]
Generation 4430: Best = 3.270363e+03 FES = 443100 Diversity = 4.4182e-03 k_t
= [1, 1, 1, 1]
Generation 4440: Best = 3.270358e+03 FES = 444100 Diversity = 5.1382e-03 k_t
= [4, 2, 4, 3]
Generation 4450: Best = 3.270353e+03 FES = 445100 Diversity = 5.6336e-03 k_t
= [1, 1, 1, 1]
Generation 4460: Best = 3.270350e+03 FES = 446100 Diversity = 4.9562e-03 k_t
= [3, 3, 4, 4]
Generation 4470: Best = 3.270345e+03 FES = 447100 Diversity = 6.0373e-03 k_t
= [1, 1, 1, 1]
Generation 4480: Best = 3.270336e+03 FES = 448100 Diversity = 6.6760e-03 k_t
= [3, 2, 4, 4]
Generation 4490: Best = 3.270331e+03 FES = 449100 Diversity = 6.1964e-03 k_t
= [1, 1, 1, 1]
Generation 4500: Best = 3.270322e+03 FES = 450100 Diversity = 7.4376e-03 k_t
= [4, 4, 2, 3]
Generation 4510: Best = 3.270311e+03 FES = 451100 Diversity = 8.4052e-03 k_t
= [1, 1, 1, 1]
Generation 4520: Best = 3.270301e+03 FES = 452100 Diversity = 8.5175e-03 k_t
= [2, 4, 4, 2]
Generation 4530: Best = 3.270293e+03 FES = 453100 Diversity = 8.0194e-03 k_t
= [1, 1, 1, 1]
Generation 4540: Best = 3.270285e+03 FES = 454100 Diversity = 6.2310e-03 k_t

```

```

= [3, 2, 2, 4]
Generation 4550: Best = 3.270277e+03 FES = 455100 Diversity = 6.1642e-03 k_t
= [1, 1, 1, 1]
Generation 4560: Best = 3.270271e+03 FES = 456100 Diversity = 6.3504e-03 k_t
= [2, 3, 3, 3]
Generation 4570: Best = 3.270266e+03 FES = 457100 Diversity = 5.8919e-03 k_t
= [1, 1, 1, 1]
Generation 4580: Best = 3.270261e+03 FES = 458100 Diversity = 5.9629e-03 k_t
= [4, 3, 2, 3]
Generation 4590: Best = 3.270255e+03 FES = 459100 Diversity = 5.2191e-03 k_t
= [1, 1, 1, 1]
Generation 4600: Best = 3.270248e+03 FES = 460100 Diversity = 6.4531e-03 k_t
= [3, 2, 2, 4]
Generation 4610: Best = 3.270243e+03 FES = 461100 Diversity = 5.3326e-03 k_t
= [1, 1, 1, 1]
Generation 4620: Best = 3.270239e+03 FES = 462100 Diversity = 5.2123e-03 k_t
= [4, 3, 2, 3]
Generation 4630: Best = 3.270234e+03 FES = 463100 Diversity = 4.6130e-03 k_t
= [1, 1, 1, 1]
Generation 4640: Best = 3.270232e+03 FES = 464100 Diversity = 3.8988e-03 k_t
= [2, 4, 4, 3]
Generation 4650: Best = 3.270229e+03 FES = 465100 Diversity = 3.7834e-03 k_t
= [1, 1, 1, 1]
Generation 4660: Best = 3.270227e+03 FES = 466100 Diversity = 3.4926e-03 k_t
= [3, 4, 4, 3]
Generation 4670: Best = 3.270224e+03 FES = 467100 Diversity = 3.5927e-03 k_t
= [1, 1, 1, 1]
Generation 4680: Best = 3.270222e+03 FES = 468100 Diversity = 3.3340e-03 k_t
= [3, 4, 4, 2]
Generation 4690: Best = 3.270220e+03 FES = 469100 Diversity = 2.7504e-03 k_t
= [1, 1, 1, 1]
Generation 4700: Best = 3.270219e+03 FES = 470100 Diversity = 2.8293e-03 k_t
= [3, 3, 2, 3]
Generation 4710: Best = 3.270217e+03 FES = 471100 Diversity = 2.9289e-03 k_t
= [1, 1, 1, 1]
Generation 4720: Best = 3.270216e+03 FES = 472100 Diversity = 2.6916e-03 k_t
= [3, 2, 2, 2]
Generation 4730: Best = 3.270215e+03 FES = 473100 Diversity = 2.5500e-03 k_t
= [1, 1, 1, 1]
Generation 4740: Best = 3.270214e+03 FES = 474100 Diversity = 2.4968e-03 k_t
= [3, 2, 2, 4]
Generation 4750: Best = 3.270213e+03 FES = 475100 Diversity = 2.4089e-03 k_t
= [1, 1, 1, 1]
Generation 4760: Best = 3.270212e+03 FES = 476100 Diversity = 2.4178e-03 k_t
= [4, 2, 3, 3]
Generation 4770: Best = 3.270211e+03 FES = 477100 Diversity = 2.1670e-03 k_t
= [1, 1, 1, 1]
Generation 4780: Best = 3.270211e+03 FES = 478100 Diversity = 1.8691e-03 k_t
= [3, 2, 3, 2]
Generation 4790: Best = 3.270210e+03 FES = 479100 Diversity = 1.6412e-03 k_t
= [1, 1, 1, 1]
Generation 4800: Best = 3.270210e+03 FES = 480100 Diversity = 1.5460e-03 k_t
= [3, 4, 4, 4]
Generation 4810: Best = 3.270209e+03 FES = 481100 Diversity = 1.3537e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 4820: Best = 3.270209e+03 FES = 482100 Diversity = 1.2048e-03 k_t
= [3, 4, 4, 3]
Generation 4830: Best = 3.270209e+03 FES = 483100 Diversity = 1.1635e-03 k_t
= [1, 1, 1, 1]
Generation 4840: Best = 3.270208e+03 FES = 484100 Diversity = 1.3630e-03 k_t
= [3, 3, 3, 2]
Generation 4850: Best = 3.270208e+03 FES = 485100 Diversity = 1.5021e-03 k_t
= [1, 1, 1, 1]
Generation 4860: Best = 3.270207e+03 FES = 486100 Diversity = 1.4335e-03 k_t
= [3, 3, 2, 2]
Generation 4870: Best = 3.270207e+03 FES = 487100 Diversity = 1.6709e-03 k_t
= [1, 1, 1, 1]
Generation 4880: Best = 3.270206e+03 FES = 488100 Diversity = 2.0815e-03 k_t
= [3, 2, 4, 4]
Generation 4890: Best = 3.270205e+03 FES = 489100 Diversity = 1.9986e-03 k_t
= [1, 1, 1, 1]
Generation 4900: Best = 3.270205e+03 FES = 490100 Diversity = 2.0455e-03 k_t
= [4, 2, 4, 2]
Generation 4910: Best = 3.270204e+03 FES = 491100 Diversity = 1.7976e-03 k_t
= [1, 1, 1, 1]
Generation 4920: Best = 3.270203e+03 FES = 492100 Diversity = 1.6197e-03 k_t
= [2, 3, 3, 3]
Generation 4930: Best = 3.270203e+03 FES = 493100 Diversity = 1.6610e-03 k_t
= [1, 1, 1, 1]
Generation 4940: Best = 3.270202e+03 FES = 494100 Diversity = 1.5499e-03 k_t
= [4, 4, 2, 4]
Generation 4950: Best = 3.270202e+03 FES = 495100 Diversity = 1.1315e-03 k_t
= [1, 1, 1, 1]
Generation 4960: Best = 3.270202e+03 FES = 496100 Diversity = 1.0736e-03 k_t
= [4, 3, 2, 3]
Generation 4970: Best = 3.270202e+03 FES = 497100 Diversity = 1.0788e-03 k_t
= [1, 1, 1, 1]
Generation 4980: Best = 3.270202e+03 FES = 498100 Diversity = 1.1656e-03 k_t
= [2, 3, 3, 3]
Generation 4990: Best = 3.270201e+03 FES = 499100 Diversity = 1.0892e-03 k_t
= [1, 1, 1, 1]
Generation 5000: Best = 3.270201e+03 FES = 500100 Diversity = 9.2450e-04 k_t
= [3, 2, 2, 4]
Generation 5010: Best = 3.270201e+03 FES = 501100 Diversity = 1.0330e-03 k_t
= [1, 1, 1, 1]
Generation 5020: Best = 3.270201e+03 FES = 502100 Diversity = 8.4149e-04 k_t
= [3, 4, 4, 3]
Generation 5030: Best = 3.270201e+03 FES = 503100 Diversity = 7.6998e-04 k_t
= [1, 1, 1, 1]
Generation 5040: Best = 3.270201e+03 FES = 504100 Diversity = 7.0813e-04 k_t
= [4, 4, 4, 3]
Generation 5050: Best = 3.270201e+03 FES = 505100 Diversity = 6.4498e-04 k_t
= [1, 1, 1, 1]
Generation 5060: Best = 3.270200e+03 FES = 506100 Diversity = 6.4546e-04 k_t
= [4, 4, 2, 3]
Generation 5070: Best = 3.270200e+03 FES = 507100 Diversity = 6.4444e-04 k_t
= [1, 1, 1, 1]
Generation 5080: Best = 3.270200e+03 FES = 508100 Diversity = 7.1975e-04 k_t

```

```

= [4, 3, 3, 2]
Generation 5090: Best = 3.270200e+03 FES = 509100 Diversity = 8.3813e-04 k_t
= [1, 1, 1, 1]
Generation 5100: Best = 3.270200e+03 FES = 510100 Diversity = 8.2709e-04 k_t
= [4, 4, 3, 2]
Generation 5110: Best = 3.270200e+03 FES = 511100 Diversity = 7.3692e-04 k_t
= [1, 1, 1, 1]
Generation 5120: Best = 3.270200e+03 FES = 512100 Diversity = 6.1054e-04 k_t
= [2, 4, 4, 3]
Generation 5130: Best = 3.270200e+03 FES = 513100 Diversity = 6.4621e-04 k_t
= [1, 1, 1, 1]
Generation 5140: Best = 3.270200e+03 FES = 514100 Diversity = 6.0892e-04 k_t
= [2, 3, 3, 4]
Generation 5150: Best = 3.270200e+03 FES = 515100 Diversity = 5.9791e-04 k_t
= [1, 1, 1, 1]
Generation 5160: Best = 3.270200e+03 FES = 516100 Diversity = 6.9794e-04 k_t
= [2, 3, 3, 3]
Generation 5170: Best = 3.270199e+03 FES = 517100 Diversity = 7.9680e-04 k_t
= [1, 1, 1, 1]
Generation 5180: Best = 3.270199e+03 FES = 518100 Diversity = 7.8444e-04 k_t
= [4, 2, 2, 4]
Generation 5190: Best = 3.270199e+03 FES = 519100 Diversity = 7.4681e-04 k_t
= [1, 1, 1, 1]
Generation 5200: Best = 3.270199e+03 FES = 520100 Diversity = 9.8563e-04 k_t
= [4, 2, 3, 3]
Generation 5210: Best = 3.270199e+03 FES = 521100 Diversity = 8.6556e-04 k_t
= [1, 1, 1, 1]
Generation 5220: Best = 3.270199e+03 FES = 522100 Diversity = 7.6361e-04 k_t
= [3, 2, 4, 3]
Generation 5230: Best = 3.270199e+03 FES = 523100 Diversity = 6.4938e-04 k_t
= [1, 1, 1, 1]
Generation 5240: Best = 3.270199e+03 FES = 524100 Diversity = 6.3685e-04 k_t
= [4, 3, 4, 3]
Generation 5250: Best = 3.270198e+03 FES = 525100 Diversity = 5.7568e-04 k_t
= [1, 1, 1, 1]
Generation 5260: Best = 3.270198e+03 FES = 526100 Diversity = 5.3604e-04 k_t
= [3, 2, 3, 3]
Generation 5270: Best = 3.270198e+03 FES = 527100 Diversity = 6.5148e-04 k_t
= [1, 1, 1, 1]
Generation 5280: Best = 3.270198e+03 FES = 528100 Diversity = 5.7551e-04 k_t
= [2, 4, 2, 3]
Generation 5290: Best = 3.270198e+03 FES = 529100 Diversity = 4.6558e-04 k_t
= [1, 1, 1, 1]
Generation 5300: Best = 3.270198e+03 FES = 530100 Diversity = 4.6084e-04 k_t
= [3, 2, 2, 4]
Generation 5310: Best = 3.270198e+03 FES = 531100 Diversity = 4.8547e-04 k_t
= [1, 1, 1, 1]
Generation 5320: Best = 3.270198e+03 FES = 532100 Diversity = 4.6700e-04 k_t
= [2, 2, 2, 3]
Generation 5330: Best = 3.270198e+03 FES = 533100 Diversity = 4.3536e-04 k_t
= [1, 1, 1, 1]
Generation 5340: Best = 3.270198e+03 FES = 534100 Diversity = 5.1368e-04 k_t
= [2, 2, 2, 3]
Generation 5350: Best = 3.270198e+03 FES = 535100 Diversity = 4.4885e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 5360: Best = 3.270198e+03  FES = 536100  Diversity = 4.4782e-04  k_t
= [4, 3, 4, 4]
Generation 5370: Best = 3.270198e+03  FES = 537100  Diversity = 4.5416e-04  k_t
= [1, 1, 1, 1]
Generation 5380: Best = 3.270198e+03  FES = 538100  Diversity = 4.7595e-04  k_t
= [2, 4, 4, 3]
Generation 5390: Best = 3.270198e+03  FES = 539100  Diversity = 5.3891e-04  k_t
= [1, 1, 1, 1]
Generation 5400: Best = 3.270198e+03  FES = 540100  Diversity = 5.7302e-04  k_t
= [4, 4, 2, 2]
Generation 5410: Best = 3.270198e+03  FES = 541100  Diversity = 5.9500e-04  k_t
= [1, 1, 1, 1]
Generation 5420: Best = 3.270198e+03  FES = 542100  Diversity = 6.3204e-04  k_t
= [3, 4, 4, 3]
Generation 5430: Best = 3.270198e+03  FES = 543100  Diversity = 5.0191e-04  k_t
= [1, 1, 1, 1]
Generation 5440: Best = 3.270198e+03  FES = 544100  Diversity = 5.1974e-04  k_t
= [4, 4, 4, 3]
Generation 5450: Best = 3.270198e+03  FES = 545100  Diversity = 4.5627e-04  k_t
= [1, 1, 1, 1]
Generation 5460: Best = 3.270198e+03  FES = 546100  Diversity = 4.3753e-04  k_t
= [2, 4, 4, 4]
Generation 5470: Best = 3.270197e+03  FES = 547100  Diversity = 4.2495e-04  k_t
= [1, 1, 1, 1]
Generation 5480: Best = 3.270197e+03  FES = 548100  Diversity = 3.9622e-04  k_t
= [2, 2, 2, 2]
Generation 5490: Best = 3.270197e+03  FES = 549100  Diversity = 4.9561e-04  k_t
= [1, 1, 1, 1]
Generation 5500: Best = 3.270197e+03  FES = 550100  Diversity = 4.6826e-04  k_t
= [4, 2, 2, 4]
Generation 5510: Best = 3.270197e+03  FES = 551100  Diversity = 4.0184e-04  k_t
= [1, 1, 1, 1]
Generation 5520: Best = 3.270197e+03  FES = 552100  Diversity = 4.4578e-04  k_t
= [4, 2, 2, 2]
Generation 5530: Best = 3.270197e+03  FES = 553100  Diversity = 4.7802e-04  k_t
= [1, 1, 1, 1]
Generation 5540: Best = 3.270197e+03  FES = 554100  Diversity = 5.2155e-04  k_t
= [2, 2, 2, 3]
Generation 5550: Best = 3.270197e+03  FES = 555100  Diversity = 5.4388e-04  k_t
= [1, 1, 1, 1]
Generation 5560: Best = 3.270197e+03  FES = 556100  Diversity = 7.4638e-04  k_t
= [3, 3, 3, 3]
Generation 5570: Best = 3.270197e+03  FES = 557100  Diversity = 7.6691e-04  k_t
= [1, 1, 1, 1]
Generation 5580: Best = 3.270197e+03  FES = 558100  Diversity = 6.0969e-04  k_t
= [2, 3, 4, 2]
Generation 5590: Best = 3.270197e+03  FES = 559100  Diversity = 6.8973e-04  k_t
= [1, 1, 1, 1]
Generation 5600: Best = 3.270197e+03  FES = 560100  Diversity = 6.9143e-04  k_t
= [2, 2, 3, 4]
Generation 5610: Best = 3.270197e+03  FES = 561100  Diversity = 8.4073e-04  k_t
= [1, 1, 1, 1]
Generation 5620: Best = 3.270197e+03  FES = 562100  Diversity = 6.3324e-04  k_t

```

```

= [3, 2, 4, 4]
Generation 5630: Best = 3.270196e+03 FES = 563100 Diversity = 6.0238e-04 k_t
= [1, 1, 1, 1]
Generation 5640: Best = 3.270196e+03 FES = 564100 Diversity = 7.1784e-04 k_t
= [3, 3, 3, 2]
Generation 5650: Best = 3.270196e+03 FES = 565100 Diversity = 8.1786e-04 k_t
= [1, 1, 1, 1]
Generation 5660: Best = 3.270196e+03 FES = 566100 Diversity = 7.7240e-04 k_t
= [3, 3, 3, 4]
Generation 5670: Best = 3.270196e+03 FES = 567100 Diversity = 7.9530e-04 k_t
= [1, 1, 1, 1]
Generation 5680: Best = 3.270196e+03 FES = 568100 Diversity = 7.0333e-04 k_t
= [4, 3, 4, 3]
Generation 5690: Best = 3.270196e+03 FES = 569100 Diversity = 7.1667e-04 k_t
= [1, 1, 1, 1]
Generation 5700: Best = 3.270196e+03 FES = 570100 Diversity = 6.7468e-04 k_t
= [4, 4, 4, 2]
Generation 5710: Best = 3.270196e+03 FES = 571100 Diversity = 8.6363e-04 k_t
= [1, 1, 1, 1]
Generation 5720: Best = 3.270195e+03 FES = 572100 Diversity = 1.0537e-03 k_t
= [2, 3, 2, 3]
Generation 5730: Best = 3.270195e+03 FES = 573100 Diversity = 1.1492e-03 k_t
= [1, 1, 1, 1]
Generation 5740: Best = 3.270195e+03 FES = 574100 Diversity = 1.1506e-03 k_t
= [4, 2, 2, 2]
Generation 5750: Best = 3.270195e+03 FES = 575100 Diversity = 1.3267e-03 k_t
= [1, 1, 1, 1]
Generation 5760: Best = 3.270194e+03 FES = 576100 Diversity = 1.3104e-03 k_t
= [3, 2, 4, 2]
Generation 5770: Best = 3.270194e+03 FES = 577100 Diversity = 1.2197e-03 k_t
= [1, 1, 1, 1]
Generation 5780: Best = 3.270193e+03 FES = 578100 Diversity = 1.5646e-03 k_t
= [2, 2, 4, 3]
Generation 5790: Best = 3.270193e+03 FES = 579100 Diversity = 1.9930e-03 k_t
= [1, 1, 1, 1]
Generation 5800: Best = 3.270191e+03 FES = 580100 Diversity = 2.2651e-03 k_t
= [2, 2, 4, 2]
Generation 5810: Best = 3.270191e+03 FES = 581100 Diversity = 1.9633e-03 k_t
= [1, 1, 1, 1]
Generation 5820: Best = 3.270190e+03 FES = 582100 Diversity = 2.0858e-03 k_t
= [2, 3, 4, 4]
Generation 5830: Best = 3.270188e+03 FES = 583100 Diversity = 2.9194e-03 k_t
= [1, 1, 1, 1]
Generation 5840: Best = 3.270186e+03 FES = 584100 Diversity = 3.2771e-03 k_t
= [4, 2, 3, 2]
Generation 5850: Best = 3.270185e+03 FES = 585100 Diversity = 3.0073e-03 k_t
= [1, 1, 1, 1]
Generation 5860: Best = 3.270183e+03 FES = 586100 Diversity = 2.6315e-03 k_t
= [4, 3, 2, 4]
Generation 5870: Best = 3.270182e+03 FES = 587100 Diversity = 2.2177e-03 k_t
= [1, 1, 1, 1]
Generation 5880: Best = 3.270181e+03 FES = 588100 Diversity = 2.2771e-03 k_t
= [3, 2, 3, 2]
Generation 5890: Best = 3.270180e+03 FES = 589100 Diversity = 2.2852e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 5900: Best = 3.270180e+03 FES = 590100 Diversity = 1.7448e-03 k_t
= [2, 4, 4, 2]
Generation 5910: Best = 3.270179e+03 FES = 591100 Diversity = 1.7211e-03 k_t
= [1, 1, 1, 1]
Generation 5920: Best = 3.270178e+03 FES = 592100 Diversity = 1.9984e-03 k_t
= [2, 3, 4, 3]
Generation 5930: Best = 3.270178e+03 FES = 593100 Diversity = 2.1273e-03 k_t
= [1, 1, 1, 1]
Generation 5940: Best = 3.270176e+03 FES = 594100 Diversity = 2.8022e-03 k_t
= [3, 4, 4, 3]
Generation 5950: Best = 3.270174e+03 FES = 595100 Diversity = 3.2382e-03 k_t
= [1, 1, 1, 1]
Generation 5960: Best = 3.270173e+03 FES = 596100 Diversity = 2.8691e-03 k_t
= [2, 3, 2, 4]
Generation 5970: Best = 3.270172e+03 FES = 597100 Diversity = 2.3457e-03 k_t
= [1, 1, 1, 1]
Generation 5980: Best = 3.270171e+03 FES = 598100 Diversity = 2.0910e-03 k_t
= [3, 4, 4, 2]
Generation 5990: Best = 3.270170e+03 FES = 599100 Diversity = 1.9239e-03 k_t
= [1, 1, 1, 1]
Generation 6000: Best = 3.270170e+03 FES = 600100 Diversity = 2.1064e-03 k_t
= [3, 4, 4, 3]
Generation 6010: Best = 3.270169e+03 FES = 601100 Diversity = 2.4958e-03 k_t
= [1, 1, 1, 1]
Generation 6020: Best = 3.270168e+03 FES = 602100 Diversity = 2.5950e-03 k_t
= [4, 2, 4, 3]
Generation 6030: Best = 3.270167e+03 FES = 603100 Diversity = 2.2401e-03 k_t
= [1, 1, 1, 1]
Generation 6040: Best = 3.270166e+03 FES = 604100 Diversity = 1.9561e-03 k_t
= [3, 3, 2, 2]
Generation 6050: Best = 3.270166e+03 FES = 605100 Diversity = 1.9815e-03 k_t
= [1, 1, 1, 1]
Generation 6060: Best = 3.270165e+03 FES = 606100 Diversity = 1.8457e-03 k_t
= [3, 3, 2, 3]
Generation 6070: Best = 3.270164e+03 FES = 607100 Diversity = 1.7850e-03 k_t
= [1, 1, 1, 1]
Generation 6080: Best = 3.270164e+03 FES = 608100 Diversity = 1.6572e-03 k_t
= [4, 2, 4, 4]
Generation 6090: Best = 3.270163e+03 FES = 609100 Diversity = 1.4719e-03 k_t
= [1, 1, 1, 1]
Generation 6100: Best = 3.270163e+03 FES = 610100 Diversity = 1.4439e-03 k_t
= [4, 4, 2, 2]
Generation 6110: Best = 3.270162e+03 FES = 611100 Diversity = 1.7988e-03 k_t
= [1, 1, 1, 1]
Generation 6120: Best = 3.270162e+03 FES = 612100 Diversity = 1.3757e-03 k_t
= [3, 2, 4, 3]
Generation 6130: Best = 3.270162e+03 FES = 613100 Diversity = 1.2753e-03 k_t
= [1, 1, 1, 1]
Generation 6140: Best = 3.270162e+03 FES = 614100 Diversity = 1.3388e-03 k_t
= [2, 4, 2, 3]
Generation 6150: Best = 3.270161e+03 FES = 615100 Diversity = 1.0676e-03 k_t
= [1, 1, 1, 1]
Generation 6160: Best = 3.270161e+03 FES = 616100 Diversity = 9.5667e-04 k_t

```



```

= [4, 2, 4, 4]
Generation 6170: Best = 3.270161e+03 FES = 617100 Diversity = 7.1882e-04 k_t
= [1, 1, 1, 1]
Generation 6180: Best = 3.270161e+03 FES = 618100 Diversity = 6.9536e-04 k_t
= [4, 2, 2, 4]
Generation 6190: Best = 3.270161e+03 FES = 619100 Diversity = 7.2662e-04 k_t
= [1, 1, 1, 1]
Generation 6200: Best = 3.270161e+03 FES = 620100 Diversity = 6.8050e-04 k_t
= [2, 2, 2, 4]
Generation 6210: Best = 3.270161e+03 FES = 621100 Diversity = 6.6231e-04 k_t
= [1, 1, 1, 1]
Generation 6220: Best = 3.270161e+03 FES = 622100 Diversity = 6.8446e-04 k_t
= [2, 4, 3, 4]
Generation 6230: Best = 3.270161e+03 FES = 623100 Diversity = 6.6713e-04 k_t
= [1, 1, 1, 1]
Generation 6240: Best = 3.270161e+03 FES = 624100 Diversity = 6.2246e-04 k_t
= [2, 3, 3, 2]
Generation 6250: Best = 3.270161e+03 FES = 625100 Diversity = 7.7032e-04 k_t
= [1, 1, 1, 1]
Generation 6260: Best = 3.270160e+03 FES = 626100 Diversity = 7.5854e-04 k_t
= [3, 4, 3, 4]
Generation 6270: Best = 3.270160e+03 FES = 627100 Diversity = 6.5981e-04 k_t
= [1, 1, 1, 1]
Generation 6280: Best = 3.270160e+03 FES = 628100 Diversity = 8.4187e-04 k_t
= [3, 4, 2, 2]
Generation 6290: Best = 3.270160e+03 FES = 629100 Diversity = 1.0200e-03 k_t
= [1, 1, 1, 1]
Generation 6300: Best = 3.270160e+03 FES = 630100 Diversity = 1.2476e-03 k_t
= [2, 3, 4, 4]
Generation 6310: Best = 3.270160e+03 FES = 631100 Diversity = 1.3118e-03 k_t
= [1, 1, 1, 1]
Generation 6320: Best = 3.270159e+03 FES = 632100 Diversity = 1.3711e-03 k_t
= [2, 4, 3, 4]
Generation 6330: Best = 3.270159e+03 FES = 633100 Diversity = 1.4355e-03 k_t
= [1, 1, 1, 1]
Generation 6340: Best = 3.270158e+03 FES = 634100 Diversity = 2.0591e-03 k_t
= [4, 2, 2, 3]
Generation 6350: Best = 3.270157e+03 FES = 635100 Diversity = 2.4909e-03 k_t
= [1, 1, 1, 1]
Generation 6360: Best = 3.270155e+03 FES = 636100 Diversity = 2.9264e-03 k_t
= [4, 2, 4, 4]
Generation 6370: Best = 3.270153e+03 FES = 637100 Diversity = 3.5496e-03 k_t
= [1, 1, 1, 1]
Generation 6380: Best = 3.270151e+03 FES = 638100 Diversity = 3.2911e-03 k_t
= [2, 3, 3, 4]
Generation 6390: Best = 3.270148e+03 FES = 639100 Diversity = 3.4861e-03 k_t
= [1, 1, 1, 1]
Generation 6400: Best = 3.270147e+03 FES = 640100 Diversity = 3.1377e-03 k_t
= [2, 4, 4, 2]
Generation 6410: Best = 3.270145e+03 FES = 641100 Diversity = 3.4448e-03 k_t
= [1, 1, 1, 1]
Generation 6420: Best = 3.270143e+03 FES = 642100 Diversity = 3.5671e-03 k_t
= [3, 4, 2, 2]
Generation 6430: Best = 3.270141e+03 FES = 643100 Diversity = 3.6876e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 6440: Best = 3.270140e+03 FES = 644100 Diversity = 3.2743e-03 k_t
= [2, 2, 3, 2]
Generation 6450: Best = 3.270138e+03 FES = 645100 Diversity = 3.5413e-03 k_t
= [1, 1, 1, 1]
Generation 6460: Best = 3.270136e+03 FES = 646100 Diversity = 3.4920e-03 k_t
= [3, 4, 3, 3]
Generation 6470: Best = 3.270134e+03 FES = 647100 Diversity = 3.6726e-03 k_t
= [1, 1, 1, 1]
Generation 6480: Best = 3.270132e+03 FES = 648100 Diversity = 2.9447e-03 k_t
= [2, 3, 3, 4]
Generation 6490: Best = 3.270131e+03 FES = 649100 Diversity = 2.8733e-03 k_t
= [1, 1, 1, 1]
Generation 6500: Best = 3.270129e+03 FES = 650100 Diversity = 3.1167e-03 k_t
= [4, 3, 4, 4]
Generation 6510: Best = 3.270126e+03 FES = 651100 Diversity = 4.1535e-03 k_t
= [1, 1, 1, 1]
Generation 6520: Best = 3.270124e+03 FES = 652100 Diversity = 3.8638e-03 k_t
= [2, 3, 4, 2]
Generation 6530: Best = 3.270120e+03 FES = 653100 Diversity = 4.1565e-03 k_t
= [1, 1, 1, 1]
Generation 6540: Best = 3.270116e+03 FES = 654100 Diversity = 5.6726e-03 k_t
= [3, 3, 2, 4]
Generation 6550: Best = 3.270111e+03 FES = 655100 Diversity = 4.9530e-03 k_t
= [1, 1, 1, 1]
Generation 6560: Best = 3.270109e+03 FES = 656100 Diversity = 4.2975e-03 k_t
= [2, 4, 4, 3]
Generation 6570: Best = 3.270106e+03 FES = 657100 Diversity = 4.2060e-03 k_t
= [1, 1, 1, 1]
Generation 6580: Best = 3.270103e+03 FES = 658100 Diversity = 3.7384e-03 k_t
= [3, 4, 4, 2]
Generation 6590: Best = 3.270101e+03 FES = 659100 Diversity = 3.2333e-03 k_t
= [1, 1, 1, 1]
Generation 6600: Best = 3.270100e+03 FES = 660100 Diversity = 2.6109e-03 k_t
= [4, 4, 4, 3]
Generation 6610: Best = 3.270098e+03 FES = 661100 Diversity = 2.4259e-03 k_t
= [1, 1, 1, 1]
Generation 6620: Best = 3.270098e+03 FES = 662100 Diversity = 2.2299e-03 k_t
= [4, 3, 4, 3]
Generation 6630: Best = 3.270096e+03 FES = 663100 Diversity = 2.4949e-03 k_t
= [1, 1, 1, 1]
Generation 6640: Best = 3.270096e+03 FES = 664100 Diversity = 1.9805e-03 k_t
= [4, 4, 4, 4]
Generation 6650: Best = 3.270095e+03 FES = 665100 Diversity = 1.8954e-03 k_t
= [1, 1, 1, 1]
Generation 6660: Best = 3.270094e+03 FES = 666100 Diversity = 1.9118e-03 k_t
= [2, 2, 4, 3]
Generation 6670: Best = 3.270094e+03 FES = 667100 Diversity = 1.7407e-03 k_t
= [1, 1, 1, 1]
Generation 6680: Best = 3.270093e+03 FES = 668100 Diversity = 1.5937e-03 k_t
= [2, 2, 2, 2]
Generation 6690: Best = 3.270093e+03 FES = 669100 Diversity = 1.4764e-03 k_t
= [1, 1, 1, 1]
Generation 6700: Best = 3.270093e+03 FES = 670100 Diversity = 1.3590e-03 k_t

```

```

= [4, 2, 3, 2]
Generation 6710: Best = 3.270092e+03 FES = 671100 Diversity = 1.2050e-03 k_t
= [1, 1, 1, 1]
Generation 6720: Best = 3.270092e+03 FES = 672100 Diversity = 1.3278e-03 k_t
= [4, 2, 2, 2]
Generation 6730: Best = 3.270092e+03 FES = 673100 Diversity = 1.3067e-03 k_t
= [1, 1, 1, 1]
Generation 6740: Best = 3.270091e+03 FES = 674100 Diversity = 1.1960e-03 k_t
= [2, 3, 2, 2]
Generation 6750: Best = 3.270091e+03 FES = 675100 Diversity = 1.0017e-03 k_t
= [1, 1, 1, 1]
Generation 6760: Best = 3.270091e+03 FES = 676100 Diversity = 9.1202e-04 k_t
= [4, 3, 3, 2]
Generation 6770: Best = 3.270091e+03 FES = 677100 Diversity = 9.4053e-04 k_t
= [1, 1, 1, 1]
Generation 6780: Best = 3.270091e+03 FES = 678100 Diversity = 8.6657e-04 k_t
= [4, 3, 4, 3]
Generation 6790: Best = 3.270091e+03 FES = 679100 Diversity = 8.6048e-04 k_t
= [1, 1, 1, 1]
Generation 6800: Best = 3.270090e+03 FES = 680100 Diversity = 8.8607e-04 k_t
= [2, 3, 3, 3]
Generation 6810: Best = 3.270090e+03 FES = 681100 Diversity = 9.3611e-04 k_t
= [1, 1, 1, 1]
Generation 6820: Best = 3.270090e+03 FES = 682100 Diversity = 8.7284e-04 k_t
= [3, 3, 3, 4]
Generation 6830: Best = 3.270090e+03 FES = 683100 Diversity = 8.6832e-04 k_t
= [1, 1, 1, 1]
Generation 6840: Best = 3.270090e+03 FES = 684100 Diversity = 8.1485e-04 k_t
= [4, 3, 2, 3]
Generation 6850: Best = 3.270090e+03 FES = 685100 Diversity = 8.5999e-04 k_t
= [1, 1, 1, 1]
Generation 6860: Best = 3.270090e+03 FES = 686100 Diversity = 7.4184e-04 k_t
= [4, 4, 4, 2]
Generation 6870: Best = 3.270090e+03 FES = 687100 Diversity = 8.5104e-04 k_t
= [1, 1, 1, 1]
Generation 6880: Best = 3.270089e+03 FES = 688100 Diversity = 9.4372e-04 k_t
= [4, 2, 3, 3]
Generation 6890: Best = 3.270089e+03 FES = 689100 Diversity = 9.2230e-04 k_t
= [1, 1, 1, 1]
Generation 6900: Best = 3.270089e+03 FES = 690100 Diversity = 8.3542e-04 k_t
= [3, 2, 4, 3]
Generation 6910: Best = 3.270089e+03 FES = 691100 Diversity = 8.2906e-04 k_t
= [1, 1, 1, 1]
Generation 6920: Best = 3.270089e+03 FES = 692100 Diversity = 8.2749e-04 k_t
= [4, 3, 4, 3]
Generation 6930: Best = 3.270089e+03 FES = 693100 Diversity = 7.2393e-04 k_t
= [1, 1, 1, 1]
Generation 6940: Best = 3.270089e+03 FES = 694100 Diversity = 6.6434e-04 k_t
= [3, 2, 2, 3]
Generation 6950: Best = 3.270089e+03 FES = 695100 Diversity = 6.7175e-04 k_t
= [1, 1, 1, 1]
Generation 6960: Best = 3.270089e+03 FES = 696100 Diversity = 6.9658e-04 k_t
= [3, 2, 4, 3]
Generation 6970: Best = 3.270088e+03 FES = 697100 Diversity = 8.0579e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 6980: Best = 3.270088e+03 FES = 698100 Diversity = 6.8567e-04 k_t
= [4, 2, 4, 4]
Generation 6990: Best = 3.270088e+03 FES = 699100 Diversity = 6.4617e-04 k_t
= [1, 1, 1, 1]
Generation 7000: Best = 3.270088e+03 FES = 700100 Diversity = 7.0197e-04 k_t
= [3, 4, 4, 2]
Generation 7010: Best = 3.270088e+03 FES = 701100 Diversity = 5.8865e-04 k_t
= [1, 1, 1, 1]
Generation 7020: Best = 3.270088e+03 FES = 702100 Diversity = 6.5182e-04 k_t
= [2, 4, 2, 3]
Generation 7030: Best = 3.270088e+03 FES = 703100 Diversity = 6.7159e-04 k_t
= [1, 1, 1, 1]
Generation 7040: Best = 3.270088e+03 FES = 704100 Diversity = 6.2275e-04 k_t
= [3, 4, 2, 4]
Generation 7050: Best = 3.270088e+03 FES = 705100 Diversity = 4.8278e-04 k_t
= [1, 1, 1, 1]
Generation 7060: Best = 3.270088e+03 FES = 706100 Diversity = 5.4283e-04 k_t
= [4, 2, 3, 4]
Generation 7070: Best = 3.270088e+03 FES = 707100 Diversity = 6.4826e-04 k_t
= [1, 1, 1, 1]
Generation 7080: Best = 3.270088e+03 FES = 708100 Diversity = 5.6554e-04 k_t
= [2, 2, 3, 2]
Generation 7090: Best = 3.270088e+03 FES = 709100 Diversity = 5.6725e-04 k_t
= [1, 1, 1, 1]
Generation 7100: Best = 3.270088e+03 FES = 710100 Diversity = 6.8627e-04 k_t
= [4, 3, 4, 4]
Generation 7110: Best = 3.270087e+03 FES = 711100 Diversity = 6.8338e-04 k_t
= [1, 1, 1, 1]
Generation 7120: Best = 3.270087e+03 FES = 712100 Diversity = 6.5132e-04 k_t
= [3, 4, 2, 2]
Generation 7130: Best = 3.270087e+03 FES = 713100 Diversity = 5.7689e-04 k_t
= [1, 1, 1, 1]
Generation 7140: Best = 3.270087e+03 FES = 714100 Diversity = 7.1535e-04 k_t
= [2, 4, 2, 3]
Generation 7150: Best = 3.270087e+03 FES = 715100 Diversity = 6.9537e-04 k_t
= [1, 1, 1, 1]
Generation 7160: Best = 3.270087e+03 FES = 716100 Diversity = 6.5071e-04 k_t
= [2, 3, 3, 3]
Generation 7170: Best = 3.270087e+03 FES = 717100 Diversity = 7.8831e-04 k_t
= [1, 1, 1, 1]
Generation 7180: Best = 3.270087e+03 FES = 718100 Diversity = 8.2121e-04 k_t
= [3, 4, 2, 3]
Generation 7190: Best = 3.270087e+03 FES = 719100 Diversity = 8.2212e-04 k_t
= [1, 1, 1, 1]
Generation 7200: Best = 3.270087e+03 FES = 720100 Diversity = 9.4430e-04 k_t
= [4, 4, 4, 4]
Generation 7210: Best = 3.270086e+03 FES = 721100 Diversity = 9.0158e-04 k_t
= [1, 1, 1, 1]
Generation 7220: Best = 3.270086e+03 FES = 722100 Diversity = 9.6030e-04 k_t
= [3, 4, 3, 2]
Generation 7230: Best = 3.270086e+03 FES = 723100 Diversity = 1.0562e-03 k_t
= [1, 1, 1, 1]
Generation 7240: Best = 3.270086e+03 FES = 724100 Diversity = 9.8513e-04 k_t

```

```

= [3, 2, 4, 3]
Generation 7250: Best = 3.270086e+03 FES = 725100 Diversity = 1.0632e-03 k_t
= [1, 1, 1, 1]
Generation 7260: Best = 3.270085e+03 FES = 726100 Diversity = 1.2671e-03 k_t
= [2, 4, 3, 4]
Generation 7270: Best = 3.270085e+03 FES = 727100 Diversity = 1.3031e-03 k_t
= [1, 1, 1, 1]
Generation 7280: Best = 3.270085e+03 FES = 728100 Diversity = 1.5978e-03 k_t
= [2, 4, 2, 2]
Generation 7290: Best = 3.270084e+03 FES = 729100 Diversity = 1.9592e-03 k_t
= [1, 1, 1, 1]
Generation 7300: Best = 3.270083e+03 FES = 730100 Diversity = 2.0848e-03 k_t
= [3, 3, 4, 2]
Generation 7310: Best = 3.270083e+03 FES = 731100 Diversity = 2.1294e-03 k_t
= [1, 1, 1, 1]
Generation 7320: Best = 3.270082e+03 FES = 732100 Diversity = 1.9481e-03 k_t
= [3, 4, 3, 3]
Generation 7330: Best = 3.270081e+03 FES = 733100 Diversity = 2.5718e-03 k_t
= [1, 1, 1, 1]
Generation 7340: Best = 3.270079e+03 FES = 734100 Diversity = 2.9760e-03 k_t
= [3, 4, 2, 2]
Generation 7350: Best = 3.270077e+03 FES = 735100 Diversity = 2.9117e-03 k_t
= [1, 1, 1, 1]
Generation 7360: Best = 3.270074e+03 FES = 736100 Diversity = 3.7300e-03 k_t
= [4, 3, 3, 3]
Generation 7370: Best = 3.270071e+03 FES = 737100 Diversity = 4.0704e-03 k_t
= [1, 1, 1, 1]
Generation 7380: Best = 3.270066e+03 FES = 738100 Diversity = 4.5727e-03 k_t
= [3, 3, 2, 4]
Generation 7390: Best = 3.270059e+03 FES = 739100 Diversity = 5.6076e-03 k_t
= [1, 1, 1, 1]
Generation 7400: Best = 3.270051e+03 FES = 740100 Diversity = 6.6319e-03 k_t
= [4, 4, 4, 3]
Generation 7410: Best = 3.270043e+03 FES = 741100 Diversity = 6.5390e-03 k_t
= [1, 1, 1, 1]
Generation 7420: Best = 3.270034e+03 FES = 742100 Diversity = 7.0602e-03 k_t
= [2, 4, 2, 3]
Generation 7430: Best = 3.270026e+03 FES = 743100 Diversity = 6.1840e-03 k_t
= [1, 1, 1, 1]
Generation 7440: Best = 3.270022e+03 FES = 744100 Diversity = 5.3070e-03 k_t
= [2, 3, 4, 4]
Generation 7450: Best = 3.270018e+03 FES = 745100 Diversity = 4.9682e-03 k_t
= [1, 1, 1, 1]
Generation 7460: Best = 3.270013e+03 FES = 746100 Diversity = 5.2587e-03 k_t
= [3, 2, 3, 4]
Generation 7470: Best = 3.270010e+03 FES = 747100 Diversity = 4.1151e-03 k_t
= [1, 1, 1, 1]
Generation 7480: Best = 3.270007e+03 FES = 748100 Diversity = 4.0223e-03 k_t
= [3, 2, 3, 3]
Generation 7490: Best = 3.270004e+03 FES = 749100 Diversity = 3.6707e-03 k_t
= [1, 1, 1, 1]
Generation 7500: Best = 3.270002e+03 FES = 750100 Diversity = 3.5677e-03 k_t
= [4, 3, 3, 2]
Generation 7510: Best = 3.269999e+03 FES = 751100 Diversity = 3.8879e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 7520: Best = 3.269998e+03 FES = 752100 Diversity = 3.2281e-03 k_t
= [2, 4, 4, 3]
Generation 7530: Best = 3.269996e+03 FES = 753100 Diversity = 3.2208e-03 k_t
= [1, 1, 1, 1]
Generation 7540: Best = 3.269994e+03 FES = 754100 Diversity = 3.5050e-03 k_t
= [2, 2, 4, 3]
Generation 7550: Best = 3.269991e+03 FES = 755100 Diversity = 3.8365e-03 k_t
= [1, 1, 1, 1]
Generation 7560: Best = 3.269989e+03 FES = 756100 Diversity = 3.3905e-03 k_t
= [2, 3, 3, 3]
Generation 7570: Best = 3.269988e+03 FES = 757100 Diversity = 2.8845e-03 k_t
= [1, 1, 1, 1]
Generation 7580: Best = 3.269987e+03 FES = 758100 Diversity = 2.8204e-03 k_t
= [2, 2, 3, 3]
Generation 7590: Best = 3.269986e+03 FES = 759100 Diversity = 2.3124e-03 k_t
= [1, 1, 1, 1]
Generation 7600: Best = 3.269985e+03 FES = 760100 Diversity = 2.0829e-03 k_t
= [2, 4, 2, 2]
Generation 7610: Best = 3.269984e+03 FES = 761100 Diversity = 2.0809e-03 k_t
= [1, 1, 1, 1]
Generation 7620: Best = 3.269983e+03 FES = 762100 Diversity = 2.2563e-03 k_t
= [4, 3, 4, 3]
Generation 7630: Best = 3.269982e+03 FES = 763100 Diversity = 2.1745e-03 k_t
= [1, 1, 1, 1]
Generation 7640: Best = 3.269981e+03 FES = 764100 Diversity = 2.0258e-03 k_t
= [4, 2, 4, 3]
Generation 7650: Best = 3.269980e+03 FES = 765100 Diversity = 2.0731e-03 k_t
= [1, 1, 1, 1]
Generation 7660: Best = 3.269980e+03 FES = 766100 Diversity = 2.0492e-03 k_t
= [2, 3, 2, 4]
Generation 7670: Best = 3.269979e+03 FES = 767100 Diversity = 1.6267e-03 k_t
= [1, 1, 1, 1]
Generation 7680: Best = 3.269979e+03 FES = 768100 Diversity = 1.4313e-03 k_t
= [4, 4, 3, 4]
Generation 7690: Best = 3.269978e+03 FES = 769100 Diversity = 1.5809e-03 k_t
= [1, 1, 1, 1]
Generation 7700: Best = 3.269978e+03 FES = 770100 Diversity = 1.5737e-03 k_t
= [3, 2, 2, 4]
Generation 7710: Best = 3.269977e+03 FES = 771100 Diversity = 1.9109e-03 k_t
= [1, 1, 1, 1]
Generation 7720: Best = 3.269977e+03 FES = 772100 Diversity = 1.8685e-03 k_t
= [2, 4, 4, 2]
Generation 7730: Best = 3.269976e+03 FES = 773100 Diversity = 1.7656e-03 k_t
= [1, 1, 1, 1]
Generation 7740: Best = 3.269975e+03 FES = 774100 Diversity = 1.6593e-03 k_t
= [4, 3, 4, 2]
Generation 7750: Best = 3.269975e+03 FES = 775100 Diversity = 1.8452e-03 k_t
= [1, 1, 1, 1]
Generation 7760: Best = 3.269974e+03 FES = 776100 Diversity = 1.8151e-03 k_t
= [2, 2, 2, 3]
Generation 7770: Best = 3.269974e+03 FES = 777100 Diversity = 1.8102e-03 k_t
= [1, 1, 1, 1]
Generation 7780: Best = 3.269973e+03 FES = 778100 Diversity = 2.1823e-03 k_t

```

```

= [3, 2, 4, 3]
Generation 7790: Best = 3.269972e+03 FES = 779100 Diversity = 2.1025e-03 k_t
= [1, 1, 1, 1]
Generation 7800: Best = 3.269971e+03 FES = 780100 Diversity = 2.5708e-03 k_t
= [3, 4, 4, 2]
Generation 7810: Best = 3.269970e+03 FES = 781100 Diversity = 2.4235e-03 k_t
= [1, 1, 1, 1]
Generation 7820: Best = 3.269968e+03 FES = 782100 Diversity = 2.5850e-03 k_t
= [2, 4, 2, 3]
Generation 7830: Best = 3.269967e+03 FES = 783100 Diversity = 2.8530e-03 k_t
= [1, 1, 1, 1]
Generation 7840: Best = 3.269966e+03 FES = 784100 Diversity = 2.7064e-03 k_t
= [4, 4, 4, 3]
Generation 7850: Best = 3.269964e+03 FES = 785100 Diversity = 2.8561e-03 k_t
= [1, 1, 1, 1]
Generation 7860: Best = 3.269963e+03 FES = 786100 Diversity = 2.9499e-03 k_t
= [4, 2, 3, 2]
Generation 7870: Best = 3.269961e+03 FES = 787100 Diversity = 2.8855e-03 k_t
= [1, 1, 1, 1]
Generation 7880: Best = 3.269958e+03 FES = 788100 Diversity = 4.1306e-03 k_t
= [2, 4, 2, 4]
Generation 7890: Best = 3.269954e+03 FES = 789100 Diversity = 4.9835e-03 k_t
= [1, 1, 1, 1]
Generation 7900: Best = 3.269950e+03 FES = 790100 Diversity = 4.7198e-03 k_t
= [3, 4, 2, 4]
Generation 7910: Best = 3.269946e+03 FES = 791100 Diversity = 5.0757e-03 k_t
= [1, 1, 1, 1]
Generation 7920: Best = 3.269942e+03 FES = 792100 Diversity = 4.9501e-03 k_t
= [3, 2, 3, 3]
Generation 7930: Best = 3.269939e+03 FES = 793100 Diversity = 4.8644e-03 k_t
= [1, 1, 1, 1]
Generation 7940: Best = 3.269936e+03 FES = 794100 Diversity = 4.3197e-03 k_t
= [2, 4, 4, 4]
Generation 7950: Best = 3.269932e+03 FES = 795100 Diversity = 3.6185e-03 k_t
= [1, 1, 1, 1]
Generation 7960: Best = 3.269930e+03 FES = 796100 Diversity = 3.6949e-03 k_t
= [3, 3, 3, 3]
Generation 7970: Best = 3.269928e+03 FES = 797100 Diversity = 3.6479e-03 k_t
= [1, 1, 1, 1]
Generation 7980: Best = 3.269926e+03 FES = 798100 Diversity = 3.2077e-03 k_t
= [4, 4, 3, 4]
Generation 7990: Best = 3.269925e+03 FES = 799100 Diversity = 2.9379e-03 k_t
= [1, 1, 1, 1]
Generation 8000: Best = 3.269924e+03 FES = 800100 Diversity = 2.3077e-03 k_t
= [4, 4, 2, 2]
Generation 8010: Best = 3.269923e+03 FES = 801100 Diversity = 2.0060e-03 k_t
= [1, 1, 1, 1]
Generation 8020: Best = 3.269922e+03 FES = 802100 Diversity = 1.7246e-03 k_t
= [4, 3, 2, 4]
Generation 8030: Best = 3.269921e+03 FES = 803100 Diversity = 2.1896e-03 k_t
= [1, 1, 1, 1]
Generation 8040: Best = 3.269920e+03 FES = 804100 Diversity = 2.3136e-03 k_t
= [2, 3, 2, 4]
Generation 8050: Best = 3.269919e+03 FES = 805100 Diversity = 2.5124e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 8060: Best = 3.269918e+03 FES = 806100 Diversity = 2.5259e-03 k_t
= [2, 3, 3, 4]
Generation 8070: Best = 3.269917e+03 FES = 807100 Diversity = 2.5785e-03 k_t
= [1, 1, 1, 1]
Generation 8080: Best = 3.269916e+03 FES = 808100 Diversity = 2.1924e-03 k_t
= [3, 2, 4, 4]
Generation 8090: Best = 3.269916e+03 FES = 809100 Diversity = 1.7936e-03 k_t
= [1, 1, 1, 1]
Generation 8100: Best = 3.269915e+03 FES = 810100 Diversity = 1.3953e-03 k_t
= [2, 2, 4, 2]
Generation 8110: Best = 3.269915e+03 FES = 811100 Diversity = 1.4580e-03 k_t
= [1, 1, 1, 1]
Generation 8120: Best = 3.269915e+03 FES = 812100 Diversity = 1.5372e-03 k_t
= [2, 4, 4, 3]
Generation 8130: Best = 3.269914e+03 FES = 813100 Diversity = 1.5510e-03 k_t
= [1, 1, 1, 1]
Generation 8140: Best = 3.269914e+03 FES = 814100 Diversity = 1.5598e-03 k_t
= [4, 4, 4, 3]
Generation 8150: Best = 3.269913e+03 FES = 815100 Diversity = 1.5168e-03 k_t
= [1, 1, 1, 1]
Generation 8160: Best = 3.269913e+03 FES = 816100 Diversity = 1.4941e-03 k_t
= [4, 2, 4, 3]
Generation 8170: Best = 3.269913e+03 FES = 817100 Diversity = 1.3209e-03 k_t
= [1, 1, 1, 1]
Generation 8180: Best = 3.269912e+03 FES = 818100 Diversity = 1.2204e-03 k_t
= [4, 4, 4, 4]
Generation 8190: Best = 3.269912e+03 FES = 819100 Diversity = 1.3937e-03 k_t
= [1, 1, 1, 1]
Generation 8200: Best = 3.269912e+03 FES = 820100 Diversity = 1.1720e-03 k_t
= [4, 3, 4, 3]
Generation 8210: Best = 3.269912e+03 FES = 821100 Diversity = 1.1910e-03 k_t
= [1, 1, 1, 1]
Generation 8220: Best = 3.269911e+03 FES = 822100 Diversity = 1.1593e-03 k_t
= [2, 2, 3, 2]
Generation 8230: Best = 3.269911e+03 FES = 823100 Diversity = 1.2243e-03 k_t
= [1, 1, 1, 1]
Generation 8240: Best = 3.269911e+03 FES = 824100 Diversity = 1.2461e-03 k_t
= [4, 2, 2, 2]
Generation 8250: Best = 3.269911e+03 FES = 825100 Diversity = 1.1488e-03 k_t
= [1, 1, 1, 1]
Generation 8260: Best = 3.269910e+03 FES = 826100 Diversity = 9.8647e-04 k_t
= [2, 3, 4, 2]
Generation 8270: Best = 3.269910e+03 FES = 827100 Diversity = 7.8382e-04 k_t
= [1, 1, 1, 1]
Generation 8280: Best = 3.269910e+03 FES = 828100 Diversity = 7.8644e-04 k_t
= [4, 4, 4, 3]
Generation 8290: Best = 3.269910e+03 FES = 829100 Diversity = 8.0016e-04 k_t
= [1, 1, 1, 1]
Generation 8300: Best = 3.269910e+03 FES = 830100 Diversity = 8.3595e-04 k_t
= [3, 4, 4, 3]
Generation 8310: Best = 3.269910e+03 FES = 831100 Diversity = 7.7630e-04 k_t
= [1, 1, 1, 1]
Generation 8320: Best = 3.269910e+03 FES = 832100 Diversity = 8.5584e-04 k_t

```



```

= [4, 2, 2, 4]
Generation 8330: Best = 3.269909e+03 FES = 833100 Diversity = 8.6357e-04 k_t
= [1, 1, 1, 1]
Generation 8340: Best = 3.269909e+03 FES = 834100 Diversity = 7.7152e-04 k_t
= [4, 3, 2, 3]
Generation 8350: Best = 3.269909e+03 FES = 835100 Diversity = 7.9809e-04 k_t
= [1, 1, 1, 1]
Generation 8360: Best = 3.269909e+03 FES = 836100 Diversity = 6.7813e-04 k_t
= [4, 3, 2, 4]
Generation 8370: Best = 3.269909e+03 FES = 837100 Diversity = 5.4327e-04 k_t
= [1, 1, 1, 1]
Generation 8380: Best = 3.269909e+03 FES = 838100 Diversity = 6.2163e-04 k_t
= [3, 4, 4, 4]
Generation 8390: Best = 3.269909e+03 FES = 839100 Diversity = 5.0198e-04 k_t
= [1, 1, 1, 1]
Generation 8400: Best = 3.269909e+03 FES = 840100 Diversity = 5.2965e-04 k_t
= [2, 3, 2, 4]
Generation 8410: Best = 3.269909e+03 FES = 841100 Diversity = 5.4013e-04 k_t
= [1, 1, 1, 1]
Generation 8420: Best = 3.269909e+03 FES = 842100 Diversity = 4.9017e-04 k_t
= [4, 2, 4, 2]
Generation 8430: Best = 3.269909e+03 FES = 843100 Diversity = 5.6436e-04 k_t
= [1, 1, 1, 1]
Generation 8440: Best = 3.269909e+03 FES = 844100 Diversity = 5.8202e-04 k_t
= [4, 3, 4, 2]
Generation 8450: Best = 3.269908e+03 FES = 845100 Diversity = 5.6703e-04 k_t
= [1, 1, 1, 1]
Generation 8460: Best = 3.269908e+03 FES = 846100 Diversity = 5.4701e-04 k_t
= [3, 3, 2, 4]
Generation 8470: Best = 3.269908e+03 FES = 847100 Diversity = 4.8338e-04 k_t
= [1, 1, 1, 1]
Generation 8480: Best = 3.269908e+03 FES = 848100 Diversity = 4.7007e-04 k_t
= [2, 2, 2, 3]
Generation 8490: Best = 3.269908e+03 FES = 849100 Diversity = 4.0538e-04 k_t
= [1, 1, 1, 1]
Generation 8500: Best = 3.269908e+03 FES = 850100 Diversity = 4.5344e-04 k_t
= [4, 2, 4, 4]
Generation 8510: Best = 3.269908e+03 FES = 851100 Diversity = 4.0915e-04 k_t
= [1, 1, 1, 1]
Generation 8520: Best = 3.269908e+03 FES = 852100 Diversity = 4.9549e-04 k_t
= [2, 3, 2, 2]
Generation 8530: Best = 3.269908e+03 FES = 853100 Diversity = 5.0240e-04 k_t
= [1, 1, 1, 1]
Generation 8540: Best = 3.269908e+03 FES = 854100 Diversity = 4.8389e-04 k_t
= [4, 3, 3, 3]
Generation 8550: Best = 3.269908e+03 FES = 855100 Diversity = 4.1779e-04 k_t
= [1, 1, 1, 1]
Generation 8560: Best = 3.269908e+03 FES = 856100 Diversity = 5.2467e-04 k_t
= [2, 2, 4, 3]
Generation 8570: Best = 3.269908e+03 FES = 857100 Diversity = 4.9748e-04 k_t
= [1, 1, 1, 1]
Generation 8580: Best = 3.269908e+03 FES = 858100 Diversity = 6.7104e-04 k_t
= [4, 2, 2, 4]
Generation 8590: Best = 3.269908e+03 FES = 859100 Diversity = 9.1055e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 8600: Best = 3.269908e+03 FES = 860100 Diversity = 8.8568e-04 k_t
= [4, 4, 3, 4]
Generation 8610: Best = 3.269907e+03 FES = 861100 Diversity = 8.4938e-04 k_t
= [1, 1, 1, 1]
Generation 8620: Best = 3.269907e+03 FES = 862100 Diversity = 9.5109e-04 k_t
= [2, 3, 3, 2]
Generation 8630: Best = 3.269907e+03 FES = 863100 Diversity = 7.6612e-04 k_t
= [1, 1, 1, 1]
Generation 8640: Best = 3.269907e+03 FES = 864100 Diversity = 5.5550e-04 k_t
= [3, 4, 2, 2]
Generation 8650: Best = 3.269907e+03 FES = 865100 Diversity = 4.8612e-04 k_t
= [1, 1, 1, 1]
Generation 8660: Best = 3.269907e+03 FES = 866100 Diversity = 6.0269e-04 k_t
= [4, 2, 3, 2]
Generation 8670: Best = 3.269907e+03 FES = 867100 Diversity = 7.4223e-04 k_t
= [1, 1, 1, 1]
Generation 8680: Best = 3.269907e+03 FES = 868100 Diversity = 8.9190e-04 k_t
= [4, 4, 2, 4]
Generation 8690: Best = 3.269907e+03 FES = 869100 Diversity = 8.4923e-04 k_t
= [1, 1, 1, 1]
Generation 8700: Best = 3.269906e+03 FES = 870100 Diversity = 7.6711e-04 k_t
= [3, 3, 4, 2]
Generation 8710: Best = 3.269906e+03 FES = 871100 Diversity = 8.0699e-04 k_t
= [1, 1, 1, 1]
Generation 8720: Best = 3.269906e+03 FES = 872100 Diversity = 7.2872e-04 k_t
= [3, 3, 4, 4]
Generation 8730: Best = 3.269906e+03 FES = 873100 Diversity = 9.2703e-04 k_t
= [1, 1, 1, 1]
Generation 8740: Best = 3.269906e+03 FES = 874100 Diversity = 1.0050e-03 k_t
= [2, 4, 3, 3]
Generation 8750: Best = 3.269906e+03 FES = 875100 Diversity = 9.9341e-04 k_t
= [1, 1, 1, 1]
Generation 8760: Best = 3.269906e+03 FES = 876100 Diversity = 1.0471e-03 k_t
= [2, 3, 3, 2]
Generation 8770: Best = 3.269905e+03 FES = 877100 Diversity = 1.1659e-03 k_t
= [1, 1, 1, 1]
Generation 8780: Best = 3.269905e+03 FES = 878100 Diversity = 1.0862e-03 k_t
= [3, 4, 3, 2]
Generation 8790: Best = 3.269905e+03 FES = 879100 Diversity = 1.3524e-03 k_t
= [1, 1, 1, 1]
Generation 8800: Best = 3.269904e+03 FES = 880100 Diversity = 1.3964e-03 k_t
= [4, 4, 3, 4]
Generation 8810: Best = 3.269904e+03 FES = 881100 Diversity = 1.2066e-03 k_t
= [1, 1, 1, 1]
Generation 8820: Best = 3.269904e+03 FES = 882100 Diversity = 1.1386e-03 k_t
= [3, 3, 4, 4]
Generation 8830: Best = 3.269904e+03 FES = 883100 Diversity = 1.1075e-03 k_t
= [1, 1, 1, 1]
Generation 8840: Best = 3.269903e+03 FES = 884100 Diversity = 1.0606e-03 k_t
= [4, 4, 2, 3]
Generation 8850: Best = 3.269903e+03 FES = 885100 Diversity = 9.8182e-04 k_t
= [1, 1, 1, 1]
Generation 8860: Best = 3.269903e+03 FES = 886100 Diversity = 7.8049e-04 k_t

```

```

= [4, 3, 3, 2]
Generation 8870: Best = 3.269903e+03 FES = 887100 Diversity = 8.8169e-04 k_t
= [1, 1, 1, 1]
Generation 8880: Best = 3.269903e+03 FES = 888100 Diversity = 8.4829e-04 k_t
= [3, 2, 3, 2]
Generation 8890: Best = 3.269903e+03 FES = 889100 Diversity = 9.0378e-04 k_t
= [1, 1, 1, 1]
Generation 8900: Best = 3.269902e+03 FES = 890100 Diversity = 7.3266e-04 k_t
= [3, 4, 4, 3]
Generation 8910: Best = 3.269902e+03 FES = 891100 Diversity = 7.9187e-04 k_t
= [1, 1, 1, 1]
Generation 8920: Best = 3.269902e+03 FES = 892100 Diversity = 7.6016e-04 k_t
= [3, 3, 2, 2]
Generation 8930: Best = 3.269902e+03 FES = 893100 Diversity = 7.2551e-04 k_t
= [1, 1, 1, 1]
Generation 8940: Best = 3.269902e+03 FES = 894100 Diversity = 7.6723e-04 k_t
= [2, 2, 2, 4]
Generation 8950: Best = 3.269902e+03 FES = 895100 Diversity = 7.1196e-04 k_t
= [1, 1, 1, 1]
Generation 8960: Best = 3.269902e+03 FES = 896100 Diversity = 7.2124e-04 k_t
= [3, 3, 2, 2]
Generation 8970: Best = 3.269902e+03 FES = 897100 Diversity = 7.3062e-04 k_t
= [1, 1, 1, 1]
Generation 8980: Best = 3.269902e+03 FES = 898100 Diversity = 8.2940e-04 k_t
= [4, 3, 2, 2]
Generation 8990: Best = 3.269902e+03 FES = 899100 Diversity = 7.9910e-04 k_t
= [1, 1, 1, 1]
Generation 9000: Best = 3.269901e+03 FES = 900100 Diversity = 7.9532e-04 k_t
= [3, 2, 4, 2]
Generation 9010: Best = 3.269901e+03 FES = 901100 Diversity = 8.9285e-04 k_t
= [1, 1, 1, 1]
Generation 9020: Best = 3.269901e+03 FES = 902100 Diversity = 9.9841e-04 k_t
= [4, 3, 3, 3]
Generation 9030: Best = 3.269901e+03 FES = 903100 Diversity = 1.2814e-03 k_t
= [1, 1, 1, 1]
Generation 9040: Best = 3.269901e+03 FES = 904100 Diversity = 1.1534e-03 k_t
= [4, 2, 2, 2]
Generation 9050: Best = 3.269900e+03 FES = 905100 Diversity = 1.2383e-03 k_t
= [1, 1, 1, 1]
Generation 9060: Best = 3.269900e+03 FES = 906100 Diversity = 1.1200e-03 k_t
= [3, 3, 4, 4]
Generation 9070: Best = 3.269900e+03 FES = 907100 Diversity = 9.2742e-04 k_t
= [1, 1, 1, 1]
Generation 9080: Best = 3.269900e+03 FES = 908100 Diversity = 8.1304e-04 k_t
= [3, 3, 4, 3]
Generation 9090: Best = 3.269900e+03 FES = 909100 Diversity = 1.0272e-03 k_t
= [1, 1, 1, 1]
Generation 9100: Best = 3.269900e+03 FES = 910100 Diversity = 9.1328e-04 k_t
= [4, 2, 3, 4]
Generation 9110: Best = 3.269900e+03 FES = 911100 Diversity = 8.5305e-04 k_t
= [1, 1, 1, 1]
Generation 9120: Best = 3.269899e+03 FES = 912100 Diversity = 8.2417e-04 k_t
= [3, 3, 2, 2]
Generation 9130: Best = 3.269899e+03 FES = 913100 Diversity = 8.2726e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 9140: Best = 3.269899e+03 FES = 914100 Diversity = 8.4948e-04 k_t
= [2, 2, 4, 4]
Generation 9150: Best = 3.269899e+03 FES = 915100 Diversity = 8.1602e-04 k_t
= [1, 1, 1, 1]
Generation 9160: Best = 3.269899e+03 FES = 916100 Diversity = 1.1385e-03 k_t
= [2, 3, 2, 2]
Generation 9170: Best = 3.269898e+03 FES = 917100 Diversity = 1.3646e-03 k_t
= [1, 1, 1, 1]
Generation 9180: Best = 3.269898e+03 FES = 918100 Diversity = 1.5223e-03 k_t
= [2, 2, 4, 4]
Generation 9190: Best = 3.269898e+03 FES = 919100 Diversity = 1.5139e-03 k_t
= [1, 1, 1, 1]
Generation 9200: Best = 3.269897e+03 FES = 920100 Diversity = 1.9787e-03 k_t
= [2, 2, 2, 3]
Generation 9210: Best = 3.269895e+03 FES = 921100 Diversity = 2.8368e-03 k_t
= [1, 1, 1, 1]
Generation 9220: Best = 3.269894e+03 FES = 922100 Diversity = 2.5588e-03 k_t
= [2, 4, 2, 3]
Generation 9230: Best = 3.269893e+03 FES = 923100 Diversity = 2.6831e-03 k_t
= [1, 1, 1, 1]
Generation 9240: Best = 3.269892e+03 FES = 924100 Diversity = 2.0844e-03 k_t
= [3, 4, 3, 2]
Generation 9250: Best = 3.269892e+03 FES = 925100 Diversity = 1.8747e-03 k_t
= [1, 1, 1, 1]
Generation 9260: Best = 3.269891e+03 FES = 926100 Diversity = 1.7951e-03 k_t
= [2, 2, 2, 4]
Generation 9270: Best = 3.269891e+03 FES = 927100 Diversity = 1.6251e-03 k_t
= [1, 1, 1, 1]
Generation 9280: Best = 3.269890e+03 FES = 928100 Diversity = 1.5600e-03 k_t
= [2, 3, 4, 4]
Generation 9290: Best = 3.269890e+03 FES = 929100 Diversity = 1.5959e-03 k_t
= [1, 1, 1, 1]
Generation 9300: Best = 3.269889e+03 FES = 930100 Diversity = 1.8050e-03 k_t
= [4, 2, 2, 2]
Generation 9310: Best = 3.269888e+03 FES = 931100 Diversity = 2.1318e-03 k_t
= [1, 1, 1, 1]
Generation 9320: Best = 3.269888e+03 FES = 932100 Diversity = 1.8789e-03 k_t
= [2, 2, 4, 3]
Generation 9330: Best = 3.269887e+03 FES = 933100 Diversity = 2.0502e-03 k_t
= [1, 1, 1, 1]
Generation 9340: Best = 3.269886e+03 FES = 934100 Diversity = 1.7076e-03 k_t
= [3, 3, 2, 3]
Generation 9350: Best = 3.269886e+03 FES = 935100 Diversity = 1.7362e-03 k_t
= [1, 1, 1, 1]
Generation 9360: Best = 3.269885e+03 FES = 936100 Diversity = 1.7205e-03 k_t
= [2, 4, 2, 3]
Generation 9370: Best = 3.269885e+03 FES = 937100 Diversity = 1.4219e-03 k_t
= [1, 1, 1, 1]
Generation 9380: Best = 3.269885e+03 FES = 938100 Diversity = 1.2821e-03 k_t
= [3, 3, 2, 2]
Generation 9390: Best = 3.269885e+03 FES = 939100 Diversity = 9.9206e-04 k_t
= [1, 1, 1, 1]
Generation 9400: Best = 3.269885e+03 FES = 940100 Diversity = 8.3478e-04 k_t

```

```

= [3, 4, 2, 2]
Generation 9410: Best = 3.269885e+03 FES = 941100 Diversity = 5.7789e-04 k_t
= [1, 1, 1, 1]
Generation 9420: Best = 3.269885e+03 FES = 942100 Diversity = 5.9168e-04 k_t
= [3, 3, 4, 2]
Generation 9430: Best = 3.269884e+03 FES = 943100 Diversity = 4.3962e-04 k_t
= [1, 1, 1, 1]
Generation 9440: Best = 3.269884e+03 FES = 944100 Diversity = 3.9442e-04 k_t
= [3, 3, 2, 2]
Generation 9450: Best = 3.269884e+03 FES = 945100 Diversity = 3.5036e-04 k_t
= [1, 1, 1, 1]
Generation 9460: Best = 3.269884e+03 FES = 946100 Diversity = 3.0847e-04 k_t
= [4, 3, 2, 4]
Generation 9470: Best = 3.269884e+03 FES = 947100 Diversity = 3.8488e-04 k_t
= [1, 1, 1, 1]
Generation 9480: Best = 3.269884e+03 FES = 948100 Diversity = 3.8025e-04 k_t
= [4, 2, 2, 4]
Generation 9490: Best = 3.269884e+03 FES = 949100 Diversity = 3.6283e-04 k_t
= [1, 1, 1, 1]
Generation 9500: Best = 3.269884e+03 FES = 950100 Diversity = 3.0145e-04 k_t
= [3, 4, 2, 3]
Generation 9510: Best = 3.269884e+03 FES = 951100 Diversity = 3.1938e-04 k_t
= [1, 1, 1, 1]
Generation 9520: Best = 3.269884e+03 FES = 952100 Diversity = 2.6808e-04 k_t
= [3, 2, 3, 2]
Generation 9530: Best = 3.269884e+03 FES = 953100 Diversity = 2.3785e-04 k_t
= [1, 1, 1, 1]
Generation 9540: Best = 3.269884e+03 FES = 954100 Diversity = 2.2273e-04 k_t
= [4, 4, 4, 4]
Generation 9550: Best = 3.269884e+03 FES = 955100 Diversity = 1.8933e-04 k_t
= [1, 1, 1, 1]
Generation 9560: Best = 3.269884e+03 FES = 956100 Diversity = 2.2332e-04 k_t
= [2, 4, 4, 3]
Generation 9570: Best = 3.269884e+03 FES = 957100 Diversity = 2.2716e-04 k_t
= [1, 1, 1, 1]
Generation 9580: Best = 3.269884e+03 FES = 958100 Diversity = 2.2387e-04 k_t
= [3, 2, 3, 2]
Generation 9590: Best = 3.269884e+03 FES = 959100 Diversity = 2.1882e-04 k_t
= [1, 1, 1, 1]
Generation 9600: Best = 3.269884e+03 FES = 960100 Diversity = 2.5675e-04 k_t
= [2, 3, 4, 3]
Generation 9610: Best = 3.269884e+03 FES = 961100 Diversity = 2.6889e-04 k_t
= [1, 1, 1, 1]
Generation 9620: Best = 3.269884e+03 FES = 962100 Diversity = 2.4827e-04 k_t
= [3, 3, 2, 3]
Generation 9630: Best = 3.269884e+03 FES = 963100 Diversity = 2.0784e-04 k_t
= [1, 1, 1, 1]
Generation 9640: Best = 3.269884e+03 FES = 964100 Diversity = 2.5789e-04 k_t
= [2, 2, 2, 4]
Generation 9650: Best = 3.269884e+03 FES = 965100 Diversity = 2.5721e-04 k_t
= [1, 1, 1, 1]
Generation 9660: Best = 3.269884e+03 FES = 966100 Diversity = 2.3413e-04 k_t
= [4, 4, 2, 4]
Generation 9670: Best = 3.269884e+03 FES = 967100 Diversity = 2.1247e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 9680: Best = 3.269884e+03 FES = 968100 Diversity = 1.8909e-04 k_t
= [3, 2, 3, 2]
Generation 9690: Best = 3.269884e+03 FES = 969100 Diversity = 2.0145e-04 k_t
= [1, 1, 1, 1]
Generation 9700: Best = 3.269884e+03 FES = 970100 Diversity = 1.9454e-04 k_t
= [3, 2, 3, 3]
Generation 9710: Best = 3.269884e+03 FES = 971100 Diversity = 2.0537e-04 k_t
= [1, 1, 1, 1]
Generation 9720: Best = 3.269884e+03 FES = 972100 Diversity = 1.9163e-04 k_t
= [3, 2, 4, 2]
Generation 9730: Best = 3.269884e+03 FES = 973100 Diversity = 2.0401e-04 k_t
= [1, 1, 1, 1]
Generation 9740: Best = 3.269884e+03 FES = 974100 Diversity = 1.7628e-04 k_t
= [4, 4, 4, 4]
Generation 9750: Best = 3.269884e+03 FES = 975100 Diversity = 1.4296e-04 k_t
= [1, 1, 1, 1]
Generation 9760: Best = 3.269884e+03 FES = 976100 Diversity = 1.2871e-04 k_t
= [3, 2, 2, 3]
Generation 9770: Best = 3.269884e+03 FES = 977100 Diversity = 1.0093e-04 k_t
= [1, 1, 1, 1]
Generation 9780: Best = 3.269884e+03 FES = 978100 Diversity = 9.7764e-05 k_t
= [2, 4, 3, 3]
Generation 9790: Best = 3.269884e+03 FES = 979100 Diversity = 8.8675e-05 k_t
= [1, 1, 1, 1]
Generation 9800: Best = 3.269884e+03 FES = 980100 Diversity = 6.9340e-05 k_t
= [4, 2, 3, 2]
Generation 9810: Best = 3.269884e+03 FES = 981100 Diversity = 5.9211e-05 k_t
= [1, 1, 1, 1]
Generation 9820: Best = 3.269884e+03 FES = 982100 Diversity = 5.8941e-05 k_t
= [2, 3, 2, 3]
Generation 9830: Best = 3.269884e+03 FES = 983100 Diversity = 5.2203e-05 k_t
= [1, 1, 1, 1]
Generation 9840: Best = 3.269884e+03 FES = 984100 Diversity = 4.8836e-05 k_t
= [3, 3, 2, 4]
Generation 9850: Best = 3.269884e+03 FES = 985100 Diversity = 4.6209e-05 k_t
= [1, 1, 1, 1]
Generation 9860: Best = 3.269884e+03 FES = 986100 Diversity = 5.1533e-05 k_t
= [3, 3, 3, 3]
Generation 9870: Best = 3.269884e+03 FES = 987100 Diversity = 5.8373e-05 k_t
= [1, 1, 1, 1]
Generation 9880: Best = 3.269884e+03 FES = 988100 Diversity = 5.8887e-05 k_t
= [4, 4, 4, 2]
Generation 9890: Best = 3.269884e+03 FES = 989100 Diversity = 6.1666e-05 k_t
= [1, 1, 1, 1]
Generation 9900: Best = 3.269884e+03 FES = 990100 Diversity = 5.6476e-05 k_t
= [2, 2, 3, 3]
Generation 9910: Best = 3.269884e+03 FES = 991100 Diversity = 6.7956e-05 k_t
= [1, 1, 1, 1]
Generation 9920: Best = 3.269884e+03 FES = 992100 Diversity = 6.7919e-05 k_t
= [4, 2, 2, 2]
Generation 9930: Best = 3.269884e+03 FES = 993100 Diversity = 7.2401e-05 k_t
= [1, 1, 1, 1]
Generation 9940: Best = 3.269884e+03 FES = 994100 Diversity = 7.0474e-05 k_t

```

```

= [2, 4, 3, 3]
Generation 9950: Best = 3.269884e+03  FES = 995100  Diversity = 6.4507e-05  k_t
= [1, 1, 1, 1]
Generation 9960: Best = 3.269884e+03  FES = 996100  Diversity = 5.8162e-05  k_t
= [4, 2, 3, 2]
Generation 9970: Best = 3.269884e+03  FES = 997100  Diversity = 3.8722e-05  k_t
= [1, 1, 1, 1]
Generation 9980: Best = 3.269884e+03  FES = 998100  Diversity = 2.9868e-05  k_t
= [3, 2, 4, 4]
Generation 9990: Best = 3.269884e+03  FES = 999100  Diversity = 2.2939e-05  k_t
= [1, 1, 1, 1]

```

```

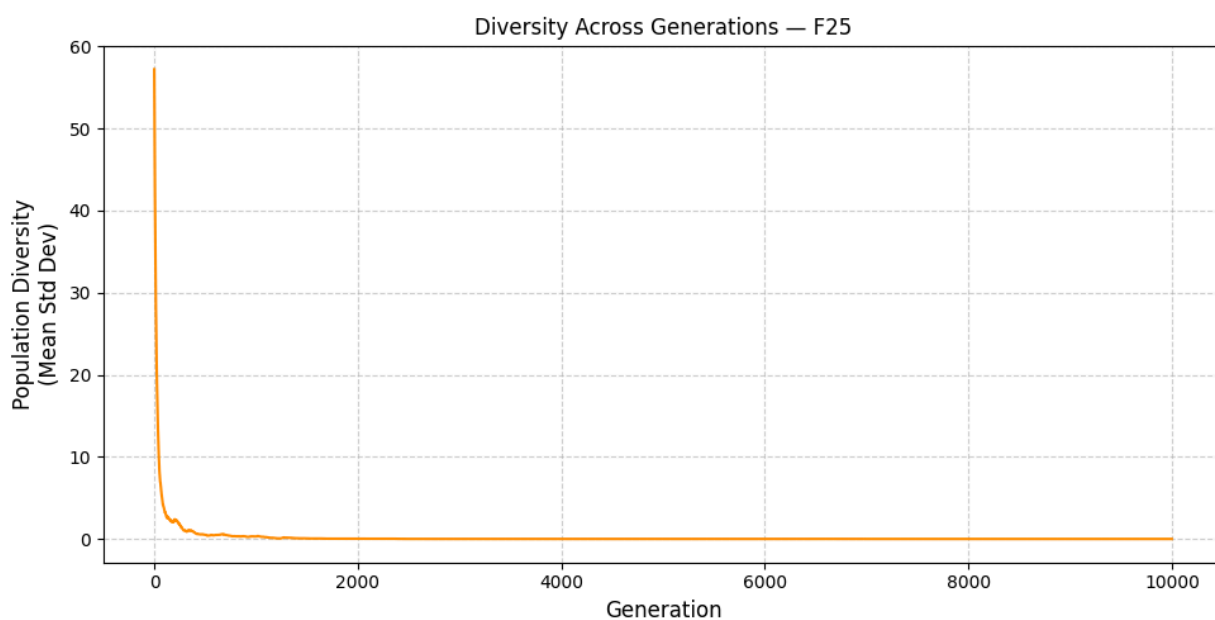
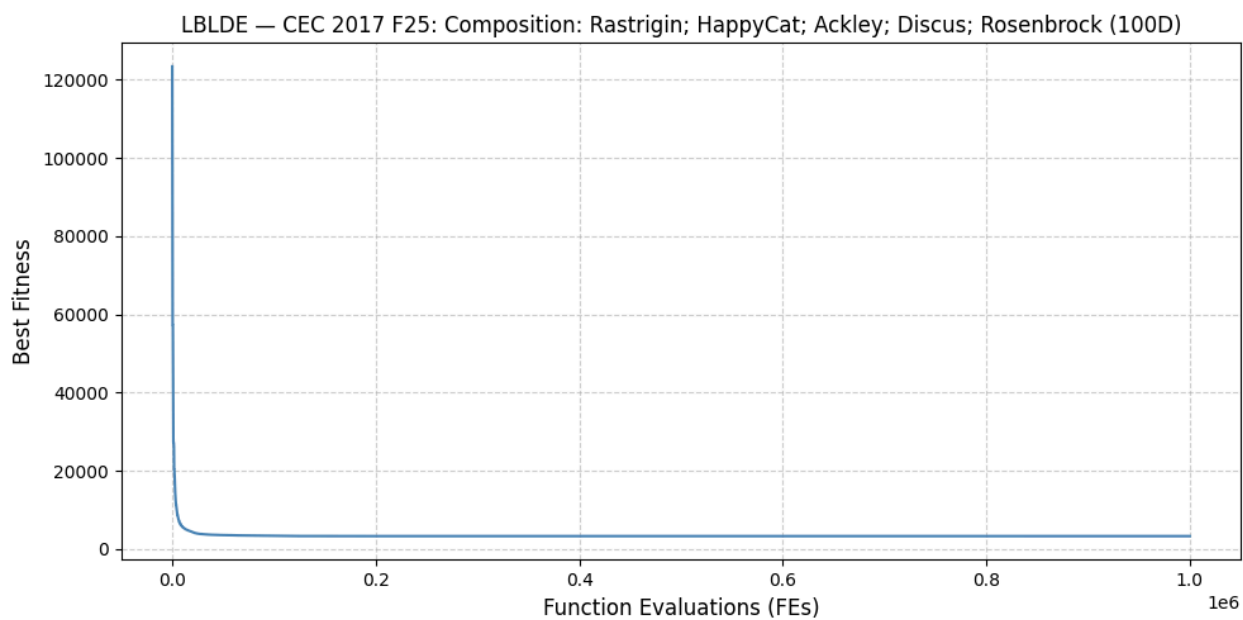
Optimization completed!
Final Best Fitness = 3.269884e+03
Total Function Evaluations = 1000000

```

```

=====
=
Results:
  Best Fitness           : 3.269884e+03
  Global Optimum         : 2.500000e+03
  Error (|best-optimum|) : 7.698841e+02
  Best Solution (5 dim)  : [-10.37478327  -9.14557131  34.08340495  10.5065264
-16.96108545]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```



Graph saved as convergence_F25.png

```
In [50]: # =====
# F26 — Composition Function 6 (Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin)
# =====
run_and_plot('F26', make_wrapper(composition.f26), 2600.0, 'Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrigin')
```



```

=====
=
LBLDE_s on CEC 2017 F26 – Single Run
=====
=
Function      : Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrig
in (100D)
Bounds       : [-100, 100]
Global optimum : 2600.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 1.023547e+05 | Diversity = 5.7220e+01
Generation 10: Best = 5.097418e+04 FES = 1100 Diversity = 4.5543e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 3.615338e+04 FES = 2100 Diversity = 3.5123e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 2.800213e+04 FES = 3100 Diversity = 2.8734e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 2.236358e+04 FES = 4100 Diversity = 2.2141e+01 k_t
= [4, 3, 4, 2]
Generation 50: Best = 1.907074e+04 FES = 5100 Diversity = 1.8899e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.860707e+04 FES = 6100 Diversity = 1.5336e+01 k_t
= [4, 3, 4, 3]
Generation 70: Best = 1.799454e+04 FES = 7100 Diversity = 1.3919e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.737641e+04 FES = 8100 Diversity = 1.2976e+01 k_t
= [4, 3, 2, 3]
Generation 90: Best = 1.722964e+04 FES = 9100 Diversity = 1.2530e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.679678e+04 FES = 10100 Diversity = 1.1957e+01 k_t
= [3, 3, 2, 3]
Generation 110: Best = 1.612167e+04 FES = 11100 Diversity = 1.1443e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.573092e+04 FES = 12100 Diversity = 1.1024e+01 k_t
= [4, 4, 4, 2]
Generation 130: Best = 1.561330e+04 FES = 13100 Diversity = 1.1042e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.512932e+04 FES = 14100 Diversity = 1.0776e+01 k_t
= [4, 2, 2, 4]
Generation 150: Best = 1.512932e+04 FES = 15100 Diversity = 1.0745e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.480203e+04 FES = 16100 Diversity = 1.0558e+01 k_t
= [3, 3, 2, 2]
Generation 170: Best = 1.473012e+04 FES = 17100 Diversity = 1.0313e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 1.473012e+04 FES = 18100 Diversity = 1.0164e+01 k_t
= [3, 2, 4, 3]
Generation 190: Best = 1.431697e+04 FES = 19100 Diversity = 1.0127e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.403528e+04 FES = 20100 Diversity = 9.9124e+00 k_t
= [3, 2, 3, 4]
Generation 210: Best = 1.389214e+04 FES = 21100 Diversity = 9.8418e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.386404e+04 FES = 22100 Diversity = 9.6773e+00 k_t

```

```

= [3, 3, 4, 3]
Generation 230: Best = 1.378525e+04 FES = 23100 Diversity = 9.5891e+00 k_t
= [1, 1, 1, 1]
Generation 240: Best = 1.378525e+04 FES = 24100 Diversity = 9.3896e+00 k_t
= [4, 2, 4, 4]
Generation 250: Best = 1.378525e+04 FES = 25100 Diversity = 9.3742e+00 k_t
= [1, 1, 1, 1]
Generation 260: Best = 1.375856e+04 FES = 26100 Diversity = 9.3337e+00 k_t
= [3, 2, 3, 3]
Generation 270: Best = 1.375856e+04 FES = 27100 Diversity = 9.2383e+00 k_t
= [1, 1, 1, 1]
Generation 280: Best = 1.375856e+04 FES = 28100 Diversity = 9.1989e+00 k_t
= [3, 4, 2, 2]
Generation 290: Best = 1.375856e+04 FES = 29100 Diversity = 9.2004e+00 k_t
= [1, 1, 1, 1]
Generation 300: Best = 1.375856e+04 FES = 30100 Diversity = 9.1202e+00 k_t
= [3, 3, 2, 4]
Generation 310: Best = 1.362399e+04 FES = 31100 Diversity = 9.0741e+00 k_t
= [1, 1, 1, 1]
Generation 320: Best = 1.362399e+04 FES = 32100 Diversity = 9.0816e+00 k_t
= [2, 2, 3, 2]
Generation 330: Best = 1.362399e+04 FES = 33100 Diversity = 9.1930e+00 k_t
= [1, 1, 1, 1]
Generation 340: Best = 1.362399e+04 FES = 34100 Diversity = 9.2265e+00 k_t
= [4, 3, 2, 2]
Generation 350: Best = 1.362399e+04 FES = 35100 Diversity = 9.2250e+00 k_t
= [1, 1, 1, 1]
Generation 360: Best = 1.362399e+04 FES = 36100 Diversity = 9.2187e+00 k_t
= [2, 4, 3, 2]
Generation 370: Best = 1.341649e+04 FES = 37100 Diversity = 9.1930e+00 k_t
= [1, 1, 1, 1]
Generation 380: Best = 1.322868e+04 FES = 38100 Diversity = 9.1493e+00 k_t
= [3, 4, 3, 2]
Generation 390: Best = 1.322868e+04 FES = 39100 Diversity = 9.0692e+00 k_t
= [1, 1, 1, 1]
Generation 400: Best = 1.322868e+04 FES = 40100 Diversity = 8.9896e+00 k_t
= [2, 4, 2, 4]
Generation 410: Best = 1.322868e+04 FES = 41100 Diversity = 8.9312e+00 k_t
= [1, 1, 1, 1]
Generation 420: Best = 1.293041e+04 FES = 42100 Diversity = 8.7804e+00 k_t
= [4, 3, 4, 3]
Generation 430: Best = 1.293041e+04 FES = 43100 Diversity = 8.7193e+00 k_t
= [1, 1, 1, 1]
Generation 440: Best = 1.293041e+04 FES = 44100 Diversity = 8.5946e+00 k_t
= [2, 3, 4, 2]
Generation 450: Best = 1.291026e+04 FES = 45100 Diversity = 8.5055e+00 k_t
= [1, 1, 1, 1]
Generation 460: Best = 1.291026e+04 FES = 46100 Diversity = 8.4039e+00 k_t
= [2, 2, 2, 2]
Generation 470: Best = 1.277029e+04 FES = 47100 Diversity = 8.3615e+00 k_t
= [1, 1, 1, 1]
Generation 480: Best = 1.277029e+04 FES = 48100 Diversity = 8.3510e+00 k_t
= [2, 2, 4, 4]
Generation 490: Best = 1.270144e+04 FES = 49100 Diversity = 8.3221e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 1.264817e+04 FES = 50100 Diversity = 8.2947e+00 k_t
= [2, 2, 2, 4]
Generation 510: Best = 1.246682e+04 FES = 51100 Diversity = 8.2538e+00 k_t
= [1, 1, 1, 1]
Generation 520: Best = 1.218522e+04 FES = 52100 Diversity = 8.1350e+00 k_t
= [2, 2, 3, 3]
Generation 530: Best = 1.203291e+04 FES = 53100 Diversity = 8.0661e+00 k_t
= [1, 1, 1, 1]
Generation 540: Best = 1.190906e+04 FES = 54100 Diversity = 8.0214e+00 k_t
= [3, 2, 3, 2]
Generation 550: Best = 1.181924e+04 FES = 55100 Diversity = 7.9582e+00 k_t
= [1, 1, 1, 1]
Generation 560: Best = 1.180651e+04 FES = 56100 Diversity = 7.9763e+00 k_t
= [4, 2, 2, 2]
Generation 570: Best = 1.136832e+04 FES = 57100 Diversity = 7.9652e+00 k_t
= [1, 1, 1, 1]
Generation 580: Best = 1.126790e+04 FES = 58100 Diversity = 7.8649e+00 k_t
= [3, 4, 2, 3]
Generation 590: Best = 1.126790e+04 FES = 59100 Diversity = 7.6508e+00 k_t
= [1, 1, 1, 1]
Generation 600: Best = 1.114034e+04 FES = 60100 Diversity = 7.6326e+00 k_t
= [4, 3, 3, 4]
Generation 610: Best = 1.082387e+04 FES = 61100 Diversity = 7.5645e+00 k_t
= [1, 1, 1, 1]
Generation 620: Best = 1.071035e+04 FES = 62100 Diversity = 7.3881e+00 k_t
= [2, 4, 4, 2]
Generation 630: Best = 1.071035e+04 FES = 63100 Diversity = 7.0760e+00 k_t
= [1, 1, 1, 1]
Generation 640: Best = 1.051374e+04 FES = 64100 Diversity = 6.8540e+00 k_t
= [2, 3, 2, 3]
Generation 650: Best = 1.037353e+04 FES = 65100 Diversity = 6.4665e+00 k_t
= [1, 1, 1, 1]
Generation 660: Best = 1.034682e+04 FES = 66100 Diversity = 5.5650e+00 k_t
= [3, 4, 3, 2]
Generation 670: Best = 1.034682e+04 FES = 67100 Diversity = 4.6161e+00 k_t
= [1, 1, 1, 1]
Generation 680: Best = 1.026231e+04 FES = 68100 Diversity = 3.3298e+00 k_t
= [4, 2, 2, 4]
Generation 690: Best = 1.005491e+04 FES = 69100 Diversity = 2.3162e+00 k_t
= [1, 1, 1, 1]
Generation 700: Best = 9.767272e+03 FES = 70100 Diversity = 1.6859e+00 k_t
= [4, 2, 2, 4]
Generation 710: Best = 9.578533e+03 FES = 71100 Diversity = 1.4263e+00 k_t
= [1, 1, 1, 1]
Generation 720: Best = 9.173024e+03 FES = 72100 Diversity = 1.1689e+00 k_t
= [4, 2, 4, 3]
Generation 730: Best = 8.853125e+03 FES = 73100 Diversity = 8.8740e-01 k_t
= [1, 1, 1, 1]
Generation 740: Best = 8.693608e+03 FES = 74100 Diversity = 6.7415e-01 k_t
= [3, 4, 2, 3]
Generation 750: Best = 8.518757e+03 FES = 75100 Diversity = 5.1539e-01 k_t
= [1, 1, 1, 1]
Generation 760: Best = 8.394467e+03 FES = 76100 Diversity = 4.0847e-01 k_t

```

```

= [3, 2, 3, 2]
Generation 770: Best = 8.316665e+03 FES = 77100 Diversity = 3.4184e-01 k_t
= [1, 1, 1, 1]
Generation 780: Best = 8.220454e+03 FES = 78100 Diversity = 2.9561e-01 k_t
= [3, 4, 4, 4]
Generation 790: Best = 8.169225e+03 FES = 79100 Diversity = 2.5907e-01 k_t
= [1, 1, 1, 1]
Generation 800: Best = 8.114778e+03 FES = 80100 Diversity = 2.5293e-01 k_t
= [2, 3, 4, 3]
Generation 810: Best = 8.081016e+03 FES = 81100 Diversity = 2.1858e-01 k_t
= [1, 1, 1, 1]
Generation 820: Best = 8.049459e+03 FES = 82100 Diversity = 1.9048e-01 k_t
= [4, 4, 2, 2]
Generation 830: Best = 8.025483e+03 FES = 83100 Diversity = 1.7311e-01 k_t
= [1, 1, 1, 1]
Generation 840: Best = 8.005772e+03 FES = 84100 Diversity = 1.6119e-01 k_t
= [4, 2, 2, 4]
Generation 850: Best = 7.989858e+03 FES = 85100 Diversity = 1.4452e-01 k_t
= [1, 1, 1, 1]
Generation 860: Best = 7.973748e+03 FES = 86100 Diversity = 1.2818e-01 k_t
= [3, 4, 3, 4]
Generation 870: Best = 7.958957e+03 FES = 87100 Diversity = 1.2021e-01 k_t
= [1, 1, 1, 1]
Generation 880: Best = 7.945175e+03 FES = 88100 Diversity = 1.1072e-01 k_t
= [2, 3, 2, 2]
Generation 890: Best = 7.937163e+03 FES = 89100 Diversity = 1.0222e-01 k_t
= [1, 1, 1, 1]
Generation 900: Best = 7.933096e+03 FES = 90100 Diversity = 8.7615e-02 k_t
= [3, 2, 2, 3]
Generation 910: Best = 7.929647e+03 FES = 91100 Diversity = 7.5214e-02 k_t
= [1, 1, 1, 1]
Generation 920: Best = 7.925725e+03 FES = 92100 Diversity = 6.3738e-02 k_t
= [2, 2, 4, 3]
Generation 930: Best = 7.923143e+03 FES = 93100 Diversity = 5.7605e-02 k_t
= [1, 1, 1, 1]
Generation 940: Best = 7.921083e+03 FES = 94100 Diversity = 5.0514e-02 k_t
= [3, 2, 4, 3]
Generation 950: Best = 7.920039e+03 FES = 95100 Diversity = 4.5212e-02 k_t
= [1, 1, 1, 1]
Generation 960: Best = 7.918835e+03 FES = 96100 Diversity = 4.1501e-02 k_t
= [4, 2, 3, 4]
Generation 970: Best = 7.917631e+03 FES = 97100 Diversity = 3.5688e-02 k_t
= [1, 1, 1, 1]
Generation 980: Best = 7.916867e+03 FES = 98100 Diversity = 3.1568e-02 k_t
= [3, 4, 2, 3]
Generation 990: Best = 7.916194e+03 FES = 99100 Diversity = 2.7366e-02 k_t
= [1, 1, 1, 1]
Generation 1000: Best = 7.915410e+03 FES = 100100 Diversity = 2.5607e-02 k_t
= [2, 2, 3, 4]
Generation 1010: Best = 7.914994e+03 FES = 101100 Diversity = 2.3289e-02 k_t
= [1, 1, 1, 1]
Generation 1020: Best = 7.914441e+03 FES = 102100 Diversity = 2.1599e-02 k_t
= [4, 2, 2, 4]
Generation 1030: Best = 7.914083e+03 FES = 103100 Diversity = 1.9773e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 1040: Best = 7.913841e+03 FES = 104100 Diversity = 1.7940e-02 k_t
= [3, 3, 3, 2]
Generation 1050: Best = 7.913516e+03 FES = 105100 Diversity = 1.6464e-02 k_t
= [1, 1, 1, 1]
Generation 1060: Best = 7.913339e+03 FES = 106100 Diversity = 1.4442e-02 k_t
= [4, 3, 2, 3]
Generation 1070: Best = 7.913217e+03 FES = 107100 Diversity = 1.2810e-02 k_t
= [1, 1, 1, 1]
Generation 1080: Best = 7.913092e+03 FES = 108100 Diversity = 1.1389e-02 k_t
= [4, 3, 3, 4]
Generation 1090: Best = 7.912983e+03 FES = 109100 Diversity = 9.7300e-03 k_t
= [1, 1, 1, 1]
Generation 1100: Best = 7.912903e+03 FES = 110100 Diversity = 8.6035e-03 k_t
= [3, 3, 4, 3]
Generation 1110: Best = 7.912851e+03 FES = 111100 Diversity = 7.6516e-03 k_t
= [1, 1, 1, 1]
Generation 1120: Best = 7.912781e+03 FES = 112100 Diversity = 6.8707e-03 k_t
= [4, 2, 4, 4]
Generation 1130: Best = 7.912756e+03 FES = 113100 Diversity = 6.1123e-03 k_t
= [1, 1, 1, 1]
Generation 1140: Best = 7.912735e+03 FES = 114100 Diversity = 5.3354e-03 k_t
= [2, 4, 2, 2]
Generation 1150: Best = 7.912716e+03 FES = 115100 Diversity = 4.6032e-03 k_t
= [1, 1, 1, 1]
Generation 1160: Best = 7.912704e+03 FES = 116100 Diversity = 3.8337e-03 k_t
= [4, 3, 3, 2]
Generation 1170: Best = 7.912692e+03 FES = 117100 Diversity = 3.2907e-03 k_t
= [1, 1, 1, 1]
Generation 1180: Best = 7.912680e+03 FES = 118100 Diversity = 2.9950e-03 k_t
= [3, 4, 2, 4]
Generation 1190: Best = 7.912670e+03 FES = 119100 Diversity = 2.7647e-03 k_t
= [1, 1, 1, 1]
Generation 1200: Best = 7.912663e+03 FES = 120100 Diversity = 2.5910e-03 k_t
= [4, 3, 4, 4]
Generation 1210: Best = 7.912655e+03 FES = 121100 Diversity = 2.5724e-03 k_t
= [1, 1, 1, 1]
Generation 1220: Best = 7.912650e+03 FES = 122100 Diversity = 2.4799e-03 k_t
= [3, 3, 3, 2]
Generation 1230: Best = 7.912646e+03 FES = 123100 Diversity = 2.2307e-03 k_t
= [1, 1, 1, 1]
Generation 1240: Best = 7.912641e+03 FES = 124100 Diversity = 2.0644e-03 k_t
= [4, 3, 2, 4]
Generation 1250: Best = 7.912639e+03 FES = 125100 Diversity = 1.8294e-03 k_t
= [1, 1, 1, 1]
Generation 1260: Best = 7.912637e+03 FES = 126100 Diversity = 1.6430e-03 k_t
= [4, 3, 4, 2]
Generation 1270: Best = 7.912634e+03 FES = 127100 Diversity = 1.5408e-03 k_t
= [1, 1, 1, 1]
Generation 1280: Best = 7.912633e+03 FES = 128100 Diversity = 1.4308e-03 k_t
= [2, 2, 4, 2]
Generation 1290: Best = 7.912632e+03 FES = 129100 Diversity = 1.1669e-03 k_t
= [1, 1, 1, 1]
Generation 1300: Best = 7.912631e+03 FES = 130100 Diversity = 1.0076e-03 k_t

```

```

= [2, 4, 3, 4]
Generation 1310: Best = 7.912630e+03 FES = 131100 Diversity = 9.0901e-04 k_t
= [1, 1, 1, 1]
Generation 1320: Best = 7.912628e+03 FES = 132100 Diversity = 8.2991e-04 k_t
= [3, 2, 2, 2]
Generation 1330: Best = 7.912628e+03 FES = 133100 Diversity = 7.7639e-04 k_t
= [1, 1, 1, 1]
Generation 1340: Best = 7.912627e+03 FES = 134100 Diversity = 6.6980e-04 k_t
= [2, 3, 2, 2]
Generation 1350: Best = 7.912627e+03 FES = 135100 Diversity = 6.5439e-04 k_t
= [1, 1, 1, 1]
Generation 1360: Best = 7.912626e+03 FES = 136100 Diversity = 6.0048e-04 k_t
= [4, 2, 3, 4]
Generation 1370: Best = 7.912626e+03 FES = 137100 Diversity = 5.6389e-04 k_t
= [1, 1, 1, 1]
Generation 1380: Best = 7.912626e+03 FES = 138100 Diversity = 5.1020e-04 k_t
= [3, 2, 3, 3]
Generation 1390: Best = 7.912626e+03 FES = 139100 Diversity = 4.7501e-04 k_t
= [1, 1, 1, 1]
Generation 1400: Best = 7.912625e+03 FES = 140100 Diversity = 4.3298e-04 k_t
= [2, 3, 2, 4]
Generation 1410: Best = 7.912625e+03 FES = 141100 Diversity = 4.0343e-04 k_t
= [1, 1, 1, 1]
Generation 1420: Best = 7.912625e+03 FES = 142100 Diversity = 3.5408e-04 k_t
= [2, 2, 3, 3]
Generation 1430: Best = 7.912625e+03 FES = 143100 Diversity = 3.4276e-04 k_t
= [1, 1, 1, 1]
Generation 1440: Best = 7.912625e+03 FES = 144100 Diversity = 3.1704e-04 k_t
= [3, 3, 2, 2]
Generation 1450: Best = 7.912625e+03 FES = 145100 Diversity = 2.7728e-04 k_t
= [1, 1, 1, 1]
Generation 1460: Best = 7.912625e+03 FES = 146100 Diversity = 2.5921e-04 k_t
= [4, 3, 4, 2]
Generation 1470: Best = 7.912625e+03 FES = 147100 Diversity = 2.2573e-04 k_t
= [1, 1, 1, 1]
Generation 1480: Best = 7.912625e+03 FES = 148100 Diversity = 2.1245e-04 k_t
= [4, 2, 4, 3]
Generation 1490: Best = 7.912625e+03 FES = 149100 Diversity = 2.1911e-04 k_t
= [1, 1, 1, 1]
Generation 1500: Best = 7.912624e+03 FES = 150100 Diversity = 1.9734e-04 k_t
= [2, 3, 3, 3]
Generation 1510: Best = 7.912624e+03 FES = 151100 Diversity = 1.9039e-04 k_t
= [1, 1, 1, 1]
Generation 1520: Best = 7.912624e+03 FES = 152100 Diversity = 1.6348e-04 k_t
= [3, 2, 4, 4]
Generation 1530: Best = 7.912624e+03 FES = 153100 Diversity = 1.5598e-04 k_t
= [1, 1, 1, 1]
Generation 1540: Best = 7.912624e+03 FES = 154100 Diversity = 1.3824e-04 k_t
= [3, 4, 4, 3]
Generation 1550: Best = 7.912624e+03 FES = 155100 Diversity = 1.2523e-04 k_t
= [1, 1, 1, 1]
Generation 1560: Best = 7.912624e+03 FES = 156100 Diversity = 1.1653e-04 k_t
= [4, 4, 4, 3]
Generation 1570: Best = 7.912624e+03 FES = 157100 Diversity = 1.0048e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 1580: Best = 7.912624e+03 FES = 158100 Diversity = 9.2550e-05 k_t
= [4, 4, 3, 4]
Generation 1590: Best = 7.912624e+03 FES = 159100 Diversity = 8.5883e-05 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 7.912624e+03 FES = 160100 Diversity = 8.2133e-05 k_t
= [2, 3, 4, 4]
Generation 1610: Best = 7.912624e+03 FES = 161100 Diversity = 8.2269e-05 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 7.912624e+03 FES = 162100 Diversity = 7.7001e-05 k_t
= [2, 2, 3, 2]
Generation 1630: Best = 7.912624e+03 FES = 163100 Diversity = 7.7669e-05 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 7.912624e+03 FES = 164100 Diversity = 6.9825e-05 k_t
= [2, 2, 3, 4]
Generation 1650: Best = 7.912624e+03 FES = 165100 Diversity = 6.6205e-05 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 7.912624e+03 FES = 166100 Diversity = 5.9674e-05 k_t
= [2, 4, 3, 2]
Generation 1670: Best = 7.912624e+03 FES = 167100 Diversity = 5.4739e-05 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 7.912624e+03 FES = 168100 Diversity = 4.7644e-05 k_t
= [2, 4, 4, 2]
Generation 1690: Best = 7.912624e+03 FES = 169100 Diversity = 4.0113e-05 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 7.912624e+03 FES = 170100 Diversity = 3.7492e-05 k_t
= [2, 4, 2, 4]
Generation 1710: Best = 7.912624e+03 FES = 171100 Diversity = 3.1447e-05 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 7.912624e+03 FES = 172100 Diversity = 2.5843e-05 k_t
= [3, 2, 3, 4]
Generation 1730: Best = 7.912624e+03 FES = 173100 Diversity = 2.1789e-05 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 7.912624e+03 FES = 174100 Diversity = 1.7991e-05 k_t
= [2, 4, 2, 3]
Generation 1750: Best = 7.912624e+03 FES = 175100 Diversity = 1.7160e-05 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 7.912624e+03 FES = 176100 Diversity = 1.6600e-05 k_t
= [3, 2, 4, 3]
Generation 1770: Best = 7.912624e+03 FES = 177100 Diversity = 1.4254e-05 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 7.912624e+03 FES = 178100 Diversity = 1.3881e-05 k_t
= [3, 4, 4, 3]
Generation 1790: Best = 7.912624e+03 FES = 179100 Diversity = 1.3956e-05 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 7.912624e+03 FES = 180100 Diversity = 1.3505e-05 k_t
= [2, 2, 4, 4]
Generation 1810: Best = 7.912624e+03 FES = 181100 Diversity = 1.1573e-05 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 7.912624e+03 FES = 182100 Diversity = 9.5884e-06 k_t
= [3, 3, 4, 3]
Generation 1830: Best = 7.912624e+03 FES = 183100 Diversity = 7.9582e-06 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 7.912624e+03 FES = 184100 Diversity = 6.7576e-06 k_t

```

```

= [4, 3, 2, 3]
Generation 1850: Best = 7.912624e+03 FES = 185100 Diversity = 5.5915e-06 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 7.912624e+03 FES = 186100 Diversity = 5.0380e-06 k_t
= [4, 4, 2, 3]
Generation 1870: Best = 7.912624e+03 FES = 187100 Diversity = 4.4205e-06 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 7.912624e+03 FES = 188100 Diversity = 4.2694e-06 k_t
= [4, 4, 3, 2]
Generation 1890: Best = 7.912624e+03 FES = 189100 Diversity = 3.8560e-06 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 7.912624e+03 FES = 190100 Diversity = 3.4264e-06 k_t
= [4, 3, 2, 4]
Generation 1910: Best = 7.912624e+03 FES = 191100 Diversity = 3.1231e-06 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 7.912624e+03 FES = 192100 Diversity = 2.9222e-06 k_t
= [4, 3, 4, 2]
Generation 1930: Best = 7.912624e+03 FES = 193100 Diversity = 2.6428e-06 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 7.912624e+03 FES = 194100 Diversity = 2.4178e-06 k_t
= [4, 4, 2, 3]
Generation 1950: Best = 7.912624e+03 FES = 195100 Diversity = 2.2532e-06 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 7.912624e+03 FES = 196100 Diversity = 2.2550e-06 k_t
= [2, 4, 3, 3]
Generation 1970: Best = 7.912624e+03 FES = 197100 Diversity = 2.0637e-06 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 7.912624e+03 FES = 198100 Diversity = 2.0516e-06 k_t
= [3, 2, 4, 4]
Generation 1990: Best = 7.912624e+03 FES = 199100 Diversity = 2.1130e-06 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 7.912624e+03 FES = 200100 Diversity = 2.0553e-06 k_t
= [3, 3, 4, 2]
Generation 2010: Best = 7.912624e+03 FES = 201100 Diversity = 1.8785e-06 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 7.912624e+03 FES = 202100 Diversity = 1.7528e-06 k_t
= [2, 3, 2, 2]
Generation 2030: Best = 7.912624e+03 FES = 203100 Diversity = 1.4848e-06 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 7.912624e+03 FES = 204100 Diversity = 1.2219e-06 k_t
= [4, 2, 3, 2]
Generation 2050: Best = 7.912624e+03 FES = 205100 Diversity = 1.0912e-06 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 7.912624e+03 FES = 206100 Diversity = 1.0522e-06 k_t
= [3, 2, 4, 4]
Generation 2070: Best = 7.912624e+03 FES = 207100 Diversity = 9.7023e-07 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 7.912624e+03 FES = 208100 Diversity = 8.4628e-07 k_t
= [2, 3, 2, 2]
Generation 2090: Best = 7.912624e+03 FES = 209100 Diversity = 7.3408e-07 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 7.912624e+03 FES = 210100 Diversity = 6.7182e-07 k_t
= [4, 3, 2, 2]
Generation 2110: Best = 7.912624e+03 FES = 211100 Diversity = 5.7814e-07 k_t

```



```

= [1, 1, 1, 1]
Generation 2120: Best = 7.912624e+03 FES = 212100 Diversity = 4.8884e-07 k_t
= [2, 2, 4, 4]
Generation 2130: Best = 7.912624e+03 FES = 213100 Diversity = 4.6555e-07 k_t
= [1, 1, 1, 1]
Generation 2140: Best = 7.912624e+03 FES = 214100 Diversity = 4.3568e-07 k_t
= [4, 4, 3, 3]
Generation 2150: Best = 7.912624e+03 FES = 215100 Diversity = 3.7982e-07 k_t
= [1, 1, 1, 1]
Generation 2160: Best = 7.912624e+03 FES = 216100 Diversity = 3.5906e-07 k_t
= [2, 2, 3, 2]
Generation 2170: Best = 7.912624e+03 FES = 217100 Diversity = 3.2556e-07 k_t
= [1, 1, 1, 1]
Generation 2180: Best = 7.912624e+03 FES = 218100 Diversity = 2.8373e-07 k_t
= [2, 4, 4, 2]
Generation 2190: Best = 7.912624e+03 FES = 219100 Diversity = 2.6489e-07 k_t
= [1, 1, 1, 1]
Generation 2200: Best = 7.912624e+03 FES = 220100 Diversity = 2.2997e-07 k_t
= [4, 4, 2, 2]
Generation 2210: Best = 7.912624e+03 FES = 221100 Diversity = 2.1955e-07 k_t
= [1, 1, 1, 1]
Generation 2220: Best = 7.912624e+03 FES = 222100 Diversity = 2.0769e-07 k_t
= [2, 3, 4, 4]
Generation 2230: Best = 7.912624e+03 FES = 223100 Diversity = 1.9975e-07 k_t
= [1, 1, 1, 1]
Generation 2240: Best = 7.912624e+03 FES = 224100 Diversity = 1.8008e-07 k_t
= [3, 2, 2, 3]
Generation 2250: Best = 7.912624e+03 FES = 225100 Diversity = 1.5238e-07 k_t
= [1, 1, 1, 1]
Generation 2260: Best = 7.912624e+03 FES = 226100 Diversity = 1.3899e-07 k_t
= [3, 3, 3, 2]
Generation 2270: Best = 7.912624e+03 FES = 227100 Diversity = 1.3791e-07 k_t
= [1, 1, 1, 1]
Generation 2280: Best = 7.912624e+03 FES = 228100 Diversity = 1.2073e-07 k_t
= [4, 4, 2, 2]
Generation 2290: Best = 7.912624e+03 FES = 229100 Diversity = 1.2491e-07 k_t
= [1, 1, 1, 1]
Generation 2300: Best = 7.912624e+03 FES = 230100 Diversity = 1.0870e-07 k_t
= [4, 4, 2, 4]
Generation 2310: Best = 7.912624e+03 FES = 231100 Diversity = 1.0456e-07 k_t
= [1, 1, 1, 1]
Generation 2320: Best = 7.912624e+03 FES = 232100 Diversity = 1.0177e-07 k_t
= [4, 4, 3, 4]
Generation 2330: Best = 7.912624e+03 FES = 233100 Diversity = 1.0222e-07 k_t
= [1, 1, 1, 1]
Generation 2340: Best = 7.912624e+03 FES = 234100 Diversity = 9.4664e-08 k_t
= [3, 3, 3, 4]
Generation 2350: Best = 7.912624e+03 FES = 235100 Diversity = 9.3874e-08 k_t
= [1, 1, 1, 1]
Generation 2360: Best = 7.912624e+03 FES = 236100 Diversity = 9.0344e-08 k_t
= [2, 3, 4, 4]
Generation 2370: Best = 7.912624e+03 FES = 237100 Diversity = 8.6162e-08 k_t
= [1, 1, 1, 1]
Generation 2380: Best = 7.912624e+03 FES = 238100 Diversity = 8.2708e-08 k_t

```

```

= [2, 4, 4, 4]
Generation 2390: Best = 7.912624e+03 FES = 239100 Diversity = 7.8596e-08 k_t
= [1, 1, 1, 1]
Generation 2400: Best = 7.912624e+03 FES = 240100 Diversity = 7.5413e-08 k_t
= [4, 4, 2, 4]
Generation 2410: Best = 7.912624e+03 FES = 241100 Diversity = 7.4253e-08 k_t
= [1, 1, 1, 1]
Generation 2420: Best = 7.912624e+03 FES = 242100 Diversity = 7.1781e-08 k_t
= [3, 2, 2, 4]
Generation 2430: Best = 7.912624e+03 FES = 243100 Diversity = 7.0059e-08 k_t
= [1, 1, 1, 1]
Generation 2440: Best = 7.912624e+03 FES = 244100 Diversity = 6.6575e-08 k_t
= [4, 4, 4, 3]
Generation 2450: Best = 7.912624e+03 FES = 245100 Diversity = 6.4627e-08 k_t
= [1, 1, 1, 1]
Generation 2460: Best = 7.912624e+03 FES = 246100 Diversity = 6.3042e-08 k_t
= [2, 4, 2, 2]
Generation 2470: Best = 7.912624e+03 FES = 247100 Diversity = 6.3323e-08 k_t
= [1, 1, 1, 1]
Generation 2480: Best = 7.912624e+03 FES = 248100 Diversity = 6.2285e-08 k_t
= [2, 2, 3, 2]
Generation 2490: Best = 7.912624e+03 FES = 249100 Diversity = 6.2465e-08 k_t
= [1, 1, 1, 1]
Generation 2500: Best = 7.912624e+03 FES = 250100 Diversity = 6.2116e-08 k_t
= [4, 2, 3, 3]
Generation 2510: Best = 7.912624e+03 FES = 251100 Diversity = 6.0803e-08 k_t
= [1, 1, 1, 1]
Generation 2520: Best = 7.912624e+03 FES = 252100 Diversity = 5.9828e-08 k_t
= [4, 4, 3, 4]
Generation 2530: Best = 7.912624e+03 FES = 253100 Diversity = 5.9858e-08 k_t
= [1, 1, 1, 1]
Generation 2540: Best = 7.912624e+03 FES = 254100 Diversity = 5.9252e-08 k_t
= [3, 2, 3, 2]
Generation 2550: Best = 7.912624e+03 FES = 255100 Diversity = 5.9516e-08 k_t
= [1, 1, 1, 1]
Generation 2560: Best = 7.912624e+03 FES = 256100 Diversity = 6.0346e-08 k_t
= [3, 4, 4, 4]
Generation 2570: Best = 7.912624e+03 FES = 257100 Diversity = 6.0767e-08 k_t
= [1, 1, 1, 1]
Generation 2580: Best = 7.912624e+03 FES = 258100 Diversity = 6.0889e-08 k_t
= [2, 4, 4, 2]
Generation 2590: Best = 7.912624e+03 FES = 259100 Diversity = 6.1375e-08 k_t
= [1, 1, 1, 1]
Generation 2600: Best = 7.912624e+03 FES = 260100 Diversity = 6.1401e-08 k_t
= [4, 2, 2, 4]
Generation 2610: Best = 7.912624e+03 FES = 261100 Diversity = 6.1558e-08 k_t
= [1, 1, 1, 1]
Generation 2620: Best = 7.912624e+03 FES = 262100 Diversity = 6.2091e-08 k_t
= [2, 2, 4, 3]
Generation 2630: Best = 7.912624e+03 FES = 263100 Diversity = 6.2061e-08 k_t
= [1, 1, 1, 1]
Generation 2640: Best = 7.912624e+03 FES = 264100 Diversity = 6.2403e-08 k_t
= [3, 3, 4, 2]
Generation 2650: Best = 7.912624e+03 FES = 265100 Diversity = 6.2831e-08 k_t

```

```

= [1, 1, 1, 1]
Generation 2660: Best = 7.912624e+03  FES = 266100  Diversity = 6.2953e-08  k_t
= [2, 3, 2, 3]
Generation 2670: Best = 7.912624e+03  FES = 267100  Diversity = 6.3308e-08  k_t
= [1, 1, 1, 1]
Generation 2680: Best = 7.912624e+03  FES = 268100  Diversity = 6.3544e-08  k_t
= [4, 2, 2, 3]
Generation 2690: Best = 7.912624e+03  FES = 269100  Diversity = 6.3330e-08  k_t
= [1, 1, 1, 1]
Generation 2700: Best = 7.912624e+03  FES = 270100  Diversity = 6.3431e-08  k_t
= [2, 4, 4, 2]
Generation 2710: Best = 7.912624e+03  FES = 271100  Diversity = 6.3760e-08  k_t
= [1, 1, 1, 1]
Generation 2720: Best = 7.912624e+03  FES = 272100  Diversity = 6.3640e-08  k_t
= [2, 2, 3, 2]
Generation 2730: Best = 7.912624e+03  FES = 273100  Diversity = 6.3448e-08  k_t
= [1, 1, 1, 1]
Generation 2740: Best = 7.912624e+03  FES = 274100  Diversity = 6.3206e-08  k_t
= [4, 3, 2, 3]
Generation 2750: Best = 7.912624e+03  FES = 275100  Diversity = 6.3371e-08  k_t
= [1, 1, 1, 1]
Generation 2760: Best = 7.912624e+03  FES = 276100  Diversity = 6.3542e-08  k_t
= [2, 3, 2, 3]
Generation 2770: Best = 7.912624e+03  FES = 277100  Diversity = 6.3798e-08  k_t
= [1, 1, 1, 1]
Generation 2780: Best = 7.912624e+03  FES = 278100  Diversity = 6.3597e-08  k_t
= [3, 2, 3, 2]
Generation 2790: Best = 7.912624e+03  FES = 279100  Diversity = 6.4165e-08  k_t
= [1, 1, 1, 1]
Generation 2800: Best = 7.912624e+03  FES = 280100  Diversity = 6.4503e-08  k_t
= [4, 2, 2, 3]
Generation 2810: Best = 7.912624e+03  FES = 281100  Diversity = 6.4756e-08  k_t
= [1, 1, 1, 1]
Generation 2820: Best = 7.912624e+03  FES = 282100  Diversity = 6.4788e-08  k_t
= [2, 4, 2, 2]
Generation 2830: Best = 7.912624e+03  FES = 283100  Diversity = 6.4731e-08  k_t
= [1, 1, 1, 1]
Generation 2840: Best = 7.912624e+03  FES = 284100  Diversity = 6.5041e-08  k_t
= [2, 4, 2, 3]
Generation 2850: Best = 7.912624e+03  FES = 285100  Diversity = 6.4541e-08  k_t
= [1, 1, 1, 1]
Generation 2860: Best = 7.912624e+03  FES = 286100  Diversity = 6.3970e-08  k_t
= [2, 3, 3, 4]
Generation 2870: Best = 7.912624e+03  FES = 287100  Diversity = 6.4293e-08  k_t
= [1, 1, 1, 1]
Generation 2880: Best = 7.912624e+03  FES = 288100  Diversity = 6.5005e-08  k_t
= [3, 4, 2, 2]
Generation 2890: Best = 7.912624e+03  FES = 289100  Diversity = 6.5187e-08  k_t
= [1, 1, 1, 1]
Generation 2900: Best = 7.912624e+03  FES = 290100  Diversity = 6.4176e-08  k_t
= [4, 3, 4, 4]
Generation 2910: Best = 7.912624e+03  FES = 291100  Diversity = 6.4380e-08  k_t
= [1, 1, 1, 1]
Generation 2920: Best = 7.912624e+03  FES = 292100  Diversity = 6.4608e-08  k_t

```

```

= [3, 3, 2, 2]
Generation 2930: Best = 7.912624e+03 FES = 293100 Diversity = 6.4259e-08 k_t
= [1, 1, 1, 1]
Generation 2940: Best = 7.912624e+03 FES = 294100 Diversity = 6.3711e-08 k_t
= [3, 2, 3, 2]
Generation 2950: Best = 7.912624e+03 FES = 295100 Diversity = 6.3993e-08 k_t
= [1, 1, 1, 1]
Generation 2960: Best = 7.912624e+03 FES = 296100 Diversity = 6.3930e-08 k_t
= [2, 2, 4, 3]
Generation 2970: Best = 7.912624e+03 FES = 297100 Diversity = 6.3419e-08 k_t
= [1, 1, 1, 1]
Generation 2980: Best = 7.912624e+03 FES = 298100 Diversity = 6.2759e-08 k_t
= [2, 2, 2, 2]
Generation 2990: Best = 7.912624e+03 FES = 299100 Diversity = 6.2302e-08 k_t
= [1, 1, 1, 1]
Generation 3000: Best = 7.912624e+03 FES = 300100 Diversity = 6.2790e-08 k_t
= [2, 2, 4, 2]
Generation 3010: Best = 7.912624e+03 FES = 301100 Diversity = 6.2612e-08 k_t
= [1, 1, 1, 1]
Generation 3020: Best = 7.912624e+03 FES = 302100 Diversity = 6.2589e-08 k_t
= [2, 3, 2, 4]
Generation 3030: Best = 7.912624e+03 FES = 303100 Diversity = 6.2615e-08 k_t
= [1, 1, 1, 1]
Generation 3040: Best = 7.912624e+03 FES = 304100 Diversity = 6.1703e-08 k_t
= [3, 4, 2, 2]
Generation 3050: Best = 7.912624e+03 FES = 305100 Diversity = 6.2318e-08 k_t
= [1, 1, 1, 1]
Generation 3060: Best = 7.912624e+03 FES = 306100 Diversity = 6.2038e-08 k_t
= [4, 3, 4, 3]
Generation 3070: Best = 7.912624e+03 FES = 307100 Diversity = 6.1992e-08 k_t
= [1, 1, 1, 1]
Generation 3080: Best = 7.912624e+03 FES = 308100 Diversity = 6.2262e-08 k_t
= [2, 4, 2, 2]
Generation 3090: Best = 7.912624e+03 FES = 309100 Diversity = 6.2003e-08 k_t
= [1, 1, 1, 1]
Generation 3100: Best = 7.912624e+03 FES = 310100 Diversity = 6.2486e-08 k_t
= [2, 4, 4, 2]
Generation 3110: Best = 7.912624e+03 FES = 311100 Diversity = 6.2016e-08 k_t
= [1, 1, 1, 1]
Generation 3120: Best = 7.912624e+03 FES = 312100 Diversity = 6.1732e-08 k_t
= [3, 4, 4, 4]
Generation 3130: Best = 7.912624e+03 FES = 313100 Diversity = 6.1357e-08 k_t
= [1, 1, 1, 1]
Generation 3140: Best = 7.912624e+03 FES = 314100 Diversity = 6.1297e-08 k_t
= [2, 4, 2, 3]
Generation 3150: Best = 7.912624e+03 FES = 315100 Diversity = 6.1364e-08 k_t
= [1, 1, 1, 1]
Generation 3160: Best = 7.912624e+03 FES = 316100 Diversity = 6.1590e-08 k_t
= [2, 3, 2, 2]
Generation 3170: Best = 7.912624e+03 FES = 317100 Diversity = 6.1419e-08 k_t
= [1, 1, 1, 1]
Generation 3180: Best = 7.912624e+03 FES = 318100 Diversity = 6.1829e-08 k_t
= [4, 4, 4, 3]
Generation 3190: Best = 7.912624e+03 FES = 319100 Diversity = 6.1806e-08 k_t

```

```

= [1, 1, 1, 1]
Generation 3200: Best = 7.912624e+03 FES = 320100 Diversity = 6.1980e-08 k_t
= [2, 3, 3, 3]
Generation 3210: Best = 7.912624e+03 FES = 321100 Diversity = 6.2316e-08 k_t
= [1, 1, 1, 1]
Generation 3220: Best = 7.912624e+03 FES = 322100 Diversity = 6.2339e-08 k_t
= [4, 3, 4, 2]
Generation 3230: Best = 7.912624e+03 FES = 323100 Diversity = 6.1874e-08 k_t
= [1, 1, 1, 1]
Generation 3240: Best = 7.912624e+03 FES = 324100 Diversity = 6.1866e-08 k_t
= [2, 4, 4, 2]
Generation 3250: Best = 7.912624e+03 FES = 325100 Diversity = 6.2022e-08 k_t
= [1, 1, 1, 1]
Generation 3260: Best = 7.912624e+03 FES = 326100 Diversity = 6.1862e-08 k_t
= [2, 2, 2, 2]
Generation 3270: Best = 7.912624e+03 FES = 327100 Diversity = 6.2085e-08 k_t
= [1, 1, 1, 1]
Generation 3280: Best = 7.912624e+03 FES = 328100 Diversity = 6.2089e-08 k_t
= [4, 4, 3, 2]
Generation 3290: Best = 7.912624e+03 FES = 329100 Diversity = 6.2666e-08 k_t
= [1, 1, 1, 1]
Generation 3300: Best = 7.912624e+03 FES = 330100 Diversity = 6.2130e-08 k_t
= [4, 4, 2, 4]
Generation 3310: Best = 7.912624e+03 FES = 331100 Diversity = 6.2341e-08 k_t
= [1, 1, 1, 1]
Generation 3320: Best = 7.912624e+03 FES = 332100 Diversity = 6.2940e-08 k_t
= [4, 4, 4, 3]
Generation 3330: Best = 7.912624e+03 FES = 333100 Diversity = 6.3223e-08 k_t
= [1, 1, 1, 1]
Generation 3340: Best = 7.912624e+03 FES = 334100 Diversity = 6.3099e-08 k_t
= [4, 3, 2, 3]
Generation 3350: Best = 7.912624e+03 FES = 335100 Diversity = 6.3450e-08 k_t
= [1, 1, 1, 1]
Generation 3360: Best = 7.912624e+03 FES = 336100 Diversity = 6.3712e-08 k_t
= [3, 4, 4, 4]
Generation 3370: Best = 7.912624e+03 FES = 337100 Diversity = 6.3836e-08 k_t
= [1, 1, 1, 1]
Generation 3380: Best = 7.912624e+03 FES = 338100 Diversity = 6.3548e-08 k_t
= [4, 2, 4, 2]
Generation 3390: Best = 7.912624e+03 FES = 339100 Diversity = 6.3636e-08 k_t
= [1, 1, 1, 1]
Generation 3400: Best = 7.912624e+03 FES = 340100 Diversity = 6.3499e-08 k_t
= [2, 2, 2, 3]
Generation 3410: Best = 7.912624e+03 FES = 341100 Diversity = 6.4038e-08 k_t
= [1, 1, 1, 1]
Generation 3420: Best = 7.912624e+03 FES = 342100 Diversity = 6.4415e-08 k_t
= [4, 2, 4, 2]
Generation 3430: Best = 7.912624e+03 FES = 343100 Diversity = 6.4432e-08 k_t
= [1, 1, 1, 1]
Generation 3440: Best = 7.912624e+03 FES = 344100 Diversity = 6.4081e-08 k_t
= [2, 4, 3, 4]
Generation 3450: Best = 7.912624e+03 FES = 345100 Diversity = 6.3989e-08 k_t
= [1, 1, 1, 1]
Generation 3460: Best = 7.912624e+03 FES = 346100 Diversity = 6.4427e-08 k_t

```

```

= [2, 4, 3, 3]
Generation 3470: Best = 7.912624e+03 FES = 347100 Diversity = 6.3765e-08 k_t
= [1, 1, 1, 1]
Generation 3480: Best = 7.912624e+03 FES = 348100 Diversity = 6.4320e-08 k_t
= [4, 2, 4, 3]
Generation 3490: Best = 7.912624e+03 FES = 349100 Diversity = 6.4132e-08 k_t
= [1, 1, 1, 1]
Generation 3500: Best = 7.912624e+03 FES = 350100 Diversity = 6.3397e-08 k_t
= [2, 3, 3, 2]
Generation 3510: Best = 7.912624e+03 FES = 351100 Diversity = 6.3282e-08 k_t
= [1, 1, 1, 1]
Generation 3520: Best = 7.912624e+03 FES = 352100 Diversity = 6.3338e-08 k_t
= [2, 2, 4, 3]
Generation 3530: Best = 7.912624e+03 FES = 353100 Diversity = 6.2998e-08 k_t
= [1, 1, 1, 1]
Generation 3540: Best = 7.912624e+03 FES = 354100 Diversity = 6.2849e-08 k_t
= [2, 3, 2, 3]
Generation 3550: Best = 7.912624e+03 FES = 355100 Diversity = 6.2630e-08 k_t
= [1, 1, 1, 1]
Generation 3560: Best = 7.912624e+03 FES = 356100 Diversity = 6.2728e-08 k_t
= [2, 4, 3, 3]
Generation 3570: Best = 7.912624e+03 FES = 357100 Diversity = 6.2475e-08 k_t
= [1, 1, 1, 1]
Generation 3580: Best = 7.912624e+03 FES = 358100 Diversity = 6.2136e-08 k_t
= [3, 4, 3, 3]
Generation 3590: Best = 7.912624e+03 FES = 359100 Diversity = 6.2015e-08 k_t
= [1, 1, 1, 1]
Generation 3600: Best = 7.912624e+03 FES = 360100 Diversity = 6.1576e-08 k_t
= [3, 2, 4, 4]
Generation 3610: Best = 7.912624e+03 FES = 361100 Diversity = 6.1699e-08 k_t
= [1, 1, 1, 1]
Generation 3620: Best = 7.912624e+03 FES = 362100 Diversity = 6.1562e-08 k_t
= [2, 3, 2, 3]
Generation 3630: Best = 7.912624e+03 FES = 363100 Diversity = 6.1368e-08 k_t
= [1, 1, 1, 1]
Generation 3640: Best = 7.912624e+03 FES = 364100 Diversity = 6.1296e-08 k_t
= [3, 2, 3, 4]
Generation 3650: Best = 7.912624e+03 FES = 365100 Diversity = 6.1092e-08 k_t
= [1, 1, 1, 1]
Generation 3660: Best = 7.912624e+03 FES = 366100 Diversity = 6.1248e-08 k_t
= [3, 2, 2, 2]
Generation 3670: Best = 7.912624e+03 FES = 367100 Diversity = 6.1618e-08 k_t
= [1, 1, 1, 1]
Generation 3680: Best = 7.912624e+03 FES = 368100 Diversity = 6.1249e-08 k_t
= [2, 4, 3, 2]
Generation 3690: Best = 7.912624e+03 FES = 369100 Diversity = 6.1159e-08 k_t
= [1, 1, 1, 1]
Generation 3700: Best = 7.912624e+03 FES = 370100 Diversity = 6.0990e-08 k_t
= [2, 4, 3, 4]
Generation 3710: Best = 7.912624e+03 FES = 371100 Diversity = 6.0729e-08 k_t
= [1, 1, 1, 1]
Generation 3720: Best = 7.912624e+03 FES = 372100 Diversity = 6.0517e-08 k_t
= [2, 2, 3, 2]
Generation 3730: Best = 7.912624e+03 FES = 373100 Diversity = 6.0744e-08 k_t

```

```

= [1, 1, 1, 1]
Generation 3740: Best = 7.912624e+03 FES = 374100 Diversity = 6.0729e-08 k_t
= [2, 3, 3, 2]
Generation 3750: Best = 7.912624e+03 FES = 375100 Diversity = 6.0752e-08 k_t
= [1, 1, 1, 1]
Generation 3760: Best = 7.912624e+03 FES = 376100 Diversity = 6.1082e-08 k_t
= [2, 4, 4, 4]
Generation 3770: Best = 7.912624e+03 FES = 377100 Diversity = 6.0804e-08 k_t
= [1, 1, 1, 1]
Generation 3780: Best = 7.912624e+03 FES = 378100 Diversity = 6.0288e-08 k_t
= [3, 4, 3, 3]
Generation 3790: Best = 7.912624e+03 FES = 379100 Diversity = 6.0026e-08 k_t
= [1, 1, 1, 1]
Generation 3800: Best = 7.912624e+03 FES = 380100 Diversity = 5.9563e-08 k_t
= [2, 4, 3, 4]
Generation 3810: Best = 7.912624e+03 FES = 381100 Diversity = 5.9347e-08 k_t
= [1, 1, 1, 1]
Generation 3820: Best = 7.912624e+03 FES = 382100 Diversity = 5.9120e-08 k_t
= [4, 3, 2, 4]
Generation 3830: Best = 7.912624e+03 FES = 383100 Diversity = 5.9164e-08 k_t
= [1, 1, 1, 1]
Generation 3840: Best = 7.912624e+03 FES = 384100 Diversity = 5.9013e-08 k_t
= [2, 3, 2, 3]
Generation 3850: Best = 7.912624e+03 FES = 385100 Diversity = 5.8849e-08 k_t
= [1, 1, 1, 1]
Generation 3860: Best = 7.912624e+03 FES = 386100 Diversity = 5.8901e-08 k_t
= [4, 3, 4, 2]
Generation 3870: Best = 7.912624e+03 FES = 387100 Diversity = 5.8937e-08 k_t
= [1, 1, 1, 1]
Generation 3880: Best = 7.912624e+03 FES = 388100 Diversity = 5.9319e-08 k_t
= [3, 3, 3, 2]
Generation 3890: Best = 7.912624e+03 FES = 389100 Diversity = 5.9115e-08 k_t
= [1, 1, 1, 1]
Generation 3900: Best = 7.912624e+03 FES = 390100 Diversity = 5.9093e-08 k_t
= [3, 4, 4, 4]
Generation 3910: Best = 7.912624e+03 FES = 391100 Diversity = 5.8890e-08 k_t
= [1, 1, 1, 1]
Generation 3920: Best = 7.912624e+03 FES = 392100 Diversity = 5.8656e-08 k_t
= [4, 3, 2, 4]
Generation 3930: Best = 7.912624e+03 FES = 393100 Diversity = 5.8768e-08 k_t
= [1, 1, 1, 1]
Generation 3940: Best = 7.912624e+03 FES = 394100 Diversity = 5.8731e-08 k_t
= [2, 4, 3, 2]
Generation 3950: Best = 7.912624e+03 FES = 395100 Diversity = 5.8482e-08 k_t
= [1, 1, 1, 1]
Generation 3960: Best = 7.912624e+03 FES = 396100 Diversity = 5.8393e-08 k_t
= [3, 2, 3, 2]
Generation 3970: Best = 7.912624e+03 FES = 397100 Diversity = 5.8211e-08 k_t
= [1, 1, 1, 1]
Generation 3980: Best = 7.912624e+03 FES = 398100 Diversity = 5.8484e-08 k_t
= [2, 2, 2, 2]
Generation 3990: Best = 7.912624e+03 FES = 399100 Diversity = 5.8849e-08 k_t
= [1, 1, 1, 1]
Generation 4000: Best = 7.912624e+03 FES = 400100 Diversity = 5.8960e-08 k_t

```

```

= [4, 2, 3, 2]
Generation 4010: Best = 7.912624e+03 FES = 401100 Diversity = 5.8809e-08 k_t
= [1, 1, 1, 1]
Generation 4020: Best = 7.912624e+03 FES = 402100 Diversity = 5.8771e-08 k_t
= [2, 4, 3, 4]
Generation 4030: Best = 7.912624e+03 FES = 403100 Diversity = 5.8585e-08 k_t
= [1, 1, 1, 1]
Generation 4040: Best = 7.912624e+03 FES = 404100 Diversity = 5.8426e-08 k_t
= [2, 2, 2, 4]
Generation 4050: Best = 7.912624e+03 FES = 405100 Diversity = 5.7420e-08 k_t
= [1, 1, 1, 1]
Generation 4060: Best = 7.912624e+03 FES = 406100 Diversity = 5.6375e-08 k_t
= [3, 4, 3, 4]
Generation 4070: Best = 7.912624e+03 FES = 407100 Diversity = 5.5398e-08 k_t
= [1, 1, 1, 1]
Generation 4080: Best = 7.912624e+03 FES = 408100 Diversity = 5.3963e-08 k_t
= [4, 3, 4, 2]
Generation 4090: Best = 7.912624e+03 FES = 409100 Diversity = 5.2916e-08 k_t
= [1, 1, 1, 1]
Generation 4100: Best = 7.912624e+03 FES = 410100 Diversity = 5.1134e-08 k_t
= [3, 2, 3, 2]
Generation 4110: Best = 7.912624e+03 FES = 411100 Diversity = 4.9906e-08 k_t
= [1, 1, 1, 1]
Generation 4120: Best = 7.912624e+03 FES = 412100 Diversity = 4.8948e-08 k_t
= [3, 3, 2, 3]
Generation 4130: Best = 7.912624e+03 FES = 413100 Diversity = 4.7699e-08 k_t
= [1, 1, 1, 1]
Generation 4140: Best = 7.912624e+03 FES = 414100 Diversity = 4.4914e-08 k_t
= [2, 3, 2, 4]
Generation 4150: Best = 7.912624e+03 FES = 415100 Diversity = 4.4072e-08 k_t
= [1, 1, 1, 1]
Generation 4160: Best = 7.912624e+03 FES = 416100 Diversity = 4.3794e-08 k_t
= [2, 2, 3, 4]
Generation 4170: Best = 7.912624e+03 FES = 417100 Diversity = 4.4471e-08 k_t
= [1, 1, 1, 1]
Generation 4180: Best = 7.912624e+03 FES = 418100 Diversity = 4.3920e-08 k_t
= [2, 4, 2, 4]
Generation 4190: Best = 7.912624e+03 FES = 419100 Diversity = 4.3416e-08 k_t
= [1, 1, 1, 1]
Generation 4200: Best = 7.912624e+03 FES = 420100 Diversity = 4.4655e-08 k_t
= [3, 2, 3, 2]
Generation 4210: Best = 7.912624e+03 FES = 421100 Diversity = 4.5132e-08 k_t
= [1, 1, 1, 1]
Generation 4220: Best = 7.912624e+03 FES = 422100 Diversity = 4.5049e-08 k_t
= [2, 2, 3, 3]
Generation 4230: Best = 7.912624e+03 FES = 423100 Diversity = 4.5148e-08 k_t
= [1, 1, 1, 1]
Generation 4240: Best = 7.912624e+03 FES = 424100 Diversity = 4.5098e-08 k_t
= [4, 3, 3, 2]
Generation 4250: Best = 7.912624e+03 FES = 425100 Diversity = 4.5015e-08 k_t
= [1, 1, 1, 1]
Generation 4260: Best = 7.912624e+03 FES = 426100 Diversity = 4.4921e-08 k_t
= [4, 2, 3, 3]
Generation 4270: Best = 7.912624e+03 FES = 427100 Diversity = 4.4464e-08 k_t

```



```

= [1, 1, 1, 1]
Generation 4280: Best = 7.912624e+03 FES = 428100 Diversity = 4.5249e-08 k_t
= [3, 4, 3, 4]
Generation 4290: Best = 7.912624e+03 FES = 429100 Diversity = 4.5108e-08 k_t
= [1, 1, 1, 1]
Generation 4300: Best = 7.912624e+03 FES = 430100 Diversity = 4.3981e-08 k_t
= [2, 4, 2, 4]
Generation 4310: Best = 7.912624e+03 FES = 431100 Diversity = 4.4600e-08 k_t
= [1, 1, 1, 1]
Generation 4320: Best = 7.912624e+03 FES = 432100 Diversity = 4.5384e-08 k_t
= [4, 4, 2, 4]
Generation 4330: Best = 7.912624e+03 FES = 433100 Diversity = 4.5244e-08 k_t
= [1, 1, 1, 1]
Generation 4340: Best = 7.912624e+03 FES = 434100 Diversity = 4.5200e-08 k_t
= [4, 2, 3, 3]
Generation 4350: Best = 7.912624e+03 FES = 435100 Diversity = 4.3851e-08 k_t
= [1, 1, 1, 1]
Generation 4360: Best = 7.912624e+03 FES = 436100 Diversity = 4.4660e-08 k_t
= [3, 4, 2, 4]
Generation 4370: Best = 7.912624e+03 FES = 437100 Diversity = 4.4351e-08 k_t
= [1, 1, 1, 1]
Generation 4380: Best = 7.912624e+03 FES = 438100 Diversity = 4.0907e-08 k_t
= [2, 2, 3, 3]
Generation 4390: Best = 7.912624e+03 FES = 439100 Diversity = 2.9098e-08 k_t
= [1, 1, 1, 1]
Generation 4400: Best = 7.912624e+03 FES = 440100 Diversity = 2.7742e-08 k_t
= [4, 3, 2, 2]
Generation 4410: Best = 7.912624e+03 FES = 441100 Diversity = 2.6531e-08 k_t
= [1, 1, 1, 1]
Generation 4420: Best = 7.912624e+03 FES = 442100 Diversity = 2.4324e-08 k_t
= [2, 4, 3, 4]
Generation 4430: Best = 7.912624e+03 FES = 443100 Diversity = 2.1487e-08 k_t
= [1, 1, 1, 1]
Generation 4440: Best = 7.912624e+03 FES = 444100 Diversity = 1.3059e-08 k_t
= [4, 4, 2, 4]
Generation 4450: Best = 7.912624e+03 FES = 445100 Diversity = 1.1198e-08 k_t
= [1, 1, 1, 1]
Generation 4460: Best = 7.912624e+03 FES = 446100 Diversity = 1.1521e-08 k_t
= [3, 3, 4, 4]
Generation 4470: Best = 7.912624e+03 FES = 447100 Diversity = 1.1399e-08 k_t
= [1, 1, 1, 1]
Generation 4480: Best = 7.912624e+03 FES = 448100 Diversity = 7.9579e-09 k_t
= [2, 4, 2, 2]
Generation 4490: Best = 7.912624e+03 FES = 449100 Diversity = 7.8415e-09 k_t
= [1, 1, 1, 1]
Generation 4500: Best = 7.912624e+03 FES = 450100 Diversity = 7.5667e-09 k_t
= [3, 2, 4, 3]
Generation 4510: Best = 7.912624e+03 FES = 451100 Diversity = 7.8380e-09 k_t
= [1, 1, 1, 1]
Generation 4520: Best = 7.912624e+03 FES = 452100 Diversity = 7.2752e-09 k_t
= [2, 2, 4, 2]
Generation 4530: Best = 7.912624e+03 FES = 453100 Diversity = 7.4392e-09 k_t
= [1, 1, 1, 1]
Generation 4540: Best = 7.912624e+03 FES = 454100 Diversity = 7.6326e-09 k_t

```

```

= [3, 2, 2, 4]
Generation 4550: Best = 7.912624e+03 FES = 455100 Diversity = 7.3801e-09 k_t
= [1, 1, 1, 1]
Generation 4560: Best = 7.912624e+03 FES = 456100 Diversity = 7.7034e-09 k_t
= [2, 3, 2, 2]
Generation 4570: Best = 7.912624e+03 FES = 457100 Diversity = 7.5233e-09 k_t
= [1, 1, 1, 1]
Generation 4580: Best = 7.912624e+03 FES = 458100 Diversity = 6.9919e-09 k_t
= [4, 4, 2, 2]
Generation 4590: Best = 7.912624e+03 FES = 459100 Diversity = 5.1215e-09 k_t
= [1, 1, 1, 1]
Generation 4600: Best = 7.912624e+03 FES = 460100 Diversity = 2.9010e-09 k_t
= [3, 3, 3, 2]
Generation 4610: Best = 7.912624e+03 FES = 461100 Diversity = 2.2252e-09 k_t
= [1, 1, 1, 1]
Generation 4620: Best = 7.912624e+03 FES = 462100 Diversity = 2.7683e-09 k_t
= [4, 2, 2, 2]
Generation 4630: Best = 7.912624e+03 FES = 463100 Diversity = 2.7898e-09 k_t
= [1, 1, 1, 1]
Generation 4640: Best = 7.912624e+03 FES = 464100 Diversity = 2.5612e-09 k_t
= [2, 3, 2, 3]
Generation 4650: Best = 7.912624e+03 FES = 465100 Diversity = 2.5857e-09 k_t
= [1, 1, 1, 1]
Generation 4660: Best = 7.912624e+03 FES = 466100 Diversity = 2.3792e-09 k_t
= [4, 3, 4, 4]
Generation 4670: Best = 7.912624e+03 FES = 467100 Diversity = 2.8744e-09 k_t
= [1, 1, 1, 1]
Generation 4680: Best = 7.912624e+03 FES = 468100 Diversity = 2.4921e-09 k_t
= [4, 3, 2, 3]
Generation 4690: Best = 7.912624e+03 FES = 469100 Diversity = 2.6698e-09 k_t
= [1, 1, 1, 1]
Generation 4700: Best = 7.912624e+03 FES = 470100 Diversity = 2.7238e-09 k_t
= [3, 2, 2, 3]
Generation 4710: Best = 7.912624e+03 FES = 471100 Diversity = 2.2127e-09 k_t
= [1, 1, 1, 1]
Generation 4720: Best = 7.912624e+03 FES = 472100 Diversity = 2.2989e-09 k_t
= [2, 2, 4, 4]
Generation 4730: Best = 7.912624e+03 FES = 473100 Diversity = 2.4799e-09 k_t
= [1, 1, 1, 1]
Generation 4740: Best = 7.912624e+03 FES = 474100 Diversity = 2.2750e-09 k_t
= [3, 2, 2, 2]
Generation 4750: Best = 7.912624e+03 FES = 475100 Diversity = 2.3384e-09 k_t
= [1, 1, 1, 1]
Generation 4760: Best = 7.912624e+03 FES = 476100 Diversity = 2.6385e-09 k_t
= [3, 3, 4, 4]
Generation 4770: Best = 7.912624e+03 FES = 477100 Diversity = 2.4895e-09 k_t
= [1, 1, 1, 1]
Generation 4780: Best = 7.912624e+03 FES = 478100 Diversity = 2.5861e-09 k_t
= [3, 3, 3, 2]
Generation 4790: Best = 7.912624e+03 FES = 479100 Diversity = 2.2531e-09 k_t
= [1, 1, 1, 1]
Generation 4800: Best = 7.912624e+03 FES = 480100 Diversity = 2.5981e-09 k_t
= [4, 2, 3, 2]
Generation 4810: Best = 7.912624e+03 FES = 481100 Diversity = 2.2584e-09 k_t

```

```

= [1, 1, 1, 1]
Generation 4820: Best = 7.912624e+03 FES = 482100 Diversity = 2.4935e-09 k_t
= [2, 3, 3, 3]
Generation 4830: Best = 7.912624e+03 FES = 483100 Diversity = 2.2388e-09 k_t
= [1, 1, 1, 1]
Generation 4840: Best = 7.912624e+03 FES = 484100 Diversity = 2.7770e-09 k_t
= [2, 3, 2, 3]
Generation 4850: Best = 7.912624e+03 FES = 485100 Diversity = 2.3729e-09 k_t
= [1, 1, 1, 1]
Generation 4860: Best = 7.912624e+03 FES = 486100 Diversity = 2.3739e-09 k_t
= [3, 3, 2, 3]
Generation 4870: Best = 7.912624e+03 FES = 487100 Diversity = 2.8960e-09 k_t
= [1, 1, 1, 1]
Generation 4880: Best = 7.912624e+03 FES = 488100 Diversity = 2.7573e-09 k_t
= [4, 2, 4, 2]
Generation 4890: Best = 7.912624e+03 FES = 489100 Diversity = 2.2616e-09 k_t
= [1, 1, 1, 1]
Generation 4900: Best = 7.912624e+03 FES = 490100 Diversity = 2.3288e-09 k_t
= [4, 2, 3, 2]
Generation 4910: Best = 7.912624e+03 FES = 491100 Diversity = 2.7022e-09 k_t
= [1, 1, 1, 1]
Generation 4920: Best = 7.912624e+03 FES = 492100 Diversity = 2.8237e-09 k_t
= [3, 4, 4, 2]
Generation 4930: Best = 7.912624e+03 FES = 493100 Diversity = 2.7790e-09 k_t
= [1, 1, 1, 1]
Generation 4940: Best = 7.912624e+03 FES = 494100 Diversity = 2.8649e-09 k_t
= [3, 2, 2, 4]
Generation 4950: Best = 7.912624e+03 FES = 495100 Diversity = 2.6977e-09 k_t
= [1, 1, 1, 1]
Generation 4960: Best = 7.912624e+03 FES = 496100 Diversity = 2.4827e-09 k_t
= [4, 2, 2, 2]
Generation 4970: Best = 7.912624e+03 FES = 497100 Diversity = 2.8258e-09 k_t
= [1, 1, 1, 1]
Generation 4980: Best = 7.912624e+03 FES = 498100 Diversity = 2.4598e-09 k_t
= [4, 4, 2, 3]
Generation 4990: Best = 7.912624e+03 FES = 499100 Diversity = 2.1802e-09 k_t
= [1, 1, 1, 1]
Generation 5000: Best = 7.912624e+03 FES = 500100 Diversity = 2.6736e-09 k_t
= [2, 4, 4, 2]
Generation 5010: Best = 7.912624e+03 FES = 501100 Diversity = 2.6915e-09 k_t
= [1, 1, 1, 1]
Generation 5020: Best = 7.912624e+03 FES = 502100 Diversity = 2.4660e-09 k_t
= [2, 4, 3, 2]
Generation 5030: Best = 7.912624e+03 FES = 503100 Diversity = 2.9149e-09 k_t
= [1, 1, 1, 1]
Generation 5040: Best = 7.912624e+03 FES = 504100 Diversity = 2.6795e-09 k_t
= [2, 4, 4, 4]
Generation 5050: Best = 7.912624e+03 FES = 505100 Diversity = 2.5274e-09 k_t
= [1, 1, 1, 1]
Generation 5060: Best = 7.912624e+03 FES = 506100 Diversity = 2.7589e-09 k_t
= [4, 4, 2, 2]
Generation 5070: Best = 7.912624e+03 FES = 507100 Diversity = 2.6734e-09 k_t
= [1, 1, 1, 1]
Generation 5080: Best = 7.912624e+03 FES = 508100 Diversity = 2.3786e-09 k_t

```

```

= [2, 4, 2, 4]
Generation 5090: Best = 7.912624e+03 FES = 509100 Diversity = 2.6025e-09 k_t
= [1, 1, 1, 1]
Generation 5100: Best = 7.912624e+03 FES = 510100 Diversity = 2.4512e-09 k_t
= [4, 2, 4, 3]
Generation 5110: Best = 7.912624e+03 FES = 511100 Diversity = 2.3162e-09 k_t
= [1, 1, 1, 1]
Generation 5120: Best = 7.912624e+03 FES = 512100 Diversity = 2.6348e-09 k_t
= [2, 3, 4, 4]
Generation 5130: Best = 7.912624e+03 FES = 513100 Diversity = 2.6620e-09 k_t
= [1, 1, 1, 1]
Generation 5140: Best = 7.912624e+03 FES = 514100 Diversity = 2.5969e-09 k_t
= [2, 3, 3, 4]
Generation 5150: Best = 7.912624e+03 FES = 515100 Diversity = 2.5761e-09 k_t
= [1, 1, 1, 1]
Generation 5160: Best = 7.912624e+03 FES = 516100 Diversity = 2.7295e-09 k_t
= [2, 2, 2, 3]
Generation 5170: Best = 7.912624e+03 FES = 517100 Diversity = 2.6315e-09 k_t
= [1, 1, 1, 1]
Generation 5180: Best = 7.912624e+03 FES = 518100 Diversity = 2.4334e-09 k_t
= [3, 2, 2, 3]
Generation 5190: Best = 7.912624e+03 FES = 519100 Diversity = 2.6403e-09 k_t
= [1, 1, 1, 1]
Generation 5200: Best = 7.912624e+03 FES = 520100 Diversity = 2.6005e-09 k_t
= [4, 2, 3, 2]
Generation 5210: Best = 7.912624e+03 FES = 521100 Diversity = 2.2846e-09 k_t
= [1, 1, 1, 1]
Generation 5220: Best = 7.912624e+03 FES = 522100 Diversity = 2.4451e-09 k_t
= [4, 2, 3, 2]
Generation 5230: Best = 7.912624e+03 FES = 523100 Diversity = 7.6500e-10 k_t
= [1, 1, 1, 1]
Generation 5240: Best = 7.912624e+03 FES = 524100 Diversity = 1.3755e-10 k_t
= [3, 2, 4, 3]
Generation 5250: Best = 7.912624e+03 FES = 525100 Diversity = 7.4098e-11 k_t
= [1, 1, 1, 1]
Generation 5260: Best = 7.912624e+03 FES = 526100 Diversity = 6.8973e-11 k_t
= [4, 3, 4, 2]
Generation 5270: Best = 7.912624e+03 FES = 527100 Diversity = 7.4098e-11 k_t
= [1, 1, 1, 1]
Generation 5280: Best = 7.912624e+03 FES = 528100 Diversity = 6.3302e-11 k_t
= [2, 2, 4, 3]
Generation 5290: Best = 7.912624e+03 FES = 529100 Diversity = 6.8973e-11 k_t
= [1, 1, 1, 1]
Generation 5300: Best = 7.912624e+03 FES = 530100 Diversity = 6.8973e-11 k_t
= [3, 2, 2, 2]
Generation 5310: Best = 7.912624e+03 FES = 531100 Diversity = 4.9559e-11 k_t
= [1, 1, 1, 1]
Generation 5320: Best = 7.912624e+03 FES = 532100 Diversity = 6.3302e-11 k_t
= [2, 2, 4, 4]
Generation 5330: Best = 7.912624e+03 FES = 533100 Diversity = 5.6922e-11 k_t
= [1, 1, 1, 1]
Generation 5340: Best = 7.912624e+03 FES = 534100 Diversity = 5.6922e-11 k_t
= [3, 4, 2, 2]
Generation 5350: Best = 7.912624e+03 FES = 535100 Diversity = 6.3302e-11 k_t

```

```

= [1, 1, 1, 1]
Generation 5360: Best = 7.912624e+03 FES = 536100 Diversity = 6.8973e-11 k_t
= [4, 4, 4, 2]
Generation 5370: Best = 7.912624e+03 FES = 537100 Diversity = 5.6922e-11 k_t
= [1, 1, 1, 1]
Generation 5380: Best = 7.912624e+03 FES = 538100 Diversity = 4.0683e-11 k_t
= [2, 4, 4, 3]
Generation 5390: Best = 7.912624e+03 FES = 539100 Diversity = 2.8929e-11 k_t
= [1, 1, 1, 1]
Generation 5400: Best = 7.912624e+03 FES = 540100 Diversity = 2.8929e-11 k_t
= [3, 4, 4, 4]
Generation 5410: Best = 7.912624e+03 FES = 541100 Diversity = 2.8929e-11 k_t
= [1, 1, 1, 1]
Generation 5420: Best = 7.912624e+03 FES = 542100 Diversity = 4.0683e-11 k_t
= [4, 2, 4, 2]
Generation 5430: Best = 7.912624e+03 FES = 543100 Diversity = 4.0683e-11 k_t
= [1, 1, 1, 1]
Generation 5440: Best = 7.912624e+03 FES = 544100 Diversity = 5.6444e-14 k_t
= [3, 3, 3, 4]
Generation 5450: Best = 7.912624e+03 FES = 545100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5460: Best = 7.912624e+03 FES = 546100 Diversity = 5.6444e-14 k_t
= [4, 2, 3, 2]
Generation 5470: Best = 7.912624e+03 FES = 547100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5480: Best = 7.912624e+03 FES = 548100 Diversity = 5.6444e-14 k_t
= [4, 4, 3, 2]
Generation 5490: Best = 7.912624e+03 FES = 549100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5500: Best = 7.912624e+03 FES = 550100 Diversity = 5.6444e-14 k_t
= [3, 3, 4, 2]
Generation 5510: Best = 7.912624e+03 FES = 551100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5520: Best = 7.912624e+03 FES = 552100 Diversity = 5.6444e-14 k_t
= [3, 4, 2, 2]
Generation 5530: Best = 7.912624e+03 FES = 553100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5540: Best = 7.912624e+03 FES = 554100 Diversity = 5.6444e-14 k_t
= [3, 2, 2, 2]
Generation 5550: Best = 7.912624e+03 FES = 555100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5560: Best = 7.912624e+03 FES = 556100 Diversity = 5.6444e-14 k_t
= [2, 4, 4, 4]
Generation 5570: Best = 7.912624e+03 FES = 557100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5580: Best = 7.912624e+03 FES = 558100 Diversity = 5.6444e-14 k_t
= [4, 4, 2, 4]
Generation 5590: Best = 7.912624e+03 FES = 559100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5600: Best = 7.912624e+03 FES = 560100 Diversity = 5.6444e-14 k_t
= [4, 4, 4, 4]
Generation 5610: Best = 7.912624e+03 FES = 561100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5620: Best = 7.912624e+03 FES = 562100 Diversity = 5.6444e-14 k_t

```

```

= [2, 3, 2, 4]
Generation 5630: Best = 7.912624e+03 FES = 563100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5640: Best = 7.912624e+03 FES = 564100 Diversity = 5.6444e-14 k_t
= [2, 3, 4, 3]
Generation 5650: Best = 7.912624e+03 FES = 565100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5660: Best = 7.912624e+03 FES = 566100 Diversity = 5.6444e-14 k_t
= [4, 3, 2, 2]
Generation 5670: Best = 7.912624e+03 FES = 567100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5680: Best = 7.912624e+03 FES = 568100 Diversity = 5.6444e-14 k_t
= [3, 3, 3, 3]
Generation 5690: Best = 7.912624e+03 FES = 569100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5700: Best = 7.912624e+03 FES = 570100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 4]
Generation 5710: Best = 7.912624e+03 FES = 571100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5720: Best = 7.912624e+03 FES = 572100 Diversity = 5.6444e-14 k_t
= [4, 2, 2, 4]
Generation 5730: Best = 7.912624e+03 FES = 573100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5740: Best = 7.912624e+03 FES = 574100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 3]
Generation 5750: Best = 7.912624e+03 FES = 575100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5760: Best = 7.912624e+03 FES = 576100 Diversity = 5.6444e-14 k_t
= [4, 3, 3, 4]
Generation 5770: Best = 7.912624e+03 FES = 577100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5780: Best = 7.912624e+03 FES = 578100 Diversity = 5.6444e-14 k_t
= [3, 2, 2, 2]
Generation 5790: Best = 7.912624e+03 FES = 579100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5800: Best = 7.912624e+03 FES = 580100 Diversity = 5.6444e-14 k_t
= [2, 2, 3, 2]
Generation 5810: Best = 7.912624e+03 FES = 581100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5820: Best = 7.912624e+03 FES = 582100 Diversity = 5.6444e-14 k_t
= [3, 3, 3, 2]
Generation 5830: Best = 7.912624e+03 FES = 583100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5840: Best = 7.912624e+03 FES = 584100 Diversity = 5.6444e-14 k_t
= [3, 2, 3, 4]
Generation 5850: Best = 7.912624e+03 FES = 585100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5860: Best = 7.912624e+03 FES = 586100 Diversity = 5.6444e-14 k_t
= [3, 2, 2, 2]
Generation 5870: Best = 7.912624e+03 FES = 587100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5880: Best = 7.912624e+03 FES = 588100 Diversity = 5.6444e-14 k_t
= [2, 4, 4, 2]
Generation 5890: Best = 7.912624e+03 FES = 589100 Diversity = 5.6444e-14 k_t

```

```

= [1, 1, 1, 1]
Generation 5900: Best = 7.912624e+03 FES = 590100 Diversity = 5.6444e-14 k_t
= [2, 3, 3, 3]
Generation 5910: Best = 7.912624e+03 FES = 591100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5920: Best = 7.912624e+03 FES = 592100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 2]
Generation 5930: Best = 7.912624e+03 FES = 593100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5940: Best = 7.912624e+03 FES = 594100 Diversity = 5.6444e-14 k_t
= [3, 3, 4, 3]
Generation 5950: Best = 7.912624e+03 FES = 595100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5960: Best = 7.912624e+03 FES = 596100 Diversity = 5.6444e-14 k_t
= [3, 2, 4, 3]
Generation 5970: Best = 7.912624e+03 FES = 597100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 5980: Best = 7.912624e+03 FES = 598100 Diversity = 5.6444e-14 k_t
= [4, 4, 3, 3]
Generation 5990: Best = 7.912624e+03 FES = 599100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6000: Best = 7.912624e+03 FES = 600100 Diversity = 5.6444e-14 k_t
= [2, 3, 3, 2]
Generation 6010: Best = 7.912624e+03 FES = 601100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6020: Best = 7.912624e+03 FES = 602100 Diversity = 5.6444e-14 k_t
= [3, 2, 3, 2]
Generation 6030: Best = 7.912624e+03 FES = 603100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6040: Best = 7.912624e+03 FES = 604100 Diversity = 5.6444e-14 k_t
= [2, 4, 4, 2]
Generation 6050: Best = 7.912624e+03 FES = 605100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6060: Best = 7.912624e+03 FES = 606100 Diversity = 5.6444e-14 k_t
= [4, 4, 4, 3]
Generation 6070: Best = 7.912624e+03 FES = 607100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6080: Best = 7.912624e+03 FES = 608100 Diversity = 5.6444e-14 k_t
= [3, 4, 2, 3]
Generation 6090: Best = 7.912624e+03 FES = 609100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6100: Best = 7.912624e+03 FES = 610100 Diversity = 5.6444e-14 k_t
= [3, 3, 4, 2]
Generation 6110: Best = 7.912624e+03 FES = 611100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6120: Best = 7.912624e+03 FES = 612100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 3]
Generation 6130: Best = 7.912624e+03 FES = 613100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6140: Best = 7.912624e+03 FES = 614100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 2]
Generation 6150: Best = 7.912624e+03 FES = 615100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6160: Best = 7.912624e+03 FES = 616100 Diversity = 5.6444e-14 k_t

```

```

= [2, 2, 4, 4]
Generation 6170: Best = 7.912624e+03  FES = 617100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6180: Best = 7.912624e+03  FES = 618100  Diversity = 5.6444e-14  k_t
= [2, 3, 3, 2]
Generation 6190: Best = 7.912624e+03  FES = 619100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6200: Best = 7.912624e+03  FES = 620100  Diversity = 5.6444e-14  k_t
= [4, 4, 3, 2]
Generation 6210: Best = 7.912624e+03  FES = 621100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6220: Best = 7.912624e+03  FES = 622100  Diversity = 5.6444e-14  k_t
= [2, 4, 2, 3]
Generation 6230: Best = 7.912624e+03  FES = 623100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6240: Best = 7.912624e+03  FES = 624100  Diversity = 5.6444e-14  k_t
= [2, 4, 2, 3]
Generation 6250: Best = 7.912624e+03  FES = 625100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6260: Best = 7.912624e+03  FES = 626100  Diversity = 5.6444e-14  k_t
= [3, 2, 2, 4]
Generation 6270: Best = 7.912624e+03  FES = 627100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6280: Best = 7.912624e+03  FES = 628100  Diversity = 5.6444e-14  k_t
= [2, 3, 2, 4]
Generation 6290: Best = 7.912624e+03  FES = 629100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6300: Best = 7.912624e+03  FES = 630100  Diversity = 5.6444e-14  k_t
= [2, 3, 3, 4]
Generation 6310: Best = 7.912624e+03  FES = 631100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6320: Best = 7.912624e+03  FES = 632100  Diversity = 5.6444e-14  k_t
= [4, 2, 3, 2]
Generation 6330: Best = 7.912624e+03  FES = 633100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6340: Best = 7.912624e+03  FES = 634100  Diversity = 5.6444e-14  k_t
= [4, 2, 3, 3]
Generation 6350: Best = 7.912624e+03  FES = 635100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6360: Best = 7.912624e+03  FES = 636100  Diversity = 5.6444e-14  k_t
= [4, 4, 3, 3]
Generation 6370: Best = 7.912624e+03  FES = 637100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6380: Best = 7.912624e+03  FES = 638100  Diversity = 5.6444e-14  k_t
= [3, 4, 3, 2]
Generation 6390: Best = 7.912624e+03  FES = 639100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6400: Best = 7.912624e+03  FES = 640100  Diversity = 5.6444e-14  k_t
= [2, 4, 3, 4]
Generation 6410: Best = 7.912624e+03  FES = 641100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6420: Best = 7.912624e+03  FES = 642100  Diversity = 5.6444e-14  k_t
= [2, 2, 3, 2]
Generation 6430: Best = 7.912624e+03  FES = 643100  Diversity = 5.6444e-14  k_t

```



```

= [1, 1, 1, 1]
Generation 6440: Best = 7.912624e+03 FES = 644100 Diversity = 5.6444e-14 k_t
= [3, 2, 2, 4]
Generation 6450: Best = 7.912624e+03 FES = 645100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6460: Best = 7.912624e+03 FES = 646100 Diversity = 5.6444e-14 k_t
= [3, 3, 2, 4]
Generation 6470: Best = 7.912624e+03 FES = 647100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6480: Best = 7.912624e+03 FES = 648100 Diversity = 5.6444e-14 k_t
= [4, 2, 4, 2]
Generation 6490: Best = 7.912624e+03 FES = 649100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6500: Best = 7.912624e+03 FES = 650100 Diversity = 5.6444e-14 k_t
= [3, 3, 4, 3]
Generation 6510: Best = 7.912624e+03 FES = 651100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6520: Best = 7.912624e+03 FES = 652100 Diversity = 5.6444e-14 k_t
= [4, 2, 2, 2]
Generation 6530: Best = 7.912624e+03 FES = 653100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6540: Best = 7.912624e+03 FES = 654100 Diversity = 5.6444e-14 k_t
= [4, 2, 3, 3]
Generation 6550: Best = 7.912624e+03 FES = 655100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6560: Best = 7.912624e+03 FES = 656100 Diversity = 5.6444e-14 k_t
= [3, 3, 2, 3]
Generation 6570: Best = 7.912624e+03 FES = 657100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6580: Best = 7.912624e+03 FES = 658100 Diversity = 5.6444e-14 k_t
= [3, 2, 2, 4]
Generation 6590: Best = 7.912624e+03 FES = 659100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6600: Best = 7.912624e+03 FES = 660100 Diversity = 5.6444e-14 k_t
= [4, 2, 4, 3]
Generation 6610: Best = 7.912624e+03 FES = 661100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6620: Best = 7.912624e+03 FES = 662100 Diversity = 5.6444e-14 k_t
= [2, 3, 2, 4]
Generation 6630: Best = 7.912624e+03 FES = 663100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6640: Best = 7.912624e+03 FES = 664100 Diversity = 5.6444e-14 k_t
= [2, 2, 3, 4]
Generation 6650: Best = 7.912624e+03 FES = 665100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6660: Best = 7.912624e+03 FES = 666100 Diversity = 5.6444e-14 k_t
= [3, 4, 3, 3]
Generation 6670: Best = 7.912624e+03 FES = 667100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6680: Best = 7.912624e+03 FES = 668100 Diversity = 5.6444e-14 k_t
= [2, 3, 4, 3]
Generation 6690: Best = 7.912624e+03 FES = 669100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 6700: Best = 7.912624e+03 FES = 670100 Diversity = 5.6444e-14 k_t

```

```

= [3, 3, 3, 2]
Generation 6710: Best = 7.912624e+03  FES = 671100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6720: Best = 7.912624e+03  FES = 672100  Diversity = 5.6444e-14  k_t
= [2, 3, 4, 4]
Generation 6730: Best = 7.912624e+03  FES = 673100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6740: Best = 7.912624e+03  FES = 674100  Diversity = 5.6444e-14  k_t
= [2, 4, 4, 4]
Generation 6750: Best = 7.912624e+03  FES = 675100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6760: Best = 7.912624e+03  FES = 676100  Diversity = 5.6444e-14  k_t
= [3, 2, 2, 2]
Generation 6770: Best = 7.912624e+03  FES = 677100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6780: Best = 7.912624e+03  FES = 678100  Diversity = 5.6444e-14  k_t
= [3, 2, 2, 4]
Generation 6790: Best = 7.912624e+03  FES = 679100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6800: Best = 7.912624e+03  FES = 680100  Diversity = 5.6444e-14  k_t
= [4, 2, 3, 2]
Generation 6810: Best = 7.912624e+03  FES = 681100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6820: Best = 7.912624e+03  FES = 682100  Diversity = 5.6444e-14  k_t
= [2, 2, 2, 4]
Generation 6830: Best = 7.912624e+03  FES = 683100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6840: Best = 7.912624e+03  FES = 684100  Diversity = 5.6444e-14  k_t
= [3, 3, 2, 4]
Generation 6850: Best = 7.912624e+03  FES = 685100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6860: Best = 7.912624e+03  FES = 686100  Diversity = 5.6444e-14  k_t
= [3, 2, 2, 4]
Generation 6870: Best = 7.912624e+03  FES = 687100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6880: Best = 7.912624e+03  FES = 688100  Diversity = 5.6444e-14  k_t
= [4, 2, 3, 3]
Generation 6890: Best = 7.912624e+03  FES = 689100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6900: Best = 7.912624e+03  FES = 690100  Diversity = 5.6444e-14  k_t
= [4, 4, 2, 2]
Generation 6910: Best = 7.912624e+03  FES = 691100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6920: Best = 7.912624e+03  FES = 692100  Diversity = 5.6444e-14  k_t
= [2, 4, 4, 4]
Generation 6930: Best = 7.912624e+03  FES = 693100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6940: Best = 7.912624e+03  FES = 694100  Diversity = 5.6444e-14  k_t
= [3, 3, 3, 3]
Generation 6950: Best = 7.912624e+03  FES = 695100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 6960: Best = 7.912624e+03  FES = 696100  Diversity = 5.6444e-14  k_t
= [3, 3, 3, 3]
Generation 6970: Best = 7.912624e+03  FES = 697100  Diversity = 5.6444e-14  k_t

```

```

= [1, 1, 1, 1]
Generation 6980: Best = 7.912624e+03 FES = 698100 Diversity = 5.6444e-14 k_t
= [4, 3, 3, 2]
Generation 6990: Best = 7.912624e+03 FES = 699100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7000: Best = 7.912624e+03 FES = 700100 Diversity = 5.6444e-14 k_t
= [2, 2, 3, 4]
Generation 7010: Best = 7.912624e+03 FES = 701100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7020: Best = 7.912624e+03 FES = 702100 Diversity = 5.6444e-14 k_t
= [3, 2, 3, 3]
Generation 7030: Best = 7.912624e+03 FES = 703100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7040: Best = 7.912624e+03 FES = 704100 Diversity = 5.6444e-14 k_t
= [4, 3, 4, 4]
Generation 7050: Best = 7.912624e+03 FES = 705100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7060: Best = 7.912624e+03 FES = 706100 Diversity = 5.6444e-14 k_t
= [3, 2, 2, 4]
Generation 7070: Best = 7.912624e+03 FES = 707100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7080: Best = 7.912624e+03 FES = 708100 Diversity = 5.6444e-14 k_t
= [4, 3, 4, 2]
Generation 7090: Best = 7.912624e+03 FES = 709100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7100: Best = 7.912624e+03 FES = 710100 Diversity = 5.6444e-14 k_t
= [2, 4, 4, 4]
Generation 7110: Best = 7.912624e+03 FES = 711100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7120: Best = 7.912624e+03 FES = 712100 Diversity = 5.6444e-14 k_t
= [4, 4, 2, 2]
Generation 7130: Best = 7.912624e+03 FES = 713100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7140: Best = 7.912624e+03 FES = 714100 Diversity = 5.6444e-14 k_t
= [4, 4, 3, 4]
Generation 7150: Best = 7.912624e+03 FES = 715100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7160: Best = 7.912624e+03 FES = 716100 Diversity = 5.6444e-14 k_t
= [4, 2, 2, 4]
Generation 7170: Best = 7.912624e+03 FES = 717100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7180: Best = 7.912624e+03 FES = 718100 Diversity = 5.6444e-14 k_t
= [3, 3, 4, 4]
Generation 7190: Best = 7.912624e+03 FES = 719100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7200: Best = 7.912624e+03 FES = 720100 Diversity = 5.6444e-14 k_t
= [4, 3, 2, 3]
Generation 7210: Best = 7.912624e+03 FES = 721100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7220: Best = 7.912624e+03 FES = 722100 Diversity = 5.6444e-14 k_t
= [4, 2, 2, 2]
Generation 7230: Best = 7.912624e+03 FES = 723100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7240: Best = 7.912624e+03 FES = 724100 Diversity = 5.6444e-14 k_t

```

```

= [2, 4, 2, 3]
Generation 7250: Best = 7.912624e+03 FES = 725100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7260: Best = 7.912624e+03 FES = 726100 Diversity = 5.6444e-14 k_t
= [3, 4, 3, 2]
Generation 7270: Best = 7.912624e+03 FES = 727100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7280: Best = 7.912624e+03 FES = 728100 Diversity = 5.6444e-14 k_t
= [2, 3, 4, 4]
Generation 7290: Best = 7.912624e+03 FES = 729100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7300: Best = 7.912624e+03 FES = 730100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 3]
Generation 7310: Best = 7.912624e+03 FES = 731100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7320: Best = 7.912624e+03 FES = 732100 Diversity = 5.6444e-14 k_t
= [2, 3, 2, 4]
Generation 7330: Best = 7.912624e+03 FES = 733100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7340: Best = 7.912624e+03 FES = 734100 Diversity = 5.6444e-14 k_t
= [4, 4, 4, 3]
Generation 7350: Best = 7.912624e+03 FES = 735100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7360: Best = 7.912624e+03 FES = 736100 Diversity = 5.6444e-14 k_t
= [4, 4, 3, 4]
Generation 7370: Best = 7.912624e+03 FES = 737100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7380: Best = 7.912624e+03 FES = 738100 Diversity = 5.6444e-14 k_t
= [2, 3, 4, 3]
Generation 7390: Best = 7.912624e+03 FES = 739100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7400: Best = 7.912624e+03 FES = 740100 Diversity = 5.6444e-14 k_t
= [2, 3, 2, 2]
Generation 7410: Best = 7.912624e+03 FES = 741100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7420: Best = 7.912624e+03 FES = 742100 Diversity = 5.6444e-14 k_t
= [2, 4, 2, 4]
Generation 7430: Best = 7.912624e+03 FES = 743100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7440: Best = 7.912624e+03 FES = 744100 Diversity = 5.6444e-14 k_t
= [4, 3, 2, 4]
Generation 7450: Best = 7.912624e+03 FES = 745100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7460: Best = 7.912624e+03 FES = 746100 Diversity = 5.6444e-14 k_t
= [3, 3, 2, 4]
Generation 7470: Best = 7.912624e+03 FES = 747100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7480: Best = 7.912624e+03 FES = 748100 Diversity = 5.6444e-14 k_t
= [2, 2, 2, 2]
Generation 7490: Best = 7.912624e+03 FES = 749100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7500: Best = 7.912624e+03 FES = 750100 Diversity = 5.6444e-14 k_t
= [4, 3, 2, 4]
Generation 7510: Best = 7.912624e+03 FES = 751100 Diversity = 5.6444e-14 k_t

```

```

= [1, 1, 1, 1]
Generation 7520: Best = 7.912624e+03 FES = 752100 Diversity = 5.6444e-14 k_t
= [3, 4, 4, 2]
Generation 7530: Best = 7.912624e+03 FES = 753100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7540: Best = 7.912624e+03 FES = 754100 Diversity = 5.6444e-14 k_t
= [3, 3, 3, 3]
Generation 7550: Best = 7.912624e+03 FES = 755100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7560: Best = 7.912624e+03 FES = 756100 Diversity = 5.6444e-14 k_t
= [2, 2, 4, 4]
Generation 7570: Best = 7.912624e+03 FES = 757100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7580: Best = 7.912624e+03 FES = 758100 Diversity = 5.6444e-14 k_t
= [4, 3, 3, 4]
Generation 7590: Best = 7.912624e+03 FES = 759100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7600: Best = 7.912624e+03 FES = 760100 Diversity = 5.6444e-14 k_t
= [4, 3, 4, 2]
Generation 7610: Best = 7.912624e+03 FES = 761100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7620: Best = 7.912624e+03 FES = 762100 Diversity = 5.6444e-14 k_t
= [4, 3, 4, 2]
Generation 7630: Best = 7.912624e+03 FES = 763100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7640: Best = 7.912624e+03 FES = 764100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 2]
Generation 7650: Best = 7.912624e+03 FES = 765100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7660: Best = 7.912624e+03 FES = 766100 Diversity = 5.6444e-14 k_t
= [3, 2, 4, 2]
Generation 7670: Best = 7.912624e+03 FES = 767100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7680: Best = 7.912624e+03 FES = 768100 Diversity = 5.6444e-14 k_t
= [2, 2, 4, 4]
Generation 7690: Best = 7.912624e+03 FES = 769100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7700: Best = 7.912624e+03 FES = 770100 Diversity = 5.6444e-14 k_t
= [3, 3, 3, 2]
Generation 7710: Best = 7.912624e+03 FES = 771100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7720: Best = 7.912624e+03 FES = 772100 Diversity = 5.6444e-14 k_t
= [4, 4, 4, 4]
Generation 7730: Best = 7.912624e+03 FES = 773100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7740: Best = 7.912624e+03 FES = 774100 Diversity = 5.6444e-14 k_t
= [2, 3, 2, 4]
Generation 7750: Best = 7.912624e+03 FES = 775100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7760: Best = 7.912624e+03 FES = 776100 Diversity = 5.6444e-14 k_t
= [2, 4, 4, 4]
Generation 7770: Best = 7.912624e+03 FES = 777100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7780: Best = 7.912624e+03 FES = 778100 Diversity = 5.6444e-14 k_t

```

```

= [4, 4, 4, 4]
Generation 7790: Best = 7.912624e+03 FES = 779100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7800: Best = 7.912624e+03 FES = 780100 Diversity = 5.6444e-14 k_t
= [3, 2, 3, 4]
Generation 7810: Best = 7.912624e+03 FES = 781100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7820: Best = 7.912624e+03 FES = 782100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 3]
Generation 7830: Best = 7.912624e+03 FES = 783100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7840: Best = 7.912624e+03 FES = 784100 Diversity = 5.6444e-14 k_t
= [2, 3, 4, 2]
Generation 7850: Best = 7.912624e+03 FES = 785100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7860: Best = 7.912624e+03 FES = 786100 Diversity = 5.6444e-14 k_t
= [3, 3, 4, 4]
Generation 7870: Best = 7.912624e+03 FES = 787100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7880: Best = 7.912624e+03 FES = 788100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 3]
Generation 7890: Best = 7.912624e+03 FES = 789100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7900: Best = 7.912624e+03 FES = 790100 Diversity = 5.6444e-14 k_t
= [4, 4, 2, 3]
Generation 7910: Best = 7.912624e+03 FES = 791100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7920: Best = 7.912624e+03 FES = 792100 Diversity = 5.6444e-14 k_t
= [4, 3, 3, 4]
Generation 7930: Best = 7.912624e+03 FES = 793100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7940: Best = 7.912624e+03 FES = 794100 Diversity = 5.6444e-14 k_t
= [3, 3, 4, 3]
Generation 7950: Best = 7.912624e+03 FES = 795100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7960: Best = 7.912624e+03 FES = 796100 Diversity = 5.6444e-14 k_t
= [4, 4, 4, 2]
Generation 7970: Best = 7.912624e+03 FES = 797100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 7980: Best = 7.912624e+03 FES = 798100 Diversity = 5.6444e-14 k_t
= [2, 2, 2, 3]
Generation 7990: Best = 7.912624e+03 FES = 799100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8000: Best = 7.912624e+03 FES = 800100 Diversity = 5.6444e-14 k_t
= [2, 3, 2, 3]
Generation 8010: Best = 7.912624e+03 FES = 801100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8020: Best = 7.912624e+03 FES = 802100 Diversity = 5.6444e-14 k_t
= [2, 4, 4, 2]
Generation 8030: Best = 7.912624e+03 FES = 803100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8040: Best = 7.912624e+03 FES = 804100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 3]
Generation 8050: Best = 7.912624e+03 FES = 805100 Diversity = 5.6444e-14 k_t

```

```

= [1, 1, 1, 1]
Generation 8060: Best = 7.912624e+03 FES = 806100 Diversity = 5.6444e-14 k_t
= [4, 2, 4, 2]
Generation 8070: Best = 7.912624e+03 FES = 807100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8080: Best = 7.912624e+03 FES = 808100 Diversity = 5.6444e-14 k_t
= [3, 4, 2, 2]
Generation 8090: Best = 7.912624e+03 FES = 809100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8100: Best = 7.912624e+03 FES = 810100 Diversity = 5.6444e-14 k_t
= [4, 2, 4, 4]
Generation 8110: Best = 7.912624e+03 FES = 811100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8120: Best = 7.912624e+03 FES = 812100 Diversity = 5.6444e-14 k_t
= [3, 3, 3, 4]
Generation 8130: Best = 7.912624e+03 FES = 813100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8140: Best = 7.912624e+03 FES = 814100 Diversity = 5.6444e-14 k_t
= [3, 4, 3, 2]
Generation 8150: Best = 7.912624e+03 FES = 815100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8160: Best = 7.912624e+03 FES = 816100 Diversity = 5.6444e-14 k_t
= [2, 3, 3, 4]
Generation 8170: Best = 7.912624e+03 FES = 817100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8180: Best = 7.912624e+03 FES = 818100 Diversity = 5.6444e-14 k_t
= [2, 2, 2, 4]
Generation 8190: Best = 7.912624e+03 FES = 819100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8200: Best = 7.912624e+03 FES = 820100 Diversity = 5.6444e-14 k_t
= [2, 2, 3, 2]
Generation 8210: Best = 7.912624e+03 FES = 821100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8220: Best = 7.912624e+03 FES = 822100 Diversity = 5.6444e-14 k_t
= [4, 3, 3, 3]
Generation 8230: Best = 7.912624e+03 FES = 823100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8240: Best = 7.912624e+03 FES = 824100 Diversity = 5.6444e-14 k_t
= [4, 4, 4, 3]
Generation 8250: Best = 7.912624e+03 FES = 825100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8260: Best = 7.912624e+03 FES = 826100 Diversity = 5.6444e-14 k_t
= [2, 2, 4, 2]
Generation 8270: Best = 7.912624e+03 FES = 827100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8280: Best = 7.912624e+03 FES = 828100 Diversity = 5.6444e-14 k_t
= [2, 4, 4, 3]
Generation 8290: Best = 7.912624e+03 FES = 829100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8300: Best = 7.912624e+03 FES = 830100 Diversity = 5.6444e-14 k_t
= [4, 2, 2, 2]
Generation 8310: Best = 7.912624e+03 FES = 831100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8320: Best = 7.912624e+03 FES = 832100 Diversity = 5.6444e-14 k_t

```

```

= [3, 2, 3, 3]
Generation 8330: Best = 7.912624e+03 FES = 833100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8340: Best = 7.912624e+03 FES = 834100 Diversity = 5.6444e-14 k_t
= [2, 3, 4, 3]
Generation 8350: Best = 7.912624e+03 FES = 835100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8360: Best = 7.912624e+03 FES = 836100 Diversity = 5.6444e-14 k_t
= [4, 4, 4, 3]
Generation 8370: Best = 7.912624e+03 FES = 837100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8380: Best = 7.912624e+03 FES = 838100 Diversity = 5.6444e-14 k_t
= [2, 2, 3, 3]
Generation 8390: Best = 7.912624e+03 FES = 839100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8400: Best = 7.912624e+03 FES = 840100 Diversity = 5.6444e-14 k_t
= [4, 4, 3, 2]
Generation 8410: Best = 7.912624e+03 FES = 841100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8420: Best = 7.912624e+03 FES = 842100 Diversity = 5.6444e-14 k_t
= [2, 4, 2, 4]
Generation 8430: Best = 7.912624e+03 FES = 843100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8440: Best = 7.912624e+03 FES = 844100 Diversity = 5.6444e-14 k_t
= [3, 3, 3, 3]
Generation 8450: Best = 7.912624e+03 FES = 845100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8460: Best = 7.912624e+03 FES = 846100 Diversity = 5.6444e-14 k_t
= [4, 2, 3, 4]
Generation 8470: Best = 7.912624e+03 FES = 847100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8480: Best = 7.912624e+03 FES = 848100 Diversity = 5.6444e-14 k_t
= [4, 3, 4, 2]
Generation 8490: Best = 7.912624e+03 FES = 849100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8500: Best = 7.912624e+03 FES = 850100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 3]
Generation 8510: Best = 7.912624e+03 FES = 851100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8520: Best = 7.912624e+03 FES = 852100 Diversity = 5.6444e-14 k_t
= [4, 4, 4, 3]
Generation 8530: Best = 7.912624e+03 FES = 853100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8540: Best = 7.912624e+03 FES = 854100 Diversity = 5.6444e-14 k_t
= [4, 3, 3, 3]
Generation 8550: Best = 7.912624e+03 FES = 855100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8560: Best = 7.912624e+03 FES = 856100 Diversity = 5.6444e-14 k_t
= [2, 2, 2, 2]
Generation 8570: Best = 7.912624e+03 FES = 857100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8580: Best = 7.912624e+03 FES = 858100 Diversity = 5.6444e-14 k_t
= [2, 4, 2, 4]
Generation 8590: Best = 7.912624e+03 FES = 859100 Diversity = 5.6444e-14 k_t

```



```

= [1, 1, 1, 1]
Generation 8600: Best = 7.912624e+03 FES = 860100 Diversity = 5.6444e-14 k_t
= [3, 4, 4, 4]
Generation 8610: Best = 7.912624e+03 FES = 861100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8620: Best = 7.912624e+03 FES = 862100 Diversity = 5.6444e-14 k_t
= [2, 3, 3, 4]
Generation 8630: Best = 7.912624e+03 FES = 863100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8640: Best = 7.912624e+03 FES = 864100 Diversity = 5.6444e-14 k_t
= [2, 3, 3, 3]
Generation 8650: Best = 7.912624e+03 FES = 865100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8660: Best = 7.912624e+03 FES = 866100 Diversity = 5.6444e-14 k_t
= [4, 4, 2, 3]
Generation 8670: Best = 7.912624e+03 FES = 867100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8680: Best = 7.912624e+03 FES = 868100 Diversity = 5.6444e-14 k_t
= [2, 2, 4, 4]
Generation 8690: Best = 7.912624e+03 FES = 869100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8700: Best = 7.912624e+03 FES = 870100 Diversity = 5.6444e-14 k_t
= [2, 2, 3, 4]
Generation 8710: Best = 7.912624e+03 FES = 871100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8720: Best = 7.912624e+03 FES = 872100 Diversity = 5.6444e-14 k_t
= [4, 2, 3, 3]
Generation 8730: Best = 7.912624e+03 FES = 873100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8740: Best = 7.912624e+03 FES = 874100 Diversity = 5.6444e-14 k_t
= [3, 4, 4, 4]
Generation 8750: Best = 7.912624e+03 FES = 875100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8760: Best = 7.912624e+03 FES = 876100 Diversity = 5.6444e-14 k_t
= [3, 2, 3, 3]
Generation 8770: Best = 7.912624e+03 FES = 877100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8780: Best = 7.912624e+03 FES = 878100 Diversity = 5.6444e-14 k_t
= [2, 2, 4, 3]
Generation 8790: Best = 7.912624e+03 FES = 879100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8800: Best = 7.912624e+03 FES = 880100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 3]
Generation 8810: Best = 7.912624e+03 FES = 881100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8820: Best = 7.912624e+03 FES = 882100 Diversity = 5.6444e-14 k_t
= [4, 3, 2, 3]
Generation 8830: Best = 7.912624e+03 FES = 883100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8840: Best = 7.912624e+03 FES = 884100 Diversity = 5.6444e-14 k_t
= [3, 3, 3, 2]
Generation 8850: Best = 7.912624e+03 FES = 885100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8860: Best = 7.912624e+03 FES = 886100 Diversity = 5.6444e-14 k_t

```

```

= [2, 4, 4, 4]
Generation 8870: Best = 7.912624e+03 FES = 887100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8880: Best = 7.912624e+03 FES = 888100 Diversity = 5.6444e-14 k_t
= [2, 3, 3, 4]
Generation 8890: Best = 7.912624e+03 FES = 889100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8900: Best = 7.912624e+03 FES = 890100 Diversity = 5.6444e-14 k_t
= [3, 2, 4, 3]
Generation 8910: Best = 7.912624e+03 FES = 891100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8920: Best = 7.912624e+03 FES = 892100 Diversity = 5.6444e-14 k_t
= [2, 4, 2, 3]
Generation 8930: Best = 7.912624e+03 FES = 893100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8940: Best = 7.912624e+03 FES = 894100 Diversity = 5.6444e-14 k_t
= [3, 3, 4, 2]
Generation 8950: Best = 7.912624e+03 FES = 895100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8960: Best = 7.912624e+03 FES = 896100 Diversity = 5.6444e-14 k_t
= [4, 3, 2, 4]
Generation 8970: Best = 7.912624e+03 FES = 897100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 8980: Best = 7.912624e+03 FES = 898100 Diversity = 5.6444e-14 k_t
= [2, 2, 4, 2]
Generation 8990: Best = 7.912624e+03 FES = 899100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9000: Best = 7.912624e+03 FES = 900100 Diversity = 5.6444e-14 k_t
= [3, 2, 4, 3]
Generation 9010: Best = 7.912624e+03 FES = 901100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9020: Best = 7.912624e+03 FES = 902100 Diversity = 5.6444e-14 k_t
= [2, 4, 4, 2]
Generation 9030: Best = 7.912624e+03 FES = 903100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9040: Best = 7.912624e+03 FES = 904100 Diversity = 5.6444e-14 k_t
= [3, 2, 4, 4]
Generation 9050: Best = 7.912624e+03 FES = 905100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9060: Best = 7.912624e+03 FES = 906100 Diversity = 5.6444e-14 k_t
= [3, 2, 2, 3]
Generation 9070: Best = 7.912624e+03 FES = 907100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9080: Best = 7.912624e+03 FES = 908100 Diversity = 5.6444e-14 k_t
= [2, 3, 3, 4]
Generation 9090: Best = 7.912624e+03 FES = 909100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9100: Best = 7.912624e+03 FES = 910100 Diversity = 5.6444e-14 k_t
= [3, 3, 4, 3]
Generation 9110: Best = 7.912624e+03 FES = 911100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9120: Best = 7.912624e+03 FES = 912100 Diversity = 5.6444e-14 k_t
= [2, 4, 3, 2]
Generation 9130: Best = 7.912624e+03 FES = 913100 Diversity = 5.6444e-14 k_t

```

```

= [1, 1, 1, 1]
Generation 9140: Best = 7.912624e+03  FES = 914100  Diversity = 5.6444e-14  k_t
= [3, 3, 4, 2]
Generation 9150: Best = 7.912624e+03  FES = 915100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9160: Best = 7.912624e+03  FES = 916100  Diversity = 5.6444e-14  k_t
= [3, 4, 3, 3]
Generation 9170: Best = 7.912624e+03  FES = 917100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9180: Best = 7.912624e+03  FES = 918100  Diversity = 5.6444e-14  k_t
= [3, 2, 3, 2]
Generation 9190: Best = 7.912624e+03  FES = 919100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9200: Best = 7.912624e+03  FES = 920100  Diversity = 5.6444e-14  k_t
= [3, 4, 4, 2]
Generation 9210: Best = 7.912624e+03  FES = 921100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9220: Best = 7.912624e+03  FES = 922100  Diversity = 5.6444e-14  k_t
= [2, 4, 3, 3]
Generation 9230: Best = 7.912624e+03  FES = 923100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9240: Best = 7.912624e+03  FES = 924100  Diversity = 5.6444e-14  k_t
= [2, 4, 2, 3]
Generation 9250: Best = 7.912624e+03  FES = 925100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9260: Best = 7.912624e+03  FES = 926100  Diversity = 5.6444e-14  k_t
= [2, 3, 3, 2]
Generation 9270: Best = 7.912624e+03  FES = 927100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9280: Best = 7.912624e+03  FES = 928100  Diversity = 5.6444e-14  k_t
= [3, 4, 2, 2]
Generation 9290: Best = 7.912624e+03  FES = 929100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9300: Best = 7.912624e+03  FES = 930100  Diversity = 5.6444e-14  k_t
= [4, 3, 4, 4]
Generation 9310: Best = 7.912624e+03  FES = 931100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9320: Best = 7.912624e+03  FES = 932100  Diversity = 5.6444e-14  k_t
= [2, 4, 4, 3]
Generation 9330: Best = 7.912624e+03  FES = 933100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9340: Best = 7.912624e+03  FES = 934100  Diversity = 5.6444e-14  k_t
= [2, 3, 3, 2]
Generation 9350: Best = 7.912624e+03  FES = 935100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9360: Best = 7.912624e+03  FES = 936100  Diversity = 5.6444e-14  k_t
= [2, 3, 2, 4]
Generation 9370: Best = 7.912624e+03  FES = 937100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9380: Best = 7.912624e+03  FES = 938100  Diversity = 5.6444e-14  k_t
= [3, 3, 2, 4]
Generation 9390: Best = 7.912624e+03  FES = 939100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9400: Best = 7.912624e+03  FES = 940100  Diversity = 5.6444e-14  k_t

```

```

= [3, 4, 2, 2]
Generation 9410: Best = 7.912624e+03  FES = 941100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9420: Best = 7.912624e+03  FES = 942100  Diversity = 5.6444e-14  k_t
= [4, 4, 4, 3]
Generation 9430: Best = 7.912624e+03  FES = 943100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9440: Best = 7.912624e+03  FES = 944100  Diversity = 5.6444e-14  k_t
= [4, 2, 2, 2]
Generation 9450: Best = 7.912624e+03  FES = 945100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9460: Best = 7.912624e+03  FES = 946100  Diversity = 5.6444e-14  k_t
= [3, 2, 3, 2]
Generation 9470: Best = 7.912624e+03  FES = 947100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9480: Best = 7.912624e+03  FES = 948100  Diversity = 5.6444e-14  k_t
= [3, 3, 4, 2]
Generation 9490: Best = 7.912624e+03  FES = 949100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9500: Best = 7.912624e+03  FES = 950100  Diversity = 5.6444e-14  k_t
= [2, 2, 2, 4]
Generation 9510: Best = 7.912624e+03  FES = 951100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9520: Best = 7.912624e+03  FES = 952100  Diversity = 5.6444e-14  k_t
= [4, 2, 3, 2]
Generation 9530: Best = 7.912624e+03  FES = 953100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9540: Best = 7.912624e+03  FES = 954100  Diversity = 5.6444e-14  k_t
= [3, 3, 2, 3]
Generation 9550: Best = 7.912624e+03  FES = 955100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9560: Best = 7.912624e+03  FES = 956100  Diversity = 5.6444e-14  k_t
= [4, 3, 2, 2]
Generation 9570: Best = 7.912624e+03  FES = 957100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9580: Best = 7.912624e+03  FES = 958100  Diversity = 5.6444e-14  k_t
= [2, 3, 2, 4]
Generation 9590: Best = 7.912624e+03  FES = 959100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9600: Best = 7.912624e+03  FES = 960100  Diversity = 5.6444e-14  k_t
= [4, 3, 4, 3]
Generation 9610: Best = 7.912624e+03  FES = 961100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9620: Best = 7.912624e+03  FES = 962100  Diversity = 5.6444e-14  k_t
= [3, 3, 3, 4]
Generation 9630: Best = 7.912624e+03  FES = 963100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9640: Best = 7.912624e+03  FES = 964100  Diversity = 5.6444e-14  k_t
= [3, 3, 3, 2]
Generation 9650: Best = 7.912624e+03  FES = 965100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9660: Best = 7.912624e+03  FES = 966100  Diversity = 5.6444e-14  k_t
= [3, 2, 3, 2]
Generation 9670: Best = 7.912624e+03  FES = 967100  Diversity = 5.6444e-14  k_t

```

```

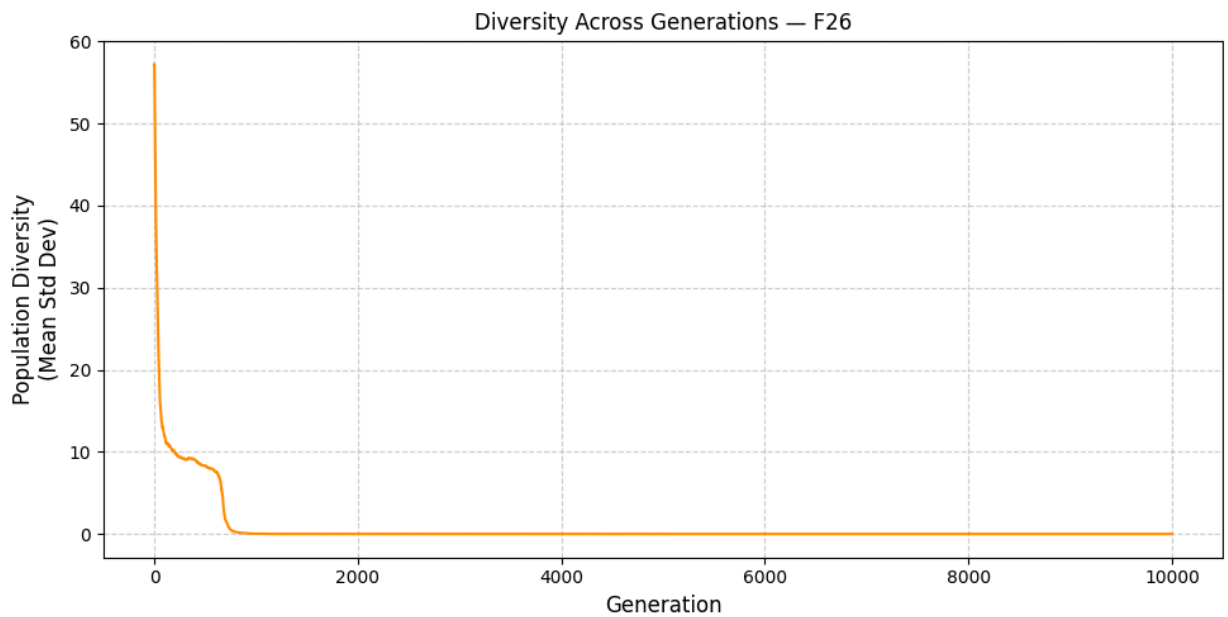
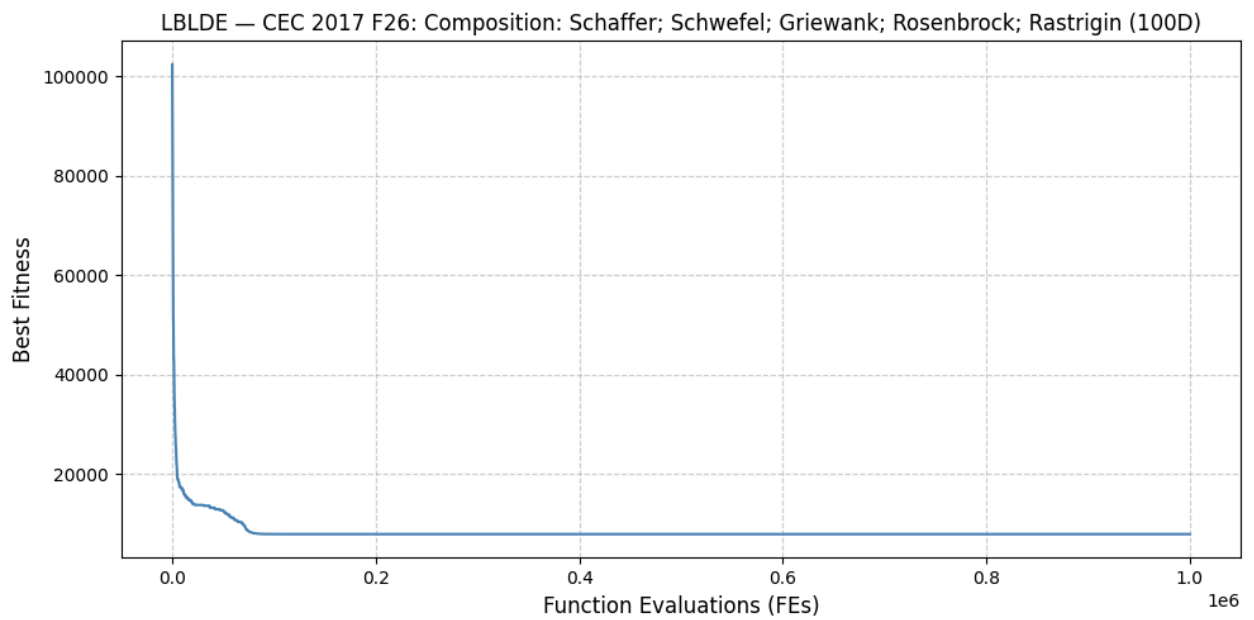
= [1, 1, 1, 1]
Generation 9680: Best = 7.912624e+03 FES = 968100 Diversity = 5.6444e-14 k_t
= [2, 2, 4, 3]
Generation 9690: Best = 7.912624e+03 FES = 969100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9700: Best = 7.912624e+03 FES = 970100 Diversity = 5.6444e-14 k_t
= [4, 2, 2, 4]
Generation 9710: Best = 7.912624e+03 FES = 971100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9720: Best = 7.912624e+03 FES = 972100 Diversity = 5.6444e-14 k_t
= [4, 3, 4, 4]
Generation 9730: Best = 7.912624e+03 FES = 973100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9740: Best = 7.912624e+03 FES = 974100 Diversity = 5.6444e-14 k_t
= [2, 3, 3, 3]
Generation 9750: Best = 7.912624e+03 FES = 975100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9760: Best = 7.912624e+03 FES = 976100 Diversity = 5.6444e-14 k_t
= [4, 4, 4, 4]
Generation 9770: Best = 7.912624e+03 FES = 977100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9780: Best = 7.912624e+03 FES = 978100 Diversity = 5.6444e-14 k_t
= [3, 4, 2, 4]
Generation 9790: Best = 7.912624e+03 FES = 979100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9800: Best = 7.912624e+03 FES = 980100 Diversity = 5.6444e-14 k_t
= [2, 2, 4, 4]
Generation 9810: Best = 7.912624e+03 FES = 981100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9820: Best = 7.912624e+03 FES = 982100 Diversity = 5.6444e-14 k_t
= [2, 4, 4, 4]
Generation 9830: Best = 7.912624e+03 FES = 983100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9840: Best = 7.912624e+03 FES = 984100 Diversity = 5.6444e-14 k_t
= [3, 3, 2, 2]
Generation 9850: Best = 7.912624e+03 FES = 985100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9860: Best = 7.912624e+03 FES = 986100 Diversity = 5.6444e-14 k_t
= [4, 2, 2, 3]
Generation 9870: Best = 7.912624e+03 FES = 987100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9880: Best = 7.912624e+03 FES = 988100 Diversity = 5.6444e-14 k_t
= [3, 4, 3, 4]
Generation 9890: Best = 7.912624e+03 FES = 989100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9900: Best = 7.912624e+03 FES = 990100 Diversity = 5.6444e-14 k_t
= [4, 2, 4, 3]
Generation 9910: Best = 7.912624e+03 FES = 991100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9920: Best = 7.912624e+03 FES = 992100 Diversity = 5.6444e-14 k_t
= [3, 4, 2, 2]
Generation 9930: Best = 7.912624e+03 FES = 993100 Diversity = 5.6444e-14 k_t
= [1, 1, 1, 1]
Generation 9940: Best = 7.912624e+03 FES = 994100 Diversity = 5.6444e-14 k_t

```

```
= [4, 3, 2, 2]
Generation 9950: Best = 7.912624e+03  FES = 995100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9960: Best = 7.912624e+03  FES = 996100  Diversity = 5.6444e-14  k_t
= [3, 3, 2, 3]
Generation 9970: Best = 7.912624e+03  FES = 997100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
Generation 9980: Best = 7.912624e+03  FES = 998100  Diversity = 5.6444e-14  k_t
= [2, 4, 3, 3]
Generation 9990: Best = 7.912624e+03  FES = 999100  Diversity = 5.6444e-14  k_t
= [1, 1, 1, 1]
```

Optimization completed!
Final Best Fitness = 7.912624e+03
Total Function Evaluations = 1000000

```
=====
=
Results:
  Best Fitness      : 7.912624e+03
  Global Optimum    : 2.600000e+03
  Error (|best-optimum|) : 5.312624e+03
  Best Solution (5 dim) : [ -7.25763221 -72.0570703 -27.09150451 -65.5603359
-8.77470993]
  D^(0) (init. diversity): 5.7220e+01
=====
=
```



Graph saved as convergence_F26.png

```
In [51]: # =====
# F27 — Composition Function 7 (HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic)
# =====
run_and_plot('F27', make_wrapper(composition.f27), 2700.0, 'Composition: HGBat')
```

```

=====
=
LBLDE_s on CEC 2017 F27 – Single Run
=====
=
Function      : Composition: HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic;
Schaffer (100D)
Bounds       : [-100, 100]
Global optimum : 2700.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 1.447050e+04 | Diversity = 5.7220e+01
Generation 10: Best = 9.939217e+03 FES = 1100 Diversity = 5.0224e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 7.526597e+03 FES = 2100 Diversity = 4.6464e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 5.845853e+03 FES = 3100 Diversity = 4.1254e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 5.331866e+03 FES = 4100 Diversity = 3.5847e+01 k_t
= [4, 3, 4, 2]
Generation 50: Best = 5.175492e+03 FES = 5100 Diversity = 3.2138e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 4.847223e+03 FES = 6100 Diversity = 2.8687e+01 k_t
= [4, 4, 2, 3]
Generation 70: Best = 4.584629e+03 FES = 7100 Diversity = 2.5460e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 4.456302e+03 FES = 8100 Diversity = 2.1772e+01 k_t
= [4, 3, 2, 3]
Generation 90: Best = 4.332801e+03 FES = 9100 Diversity = 2.0277e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 4.218228e+03 FES = 10100 Diversity = 1.7963e+01 k_t
= [3, 3, 2, 3]
Generation 110: Best = 4.088238e+03 FES = 11100 Diversity = 1.5259e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 4.026838e+03 FES = 12100 Diversity = 1.2845e+01 k_t
= [4, 2, 2, 3]
Generation 130: Best = 3.971500e+03 FES = 13100 Diversity = 1.1762e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 3.919923e+03 FES = 14100 Diversity = 1.0552e+01 k_t
= [4, 4, 3, 4]
Generation 150: Best = 3.918798e+03 FES = 15100 Diversity = 9.6715e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 3.880090e+03 FES = 16100 Diversity = 9.3082e+00 k_t
= [3, 3, 4, 4]
Generation 170: Best = 3.850579e+03 FES = 17100 Diversity = 8.7899e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.833667e+03 FES = 18100 Diversity = 8.6125e+00 k_t
= [2, 3, 3, 2]
Generation 190: Best = 3.793201e+03 FES = 19100 Diversity = 8.3615e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 3.767135e+03 FES = 20100 Diversity = 7.9892e+00 k_t
= [2, 3, 3, 2]
Generation 210: Best = 3.741172e+03 FES = 21100 Diversity = 7.9531e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 3.708297e+03 FES = 22100 Diversity = 7.9348e+00 k_t

```



```

= [4, 3, 4, 3]
Generation 230: Best = 3.687678e+03 FES = 23100 Diversity = 7.5643e+00 k_t
= [1, 1, 1, 1]
Generation 240: Best = 3.673258e+03 FES = 24100 Diversity = 7.6456e+00 k_t
= [4, 3, 3, 4]
Generation 250: Best = 3.667750e+03 FES = 25100 Diversity = 7.7980e+00 k_t
= [1, 1, 1, 1]
Generation 260: Best = 3.648325e+03 FES = 26100 Diversity = 7.4459e+00 k_t
= [2, 3, 4, 3]
Generation 270: Best = 3.623430e+03 FES = 27100 Diversity = 7.1033e+00 k_t
= [1, 1, 1, 1]
Generation 280: Best = 3.612885e+03 FES = 28100 Diversity = 6.4521e+00 k_t
= [4, 2, 2, 2]
Generation 290: Best = 3.599329e+03 FES = 29100 Diversity = 6.1746e+00 k_t
= [1, 1, 1, 1]
Generation 300: Best = 3.593590e+03 FES = 30100 Diversity = 5.6853e+00 k_t
= [4, 3, 3, 3]
Generation 310: Best = 3.579031e+03 FES = 31100 Diversity = 5.4389e+00 k_t
= [1, 1, 1, 1]
Generation 320: Best = 3.565079e+03 FES = 32100 Diversity = 5.2566e+00 k_t
= [3, 4, 4, 2]
Generation 330: Best = 3.556469e+03 FES = 33100 Diversity = 4.9881e+00 k_t
= [1, 1, 1, 1]
Generation 340: Best = 3.544339e+03 FES = 34100 Diversity = 5.0500e+00 k_t
= [3, 2, 3, 3]
Generation 350: Best = 3.532466e+03 FES = 35100 Diversity = 4.6553e+00 k_t
= [1, 1, 1, 1]
Generation 360: Best = 3.529391e+03 FES = 36100 Diversity = 4.3644e+00 k_t
= [2, 3, 4, 4]
Generation 370: Best = 3.520425e+03 FES = 37100 Diversity = 4.0061e+00 k_t
= [1, 1, 1, 1]
Generation 380: Best = 3.501127e+03 FES = 38100 Diversity = 3.8014e+00 k_t
= [2, 3, 3, 2]
Generation 390: Best = 3.499037e+03 FES = 39100 Diversity = 3.4767e+00 k_t
= [1, 1, 1, 1]
Generation 400: Best = 3.494230e+03 FES = 40100 Diversity = 3.2977e+00 k_t
= [2, 2, 2, 2]
Generation 410: Best = 3.489735e+03 FES = 41100 Diversity = 3.1430e+00 k_t
= [1, 1, 1, 1]
Generation 420: Best = 3.487497e+03 FES = 42100 Diversity = 2.8736e+00 k_t
= [2, 2, 3, 2]
Generation 430: Best = 3.477532e+03 FES = 43100 Diversity = 2.5521e+00 k_t
= [1, 1, 1, 1]
Generation 440: Best = 3.470942e+03 FES = 44100 Diversity = 2.2981e+00 k_t
= [2, 4, 2, 3]
Generation 450: Best = 3.465463e+03 FES = 45100 Diversity = 2.0697e+00 k_t
= [1, 1, 1, 1]
Generation 460: Best = 3.461307e+03 FES = 46100 Diversity = 1.9419e+00 k_t
= [3, 4, 4, 2]
Generation 470: Best = 3.453381e+03 FES = 47100 Diversity = 1.7606e+00 k_t
= [1, 1, 1, 1]
Generation 480: Best = 3.450142e+03 FES = 48100 Diversity = 1.5459e+00 k_t
= [3, 2, 3, 2]
Generation 490: Best = 3.442327e+03 FES = 49100 Diversity = 1.3892e+00 k_t

```

```

= [1, 1, 1, 1]
Generation 500: Best = 3.438366e+03 FES = 50100 Diversity = 1.2123e+00 k_t
= [2, 3, 3, 4]
Generation 510: Best = 3.433495e+03 FES = 51100 Diversity = 1.0795e+00 k_t
= [1, 1, 1, 1]
Generation 520: Best = 3.429857e+03 FES = 52100 Diversity = 9.7095e-01 k_t
= [4, 2, 2, 2]
Generation 530: Best = 3.427606e+03 FES = 53100 Diversity = 8.5615e-01 k_t
= [1, 1, 1, 1]
Generation 540: Best = 3.424493e+03 FES = 54100 Diversity = 7.9173e-01 k_t
= [4, 4, 3, 2]
Generation 550: Best = 3.421174e+03 FES = 55100 Diversity = 7.3683e-01 k_t
= [1, 1, 1, 1]
Generation 560: Best = 3.419472e+03 FES = 56100 Diversity = 6.8382e-01 k_t
= [4, 4, 4, 4]
Generation 570: Best = 3.417754e+03 FES = 57100 Diversity = 6.3651e-01 k_t
= [1, 1, 1, 1]
Generation 580: Best = 3.416083e+03 FES = 58100 Diversity = 5.8798e-01 k_t
= [4, 4, 2, 4]
Generation 590: Best = 3.414768e+03 FES = 59100 Diversity = 5.5473e-01 k_t
= [1, 1, 1, 1]
Generation 600: Best = 3.413692e+03 FES = 60100 Diversity = 5.0121e-01 k_t
= [2, 2, 4, 2]
Generation 610: Best = 3.412317e+03 FES = 61100 Diversity = 4.4221e-01 k_t
= [1, 1, 1, 1]
Generation 620: Best = 3.410542e+03 FES = 62100 Diversity = 3.9845e-01 k_t
= [2, 3, 2, 2]
Generation 630: Best = 3.409220e+03 FES = 63100 Diversity = 3.8785e-01 k_t
= [1, 1, 1, 1]
Generation 640: Best = 3.408021e+03 FES = 64100 Diversity = 3.7052e-01 k_t
= [3, 4, 3, 4]
Generation 650: Best = 3.407052e+03 FES = 65100 Diversity = 3.4880e-01 k_t
= [1, 1, 1, 1]
Generation 660: Best = 3.406273e+03 FES = 66100 Diversity = 3.0627e-01 k_t
= [3, 4, 3, 2]
Generation 670: Best = 3.405620e+03 FES = 67100 Diversity = 2.6939e-01 k_t
= [1, 1, 1, 1]
Generation 680: Best = 3.404849e+03 FES = 68100 Diversity = 2.4640e-01 k_t
= [4, 4, 3, 4]
Generation 690: Best = 3.404528e+03 FES = 69100 Diversity = 2.2296e-01 k_t
= [1, 1, 1, 1]
Generation 700: Best = 3.403768e+03 FES = 70100 Diversity = 2.0863e-01 k_t
= [3, 2, 2, 4]
Generation 710: Best = 3.403231e+03 FES = 71100 Diversity = 2.1906e-01 k_t
= [1, 1, 1, 1]
Generation 720: Best = 3.402771e+03 FES = 72100 Diversity = 2.3847e-01 k_t
= [2, 2, 4, 4]
Generation 730: Best = 3.402288e+03 FES = 73100 Diversity = 2.3658e-01 k_t
= [1, 1, 1, 1]
Generation 740: Best = 3.402250e+03 FES = 74100 Diversity = 2.0654e-01 k_t
= [2, 4, 4, 4]
Generation 750: Best = 3.401942e+03 FES = 75100 Diversity = 1.7586e-01 k_t
= [1, 1, 1, 1]
Generation 760: Best = 3.401759e+03 FES = 76100 Diversity = 1.4999e-01 k_t

```

```

= [3, 4, 4, 4]
Generation 770: Best = 3.401511e+03 FES = 77100 Diversity = 1.3204e-01 k_t
= [1, 1, 1, 1]
Generation 780: Best = 3.401401e+03 FES = 78100 Diversity = 1.1364e-01 k_t
= [4, 4, 3, 4]
Generation 790: Best = 3.401303e+03 FES = 79100 Diversity = 1.0246e-01 k_t
= [1, 1, 1, 1]
Generation 800: Best = 3.401215e+03 FES = 80100 Diversity = 8.8160e-02 k_t
= [2, 3, 4, 2]
Generation 810: Best = 3.401139e+03 FES = 81100 Diversity = 7.8498e-02 k_t
= [1, 1, 1, 1]
Generation 820: Best = 3.401084e+03 FES = 82100 Diversity = 7.5026e-02 k_t
= [3, 3, 4, 2]
Generation 830: Best = 3.401038e+03 FES = 83100 Diversity = 6.8927e-02 k_t
= [1, 1, 1, 1]
Generation 840: Best = 3.401006e+03 FES = 84100 Diversity = 5.9990e-02 k_t
= [2, 4, 4, 2]
Generation 850: Best = 3.400976e+03 FES = 85100 Diversity = 5.5612e-02 k_t
= [1, 1, 1, 1]
Generation 860: Best = 3.400941e+03 FES = 86100 Diversity = 5.0665e-02 k_t
= [4, 4, 2, 2]
Generation 870: Best = 3.400918e+03 FES = 87100 Diversity = 4.4054e-02 k_t
= [1, 1, 1, 1]
Generation 880: Best = 3.400898e+03 FES = 88100 Diversity = 4.0615e-02 k_t
= [4, 3, 4, 3]
Generation 890: Best = 3.400878e+03 FES = 89100 Diversity = 3.5874e-02 k_t
= [1, 1, 1, 1]
Generation 900: Best = 3.400863e+03 FES = 90100 Diversity = 3.4350e-02 k_t
= [2, 4, 4, 2]
Generation 910: Best = 3.400850e+03 FES = 91100 Diversity = 3.4500e-02 k_t
= [1, 1, 1, 1]
Generation 920: Best = 3.400842e+03 FES = 92100 Diversity = 3.2445e-02 k_t
= [2, 2, 3, 4]
Generation 930: Best = 3.400829e+03 FES = 93100 Diversity = 3.0881e-02 k_t
= [1, 1, 1, 1]
Generation 940: Best = 3.400822e+03 FES = 94100 Diversity = 3.1467e-02 k_t
= [2, 4, 4, 2]
Generation 950: Best = 3.400814e+03 FES = 95100 Diversity = 2.8299e-02 k_t
= [1, 1, 1, 1]
Generation 960: Best = 3.400806e+03 FES = 96100 Diversity = 2.6794e-02 k_t
= [3, 3, 2, 4]
Generation 970: Best = 3.400800e+03 FES = 97100 Diversity = 2.5071e-02 k_t
= [1, 1, 1, 1]
Generation 980: Best = 3.400793e+03 FES = 98100 Diversity = 2.4472e-02 k_t
= [2, 2, 2, 2]
Generation 990: Best = 3.400787e+03 FES = 99100 Diversity = 2.2989e-02 k_t
= [1, 1, 1, 1]
Generation 1000: Best = 3.400781e+03 FES = 100100 Diversity = 2.2161e-02 k_t
= [3, 3, 4, 4]
Generation 1010: Best = 3.400772e+03 FES = 101100 Diversity = 2.3614e-02 k_t
= [1, 1, 1, 1]
Generation 1020: Best = 3.400767e+03 FES = 102100 Diversity = 2.3952e-02 k_t
= [2, 4, 4, 2]
Generation 1030: Best = 3.400762e+03 FES = 103100 Diversity = 2.3065e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 1040: Best = 3.400759e+03 FES = 104100 Diversity = 2.2333e-02 k_t
= [3, 2, 4, 4]
Generation 1050: Best = 3.400756e+03 FES = 105100 Diversity = 2.0575e-02 k_t
= [1, 1, 1, 1]
Generation 1060: Best = 3.400751e+03 FES = 106100 Diversity = 2.0374e-02 k_t
= [2, 3, 4, 4]
Generation 1070: Best = 3.400748e+03 FES = 107100 Diversity = 2.0214e-02 k_t
= [1, 1, 1, 1]
Generation 1080: Best = 3.400745e+03 FES = 108100 Diversity = 1.9852e-02 k_t
= [4, 4, 2, 4]
Generation 1090: Best = 3.400740e+03 FES = 109100 Diversity = 1.9743e-02 k_t
= [1, 1, 1, 1]
Generation 1100: Best = 3.400736e+03 FES = 110100 Diversity = 1.9735e-02 k_t
= [4, 4, 2, 2]
Generation 1110: Best = 3.400728e+03 FES = 111100 Diversity = 2.2145e-02 k_t
= [1, 1, 1, 1]
Generation 1120: Best = 3.400723e+03 FES = 112100 Diversity = 2.2487e-02 k_t
= [4, 3, 2, 4]
Generation 1130: Best = 3.400718e+03 FES = 113100 Diversity = 2.1725e-02 k_t
= [1, 1, 1, 1]
Generation 1140: Best = 3.400715e+03 FES = 114100 Diversity = 2.0666e-02 k_t
= [2, 2, 3, 2]
Generation 1150: Best = 3.400711e+03 FES = 115100 Diversity = 1.9325e-02 k_t
= [1, 1, 1, 1]
Generation 1160: Best = 3.400708e+03 FES = 116100 Diversity = 1.8476e-02 k_t
= [3, 2, 4, 4]
Generation 1170: Best = 3.400705e+03 FES = 117100 Diversity = 1.7342e-02 k_t
= [1, 1, 1, 1]
Generation 1180: Best = 3.400701e+03 FES = 118100 Diversity = 1.7053e-02 k_t
= [4, 3, 4, 2]
Generation 1190: Best = 3.400700e+03 FES = 119100 Diversity = 1.6337e-02 k_t
= [1, 1, 1, 1]
Generation 1200: Best = 3.400699e+03 FES = 120100 Diversity = 1.5130e-02 k_t
= [3, 4, 2, 2]
Generation 1210: Best = 3.400697e+03 FES = 121100 Diversity = 1.3671e-02 k_t
= [1, 1, 1, 1]
Generation 1220: Best = 3.400696e+03 FES = 122100 Diversity = 1.2777e-02 k_t
= [3, 4, 2, 4]
Generation 1230: Best = 3.400694e+03 FES = 123100 Diversity = 1.2658e-02 k_t
= [1, 1, 1, 1]
Generation 1240: Best = 3.400692e+03 FES = 124100 Diversity = 1.2479e-02 k_t
= [4, 3, 2, 2]
Generation 1250: Best = 3.400691e+03 FES = 125100 Diversity = 1.1570e-02 k_t
= [1, 1, 1, 1]
Generation 1260: Best = 3.400690e+03 FES = 126100 Diversity = 1.1468e-02 k_t
= [2, 4, 2, 4]
Generation 1270: Best = 3.400689e+03 FES = 127100 Diversity = 1.1216e-02 k_t
= [1, 1, 1, 1]
Generation 1280: Best = 3.400688e+03 FES = 128100 Diversity = 1.0572e-02 k_t
= [4, 3, 2, 3]
Generation 1290: Best = 3.400688e+03 FES = 129100 Diversity = 1.0008e-02 k_t
= [1, 1, 1, 1]
Generation 1300: Best = 3.400687e+03 FES = 130100 Diversity = 8.9714e-03 k_t

```

```

= [3, 2, 4, 4]
Generation 1310: Best = 3.400687e+03 FES = 131100 Diversity = 8.7304e-03 k_t
= [1, 1, 1, 1]
Generation 1320: Best = 3.400686e+03 FES = 132100 Diversity = 7.6862e-03 k_t
= [4, 2, 3, 2]
Generation 1330: Best = 3.400686e+03 FES = 133100 Diversity = 6.9283e-03 k_t
= [1, 1, 1, 1]
Generation 1340: Best = 3.400685e+03 FES = 134100 Diversity = 6.4321e-03 k_t
= [3, 4, 2, 2]
Generation 1350: Best = 3.400685e+03 FES = 135100 Diversity = 5.8804e-03 k_t
= [1, 1, 1, 1]
Generation 1360: Best = 3.400685e+03 FES = 136100 Diversity = 5.5877e-03 k_t
= [2, 2, 2, 4]
Generation 1370: Best = 3.400684e+03 FES = 137100 Diversity = 5.2213e-03 k_t
= [1, 1, 1, 1]
Generation 1380: Best = 3.400684e+03 FES = 138100 Diversity = 5.3056e-03 k_t
= [3, 2, 4, 3]
Generation 1390: Best = 3.400684e+03 FES = 139100 Diversity = 5.3597e-03 k_t
= [1, 1, 1, 1]
Generation 1400: Best = 3.400684e+03 FES = 140100 Diversity = 5.2426e-03 k_t
= [3, 2, 3, 2]
Generation 1410: Best = 3.400683e+03 FES = 141100 Diversity = 5.1305e-03 k_t
= [1, 1, 1, 1]
Generation 1420: Best = 3.400683e+03 FES = 142100 Diversity = 4.9967e-03 k_t
= [4, 3, 3, 2]
Generation 1430: Best = 3.400683e+03 FES = 143100 Diversity = 4.7492e-03 k_t
= [1, 1, 1, 1]
Generation 1440: Best = 3.400682e+03 FES = 144100 Diversity = 4.9605e-03 k_t
= [2, 4, 3, 2]
Generation 1450: Best = 3.400682e+03 FES = 145100 Diversity = 5.0078e-03 k_t
= [1, 1, 1, 1]
Generation 1460: Best = 3.400682e+03 FES = 146100 Diversity = 4.7788e-03 k_t
= [2, 3, 3, 4]
Generation 1470: Best = 3.400682e+03 FES = 147100 Diversity = 4.5373e-03 k_t
= [1, 1, 1, 1]
Generation 1480: Best = 3.400682e+03 FES = 148100 Diversity = 4.3467e-03 k_t
= [3, 2, 2, 3]
Generation 1490: Best = 3.400681e+03 FES = 149100 Diversity = 4.3060e-03 k_t
= [1, 1, 1, 1]
Generation 1500: Best = 3.400681e+03 FES = 150100 Diversity = 4.3245e-03 k_t
= [3, 3, 4, 3]
Generation 1510: Best = 3.400681e+03 FES = 151100 Diversity = 4.2140e-03 k_t
= [1, 1, 1, 1]
Generation 1520: Best = 3.400681e+03 FES = 152100 Diversity = 4.0593e-03 k_t
= [3, 2, 4, 4]
Generation 1530: Best = 3.400681e+03 FES = 153100 Diversity = 3.8564e-03 k_t
= [1, 1, 1, 1]
Generation 1540: Best = 3.400681e+03 FES = 154100 Diversity = 3.6069e-03 k_t
= [2, 4, 4, 2]
Generation 1550: Best = 3.400681e+03 FES = 155100 Diversity = 3.3121e-03 k_t
= [1, 1, 1, 1]
Generation 1560: Best = 3.400681e+03 FES = 156100 Diversity = 3.2726e-03 k_t
= [3, 3, 3, 4]
Generation 1570: Best = 3.400680e+03 FES = 157100 Diversity = 3.2753e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 1580: Best = 3.400680e+03 FES = 158100 Diversity = 3.2741e-03 k_t
= [4, 2, 3, 2]
Generation 1590: Best = 3.400680e+03 FES = 159100 Diversity = 3.1649e-03 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 3.400680e+03 FES = 160100 Diversity = 3.1232e-03 k_t
= [4, 2, 4, 4]
Generation 1610: Best = 3.400680e+03 FES = 161100 Diversity = 3.0755e-03 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 3.400680e+03 FES = 162100 Diversity = 2.9530e-03 k_t
= [4, 4, 4, 3]
Generation 1630: Best = 3.400680e+03 FES = 163100 Diversity = 2.9533e-03 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 3.400680e+03 FES = 164100 Diversity = 2.9659e-03 k_t
= [2, 3, 4, 4]
Generation 1650: Best = 3.400680e+03 FES = 165100 Diversity = 2.9838e-03 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 3.400680e+03 FES = 166100 Diversity = 2.9407e-03 k_t
= [4, 2, 3, 2]
Generation 1670: Best = 3.400680e+03 FES = 167100 Diversity = 2.9624e-03 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 3.400680e+03 FES = 168100 Diversity = 2.8248e-03 k_t
= [4, 3, 4, 2]
Generation 1690: Best = 3.400679e+03 FES = 169100 Diversity = 2.6128e-03 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 3.400679e+03 FES = 170100 Diversity = 2.5121e-03 k_t
= [2, 3, 3, 2]
Generation 1710: Best = 3.400679e+03 FES = 171100 Diversity = 2.5920e-03 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 3.400679e+03 FES = 172100 Diversity = 2.5275e-03 k_t
= [2, 3, 3, 3]
Generation 1730: Best = 3.400679e+03 FES = 173100 Diversity = 2.3441e-03 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 3.400679e+03 FES = 174100 Diversity = 2.3139e-03 k_t
= [4, 2, 3, 2]
Generation 1750: Best = 3.400679e+03 FES = 175100 Diversity = 2.4141e-03 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 3.400679e+03 FES = 176100 Diversity = 2.5696e-03 k_t
= [4, 4, 3, 4]
Generation 1770: Best = 3.400679e+03 FES = 177100 Diversity = 2.7478e-03 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 3.400679e+03 FES = 178100 Diversity = 2.8018e-03 k_t
= [3, 2, 2, 3]
Generation 1790: Best = 3.400679e+03 FES = 179100 Diversity = 2.6729e-03 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 3.400679e+03 FES = 180100 Diversity = 2.7140e-03 k_t
= [4, 2, 3, 3]
Generation 1810: Best = 3.400679e+03 FES = 181100 Diversity = 2.8665e-03 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 3.400679e+03 FES = 182100 Diversity = 2.7787e-03 k_t
= [4, 3, 3, 4]
Generation 1830: Best = 3.400678e+03 FES = 183100 Diversity = 2.7891e-03 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 3.400678e+03 FES = 184100 Diversity = 3.0522e-03 k_t

```

```

= [2, 2, 4, 3]
Generation 1850: Best = 3.400678e+03 FES = 185100 Diversity = 3.3223e-03 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 3.400678e+03 FES = 186100 Diversity = 3.2823e-03 k_t
= [2, 4, 3, 2]
Generation 1870: Best = 3.400678e+03 FES = 187100 Diversity = 3.1991e-03 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 3.400678e+03 FES = 188100 Diversity = 3.2270e-03 k_t
= [3, 3, 3, 4]
Generation 1890: Best = 3.400678e+03 FES = 189100 Diversity = 3.1881e-03 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 3.400678e+03 FES = 190100 Diversity = 2.9958e-03 k_t
= [4, 3, 3, 2]
Generation 1910: Best = 3.400678e+03 FES = 191100 Diversity = 3.0690e-03 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 3.400677e+03 FES = 192100 Diversity = 3.1500e-03 k_t
= [4, 3, 3, 4]
Generation 1930: Best = 3.400677e+03 FES = 193100 Diversity = 3.1582e-03 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 3.400677e+03 FES = 194100 Diversity = 3.2159e-03 k_t
= [2, 2, 2, 2]
Generation 1950: Best = 3.400677e+03 FES = 195100 Diversity = 3.1941e-03 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 3.400677e+03 FES = 196100 Diversity = 3.3679e-03 k_t
= [2, 2, 3, 4]
Generation 1970: Best = 3.400677e+03 FES = 197100 Diversity = 3.3033e-03 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 3.400677e+03 FES = 198100 Diversity = 3.4547e-03 k_t
= [2, 3, 4, 3]
Generation 1990: Best = 3.400677e+03 FES = 199100 Diversity = 3.6477e-03 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 3.400676e+03 FES = 200100 Diversity = 3.5959e-03 k_t
= [3, 4, 4, 3]
Generation 2010: Best = 3.400676e+03 FES = 201100 Diversity = 3.5336e-03 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 3.400676e+03 FES = 202100 Diversity = 3.5491e-03 k_t
= [3, 2, 4, 3]
Generation 2030: Best = 3.400676e+03 FES = 203100 Diversity = 3.4775e-03 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 3.400676e+03 FES = 204100 Diversity = 3.2064e-03 k_t
= [4, 3, 3, 2]
Generation 2050: Best = 3.400676e+03 FES = 205100 Diversity = 2.9872e-03 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 3.400676e+03 FES = 206100 Diversity = 2.6570e-03 k_t
= [3, 4, 4, 4]
Generation 2070: Best = 3.400676e+03 FES = 207100 Diversity = 2.7329e-03 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 3.400676e+03 FES = 208100 Diversity = 2.7441e-03 k_t
= [2, 2, 3, 4]
Generation 2090: Best = 3.400676e+03 FES = 209100 Diversity = 2.6747e-03 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 3.400676e+03 FES = 210100 Diversity = 2.5462e-03 k_t
= [4, 2, 4, 3]
Generation 2110: Best = 3.400676e+03 FES = 211100 Diversity = 2.4023e-03 k_t

```

```

= [1, 1, 1, 1]
Generation 2120: Best = 3.400675e+03  FES = 212100  Diversity = 2.3847e-03  k_t
= [2, 4, 3, 3]
Generation 2130: Best = 3.400675e+03  FES = 213100  Diversity = 2.2910e-03  k_t
= [1, 1, 1, 1]
Generation 2140: Best = 3.400675e+03  FES = 214100  Diversity = 2.5742e-03  k_t
= [4, 3, 3, 2]
Generation 2150: Best = 3.400675e+03  FES = 215100  Diversity = 2.6505e-03  k_t
= [1, 1, 1, 1]
Generation 2160: Best = 3.400675e+03  FES = 216100  Diversity = 2.5831e-03  k_t
= [4, 3, 2, 2]
Generation 2170: Best = 3.400675e+03  FES = 217100  Diversity = 2.6148e-03  k_t
= [1, 1, 1, 1]
Generation 2180: Best = 3.400675e+03  FES = 218100  Diversity = 2.8163e-03  k_t
= [2, 2, 4, 3]
Generation 2190: Best = 3.400675e+03  FES = 219100  Diversity = 3.0714e-03  k_t
= [1, 1, 1, 1]
Generation 2200: Best = 3.400675e+03  FES = 220100  Diversity = 2.9917e-03  k_t
= [4, 4, 3, 3]
Generation 2210: Best = 3.400675e+03  FES = 221100  Diversity = 3.0728e-03  k_t
= [1, 1, 1, 1]
Generation 2220: Best = 3.400675e+03  FES = 222100  Diversity = 3.0759e-03  k_t
= [3, 4, 3, 4]
Generation 2230: Best = 3.400675e+03  FES = 223100  Diversity = 2.8676e-03  k_t
= [1, 1, 1, 1]
Generation 2240: Best = 3.400675e+03  FES = 224100  Diversity = 2.6884e-03  k_t
= [2, 3, 2, 2]
Generation 2250: Best = 3.400674e+03  FES = 225100  Diversity = 2.5714e-03  k_t
= [1, 1, 1, 1]
Generation 2260: Best = 3.400674e+03  FES = 226100  Diversity = 2.4137e-03  k_t
= [4, 3, 3, 4]
Generation 2270: Best = 3.400674e+03  FES = 227100  Diversity = 2.2509e-03  k_t
= [1, 1, 1, 1]
Generation 2280: Best = 3.400674e+03  FES = 228100  Diversity = 2.1455e-03  k_t
= [4, 3, 2, 3]
Generation 2290: Best = 3.400674e+03  FES = 229100  Diversity = 2.0224e-03  k_t
= [1, 1, 1, 1]
Generation 2300: Best = 3.400674e+03  FES = 230100  Diversity = 1.8650e-03  k_t
= [2, 2, 3, 4]
Generation 2310: Best = 3.400674e+03  FES = 231100  Diversity = 1.8169e-03  k_t
= [1, 1, 1, 1]
Generation 2320: Best = 3.400674e+03  FES = 232100  Diversity = 1.7108e-03  k_t
= [3, 3, 4, 2]
Generation 2330: Best = 3.400674e+03  FES = 233100  Diversity = 1.6747e-03  k_t
= [1, 1, 1, 1]
Generation 2340: Best = 3.400674e+03  FES = 234100  Diversity = 1.7133e-03  k_t
= [3, 2, 2, 2]
Generation 2350: Best = 3.400674e+03  FES = 235100  Diversity = 1.6045e-03  k_t
= [1, 1, 1, 1]
Generation 2360: Best = 3.400674e+03  FES = 236100  Diversity = 1.6195e-03  k_t
= [3, 2, 3, 2]
Generation 2370: Best = 3.400674e+03  FES = 237100  Diversity = 1.6214e-03  k_t
= [1, 1, 1, 1]
Generation 2380: Best = 3.400674e+03  FES = 238100  Diversity = 1.6757e-03  k_t

```



```

= [3, 3, 3, 2]
Generation 2390: Best = 3.400674e+03 FES = 239100 Diversity = 1.6982e-03 k_t
= [1, 1, 1, 1]
Generation 2400: Best = 3.400674e+03 FES = 240100 Diversity = 1.7150e-03 k_t
= [3, 3, 2, 3]
Generation 2410: Best = 3.400674e+03 FES = 241100 Diversity = 1.7052e-03 k_t
= [1, 1, 1, 1]
Generation 2420: Best = 3.400674e+03 FES = 242100 Diversity = 1.6967e-03 k_t
= [3, 4, 3, 4]
Generation 2430: Best = 3.400674e+03 FES = 243100 Diversity = 1.6527e-03 k_t
= [1, 1, 1, 1]
Generation 2440: Best = 3.400674e+03 FES = 244100 Diversity = 1.5638e-03 k_t
= [3, 4, 4, 2]
Generation 2450: Best = 3.400674e+03 FES = 245100 Diversity = 1.4253e-03 k_t
= [1, 1, 1, 1]
Generation 2460: Best = 3.400674e+03 FES = 246100 Diversity = 1.3694e-03 k_t
= [3, 3, 4, 3]
Generation 2470: Best = 3.400674e+03 FES = 247100 Diversity = 1.2962e-03 k_t
= [1, 1, 1, 1]
Generation 2480: Best = 3.400674e+03 FES = 248100 Diversity = 1.2656e-03 k_t
= [2, 4, 3, 3]
Generation 2490: Best = 3.400674e+03 FES = 249100 Diversity = 1.2338e-03 k_t
= [1, 1, 1, 1]
Generation 2500: Best = 3.400674e+03 FES = 250100 Diversity = 1.2297e-03 k_t
= [3, 3, 2, 3]
Generation 2510: Best = 3.400674e+03 FES = 251100 Diversity = 1.2361e-03 k_t
= [1, 1, 1, 1]
Generation 2520: Best = 3.400674e+03 FES = 252100 Diversity = 1.1916e-03 k_t
= [4, 3, 4, 4]
Generation 2530: Best = 3.400674e+03 FES = 253100 Diversity = 1.1446e-03 k_t
= [1, 1, 1, 1]
Generation 2540: Best = 3.400674e+03 FES = 254100 Diversity = 1.0698e-03 k_t
= [3, 4, 2, 4]
Generation 2550: Best = 3.400674e+03 FES = 255100 Diversity = 9.6900e-04 k_t
= [1, 1, 1, 1]
Generation 2560: Best = 3.400674e+03 FES = 256100 Diversity = 9.3925e-04 k_t
= [4, 2, 3, 2]
Generation 2570: Best = 3.400674e+03 FES = 257100 Diversity = 9.2377e-04 k_t
= [1, 1, 1, 1]
Generation 2580: Best = 3.400674e+03 FES = 258100 Diversity = 9.0571e-04 k_t
= [3, 2, 2, 3]
Generation 2590: Best = 3.400674e+03 FES = 259100 Diversity = 8.7020e-04 k_t
= [1, 1, 1, 1]
Generation 2600: Best = 3.400674e+03 FES = 260100 Diversity = 8.4186e-04 k_t
= [4, 2, 4, 2]
Generation 2610: Best = 3.400674e+03 FES = 261100 Diversity = 8.2537e-04 k_t
= [1, 1, 1, 1]
Generation 2620: Best = 3.400674e+03 FES = 262100 Diversity = 8.8064e-04 k_t
= [3, 2, 4, 2]
Generation 2630: Best = 3.400674e+03 FES = 263100 Diversity = 8.6580e-04 k_t
= [1, 1, 1, 1]
Generation 2640: Best = 3.400674e+03 FES = 264100 Diversity = 9.1051e-04 k_t
= [3, 3, 3, 2]
Generation 2650: Best = 3.400674e+03 FES = 265100 Diversity = 8.7235e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 2660: Best = 3.400674e+03 FES = 266100 Diversity = 8.9466e-04 k_t
= [2, 3, 2, 2]
Generation 2670: Best = 3.400674e+03 FES = 267100 Diversity = 9.2451e-04 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 3.400674e+03 FES = 268100 Diversity = 9.6411e-04 k_t
= [4, 4, 2, 3]
Generation 2690: Best = 3.400674e+03 FES = 269100 Diversity = 1.0227e-03 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 3.400674e+03 FES = 270100 Diversity = 1.0406e-03 k_t
= [2, 3, 4, 3]
Generation 2710: Best = 3.400674e+03 FES = 271100 Diversity = 1.0171e-03 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 3.400674e+03 FES = 272100 Diversity = 1.0087e-03 k_t
= [3, 4, 2, 2]
Generation 2730: Best = 3.400674e+03 FES = 273100 Diversity = 1.0857e-03 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 3.400674e+03 FES = 274100 Diversity = 1.0726e-03 k_t
= [2, 2, 2, 4]
Generation 2750: Best = 3.400674e+03 FES = 275100 Diversity = 1.0970e-03 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 3.400674e+03 FES = 276100 Diversity = 1.0649e-03 k_t
= [4, 4, 4, 2]
Generation 2770: Best = 3.400674e+03 FES = 277100 Diversity = 1.0583e-03 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 3.400674e+03 FES = 278100 Diversity = 1.0275e-03 k_t
= [3, 2, 2, 4]
Generation 2790: Best = 3.400674e+03 FES = 279100 Diversity = 9.6812e-04 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 3.400674e+03 FES = 280100 Diversity = 9.7081e-04 k_t
= [3, 4, 3, 2]
Generation 2810: Best = 3.400674e+03 FES = 281100 Diversity = 9.3671e-04 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 3.400674e+03 FES = 282100 Diversity = 8.8059e-04 k_t
= [3, 2, 2, 2]
Generation 2830: Best = 3.400674e+03 FES = 283100 Diversity = 8.8731e-04 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 3.400674e+03 FES = 284100 Diversity = 8.7113e-04 k_t
= [2, 2, 3, 2]
Generation 2850: Best = 3.400674e+03 FES = 285100 Diversity = 8.5784e-04 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 3.400674e+03 FES = 286100 Diversity = 7.8493e-04 k_t
= [2, 2, 2, 4]
Generation 2870: Best = 3.400674e+03 FES = 287100 Diversity = 7.2094e-04 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 3.400674e+03 FES = 288100 Diversity = 6.8038e-04 k_t
= [2, 4, 2, 2]
Generation 2890: Best = 3.400674e+03 FES = 289100 Diversity = 7.1661e-04 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 3.400674e+03 FES = 290100 Diversity = 7.3610e-04 k_t
= [4, 2, 4, 4]
Generation 2910: Best = 3.400674e+03 FES = 291100 Diversity = 7.3208e-04 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 3.400674e+03 FES = 292100 Diversity = 7.0300e-04 k_t

```

```

= [4, 3, 4, 3]
Generation 2930: Best = 3.400673e+03 FES = 293100 Diversity = 7.2324e-04 k_t
= [1, 1, 1, 1]
Generation 2940: Best = 3.400673e+03 FES = 294100 Diversity = 7.7126e-04 k_t
= [2, 3, 2, 2]
Generation 2950: Best = 3.400673e+03 FES = 295100 Diversity = 7.7727e-04 k_t
= [1, 1, 1, 1]
Generation 2960: Best = 3.400673e+03 FES = 296100 Diversity = 7.7478e-04 k_t
= [3, 3, 2, 3]
Generation 2970: Best = 3.400673e+03 FES = 297100 Diversity = 7.7317e-04 k_t
= [1, 1, 1, 1]
Generation 2980: Best = 3.400673e+03 FES = 298100 Diversity = 7.7602e-04 k_t
= [3, 3, 3, 2]
Generation 2990: Best = 3.400673e+03 FES = 299100 Diversity = 7.5424e-04 k_t
= [1, 1, 1, 1]
Generation 3000: Best = 3.400673e+03 FES = 300100 Diversity = 7.3936e-04 k_t
= [3, 2, 2, 4]
Generation 3010: Best = 3.400673e+03 FES = 301100 Diversity = 6.7260e-04 k_t
= [1, 1, 1, 1]
Generation 3020: Best = 3.400673e+03 FES = 302100 Diversity = 6.2170e-04 k_t
= [2, 4, 2, 2]
Generation 3030: Best = 3.400673e+03 FES = 303100 Diversity = 5.7302e-04 k_t
= [1, 1, 1, 1]
Generation 3040: Best = 3.400673e+03 FES = 304100 Diversity = 5.1560e-04 k_t
= [4, 4, 2, 2]
Generation 3050: Best = 3.400673e+03 FES = 305100 Diversity = 4.7142e-04 k_t
= [1, 1, 1, 1]
Generation 3060: Best = 3.400673e+03 FES = 306100 Diversity = 4.6431e-04 k_t
= [4, 2, 3, 4]
Generation 3070: Best = 3.400673e+03 FES = 307100 Diversity = 4.7984e-04 k_t
= [1, 1, 1, 1]
Generation 3080: Best = 3.400673e+03 FES = 308100 Diversity = 4.8097e-04 k_t
= [2, 3, 2, 4]
Generation 3090: Best = 3.400673e+03 FES = 309100 Diversity = 4.6318e-04 k_t
= [1, 1, 1, 1]
Generation 3100: Best = 3.400673e+03 FES = 310100 Diversity = 4.8080e-04 k_t
= [3, 2, 3, 2]
Generation 3110: Best = 3.400673e+03 FES = 311100 Diversity = 4.7655e-04 k_t
= [1, 1, 1, 1]
Generation 3120: Best = 3.400673e+03 FES = 312100 Diversity = 4.6394e-04 k_t
= [3, 4, 3, 3]
Generation 3130: Best = 3.400673e+03 FES = 313100 Diversity = 4.5660e-04 k_t
= [1, 1, 1, 1]
Generation 3140: Best = 3.400673e+03 FES = 314100 Diversity = 4.6662e-04 k_t
= [3, 4, 3, 3]
Generation 3150: Best = 3.400673e+03 FES = 315100 Diversity = 4.9559e-04 k_t
= [1, 1, 1, 1]
Generation 3160: Best = 3.400673e+03 FES = 316100 Diversity = 5.3058e-04 k_t
= [4, 3, 3, 3]
Generation 3170: Best = 3.400673e+03 FES = 317100 Diversity = 5.5742e-04 k_t
= [1, 1, 1, 1]
Generation 3180: Best = 3.400673e+03 FES = 318100 Diversity = 5.4966e-04 k_t
= [4, 3, 2, 3]
Generation 3190: Best = 3.400673e+03 FES = 319100 Diversity = 5.3029e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 3200: Best = 3.400673e+03 FES = 320100 Diversity = 5.2124e-04 k_t
= [2, 2, 3, 2]
Generation 3210: Best = 3.400673e+03 FES = 321100 Diversity = 4.9622e-04 k_t
= [1, 1, 1, 1]
Generation 3220: Best = 3.400673e+03 FES = 322100 Diversity = 4.8686e-04 k_t
= [4, 2, 3, 2]
Generation 3230: Best = 3.400673e+03 FES = 323100 Diversity = 4.8116e-04 k_t
= [1, 1, 1, 1]
Generation 3240: Best = 3.400673e+03 FES = 324100 Diversity = 4.8907e-04 k_t
= [3, 2, 3, 2]
Generation 3250: Best = 3.400673e+03 FES = 325100 Diversity = 4.7518e-04 k_t
= [1, 1, 1, 1]
Generation 3260: Best = 3.400673e+03 FES = 326100 Diversity = 4.9871e-04 k_t
= [3, 2, 3, 3]
Generation 3270: Best = 3.400673e+03 FES = 327100 Diversity = 5.1826e-04 k_t
= [1, 1, 1, 1]
Generation 3280: Best = 3.400673e+03 FES = 328100 Diversity = 5.5756e-04 k_t
= [4, 4, 3, 3]
Generation 3290: Best = 3.400673e+03 FES = 329100 Diversity = 5.5028e-04 k_t
= [1, 1, 1, 1]
Generation 3300: Best = 3.400673e+03 FES = 330100 Diversity = 5.5179e-04 k_t
= [4, 3, 3, 4]
Generation 3310: Best = 3.400673e+03 FES = 331100 Diversity = 5.3305e-04 k_t
= [1, 1, 1, 1]
Generation 3320: Best = 3.400673e+03 FES = 332100 Diversity = 5.2208e-04 k_t
= [2, 2, 3, 3]
Generation 3330: Best = 3.400673e+03 FES = 333100 Diversity = 4.9361e-04 k_t
= [1, 1, 1, 1]
Generation 3340: Best = 3.400673e+03 FES = 334100 Diversity = 5.0363e-04 k_t
= [2, 2, 3, 3]
Generation 3350: Best = 3.400673e+03 FES = 335100 Diversity = 4.9491e-04 k_t
= [1, 1, 1, 1]
Generation 3360: Best = 3.400673e+03 FES = 336100 Diversity = 4.6542e-04 k_t
= [2, 4, 4, 4]
Generation 3370: Best = 3.400673e+03 FES = 337100 Diversity = 4.3978e-04 k_t
= [1, 1, 1, 1]
Generation 3380: Best = 3.400673e+03 FES = 338100 Diversity = 4.2218e-04 k_t
= [2, 4, 3, 2]
Generation 3390: Best = 3.400673e+03 FES = 339100 Diversity = 4.0782e-04 k_t
= [1, 1, 1, 1]
Generation 3400: Best = 3.400673e+03 FES = 340100 Diversity = 4.0375e-04 k_t
= [4, 4, 2, 2]
Generation 3410: Best = 3.400673e+03 FES = 341100 Diversity = 3.9277e-04 k_t
= [1, 1, 1, 1]
Generation 3420: Best = 3.400673e+03 FES = 342100 Diversity = 3.8068e-04 k_t
= [2, 4, 2, 2]
Generation 3430: Best = 3.400673e+03 FES = 343100 Diversity = 3.9604e-04 k_t
= [1, 1, 1, 1]
Generation 3440: Best = 3.400673e+03 FES = 344100 Diversity = 3.7845e-04 k_t
= [3, 4, 2, 4]
Generation 3450: Best = 3.400673e+03 FES = 345100 Diversity = 3.6235e-04 k_t
= [1, 1, 1, 1]
Generation 3460: Best = 3.400673e+03 FES = 346100 Diversity = 3.4930e-04 k_t

```

```

= [2, 3, 2, 2]
Generation 3470: Best = 3.400673e+03 FES = 347100 Diversity = 3.3350e-04 k_t
= [1, 1, 1, 1]
Generation 3480: Best = 3.400673e+03 FES = 348100 Diversity = 3.4173e-04 k_t
= [2, 2, 2, 2]
Generation 3490: Best = 3.400673e+03 FES = 349100 Diversity = 3.3713e-04 k_t
= [1, 1, 1, 1]
Generation 3500: Best = 3.400673e+03 FES = 350100 Diversity = 3.5867e-04 k_t
= [3, 2, 4, 2]
Generation 3510: Best = 3.400673e+03 FES = 351100 Diversity = 3.4955e-04 k_t
= [1, 1, 1, 1]
Generation 3520: Best = 3.400673e+03 FES = 352100 Diversity = 3.5099e-04 k_t
= [2, 3, 4, 2]
Generation 3530: Best = 3.400673e+03 FES = 353100 Diversity = 3.4567e-04 k_t
= [1, 1, 1, 1]
Generation 3540: Best = 3.400673e+03 FES = 354100 Diversity = 3.3177e-04 k_t
= [4, 3, 3, 2]
Generation 3550: Best = 3.400673e+03 FES = 355100 Diversity = 3.2479e-04 k_t
= [1, 1, 1, 1]
Generation 3560: Best = 3.400673e+03 FES = 356100 Diversity = 3.2115e-04 k_t
= [2, 3, 2, 4]
Generation 3570: Best = 3.400673e+03 FES = 357100 Diversity = 3.0042e-04 k_t
= [1, 1, 1, 1]
Generation 3580: Best = 3.400673e+03 FES = 358100 Diversity = 3.3996e-04 k_t
= [4, 3, 4, 4]
Generation 3590: Best = 3.400673e+03 FES = 359100 Diversity = 3.2360e-04 k_t
= [1, 1, 1, 1]
Generation 3600: Best = 3.400673e+03 FES = 360100 Diversity = 3.2495e-04 k_t
= [2, 4, 2, 2]
Generation 3610: Best = 3.400673e+03 FES = 361100 Diversity = 3.2757e-04 k_t
= [1, 1, 1, 1]
Generation 3620: Best = 3.400673e+03 FES = 362100 Diversity = 3.2674e-04 k_t
= [3, 2, 2, 4]
Generation 3630: Best = 3.400673e+03 FES = 363100 Diversity = 3.0528e-04 k_t
= [1, 1, 1, 1]
Generation 3640: Best = 3.400673e+03 FES = 364100 Diversity = 2.9422e-04 k_t
= [3, 4, 2, 3]
Generation 3650: Best = 3.400673e+03 FES = 365100 Diversity = 3.0301e-04 k_t
= [1, 1, 1, 1]
Generation 3660: Best = 3.400673e+03 FES = 366100 Diversity = 3.1207e-04 k_t
= [2, 2, 3, 3]
Generation 3670: Best = 3.400673e+03 FES = 367100 Diversity = 3.0870e-04 k_t
= [1, 1, 1, 1]
Generation 3680: Best = 3.400673e+03 FES = 368100 Diversity = 3.0490e-04 k_t
= [3, 4, 3, 4]
Generation 3690: Best = 3.400673e+03 FES = 369100 Diversity = 2.9506e-04 k_t
= [1, 1, 1, 1]
Generation 3700: Best = 3.400673e+03 FES = 370100 Diversity = 2.8916e-04 k_t
= [3, 4, 4, 4]
Generation 3710: Best = 3.400673e+03 FES = 371100 Diversity = 2.8801e-04 k_t
= [1, 1, 1, 1]
Generation 3720: Best = 3.400673e+03 FES = 372100 Diversity = 2.9318e-04 k_t
= [2, 4, 2, 4]
Generation 3730: Best = 3.400673e+03 FES = 373100 Diversity = 2.8781e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 3740: Best = 3.400673e+03 FES = 374100 Diversity = 2.7800e-04 k_t
= [2, 2, 3, 2]
Generation 3750: Best = 3.400673e+03 FES = 375100 Diversity = 2.7447e-04 k_t
= [1, 1, 1, 1]
Generation 3760: Best = 3.400673e+03 FES = 376100 Diversity = 2.6826e-04 k_t
= [2, 3, 2, 2]
Generation 3770: Best = 3.400673e+03 FES = 377100 Diversity = 2.6460e-04 k_t
= [1, 1, 1, 1]
Generation 3780: Best = 3.400673e+03 FES = 378100 Diversity = 2.4909e-04 k_t
= [4, 3, 4, 2]
Generation 3790: Best = 3.400673e+03 FES = 379100 Diversity = 2.4983e-04 k_t
= [1, 1, 1, 1]
Generation 3800: Best = 3.400673e+03 FES = 380100 Diversity = 2.6058e-04 k_t
= [2, 3, 3, 4]
Generation 3810: Best = 3.400673e+03 FES = 381100 Diversity = 2.5677e-04 k_t
= [1, 1, 1, 1]
Generation 3820: Best = 3.400673e+03 FES = 382100 Diversity = 2.5276e-04 k_t
= [4, 3, 3, 4]
Generation 3830: Best = 3.400673e+03 FES = 383100 Diversity = 2.4766e-04 k_t
= [1, 1, 1, 1]
Generation 3840: Best = 3.400673e+03 FES = 384100 Diversity = 2.6821e-04 k_t
= [2, 2, 4, 3]
Generation 3850: Best = 3.400673e+03 FES = 385100 Diversity = 2.6711e-04 k_t
= [1, 1, 1, 1]
Generation 3860: Best = 3.400673e+03 FES = 386100 Diversity = 2.6114e-04 k_t
= [2, 4, 3, 2]
Generation 3870: Best = 3.400673e+03 FES = 387100 Diversity = 2.6070e-04 k_t
= [1, 1, 1, 1]
Generation 3880: Best = 3.400673e+03 FES = 388100 Diversity = 2.6116e-04 k_t
= [2, 4, 4, 4]
Generation 3890: Best = 3.400673e+03 FES = 389100 Diversity = 2.6584e-04 k_t
= [1, 1, 1, 1]
Generation 3900: Best = 3.400673e+03 FES = 390100 Diversity = 2.7295e-04 k_t
= [4, 3, 4, 2]
Generation 3910: Best = 3.400673e+03 FES = 391100 Diversity = 2.6256e-04 k_t
= [1, 1, 1, 1]
Generation 3920: Best = 3.400673e+03 FES = 392100 Diversity = 2.6385e-04 k_t
= [2, 2, 4, 3]
Generation 3930: Best = 3.400673e+03 FES = 393100 Diversity = 2.5580e-04 k_t
= [1, 1, 1, 1]
Generation 3940: Best = 3.400673e+03 FES = 394100 Diversity = 2.4149e-04 k_t
= [3, 4, 3, 3]
Generation 3950: Best = 3.400673e+03 FES = 395100 Diversity = 2.3686e-04 k_t
= [1, 1, 1, 1]
Generation 3960: Best = 3.400673e+03 FES = 396100 Diversity = 2.1637e-04 k_t
= [4, 2, 2, 4]
Generation 3970: Best = 3.400673e+03 FES = 397100 Diversity = 2.1646e-04 k_t
= [1, 1, 1, 1]
Generation 3980: Best = 3.400673e+03 FES = 398100 Diversity = 2.2711e-04 k_t
= [2, 3, 3, 4]
Generation 3990: Best = 3.400673e+03 FES = 399100 Diversity = 2.3745e-04 k_t
= [1, 1, 1, 1]
Generation 4000: Best = 3.400673e+03 FES = 400100 Diversity = 2.5109e-04 k_t

```

```

= [2, 2, 4, 2]
Generation 4010: Best = 3.400673e+03 FES = 401100 Diversity = 2.5693e-04 k_t
= [1, 1, 1, 1]
Generation 4020: Best = 3.400673e+03 FES = 402100 Diversity = 2.4335e-04 k_t
= [2, 4, 2, 4]
Generation 4030: Best = 3.400673e+03 FES = 403100 Diversity = 2.3641e-04 k_t
= [1, 1, 1, 1]
Generation 4040: Best = 3.400673e+03 FES = 404100 Diversity = 2.3583e-04 k_t
= [4, 3, 3, 4]
Generation 4050: Best = 3.400673e+03 FES = 405100 Diversity = 2.2413e-04 k_t
= [1, 1, 1, 1]
Generation 4060: Best = 3.400673e+03 FES = 406100 Diversity = 2.4991e-04 k_t
= [2, 2, 2, 2]
Generation 4070: Best = 3.400673e+03 FES = 407100 Diversity = 2.5311e-04 k_t
= [1, 1, 1, 1]
Generation 4080: Best = 3.400673e+03 FES = 408100 Diversity = 2.4739e-04 k_t
= [2, 4, 4, 2]
Generation 4090: Best = 3.400673e+03 FES = 409100 Diversity = 2.6718e-04 k_t
= [1, 1, 1, 1]
Generation 4100: Best = 3.400673e+03 FES = 410100 Diversity = 2.4852e-04 k_t
= [3, 2, 3, 4]
Generation 4110: Best = 3.400673e+03 FES = 411100 Diversity = 2.5078e-04 k_t
= [1, 1, 1, 1]
Generation 4120: Best = 3.400673e+03 FES = 412100 Diversity = 2.3976e-04 k_t
= [4, 3, 3, 3]
Generation 4130: Best = 3.400673e+03 FES = 413100 Diversity = 2.3931e-04 k_t
= [1, 1, 1, 1]
Generation 4140: Best = 3.400673e+03 FES = 414100 Diversity = 2.3556e-04 k_t
= [3, 2, 4, 2]
Generation 4150: Best = 3.400673e+03 FES = 415100 Diversity = 2.3883e-04 k_t
= [1, 1, 1, 1]
Generation 4160: Best = 3.400673e+03 FES = 416100 Diversity = 2.4137e-04 k_t
= [3, 3, 3, 4]
Generation 4170: Best = 3.400673e+03 FES = 417100 Diversity = 2.4638e-04 k_t
= [1, 1, 1, 1]
Generation 4180: Best = 3.400673e+03 FES = 418100 Diversity = 2.4609e-04 k_t
= [2, 2, 3, 2]
Generation 4190: Best = 3.400673e+03 FES = 419100 Diversity = 2.3993e-04 k_t
= [1, 1, 1, 1]
Generation 4200: Best = 3.400673e+03 FES = 420100 Diversity = 2.3539e-04 k_t
= [2, 4, 2, 2]
Generation 4210: Best = 3.400673e+03 FES = 421100 Diversity = 2.4939e-04 k_t
= [1, 1, 1, 1]
Generation 4220: Best = 3.400673e+03 FES = 422100 Diversity = 2.4033e-04 k_t
= [3, 3, 4, 4]
Generation 4230: Best = 3.400673e+03 FES = 423100 Diversity = 2.6186e-04 k_t
= [1, 1, 1, 1]
Generation 4240: Best = 3.400673e+03 FES = 424100 Diversity = 2.4950e-04 k_t
= [4, 2, 4, 4]
Generation 4250: Best = 3.400673e+03 FES = 425100 Diversity = 2.5314e-04 k_t
= [1, 1, 1, 1]
Generation 4260: Best = 3.400673e+03 FES = 426100 Diversity = 2.5285e-04 k_t
= [4, 3, 2, 3]
Generation 4270: Best = 3.400673e+03 FES = 427100 Diversity = 2.3566e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 4280: Best = 3.400673e+03 FES = 428100 Diversity = 2.1289e-04 k_t
= [2, 2, 2, 2]
Generation 4290: Best = 3.400673e+03 FES = 429100 Diversity = 1.9122e-04 k_t
= [1, 1, 1, 1]
Generation 4300: Best = 3.400673e+03 FES = 430100 Diversity = 1.9097e-04 k_t
= [2, 4, 4, 3]
Generation 4310: Best = 3.400673e+03 FES = 431100 Diversity = 1.8990e-04 k_t
= [1, 1, 1, 1]
Generation 4320: Best = 3.400673e+03 FES = 432100 Diversity = 1.8272e-04 k_t
= [4, 4, 4, 3]
Generation 4330: Best = 3.400673e+03 FES = 433100 Diversity = 1.8486e-04 k_t
= [1, 1, 1, 1]
Generation 4340: Best = 3.400673e+03 FES = 434100 Diversity = 1.7552e-04 k_t
= [4, 2, 4, 4]
Generation 4350: Best = 3.400673e+03 FES = 435100 Diversity = 1.6807e-04 k_t
= [1, 1, 1, 1]
Generation 4360: Best = 3.400673e+03 FES = 436100 Diversity = 1.6054e-04 k_t
= [4, 3, 4, 3]
Generation 4370: Best = 3.400673e+03 FES = 437100 Diversity = 1.4993e-04 k_t
= [1, 1, 1, 1]
Generation 4380: Best = 3.400673e+03 FES = 438100 Diversity = 1.4384e-04 k_t
= [2, 2, 3, 4]
Generation 4390: Best = 3.400673e+03 FES = 439100 Diversity = 1.4959e-04 k_t
= [1, 1, 1, 1]
Generation 4400: Best = 3.400673e+03 FES = 440100 Diversity = 1.5350e-04 k_t
= [3, 4, 2, 4]
Generation 4410: Best = 3.400673e+03 FES = 441100 Diversity = 1.4579e-04 k_t
= [1, 1, 1, 1]
Generation 4420: Best = 3.400673e+03 FES = 442100 Diversity = 1.4472e-04 k_t
= [3, 2, 2, 3]
Generation 4430: Best = 3.400673e+03 FES = 443100 Diversity = 1.4061e-04 k_t
= [1, 1, 1, 1]
Generation 4440: Best = 3.400673e+03 FES = 444100 Diversity = 1.3634e-04 k_t
= [3, 2, 4, 2]
Generation 4450: Best = 3.400673e+03 FES = 445100 Diversity = 1.3340e-04 k_t
= [1, 1, 1, 1]
Generation 4460: Best = 3.400673e+03 FES = 446100 Diversity = 1.3471e-04 k_t
= [4, 2, 4, 2]
Generation 4470: Best = 3.400673e+03 FES = 447100 Diversity = 1.3207e-04 k_t
= [1, 1, 1, 1]
Generation 4480: Best = 3.400673e+03 FES = 448100 Diversity = 1.2690e-04 k_t
= [4, 2, 4, 2]
Generation 4490: Best = 3.400673e+03 FES = 449100 Diversity = 1.2463e-04 k_t
= [1, 1, 1, 1]
Generation 4500: Best = 3.400673e+03 FES = 450100 Diversity = 1.2329e-04 k_t
= [4, 3, 3, 4]
Generation 4510: Best = 3.400673e+03 FES = 451100 Diversity = 1.2556e-04 k_t
= [1, 1, 1, 1]
Generation 4520: Best = 3.400673e+03 FES = 452100 Diversity = 1.2085e-04 k_t
= [4, 2, 3, 2]
Generation 4530: Best = 3.400673e+03 FES = 453100 Diversity = 1.1476e-04 k_t
= [1, 1, 1, 1]
Generation 4540: Best = 3.400673e+03 FES = 454100 Diversity = 1.0931e-04 k_t

```



```

= [3, 2, 3, 2]
Generation 4550: Best = 3.400673e+03 FES = 455100 Diversity = 1.0402e-04 k_t
= [1, 1, 1, 1]
Generation 4560: Best = 3.400673e+03 FES = 456100 Diversity = 9.9609e-05 k_t
= [2, 3, 3, 4]
Generation 4570: Best = 3.400673e+03 FES = 457100 Diversity = 1.0915e-04 k_t
= [1, 1, 1, 1]
Generation 4580: Best = 3.400673e+03 FES = 458100 Diversity = 1.1170e-04 k_t
= [2, 2, 2, 4]
Generation 4590: Best = 3.400673e+03 FES = 459100 Diversity = 1.2783e-04 k_t
= [1, 1, 1, 1]
Generation 4600: Best = 3.400673e+03 FES = 460100 Diversity = 1.3017e-04 k_t
= [3, 3, 4, 2]
Generation 4610: Best = 3.400673e+03 FES = 461100 Diversity = 1.3508e-04 k_t
= [1, 1, 1, 1]
Generation 4620: Best = 3.400673e+03 FES = 462100 Diversity = 1.3234e-04 k_t
= [4, 2, 3, 3]
Generation 4630: Best = 3.400673e+03 FES = 463100 Diversity = 1.2728e-04 k_t
= [1, 1, 1, 1]
Generation 4640: Best = 3.400673e+03 FES = 464100 Diversity = 1.2854e-04 k_t
= [4, 2, 3, 3]
Generation 4650: Best = 3.400673e+03 FES = 465100 Diversity = 1.3772e-04 k_t
= [1, 1, 1, 1]
Generation 4660: Best = 3.400673e+03 FES = 466100 Diversity = 1.4109e-04 k_t
= [3, 3, 4, 3]
Generation 4670: Best = 3.400673e+03 FES = 467100 Diversity = 1.4195e-04 k_t
= [1, 1, 1, 1]
Generation 4680: Best = 3.400673e+03 FES = 468100 Diversity = 1.4057e-04 k_t
= [2, 3, 2, 3]
Generation 4690: Best = 3.400673e+03 FES = 469100 Diversity = 1.4338e-04 k_t
= [1, 1, 1, 1]
Generation 4700: Best = 3.400673e+03 FES = 470100 Diversity = 1.3190e-04 k_t
= [3, 3, 3, 4]
Generation 4710: Best = 3.400673e+03 FES = 471100 Diversity = 1.4478e-04 k_t
= [1, 1, 1, 1]
Generation 4720: Best = 3.400673e+03 FES = 472100 Diversity = 1.4204e-04 k_t
= [4, 2, 4, 2]
Generation 4730: Best = 3.400673e+03 FES = 473100 Diversity = 1.3489e-04 k_t
= [1, 1, 1, 1]
Generation 4740: Best = 3.400673e+03 FES = 474100 Diversity = 1.3103e-04 k_t
= [2, 4, 2, 4]
Generation 4750: Best = 3.400673e+03 FES = 475100 Diversity = 1.2854e-04 k_t
= [1, 1, 1, 1]
Generation 4760: Best = 3.400673e+03 FES = 476100 Diversity = 1.2918e-04 k_t
= [3, 2, 3, 3]
Generation 4770: Best = 3.400673e+03 FES = 477100 Diversity = 1.2671e-04 k_t
= [1, 1, 1, 1]
Generation 4780: Best = 3.400673e+03 FES = 478100 Diversity = 1.3436e-04 k_t
= [2, 4, 2, 3]
Generation 4790: Best = 3.400673e+03 FES = 479100 Diversity = 1.3560e-04 k_t
= [1, 1, 1, 1]
Generation 4800: Best = 3.400673e+03 FES = 480100 Diversity = 1.2973e-04 k_t
= [2, 4, 3, 2]
Generation 4810: Best = 3.400673e+03 FES = 481100 Diversity = 1.2000e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 4820: Best = 3.400673e+03 FES = 482100 Diversity = 1.2551e-04 k_t
= [2, 2, 4, 2]
Generation 4830: Best = 3.400673e+03 FES = 483100 Diversity = 1.2565e-04 k_t
= [1, 1, 1, 1]
Generation 4840: Best = 3.400673e+03 FES = 484100 Diversity = 1.2326e-04 k_t
= [4, 2, 4, 4]
Generation 4850: Best = 3.400673e+03 FES = 485100 Diversity = 1.1663e-04 k_t
= [1, 1, 1, 1]
Generation 4860: Best = 3.400673e+03 FES = 486100 Diversity = 1.2092e-04 k_t
= [4, 3, 2, 3]
Generation 4870: Best = 3.400673e+03 FES = 487100 Diversity = 1.1655e-04 k_t
= [1, 1, 1, 1]
Generation 4880: Best = 3.400673e+03 FES = 488100 Diversity = 1.1482e-04 k_t
= [3, 3, 4, 4]
Generation 4890: Best = 3.400673e+03 FES = 489100 Diversity = 1.1708e-04 k_t
= [1, 1, 1, 1]
Generation 4900: Best = 3.400673e+03 FES = 490100 Diversity = 1.1952e-04 k_t
= [3, 2, 4, 3]
Generation 4910: Best = 3.400673e+03 FES = 491100 Diversity = 1.1745e-04 k_t
= [1, 1, 1, 1]
Generation 4920: Best = 3.400673e+03 FES = 492100 Diversity = 1.2045e-04 k_t
= [3, 2, 2, 4]
Generation 4930: Best = 3.400673e+03 FES = 493100 Diversity = 1.2103e-04 k_t
= [1, 1, 1, 1]
Generation 4940: Best = 3.400673e+03 FES = 494100 Diversity = 1.2024e-04 k_t
= [2, 4, 3, 4]
Generation 4950: Best = 3.400673e+03 FES = 495100 Diversity = 1.1771e-04 k_t
= [1, 1, 1, 1]
Generation 4960: Best = 3.400673e+03 FES = 496100 Diversity = 1.1857e-04 k_t
= [4, 3, 4, 4]
Generation 4970: Best = 3.400673e+03 FES = 497100 Diversity = 1.1697e-04 k_t
= [1, 1, 1, 1]
Generation 4980: Best = 3.400673e+03 FES = 498100 Diversity = 1.1987e-04 k_t
= [3, 4, 4, 4]
Generation 4990: Best = 3.400673e+03 FES = 499100 Diversity = 1.1724e-04 k_t
= [1, 1, 1, 1]
Generation 5000: Best = 3.400673e+03 FES = 500100 Diversity = 1.1391e-04 k_t
= [2, 3, 4, 2]
Generation 5010: Best = 3.400673e+03 FES = 501100 Diversity = 1.0992e-04 k_t
= [1, 1, 1, 1]
Generation 5020: Best = 3.400673e+03 FES = 502100 Diversity = 1.0317e-04 k_t
= [2, 2, 3, 3]
Generation 5030: Best = 3.400673e+03 FES = 503100 Diversity = 1.0049e-04 k_t
= [1, 1, 1, 1]
Generation 5040: Best = 3.400673e+03 FES = 504100 Diversity = 1.0245e-04 k_t
= [4, 3, 2, 3]
Generation 5050: Best = 3.400673e+03 FES = 505100 Diversity = 9.6191e-05 k_t
= [1, 1, 1, 1]
Generation 5060: Best = 3.400673e+03 FES = 506100 Diversity = 9.2867e-05 k_t
= [2, 4, 4, 3]
Generation 5070: Best = 3.400673e+03 FES = 507100 Diversity = 9.2658e-05 k_t
= [1, 1, 1, 1]
Generation 5080: Best = 3.400673e+03 FES = 508100 Diversity = 9.3170e-05 k_t

```

```

= [2, 2, 3, 3]
Generation 5090: Best = 3.400673e+03 FES = 509100 Diversity = 8.9490e-05 k_t
= [1, 1, 1, 1]
Generation 5100: Best = 3.400673e+03 FES = 510100 Diversity = 8.6011e-05 k_t
= [4, 4, 2, 4]
Generation 5110: Best = 3.400673e+03 FES = 511100 Diversity = 9.4484e-05 k_t
= [1, 1, 1, 1]
Generation 5120: Best = 3.400673e+03 FES = 512100 Diversity = 9.8458e-05 k_t
= [3, 4, 4, 3]
Generation 5130: Best = 3.400673e+03 FES = 513100 Diversity = 1.0124e-04 k_t
= [1, 1, 1, 1]
Generation 5140: Best = 3.400673e+03 FES = 514100 Diversity = 1.0302e-04 k_t
= [4, 2, 3, 2]
Generation 5150: Best = 3.400673e+03 FES = 515100 Diversity = 1.0189e-04 k_t
= [1, 1, 1, 1]
Generation 5160: Best = 3.400673e+03 FES = 516100 Diversity = 9.2309e-05 k_t
= [4, 4, 3, 3]
Generation 5170: Best = 3.400673e+03 FES = 517100 Diversity = 8.5984e-05 k_t
= [1, 1, 1, 1]
Generation 5180: Best = 3.400673e+03 FES = 518100 Diversity = 9.5237e-05 k_t
= [3, 2, 4, 2]
Generation 5190: Best = 3.400673e+03 FES = 519100 Diversity = 9.4347e-05 k_t
= [1, 1, 1, 1]
Generation 5200: Best = 3.400673e+03 FES = 520100 Diversity = 9.5764e-05 k_t
= [2, 4, 4, 3]
Generation 5210: Best = 3.400673e+03 FES = 521100 Diversity = 1.0150e-04 k_t
= [1, 1, 1, 1]
Generation 5220: Best = 3.400673e+03 FES = 522100 Diversity = 9.8477e-05 k_t
= [2, 4, 2, 3]
Generation 5230: Best = 3.400673e+03 FES = 523100 Diversity = 9.2856e-05 k_t
= [1, 1, 1, 1]
Generation 5240: Best = 3.400673e+03 FES = 524100 Diversity = 9.6015e-05 k_t
= [3, 3, 4, 4]
Generation 5250: Best = 3.400673e+03 FES = 525100 Diversity = 9.5452e-05 k_t
= [1, 1, 1, 1]
Generation 5260: Best = 3.400673e+03 FES = 526100 Diversity = 1.0172e-04 k_t
= [3, 4, 3, 2]
Generation 5270: Best = 3.400673e+03 FES = 527100 Diversity = 1.0913e-04 k_t
= [1, 1, 1, 1]
Generation 5280: Best = 3.400673e+03 FES = 528100 Diversity = 1.0572e-04 k_t
= [2, 4, 2, 3]
Generation 5290: Best = 3.400673e+03 FES = 529100 Diversity = 1.0793e-04 k_t
= [1, 1, 1, 1]
Generation 5300: Best = 3.400673e+03 FES = 530100 Diversity = 1.0888e-04 k_t
= [2, 4, 3, 4]
Generation 5310: Best = 3.400673e+03 FES = 531100 Diversity = 1.0633e-04 k_t
= [1, 1, 1, 1]
Generation 5320: Best = 3.400673e+03 FES = 532100 Diversity = 1.0287e-04 k_t
= [4, 4, 2, 2]
Generation 5330: Best = 3.400673e+03 FES = 533100 Diversity = 1.1061e-04 k_t
= [1, 1, 1, 1]
Generation 5340: Best = 3.400673e+03 FES = 534100 Diversity = 1.2466e-04 k_t
= [3, 4, 3, 4]
Generation 5350: Best = 3.400673e+03 FES = 535100 Diversity = 1.3277e-04 k_t

```

```

= [1, 1, 1, 1]
Generation 5360: Best = 3.400673e+03 FES = 536100 Diversity = 1.3189e-04 k_t
= [3, 4, 3, 4]
Generation 5370: Best = 3.400673e+03 FES = 537100 Diversity = 1.2978e-04 k_t
= [1, 1, 1, 1]
Generation 5380: Best = 3.400673e+03 FES = 538100 Diversity = 1.2484e-04 k_t
= [3, 3, 2, 4]
Generation 5390: Best = 3.400673e+03 FES = 539100 Diversity = 1.2167e-04 k_t
= [1, 1, 1, 1]
Generation 5400: Best = 3.400673e+03 FES = 540100 Diversity = 1.2013e-04 k_t
= [2, 3, 4, 4]
Generation 5410: Best = 3.400673e+03 FES = 541100 Diversity = 1.1637e-04 k_t
= [1, 1, 1, 1]
Generation 5420: Best = 3.400673e+03 FES = 542100 Diversity = 1.1479e-04 k_t
= [4, 4, 2, 3]
Generation 5430: Best = 3.400673e+03 FES = 543100 Diversity = 1.1105e-04 k_t
= [1, 1, 1, 1]
Generation 5440: Best = 3.400673e+03 FES = 544100 Diversity = 1.2231e-04 k_t
= [2, 3, 3, 4]
Generation 5450: Best = 3.400673e+03 FES = 545100 Diversity = 1.2215e-04 k_t
= [1, 1, 1, 1]
Generation 5460: Best = 3.400673e+03 FES = 546100 Diversity = 1.2016e-04 k_t
= [4, 3, 2, 4]
Generation 5470: Best = 3.400673e+03 FES = 547100 Diversity = 1.2011e-04 k_t
= [1, 1, 1, 1]
Generation 5480: Best = 3.400673e+03 FES = 548100 Diversity = 1.1357e-04 k_t
= [2, 3, 3, 4]
Generation 5490: Best = 3.400673e+03 FES = 549100 Diversity = 1.1453e-04 k_t
= [1, 1, 1, 1]
Generation 5500: Best = 3.400673e+03 FES = 550100 Diversity = 1.0927e-04 k_t
= [2, 3, 2, 2]
Generation 5510: Best = 3.400673e+03 FES = 551100 Diversity = 1.0449e-04 k_t
= [1, 1, 1, 1]
Generation 5520: Best = 3.400673e+03 FES = 552100 Diversity = 1.0427e-04 k_t
= [3, 3, 3, 3]
Generation 5530: Best = 3.400673e+03 FES = 553100 Diversity = 1.0230e-04 k_t
= [1, 1, 1, 1]
Generation 5540: Best = 3.400673e+03 FES = 554100 Diversity = 1.0391e-04 k_t
= [3, 2, 4, 4]
Generation 5550: Best = 3.400673e+03 FES = 555100 Diversity = 1.0304e-04 k_t
= [1, 1, 1, 1]
Generation 5560: Best = 3.400673e+03 FES = 556100 Diversity = 9.9301e-05 k_t
= [3, 4, 2, 2]
Generation 5570: Best = 3.400673e+03 FES = 557100 Diversity = 1.0028e-04 k_t
= [1, 1, 1, 1]
Generation 5580: Best = 3.400673e+03 FES = 558100 Diversity = 9.9320e-05 k_t
= [3, 3, 2, 3]
Generation 5590: Best = 3.400673e+03 FES = 559100 Diversity = 1.0089e-04 k_t
= [1, 1, 1, 1]
Generation 5600: Best = 3.400673e+03 FES = 560100 Diversity = 9.8524e-05 k_t
= [2, 2, 3, 2]
Generation 5610: Best = 3.400673e+03 FES = 561100 Diversity = 9.5699e-05 k_t
= [1, 1, 1, 1]
Generation 5620: Best = 3.400673e+03 FES = 562100 Diversity = 9.5671e-05 k_t

```

```

= [2, 2, 3, 3]
Generation 5630: Best = 3.400673e+03 FES = 563100 Diversity = 9.5818e-05 k_t
= [1, 1, 1, 1]
Generation 5640: Best = 3.400673e+03 FES = 564100 Diversity = 9.2037e-05 k_t
= [4, 4, 3, 3]
Generation 5650: Best = 3.400673e+03 FES = 565100 Diversity = 8.6191e-05 k_t
= [1, 1, 1, 1]
Generation 5660: Best = 3.400673e+03 FES = 566100 Diversity = 8.3978e-05 k_t
= [4, 2, 4, 2]
Generation 5670: Best = 3.400673e+03 FES = 567100 Diversity = 7.9534e-05 k_t
= [1, 1, 1, 1]
Generation 5680: Best = 3.400673e+03 FES = 568100 Diversity = 8.0217e-05 k_t
= [2, 3, 4, 3]
Generation 5690: Best = 3.400673e+03 FES = 569100 Diversity = 7.6771e-05 k_t
= [1, 1, 1, 1]
Generation 5700: Best = 3.400673e+03 FES = 570100 Diversity = 7.2948e-05 k_t
= [4, 4, 3, 4]
Generation 5710: Best = 3.400673e+03 FES = 571100 Diversity = 6.9893e-05 k_t
= [1, 1, 1, 1]
Generation 5720: Best = 3.400673e+03 FES = 572100 Diversity = 7.7610e-05 k_t
= [4, 4, 4, 2]
Generation 5730: Best = 3.400673e+03 FES = 573100 Diversity = 8.6439e-05 k_t
= [1, 1, 1, 1]
Generation 5740: Best = 3.400673e+03 FES = 574100 Diversity = 9.0682e-05 k_t
= [3, 3, 4, 3]
Generation 5750: Best = 3.400673e+03 FES = 575100 Diversity = 9.5554e-05 k_t
= [1, 1, 1, 1]
Generation 5760: Best = 3.400673e+03 FES = 576100 Diversity = 9.1547e-05 k_t
= [2, 3, 3, 3]
Generation 5770: Best = 3.400673e+03 FES = 577100 Diversity = 8.7524e-05 k_t
= [1, 1, 1, 1]
Generation 5780: Best = 3.400673e+03 FES = 578100 Diversity = 8.4837e-05 k_t
= [2, 4, 4, 3]
Generation 5790: Best = 3.400673e+03 FES = 579100 Diversity = 7.8116e-05 k_t
= [1, 1, 1, 1]
Generation 5800: Best = 3.400673e+03 FES = 580100 Diversity = 7.0188e-05 k_t
= [2, 3, 4, 4]
Generation 5810: Best = 3.400673e+03 FES = 581100 Diversity = 6.9065e-05 k_t
= [1, 1, 1, 1]
Generation 5820: Best = 3.400673e+03 FES = 582100 Diversity = 6.6862e-05 k_t
= [3, 4, 3, 3]
Generation 5830: Best = 3.400673e+03 FES = 583100 Diversity = 6.5538e-05 k_t
= [1, 1, 1, 1]
Generation 5840: Best = 3.400673e+03 FES = 584100 Diversity = 6.4619e-05 k_t
= [3, 2, 3, 3]
Generation 5850: Best = 3.400673e+03 FES = 585100 Diversity = 6.2337e-05 k_t
= [1, 1, 1, 1]
Generation 5860: Best = 3.400673e+03 FES = 586100 Diversity = 6.0970e-05 k_t
= [3, 4, 3, 4]
Generation 5870: Best = 3.400673e+03 FES = 587100 Diversity = 6.0013e-05 k_t
= [1, 1, 1, 1]
Generation 5880: Best = 3.400673e+03 FES = 588100 Diversity = 5.5829e-05 k_t
= [3, 4, 2, 4]
Generation 5890: Best = 3.400673e+03 FES = 589100 Diversity = 5.3475e-05 k_t

```

```

= [1, 1, 1, 1]
Generation 5900: Best = 3.400673e+03 FES = 590100 Diversity = 5.1774e-05 k_t
= [2, 4, 2, 3]
Generation 5910: Best = 3.400673e+03 FES = 591100 Diversity = 4.9787e-05 k_t
= [1, 1, 1, 1]
Generation 5920: Best = 3.400673e+03 FES = 592100 Diversity = 5.1966e-05 k_t
= [3, 3, 3, 4]
Generation 5930: Best = 3.400673e+03 FES = 593100 Diversity = 5.4351e-05 k_t
= [1, 1, 1, 1]
Generation 5940: Best = 3.400673e+03 FES = 594100 Diversity = 5.4451e-05 k_t
= [4, 4, 4, 2]
Generation 5950: Best = 3.400673e+03 FES = 595100 Diversity = 5.2183e-05 k_t
= [1, 1, 1, 1]
Generation 5960: Best = 3.400673e+03 FES = 596100 Diversity = 5.0955e-05 k_t
= [2, 2, 4, 4]
Generation 5970: Best = 3.400673e+03 FES = 597100 Diversity = 5.0915e-05 k_t
= [1, 1, 1, 1]
Generation 5980: Best = 3.400673e+03 FES = 598100 Diversity = 4.8909e-05 k_t
= [4, 2, 3, 2]
Generation 5990: Best = 3.400673e+03 FES = 599100 Diversity = 4.6956e-05 k_t
= [1, 1, 1, 1]
Generation 6000: Best = 3.400673e+03 FES = 600100 Diversity = 4.4373e-05 k_t
= [3, 3, 2, 2]
Generation 6010: Best = 3.400673e+03 FES = 601100 Diversity = 4.1986e-05 k_t
= [1, 1, 1, 1]
Generation 6020: Best = 3.400673e+03 FES = 602100 Diversity = 4.3958e-05 k_t
= [3, 2, 4, 2]
Generation 6030: Best = 3.400673e+03 FES = 603100 Diversity = 4.5305e-05 k_t
= [1, 1, 1, 1]
Generation 6040: Best = 3.400673e+03 FES = 604100 Diversity = 4.8031e-05 k_t
= [3, 4, 3, 3]
Generation 6050: Best = 3.400673e+03 FES = 605100 Diversity = 4.8768e-05 k_t
= [1, 1, 1, 1]
Generation 6060: Best = 3.400673e+03 FES = 606100 Diversity = 4.8263e-05 k_t
= [4, 3, 3, 4]
Generation 6070: Best = 3.400673e+03 FES = 607100 Diversity = 5.4297e-05 k_t
= [1, 1, 1, 1]
Generation 6080: Best = 3.400673e+03 FES = 608100 Diversity = 5.5004e-05 k_t
= [3, 4, 3, 2]
Generation 6090: Best = 3.400673e+03 FES = 609100 Diversity = 5.4056e-05 k_t
= [1, 1, 1, 1]
Generation 6100: Best = 3.400673e+03 FES = 610100 Diversity = 5.5212e-05 k_t
= [3, 3, 3, 4]
Generation 6110: Best = 3.400673e+03 FES = 611100 Diversity = 5.4842e-05 k_t
= [1, 1, 1, 1]
Generation 6120: Best = 3.400673e+03 FES = 612100 Diversity = 5.5913e-05 k_t
= [4, 4, 4, 3]
Generation 6130: Best = 3.400673e+03 FES = 613100 Diversity = 5.3432e-05 k_t
= [1, 1, 1, 1]
Generation 6140: Best = 3.400673e+03 FES = 614100 Diversity = 5.2914e-05 k_t
= [2, 3, 3, 4]
Generation 6150: Best = 3.400673e+03 FES = 615100 Diversity = 6.2856e-05 k_t
= [1, 1, 1, 1]
Generation 6160: Best = 3.400673e+03 FES = 616100 Diversity = 6.7226e-05 k_t

```

```

= [3, 3, 3, 2]
Generation 6170: Best = 3.400673e+03 FES = 617100 Diversity = 6.5916e-05 k_t
= [1, 1, 1, 1]
Generation 6180: Best = 3.400673e+03 FES = 618100 Diversity = 6.7112e-05 k_t
= [2, 2, 2, 2]
Generation 6190: Best = 3.400673e+03 FES = 619100 Diversity = 6.7057e-05 k_t
= [1, 1, 1, 1]
Generation 6200: Best = 3.400673e+03 FES = 620100 Diversity = 6.1547e-05 k_t
= [4, 2, 3, 4]
Generation 6210: Best = 3.400673e+03 FES = 621100 Diversity = 5.7387e-05 k_t
= [1, 1, 1, 1]
Generation 6220: Best = 3.400673e+03 FES = 622100 Diversity = 5.3335e-05 k_t
= [2, 3, 2, 2]
Generation 6230: Best = 3.400673e+03 FES = 623100 Diversity = 5.0415e-05 k_t
= [1, 1, 1, 1]
Generation 6240: Best = 3.400673e+03 FES = 624100 Diversity = 4.8306e-05 k_t
= [4, 2, 2, 3]
Generation 6250: Best = 3.400673e+03 FES = 625100 Diversity = 4.6932e-05 k_t
= [1, 1, 1, 1]
Generation 6260: Best = 3.400673e+03 FES = 626100 Diversity = 4.2928e-05 k_t
= [4, 3, 4, 2]
Generation 6270: Best = 3.400673e+03 FES = 627100 Diversity = 4.1031e-05 k_t
= [1, 1, 1, 1]
Generation 6280: Best = 3.400673e+03 FES = 628100 Diversity = 4.0879e-05 k_t
= [2, 3, 3, 2]
Generation 6290: Best = 3.400673e+03 FES = 629100 Diversity = 4.3003e-05 k_t
= [1, 1, 1, 1]
Generation 6300: Best = 3.400673e+03 FES = 630100 Diversity = 4.2583e-05 k_t
= [2, 3, 3, 4]
Generation 6310: Best = 3.400673e+03 FES = 631100 Diversity = 4.2161e-05 k_t
= [1, 1, 1, 1]
Generation 6320: Best = 3.400673e+03 FES = 632100 Diversity = 4.1591e-05 k_t
= [4, 2, 4, 4]
Generation 6330: Best = 3.400673e+03 FES = 633100 Diversity = 4.1167e-05 k_t
= [1, 1, 1, 1]
Generation 6340: Best = 3.400673e+03 FES = 634100 Diversity = 3.9594e-05 k_t
= [2, 2, 4, 2]
Generation 6350: Best = 3.400673e+03 FES = 635100 Diversity = 3.9923e-05 k_t
= [1, 1, 1, 1]
Generation 6360: Best = 3.400673e+03 FES = 636100 Diversity = 4.0760e-05 k_t
= [3, 2, 2, 2]
Generation 6370: Best = 3.400673e+03 FES = 637100 Diversity = 4.0717e-05 k_t
= [1, 1, 1, 1]
Generation 6380: Best = 3.400673e+03 FES = 638100 Diversity = 3.8307e-05 k_t
= [2, 2, 4, 4]
Generation 6390: Best = 3.400673e+03 FES = 639100 Diversity = 3.6387e-05 k_t
= [1, 1, 1, 1]
Generation 6400: Best = 3.400673e+03 FES = 640100 Diversity = 3.4727e-05 k_t
= [4, 4, 3, 3]
Generation 6410: Best = 3.400673e+03 FES = 641100 Diversity = 3.1890e-05 k_t
= [1, 1, 1, 1]
Generation 6420: Best = 3.400673e+03 FES = 642100 Diversity = 2.9619e-05 k_t
= [2, 2, 4, 2]
Generation 6430: Best = 3.400673e+03 FES = 643100 Diversity = 2.8630e-05 k_t

```

```

= [1, 1, 1, 1]
Generation 6440: Best = 3.400673e+03 FES = 644100 Diversity = 2.8730e-05 k_t
= [2, 3, 4, 3]
Generation 6450: Best = 3.400673e+03 FES = 645100 Diversity = 2.7862e-05 k_t
= [1, 1, 1, 1]
Generation 6460: Best = 3.400673e+03 FES = 646100 Diversity = 2.7682e-05 k_t
= [2, 4, 4, 2]
Generation 6470: Best = 3.400673e+03 FES = 647100 Diversity = 2.7993e-05 k_t
= [1, 1, 1, 1]
Generation 6480: Best = 3.400673e+03 FES = 648100 Diversity = 2.8135e-05 k_t
= [4, 4, 2, 4]
Generation 6490: Best = 3.400673e+03 FES = 649100 Diversity = 2.7739e-05 k_t
= [1, 1, 1, 1]
Generation 6500: Best = 3.400673e+03 FES = 650100 Diversity = 2.6359e-05 k_t
= [3, 2, 3, 3]
Generation 6510: Best = 3.400673e+03 FES = 651100 Diversity = 2.5176e-05 k_t
= [1, 1, 1, 1]
Generation 6520: Best = 3.400673e+03 FES = 652100 Diversity = 2.6360e-05 k_t
= [2, 3, 3, 3]
Generation 6530: Best = 3.400673e+03 FES = 653100 Diversity = 2.6970e-05 k_t
= [1, 1, 1, 1]
Generation 6540: Best = 3.400673e+03 FES = 654100 Diversity = 2.5522e-05 k_t
= [4, 4, 4, 4]
Generation 6550: Best = 3.400673e+03 FES = 655100 Diversity = 2.6605e-05 k_t
= [1, 1, 1, 1]
Generation 6560: Best = 3.400673e+03 FES = 656100 Diversity = 2.7005e-05 k_t
= [3, 2, 4, 2]
Generation 6570: Best = 3.400673e+03 FES = 657100 Diversity = 2.8646e-05 k_t
= [1, 1, 1, 1]
Generation 6580: Best = 3.400673e+03 FES = 658100 Diversity = 2.8938e-05 k_t
= [2, 2, 2, 2]
Generation 6590: Best = 3.400673e+03 FES = 659100 Diversity = 3.0349e-05 k_t
= [1, 1, 1, 1]
Generation 6600: Best = 3.400673e+03 FES = 660100 Diversity = 3.1651e-05 k_t
= [3, 4, 3, 3]
Generation 6610: Best = 3.400673e+03 FES = 661100 Diversity = 3.2537e-05 k_t
= [1, 1, 1, 1]
Generation 6620: Best = 3.400673e+03 FES = 662100 Diversity = 3.1183e-05 k_t
= [2, 3, 4, 3]
Generation 6630: Best = 3.400673e+03 FES = 663100 Diversity = 3.0258e-05 k_t
= [1, 1, 1, 1]
Generation 6640: Best = 3.400673e+03 FES = 664100 Diversity = 3.2081e-05 k_t
= [2, 3, 4, 2]
Generation 6650: Best = 3.400673e+03 FES = 665100 Diversity = 3.1517e-05 k_t
= [1, 1, 1, 1]
Generation 6660: Best = 3.400673e+03 FES = 666100 Diversity = 3.0472e-05 k_t
= [4, 2, 4, 3]
Generation 6670: Best = 3.400673e+03 FES = 667100 Diversity = 3.0064e-05 k_t
= [1, 1, 1, 1]
Generation 6680: Best = 3.400673e+03 FES = 668100 Diversity = 2.9709e-05 k_t
= [2, 2, 3, 4]
Generation 6690: Best = 3.400673e+03 FES = 669100 Diversity = 2.9678e-05 k_t
= [1, 1, 1, 1]
Generation 6700: Best = 3.400673e+03 FES = 670100 Diversity = 2.9426e-05 k_t

```



```

= [3, 2, 3, 3]
Generation 6710: Best = 3.400673e+03 FES = 671100 Diversity = 3.0732e-05 k_t
= [1, 1, 1, 1]
Generation 6720: Best = 3.400673e+03 FES = 672100 Diversity = 3.0353e-05 k_t
= [2, 4, 4, 4]
Generation 6730: Best = 3.400673e+03 FES = 673100 Diversity = 3.4264e-05 k_t
= [1, 1, 1, 1]
Generation 6740: Best = 3.400673e+03 FES = 674100 Diversity = 3.6179e-05 k_t
= [2, 3, 2, 3]
Generation 6750: Best = 3.400673e+03 FES = 675100 Diversity = 3.7085e-05 k_t
= [1, 1, 1, 1]
Generation 6760: Best = 3.400673e+03 FES = 676100 Diversity = 4.0959e-05 k_t
= [4, 4, 2, 2]
Generation 6770: Best = 3.400673e+03 FES = 677100 Diversity = 4.4298e-05 k_t
= [1, 1, 1, 1]
Generation 6780: Best = 3.400673e+03 FES = 678100 Diversity = 4.5811e-05 k_t
= [2, 2, 4, 2]
Generation 6790: Best = 3.400673e+03 FES = 679100 Diversity = 4.6073e-05 k_t
= [1, 1, 1, 1]
Generation 6800: Best = 3.400673e+03 FES = 680100 Diversity = 4.6954e-05 k_t
= [2, 4, 4, 4]
Generation 6810: Best = 3.400673e+03 FES = 681100 Diversity = 5.0517e-05 k_t
= [1, 1, 1, 1]
Generation 6820: Best = 3.400673e+03 FES = 682100 Diversity = 5.6313e-05 k_t
= [3, 2, 4, 2]
Generation 6830: Best = 3.400673e+03 FES = 683100 Diversity = 5.5634e-05 k_t
= [1, 1, 1, 1]
Generation 6840: Best = 3.400673e+03 FES = 684100 Diversity = 5.5453e-05 k_t
= [2, 3, 3, 2]
Generation 6850: Best = 3.400673e+03 FES = 685100 Diversity = 5.3766e-05 k_t
= [1, 1, 1, 1]
Generation 6860: Best = 3.400673e+03 FES = 686100 Diversity = 5.4854e-05 k_t
= [3, 3, 2, 3]
Generation 6870: Best = 3.400673e+03 FES = 687100 Diversity = 6.1433e-05 k_t
= [1, 1, 1, 1]
Generation 6880: Best = 3.400673e+03 FES = 688100 Diversity = 6.2178e-05 k_t
= [2, 2, 2, 3]
Generation 6890: Best = 3.400673e+03 FES = 689100 Diversity = 6.1311e-05 k_t
= [1, 1, 1, 1]
Generation 6900: Best = 3.400673e+03 FES = 690100 Diversity = 6.0161e-05 k_t
= [2, 4, 2, 2]
Generation 6910: Best = 3.400673e+03 FES = 691100 Diversity = 6.1332e-05 k_t
= [1, 1, 1, 1]
Generation 6920: Best = 3.400673e+03 FES = 692100 Diversity = 6.0564e-05 k_t
= [4, 3, 2, 2]
Generation 6930: Best = 3.400673e+03 FES = 693100 Diversity = 5.9224e-05 k_t
= [1, 1, 1, 1]
Generation 6940: Best = 3.400673e+03 FES = 694100 Diversity = 5.8091e-05 k_t
= [3, 4, 4, 3]
Generation 6950: Best = 3.400673e+03 FES = 695100 Diversity = 5.4197e-05 k_t
= [1, 1, 1, 1]
Generation 6960: Best = 3.400673e+03 FES = 696100 Diversity = 5.4231e-05 k_t
= [4, 3, 3, 3]
Generation 6970: Best = 3.400673e+03 FES = 697100 Diversity = 5.1468e-05 k_t

```

```

= [1, 1, 1, 1]
Generation 6980: Best = 3.400673e+03 FES = 698100 Diversity = 4.7065e-05 k_t
= [3, 3, 3, 2]
Generation 6990: Best = 3.400673e+03 FES = 699100 Diversity = 4.6671e-05 k_t
= [1, 1, 1, 1]
Generation 7000: Best = 3.400673e+03 FES = 700100 Diversity = 4.5900e-05 k_t
= [2, 4, 3, 3]
Generation 7010: Best = 3.400673e+03 FES = 701100 Diversity = 4.3547e-05 k_t
= [1, 1, 1, 1]
Generation 7020: Best = 3.400673e+03 FES = 702100 Diversity = 4.1948e-05 k_t
= [2, 2, 4, 3]
Generation 7030: Best = 3.400673e+03 FES = 703100 Diversity = 4.1521e-05 k_t
= [1, 1, 1, 1]
Generation 7040: Best = 3.400673e+03 FES = 704100 Diversity = 4.0430e-05 k_t
= [2, 4, 4, 3]
Generation 7050: Best = 3.400673e+03 FES = 705100 Diversity = 3.8774e-05 k_t
= [1, 1, 1, 1]
Generation 7060: Best = 3.400673e+03 FES = 706100 Diversity = 3.8077e-05 k_t
= [4, 4, 3, 3]
Generation 7070: Best = 3.400673e+03 FES = 707100 Diversity = 3.9874e-05 k_t
= [1, 1, 1, 1]
Generation 7080: Best = 3.400673e+03 FES = 708100 Diversity = 3.9736e-05 k_t
= [4, 4, 3, 3]
Generation 7090: Best = 3.400673e+03 FES = 709100 Diversity = 4.0505e-05 k_t
= [1, 1, 1, 1]
Generation 7100: Best = 3.400673e+03 FES = 710100 Diversity = 3.8538e-05 k_t
= [4, 4, 3, 4]
Generation 7110: Best = 3.400673e+03 FES = 711100 Diversity = 3.8167e-05 k_t
= [1, 1, 1, 1]
Generation 7120: Best = 3.400673e+03 FES = 712100 Diversity = 3.5493e-05 k_t
= [2, 3, 3, 4]
Generation 7130: Best = 3.400673e+03 FES = 713100 Diversity = 3.5565e-05 k_t
= [1, 1, 1, 1]
Generation 7140: Best = 3.400673e+03 FES = 714100 Diversity = 3.3397e-05 k_t
= [4, 3, 3, 2]
Generation 7150: Best = 3.400673e+03 FES = 715100 Diversity = 3.0043e-05 k_t
= [1, 1, 1, 1]
Generation 7160: Best = 3.400673e+03 FES = 716100 Diversity = 2.9254e-05 k_t
= [2, 3, 4, 3]
Generation 7170: Best = 3.400673e+03 FES = 717100 Diversity = 2.8684e-05 k_t
= [1, 1, 1, 1]
Generation 7180: Best = 3.400673e+03 FES = 718100 Diversity = 2.7732e-05 k_t
= [2, 3, 4, 2]
Generation 7190: Best = 3.400673e+03 FES = 719100 Diversity = 2.6869e-05 k_t
= [1, 1, 1, 1]
Generation 7200: Best = 3.400673e+03 FES = 720100 Diversity = 2.4614e-05 k_t
= [4, 2, 2, 4]
Generation 7210: Best = 3.400673e+03 FES = 721100 Diversity = 2.5203e-05 k_t
= [1, 1, 1, 1]
Generation 7220: Best = 3.400673e+03 FES = 722100 Diversity = 2.5335e-05 k_t
= [2, 3, 2, 3]
Generation 7230: Best = 3.400673e+03 FES = 723100 Diversity = 2.5267e-05 k_t
= [1, 1, 1, 1]
Generation 7240: Best = 3.400673e+03 FES = 724100 Diversity = 2.5022e-05 k_t

```

```

= [2, 4, 4, 4]
Generation 7250: Best = 3.400673e+03 FES = 725100 Diversity = 2.5615e-05 k_t
= [1, 1, 1, 1]
Generation 7260: Best = 3.400673e+03 FES = 726100 Diversity = 2.5193e-05 k_t
= [2, 4, 2, 2]
Generation 7270: Best = 3.400673e+03 FES = 727100 Diversity = 2.4212e-05 k_t
= [1, 1, 1, 1]
Generation 7280: Best = 3.400673e+03 FES = 728100 Diversity = 2.4243e-05 k_t
= [3, 4, 3, 2]
Generation 7290: Best = 3.400673e+03 FES = 729100 Diversity = 2.3546e-05 k_t
= [1, 1, 1, 1]
Generation 7300: Best = 3.400673e+03 FES = 730100 Diversity = 2.3295e-05 k_t
= [4, 2, 3, 4]
Generation 7310: Best = 3.400673e+03 FES = 731100 Diversity = 2.3328e-05 k_t
= [1, 1, 1, 1]
Generation 7320: Best = 3.400673e+03 FES = 732100 Diversity = 2.2869e-05 k_t
= [3, 2, 3, 4]
Generation 7330: Best = 3.400673e+03 FES = 733100 Diversity = 2.2741e-05 k_t
= [1, 1, 1, 1]
Generation 7340: Best = 3.400673e+03 FES = 734100 Diversity = 2.1326e-05 k_t
= [2, 2, 3, 3]
Generation 7350: Best = 3.400673e+03 FES = 735100 Diversity = 2.0529e-05 k_t
= [1, 1, 1, 1]
Generation 7360: Best = 3.400673e+03 FES = 736100 Diversity = 2.0614e-05 k_t
= [4, 2, 2, 4]
Generation 7370: Best = 3.400673e+03 FES = 737100 Diversity = 2.0966e-05 k_t
= [1, 1, 1, 1]
Generation 7380: Best = 3.400673e+03 FES = 738100 Diversity = 2.0876e-05 k_t
= [2, 4, 3, 4]
Generation 7390: Best = 3.400673e+03 FES = 739100 Diversity = 2.0784e-05 k_t
= [1, 1, 1, 1]
Generation 7400: Best = 3.400673e+03 FES = 740100 Diversity = 2.0021e-05 k_t
= [4, 3, 2, 4]
Generation 7410: Best = 3.400673e+03 FES = 741100 Diversity = 1.9047e-05 k_t
= [1, 1, 1, 1]
Generation 7420: Best = 3.400673e+03 FES = 742100 Diversity = 1.7502e-05 k_t
= [4, 3, 2, 4]
Generation 7430: Best = 3.400673e+03 FES = 743100 Diversity = 1.7360e-05 k_t
= [1, 1, 1, 1]
Generation 7440: Best = 3.400673e+03 FES = 744100 Diversity = 1.6758e-05 k_t
= [4, 2, 4, 3]
Generation 7450: Best = 3.400673e+03 FES = 745100 Diversity = 1.6197e-05 k_t
= [1, 1, 1, 1]
Generation 7460: Best = 3.400673e+03 FES = 746100 Diversity = 1.6677e-05 k_t
= [2, 3, 2, 4]
Generation 7470: Best = 3.400673e+03 FES = 747100 Diversity = 1.5728e-05 k_t
= [1, 1, 1, 1]
Generation 7480: Best = 3.400673e+03 FES = 748100 Diversity = 1.6174e-05 k_t
= [4, 4, 4, 3]
Generation 7490: Best = 3.400673e+03 FES = 749100 Diversity = 1.6781e-05 k_t
= [1, 1, 1, 1]
Generation 7500: Best = 3.400673e+03 FES = 750100 Diversity = 1.7479e-05 k_t
= [3, 2, 2, 4]
Generation 7510: Best = 3.400673e+03 FES = 751100 Diversity = 1.8109e-05 k_t

```

```

= [1, 1, 1, 1]
Generation 7520: Best = 3.400673e+03 FES = 752100 Diversity = 1.8867e-05 k_t
= [2, 3, 4, 3]
Generation 7530: Best = 3.400673e+03 FES = 753100 Diversity = 2.0606e-05 k_t
= [1, 1, 1, 1]
Generation 7540: Best = 3.400673e+03 FES = 754100 Diversity = 2.3493e-05 k_t
= [2, 3, 2, 2]
Generation 7550: Best = 3.400673e+03 FES = 755100 Diversity = 2.3087e-05 k_t
= [1, 1, 1, 1]
Generation 7560: Best = 3.400673e+03 FES = 756100 Diversity = 2.1509e-05 k_t
= [2, 2, 2, 4]
Generation 7570: Best = 3.400673e+03 FES = 757100 Diversity = 2.0608e-05 k_t
= [1, 1, 1, 1]
Generation 7580: Best = 3.400673e+03 FES = 758100 Diversity = 2.0845e-05 k_t
= [2, 2, 2, 3]
Generation 7590: Best = 3.400673e+03 FES = 759100 Diversity = 2.1263e-05 k_t
= [1, 1, 1, 1]
Generation 7600: Best = 3.400673e+03 FES = 760100 Diversity = 2.1551e-05 k_t
= [3, 4, 2, 3]
Generation 7610: Best = 3.400673e+03 FES = 761100 Diversity = 2.0305e-05 k_t
= [1, 1, 1, 1]
Generation 7620: Best = 3.400673e+03 FES = 762100 Diversity = 2.0823e-05 k_t
= [4, 3, 2, 3]
Generation 7630: Best = 3.400673e+03 FES = 763100 Diversity = 2.0969e-05 k_t
= [1, 1, 1, 1]
Generation 7640: Best = 3.400673e+03 FES = 764100 Diversity = 2.1427e-05 k_t
= [4, 2, 4, 3]
Generation 7650: Best = 3.400673e+03 FES = 765100 Diversity = 2.0039e-05 k_t
= [1, 1, 1, 1]
Generation 7660: Best = 3.400673e+03 FES = 766100 Diversity = 1.9080e-05 k_t
= [3, 4, 3, 4]
Generation 7670: Best = 3.400673e+03 FES = 767100 Diversity = 2.1217e-05 k_t
= [1, 1, 1, 1]
Generation 7680: Best = 3.400673e+03 FES = 768100 Diversity = 2.1527e-05 k_t
= [3, 3, 3, 4]
Generation 7690: Best = 3.400673e+03 FES = 769100 Diversity = 1.9985e-05 k_t
= [1, 1, 1, 1]
Generation 7700: Best = 3.400673e+03 FES = 770100 Diversity = 2.0231e-05 k_t
= [3, 2, 3, 2]
Generation 7710: Best = 3.400673e+03 FES = 771100 Diversity = 2.0761e-05 k_t
= [1, 1, 1, 1]
Generation 7720: Best = 3.400673e+03 FES = 772100 Diversity = 2.0661e-05 k_t
= [3, 4, 2, 2]
Generation 7730: Best = 3.400673e+03 FES = 773100 Diversity = 2.0096e-05 k_t
= [1, 1, 1, 1]
Generation 7740: Best = 3.400673e+03 FES = 774100 Diversity = 2.0244e-05 k_t
= [4, 2, 4, 3]
Generation 7750: Best = 3.400673e+03 FES = 775100 Diversity = 2.0469e-05 k_t
= [1, 1, 1, 1]
Generation 7760: Best = 3.400673e+03 FES = 776100 Diversity = 2.0069e-05 k_t
= [3, 2, 2, 2]
Generation 7770: Best = 3.400673e+03 FES = 777100 Diversity = 2.2380e-05 k_t
= [1, 1, 1, 1]
Generation 7780: Best = 3.400673e+03 FES = 778100 Diversity = 2.1377e-05 k_t

```

```

= [3, 3, 2, 4]
Generation 7790: Best = 3.400673e+03 FES = 779100 Diversity = 2.1484e-05 k_t
= [1, 1, 1, 1]
Generation 7800: Best = 3.400673e+03 FES = 780100 Diversity = 2.1554e-05 k_t
= [2, 4, 2, 2]
Generation 7810: Best = 3.400673e+03 FES = 781100 Diversity = 2.1333e-05 k_t
= [1, 1, 1, 1]
Generation 7820: Best = 3.400673e+03 FES = 782100 Diversity = 2.2314e-05 k_t
= [4, 4, 4, 4]
Generation 7830: Best = 3.400673e+03 FES = 783100 Diversity = 2.3375e-05 k_t
= [1, 1, 1, 1]
Generation 7840: Best = 3.400673e+03 FES = 784100 Diversity = 2.2714e-05 k_t
= [3, 4, 2, 2]
Generation 7850: Best = 3.400673e+03 FES = 785100 Diversity = 2.2279e-05 k_t
= [1, 1, 1, 1]
Generation 7860: Best = 3.400673e+03 FES = 786100 Diversity = 2.1840e-05 k_t
= [4, 3, 2, 4]
Generation 7870: Best = 3.400673e+03 FES = 787100 Diversity = 2.1597e-05 k_t
= [1, 1, 1, 1]
Generation 7880: Best = 3.400673e+03 FES = 788100 Diversity = 2.2301e-05 k_t
= [3, 3, 3, 4]
Generation 7890: Best = 3.400673e+03 FES = 789100 Diversity = 2.1732e-05 k_t
= [1, 1, 1, 1]
Generation 7900: Best = 3.400673e+03 FES = 790100 Diversity = 2.0753e-05 k_t
= [4, 2, 4, 4]
Generation 7910: Best = 3.400673e+03 FES = 791100 Diversity = 2.0456e-05 k_t
= [1, 1, 1, 1]
Generation 7920: Best = 3.400673e+03 FES = 792100 Diversity = 2.0237e-05 k_t
= [3, 3, 4, 3]
Generation 7930: Best = 3.400673e+03 FES = 793100 Diversity = 2.1042e-05 k_t
= [1, 1, 1, 1]
Generation 7940: Best = 3.400673e+03 FES = 794100 Diversity = 2.2236e-05 k_t
= [4, 4, 4, 4]
Generation 7950: Best = 3.400673e+03 FES = 795100 Diversity = 2.1855e-05 k_t
= [1, 1, 1, 1]
Generation 7960: Best = 3.400673e+03 FES = 796100 Diversity = 2.1796e-05 k_t
= [3, 4, 2, 4]
Generation 7970: Best = 3.400673e+03 FES = 797100 Diversity = 2.1176e-05 k_t
= [1, 1, 1, 1]
Generation 7980: Best = 3.400673e+03 FES = 798100 Diversity = 2.0891e-05 k_t
= [3, 2, 4, 4]
Generation 7990: Best = 3.400673e+03 FES = 799100 Diversity = 1.9665e-05 k_t
= [1, 1, 1, 1]
Generation 8000: Best = 3.400673e+03 FES = 800100 Diversity = 1.8723e-05 k_t
= [2, 3, 3, 2]
Generation 8010: Best = 3.400673e+03 FES = 801100 Diversity = 1.8239e-05 k_t
= [1, 1, 1, 1]
Generation 8020: Best = 3.400673e+03 FES = 802100 Diversity = 1.8542e-05 k_t
= [3, 4, 3, 3]
Generation 8030: Best = 3.400673e+03 FES = 803100 Diversity = 1.8463e-05 k_t
= [1, 1, 1, 1]
Generation 8040: Best = 3.400673e+03 FES = 804100 Diversity = 1.8767e-05 k_t
= [2, 3, 2, 3]
Generation 8050: Best = 3.400673e+03 FES = 805100 Diversity = 1.8995e-05 k_t

```

```

= [1, 1, 1, 1]
Generation 8060: Best = 3.400673e+03 FES = 806100 Diversity = 1.9001e-05 k_t
= [3, 2, 4, 2]
Generation 8070: Best = 3.400673e+03 FES = 807100 Diversity = 1.7558e-05 k_t
= [1, 1, 1, 1]
Generation 8080: Best = 3.400673e+03 FES = 808100 Diversity = 1.5743e-05 k_t
= [3, 3, 2, 4]
Generation 8090: Best = 3.400673e+03 FES = 809100 Diversity = 1.4400e-05 k_t
= [1, 1, 1, 1]
Generation 8100: Best = 3.400673e+03 FES = 810100 Diversity = 1.3267e-05 k_t
= [3, 3, 4, 4]
Generation 8110: Best = 3.400673e+03 FES = 811100 Diversity = 1.2403e-05 k_t
= [1, 1, 1, 1]
Generation 8120: Best = 3.400673e+03 FES = 812100 Diversity = 1.2069e-05 k_t
= [4, 4, 4, 2]
Generation 8130: Best = 3.400673e+03 FES = 813100 Diversity = 1.1062e-05 k_t
= [1, 1, 1, 1]
Generation 8140: Best = 3.400673e+03 FES = 814100 Diversity = 1.1192e-05 k_t
= [2, 3, 4, 3]
Generation 8150: Best = 3.400673e+03 FES = 815100 Diversity = 1.1244e-05 k_t
= [1, 1, 1, 1]
Generation 8160: Best = 3.400673e+03 FES = 816100 Diversity = 1.0457e-05 k_t
= [4, 4, 4, 2]
Generation 8170: Best = 3.400673e+03 FES = 817100 Diversity = 1.0508e-05 k_t
= [1, 1, 1, 1]
Generation 8180: Best = 3.400673e+03 FES = 818100 Diversity = 1.0311e-05 k_t
= [3, 4, 3, 3]
Generation 8190: Best = 3.400673e+03 FES = 819100 Diversity = 1.0666e-05 k_t
= [1, 1, 1, 1]
Generation 8200: Best = 3.400673e+03 FES = 820100 Diversity = 1.1059e-05 k_t
= [4, 4, 3, 3]
Generation 8210: Best = 3.400673e+03 FES = 821100 Diversity = 1.0392e-05 k_t
= [1, 1, 1, 1]
Generation 8220: Best = 3.400673e+03 FES = 822100 Diversity = 1.0220e-05 k_t
= [4, 4, 4, 4]
Generation 8230: Best = 3.400673e+03 FES = 823100 Diversity = 1.0378e-05 k_t
= [1, 1, 1, 1]
Generation 8240: Best = 3.400673e+03 FES = 824100 Diversity = 1.1311e-05 k_t
= [3, 4, 4, 2]
Generation 8250: Best = 3.400673e+03 FES = 825100 Diversity = 1.1905e-05 k_t
= [1, 1, 1, 1]
Generation 8260: Best = 3.400673e+03 FES = 826100 Diversity = 1.1689e-05 k_t
= [3, 3, 3, 2]
Generation 8270: Best = 3.400673e+03 FES = 827100 Diversity = 1.1837e-05 k_t
= [1, 1, 1, 1]
Generation 8280: Best = 3.400673e+03 FES = 828100 Diversity = 1.1604e-05 k_t
= [2, 4, 3, 3]
Generation 8290: Best = 3.400673e+03 FES = 829100 Diversity = 1.1800e-05 k_t
= [1, 1, 1, 1]
Generation 8300: Best = 3.400673e+03 FES = 830100 Diversity = 1.2042e-05 k_t
= [3, 3, 4, 4]
Generation 8310: Best = 3.400673e+03 FES = 831100 Diversity = 1.1916e-05 k_t
= [1, 1, 1, 1]
Generation 8320: Best = 3.400673e+03 FES = 832100 Diversity = 1.2010e-05 k_t

```

```

= [3, 4, 2, 3]
Generation 8330: Best = 3.400673e+03 FES = 833100 Diversity = 1.2283e-05 k_t
= [1, 1, 1, 1]
Generation 8340: Best = 3.400673e+03 FES = 834100 Diversity = 1.2294e-05 k_t
= [3, 3, 2, 3]
Generation 8350: Best = 3.400673e+03 FES = 835100 Diversity = 1.2779e-05 k_t
= [1, 1, 1, 1]
Generation 8360: Best = 3.400673e+03 FES = 836100 Diversity = 1.2817e-05 k_t
= [2, 2, 2, 3]
Generation 8370: Best = 3.400673e+03 FES = 837100 Diversity = 1.2671e-05 k_t
= [1, 1, 1, 1]
Generation 8380: Best = 3.400673e+03 FES = 838100 Diversity = 1.3014e-05 k_t
= [3, 2, 2, 2]
Generation 8390: Best = 3.400673e+03 FES = 839100 Diversity = 1.3921e-05 k_t
= [1, 1, 1, 1]
Generation 8400: Best = 3.400673e+03 FES = 840100 Diversity = 1.3697e-05 k_t
= [3, 3, 3, 2]
Generation 8410: Best = 3.400673e+03 FES = 841100 Diversity = 1.3446e-05 k_t
= [1, 1, 1, 1]
Generation 8420: Best = 3.400673e+03 FES = 842100 Diversity = 1.3644e-05 k_t
= [2, 4, 2, 3]
Generation 8430: Best = 3.400673e+03 FES = 843100 Diversity = 1.4465e-05 k_t
= [1, 1, 1, 1]
Generation 8440: Best = 3.400673e+03 FES = 844100 Diversity = 1.4976e-05 k_t
= [4, 2, 3, 3]
Generation 8450: Best = 3.400673e+03 FES = 845100 Diversity = 1.4562e-05 k_t
= [1, 1, 1, 1]
Generation 8460: Best = 3.400673e+03 FES = 846100 Diversity = 1.4777e-05 k_t
= [2, 4, 2, 3]
Generation 8470: Best = 3.400673e+03 FES = 847100 Diversity = 1.5099e-05 k_t
= [1, 1, 1, 1]
Generation 8480: Best = 3.400673e+03 FES = 848100 Diversity = 1.4837e-05 k_t
= [4, 3, 3, 3]
Generation 8490: Best = 3.400673e+03 FES = 849100 Diversity = 1.5051e-05 k_t
= [1, 1, 1, 1]
Generation 8500: Best = 3.400673e+03 FES = 850100 Diversity = 1.4827e-05 k_t
= [3, 2, 4, 4]
Generation 8510: Best = 3.400673e+03 FES = 851100 Diversity = 1.3098e-05 k_t
= [1, 1, 1, 1]
Generation 8520: Best = 3.400673e+03 FES = 852100 Diversity = 1.2459e-05 k_t
= [2, 3, 4, 3]
Generation 8530: Best = 3.400673e+03 FES = 853100 Diversity = 1.2557e-05 k_t
= [1, 1, 1, 1]
Generation 8540: Best = 3.400673e+03 FES = 854100 Diversity = 1.2118e-05 k_t
= [3, 2, 4, 3]
Generation 8550: Best = 3.400673e+03 FES = 855100 Diversity = 1.1528e-05 k_t
= [1, 1, 1, 1]
Generation 8560: Best = 3.400673e+03 FES = 856100 Diversity = 1.0778e-05 k_t
= [4, 2, 4, 2]
Generation 8570: Best = 3.400673e+03 FES = 857100 Diversity = 1.0278e-05 k_t
= [1, 1, 1, 1]
Generation 8580: Best = 3.400673e+03 FES = 858100 Diversity = 1.0767e-05 k_t
= [3, 2, 2, 2]
Generation 8590: Best = 3.400673e+03 FES = 859100 Diversity = 1.0393e-05 k_t

```

```

= [1, 1, 1, 1]
Generation 8600: Best = 3.400673e+03 FES = 860100 Diversity = 1.0189e-05 k_t
= [4, 2, 2, 2]
Generation 8610: Best = 3.400673e+03 FES = 861100 Diversity = 1.0233e-05 k_t
= [1, 1, 1, 1]
Generation 8620: Best = 3.400673e+03 FES = 862100 Diversity = 1.0174e-05 k_t
= [2, 4, 2, 3]
Generation 8630: Best = 3.400673e+03 FES = 863100 Diversity = 1.0652e-05 k_t
= [1, 1, 1, 1]
Generation 8640: Best = 3.400673e+03 FES = 864100 Diversity = 1.0622e-05 k_t
= [3, 2, 2, 2]
Generation 8650: Best = 3.400673e+03 FES = 865100 Diversity = 1.0083e-05 k_t
= [1, 1, 1, 1]
Generation 8660: Best = 3.400673e+03 FES = 866100 Diversity = 9.8901e-06 k_t
= [4, 3, 4, 4]
Generation 8670: Best = 3.400673e+03 FES = 867100 Diversity = 1.0369e-05 k_t
= [1, 1, 1, 1]
Generation 8680: Best = 3.400673e+03 FES = 868100 Diversity = 1.1035e-05 k_t
= [4, 2, 3, 3]
Generation 8690: Best = 3.400673e+03 FES = 869100 Diversity = 1.0691e-05 k_t
= [1, 1, 1, 1]
Generation 8700: Best = 3.400673e+03 FES = 870100 Diversity = 1.0450e-05 k_t
= [2, 3, 3, 4]
Generation 8710: Best = 3.400673e+03 FES = 871100 Diversity = 1.1361e-05 k_t
= [1, 1, 1, 1]
Generation 8720: Best = 3.400673e+03 FES = 872100 Diversity = 1.1840e-05 k_t
= [4, 2, 2, 2]
Generation 8730: Best = 3.400673e+03 FES = 873100 Diversity = 1.1393e-05 k_t
= [1, 1, 1, 1]
Generation 8740: Best = 3.400673e+03 FES = 874100 Diversity = 1.1161e-05 k_t
= [3, 4, 2, 4]
Generation 8750: Best = 3.400673e+03 FES = 875100 Diversity = 1.1185e-05 k_t
= [1, 1, 1, 1]
Generation 8760: Best = 3.400673e+03 FES = 876100 Diversity = 1.1048e-05 k_t
= [3, 2, 4, 2]
Generation 8770: Best = 3.400673e+03 FES = 877100 Diversity = 1.1404e-05 k_t
= [1, 1, 1, 1]
Generation 8780: Best = 3.400673e+03 FES = 878100 Diversity = 1.1277e-05 k_t
= [4, 3, 4, 3]
Generation 8790: Best = 3.400673e+03 FES = 879100 Diversity = 1.1324e-05 k_t
= [1, 1, 1, 1]
Generation 8800: Best = 3.400673e+03 FES = 880100 Diversity = 1.2201e-05 k_t
= [4, 4, 4, 4]
Generation 8810: Best = 3.400673e+03 FES = 881100 Diversity = 1.2606e-05 k_t
= [1, 1, 1, 1]
Generation 8820: Best = 3.400673e+03 FES = 882100 Diversity = 1.3156e-05 k_t
= [3, 3, 3, 4]
Generation 8830: Best = 3.400673e+03 FES = 883100 Diversity = 1.3095e-05 k_t
= [1, 1, 1, 1]
Generation 8840: Best = 3.400673e+03 FES = 884100 Diversity = 1.3331e-05 k_t
= [4, 4, 2, 3]
Generation 8850: Best = 3.400673e+03 FES = 885100 Diversity = 1.4861e-05 k_t
= [1, 1, 1, 1]
Generation 8860: Best = 3.400673e+03 FES = 886100 Diversity = 1.5038e-05 k_t

```



```

= [3, 4, 3, 2]
Generation 8870: Best = 3.400673e+03 FES = 887100 Diversity = 1.4486e-05 k_t
= [1, 1, 1, 1]
Generation 8880: Best = 3.400673e+03 FES = 888100 Diversity = 1.3693e-05 k_t
= [3, 2, 4, 3]
Generation 8890: Best = 3.400673e+03 FES = 889100 Diversity = 1.3179e-05 k_t
= [1, 1, 1, 1]
Generation 8900: Best = 3.400673e+03 FES = 890100 Diversity = 1.3777e-05 k_t
= [3, 3, 3, 3]
Generation 8910: Best = 3.400673e+03 FES = 891100 Diversity = 1.3209e-05 k_t
= [1, 1, 1, 1]
Generation 8920: Best = 3.400673e+03 FES = 892100 Diversity = 1.2356e-05 k_t
= [4, 4, 2, 2]
Generation 8930: Best = 3.400673e+03 FES = 893100 Diversity = 1.2152e-05 k_t
= [1, 1, 1, 1]
Generation 8940: Best = 3.400673e+03 FES = 894100 Diversity = 1.2213e-05 k_t
= [3, 4, 4, 3]
Generation 8950: Best = 3.400673e+03 FES = 895100 Diversity = 1.2022e-05 k_t
= [1, 1, 1, 1]
Generation 8960: Best = 3.400673e+03 FES = 896100 Diversity = 1.2151e-05 k_t
= [4, 4, 3, 2]
Generation 8970: Best = 3.400673e+03 FES = 897100 Diversity = 1.1677e-05 k_t
= [1, 1, 1, 1]
Generation 8980: Best = 3.400673e+03 FES = 898100 Diversity = 1.1392e-05 k_t
= [3, 2, 3, 3]
Generation 8990: Best = 3.400673e+03 FES = 899100 Diversity = 1.1293e-05 k_t
= [1, 1, 1, 1]
Generation 9000: Best = 3.400673e+03 FES = 900100 Diversity = 1.1358e-05 k_t
= [3, 2, 2, 3]
Generation 9010: Best = 3.400673e+03 FES = 901100 Diversity = 1.2609e-05 k_t
= [1, 1, 1, 1]
Generation 9020: Best = 3.400673e+03 FES = 902100 Diversity = 1.4526e-05 k_t
= [3, 3, 2, 4]
Generation 9030: Best = 3.400673e+03 FES = 903100 Diversity = 1.5412e-05 k_t
= [1, 1, 1, 1]
Generation 9040: Best = 3.400673e+03 FES = 904100 Diversity = 1.6063e-05 k_t
= [2, 3, 3, 4]
Generation 9050: Best = 3.400673e+03 FES = 905100 Diversity = 1.7854e-05 k_t
= [1, 1, 1, 1]
Generation 9060: Best = 3.400673e+03 FES = 906100 Diversity = 1.9804e-05 k_t
= [4, 4, 2, 2]
Generation 9070: Best = 3.400673e+03 FES = 907100 Diversity = 2.0924e-05 k_t
= [1, 1, 1, 1]
Generation 9080: Best = 3.400673e+03 FES = 908100 Diversity = 2.2825e-05 k_t
= [2, 2, 2, 2]
Generation 9090: Best = 3.400673e+03 FES = 909100 Diversity = 2.3114e-05 k_t
= [1, 1, 1, 1]
Generation 9100: Best = 3.400673e+03 FES = 910100 Diversity = 2.4081e-05 k_t
= [2, 3, 4, 4]
Generation 9110: Best = 3.400673e+03 FES = 911100 Diversity = 2.6487e-05 k_t
= [1, 1, 1, 1]
Generation 9120: Best = 3.400673e+03 FES = 912100 Diversity = 2.6330e-05 k_t
= [2, 4, 2, 2]
Generation 9130: Best = 3.400673e+03 FES = 913100 Diversity = 2.7087e-05 k_t

```

```

= [1, 1, 1, 1]
Generation 9140: Best = 3.400673e+03 FES = 914100 Diversity = 2.5658e-05 k_t
= [3, 4, 4, 3]
Generation 9150: Best = 3.400673e+03 FES = 915100 Diversity = 2.5259e-05 k_t
= [1, 1, 1, 1]
Generation 9160: Best = 3.400673e+03 FES = 916100 Diversity = 2.5324e-05 k_t
= [2, 2, 2, 3]
Generation 9170: Best = 3.400673e+03 FES = 917100 Diversity = 2.4508e-05 k_t
= [1, 1, 1, 1]
Generation 9180: Best = 3.400673e+03 FES = 918100 Diversity = 2.4608e-05 k_t
= [3, 2, 3, 3]
Generation 9190: Best = 3.400673e+03 FES = 919100 Diversity = 2.3646e-05 k_t
= [1, 1, 1, 1]
Generation 9200: Best = 3.400673e+03 FES = 920100 Diversity = 2.3914e-05 k_t
= [4, 2, 4, 3]
Generation 9210: Best = 3.400673e+03 FES = 921100 Diversity = 2.2677e-05 k_t
= [1, 1, 1, 1]
Generation 9220: Best = 3.400673e+03 FES = 922100 Diversity = 2.2955e-05 k_t
= [2, 3, 2, 3]
Generation 9230: Best = 3.400673e+03 FES = 923100 Diversity = 2.3455e-05 k_t
= [1, 1, 1, 1]
Generation 9240: Best = 3.400673e+03 FES = 924100 Diversity = 2.3618e-05 k_t
= [4, 4, 3, 4]
Generation 9250: Best = 3.400673e+03 FES = 925100 Diversity = 2.1711e-05 k_t
= [1, 1, 1, 1]
Generation 9260: Best = 3.400673e+03 FES = 926100 Diversity = 2.1378e-05 k_t
= [4, 3, 4, 4]
Generation 9270: Best = 3.400673e+03 FES = 927100 Diversity = 2.1011e-05 k_t
= [1, 1, 1, 1]
Generation 9280: Best = 3.400673e+03 FES = 928100 Diversity = 1.9686e-05 k_t
= [3, 4, 4, 3]
Generation 9290: Best = 3.400673e+03 FES = 929100 Diversity = 1.8706e-05 k_t
= [1, 1, 1, 1]
Generation 9300: Best = 3.400673e+03 FES = 930100 Diversity = 1.7887e-05 k_t
= [3, 4, 2, 4]
Generation 9310: Best = 3.400673e+03 FES = 931100 Diversity = 1.7984e-05 k_t
= [1, 1, 1, 1]
Generation 9320: Best = 3.400673e+03 FES = 932100 Diversity = 1.7370e-05 k_t
= [4, 3, 4, 4]
Generation 9330: Best = 3.400673e+03 FES = 933100 Diversity = 1.5767e-05 k_t
= [1, 1, 1, 1]
Generation 9340: Best = 3.400673e+03 FES = 934100 Diversity = 1.6255e-05 k_t
= [4, 3, 3, 3]
Generation 9350: Best = 3.400673e+03 FES = 935100 Diversity = 1.5819e-05 k_t
= [1, 1, 1, 1]
Generation 9360: Best = 3.400673e+03 FES = 936100 Diversity = 1.5277e-05 k_t
= [3, 2, 2, 3]
Generation 9370: Best = 3.400673e+03 FES = 937100 Diversity = 1.6264e-05 k_t
= [1, 1, 1, 1]
Generation 9380: Best = 3.400673e+03 FES = 938100 Diversity = 1.6717e-05 k_t
= [3, 4, 3, 2]
Generation 9390: Best = 3.400673e+03 FES = 939100 Diversity = 1.6157e-05 k_t
= [1, 1, 1, 1]
Generation 9400: Best = 3.400673e+03 FES = 940100 Diversity = 1.4309e-05 k_t

```

```

= [3, 2, 2, 2]
Generation 9410: Best = 3.400673e+03 FES = 941100 Diversity = 1.3502e-05 k_t
= [1, 1, 1, 1]
Generation 9420: Best = 3.400673e+03 FES = 942100 Diversity = 1.3042e-05 k_t
= [4, 2, 3, 4]
Generation 9430: Best = 3.400673e+03 FES = 943100 Diversity = 1.2420e-05 k_t
= [1, 1, 1, 1]
Generation 9440: Best = 3.400673e+03 FES = 944100 Diversity = 1.1921e-05 k_t
= [3, 2, 3, 2]
Generation 9450: Best = 3.400673e+03 FES = 945100 Diversity = 1.2384e-05 k_t
= [1, 1, 1, 1]
Generation 9460: Best = 3.400673e+03 FES = 946100 Diversity = 1.3665e-05 k_t
= [4, 3, 2, 3]
Generation 9470: Best = 3.400673e+03 FES = 947100 Diversity = 1.3389e-05 k_t
= [1, 1, 1, 1]
Generation 9480: Best = 3.400673e+03 FES = 948100 Diversity = 1.3602e-05 k_t
= [4, 4, 3, 2]
Generation 9490: Best = 3.400673e+03 FES = 949100 Diversity = 1.4020e-05 k_t
= [1, 1, 1, 1]
Generation 9500: Best = 3.400673e+03 FES = 950100 Diversity = 1.4459e-05 k_t
= [2, 3, 2, 4]
Generation 9510: Best = 3.400673e+03 FES = 951100 Diversity = 1.5141e-05 k_t
= [1, 1, 1, 1]
Generation 9520: Best = 3.400673e+03 FES = 952100 Diversity = 1.5384e-05 k_t
= [3, 4, 4, 3]
Generation 9530: Best = 3.400673e+03 FES = 953100 Diversity = 1.6665e-05 k_t
= [1, 1, 1, 1]
Generation 9540: Best = 3.400673e+03 FES = 954100 Diversity = 1.7083e-05 k_t
= [3, 3, 2, 2]
Generation 9550: Best = 3.400673e+03 FES = 955100 Diversity = 1.7057e-05 k_t
= [1, 1, 1, 1]
Generation 9560: Best = 3.400673e+03 FES = 956100 Diversity = 1.8338e-05 k_t
= [3, 4, 4, 3]
Generation 9570: Best = 3.400673e+03 FES = 957100 Diversity = 1.8992e-05 k_t
= [1, 1, 1, 1]
Generation 9580: Best = 3.400673e+03 FES = 958100 Diversity = 1.8674e-05 k_t
= [4, 3, 2, 3]
Generation 9590: Best = 3.400673e+03 FES = 959100 Diversity = 1.8700e-05 k_t
= [1, 1, 1, 1]
Generation 9600: Best = 3.400673e+03 FES = 960100 Diversity = 1.9511e-05 k_t
= [3, 4, 2, 3]
Generation 9610: Best = 3.400673e+03 FES = 961100 Diversity = 1.9698e-05 k_t
= [1, 1, 1, 1]
Generation 9620: Best = 3.400673e+03 FES = 962100 Diversity = 2.1667e-05 k_t
= [2, 3, 3, 3]
Generation 9630: Best = 3.400673e+03 FES = 963100 Diversity = 2.2954e-05 k_t
= [1, 1, 1, 1]
Generation 9640: Best = 3.400673e+03 FES = 964100 Diversity = 2.2848e-05 k_t
= [2, 3, 2, 2]
Generation 9650: Best = 3.400673e+03 FES = 965100 Diversity = 2.3921e-05 k_t
= [1, 1, 1, 1]
Generation 9660: Best = 3.400673e+03 FES = 966100 Diversity = 2.5227e-05 k_t
= [4, 4, 3, 3]
Generation 9670: Best = 3.400673e+03 FES = 967100 Diversity = 2.4237e-05 k_t

```

```

= [1, 1, 1, 1]
Generation 9680: Best = 3.400673e+03 FES = 968100 Diversity = 2.2710e-05 k_t
= [2, 2, 4, 2]
Generation 9690: Best = 3.400673e+03 FES = 969100 Diversity = 2.1987e-05 k_t
= [1, 1, 1, 1]
Generation 9700: Best = 3.400673e+03 FES = 970100 Diversity = 2.1151e-05 k_t
= [2, 2, 2, 2]
Generation 9710: Best = 3.400673e+03 FES = 971100 Diversity = 2.0431e-05 k_t
= [1, 1, 1, 1]
Generation 9720: Best = 3.400673e+03 FES = 972100 Diversity = 2.1076e-05 k_t
= [4, 4, 4, 4]
Generation 9730: Best = 3.400673e+03 FES = 973100 Diversity = 2.2003e-05 k_t
= [1, 1, 1, 1]
Generation 9740: Best = 3.400673e+03 FES = 974100 Diversity = 2.0930e-05 k_t
= [3, 4, 3, 3]
Generation 9750: Best = 3.400673e+03 FES = 975100 Diversity = 2.2582e-05 k_t
= [1, 1, 1, 1]
Generation 9760: Best = 3.400673e+03 FES = 976100 Diversity = 2.1780e-05 k_t
= [4, 3, 3, 4]
Generation 9770: Best = 3.400673e+03 FES = 977100 Diversity = 2.2453e-05 k_t
= [1, 1, 1, 1]
Generation 9780: Best = 3.400673e+03 FES = 978100 Diversity = 2.0853e-05 k_t
= [4, 4, 2, 4]
Generation 9790: Best = 3.400673e+03 FES = 979100 Diversity = 2.0056e-05 k_t
= [1, 1, 1, 1]
Generation 9800: Best = 3.400673e+03 FES = 980100 Diversity = 2.0165e-05 k_t
= [4, 3, 3, 4]
Generation 9810: Best = 3.400673e+03 FES = 981100 Diversity = 2.0213e-05 k_t
= [1, 1, 1, 1]
Generation 9820: Best = 3.400673e+03 FES = 982100 Diversity = 2.0233e-05 k_t
= [3, 3, 4, 2]
Generation 9830: Best = 3.400673e+03 FES = 983100 Diversity = 1.8330e-05 k_t
= [1, 1, 1, 1]
Generation 9840: Best = 3.400673e+03 FES = 984100 Diversity = 1.8207e-05 k_t
= [3, 2, 3, 2]
Generation 9850: Best = 3.400673e+03 FES = 985100 Diversity = 1.7478e-05 k_t
= [1, 1, 1, 1]
Generation 9860: Best = 3.400673e+03 FES = 986100 Diversity = 1.7801e-05 k_t
= [3, 2, 4, 2]
Generation 9870: Best = 3.400673e+03 FES = 987100 Diversity = 1.6335e-05 k_t
= [1, 1, 1, 1]
Generation 9880: Best = 3.400673e+03 FES = 988100 Diversity = 1.6047e-05 k_t
= [2, 2, 3, 3]
Generation 9890: Best = 3.400673e+03 FES = 989100 Diversity = 1.6528e-05 k_t
= [1, 1, 1, 1]
Generation 9900: Best = 3.400673e+03 FES = 990100 Diversity = 1.5568e-05 k_t
= [4, 3, 4, 2]
Generation 9910: Best = 3.400673e+03 FES = 991100 Diversity = 1.5156e-05 k_t
= [1, 1, 1, 1]
Generation 9920: Best = 3.400673e+03 FES = 992100 Diversity = 1.5451e-05 k_t
= [3, 2, 2, 4]
Generation 9930: Best = 3.400673e+03 FES = 993100 Diversity = 1.5285e-05 k_t
= [1, 1, 1, 1]
Generation 9940: Best = 3.400673e+03 FES = 994100 Diversity = 1.5845e-05 k_t

```

```

= [3, 3, 3, 4]
Generation 9950: Best = 3.400673e+03  FES = 995100  Diversity = 1.6503e-05  k_t
= [1, 1, 1, 1]
Generation 9960: Best = 3.400673e+03  FES = 996100  Diversity = 1.7094e-05  k_t
= [4, 3, 2, 3]
Generation 9970: Best = 3.400673e+03  FES = 997100  Diversity = 1.6924e-05  k_t
= [1, 1, 1, 1]
Generation 9980: Best = 3.400673e+03  FES = 998100  Diversity = 1.6733e-05  k_t
= [4, 4, 4, 2]
Generation 9990: Best = 3.400673e+03  FES = 999100  Diversity = 1.8852e-05  k_t
= [1, 1, 1, 1]

```

```

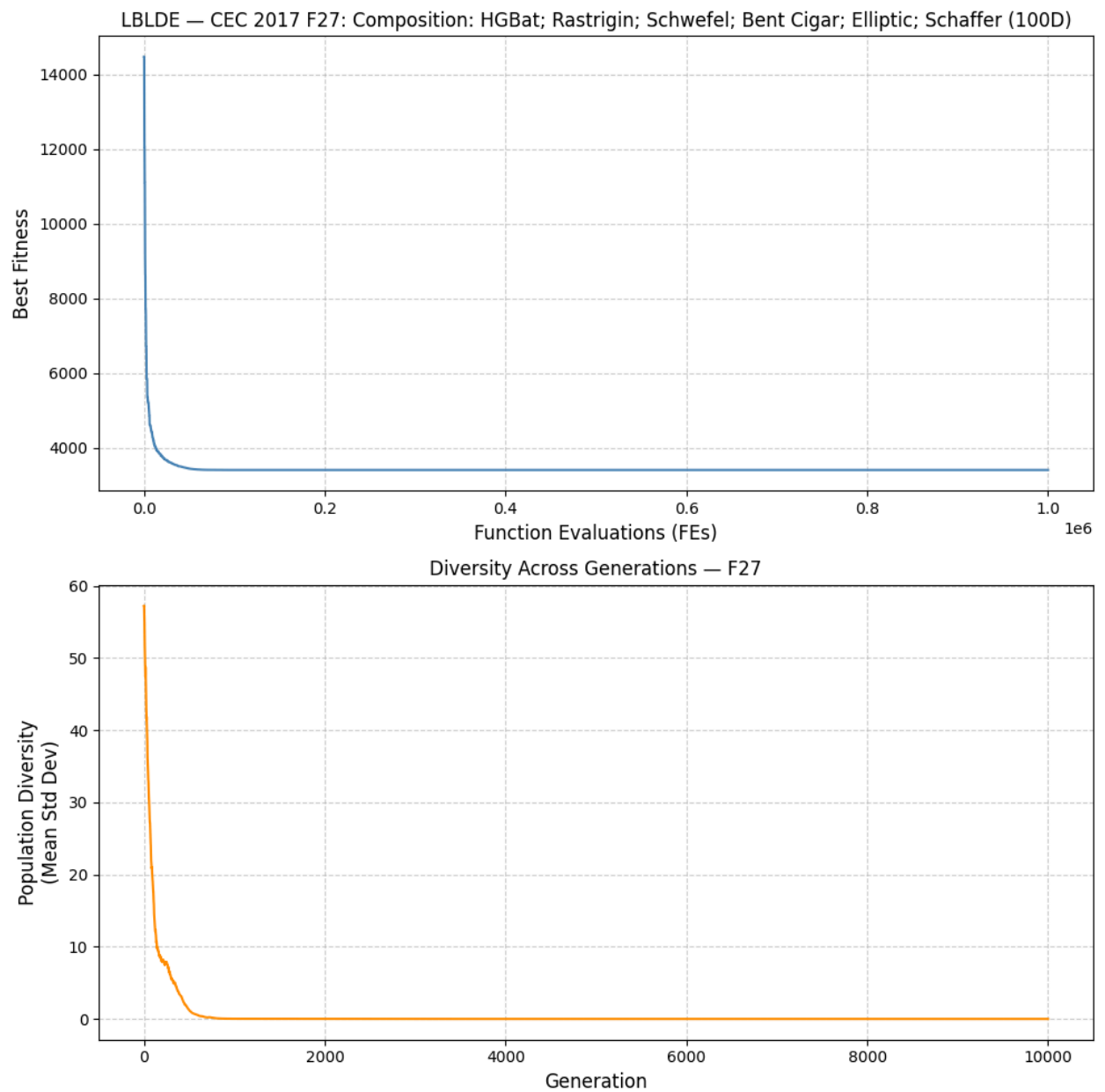
Optimization completed!
Final Best Fitness = 3.400673e+03
Total Function Evaluations = 1000000

```

```

=====
=
Results:
  Best Fitness      : 3.400673e+03
  Global Optimum    : 2.700000e+03
  Error (|best-optimum|) : 7.006733e+02
  Best Solution (5 dim) : [ 81.78496598  68.99397232 -98.48344103 -26.49501026
-96.30667678]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```



Graph saved as convergence_F27.png

```
In [52]: # =====
# F28 — Composition Function 8 (Ackley; Griewank; Discus; Rosenbrock; HappyCat)
# =====
run_and_plot('F28', make_wrapper(composition.f28), 2800.0, 'Composition: Ackle
```

```

=====
=
LBLDE_s on CEC 2017 F28 – Single Run
=====
=
Function      : Composition: Ackley; Griewank; Discus; Rosenbrock; HappyCat; S
chaffer (100D)
Bounds       : [-100, 100]
Global optimum : 2800.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 6.506878e+04 | Diversity = 5.7220e+01
Generation 10: Best = 3.695019e+04 FES = 1100 Diversity = 4.1281e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 2.231415e+04 FES = 2100 Diversity = 2.9596e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 1.838927e+04 FES = 3100 Diversity = 2.0595e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.411625e+04 FES = 4100 Diversity = 1.4328e+01 k_t
= [2, 3, 4, 3]
Generation 50: Best = 1.173105e+04 FES = 5100 Diversity = 1.1171e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.050079e+04 FES = 6100 Diversity = 8.6684e+00 k_t
= [3, 4, 2, 3]
Generation 70: Best = 9.543121e+03 FES = 7100 Diversity = 6.8448e+00 k_t
= [1, 1, 1, 1]
Generation 80: Best = 8.986660e+03 FES = 8100 Diversity = 5.7250e+00 k_t
= [4, 3, 2, 3]
Generation 90: Best = 8.411274e+03 FES = 9100 Diversity = 5.0683e+00 k_t
= [1, 1, 1, 1]
Generation 100: Best = 7.997278e+03 FES = 10100 Diversity = 4.5581e+00 k_t
= [3, 3, 2, 3]
Generation 110: Best = 7.667659e+03 FES = 11100 Diversity = 4.1881e+00 k_t
= [1, 1, 1, 1]
Generation 120: Best = 7.336227e+03 FES = 12100 Diversity = 3.8951e+00 k_t
= [4, 3, 2, 2]
Generation 130: Best = 6.862740e+03 FES = 13100 Diversity = 3.7321e+00 k_t
= [1, 1, 1, 1]
Generation 140: Best = 6.644533e+03 FES = 14100 Diversity = 3.7013e+00 k_t
= [2, 2, 3, 3]
Generation 150: Best = 6.352982e+03 FES = 15100 Diversity = 3.5936e+00 k_t
= [1, 1, 1, 1]
Generation 160: Best = 6.146947e+03 FES = 16100 Diversity = 3.5107e+00 k_t
= [3, 4, 3, 4]
Generation 170: Best = 5.947401e+03 FES = 17100 Diversity = 3.3899e+00 k_t
= [1, 1, 1, 1]
Generation 180: Best = 5.739164e+03 FES = 18100 Diversity = 3.2022e+00 k_t
= [4, 4, 2, 4]
Generation 190: Best = 5.502238e+03 FES = 19100 Diversity = 3.2154e+00 k_t
= [1, 1, 1, 1]
Generation 200: Best = 5.274195e+03 FES = 20100 Diversity = 3.0277e+00 k_t
= [3, 4, 3, 4]
Generation 210: Best = 5.202750e+03 FES = 21100 Diversity = 2.8843e+00 k_t
= [1, 1, 1, 1]
Generation 220: Best = 5.039266e+03 FES = 22100 Diversity = 2.5791e+00 k_t

```

```

= [3, 4, 2, 4]
Generation 230: Best = 4.945075e+03 FES = 23100 Diversity = 2.2770e+00 k_t
= [1, 1, 1, 1]
Generation 240: Best = 4.833805e+03 FES = 24100 Diversity = 2.2890e+00 k_t
= [4, 2, 4, 3]
Generation 250: Best = 4.735712e+03 FES = 25100 Diversity = 2.2292e+00 k_t
= [1, 1, 1, 1]
Generation 260: Best = 4.643534e+03 FES = 26100 Diversity = 2.2412e+00 k_t
= [2, 2, 3, 4]
Generation 270: Best = 4.557076e+03 FES = 27100 Diversity = 2.1032e+00 k_t
= [1, 1, 1, 1]
Generation 280: Best = 4.479828e+03 FES = 28100 Diversity = 2.0442e+00 k_t
= [4, 2, 3, 4]
Generation 290: Best = 4.417263e+03 FES = 29100 Diversity = 1.8648e+00 k_t
= [1, 1, 1, 1]
Generation 300: Best = 4.376781e+03 FES = 30100 Diversity = 1.9539e+00 k_t
= [4, 2, 4, 4]
Generation 310: Best = 4.299756e+03 FES = 31100 Diversity = 1.9561e+00 k_t
= [1, 1, 1, 1]
Generation 320: Best = 4.241663e+03 FES = 32100 Diversity = 1.8322e+00 k_t
= [3, 3, 2, 4]
Generation 330: Best = 4.177857e+03 FES = 33100 Diversity = 1.8486e+00 k_t
= [1, 1, 1, 1]
Generation 340: Best = 4.138871e+03 FES = 34100 Diversity = 1.6964e+00 k_t
= [3, 2, 4, 4]
Generation 350: Best = 4.106492e+03 FES = 35100 Diversity = 1.5959e+00 k_t
= [1, 1, 1, 1]
Generation 360: Best = 4.070838e+03 FES = 36100 Diversity = 1.5561e+00 k_t
= [4, 4, 4, 2]
Generation 370: Best = 4.039075e+03 FES = 37100 Diversity = 1.4412e+00 k_t
= [1, 1, 1, 1]
Generation 380: Best = 4.010486e+03 FES = 38100 Diversity = 1.3123e+00 k_t
= [2, 2, 2, 2]
Generation 390: Best = 3.992169e+03 FES = 39100 Diversity = 1.1330e+00 k_t
= [1, 1, 1, 1]
Generation 400: Best = 3.968688e+03 FES = 40100 Diversity = 1.0504e+00 k_t
= [4, 4, 4, 2]
Generation 410: Best = 3.944721e+03 FES = 41100 Diversity = 1.0645e+00 k_t
= [1, 1, 1, 1]
Generation 420: Best = 3.924840e+03 FES = 42100 Diversity = 1.0679e+00 k_t
= [3, 4, 2, 2]
Generation 430: Best = 3.908291e+03 FES = 43100 Diversity = 1.0944e+00 k_t
= [1, 1, 1, 1]
Generation 440: Best = 3.890385e+03 FES = 44100 Diversity = 1.0850e+00 k_t
= [4, 2, 4, 4]
Generation 450: Best = 3.870524e+03 FES = 45100 Diversity = 1.1225e+00 k_t
= [1, 1, 1, 1]
Generation 460: Best = 3.856726e+03 FES = 46100 Diversity = 1.0760e+00 k_t
= [2, 3, 3, 2]
Generation 470: Best = 3.839670e+03 FES = 47100 Diversity = 1.0875e+00 k_t
= [1, 1, 1, 1]
Generation 480: Best = 3.824475e+03 FES = 48100 Diversity = 1.0799e+00 k_t
= [3, 2, 3, 4]
Generation 490: Best = 3.802817e+03 FES = 49100 Diversity = 1.1675e+00 k_t

```



```

= [1, 1, 1, 1]
Generation 500: Best = 3.793897e+03 FES = 50100 Diversity = 1.1001e+00 k_t
= [4, 2, 2, 4]
Generation 510: Best = 3.780070e+03 FES = 51100 Diversity = 1.0396e+00 k_t
= [1, 1, 1, 1]
Generation 520: Best = 3.767945e+03 FES = 52100 Diversity = 9.3019e-01 k_t
= [3, 4, 2, 3]
Generation 530: Best = 3.757040e+03 FES = 53100 Diversity = 8.5587e-01 k_t
= [1, 1, 1, 1]
Generation 540: Best = 3.748530e+03 FES = 54100 Diversity = 7.8852e-01 k_t
= [4, 2, 3, 2]
Generation 550: Best = 3.739758e+03 FES = 55100 Diversity = 8.0203e-01 k_t
= [1, 1, 1, 1]
Generation 560: Best = 3.733100e+03 FES = 56100 Diversity = 7.6793e-01 k_t
= [4, 2, 2, 2]
Generation 570: Best = 3.724220e+03 FES = 57100 Diversity = 7.1030e-01 k_t
= [1, 1, 1, 1]
Generation 580: Best = 3.717757e+03 FES = 58100 Diversity = 6.8371e-01 k_t
= [4, 3, 4, 4]
Generation 590: Best = 3.710800e+03 FES = 59100 Diversity = 6.7555e-01 k_t
= [1, 1, 1, 1]
Generation 600: Best = 3.705873e+03 FES = 60100 Diversity = 6.7189e-01 k_t
= [4, 2, 3, 3]
Generation 610: Best = 3.700341e+03 FES = 61100 Diversity = 6.3371e-01 k_t
= [1, 1, 1, 1]
Generation 620: Best = 3.695806e+03 FES = 62100 Diversity = 6.2624e-01 k_t
= [2, 3, 3, 3]
Generation 630: Best = 3.689816e+03 FES = 63100 Diversity = 6.3718e-01 k_t
= [1, 1, 1, 1]
Generation 640: Best = 3.684622e+03 FES = 64100 Diversity = 6.2242e-01 k_t
= [3, 4, 3, 3]
Generation 650: Best = 3.678772e+03 FES = 65100 Diversity = 6.1607e-01 k_t
= [1, 1, 1, 1]
Generation 660: Best = 3.672398e+03 FES = 66100 Diversity = 6.1344e-01 k_t
= [3, 4, 4, 2]
Generation 670: Best = 3.669377e+03 FES = 67100 Diversity = 5.7186e-01 k_t
= [1, 1, 1, 1]
Generation 680: Best = 3.663243e+03 FES = 68100 Diversity = 6.1234e-01 k_t
= [3, 4, 4, 4]
Generation 690: Best = 3.656671e+03 FES = 69100 Diversity = 6.4590e-01 k_t
= [1, 1, 1, 1]
Generation 700: Best = 3.652057e+03 FES = 70100 Diversity = 6.7206e-01 k_t
= [4, 3, 2, 3]
Generation 710: Best = 3.647449e+03 FES = 71100 Diversity = 6.2687e-01 k_t
= [1, 1, 1, 1]
Generation 720: Best = 3.642898e+03 FES = 72100 Diversity = 5.8298e-01 k_t
= [3, 2, 4, 3]
Generation 730: Best = 3.638015e+03 FES = 73100 Diversity = 5.8637e-01 k_t
= [1, 1, 1, 1]
Generation 740: Best = 3.634313e+03 FES = 74100 Diversity = 5.6964e-01 k_t
= [2, 2, 3, 3]
Generation 750: Best = 3.629732e+03 FES = 75100 Diversity = 5.7056e-01 k_t
= [1, 1, 1, 1]
Generation 760: Best = 3.625032e+03 FES = 76100 Diversity = 6.0179e-01 k_t

```

```

= [3, 4, 4, 2]
Generation 770: Best = 3.621917e+03  FES = 77100  Diversity = 6.2236e-01  k_t
= [1, 1, 1, 1]
Generation 780: Best = 3.617881e+03  FES = 78100  Diversity = 5.7678e-01  k_t
= [4, 2, 4, 4]
Generation 790: Best = 3.614589e+03  FES = 79100  Diversity = 5.2133e-01  k_t
= [1, 1, 1, 1]
Generation 800: Best = 3.610994e+03  FES = 80100  Diversity = 4.9051e-01  k_t
= [3, 4, 4, 4]
Generation 810: Best = 3.607940e+03  FES = 81100  Diversity = 4.8629e-01  k_t
= [1, 1, 1, 1]
Generation 820: Best = 3.605129e+03  FES = 82100  Diversity = 4.5072e-01  k_t
= [4, 4, 4, 2]
Generation 830: Best = 3.602169e+03  FES = 83100  Diversity = 4.6476e-01  k_t
= [1, 1, 1, 1]
Generation 840: Best = 3.599118e+03  FES = 84100  Diversity = 4.5643e-01  k_t
= [2, 3, 3, 4]
Generation 850: Best = 3.597021e+03  FES = 85100  Diversity = 4.4730e-01  k_t
= [1, 1, 1, 1]
Generation 860: Best = 3.594173e+03  FES = 86100  Diversity = 4.4266e-01  k_t
= [2, 4, 2, 3]
Generation 870: Best = 3.591727e+03  FES = 87100  Diversity = 4.4249e-01  k_t
= [1, 1, 1, 1]
Generation 880: Best = 3.589165e+03  FES = 88100  Diversity = 4.6399e-01  k_t
= [3, 2, 2, 3]
Generation 890: Best = 3.586719e+03  FES = 89100  Diversity = 4.7036e-01  k_t
= [1, 1, 1, 1]
Generation 900: Best = 3.583378e+03  FES = 90100  Diversity = 4.6779e-01  k_t
= [3, 3, 2, 4]
Generation 910: Best = 3.580722e+03  FES = 91100  Diversity = 4.4517e-01  k_t
= [1, 1, 1, 1]
Generation 920: Best = 3.578595e+03  FES = 92100  Diversity = 4.3321e-01  k_t
= [4, 2, 3, 4]
Generation 930: Best = 3.575650e+03  FES = 93100  Diversity = 4.2769e-01  k_t
= [1, 1, 1, 1]
Generation 940: Best = 3.573379e+03  FES = 94100  Diversity = 4.2965e-01  k_t
= [2, 4, 4, 2]
Generation 950: Best = 3.571395e+03  FES = 95100  Diversity = 4.0856e-01  k_t
= [1, 1, 1, 1]
Generation 960: Best = 3.567729e+03  FES = 96100  Diversity = 4.3513e-01  k_t
= [2, 2, 2, 2]
Generation 970: Best = 3.565757e+03  FES = 97100  Diversity = 4.4550e-01  k_t
= [1, 1, 1, 1]
Generation 980: Best = 3.563253e+03  FES = 98100  Diversity = 4.4465e-01  k_t
= [3, 3, 4, 3]
Generation 990: Best = 3.560813e+03  FES = 99100  Diversity = 4.4383e-01  k_t
= [1, 1, 1, 1]
Generation 1000: Best = 3.558494e+03  FES = 100100  Diversity = 4.5085e-01  k_t
= [3, 4, 4, 2]
Generation 1010: Best = 3.556100e+03  FES = 101100  Diversity = 4.5730e-01  k_t
= [1, 1, 1, 1]
Generation 1020: Best = 3.553834e+03  FES = 102100  Diversity = 4.7133e-01  k_t
= [4, 2, 2, 4]
Generation 1030: Best = 3.551525e+03  FES = 103100  Diversity = 4.7736e-01  k_t

```

```

= [1, 1, 1, 1]
Generation 1040: Best = 3.548626e+03 FES = 104100 Diversity = 4.5780e-01 k_t
= [4, 3, 2, 2]
Generation 1050: Best = 3.546427e+03 FES = 105100 Diversity = 4.5053e-01 k_t
= [1, 1, 1, 1]
Generation 1060: Best = 3.544251e+03 FES = 106100 Diversity = 4.7506e-01 k_t
= [4, 3, 4, 3]
Generation 1070: Best = 3.541486e+03 FES = 107100 Diversity = 4.8562e-01 k_t
= [1, 1, 1, 1]
Generation 1080: Best = 3.538825e+03 FES = 108100 Diversity = 4.5673e-01 k_t
= [3, 2, 3, 3]
Generation 1090: Best = 3.537206e+03 FES = 109100 Diversity = 4.6069e-01 k_t
= [1, 1, 1, 1]
Generation 1100: Best = 3.534865e+03 FES = 110100 Diversity = 4.3824e-01 k_t
= [3, 4, 4, 4]
Generation 1110: Best = 3.533027e+03 FES = 111100 Diversity = 3.9097e-01 k_t
= [1, 1, 1, 1]
Generation 1120: Best = 3.531238e+03 FES = 112100 Diversity = 3.7417e-01 k_t
= [2, 3, 2, 4]
Generation 1130: Best = 3.529672e+03 FES = 113100 Diversity = 3.7987e-01 k_t
= [1, 1, 1, 1]
Generation 1140: Best = 3.528161e+03 FES = 114100 Diversity = 3.5987e-01 k_t
= [2, 3, 2, 2]
Generation 1150: Best = 3.526634e+03 FES = 115100 Diversity = 3.5614e-01 k_t
= [1, 1, 1, 1]
Generation 1160: Best = 3.524998e+03 FES = 116100 Diversity = 3.4031e-01 k_t
= [4, 3, 3, 3]
Generation 1170: Best = 3.523684e+03 FES = 117100 Diversity = 3.3026e-01 k_t
= [1, 1, 1, 1]
Generation 1180: Best = 3.522256e+03 FES = 118100 Diversity = 3.3313e-01 k_t
= [4, 2, 4, 3]
Generation 1190: Best = 3.520951e+03 FES = 119100 Diversity = 3.2922e-01 k_t
= [1, 1, 1, 1]
Generation 1200: Best = 3.519556e+03 FES = 120100 Diversity = 3.2750e-01 k_t
= [2, 2, 2, 4]
Generation 1210: Best = 3.518407e+03 FES = 121100 Diversity = 3.2246e-01 k_t
= [1, 1, 1, 1]
Generation 1220: Best = 3.516701e+03 FES = 122100 Diversity = 3.4296e-01 k_t
= [4, 4, 2, 3]
Generation 1230: Best = 3.514864e+03 FES = 123100 Diversity = 3.5128e-01 k_t
= [1, 1, 1, 1]
Generation 1240: Best = 3.513512e+03 FES = 124100 Diversity = 3.7746e-01 k_t
= [4, 2, 2, 2]
Generation 1250: Best = 3.512017e+03 FES = 125100 Diversity = 3.6194e-01 k_t
= [1, 1, 1, 1]
Generation 1260: Best = 3.510318e+03 FES = 126100 Diversity = 3.4675e-01 k_t
= [3, 2, 4, 4]
Generation 1270: Best = 3.509333e+03 FES = 127100 Diversity = 3.1907e-01 k_t
= [1, 1, 1, 1]
Generation 1280: Best = 3.507994e+03 FES = 128100 Diversity = 2.7860e-01 k_t
= [3, 4, 3, 3]
Generation 1290: Best = 3.507077e+03 FES = 129100 Diversity = 2.6637e-01 k_t
= [1, 1, 1, 1]
Generation 1300: Best = 3.505772e+03 FES = 130100 Diversity = 2.6476e-01 k_t

```

```

= [4, 2, 4, 4]
Generation 1310: Best = 3.505035e+03 FES = 131100 Diversity = 2.5295e-01 k_t
= [1, 1, 1, 1]
Generation 1320: Best = 3.504100e+03 FES = 132100 Diversity = 2.4221e-01 k_t
= [4, 2, 3, 2]
Generation 1330: Best = 3.503251e+03 FES = 133100 Diversity = 2.5046e-01 k_t
= [1, 1, 1, 1]
Generation 1340: Best = 3.502074e+03 FES = 134100 Diversity = 2.6116e-01 k_t
= [4, 4, 3, 2]
Generation 1350: Best = 3.501097e+03 FES = 135100 Diversity = 2.8103e-01 k_t
= [1, 1, 1, 1]
Generation 1360: Best = 3.500082e+03 FES = 136100 Diversity = 2.7836e-01 k_t
= [4, 2, 4, 2]
Generation 1370: Best = 3.498830e+03 FES = 137100 Diversity = 2.7565e-01 k_t
= [1, 1, 1, 1]
Generation 1380: Best = 3.498026e+03 FES = 138100 Diversity = 2.4462e-01 k_t
= [3, 3, 4, 4]
Generation 1390: Best = 3.496909e+03 FES = 139100 Diversity = 2.6513e-01 k_t
= [1, 1, 1, 1]
Generation 1400: Best = 3.495582e+03 FES = 140100 Diversity = 2.9090e-01 k_t
= [4, 4, 4, 3]
Generation 1410: Best = 3.494273e+03 FES = 141100 Diversity = 2.8971e-01 k_t
= [1, 1, 1, 1]
Generation 1420: Best = 3.493470e+03 FES = 142100 Diversity = 2.8177e-01 k_t
= [3, 4, 3, 2]
Generation 1430: Best = 3.492263e+03 FES = 143100 Diversity = 2.6976e-01 k_t
= [1, 1, 1, 1]
Generation 1440: Best = 3.491342e+03 FES = 144100 Diversity = 2.5344e-01 k_t
= [3, 2, 3, 3]
Generation 1450: Best = 3.490250e+03 FES = 145100 Diversity = 2.6562e-01 k_t
= [1, 1, 1, 1]
Generation 1460: Best = 3.489257e+03 FES = 146100 Diversity = 2.4990e-01 k_t
= [4, 3, 3, 2]
Generation 1470: Best = 3.488292e+03 FES = 147100 Diversity = 2.5307e-01 k_t
= [1, 1, 1, 1]
Generation 1480: Best = 3.487388e+03 FES = 148100 Diversity = 2.4260e-01 k_t
= [3, 3, 3, 3]
Generation 1490: Best = 3.486612e+03 FES = 149100 Diversity = 2.1245e-01 k_t
= [1, 1, 1, 1]
Generation 1500: Best = 3.485621e+03 FES = 150100 Diversity = 2.1695e-01 k_t
= [2, 4, 4, 4]
Generation 1510: Best = 3.484718e+03 FES = 151100 Diversity = 2.2757e-01 k_t
= [1, 1, 1, 1]
Generation 1520: Best = 3.483809e+03 FES = 152100 Diversity = 2.2103e-01 k_t
= [2, 2, 2, 2]
Generation 1530: Best = 3.482840e+03 FES = 153100 Diversity = 2.5619e-01 k_t
= [1, 1, 1, 1]
Generation 1540: Best = 3.481587e+03 FES = 154100 Diversity = 2.9888e-01 k_t
= [3, 4, 3, 2]
Generation 1550: Best = 3.480556e+03 FES = 155100 Diversity = 2.9831e-01 k_t
= [1, 1, 1, 1]
Generation 1560: Best = 3.479394e+03 FES = 156100 Diversity = 2.9198e-01 k_t
= [3, 3, 3, 4]
Generation 1570: Best = 3.478189e+03 FES = 157100 Diversity = 2.7821e-01 k_t

```

```

= [1, 1, 1, 1]
Generation 1580: Best = 3.477033e+03 FES = 158100 Diversity = 2.6134e-01 k_t
= [4, 4, 3, 4]
Generation 1590: Best = 3.476040e+03 FES = 159100 Diversity = 2.4753e-01 k_t
= [1, 1, 1, 1]
Generation 1600: Best = 3.475468e+03 FES = 160100 Diversity = 2.1806e-01 k_t
= [3, 4, 3, 4]
Generation 1610: Best = 3.474890e+03 FES = 161100 Diversity = 1.9727e-01 k_t
= [1, 1, 1, 1]
Generation 1620: Best = 3.474210e+03 FES = 162100 Diversity = 1.8840e-01 k_t
= [3, 4, 2, 2]
Generation 1630: Best = 3.473682e+03 FES = 163100 Diversity = 1.6876e-01 k_t
= [1, 1, 1, 1]
Generation 1640: Best = 3.473088e+03 FES = 164100 Diversity = 1.7327e-01 k_t
= [4, 2, 4, 2]
Generation 1650: Best = 3.472505e+03 FES = 165100 Diversity = 1.8240e-01 k_t
= [1, 1, 1, 1]
Generation 1660: Best = 3.471588e+03 FES = 166100 Diversity = 1.9190e-01 k_t
= [3, 3, 2, 4]
Generation 1670: Best = 3.469361e+03 FES = 167100 Diversity = 2.9192e-01 k_t
= [1, 1, 1, 1]
Generation 1680: Best = 3.467841e+03 FES = 168100 Diversity = 3.1251e-01 k_t
= [3, 3, 2, 3]
Generation 1690: Best = 3.466448e+03 FES = 169100 Diversity = 3.3035e-01 k_t
= [1, 1, 1, 1]
Generation 1700: Best = 3.465068e+03 FES = 170100 Diversity = 3.0770e-01 k_t
= [2, 4, 3, 2]
Generation 1710: Best = 3.463602e+03 FES = 171100 Diversity = 2.9415e-01 k_t
= [1, 1, 1, 1]
Generation 1720: Best = 3.462667e+03 FES = 172100 Diversity = 2.7315e-01 k_t
= [2, 2, 4, 3]
Generation 1730: Best = 3.461488e+03 FES = 173100 Diversity = 2.9228e-01 k_t
= [1, 1, 1, 1]
Generation 1740: Best = 3.460443e+03 FES = 174100 Diversity = 2.8581e-01 k_t
= [3, 3, 3, 3]
Generation 1750: Best = 3.458830e+03 FES = 175100 Diversity = 2.9383e-01 k_t
= [1, 1, 1, 1]
Generation 1760: Best = 3.457675e+03 FES = 176100 Diversity = 2.7871e-01 k_t
= [3, 2, 4, 2]
Generation 1770: Best = 3.456542e+03 FES = 177100 Diversity = 2.7342e-01 k_t
= [1, 1, 1, 1]
Generation 1780: Best = 3.455511e+03 FES = 178100 Diversity = 2.3846e-01 k_t
= [4, 3, 2, 2]
Generation 1790: Best = 3.454581e+03 FES = 179100 Diversity = 2.2185e-01 k_t
= [1, 1, 1, 1]
Generation 1800: Best = 3.453249e+03 FES = 180100 Diversity = 2.5453e-01 k_t
= [2, 2, 4, 4]
Generation 1810: Best = 3.452128e+03 FES = 181100 Diversity = 2.4775e-01 k_t
= [1, 1, 1, 1]
Generation 1820: Best = 3.451116e+03 FES = 182100 Diversity = 2.4483e-01 k_t
= [2, 2, 4, 2]
Generation 1830: Best = 3.450277e+03 FES = 183100 Diversity = 2.6294e-01 k_t
= [1, 1, 1, 1]
Generation 1840: Best = 3.448999e+03 FES = 184100 Diversity = 2.5264e-01 k_t

```

```

= [3, 3, 2, 2]
Generation 1850: Best = 3.447974e+03 FES = 185100 Diversity = 2.5841e-01 k_t
= [1, 1, 1, 1]
Generation 1860: Best = 3.446983e+03 FES = 186100 Diversity = 2.6770e-01 k_t
= [3, 4, 2, 2]
Generation 1870: Best = 3.446107e+03 FES = 187100 Diversity = 2.5489e-01 k_t
= [1, 1, 1, 1]
Generation 1880: Best = 3.445398e+03 FES = 188100 Diversity = 2.4365e-01 k_t
= [4, 4, 4, 3]
Generation 1890: Best = 3.444651e+03 FES = 189100 Diversity = 2.2386e-01 k_t
= [1, 1, 1, 1]
Generation 1900: Best = 3.443657e+03 FES = 190100 Diversity = 2.1643e-01 k_t
= [3, 4, 2, 3]
Generation 1910: Best = 3.443013e+03 FES = 191100 Diversity = 2.1264e-01 k_t
= [1, 1, 1, 1]
Generation 1920: Best = 3.442219e+03 FES = 192100 Diversity = 2.0847e-01 k_t
= [2, 4, 3, 2]
Generation 1930: Best = 3.441719e+03 FES = 193100 Diversity = 1.9610e-01 k_t
= [1, 1, 1, 1]
Generation 1940: Best = 3.441003e+03 FES = 194100 Diversity = 1.8495e-01 k_t
= [2, 4, 2, 3]
Generation 1950: Best = 3.440548e+03 FES = 195100 Diversity = 1.7079e-01 k_t
= [1, 1, 1, 1]
Generation 1960: Best = 3.440055e+03 FES = 196100 Diversity = 1.8194e-01 k_t
= [2, 3, 3, 3]
Generation 1970: Best = 3.439491e+03 FES = 197100 Diversity = 1.8276e-01 k_t
= [1, 1, 1, 1]
Generation 1980: Best = 3.438909e+03 FES = 198100 Diversity = 1.7171e-01 k_t
= [2, 4, 3, 2]
Generation 1990: Best = 3.438510e+03 FES = 199100 Diversity = 1.4486e-01 k_t
= [1, 1, 1, 1]
Generation 2000: Best = 3.438177e+03 FES = 200100 Diversity = 1.4318e-01 k_t
= [3, 2, 2, 3]
Generation 2010: Best = 3.437809e+03 FES = 201100 Diversity = 1.4786e-01 k_t
= [1, 1, 1, 1]
Generation 2020: Best = 3.437477e+03 FES = 202100 Diversity = 1.3478e-01 k_t
= [3, 2, 2, 3]
Generation 2030: Best = 3.437117e+03 FES = 203100 Diversity = 1.2123e-01 k_t
= [1, 1, 1, 1]
Generation 2040: Best = 3.436912e+03 FES = 204100 Diversity = 1.1956e-01 k_t
= [3, 2, 3, 4]
Generation 2050: Best = 3.436608e+03 FES = 205100 Diversity = 1.1767e-01 k_t
= [1, 1, 1, 1]
Generation 2060: Best = 3.436346e+03 FES = 206100 Diversity = 1.0334e-01 k_t
= [4, 4, 4, 2]
Generation 2070: Best = 3.436123e+03 FES = 207100 Diversity = 1.1695e-01 k_t
= [1, 1, 1, 1]
Generation 2080: Best = 3.435939e+03 FES = 208100 Diversity = 1.0974e-01 k_t
= [4, 2, 4, 3]
Generation 2090: Best = 3.435737e+03 FES = 209100 Diversity = 1.0353e-01 k_t
= [1, 1, 1, 1]
Generation 2100: Best = 3.435581e+03 FES = 210100 Diversity = 9.9280e-02 k_t
= [2, 2, 3, 4]
Generation 2110: Best = 3.435425e+03 FES = 211100 Diversity = 8.4517e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 2120: Best = 3.435314e+03 FES = 212100 Diversity = 7.9833e-02 k_t
= [2, 2, 4, 2]
Generation 2130: Best = 3.435195e+03 FES = 213100 Diversity = 7.8701e-02 k_t
= [1, 1, 1, 1]
Generation 2140: Best = 3.435048e+03 FES = 214100 Diversity = 7.4388e-02 k_t
= [4, 2, 2, 2]
Generation 2150: Best = 3.434971e+03 FES = 215100 Diversity = 7.3181e-02 k_t
= [1, 1, 1, 1]
Generation 2160: Best = 3.434818e+03 FES = 216100 Diversity = 8.2520e-02 k_t
= [4, 4, 2, 3]
Generation 2170: Best = 3.434699e+03 FES = 217100 Diversity = 8.4582e-02 k_t
= [1, 1, 1, 1]
Generation 2180: Best = 3.434565e+03 FES = 218100 Diversity = 8.2827e-02 k_t
= [3, 2, 2, 3]
Generation 2190: Best = 3.434427e+03 FES = 219100 Diversity = 8.1456e-02 k_t
= [1, 1, 1, 1]
Generation 2200: Best = 3.434283e+03 FES = 220100 Diversity = 7.7361e-02 k_t
= [4, 2, 2, 4]
Generation 2210: Best = 3.434140e+03 FES = 221100 Diversity = 8.1226e-02 k_t
= [1, 1, 1, 1]
Generation 2220: Best = 3.433983e+03 FES = 222100 Diversity = 8.2532e-02 k_t
= [4, 3, 2, 2]
Generation 2230: Best = 3.433843e+03 FES = 223100 Diversity = 8.3348e-02 k_t
= [1, 1, 1, 1]
Generation 2240: Best = 3.433704e+03 FES = 224100 Diversity = 7.5115e-02 k_t
= [3, 2, 4, 4]
Generation 2250: Best = 3.433550e+03 FES = 225100 Diversity = 7.3622e-02 k_t
= [1, 1, 1, 1]
Generation 2260: Best = 3.433413e+03 FES = 226100 Diversity = 8.1345e-02 k_t
= [3, 2, 4, 2]
Generation 2270: Best = 3.433275e+03 FES = 227100 Diversity = 8.3155e-02 k_t
= [1, 1, 1, 1]
Generation 2280: Best = 3.433109e+03 FES = 228100 Diversity = 8.4099e-02 k_t
= [2, 4, 2, 3]
Generation 2290: Best = 3.432943e+03 FES = 229100 Diversity = 8.2358e-02 k_t
= [1, 1, 1, 1]
Generation 2300: Best = 3.432764e+03 FES = 230100 Diversity = 8.6982e-02 k_t
= [2, 4, 3, 4]
Generation 2310: Best = 3.432612e+03 FES = 231100 Diversity = 8.3858e-02 k_t
= [1, 1, 1, 1]
Generation 2320: Best = 3.432439e+03 FES = 232100 Diversity = 7.9838e-02 k_t
= [2, 3, 4, 4]
Generation 2330: Best = 3.432275e+03 FES = 233100 Diversity = 8.2766e-02 k_t
= [1, 1, 1, 1]
Generation 2340: Best = 3.432139e+03 FES = 234100 Diversity = 8.7204e-02 k_t
= [4, 4, 2, 2]
Generation 2350: Best = 3.431940e+03 FES = 235100 Diversity = 9.2186e-02 k_t
= [1, 1, 1, 1]
Generation 2360: Best = 3.431805e+03 FES = 236100 Diversity = 7.8756e-02 k_t
= [4, 3, 4, 2]
Generation 2370: Best = 3.431654e+03 FES = 237100 Diversity = 8.4775e-02 k_t
= [1, 1, 1, 1]
Generation 2380: Best = 3.431519e+03 FES = 238100 Diversity = 7.9991e-02 k_t

```

```

= [3, 3, 3, 4]
Generation 2390: Best = 3.431408e+03 FES = 239100 Diversity = 6.8877e-02 k_t
= [1, 1, 1, 1]
Generation 2400: Best = 3.431318e+03 FES = 240100 Diversity = 6.6076e-02 k_t
= [4, 3, 4, 2]
Generation 2410: Best = 3.431213e+03 FES = 241100 Diversity = 6.3417e-02 k_t
= [1, 1, 1, 1]
Generation 2420: Best = 3.431047e+03 FES = 242100 Diversity = 7.3031e-02 k_t
= [3, 3, 2, 3]
Generation 2430: Best = 3.430904e+03 FES = 243100 Diversity = 8.2268e-02 k_t
= [1, 1, 1, 1]
Generation 2440: Best = 3.430759e+03 FES = 244100 Diversity = 7.1472e-02 k_t
= [3, 4, 2, 4]
Generation 2450: Best = 3.430592e+03 FES = 245100 Diversity = 7.8820e-02 k_t
= [1, 1, 1, 1]
Generation 2460: Best = 3.430487e+03 FES = 246100 Diversity = 7.5177e-02 k_t
= [2, 4, 4, 3]
Generation 2470: Best = 3.430329e+03 FES = 247100 Diversity = 7.8924e-02 k_t
= [1, 1, 1, 1]
Generation 2480: Best = 3.430160e+03 FES = 248100 Diversity = 8.4339e-02 k_t
= [4, 2, 4, 3]
Generation 2490: Best = 3.430037e+03 FES = 249100 Diversity = 7.9584e-02 k_t
= [1, 1, 1, 1]
Generation 2500: Best = 3.429893e+03 FES = 250100 Diversity = 8.1147e-02 k_t
= [4, 3, 2, 2]
Generation 2510: Best = 3.429737e+03 FES = 251100 Diversity = 7.5480e-02 k_t
= [1, 1, 1, 1]
Generation 2520: Best = 3.429607e+03 FES = 252100 Diversity = 7.7618e-02 k_t
= [4, 2, 2, 2]
Generation 2530: Best = 3.429441e+03 FES = 253100 Diversity = 7.8646e-02 k_t
= [1, 1, 1, 1]
Generation 2540: Best = 3.429322e+03 FES = 254100 Diversity = 8.2055e-02 k_t
= [4, 4, 2, 3]
Generation 2550: Best = 3.429221e+03 FES = 255100 Diversity = 7.4976e-02 k_t
= [1, 1, 1, 1]
Generation 2560: Best = 3.429110e+03 FES = 256100 Diversity = 6.3483e-02 k_t
= [4, 2, 3, 4]
Generation 2570: Best = 3.429006e+03 FES = 257100 Diversity = 6.1029e-02 k_t
= [1, 1, 1, 1]
Generation 2580: Best = 3.428889e+03 FES = 258100 Diversity = 6.6122e-02 k_t
= [4, 2, 4, 4]
Generation 2590: Best = 3.428802e+03 FES = 259100 Diversity = 6.4447e-02 k_t
= [1, 1, 1, 1]
Generation 2600: Best = 3.428654e+03 FES = 260100 Diversity = 7.5548e-02 k_t
= [3, 3, 2, 4]
Generation 2610: Best = 3.428494e+03 FES = 261100 Diversity = 7.8384e-02 k_t
= [1, 1, 1, 1]
Generation 2620: Best = 3.428361e+03 FES = 262100 Diversity = 7.2036e-02 k_t
= [2, 4, 2, 2]
Generation 2630: Best = 3.428209e+03 FES = 263100 Diversity = 7.6437e-02 k_t
= [1, 1, 1, 1]
Generation 2640: Best = 3.428043e+03 FES = 264100 Diversity = 7.7784e-02 k_t
= [2, 3, 4, 3]
Generation 2650: Best = 3.427902e+03 FES = 265100 Diversity = 7.7151e-02 k_t

```



```

= [1, 1, 1, 1]
Generation 2660: Best = 3.427748e+03 FES = 266100 Diversity = 7.6149e-02 k_t
= [3, 2, 4, 3]
Generation 2670: Best = 3.427633e+03 FES = 267100 Diversity = 7.8617e-02 k_t
= [1, 1, 1, 1]
Generation 2680: Best = 3.427490e+03 FES = 268100 Diversity = 8.2721e-02 k_t
= [4, 4, 2, 3]
Generation 2690: Best = 3.427347e+03 FES = 269100 Diversity = 8.2426e-02 k_t
= [1, 1, 1, 1]
Generation 2700: Best = 3.427207e+03 FES = 270100 Diversity = 8.0368e-02 k_t
= [2, 4, 4, 2]
Generation 2710: Best = 3.427076e+03 FES = 271100 Diversity = 8.1274e-02 k_t
= [1, 1, 1, 1]
Generation 2720: Best = 3.426936e+03 FES = 272100 Diversity = 8.5056e-02 k_t
= [4, 2, 4, 2]
Generation 2730: Best = 3.426794e+03 FES = 273100 Diversity = 7.7581e-02 k_t
= [1, 1, 1, 1]
Generation 2740: Best = 3.426663e+03 FES = 274100 Diversity = 7.2004e-02 k_t
= [2, 2, 3, 3]
Generation 2750: Best = 3.426548e+03 FES = 275100 Diversity = 6.7238e-02 k_t
= [1, 1, 1, 1]
Generation 2760: Best = 3.426445e+03 FES = 276100 Diversity = 6.6692e-02 k_t
= [3, 4, 4, 3]
Generation 2770: Best = 3.426315e+03 FES = 277100 Diversity = 6.2633e-02 k_t
= [1, 1, 1, 1]
Generation 2780: Best = 3.426144e+03 FES = 278100 Diversity = 6.8990e-02 k_t
= [3, 2, 3, 4]
Generation 2790: Best = 3.426015e+03 FES = 279100 Diversity = 7.3842e-02 k_t
= [1, 1, 1, 1]
Generation 2800: Best = 3.425882e+03 FES = 280100 Diversity = 7.3744e-02 k_t
= [4, 2, 2, 3]
Generation 2810: Best = 3.425700e+03 FES = 281100 Diversity = 8.1535e-02 k_t
= [1, 1, 1, 1]
Generation 2820: Best = 3.425567e+03 FES = 282100 Diversity = 6.9586e-02 k_t
= [4, 2, 4, 2]
Generation 2830: Best = 3.425463e+03 FES = 283100 Diversity = 6.7621e-02 k_t
= [1, 1, 1, 1]
Generation 2840: Best = 3.425348e+03 FES = 284100 Diversity = 6.2908e-02 k_t
= [2, 4, 3, 4]
Generation 2850: Best = 3.425226e+03 FES = 285100 Diversity = 7.2380e-02 k_t
= [1, 1, 1, 1]
Generation 2860: Best = 3.425069e+03 FES = 286100 Diversity = 7.9728e-02 k_t
= [4, 4, 4, 3]
Generation 2870: Best = 3.424939e+03 FES = 287100 Diversity = 7.0739e-02 k_t
= [1, 1, 1, 1]
Generation 2880: Best = 3.424843e+03 FES = 288100 Diversity = 6.3515e-02 k_t
= [3, 2, 4, 4]
Generation 2890: Best = 3.424721e+03 FES = 289100 Diversity = 5.5666e-02 k_t
= [1, 1, 1, 1]
Generation 2900: Best = 3.424650e+03 FES = 290100 Diversity = 5.5448e-02 k_t
= [4, 2, 2, 4]
Generation 2910: Best = 3.424535e+03 FES = 291100 Diversity = 5.8764e-02 k_t
= [1, 1, 1, 1]
Generation 2920: Best = 3.424445e+03 FES = 292100 Diversity = 5.1751e-02 k_t

```

```

= [3, 2, 3, 2]
Generation 2930: Best = 3.424319e+03 FES = 293100 Diversity = 6.4940e-02 k_t
= [1, 1, 1, 1]
Generation 2940: Best = 3.424154e+03 FES = 294100 Diversity = 7.8805e-02 k_t
= [3, 2, 3, 4]
Generation 2950: Best = 3.424034e+03 FES = 295100 Diversity = 6.6458e-02 k_t
= [1, 1, 1, 1]
Generation 2960: Best = 3.423892e+03 FES = 296100 Diversity = 7.3953e-02 k_t
= [3, 2, 4, 4]
Generation 2970: Best = 3.423763e+03 FES = 297100 Diversity = 7.3192e-02 k_t
= [1, 1, 1, 1]
Generation 2980: Best = 3.423646e+03 FES = 298100 Diversity = 6.6046e-02 k_t
= [4, 3, 4, 4]
Generation 2990: Best = 3.423485e+03 FES = 299100 Diversity = 6.7298e-02 k_t
= [1, 1, 1, 1]
Generation 3000: Best = 3.423326e+03 FES = 300100 Diversity = 7.7909e-02 k_t
= [4, 4, 3, 4]
Generation 3010: Best = 3.423149e+03 FES = 301100 Diversity = 7.6696e-02 k_t
= [1, 1, 1, 1]
Generation 3020: Best = 3.423030e+03 FES = 302100 Diversity = 6.9203e-02 k_t
= [3, 4, 4, 4]
Generation 3030: Best = 3.422922e+03 FES = 303100 Diversity = 7.3901e-02 k_t
= [1, 1, 1, 1]
Generation 3040: Best = 3.422773e+03 FES = 304100 Diversity = 7.3263e-02 k_t
= [2, 4, 4, 4]
Generation 3050: Best = 3.422648e+03 FES = 305100 Diversity = 7.0960e-02 k_t
= [1, 1, 1, 1]
Generation 3060: Best = 3.422513e+03 FES = 306100 Diversity = 6.7200e-02 k_t
= [2, 4, 2, 2]
Generation 3070: Best = 3.422377e+03 FES = 307100 Diversity = 7.3011e-02 k_t
= [1, 1, 1, 1]
Generation 3080: Best = 3.422280e+03 FES = 308100 Diversity = 6.3014e-02 k_t
= [2, 3, 3, 2]
Generation 3090: Best = 3.422169e+03 FES = 309100 Diversity = 5.6882e-02 k_t
= [1, 1, 1, 1]
Generation 3100: Best = 3.422092e+03 FES = 310100 Diversity = 4.9870e-02 k_t
= [4, 2, 3, 2]
Generation 3110: Best = 3.421979e+03 FES = 311100 Diversity = 6.0842e-02 k_t
= [1, 1, 1, 1]
Generation 3120: Best = 3.421879e+03 FES = 312100 Diversity = 5.5801e-02 k_t
= [3, 4, 3, 4]
Generation 3130: Best = 3.421754e+03 FES = 313100 Diversity = 6.3116e-02 k_t
= [1, 1, 1, 1]
Generation 3140: Best = 3.421642e+03 FES = 314100 Diversity = 6.5731e-02 k_t
= [4, 3, 4, 2]
Generation 3150: Best = 3.421543e+03 FES = 315100 Diversity = 6.3520e-02 k_t
= [1, 1, 1, 1]
Generation 3160: Best = 3.421395e+03 FES = 316100 Diversity = 6.7070e-02 k_t
= [4, 4, 4, 4]
Generation 3170: Best = 3.421300e+03 FES = 317100 Diversity = 6.2429e-02 k_t
= [1, 1, 1, 1]
Generation 3180: Best = 3.421207e+03 FES = 318100 Diversity = 5.4138e-02 k_t
= [4, 4, 2, 2]
Generation 3190: Best = 3.421102e+03 FES = 319100 Diversity = 6.2147e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 3200: Best = 3.420947e+03 FES = 320100 Diversity = 6.8362e-02 k_t
= [3, 2, 2, 2]
Generation 3210: Best = 3.420806e+03 FES = 321100 Diversity = 6.7866e-02 k_t
= [1, 1, 1, 1]
Generation 3220: Best = 3.420697e+03 FES = 322100 Diversity = 5.8523e-02 k_t
= [3, 2, 3, 4]
Generation 3230: Best = 3.420546e+03 FES = 323100 Diversity = 6.6670e-02 k_t
= [1, 1, 1, 1]
Generation 3240: Best = 3.420409e+03 FES = 324100 Diversity = 6.5849e-02 k_t
= [2, 4, 2, 2]
Generation 3250: Best = 3.420256e+03 FES = 325100 Diversity = 6.8968e-02 k_t
= [1, 1, 1, 1]
Generation 3260: Best = 3.420153e+03 FES = 326100 Diversity = 6.3069e-02 k_t
= [3, 4, 3, 4]
Generation 3270: Best = 3.420033e+03 FES = 327100 Diversity = 6.0671e-02 k_t
= [1, 1, 1, 1]
Generation 3280: Best = 3.419888e+03 FES = 328100 Diversity = 6.0212e-02 k_t
= [3, 4, 4, 2]
Generation 3290: Best = 3.419807e+03 FES = 329100 Diversity = 5.4406e-02 k_t
= [1, 1, 1, 1]
Generation 3300: Best = 3.419715e+03 FES = 330100 Diversity = 4.9968e-02 k_t
= [4, 2, 4, 4]
Generation 3310: Best = 3.419616e+03 FES = 331100 Diversity = 5.0129e-02 k_t
= [1, 1, 1, 1]
Generation 3320: Best = 3.419525e+03 FES = 332100 Diversity = 4.9032e-02 k_t
= [2, 2, 4, 3]
Generation 3330: Best = 3.419422e+03 FES = 333100 Diversity = 5.0811e-02 k_t
= [1, 1, 1, 1]
Generation 3340: Best = 3.419309e+03 FES = 334100 Diversity = 5.4433e-02 k_t
= [2, 4, 2, 3]
Generation 3350: Best = 3.419190e+03 FES = 335100 Diversity = 5.6018e-02 k_t
= [1, 1, 1, 1]
Generation 3360: Best = 3.419080e+03 FES = 336100 Diversity = 5.6143e-02 k_t
= [2, 3, 4, 2]
Generation 3370: Best = 3.418975e+03 FES = 337100 Diversity = 5.1207e-02 k_t
= [1, 1, 1, 1]
Generation 3380: Best = 3.418895e+03 FES = 338100 Diversity = 5.2670e-02 k_t
= [4, 2, 2, 3]
Generation 3390: Best = 3.418810e+03 FES = 339100 Diversity = 4.5174e-02 k_t
= [1, 1, 1, 1]
Generation 3400: Best = 3.418713e+03 FES = 340100 Diversity = 5.0419e-02 k_t
= [4, 2, 2, 3]
Generation 3410: Best = 3.418623e+03 FES = 341100 Diversity = 5.2257e-02 k_t
= [1, 1, 1, 1]
Generation 3420: Best = 3.418524e+03 FES = 342100 Diversity = 5.8981e-02 k_t
= [4, 4, 4, 2]
Generation 3430: Best = 3.418398e+03 FES = 343100 Diversity = 5.7338e-02 k_t
= [1, 1, 1, 1]
Generation 3440: Best = 3.418305e+03 FES = 344100 Diversity = 5.2885e-02 k_t
= [4, 4, 2, 3]
Generation 3450: Best = 3.418194e+03 FES = 345100 Diversity = 5.5988e-02 k_t
= [1, 1, 1, 1]
Generation 3460: Best = 3.418049e+03 FES = 346100 Diversity = 5.6908e-02 k_t

```

```

= [2, 3, 2, 4]
Generation 3470: Best = 3.417899e+03 FES = 347100 Diversity = 5.5618e-02 k_t
= [1, 1, 1, 1]
Generation 3480: Best = 3.417778e+03 FES = 348100 Diversity = 5.2711e-02 k_t
= [2, 4, 2, 2]
Generation 3490: Best = 3.417644e+03 FES = 349100 Diversity = 5.9342e-02 k_t
= [1, 1, 1, 1]
Generation 3500: Best = 3.417504e+03 FES = 350100 Diversity = 7.0346e-02 k_t
= [3, 4, 2, 3]
Generation 3510: Best = 3.417329e+03 FES = 351100 Diversity = 6.9397e-02 k_t
= [1, 1, 1, 1]
Generation 3520: Best = 3.417178e+03 FES = 352100 Diversity = 6.6519e-02 k_t
= [4, 2, 3, 4]
Generation 3530: Best = 3.417030e+03 FES = 353100 Diversity = 6.8841e-02 k_t
= [1, 1, 1, 1]
Generation 3540: Best = 3.416862e+03 FES = 354100 Diversity = 7.5564e-02 k_t
= [4, 3, 4, 2]
Generation 3550: Best = 3.416636e+03 FES = 355100 Diversity = 6.8096e-02 k_t
= [1, 1, 1, 1]
Generation 3560: Best = 3.416475e+03 FES = 356100 Diversity = 6.4931e-02 k_t
= [2, 2, 2, 4]
Generation 3570: Best = 3.416287e+03 FES = 357100 Diversity = 6.5401e-02 k_t
= [1, 1, 1, 1]
Generation 3580: Best = 3.416081e+03 FES = 358100 Diversity = 7.7433e-02 k_t
= [2, 3, 4, 2]
Generation 3590: Best = 3.415927e+03 FES = 359100 Diversity = 6.8231e-02 k_t
= [1, 1, 1, 1]
Generation 3600: Best = 3.415734e+03 FES = 360100 Diversity = 7.5019e-02 k_t
= [3, 2, 2, 2]
Generation 3610: Best = 3.415621e+03 FES = 361100 Diversity = 6.1008e-02 k_t
= [1, 1, 1, 1]
Generation 3620: Best = 3.415519e+03 FES = 362100 Diversity = 5.1956e-02 k_t
= [2, 4, 4, 2]
Generation 3630: Best = 3.415380e+03 FES = 363100 Diversity = 5.7979e-02 k_t
= [1, 1, 1, 1]
Generation 3640: Best = 3.415238e+03 FES = 364100 Diversity = 6.7194e-02 k_t
= [2, 4, 2, 2]
Generation 3650: Best = 3.415060e+03 FES = 365100 Diversity = 6.4355e-02 k_t
= [1, 1, 1, 1]
Generation 3660: Best = 3.414970e+03 FES = 366100 Diversity = 5.0466e-02 k_t
= [2, 3, 2, 3]
Generation 3670: Best = 3.414857e+03 FES = 367100 Diversity = 4.4507e-02 k_t
= [1, 1, 1, 1]
Generation 3680: Best = 3.414736e+03 FES = 368100 Diversity = 4.7841e-02 k_t
= [2, 3, 2, 4]
Generation 3690: Best = 3.414633e+03 FES = 369100 Diversity = 4.8441e-02 k_t
= [1, 1, 1, 1]
Generation 3700: Best = 3.414512e+03 FES = 370100 Diversity = 5.5538e-02 k_t
= [2, 3, 3, 4]
Generation 3710: Best = 3.414417e+03 FES = 371100 Diversity = 5.2994e-02 k_t
= [1, 1, 1, 1]
Generation 3720: Best = 3.414288e+03 FES = 372100 Diversity = 5.1334e-02 k_t
= [2, 4, 2, 4]
Generation 3730: Best = 3.414119e+03 FES = 373100 Diversity = 6.4270e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 3740: Best = 3.413948e+03 FES = 374100 Diversity = 6.2353e-02 k_t
= [4, 4, 3, 3]
Generation 3750: Best = 3.413794e+03 FES = 375100 Diversity = 5.9104e-02 k_t
= [1, 1, 1, 1]
Generation 3760: Best = 3.413628e+03 FES = 376100 Diversity = 7.0610e-02 k_t
= [3, 2, 4, 3]
Generation 3770: Best = 3.413450e+03 FES = 377100 Diversity = 7.1043e-02 k_t
= [1, 1, 1, 1]
Generation 3780: Best = 3.413280e+03 FES = 378100 Diversity = 6.8319e-02 k_t
= [4, 4, 2, 4]
Generation 3790: Best = 3.413172e+03 FES = 379100 Diversity = 6.5609e-02 k_t
= [1, 1, 1, 1]
Generation 3800: Best = 3.412961e+03 FES = 380100 Diversity = 7.0227e-02 k_t
= [4, 4, 2, 4]
Generation 3810: Best = 3.412794e+03 FES = 381100 Diversity = 6.9940e-02 k_t
= [1, 1, 1, 1]
Generation 3820: Best = 3.412674e+03 FES = 382100 Diversity = 6.3499e-02 k_t
= [2, 4, 4, 2]
Generation 3830: Best = 3.412489e+03 FES = 383100 Diversity = 6.9406e-02 k_t
= [1, 1, 1, 1]
Generation 3840: Best = 3.412267e+03 FES = 384100 Diversity = 6.5966e-02 k_t
= [3, 2, 3, 4]
Generation 3850: Best = 3.412044e+03 FES = 385100 Diversity = 7.6551e-02 k_t
= [1, 1, 1, 1]
Generation 3860: Best = 3.411893e+03 FES = 386100 Diversity = 7.6304e-02 k_t
= [3, 3, 2, 4]
Generation 3870: Best = 3.411759e+03 FES = 387100 Diversity = 6.1830e-02 k_t
= [1, 1, 1, 1]
Generation 3880: Best = 3.411632e+03 FES = 388100 Diversity = 6.2889e-02 k_t
= [4, 3, 4, 3]
Generation 3890: Best = 3.411475e+03 FES = 389100 Diversity = 6.6680e-02 k_t
= [1, 1, 1, 1]
Generation 3900: Best = 3.411347e+03 FES = 390100 Diversity = 6.3414e-02 k_t
= [2, 3, 4, 3]
Generation 3910: Best = 3.411169e+03 FES = 391100 Diversity = 6.9055e-02 k_t
= [1, 1, 1, 1]
Generation 3920: Best = 3.410970e+03 FES = 392100 Diversity = 7.0858e-02 k_t
= [3, 3, 2, 4]
Generation 3930: Best = 3.410775e+03 FES = 393100 Diversity = 7.9212e-02 k_t
= [1, 1, 1, 1]
Generation 3940: Best = 3.410546e+03 FES = 394100 Diversity = 8.1614e-02 k_t
= [3, 4, 4, 3]
Generation 3950: Best = 3.410378e+03 FES = 395100 Diversity = 7.8209e-02 k_t
= [1, 1, 1, 1]
Generation 3960: Best = 3.410164e+03 FES = 396100 Diversity = 7.7373e-02 k_t
= [2, 4, 2, 3]
Generation 3970: Best = 3.409982e+03 FES = 397100 Diversity = 6.9611e-02 k_t
= [1, 1, 1, 1]
Generation 3980: Best = 3.409825e+03 FES = 398100 Diversity = 5.9217e-02 k_t
= [4, 3, 3, 4]
Generation 3990: Best = 3.409697e+03 FES = 399100 Diversity = 6.0753e-02 k_t
= [1, 1, 1, 1]
Generation 4000: Best = 3.409528e+03 FES = 400100 Diversity = 6.7126e-02 k_t

```

```

= [4, 4, 3, 3]
Generation 4010: Best = 3.409365e+03 FES = 401100 Diversity = 6.1606e-02 k_t
= [1, 1, 1, 1]
Generation 4020: Best = 3.409142e+03 FES = 402100 Diversity = 7.6427e-02 k_t
= [3, 3, 2, 3]
Generation 4030: Best = 3.408969e+03 FES = 403100 Diversity = 6.7968e-02 k_t
= [1, 1, 1, 1]
Generation 4040: Best = 3.408806e+03 FES = 404100 Diversity = 6.3835e-02 k_t
= [4, 4, 4, 2]
Generation 4050: Best = 3.408612e+03 FES = 405100 Diversity = 6.3256e-02 k_t
= [1, 1, 1, 1]
Generation 4060: Best = 3.408408e+03 FES = 406100 Diversity = 7.1461e-02 k_t
= [2, 4, 2, 2]
Generation 4070: Best = 3.408193e+03 FES = 407100 Diversity = 7.6581e-02 k_t
= [1, 1, 1, 1]
Generation 4080: Best = 3.408010e+03 FES = 408100 Diversity = 8.1080e-02 k_t
= [3, 3, 2, 3]
Generation 4090: Best = 3.407864e+03 FES = 409100 Diversity = 7.9537e-02 k_t
= [1, 1, 1, 1]
Generation 4100: Best = 3.407710e+03 FES = 410100 Diversity = 6.2670e-02 k_t
= [4, 3, 4, 4]
Generation 4110: Best = 3.407574e+03 FES = 411100 Diversity = 5.8426e-02 k_t
= [1, 1, 1, 1]
Generation 4120: Best = 3.407410e+03 FES = 412100 Diversity = 5.9801e-02 k_t
= [4, 3, 3, 4]
Generation 4130: Best = 3.407268e+03 FES = 413100 Diversity = 5.7580e-02 k_t
= [1, 1, 1, 1]
Generation 4140: Best = 3.407044e+03 FES = 414100 Diversity = 6.5601e-02 k_t
= [4, 3, 2, 4]
Generation 4150: Best = 3.406915e+03 FES = 415100 Diversity = 6.1494e-02 k_t
= [1, 1, 1, 1]
Generation 4160: Best = 3.406682e+03 FES = 416100 Diversity = 7.3647e-02 k_t
= [2, 2, 4, 4]
Generation 4170: Best = 3.406477e+03 FES = 417100 Diversity = 7.2811e-02 k_t
= [1, 1, 1, 1]
Generation 4180: Best = 3.406168e+03 FES = 418100 Diversity = 8.8405e-02 k_t
= [4, 2, 3, 2]
Generation 4190: Best = 3.405714e+03 FES = 419100 Diversity = 9.8192e-02 k_t
= [1, 1, 1, 1]
Generation 4200: Best = 3.405400e+03 FES = 420100 Diversity = 1.0020e-01 k_t
= [2, 3, 3, 2]
Generation 4210: Best = 3.405007e+03 FES = 421100 Diversity = 1.0432e-01 k_t
= [1, 1, 1, 1]
Generation 4220: Best = 3.404661e+03 FES = 422100 Diversity = 1.0176e-01 k_t
= [3, 3, 2, 2]
Generation 4230: Best = 3.404343e+03 FES = 423100 Diversity = 1.0211e-01 k_t
= [1, 1, 1, 1]
Generation 4240: Best = 3.404095e+03 FES = 424100 Diversity = 8.3893e-02 k_t
= [4, 3, 4, 2]
Generation 4250: Best = 3.403911e+03 FES = 425100 Diversity = 7.6191e-02 k_t
= [1, 1, 1, 1]
Generation 4260: Best = 3.403752e+03 FES = 426100 Diversity = 6.7562e-02 k_t
= [4, 2, 4, 2]
Generation 4270: Best = 3.403631e+03 FES = 427100 Diversity = 5.6715e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 4280: Best = 3.403453e+03 FES = 428100 Diversity = 6.9103e-02 k_t
= [2, 4, 2, 3]
Generation 4290: Best = 3.403269e+03 FES = 429100 Diversity = 7.2093e-02 k_t
= [1, 1, 1, 1]
Generation 4300: Best = 3.403126e+03 FES = 430100 Diversity = 6.7199e-02 k_t
= [2, 3, 3, 2]
Generation 4310: Best = 3.402952e+03 FES = 431100 Diversity = 8.1989e-02 k_t
= [1, 1, 1, 1]
Generation 4320: Best = 3.402809e+03 FES = 432100 Diversity = 7.1940e-02 k_t
= [2, 3, 3, 3]
Generation 4330: Best = 3.402606e+03 FES = 433100 Diversity = 8.1812e-02 k_t
= [1, 1, 1, 1]
Generation 4340: Best = 3.402372e+03 FES = 434100 Diversity = 8.0811e-02 k_t
= [2, 2, 2, 4]
Generation 4350: Best = 3.402085e+03 FES = 435100 Diversity = 9.3291e-02 k_t
= [1, 1, 1, 1]
Generation 4360: Best = 3.401787e+03 FES = 436100 Diversity = 8.9032e-02 k_t
= [3, 3, 3, 2]
Generation 4370: Best = 3.401594e+03 FES = 437100 Diversity = 7.8156e-02 k_t
= [1, 1, 1, 1]
Generation 4380: Best = 3.401416e+03 FES = 438100 Diversity = 6.9289e-02 k_t
= [4, 4, 3, 3]
Generation 4390: Best = 3.401235e+03 FES = 439100 Diversity = 6.3412e-02 k_t
= [1, 1, 1, 1]
Generation 4400: Best = 3.401108e+03 FES = 440100 Diversity = 5.7849e-02 k_t
= [4, 2, 4, 3]
Generation 4410: Best = 3.400990e+03 FES = 441100 Diversity = 5.9407e-02 k_t
= [1, 1, 1, 1]
Generation 4420: Best = 3.400858e+03 FES = 442100 Diversity = 5.5900e-02 k_t
= [4, 3, 2, 4]
Generation 4430: Best = 3.400752e+03 FES = 443100 Diversity = 5.6215e-02 k_t
= [1, 1, 1, 1]
Generation 4440: Best = 3.400631e+03 FES = 444100 Diversity = 5.0935e-02 k_t
= [4, 3, 2, 4]
Generation 4450: Best = 3.400517e+03 FES = 445100 Diversity = 5.2327e-02 k_t
= [1, 1, 1, 1]
Generation 4460: Best = 3.400417e+03 FES = 446100 Diversity = 4.9499e-02 k_t
= [2, 2, 3, 3]
Generation 4470: Best = 3.400311e+03 FES = 447100 Diversity = 5.2301e-02 k_t
= [1, 1, 1, 1]
Generation 4480: Best = 3.400207e+03 FES = 448100 Diversity = 6.0945e-02 k_t
= [3, 3, 3, 2]
Generation 4490: Best = 3.400110e+03 FES = 449100 Diversity = 4.8028e-02 k_t
= [1, 1, 1, 1]
Generation 4500: Best = 3.400045e+03 FES = 450100 Diversity = 4.2579e-02 k_t
= [2, 2, 3, 3]
Generation 4510: Best = 3.399972e+03 FES = 451100 Diversity = 4.6591e-02 k_t
= [1, 1, 1, 1]
Generation 4520: Best = 3.399913e+03 FES = 452100 Diversity = 4.1259e-02 k_t
= [3, 4, 3, 2]
Generation 4530: Best = 3.399833e+03 FES = 453100 Diversity = 4.3677e-02 k_t
= [1, 1, 1, 1]
Generation 4540: Best = 3.399768e+03 FES = 454100 Diversity = 4.3578e-02 k_t

```

```

= [2, 2, 3, 4]
Generation 4550: Best = 3.399677e+03 FES = 455100 Diversity = 4.4496e-02 k_t
= [1, 1, 1, 1]
Generation 4560: Best = 3.399555e+03 FES = 456100 Diversity = 5.8832e-02 k_t
= [2, 2, 4, 4]
Generation 4570: Best = 3.399445e+03 FES = 457100 Diversity = 5.9324e-02 k_t
= [1, 1, 1, 1]
Generation 4580: Best = 3.399344e+03 FES = 458100 Diversity = 5.2509e-02 k_t
= [2, 4, 2, 3]
Generation 4590: Best = 3.399210e+03 FES = 459100 Diversity = 5.9857e-02 k_t
= [1, 1, 1, 1]
Generation 4600: Best = 3.399121e+03 FES = 460100 Diversity = 5.2365e-02 k_t
= [4, 3, 3, 4]
Generation 4610: Best = 3.399051e+03 FES = 461100 Diversity = 4.2113e-02 k_t
= [1, 1, 1, 1]
Generation 4620: Best = 3.398965e+03 FES = 462100 Diversity = 4.9363e-02 k_t
= [4, 2, 4, 3]
Generation 4630: Best = 3.398872e+03 FES = 463100 Diversity = 5.0266e-02 k_t
= [1, 1, 1, 1]
Generation 4640: Best = 3.398773e+03 FES = 464100 Diversity = 5.5742e-02 k_t
= [3, 3, 2, 4]
Generation 4650: Best = 3.398643e+03 FES = 465100 Diversity = 5.7296e-02 k_t
= [1, 1, 1, 1]
Generation 4660: Best = 3.398515e+03 FES = 466100 Diversity = 5.4185e-02 k_t
= [2, 3, 4, 4]
Generation 4670: Best = 3.398387e+03 FES = 467100 Diversity = 5.5215e-02 k_t
= [1, 1, 1, 1]
Generation 4680: Best = 3.398226e+03 FES = 468100 Diversity = 6.5470e-02 k_t
= [2, 2, 3, 3]
Generation 4690: Best = 3.398078e+03 FES = 469100 Diversity = 6.1343e-02 k_t
= [1, 1, 1, 1]
Generation 4700: Best = 3.397929e+03 FES = 470100 Diversity = 5.8496e-02 k_t
= [2, 3, 4, 3]
Generation 4710: Best = 3.397774e+03 FES = 471100 Diversity = 6.6302e-02 k_t
= [1, 1, 1, 1]
Generation 4720: Best = 3.397595e+03 FES = 472100 Diversity = 6.9122e-02 k_t
= [2, 4, 3, 2]
Generation 4730: Best = 3.397431e+03 FES = 473100 Diversity = 6.8759e-02 k_t
= [1, 1, 1, 1]
Generation 4740: Best = 3.397252e+03 FES = 474100 Diversity = 7.7453e-02 k_t
= [2, 3, 4, 4]
Generation 4750: Best = 3.397091e+03 FES = 475100 Diversity = 7.4275e-02 k_t
= [1, 1, 1, 1]
Generation 4760: Best = 3.396927e+03 FES = 476100 Diversity = 6.7557e-02 k_t
= [3, 3, 2, 3]
Generation 4770: Best = 3.396798e+03 FES = 477100 Diversity = 5.7033e-02 k_t
= [1, 1, 1, 1]
Generation 4780: Best = 3.396652e+03 FES = 478100 Diversity = 5.4677e-02 k_t
= [3, 3, 4, 4]
Generation 4790: Best = 3.396490e+03 FES = 479100 Diversity = 6.6639e-02 k_t
= [1, 1, 1, 1]
Generation 4800: Best = 3.396382e+03 FES = 480100 Diversity = 5.9456e-02 k_t
= [3, 3, 4, 2]
Generation 4810: Best = 3.396243e+03 FES = 481100 Diversity = 5.9202e-02 k_t

```



```

= [1, 1, 1, 1]
Generation 4820: Best = 3.396030e+03 FES = 482100 Diversity = 7.4709e-02 k_t
= [3, 2, 2, 4]
Generation 4830: Best = 3.395838e+03 FES = 483100 Diversity = 7.5991e-02 k_t
= [1, 1, 1, 1]
Generation 4840: Best = 3.395730e+03 FES = 484100 Diversity = 6.6791e-02 k_t
= [4, 2, 3, 4]
Generation 4850: Best = 3.395593e+03 FES = 485100 Diversity = 6.3815e-02 k_t
= [1, 1, 1, 1]
Generation 4860: Best = 3.395507e+03 FES = 486100 Diversity = 5.7502e-02 k_t
= [2, 4, 3, 4]
Generation 4870: Best = 3.395370e+03 FES = 487100 Diversity = 6.5207e-02 k_t
= [1, 1, 1, 1]
Generation 4880: Best = 3.395262e+03 FES = 488100 Diversity = 5.4155e-02 k_t
= [3, 3, 2, 2]
Generation 4890: Best = 3.395162e+03 FES = 489100 Diversity = 5.5359e-02 k_t
= [1, 1, 1, 1]
Generation 4900: Best = 3.395042e+03 FES = 490100 Diversity = 4.9851e-02 k_t
= [4, 3, 2, 2]
Generation 4910: Best = 3.394889e+03 FES = 491100 Diversity = 5.9808e-02 k_t
= [1, 1, 1, 1]
Generation 4920: Best = 3.394748e+03 FES = 492100 Diversity = 6.0020e-02 k_t
= [2, 4, 2, 4]
Generation 4930: Best = 3.394597e+03 FES = 493100 Diversity = 6.4319e-02 k_t
= [1, 1, 1, 1]
Generation 4940: Best = 3.394498e+03 FES = 494100 Diversity = 6.3007e-02 k_t
= [3, 2, 3, 3]
Generation 4950: Best = 3.394371e+03 FES = 495100 Diversity = 5.7070e-02 k_t
= [1, 1, 1, 1]
Generation 4960: Best = 3.394280e+03 FES = 496100 Diversity = 4.8451e-02 k_t
= [3, 2, 3, 4]
Generation 4970: Best = 3.394210e+03 FES = 497100 Diversity = 4.6028e-02 k_t
= [1, 1, 1, 1]
Generation 4980: Best = 3.394115e+03 FES = 498100 Diversity = 5.4882e-02 k_t
= [3, 4, 3, 4]
Generation 4990: Best = 3.393973e+03 FES = 499100 Diversity = 5.9233e-02 k_t
= [1, 1, 1, 1]
Generation 5000: Best = 3.393824e+03 FES = 500100 Diversity = 5.2991e-02 k_t
= [2, 2, 3, 4]
Generation 5010: Best = 3.393718e+03 FES = 501100 Diversity = 4.9379e-02 k_t
= [1, 1, 1, 1]
Generation 5020: Best = 3.393611e+03 FES = 502100 Diversity = 5.6282e-02 k_t
= [3, 2, 3, 2]
Generation 5030: Best = 3.393485e+03 FES = 503100 Diversity = 5.8533e-02 k_t
= [1, 1, 1, 1]
Generation 5040: Best = 3.393369e+03 FES = 504100 Diversity = 5.7338e-02 k_t
= [4, 3, 3, 2]
Generation 5050: Best = 3.393258e+03 FES = 505100 Diversity = 5.6325e-02 k_t
= [1, 1, 1, 1]
Generation 5060: Best = 3.393165e+03 FES = 506100 Diversity = 4.9239e-02 k_t
= [4, 3, 3, 4]
Generation 5070: Best = 3.393069e+03 FES = 507100 Diversity = 4.9657e-02 k_t
= [1, 1, 1, 1]
Generation 5080: Best = 3.393004e+03 FES = 508100 Diversity = 4.3683e-02 k_t

```

```

= [4, 3, 4, 2]
Generation 5090: Best = 3.392943e+03 FES = 509100 Diversity = 3.8808e-02 k_t
= [1, 1, 1, 1]
Generation 5100: Best = 3.392886e+03 FES = 510100 Diversity = 3.7887e-02 k_t
= [4, 2, 2, 4]
Generation 5110: Best = 3.392840e+03 FES = 511100 Diversity = 3.8322e-02 k_t
= [1, 1, 1, 1]
Generation 5120: Best = 3.392788e+03 FES = 512100 Diversity = 3.9485e-02 k_t
= [2, 2, 3, 2]
Generation 5130: Best = 3.392724e+03 FES = 513100 Diversity = 3.9161e-02 k_t
= [1, 1, 1, 1]
Generation 5140: Best = 3.392672e+03 FES = 514100 Diversity = 4.0237e-02 k_t
= [2, 4, 4, 2]
Generation 5150: Best = 3.392587e+03 FES = 515100 Diversity = 4.7801e-02 k_t
= [1, 1, 1, 1]
Generation 5160: Best = 3.392530e+03 FES = 516100 Diversity = 4.2618e-02 k_t
= [2, 3, 2, 2]
Generation 5170: Best = 3.392468e+03 FES = 517100 Diversity = 4.2491e-02 k_t
= [1, 1, 1, 1]
Generation 5180: Best = 3.392408e+03 FES = 518100 Diversity = 4.4802e-02 k_t
= [3, 4, 2, 3]
Generation 5190: Best = 3.392348e+03 FES = 519100 Diversity = 4.0401e-02 k_t
= [1, 1, 1, 1]
Generation 5200: Best = 3.392275e+03 FES = 520100 Diversity = 4.2459e-02 k_t
= [2, 2, 3, 2]
Generation 5210: Best = 3.392220e+03 FES = 521100 Diversity = 4.2109e-02 k_t
= [1, 1, 1, 1]
Generation 5220: Best = 3.392152e+03 FES = 522100 Diversity = 4.0195e-02 k_t
= [4, 3, 4, 4]
Generation 5230: Best = 3.392093e+03 FES = 523100 Diversity = 4.2250e-02 k_t
= [1, 1, 1, 1]
Generation 5240: Best = 3.392016e+03 FES = 524100 Diversity = 5.4357e-02 k_t
= [4, 4, 2, 3]
Generation 5250: Best = 3.391949e+03 FES = 525100 Diversity = 4.6623e-02 k_t
= [1, 1, 1, 1]
Generation 5260: Best = 3.391902e+03 FES = 526100 Diversity = 4.4349e-02 k_t
= [4, 4, 4, 3]
Generation 5270: Best = 3.391834e+03 FES = 527100 Diversity = 3.9201e-02 k_t
= [1, 1, 1, 1]
Generation 5280: Best = 3.391779e+03 FES = 528100 Diversity = 3.6923e-02 k_t
= [3, 4, 3, 2]
Generation 5290: Best = 3.391741e+03 FES = 529100 Diversity = 3.4672e-02 k_t
= [1, 1, 1, 1]
Generation 5300: Best = 3.391671e+03 FES = 530100 Diversity = 4.2402e-02 k_t
= [2, 4, 3, 3]
Generation 5310: Best = 3.391630e+03 FES = 531100 Diversity = 4.0140e-02 k_t
= [1, 1, 1, 1]
Generation 5320: Best = 3.391581e+03 FES = 532100 Diversity = 3.8601e-02 k_t
= [3, 2, 3, 2]
Generation 5330: Best = 3.391528e+03 FES = 533100 Diversity = 3.8378e-02 k_t
= [1, 1, 1, 1]
Generation 5340: Best = 3.391477e+03 FES = 534100 Diversity = 4.0192e-02 k_t
= [4, 3, 4, 4]
Generation 5350: Best = 3.391429e+03 FES = 535100 Diversity = 4.2784e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 5360: Best = 3.391354e+03 FES = 536100 Diversity = 4.5891e-02 k_t
= [3, 3, 3, 4]
Generation 5370: Best = 3.391285e+03 FES = 537100 Diversity = 4.3893e-02 k_t
= [1, 1, 1, 1]
Generation 5380: Best = 3.391236e+03 FES = 538100 Diversity = 3.6973e-02 k_t
= [3, 2, 4, 2]
Generation 5390: Best = 3.391191e+03 FES = 539100 Diversity = 3.6583e-02 k_t
= [1, 1, 1, 1]
Generation 5400: Best = 3.391142e+03 FES = 540100 Diversity = 3.5856e-02 k_t
= [4, 3, 3, 3]
Generation 5410: Best = 3.391085e+03 FES = 541100 Diversity = 3.8702e-02 k_t
= [1, 1, 1, 1]
Generation 5420: Best = 3.391009e+03 FES = 542100 Diversity = 4.1858e-02 k_t
= [2, 4, 2, 3]
Generation 5430: Best = 3.390937e+03 FES = 543100 Diversity = 4.1170e-02 k_t
= [1, 1, 1, 1]
Generation 5440: Best = 3.390891e+03 FES = 544100 Diversity = 3.8796e-02 k_t
= [4, 4, 4, 3]
Generation 5450: Best = 3.390816e+03 FES = 545100 Diversity = 3.9771e-02 k_t
= [1, 1, 1, 1]
Generation 5460: Best = 3.390775e+03 FES = 546100 Diversity = 3.0471e-02 k_t
= [2, 4, 4, 2]
Generation 5470: Best = 3.390722e+03 FES = 547100 Diversity = 2.9577e-02 k_t
= [1, 1, 1, 1]
Generation 5480: Best = 3.390670e+03 FES = 548100 Diversity = 4.1830e-02 k_t
= [3, 4, 3, 3]
Generation 5490: Best = 3.390593e+03 FES = 549100 Diversity = 4.8116e-02 k_t
= [1, 1, 1, 1]
Generation 5500: Best = 3.390527e+03 FES = 550100 Diversity = 4.1764e-02 k_t
= [3, 2, 3, 2]
Generation 5510: Best = 3.390464e+03 FES = 551100 Diversity = 3.6351e-02 k_t
= [1, 1, 1, 1]
Generation 5520: Best = 3.390389e+03 FES = 552100 Diversity = 3.9157e-02 k_t
= [3, 3, 2, 2]
Generation 5530: Best = 3.390326e+03 FES = 553100 Diversity = 3.7056e-02 k_t
= [1, 1, 1, 1]
Generation 5540: Best = 3.390276e+03 FES = 554100 Diversity = 3.4754e-02 k_t
= [4, 4, 4, 3]
Generation 5550: Best = 3.390214e+03 FES = 555100 Diversity = 3.9817e-02 k_t
= [1, 1, 1, 1]
Generation 5560: Best = 3.390163e+03 FES = 556100 Diversity = 3.8505e-02 k_t
= [3, 2, 2, 3]
Generation 5570: Best = 3.390099e+03 FES = 557100 Diversity = 4.0666e-02 k_t
= [1, 1, 1, 1]
Generation 5580: Best = 3.390030e+03 FES = 558100 Diversity = 4.2016e-02 k_t
= [2, 3, 4, 3]
Generation 5590: Best = 3.389925e+03 FES = 559100 Diversity = 5.1096e-02 k_t
= [1, 1, 1, 1]
Generation 5600: Best = 3.389831e+03 FES = 560100 Diversity = 5.4238e-02 k_t
= [4, 3, 2, 3]
Generation 5610: Best = 3.389726e+03 FES = 561100 Diversity = 4.7822e-02 k_t
= [1, 1, 1, 1]
Generation 5620: Best = 3.389651e+03 FES = 562100 Diversity = 4.3631e-02 k_t

```

```

= [3, 3, 2, 4]
Generation 5630: Best = 3.389575e+03  FES = 563100  Diversity = 4.4791e-02  k_t
= [1, 1, 1, 1]
Generation 5640: Best = 3.389519e+03  FES = 564100  Diversity = 4.0899e-02  k_t
= [4, 3, 2, 2]
Generation 5650: Best = 3.389437e+03  FES = 565100  Diversity = 4.3953e-02  k_t
= [1, 1, 1, 1]
Generation 5660: Best = 3.389359e+03  FES = 566100  Diversity = 3.8086e-02  k_t
= [4, 3, 2, 3]
Generation 5670: Best = 3.389302e+03  FES = 567100  Diversity = 3.9748e-02  k_t
= [1, 1, 1, 1]
Generation 5680: Best = 3.389235e+03  FES = 568100  Diversity = 3.7672e-02  k_t
= [4, 2, 2, 4]
Generation 5690: Best = 3.389186e+03  FES = 569100  Diversity = 3.9120e-02  k_t
= [1, 1, 1, 1]
Generation 5700: Best = 3.389115e+03  FES = 570100  Diversity = 4.4907e-02  k_t
= [2, 3, 3, 4]
Generation 5710: Best = 3.389062e+03  FES = 571100  Diversity = 3.9724e-02  k_t
= [1, 1, 1, 1]
Generation 5720: Best = 3.389023e+03  FES = 572100  Diversity = 3.1061e-02  k_t
= [4, 2, 4, 2]
Generation 5730: Best = 3.388984e+03  FES = 573100  Diversity = 3.2005e-02  k_t
= [1, 1, 1, 1]
Generation 5740: Best = 3.388946e+03  FES = 574100  Diversity = 3.3825e-02  k_t
= [2, 3, 2, 2]
Generation 5750: Best = 3.388877e+03  FES = 575100  Diversity = 4.0760e-02  k_t
= [1, 1, 1, 1]
Generation 5760: Best = 3.388800e+03  FES = 576100  Diversity = 4.5165e-02  k_t
= [2, 3, 2, 3]
Generation 5770: Best = 3.388723e+03  FES = 577100  Diversity = 5.7444e-02  k_t
= [1, 1, 1, 1]
Generation 5780: Best = 3.388580e+03  FES = 578100  Diversity = 6.9953e-02  k_t
= [2, 2, 2, 3]
Generation 5790: Best = 3.388317e+03  FES = 579100  Diversity = 8.6869e-02  k_t
= [1, 1, 1, 1]
Generation 5800: Best = 3.387995e+03  FES = 580100  Diversity = 9.4293e-02  k_t
= [2, 4, 3, 3]
Generation 5810: Best = 3.387691e+03  FES = 581100  Diversity = 9.9802e-02  k_t
= [1, 1, 1, 1]
Generation 5820: Best = 3.387374e+03  FES = 582100  Diversity = 9.7987e-02  k_t
= [4, 2, 2, 2]
Generation 5830: Best = 3.386984e+03  FES = 583100  Diversity = 1.0272e-01  k_t
= [1, 1, 1, 1]
Generation 5840: Best = 3.386553e+03  FES = 584100  Diversity = 1.2001e-01  k_t
= [4, 4, 3, 4]
Generation 5850: Best = 3.385971e+03  FES = 585100  Diversity = 1.2342e-01  k_t
= [1, 1, 1, 1]
Generation 5860: Best = 3.385547e+03  FES = 586100  Diversity = 1.1840e-01  k_t
= [2, 3, 2, 3]
Generation 5870: Best = 3.385064e+03  FES = 587100  Diversity = 1.3134e-01  k_t
= [1, 1, 1, 1]
Generation 5880: Best = 3.384705e+03  FES = 588100  Diversity = 9.8441e-02  k_t
= [3, 2, 3, 4]
Generation 5890: Best = 3.384314e+03  FES = 589100  Diversity = 9.8729e-02  k_t

```

```

= [1, 1, 1, 1]
Generation 5900: Best = 3.384002e+03 FES = 590100 Diversity = 9.0574e-02 k_t
= [3, 2, 3, 4]
Generation 5910: Best = 3.383642e+03 FES = 591100 Diversity = 1.0102e-01 k_t
= [1, 1, 1, 1]
Generation 5920: Best = 3.383025e+03 FES = 592100 Diversity = 1.1740e-01 k_t
= [2, 3, 2, 4]
Generation 5930: Best = 3.382245e+03 FES = 593100 Diversity = 1.3523e-01 k_t
= [1, 1, 1, 1]
Generation 5940: Best = 3.381215e+03 FES = 594100 Diversity = 1.6047e-01 k_t
= [4, 2, 2, 3]
Generation 5950: Best = 3.380709e+03 FES = 595100 Diversity = 1.3535e-01 k_t
= [1, 1, 1, 1]
Generation 5960: Best = 3.380105e+03 FES = 596100 Diversity = 1.1309e-01 k_t
= [4, 3, 4, 4]
Generation 5970: Best = 3.379539e+03 FES = 597100 Diversity = 1.2834e-01 k_t
= [1, 1, 1, 1]
Generation 5980: Best = 3.378963e+03 FES = 598100 Diversity = 1.2589e-01 k_t
= [3, 4, 4, 2]
Generation 5990: Best = 3.378457e+03 FES = 599100 Diversity = 1.1806e-01 k_t
= [1, 1, 1, 1]
Generation 6000: Best = 3.377905e+03 FES = 600100 Diversity = 1.1598e-01 k_t
= [3, 3, 2, 2]
Generation 6010: Best = 3.377322e+03 FES = 601100 Diversity = 1.2922e-01 k_t
= [1, 1, 1, 1]
Generation 6020: Best = 3.376869e+03 FES = 602100 Diversity = 1.2001e-01 k_t
= [4, 4, 4, 2]
Generation 6030: Best = 3.376301e+03 FES = 603100 Diversity = 1.2713e-01 k_t
= [1, 1, 1, 1]
Generation 6040: Best = 3.375701e+03 FES = 604100 Diversity = 1.1065e-01 k_t
= [3, 3, 3, 3]
Generation 6050: Best = 3.375240e+03 FES = 605100 Diversity = 1.0614e-01 k_t
= [1, 1, 1, 1]
Generation 6060: Best = 3.374783e+03 FES = 606100 Diversity = 1.2056e-01 k_t
= [4, 4, 3, 2]
Generation 6070: Best = 3.374207e+03 FES = 607100 Diversity = 1.4012e-01 k_t
= [1, 1, 1, 1]
Generation 6080: Best = 3.373454e+03 FES = 608100 Diversity = 1.4782e-01 k_t
= [2, 2, 3, 4]
Generation 6090: Best = 3.372820e+03 FES = 609100 Diversity = 1.5228e-01 k_t
= [1, 1, 1, 1]
Generation 6100: Best = 3.372295e+03 FES = 610100 Diversity = 1.2158e-01 k_t
= [4, 2, 3, 3]
Generation 6110: Best = 3.371790e+03 FES = 611100 Diversity = 1.2414e-01 k_t
= [1, 1, 1, 1]
Generation 6120: Best = 3.371329e+03 FES = 612100 Diversity = 1.0948e-01 k_t
= [4, 3, 3, 3]
Generation 6130: Best = 3.370692e+03 FES = 613100 Diversity = 1.2380e-01 k_t
= [1, 1, 1, 1]
Generation 6140: Best = 3.370129e+03 FES = 614100 Diversity = 1.3681e-01 k_t
= [3, 3, 4, 3]
Generation 6150: Best = 3.369722e+03 FES = 615100 Diversity = 1.1237e-01 k_t
= [1, 1, 1, 1]
Generation 6160: Best = 3.369256e+03 FES = 616100 Diversity = 1.1781e-01 k_t

```

```

= [4, 4, 2, 3]
Generation 6170: Best = 3.368673e+03 FES = 617100 Diversity = 1.1896e-01 k_t
= [1, 1, 1, 1]
Generation 6180: Best = 3.368071e+03 FES = 618100 Diversity = 1.1727e-01 k_t
= [3, 2, 3, 2]
Generation 6190: Best = 3.367456e+03 FES = 619100 Diversity = 1.1175e-01 k_t
= [1, 1, 1, 1]
Generation 6200: Best = 3.366899e+03 FES = 620100 Diversity = 1.3264e-01 k_t
= [4, 3, 4, 3]
Generation 6210: Best = 3.366256e+03 FES = 621100 Diversity = 1.3561e-01 k_t
= [1, 1, 1, 1]
Generation 6220: Best = 3.365673e+03 FES = 622100 Diversity = 1.3459e-01 k_t
= [4, 3, 4, 2]
Generation 6230: Best = 3.365049e+03 FES = 623100 Diversity = 1.2615e-01 k_t
= [1, 1, 1, 1]
Generation 6240: Best = 3.364381e+03 FES = 624100 Diversity = 1.2171e-01 k_t
= [3, 3, 4, 2]
Generation 6250: Best = 3.363911e+03 FES = 625100 Diversity = 1.1253e-01 k_t
= [1, 1, 1, 1]
Generation 6260: Best = 3.363452e+03 FES = 626100 Diversity = 1.1172e-01 k_t
= [2, 3, 2, 4]
Generation 6270: Best = 3.362971e+03 FES = 627100 Diversity = 1.0977e-01 k_t
= [1, 1, 1, 1]
Generation 6280: Best = 3.362500e+03 FES = 628100 Diversity = 1.1059e-01 k_t
= [3, 4, 4, 2]
Generation 6290: Best = 3.361882e+03 FES = 629100 Diversity = 1.2625e-01 k_t
= [1, 1, 1, 1]
Generation 6300: Best = 3.361125e+03 FES = 630100 Diversity = 1.4480e-01 k_t
= [4, 4, 3, 4]
Generation 6310: Best = 3.360746e+03 FES = 631100 Diversity = 1.2183e-01 k_t
= [1, 1, 1, 1]
Generation 6320: Best = 3.360357e+03 FES = 632100 Diversity = 1.1108e-01 k_t
= [3, 2, 4, 2]
Generation 6330: Best = 3.359882e+03 FES = 633100 Diversity = 9.9137e-02 k_t
= [1, 1, 1, 1]
Generation 6340: Best = 3.359517e+03 FES = 634100 Diversity = 9.7372e-02 k_t
= [2, 3, 4, 4]
Generation 6350: Best = 3.359209e+03 FES = 635100 Diversity = 1.0313e-01 k_t
= [1, 1, 1, 1]
Generation 6360: Best = 3.358949e+03 FES = 636100 Diversity = 8.4999e-02 k_t
= [3, 3, 2, 3]
Generation 6370: Best = 3.358736e+03 FES = 637100 Diversity = 7.4359e-02 k_t
= [1, 1, 1, 1]
Generation 6380: Best = 3.358484e+03 FES = 638100 Diversity = 7.0635e-02 k_t
= [4, 2, 3, 2]
Generation 6390: Best = 3.358258e+03 FES = 639100 Diversity = 6.6099e-02 k_t
= [1, 1, 1, 1]
Generation 6400: Best = 3.358076e+03 FES = 640100 Diversity = 6.9569e-02 k_t
= [4, 3, 2, 4]
Generation 6410: Best = 3.357877e+03 FES = 641100 Diversity = 7.4084e-02 k_t
= [1, 1, 1, 1]
Generation 6420: Best = 3.357630e+03 FES = 642100 Diversity = 6.9561e-02 k_t
= [3, 2, 2, 4]
Generation 6430: Best = 3.357362e+03 FES = 643100 Diversity = 8.5996e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 6440: Best = 3.357189e+03 FES = 644100 Diversity = 8.1887e-02 k_t
= [3, 2, 2, 2]
Generation 6450: Best = 3.356906e+03 FES = 645100 Diversity = 8.5191e-02 k_t
= [1, 1, 1, 1]
Generation 6460: Best = 3.356700e+03 FES = 646100 Diversity = 7.5758e-02 k_t
= [4, 2, 2, 2]
Generation 6470: Best = 3.356485e+03 FES = 647100 Diversity = 7.9998e-02 k_t
= [1, 1, 1, 1]
Generation 6480: Best = 3.356328e+03 FES = 648100 Diversity = 6.5478e-02 k_t
= [2, 2, 3, 2]
Generation 6490: Best = 3.356194e+03 FES = 649100 Diversity = 6.3052e-02 k_t
= [1, 1, 1, 1]
Generation 6500: Best = 3.356029e+03 FES = 650100 Diversity = 6.3940e-02 k_t
= [4, 3, 4, 3]
Generation 6510: Best = 3.355894e+03 FES = 651100 Diversity = 5.7827e-02 k_t
= [1, 1, 1, 1]
Generation 6520: Best = 3.355741e+03 FES = 652100 Diversity = 5.9063e-02 k_t
= [2, 2, 3, 2]
Generation 6530: Best = 3.355525e+03 FES = 653100 Diversity = 7.4134e-02 k_t
= [1, 1, 1, 1]
Generation 6540: Best = 3.355266e+03 FES = 654100 Diversity = 6.6831e-02 k_t
= [4, 2, 2, 3]
Generation 6550: Best = 3.355069e+03 FES = 655100 Diversity = 7.3954e-02 k_t
= [1, 1, 1, 1]
Generation 6560: Best = 3.354755e+03 FES = 656100 Diversity = 8.6933e-02 k_t
= [4, 3, 4, 2]
Generation 6570: Best = 3.354364e+03 FES = 657100 Diversity = 9.4546e-02 k_t
= [1, 1, 1, 1]
Generation 6580: Best = 3.354044e+03 FES = 658100 Diversity = 9.0037e-02 k_t
= [2, 2, 4, 3]
Generation 6590: Best = 3.353711e+03 FES = 659100 Diversity = 9.6019e-02 k_t
= [1, 1, 1, 1]
Generation 6600: Best = 3.353395e+03 FES = 660100 Diversity = 1.0490e-01 k_t
= [4, 4, 4, 4]
Generation 6610: Best = 3.353126e+03 FES = 661100 Diversity = 8.1363e-02 k_t
= [1, 1, 1, 1]
Generation 6620: Best = 3.353006e+03 FES = 662100 Diversity = 5.7214e-02 k_t
= [2, 2, 2, 3]
Generation 6630: Best = 3.352832e+03 FES = 663100 Diversity = 5.7422e-02 k_t
= [1, 1, 1, 1]
Generation 6640: Best = 3.352663e+03 FES = 664100 Diversity = 5.9907e-02 k_t
= [3, 4, 3, 4]
Generation 6650: Best = 3.352509e+03 FES = 665100 Diversity = 6.1717e-02 k_t
= [1, 1, 1, 1]
Generation 6660: Best = 3.352343e+03 FES = 666100 Diversity = 6.0798e-02 k_t
= [4, 4, 4, 2]
Generation 6670: Best = 3.352189e+03 FES = 667100 Diversity = 6.2116e-02 k_t
= [1, 1, 1, 1]
Generation 6680: Best = 3.352009e+03 FES = 668100 Diversity = 6.2519e-02 k_t
= [3, 4, 3, 3]
Generation 6690: Best = 3.351881e+03 FES = 669100 Diversity = 5.6249e-02 k_t
= [1, 1, 1, 1]
Generation 6700: Best = 3.351719e+03 FES = 670100 Diversity = 6.1038e-02 k_t

```

```

= [4, 4, 3, 2]
Generation 6710: Best = 3.351557e+03  FES = 671100  Diversity = 6.3847e-02  k_t
= [1, 1, 1, 1]
Generation 6720: Best = 3.351368e+03  FES = 672100  Diversity = 6.5436e-02  k_t
= [4, 4, 2, 4]
Generation 6730: Best = 3.351187e+03  FES = 673100  Diversity = 6.5096e-02  k_t
= [1, 1, 1, 1]
Generation 6740: Best = 3.351025e+03  FES = 674100  Diversity = 6.8649e-02  k_t
= [2, 3, 3, 4]
Generation 6750: Best = 3.350908e+03  FES = 675100  Diversity = 5.9628e-02  k_t
= [1, 1, 1, 1]
Generation 6760: Best = 3.350820e+03  FES = 676100  Diversity = 5.2494e-02  k_t
= [2, 2, 4, 4]
Generation 6770: Best = 3.350736e+03  FES = 677100  Diversity = 4.4792e-02  k_t
= [1, 1, 1, 1]
Generation 6780: Best = 3.350673e+03  FES = 678100  Diversity = 4.3822e-02  k_t
= [2, 3, 3, 3]
Generation 6790: Best = 3.350581e+03  FES = 679100  Diversity = 4.4243e-02  k_t
= [1, 1, 1, 1]
Generation 6800: Best = 3.350507e+03  FES = 680100  Diversity = 4.1998e-02  k_t
= [4, 4, 3, 4]
Generation 6810: Best = 3.350444e+03  FES = 681100  Diversity = 4.3186e-02  k_t
= [1, 1, 1, 1]
Generation 6820: Best = 3.350394e+03  FES = 682100  Diversity = 3.7823e-02  k_t
= [2, 3, 4, 2]
Generation 6830: Best = 3.350331e+03  FES = 683100  Diversity = 4.0953e-02  k_t
= [1, 1, 1, 1]
Generation 6840: Best = 3.350282e+03  FES = 684100  Diversity = 3.7004e-02  k_t
= [4, 2, 4, 4]
Generation 6850: Best = 3.350231e+03  FES = 685100  Diversity = 3.6844e-02  k_t
= [1, 1, 1, 1]
Generation 6860: Best = 3.350175e+03  FES = 686100  Diversity = 3.5635e-02  k_t
= [2, 3, 2, 3]
Generation 6870: Best = 3.350134e+03  FES = 687100  Diversity = 3.4145e-02  k_t
= [1, 1, 1, 1]
Generation 6880: Best = 3.350085e+03  FES = 688100  Diversity = 3.5645e-02  k_t
= [3, 3, 2, 4]
Generation 6890: Best = 3.350046e+03  FES = 689100  Diversity = 3.6168e-02  k_t
= [1, 1, 1, 1]
Generation 6900: Best = 3.350010e+03  FES = 690100  Diversity = 3.1394e-02  k_t
= [4, 2, 2, 4]
Generation 6910: Best = 3.349979e+03  FES = 691100  Diversity = 3.4657e-02  k_t
= [1, 1, 1, 1]
Generation 6920: Best = 3.349937e+03  FES = 692100  Diversity = 3.8840e-02  k_t
= [3, 2, 4, 3]
Generation 6930: Best = 3.349891e+03  FES = 693100  Diversity = 3.6643e-02  k_t
= [1, 1, 1, 1]
Generation 6940: Best = 3.349854e+03  FES = 694100  Diversity = 2.9600e-02  k_t
= [2, 3, 2, 3]
Generation 6950: Best = 3.349820e+03  FES = 695100  Diversity = 2.6542e-02  k_t
= [1, 1, 1, 1]
Generation 6960: Best = 3.349788e+03  FES = 696100  Diversity = 2.9252e-02  k_t
= [4, 4, 2, 3]
Generation 6970: Best = 3.349752e+03  FES = 697100  Diversity = 3.1647e-02  k_t

```



```

= [1, 1, 1, 1]
Generation 6980: Best = 3.349711e+03 FES = 698100 Diversity = 3.2860e-02 k_t
= [2, 3, 3, 3]
Generation 6990: Best = 3.349673e+03 FES = 699100 Diversity = 3.4019e-02 k_t
= [1, 1, 1, 1]
Generation 7000: Best = 3.349639e+03 FES = 700100 Diversity = 2.9726e-02 k_t
= [4, 2, 4, 4]
Generation 7010: Best = 3.349605e+03 FES = 701100 Diversity = 3.1751e-02 k_t
= [1, 1, 1, 1]
Generation 7020: Best = 3.349557e+03 FES = 702100 Diversity = 3.0712e-02 k_t
= [2, 4, 4, 2]
Generation 7030: Best = 3.349515e+03 FES = 703100 Diversity = 3.0537e-02 k_t
= [1, 1, 1, 1]
Generation 7040: Best = 3.349479e+03 FES = 704100 Diversity = 3.3244e-02 k_t
= [2, 3, 4, 4]
Generation 7050: Best = 3.349433e+03 FES = 705100 Diversity = 3.5265e-02 k_t
= [1, 1, 1, 1]
Generation 7060: Best = 3.349388e+03 FES = 706100 Diversity = 3.3573e-02 k_t
= [2, 2, 2, 4]
Generation 7070: Best = 3.349355e+03 FES = 707100 Diversity = 3.2582e-02 k_t
= [1, 1, 1, 1]
Generation 7080: Best = 3.349312e+03 FES = 708100 Diversity = 3.6045e-02 k_t
= [3, 4, 4, 2]
Generation 7090: Best = 3.349274e+03 FES = 709100 Diversity = 3.5623e-02 k_t
= [1, 1, 1, 1]
Generation 7100: Best = 3.349244e+03 FES = 710100 Diversity = 3.4117e-02 k_t
= [2, 2, 4, 2]
Generation 7110: Best = 3.349205e+03 FES = 711100 Diversity = 3.3695e-02 k_t
= [1, 1, 1, 1]
Generation 7120: Best = 3.349177e+03 FES = 712100 Diversity = 2.7488e-02 k_t
= [4, 4, 3, 4]
Generation 7130: Best = 3.349147e+03 FES = 713100 Diversity = 2.7424e-02 k_t
= [1, 1, 1, 1]
Generation 7140: Best = 3.349112e+03 FES = 714100 Diversity = 3.2288e-02 k_t
= [3, 4, 4, 4]
Generation 7150: Best = 3.349085e+03 FES = 715100 Diversity = 2.9531e-02 k_t
= [1, 1, 1, 1]
Generation 7160: Best = 3.349063e+03 FES = 716100 Diversity = 2.5621e-02 k_t
= [2, 3, 4, 3]
Generation 7170: Best = 3.349041e+03 FES = 717100 Diversity = 2.2411e-02 k_t
= [1, 1, 1, 1]
Generation 7180: Best = 3.349026e+03 FES = 718100 Diversity = 2.1383e-02 k_t
= [4, 2, 2, 4]
Generation 7190: Best = 3.349004e+03 FES = 719100 Diversity = 2.3564e-02 k_t
= [1, 1, 1, 1]
Generation 7200: Best = 3.348980e+03 FES = 720100 Diversity = 2.5860e-02 k_t
= [3, 4, 3, 2]
Generation 7210: Best = 3.348962e+03 FES = 721100 Diversity = 2.3386e-02 k_t
= [1, 1, 1, 1]
Generation 7220: Best = 3.348938e+03 FES = 722100 Diversity = 2.3106e-02 k_t
= [3, 4, 4, 3]
Generation 7230: Best = 3.348919e+03 FES = 723100 Diversity = 2.4874e-02 k_t
= [1, 1, 1, 1]
Generation 7240: Best = 3.348889e+03 FES = 724100 Diversity = 2.6362e-02 k_t

```

```

= [3, 2, 3, 3]
Generation 7250: Best = 3.348864e+03 FES = 725100 Diversity = 2.7282e-02 k_t
= [1, 1, 1, 1]
Generation 7260: Best = 3.348835e+03 FES = 726100 Diversity = 2.7395e-02 k_t
= [3, 4, 2, 2]
Generation 7270: Best = 3.348798e+03 FES = 727100 Diversity = 3.1676e-02 k_t
= [1, 1, 1, 1]
Generation 7280: Best = 3.348767e+03 FES = 728100 Diversity = 2.8512e-02 k_t
= [2, 2, 2, 2]
Generation 7290: Best = 3.348726e+03 FES = 729100 Diversity = 3.1404e-02 k_t
= [1, 1, 1, 1]
Generation 7300: Best = 3.348694e+03 FES = 730100 Diversity = 2.9798e-02 k_t
= [2, 2, 4, 4]
Generation 7310: Best = 3.348667e+03 FES = 731100 Diversity = 2.6792e-02 k_t
= [1, 1, 1, 1]
Generation 7320: Best = 3.348638e+03 FES = 732100 Diversity = 2.8479e-02 k_t
= [4, 2, 4, 4]
Generation 7330: Best = 3.348600e+03 FES = 733100 Diversity = 3.1463e-02 k_t
= [1, 1, 1, 1]
Generation 7340: Best = 3.348560e+03 FES = 734100 Diversity = 3.3874e-02 k_t
= [2, 2, 4, 2]
Generation 7350: Best = 3.348518e+03 FES = 735100 Diversity = 3.4447e-02 k_t
= [1, 1, 1, 1]
Generation 7360: Best = 3.348484e+03 FES = 736100 Diversity = 2.7943e-02 k_t
= [2, 4, 3, 3]
Generation 7370: Best = 3.348448e+03 FES = 737100 Diversity = 3.5426e-02 k_t
= [1, 1, 1, 1]
Generation 7380: Best = 3.348406e+03 FES = 738100 Diversity = 3.2783e-02 k_t
= [4, 2, 3, 3]
Generation 7390: Best = 3.348375e+03 FES = 739100 Diversity = 3.0199e-02 k_t
= [1, 1, 1, 1]
Generation 7400: Best = 3.348348e+03 FES = 740100 Diversity = 2.7609e-02 k_t
= [3, 3, 3, 4]
Generation 7410: Best = 3.348325e+03 FES = 741100 Diversity = 2.5165e-02 k_t
= [1, 1, 1, 1]
Generation 7420: Best = 3.348307e+03 FES = 742100 Diversity = 2.1216e-02 k_t
= [4, 2, 2, 3]
Generation 7430: Best = 3.348287e+03 FES = 743100 Diversity = 2.0731e-02 k_t
= [1, 1, 1, 1]
Generation 7440: Best = 3.348267e+03 FES = 744100 Diversity = 1.8939e-02 k_t
= [2, 3, 2, 3]
Generation 7450: Best = 3.348238e+03 FES = 745100 Diversity = 2.3726e-02 k_t
= [1, 1, 1, 1]
Generation 7460: Best = 3.348194e+03 FES = 746100 Diversity = 2.9256e-02 k_t
= [2, 4, 2, 4]
Generation 7470: Best = 3.348148e+03 FES = 747100 Diversity = 3.6819e-02 k_t
= [1, 1, 1, 1]
Generation 7480: Best = 3.348106e+03 FES = 748100 Diversity = 3.3985e-02 k_t
= [2, 3, 2, 4]
Generation 7490: Best = 3.348079e+03 FES = 749100 Diversity = 2.8744e-02 k_t
= [1, 1, 1, 1]
Generation 7500: Best = 3.348056e+03 FES = 750100 Diversity = 2.5469e-02 k_t
= [3, 3, 3, 4]
Generation 7510: Best = 3.348032e+03 FES = 751100 Diversity = 2.4908e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 7520: Best = 3.348009e+03 FES = 752100 Diversity = 2.4169e-02 k_t
= [4, 3, 4, 4]
Generation 7530: Best = 3.347979e+03 FES = 753100 Diversity = 2.7743e-02 k_t
= [1, 1, 1, 1]
Generation 7540: Best = 3.347946e+03 FES = 754100 Diversity = 3.1659e-02 k_t
= [3, 4, 4, 4]
Generation 7550: Best = 3.347918e+03 FES = 755100 Diversity = 2.5810e-02 k_t
= [1, 1, 1, 1]
Generation 7560: Best = 3.347887e+03 FES = 756100 Diversity = 2.6346e-02 k_t
= [2, 4, 3, 2]
Generation 7570: Best = 3.347831e+03 FES = 757100 Diversity = 3.6422e-02 k_t
= [1, 1, 1, 1]
Generation 7580: Best = 3.347778e+03 FES = 758100 Diversity = 3.0348e-02 k_t
= [3, 3, 3, 3]
Generation 7590: Best = 3.347728e+03 FES = 759100 Diversity = 3.2854e-02 k_t
= [1, 1, 1, 1]
Generation 7600: Best = 3.347691e+03 FES = 760100 Diversity = 3.3198e-02 k_t
= [2, 3, 2, 4]
Generation 7610: Best = 3.347641e+03 FES = 761100 Diversity = 3.6265e-02 k_t
= [1, 1, 1, 1]
Generation 7620: Best = 3.347610e+03 FES = 762100 Diversity = 2.9779e-02 k_t
= [3, 4, 2, 2]
Generation 7630: Best = 3.347574e+03 FES = 763100 Diversity = 2.8173e-02 k_t
= [1, 1, 1, 1]
Generation 7640: Best = 3.347536e+03 FES = 764100 Diversity = 2.7443e-02 k_t
= [4, 4, 4, 3]
Generation 7650: Best = 3.347497e+03 FES = 765100 Diversity = 2.9583e-02 k_t
= [1, 1, 1, 1]
Generation 7660: Best = 3.347470e+03 FES = 766100 Diversity = 2.4177e-02 k_t
= [2, 4, 2, 4]
Generation 7670: Best = 3.347425e+03 FES = 767100 Diversity = 3.4761e-02 k_t
= [1, 1, 1, 1]
Generation 7680: Best = 3.347396e+03 FES = 768100 Diversity = 3.0446e-02 k_t
= [3, 2, 3, 3]
Generation 7690: Best = 3.347371e+03 FES = 769100 Diversity = 2.7378e-02 k_t
= [1, 1, 1, 1]
Generation 7700: Best = 3.347343e+03 FES = 770100 Diversity = 2.5553e-02 k_t
= [3, 4, 2, 4]
Generation 7710: Best = 3.347318e+03 FES = 771100 Diversity = 2.6621e-02 k_t
= [1, 1, 1, 1]
Generation 7720: Best = 3.347297e+03 FES = 772100 Diversity = 2.3691e-02 k_t
= [4, 3, 2, 2]
Generation 7730: Best = 3.347270e+03 FES = 773100 Diversity = 2.6716e-02 k_t
= [1, 1, 1, 1]
Generation 7740: Best = 3.347244e+03 FES = 774100 Diversity = 2.5211e-02 k_t
= [3, 3, 2, 4]
Generation 7750: Best = 3.347213e+03 FES = 775100 Diversity = 2.6744e-02 k_t
= [1, 1, 1, 1]
Generation 7760: Best = 3.347194e+03 FES = 776100 Diversity = 2.2003e-02 k_t
= [4, 3, 2, 2]
Generation 7770: Best = 3.347172e+03 FES = 777100 Diversity = 2.2062e-02 k_t
= [1, 1, 1, 1]
Generation 7780: Best = 3.347152e+03 FES = 778100 Diversity = 2.2093e-02 k_t

```

```

= [4, 3, 2, 2]
Generation 7790: Best = 3.347138e+03 FES = 779100 Diversity = 1.9733e-02 k_t
= [1, 1, 1, 1]
Generation 7800: Best = 3.347125e+03 FES = 780100 Diversity = 1.6673e-02 k_t
= [4, 4, 3, 3]
Generation 7810: Best = 3.347111e+03 FES = 781100 Diversity = 1.8998e-02 k_t
= [1, 1, 1, 1]
Generation 7820: Best = 3.347099e+03 FES = 782100 Diversity = 1.9800e-02 k_t
= [2, 4, 4, 3]
Generation 7830: Best = 3.347081e+03 FES = 783100 Diversity = 2.1865e-02 k_t
= [1, 1, 1, 1]
Generation 7840: Best = 3.347058e+03 FES = 784100 Diversity = 2.3986e-02 k_t
= [4, 3, 2, 3]
Generation 7850: Best = 3.347046e+03 FES = 785100 Diversity = 1.8849e-02 k_t
= [1, 1, 1, 1]
Generation 7860: Best = 3.347033e+03 FES = 786100 Diversity = 1.9898e-02 k_t
= [3, 3, 4, 3]
Generation 7870: Best = 3.347019e+03 FES = 787100 Diversity = 1.9049e-02 k_t
= [1, 1, 1, 1]
Generation 7880: Best = 3.347007e+03 FES = 788100 Diversity = 1.6806e-02 k_t
= [2, 3, 2, 3]
Generation 7890: Best = 3.346994e+03 FES = 789100 Diversity = 1.8777e-02 k_t
= [1, 1, 1, 1]
Generation 7900: Best = 3.346976e+03 FES = 790100 Diversity = 2.2193e-02 k_t
= [4, 2, 3, 3]
Generation 7910: Best = 3.346959e+03 FES = 791100 Diversity = 1.9588e-02 k_t
= [1, 1, 1, 1]
Generation 7920: Best = 3.346946e+03 FES = 792100 Diversity = 1.7200e-02 k_t
= [2, 2, 4, 3]
Generation 7930: Best = 3.346934e+03 FES = 793100 Diversity = 1.7163e-02 k_t
= [1, 1, 1, 1]
Generation 7940: Best = 3.346912e+03 FES = 794100 Diversity = 2.3889e-02 k_t
= [4, 4, 3, 2]
Generation 7950: Best = 3.346883e+03 FES = 795100 Diversity = 2.3833e-02 k_t
= [1, 1, 1, 1]
Generation 7960: Best = 3.346836e+03 FES = 796100 Diversity = 3.3199e-02 k_t
= [2, 4, 4, 4]
Generation 7970: Best = 3.346800e+03 FES = 797100 Diversity = 2.8466e-02 k_t
= [1, 1, 1, 1]
Generation 7980: Best = 3.346768e+03 FES = 798100 Diversity = 2.8388e-02 k_t
= [4, 4, 2, 2]
Generation 7990: Best = 3.346736e+03 FES = 799100 Diversity = 2.9364e-02 k_t
= [1, 1, 1, 1]
Generation 8000: Best = 3.346711e+03 FES = 800100 Diversity = 2.6372e-02 k_t
= [2, 2, 4, 2]
Generation 8010: Best = 3.346677e+03 FES = 801100 Diversity = 2.9655e-02 k_t
= [1, 1, 1, 1]
Generation 8020: Best = 3.346637e+03 FES = 802100 Diversity = 2.7867e-02 k_t
= [3, 2, 2, 4]
Generation 8030: Best = 3.346598e+03 FES = 803100 Diversity = 2.7617e-02 k_t
= [1, 1, 1, 1]
Generation 8040: Best = 3.346562e+03 FES = 804100 Diversity = 2.9499e-02 k_t
= [2, 2, 2, 3]
Generation 8050: Best = 3.346525e+03 FES = 805100 Diversity = 3.0031e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 8060: Best = 3.346490e+03 FES = 806100 Diversity = 3.0553e-02 k_t
= [4, 2, 4, 4]
Generation 8070: Best = 3.346456e+03 FES = 807100 Diversity = 2.6794e-02 k_t
= [1, 1, 1, 1]
Generation 8080: Best = 3.346428e+03 FES = 808100 Diversity = 2.4886e-02 k_t
= [3, 3, 2, 3]
Generation 8090: Best = 3.346406e+03 FES = 809100 Diversity = 2.2781e-02 k_t
= [1, 1, 1, 1]
Generation 8100: Best = 3.346380e+03 FES = 810100 Diversity = 2.2656e-02 k_t
= [4, 4, 2, 3]
Generation 8110: Best = 3.346367e+03 FES = 811100 Diversity = 2.1323e-02 k_t
= [1, 1, 1, 1]
Generation 8120: Best = 3.346353e+03 FES = 812100 Diversity = 2.0163e-02 k_t
= [3, 2, 4, 3]
Generation 8130: Best = 3.346335e+03 FES = 813100 Diversity = 2.1140e-02 k_t
= [1, 1, 1, 1]
Generation 8140: Best = 3.346321e+03 FES = 814100 Diversity = 1.9850e-02 k_t
= [3, 3, 3, 4]
Generation 8150: Best = 3.346296e+03 FES = 815100 Diversity = 2.5340e-02 k_t
= [1, 1, 1, 1]
Generation 8160: Best = 3.346282e+03 FES = 816100 Diversity = 1.9918e-02 k_t
= [3, 3, 2, 3]
Generation 8170: Best = 3.346263e+03 FES = 817100 Diversity = 1.9842e-02 k_t
= [1, 1, 1, 1]
Generation 8180: Best = 3.346249e+03 FES = 818100 Diversity = 1.8703e-02 k_t
= [4, 3, 4, 4]
Generation 8190: Best = 3.346231e+03 FES = 819100 Diversity = 1.8157e-02 k_t
= [1, 1, 1, 1]
Generation 8200: Best = 3.346218e+03 FES = 820100 Diversity = 1.8415e-02 k_t
= [4, 2, 4, 4]
Generation 8210: Best = 3.346201e+03 FES = 821100 Diversity = 1.9456e-02 k_t
= [1, 1, 1, 1]
Generation 8220: Best = 3.346186e+03 FES = 822100 Diversity = 1.9505e-02 k_t
= [2, 3, 2, 2]
Generation 8230: Best = 3.346171e+03 FES = 823100 Diversity = 1.9369e-02 k_t
= [1, 1, 1, 1]
Generation 8240: Best = 3.346149e+03 FES = 824100 Diversity = 2.3037e-02 k_t
= [4, 4, 4, 4]
Generation 8250: Best = 3.346131e+03 FES = 825100 Diversity = 1.8039e-02 k_t
= [1, 1, 1, 1]
Generation 8260: Best = 3.346113e+03 FES = 826100 Diversity = 2.0355e-02 k_t
= [2, 4, 2, 4]
Generation 8270: Best = 3.346091e+03 FES = 827100 Diversity = 2.2909e-02 k_t
= [1, 1, 1, 1]
Generation 8280: Best = 3.346065e+03 FES = 828100 Diversity = 2.5725e-02 k_t
= [4, 3, 3, 3]
Generation 8290: Best = 3.346041e+03 FES = 829100 Diversity = 2.4189e-02 k_t
= [1, 1, 1, 1]
Generation 8300: Best = 3.346024e+03 FES = 830100 Diversity = 2.1785e-02 k_t
= [4, 2, 3, 2]
Generation 8310: Best = 3.346012e+03 FES = 831100 Diversity = 1.8727e-02 k_t
= [1, 1, 1, 1]
Generation 8320: Best = 3.345999e+03 FES = 832100 Diversity = 1.7431e-02 k_t

```

```

= [3, 4, 4, 3]
Generation 8330: Best = 3.345988e+03 FES = 833100 Diversity = 1.8594e-02 k_t
= [1, 1, 1, 1]
Generation 8340: Best = 3.345972e+03 FES = 834100 Diversity = 1.8762e-02 k_t
= [4, 3, 2, 4]
Generation 8350: Best = 3.345959e+03 FES = 835100 Diversity = 1.6516e-02 k_t
= [1, 1, 1, 1]
Generation 8360: Best = 3.345943e+03 FES = 836100 Diversity = 1.9046e-02 k_t
= [3, 3, 4, 4]
Generation 8370: Best = 3.345929e+03 FES = 837100 Diversity = 2.0748e-02 k_t
= [1, 1, 1, 1]
Generation 8380: Best = 3.345914e+03 FES = 838100 Diversity = 2.2532e-02 k_t
= [2, 4, 2, 3]
Generation 8390: Best = 3.345901e+03 FES = 839100 Diversity = 2.0307e-02 k_t
= [1, 1, 1, 1]
Generation 8400: Best = 3.345889e+03 FES = 840100 Diversity = 2.0251e-02 k_t
= [3, 3, 4, 3]
Generation 8410: Best = 3.345880e+03 FES = 841100 Diversity = 1.7396e-02 k_t
= [1, 1, 1, 1]
Generation 8420: Best = 3.345864e+03 FES = 842100 Diversity = 1.8904e-02 k_t
= [4, 2, 2, 3]
Generation 8430: Best = 3.345850e+03 FES = 843100 Diversity = 1.8331e-02 k_t
= [1, 1, 1, 1]
Generation 8440: Best = 3.345835e+03 FES = 844100 Diversity = 1.9270e-02 k_t
= [2, 4, 2, 2]
Generation 8450: Best = 3.345816e+03 FES = 845100 Diversity = 2.1077e-02 k_t
= [1, 1, 1, 1]
Generation 8460: Best = 3.345804e+03 FES = 846100 Diversity = 1.7142e-02 k_t
= [3, 3, 4, 4]
Generation 8470: Best = 3.345795e+03 FES = 847100 Diversity = 1.4969e-02 k_t
= [1, 1, 1, 1]
Generation 8480: Best = 3.345786e+03 FES = 848100 Diversity = 1.5835e-02 k_t
= [2, 2, 3, 3]
Generation 8490: Best = 3.345776e+03 FES = 849100 Diversity = 1.6480e-02 k_t
= [1, 1, 1, 1]
Generation 8500: Best = 3.345769e+03 FES = 850100 Diversity = 1.5844e-02 k_t
= [3, 2, 3, 4]
Generation 8510: Best = 3.345757e+03 FES = 851100 Diversity = 1.6989e-02 k_t
= [1, 1, 1, 1]
Generation 8520: Best = 3.345745e+03 FES = 852100 Diversity = 1.7849e-02 k_t
= [4, 4, 4, 4]
Generation 8530: Best = 3.345731e+03 FES = 853100 Diversity = 1.7680e-02 k_t
= [1, 1, 1, 1]
Generation 8540: Best = 3.345719e+03 FES = 854100 Diversity = 1.7970e-02 k_t
= [4, 3, 4, 2]
Generation 8550: Best = 3.345699e+03 FES = 855100 Diversity = 2.2987e-02 k_t
= [1, 1, 1, 1]
Generation 8560: Best = 3.345680e+03 FES = 856100 Diversity = 2.4216e-02 k_t
= [2, 2, 2, 3]
Generation 8570: Best = 3.345662e+03 FES = 857100 Diversity = 2.0258e-02 k_t
= [1, 1, 1, 1]
Generation 8580: Best = 3.345651e+03 FES = 858100 Diversity = 1.8899e-02 k_t
= [3, 4, 4, 3]
Generation 8590: Best = 3.345639e+03 FES = 859100 Diversity = 1.6377e-02 k_t

```

```

= [1, 1, 1, 1]
Generation 8600: Best = 3.345627e+03 FES = 860100 Diversity = 1.5571e-02 k_t
= [3, 4, 4, 3]
Generation 8610: Best = 3.345613e+03 FES = 861100 Diversity = 1.6388e-02 k_t
= [1, 1, 1, 1]
Generation 8620: Best = 3.345603e+03 FES = 862100 Diversity = 1.4267e-02 k_t
= [4, 2, 4, 3]
Generation 8630: Best = 3.345592e+03 FES = 863100 Diversity = 1.5649e-02 k_t
= [1, 1, 1, 1]
Generation 8640: Best = 3.345583e+03 FES = 864100 Diversity = 1.5245e-02 k_t
= [4, 3, 3, 4]
Generation 8650: Best = 3.345573e+03 FES = 865100 Diversity = 1.4784e-02 k_t
= [1, 1, 1, 1]
Generation 8660: Best = 3.345565e+03 FES = 866100 Diversity = 1.2090e-02 k_t
= [2, 4, 4, 2]
Generation 8670: Best = 3.345557e+03 FES = 867100 Diversity = 1.1568e-02 k_t
= [1, 1, 1, 1]
Generation 8680: Best = 3.345550e+03 FES = 868100 Diversity = 1.2719e-02 k_t
= [2, 3, 4, 2]
Generation 8690: Best = 3.345547e+03 FES = 869100 Diversity = 9.6545e-03 k_t
= [1, 1, 1, 1]
Generation 8700: Best = 3.345543e+03 FES = 870100 Diversity = 8.9214e-03 k_t
= [2, 2, 4, 4]
Generation 8710: Best = 3.345538e+03 FES = 871100 Diversity = 9.6944e-03 k_t
= [1, 1, 1, 1]
Generation 8720: Best = 3.345534e+03 FES = 872100 Diversity = 8.7825e-03 k_t
= [4, 3, 2, 2]
Generation 8730: Best = 3.345529e+03 FES = 873100 Diversity = 1.0484e-02 k_t
= [1, 1, 1, 1]
Generation 8740: Best = 3.345525e+03 FES = 874100 Diversity = 9.3857e-03 k_t
= [2, 3, 3, 2]
Generation 8750: Best = 3.345520e+03 FES = 875100 Diversity = 1.0254e-02 k_t
= [1, 1, 1, 1]
Generation 8760: Best = 3.345516e+03 FES = 876100 Diversity = 1.0064e-02 k_t
= [2, 4, 3, 3]
Generation 8770: Best = 3.345511e+03 FES = 877100 Diversity = 1.1186e-02 k_t
= [1, 1, 1, 1]
Generation 8780: Best = 3.345503e+03 FES = 878100 Diversity = 1.3142e-02 k_t
= [2, 2, 4, 4]
Generation 8790: Best = 3.345493e+03 FES = 879100 Diversity = 1.3714e-02 k_t
= [1, 1, 1, 1]
Generation 8800: Best = 3.345488e+03 FES = 880100 Diversity = 1.2004e-02 k_t
= [3, 2, 4, 4]
Generation 8810: Best = 3.345484e+03 FES = 881100 Diversity = 1.1632e-02 k_t
= [1, 1, 1, 1]
Generation 8820: Best = 3.345480e+03 FES = 882100 Diversity = 1.0659e-02 k_t
= [2, 4, 3, 4]
Generation 8830: Best = 3.345475e+03 FES = 883100 Diversity = 1.1650e-02 k_t
= [1, 1, 1, 1]
Generation 8840: Best = 3.345470e+03 FES = 884100 Diversity = 1.1402e-02 k_t
= [4, 4, 4, 3]
Generation 8850: Best = 3.345466e+03 FES = 885100 Diversity = 1.1614e-02 k_t
= [1, 1, 1, 1]
Generation 8860: Best = 3.345462e+03 FES = 886100 Diversity = 9.8239e-03 k_t

```

```

= [2, 4, 4, 4]
Generation 8870: Best = 3.345456e+03 FES = 887100 Diversity = 1.2178e-02 k_t
= [1, 1, 1, 1]
Generation 8880: Best = 3.345448e+03 FES = 888100 Diversity = 1.4779e-02 k_t
= [4, 2, 2, 4]
Generation 8890: Best = 3.345437e+03 FES = 889100 Diversity = 1.5853e-02 k_t
= [1, 1, 1, 1]
Generation 8900: Best = 3.345430e+03 FES = 890100 Diversity = 1.3497e-02 k_t
= [4, 3, 4, 4]
Generation 8910: Best = 3.345424e+03 FES = 891100 Diversity = 1.1893e-02 k_t
= [1, 1, 1, 1]
Generation 8920: Best = 3.345416e+03 FES = 892100 Diversity = 1.4314e-02 k_t
= [4, 4, 2, 4]
Generation 8930: Best = 3.345408e+03 FES = 893100 Diversity = 1.4218e-02 k_t
= [1, 1, 1, 1]
Generation 8940: Best = 3.345400e+03 FES = 894100 Diversity = 1.4269e-02 k_t
= [3, 4, 3, 4]
Generation 8950: Best = 3.345395e+03 FES = 895100 Diversity = 1.3070e-02 k_t
= [1, 1, 1, 1]
Generation 8960: Best = 3.345390e+03 FES = 896100 Diversity = 1.1002e-02 k_t
= [3, 2, 2, 2]
Generation 8970: Best = 3.345385e+03 FES = 897100 Diversity = 1.2876e-02 k_t
= [1, 1, 1, 1]
Generation 8980: Best = 3.345380e+03 FES = 898100 Diversity = 1.2190e-02 k_t
= [4, 4, 2, 4]
Generation 8990: Best = 3.345374e+03 FES = 899100 Diversity = 1.1297e-02 k_t
= [1, 1, 1, 1]
Generation 9000: Best = 3.345372e+03 FES = 900100 Diversity = 8.4486e-03 k_t
= [2, 3, 2, 2]
Generation 9010: Best = 3.345368e+03 FES = 901100 Diversity = 8.8099e-03 k_t
= [1, 1, 1, 1]
Generation 9020: Best = 3.345364e+03 FES = 902100 Diversity = 9.4931e-03 k_t
= [2, 4, 4, 3]
Generation 9030: Best = 3.345360e+03 FES = 903100 Diversity = 9.7363e-03 k_t
= [1, 1, 1, 1]
Generation 9040: Best = 3.345357e+03 FES = 904100 Diversity = 8.2439e-03 k_t
= [3, 4, 3, 3]
Generation 9050: Best = 3.345352e+03 FES = 905100 Diversity = 1.0239e-02 k_t
= [1, 1, 1, 1]
Generation 9060: Best = 3.345346e+03 FES = 906100 Diversity = 1.1411e-02 k_t
= [3, 2, 3, 4]
Generation 9070: Best = 3.345341e+03 FES = 907100 Diversity = 1.1074e-02 k_t
= [1, 1, 1, 1]
Generation 9080: Best = 3.345337e+03 FES = 908100 Diversity = 9.2888e-03 k_t
= [2, 3, 4, 3]
Generation 9090: Best = 3.345334e+03 FES = 909100 Diversity = 8.9408e-03 k_t
= [1, 1, 1, 1]
Generation 9100: Best = 3.345330e+03 FES = 910100 Diversity = 8.8078e-03 k_t
= [2, 4, 2, 2]
Generation 9110: Best = 3.345328e+03 FES = 911100 Diversity = 8.3499e-03 k_t
= [1, 1, 1, 1]
Generation 9120: Best = 3.345326e+03 FES = 912100 Diversity = 6.1894e-03 k_t
= [2, 2, 3, 2]
Generation 9130: Best = 3.345324e+03 FES = 913100 Diversity = 6.3267e-03 k_t

```



```

= [1, 1, 1, 1]
Generation 9140: Best = 3.345322e+03 FES = 914100 Diversity = 6.5276e-03 k_t
= [4, 4, 3, 4]
Generation 9150: Best = 3.345319e+03 FES = 915100 Diversity = 7.8084e-03 k_t
= [1, 1, 1, 1]
Generation 9160: Best = 3.345317e+03 FES = 916100 Diversity = 7.7366e-03 k_t
= [3, 2, 4, 4]
Generation 9170: Best = 3.345314e+03 FES = 917100 Diversity = 7.2797e-03 k_t
= [1, 1, 1, 1]
Generation 9180: Best = 3.345312e+03 FES = 918100 Diversity = 7.7449e-03 k_t
= [4, 2, 4, 4]
Generation 9190: Best = 3.345309e+03 FES = 919100 Diversity = 7.5665e-03 k_t
= [1, 1, 1, 1]
Generation 9200: Best = 3.345307e+03 FES = 920100 Diversity = 6.3991e-03 k_t
= [2, 4, 3, 2]
Generation 9210: Best = 3.345305e+03 FES = 921100 Diversity = 7.3981e-03 k_t
= [1, 1, 1, 1]
Generation 9220: Best = 3.345302e+03 FES = 922100 Diversity = 8.1338e-03 k_t
= [2, 4, 2, 4]
Generation 9230: Best = 3.345297e+03 FES = 923100 Diversity = 1.1227e-02 k_t
= [1, 1, 1, 1]
Generation 9240: Best = 3.345292e+03 FES = 924100 Diversity = 1.1613e-02 k_t
= [2, 4, 3, 4]
Generation 9250: Best = 3.345286e+03 FES = 925100 Diversity = 1.1230e-02 k_t
= [1, 1, 1, 1]
Generation 9260: Best = 3.345280e+03 FES = 926100 Diversity = 1.0619e-02 k_t
= [3, 2, 3, 2]
Generation 9270: Best = 3.345273e+03 FES = 927100 Diversity = 1.2972e-02 k_t
= [1, 1, 1, 1]
Generation 9280: Best = 3.345265e+03 FES = 928100 Diversity = 1.1860e-02 k_t
= [3, 4, 2, 4]
Generation 9290: Best = 3.345259e+03 FES = 929100 Diversity = 1.0238e-02 k_t
= [1, 1, 1, 1]
Generation 9300: Best = 3.345254e+03 FES = 930100 Diversity = 1.2491e-02 k_t
= [3, 2, 2, 4]
Generation 9310: Best = 3.345248e+03 FES = 931100 Diversity = 1.0709e-02 k_t
= [1, 1, 1, 1]
Generation 9320: Best = 3.345242e+03 FES = 932100 Diversity = 1.1971e-02 k_t
= [2, 4, 3, 4]
Generation 9330: Best = 3.345238e+03 FES = 933100 Diversity = 1.0334e-02 k_t
= [1, 1, 1, 1]
Generation 9340: Best = 3.345234e+03 FES = 934100 Diversity = 8.9161e-03 k_t
= [4, 3, 2, 2]
Generation 9350: Best = 3.345231e+03 FES = 935100 Diversity = 8.8863e-03 k_t
= [1, 1, 1, 1]
Generation 9360: Best = 3.345226e+03 FES = 936100 Diversity = 1.0075e-02 k_t
= [3, 2, 4, 4]
Generation 9370: Best = 3.345223e+03 FES = 937100 Diversity = 1.0366e-02 k_t
= [1, 1, 1, 1]
Generation 9380: Best = 3.345219e+03 FES = 938100 Diversity = 8.2581e-03 k_t
= [4, 2, 2, 4]
Generation 9390: Best = 3.345216e+03 FES = 939100 Diversity = 8.1469e-03 k_t
= [1, 1, 1, 1]
Generation 9400: Best = 3.345211e+03 FES = 940100 Diversity = 1.0335e-02 k_t

```

```

= [4, 4, 3, 3]
Generation 9410: Best = 3.345207e+03 FES = 941100 Diversity = 9.0588e-03 k_t
= [1, 1, 1, 1]
Generation 9420: Best = 3.345204e+03 FES = 942100 Diversity = 8.9944e-03 k_t
= [3, 3, 4, 4]
Generation 9430: Best = 3.345200e+03 FES = 943100 Diversity = 9.7650e-03 k_t
= [1, 1, 1, 1]
Generation 9440: Best = 3.345195e+03 FES = 944100 Diversity = 1.0521e-02 k_t
= [3, 3, 3, 3]
Generation 9450: Best = 3.345190e+03 FES = 945100 Diversity = 9.7190e-03 k_t
= [1, 1, 1, 1]
Generation 9460: Best = 3.345186e+03 FES = 946100 Diversity = 8.9397e-03 k_t
= [4, 3, 3, 3]
Generation 9470: Best = 3.345182e+03 FES = 947100 Diversity = 9.3440e-03 k_t
= [1, 1, 1, 1]
Generation 9480: Best = 3.345178e+03 FES = 948100 Diversity = 9.4314e-03 k_t
= [3, 2, 4, 2]
Generation 9490: Best = 3.345175e+03 FES = 949100 Diversity = 9.0511e-03 k_t
= [1, 1, 1, 1]
Generation 9500: Best = 3.345170e+03 FES = 950100 Diversity = 8.5989e-03 k_t
= [3, 3, 2, 2]
Generation 9510: Best = 3.345166e+03 FES = 951100 Diversity = 9.8541e-03 k_t
= [1, 1, 1, 1]
Generation 9520: Best = 3.345163e+03 FES = 952100 Diversity = 9.5507e-03 k_t
= [4, 4, 3, 2]
Generation 9530: Best = 3.345160e+03 FES = 953100 Diversity = 9.9382e-03 k_t
= [1, 1, 1, 1]
Generation 9540: Best = 3.345156e+03 FES = 954100 Diversity = 9.6253e-03 k_t
= [3, 3, 2, 2]
Generation 9550: Best = 3.345154e+03 FES = 955100 Diversity = 7.5706e-03 k_t
= [1, 1, 1, 1]
Generation 9560: Best = 3.345151e+03 FES = 956100 Diversity = 8.2457e-03 k_t
= [3, 3, 3, 2]
Generation 9570: Best = 3.345148e+03 FES = 957100 Diversity = 8.5052e-03 k_t
= [1, 1, 1, 1]
Generation 9580: Best = 3.345145e+03 FES = 958100 Diversity = 7.6708e-03 k_t
= [3, 2, 4, 2]
Generation 9590: Best = 3.345143e+03 FES = 959100 Diversity = 7.1000e-03 k_t
= [1, 1, 1, 1]
Generation 9600: Best = 3.345140e+03 FES = 960100 Diversity = 7.5416e-03 k_t
= [3, 2, 4, 3]
Generation 9610: Best = 3.345138e+03 FES = 961100 Diversity = 6.8105e-03 k_t
= [1, 1, 1, 1]
Generation 9620: Best = 3.345135e+03 FES = 962100 Diversity = 7.9334e-03 k_t
= [2, 2, 4, 3]
Generation 9630: Best = 3.345133e+03 FES = 963100 Diversity = 7.2090e-03 k_t
= [1, 1, 1, 1]
Generation 9640: Best = 3.345129e+03 FES = 964100 Diversity = 9.0196e-03 k_t
= [3, 4, 4, 3]
Generation 9650: Best = 3.345127e+03 FES = 965100 Diversity = 6.6825e-03 k_t
= [1, 1, 1, 1]
Generation 9660: Best = 3.345125e+03 FES = 966100 Diversity = 6.7584e-03 k_t
= [4, 3, 4, 2]
Generation 9670: Best = 3.345123e+03 FES = 967100 Diversity = 7.4131e-03 k_t

```

```

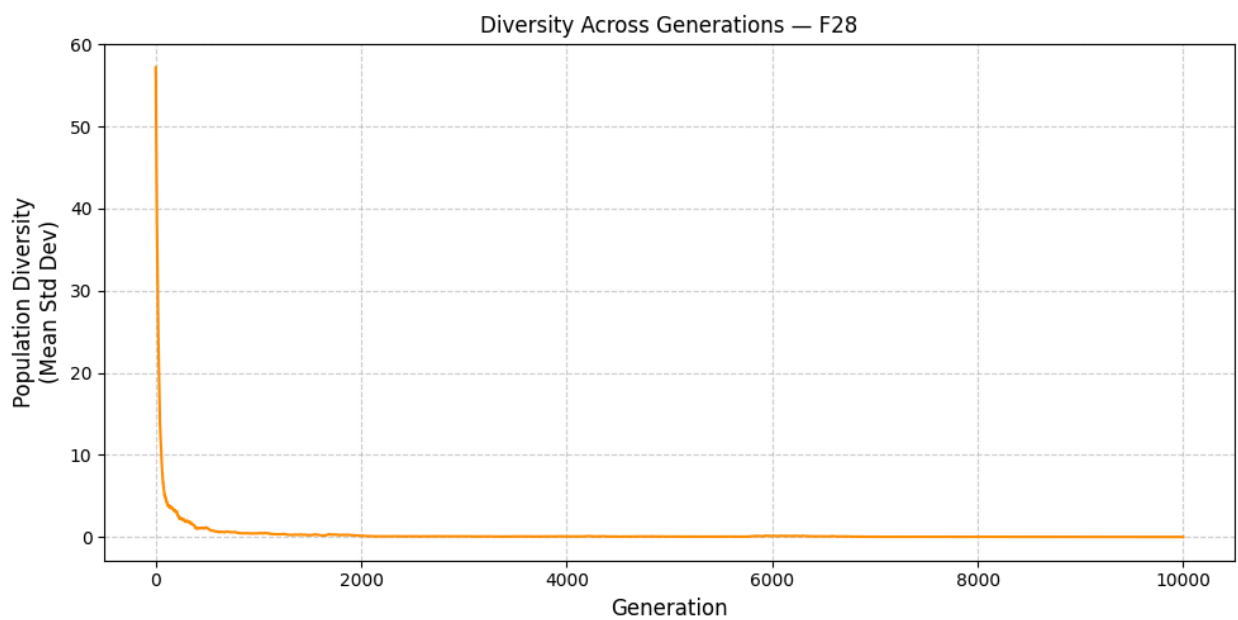
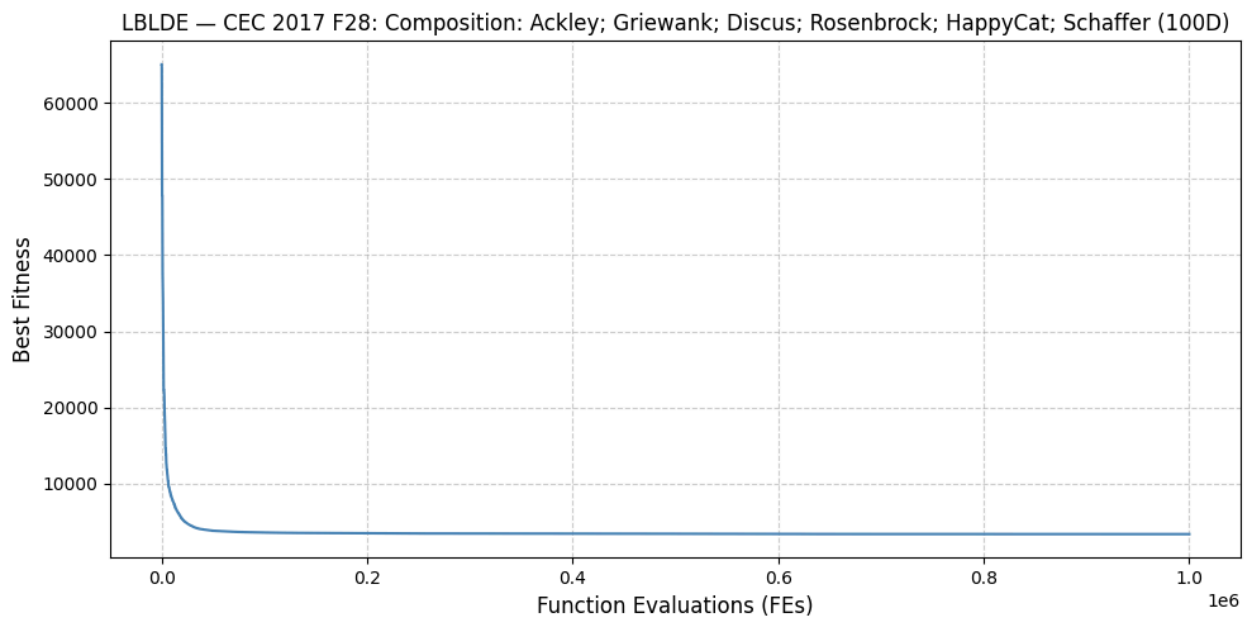
= [1, 1, 1, 1]
Generation 9680: Best = 3.345121e+03 FES = 968100 Diversity = 6.3571e-03 k_t
= [4, 3, 4, 2]
Generation 9690: Best = 3.345120e+03 FES = 969100 Diversity = 6.1245e-03 k_t
= [1, 1, 1, 1]
Generation 9700: Best = 3.345118e+03 FES = 970100 Diversity = 6.3280e-03 k_t
= [4, 3, 3, 2]
Generation 9710: Best = 3.345116e+03 FES = 971100 Diversity = 6.6468e-03 k_t
= [1, 1, 1, 1]
Generation 9720: Best = 3.345114e+03 FES = 972100 Diversity = 6.9535e-03 k_t
= [3, 3, 4, 2]
Generation 9730: Best = 3.345112e+03 FES = 973100 Diversity = 6.7146e-03 k_t
= [1, 1, 1, 1]
Generation 9740: Best = 3.345110e+03 FES = 974100 Diversity = 6.2867e-03 k_t
= [3, 4, 2, 4]
Generation 9750: Best = 3.345109e+03 FES = 975100 Diversity = 6.0710e-03 k_t
= [1, 1, 1, 1]
Generation 9760: Best = 3.345107e+03 FES = 976100 Diversity = 5.7942e-03 k_t
= [3, 2, 2, 2]
Generation 9770: Best = 3.345105e+03 FES = 977100 Diversity = 6.8036e-03 k_t
= [1, 1, 1, 1]
Generation 9780: Best = 3.345103e+03 FES = 978100 Diversity = 6.8889e-03 k_t
= [2, 2, 2, 2]
Generation 9790: Best = 3.345102e+03 FES = 979100 Diversity = 5.2092e-03 k_t
= [1, 1, 1, 1]
Generation 9800: Best = 3.345100e+03 FES = 980100 Diversity = 5.1454e-03 k_t
= [3, 3, 4, 2]
Generation 9810: Best = 3.345099e+03 FES = 981100 Diversity = 5.1052e-03 k_t
= [1, 1, 1, 1]
Generation 9820: Best = 3.345097e+03 FES = 982100 Diversity = 5.8114e-03 k_t
= [2, 2, 3, 4]
Generation 9830: Best = 3.345095e+03 FES = 983100 Diversity = 6.0700e-03 k_t
= [1, 1, 1, 1]
Generation 9840: Best = 3.345094e+03 FES = 984100 Diversity = 5.2579e-03 k_t
= [4, 2, 3, 3]
Generation 9850: Best = 3.345093e+03 FES = 985100 Diversity = 5.5125e-03 k_t
= [1, 1, 1, 1]
Generation 9860: Best = 3.345091e+03 FES = 986100 Diversity = 6.3909e-03 k_t
= [4, 2, 2, 2]
Generation 9870: Best = 3.345090e+03 FES = 987100 Diversity = 6.6622e-03 k_t
= [1, 1, 1, 1]
Generation 9880: Best = 3.345088e+03 FES = 988100 Diversity = 6.0355e-03 k_t
= [3, 2, 3, 2]
Generation 9890: Best = 3.345087e+03 FES = 989100 Diversity = 4.6745e-03 k_t
= [1, 1, 1, 1]
Generation 9900: Best = 3.345086e+03 FES = 990100 Diversity = 4.7841e-03 k_t
= [4, 3, 4, 3]
Generation 9910: Best = 3.345085e+03 FES = 991100 Diversity = 4.2599e-03 k_t
= [1, 1, 1, 1]
Generation 9920: Best = 3.345084e+03 FES = 992100 Diversity = 4.0518e-03 k_t
= [4, 3, 2, 4]
Generation 9930: Best = 3.345083e+03 FES = 993100 Diversity = 4.4836e-03 k_t
= [1, 1, 1, 1]
Generation 9940: Best = 3.345082e+03 FES = 994100 Diversity = 4.4381e-03 k_t

```

```
= [4, 2, 4, 3]
Generation 9950: Best = 3.345081e+03  FES = 995100  Diversity = 5.4153e-03  k_t
= [1, 1, 1, 1]
Generation 9960: Best = 3.345080e+03  FES = 996100  Diversity = 6.0177e-03  k_t
= [3, 4, 2, 2]
Generation 9970: Best = 3.345078e+03  FES = 997100  Diversity = 6.2782e-03  k_t
= [1, 1, 1, 1]
Generation 9980: Best = 3.345077e+03  FES = 998100  Diversity = 5.6702e-03  k_t
= [4, 2, 3, 3]
Generation 9990: Best = 3.345076e+03  FES = 999100  Diversity = 5.8651e-03  k_t
= [1, 1, 1, 1]
```

```
Optimization completed!
Final Best Fitness = 3.345075e+03
Total Function Evaluations = 1000000
```

```
=====
=
Results:
  Best Fitness          : 3.345075e+03
  Global Optimum        : 2.800000e+03
  Error (|best-optimum|) : 5.450748e+02
  Best Solution (5 dim) : [-61.87071001 -26.30407083 -40.93427455  41.89852526
-51.23502276]
  D^(0) (init. diversity): 5.7220e+01
=====
=
```



Graph saved as convergence_F28.png

```
In [53]: # =====
# F29 — Composition Function 9 (F15; F16; F17)
# =====
run_and_plot('F29', make_wrapper(composition.f29), 2900.0, 'Composition: F15;
```

```

=====
=
LBLDE_s on CEC 2017 F29 – Single Run
=====
=
Function      : Composition: F15; F16; F17 (100D)
Bounds       : [-100, 100]
Global optimum : 2900.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 1.114421e+07 | Diversity = 5.7220e+01
Generation 10: Best = 4.152187e+04 FES = 1100 Diversity = 4.5704e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 1.709025e+04 FES = 2100 Diversity = 3.8098e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 1.447384e+04 FES = 3100 Diversity = 3.2043e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 1.295081e+04 FES = 4100 Diversity = 2.7244e+01 k_t
= [2, 3, 4, 3]
Generation 50: Best = 1.207224e+04 FES = 5100 Diversity = 2.5408e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 1.157280e+04 FES = 6100 Diversity = 2.4069e+01 k_t
= [3, 4, 2, 3]
Generation 70: Best = 1.121522e+04 FES = 7100 Diversity = 2.3388e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.121522e+04 FES = 8100 Diversity = 2.3149e+01 k_t
= [4, 3, 2, 3]
Generation 90: Best = 1.121522e+04 FES = 9100 Diversity = 2.2698e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 1.114359e+04 FES = 10100 Diversity = 2.2602e+01 k_t
= [2, 4, 4, 2]
Generation 110: Best = 1.066425e+04 FES = 11100 Diversity = 2.2540e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 1.057616e+04 FES = 12100 Diversity = 2.2476e+01 k_t
= [3, 3, 4, 4]
Generation 130: Best = 1.057072e+04 FES = 13100 Diversity = 2.2538e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.038627e+04 FES = 14100 Diversity = 2.2765e+01 k_t
= [3, 4, 2, 3]
Generation 150: Best = 1.022154e+04 FES = 15100 Diversity = 2.2981e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 1.021502e+04 FES = 16100 Diversity = 2.3101e+01 k_t
= [3, 4, 2, 2]
Generation 170: Best = 9.747807e+03 FES = 17100 Diversity = 2.2905e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 9.747807e+03 FES = 18100 Diversity = 2.3367e+01 k_t
= [3, 3, 3, 3]
Generation 190: Best = 9.747807e+03 FES = 19100 Diversity = 2.3465e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 9.605481e+03 FES = 20100 Diversity = 2.3713e+01 k_t
= [2, 3, 3, 4]
Generation 210: Best = 9.557161e+03 FES = 21100 Diversity = 2.3769e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 9.474034e+03 FES = 22100 Diversity = 2.3769e+01 k_t
= [2, 2, 4, 4]

```

Generation 230:	Best = 8.937619e+03	FES = 23100	Diversity = 2.3763e+01	k_t = [1, 1, 1, 1]
Generation 240:	Best = 8.871076e+03	FES = 24100	Diversity = 2.3987e+01	k_t = [2, 2, 4, 4]
Generation 250:	Best = 8.822732e+03	FES = 25100	Diversity = 2.3989e+01	k_t = [1, 1, 1, 1]
Generation 260:	Best = 8.822732e+03	FES = 26100	Diversity = 2.4070e+01	k_t = [3, 4, 4, 4]
Generation 270:	Best = 8.725964e+03	FES = 27100	Diversity = 2.4109e+01	k_t = [1, 1, 1, 1]
Generation 280:	Best = 8.725964e+03	FES = 28100	Diversity = 2.4183e+01	k_t = [4, 3, 3, 3]
Generation 290:	Best = 8.725964e+03	FES = 29100	Diversity = 2.4370e+01	k_t = [1, 1, 1, 1]
Generation 300:	Best = 8.668760e+03	FES = 30100	Diversity = 2.4327e+01	k_t = [2, 2, 4, 2]
Generation 310:	Best = 8.595559e+03	FES = 31100	Diversity = 2.4284e+01	k_t = [1, 1, 1, 1]
Generation 320:	Best = 8.595559e+03	FES = 32100	Diversity = 2.4393e+01	k_t = [3, 2, 4, 4]
Generation 330:	Best = 8.595559e+03	FES = 33100	Diversity = 2.4493e+01	k_t = [1, 1, 1, 1]
Generation 340:	Best = 8.595559e+03	FES = 34100	Diversity = 2.4558e+01	k_t = [2, 2, 4, 4]
Generation 350:	Best = 8.306460e+03	FES = 35100	Diversity = 2.4491e+01	k_t = [1, 1, 1, 1]
Generation 360:	Best = 8.263849e+03	FES = 36100	Diversity = 2.4515e+01	k_t = [4, 4, 3, 2]
Generation 370:	Best = 8.257929e+03	FES = 37100	Diversity = 2.4649e+01	k_t = [1, 1, 1, 1]
Generation 380:	Best = 7.998902e+03	FES = 38100	Diversity = 2.4919e+01	k_t = [4, 3, 3, 3]
Generation 390:	Best = 7.990981e+03	FES = 39100	Diversity = 2.5013e+01	k_t = [1, 1, 1, 1]
Generation 400:	Best = 7.990981e+03	FES = 40100	Diversity = 2.4957e+01	k_t = [4, 2, 3, 4]
Generation 410:	Best = 7.919046e+03	FES = 41100	Diversity = 2.4866e+01	k_t = [1, 1, 1, 1]
Generation 420:	Best = 7.919046e+03	FES = 42100	Diversity = 2.4876e+01	k_t = [3, 4, 2, 3]
Generation 430:	Best = 7.892051e+03	FES = 43100	Diversity = 2.4957e+01	k_t = [1, 1, 1, 1]
Generation 440:	Best = 7.748430e+03	FES = 44100	Diversity = 2.4979e+01	k_t = [3, 2, 3, 3]
Generation 450:	Best = 7.740117e+03	FES = 45100	Diversity = 2.4927e+01	k_t = [1, 1, 1, 1]
Generation 460:	Best = 7.740117e+03	FES = 46100	Diversity = 2.4883e+01	k_t = [4, 4, 3, 4]
Generation 470:	Best = 7.595759e+03	FES = 47100	Diversity = 2.4928e+01	k_t = [1, 1, 1, 1]
Generation 480:	Best = 7.595759e+03	FES = 48100	Diversity = 2.4834e+01	k_t = [3, 2, 3, 2]
Generation 490:	Best = 7.334763e+03	FES = 49100	Diversity = 2.4900e+01	k_t = [1, 1, 1, 1]

Generation 500:	Best = 7.326901e+03	FES = 50100	Diversity = 2.4841e+01	k_t
= [2, 3, 2, 2]				
Generation 510:	Best = 7.326901e+03	FES = 51100	Diversity = 2.4857e+01	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 7.326901e+03	FES = 52100	Diversity = 2.4817e+01	k_t
= [4, 2, 3, 4]				
Generation 530:	Best = 7.326901e+03	FES = 53100	Diversity = 2.4784e+01	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 7.326901e+03	FES = 54100	Diversity = 2.4642e+01	k_t
= [4, 4, 4, 4]				
Generation 550:	Best = 7.326901e+03	FES = 55100	Diversity = 2.4680e+01	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 7.299687e+03	FES = 56100	Diversity = 2.4621e+01	k_t
= [4, 3, 2, 3]				
Generation 570:	Best = 7.251676e+03	FES = 57100	Diversity = 2.4611e+01	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 7.245485e+03	FES = 58100	Diversity = 2.4716e+01	k_t
= [2, 2, 2, 3]				
Generation 590:	Best = 7.245485e+03	FES = 59100	Diversity = 2.4583e+01	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 7.245485e+03	FES = 60100	Diversity = 2.4534e+01	k_t
= [2, 2, 3, 4]				
Generation 610:	Best = 7.245485e+03	FES = 61100	Diversity = 2.4407e+01	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 7.245485e+03	FES = 62100	Diversity = 2.4518e+01	k_t
= [3, 3, 3, 3]				
Generation 630:	Best = 7.245485e+03	FES = 63100	Diversity = 2.4372e+01	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 7.215983e+03	FES = 64100	Diversity = 2.4282e+01	k_t
= [2, 2, 2, 4]				
Generation 650:	Best = 7.215983e+03	FES = 65100	Diversity = 2.4355e+01	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 7.215983e+03	FES = 66100	Diversity = 2.4402e+01	k_t
= [2, 2, 4, 3]				
Generation 670:	Best = 7.215983e+03	FES = 67100	Diversity = 2.4380e+01	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 7.189416e+03	FES = 68100	Diversity = 2.4362e+01	k_t
= [4, 4, 2, 3]				
Generation 690:	Best = 7.189416e+03	FES = 69100	Diversity = 2.4354e+01	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 7.189416e+03	FES = 70100	Diversity = 2.4376e+01	k_t
= [2, 4, 4, 2]				
Generation 710:	Best = 7.127150e+03	FES = 71100	Diversity = 2.4306e+01	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 7.127150e+03	FES = 72100	Diversity = 2.4247e+01	k_t
= [2, 2, 3, 3]				
Generation 730:	Best = 7.127150e+03	FES = 73100	Diversity = 2.4309e+01	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 6.967830e+03	FES = 74100	Diversity = 2.4268e+01	k_t
= [3, 4, 3, 3]				
Generation 750:	Best = 6.967830e+03	FES = 75100	Diversity = 2.4280e+01	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 6.967830e+03	FES = 76100	Diversity = 2.4199e+01	k_t
= [4, 2, 4, 4]				

Generation 770:	Best = 6.912361e+03	FES = 77100	Diversity = 2.4113e+01	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 6.905783e+03	FES = 78100	Diversity = 2.4152e+01	k_t
= [3, 4, 2, 2]				
Generation 790:	Best = 6.905783e+03	FES = 79100	Diversity = 2.4070e+01	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 6.905783e+03	FES = 80100	Diversity = 2.4004e+01	k_t
= [4, 4, 3, 3]				
Generation 810:	Best = 6.905783e+03	FES = 81100	Diversity = 2.3927e+01	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 6.905783e+03	FES = 82100	Diversity = 2.3803e+01	k_t
= [4, 2, 3, 3]				
Generation 830:	Best = 6.897360e+03	FES = 83100	Diversity = 2.3796e+01	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 6.897360e+03	FES = 84100	Diversity = 2.3788e+01	k_t
= [4, 4, 4, 2]				
Generation 850:	Best = 6.878413e+03	FES = 85100	Diversity = 2.3784e+01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 6.878413e+03	FES = 86100	Diversity = 2.3695e+01	k_t
= [4, 4, 4, 3]				
Generation 870:	Best = 6.728486e+03	FES = 87100	Diversity = 2.3749e+01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 6.724750e+03	FES = 88100	Diversity = 2.3776e+01	k_t
= [3, 3, 4, 4]				
Generation 890:	Best = 6.696327e+03	FES = 89100	Diversity = 2.3920e+01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 6.561197e+03	FES = 90100	Diversity = 2.4077e+01	k_t
= [4, 4, 4, 3]				
Generation 910:	Best = 6.539923e+03	FES = 91100	Diversity = 2.3936e+01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 6.539923e+03	FES = 92100	Diversity = 2.3742e+01	k_t
= [2, 2, 4, 2]				
Generation 930:	Best = 6.539923e+03	FES = 93100	Diversity = 2.4033e+01	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 6.539923e+03	FES = 94100	Diversity = 2.4180e+01	k_t
= [4, 3, 3, 3]				
Generation 950:	Best = 6.480384e+03	FES = 95100	Diversity = 2.4154e+01	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 6.428224e+03	FES = 96100	Diversity = 2.3908e+01	k_t
= [2, 3, 4, 2]				
Generation 970:	Best = 6.342514e+03	FES = 97100	Diversity = 2.3934e+01	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 6.341054e+03	FES = 98100	Diversity = 2.3852e+01	k_t
= [4, 2, 2, 3]				
Generation 990:	Best = 6.341054e+03	FES = 99100	Diversity = 2.3664e+01	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 6.132958e+03	FES = 100100	Diversity = 2.3840e+01	k_t
= [2, 2, 2, 3]				
Generation 1010:	Best = 6.132958e+03	FES = 101100	Diversity = 2.3789e+01	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 6.069539e+03	FES = 102100	Diversity = 2.3644e+01	k_t
= [3, 2, 2, 4]				
Generation 1030:	Best = 5.904832e+03	FES = 103100	Diversity = 2.3302e+01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 5.858883e+03 = [3, 2, 2, 3]	FES = 104100	Diversity = 2.3118e+01	k_t
Generation 1050: Best = 5.858883e+03 = [1, 1, 1, 1]	FES = 105100	Diversity = 2.3107e+01	k_t
Generation 1060: Best = 5.707396e+03 = [3, 2, 3, 4]	FES = 106100	Diversity = 2.2950e+01	k_t
Generation 1070: Best = 5.707396e+03 = [1, 1, 1, 1]	FES = 107100	Diversity = 2.3092e+01	k_t
Generation 1080: Best = 5.648277e+03 = [2, 4, 3, 2]	FES = 108100	Diversity = 2.2844e+01	k_t
Generation 1090: Best = 5.606277e+03 = [1, 1, 1, 1]	FES = 109100	Diversity = 2.2403e+01	k_t
Generation 1100: Best = 5.504139e+03 = [2, 2, 3, 3]	FES = 110100	Diversity = 2.1396e+01	k_t
Generation 1110: Best = 5.492981e+03 = [1, 1, 1, 1]	FES = 111100	Diversity = 2.0207e+01	k_t
Generation 1120: Best = 5.432056e+03 = [2, 2, 2, 2]	FES = 112100	Diversity = 1.8637e+01	k_t
Generation 1130: Best = 5.412870e+03 = [1, 1, 1, 1]	FES = 113100	Diversity = 1.7515e+01	k_t
Generation 1140: Best = 5.380858e+03 = [4, 4, 4, 4]	FES = 114100	Diversity = 1.6331e+01	k_t
Generation 1150: Best = 5.277346e+03 = [1, 1, 1, 1]	FES = 115100	Diversity = 1.4462e+01	k_t
Generation 1160: Best = 5.237666e+03 = [4, 3, 2, 2]	FES = 116100	Diversity = 1.3850e+01	k_t
Generation 1170: Best = 5.215147e+03 = [1, 1, 1, 1]	FES = 117100	Diversity = 1.3312e+01	k_t
Generation 1180: Best = 5.193682e+03 = [2, 2, 4, 3]	FES = 118100	Diversity = 1.3008e+01	k_t
Generation 1190: Best = 5.193682e+03 = [1, 1, 1, 1]	FES = 119100	Diversity = 1.2004e+01	k_t
Generation 1200: Best = 5.153070e+03 = [4, 3, 3, 2]	FES = 120100	Diversity = 1.0524e+01	k_t
Generation 1210: Best = 5.153070e+03 = [1, 1, 1, 1]	FES = 121100	Diversity = 1.0255e+01	k_t
Generation 1220: Best = 5.073715e+03 = [3, 4, 2, 2]	FES = 122100	Diversity = 9.6397e+00	k_t
Generation 1230: Best = 5.073715e+03 = [1, 1, 1, 1]	FES = 123100	Diversity = 9.4288e+00	k_t
Generation 1240: Best = 5.073715e+03 = [3, 2, 4, 3]	FES = 124100	Diversity = 9.0555e+00	k_t
Generation 1250: Best = 5.073715e+03 = [1, 1, 1, 1]	FES = 125100	Diversity = 8.9840e+00	k_t
Generation 1260: Best = 5.073715e+03 = [4, 2, 4, 3]	FES = 126100	Diversity = 8.8331e+00	k_t
Generation 1270: Best = 5.069600e+03 = [1, 1, 1, 1]	FES = 127100	Diversity = 8.6788e+00	k_t
Generation 1280: Best = 5.047107e+03 = [4, 3, 3, 4]	FES = 128100	Diversity = 8.5289e+00	k_t
Generation 1290: Best = 5.040051e+03 = [1, 1, 1, 1]	FES = 129100	Diversity = 8.4052e+00	k_t
Generation 1300: Best = 5.040051e+03 = [2, 2, 3, 3]	FES = 130100	Diversity = 8.2793e+00	k_t

Generation 1310: Best = 5.034232e+03 = [1, 1, 1, 1]	FES = 131100	Diversity = 8.1571e+00	k_t
Generation 1320: Best = 5.021193e+03 = [4, 2, 2, 4]	FES = 132100	Diversity = 7.9941e+00	k_t
Generation 1330: Best = 4.996490e+03 = [1, 1, 1, 1]	FES = 133100	Diversity = 7.9125e+00	k_t
Generation 1340: Best = 4.996490e+03 = [3, 2, 3, 3]	FES = 134100	Diversity = 7.8636e+00	k_t
Generation 1350: Best = 4.984109e+03 = [1, 1, 1, 1]	FES = 135100	Diversity = 7.8190e+00	k_t
Generation 1360: Best = 4.983906e+03 = [3, 2, 2, 3]	FES = 136100	Diversity = 7.6379e+00	k_t
Generation 1370: Best = 4.976587e+03 = [1, 1, 1, 1]	FES = 137100	Diversity = 7.4461e+00	k_t
Generation 1380: Best = 4.976587e+03 = [3, 4, 4, 2]	FES = 138100	Diversity = 7.3271e+00	k_t
Generation 1390: Best = 4.976373e+03 = [1, 1, 1, 1]	FES = 139100	Diversity = 7.2959e+00	k_t
Generation 1400: Best = 4.970721e+03 = [2, 3, 2, 3]	FES = 140100	Diversity = 7.2398e+00	k_t
Generation 1410: Best = 4.967412e+03 = [1, 1, 1, 1]	FES = 141100	Diversity = 7.1335e+00	k_t
Generation 1420: Best = 4.958836e+03 = [2, 2, 2, 2]	FES = 142100	Diversity = 7.0252e+00	k_t
Generation 1430: Best = 4.954034e+03 = [1, 1, 1, 1]	FES = 143100	Diversity = 6.9476e+00	k_t
Generation 1440: Best = 4.954034e+03 = [2, 4, 2, 2]	FES = 144100	Diversity = 6.9108e+00	k_t
Generation 1450: Best = 4.954034e+03 = [1, 1, 1, 1]	FES = 145100	Diversity = 6.9060e+00	k_t
Generation 1460: Best = 4.952968e+03 = [3, 4, 4, 4]	FES = 146100	Diversity = 6.8932e+00	k_t
Generation 1470: Best = 4.950700e+03 = [1, 1, 1, 1]	FES = 147100	Diversity = 6.8788e+00	k_t
Generation 1480: Best = 4.931790e+03 = [4, 2, 2, 4]	FES = 148100	Diversity = 6.8593e+00	k_t
Generation 1490: Best = 4.930881e+03 = [1, 1, 1, 1]	FES = 149100	Diversity = 6.7039e+00	k_t
Generation 1500: Best = 4.920530e+03 = [2, 4, 3, 3]	FES = 150100	Diversity = 6.6702e+00	k_t
Generation 1510: Best = 4.912738e+03 = [1, 1, 1, 1]	FES = 151100	Diversity = 6.5993e+00	k_t
Generation 1520: Best = 4.912738e+03 = [4, 3, 3, 2]	FES = 152100	Diversity = 6.5770e+00	k_t
Generation 1530: Best = 4.912738e+03 = [1, 1, 1, 1]	FES = 153100	Diversity = 6.5455e+00	k_t
Generation 1540: Best = 4.908095e+03 = [4, 2, 2, 3]	FES = 154100	Diversity = 6.5505e+00	k_t
Generation 1550: Best = 4.908095e+03 = [1, 1, 1, 1]	FES = 155100	Diversity = 6.5533e+00	k_t
Generation 1560: Best = 4.908095e+03 = [3, 3, 2, 2]	FES = 156100	Diversity = 6.5460e+00	k_t
Generation 1570: Best = 4.908095e+03 = [1, 1, 1, 1]	FES = 157100	Diversity = 6.5644e+00	k_t

Generation 1580: Best = 4.908095e+03 = [4, 4, 2, 4]	FES = 158100	Diversity = 6.5510e+00	k_t
Generation 1590: Best = 4.903949e+03 = [1, 1, 1, 1]	FES = 159100	Diversity = 6.4956e+00	k_t
Generation 1600: Best = 4.903949e+03 = [4, 3, 3, 3]	FES = 160100	Diversity = 6.4714e+00	k_t
Generation 1610: Best = 4.903257e+03 = [1, 1, 1, 1]	FES = 161100	Diversity = 6.4600e+00	k_t
Generation 1620: Best = 4.891097e+03 = [2, 4, 2, 3]	FES = 162100	Diversity = 6.4414e+00	k_t
Generation 1630: Best = 4.883254e+03 = [1, 1, 1, 1]	FES = 163100	Diversity = 6.4272e+00	k_t
Generation 1640: Best = 4.879250e+03 = [3, 3, 4, 4]	FES = 164100	Diversity = 6.4182e+00	k_t
Generation 1650: Best = 4.879250e+03 = [1, 1, 1, 1]	FES = 165100	Diversity = 6.4051e+00	k_t
Generation 1660: Best = 4.866639e+03 = [2, 4, 4, 2]	FES = 166100	Diversity = 6.3853e+00	k_t
Generation 1670: Best = 4.866639e+03 = [1, 1, 1, 1]	FES = 167100	Diversity = 6.3598e+00	k_t
Generation 1680: Best = 4.863898e+03 = [2, 4, 2, 2]	FES = 168100	Diversity = 6.3600e+00	k_t
Generation 1690: Best = 4.863898e+03 = [1, 1, 1, 1]	FES = 169100	Diversity = 6.3439e+00	k_t
Generation 1700: Best = 4.863898e+03 = [2, 2, 4, 3]	FES = 170100	Diversity = 6.3516e+00	k_t
Generation 1710: Best = 4.863898e+03 = [1, 1, 1, 1]	FES = 171100	Diversity = 6.3189e+00	k_t
Generation 1720: Best = 4.862517e+03 = [4, 4, 3, 4]	FES = 172100	Diversity = 6.3194e+00	k_t
Generation 1730: Best = 4.862517e+03 = [1, 1, 1, 1]	FES = 173100	Diversity = 6.2910e+00	k_t
Generation 1740: Best = 4.862517e+03 = [3, 2, 4, 2]	FES = 174100	Diversity = 6.2475e+00	k_t
Generation 1750: Best = 4.858126e+03 = [1, 1, 1, 1]	FES = 175100	Diversity = 6.2334e+00	k_t
Generation 1760: Best = 4.858126e+03 = [3, 4, 2, 2]	FES = 176100	Diversity = 6.1885e+00	k_t
Generation 1770: Best = 4.856972e+03 = [1, 1, 1, 1]	FES = 177100	Diversity = 6.1206e+00	k_t
Generation 1780: Best = 4.856972e+03 = [2, 4, 2, 3]	FES = 178100	Diversity = 6.0855e+00	k_t
Generation 1790: Best = 4.856972e+03 = [1, 1, 1, 1]	FES = 179100	Diversity = 6.0871e+00	k_t
Generation 1800: Best = 4.855787e+03 = [4, 3, 3, 2]	FES = 180100	Diversity = 6.0601e+00	k_t
Generation 1810: Best = 4.855787e+03 = [1, 1, 1, 1]	FES = 181100	Diversity = 5.9173e+00	k_t
Generation 1820: Best = 4.855787e+03 = [3, 4, 3, 2]	FES = 182100	Diversity = 5.8730e+00	k_t
Generation 1830: Best = 4.855130e+03 = [1, 1, 1, 1]	FES = 183100	Diversity = 5.9305e+00	k_t
Generation 1840: Best = 4.854954e+03 = [3, 3, 3, 4]	FES = 184100	Diversity = 5.8659e+00	k_t

Generation 1850: Best = 4.854618e+03 = [1, 1, 1, 1]	FES = 185100	Diversity = 5.8528e+00	k_t
Generation 1860: Best = 4.854618e+03 = [4, 3, 4, 4]	FES = 186100	Diversity = 5.8310e+00	k_t
Generation 1870: Best = 4.850406e+03 = [1, 1, 1, 1]	FES = 187100	Diversity = 5.7358e+00	k_t
Generation 1880: Best = 4.848690e+03 = [4, 3, 4, 2]	FES = 188100	Diversity = 5.7305e+00	k_t
Generation 1890: Best = 4.848690e+03 = [1, 1, 1, 1]	FES = 189100	Diversity = 5.7082e+00	k_t
Generation 1900: Best = 4.841789e+03 = [3, 4, 3, 3]	FES = 190100	Diversity = 5.6838e+00	k_t
Generation 1910: Best = 4.841789e+03 = [1, 1, 1, 1]	FES = 191100	Diversity = 5.6279e+00	k_t
Generation 1920: Best = 4.839507e+03 = [2, 3, 3, 3]	FES = 192100	Diversity = 5.6168e+00	k_t
Generation 1930: Best = 4.837885e+03 = [1, 1, 1, 1]	FES = 193100	Diversity = 5.5948e+00	k_t
Generation 1940: Best = 4.836890e+03 = [2, 3, 2, 3]	FES = 194100	Diversity = 5.5383e+00	k_t
Generation 1950: Best = 4.836885e+03 = [1, 1, 1, 1]	FES = 195100	Diversity = 5.5798e+00	k_t
Generation 1960: Best = 4.835441e+03 = [2, 4, 2, 3]	FES = 196100	Diversity = 5.5485e+00	k_t
Generation 1970: Best = 4.835441e+03 = [1, 1, 1, 1]	FES = 197100	Diversity = 5.5428e+00	k_t
Generation 1980: Best = 4.834026e+03 = [2, 3, 3, 4]	FES = 198100	Diversity = 5.5092e+00	k_t
Generation 1990: Best = 4.827821e+03 = [1, 1, 1, 1]	FES = 199100	Diversity = 5.4703e+00	k_t
Generation 2000: Best = 4.827821e+03 = [4, 4, 2, 4]	FES = 200100	Diversity = 5.4026e+00	k_t
Generation 2010: Best = 4.814984e+03 = [1, 1, 1, 1]	FES = 201100	Diversity = 5.2595e+00	k_t
Generation 2020: Best = 4.812204e+03 = [3, 4, 3, 2]	FES = 202100	Diversity = 5.1589e+00	k_t
Generation 2030: Best = 4.806244e+03 = [1, 1, 1, 1]	FES = 203100	Diversity = 5.0923e+00	k_t
Generation 2040: Best = 4.805069e+03 = [2, 2, 4, 3]	FES = 204100	Diversity = 5.1177e+00	k_t
Generation 2050: Best = 4.802996e+03 = [1, 1, 1, 1]	FES = 205100	Diversity = 5.1044e+00	k_t
Generation 2060: Best = 4.801795e+03 = [2, 2, 2, 3]	FES = 206100	Diversity = 5.0383e+00	k_t
Generation 2070: Best = 4.801795e+03 = [1, 1, 1, 1]	FES = 207100	Diversity = 5.0219e+00	k_t
Generation 2080: Best = 4.800140e+03 = [4, 4, 3, 2]	FES = 208100	Diversity = 4.9405e+00	k_t
Generation 2090: Best = 4.796489e+03 = [1, 1, 1, 1]	FES = 209100	Diversity = 4.9153e+00	k_t
Generation 2100: Best = 4.796489e+03 = [3, 2, 2, 2]	FES = 210100	Diversity = 4.8200e+00	k_t
Generation 2110: Best = 4.794685e+03 = [1, 1, 1, 1]	FES = 211100	Diversity = 4.7492e+00	k_t

Generation 2120: Best = 4.793920e+03 = [4, 2, 3, 2]	FES = 212100	Diversity = 4.6375e+00	k_t
Generation 2130: Best = 4.793920e+03 = [1, 1, 1, 1]	FES = 213100	Diversity = 4.6480e+00	k_t
Generation 2140: Best = 4.793811e+03 = [4, 4, 3, 3]	FES = 214100	Diversity = 4.4843e+00	k_t
Generation 2150: Best = 4.791541e+03 = [1, 1, 1, 1]	FES = 215100	Diversity = 4.4667e+00	k_t
Generation 2160: Best = 4.791541e+03 = [3, 3, 3, 2]	FES = 216100	Diversity = 4.4063e+00	k_t
Generation 2170: Best = 4.790364e+03 = [1, 1, 1, 1]	FES = 217100	Diversity = 4.3890e+00	k_t
Generation 2180: Best = 4.789535e+03 = [3, 4, 2, 4]	FES = 218100	Diversity = 4.1949e+00	k_t
Generation 2190: Best = 4.788246e+03 = [1, 1, 1, 1]	FES = 219100	Diversity = 4.0924e+00	k_t
Generation 2200: Best = 4.784687e+03 = [3, 3, 4, 2]	FES = 220100	Diversity = 4.0825e+00	k_t
Generation 2210: Best = 4.781552e+03 = [1, 1, 1, 1]	FES = 221100	Diversity = 3.9647e+00	k_t
Generation 2220: Best = 4.777188e+03 = [2, 2, 2, 2]	FES = 222100	Diversity = 3.8977e+00	k_t
Generation 2230: Best = 4.776623e+03 = [1, 1, 1, 1]	FES = 223100	Diversity = 3.8394e+00	k_t
Generation 2240: Best = 4.773813e+03 = [2, 4, 2, 3]	FES = 224100	Diversity = 3.6854e+00	k_t
Generation 2250: Best = 4.770966e+03 = [1, 1, 1, 1]	FES = 225100	Diversity = 3.6439e+00	k_t
Generation 2260: Best = 4.770966e+03 = [3, 2, 4, 3]	FES = 226100	Diversity = 3.4684e+00	k_t
Generation 2270: Best = 4.768578e+03 = [1, 1, 1, 1]	FES = 227100	Diversity = 3.4269e+00	k_t
Generation 2280: Best = 4.763077e+03 = [4, 3, 2, 2]	FES = 228100	Diversity = 3.3388e+00	k_t
Generation 2290: Best = 4.760686e+03 = [1, 1, 1, 1]	FES = 229100	Diversity = 3.1507e+00	k_t
Generation 2300: Best = 4.759077e+03 = [3, 4, 2, 3]	FES = 230100	Diversity = 2.8318e+00	k_t
Generation 2310: Best = 4.759077e+03 = [1, 1, 1, 1]	FES = 231100	Diversity = 2.6269e+00	k_t
Generation 2320: Best = 4.758657e+03 = [2, 2, 4, 2]	FES = 232100	Diversity = 2.6212e+00	k_t
Generation 2330: Best = 4.758173e+03 = [1, 1, 1, 1]	FES = 233100	Diversity = 2.6225e+00	k_t
Generation 2340: Best = 4.756463e+03 = [2, 4, 3, 4]	FES = 234100	Diversity = 2.4760e+00	k_t
Generation 2350: Best = 4.756463e+03 = [1, 1, 1, 1]	FES = 235100	Diversity = 2.3363e+00	k_t
Generation 2360: Best = 4.750661e+03 = [3, 3, 2, 3]	FES = 236100	Diversity = 2.3067e+00	k_t
Generation 2370: Best = 4.750204e+03 = [1, 1, 1, 1]	FES = 237100	Diversity = 2.2952e+00	k_t
Generation 2380: Best = 4.750204e+03 = [4, 3, 2, 4]	FES = 238100	Diversity = 2.2544e+00	k_t

Generation 2390: Best = 4.750204e+03 = [1, 1, 1, 1]	FES = 239100	Diversity = 2.2202e+00	k_t
Generation 2400: Best = 4.749595e+03 = [2, 2, 3, 3]	FES = 240100	Diversity = 2.2118e+00	k_t
Generation 2410: Best = 4.748492e+03 = [1, 1, 1, 1]	FES = 241100	Diversity = 2.1808e+00	k_t
Generation 2420: Best = 4.748220e+03 = [4, 3, 2, 4]	FES = 242100	Diversity = 2.1416e+00	k_t
Generation 2430: Best = 4.747829e+03 = [1, 1, 1, 1]	FES = 243100	Diversity = 2.1136e+00	k_t
Generation 2440: Best = 4.747168e+03 = [4, 2, 4, 4]	FES = 244100	Diversity = 2.0859e+00	k_t
Generation 2450: Best = 4.745523e+03 = [1, 1, 1, 1]	FES = 245100	Diversity = 2.0740e+00	k_t
Generation 2460: Best = 4.740031e+03 = [3, 4, 3, 4]	FES = 246100	Diversity = 2.0769e+00	k_t
Generation 2470: Best = 4.737083e+03 = [1, 1, 1, 1]	FES = 247100	Diversity = 2.0078e+00	k_t
Generation 2480: Best = 4.737083e+03 = [4, 3, 2, 3]	FES = 248100	Diversity = 1.9653e+00	k_t
Generation 2490: Best = 4.737083e+03 = [1, 1, 1, 1]	FES = 249100	Diversity = 1.9601e+00	k_t
Generation 2500: Best = 4.731024e+03 = [2, 3, 3, 2]	FES = 250100	Diversity = 1.9241e+00	k_t
Generation 2510: Best = 4.731024e+03 = [1, 1, 1, 1]	FES = 251100	Diversity = 1.9063e+00	k_t
Generation 2520: Best = 4.730531e+03 = [4, 4, 4, 4]	FES = 252100	Diversity = 1.8903e+00	k_t
Generation 2530: Best = 4.728455e+03 = [1, 1, 1, 1]	FES = 253100	Diversity = 1.8120e+00	k_t
Generation 2540: Best = 4.728455e+03 = [4, 4, 2, 4]	FES = 254100	Diversity = 1.7778e+00	k_t
Generation 2550: Best = 4.728455e+03 = [1, 1, 1, 1]	FES = 255100	Diversity = 1.7676e+00	k_t
Generation 2560: Best = 4.728455e+03 = [3, 4, 2, 2]	FES = 256100	Diversity = 1.7590e+00	k_t
Generation 2570: Best = 4.728455e+03 = [1, 1, 1, 1]	FES = 257100	Diversity = 1.7554e+00	k_t
Generation 2580: Best = 4.727782e+03 = [3, 2, 4, 4]	FES = 258100	Diversity = 1.6980e+00	k_t
Generation 2590: Best = 4.726120e+03 = [1, 1, 1, 1]	FES = 259100	Diversity = 1.6939e+00	k_t
Generation 2600: Best = 4.724052e+03 = [3, 2, 4, 3]	FES = 260100	Diversity = 1.6056e+00	k_t
Generation 2610: Best = 4.723959e+03 = [1, 1, 1, 1]	FES = 261100	Diversity = 1.5743e+00	k_t
Generation 2620: Best = 4.722779e+03 = [2, 4, 3, 2]	FES = 262100	Diversity = 1.5807e+00	k_t
Generation 2630: Best = 4.722779e+03 = [1, 1, 1, 1]	FES = 263100	Diversity = 1.5382e+00	k_t
Generation 2640: Best = 4.722779e+03 = [2, 4, 4, 2]	FES = 264100	Diversity = 1.5287e+00	k_t
Generation 2650: Best = 4.722779e+03 = [1, 1, 1, 1]	FES = 265100	Diversity = 1.5138e+00	k_t

Generation 2660: Best = 4.722711e+03 = [3, 4, 3, 2]	FES = 266100	Diversity = 1.5096e+00	k_t
Generation 2670: Best = 4.722152e+03 = [1, 1, 1, 1]	FES = 267100	Diversity = 1.4678e+00	k_t
Generation 2680: Best = 4.721600e+03 = [4, 4, 3, 3]	FES = 268100	Diversity = 1.4313e+00	k_t
Generation 2690: Best = 4.721512e+03 = [1, 1, 1, 1]	FES = 269100	Diversity = 1.3837e+00	k_t
Generation 2700: Best = 4.719252e+03 = [3, 3, 2, 2]	FES = 270100	Diversity = 1.3712e+00	k_t
Generation 2710: Best = 4.719252e+03 = [1, 1, 1, 1]	FES = 271100	Diversity = 1.3494e+00	k_t
Generation 2720: Best = 4.718831e+03 = [4, 3, 2, 3]	FES = 272100	Diversity = 1.3258e+00	k_t
Generation 2730: Best = 4.718090e+03 = [1, 1, 1, 1]	FES = 273100	Diversity = 1.3069e+00	k_t
Generation 2740: Best = 4.718090e+03 = [3, 2, 2, 2]	FES = 274100	Diversity = 1.2488e+00	k_t
Generation 2750: Best = 4.718061e+03 = [1, 1, 1, 1]	FES = 275100	Diversity = 1.2423e+00	k_t
Generation 2760: Best = 4.718061e+03 = [2, 4, 4, 4]	FES = 276100	Diversity = 1.2377e+00	k_t
Generation 2770: Best = 4.713959e+03 = [1, 1, 1, 1]	FES = 277100	Diversity = 1.2028e+00	k_t
Generation 2780: Best = 4.713959e+03 = [3, 4, 3, 2]	FES = 278100	Diversity = 1.1934e+00	k_t
Generation 2790: Best = 4.713951e+03 = [1, 1, 1, 1]	FES = 279100	Diversity = 1.1935e+00	k_t
Generation 2800: Best = 4.713892e+03 = [4, 4, 4, 2]	FES = 280100	Diversity = 1.1822e+00	k_t
Generation 2810: Best = 4.713114e+03 = [1, 1, 1, 1]	FES = 281100	Diversity = 1.1857e+00	k_t
Generation 2820: Best = 4.712430e+03 = [2, 2, 2, 2]	FES = 282100	Diversity = 1.1877e+00	k_t
Generation 2830: Best = 4.712303e+03 = [1, 1, 1, 1]	FES = 283100	Diversity = 1.1847e+00	k_t
Generation 2840: Best = 4.711911e+03 = [2, 4, 2, 3]	FES = 284100	Diversity = 1.1620e+00	k_t
Generation 2850: Best = 4.711911e+03 = [1, 1, 1, 1]	FES = 285100	Diversity = 1.1519e+00	k_t
Generation 2860: Best = 4.711798e+03 = [2, 3, 3, 3]	FES = 286100	Diversity = 1.1387e+00	k_t
Generation 2870: Best = 4.711471e+03 = [1, 1, 1, 1]	FES = 287100	Diversity = 1.1404e+00	k_t
Generation 2880: Best = 4.711471e+03 = [2, 4, 4, 3]	FES = 288100	Diversity = 1.1377e+00	k_t
Generation 2890: Best = 4.711375e+03 = [1, 1, 1, 1]	FES = 289100	Diversity = 1.1255e+00	k_t
Generation 2900: Best = 4.710764e+03 = [3, 2, 2, 2]	FES = 290100	Diversity = 1.1175e+00	k_t
Generation 2910: Best = 4.709865e+03 = [1, 1, 1, 1]	FES = 291100	Diversity = 1.1095e+00	k_t
Generation 2920: Best = 4.708279e+03 = [2, 3, 3, 2]	FES = 292100	Diversity = 1.0910e+00	k_t

Generation 2930: Best = 4.708279e+03 = [1, 1, 1, 1]	FES = 293100	Diversity = 1.0654e+00	k_t
Generation 2940: Best = 4.708226e+03 = [4, 4, 4, 2]	FES = 294100	Diversity = 1.0541e+00	k_t
Generation 2950: Best = 4.708226e+03 = [1, 1, 1, 1]	FES = 295100	Diversity = 1.0463e+00	k_t
Generation 2960: Best = 4.707711e+03 = [4, 4, 2, 4]	FES = 296100	Diversity = 1.0422e+00	k_t
Generation 2970: Best = 4.707711e+03 = [1, 1, 1, 1]	FES = 297100	Diversity = 1.0283e+00	k_t
Generation 2980: Best = 4.707711e+03 = [2, 2, 3, 2]	FES = 298100	Diversity = 1.0222e+00	k_t
Generation 2990: Best = 4.707711e+03 = [1, 1, 1, 1]	FES = 299100	Diversity = 1.0137e+00	k_t
Generation 3000: Best = 4.707711e+03 = [2, 4, 2, 2]	FES = 300100	Diversity = 1.0103e+00	k_t
Generation 3010: Best = 4.706932e+03 = [1, 1, 1, 1]	FES = 301100	Diversity = 9.8737e-01	k_t
Generation 3020: Best = 4.706932e+03 = [4, 2, 3, 4]	FES = 302100	Diversity = 9.8053e-01	k_t
Generation 3030: Best = 4.706932e+03 = [1, 1, 1, 1]	FES = 303100	Diversity = 9.7252e-01	k_t
Generation 3040: Best = 4.704790e+03 = [4, 3, 2, 4]	FES = 304100	Diversity = 9.6869e-01	k_t
Generation 3050: Best = 4.704730e+03 = [1, 1, 1, 1]	FES = 305100	Diversity = 9.6632e-01	k_t
Generation 3060: Best = 4.704655e+03 = [4, 2, 3, 4]	FES = 306100	Diversity = 9.5442e-01	k_t
Generation 3070: Best = 4.704655e+03 = [1, 1, 1, 1]	FES = 307100	Diversity = 9.5150e-01	k_t
Generation 3080: Best = 4.704619e+03 = [3, 2, 2, 3]	FES = 308100	Diversity = 9.4330e-01	k_t
Generation 3090: Best = 4.704535e+03 = [1, 1, 1, 1]	FES = 309100	Diversity = 9.3151e-01	k_t
Generation 3100: Best = 4.704535e+03 = [4, 3, 3, 3]	FES = 310100	Diversity = 9.2941e-01	k_t
Generation 3110: Best = 4.704535e+03 = [1, 1, 1, 1]	FES = 311100	Diversity = 9.1644e-01	k_t
Generation 3120: Best = 4.704535e+03 = [2, 4, 4, 2]	FES = 312100	Diversity = 9.1360e-01	k_t
Generation 3130: Best = 4.704535e+03 = [1, 1, 1, 1]	FES = 313100	Diversity = 9.0928e-01	k_t
Generation 3140: Best = 4.704535e+03 = [2, 4, 3, 3]	FES = 314100	Diversity = 8.9628e-01	k_t
Generation 3150: Best = 4.704317e+03 = [1, 1, 1, 1]	FES = 315100	Diversity = 8.8904e-01	k_t
Generation 3160: Best = 4.703956e+03 = [2, 4, 3, 2]	FES = 316100	Diversity = 8.8396e-01	k_t
Generation 3170: Best = 4.703749e+03 = [1, 1, 1, 1]	FES = 317100	Diversity = 8.8378e-01	k_t
Generation 3180: Best = 4.703293e+03 = [3, 3, 2, 3]	FES = 318100	Diversity = 8.6683e-01	k_t
Generation 3190: Best = 4.703248e+03 = [1, 1, 1, 1]	FES = 319100	Diversity = 8.7025e-01	k_t

Generation 3200: Best = 4.703016e+03 = [2, 4, 4, 4]	FES = 320100	Diversity = 8.6443e-01	k_t
Generation 3210: Best = 4.703016e+03 = [1, 1, 1, 1]	FES = 321100	Diversity = 8.6074e-01	k_t
Generation 3220: Best = 4.702977e+03 = [4, 4, 4, 2]	FES = 322100	Diversity = 8.4267e-01	k_t
Generation 3230: Best = 4.702977e+03 = [1, 1, 1, 1]	FES = 323100	Diversity = 8.3631e-01	k_t
Generation 3240: Best = 4.701744e+03 = [2, 4, 3, 2]	FES = 324100	Diversity = 8.2699e-01	k_t
Generation 3250: Best = 4.701719e+03 = [1, 1, 1, 1]	FES = 325100	Diversity = 8.1724e-01	k_t
Generation 3260: Best = 4.701719e+03 = [3, 2, 4, 2]	FES = 326100	Diversity = 7.9828e-01	k_t
Generation 3270: Best = 4.701719e+03 = [1, 1, 1, 1]	FES = 327100	Diversity = 8.0073e-01	k_t
Generation 3280: Best = 4.701335e+03 = [4, 2, 2, 4]	FES = 328100	Diversity = 7.9352e-01	k_t
Generation 3290: Best = 4.701335e+03 = [1, 1, 1, 1]	FES = 329100	Diversity = 7.8929e-01	k_t
Generation 3300: Best = 4.701335e+03 = [4, 2, 3, 3]	FES = 330100	Diversity = 7.8296e-01	k_t
Generation 3310: Best = 4.701335e+03 = [1, 1, 1, 1]	FES = 331100	Diversity = 7.7897e-01	k_t
Generation 3320: Best = 4.701028e+03 = [2, 3, 2, 2]	FES = 332100	Diversity = 7.5822e-01	k_t
Generation 3330: Best = 4.701028e+03 = [1, 1, 1, 1]	FES = 333100	Diversity = 7.4902e-01	k_t
Generation 3340: Best = 4.700863e+03 = [2, 4, 2, 2]	FES = 334100	Diversity = 7.4600e-01	k_t
Generation 3350: Best = 4.700807e+03 = [1, 1, 1, 1]	FES = 335100	Diversity = 7.3256e-01	k_t
Generation 3360: Best = 4.700531e+03 = [4, 2, 3, 2]	FES = 336100	Diversity = 7.3083e-01	k_t
Generation 3370: Best = 4.700531e+03 = [1, 1, 1, 1]	FES = 337100	Diversity = 7.2688e-01	k_t
Generation 3380: Best = 4.700531e+03 = [4, 4, 2, 4]	FES = 338100	Diversity = 7.2569e-01	k_t
Generation 3390: Best = 4.700260e+03 = [1, 1, 1, 1]	FES = 339100	Diversity = 7.2514e-01	k_t
Generation 3400: Best = 4.700260e+03 = [4, 2, 3, 3]	FES = 340100	Diversity = 7.1344e-01	k_t
Generation 3410: Best = 4.699604e+03 = [1, 1, 1, 1]	FES = 341100	Diversity = 7.0147e-01	k_t
Generation 3420: Best = 4.698788e+03 = [2, 2, 2, 2]	FES = 342100	Diversity = 6.9903e-01	k_t
Generation 3430: Best = 4.698639e+03 = [1, 1, 1, 1]	FES = 343100	Diversity = 6.9448e-01	k_t
Generation 3440: Best = 4.698617e+03 = [3, 2, 4, 3]	FES = 344100	Diversity = 6.7917e-01	k_t
Generation 3450: Best = 4.698617e+03 = [1, 1, 1, 1]	FES = 345100	Diversity = 6.8014e-01	k_t
Generation 3460: Best = 4.698617e+03 = [3, 3, 4, 4]	FES = 346100	Diversity = 6.7281e-01	k_t

Generation 3470: Best = 4.698598e+03 = [1, 1, 1, 1]	FES = 347100	Diversity = 6.7133e-01	k_t
Generation 3480: Best = 4.698534e+03 = [3, 3, 2, 3]	FES = 348100	Diversity = 6.5787e-01	k_t
Generation 3490: Best = 4.698497e+03 = [1, 1, 1, 1]	FES = 349100	Diversity = 6.4990e-01	k_t
Generation 3500: Best = 4.698497e+03 = [3, 2, 3, 4]	FES = 350100	Diversity = 6.3988e-01	k_t
Generation 3510: Best = 4.698497e+03 = [1, 1, 1, 1]	FES = 351100	Diversity = 6.3012e-01	k_t
Generation 3520: Best = 4.698418e+03 = [4, 4, 4, 4]	FES = 352100	Diversity = 6.2184e-01	k_t
Generation 3530: Best = 4.698019e+03 = [1, 1, 1, 1]	FES = 353100	Diversity = 6.1002e-01	k_t
Generation 3540: Best = 4.697932e+03 = [4, 2, 3, 2]	FES = 354100	Diversity = 6.0089e-01	k_t
Generation 3550: Best = 4.697932e+03 = [1, 1, 1, 1]	FES = 355100	Diversity = 6.0887e-01	k_t
Generation 3560: Best = 4.697932e+03 = [2, 3, 4, 4]	FES = 356100	Diversity = 6.0234e-01	k_t
Generation 3570: Best = 4.697714e+03 = [1, 1, 1, 1]	FES = 357100	Diversity = 5.9993e-01	k_t
Generation 3580: Best = 4.697535e+03 = [4, 4, 3, 4]	FES = 358100	Diversity = 5.9145e-01	k_t
Generation 3590: Best = 4.696751e+03 = [1, 1, 1, 1]	FES = 359100	Diversity = 5.7634e-01	k_t
Generation 3600: Best = 4.696354e+03 = [2, 3, 3, 4]	FES = 360100	Diversity = 5.6939e-01	k_t
Generation 3610: Best = 4.696297e+03 = [1, 1, 1, 1]	FES = 361100	Diversity = 5.5683e-01	k_t
Generation 3620: Best = 4.696224e+03 = [3, 3, 4, 4]	FES = 362100	Diversity = 5.3977e-01	k_t
Generation 3630: Best = 4.695699e+03 = [1, 1, 1, 1]	FES = 363100	Diversity = 5.2991e-01	k_t
Generation 3640: Best = 4.695447e+03 = [4, 4, 3, 4]	FES = 364100	Diversity = 5.2406e-01	k_t
Generation 3650: Best = 4.695359e+03 = [1, 1, 1, 1]	FES = 365100	Diversity = 5.1931e-01	k_t
Generation 3660: Best = 4.695109e+03 = [3, 2, 2, 2]	FES = 366100	Diversity = 4.8667e-01	k_t
Generation 3670: Best = 4.694690e+03 = [1, 1, 1, 1]	FES = 367100	Diversity = 4.6948e-01	k_t
Generation 3680: Best = 4.694548e+03 = [3, 4, 2, 4]	FES = 368100	Diversity = 4.4571e-01	k_t
Generation 3690: Best = 4.694548e+03 = [1, 1, 1, 1]	FES = 369100	Diversity = 4.2862e-01	k_t
Generation 3700: Best = 4.694548e+03 = [3, 3, 2, 4]	FES = 370100	Diversity = 4.0393e-01	k_t
Generation 3710: Best = 4.694306e+03 = [1, 1, 1, 1]	FES = 371100	Diversity = 3.7879e-01	k_t
Generation 3720: Best = 4.694306e+03 = [3, 2, 4, 2]	FES = 372100	Diversity = 3.5418e-01	k_t
Generation 3730: Best = 4.694266e+03 = [1, 1, 1, 1]	FES = 373100	Diversity = 3.5086e-01	k_t

Generation 3740: Best = 4.694015e+03 = [3, 3, 2, 3]	FES = 374100	Diversity = 3.2414e-01	k_t
Generation 3750: Best = 4.693930e+03 = [1, 1, 1, 1]	FES = 375100	Diversity = 3.1552e-01	k_t
Generation 3760: Best = 4.693930e+03 = [3, 2, 2, 2]	FES = 376100	Diversity = 3.0024e-01	k_t
Generation 3770: Best = 4.693875e+03 = [1, 1, 1, 1]	FES = 377100	Diversity = 2.8475e-01	k_t
Generation 3780: Best = 4.693259e+03 = [3, 4, 3, 3]	FES = 378100	Diversity = 2.6586e-01	k_t
Generation 3790: Best = 4.693259e+03 = [1, 1, 1, 1]	FES = 379100	Diversity = 2.5320e-01	k_t
Generation 3800: Best = 4.693176e+03 = [2, 2, 3, 2]	FES = 380100	Diversity = 2.4410e-01	k_t
Generation 3810: Best = 4.692840e+03 = [1, 1, 1, 1]	FES = 381100	Diversity = 2.3897e-01	k_t
Generation 3820: Best = 4.692840e+03 = [3, 3, 4, 3]	FES = 382100	Diversity = 2.2671e-01	k_t
Generation 3830: Best = 4.692738e+03 = [1, 1, 1, 1]	FES = 383100	Diversity = 2.2181e-01	k_t
Generation 3840: Best = 4.692525e+03 = [3, 4, 3, 3]	FES = 384100	Diversity = 2.0920e-01	k_t
Generation 3850: Best = 4.692524e+03 = [1, 1, 1, 1]	FES = 385100	Diversity = 2.0609e-01	k_t
Generation 3860: Best = 4.692479e+03 = [3, 4, 4, 3]	FES = 386100	Diversity = 1.8755e-01	k_t
Generation 3870: Best = 4.691931e+03 = [1, 1, 1, 1]	FES = 387100	Diversity = 1.8839e-01	k_t
Generation 3880: Best = 4.691914e+03 = [3, 4, 2, 4]	FES = 388100	Diversity = 1.7680e-01	k_t
Generation 3890: Best = 4.691801e+03 = [1, 1, 1, 1]	FES = 389100	Diversity = 1.5631e-01	k_t
Generation 3900: Best = 4.691793e+03 = [3, 4, 2, 2]	FES = 390100	Diversity = 1.5573e-01	k_t
Generation 3910: Best = 4.691674e+03 = [1, 1, 1, 1]	FES = 391100	Diversity = 1.5161e-01	k_t
Generation 3920: Best = 4.691559e+03 = [4, 2, 3, 2]	FES = 392100	Diversity = 1.4792e-01	k_t
Generation 3930: Best = 4.691436e+03 = [1, 1, 1, 1]	FES = 393100	Diversity = 1.4422e-01	k_t
Generation 3940: Best = 4.691362e+03 = [3, 4, 2, 4]	FES = 394100	Diversity = 1.3986e-01	k_t
Generation 3950: Best = 4.691245e+03 = [1, 1, 1, 1]	FES = 395100	Diversity = 1.3300e-01	k_t
Generation 3960: Best = 4.690886e+03 = [2, 2, 2, 3]	FES = 396100	Diversity = 1.2445e-01	k_t
Generation 3970: Best = 4.690635e+03 = [1, 1, 1, 1]	FES = 397100	Diversity = 1.1761e-01	k_t
Generation 3980: Best = 4.690578e+03 = [2, 2, 3, 4]	FES = 398100	Diversity = 1.0760e-01	k_t
Generation 3990: Best = 4.690488e+03 = [1, 1, 1, 1]	FES = 399100	Diversity = 9.8542e-02	k_t
Generation 4000: Best = 4.690413e+03 = [2, 4, 3, 3]	FES = 400100	Diversity = 9.4894e-02	k_t

Generation 4010: Best = 4.690273e+03 = [1, 1, 1, 1]	FES = 401100	Diversity = 8.8712e-02	k_t
Generation 4020: Best = 4.690025e+03 = [4, 3, 4, 3]	FES = 402100	Diversity = 8.1406e-02	k_t
Generation 4030: Best = 4.689925e+03 = [1, 1, 1, 1]	FES = 403100	Diversity = 7.2477e-02	k_t
Generation 4040: Best = 4.689824e+03 = [2, 4, 4, 2]	FES = 404100	Diversity = 6.4143e-02	k_t
Generation 4050: Best = 4.689415e+03 = [1, 1, 1, 1]	FES = 405100	Diversity = 5.1762e-02	k_t
Generation 4060: Best = 4.689382e+03 = [4, 4, 3, 2]	FES = 406100	Diversity = 4.6676e-02	k_t
Generation 4070: Best = 4.689164e+03 = [1, 1, 1, 1]	FES = 407100	Diversity = 3.8988e-02	k_t
Generation 4080: Best = 4.689045e+03 = [2, 4, 3, 4]	FES = 408100	Diversity = 3.4696e-02	k_t
Generation 4090: Best = 4.688838e+03 = [1, 1, 1, 1]	FES = 409100	Diversity = 3.3058e-02	k_t
Generation 4100: Best = 4.688765e+03 = [4, 2, 4, 3]	FES = 410100	Diversity = 3.1771e-02	k_t
Generation 4110: Best = 4.688692e+03 = [1, 1, 1, 1]	FES = 411100	Diversity = 3.0757e-02	k_t
Generation 4120: Best = 4.688614e+03 = [4, 3, 3, 2]	FES = 412100	Diversity = 2.9614e-02	k_t
Generation 4130: Best = 4.688523e+03 = [1, 1, 1, 1]	FES = 413100	Diversity = 2.8604e-02	k_t
Generation 4140: Best = 4.688441e+03 = [4, 2, 4, 2]	FES = 414100	Diversity = 2.7883e-02	k_t
Generation 4150: Best = 4.688376e+03 = [1, 1, 1, 1]	FES = 415100	Diversity = 2.8083e-02	k_t
Generation 4160: Best = 4.688337e+03 = [3, 4, 4, 4]	FES = 416100	Diversity = 2.7804e-02	k_t
Generation 4170: Best = 4.688314e+03 = [1, 1, 1, 1]	FES = 417100	Diversity = 2.7955e-02	k_t
Generation 4180: Best = 4.688240e+03 = [2, 2, 2, 4]	FES = 418100	Diversity = 2.8401e-02	k_t
Generation 4190: Best = 4.688148e+03 = [1, 1, 1, 1]	FES = 419100	Diversity = 2.7219e-02	k_t
Generation 4200: Best = 4.688115e+03 = [4, 3, 3, 3]	FES = 420100	Diversity = 2.6027e-02	k_t
Generation 4210: Best = 4.688070e+03 = [1, 1, 1, 1]	FES = 421100	Diversity = 2.6662e-02	k_t
Generation 4220: Best = 4.688057e+03 = [2, 4, 2, 3]	FES = 422100	Diversity = 2.6671e-02	k_t
Generation 4230: Best = 4.687995e+03 = [1, 1, 1, 1]	FES = 423100	Diversity = 2.6379e-02	k_t
Generation 4240: Best = 4.687931e+03 = [3, 2, 3, 2]	FES = 424100	Diversity = 2.6563e-02	k_t
Generation 4250: Best = 4.687859e+03 = [1, 1, 1, 1]	FES = 425100	Diversity = 2.6854e-02	k_t
Generation 4260: Best = 4.687850e+03 = [3, 4, 2, 3]	FES = 426100	Diversity = 2.6581e-02	k_t
Generation 4270: Best = 4.687802e+03 = [1, 1, 1, 1]	FES = 427100	Diversity = 2.6861e-02	k_t

Generation 4280: Best = 4.687785e+03 = [3, 3, 4, 3]	FES = 428100	Diversity = 2.7069e-02	k_t
Generation 4290: Best = 4.687754e+03 = [1, 1, 1, 1]	FES = 429100	Diversity = 2.6404e-02	k_t
Generation 4300: Best = 4.687709e+03 = [4, 2, 2, 4]	FES = 430100	Diversity = 2.6193e-02	k_t
Generation 4310: Best = 4.687646e+03 = [1, 1, 1, 1]	FES = 431100	Diversity = 2.6169e-02	k_t
Generation 4320: Best = 4.687609e+03 = [4, 2, 3, 4]	FES = 432100	Diversity = 2.5538e-02	k_t
Generation 4330: Best = 4.687580e+03 = [1, 1, 1, 1]	FES = 433100	Diversity = 2.5392e-02	k_t
Generation 4340: Best = 4.687526e+03 = [2, 2, 2, 4]	FES = 434100	Diversity = 2.4478e-02	k_t
Generation 4350: Best = 4.687511e+03 = [1, 1, 1, 1]	FES = 435100	Diversity = 2.4141e-02	k_t
Generation 4360: Best = 4.687470e+03 = [4, 3, 3, 3]	FES = 436100	Diversity = 2.3569e-02	k_t
Generation 4370: Best = 4.687445e+03 = [1, 1, 1, 1]	FES = 437100	Diversity = 2.2981e-02	k_t
Generation 4380: Best = 4.687405e+03 = [3, 2, 4, 3]	FES = 438100	Diversity = 2.2809e-02	k_t
Generation 4390: Best = 4.687378e+03 = [1, 1, 1, 1]	FES = 439100	Diversity = 2.3079e-02	k_t
Generation 4400: Best = 4.687333e+03 = [3, 3, 2, 2]	FES = 440100	Diversity = 2.3398e-02	k_t
Generation 4410: Best = 4.687300e+03 = [1, 1, 1, 1]	FES = 441100	Diversity = 2.3409e-02	k_t
Generation 4420: Best = 4.687265e+03 = [2, 2, 3, 3]	FES = 442100	Diversity = 2.3915e-02	k_t
Generation 4430: Best = 4.687247e+03 = [1, 1, 1, 1]	FES = 443100	Diversity = 2.4255e-02	k_t
Generation 4440: Best = 4.687207e+03 = [2, 4, 4, 4]	FES = 444100	Diversity = 2.3827e-02	k_t
Generation 4450: Best = 4.687207e+03 = [1, 1, 1, 1]	FES = 445100	Diversity = 2.3562e-02	k_t
Generation 4460: Best = 4.687174e+03 = [3, 4, 3, 3]	FES = 446100	Diversity = 2.3208e-02	k_t
Generation 4470: Best = 4.687104e+03 = [1, 1, 1, 1]	FES = 447100	Diversity = 2.2696e-02	k_t
Generation 4480: Best = 4.687099e+03 = [3, 2, 3, 3]	FES = 448100	Diversity = 2.2388e-02	k_t
Generation 4490: Best = 4.687064e+03 = [1, 1, 1, 1]	FES = 449100	Diversity = 2.2512e-02	k_t
Generation 4500: Best = 4.687053e+03 = [4, 4, 2, 3]	FES = 450100	Diversity = 2.2678e-02	k_t
Generation 4510: Best = 4.687024e+03 = [1, 1, 1, 1]	FES = 451100	Diversity = 2.2888e-02	k_t
Generation 4520: Best = 4.686995e+03 = [4, 2, 2, 4]	FES = 452100	Diversity = 2.2260e-02	k_t
Generation 4530: Best = 4.686954e+03 = [1, 1, 1, 1]	FES = 453100	Diversity = 2.1925e-02	k_t
Generation 4540: Best = 4.686943e+03 = [3, 2, 4, 4]	FES = 454100	Diversity = 2.2241e-02	k_t

Generation 4550: Best = 4.686912e+03 = [1, 1, 1, 1]	FES = 455100	Diversity = 2.2197e-02	k_t
Generation 4560: Best = 4.686910e+03 = [3, 2, 4, 3]	FES = 456100	Diversity = 2.2740e-02	k_t
Generation 4570: Best = 4.686872e+03 = [1, 1, 1, 1]	FES = 457100	Diversity = 2.2568e-02	k_t
Generation 4580: Best = 4.686841e+03 = [3, 4, 3, 4]	FES = 458100	Diversity = 2.2833e-02	k_t
Generation 4590: Best = 4.686803e+03 = [1, 1, 1, 1]	FES = 459100	Diversity = 2.2034e-02	k_t
Generation 4600: Best = 4.686787e+03 = [4, 2, 3, 4]	FES = 460100	Diversity = 2.1450e-02	k_t
Generation 4610: Best = 4.686733e+03 = [1, 1, 1, 1]	FES = 461100	Diversity = 2.1352e-02	k_t
Generation 4620: Best = 4.686720e+03 = [3, 3, 4, 2]	FES = 462100	Diversity = 2.0724e-02	k_t
Generation 4630: Best = 4.686709e+03 = [1, 1, 1, 1]	FES = 463100	Diversity = 1.9731e-02	k_t
Generation 4640: Best = 4.686656e+03 = [2, 3, 4, 3]	FES = 464100	Diversity = 1.7693e-02	k_t
Generation 4650: Best = 4.686618e+03 = [1, 1, 1, 1]	FES = 465100	Diversity = 1.5516e-02	k_t
Generation 4660: Best = 4.686576e+03 = [3, 3, 2, 3]	FES = 466100	Diversity = 1.3078e-02	k_t
Generation 4670: Best = 4.686559e+03 = [1, 1, 1, 1]	FES = 467100	Diversity = 1.1859e-02	k_t
Generation 4680: Best = 4.686519e+03 = [2, 3, 2, 2]	FES = 468100	Diversity = 1.1285e-02	k_t
Generation 4690: Best = 4.686504e+03 = [1, 1, 1, 1]	FES = 469100	Diversity = 1.0571e-02	k_t
Generation 4700: Best = 4.686445e+03 = [4, 4, 3, 2]	FES = 470100	Diversity = 1.0331e-02	k_t
Generation 4710: Best = 4.686421e+03 = [1, 1, 1, 1]	FES = 471100	Diversity = 1.0043e-02	k_t
Generation 4720: Best = 4.686396e+03 = [4, 3, 3, 2]	FES = 472100	Diversity = 1.0469e-02	k_t
Generation 4730: Best = 4.686370e+03 = [1, 1, 1, 1]	FES = 473100	Diversity = 1.0375e-02	k_t
Generation 4740: Best = 4.686317e+03 = [3, 4, 2, 2]	FES = 474100	Diversity = 1.0054e-02	k_t
Generation 4750: Best = 4.686293e+03 = [1, 1, 1, 1]	FES = 475100	Diversity = 1.0290e-02	k_t
Generation 4760: Best = 4.686265e+03 = [3, 3, 4, 4]	FES = 476100	Diversity = 1.0807e-02	k_t
Generation 4770: Best = 4.686218e+03 = [1, 1, 1, 1]	FES = 477100	Diversity = 1.0776e-02	k_t
Generation 4780: Best = 4.686197e+03 = [3, 4, 3, 2]	FES = 478100	Diversity = 1.0742e-02	k_t
Generation 4790: Best = 4.686184e+03 = [1, 1, 1, 1]	FES = 479100	Diversity = 1.0424e-02	k_t
Generation 4800: Best = 4.686147e+03 = [3, 4, 2, 3]	FES = 480100	Diversity = 1.0128e-02	k_t
Generation 4810: Best = 4.686131e+03 = [1, 1, 1, 1]	FES = 481100	Diversity = 9.8135e-03	k_t

Generation 4820: Best = 4.686117e+03 = [2, 4, 3, 3]	FES = 482100	Diversity = 9.0527e-03	k_t
Generation 4830: Best = 4.686080e+03 = [1, 1, 1, 1]	FES = 483100	Diversity = 8.7535e-03	k_t
Generation 4840: Best = 4.686066e+03 = [4, 3, 2, 2]	FES = 484100	Diversity = 8.2737e-03	k_t
Generation 4850: Best = 4.686042e+03 = [1, 1, 1, 1]	FES = 485100	Diversity = 8.2208e-03	k_t
Generation 4860: Best = 4.686017e+03 = [4, 4, 2, 2]	FES = 486100	Diversity = 8.1251e-03	k_t
Generation 4870: Best = 4.685999e+03 = [1, 1, 1, 1]	FES = 487100	Diversity = 8.0589e-03	k_t
Generation 4880: Best = 4.685969e+03 = [4, 2, 2, 3]	FES = 488100	Diversity = 8.3565e-03	k_t
Generation 4890: Best = 4.685945e+03 = [1, 1, 1, 1]	FES = 489100	Diversity = 8.1268e-03	k_t
Generation 4900: Best = 4.685917e+03 = [4, 4, 4, 3]	FES = 490100	Diversity = 8.2310e-03	k_t
Generation 4910: Best = 4.685909e+03 = [1, 1, 1, 1]	FES = 491100	Diversity = 8.3417e-03	k_t
Generation 4920: Best = 4.685887e+03 = [3, 2, 4, 4]	FES = 492100	Diversity = 8.2929e-03	k_t
Generation 4930: Best = 4.685865e+03 = [1, 1, 1, 1]	FES = 493100	Diversity = 8.3941e-03	k_t
Generation 4940: Best = 4.685843e+03 = [4, 2, 2, 2]	FES = 494100	Diversity = 8.7288e-03	k_t
Generation 4950: Best = 4.685819e+03 = [1, 1, 1, 1]	FES = 495100	Diversity = 9.5391e-03	k_t
Generation 4960: Best = 4.685807e+03 = [4, 2, 3, 4]	FES = 496100	Diversity = 9.4086e-03	k_t
Generation 4970: Best = 4.685792e+03 = [1, 1, 1, 1]	FES = 497100	Diversity = 9.4879e-03	k_t
Generation 4980: Best = 4.685765e+03 = [2, 2, 3, 4]	FES = 498100	Diversity = 9.6906e-03	k_t
Generation 4990: Best = 4.685736e+03 = [1, 1, 1, 1]	FES = 499100	Diversity = 1.1021e-02	k_t
Generation 5000: Best = 4.685723e+03 = [3, 2, 3, 2]	FES = 500100	Diversity = 1.0919e-02	k_t
Generation 5010: Best = 4.685702e+03 = [1, 1, 1, 1]	FES = 501100	Diversity = 1.1173e-02	k_t
Generation 5020: Best = 4.685662e+03 = [3, 2, 2, 4]	FES = 502100	Diversity = 1.0618e-02	k_t
Generation 5030: Best = 4.685643e+03 = [1, 1, 1, 1]	FES = 503100	Diversity = 1.0215e-02	k_t
Generation 5040: Best = 4.685619e+03 = [3, 2, 4, 2]	FES = 504100	Diversity = 9.8604e-03	k_t
Generation 5050: Best = 4.685609e+03 = [1, 1, 1, 1]	FES = 505100	Diversity = 9.5608e-03	k_t
Generation 5060: Best = 4.685593e+03 = [3, 2, 4, 2]	FES = 506100	Diversity = 9.1265e-03	k_t
Generation 5070: Best = 4.685574e+03 = [1, 1, 1, 1]	FES = 507100	Diversity = 9.1216e-03	k_t
Generation 5080: Best = 4.685559e+03 = [2, 4, 3, 3]	FES = 508100	Diversity = 9.3324e-03	k_t

Generation 5090: Best = 4.685535e+03 = [1, 1, 1, 1]	FES = 509100	Diversity = 9.0099e-03	k_t
Generation 5100: Best = 4.685510e+03 = [3, 2, 3, 3]	FES = 510100	Diversity = 8.3346e-03	k_t
Generation 5110: Best = 4.685490e+03 = [1, 1, 1, 1]	FES = 511100	Diversity = 8.3442e-03	k_t
Generation 5120: Best = 4.685473e+03 = [2, 2, 3, 3]	FES = 512100	Diversity = 7.9504e-03	k_t
Generation 5130: Best = 4.685450e+03 = [1, 1, 1, 1]	FES = 513100	Diversity = 7.5475e-03	k_t
Generation 5140: Best = 4.685434e+03 = [3, 2, 3, 2]	FES = 514100	Diversity = 7.7667e-03	k_t
Generation 5150: Best = 4.685416e+03 = [1, 1, 1, 1]	FES = 515100	Diversity = 7.6288e-03	k_t
Generation 5160: Best = 4.685402e+03 = [3, 4, 4, 2]	FES = 516100	Diversity = 7.4704e-03	k_t
Generation 5170: Best = 4.685388e+03 = [1, 1, 1, 1]	FES = 517100	Diversity = 6.6486e-03	k_t
Generation 5180: Best = 4.685370e+03 = [3, 4, 3, 2]	FES = 518100	Diversity = 6.8297e-03	k_t
Generation 5190: Best = 4.685355e+03 = [1, 1, 1, 1]	FES = 519100	Diversity = 6.9045e-03	k_t
Generation 5200: Best = 4.685345e+03 = [3, 3, 2, 4]	FES = 520100	Diversity = 6.8379e-03	k_t
Generation 5210: Best = 4.685321e+03 = [1, 1, 1, 1]	FES = 521100	Diversity = 7.2042e-03	k_t
Generation 5220: Best = 4.685300e+03 = [3, 3, 2, 4]	FES = 522100	Diversity = 7.4903e-03	k_t
Generation 5230: Best = 4.685283e+03 = [1, 1, 1, 1]	FES = 523100	Diversity = 7.7950e-03	k_t
Generation 5240: Best = 4.685280e+03 = [4, 2, 4, 4]	FES = 524100	Diversity = 7.6022e-03	k_t
Generation 5250: Best = 4.685268e+03 = [1, 1, 1, 1]	FES = 525100	Diversity = 7.6550e-03	k_t
Generation 5260: Best = 4.685253e+03 = [3, 3, 4, 3]	FES = 526100	Diversity = 7.6876e-03	k_t
Generation 5270: Best = 4.685214e+03 = [1, 1, 1, 1]	FES = 527100	Diversity = 7.5489e-03	k_t
Generation 5280: Best = 4.685209e+03 = [4, 2, 2, 2]	FES = 528100	Diversity = 7.3061e-03	k_t
Generation 5290: Best = 4.685185e+03 = [1, 1, 1, 1]	FES = 529100	Diversity = 7.7420e-03	k_t
Generation 5300: Best = 4.685165e+03 = [4, 4, 3, 4]	FES = 530100	Diversity = 8.6228e-03	k_t
Generation 5310: Best = 4.685152e+03 = [1, 1, 1, 1]	FES = 531100	Diversity = 8.4967e-03	k_t
Generation 5320: Best = 4.685124e+03 = [3, 4, 2, 3]	FES = 532100	Diversity = 8.6837e-03	k_t
Generation 5330: Best = 4.685109e+03 = [1, 1, 1, 1]	FES = 533100	Diversity = 8.4152e-03	k_t
Generation 5340: Best = 4.685078e+03 = [2, 2, 4, 4]	FES = 534100	Diversity = 8.3370e-03	k_t
Generation 5350: Best = 4.685060e+03 = [1, 1, 1, 1]	FES = 535100	Diversity = 7.8175e-03	k_t

Generation 5360: Best = 4.685038e+03 = [2, 3, 2, 4]	FES = 536100	Diversity = 7.9380e-03	k_t
Generation 5370: Best = 4.685026e+03 = [1, 1, 1, 1]	FES = 537100	Diversity = 7.9227e-03	k_t
Generation 5380: Best = 4.685000e+03 = [4, 4, 3, 3]	FES = 538100	Diversity = 8.2839e-03	k_t
Generation 5390: Best = 4.684982e+03 = [1, 1, 1, 1]	FES = 539100	Diversity = 7.9869e-03	k_t
Generation 5400: Best = 4.684958e+03 = [3, 3, 4, 4]	FES = 540100	Diversity = 7.6063e-03	k_t
Generation 5410: Best = 4.684938e+03 = [1, 1, 1, 1]	FES = 541100	Diversity = 7.3015e-03	k_t
Generation 5420: Best = 4.684932e+03 = [3, 3, 3, 3]	FES = 542100	Diversity = 7.2619e-03	k_t
Generation 5430: Best = 4.684913e+03 = [1, 1, 1, 1]	FES = 543100	Diversity = 6.6195e-03	k_t
Generation 5440: Best = 4.684902e+03 = [4, 2, 2, 2]	FES = 544100	Diversity = 6.4699e-03	k_t
Generation 5450: Best = 4.684890e+03 = [1, 1, 1, 1]	FES = 545100	Diversity = 6.5220e-03	k_t
Generation 5460: Best = 4.684872e+03 = [3, 4, 3, 4]	FES = 546100	Diversity = 6.9137e-03	k_t
Generation 5470: Best = 4.684852e+03 = [1, 1, 1, 1]	FES = 547100	Diversity = 6.7461e-03	k_t
Generation 5480: Best = 4.684845e+03 = [4, 4, 2, 4]	FES = 548100	Diversity = 6.4183e-03	k_t
Generation 5490: Best = 4.684828e+03 = [1, 1, 1, 1]	FES = 549100	Diversity = 6.1801e-03	k_t
Generation 5500: Best = 4.684808e+03 = [3, 4, 3, 2]	FES = 550100	Diversity = 6.7858e-03	k_t
Generation 5510: Best = 4.684796e+03 = [1, 1, 1, 1]	FES = 551100	Diversity = 6.5561e-03	k_t
Generation 5520: Best = 4.684778e+03 = [2, 2, 4, 2]	FES = 552100	Diversity = 6.5795e-03	k_t
Generation 5530: Best = 4.684768e+03 = [1, 1, 1, 1]	FES = 553100	Diversity = 6.6580e-03	k_t
Generation 5540: Best = 4.684748e+03 = [3, 2, 4, 3]	FES = 554100	Diversity = 6.7965e-03	k_t
Generation 5550: Best = 4.684727e+03 = [1, 1, 1, 1]	FES = 555100	Diversity = 6.9446e-03	k_t
Generation 5560: Best = 4.684714e+03 = [4, 3, 4, 2]	FES = 556100	Diversity = 7.0959e-03	k_t
Generation 5570: Best = 4.684713e+03 = [1, 1, 1, 1]	FES = 557100	Diversity = 6.9612e-03	k_t
Generation 5580: Best = 4.684698e+03 = [4, 2, 4, 4]	FES = 558100	Diversity = 6.7002e-03	k_t
Generation 5590: Best = 4.684678e+03 = [1, 1, 1, 1]	FES = 559100	Diversity = 6.3704e-03	k_t
Generation 5600: Best = 4.684665e+03 = [4, 3, 2, 2]	FES = 560100	Diversity = 6.0727e-03	k_t
Generation 5610: Best = 4.684660e+03 = [1, 1, 1, 1]	FES = 561100	Diversity = 5.8834e-03	k_t
Generation 5620: Best = 4.684651e+03 = [4, 3, 4, 4]	FES = 562100	Diversity = 5.7086e-03	k_t

Generation 5630: Best = 4.684642e+03 = [1, 1, 1, 1]	FES = 563100	Diversity = 5.4839e-03	k_t
Generation 5640: Best = 4.684633e+03 = [3, 3, 2, 4]	FES = 564100	Diversity = 5.4550e-03	k_t
Generation 5650: Best = 4.684624e+03 = [1, 1, 1, 1]	FES = 565100	Diversity = 5.5532e-03	k_t
Generation 5660: Best = 4.684621e+03 = [4, 4, 2, 2]	FES = 566100	Diversity = 5.3206e-03	k_t
Generation 5670: Best = 4.684614e+03 = [1, 1, 1, 1]	FES = 567100	Diversity = 5.2609e-03	k_t
Generation 5680: Best = 4.684602e+03 = [3, 3, 2, 2]	FES = 568100	Diversity = 5.1459e-03	k_t
Generation 5690: Best = 4.684598e+03 = [1, 1, 1, 1]	FES = 569100	Diversity = 4.9618e-03	k_t
Generation 5700: Best = 4.684588e+03 = [2, 3, 4, 4]	FES = 570100	Diversity = 4.7190e-03	k_t
Generation 5710: Best = 4.684581e+03 = [1, 1, 1, 1]	FES = 571100	Diversity = 5.0666e-03	k_t
Generation 5720: Best = 4.684571e+03 = [4, 3, 3, 4]	FES = 572100	Diversity = 4.9144e-03	k_t
Generation 5730: Best = 4.684567e+03 = [1, 1, 1, 1]	FES = 573100	Diversity = 4.8909e-03	k_t
Generation 5740: Best = 4.684556e+03 = [3, 2, 2, 2]	FES = 574100	Diversity = 5.1341e-03	k_t
Generation 5750: Best = 4.684542e+03 = [1, 1, 1, 1]	FES = 575100	Diversity = 5.2650e-03	k_t
Generation 5760: Best = 4.684536e+03 = [4, 4, 4, 3]	FES = 576100	Diversity = 5.4406e-03	k_t
Generation 5770: Best = 4.684530e+03 = [1, 1, 1, 1]	FES = 577100	Diversity = 5.5036e-03	k_t
Generation 5780: Best = 4.684518e+03 = [2, 3, 4, 4]	FES = 578100	Diversity = 5.3711e-03	k_t
Generation 5790: Best = 4.684504e+03 = [1, 1, 1, 1]	FES = 579100	Diversity = 5.4725e-03	k_t
Generation 5800: Best = 4.684497e+03 = [3, 2, 2, 3]	FES = 580100	Diversity = 5.3205e-03	k_t
Generation 5810: Best = 4.684492e+03 = [1, 1, 1, 1]	FES = 581100	Diversity = 4.7412e-03	k_t
Generation 5820: Best = 4.684480e+03 = [4, 3, 3, 3]	FES = 582100	Diversity = 4.7372e-03	k_t
Generation 5830: Best = 4.684473e+03 = [1, 1, 1, 1]	FES = 583100	Diversity = 4.7113e-03	k_t
Generation 5840: Best = 4.684459e+03 = [3, 4, 3, 2]	FES = 584100	Diversity = 4.5863e-03	k_t
Generation 5850: Best = 4.684454e+03 = [1, 1, 1, 1]	FES = 585100	Diversity = 4.7282e-03	k_t
Generation 5860: Best = 4.684446e+03 = [2, 2, 2, 2]	FES = 586100	Diversity = 4.8112e-03	k_t
Generation 5870: Best = 4.684435e+03 = [1, 1, 1, 1]	FES = 587100	Diversity = 5.2936e-03	k_t
Generation 5880: Best = 4.684424e+03 = [3, 2, 4, 4]	FES = 588100	Diversity = 5.4723e-03	k_t
Generation 5890: Best = 4.684416e+03 = [1, 1, 1, 1]	FES = 589100	Diversity = 5.6089e-03	k_t

Generation 5900: Best = 4.684406e+03 = [2, 2, 3, 2]	FES = 590100	Diversity = 5.6953e-03	k_t
Generation 5910: Best = 4.684395e+03 = [1, 1, 1, 1]	FES = 591100	Diversity = 5.8950e-03	k_t
Generation 5920: Best = 4.684375e+03 = [2, 2, 4, 2]	FES = 592100	Diversity = 6.1617e-03	k_t
Generation 5930: Best = 4.684363e+03 = [1, 1, 1, 1]	FES = 593100	Diversity = 6.5212e-03	k_t
Generation 5940: Best = 4.684349e+03 = [4, 4, 3, 3]	FES = 594100	Diversity = 6.9805e-03	k_t
Generation 5950: Best = 4.684334e+03 = [1, 1, 1, 1]	FES = 595100	Diversity = 6.9361e-03	k_t
Generation 5960: Best = 4.684320e+03 = [4, 2, 3, 3]	FES = 596100	Diversity = 6.9332e-03	k_t
Generation 5970: Best = 4.684315e+03 = [1, 1, 1, 1]	FES = 597100	Diversity = 6.7083e-03	k_t
Generation 5980: Best = 4.684299e+03 = [4, 3, 4, 4]	FES = 598100	Diversity = 6.5600e-03	k_t
Generation 5990: Best = 4.684291e+03 = [1, 1, 1, 1]	FES = 599100	Diversity = 6.4376e-03	k_t
Generation 6000: Best = 4.684277e+03 = [3, 2, 2, 2]	FES = 600100	Diversity = 6.0167e-03	k_t
Generation 6010: Best = 4.684263e+03 = [1, 1, 1, 1]	FES = 601100	Diversity = 5.5839e-03	k_t
Generation 6020: Best = 4.684254e+03 = [4, 4, 4, 2]	FES = 602100	Diversity = 5.5229e-03	k_t
Generation 6030: Best = 4.684249e+03 = [1, 1, 1, 1]	FES = 603100	Diversity = 5.5790e-03	k_t
Generation 6040: Best = 4.684248e+03 = [2, 2, 2, 4]	FES = 604100	Diversity = 5.7506e-03	k_t
Generation 6050: Best = 4.684236e+03 = [1, 1, 1, 1]	FES = 605100	Diversity = 5.5182e-03	k_t
Generation 6060: Best = 4.684229e+03 = [2, 2, 2, 2]	FES = 606100	Diversity = 5.6256e-03	k_t
Generation 6070: Best = 4.684219e+03 = [1, 1, 1, 1]	FES = 607100	Diversity = 5.6421e-03	k_t
Generation 6080: Best = 4.684209e+03 = [3, 3, 3, 4]	FES = 608100	Diversity = 5.4709e-03	k_t
Generation 6090: Best = 4.684203e+03 = [1, 1, 1, 1]	FES = 609100	Diversity = 5.1705e-03	k_t
Generation 6100: Best = 4.684193e+03 = [2, 3, 3, 2]	FES = 610100	Diversity = 5.2734e-03	k_t
Generation 6110: Best = 4.684187e+03 = [1, 1, 1, 1]	FES = 611100	Diversity = 5.1410e-03	k_t
Generation 6120: Best = 4.684181e+03 = [2, 2, 3, 4]	FES = 612100	Diversity = 5.0803e-03	k_t
Generation 6130: Best = 4.684173e+03 = [1, 1, 1, 1]	FES = 613100	Diversity = 4.7647e-03	k_t
Generation 6140: Best = 4.684166e+03 = [4, 3, 2, 3]	FES = 614100	Diversity = 4.1658e-03	k_t
Generation 6150: Best = 4.684159e+03 = [1, 1, 1, 1]	FES = 615100	Diversity = 4.2020e-03	k_t
Generation 6160: Best = 4.684154e+03 = [2, 4, 3, 3]	FES = 616100	Diversity = 4.0644e-03	k_t

Generation 6170: Best = 4.684145e+03 = [1, 1, 1, 1]	FES = 617100	Diversity = 4.4766e-03	k_t
Generation 6180: Best = 4.684141e+03 = [2, 4, 3, 4]	FES = 618100	Diversity = 4.7152e-03	k_t
Generation 6190: Best = 4.684137e+03 = [1, 1, 1, 1]	FES = 619100	Diversity = 4.5311e-03	k_t
Generation 6200: Best = 4.684131e+03 = [2, 2, 2, 3]	FES = 620100	Diversity = 4.5280e-03	k_t
Generation 6210: Best = 4.684123e+03 = [1, 1, 1, 1]	FES = 621100	Diversity = 4.5127e-03	k_t
Generation 6220: Best = 4.684117e+03 = [4, 3, 4, 2]	FES = 622100	Diversity = 4.3481e-03	k_t
Generation 6230: Best = 4.684113e+03 = [1, 1, 1, 1]	FES = 623100	Diversity = 4.2270e-03	k_t
Generation 6240: Best = 4.684110e+03 = [3, 2, 4, 3]	FES = 624100	Diversity = 4.1323e-03	k_t
Generation 6250: Best = 4.684104e+03 = [1, 1, 1, 1]	FES = 625100	Diversity = 4.0565e-03	k_t
Generation 6260: Best = 4.684099e+03 = [4, 3, 4, 3]	FES = 626100	Diversity = 4.1595e-03	k_t
Generation 6270: Best = 4.684095e+03 = [1, 1, 1, 1]	FES = 627100	Diversity = 4.0802e-03	k_t
Generation 6280: Best = 4.684090e+03 = [4, 4, 2, 2]	FES = 628100	Diversity = 4.1508e-03	k_t
Generation 6290: Best = 4.684086e+03 = [1, 1, 1, 1]	FES = 629100	Diversity = 4.0505e-03	k_t
Generation 6300: Best = 4.684081e+03 = [2, 3, 2, 2]	FES = 630100	Diversity = 3.9409e-03	k_t
Generation 6310: Best = 4.684077e+03 = [1, 1, 1, 1]	FES = 631100	Diversity = 3.8799e-03	k_t
Generation 6320: Best = 4.684072e+03 = [3, 2, 2, 3]	FES = 632100	Diversity = 3.8311e-03	k_t
Generation 6330: Best = 4.684069e+03 = [1, 1, 1, 1]	FES = 633100	Diversity = 3.7473e-03	k_t
Generation 6340: Best = 4.684065e+03 = [3, 4, 2, 2]	FES = 634100	Diversity = 3.6470e-03	k_t
Generation 6350: Best = 4.684058e+03 = [1, 1, 1, 1]	FES = 635100	Diversity = 3.7603e-03	k_t
Generation 6360: Best = 4.684050e+03 = [2, 3, 3, 2]	FES = 636100	Diversity = 4.3651e-03	k_t
Generation 6370: Best = 4.684044e+03 = [1, 1, 1, 1]	FES = 637100	Diversity = 4.3489e-03	k_t
Generation 6380: Best = 4.684042e+03 = [2, 4, 3, 4]	FES = 638100	Diversity = 4.3195e-03	k_t
Generation 6390: Best = 4.684034e+03 = [1, 1, 1, 1]	FES = 639100	Diversity = 4.2017e-03	k_t
Generation 6400: Best = 4.684027e+03 = [2, 2, 4, 2]	FES = 640100	Diversity = 4.5744e-03	k_t
Generation 6410: Best = 4.684021e+03 = [1, 1, 1, 1]	FES = 641100	Diversity = 4.7462e-03	k_t
Generation 6420: Best = 4.684013e+03 = [4, 3, 2, 4]	FES = 642100	Diversity = 4.9230e-03	k_t
Generation 6430: Best = 4.684010e+03 = [1, 1, 1, 1]	FES = 643100	Diversity = 5.0585e-03	k_t

Generation 6440: Best = 4.683998e+03 = [2, 2, 3, 4]	FES = 644100	Diversity = 4.6823e-03	k_t
Generation 6450: Best = 4.683993e+03 = [1, 1, 1, 1]	FES = 645100	Diversity = 4.3515e-03	k_t
Generation 6460: Best = 4.683990e+03 = [2, 4, 4, 3]	FES = 646100	Diversity = 4.3881e-03	k_t
Generation 6470: Best = 4.683984e+03 = [1, 1, 1, 1]	FES = 647100	Diversity = 4.3956e-03	k_t
Generation 6480: Best = 4.683977e+03 = [2, 2, 4, 3]	FES = 648100	Diversity = 4.4575e-03	k_t
Generation 6490: Best = 4.683972e+03 = [1, 1, 1, 1]	FES = 649100	Diversity = 4.7371e-03	k_t
Generation 6500: Best = 4.683960e+03 = [4, 4, 2, 4]	FES = 650100	Diversity = 4.7411e-03	k_t
Generation 6510: Best = 4.683952e+03 = [1, 1, 1, 1]	FES = 651100	Diversity = 4.9747e-03	k_t
Generation 6520: Best = 4.683950e+03 = [2, 3, 3, 4]	FES = 652100	Diversity = 5.0074e-03	k_t
Generation 6530: Best = 4.683942e+03 = [1, 1, 1, 1]	FES = 653100	Diversity = 4.8060e-03	k_t
Generation 6540: Best = 4.683936e+03 = [4, 3, 4, 2]	FES = 654100	Diversity = 4.7618e-03	k_t
Generation 6550: Best = 4.683928e+03 = [1, 1, 1, 1]	FES = 655100	Diversity = 4.5227e-03	k_t
Generation 6560: Best = 4.683924e+03 = [2, 2, 4, 4]	FES = 656100	Diversity = 4.2748e-03	k_t
Generation 6570: Best = 4.683921e+03 = [1, 1, 1, 1]	FES = 657100	Diversity = 4.4067e-03	k_t
Generation 6580: Best = 4.683913e+03 = [4, 3, 2, 4]	FES = 658100	Diversity = 4.5641e-03	k_t
Generation 6590: Best = 4.683905e+03 = [1, 1, 1, 1]	FES = 659100	Diversity = 4.7721e-03	k_t
Generation 6600: Best = 4.683897e+03 = [4, 3, 2, 3]	FES = 660100	Diversity = 4.9974e-03	k_t
Generation 6610: Best = 4.683889e+03 = [1, 1, 1, 1]	FES = 661100	Diversity = 4.9319e-03	k_t
Generation 6620: Best = 4.683886e+03 = [2, 4, 4, 2]	FES = 662100	Diversity = 4.7242e-03	k_t
Generation 6630: Best = 4.683877e+03 = [1, 1, 1, 1]	FES = 663100	Diversity = 4.5274e-03	k_t
Generation 6640: Best = 4.683867e+03 = [2, 3, 3, 2]	FES = 664100	Diversity = 4.3657e-03	k_t
Generation 6650: Best = 4.683865e+03 = [1, 1, 1, 1]	FES = 665100	Diversity = 4.1026e-03	k_t
Generation 6660: Best = 4.683859e+03 = [3, 3, 2, 3]	FES = 666100	Diversity = 4.0047e-03	k_t
Generation 6670: Best = 4.683852e+03 = [1, 1, 1, 1]	FES = 667100	Diversity = 3.9436e-03	k_t
Generation 6680: Best = 4.683845e+03 = [2, 2, 4, 2]	FES = 668100	Diversity = 3.9608e-03	k_t
Generation 6690: Best = 4.683841e+03 = [1, 1, 1, 1]	FES = 669100	Diversity = 4.0070e-03	k_t
Generation 6700: Best = 4.683834e+03 = [3, 2, 3, 4]	FES = 670100	Diversity = 3.9350e-03	k_t

Generation 6710: Best = 4.683827e+03 = [1, 1, 1, 1]	FES = 671100	Diversity = 3.9730e-03	k_t
Generation 6720: Best = 4.683822e+03 = [3, 2, 4, 2]	FES = 672100	Diversity = 4.0559e-03	k_t
Generation 6730: Best = 4.683816e+03 = [1, 1, 1, 1]	FES = 673100	Diversity = 4.1456e-03	k_t
Generation 6740: Best = 4.683810e+03 = [2, 4, 3, 2]	FES = 674100	Diversity = 4.2056e-03	k_t
Generation 6750: Best = 4.683803e+03 = [1, 1, 1, 1]	FES = 675100	Diversity = 4.0982e-03	k_t
Generation 6760: Best = 4.683799e+03 = [4, 4, 4, 4]	FES = 676100	Diversity = 4.1931e-03	k_t
Generation 6770: Best = 4.683792e+03 = [1, 1, 1, 1]	FES = 677100	Diversity = 3.8707e-03	k_t
Generation 6780: Best = 4.683787e+03 = [4, 2, 2, 2]	FES = 678100	Diversity = 3.9520e-03	k_t
Generation 6790: Best = 4.683784e+03 = [1, 1, 1, 1]	FES = 679100	Diversity = 3.9078e-03	k_t
Generation 6800: Best = 4.683778e+03 = [4, 2, 3, 2]	FES = 680100	Diversity = 4.0194e-03	k_t
Generation 6810: Best = 4.683776e+03 = [1, 1, 1, 1]	FES = 681100	Diversity = 3.8912e-03	k_t
Generation 6820: Best = 4.683770e+03 = [2, 2, 4, 4]	FES = 682100	Diversity = 3.7638e-03	k_t
Generation 6830: Best = 4.683767e+03 = [1, 1, 1, 1]	FES = 683100	Diversity = 3.4574e-03	k_t
Generation 6840: Best = 4.683763e+03 = [4, 4, 2, 3]	FES = 684100	Diversity = 3.4606e-03	k_t
Generation 6850: Best = 4.683757e+03 = [1, 1, 1, 1]	FES = 685100	Diversity = 3.6512e-03	k_t
Generation 6860: Best = 4.683751e+03 = [2, 4, 2, 4]	FES = 686100	Diversity = 3.9086e-03	k_t
Generation 6870: Best = 4.683746e+03 = [1, 1, 1, 1]	FES = 687100	Diversity = 4.0056e-03	k_t
Generation 6880: Best = 4.683740e+03 = [2, 4, 3, 3]	FES = 688100	Diversity = 4.1207e-03	k_t
Generation 6890: Best = 4.683735e+03 = [1, 1, 1, 1]	FES = 689100	Diversity = 4.2997e-03	k_t
Generation 6900: Best = 4.683729e+03 = [2, 2, 3, 2]	FES = 690100	Diversity = 4.2081e-03	k_t
Generation 6910: Best = 4.683724e+03 = [1, 1, 1, 1]	FES = 691100	Diversity = 4.2940e-03	k_t
Generation 6920: Best = 4.683722e+03 = [4, 4, 2, 2]	FES = 692100	Diversity = 4.2220e-03	k_t
Generation 6930: Best = 4.683715e+03 = [1, 1, 1, 1]	FES = 693100	Diversity = 4.0579e-03	k_t
Generation 6940: Best = 4.683712e+03 = [4, 3, 4, 4]	FES = 694100	Diversity = 3.9615e-03	k_t
Generation 6950: Best = 4.683708e+03 = [1, 1, 1, 1]	FES = 695100	Diversity = 3.8714e-03	k_t
Generation 6960: Best = 4.683704e+03 = [2, 2, 4, 2]	FES = 696100	Diversity = 3.6186e-03	k_t
Generation 6970: Best = 4.683702e+03 = [1, 1, 1, 1]	FES = 697100	Diversity = 3.3096e-03	k_t

Generation 6980: Best = 4.683698e+03 = [4, 2, 4, 3]	FES = 698100	Diversity = 3.3258e-03	k_t
Generation 6990: Best = 4.683696e+03 = [1, 1, 1, 1]	FES = 699100	Diversity = 3.1773e-03	k_t
Generation 7000: Best = 4.683694e+03 = [2, 2, 3, 4]	FES = 700100	Diversity = 2.9890e-03	k_t
Generation 7010: Best = 4.683691e+03 = [1, 1, 1, 1]	FES = 701100	Diversity = 2.9151e-03	k_t
Generation 7020: Best = 4.683687e+03 = [4, 2, 2, 3]	FES = 702100	Diversity = 2.7479e-03	k_t
Generation 7030: Best = 4.683685e+03 = [1, 1, 1, 1]	FES = 703100	Diversity = 2.8571e-03	k_t
Generation 7040: Best = 4.683683e+03 = [3, 3, 4, 2]	FES = 704100	Diversity = 2.7555e-03	k_t
Generation 7050: Best = 4.683681e+03 = [1, 1, 1, 1]	FES = 705100	Diversity = 2.7120e-03	k_t
Generation 7060: Best = 4.683677e+03 = [2, 3, 3, 4]	FES = 706100	Diversity = 2.6967e-03	k_t
Generation 7070: Best = 4.683674e+03 = [1, 1, 1, 1]	FES = 707100	Diversity = 2.7079e-03	k_t
Generation 7080: Best = 4.683673e+03 = [2, 3, 2, 4]	FES = 708100	Diversity = 2.7714e-03	k_t
Generation 7090: Best = 4.683670e+03 = [1, 1, 1, 1]	FES = 709100	Diversity = 2.7351e-03	k_t
Generation 7100: Best = 4.683669e+03 = [3, 2, 2, 4]	FES = 710100	Diversity = 2.6314e-03	k_t
Generation 7110: Best = 4.683666e+03 = [1, 1, 1, 1]	FES = 711100	Diversity = 2.4981e-03	k_t
Generation 7120: Best = 4.683664e+03 = [3, 2, 4, 2]	FES = 712100	Diversity = 2.4206e-03	k_t
Generation 7130: Best = 4.683662e+03 = [1, 1, 1, 1]	FES = 713100	Diversity = 2.3809e-03	k_t
Generation 7140: Best = 4.683660e+03 = [2, 2, 3, 2]	FES = 714100	Diversity = 2.3988e-03	k_t
Generation 7150: Best = 4.683658e+03 = [1, 1, 1, 1]	FES = 715100	Diversity = 2.3262e-03	k_t
Generation 7160: Best = 4.683657e+03 = [3, 3, 3, 3]	FES = 716100	Diversity = 2.4111e-03	k_t
Generation 7170: Best = 4.683655e+03 = [1, 1, 1, 1]	FES = 717100	Diversity = 2.3918e-03	k_t
Generation 7180: Best = 4.683652e+03 = [4, 2, 3, 2]	FES = 718100	Diversity = 2.3942e-03	k_t
Generation 7190: Best = 4.683650e+03 = [1, 1, 1, 1]	FES = 719100	Diversity = 2.4358e-03	k_t
Generation 7200: Best = 4.683648e+03 = [4, 3, 2, 4]	FES = 720100	Diversity = 2.5418e-03	k_t
Generation 7210: Best = 4.683645e+03 = [1, 1, 1, 1]	FES = 721100	Diversity = 2.5329e-03	k_t
Generation 7220: Best = 4.683643e+03 = [2, 3, 4, 2]	FES = 722100	Diversity = 2.4533e-03	k_t
Generation 7230: Best = 4.683642e+03 = [1, 1, 1, 1]	FES = 723100	Diversity = 2.4873e-03	k_t
Generation 7240: Best = 4.683640e+03 = [3, 3, 3, 2]	FES = 724100	Diversity = 2.5185e-03	k_t

Generation 7250: Best = 4.683637e+03 = [1, 1, 1, 1]	FES = 725100	Diversity = 2.4374e-03	k_t
Generation 7260: Best = 4.683635e+03 = [2, 4, 2, 2]	FES = 726100	Diversity = 2.5565e-03	k_t
Generation 7270: Best = 4.683632e+03 = [1, 1, 1, 1]	FES = 727100	Diversity = 2.6947e-03	k_t
Generation 7280: Best = 4.683630e+03 = [2, 3, 4, 3]	FES = 728100	Diversity = 2.7325e-03	k_t
Generation 7290: Best = 4.683626e+03 = [1, 1, 1, 1]	FES = 729100	Diversity = 2.9305e-03	k_t
Generation 7300: Best = 4.683621e+03 = [2, 3, 3, 4]	FES = 730100	Diversity = 3.1330e-03	k_t
Generation 7310: Best = 4.683619e+03 = [1, 1, 1, 1]	FES = 731100	Diversity = 3.0731e-03	k_t
Generation 7320: Best = 4.683616e+03 = [2, 4, 4, 3]	FES = 732100	Diversity = 2.8308e-03	k_t
Generation 7330: Best = 4.683614e+03 = [1, 1, 1, 1]	FES = 733100	Diversity = 2.6788e-03	k_t
Generation 7340: Best = 4.683611e+03 = [3, 2, 4, 2]	FES = 734100	Diversity = 2.7433e-03	k_t
Generation 7350: Best = 4.683607e+03 = [1, 1, 1, 1]	FES = 735100	Diversity = 2.6937e-03	k_t
Generation 7360: Best = 4.683605e+03 = [3, 3, 3, 2]	FES = 736100	Diversity = 2.9934e-03	k_t
Generation 7370: Best = 4.683602e+03 = [1, 1, 1, 1]	FES = 737100	Diversity = 3.1689e-03	k_t
Generation 7380: Best = 4.683599e+03 = [2, 2, 4, 2]	FES = 738100	Diversity = 3.0682e-03	k_t
Generation 7390: Best = 4.683596e+03 = [1, 1, 1, 1]	FES = 739100	Diversity = 2.9720e-03	k_t
Generation 7400: Best = 4.683594e+03 = [2, 3, 3, 3]	FES = 740100	Diversity = 2.9582e-03	k_t
Generation 7410: Best = 4.683591e+03 = [1, 1, 1, 1]	FES = 741100	Diversity = 2.9245e-03	k_t
Generation 7420: Best = 4.683590e+03 = [2, 3, 3, 2]	FES = 742100	Diversity = 2.9264e-03	k_t
Generation 7430: Best = 4.683587e+03 = [1, 1, 1, 1]	FES = 743100	Diversity = 2.8986e-03	k_t
Generation 7440: Best = 4.683585e+03 = [3, 3, 4, 2]	FES = 744100	Diversity = 2.8673e-03	k_t
Generation 7450: Best = 4.683581e+03 = [1, 1, 1, 1]	FES = 745100	Diversity = 3.0303e-03	k_t
Generation 7460: Best = 4.683579e+03 = [4, 3, 2, 3]	FES = 746100	Diversity = 2.9210e-03	k_t
Generation 7470: Best = 4.683575e+03 = [1, 1, 1, 1]	FES = 747100	Diversity = 2.9690e-03	k_t
Generation 7480: Best = 4.683573e+03 = [2, 4, 2, 2]	FES = 748100	Diversity = 2.8380e-03	k_t
Generation 7490: Best = 4.683571e+03 = [1, 1, 1, 1]	FES = 749100	Diversity = 2.8636e-03	k_t
Generation 7500: Best = 4.683568e+03 = [3, 2, 2, 3]	FES = 750100	Diversity = 2.7871e-03	k_t
Generation 7510: Best = 4.683566e+03 = [1, 1, 1, 1]	FES = 751100	Diversity = 2.5824e-03	k_t

Generation 7520: Best = 4.683564e+03 = [4, 2, 4, 3]	FES = 752100	Diversity = 2.5895e-03	k_t
Generation 7530: Best = 4.683560e+03 = [1, 1, 1, 1]	FES = 753100	Diversity = 2.5859e-03	k_t
Generation 7540: Best = 4.683555e+03 = [4, 3, 2, 4]	FES = 754100	Diversity = 2.8686e-03	k_t
Generation 7550: Best = 4.683546e+03 = [1, 1, 1, 1]	FES = 755100	Diversity = 3.3270e-03	k_t
Generation 7560: Best = 4.683536e+03 = [2, 4, 4, 2]	FES = 756100	Diversity = 4.2336e-03	k_t
Generation 7570: Best = 4.683523e+03 = [1, 1, 1, 1]	FES = 757100	Diversity = 4.9813e-03	k_t
Generation 7580: Best = 4.683507e+03 = [4, 4, 3, 4]	FES = 758100	Diversity = 5.4465e-03	k_t
Generation 7590: Best = 4.683500e+03 = [1, 1, 1, 1]	FES = 759100	Diversity = 5.6557e-03	k_t
Generation 7600: Best = 4.683486e+03 = [2, 4, 2, 3]	FES = 760100	Diversity = 5.9341e-03	k_t
Generation 7610: Best = 4.683476e+03 = [1, 1, 1, 1]	FES = 761100	Diversity = 6.0099e-03	k_t
Generation 7620: Best = 4.683469e+03 = [2, 2, 3, 4]	FES = 762100	Diversity = 6.1996e-03	k_t
Generation 7630: Best = 4.683456e+03 = [1, 1, 1, 1]	FES = 763100	Diversity = 6.0015e-03	k_t
Generation 7640: Best = 4.683452e+03 = [3, 3, 2, 4]	FES = 764100	Diversity = 5.9846e-03	k_t
Generation 7650: Best = 4.683447e+03 = [1, 1, 1, 1]	FES = 765100	Diversity = 5.6716e-03	k_t
Generation 7660: Best = 4.683440e+03 = [4, 4, 4, 4]	FES = 766100	Diversity = 5.3312e-03	k_t
Generation 7670: Best = 4.683436e+03 = [1, 1, 1, 1]	FES = 767100	Diversity = 4.9274e-03	k_t
Generation 7680: Best = 4.683431e+03 = [3, 3, 2, 2]	FES = 768100	Diversity = 4.5066e-03	k_t
Generation 7690: Best = 4.683429e+03 = [1, 1, 1, 1]	FES = 769100	Diversity = 4.3433e-03	k_t
Generation 7700: Best = 4.683425e+03 = [2, 4, 3, 4]	FES = 770100	Diversity = 4.1392e-03	k_t
Generation 7710: Best = 4.683423e+03 = [1, 1, 1, 1]	FES = 771100	Diversity = 3.9069e-03	k_t
Generation 7720: Best = 4.683419e+03 = [2, 3, 4, 3]	FES = 772100	Diversity = 3.5351e-03	k_t
Generation 7730: Best = 4.683417e+03 = [1, 1, 1, 1]	FES = 773100	Diversity = 3.2879e-03	k_t
Generation 7740: Best = 4.683415e+03 = [2, 2, 4, 3]	FES = 774100	Diversity = 3.1937e-03	k_t
Generation 7750: Best = 4.683414e+03 = [1, 1, 1, 1]	FES = 775100	Diversity = 2.9565e-03	k_t
Generation 7760: Best = 4.683410e+03 = [2, 4, 4, 3]	FES = 776100	Diversity = 2.8963e-03	k_t
Generation 7770: Best = 4.683406e+03 = [1, 1, 1, 1]	FES = 777100	Diversity = 3.0590e-03	k_t
Generation 7780: Best = 4.683404e+03 = [2, 4, 2, 2]	FES = 778100	Diversity = 3.0760e-03	k_t

Generation 7790: Best = 4.683402e+03 = [1, 1, 1, 1]	FES = 779100	Diversity = 2.8516e-03	k_t
Generation 7800: Best = 4.683400e+03 = [2, 3, 4, 4]	FES = 780100	Diversity = 3.0226e-03	k_t
Generation 7810: Best = 4.683398e+03 = [1, 1, 1, 1]	FES = 781100	Diversity = 3.0416e-03	k_t
Generation 7820: Best = 4.683397e+03 = [4, 2, 4, 2]	FES = 782100	Diversity = 2.9376e-03	k_t
Generation 7830: Best = 4.683394e+03 = [1, 1, 1, 1]	FES = 783100	Diversity = 2.9471e-03	k_t
Generation 7840: Best = 4.683391e+03 = [2, 2, 4, 4]	FES = 784100	Diversity = 3.0403e-03	k_t
Generation 7850: Best = 4.683390e+03 = [1, 1, 1, 1]	FES = 785100	Diversity = 3.0015e-03	k_t
Generation 7860: Best = 4.683386e+03 = [3, 3, 3, 4]	FES = 786100	Diversity = 2.9876e-03	k_t
Generation 7870: Best = 4.683386e+03 = [1, 1, 1, 1]	FES = 787100	Diversity = 2.8456e-03	k_t
Generation 7880: Best = 4.683383e+03 = [2, 2, 3, 3]	FES = 788100	Diversity = 2.8294e-03	k_t
Generation 7890: Best = 4.683382e+03 = [1, 1, 1, 1]	FES = 789100	Diversity = 2.6055e-03	k_t
Generation 7900: Best = 4.683381e+03 = [4, 4, 4, 3]	FES = 790100	Diversity = 2.5118e-03	k_t
Generation 7910: Best = 4.683380e+03 = [1, 1, 1, 1]	FES = 791100	Diversity = 2.5244e-03	k_t
Generation 7920: Best = 4.683377e+03 = [3, 4, 3, 3]	FES = 792100	Diversity = 2.4205e-03	k_t
Generation 7930: Best = 4.683376e+03 = [1, 1, 1, 1]	FES = 793100	Diversity = 2.3811e-03	k_t
Generation 7940: Best = 4.683374e+03 = [4, 2, 3, 4]	FES = 794100	Diversity = 2.1417e-03	k_t
Generation 7950: Best = 4.683372e+03 = [1, 1, 1, 1]	FES = 795100	Diversity = 2.2535e-03	k_t
Generation 7960: Best = 4.683370e+03 = [2, 3, 4, 3]	FES = 796100	Diversity = 2.2905e-03	k_t
Generation 7970: Best = 4.683368e+03 = [1, 1, 1, 1]	FES = 797100	Diversity = 2.3695e-03	k_t
Generation 7980: Best = 4.683366e+03 = [4, 4, 4, 4]	FES = 798100	Diversity = 2.3157e-03	k_t
Generation 7990: Best = 4.683365e+03 = [1, 1, 1, 1]	FES = 799100	Diversity = 2.1564e-03	k_t
Generation 8000: Best = 4.683364e+03 = [3, 3, 4, 3]	FES = 800100	Diversity = 2.0946e-03	k_t
Generation 8010: Best = 4.683363e+03 = [1, 1, 1, 1]	FES = 801100	Diversity = 2.0408e-03	k_t
Generation 8020: Best = 4.683359e+03 = [2, 2, 3, 2]	FES = 802100	Diversity = 2.1609e-03	k_t
Generation 8030: Best = 4.683358e+03 = [1, 1, 1, 1]	FES = 803100	Diversity = 2.1919e-03	k_t
Generation 8040: Best = 4.683357e+03 = [2, 4, 2, 4]	FES = 804100	Diversity = 2.2133e-03	k_t
Generation 8050: Best = 4.683355e+03 = [1, 1, 1, 1]	FES = 805100	Diversity = 2.1250e-03	k_t

Generation 8060: Best = 4.683354e+03 = [3, 4, 4, 4]	FES = 806100	Diversity = 2.4145e-03	k_t
Generation 8070: Best = 4.683352e+03 = [1, 1, 1, 1]	FES = 807100	Diversity = 2.4407e-03	k_t
Generation 8080: Best = 4.683349e+03 = [4, 4, 4, 3]	FES = 808100	Diversity = 2.6521e-03	k_t
Generation 8090: Best = 4.683348e+03 = [1, 1, 1, 1]	FES = 809100	Diversity = 2.7129e-03	k_t
Generation 8100: Best = 4.683345e+03 = [4, 2, 4, 2]	FES = 810100	Diversity = 2.6988e-03	k_t
Generation 8110: Best = 4.683343e+03 = [1, 1, 1, 1]	FES = 811100	Diversity = 2.7068e-03	k_t
Generation 8120: Best = 4.683341e+03 = [2, 2, 4, 4]	FES = 812100	Diversity = 2.5647e-03	k_t
Generation 8130: Best = 4.683339e+03 = [1, 1, 1, 1]	FES = 813100	Diversity = 2.4328e-03	k_t
Generation 8140: Best = 4.683337e+03 = [2, 2, 2, 2]	FES = 814100	Diversity = 2.4282e-03	k_t
Generation 8150: Best = 4.683336e+03 = [1, 1, 1, 1]	FES = 815100	Diversity = 2.3246e-03	k_t
Generation 8160: Best = 4.683335e+03 = [3, 4, 2, 3]	FES = 816100	Diversity = 2.2539e-03	k_t
Generation 8170: Best = 4.683334e+03 = [1, 1, 1, 1]	FES = 817100	Diversity = 2.2862e-03	k_t
Generation 8180: Best = 4.683331e+03 = [2, 4, 3, 2]	FES = 818100	Diversity = 2.2512e-03	k_t
Generation 8190: Best = 4.683331e+03 = [1, 1, 1, 1]	FES = 819100	Diversity = 2.0990e-03	k_t
Generation 8200: Best = 4.683330e+03 = [4, 3, 4, 4]	FES = 820100	Diversity = 2.2295e-03	k_t
Generation 8210: Best = 4.683328e+03 = [1, 1, 1, 1]	FES = 821100	Diversity = 2.1819e-03	k_t
Generation 8220: Best = 4.683326e+03 = [4, 4, 4, 2]	FES = 822100	Diversity = 2.1941e-03	k_t
Generation 8230: Best = 4.683326e+03 = [1, 1, 1, 1]	FES = 823100	Diversity = 2.2475e-03	k_t
Generation 8240: Best = 4.683324e+03 = [3, 2, 4, 2]	FES = 824100	Diversity = 2.0615e-03	k_t
Generation 8250: Best = 4.683324e+03 = [1, 1, 1, 1]	FES = 825100	Diversity = 1.9384e-03	k_t
Generation 8260: Best = 4.683322e+03 = [2, 2, 2, 4]	FES = 826100	Diversity = 1.8997e-03	k_t
Generation 8270: Best = 4.683322e+03 = [1, 1, 1, 1]	FES = 827100	Diversity = 1.8601e-03	k_t
Generation 8280: Best = 4.683321e+03 = [4, 3, 3, 4]	FES = 828100	Diversity = 1.8294e-03	k_t
Generation 8290: Best = 4.683319e+03 = [1, 1, 1, 1]	FES = 829100	Diversity = 1.8222e-03	k_t
Generation 8300: Best = 4.683319e+03 = [4, 2, 3, 3]	FES = 830100	Diversity = 1.8746e-03	k_t
Generation 8310: Best = 4.683318e+03 = [1, 1, 1, 1]	FES = 831100	Diversity = 1.9703e-03	k_t
Generation 8320: Best = 4.683317e+03 = [3, 4, 3, 3]	FES = 832100	Diversity = 1.8915e-03	k_t

Generation 8330: Best = 4.683316e+03 = [1, 1, 1, 1]	FES = 833100	Diversity = 1.8901e-03	k_t
Generation 8340: Best = 4.683315e+03 = [3, 3, 3, 3]	FES = 834100	Diversity = 1.8193e-03	k_t
Generation 8350: Best = 4.683315e+03 = [1, 1, 1, 1]	FES = 835100	Diversity = 1.8280e-03	k_t
Generation 8360: Best = 4.683314e+03 = [2, 4, 2, 2]	FES = 836100	Diversity = 1.6682e-03	k_t
Generation 8370: Best = 4.683313e+03 = [1, 1, 1, 1]	FES = 837100	Diversity = 1.6133e-03	k_t
Generation 8380: Best = 4.683312e+03 = [3, 4, 3, 3]	FES = 838100	Diversity = 1.6389e-03	k_t
Generation 8390: Best = 4.683312e+03 = [1, 1, 1, 1]	FES = 839100	Diversity = 1.5349e-03	k_t
Generation 8400: Best = 4.683310e+03 = [3, 2, 3, 2]	FES = 840100	Diversity = 1.6141e-03	k_t
Generation 8410: Best = 4.683309e+03 = [1, 1, 1, 1]	FES = 841100	Diversity = 1.6713e-03	k_t
Generation 8420: Best = 4.683309e+03 = [3, 2, 4, 2]	FES = 842100	Diversity = 1.6528e-03	k_t
Generation 8430: Best = 4.683308e+03 = [1, 1, 1, 1]	FES = 843100	Diversity = 1.6357e-03	k_t
Generation 8440: Best = 4.683307e+03 = [3, 2, 4, 2]	FES = 844100	Diversity = 1.6202e-03	k_t
Generation 8450: Best = 4.683306e+03 = [1, 1, 1, 1]	FES = 845100	Diversity = 1.5813e-03	k_t
Generation 8460: Best = 4.683305e+03 = [2, 4, 2, 2]	FES = 846100	Diversity = 1.4650e-03	k_t
Generation 8470: Best = 4.683304e+03 = [1, 1, 1, 1]	FES = 847100	Diversity = 1.4705e-03	k_t
Generation 8480: Best = 4.683304e+03 = [2, 2, 3, 2]	FES = 848100	Diversity = 1.5468e-03	k_t
Generation 8490: Best = 4.683302e+03 = [1, 1, 1, 1]	FES = 849100	Diversity = 1.4722e-03	k_t
Generation 8500: Best = 4.683302e+03 = [2, 4, 2, 3]	FES = 850100	Diversity = 1.4259e-03	k_t
Generation 8510: Best = 4.683301e+03 = [1, 1, 1, 1]	FES = 851100	Diversity = 1.4163e-03	k_t
Generation 8520: Best = 4.683300e+03 = [4, 3, 3, 2]	FES = 852100	Diversity = 1.4269e-03	k_t
Generation 8530: Best = 4.683299e+03 = [1, 1, 1, 1]	FES = 853100	Diversity = 1.4215e-03	k_t
Generation 8540: Best = 4.683299e+03 = [4, 2, 2, 4]	FES = 854100	Diversity = 1.3637e-03	k_t
Generation 8550: Best = 4.683298e+03 = [1, 1, 1, 1]	FES = 855100	Diversity = 1.3813e-03	k_t
Generation 8560: Best = 4.683297e+03 = [4, 2, 3, 4]	FES = 856100	Diversity = 1.3739e-03	k_t
Generation 8570: Best = 4.683297e+03 = [1, 1, 1, 1]	FES = 857100	Diversity = 1.4337e-03	k_t
Generation 8580: Best = 4.683296e+03 = [4, 3, 3, 2]	FES = 858100	Diversity = 1.4699e-03	k_t
Generation 8590: Best = 4.683294e+03 = [1, 1, 1, 1]	FES = 859100	Diversity = 1.4637e-03	k_t

Generation 8600: Best = 4.683294e+03	FES = 860100	Diversity = 1.4215e-03	k_t
= [2, 3, 3, 3]			
Generation 8610: Best = 4.683293e+03	FES = 861100	Diversity = 1.3872e-03	k_t
= [1, 1, 1, 1]			
Generation 8620: Best = 4.683293e+03	FES = 862100	Diversity = 1.3191e-03	k_t
= [4, 3, 3, 2]			
Generation 8630: Best = 4.683292e+03	FES = 863100	Diversity = 1.2964e-03	k_t
= [1, 1, 1, 1]			
Generation 8640: Best = 4.683292e+03	FES = 864100	Diversity = 1.3302e-03	k_t
= [4, 3, 4, 3]			
Generation 8650: Best = 4.683291e+03	FES = 865100	Diversity = 1.3238e-03	k_t
= [1, 1, 1, 1]			
Generation 8660: Best = 4.683290e+03	FES = 866100	Diversity = 1.3516e-03	k_t
= [4, 3, 2, 2]			
Generation 8670: Best = 4.683290e+03	FES = 867100	Diversity = 1.4144e-03	k_t
= [1, 1, 1, 1]			
Generation 8680: Best = 4.683289e+03	FES = 868100	Diversity = 1.2767e-03	k_t
= [4, 2, 4, 4]			
Generation 8690: Best = 4.683287e+03	FES = 869100	Diversity = 1.2598e-03	k_t
= [1, 1, 1, 1]			
Generation 8700: Best = 4.683287e+03	FES = 870100	Diversity = 1.4134e-03	k_t
= [4, 2, 2, 4]			
Generation 8710: Best = 4.683286e+03	FES = 871100	Diversity = 1.4618e-03	k_t
= [1, 1, 1, 1]			
Generation 8720: Best = 4.683285e+03	FES = 872100	Diversity = 1.5356e-03	k_t
= [2, 2, 4, 2]			
Generation 8730: Best = 4.683284e+03	FES = 873100	Diversity = 1.6193e-03	k_t
= [1, 1, 1, 1]			
Generation 8740: Best = 4.683283e+03	FES = 874100	Diversity = 1.6454e-03	k_t
= [3, 3, 2, 4]			
Generation 8750: Best = 4.683282e+03	FES = 875100	Diversity = 1.7893e-03	k_t
= [1, 1, 1, 1]			
Generation 8760: Best = 4.683281e+03	FES = 876100	Diversity = 1.7427e-03	k_t
= [4, 4, 3, 3]			
Generation 8770: Best = 4.683280e+03	FES = 877100	Diversity = 1.7685e-03	k_t
= [1, 1, 1, 1]			
Generation 8780: Best = 4.683279e+03	FES = 878100	Diversity = 1.7649e-03	k_t
= [2, 2, 4, 4]			
Generation 8790: Best = 4.683279e+03	FES = 879100	Diversity = 1.6983e-03	k_t
= [1, 1, 1, 1]			
Generation 8800: Best = 4.683278e+03	FES = 880100	Diversity = 1.6630e-03	k_t
= [2, 3, 3, 2]			
Generation 8810: Best = 4.683276e+03	FES = 881100	Diversity = 1.6453e-03	k_t
= [1, 1, 1, 1]			
Generation 8820: Best = 4.683276e+03	FES = 882100	Diversity = 1.5949e-03	k_t
= [2, 2, 2, 2]			
Generation 8830: Best = 4.683275e+03	FES = 883100	Diversity = 1.6252e-03	k_t
= [1, 1, 1, 1]			
Generation 8840: Best = 4.683274e+03	FES = 884100	Diversity = 1.6794e-03	k_t
= [2, 3, 4, 4]			
Generation 8850: Best = 4.683273e+03	FES = 885100	Diversity = 1.5285e-03	k_t
= [1, 1, 1, 1]			
Generation 8860: Best = 4.683273e+03	FES = 886100	Diversity = 1.4954e-03	k_t
= [3, 4, 2, 2]			

Generation 8870: Best = 4.683272e+03 = [1, 1, 1, 1]	FES = 887100	Diversity = 1.4088e-03	k_t
Generation 8880: Best = 4.683271e+03 = [2, 3, 2, 4]	FES = 888100	Diversity = 1.5052e-03	k_t
Generation 8890: Best = 4.683270e+03 = [1, 1, 1, 1]	FES = 889100	Diversity = 1.3989e-03	k_t
Generation 8900: Best = 4.683269e+03 = [3, 4, 4, 3]	FES = 890100	Diversity = 1.4497e-03	k_t
Generation 8910: Best = 4.683268e+03 = [1, 1, 1, 1]	FES = 891100	Diversity = 1.5425e-03	k_t
Generation 8920: Best = 4.683267e+03 = [3, 3, 2, 3]	FES = 892100	Diversity = 1.5499e-03	k_t
Generation 8930: Best = 4.683267e+03 = [1, 1, 1, 1]	FES = 893100	Diversity = 1.5062e-03	k_t
Generation 8940: Best = 4.683266e+03 = [3, 2, 3, 4]	FES = 894100	Diversity = 1.4821e-03	k_t
Generation 8950: Best = 4.683265e+03 = [1, 1, 1, 1]	FES = 895100	Diversity = 1.4933e-03	k_t
Generation 8960: Best = 4.683264e+03 = [4, 3, 2, 2]	FES = 896100	Diversity = 1.5557e-03	k_t
Generation 8970: Best = 4.683263e+03 = [1, 1, 1, 1]	FES = 897100	Diversity = 1.6050e-03	k_t
Generation 8980: Best = 4.683262e+03 = [4, 2, 2, 2]	FES = 898100	Diversity = 1.5410e-03	k_t
Generation 8990: Best = 4.683262e+03 = [1, 1, 1, 1]	FES = 899100	Diversity = 1.4874e-03	k_t
Generation 9000: Best = 4.683261e+03 = [3, 3, 4, 3]	FES = 900100	Diversity = 1.3871e-03	k_t
Generation 9010: Best = 4.683260e+03 = [1, 1, 1, 1]	FES = 901100	Diversity = 1.3649e-03	k_t
Generation 9020: Best = 4.683260e+03 = [3, 4, 3, 2]	FES = 902100	Diversity = 1.3392e-03	k_t
Generation 9030: Best = 4.683259e+03 = [1, 1, 1, 1]	FES = 903100	Diversity = 1.3201e-03	k_t
Generation 9040: Best = 4.683258e+03 = [4, 4, 2, 2]	FES = 904100	Diversity = 1.2599e-03	k_t
Generation 9050: Best = 4.683257e+03 = [1, 1, 1, 1]	FES = 905100	Diversity = 1.3736e-03	k_t
Generation 9060: Best = 4.683257e+03 = [3, 3, 2, 3]	FES = 906100	Diversity = 1.4606e-03	k_t
Generation 9070: Best = 4.683256e+03 = [1, 1, 1, 1]	FES = 907100	Diversity = 1.4517e-03	k_t
Generation 9080: Best = 4.683256e+03 = [3, 4, 4, 4]	FES = 908100	Diversity = 1.4277e-03	k_t
Generation 9090: Best = 4.683254e+03 = [1, 1, 1, 1]	FES = 909100	Diversity = 1.4528e-03	k_t
Generation 9100: Best = 4.683254e+03 = [3, 2, 2, 4]	FES = 910100	Diversity = 1.4825e-03	k_t
Generation 9110: Best = 4.683253e+03 = [1, 1, 1, 1]	FES = 911100	Diversity = 1.3913e-03	k_t
Generation 9120: Best = 4.683253e+03 = [3, 2, 3, 3]	FES = 912100	Diversity = 1.4314e-03	k_t
Generation 9130: Best = 4.683252e+03 = [1, 1, 1, 1]	FES = 913100	Diversity = 1.3432e-03	k_t

Generation 9140: Best = 4.683251e+03 = [2, 4, 4, 4]	FES = 914100	Diversity = 1.2869e-03	k_t
Generation 9150: Best = 4.683251e+03 = [1, 1, 1, 1]	FES = 915100	Diversity = 1.2989e-03	k_t
Generation 9160: Best = 4.683250e+03 = [4, 3, 3, 3]	FES = 916100	Diversity = 1.2513e-03	k_t
Generation 9170: Best = 4.683249e+03 = [1, 1, 1, 1]	FES = 917100	Diversity = 1.2160e-03	k_t
Generation 9180: Best = 4.683249e+03 = [4, 2, 3, 2]	FES = 918100	Diversity = 1.2001e-03	k_t
Generation 9190: Best = 4.683248e+03 = [1, 1, 1, 1]	FES = 919100	Diversity = 1.2032e-03	k_t
Generation 9200: Best = 4.683248e+03 = [2, 2, 3, 3]	FES = 920100	Diversity = 1.1595e-03	k_t
Generation 9210: Best = 4.683247e+03 = [1, 1, 1, 1]	FES = 921100	Diversity = 1.1071e-03	k_t
Generation 9220: Best = 4.683247e+03 = [4, 4, 3, 3]	FES = 922100	Diversity = 1.0677e-03	k_t
Generation 9230: Best = 4.683246e+03 = [1, 1, 1, 1]	FES = 923100	Diversity = 1.0337e-03	k_t
Generation 9240: Best = 4.683246e+03 = [3, 4, 2, 4]	FES = 924100	Diversity = 1.0481e-03	k_t
Generation 9250: Best = 4.683245e+03 = [1, 1, 1, 1]	FES = 925100	Diversity = 1.0497e-03	k_t
Generation 9260: Best = 4.683245e+03 = [3, 3, 2, 2]	FES = 926100	Diversity = 1.0434e-03	k_t
Generation 9270: Best = 4.683244e+03 = [1, 1, 1, 1]	FES = 927100	Diversity = 1.0441e-03	k_t
Generation 9280: Best = 4.683244e+03 = [4, 4, 3, 2]	FES = 928100	Diversity = 1.0444e-03	k_t
Generation 9290: Best = 4.683243e+03 = [1, 1, 1, 1]	FES = 929100	Diversity = 1.0581e-03	k_t
Generation 9300: Best = 4.683243e+03 = [2, 3, 2, 2]	FES = 930100	Diversity = 1.1203e-03	k_t
Generation 9310: Best = 4.683242e+03 = [1, 1, 1, 1]	FES = 931100	Diversity = 1.1570e-03	k_t
Generation 9320: Best = 4.683242e+03 = [3, 2, 4, 2]	FES = 932100	Diversity = 1.1662e-03	k_t
Generation 9330: Best = 4.683241e+03 = [1, 1, 1, 1]	FES = 933100	Diversity = 1.1072e-03	k_t
Generation 9340: Best = 4.683241e+03 = [4, 3, 4, 4]	FES = 934100	Diversity = 1.1819e-03	k_t
Generation 9350: Best = 4.683241e+03 = [1, 1, 1, 1]	FES = 935100	Diversity = 1.1482e-03	k_t
Generation 9360: Best = 4.683240e+03 = [3, 2, 2, 4]	FES = 936100	Diversity = 1.0816e-03	k_t
Generation 9370: Best = 4.683240e+03 = [1, 1, 1, 1]	FES = 937100	Diversity = 1.0654e-03	k_t
Generation 9380: Best = 4.683239e+03 = [2, 4, 2, 4]	FES = 938100	Diversity = 1.0286e-03	k_t
Generation 9390: Best = 4.683239e+03 = [1, 1, 1, 1]	FES = 939100	Diversity = 9.8984e-04	k_t
Generation 9400: Best = 4.683239e+03 = [4, 2, 3, 2]	FES = 940100	Diversity = 1.0132e-03	k_t

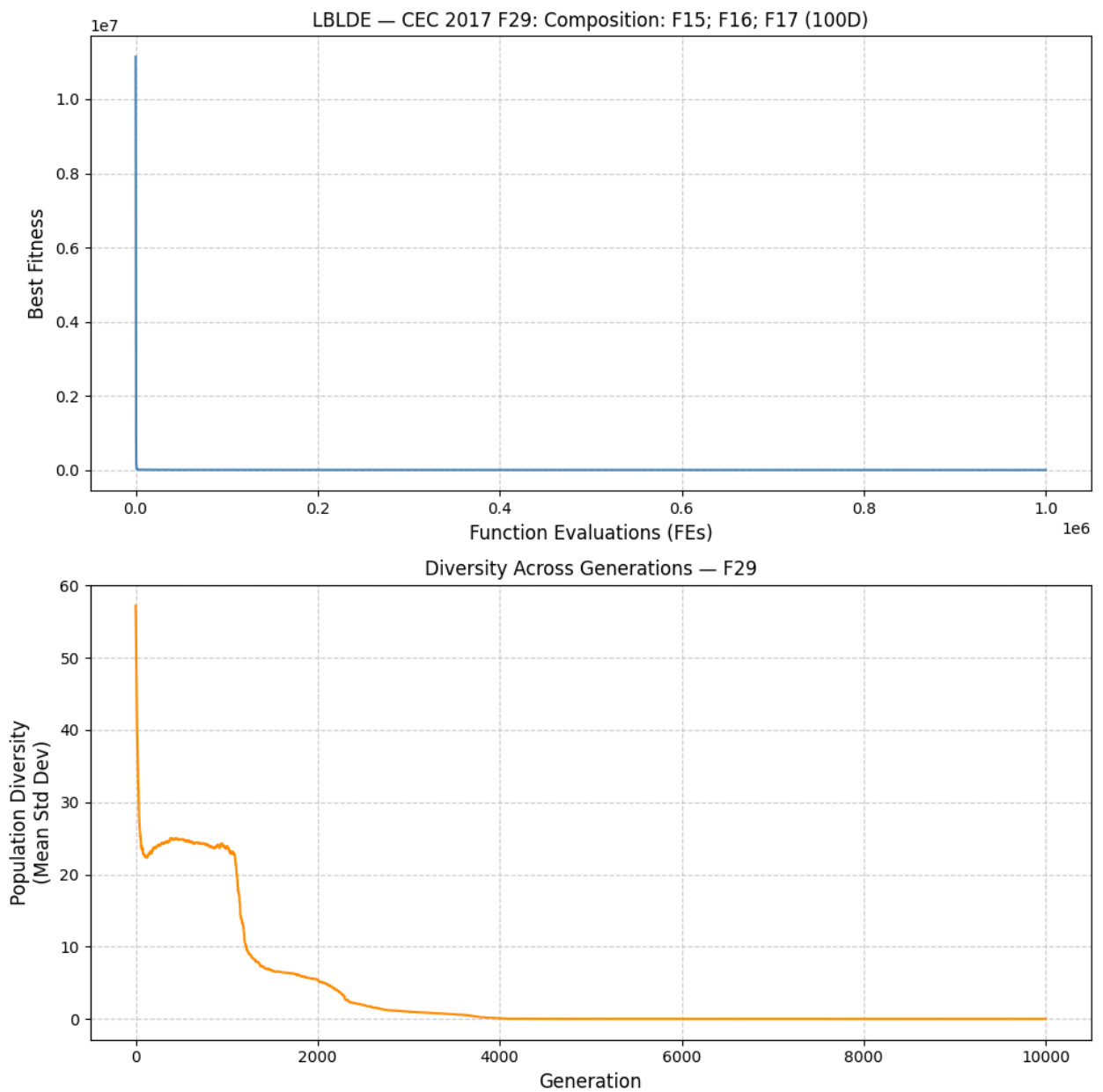
Generation 9410: Best = 4.683238e+03 = [1, 1, 1, 1]	FES = 941100	Diversity = 1.0067e-03	k_t
Generation 9420: Best = 4.683238e+03 = [4, 4, 2, 4]	FES = 942100	Diversity = 9.7532e-04	k_t
Generation 9430: Best = 4.683237e+03 = [1, 1, 1, 1]	FES = 943100	Diversity = 9.8954e-04	k_t
Generation 9440: Best = 4.683237e+03 = [4, 4, 3, 2]	FES = 944100	Diversity = 9.8819e-04	k_t
Generation 9450: Best = 4.683237e+03 = [1, 1, 1, 1]	FES = 945100	Diversity = 9.2108e-04	k_t
Generation 9460: Best = 4.683237e+03 = [2, 3, 3, 2]	FES = 946100	Diversity = 9.3729e-04	k_t
Generation 9470: Best = 4.683236e+03 = [1, 1, 1, 1]	FES = 947100	Diversity = 1.0003e-03	k_t
Generation 9480: Best = 4.683236e+03 = [4, 4, 4, 2]	FES = 948100	Diversity = 1.0688e-03	k_t
Generation 9490: Best = 4.683236e+03 = [1, 1, 1, 1]	FES = 949100	Diversity = 1.0617e-03	k_t
Generation 9500: Best = 4.683235e+03 = [2, 3, 4, 4]	FES = 950100	Diversity = 1.1038e-03	k_t
Generation 9510: Best = 4.683235e+03 = [1, 1, 1, 1]	FES = 951100	Diversity = 1.1069e-03	k_t
Generation 9520: Best = 4.683235e+03 = [3, 2, 4, 4]	FES = 952100	Diversity = 1.0771e-03	k_t
Generation 9530: Best = 4.683234e+03 = [1, 1, 1, 1]	FES = 953100	Diversity = 1.1337e-03	k_t
Generation 9540: Best = 4.683234e+03 = [2, 4, 4, 2]	FES = 954100	Diversity = 1.2152e-03	k_t
Generation 9550: Best = 4.683233e+03 = [1, 1, 1, 1]	FES = 955100	Diversity = 1.2390e-03	k_t
Generation 9560: Best = 4.683233e+03 = [3, 3, 2, 2]	FES = 956100	Diversity = 1.2539e-03	k_t
Generation 9570: Best = 4.683232e+03 = [1, 1, 1, 1]	FES = 957100	Diversity = 1.1966e-03	k_t
Generation 9580: Best = 4.683232e+03 = [4, 3, 2, 4]	FES = 958100	Diversity = 1.1353e-03	k_t
Generation 9590: Best = 4.683232e+03 = [1, 1, 1, 1]	FES = 959100	Diversity = 1.1361e-03	k_t
Generation 9600: Best = 4.683231e+03 = [3, 4, 2, 4]	FES = 960100	Diversity = 1.1332e-03	k_t
Generation 9610: Best = 4.683231e+03 = [1, 1, 1, 1]	FES = 961100	Diversity = 1.1342e-03	k_t
Generation 9620: Best = 4.683230e+03 = [4, 3, 2, 4]	FES = 962100	Diversity = 1.1202e-03	k_t
Generation 9630: Best = 4.683230e+03 = [1, 1, 1, 1]	FES = 963100	Diversity = 1.0515e-03	k_t
Generation 9640: Best = 4.683230e+03 = [2, 4, 4, 3]	FES = 964100	Diversity = 1.0244e-03	k_t
Generation 9650: Best = 4.683230e+03 = [1, 1, 1, 1]	FES = 965100	Diversity = 9.9792e-04	k_t
Generation 9660: Best = 4.683229e+03 = [4, 2, 2, 2]	FES = 966100	Diversity = 9.5146e-04	k_t
Generation 9670: Best = 4.683229e+03 = [1, 1, 1, 1]	FES = 967100	Diversity = 9.8634e-04	k_t

Generation 9680: Best = 4.683229e+03 = [3, 2, 3, 2]	FES = 968100	Diversity = 1.0098e-03	k_t
Generation 9690: Best = 4.683228e+03 = [1, 1, 1, 1]	FES = 969100	Diversity = 1.0130e-03	k_t
Generation 9700: Best = 4.683228e+03 = [2, 2, 3, 3]	FES = 970100	Diversity = 9.2681e-04	k_t
Generation 9710: Best = 4.683228e+03 = [1, 1, 1, 1]	FES = 971100	Diversity = 9.0250e-04	k_t
Generation 9720: Best = 4.683228e+03 = [2, 4, 2, 4]	FES = 972100	Diversity = 9.0521e-04	k_t
Generation 9730: Best = 4.683227e+03 = [1, 1, 1, 1]	FES = 973100	Diversity = 9.9604e-04	k_t
Generation 9740: Best = 4.683227e+03 = [3, 4, 3, 4]	FES = 974100	Diversity = 1.0286e-03	k_t
Generation 9750: Best = 4.683227e+03 = [1, 1, 1, 1]	FES = 975100	Diversity = 9.8982e-04	k_t
Generation 9760: Best = 4.683226e+03 = [3, 3, 4, 3]	FES = 976100	Diversity = 9.9006e-04	k_t
Generation 9770: Best = 4.683226e+03 = [1, 1, 1, 1]	FES = 977100	Diversity = 1.0170e-03	k_t
Generation 9780: Best = 4.683226e+03 = [3, 2, 2, 3]	FES = 978100	Diversity = 1.0535e-03	k_t
Generation 9790: Best = 4.683226e+03 = [1, 1, 1, 1]	FES = 979100	Diversity = 1.0375e-03	k_t
Generation 9800: Best = 4.683225e+03 = [4, 4, 4, 2]	FES = 980100	Diversity = 1.1153e-03	k_t
Generation 9810: Best = 4.683225e+03 = [1, 1, 1, 1]	FES = 981100	Diversity = 1.1055e-03	k_t
Generation 9820: Best = 4.683224e+03 = [2, 2, 2, 4]	FES = 982100	Diversity = 1.1135e-03	k_t
Generation 9830: Best = 4.683224e+03 = [1, 1, 1, 1]	FES = 983100	Diversity = 1.1618e-03	k_t
Generation 9840: Best = 4.683224e+03 = [3, 3, 3, 4]	FES = 984100	Diversity = 1.1473e-03	k_t
Generation 9850: Best = 4.683223e+03 = [1, 1, 1, 1]	FES = 985100	Diversity = 1.1534e-03	k_t
Generation 9860: Best = 4.683223e+03 = [2, 2, 2, 4]	FES = 986100	Diversity = 1.1386e-03	k_t
Generation 9870: Best = 4.683222e+03 = [1, 1, 1, 1]	FES = 987100	Diversity = 1.1121e-03	k_t
Generation 9880: Best = 4.683222e+03 = [3, 2, 3, 4]	FES = 988100	Diversity = 1.1338e-03	k_t
Generation 9890: Best = 4.683221e+03 = [1, 1, 1, 1]	FES = 989100	Diversity = 1.1395e-03	k_t
Generation 9900: Best = 4.683221e+03 = [4, 4, 4, 4]	FES = 990100	Diversity = 1.0909e-03	k_t
Generation 9910: Best = 4.683221e+03 = [1, 1, 1, 1]	FES = 991100	Diversity = 1.0796e-03	k_t
Generation 9920: Best = 4.683220e+03 = [4, 4, 3, 2]	FES = 992100	Diversity = 9.5153e-04	k_t
Generation 9930: Best = 4.683220e+03 = [1, 1, 1, 1]	FES = 993100	Diversity = 8.7548e-04	k_t
Generation 9940: Best = 4.683220e+03 = [3, 4, 4, 3]	FES = 994100	Diversity = 9.2463e-04	k_t

Generation 9950: Best = 4.683219e+03 FES = 995100 Diversity = 9.7452e-04 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 4.683219e+03 FES = 996100 Diversity = 9.8957e-04 k_t
= [4, 4, 4, 3]
Generation 9970: Best = 4.683219e+03 FES = 997100 Diversity = 1.0288e-03 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 4.683218e+03 FES = 998100 Diversity = 1.0479e-03 k_t
= [4, 3, 2, 4]
Generation 9990: Best = 4.683218e+03 FES = 999100 Diversity = 1.0734e-03 k_t
= [1, 1, 1, 1]

Optimization completed!
Final Best Fitness = 4.683217e+03
Total Function Evaluations = 1000000

=====
=
Results:
Best Fitness : 4.683217e+03
Global Optimum : 2.900000e+03
Error (|best-optimum|) : 1.783217e+03
Best Solution (5 dim) : [0.7868291 -47.60016444 -7.04219765 93.97346504
53.4489426]
D^(0) (init. diversity): 5.7220e+01
=====



Graph saved as convergence_F29.png

```
In [54]: # =====
# F30 — Composition Function 10 (F15; F18; F19)
# =====
run_and_plot('F30', make_wrapper(composition.f30), 3000.0, 'Composition: F15;
```

```

=====
=
LBLDE_s on CEC 2017 F30 – Single Run
=====
=
Function      : Composition: F15; F18; F19 (100D)
Bounds       : [-100, 100]
Global optimum : 3000.0
Initial D^(0) = 5.7220e+01
Generation 0: Best Fitness = 7.713945e+11 | Diversity = 5.7220e+01
Generation 10: Best = 1.616180e+11 FES = 1100 Diversity = 4.3261e+01 k_t
= [1, 1, 1, 1]
Generation 20: Best = 5.834694e+10 FES = 2100 Diversity = 3.6828e+01 k_t
= [4, 2, 2, 3]
Generation 30: Best = 1.642589e+10 FES = 3100 Diversity = 3.0999e+01 k_t
= [1, 1, 1, 1]
Generation 40: Best = 4.681066e+09 FES = 4100 Diversity = 2.7773e+01 k_t
= [4, 3, 4, 2]
Generation 50: Best = 1.354025e+09 FES = 5100 Diversity = 2.6386e+01 k_t
= [1, 1, 1, 1]
Generation 60: Best = 6.760561e+08 FES = 6100 Diversity = 2.4240e+01 k_t
= [4, 4, 2, 3]
Generation 70: Best = 3.342003e+08 FES = 7100 Diversity = 2.2487e+01 k_t
= [1, 1, 1, 1]
Generation 80: Best = 1.756941e+08 FES = 8100 Diversity = 2.1235e+01 k_t
= [4, 2, 4, 4]
Generation 90: Best = 1.071104e+08 FES = 9100 Diversity = 1.8198e+01 k_t
= [1, 1, 1, 1]
Generation 100: Best = 5.859093e+07 FES = 10100 Diversity = 1.6988e+01 k_t
= [4, 2, 4, 3]
Generation 110: Best = 4.089290e+07 FES = 11100 Diversity = 1.6456e+01 k_t
= [1, 1, 1, 1]
Generation 120: Best = 2.630923e+07 FES = 12100 Diversity = 1.6121e+01 k_t
= [4, 2, 2, 3]
Generation 130: Best = 1.870907e+07 FES = 13100 Diversity = 1.5181e+01 k_t
= [1, 1, 1, 1]
Generation 140: Best = 1.308741e+07 FES = 14100 Diversity = 1.4028e+01 k_t
= [4, 4, 4, 4]
Generation 150: Best = 9.435321e+06 FES = 15100 Diversity = 1.3931e+01 k_t
= [1, 1, 1, 1]
Generation 160: Best = 7.289772e+06 FES = 16100 Diversity = 1.3426e+01 k_t
= [4, 3, 3, 4]
Generation 170: Best = 4.639709e+06 FES = 17100 Diversity = 1.2659e+01 k_t
= [1, 1, 1, 1]
Generation 180: Best = 3.707537e+06 FES = 18100 Diversity = 1.2222e+01 k_t
= [4, 4, 3, 4]
Generation 190: Best = 2.707053e+06 FES = 19100 Diversity = 1.2093e+01 k_t
= [1, 1, 1, 1]
Generation 200: Best = 1.916642e+06 FES = 20100 Diversity = 1.1663e+01 k_t
= [4, 4, 4, 2]
Generation 210: Best = 1.581604e+06 FES = 21100 Diversity = 1.1306e+01 k_t
= [1, 1, 1, 1]
Generation 220: Best = 1.227729e+06 FES = 22100 Diversity = 1.0606e+01 k_t
= [3, 3, 4, 2]

```

Generation 230:	Best = 1.076770e+06	FES = 23100	Diversity = 9.8529e+00	k_t = [1, 1, 1, 1]
Generation 240:	Best = 8.688956e+05	FES = 24100	Diversity = 9.0200e+00	k_t = [3, 3, 2, 2]
Generation 250:	Best = 7.463500e+05	FES = 25100	Diversity = 8.8247e+00	k_t = [1, 1, 1, 1]
Generation 260:	Best = 6.075678e+05	FES = 26100	Diversity = 8.4688e+00	k_t = [4, 2, 4, 3]
Generation 270:	Best = 5.814862e+05	FES = 27100	Diversity = 7.9014e+00	k_t = [1, 1, 1, 1]
Generation 280:	Best = 5.141678e+05	FES = 28100	Diversity = 7.5445e+00	k_t = [4, 2, 4, 3]
Generation 290:	Best = 4.957564e+05	FES = 29100	Diversity = 7.3662e+00	k_t = [1, 1, 1, 1]
Generation 300:	Best = 4.364204e+05	FES = 30100	Diversity = 7.2772e+00	k_t = [3, 4, 2, 4]
Generation 310:	Best = 4.104883e+05	FES = 31100	Diversity = 7.4160e+00	k_t = [1, 1, 1, 1]
Generation 320:	Best = 3.649463e+05	FES = 32100	Diversity = 7.1582e+00	k_t = [3, 4, 2, 2]
Generation 330:	Best = 3.369153e+05	FES = 33100	Diversity = 7.0713e+00	k_t = [1, 1, 1, 1]
Generation 340:	Best = 3.091294e+05	FES = 34100	Diversity = 6.7679e+00	k_t = [3, 3, 3, 4]
Generation 350:	Best = 2.741001e+05	FES = 35100	Diversity = 6.6352e+00	k_t = [1, 1, 1, 1]
Generation 360:	Best = 2.443741e+05	FES = 36100	Diversity = 6.0900e+00	k_t = [3, 4, 2, 2]
Generation 370:	Best = 2.430558e+05	FES = 37100	Diversity = 5.7782e+00	k_t = [1, 1, 1, 1]
Generation 380:	Best = 2.142968e+05	FES = 38100	Diversity = 5.2465e+00	k_t = [2, 3, 3, 2]
Generation 390:	Best = 2.086921e+05	FES = 39100	Diversity = 5.1573e+00	k_t = [1, 1, 1, 1]
Generation 400:	Best = 2.029150e+05	FES = 40100	Diversity = 4.7222e+00	k_t = [4, 4, 4, 4]
Generation 410:	Best = 1.925047e+05	FES = 41100	Diversity = 4.4303e+00	k_t = [1, 1, 1, 1]
Generation 420:	Best = 1.769567e+05	FES = 42100	Diversity = 4.2339e+00	k_t = [4, 3, 2, 4]
Generation 430:	Best = 1.667143e+05	FES = 43100	Diversity = 4.2170e+00	k_t = [1, 1, 1, 1]
Generation 440:	Best = 1.596208e+05	FES = 44100	Diversity = 3.7445e+00	k_t = [3, 4, 2, 2]
Generation 450:	Best = 1.539089e+05	FES = 45100	Diversity = 3.5944e+00	k_t = [1, 1, 1, 1]
Generation 460:	Best = 1.469015e+05	FES = 46100	Diversity = 3.3871e+00	k_t = [2, 4, 4, 4]
Generation 470:	Best = 1.332701e+05	FES = 47100	Diversity = 3.1367e+00	k_t = [1, 1, 1, 1]
Generation 480:	Best = 1.317987e+05	FES = 48100	Diversity = 2.9987e+00	k_t = [4, 3, 3, 2]
Generation 490:	Best = 1.275917e+05	FES = 49100	Diversity = 2.9201e+00	k_t = [1, 1, 1, 1]

Generation 500:	Best = 1.223960e+05	FES = 50100	Diversity = 2.7242e+00	k_t
= [4, 4, 3, 2]				
Generation 510:	Best = 1.162977e+05	FES = 51100	Diversity = 2.7341e+00	k_t
= [1, 1, 1, 1]				
Generation 520:	Best = 1.138367e+05	FES = 52100	Diversity = 2.7979e+00	k_t
= [4, 4, 3, 2]				
Generation 530:	Best = 1.098594e+05	FES = 53100	Diversity = 2.6465e+00	k_t
= [1, 1, 1, 1]				
Generation 540:	Best = 1.027338e+05	FES = 54100	Diversity = 2.5460e+00	k_t
= [4, 2, 3, 2]				
Generation 550:	Best = 9.822225e+04	FES = 55100	Diversity = 2.6431e+00	k_t
= [1, 1, 1, 1]				
Generation 560:	Best = 9.122777e+04	FES = 56100	Diversity = 2.6899e+00	k_t
= [2, 2, 2, 4]				
Generation 570:	Best = 8.896550e+04	FES = 57100	Diversity = 2.6864e+00	k_t
= [1, 1, 1, 1]				
Generation 580:	Best = 8.438880e+04	FES = 58100	Diversity = 2.6166e+00	k_t
= [3, 2, 4, 3]				
Generation 590:	Best = 7.900595e+04	FES = 59100	Diversity = 2.5475e+00	k_t
= [1, 1, 1, 1]				
Generation 600:	Best = 7.722254e+04	FES = 60100	Diversity = 2.3358e+00	k_t
= [4, 2, 2, 4]				
Generation 610:	Best = 7.508057e+04	FES = 61100	Diversity = 2.3351e+00	k_t
= [1, 1, 1, 1]				
Generation 620:	Best = 7.129883e+04	FES = 62100	Diversity = 2.2880e+00	k_t
= [4, 4, 4, 2]				
Generation 630:	Best = 6.990548e+04	FES = 63100	Diversity = 2.1365e+00	k_t
= [1, 1, 1, 1]				
Generation 640:	Best = 6.743975e+04	FES = 64100	Diversity = 2.1408e+00	k_t
= [3, 4, 3, 2]				
Generation 650:	Best = 6.653996e+04	FES = 65100	Diversity = 2.0566e+00	k_t
= [1, 1, 1, 1]				
Generation 660:	Best = 6.423701e+04	FES = 66100	Diversity = 1.9865e+00	k_t
= [4, 2, 2, 4]				
Generation 670:	Best = 6.200567e+04	FES = 67100	Diversity = 2.0532e+00	k_t
= [1, 1, 1, 1]				
Generation 680:	Best = 6.027508e+04	FES = 68100	Diversity = 2.1063e+00	k_t
= [3, 4, 2, 3]				
Generation 690:	Best = 5.865508e+04	FES = 69100	Diversity = 2.1386e+00	k_t
= [1, 1, 1, 1]				
Generation 700:	Best = 5.731973e+04	FES = 70100	Diversity = 2.1649e+00	k_t
= [3, 4, 3, 2]				
Generation 710:	Best = 5.611747e+04	FES = 71100	Diversity = 2.1117e+00	k_t
= [1, 1, 1, 1]				
Generation 720:	Best = 5.380264e+04	FES = 72100	Diversity = 2.0117e+00	k_t
= [3, 2, 2, 2]				
Generation 730:	Best = 5.276111e+04	FES = 73100	Diversity = 1.9148e+00	k_t
= [1, 1, 1, 1]				
Generation 740:	Best = 5.135845e+04	FES = 74100	Diversity = 1.8921e+00	k_t
= [2, 2, 2, 3]				
Generation 750:	Best = 4.992775e+04	FES = 75100	Diversity = 1.7601e+00	k_t
= [1, 1, 1, 1]				
Generation 760:	Best = 4.837087e+04	FES = 76100	Diversity = 1.7226e+00	k_t
= [2, 4, 3, 2]				

Generation 770:	Best = 4.763452e+04	FES = 77100	Diversity = 1.6174e+00	k_t
= [1, 1, 1, 1]				
Generation 780:	Best = 4.666469e+04	FES = 78100	Diversity = 1.5536e+00	k_t
= [2, 4, 4, 2]				
Generation 790:	Best = 4.549856e+04	FES = 79100	Diversity = 1.4453e+00	k_t
= [1, 1, 1, 1]				
Generation 800:	Best = 4.470981e+04	FES = 80100	Diversity = 1.3593e+00	k_t
= [2, 3, 2, 3]				
Generation 810:	Best = 4.382125e+04	FES = 81100	Diversity = 1.1815e+00	k_t
= [1, 1, 1, 1]				
Generation 820:	Best = 4.317995e+04	FES = 82100	Diversity = 1.1238e+00	k_t
= [2, 2, 2, 2]				
Generation 830:	Best = 4.230576e+04	FES = 83100	Diversity = 1.0731e+00	k_t
= [1, 1, 1, 1]				
Generation 840:	Best = 4.188162e+04	FES = 84100	Diversity = 1.0096e+00	k_t
= [2, 2, 2, 3]				
Generation 850:	Best = 4.151555e+04	FES = 85100	Diversity = 9.2366e-01	k_t
= [1, 1, 1, 1]				
Generation 860:	Best = 4.104822e+04	FES = 86100	Diversity = 9.0900e-01	k_t
= [4, 3, 2, 3]				
Generation 870:	Best = 4.051107e+04	FES = 87100	Diversity = 9.3392e-01	k_t
= [1, 1, 1, 1]				
Generation 880:	Best = 4.023236e+04	FES = 88100	Diversity = 9.1402e-01	k_t
= [3, 2, 2, 2]				
Generation 890:	Best = 3.955046e+04	FES = 89100	Diversity = 8.9475e-01	k_t
= [1, 1, 1, 1]				
Generation 900:	Best = 3.894135e+04	FES = 90100	Diversity = 9.0793e-01	k_t
= [3, 2, 4, 3]				
Generation 910:	Best = 3.855742e+04	FES = 91100	Diversity = 9.8671e-01	k_t
= [1, 1, 1, 1]				
Generation 920:	Best = 3.781497e+04	FES = 92100	Diversity = 1.0069e+00	k_t
= [2, 4, 4, 2]				
Generation 930:	Best = 3.702555e+04	FES = 93100	Diversity = 1.0048e+00	k_t
= [1, 1, 1, 1]				
Generation 940:	Best = 3.669552e+04	FES = 94100	Diversity = 1.0476e+00	k_t
= [2, 2, 3, 2]				
Generation 950:	Best = 3.563688e+04	FES = 95100	Diversity = 1.0979e+00	k_t
= [1, 1, 1, 1]				
Generation 960:	Best = 3.520668e+04	FES = 96100	Diversity = 1.1061e+00	k_t
= [3, 2, 2, 2]				
Generation 970:	Best = 3.432481e+04	FES = 97100	Diversity = 1.0976e+00	k_t
= [1, 1, 1, 1]				
Generation 980:	Best = 3.399319e+04	FES = 98100	Diversity = 1.0350e+00	k_t
= [2, 3, 4, 2]				
Generation 990:	Best = 3.335199e+04	FES = 99100	Diversity = 1.0735e+00	k_t
= [1, 1, 1, 1]				
Generation 1000:	Best = 3.278009e+04	FES = 100100	Diversity = 1.0665e+00	k_t
= [4, 4, 3, 4]				
Generation 1010:	Best = 3.229292e+04	FES = 101100	Diversity = 1.0498e+00	k_t
= [1, 1, 1, 1]				
Generation 1020:	Best = 3.186653e+04	FES = 102100	Diversity = 1.0120e+00	k_t
= [3, 2, 2, 2]				
Generation 1030:	Best = 3.122695e+04	FES = 103100	Diversity = 9.4703e-01	k_t
= [1, 1, 1, 1]				

Generation 1040: Best = 3.096086e+04 = [4, 3, 3, 2]	FES = 104100	Diversity = 9.2038e-01	k_t
Generation 1050: Best = 3.072805e+04 = [1, 1, 1, 1]	FES = 105100	Diversity = 8.9848e-01	k_t
Generation 1060: Best = 3.051598e+04 = [4, 4, 4, 3]	FES = 106100	Diversity = 8.9133e-01	k_t
Generation 1070: Best = 2.997857e+04 = [1, 1, 1, 1]	FES = 107100	Diversity = 8.8640e-01	k_t
Generation 1080: Best = 2.962722e+04 = [4, 4, 3, 2]	FES = 108100	Diversity = 8.8234e-01	k_t
Generation 1090: Best = 2.950624e+04 = [1, 1, 1, 1]	FES = 109100	Diversity = 8.8594e-01	k_t
Generation 1100: Best = 2.809650e+04 = [2, 2, 3, 2]	FES = 110100	Diversity = 1.0657e+00	k_t
Generation 1110: Best = 2.788980e+04 = [1, 1, 1, 1]	FES = 111100	Diversity = 1.0631e+00	k_t
Generation 1120: Best = 2.762513e+04 = [3, 2, 2, 2]	FES = 112100	Diversity = 8.1813e-01	k_t
Generation 1130: Best = 2.748637e+04 = [1, 1, 1, 1]	FES = 113100	Diversity = 7.2931e-01	k_t
Generation 1140: Best = 2.711133e+04 = [4, 3, 4, 2]	FES = 114100	Diversity = 7.0956e-01	k_t
Generation 1150: Best = 2.698071e+04 = [1, 1, 1, 1]	FES = 115100	Diversity = 6.4667e-01	k_t
Generation 1160: Best = 2.673588e+04 = [3, 3, 2, 2]	FES = 116100	Diversity = 6.1903e-01	k_t
Generation 1170: Best = 2.648385e+04 = [1, 1, 1, 1]	FES = 117100	Diversity = 5.5982e-01	k_t
Generation 1180: Best = 2.634563e+04 = [4, 2, 3, 3]	FES = 118100	Diversity = 5.4820e-01	k_t
Generation 1190: Best = 2.614882e+04 = [1, 1, 1, 1]	FES = 119100	Diversity = 5.4054e-01	k_t
Generation 1200: Best = 2.585905e+04 = [4, 2, 2, 3]	FES = 120100	Diversity = 5.7473e-01	k_t
Generation 1210: Best = 2.572275e+04 = [1, 1, 1, 1]	FES = 121100	Diversity = 5.7687e-01	k_t
Generation 1220: Best = 2.549541e+04 = [3, 4, 4, 3]	FES = 122100	Diversity = 5.7885e-01	k_t
Generation 1230: Best = 2.523682e+04 = [1, 1, 1, 1]	FES = 123100	Diversity = 5.8728e-01	k_t
Generation 1240: Best = 2.503092e+04 = [4, 2, 3, 2]	FES = 124100	Diversity = 5.6506e-01	k_t
Generation 1250: Best = 2.496251e+04 = [1, 1, 1, 1]	FES = 125100	Diversity = 5.4149e-01	k_t
Generation 1260: Best = 2.484702e+04 = [4, 4, 3, 4]	FES = 126100	Diversity = 4.9654e-01	k_t
Generation 1270: Best = 2.469300e+04 = [1, 1, 1, 1]	FES = 127100	Diversity = 5.0924e-01	k_t
Generation 1280: Best = 2.449509e+04 = [2, 3, 4, 4]	FES = 128100	Diversity = 5.0661e-01	k_t
Generation 1290: Best = 2.426455e+04 = [1, 1, 1, 1]	FES = 129100	Diversity = 5.2669e-01	k_t
Generation 1300: Best = 2.406931e+04 = [3, 3, 3, 4]	FES = 130100	Diversity = 5.2111e-01	k_t

Generation 1310: Best = 2.400026e+04 = [1, 1, 1, 1]	FES = 131100	Diversity = 5.0725e-01	k_t
Generation 1320: Best = 2.371949e+04 = [4, 4, 3, 3]	FES = 132100	Diversity = 5.3784e-01	k_t
Generation 1330: Best = 2.348372e+04 = [1, 1, 1, 1]	FES = 133100	Diversity = 5.5495e-01	k_t
Generation 1340: Best = 2.329508e+04 = [4, 3, 2, 3]	FES = 134100	Diversity = 5.7715e-01	k_t
Generation 1350: Best = 2.312994e+04 = [1, 1, 1, 1]	FES = 135100	Diversity = 5.8770e-01	k_t
Generation 1360: Best = 2.297640e+04 = [4, 2, 3, 4]	FES = 136100	Diversity = 5.4231e-01	k_t
Generation 1370: Best = 2.268581e+04 = [1, 1, 1, 1]	FES = 137100	Diversity = 5.6423e-01	k_t
Generation 1380: Best = 2.246875e+04 = [4, 3, 2, 4]	FES = 138100	Diversity = 6.1043e-01	k_t
Generation 1390: Best = 2.232786e+04 = [1, 1, 1, 1]	FES = 139100	Diversity = 6.5182e-01	k_t
Generation 1400: Best = 2.209997e+04 = [3, 3, 4, 4]	FES = 140100	Diversity = 6.3418e-01	k_t
Generation 1410: Best = 2.197143e+04 = [1, 1, 1, 1]	FES = 141100	Diversity = 6.0080e-01	k_t
Generation 1420: Best = 2.183280e+04 = [4, 2, 3, 4]	FES = 142100	Diversity = 6.4093e-01	k_t
Generation 1430: Best = 2.164606e+04 = [1, 1, 1, 1]	FES = 143100	Diversity = 6.5975e-01	k_t
Generation 1440: Best = 2.146682e+04 = [2, 3, 4, 2]	FES = 144100	Diversity = 6.8588e-01	k_t
Generation 1450: Best = 2.135866e+04 = [1, 1, 1, 1]	FES = 145100	Diversity = 6.6451e-01	k_t
Generation 1460: Best = 2.120319e+04 = [3, 2, 3, 3]	FES = 146100	Diversity = 6.1759e-01	k_t
Generation 1470: Best = 2.115977e+04 = [1, 1, 1, 1]	FES = 147100	Diversity = 6.1070e-01	k_t
Generation 1480: Best = 2.102093e+04 = [4, 3, 4, 3]	FES = 148100	Diversity = 5.7043e-01	k_t
Generation 1490: Best = 2.096359e+04 = [1, 1, 1, 1]	FES = 149100	Diversity = 5.1971e-01	k_t
Generation 1500: Best = 2.081970e+04 = [3, 2, 4, 2]	FES = 150100	Diversity = 4.9304e-01	k_t
Generation 1510: Best = 2.072525e+04 = [1, 1, 1, 1]	FES = 151100	Diversity = 4.5056e-01	k_t
Generation 1520: Best = 2.064228e+04 = [4, 4, 4, 4]	FES = 152100	Diversity = 4.7551e-01	k_t
Generation 1530: Best = 2.054200e+04 = [1, 1, 1, 1]	FES = 153100	Diversity = 4.7430e-01	k_t
Generation 1540: Best = 2.043154e+04 = [2, 4, 2, 4]	FES = 154100	Diversity = 5.0144e-01	k_t
Generation 1550: Best = 2.038223e+04 = [1, 1, 1, 1]	FES = 155100	Diversity = 4.8088e-01	k_t
Generation 1560: Best = 2.030540e+04 = [3, 3, 3, 2]	FES = 156100	Diversity = 4.7178e-01	k_t
Generation 1570: Best = 2.023480e+04 = [1, 1, 1, 1]	FES = 157100	Diversity = 4.2858e-01	k_t

Generation 1580: Best = 2.018155e+04 = [2, 4, 4, 2]	FES = 158100	Diversity = 4.1835e-01	k_t
Generation 1590: Best = 2.009893e+04 = [1, 1, 1, 1]	FES = 159100	Diversity = 4.5005e-01	k_t
Generation 1600: Best = 2.005475e+04 = [2, 3, 2, 3]	FES = 160100	Diversity = 4.5399e-01	k_t
Generation 1610: Best = 2.000620e+04 = [1, 1, 1, 1]	FES = 161100	Diversity = 4.3219e-01	k_t
Generation 1620: Best = 1.996003e+04 = [2, 4, 3, 3]	FES = 162100	Diversity = 4.2308e-01	k_t
Generation 1630: Best = 1.992771e+04 = [1, 1, 1, 1]	FES = 163100	Diversity = 4.0188e-01	k_t
Generation 1640: Best = 1.983487e+04 = [2, 3, 3, 2]	FES = 164100	Diversity = 4.0877e-01	k_t
Generation 1650: Best = 1.977295e+04 = [1, 1, 1, 1]	FES = 165100	Diversity = 4.3289e-01	k_t
Generation 1660: Best = 1.973280e+04 = [3, 3, 4, 4]	FES = 166100	Diversity = 4.5597e-01	k_t
Generation 1670: Best = 1.966334e+04 = [1, 1, 1, 1]	FES = 167100	Diversity = 4.7602e-01	k_t
Generation 1680: Best = 1.955044e+04 = [2, 2, 2, 4]	FES = 168100	Diversity = 4.9347e-01	k_t
Generation 1690: Best = 1.949162e+04 = [1, 1, 1, 1]	FES = 169100	Diversity = 5.0470e-01	k_t
Generation 1700: Best = 1.942923e+04 = [3, 3, 2, 3]	FES = 170100	Diversity = 4.9126e-01	k_t
Generation 1710: Best = 1.933662e+04 = [1, 1, 1, 1]	FES = 171100	Diversity = 5.1087e-01	k_t
Generation 1720: Best = 1.923512e+04 = [3, 2, 3, 3]	FES = 172100	Diversity = 5.5190e-01	k_t
Generation 1730: Best = 1.912000e+04 = [1, 1, 1, 1]	FES = 173100	Diversity = 5.6672e-01	k_t
Generation 1740: Best = 1.897914e+04 = [4, 4, 4, 4]	FES = 174100	Diversity = 6.3701e-01	k_t
Generation 1750: Best = 1.855886e+04 = [1, 1, 1, 1]	FES = 175100	Diversity = 7.9350e-01	k_t
Generation 1760: Best = 1.827802e+04 = [4, 3, 3, 3]	FES = 176100	Diversity = 1.0351e+00	k_t
Generation 1770: Best = 1.824865e+04 = [1, 1, 1, 1]	FES = 177100	Diversity = 9.1887e-01	k_t
Generation 1780: Best = 1.814293e+04 = [3, 2, 3, 3]	FES = 178100	Diversity = 8.1137e-01	k_t
Generation 1790: Best = 1.807774e+04 = [1, 1, 1, 1]	FES = 179100	Diversity = 6.7148e-01	k_t
Generation 1800: Best = 1.794809e+04 = [3, 2, 4, 4]	FES = 180100	Diversity = 6.3039e-01	k_t
Generation 1810: Best = 1.794284e+04 = [1, 1, 1, 1]	FES = 181100	Diversity = 5.8244e-01	k_t
Generation 1820: Best = 1.791156e+04 = [3, 4, 4, 4]	FES = 182100	Diversity = 5.3188e-01	k_t
Generation 1830: Best = 1.785829e+04 = [1, 1, 1, 1]	FES = 183100	Diversity = 4.8238e-01	k_t
Generation 1840: Best = 1.783276e+04 = [2, 4, 2, 3]	FES = 184100	Diversity = 4.2903e-01	k_t

Generation 1850: Best = 1.780230e+04 = [1, 1, 1, 1]	FES = 185100	Diversity = 4.0387e-01	k_t
Generation 1860: Best = 1.777281e+04 = [2, 2, 4, 2]	FES = 186100	Diversity = 3.6896e-01	k_t
Generation 1870: Best = 1.775552e+04 = [1, 1, 1, 1]	FES = 187100	Diversity = 3.4499e-01	k_t
Generation 1880: Best = 1.772501e+04 = [2, 4, 4, 2]	FES = 188100	Diversity = 3.1467e-01	k_t
Generation 1890: Best = 1.769483e+04 = [1, 1, 1, 1]	FES = 189100	Diversity = 3.1659e-01	k_t
Generation 1900: Best = 1.766449e+04 = [3, 3, 2, 2]	FES = 190100	Diversity = 3.1565e-01	k_t
Generation 1910: Best = 1.765182e+04 = [1, 1, 1, 1]	FES = 191100	Diversity = 3.0531e-01	k_t
Generation 1920: Best = 1.761506e+04 = [4, 4, 3, 4]	FES = 192100	Diversity = 3.2336e-01	k_t
Generation 1930: Best = 1.758304e+04 = [1, 1, 1, 1]	FES = 193100	Diversity = 3.2726e-01	k_t
Generation 1940: Best = 1.756141e+04 = [4, 2, 4, 3]	FES = 194100	Diversity = 3.4152e-01	k_t
Generation 1950: Best = 1.754186e+04 = [1, 1, 1, 1]	FES = 195100	Diversity = 3.4502e-01	k_t
Generation 1960: Best = 1.751516e+04 = [4, 4, 2, 2]	FES = 196100	Diversity = 3.4278e-01	k_t
Generation 1970: Best = 1.747951e+04 = [1, 1, 1, 1]	FES = 197100	Diversity = 3.4367e-01	k_t
Generation 1980: Best = 1.743553e+04 = [2, 2, 2, 3]	FES = 198100	Diversity = 3.5168e-01	k_t
Generation 1990: Best = 1.738904e+04 = [1, 1, 1, 1]	FES = 199100	Diversity = 3.6303e-01	k_t
Generation 2000: Best = 1.737189e+04 = [3, 4, 3, 3]	FES = 200100	Diversity = 3.7809e-01	k_t
Generation 2010: Best = 1.732565e+04 = [1, 1, 1, 1]	FES = 201100	Diversity = 3.8287e-01	k_t
Generation 2020: Best = 1.732357e+04 = [2, 4, 3, 3]	FES = 202100	Diversity = 3.8322e-01	k_t
Generation 2030: Best = 1.728597e+04 = [1, 1, 1, 1]	FES = 203100	Diversity = 3.6449e-01	k_t
Generation 2040: Best = 1.725384e+04 = [2, 2, 2, 3]	FES = 204100	Diversity = 3.5551e-01	k_t
Generation 2050: Best = 1.724702e+04 = [1, 1, 1, 1]	FES = 205100	Diversity = 3.6239e-01	k_t
Generation 2060: Best = 1.721895e+04 = [4, 2, 3, 4]	FES = 206100	Diversity = 3.8004e-01	k_t
Generation 2070: Best = 1.720047e+04 = [1, 1, 1, 1]	FES = 207100	Diversity = 3.8318e-01	k_t
Generation 2080: Best = 1.716881e+04 = [2, 4, 3, 3]	FES = 208100	Diversity = 3.8449e-01	k_t
Generation 2090: Best = 1.715143e+04 = [1, 1, 1, 1]	FES = 209100	Diversity = 3.7731e-01	k_t
Generation 2100: Best = 1.713693e+04 = [4, 4, 3, 2]	FES = 210100	Diversity = 3.5193e-01	k_t
Generation 2110: Best = 1.712213e+04 = [1, 1, 1, 1]	FES = 211100	Diversity = 3.2935e-01	k_t

Generation 2120: Best = 1.710941e+04 = [3, 4, 2, 3]	FES = 212100	Diversity = 3.3003e-01	k_t
Generation 2130: Best = 1.710162e+04 = [1, 1, 1, 1]	FES = 213100	Diversity = 2.9783e-01	k_t
Generation 2140: Best = 1.709016e+04 = [4, 3, 3, 4]	FES = 214100	Diversity = 2.7827e-01	k_t
Generation 2150: Best = 1.707885e+04 = [1, 1, 1, 1]	FES = 215100	Diversity = 2.7427e-01	k_t
Generation 2160: Best = 1.707362e+04 = [3, 4, 4, 4]	FES = 216100	Diversity = 2.7430e-01	k_t
Generation 2170: Best = 1.706881e+04 = [1, 1, 1, 1]	FES = 217100	Diversity = 2.8026e-01	k_t
Generation 2180: Best = 1.706069e+04 = [2, 3, 2, 4]	FES = 218100	Diversity = 2.7415e-01	k_t
Generation 2190: Best = 1.705192e+04 = [1, 1, 1, 1]	FES = 219100	Diversity = 2.6347e-01	k_t
Generation 2200: Best = 1.704925e+04 = [2, 4, 4, 2]	FES = 220100	Diversity = 2.7748e-01	k_t
Generation 2210: Best = 1.704378e+04 = [1, 1, 1, 1]	FES = 221100	Diversity = 2.7394e-01	k_t
Generation 2220: Best = 1.702802e+04 = [3, 3, 3, 2]	FES = 222100	Diversity = 2.7810e-01	k_t
Generation 2230: Best = 1.702106e+04 = [1, 1, 1, 1]	FES = 223100	Diversity = 2.7292e-01	k_t
Generation 2240: Best = 1.700782e+04 = [3, 2, 2, 3]	FES = 224100	Diversity = 2.6652e-01	k_t
Generation 2250: Best = 1.700080e+04 = [1, 1, 1, 1]	FES = 225100	Diversity = 2.7562e-01	k_t
Generation 2260: Best = 1.698251e+04 = [2, 2, 3, 4]	FES = 226100	Diversity = 2.8665e-01	k_t
Generation 2270: Best = 1.697307e+04 = [1, 1, 1, 1]	FES = 227100	Diversity = 2.9490e-01	k_t
Generation 2280: Best = 1.697100e+04 = [4, 4, 4, 3]	FES = 228100	Diversity = 3.0185e-01	k_t
Generation 2290: Best = 1.695149e+04 = [1, 1, 1, 1]	FES = 229100	Diversity = 2.9498e-01	k_t
Generation 2300: Best = 1.694100e+04 = [2, 2, 2, 3]	FES = 230100	Diversity = 2.8745e-01	k_t
Generation 2310: Best = 1.693174e+04 = [1, 1, 1, 1]	FES = 231100	Diversity = 2.8457e-01	k_t
Generation 2320: Best = 1.691649e+04 = [3, 4, 2, 4]	FES = 232100	Diversity = 2.9032e-01	k_t
Generation 2330: Best = 1.690193e+04 = [1, 1, 1, 1]	FES = 233100	Diversity = 2.9691e-01	k_t
Generation 2340: Best = 1.688970e+04 = [4, 2, 4, 3]	FES = 234100	Diversity = 2.9947e-01	k_t
Generation 2350: Best = 1.686840e+04 = [1, 1, 1, 1]	FES = 235100	Diversity = 3.0010e-01	k_t
Generation 2360: Best = 1.684241e+04 = [2, 2, 3, 2]	FES = 236100	Diversity = 3.1616e-01	k_t
Generation 2370: Best = 1.680387e+04 = [1, 1, 1, 1]	FES = 237100	Diversity = 3.2431e-01	k_t
Generation 2380: Best = 1.677983e+04 = [2, 2, 2, 4]	FES = 238100	Diversity = 3.1549e-01	k_t

Generation 2390: Best = 1.676502e+04 = [1, 1, 1, 1]	FES = 239100	Diversity = 3.3150e-01	k_t
Generation 2400: Best = 1.673959e+04 = [3, 4, 4, 2]	FES = 240100	Diversity = 3.4604e-01	k_t
Generation 2410: Best = 1.671383e+04 = [1, 1, 1, 1]	FES = 241100	Diversity = 3.5430e-01	k_t
Generation 2420: Best = 1.670443e+04 = [4, 3, 3, 3]	FES = 242100	Diversity = 3.6517e-01	k_t
Generation 2430: Best = 1.668243e+04 = [1, 1, 1, 1]	FES = 243100	Diversity = 3.4394e-01	k_t
Generation 2440: Best = 1.667244e+04 = [4, 2, 4, 2]	FES = 244100	Diversity = 3.3976e-01	k_t
Generation 2450: Best = 1.664570e+04 = [1, 1, 1, 1]	FES = 245100	Diversity = 3.3951e-01	k_t
Generation 2460: Best = 1.664279e+04 = [3, 2, 4, 4]	FES = 246100	Diversity = 3.1589e-01	k_t
Generation 2470: Best = 1.661761e+04 = [1, 1, 1, 1]	FES = 247100	Diversity = 3.1001e-01	k_t
Generation 2480: Best = 1.660503e+04 = [2, 3, 4, 2]	FES = 248100	Diversity = 3.1231e-01	k_t
Generation 2490: Best = 1.659722e+04 = [1, 1, 1, 1]	FES = 249100	Diversity = 3.1047e-01	k_t
Generation 2500: Best = 1.659012e+04 = [4, 2, 2, 2]	FES = 250100	Diversity = 3.1714e-01	k_t
Generation 2510: Best = 1.656415e+04 = [1, 1, 1, 1]	FES = 251100	Diversity = 3.0918e-01	k_t
Generation 2520: Best = 1.656001e+04 = [4, 3, 2, 3]	FES = 252100	Diversity = 3.0631e-01	k_t
Generation 2530: Best = 1.654981e+04 = [1, 1, 1, 1]	FES = 253100	Diversity = 3.0730e-01	k_t
Generation 2540: Best = 1.654086e+04 = [2, 4, 2, 3]	FES = 254100	Diversity = 3.0781e-01	k_t
Generation 2550: Best = 1.651848e+04 = [1, 1, 1, 1]	FES = 255100	Diversity = 3.1245e-01	k_t
Generation 2560: Best = 1.651687e+04 = [4, 4, 4, 3]	FES = 256100	Diversity = 3.0692e-01	k_t
Generation 2570: Best = 1.650518e+04 = [1, 1, 1, 1]	FES = 257100	Diversity = 3.0353e-01	k_t
Generation 2580: Best = 1.649593e+04 = [2, 2, 4, 2]	FES = 258100	Diversity = 3.0271e-01	k_t
Generation 2590: Best = 1.649252e+04 = [1, 1, 1, 1]	FES = 259100	Diversity = 2.9801e-01	k_t
Generation 2600: Best = 1.648052e+04 = [3, 4, 4, 4]	FES = 260100	Diversity = 2.9167e-01	k_t
Generation 2610: Best = 1.647754e+04 = [1, 1, 1, 1]	FES = 261100	Diversity = 2.9145e-01	k_t
Generation 2620: Best = 1.647303e+04 = [3, 2, 3, 2]	FES = 262100	Diversity = 2.8973e-01	k_t
Generation 2630: Best = 1.646715e+04 = [1, 1, 1, 1]	FES = 263100	Diversity = 2.8187e-01	k_t
Generation 2640: Best = 1.645859e+04 = [4, 4, 3, 2]	FES = 264100	Diversity = 2.9771e-01	k_t
Generation 2650: Best = 1.645439e+04 = [1, 1, 1, 1]	FES = 265100	Diversity = 2.9692e-01	k_t

Generation 2660: Best = 1.644897e+04 = [2, 3, 2, 4]	FES = 266100	Diversity = 2.8566e-01	k_t
Generation 2670: Best = 1.643591e+04 = [1, 1, 1, 1]	FES = 267100	Diversity = 2.7751e-01	k_t
Generation 2680: Best = 1.642711e+04 = [3, 3, 3, 3]	FES = 268100	Diversity = 2.7141e-01	k_t
Generation 2690: Best = 1.641612e+04 = [1, 1, 1, 1]	FES = 269100	Diversity = 2.8567e-01	k_t
Generation 2700: Best = 1.640303e+04 = [4, 2, 4, 2]	FES = 270100	Diversity = 2.7568e-01	k_t
Generation 2710: Best = 1.640025e+04 = [1, 1, 1, 1]	FES = 271100	Diversity = 2.7977e-01	k_t
Generation 2720: Best = 1.638482e+04 = [3, 3, 3, 4]	FES = 272100	Diversity = 2.7717e-01	k_t
Generation 2730: Best = 1.637966e+04 = [1, 1, 1, 1]	FES = 273100	Diversity = 2.7213e-01	k_t
Generation 2740: Best = 1.636435e+04 = [3, 4, 4, 3]	FES = 274100	Diversity = 2.7320e-01	k_t
Generation 2750: Best = 1.635718e+04 = [1, 1, 1, 1]	FES = 275100	Diversity = 2.8222e-01	k_t
Generation 2760: Best = 1.634467e+04 = [3, 3, 3, 3]	FES = 276100	Diversity = 2.6860e-01	k_t
Generation 2770: Best = 1.628686e+04 = [1, 1, 1, 1]	FES = 277100	Diversity = 3.4295e-01	k_t
Generation 2780: Best = 1.625781e+04 = [3, 3, 3, 2]	FES = 278100	Diversity = 3.3011e-01	k_t
Generation 2790: Best = 1.619782e+04 = [1, 1, 1, 1]	FES = 279100	Diversity = 3.8616e-01	k_t
Generation 2800: Best = 1.614360e+04 = [2, 4, 3, 3]	FES = 280100	Diversity = 4.6605e-01	k_t
Generation 2810: Best = 1.608613e+04 = [1, 1, 1, 1]	FES = 281100	Diversity = 4.9541e-01	k_t
Generation 2820: Best = 1.606409e+04 = [3, 4, 4, 3]	FES = 282100	Diversity = 5.7839e-01	k_t
Generation 2830: Best = 1.606385e+04 = [1, 1, 1, 1]	FES = 283100	Diversity = 5.7073e-01	k_t
Generation 2840: Best = 1.604764e+04 = [2, 4, 3, 3]	FES = 284100	Diversity = 5.3709e-01	k_t
Generation 2850: Best = 1.602205e+04 = [1, 1, 1, 1]	FES = 285100	Diversity = 4.6177e-01	k_t
Generation 2860: Best = 1.600316e+04 = [3, 3, 4, 2]	FES = 286100	Diversity = 4.0539e-01	k_t
Generation 2870: Best = 1.599632e+04 = [1, 1, 1, 1]	FES = 287100	Diversity = 3.7818e-01	k_t
Generation 2880: Best = 1.598133e+04 = [3, 2, 4, 2]	FES = 288100	Diversity = 3.5496e-01	k_t
Generation 2890: Best = 1.597957e+04 = [1, 1, 1, 1]	FES = 289100	Diversity = 3.1951e-01	k_t
Generation 2900: Best = 1.597089e+04 = [4, 3, 3, 4]	FES = 290100	Diversity = 2.9506e-01	k_t
Generation 2910: Best = 1.596269e+04 = [1, 1, 1, 1]	FES = 291100	Diversity = 2.7935e-01	k_t
Generation 2920: Best = 1.595586e+04 = [3, 3, 2, 2]	FES = 292100	Diversity = 2.5829e-01	k_t

Generation 2930: Best = 1.594881e+04 = [1, 1, 1, 1]	FES = 293100	Diversity = 2.4796e-01	k_t
Generation 2940: Best = 1.593604e+04 = [3, 3, 3, 2]	FES = 294100	Diversity = 2.4789e-01	k_t
Generation 2950: Best = 1.593166e+04 = [1, 1, 1, 1]	FES = 295100	Diversity = 2.4478e-01	k_t
Generation 2960: Best = 1.592740e+04 = [2, 2, 2, 2]	FES = 296100	Diversity = 2.5856e-01	k_t
Generation 2970: Best = 1.592171e+04 = [1, 1, 1, 1]	FES = 297100	Diversity = 2.4132e-01	k_t
Generation 2980: Best = 1.591620e+04 = [3, 2, 2, 4]	FES = 298100	Diversity = 2.4014e-01	k_t
Generation 2990: Best = 1.591155e+04 = [1, 1, 1, 1]	FES = 299100	Diversity = 2.3232e-01	k_t
Generation 3000: Best = 1.590703e+04 = [2, 3, 4, 4]	FES = 300100	Diversity = 2.2912e-01	k_t
Generation 3010: Best = 1.590148e+04 = [1, 1, 1, 1]	FES = 301100	Diversity = 2.1855e-01	k_t
Generation 3020: Best = 1.589977e+04 = [3, 4, 4, 4]	FES = 302100	Diversity = 2.2345e-01	k_t
Generation 3030: Best = 1.589418e+04 = [1, 1, 1, 1]	FES = 303100	Diversity = 2.2200e-01	k_t
Generation 3040: Best = 1.589147e+04 = [2, 3, 2, 4]	FES = 304100	Diversity = 2.2344e-01	k_t
Generation 3050: Best = 1.588969e+04 = [1, 1, 1, 1]	FES = 305100	Diversity = 2.2887e-01	k_t
Generation 3060: Best = 1.588759e+04 = [2, 2, 4, 2]	FES = 306100	Diversity = 2.2869e-01	k_t
Generation 3070: Best = 1.588038e+04 = [1, 1, 1, 1]	FES = 307100	Diversity = 2.4994e-01	k_t
Generation 3080: Best = 1.587593e+04 = [2, 3, 4, 3]	FES = 308100	Diversity = 2.6857e-01	k_t
Generation 3090: Best = 1.587409e+04 = [1, 1, 1, 1]	FES = 309100	Diversity = 2.7134e-01	k_t
Generation 3100: Best = 1.586965e+04 = [4, 4, 3, 2]	FES = 310100	Diversity = 2.8041e-01	k_t
Generation 3110: Best = 1.586050e+04 = [1, 1, 1, 1]	FES = 311100	Diversity = 2.7026e-01	k_t
Generation 3120: Best = 1.584893e+04 = [2, 2, 3, 3]	FES = 312100	Diversity = 2.9113e-01	k_t
Generation 3130: Best = 1.584365e+04 = [1, 1, 1, 1]	FES = 313100	Diversity = 3.2529e-01	k_t
Generation 3140: Best = 1.582359e+04 = [2, 4, 2, 2]	FES = 314100	Diversity = 3.5505e-01	k_t
Generation 3150: Best = 1.580871e+04 = [1, 1, 1, 1]	FES = 315100	Diversity = 3.6503e-01	k_t
Generation 3160: Best = 1.580538e+04 = [4, 2, 3, 2]	FES = 316100	Diversity = 3.2495e-01	k_t
Generation 3170: Best = 1.580100e+04 = [1, 1, 1, 1]	FES = 317100	Diversity = 3.0831e-01	k_t
Generation 3180: Best = 1.579664e+04 = [3, 3, 4, 2]	FES = 318100	Diversity = 2.7925e-01	k_t
Generation 3190: Best = 1.579330e+04 = [1, 1, 1, 1]	FES = 319100	Diversity = 2.6055e-01	k_t

Generation 3200: Best = 1.579108e+04 = [4, 4, 2, 4]	FES = 320100	Diversity = 2.2635e-01	k_t
Generation 3210: Best = 1.578591e+04 = [1, 1, 1, 1]	FES = 321100	Diversity = 2.1200e-01	k_t
Generation 3220: Best = 1.578041e+04 = [2, 3, 4, 2]	FES = 322100	Diversity = 1.9751e-01	k_t
Generation 3230: Best = 1.578041e+04 = [1, 1, 1, 1]	FES = 323100	Diversity = 1.9348e-01	k_t
Generation 3240: Best = 1.577946e+04 = [4, 2, 4, 3]	FES = 324100	Diversity = 1.7845e-01	k_t
Generation 3250: Best = 1.577876e+04 = [1, 1, 1, 1]	FES = 325100	Diversity = 1.7152e-01	k_t
Generation 3260: Best = 1.577241e+04 = [4, 3, 2, 3]	FES = 326100	Diversity = 1.6884e-01	k_t
Generation 3270: Best = 1.577190e+04 = [1, 1, 1, 1]	FES = 327100	Diversity = 1.6756e-01	k_t
Generation 3280: Best = 1.577144e+04 = [3, 4, 2, 4]	FES = 328100	Diversity = 1.6418e-01	k_t
Generation 3290: Best = 1.577014e+04 = [1, 1, 1, 1]	FES = 329100	Diversity = 1.6498e-01	k_t
Generation 3300: Best = 1.576899e+04 = [2, 4, 3, 3]	FES = 330100	Diversity = 1.6267e-01	k_t
Generation 3310: Best = 1.576629e+04 = [1, 1, 1, 1]	FES = 331100	Diversity = 1.6439e-01	k_t
Generation 3320: Best = 1.576383e+04 = [3, 4, 3, 2]	FES = 332100	Diversity = 1.6592e-01	k_t
Generation 3330: Best = 1.576150e+04 = [1, 1, 1, 1]	FES = 333100	Diversity = 1.6510e-01	k_t
Generation 3340: Best = 1.575548e+04 = [4, 3, 2, 2]	FES = 334100	Diversity = 1.6243e-01	k_t
Generation 3350: Best = 1.575486e+04 = [1, 1, 1, 1]	FES = 335100	Diversity = 1.6692e-01	k_t
Generation 3360: Best = 1.575225e+04 = [2, 2, 4, 2]	FES = 336100	Diversity = 1.6406e-01	k_t
Generation 3370: Best = 1.575225e+04 = [1, 1, 1, 1]	FES = 337100	Diversity = 1.6599e-01	k_t
Generation 3380: Best = 1.574739e+04 = [4, 2, 3, 4]	FES = 338100	Diversity = 1.6947e-01	k_t
Generation 3390: Best = 1.574715e+04 = [1, 1, 1, 1]	FES = 339100	Diversity = 1.7950e-01	k_t
Generation 3400: Best = 1.574518e+04 = [3, 4, 4, 4]	FES = 340100	Diversity = 1.9021e-01	k_t
Generation 3410: Best = 1.574120e+04 = [1, 1, 1, 1]	FES = 341100	Diversity = 2.0173e-01	k_t
Generation 3420: Best = 1.573863e+04 = [3, 3, 3, 4]	FES = 342100	Diversity = 2.1436e-01	k_t
Generation 3430: Best = 1.572961e+04 = [1, 1, 1, 1]	FES = 343100	Diversity = 2.2485e-01	k_t
Generation 3440: Best = 1.572708e+04 = [3, 4, 2, 2]	FES = 344100	Diversity = 2.2223e-01	k_t
Generation 3450: Best = 1.572671e+04 = [1, 1, 1, 1]	FES = 345100	Diversity = 2.1953e-01	k_t
Generation 3460: Best = 1.572278e+04 = [3, 4, 3, 4]	FES = 346100	Diversity = 2.1248e-01	k_t

Generation 3470: Best = 1.571533e+04 = [1, 1, 1, 1]	FES = 347100	Diversity = 2.1185e-01	k_t
Generation 3480: Best = 1.571532e+04 = [2, 2, 4, 3]	FES = 348100	Diversity = 2.1005e-01	k_t
Generation 3490: Best = 1.571409e+04 = [1, 1, 1, 1]	FES = 349100	Diversity = 2.0939e-01	k_t
Generation 3500: Best = 1.571230e+04 = [4, 2, 2, 3]	FES = 350100	Diversity = 2.0865e-01	k_t
Generation 3510: Best = 1.570881e+04 = [1, 1, 1, 1]	FES = 351100	Diversity = 2.0444e-01	k_t
Generation 3520: Best = 1.570879e+04 = [4, 2, 2, 3]	FES = 352100	Diversity = 2.0664e-01	k_t
Generation 3530: Best = 1.570640e+04 = [1, 1, 1, 1]	FES = 353100	Diversity = 2.0790e-01	k_t
Generation 3540: Best = 1.570372e+04 = [3, 2, 2, 3]	FES = 354100	Diversity = 2.0700e-01	k_t
Generation 3550: Best = 1.570309e+04 = [1, 1, 1, 1]	FES = 355100	Diversity = 2.0260e-01	k_t
Generation 3560: Best = 1.570207e+04 = [2, 2, 3, 4]	FES = 356100	Diversity = 2.0126e-01	k_t
Generation 3570: Best = 1.570190e+04 = [1, 1, 1, 1]	FES = 357100	Diversity = 1.9783e-01	k_t
Generation 3580: Best = 1.570185e+04 = [4, 4, 2, 4]	FES = 358100	Diversity = 1.9521e-01	k_t
Generation 3590: Best = 1.569856e+04 = [1, 1, 1, 1]	FES = 359100	Diversity = 1.9536e-01	k_t
Generation 3600: Best = 1.569738e+04 = [2, 4, 2, 2]	FES = 360100	Diversity = 1.9726e-01	k_t
Generation 3610: Best = 1.569596e+04 = [1, 1, 1, 1]	FES = 361100	Diversity = 2.0319e-01	k_t
Generation 3620: Best = 1.569527e+04 = [4, 4, 3, 3]	FES = 362100	Diversity = 2.0206e-01	k_t
Generation 3630: Best = 1.569527e+04 = [1, 1, 1, 1]	FES = 363100	Diversity = 2.0139e-01	k_t
Generation 3640: Best = 1.568361e+04 = [3, 3, 2, 2]	FES = 364100	Diversity = 2.2779e-01	k_t
Generation 3650: Best = 1.567040e+04 = [1, 1, 1, 1]	FES = 365100	Diversity = 3.0753e-01	k_t
Generation 3660: Best = 1.565294e+04 = [2, 3, 2, 3]	FES = 366100	Diversity = 3.2335e-01	k_t
Generation 3670: Best = 1.564952e+04 = [1, 1, 1, 1]	FES = 367100	Diversity = 2.9724e-01	k_t
Generation 3680: Best = 1.564520e+04 = [3, 4, 4, 2]	FES = 368100	Diversity = 2.6808e-01	k_t
Generation 3690: Best = 1.564124e+04 = [1, 1, 1, 1]	FES = 369100	Diversity = 2.5318e-01	k_t
Generation 3700: Best = 1.563971e+04 = [2, 4, 4, 2]	FES = 370100	Diversity = 2.3378e-01	k_t
Generation 3710: Best = 1.563453e+04 = [1, 1, 1, 1]	FES = 371100	Diversity = 2.1570e-01	k_t
Generation 3720: Best = 1.563330e+04 = [3, 3, 3, 3]	FES = 372100	Diversity = 2.0282e-01	k_t
Generation 3730: Best = 1.563111e+04 = [1, 1, 1, 1]	FES = 373100	Diversity = 1.9623e-01	k_t

Generation 3740: Best = 1.562987e+04 = [2, 3, 2, 3]	FES = 374100	Diversity = 1.9177e-01	k_t
Generation 3750: Best = 1.562769e+04 = [1, 1, 1, 1]	FES = 375100	Diversity = 1.9140e-01	k_t
Generation 3760: Best = 1.562769e+04 = [4, 3, 2, 3]	FES = 376100	Diversity = 1.8527e-01	k_t
Generation 3770: Best = 1.562754e+04 = [1, 1, 1, 1]	FES = 377100	Diversity = 1.8718e-01	k_t
Generation 3780: Best = 1.562664e+04 = [4, 4, 3, 2]	FES = 378100	Diversity = 1.8077e-01	k_t
Generation 3790: Best = 1.562631e+04 = [1, 1, 1, 1]	FES = 379100	Diversity = 1.8245e-01	k_t
Generation 3800: Best = 1.562452e+04 = [2, 4, 4, 3]	FES = 380100	Diversity = 1.8094e-01	k_t
Generation 3810: Best = 1.562412e+04 = [1, 1, 1, 1]	FES = 381100	Diversity = 1.7716e-01	k_t
Generation 3820: Best = 1.562067e+04 = [4, 2, 2, 4]	FES = 382100	Diversity = 1.7770e-01	k_t
Generation 3830: Best = 1.562067e+04 = [1, 1, 1, 1]	FES = 383100	Diversity = 1.7628e-01	k_t
Generation 3840: Best = 1.562067e+04 = [3, 4, 2, 3]	FES = 384100	Diversity = 1.7253e-01	k_t
Generation 3850: Best = 1.561868e+04 = [1, 1, 1, 1]	FES = 385100	Diversity = 1.7155e-01	k_t
Generation 3860: Best = 1.561861e+04 = [4, 2, 4, 2]	FES = 386100	Diversity = 1.6938e-01	k_t
Generation 3870: Best = 1.561861e+04 = [1, 1, 1, 1]	FES = 387100	Diversity = 1.6823e-01	k_t
Generation 3880: Best = 1.561843e+04 = [2, 2, 3, 3]	FES = 388100	Diversity = 1.6819e-01	k_t
Generation 3890: Best = 1.561842e+04 = [1, 1, 1, 1]	FES = 389100	Diversity = 1.6976e-01	k_t
Generation 3900: Best = 1.561768e+04 = [2, 2, 2, 2]	FES = 390100	Diversity = 1.6714e-01	k_t
Generation 3910: Best = 1.561761e+04 = [1, 1, 1, 1]	FES = 391100	Diversity = 1.6591e-01	k_t
Generation 3920: Best = 1.561737e+04 = [4, 3, 4, 3]	FES = 392100	Diversity = 1.6557e-01	k_t
Generation 3930: Best = 1.561592e+04 = [1, 1, 1, 1]	FES = 393100	Diversity = 1.6298e-01	k_t
Generation 3940: Best = 1.561529e+04 = [4, 3, 2, 3]	FES = 394100	Diversity = 1.6013e-01	k_t
Generation 3950: Best = 1.561529e+04 = [1, 1, 1, 1]	FES = 395100	Diversity = 1.5900e-01	k_t
Generation 3960: Best = 1.561510e+04 = [3, 2, 3, 4]	FES = 396100	Diversity = 1.6133e-01	k_t
Generation 3970: Best = 1.561510e+04 = [1, 1, 1, 1]	FES = 397100	Diversity = 1.6080e-01	k_t
Generation 3980: Best = 1.561416e+04 = [4, 3, 2, 2]	FES = 398100	Diversity = 1.6236e-01	k_t
Generation 3990: Best = 1.561367e+04 = [1, 1, 1, 1]	FES = 399100	Diversity = 1.5843e-01	k_t
Generation 4000: Best = 1.561192e+04 = [2, 2, 3, 4]	FES = 400100	Diversity = 1.5873e-01	k_t

Generation 4010: Best = 1.561192e+04 = [1, 1, 1, 1]	FES = 401100	Diversity = 1.5622e-01	k_t
Generation 4020: Best = 1.561187e+04 = [2, 2, 4, 2]	FES = 402100	Diversity = 1.5648e-01	k_t
Generation 4030: Best = 1.561187e+04 = [1, 1, 1, 1]	FES = 403100	Diversity = 1.5853e-01	k_t
Generation 4040: Best = 1.561145e+04 = [2, 4, 2, 3]	FES = 404100	Diversity = 1.5910e-01	k_t
Generation 4050: Best = 1.561068e+04 = [1, 1, 1, 1]	FES = 405100	Diversity = 1.5680e-01	k_t
Generation 4060: Best = 1.560976e+04 = [2, 2, 3, 4]	FES = 406100	Diversity = 1.5754e-01	k_t
Generation 4070: Best = 1.560928e+04 = [1, 1, 1, 1]	FES = 407100	Diversity = 1.5925e-01	k_t
Generation 4080: Best = 1.560928e+04 = [2, 4, 4, 2]	FES = 408100	Diversity = 1.6072e-01	k_t
Generation 4090: Best = 1.560890e+04 = [1, 1, 1, 1]	FES = 409100	Diversity = 1.5849e-01	k_t
Generation 4100: Best = 1.560785e+04 = [4, 3, 4, 3]	FES = 410100	Diversity = 1.6123e-01	k_t
Generation 4110: Best = 1.560785e+04 = [1, 1, 1, 1]	FES = 411100	Diversity = 1.5730e-01	k_t
Generation 4120: Best = 1.560671e+04 = [3, 4, 2, 3]	FES = 412100	Diversity = 1.5763e-01	k_t
Generation 4130: Best = 1.560670e+04 = [1, 1, 1, 1]	FES = 413100	Diversity = 1.5793e-01	k_t
Generation 4140: Best = 1.560647e+04 = [4, 3, 2, 3]	FES = 414100	Diversity = 1.5803e-01	k_t
Generation 4150: Best = 1.560628e+04 = [1, 1, 1, 1]	FES = 415100	Diversity = 1.6001e-01	k_t
Generation 4160: Best = 1.560628e+04 = [4, 2, 2, 4]	FES = 416100	Diversity = 1.6072e-01	k_t
Generation 4170: Best = 1.560526e+04 = [1, 1, 1, 1]	FES = 417100	Diversity = 1.5909e-01	k_t
Generation 4180: Best = 1.560485e+04 = [2, 2, 4, 2]	FES = 418100	Diversity = 1.6063e-01	k_t
Generation 4190: Best = 1.560458e+04 = [1, 1, 1, 1]	FES = 419100	Diversity = 1.6122e-01	k_t
Generation 4200: Best = 1.560458e+04 = [4, 4, 3, 3]	FES = 420100	Diversity = 1.6062e-01	k_t
Generation 4210: Best = 1.560420e+04 = [1, 1, 1, 1]	FES = 421100	Diversity = 1.6162e-01	k_t
Generation 4220: Best = 1.560417e+04 = [2, 4, 4, 2]	FES = 422100	Diversity = 1.6083e-01	k_t
Generation 4230: Best = 1.560288e+04 = [1, 1, 1, 1]	FES = 423100	Diversity = 1.5688e-01	k_t
Generation 4240: Best = 1.560252e+04 = [2, 4, 4, 3]	FES = 424100	Diversity = 1.5602e-01	k_t
Generation 4250: Best = 1.560252e+04 = [1, 1, 1, 1]	FES = 425100	Diversity = 1.5730e-01	k_t
Generation 4260: Best = 1.560240e+04 = [3, 3, 3, 4]	FES = 426100	Diversity = 1.5665e-01	k_t
Generation 4270: Best = 1.560177e+04 = [1, 1, 1, 1]	FES = 427100	Diversity = 1.5634e-01	k_t

Generation 4280: Best = 1.560145e+04 = [4, 3, 4, 4]	FES = 428100	Diversity = 1.5565e-01	k_t
Generation 4290: Best = 1.560063e+04 = [1, 1, 1, 1]	FES = 429100	Diversity = 1.5618e-01	k_t
Generation 4300: Best = 1.559989e+04 = [3, 2, 3, 2]	FES = 430100	Diversity = 1.5683e-01	k_t
Generation 4310: Best = 1.559989e+04 = [1, 1, 1, 1]	FES = 431100	Diversity = 1.5504e-01	k_t
Generation 4320: Best = 1.559989e+04 = [3, 3, 2, 4]	FES = 432100	Diversity = 1.5605e-01	k_t
Generation 4330: Best = 1.559985e+04 = [1, 1, 1, 1]	FES = 433100	Diversity = 1.5450e-01	k_t
Generation 4340: Best = 1.559971e+04 = [2, 3, 2, 2]	FES = 434100	Diversity = 1.5536e-01	k_t
Generation 4350: Best = 1.559971e+04 = [1, 1, 1, 1]	FES = 435100	Diversity = 1.5465e-01	k_t
Generation 4360: Best = 1.559949e+04 = [4, 3, 4, 2]	FES = 436100	Diversity = 1.5374e-01	k_t
Generation 4370: Best = 1.559920e+04 = [1, 1, 1, 1]	FES = 437100	Diversity = 1.5298e-01	k_t
Generation 4380: Best = 1.559920e+04 = [3, 3, 2, 2]	FES = 438100	Diversity = 1.5375e-01	k_t
Generation 4390: Best = 1.559869e+04 = [1, 1, 1, 1]	FES = 439100	Diversity = 1.5189e-01	k_t
Generation 4400: Best = 1.559855e+04 = [2, 2, 2, 3]	FES = 440100	Diversity = 1.4879e-01	k_t
Generation 4410: Best = 1.559752e+04 = [1, 1, 1, 1]	FES = 441100	Diversity = 1.4916e-01	k_t
Generation 4420: Best = 1.559752e+04 = [3, 4, 2, 4]	FES = 442100	Diversity = 1.4565e-01	k_t
Generation 4430: Best = 1.559745e+04 = [1, 1, 1, 1]	FES = 443100	Diversity = 1.4349e-01	k_t
Generation 4440: Best = 1.559745e+04 = [3, 3, 2, 4]	FES = 444100	Diversity = 1.4238e-01	k_t
Generation 4450: Best = 1.559721e+04 = [1, 1, 1, 1]	FES = 445100	Diversity = 1.3889e-01	k_t
Generation 4460: Best = 1.559581e+04 = [3, 2, 4, 4]	FES = 446100	Diversity = 1.4003e-01	k_t
Generation 4470: Best = 1.559504e+04 = [1, 1, 1, 1]	FES = 447100	Diversity = 1.3971e-01	k_t
Generation 4480: Best = 1.559504e+04 = [4, 3, 3, 4]	FES = 448100	Diversity = 1.3821e-01	k_t
Generation 4490: Best = 1.559504e+04 = [1, 1, 1, 1]	FES = 449100	Diversity = 1.3833e-01	k_t
Generation 4500: Best = 1.559504e+04 = [4, 3, 4, 3]	FES = 450100	Diversity = 1.3545e-01	k_t
Generation 4510: Best = 1.559466e+04 = [1, 1, 1, 1]	FES = 451100	Diversity = 1.3559e-01	k_t
Generation 4520: Best = 1.559384e+04 = [2, 4, 2, 4]	FES = 452100	Diversity = 1.3401e-01	k_t
Generation 4530: Best = 1.559378e+04 = [1, 1, 1, 1]	FES = 453100	Diversity = 1.3466e-01	k_t
Generation 4540: Best = 1.559333e+04 = [3, 2, 2, 3]	FES = 454100	Diversity = 1.3431e-01	k_t

Generation 4550: Best = 1.559270e+04 = [1, 1, 1, 1]	FES = 455100	Diversity = 1.3434e-01	k_t
Generation 4560: Best = 1.559247e+04 = [2, 2, 2, 2]	FES = 456100	Diversity = 1.3254e-01	k_t
Generation 4570: Best = 1.559166e+04 = [1, 1, 1, 1]	FES = 457100	Diversity = 1.3129e-01	k_t
Generation 4580: Best = 1.559118e+04 = [2, 3, 2, 4]	FES = 458100	Diversity = 1.3177e-01	k_t
Generation 4590: Best = 1.559096e+04 = [1, 1, 1, 1]	FES = 459100	Diversity = 1.2852e-01	k_t
Generation 4600: Best = 1.559086e+04 = [2, 4, 3, 4]	FES = 460100	Diversity = 1.2771e-01	k_t
Generation 4610: Best = 1.559000e+04 = [1, 1, 1, 1]	FES = 461100	Diversity = 1.2468e-01	k_t
Generation 4620: Best = 1.558951e+04 = [4, 3, 3, 3]	FES = 462100	Diversity = 1.2320e-01	k_t
Generation 4630: Best = 1.558903e+04 = [1, 1, 1, 1]	FES = 463100	Diversity = 1.2231e-01	k_t
Generation 4640: Best = 1.558774e+04 = [2, 4, 4, 4]	FES = 464100	Diversity = 1.1316e-01	k_t
Generation 4650: Best = 1.558647e+04 = [1, 1, 1, 1]	FES = 465100	Diversity = 1.0049e-01	k_t
Generation 4660: Best = 1.558574e+04 = [4, 4, 3, 2]	FES = 466100	Diversity = 9.7130e-02	k_t
Generation 4670: Best = 1.558531e+04 = [1, 1, 1, 1]	FES = 467100	Diversity = 8.8590e-02	k_t
Generation 4680: Best = 1.558349e+04 = [2, 3, 3, 3]	FES = 468100	Diversity = 7.9887e-02	k_t
Generation 4690: Best = 1.558325e+04 = [1, 1, 1, 1]	FES = 469100	Diversity = 6.4205e-02	k_t
Generation 4700: Best = 1.558200e+04 = [3, 4, 3, 3]	FES = 470100	Diversity = 5.9496e-02	k_t
Generation 4710: Best = 1.558117e+04 = [1, 1, 1, 1]	FES = 471100	Diversity = 5.7026e-02	k_t
Generation 4720: Best = 1.558020e+04 = [2, 3, 2, 3]	FES = 472100	Diversity = 5.2245e-02	k_t
Generation 4730: Best = 1.557937e+04 = [1, 1, 1, 1]	FES = 473100	Diversity = 4.8638e-02	k_t
Generation 4740: Best = 1.557795e+04 = [4, 4, 3, 2]	FES = 474100	Diversity = 4.4867e-02	k_t
Generation 4750: Best = 1.557723e+04 = [1, 1, 1, 1]	FES = 475100	Diversity = 4.5467e-02	k_t
Generation 4760: Best = 1.557666e+04 = [3, 4, 4, 3]	FES = 476100	Diversity = 4.1935e-02	k_t
Generation 4770: Best = 1.557585e+04 = [1, 1, 1, 1]	FES = 477100	Diversity = 4.0145e-02	k_t
Generation 4780: Best = 1.557508e+04 = [2, 3, 3, 3]	FES = 478100	Diversity = 3.6761e-02	k_t
Generation 4790: Best = 1.557424e+04 = [1, 1, 1, 1]	FES = 479100	Diversity = 3.6553e-02	k_t
Generation 4800: Best = 1.557276e+04 = [4, 4, 4, 2]	FES = 480100	Diversity = 3.5999e-02	k_t
Generation 4810: Best = 1.557212e+04 = [1, 1, 1, 1]	FES = 481100	Diversity = 3.5350e-02	k_t

Generation 4820: Best = 1.557150e+04 = [2, 2, 2, 4]	FES = 482100	Diversity = 3.4575e-02	k_t
Generation 4830: Best = 1.557060e+04 = [1, 1, 1, 1]	FES = 483100	Diversity = 3.4351e-02	k_t
Generation 4840: Best = 1.556970e+04 = [3, 4, 4, 2]	FES = 484100	Diversity = 3.3848e-02	k_t
Generation 4850: Best = 1.556886e+04 = [1, 1, 1, 1]	FES = 485100	Diversity = 3.2766e-02	k_t
Generation 4860: Best = 1.556851e+04 = [2, 4, 4, 3]	FES = 486100	Diversity = 3.2884e-02	k_t
Generation 4870: Best = 1.556785e+04 = [1, 1, 1, 1]	FES = 487100	Diversity = 3.2850e-02	k_t
Generation 4880: Best = 1.556741e+04 = [3, 4, 4, 2]	FES = 488100	Diversity = 3.0883e-02	k_t
Generation 4890: Best = 1.556695e+04 = [1, 1, 1, 1]	FES = 489100	Diversity = 3.0508e-02	k_t
Generation 4900: Best = 1.556650e+04 = [3, 3, 3, 3]	FES = 490100	Diversity = 2.6467e-02	k_t
Generation 4910: Best = 1.556619e+04 = [1, 1, 1, 1]	FES = 491100	Diversity = 2.3415e-02	k_t
Generation 4920: Best = 1.556590e+04 = [3, 3, 3, 3]	FES = 492100	Diversity = 2.1313e-02	k_t
Generation 4930: Best = 1.556575e+04 = [1, 1, 1, 1]	FES = 493100	Diversity = 2.0627e-02	k_t
Generation 4940: Best = 1.556553e+04 = [2, 4, 4, 4]	FES = 494100	Diversity = 1.8420e-02	k_t
Generation 4950: Best = 1.556531e+04 = [1, 1, 1, 1]	FES = 495100	Diversity = 1.6155e-02	k_t
Generation 4960: Best = 1.556508e+04 = [3, 3, 4, 2]	FES = 496100	Diversity = 1.5499e-02	k_t
Generation 4970: Best = 1.556496e+04 = [1, 1, 1, 1]	FES = 497100	Diversity = 1.4868e-02	k_t
Generation 4980: Best = 1.556472e+04 = [4, 4, 4, 2]	FES = 498100	Diversity = 1.4922e-02	k_t
Generation 4990: Best = 1.556452e+04 = [1, 1, 1, 1]	FES = 499100	Diversity = 1.6152e-02	k_t
Generation 5000: Best = 1.556420e+04 = [4, 2, 2, 4]	FES = 500100	Diversity = 1.5403e-02	k_t
Generation 5010: Best = 1.556408e+04 = [1, 1, 1, 1]	FES = 501100	Diversity = 1.5716e-02	k_t
Generation 5020: Best = 1.556373e+04 = [4, 4, 3, 3]	FES = 502100	Diversity = 1.5927e-02	k_t
Generation 5030: Best = 1.556358e+04 = [1, 1, 1, 1]	FES = 503100	Diversity = 1.6264e-02	k_t
Generation 5040: Best = 1.556326e+04 = [4, 4, 2, 2]	FES = 504100	Diversity = 1.6988e-02	k_t
Generation 5050: Best = 1.556284e+04 = [1, 1, 1, 1]	FES = 505100	Diversity = 1.8418e-02	k_t
Generation 5060: Best = 1.556236e+04 = [2, 4, 2, 4]	FES = 506100	Diversity = 1.9908e-02	k_t
Generation 5070: Best = 1.556204e+04 = [1, 1, 1, 1]	FES = 507100	Diversity = 2.2021e-02	k_t
Generation 5080: Best = 1.556157e+04 = [3, 2, 4, 2]	FES = 508100	Diversity = 1.9782e-02	k_t

Generation 5090: Best = 1.556145e+04 = [1, 1, 1, 1]	FES = 509100	Diversity = 1.7984e-02	k_t
Generation 5100: Best = 1.556124e+04 = [3, 4, 2, 4]	FES = 510100	Diversity = 1.5687e-02	k_t
Generation 5110: Best = 1.556105e+04 = [1, 1, 1, 1]	FES = 511100	Diversity = 1.5383e-02	k_t
Generation 5120: Best = 1.556074e+04 = [3, 2, 3, 4]	FES = 512100	Diversity = 1.4624e-02	k_t
Generation 5130: Best = 1.556050e+04 = [1, 1, 1, 1]	FES = 513100	Diversity = 1.4516e-02	k_t
Generation 5140: Best = 1.556034e+04 = [4, 3, 2, 2]	FES = 514100	Diversity = 1.3532e-02	k_t
Generation 5150: Best = 1.556018e+04 = [1, 1, 1, 1]	FES = 515100	Diversity = 1.3189e-02	k_t
Generation 5160: Best = 1.556000e+04 = [3, 4, 3, 2]	FES = 516100	Diversity = 1.1745e-02	k_t
Generation 5170: Best = 1.555985e+04 = [1, 1, 1, 1]	FES = 517100	Diversity = 1.1269e-02	k_t
Generation 5180: Best = 1.555971e+04 = [2, 4, 2, 4]	FES = 518100	Diversity = 1.0582e-02	k_t
Generation 5190: Best = 1.555958e+04 = [1, 1, 1, 1]	FES = 519100	Diversity = 1.0596e-02	k_t
Generation 5200: Best = 1.555948e+04 = [3, 3, 3, 2]	FES = 520100	Diversity = 1.0437e-02	k_t
Generation 5210: Best = 1.555931e+04 = [1, 1, 1, 1]	FES = 521100	Diversity = 9.9920e-03	k_t
Generation 5220: Best = 1.555908e+04 = [4, 2, 4, 3]	FES = 522100	Diversity = 1.0282e-02	k_t
Generation 5230: Best = 1.555902e+04 = [1, 1, 1, 1]	FES = 523100	Diversity = 9.4852e-03	k_t
Generation 5240: Best = 1.555890e+04 = [2, 2, 4, 2]	FES = 524100	Diversity = 9.3199e-03	k_t
Generation 5250: Best = 1.555865e+04 = [1, 1, 1, 1]	FES = 525100	Diversity = 8.9793e-03	k_t
Generation 5260: Best = 1.555847e+04 = [4, 4, 3, 3]	FES = 526100	Diversity = 9.1619e-03	k_t
Generation 5270: Best = 1.555837e+04 = [1, 1, 1, 1]	FES = 527100	Diversity = 9.3029e-03	k_t
Generation 5280: Best = 1.555814e+04 = [2, 3, 2, 2]	FES = 528100	Diversity = 9.6061e-03	k_t
Generation 5290: Best = 1.555794e+04 = [1, 1, 1, 1]	FES = 529100	Diversity = 1.0167e-02	k_t
Generation 5300: Best = 1.555778e+04 = [4, 3, 3, 2]	FES = 530100	Diversity = 1.0012e-02	k_t
Generation 5310: Best = 1.555760e+04 = [1, 1, 1, 1]	FES = 531100	Diversity = 9.2467e-03	k_t
Generation 5320: Best = 1.555746e+04 = [2, 3, 4, 4]	FES = 532100	Diversity = 1.0358e-02	k_t
Generation 5330: Best = 1.555722e+04 = [1, 1, 1, 1]	FES = 533100	Diversity = 9.7879e-03	k_t
Generation 5340: Best = 1.555710e+04 = [4, 2, 2, 2]	FES = 534100	Diversity = 9.0924e-03	k_t
Generation 5350: Best = 1.555699e+04 = [1, 1, 1, 1]	FES = 535100	Diversity = 8.8946e-03	k_t

Generation 5360: Best = 1.555686e+04 = [4, 3, 2, 3]	FES = 536100	Diversity = 8.5231e-03	k_t
Generation 5370: Best = 1.555676e+04 = [1, 1, 1, 1]	FES = 537100	Diversity = 8.5274e-03	k_t
Generation 5380: Best = 1.555662e+04 = [2, 4, 4, 2]	FES = 538100	Diversity = 8.3153e-03	k_t
Generation 5390: Best = 1.555651e+04 = [1, 1, 1, 1]	FES = 539100	Diversity = 8.0561e-03	k_t
Generation 5400: Best = 1.555638e+04 = [2, 2, 3, 3]	FES = 540100	Diversity = 7.5171e-03	k_t
Generation 5410: Best = 1.555629e+04 = [1, 1, 1, 1]	FES = 541100	Diversity = 7.5999e-03	k_t
Generation 5420: Best = 1.555613e+04 = [4, 2, 4, 4]	FES = 542100	Diversity = 7.2994e-03	k_t
Generation 5430: Best = 1.555602e+04 = [1, 1, 1, 1]	FES = 543100	Diversity = 7.0334e-03	k_t
Generation 5440: Best = 1.555594e+04 = [4, 3, 2, 3]	FES = 544100	Diversity = 6.6485e-03	k_t
Generation 5450: Best = 1.555586e+04 = [1, 1, 1, 1]	FES = 545100	Diversity = 6.0022e-03	k_t
Generation 5460: Best = 1.555580e+04 = [2, 4, 4, 3]	FES = 546100	Diversity = 5.5032e-03	k_t
Generation 5470: Best = 1.555573e+04 = [1, 1, 1, 1]	FES = 547100	Diversity = 5.0434e-03	k_t
Generation 5480: Best = 1.555567e+04 = [2, 4, 4, 4]	FES = 548100	Diversity = 4.9435e-03	k_t
Generation 5490: Best = 1.555561e+04 = [1, 1, 1, 1]	FES = 549100	Diversity = 4.7966e-03	k_t
Generation 5500: Best = 1.555554e+04 = [4, 2, 3, 2]	FES = 550100	Diversity = 4.6942e-03	k_t
Generation 5510: Best = 1.555548e+04 = [1, 1, 1, 1]	FES = 551100	Diversity = 4.4553e-03	k_t
Generation 5520: Best = 1.555540e+04 = [2, 3, 4, 3]	FES = 552100	Diversity = 4.0946e-03	k_t
Generation 5530: Best = 1.555534e+04 = [1, 1, 1, 1]	FES = 553100	Diversity = 4.0084e-03	k_t
Generation 5540: Best = 1.555529e+04 = [4, 2, 4, 4]	FES = 554100	Diversity = 4.0401e-03	k_t
Generation 5550: Best = 1.555522e+04 = [1, 1, 1, 1]	FES = 555100	Diversity = 4.3245e-03	k_t
Generation 5560: Best = 1.555516e+04 = [3, 3, 2, 3]	FES = 556100	Diversity = 4.3379e-03	k_t
Generation 5570: Best = 1.555509e+04 = [1, 1, 1, 1]	FES = 557100	Diversity = 4.8306e-03	k_t
Generation 5580: Best = 1.555503e+04 = [4, 4, 3, 4]	FES = 558100	Diversity = 4.5316e-03	k_t
Generation 5590: Best = 1.555498e+04 = [1, 1, 1, 1]	FES = 559100	Diversity = 4.2480e-03	k_t
Generation 5600: Best = 1.555494e+04 = [3, 2, 2, 4]	FES = 560100	Diversity = 4.0630e-03	k_t
Generation 5610: Best = 1.555491e+04 = [1, 1, 1, 1]	FES = 561100	Diversity = 3.9799e-03	k_t
Generation 5620: Best = 1.555489e+04 = [4, 4, 3, 2]	FES = 562100	Diversity = 3.9218e-03	k_t

Generation 5630: Best = 1.555483e+04 = [1, 1, 1, 1]	FES = 563100	Diversity = 3.9550e-03	k_t
Generation 5640: Best = 1.555480e+04 = [2, 4, 2, 4]	FES = 564100	Diversity = 3.8865e-03	k_t
Generation 5650: Best = 1.555474e+04 = [1, 1, 1, 1]	FES = 565100	Diversity = 4.0055e-03	k_t
Generation 5660: Best = 1.555469e+04 = [4, 2, 3, 4]	FES = 566100	Diversity = 4.1152e-03	k_t
Generation 5670: Best = 1.555464e+04 = [1, 1, 1, 1]	FES = 567100	Diversity = 3.9699e-03	k_t
Generation 5680: Best = 1.555461e+04 = [2, 2, 3, 4]	FES = 568100	Diversity = 4.0386e-03	k_t
Generation 5690: Best = 1.555458e+04 = [1, 1, 1, 1]	FES = 569100	Diversity = 4.2562e-03	k_t
Generation 5700: Best = 1.555456e+04 = [4, 2, 3, 2]	FES = 570100	Diversity = 3.9374e-03	k_t
Generation 5710: Best = 1.555452e+04 = [1, 1, 1, 1]	FES = 571100	Diversity = 4.0169e-03	k_t
Generation 5720: Best = 1.555449e+04 = [3, 3, 3, 2]	FES = 572100	Diversity = 3.7334e-03	k_t
Generation 5730: Best = 1.555445e+04 = [1, 1, 1, 1]	FES = 573100	Diversity = 3.7619e-03	k_t
Generation 5740: Best = 1.555440e+04 = [4, 3, 4, 3]	FES = 574100	Diversity = 4.0785e-03	k_t
Generation 5750: Best = 1.555438e+04 = [1, 1, 1, 1]	FES = 575100	Diversity = 3.9985e-03	k_t
Generation 5760: Best = 1.555434e+04 = [4, 2, 3, 3]	FES = 576100	Diversity = 3.8829e-03	k_t
Generation 5770: Best = 1.555430e+04 = [1, 1, 1, 1]	FES = 577100	Diversity = 3.7873e-03	k_t
Generation 5780: Best = 1.555427e+04 = [4, 3, 4, 2]	FES = 578100	Diversity = 3.5884e-03	k_t
Generation 5790: Best = 1.555424e+04 = [1, 1, 1, 1]	FES = 579100	Diversity = 3.0985e-03	k_t
Generation 5800: Best = 1.555422e+04 = [4, 4, 4, 4]	FES = 580100	Diversity = 2.9719e-03	k_t
Generation 5810: Best = 1.555419e+04 = [1, 1, 1, 1]	FES = 581100	Diversity = 3.0599e-03	k_t
Generation 5820: Best = 1.555416e+04 = [4, 2, 3, 3]	FES = 582100	Diversity = 3.0924e-03	k_t
Generation 5830: Best = 1.555414e+04 = [1, 1, 1, 1]	FES = 583100	Diversity = 3.2529e-03	k_t
Generation 5840: Best = 1.555410e+04 = [3, 4, 2, 4]	FES = 584100	Diversity = 3.5197e-03	k_t
Generation 5850: Best = 1.555405e+04 = [1, 1, 1, 1]	FES = 585100	Diversity = 4.1956e-03	k_t
Generation 5860: Best = 1.555402e+04 = [3, 2, 3, 4]	FES = 586100	Diversity = 3.8436e-03	k_t
Generation 5870: Best = 1.555400e+04 = [1, 1, 1, 1]	FES = 587100	Diversity = 3.6176e-03	k_t
Generation 5880: Best = 1.555396e+04 = [2, 2, 3, 4]	FES = 588100	Diversity = 3.5243e-03	k_t
Generation 5890: Best = 1.555396e+04 = [1, 1, 1, 1]	FES = 589100	Diversity = 3.3017e-03	k_t

Generation 5900: Best = 1.555393e+04 = [2, 2, 3, 2]	FES = 590100	Diversity = 3.2576e-03	k_t
Generation 5910: Best = 1.555392e+04 = [1, 1, 1, 1]	FES = 591100	Diversity = 3.1186e-03	k_t
Generation 5920: Best = 1.555390e+04 = [2, 4, 2, 3]	FES = 592100	Diversity = 2.9855e-03	k_t
Generation 5930: Best = 1.555388e+04 = [1, 1, 1, 1]	FES = 593100	Diversity = 2.7295e-03	k_t
Generation 5940: Best = 1.555386e+04 = [4, 3, 4, 2]	FES = 594100	Diversity = 2.6559e-03	k_t
Generation 5950: Best = 1.555385e+04 = [1, 1, 1, 1]	FES = 595100	Diversity = 2.4608e-03	k_t
Generation 5960: Best = 1.555384e+04 = [4, 3, 2, 3]	FES = 596100	Diversity = 2.3632e-03	k_t
Generation 5970: Best = 1.555383e+04 = [1, 1, 1, 1]	FES = 597100	Diversity = 2.1878e-03	k_t
Generation 5980: Best = 1.555381e+04 = [4, 3, 2, 2]	FES = 598100	Diversity = 2.2185e-03	k_t
Generation 5990: Best = 1.555380e+04 = [1, 1, 1, 1]	FES = 599100	Diversity = 2.0372e-03	k_t
Generation 6000: Best = 1.555379e+04 = [3, 2, 2, 2]	FES = 600100	Diversity = 1.9636e-03	k_t
Generation 6010: Best = 1.555378e+04 = [1, 1, 1, 1]	FES = 601100	Diversity = 1.9325e-03	k_t
Generation 6020: Best = 1.555376e+04 = [2, 3, 2, 2]	FES = 602100	Diversity = 2.1449e-03	k_t
Generation 6030: Best = 1.555374e+04 = [1, 1, 1, 1]	FES = 603100	Diversity = 2.2774e-03	k_t
Generation 6040: Best = 1.555373e+04 = [4, 2, 2, 2]	FES = 604100	Diversity = 2.2823e-03	k_t
Generation 6050: Best = 1.555371e+04 = [1, 1, 1, 1]	FES = 605100	Diversity = 2.3759e-03	k_t
Generation 6060: Best = 1.555370e+04 = [2, 4, 2, 3]	FES = 606100	Diversity = 2.6207e-03	k_t
Generation 6070: Best = 1.555369e+04 = [1, 1, 1, 1]	FES = 607100	Diversity = 2.4548e-03	k_t
Generation 6080: Best = 1.555368e+04 = [2, 3, 3, 3]	FES = 608100	Diversity = 2.4117e-03	k_t
Generation 6090: Best = 1.555367e+04 = [1, 1, 1, 1]	FES = 609100	Diversity = 1.9897e-03	k_t
Generation 6100: Best = 1.555366e+04 = [3, 3, 3, 2]	FES = 610100	Diversity = 1.7937e-03	k_t
Generation 6110: Best = 1.555365e+04 = [1, 1, 1, 1]	FES = 611100	Diversity = 1.7294e-03	k_t
Generation 6120: Best = 1.555364e+04 = [4, 2, 2, 3]	FES = 612100	Diversity = 1.9828e-03	k_t
Generation 6130: Best = 1.555363e+04 = [1, 1, 1, 1]	FES = 613100	Diversity = 2.0047e-03	k_t
Generation 6140: Best = 1.555362e+04 = [3, 3, 3, 2]	FES = 614100	Diversity = 2.1851e-03	k_t
Generation 6150: Best = 1.555361e+04 = [1, 1, 1, 1]	FES = 615100	Diversity = 2.3483e-03	k_t
Generation 6160: Best = 1.555360e+04 = [4, 4, 3, 2]	FES = 616100	Diversity = 2.1208e-03	k_t

Generation 6170: Best = 1.555358e+04 = [1, 1, 1, 1]	FES = 617100	Diversity = 1.8782e-03	k_t
Generation 6180: Best = 1.555358e+04 = [4, 3, 4, 3]	FES = 618100	Diversity = 2.0078e-03	k_t
Generation 6190: Best = 1.555357e+04 = [1, 1, 1, 1]	FES = 619100	Diversity = 2.0135e-03	k_t
Generation 6200: Best = 1.555356e+04 = [2, 3, 4, 2]	FES = 620100	Diversity = 1.8419e-03	k_t
Generation 6210: Best = 1.555355e+04 = [1, 1, 1, 1]	FES = 621100	Diversity = 1.7631e-03	k_t
Generation 6220: Best = 1.555354e+04 = [4, 4, 4, 3]	FES = 622100	Diversity = 1.7462e-03	k_t
Generation 6230: Best = 1.555354e+04 = [1, 1, 1, 1]	FES = 623100	Diversity = 1.6966e-03	k_t
Generation 6240: Best = 1.555353e+04 = [4, 3, 3, 3]	FES = 624100	Diversity = 1.7691e-03	k_t
Generation 6250: Best = 1.555352e+04 = [1, 1, 1, 1]	FES = 625100	Diversity = 1.6635e-03	k_t
Generation 6260: Best = 1.555351e+04 = [3, 3, 3, 2]	FES = 626100	Diversity = 1.6320e-03	k_t
Generation 6270: Best = 1.555351e+04 = [1, 1, 1, 1]	FES = 627100	Diversity = 1.4636e-03	k_t
Generation 6280: Best = 1.555350e+04 = [3, 3, 2, 3]	FES = 628100	Diversity = 1.3794e-03	k_t
Generation 6290: Best = 1.555350e+04 = [1, 1, 1, 1]	FES = 629100	Diversity = 1.3792e-03	k_t
Generation 6300: Best = 1.555349e+04 = [4, 2, 4, 2]	FES = 630100	Diversity = 1.3199e-03	k_t
Generation 6310: Best = 1.555349e+04 = [1, 1, 1, 1]	FES = 631100	Diversity = 1.3758e-03	k_t
Generation 6320: Best = 1.555348e+04 = [2, 3, 2, 4]	FES = 632100	Diversity = 1.4277e-03	k_t
Generation 6330: Best = 1.555348e+04 = [1, 1, 1, 1]	FES = 633100	Diversity = 1.4372e-03	k_t
Generation 6340: Best = 1.555347e+04 = [4, 3, 2, 3]	FES = 634100	Diversity = 1.3939e-03	k_t
Generation 6350: Best = 1.555347e+04 = [1, 1, 1, 1]	FES = 635100	Diversity = 1.2240e-03	k_t
Generation 6360: Best = 1.555347e+04 = [2, 4, 4, 4]	FES = 636100	Diversity = 1.1029e-03	k_t
Generation 6370: Best = 1.555346e+04 = [1, 1, 1, 1]	FES = 637100	Diversity = 1.1257e-03	k_t
Generation 6380: Best = 1.555346e+04 = [4, 4, 3, 4]	FES = 638100	Diversity = 1.1819e-03	k_t
Generation 6390: Best = 1.555345e+04 = [1, 1, 1, 1]	FES = 639100	Diversity = 1.0725e-03	k_t
Generation 6400: Best = 1.555345e+04 = [2, 3, 2, 3]	FES = 640100	Diversity = 1.1141e-03	k_t
Generation 6410: Best = 1.555345e+04 = [1, 1, 1, 1]	FES = 641100	Diversity = 1.2243e-03	k_t
Generation 6420: Best = 1.555344e+04 = [3, 3, 4, 2]	FES = 642100	Diversity = 1.3012e-03	k_t
Generation 6430: Best = 1.555344e+04 = [1, 1, 1, 1]	FES = 643100	Diversity = 1.2830e-03	k_t

Generation 6440: Best = 1.555343e+04 = [2, 4, 3, 4]	FES = 644100	Diversity = 1.3051e-03	k_t
Generation 6450: Best = 1.555343e+04 = [1, 1, 1, 1]	FES = 645100	Diversity = 1.3554e-03	k_t
Generation 6460: Best = 1.555343e+04 = [4, 4, 4, 2]	FES = 646100	Diversity = 1.4279e-03	k_t
Generation 6470: Best = 1.555342e+04 = [1, 1, 1, 1]	FES = 647100	Diversity = 1.2946e-03	k_t
Generation 6480: Best = 1.555341e+04 = [4, 2, 3, 2]	FES = 648100	Diversity = 1.3065e-03	k_t
Generation 6490: Best = 1.555341e+04 = [1, 1, 1, 1]	FES = 649100	Diversity = 1.3005e-03	k_t
Generation 6500: Best = 1.555341e+04 = [2, 3, 3, 3]	FES = 650100	Diversity = 1.2330e-03	k_t
Generation 6510: Best = 1.555340e+04 = [1, 1, 1, 1]	FES = 651100	Diversity = 1.0874e-03	k_t
Generation 6520: Best = 1.555340e+04 = [4, 3, 4, 2]	FES = 652100	Diversity = 9.5532e-04	k_t
Generation 6530: Best = 1.555339e+04 = [1, 1, 1, 1]	FES = 653100	Diversity = 9.3008e-04	k_t
Generation 6540: Best = 1.555339e+04 = [3, 2, 3, 3]	FES = 654100	Diversity = 8.9793e-04	k_t
Generation 6550: Best = 1.555339e+04 = [1, 1, 1, 1]	FES = 655100	Diversity = 9.3320e-04	k_t
Generation 6560: Best = 1.555338e+04 = [2, 4, 2, 4]	FES = 656100	Diversity = 9.1626e-04	k_t
Generation 6570: Best = 1.555338e+04 = [1, 1, 1, 1]	FES = 657100	Diversity = 9.5365e-04	k_t
Generation 6580: Best = 1.555337e+04 = [4, 3, 4, 4]	FES = 658100	Diversity = 1.0548e-03	k_t
Generation 6590: Best = 1.555336e+04 = [1, 1, 1, 1]	FES = 659100	Diversity = 1.4335e-03	k_t
Generation 6600: Best = 1.555336e+04 = [3, 3, 2, 3]	FES = 660100	Diversity = 1.5774e-03	k_t
Generation 6610: Best = 1.555335e+04 = [1, 1, 1, 1]	FES = 661100	Diversity = 1.5907e-03	k_t
Generation 6620: Best = 1.555334e+04 = [3, 4, 2, 4]	FES = 662100	Diversity = 1.5814e-03	k_t
Generation 6630: Best = 1.555334e+04 = [1, 1, 1, 1]	FES = 663100	Diversity = 1.6573e-03	k_t
Generation 6640: Best = 1.555333e+04 = [3, 2, 3, 2]	FES = 664100	Diversity = 1.5694e-03	k_t
Generation 6650: Best = 1.555333e+04 = [1, 1, 1, 1]	FES = 665100	Diversity = 1.5912e-03	k_t
Generation 6660: Best = 1.555332e+04 = [4, 2, 2, 4]	FES = 666100	Diversity = 1.7224e-03	k_t
Generation 6670: Best = 1.555332e+04 = [1, 1, 1, 1]	FES = 667100	Diversity = 1.5585e-03	k_t
Generation 6680: Best = 1.555331e+04 = [4, 4, 4, 3]	FES = 668100	Diversity = 1.4922e-03	k_t
Generation 6690: Best = 1.555330e+04 = [1, 1, 1, 1]	FES = 669100	Diversity = 1.2037e-03	k_t
Generation 6700: Best = 1.555330e+04 = [4, 3, 4, 4]	FES = 670100	Diversity = 1.3523e-03	k_t

Generation 6710: Best = 1.555329e+04 = [1, 1, 1, 1]	FES = 671100	Diversity = 1.4052e-03	k_t
Generation 6720: Best = 1.555328e+04 = [4, 4, 3, 4]	FES = 672100	Diversity = 1.3996e-03	k_t
Generation 6730: Best = 1.555328e+04 = [1, 1, 1, 1]	FES = 673100	Diversity = 1.3317e-03	k_t
Generation 6740: Best = 1.555327e+04 = [2, 3, 3, 2]	FES = 674100	Diversity = 1.2748e-03	k_t
Generation 6750: Best = 1.555327e+04 = [1, 1, 1, 1]	FES = 675100	Diversity = 1.2409e-03	k_t
Generation 6760: Best = 1.555326e+04 = [2, 2, 4, 3]	FES = 676100	Diversity = 1.3489e-03	k_t
Generation 6770: Best = 1.555326e+04 = [1, 1, 1, 1]	FES = 677100	Diversity = 1.3279e-03	k_t
Generation 6780: Best = 1.555325e+04 = [2, 4, 4, 2]	FES = 678100	Diversity = 1.2053e-03	k_t
Generation 6790: Best = 1.555325e+04 = [1, 1, 1, 1]	FES = 679100	Diversity = 1.2730e-03	k_t
Generation 6800: Best = 1.555325e+04 = [3, 4, 4, 2]	FES = 680100	Diversity = 1.1161e-03	k_t
Generation 6810: Best = 1.555324e+04 = [1, 1, 1, 1]	FES = 681100	Diversity = 1.1256e-03	k_t
Generation 6820: Best = 1.555324e+04 = [4, 4, 2, 2]	FES = 682100	Diversity = 9.8381e-04	k_t
Generation 6830: Best = 1.555324e+04 = [1, 1, 1, 1]	FES = 683100	Diversity = 9.7425e-04	k_t
Generation 6840: Best = 1.555323e+04 = [2, 4, 3, 2]	FES = 684100	Diversity = 8.7750e-04	k_t
Generation 6850: Best = 1.555323e+04 = [1, 1, 1, 1]	FES = 685100	Diversity = 8.4742e-04	k_t
Generation 6860: Best = 1.555322e+04 = [2, 4, 2, 2]	FES = 686100	Diversity = 8.7744e-04	k_t
Generation 6870: Best = 1.555322e+04 = [1, 1, 1, 1]	FES = 687100	Diversity = 1.0993e-03	k_t
Generation 6880: Best = 1.555322e+04 = [4, 4, 2, 3]	FES = 688100	Diversity = 1.1642e-03	k_t
Generation 6890: Best = 1.555321e+04 = [1, 1, 1, 1]	FES = 689100	Diversity = 1.3416e-03	k_t
Generation 6900: Best = 1.555320e+04 = [2, 4, 4, 3]	FES = 690100	Diversity = 1.2886e-03	k_t
Generation 6910: Best = 1.555320e+04 = [1, 1, 1, 1]	FES = 691100	Diversity = 1.3359e-03	k_t
Generation 6920: Best = 1.555319e+04 = [4, 4, 2, 3]	FES = 692100	Diversity = 1.3098e-03	k_t
Generation 6930: Best = 1.555319e+04 = [1, 1, 1, 1]	FES = 693100	Diversity = 1.1815e-03	k_t
Generation 6940: Best = 1.555319e+04 = [2, 2, 3, 2]	FES = 694100	Diversity = 1.1239e-03	k_t
Generation 6950: Best = 1.555318e+04 = [1, 1, 1, 1]	FES = 695100	Diversity = 1.1075e-03	k_t
Generation 6960: Best = 1.555318e+04 = [3, 2, 2, 2]	FES = 696100	Diversity = 1.0329e-03	k_t
Generation 6970: Best = 1.555317e+04 = [1, 1, 1, 1]	FES = 697100	Diversity = 8.4269e-04	k_t

Generation 6980: Best = 1.555317e+04 = [2, 2, 4, 3]	FES = 698100	Diversity = 7.5597e-04	k_t
Generation 6990: Best = 1.555317e+04 = [1, 1, 1, 1]	FES = 699100	Diversity = 7.6924e-04	k_t
Generation 7000: Best = 1.555317e+04 = [2, 3, 3, 3]	FES = 700100	Diversity = 7.6426e-04	k_t
Generation 7010: Best = 1.555316e+04 = [1, 1, 1, 1]	FES = 701100	Diversity = 7.6783e-04	k_t
Generation 7020: Best = 1.555316e+04 = [4, 2, 3, 3]	FES = 702100	Diversity = 8.4406e-04	k_t
Generation 7030: Best = 1.555316e+04 = [1, 1, 1, 1]	FES = 703100	Diversity = 8.7985e-04	k_t
Generation 7040: Best = 1.555316e+04 = [2, 2, 4, 4]	FES = 704100	Diversity = 9.1669e-04	k_t
Generation 7050: Best = 1.555315e+04 = [1, 1, 1, 1]	FES = 705100	Diversity = 9.8392e-04	k_t
Generation 7060: Best = 1.555315e+04 = [3, 4, 4, 2]	FES = 706100	Diversity = 1.0424e-03	k_t
Generation 7070: Best = 1.555314e+04 = [1, 1, 1, 1]	FES = 707100	Diversity = 1.0216e-03	k_t
Generation 7080: Best = 1.555314e+04 = [2, 3, 4, 4]	FES = 708100	Diversity = 9.7013e-04	k_t
Generation 7090: Best = 1.555314e+04 = [1, 1, 1, 1]	FES = 709100	Diversity = 9.2457e-04	k_t
Generation 7100: Best = 1.555313e+04 = [2, 4, 3, 4]	FES = 710100	Diversity = 8.4933e-04	k_t
Generation 7110: Best = 1.555313e+04 = [1, 1, 1, 1]	FES = 711100	Diversity = 9.3547e-04	k_t
Generation 7120: Best = 1.555312e+04 = [3, 3, 4, 3]	FES = 712100	Diversity = 9.6638e-04	k_t
Generation 7130: Best = 1.555312e+04 = [1, 1, 1, 1]	FES = 713100	Diversity = 9.3263e-04	k_t
Generation 7140: Best = 1.555311e+04 = [3, 2, 3, 2]	FES = 714100	Diversity = 8.6751e-04	k_t
Generation 7150: Best = 1.555311e+04 = [1, 1, 1, 1]	FES = 715100	Diversity = 8.6956e-04	k_t
Generation 7160: Best = 1.555310e+04 = [3, 2, 3, 2]	FES = 716100	Diversity = 1.0950e-03	k_t
Generation 7170: Best = 1.555310e+04 = [1, 1, 1, 1]	FES = 717100	Diversity = 1.0611e-03	k_t
Generation 7180: Best = 1.555310e+04 = [2, 2, 3, 3]	FES = 718100	Diversity = 9.2730e-04	k_t
Generation 7190: Best = 1.555310e+04 = [1, 1, 1, 1]	FES = 719100	Diversity = 1.0623e-03	k_t
Generation 7200: Best = 1.555309e+04 = [4, 3, 3, 2]	FES = 720100	Diversity = 1.0925e-03	k_t
Generation 7210: Best = 1.555309e+04 = [1, 1, 1, 1]	FES = 721100	Diversity = 1.1263e-03	k_t
Generation 7220: Best = 1.555309e+04 = [3, 4, 3, 4]	FES = 722100	Diversity = 1.1431e-03	k_t
Generation 7230: Best = 1.555308e+04 = [1, 1, 1, 1]	FES = 723100	Diversity = 9.8497e-04	k_t
Generation 7240: Best = 1.555308e+04 = [2, 3, 3, 4]	FES = 724100	Diversity = 9.4336e-04	k_t

Generation 7250: Best = 1.555308e+04 = [1, 1, 1, 1]	FES = 725100	Diversity = 9.8632e-04	k_t
Generation 7260: Best = 1.555308e+04 = [2, 3, 2, 3]	FES = 726100	Diversity = 8.3677e-04	k_t
Generation 7270: Best = 1.555308e+04 = [1, 1, 1, 1]	FES = 727100	Diversity = 7.1660e-04	k_t
Generation 7280: Best = 1.555307e+04 = [4, 2, 3, 3]	FES = 728100	Diversity = 7.2743e-04	k_t
Generation 7290: Best = 1.555307e+04 = [1, 1, 1, 1]	FES = 729100	Diversity = 7.7637e-04	k_t
Generation 7300: Best = 1.555307e+04 = [3, 2, 3, 4]	FES = 730100	Diversity = 7.7172e-04	k_t
Generation 7310: Best = 1.555307e+04 = [1, 1, 1, 1]	FES = 731100	Diversity = 7.7734e-04	k_t
Generation 7320: Best = 1.555306e+04 = [3, 4, 3, 3]	FES = 732100	Diversity = 7.5597e-04	k_t
Generation 7330: Best = 1.555306e+04 = [1, 1, 1, 1]	FES = 733100	Diversity = 7.1298e-04	k_t
Generation 7340: Best = 1.555306e+04 = [2, 4, 4, 4]	FES = 734100	Diversity = 7.1333e-04	k_t
Generation 7350: Best = 1.555306e+04 = [1, 1, 1, 1]	FES = 735100	Diversity = 7.2975e-04	k_t
Generation 7360: Best = 1.555305e+04 = [2, 3, 2, 2]	FES = 736100	Diversity = 7.6527e-04	k_t
Generation 7370: Best = 1.555305e+04 = [1, 1, 1, 1]	FES = 737100	Diversity = 8.3459e-04	k_t
Generation 7380: Best = 1.555305e+04 = [3, 2, 4, 2]	FES = 738100	Diversity = 8.3956e-04	k_t
Generation 7390: Best = 1.555304e+04 = [1, 1, 1, 1]	FES = 739100	Diversity = 8.3692e-04	k_t
Generation 7400: Best = 1.555304e+04 = [4, 2, 2, 3]	FES = 740100	Diversity = 8.0850e-04	k_t
Generation 7410: Best = 1.555304e+04 = [1, 1, 1, 1]	FES = 741100	Diversity = 1.0351e-03	k_t
Generation 7420: Best = 1.555303e+04 = [3, 2, 4, 3]	FES = 742100	Diversity = 9.6222e-04	k_t
Generation 7430: Best = 1.555303e+04 = [1, 1, 1, 1]	FES = 743100	Diversity = 1.0630e-03	k_t
Generation 7440: Best = 1.555303e+04 = [4, 4, 2, 3]	FES = 744100	Diversity = 1.2098e-03	k_t
Generation 7450: Best = 1.555302e+04 = [1, 1, 1, 1]	FES = 745100	Diversity = 1.3662e-03	k_t
Generation 7460: Best = 1.555301e+04 = [2, 3, 2, 3]	FES = 746100	Diversity = 1.4987e-03	k_t
Generation 7470: Best = 1.555301e+04 = [1, 1, 1, 1]	FES = 747100	Diversity = 1.2966e-03	k_t
Generation 7480: Best = 1.555300e+04 = [3, 3, 2, 4]	FES = 748100	Diversity = 1.2568e-03	k_t
Generation 7490: Best = 1.555300e+04 = [1, 1, 1, 1]	FES = 749100	Diversity = 1.2339e-03	k_t
Generation 7500: Best = 1.555299e+04 = [3, 4, 3, 2]	FES = 750100	Diversity = 1.1118e-03	k_t
Generation 7510: Best = 1.555299e+04 = [1, 1, 1, 1]	FES = 751100	Diversity = 1.0939e-03	k_t

Generation 7520: Best = 1.555299e+04 = [3, 4, 3, 4]	FES = 752100	Diversity = 1.0549e-03	k_t
Generation 7530: Best = 1.555298e+04 = [1, 1, 1, 1]	FES = 753100	Diversity = 9.0854e-04	k_t
Generation 7540: Best = 1.555298e+04 = [4, 4, 2, 3]	FES = 754100	Diversity = 9.1235e-04	k_t
Generation 7550: Best = 1.555298e+04 = [1, 1, 1, 1]	FES = 755100	Diversity = 9.2734e-04	k_t
Generation 7560: Best = 1.555297e+04 = [2, 4, 2, 2]	FES = 756100	Diversity = 8.7782e-04	k_t
Generation 7570: Best = 1.555297e+04 = [1, 1, 1, 1]	FES = 757100	Diversity = 8.7454e-04	k_t
Generation 7580: Best = 1.555296e+04 = [3, 3, 2, 3]	FES = 758100	Diversity = 1.0015e-03	k_t
Generation 7590: Best = 1.555296e+04 = [1, 1, 1, 1]	FES = 759100	Diversity = 9.6243e-04	k_t
Generation 7600: Best = 1.555296e+04 = [4, 2, 2, 3]	FES = 760100	Diversity = 9.7306e-04	k_t
Generation 7610: Best = 1.555295e+04 = [1, 1, 1, 1]	FES = 761100	Diversity = 9.9686e-04	k_t
Generation 7620: Best = 1.555295e+04 = [2, 4, 3, 4]	FES = 762100	Diversity = 1.1207e-03	k_t
Generation 7630: Best = 1.555295e+04 = [1, 1, 1, 1]	FES = 763100	Diversity = 1.0435e-03	k_t
Generation 7640: Best = 1.555294e+04 = [3, 4, 4, 3]	FES = 764100	Diversity = 9.3274e-04	k_t
Generation 7650: Best = 1.555294e+04 = [1, 1, 1, 1]	FES = 765100	Diversity = 8.6087e-04	k_t
Generation 7660: Best = 1.555294e+04 = [4, 4, 2, 2]	FES = 766100	Diversity = 9.4367e-04	k_t
Generation 7670: Best = 1.555293e+04 = [1, 1, 1, 1]	FES = 767100	Diversity = 9.0059e-04	k_t
Generation 7680: Best = 1.555293e+04 = [2, 2, 3, 4]	FES = 768100	Diversity = 9.1217e-04	k_t
Generation 7690: Best = 1.555293e+04 = [1, 1, 1, 1]	FES = 769100	Diversity = 9.7124e-04	k_t
Generation 7700: Best = 1.555292e+04 = [4, 4, 3, 3]	FES = 770100	Diversity = 9.7601e-04	k_t
Generation 7710: Best = 1.555292e+04 = [1, 1, 1, 1]	FES = 771100	Diversity = 9.6123e-04	k_t
Generation 7720: Best = 1.555292e+04 = [4, 4, 4, 4]	FES = 772100	Diversity = 1.0157e-03	k_t
Generation 7730: Best = 1.555292e+04 = [1, 1, 1, 1]	FES = 773100	Diversity = 9.1398e-04	k_t
Generation 7740: Best = 1.555291e+04 = [2, 3, 2, 2]	FES = 774100	Diversity = 9.1206e-04	k_t
Generation 7750: Best = 1.555291e+04 = [1, 1, 1, 1]	FES = 775100	Diversity = 8.1650e-04	k_t
Generation 7760: Best = 1.555291e+04 = [2, 3, 2, 4]	FES = 776100	Diversity = 8.2174e-04	k_t
Generation 7770: Best = 1.555290e+04 = [1, 1, 1, 1]	FES = 777100	Diversity = 9.0670e-04	k_t
Generation 7780: Best = 1.555290e+04 = [3, 4, 2, 3]	FES = 778100	Diversity = 8.0724e-04	k_t

Generation 7790: Best = 1.555290e+04 = [1, 1, 1, 1]	FES = 779100	Diversity = 7.8949e-04	k_t
Generation 7800: Best = 1.555290e+04 = [3, 2, 4, 3]	FES = 780100	Diversity = 8.3253e-04	k_t
Generation 7810: Best = 1.555289e+04 = [1, 1, 1, 1]	FES = 781100	Diversity = 8.8018e-04	k_t
Generation 7820: Best = 1.555289e+04 = [3, 4, 3, 2]	FES = 782100	Diversity = 8.7476e-04	k_t
Generation 7830: Best = 1.555289e+04 = [1, 1, 1, 1]	FES = 783100	Diversity = 8.6754e-04	k_t
Generation 7840: Best = 1.555289e+04 = [4, 3, 4, 4]	FES = 784100	Diversity = 7.8624e-04	k_t
Generation 7850: Best = 1.555288e+04 = [1, 1, 1, 1]	FES = 785100	Diversity = 8.8177e-04	k_t
Generation 7860: Best = 1.555288e+04 = [4, 4, 2, 3]	FES = 786100	Diversity = 8.7020e-04	k_t
Generation 7870: Best = 1.555288e+04 = [1, 1, 1, 1]	FES = 787100	Diversity = 8.3893e-04	k_t
Generation 7880: Best = 1.555288e+04 = [3, 2, 3, 2]	FES = 788100	Diversity = 8.1921e-04	k_t
Generation 7890: Best = 1.555287e+04 = [1, 1, 1, 1]	FES = 789100	Diversity = 8.6913e-04	k_t
Generation 7900: Best = 1.555287e+04 = [2, 4, 2, 4]	FES = 790100	Diversity = 9.5220e-04	k_t
Generation 7910: Best = 1.555287e+04 = [1, 1, 1, 1]	FES = 791100	Diversity = 9.7203e-04	k_t
Generation 7920: Best = 1.555286e+04 = [3, 4, 3, 2]	FES = 792100	Diversity = 9.3375e-04	k_t
Generation 7930: Best = 1.555286e+04 = [1, 1, 1, 1]	FES = 793100	Diversity = 9.4506e-04	k_t
Generation 7940: Best = 1.555285e+04 = [4, 2, 4, 3]	FES = 794100	Diversity = 9.5372e-04	k_t
Generation 7950: Best = 1.555285e+04 = [1, 1, 1, 1]	FES = 795100	Diversity = 1.0945e-03	k_t
Generation 7960: Best = 1.555284e+04 = [4, 4, 4, 3]	FES = 796100	Diversity = 1.0753e-03	k_t
Generation 7970: Best = 1.555284e+04 = [1, 1, 1, 1]	FES = 797100	Diversity = 8.6204e-04	k_t
Generation 7980: Best = 1.555284e+04 = [3, 3, 2, 2]	FES = 798100	Diversity = 9.7993e-04	k_t
Generation 7990: Best = 1.555283e+04 = [1, 1, 1, 1]	FES = 799100	Diversity = 9.4978e-04	k_t
Generation 8000: Best = 1.555283e+04 = [2, 3, 3, 4]	FES = 800100	Diversity = 9.3544e-04	k_t
Generation 8010: Best = 1.555283e+04 = [1, 1, 1, 1]	FES = 801100	Diversity = 9.3209e-04	k_t
Generation 8020: Best = 1.555282e+04 = [2, 3, 4, 2]	FES = 802100	Diversity = 7.2065e-04	k_t
Generation 8030: Best = 1.555282e+04 = [1, 1, 1, 1]	FES = 803100	Diversity = 7.4016e-04	k_t
Generation 8040: Best = 1.555282e+04 = [2, 3, 2, 4]	FES = 804100	Diversity = 8.3774e-04	k_t
Generation 8050: Best = 1.555282e+04 = [1, 1, 1, 1]	FES = 805100	Diversity = 8.0956e-04	k_t

Generation 8060: Best = 1.555281e+04 = [4, 3, 3, 4]	FES = 806100	Diversity = 7.7455e-04	k_t
Generation 8070: Best = 1.555281e+04 = [1, 1, 1, 1]	FES = 807100	Diversity = 6.0866e-04	k_t
Generation 8080: Best = 1.555281e+04 = [3, 4, 2, 3]	FES = 808100	Diversity = 6.0222e-04	k_t
Generation 8090: Best = 1.555281e+04 = [1, 1, 1, 1]	FES = 809100	Diversity = 5.6956e-04	k_t
Generation 8100: Best = 1.555281e+04 = [3, 3, 4, 3]	FES = 810100	Diversity = 5.6525e-04	k_t
Generation 8110: Best = 1.555280e+04 = [1, 1, 1, 1]	FES = 811100	Diversity = 5.3024e-04	k_t
Generation 8120: Best = 1.555280e+04 = [4, 4, 4, 4]	FES = 812100	Diversity = 5.8335e-04	k_t
Generation 8130: Best = 1.555280e+04 = [1, 1, 1, 1]	FES = 813100	Diversity = 6.7408e-04	k_t
Generation 8140: Best = 1.555280e+04 = [3, 2, 4, 3]	FES = 814100	Diversity = 6.8045e-04	k_t
Generation 8150: Best = 1.555280e+04 = [1, 1, 1, 1]	FES = 815100	Diversity = 6.2196e-04	k_t
Generation 8160: Best = 1.555279e+04 = [2, 4, 4, 4]	FES = 816100	Diversity = 5.8549e-04	k_t
Generation 8170: Best = 1.555279e+04 = [1, 1, 1, 1]	FES = 817100	Diversity = 6.3231e-04	k_t
Generation 8180: Best = 1.555279e+04 = [3, 3, 3, 3]	FES = 818100	Diversity = 6.8184e-04	k_t
Generation 8190: Best = 1.555279e+04 = [1, 1, 1, 1]	FES = 819100	Diversity = 7.0660e-04	k_t
Generation 8200: Best = 1.555278e+04 = [4, 2, 2, 3]	FES = 820100	Diversity = 7.6843e-04	k_t
Generation 8210: Best = 1.555278e+04 = [1, 1, 1, 1]	FES = 821100	Diversity = 7.8444e-04	k_t
Generation 8220: Best = 1.555278e+04 = [4, 3, 2, 4]	FES = 822100	Diversity = 9.5404e-04	k_t
Generation 8230: Best = 1.555278e+04 = [1, 1, 1, 1]	FES = 823100	Diversity = 8.6045e-04	k_t
Generation 8240: Best = 1.555278e+04 = [2, 4, 3, 2]	FES = 824100	Diversity = 7.1055e-04	k_t
Generation 8250: Best = 1.555277e+04 = [1, 1, 1, 1]	FES = 825100	Diversity = 6.2997e-04	k_t
Generation 8260: Best = 1.555277e+04 = [2, 4, 4, 2]	FES = 826100	Diversity = 6.4986e-04	k_t
Generation 8270: Best = 1.555277e+04 = [1, 1, 1, 1]	FES = 827100	Diversity = 6.1285e-04	k_t
Generation 8280: Best = 1.555277e+04 = [3, 4, 2, 3]	FES = 828100	Diversity = 6.3639e-04	k_t
Generation 8290: Best = 1.555276e+04 = [1, 1, 1, 1]	FES = 829100	Diversity = 5.3449e-04	k_t
Generation 8300: Best = 1.555276e+04 = [4, 2, 4, 3]	FES = 830100	Diversity = 5.9418e-04	k_t
Generation 8310: Best = 1.555276e+04 = [1, 1, 1, 1]	FES = 831100	Diversity = 5.8502e-04	k_t
Generation 8320: Best = 1.555276e+04 = [4, 3, 4, 3]	FES = 832100	Diversity = 6.4165e-04	k_t

Generation 8330: Best = 1.555276e+04 = [1, 1, 1, 1]	FES = 833100	Diversity = 5.9965e-04	k_t
Generation 8340: Best = 1.555275e+04 = [3, 2, 4, 3]	FES = 834100	Diversity = 5.6553e-04	k_t
Generation 8350: Best = 1.555275e+04 = [1, 1, 1, 1]	FES = 835100	Diversity = 5.6453e-04	k_t
Generation 8360: Best = 1.555275e+04 = [2, 2, 2, 4]	FES = 836100	Diversity = 5.0181e-04	k_t
Generation 8370: Best = 1.555275e+04 = [1, 1, 1, 1]	FES = 837100	Diversity = 5.1473e-04	k_t
Generation 8380: Best = 1.555275e+04 = [3, 4, 2, 2]	FES = 838100	Diversity = 6.2468e-04	k_t
Generation 8390: Best = 1.555275e+04 = [1, 1, 1, 1]	FES = 839100	Diversity = 6.7259e-04	k_t
Generation 8400: Best = 1.555274e+04 = [4, 3, 4, 3]	FES = 840100	Diversity = 7.5633e-04	k_t
Generation 8410: Best = 1.555274e+04 = [1, 1, 1, 1]	FES = 841100	Diversity = 7.2275e-04	k_t
Generation 8420: Best = 1.555274e+04 = [3, 2, 4, 4]	FES = 842100	Diversity = 8.9851e-04	k_t
Generation 8430: Best = 1.555274e+04 = [1, 1, 1, 1]	FES = 843100	Diversity = 7.9033e-04	k_t
Generation 8440: Best = 1.555273e+04 = [3, 4, 3, 3]	FES = 844100	Diversity = 7.9416e-04	k_t
Generation 8450: Best = 1.555273e+04 = [1, 1, 1, 1]	FES = 845100	Diversity = 8.2222e-04	k_t
Generation 8460: Best = 1.555273e+04 = [2, 4, 2, 2]	FES = 846100	Diversity = 7.7100e-04	k_t
Generation 8470: Best = 1.555272e+04 = [1, 1, 1, 1]	FES = 847100	Diversity = 6.3786e-04	k_t
Generation 8480: Best = 1.555272e+04 = [2, 2, 3, 2]	FES = 848100	Diversity = 6.3469e-04	k_t
Generation 8490: Best = 1.555272e+04 = [1, 1, 1, 1]	FES = 849100	Diversity = 5.9398e-04	k_t
Generation 8500: Best = 1.555272e+04 = [4, 2, 3, 2]	FES = 850100	Diversity = 6.3402e-04	k_t
Generation 8510: Best = 1.555271e+04 = [1, 1, 1, 1]	FES = 851100	Diversity = 5.9422e-04	k_t
Generation 8520: Best = 1.555271e+04 = [3, 4, 3, 2]	FES = 852100	Diversity = 6.1292e-04	k_t
Generation 8530: Best = 1.555271e+04 = [1, 1, 1, 1]	FES = 853100	Diversity = 5.7927e-04	k_t
Generation 8540: Best = 1.555271e+04 = [4, 2, 4, 2]	FES = 854100	Diversity = 6.2697e-04	k_t
Generation 8550: Best = 1.555270e+04 = [1, 1, 1, 1]	FES = 855100	Diversity = 6.5789e-04	k_t
Generation 8560: Best = 1.555270e+04 = [3, 2, 3, 2]	FES = 856100	Diversity = 8.0071e-04	k_t
Generation 8570: Best = 1.555269e+04 = [1, 1, 1, 1]	FES = 857100	Diversity = 8.2159e-04	k_t
Generation 8580: Best = 1.555269e+04 = [3, 4, 2, 3]	FES = 858100	Diversity = 8.8944e-04	k_t
Generation 8590: Best = 1.555268e+04 = [1, 1, 1, 1]	FES = 859100	Diversity = 1.0860e-03	k_t

Generation 8600: Best = 1.555268e+04 = [4, 2, 3, 4]	FES = 860100	Diversity = 1.1683e-03	k_t
Generation 8610: Best = 1.555267e+04 = [1, 1, 1, 1]	FES = 861100	Diversity = 9.7453e-04	k_t
Generation 8620: Best = 1.555267e+04 = [2, 2, 2, 4]	FES = 862100	Diversity = 8.0972e-04	k_t
Generation 8630: Best = 1.555266e+04 = [1, 1, 1, 1]	FES = 863100	Diversity = 9.1832e-04	k_t
Generation 8640: Best = 1.555266e+04 = [4, 2, 4, 4]	FES = 864100	Diversity = 8.9357e-04	k_t
Generation 8650: Best = 1.555265e+04 = [1, 1, 1, 1]	FES = 865100	Diversity = 8.5880e-04	k_t
Generation 8660: Best = 1.555265e+04 = [4, 4, 3, 2]	FES = 866100	Diversity = 9.4967e-04	k_t
Generation 8670: Best = 1.555265e+04 = [1, 1, 1, 1]	FES = 867100	Diversity = 8.9447e-04	k_t
Generation 8680: Best = 1.555264e+04 = [4, 3, 4, 4]	FES = 868100	Diversity = 7.2642e-04	k_t
Generation 8690: Best = 1.555264e+04 = [1, 1, 1, 1]	FES = 869100	Diversity = 6.8532e-04	k_t
Generation 8700: Best = 1.555264e+04 = [2, 3, 4, 2]	FES = 870100	Diversity = 6.5857e-04	k_t
Generation 8710: Best = 1.555263e+04 = [1, 1, 1, 1]	FES = 871100	Diversity = 6.7158e-04	k_t
Generation 8720: Best = 1.555263e+04 = [3, 2, 2, 3]	FES = 872100	Diversity = 7.0244e-04	k_t
Generation 8730: Best = 1.555263e+04 = [1, 1, 1, 1]	FES = 873100	Diversity = 7.1820e-04	k_t
Generation 8740: Best = 1.555263e+04 = [4, 2, 3, 2]	FES = 874100	Diversity = 7.9807e-04	k_t
Generation 8750: Best = 1.555262e+04 = [1, 1, 1, 1]	FES = 875100	Diversity = 8.2733e-04	k_t
Generation 8760: Best = 1.555262e+04 = [3, 4, 2, 4]	FES = 876100	Diversity = 9.4868e-04	k_t
Generation 8770: Best = 1.555261e+04 = [1, 1, 1, 1]	FES = 877100	Diversity = 8.7380e-04	k_t
Generation 8780: Best = 1.555261e+04 = [2, 2, 2, 4]	FES = 878100	Diversity = 7.5865e-04	k_t
Generation 8790: Best = 1.555261e+04 = [1, 1, 1, 1]	FES = 879100	Diversity = 8.0234e-04	k_t
Generation 8800: Best = 1.555260e+04 = [3, 2, 2, 2]	FES = 880100	Diversity = 7.7063e-04	k_t
Generation 8810: Best = 1.555260e+04 = [1, 1, 1, 1]	FES = 881100	Diversity = 7.0700e-04	k_t
Generation 8820: Best = 1.555260e+04 = [3, 4, 4, 3]	FES = 882100	Diversity = 6.3004e-04	k_t
Generation 8830: Best = 1.555259e+04 = [1, 1, 1, 1]	FES = 883100	Diversity = 5.9174e-04	k_t
Generation 8840: Best = 1.555259e+04 = [4, 3, 4, 2]	FES = 884100	Diversity = 7.3349e-04	k_t
Generation 8850: Best = 1.555259e+04 = [1, 1, 1, 1]	FES = 885100	Diversity = 8.5890e-04	k_t
Generation 8860: Best = 1.555258e+04 = [4, 2, 2, 4]	FES = 886100	Diversity = 7.4639e-04	k_t

Generation 8870: Best = 1.555258e+04 = [1, 1, 1, 1]	FES = 887100	Diversity = 7.4495e-04	k_t
Generation 8880: Best = 1.555258e+04 = [4, 3, 3, 2]	FES = 888100	Diversity = 8.5870e-04	k_t
Generation 8890: Best = 1.555258e+04 = [1, 1, 1, 1]	FES = 889100	Diversity = 8.0204e-04	k_t
Generation 8900: Best = 1.555257e+04 = [4, 2, 2, 4]	FES = 890100	Diversity = 7.2669e-04	k_t
Generation 8910: Best = 1.555257e+04 = [1, 1, 1, 1]	FES = 891100	Diversity = 5.3983e-04	k_t
Generation 8920: Best = 1.555257e+04 = [4, 3, 2, 3]	FES = 892100	Diversity = 5.2142e-04	k_t
Generation 8930: Best = 1.555257e+04 = [1, 1, 1, 1]	FES = 893100	Diversity = 4.9958e-04	k_t
Generation 8940: Best = 1.555256e+04 = [3, 3, 2, 3]	FES = 894100	Diversity = 5.7182e-04	k_t
Generation 8950: Best = 1.555256e+04 = [1, 1, 1, 1]	FES = 895100	Diversity = 5.4367e-04	k_t
Generation 8960: Best = 1.555256e+04 = [4, 3, 2, 3]	FES = 896100	Diversity = 5.0353e-04	k_t
Generation 8970: Best = 1.555256e+04 = [1, 1, 1, 1]	FES = 897100	Diversity = 5.2051e-04	k_t
Generation 8980: Best = 1.555255e+04 = [3, 4, 3, 4]	FES = 898100	Diversity = 5.3303e-04	k_t
Generation 8990: Best = 1.555255e+04 = [1, 1, 1, 1]	FES = 899100	Diversity = 5.6600e-04	k_t
Generation 9000: Best = 1.555255e+04 = [3, 4, 4, 3]	FES = 900100	Diversity = 5.3123e-04	k_t
Generation 9010: Best = 1.555255e+04 = [1, 1, 1, 1]	FES = 901100	Diversity = 5.5363e-04	k_t
Generation 9020: Best = 1.555255e+04 = [3, 3, 3, 4]	FES = 902100	Diversity = 6.1532e-04	k_t
Generation 9030: Best = 1.555254e+04 = [1, 1, 1, 1]	FES = 903100	Diversity = 6.9352e-04	k_t
Generation 9040: Best = 1.555254e+04 = [3, 3, 4, 2]	FES = 904100	Diversity = 5.4963e-04	k_t
Generation 9050: Best = 1.555254e+04 = [1, 1, 1, 1]	FES = 905100	Diversity = 5.9490e-04	k_t
Generation 9060: Best = 1.555253e+04 = [4, 2, 2, 4]	FES = 906100	Diversity = 7.7357e-04	k_t
Generation 9070: Best = 1.555253e+04 = [1, 1, 1, 1]	FES = 907100	Diversity = 9.2172e-04	k_t
Generation 9080: Best = 1.555252e+04 = [2, 2, 3, 4]	FES = 908100	Diversity = 1.0242e-03	k_t
Generation 9090: Best = 1.555252e+04 = [1, 1, 1, 1]	FES = 909100	Diversity = 1.0547e-03	k_t
Generation 9100: Best = 1.555251e+04 = [3, 2, 2, 3]	FES = 910100	Diversity = 1.0284e-03	k_t
Generation 9110: Best = 1.555251e+04 = [1, 1, 1, 1]	FES = 911100	Diversity = 7.5626e-04	k_t
Generation 9120: Best = 1.555251e+04 = [2, 2, 2, 3]	FES = 912100	Diversity = 6.3297e-04	k_t
Generation 9130: Best = 1.555251e+04 = [1, 1, 1, 1]	FES = 913100	Diversity = 6.0863e-04	k_t

Generation 9140: Best = 1.555250e+04 = [2, 3, 2, 3]	FES = 914100	Diversity = 6.8575e-04	k_t
Generation 9150: Best = 1.555250e+04 = [1, 1, 1, 1]	FES = 915100	Diversity = 6.0239e-04	k_t
Generation 9160: Best = 1.555250e+04 = [2, 2, 2, 3]	FES = 916100	Diversity = 5.1298e-04	k_t
Generation 9170: Best = 1.555250e+04 = [1, 1, 1, 1]	FES = 917100	Diversity = 4.9023e-04	k_t
Generation 9180: Best = 1.555250e+04 = [2, 2, 2, 4]	FES = 918100	Diversity = 5.0453e-04	k_t
Generation 9190: Best = 1.555250e+04 = [1, 1, 1, 1]	FES = 919100	Diversity = 5.1820e-04	k_t
Generation 9200: Best = 1.555249e+04 = [2, 2, 4, 3]	FES = 920100	Diversity = 6.0841e-04	k_t
Generation 9210: Best = 1.555249e+04 = [1, 1, 1, 1]	FES = 921100	Diversity = 4.9226e-04	k_t
Generation 9220: Best = 1.555249e+04 = [3, 2, 4, 3]	FES = 922100	Diversity = 5.0708e-04	k_t
Generation 9230: Best = 1.555249e+04 = [1, 1, 1, 1]	FES = 923100	Diversity = 5.4278e-04	k_t
Generation 9240: Best = 1.555249e+04 = [4, 4, 4, 4]	FES = 924100	Diversity = 5.2340e-04	k_t
Generation 9250: Best = 1.555248e+04 = [1, 1, 1, 1]	FES = 925100	Diversity = 6.5008e-04	k_t
Generation 9260: Best = 1.555248e+04 = [2, 3, 4, 2]	FES = 926100	Diversity = 6.0613e-04	k_t
Generation 9270: Best = 1.555248e+04 = [1, 1, 1, 1]	FES = 927100	Diversity = 4.9829e-04	k_t
Generation 9280: Best = 1.555248e+04 = [3, 2, 4, 2]	FES = 928100	Diversity = 5.0891e-04	k_t
Generation 9290: Best = 1.555247e+04 = [1, 1, 1, 1]	FES = 929100	Diversity = 5.6373e-04	k_t
Generation 9300: Best = 1.555247e+04 = [4, 4, 3, 3]	FES = 930100	Diversity = 7.2487e-04	k_t
Generation 9310: Best = 1.555247e+04 = [1, 1, 1, 1]	FES = 931100	Diversity = 7.5357e-04	k_t
Generation 9320: Best = 1.555247e+04 = [3, 2, 4, 4]	FES = 932100	Diversity = 7.3265e-04	k_t
Generation 9330: Best = 1.555246e+04 = [1, 1, 1, 1]	FES = 933100	Diversity = 7.2882e-04	k_t
Generation 9340: Best = 1.555246e+04 = [2, 2, 4, 4]	FES = 934100	Diversity = 6.6992e-04	k_t
Generation 9350: Best = 1.555246e+04 = [1, 1, 1, 1]	FES = 935100	Diversity = 6.4564e-04	k_t
Generation 9360: Best = 1.555246e+04 = [3, 3, 3, 3]	FES = 936100	Diversity = 7.0836e-04	k_t
Generation 9370: Best = 1.555245e+04 = [1, 1, 1, 1]	FES = 937100	Diversity = 7.2007e-04	k_t
Generation 9380: Best = 1.555245e+04 = [3, 4, 4, 4]	FES = 938100	Diversity = 7.6500e-04	k_t
Generation 9390: Best = 1.555245e+04 = [1, 1, 1, 1]	FES = 939100	Diversity = 7.9046e-04	k_t
Generation 9400: Best = 1.555244e+04 = [3, 2, 3, 4]	FES = 940100	Diversity = 9.6133e-04	k_t

Generation 9410: Best = 1.555244e+04 = [1, 1, 1, 1]	FES = 941100	Diversity = 8.5050e-04	k_t
Generation 9420: Best = 1.555244e+04 = [4, 2, 4, 3]	FES = 942100	Diversity = 7.9891e-04	k_t
Generation 9430: Best = 1.555243e+04 = [1, 1, 1, 1]	FES = 943100	Diversity = 7.5070e-04	k_t
Generation 9440: Best = 1.555243e+04 = [3, 4, 4, 3]	FES = 944100	Diversity = 6.5049e-04	k_t
Generation 9450: Best = 1.555243e+04 = [1, 1, 1, 1]	FES = 945100	Diversity = 5.8726e-04	k_t
Generation 9460: Best = 1.555243e+04 = [4, 3, 2, 4]	FES = 946100	Diversity = 4.4855e-04	k_t
Generation 9470: Best = 1.555243e+04 = [1, 1, 1, 1]	FES = 947100	Diversity = 4.2209e-04	k_t
Generation 9480: Best = 1.555243e+04 = [2, 4, 4, 3]	FES = 948100	Diversity = 4.6722e-04	k_t
Generation 9490: Best = 1.555243e+04 = [1, 1, 1, 1]	FES = 949100	Diversity = 4.4736e-04	k_t
Generation 9500: Best = 1.555243e+04 = [2, 4, 2, 2]	FES = 950100	Diversity = 4.6792e-04	k_t
Generation 9510: Best = 1.555243e+04 = [1, 1, 1, 1]	FES = 951100	Diversity = 4.7793e-04	k_t
Generation 9520: Best = 1.555242e+04 = [3, 2, 2, 3]	FES = 952100	Diversity = 4.6280e-04	k_t
Generation 9530: Best = 1.555242e+04 = [1, 1, 1, 1]	FES = 953100	Diversity = 5.2258e-04	k_t
Generation 9540: Best = 1.555242e+04 = [2, 2, 3, 2]	FES = 954100	Diversity = 5.1275e-04	k_t
Generation 9550: Best = 1.555242e+04 = [1, 1, 1, 1]	FES = 955100	Diversity = 3.7617e-04	k_t
Generation 9560: Best = 1.555242e+04 = [2, 3, 4, 2]	FES = 956100	Diversity = 4.5681e-04	k_t
Generation 9570: Best = 1.555242e+04 = [1, 1, 1, 1]	FES = 957100	Diversity = 6.4186e-04	k_t
Generation 9580: Best = 1.555241e+04 = [4, 3, 3, 3]	FES = 958100	Diversity = 6.3558e-04	k_t
Generation 9590: Best = 1.555241e+04 = [1, 1, 1, 1]	FES = 959100	Diversity = 5.0305e-04	k_t
Generation 9600: Best = 1.555241e+04 = [4, 2, 4, 2]	FES = 960100	Diversity = 5.3569e-04	k_t
Generation 9610: Best = 1.555241e+04 = [1, 1, 1, 1]	FES = 961100	Diversity = 5.3603e-04	k_t
Generation 9620: Best = 1.555241e+04 = [2, 3, 3, 2]	FES = 962100	Diversity = 5.6560e-04	k_t
Generation 9630: Best = 1.555240e+04 = [1, 1, 1, 1]	FES = 963100	Diversity = 5.8027e-04	k_t
Generation 9640: Best = 1.555240e+04 = [2, 3, 4, 4]	FES = 964100	Diversity = 5.6913e-04	k_t
Generation 9650: Best = 1.555240e+04 = [1, 1, 1, 1]	FES = 965100	Diversity = 5.0165e-04	k_t
Generation 9660: Best = 1.555240e+04 = [3, 4, 2, 2]	FES = 966100	Diversity = 4.9348e-04	k_t
Generation 9670: Best = 1.555240e+04 = [1, 1, 1, 1]	FES = 967100	Diversity = 4.5935e-04	k_t

Generation 9680: Best = 1.555239e+04 = [3, 2, 4, 3]	FES = 968100	Diversity = 4.8001e-04	k_t
Generation 9690: Best = 1.555239e+04 = [1, 1, 1, 1]	FES = 969100	Diversity = 6.3044e-04	k_t
Generation 9700: Best = 1.555239e+04 = [2, 3, 2, 4]	FES = 970100	Diversity = 6.5489e-04	k_t
Generation 9710: Best = 1.555239e+04 = [1, 1, 1, 1]	FES = 971100	Diversity = 5.6703e-04	k_t
Generation 9720: Best = 1.555238e+04 = [4, 3, 2, 4]	FES = 972100	Diversity = 5.5183e-04	k_t
Generation 9730: Best = 1.555238e+04 = [1, 1, 1, 1]	FES = 973100	Diversity = 4.7918e-04	k_t
Generation 9740: Best = 1.555238e+04 = [4, 3, 4, 4]	FES = 974100	Diversity = 5.7473e-04	k_t
Generation 9750: Best = 1.555237e+04 = [1, 1, 1, 1]	FES = 975100	Diversity = 7.2213e-04	k_t
Generation 9760: Best = 1.555237e+04 = [4, 4, 3, 4]	FES = 976100	Diversity = 6.0700e-04	k_t
Generation 9770: Best = 1.555237e+04 = [1, 1, 1, 1]	FES = 977100	Diversity = 5.0347e-04	k_t
Generation 9780: Best = 1.555237e+04 = [2, 2, 4, 2]	FES = 978100	Diversity = 5.1952e-04	k_t
Generation 9790: Best = 1.555237e+04 = [1, 1, 1, 1]	FES = 979100	Diversity = 5.0126e-04	k_t
Generation 9800: Best = 1.555236e+04 = [3, 4, 3, 3]	FES = 980100	Diversity = 4.9098e-04	k_t
Generation 9810: Best = 1.555236e+04 = [1, 1, 1, 1]	FES = 981100	Diversity = 7.6226e-04	k_t
Generation 9820: Best = 1.555236e+04 = [3, 3, 4, 3]	FES = 982100	Diversity = 6.5126e-04	k_t
Generation 9830: Best = 1.555235e+04 = [1, 1, 1, 1]	FES = 983100	Diversity = 5.6119e-04	k_t
Generation 9840: Best = 1.555235e+04 = [2, 2, 2, 4]	FES = 984100	Diversity = 6.6522e-04	k_t
Generation 9850: Best = 1.555235e+04 = [1, 1, 1, 1]	FES = 985100	Diversity = 6.3573e-04	k_t
Generation 9860: Best = 1.555235e+04 = [2, 3, 4, 3]	FES = 986100	Diversity = 5.7213e-04	k_t
Generation 9870: Best = 1.555234e+04 = [1, 1, 1, 1]	FES = 987100	Diversity = 5.5331e-04	k_t
Generation 9880: Best = 1.555234e+04 = [4, 2, 3, 2]	FES = 988100	Diversity = 5.9915e-04	k_t
Generation 9890: Best = 1.555234e+04 = [1, 1, 1, 1]	FES = 989100	Diversity = 4.8217e-04	k_t
Generation 9900: Best = 1.555234e+04 = [3, 2, 4, 4]	FES = 990100	Diversity = 4.9682e-04	k_t
Generation 9910: Best = 1.555234e+04 = [1, 1, 1, 1]	FES = 991100	Diversity = 4.2841e-04	k_t
Generation 9920: Best = 1.555233e+04 = [2, 3, 2, 3]	FES = 992100	Diversity = 3.3006e-04	k_t
Generation 9930: Best = 1.555233e+04 = [1, 1, 1, 1]	FES = 993100	Diversity = 2.9140e-04	k_t
Generation 9940: Best = 1.555233e+04 = [3, 3, 3, 3]	FES = 994100	Diversity = 3.1591e-04	k_t

Generation 9950: Best = 1.555233e+04 FES = 995100 Diversity = 2.7041e-04 k_t
= [1, 1, 1, 1]
Generation 9960: Best = 1.555233e+04 FES = 996100 Diversity = 2.2710e-04 k_t
= [4, 3, 4, 4]
Generation 9970: Best = 1.555233e+04 FES = 997100 Diversity = 2.5391e-04 k_t
= [1, 1, 1, 1]
Generation 9980: Best = 1.555233e+04 FES = 998100 Diversity = 2.8206e-04 k_t
= [2, 3, 4, 4]
Generation 9990: Best = 1.555233e+04 FES = 999100 Diversity = 2.9143e-04 k_t
= [1, 1, 1, 1]

Optimization completed!

Final Best Fitness = 1.555233e+04

Total Function Evaluations = 1000000

=====
=

Results:

Best Fitness : 1.555233e+04

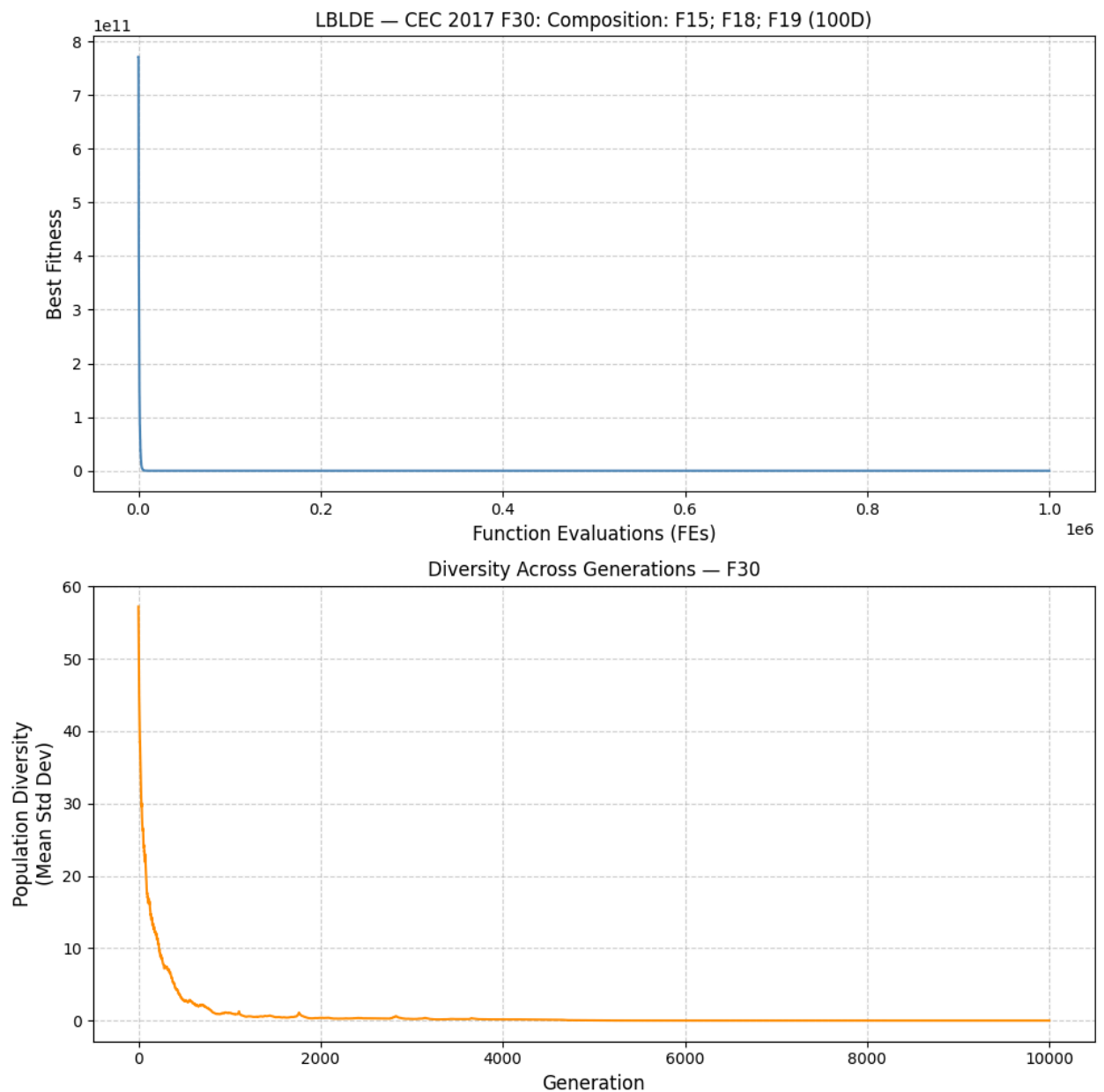
Global Optimum : 3.000000e+03

Error (|best-optimum|) : 1.255233e+04

Best Solution (5 dim) : [-99.99998968 99.99999801 49.52926243 12.41128138
-99.99998663]

D^(0) (init. diversity): 5.7220e+01

=====
=



Graph saved as convergence_F30.png

Section 7. Compiled Results

Print out a summary of all results in a tabular format

```
In [55]: print('=' * 80)
print('SUMMARY — LBLDE on CEC 2017 (100D), Single Run, Seed=42')
print('=' * 80)
print(f'{"Function":<10} {"Best Fitness":>18} {"Global Optimum":>18} {"Error |')
print('-' * 70)
for fn in [f'F{i}' for i in range(1, 31)]:
```

```

if fn not in all_results:
    continue
r = all_results[fn]
if r['best_fit'] is None:
    print(f'{fn:<10} {"N/A (deprecated)":>18} {r["f_global"]:>18.2f} {"N/A"}')
else:
    print(f'{fn:<10} {r["best_fit"]:>18.6e} {r["f_global"]:>18.2f} {r["error"]:>18.2f}')
print('=' * 80)

```

```

=====
=
SUMMARY – LBLDE on CEC 2017 (100D), Single Run, Seed=42
=====
=

```

Function	Best Fitness	Global Optimum	Error best-opt
F1	3.252209e+03	100.00	3.152209e+03
F2	6.557252e+28	200.00	6.557252e+28
F3	1.840295e+03	300.00	1.540295e+03
F4	5.680283e+02	400.00	1.680283e+02
F5	7.805774e+02	500.00	2.805774e+02
F6	6.158432e+02	600.00	1.584323e+01
F7	1.147479e+03	700.00	4.474785e+02
F8	1.054807e+03	800.00	2.548073e+02
F9	1.381655e+03	900.00	4.816550e+02
F10	1.381721e+04	1000.00	1.281721e+04
F11	1.312781e+03	1100.00	2.127808e+02
F12	1.165080e+06	1200.00	1.163880e+06
F13	4.163566e+03	1300.00	2.863566e+03
F14	1.923153e+04	1400.00	1.783153e+04
F15	2.664671e+03	1500.00	1.164671e+03
F16	5.202516e+03	1600.00	3.602516e+03
F17	3.857769e+03	1700.00	2.157769e+03
F18	7.212491e+04	1800.00	7.032491e+04
F19	4.642459e+03	1900.00	2.742459e+03
F20	3.597588e+03	2000.00	1.597588e+03
F21	2.508546e+03	2100.00	4.085460e+02
F22	1.365609e+04	2200.00	1.145609e+04
F23	3.005528e+03	2300.00	7.055284e+02
F24	3.463246e+03	2400.00	1.063246e+03
F25	3.269884e+03	2500.00	7.698841e+02
F26	7.912624e+03	2600.00	5.312624e+03
F27	3.400673e+03	2700.00	7.006733e+02
F28	3.345075e+03	2800.00	5.450748e+02
F29	4.683217e+03	2900.00	1.783217e+03
F30	1.555233e+04	3000.00	1.255233e+04

```

=====
=

```

In [56]: !zip plots.zip convergence_F{1..30}.png

adding: convergence_F1.png (deflated 23%)
adding: convergence_F2.png (deflated 22%)
adding: convergence_F3.png (deflated 20%)
adding: convergence_F4.png (deflated 22%)
adding: convergence_F5.png (deflated 20%)
adding: convergence_F6.png (deflated 20%)
adding: convergence_F7.png (deflated 21%)
adding: convergence_F8.png (deflated 20%)
adding: convergence_F9.png (deflated 21%)
adding: convergence_F10.png (deflated 20%)
adding: convergence_F11.png (deflated 20%)
adding: convergence_F12.png (deflated 19%)
adding: convergence_F13.png (deflated 22%)
adding: convergence_F14.png (deflated 14%)
adding: convergence_F15.png (deflated 21%)
adding: convergence_F16.png (deflated 20%)
adding: convergence_F17.png (deflated 21%)
adding: convergence_F18.png (deflated 16%)
adding: convergence_F19.png (deflated 21%)
adding: convergence_F20.png (deflated 19%)
adding: convergence_F21.png (deflated 21%)
adding: convergence_F22.png (deflated 20%)
adding: convergence_F23.png (deflated 20%)
adding: convergence_F24.png (deflated 21%)
adding: convergence_F25.png (deflated 22%)
adding: convergence_F26.png (deflated 21%)
adding: convergence_F27.png (deflated 21%)
adding: convergence_F28.png (deflated 20%)
adding: convergence_F29.png (deflated 22%)
adding: convergence_F30.png (deflated 22%)